

Tensor Network Theory:

A formalization blueprint

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Chapter 1

Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices $A = \{A^i\}_{i=0}^{d-1}$, which assigns to each chain length N the vector $|V^{(N)}(A)\rangle$ with coefficients indexed by $\sigma = (i_1, \dots, i_N)$,

$$V^{(N)}(A)_\sigma = \text{tr}(A^{i_1} \dots A^{i_N}). \quad (1.1)$$

The main argument, for translation-invariant tensors, follows the canonical-form and basis-of-normal-tensors argument of [11, 16], with the single-block injective case anchored in the earlier representation theorem of [193].

Notation

$\mathbb{C}, \mathbb{N}, \mathbb{Z}$	complex numbers, natural numbers, integers
$M_D(\mathbb{C}), \text{GL}_D(\mathbb{C})$	complex matrices and invertible complex matrices
$\text{tr}(X), \text{span}_{\mathbb{C}}(S)$	trace and complex linear span
$A = \{A^i\}_{i=0}^{d-1}$	a translation-invariant MPS tensor
$w = (i_1, \dots, i_L), A^w$	a word and the product $A^{i_1} \dots A^{i_L}$
$ V^{(N)}(A)\rangle$	the length- N matrix product vector
$V^{(N)}(A)_\sigma$	the coefficient of $ V^{(N)}(A)\rangle$ at σ
$O_{AB}(N)$	the bilinear overlap $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$
$\mathcal{E}_A(X)$	the transfer map $\sum_i A^i X (A^i)^\dagger$
$A^{[L]}$	the length- L blocked tensor

In canonical form one writes

$$A^i = \bigoplus_{j=1}^g \mu_j A_j^i, \quad (1.2)$$

$$B^i = \bigoplus_{k=1}^h \nu_k B_k^i. \quad (1.3)$$

The blocks in (1.2) and (1.3) are normal tensors. If $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every positive chain length N , then the bases of normal tensors have the same size. After relabelling the blocks of B , each matched pair satisfies

$$B_j^i = \zeta_j X_j A_j^i X_j^{-1}, \quad (1.4)$$

for some invertible matrix X_j and nonzero scalar ζ_j , and the equal case compares the coefficient power sums attached to the relabelled bases. The block gauges in (1.4) and the weight comparison then combine into a single gauge conjugacy of the two weighted block-diagonal tensors in (1.2) and (1.3). For a single injective block this reduces to the conclusion $B^i = XA^iX^{-1}$ for all physical indices i , the single-block specialization of (1.4).

Chapters 2–9 carry out this argument. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and site-periodic tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4 and 5 develop the Perron–Frobenius and overlap-decay tools used to control normal blocks. The general theory of Schwarz inequalities and multiplicative domains is developed in the companion quantum-channel volume [182, Chapter “Schwarz Inequalities and Multiplicative Domains”]. Chapters 6 and 7 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapter 7 includes the invariant-projection splitting. Chapter 8 supplies bases of normal tensors and the power-sum comparison.

Chapter 10 uses the Fundamental Theorem to obtain virtual symmetry gauges, projective bond representations, and string-order criteria; its Kraus-mixing step uses the rectangular Kraus-freedom theorem from the companion quantum-channel volume [182, “Rectangular Kraus freedom”].

Beyond Chapter 10, subsequent chapters develop parent Hamiltonians (Chapter 11), correlation decay (Chapter 14), concrete examples (Chapter 15), matrix product density operators (Chapter 16), renormalization fixed points (Chapters 17 and 20), matrix product unitaries (Chapter 18), periodic tensors (Chapter 22), the algebraic Fundamental Theorem (Chapter 23), two-dimensional tensor networks (PEPS, Chapter 24), and the asymmetric Fundamental Theorem (Chapter 25). General quantum-channel representations, positive-map examples, dynamical semigroups, operator convexity, and entropy are developed in the companion quantum-channel volume [182, Chapters “Quantum Channels and Positive Maps,” “Channel Representations and Normal Forms,” “Positive but Not Completely Positive Maps,” “Operator Convexity and Jensen Inequalities,” “Quantum Dynamical Semigroups,” and “Quantum Entropy”].

Chapter 2

Matrix Product Vectors

Fix a physical dimension d and a bond dimension D . All tensors here are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1.1 (MPS tensor). [Lean](#) [✓ Stmt](#) A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices $\{A^i\}_{i=0}^{d-1}$, where $A^i \in M_D(\mathbb{C})$, indexed by a physical index $i \in \{0, \dots, d-1\}$. Such a tensor defines an MPV family. Diagrammatically,

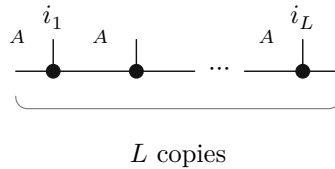


The black node denotes the tensor A , the horizontal legs are virtual, and the upper leg is the physical index i .

Definition 2.1.2 (Word evaluation). [Lean](#) [✓ Stmt](#) Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}). \quad (2.1)$$

The empty word evaluates to the identity, $A^\emptyset = \mathbb{1}_D$. In tensor-network notation,



in which each black node denotes the same local tensor A , the virtual legs remain open, and the physical legs are labelled by the word $(i_1, \dots, i_L)$.

Lemma 2.1.3 (Word evaluation along a listed configuration). [Lean](#) [✓ Stmt](#) For a configuration $\sigma = (\sigma_0, \dots, \sigma_{N-1})$, let $w_\sigma = [\sigma_0, \dots, \sigma_{N-1}]$. Then $A^{w_\sigma} = A^{\sigma_0} A^{\sigma_1} \dots A^{\sigma_{N-1}}$.

Proof. ✓ Proof This follows by induction on N from the recursive definition of word evaluation. $\square$

Lemma 2.1.4 (Multiplicativity of word evaluation). Lean ✓ Stmt For any words w_1, w_2 , $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$.

Proof. ✓ Proof The claim follows from the definition of A^w . $\square$

Lemma 2.1.5 (Centre of $M_D(\mathbb{C})$). Lean ✓ Stmt If a matrix $Z \in M_D(\mathbb{C})$ commutes with every element of $M_D(\mathbb{C})$, then there exists $\lambda \in \mathbb{C}$ such that $Z = \lambda \mathbb{1}_D$.

Proof. ✓ Proof Since Z commutes with every matrix unit, all off-diagonal entries of Z vanish and all diagonal entries are equal. $\square$

Lemma 2.1.6 (Rectangular amplified commutant). Lean ✓ Stmt Let $(A_i)_{i \in I}$ span $M_D(\mathbb{C})$, and let $C : \mathbb{C}^{m_2} \otimes \mathbb{C}^D \rightarrow \mathbb{C}^{m_1} \otimes \mathbb{C}^D$ satisfy, for every $i \in I$,

$$(\mathbb{1}_{m_1} \otimes A_i)C = C(\mathbb{1}_{m_2} \otimes A_i).$$

Then there is a matrix $F : \mathbb{C}^{m_2} \rightarrow \mathbb{C}^{m_1}$ such that $C = F \otimes \mathbb{1}_D$.

Proof. ✓ Proof Each (a, b) multiplicity block of C commutes with every A_i . Since the A_i span the full matrix algebra, Lemma 2.1.5 makes this block a scalar multiple of the identity. These scalars are the entries of F . $\square$

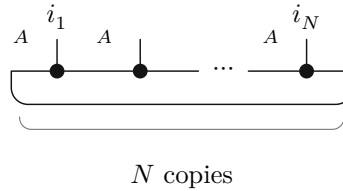
Definition 2.1.7 (Matrix product vector). Lean ✓ Stmt The *matrix product vector* (MPV) of a tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}. \quad (2.2)$$

Equivalently, for a configuration $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$, we write

$$V^{(N)}(A)_\sigma := \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}). \quad (2.3)$$

The coefficient function $\sigma \mapsto V^{(N)}(A)_\sigma$ gives the components of the vector in (2.2); the displayed ket is the corresponding vector in $(\mathbb{C}^d)^{\otimes N}$. For a general word w , we also write $c_w(A) := \text{tr}(A^w)$. The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$. The coefficient $V^{(N)}(A)_\sigma$ is the periodic contraction



of N copies of the local tensor A , with the outer virtual legs closed by the trace.

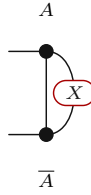
2.2 Transfer maps and quantum-channel interfaces

Transfer maps connect matrix product states to completely positive maps and quantum channels. The underlying channel theory is developed in the companion volume [182, “Positive map” and “Quantum channel”].

Definition 2.2.1 (Transfer map). Lean ✓ Stmt The *transfer map* associated to a tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger. \quad (2.4)$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes A , the lower node denotes $A^\dagger$, and the physical index is summed over between them.

Remark 2.2.2. ✓ Stmt The transfer map is automatically positive: if $X \geq 0$ then $\mathcal{E}_A(X) \geq 0$, because each summand $A^i X (A^i)^\dagger$ is positive semidefinite. The abstract framework is developed in the companion quantum-channel volume [182, “Positive map”].

Theorem 2.2.3 (Transfer map is CP). Lean ✓ Stmt The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive.

Proof. ✓ Proof It is already written in the Kraus form (established in the companion quantum-channel volume [182, “Completely positive map”]), with Kraus operators $\{A^i\}$. $\square$

Theorem 2.2.4 (Transfer matrix of an MPS transfer map). Lean ✓ Stmt For an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$, the matrix representing $\mathcal{E}_A$ under vectorization is

$$K_A = \sum_{i=0}^{d-1} \overline{A^i} \otimes A^i.$$

Proof. ✓ Proof Apply the standard vectorization identity $\text{vec}(AXB) = (B^\top \otimes A) \text{vec}(X)$ to each summand $A^i X (A^i)^\dagger$ and sum over the physical index. $\square$

Lemma 2.2.5 (Trace adjointness of a transfer map). Lean ✓ Stmt For an MPS tensor A and bond matrices $\sigma, \rho \in M_D(\mathbb{C})$,

$$\text{tr}(\sigma \mathcal{E}_A(\rho)) = \text{tr}(\mathcal{E}_{A^\dagger}(\sigma) \rho),$$

where $\mathcal{E}_{A^\dagger}$ is the transfer map with Kraus operators $(A^i)^\dagger$.

Proof. ✓ Proof Apply trace adjointness for Kraus maps [182, “Trace adjointness of a Kraus map”] to the Kraus family $\{A^i\}$. $\square$

Lemma 2.2.6 (Transfer map under scalar multiplication). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor and let $w \in \mathbb{C}$. Then, for every bond matrix X ,

$$\mathcal{E}_{wA}(X) = w\bar{w} \mathcal{E}_A(X).$$

Proof. [✓ Proof](#) Expand the transfer map and pull the scalar w and its conjugate out of each summand. $\square$

Lemma 2.2.7 (Left-canonical transfer maps preserve trace). [Lean](#) [✓ Stmt](#) If A is left canonical, then, for every bond matrix X ,

$$\text{tr}(\mathcal{E}_A(X)) = \text{tr}(X).$$

Proof. [✓ Proof](#) Cyclicity of trace gives

$$\text{tr}(\mathcal{E}_A(X)) = \text{tr}\left(\left(\sum_i (A^i)^\dagger A^i\right) X\right) = \text{tr}(X),$$

since left canonicity makes the sum in parentheses equal to the identity. $\square$

Theorem 2.2.8 (Transfer map is a channel). [Lean](#) [✓ Stmt](#) If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).

Proof. [✓ Proof](#) Complete positivity is Theorem 2.2.3. Trace preservation follows by cyclicity and the normalization hypothesis:

$$\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}\left(\left(\sum_i (A^i)^\dagger A^i\right) X\right) = \text{tr}(X).$$

$\square$

Theorem 2.2.9 (Positive-definite transport by a coisometry). [Lean](#) [✓ Stmt](#) Let $A > 0$ be a square complex matrix and let V be a possibly rectangular matrix satisfying $VV^\dagger = \mathbb{1}$. Then

$$VAV^\dagger > 0. \tag{2.5}$$

Proof. [✓ Proof](#) For a row vector x , one has

$$x(VAV^\dagger)x^\dagger = (xV)A(xV)^\dagger. \tag{2.6}$$

Moreover, $xV = 0$ implies

$$x = x(VV^\dagger) = 0. \tag{2.7}$$

Thus every nonzero x gives $xV \neq 0$, and positive definiteness of A makes the displayed quadratic form strictly positive. $\square$

2.3 Site-periodic tensor families

Definition 2.3.1 (Site-periodic tensor family). [Lean](#) [✓ Stmt](#) For a fixed period m , a *site-periodic tensor family* is a family of local tensors indexed by $\{0, \dots, m-1\}$, with local physical dimension d and bond dimension D .

Definition 2.3.2 (Periodic same-state and gauge-equivalence relations). [Lean](#) [Lean](#) [✓ Stmt](#) For site-periodic tensors A, B at fixed period m :

- equality of the induced site-periodic states;
- cyclic gauge equivalence.

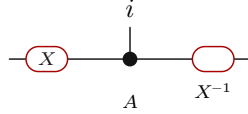
Lemma 2.3.3 (Equivalence structures for periodic relations). [Lean](#) [Lean](#) [✓ Stmt](#) Both periodic relations are equivalence relations (reflexive, symmetric, transitive).

2.4 Gauge equivalence and same MPV

Definition 2.4.1 (Gauge equivalence). [Lean](#) [✓ Stmt](#) Two tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that, for every $i \in \{0, \dots, d-1\}$,

$$B^i = X A^i X^{-1}. \quad (2.8)$$

In tensor-network notation, (2.8) is represented by



where the black node denotes the tensor A , the upper leg is the physical index i , and the two red side nodes denote the gauge matrices acting on the virtual legs.

Definition 2.4.2 (Same MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B (of the same bond dimension) generate the same MPV family if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size N .

Definition 2.4.3 (Same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors A and B of possibly different bond dimensions D_1 and D_2 , we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every word length N , including the empty word. This algebraic relation is used for exact direct-sum identities.

Definition 2.4.4 (Positive-length same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors A and B of possibly different bond dimensions, we write $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N > 0$.

Definition 2.4.5 (Rectangular MPS reduction). [Lean](#) [✓ Stmt](#) Let A and B have the same physical alphabet and respective bond dimensions D_A and D_B . A reduction from B to A is a pair $V \in \mathbb{C}^{D_A \times D_B}$ and $W \in \mathbb{C}^{D_B \times D_A}$ such that

$$VW = 1_{D_A}, \quad (2.9)$$

$$VB^{\mathbf{a}}W = A^{\mathbf{a}}. \quad (2.10)$$

for every word $\mathbf{a}$, including the empty word. This is the reduction of [187, Proposition 20 and the following definition].

Theorem 2.4.6 (Basic rectangular reduction identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *A rectangular reduction satisfies the retraction identity (2.9) and the word identity (2.10). Conversely, the all-word identity alone implies $VW = 1_{D_A}$ by taking the empty word. Moreover, the target bond dimension cannot exceed the source bond dimension: $D_A \leq D_B$.*

Proof. [✓ Proof](#) The forward statements are the two defining clauses. For the converse, $A^\emptyset = 1_{D_A}$ and $B^\emptyset = 1_{D_B}$, so the empty-word instance reads $V1_{D_B}W = 1_{D_A}$. Since V has a right inverse, it is surjective, and finite-dimensional linear algebra gives $D_A \leq D_B$. $\square$

Remark 2.4.7. [✓ Stmt](#) Definition 2.4.2 specializes Definition 2.4.3 to equal bond dimensions $D_1 = D_2 = D$.

Definition 2.4.8 (Nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B (of possibly different bond dimensions) generate nonzero proportional MPV families if for every $N > 0$ there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. This is the projective reading of the proportionality hypothesis in [11, Theorem 2.10].

Definition 2.4.9 (Eventually nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors A and B are eventually nonzero proportional if, for all sufficiently large N , there is a nonzero scalar $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle$. This auxiliary form is used with the eventual linear-independence statement Lemma E.1.7.

Lemma 2.4.10 (Elementary nonzero proportionality rules). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Nonzero proportionality implies eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar 1, at every positive length.*

Proof. [✓ Proof](#) The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar c_N at each length, and equality is the case $c_N = 1$. $\square$

Lemma 2.4.11 (Coefficient isolation after complementary matching). [Lean](#) [✓ Stmt](#) *Let R be a ring, let V be an R -module, and let I and K be finite index sets. Let $T \subseteq I$, and let $\{u_i\}_{i \in I}$ and $\{v_k\}_{k \in K}$ be vectors in V . Suppose that, for every $i \notin T$, there are a scalar α_i and an index $\phi(i) \in K$ such that $u_i = \alpha_i v_{\phi(i)}$. If*

$$\sum_{i \in I} a_i u_i = \sum_{k \in K} b_k v_k$$

and the coproduct-indexed family $w: T \amalg K \rightarrow V$, defined by $w(\text{inl}(i)) = u_i$ and $w(\text{inr}(k)) = v_k$, is linearly independent, then $a_i = 0$ for every $i \in T$.

Proof. [✓ Proof](#) Substitute $u_i = \alpha_i v_{\phi(i)}$ for $i \notin T$ and collect the resulting coefficients over the fibres of ϕ . The given equality is then a linear relation in the combined family. Linear independence makes every coefficient vanish, in particular each a_i with $i \in T$. $\square$

Lemma 2.4.12 (Scaling of word evaluation). [Lean](#) [✓ Stmt](#) *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then, for any word w , $(\zeta A)^w = \zeta^{|w|} A^w$, where $|w|$ is the length of w .*

Proof. [✓ Proof](#) Induction on w : the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ . $\square$

Lemma 2.4.13 (Scaling of MPV components). [Lean](#) [✓ Stmt](#) *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then, for every N and every configuration $\sigma = (i_1, \dots, i_N)$, $V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma$.*

Proof. ✓ Proof Apply Lemma 2.4.12 to the word $w = (i_1, \dots, i_N)$ of length N , then take the trace; the scalar ζ^N pulls out by linearity. $\square$

Definition 2.4.14 (Gauge-phase equivalence). Lean Lean ✓ Stmt Two tensors A and B are *gauge-phase equivalent* if there exist $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that, for every physical index i ,

$$B^i = \zeta X A^i X^{-1}. \quad (2.11)$$

Proof. ✓ Proof Gauge equivalence is the special case $\zeta = 1$. $\square$

Definition 2.4.15 (Unit gauge-phase equivalence). Lean ✓ Stmt Two tensors are *unit gauge-phase equivalent* when the gauge-phase equation holds with $|\zeta| = 1$.

Lemma 2.4.16 (Unit gauge phase implies gauge phase). Lean ✓ Stmt *Unit gauge-phase equivalence implies gauge-phase equivalence.*

Proof. ✓ Proof A unit-modulus scalar is nonzero. $\square$

Lemma 2.4.17 (Gauge-phase equivalence across equal bond dimensions). Lean Lean Lean ✓ Stmt *Gauge-phase equivalence is symmetric after identifying equal bond dimensions. Moreover, suppose that A , B , and C have identified bond dimensions, that X_1 and X_2 are invertible, that $\zeta_1, \zeta_2 \neq 0$, and that, for every physical index i ,*

$$\begin{aligned} B^i &= \zeta_1 X_1 A^i X_1^{-1}, \\ C^i &= \zeta_2 X_2 A^i X_2^{-1}. \end{aligned}$$

Then

$$C^i = (\zeta_2 \zeta_1^{-1}) (X_2 X_1^{-1}) B^i (X_1 X_2^{-1}), \quad (2.12)$$

so B and C are gauge-phase equivalent after the corresponding identification of their bond dimensions.

Proof. ✓ Proof Symmetry follows by replacing (X, ζ) with (X^{-1}, ζ^{-1}) . Formula (2.12) follows by substituting the first gauge formula into the second. $\square$

Lemma 2.4.18 (Gauge covariance of word evaluation). Lean ✓ Stmt *If $B^i = X A^i X^{-1}$ for all i , then $B^w = X A^w X^{-1}$ for every word w .*

Proof. ✓ Proof Induction on the word length, using $XX^{-1} = \mathbb{1}$. $\square$

Theorem 2.4.19 (Gauge invariance of MPVs). Lean ✓ Stmt *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. ✓ Proof Let $B^i = X A^i X^{-1}$. By Lemma 2.4.18, $B^w = X A^w X^{-1}$ for every word w . Then $\text{tr}(B^w) = \text{tr}(X A^w X^{-1}) = \text{tr}(A^w)$ by cyclicity of the trace. $\square$

Theorem 2.4.20 (Gauge-phase scaling of MPVs). Lean ✓ Stmt *If $B^i = \zeta X A^i X^{-1}$ for every physical index i , then for every system size N and configuration σ one has $V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$.*

Proof. ✓ Proof Combine gauge covariance of word evaluation with scalar scaling, then take traces. $\square$

2.5 Injectivity and normality

Definition 2.5.1 (Injectivity). [Lean](#) [✓ Stmt](#) A tensor A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Theorem 2.5.2 (Injectivity implies irreducibility). [Lean](#) [✓ Stmt](#) If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ is irreducible.

Proof. [✓ Proof](#) If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{1}$. $\square$

The inverse-tensor characterization of injectivity says that A is injective exactly when there is a tensor A^{-1} such that

$$\sum_i (A^i)_{\alpha,\beta} (A^{-1,i})_{\alpha',\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}. \quad (2.13)$$

Thus (2.13) identifies the two pairs of virtual indices when A^{-1} is contracted with A through their common physical index. In shorthand, $A^{-1}A = \mathbb{1}$. This is the identity used in [11, Section 2.3, lines 324–331].

Definition 2.5.3 (L -block injectivity). [Lean](#) [✓ Stmt](#) A tensor A is *L -block injective* if

$$\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C}). \quad (2.14)$$

An injective tensor is 1-block injective.

Lemma 2.5.4 (One-site injectivity gives one-block injectivity). [Lean](#) [✓ Stmt](#) If the one-site matrices $\{A^i\}_{i=0}^{d-1}$ span $M_D(\mathbb{C})$, then the length-one word products span $M_D(\mathbb{C})$, so A is 1-block injective.

Proof. [✓ Proof](#) The length-one words are exactly the letters $i \in \{0, \dots, d-1\}$, and evaluating a length-one word gives the matrix A^i . Thus the length-one word span is the same span as in one-site injectivity. $\square$

Definition 2.5.5 (Normal tensor). [Lean](#) [✓ Stmt](#) A tensor A is *normal* [16] if there exists a positive integer L_0 such that A is L_0 -block injective.

Lemma 2.5.6 (Gauge invariance of block injectivity). [Lean](#) [✓ Stmt](#) If A and B are gauge equivalent and A is L -block injective, then B is L -block injective.

Proof. [✓ Proof](#) Conjugation by an invertible matrix maps the span of the length- L words of A onto the span of the length- L words of B . Since conjugation is an automorphism of the full matrix algebra, fullness of the first span implies fullness of the second. $\square$

Theorem 2.5.7 (Gauge transport of normality). [Lean](#) [✓ Stmt](#) If A and B are gauge equivalent and A is normal, then B is normal.

Proof. [✓ Proof](#) Apply Lemma 2.5.6 at a blocking length for which A is block-injective. $\square$

Remark 2.5.8. [✓ Stmt](#) The algebraic characterization of normality in Definition 2.5.5 (eventual block injectivity) is equivalent to the spectral characterization of [16, Definition IV.1] (primitivity of the trace-preserving normalized transfer map $\mathcal{E}_A$); see [200, Proposition 3]. In this development, the equivalence is established by Lemma 6.7.2.1, Theorem 6.7.4.4, Theorem 6.8.2, and Theorem 7.9.2.1.

2.6 Canonical form

Definition 2.6.1 (Block-injective canonical form). Lean ✓ Stmt Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i. \quad (2.15)$$

This block-injective formulation of the canonical form is used in [193]; see also [16] for the normal canonical form.

Remark 2.6.2. ✓ Stmt The block-injective canonical form requires each block to be injective (Definition 2.5.1), which is stronger than the normal canonical form of [16] where blocks are normal in the spectral sense. Chapter 7 develops the normal canonical form used in the multi-block theory.

Definition 2.6.3 (Block projection). Lean ✓ Stmt For a finite direct sum $\bigoplus_k \mathbb{C}^{D_k}$, the block projection P_j is the matrix that is the identity on the j -th summand and zero on all other summands.

Definition 2.6.4 (Block-diagonal matrix). Lean ✓ Stmt A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if it is a direct sum of square matrices, one on each summand.

Lemma 2.6.5 (Off-block characterization). Lean ✓ Stmt A matrix on $\bigoplus_k \mathbb{C}^{D_k}$ is block diagonal if and only if every entry from one summand to a distinct summand is zero.

Proof. ✓ Proof The forward direction follows from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix. $\square$

Lemma 2.6.6 (Commutation with block projections). Lean ✓ Stmt If a matrix on $\bigoplus_k \mathbb{C}^{D_k}$ commutes with every block projection P_j , then it is block diagonal.

Proof. ✓ Proof Compare the (i, j) block of the identity $XP_j = P_jX$ for $i \neq j$. The right multiplication by P_j keeps that block, while the left multiplication by P_j kills it. Hence every off-block entry is zero, and Lemma 2.6.5 applies. $\square$

Definition 2.6.7 (Block-diagonal tensor from blocks). Lean ✓ Stmt Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the associated *block-diagonal tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

Lemma 2.6.8 (Word evaluation of a block-diagonal tensor). [Lean] [✓ Stmt] For a word w , the word evaluation of a block-diagonal tensor remains block diagonal:

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w. \quad (2.16)$$

Proof. [✓ Proof] Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor μ_k on the k -th block. $\square$

Theorem 2.6.9 (MPV decomposition). [Lean] [✓ Stmt] The MPV of a block-diagonal tensor decomposes as

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle. \quad (2.17)$$

Equivalently, for each configuration σ one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma.$$

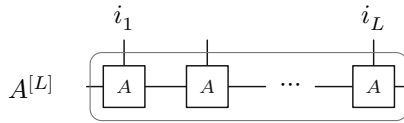
Proof. [✓ Proof] By Lemma 2.6.8, the full word evaluation is block diagonal with blocks $\mu_k^N A_k^w$ for a word w of length N . Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$. $\square$

2.7 Blocking

Definition 2.7.1 (Blocked tensor). [Lean] [✓ Stmt] Given a tensor A and a blocking length L , the L -blocked tensor is the tensor with physical index set $\{0, \dots, d-1\}^L$ (hence physical dimension d^L) and the same bond dimension D , whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}. \quad (2.18)$$

Blocking coarse-grains L neighbouring sites into one tensor: the inner virtual bonds are contracted, the L physical legs merge into a single composite index, and the two outer bonds remain as the bond of $A^{[L]}$.



Lemma 2.7.2 (Blocked word evaluation). [Lean] [Lean] [Lean] [✓ Stmt] If $w = (J_1, \dots, J_m)$ is a word in the blocked alphabet, where each J_k is an L -tuple in $\{0, \dots, d-1\}^L$, then concatenating the block words gives a word $\tilde{w}$ of length mL in the original alphabet and $(A^{[L]})^w = A^{\tilde{w}}$.

Proof. [✓ Proof] Induct on the blocked word w and split off the first block word. Lemma 2.1.4 turns concatenation of blocks into multiplication of the corresponding evaluations. $\square$

Lemma 2.7.3 (Blocking preserves MPV coefficients). [Lean] [✓ Stmt] For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system, where each σ_k is an L -tuple in $\{0, \dots, d-1\}^L$, there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that $V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}$.

Proof. ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.7.2). $\square$

Theorem 2.7.4 (Blocking preserves same MPV). Lean ✓ Stmt *If A and B generate the same MPV family, then for any blocking length L , the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.*

Proof. ✓ Proof By Lemma 2.7.3, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the corresponding concatenated configuration. The same holds for $B^{[L]}$ (with the same concatenated configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. $\square$

2.8 MPV overlap

Definition 2.8.1 (MPV as Hilbert-space vector). Lean ✓ Stmt We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ -component $V^{(N)}(A)_\sigma$ as in Definition 2.1.7. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.8.2 (MPV inner product). Lean ✓ Stmt The *MPV inner product* between two tensors A and B at system size N is

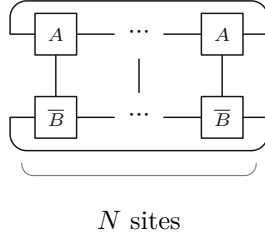
$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma, \quad (2.19)$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.8.3 (MPV overlap). Lean ✓ Stmt The *MPV overlap* between two tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}. \quad (2.20)$$

Diagrammatically, the overlap contracts the MPV ring of A against the conjugate ring of B along their shared physical indices, both virtual rings closed by the trace:



This pairing is linear in the first MPV coefficient and conjugate-linear in the second. By Lemma 2.8.4, it is the complex conjugate of the Hilbert-space inner product from Definition 2.8.2.

Lemma 2.8.4 (Overlap equals conjugate inner product). Lean ✓ Stmt $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}$.

Proof. ✓ Proof The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. $\square$

Lemma 2.8.5 (Overlap determined by positive-length MPV). Lean ✓ Stmt If $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$ and $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$ for all $N > 0$ and all σ , then $O_{AB}(N) = O_{A'B'}(N)$ for all positive N .

Proof. ✓ Proof Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical. □

Chapter 3

The Single-Block Fundamental Theorem

An injective MPS tensor that generates the same matrix-product vectors as a second tensor of equal bond dimension is conjugate to it [16, Corollary IV.5]: there is an invertible X with $B^i = XA^iX^{-1}$ for every physical index i . The argument is purely algebraic. Equality of the matrix-product vectors gives $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w ; the length-two and length-three traces produce a linear map T with $T(A^i) = B^i$ that is multiplicative and nonzero; a nonzero multiplicative endomorphism of the simple algebra $M_D(\mathbb{C})$ is a unital automorphism; and by Skolem–Noether every automorphism of $M_D(\mathbb{C})$ is conjugation by an invertible X . Evaluating at A^i closes the proof. The same conjugation, one normal block at a time, is the central step of the multi-block theorem in Chapter 9.

3.1 The multiplicative linear extension

Nondegeneracy of the trace pairing on $M_D(\mathbb{C})$ is a generic algebra fact, proved in the companion quantum-channel volume [182, “Nondegeneracy of the trace pairing”].

Lemma 3.1.1 (Same matrix-product vectors give trace agreement). [Lean](#) [✓ Stmt](#) *If A and B generate the same matrix-product vectors, then $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w .*

Proof. [✓ Proof](#) By (2.3), the configuration $\sigma = w$ gives $V^{(|w|)}(A)_w = \text{tr}(A^w)$. Equality of the matrix-product vectors gives $V^{(N)}(A) = V^{(N)}(B)$ for all N , so $\text{tr}(A^w) = \text{tr}(B^w)$. $\square$

The trace pairing against the family $\{A^i\}$ connects word traces to the algebra structure of $M_D(\mathbb{C})$.

Definition 3.1.2 (Trace pairing map). [Lean](#) [✓ Stmt](#) For a tensor A , the trace pairing map $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ is defined by

$$\Phi_A(M)_i = \text{tr}(MA^i). \quad (3.1)$$

Theorem 3.1.3 (The trace pairing map is injective for injective tensors). [Lean](#) [Lean](#) [✓ Stmt](#) *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. [✓ Proof](#) If $\Phi_A(M) = 0$, then (3.1) gives $\text{tr}(MA^i) = 0$ for every i . The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\text{tr}(MN) = 0$ for all N , and $M = 0$ by the corresponding theorem in the companion quantum-channel volume [182, “Nondegeneracy of the trace pairing”]. $\square$

Lemma 3.1.4 (Injectivity transfers across a range inclusion). Lean ✓ Stmt *Let $\Phi, \Psi : V \rightarrow W$ be $\mathbb{C}$ -linear maps with V finite-dimensional. If $\ker \Phi = \{0\}$ and $\text{range } \Phi \subseteq \text{range } \Psi$, then $\ker \Psi = \{0\}$.*

Proof. ✓ Proof By rank-nullity, $\ker \Phi = \{0\}$ gives $\dim \text{range } \Phi = \dim V$, and the range inclusion gives

$$\dim V = \dim \text{range } \Phi \leq \dim \text{range } \Psi \leq \dim V.$$

so $\dim \text{range } \Psi = \dim V$ and $\ker \Psi = \{0\}$. □

Theorem 3.1.5 (The linear extension exists and is unique). Lean ✓ Stmt *If A is injective and A, B generate the same matrix-product vectors, then there is a unique linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ with $T(A^i) = B^i$ for all i .*

Proof. ✓ Proof Length-two trace agreement gives $\Phi_A(A^i) = \Phi_B(B^i)$ for every i . The $\{A^i\}$ span $M_D(\mathbb{C})$, so $\text{range } \Phi_A \subseteq \text{range } \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.1.3), Lemma 3.1.4 gives $\ker \Phi_B = \{0\}$. Let g be a left inverse of Φ_B and set $T = g \circ \Phi_A$; then

$$T(A^i) = g(\Phi_A(A^i)) = g(\Phi_B(B^i)) = B^i.$$

Uniqueness holds because the $\{A^i\}$ span $M_D(\mathbb{C})$. □

Theorem 3.1.6 (The linear extension is multiplicative). Lean ✓ Stmt *Under the hypotheses of Theorem 3.1.5, the map T satisfies $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Length-two trace agreement gives $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$; since the $\{A^i\}$ span $M_D(\mathbb{C})$, this extends to $\Phi_B \circ T = \Phi_A$, so $\text{range } \Phi_A = \text{range } (\Phi_B \circ T) \subseteq \text{range } \Phi_B$. With $\ker \Phi_A = \{0\}$ (Theorem 3.1.3), Lemma 3.1.4 makes Φ_B injective. Length-three trace agreement gives, for all i, j , and k ,

$$\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k.$$

Hence injectivity of Φ_B yields $T(A^i A^j) = B^i B^j$. Extend in two stages: for fixed j , the maps $M \mapsto T(MA^j)$ and $M \mapsto T(M)B^j$ agree on the spanning $\{A^i\}$, hence $T(MA^j) = T(M)B^j$ for all M ; then for fixed M , the maps $N \mapsto T(MN)$ and $N \mapsto T(M)T(N)$ agree on the spanning $\{A^j\}$, hence $T(MN) = T(M)T(N)$ for all N . □

3.2 Inner automorphism and the single-block theorem

A nonzero multiplicative linear map on $M_D(\mathbb{C})$ is bijective by simplicity, preserves the unit by surjectivity, and is given by conjugation by an invertible matrix by the Skolem–Noether theorem.

Theorem 3.2.1 (A nonzero multiplicative endomorphism of $M_D(\mathbb{C})$ is bijective). Lean ✓ Stmt *A nonzero multiplicative $\mathbb{C}$ -linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. The algebra $M_D(\mathbb{C})$ is simple, so the kernel is $\{0\}$ or all of $M_D(\mathbb{C})$; the latter forces the map to be zero. Hence the map is injective, and surjective by finite-dimensionality. □

Lemma 3.2.2 (A surjective multiplicative map preserves the unit). Lean ✓ Stmt *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a surjective multiplicative $\mathbb{C}$ -linear map, then $T(1) = 1$, so T is a $\mathbb{C}$ -algebra homomorphism.*

Proof. ✓ Proof Choose X with $T(X) = \mathbb{1}$. Then

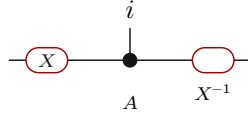
$$T(\mathbb{1}) = T(X)T(\mathbb{1}) = T(X\mathbb{1}) = T(X) = \mathbb{1}.$$

□

Lemma 3.2.3 (An injective tensor forces a nonzero partner). Lean ✓ Stmt *Let $D \geq 1$. If A is injective and A, B generate the same matrix-product vectors, then the B^i are not all zero.*

Proof. ✓ Proof If $B^i = 0$ for every i , then Lemma 3.1.1 on length-one words gives $\text{tr}(A^i) = 0$ for all i . The $\{A^i\}$ span $M_D(\mathbb{C})$, so the trace functional vanishes identically, contradicting $\text{tr}(\mathbb{1}) = D \neq 0$. □

Theorem 3.2.4 (Single-block Fundamental Theorem). Lean ✓ Stmt *Let A be an injective MPS tensor and B a tensor of the same bond dimension. If A and B generate the same matrix-product vectors, then there is $X \in \text{GL}_D(\mathbb{C})$ with $B^i = XA^iX^{-1}$ for all i . In tensor-network notation the gauge conjugates the local tensor on its virtual legs, with the physical index i on the upper leg:*



Proof. ✓ Proof Theorem 3.1.5 gives the unique linear T with $T(A^i) = B^i$, and Theorem 3.1.6 makes it multiplicative. It is nonzero: otherwise $B^i = 0$ for all i , against Lemma 3.2.3. By Theorem 3.2.1, T is bijective, and by Lemma 3.2.2 it fixes $\mathbb{1}$, so T is a $\mathbb{C}$ -algebra automorphism. The Skolem–Noether theorem [182, “Skolem–Noether”] gives $X \in \text{GL}_D(\mathbb{C})$ with $T(M) = XM X^{-1}$; evaluating at $M = A^i$ gives $B^i = XA^iX^{-1}$. □

Theorem 3.2.5 (Single-block Fundamental Theorem up to a phase). Lean Lean ✓ Stmt *Let A be an injective MPS tensor with $D \geq 1$ and B a tensor of the same bond dimension. Suppose that for every $N \geq 1$ there is $c_N \in \mathbb{C}$ with $|c_N| = 1$ and $|V^{(N)}(B)\rangle = c_N |V^{(N)}(A)\rangle$. Then there are $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and $X \in \text{GL}_D(\mathbb{C})$ with $B^i = \lambda XA^iX^{-1}$ for all i , and $c_N = \lambda^N$. This is the single-block case of the Fundamental Theorem of matrix product vectors [11, Theorem 2.10], with unimodular proportionality constants.*

Proof. ✓ Proof Theorem 25.7.2, applied to B and the single normal target A , gives nonzero $\mu_1, \dots, \mu_n$ with $|V^{(N)}(B)\rangle = (\sum_k \mu_k^N) |V^{(N)}(A)\rangle$ for every $N \geq 1$. Since $|V^{(N)}(A)\rangle \neq 0$, $c_N = \sum_k \mu_k^N$, and Theorem 26.4.5 gives $c_N = \lambda^N$ with $|\lambda| = 1$. Then $\lambda^{-1}B$ and A generate the same matrix product vectors, and Theorem 3.2.4 gives $\lambda^{-1}B^i = XA^iX^{-1}$. □

Chapter 4

Perron–Frobenius Theory for Channels and Transfer Maps

Quantum Perron–Frobenius theory governs the fixed points and spectral structure of positive and completely positive maps. For injective and irreducible transfer maps, positive fixed points are unique up to scaling and strictly positive definite, determining the canonical gauges. The ergodicity of irreducible channels and the spectral characterization of irreducibility are treated in the companion volume [182, “Cesàro convergence for irreducible channels” and “Irreducibility iff the restricted CP spectral properties hold”], following Wolf [212, Chapter 6] and Evans [159].

4.1 Positive definiteness

Theorem 4.1.1 (Transfer-map PSD fixed point — irreducible version). [Lean](#) [✓ Stmt](#) *Let A be an MPS tensor whose transfer map $\mathcal{E}_A$ is irreducible. If $\rho \geq 0$, $\rho \neq 0$, and $\mathcal{E}_A(\rho) = \rho$, then ρ is positive definite.*

Proof. [✓ Proof](#) The transfer map is completely positive, so Theorem [182, “PSD fixed point is positive definite — irreducible version”] applies. $\square$

Theorem 4.1.2 (PSD fixed point is positive definite). [Lean](#) [✓ Stmt](#) *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of $\mathcal{E}_A$. Then ρ is positive definite.*

Proof. [✓ Proof](#) Injectivity implies irreducibility of the transfer map. Apply Theorem 4.1.1. $\square$

4.2 Uniqueness

Theorem 4.2.1 (Irreducible fixed-point uniqueness). [Lean](#) [✓ Stmt](#) *Let A be an MPS tensor whose transfer map is irreducible. If $\rho, \sigma \geq 0$ are fixed points of $\mathcal{E}_A$ and $\rho \neq 0$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. [✓ Proof](#) The transfer map is completely positive, so Theorem [182, “Irreducible fixed-point uniqueness — CP version”] applies. $\square$

Theorem 4.2.2 (Uniqueness of PSD fixed point). [Lean](#) [✓ Stmt](#) *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. [✓ Proof](#) Injectivity implies irreducibility of the transfer map. Apply Theorem 4.2.1. $\square$

4.3 Existence and the Perron–Frobenius theorem

Theorem 4.3.1 (Existence of PSD fixed point). Lean ✓ Stmt Assume $D \geq 1$. Let A be an MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so that $\mathcal{E}_A$ is trace-preserving. Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.

Proof. ✓ Proof Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 2.2.8). The Cesàro fixed-point theorem (the corresponding theorem in the companion quantum-channel volume [182, “Cesàro fixed-point theorem”]) then provides the fixed point; injectivity is not needed for this step. $\square$

Theorem 4.3.2 (Quantum Perron–Frobenius). Lean ✓ Stmt Assume $D \geq 1$. Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then the transfer map $\mathcal{E}_A$ has a unique positive semidefinite fixed point ρ up to scaling, and ρ is positive definite.

Proof. ✓ Proof Combine Theorems 4.3.1, 4.1.2, and 4.2.2. $\square$

Theorem 4.3.3 (Quantum Perron–Frobenius for irreducible transfer maps). Lean ✓ Stmt Assume $D \geq 1$. Let A be an MPS tensor such that its transfer map is irreducible and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so that $\mathcal{E}_A$ is trace-preserving. Then $\mathcal{E}_A$ has a unique positive definite fixed point, up to scalar multiple.

Proof. ✓ Proof Existence is the channel fixed-point theorem (Theorem 4.3.1). Irreducibility upgrades the fixed point to positive definite, and Theorem 4.2.1 gives uniqueness up to scalar multiple. $\square$

4.4 Right- and left-canonical gauges

Definition 4.4.1 (Left-canonical normalization). Lean ✓ Stmt An MPS tensor $A = (A^i)_i$ is left-canonical when

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Equivalently, its Kraus map is trace-preserving.

Theorem 4.4.2 (Left-canonical (trace-preserving) gauge). Lean Lean ✓ Stmt Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = S A^i S^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization.

Proof. ✓ Proof Conjugating each summand gives $(S A^i S^{-1})^\dagger (S A^i S^{-1}) = (S^\dagger)^{-1} (A^i)^\dagger \sigma A^i S^{-1}$. The adjoint fixed-point substitution leaves $(S^\dagger)^{-1} \sigma S^{-1}$, and $\sigma = S^\dagger S$ cancels the outer factors. See Section A.1 for the full calculation. $\square$

Theorem 4.4.3 (Square-root trace-preserving gauge from an adjoint fixed point). Lean ✓ Stmt Let σ be positive definite and satisfy $\sum_i (A^i)^\dagger \sigma A^i = \sigma$. Define

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2}. \tag{4.1}$$

Then B is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

Proof. ✓ Proof

This is the specialization $K_i = A^i$ of Theorem [182, “Trace-preserving gauge from an adjoint fixed point, Kraus form”]. Each conjugated summand is $(B^i)^\dagger B^i = \sigma^{-1/2} (A^i)^\dagger \sigma A^i \sigma^{-1/2}$. Substituting $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ and cancelling the outer inverse square roots gives $\mathbb{1}$. The complete calculation appears in Section A.1. $\square$

Theorem 4.4.4 (Square-root trace-preserving gauge from an adjoint eigenvector). Lean ✓ Stmt *Let σ be positive definite and satisfy $\sum_i (A^i)^\dagger \sigma A^i = r\sigma$ for $r > 0$. Define*

$$B^i = \sigma^{1/2} (r^{-1/2} A^i) \sigma^{-1/2}. \quad (4.2)$$

Then B is trace-preserving: $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

Proof. ✓ Proof Rescale A by $r^{-1/2}$. The adjoint transfer map is quadratic in the tensor, so the eigenvector equation becomes a fixed-point equation. Theorem 4.4.3 then gives the result. $\square$

Remark 4.4.5. ✓ Stmt The right-canonical (unital) gauge and the left-canonical (trace-preserving) gauge arise from different similarity transformations: the former uses a fixed point of the transfer map $\mathcal{E}_A$, whereas the latter uses a fixed point of the adjoint map $\mathcal{E}_A^\dagger$. In general, a transfer map cannot be brought into simultaneously unital and trace-preserving form by a single similarity transform.

4.5 Perron–Frobenius eigenvector existence

Every irreducible MPS tensor admits a positive definite eigenvector for its adjoint transfer map with positive eigenvalue, which yields a trace-preserving (left-canonical) gauge. The underlying fixed-point and positive-definiteness theorems follow Wolf [212, Theorems 6.3 and 6.5] and Evans [159]; the gauge construction follows Cirac et al. [11, Appendix A].

Theorem 4.5.1 (Irreducibility of the transfer map of an irreducible tensor). Lean ✓ Stmt *If an MPS tensor A is irreducible, then its transfer map $\mathcal{E}_A$ is irreducible.*

Proof. ✓ Proof If P is a nontrivial invariant projection for $\mathcal{E}_A$, then

$$(\mathbb{1} - P)A^i P = 0$$

for every i , which is exactly a nontrivial invariant projection for the family $\{A^i\}$, contradicting irreducibility of A . $\square$

Theorem 4.5.2 (Positive-definite adjoint eigenvector for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive definite matrix σ and a positive real $r > 0$ such that $\mathcal{E}_A^\dagger(\sigma) = r\sigma$.*

This combines the eigenvector-existence part of [212, Theorem 6.5] with the irreducible upgrade to positive definiteness ([212, Theorem 6.3(2)–(3)]); the application to the adjoint transfer map follows [11, Appendix A].

Proof. ✓ Proof By Theorem 4.5.1, $\mathcal{E}_A$ is irreducible, and since $\mathcal{K}_A = \mathcal{E}_A$ (the corresponding theorem in the companion quantum-channel volume [182, “Kraus-map form of the transfer map”]), its Kraus map $\mathcal{K}_A$ is irreducible too (the corresponding theorem in the companion quantum-channel volume [182, “Irreducibility in Kraus-map notation”]). Applying Theorem [182, “Positive-definite adjoint eigenvector for irreducible Kraus families”] to $K_i = A^i$ gives a positive definite σ and a positive real $r > 0$ with $\sum_i (A^i)^\dagger \sigma A^i = r\sigma$, i.e. $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. $\square$

Theorem 4.5.3 (TP-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r , a positive definite matrix σ , and a tensor B gauge-equivalent to $r^{-1/2}A$ such that*

$$B^i = \sigma^{1/2}(r^{-1/2}A^i)\sigma^{-1/2}, \quad (4.3)$$

and $\sum_i (B^i)^\dagger B^i = \mathbb{1}$.

This corresponds to the spectral rescaling and trace-preserving gauge construction of [11, Appendix A].

Proof. ✓ Proof By Theorem 4.5.2, obtain σ positive definite and $r > 0$ with $\mathcal{E}_A^\dagger(\sigma) = r\sigma$. Set $c = r^{-1/2}$ and $A'^i = cA^i$. Then $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$. By the square-root trace-preserving eigenvector gauge construction (Theorem 4.4.4), the tensor defined by (4.3) satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$. Lemma A.1.1 gives the required gauge equivalence to $r^{-1/2}A$. $\square$

Theorem 4.5.4 (Unital-gauge existence for irreducible tensors). Lean ✓ Stmt *Let A be an irreducible MPS tensor with $D > 0$ and some $A^i \neq 0$. Then there exist a positive real r , a positive definite matrix ρ , and a tensor B gauge-equivalent to $r^{-1/2}A$ such that*

$$B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}, \quad (4.4)$$

and $\sum_i B^i (B^i)^\dagger = \mathbb{1}$.

This corresponds to the Perron–Frobenius unital gauge of [193, Theorem 4] for an irreducible block.

Proof. ✓ Proof Apply Theorem 4.5.2 to the conjugate-transposed Kraus family. Irreducibility transfers to that family, and the nonzero Kraus hypothesis is preserved by taking adjoints. Translating the resulting adjoint eigenvector equation back gives $\mathcal{E}_A(\rho) = r\rho$ with ρ positive definite and $r > 0$. The unital gauge theorem then gives the representative (4.4), and the same square-root similarity gives gauge equivalence to $r^{-1/2}A$. $\square$

Chapter 5

Peripheral Channel Structure and Transfer-Operator Gaps

The spectral theory of finite-dimensional quantum channels and MPS transfer maps rests on two principal structures, following Pérez-García et al. [193] and Wolf [212]: the peripheral spectrum and the mixed transfer operator.

First, for an irreducible unital Schwarz map with a faithful adjoint fixed point, the peripheral eigenvalues on the unit circle form a finite cyclic group (Theorem [182, “Peripheral eigenvalues: cyclic structure for Kraus maps”]). Under trace preservation, the order of this group divides the bond dimension D (Theorem [182, “Transfer-map peripheral eigenvalues form a cyclic group”]). Unitary conjugation and cyclic corner projections yield the periodicity-free cyclic-sector decomposition (Theorem [182, “Cyclic decomposition for irreducible finite Kraus maps”]) used in the canonical-form reduction of Chapter 7.

Second, the *mixed transfer operator* F_{AB} of two MPS tensors A and B governs the asymptotic overlap of matrix product vectors. Under trace-preserving normalization, the spectral radius satisfies $\rho_{\text{spec}}(F_{AB}) \leq 1$ (Theorem 5.3.1); for injective (or irreducible trace-preserving) tensors, equality $\rho_{\text{spec}}(F_{AB}) = 1$ forces A and B to be gauge-phase equivalent (Theorem 5.3.4), while unequal bond dimensions force a strict gap $\rho_{\text{spec}}(F_{AB}) < 1$ (Theorem 5.4.2). Because the MPV overlap $O_{AB}(N)$ equals the operator trace $\text{Tr}(F_{AB}^N)$ (Theorem 5.2.1), these spectral gaps imply the overlap-decay theorems (Theorem 5.5.5 and its rectangular and irreducible-trace-preserving variants). For a single primitive tensor whose transfer map has trivial peripheral spectrum away from the fixed-point projection, the complementary gap yields the rank-one Peron limit $\text{Tr}(\mathcal{E}_A^n) \rightarrow 1$ (Theorem 5.10.1) and primitive self-overlap convergence $O_{AA}(N) \rightarrow 1$ (Theorem 5.10.2).

Throughout, $\rho_{\text{spec}}(\cdot)$ denotes the spectral radius of a linear endomorphism of a finite-dimensional matrix space: $\rho_{\text{spec}}(F_{AB}) < 1$ for a mixed transfer operator and $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ for a transfer map after subtracting its fixed-point projection P . The unqualified phrase *spectral gap* is reserved for the energy gap of a many-body Hamiltonian.

The algebraic mixed-transfer identities hold without normalization. The subsequent spectral-radius estimates assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}. \quad (5.1)$$

Right-canonical (unital) and left-canonical (trace-preserving) gauges are distinct similarity transformations of a tensor (Remark 5.3.5), used according to whether the fixed point of the transfer

map or that of its adjoint governs the transformation.

Supporting proofs for the mixed-transfer identities, operator-norm estimates, gauge intertwiners, and cyclic-decomposition details appear in Appendix B.

5.1 Mixed transfer operator

Definition 5.1.1 (Mixed transfer operator). ✓ Stmt The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$F_{AB}(X) := \sum_{i=0}^{d-1} A^i X(B^i)^\dagger. \quad (5.2)$$

Definition 5.1.2 (Rectangular mixed transfer operator). Lean ✓ Stmt For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 , possibly with $D_1 \neq D_2$, the *rectangular mixed transfer operator* is the map $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\text{rect}}(X) := \sum_{i=0}^{d-1} A^i X(B^i)^\dagger. \quad (5.3)$$

The self-transfer identity $F_{AA} = \mathcal{E}_A$, the iterated word expansion of F_{AB} , and the trace-expansion lemma over matrix units are proved in Section B.1.

5.2 MPV overlap as transfer trace

The overlap $O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$ (2.20) coincides with the operator trace of the N -th power of the mixed transfer operator.

Lemma 5.2.1 (Overlap equals transfer trace). Lean ✓ Stmt For tensors A and B of bond dimension $D \geq 1$,

$$O_{AB}(N) = \text{Tr}(F_{AB}^N). \quad (5.4)$$

Proof. ✓ Proof Specialize Lemma 5.2.2 to $D_1 = D_2 = D$. □

Lemma 5.2.2 (Rectangular overlap as transfer trace). Lean ✓ Stmt For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$,

$$O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N). \quad (5.5)$$

Proof. ✓ Proof Section B.1 expands the operator trace over the rectangular matrix-unit basis E_{pq} , $0 \leq p \leq D_1 - 1$, $0 \leq q \leq D_2 - 1$ (Lemma [182, “Trace expansion over matrix units”]), and the transfer power by words (Lemma B.1.2); reassembling the sums over matrix-unit indices gives

$$\text{Tr}((F_{AB}^{\text{rect}})^N) = \sum_{\sigma} \text{tr}(A^\sigma) \overline{\text{tr}(B^\sigma)} = \sum_{\sigma} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N).$$

□

5.3 Eigenvalue bound and transfer-operator gap

For trace-preserving Kraus maps, every eigenvalue of the rectangular mixed transfer operator has modulus at most 1. When the spectral radius reaches 1, Kadison–Schwarz rigidity ([212, Theorem 5.3]) applied to a block embedding of the eigenvector forces gauge-phase equivalence between two injective (or irreducible trace-preserving) tensors.

Theorem 5.3.1 (Eigenvalue bound). [Lean] [✓ Stmt] *Let A and B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \leq 1$.*

Proof. [✓ Proof] This is the equal-bond-dimension case $D_1 = D_2 = D$ of Theorem 5.3.2. The argument uses only the trace-preserving normalization (5.1) and the general eigenvalue bound for trace-preserving Kraus maps; it does not appeal to Perron–Frobenius theory. $\square$

Theorem 5.3.2 (Rectangular eigenvalue bound). [Lean] [✓ Stmt] *Let A and B be normalized MPS tensors of bond dimensions $D_1 \geq 1$ and $D_2 \geq 1$. Every eigenvalue μ of F_{AB}^{rect} satisfies $|\mu| \leq 1$.*

Proof. [✓ Proof] This is the MPS formulation of the corresponding theorem in the companion quantum-channel volume [182, “Eigenvalue bound for a trace-preserving mixed Kraus map”]: the map F_{AB}^{rect} is the mixed Kraus map of the two matrix families, and the normalizations of A and B are exactly its two trace-preserving hypotheses. $\square$

Lemma 5.3.3 (Spectral radius bound). [Lean] [✓ Stmt] *For normalized tensors, $\rho_{\text{spec}}(F_{AB}) \leq 1$.*

Proof. [✓ Proof] The spectral radius is the maximum of $|\mu|$ over the eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 5.3.1. $\square$

Theorem 5.3.4 (Spectral radius ≥ 1 implies gauge-phase equivalence). [Lean] [✓ Stmt] *Let A and B be injective normalized MPS tensors. If $\rho_{\text{spec}}(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. [✓ Proof] Since $\rho_{\text{spec}}(F_{AB}) \leq 1$ by Lemma 5.3.3, the hypothesis gives an eigenvalue μ with $|\mu| = 1$ and a nonzero eigenvector X satisfying $F_{AB}(X) = \mu X$. The proof proceeds in five steps, elaborated in Section B.4.

Step 1 (Separate gauging). Choose positive definite fixed points ρ_A and ρ_B of $\mathcal{E}_A$ and $\mathcal{E}_B$ by Theorem 4.3.2. Gauge the tensors separately to unital Kraus families by Theorem [182, “Right-canonical (unital) gauge”]: $A'^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$ and similarly for B . Set $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$.

Step 2 (Block Kraus family and off-diagonal embedding). Form

$$C^i := \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}, \quad Y := \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}.$$

The matrices $\{C^i\}$ form a unital Kraus family on $M_{2D}(\mathbb{C})$, and $E_C(Y) = \mu Y$.

Step 3 (Kadison–Schwarz equality and Kraus relations). The positive definite fixed points of the adjoint maps give a faithful weighted trace for the block map. Applying the Kadison–Schwarz inequality to Y and pairing its positive semidefinite gap with this fixed point shows that the gap vanishes. The Kraus relation of the corresponding theorem in the companion quantum-channel volume [182, “KS equality implies Kraus commutation”] then gives $Y(C^i)^\dagger = \mu(C^i)^\dagger Y$. Applying the same argument to the complementary product gives $C^i Y = \mu Y C^i$.

Step 4 (Intertwining identity and invertibility). Expanding the block relations from Step 3 gives

$$X'(B'^i)^\dagger = \mu(A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i.$$

The corresponding multiplicative identities make $X'^\dagger X'$ and $X'X'^\dagger$ nonzero positive semidefinite fixed points of the appropriate transfer maps. Injectivity upgrades them to positive definite matrices, so X' is invertible.

Step 5 (Skolem–Noether conclusion). The intertwining identity and invertibility of X' give an algebra isomorphism between the full matrix algebras generated by the two families. By Theorem [182, “Skolem–Noether”], this isomorphism is inner. Undoing the separate gauges gives $B^i = \bar{\mu} X A^i X^{-1}$ for every i , which is the gauge-phase relation (2.11).

The gauging is applied separately to $\mathcal{E}_A$ and $\mathcal{E}_B$; the mixed transfer map F_{AB} need not be simultaneously unital and trace-preserving. $\square$

Remark 5.3.5 (Two gauge conventions). ✓ Stmt The right-canonical gauge used here, $A'^i = \rho^{-1/2} A^i \rho^{1/2}$, makes $\{A'^i\}$ a unital Kraus family: $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$. The convention $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$ makes the family trace-preserving: $\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$. Both conventions appear in practice: the unital form is used in the transfer-operator gap proof above, while the trace-preserving form is used in the canonical-form reduction of Chapter 7.

Theorem 5.3.6 (Strict transfer-operator gap). Lean ✓ Stmt *Let A and B be injective normalized MPS tensors that are not gauge-phase equivalent. Then $\rho_{\text{spec}}(F_{AB}) < 1$.*

Proof. ✓ Proof Lemma 5.3.3 gives $\rho_{\text{spec}}(F_{AB}) \leq 1$. Equality would imply gauge-phase equivalence by Theorem 5.3.4, contradicting the hypothesis. $\square$

5.4 Transfer-operator gap — rectangular case

Lemma 5.4.1 (Rectangular intertwining forces equal bond dimension). Lean ✓ Stmt *Let A and B be injective tensors of bond dimensions D_1 and D_2 . If there are $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ and $\mu \in \mathbb{C}$ such that, for every physical index i ,*

$$X(B^i)^\dagger = \mu(A^i)^\dagger X, \quad (5.6)$$

$$A^i X = \mu X B^i, \quad (5.7)$$

then $D_1 = D_2$.

Proof. ✓ Proof Relation (5.6) shows that $\ker(X)$ is invariant under every $(B^i)^\dagger$: if $Xv = 0$, then $X(B^i)^\dagger v = \mu(A^i)^\dagger Xv = 0$. Since the $\{B^i\}$ span the full matrix algebra, so do the $\{(B^i)^\dagger\}$ (conjugate transposition is a conjugate-linear bijection and therefore preserves complex linear spans), so $\ker(X)$ is invariant under every matrix; injectivity of the resulting map forces $\ker(X) = 0$, giving $D_2 \leq D_1$. Taking adjoints of (5.7) gives $X^\dagger(A^i)^\dagger = \bar{\mu}(B^i)^\dagger X^\dagger$, so the same argument applied to $X^\dagger$ shows $\ker(X^\dagger)$ is invariant under every $(A^i)^\dagger$ and hence trivial, giving $D_1 \leq D_2$. $\square$

Theorem 5.4.2 (Rectangular transfer-operator gap). Lean ✓ Stmt *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2 , with $D_1 \neq D_2$. Then $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$.*

Proof. ✓ Proof Theorem 5.3.2 gives $|\mu| \leq 1$ for every eigenvalue of F_{AB}^{rect} . If its spectral radius were 1, there would be a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$.

Gauge A and B to unital families A' and B' and transport X to a nonzero rectangular intertwiner X' satisfying (B.2). Lemma 5.4.1 would then give $D_1 = D_2$, contradicting the hypothesis. $\square$

Theorem 5.4.3 (Rectangular transfer powers below unit spectral radius). [Lean](#) [✓ Stmt](#) *Let A and B have bond dimensions D_1 and D_2 , respectively. If $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$, then, for every $X \in M_{D_1 \times D_2}(\mathbb{C})$,*

$$(F_{AB}^{\text{rect}})^n(X) \rightarrow 0.$$

This is the decay mechanism for the off-diagonal transfer operators in [11, Appendix B, equations (B.3)–(B.4) and the paragraph following equation (B.6)].

Proof. [✓ Proof](#)

The Gelfand formula implies convergence of $(F_{AB}^{\text{rect}})^n$ to zero in operator norm. Evaluation at a fixed rectangular matrix is continuous, and therefore preserves this limit. $\square$

Theorem 5.4.4 (Rectangular overlap decay). [Lean](#) [✓ Stmt](#) *If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) The gap $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ from Theorem 5.4.2 and Lemma [182, “Powers vanish below unit spectral radius”] imply operator-norm convergence $(F_{AB}^{\text{rect}})^N \rightarrow 0$, so its operator trace tends to zero. The identity (5.5) therefore gives $O_{AB}(N) \rightarrow 0$. $\square$

5.5 MPV overlap decay

The spectral radius on complex Banach algebras satisfies two standard identities used in the overlap and fixed-point analysis.

Theorem 5.5.1 (Spectral radius under nonzero scaling). [Lean](#) [✓ Stmt](#) *Let a belong to a complex Banach algebra and let $c \in \mathbb{C} \setminus \{0\}$. Then*

$$\rho_{\text{spec}}(ca) = |c| \rho_{\text{spec}}(a).$$

Proof. [✓ Proof](#) Multiplication by c sends the spectrum of a onto the spectrum of ca . Taking the supremum of the moduli of the spectral values gives the formula. $\square$

Theorem 5.5.2 (Spectral radius of a nonzero idempotent). [Lean](#) [✓ Stmt](#) *Let a be a nonzero element of a complex normed algebra. If $a^2 = a$, then*

$$\rho_{\text{spec}}(a) = 1.$$

Proof. [✓ Proof](#) The identity $a^2 = a$ gives $a(a - 1) = 0$. Since $a \neq 0$, the element $a - 1$ is not invertible, so 1 belongs to the spectrum of a . Every spectral value λ satisfies $\lambda^2 = \lambda$, hence the spectrum is contained in $\{0, 1\}$. $\square$

The Banach-algebra fact that powers below unit spectral radius vanish is proved in Section B.5.

Lemma 5.5.3 (An idempotent below unit spectral radius vanishes). [Lean](#) [✓ Stmt](#) *Let a be an element of a complex Banach algebra with $a^2 = a$ and $\rho_{\text{spec}}(a) < 1$. Then $a = 0$.*

Proof. [✓ Proof](#)

Lemma [182, “Powers vanish below unit spectral radius”] gives $a^n \rightarrow 0$. Idempotence gives $a^{n+1} = a$ for every n , so uniqueness of limits yields $a = 0$. $\square$

Theorem 5.5.4 (Transfer powers converge to zero). [Lean](#) [✓ Stmt](#) *Let A and B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) The gap $\rho_{\text{spec}}(F_{AB}) < 1$ from Theorem 5.3.6 and the equal-bond-dimension case of Theorem 5.4.3 give $F_{AB}^n(X) \rightarrow 0$ for every X . $\square$

Theorem 5.5.5 (Overlap decay). [Lean](#) [✓ Stmt](#) *Let A and B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) By (5.4) and the trace-expansion formula of [182, “Trace expansion over matrix units”],

$$O_{AB}(N) = \text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}.$$

Theorem 5.5.4 makes every summand tend to zero. Since the sum is finite, $O_{AB}(N) \rightarrow 0$. $\square$

Cross-correlation decay between distinct blocks and the persistence of self-correlation at a fixed point, used in the canonical-form separation of Chapter 7, are established in Section B.5.

5.6 Transfer-operator gap under irreducible-TP hypotheses

Under the trace-preserving normalization (5.1), the transfer-operator gap and overlap-decay theorems extend from injective tensors to irreducible tensors, following Cirac et al. [11] and Wolf [212, Section 6.2]. In place of the full matrix algebra span, the argument uses the strict positive definiteness of the Perron–Frobenius fixed point.

Theorem 5.6.1 (Gauge-equivalence from irreducible-TP spectral radius). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized MPS tensors of bond dimension D . If $\rho_{\text{spec}}(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. [✓ Proof](#) Choose a modulus-one eigenvector as in Theorem 5.3.4. The proof follows its five steps, with one change. Separate gauging, the block Kraus family, and the Kadison–Schwarz equality proceed as before. The equality makes the relevant positive products of the transported intertwiner nonzero positive semidefinite fixed points. Under irreducibility, Theorem 4.1.1 makes these fixed points positive definite, so the transported intertwiner is invertible. Lemma B.4.1 supplies (B.2), and Lemma B.4.2 then gives gauge-phase equivalence. $\square$

Theorem 5.6.2 (Strict transfer-operator gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then $\rho_{\text{spec}}(F_{AB}) < 1$.*

Proof. [✓ Proof](#) Lemma 5.3.3 gives $\rho_{\text{spec}}(F_{AB}) \leq 1$, while the contrapositive of Theorem 5.6.1 excludes equality. $\square$

Theorem 5.6.3 (Overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) Theorem 5.6.2 gives $\rho_{\text{spec}}(F_{AB}) < 1$, so $F_{AB}^N \rightarrow 0$. The overlap–trace identity (5.4) gives $O_{AB}(N) \rightarrow 0$. $\square$

Theorem 5.6.4 (Rectangular transfer-operator gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible trace-preserving normalized tensors of bond dimensions D_1 and D_2 , respectively, with $D_1 \neq D_2$. Then $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$.*

Proof. [✓ Proof](#) Suppose instead that $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$, and choose a modulus-one eigenvector $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$. Gauge A and B separately to unital families A' and B' and transport X to $X' \neq 0$. The matrix $X'X'^{\dagger}$ is a nonzero positive semidefinite fixed point of the appropriate transfer map on $M_{D_1}(\mathbb{C})$, so Theorem 4.1.1 makes it positive definite. Thus X' has full row rank and $D_1 \leq D_2$. Similarly, $X'^{\dagger}X'$ is positive definite, so X' has full column rank and $D_2 \leq D_1$. Hence $D_1 = D_2$, a contradiction. $\square$

Theorem 5.6.5 (Rectangular overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) *If $D_1 \neq D_2$ and both tensors are irreducible and trace-preserving normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. [✓ Proof](#) Theorem 5.6.4 gives $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$, so $(F_{AB}^{\text{rect}})^N \rightarrow 0$. The rectangular overlap-trace identity (5.5) gives $O_{AB}(N) \rightarrow 0$. $\square$

5.7 Overlap rigidity for normal blocks

A non-decaying overlap between irreducible tensors forces unit spectral radius for their mixed transfer operator, establishing gauge-phase equivalence.

Lemma 5.7.1 (Mixed-transfer spectral radius from unit-modulus overlap). [Lean](#) [✓ Stmt](#) *Let A and B have the same bond dimension. If $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$, then the mixed transfer map has spectral radius at least 1.*

Proof. [✓ Proof](#) If the mixed-transfer spectral radius were less than 1, the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace. By (5.4), the overlap would tend to zero, contradicting the hypothesis. $\square$

Theorem 5.7.2 (Gauge-phase equivalence from unit-modulus overlap). [Lean](#) [✓ Stmt](#) *Let A and B be irreducible MPS tensors of the same bond dimension with $\sum_i (A^i)^{\dagger} A^i = \mathbb{1}$ and $\sum_i (B^i)^{\dagger} B^i = \mathbb{1}$. If $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$, then A and B are gauge-phase equivalent.*

Proof. [✓ Proof](#) By Lemma 5.7.1, the mixed-transfer spectral radius is at least 1, and Theorem 5.6.1 then gives gauge-phase equivalence. $\square$

5.8 Peripheral spectral refinements

The results in this section concerning peripheral eigenvalues, cyclic projections, and peripheral eigenspaces are stated for completely positive positive finite Kraus maps, rather than the general Schwarz-map setting of Wolf Theorem 6.6; see [17].

5.8.1 Cyclic decomposition of irreducible finite Kraus maps

The cyclic decomposition proceeds by normalizing a peripheral eigenvalue to a unitary eigenvector, constructing the discrete Fourier spectral projections, and restricting the map to the resulting invariant corners $PM_D(\mathbb{C})P \cong M_n(\mathbb{C})$; details appear in Section B.7.

Lemma 5.8.1.1 (Cyclic shift of the Kraus operators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $P_0, \dots, P_{m-1}$ be orthogonal projections summing to the identity and cyclically permuted by a Kraus map, $\sum_v K_v P_{k+1} K_v^\dagger = P_k$. Then the projections are mutually orthogonal, none of them vanishes, and each Kraus operator intertwines consecutive projections:

$$K_v P_{k+1} = P_k K_v. \quad (5.8)$$

Proof. [✓ Proof](#)

Compressing $\sum_v K_v P_{k+1} K_v^\dagger = P_k$ by $\mathbb{1} - P_k$ exhibits a vanishing sum of positive semidefinite matrices, so each summand vanishes: $(\mathbb{1} - P_k) K_v P_{k+1} = 0$. Mutual orthogonality follows by compressing the resolution of the identity by a fixed P_k , another vanishing sum of positive semidefinite matrices. Expanding $K_v P_{k+1}$ and $P_k K_v$ through the resolution of the identity, the off-diagonal compressions vanish by orthogonality, leaving (5.8). If some projection vanished, the cyclic action would propagate the vanishing around the cycle, contradicting the resolution of the identity. $\square$

Theorem 5.8.1.2 (Cyclic shift along a Kraus word). [Lean](#) [✓ Stmt](#) Under the hypotheses of Lemma 5.8.1.1, a product of ℓ Kraus operators moves each cyclic projection back by ℓ steps. In particular, every product of m Kraus operators commutes with every cyclic projection.

Proof. [✓ Proof](#) Induct on the length of the word using (5.8). $\square$

The primitivity and irreducibility of the restricted sector dynamics, the invariance of each cyclic corner under the period, and the closure of the peripheral eigenvalues under products and inverses are proved in Section B.7.

5.9 Self-overlap and the period-one criterion

Theorem 5.9.1 (Self-overlap along peripheral-period multiples). [Lean](#) [✓ Stmt](#) Let $D > 0$ and $m > 0$. Let A be a left-canonical irreducible MPS tensor with bond dimension D , and suppose that the peripheral eigenvalues of its transfer map are exactly $\{z \in \mathbb{C} : z^m = 1\}$. Then $O_{AA}(mn) \rightarrow m$.

Proof. [✓ Proof](#) Apply Theorem 7.11.1 to the period- m blocking $A^{[m]}$. It gives m nonzero cyclic-sector tensors $C_0, \dots, C_{m-1}$, each left-canonical, whose unit-weight direct sum has the same MPV family as $A^{[m]}$. The cyclic-sector relations make every C_u primitive and irreducible. Distinct sectors are not gauge-phase equivalent. Hence the primitive self-overlap limit and the equal- and unequal-dimension overlap-decay theorems give

$$O_{C_u C_v}(n) \rightarrow \begin{cases} 1, & u = v, \\ 0, & u \neq v. \end{cases}$$

Expanding the overlap of the unit-weight direct sum and taking the finite sum of these limits yields

$$O_{A^{[m]} A^{[m]}}(n) = \sum_{u,v \in \mathbb{Z}_m} O_{C_u C_v}(n) \rightarrow m.$$

Finally, blocking identifies $O_{A^{[m]} A^{[m]}}(n) = O_{AA}(mn)$, which proves the claim. $\square$

Theorem 5.9.2 (Primitivity and normality from normalized self-overlap). [Lean](#) [✓ Stmt](#) Let A be an irreducible left-canonical tensor of positive bond dimension. If $O_{AA}(N) \rightarrow 1$, then the transfer map of A is primitive and A is normal.

Proof. ✓ **Proof** Let m be the peripheral period. Along its multiples, $O_{AA}(mn) \rightarrow m$. The assumed full limit gives $O_{AA}(mn) \rightarrow 1$, hence $m = 1$. The transfer map is therefore primitive, and irreducibility and left-canonicity give normality. $\square$

Theorem 5.9.3 (Normality from normalized self-overlap). Lean ✓ **Stmt** *Let A be an irreducible left-canonical tensor of positive bond dimension. If $O_{AA}(N) \rightarrow 1$, then A is normal.*

Proof. ✓ **Proof** This is the normality conclusion of Theorem 5.9.2. $\square$

5.10 Primitive overlap convergence (complementary transfer-map gap form)

For a primitive quantum channel or transfer map, removing the rank-one Perron projection P leaves a complementary map $E - P$ with spectral radius strictly less than 1 [212, Theorem 6.7]. Trace convergence and self-overlap asymptotics follow from this spectral gap, with the rank-one projection $P = P_\rho$ and its trace properties developed in Section B.6 following [182, “Complementary transfer-map gap for primitive maps”].

Theorem 5.10.1 (Complementary transfer-map gap implies trace convergence to 1). Lean ✓ **Stmt** *Let E be trace-preserving with fixed point ρ , where $\text{tr}(\rho) \neq 0$, and let P be the fixed-point projection of the corresponding definition in the companion quantum-channel volume [182, “Fixed-point projection”]. If $\rho_{\text{spec}}(E - P) < 1$, then $\text{Tr}(E^n) \rightarrow 1$ as $n \rightarrow \infty$.*

Proof. ✓ **Proof**

By the power decomposition (established in the companion quantum-channel volume [182, “Power decomposition”]), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho_{\text{spec}}(E - P) < 1$, the Gelfand formula gives $(E - P)^n \rightarrow 0$, so $\text{Tr}(E^n) \rightarrow \text{Tr}(P)$. Lemma [182, “Trace of the fixed-point projection”] (Section B.6) gives $\text{Tr}(P) = 1$. $\square$

Theorem 5.10.2 (Self-overlap convergence from a complementary transfer-map gap). Lean ✓ **Stmt** *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$, where P is the fixed-point projection, then $O_{AA}(N) \rightarrow 1$ as $N \rightarrow \infty$.*

Proof. ✓ **Proof** By the overlap-trace identity (5.4) with $A = B$, one has $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$. By Theorem 5.10.1, $\text{Tr}(\mathcal{E}_A^N) \rightarrow 1$. $\square$

Chapter 6

Wielandt Bound

Quantum Wielandt theory, established by Sanz et al. [200], bounds the word length required for products of local MPS matrices to span the full matrix algebra $M_D(\mathbb{C})$. For a primitive normalized tensor with bond dimension D , the uniform vector-spreading index $q(\mathcal{E}_A)$ and the injectivity length $\iota(A)$ satisfy $\iota(A) \leq (D^2 - \text{kr}(A) + 1)D^2$, with the cumulative bound $T_{D^2}(A) = M_D(\mathbb{C})$ serving as an intermediate step.

6.1 Cumulative span

The exact-word identities used in this section are proved in Section C.1. The stabilization and spectral linear algebra behind the cumulative estimates are collected in Section C.2.

Definition 6.1.1 (Word span). [Lean](#) [✓ Stmt](#) The *word span* at length n is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [200, Section II].

Definition 6.1.2 (Cumulative span). [Lean](#) [✓ Stmt](#) The *cumulative span* at level n is $T_n(A) = S_0(A) + S_1(A) + \dots + S_n(A)$.

Definition 6.1.3 (Fixed-length vector span of an MPS tensor). [Lean](#) [✓ Stmt](#)

For an MPS tensor A , its fixed-length vector span is the fixed-length vector span of the underlying finite matrix family.

6.1.1 Paper primitivity and indices

For the quantitative bounds, write $d' = \dim S_1(A)$ for the number of linearly independent Kraus operators. The source definitions and Proposition 3 of [200] distinguish uniform vector spreading, eventual exact-word spanning, and spectral strong irreducibility.

Definition 6.1.1.1 (Kraus rank). [Lean](#) [✓ Stmt](#) The *Kraus rank* is $\text{kr}(A) = \dim S_1(A)$.

Definition 6.1.1.2 (Eventually full Kraus rank). [Lean](#) [✓ Stmt](#) The Kraus family has *eventually full Kraus rank* if there is a positive integer i such that $S_i(A) = M_D(\mathbb{C})$.

Definition 6.1.1.3 (Paper primitivity). [Lean](#) [✓ Stmt](#)

The transfer map is *primitive in the sense of [200]* if there is a positive integer q such that

$$H_q(A, \varphi) = \mathbb{C}^D \quad (6.1)$$

for every nonzero $\varphi \in \mathbb{C}^D$. The same q works for all nonzero vectors.

Definition 6.1.1.4 (Full-Kraus-rank index). [Lean](#) [✓ Stmt](#) The *full-Kraus-rank index* is

$$\iota(A) = \min\{n \geq 1 : S_n(A) = M_D(\mathbb{C})\}.$$

If the set is empty, its value is defined to be zero.

Definition 6.1.1.5 (Primitivity index). [Lean](#) [✓ Stmt](#) The *primitivity index* is

$$q(\mathcal{E}_A) = \min\{q \geq 1 : H_q(A, \varphi) = \mathbb{C}^D \text{ for every } \varphi \neq 0\}.$$

If the set is empty, its value is defined to be zero.

Definition 6.1.1.6 (Strengthened strong irreducibility). [Lean](#) [✓ Stmt](#) An MPS tensor A is *strongly irreducible* if there exists a matrix $\rho > 0$ such that

$$\begin{aligned} \mathcal{E}_A(\rho) &= \rho, \\ \text{spec}_{\text{per}}(\mathcal{E}_A) &= \{1\}, \end{aligned}$$

and $\mathcal{E}_A$ is irreducible. This formulation includes explicit irreducibility alongside the peripheral spectrum and fixed-point condition of Sanz et al. [200, Proposition 3(c)]; see [97].

The next seven results use the following notation. For nonnegative integers d and D , let $K = \{K^i\}_{i=0}^{d-1}$ be a finite family of $D \times D$ matrices, and let $\mathcal{E}_K$ be its Kraus map.

Theorem 6.1.1.7 (Powers of a transfer-map eigenmatrix). [Lean](#) [✓ Stmt](#) If $\mathcal{E}_A(X) = \mu X$, then $\mathcal{E}_A^n(X) = \mu^n X$ for every $n \geq 0$.

Proof. [✓ Proof](#) This is the corresponding theorem in the companion quantum-channel volume [182, “Powers on an eigenspace”] for the eigenspace of μ . $\square$

Theorem 6.1.1.8 (Hermitian power-fixed point from a peripheral eigenvector). [Lean](#) [✓ Stmt](#) Let A be a normalized MPS tensor, and suppose $\mathcal{E}_A(X) = \mu X$ with $X \neq 0$, $\mu \neq 1$, and $\mu^p = 1$. Then there is a nonzero Hermitian matrix H such that

$$\begin{aligned} \text{tr}(H) &= 0, \\ \mathcal{E}_A^p(H) &= H. \end{aligned}$$

Moreover, H is not positive semidefinite.

Proof. [✓ Proof](#)

Apply Theorem [182, “Hermitian power-fixed point from a finite-order eigenmatrix”] to the transfer channel. The resulting matrix is nonzero, Hermitian, trace zero, and fixed by $\mathcal{E}_A^p$. A nonzero positive-semidefinite matrix has positive trace, so this matrix is not positive semidefinite. $\square$

Theorem 6.1.1.9 (Uniqueness of power-fixed positive matrices). [Lean](#) [✓ Stmt](#) Suppose there is a length $q \geq 0$ such that $H_q(A, \varphi) = \mathbb{C}^D$ for every nonzero $\varphi \in \mathbb{C}^D$. Let $p > 0$. If ρ and σ are nonzero positive-semidefinite fixed points of $\mathcal{E}_A^p$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.

Proof. ✓ Proof

This is Theorem [182, “Uniqueness of power-fixed positive matrices for a Kraus map”] for the Kraus family formed by the matrices of the tensor. $\square$

Theorem 6.1.1.10 (Kraus-span equivalence). Lean ✓ Stmt *Under normalization, primitivity in Wolf’s sense is equivalent to eventual full Kraus rank.*

Theorem 6.1.1.11 (Combined primitivity equivalence). Lean ✓ Stmt *Under normalization, primitivity in Wolf’s sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.*

Remark 6.1.1.12 (Eventually full Kraus rank and normality). Lean ✓ Stmt *Eventual full Kraus rank is equivalent to normality. The remaining pairwise primitivity equivalences are stated in Theorem 6.1.1.15.*

Theorem 6.1.1.13 (Eventual full Kraus rank gives paper primitivity). Lean ✓ Stmt *If $S_i(A) = M_D(\mathbb{C})$ for some positive i , then $H_i(A, \varphi) = \mathbb{C}^D$ for every $\varphi \neq 0$, so A is primitive in the sense of Definition 6.1.1.3. No normalization is imposed on A .*

Proof. ✓ Proof Applying every matrix of $M_D(\mathbb{C})$ to a nonzero vector φ gives all of $\mathbb{C}^D$, and $S_i(A)\varphi$ is the span of the length- i words applied to φ . $\square$

Theorem 6.1.1.14 (Paper primitivity gives strong irreducibility). Lean ✓ Stmt *Let $D > 0$ and suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. If A is primitive in the sense of Definition 6.1.1.3, then $\mathcal{E}_A$ is strongly irreducible.*

Proof. ✓ Proof Regard the matrices of A as a finite Kraus family. Theorems [182, “Positive-definite fixed point from vector spreading”] and [182, “Primitivity from vector spreading”] give a positive-definite fixed point and show that 1 is the only peripheral eigenvalue, and Theorem [182, “Irreducibility from vector spreading”] gives irreducibility of $\mathcal{E}_A$. Together these are the three conditions of Definition 6.1.1.6. $\square$

Theorem 6.1.1.15 (Strengthened form of the primitivity equivalence). Lean Lean Lean ✓ Stmt *Let $D > 0$ and suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. The following are equivalent:*

1. *there is a positive q for which $H_q(A, \varphi) = \mathbb{C}^D$ for every $\varphi \neq 0$;*
2. *there is a positive i for which $S_i(A) = M_D(\mathbb{C})$;*
3. *$\mathcal{E}_A$ is strongly irreducible.*

Proposition 3 of [200] states the same equivalence with item (3) replaced by the literal source condition: 1 is the only peripheral eigenvalue, and its one-dimensional eigenspace is generated by a positive-definite fixed point. Here item (3) includes the additional irreducibility condition.

Proof. ✓ Proof Theorem 6.1.1.13 is (2) $\Rightarrow$ (1), and Theorem 6.1.1.14 is (1) $\Rightarrow$ (3).

For (3) $\Rightarrow$ (2), write $\mathcal{E}_A^m = P_\rho + (\mathcal{E}_A - P_\rho)^m$. The complementary powers tend to zero. If $S_m(A) \neq M_D(\mathbb{C})$ for every m , choose a nonzero matrix B orthogonal to $S_m(A)$. The associated trace pairing vanishes on the left, while positive definiteness gives a lower bound $c\|B\|^2$ for the P_ρ term and the complementary term is $o(\|B\|^2)$. For large m this is a contradiction. $\square$

Theorem 6.1.1.16 (Normalized normal tensors are strongly irreducible). Lean Lean ✓ Stmt *Let $D > 0$ and suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. If A has eventually full Kraus rank, or equivalently is normal, then $\mathcal{E}_A$ is strongly irreducible.*

Proof. ✓ Proof Normality is eventual full Kraus rank by Remark 6.1.1.12. Theorem 6.1.1.13 then gives paper primitivity, and Theorem 6.1.1.14 gives strong irreducibility. $\square$

Theorem 6.1.1.17 (Primitive-channel specialization of the index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 6.1.1.15, paper primitivity implies*

$$q(\mathcal{E}_A) \leq \iota(A).$$

This is the primitive-channel specialization of the universal inequality $q(\mathcal{E}) \leq \iota$ of Sanz et al. [200, Proposition 1]; see [97].

Proof. ✓ Proof By Theorem 6.1.1.15, $\iota(A)$ is a positive length with $S_{\iota(A)}(A) = M_D(\mathbb{C})$. Therefore $H_{\iota(A)}(A, \varphi) = \mathbb{C}^D$ for every $\varphi \neq 0$, so the defining minimum for $q(\mathcal{E}_A)$ is at most $\iota(A)$. $\square$

Theorem 6.1.1.18 (Unique fixed points of full rank). Lean ✓ Stmt *Let $D > 0$ and suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. If there is a positive n for which $S_n(A) = M_D(\mathbb{C})$, then there is a unique positive definite density matrix $\rho \in M_D(\mathbb{C})$ such that every fixed point of $\mathcal{E}_A$ is a scalar multiple of ρ .*

This is [212, Theorem 6.15].

Proof. ✓ Proof By Theorem 6.1.1.15, the span hypothesis makes $\mathcal{E}_A$ strongly irreducible, giving a positive-definite fixed point ρ_0 together with peripheral primitivity (unique peripheral eigenvalue 1) and irreducibility of $\mathcal{E}_A$. Peripheral primitivity and irreducibility of $\mathcal{E}_A$ together give the complementary transfer-map gap at ρ_0 (Lemma C.5.5): the restriction of $\mathcal{E}_A$ off the fixed-point projection at ρ_0 has spectral radius strictly less than one. This gap forces every fixed point of $\mathcal{E}_A$ to be a scalar multiple of ρ_0 ; rescaling ρ_0 by the inverse of its (positive) trace gives the unique density-matrix representative ρ .

The source establishes this by showing that $\mathcal{E}_A^n(|\psi\rangle\langle\psi|)$ has full rank for each pure state and ruling out multidimensional fixed spaces via Corollary 6.5 of [212]. The present argument proceeds via the Proposition 3 equivalence (Theorem 6.1.1.15). $\square$

Lemma 6.1.1.19 (Monotonicity of cumulative span). Lean ✓ Stmt *For every n , one has $T_n(A) \leq T_{n+1}(A)$.*

Proof. ✓ Proof The span T_{n+1} includes all words of length at most $n + 1$ and therefore contains all words of length at most n . $\square$

Lemma 6.1.1.20 (Dimension bound). Lean ✓ Stmt *For every n , one has $\dim T_n(A) \leq D^2$.*

Proof. ✓ Proof Since $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$, the result follows. $\square$

Theorem 6.1.1.21 (Cumulative span stabilisation). Lean ✓ Stmt *If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for every $m \geq n$.*

Proof. ✓ Proof If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2} , so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for every $m > n$, hence $T_m = T_n$. $\square$

Theorem 6.1.1.22 (Dimension growth). Lean ✓ Stmt *If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.*

Proof. ✓ Proof The inclusion $T_n \leq T_{n+1}$ from Lemma 6.1.1.19, together with $T_n \neq T_{n+1}$, gives strict subspace inclusion and hence strict dimension growth. $\square$

Lemma 6.1.1.23 (Word span characterises block injectivity). [Lean](#) [✓ Stmt](#) One has $S_L(A) = M_D(\mathbb{C})$ if and only if A is L -block injective.

Proof. [✓ Proof](#) Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. $\square$

Theorem 6.1.1.24 (Positive-length propagation of block injectivity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every n ,

$$S_{n+1}(A) = \text{span}_{\mathbb{C}}\{A^i\}_{i=0}^{d-1} S_n(A).$$

Consequently, if $L > 0$ and A is L -block injective, then A is m -block injective for every $m \geq L$. This combines the block-injectivity discussion in [11, Section II] with [11, Appendix C.3, Lemma L].

Proof. [✓ Proof](#) Every word of length $n + 1$ factors into its first letter and a word of length n , which proves the displayed equality by bilinearity. Write $L = n + 1$. Since $S_L(A) = M_D(\mathbb{C})$, one has $S_n(A) \leq S_L(A)$. Hence

$$M_D(\mathbb{C}) = S_L(A) = \text{span}_{\mathbb{C}}\{A^i\}_{i=0}^{d-1} S_n(A) \leq \text{span}_{\mathbb{C}}\{A^i\}_{i=0}^{d-1} S_L(A) = S_{L+1}(A).$$

Iterating this implication gives $S_m(A) = M_D(\mathbb{C})$ for every $m \geq L$. $\square$

6.2 Nonzero trace product

Theorem 6.2.1 (Cumulative span reaches $M_D(\mathbb{C})$). [Lean](#) [✓ Stmt](#) Let $D > 0$. If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$. This cumulative estimate corresponds to the intermediate step in the proof of [200, Lemma 1].

Proof. [✓ Proof](#) This is Theorem C.2.4. $\square$

Theorem 6.2.2 (Cumulative span reaches $M_D(\mathbb{C})$ — sharp bound). [Lean](#) [✓ Stmt](#) Let $D > 0$. If A is normal, then $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$, where $d' = \dim S_1(A) = \text{kr}(A)$. This is the cumulative-span estimate used to prove the sharp trace conclusion in [200, Lemma 1].

Proof. [✓ Proof](#) If $T_n(A) \neq T_{n+1}(A)$, then the dimension strictly increases. Since $S_1(A) \subseteq T_1(A)$ and $\dim S_1(A) = d'$, one has $\dim T_1(A) \geq d'$. After at most $D^2 - d'$ additional strict-growth steps, the dimension reaches $D^2 = \dim M_D(\mathbb{C})$. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality. Hence $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$. $\square$

Theorem 6.2.3 (Eigenvector extraction for an MPS tensor). [Lean](#) [✓ Stmt](#) The conclusion of Theorem [182, “Eigenvector extraction from a full cumulative span”] holds for the matrix family of an MPS tensor whenever its cumulative span at level N is the full matrix algebra.

Proof. [✓ Proof](#)

Apply the finite-family theorem to the matrices of the tensor. $\square$

Theorem 6.2.4 (Sharp nonzero trace product for a primitive channel). [Lean](#) [✓ Stmt](#) Let $D > 0$, suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume that A is primitive in the sense of Definition 6.1.1.3. Then there is a word w with

$$1 \leq |w| \leq D^2 - \text{kr}(A) + 1, \\ \text{tr}(A^w) \neq 0.$$

This is the sharp trace conclusion of [200, Lemma 1].

Proof. ✓ Proof Theorem 6.1.1.15 gives eventual full Kraus rank, hence normality. Put $b = D^2 - \text{kr}(A) + 1$ and let V_n be the span of A^w with $1 \leq |w| \leq n$. Then $S_1(A) \subseteq V_1$, so $\dim V_1 \geq \text{kr}(A)$. If $V_n = V_{n+1}$ for some $n \geq 1$, then V_n is closed under left multiplication by every A^i . It therefore contains every positive-length word product; normality forces $V_n = M_D(\mathbb{C})$. Thus, until the full matrix algebra is reached, the dimensions increase strictly, and the bound on $\dim M_D(\mathbb{C})$ gives $V_b = M_D(\mathbb{C})$. Express $\mathbb{1}$ as a linear combination of the positive-length words defining V_b . Since $\text{tr}(\mathbb{1}) = D \neq 0$, at least one has nonzero trace. $\square$

Theorem 6.2.5 (Sharp nonzero trace product). Lean ✓ Stmt *Let $D > 0$. If A is normal, then there exists a word w of length at most $D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This normality variant has the same sharp length bound as [200, Lemma 1]; the source hypotheses are stated in Theorem 6.2.4.*

Proof. ✓ Proof By Theorem 6.2.2, the identity $\mathbb{1}$ lies in $T_{D^2-d'+1}(A)$ and is therefore a linear combination of word products of length at most $D^2 - d' + 1$. At least one such product has nonzero trace. $\square$

Theorem 6.2.6 (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If A is normal and $D \geq 1$, then there exists a word w of positive length $1 \leq |w| \leq D^2 - \text{kr}(A) + 1$ such that $\text{tr}(A^w) \neq 0$. This strengthens Theorem 6.2.5 by requiring $|w| \geq 1$, a feature needed for the blocking argument in the general Wielandt bound below.*

Proof. ✓ Proof Define the *positive-level* cumulative span $V_n = \text{span}_{\mathbb{C}}\{A^w : 1 \leq |w| \leq n\}$. The same stabilization-or-strict-growth dichotomy shows $V_{D^2-d'+1} = M_D(\mathbb{C})$: if V_n stabilises and is not $M_D(\mathbb{C})$, then V_n is closed under left multiplication by each A^i , so $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$ for every $N \geq 1$, contradicting normality. Since $V_{D^2-d'+1} = M_D(\mathbb{C})$, the identity $\mathbb{1}$ lies in $V_{D^2-d'+1}$ and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace because $\text{tr}(\mathbb{1}) = D \neq 0$ for $D \geq 1$. $\square$

6.3 Eigenvector spreading

Definition 6.3.1 (Cumulative vector span). Lean ✓ Stmt Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w \varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [200, proof of Lemma 2(a)].

Theorem 6.3.2 (Eigenvector spreading). Lean ✓ Stmt *Let A be a normal MPS tensor and let $A^{i_0} \varphi = \mu \varphi$, where $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This cumulative form supplies the dimension-growth step in [200, Lemma 2(a)].*

Proof. ✓ Proof The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D - 1$, the span stabilises by Theorem C.2.7. But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 6.2.1, so Lemma C.2.8 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by at least 1 at each step, reaching D by step $D - 1$. $\square$

Theorem 6.3.3 (Fixed-length eigenvector spreading for a primitive channel). Lean ✓ Stmt *Let $D > 0$, suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume that A is primitive in the sense of Definition 6.1.1.3. If $\varphi \neq 0$, $\mu \neq 0$, and $A^{i_0} \varphi = \mu \varphi$, then $H_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [200, Lemma 2(a)].*

Proof. [✓ Proof](#) Proposition 3, in the form of Theorem 6.1.1.15, gives eventual full Kraus rank and hence normality. The cumulative spaces $K_n(A, \varphi)$ cannot stabilise below $\mathbb{C}^D$, because every matrix eventually lies in an exact word span. Their dimensions therefore reach D by $n = D - 1$, giving $K_{D-1}(A, \varphi) = \mathbb{C}^D$. Finally, the eigenvector relation pads every shorter word to length $D - 1$ by a nonzero power of μ , so $K_{D-1}(A, \varphi) = H_{D-1}(A, \varphi)$. $\square$

6.4 Quantum Wielandt bounds

Section C.3 proves the one-step augmentation facts used in cases (2) and (3). The blocking identities used in the general case are proved in Section C.4.

Remark 6.4.1 (Explicit blocking-length estimates). [✓ Stmt](#) Theorem 6.2.1 gives the D^2 cumulative-span bound. The full-rank index bound $\iota(A) \leq (D^2 - d' + 1)D^2$ ([200, Theorem 1, case (1)]) is proved in Theorem 6.4.8 below. The left-canonical normal blocked-injectivity estimate D^4 is proved below in Theorem 6.5.5. A sharper $3D^5$ canonical-form block-separation estimate, developed in the full blueprint, combines this D^4 estimate with the direct-sum argument of [193].

Theorem 6.4.2 (Wielandt case (2), one-step invertible element). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 6.1.1.3. If $S_1(A)$ contains an invertible matrix X , then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}). \quad (6.2)$$

In the terminology of [200], this is the case where the first application space $S_1(A)$ contains an invertible operator. This is the exact word-span form of [200, Theorem 1, case (2)].

Proof. [✓ Proof](#) Adjoin X as an extra first Kraus operator, obtaining $\tilde{A}$. Theorem 6.1.1.15 gives normality of A , hence normality of $\tilde{A}$; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of $\tilde{A}$ is invertible, so Theorem C.3.5 gives $S_{D^2 - \text{kr}(\tilde{A}) + 1}(\tilde{A}) = M_D(\mathbb{C})$. Transfer this equality back to A . $\square$

Corollary 6.4.3 (Wielandt case (2), index bound). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 6.4.2, one has $\iota(A) \leq D^2 - \text{kr}(A) + 1$, where $\iota(A) = \min\{n \geq 1 : S_n(A) = M_D(\mathbb{C})\}$.

Proof. [✓ Proof](#) The word-span equality (6.2) gives an admissible value in the defining minimum for $\iota(A)$. $\square$

Lemma 6.4.4 (Rank-one extraction from a non-invertible eigenvector). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a normal MPS tensor with bond dimension $D \geq 1$. Suppose that a Kraus operator A^{i_0} is non-invertible and that there exist a nonzero vector $\varphi \in \mathbb{C}^D$ and a nonzero scalar μ such that $A^{i_0}\varphi = \mu\varphi$. Then, for every $\psi \in \mathbb{C}^D$, $|\varphi\rangle\langle\psi| \in S_{D^2 - D + 1}(A)$.

Proof. [✓ Proof](#) Let $\mathcal{L}(M) = A^{i_0}M$, and choose r so that $\mathcal{L}^r$ annihilates the generalized zero-eigenspace of $\mathcal{L}$; equivalently, $\text{range}(\mathcal{L}^r)$ is its stabilized nonzero-spectral part. The dimension-growth argument gives a level n_0 for which

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\}.$$

Since $A^{i_0}\varphi = \mu\varphi$ and $\mu \neq 0$, iterating gives $(A^{i_0})^r\varphi = \mu^r\varphi$ and hence $\varphi = \mu^{-r}(A^{i_0})^r\varphi \in \text{range}((A^{i_0})^r)$. Therefore, for every $\psi \in \mathbb{C}^D$, $|\varphi\rangle\langle\psi| \in S_{r+n_0}(A)$. The corresponding Fitting-index estimate gives $r + n_0 \leq D^2 - D + 1$. Since

$$A^{i_0}|\varphi\rangle\langle\psi| = |A^{i_0}\varphi\rangle\langle\psi| = \mu|\varphi\rangle\langle\psi|,$$

one has $|\varphi\rangle\langle\psi| = \mu^{-1}A^{i_0}|\varphi\rangle\langle\psi|$. Thus $|\varphi\rangle\langle\psi| \in S_k(A)$ implies $|\varphi\rangle\langle\psi| \in S_{k+1}(A)$, because left multiplication by A^{i_0} maps $S_k(A)$ into $S_{k+1}(A)$. Iterating from $k = r + n_0$ and using $r + n_0 \leq D^2 - D + 1$ gives $|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A)$. $\square$

Theorem 6.4.5 (Sharp rank-one extraction for a primitive channel). Lean ✓ Stmt *Let $D > 0$, suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and assume that A is primitive in the sense of Definition 6.1.1.3. Suppose A^{i_0} is not invertible and $A^{i_0}\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. Then, for every $\psi \in \mathbb{C}^D$,*

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

This is the quantitative rank-one conclusion of [200, Lemma 2(b)].

Proof. ✓ Proof Theorem 6.1.1.15 first gives normality. Apply the Fitting decomposition to left multiplication $\mathcal{L}(M) = A^{i_0}M$. Choose r so that $\mathcal{L}^r$ kills its generalized zero-eigenspace. Dimension growth then gives n_0 with

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\},$$

$$r + n_0 \leq D^2 - D + 1.$$

Since $(A^{i_0})^r \varphi = \mu^r \varphi$ and $\mu \neq 0$, the vector φ lies in the range of $(A^{i_0})^r$. Hence $|\varphi\rangle\langle\psi| \in S_{r+n_0}(A)$ for every ψ . The identity $|\varphi\rangle\langle\psi| = \mu^{-1}A^{i_0}|\varphi\rangle\langle\psi|$ pads this element to the exact length $D^2 - D + 1$. $\square$

Theorem 6.4.6 (Wielandt case (3), one-step non-invertible element). Lean ✓ Stmt *Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 6.1.1.3. If $S_1(A)$ contains a non-invertible matrix X with a nonzero eigenvalue, then $S_{D^2}(A) = M_D(\mathbb{C})$. More explicitly, it is enough to have $X\varphi = \mu\varphi$ with $\varphi \neq 0$ and $\mu \neq 0$. In the terminology of [200], this is the case where the first application space $S_1(A)$ contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [200, Theorem 1, case (3)].*

Proof. ✓ Proof Theorem 6.1.1.15 gives normality. Adjoin X as an extra first Kraus operator. The augmented tensor remains normal and has the same exact word spans as A . The eigenvector relation for X is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length $D-1$, and Lemma 6.4.4 gives the rank-one operators $|\varphi\rangle\langle\psi|$ at length $D^2 - D + 1$. Multiplying these two fixed-length spans gives S_{D^2} for the augmented tensor, hence for A . $\square$

Corollary 6.4.7 (Wielandt case (3), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 6.4.6, one has $\iota(A) \leq D^2$.*

Proof. ✓ Proof The equality $S_{D^2}(A) = M_D(\mathbb{C})$ gives an admissible value in the defining minimum for $\iota(A)$. $\square$

Theorem 6.4.8 (Wielandt's inequality — general bound). Lean ✓ Stmt *Let A be a normalized MPS tensor with $D > 0$, $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, that is primitive in the sense of Definition 6.1.1.3. Then*

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1)D^2,$$

where $\iota(A) = \min\{n \geq 1 : S_n(A) = M_D(\mathbb{C})\}$ is the full-Kraus-rank index. This is [200, Theorem 1, case (1)].

Proof. ✓ Proof Theorem 6.1.1.15 gives normality. By Theorem 6.2.4, there exists a *positive-length* word w with $|w| \leq D^2 - d' + 1$ and $\text{tr}(A^w) \neq 0$, where $d' = \text{kr}(A)$. The matrix A^w has nonzero trace, hence a nonzero eigenvalue μ and eigenvector φ by Theorem [182, “Nonzero trace implies eigenvector”]. Set $n = |w|$ and let $B = A^{[n]}$ be the blocked tensor. Then B is normal by Theorem C.4.1, and the encoding of w as a single block index gives $B^{i_0} = A^w$.

1. If A^w is invertible, then $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2n}(A) = M_D(\mathbb{C})$.
2. If A^w is non-invertible, then the eigenvector φ spans a proper subspace, and the rank-one extraction argument gives $S_{D^2}(B) = M_D(\mathbb{C})$, whence $S_{D^2n}(A) = M_D(\mathbb{C})$.

In the invertible case, the invertible Kraus operator B^{i_0} allows direct dimension growth, giving $S_{D^2-k_B+1}(B) = M_D(\mathbb{C})$, where $k_B = \text{kr}(B)$. In the non-invertible case, the bound $S_{D^2}(B) = M_D(\mathbb{C})$ follows from the rank-one extraction argument of [200, Lemma 2(b)]. In both cases, $\iota(A) \leq D^2n \leq D^2(D^2 - d' + 1)$. $\square$

Remark 6.4.9 (Quantitative and qualitative Wielandt bounds). ✓ Stmt The quantitative bound $\iota(A) \leq (D^2 - d' + 1)D^2$ ([200, Theorem 1, case (1)]) gives an explicit upper bound on the injectivity length. For the Fundamental Theorem of MPS, the qualitative assertion that $S_N(A) = M_D(\mathbb{C})$ for some finite N suffices.

6.5 Fixed-length matrix spanning

Eigenvector spreading establishes the cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$ ([200, Lemma 2(a)]). To obtain block injectivity, this cumulative span is promoted to an exact fixed-length matrix span $S_{n_0}(A) = M_D(\mathbb{C})$. Supporting word-product results appear in Section C.1, and the blocked fixed-length construction appears in Section C.4.

Lemma 6.5.1 (Eigenvector padding). Lean ✓ Stmt Suppose $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$. Then $K_n(A, \varphi) = H_n(A, \varphi)$ for every n . In particular, if $K_n(A, \varphi) = \mathbb{C}^D$, then $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. ✓ Proof Any word w with $|w| \leq n$ can be padded to exact length n by appending $k = n - |w|$ copies of the index i_0 . The eigenvector relation gives $A^{w-i_0^k}\varphi = \mu^k A^w\varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n . The reverse inclusion $H_n \leq K_n$ is immediate. $\square$

Theorem 6.5.2 (Fixed-length vector spanning implies matrix spanning). Lean ✓ Stmt Assume:

1. $H_n(A, \varphi) = \mathbb{C}^D$ (length- n word products applied to φ span all of $\mathbb{C}^D$), and
2. for each standard basis vector e_j , the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. ✓ Proof Fix a matrix unit E_{ij} . By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i|\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij}.$$

Lemma C.1.1 therefore gives $E_{ij} \in S_{n+m}(A)$. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 6.5.3 (Block injectivity and blocked-tensor injectivity). Lean ✓ Stmt A tensor A is L -block injective if and only if its L -blocked tensor $A^{[L]}$ is injective.

Proof. ✓ Proof The blocked physical alphabet is a reindexing of words of length L . Decoding blocked indices therefore identifies the two spanning families, and hence the two injectivity conditions are equivalent. $\square$

Lemma 6.5.4 (Blocking transfer). Lean ✓ Stmt *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$. In particular, A is L -block injective if and only if $A^{[L]}$ is one-site injective.*

Proof. ✓ Proof By the blocked-tensor definition (2.18), each length- n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length- nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. For $n = 1$, the d^L blocked letters are indexed by the length- L words of A , giving $S_1(A^{[L]}) = S_L(A)$, which proves the stated equivalence. $\square$

Theorem 6.5.5 (Bounded injective blocking for left-canonical normal tensors). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then there is a positive blocking length $L \leq D^4$ such that the blocked tensor $A^{[L]}$ is one-site injective.*

This is the blocked-injectivity form of the quantum Wielandt input used in the Fundamental Theorem for translation-invariant tensors. It corresponds to the statement in [11, Section II] that every normal tensor becomes injective after at most D^4 blockings, using the index estimate of [200, Theorem 1]. The left-canonical/trace-preserving hypothesis is included because this is the canonical-form context in which the blocked-injectivity input is used.

Proof. ✓ Proof Normality is eventual full Kraus rank, so Theorem 6.1.1.15 gives paper primitivity. Theorem 6.4.8 then yields

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1)D^2.$$

Trace preservation gives a nonzero Kraus operator, hence $\text{kr}(A) \geq 1$ and the right-hand side is at most D^4 . Set $L = \iota(A)$. By definition, $L > 0$ and $S_L(A) = M_D(\mathbb{C})$. Lemma 6.5.4 identifies the latter equality with one-site injectivity of $A^{[L]}$. $\square$

Theorem 6.5.6 (Exact quantum-Wielandt blocking length). Lean ✓ Stmt *Let A be a left-canonical normal MPS tensor with bond dimension $D > 0$. Then $S_{D^4}(A) = M_D(\mathbb{C})$; equivalently, A is D^4 -block injective.*

Proof. ✓ Proof Choose L with $0 < L \leq D^4$ and $S_L(A) = M_D(\mathbb{C})$, and write $D^4 = L + k$. Induction on k , using positive-length propagation at each step, gives $S_{L+k}(A) = M_D(\mathbb{C})$. $\square$

Theorem 6.5.7 (Exact block injectivity at positive multiples). Lean ✓ Stmt *If A is N -block injective and $m > 0$, then A is (mN) -block injective.*

Proof. ✓ Proof Induct on m . Theorem C.1.2 gives $S_{mN+N}(A) = S_{mN}(A)S_N(A)$, and the product of the full matrix algebra with itself is again the full matrix algebra. $\square$

Theorem 6.5.8 (Common exact block-injectivity length). Lean ✓ Stmt *Let $(A_j)_{j=1}^g$ be a finite family of positive-bond-dimension, left-canonical normal tensors. There is a positive integer L such that every A_j is L -block injective.*

Proof. ✓ Proof Apply Theorem 6.5.5 to each block. The product of the finitely many positive lengths is a common positive multiple, and exact block injectivity persists under positive multiples. $\square$

Corollary 6.5.9 (Uniform injective blocking for separated normal-canonical families). [Lean](#) [Lean](#) [✓ Stmt](#) Let $(A_x)_{x \in X}$ be a finite separated normal-canonical family whose blocks have positive virtual bond dimension. Then there is a positive integer L such that every blocked tensor $A_x^{[L]}$ is one-site injective. The same conclusion holds simultaneously for two finite separated normal-canonical families whose blocks have positive virtual bond dimension.

Here the family $(A_x)_{x \in X}$ is a basis of normal-canonical representatives (Definition 7.9.1.2) with distinct blocks; see [29].

Proof. [✓ Proof](#) Apply Theorem 6.5.5 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking. $\square$

Remark 6.5.10 (Rank-one extraction and fixed-length spanning). [✓ Stmt](#) The quantitative rank-one result of Sanz et al. [200, Lemma 2(b)] is Theorem 6.4.5. Combined with fixed-length vector spreading, it yields matrix spanning via Theorem 6.5.2. Alternatively, the blocked construction in Theorem C.4.6 extracts a rank-one operator directly from normality without tracking explicit exponents.

Remark 6.5.11 (Exact-length spanning and block injectivity). The cumulative bound $T_{D^2}(A) = M_D(\mathbb{C})$ (Theorem 6.2.1) yields exact-length spanning $S_N(A) = M_D(\mathbb{C})$ of [200, Theorem 1] under identity padding or blocked construction (Theorem C.4.6). This exact-length spanning is equivalent to injectivity of Γ_N and one-site injectivity of the blocked tensor $A^{[N]}$ (Lemma 6.5.4).

The blocked normality and rank-one constructions used below are proved in Section C.4.

6.6 Primitive MPS tensors

Definition 6.6.1 (Primitive MPS tensor). [Lean](#) [✓ Stmt](#) Assume $D > 0$. A tensor A together with a matrix $\rho \in M_D(\mathbb{C})$ is *primitive* if

$$\begin{aligned} \sum_i (A^i)^\dagger A^i &= \mathbb{1}, \\ \rho &\geq 0, \quad \rho \neq 0, \quad \mathcal{E}_A(\rho) = \rho, \\ \rho_{\text{spec}}(\mathcal{E}_A - P) &< 1, \end{aligned}$$

where P is the fixed-point projection associated to ρ . When the choice of ρ is irrelevant, we simply say that A is a primitive MPS tensor. This condition combines a complementary spectral gap with a nonzero positive semidefinite fixed point. When $\rho > 0$, this implies strong irreducibility and uniform vector spreading (Definition 6.1.1.3) by Theorem 6.1.1.15.

Theorem 6.6.2 (Normalizing the primitive fixed point to unit trace). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The fixed-point projection P depends only on the ray through ρ : for every nonzero scalar c with $\text{tr}(c\rho) \neq 0$, the projection associated to $c\rho$ equals the projection associated to ρ . Write

$$\sigma = \frac{\rho}{\text{tr}(\rho)} \tag{6.3}$$

for the normalized representative of the ray through a matrix of nonzero trace. Then $\text{tr}(\sigma) = 1$; if $\rho > 0$ then also $\sigma > 0$; and if A is primitive with positive-definite fixed point ρ , then A is primitive with fixed point σ .

Proof. ✓ Proof Both projections send X to $\text{tr}(X)\rho/\text{tr}(\rho)$, so they agree. The trace of σ is $\text{tr}(\rho)/\text{tr}(\rho) = 1$, and a positive-definite matrix stays positive definite under multiplication by the positive real $\text{tr}(\rho)^{-1}$. For the gap certificate, the trace-preservation clause does not involve ρ ; positivity, nonvanishing and the fixed-point equation are preserved by a positive real rescaling; and the complementary spectral radius is unchanged because the two projections coincide. $\square$

6.7 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 6.6.1: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary map has spectral radius < 1 .

6.7.1 From primitivity to normality

The primitivity condition of Definition 6.6.1 combines TP normalization, a nonzero PSD fixed point, and a complementary transfer-map gap. It is connected to normality through the following results.

Remark 6.7.1.1 (Complementary gap and uniform spreading). ✓ Stmt When the fixed point ρ in Definition 6.6.1 is strictly positive definite, irreducibility holds and the peripheral spectrum is $\{1\}$, so the tensor is strongly irreducible (Definition 6.1.1.6). By Theorem 6.1.1.15, this is equivalent to uniform vector spreading (Definition 6.1.1.3) and eventual full Kraus rank.

6.7.2 Primitive implies irreducible

Lemma 6.7.2.1 (Primitive tensors with positive-definite fixed point are irreducible). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ and ρ is positive definite, then A is an irreducible tensor. This is part of [200, Proposition 3] (see also [11, Section 2.3]).*

Proof. ✓ Proof By contrapositive. Assume A has an invariant projection P ($P \neq 0$, $P \neq 1$, $(1 - P)A^i P = 0$ for all i). Set $\sigma = P\rho P$.

1. *Invariance of σ under $\mathcal{E}_A^n$:* the projection relation $(1 - P)A^i P = 0$ implies $(1 - P)\mathcal{E}_A^n(\sigma) = 0$ for all n (by induction on n).
2. *Convergence:* By Lemma C.5.6, $\mathcal{E}_A^n(\sigma) \rightarrow \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Hence $(1 - P)\frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho = 0$.
3. *Nonzero scalar:* Since $P \neq 0$ and ρ is positive definite, $\text{tr}(P\rho P) > 0$, so $\frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \neq 0$. Therefore $(1 - P)\rho = 0$.
4. *Contradiction:* Since ρ is positive definite, it is a unit in $M_D(\mathbb{C})$. From $(1 - P)\rho = 0$ we get $1 - P = 0$, i.e. $P = 1$, contradicting $P \neq 1$.

$\square$

6.7.3 From irreducibility to full spanning via Burnside's theorem

The irreducibility condition (Definition 7.1.1.1) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of $\mathbb{C}^D$ under a finite matrix family.

Definition 6.7.3.1 (Algebra span). [Lean](#) [✓ Stmt](#) For a finite matrix family A , let $\text{alg}(A)$ be the unital $\mathbb{C}$ -subalgebra of $M_D(\mathbb{C})$ generated by the matrices $\{A^i\}_{i=0}^{d-1}$: $\text{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C})$.

Definition 6.7.3.2 (Invariant submodule). [Lean](#) [✓ Stmt](#) A subspace $W \leq \mathbb{C}^D$ is *A-invariant* if $A^i W \subseteq W$ for every i .

Definition 6.7.3.3 (Irreducible action). [Lean](#) [✓ Stmt](#) The matrices in A act *irreducibly* on $\mathbb{C}^D$ if every A -invariant subspace is trivial, i.e. equal to 0 or to $\mathbb{C}^D$.

Theorem 6.7.3.4 (Irreducible tensor implies irreducible action). [Lean](#) [✓ Stmt](#) If A is irreducible as a tensor (Definition 7.1.1.1), then the Kraus operators act irreducibly on $\mathbb{C}^D$ (Definition 6.7.3.3).

Proof. [✓ Proof](#) Let $W \leq \mathbb{C}^D$ be an A -invariant subspace. In finite dimension the orthogonal projection P onto W exists. Invariance of W implies $(\mathbb{1} - P)A^i P = 0$ for all i . Thus P is a nontrivial invariant projection unless $W = 0$ or $W = \mathbb{C}^D$, contradicting irreducibility of the tensor. $\square$

Theorem 6.7.3.5 (Burnside's theorem for matrix algebras). [Lean](#) [✓ Stmt](#) Assume $D > 0$. If A acts irreducibly on $\mathbb{C}^D$, then the generated unital subalgebra is the full matrix algebra: $\text{alg}(A) = M_D(\mathbb{C})$. This is the complex finite-dimensional case of Burnside's theorem (equivalently, Jacobson's density theorem); see [172].

Proof. [✓ Proof](#) Identify $\text{alg}(A)$ with its image under the algebra equivalence $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathbb{C}^D)$. The irreducible-action hypothesis makes $\mathbb{C}^D$ a simple module for this algebra. Over the algebraically closed field $\mathbb{C}$, Schur's lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. $\square$

Remark 6.7.3.6. [✓ Stmt](#) Burnside's theorem ensures that every irreducible subalgebra of $M_D(\mathbb{C})$ coincides with the full matrix algebra $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$, providing the algebraic foundation for matrix spanning.

6.7.4 From primitivity to strong irreducibility and normality

The complementary transfer-map gap of Definition 6.6.1 implies normality (eventual full Kraus rank) via the convergence of transfer-map iterates to the fixed-point projection, without requiring an identity element in $S_1(A)$.

Theorem 6.7.4.1 (Positive definiteness of the fixed point under map irreducibility). [Lean](#) [✓ Stmt](#) Let $D > 0$, let (A, ρ) be primitive in the sense of Definition 6.6.1, and let $\mathcal{E}_A$ be irreducible. Then $\rho > 0$.

Proof. [✓ Proof](#) Let P be the support projection of ρ . Since $\mathcal{E}_A$ is completely positive and fixes ρ , the support of $\mathcal{E}_A(X)$ lies in the support of ρ for every $X \geq 0$ supported in P , so $\mathcal{E}_A$ maps the corner $PM_D(\mathbb{C})P$ into itself. Irreducibility forces $P = 0$ or $P = \mathbb{1}$, and $P \neq 0$ because $\rho \neq 0$; hence $P = \mathbb{1}$ and $\rho > 0$. $\square$

Theorem 6.7.4.2 (Strongly irreducible tensors are primitive with a positive-definite fixed point). [Lean](#) [✓ Stmt](#) Let $D > 0$ and suppose $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. If $\mathcal{E}_A$ is strongly irreducible in the sense of Definition 6.1.1.6, then there is a matrix $\rho > 0$ such that (A, ρ) is primitive in the sense of Definition 6.6.1.

Proof. ✓ Proof Irreducibility of $\mathcal{E}_A$ gives irreducibility of the tensor A by the last clause of Lemma D.4.9: if P were a nontrivial invariant projection of A , then $\mathcal{E}_A$ would map the corner $PM_D(\mathbb{C})P$ into itself. Tensor irreducibility, the trace-preserving normalization, and the peripheral spectrum $\{1\}$ then give primitive fixed-point data (A, ρ) with $\rho \geq 0$ nonzero by the first clause of the same lemma. Finally, Theorem 6.7.4.1 makes ρ positive definite because $\mathcal{E}_A$ is irreducible. $\square$

Lemma 6.7.4.3 (Primitive tensors with positive-definite fixed point are strongly irreducible).

Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ , then it satisfies the strengthened condition in Definition 6.1.1.6: $\mathcal{E}_A$ is irreducible, the fixed point ρ is positive definite, and the peripheral spectrum of $\mathcal{E}_A$ is $\{1\}$. The source clause is [200, Proposition 3(c)]; the definition used here records irreducibility separately.*

Proof. ✓ Proof Lemma C.5.7 gives irreducibility of $\mathcal{E}_A$. The complementary-gap hypothesis gives peripheral primitivity directly: the only eigenvalue of modulus one is 1. Together with the positive-definite fixed point ρ , these are exactly the three conditions in Definition 6.1.1.6. The complementary-gap consequences are proved in Section C.5. $\square$

Theorem 6.7.4.4 (Primitive MPS with positive-definite fixed point is normal). Lean ✓ Stmt *If A is a primitive MPS tensor with positive-definite fixed point ρ , then A is normal (eventually full Kraus rank).*

Proof. ✓ Proof By Lemma 6.7.4.3, A is strongly irreducible. Theorem 6.1.1.15 identifies strong irreducibility with eventual full Kraus rank, hence with normality. $\square$

6.8 One-step padding and exact word spans

Cumulative spanning $T_N(A) = M_D(\mathbb{C})$ is weaker than exact-length spanning $S_L(A) = M_D(\mathbb{C})$. When $\mathbb{1} \in S_1(A)$, products of shorter words pad directly to a common length. Supporting proofs appear in Section C.6.

Theorem 6.8.1 (Homogeneous word-span propagation under normalization). Lean Lean Lean ✓ Stmt *Suppose*

$$\sum_i A^i (A^i)^\dagger = \mathbb{1}, \tag{6.4}$$

and $S_L(A) = M_D(\mathbb{C})$. Then $S_m(A) = M_D(\mathbb{C})$ for every $m \geq L$. Equivalently, L -block injectivity propagates to every larger homogeneous length.

Proof. ✓ Proof If $S_n(A) = M_D(\mathbb{C})$, then for every $X \in M_D(\mathbb{C})$, substituting (6.4) gives

$$X = \left(\sum_i A^i (A^i)^\dagger \right) X = \sum_i A^i ((A^i)^\dagger X) \in S_1(A) S_n(A) \subseteq S_{n+1}(A).$$

Indeed, $(A^i)^\dagger X \in S_n(A)$ because $S_n(A) = M_D(\mathbb{C})$. Thus $S_{n+1}(A) = M_D(\mathbb{C})$. Iterating this implication gives the result for every $m \geq L$. $\square$

Theorem 6.8.2 (Algebra span plus one-step padding implies normality). Lean ✓ Stmt *If $\text{alg}(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. ✓ Proof Since $\text{alg}(A) = M_D(\mathbb{C})$, the ascending chain condition for finite-dimensional subspaces gives an N such that $T_N(A) = M_D(\mathbb{C})$. Apply Theorem C.6.5. $\square$

Remark 6.8.3 (Normal canonical form and blocked primitivity). The normal-canonical form developed in Chapters 7–8 operates directly on irreducible trace-preserving blocks whose blocked transfer maps are primitive, following Cirac et al. [11], without requiring an identity element in the single-site word span.

6.9 Block injectivity from TP/primitive/irreducible blocks

Theorem 6.9.1 (TP-primitive irreducible tensor is injective after blocking). Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$. Suppose that A is tensor-irreducible and that its transfer map is primitive. Then there exists a positive blocking length L such that $A^{[L]}$ is one-site injective.*

Proof. ✓ Proof If $D = 1$, trace preservation gives a nonzero one-site Kraus matrix, and every nonzero scalar matrix spans $M_1(\mathbb{C})$. Hence A is one-site injective, so Lemma 2.5.4 gives the positive witness $L = 1$. Suppose now that $D \geq 2$. Theorem 7.11.1.1 gives a block-injectivity witness L . If $L = 0$, Corollary C.1.4 would force $D = 1$, a contradiction. Thus $L > 0$, and the blocked-chain equivalence identifies this L -block injectivity with one-site injectivity of the blocked tensor $A^{[L]}$. □

Remark 6.9.2. Blocking preserves normality by Theorem C.4.1, proved in Section C.4.

Chapter 7

Canonical Form Reduction

The reduction of a translation-invariant matrix product state tensor to a weighted direct sum of primitive blocks follows [193], [11], [16], and [212, Chapter 6]. The transfer map of an arbitrary tensor may be reducible and periodic. The reduction decomposes the bond space into irreducible subspaces. For each irreducible block in left-canonical form, powering the transfer map collapses the peripheral spectrum to roots of unity, whose cyclic projections decompose the tensor into primitive, tensor-irreducible corners. When the transfer map has period p , [193, Theorem 5] expresses the state vector for $p \mid N$ as a sum of p cyclic component chains, and gives zero when $p \nmid N$.

Definition 7.1 (Periodic projector-component chain). Lean Lean Lean ✓ Stmt Let $A = (A^i)_{i=0}^{d-1}$ have bond dimension D , let $(Q_u)_{u \in \mathbb{Z}/p\mathbb{Z}}$ be cyclic projections, and fix a starting sector u . For each $0 \leq j < N$, the corresponding chain of length N has local matrices

$$B_{u,j}^i = Q_{u+j} A^i Q_{u+j+1}, \quad (7.1)$$

where sector indices are read modulo p . It has the same bond dimension D as A and repeats with period p .

Theorem 7.2 (Boundary traces and component-chain coefficients). Lean Lean ✓ Stmt Suppose that the orthogonal projections satisfy $\sum_u Q_u = \mathbb{1}$ and $Q_u A^i = A^i Q_{u+1}$. For a word $\sigma = (i_0, \dots, i_{N-1})$, define the boundary product

$$F_u(\sigma) = Q_u A^{i_0} Q_{u+1} A^{i_1} \dots Q_{u+N-1} A^{i_{N-1}} Q_{u+N}. \quad (7.2)$$

At every length $N \geq 0$,

$$V^{(N)}(A)_\sigma = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} \text{tr} F_u(\sigma). \quad (7.3)$$

If $N > 0$, then $\text{tr} F_u(\sigma)$ is the coefficient of the component chain B_u from Definition 7.1. At $N = 0$, the boundary product is $F_u(\emptyset) = Q_u$, so

$$\text{tr}(\mathbb{1}) = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} \text{tr}(Q_u). \quad (7.4)$$

Proof. ✓ Proof The cyclic relation and orthogonality give $F_u(\sigma) = Q_u A^{i_0} \dots A^{i_{N-1}}$. Hence

$$\text{tr}(A^{i_0} \dots A^{i_{N-1}}) = \text{tr} \left(\left(\sum_u Q_u \right) A^{i_0} \dots A^{i_{N-1}} \right) = \sum_u \text{tr} F_u(\sigma). \quad (7.5)$$

For $N > 0$, multiplying the local matrices in (7.1) gives $F_u(\sigma)$, proving the coefficient identity. For $N = 0$, the same trace calculation reduces to (7.4). $\square$

Theorem 7.3 (Normalized periodic state-vector decomposition). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *This is a normalized form of the state-vector conclusion in [193, Theorem 5]. Let A be an irreducible left-canonical tensor whose peripheral transfer spectrum is the set of p -th roots of unity. If $p \mid N$, there are orthogonal cyclic projections $(Q_u)_{u \in \mathbb{Z}/p\mathbb{Z}}$ such that $\sum_u Q_u = \mathbb{1}$, $Q_u A^i = A^i Q_{u+1}$, and $Q_{u+N} = Q_u$. For every ring length $N \geq 0$, its coefficients satisfy the boundary-aware decomposition*

$$V^{(N)}(A)_\sigma = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} \text{tr } F_u(\sigma). \quad (7.6)$$

For $N > 0$, this becomes the source's decomposition into the p explicit component chains:

$$|V^{(N)}(A)\rangle = \sum_{u \in \mathbb{Z}/p\mathbb{Z}} |V^{(N)}(B_u)\rangle. \quad (7.7)$$

Each B_u is a p -periodic MPS of bond dimension D with local matrices given by (7.1). At $N = 0$, (7.6) reduces to the boundary traces $\text{tr}(Q_u)$ from (7.4). If $p \nmid N$, then $|V^{(N)}(A)\rangle = 0$.

Proof. [✓ Proof](#) Choose the cyclic projections supplied by periodicity. If $p \mid N$, then $Q_{u+N} = Q_u$, and Theorem 7.2 gives (7.6) at every length. For $N > 0$, the explicit component-chain decomposition follows by inserting $\sum_u Q_u = \mathbb{1}$ and multiplying the local corner matrices; this proves (7.7). The separate coefficient identity in Theorem 7.2 identifies its summands with the boundary traces. If $p \nmid N$, then $Q_{u+N} \neq Q_u$; cyclicity of the trace and orthogonality give $\text{tr } F_u(\sigma) = 0$ for every u and σ . $\square$

Two normalizations appear. The unital orientation $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ is used in [193, Theorem 4]; the trace-preserving (left-canonical) orientation $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ is used in the TP-gauge and after-blocking reductions below. They exchange under the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, whose transfer map is the Frobenius adjoint of $\mathcal{E}_A$.

7.1 Invariant-subspace decomposition and irreducible blocks

The block decomposition rests on one principle: a positive semidefinite fixed point of $\mathcal{E}_A$ that is not full rank has a proper invariant support, which block-diagonalizes the Kraus operators without changing the MPV family. Iterating the resulting splitting decomposes any tensor into irreducible blocks.

Lemma 7.1.1 (Lower-triangular Kraus condition implies corner invariance). [Lean](#) [✓ Stmt](#) *Let A be an MPS tensor and let P be an orthogonal projection. If $(\mathbb{1} - P)A^i P = 0$ for every physical index i , then $P\mathcal{E}_A(PXP)P = \mathcal{E}_A(PXP)$ for every $X \in M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) Expand the transfer map using (2.4) and use $(\mathbb{1} - P)A^i P = 0$ to reduce each Kraus term to the P -corner. $\square$

Theorem 7.1.2 (Invariant projection gives a two-block decomposition). [Lean](#) [✓ Stmt](#) *Suppose P is an orthogonal projection satisfying $(\mathbb{1} - P)A^i P = 0$ for every physical index i . Then there exist n, m with $n + m = D$ and MPS tensors A_1, A_2 of bond dimensions n and m such that A and the two-block tensor $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.4.3.*

Proof. ✓ Proof Diagonalize P by a unitary U . Since $P^2 = P$, the diagonal form of P has only 0- and 1-eigenvalues, so the basis splits as $n + m = D$. Conjugating A by U preserves the MPV family by Theorem D.1.1. In this basis the relation $(\mathbb{1} - P)A^iP = 0$ makes each letter upper block triangular. Write the conjugated letters as

$$B^i = \begin{pmatrix} A_1^i & * \\ 0 & A_2^i \end{pmatrix}.$$

Since $B^{i_1} \dots B^{i_N}$ is again upper block triangular, its trace satisfies

$$\text{tr}(B^{i_1} \dots B^{i_N}) = \text{tr}(A_1^{i_1} \dots A_1^{i_N}) + \text{tr}(A_2^{i_1} \dots A_2^{i_N}). \quad (7.8)$$

Hence the block-diagonal tensor $A_1 \oplus A_2$ has the same MPV coefficients as the conjugated tensor. $\square$

Definition 7.1.3 (Invariant projection condition). Lean ✓ Stmt An MPS tensor A has an *invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^iP = 0$ for all i . A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 7.1.1.1).

Theorem 7.1.4 (PSD fixed point gives invariant projection). Lean ✓ Stmt Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be its support projection. Then $(\mathbb{1} - P)A^iP = 0$ for all i : the range of P is invariant under every A^i .

Proof. ✓ Proof From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i\rho(A^i)^\dagger$, one gets $(\mathbb{1} - P)A^i\rho(A^i)^\dagger(\mathbb{1} - P) = 0$ for each i . Since ρ is supported on the range of P , this forces $(\mathbb{1} - P)A^iP = 0$.

This is the finite-dimensional case of [212, Proposition 6.10]: the support projection of any stationary state is sub-harmonic. The equivalence between irreducibility and absence of nontrivial invariant projections is [212, Theorem 6.2]; see also [16, Section IV.A.1]. $\square$

Theorem 7.1.5 (Two-block decomposition). Lean ✓ Stmt If $\mathcal{E}_A$ has a nonzero PSD fixed point ρ that is not positive definite, then there exist MPS tensors A_1, A_2 of smaller bond dimensions with $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$.

Proof. ✓ Proof The support projection P of ρ is neither 0 nor $\mathbb{1}$, since $\rho \neq 0$ and ρ is not positive definite. Theorem 7.1.4 gives $(\mathbb{1} - P)A^iP = 0$, so a unitary change of basis diagonalizing P puts A^i in upper-triangular block form with diagonal blocks B_{11}^i, B_{22}^i . Theorem D.1.1 preserves the MPV under the conjugation, and the trace identity (7.8) shows that deleting the off-diagonal blocks preserves all MPV coefficients. $\square$

7.1.1 Iterated reduction to irreducible blocks

Iterating the two-block splitting decomposes any tensor into irreducible blocks of strictly smaller bond dimensions.

Definition 7.1.1.1 (Irreducible tensor). Lean ✓ Stmt An MPS tensor A is *irreducible* if it has no nontrivial invariant projection.

Theorem 7.1.1.2 (Irreducible block decomposition). Lean ✓ Stmt Every MPS tensor A admits a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ with each A_k irreducible, $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$, and $\sum_{k=1}^r D_k = D$.

Proof. ✓ **Proof** By strong induction on D . If A is irreducible, take $r = 1$. Otherwise a nontrivial invariant projection P gives the upper-triangular splitting of Theorem 7.1.5; deleting the off-diagonal part preserves the MPV family and yields two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block.

Both [193, Theorem 4] and [16, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting. $\square$

7.2 Nonzero blocks and the trace-preserving gauge

In an irreducible block decomposition some blocks may have all-zero Kraus operators. Such a block contributes nothing to a matrix product vector at any positive length and is therefore discarded. The structural direct-sum identity implies that the retained block dimensions satisfy $\sum_k D_k \leq D$ without using an empty-word coefficient. Each retained block is then placed in the trace-preserving gauge.

Theorem 7.2.1 (Blockwise TP-gauge normalization of irreducible blocks). Lean ✓ **Stmt** Suppose A generates the same MPV family as $\bigoplus_{k=1}^r B_k$, where each B_k is irreducible with at least one nonzero Kraus operator. Then there are positive real weights $(\mu_k)_{k=1}^r$ and blocks $(C_k)_{k=1}^r$ such that, for every N and every σ ,

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k C_k\right)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(C_k)_\sigma, \quad (7.9)$$

each C_k is irreducible and left-canonical, $\sum_i (C_k^i)^\dagger C_k^i = \mathbb{1}$, with positive bond dimension.

Proof. ✓ **Proof** The Perron–Frobenius TP-gauge construction is applied to each irreducible block using the gauge (4.3). Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the weights μ_k . $\square$

7.2.1 Trace-preserving gauge for arbitrary tensors

Theorem 7.2.1.1 (TP-gauge reduction for arbitrary tensors). Lean ✓ **Stmt** For any MPS tensor A , there exist nontrivial blocks $(B_k)_{k=1}^r$ with positive real weights $(\mu_k)_{k=1}^r$, each block irreducible, left-canonical ($\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$), and of positive bond dimension, such that, for every $N \geq 1$ and every σ ,

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k B_k\right)_\sigma, \quad (7.10)$$

and $\sum_{k=1}^r D_k \leq D$.

Proof. ✓ **Proof** Theorem D.1.3 supplies the nonzero irreducible blocks $(C_k)_{k=1}^r$, with $V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma$ for every $N \geq 1$ and every σ . Applying the Perron–Frobenius TP gauge (4.3) to each C_k gives a left-canonical block B_k with weight μ_k , and

$$V^{(N)}\left(\bigoplus_{k=1}^r C_k\right)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k B_k\right)_\sigma.$$

The gauge preserves every block dimension, so the structural bound $\sum_k D_k \leq D$ is unchanged. $\square$

7.3 Blocking, period removal, and primitive blocks

Blocking by an integer p carries MPV equalities, weights, transfer maps, and identifications of the blocked physical alphabet. For an irreducible trace-preserving block, powering first collapses the peripheral eigenvalue set to $\{1\}$; the powered map can nevertheless split into several cyclic corners. The reduction compresses to those corners and proves primitivity there. A later least-common-multiple step places a finite family of the resulting sectors over one blocked alphabet.

7.3.1 Period removal by cyclic sectors

The period-removing blocking of [11, Section 2.3] has four distinct steps. Power the transfer map by its period, obtain the adjoint-fixed cyclic projections, compress to their corners, and prove that each corner restriction is irreducible. Only then does the peripheral set $\{1\}$ imply primitivity of the sector maps. The cyclic projections come from [212, Theorem 6.6].

Theorem 7.3.1.1 (Period removal by blocking). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period-removing length $m \geq 1$ such that $A^{[m]}$ admits:*

1. left-canonical sector tensors $(C_k)_{k=1}^m$ of nonzero bond dimension,
2. a unit-weight direct-sum MPV decomposition $A^{[m]} \sim \bigoplus_{k=1}^m C_k$,
3. orthogonal projections $(P_k)_{k=1}^m$ summing to $\mathbb{1}$ with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$,
4. commutation relations $P_k(A^{[m]})^i = (A^{[m]})^i P_k$ for every blocked letter i ,
5. the sector trace formula $f_{C_k}(\sigma) = \text{tr}(P_k(A^{[m]})^{\sigma_1} \dots (A^{[m]})^{\sigma_N})$,
6. linear isomorphisms onto the compression corners $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$,
7. transfer-intertwining identities $\varphi_k(\mathcal{E}_{C_k^\dagger}(X)) = \mathcal{E}_{(P_k A^{[m]})^\dagger}(\varphi_k(X))$,
8. multiplicativity and adjoint compatibility $\varphi_k(XY) = \varphi_k(X)\varphi_k(Y)$ and $\varphi_k(X^\dagger) = \varphi_k(X)^\dagger$.

This is the cyclic-sector part of the period-removal step of [11, Section 2.3]; the projections P_k come from the peripheral spectrum of the adjoint transfer map ([212, Theorem 6.6]).

Proof. ✓ Proof The peripheral cyclic-structure theorem identifies the period and its primitive root. The cyclic decomposition of the adjoint transfer map gives projections (P_k) with $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating the cyclic relation m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and the blocked-transfer identity $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$ identifies these with the adjoint transfer map of $A^{[m]}$. Apply the block decomposition from adjoint-fixed projections. $\square$

Theorem 7.3.1.2 (Unconditional primitive blocked sectors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum $\{\gamma^j : 0 \leq j < m\}$ for a primitive m th root of unity γ , with blocked cyclic-sector decomposition $(C_u)_{u=1}^m, (P_u)_{u=1}^m$, and linear isomorphisms $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$. Assume explicitly that the P_u are orthogonal projections summing to $\mathbb{1}$ and satisfying $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$, that each sector has nonzero bond dimension, and that the compression maps satisfy*

$$\begin{aligned} \varphi_u(\mathcal{E}_{C_u^\dagger}(X)) &= \mathcal{E}_{(P_u A^{[m]})^\dagger}(\varphi_u(X)), \\ \varphi_u(XY) &= \varphi_u(X)\varphi_u(Y), \\ \varphi_u(X^\dagger) &= \varphi_u(X)^\dagger. \end{aligned} \tag{7.11}$$

Here $P_u A^{[m]}$ denotes the blocked tensor with letters $P_u(A^{[m]})^\alpha$. Then every sector tensor satisfies $\sigma_\partial(\mathcal{E}_{C_u}) = \{1\}$ and is tensor-irreducible.

Proof. ✓ Proof Theorem 7.4.3 gives irreducibility of $(\mathcal{E}_A^\dagger)^m$ on each corner P_u . The compression equivalences in (7.11) intertwine the adjoint transfer map of C_u with the corresponding corner restriction of $(\mathcal{E}_A^\dagger)^m$. The peripheral spectrum of the blocked adjoint map is

$$\sigma_\partial((\mathcal{E}_A^\dagger)^m) = \{(\gamma^j)^m : 0 \leq j < m\} = \{1\}, \quad (7.12)$$

because γ is a primitive m th root of unity. Hence each nonzero irreducible corner restriction is primitive. Primitivity and irreducibility are preserved across φ_u ; then use Theorem D.4.4 to pass from the adjoint blocked map to the primal transfer map of C_u . $\square$

Theorem 7.3.1.3 (Period removal with primitive irreducible sectors). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor. Then there is a period $m \geq 1$ and a unit-weight decomposition of $A^{[m]}$ into sector blocks $(C_k)_{k=1}^m$, each of which is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension.*

Proof. ✓ Proof Left-canonicity makes $K_i := (A^i)^\dagger$ unital, irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point; the cyclic peripheral-spectrum theorem and Theorem 7.3.1.1 then produce the cyclic-sector decomposition. Keep the projections and compression equivalences, and apply Theorem 7.3.1.2 to the sector blocks. $\square$

Theorem 7.3.1.4 (Common blocking length for cyclic sectors). Lean ✓ Stmt *If two tensors generate the same MPV family at every positive length, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length p . For every $N \geq 1$ and every blocked σ , each blocked tensor equals its powered nonzero part:*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\left(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\right)_\sigma, \quad (7.13)$$

$$V^{(N)}(B^{[p]})_\sigma = V^{(N)}\left(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]}\right)_\sigma. \quad (7.14)$$

The two powered nonzero parts agree:

$$V^{(N)}\left(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]}\right)_\sigma = V^{(N)}\left(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]}\right)_\sigma. \quad (7.15)$$

Proof. ✓ Proof Start from Theorem D.5.4. With p_A, p_B the least common multiples of the period-removal lengths on the two sides, use $p = \text{lcm}(p_A, p_B)$. The prescribed-common-multiple lemma constructs both one-sided sector families at this length, and the positive-length blocking lemmas carry the one-sided tensor/nonzero-part equalities over to the common alphabet, giving (7.13) and (7.14). The two-sided blocking lemma applied to (D.16) gives (7.15). $\square$

7.4 Cyclic-sector irreducibility

The powered transfer map is analyzed on each corner projection before the sector families are placed over a common blocked alphabet.

Theorem 7.4.1 (Fixed-point upgrade on an invariant corner). [Lean](#) [✓ Stmt](#) Let A have positive bond dimension, suppose that $\mathcal{E}_A$ is irreducible, and assume $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Let Q be an orthogonal projection. If $(\mathcal{E}_{A^\dagger})^m$ preserves the corner $QM_D(\mathbb{C})Q$, then

$$(\mathcal{E}_{A^\dagger})^m(Q) = Q.$$

Proof. [✓ Proof](#) Choose a positive-definite fixed point ρ of $\mathcal{E}_A$. The weighted trace $X \mapsto \text{tr}(\rho X)$ is invariant under $\mathcal{E}_{A^\dagger}$ and its powers. Corner preservation gives

$$Q - (\mathcal{E}_{A^\dagger})^m(Q) = Q(\mathcal{E}_{A^\dagger})^m(\mathbb{1} - Q)Q \geq 0.$$

The invariant weighted trace of this positive semidefinite difference is zero. Since $\rho > 0$, the difference vanishes. $\square$

Theorem 7.4.2 (Fixed-point algebra rigidity on cyclic sectors). [Lean](#) [✓ Stmt](#) Let A be an irreducible MPS tensor with positive bond dimension and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Let $(P_k)_{k \in \mathbb{Z}/m\mathbb{Z}}$ be orthogonal projections such that

$$\sum_k P_k = \mathbb{1}, \quad \mathcal{E}_{A^\dagger}(P_{k+1}) = P_k.$$

If $X, Y \in P_k M_D(\mathbb{C}) P_k$ are fixed by $(\mathcal{E}_{A^\dagger})^m$, then

$$\mathcal{E}_{A^\dagger}(XY) = \mathcal{E}_{A^\dagger}(X)\mathcal{E}_{A^\dagger}(Y).$$

Proof. [✓ Proof](#) Irreducibility passes to the adjoint transfer map. The cyclic relation gives equality in both Kadison–Schwarz inequalities:

$$\begin{aligned} \mathcal{E}_{A^\dagger}(P_k^\dagger P_k) &= \mathcal{E}_{A^\dagger}(P_k)^\dagger \mathcal{E}_{A^\dagger}(P_k), \\ \mathcal{E}_{A^\dagger}(P_k P_k^\dagger) &= \mathcal{E}_{A^\dagger}(P_k) \mathcal{E}_{A^\dagger}(P_k)^\dagger. \end{aligned}$$

Hence every P_k lies in the multiplicative domain, so for every Z ,

$$\begin{aligned} \mathcal{E}_{A^\dagger}(P_k Z) &= \mathcal{E}_{A^\dagger}(P_k) \mathcal{E}_{A^\dagger}(Z), \\ \mathcal{E}_{A^\dagger}(Z P_k) &= \mathcal{E}_{A^\dagger}(Z) \mathcal{E}_{A^\dagger}(P_k). \end{aligned}$$

If $X \in P_k M_D(\mathbb{C}) P_k$ is fixed by $(\mathcal{E}_{A^\dagger})^m$, then its orbit sum

$$R_X = \sum_{\ell=0}^{m-1} (\mathcal{E}_{A^\dagger})^\ell(X)$$

is fixed by $\mathcal{E}_{A^\dagger}$. By Theorem 7.8.1, $R_X = c\mathbb{1}$. Orthogonality of the cyclic sectors and the multiplicative-domain identities give $X = P_k R_X P_k = c P_k$. Similarly, $Y = c' P_k$. Since $\mathcal{E}_{A^\dagger}(P_k) = P_{k-1}$, both sides of the required identity equal $cc' P_{k-1}$. $\square$

Lemma 7.4.3 (Sector irreducibility for the adjoint blocked map). [Lean](#) [✓ Stmt](#) Let A be a left-canonical irreducible MPS tensor with $D > 0$, let m be the period of its cyclic-sector decomposition, and let $(P_u)_{u=1}^m$ be the cyclic projections of the adjoint transfer map with $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ and $\sum_u P_u = \mathbb{1}$. Then the restriction of $(\mathcal{E}_{A^\dagger})^m$ to each corner $P_u M_D(\mathbb{C}) P_u$ is irreducible.

Definition 7.4.4 (Primitive irreducible cyclic-sector decomposition). [Lean](#) [✓ Stmt](#) A has primitive irreducible cyclic sectors of period $m \geq 1$ if there is a family $(C_k)_{k=1}^m$ with $A^{[m]} \sim \bigoplus_{k=1}^m C_k$ with unit weights, each C_k left-canonical, with primitive transfer map, tensor-irreducible, and of positive bond dimension.

7.5 Translation-invariant canonical form

Definition 7.5.1 (Unnormalized PGVWC07 canonical decomposition at positive lengths). [Lean](#)

[✓ Stmt](#) Consider a decomposition of a tensor A into finitely many blocks $(A_k)_{k=1}^r$ of positive bond dimensions and positive real weights $(a_k)_{k=1}^r$. Each block is unital, has only scalar transfer-map fixed points, and has a diagonal positive-definite fixed point Λ_k for the adjoint transfer map. Moreover,

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r a_k A_k\right)_\sigma \quad (N > 0),$$

and the retained bond dimensions satisfy $\sum_k \dim(A_k) \leq D$.

Theorem 7.5.2 (Existence of the PGVWC07 canonical decomposition). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Every MPS tensor admits a decomposition satisfying Definition 7.5.1.

Proof. [✓ Proof](#) Follow [193, Theorem 4]. The nonzero irreducible decomposition is obtained by strict dimension induction. On a nonzero current block, Perron–Frobenius gives a positive semidefinite eigenvector at a positive spectral radius. If it is singular, positivity makes its support invariant: $(\mathbb{1} - P)A^i P = 0$. After diagonalizing P , every letter is upper block triangular and the finite-ring trace splits as

$$\mathrm{tr}(A^{i_1} \dots A^{i_N}) = \mathrm{tr}(A_1^{i_1} \dots A_1^{i_N}) + \mathrm{tr}(A_2^{i_1} \dots A_2^{i_N}).$$

Discard all-zero blocks and repeat on the two strictly smaller diagonal blocks. This gives the finite nonzero irreducible family of Theorem D.1.3.

For each retained irreducible block, divide by the square root of its positive spectral radius and conjugate by the square root of its positive-definite Perron eigenvector. Theorem 4.5.4 gives the unital identity

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}.$$

Nonzero scalar rescaling preserves irreducibility. The proof passes from tensor irreducibility to irreducibility of the matrix action, carries invariant subspaces through the gauge similarity, and returns to tensor irreducibility using Theorem 6.7.3.4 and Lemma C.2.2. Hence Theorem 7.8.1 makes every fixed point scalar. Theorem 7.10.2 then diagonalizes a positive-definite dual fixed point by a unitary without changing unitality.

Apply this single-block construction to every member of the finite family. The square roots of the block spectral radii become positive weights, gauge invariance preserves every positive-length MPV coefficient, and the block dimensions are unchanged. Combining this blockwise result with the initial decomposition yields the arbitrary-input unnormalized form and its bound $\sum_k \dim(A_k) \leq D$. $\square$

Theorem 7.5.3 (Normalization of a nonempty PGVWC07 decomposition). [Lean](#) [Lean](#) [✓ Stmt](#)

Let A_k, a_k satisfy Definition 7.5.1. If some positive-length MPV coefficient of the original tensor is nonzero, then the decomposition has at least one block. For a nonempty block family, define

$$s := \max_{1 \leq k \leq r} a_k, \\ \nu_k := \frac{a_k}{s}.$$

Then $s > 0$, every ν_k is positive, $|\nu_k| \leq 1$, and $|\nu_k| = 1$ for at least one k . For every $N > 0$,

$$V^{(N)}(A)_\sigma = s^N V^{(N)}\left(\bigoplus_k \nu_k A_k\right)_\sigma. \quad (7.16)$$

Proof. [✓ Proof](#) If the decomposition had no blocks, its weighted block tensor would have zero MPV coefficients at every positive length, contrary to the assumed nonzero coefficient. Thus $r > 0$. Since the finite family (a_k) is nonempty and every $a_k > 0$, its maximum s is positive. The definition $\nu_k = a_k/s$ gives $0 < \nu_k \leq 1$, and any maximizing index has $\nu_k = 1$. Finally, $a_k = s\nu_k$ for every k , so homogeneity of a length- N matrix product vector gives (7.16). $\square$

Theorem 7.5.4 (Normalized exact form for a nonzero MPV family). [Lean](#) [✓ Stmt](#) Suppose that $V^{(N)}(A)_\sigma \neq 0$ for some $N > 0$ and some word σ . Then there are a scalar $s > 0$ and a nonempty finite family of positive-bond-dimension blocks $(A_k)_{k=1}^r$ with positive weights $(\mu_k)_{k=1}^r$ such that $s^{-1}A$ and $\bigoplus_k \mu_k A_k$ have identical MPV coefficients at every positive length. The weights satisfy $|\mu_k| \leq 1$, with equality for at least one block, and

$$\begin{aligned} \sum_i A_k^i (A_k^i)^\dagger &= \mathbb{1}, \\ \sum_i (A_k^i)^\dagger \Lambda_k A_k^i &= \Lambda_k \end{aligned} \quad (7.17)$$

for a diagonal positive-definite matrix Λ_k . Every fixed point of $\mathcal{E}_{A_k}$ is a scalar multiple of $\mathbb{1}$, and $\sum_k \dim(A_k) \leq D$.

Proof. [✓ Proof](#) Choose the blocks and weights from Theorem 7.5.2. The assumed nonzero coefficient makes its block family nonempty, so Theorem 7.5.3 supplies $s > 0$ and the normalized positive weights. Equation (7.16) and homogeneity give

$$V^{(N)}(s^{-1}A)_\sigma = s^{-N} V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_k \mu_k A_k\right)_\sigma.$$

The decomposition satisfies all required block conditions and the bond-dimension bound. $\square$

Theorem 7.5.5 (Zero/nonzero alternative for the normalized exact form). [Lean](#) [✓ Stmt](#) For every MPS tensor A , one of the following alternatives holds:

1. $V^{(N)}(A)_\sigma = 0$ for every $N > 0$ and every word σ ;
2. there are a scalar $s > 0$ and a nonempty normalized block family satisfying all conclusions of Theorem 7.5.4.

Proof. [✓ Proof](#) Split according to whether there exist $N > 0$ and σ with $V^{(N)}(A)_\sigma \neq 0$. In that case apply Theorem 7.5.4. If no such pair exists, every positive-length coefficient vanishes. $\square$

Theorem 7.5.6 (Translation-invariant canonical form). [Lean](#) [✓ Stmt](#) Let A be a translation-invariant MPS representation of bond dimension D . There is a positive scalar s and a finite family of blocks A_k , possibly empty, such that the globally rescaled tensor $s^{-1}A$ has the same MPV coefficients on every positive-length ring as $\bigoplus_k \mu_k A_k^i$. The weights are positive and satisfy $1 \geq |\mu_k|$; if the family is nonempty, then some weight has norm 1. For each block,

$$\begin{aligned} \sum_i A_k^i (A_k^i)^\dagger &= \mathbb{1}, \\ \sum_i (A_k^i)^\dagger \Lambda_k A_k^i &= \Lambda_k. \end{aligned} \quad (7.18)$$

Here Λ_k is diagonal, positive, and full-rank, and each block has positive bond dimension. Moreover, the only fixed points of $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$ are the scalar multiples of $\mathbb{1}$: for every X , the identity $\mathcal{E}_{A_k}(X) = X$ implies that there exists $c \in \mathbb{C}$ such that $X = c\mathbb{1}$. The total bond dimension of the decomposition is at most D .

This is the positive-length form of [193, Theorem 4], with the global spectral-radius normalization step made explicit.

Proof. ✓ Proof Apply Theorem 7.5.5. If every positive-length coefficient vanishes, take $s = 1$ and the empty retained family; the block conditions and bond-dimension bound are vacuous, and the MPV identity is the zero identity. In the nonzero branch, use the nonempty normalized family supplied by the second alternative. $\square$

Remark 7.5.7 (CPSV blocked canonical form (source statement)). The blocked canonical-form reduction of [11, Section 2.3] and [16, Section IV.A] states that an arbitrary translation-invariant MPS tensor becomes, after blocking, a weighted family of normal tensors, up to a summand that vanishes on every positive-length ring. Four statements must be distinguished. Theorem 7.5.6 gives the unconditional PGVWC07 normalized exact form before period removal. Theorem 7.11.1.2 gives the prepared trace-preserving primitive-block reduction. For a family with a nonzero positive-length coefficient, Theorem E.3.7 normalizes the weights and Theorem 8.6.3.14 gives the normalized BNT form, including its per-site scalar. If every positive-length coefficient vanishes, there is no nonempty family from which to choose a unit-modulus weight; this exceptional zero family is represented by the empty retained family instead.

7.6 Normal tensors and canonical forms

The definitions of *normal tensor* and *canonical form* follow [11]. For a transfer map $\mathcal{E}$ with spectral radius $r(\mathcal{E})$, write

$$\sigma_\partial(\mathcal{E}) = \{\lambda \in \sigma(\mathcal{E}) : |\lambda| = r(\mathcal{E})\} \quad (7.19)$$

for its peripheral spectrum. The weight moduli are ordered non-increasingly, $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r| > 0$; equal-modulus and repeated sectors are retained and compared by the sector coefficients in Chapter 8.

Definition 7.6.1 (CPSV normal tensor (NT)). Lean ✓ Stmt A tensor A of physical dimension d and bond dimension D is a *normal tensor* if, after the spectral-radius normalization of [11], (i) A admits no nontrivial invariant orthogonal projection and (ii) the associated CPM $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ has spectral radius $r(\mathcal{E}_A) = 1$ and peripheral spectrum $\sigma_\partial(\mathcal{E}_A) = \{1\}$.

Remark 7.6.2. Condition (i) cannot be dropped. A reducible tensor may have peripheral spectrum $\{1\}$ while still admitting a nontrivial invariant projection, so condition (ii) alone does not characterize normality.

Theorem 7.6.3 (Algebraically normal left-canonical tensors are CPSV normal). Lean ✓ Stmt Let $D > 0$. If A is algebraically normal and left-canonical, then

$$A \text{ is a CPSV normal tensor.} \quad (7.20)$$

Proof. ✓ Proof Algebraic normality together with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ makes the transfer map strongly irreducible. It consequently has a positive-definite fixed point, no nontrivial invariant projection, and peripheral spectrum $\{1\}$. The fixed-point eigenvalue and left-canonical normalization give $r(\mathcal{E}_A) = 1$. Thus no scalar rescaling remains, and the two conditions in Definition 7.6.1 hold for A itself. $\square$

Theorem 7.6.4 (Normality and left-canonical normalization under entrywise conjugation).

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a tensor or matrix, an overline denotes entrywise conjugation. If A and B are gauge equivalent through X , then $\overline{A}$ and $\overline{B}$ are gauge equivalent through $\overline{X}$. If A is left-canonical, then $\overline{A}$ is left-canonical. If A is algebraically normal, then $\overline{A}$ is algebraically normal. Independently, if A is a CPSV normal tensor, then $\overline{A}$ is a CPSV normal tensor.

Proof. [✓ Proof](#) Entrywise conjugating $B^i = XA^iX^{-1}$ gives $\overline{B^i} = \overline{X} \overline{A^i} (\overline{X})^{-1}$. Conjugating the left-canonical identity gives

$$\sum_i (\overline{A^i})^\dagger \overline{A^i} = \overline{\sum_i (A^i)^\dagger A^i} = \overline{\mathbb{1}} = \mathbb{1}. \quad (7.21)$$

Entrywise conjugation maps the span of the length- N products of A onto the span of those of $\overline{A}$, so it preserves positive-length block injectivity and hence algebraic normality. For the normal-tensor assertion, the transfer map of $\overline{A}$ is the anti-linear conjugate $\sigma \circ \mathcal{E}_A \circ \sigma$ by (7.22). By the corresponding theorem in the companion quantum-channel volume [182, “Entrywise conjugation transports peripheral eigenvalues”] this conjugation preserves the spectral radius and primitivity, and by Theorem 7.6.6 entrywise conjugation preserves irreducibility. Hence both clauses of Definition 7.6.1 hold for $\overline{A}$. $\square$

Theorem 7.6.5 (Transfer maps and positive diagonal matrices under entrywise conjugation).

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let σ denote entrywise conjugation. For every tensor A , as operators on the matrix space,

$$\mathcal{E}_{\overline{A}} = \sigma \circ \mathcal{E}_A \circ \sigma; \quad (7.22)$$

evaluating at $\overline{X}$ gives, for every matrix X ,

$$\mathcal{E}_{\overline{A}}(\overline{X}) = \overline{\mathcal{E}_A(X)}. \quad (7.23)$$

If Λ is positive definite and diagonal, then

$$\overline{\Lambda} = \Lambda. \quad (7.24)$$

Proof. [✓ Proof](#) Entrywise conjugation commutes with finite sums, matrix products, and conjugate transposition, which gives (7.22) on every matrix, and (7.23) follows since σ is an involution. The diagonal entries of a positive-definite complex matrix are real, while all off-diagonal entries of a diagonal matrix vanish. This proves (7.24). $\square$

Theorem 7.6.6 (Irreducibility under entrywise conjugation). [Lean](#) [✓ Stmt](#) If A is irreducible, then $\overline{A}$ is irreducible.

Proof. [✓ Proof](#) Suppose P is a nontrivial invariant orthogonal projection for $\overline{A}$, so $(\mathbb{1} - P) \overline{A^i} P = 0$ for every i . Conjugating entrywise gives $(\mathbb{1} - \overline{P}) A^i \overline{P} = 0$, and $\overline{P}$ is again a Hermitian idempotent distinct from 0 and $\mathbb{1}$. Hence $\overline{P}$ is a nontrivial invariant orthogonal projection for A , contradicting the irreducibility of A . $\square$

Definition 7.6.7 (CPSV canonical form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A have bond dimension D . It is in the canonical form of [11, Section 2.3] if there are positive integers $D_1, \dots, D_r$, normal tensors A_k of bond dimension D_k , nonzero weights $\mu_k \in \mathbb{C}$, and, writing $S = \sum_k D_k$, a matrix $U \in \mathbb{C}^{S \times D}$ such that $S \leq D$ and, for every physical index i ,

$$UU^\dagger = \mathbb{1}_S, \quad A^i = U^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) U. \quad (\text{CF})$$

Thus the complement of the retained direct sum consists literally of zero bond-space coordinates. In particular, $r = 0$ is possible only when $A = 0$.

The convention in line 246 of [11] is a separate normalization of these data, not part of canonical-form membership:

$$\begin{aligned} |\mu_k| &\leq 1 \quad \text{for every } k, \\ A \neq 0 &\implies \exists k, |\mu_k| = 1. \end{aligned} \tag{7.25}$$

Every weight μ_k is nonzero. This is the reading of [11, Section 2.3] adopted here: the remark in line 219 there places the degenerate case in the block dimensions ($D_k = 0$), and lines 224–225 choose each μ_k so that the transfer map of $\mu_k A_k$ has spectral radius one, which a vanishing weight cannot satisfy. The reading loses nothing, since if $\mu_{k_0} = 0$ then $\mu_{k_0} A_{k_0}^i = 0$ is the zero matrix of size D_{k_0} and

$$\bigoplus_{k=1}^r \mu_k A_k^i = \bigoplus_{k \neq k_0} \mu_k A_k^i \oplus 0_{D_{k_0}}, \tag{7.26}$$

so A^i is the same coisometric image of the direct sum over the remaining indices, with the D_{k_0} coordinates of the omitted summand moved to the zero complement. No ordering or separation condition is imposed.

The basis-of-normal-tensors refinement is Definition 8.1.1; it is not part of canonical form itself.

Theorem 7.6.8 (Physical relabeling preserves CPSV normal and canonical form). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $e : \{0, \dots, d' - 1\} \rightarrow \{0, \dots, d - 1\}$ be a bijection and set

$$(e^* A)^j = A^{e(j)}. \tag{7.27}$$

The transfer maps of A and $e^* A$ agree. In particular, physical relabeling preserves left-canonical normalization and CPSV normality. If A has canonical-form data

$$A^i = U^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) U,$$

then $e^* A$ has the same weights and coisometry and the relabeled normal blocks:

$$(e^* A)^j = U^\dagger \left(\bigoplus_{k=1}^r \mu_k (e^* A_k)^j \right) U. \tag{7.28}$$

Proof. [✓ Proof](#) A bijection of the physical alphabet only permutes the summands in $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$. Hence the transfer map, its spectral radius, its peripheral spectrum, and its invariant orthogonal projections are unchanged. Formula (7.28) is obtained from Formula (CF) by replacing i with $e(j)$. $\square$

Theorem 7.6.9 (Transfer maps under a bond-space gauge). [Lean](#) [✓ Stmt](#) If $B^i = X A^i X^{-1}$ for every physical index i , then there is an invertible matrix C such that

$$\mathcal{E}_B(Y) = C^{-1} \mathcal{E}_A(C Y C^\dagger) (C^\dagger)^{-1}.$$

Proof. [✓ Proof](#) Expand both transfer maps and take $C = X^{-1}$. $\square$

Theorem 7.6.10 (Gauge invariance of transfer quasi-idempotence). [Lean](#) [✓ Stmt](#) Let A and B be gauge-equivalent MPS tensors and let $c \in \mathbb{C}$. Then

$$\mathcal{E}_B^2 = c\mathcal{E}_B \iff \mathcal{E}_A^2 = c\mathcal{E}_A.$$

Proof. [✓ Proof](#) Theorem 7.6.9 identifies the two transfer maps by an algebra automorphism. Such an automorphism preserves composition and scalar multiplication, and therefore preserves the displayed equation. $\square$

Theorem 7.6.11 (Spectral radius under congruence similarity). [Lean](#) [✓ Stmt](#) For an invertible matrix C , the linear maps

$$E \quad \text{and} \quad Y \mapsto C^{-1}E(CYC^\dagger)(C^\dagger)^{-1}$$

have the same spectral radius.

Proof. [✓ Proof](#) The two maps are conjugate by the invertible linear transformation $Y \mapsto CYC^\dagger$, and therefore have the same spectrum. $\square$

Theorem 7.6.12 (Primitivity under congruence similarity). [Lean](#) [✓ Stmt](#) For an invertible matrix C , a linear map E is primitive if and only if

$$Y \mapsto C^{-1}E(CYC^\dagger)(C^\dagger)^{-1}$$

is primitive.

Proof. [✓ Proof](#) Congruence similarity is an algebra isomorphism on the endomorphism algebra and hence preserves the peripheral spectrum. $\square$

Theorem 7.6.13 (Gauge transport of CPSV normality). [Lean](#) [✓ Stmt](#) Let X be invertible and suppose

$$B^i = XA^iX^{-1}$$

for every physical index i . If B is a CPSV normal tensor, then A is a CPSV normal tensor.

Proof. [✓ Proof](#) With $S_X(Y) = XYX^\dagger$, the transfer maps satisfy

$$\mathcal{E}_B = S_X \circ \mathcal{E}_A \circ S_X^{-1}. \quad (7.29)$$

Similarity preserves the spectral radius and the peripheral spectrum. For irreducibility, start with an invariant orthogonal projection Q for one transfer map and form the support projection of the corresponding positive congruence $XQX^\dagger$ (using X^{-1} in the reverse direction). The intertwining in Formula (7.29) makes this support projection invariant for the other transfer map. Irreducibility forces its support to be zero or full, and invertibility of X then forces $Q = 0$ or $Q = 1$. Thus both conditions in Definition 7.6.1 are transported across the gauge. $\square$

Theorem 7.6.14 (Positive blocking preserves CPSV normal tensors). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a CPSV normal tensor and let $L \geq 1$. Then the blocked tensor $A^{[L]}$ is a CPSV normal tensor.

Proof. [✓ Proof](#) Choose a trace-preserving gauge $B^i = XA^iX^{-1}$. Positive blocking gives

$$(B^{[L]})^I = X(A^{[L]})^IX^{-1}. \quad (7.30)$$

The trace-preserving primitive and irreducible structure of B is preserved by positive blocking, so Theorem 7.6.3 makes $B^{[L]}$ CPSV normal. Gauge transport (Theorem 7.6.13) then gives CPSV normality of $A^{[L]}$. $\square$

Proof. ✓ Proof Induction on a nonempty physical word w of length N , using (CF) at each letter and $UU^\dagger = \mathbb{1}_S$ between adjacent letters, gives

$$A^w = U^\dagger \left(\bigoplus_{k=1}^r \mu_k^N A_k^w \right) U. \quad (7.36)$$

Applying cyclicity of the trace and $UU^\dagger = \mathbb{1}_S$ to (7.36) gives

$$\mathrm{tr}(A^w) = \mathrm{tr} \left(\left(\bigoplus_{k=1}^r \mu_k^N A_k^w \right) UU^\dagger \right) = \mathrm{tr} \left(\bigoplus_{k=1}^r \mu_k^N A_k^w \right). \quad (7.37)$$

Applying Theorem 2.6.9 to the right-hand side of (7.37) yields $\mathrm{tr}(A^w) = \sum_{k=1}^r \mu_k^N \mathrm{tr}(A_k^w)$, which is (7.35). $\square$

Theorem 7.6.18 (BNT characterization for CPSV canonical form). Lean ✓ Stmt *Let A have canonical-form data as in (CF), and let $(B_j)_{j=1}^g$ have positive bond dimensions. The family $(B_j)_{j=1}^g$ is a basis of normal tensors for A if and only if*

$$\begin{aligned} & (\forall j, B_j \text{ is normal}) \\ & \wedge (\forall k, \exists j, X_k, \phi_k, \forall i, A_k^i = e^{i\phi_k} X_k B_j^i X_k^{-1}) \\ & \wedge (\forall j \neq j', \nexists X, \phi, \forall i, B_{j'}^i = e^{i\phi} X B_j^i X^{-1}). \end{aligned} \quad (7.38)$$

Each displayed conjugacy also requires the two bond dimensions to agree, and every scalar $e^{i\phi}$ has modulus one. This is [11, Proposition 2.7 and Appendix A], specialized to the literal canonical-form data.

Proof. ✓ Proof The canonical-form data supply normality of every A_k and the positive-length identity $|V^{(N)}(A)| = \sum_k \mu_k^N |V^{(N)}(A_k)|$. Substitution of these two facts into Theorem 8.1.4 gives (7.38). $\square$

Combining the orthogonal ambient embeddings from Theorem 7.7.5 into a single coisometry yields Definition 7.6.7.

Remark 7.6.19 (Relation to stronger normality conditions). An irreducible tensor whose transfer map has spectral radius one and is primitive is normal in the sense of Definition 7.6.1. The equivalence with algebraic normality (Definition 2.5.5) is established in Chapter 8 and Chapter 6.

7.6.1 Normalization conventions

Remark 7.6.1.1 (Scalar normalization conventions). The reductions use four scalar facts: scaling preserves injectivity; the transfer map under a scalar ζ rescales by $|\zeta|^2$; unit-modulus phase rescaling preserves $\sum_i (A^i)^\dagger A^i = \mathbb{1}$; and the MPV of a weighted block-diagonal tensor factorizes as $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$, with $\eta_k := \mu_k / |\mu_k|$.

Left-canonical normalization is required before the blocks can be compared. Without it, the implication $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k, \mathcal{V}(A_k) = \mathcal{V}(B_k)$ already fails for two scalar blocks: for $\mu = (1, 2)$, $A = (1, 3/2)$, and $B = (3, 1/2)$, the two direct sums generate identical MPV families but the first blocks differ at $N = 1$. Left-canonical normalization removes this particular ambiguity; the sectors are then matched, up to a permutation, by the exact fixed-length linear independence of Chapter 8, which requires no ordering of the weight moduli.

7.7 Projector criterion for canonical decomposition

Definition 7.7.1 (Invariant-projector closure). Lean ✓ Stmt An MPS tensor A has *invariant-projector closure* if every orthogonal projection P satisfying $PA^i = PA^iP$ for all i also satisfies $A^iP = PA^iP$ for all i . This is the projector hypothesis in the sufficient condition for canonical form of [11, Section 2].

Theorem 7.7.2 (Projector closure from a faithful stationary weight). Lean Lean ✓ Stmt Let A be left-canonical, and suppose its transfer map has a positive-definite fixed point ρ . Every orthogonal projection P satisfying $(1 - P)A^iP = 0$ for all i commutes with every A^i . Consequently A has invariant-projector closure. This is a sufficient stationary-weight criterion, obtained from [212, Proposition 6.11 and Theorem 6.13].

Proof. ✓ Proof Write E for the transfer map and E^* for its trace adjoint. Invariance and normalization give $PE^*(P) = P = E^*(P)P$, whence

$$E^*(P) - P = (1 - P)E^*(P)(1 - P) \geq 0.$$

Stationarity gives $\text{tr}(\rho(E^*(P) - P)) = \text{tr}((E(\rho) - \rho)P) = 0$. Faithfulness of the positive weight implies $E^*(P) = P$. Since $P^\dagger P = P$, the fixed-point commutant criterion gives $PA^i = A^iP$. Applying this conclusion to $1 - P$ gives the stated projector closure in the opposite orientation. $\square$

Theorem 7.7.3 (Decomposition into irreducible corners under projector closure). Lean ✓ Stmt Suppose A has invariant-projector closure. Then there are isometries $V_1, \dots, V_r$ with $V_k^\dagger V_k = 1$, pairwise orthogonal ranges, and $\sum_k V_k V_k^\dagger = 1$, such that $A^i V_k = V_k A_k^i$ for the corner tensors $A_k^i = V_k^\dagger A^i V_k$, every corner tensor A_k is irreducible, and $\mathcal{V}(A) = \mathcal{V}(\bigoplus_{k=1}^r A_k)$.

Proof. ✓ Proof Proceed by strong induction on the bond dimension. If A is irreducible, the single corner $V = 1$ suffices. Otherwise, A has a nontrivial invariant projection P , which commutes with every A^i by Theorem D.2.1. Factoring $P = V_1 V_1^\dagger$ and $1 - P = V_2 V_2^\dagger$ through support isometries by Lemma D.2.3 splits A into two corner tensors of strictly smaller bond dimension. For $j = 1, 2$, commutation gives $A^i V_j = V_j (V_j^\dagger A^i V_j)$, and each corner inherits invariant-projector closure by Lemma D.2.4. The two recursively obtained families are combined. Ranges from different corners are orthogonal because

$$V_1^\dagger V_2 = V_1^\dagger P(1 - P)V_2 = 0.$$

Equality of matrix product vectors follows from Lemma D.2.6. $\square$

Definition 7.7.4 (No nontrivial periodic vectors). Lean ✓ Stmt An MPS tensor A has *no nontrivial p -periodic vectors* if, for every isometry V with $V^\dagger V = 1$ and $A^i V = V B^i$ for all i , with B irreducible (equivalently, for every restriction B of A to a minimal invariant subspace), and for every $\rho > 0$ and $r > 0$ with $\mathcal{E}_B(\rho) = r\rho$, every eigenvalue of $\mathcal{E}_B$ of modulus r equals r . The eigenvalue r is the spectral radius of $\mathcal{E}_B$ (Theorem [182, “Spectral radius is the Perron eigenvalue”]), so the condition states that no block of the invariant-subspace decomposition of [11, Section 2], rescaled to spectral radius one, has a peripheral eigenvalue $e^{2\pi i q/p} \neq 1$. Eigenvectors with such eigenvalues are the p -periodic vectors of [11, Section 2].

Theorem 7.7.5 (Sufficient condition for canonical form). Lean Lean ✓ Stmt Suppose A has invariant-projector closure and no nontrivial p -periodic vectors. Then there are normal tensors $A_1, \dots, A_r$ of positive bond dimensions $D_1, \dots, D_r$, with $\sum_k D_k \leq D$, and nonzero weights

$\mu_1, \dots, \mu_r$ such that

$$\mathcal{V}^+(A) = \mathcal{V}^+\left(\bigoplus_{k=1}^r \mu_k A_k\right). \quad (7.39)$$

Moreover, there are isometries $V_k : \mathbb{C}^{D_k} \rightarrow \mathbb{C}^D$ with pairwise orthogonal ranges such that

$$\begin{aligned} A^i V_k &= V_k (\mu_k A_k^i), \\ A^i &= \sum_k V_k (\mu_k A_k^i) V_k^\dagger. \end{aligned} \quad (7.40)$$

This is the sufficient condition for canonical form of [11, Section 2], with that paper's convention $\sum_k D_k \leq D$ allowing zero blocks: a corner of A carrying the zero tensor cannot be rescaled to spectral radius one and is omitted because it vanishes at every positive length.

Proof. ✓ Proof Decompose A into irreducible corners B_k along isometries V_k by Theorem 7.7.3; every matrix product vector coefficient of A is the sum of those of the corners. A corner whose matrices all vanish contributes nothing at positive length and is dropped. Every remaining corner has a Perron eigenpair $\mathcal{E}_{B_k}(\rho_k) = r_k \rho_k$ by Lemma D.3.1. The absence of p -periodic vectors, applied to the corner B_k , gives that every eigenvalue of $\mathcal{E}_{B_k}$ of modulus r_k equals r_k , so $A_k := r_k^{-1/2} B_k$ is normal by Lemma D.3.2. Setting $\mu_k := r_k^{1/2}$ gives $\mu_k A_k = B_k$ at the level of matrices, so for every word w of positive length N the weight cancels:

$$\mu_k^N \operatorname{tr}(A_k^w) = (r_k^{1/2})^N (r_k^{-1/2})^N \operatorname{tr}(B_k^w) = \operatorname{tr}(B_k^w).$$

Summing over k gives (7.39), while the corner reconstruction gives (7.40). The retained dimensions satisfy $\sum_k D_k \leq \sum_k \operatorname{tr}(V_k V_k^\dagger) = \operatorname{tr}(\mathbb{1}) = D$. $\square$

Theorem 7.7.6 (Non-scalar fixed point gives a further split). Lean Lean ✓ Stmt Suppose A is unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If X is a Hermitian fixed point of $\mathcal{E}_A$ which is not a scalar multiple of $\mathbb{1}$, then there is a nonzero PSD fixed point ρ which is not positive definite, and consequently A admits a two-block MPV-equivalent decomposition with both block dimensions strictly smaller than D . This is the fixed-point shift and further decomposition step in [193, Theorem 4].

Proof. ✓ Proof Let $\lambda_{\max}$ be the largest eigenvalue of X . Then $\rho := \lambda_{\max} \mathbb{1} - X$ is positive semidefinite, singular, and nonzero because X is not scalar. Unitality and the fixed-point equation for X give $\mathcal{E}_A(\rho) = \rho$. Apply Theorem 7.1.5. The non-scalar hypothesis is needed because scalar multiples of $\mathbb{1}$ are fixed by every unital map and yield no nontrivial support projection. $\square$

7.8 Fixed points and canonical gauges of irreducible blocks

Theorem 7.8.1 (Scalar fixed points in an irreducible unital block). Lean ✓ Stmt If A is irreducible and unital, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and $\mathcal{E}_A(X) = X$, then X is a scalar multiple of $\mathbb{1}$. This is the final fixed-point assertion in the proof of [193, Theorem 4].

Proof. ✓ Proof Tensor irreducibility gives irreducibility of the completely positive transfer map. By Wolf's Theorem 6.6 for irreducible unital Kraus maps, every fixed point is scalar. $\square$

Theorem 7.8.2 (Unital gauge from a full-rank Perron–Frobenius eigenvector). Lean Lean Lean ✓ Stmt Let ρ be positive definite with $\mathcal{E}_A(\rho) = r\rho$ for some $r > 0$. For $B^i := r^{-1/2} \rho^{-1/2} A^i \rho^{1/2}$,

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}. \quad (7.41)$$

If $r = 1$, then the unscaled gauge $B^i := \rho^{-1/2} A^i \rho^{1/2}$ is unital and generates the same MPV family as A . This is the spectral-radius normalization and full-rank fixed-point gauge of [193, Theorem 4].

Proof. ✓ Proof The positive square root of ρ is invertible. Substituting $B^i = r^{-1/2} \rho^{-1/2} A^i \rho^{1/2}$ and using $\mathcal{E}_A(\rho) = r\rho$ gives

$$\sum_i B^i (B^i)^\dagger = r^{-1} \rho^{-1/2} \mathcal{E}_A(\rho) \rho^{-1/2} = \mathbb{1},$$

which is (7.41). For $r = 1$, the change from A to B is a similarity, so cyclicity of the trace gives equality of all MPV coefficients. $\square$

Definition 7.8.3 (CPSV canonical form II). Lean Lean Lean ✓ Stmt In canonical form II of [11, Appendix A], there are positive retained dimensions D_k , normal tensors A_k , weights μ_k , and, for $S = \sum_k D_k \leq D$, a matrix $U \in \mathbb{C}^{S \times D}$ such that, for every physical index i ,

$$\begin{aligned} UU^\dagger &= \mathbb{1}_S, \\ A^i &= U^\dagger \left(\bigoplus_k \mu_k A_k^i \right) U. \end{aligned} \tag{7.42}$$

Thus omitted bond-space coordinates are literally zero. Every retained block additionally satisfies

$$\begin{aligned} \sum_i (A_k^i)^\dagger A_k^i &= \mathbb{1}_{D_k}, \\ \mathcal{E}_{A_k}(\Lambda_k) &= \Lambda_k, \end{aligned}$$

where Λ_k is diagonal and positive definite. The two displayed equations are the only additional blockwise conditions; no ordering, injectivity, gauges, or basis-of-normal-tensors assumptions are included. In particular, a normal left-canonical tensor of positive bond dimension with a positive definite diagonal fixed matrix has a one-block presentation with weight one and no omitted bond coordinates. The line-246 convention (7.25) remains a separate condition on the underlying canonical-form data.

Theorem 7.8.4 (Canonical form admits a canonical-form-II gauge). Lean Lean Lean ✓ Stmt Let A be a tensor in CPSV canonical form, with ambient bond dimension D . Then there is a tensor B of the same bond dimension and an $G \in \text{GL}(D, \mathbb{C})$ such that, for every physical index i ,

$$B^i = G A^i G^{-1}. \tag{7.43}$$

Moreover, B is in canonical form II and the total retained block dimension is unchanged. This is the nonsingular gauge assertion of [11, Appendix A].

Proof. ✓ Proof Write the retained part of the canonical-form decomposition as $C^i = \bigoplus_k \mu_k A_k^i$, and let $U \in \mathbb{C}^{S \times D}$ satisfy $UU^\dagger = \mathbb{1}_S$ and $A^i = U^\dagger C^i U$. For each normal block, first choose its pure trace-preserving Perron gauge and then diagonalize its positive definite fixed point by a unitary. This gives $X_k \in \text{GL}(D_k, \mathbb{C})$ and blocks B_k satisfying the two canonical-form-II identities. Put $X = \bigoplus_k X_k$ and extend it to the omitted zero coordinates by

$$G = U^\dagger X U + (\mathbb{1}_D - U^\dagger U). \tag{7.44}$$

Its inverse is

$$G^{-1} = U^\dagger X^{-1} U + (\mathbb{1}_D - U^\dagger U). \tag{7.45}$$

The identity $UU^\dagger = \mathbb{1}_S$ gives $GG^{-1} = G^{-1}G = \mathbb{1}_D$. It also gives

$$GU^\dagger = U^\dagger X, \quad (7.46)$$

and

$$UG^{-1} = X^{-1}U. \quad (7.47)$$

Therefore

$$GA^iG^{-1} = U^\dagger XC^iX^{-1}U = U^\dagger \left(\bigoplus_k \mu_k B_k^i \right) U, \quad (7.48)$$

which has the same ambient reconstruction and satisfies canonical form II. $\square$

Theorem 7.8.5 (Gauge equivalence under coisometric reconstruction). [Lean](#) [✓ Stmt](#) *Let A and B have common bond dimension D , and let C and E have common retained bond dimension n . Suppose that $U_A, U_B \in \mathbb{C}^{n \times D}$ satisfy $U_A U_A^\dagger = U_B U_B^\dagger = \mathbb{1}_n$ and, for every physical index i ,*

$$A^i = U_A^\dagger C^i U_A. \quad (7.49)$$

$$B^i = U_B^\dagger E^i U_B. \quad (7.50)$$

If C and E are gauge equivalent, then A and B are gauge equivalent.

Proof. [✓ Proof](#) Choose $X \in \text{GL}(n, \mathbb{C})$ such that, for every physical index i , $E^i = XC^iX^{-1}$. Equality of the Gram matrices $U_A U_A^\dagger$ and $U_B U_B^\dagger$ gives a unitary $W \in \mathbb{C}^{D \times D}$ with

$$U_B^\dagger = W U_A^\dagger. \quad (7.51)$$

$$U_B = U_A W^\dagger. \quad (7.52)$$

Extend the retained gauge through U_A by

$$G = U_A^\dagger X U_A + (\mathbb{1}_D - U_A^\dagger U_A). \quad (7.53)$$

Its inverse is

$$G^{-1} = U_A^\dagger X^{-1} U_A + (\mathbb{1}_D - U_A^\dagger U_A). \quad (7.54)$$

The coisometry identity gives $GU_A^\dagger = U_A^\dagger X$ and $U_A G^{-1} = X^{-1} U_A$. Hence, with $Y = WG$,

$$B^i = W U_A^\dagger E^i U_A W^\dagger = W G A^i G^{-1} W^\dagger = Y A^i Y^{-1}. \quad (7.55)$$

$\square$

Theorem 7.8.6 (Diagonal Perron normalization for an irreducible TP tensor). [Lean](#) [✓ Stmt](#) *Let A be an irreducible MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\sum_i (B^i)^\dagger B^i = \mathbb{1},$$

$$\mathcal{E}_B(\Lambda) = \Lambda.$$

This establishes the diagonal Perron normalization required in canonical form II.

Proof. [✓ Proof](#) The TP hypothesis makes $\mathcal{E}_A$ a channel with a nonzero PSD fixed point. Irreducibility makes $\mathcal{E}_A$ an irreducible CP map, and Theorem 4.1.1 upgrades the fixed point to positive definite. The spectral theorem diagonalizes it by a unitary conjugation, which preserves the TP normalization. $\square$

7.9 Normal canonical form

The spectral normal canonical form equips each irreducible block with left-canonical normalization and primitivity. Supporting overlap consequences are collected in Section D.6 and Section 5.7.

Recall from Definition 2.8.3 and Lemma 5.2.1 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \text{Tr}(F_{AB}^N). \quad (7.56)$$

This notation enters the self-overlap clause of the canonical-form conditions below.

Remark 7.9.1 (Normalization convention). [193, Theorem 4] uses the unital gauge $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$; here the blocks are in the dual TP gauge $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$. A positive-definite fixed point of the adjoint transfer map provides the change of gauge between the two (Theorem 4.4.2).

Definition 7.9.2 (Normal canonical form conditions). Lean ✓ Stmt Scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfy the *normal canonical form conditions* if:

1. each A_k is irreducible,
2. each A_k satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. each $\mathcal{E}_{A_k}$ is primitive (equivalently, $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$),
4. $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
5. all $\mu_k \neq 0$,
6. all bond dimensions are positive.

Here “normal” is the spectral sense of [11], which by [200, Proposition 3] is equivalent to the eventual block-injectivity formulation of Definition 2.5.5.

7.9.1 Separated one-copy normal canonical forms

The following specialization attaches BNT separation to the one-copy normal canonical form. Repeated representatives and their power-sum coefficients are instead recorded by the sector decomposition of Chapter 8.

Definition 7.9.1.1 (No gauge-phase-equivalent distinct blocks). Lean ✓ Stmt A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

Definition 7.9.1.2 (Normal canonical form with BNT separation). Lean ✓ Stmt A family $(\mu_k, A_k)_{k=1}^r$ is in *normal canonical form with BNT separation* if it satisfies Definition 7.9.2 and distinct blocks of equal bond dimension are not gauge-phase equivalent. The weight moduli satisfy $|\mu_1| \geq \dots \geq |\mu_r| > 0$. Equal-modulus blocks are permitted; repeated copies of a single basis tensor are grouped into sectors in Chapter 8.

The off-diagonal decay needed for the BNT criterion follows from the separation condition together with the overlap-decay theorems of the preceding chapter.

Theorem 7.9.1.3 (Cross-overlap decay from explicit BNT hypotheses). Lean ✓ Stmt Let $(A_j)_{j=1}^r$ be a family in which each A_j is injective and left-canonical, $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$, and distinct equal-dimension blocks are not gauge-phase equivalent. Then $O_{A_j A_k}(N) \rightarrow 0$ for every pair $j \neq k$.

Proof. ✓ **Proof** If the bond dimensions differ, Theorem 5.4.4 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 5.5.5 gives the decay. $\square$

Lemma 7.9.1.4 (BNT from explicit blockwise hypotheses). Lean ✓ **Stmt** *Let $(\mu_k, A_k)_{k=1}^r$ be a weighted block family. Assume each A_k is injective and left-canonical, $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, that $O_{A_k A_k}(N) \rightarrow 1$ for each k , and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. ✓ **Proof** Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of $\bigoplus_k \mu_k A_k$ in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 7.9.1.3 supplies the off-diagonal decay. Therefore Lemma E.1.7 gives the eventual linear independence required in Definition 8.1.1. $\square$

The normal-canonical formulation encodes normality spectrally (primitive transfer map), the BNT definition algebraically (eventual block injectivity).

Remark 7.9.1.5 (Comparing the two normality conditions). The spectral normality of Definition 7.9.2 and the algebraic normality of Definition 2.5.5 agree: the Wielandt theory of Chapter 6 supplies the implication from primitivity to eventual block injectivity used here.

Lemma 7.9.1.6 (Restricted normal-CF-BNT hypotheses give a BNT). Lean ✓ **Stmt** *On this one-copy surface, if a family is in normal canonical form with BNT separation and each block is also normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor. The blocks here are distinct representatives.*

Proof. ✓ **Proof** The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 8.1.1. $\square$

Remark 7.9.1.7. See Theorem 5.6.1 for the modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

7.9.2 Canonical form from primitivity

A normal block needs a primitive transfer map; an irreducible TP block may still have a cyclic peripheral spectrum. Raising the map to its period sends every peripheral eigenvalue to 1, but eigenvalue 1 can then have several cyclic corners and need not be simple on the full powered space. Period removal therefore proceeds in this order: power the map, obtain the adjoint-fixed cyclic projections, compress to their corners, prove each corner irreducible, and only then infer primitivity of the sector maps. This is the blocking step of [11, Section 2.3].

Theorem 7.9.2.1 (Blocking yields a peripheral-spectrum primitive transfer map). Lean ✓ **Stmt** *Let A be an irreducible MPS tensor with $D > 0$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$.*

Proof. ✓ **Proof** Pass to the conjugate-transposed Kraus family $K_i := (A^i)^\dagger$, which is unital with irreducible Kraus map. The TP fixed-point theorem for $\mathcal{E}_A$ provides a positive definite fixed point for the adjoint of K , so the corresponding theorem in the companion quantum-channel volume [182, “Roots of unity via adjoint fixed point”] shows that every peripheral eigenvalue is a root of unity. Since $\sigma_\partial(\mathcal{E}_A)$ is finite (the corresponding theorem in the companion quantum-channel volume [182, “Finiteness of peripheral eigenvalues”]), the corresponding lemma in the

companion quantum-channel volume [182, “Common exponent for a finite set of roots of unity”] gives a common exponent p with $\mu^p = 1$ for every $\mu \in \sigma_\partial(\mathcal{E}_A)$, and the corresponding lemma in the companion quantum-channel volume [182, “Powering collapses peripheral eigenvalues”] gives $\sigma_\partial(\mathcal{E}_A^p) = \{1\}$. Blocking by p takes the p th power of the transfer map, so $\mathcal{E}_{A[p]}$ is primitive in the peripheral-spectrum sense of the corresponding definition in the companion quantum-channel volume [182, “Primitive map”]. The blocked map need not remain irreducible: eigenvalue 1 may have several cyclic corners. The adjoint-fixed projections are therefore compressed before irreducibility and primitivity are asserted sector by sector in Theorem 7.3.1.3. This is the periodicity-removal step of [11, Section 2.3] for an irreducible block already in TP gauge. $\square$

7.10 Left-canonical and dual diagonalization of irreducible blocks

Each irreducible block in trace-preserving gauge admits a unitary normalization with diagonal positive Perron data. This is the normalization component of CFII (Definition 7.8.3); the full CFII assertion also requires the CPSV canonical-form and normal-block hypotheses. The full PGVWC07 canonical form is Theorem 7.5.6.

Theorem 7.10.1 (Diagonal Perron data for an irreducible TP block). Lean Stmt *Let A be an irreducible MPS tensor in TP gauge, $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that:*

1. $B^i := U^\dagger A^i U$ is TP,
2. $\mathcal{E}_B(\Lambda) = \Lambda$,
3. Λ is diagonal and positive definite,
4. A and B generate the same MPV family.

These are the two normalization identities in the canonical form II definition of [11, Appendix A].

Proof. Proof Theorem 7.8.6 provides the unitary and the diagonal positive-definite fixed point, and unitary conjugation preserves trace preservation and the MPV family (Theorem D.1.1). $\square$

Theorem 7.10.2 (Dual diagonalization for irreducible unital blocks). Lean Stmt *Let A be an irreducible MPS tensor in unital gauge $\sum_i A^i (A^i)^\dagger = \mathbb{1}$, with $D > 0$. Then there exist a unitary U and a diagonal positive definite Λ such that, for $B^i := U^\dagger A^i U$,*

$$\begin{aligned} \sum_i B^i (B^i)^\dagger &= \mathbb{1}, \\ \sum_i (B^i)^\dagger \Lambda B^i &= \Lambda. \end{aligned} \tag{7.57}$$

Moreover, A and B generate the same MPV family. This is the dual-map fixed-point and unitary-diagonalization step of [193, Theorem 4].

Proof. Proof Apply Theorem 7.10.1 to the conjugate-transposed family $i \mapsto (A^i)^\dagger$: unitality of A is trace preservation for this family, whose transfer map is the dual of $\mathcal{E}_A$. Translating back gives (7.57). $\square$

7.11 Cyclic-sector isometries and primitive reduction

Theorem 7.11.1 (Spectral cyclic sectors with support isometries). Lean ✓ Stmt *Let A be a left-canonical irreducible tensor, and suppose the adjoint transfer map $\mathcal{E}_{A^\dagger}$ has a positive-definite fixed point, is irreducible as a positive map, and has peripheral spectrum*

$$\sigma_\partial(\mathcal{E}_{A^\dagger}) = \{\gamma^j : j \in \mathbb{Z}/m\mathbb{Z}\}, \quad (7.58)$$

where γ is a primitive m th root of unity and $m \geq 1$. Then the spectral cyclic-sector decomposition of $A^{[m]}$ has compression maps which may be chosen with rectangular support isometries V_k satisfying

$$\begin{aligned} V_k^\dagger V_k &= \mathbb{1}, \\ V_k V_k^\dagger &= P_k, \\ \varphi_k(X) &= V_k X V_k^\dagger. \end{aligned} \quad (7.59)$$

These isometries also realize the blocked corner letters $\varphi_k(C_k^\alpha) = P_k (A^{[m]})^\alpha P_k$.

Proof. ✓ Proof Theorem [182, “Cyclic decomposition for irreducible finite Kraus maps”], applied to $\mathcal{E}_{A^\dagger}$, provides projections P_k satisfying $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$. Iterating m times gives $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$, and $(\mathcal{E}_{A^\dagger})^m = \mathcal{E}_{(A^{[m]})^\dagger}$ shows that each P_k is fixed by the adjoint transfer map of $A^{[m]}$. Applying Theorem D.5.3 to that blocked tensor gives the compressed sector tensors and the support isometries in (7.59). □

7.11.1 Reduction to primitive blocks

A left-canonical primitive irreducible block is normal, and an arbitrary tensor reduces, after blocking, to a weighted family of left-canonical primitive blocks whose total bond dimension is bounded by the original bond dimension. For two tensors with the same MPV family, the cyclic sectors can be expressed over a common blocked alphabet.

Theorem 7.11.1.1 (TP-primitive irreducible tensor is normal). Lean ✓ Stmt *Let A be a left-canonical MPS tensor with $D > 0$, tensor-irreducible, and with peripheral spectrum $\{1\}$. Tensor irreducibility implies irreducibility of the completely positive transfer map, so A is normal.*

Proof. ✓ Proof The primitive peripheral spectrum, tensor irreducibility, and left-canonical normalization give primitive fixed-point data. The fixed point is positive definite by Lemma D.4.9. Theorem 6.7.4.4 then gives normality. □

An arbitrary translation-invariant tensor reduces, after blocking, to a weighted direct sum of peripheral-spectrum primitive blocks. These globally blocked pieces need not remain irreducible; the cyclic-sector decomposition below refines them into primitive irreducible blocks.

Theorem 7.11.1.2 (Arbitrary tensor to primitive block decomposition). Lean ✓ Stmt *For any MPS tensor A , there exist $p \geq 1$, nonzero weights $(\mu_k)_{k=1}^r$, and blocks $(B_k)_{k=1}^r$ such that, for every blocked σ of positive length N ,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r \mu_k B_k\right)_\sigma. \quad (7.60)$$

The retained block dimensions satisfy $\sum_{k=1}^r D_k \leq D$. Each block is left-canonical and primitive:

$$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1},$$

$$\sigma_\partial(\mathcal{E}_{B_k}) = \{1\}.$$

Each bond dimension is positive.

Proof. ✓ **Proof** Theorem 7.2.1.1 supplies the left-canonical nontrivial blocks, the positive-length equality, and the bound $\sum_k D_k \leq D$. Apply Theorem D.4.2 directly to this finite irreducible family. Its proof first applies Theorem 7.9.2.1 to each block and then takes the least common multiple of the resulting periods through Theorem D.4.1. This gives one positive length p at which every blocked transfer map is primitive. Lemma D.4.18 keeps the blocked weights nonzero, and Lemma D.4.16 carries the positive-length equality through blocking, giving (7.60). $\square$

Theorem 7.11.1.3 (Unconditional common primitive block decompositions). Lean ✓ **Stmt** Let A and B have the same MPV family at every positive length. Then there is a positive blocking length at which both blocked tensors have the same positive-length MPV family as weighted direct sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two weighted nonzero-sector families have the same positive-length MPV family.

Proof. ✓ **Proof** Apply Theorem 7.3.1.4 to choose the common blocking length and the two common cyclic-sector families. For either family, write G_k for its cyclic-sector blocks and (ν_x, C_x) for the weights and blocks of the flattened common-sector family from Lemma D.5.10. The common-alphabet nonzero-part identity gives, for every length N and blocked word σ ,

$$V^{(N)}\left(\bigoplus_k \mu_k^p G_k^{[p]}\right)_\sigma = V^{(N)}\left(\bigoplus_x \nu_x C_x\right)_\sigma.$$

Applying this identity on the two sides of the comparison yields a positive length p and weighted primitive block families such that, for every $N \geq 1$ and blocked word σ ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}\left(\bigoplus_x \mu_x^A A_x\right)_\sigma,$$

$$V^{(N)}(B^{[p]})_\sigma = V^{(N)}\left(\bigoplus_y \mu_y^B B_y\right)_\sigma.$$

The two weighted nonzero parts are equal at every positive length. The structural properties and the nonvanishing of the weights come from the common-sector families. $\square$

Chapter 8

Basis of Normal Tensors

A *basis of normal tensors* (BNT) for an MPS tensor A is a finite family of normal blocks $\{A_j\}$ whose MPV states $\{|V^{(N)}(A_j)\rangle\}$ span $|V^{(N)}(A)\rangle$ at every positive length and are eventually linearly independent. Every coefficient of a canonical form is nonzero, so equal MPV families force a bijective gauge-phase matching $B_k = e^{i\phi_k} X_k A_j X_k^{-1}$, and the sector coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ recover the per-copy weight multisets by Newton–Girard. The references are [193], [11], and [16].

8.1 Bases of normal tensors

Definition 8.1.1 (Basis of Normal Tensors (BNT)). Lean ✓ Stmt A *basis of normal tensors* (BNT) for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that each A_j is normal (Definition 2.5.5); at each system size $N \geq 1$ the MPV of A_{tot} lies in the span of the block MPVs $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$; and for sufficiently large N those block MPVs are linearly independent [16, Definition IV.2]. In the canonical-form characterization of [11, Section 2.3], a BNT is a minimal family of representatives for $\tilde{A}_k^i = e^{i\phi_k} X_k A_j X_k^{-1}$ among the normal blocks.

Definition 8.1.2 (Basis of normal tensors, coefficient form). Lean ✓ Stmt In the coefficient form of a basis of normal tensors, each block A_j is a normal tensor in the spectral sense (Definition 7.6.1), the MPV of A is given at every positive length by coefficients $c_j^{(N)} \in \mathbb{C}$ with $|V^{(N)}(A)\rangle = \sum_{j=1}^g c_j^{(N)} |V^{(N)}(A_j)\rangle$, and the block MPVs $\{|V^{(N)}(A_j)\rangle\}$ are eventually linearly independent [11, Section 2.3].

Theorem 8.1.3 (Distinct BNT members are not gauge-phase equivalent). Lean ✓ Stmt *Distinct members of a basis of normal tensors are not gauge-phase equivalent after identifying equal bond dimensions.*

Proof. ✓ Proof At every sufficiently large length the matrix product vectors of the basis members are linearly independent. A gauge-phase equivalence between two distinct members would give $|V^{(N)}(A_k)\rangle = \zeta^N |V^{(N)}(A_j)\rangle$ for a nonzero scalar ζ . Thus one vector would be a scalar multiple of the other, contradicting linear independence. $\square$

Theorem 8.1.4 (Characterization by canonical-form representatives). [Lean](#) [Lean](#) [Stmt](#) Let A have a canonical-form MPV expansion

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(\tilde{A}_k)\rangle$$

at every positive length, where the tensors $\tilde{A}_k$ are normal tensors of positive bond dimension and every coefficient μ_k is nonzero. Let $(A_j)_{j=1}^g$ be tensors of positive bond dimension. Then $(A_j)_{j=1}^g$ is a basis of normal tensors for A if and only if:

1. every A_j is normal;
2. for every k there are an index j , an invertible matrix X_k , and a phase $e^{i\phi_k}$ such that $\tilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$ for every physical index i ;
3. no two distinct tensors A_j and $A_{j'}$ are related in this way.

This is [11, Proposition 2.7], read with the nonzero-coefficient convention of Definition 7.6.7.

Proof. [Proof](#) Group the summands into the MPV phase classes of Definition 8.6.1.7, and choose one representative from each class. By Theorem 8.1.3, eventual linear independence of a BNT excludes gauge-phase equivalence between two distinct A_j .

Suppose that some phase-class representative is not gauge-phase equivalent to any A_j . Partition all phase classes into the unmatched classes T and the remaining classes. Every remaining representative is a nonzero scalar multiple of some $|V^{(N)}(A_j)\rangle$ at each positive length. Distinct unmatched representatives have vanishing mutual overlaps, and their overlaps with every A_j also vanish. Explicitly, for $t \neq t'$ in T and every j ,

$$\begin{aligned} O_{\tilde{A}_t \tilde{A}_{t'}}(N) &\longrightarrow 0, \\ O_{\tilde{A}_t A_j}(N) &\longrightarrow 0. \end{aligned}$$

By Lemma E.1.7, the combined family

$$\{|V^{(N)}(\tilde{A}_t)\rangle : t \in T\} \cup \{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g\}$$

is linearly independent for all sufficiently large N . Comparing the canonical-form expansion with the BNT expansion gives a relation

$$\sum_{t \in T} c_{N,t} |V^{(N)}(\tilde{A}_t)\rangle + \sum_{j=1}^g \alpha_{N,j} |V^{(N)}(A_j)\rangle = 0.$$

Linear independence forces every $c_{N,t}$ to vanish. Hence the coefficient of every unmatched phase class vanishes for all sufficiently large N . For an unmatched class t , let q range over its canonical-form summands, write μ_q for the corresponding canonical-form weight, and write ζ_q for the phase relating that summand to the chosen representative. Then $c_{N,t} = \sum_{q \in t} (\zeta_q \mu_q)^N$. Every base $\zeta_q \mu_q$ is nonzero because every coefficient is nonzero. Eventual vanishing of this finite geometric sum would imply vanishing at $N = 0$, where its value is the positive number of summands in the class. This contradiction proves coverage. The normal-tensor phase theorem turns the resulting MPV phase relation into the stated gauge-phase relation.

Conversely, coverage gives, for each canonical-form summand, $|V^{(N)}(\tilde{A}_k)\rangle = \zeta_k^N |V^{(N)}(A_{j(k)})\rangle$. Hence

$$|V^{(N)}(A)\rangle = \sum_j \left(\sum_{k:j(k)=j} (\mu_k \zeta_k)^N \right) |V^{(N)}(A_j)\rangle.$$

This is the required BNT span at every positive length. Pairwise gauge-phase distinction and normality give eventual linear independence, completing the converse. $\square$

8.2 Normal blocks and overlap rigidity

Spectral normality implies eventual block injectivity, and the overlap dichotomy determines bond-dimension and gauge-phase matching between normal blocks.

Theorem 8.2.1 (Normality forces positive bond dimension). [Lean](#) [✓ Stmt](#) *Every normal tensor has positive bond dimension.*

Proof. [✓ Proof](#) If the bond dimension were zero, the bond-matrix space and its endomorphism algebra would each contain only zero. The transfer operator would therefore have spectral radius zero, contradicting the spectral-radius-one clause in the definition of a normal tensor. $\square$

Theorem 8.2.2 (Pure Perron normalization at spectral radius one). [Lean](#) [✓ Stmt](#) *Let A be irreducible and suppose that the spectral radius of $\mathcal{E}_A$ is one. There is a positive definite matrix σ satisfying*

$$\sum_i (A^i)^\dagger \sigma A^i = \sigma. \quad (8.1)$$

The tensor

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2} \quad (8.2)$$

is left-canonical and irreducible, and A and B are related by a pure invertible similarity.

Proof. [✓ Proof](#) Positive-definite right and left Perron eigenvectors have the same positive eigenvalue by the trace pairing. The spectral-radius identity makes this eigenvalue equal to one. Thus the left Perron eigenvector is an adjoint fixed point, so its square-root gauge is trace-preserving. Similarity preserves irreducibility. $\square$

Theorem 8.2.3 (Spectral normality gives a pure trace-preserving gauge). [Lean](#) [✓ Stmt](#) *Let A be a normal tensor in the sense of [11, Section 2.2]. Its bond dimension is necessarily positive, and there is a positive definite matrix σ such that the gauge obtained from σ is trace-preserving, primitive, and irreducible. No scalar rescaling is needed.*

Proof. [✓ Proof](#) Pure Perron normalization gives a trace-preserving gauge by a pure similarity. Irreducibility and primitivity are invariant under this similarity. $\square$

Theorem 8.2.4 (Spectral normality gives eventual block injectivity). [Lean](#) [✓ Stmt](#) *Every normal tensor in the sense of [11, Section 2.2] is injective after blocking by some finite length; its positive bond dimension follows from normality.*

Proof. [✓ Proof](#) Apply Theorem 8.2.3. The trace-preserving, primitive, irreducible representative is eventually block-injective, and gauge invariance transports this conclusion back to the original tensor. $\square$

Theorem 8.2.5 (A coefficient-form BNT is an algebraic BNT). [Lean](#) [✓ Stmt](#) If $(A_j)_{j=1}^g$ is a coefficient-form basis of normal tensors for A , then $(A_j)_{j=1}^g$ is an algebraic basis of normal tensors for A .

Proof. [✓ Proof](#) Every block has positive bond dimension, so spectral normality implies eventual block injectivity for each block. The spanning identities and eventual linear independence are unchanged. $\square$

Theorem 8.2.6 (Non-decaying rectangular overlap determines the bond dimension). [Lean](#) [✓ Stmt](#) Let A and B be irreducible trace-preserving tensors of positive, possibly different, bond dimensions. If $|O_{AB}(N)| \rightarrow 1$, then their bond dimensions are equal.

Proof. [✓ Proof](#) If the bond dimensions differed, the rectangular overlap-decay theorem would give $O_{AB}(N) \rightarrow 0$, and hence $|O_{AB}(N)| \rightarrow 0$, contradicting the assumed limit. $\square$

Theorem 8.2.7 (A normal tensor has a nonzero letter). [Lean](#) [✓ Stmt](#) Let A be a CPSV normal tensor of positive bond dimension. Then $A^i \neq 0$ for some physical letter i .

Proof. [✓ Proof](#) If every letter vanished, then the transfer map would vanish and its spectral radius would be zero, contradicting the spectral-radius-one normalization of a CPSV normal tensor. $\square$

Theorem 8.2.8 (Phase-equivalent normal tensors are gauge-phase equivalent). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Suppose that, for some nonzero $\zeta \in \mathbb{C}$, $|V^{(N)}(B)\rangle = \zeta^N |V^{(N)}(A)\rangle$ for $N \geq 0$. Then the bond dimensions are equal and there is an invertible matrix X and a scalar η of unit modulus such that $B^i = \eta X A^i X^{-1}$ for every physical letter i . This is the gauge recovery used in [11, Appendix A].

Proof. [✓ Proof](#) Apply Theorem 8.2.3 to both tensors. Gauge invariance of matrix product vectors preserves the phase relation, whose self-overlap identities imply unit modulus for the phase. The resulting rectangular cross-overlap has norm tending to one. If the bond dimensions differed, Theorem 8.2.6 would give a contradiction. Thus the bond dimensions agree, and the unit-overlap rigidity theorem gives a gauge-phase equivalence between the trace-preserving representatives. Conjugating by the two Perron gauges gives the asserted relation for A and B . $\square$

Theorem 8.2.9 (Normal-tensor overlap dichotomy). [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Their self-overlaps satisfy

$$\begin{aligned} O_{AA}(N) &\rightarrow 1, \\ O_{BB}(N) &\rightarrow 1. \end{aligned}$$

Moreover, either $O_{AB}(N) \rightarrow 0$, or their bond dimensions are equal; call the common value D . Then there are $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C}$, with $|\zeta| = 1$, such that $B^i = \zeta X A^i X^{-1}$ for every physical letter i . In the latter case, $|O_{AB}(N)| \rightarrow 1$. This is [11, Lemma A.2].

Proof. [✓ Proof](#) Put both tensors in their pure trace-preserving gauges. If their bond dimensions differ, rectangular overlap decay gives $O_{AB}(N) \rightarrow 0$. If their dimensions agree but they are not gauge-phase equivalent, the equal-dimension overlap-decay theorem gives the same conclusion. In the remaining case, $B^i = \zeta X A^i X^{-1}$. By Theorem 8.2.8, $|\zeta| = 1$. Hence

$$\begin{aligned} O_{AB}(N) &= \bar{\zeta}^N O_{AA}(N), \\ |O_{AB}(N)| &= |O_{AA}(N)| \rightarrow 1. \end{aligned}$$

$\square$

Theorem 8.2.10 (Exact phase alternative for normal tensors). [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Then either $O_{AB}(N) \rightarrow 0$, or there is a phase $\zeta \in \mathbb{C}$, with $|\zeta| = 1$, such that $|V^{(N)}(B)\rangle = \zeta^N |V^{(N)}(A)\rangle$ for every chain length N . This is [11, Corollary A.3].

Proof. [✓ Proof](#) Apply Theorem 8.2.9. In its second alternative, gauge-phase covariance gives $V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$ for every configuration σ , and hence the asserted equality of vectors. $\square$

Theorem 8.2.11 (Unit overlap from eventual proportionality). [Lean](#) [✓ Stmt](#) Let A and B be tensors, possibly with different bond dimensions, whose self-overlaps tend to one. Suppose that, for all sufficiently large N , there is $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. Then $|O_{AB}(N)| \rightarrow 1$.

Proof. [✓ Proof](#) The proportionality relation gives

$$O_{AB}(N) = c_N O_{BB}(N), \quad (8.3)$$

$$O_{AA}(N) = |c_N|^2 O_{BB}(N). \quad (8.4)$$

Since both self-overlaps tend to one, (8.4) gives $|c_N|^2 \rightarrow 1$ and therefore $|c_N| \rightarrow 1$. Substituting into (8.3) now gives $|O_{AB}(N)| \rightarrow 1$. $\square$

Theorem 8.2.12 (Geometric proportionality phase for normal tensors). [Lean](#) [✓ Stmt](#) Let A and B be normal tensors of positive, possibly different, bond dimensions. Suppose that, for all sufficiently large N , their matrix product vectors are proportional by a nonzero scalar. Then there is $a \in \mathbb{C}$, with $|a| = 1$, such that $|V^{(N)}(A)\rangle = a^N |V^{(N)}(B)\rangle$ for every positive N . This is the one-normal-block consequence of [11, Corollary A.3 and the proof of Theorem 2.10].

Proof. [✓ Proof](#) Normality gives $O_{AA}(N) \rightarrow 1$ and $O_{BB}(N) \rightarrow 1$, so Theorem 8.2.11 gives $|O_{AB}(N)| \rightarrow 1$. The first alternative in Theorem 8.2.10 gives $O_{AB}(N) \rightarrow 0$, hence $|O_{AB}(N)| \rightarrow 0$, a contradiction. In the remaining alternative, $|V^{(N)}(B)\rangle = \zeta^N |V^{(N)}(A)\rangle$ with $|\zeta| = 1$; take $a = \zeta^{-1}$. $\square$

Theorem 8.2.13 (Separated normal tensors are eventually independent). [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=1}^g$ be a finite family of positive-bond-dimension normal tensors, no two of which are gauge-phase equivalent after identifying equal bond dimensions. Then the matrix product vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ are linearly independent for all sufficiently large N , as in [11, Appendix A].

Proof. [✓ Proof](#) Put every tensor in the pure trace-preserving gauge of Theorem 8.2.3. Distinct gauge-phase classes have overlaps tending to zero, while each self-overlap tends to one. The finite Gram matrix therefore tends to the identity and is invertible for all sufficiently large lengths. Gauge invariance of matrix product vectors gives the conclusion for the original tensors. $\square$

8.3 Block-injective consequences for BNT families

The overlap separation of normal blocks yields a simultaneous product-algebra span after blocking.

Definition 8.3.1 (Fixed-length block-injective canonical-form condition). [Lean](#) [Lean](#) [✓ Stmt](#) For a fixed blocked length m , the fixed-length form of the block-injective canonical-form condition is

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Equivalently,

$$\forall X = (X_j)_j \in \bigoplus_{j=1}^g M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_{|\omega|=m}, \quad (X_j)_{j=1}^g = \sum_{|\omega|=m} c_\omega(X) (A_1^\omega, \dots, A_g^\omega).$$

This is the fixed-length form of the block-injective canonical-form condition of [16, lines 2113–2117].

Theorem 8.3.2 (Gauge invariance of simultaneous fixed-length product spans). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be tensor families with the same physical dimension and the same bond dimension D_j in each labelled block. Suppose that for every j there is an arbitrary invertible matrix $X_j \in \text{GL}_{D_j}(\mathbb{C})$ such that, for every physical index i ,

$$B_j^i = X_j A_j^i X_j^{-1}.$$

Then, at every fixed length L ,

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}) \iff \text{span}\{(B_1^\omega, \dots, B_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Both directions hold for general invertible gauges without unitarity constraints, allowing span certificates in left-canonical form to be transported to unital gauges.

Proof. [Stmnt](#) [Proof](#) Given a target tuple $(M_j)_j$, conjugate it backwards to $(X_j^{-1} M_j X_j)_j$. If the word tuples of $(A_j)_j$ span the product algebra, one common coefficient family $(c_\omega)_\omega$ expresses this conjugated tuple. Gauge covariance of word evaluation then gives

$$\sum_{|\omega|=L} c_\omega B_j^\omega = X_j \left(\sum_{|\omega|=L} c_\omega A_j^\omega \right) X_j^{-1} = M_j \quad (8.5)$$

for every j . Thus the $(B_j)_j$ word tuples span the product algebra. Replacing every X_j by X_j^{-1} proves the reverse implication. $\square$

Lemma 8.3.3 (Direct-sum block injectivity at the source length). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $(A_j)_{j=1}^g$ be irreducible left-canonical tensors with normalized self-overlaps and with no gauge-phase-equivalent distinct blocks. Suppose $g \geq 2$, $L_0 > 0$, and every A_j is block-injective at lengths L_0 , $L_0 + 1$, and $3(L_0 + 1)$. Then

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = 3(g-1)(L_0 + 1)\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

This is the direct-sum induction length in [193, lines 1346–1408].

Proof. [Stmnt](#) [Proof](#) Lemma E.3.1 gives homogeneous pair-trace separation at $S = 3(L_0 + 1)$. For a chosen anchor block, pair separation prescribes an arbitrary matrix there and zero on one other block. Multiplying $g - 1$ such homogeneous combinations removes the other blocks one at a time. Induction on the number of removed blocks therefore realizes every tuple supported on the anchor at length $(g - 1)S$. Summing over the anchors gives the full product algebra at $3(g - 1)(L_0 + 1)$. $\square$

Theorem 8.3.4 (Simultaneous one-letter span after blocking a BNT). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ *Stmt* Let $(A_j)_{j=1}^g$ be a basis of normal tensors, with bond dimensions D_j . There is a positive integer L such that the one-letter evaluations of the blocked tensors span the full product algebra:

$$\text{span}_{\mathbb{C}} \{((A_1^{[L]})^i, \dots, (A_g^{[L]})^i) : 0 \leq i < d^L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Moreover, there is L_0 such that, for every $L \geq L_0$, the unblocked word tuples have the simultaneous span

$$\text{span}_{\mathbb{C}} \{(A_1^\omega, \dots, A_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

This establishes simultaneous block injectivity following [11, lines 317–345].

Proof. $\checkmark$ *Proof* Put each BNT representative in its trace-preserving Perron gauge. The gauged family is irreducible and left-canonical, has normalized self-overlaps, and remains gauge-phase separated. Choose a common positive length p at which every gauged block is block-injective. Propagation of homogeneous word spans gives block injectivity at $p+1$ and $3(p+1)$.

Lemma 8.3.3 gives pairwise separation at length $S = 3(p+1)$. Write $g \dot{-} 1 = \max\{g-1, 0\}$. Multiplying one separator for each of the other blocks gives a selector of length $(g \dot{-} 1)S$ for every block. Transport these selectors back from the Perron gauges. Gauge invariance also gives block injectivity of every original block at length p . Concatenating the injective prefix with the selectors yields the full product-algebra span at

$$L = p + (g \dot{-} 1)S = p + 3(g \dot{-} 1)(p+1) > 0.$$

The truncated factor makes the same construction valid for empty and singleton families. The same span provides selectors and block injectivity at length L . For every sufficiently large n , concatenate an $(n-L)$ -letter injective prefix with these selectors to obtain the full product-algebra span at length n . Finally, length- L words of A_j are the one-letter evaluations of $A_j^{[L]}$, which proves the blocked statement. $\square$

Theorem 8.3.5 (Sharp simultaneous span under a total-dimension bound). [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ *Stmt* Let $(A_j)_{j=1}^g$ be pairwise inequivalent normalized normal tensors with bond dimensions D_j ; in particular, they may form a basis of normal tensors. If $K > 0$ and $\sum_j D_j \leq K$, then there is a positive integer L with $L+1 \leq 3K^5$ such that

$$\text{span}_{\mathbb{C}} \{(A_1^\omega, \dots, A_g^\omega) : |\omega| = L\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Proof. $\checkmark$ *Proof* Put every representative in its trace-preserving Perron gauge. Since $D_j \leq \sum_k D_k \leq K$, quantum Wielandt and homogeneous-span propagation make every representative block-injective at K^4 , K^4+1 , and $3(K^4+1)$. Conjugation preserves the simultaneous word span.

For $g=0$, take $L=1$; the product algebra has only its zero element. For $g=1$, take $L=K^4$. If $g \geq 2$, the three-block separation theorem gives

$$L = 3(g-1)(K^4+1).$$

Positivity of the D_j gives $g = \sum_j 1 \leq \sum_j D_j \leq K$, and hence $L+1 \leq 3(K-1)(K^4+1)+1 \leq 3K^5$. For the last inequality, use $(K-1)(K^4+1) < K^5$, which follows from $K-1 < K^4$. The same bound includes $L+1$ when g is zero or one. $\square$

Theorem 8.3.6 (Total dimension of the BNT representatives). [Lean](#) [✓ Stmt](#) *Let literal canonical-form data of ambient bond dimension D have phase classes j and chosen representatives of dimensions D_j . Then*

$$\sum_j D_j \leq D.$$

Proof. [✓ Proof](#) In every phase class, choose its first copy. The class-copy equivalence sends these choices to distinct displayed block indices, and the first copy is the chosen representative. Thus the sum of the representative dimensions is bounded by the sum over all displayed blocks, which is at most D by the canonical-form dimension bound. $\square$

Theorem 8.3.7 (Sharp block-injective form for literal CPSV canonical form). [Lean](#) [✓ Stmt](#) *Let A be in literal CPSV canonical form with positive ambient bond dimension D . There is a basis of normal tensors $(A_j)_{j=1}^g$ and a positive integer $L \leq 3D^5$ such that*

$$\text{span}_{\mathbb{C}} \left\{ ((A_1^{[L]})^i, \dots, (A_g^{[L]})^i) : 0 \leq i < d^L \right\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Hence A is in block-injective canonical form after blocking at most $3D^5$ sites. This is the proposition at [11, lines 342–345].

Proof. [✓ Proof](#) Choose one representative from every matrix-product-vector phase class. These representatives form a basis of normal tensors, and their total bond dimension is at most D by Theorem 8.3.6. Apply Theorem 8.3.5 with $K = D$. A word of length L in A_j is a one-letter evaluation of $A_j^{[L]}$, so the resulting length- L span is exactly the asserted one-letter span. $\square$

Let A be a tensor in canonical form, and let $A_1, \dots, A_g$ be a basis of normal tensors of A . For suitable coefficients $\mu_{j,q}$ and invertible matrices $X_{j,q}$, write $M_j = \text{diag}(\mu_{j,1}, \dots, \mu_{j,r_j})$ and $X = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} X_{j,q}$. Then

$$A^i = X \left[\bigoplus_{j=1}^g (M_j \otimes A_j^i) \right] X^{-1} = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} A_j^i X_{j,q}^{-1}. \quad (8.6)$$

This is [11, Section 2.3, Equations (20a)–(21)]. The decomposition (8.6) is used below to separate the normal blocks.

8.4 Permutation rigidity for bases of normal tensors

When two bases of normal tensors generate the same MPV subspace for all sufficiently large positive lengths, the blocks must agree up to a permutation and per-block gauge phases. The argument turns the eventual equality of spans into an invertible change-of-basis matrix, then analyses its asymptotics through the overlaps.

Theorem 8.4.1 (Permutation rigidity for equal block counts). [Lean](#) [✓ Stmt](#) *Let $(A_j)_{j=1}^g$ and $(B_j)_{j=1}^g$ be injective families of positive bond dimension satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for all sufficiently large system sizes N , then there is a permutation π of $\{1, \dots, g\}$ such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent.*

Proof. ✓ Proof For large N , Lemma E.2.2 gives an invertible matrix U_N relating the two MPV families. Fix j and write $|V^{(N)}(B_j)\rangle = \sum_i u_{ij}^{(N)} |V^{(N)}(A_i)\rangle$. With $G_A(N) = (O_{A_i A_\ell}(N))_{i,\ell}$ and $b_j(N) = (O_{A_i B_j}(N))_i$, this gives

$$G_A(N) \overline{u_j(N)} = b_j(N).$$

Moreover,

$$O_{B_j B_j}(N) = u_j(N)^\top G_A(N) \overline{u_j(N)}.$$

If all mixed overlaps $O_{A_i B_j}(N)$ tended to 0, then $b_j(N) \rightarrow 0$; since $G_A(N) \rightarrow \mathbb{1}$, one obtains $\overline{u_j(N)} \rightarrow 0$ and hence $u_j(N) \rightarrow 0$. Thus $O_{B_j B_j}(N) \rightarrow 0$, contradicting $O_{B_j B_j}(N) \rightarrow 1$. Thus each B_j has a matched block $A_{f(j)}$ with non-decaying mixed overlap. Theorems 5.4.4 and 5.5.5 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the B -family makes f injective, hence bijective, and its inverse is the required permutation. $\square$

Lemma 8.4.2 (Permutation rigidity with asymptotically orthonormal overlaps). Lean ✓ Stmt Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective tensors of positive bond dimension satisfying the TP normalization $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ and $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$, whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If, for all sufficiently large system sizes N ,

$$\text{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \text{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then $g_A = g_B$, and there is a permutation π of the common block index set such that A_j and $B_{\pi(j)}$ have the same bond dimension and are gauge-phase equivalent for each j .

Proof. ✓ Proof By Lemma E.1.7, both families are linearly independent for all sufficiently large N . Choose N large enough that both families are linearly independent and the assumed span equality holds. The common span then has the same dimension when computed from the A -family or the B -family, so $g_A = g_B$. After identifying the two index sets, Theorem 8.4.1 yields the permutation conclusion. $\square$

The asymptotic orthonormality of overlaps matches the blocks up to a permutation and gauge phases. Section 8.8 establishes the exact sector counterpart used in the Fundamental Theorem (Chapter 9).

The overlap dichotomy also shows that the MPV states of distinct normal blocks are non-proportional at arbitrarily large lengths, which enters the exact sector matching.

Theorem 8.4.3 (Non-equivalent irreducible TP blocks are eventually non-proportional). Lean

Lean ✓ Stmt Let A and B be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound N_0 there is $N \geq N_0$ such that $|V^{(N)}(A)\rangle$ is not proportional to $|V^{(N)}(B)\rangle$. Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.

Proof. ✓ Proof Suppose that for all sufficiently large N one had $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$. Then

$$O_{AA}(N) = |c_N|^2 O_{BB}(N), \tag{8.7}$$

$$O_{AB}(N) = c_N O_{BB}(N). \tag{8.8}$$

Since $O_{AA}(N) \rightarrow 1$ and $O_{BB}(N) \rightarrow 1$, the identity (8.7) gives $|c_N| \rightarrow 1$, and hence (8.8) gives $|O_{AB}(N)| \rightarrow 1$. The irreducible trace-preserving overlap decay theorem gives $O_{AB}(N) \rightarrow 0$ when $A \sim_{\text{gp}} B$, a contradiction. $\square$

8.5 Newton–Girard identities and power-sum recovery

The coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$ that appear in the two-layer BNT expansion are power sums of the copy weights. Recovering the individual weights from these sums is a Newton–Girard problem: equal power sums determine equal characteristic polynomials, hence equal multisets of roots.

The general matrix statement is the corresponding theorem in the companion quantum-channel volume [182, “Newton–Girard trace recursion”]. It identifies a characteristic polynomial from the first n power-sum traces.

The matched-sector comparison below applies the power-sum identities to a single matched pair, after the BNT permutation and phase gauges have identified which basis tensors are compared: the coefficient $\sum_q \mu_{j,q}^N$ then equals $\sum_q (\nu_{j,q} e^{i\phi_j})^N$, and Newton–Girard recovers the weight multiset with multiplicity.

Theorem 8.5.1 (Finite-range equal power sums imply equal multisets). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ *Stmt* Let $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for $1 \leq k \leq n$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.

Two variants relax the equal-cardinality assumption. If $\alpha : \{0, \dots, m-1\} \rightarrow \mathbb{C}$ and $\beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$ have no zero entries and their power sums agree for $1 \leq k \leq \max(m, n)$, then $m = n$ and the multisets are equal. Without a nonzero hypothesis, the list form $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$ for $1 \leq N \leq \max(x_a, x_b)$ gives equality of the nonzero multisets; if it also holds at $N = 0$ then $x_a = x_b$ and the full lists agree. The nonzero variant is the one applied to the nonzero coefficients of a canonical form, which is how [11, Lemma A.5] is used.

Proof. $\checkmark$ *Proof* Apply the finite-range Newton–Girard theorem to $\text{diag}(\alpha)$ and $\text{diag}(\beta)$ and extract the roots of $\prod_i (X - \alpha_i)$ and $\prod_i (X - \beta_i)$. For unequal cardinalities, pad both lists with zeros to length $\max(m, n)$; the nonzero hypothesis makes the padded-zero counts $\max(m, n) - m$ and $\max(m, n) - n$, forcing $m = n$. Without that hypothesis, delete the zero entries first (unchanged for positive N); the equality at $N = 0$ gives $x_a = x_b$. $\square$

Corollary 8.5.2 (A nonzero finite family has a nonvanishing power sum). [Lean](#) $\checkmark$ *Stmt* Let $m \geq 1$ and let $\lambda_0, \dots, \lambda_{m-1} \in \mathbb{C}$ be nonzero. Then there is an exponent N with $1 \leq N \leq m$ such that $\sum_{q=0}^{m-1} \lambda_q^N \neq 0$. This is the consequence of the Appendix power-sum lemma used in the proof of Lemma L [11, Appendix C.3, lines 1851–1858]; the finite-range comparison is [11, Appendix A, lines 1155–1163].

Proof. $\checkmark$ *Proof* Suppose that every power sum with $1 \leq N \leq m$ vanishes. Comparing $\lambda_0, \dots, \lambda_{m-1}$ with the empty family in Theorem 8.5.1 gives $m = 0$, contradicting $m \geq 1$. $\square$

8.6 The BNT canonical form on a sector decomposition

The basis-of-normal-tensors canonical form of [11, Section 2] is defined on a sector decomposition with sector weights $\mu_{j,q}$ and coefficients $c_N^{(j)} = \sum_q \mu_{j,q}^N$.

8.6.1 Sector decompositions and phase matching

Definition 8.6.1.1 (Sector weights and sector decomposition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) $\checkmark$ *Stmt* A sector-weight tuple over g basis blocks consists of multiplicities $r_j > 0$,

nonzero weights $\mu_{j,q}$ for $q \in \{1, \dots, r_j\}$, and the sector coefficients

$$c_N^{(j)} := \sum_{q=1}^{r_j} \mu_{j,q}^N. \quad (8.9)$$

A *sector decomposition* consists of the basis blocks $(A_j)_{j=1}^g$ together with these multiplicities and weights. If A_j has bond dimension D_j , then the flattened copy family has total bond dimension

$$D_{\text{tot}} = \sum_{j=1}^g r_j D_j.$$

Definition 8.6.1.2 (Physical blocking of a sector decomposition). [Lean](#) [✓ Stmt](#) Let P be a sector decomposition with representatives A_j , multiplicities r_j , and copy weights $\mu_{j,q}$. Its physical p -site blocking has the same representative and copy indices, with

$$\begin{aligned} A_j &\mapsto A_j^{[p]}, \\ \mu_{j,q} &\mapsto \mu_{j,q}^p. \end{aligned}$$

The two-layer decomposition is [11, lines 283–301]. Blocking preserves the copy index sets and bond dimensions, replaces each flattened representative by its p -site blocking, and raises each flattened weight to the p -th power.

Theorem 8.6.1.3 (Indices, dimensions, weights, and representatives under blocking). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Physical blocking preserves the copy sets and bond dimensions. On the flattened copy index, it raises the weight to the p -th power and replaces the representative by its p -site blocking.*

Proof. [✓ Proof](#) These identities follow directly from the definition. □

Lemma 8.6.1.4 (Sector decomposition formula). [Lean](#) [✓ Stmt](#) *For a sector decomposition S with associated tensor A_{tot} , the MPV satisfies*

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^g c_N^{(j)} V^{(N)}(A_j)_\sigma. \quad (8.10)$$

Two blocks are identified for canonical-form purposes when their MPV families differ by a scalar power. The next three definitions formalize this relation, the common cover it generates between two families, and the phase classes of a single family.

Definition 8.6.1.5 (MPV phase equivalence for nonzero-weight blocks). [Lean](#) [✓ Stmt](#) Two blocks A and B are MPV-phase equivalent if there is a nonzero scalar ζ such that, for every positive system size N and every physical word, $V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$. The two bond dimensions are allowed to differ.

Definition 8.6.1.6 (Common MPV phase cover). [Lean](#) [✓ Stmt](#) For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

If $G = \bigoplus_k X_k$ is the block-diagonal gauge, then, letter by letter,

$$\bigoplus_{k=1}^r \mu_k A_k^i = GT_{\text{grp}}^i G^{-1}, \quad (8.17)$$

and

$$A^i = U^\dagger GT_{\text{grp}}^i G^{-1} U. \quad (8.18)$$

Proof. [✓ Proof](#) Enumerate the block indices by their phase classes and choose one normal representative in each class. The normal-tensor phase theorem gives the dimension equality, unit phase, and invertible gauge in (8.14). The BNT characterization proves that the representatives form a basis of normal tensors. Reindexing the dependent block coordinates gives Φ and (8.15). The exact heterogeneous block-diagonal reindexing identity identifies the grouped tensor with the representative copies in the original retained-coordinate space. Block-diagonal conjugation then gives (8.17); substitution into (8.11) gives (8.18) with the original coisometry U . In the notation of [11, Proposition 4.13], these identities are

$$\bigoplus_{k=1}^r \mu_k A_k^i = GT_{\text{grp}}^i G^{-1}, \quad (8.19)$$

$$A^i = U^\dagger GT_{\text{grp}}^i G^{-1} U, \quad (8.20)$$

$$UU^\dagger = \mathbb{1}. \quad (8.21)$$

□

Definition 8.6.1.10 (BNT sector presentation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A *BNT sector presentation* of a tensor A is a nonempty sector decomposition P such that every representative has a positive number of copies, every copy has nonzero weight, every representative C_j is normal, the representative matrix product vectors are eventually linearly independent, and, for every $N > 0$,

$$V^{(N)}(A) = V^{(N)}(P). \quad (8.22)$$

This records the nonzero canonical summands constructed at [11, lines 217–246]: repeated copies and their weights remain explicit.

Theorem 8.6.1.11 (Uniqueness of BNT presentations). [Lean](#) [✓ Stmt](#) *Two BNT presentations of the same positive-length matrix product vector family have equally many representatives. After a permutation, corresponding representatives have equal bond dimension and are related by an invertible gauge and a unit-modulus phase.*

Proof. [✓ Proof](#) Apply Theorem 8.1.4 first to the direct-sum tensor of the first presentation, using the representatives of the second as a BNT, and then with the presentations interchanged. This gives a matching map in each direction. Eventual linear independence excludes gauge-phase-equivalent distinct representatives, so both maps are injective. Since the index sets are finite, they have equal cardinality, and either matching map is a bijection. Normal-tensor gauge rigidity gives the asserted gauge-phase relation for each matched pair. □

Theorem 8.6.1.12 (Representative sector decomposition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *In the notation of Theorem 8.6.1.9, let $k(j, q)$ be*

the listed block corresponding to the class-copy pair (j, q) . The representatives C_j define a sector decomposition P_{rep} with copy weights

$$\lambda_{j,q} = \nu_{k(j,q)} \zeta_{k(j,q)}. \quad (8.23)$$

Every copy weight in P_{rep} is nonzero. The representatives have simultaneous word-tuple separation at every sufficiently large length.

For every positive length N , the grouped tensor T_{grp} of Theorem 8.6.1.9 and the representative sector tensor generate the same matrix product vector. The original tensor has the same positive-length matrix product vectors as both:

$$V^{(N)}(T_{\text{grp}}) = V^{(N)}(P_{\text{rep}}). \quad (8.24)$$

$$V^{(N)}(A) = V^{(N)}(P_{\text{rep}}). \quad (8.25)$$

Hence P_{rep} is a BNT sector presentation of A whenever at least one listed block occurs.

Proof. ✓ Proof The class-copy equivalence reindexes the listed blocks. (8.23) combines the original coefficient with the unit phase relating its block to C_j ; both factors are nonzero. The basis-of-normal-tensors condition and Theorem 8.3.4 give eventual simultaneous word-tuple separation for the representatives. Reindexing the finite sum by (j, q) gives (8.24). The original coisometric reconstruction and simultaneous block gauge preserve every positive-length closed-chain coefficient, yielding (8.25). $\square$

Definition 8.6.1.13 (BNT canonical form). Lean ✓ Stmt Let P be a sector decomposition. In addition to spanning and eventual linear independence, the canonical form specifies irreducible left-canonical representatives, self-overlap normalization, pairwise gauge-phase separation, and weight normalization. Explicitly, each basis block has positive bond dimension, is irreducible, and is left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent after identifying equal bond dimensions; and the [11, Section 2.3] canonical-form normalization holds:

$$\begin{aligned} \forall (j, q), \quad & |\mu_{j,q}| \leq 1, \\ \exists (j_*, q_*), \quad & |\mu_{j_*, q_*}| = 1. \end{aligned}$$

The MPV expansion takes the two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^g \left(\sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma, \quad (8.26)$$

matching [11, Section 2.3] and [16, Definition IV.2]. The canonical-form normalization accommodates equal-modulus and repeated-copy sectors, such as $C \oplus (-C)$ and $C \oplus (1/2)C$.

Theorem 8.6.1.14 (Physical relabeling preserves BNT canonical form). Lean Lean ✓ Stmt Let P be a BNT canonical form and let e be a bijection of its physical alphabet with another finite alphabet. Relabel every basis tensor of P by e , leaving the representative and copy labels, bond dimensions, and weights unchanged. The resulting sector decomposition is again a BNT canonical form.

Proof. ✓ Proof Physical relabeling preserves irreducibility, left-canonical normalization, normalized self-overlaps, eventual linear independence, and gauge-phase separation of the representatives. Since the weights and all sector data are unchanged, the canonical-form normalization is also preserved. $\square$

Definition 8.6.1.15 (Single-sector decomposition). [Lean](#) [✓ Stmt](#) For a tensor A , its single-sector decomposition has A as its only basis tensor, with one copy of weight 1.

Theorem 8.6.1.16 (Single-sector BNT canonical form). [Lean](#) [✓ Stmt](#) Let A have positive bond dimension, be irreducible and left-canonical, and satisfy $O_{AA}(N) \rightarrow 1$. Then its single-sector decomposition is a BNT canonical form.

Proof. [✓ Proof](#) The identity

$$O_{AA}(N) = \|V^{(N)}(A)\|^2 \rightarrow 1 \neq 0. \quad (8.27)$$

shows that $V^{(N)}(A)$ is nonzero for all sufficiently large N . A family consisting of one nonzero vector is linearly independent. All remaining conditions are immediate from the hypotheses and the unit weight. $\square$

Theorem 8.6.1.17 (Normal tensors have single-sector BNT presentations). [Lean](#) [✓ Stmt](#) Every normal tensor is gauge equivalent to a tensor whose single-sector decomposition is a BNT canonical form. The original tensor and the total tensor of this decomposition generate the same matrix product vectors at every positive length.

Proof. [✓ Proof](#) Choose the pure trace-preserving Perron gauge. Its tensor is irreducible and left-canonical, and primitivity gives self-overlap converging to one. The preceding theorem supplies the BNT canonical form. If P denotes the total tensor of its single-sector decomposition, then gauge invariance and the sector-decomposition formula give, for every $N > 0$ and word σ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma = 1^N V^{(N)}(B)_\sigma = V^{(N)}(P)_\sigma. \quad (8.28)$$

$\square$

Lemma 8.6.1.18 (Existence of a unit-modulus weight from the canonical-form normalization). [Lean](#) [✓ Stmt](#) Under the BNT canonical form, the [11, Section 2.3] normalization provides indices (j_*, q_*) with $|\mu_{j_*, q_*}| = 1$.

Proof. [✓ Proof](#) This is the unit-modulus existential in the canonical-form normalization. $\square$

Corollary 8.6.1.19 (Basis blocks of a BNT canonical form are normal). [Lean](#) [✓ Stmt](#) If P is in BNT canonical form, then every basis block P_j is normal in the algebraic sense of Definition 2.5.5.

Proof. [✓ Proof](#) Each basis block has positive bond dimension, is irreducible and left-canonical, and has normalized self-overlap converging to 1. Theorem 5.9.3 applies to each block. $\square$

Theorem 8.6.1.20 (A BNT canonical form determines an algebraic BNT). [Lean](#) [✓ Stmt](#) If P is in BNT canonical form, then its basis blocks form an algebraic basis of normal tensors for the total tensor associated with P .

Proof. [✓ Proof](#) The preceding corollary gives normality of every basis block. The sector coefficient formula (8.10) expresses the total MPV as a linear combination of the basis MPVs at each positive length, and the BNT sector data give their eventual linear independence. $\square$

Theorem 8.6.1.21 (Positive blocking preserves BNT canonical form). [Lean](#) [✓ Stmt](#) Let P be a BNT canonical form and let $p \geq 1$. Then the blocked sector decomposition $P^{[p]}$ is again a BNT canonical form.

Proof. ✓ Proof For each representative A_j , normalized self-overlap gives peripheral period one, hence a primitive transfer map. Positive blocking therefore preserves irreducibility and left-canonical normalization. Moreover, $O_{A_j^{[p]}A_j^{[p]}}(N) = O_{A_jA_j}(Np) \rightarrow 1$. Let F_N be the linear isomorphism induced by concatenating blocked words, from the blocked configuration space at length N to the original one at length Np . For every representative, $F_N|V^{(N)}(A_j^{[p]})\rangle = |V^{(Np)}(A_j)\rangle$. Hence eventual linear independence is preserved. If two blocked representatives were gauge-phase equivalent, then for some nonzero ζ and every N , $|V^{(N)}(A_k^{[p]})\rangle = \zeta^N |V^{(N)}(A_j^{[p]})\rangle$. Writing these two vectors as f_k and f_j gives $1 \cdot f_k = \zeta^N \cdot f_j$ in an eventually linearly independent family, and therefore $k = j$. Finally, $|\mu_{j,q}^p| = |\mu_{j,q}|^p \leq 1$, and a unit-modulus weight remains of unit modulus after taking its p -th power. $\square$

Theorem 8.6.1.22 (Positive blocking preserves a basis of normal tensors). Lean ✓ Stmt *If $(A_j)_{j=1}^g$ is a basis of normal tensors for A and $p \geq 1$, then $(A_j^{[p]})_{j=1}^g$ is a basis of normal tensors for $A^{[p]}$.*

Proof. ✓ Proof Positive blocking preserves normality of each representative. Fix a blocked length N . If the m -th blocked letter is $J_m = (i_{m,1}, \dots, i_{m,p})$, define the configuration reindexing τ_p by concatenation:

$$\tau_p(J_1, \dots, J_N) := (i_{1,1}, \dots, i_{1,p}, \dots, i_{N,1}, \dots, i_{N,p}). \quad (8.29)$$

For every blocked configuration σ and every representative A_j , the coefficient identities are

$$\langle \sigma | V^{(N)}(A^{[p]}) \rangle = \langle \tau_p(\sigma) | V^{(Np)}(A) \rangle, \quad (8.30)$$

$$\langle \sigma | V^{(N)}(A_j^{[p]}) \rangle = \langle \tau_p(\sigma) | V^{(Np)}(A_j) \rangle. \quad (8.31)$$

The spanning coefficients at length Np therefore give those at blocked length N , and the same configuration reindexing preserves eventual linear independence. $\square$

Theorem 8.6.1.23 (Common exact quantum-Wielandt length for BNT representatives). Lean ✓ Stmt *Let P be a BNT canonical form of total bond dimension D , with representatives A_j of bond dimensions D_j . For every $0 \leq j < g$, the representative A_j is D^4 -block injective: $S_{D^4}(A_j) = M_{D_j}(\mathbb{C})$.*

Proof. ✓ Proof The BNT hypotheses make A_j irreducible and left-canonical, with $O_{A_jA_j}(N) \rightarrow 1$. The preceding theorem therefore makes A_j normal, and quantum Wielandt gives $\tilde{S}_{D_j^4}(A_j) = M_{D_j}(\mathbb{C})$. By Theorem E.2.6, $D_j^4 \leq D^4$. Write $D^4 = D_j^4 + k$ and apply positive-length propagation k times. $\square$

Theorem 8.6.1.24 (CPSV simultaneous block-injectivity bound). Lean Lean ✓ Stmt *Let P be a BNT canonical form with total bond dimension D . There is a positive integer $N \leq 3D^5$ such that its representatives have the simultaneous fixed-length span*

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = N\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

This is the block-injectivity estimate in [11, lines 340–345].

Proof. ✓ Proof Every representative is block-injective at D^4 , and homogeneous-span propagation gives the same conclusion at $D^4 + 1$ and $3(D^4 + 1)$. If $g = 1$, take $N = D^4$. If $g \geq 2$, apply Lemma 8.3.3 with $L_0 = D^4$ and take

$$N = 3(g - 1)(D^4 + 1).$$

The positive bond dimension of every representative gives $g \leq D$. Consequently $N \leq 3D^5$; the same estimate also covers the one-block case. The unit-modulus weight required by the BNT normalization ensures $D > 0$. □

The matching arguments below read off three facts from a BNT canonical form: the raw coefficient identity, the cross-overlap decay between distinct basis blocks, and the combined-family linear independence.

Lemma 8.6.1.25 (Raw two-layer sector coefficient identity). Lean ✓ Stmt *For every sector decomposition P , length N , and basis index j , $c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N$. This is [11, Section 2.3].*

Proof. ✓ Proof The claim follows from the definition of the sector coefficient. □

Lemma 8.6.1.26 (Cross-overlap decay between distinct basis blocks). Lean ✓ Stmt *If P is a BNT canonical form and $j \neq k$, then $O_{P_j P_k}(N) \rightarrow 0$.*

Proof. ✓ Proof Case on the bond dimensions. If they differ, the dimension-mismatch overlap-decay theorem applies. If they agree, the basis-distinctness clause rules out gauge-phase equivalence, so the equal-dimension overlap-decay theorem applies. □

Lemma 8.6.1.27 (Combined-family eventual linear independence). Lean ✓ Stmt *Let P and Q each be a BNT canonical form, and suppose every mixed overlap $O_{P_j Q_k}(N)$ tends to 0. Then the union family*

$$\{|V^{(N)}(P_j)\rangle\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\rangle\}_{k=1}^{g_Q}$$

is linearly independent for all sufficiently large N . This is [11, Appendix A].

Proof. ✓ Proof Feed the per-family normalized self-overlaps, the within-family cross-overlap decay (Lemma 8.6.1.26), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma E.1.9). □

Example 8.6.1.28 ($C \oplus (1/2)C$, unequal-modulus admissibility). Lean ✓ Stmt For any tensor C , this construction forms a single-sector, two-copy decomposition with basis tensor C and raw weights $(1, 1/2)$. Thus the sector data themselves allow unequal-modulus copies. If C additionally has positive bond dimension, is irreducible and left-canonical, has normalized nonzero self-overlap, and satisfies the BNT independence clause, these data satisfy Definition 8.6.1.13; those hypotheses are not consequences of the construction alone.

8.6.2 Single-copy sector decompositions

When each sector contains a single copy ($r_j = 1$), the sector decomposition reduces to a weighted direct sum.

8.6.3 Prepared-block construction of BNT canonical forms

After blocking, an MPS tensor reduces to a direct sum of trace-preserving, primitive, irreducible, and injective blocks. Quotienting gauge-phase-equivalent blocks into sectors and normalizing weights produces the canonical form of Definition 8.6.1.13.

Theorem 8.6.3.1 (Prepared BNT blocks after blocking). Lean ✓ Stmt *Let A be an MPS tensor. Then there is a blocking length $p \geq 1$, nonzero weights μ_k , and blocks B_k over the p -blocked alphabet such that:*

1. *each block has positive bond dimension;*
2. *each block is trace-preserving, primitive, tensor-irreducible, and one-site injective;*
3. *$A^{[p]}$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family at every positive length.*

Proof. ✓ Proof Apply the common primitive irreducible block decomposition to (A, A) , giving a positive blocking length and a nonzero weighted family of trace-preserving primitive irreducible blocks with the positive-length MPV identity. Theorem E.3.2 supplies one common further blocking making every block one-site injective while preserving the other hypotheses; finally identify the iterated blocked alphabet with the direct one. $\square$

With the prepared blocks in hand, normalized weights produce a BNT canonical form by quotienting gauge-phase-equivalent blocks into sectors.

Theorem 8.6.3.2 (BNT canonical form of the phase-class quotient). Lean ✓ Stmt *Let $(B_k)_{k=1}^r$ have positive bond dimensions and suppose that every B_k is trace-preserving, primitive, and irreducible. Assume $\mu_k \neq 0$ and $|\mu_k| \leq 1$ for every k , and $|\mu_{k_*}| = 1$ for some k_* . Partition the blocks into matrix-product-vector phase classes, choose one representative from each class, and retain every class member as a copy of its representative with the corresponding phase-adjusted weight. The resulting sector decomposition is in BNT canonical form.*

Proof. ✓ Proof The chosen representatives inherit irreducibility and trace preservation, while primitivity gives normalized self-overlap. Distinct phase classes give gauge-phase-distinct representatives. The collapsed phase-class construction supplies the eventual linear independence and retains every original block as one copy. Its phase multipliers have unit modulus, so the adjusted weights still have modulus at most one, and at least one has modulus one. Thus Definition 8.6.1.13 holds. $\square$

Theorem 8.6.3.3 (BNT from normal blocks with normalized weights). Lean ✓ Stmt *Let $r \in \mathbb{N}$, and for each $k = 1, \dots, r$ let $D_k \in \mathbb{N}$, $\mu_k \in \mathbb{C}$, and let B_k be an MPS tensor of physical dimension d and bond dimension D_k . Assume:*

1. **positive bond dimensions:** $D_k > 0$ for all k ,
2. **TP normalization:** $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$,
3. **primitivity:** $\mathcal{E}_{B_k}$ is primitive for each k ,
4. **irreducibility:** each B_k is irreducible,
5. **nonzero weights:** $\mu_k \neq 0$ for all k ,
6. **bounded weights:** $|\mu_k| \leq 1$ for all k ,
7. **at least one weight has modulus one:** $|\mu_k| = 1$ for some k .

Then there exists a sector decomposition P such that P and $\bigoplus_{k=1}^r \mu_k B_k$ generate the same MPV family at every positive length, and P satisfies the basis-of-normal-tensors conditions (Definition 8.1.2) together with the multiplicity, span, and weight-bound conditions of Definition 8.6.1.13.

Proof. ✓ Proof The phase-class quotient groups gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars. Under the TP, primitive, and irreducible hypotheses the phase-class construction applies, and for distinct representatives $B_j, B_{j'}$ ($j \neq j'$) the cross-overlap decays,

$$\begin{aligned} \lim_{N \rightarrow \infty} O_{B_j B_{j'}}(N) &= 0, \\ \lim_{N \rightarrow \infty} O_{B_j B_j}(N) &= 1, \end{aligned}$$

which gives the eventual linear-independence condition with no injectivity required. The resulting sector decomposition inherits the span and multiplicity conditions, and the bounded-weight and global-unit hypotheses give the weight-norm conditions. $\square$

The BNT construction preserves total bond dimension because CF summands assigned to the same BNT representative have equal bond dimension. Indeed, suppose trace-preserving primitive irreducible NTs X and Y of positive bond dimension have MPV families related by $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ with $\zeta \neq 0$. The normalized self-overlaps give $|O_{XX}(N)| \rightarrow 1$. The phase relation gives the cross-overlap identity

$$\begin{aligned} O_{YX}(N) &= \zeta^N O_{XX}(N), \\ |O_{YX}(N)| &= |\zeta|^N |O_{XX}(N)|. \end{aligned}$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \rightarrow 1$. If their bond dimensions differed, the rectangular overlap-decay theorem would give $O_{YX}(N) \rightarrow 0$, a contradiction.

Theorem 8.6.3.4 (Gauge realization of prepared phase-equivalent NTs). Lean ✓ Stmt *Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimensions. If their matrix-product-vector families differ by a nonzero scalar power, then their bond dimensions are equal and, after identifying the bond spaces, X and Y are gauge-phase equivalent.*

Proof. ✓ Proof Lemma E.3.4 identifies the bond dimensions. If the resulting same-dimensional blocks were not gauge-phase equivalent, Theorem 8.4.3 would give a positive length at which their matrix product vectors are not proportional. This contradicts the assumed scalar-power relation. The normalized self-overlaps required there follow from Theorem C.5.2. $\square$

Lemma 8.6.3.5 (Gauge realization with the fixed phase scalar). Lean ✓ Stmt *Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimensions. Fix $\zeta \in \mathbb{C} \setminus \{0\}$ and suppose that, for every positive length N and every physical word σ ,*

$$V^{(N)}(Y)_\sigma = \zeta^N V^{(N)}(X)_\sigma. \quad (8.32)$$

Then, after identifying the bond spaces, there exists $G \in \text{GL}_{D_Y}(\mathbb{C})$ such that, for every physical index i ,

$$Y^i = \zeta G X^i G^{-1}. \quad (8.33)$$

Proof. ✓ Proof Theorem 8.6.3.4 gives an invertible G and a nonzero scalar η such that $Y^i = \eta G X^i G^{-1}$. By Theorem 2.4.20, $V^{(N)}(Y)_\sigma = \eta^N V^{(N)}(X)_\sigma$. Theorem C.5.2 gives consecutive positive lengths M and $M+1$ with nonzero coefficients of $V^{(M)}(X)$ and $V^{(M+1)}(X)$. Comparing the two scalar-power identities gives $\zeta^M = \eta^M$ and $\zeta^{M+1} = \eta^{M+1}$. Hence $\zeta = \eta$, which proves (8.33). $\square$

Theorem 8.6.3.6 (Dimension-preserving BNT form of prepared blocks). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 8.6.3.2, there is a BNT canonical-form sector decomposition P such that*

$$\begin{aligned}\mathcal{V}^+(P) &= \mathcal{V}^+\left(\bigoplus_{k=1}^r \mu_k B_k\right), \\ D_{\text{tot}}(P) &= \sum_{k=1}^r D_k.\end{aligned}\tag{8.34}$$

Proof. [✓ Proof](#) Use the phase-class sector decomposition from Theorem 8.6.3.2. Its flattened copies are in bijection with the original blocks, and phase-equivalent blocks have equal bond dimension by Theorem 8.6.3.4. Summing the copy dimensions over this bijection gives (8.34). $\square$

Theorem 8.6.3.7 (Exact reindexing of heterogeneous weighted block sums). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $e : \{1, \dots, r\} \simeq \{1, \dots, r'\}$ satisfy $D_k = D'_{e(k)}$. Let Q_k be the coordinate permutation whose rows are indexed by the primed block and whose columns are indexed by the unprimed block. Suppose that, for every block and physical index,*

$$\mu_k A_k^i = Q_k^\dagger \left(\mu'_{e(k)} B_{e(k)}^i \right) Q_k.\tag{8.35}$$

The induced equivalence E of flattened dependent block coordinates has a permutation matrix Q_E , with rows indexed by the primed coordinates and columns indexed by the unprimed coordinates. For every physical index i ,

$$\bigoplus_{k=1}^r \mu_k A_k^i = Q_E^\dagger \left(\bigoplus_{\ell=1}^{r'} \mu'_\ell B_\ell^i \right) Q_E.\tag{8.36}$$

Proof. [✓ Proof](#) Evaluate both sides on two flattened block coordinates. When their block labels agree, (8.35) gives the matrix entry. When the labels differ, both block-diagonal entries vanish. This proves (8.36). $\square$

Theorem 8.6.3.8 (Exact BNT reconstruction of prepared normal blocks). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 8.6.3.2, there are a BNT canonical-form sector decomposition P , an equivalence E of flattened bond coordinates, and $X \in \text{GL}_{D_{\text{tot}}(P)}(\mathbb{C})$. Write Q_E for the permutation matrix whose rows are indexed by the coordinates of P and whose columns are indexed by the original block coordinates. The first identity below holds, and the second holds for every physical index i :*

$$D_{\text{tot}}(P) = \sum_{k=1}^r D_k,\tag{8.37}$$

$$\bigoplus_{k=1}^r \mu_k B_k^i = Q_E^\dagger (X P^i X^{-1}) Q_E.\tag{8.38}$$

Proof. [✓ Proof](#) Partition the prepared blocks into matrix-product-vector phase classes. For the copy q in class j , retain the precise class scalar $\zeta_{j,q}$ chosen in the weight $\zeta_{j,q} \mu_{j,q}$. The normal-tensor phase theorem supplies an invertible $G_{j,q}$ with

$$B_{j,q}^i = \zeta_{j,q} G_{j,q} C_j^i G_{j,q}^{-1}.\tag{8.39}$$

Thus the original weighted block equals the corresponding sector copy after conjugation, since its sector weight is $\zeta_{j,q} \mu_{j,q}$. Assemble the $G_{j,q}$ into a block-diagonal matrix X and apply Theorem 8.6.3.7. This gives (8.38); summing the matched copy dimensions gives (8.37). $\square$

Lemma 8.6.3.9 (Gauge equivalence of direct sums from per-block gauge equivalence). [Lean](#) [Stmnt](#) If each pair (A_k, B_k) is gauge equivalent, then the weighted block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.

Proof. [Proof](#) Choose per-block gauge matrices via the hypothesis; then apply Lemma F.4.3. $\square$

Theorem 8.6.3.10 (Positive-length MPVs under coisometric reconstruction). [Lean](#) [Stmnt](#) Let A and B be MPS tensors, possibly with different bond dimensions. Suppose that U satisfies $UU^\dagger = \mathbb{1}$ and $A^i = U^\dagger B^i U$ for every physical index i . Then A and B generate the same matrix product vectors at every positive length.

Proof. [Proof](#) Multiply the reconstruction identities around a positive-length cycle. Each adjacent pair $UU^\dagger$ cancels, and cyclicity of the trace removes the remaining outer factors. $\square$

Theorem 8.6.3.11 (Sector BNT form of normalized CPSV canonical data). [Lean](#) [Stmnt](#) Let A have CPSV canonical-form data satisfying the separate normalization condition (7.25), and assume $A \neq 0$. Then there is a BNT canonical-form sector decomposition P such that

$$\begin{aligned} \mathcal{V}^+(A) &= \mathcal{V}^+(P), \\ D_{\text{tot}}(P) &= \sum_{k=1}^r D_k. \end{aligned} \tag{8.40}$$

The total dimension $D_{\text{tot}}(P)$ equals the sum of the retained block dimensions.

Proof. [Proof](#) Apply Theorem 8.6.3.12. Its reconstruction writes $A^i = U^\dagger B^i U$, where $UU^\dagger = \mathbb{1}$ and $B^i = XP^i X^{-1}$. Theorem 8.6.3.10 shows that A and B generate the same matrix product vectors at every positive length. Theorem 2.4.19 gives the same conclusion for B and P . Transitivity yields $\mathcal{V}^+(A) = \mathcal{V}^+(P)$, while the dimension identity is already part of the exact reconstruction theorem. $\square$

Theorem 8.6.3.12 (Exact coisometric BNT reconstruction). [Lean](#) [Stmnt](#) Let A have CPSV canonical-form data satisfying the separate normalization condition (7.25), and assume $A \neq 0$. There are a BNT canonical-form sector decomposition P , a coisometry $U \in \mathbb{C}^{D_{\text{tot}}(P) \times D}$, and $X \in \text{GL}_{D_{\text{tot}}(P)}(\mathbb{C})$. The first two identities below hold, and the third holds for every physical index i :

$$D_{\text{tot}}(P) = \sum_{k=1}^r D_k, \tag{8.41}$$

$$UU^\dagger = \mathbb{1}, \tag{8.42}$$

$$A^i = U^\dagger (XP^i X^{-1}) U. \tag{8.43}$$

Proof. [Proof](#) The canonical-form data give the exact reconstruction

$$A^i = V^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) V, \tag{8.44}$$

$$VV^\dagger = \mathbb{1}. \tag{8.45}$$

Put every normal block into a trace-preserving primitive irreducible gauge Y , and apply Theorem 8.6.3.8. Let Q be the permutation matrix of its flattened coordinate equivalence, so $QQ^\dagger = \mathbb{1}$.

Reindexing Y by the same equivalence gives an invertible matrix W on the bond space of P and the identities

$$QY = WQ, \quad (8.46)$$

$$Y^{-1}Q^\dagger = Q^\dagger W^{-1}. \quad (8.47)$$

If X_0 is the prepared-block gauge, set $X = W^{-1}X_0$ and $U = QV$. Substitution in (8.44) gives (8.43), while

$$UU^\dagger = Q(VV^\dagger)Q^\dagger = \mathbb{1}.$$

This proves (8.42). The block enumeration gives (8.41). $\square$

Theorem 8.6.3.13 (Equal-ambient gauge equivalence for CPSV canonical forms). [Lean](#) [✓ Stmt](#) *Let A and B have the same ambient bond dimension D . Suppose each tensor has CPSV canonical-form data, both data satisfy the separate weight normalization condition (7.25), neither tensor is zero, and $\mathcal{V}^+(A) = \mathcal{V}^+(B)$. Then A and B are gauge equivalent: there is $G \in \text{GL}_D(\mathbb{C})$ such that, for every physical index i ,*

$$B^i = GA^iG^{-1}. \quad (8.48)$$

The result applies to tensors sharing the ambient bond dimension D .

The cited sources supply the retained canonical-form comparison. The passage back from the retained tensors to the ambient tensors is the coisometric lifting of Theorem 7.8.5.

Proof. [✓ Proof](#) Apply Theorem 8.6.3.12 to obtain BNT canonical forms P, Q , coisometries U_A, U_B , and retained-space gauges X_A, X_B with

$$\begin{aligned} A^i &= U_A^\dagger X_A P^i X_A^{-1} U_A, \\ B^i &= U_B^\dagger X_B Q^i X_B^{-1} U_B. \end{aligned} \quad (8.49)$$

Theorem 8.6.3.10, gauge invariance, and the hypothesis give

$$\mathcal{V}^+(P) = \mathcal{V}^+(A) = \mathcal{V}^+(B) = \mathcal{V}^+(Q). \quad (8.50)$$

Theorem 9.4.2.1 therefore identifies the retained dimensions and gives Y such that $Q^i = YP^iY^{-1}$. After identifying the retained spaces, the two middle tensors in (8.49) are conjugate by $X_BYX_A^{-1}$. Theorem 7.8.5 aligns the two coisometries inside the common ambient space and yields (8.48). $\square$

Combining the after-blocking preparation with the weight normalization of Theorem E.3.7 gives the arbitrary-input statement.

Theorem 8.6.3.14 (Normalized BNT form after blocking for a nonzero MPV family). [Lean](#) [✓ Stmt](#) *Let A be an MPS tensor whose matrix-product family has a nonzero coefficient at some positive length. Then there exist a blocking length $p \geq 1$, a positive real m , and a sector decomposition P over the blocked alphabet in the BNT canonical form of Definition 8.6.1.13, such that for every positive length N ,*

$$V^{(N)}(A^{[p]}) = m^N V^{(N)}(P). \quad (8.51)$$

The scalar m^N is the per-site normalization cost of the line-246 weight choice, which is realized rather than assumed. The nonzero-family hypothesis is the standing convention of [11, Section 2.3]: the unit-modulus weight chosen there presupposes a nonempty weight family. The normalization and the proportional-MPV comparison are documented in [33].

Proof. ✓ Proof Choose a word w with $\text{tr}(A^w) \neq 0$. For every $p \geq 1$, there is a $q \geq 1$ such that $\text{tr}((A^w)^{pq}) \neq 0$. Otherwise Newton–Girard applied to $(A^w)^p$ and the zero matrix would show that $(A^w)^p$, and hence A^w , is nilpotent, contradicting $\text{tr}(A^w) \neq 0$. Repeating w exactly pq times gives a nonzero coefficient of the p -blocked tensor. Thus the prepared family at the blocking length supplied by Theorem 8.6.3.1 is nonempty.

Normalize its weights by their maximal modulus m (Theorem E.3.7). By the weight-scaling identity of Lemma E.3.8,

$$\begin{aligned} V^{(N)}(A^{[p]}) &= V^{(N)}\left(\bigoplus_k \mu_k B_k\right) = V^{(N)}\left(\bigoplus_k m \frac{\mu_k}{m} B_k\right), \\ &= m^N V^{(N)}\left(\bigoplus_k \frac{\mu_k}{m} B_k\right) = m^N V^{(N)}(P). \end{aligned}$$

where the last identity is the prepared supplier of Theorem 8.6.3.3 applied to the normalized weights, which gives the BNT canonical form P . $\square$

8.7 Coefficient comparison and copy-weight recovery

The exact matching of Section 8.8 produces a coefficient identity on each matched pair. By Newton–Girard power-sum recovery, this identity determines the per-copy weight multisets with multiplicity [11, Appendix A].

8.7.1 Matched-sector weight multiset equality

For a matched pair of sectors with $c_N^{(j_0)}(P) = \zeta^N c_N^{(\tilde{k}_0)}(Q)$ for all $N > N_0$ and $\zeta \neq 0$, power-sum recovery yields multiset equality of the per-copy weights, inducing an explicit copy permutation that realizes $\mu_{j,q} = \nu_{j,q} e^{i\varphi_j}$ [11, Appendix A]. In the equal-MPV case, Chapter 9 obtains this coefficient identity by expanding both total MPVs in the matched BNT basis and applying eventual linear independence (Lemma 9.2.2). For merely proportional total MPVs, a common length-dependent scalar remains in every matched coefficient identity and the scalar-free comparison need not hold; see Remark 9.3.6.

Lemma 8.7.1.1 (Matched-sector weight multiset equality). Lean ✓ Stmt Let P, Q be sector decompositions with weights $\mu_{j,q}$ and $\nu_{k,q'}$, and fix indices $j_0, \tilde{k}_0$, and a nonzero scalar $\zeta \in \mathbb{C} \setminus \{0\}$. Suppose that for every $N > N_0$,

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q). \quad (8.52)$$

Then the per-sector multiplicities agree, $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$, and the rescaled per-copy weight multisets coincide. This is [11, Appendix A] read on a single matched sector.

Proof. ✓ Proof Expand both sides of the coefficient identity (8.52) into per-copy power sums $\sum_q \mu_{j_0,q}^N$ and $\sum_{q'} \nu_{\tilde{k}_0,q'}^N$, and distribute ζ^N through the right-hand sum to obtain, for every $N > N_0$,

$$\sum_q \mu_{j_0,q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0,q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 8.5.1), using $\zeta \neq 0$ together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality. $\square$

Lemma 8.7.1.2 (Matched-sector weight-permutation extractor). Lean ✓ Stmt *Under the hypotheses of Lemma 8.7.1.1, the per-sector multiplicities agree and there is a permutation τ of the per-copy index set identifying the individual weights up to the gauge phase: for every q , $\nu_{\tilde{k}_0, \tau(q)} = \zeta^{-1} \cdot \mu_{j_0, q}$. This is the per-copy realisation of [11, Appendix A]: the multiset equality from Lemma 8.7.1.1 is upgraded to a concrete indexing between the matched P - and Q -copies.*

Proof. ✓ Proof Invoke Lemma 8.7.1.1 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality identification and cancel ζ via $\zeta \neq 0$ to convert the gauge factor on the right of the pointwise identity to ζ^{-1} on the left. $\square$

8.8 Strong existential matching by exact linear independence

The exact matching shows that equal MPV families force each basis sector of one canonical form to pair with a unique basis sector of the other [11, Appendix A].

Theorem 8.8.1 (Exact single-sector matching). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family at all positive lengths. For every sector j of P there exist a sector k of Q with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.*

Proof. ✓ Proof Equality at every positive length gives eventual nonzero proportionality with scalar 1. Apply Theorem E.2.1.1 to the chosen sector. $\square$

Theorem 8.8.2 (Strong existential matching, full-basis form). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family at every positive length. Then for every sector k of Q there exist a sector j of P with matched bond dimension $\dim_P(j) = \dim_Q(k)$, a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.*

Proof. ✓ Proof Convert equality to eventual nonzero proportionality with scalar 1 and apply Theorem E.2.1.2. $\square$

8.8.1 Bijective matching by symmetry

Applying the matching in both directions yields a bijection between the basis sectors: the basis-distinctness clause ensures injectivity both ways, so $g_a = g_b$ and $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$ on the full basis [11, Appendix A].

Theorem 8.8.1.1 (Bijective matching by symmetry). Lean ✓ Stmt *Let P, Q be two BNT canonical forms with the same MPV family at every positive length. Then there is a bijection $\beta : J(Q) \simeq J(P)$ such that for every $k \in J(Q)$ the bond dimensions match, the basis tensors $P_{\beta(k)}$ and Q_k are gauge-phase equivalent after identifying the matched bond dimensions, and the basis cross-overlap does not tend to zero.*

Proof. ✓ Proof Apply the full-basis matcher in both directions, using symmetry of the same-MPV relation. Lemma E.2.1.3 converts these two families of matches into the required bijection. $\square$

Chapter 9

Proof of the Fundamental Theorem

For BNT canonical forms P and Q , write the weighted direct sums over normal-tensor sectors as $P^i = \bigoplus_{j,q} \mu_{j,q} P_j^i$ and $Q^i = \bigoplus_{k,q} \nu_{k,q} Q_k^i$. When the generated matrix product vectors are proportional, the normal sectors are matched by a bijection β , unit phases ζ_k , and invertible matrices X_k satisfying

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

In the equal-MPV case, eventual linear independence yields the coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Finite power-sum comparison determines the copy weights, $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$ for a copy permutation τ_k . Each sector phase cancels against the inverse phase in its matched copy weight, and the block conjugacies combine into $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$, giving global gauge equivalence.

Throughout, “eventually” means “for all sufficiently large N ”. At a fixed length in that range, BNT linear independence detects exact coefficient equalities even when a weight of modulus less than 1 makes its powers decay.

9.1 Matched block maps for product algebras

The Fundamental Theorem compares the simple matrix summands under product-algebra isomorphisms, preserving the dimensions of matched blocks.

Theorem 9.1.1 (Paired component maps for products of matrix algebras). [Lean](#) [Lean](#) [Lean](#)

 **Stmt**

For a ring equivalence between products of full matrix algebras, the map induced between each pair of matched components is bijective.

Proof.  **Proof**

Apply Theorem [182, “Paired component maps are bijections”] to the matched block ideals.

□

Theorem 9.1.2 (Dimension preservation for matched matrix summands). [Lean](#) [✓ Stmt](#) *An algebra automorphism of a product of full matrix algebras preserves the dimension of every pair of summands matched by its induced permutation.*

Proof. [✓ Proof](#) Restrict the automorphism to the two matched summands. The resulting component map is a bijective complex-linear map by Theorem 9.1.1. Hence the two matrix spaces have equal complex dimensions, so their sizes have equal squares and therefore are equal. $\square$

9.2 Coefficient identities and copy weights

Let P and Q be BNT canonical forms. Their basis sectors are normalized, left-canonical, and aperiodic, and their raw copy weights satisfy the global normalization of Definition 8.6.1.13.

The proportional theorem below allows nonzero projective proportionality for all sufficiently large lengths; proportionality at every positive length is a special case.

Theorem 9.2.1 (Proportional sector matching with unit phases). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$, scalars $\zeta_k \in \mathbb{C}$, and block gauges X_k such that $|\zeta_k| = 1$ and, for every sector k and physical index i ,*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}. \quad (9.1)$$

Diagrammatically, the gauge relation (9.1) reads

with every open virtual end a matrix index. Equivalently, for every positive length N and word σ ,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

This is [11, Appendix MPV theorem statement, lines 1167–1170, and proof line 1182].

Proof. [✓ Proof](#) Theorem F.1.1 gives a bijection $\beta : J(Q) \simeq J(P)$ together with equal bond dimensions, gauge-phase equivalence, and

$$O_{P_{\beta(k)} Q_k}(N) \not\rightarrow 0$$

for every matched pair. Its proof defines $T = \{j \in J(P) : O_{P_j Q_k}(N) \rightarrow 0 \text{ for every } k\}$, substitutes exact Q -sector relations for $j \notin T$, applies linear independence to $\{P_j : j \in T\} \cup \{Q_k : k \in J(Q)\}$, and reverses direction; basis separation and finiteness turn the resulting matches into β .

Unpack each matched gauge-phase equivalence to choose an invertible matrix X_k and a nonzero scalar ζ_k . This gives $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ and, by taking traces around a positive-length word,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

Lemma F.1.2 applies to this MPV identity. Equivalently, the normalized self-overlaps satisfy

$$O_{Q_k Q_k}(N) = |\zeta_k|^{2N} O_{P_{\beta(k)} P_{\beta(k)}}(N),$$

and both outer sequences tend to 1, which forces $|\zeta_k| = 1$. $\square$

Lemma 9.2.2 (Full-basis coefficient identity from matched phases). [Lean](#) [✓ Stmt](#) Let P be a BNT canonical form, let Q be a sector decomposition, and assume P_{tot} and Q_{tot} have the same MPV family at every positive length. Suppose that there is a bijection $\beta : J(Q) \simeq J(P)$ and scalars $\zeta_k \in \mathbb{C}$ such that, for every $k \in J(Q)$, every positive length N , and every spin configuration σ of length N ,

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma. \quad (9.2)$$

Then for every $k \in J(Q)$ there is N_0 such that, for all $N > N_0$, $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$. This is the coefficient-comparison step of [11, Appendix MPV proof, lines 1187–1188] in full-basis form.

Proof. [✓ Proof](#) For every N , expand the equality of total MPVs as

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

The phase relation (9.2) gives $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$, so reindexing the Q -sum by β onto the P -basis yields

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{j \in J(P)} \zeta_{\beta^{-1}(j)}^N c_N^{(\beta^{-1}(j))}(Q) |V^{(N)}(P_j)\rangle.$$

For all sufficiently large N the P -basis MPV family $\{|V^{(N)}(P_j)\rangle\}_{j \in J(P)}$ is linearly independent, by the BNT eventual linear independence of P . Lemma E.1.10 then extracts the coefficient equalities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ from this eventual linear independence, for all sufficiently large N . $\square$

Definition 9.2.3 (Copy-weight matching for matched BNT sectors). [Lean](#) [✓ Stmt](#) Let the sectors of two decompositions P and Q be matched by $\beta : J(Q) \simeq J(P)$, with sector phases ζ_k . A copy-weight matching consists of copy permutations τ_k such that, for all sectors k and copies q , $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. This is the copy-level conclusion of [11, Appendix MPV proof, line 1188].

Definition 9.2.4 (Copy-weight matching from matched-sector coefficient identities). [Lean](#) [✓ Stmt](#) The construction takes as input a sector matching $\beta : J(Q) \simeq J(P)$, nonzero sector phases ζ_k , and the eventual matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q). \quad (9.3)$$

It returns the copy-weight matching obtained by choosing, in every sector, a copy permutation τ_k satisfying $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. This is the sectorwise construction in [11, Appendix MPV proof, line 1188].

Proof. [✓ Proof](#) Eventually the coefficient identity (9.3) in sector k reads

$$\sum_r \mu_{\beta(k), r}^N = \zeta_k^N \sum_q \nu_{k,q}^N = \sum_q (\zeta_k \nu_{k,q})^N.$$

Newton–Girard finite power-sum comparison (Lemma 8.7.1.2) applies sector by sector, producing a copy permutation τ_k with $\mu_{\beta(k), \tau_k(q)} = \zeta_k \nu_{k,q}$; since $\zeta_k \neq 0$ this is $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. $\square$

Theorem 9.2.5 (Equal-case sector and copy matching with differing bond dimensions). [Lean](#) [✓ Stmt](#) Let P and Q be BNT canonical forms, possibly with different sector counts and different bond dimensions before matching. If their total tensors generate the same MPV family at every

positive length, then there is a bijection $\beta : J(Q) \simeq J(P)$ such that each matched basis pair has equal bond dimension and is gauge-phase equivalent. The matched copy multiplicities agree, and there are unit phases ζ_k and copy bijections τ_k satisfying

$$\begin{aligned} Q_k^i &= \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}. \end{aligned}$$

This requires only the global weight normalization of Definition 8.6.1.13. This is the sector-data form of [11, Corollary 2.11 and Appendix A, lines 1182–1188].

Proof. ✓ Proof First, Theorem 8.8.1.1 gives the sector bijection, the bond-dimension equalities, and the gauge-phase relations. Choose their phases ζ_k and use the normalized self-overlaps to obtain $|\zeta_k| = 1$. Second, Lemma 9.2.2 expands the equality of the two total MPVs in the matched P -basis. Eventual linear independence gives, for every k and all sufficiently large N ,

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Third, expand both coefficients as finite power sums. Their eventual equality determines the two multisets of nonzero copy weights, so the multiplicities agree and a copy bijection τ_k realizes the identity $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. $\square$

9.3 The equal-MPV global gauge

For equal MPV families, the sector phases occur twice: in the matched tensor relation and with the inverse power in the matched copy weights. They therefore cancel before the block gauges are assembled.

Definition 9.3.1 (Flattened block-diagonal gauge). Lean Lean ✓ Stmt Given per-block gauges $X_k \in \text{GL}_{D_k}(\mathbb{C})$, form the block-diagonal element $\bigoplus_k X_k$ and then reindex the dependent block-coordinate space with the standard bond index $\{0, \dots, \sum_k D_k - 1\}$ of the assembled tensor. The result is an element of $\text{GL}_{\sum_k D_k}(\mathbb{C})$.

Theorem 9.3.2 (Global gauge from matched sectors and copy weights). Lean ✓ Stmt Suppose that the sectors of two sector decompositions P and Q are matched by a bijection $\beta : J(Q) \simeq J(P)$, with bond-dimension equalities, copy permutations τ_k , nonzero phases ζ_k , and block gauges X_k . If, for every sector k , copy q , and physical index i ,

$$\begin{aligned} Q_k^i &= \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}, \end{aligned} \tag{9.4}$$

then, in the flattened Q -coordinates,

$$\begin{aligned} Q_{\text{flat}}^i &= X P_{\text{matched}}^i X^{-1}, \\ X &= \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k). \end{aligned}$$

This is the direct-sum gauge construction of [11, Appendix MPV proof, lines 1189–1192], separated from the coefficient-comparison step that supplies the weight identities.

Proof. ✓ **Proof** For each matched copy,

$$\nu_{k,q} Q_k^i = (\zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}) (\zeta_k X_k P_{\beta(k)}^i X_k^{-1}) = \mu_{\beta(k), \tau_k(q)} X_k P_{\beta(k)}^i X_k^{-1}.$$

Thus every flattened weighted block is conjugated by its sector gauge X_k . Lemma F.4.3 combines these block conjugacies into the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. $\square$

Lemma 9.3.3 (Equal-MPV matched copy-weight witnesses). Lean ✓ **Stmnt** *Let P and Q be BNT canonical forms whose total tensors have the same MPV family at every positive length. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k , and block gauges X_k such that $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. For the same phases there is a copy-weight matching: after a permutation of the copies inside each matched sector, $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$.*

Proof. ✓ **Proof** The full-basis matching theorem gives the bijection and block gauges. The self-overlap comparison

$$\langle V_N(Q_k), V_N(Q_k) \rangle = |\zeta_k|^{2N} \langle V_N(P_{\beta(k)}), V_N(P_{\beta(k)}) \rangle$$

together with the BNT normalization limits on both sides forces $|\zeta_k| = 1$. The matched MPV phase identities also give the eventual coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$. Definition 9.2.4 applies the finite power-sum comparison in each sector and gives the copy-weight identities. $\square$

Theorem 9.3.4 (Full-basis global gauge in matched coordinates). Lean ✓ **Stmnt** *Let P and Q be BNT canonical forms whose total tensors have the same MPV family at every positive length. Then there exist:*

- a bijection $\beta : J(Q) \simeq J(P)$,
- bond-dimension equalities $\dim P_{\beta(k)} = \dim Q_k$,
- copy-number equalities $r_{\beta(k)}(P) = r_k(Q)$ and copy permutations τ_k ,
- unit-modulus phases ζ_k with $|\zeta_k| = 1$ and block gauges X_k ,

such that, for every sector k , copy q , and physical index i ,

$$\begin{aligned} Q_k^i &= \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}. \end{aligned}$$

Moreover, if $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ is the flattened block-diagonal gauge, then, for every physical index i , the flattened Q -coordinates satisfy

$$Q_{\text{flat}}^i = X P_{\text{matched}}^i X^{-1}.$$

This is the explicit direct-sum gauge construction of [11, Appendix MPV proof, lines 1189–1192].

Proof. ✓ **Proof** Lemma 9.3.3 gives β , the bond-dimension identifications, the phases ζ_k , the block gauges X_k , and copy permutations τ_k satisfying $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$. Theorem F.4.4 combines the block conjugacies into the flattened direct-sum conjugation by $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. $\square$

Theorem 9.3.5 (Literal equal-MPV gauge after sector-coordinate permutation). Lean ✓ Stmt Let P and Q be BNT canonical forms whose total tensors have the same MPV family at every positive length. Then the total bond dimensions agree; write their common value as D . There is

$$Y \in \mathrm{GL}_D(\mathbb{C}) \tag{9.5}$$

such that, for every physical index i ,

$$Q_{\mathrm{tot}}^i = Y P_{\mathrm{tot}}^i Y^{-1},$$

where P_{tot}^i and Q_{tot}^i are both regarded as matrices in $M_D(\mathbb{C})$ using the total bond-dimension equality. This is the BNT form of [11, Corollary 2.11 and Appendix A, lines 1189–1192].

Proof. ✓ Proof Apply Theorem 9.3.4 to obtain the flattened gauge after the sectors of P have been reindexed by the bijection between basis blocks of P and Q . The sector-coordinate permutation ρ , induced by this bijection, the copy permutations, and the bond-dimension equalities, identifies the reindexed P -side direct sum with the total tensor P_{tot} , transported along the total bond-dimension equality. Composing the permutation gauge for ρ with the sector-permuted gauge gives the single matrix Y of (9.5). $\square$

Remark 9.3.6 (Proportional sector matching does not determine copy weights). The proportional theorem determines the normal sectors, but it does not determine the copy weights or imply conjugacy of the total tensors. Let C be a normalized normal tensor and consider the one-sector BNT forms

$$P = C \oplus \tfrac{1}{2}C, \quad Q = C \oplus \tfrac{1}{3}C.$$

Their positive-length MPV families are

$$V^{(N)}(P) = (1 + 2^{-N})V^{(N)}(C), \quad V^{(N)}(Q) = (1 + 3^{-N})V^{(N)}(C).$$

They are nonzero and proportional with

$$c_N = \frac{1 + 2^{-N}}{1 + 3^{-N}},$$

but $(c_N)_N$ is not a geometric sequence ξ^N . For a one-dimensional choice of C , the total matrices have eigenvalue multisets $\{1, \frac{1}{2}\}$ and $\{1, \frac{1}{3}\}$, so they are not conjugate, even after multiplication by a common phase.

Thus the scalar-free coefficient identity used in copy-weight recovery is an additional hypothesis in the proportional setting, not a consequence awaiting an analytic estimate. CPSV16 Theorem 2.10 asserts the sector matching proved in Theorem 9.4.1. The copy-weight recovery and global block gauge belong to the equal-MPV Corollary 2.11, where the proportionality scalar is identically 1.

9.4 The proportional and equal cases

Theorem 9.4.1 (Multi-block fundamental theorem for normal tensors). Lean ✓ Stmt Let P and Q be BNT canonical forms (Definition 8.6.1.13), with bases of normal tensors $(P_j)_{j=1}^{g_a}$ and $(Q_k)_{k=1}^{g_b}$. Every listed sector has a positive number of copies, and every copy weight is nonzero. If the total tensors P_{tot} , Q_{tot} generate eventually nonzero proportional MPV families — for all

sufficiently large N there is a nonzero scalar c_N with $V^{(N)}(P_{\text{tot}}) = c_N V^{(N)}(Q_{\text{tot}})$ — then $g_a = g_b$. After relabelling the basis of normal tensors of P by a bijection $\beta : J(Q) \simeq J(P)$, there are scalars $\zeta_k \in \mathbb{C}$ with $|\zeta_k| = 1$ and invertible matrices Y_k such that, for every k and every physical index i ,

$$Q_k^i = \zeta_k Y_k P_{\beta(k)}^i Y_k^{-1}.$$

Thus the paired normal tensors have the same bond dimension ($\dim P_{\beta(k)} = \dim Q_k$) and the same gauge-phase class.

Proof. ✓ Proof Apply Theorem 9.2.1. Its partition-and-substitution argument gives a bijection $\beta : J(Q) \simeq J(P)$, equal bond dimensions, invertible block gauges X_k , and phases ζ_k of unit modulus. Taking $Y_k = X_k$ gives exactly the asserted relations. $\square$

9.4.1 Unitary gauges on normalized sectors

Lemma 9.4.1.1 (Phase normalization for an irreducible gauge). Lean ✓ Stmt Let A and B be left-canonical tensors of common positive bond dimension, with B irreducible. If X is invertible, $\zeta \neq 0$, and $B^i = \zeta X A^i X^{-1}$ for every i , then $|\zeta| = 1$.

Proof. ✓ Proof

Let ρ be a nonzero positive fixed point of $\mathcal{E}_A$ and set $\sigma = X\rho X^\dagger$. Then σ is nonzero and positive, and $\mathcal{E}_B(\sigma) = |\zeta|^2 \sigma$. The map $\mathcal{E}_B$ also has a nonzero positive fixed point. Its irreducibility therefore forces $|\zeta|^2 = 1$. $\square$

Lemma 9.4.1.2 (Unitary replacement with the same phase). Lean ✓ Stmt Let A and B be left-canonical irreducible tensors of common positive bond dimension satisfying $B^i = \zeta X A^i X^{-1}$ for an invertible X and a nonzero scalar ζ . There is a unitary matrix U , with the same scalar ζ , such that $|\zeta| = 1$ and $B^i = \zeta U A^i U^\dagger$ for every i .

Proof. ✓ Proof Lemma 9.4.1.1 gives $|\zeta| = 1$. Substitution into the left-canonical identities gives

$$\sum_i A^{i\dagger} (X^\dagger X) A^i = X^\dagger X.$$

Thus $X^\dagger X$ is a positive-semidefinite fixed point of the adjoint transfer map of A . Irreducibility gives $X^\dagger X = c\mathbb{1}$ for some $c > 0$, so $X^{-1} = c^{-1} X^\dagger$ and $U = c^{-1/2} X$ is unitary. Moreover,

$$U A^i U^\dagger = c^{-1} X A^i X^\dagger = X A^i X^{-1}.$$

$\square$

Corollary 9.4.1.3 (Canonical Form II unitary refinement of proportional matching). Lean ✓ Stmt Under the hypotheses of Theorem 9.4.1, there exist a bijection $\beta : J(Q) \simeq J(P)$, unitary matrices $U_k \in \text{U}(\dim Q_k)$, and phases $\zeta_k \in \mathbb{C}$ with $|\zeta_k| = 1$, such that $\dim P_{\beta(k)} = \dim Q_k$ and, for every k and every physical index i ,

$$Q_k^i = \zeta_k U_k P_{\beta(k)}^i U_k^\dagger.$$

This is the proportional part of [11, Corollary A.6, lines 1197–1199].

Proof. ✓ Proof The proportional sector-matching theorem gives a bijection β , invertible matrices X_k , and unit-modulus phases ζ_k with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. The matched tensors are left-canonical and irreducible. Hence Lemma F.8.1 gives unitary matrices U_k satisfying $X_k P_{\beta(k)}^i X_k^{-1} = U_k P_{\beta(k)}^i U_k^\dagger$, and therefore $Q_k^i = \zeta_k U_k P_{\beta(k)}^i U_k^\dagger$. $\square$

9.4.2 The equal case

Theorem 9.4.2.1 (Equal BNT canonical forms are globally conjugate). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT canonical forms (Definition 8.6.1.13) whose total tensors $P_{\text{tot}}, Q_{\text{tot}}$ generate the same MPV family for every positive system size N . Every listed sector has a positive number of copies and every copy weight is nonzero. Then their total bond dimensions coincide; writing the common value as D , there is one invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that, for every physical index i ,*

$$Q_{\text{tot}}^i = X P_{\text{tot}}^i X^{-1}.$$

Proof. [✓ Proof](#) This is Theorem 9.3.5. The BNT comparison of Chapter 8 provides the basis relabelling, the per-block gauge-phase equivalences, and the copy-weight multiset matching, which combine into the block-diagonal global gauge. $\square$

Corollary 9.4.2.2 (Canonical Form II unitary refinement in the equal case). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT canonical forms whose total tensors generate the same MPV family at every positive length. Assume every listed sector has a positive number of copies and every copy weight is nonzero, as in Definition 8.6.1.13. Their total bond dimensions agree; writing the common value as D , there is a unitary matrix $U \in \text{U}(D)$ such that, for every physical index i ,*

$$Q_{\text{tot}}^i = U P_{\text{tot}}^i U^\dagger.$$

Proof. [✓ Proof](#) For each matched sector, retain the phase in both identities

$$\begin{aligned} Q_k^i &= \zeta_k U_k P_{\beta(k)}^i U_k^\dagger, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}. \end{aligned}$$

For every matched copy (k, q) ,

$$\nu_{k,q} \zeta_k = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)} \zeta_k = \mu_{\beta(k), \tau_k(q)}.$$

Hence $Q_{\text{flat}}^i = X P_{\text{match}}^i X^{-1}$. Repeating U_k on its matched copies gives

$$\begin{aligned} X &= \bigoplus_{k,q} U_k, \\ X^\dagger X &= \mathbb{1}. \end{aligned}$$

If Π is the permutation matrix restoring the original sector and copy order, then $\Pi^\dagger \Pi = \mathbb{1}$. Hence

$$\begin{aligned} U &= X \Pi, \\ U^\dagger U &= \mathbb{1}, \end{aligned}$$

and the permutation of matched coordinates gives

$$Q_{\text{tot}}^i = U P_{\text{tot}}^i U^\dagger.$$

$\square$

Chapter 10

Symmetries, Physical String Order, and Virtual-Boundary Nondecay

On-site symmetries of matrix product states induce projective representations on the virtual bond space through the Fundamental Theorem of MPS [193, 194]. For injective tensors, invariance under a physical group action determines virtual gauge transformations whose 2-cocycles classify symmetry-protected topological (SPT) phases, while distinguishing physical endpoint string order parameters from virtual-boundary nondecay.

10.1 Virtual Local Covariance and Projective Bond Actions

When an MPS is invariant under an on-site group action, the resulting virtual gauge transformations form a projective representation on the bond space.

Definition 10.1.1 (Twisted tensor). [Lean](#) [✓ Stmt](#) Given a group representation $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ on the physical index, the *g-twisted tensor* is defined, for each $i \in \{0, \dots, d-1\}$, by

$$\tilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij} A^j. \quad (10.1)$$

Definition 10.1.2 (MPV-family invariance under physical twists). [Lean](#) [✓ Stmt](#) An MPS tensor A has *on-site symmetry* under a representation $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$ if $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$ for every $g \in G$. This equality of periodic MPV families provides the condition for constructing virtual symmetry gauges.

Lemma 10.1.3 (On-site symmetry under rescaling and gauge). [Lean](#) [Lean](#) [✓ Stmt](#) If A has on-site symmetry under U , then so does cA for every $c \in \mathbb{C}$, and so does every tensor gauge equivalent to A .

Proof. [✓ Proof](#) Twisting commutes with rescaling, $(\widetilde{cA})_g = c \tilde{A}_g$, and the N -site coefficients of cB are c^N times those of B . If $B^i = XA^iX^{-1}$, then $(\tilde{B}_g)^i = \sum_j U(g)_{ij} XA^jX^{-1} = X(\tilde{A}_g)^iX^{-1}$, so $\tilde{B}_g$ is gauge equivalent to $\tilde{A}_g$, and Theorem 2.4.19 gives $\mathcal{V}(B) = \mathcal{V}(A) = \mathcal{V}(\tilde{A}_g) = \mathcal{V}(\tilde{B}_g)$. $\square$

Lemma 10.1.4 (Functoriality of twisting). Lean ✓ Stmt *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g, \quad (10.2)$$

i.e. twisting by gh is the same as first twisting by h and then by g .

Proof. ✓ Proof Expand both sides of (10.2) using (10.1) and $U(gh) = U(g)U(h)$, then swap the order of summation. □

Lemma 10.1.5 (Identity twist). Lean ✓ Stmt *Twisting by the identity group element is trivial: $\widetilde{A}_1 = A$.*

Proof. ✓ Proof Since $U(1) = \mathbb{1}_d$, each component in (10.1) reduces to $\sum_j \delta_{ij} A^j = A^i$. □

Definition 10.1.6 (Blocked on-site action). Lean ✓ Stmt *For a block of L physical sites, the induced action is the Kronecker-power representation*

$$U^{[L]}(g) := U(g)^{\otimes L} \in \mathrm{GL}_{d^L}(\mathbb{C}).$$

Lemma 10.1.7 (Twisting commutes with blocking). Lean ✓ Stmt *For every $g \in G$ and $L \geq 0$,*

$$\widetilde{(A^{[L]})}_g^{U^{[L]}} = (\widetilde{A}_g)^{[L]}. \quad (10.3)$$

Proof. ✓ Proof The twisted tensor $\widetilde{A}_g$ is the physical rotation of A by $U(g)$, and $U^{[L]}(g) = U(g)^{\otimes L}$. Hence (10.3) is the case $W = U(g)$, $C = A$ of Theorem G.1.1.3. □

Theorem 10.1.8 (MPV-family symmetry is preserved by blocking). Lean ✓ Stmt *If A has MPV-family invariance under U , then $A^{[L]}$ has MPV-family invariance under $U^{[L]}$ for every L .*

Proof. ✓ Proof Equation (10.3) identifies the twisted blocked tensor with the blocking of the twisted tensor. Equality of MPV families is preserved by blocking, so the defining equality for A yields the defining equality for $A^{[L]}$. □

Lemma 10.1.9 (Virtual gauges survive blocking). Lean ✓ Stmt *If $R \in \mathrm{GL}_D(\mathbb{C})$ satisfies $(\widetilde{A}_g)^i = R A^i R^{-1}$ for all i , then $((\widetilde{A}^{[L]})_g^{U^{[L]}})^I = R (A^{[L]})^I R^{-1}$ for every blocked index I .*

Proof. ✓ Proof By (10.3) the left side is a product of L matrices $(\widetilde{A}_g)^{i_k} = R A^{i_k} R^{-1}$, and the inner factors $R^{-1}R$ cancel. □

Lemma 10.1.10 (Symmetric twists are gauge equivalent). Lean ✓ Stmt *If A is injective and on-site symmetric under U , then for every $g \in G$ the twisted tensor $\widetilde{A}_g$ is gauge equivalent to A .*

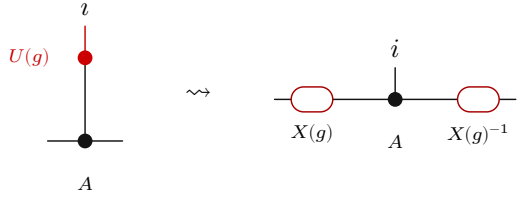
Proof. ✓ Proof On-site symmetry gives $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$, and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence. □

Remark 10.1.11 (Virtual representation via the Fundamental Theorem). On-site symmetry identifies A and $\widetilde{A}_g$ as defining the same MPV family. The single-block Fundamental Theorem yields the gauge matrices in Lemma 10.1.10; gauge uniqueness and twist functoriality then establish the projective representation of Theorem 10.1.21.

Theorem 10.1.12 (Virtual symmetry equation). [Lean](#) [✓ Stmt](#) If A is injective and on-site symmetric under U , then for every $g \in G$ there exist an invertible matrix $X(g) \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\varphi(g) \in \mathbb{C}^\times$ (in fact $\varphi = 1$ in the single-block case) such that, for every physical index i ,

$$\tilde{A}_g^i = \varphi(g) X(g) A^i X(g)^{-1}. \quad (10.4)$$

Proof. [✓ Proof](#) The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 10.1.10 to obtain $X(g)$, then set $\varphi(g) = 1$ in (10.4). $\square$

Lemma 10.1.13 (Gauge uniqueness up to scalar). [Lean](#) [✓ Stmt](#) Let A be an injective tensor. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there exists a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.

Proof. [✓ Proof](#) The matrix $Z = Y^{-1}X$ commutes with every A^i . Since A is injective, $\{A^i\}$ spans $M_D(\mathbb{C})$, so Z lies in the centre of $M_D(\mathbb{C})$. By Lemma 2.1.5, Z is a scalar matrix. Since Z is invertible, that scalar is nonzero, and $Y = u \cdot X$. $\square$

Theorem 10.1.14 (Gauge uniqueness from equality of MPV families). [Lean](#) [✓ Stmt](#) Let A be an injective tensor and assume that A and B generate the same MPV family. If $X, Y \in \text{GL}_D(\mathbb{C})$ both satisfy $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i , then there is a nonzero scalar $u \in \mathbb{C}^\times$ such that $Y = u \cdot X$.

Proof. [✓ Proof](#) This is exactly Lemma 10.1.13; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced. $\square$

Lemma 10.1.15 (Gauge-phase uniqueness). [Lean](#) [✓ Stmt](#) Let A be an injective tensor with $D \geq 1$. If $X, Y \in \text{GL}_D(\mathbb{C})$ and $c \in \mathbb{C}$ satisfy $X^{-1}A^iX = cY^{-1}A^iY$ for all i , then $c = 1$ and $X = uY$ for some $u \in \mathbb{C}^\times$ [16, Section III.A].

Proof. [✓ Proof](#) The matrix $Z = YX^{-1}$ satisfies $ZA^i = cA^iZ$ for every i , hence $ZM = cMZ$ for every matrix M by linearity. Taking $M = \mathbb{1}$ gives $(1-c)Z = 0$, so $c = 1$ because Z is invertible. Then Z commutes with $M_D(\mathbb{C})$, so it is a nonzero scalar by Lemma 2.1.5. $\square$

Lemma 10.1.16 (Gauge phases on words). [Lean](#) [✓ Stmt](#) If $X, Y \in \text{GL}_D(\mathbb{C})$ and $c \in \mathbb{C}$ satisfy $X^{-1}A^iX = cY^{-1}A^iY$ for all i , then for every word w ,

$$X^{-1}A^wX = c^{|w|} Y^{-1}A^wY. \quad (10.5)$$

Proof. [✓ Proof](#) By induction on w : inserting XX^{-1} and YY^{-1} between the letters gives $X^{-1}A^{iw}X = (X^{-1}A^iX)(X^{-1}A^wX)$, and likewise for Y . $\square$

Lemma 10.1.17 (Gauge-phase uniqueness for normal tensors). [Lean](#) [✓ Stmt](#) Let A be a normal tensor with $D \geq 1$. If $X, Y \in \text{GL}_D(\mathbb{C})$ and $c \in \mathbb{C}$ satisfy $X^{-1}A^iX = cY^{-1}A^iY$ for all i , then $c = 1$ and $X = uY$ for some $u \in \mathbb{C}^\times$ [16, Section III.A].

Proof. ✓ Proof By Lemma 10.1.16 the relation propagates to words of length N with the scalar c^N . For an injective blocking length N , Lemma 10.1.15 applied to the blocked tensor gives $X = uY$. Substituting back, $(1 - c)Y^{-1}A^iY = 0$ for every i , and some A^i is nonzero because the words of length N span the full matrix algebra; hence $c = 1$. $\square$

Definition 10.1.18 (Scalar 2-cochain). Lean ✓ Stmt A scalar 2-cochain on a group G is a function $\omega : G \times G \rightarrow \mathbb{C}^\times$.

Definition 10.1.19 (Projective representation). Lean ✓ Stmt Given a scalar 2-cochain ω , a projective representation of G on $\mathbb{C}^D$ is a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g)\rho(h) = \omega(g, h)\rho(gh). \quad (10.6)$$

Lemma 10.1.20 (Associativity forces the cocycle condition). Lean ✓ Stmt If ρ is a projective representation with factor system ω and $D > 0$, then ω satisfies the 2-cocycle identity

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k). \quad (10.7)$$

Proof. ✓ Proof Expand $\rho(g)(\rho(h)\rho(k))$ and $(\rho(g)\rho(h))\rho(k)$ using the projective multiplication law (10.6) and equate the results via matrix associativity. The hypothesis $D > 0$ allows cancellation of the invertible matrix factor, giving (10.7). $\square$

Theorem 10.1.21 (Virtual representation theorem for injective MPS). Lean ✓ Stmt Let A be an injective MPS tensor that is on-site symmetric under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. Then there exist:

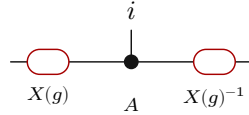
1. a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$, and
2. a map $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying, for all $g, h \in G$,

$$\rho(g)\rho(h) = \omega(g, h)\rho(gh).$$

such that, defining the gauge matrices $X(g) := \rho(g^{-1})$, one has, for every $g \in G$ and $i \in \{0, \dots, d-1\}$,

$$\tilde{A}_g^i = X(g) A^i X(g)^{-1}. \quad (10.8)$$

In tensor-network notation, for each fixed g this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular, ρ is a projective representation of G on the bond space $\mathbb{C}^D$.

Proof. ✓ Proof Since A is injective and on-site symmetric, each twisted tensor $\tilde{A}_g$ is gauge equivalent to A (Lemma 10.1.10). Choose $X(g) \in \text{GL}_D(\mathbb{C})$ satisfying (10.8) for all i .

By functoriality (Lemma 10.1.4), the product $X(h)X(g)$ also intertwines A with $\tilde{A}_{gh}$. Since $X(gh)$ does the same, gauge uniqueness (Lemma 10.1.13) gives a nonzero scalar $\omega'(h, g)$ such that

$$X(h)X(g) = \omega'(h, g)X(gh). \quad (10.9)$$

Defining $\rho(g) := X(g^{-1})$ converts (10.9) to the standard law (10.6), where $\omega(g, h) := \omega'(g^{-1}, h^{-1})$. $\square$

Corollary 10.1.22 (2-cocycle identity for the factor system). [Lean] ✓ Stmt If $D > 0$, the factor system ω from Theorem 10.1.21 satisfies the 2-cocycle identity for all $g, h, k \in G$:

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k).$$

Proof. ✓ Proof Theorem 10.1.21 gives ρ and ω . By Lemma 10.1.20, the cocycle identity (10.7) follows from associativity of matrix multiplication and invertibility of the representation matrices.

10.2 Cohomology Classes of Virtual Symmetry Cocycles

Projective representations are classified up to projective equivalence by the second cohomology group $H^2(G, \mathbb{C}^\times)$ of cocycles modulo coboundaries.

Definition 10.2.1 (Coboundary). [Lean] [Stm] A scalar 2-cocycle ω is a *coboundary* if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

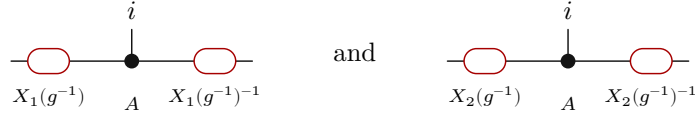
$$\omega(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}. \quad (10.10)$$

Definition 10.2.2 (Cohomologous cocycles). [Lean] [✓ Stmt] Two cocycles ω_1, ω_2 are *cohomologous* (written $\omega_1 \sim \omega_2$) if there exists $\varphi: G \rightarrow \mathbb{C}^\times$ such that, for all $g, h \in G$,

$$\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h). \quad (10.11)$$

Theorem 10.2.3 (Gauge independence of the cocycle class). Lean ✔ Stmt *If the bond dimension satisfies $D \geq 1$ and two projective representations ρ_1, ρ_2 (with cocycles ω_1, ω_2) both arise as virtual representations of the same injective MPS tensor A under an on-site symmetry U , then $\omega_2 \sim \omega_1$, and hence also $\omega_1 \sim \omega_2$ by symmetry.*

Proof. ✓ Proof Gauge uniqueness gives nonzero scalars $f(k)$ with $X_2(k) = f(k) X_1(k)$ for every physical twist k . Since the projective representation is $\rho_a(g) = X_a(g^{-1})$, setting $k = g^{-1}$ gives $\rho_2(g) = f(g^{-1})\rho_1(g)$. Equivalently, for each fixed g the two local gauges



describe the same twisted tensor, so they differ by a scalar. Set $\alpha(g) := f(g^{-1})$. Substituting $\rho_2(g) = \alpha(g)\rho_1(g)$ into the projective law (10.6) and cancelling the invertible matrix $X_1((gh)^{-1}) = \rho_1(gh)$ yields

$$\omega_2(g, h) = \alpha(g)\alpha(h)\alpha(gh)^{-1}\omega_1(g, h). \quad (10.12)$$

Thus ω_2 is the coboundary multiple of ω_1 with cochain α . This is precisely the orientation required by (10.11) for the stated relation $\omega_2 \sim \omega_1$. Equivalently,

$$\omega_1(g, h) = \alpha(g)^{-1} \alpha(h)^{-1} \alpha(gh) \omega_2(g, h).$$

so the inverse cochain also gives the symmetric conclusion $\omega_1 \sim \omega_2$.

Definition 10.2.4 (Multiplicative 2-cocycle condition). [Lean] [✓ Stmt] A scalar 2-cochain $\omega: G \times G \rightarrow \mathbb{C}^\times$ is a *2-cocycle* if it satisfies, for all $g, h, k \in G$,

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k).$$

Theorem 10.2.5 (A projective factor system is a cocycle). [Lean](#) [✓ Stmt](#) If $D > 0$ and $\rho : G \rightarrow \mathrm{GL}_D(\mathbb{C})$ obeys $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$, then ω is a $\mathbb{C}^\times$ -valued 2-cocycle.

Proof. [✓ Proof](#) Associativity gives the scalar identity of Lemma 10.1.20. Because the inclusion $\mathbb{C}^\times \hookrightarrow \mathbb{C}$ is injective, the same identity holds in $\mathbb{C}^\times$, which is precisely Definition 10.2.4. $\square$

Definition 10.2.6 (Second cohomology quotient). [Lean](#) [✓ Stmt](#) The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times) / \sim, \quad (10.13)$$

where $\sim$ is the coboundary-equivalence relation.

Definition 10.2.7 (Agreement of factor-system cohomology classes). [Lean](#) [✓ Stmt](#) Two projective representations are *projectively equivalent* when their factor systems are cohomologous in $H^2(G, \mathbb{C}^\times)$.

Theorem 10.2.8 (Factor-system class agreement). [Lean](#) [✓ Stmt](#) Let ρ_1, ρ_2 be projective representations with factor systems ω_1, ω_2 . Then

$$\rho_1 \sim_{\mathrm{proj}} \rho_2 \iff \omega_1 \sim \omega_2. \quad (10.14)$$

Proof. [✓ Proof](#) This follows from the definition of projective equivalence as cohomology-class equality for factor systems. $\square$

10.2.1 Non-triviality via the commutator phase

For an abelian symmetry group, the antisymmetric commutator phase is invariant under a change of cocycle representative, so a value different from 1 witnesses non-triviality of the class.

Definition 10.2.1.1 (Commutator phase). [Lean](#) [✓ Stmt](#) The *commutator phase* of a 2-cochain ω at a pair g, h is $\beta_\omega(g, h) := \omega(g, h)\omega(h, g)^{-1}$, defined for any $\mathbb{C}^\times$ -valued 2-cochain.

Theorem 10.2.1.2 (The commutator phase is a class invariant). [Lean](#) [✓ Stmt](#) If $\omega_1 \sim \omega_2$ and $gh = hg$, then $\beta_{\omega_1}(g, h) = \beta_{\omega_2}(g, h)$.

Proof. [✓ Proof](#) Write $\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)$ for a coboundary witness φ . Substituting into the ratio and using $gh = hg$ and the commutativity of $\mathbb{C}^\times$ gives

$$\begin{aligned} \beta_{\omega_1}(g, h) &= \frac{\varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)}{\varphi(h)\varphi(g)\varphi(hg)^{-1}\omega_2(h, g)} \\ &= \frac{\omega_2(g, h)}{\omega_2(h, g)} = \beta_{\omega_2}(g, h), \end{aligned} \quad (10.15)$$

where the φ -factors cancel since $\varphi(gh) = \varphi(hg)$. $\square$

Definition 10.2.1.3 (Non-trivial cohomology class). [Lean](#) [✓ Stmt](#) A factor system ω has a *non-trivial class* when it is not cohomologous to the constant cocycle 1.

Theorem 10.2.1.4 (Commutator-phase non-triviality test). [Lean](#) [✓ Stmt](#) If a commuting pair g, h has $\beta_\omega(g, h) \neq 1$, then ω has a non-trivial class.

Proof. [✓ Proof](#) The trivial cocycle has commutator phase 1 at every pair, so by Theorem 10.2.1.2 and (10.15), any cocycle cohomologous to it has $\beta = 1$ at every commuting pair. Contrapositively, $\beta_\omega(g, h) \neq 1$ for one commuting pair forces ω to be non-cohomologous to the trivial cocycle. $\square$

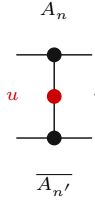
10.3 Physical Endpoint String Order and Virtual-Boundary Nondecay

String order parameters probe symmetry-protected topological phases through non-local correlators with a unitary string and endpoint operators [194]. For injective canonical matrix product states, this physical behavior provides a diagnostic correspondence with virtual-boundary non-decay and virtual local covariance.

Definition 10.3.1 (Twisted transfer map). [Lean](#) [✓ Stmt](#) Given an MPS tensor A and a physical-index matrix u , the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n' | u | n \rangle A_n X A_{n'}^\dagger. \quad (10.16)$$

Diagrammatically one inserts the physical operator u on the doubled local transfer cell:

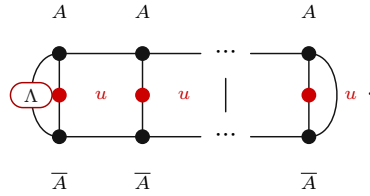


The upper and lower black nodes denote A_n and $A_{n'}^\dagger$, and the red node labelled u couples the physical legs.

Definition 10.3.2 (Stationary-boundary block-twist functional). [Lean](#) [✓ Stmt](#) The *stationary-boundary block-twist functional* is

$$T_L(A, u, \Lambda) := \text{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})). \quad (10.17)$$

In tensor-network notation this is the length- L doubled chain with a copy of u inserted on each physical rung:



L sites

The boundary matrix Λ is contracted into the virtual boundary, the red nodes labelled u sit on the physical rungs, and the right boundary is traced.

Definition 10.3.3 (Periodic finite-chain symmetry overlap). [Not ready](#) For the periodic length- L matrix product vector $|\Psi_L(A)\rangle$, the source quantity is literally

$$R_L(u) := \langle \Psi_L(A) | u^{\otimes L} | \Psi_L(A) \rangle. \quad (10.18)$$

Under thermodynamic-boundary and normalization hypotheses, the normalized stationary-boundary functional $T_L(A, u, \Lambda)$ corresponds to this periodic overlap.

Definition 10.3.4 (Physical string correlator). Lean ✓ Stmt For physical operators $x, y \in M_d(\mathbb{C})$, a twist $u \in M_d(\mathbb{C})$, and boundary state Λ , the transfer-form string correlator of [194, display SOPMP, lines 176–181] is

$$S_N(x, y, u) := \text{tr}(\Lambda \mathcal{E}_x \mathcal{E}_u^N(\mathcal{E}_y(\mathbb{1}))). \quad (10.19)$$

This is the physical-endpoint expression corresponding to $\langle \Psi | x \otimes u^{\otimes N} \otimes y | \Psi \rangle$.

Definition 10.3.5 (Fixed-endpoint physical string order). Lean ✓ Stmt For fixed physical operators x, y, u , the correlator has physical string order if there is $s > 0$ such that $|S_N(x, y, u)| \rightarrow s$.

Definition 10.3.6 (Physical endpoint string order). Lean ✓ Stmt A tensor has physical string order in the sense of [194, display SOP, lines 112–122] if there exist a unitary $u \neq \mathbb{1}$ and physical endpoint operators x, y for which Definition 10.3.5 holds.

Lemma 10.3.7 (Twisted spectral-radius bound and peripheral rigidity). Not ready *This is Lemma 1 of [194]. Let A represent a pure finitely correlated state in the canonical gauge $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_A^*(\Lambda) = \Lambda$, with $\Lambda > 0$ and $\text{tr}(\Lambda) = 1$. For every physical unitary u ,*

$$\rho(\mathcal{E}_u) \leq 1.$$

Equality holds if and only if there are a unitary V and $\theta \in [0, 2\pi)$ satisfying the following identity for every j in an eigenbasis $u = \sum_j e^{i\theta_j} |\tilde{j}\rangle \langle \tilde{j}|$:

$$V^\dagger \tilde{A}_j = e^{i(\theta - \theta_j)} \tilde{A}_j V^\dagger. \quad (10.20)$$

Moreover, $\mathcal{E}_u$ has at most one eigenvalue of modulus one.

Proof. Let $\mathcal{E}_u(V) = \lambda V$. Multiplication by $\Lambda V^\dagger$, the trace, and Cauchy–Schwarz give

$$\begin{aligned} |\lambda| \text{tr}(V \Lambda V^\dagger) &\leq \left(\sum_j \text{tr}(V \tilde{A}_j^\dagger \Lambda \tilde{A}_j V^\dagger) \right)^{1/2} \left(\sum_j \text{tr}(\tilde{A}_j^\dagger V \Lambda V^\dagger \tilde{A}_j) \right)^{1/2} \\ &= \text{tr}(V \Lambda V^\dagger). \end{aligned} \quad (10.21)$$

Faithfulness of Λ yields $|\lambda| \leq 1$. Equality in Cauchy–Schwarz is equivalent to (10.20); conversely that relation makes $e^{i\theta}$ an eigenvalue. It also gives $\mathcal{E}_A(V^\dagger V) = V^\dagger V$. The fixed space of the pure canonical transfer map is one-dimensional, so $V^\dagger V = c\mathbb{1}$ for some $c > 0$. Replacing V by $c^{-1/2}V$ makes it unitary without changing the intertwining relation.

If V, V' are peripheral eigenmatrices with eigenvalues $e^{i\theta}, e^{i\theta'}$, respectively, the equality relations give the exact identity

$$\mathcal{E}_A(V^\dagger V') = e^{i(\theta' - \theta)} V^\dagger V'.$$

Purity leaves only the peripheral eigenvalue 1, hence $\theta' = \theta$ modulo 2π . The one-dimensional fixed space then gives $V^\dagger V' \in \mathbb{C}\mathbb{1}$, proving uniqueness of the peripheral eigenvalue and its eigendirection. $\square$

Theorem 10.3.8 (Physical endpoint criterion for string order). Not ready *This is Theorem 1 of [194]. A pure finitely correlated state has physical string order if and only if there exist a unitary $\tilde{u} \neq \mathbb{1}$, a unitary virtual matrix V , and physical indices n, m such that*

$$\mathcal{E}_{\tilde{u}}(V) = V, \quad \text{tr}(V \Lambda A_n A_m^\dagger) \neq 0. \quad (10.22)$$

For an injective tensor, sufficiently long physical endpoint operators realize the corresponding virtual boundary matrices.

Proof. Lemma 10.3.7 shows that a nonzero limit requires a unique peripheral eigenvalue $e^{i\theta}$. If V and Y are the corresponding right and left eigenmatrices, then $Y = \Lambda V^\dagger$, and the leading term retains its length-dependent phase:

$$S_N(x, y, u) = e^{iN\theta} \text{tr}(Y \mathcal{E}_y(\mathbb{1})) \text{tr}(\Lambda \mathcal{E}_x(V)) + o(1).$$

In the eigenbasis $u = \sum_j e^{i\theta_j} |\tilde{j}\rangle \langle \tilde{j}|$, shift the string unitary by the peripheral phase,

$$\tilde{u} := e^{-i\theta} u = \sum_j e^{i(\theta_j - \theta)} |\tilde{j}\rangle \langle \tilde{j}|.$$

Then $\mathcal{E}_{\tilde{u}}(V) = V$. Choosing $y = x^\dagger \tilde{u}$ reduces nonvanishing to the second condition in (10.22); if it holds for n, m , take $x = |n\rangle \langle m|$. The reverse implication follows from the same phase-retaining factorization and the endpoint-realizability hypothesis. $\square$

Theorem 10.3.9 (Physical state symmetry and twisted spectral radius). Not ready *This is Theorem 2 of [194]. For a pure finitely correlated state and a unitary $u \neq \mathbb{1}$, the physical state is invariant under $u^{\otimes N}$ on every N -site reduced density operator if and only if*

$$\rho(\mathcal{E}_u) = 1.$$

For one-site-injective tensors, this physical symmetry is equivalent to virtual local covariance.

Proof. If $\rho(\mathcal{E}_u) = 1$, Lemma 10.3.7 gives a unitary V satisfying the local intertwining relation. Substitution of that relation into the adjoint transfer map, together with $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$, gives $\mathcal{E}_A^\dagger(V \Lambda V^\dagger) = V \Lambda V^\dagger$. This positive fixed point has trace one, so uniqueness gives $V \Lambda V^\dagger = \Lambda$, equivalently $V^\dagger \Lambda V = \Lambda$, which yields the physical symmetry. Conversely, physical invariance gives, for length-independent boundary tensors L, R ,

$$D^{-2} \leq \text{tr}(\varrho_N^2) = \text{tr}[L(\mathcal{E}_u \otimes \mathcal{E}_{u^\dagger})^N(R)].$$

Exponential decay would contradict this bound, so $\rho(\mathcal{E}_u) = 1$. $\square$

Definition 10.3.10 (Virtual local covariance with boundary invariance). Lean ✓ Stmt The virtual local-covariance predicate holds for A, u, Λ if there exist a unitary matrix V and a phase μ with $|\mu| = 1$ such that $V^\dagger \Lambda V = \Lambda$ and, for every physical index i ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger. \quad (10.23)$$

The unitary V intertwines the physical action of u with a virtual conjugation, and the boundary state Λ is invariant under V .

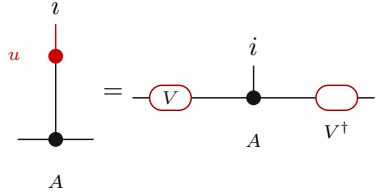
Definition 10.3.11 (Virtual-boundary nondecay). Lean ✓ Stmt For boundary matrices $X, Y \in M_D(\mathbb{C})$, set explicitly

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The virtual-boundary nondecay predicate holds for u and Λ if there exist X, Y and a positive constant $c > 0$ such that $c \leq \|R_L(u; X, Y)\|$ for every L . Thus some virtual-boundary sequence does not decay to zero.

This virtual-boundary condition asserts the existence of boundary matrices with non-vanishing transfer overlap under the twist u . The relationship with physical endpoint operators (Definition 10.3.6) is analyzed in [103].

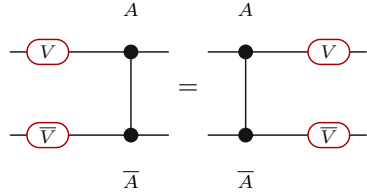
Definition 10.3.12 (Local physical intertwining). [Lean](#) [✓ Stmt](#) The local physical intertwining relation is $\sum_j u_{ij} A^j = V A^i V^\dagger$. Locally, the physical action of u can therefore be pushed to the virtual legs:



Definition 10.3.13 (Transfer-map covariance). [Lean](#) [✓ Stmt](#) Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation:

$$\mathcal{E}(V X V^\dagger) = V \mathcal{E}(X) V^\dagger. \quad (10.24)$$

In doubled-layer notation, the virtual operators slide through the local transfer cell:

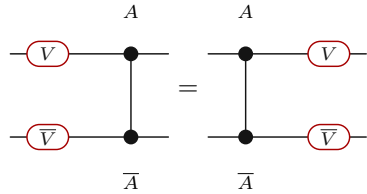


Definition 10.3.14 (Commutation in the doubled transfer picture). [Lean](#) [✓ Stmt](#) In the transfer-map language, the commutation of the doubled transfer matrix $E = \sum_i A_i \otimes \bar{A}_i$ with $V \otimes \bar{V}$ is

$$V \mathcal{E}_1(X) V^\dagger = \mathcal{E}_1(V X V^\dagger). \quad (10.25)$$

Theorem 10.3.15 (Transfer-map covariance and doubled commutation are equivalent). [Lean](#) [✓ Stmt](#) Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix V .

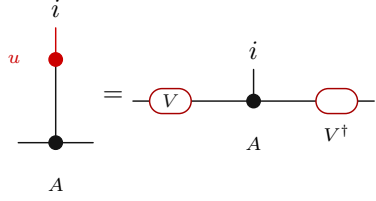
Proof. [✓ Proof](#) Both sides express the same local picture with opposite orientation:



The two formulas are the same identity read from opposite sides. □

Theorem 10.3.16 (Local physical intertwining implies transfer-map covariance). [Lean](#) [✓ Stmt](#) If V and u are unitary and the local physical intertwining relation holds, then the transfer map is covariant.

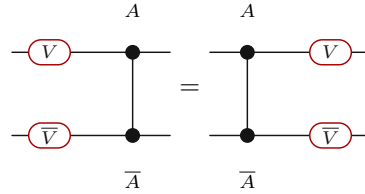
Proof. [✓ Proof](#) Starting from



one inserts $V^\dagger V = \mathbb{1}$ so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical u -boxes disappear by the unitary Kraus-mixing identity. $\square$

Theorem 10.3.17 (Transfer-map covariance implies local physical intertwining). [Lean](#) [✓ Stmt](#) Assume that A is one-site injective, $VV^\dagger = \mathbb{1}$, and the transfer map is covariant. Then there is u with $uu^\dagger = \mathbb{1}$ satisfying the local physical intertwining relation. Covariance and coisometry of V alone suffice for the conclusion.

Proof. [✓ Proof](#) Set $B^i := V A^i V^\dagger$. The transfer-map covariance relation



implies that the two d -element Kraus families B and A define the same transfer map. Rectangular Kraus freedom (the corresponding theorem in the companion quantum-channel volume [182, “Rectangular Kraus freedom”]) converts $\mathcal{E}_B = \mathcal{E}_A$ into a unitary Kraus mixing matrix u with $B^i = \sum_j u_{ij} A^j$. $\square$

Theorem 10.3.18 (Twisted-transfer eigenvalues have modulus at most one). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If $\mathcal{E}_u(X) = \lambda X$ for some nonzero $X \in M_D(\mathbb{C})$, then $|\lambda| \leq 1$.

Proof. [✓ Proof](#) By Lemma G.4.2, the twisted transfer map is the mixed transfer operator F_{A, B_u} , as in (G.6). Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. Theorem 5.3.1 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue λ . $\square$

Theorem 10.3.19 (Modulus-one twisted eigenvalue implies gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let A be an injective MPS tensor with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, and let u satisfy $uu^\dagger = \mathbb{1}$. If

$$\mathcal{E}_u(X) = \lambda X \quad (10.26)$$

for some nonzero $X \in M_D(\mathbb{C})$ with $|\lambda| = 1$, then A is gauge-phase equivalent to the companion tensor B_u .

Proof. [✓ Proof](#) By Lemma G.4.2, the twisted transfer map is the mixed transfer operator F_{A, B_u} . Because B_u is obtained from A by unitary mixing of the Kraus operators, the transfer maps of A and B_u coincide, so the two adjoint channels share a common positive definite fixed point.

Gauging both families by that fixed point puts them in a common trace-preserving gauge. In that gauge, (10.26) becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 5.6.1 yields gauge-phase equivalence. Undoing the common trace-preserving gauge gives the claim for A and B_u . $\square$

Theorem 10.3.20 (Virtual-boundary nondecay gives companion gauge-phase equivalence).

Lean **✓ Stmt** *Let A be injective, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If virtual-boundary nondecay holds for u with boundary parameter Λ , then A is gauge-phase equivalent to the companion tensor B_u .*

Proof. **✓ Proof** If $\rho(\mathcal{E}_u) < 1$, Lemma G.6.1 forces every virtual-boundary sequence to decay to zero, contradicting Definition 10.3.11. Hence $\rho(\mathcal{E}_u) \geq 1$. Theorem 10.3.18 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 10.3.19 then gives the gauge-phase equivalence. $\square$

Theorem 10.3.21 (Gauge-phase equivalence with the companion tensor yields a virtual unitary). **Lean** **✓ Stmt** *Let A be injective, assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A is gauge-phase equivalent to the companion tensor B_u , then there exist a unitary V and a phase μ with $|\mu| = 1$ such that, for every physical index i ,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger. \quad (10.27)$$

Proof. **✓ Proof** Start from an invertible intertwiner X and a nonzero gauge scalar ζ between A and B_u . The positive matrix $Q := XX^\dagger$ satisfies $\mathcal{E}_A(Q) = |\zeta|^2 Q$. Since $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, irreducibility forces the two positive eigenvalues to agree, so $|\zeta| = 1$ and Q is a positive fixed point of $\mathcal{E}_A$. Uniqueness of positive fixed points for an injective normalized tensor then makes Q a positive scalar multiple of the identity. Rescaling X by the square root of that scalar produces a unitary V , and the original intertwining relation becomes (10.27), with $|\mu| = 1$. $\square$

Theorem 10.3.22 (Virtual unitary gives a twisted-transfer eigenmatrix). **Lean** **✓ Stmt** *Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and that, for every physical index i , $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ with V unitary. Then V is an eigenmatrix of the twisted transfer map:*

$$\mathcal{E}_u(V) = \mu V. \quad (10.28)$$

Proof. **✓ Proof** Expand $\mathcal{E}_u(V)$ using (10.16), insert the intertwining relation into each local term, and use $V^\dagger V = \mathbb{1}$ together with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. $\square$

Theorem 10.3.23 (Boundary-state invariance of the virtual unitary). **Lean** **✓ Stmt** *Let A be injective, assume $uu^\dagger = \mathbb{1}$, and let Λ be a positive definite boundary state with $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. If V is a unitary and μ a phase satisfying, for every physical index i ,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger, \quad (10.29)$$

then

$$V^\dagger \Lambda V = \Lambda. \quad (10.30)$$

Proof. **✓ Proof** Relation (10.29) first shows that $V \Lambda V^\dagger$ is another positive fixed point of the adjoint transfer map. It has the same trace as Λ , so uniqueness of the normalized positive fixed point gives $V \Lambda V^\dagger = \Lambda$. Multiplying by $V^\dagger$ on the left and by V on the right yields (10.30). $\square$

Lemma 10.3.24 (Virtual local covariance implies virtual-boundary nondecay). [Lean](#) [Stmnt](#) Assume $\text{tr}(\Lambda) = 1$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If A, u, Λ satisfy virtual local covariance, then virtual-boundary nondecay holds for u .

Proof. [Proof](#) Choose the boundary matrices $X = V^\dagger$ and $Y = V$, where V is the unitary from the virtual local-covariance datum. By Theorem 10.3.22, one has $\mathcal{E}_u^L(V) = \mu^L V$. Therefore

$$R_L(u; V^\dagger, V) = \mu^L \text{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1. $\square$

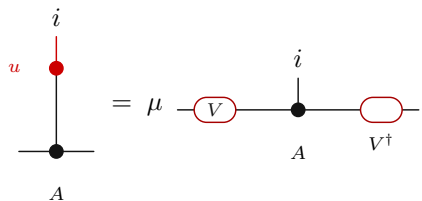
Theorem 10.3.25 (Virtual local covariance iff a unitary peripheral eigenmatrix exists). [Lean](#) [Stmnt](#) Let A be injective, and assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then virtual local covariance holds if and only if there exist a unitary $V \in M_D(\mathbb{C})$ and a phase μ with $|\mu| = 1$ such that $\mathcal{E}_u(V) = \mu V$. By Theorem 10.3.18, this witness form is equivalent to the peripheral-spectrum condition $\rho(\mathcal{E}_u) = 1$.

Proof. [Proof](#) The forward direction is Theorem 10.3.22. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 10.3.19 gives gauge-phase equivalence with the companion tensor; then Theorem 10.3.21 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 10.3.23 gives the boundary-state invariance needed for virtual local covariance. $\square$

Theorem 10.3.26 (Virtual-boundary nondecay $\Leftrightarrow$ virtual local covariance). [Lean](#) [Stmnt](#) For an injective MPS, assume $uu^\dagger = \mathbb{1}$, $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then virtual-boundary nondecay for u holds if and only if A, u, Λ satisfy virtual local covariance.

Proof. [Proof](#) If virtual-boundary nondecay holds, Theorem 10.3.20 gives a gauge-phase equivalence with the companion tensor. Theorem 10.3.21 extracts a virtual unitary and phase from that equivalence, and Theorem 10.3.23 shows that the same unitary preserves the boundary state, so one obtains virtual local covariance. Conversely, Lemma 10.3.24 turns virtual local covariance into a nondecaying virtual-boundary sequence. $\square$

Theorem 10.3.27 (Virtual unitary from virtual-boundary nondecay). [Lean](#) [Stmnt](#) For an injective MPS, let $\Lambda \in M_D(\mathbb{C})$, and assume $uu^\dagger = \mathbb{1}$ and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$. If virtual-boundary nondecay holds for u with boundary parameter Λ , then there exists a virtual unitary V and a unit-modulus scalar μ such that $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$ for all i . Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:



Proof. [Proof](#) Virtual-boundary nondecay first gives gauge-phase equivalence with the companion tensor by Theorem 10.3.20. Theorem 10.3.21 then normalizes the resulting intertwiner to a unitary and produces the phase μ . $\square$

10.4 SPT Phase Labels and Virtual-Boundary Nondecay

For one-dimensional gapped systems within the injective symmetric MPS and parent-Hamiltonian framework [204], symmetry-protected topological (SPT) phases are classified by the second cohomology group $H^2(G, \mathbb{U}(1))$ of the symmetry group, appearing as the cohomology class of the virtual projective representation.

Definition 10.4.1 (Equality of virtual cocycle labels). Lean ✓ Stmt For tensors A, B and a fixed physical representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$, the predicate called *same SPT phase* asserts the existence of factor systems $\omega_A, \omega_B : G \times G \rightarrow \mathbb{C}^\times$ and projective bond representations ρ_A, ρ_B such that

$$\begin{aligned}\tilde{A}_g^i &= \rho_A(g^{-1}) A^i \rho_A(g^{-1})^{-1}, \\ \tilde{B}_g^i &= \rho_B(g^{-1}) B^i \rho_B(g^{-1})^{-1},\end{aligned}\tag{10.31}$$

and $\omega_A \sim \omega_B$.

Remark 10.4.2 (Cohomology with $\mathbb{C}^\times$ and $\mathbb{U}(1)$ coefficients). While factor systems here take values in $\mathbb{C}^\times$, for finite groups G every class admits a unit-modulus representative, identifying $H^2(G, \mathbb{C}^\times)$ with $H^2(G, \mathbb{U}(1))$ (Theorem 19.5.10.67).

Theorem 10.4.3 (Classification along symmetric MPS parent-Hamiltonian paths). Not ready For $p = 0, 1$, let A_p be an injective MPS tensor whose parent Hamiltonian H_p has a unique ground state and is symmetric under a unitary representation U^p of G on $\mathcal{H}_p$. Choose the endpoint phase gauges. After a common finite blocking, the embedded endpoints admit a continuous symmetric MPS path with corresponding translationally invariant, finite-range parent Hamiltonians that vary smoothly and remain uniformly gapped if and only if their virtual projective representations determine the same class in $H^2(G, \mathbb{U}(1))$.

More precisely, for the forward construction there exist a path-sector Hilbert space $\mathcal{H}_{\text{path}}$ and a linear representation U^{path} such that the endpoints embed into

$$\mathcal{H} = \mathcal{H}_0 \oplus \mathcal{H}_1 \oplus \mathcal{H}_{\text{path}}, \quad U_g = U_g^0 \oplus U_g^1 \oplus U_g^{\text{path}}.$$

The converse quantifies over smooth paths of MPS with corresponding parent Hamiltonians [204, Section II.F, lines 929–953].

Proof. First deform each blocked injective MPS to its isometric form; the deformation preserves the symmetry relation

$$U_g \mathcal{P} = \mathcal{P}(V_g \otimes \overline{V}_g).$$

Along any continuous symmetric MPS path quantified in the theorem, the tensor and its induced virtual action V_g vary continuously. The cohomology class is discrete, so it cannot change along that path. This proves necessity within the MPS/parent-Hamiltonian framework.

Conversely, let the endpoint isometric forms have virtual actions V_g^0 and V_g^1 in the same cohomology class. Embed their bond spaces into the direct sum and interpolate between the maximally entangled bonds by

$$|\omega(\gamma)\rangle = (1 - \gamma) \sum_{i=1}^{D_0} |i, i\rangle + \gamma \sum_{i=D_0+1}^{D_0+D_1} |i, i\rangle.$$

The commuting parent Hamiltonians along this interpolation remain gapped, with the uniform spectral estimate of Lemma 10.4.3.14. Their symmetry is

$$(V_g^0 \oplus V_g^1) \otimes (\overline{V_g^0} \oplus \overline{V_g^1}) = \widehat{U}_g^0 \oplus \widehat{U}_g^1 \oplus \widehat{U}_g^{\text{path}}, \quad (10.32)$$

$$\widehat{U}_g^{\text{path}} = (V_g^0 \otimes \overline{V_g^1}) \oplus (V_g^1 \otimes \overline{V_g^0}).$$

Equality of the cohomology classes is exactly what makes the cross-term action $\widehat{U}^{\text{path}}$ a linear representation. Conjugating back by the endpoint isometries gives the common direct-sum physical representation and the required symmetric gapped path. $\square$

Theorem 10.4.4 (Virtual-boundary nondecay for injective MPV-symmetric tensors). [Lean](#)

[✓ Stmt](#) *Let A be an injective tensor, symmetric under a unitary representation U . Assume $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, Λ is positive definite, $\text{tr}(\Lambda) = 1$, and $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$. Then the virtual-boundary nondecay predicate holds for every group element g .*

Proof. [✓ Proof](#) The virtual representation theorem produces $\rho(g^{-1}) \in \text{GL}_D(\mathbb{C})$ satisfying $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$. A direct computation using $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ shows that $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$, so $\rho(g^{-1})$ is a nonzero eigenvector with eigenvalue 1. Theorem 10.3.19 gives gauge-phase equivalence with the companion tensor, and Theorem 10.3.21 extracts a virtual unitary and phase. By Theorem 10.3.23, this unitary preserves the boundary state, so Lemma 10.3.24 gives virtual-boundary nondecay. $\square$

Remark 10.4.5 (Virtual-boundary nondecay does not determine the SPT label). Theorem 10.4.4 applies to every injective symmetric tensor with canonical normalisations, irrespective of its cocycle class, so tensors in different SPT phases all satisfy virtual-boundary nondecay for every group element. This nondecay is equivalent to virtual local covariance (Theorem 10.3.26); the invariant that separates SPT phases is the cohomology class of the virtual representation cocycle (Theorem 10.4.3).

10.4.1 Virtual actions along symmetric paths

The interpolation in the proof of Theorem 10.4.3 first rephases the two endpoint virtual actions to a common factor system and then forms their direct sum; necessity rests on the discreteness of the class along a continuous virtual path [204, Section II.F.2].

Theorem 10.4.1.1 (Unitary rephasing of cohomologous virtual actions). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) *Multiplying each matrix V_g of a projective representation with factor system ω by a scalar $\lambda_g \neq 0$ gives a projective representation with factor system $\lambda_g \lambda_h \lambda_{gh}^{-1} \omega(g, h)$. Consequently, let V^0 be a unitary projective representation whose factor system is the pointwise phase $\omega_0/|\omega_0|$. If ω_1 is cohomologous to ω_0 , then multiplying each V_g^0 by a unit-modulus scalar gives a unitary projective representation with factor system $\omega_1/|\omega_1|$ [204, Section II.F.2, lines 886–921].*

Proof. [✓ Proof](#) Let λ witness $\omega_1 \sim \omega_0$, so that $\omega_1(g, h) = \lambda_g \lambda_h \lambda_{gh}^{-1} \omega_0(g, h)$, and put $\psi_g = \lambda_g/|\lambda_g|$. The unit-phase map $z \mapsto z/|z|$ is multiplicative, so

$$\frac{\omega_1(g, h)}{|\omega_1(g, h)|} = \psi_g \psi_h \psi_{gh}^{-1} \frac{\omega_0(g, h)}{|\omega_0(g, h)|}.$$

Hence $\psi_g V_g^0$ has factor system $\omega_1/|\omega_1|$, and multiplication by a unit-modulus scalar preserves unitarity. $\square$

Definition 10.4.1.2 (Direct sum with a common factor system). [Lean](#) [✓ Stmt](#) Let V^0 and V^1 be projective representations on $\mathbb{C}^{D_0}$ and $\mathbb{C}^{D_1}$ with the same factor system ω . Their direct sum is the projective representation on $\mathbb{C}^{D_0+D_1}$ given by

$$V_g = V_g^0 \oplus V_g^1, \quad V_g V_h = \omega(g, h) V_{gh}.$$

This is the virtual direct sum in the symmetric interpolation of [204, Section II.F.2, lines 909–921].

Theorem 10.4.1.3 (Unitarity of the virtual direct sum). [Lean](#) [✓ Stmt](#) If V_g^0 and V_g^1 are unitary, then $V_g^0 \oplus V_g^1$ is unitary.

Proof. [✓ Proof](#) The product with its adjoint is $(V_g^0 (V_g^0)^*) \oplus (V_g^1 (V_g^1)^*) = I_{D_0} \oplus I_{D_1}$. □

Theorem 10.4.1.4 (Cohomology along a fixed-dimension virtual path). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let G be any group and $D > 0$. Suppose matrices $V_\gamma(g) \in U(D)$ depend continuously on $\gamma \in [0, 1]$ for each $g \in G$ and obey $V_\gamma(g) V_\gamma(h) = \omega_\gamma(g, h) V_\gamma(gh)$. Then ω_0 and ω_1 are cohomologous as $\mathbb{C}^\times$ -valued factor systems. This theorem assumes the virtual path; obtaining it from a smooth symmetric MPS parent-Hamiltonian path, including changes of bond dimension, is a separate step in the classification theorem [204, Section II.F.2, lines 930–946].

Proof. [✓ Proof](#) Each $\det V_\gamma(g)$ is a continuous nonzero complex path, so it has a continuous D th root $\beta_\gamma(g)$ on the interval. Taking determinants in $V_\gamma(g) V_\gamma(h) = \omega_\gamma(g, h) V_\gamma(gh)$ gives $\beta_\gamma(g)^D \beta_\gamma(h)^D = \omega_\gamma(g, h)^D \beta_\gamma(gh)^D$, so the rescaled factor system

$$\eta_\gamma(g, h) = \omega_\gamma(g, h) \frac{\beta_\gamma(gh)}{\beta_\gamma(g) \beta_\gamma(h)} \quad \eta_\gamma(g, h)^D = 1$$

takes values among the D th roots of unity. It depends continuously on γ , hence is constant, and $\eta_0 = \eta_1$ reads

$$\omega_1(g, h) = \psi(g) \psi(h) \psi(gh)^{-1} \omega_0(g, h), \quad \psi(g) = \beta_1(g) / \beta_0(g).$$

□

10.4.2 Renormalization fixed points of SPT phases

Within each SPT phase a representative with zero correlation length can be built from the cocycle data [16, Section III.A]. The tensor printed in the source commutes with its own gauges and, for a nontrivial group, is not normal; the construction below is the dimer fixed point that the printed claims describe [138]. The source starts from a 2-cocycle alone. Most results below take a projective representation with that factor system as given; for a finite group, the twisted regular representation on $\mathbb{C}^G$ supplies one whose factor system lies in the class of the cocycle (Theorem 10.4.2.4, [137]). The source’s symmetry carries the phase of a 1-cocycle; the symmetry and class results below are stated for the trivial character [139]. The source’s symmetry group may also contain time-reversal and reflection symmetries, with twisted cocycles; the results below treat on-site symmetries only [136].

Definition 10.4.2.1 (SPT fixed-point tensor and its symmetry). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ be a projective representation of G on $\mathbb{C}^D$ with factor system ω , let ϕ be a character of G , and put $W_g = \rho(g^{-1})$. The SPT fixed-point tensor has physical space $\mathbb{C}^D \otimes \mathbb{C}^D$ and letters $A^{(a,b)} = D^{-1/2} |a\rangle \langle b|$, and G acts on the physical space by $U(g) = \phi(g) W_g^T \otimes W_g^{-1}$. The projective representation need not be unitary; when it is, $W_g^{-1} = W_g^\dagger$, so with $X_g = W_g^\dagger$ the conjugation $W_g A^i W_g^{-1}$ is the source’s $X_g^\dagger A^i X_g$.

Proof. ✓ Proof The two blocks have disjoint support, so the squared norm of the coefficient matrix is $D_0(1-\gamma)^2 + D_1\gamma^2$. This is positive for every real γ when $D_0, D_1 > 0$. The entries and the positive normalization factor vary continuously with γ . $\square$

Theorem 10.4.3.7 (Symmetry of the interpolating bond). Lean Lean Lean Lean ✓ Stmt Let V^0, V^1 be unitary projective representations with a common factor system, and put $V_g = V_g^0 \oplus V_g^1$. For the bond coefficient matrices $W(\gamma)$ and $\widehat{W}(\gamma)$ of Theorem 10.4.3.6,

$$V_g W(\gamma) V_g^* = W(\gamma), \quad V_g \widehat{W}(\gamma) V_g^* = \widehat{W}(\gamma).$$

These are the coefficient-matrix identities expressing invariance of the bond vector under $V_g \otimes \overline{V_g}$ [204, Section II.F.2].

Proof. ✓ Proof The matrix $W(\gamma)$ is scalar on each of the two summands, so it commutes with V_g . Unitarity gives the first identity, and scalar normalization gives the second. $\square$

Definition 10.4.3.8 (Incoming-bond coordinates). Lean Lean ✓ Stmt Let each site of a cyclic chain carry two registers of dimension D . For a configuration $s = ((L_i, R_i))_i$, define

$$e_N(s)_i = (R_{i-1}, L_i).$$

The inverse recovers $(L_i, R_i) = ((b_i)_2, (b_{i+1})_1)$. Pair coordinates may equivalently be enumerated by an alphabet of size D^2 . These are the independent-bond coordinates of [204, Section II.D.2].

Theorem 10.4.3.9 (Unitarity of bond regrouping). Lean ✓ Stmt The matrix P_N defined by $(P_N)_{s,b} = \delta_{e_N(s),b}$ is unitary. Thus it takes a bond-coordinate wavefunction to the corresponding physical-coordinate wavefunction by composition with e_N .

Proof. ✓ Proof The map e_N permutes the configuration basis. $\square$

Theorem 10.4.3.10 (Locality after bond regrouping). Lean ✓ Stmt Let $N > 0$ and let K_i act as a matrix K on incoming bond i and as the identity on every other bond. Then $P_N K_i P_N^\dagger$ acts on the registers (R_{i-1}, L_i) and as the identity on all other registers. More precisely, its matrix entries between physical configurations s, t are

$$(P_N K_i P_N^\dagger)_{s,t} = K_{e_N(s)_i, e_N(t)_i} \prod_{j \neq i} \delta_{e_N(s)_j, e_N(t)_j}.$$

Hence every one-bond interaction becomes a nearest-neighbor physical interaction (a one-site interaction when $N = 1$).

Proof. ✓ Proof Conjugation by a permutation matrix replaces the row and column configurations by their incoming-bond coordinates. The tensor-product formula for K_i gives the displayed entries. $\square$

Theorem 10.4.3.11 (Symmetry under bond regrouping). Lean Lean ✓ Stmt For matrices L, R on a register of dimension D ,

$$P_N (R \otimes L)^{\otimes N} P_N^\dagger = (L \otimes R)^{\otimes N}.$$

In particular, for the physical action $U(g) = W_g^\top \otimes W_g^{-1}$ of the SPT fixed-point tensor with trivial character (Definition 10.4.2.1), the incoming-bond action is $W_g^{-1} \otimes W_g^\top$ [204, Section II.F.2].

Theorem 10.4.3.15 (Physical parent Hamiltonian of the interpolating bonds). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Conjugate $H_N(\gamma)$ by the incoming-bond unitary P_N . In the original physical-site coordinates, the resulting Hamiltonian $P_N H_N(\gamma) P_N^*$ varies continuously with γ , has the transported product vector $P_N \hat{\omega}(\gamma)^{\otimes N}$ as its unique ground line, and satisfies

$$\langle v, (P_N H_N(\gamma) P_N^*)^2 v \rangle \geq \langle v, P_N H_N(\gamma) P_N^* v \rangle$$

for every $N > 0$ and every physical-chain vector v .

Proof. [Stmnt](#) The incoming-bond change of coordinates is unitary. It carries the product ground line and the quadratic-form inequality to physical coordinates. Continuity follows from the continuity of the finite-chain bond Hamiltonian, since the unitary conjugation is fixed. $\square$

Theorem 10.4.3.16 (On-site symmetry of the interpolating-bond parent Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $V^{(0)}$ and $V^{(1)}$ be finite-dimensional unitary projective representations with the same factor system, and let V be their direct sum. Let U be the physical action $U(g) = W_g^\top \otimes W_g^{-1}$, $W_g = V_{g^{-1}}$, of the SPT fixed-point tensor of V with trivial character (Definition 10.4.2.1); it does not depend on γ . For every group element g and every $N > 0$,

$$[(U_g)^{\otimes N}, P_N H_N(\gamma) P_N^*] = 0$$

[204, Section II.F.2].

Proof. [Stmnt](#) The block-scalar coefficient matrix of the bond is invariant under conjugation by V_g . In bond coordinates this fixes the bond vector, so each rank-one bond projection commutes with the bond action. Regrouping carries that action to $(U_g)^{\otimes N}$ and therefore carries the commutation relation to physical coordinates. $\square$

Theorem 10.4.3.17 (Spectral gap of the physical bond parent Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For every unit bond vector, and in particular for $\hat{\omega}(\gamma)$ with $D_0, D_1 > 0$, the physical parent Hamiltonian $P_N H_N(\gamma) P_N^*$ is positive semidefinite for every $N > 0$. Its spectrum contains 0, and every spectral value z satisfies $\operatorname{Re} z = 0$ or $\operatorname{Re} z \geq 1$.

Proof. [Stmnt](#) The Hamiltonian is a unitary conjugate of a sum of orthogonal projections, hence positive semidefinite. Its nonzero kernel gives the spectral value 0. For a unit eigenvector with eigenvalue λ , the quadratic-form inequality of Theorem 10.4.3.15 reads $\lambda \leq \lambda^2$; with $\lambda \geq 0$ this forces $\lambda = 0$ or $\lambda \geq 1$. $\square$

Theorem 10.4.3.18 (Two-site interaction for the bond parent Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For a bond matrix K , let $h(K)$ act as K on the middle registers (R_0, L_1) of two sites, with identity on the exterior registers. If K is an orthogonal projection, so is $h(K)$, and its Hilbert-space operator norm is at most one. For every $N \geq 2$,

$$\sum_{i=0}^{N-1} h(I - |\eta\rangle\langle\eta|)_{i,i+1} = P_N H_N(\eta) P_N^*.$$

Here the indices are cyclic and $H_N(\eta)$ is the sum of the independent-bond penalties [204, Section II.D.2].

Proof. [Stmnt](#) The two-site operator is a reordering of $I \otimes K \otimes I$. Its translate at i is the conjugate under P_N of the independent-bond term at $i + 1$. Summing and reindexing by the cyclic successor gives the identity. $\square$

Theorem 10.4.3.19 (Symmetric gapped path between the direct-sum fixed points). [Lean](#) [Lean](#)

[✓ Stmt](#) Let $D_0, D_1 > 0$, and let $V^{(0)}$ and $V^{(1)}$ be unitary projective representations with the same factor system. Set $D = D_0 + D_1$ and let U be the unitary physical action of the SPT fixed-point tensor of $V^{(0)} \oplus V^{(1)}$ with trivial character. If $\hat{\omega}(\gamma)$ is the normalized interpolating bond, define the two-site interaction to be the identity on the outer registers and $1 - |\hat{\omega}(\gamma)\rangle\langle\hat{\omega}(\gamma)|$ on the bond joining the sites. Then these interactions form a symmetric gapped path on the common physical space of dimension D^2 . Their operator norms are at most one, and the gap above the unique ground line is at least one for every $N \geq 2$ and every $\gamma \in [0, 1]$ [204, Section II.F.2].

Proof. [✓ Proof](#) Each local interaction is an orthogonal projection and depends continuously on the normalized bond vector. The sum of its translates is the physical-coordinate independent-bond parent Hamiltonian. Its spectrum has zero as the ground value and no values strictly between zero and one. The direct-sum action fixes the bond vector, so the full Hamiltonian commutes with $U_g^{\otimes N}$ for every g . $\square$

Theorem 10.4.3.20 (Weighted matrix-unit MPS as a product of bonds). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Let $N > 0$ and $D = D_0 + D_1$. Define an MPS tensor with physical letters (a, b) by $A_\gamma^{(a,b)} = \widehat{W}(\gamma)_{aa}|a\rangle\langle b|$. For a cyclic physical configuration $s = ((L_i, R_i))_i$, its matrix product coefficient equals the product of the incoming-bond amplitudes:

$$\text{tr} \left(\prod_{i=1}^N A_\gamma^{(L_i, R_i)} \right) = \prod_{i=1}^N \widehat{W}(\gamma)_{R_{i-1}, L_i}.$$

At $\gamma = 0, 1$ this identifies the positive-length physical states with the embedded endpoint bond-product states, with every nonzero incoming bond coefficient equal to $D_p^{-1/2}$ in endpoint block p . The path of tensors is continuous and symmetric under the physical action of the SPT fixed-point tensor of $V^{(0)} \oplus V^{(1)}$ with trivial character, which does not depend on γ [204, Section II.F.2]. The tensor itself may have letters with cross-block indices at an endpoint; the cyclic coefficient identity is the relevant endpoint statement.

Proof. [✓ Proof](#) The trace of a cyclic product of matrix units vanishes unless each column index equals the next row index. On that support, it is the product of the diagonal weights. Regrouping the cyclic indices gives the displayed product of bond coefficients. The diagonal weight matrix commutes with the direct-sum virtual action, so the symmetry relation of the matrix units passes to A_γ . $\square$

Theorem 10.4.3.21 (Two-site fixed-point vectors lie in the bond kernel). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let $D > 0$, let $A^{(a,b)} = D^{-1/2}|a\rangle\langle b|$ be the SPT fixed-point tensor, and set $\eta = D^{-1/2} \sum_{a=1}^D |a, a\rangle$. Then $\|\eta\| = 1$. For every boundary matrix X , the two-site open-boundary vector Ψ_X has coefficients $\Psi_X(a, b, c, d) = D^{-1} \delta_{bc} X_{da}$ and is annihilated by the interaction $h(I - |\eta\rangle\langle\eta|)$ acting on the middle registers. This bond interaction is bounded above by the canonical two-site parent interaction of A [204, Section II.D.2].

Proof. [✓ Proof](#) Multiplication of two letters gives $A^{(a,b)} A^{(c,d)} = D^{-1} \delta_{bc} |a\rangle\langle d|$. Taking the trace against X gives the coefficient formula. The middle-register vector is $\sum_b |b, b\rangle = \sqrt{D} \eta$, which lies in the kernel of $I - |\eta\rangle\langle\eta|$. Both interactions are orthogonal projections. The inclusion of their kernels therefore gives the asserted operator inequality. $\square$

10.4.4 Degeneracy of the entanglement spectrum

For an injective tensor in canonical form, the positive fixed point of the transfer map is the half-chain reduced density matrix, whose spectrum is the entanglement spectrum. A symmetry forces this fixed point to commute with the virtual gauges, and a projective representation whose factor system has a non-trivial class, and hence no one-dimensional invariant subspace, then forces every eigenvalue to be degenerate [16, Section III.A]. Here the canonical form is $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ and the fixed point is that of the adjoint map $\mathcal{E}_A^\dagger$; the reflected normalization, in which the fixed point of $\mathcal{E}_A$ itself is used, describes the same spectrum.

Lemma 10.4.4.1 (Anticommuting invertible operators force even dimension). [Lean](#) [Lean](#) [Stmt](#) *Let f, g be invertible operators on a finite-dimensional complex vector space V with $fg = -gf$. Then $\dim V$ is even. Consequently, if a matrix Λ commutes with two anticommuting invertible matrices X and Y , every eigenspace of Λ has even dimension.*

Proof. [Proof](#) Taking determinants in $fg = -gf$ gives $\det f \det g = (-1)^{\dim V} \det g \det f$ with $\det f \det g \neq 0$, so $(-1)^{\dim V} = 1$. For the second statement, X and Y map each eigenspace of Λ into itself, and their restrictions are invertible and anticommute. $\square$

Theorem 10.4.4.2 (Virtual gauges fix the boundary state). [Lean](#) [Lean](#) [Stmt](#) *Let A be injective with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, let Λ be a positive definite matrix with $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$, and let u be a unitary with $\sum_j u_{ij} A^j = \zeta X A^i X^{-1}$ for all i , where $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \neq 0$. Then $X = cW$ for a nonzero scalar c and a unitary W with $W\Lambda W^\dagger = \Lambda$. In particular $X\Lambda = \Lambda X$.*

Proof. [Proof](#) Theorem 10.3.21, applied to the physical unitary $u^\dagger$, gives a unitary V and a phase μ with $\sum_j (u^\dagger)_{ij} A^j = \mu V A^i V^\dagger$, and Theorem 10.3.23 gives $V^\dagger \Lambda V = \Lambda$ after normalizing the trace of Λ . Undoing the physical mixing yields $A^i = \mu \zeta (VX) A^i (VX)^{-1}$, so Lemma 10.1.15 makes VX a nonzero scalar. Hence X is a multiple of $W = V^\dagger$. $\square$

Lemma 10.4.4.3 (No one-dimensional eigenspace for a commutant of a non-trivial projective representation). [Lean](#) [Stmt](#) *Let ρ be a projective representation of G on $\mathbb{C}^D$ whose factor system ω is not cohomologous to the trivial one, and let Λ commute with every $\rho(g)$. Then no eigenspace of Λ is one-dimensional.*

Proof. [Proof](#) Each $\rho(g)$ maps every eigenspace of Λ into itself. If an eigenspace were spanned by one vector v , then $\rho(g)v = c_g v$ with $c_g \neq 0$, and $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$ applied to v gives $\omega(g, h) = c_g c_h c_{gh}^{-1}$, a coboundary. $\square$

Theorem 10.4.4.4 (Degeneracy of the entanglement spectrum). [Lean](#) [Stmt](#) *Let A be injective with $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, let Λ be a positive definite matrix with $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$, and let U be a unitary representation of G under which A is symmetric with virtual projective representation ρ up to nonzero scalars $\zeta(g)$, that is, $\sum_j U(g)_{ij} A^j = \zeta(g) \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$. If the factor system of ρ has a non-trivial class, no eigenspace of Λ is one-dimensional: every eigenvalue of the entanglement spectrum is degenerate [16, Section III.A].*

Proof. [Proof](#) Every $\rho(g)$ is the virtual gauge of $U(g^{-1})$ and commutes with Λ by Theorem 10.4.4.2; apply Lemma 10.4.4.3. $\square$

Theorem 10.4.4.5 (Even degeneracy for anticommuting gauges). [Lean](#) [Stmt](#) *In the setting of Theorem 10.4.4.2, let u, v be unitaries with virtual gauges X, Y , that is, $\sum_j u_{ij} A^j = \zeta X A^i X^{-1}$ and $\sum_j v_{ij} A^j = \eta Y A^i Y^{-1}$ with $\zeta, \eta \neq 0$. If $XY = -YX$, then every eigenspace of Λ has*

even dimension. This sharpens Theorem 10.4.4.4 when the projective representation contains an anticommuting pair.

Proof. ✓ Proof Both gauges commute with Λ by Theorem 10.4.4.2; apply Lemma 10.4.4.1. □

10.4.5 Time reversal, reflection, and symmetry obstructions

Time reversal acts on an MPS tensor by $A^i \mapsto \sum_j P_{ij} \bar{A}^j$, with $\bar{A}$ the entrywise conjugate and P its on-site part, and reflection acts by transposition $A^i \mapsto (A^i)^T$. Applying either symmetry twice returns the tensor up to the gauge $X\bar{X}$, which Lemma 10.1.17 forces to be a scalar [16, Section III.A].

Lemma 10.4.5.1 (Involution index from a doubled symmetry). Lean Lean Lean ✓ Stmt *Let A be normal with $D \geq 1$ and let X be unitary. If $A^i = c (X\bar{X})^\dagger A^i (X\bar{X})$ for all i , then $X\bar{X} = \pm 1$. If two unitaries X, X' satisfy $\zeta X^\dagger A^i X = \zeta' X'^\dagger A^i X'$ for all i with $\zeta \neq 0$, then $X\bar{X} = X'\bar{X}'$.*

Proof. ✓ Proof By Lemma 10.1.17 the unitary $X\bar{X}$ is a scalar σ , and $\sigma = \pm 1$ because $X\bar{X} = \sigma$ implies $\bar{X}X = \sigma$ and $\sigma = \bar{\sigma}$ with $|\sigma| = 1$. In the second statement $X = uX'$ with $|u| = 1$, so $X\bar{X} = |u|^2 X'\bar{X}'$. □

Theorem 10.4.5.2 (Time-reversal index). Lean Lean ✓ Stmt *Let A be normal with $D \geq 1$, let P satisfy $P\bar{P} = 1$, and let X be unitary with $\sum_j P_{ij} \bar{A}^j = \zeta X^\dagger A^i X$ for all i . Then $X\bar{X} = \pm 1$, and the sign is the same for every unitary X with this property [16, Section III.A].*

Proof. ✓ Proof Conjugating the relation and substituting it into itself gives $A^i = |\zeta|^2 (X\bar{X})^\dagger A^i (X\bar{X})$, because the doubled on-site action is $P\bar{P} = 1$. Apply Lemma 10.4.5.1; the phase ζ is nonzero because A has a nonzero letter. □

Theorem 10.4.5.3 (Reflection index). Lean Lean ✓ Stmt *Let A be normal with $D \geq 1$ and let X be unitary with $(A^i)^T = \zeta X^\dagger A^i X$ for all i . Then $X\bar{X} = \pm 1$, and the sign is the same for every unitary X with this property [16, Section III.A].*

Proof. ✓ Proof Transposing the relation and substituting it into itself gives $A^i = \zeta^2 (X\bar{X})^\dagger A^i (X\bar{X})$, since $X^T X^\dagger = (X\bar{X})^\dagger$ and $(X^\dagger)^T = \bar{X}$. Apply Lemma 10.4.5.1. □

Theorem 10.4.5.4 (Kramers obstruction). Lean Lean Lean ✓ Stmt *Let P satisfy $P\bar{P} = -1$, as σ_y does. No normal tensor with $D \geq 1$ admits $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C}$ with $\sum_j P_{ij} \bar{A}^j = \zeta X^{-1} A^i X$ for all i . In particular no normal spin- $\frac{1}{2}$ tensor carries the Wigner time reversal $A^i \mapsto \sum_j (\sigma_y)_{ij} \bar{A}^j$ in this form [16, Section III.A].*

Proof. ✓ Proof Applying the relation twice gives $-A^i = |\zeta|^2 (X\bar{X})^{-1} A^i (X\bar{X})$. By Lemma 10.1.17, $-|\zeta|^2 = 1$, which is impossible. □

Theorem 10.4.5.5 (Kramers obstruction for injective tensors). Lean ✓ Stmt *No injective spin- $\frac{1}{2}$ tensor with $D \geq 1$ is invariant up to a phase under the Wigner time reversal: it is not the case that for every $N \geq 1$ the matrix product vector of $\tilde{A}^i = \sum_j (\sigma_y)_{ij} \bar{A}^j$ is $c_N |\psi_N(A)\rangle$ with $|c_N| = 1$ [16, Section III.A].*

Proof. ✓ Proof By Theorem 3.2.5, $\tilde{A}^i = \lambda X A^i X^{-1}$ for an invertible X and a phase λ , which Theorem 10.4.5.4 excludes. □

Theorem 10.4.5.6 (Anticommuting physical symmetries are obstructed). [Lean](#) [Stmt](#) Let P, Q be physical matrices with $PQ = -QP$. No normal tensor with $D \geq 1$ satisfies $\sum_j P_{ij} A^j = \zeta X A^i X^{-1}$ and $\sum_j Q_{ij} A^j = \eta Y A^i Y^{-1}$ for all i , with $X, Y \in \text{GL}_D(\mathbb{C})$ and $\zeta, \eta \neq 0$.

Proof. [Proof](#) Composing the two twists in both orders gives $\eta \zeta (YX) A^i (YX)^{-1} = -\zeta \eta (XY) A^i (XY)^{-1}$, so Lemma 10.1.17 would give $-1 = 1$. $\square$

Theorem 10.4.5.7 (No normal spin- $\frac{1}{2}$ tensor is $\text{SU}(2)$ symmetric). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) No normal tensor A on a spin- $\frac{1}{2}$ site with $D \geq 1$ is invariant up to phases under the defining representation of $\text{SU}(2)$: it is not the case that for every g and every N the vector $U(g)^{\otimes N} |\psi_N(A)\rangle$ is a multiple of $|\psi_N(A)\rangle$ [16, Section III.A]. In particular A has no MPV-family invariance under $\text{SU}(2)$.

Proof. [Proof](#) Fix a length N . If $|\psi_N(A)\rangle = 0$, every twist leaves it unchanged. Otherwise write $U(g)^{\otimes N} |\psi_N(A)\rangle = c_N(g) |\psi_N(A)\rangle$; then $c_N(gh) = c_N(g)c_N(h)$, so c_N is trivial on commutators. Thus at every length N each commutator leaves $|\psi_N(A)\rangle$ unchanged. Both $i\sigma_x$ and $i\sigma_z$ are commutators in $\text{SU}(2)$: $i\sigma_x = (i\sigma_z)b_1(i\sigma_z)^{-1}b_1^{-1}$ and $i\sigma_z = (i\sigma_x)b_2(i\sigma_x)^{-1}b_2^{-1}$ with $b_1 = \frac{1}{2} \begin{pmatrix} 1-i & 1-i \\ -1-i & 1+i \end{pmatrix}$ and $b_2 = \frac{1}{2} \begin{pmatrix} 1-i & 1+i \\ -1+i & 1+i \end{pmatrix}$. So the tensor has MPV-family invariance under these two elements. Block to an odd length L at which the tensor is injective. The blocked tensor is symmetric under the L -fold Kronecker powers, and the powers of $i\sigma_x$ and $i\sigma_z$ anticommute because L is odd. The fundamental theorem supplies virtual gauges for both, which Theorem 10.4.5.6 rules out. The source argues instead through the alternation of integer and half-integer virtual spins. $\square$

Chapter 11

Parent-Hamiltonian Foundations: Injective and Normal Ground Spaces

For a translation-invariant matrix product state with tensor A , the *parent Hamiltonian* is the translation-invariant sum of local terms, each the orthogonal projector onto the complement of the length- L local space $G_L(A) = \text{im}(\Gamma_L)$, where $\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X)$ [193, 16]. We identify the N -site space with functions $\{0, \dots, d-1\}^N \rightarrow \mathbb{C}$ on the computational basis.

The local ground spaces satisfy an intersection property that ensures uniqueness of the periodic ground state $V^{(N)}(A)$ for injective and normal matrix product states. For injective tensors with positive bond dimension $D \geq 1$, system size $N \geq 2$, and interaction range $1 < L \leq N$, the parent Hamiltonian is frustration-free with unique ground state $V^{(N)}(A)$. For normal tensors with positive bond dimension $D \geq 1$ and injectivity length L_0 , uniqueness holds whenever $L_0 < L \leq N$ with $N \geq L_0 + 1$; word-span injectivity after blocking determines the boundary-closure identities that match open-interval restrictions to the periodic chain.

11.1 The N -site Hilbert space

Definition 11.1.1 (N -site Hilbert space). Lean ✓ Stmt For local dimension d and chain length N , the N -site Hilbert space is $\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \rightarrow \mathbb{C}$, identified with functions on the computational basis.

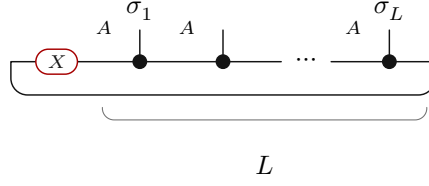
11.2 Local ground space $G_L(A)$

Fix an MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ with $A^i \in M_D(\mathbb{C})$ and a block length $L \geq 1$. For a word $w = (i_1, \dots, i_L)$, write $A^w := A^{i_1} \dots A^{i_L}$. The boundary-condition parametrization is

$$\begin{aligned} \Gamma_L: M_D(\mathbb{C}) &\longrightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}), \\ \Gamma_L(X)(\sigma) &:= \text{tr}(A^\sigma X). \end{aligned} \tag{11.1}$$

Here A^σ abbreviates $A^{\sigma(0)} \dots A^{\sigma(L-1)}$. By cyclicity of the trace, this is the same as the convention of [193], namely $\text{tr}(XA^\sigma)$ for the local space $\mathcal{G}_L^A$.

Definition 11.2.1 (Ground-space map). [Lean](#) [✓ Stmt](#) The map Γ_L in (11.1) is a $\mathbb{C}$ -linear map from $M_D(\mathbb{C})$ to $(\{0, \dots, d-1\}^L \rightarrow \mathbb{C})$. In tensor-network notation,



The chain denotes the word tensor A^σ with open physical indices $\sigma_1, \dots, \sigma_L$, and the red capsule denotes the boundary matrix X inserted on the closing virtual bond before taking the trace. The brace records the chain length L .

Lemma 11.2.2 (Evaluation formula). [Lean](#) [✓ Stmt](#) For every boundary matrix $X \in M_D(\mathbb{C})$ and every length- L word σ , one has $\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X)$.

11.2.1 The FNW boundary convention

Fannes, Nachtergaele, and Werner fix the product convention $v(\mu_1, \dots, \mu_N) = v(\mu_1) \cdots v(\mu_N)$ in [160, Equation (5.4)] and define their boundary map in [160, Equation (5.5)]. The exact identification with the map in Definition 11.2.1 uses $A_\mu = v(\mu)^\dagger$ and reverses the order of the physical sites.

Definition 11.2.1.1 (Reversal of physical configurations). [Lean](#) [✓ Stmt](#) For a length- N configuration σ , define

$$r_N(\sigma)(j) = \sigma(N-1-j), \quad (11.2)$$

where $0 \leq j < N$. This is an involutive permutation of $\{0, \dots, d-1\}^N$.

Definition 11.2.1.2 (Physical-site reversal). [Lean](#) [✓ Stmt](#) Precomposition with r_N defines a complex-linear isomorphism

$$R_N: (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}) \longrightarrow (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}). \quad (11.3)$$

This is a linear map of coefficient spaces and does not require an inner product.

Theorem 11.2.1.3 (Evaluation under physical-site reversal). [Lean](#) [✓ Stmt](#) For every coefficient function ψ and every configuration σ ,

$$(R_N \psi)(\sigma) = \psi(r_N(\sigma)). \quad (11.4)$$

Proof. [✓ Proof](#) This is the definition of precomposition by the permutation r_N . $\square$

Definition 11.2.1.4 (FNW boundary map). [Lean](#) [✓ Stmt](#) For $v = \{v(\mu)\}_{\mu=0}^{d-1}$, write $v^\sigma = v(\sigma(0)) \cdots v(\sigma(N-1))$, with the empty word equal to I_D . Define the complex-linear map $F_N^v: M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^N \rightarrow \mathbb{C})$ by

$$F_N^v(B)(\sigma) = \text{tr}(B(v^\sigma)^\dagger). \quad (11.5)$$

This is the boundary map of [160, Equation (5.5)], with the word order fixed by [160, Equation (5.4)].

Theorem 11.2.1.5 (FNW boundary-map evaluation). Lean ✓ Stmt For every $B \in M_D(\mathbb{C})$ and every length- N configuration σ ,

$$F_N^v(B)(\sigma) = \text{tr}(B(v^\sigma)^\dagger). \quad (11.6)$$

Proof. ✓ Proof Evaluate the defining linear map at B and then at σ . □

Theorem 11.2.1.6 (Reversed adjoint word). Lean ✓ Stmt Set $A_\mu = v(\mu)^\dagger$. Then

$$F_N^v(B)(\sigma) = \text{tr}(BA^{r_N(\sigma)}). \quad (11.7)$$

For $N > 0$, the word on the right is $A_{\sigma(N-1)} \cdots A_{\sigma(0)}$.

Proof. ✓ Proof Conjugate transposition reverses the product:

$$(v^\sigma)^\dagger = A^{r_N(\sigma)}. \quad (11.8)$$

Thus the site order in (11.7) comes from the adjointed word itself. □

Theorem 11.2.1.7 (FNW boundary map in the ground-space convention). Lean ✓ Stmt With the exact convention $A_\mu = v(\mu)^\dagger$, one has

$$F_N^v = R_N \circ \Gamma_N^A. \quad (11.9)$$

Proof. ✓ Proof For every $B \in M_D(\mathbb{C})$ and every configuration σ ,

$$(R_N \Gamma_N^A(B))(\sigma) = \Gamma_N^A(B)(r_N(\sigma)) \quad (11.10)$$

$$= \text{tr}(A^{r_N(\sigma)} B) \quad (11.11)$$

$$= \text{tr}(BA^{r_N(\sigma)}) \quad (11.12)$$

$$= F_N^v(B)(\sigma). \quad (11.13)$$

The ordered list associated with $r_N(\sigma)$ is the reverse of the list associated with σ , as in (11.2). Together with the adjointed word in Theorem 11.2.1.6, this gives the physical-site reversal. Trace cyclicity is used only in the third equality, where it moves the boundary matrix B across the word. □

These boundary identifications are algebraic relations between matrix spaces and coefficient functions, independent of any state-dependent boundary inner product.

Definition 11.2.1.8 (Local ground space). Lean ✓ Stmt The *local ground space* is the image $G_L(A) := \text{range}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C})$.

Lemma 11.2.1.9 (Dimension bound). Lean ✓ Stmt For every L , one has $\dim G_L(A) \leq D^2$.

Proof. ✓ Proof Since $G_L(A)$ is the range of a linear map from $M_D(\mathbb{C})$, one has $\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2$. □

Lemma 11.2.1.10 (Ambient-space dimension). Lean ✓ Stmt The ambient space of L -site tensors satisfies $\dim(\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) = d^L$.

Lemma 11.2.1.11 (Nontriviality criterion). Lean ✓ Stmt If $d^L > D^2$, then $G_L(A)$ is a proper subspace of the local Hilbert space, namely $G_L(A) \neq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C})$.

Proof. ✓ Proof If $G_L(A)$ were the whole space, its dimension would be d^L by Lemma 11.2.1.10; Lemma 11.2.1.9 would then give the contradiction $d^L \leq D^2$. □

11.3 Parent interaction and chain Hamiltonian

The canonical parent interaction at range L is

$$\begin{aligned} h_L(A) &= \mathbb{1} - \Pi_{G_L(A)} = \Pi_{G_L(A)^\perp}, \\ \ker h_L(A) &= G_L(A). \end{aligned} \tag{11.14}$$

The sources also allow any positive local interaction with this kernel; the orthogonal projector is the canonical representative used here.

Definition 11.3.1 (Local ground space under the canonical ℓ^2 identification). [Lean](#) [✓ Stmt](#) Let $\iota_L: (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) \simeq \ell^2(\{0, \dots, d-1\}^L)$ be the canonical computational-basis identification. The local ground space under this ℓ^2 identification is $G_L^{\text{ES}}(A) := \iota_L(G_L(A)) \subseteq \ell^2(\{0, \dots, d-1\}^L)$.

Lemma 11.3.2 (Membership under the canonical ℓ^2 identification). [Lean](#) [✓ Stmt](#) For $v \in \ell^2(\{0, \dots, d-1\}^L)$, one has $v \in G_L^{\text{ES}}(A)$ if and only if $\iota_L^{-1}v \in G_L(A)$.

Proof. [✓ Proof](#) This is the defining equation $G_L^{\text{ES}}(A) = \iota_L(G_L(A))$. □

Definition 11.3.3 (Hilbert-space boundary map). [Lean](#) [✓ Stmt](#) Let $U: M_D(\mathbb{C}) \rightarrow \ell^2(\{0, \dots, D-1\} \times \{0, \dots, D-1\})$ be Frobenius vectorization in column–row coordinates. The Hilbert-space boundary map is

$$\Gamma_L^{\text{ES}} := \iota_L \circ \Gamma_L \circ U^{-1}: \ell^2(\{0, \dots, D-1\} \times \{0, \dots, D-1\}) \longrightarrow \ell^2(\{0, \dots, d-1\}^L). \tag{11.15}$$

Thus Γ_L^{ES} has a Euclidean coordinate-space domain, whereas the original map Γ_L has matrix domain $M_D(\mathbb{C})$.

Theorem 11.3.4 (Boundary-map transport under Frobenius vectorization). [Lean](#) [✓ Stmt](#) Let $U: M_D(\mathbb{C}) \rightarrow \ell^2(\{0, \dots, D-1\} \times \{0, \dots, D-1\})$ be Frobenius vectorization, with the target indexed in column–row order, and let ι_L be the computational-basis identification on the physical coefficient space. Then, for every $X \in M_D(\mathbb{C})$, $\Gamma_L^{\text{ES}}(U(X)) = \iota_L(\Gamma_L(X))$.

Proof. [✓ Proof](#) By definition, Γ_L^{ES} is the composite of U^{-1} , the original boundary map Γ_L , and ι_L . Applying this composite to $U(X)$ and cancelling $U^{-1}U$ gives the claimed identity. □

Theorem 11.3.5 (Boundary map on a matrix unit). [Lean](#) [✓ Stmt](#) Let $e_{(b,a)}$ denote the Frobenius coordinate vector representing the virtual matrix unit E_{ab} . Then $\Gamma_L^{\text{ES}}(e_{(b,a)}) = \iota_L(\Gamma_L(E_{ab}))$.

Proof. [✓ Proof](#) Frobenius vectorization records a matrix entry (a, b) at the column–row coordinate (b, a) , so $U(E_{ab}) = e_{(b,a)}$. Substitute this identity into the boundary-map transport theorem. □

Theorem 11.3.6 (Exact range of the Hilbert-space boundary map). [Lean](#) [✓ Stmt](#) The range of Γ_L^{ES} is exactly $G_L^{\text{ES}}(A)$.

Proof. [✓ Proof](#) Every virtual vector is $U(X)$ for a unique matrix X , and the transport identity sends it to $\iota_L(\Gamma_L(X))$. These are exactly the vectors in $\iota_L(\text{range } \Gamma_L) = G_L^{\text{ES}}(A)$, proving both inclusions. □

Theorem 11.3.7 (Hilbert-space boundary-map injectivity). [Lean](#) [✓ Stmt](#) If A is L -block injective, then Γ_L^{ES} is injective.

Proof. [✓ Proof](#) Write each virtual vector uniquely as $U(X)$ by Frobenius vectorization. The transport identity reduces equality of their images under Γ_L^{ES} to equality of the images of the corresponding matrices under Γ_L . Block injectivity makes the latter map injective. □

Theorem 11.3.8 (Long-length Hilbert-space boundary-map injectivity). [Lean](#) [✓ Stmt](#) Suppose that $L_0 > 0$, that A is L_0 -block injective, and that $L \geq L_0$. Then Γ_L^{ES} is injective.

Proof. [✓ Proof](#) Positive-length propagation makes A block injective at length L . Theorem 11.3.7 then gives injectivity of Γ_L^{ES} . $\square$

Definition 11.3.9 (Ground-space Gram operator). [Lean](#) [✓ Stmt](#) The boundary Gram operator is

$$\mathcal{K}_L := (\Gamma_L^{\text{ES}})^\dagger \Gamma_L^{\text{ES}}. \quad (11.16)$$

The FNW convention starts from an isometry $V: \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^D$ written as $V\xi = \sum_\mu |\mu\rangle \otimes v(\mu)^\dagger \xi$ in [160, Equation (5.2)]. Equations (5.3b) and (5.3d) of that paper give $\sum_\mu v(\mu)v(\mu)^\dagger = I_D$ and $E(B) = \sum_\mu v(\mu)Bv(\mu)^\dagger$. Under the identification $A^\mu = v(\mu)^\dagger$, this map uses the adjoint tensor letters. The asymptotic identity $E^n(B) \rightarrow \text{tr}(\rho B)I_D$ in [160, Equation (5.1)] is a separate assertion and is not used in the convention identities below.

Definition 11.3.10 (FNW transfer map). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor, with $A^\mu = v(\mu)^\dagger$ in the notation of [160, Equations (5.2)–(5.3)]. Its FNW transfer map is

$$E_{\text{FNW},A}(X) := \sum_{\mu=0}^{d-1} (A^\mu)^\dagger X A^\mu = \sum_{\mu=0}^{d-1} v(\mu) X v(\mu)^\dagger. \quad (11.17)$$

Theorem 11.3.11 (FNW map as the trace-pairing adjoint). [Lean](#) [✓ Stmt](#) Write $\mathcal{E}_A^\sharp$ for the adjoint of $\mathcal{E}_A$ with respect to the bilinear trace pairing, characterized by

$$\text{tr}(\mathcal{E}_A(X)Y) = \text{tr}(X\mathcal{E}_A^\sharp(Y)). \quad (11.18)$$

Then $E_{\text{FNW},A} = \mathcal{E}_A^\sharp$. Here $\sharp$ denotes the adjoint for $(X, Y) \mapsto \text{tr}(XY)$, so the theorem is phrased as a bilinear trace-adjoint identity. For the unweighted Frobenius inner product, the same map also agrees with the Hilbert-space adjoint of $\mathcal{E}_A$. This Frobenius adjoint is generally different from the adjoint for the ρ -weighted inner product in [160, Equation (5.6)].

Proof. [✓ Proof](#) Expanding both maps and using cyclicity of trace gives

$$\text{tr}(\mathcal{E}_A(X)Y) = \sum_\mu \text{tr}(A^\mu X (A^\mu)^\dagger Y) = \sum_\mu \text{tr}(X (A^\mu)^\dagger Y A^\mu). \quad (11.19)$$

The final sum is $\text{tr}(X E_{\text{FNW},A}(Y))$. $\square$

Theorem 11.3.12 (Unitality of the FNW transfer map). [Lean](#) [✓ Stmt](#) If A is left canonical, then

$$E_{\text{FNW},A}(I_D) = I_D. \quad (11.20)$$

Under $A^\mu = v(\mu)^\dagger$, this is the normalization $\sum_\mu v(\mu)v(\mu)^\dagger = I_D$ from [160, Equation (5.3b)].

Proof. [✓ Proof](#) Left canonicity makes $\mathcal{E}_A$ trace-preserving. Therefore, for every X ,

$$\text{tr}(X E_{\text{FNW},A}(I_D)) = \text{tr}(\mathcal{E}_A(X)) = \text{tr}(X). \quad (11.21)$$

Nondegeneracy of the trace pairing gives $E_{\text{FNW},A}(I_D) = I_D$. $\square$

Theorem 11.3.13 (FNW transfer pairing identity). [Lean](#) [✓ Stmt](#) For all $\rho, X \in M_D(\mathbb{C})$,

$$\mathrm{tr}(\rho E_{\mathrm{FNW},A}(X)) = \mathrm{tr}(\mathcal{E}_A(\rho)X). \quad (11.22)$$

This is the explicit trace-pairing form of the convention $A^\mu = v(\mu)^\dagger$ for the map in [160, Equations (5.2)–(5.3)].

Proof. [✓ Proof](#) Substitute the trace-pairing adjoint identity and use cyclicity of trace. $\square$

These identities establish the algebraic equivalence between the two transfer-map conventions.

Definition 11.3.14 (Normalized limiting FNW observable map). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\rho \in M_D(\mathbb{C})$ satisfy $\mathrm{tr}(\rho) \neq 0$. The normalized rank-one map associated with ρ is

$$E_\infty^\rho(B) := \frac{\mathrm{tr}(\rho B)}{\mathrm{tr}(\rho)} I_D. \quad (11.23)$$

If $\mathrm{tr}(\rho) = 1$, this becomes $E_\infty^\rho(B) = \mathrm{tr}(\rho B) I_D$, which is the limiting map in [160, Equation (5.1)].

Theorem 11.3.15 (Limiting map as a trace-pairing adjoint). [Lean](#) [✓ Stmt](#) Define the fixed-point projection

$$P_\rho(X) := \frac{\mathrm{tr}(X)}{\mathrm{tr}(\rho)} \rho. \quad (11.24)$$

Then $E_\infty^\rho = P_\rho^\sharp$, where $\sharp$ denotes the adjoint for the bilinear trace pairing. For the trace-one Hermitian density matrix used by FNW, the same map also agrees with the Hilbert-space adjoint for the unweighted Frobenius inner product. This Frobenius adjoint is generally different from the adjoint for the ρ -weighted inner product of [160, Equation (5.6)].

Proof. [✓ Proof](#) For all $X, B \in M_D(\mathbb{C})$,

$$\mathrm{tr}(P_\rho(X)B) = \frac{\mathrm{tr}(X) \mathrm{tr}(\rho B)}{\mathrm{tr}(\rho)} = \mathrm{tr}(X E_\infty^\rho(B)). \quad (11.25)$$

Nondegeneracy of the trace pairing proves the equality of maps. $\square$

Theorem 11.3.16 (Pointwise idempotence of the limiting map). [Lean](#) [✓ Stmt](#) For every $B \in M_D(\mathbb{C})$,

$$E_\infty^\rho(E_\infty^\rho(B)) = E_\infty^\rho(B). \quad (11.26)$$

Proof. [✓ Proof](#) Substitute (11.23) and use $\mathrm{tr}(\rho I_D) = \mathrm{tr}(\rho)$. The nonzero trace cancels the normalizing denominator. $\square$

Theorem 11.3.17 (The limiting map is a projection). [Lean](#) [✓ Stmt](#) As an endomorphism of $M_D(\mathbb{C})$, the limiting map satisfies

$$(E_\infty^\rho)^2 = E_\infty^\rho. \quad (11.27)$$

Proof. [✓ Proof](#) Apply Theorem 11.3.16 to every matrix B . $\square$

Theorem 11.3.18 (Unital absorption of the limiting map). [Lean](#) [✓ Stmt](#) If A is left canonical, then the FNW observable map in (11.17) satisfies

$$E_{\mathrm{FNW},A} E_\infty^\rho = E_\infty^\rho. \quad (11.28)$$

Proof. ✓ Proof Apply Theorem 11.3.12 to obtain $E_{\text{FNW},A}(I_D) = \sum_{\mu} (A^{\mu})^{\dagger} A^{\mu} = I_D$. Hence

$$E_{\text{FNW},A}(E_{\infty}^{\rho}(B)) = \frac{\text{tr}(\rho B)}{\text{tr}(\rho)} E_{\text{FNW},A}(I_D) = E_{\infty}^{\rho}(B). \quad (11.29)$$

□

Theorem 11.3.19 (Stationary absorption by the limiting map). Lean ✓ Stmt Suppose that ρ is stationary for the Schrödinger transfer map, $\mathcal{E}_A(\rho) = \rho$. Then

$$E_{\infty}^{\rho} E_{\text{FNW},A} = E_{\infty}^{\rho}. \quad (11.30)$$

Proof. ✓ Proof Apply Theorem 11.3.13 and stationarity to get

$$\text{tr}(\rho E_{\text{FNW},A}(B)) = \text{tr}(\mathcal{E}_A(\rho)B) = \text{tr}(\rho B). \quad (11.31)$$

Substitute this equality into (11.23). □

Theorem 11.3.20 (Positive powers of the FNW remainder). Lean ✓ Stmt Suppose that A is left canonical and that $\mathcal{E}_A(\rho) = \rho$. For every integer $n \geq 1$,

$$(E_{\text{FNW},A} - E_{\infty}^{\rho})^n = E_{\text{FNW},A}^n - E_{\infty}^{\rho}. \quad (11.32)$$

The restriction $n \geq 1$ is essential. At $n = 0$, the two sides would be I and $I - E_{\infty}^{\rho}$, respectively, and $E_{\infty}^{\rho}(I_D) = I_D$.

Proof. ✓ Proof Write $E = E_{\text{FNW},A}$ and $P = E_{\infty}^{\rho}$. The identities of Theorem 11.3.17, Theorem 11.3.18, and Theorem 11.3.19 give $EP = PE = P^2 = P$, and hence $E^m P = P$ for every $m \geq 0$. The case $n = 1$ is immediate. If $(E - P)^n = E^n - P$, then

$$(E - P)^{n+1} = (E^n - P)(E - P) = E^{n+1} - E^n P - PE + P^2 = E^{n+1} - P. \quad (11.33)$$

□

Theorem 11.3.21 (Primitive-MPS remainder identity). Lean ✓ Stmt Let A be a primitive MPS tensor of nonzero bond dimension with fixed-point witness ρ . Then, for every integer $n \geq 1$,

$$(E_{\text{FNW},A} - E_{\infty}^{\rho})^n = E_{\text{FNW},A}^n - E_{\infty}^{\rho}. \quad (11.34)$$

Proof. ✓ Proof Primitivity supplies $\text{tr}(\rho) \neq 0$, left canonicity, and the stationary equation $\mathcal{E}_A(\rho) = \rho$. Apply Theorem 11.3.20. □

These algebraic relations express the spectral projection and remainder decomposition of the transfer map before introducing the ρ -weighted inner product.

Definition 11.3.22 (Operator-level Gram reshuffling). Lean Lean ✓ Stmt

For a complex-linear map $\mathcal{L}: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, let $R(\mathcal{L})$ be the matrix with entries

$$R(\mathcal{L})_{(b,a),(e,c)} := J(\mathcal{L})_{(c,e),(a,b)}. \quad (11.35)$$

The operator $\mathcal{R}(\mathcal{L})$ on $\ell^2(\{0, \dots, D-1\} \times \{0, \dots, D-1\})$ is represented by the matrix $R(\mathcal{L})$ in the standard orthonormal basis. Thus the row-column pair $((b,a),(e,c))$ is sent explicitly to the Choi row-column pair $((c,e),(a,b))$.

Theorem 11.3.23 (Gram reshuffling commutes with subtraction). [Lean](#) [✓ Stmt](#) For complex-linear maps $\mathcal{L}, \mathcal{M}: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$,

$$\mathcal{R}(\mathcal{L} - \mathcal{M}) = \mathcal{R}(\mathcal{L}) - \mathcal{R}(\mathcal{M}). \quad (11.36)$$

Proof. [✓ Proof](#) The defining matrix entries of the Choi reshuffling are linear in the map. $\square$

Theorem 11.3.24 (Matrix-unit coordinates of Gram reshuffling). [Lean](#) [✓ Stmt](#) Let $\mathcal{L}: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be complex-linear and let $\mathbf{e}_{(j,i)}$ be the standard unit vector at (j,i) . For all $a, b, c, e \in \{0, \dots, D-1\}$,

$$\langle \mathbf{e}_{(b,a)}, \mathcal{R}(\mathcal{L})\mathbf{e}_{(e,c)} \rangle_{\ell^2} = J(\mathcal{L})_{(c,e),(a,b)}. \quad (11.37)$$

This is the coordinate identity in (11.35); it is not a Hilbert–Schmidt pairing of E_{ab} with $\mathcal{L}(E_{ce})$.

Proof. [✓ Proof](#) Evaluate the matrix of $\mathcal{R}(\mathcal{L})$ on the two standard basis vectors. $\square$

Theorem 11.3.25 (Euclidean operator norm bounded by Frobenius mass). [Lean](#) [✓ Stmt](#) Let I be a finite index set and let $B = (B_{ij})_{i,j \in I}$ be a complex matrix. Then the squared operator norm of B on $\ell^2(I)$ satisfies

$$\|B\|_{\ell^2(I) \rightarrow \ell^2(I)}^2 \leq \sum_{(i,j) \in I \times I} |B_{ij}|^2. \quad (11.38)$$

Proof. [✓ Proof](#) Apply Cauchy–Schwarz to each row of Bx , sum over the rows, and take the supremum over unit vectors $x \in \ell^2(I)$. $\square$

Theorem 11.3.26 (Squared mass of one column). [Lean](#) [✓ Stmt](#) Let I be a finite index set, let $A = (A_{ij})_{i,j \in I}$ be a complex matrix, and let $j \in I$. Then

$$\sum_{i \in I} |A_{ij}|^2 \leq \|A\|_{\ell^2(I) \rightarrow \ell^2(I)}^2. \quad (11.39)$$

Proof. [✓ Proof](#) Apply A to the unit vector supported at j . Its image is the j th column of A , while the vector itself has Euclidean norm one. $\square$

Theorem 11.3.27 (Entrywise mass bounded by the L^2 operator norm). [Lean](#) [✓ Stmt](#) Let I be a finite index set and let $A = (A_{ij})_{i,j \in I}$ be a complex matrix. Then

$$\sum_{(i,j) \in I \times I} |A_{ij}|^2 \leq |I| \|A\|_{\ell^2(I) \rightarrow \ell^2(I)}^2. \quad (11.40)$$

Proof. [✓ Proof](#) Sum Theorem 11.3.26 over all columns. $\square$

Theorem 11.3.28 (L^2 operator norm of a matrix unit). [Lean](#) [✓ Stmt](#) Let I be a finite index set. For every $i, j \in I$, the matrix unit E_{ij} satisfies

$$\|E_{ij}\|_{\ell^2(I) \rightarrow \ell^2(I)} = 1. \quad (11.41)$$

Proof. [✓ Proof](#) The matrix unit sends the unit vector at j to the unit vector at i and sends every vector to a scalar multiple of that unit vector. $\square$

Theorem 11.3.29 (Entrywise Frobenius invariance of Gram reshuffling). [Lean] [✓ Stmt] For every complex-linear map $\mathcal{L}: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, the sum of squared entry norms of $R(\mathcal{L})$ equals that of the rectangular Choi matrix:

$$\sum_{p: (\{0, \dots, D-1\} \times \{0, \dots, D-1\}) \times (\{0, \dots, D-1\} \times \{0, \dots, D-1\})} \|R(\mathcal{L})_{p_1, p_2}\|^2 = \sum_{p: (\{0, \dots, D-1\} \times \{0, \dots, D-1\}) \times (\{0, \dots, D-1\} \times \{0, \dots, D-1\})} \|J(\mathcal{L})_{p_1, p_2}\|^2. \quad (11.42)$$

The two matrices differ only by a reshuffling of indices; the equality is entrywise Frobenius, not operator norm.

Proof. [✓ Proof] The map $((b, a), (e, c)) \mapsto ((c, e), (a, b))$ is an involution (bijection) on the product index space. Reindexing the sum of squared norms by this involution yields the claimed equality. □

Theorem 11.3.30 (Operator bound for a reshuffled Choi matrix). [Lean] [✓ Stmt] Let I be a finite index set and let $\mathcal{L}: \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{I \times I}$ be complex-linear. Then

$$\|\mathcal{R}(\mathcal{L})\|_{\ell^2(I \times I) \rightarrow \ell^2(I \times I)}^2 \leq \sum_{u, v \in I \times I} |J(\mathcal{L})_{u, v}|^2. \quad (11.43)$$

Proof. [✓ Proof] Apply Theorem 11.3.25 to $R(\mathcal{L})$, then use Theorem 11.3.29 to rewrite its entrywise Frobenius mass as that of $J(\mathcal{L})$. □

Theorem 11.3.31 (Dimension bound for Gram reshuffling). [Lean] [✓ Stmt] Let I be a finite index set, equip $\mathbb{C}^{I \times I}$ with its L^2 operator norm, and let $\mathcal{L}: \mathbb{C}^{I \times I} \rightarrow \mathbb{C}^{I \times I}$ be complex-linear. Denote the operator norm of $\mathcal{L}$ for these matrix norms by $\|\mathcal{L}\|_{\text{op}, L^2}$. Then

$$\|\mathcal{R}(\mathcal{L})\|_{\ell^2(I \times I) \rightarrow \ell^2(I \times I)}^2 \leq |I|^3 \|\mathcal{L}\|_{\text{op}, L^2}^2. \quad (11.44)$$

Proof. [✓ Proof] Use Theorem 11.3.30 and expand the Choi matrix in matrix units. For each of the $|I|^2$ input matrix units, Theorem 11.3.27 contributes a factor $|I|$. The operator-norm estimate for $\mathcal{L}$ and Theorem 11.3.28 bound the remaining matrix norm. □

Theorem 11.3.32 (Geometric power bound at a prescribed rate). [Lean] [✓ Stmt] Let A be a complex Banach algebra, let $a \in A$, and let $\lambda \geq 0$. If $\rho_{\text{spec}}(a) < \lambda$, then there is a constant $C_\lambda > 0$ such that

$$\|a^n\| \leq C_\lambda \lambda^n \quad (11.45)$$

for every $n \in \mathbb{N}$. This operator-theoretic estimate provides the prescribed-rate geometric decay used for the peripheral-spectrum remainder of the transfer map in [160, Lemma 5.2].

Proof. [✓ Proof] Gelfand's formula gives $N \in \mathbb{N}$ such that $\|a^n\| \leq \lambda^n$ for every $n \geq N$. Since the spectral radius is nonnegative and strictly smaller than λ , one has $\lambda > 0$. Set

$$C_\lambda = 1 + \sum_{j=0}^{N-1} \frac{\|a^j\|}{\lambda^j}. \quad (11.46)$$

The eventual estimate proves the claim for $n \geq N$, while the corresponding summand in C_λ proves it for $n < N$. □

Theorem 11.3.33 (Pointwise geometric power bound at a prescribed rate). [Lean](#) [✓ Stmt](#) Let V be a complex Banach space, let $T: V \rightarrow V$ be a bounded linear operator, and let $\lambda \geq 0$. If $\rho_{\text{spec}}(T) < \lambda$, then there is a constant $C_\lambda > 0$ such that

$$\|T^n x\| \leq C_\lambda \lambda^n \|x\| \quad (11.47)$$

for every $n \in \mathbb{N}$ and $x \in V$.

Proof. [✓ Proof](#) Apply Theorem 11.3.32 in the Banach algebra of bounded endomorphisms of V , then use $\|T^n x\| \leq \|T^n\| \|x\|$. $\square$

Theorem 11.3.34 (FNW remainder eigenvalue gap). [Lean](#) [✓ Stmt](#) Let A be a primitive MPS tensor of nonzero bond dimension with fixed-point witness ρ , and set $R_A^\rho = E_{\text{FNW},A} - E_\infty^\rho$. If ν is an eigenvalue of R_A^ρ , then

$$|\nu| < 1. \quad (11.48)$$

This is the norm-independent spectral step in [160, Lemma 5.2].

Proof. [✓ Proof](#) The remainder R_A^ρ is the bilinear trace adjoint of $\mathcal{E}_A - P_\rho$. A linear map and its trace adjoint have the same characteristic polynomial, hence the same eigenvalues. Primitivity places every eigenvalue of $\mathcal{E}_A - P_\rho$ strictly inside the unit disk. $\square$

Definition 11.3.35 (Trace-inverse factor). [Lean](#) [✓ Stmt](#) Let $\rho \in M_D(\mathbb{C})$. Define

$$c_\rho := \text{Re tr}(\rho^{-1}). \quad (11.49)$$

This is the factor multiplying the weighted remainder norm in [160, Equations (5.9)–(5.10)].

Theorem 11.3.36 (Positivity of the trace-inverse factor). [Lean](#) [✓ Stmt](#) If $\rho \in M_D(\mathbb{C})$ is positive definite and $D > 0$, then $c_\rho > 0$.

Proof. [✓ Proof](#) The inverse ρ^{-1} is positive definite, so its trace has strictly positive real part. $\square$

Definition 11.3.37 (Weighted FNW remainder data). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\rho \in M_D(\mathbb{C})$ be positive definite and let $\text{tr}(\rho) \neq 0$. Equip $M_D(\mathbb{C})$ with $\langle X, Y \rangle_\rho = \text{tr}(\rho X^\dagger Y)$. For a linear map $F: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$, write

$$\|F\|_{\text{op},\rho} := \sup_{X \neq 0} \frac{\|F(X)\|_\rho}{\|X\|_\rho}. \quad (11.50)$$

The weighted remainder spectral radius is

$$r_\rho(A) := \rho_{\text{spec}}(E_{\text{FNW},A} - E_\infty^\rho), \quad (11.51)$$

where the remainder is viewed as a bounded operator on this weighted Hilbert space.

Theorem 11.3.38 (Weighted FNW remainder spectral gap). [Lean](#) [✓ Stmt](#) Let A be primitive with nonzero bond dimension and positive-definite fixed-point witness ρ . Then

$$r_\rho(A) < 1. \quad (11.52)$$

Proof. [✓ Proof](#) In finite dimension the spectral radius is determined by the eigenvalues. Apply Theorem 11.3.34 in the weighted norm. $\square$

Definition 11.3.39 (FNW mixing quantity). [Lean](#) [✓ Stmt](#) Suppose that ρ is positive definite and $\text{tr}(\rho) = 1$. For $n \in \mathbb{N}$, define

$$a(n) := c_\rho \|E_{\text{FNW},A}^n - E_\infty^\rho\|_{\text{op},\rho}. \quad (11.53)$$

This is the trace-one transfer-remainder quantity formed from the factor and weighted norm in [160, Equations (5.9)–(5.10)].

Theorem 11.3.40 (Nonnegativity of the FNW mixing quantity). [Lean](#) [Lean](#) [✓ Stmt](#) The weighted operator norm is nonnegative, and hence so is $a(n)$ for every $n \in \mathbb{N}$.

Proof. [✓ Proof](#) A norm is nonnegative, and the trace factor c_ρ is strictly positive. $\square$

Theorem 11.3.41 (Prescribed-rate bound for the FNW mixing quantity). [Lean](#) [✓ Stmt](#) Let A be primitive with nonzero bond dimension and positive-definite trace-one fixed point ρ . Let $\lambda \geq 0$ satisfy $r_\rho(A) < \lambda$. There is a constant $C_\lambda > 0$ such that, for every integer $n \geq 1$,

$$a(n) \leq C_\lambda \lambda^n. \quad (11.54)$$

This proves the prescribed-rate transfer-remainder estimate in [160, Lemma 5.2]. The condition $\lambda < 1$ is needed only when one subsequently deduces decay to zero.

Proof. [✓ Proof](#) Apply Theorem 11.3.32 to the weighted remainder R_A^ρ . This gives $C_0 > 0$ with $\|(R_A^\rho)^n\|_{\text{op},\rho} \leq C_0 \lambda^n$. When $n \geq 1$, Theorem 11.3.21 identifies $(R_A^\rho)^n$ with $E_{\text{FNW},A}^n - E_\infty^\rho$. Multiply by $c_\rho > 0$ and take $C_\lambda = c_\rho C_0$. $\square$

The theorem does not identify C_λ with k^2 or identify the FNW auxiliary dimension k with the chosen bond dimension D .

11.3.1 The weighted FNW boundary estimate

Let $\rho \in M_D(\mathbb{C})$ be positive definite with $\text{tr}(\rho) = 1$. Equip $M_D(\mathbb{C})$ with the inner product $\langle B, C \rangle_\rho = \text{tr}(\rho B^\dagger C)$ from [160, Equation (5.6)], and let E_{ik} denote the standard matrix unit. Throughout this subsection, $A^\mu = v(\mu)^\dagger$.

Definition 11.3.1.1 (Euclidean physical-site reversal). [Lean](#) [Lean](#) [✓ Stmt](#) Precomposition with the reversal r_N defines a unitary operator

$$\mathcal{R}_N: \mathbb{C}^{d^N} \longrightarrow \mathbb{C}^{d^N}, \quad (11.55)$$

$$(\mathcal{R}_N \psi)(\sigma) = \psi(r_N(\sigma)). \quad (11.56)$$

Proof. [✓ Proof](#) A permutation of the standard coordinates preserves the Euclidean inner product, and evaluation gives (11.56). $\square$

Definition 11.3.1.2 (Weighted FNW boundary operator). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The boundary map of [160, Equation (5.5)] defines a bounded linear operator

$$F_N: (M_D(\mathbb{C}), \langle \cdot, \cdot \rangle_\rho) \longrightarrow \mathbb{C}^{d^N}, \quad (11.57)$$

$$F_N(B)(\sigma) = \text{tr}(B(v^\sigma)^\dagger). \quad (11.58)$$

In the ground-space convention,

$$F_N = \mathcal{R}_N \circ \Gamma_N^A. \quad (11.59)$$

Proof. ✓ Proof The algebraic identity $F_N^v = R_N \circ \Gamma_N^A$ from Theorem 11.2.1.7 is unchanged after equipping the domain with the weighted norm and the codomain with the Euclidean norm. The reversal becomes the unitary operator $\mathcal{R}_N$. $\square$

Theorem 11.3.1.3 (FNW boundary scalar product). Lean Lean ✓ Stmt For all $B, C \in M_D(\mathbb{C})$,

$$\langle F_N(B), F_N(C) \rangle = \sum_{i,k} \left(B^\dagger E_{\text{FNW},A}^N(E_{ik})C \right)_{ik} \quad (11.60)$$

$$= \sum_{i,k} \left(E_{\text{FNW},A}^N(B^\dagger E_{ik}C) \right)_{ik}. \quad (11.61)$$

These are the two forms of the scalar-product identity in [160, Equation (5.8)]. Identity (11.61) preserves the source orientation $B^\dagger E_{ik}C$.

Proof. ✓ Proof Expand both boundary coefficients using (11.58), interchange the finite sums, and write the two traces as sums of diagonal entries. Multiplication by the matrix unit E_{ik} gives respectively (11.60) and (11.61). $\square$

Theorem 11.3.1.4 (Exact weighted boundary defect). Lean ✓ Stmt For all $B, C \in M_D(\mathbb{C})$,

$$\langle F_N(B), F_N(C) \rangle - \langle B, C \rangle_\rho = \sum_{i,k} \left((E_{\text{FNW},A}^N - E_\infty^\rho)(B^\dagger E_{ik}C) \right)_{ik}. \quad (11.62)$$

Proof. ✓ Proof The trace-one limiting map satisfies

$$\sum_{i,k} (E_\infty^\rho(B^\dagger E_{ik}C))_{ik} = \text{tr}(\rho B^\dagger C) = \langle B, C \rangle_\rho. \quad (11.63)$$

Subtract this identity from (11.61). $\square$

Theorem 11.3.1.5 (Weighted FNW boundary near-isometry). Lean ✓ Stmt For all $B, C \in M_D(\mathbb{C})$,

$$|\langle F_N(B), F_N(C) \rangle - \langle B, C \rangle_\rho| \leq \text{Re tr}(\rho^{-1}) \|E_{\text{FNW},A}^N - E_\infty^\rho\|_{\text{op},\rho} \|B\|_\rho \|C\|_\rho. \quad (11.64)$$

This is the estimate in [160, Equation (5.9)]. It contains no factor depending only on the dimension.

Proof. ✓ Proof Regard the sum in (11.62) as the trace of the endomorphism obtained by composing the transfer remainder with $X \mapsto B^\dagger XC$. Weighted trace duality bounds this trace by $\text{Re tr}(\rho^{-1})$ times the weighted operator norm of the remainder and the two weighted boundary norms. $\square$

Theorem 11.3.1.6 (Right-boundary successor recursion). Lean ✓ Stmt For every $B \in M_D(\mathbb{C})$,

$$\|F_{N+1}(B)\|^2 = \sum_{\mu=0}^{d-1} \|F_N(BA^\mu)\|^2. \quad (11.65)$$

This recursion follows from the boundary formula in [160, Equation (5.5)] and is used in the proof of [160, Lemma 5.3]. Here $A^\mu = v(\mu)^\dagger$.

Proof. ✓ Proof Split a configuration of length $N+1$ into its first N entries and its final entry μ . The adjointed word identity gives $F_{N+1}(B)(\sigma, \mu) = F_N(BA^\mu)(\sigma)$. Summing the squared absolute values over σ and μ proves (11.65). $\square$

Theorem 11.3.1.5 and Theorem 11.3.1.6 establish (11.64) and the right-boundary recursion (11.65) for [160, Lemma 5.3]. They do not prove Lemma 6.2. In its source proof, Equation (6.5) contributes $a(m)/a_-(m)$, while the subsequent orthogonality estimates contribute $a(m)^2/a_-(m)$. Their sum is $a(m)(1 + a(m))/a_-(m)$.

11.3.2 The FNW lower boundary constant

Let $D > 0$, let $\rho \in M_D(\mathbb{C})$ be positive definite with $\text{tr}(\rho) = 1$, and equip $M_D(\mathbb{C})$ with

$$\langle B, C \rangle_\rho = \text{tr}(\rho B^\dagger C), \quad (11.66)$$

$$\|B\|_\rho^2 = \text{Re tr}(\rho B^\dagger B). \quad (11.67)$$

Write F_N for the length- N FNW boundary map and let $a(N)$ be the source mixing quantity in (11.53). This is the normalization used in [160, Equations (5.9)–(5.10)].

Theorem 11.3.2.1 (Weighted boundary near-isometry). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $B, C \in M_D(\mathbb{C})$,

$$|\langle F_N(B), F_N(C) \rangle - \langle B, C \rangle_\rho| \leq a(N) \|B\|_\rho \|C\|_\rho. \quad (11.68)$$

In particular,

$$|\|F_N(B)\|^2 - \|B\|_\rho^2| \leq a(N) \|B\|_\rho^2, \quad (11.69)$$

$$(1 - a(N)) \|B\|_\rho^2 \leq \|F_N(B)\|^2. \quad (11.70)$$

No sign assumption on $a(N)$ is needed for the last inequality.

Proof. [✓ Proof](#) The estimate in [160, Equation (5.9)] gives (11.68). Set $C = B$ and use $\langle B, B \rangle_\rho = \|B\|_\rho^2$ to obtain (11.69). Its lower half is (11.70). $\square$

Definition 11.3.2.2 (Lower boundary constant). [Lean](#) [Lean](#) [✓ Stmt](#) For $B \neq 0$, define the boundary norm-square ratio

$$r_N(B) := \frac{\|F_N(B)\|^2}{\|B\|_\rho^2}. \quad (11.71)$$

The lower boundary constant of [160, Lemma 5.3] is

$$a_-(N) := \inf_{B \in M_D(\mathbb{C}), B \neq 0} r_N(B). \quad (11.72)$$

Definition 11.3.2.3 (Weighted boundary Gram operator). [Lean](#) [✓ Stmt](#) The weighted Gram operator on the virtual matrix Hilbert space is

$$G_N := F_N^\dagger F_N. \quad (11.73)$$

Theorem 11.3.2.4 (Rayleigh characterization of the lower boundary constant). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The operator G_N is self-adjoint and positive. For every nonzero $B \in M_D(\mathbb{C})$,

$$r_N(B) = \frac{\text{Re} \langle G_N B, B \rangle_\rho}{\|B\|_\rho^2}, \quad (11.74)$$

$$a_-(N) = \inf_{B \neq 0} \frac{\text{Re} \langle G_N B, B \rangle_\rho}{\|B\|_\rho^2}. \quad (11.75)$$

Proof. [✓ Proof](#) The adjoint identity gives $\langle G_N B, B \rangle_\rho = \langle F_N B, F_N B \rangle$. Divide by $\|B\|_\rho^2$ and take the infimum over $B \neq 0$. Self-adjointness and positivity follow directly from $G_N = F_N^\dagger F_N$. $\square$

Theorem 11.3.2.5 (Least spectral value of the weighted Gram operator). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The real number $a_-(N)$ is a spectral value of G_N , and every $z \in \text{spec}(G_N)$ satisfies

$$a_-(N) \leq \text{Re } z. \quad (11.76)$$

Thus $a_-(N)$ is the least spectral value in the sense of real parts. The spectrum is a subset of $\mathbb{C}$, where no compatible order is fixed, so the assertion is stated as spectral membership together with (11.76).

Proof. [✓ Proof](#) The minimum of the Rayleigh quotient of a finite-dimensional self-adjoint operator is an eigenvalue. Conversely, if $G_N B = zB$ with $B \neq 0$, then self-adjointness makes z real and

$$a_-(N) \leq r_N(B) = \text{Re } z. \quad (11.77)$$

□

Theorem 11.3.2.6 (Least finite-domain singular value). [Lean](#) [✓ Stmt](#) Let $m = \dim_{\mathbb{C}} M_D(\mathbb{C})$. There is an index $i \in \{0, \dots, m-1\}$ such that

$$\sigma_i(F_N)^2 = a_-(N), \quad (11.78)$$

and $a_-(N) \leq \sigma_j(F_N)^2$ for every $j \in \{0, \dots, m-1\}$. The indices range only over the finite virtual domain dimension. No zero-padded sequence indexed by all natural numbers is used.

Proof. [✓ Proof](#) The squared singular values of F_N are the eigenvalues of $F_N^\dagger F_N = G_N$, indexed by an orthonormal eigenbasis of the finite-dimensional domain. Apply Theorem 11.3.2.5. □

Theorem 11.3.2.7 (Optimal quadratic lower-bound constant). [Lean](#) [Lean](#) [✓ Stmt](#) For every $B \in M_D(\mathbb{C})$,

$$a_-(N) \|B\|_\rho^2 \leq \|F_N(B)\|^2. \quad (11.79)$$

More generally, a real number c satisfies

$$c \leq a_-(N) \iff c \|B\|_\rho^2 \leq \|F_N(B)\|^2 \text{ for every } B \in M_D(\mathbb{C}). \quad (11.80)$$

Proof. [✓ Proof](#) For $B \neq 0$, divide the desired inequality by $\|B\|_\rho^2$ and use the infimum in (11.72). The zero matrix is immediate. Taking the infimum over nonzero B proves the reverse implication in (11.80). □

Theorem 11.3.2.8 (FNW lower estimate and positivity). [Lean](#) [Lean](#) [✓ Stmt](#) The lower boundary constant satisfies

$$a_-(N) \geq 1 - a(N). \quad (11.81)$$

Consequently, $a_-(N) > 0$ whenever $a(N) < 1$. This is the lower estimate in [160, Lemma 5.3]. No equality $a_-(N) = 1 - a(N)$ is asserted.

Proof. [✓ Proof](#) The bound (11.70) says that $1 - a(N)$ is a universal quadratic lower bound. Apply (11.80). If $a(N) < 1$, then $1 - a(N) > 0$, and (11.81) gives positivity. □

Theorem 11.3.2.9 (Stationary weighted right-multiplication identity). [Lean](#) [✓ Stmt](#) If $\mathcal{E}_A(\rho) = \rho$, then every $B \in M_D(\mathbb{C})$ satisfies

$$\sum_{\mu=0}^{d-1} \|BA^\mu\|_\rho^2 = \|B\|_\rho^2. \quad (11.82)$$

Proof. ✓ Proof Expand the left-hand side using (11.67), cycle the trace, and sum over μ . The result is $\text{Re tr}(\mathcal{E}_A(\rho)B^\dagger B)$, which equals the right-hand side by stationarity. $\square$

Theorem 11.3.2.10 (Monotonicity of the lower boundary constant). Lean Lean Lean ✓ Stmt If $\mathcal{E}_A(\rho) = \rho$, then

$$a_-(N+1) \geq a_-(N) \quad (11.83)$$

for every $N \in \mathbb{N}$. Hence $N \mapsto a_-(N)$ is nondecreasing. The same conclusion holds for every primitive MPS tensor with positive-definite fixed point ρ .

Proof. ✓ Proof The exact right-multiplication recursion follows from [160, Equation (5.5)] and is

$$\|F_{N+1}(B)\|^2 = \sum_{\mu=0}^{d-1} \|F_N(BA^\mu)\|^2. \quad (11.84)$$

Apply (11.79) to each summand and then use (11.82). This proves (11.83); induction gives monotonicity. $\square$

Theorem 11.3.2.11 (Positivity from block injectivity). Lean Lean ✓ Stmt Suppose that A is block injective at length L_0 . Then

$$a_-(L_0) > 0. \quad (11.85)$$

If $\mathcal{E}_A(\rho) = \rho$, then for every $m \geq L_0$,

$$a_-(m) > 0. \quad (11.86)$$

This is the positivity input used in the proof of [160, Lemma 6.2].

Proof. ✓ Proof Injectivity makes every singular value of F_{L_0} positive. The least squared singular value is $a_-(L_0)$ by Theorem 11.3.2.6. Monotonicity then gives $a_-(m) \geq a_-(L_0) > 0$ for $m \geq L_0$. $\square$

These results establish the lower-boundary assertions of [160, Lemma 5.3].

11.3.3 The linear overlap estimate in FNW Lemma 6.2

Fix consecutive block lengths ℓ, m, r . Write μ^ℓ , μ^m , and μ^r for configurations on these blocks. The physical configuration is $\mu^\ell \mu^m \mu^r$, with the first two blocks grouped before the right block. All virtual matrix families below use the ρ -weighted norm from (11.67).

Definition 11.3.3.1 (Weighted spectator families and three-block coordinates). Lean Lean ✓ Stmt For a finite configuration space S , let

$$\mathcal{H}_\rho(S) := \bigoplus_{\mu \in S}^{\ell^2} (M_D(\mathbb{C}), \|\cdot\|_\rho). \quad (11.87)$$

The canonical three-block bijection sends $((\mu^\ell, \mu^m), \mu^r)$ to the concatenated configuration $\mu^\ell \mu^m \mu^r$ and has the evident inverse on such concatenations.

Theorem 11.3.3.6 (Spectator-family norm identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Suppose that A is left canonical and $\mathcal{E}_A(\rho) = \rho$. For every $N \geq 0$ and $X \in M_D(\mathbb{C})$,

$$E_{\text{FNW},A}^N(I_D) = I_D, \quad (11.100)$$

$$\text{tr}(\rho E_{\text{FNW},A}^N(X)) = \text{tr}(\mathcal{E}_A^N(\rho)X). \quad (11.101)$$

Consequently,

$$\sum_{\mu^\ell, \mu^r} \|X_{\mu^\ell, \mu^r}\|_\rho^2 = \|\Phi\|^2, \quad (11.102)$$

$$\sum_{\mu^\ell, \mu^r} \|Y_{\mu^\ell, \mu^r}\|_\rho^2 = \|\Psi\|^2. \quad (11.103)$$

Proof. [✓ Proof](#) Unitality of $E_{\text{FNW},A}^N$ gives the first word sum. The trace-pairing identity and stationarity of ρ give the second. Summing over the remaining spectator index yields the two displayed identities. $\square$

Theorem 11.3.3.7 (Full-ground spectator identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Suppose that A is left canonical and $\mathcal{E}_A(\rho) = \rho$. Then

$$L_{\ell, m, r} \Phi_B = F_{\ell+m+r}(B) = R_{\ell, m, r} \Psi_B, \quad (11.104)$$

$$A_{\varphi, B} = B = A_{\psi, B}, \quad (11.105)$$

$$\|\Phi_B\| = \|B\|_\rho = \|\Psi_B\|. \quad (11.106)$$

Proof. [✓ Proof](#) Expand the finite words using the boundary convention from [160, Equations (5.5) and (5.8)]. Trace cyclicity gives the right overlap identity. Stationarity of ρ gives the left aggregate identity, while unitality of the FNW word map gives the right one and the two norm identities. $\square$

Theorem 11.3.3.8 (Termwise equation (5.9) estimate). [Lean](#) [Lean](#) [✓ Stmt](#) If $\text{tr}(\rho) = 1$, then

$$|\langle L_{\ell, m, r} \Phi, R_{\ell, m, r} \Psi \rangle - \langle A_\varphi, A_\psi \rangle_\rho| \leq a(m) \left(\sum_{\mu^\ell, \mu^r} \|X_{\mu^\ell, \mu^r}\|_\rho^2 \right)^{1/2} \left(\sum_{\mu^\ell, \mu^r} \|Y_{\mu^\ell, \mu^r}\|_\rho^2 \right)^{1/2}. \quad (11.107)$$

If, in addition, A is left canonical and $\mathcal{E}_A(\rho) = \rho$, then

$$|\langle L_{\ell, m, r} \Phi, R_{\ell, m, r} \Psi \rangle - \langle A_\varphi, A_\psi \rangle_\rho| \leq a(m) \|\Phi\| \|\Psi\|. \quad (11.108)$$

Proof. [✓ Proof](#) Apply (11.68) to every middle-boundary pair, sum the estimates, and use finite Cauchy–Schwarz. The identities (11.102) and (11.103) give the stated family norms. $\square$

Theorem 11.3.3.9 (Overlap norms and the lower boundary constant). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The overlap norms satisfy

$$\|L_{\ell, m, r} \Phi\|^2 = \sum_{\mu^r} \|F_{\ell+m}(\Phi(\mu^r))\|^2, \quad (11.109)$$

$$\|R_{\ell, m, r} \Psi\|^2 = \sum_{\mu^\ell} \|F_{m+r}(\Psi(\mu^\ell))\|^2. \quad (11.110)$$

The optimal lower-boundary inequality also gives

$$a_-(\ell + m)\|\Phi\|^2 \leq \|L_{\ell,m,r}\Phi\|^2, \quad (11.111)$$

$$a_-(m + r)\|\Psi\|^2 \leq \|R_{\ell,m,r}\Psi\|^2. \quad (11.112)$$

If $\mathcal{E}_A(\rho) = \rho$ and $a_-(m) > 0$, monotonicity from m to $\ell + m$ and from m to $m + r$ gives

$$\sqrt{a_-(m)}\|\Phi\| \leq \|L_{\ell,m,r}\Phi\|, \quad (11.113)$$

$$\sqrt{a_-(m)}\|\Psi\| \leq \|R_{\ell,m,r}\Psi\|, \quad (11.114)$$

$$a_-(m)\|\Phi\|\|\Psi\| \leq \|L_{\ell,m,r}\Phi\|\|R_{\ell,m,r}\Psi\|. \quad (11.115)$$

Proof. ✓ Proof Reindex each physical norm by the spectator configuration. Apply (11.79) to every summand. Monotonicity gives $a_-(m) \leq a_-(\ell + m)$ and $a_-(m) \leq a_-(m + r)$. Multiply the resulting square-root estimates. $\square$

Theorem 11.3.3.10 (FNW equation (6.5), linear contribution). Lean ✓ Stmt Suppose that $\text{tr}(\rho) = 1$, A is left canonical, $\mathcal{E}_A(\rho) = \rho$, and $a_-(m) > 0$. Then

$$|\langle L_{\ell,m,r}\Phi, R_{\ell,m,r}\Psi \rangle - \langle A_\varphi, A_\psi \rangle_\rho| \leq \frac{a(m)}{a_-(m)} \|L_{\ell,m,r}\Phi\| \|R_{\ell,m,r}\Psi\|. \quad (11.116)$$

This is the linear estimate in [160, Equation (6.5)]. Positivity of the actual lower boundary constant is assumed directly.

Proof. ✓ Proof Divide (11.115) by $a_-(m)$ and substitute the resulting family-norm bound into (11.108). $\square$

11.3.4 The quadratic aggregate contribution in FNW Lemma 6.2

Definition 11.3.4.1 (Full boundary range). Lean ✓ Stmt At length N , let

$$\mathcal{G}_N := \text{ran } F_N. \quad (11.117)$$

This is the full open-chain ground-space range used in the orthogonality step of [160, Lemma 6.2].

Theorem 11.3.4.2 (Unconverted aggregate estimates). Lean Lean Lean Lean ✓ Stmt Suppose that $\text{tr}(\rho) = 1$, A is left canonical, $\mathcal{E}_A(\rho) = \rho$, and both $L_{\ell,m,r}\Phi$ and $R_{\ell,m,r}\Psi$ are orthogonal to $\mathcal{G}_{\ell+m+r}$. Then, for every $B \in M_D(\mathbb{C})$,

$$|\langle A_\varphi, B \rangle_\rho| \leq a(m)\|\Phi\|\|B\|_\rho, \quad (11.118)$$

$$|\langle B, A_\psi \rangle_\rho| \leq a(m)\|B\|_\rho\|\Psi\|. \quad (11.119)$$

Consequently,

$$\|A_\varphi\|_\rho \leq a(m)\|\Phi\|, \quad (11.120)$$

$$\|A_\psi\|_\rho \leq a(m)\|\Psi\|. \quad (11.121)$$

Proof. ✓ Proof In the estimate preceding the lower-boundary conversion in [160, Equation (6.5)], test each overlap vector against the matching full-ground spectator family from Theorem 11.3.3.7. Orthogonality gives the left identity

$$\langle L_{\ell,m,r}\Phi, F_{\ell+m+r}(B) \rangle = 0. \quad (11.122)$$

In the right case, it gives

$$\langle F_{\ell+m+r}(B), R_{\ell,m,r}\Psi \rangle = 0. \quad (11.123)$$

Applying equation (5.9) with (11.122) and (11.123) leaves the corresponding aggregate pairings with the stated bounds. Test once more against each aggregate itself to obtain the two norm estimates. $\square$

Theorem 11.3.4.3 (One-sided aggregate conversion). [Lean](#) [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 11.3.4.2, if $a_-(m) > 0$, then*

$$\|A_\varphi\|_\rho \leq \frac{a(m)}{\sqrt{a_-(m)}} \|L_{\ell,m,r}\Phi\|, \quad (11.124)$$

$$\|A_\psi\|_\rho \leq \frac{a(m)}{\sqrt{a_-(m)}} \|R_{\ell,m,r}\Psi\|. \quad (11.125)$$

Proof. [✓ Proof](#) Apply only the matching one-sided square-root inequality (11.113) or (11.114) to each family bound. $\square$

Theorem 11.3.4.4 (Quadratic aggregate contribution). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 11.3.4.2, if $a_-(m) > 0$, then*

$$|\langle A_\varphi, A_\psi \rangle_\rho| \leq \frac{a(m)^2}{a_-(m)} \|L_{\ell,m,r}\Phi\| \|R_{\ell,m,r}\Psi\|. \quad (11.126)$$

This is the quadratic contribution in the proof of [160, Lemma 6.2].

Proof. [✓ Proof](#) Apply Cauchy–Schwarz to the aggregate pairing and multiply the two one-sided bounds. Since $\sqrt{a_-(m)}^2 = a_-(m)$, the coefficient is exactly $a(m)^2/a_-(m)$. $\square$

Theorem 11.3.4.5 (Linear and quadratic overlap bound). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorems 11.3.3.10 and 11.3.4.4,*

$$|\langle L_{\ell,m,r}\Phi, R_{\ell,m,r}\Psi \rangle| \leq \left(\frac{a(m)}{a_-(m)} + \frac{a(m)^2}{a_-(m)} \right) \|L_{\ell,m,r}\Phi\| \|R_{\ell,m,r}\Psi\|. \quad (11.127)$$

The two source contributions are deliberately left unfactored.

Proof. [✓ Proof](#) Insert the aggregate pairing and apply the triangle inequality. The difference term is bounded by Equation (6.5), and the aggregate term is bounded by Theorem 11.3.4.4. $\square$

Write $P_{r;m,\ell}^{\text{tail}}$ and $P_{r+m;\ell}^{\text{left}}$ for the orthogonal projectors onto the reassociated tail-boundary range and the left-boundary range, respectively, and write $G_{r+m+\ell}$ for the orthogonal projector onto the full ground space.

Theorem 11.3.4.6 (FNW projector defect with separate contributions). [Lean](#) [✓ Stmt](#) *Let A have nonzero bond dimension, let ρ be positive definite with $\text{tr}(\rho) = 1$, and suppose that A is left canonical and $\mathcal{E}_A(\rho) = \rho$. Suppose that $L_0 \geq 1$ is an injective length. For every $r \geq 0$, every $\ell \geq 1$ and every $m \geq L_0$,*

$$\|P_{r;m,\ell}^{\text{tail}} P_{r+m;\ell}^{\text{left}} - G_{r+m+\ell}\| \leq \frac{a(m)}{a_-(m)} + \frac{a(m)^2}{a_-(m)}. \quad (11.128)$$

Proof. ✓ Proof In the source coordinates, let U and V be the two overlap ranges and let W be the full boundary range. Then $W \subseteq U \cap V$. For $x \in U \cap W^\perp$ and $y \in V \cap W^\perp$, Theorem 11.3.4.5 gives

$$|\langle x, y \rangle| \leq \left(\frac{a(m)}{a_-(m)} + \frac{a(m)^2}{a_-(m)} \right) \|x\| \|y\|. \quad (11.129)$$

Projecting a vector first onto $V \cap W^\perp$ and then onto $U \cap W^\perp$ converts this relative-overlap bound into the same operator-norm bound for $P_U P_V - P_W$. Global reversal is unitary and sends U , V , and W to the three ranges in the statement, so it preserves the norm. $\square$

Theorem 11.3.4.7 (FNW projector-defect coefficient). Lean ✓ Stmt *Under the hypotheses of Theorem 11.3.4.6,*

$$\|P_{r;m,\ell}^{\text{tail}} P_{r+m;\ell}^{\text{left}} - G_{r+m+\ell}\| \leq \frac{a(m)(1+a(m))}{a_-(m)}. \quad (11.130)$$

This is the coefficient in [160, Lemma 6.2].

Proof. ✓ Proof First apply Theorem 11.3.4.6, which retains the two source contributions. Then use

$$\frac{a(m)}{a_-(m)} + \frac{a(m)^2}{a_-(m)} = \frac{a(m)(1+a(m))}{a_-(m)}. \quad (11.131)$$

$\square$

Theorem 11.3.4.8 (Primitive FNW projector-defect estimate). Lean ✓ Stmt *Let A be primitive with positive-definite fixed point ρ and $\text{tr}(\rho) = 1$. Then there is an injective length $L_0 \geq 1$ such that the bound (11.128) holds whenever $m \geq L_0$ and $\ell \geq 1$, for every $r \geq 0$.*

Proof. ✓ Proof Choose a positive injective length from primitivity. Left canonicity and the fixed-point equation are part of the primitive hypotheses, so Theorem 11.3.4.6 applies. $\square$

11.4 The geometric form of the FNW projector estimate

Nachtergaele imports the projector estimate of Theorem 11.3.4.8 in the rational geometric form

$$\|G_{\Lambda_n} G_{\Lambda_{n+1} \setminus \Lambda_{n-l}} - G_{\Lambda_{n+1}}\| \leq c \lambda^l \frac{1 + c \lambda^l}{1 - c \lambda^l}, \quad (11.132)$$

stated in [191, lines 1180–1194] and again for the best overlap constant A_m^α in [191, Equation (6.1)]. This bound follows from the FNW coefficient $a(m)(1+a(m))/a_-(m)$, the lower-boundary inequality $1 - a(m) \leq a_-(m)$, and the prescribed-rate estimate $a(m) \leq c \lambda^m$.

The source's rate condition is now established by Theorem 12.8.45: every positive rate exceeding the moduli of the nonunit observable-transfer eigenvalues also exceeds the weighted spectral radius of the transfer remainder. In particular, the geometric estimate is available for every source-admissible rate $0 < \lambda < 1$, with a positive rate-dependent prefactor.

The source's simultaneous assertions that $c = k^2$, the square of the auxiliary dimension, and that every admissible rate may be prescribed require a correction at the fixed intersection threshold. Theorem 12.8.1.16 gives a physical counterexample. The valid replacement retains the rate-dependent prefactor, or permits any prescribed positive prefactor after an additional onset depending on the state, rate, and prefactor, as proved in Theorem 12.8.47. See [83].

Theorem 11.4.1 (Geometric FNW projector defect at a fixed length). [Lean](#) [Stmt](#) Under the hypotheses of Theorem 11.3.4.6, suppose in addition that $a(m) \leq c\lambda^m < 1$. Then

$$\|P_{r;m,\ell}^{\text{tail}} P_{r+m;\ell}^{\text{left}} - G_{r+m+\ell}\| \leq \frac{c\lambda^m(1+c\lambda^m)}{1-c\lambda^m}. \quad (11.133)$$

Proof. [Proof](#) The FNW coefficient is $a(m)(1+a(m))/a_-(m)$. Equation (5.9) gives $1-a(m) \leq a_-(m)$, and $a(m) \leq c\lambda^m < 1$ makes the left-hand side strictly positive, so the denominator may be replaced by $1-c\lambda^m$. The numerator is nondecreasing in $a(m)$ on the nonnegative reals, so it is at most $c\lambda^m(1+c\lambda^m)$. $\square$

Theorem 11.4.2 (Geometric FNW projector defect for a primitive tensor). [Lean](#) [Stmt](#) Let A be primitive with positive-definite fixed point ρ and nonzero bond dimension. There are constants $c > 0$ and $0 < \lambda < 1$ and an injective length $L_0 \geq 1$ such that, whenever $m \geq L_0$ and $c\lambda^m < 1$, the bound (11.133) holds for every prefix length r and every $\ell \geq 1$.

Proof. [Proof](#) Rescale the fixed point to unit trace; this changes neither the tensor nor the primitivity certificate. The weighted remainder has spectral radius below one, so there is a rate λ strictly between that radius and one; the prescribed-rate estimate supplies $c > 0$ with $a(m) \leq c\lambda^m$ for every $m \geq 1$. Normality supplies a positive injective length. Apply the fixed-length geometric estimate. $\square$

Theorem 11.4.3 (Open-chain form of the geometric FNW projector defect). [Lean](#) [Lean](#) [Stmt](#) A one-site suffix increment needs no reassociation, so the open-chain projector defect at overlap length l and prefix length K equals the whole-increment defect with $r = K$, $m = l$, and $\ell = 1$. Consequently, for a primitive tensor with positive-definite fixed point there are $c > 0$, $0 < \lambda < 1$, and an injective length $L_0 \geq 1$ such that, whenever $l \geq L_0$ and $c\lambda^l < 1$, for every prefix length K ,

$$\|P_{K,l+1}^{\text{tail}} P_{K+l}^{\text{left}} - P_{\mathcal{G}_{K+l+1}^{\text{ES}}(A)}\| \leq \frac{c\lambda^l(1+c\lambda^l)}{1-c\lambda^l}. \quad (11.134)$$

This establishes the projector defect bound (11.132) for primitive matrix product states.

Proof. [Proof](#) The two projectors of the whole-increment display at $\ell = 1$ are the two projectors of the open-chain display. Indeed

$$\text{ran } \tilde{\Gamma}_{K;l,1}^{\text{tail}} = \text{ran } \Gamma_{K,l+1}^{\text{tail}} = \mathcal{V}_{K,l+1}(A), \quad \text{ran } \Gamma_{K+l,1}^{\text{left}} = \mathcal{U}_{K+l}(A), \quad (11.135)$$

where $\text{ran } \tilde{\Gamma}_{K;l,1}^{\text{tail}} = \text{ran } \Gamma_{K,l+1}^{\text{tail}}$ follows by (12.210) and the remaining subspace identifications hold by Theorem 12.3.5.17. Taking orthogonal projectors onto the three subspaces gives

$$P_{K;l,1}^{\text{tail}} = P_{K,l+1}^{\text{tail}}(A), \quad P_{K+l,1}^{\text{left}} = P_{K+l}^{\text{left}}(A), \quad (11.136)$$

so the two defect operators are equal. Substituting $\ell = 1$ in the geometric estimate gives the display. $\square$

Theorem 11.4.4 (Geometric power bound below unit spectral radius). [Lean](#) [Stmt](#) Let V be a complex Banach space and let $T: V \rightarrow V$ be a bounded linear operator. If $\rho_{\text{spec}}(T) < 1$, then there are real constants C, r with $C > 0$ and $0 < r < 1$ such that

$$\|T^n\| \leq Cr^n \quad (11.137)$$

for every $n \in \mathbb{N}$.

Proof. [✓ Proof](#) Choose r strictly between the spectral radius and one, then apply Theorem 11.3.32 with $\lambda = r$. $\square$

Theorem 11.4.5 (Pointwise geometric power bound below unit spectral radius). [Lean](#) [✓ Stmt](#) Let V be a complex Banach space and let $T: V \rightarrow V$ be a bounded linear operator. If $\rho_{\text{spec}}(T) < 1$, then there are real constants C, r with $C > 0$ and $0 < r < 1$ such that

$$\|T^n x\| \leq Cr^n \|x\| \quad (11.138)$$

for every $n \in \mathbb{N}$ and $x \in V$.

Proof. [✓ Proof](#) Apply Theorem 11.4.4 and the defining operator-norm inequality to $T^n x$. $\square$

Theorem 11.4.6 (Transfer–Gram Choi reshuffling). [Lean](#) [✓ Stmt](#) Let $\mathbf{e}_{(j,i)} := \text{single}_{(j,i)}(1)$ denote the standard unit vector in $\ell^2(\{0, \dots, D-1\} \times \{0, \dots, D-1\})$, and let $\mathcal{E}_A$ be the transfer map. Then

$$\langle \mathbf{e}_{(b,a)}, \mathcal{K}_L(\mathbf{e}_{(e,c)}) \rangle_{\ell^2} = J(\mathcal{E}_A^L)_{(c,e),(a,b)}. \quad (11.139)$$

Under Frobenius vectorization, $\mathbf{e}_{(b,a)} = U(E_{ab})$ and $\mathbf{e}_{(e,c)} = U(E_{ce})$, explaining the reversed coordinate order. Thus the Gram operator is the indicated Choi reshuffling; in particular it is not obtained by replacing the right-hand side with $\langle E_{ab}, \mathcal{E}_A^L(E_{ce}) \rangle_{\text{HS}}$.

Proof. [✓ Proof](#) Move the adjoint in $\mathcal{K}_L = (\Gamma_L^{\text{ES}})^\dagger \Gamma_L^{\text{ES}}$ across the inner product and expand the two boundary-map coefficients. For each physical word σ , the resulting summand is $(A^\sigma)_{ec} \overline{(A^\sigma)_{ba}}$. On the other hand, expanding the transfer power on E_{ca} gives

$$(\mathcal{E}_A^L(E_{ca}))_{eb} = \sum_{\sigma} (A^\sigma)_{ec} \overline{(A^\sigma)_{ba}}. \quad (11.140)$$

The defining Choi indexing $J(\mathcal{L})_{(c,e),(a,b)} = (\mathcal{L}(E_{ca}))_{eb}$ now yields the stated reshuffling. $\square$

Theorem 11.4.7 (Ground-space Gram operator as a reshuffled Choi matrix). [Lean](#) [✓ Stmt](#) The ground-space Gram operator is exactly the operator-level reshuffling of the Choi matrix of the iterated transfer map:

$$\mathcal{K}_L = \mathcal{R}(\mathcal{E}_A^L). \quad (11.141)$$

The equality is an equality of operators on $\ell^2(\{0, \dots, D-1\} \times \{0, \dots, D-1\})$. Its entries are those in (11.37), rather than Hilbert–Schmidt pairings with $\mathcal{E}_A^L$.

Proof. [✓ Proof](#) The two operators have equal matrix coefficients on every pair of standard basis vectors. Hence they are equal. $\square$

Theorem 11.4.8 (Geometric convergence of the ground-space Gram operator). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor of bond dimension D , let $\rho \in M_D(\mathbb{C})$ satisfy $\text{tr}(\rho) \neq 0$, and let P_ρ be the fixed-point projection

$$P_\rho(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho. \quad (11.142)$$

Suppose that $\mathcal{E}_A$ is trace-preserving, $\mathcal{E}_A(\rho) = \rho$, and

$$\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1. \quad (11.143)$$

Then there are real constants C, r with $C > 0$ and $0 < r < 1$ such that, for every $n \geq 1$,

$$\|\mathcal{K}_n - \mathcal{R}(P_\rho)\|^2 \leq D^3(Cr^n)^2. \quad (11.144)$$

No positivity, injectivity, or uniqueness hypothesis is imposed.

Proof. ✓ Proof Write $N = \mathcal{E}_A - P_\rho$. The power decomposition gives $\mathcal{E}_A^n = P_\rho + N^n$ for $n \geq 1$, while Theorem 11.4.4 gives $\|N^n\| \leq Cr^n$. Use the exact Gram reshuffling identity and Theorem 11.3.23 to rewrite the left-hand side as $\|\mathcal{R}(N^n)\|^2$, then apply Theorem 11.3.31. $\square$

Theorem 11.4.9 (Primitive-MPS geometric Gram convergence). Lean ✓ Stmt *Let A be a primitive MPS tensor with fixed-point witness ρ . Then there are real constants C, r with $C > 0$ and $0 < r < 1$ such that, for every $n \geq 1$,*

$$\|\mathcal{K}_n - \mathcal{R}(P_\rho)\|^2 \leq D^3(Cr^n)^2. \quad (11.145)$$

Proof. ✓ Proof Positive semidefiniteness and nonvanishing of ρ imply $\text{tr}(\rho) \neq 0$. The trace-preserving identity, fixed-point equation, and complementary spectral-radius bound are the remaining fields of the primitive-MPS hypothesis, so Theorem 11.4.8 applies. $\square$

Theorem 11.4.10 (Ground-space Gram convergence). Lean ✓ Stmt *Let A be an MPS tensor of bond dimension D , let $\rho \in M_D(\mathbb{C})$ satisfy $\text{tr}(\rho) \neq 0$, and let P_ρ be the corresponding fixed-point projection. If $\mathcal{E}_A$ is trace-preserving, $\mathcal{E}_A(\rho) = \rho$, and $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$, then*

$$\mathcal{K}_n \longrightarrow \mathcal{R}(P_\rho) \quad (11.146)$$

in operator norm as $n \rightarrow \infty$.

Proof. ✓ Proof The right-hand side of (11.144) tends to zero because $0 < r < 1$. Taking square roots gives the asserted norm convergence. $\square$

Theorem 11.4.11 (Primitive-MPS ground-space Gram convergence). Lean ✓ Stmt *If A is a primitive MPS tensor with fixed-point witness ρ , then*

$$\mathcal{K}_n \longrightarrow \mathcal{R}(P_\rho) \quad (11.147)$$

in operator norm as $n \rightarrow \infty$.

Proof. ✓ Proof The primitive-MPS data supply the nonzero trace, trace preservation, fixed-point equation, and complementary spectral-radius hypothesis of Theorem 11.4.10. $\square$

The limit in Theorem 11.4.10 is the reshuffled fixed-point projection $\mathcal{R}(P_\rho)$, so the limiting metric is generally nonidentity. Under a positive-definite fixed-point hypothesis, the inverse estimates below perturb around $\mathcal{R}(P_\rho)$ rather than around the identity.

Theorem 11.4.12 (Exact limiting Gram metric). Lean Lean Lean ✓ Stmt *Let $\rho \in M_D(\mathbb{C})$ have nonzero trace and define $P_\rho(X) := (\text{tr } X / \text{tr } \rho)\rho$. Then*

$$R(P_\rho) = (\text{tr } \rho)^{-1}(\rho^\top \otimes I_D), \quad (11.148)$$

$$\mathcal{R}(P_\rho)U(X) = U((\text{tr } \rho)^{-1}X\rho). \quad (11.149)$$

Proof. ✓ Proof For matrix units E_{ca} , direct evaluation gives

$$R(P_\rho)_{(b,a),(e,c)} = (\operatorname{tr} \rho)^{-1} \rho_{eb} \delta_{ca}. \quad (11.150)$$

These are exactly the entries in (11.148). Applying that Kronecker product to the column–row vectorization of X gives (11.149). $\square$

Theorem 11.4.13 (Invertibility of the limiting Gram metric). Lean Lean Lean ✓ Stmt *If $D > 0$ and $\rho \in M_D(\mathbb{C})$ is positive definite, then $R(P_\rho)$ is positive definite and $\mathcal{R}(P_\rho)$ is invertible, hence injective.*

Proof. ✓ Proof Positive definiteness gives $\operatorname{tr} \rho > 0$. By (11.148),

$$R(P_\rho) = (\operatorname{tr} \rho)^{-1} (\rho^\top \otimes I_D) > 0. \quad (11.151)$$

Hence its associated operator $\mathcal{R}(P_\rho)$ is invertible. $\square$

Theorem 11.4.14 (Arbitrary-base Neumann inverse bound). Lean Lean ✓ Stmt *Let K_0 be invertible in a normed ring with convergent geometric series and $\|I\| = 1$. If $a < 1$ and*

$$\|K_0^{-1}(K - K_0)\| \leq a, \quad (11.152)$$

then K is invertible and

$$\|K^{-1}\| \leq (1 - a)^{-1} \|K_0^{-1}\|. \quad (11.153)$$

The same conclusion follows from $\|K_0^{-1}\| \|K - K_0\| \leq a$.

Proof. ✓ Proof Put $E := K_0^{-1}(K - K_0)$. Then $K = K_0(I + E)$ and $\|E\| < 1$, so

$$K^{-1} = (I + E)^{-1} K_0^{-1}, \quad (11.154)$$

$$\|K^{-1}\| \leq \|(I + E)^{-1}\| \|K_0^{-1}\| \leq (1 - a)^{-1} \|K_0^{-1}\|. \quad (11.155)$$

Submultiplicativity reduces the product-form hypothesis to (11.152). $\square$

Theorem 11.4.15 (Arbitrary-base inverse-displacement bound). Lean Lean ✓ Stmt *Let K_0 be invertible in a normed ring with convergent geometric series and $\|I\| = 1$. If $a < 1$ and (11.152) holds, then*

$$\|K^{-1} - K_0^{-1}\| \leq (1 - a)^{-1} a \|K_0^{-1}\|. \quad (11.156)$$

The same estimate follows from $\|K_0^{-1}\| \|K - K_0\| \leq a$.

Proof. ✓ Proof With $E := K_0^{-1}(K - K_0)$, the resolvent identity is

$$K^{-1} - K_0^{-1} = -(I + E)^{-1} E K_0^{-1}. \quad (11.157)$$

Since $\|(I + E)^{-1}\| \leq (1 - a)^{-1}$ and $\|E\| \leq a$, taking norms gives (11.156). Submultiplicativity again gives the product-form variant. $\square$

Theorem 11.4.16 (Continuity of inversion at an invertible limit). Lean ✓ Stmt *Let K_i be a net in a normed ring with convergent geometric series. If $K_i \rightarrow K_0$ and K_0 is invertible, then*

$$K_i^{-1} \rightarrow K_0^{-1}. \quad (11.158)$$

Proof. ✓ Proof Write K_0 as the image of a unit. Continuity of inversion on the unit group, followed by the convergence $K_i \rightarrow K_0$, gives the result. $\square$

Definition 11.4.17 (Inverse Gram operator). Lean ✓ Stmt For an injective continuous linear map T with finite-dimensional domain, let S_T be the continuous endomorphism induced by the inverse linear equivalence of the Gram endomorphism $T^\dagger T$.

Theorem 11.4.18 (Inverse-Gram identities). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $\mathbb{K}$ be a real or complex scalar field, let E be a finite-dimensional complete $\mathbb{K}$ -Hilbert space, and let F be a complete $\mathbb{K}$ -Hilbert space. If $T: E \rightarrow F$ is an injective continuous linear map, then the inverse Gram operator is a two-sided inverse of the Gram endomorphism:

$$S_T(T^\dagger T) = I_E, \quad (11.159)$$

$$(T^\dagger T)S_T = I_E. \quad (11.160)$$

Consequently, S_T is both the canonical inverse and the ring inverse of $T^\dagger T$. The two pseudoinverse factors satisfy

$$\|TS_T\| = \sqrt{\|S_T\|}, \quad (11.161)$$

$$\|S_T T^\dagger\| = \sqrt{\|S_T\|}. \quad (11.162)$$

Proof. ✓ Proof By definition, S_T is induced by the inverse linear equivalence of the injective endomorphism $T^\dagger T$. The inverse-equivalence identities give (11.159) and (11.160). Together these equations characterize the canonical inverse of a continuous linear endomorphism, which agrees with its inverse in the endomorphism ring. For the norm identities, use the C^* -identity and the two inverse equations to compute the squares of $\|TS_T\|$ and $\|S_T T^\dagger\|$. $\square$

For a positive-definite fixed point ρ , Theorem 11.4.19 provides quantitative inverse bounds for the limiting metric $\mathcal{R}(P_\rho)$.

Theorem 11.4.19 (Pointwise geometric Gram inverse bounds). Lean ✓ Stmt Let A be a primitive MPS tensor of nonzero bond dimension whose fixed-point witness ρ is positive definite. Put

$$\mathcal{K}_\infty = \mathcal{R}(P_\rho), \quad \mathcal{J}_\infty = \mathcal{K}_\infty^{-1}. \quad (11.163)$$

There are constants $g, c, r > 0$ with $r < 1$ and $c = \|\mathcal{J}_\infty\|g$ such that, for every $n \geq 1$,

$$\|\mathcal{K}_n - \mathcal{K}_\infty\| \leq gr^n. \quad (11.164)$$

Whenever $cr^n < 1$, the operator $\mathcal{K}_n$ is invertible and

$$\|\mathcal{K}_n^{-1}\| \leq \frac{\|\mathcal{J}_\infty\|}{1 - cr^n}, \quad (11.165)$$

$$\|\mathcal{K}_n^{-1} - \mathcal{J}_\infty\| \leq \frac{cr^n}{1 - cr^n} \|\mathcal{J}_\infty\|. \quad (11.166)$$

Proof. ✓ Proof Start from the squared estimate $\|\mathcal{K}_n - \mathcal{K}_\infty\|^2 \leq D^3(Cr^n)^2$. Taking square roots gives $g = \sqrt{D^3} C$ without any additional dimension loss, and hence (11.164). Positive definiteness makes $\mathcal{K}_\infty$ invertible. The relative error is at most $\|\mathcal{J}_\infty\|gr^n = cr^n$, so the two arbitrary-base Neumann estimates give (11.165) and (11.166). $\square$

Theorem 11.4.20 (Pointwise geometric inverse-Gram bounds). Lean ✓ Stmt Under the hypotheses of Theorem 11.4.19, there are constants $g, c, r > 0$ with $r < 1$ and $c = \|\mathcal{J}_\infty\|g$ such that (11.164) holds for every $n \geq 1$. If $cr^n < 1$, then the physical boundary map Γ_n^{ES} is injective, its inverse Gram operator equals $\mathcal{K}_n^{-1}$, and

$$\|S_{\Gamma_n^{\text{ES}}}\| \leq \frac{\|\mathcal{J}_\infty\|}{1 - cr^n}, \quad (11.167)$$

$$\|S_{\Gamma_n^{\text{ES}}} - \mathcal{J}_\infty\| \leq \frac{cr^n}{1 - cr^n} \|\mathcal{J}_\infty\|. \quad (11.168)$$

Proof. ✓ Proof Invertibility of $(\Gamma_n^{\text{ES}})^\dagger \Gamma_n^{\text{ES}} = \mathcal{K}_n$ implies injectivity of Γ_n^{ES} . The inverse-Gram identities of Theorem 11.4.18 identify $S_{\Gamma_n^{\text{ES}}}$ with $\mathcal{K}_n^{-1}$, so the pointwise bounds of Theorem 11.4.19 apply unchanged. $\square$

Theorem 11.4.21 (Geometric smallness at the C3 correction lengths). Lean ✓ Stmt Let $c > 0$, let $0 < r < 1$, and let $l, K \in \mathbb{N}$. If $cr^l < 1$, then

$$cr^{l+1} < 1, \quad (11.169)$$

$$cr^{K+l} < 1, \quad (11.170)$$

$$cr^{K+l+1} < 1. \quad (11.171)$$

Proof. ✓ Proof Since $0 < r < 1$, the sequence r^m is nonincreasing. Each of $l+1$, $K+l$, and $K+l+1$ is at least l , so every displayed inequality follows from $cr^l < 1$. $\square$

Theorem 11.4.22 (Eventual inverse-Gram control in the limiting metric). Lean ✓ Stmt Let A be a primitive MPS tensor whose fixed-point witness ρ is positive definite, and put $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$. For every $0 < a < 1$, all sufficiently large n satisfy

$$\begin{aligned} \mathcal{K}_n &\in \text{GL}, \\ \|\mathcal{K}_n^{-1} - \mathcal{K}_\infty^{-1}\| &\leq \frac{a}{1-a} \|\mathcal{K}_\infty^{-1}\|. \end{aligned} \quad (11.172)$$

Proof. ✓ Proof The positive-definite fixed point makes $\mathcal{K}_\infty$ invertible. Gram convergence implies that eventually

$$\|\mathcal{K}_\infty^{-1}\| \|\mathcal{K}_n - \mathcal{K}_\infty\| \leq a. \quad (11.173)$$

Factoring $\mathcal{K}_n$ around the invertible base $\mathcal{K}_\infty$ and applying the arbitrary-base Neumann estimate gives (11.172). $\square$

Theorem 11.4.23 (Convergence of finite-volume inverse Gram operators). Lean ✓ Stmt Under the same primitive-MPS and positive-definite fixed-point hypotheses,

$$\mathcal{K}_n^{-1} \longrightarrow \mathcal{K}_\infty^{-1} \quad (11.174)$$

in operator norm.

Proof. ✓ Proof Inversion is continuous at the invertible operator $\mathcal{K}_\infty$, so the claim follows from $\mathcal{K}_n \rightarrow \mathcal{K}_\infty$. $\square$

Theorem 11.4.24 (Eventual injectivity and inverse-Gram control). Lean ✓ Stmt Under the same hypotheses, for every $0 < a < 1$ and all sufficiently large n , the physical boundary map Γ_n^{ES} is injective and its inverse Gram operator is $\mathcal{K}_n^{-1}$, satisfying

$$\|S_{\Gamma_n^{\text{ES}}} - \mathcal{K}_\infty^{-1}\| \leq \frac{a}{1-a} \|\mathcal{K}_\infty^{-1}\|. \quad (11.175)$$

Proof. ✓ Proof Eventual invertibility of $(\Gamma_n^{\text{ES}})^\dagger \Gamma_n^{\text{ES}}$ implies eventual injectivity of Γ_n^{ES} . The inverse-Gram identities of Theorem 11.4.18 then identify $S_{\Gamma_n^{\text{ES}}}$ with $\mathcal{K}_n^{-1}$, and the estimate of Theorem 11.4.22 applies. $\square$

Definition 11.4.25 (Injective range-projector candidate). Lean ✓ Stmt For injective T , define the operator $Q_T := TS_T T^\dagger$ on its codomain.

Theorem 11.4.26 (Injective range-projector formula). Lean ✓ Stmt If T is an injective map from a finite-dimensional Hilbert space into a complete Hilbert space, then

$$P_{\text{range } T} = T(T^\dagger T)^{-1} T^\dagger. \quad (11.176)$$

Proof. ✓ Proof The vector $Q_T x$ lies in $\text{range } T$. Since $T^\dagger Q_T x = (T^\dagger T) S_T T^\dagger x = T^\dagger x$, the adjoint identity gives $\langle x - Q_T x, T y \rangle = 0$ for every y . Thus $x - Q_T x$ is orthogonal to $\text{range } T$, and the range-orthogonal decomposition characterizing the orthogonal projector gives $Q_T = P_{\text{range } T}$. $\square$

Theorem 11.4.27 (Physical ground-space range projector). Lean ✓ Stmt If A is L -block injective, then the injective range projector of Γ_L^{ES} is the physical orthogonal projector onto the local ground space:

$$Q_{\Gamma_L^{\text{ES}}} = P_{G_L^{\text{ES}}(A)}. \quad (11.177)$$

In particular, this is the physical range projector on the local Hilbert space, not the virtual transfer fixed-point projector.

Proof. ✓ Proof The general injective range-projector theorem identifies $Q_{\Gamma_L^{\text{ES}}}$ with the orthogonal projector onto $\text{range}(\Gamma_L^{\text{ES}})$. The exact-range theorem identifies this range with $G_L^{\text{ES}}(A)$. $\square$

Theorem 11.4.28 (Physical ground-space inverse-Gram projector formula). Lean ✓ Stmt Suppose that $L_0 > 0$, that A is L_0 -block injective, and that $L \geq L_0$. Then the physical orthogonal projector onto the local ground space is

$$P_{G_L^{\text{ES}}(A)} = \Gamma_L^{\text{ES}} ((\Gamma_L^{\text{ES}})^\dagger \Gamma_L^{\text{ES}})^{-1} (\Gamma_L^{\text{ES}})^\dagger. \quad (11.178)$$

Thus the displayed operator is the physical range projector on $\ell^2(\{0, \dots, d-1\}^L)$.

Proof. ✓ Proof Positive-length propagation makes A block injective at length L . Apply the physical range-projector identity at that length and unfold its inverse-Gram expression. $\square$

Theorem 11.4.29 (Neumann inverse estimate). Lean ✓ Stmt If $\|t\| \leq a < 1$ in a normed ring with convergent geometric series and $\|1\| = 1$, then

$$\|(1-t)^{-1}\| \leq (1-a)^{-1}. \quad (11.179)$$

Proof. ✓ Proof Since $\|t\| \leq a < 1$, the Neumann series converges and $(1-t)^{-1} = \sum_{n \geq 0} t^n$. The triangle inequality and submultiplicativity give

$$\left\| \sum_{n \geq 0} t^n \right\| \leq \sum_{n \geq 0} \|t\|^n = (1 - \|t\|)^{-1} \leq (1-a)^{-1}, \quad (11.180)$$

where the last step uses monotonicity of the reciprocal on the positive reals. $\square$

Theorem 11.4.30 (Quantitative inverse-Gram estimate). [Lean](#) [✓ Stmt](#) Let $\mathbb{K}$ be a real or complex scalar field, let E be a finite-dimensional complete $\mathbb{K}$ -Hilbert space, and let F be a complete $\mathbb{K}$ -Hilbert space. Let $T: E \rightarrow F$ be an injective continuous linear map. If $a \in \mathbb{R}$ satisfies $a < 1$ and

$$\|I_E - T^\dagger T\| \leq a, \quad (11.181)$$

then

$$\|S_T\| \leq (1 - a)^{-1}. \quad (11.182)$$

Proof. [✓ Proof](#) If E is trivial, then $S_T = 0$ and the bound follows from $a < 1$. Otherwise, identify S_T with the ring inverse of $T^\dagger T$ and apply Theorem 11.4.29 to $t = I_E - T^\dagger T$. $\square$

Definition 11.4.31 (Canonical parent interaction). [Lean](#) [✓ Stmt](#) The parent interaction at range L is the canonical orthogonal projector specified in (11.14).

Lemma 11.4.32 (Parent interaction is idempotent). [Lean](#) [✓ Stmt](#) The local parent interaction is a projector: $h_L(A)^2 = h_L(A)$.

Proof. [✓ Proof](#) This is the idempotency of the orthogonal projector onto $G_L(A)^\perp$, transported from the Hilbert-space realization of the local coefficient space. $\square$

Lemma 11.4.33 (Parent interaction annihilates ground-space elements). [Lean](#) [✓ Stmt](#) For any $v \in G_L(A)$, one has $h_L(A)v = 0$.

Proof. [✓ Proof](#) Since $h_L(A)$ is the orthogonal projector onto $G_L(A)^\perp$, it annihilates every element of $G_L(A)$. $\square$

Definition 11.4.34 (Restriction to a cyclic interval). [Lean](#) [✓ Stmt](#) For $\sigma \in \{0, \dots, d-1\}^N$ and a starting site i , write $\sigma|_{[i, i+L-1]} \in \{0, \dots, d-1\}^L$ for the word defined by $(\sigma|_{[i, i+L-1]})(r) = \sigma(i + r \bmod N)$ for $0 \leq r < L$.

Definition 11.4.35 (Prescribing a cyclic interval). [Lean](#) [✓ Stmt](#) If $L \leq N$ and $\tau \in \{0, \dots, d-1\}^L$, write $\sigma^{[i, i+L-1] \leftarrow \tau}$ for the configuration defined by

$$\sigma^{[i, i+L-1] \leftarrow \tau}(k) = \begin{cases} \tau(r), & k \equiv i + r \pmod{N} \text{ for some } 0 \leq r < L, \\ \sigma(k), & \text{otherwise.} \end{cases}$$

The index r in the first case is unique because $L \leq N$.

Lemma 11.4.36 (Extracting a replaced window). [Lean](#) [✓ Stmt](#) If $L \leq N$, then restricting to the same cyclic interval after prescribing its values recovers the inserted word: $(\sigma^{[i, i+L-1] \leftarrow \tau})|_{[i, i+L-1]} = \tau$.

Lemma 11.4.37 (Replacing an extracted window). [Lean](#) [✓ Stmt](#) If $L \leq N$, then prescribing on a cyclic interval the values already carried by σ leaves σ unchanged: $\sigma^{[i, i+L-1] \leftarrow \sigma|_{[i, i+L-1]}} = \sigma$.

Proof. [✓ Proof](#) Case-split on whether the offset $(k - i + N) \bmod N$ lies in $[0, L)$: inside the cyclic interval the prescribed value is the original value of σ ; outside the interval both sides already agree with σ . $\square$

Lemma 11.4.38 (Replacing the same window twice). [Lean](#) [✓ Stmt](#) If $L \leq N$, then two prescriptions on the same cyclic interval reduce to the second prescription:

$$(\sigma^{[i, i+L-1] \leftarrow \tau})^{[i, i+L-1] \leftarrow v} = \sigma^{[i, i+L-1] \leftarrow v}.$$

Proof. ✓ Proof On the cyclic interval both sides take the value prescribed by v . Away from that interval neither second prescription changes the value. $\square$

For a periodic chain of length N , the parent Hamiltonian is the sum of translated copies of the local interaction.

Definition 11.4.39 (Translated local term). Lean ✓ Stmt For each site index $i \in \{0, \dots, N-1\}$, the translated local term is determined by

$$(h_i \psi)(\sigma) = \begin{cases} [h_L(A) (\tau \mapsto \psi(\sigma^{[i, i+L-1] \leftarrow \tau}))] (\sigma|_{[i, i+L-1]}), & L \leq N, \\ 0, & L > N. \end{cases} \quad (11.183)$$

Lemma 11.4.40 (Pointwise form of a translated local term). Lean ✓ Stmt If $L \leq N$, then the translated term is obtained by applying the local parent interaction on the chosen cyclic window:

$$(h_i(A, L) \psi)(\sigma) = [h_L(A) (\tau \mapsto \psi(\sigma^{[i, i+L-1] \leftarrow \tau}))] (\sigma|_{[i, i+L-1]}).$$

Proof. ✓ Proof This is the first case in (11.183). $\square$

Lemma 11.4.41 (Translated local terms are idempotent). Lean ✓ Stmt For every site i , one has $h_i(A, L)^2 = h_i(A, L)$.

Proof. ✓ Proof If $L > N$, the translated term is zero. Suppose $L \leq N$ and put $f_{\sigma, i}(\tau) := \psi(\sigma^{[i, i+L-1] \leftarrow \tau})$. The pointwise formula for translated terms, together with Lemmas 11.4.36 and 11.4.38, gives

$$\begin{aligned} (h_i(A, L)^2 \psi)(\sigma) &= (h_L(A)^2 f_{\sigma, i}) (\sigma|_{[i, i+L-1]}) \\ &= (h_L(A) f_{\sigma, i}) (\sigma|_{[i, i+L-1]}) \\ &= (h_i(A, L) \psi)(\sigma), \end{aligned}$$

using $h_L(A)^2 = h_L(A)$ by Lemma 11.4.32. $\square$

Definition 11.4.42 (Parent Hamiltonian). Lean ✓ Stmt The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i. \quad (11.184)$$

Definition 11.4.43 (Frustration-free condition). Lean ✓ Stmt A vector ψ is a frustration-free ground state of the parent Hamiltonian when each local interaction annihilates it, that is, when $h_i \psi = 0$ for every $i \in \{0, \dots, N-1\}$.

Lemma 11.4.44 (Periodic MPS vector window membership). Lean ✓ Stmt For any window of L consecutive sites starting at position i in an N -site periodic chain with $L \leq N$, the restriction of the periodic MPS vector $V^{(N)}(A)$ to that window lies in $G_L(A)$.

Proof. ✓ Proof The boundary matrix is the product of the A -matrices on the complement sites. Trace cyclicity rotates the full N -site product so that the window indices come first, matching the definition of the ground-space map. $\square$

Lemma 11.4.45 (Local term annihilates the periodic MPS vector). Lean ✓ Stmt For each site i , one has $h_i V^{(N)}(A) = 0$.

Proof. ✓ Proof This combines periodic-MPS-vector window membership with the fact that the parent interaction annihilates ground-space elements. $\square$

Lemma 11.4.46 (Parent Hamiltonian annihilates the periodic MPS vector). Lean ✓ Stmt For every N with $L \leq N$, the periodic MPS vector satisfies $H_N(A, L)V^{(N)}(A) = 0$.

Proof. ✓ Proof This is a sum of zeros, since each local term annihilates the periodic MPS vector. $\square$

Lemma 11.4.47 (Frustration-freeness of the periodic MPS vector). Lean ✓ Stmt The periodic MPS vector is a frustration-free ground state: for every site i , one has $h_i V^{(N)}(A) = 0$.

Proof. ✓ Proof The claim follows from Lemma 11.4.45. $\square$

11.5 Intersection property and unique ground state

11.5.1 Restriction maps

Given a state ψ on $L + 1$ sites, we define two restriction maps.

Remark 11.5.1.1 (Word and restriction identities). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction maps are the associated linear maps on the state spaces under the computational-basis identification, together with their pointwise formulas.

Definition 11.5.1.2 (Left restriction). Lean ✓ Stmt Fixing the last physical index j gives the state $\sigma \mapsto \psi(\sigma_1, \dots, \sigma_L, j)$.

Definition 11.5.1.3 (Right restriction). Lean ✓ Stmt Fixing the first physical index i gives the state $\sigma \mapsto \psi(i, \sigma_1, \dots, \sigma_L)$.

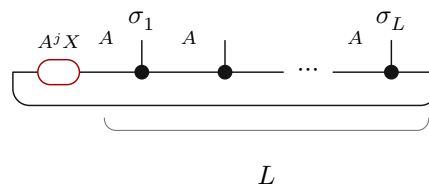
Definition 11.5.1.4 (Left ground membership). Lean ✓ Stmt A state ψ on $L + 1$ sites satisfies the left ground condition if, for every j , the state obtained by fixing the last index to j lies in $G_L(A)$.

Definition 11.5.1.5 (Right ground membership). Lean ✓ Stmt A state ψ on $L + 1$ sites satisfies the right ground condition if, for every i , the state obtained by fixing the first index to i lies in $G_L(A)$.

11.5.2 Forward direction: ground space restricts

Lemma 11.5.2.1 (Left restriction). Lean ✓ Stmt If $\psi \in G_{L+1}(A)$, i.e. if $\psi(\sigma) = \text{tr}(A^\sigma X)$ for some X , then, for each j , the state obtained by fixing the last index to j lies in $G_L(A)$ with witness $A^j X$.

Proof. ✓ Proof The restriction simply absorbs the last tensor into the boundary matrix:



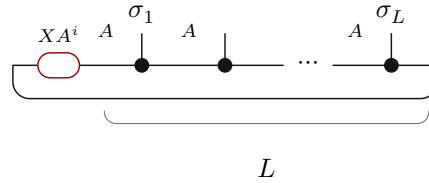
Direct computation gives

$$\psi(\sigma_1, \dots, \sigma_L, j) = \text{tr}(A^{\sigma_1} \dots A^{\sigma_L} A^j X) = \Gamma_L(A^j X)(\sigma).$$

□

Lemma 11.5.2.2 (Right restriction). [Lean](#) [✓ Stmt](#) If $\psi \in G_{L+1}(A)$ with $\psi(\sigma) = \text{tr}(A^\sigma X)$, then, for each i , the state obtained by fixing the first index to i lies in $G_L(A)$ with witness XA^i .

Proof. [✓ Proof](#) On the right restriction, one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity,

$$\psi(i, \sigma_1, \dots, \sigma_L) = \text{tr}(A^i A^\sigma X) = \text{tr}(A^\sigma X A^i) = \Gamma_L(XA^i)(\sigma).$$

□

11.5.3 Injectivity of the ground-space map

Theorem 11.5.3.1 (Ground-space map injectivity). [Lean](#) [✓ Stmt](#) If A is injective and $L \geq 1$, then Γ_L is injective.

Proof. [✓ Proof](#) If $\Gamma_L(X) = 0$, then $\text{tr}(A^\sigma X) = 0$ for every length- L word σ . Since A is injective, the products $\{A^\sigma\}_\sigma$ span $M_D(\mathbb{C})$, so the trace pairing gives $X = 0$. □

Theorem 11.5.3.2 (Ground-space map injectivity from word span). [Lean](#) [✓ Stmt](#) If the length- L products $\{A^\sigma : |\sigma| = L\}$ span $M_D(\mathbb{C})$, then Γ_L is injective.

Proof. [✓ Proof](#) If $\Gamma_L(X) = 0$, then $\text{tr}(A^\sigma X) = 0$ for every word σ of length L . The spanning hypothesis turns this into $\text{tr}(MX) = 0$ for every $M \in M_D(\mathbb{C})$, and the trace pairing gives $X = 0$. □

Theorem 11.5.3.3 (Ground-space map injectivity after blocking). [Lean](#) [✓ Stmt](#) If A is L_0 -block injective, then Γ_{L_0} is injective.

Proof. [✓ Proof](#) Lemma 6.1.1.23 identifies L_0 -block injectivity with the word-span hypothesis of Theorem 11.5.3.2. □

Theorem 11.5.3.4 (Ground-space dimension). [Lean](#) [✓ Stmt](#) If A is injective and $L \geq 1$, then $\dim G_L(A) = D^2$.

Proof. [✓ Proof](#) The upper bound $\dim G_L(A) \leq D^2$ is Lemma 11.2.1.9. Injectivity of Γ_L gives $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$. □

11.5.4 The intersection property

The following two statements hold for every tensor A ; injectivity enters only through the choice of the matrices R_j in the second one.

Theorem 11.5.4.1 (Overlap of one-site restrictions). [Lean](#) [✓ Stmt](#) Let ψ be a vector on $K + 2$ sites and let $Y_i, Z_j \in M_D(\mathbb{C})$ satisfy $\psi(i, \tau) = \Gamma_{K+1}(Y_i)(\tau)$ and $\psi(\tau, j) = \Gamma_{K+1}(Z_j)(\tau)$ for all letters i, j and all words τ of length $K + 1$. Then

$$\Gamma_K(A^j Y_i) = \Gamma_K(Z_j A^i)$$

for all letters i, j .

Proof. [✓ Proof](#) For a word σ of length K , cyclicity of the trace gives

$$\Gamma_K(A^j Y_i)(\sigma) = \text{tr}(A^\sigma A^j Y_i) = \psi(i, \sigma, j) = \text{tr}(A^i A^\sigma Z_j) = \Gamma_K(Z_j A^i)(\sigma).$$

□

Theorem 11.5.4.2 (Growing back from intertwined boundary matrices). [Lean](#) [✓ Stmt](#) Let ψ be a vector on $K + 1$ sites and let $Y_i, Z_j \in M_D(\mathbb{C})$ satisfy $\psi(i, \tau) = \Gamma_K(Y_i)(\tau)$ for all letters i and words τ of length K , and $A^j Y_i = Z_j A^i$ for all letters i, j . If matrices $R_j \in M_D(\mathbb{C})$ satisfy $\sum_j R_j A^j = \mathbb{1}$, then $\psi \in G_{K+1}(A)$.

Proof. [✓ Proof](#) Put $X = \sum_j R_j Z_j$. Then

$$\begin{aligned} Y_i &= \sum_j R_j A^j Y_i = \sum_j R_j Z_j A^i = X A^i, \\ \psi(i, \tau) &= \text{tr}(A^\tau X A^i) = \text{tr}(A^i A^\tau X) = \Gamma_{K+1}(X)(i, \tau), \end{aligned}$$

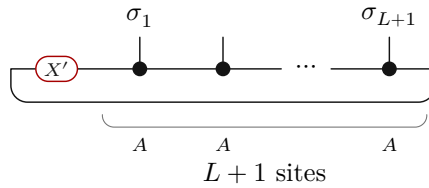
so $\psi = \Gamma_{K+1}(X)$.

□

Theorem 11.5.4.3 (One inclusion of the intersection property). [Lean](#) [✓ Stmt](#) If A is injective and $L \geq 2$, then the nontrivial inclusion in the intersection property holds:

$$G_L^{\text{left}} \cap G_L^{\text{right}} \subseteq G_{L+1}(A).$$

Proof. [✓ Proof](#) Comparing the left and right restrictions on their common $(L - 1)$ -site overlap determines a boundary matrix X' that reconstructs the full $(L + 1)$ -site word:



1. From the right ground condition, for each physical index i there exists a unique $Y_i \in M_D(\mathbb{C})$, by injectivity of Γ_L , such that $\psi(i, \sigma) = \text{tr}(A^\sigma Y_i)$.
2. Similarly, from the left ground condition, for each j there exists a unique Z_j such that $\psi(\sigma, j) = \text{tr}(A^\sigma Z_j)$.

3. Matching on the $(L-1)$ -site overlap (Theorem 11.5.4.1) and using injectivity of Γ_{L-1} gives $A^j Y_i = Z_j A^i$ for all i, j .
4. Since A is injective, the matrices $\{A^j\}$ span $M_D(\mathbb{C})$ (Definition 2.5.1), so $\mathbb{1} = \sum_j c_j A^j$ for some coefficients $c_j \in \mathbb{C}$. Thus, with $X' := \sum_j c_j Z_j$, $Y_i = \mathbb{1} Y_i = \sum_j c_j A^j Y_i = \sum_j c_j Z_j A^i = X' A^i$. This is Theorem 11.5.4.2 with $R_j = c_j \mathbb{1}$.
5. By trace cyclicity, $\psi(\sigma) = \text{tr}(A^\sigma X')$, so $\psi \in G_{L+1}(A)$.

☐

Theorem 11.5.4.4 (Intersection property). [Lean] [✓ Stmt] For injective A and $L \geq 2$, $G_L^{\text{left}} \cap G_L^{\text{right}} = G_{L+1}(A)$, equivalently, $\psi \in G_{L+1}(A)$ if and only if $\psi \in G_L^{\text{left}} \cap G_L^{\text{right}}$.

Proof. ✓ Proof The claim follows from the two forward-direction lemmas and Theorem 11.5.4.3. □

11.5.5 Unique ground state

Contiguous and cyclic interval restrictions

Remark 11.5.5.1 (Contiguous interval restrictions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 **Stmnt** For a non-wrapping interval $[s, s + M - 1] \subseteq \{0, \dots, N - 1\}$ and a full configuration ρ supplying the complementary values, the restriction map is

$$(\mathrm{Res}_{s,M}^\rho \psi)(\omega) = \psi(\rho^{[s,s+M-1] \leftarrow \omega}).$$

These identities include the boundary cases $s = 0$ and $M = N$, the deletion of the first or last site, and the composition of fixing K prefix sites followed by restricting to $[s + K, s + K + L - 1]$.

Remark 11.5.5.2 (Cyclic interval restrictions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) On the periodic chain, for a full configuration ρ supplying the complementary values, put

$$\begin{aligned}\delta_i(k) &:= (k + N - i) \bmod N, \\ \rho_{i,L,\omega}^{\text{cyc}}(k) &:= \begin{cases} \omega(\delta_i(k)), & \delta_i(k) < L, \\ \rho(k), & \delta_i(k) \geq L. \end{cases}\end{aligned}$$

The restriction to the cyclic interval beginning at i is

$$(\mathrm{Res}_{i,L}^{\rho} \psi)(\omega) = \psi(\rho_{i,L,\omega}^{\mathrm{cyc}}),$$

with all site labels read modulo N . If $L \leq N$, this is the same configuration as $\rho^{[i, i+L-1] \leftarrow \omega}$. If two boundary conditions agree whenever $\delta_i(k) \geq L$, their restrictions are equal. In particular, filling the interval inside the outside configuration leaves the restriction unchanged:

$$\mathrm{Res}_{i,L}^{\rho_{i,L,\omega}^{\mathrm{cyc}}} \psi = \mathrm{Res}_{i,L}^{\rho} \psi.$$

For every $r < L$, the inserted word is recovered as $\rho_{i,L,\omega}^{\text{cyc}}(i+r) = \omega(r)$ with site labels taken modulo N . Reconstructing a configuration from its local restriction returns the original configuration: taking $\rho = \sigma$ and $\omega(r) = \sigma(i+r)$ for $r < L$,

$$\sigma_{i, I, \omega}^{\text{cyc}} = \sigma.$$

a full configuration τ supplying the complementary values, and a boundary matrix X , the non-wrapping restriction of the open-boundary ground-space vector $\Gamma_N(X)$ to the window $[s, s+L-1]$ is again an open-boundary ground-space vector,

$$\text{Res}_{s,L}^\tau \Gamma_N(X) \in G_L(A).$$

Both outside words are absorbed into the boundary matrix: writing $w_L = (\tau_0, \dots, \tau_{s-1})$ and $w_R = (\tau_{s+L}, \dots, \tau_{N-1})$, word multiplicativity and cyclicity of the trace give, for any window word ω ,

$$\text{tr}(A^{w_L} A^\omega A^{w_R} X) = \text{tr}(A^\omega A^{w_R} X A^{w_L}),$$

so the restriction is $\Gamma_L(A^{w_R} X A^{w_L})$. This two-sided absorption generalizes Lemma 11.5.5.7, whose suffix restriction absorbs only the prefix.

Lemma 11.5.5.9 (Left-tail compatibility). Lean ✓ Stmt Assume ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every fixed prefix of length K . Then there exist matrices $(Z_j)_j$ and $(Y_u)_u$ with the two trace representations below and, for all j and u , the compatibility relation

$$\begin{aligned} \psi(\sigma, j) &= \text{tr}(A^\sigma Z_j), \\ \psi(u, \tau) &= \text{tr}(A^\tau Y_u), \\ Z_j A^u &= A^j Y_u. \end{aligned}$$

Proof. ✓ Proof Choose Z_j from the long left-window condition and Y_u from the suffix ground condition. The two restrictions give the same vector in $G_{L_0}(A)$, namely $\Gamma_{L_0}(Z_j A^u) = \Gamma_{L_0}(A^j Y_u)$. Injectivity of Γ_{L_0} , obtained from L_0 -block injectivity, gives $Z_j A^u = A^j Y_u$. $\square$

Theorem 11.5.5.10 (Inverting step for word compatibility). Lean ✓ Stmt Assume A is L_0 -block-injective with $L_0 > 0$. If matrices $(Z_j)_j$ satisfy $Z_j A^\sigma = A^j Y_\sigma$ for every word σ of length K , then there exists a matrix X such that $Z_j = A^j X$ for all j .

Proof. ✓ Proof For $K = 1$, the hypothesis gives $Z_j A^i = A^j Y_i$ for every letter i . Every blocked word of length L_0 is nonempty, so this identity extends to the coefficients of the blocked tensor $A^{[L_0]}$. Since $A^{[L_0]}$ is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields $Z_j = A^j X$ for a common matrix X . For larger K , strip the first letter: for each i , the matrices $Z_j A^i$ satisfy the same compatibility hypothesis with words of length $K - 1$, so the induction hypothesis reduces the problem to the case $K = 1$. $\square$

Theorem 11.5.5.11 (Growing-back step at the injectivity length). Lean ✓ Stmt Assume A is L_0 -block-injective with $L_0 > 0$. If an element ψ on $K + L_0 + 1$ sites satisfies the left ground condition on the first $K + L_0$ sites and the suffix ground condition on every prefix of length K , then $\psi \in G_{K+L_0+1}(A)$.

Proof. ✓ Proof Lemma 11.5.5.9 produces matrices $(Z_j)_j$ and $(Y_u)_u$ with $Z_j A^u = A^j Y_u$ for every prefix word u of length K . Theorem 11.5.5.10 gives a common right factor X such that $Z_j = A^j X$ for all j . Therefore, each last-site restriction of ψ agrees with the corresponding last-site restriction of the ground-space vector determined by X . Since every configuration is obtained by adjoining its final letter to its initial $(K + L_0)$ -tuple, the two states coincide, so ψ lies in $G_{K+L_0+1}(A)$. $\square$

Definition 11.5.5.12 (Open-chain left and tail ground subspaces). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

For every $L \in \mathbb{N}$, let $\mathcal{U}_L(A)$ be the subspace of the $(L+1)$ -site Hilbert space whose corresponding functions satisfy the left ground condition: every fixed-final-site restriction belongs to $G_L(A)$. For every $K, L \in \mathbb{N}$, let $\mathcal{V}_{K,L}(A)$ be the subspace of the $(K+L)$ -site Hilbert space whose corresponding functions satisfy the tail ground condition: every fixed-prefix restriction belongs to $G_L(A)$. These definitions include $L = 0$. Let $P_L^{\text{left}}(A)$ and $P_{K,L}^{\text{tail}}(A)$ be the orthogonal projectors onto $\mathcal{U}_L(A)$ and $\mathcal{V}_{K,L}(A)$, respectively.

Theorem 11.5.5.13 (Membership in the open-chain left ground subspace). [Lean](#) [✓ Stmt](#) *Under the canonical ℓ^2 identification, a Hilbert-space vector belongs to $\mathcal{U}_L(A)$ exactly when the corresponding function satisfies the left ground condition: every fixed-final-site restriction belongs to $G_L(A)$.*

Proof. [✓ Proof](#) Transport through the canonical ℓ^2 equivalence and unfold the left-site restriction condition. $\square$

Theorem 11.5.5.14 (Membership in the open-chain tail ground subspace). [Lean](#) [✓ Stmt](#) *Under the canonical ℓ^2 identification, a Hilbert-space vector belongs to $\mathcal{V}_{K,L}(A)$ exactly when the corresponding function satisfies the tail ground condition: every fixed-prefix restriction belongs to $G_L(A)$.*

Proof. [✓ Proof](#) Transport through the canonical ℓ^2 equivalence and unfold the tail restriction condition. $\square$

Theorem 11.5.5.15 (Open-chain left–tail ground-space intersection). [Lean](#) [✓ Stmt](#) *If $D \geq 1$, A is L_0 -block-injective, and $L_0 > 0$, then*

$$\mathcal{U}_{K+L_0}(A) \cap \mathcal{V}_{K,L_0+1}(A) = \mathcal{G}_{K+L_0+1}^{\text{ES}}(A).$$

In Nachtergaele’s notation, with $n = K + L_0$ and $l = L_0$, this is the common range identity for G_{Λ_n} and $G_{\Lambda_{n+1} \setminus \Lambda_{n-l}}$ inside Λ_{n+1} .

Proof. [✓ Proof](#) The forward inclusion is the growing-back step at the injectivity length. Conversely, a vector in the longer ground space remains in the ground space after every left or tail restriction. $\square$

Theorem 11.5.5.16 (Open-chain iteration of the intersection property). [Lean](#) [✓ Stmt](#) *Let A be L_0 -block-injective with $D \geq 1$ and $L_0 > 0$. If an N -site state satisfies the ground-space constraint on every non-wrapping contiguous interval of length $L_0 + 1$, for every fixed choice of the complementary sites, with $N \geq L_0 + 1$, then it lies in $G_N(A)$.*

Proof. [✓ Proof](#) Induct on the extra length beyond $L_0 + 1$. The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 11.5.5.5 and Remark 11.5.5.1 identify every fixed-prefix suffix with one of the assumed length- $L_0 + 1$ intervals. Theorem 11.5.5.11 then grows back the rightmost site. $\square$

Lemma 11.5.5.17 (Open-chain iteration for a sequence of subspaces). [Lean](#) [✓ Stmt](#) *Let $S_m \subseteq (\mathbb{C}^d)^{\otimes m}$ be subspaces, and let $0 < L \leq N$. Suppose that every non-wrapping length- L interval of ψ , for every fixed choice of the complementary sites, lies in S_L . Suppose also that, for every $m \geq L$, $(\mathbb{C}^d \otimes S_m) \cap (S_m \otimes \mathbb{C}^d) = S_{m+1}$. Then $\psi \in S_N$.*

Proof. [✓ Proof](#) Induct on the length m of a non-wrapping interval. The case $m = L$ is the assumed local condition. If all length- m intervals lie in S_m , then the two one-site restrictions of a length- $(m+1)$ interval are adjacent length- m intervals. Hence the length- $(m+1)$ interval lies in $(\mathbb{C}^d \otimes S_m) \cap (S_m \otimes \mathbb{C}^d)$, and the displayed identity places it in S_{m+1} . Taking $m = N$ and the interval starting at the first site gives $\psi \in S_N$. $\square$

Lemma 11.5.5.18 (Periodic constraints and propagated subspaces). [Lean](#) [✓ Stmt](#) *Let $S_m \subseteq (\mathbb{C}^d)^{\otimes m}$ be subspaces, and let $0 < L \leq N$. Suppose that $G_L(A) \subseteq S_L$. Suppose also that, for every $m \geq L$, $(\mathbb{C}^d \otimes S_m) \cap (S_m \otimes \mathbb{C}^d) = S_{m+1}$. Then the periodic local constraints imply $\mathcal{G}_{N,L}(A) \subseteq S_N$.*

Proof. [✓ Proof](#) If $\psi \in \mathcal{G}_{N,L}(A)$, then every length- L cyclic interval of ψ lies in $G_L(A)$, and hence in S_L . A non-wrapping interval is the cyclic interval starting at the same site. Therefore, all non-wrapping length- L intervals lie in S_L , and Lemma 11.5.5.17 gives $\psi \in S_N$. $\square$

Lemma 11.5.5.19 (Cyclic-window range antitonicity). [Lean](#) [Lean](#) [✓ Stmt](#) *On a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if $L' \leq L \leq N$, then $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$.*

Proof. [✓ Proof](#) For every cyclic window vector $v \in G_{L+1}(A)$ and every last letter b , one has $v|_{\text{last}=b} \in G_L(A)$. Thus $\mathcal{G}_{N,L+1}(A) \subseteq \mathcal{G}_{N,L}(A)$, and iteration gives the general antitonicity. $\square$

Theorem 11.5.5.20 (Open-chain containment from cyclic local constraints). [Lean](#) [✓ Stmt](#) *Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$. If an N -site periodic-chain state satisfies all cyclic ground-space constraints at some range $L_0 + 1 \leq L \leq N$, then it lies in the open-chain ground space $G_N(A)$.*

Proof. [✓ Proof](#) Lemma 11.5.5.19 reduces the cyclic hypotheses to range $L_0 + 1$. On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 11.5.5.16 applies. $\square$

11.6 Transport under physical blocking

For a tensor $A = (A^i)_{i=0}^{d-1}$ and a block length p , write $A^{[p]}$ for the tensor whose physical matrices are the consecutive products $A^{i_1} \dots A^{i_p}$. We distinguish the number N of blocked sites from the number Np of original sites.

Definition 11.6.1 (Blocked configuration isometry). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Grouping each consecutive p -tuple of physical indices defines a canonical computational-basis unitary satisfying

$$W_{N,p} : \ell^2(\{0, \dots, d^p - 1\}^N) \longrightarrow \ell^2(\{0, \dots, d - 1\}^{Np}). \quad (11.185)$$

Its forward and inverse coordinate formulas are the corresponding reindexings by the blocked-configuration equivalence and its inverse.

Lemma 11.6.2 (Boundary-map transport under blocking). [Lean](#) [Lean](#) [✓ Stmt](#) *For every $x \in \ell^2(\{0, \dots, D-1\} \times \{0, \dots, D-1\})$, the Euclidean boundary maps intertwine with the blocked-configuration isometry:*

$$W_{N,p} \Gamma_N^{\text{ES}, A^{[p]}}(x) = \Gamma_{Np}^{\text{ES}, A}(x). \quad (11.186)$$

Proof. ✓ Proof For a blocked configuration $J = (J_1, \dots, J_N)$, its flattened word σ , and a boundary matrix $X \in M_D(\mathbb{C})$, blocked-word evaluation gives

$$(A^{[p]})^J = A^{\sigma_1} \dots A^{\sigma_{Np}}, \quad (11.187)$$

$$\Gamma_N^{A^{[p]}}(X)(J) = \text{tr}(A^{\sigma_1} \dots A^{\sigma_{Np}} X) = \Gamma_{Np}^A(X)(\sigma). \quad (11.188)$$

Applying the Frobenius transport formula to the coefficient identity yields (11.186). $\square$

Lemma 11.6.3 (Local spaces under coisometric reconstruction). Lean Lean ✓ Stmt Let $A = (A^i)$ and $C = (C^i)$ have the same physical alphabet, and suppose there is a coisometry $U : \mathbb{C}^D \rightarrow \mathbb{C}^n$ such that $UU^\dagger = I_n$ and $A^i = U^\dagger C^i U$ for every i . Then, for every $L \geq 1$,

$$G_L(A) = G_L(C). \quad (11.189)$$

In particular, if $(A_j)_{j=1}^g$ are the distinct BNT representatives selected from literal canonical-form data for A , then, for $p, L \geq 1$,

$$G_L(A^{[p]}) = \bigvee_{j=1}^g G_L(A_j^{[p]}). \quad (11.190)$$

Identity (11.189) is a derived trace identity for an exact coisometric reconstruction. It is not the literal direct-sum canonical-form equation of CPSV16.

Proof. ✓ Proof For every word w of positive length, $A^w = U^\dagger C^w U$. Cyclicity of the trace gives

$$\begin{aligned} \Gamma_L^A(X) &= \Gamma_L^C(UXU^\dagger), \\ \Gamma_L^C(Y) &= \Gamma_L^A(U^\dagger Y U), \end{aligned} \quad (11.191)$$

where (11.191) uses $UU^\dagger = I_n$. This proves (11.189). For literal canonical-form data, apply this equality to the retained direct sum. Theorem 13.3.6 gives the sum over all displayed normal blocks. Blocks in one BNT phase class differ by a nonzero phase and an invertible gauge, so their local spaces agree; the sum therefore reduces to the distinct representatives, which proves (11.190). $\square$

Theorem 11.6.4 (A simultaneous one-letter span gives nearest-neighbor ground spaces). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $B = \bigoplus_{j=1}^g \mu_j A_j$, where every μ_j is nonzero and every bond dimension is positive. Suppose that the simultaneous one-letter evaluations span the product algebra:

$$\text{span}_{\mathbb{C}} \{(A_1^i, \dots, A_g^i) : 0 \leq i < d\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}). \quad (11.192)$$

Then, for every $N > 2$,

$$\mathcal{G}_{N,2}(B) = \bigvee_{j=1}^g \mathcal{G}_{N,2}(A_j), \quad (11.193)$$

$$\ker H_2^{(N)}(B) = \text{span} \{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g\}. \quad (11.194)$$

Proof. ✓ Proof Equation (11.192) supplies common coefficients c_i satisfying $\sum_i c_i A_j^i = I_{D_j}$ for every j . In the restriction-intersection argument of the proof of [193, Theorem 12], these coefficients replace the right-canonical identity and yield (11.193). This conditional variant and its relation to the source proof are documented in [36]. The same span makes every A_j injective, so Theorem 11.11.2.1 identifies its periodic chain space with $\text{span}\{|V^{(N)}(A_j)\rangle\}$. The block-diagonal comparison of Lemma 12.1.25 then gives (11.194). $\square$

Lemma 11.6.5 (The spanning equation depends only on the local space). [Lean](#) [Lean](#) [✓ Stmt](#) Let B and C have the same physical alphabet and satisfy $G_L(B) = G_L(C)$. Their periodic chain spaces agree for every N :

$$\mathcal{G}_{N,L}(B) = \mathcal{G}_{N,L}(C). \quad (11.195)$$

Their parent-Hamiltonian kernels agree for every $N > L$:

$$\ker H_L^{(N)}(B) = \ker H_L^{(N)}(C). \quad (11.196)$$

Consequently, any fixed BNT-vector family satisfying the parent-Hamiltonian spanning equation for C also satisfies it for B .

Proof. [✓ Proof](#) When $N = 0$ or $L > N$, both chain spaces are $\top$ by definition. Otherwise $0 < N$ and $L \leq N$. Every translated local constraint is then the inverse image of the same subspace $G_L(B) = G_L(C)$, so their intersections are equal. This gives (11.195) for every N . When $N > L$, Theorem 12.3.17 identifies those intersections with the two parent-Hamiltonian kernels and gives (11.196). $\square$

Theorem 11.6.6 (Original-lattice parent-Hamiltonian range). [Lean](#) [✓ Stmt](#) Let A be in literal CPSV canonical form with positive ambient bond dimension D , and suppose that the physical dimension satisfies $1 < d$. There are a basis of normal tensors $(A_j)_{j=1}^g$ and a positive integer L such that

$$D^2 < d^L, \quad L \leq 3D^5, \quad (11.197)$$

and, for every original periodic chain of length $N > L$,

$$\ker H_L^{(N)}(A) = \text{span} \{ |V^{(N)}(A_j)\rangle : 1 \leq j \leq g \}. \quad (11.198)$$

Thus the orthogonal complement of the local MPS space is nonzero, and the corresponding bounded-range interaction is a parent Hamiltonian. This is the construction of [41, Definition 3.9 and line 527], with its condition $d^L > D^2$ made explicit.

Proof. [✓ Proof](#) Choose one normal representative from each BNT phase class and put each representative in a Perron gauge for which

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}, \quad \sum_a A_a^{j\dagger} \Lambda_j A_a^j = \Lambda_j$$

with Λ_j positive definite. Their dimensions satisfy $\sum_j D_j \leq D$, so $g \leq D$, and quantum Wielandt makes every representative block-injective at the common length $L_0 = D^4$.

Suppose first that $g \geq 2$. Put

$$R = 3(g-1)(L_0+1) + 1, \quad L_{\text{aux}} = 3(g-1)(L_0+1) + L_0.$$

If $N > L_{\text{aux}}$, then $N \geq R + L_0$. The source-normalized form of [193, Theorem 12] therefore identifies the range- R parent-Hamiltonian kernel with the BNT span. Since $R \leq L_{\text{aux}}$, cyclic-window antitonicity and the usual frustration-free inclusion give

$$\ker H_{L_{\text{aux}}}^{(N)} \subseteq \ker H_R^{(N)} = \text{span} \{ |V^{(N)}(A_j)\rangle : 1 \leq j \leq g \} \subseteq \ker H_{L_{\text{aux}}}^{(N)}.$$

Thus the range- L_{aux} kernel is exactly the BNT span for every $N > L_{\text{aux}}$, and

$$L_{\text{aux}} \leq 3(D-1)(D^4+1) + D^4 \leq 3D^5.$$

If $g = 1$, the same spanning conclusion holds at $L_{\text{aux}} = D^4 + 1$ by the normal-tensor intersection and closure property reviewed in [16, Section IV.C]. If $g = 0$, the one-site block-diagonal spanning result gives it at $L_{\text{aux}} = 2$; this is a formal boundary extension of the source argument.

In every case enlarge the interaction range to the common value $L = 3D^5$. For every $N > L$, range antitonicity and the frustration-free inclusion give the second kernel sandwich

$$\ker H_L^{(N)} \subseteq \ker H_{L_{\text{aux}}}^{(N)} = \text{span}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g\} \subseteq \ker H_L^{(N)}.$$

Hence the all- $N > L$ spanning equation is preserved. Since $D > 0$ and $d \geq 2$,

$$D^2 < 2^{2D} \leq d^{2D} \leq d^{3D^5} = d^L.$$

Exact coisometric reconstruction identifies the local space of the representative direct sum with $G_L(A)$ and transfers the equality to the original tensor. The chosen range is positive and satisfies $L \leq 3D^5$ by equality. $\square$

Theorem 11.6.7 (Sharp blocking gives a nearest-neighbor parent Hamiltonian). [Lean](#) [Stmt](#)
Let A be in literal CPSV canonical form with positive ambient bond dimension D . There are a basis of normal tensors $(A_j)_{j=1}^g$ and an integer p satisfying $1 \leq p \leq 3D^5$ such that $(A_j^{[p]})_{j=1}^g$ is a basis of normal tensors for $A^{[p]}$ and, for every $N > 2$,

$$\ker H_2^{(N)}(A^{[p]}) = \text{span}\{|V^{(N)}(A_j^{[p]})\rangle : 1 \leq j \leq g\}. \quad (11.199)$$

Thus the actual blocked tensor has a nearest-neighbor parent Hamiltonian on the blocked lattice. This is distinct from Theorem 11.6.6, which supplies a nonzero bounded-range interaction and the parent-Hamiltonian spanning identity on every sufficiently long original chain. The two conclusions and the argument closing the short-ring window are recorded in [31].

Proof. [Proof](#) Choose p so that the simultaneous one-letter evaluations of the blocked normal representatives span the product of their full matrix algebras. Positive blocking preserves the BNT property. Set $C^i = \bigoplus_{j=1}^g (A_j^{[p]})^i$. Then Theorem 11.6.4 gives, for every $N > 2$,

$$\ker H_2^{(N)}(C) = \text{span}\{|V^{(N)}(A_j^{[p]})\rangle : 1 \leq j \leq g\}. \quad (11.200)$$

After the repeated phase-gauge copies and the ambient coisometry are removed, Lemma 11.6.3 gives

$$G_2(A^{[p]}) = \bigvee_{j=1}^g G_2(A_j^{[p]}) = G_2(C). \quad (11.201)$$

Lemma 11.6.5 identifies the two parent-Hamiltonian kernels, so (11.200) becomes (11.199). $\square$

Theorem 11.6.8 (Local ground-space transport under blocking). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Stmt](#) *The Euclidean local space of the blocked tensor is carried exactly to the original Euclidean local space:*

$$W_{N,p} G_N^{\text{ES}}(A^{[p]}) = G_{Np}^{\text{ES}}(A). \quad (11.202)$$

Hence their orthogonal projectors satisfy

$$P_{G_{Np}^{\text{ES}}(A)} = W_{N,p} P_{G_N^{\text{ES}}(A^{[p]})} W_{N,p}^*. \quad (11.203)$$

These statements include $N = 0$ or $p = 0$; in those cases the same explicit configuration isometry and boundary-map calculation apply. In particular, the projector identity holds for three blocked sites and $3p$ original sites. These identities concern the full local ground spaces and their orthogonal projectors under blocking.

Proof. ✓ Proof The exact range identification for the Euclidean boundary map and (11.186) give

$$W_{N,p} \text{range}\left(\Gamma_N^{\text{ES}, A^{[p]}}\right) = \text{range}\left(\Gamma_{Np}^{\text{ES}, A}\right), \quad (11.204)$$

which is (11.202). For every subspace K , orthogonal projection under the unitary $W_{N,p}$ satisfies

$$P_{W_{N,p}(K)} = W_{N,p} P_K W_{N,p}^*. \quad (11.205)$$

Taking $K = G_N^{\text{ES}}(A^{[p]})$ gives (11.203). □

11.7 Ground-space correspondence under an invertible on-site map

An invertible map Λ on the physical space $\mathbb{C}^d$ acts on a tensor by mixing its physical matrices, $A^i \mapsto \sum_j \Lambda_{ij} A^j$ [16, Section IV.C]. Under this action, the deformed local interaction $h' = (\Lambda^{-1\dagger})^{\otimes L} h (\Lambda^{-1})^{\otimes L}$ is a parent interaction for the deformed tensor, and the ground spaces of the two chain Hamiltonians correspond one-to-one through $\Lambda^{\otimes N}$.

Scope restriction (periodic-chain specialization): The ground-space correspondence under physical deformation is stated in [16, Section IV.C] for tensor network states on a general graph; here it is formalized for matrix product states on the periodic chain. Documented in [127].

Definition 11.7.1 (Physical rotation of a tensor). ✓ Stmt For a matrix $\Lambda \in M_d(\mathbb{C})$ and a tensor $A = (A^i)_{i=0}^{d-1}$, the *physically rotated tensor* ΛA has matrices

$$(\Lambda A)^i = \sum_{j=0}^{d-1} \Lambda_{ij} A^j. \quad (11.206)$$

Lemma 11.7.2 (Rotation by the identity). Lean ✓ Stmt Rotation by the identity is trivial: $\mathbb{1}A = A$.

Proof. ✓ Proof Since $\mathbb{1}_{ij} = \delta_{ij}$, each component in (11.206) reduces to $\sum_j \delta_{ij} A^j = A^i$. □

Lemma 11.7.3 (Composition of rotations). Lean ✓ Stmt For $\Lambda, \Lambda' \in M_d(\mathbb{C})$,

$$\Lambda(\Lambda' A) = (\Lambda\Lambda')A. \quad (11.207)$$

Proof. ✓ Proof Expand both sides of (11.207) by (11.206) and swap the order of summation:

$$(\Lambda(\Lambda' A))^i = \sum_j \Lambda_{ij} \sum_k \Lambda'_{jk} A^k = \sum_k \left(\sum_j \Lambda_{ij} \Lambda'_{jk} \right) A^k = ((\Lambda\Lambda')A)^i.$$

□

Definition 11.7.4 (Tensor power of an on-site map). [Lean](#) [Lean](#) [✓ Stmt](#) For $\Lambda \in M_d(\mathbb{C})$ and $L \geq 0$, the tensor power $\Lambda^{\otimes L}$ is the matrix on L -site configurations with entries

$$(\Lambda^{\otimes L})_{\sigma, \tau} = \prod_{n=0}^{L-1} \Lambda_{\sigma_n \tau_n}. \quad (11.208)$$

It acts on the L -site space $\{0, \dots, d-1\}^L \rightarrow \mathbb{C}$ by matrix multiplication and on its Hilbert-space realization $\ell^2(\{0, \dots, d-1\}^L)$ in the same way.

Lemma 11.7.5 (Multiplicativity of the tensor power). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $\Lambda, \Lambda' \in M_d(\mathbb{C})$ one has $(\Lambda\Lambda')^{\otimes L} = \Lambda^{\otimes L} \Lambda'^{\otimes L}$, $\mathbb{1}^{\otimes L} = \mathbb{1}$, and $(\Lambda^{\otimes L})^\dagger = (\Lambda^\dagger)^{\otimes L}$. If Λ is invertible, then $(\Lambda^{-1})^{\otimes L}$ is a two-sided inverse of $\Lambda^{\otimes L}$.

Proof. [✓ Proof](#) Reading the blocked physical index of L consecutive sites as an L -site configuration identifies $\Lambda^{\otimes L}$ with the Kronecker lift of Λ through blocking, for which multiplicativity, the identity, and the conjugate transpose are known. The inverse statements follow from multiplicativity applied to $\Lambda^{-1}\Lambda = \Lambda\Lambda^{-1} = \mathbb{1}$. $\square$

Lemma 11.7.6 (State and boundary map of the rotated tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every word σ of length N ,

$$(\Lambda A)^\sigma = \sum_{\tau} \left(\prod_{n=0}^{N-1} \Lambda_{\sigma_n \tau_n} \right) A^\tau, \quad (11.209)$$

so that $|V^{(N)}(\Lambda A)\rangle = \Lambda^{\otimes N} |V^{(N)}(A)\rangle$ and $\Gamma_L^{\Lambda A}(X) = \Lambda^{\otimes L} \Gamma_L^A(X)$ for every boundary matrix $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Expand each factor of $(\Lambda A)^{\sigma_1} \dots (\Lambda A)^{\sigma_N}$ by (11.206) and collect the coefficient of A^τ , by induction on N . Taking the trace against X gives the boundary-map identity, and $X = \mathbb{1}$ gives the state identity. $\square$

Theorem 11.7.7 (Local ground space of the rotated tensor). [Lean](#) [Lean](#) [✓ Stmt](#) For every $\Lambda \in M_d(\mathbb{C})$ and $L \geq 0$,

$$G_L(\Lambda A) = \Lambda^{\otimes L} G_L(A), \quad (11.210)$$

and the same identity holds for the Hilbert-space realizations of the two local ground spaces.

Proof. [✓ Proof](#) By Lemma 11.7.6, the boundary map of ΛA is the composition of $\Lambda^{\otimes L}$ with the boundary map of A , so its range is the image of $G_L(A)$ under $\Lambda^{\otimes L}$. $\square$

Definition 11.7.8 (Parent interaction). [Lean](#) [✓ Stmt](#) Following [16, Section IV.C, Definition (Parent Hamiltonian)], a *parent interaction* on L sites for the tensor A is a hermitian positive semidefinite operator $h \geq 0$ on $\ell^2(\{0, \dots, d-1\}^L)$ whose kernel equals $G_L(A)$.

Theorem 11.7.9 (The canonical parent interaction is a parent interaction). [Lean](#) [✓ Stmt](#) The canonical parent interaction $P_L^{\text{ES}}(A)$, the orthogonal projector onto $G_L(A)^\perp$, is a parent interaction on L sites for A .

Proof. [✓ Proof](#) An orthogonal projector is hermitian and positive semidefinite (Lemma 12.3.6), and by Lemma 12.3.10 a vector is annihilated by the projector onto $G_L(A)^\perp$ exactly when it lies in $G_L(A)$, so $\ker P_L^{\text{ES}}(A) = G_L(A)$. $\square$

Definition 11.7.10 (Deformed interaction). [Lean](#) [✓ Stmt](#) For $\Lambda \in M_d(\mathbb{C})$ and an operator h on $\ell^2(\{0, \dots, d-1\}^L)$, the *deformed interaction* is

$$h' = (\Lambda^{-1})^{\otimes L} h (\Lambda^{-1})^{\otimes L}. \quad (11.211)$$

This is the operator written in [16, Section IV.C] as $h' = (A^{-1} \otimes A^{-1})^\dagger h (A^{-1} \otimes A^{-1})$ for the two-site interaction of an isometric tensor network, and as $h' = (\Lambda^{-1})^{\otimes k} h (\Lambda^{-1})^{\otimes k}$ for a general physical perturbation.

Theorem 11.7.11 (The deformed interaction is a parent interaction). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\Lambda \in M_d(\mathbb{C})$ be invertible and let h be a parent interaction on L sites for A . Then the deformed interaction h' of (11.211) is a parent interaction on L sites for ΛA : $h' \geq 0$ and $\ker h' = G_L(\Lambda A)$.

Proof. [✓ Proof](#) Write $S = (\Lambda^{-1})^{\otimes L}$. By Lemma 11.7.5, $(\Lambda^{-1})^{\otimes L} = S^\dagger$, so $h' = S^\dagger h S$ is positive semidefinite. Since S and $S^\dagger$ are invertible with inverses $\Lambda^{\otimes L}$ and $(\Lambda^\dagger)^{\otimes L}$,

$$h' \Phi = 0 \iff h S \Phi = 0 \iff S \Phi \in G_L(A) \iff \Phi \in \Lambda^{\otimes L} G_L(A) = G_L(\Lambda A), \quad (11.212)$$

where the last equality is (11.210). $\square$

Lemma 11.7.12 (Window factorization of the tensor power). [Lean](#) [✓ Stmt](#) Let $0 < N, L \leq N$, let $i \in \{0, \dots, N-1\}$, and let $\tau \in \{0, \dots, d-1\}^N$ be an outside configuration. For every $\phi: \{0, \dots, d-1\}^N \rightarrow \mathbb{C}$, the restriction of $\Lambda^{\otimes N} \phi$ to the cyclic window $[i, i+L-1]$ with outside configuration τ is

$$(\Lambda^{\otimes N} \phi)|_{[i, i+L-1], \tau} = \Lambda^{\otimes L} \sum_{\omega} c_{\tau}(\omega) \phi|_{[i, i+L-1], \omega}, \quad (11.213)$$

where the sum runs over the configurations $\omega \in \{0, \dots, d-1\}^N$ that agree with τ on the window, and $c_{\tau}(\omega) = \prod_{k \notin [i, i+L-1]} \Lambda_{\tau_k \omega_k}$ is the product of the entries of Λ over the sites outside the window.

Proof. [✓ Proof](#) Write $W = [i, i+L-1]$ for the window. For a configuration $\omega \in \{0, \dots, d-1\}^N$ let $\omega_W \in \{0, \dots, d-1\}^L$ be the word read off the window, $(\omega_W)_r = \omega_{i+r}$, and for a window word $\sigma' \in \{0, \dots, d-1\}^L$ let $\omega[\sigma'] \in \{0, \dots, d-1\}^N$ be the configuration that reads σ' on the window and agrees with ω outside it, so that $(\phi|_{W, \omega})(\sigma') = \phi(\omega[\sigma'])$, $(\omega[\sigma'])_W = \sigma'$, $\omega[\sigma']|_{W^c} = \omega$, and $c_{\tau}(\omega[\sigma']) = c_{\tau}(\omega)$. Evaluate both sides of (11.213) at a window word σ . On the left, the product over all sites in (11.208) splits into the product over the window, where $\tau[\sigma]$ reads σ , and the product over its complement, where $\tau[\sigma]$ agrees with τ :

$$(\Lambda^{\otimes N} \phi)(\tau[\sigma]) = \sum_{\rho} \left(\prod_{k=0}^{N-1} \Lambda_{\tau[\sigma]_k \rho_k} \right) \phi(\rho) = \sum_{\rho} \left(\prod_{r=0}^{L-1} \Lambda_{\sigma_r \rho_{i+r}} \right) c_{\tau}(\rho) \phi(\rho). \quad (11.214)$$

On the right, with $[\cdot]$ equal to 1 when the enclosed condition holds and to 0 otherwise,

$$\sum_{\sigma'} \left(\prod_{r=0}^{L-1} \Lambda_{\sigma_r \sigma'_r} \right) \sum_{\omega} [\omega_W = \tau_W] c_{\tau}(\omega) \phi(\omega[\sigma']) = \sum_{\sigma'} \sum_{\omega} \left(\prod_{r=0}^{L-1} \Lambda_{\sigma_r \omega_{i+r}} \right) [\sigma' = \tau_W] c_{\tau}(\omega) \phi(\omega) \quad (11.215)$$

$$= \sum_{\omega} \left(\prod_{r=0}^{L-1} \Lambda_{\sigma_r \omega_{i+r}} \right) c_{\tau}(\omega) \phi(\omega), \quad (11.216)$$

where (11.215) reindexes the double sum by the involution

$$(\sigma', \omega) \mapsto (\omega_W, \omega[\sigma']) \quad (11.217)$$

on pairs of a window word and a configuration, and (11.216) collapses the sum over σ' to the single term $\sigma' = \tau_W$. The right sides of (11.214) and (11.216) agree term by term with $\rho = \omega$. $\square$

Theorem 11.7.13 (Ground-space correspondence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\Lambda \in M_d(\mathbb{C})$ be invertible. Then, for all L and N ,*

$$\mathcal{G}_{N,L}(\Lambda A) = \Lambda^{\otimes N} \mathcal{G}_{N,L}(A), \quad (11.218)$$

and for $0 < N$ and $L \leq N$ the kernels of the parent Hamiltonians satisfy $\ker H_N(\Lambda A) = \Lambda^{\otimes N} \ker H_N(A)$. This is the matrix-product-state instance, on the periodic chain, of the one-to-one correspondence, through $R^{\otimes N}$, between the ground states of the parent Hamiltonians of $|\psi\rangle$ and $|\psi'\rangle = R^{\otimes N}|\psi\rangle$ for an invertible map R stated in [16, Section IV.C] for tensor network states on a general graph. Without invertibility, the inclusion $\Lambda^{\otimes N} \mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L}(\Lambda A)$ still holds.

Proof. [✓ Proof](#) Let $\phi \in \mathcal{G}_{N,L}(A)$, so that every window restriction of ϕ lies in $G_L(A)$. By (11.213), every window restriction of $\Lambda^{\otimes N} \phi$ is $\Lambda^{\otimes L}$ applied to a linear combination of window restrictions of ϕ , hence lies in $\Lambda^{\otimes L} G_L(A) = G_L(\Lambda A)$ by (11.210). This proves the inclusion $\Lambda^{\otimes N} \mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L}(\Lambda A)$ for every Λ . Applying it to Λ^{-1} and ΛA gives $(\Lambda^{-1})^{\otimes N} \mathcal{G}_{N,L}(\Lambda A) \subseteq \mathcal{G}_{N,L}(A)$, and composing with $\Lambda^{\otimes N}$ gives the reverse inclusion. The kernel statement follows from Theorem 12.3.17. $\square$

Theorem 11.7.14 (Uniqueness of the ground state is preserved). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\Lambda \in M_d(\mathbb{C})$ be invertible. If $\dim \mathcal{G}_{N,L}(A) = 1$, then $\dim \mathcal{G}_{N,L}(\Lambda A) = 1$.*

Proof. [✓ Proof](#) By (11.218), $\mathcal{G}_{N,L}(\Lambda A)$ is the image of $\mathcal{G}_{N,L}(A)$ under the linear automorphism $\Lambda^{\otimes N}$, which preserves dimension. $\square$

11.8 Periodic-boundary comparison for injective tensors

On the periodic ring, the two cyclic supports that cross the periodic-boundary cut expose the same complementary word, and matching their boundary matrices forces X to commute with every A^j via the boundary-crossing compatibilities $C_\tau^+ A^j X = Y_\tau A^j$ and $X A^j C_\tau^- = A^j Y_\tau$.

Lemma 11.8.1 (Boundary-crossing compatibility at the last site). [Lean](#) [✓ Stmt](#) *Assume A is L_0 -block-injective with $L_0 > 0$ and $M \geq L_0$. If the reduced cyclic interval crossing the last site has boundary matrices Y_τ , then, for every physical letter j , one has*

$$C_\tau^+ A^j X = Y_\tau A^j. \quad (11.219)$$

Here C_τ^+ is the complementary word seen by that cyclic interval.

Proof. [✓ Proof](#) The trace identity for this cyclic interval is tested against all words of length L_0 . Injectivity of Γ_{L_0} , which follows from L_0 -block injectivity, turns these test-word identities into (11.219). $\square$

Lemma 11.8.2 (Boundary-crossing compatibility at the second boundary position). [Lean](#) [✓ Stmt](#) *Assume A is L_0 -block-injective with $L_0 > 0$ and $M \geq L_0$. If the second boundary-crossing reduced cyclic interval has boundary matrices Y_τ , then $X A^j C_\tau^- = A^j Y_\tau$ for every physical letter j , where C_τ^- is the complementary word seen from the second cyclic position.*

Proof. [✓ Proof](#) Factor the cyclic word at the second cyclic position, rotate the trace, and use injectivity of Γ_{L_0} to remove the length- L_0 test word. $\square$

Theorem 11.8.3 (The two boundary-crossing compatibilities). [Lean](#) [✓ Stmt](#) *Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$. Suppose $\psi \in \mathcal{G}_{N,L}(A)$, $N \geq 2$, $L_0 < L \leq N$, and $\psi = \Gamma_N(X)$ for a boundary matrix X . Then there are two families of boundary matrices Y_τ^+ and Y_τ^- , indexed by fixed complement values τ , such that, for every physical letter j ,*

$$\begin{aligned} C_\tau^+ A^j X &= Y_\tau^+ A^j, \\ X A^j C_\tau^- &= A^j Y_\tau^-. \end{aligned} \tag{11.220}$$

Here C_τ^+ and C_τ^- are the complementary words exposed by the two boundary-crossing cyclic intervals, both of reduced length $L_0 + 1$.

Proof. [✓ Proof](#) First use Lemma 11.5.5.19 to reduce the cyclic constraints from range L to range $L_0 + 1$. Then apply the two cyclic-interval compatibility lemmas at the two boundary positions to obtain the one-sided identities in (11.220). $\square$

Lemma 11.8.4 (Boundary-crossing equations for $\Gamma_{M+1}(X)$). [Lean](#) [✓ Stmt](#) *Let A be L_0 -block-injective with $L_0 > 0$, $D \geq 1$, and $L_0 \leq M$. Suppose $\psi = \Gamma_{M+1}(X)$, and suppose matrices $Y_i(\tau)$ represent the length- $(L_0 + 1)$ cyclic restrictions $\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. Then, for every physical letter j and boundary condition τ ,*

$$\begin{aligned} A^{\tau_{L_0}} \dots A^{\tau_{M-1}} A^j X &= Y_M(\tau) A^j, \\ X A^j A^{\tau_1} \dots A^{\tau_{M-L_0}} &= A^j Y_{M+1-L_0}(\tau). \end{aligned} \tag{11.221}$$

Proof. [✓ Proof](#) Substitute $\psi = \Gamma_{M+1}(X)$ in the two cyclic restrictions at the last site and at the second boundary-crossing support. Lemmas 11.8.1 and 11.8.2 then give (11.221). $\square$

Lemma 11.8.5 (One-sided products for a common complement word). [Lean](#) [✓ Stmt](#) *Let A be L_0 -block-injective with $L_0 > 0$, $D \geq 1$, and $L_0 \leq M$. Suppose $\psi = \Gamma_{M+1}(X)$, and suppose matrices $Y_i(\tau)$ represent the length- $(L_0 + 1)$ cyclic restrictions $\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. For every boundary letter η , every physical letter j , and every complementary word μ ,*

$$\begin{aligned} Y_M(\tau_\eta^+(\mu)) A^j &= A^\mu A^j X, \\ X A^j A^\mu &= A^j Y_{M+1-L_0}(\tau_\eta^-(\mu)). \end{aligned}$$

Proof. [✓ Proof](#) Substitute $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$ into Lemma 11.8.4. The exposed-complement identities of Lemma 11.8.6 identify both complementary products with A^μ . $\square$

Lemma 11.8.6 (Common complement word for the two boundary supports). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *On a chain of length $N = M + 1$, with $L_0 \leq M$, fix a physical letter η used outside the complementary sites and a word μ of length $M + 1 - (L_0 + 1)$. Write $\tau_\eta^+(\mu)$ for the boundary condition at the support crossing the last site and $\tau_\eta^-(\mu)$ for the boundary condition at the second boundary-crossing support. Their exposed complements are both exactly μ .*

Proof. [✓ Proof](#) Put the same letter η on the remaining sites. Fill the physical sites $L_0, \dots, M - 1$ with μ for the support crossing the last site, and fill the sites $1, \dots, M - L_0$ with μ for the second boundary-crossing support. The two complement-extraction identities follow from the index arithmetic. The same outside-site calculation gives $\text{Res}_{M, L_0+1}^\rho \psi = \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)} \psi$ for any ρ with $\rho_{k+L_0} = \mu_k$, and $\text{Res}_{M+1-L_0, L_0+1}^\rho \psi = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)} \psi$ for any ρ with $\rho_{k+1} = \mu_k$. $\square$

Proof. [✓ Proof](#) Fix j and σ . Lemma 11.8.7 gives

$$Y_M(\rho_{j,\sigma}^+)A^j = Y_M(\tau_\eta^+(\mu))A^j, \quad (11.222)$$

$$Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j. \quad (11.223)$$

Multiply (11.222) and (11.223) on the right by A^σ and use the assumed product equation for the chosen pair (j, σ) . $\square$

Lemma 11.8.10 (Endpoint words for the periodic-boundary windows). [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M + 1$. Suppose that matrices $Y_r(\rho)$, for $0 \leq r \leq L_0 - 1$, represent the length- $(L_0 + 1)$ cyclic restrictions beginning at the sites $M + 1 - L_0 + r$: $\text{Res}_{M+1-L_0+r, L_0+1}^\rho \psi = \Gamma_{L_0+1}(Y_r(\rho))$. Then

$$Y_0(\rho)A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} = A^{\rho_1} \dots A^{\rho_{L_0-1}}Y_{L_0-1}(\rho).$$

For $L_0 = 1$, both products are empty.

Proof. [✓ Proof](#) Use Theorem 11.5.5.3 for the $L_0 - 1$ adjacent restrictions starting at $i_0 = M + 1 - L_0$. For $0 \leq r < L_0 - 1$, the two endpoint words are determined by

$$\begin{aligned} i_0 + r &= M + 1 - L_0 + r, \\ i_0 + r + L_0 + 1 &\equiv r + 1 \pmod{M + 1}. \end{aligned}$$

Substituting these two words in the iterated transport identity gives the displayed product equation. $\square$

Lemma 11.8.11 (Boundary-condition product for the periodic-boundary windows). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M + 1$, and suppose matrices $Y_i(\rho)$ represent the length- $(L_0 + 1)$ cyclic restrictions $\text{Res}_{i, L_0+1}^\rho(\psi) = \Gamma_{L_0+1}(Y_i(\rho))$. Then, for every boundary condition ρ ,

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} = A^{\rho_1} \dots A^{\rho_{L_0-1}}Y_M(\rho).$$

For $L_0 = 1$, both products are empty. The same equation remains true after multiplying both sides on the right by any matrix R .

Proof. [✓ Proof](#) Apply Theorem 11.5.5.3 to the $L_0 - 1$ restrictions beginning at $M + 1 - L_0 + r$, with $Y_r(\rho) = Y_{M+1-L_0+r}(\rho)$. With $i_0 = M + 1 - L_0$, the site labels satisfy

$$\begin{aligned} i_0 + r &= M + 1 - L_0 + r, \\ i_0 + r + L_0 + 1 &\equiv r + 1 \pmod{M + 1}. \end{aligned}$$

Hence, for $r = 0, \dots, L_0 - 2$,

$$Y_{M+1-L_0+r}(\rho)A^{\rho_{M+1-L_0+r}} = A^{\rho_{r+1}}Y_{M+2-L_0+r}(\rho).$$

Multiplying these $L_0 - 1$ equations from left to right gives the displayed product equation. The right-multiplied form follows by multiplying this equality by R . $\square$

Lemma 11.8.12 (Transport followed by the one-sided boundary equation). [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, and let $L_0 \leq M$. Let $\psi = \Gamma_{M+1}(X)$, and suppose

that matrices $Y_i(\rho)$ represent all length- (L_0+1) cyclic restrictions of ψ . Then, for every boundary condition ρ and every physical letter j ,

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}}A^j = A^{\rho_1} \dots A^{\rho_{L_0-1}}A^{\rho_{L_0}} \dots A^{\rho_{M-1}}A^jX.$$

For $\rho = \tau_\eta^-(\mu)$, this is the transport identity supplied by the adjacent-window argument, with the single factor A^j following $Y_{M+1-L_0}(\tau_\eta^-(\mu))$. The padded identity, whose corresponding factor is A^jA^σ , is supplied separately.

Proof. [✓ Proof](#) Lemma 11.8.11 gives

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}}A^j = A^{\rho_1} \dots A^{\rho_{L_0-1}}Y_M(\rho)A^j.$$

The one-sided equation for the window beginning at M gives $Y_M(\rho)A^j = A^{\rho_{L_0}} \dots A^{\rho_{M-1}}A^jX$. Substitution gives the displayed formula. $\square$

Lemma 11.8.13 (Long boundary-condition product for the periodic-boundary windows). [Lean](#)

[✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M+1$, and suppose matrices $Y_i(\rho)$ represent the length- (L_0+1) cyclic restrictions $\text{Res}_{i, L_0+1}^\rho(\psi) = \Gamma_{L_0+1}(Y_i(\rho))$. Then, for every boundary condition ρ ,

$$Y_M(\rho)A^{\rho_M}A^{\rho_0} \dots A^{\rho_{M-L_0}} = A^{\rho_{L_0}} \dots A^{\rho_M}A^{\rho_0}Y_{M+1-L_0}(\rho).$$

Each side has $M+2-L_0$ one-site factors, with site labels read modulo $M+1$.

Proof. [✓ Proof](#) Apply Theorem 11.5.5.3 to the $M+2-L_0$ restrictions beginning at $M+r$ modulo $M+1$. With $i_0 = M$, the site labels satisfy

$$\begin{aligned} i_0 + r &\equiv M + r \pmod{M+1}, \\ i_0 + r + L_0 + 1 &\equiv L_0 + r \pmod{M+1}. \end{aligned}$$

The final window is $i_0 + M + 2 - L_0 \equiv M + 1 - L_0 \pmod{M+1}$. Substituting these site labels in the iterated product identity gives the displayed equation. $\square$

Lemma 11.8.14 (One matrix Y_μ from compared boundary matrices). [Lean](#) [✓ Stmt](#) Fix a physical letter η used outside the complementary sites. Suppose the two one-sided boundary matrix families Y^+ and Y^- have been extracted from the cyclic windows. If, for every word μ on the complementary sites, these boundary matrices agree on the boundary conditions built from η , $Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$, then there is a single family Y_μ satisfying

$$\begin{aligned} A^\mu A^b X &= Y_\mu A^b, \\ X A^a A^\mu &= A^a Y_\mu \end{aligned}$$

for all letters a, b .

Proof. [✓ Proof](#) Define Y_μ to be the first boundary matrix evaluated on the boundary condition $\tau_\eta^+(\mu)$. Lemma 11.8.6 rewrites the first cyclic-window complement as μ , while the assumed comparison at $\tau_\eta^-(\mu)$ rewrites the second boundary matrix to the same Y_μ . $\square$

Lemma 11.8.15 (Commutation from two one-sided equations). [Lean](#) [✓ Stmt](#) Suppose a single family of boundary matrices Y_μ , indexed by words μ on the complementary sites, satisfies the two one-sided identities

$$A^\mu A^b X = Y_\mu A^b, \quad (11.224)$$

$$X A^a A^\mu = A^a Y_\mu. \quad (11.225)$$

for every pair of physical letters a, b . Then X commutes with every word product $A^a A^\mu A^b$ obtained by adjoining one letter on each side of the word A^μ .

Proof. [✓ Proof](#) Multiplying (11.225) on the right by A^b and (11.224) on the left by A^a gives $X A^a A^\mu A^b = A^a Y_\mu A^b = A^a A^\mu A^b X$. $\square$

Theorem 11.8.16 (Boundary matrix commutation from periodic closure). [Lean](#) [✓ Stmt](#) Let A be injective with $D \geq 1$, and let $\psi = \Gamma_N(X)$ for some boundary matrix $X \in M_D(\mathbb{C})$. If every cyclic restriction of length L of ψ lies in $G_L(A)$, with $N \geq 2$ and $1 < L \leq N$, then $X A^j = A^j X$ for every $j = 0, \dots, d-1$.

Proof. [✓ Proof](#) The periodic boundary condition produces matrix identities of the form $X M W = M Y$ for arbitrary test matrices M and complementary words W . Spanning first over the length- $(L-1)$ tails and then over the complementary words, injectivity forces $(X M - M X) W = 0$ for every W , hence $X M = M X$ for every matrix M . In particular, X commutes with each letter A^j . $\square$

11.9 Periodic unique ground state

For an injective tensor with $D \geq 1$, $1 < L \leq N$, and $N \geq 2$ (or an L_0 -block-injective tensor with $D \geq 1$, $L_0 > 0$, $L_0 < L \leq N$, and $N \geq 2$), the periodic matrix product vector $V^{(N)}(A) = \Gamma_N(\mathbb{1})$ satisfies every cyclic window constraint $\mathcal{G}_{N,L}(A)$. Together with the boundary-matrix commutation $X A^j = A^j X$, this establishes that the periodic ground space is spanned by $V^{(N)}(A)$.

Definition 11.9.1 (Span of the periodic MPS vector). [Lean](#) [✓ Stmt](#) The subspace spanned by the MPS vector $V^{(N)}(A)(\sigma) = \text{tr}(A^{\sigma_0} \dots A^{\sigma_{N-1}})$.

Lemma 11.9.2 (The periodic MPS vector is the ground-space map at the identity). [Lean](#) [✓ Stmt](#) For every chain length N , one has $V^{(N)}(A) = \Gamma_N(\mathbb{1})$.

Lemma 11.9.3 (The periodic MPS vector lies in the local ground space). [Lean](#) [✓ Stmt](#) $V^{(N)}(A) \in G_N(A)$ for every N , with boundary matrix $X = \mathbb{1}$.

Proof. [✓ Proof](#) For every σ , $V^{(N)}(A)(\sigma) = \text{tr}(A^\sigma) = \text{tr}(A^\sigma \mathbb{1}) = \Gamma_N(\mathbb{1})(\sigma)$. $\square$

Definition 11.9.4 (One-dimensional ground space). [Lean](#) [✓ Stmt](#) A finite-dimensional ground space S is one-dimensional if $\dim S = 1$.

Theorem 11.9.5 (One-dimensionality criterion). [Lean](#) [✓ Stmt](#) For finite-dimensional S , $\dim S = 1$ if and only if there exists a nonzero $\psi_0 \in S$ such that every $\psi \in S$ is of the form $c \psi_0$.

Proof. [✓ Proof](#) Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. $\square$

Definition 11.9.6 (Periodic parent-Hamiltonian ground space). [Lean](#) [✓ Stmt](#) For $N > 0$ and $L \leq N$, define the *periodic parent-Hamiltonian ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}) : \forall i \in \{0, \dots, N-1\}, \psi|_{[i, i+L-1]} \in G_L(A)\}.$$

Here the condition $\psi|_{[i, i+L-1]} \in G_L(A)$ means that, for every choice of physical indices outside the window, the restriction of ψ to that window lies in $G_L(A)$. The one-index notation $G_L(A)$ denotes the local ground space $\mathcal{G}_L$ of [16, Section IV.C]. The two-index notation $\mathcal{G}_{N,L}(A)$ used here denotes the periodic intersection of those local constraints over all translated length- L windows on the N -site ring. When $N = 0$ or $L > N$, set $\mathcal{G}_{N,L}(A) := \top$ by convention.

Lemma 11.9.7 (Cyclic-window representation matrices inside the periodic ground space). [Lean](#)

[✓ Stmt](#) If $0 < N$, $L \leq N$, and $\psi \in \mathcal{G}_{N,L}(A)$, then, for every cyclic window and every outside configuration, there is a boundary matrix Y such that the corresponding restricted L -site state is $\Gamma_L(Y)$.

Proof. [✓ Proof](#) This is the defining cyclic-window condition for membership in $\mathcal{G}_{N,L}(A)$, together with the definition of the local ground space as the range of Γ_L . $\square$

Lemma 11.9.8 (The periodic MPS vector satisfies every cyclic window constraint). [Lean](#) [✓ Stmt](#)

If $0 < N$ and $L \leq N$, then $V^{(N)}(A) \in \mathcal{G}_{N,L}(A)$.

Theorem 11.9.9 (Periodic MPS vector nonvanishing for block-injective tensors). [Lean](#) [✓ Stmt](#)

If A is L_0 -block-injective with $L_0 > 0$, $D \geq 1$, and $N \geq L_0 + 1$, then $V^{(N)}(A) \neq 0$ in the N -site space.

Proof. [✓ Proof](#) Suppose for contradiction that $V^{(N)}(A) = 0$, i.e., $\text{tr}(A^\sigma) = 0$ for every word σ of length N . For any words w and u with $|w| = N - L_0$ and $|u| = L_0$, we have $\text{tr}(A^w A^u) = 0$. Since A is L_0 -block-injective, $\{A^u : |u| = L_0\}$ spans $M_D(\mathbb{C})$. By nondegeneracy of the trace pairing, $A^w = 0$ for every $|w| = N - L_0$. Repeatedly strip a length- L_0 suffix: if every product of length $m + L_0$ vanishes, then the spanning property of the length- L_0 products implies that every product of length m vanishes. Eventually all products of some length $r < L_0$ vanish. If $r = 0$, this gives $1 = 0$. If $r > 0$, every length- L_0 product has a zero length- r prefix, so all such products vanish, contradicting that they span $M_D(\mathbb{C})$. $\square$

11.10 Injectivity-length reduction for the closure property

Following [16, Section IV.C], commutation or annihilation relations for length- m word products can be reduced to the injectivity length L_0 by word-spanning arguments. Throughout, $A^w := A^{i_1} \dots A^{i_L}$ denotes the product along a word $w = (i_1, \dots, i_L)$, Γ_N sends a boundary matrix to its open-chain vector, and A being L_0 -block-injective means $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$, so a relation valid on every length- L_0 product extends by linearity to all of $M_D(\mathbb{C})$.

Theorem 11.10.1 (Common right factor from suffix equations). [Lean](#) [✓ Stmt](#) Assume A is L_0 -block-injective with $L_0 > 0$. Let $\{Z_u\}_{|u|=L_0}$ be a family indexed by words of length L_0 . If, for some $K \geq 0$ and every suffix word w of length K , there exists $Y_w \in M_D(\mathbb{C})$ such that $Z_u A^w = A^u Y_w$ for every word u of length L_0 , then there exists $X \in M_D(\mathbb{C})$ with $Z_u = A^u X$ for every word u of length L_0 .

Proof. [✓ Proof](#) Induct on K . For $K = 0$, the empty suffix gives $Z_u = A^u Y_\emptyset$. For the successor step, fix the first physical index i and apply the induction hypothesis to the shortened suffix. This gives a matrix X_i satisfying $Z_u A^i = A^u X_i$ for every word u of length L_0 . Extending this

letter equation to all length- L_0 words and decomposing $\mathbb{1}$ in the spanning family $\{A^u : |u| = L_0\}$ yields a common factor X . $\square$

Theorem 11.10.2 (Common boundary matrix from spanning word identities). [Lean](#) [✓ Stmt](#)
Let A be a tensor whose length- K word products span $M_D(\mathbb{C})$. Let $\{F_b\}_{b \in B}$ and $\{Z_b\}_{b \in B}$ be two families of matrices. If, for every length- K word w , there is a matrix Y_w such that $Z_b A^w = F_b Y_w$ for every $b \in B$, then there is a single matrix Y such that $Z_b = F_b Y$ for every $b \in B$.

Proof. [✓ Proof](#) Suppose $Z_b M_1 = F_b Y_1$ and $Z_b M_2 = F_b Y_2$ for every $b \in B$. Then $Z_b(M_1 + M_2) = Z_b M_1 + Z_b M_2 = F_b(Y_1 + Y_2)$. For a scalar c , one similarly has $Z_b(cM_1) = cZ_b M_1 = F_b(cY_1)$. Thus the identity holds for every matrix in the span of the length- K word products. Since this span is all of $M_D(\mathbb{C})$, apply it to the identity matrix. $\square$

Theorem 11.10.3 (Injectivity-length commutation from length- m commutation). [Lean](#) [✓ Stmt](#)
If A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m product A^w , then X already commutes with every length- L_0 product A^u .

Proof. [✓ Proof](#) Write each length- m word as a concatenation uw with $|u| = L_0$ and $|w| = m - L_0$. The hypothesis gives $X A^u A^w = A^u A^w X = A^u (A^w X)$. Thus the family $Z_u := X A^u$ satisfies the hypothesis of Theorem 11.10.1. One obtains $X A^u = A^u R$ for all $|u| = L_0$. Decomposing $\mathbb{1}$ in the span of the length- L_0 words shows $R = X$, hence $X A^u = A^u X$. $\square$

Theorem 11.10.4 (Centrality from length- m commutation). [Lean](#) [✓ Stmt](#) *If A is L_0 -block-injective with $L_0 > 0$, $m \geq L_0$, and $X \in M_D(\mathbb{C})$ commutes with every length- m word product, then X commutes with every matrix in $M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) By Theorem 11.10.3, X commutes with every length- L_0 word product. These products span $M_D(\mathbb{C})$ by block injectivity, so linearity gives $XM = MX$ for every $M \in M_D(\mathbb{C})$. $\square$

Lemma 11.10.5 (Amplifying fixed-length word commutation). [Lean](#) [✓ Stmt](#) *If a boundary matrix X commutes with every word product of length m , then it commutes with every word product whose length is a multiple qm .*

Proof. [✓ Proof](#) Proceed by induction on q . For $q = 0$, the claim is trivial. For the inductive step, let a word of length $(q+1)m$ be given and write it as a length- m prefix followed by a length- qm suffix. The prefix commutes with X by hypothesis, and the suffix commutes by the induction hypothesis. $\square$

Lemma 11.10.6 (Padded complement annihilation). [Lean](#) [Lean](#) [✓ Stmt](#) *If A is L_0 -block-injective, $q \geq 1$, and $k \leq qL_0$, then the two one-sided annihilation statements are*

$$\begin{aligned} Z A^w &= 0 \quad (|w| = k) &\implies & Z = 0, \\ A^w Z &= 0 \quad (|w| = k) &\implies & Z = 0. \end{aligned}$$

The second implication is the left-handed form.

Proof. [✓ Proof](#) The hypothesis $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$ implies the same spanning statement at every positive multiple qL_0 . Each length- qL_0 word factors as a length- k prefix followed by padding, so Z annihilates all generators of the full span and hence annihilates the identity. For the left-handed form, factor each length- qL_0 word as a prefix followed by a length- k suffix and use the same spanning argument. $\square$

Lemma 11.10.7 (One-sided boundary matrices are unique). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$. If two boundary matrices have the same product with every one-site tensor on one fixed side, then the two boundary matrices are equal.

Proof. [✓ Proof](#) If $Y_1 A^j = Y_2 A^j$ for every physical index j , then induction on the word gives $Y_1 A^w = Y_2 A^w$ for every nonempty word w . Since A is L_0 -block-injective, $\text{span}\{A^w : |w| = L_0\} = M_D(\mathbb{C})$. Thus the left multiplication maps $M \mapsto Y_1 M$ and $M \mapsto Y_2 M$ are equal on $M_D(\mathbb{C})$, and evaluating them on the identity gives $Y_1 = Y_2$. The other one-sided statement follows by the same argument, with right multiplication in place of left multiplication. $\square$

Lemma 11.10.8 (Generator commutation from length- m commutation). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 11.10.4, one has $XA^i = A^i X$ for every physical index i .

Proof. [✓ Proof](#) Apply Theorem 11.10.4 with $M = A^i$. $\square$

Theorem 11.10.9 (MPS-line containment from length- m commutation). [Lean](#) [✓ Stmt](#) Let A be injective after blocking $L_0 > 0$ sites, and let $m \geq L_0$. If a boundary matrix X commutes with every length- m word product, then $\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}$.

Proof. [✓ Proof](#) Theorem 11.10.4 makes X commute with the whole matrix algebra. The center of $M_D(\mathbb{C})$ consists of scalar matrices, so $X = c\mathbb{1}$, and therefore $\Gamma_N(X) = cV^{(N)}(A)$. $\square$

Theorem 11.10.10 (MPS-line containment from positive-length commutation). [Lean](#) [✓ Stmt](#) Let A be injective after blocking $L_0 > 0$ sites. If X commutes with every word product of some positive length m , then $\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}$.

Proof. [✓ Proof](#) Lemma 11.10.5 amplifies the length- m commutation hypothesis to length $L_0 m \geq L_0$. Then Theorem 11.10.9 applies. $\square$

Theorem 11.10.11 (MPS-line containment from two one-sided equations). [Lean](#) [✓ Stmt](#) Let A be injective after blocking $L_0 > 0$ sites. Suppose there are matrices Y_μ such that

$$\begin{aligned} A^\mu A^b X &= Y_\mu A^b, \\ X A^a A^\mu &= A^a Y_\mu, \end{aligned}$$

for every pair of physical letters a, b . Then $\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}$.

Proof. [✓ Proof](#) Lemma 11.8.15 gives commutation with all words of length $|\mu| + 2$, a positive length. The positive-length containment theorem then applies. $\square$

Theorem 11.10.12 (Boundary equality implies containment in the MPS line). [Lean](#) [✓ Stmt](#) Let A be injective after blocking $L_0 > 0$ sites, and fix a physical letter η used outside the complementary sites. Suppose that, for every boundary condition τ and every physical letter j ,

$$\begin{aligned} C_\tau^+ A^j X &= Y_\tau^+ A^j, \\ X A^j C_\tau^- &= A^j Y_\tau^-, \end{aligned}$$

where C_τ^+ and C_τ^- are the complementary words exposed by the two boundary-crossing cyclic windows. Suppose also that the two boundary matrices satisfy $Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$ for every word μ on the complementary sites. Then $\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}$.

Proof. ✓ Proof Lemma 11.8.14 gives a family Y_μ satisfying

$$\begin{aligned} A^\mu A^b X &= Y_\mu A^b, \\ X A^a A^\mu &= A^a Y_\mu. \end{aligned}$$

Theorem 11.10.11 applies. □

Theorem 11.10.13 (Closure-property step for periodic chains). Lean ✓ Stmt *Let A be injective after blocking $L_0 > 0$ sites and let $D \geq 1$. Assume $N \geq 2$ and $L_0 < L \leq N$. Suppose that, for every physical letter η used in the two boundary conditions and every boundary matrix obtained from an N -site chain ground state, the identities*

$$\begin{aligned} A^\mu A^j X &= Y_{\tau_\eta^+(\mu)}^+ A^j, \\ X A^j A^\mu &= A^j Y_{\tau_\eta^-(\mu)}^-, \end{aligned}$$

hold for every physical letter j and every complementary word μ , and are supplemented, for every μ , by $Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$. Then $\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}$.

Proof. ✓ Proof The cyclic-to-open-chain reduction writes any chain ground state as $\Gamma_N(X)$. The two boundary-crossing cyclic windows give

$$\begin{aligned} A^\mu A^j X &= Y_{\tau_\eta^+(\mu)}^+ A^j, \\ X A^j A^\mu &= A^j Y_{\tau_\eta^-(\mu)}^-, \\ Y_{\tau_\eta^+(\mu)}^+ &= Y_{\tau_\eta^-(\mu)}^-. \end{aligned}$$

Theorem 11.10.12 finishes. □

11.11 Periodic-boundary comparison for normal tensors

For normal tensors, the periodic-boundary argument closes open intervals into periodic chains using boundary-crossing restrictions [16, Section IV.C]. For a vector $\psi = \Gamma_{M+1}(X)$, the closure property requires that the two boundary-crossing restrictions $\text{Res}_{M,L_0+1}^{\tau_\eta^+(\mu)}(\psi)$ and $\text{Res}_{M+1-L_0,L_0+1}^{\tau_\eta^-(\mu)}(\psi)$ coincide. The boundary conditions $\tau_\eta^\pm(\mu)$ parametrize these restrictions, and the matrices $Y_i(\tau)$ satisfy $\text{Res}_{i,L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. Once the one-sided boundary products are known, equality of the restrictions follows from the boundary-matrix commutation relation $X A^j = A^j X$ for every physical letter j . The block-window identity $X A^\alpha A^\nu = A^\alpha Y_\nu$ is derived from the cyclic-window constraints crossing the periodic cut.

Lemma 11.11.1 (Block-to-letter translation of the boundary block matrix equation). Lean ✓ Stmt *Let A be an MPS tensor and let $L_0, K \geq 0$. Let X be a matrix, and let Y_{c_b} be a matrix for each iterated block index c_b of the complement. Suppose that, for every block letter b of the alphabet $\{0, \dots, d-1\}^{L_0}$ and every iterated block index c_b of the complement, with $w(b)$ the length- L_0 word of b and $\widetilde{w(c_b)}$ the length- $L_0 K$ complement word obtained by concatenating the blocks of c_b , $X A^{w(b)} A^{\widetilde{w(c_b)}} = A^{w(b)} Y_{c_b}$. Then there is a family Y'_c , indexed by the length- $L_0 K$ words c , with $X A^s A^c = A^s Y'_c$ for every length- L_0 word s and every length- $L_0 K$ word c .*

Proof. ✓ Proof Each length- L_0 word s is the word $w(b)$ of its block letter b , and each length- $L_0 K$ word c regroups into K blocks, recovering it as the concatenated complement word $\widetilde{w(c_b)}$ of an iterated block index c_b ; both identifications are bijective. Substituting these representations into the hypothesis and setting $Y'_c = Y_{c_b}$ gives the asserted equation for every s and c . □

Lemma 11.11.2 (Boundary matrix commutation from block-window equations). [Lean](#) [Stmnt](#) Let A be L_0 -block-injective with $L_0 > 0$. Let X be a boundary matrix, and let Y_ν be a matrix for each complementary word ν of length K . Suppose that, for every word u of length L_0 and every word v of length K , $XA^uA^v = A^uY_\nu$. Then, for every physical letter j , $XA^j = A^jX$. This establishes the boundary-matrix commutation step of [16, Section IV.C, lines 2049–2090].

Proof. [Stmnt](#) By Lemma 6.1.1.23, $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$. Hence, for every $M \in M_D(\mathbb{C})$ and every complementary word ν of length K , the displayed hypothesis extends by linearity in the length- L_0 block to $XMA^\nu = MY_\nu$. Taking $M = \mathbb{1}$ gives

$$Y_\nu = XA^\nu. \quad (11.226)$$

Taking $M = A^j$ and substituting (11.226) gives $(XA^j - A^jX)A^\nu = 0$ for every word ν of length K . Since $K \leq (K+1)L_0$ and $K+1 \geq 1$, Lemma 11.10.6 gives $XA^j - A^jX = 0$. $\square$

Lemma 11.11.3 (Trace identity from the boundary-crossing cyclic window). [Lean](#) [Stmnt](#) Let A be a tensor with virtual dimension at least one, and let $L_0 > 0$ and $L_0 < M$. Let $\psi = \Gamma_{M+1}(X)$ and assume that every cyclic restriction of length $L_0 + 1$ belongs to $G_{L_0+1}(A)$. Then there are boundary matrices Y_ν , indexed by nonempty complementary words ν of length $M+1 - (L_0+1)$, such that, for every physical letter j and every word α of length L_0 ,

$$\text{tr}(A^jXA^\alpha A^\nu) = \text{tr}(A^jA^\alpha Y_\nu).$$

Proof. [Stmnt](#) Let Y_ν be the matrix representing the cyclic restriction whose window begins at the last site. The cyclic word is $\alpha\nu j$ after deleting the first letter from the window and reading the complement, so the ground-space representation and cyclicity of the trace give

$$\text{tr}(A^jXA^\alpha A^\nu) = \text{tr}(A^\alpha A^\nu A^jX) = \text{tr}(A^jA^\alpha Y_\nu).$$

$\square$

Lemma 11.11.4 (One-site-injective boundary block-window identity). [Lean](#) [Stmnt](#) Let A be a tensor with virtual dimension at least one whose single-site matrices span the full matrix algebra. Let $L_0 > 0$ and $L_0 < M$, and let $\psi = \Gamma_{M+1}(X)$. If every cyclic restriction of length $L_0 + 1$ belongs to $G_{L_0+1}(A)$, then there are boundary matrices Y_ν , indexed by nonempty complementary words ν of length $M+1 - (L_0+1)$, such that, for every word α of length L_0 , $XA^\alpha A^\nu = A^\alpha Y_\nu$.

Proof. [Stmnt](#) For every physical letter j , Lemma 11.11.3 gives

$$\text{tr}(A^jXA^\alpha A^\nu) = \text{tr}(A^jA^\alpha Y_\nu). \quad (11.227)$$

Put $Z = XA^\alpha A^\nu - A^\alpha Y_\nu$. For every physical letter j , (11.227) gives $\text{tr}(A^jZ) = 0$. Since the matrices A^j span the full matrix algebra, the trace pairing gives $(\forall j, \text{tr}(A^jZ) = 0) \implies Z = 0$. Hence $XA^\alpha A^\nu = A^\alpha Y_\nu$. $\square$

Theorem 11.11.5 (Trace identities with length- L_0 probes from cyclic-window constraints). [Lean](#) [Stmnt](#) Let A be L_0 -block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let $\psi = \Gamma_{M+1}(X)$ and assume that every cyclic restriction of length $L_0 + 1$ belongs to $G_{L_0+1}(A)$. Then there are boundary matrices Y_ν , indexed by nonempty complementary words ν of length $M+1 - (L_0+1)$, such that, for every word α of length L_0 and every word β of length L_0 ,

$$\text{tr}(A^\beta XA^\alpha A^\nu) = \text{tr}(A^\beta A^\alpha Y_\nu).$$

Proof. ✓ Proof Choose representation matrices for every cyclic restriction of length $L_0 + 1$. Fix α and ν . The boundary-crossing equations move the matrix X through the first boundary letter. In local boundary notation, this has the form

$$XA^{\rho_0}A^{\rho_1\cdots\rho_{M-L_0}} = A^{\rho_0}Y_{M+1-L_0}(\rho).$$

The adjacent boundary-window product then transports this comparison through the remaining $L_0 - 1$ boundary letters:

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}\cdots\rho_{M-1}} = A^{\rho_1\cdots\rho_{L_0-1}}Y_M(\rho).$$

Finally, the outside-label uniqueness lemma identifies the final boundary matrix with the matrix Y_ν attached to the complementary word ν . Thus $XA^\alpha A^\nu = A^\alpha Y_\nu$. Multiplying on the left by A^β and taking the trace gives the asserted identity for every length- L_0 word β . $\square$

Theorem 11.11.6 (Length-word trace separation for the boundary block window). Lean ✓ Stmt *Let A be L_0 -block-injective. Let X be a boundary matrix, and let Y_ν be a matrix for each complementary word ν of length K . Suppose that, for every word α of length L_0 , every word β of length L_0 , and every complementary word ν ,*

$$\text{tr}(A^\beta XA^\alpha A^\nu) = \text{tr}(A^\beta A^\alpha Y_\nu).$$

Then, for every word α of length L_0 and every complementary word ν , $XA^\alpha A^\nu = A^\alpha Y_\nu$.

Proof. ✓ Proof The trace identities say exactly that, for each fixed pair (α, ν) ,

$$\Gamma_{L_0}(XA^\alpha A^\nu) = \Gamma_{L_0}(A^\alpha Y_\nu).$$

Since A is L_0 -block-injective, Γ_{L_0} is injective by Theorem 11.5.3.3. Hence $XA^\alpha A^\nu = A^\alpha Y_\nu$. $\square$

Lemma 11.11.7 (Matrix identity $XA^\alpha A^\nu = A^\alpha Y_\nu$ from cyclic-window constraints). Lean ✓ Stmt *Let A be L_0 -block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let $\psi = \Gamma_{M+1}(X)$ and assume that every cyclic restriction of length $L_0 + 1$ belongs to $G_{L_0+1}(A)$. Then there are boundary matrices Y_ν , indexed by nonempty complementary words ν of length $M + 1 - (L_0 + 1)$, such that, for every word α of length L_0 , $XA^\alpha A^\nu = A^\alpha Y_\nu$.*

Proof. ✓ Proof Theorem 11.11.5 gives, for every word β of length L_0 ,

$$\text{tr}(A^\beta XA^\alpha A^\nu) = \text{tr}(A^\beta A^\alpha Y_\nu).$$

Theorem 11.11.6 then gives the displayed matrix equation. $\square$

Lemma 11.11.8 (Boundary restrictions from commutation and one-sided products). Lean ✓ Stmt *Let A be L_0 -block-injective with $L_0 > 0$. Let X be a boundary matrix, let W_η and V_η be matrix families indexed by a physical letter η , and let μ be a word. Suppose that, for every physical letter j , $XA^j = A^jX$. Suppose also that, for every η and every j ,*

$$\begin{aligned} W_\eta A^j &= A^\mu A^j X, \\ XA^j A^\mu &= A^j V_\eta. \end{aligned}$$

Then, for every η and every j , $W_\eta A^j = V_\eta A^j$.

Proof. [✓ Proof](#) The commutation relation $XA^j = A^jX$ extends by induction to $XA^w = A^wX$ for every word w . Hence the second one-sided equation gives, for every k , $A^kV_\eta = XA^kA^\mu = A^kXA^\mu = A^kA^\mu X$. Lemma 11.10.7 gives $V_\eta = A^\mu X$. The first one-sided equation then gives $W_\eta A^j = A^\mu A^j X = A^\mu X A^j = V_\eta A^j$. $\square$

Lemma 11.11.9 (Products with one-site tensors determine the boundary matrix). [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$. Fix two matrix families Y^+ and Y^- and the two reindexed boundary conditions $\tau_\eta^+(\mu)$ and $\tau_\eta^-(\mu)$. If, for every physical letter j , $Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j$, then $Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$.

Proof. [✓ Proof](#) Apply Lemma 11.10.7 to the two matrices $Y_{\tau_\eta^+(\mu)}^+$ and $Y_{\tau_\eta^-(\mu)}^-$. $\square$

Lemma 11.11.10 (Outside labels determine boundary-restriction matrices). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$ and $L_0 \leq M$. If an outside configuration ρ has the same word μ as $\tau_\eta^+(\mu)$ on the sites outside the last-site cyclic window, then the two matrices representing that restriction agree: $Y_\rho = Y_{\tau_\eta^+(\mu)}$. The same assertion holds for the second boundary-crossing window: if ρ has the same outside word as $\tau_\eta^-(\mu)$, then $Y_\rho = Y_{\tau_\eta^-(\mu)}$. This is the boundary-matrix independence step used to choose one representative outside configuration for each outside word in the closure-property matrix comparison.

Proof. [✓ Proof](#) Equality of the cyclic restrictions gives, for every physical letter j , $Y_\rho A^j = Y_{\tau_\eta^+(\mu)} A^j$. Lemma 11.10.7 identifies the two matrices from these one-site products. $\square$

Lemma 11.11.11 (First-letter restrictions determine a vector). [Lean](#) [✓ Stmt](#) Let R_j be restriction to first letter j :

$$(R_j \phi)(\sigma_1, \dots, \sigma_L) = \phi(j, \sigma_1, \dots, \sigma_L).$$

For $\phi, \psi \in (\mathbb{C}^d)^{\otimes(L+1)}$,

$$(\forall j, R_j \phi = R_j \psi) \implies \phi = \psi.$$

Proof. [✓ Proof](#) At $(\sigma_0, \sigma_1, \dots, \sigma_L)$, the required equality is

$$\phi(\sigma_0, \sigma_1, \dots, \sigma_L) = (R_{\sigma_0} \phi)(\sigma_1, \dots, \sigma_L) = (R_{\sigma_0} \psi)(\sigma_1, \dots, \sigma_L) = \psi(\sigma_0, \sigma_1, \dots, \sigma_L).$$

$\square$

Lemma 11.11.12 (Auxiliary restriction equality for the closure property). [Lean](#) [✓ Stmt](#) Let $L_0 > 0$ and $L_0 \leq M$, and set

$$\begin{aligned} B_{\eta, \mu}^+(\psi) &= \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi), \\ B_{\eta, \mu}^-(\psi) &= \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi). \end{aligned}$$

With R_j as in Lemma 11.11.11,

$$(\forall j, R_j B_{\eta, \mu}^+(\psi) = R_j B_{\eta, \mu}^-(\psi)) \implies B_{\eta, \mu}^+(\psi) = B_{\eta, \mu}^-(\psi).$$

Proof. [✓ Proof](#) Apply Lemma 11.11.11 to the two length- (L_0+1) restrictions, using the displayed family of first-letter equalities. $\square$

Lemma 11.11.13 (Boundary-crossing restriction from right products). [Lean](#) [Stmt](#) Let A be an MPS tensor, let $L_0 > 0$, and let $L_0 \leq M$. Suppose the two boundary-crossing restrictions of length $L_0 + 1$ are represented by boundary matrices $Y_{\eta,\mu}^+$ and $Y_{\eta,\mu}^-$:

$$\begin{aligned} B_{\eta,\mu}^+(\psi) &= \Gamma_{L_0+1}(Y_{\eta,\mu}^+), \\ B_{\eta,\mu}^-(\psi) &= \Gamma_{L_0+1}(Y_{\eta,\mu}^-). \end{aligned}$$

If, for every physical letter j , $Y_{\eta,\mu}^+ A^j = Y_{\eta,\mu}^- A^j$, then $B_{\eta,\mu}^+(\psi) = B_{\eta,\mu}^-(\psi)$.

Proof. [Proof](#) Fix j and take the first-letter restriction of the two displayed length- $(L_0 + 1)$ restrictions. These are $\Gamma_{L_0}(Y_{\eta,\mu}^+ A^j)$ and $\Gamma_{L_0}(Y_{\eta,\mu}^- A^j)$. The one-site product equality gives, for every j ,

$$\Gamma_{L_0}(Y_{\eta,\mu}^+ A^j) = \Gamma_{L_0}(Y_{\eta,\mu}^- A^j).$$

Lemma 11.11.11 identifies the two original restrictions. □

Lemma 11.11.14 (Boundary restrictions from first-letter products). [Lean](#) [Stmt](#) Let $L_0 > 0$ and $L_0 \leq M$. Let $Y_i(\tau)$ represent the length- $(L_0 + 1)$ cyclic restrictions of ψ . Fix a complementary word μ . If, for every boundary letter η and physical letter j , $Y_M(\tau_\eta^+(\mu)) A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j$, then, for every η ,

$$\text{Res}_{M,L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0,L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Proof. [Proof](#) Apply Lemma 11.11.13 for each boundary letter η , using the corresponding first-letter product equations. □

Lemma 11.11.15 (Equality of the two restrictions crossing the periodic cut). [Lean](#) [Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, let $L_0 < M$, and let

$$\begin{aligned} \psi &\in \mathcal{G}_{M+1,L_0+1}(A), \\ \psi &= \Gamma_{M+1}(X). \end{aligned}$$

For every boundary letter η and complementary word μ , the two restrictions crossing the periodic cut satisfy

$$\text{Res}_{M,L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0,L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Proof. [Proof](#) The cyclic-window representation lemma gives matrices $Y_M(\tau_\eta^+(\mu))$ and $Y_{M+1-L_0}(\tau_\eta^-(\mu))$ such that

$$\begin{aligned} \text{Res}_{M,L_0+1}^{\tau_\eta^+(\mu)}(\psi) &= \Gamma_{L_0+1}(Y_M(\tau_\eta^+(\mu))), \\ \text{Res}_{M+1-L_0,L_0+1}^{\tau_\eta^-(\mu)}(\psi) &= \Gamma_{L_0+1}(Y_{M+1-L_0}(\tau_\eta^-(\mu))). \end{aligned}$$

Theorem 11.11.1.8 gives, for every j , $Y_M(\tau_\eta^+(\mu)) A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j$. Lemma 11.11.13 identifies the two length- $(L_0 + 1)$ restrictions. □

Lemma 11.11.16 (Boundary-matrix product from boundary-crossing restrictions). [Lean](#) [Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length

$M+1$, and let $Y_i(\tau)$ be matrices satisfying $\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. Suppose that, for every outside letter η ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi). \quad (11.228)$$

Then there are boundary conditions $\rho_{j, \sigma}^+$ and $\rho_{j, \sigma}^-$ such that, for every complementary position k ,

$$\begin{aligned} \rho_{j, \sigma}^+(k + L_0) &= \mu_k, \\ \rho_{j, \sigma}^-(k + 1) &= \mu_k, \end{aligned}$$

and, for every physical letter j and every word σ of length L_0 ,

$$Y_M(\rho_{j, \sigma}^+) A^j A^\sigma = Y_{M+1-L_0}(\rho_{j, \sigma}^-) A^j A^\sigma.$$

Proof. ✓ Proof For the product indexed by j and σ , specialize (11.228) at the outside letter $\eta = j$. Take $\rho_{j, \sigma}^+ = \tau_j^+(\mu)$ and $\rho_{j, \sigma}^- = \tau_j^-(\mu)$. The complement equations are the two identities of Lemma 11.8.6. Fix j . Applying the first-letter restriction to the displayed equality gives

$$\Gamma_{L_0}(Y_M(\tau_j^+(\mu)) A^j) = \Gamma_{L_0}(Y_{M+1-L_0}(\tau_j^-(\mu)) A^j),$$

where the two sides are identified by Theorem 11.5.5.3. By Theorem 11.5.3.3, $Y_M(\tau_j^+(\mu)) A^j = Y_{M+1-L_0}(\tau_j^-(\mu)) A^j$. Right multiplication by A^σ gives the asserted product equation. $\square$

Lemma 11.11.17 (Boundary-matrix product from right products). Lean ✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M+1$, and let $Y_i(\tau)$ be matrices satisfying $\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. Fix a complementary word μ of length $M+1-(L_0+1)$. Suppose that, for every boundary letter η and every physical letter j , $Y_M(\tau_\eta^+(\mu)) A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j$. Then there are boundary conditions $\rho_{j, \sigma}^+$ and $\rho_{j, \sigma}^-$ such that, for every complementary position k ,

$$\begin{aligned} \rho_{j, \sigma}^+(k + L_0) &= \mu_k, \\ \rho_{j, \sigma}^-(k + 1) &= \mu_k, \end{aligned}$$

and, for every physical letter j and every word σ of length L_0 ,

$$Y_M(\rho_{j, \sigma}^+) A^j A^\sigma = Y_{M+1-L_0}(\rho_{j, \sigma}^-) A^j A^\sigma.$$

Proof. ✓ Proof For each boundary letter η , Lemma 11.11.13 gives

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Applying these equalities, Lemma 11.11.16 gives the displayed boundary conditions and product equation. $\square$

Lemma 11.11.18 (Boundary-crossing comparison after multiplication by words). Lean ✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Fix a complementary word μ , a matrix X , and a family of matrices $Y_i(\tau)$. Suppose that, for every boundary letter η , every physical letter j , and every word σ of length L_0 ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Then, for every η and j , $Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j = A^\mu A^j X$.

Proof. ✓ Proof Fix η and j , and set $Z_{\eta,j} = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j - A^\mu A^j X$. The hypothesis gives $Z_{\eta,j}A^\sigma = 0$ for every σ with $|\sigma| = L_0$. Since A is L_0 -block-injective, the products A^σ with $|\sigma| = L_0$ span the full matrix algebra. Applying Lemma 11.10.6 gives $Z_{\eta,j} = 0$. $\square$

Lemma 11.11.19 (Boundary-crossing comparison after left multiplication by words). Lean
✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Fix a complementary word μ , a matrix X , and a family of matrices $Y_i(\tau)$. Suppose that, for every boundary letter η , every physical letter j , and every pair of words σ, α of length L_0 ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then, for every η, j , and σ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Proof. ✓ Proof Fix η, j , and σ , and set

$$Z_{\eta,j,\sigma} = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma - A^\mu A^j X A^\sigma.$$

The hypothesis says that $A^\alpha Z_{\eta,j,\sigma} = 0$ for every word α of length L_0 . The left-handed form of Lemma 11.10.6 gives $Z_{\eta,j,\sigma} = 0$. $\square$

Lemma 11.11.20 (Boundary-matrix product from the boundary-crossing comparison). Lean
✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M+1$, and let $Y_i(\tau)$ be matrices satisfying $\text{Res}_{i,L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. Fix a complementary word μ and a matrix X . Suppose that, for every boundary letter η and every physical letter j , $Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$. Suppose also that, for every η, j , and every word σ of length L_0 ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Then there are boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$ such that, for every complementary position k ,

$$\begin{aligned} \rho_{j,\sigma}^+(k + L_0) &= \mu_k, \\ \rho_{j,\sigma}^-(k + 1) &= \mu_k, \end{aligned}$$

and, for every physical letter j and every word σ of length L_0 ,

$$Y_M(\rho_{j,\sigma}^+)A^j A^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j A^\sigma.$$

Proof. ✓ Proof Lemma 11.11.18 gives, for every η and j , $Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j = A^\mu A^j X$. Combining this equation with $Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$ gives $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$. Lemma 11.11.17 then gives the displayed boundary conditions and product equation. $\square$

Lemma 11.11.21 (Boundary-matrix product from the left-multiplied boundary-crossing comparison). Lean ✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$, and let $L_0 \leq M$. Let ψ be a state on the chain of length $M+1$, and let $Y_i(\tau)$ be matrices satisfying $\text{Res}_{i,L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. Fix a complementary word μ and a matrix X . Suppose that, for every boundary letter η and every physical letter j , $Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$. Suppose also that, for every η, j , and every pair of words σ, α of length L_0 ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then there are boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$ satisfying the complementary-word equations

$$\begin{aligned}\rho_{j,\sigma}^+(k + L_0) &= \mu_k, \\ \rho_{j,\sigma}^-(k + 1) &= \mu_k,\end{aligned}$$

and the product equation

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

Proof. ✓ Proof Lemma 11.11.19 gives

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

Lemma 11.11.20 then gives the displayed boundary conditions and product equation. □

Theorem 11.11.22 (Boundary-matrix product from left-multiplied equations). Lean ✓ Stmt
Let A be L_0 -block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let $\psi = \Gamma_{M+1}(X)$ be an open-chain representation, and let $Y_i(\tau)$ represent the length- $(L_0 + 1)$ cyclic restrictions of ψ . Fix a complementary word μ . Suppose that, for every boundary letter η , physical letter j , and length- L_0 words σ, α ,

$$A^\alpha \left(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma \right) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then there are boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$ satisfying the complementary-word equations

$$\begin{aligned}\rho_{j,\sigma}^+(k + L_0) &= \mu_k, \\ \rho_{j,\sigma}^-(k + 1) &= \mu_k,\end{aligned}$$

and the product equation

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

Proof. ✓ Proof Lemma 11.8.5 gives the equation for the boundary-crossing window beginning at M , $Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$. The assumed left-multiplied coordinate equation and Lemma 11.11.21 then give the displayed boundary conditions and product equation. □

Theorem 11.11.23 (Boundary restrictions from the left-multiplied comparison). Lean ✓ Stmt
Let A be L_0 -block-injective with $L_0 > 0$, $L_0 \leq M$, and $D \geq 1$. Let $\psi = \Gamma_{M+1}(X)$ be an open-chain representation, and let $Y_i(\tau)$ represent the length- $(L_0 + 1)$ cyclic restrictions of ψ . Fix a complementary word μ . Suppose that, for every boundary letter η , physical letter j , and length- L_0 words α, σ ,

$$A^\alpha \left(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma \right) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then, for every boundary letter η ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Proof. ✓ Proof The assumed equations and Lemma 11.11.19 give, for every η, j , and σ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

Lemma 11.11.18 removes the right word: $Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j = A^\mu A^j X$. The equation for the boundary-crossing window beginning at M , from Lemma 11.8.5, gives $Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$. Hence the first-letter products agree, and Lemma 11.11.14 identifies the two restrictions. □

Theorem 11.11.24 (Equality of the boundary-crossing restrictions). [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let $\psi = \Gamma_{M+1}(X)$ be an open-chain representation. Assume that, for all i and τ , $\text{Res}_{i, L_0+1}^\tau(\psi) \in G_{L_0+1}(A)$. In local coordinate notation, fix a nonempty word μ on the sites outside the two cyclic windows. For every outside letter η , choose the two outside configurations so that, for each index k of this outside word,

$$\begin{aligned}\tau_\eta^+(\mu)_{k+L_0} &= \mu_k, \\ \tau_\eta^-(\mu)_{k+1} &= \mu_k,\end{aligned}$$

and assign the letter η to the remaining sites, following the coordinate parametrization of the periodic-boundary comparison in [16, Section IV.C, lines 2078–2079]. The corresponding boundary-crossing restriction equality is

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Proof. [✓ Proof](#) Choose matrices $Y_i(\tau)$ representing the length- (L_0+1) restrictions. The one-sided boundary-product lemma gives, for every outside letter η and physical letter j ,

$$\begin{aligned}Y_M(\tau_\eta^+(\mu))A^j &= A^\mu A^j X, \\ XA^j A^\mu &= A^j Y_{M+1-L_0}(\tau_\eta^-(\mu)).\end{aligned}$$

Lemma 11.11.7 gives matrices Y_ν satisfying, for every word α of length L_0 and every complementary word ν , $XA^\alpha A^\nu = A^\alpha Y_\nu$. Lemma 11.11.2 then gives, for every physical letter k , the commutation identity $XA^k = A^k X$. Lemma 11.11.8 gives $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$. Lemma 11.11.14 then identifies the two boundary-crossing restrictions, giving the periodic-boundary comparison of [16, lines 2078–2079]. $\square$

11.11.1 Reverse comparison for boundary-crossing restrictions

Conversely, equality of the two cyclic restrictions yields the first-letter product equation $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$ and the left-multiplied boundary comparison.

Lemma 11.11.1.1 (First-letter products from equal boundary restrictions). [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$ and $L_0 \leq M$. Let ψ be a vector on $M+1$ sites, and suppose the two length- (L_0+1) boundary restrictions are represented by $\Gamma_{L_0+1}(Y_M(\tau_\eta^+(\mu)))$ and $\Gamma_{L_0+1}(Y_{M+1-L_0}(\tau_\eta^-(\mu)))$. If these two restrictions are equal, then for every physical letter j , $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$.

Proof. [✓ Proof](#) Restricting both sides to the first physical letter j gives

$$\Gamma_{L_0}(Y_M(\tau_\eta^+(\mu))A^j) = \Gamma_{L_0}(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j).$$

Since A is L_0 -block-injective, Γ_{L_0} is injective, and the displayed product equality follows. $\square$

Lemma 11.11.1.2 (Left-multiplied comparison from equal boundary restrictions). [Lean](#) [✓ Stmt](#) Let A be L_0 -block-injective with $L_0 > 0$ and $L_0 \leq M$. Let ψ be a vector on $M+1$ sites, and let $Y_i(\tau)$ represent the length- (L_0+1) cyclic restrictions of ψ . Assume that for every boundary letter η ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Then for every boundary letter η , physical letter j , and length- L_0 words α, σ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

Proof. ✓ **Proof** Lemma 11.11.1.1 gives $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$. Multiplying on the left by A^α and on the right by A^σ gives the desired comparison. $\square$

Theorem 11.11.1.3 (Left-multiplied comparison of boundary matrices). Lean ✓ **Stmt** Let A be L_0 -block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let $\psi = \Gamma_{M+1}(X)$ be an open-chain representation, and let $Y_i(\tau)$ represent the length- (L_0+1) cyclic restrictions of ψ . Fix a nonempty complementary word μ . Then for every boundary letter η , physical letter j , and length- L_0 words α, σ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

Proof. ✓ **Proof** The matrices $Y_i(\tau)$ give the local ground-space representations needed in Lemma 11.11.1.2. Once that boundary-restriction equality is proved, it gives

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Applying the left-multiplied comparison for equal boundary restrictions gives

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

$\square$

Theorem 11.11.1.4 (Left-multiplied coordinate comparison at the periodic boundary). Lean ✓ **Stmt** Let A be L_0 -block-injective with $L_0 > 0$, $L_0 < M$, and $D \geq 1$. Let $\psi = \Gamma_{M+1}(X)$ be an open-chain representation, and let $Y_i(\tau)$ represent the length- (L_0+1) cyclic restrictions of ψ . Fix a nonempty complementary word μ . Then for every boundary letter η , physical letter j , and length- L_0 words α, σ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Proof. ✓ **Proof** Once the comparison of the two cyclic restrictions is available, it gives

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

The one-sided equation for the boundary-crossing window beginning at M gives $Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$. Substitution gives the displayed identity. $\square$

Lemma 11.11.1.5 (Letterwise comparison of boundary-crossing restrictions). Lean ✓ **Stmt** Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, let $L_0 < M$, and let $\psi \in \mathcal{G}_{M+1, L_0+1}(A)$ with $\psi = \Gamma_{M+1}(X)$. Let R_j denote restriction to the first physical letter j . For every outside letter η , complementary word μ , and physical letter j , one has

$$R_j \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = R_j \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

This provides the single-letter product form of the periodic-boundary comparison in [16, Section IV.C, lines 2078–2079].

Proof. ✓ **Proof** Let $Y_i(\tau)$ be the representation matrices supplied by Lemma 11.9.7. By Theorem 11.11.1.8, $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$. Fixing the first physical letter in the two length- (L_0+1) windows gives

$$\begin{aligned} R_j \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) &= \Gamma_{L_0}(Y_M(\tau_\eta^+(\mu))A^j), \\ R_j \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi) &= \Gamma_{L_0}(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j). \end{aligned}$$

The two restrictions are therefore equal. $\square$

Theorem 11.11.1.6 (Boundary-matrix product for the two boundary-crossing restrictions).

Lean ✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$. Let ψ be a state on the chain of length $M + 1$ with open-chain representation $\psi = \Gamma_{M+1}(X)$. Let $Y_i(\tau)$ be matrices satisfying $\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. If $L_0 < M$, then for every nonempty complementary word μ there are boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$, depending on the physical letter j and the word σ of length L_0 , such that, for every complementary position k ,

$$\begin{aligned}\rho_{j,\sigma}^+(k + L_0) &= \mu_k, \\ \rho_{j,\sigma}^-(k + 1) &= \mu_k,\end{aligned}$$

and, for every j and σ ,

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

Proof. ✓ Proof Theorem 11.11.1.4 gives, for every boundary letter η , physical letter j , and words α, σ of length L_0 ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Theorem 11.11.22 applies this left-multiplied comparison together with the one-sided boundary products and gives the displayed boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$, including

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

□

Theorem 11.11.1.7 (Product equality at the periodic boundary). Lean ✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$. Let $\psi \in \mathcal{G}_{M+1, L_0+1}(A)$ have an open-chain representation $\psi = \Gamma_{M+1}(X)$. Let $Y_i(\tau)$ be matrices satisfying $\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$. If $L_0 < M$, then for every boundary letter η , complementary word μ , physical letter j , and word σ of length L_0 ,

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma.$$

Proof. ✓ Proof Theorem 11.11.1.6 gives boundary conditions $\rho_{j,\sigma}^+$ and $\rho_{j,\sigma}^-$ satisfying

$$\begin{aligned}\rho_{j,\sigma}^+(k + L_0) &= \mu_k, \\ \rho_{j,\sigma}^-(k + 1) &= \mu_k,\end{aligned}$$

and

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

Theorem 11.8.9 replaces these auxiliary conditions by the displayed boundary conditions:

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma,$$

which is the product equality needed for the periodic-boundary comparison. □

Theorem 11.11.1.8 (Right-product equality at the periodic boundary). Lean ✓ Stmt Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$. Let $\psi \in \mathcal{G}_{M+1, L_0+1}(A)$ have an open-chain representation $\psi = \Gamma_{M+1}(X)$. Let $Y_i(\tau)$ be matrices satisfying $\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau))$.

If $L_0 < M$, then for every letter η used in the two boundary conditions, every word μ on the complementary sites, and every physical letter j ,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

This is the one-site product equality used to compare the two boundary matrices in the periodic-boundary comparison.

Proof. ✓ **Proof** Fix j and set

$$Z_j = Y_M(\tau_\eta^+(\mu))A^j - Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Theorem 11.11.1.7 gives, after subtracting the two sides, $Z_j A^\sigma = 0$ for every word σ of length L_0 . Since A is L_0 -block-injective, $\text{span}\{A^\sigma : |\sigma| = L_0\} = M_D(\mathbb{C})$. Applying Lemma 11.10.6 gives $Z_j = 0$, hence

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

□

Theorem 11.11.1.9 (Equality of the two boundary-matrix families). Lean ✓ **Stmt** Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, let $N \geq 2$, $L_0 + 1 < N$, and $L_0 < L \leq N$, and let $\psi \in \mathcal{G}_{N,L}(A)$ have an open-chain representation $\psi = \Gamma_N(X)$. Suppose Y^+ and Y^- are two families of matrices obtained from the two boundary-crossing cyclic-window constraints, and that they satisfy, for every physical letter j and every boundary condition τ ,

$$\begin{aligned} A^{\tau_{L_0} \cdots \tau_{N-2}} A^j X &= Y_\tau^+ A^j, \\ X A^j A^{\tau_1 \cdots \tau_{N-L_0-1}} &= A^j Y_\tau^-. \end{aligned}$$

For every physical letter $\eta \in \{0, \dots, d-1\}$ and every word μ on the complementary sites, define two boundary conditions by

$$\begin{aligned} \tau_\eta^+(\mu)_i &= \begin{cases} \mu_{i-L_0}, & L_0 \leq i < N-1, \\ \eta, & \text{otherwise,} \end{cases} \\ \tau_\eta^-(\mu)_i &= \begin{cases} \mu_{i-1}, & 1 \leq i < N-L_0, \\ \eta, & \text{otherwise.} \end{cases} \end{aligned}$$

Then the two families agree on these two boundary conditions: $Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$. This is the boundary-matrix comparison required for equality of the two restrictions crossing the periodic cut.

Proof. ✓ **Proof** The reduced cyclic-window compatibilities at the boundary give, for every physical letter j ,

$$\begin{aligned} A^\mu A^j X &= Y_{\tau_\eta^+(\mu)}^+ A^j, \\ X A^j A^\mu &= A^j Y_{\tau_\eta^-(\mu)}^-. \end{aligned}$$

By Lemma 11.11.9, it is enough to prove, for every j ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j.$$

Write $N = M + 1$. The two boundary-crossing supports start at M and $M + 1 - L_0$. Let $Y_i(\tau)$ denote the cyclic-window representation matrix given by Lemma 11.9.7 at the window beginning at i . The first step identifies the two abstract boundary-condition families with the two boundary-restriction matrices:

$$\begin{aligned} Y_{\tau_\eta^+(\mu)}^+ &= Y_M(\tau_\eta^+(\mu)), \\ Y_{\tau_\eta^-(\mu)}^- &= Y_{M+1-L_0}(\tau_\eta^-(\mu)). \end{aligned} \quad (11.229)$$

Together with (11.229), Theorem 11.11.1.8 gives, for every physical letter j ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j.$$

Lemma 11.11.9 then gives the desired equality of boundary matrices. $\square$

Theorem 11.11.1.10 (Change of cut on the full ring). [Lean](#) [✓ Stmt](#) *Let A be any tensor. If*

$$T_{L_0}(\Gamma_{L_0+1}(X)) = \Gamma_{L_0+1}(Y),$$

then $\Gamma_{L_0}(XA^j) = \Gamma_{L_0}(A^jY)$ for every physical letter j .

Proof. [✓ Proof](#) Evaluate the translated equality on a word βj , where $|\beta| = L_0$. Cyclic translation exchanges the two parts of the word, and cyclicity of the trace gives

$$\Gamma_{L_0}(XA^j)(\beta) = \text{tr}(A^j A^\beta X) = \text{tr}(A^\beta A^j Y) = \Gamma_{L_0}(A^j Y)(\beta).$$

$\square$

Theorem 11.11.1.11 (One-step full-ring boundary comparison). [Lean](#) [✓ Stmt](#) *Let A be L_0 -block-injective with $L_0 > 0$. If*

$$T_{L_0}(\Gamma_{L_0+1}(X)) = \Gamma_{L_0+1}(Y),$$

then $XA^j = A^jY$ for every physical letter j .

Proof. [✓ Proof](#) Theorem 11.11.1.10 gives

$$\Gamma_{L_0}(XA^j) = \Gamma_{L_0}(A^jY),$$

and injectivity of Γ_{L_0} gives $XA^j = A^jY$. $\square$

Theorem 11.11.1.12 (Full-ring boundary comparison). [Lean](#) [✓ Stmt](#) *Let A be L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, and let $\psi \in \mathcal{G}_{L_0+1, L_0+1}(A)$ satisfy $\psi = \Gamma_{L_0+1}(X)$. Then $XA^j = A^jX$ for every physical letter j .*

Proof. [✓ Proof](#) Write s for the cut shift by L_0 and Y_i for the boundary matrix of the translated state at cut i . Translation invariance of the chain space, the cyclic open-chain reduction, and the composition law supply compatible matrices at successive cuts. Theorem 11.11.1.11 gives the single-letter comparison, and induction on the word w gives

$$Y_i A^w = A^w Y_{|w|s+i}.$$

Injectivity of the propagated length- $(L_0 + 1)$ boundary map gives $Y_0 = X$. Since $(L_0 + 1)s = 0$, every full-ring word ω satisfies $XA^\omega = A^\omega X$. Lemma 11.10.8 then gives $XA^j = A^jX$ for every letter j . $\square$

Theorem 11.11.1.13 (Minimal-ring ground-space containment). [Lean](#) [✓ Stmt](#) *If A is L_0 -block-injective with $L_0 > 0$ and $D \geq 1$, then*

$$\mathcal{G}_{L_0+1, L_0+1}(A) \subseteq \text{span}\{V^{(L_0+1)}(A)\}.$$

Proof. [✓ Proof](#) Let $\psi \in \mathcal{G}_{L_0+1, L_0+1}(A)$. The cyclic open-chain reduction gives $\psi = \Gamma_{L_0+1}(X)$. Theorem 11.11.1.2 shows that X commutes with every letter, and hence with every word of length L_0 . The long-word commutation theorem then places ψ in $\text{span}\{V^{(L_0+1)}(A)\}$. $\square$

11.11.2 Normal unique-ground-state consequences

For injective tensors with $N \geq 2$ and $1 < L \leq N$, the chain ground space equals $\text{span}\{V^{(N)}(A)\}$, ensuring a unique ground state for the parent Hamiltonian. For normal tensors, uniqueness holds for $N \geq 2$, $L_0 + 1 \leq N$, and $L_0 < L \leq N$ by combining the boundary-crossing comparison with the open-chain range reduction. In the minimal-ring case $N = L_0 + 1$, the full-ring change-of-cut theorem replaces the complementary-word comparison.

Theorem 11.11.2.1 (Chain ground space is spanned by the MPS vector in the injective case). [Lean](#) [✓ Stmt](#) *If A is injective, $D \geq 1$, $N \geq 2$, and $1 < L \leq N$, then $\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}$.*

Proof. [✓ Proof](#) Let $\psi \in \mathcal{G}_{N,L}(A)$. The non-wrapping window conditions and Lemma 11.5.5.4 imply $\psi \in G_N(A)$, so $\psi = \Gamma_N(X)$ for some boundary matrix X . The periodic boundary condition now applies Theorem 11.8.16 and forces X to commute with every letter A^j . Since A is injective, the letters generate the full matrix algebra, hence X is scalar. Therefore ψ is proportional to $V^{(N)}(A)$, while the reverse inclusion is Lemma 11.9.8. $\square$

Theorem 11.11.2.2 (Normal tensor: $\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}$). [Lean](#) [✓ Stmt](#) *If A is normal, $D \geq 1$, and injective after blocking $L_0 > 0$ sites, with $N \geq 2$, $L_0 + 1 \leq N$, and $L_0 < L \leq N$, then $\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}$.*

Proof. [✓ Proof](#) If $N = L_0 + 1$, range antitonicity gives

$$\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N, L_0+1}(A),$$

and Theorem 11.11.1.13 applies to the minimal ring. If $N > L_0 + 1$, the open-chain containment gives $\psi = \Gamma_N(X)$. The two boundary-crossing cyclic-window constraints give

$$\begin{aligned} A^\mu A^j X &= Y_{\tau_\eta^+(\mu)}^+ A^j, \\ X A^j A^\mu &= A^j Y_{\tau_\eta^-(\mu)}^-. \end{aligned}$$

The boundary-crossing comparison is $Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-$. These equations imply $\psi \in \text{span}\{V^{(N)}(A)\}$. $\square$

Theorem 11.11.2.3 (Chain ground space is spanned by the MPS vector in the normal case). [Lean](#) [✓ Stmt](#) *If A is normal, $D \geq 1$, and injective after blocking $L_0 > 0$ sites, with $N \geq 2$, $L_0 + 1 \leq N$, and $L_0 < L \leq N$, then $\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}$.*

Theorem 11.11.2.4 (Unique ground state throughout the block-injective interaction range). [Lean](#) [✓ Stmt](#) *If $D \geq 1$, A is injective after blocking $L_0 > 0$ sites, and $L_0 < L \leq N$, then the parent Hamiltonian of interaction range L has a unique periodic ground state. These inequalities imply $N \geq 2$ and $N \geq L_0 + 1$, extending the finite-size range $N \geq 2L_0$ of [193, Theorem 10].*

Proof. ✓ **Proof** Block injectivity at the positive length L_0 makes A normal. Hence Theorem 11.11.2.3 gives

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}. \quad (11.230)$$

The block-injectivity nonvanishing theorem gives

$$V^{(N)}(A) \neq 0. \quad (11.231)$$

Therefore

$$\dim \mathcal{G}_{N,L}(A) = \dim \text{span}\{V^{(N)}(A)\} = 1. \quad (11.232)$$

□

Theorem 11.11.2.5 (Unique ground state on the periodic chain). Lean ✓ **Stmt** *For an injective tensor A with $D \geq 1$, if $N \geq 2$ and $1 < L \leq N$, then the periodic chain ground space is one-dimensional.*

Proof. ✓ **Proof** By Theorem 11.11.2.1, the ground space is the line $\text{span}\{V^{(N)}(A)\}$. It remains to show that the MPV is nonzero. If $V^{(N)}(A) = 0$, then Lemma 11.9.2 gives $\Gamma_N(\mathbb{1}) = \Gamma_N(0)$. Since A is injective and $N > 0$, Theorem 11.5.3.1 forces $\mathbb{1} = 0$, a contradiction. Therefore the spanning line is one-dimensional. □

Theorem 11.11.2.6 (Unique ground state after blocking). Lean ✓ **Stmt** *If A becomes injective after blocking $L_0 > 0$ sites and $D \geq 1$, the parent Hamiltonian with interaction range $2L_0$ has a unique ground state on every periodic chain with $N \geq 2L_0$. In particular, this includes $L_0 = 1$ and $N = 2$.*

Proof. ✓ **Proof** Since $L_0 > 0$, one has $L_0 < 2L_0$. The result is therefore the case $L = 2L_0$ of Theorem 11.11.2.4. □

Theorem 11.11.2.7 (Unique ground state for normal tensors at range $L_0 + 1$). Lean ✓ **Stmt** *If A is normal, becomes injective after blocking $L_0 > 0$ sites, and $D \geq 1$, then the parent Hamiltonian with interaction range $L_0 + 1$ has a unique ground state on every periodic chain with $L_0 + 1 \leq N$.*

Proof. ✓ **Proof** The inequalities $L_0 < L_0 + 1 \leq N$ put this statement in the range of Theorem 11.11.2.4. □

11.12 Half-chain Schmidt spectrum and the dual fixed point

On a ring of $2L$ sites split into two halves of L sites each, the matrix product vector splits at the two boundaries into a pair of half-chain families whose overlaps are matrix elements of the L -fold transfer map [193, Theorem 6]. The unnormalized half-chain reduced density matrix factors through the half-chain Gram matrix. Away from zero, its spectrum matches that of a boundary comparison matrix of size D^2 , which, under the spectral gap condition $\rho_{\text{spec}}(\mathcal{E}_A - P_\Lambda) < 1$, converges entrywise, geometrically in L , to the diagonal matrix with entries $\lambda_\alpha \lambda_\beta / \text{tr}(\Lambda)^2$ determined by the fixed point Λ .

The source states its canonical form with a trace-preserving dual map and a positive diagonal fixed point Λ of that dual. Here the trace-preserving direction is placed on the transfer map itself and Λ is its fixed point; the two conventions are exchanged by passing to the conjugate-transposed tensor.

Definition 11.12.1 (Half-chain boundary families). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a tensor A and a half-chain length L , the *half-chain family* is the matrix Ψ with rows indexed by configurations $\sigma \in \{0, \dots, d-1\}^L$ and columns indexed by boundary pairs (α, β) , whose column at (α, β) is the half-chain vector

$$|\Psi_{\alpha, \beta}\rangle = \sum_{\sigma} \langle \alpha | A^{\sigma} | \beta \rangle |\sigma\rangle. \quad (11.233)$$

The *complementary family* Φ is the matrix with rows indexed by boundary pairs and columns by configurations, whose row at (α, β) is the vector $|\Psi_{\beta, \alpha}\rangle$ with the two boundary indices swapped.

Definition 11.12.2 (Half-chain coefficient kernel). [Lean](#) [Lean](#) [✓ Stmt](#) The *coefficient kernel* of the ring of $2L$ sites is the matrix over the two half-chain configuration spaces whose entry at (σ, τ) is the matrix product vector component $V^{(2L)}(A)_{\sigma\tau} = \text{tr}(A^{\sigma} A^{\tau})$ of the joined configuration.

Lemma 11.12.3 (Splitting of the ring state at the two boundaries). [Lean](#) [✓ Stmt](#) The coefficient kernel factors as the product $\Psi\Phi$ of the half-chain family with its complementary family. Equivalently,

$$|\psi\rangle = \sum_{\alpha, \beta} |\Psi_{\alpha, \beta}\rangle |\Psi_{\beta, \alpha}\rangle. \quad (11.234)$$

Proof. [✓ Proof](#) Insert the resolution of the identity $\sum_{\beta} |\beta\rangle\langle\beta|$ at the two cut bonds of the ring trace: $\text{tr}(A^{\sigma} A^{\tau}) = \sum_{\alpha, \beta} \langle \alpha | A^{\sigma} | \beta \rangle \langle \beta | A^{\tau} | \alpha \rangle$. $\square$

Definition 11.12.4 (Half-chain reduced density matrix). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The *unnormalized half-chain reduced density matrix* of the ring state on $2L$ sites is $\rho_L = KK^{\dagger}$ with K the coefficient kernel. Entrywise this is the partial trace of the unnormalized ring state over the complementary half,

$$(\rho_L)_{\sigma\sigma'} = \sum_{\tau} V^{(2L)}(A)_{\sigma\tau} \overline{V^{(2L)}(A)_{\sigma'\tau}}, \quad (11.235)$$

and ρ_L is positive semidefinite.

Definition 11.12.5 (Half-chain Gram matrix). [Lean](#) [✓ Stmt](#) The *half-chain Gram matrix* is $G = \Psi^{\dagger}\Psi$, the matrix of overlaps $G_{(\alpha, \beta), (\alpha', \beta')} = \langle \Psi_{\alpha, \beta} | \Psi_{\alpha', \beta'} \rangle$ of the half-chain vectors.

Lemma 11.12.6 (Half-chain overlaps through the transfer map). [Lean](#) [✓ Stmt](#) The half-chain overlaps are matrix elements of the L -fold transfer map on matrix units:

$$\langle \Psi_{\alpha, \beta} | \Psi_{\alpha', \beta'} \rangle = \langle \alpha' | \mathcal{E}_A^L(|\beta'\rangle\langle\beta|) | \alpha \rangle. \quad (11.236)$$

Proof. [✓ Proof](#) Expand the overlap as a sum over configurations of products $\langle \alpha' | A^{\sigma} | \beta' \rangle \overline{\langle \alpha | A^{\sigma} | \beta \rangle}$ and recognize the entry of $\sum_{\sigma} A^{\sigma} | \beta' \rangle \langle \beta | (A^{\sigma})^{\dagger}$, which is the L -fold transfer map applied to the matrix unit. $\square$

Theorem 11.12.7 (Half-chain Schmidt factorization). [Lean](#) [Lean](#) [✓ Stmt](#) Let S denote the swap of the two boundary indices. The unnormalized half-chain reduced density matrix factors exactly as

$$\rho_L = \Psi (SG^{\dagger}S) \Psi^{\dagger}, \quad (11.237)$$

where $SG^{\dagger}S$ is the Gram matrix of the complementary family.

Proof. ✓ Proof By the splitting $K = \Psi\Phi$ of Lemma 11.12.3, $\rho_L = \Psi(\Phi\Phi^\dagger)\Psi^\dagger$, and the Gram matrix $\Phi\Phi^\dagger$ of the complementary family is the transpose of the boundary-swapped half-chain Gram matrix. $\square$

Theorem 11.12.8 (Boundary comparison of the half-chain spectrum). Lean ✓ Stmt *The characteristic polynomials of the half-chain reduced density matrix and of the boundary comparison matrix $(SG^\top S)G$ of size D^2 agree after matching the trivial kernel factors:*

$$X^{D^2} \chi_{\rho_L}(X) = X^{d^L} \chi_{(SG^\top S)G}(X). \quad (11.238)$$

In particular the nonzero spectrum of ρ_L , with multiplicities, is that of the boundary comparison matrix.

Proof. ✓ Proof Apply the rectangular cyclicity of the characteristic polynomial to the factorization (11.237), regrouping the product as Ψ times $(SG^\top S)\Psi^\dagger$; the reversed product is $(SG^\top S)\Psi^\dagger\Psi = (SG^\top S)G$. $\square$

Theorem 11.12.9 (Geometric transfer error of the half-chain overlaps). Lean ✓ Stmt *Let A have trace-preserving transfer map $\mathcal{E}_A$ with fixed point Λ , $\text{tr}(\Lambda) \neq 0$, and suppose*

$$\rho_{\text{spec}}(\mathcal{E}_A - P_\Lambda) < 1, \quad (11.239)$$

where P_Λ is the fixed-point projection. Then there are constants $C > 0$ and $0 < r < 1$ such that for every $L \geq 1$ and all boundary pairs,

$$\left| G_{(\alpha, \beta), (\alpha', \beta')} - \langle \alpha' | P_\Lambda (|\beta'\rangle\langle\beta|) | \alpha \rangle \right| \leq Cr^L. \quad (11.240)$$

The hypothesis (11.239) is the spectral gap condition of [193, Theorem 6] in the trace-preserving gauge: apart from the simple eigenvalue one, the spectrum of $\mathcal{E}_A$ lies strictly inside the unit disc. The rate r is chosen strictly between the spectral radius of $\mathcal{E}_A - P_\Lambda$ and one.

Proof. ✓ Proof Write $\mathcal{E}_A^L = P_\Lambda + (\mathcal{E}_A - P_\Lambda)^L$ for $L \geq 1$ by the power decomposition. By (11.236) the deviation of a Gram entry from the fixed-point projection entry is an entry of $(\mathcal{E}_A - P_\Lambda)^L$ applied to a matrix unit, bounded by the operator norm of the L -th power, which decays geometrically below unit spectral radius. $\square$

Lemma 11.12.10 (Diagonal limit of the half-chain overlaps). Lean ✓ Stmt *For a diagonal fixed point $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_D)$ with $\text{tr}(\Lambda) \neq 0$, the limiting overlaps form a diagonal matrix in the boundary pairs:*

$$\langle \alpha' | P_\Lambda (|\beta'\rangle\langle\beta|) | \alpha \rangle = \frac{\lambda_\alpha}{\text{tr}(\Lambda)} \delta_{\alpha, \alpha'} \delta_{\beta, \beta'}. \quad (11.241)$$

For $\text{tr}(\Lambda) = 1$ this is the display $\lambda_\alpha \delta_{\alpha, \alpha'} \delta_{\beta, \beta'}$ of [193].

Proof. ✓ Proof Evaluate the fixed-point projection on the matrix unit: its trace is $\delta_{\beta, \beta'}$, and the diagonal entry of Λ at (α', α) is $\lambda_\alpha \delta_{\alpha, \alpha'}$. $\square$

Theorem 11.12.11 (Geometric convergence to the doubled fixed-point spectrum). Lean ✓ Stmt *Let A have trace-preserving transfer map with diagonal fixed point $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_D)$, $\text{tr}(\Lambda) \neq 0$, satisfying the spectral condition (11.239). Then there are constants $C > 0$ and $0 < r < 1$ such that for every $L \geq 1$ the boundary comparison matrix satisfies, entrywise,*

$$\left| ((SG^\top S)G)_{(\alpha, \beta), (\alpha', \beta')} - \frac{\lambda_\alpha \lambda_\beta}{\text{tr}(\Lambda)^2} \delta_{\alpha, \alpha'} \delta_{\beta, \beta'} \right| \leq Cr^L. \quad (11.242)$$

The limiting matrix is the diagonal matrix $\Lambda \otimes \Lambda$ up to the trace normalization. Together with Theorem 11.12.8, the nonzero spectrum of the half-chain reduced density matrix on the ring of $2L$ sites is that of a $D^2 \times D^2$ matrix converging entrywise, geometrically in L , to $\Lambda \otimes \Lambda / \text{tr}(\Lambda)^2$. The remaining passage to the convergence of the eigenvalues themselves is the continuity of the spectrum of a matrix in its entries and is not part of this section.

Proof. ✓ Proof Both factors of the comparison matrix are entrywise within Cr^L of their diagonal limits by Theorem 11.12.9 and Lemma 11.12.10, the swapped-transposed factor because swapping and transposing permute entries and exchange the two diagonal weights. The limit entries are bounded, so the product deviates from the product of the limits by at most a constant multiple of r^L , with constant $D^2(C(B+C) + BC) + 1$ for an entry bound B on the limits. $\square$

Commuting Parent Hamiltonians and Spectral Gaps

12.1 Commuting Local Terms for Parent Hamiltonians

Definition 12.1.1 (Three-coordinate identification). [Lean](#) [✓ Stmt](#) For every type α , evaluation at the three coordinates gives the canonical equivalence

Its forward map is

Definition 12.1.2 (Overlapping two-site supports). On a three-site space $\mathcal{H}_A \otimes \mathcal{H}_X \otimes \mathcal{H}_B$, let Q_{AX} and Q_{XB} be two-site projectors. Their local actions on the three-site space are the operators

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The algebraic part of the overlapping nearest-neighbor commutation condition is

$$\begin{aligned} Q_{AX}^2 &= Q_{AX}, \\ Q_{XB}^2 &= Q_{XB}, \\ Q_{AX}Q_{XB} &= Q_{XB}Q_{AX}, \end{aligned}$$

after these two local actions are viewed on the common three-site space. This is the local-support part of the parent commuting Hamiltonian condition in [11, Definition D.2]. The cyclic windows beginning at i and $i + 1$ realize the two supports AX and XB , and hence overlap at the middle site.

The source local condition also fixes a three-site subspace K_{AXB} and imposes the kernel-intersection equation $K_{AXB} = \ker((Q_{AX})_{AX}) \cap \ker((Q_{XB})_{XB})$. Forgetting this intersection equation gives the algebraic overlapping two-site commutation condition displayed above.

The two lift maps also distribute over complementary projectors:

$$\begin{aligned} (1 - Q_{AX})_{AX} &= 1 - (Q_{AX})_{AX}, \\ (1 - Q_{XB})_{XB} &= 1 - (Q_{XB})_{XB}. \end{aligned}$$

Consequently, if the projectors Q_{AX} and Q_{XB} commute on the three-site space, then their complementary projectors $P_{AX} = 1 - Q_{AX}$ and $P_{XB} = 1 - Q_{XB}$ commute after the same lifts. This is the elementary algebraic passage used when one moves between the Q -projectors of [11, Appendix D.2] and the complementary ground-space projectors P . Since the original two projectors are idempotent, the complementary projectors are idempotent as well; hence the pair P_{AX}, P_{XB} satisfies the same idempotent-and-commutation condition on the three-site space.

Theorem 12.1.3 (Matrices of the two adjacent factor actions). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{H} = \mathbb{C}^d$, and let Q be an endomorphism of $\mathcal{H} \otimes \mathcal{H}$. Under the canonical identifications of two- and three-site configurations with pairs and right-associated triples, the matrices of its two adjacent actions are*

$$\begin{aligned} [Q_{AX}] &= [Q] \otimes I_d, \\ [Q_{XB}] &= I_d \otimes [Q]. \end{aligned}$$

Equivalently, for all $a, x, b, a', x', b' \in \{0, \dots, d-1\}$,

$$\begin{aligned} [Q_{AX}]_{(a,x,b),(a',x',b')} &= [Q]_{(a,x),(a',x')} \delta_{b,b'}, \\ [Q_{XB}]_{(a,x,b),(a',x',b')} &= \delta_{a,a'} [Q]_{(x,b),(x',b')}. \end{aligned}$$

This is the homogeneous three-single-site specialization of the factor actions on $\mathcal{H}_A \otimes \mathcal{H}_X \otimes \mathcal{H}_B$ in [11, Definition D.2]; here $\mathcal{H}_A = \mathcal{H}_X = \mathcal{H}_B = \mathcal{H}$. No assertion is made for unequal factor spaces.

Proof. [✓ Proof](#) On the first pair, the final coordinate is unchanged, which contributes $\delta_{b,b'}$ to the matrix coefficient. On the final pair, the first coordinate is unchanged, which contributes $\delta_{a,a'}$. $\square$

Theorem 12.1.4 (Three-site translated local terms). [Lean](#) [Lean](#) [✓ Stmt](#) *Put $q(A) = 1 - P_{G_2(A)}$ for the canonical two-site parent interaction. On the three-site space $\mathcal{H}_A \otimes \mathcal{H}_X \otimes \mathcal{H}_B$, with sites $0, 1, 2$ labelled A, X, B , the two adjacent translated length-two parent interactions are the two local actions*

$$\begin{aligned} h_0^{(3)}(A, 2) &= q(A)_{AX}, \\ h_1^{(3)}(A, 2) &= q(A)_{XB}. \end{aligned}$$

This identifies the two translated parent interactions with the local actions in [11, Definition D.2].

Proof. ✓ **Proof** Let $\sigma = (a, x, b)$ be a three-site configuration. For the first translated term, the length-two window and the corresponding replacement satisfy

$$\begin{aligned}\sigma_{\{0,1\}} &= (a, x) = \sigma_{AX}, \\ \sigma^{\{0,1\} \leftarrow (\alpha, \xi)} &= (\alpha, \xi, b) = \sigma^{AX \leftarrow (\alpha, \xi)}.\end{aligned}$$

For the second translated term,

$$\begin{aligned}\sigma_{\{1,2\}} &= (x, b) = \sigma_{XB}, \\ \sigma^{\{1,2\} \leftarrow (\xi, \beta)} &= (a, \xi, \beta) = \sigma^{XB \leftarrow (\xi, \beta)}.\end{aligned}$$

Substituting these two pairs of identities into the definition of the translated parent term gives the two displayed equations. $\square$

Theorem 12.1.5 (Three-site commutation from overlapping supports). Lean ✓ **Stmt** Suppose the canonical two-site parent interaction $q(A) = 1 - P_{G_2(A)}$ satisfies the overlapping AX/XB commutation condition of Definition 12.1.2. Then, on the three-site chain, $h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2)$. This only identifies the first two translated local terms with the lifted AX and XB actions; the construction of the [11, Appendix D.2] projectors associated with the [11, Appendix B] basic-vector form is a separate step.

Proof. ✓ **Proof** By Theorem 12.1.4, the two translated local terms are $q(A)_{AX}$ and $q(A)_{XB}$. The assumed overlapping support condition says exactly that these two lifted endomorphisms commute. $\square$

Theorem 12.1.6 (Three-site commutation from two-site representatives). Lean ✓ **Stmt** Let Q_{AX} and Q_{XB} be two-site projectors whose lifted AX and XB actions commute. Suppose that $q(A) = Q_{AX}$ and $q(A) = Q_{XB}$. Then the first two translated length-two local terms on the three-site chain commute.

Proof. ✓ **Proof** The lift identities identify the translated terms with the lifted actions of the two projectors:

$$\begin{aligned}h_0^{(3)}(A, 2) &= (Q_{AX})_{AX}, \\ h_1^{(3)}(A, 2) &= (Q_{XB})_{XB}.\end{aligned}$$

The overlapping-support hypothesis is the equation $(Q_{AX})_{AX}(Q_{XB})_{XB} = (Q_{XB})_{XB}(Q_{AX})_{AX}$. Substituting the two lifted identities gives the desired commutation of $h_0^{(3)}(A, 2)$ and $h_1^{(3)}(A, 2)$. $\square$

Theorem 12.1.7 (Three-site commutation from source projectors). Lean ✓ **Stmt** Suppose the projectors Q_{AX} and Q_{XB} satisfy the [11, Definition D.2] local parent-commuting condition on AXB , and suppose that the canonical two-site parent interaction is identified, as a two-site operator, with the two source projectors: $q(A) = Q_{AX}$ and $q(A) = Q_{XB}$. After lifting these two equalities to the AX and XB faces, the first two translated length-two local terms on the three-site chain commute. Thus, $h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2)$.

Proof. ✓ **Proof** The lift identities of Theorem 12.1.4 identify the translated terms with $(Q_{AX})_{AX}$ and $(Q_{XB})_{XB}$. After forgetting the kernel-intersection equation, the [11, Definition D.2] condition says exactly that these two lifted projectors commute. $\square$

Theorem 12.1.8 (Three-site commutation from complementary Appendix projectors). [Lean](#)

[Lean](#) [✓ Stmt](#) Suppose the Appendix D.2 projectors Q_{AX} and Q_{XB} satisfy the local parent-commuting condition on AXB , and suppose that the canonical two-site parent interaction is their complement on both faces: $q(A) = 1 - Q_{AX} = 1 - Q_{XB}$. Then the first two translated length-two local terms on the three-site chain commute, $h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2)$.

Proof. [✓ Proof](#) The lift identities of Theorem 12.1.4 identify the translated terms with $(1 - Q_{AX})_{AX}$ and $(1 - Q_{XB})_{XB}$. The Appendix D.2 condition implies the overlapping two-site commutation condition after forgetting the kernel-intersection equation. Definition 12.1.2 then gives commutation of the complementary lifted projectors, hence the displayed commutation relation. $\square$

Definition 12.1.9 (Commuting local terms for a parent Hamiltonian). [Lean](#) [✓ Stmt](#) The parent Hamiltonian with block length L on N sites has *commuting* local terms if $h_i h_j = h_j h_i$ for all $i, j \in \{0, \dots, N-1\}$. This records the translated commutation equation $[\tau_j(P_L), P_L] = 0$ from [11, Definition 3.9]; the same source definition also requires the ground space to be spanned, for $N > L$, by the periodic MPS vectors of the BNT components.

Definition 12.1.10 (Nearest-neighbor commuting parent Hamiltonian). [Lean](#) [✓ Stmt](#) The nearest-neighbor commuting condition is the length-2 specialization of the preceding commutation equation. For every $i, j \in \{0, \dots, N-1\}$, one has $h_i h_j = h_j h_i$. The source definition [11, Definition 3.9] also requires the corresponding parent Hamiltonian to have $L = 2$ and to have the prescribed ground-space spanning property for $N > 2$.

Lemma 12.1.11 (Overlap reduction for commuting local terms). [Lean](#) [Lean](#) [✓ Stmt](#) Let $L \leq N$. Suppose that $h_i h_j = h_j h_i$ whenever the length- L cyclic supports of the two local terms meet. Then the parent Hamiltonian has commuting local terms: $h_i h_j = h_j h_i$ for all i, j . In the case $L = 2$, this gives the nearest-neighbor commuting condition.

Proof. [✓ Proof](#) If the two cyclic supports meet, the hypothesis gives the commutation relation. If they are disjoint, the corresponding local terms commute by locality. $\square$

Lemma 12.1.12 (Length-two cyclic-overlap alternatives). [Lean](#) [✓ Stmt](#) For $i, j \in \mathbb{Z}/N\mathbb{Z}$, if the two length-two cyclic supports starting at i and j overlap, then

$$\begin{aligned} & j = i, \\ \text{or} \quad & j \equiv i + 1 \pmod{N}, \\ \text{or} \quad & i \equiv j + 1 \pmod{N}. \end{aligned} \tag{12.3}$$

Proof. [✓ Proof](#) Expanding each length-two cyclic support gives shifts $r_i, r_j \in \{0, 1\}$ and an equality of sites $i + r_i \equiv j + r_j \pmod{N}$. The four possible pairs (r_i, r_j) give exactly the three alternatives in (12.3). $\square$

Lemma 12.1.13 (Adjacent two-site translates suffice for nearest-neighbor commutation). [Lean](#) [✓ Stmt](#) Let $N \geq 2$. Suppose that, for every $i \in \mathbb{Z}/N\mathbb{Z}$, the translated length-two local term commutes with its cyclic right neighbor:

$$h_i(A, 2)h_{i+1}(A, 2) = h_{i+1}(A, 2)h_i(A, 2). \tag{12.4}$$

Then all translated length-two local terms commute.

Proof. ✓ **Proof** By Lemma 12.1.12, overlapping length-two cyclic supports force the three alternatives in (12.3). The equal-start case gives $h_i(A, 2) = h_j(A, 2)$. The two nearest-neighbor cases are (12.4) and its transpose. The overlap-reduction lemma supplies the remaining disjoint pairs by locality. $\square$

Definition 12.1.14 (Commutation and zero-energy equations for nearest-neighbor local terms).

Lean ✓ **Stmnt** This condition records the commutation equation and the zero-energy equation for the nearest-neighbor local terms:

$$\begin{aligned} h_i h_j &= h_j h_i, \\ h_i V^{(N)}(A) &= 0. \end{aligned}$$

This is the zero-energy part of the condition appearing in [11, Theorem 3.10(iii)]: the MPS vector has zero energy for the translated two-site local terms, and those terms commute. The missing source assertion is that the whole parent-Hamiltonian ground space is spanned by the corresponding periodic MPS vectors from the BNT components, as recorded in Definition 12.1.15.

Definition 12.1.15 (Parent-Hamiltonian ground-space spanning condition). Lean ✓ **Stmnt** Let

B be a canonical-form tensor with BNT components $A_1, \dots, A_g$. The source parent-Hamiltonian spanning condition is that, for every $N > L$, $\ker H_L^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$. This is the ground-space clause in [11, Definition 3.9]. It is separate from the commutation equation $[\tau_j(P_L), P_L] = 0$ and from the zero-energy equation for a single periodic MPS vector.

Definition 12.1.16 (Nearest-neighbor commuting parent-Hamiltonian ground-space condition).

Lean ✓ **Stmnt** For a fixed chain length N , this condition is the conjunction of the nearest-neighbor commutation equation, the zero-energy equation for the MPS vector, and the ground-space spanning equation:

$$\begin{aligned} h_i h_j &= h_j h_i, \\ h_i V^{(N)}(B) &= 0, \\ \ker H_2^{(N)}(B) &= \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}. \end{aligned}$$

The source statement in [11, Theorem 3.10(iii)] quantifies this condition over all $N > 2$.

Definition 12.1.17 (All-chain nearest-neighbor commuting parent-Hamiltonian ground-space condition).

Lean ✓ **Stmnt** This is the source quantification in [11, Theorem 3.10(iii)]. For every $N > 2$, the fixed-chain condition of Definition 12.1.16 holds:

$$\begin{aligned} h_i h_j &= h_j h_i, \\ h_i V^{(N)}(B) &= 0, \\ \ker H_2^{(N)}(B) &= \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}. \end{aligned}$$

This condition records the ground-space spanning equation, but does not prove it for a concrete canonical-form tensor.

Remark 12.1.18 (Elementary consequences of the commuting definition). Lean Lean Lean

Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ **Stmnt** The nearest-neighbor condition is the length-2 specialization of the general commuting condition. The ground-vector condition combines this commuting equation with the frustration-free equation for $V^{(N)}(A)$. For $N \geq 2$, Lemma 11.4.47 with $L = 2$ gives $h_i V^{(N)}(A) = 0$ for every site i . The

fixed-chain ground-space condition also contains $\ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$, and quantifying this condition over all $N > 2$ gives the nearest-neighbor case of the spanning condition above. Conversely, the all-chain condition is exactly the conjunction of the all-chain ground-vector equations and the nearest-neighbor instance of the spanning condition above. This is precisely the condition named in Definition 12.1.17. Since the displayed zero-energy equation follows from the parent-Hamiltonian construction for $N > 2$, independently of the commutation and spanning equations, this is equivalently

$$\begin{aligned} \forall N > 2, \quad & \left[\begin{array}{l} \forall i, j, \quad h_i^{(N)} h_j^{(N)} = h_j^{(N)} h_i^{(N)}, \\ \forall i, \quad h_i^{(N)} V^{(N)}(B) = 0, \\ \ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \end{array} \right] \\ \iff & \left[\begin{array}{l} \forall N > 2, \quad \forall i, j, \quad h_i^{(N)} h_j^{(N)} = h_j^{(N)} h_i^{(N)}, \\ \forall N > 2, \quad \ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}. \end{array} \right]. \end{aligned}$$

Theorem 12.1.19 (Chain ground-space vectors are frustration-free). [Lean](#) [Stmt](#) Assume $N > 0$ and $L \leq N$. Let ψ be a vector in the periodic chain ground space $G_L^{(N)}(A)$. Then ψ satisfies all local zero-energy equations: for every site i , one has $h_i(A, L)\psi = 0$.

Proof. [Proof](#) Write $R_{i,\eta}\psi$ for the cyclic length- L restriction of ψ at site i , with the outside configuration fixed by η . Membership in $G_L^{(N)}(A)$ gives $R_{i,\eta}\psi \in G_L(A) = \ker h_L(A)$ for every i and η . Hence $h_L(A)R_{i,\eta}\psi = 0$. For a full configuration σ , taking $\eta = \sigma|_{\{i, \dots, i+L-1\}^c}$ gives

$$(h_i(A, L)\psi)(\sigma) = (h_L(A)R_{i,\eta}\psi)(\sigma|_{\{i, \dots, i+L-1\}}) = 0.$$

Thus $h_i(A, L)\psi = 0$ for every i . □

Lemma 12.1.20 (Zero-energy BNT vectors lie in the parent-Hamiltonian kernel). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let B be a canonical-form tensor with BNT components $A_1, \dots, A_g$. If every component vector has zero energy for the parent Hamiltonian, $H_L^{(N)}(B)V^{(N)}(A_j) = 0$ for $1 \leq j \leq g$, then

$$\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \subseteq \ker H_L^{(N)}(B).$$

It is enough to assume the local frustration-free equations $h_i V^{(N)}(A_j) = 0$ for every i and j . For $N > 0$ and $L \leq N$, in the block-diagonal case $B = \bigoplus_j \mu_j A_j$, with every $\mu_j \neq 0$, the same inclusion follows because each local space $G_L(A_j)$ is contained in $G_L(B)$. This establishes the forward inclusion $\text{span}\{V^{(N)}(A_j)\}_j \subseteq \ker H_L^{(N)}(B)$ of [11, Definition 3.9].

Proof. [Proof](#) The kernel is a linear subspace. Hence it is enough to check the generators $V^{(N)}(A_j)$. The first hypothesis gives this directly. Under the local frustration-free equations,

$$H_L^{(N)}(B)V^{(N)}(A_j) = \sum_i h_i V^{(N)}(A_j) = 0.$$

For a block-diagonal tensor, write $R_{i,\tau}$ for the cyclic length- L restriction at site i , with the outside configuration fixed by τ . Then $R_{i,\tau}V^{(N)}(A_j) \in G_L(A_j) \subseteq G_L(B)$. The preceding theorem therefore gives $h_i(B, L)V^{(N)}(A_j) = 0$ for every i . □

Lemma 12.1.21 (Kernel containment from chain-space splitting). [Lean](#) [✓ Stmt](#) Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$. Suppose $N > 0$, $L \leq N$, and the periodic chain constraints for B split into the block constraints:

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=0}^{r-1} \mathcal{G}_{N,L}(A_j).$$

If every block chain space is contained in its periodic MPS line,

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathbb{C}V^{(N)}(A_j) \quad (0 \leq j < r),$$

then the zero-energy space of the parent Hamiltonian is contained in the BNT vector span:

$$\ker H_L^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

Proof. [✓ Proof](#) Theorem 12.3.14 gives $\ker H_L^{(N)}(B) \subseteq \mathcal{G}_{N,L}(B)$. Together with the splitting hypothesis,

$$\ker H_L^{(N)}(B) \subseteq \mathcal{G}_{N,L}(B) \subseteq \bigvee_j \mathcal{G}_{N,L}(A_j).$$

The blockwise containments give

$$\bigvee_j \mathcal{G}_{N,L}(A_j) \subseteq \bigvee_j \mathbb{C}V^{(N)}(A_j),$$

and Lemma 13.1.2 gives

$$\bigvee_j \mathbb{C}V^{(N)}(A_j) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

□

Lemma 12.1.22 (Kernel containment from the block-diagonal boundary comparison). [Lean](#) [✓ Stmt](#) Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$, with every $\mu_j \neq 0$. Suppose the physical alphabet and all block dimensions are nonempty. Assume that the block family is irreducible, left-canonical, has normalized self-overlap, and has no two blocks gauge-phase equivalent. Suppose also that each A_j is L_0 -block-injective, that $L_0 > 0$, and that $\sum_a A_a^j (A_a^j)^\dagger = I$ ($0 \leq j < r$). Let $N > 0$, $L > 0$, $L \leq N$, $L + L_0 \leq N$, and $(L_0 + 1) + (r-1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1 \leq L$. Assume the opened-boundary C^j, D^j comparison: whenever $\psi \in \mathcal{G}_{N,L}(B)$ and $\psi = \Gamma_N^B(\bigoplus_j X_j)$, there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every ρ , and every β , $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. If every block chain space is contained in its periodic MPS line,

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathbb{C}V^{(N)}(A_j) \quad (0 \leq j < r),$$

then

$$\ker H_L^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

Proof. [✓ Proof](#) Lemma 13.3.2.9 gives $\mathcal{G}_{N,L}(B) = \bigvee_{j=0}^{r-1} \mathcal{G}_{N,L}(A_j)$ from the stated block hypotheses and the opened-boundary comparison. Lemma 12.1.21 then applies to this equality and to the blockwise containments. □

Lemma 12.1.23 (Kernel containment from boundary-crossing comparisons). [Lean](#) [✓ Stmt](#) Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$, with every $\mu_j \neq 0$. Suppose the physical alphabet and all block dimensions are nonempty. Assume that the block family is irreducible, left-canonical, has normalized self-overlap, and has no two blocks gauge-phase equivalent. Suppose also that each A_j is L_0 -block-injective, that $L_0 > 0$, and that $\sum_a A_a^j (A_a^j)^\dagger = I$ ($0 \leq j < r$). Let $N > 0$, $L > 0$, $L \leq N$, $L + L_0 \leq N$, and $(L_0 + 1) + (r - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1 \leq L$. Assume that, for every boundary-crossing interval beginning at i , the simultaneous products $w \mapsto (A_w^1, \dots, A_w^r)$, $|w| = N - i$, span the product algebra $\prod_j M_{D_j}(\mathbb{C})$. If every block chain space is contained in its periodic MPS line,

$$\mathcal{G}_{N,L}(A_j) \subseteq \text{CV}^{(N)}(A_j) \quad (0 \leq j < r),$$

then

$$\ker H_L^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r - 1\}.$$

Proof. [✓ Proof](#) Lemma 13.3.2.11 gives $\mathcal{G}_{N,L}(B) = \bigvee_{j=0}^{r-1} \mathcal{G}_{N,L}(A_j)$ from the block hypotheses, the local boundary-crossing constraints, and the stated tail-word span. Lemma 12.1.21 then applies to this equality and to the blockwise containments. $\square$

Lemma 12.1.24 (Block-diagonal spanning is equivalent to the reverse inclusion). [Lean](#) [Lean](#) [✓ Stmt](#) Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$, with every $\mu_j \neq 0$. Then, for every $N > L$, the source parent-Hamiltonian spanning condition $\ker H_L^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r - 1\}$ is equivalent to the reverse containment, for the same N :

$$\ker H_L^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r - 1\}.$$

In the nearest-neighbor case $L = 2$, the all-chain source condition of Definition 12.1.17 is equivalent to the conjunction

$$\begin{aligned} & (\forall N > 2, \forall i, j \in \{0, \dots, N - 1\}, h_i(B, 2)h_j(B, 2) = h_j(B, 2)h_i(B, 2)) \\ & \wedge \left(\forall N > 2, \ker H_2^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r - 1\} \right). \end{aligned}$$

Proof. [✓ Proof](#) The forward direction of the displayed equality gives the containment by rewriting the kernel. Conversely, for $N > L$, the forward containment gives

$$\text{span}\{V^{(N)}(A_j) : j = 0, \dots, r - 1\} \subseteq \ker H_L^{(N)}(B),$$

and the assumed reverse containment gives equality. For $L = 2$, the all-chain nearest-neighbor condition of Definition 12.1.17 decomposes as the conjunction

$$\begin{aligned} & (\forall N > 2, \forall i, j \in \{0, \dots, N - 1\}, h_i(B, 2)h_j(B, 2) = h_j(B, 2)h_i(B, 2)) \\ & \wedge \left(\forall N > 2, \ker H_2^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r - 1\} \right). \end{aligned}$$

The second conjunct is the reverse containment just identified with the source spanning clause. $\square$

Lemma 12.1.25 (Chain-space comparison gives the block-diagonal spanning equation). [Lean](#) [Lean](#) [✓ Stmt](#) Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$, where $\mu_j \neq 0$. Suppose that, for every $N > L$, $G_L^{(N)}(B) =$

$\bigvee_{j=0}^{r-1} \text{span}\{V^{(N)}(A_j)\}$. Then the source parent-Hamiltonian spanning equation holds: $\ker H_N(B, L) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}$. It is enough to prove the two equations

$$G_L^{(N)}(B) = \bigvee_{j=0}^{r-1} G_L^{(N)}(A_j),$$

$$G_L^{(N)}(A_j) = \text{span}\{V^{(N)}(A_j)\}.$$

Proof. ✓ Proof The finite-volume identity gives $\ker H_N(B, L) = G_L^{(N)}(B)$. The assumed chain-space comparison therefore rewrites the kernel as $\bigvee_{j=0}^{r-1} \text{span}\{V^{(N)}(A_j)\}$. The latter supremum is the BNT vector span. Conversely, Lemma 12.1.20 gives $\text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\} \subseteq \ker H_N(B, L)$, so the displayed equation is the source spanning clause. If the comparison is first stated as $G_L^{(N)}(B) = \bigvee_j G_L^{(N)}(A_j)$, the single-block equations $G_L^{(N)}(A_j) = \text{span}\{V^{(N)}(A_j)\}$ give the previous hypothesis. $\square$

Lemma 12.1.26 (Gauge invariance of fixed points and parent interactions). Lean Lean Lean ✓ Stmt *A virtual similarity preserves the physical blocking relation and hence transfer idempotence. It also preserves every local MPS space, so the canonical parent interaction and all its translated local terms are unchanged.*

Proof. ✓ Proof Suppose $B^i = XA^iX^{-1}$ for every physical letter. If the blocking relation for A is $A^{i_1}A^{i_2} = \sum_j U_{i_1, i_2}^j A^j$, then conjugation by X gives the same relation for B . Hence transfer idempotence is invariant under the similarity. At every length L , the two tensors have the same local MPS space, so $G_L(A) = G_L(B)$ and $h_i^{(N)}(A) = h_i^{(N)}(B)$. The parent-interaction identities and the commutation equations follow. $\square$

Theorem 12.1.27 (Renormalization fixed points have commuting nearest-neighbor local terms). Lean ✓ Stmt *If A has nonzero virtual dimension, is a normal tensor in the sense of [11], and is a renormalization fixed point, then for every chain length $N > 2$ the two-site local terms commute. This theorem records the commutation part of the nearest-neighbor source condition, not the ground-space spanning part.*

Proof. ✓ Proof Put A into its trace-preserving gauge. This is a pure similarity, since the normal-tensor normalization fixes the transfer spectral radius at one. Similarity preserves the physical blocking relation and the local MPS spaces, hence also the canonical parent interactions. In the trace-preserving gauge, the Appendix B virtual-bond projectors give $h_0^{(3)}h_1^{(3)} = h_1^{(3)}h_0^{(3)}$. Cyclic transport gives, for every $N > 2$ and every site i , $h_i^{(N)}h_{i+1}^{(N)} = h_{i+1}^{(N)}h_i^{(N)}$. For every pair i, j whose two-site supports are disjoint, locality gives $h_i^{(N)}h_j^{(N)} = h_j^{(N)}h_i^{(N)}$. These cases exhaust all pairs of two-site terms on the periodic chain. Transporting the parent interactions back proves the claim. $\square$

Theorem 12.1.28 (Renormalization fixed points satisfy the NNCPH commutation and zero-energy equations). Lean ✓ Stmt *If A has nonzero virtual dimension, is a normal tensor in the sense of [11], and is a renormalization fixed point, then for every chain length $N > 2$ the MPS vector $V^{(N)}(A)$ satisfies the commutation and zero-energy equations for nearest-neighbor local terms:*

$$h_i h_j = h_j h_i, \tag{12.5}$$

$$h_i V^{(N)}(A) = 0. \tag{12.6}$$

The zero-energy equation (12.6) is the parent-Hamiltonian frustration-freeness statement.

Proof. ✓ Proof Commutation (12.5) is Theorem 12.1.27. Annihilation (12.6) is Lemma 11.4.47 with $L = 2$. $\square$

Theorem 12.1.29 (Nearest-neighbor parent ground spaces for a single normal RFP tensor). Lean Lean ✓ Stmt *Let A have positive bond dimension and be a normal renormalization fixed-point tensor in the sense of [11]. If the canonical-form tensor consists of the single normal sector A , then for every $N > 2$, $\ker H_2^{(N)}(A) = \text{span}\{V^{(N)}(A)\}$. Its translated two-site interactions also commute and annihilate $V^{(N)}(A)$. Hence the singleton family satisfies the all-chain nearest-neighbor commuting parent-Hamiltonian ground-space condition.*

Proof. ✓ Proof Spectral normality gives eventual block injectivity. The fixed-point equation then implies that the one-site matrices already span the full matrix algebra. Thus A is injective. For $N > 2$, the finite-volume kernel identity and injective periodic-chain theorem give

$$\ker H_2^{(N)}(A) = \mathcal{G}_{N,2}(A) = \text{span}\{V^{(N)}(A)\}.$$

The span of the matrix product vectors of the singleton family is the last space in this equality. Theorem 12.1.28 supplies commutation and zero energy on every chain of length greater than two. $\square$

Theorem 12.1.30 (Commuting nearest-neighbor terms for distinct multiplicity-one RFP sectors). Lean ✓ Stmt *Let $A_1, \dots, A_g$ be the distinct basis tensors of a BNT canonical form, and let $B = \bigoplus_{j=1}^g A_j$ contain one copy of every basis tensor. If B is a renormalization fixed point, then for every $N > 2$ its translated two-site parent interactions commute pairwise. Thus B satisfies the nearest-neighbor commuting parent-Hamiltonian condition.*

Proof. ✓ Proof The fixed-point equation gives residual tensors U_j whose matrix-entry columns form one orthonormal family on the disjoint union of the sector matrix spaces. On the tensor square of this disjoint union, let Q be the sum of the sector bond projectors, extended by zero between unequal sectors. Write L_j and R_j for the left and right overlapping lifts of the sector- j bond projector. The single-sector relation and the mixed-sector orthogonality give

$$\begin{aligned} L_j R_j &= R_j L_j, \\ L_j R_k &= 0 = R_j L_k \end{aligned} \tag{12.7}$$

for $j \neq k$, where (12.7) follows from the shared one-site factor $E_j^* E_k = 0$ for the canonical summand inclusions E_j . Summing over the sectors shows that the two adjacent placements of Q commute.

If U is the common isometry, then transport gives the identity $(U \otimes U)Q(U^* \otimes U^*) = P_{G_2(B)}$, since the range on either side is $\bigvee_j G_2(A_j) = G_2(B)$. Hence it is the complement of the canonical two-site parent interaction of B . The two adjacent three-site parent terms therefore commute. Cyclic transport gives adjacent commutation on every chain with $N > 2$; disjoint local terms commute by locality, which proves the claim. $\square$

Theorem 12.1.31 (Nearest-neighbor parent ground spaces for distinct multiplicity-one RFP sectors). Lean ✓ Stmt *Let $A_1, \dots, A_g$ be the distinct basis tensors of a BNT canonical form, and let $B = \bigoplus_{j=1}^g A_j$ contain one copy of every basis tensor. If B is a renormalization fixed point, then for every $N > 2$,*

$$\ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

This establishes the multiplicity-one distinct-sector case of [11, Theorem 3.10].

Proof. ✓ Proof The residual isometry equations imply that the simultaneous one-site matrices $(A_j^i)_j$ span the product of the block matrix algebras. Hence the same is true for every positive word length. The one-step restriction intersection therefore gives $\mathcal{G}_{N,2}(B) \subseteq \bigvee_j \mathcal{G}_N(A_j)$. Comparing this block decomposition before and after moving the periodic cut by one site shows that every block boundary matrix commutes with the corresponding one-site matrices. Thus each block component lies in $\mathcal{G}_{N,2}(A_j)$, and $\mathcal{G}_{N,2}(B) = \bigvee_j \mathcal{G}_{N,2}(A_j)$. Each block is a normal renormalization fixed point, so one-site injectivity and the injective periodic-chain theorem identify its chain space with the line spanned by $V^{(N)}(A_j)$. The finite-volume kernel identity completes the proof. $\square$

Theorem 12.1.32 (Forward NNCPH ground spaces for distinct multiplicity-one RFP sectors).

Lean ✓ Stmt Let $A_1, \dots, A_g$ be the distinct basis tensors of a BNT canonical form, and let $B = \bigoplus_{j=1}^g A_j$ contain one copy of every basis tensor. If B is a renormalization fixed point, then for every $N > 2$ its translated two-site parent interactions commute and

$$H_2^{(N)}(B)V^{(N)}(B) = 0,$$

$$\ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Hence B satisfies the all-chain nearest-neighbor commuting parent-Hamiltonian ground-space condition, establishing the multiplicity-one distinct-sector forward implication of [11, Theorem 3.10].

Proof. ✓ Proof Theorem 12.1.30 gives pairwise commutation of the translated two-site interactions for every $N > 2$. The parent-Hamiltonian construction gives the zero-energy equation. Theorem 12.1.31 identifies its kernel with the span of the periodic matrix product vectors of the distinct sectors. These are the three clauses of the all-chain condition. $\square$

Theorem 12.1.33 (Cyclic trace expansion for a closed MPS chain). Lean ✓ Stmt For a nonempty closed chain, the trace of the word $A^{\sigma_0} \dots A^{\sigma_{L-1}}$ is the sum over cyclic virtual-index configurations:

$$\text{tr}(A^{\sigma_0} \dots A^{\sigma_{L-1}}) = \sum_{\alpha_0, \dots, \alpha_{L-1}} \prod_{t=0}^{L-1} (A^{\sigma_t})_{\alpha_t, \alpha_{t+1}}, \quad \alpha_L = \alpha_0.$$

Proof. ✓ Proof The trace is the diagonal sum:

$$\text{tr}(A^{\sigma_0} \dots A^{\sigma_{L-1}}) = \sum_{\alpha_0} (A^{\sigma_0} \dots A^{\sigma_{L-1}})_{\alpha_0, \alpha_0}.$$

For prescribed endpoints, the matrix product entry expands as

$$(A^{\sigma_0} \dots A^{\sigma_{L-1}})_{\alpha_0, \alpha_L} = \sum_{\alpha_1, \dots, \alpha_{L-1}} \prod_{t=0}^{L-1} (A^{\sigma_t})_{\alpha_t, \alpha_{t+1}}.$$

Substituting $\alpha_L = \alpha_0$ in the trace gives the cyclic formula. $\square$

Definition 12.1.34 (Physical isometry in the basic-vector expression). Lean Lean ✓ Stmt Suppose that A is equipped with Appendix B structural data (X, Λ, U) : the matrix X is invertible, the diagonal entries of Λ are strictly positive, $A^i = X \Lambda U^i X^{-1}$, and U satisfies the pair-index isometry equation below. Let $\mathcal{H}_a$ and $\mathcal{H}_b$ have bases indexed by $\alpha, \beta \in \{0, \dots, D-1\}$. The coefficients $U_{\alpha, \beta}^i$ define maps

$$U: \mathcal{H}_a \otimes \mathcal{H}_b \longrightarrow \mathcal{H}_{\text{phys}},$$

$$(Uv)_i = \sum_{\alpha, \beta} U_{\alpha, \beta}^i v_{\alpha, \beta},$$

and

$$(U^*\psi)_{\alpha,\beta} = \sum_i \overline{U_{\alpha,\beta}^i} \psi_i.$$

Theorem 12.1.35 (The physical map is an isometric embedding). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be equipped with Appendix B structural data (X, Λ, U) as in Definition 12.1.34. Its pair-index isometry equation*

$$\sum_i \overline{U_{\alpha,\beta}^i} U_{\alpha',\beta'}^i = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}$$

*implies $U^*U = 1$. In particular, U is injective.*

Proof. [✓ Proof](#) Expanding the coefficient of U^*Uv at (α, β) and interchanging the two finite sums gives

$$(U^*Uv)_{\alpha,\beta} = \sum_{\alpha',\beta'} \left(\sum_i \overline{U_{\alpha,\beta}^i} U_{\alpha',\beta'}^i \right) v_{\alpha',\beta'} = v_{\alpha,\beta}.$$

A map with a left inverse is injective. □

Definition 12.1.36 (Tensor powers of the physical isometry). [Lean](#) [Lean](#) [✓ Stmt](#) For every $N \geq 0$, the coefficients of U define

$$U^{\otimes N}: (\mathcal{H}_a \otimes \mathcal{H}_b)^{\otimes N} \longrightarrow \mathcal{H}_{\text{phys}}^{\otimes N},$$

with

$$(U^{\otimes N}v)_\sigma = \sum_{p_0, \dots, p_{N-1}} \left(\prod_{t=0}^{N-1} U_{(p_t)_a, (p_t)_b}^{\sigma_t} \right) v_{p_0, \dots, p_{N-1}}.$$

The conjugate coefficient map is

$$(U^{*\otimes N}\psi)_p = \sum_\sigma \left(\prod_{t=0}^{N-1} \overline{U_{(p_t)_a, (p_t)_b}^{\sigma_t}} \right) \psi_\sigma.$$

This is the tensor power occurring in the basic-vector expression [11, Equation (3.17)].

Theorem 12.1.37 (The tensor powers are isometric embeddings). [Lean](#) [Lean](#) [✓ Stmt](#) *For every $N \geq 0$, one has $U^{*\otimes N}U^{\otimes N} = 1$. Hence $U^{\otimes N}$ is injective.*

Proof. [✓ Proof](#) For virtual-pair configurations p and q , interchange the two finite sums and factor the sum over physical configurations site by site:

$$\sum_\sigma \prod_t \overline{U_{(p_t)_a, (p_t)_b}^{\sigma_t}} U_{(q_t)_a, (q_t)_b}^{\sigma_t} = \prod_t \sum_i \overline{U_{(p_t)_a, (p_t)_b}^i} U_{(q_t)_a, (q_t)_b}^i.$$

The pair-index isometry equation turns the last product into $\prod_t \delta_{p_t, q_t} = \delta_{p, q}$. Therefore, the coefficient at p is v_p . A map with a left inverse is injective. □

Theorem 12.1.38 (Appendix B basis change and the parent interaction). [Lean](#) [✓ Stmt](#) *Let $A^i = X\Lambda U^i X^{-1}$ be the Appendix B structural form, and write $A_{\text{core}}^i = \Lambda U^i$. The canonical local parent interaction is unchanged by this basis change: $q_L(A) = q_L(A_{\text{core}})$.*

Proof. [✓ Proof](#) The basis change gives a gauge equivalence between A_{core} and A , hence equality of their length- L local ground spaces. The parent interaction is the orthogonal complement projector of this local ground space. $\square$

Definition 12.1.39 (Appendix B two-site basic support). [Lean](#) [✓ Stmt](#) For Appendix B structural data, the two-site basic support is $G_2(A_{\text{core}}) = G_2(\Lambda U)$.

Definition 12.1.40 (Two-site bond insertion and basic-vector embedding). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Put $|\varphi\rangle = \sum_b \lambda_b |b, b\rangle$. For outer-boundary coefficients $v_{a,c}$, insertion of this vector on the middle virtual bond is the map I_φ defined by

$$(I_\varphi v)_{(a,b),(b',c)} = \delta_{b,b'} \lambda_b v_{a,c}.$$

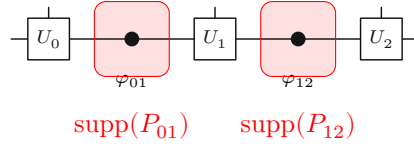
The two-site physical map is $U^{\otimes 2} I_\varphi$. A boundary matrix Y determines outer-boundary coefficients by $(v_Y)_{a,c} = \lambda_a Y_{c,a}$. Opening the outer virtual bond in the two-site contraction of [11, Equations (3.17)–(3.18)] gives these coefficients.

Definition 12.1.41 (Virtual bond projector and its adjacent placements). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Write $s = \langle \varphi, \varphi \rangle = \sum_b \lambda_b^2$. The orthogonal projector onto $\mathbb{C}\varphi$ has pair-index coefficients

$$(P_\varphi v)_{b,b'} = \delta_{b,b'} \frac{\lambda_b}{s} \sum_k \lambda_k v_{k,k}.$$

On three virtual site pairs, let P_{01} place this projector on the bond joining the right index of site zero to the left index of site one, and let P_{12} place it on the next bond.

Theorem 12.1.42 (Commutation of the adjacent virtual bond projectors). [Lean](#) [✓ Stmt](#) *The two adjacent placements commute: $P_{01}P_{12} = P_{12}P_{01}$. The following support-location schematic marks the two bonds on which the projectors act; it does not depict the doubled operator network for the equality above.*



Proof. [✓ Proof](#) The contracted bond coordinates are disjoint. Thus replacing the first bond by the diagonal basis vector $|k, k\rangle$ and the second by $|\ell, \ell\rangle$ is independent of the order. Expanding both rank-one projectors gives the same double sum over (k, ℓ) . $\square$

Theorem 12.1.43 (Surjectivity of the weighted boundary parametrization). [Lean](#) [✓ Stmt](#) *Strict positivity of the λ_a makes $Y \mapsto v_Y$ surjective.*

Proof. [✓ Proof](#) For prescribed coefficients $v_{a,c}$, take $Y_{c,a} = \lambda_a^{-1} v_{a,c}$. $\square$

Theorem 12.1.44 (The two-site basic-vector image). [Lean](#) [Lean](#) [✓ Stmt](#) *Opening the outer virtual bond in the two-site contraction of [11, Equations (3.17)–(3.18)] gives the following local-support family. For every boundary matrix Y , $U^{\otimes 2} I_\varphi(v_Y) = \Gamma_2(\Lambda U)(Y)$. Consequently, $\text{ran}(U^{\otimes 2} I_\varphi) = G_2(\Lambda U)$.*

Proof. [✓ Proof](#) At a two-site physical configuration (i, j) , the bond insertion gives $\sum_{a,b,c} U_{a,b}^i U_{b,c}^j \lambda_b \lambda_a Y_{c,a}$. Expanding $\text{tr}((\Lambda U^i)(\Lambda U^j)Y)$ gives the same sum, so the two maps agree after the weighted boundary parametrization. Since that parametrization is surjective, their ranges are equal. $\square$

Definition 12.1.45 (Two-site basic support projector). [Lean](#) [✓ Stmt](#) Let P_2 be the orthogonal projector onto $G_2(\Lambda U)$, as in the local parent-space construction of [11, Definition 3.9].

Theorem 12.1.46 (Range and complement of the two-site basic support projector). [Lean](#) [✓ Stmt](#) The projector P_2 satisfies

$$\begin{aligned} P_2^2 &= P_2, \\ \text{ran}(P_2) &= \text{ran}(U^{\otimes 2} I_\varphi) = G_2(\Lambda U), \\ P_2 &= 1 - q_2(\Lambda U). \end{aligned}$$

In particular, $q_2(\Lambda U)$ annihilates the image of $U^{\otimes 2} I_\varphi$, while P_2 fixes that image pointwise.

Proof. [✓ Proof](#) Orthogonal projection onto $G_2(\Lambda U)$ has that subspace as its range. The preceding theorem identifies this range with $\text{ran}(U^{\otimes 2} I_\varphi)$. Orthogonal projection onto the complementary subspace is $q_2(\Lambda U)$, hence $P_2 = 1 - q_2(\Lambda U)$. Since $q_2(\Lambda U)^2 = q_2(\Lambda U)$,

$$P_2^2 = (1 - q_2(\Lambda U))^2 = 1 - q_2(\Lambda U) = P_2.$$

If ψ lies in $\text{ran}(U^{\otimes 2} I_\varphi)$, then $\psi \in G_2(\Lambda U) = \ker q_2(\Lambda U)$, and therefore

$$\begin{aligned} q_2(\Lambda U)\psi &= 0, \\ P_2\psi &= (1 - q_2(\Lambda U))\psi = \psi. \end{aligned}$$

□

Definition 12.1.47 (Two-site support of a residual isometry family). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{V} = \bigsqcup_j \{j\} \times [D_j] \times [D_j]$ be the disjoint union of the within-sector virtual pairs, and let $V: \mathcal{V} \rightarrow \mathbb{C}^d$ be the common physical map whose matrix elements are $V_{i,(j,\alpha,\beta)} = (U_j^i)_{\alpha,\beta}$. For each sector insert the vector $|\varphi_j\rangle = \sum_b \lambda_{j,b} |b, b\rangle$ on the middle virtual bond, apply V on both sites, and sum the resulting physical vectors. This defines the equal-sector two-site embedding

$$I_{\text{res}}: \prod_j \mathbb{C}^{D_j \times D_j} \longrightarrow (\mathbb{C}^d)^{\otimes 2}.$$

Unequal-sector virtual bonds do not occur. The corresponding physical support operator $P_{2,\text{res}}$ is the orthogonal projector onto $G_2(\bigoplus_j B_j)$.

Theorem 12.1.48 (Range of the residual-family bond embedding). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The common physical map is an isometry, $V^*V = 1$. Moreover, the equal-sector bond embedding and the physical support projector satisfy

$$\text{ran}(I_{\text{res}}) = \sum_j G_2(B_j) = G_2\left(\bigoplus_j B_j\right), \quad (12.8)$$

$$\text{ran}(P_{2,\text{res}}) = \text{ran}(I_{\text{res}}), \quad (12.9)$$

and $P_{2,\text{res}} = 1 - q_2\left(\bigoplus_j B_j\right)$. If the B_j form a basis of normal tensors and their direct sum is a renormalization fixed point, the sectorwise Appendix B decompositions and the common residual isometry required above exist.

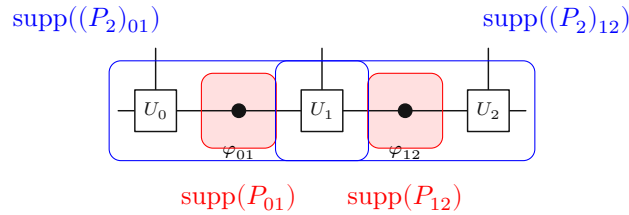
Proof. ✓ Proof The isometry relation $V^*V = 1$ is the one-letter Gram identity on the disjoint union of the within-sector virtual pairs. In each sector, the bond insertion has range $G_2(B_j)$; allowing independent outer-boundary coefficients and summing over sectors therefore gives $\sum_j G_2(B_j)$. The local ground space of a direct sum is this sum, which proves (12.8) and (12.9). Orthogonal projection onto that space is complementary to its parent interaction. Finally, the residual-isometry form of the cross-block fixed-point structure supplies the simultaneous sector decompositions. $\square$

Theorem 12.1.49 (Commutation of adjacent two-site basic support projectors). Lean ✓ Stmt Let $(P_2)_{01}$ and $(P_2)_{12}$ be the two adjacent placements of the two-site basic support projector on three physical sites. Then $(P_2)_{01}(P_2)_{12} = (P_2)_{12}(P_2)_{01}$. This is the local three-site consequence of the parent-space construction and the basic-vector form in [11, Definition 3.9, Equations (3.16)–(3.18), and Appendix B].

Proof. ✓ Proof The one-site map U is an isometry. The preceding range identity therefore identifies the physical projector with the isometric transport of the virtual bond projector: $P_2 = U^{\otimes 2} P_\varphi (U^*)^{\otimes 2}$. On three sites, multiplication of the two adjacent transports cancels the middle factors by $U^*U = 1$. Consequently their products are

$$\begin{aligned} U^{\otimes 3} P_{01} P_{12} (U^*)^{\otimes 3}, \\ U^{\otimes 3} P_{12} P_{01} (U^*)^{\otimes 3}, \end{aligned}$$

respectively. They agree by Theorem 12.1.42. The following support-location schematic records the one-site embedding and the two overlapping physical supports. It does not depict the doubled $U \cdots U^*$ operator networks in the identities above.



$\square$

Definition 12.1.50 (Appendix B AX coefficient representative). Lean ✓ Stmt The AX coefficient-space representative is, before the AX-lift, the canonical two-site parent interaction of the core tensor, $\widehat{Q}_{AX} = q_2(A_{\text{core}})$. The source projector Q_{AX} of Definition D.2 acts on $\mathcal{H}_A \otimes \mathcal{H}_X$. The displayed operator is the two-site coefficient representative that will be placed on the AX face; it is not by itself the construction of that source projector.

Definition 12.1.51 (Appendix B XB coefficient representative). Lean ✓ Stmt The XB coefficient-space representative is, before the XB-lift, the same coefficient-space representative, $\widehat{Q}_{XB} = \widehat{Q}_{AX}$. The source projector Q_{XB} of Definition D.2 acts on $\mathcal{H}_X \otimes \mathcal{H}_B$. The displayed operator is the coefficient representative later placed on the XB face.

Theorem 12.1.52 (The (AX) factor action). Lean ✓ Stmt On the homogeneous three-single-site coefficient space $(\mathbb{C}^d)^{\otimes 3}$, the hatted Appendix B (AX) representative acts as $[\widehat{Q}_{AX}]_{AX} = [\widehat{Q}_{AX}] \otimes I_d$. This equation gives the coefficient-factor specialization of the left local action in [11, Definition D.2]. It does not construct a projector for an arbitrary factorization $\mathcal{H}_A \otimes \mathcal{H}_X \otimes \mathcal{H}_B$.

Proof. ✓ Proof Apply the left identity in Theorem 12.1.3 to $\widehat{Q}_{AX}$. □

Theorem 12.1.53 (The (XB) factor action). Lean ✓ Stmt *On the homogeneous three-single-site coefficient space $(\mathbb{C}^d)^{\otimes 3}$, the hatted Appendix B (XB) representative acts as $[\widehat{Q}_{XB}]_{XB} = I_d \otimes [\widehat{Q}_{XB}]$. This equation gives the coefficient-factor specialization of the right local action in [11, Definition D.2]. It does not construct a projector for an arbitrary factorization $\mathcal{H}_A \otimes \mathcal{H}_X \otimes \mathcal{H}_B$.*

Proof. ✓ Proof Apply the right identity in Theorem 12.1.3 to $\widehat{Q}_{XB}$. □

Lemma 12.1.54 (Appendix B coefficient representatives and the parent interaction). Lean Lean ✓ Stmt *As coefficient-space representatives, $\widehat{Q}_{AX} = q_2(A)$ and $\widehat{Q}_{XB} = q_2(A)$.*

Proof. ✓ Proof Both operators are $q_2(A_{\text{core}})$, and the Appendix B basis change leaves q_2 unchanged. □

Lemma 12.1.55 (Appendix B two-site representative idempotents). Lean Lean ✓ Stmt *The two Appendix B two-site coefficient representatives are idempotent:*

$$\begin{aligned}\widehat{Q}_{AX}^2 &= \widehat{Q}_{AX}, \\ \widehat{Q}_{XB}^2 &= \widehat{Q}_{XB}.\end{aligned}$$

Proof. ✓ Proof By the preceding identification, both representatives are the canonical two-site parent interaction $q_2(A)$. The latter is an orthogonal projector, hence is idempotent. □

Theorem 12.1.56 (Appendix B overlapping two-site commutation). Lean ✓ Stmt *The two Appendix B coefficient representatives satisfy*

$$\begin{aligned}\widehat{Q}_{AX}^2 &= \widehat{Q}_{AX}, \\ \widehat{Q}_{XB}^2 &= \widehat{Q}_{XB},\end{aligned}$$

and their adjacent actions on the common three-site coefficient space commute:

$$(\widehat{Q}_{AX})_{AX}(\widehat{Q}_{XB})_{XB} = (\widehat{Q}_{XB})_{XB}(\widehat{Q}_{AX})_{AX}.$$

Thus they satisfy the algebraic overlapping two-site condition of Definition 12.1.2, providing the coefficient-space analog of the commutation condition in [11, Definition D.2].

Proof. ✓ Proof Both coefficient representatives are the complement $1 - P_2$ of the two-site basic support projector. Their idempotency was established in Lemma 12.1.55. Since the two adjacent lifts of P_2 commute, their complementary lifts commute as well. □

Theorem 12.1.57 (Three-site intersection of adjacent parent kernels). Lean Lean ✓ Stmt *If A is injective, then $G_3(A) = \ker(q_2(A)_{AX}) \cap \ker(q_2(A)_{XB})$. More generally, on $L + 1$ sites, simultaneous annihilation by the two adjacent length- L parent interactions is equivalent to membership in $G_{L+1}(A)$ whenever A is injective and $L > 1$.*

Proof. ✓ Proof By Theorem 11.5.4.3, a three-site vector belongs to $G_3(A)$ exactly when its restrictions to both adjacent pairs belong to $G_2(A)$. Since $\ker q_2(A) = G_2(A)$, this is equivalent to annihilation by both adjacent terms. Theorem 12.1.4 identifies these terms with the two displayed lifts. □

Theorem 12.1.58 (Appendix B coefficient-space analog of the Appendix D kernel datum).

[Lean](#) [Lean](#) [✓ Stmt](#) Let A be injective and let an Appendix B structural datum be fixed. Put

$$\begin{aligned}\widehat{K}_{AXB} &= G_3(A), \\ \widehat{Q}_{AX} &= q_2(A), \\ \widehat{Q}_{XB} &= q_2(A),\end{aligned}$$

with the two copies of $q_2(A)$ lifted to the indicated faces of the three-site coefficient space. Then

$$\begin{aligned}[(\widehat{Q}_{AX})_{AX}, (\widehat{Q}_{XB})_{XB}] &= 0, \\ \widehat{K}_{AXB} &= \ker((\widehat{Q}_{AX})_{AX}) \cap \ker((\widehat{Q}_{XB})_{XB}).\end{aligned}$$

These are the coefficient-space analogs of the idempotence, commutation, and kernel-intersection equations in [11, Definition D.2]. The hatted operators are not identified here with the source projectors on $\mathcal{H}_A \otimes \mathcal{H}_X$ and $\mathcal{H}_X \otimes \mathcal{H}_B$. For a normal left-canonical RFP tensor, injectivity follows from the RFP structural theorem and is not an additional hypothesis.

Proof. [✓ Proof](#) Idempotency and commutation are Theorem 12.1.56. The two coefficient representatives are $q_2(A)$, and Theorem 12.1.57 supplies the remaining kernel equation. $\square$

Theorem 12.1.59 (Transport of the three-site commutator). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the first two length-two parent interactions commute on the three-site chain: $h_0^{(3)} h_1^{(3)} = h_1^{(3)} h_0^{(3)}$. Then, for every $N > 2$ and every $i \in \mathbb{Z}/N\mathbb{Z}$, the two adjacent translates commute: $h_i^{(N)} h_{i+1}^{(N)} = h_{i+1}^{(N)} h_i^{(N)}$. For each exterior configuration τ , let $R_{i,\tau}^{(3)}$ restrict to the cyclic three-site window beginning at i . It intertwines the two N -site terms with the two three-site terms:

$$\begin{aligned}R_{i,\tau}^{(3)} h_i^{(N)} &= h_0^{(3)} R_{i,\tau}^{(3)}, \\ R_{i,\tau}^{(3)} h_{i+1}^{(N)} &= h_1^{(3)} R_{i,\tau}^{(3)}.\end{aligned}$$

Thus the same local AXB calculation is placed at every cyclic translate through the two restriction identities above. The hypothesis $N > 2$ is essential to this transport: at $N = 2$ the two cyclic windows contain the same sites in opposite orders. This case is not part of [11, Theorem 3.10].

Proof. [✓ Proof](#) Restrict an N -site vector to the cyclic window $(i, i+1, i+2)$ while holding all other physical indices fixed. Window extraction and replacement identify the two restricted local terms with $h_0^{(3)}$ and $h_1^{(3)}$. Apply the three-site commutator to the restricted vector and then evaluate at the original three-site configuration. Replacing a window by the values extracted from that configuration recovers the original N -site configuration. $\square$

Theorem 12.1.60 (Appendix B adjacent commutation on periodic chains). [Lean](#) [✓ Stmt](#) For every Appendix B structural datum and every periodic chain of length $N > 2$, the adjacent translated length-two parent interactions commute.

Proof. [✓ Proof](#) Theorem 12.1.56 gives the three-site commutator, and Theorem 12.1.59 places it at every adjacent pair on the periodic chain. $\square$

Theorem 12.1.61 (Left-canonical RFP commutation and zero energy). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A normal left-canonical renormalization fixed-point tensor of nonzero virtual dimension has pairwise commuting length-two parent interactions on every periodic chain of length $N > 2$, and its periodic matrix product vector is annihilated by every such interaction. The removal of the left-canonical hypothesis and the ground-space spanning equation in [11, Definition 3.9] remain separate.

Proof. [✓ Proof](#) The Appendix B structural theorem supplies a datum whose adjacent translated parent interactions commute by Theorem 12.1.60. Lemma 12.1.13 gives pairwise commutation directly. The standard frustration-free identity shows that each interaction annihilates the periodic matrix product vector. $\square$

Lemma 12.1.62 (Appendix B overlapping condition from the lifted commutator). [Lean](#) [✓ Stmt](#)
Suppose the lifted AX and XB representatives commute:

$$(\widehat{Q}_{AX})_{AX}(\widehat{Q}_{XB})_{XB} = (\widehat{Q}_{XB})_{XB}(\widehat{Q}_{AX})_{AX}.$$

Then $\widehat{Q}_{AX}$ and $\widehat{Q}_{XB}$ satisfy the algebraic overlapping two-site condition of Definition 12.1.2.

Proof. [✓ Proof](#) The idempotency clauses are supplied by Lemma 12.1.55. The only remaining clause in Definition 12.1.2 is exactly the displayed lifted commutator. $\square$

Lemma 12.1.63 (Definition D.2 projectors identified with Appendix B representatives). [Lean](#)

[Lean](#) [✓ Stmt](#) *Let Q_{AX} and Q_{XB} be the two-site coefficient-space transports of source projectors satisfying the Definition D.2 local condition on AXB . If these transported projectors agree with the Appendix B coefficient representatives,*

$$\begin{aligned} Q_{AX} &= \widehat{Q}_{AX}, \\ Q_{XB} &= \widehat{Q}_{XB}, \end{aligned}$$

then $\widehat{Q}_{AX}$ and $\widehat{Q}_{XB}$ satisfy the algebraic overlapping two-site condition of Definition 12.1.2. Consequently, $h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2)$.

Proof. [✓ Proof](#) Rewriting the two projectors in the Definition D.2 condition as $\widehat{Q}_{AX}$ and $\widehat{Q}_{XB}$ gives the algebraic overlapping condition. The three-site commutation is then Lemma 12.1.65. $\square$

Lemma 12.1.64 (Three-site lifts of the Appendix B coefficient representatives). [Lean](#) [Lean](#)

[✓ Stmt](#) *On the three-site A, X, B window, the two adjacent length-two local terms are the two local lifts*

$$\begin{aligned} h_0^{(3)}(A, 2) &= (\widehat{Q}_{AX})_{AX}, \\ h_1^{(3)}(A, 2) &= (\widehat{Q}_{XB})_{XB}. \end{aligned}$$

Proof. [✓ Proof](#) Apply the three-site local-term identities to $q_2(A)$, and replace $q_2(A)$ by $\widehat{Q}_{AX}$ or $\widehat{Q}_{XB}$ using the preceding two-site identifications. $\square$

Lemma 12.1.65 (Appendix B overlapping condition gives three-site local-term commutation).

[Lean](#) [✓ Stmt](#) *If the Appendix B AX and XB coefficient representatives satisfy the overlapping two-site commutation datum*

$$\begin{aligned} \widehat{Q}_{AX}^2 &= \widehat{Q}_{AX}, \\ \widehat{Q}_{XB}^2 &= \widehat{Q}_{XB}, \\ (\widehat{Q}_{AX})_{AX}(\widehat{Q}_{XB})_{XB} &= (\widehat{Q}_{XB})_{XB}(\widehat{Q}_{AX})_{AX}, \end{aligned}$$

then the corresponding first two translated length-two local terms on the three-site window commute: $h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2)$.

Proof. ✓ Proof By Lemma 12.1.64, the two translated local terms are exactly the AX and XB lifts of $\widehat{Q}_{AX}$ and $\widehat{Q}_{XB}$. The displayed equality is therefore the commutation clause of the overlapping two-site datum. $\square$

Lemma 12.1.66 (Appendix B lifted commutator gives three-site local-term commutation). Lean Lean ✓ Stmt *If the Appendix B AX and XB coefficient representatives satisfy the lifted Definition D.2 commutator*

$$(\widehat{Q}_{AX})_{AX}(\widehat{Q}_{XB})_{XB} = (\widehat{Q}_{XB})_{XB}(\widehat{Q}_{AX})_{AX},$$

then the corresponding first two translated length-two local terms on the three-site window commute: $h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2)$. The same conclusion holds if one supplies a Definition D.2-shaped local condition whose two operators are these coefficient-space representatives, rather than the source projectors themselves.

Proof. ✓ Proof The lifted commutator, together with the Appendix B idempotency already recorded above, gives the overlapping two-site commutation datum. A supplied Definition D.2-shaped local condition on the coefficient representatives implies the same conclusion by retaining its commutation clause and not using the kernel-intersection condition

$$K_{AXB} = \ker((\widehat{Q}_{AX})_{AX}) \cap \ker((\widehat{Q}_{XB})_{XB}).$$

$\square$

Remark 12.1.67 (Basic-vector coefficients and nearest-neighbor local-term commutation). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The Appendix B structural form supplies $A^i = X\Lambda U^i X^{-1}$. In the source unit pair-index convention, the residual tensor satisfies

$$\sum_i \overline{(U^i)_{\alpha, \beta}} (U^i)_{\alpha', \beta'} = \delta_{\alpha, \alpha'} \delta_{\beta, \beta'}.$$

Therefore the residual core tensor ΛU^i satisfies the weighted pair-index relation

$$\sum_i \overline{(\Lambda U^i)_{\alpha, \beta}} (\Lambda U^i)_{\alpha', \beta'} = \lambda_\alpha \lambda_{\alpha'} \delta_{\alpha, \alpha'} \delta_{\beta, \beta'}.$$

This is an orthogonality relation after summing over the physical letter i . It is not a coefficient identity for a fixed physical word σ . The coefficient associated to the source basic-vector formula is

$$C_L(\sigma) = \sum_{\alpha_0, \dots, \alpha_{L-1}} \prod_{t=0}^{L-1} \Lambda_{\alpha_t} (U^{\sigma_t})_{\alpha_t, \alpha_{t+1}}, \quad \alpha_L = \alpha_0.$$

This is the coefficient expression for the source vector

$$\begin{aligned} |V^{(L)}(A_j)\rangle &= U^{\otimes L} |\varphi_j\rangle^{\otimes L}, \\ |\varphi_j\rangle &= \sum_m \lambda_m |m, m\rangle, \end{aligned}$$

where φ_j is shared by b_t and a_{t+1} and U acts on (a_t, b_t) . At length two, this cyclic coefficient is the structural two-site amplitude: $C_2(\sigma) = \phi_2(\sigma)$. The change of basis does not alter either the local spaces G_L or the periodic MPS coefficients: $G_L(A) = G_L(\Lambda U)$. It therefore also leaves

the canonical parent interaction unchanged: $q_L(A) = q_L(\Lambda U)$. At length two, the Appendix B two-site basic support is $G_2(\Lambda U)$, and the coefficient-space representatives $\widehat{Q}_{AX}$ and $\widehat{Q}_{XB}$ are the two copies of $q_2(\Lambda U)$ before they are lifted to the AX and XB faces. Hence

$$\begin{aligned}\widehat{Q}_{AX} &= q_2(A), \\ \widehat{Q}_{XB} &= q_2(A),\end{aligned}$$

as coefficient-space representatives. On the three-site window their two lifts give the local identifications

$$\begin{aligned}h_0^{(3)}(A, 2) &= (\widehat{Q}_{AX})_{AX}, \\ h_1^{(3)}(A, 2) &= (\widehat{Q}_{XB})_{XB}.\end{aligned}$$

Consequently, for every chain length $L \geq 1$, the original tensor has the same cyclic coefficient formula: $V^{(L)}(A)_\sigma = C_L(\sigma)$. In particular, the corresponding two-site amplitude is

$$\phi_2(a, b) = V^{(2)}(A)(a, b) = V^{(2)}(\Lambda U)(a, b).$$

The source places each virtual pair across neighboring sites and applies an on-site physical isometry, as in [11, equations (3.17)–(3.18) and Appendix B]. On a three-site chain, the two adjacent length-two local terms are the AX and XB actions of the canonical two-site parent interaction. Theorem 12.1.56 constructs the adjacent physical support projectors from the basic-vector expression and proves the overlapping relation

$$(\widehat{Q}_{AX})_{AX}(\widehat{Q}_{XB})_{XB} = (\widehat{Q}_{XB})_{XB}(\widehat{Q}_{AX})_{AX}.$$

Theorem 12.1.59 transports this identity around a periodic chain. Consequently, for every $N > 2$ and every pair of sites $i, j \in \mathbb{Z}/N\mathbb{Z}$, $h_i(A, 2)h_j(A, 2) = h_j(A, 2)h_i(A, 2)$. Lemma 12.1.13 gives this pairwise commutation directly from adjacent commutation. Together with Lemma 11.4.47, this gives the zero-energy equation for $V^{(N)}(A)$. Neither conclusion requires a disjoint physical-pair factorization. The full source implication additionally includes the ground-space spanning equation. The coefficient-space analog of the local kernel-intersection equation is Theorem 12.1.58; identifying its hatted maps with the source projectors remains separate and is not needed for the chain commutator.

Remark 12.1.68 (Cyclic contraction and disjoint physical pairs). The unit pair-index isometry and cyclic coefficient formula in Remark 12.1.67 do not imply factorization of a fixed physical-word coefficient over disjoint physical pairs. For example, take virtual indices 0, 1, physical labels (a, b) , and $U_{x,y}^{(a,b)} = \delta_{x,a}\delta_{y,b}$, with $A^{(a,b)} = \lambda_a E_{a,b}$ and $\lambda_0, \lambda_1 > 0$. The pair-index isometry holds. For the word $\sigma = ((0, 1), (1, 0), (1, 0), (0, 1))$, the cyclic coefficient is

$$V^{(4)}(A)_\sigma = C_4(\sigma) = 0.$$

In contrast, the product over disjoint physical pairs is

$$\prod_{p=0}^1 \phi_2(\sigma_{2p}, \sigma_{2p+1}) = (\lambda_0 \lambda_1)(\lambda_1 \lambda_0) = \lambda_0^2 \lambda_1^2 > 0.$$

Theorem 12.1.69 (Eventual dimension of nearest-neighbor parent ground spaces). [Lean](#) [Stmt](#)
Let P be a sector decomposition with representatives $A_1, \dots, A_g$ and total tensor B . Suppose that the vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ are linearly independent for all sufficiently large N , and that B satisfies the all-chain nearest-neighbor commuting parent-Hamiltonian ground-space condition with these representatives. Then there is N_0 such that, for every $N > N_0$, $\dim_{\mathbb{C}} \ker H_2^{(N)}(B) = g$. Thus the ground-state degeneracy is eventually constant, as required in [4, Section IV].

Proof. ✓ Proof Choose $N_0 \geq 2$ beyond the linear-independence threshold. For $N > N_0$, the ground-space spanning equation gives $\ker H_2^{(N)}(B) = \text{span}\{|V^{(N)}(A_j)\rangle : j = 1, \dots, g\}$. Identifying these vectors with their coefficient functions preserves linear independence. The dimension of their span is therefore g . $\square$

Definition 12.1.70 (Two-site parent-interaction matrix). Lean ✓ Stmt Let A be a matrix product tensor. Write its canonical two-site parent interaction in the ordered-pair basis of the two-site physical Hilbert space, and denote the resulting matrix by Q_A .

Theorem 12.1.71 (Hermitian commuting overlapping parent interactions). Lean ✓ Stmt Suppose that the canonical length-two parent terms of A commute on a three-site chain. Then Q_A is Hermitian, and its natural placements on the first and last pairs of a three-fold tensor product commute:

$$(Q_A \otimes \mathbb{1})(\mathbb{1} \otimes Q_A) = (\mathbb{1} \otimes Q_A)(Q_A \otimes \mathbb{1}).$$

Proof. ✓ Proof The parent interaction is the orthogonal projector onto the orthogonal complement of the local matrix product space, hence is Hermitian. Transporting the adjacent three-site parent-term commutator to the common left-associated tensor-product basis gives the displayed identity. $\square$

Theorem 12.1.72 (Spatial block actions of the two-site parent interaction). Lean ✓ Stmt Under the preceding hypothesis, there are positive integers d_q, m_q , a unitary identification

$$\mathcal{H} \cong \bigoplus_q \mathcal{H}_{q,l} \otimes \mathcal{H}_{q,r},$$

and Hermitian matrices R_q on $\mathcal{H} \otimes \mathcal{H}_{q,l}$ and S_q on $\mathcal{H}_{q,r} \otimes \mathcal{H}$ such that, after conjugating the middle site by this unitary, the interaction decomposes as

$$Q_A = \bigoplus_q (R_q \otimes \mathbb{1}_{q,r})$$

on the first pair and

$$Q_A = \bigoplus_q (\mathbb{1}_{q,l} \otimes S_q)$$

on the last pair. This is the spatial decomposition of [4, Lemma 2.1] for the canonical parent interaction.

Proof. ✓ Proof Apply the finite-dimensional spatial decomposition for two Hermitian overlapping operators to Q_A in both positions, using Theorem 12.1.71. $\square$

Definition 12.1.73 (Finite sector graph). Lean Lean Lean Lean Lean ✓ Stmt Suppose that a one-site unitary identifies the physical Hilbert space with a finite direct sum

$$\mathcal{H} \cong \bigoplus_a \mathcal{H}_{a,l} \otimes \mathcal{H}_{a,r},$$

and that in these coordinates the complementary two-site interaction is $\bigoplus_{a,b} \mathbb{1}_{a,l} \otimes P_{a,b} \otimes \mathbb{1}_{b,r}$, where each $P_{a,b}$ is an orthogonal projector. The directed edge $a \rightarrow b$ is present precisely when $\text{ran } P_{a,b}$ is nonzero. An ordered cycle of length N is a sequence $(a_0, \dots, a_{N-1})$ for which every cyclically adjacent edge $a_n \rightarrow a_{n+1}$ is present. Cyclic rotations are not identified. This is the graph following equation (2) of [4, Section III].

Theorem 12.1.74 (Symmetric spatial form of a commuting positive interaction). [Lean](#) [Lean](#) [✓ Stmt](#) Let $B \geq 0$ be a two-site operator whose overlapping placements commute: $(B \otimes \mathbb{1})(\mathbb{1} \otimes B) = (\mathbb{1} \otimes B)(B \otimes \mathbb{1})$. There are positive integers $d_{q,l}, d_{q,r}$, a one-site unitary U , and positive semidefinite operators $\eta_{q,h}$ on $\mathcal{H}_{q,r} \otimes \mathcal{H}_{h,l}$ such that

$$(U \otimes U)^* B (U \otimes U) = \bigoplus_{q,h} \mathbb{1}_{q,l} \otimes \eta_{q,h} \otimes \mathbb{1}_{h,r}.$$

This is the symmetric two-sided form obtained from [4, Lemma 2.1 and Section III, equation (2)].

Proof. [✓ Proof](#) Apply the spatial decomposition to the two overlapping placements of B . In the resulting coordinates, matrix entries vanish unless both the left and right sector labels agree. For fixed sectors q, h , choose unit vectors $l_0 \in \mathcal{H}_{q,l}$ and $r_0 \in \mathcal{H}_{h,r}$ and define

$$\begin{aligned} \langle r, l | \eta_{q,h} | r', l' \rangle &= \langle l_0, r; l, r_0 | \widetilde{B} | l_0, r'; l', r_0 \rangle, \\ \widetilde{B} &= (U \otimes U)^* B (U \otimes U). \end{aligned}$$

The left block action makes the right outer factor an identity, while the right block action makes the left outer factor an identity. Hence the displayed matrix element is independent of the chosen outer coordinates and gives the stated direct sum. Each $\eta_{q,h}$ is a compression of the positive semidefinite matrix $\widetilde{B}$, so it is positive semidefinite. $\square$

Theorem 12.1.75 (Sector graph of a commuting parent interaction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the canonical length-two parent terms of A commute on a three-site chain. Then the complementary two-site interaction admits a finite sector graph as in Definition 12.1.73.

Proof. [✓ Proof](#) The canonical two-site parent interaction Q_A is an orthogonal projector. Thus $\mathbb{1} - Q_A$ is a positive semidefinite orthogonal projector. The overlapping placements of $\mathbb{1} - Q_A$ commute because those of Q_A commute. Apply Theorem 12.1.74 to obtain

$$(U \otimes U)^* (\mathbb{1} - Q_A) (U \otimes U) = \bigoplus_{q,h} \mathbb{1}_{q,l} \otimes \eta_{q,h} \otimes \mathbb{1}_{h,r}.$$

The left-hand side is idempotent, so every block on the right is idempotent. Evaluating the two outer identity factors at fixed unit vectors gives $\eta_{q,h}^2 = \eta_{q,h}$. Since $\eta_{q,h} \geq 0$, each $\eta_{q,h}$ is an orthogonal projector and therefore defines the required edge projection. $\square$

Theorem 12.1.76 (Ordered-cycle ground-space dimension formula). [Lean](#) [✓ Stmt](#) For every $N \geq 2$, the periodic ground-space dimension is

$$\dim \ker H_N = \sum_{(a_0, \dots, a_{N-1})} \prod_{n=0}^{N-1} \dim \operatorname{ran} P_{a_n, a_{n+1}},$$

where the sum is over ordered cycles and the indices are read modulo N . This is equations (3)–(4) of [4, Section III] for chains of length at least two. The one-site periodic interaction requires a separate convention.

Proof. [✓ Proof](#) The translated parent interactions commute. Hence the ordered product of their complementary orthogonal projectors is the projector onto $\ker H_N$. In the cyclic sector coordinates this product is block diagonal, with the block indexed by $(a_0, \dots, a_{N-1})$ equal to the tensor product of the adjacent projectors $P_{a_n, a_{n+1}}$. Taking the trace gives the product of their ranks. A sequence containing a missing edge contributes zero, leaving exactly the displayed ordered-cycle sum. $\square$

Theorem 12.1.77 (Period-two ordered-cycle ground-space dimension formula). [Lean](#) [✓ Stmt](#)
At period two, the ground-space dimension is

$$\dim \ker H_2 = \sum_{\substack{a \rightarrow b \\ b \rightarrow a}} \dim \operatorname{ran} P_{a,b} \dim \operatorname{ran} P_{b,a}.$$

The sum is over ordered pairs: if $a \neq b$, then (a,b) and (b,a) are distinct summands. This is the length-two convention stated after equation (4) of [4, Section III].

Proof. [✓ Proof](#) Specialize the ordered-cycle formula to $N = 2$. Its two cyclic factors are $P_{a,b}$ and $P_{b,a}$, which gives the displayed product of ranks. $\square$

Theorem 12.1.78 (Two-site parent Hamiltonian on a single-site chain). [Lean](#) [Lean](#) [✓ Stmt](#) *Under the convention that an L -site interaction vanishes for $L > N$, the two-site parent Hamiltonian on a one-site periodic chain satisfies*

$$\begin{aligned} H_2^{(1)} &= 0, \\ \ker H_2^{(1)} &= \mathbb{C}^d. \end{aligned}$$

Proof. [✓ Proof](#) Since $2 > 1$, the interaction vanishes by convention, and the kernel of the zero operator is the full one-site Hilbert space. $\square$

Theorem 12.1.79 (Classification of eventually constant weighted sector graphs). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let G be a finite directed graph whose edge weights are non-negative integers, and let*

$$s_N = \sum_{(a_0, \dots, a_{N-1})} \prod_{n=0}^{N-1} w(a_n, a_{n+1})$$

be the weighted sum over ordered cyclic vertex sequences. Suppose that s_N is constant for all sufficiently large N . Then every edge that belongs to a directed cycle has weight one, every directed cycle is a repetition of a single loop, and, for every $N > 0$, $s_N = \#\{a : w(a, a) > 0\}$. This is the comparison in [4, Section IV].

Proof. [✓ Proof](#) Choose a sufficiently large N such that the sums at lengths N , $N+1$, and $N(N+1)$ have their common eventual value. Repeating each ordered N -cycle $N+1$ times gives an injection into the cycles of length $N(N+1)$, while repeating each ordered $(N+1)$ -cycle N times gives a second injection. Comparison of the three weighted sums shows first that every positive cyclic weight is one. Both injections must then be exhaustive. Hence a cycle at the common length has periods N and $N+1$; comparison of adjacent entries gives period one. Repeating an arbitrary shorter cycle to a sufficiently large length proves that it is constant. The only surviving ordered cycles at any positive length are therefore the constant words supported on positive loops, each of weight one. $\square$

Corollary 12.1.80 (Parent-ground-space dimension is the loop number). [Lean](#) [Lean](#) [✓ Stmt](#) *If the parent-ground-space dimension is eventually constant, then for every $N > 2$ it equals the number of positive loops in the sector graph. In particular, this conclusion holds under the BNT and all-chain nearest-neighbor parent-ground-space hypotheses of Theorem 12.1.69.*

Proof. [✓ Proof](#) The ordered-cycle dimension formula identifies the eventual parent-ground-space dimension with the eventual weighted-cycle sum. The preceding theorem identifies that sum, at every positive length, with the loop number. Transport between coefficient and Euclidean coordinates preserves the dimension of the ground space. $\square$

Definition 12.1.81 (Tensor associated with a positive Beigi loop). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $a \rightarrow a$ be a positive loop and choose a nonzero vector $\varphi_a \in \text{ran } P_{a,a}$. In sector coordinates define the matrix product tensor $(C_a^{(r,l)})_{x,y} = \delta_{x,l} \varphi_a(r,y)$ and set it to zero outside sector a . The physical loop tensor is the image of C_a under the one-site unitary of the spatial decomposition. Its associated cyclic product pairs the right factor at site n with the left factor at site $n+1$. The corresponding N -site product state is denoted by $\Psi_a^{(N)} \in (\mathbb{C}^d)^{\otimes N}$.

Definition 12.1.82 (Schmidt-support tensor associated with a positive Beigi loop). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\Phi_a = (\varphi_a(r,l))_{r,l}$ and write $r_a = \text{rank } \Phi_a$. Choose a rank factorization

$$\begin{aligned}\Phi_a &= X_a Y_a, \\ X_a &\in M_{D_a^R, r_a}(\mathbb{C}), \\ Y_a &\in M_{r_a, D_a^L}(\mathbb{C}).\end{aligned}$$

The tensor on the minimal Schmidt support is

$$(\tilde{C}_a^{(r,l)})_{\alpha,\beta} = (Y_a)_{\alpha,l} (X_a)_{r,\beta},$$

again extended by zero outside sector a . Its physical tensor is its image under the same one-site unitary.

Definition 12.1.83 (Normalized minimal tensor associated with a positive Beigi loop). [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let $\nu_a = \|\varphi_a\|_2 > 0$, where the norm is the Hilbert-space norm of the bond vector, or equivalently the Frobenius norm of Φ_a . In either coordinate system, the normalized representative is $\hat{C}_a = \nu_a^{-1} \tilde{C}_a$. The raw coordinate and physical tensors are retained separately.

Lemma 12.1.84 (Positive Schmidt dimension of a Beigi loop). [Lean](#) [✓ Stmt](#) *The Schmidt rank r_a of the nonzero loop bond vector is positive.*

Proof. [✓ Proof](#) If $r_a = 0$, then the intermediate index set in the factorization $\Phi_a = X_a Y_a$ is empty. Every entry of $X_a Y_a$ is therefore zero, contrary to $\varphi_a \neq 0$. $\square$

Theorem 12.1.85 (Injectivity of the minimal Beigi loop tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The tensor $\hat{C}_a$ on the Schmidt support is injective, as is its image under the one-site unitary. In particular, the physical tensor is normal.*

Proof. [✓ Proof](#) In a rank factorization $\Phi_a = X_a Y_a$ through $r_a = \text{rank } \Phi_a$, the columns of Y_a and the rows of X_a each span $\mathbb{C}^{r_a}$. Their pairwise outer products therefore span $M_{r_a}(\mathbb{C})$, and these outer products are precisely the loop-sector letters of $\tilde{C}_a$. A unitary change of physical coordinates preserves this span. One-site injectivity implies normality. $\square$

Theorem 12.1.86 (Transfer fixed point of a normalized Beigi loop tensor). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The transfer map of the unnormalized coordinate tensor is*

$$\begin{aligned}\mathcal{E}_a(Z) &= \text{tr}(X_a^* X_a Z) Y_a Y_a^*, \\ \mathcal{E}_a^2 &= \nu_a^2 \mathcal{E}_a.\end{aligned}$$

Hence the transfer map of $\hat{C}_a$ is idempotent. This remains true after the physical unitary rotation. The normalized physical tensor is injective and therefore normal.

Proof. ✓ Proof Summing the Kraus terms over the loop-sector coordinates gives the stated rank-one formula. Cyclicity of the trace and $\Phi_a = X_a Y_a$ give $\text{tr}(X_a^* X_a Y_a Y_a^*) = \text{tr}(\Phi_a^* \Phi_a) = \nu_a^2$. Scaling every letter by ν_a^{-1} scales the transfer map by ν_a^{-2} , which makes it idempotent. A unitary change of physical coordinates leaves the transfer map invariant, and the nonzero scalar preserves injectivity. $\square$

Theorem 12.1.87 (Periodic vector of a positive Beigi loop). Lean Lean Lean Lean Lean

✓ Stmt For every $N > 0$, the periodic matrix product vector of C_a has coefficient $\prod_{n=0}^{N-1} \varphi_a(r_n, l_{n+1})$ on a cyclic configuration in sector a , and coefficient zero on every other sector configuration. Consequently, the periodic vector of the physical loop tensor is exactly the sitewise-unitary image of this cyclic product-of-pairs state. The same identities hold for the tensor on the minimal Schmidt support. For the normalized minimal tensor it is $\nu_a^{-N} \Psi_a^{(N)}$.

Proof. ✓ Proof Write D_a^L for the left-factor dimension of sector a . Expanding the closed matrix product on a sector configuration $((r_n, l_n))_{n=0}^{N-1}$ gives

$$\sum_{g \in \{0, \dots, D_a^L - 1\}^N} \prod_{n=0}^{N-1} \delta_{g_n, l_n} \varphi_a(r_n, g_{n+1}) = \prod_{n=0}^{N-1} \varphi_a(r_n, l_{n+1}),$$

where all site labels are read cyclically. Indeed, the Kronecker deltas leave only the assignment $g_n = l_n$.

For the minimal tensor, let $S^{(r,l)}$ be the rectangular matrix whose only nonzero row is row l , equal there to row r of X_a . Then $C_a^{(r,l)} = S^{(r,l)} Y_a$ and $\tilde{C}_a^{(r,l)} = Y_a S^{(r,l)}$. For every nonempty word there is therefore a rectangular matrix R such that its two products are $R Y_a$ and $Y_a R$, respectively. Cyclicity of the trace gives $\text{tr}(R Y_a) = \text{tr}(Y_a R)$, proving equality of all positive-length periodic vectors.

If A_a denotes the physical loop tensor and U the one-site unitary, then for a physical configuration $s = (s_0, \dots, s_{N-1})$ one has

$$\text{tr}(A_a^{s_0} \dots A_a^{s_{N-1}}) = \sum_{t \in \{0, \dots, d-1\}^N} \left(\prod_{n=0}^{N-1} U_{s_n, t_n} \right) \text{tr}(C_a^{t_0} \dots C_a^{t_{N-1}}).$$

Substitution of the preceding cyclic coefficient formula gives the stated sitewise-unitary image. $\square$

Theorem 12.1.88 (Spatial conjugation of complementary interaction products). Lean ✓ Stmt

Suppose that A is equipped with Beigi sector data, with one-site spatial unitary U . For $N \geq 2$, let Q_N be the product of the translated complementary parent interactions and let $\tilde{Q}_N$ be the corresponding product in sector coordinates. Then $(U^*)^{\otimes N} Q_N U^{\otimes N} = \tilde{Q}_N$.

Proof. ✓ Proof The two-site coordinate identity gives $(U^* \otimes U^*)(\mathbb{1} - h)(U \otimes U) = \tilde{Q}$ for one bond. Conjugation by $(U^*)^{\otimes N}$ carries every translated copy of this bond identity to the corresponding translated sector interaction. Since unitary conjugation distributes over an ordered product,

$$(U^*)^{\otimes N} \left(\prod_{n=0}^{N-1} Q_n \right) U^{\otimes N} = \prod_{n=0}^{N-1} ((U^*)^{\otimes N} Q_n U^{\otimes N}) = \prod_{n=0}^{N-1} \tilde{Q}_n = \tilde{Q}_N,$$

where Q_n and $\tilde{Q}_n$ are the translated physical and sector complementary interactions, respectively. $\square$

Theorem 12.1.89 (Fixed complementary-product vectors have zero parent energy). [Lean](#)
[✓ Stmt](#) Suppose that A is equipped with Beigi sector data. For $N \geq 2$, let $h_0, \dots, h_{N-1}$ be its translated length-two parent interactions and set $Q_N = \prod_{n=0}^{N-1} (\mathbb{1} - h_n)$. Every vector fixed by Q_N belongs to the kernel of the periodic parent Hamiltonian $H_N = \sum_n h_n$.

Proof. [✓ Proof](#) Under the Euclidean identification E , the matrix product of the complementary interactions is carried to

$$Q_N^E = E^{-1} Q_N E = \prod_{n=0}^{N-1} (\mathbb{1} - h_n^E).$$

The operators h_n^E are orthogonal projections. Beigi sector data supply pairwise commutativity, which Euclidean transport preserves. For all $0 \leq m, n < N$, one has $h_m^E h_n^E = h_n^E h_m^E$. For any commuting orthogonal projections $P_0, \dots, P_{N-1}$,

$$\prod_{n=0}^{N-1} (\mathbb{1} - P_n) = \text{proj}_{\bigcap_{n=0}^{N-1} \ker P_n}.$$

Applying this identity to the h_n^E gives

$$\begin{aligned} Q_N^E &= \text{proj}_{\bigcap_{n=0}^{N-1} \ker h_n^E}, \\ \bigcap_{n=0}^{N-1} \ker h_n^E &= \ker \left(\sum_{n=0}^{N-1} h_n^E \right) = \ker H_N^E. \end{aligned}$$

Thus $Q_N v = v$ implies $E^{-1}v \in \ker H_N^E$, and conjugating back by E gives $v \in \ker H_N$. $\square$

Theorem 12.1.90 (A positive Beigi loop gives a parent ground state). [Lean](#) [Lean](#) [✓ Stmt](#) Let $a \rightarrow a$ be a positive loop. For every $N \geq 2$, its physical cyclic product-of-pairs state belongs to the kernel of the length-two periodic parent Hamiltonian. Equivalently, its Euclidean representative belongs to the parent ground space.

Proof. [✓ Proof](#) In sector coordinates, the complete product $\tilde{Q}_N$ of the complementary two-site interactions is block diagonal over cyclic sector sequences. Write $\Phi_a(k; x)$ for the coefficient of the loop cyclic product in the block indexed by k . Its support lies in sector a , so for every $k \neq (a, \dots, a)$ and every edge-index tuple x , one has $\Phi_a(k; x) = 0$. Thus only the constant cyclic sector sequence contributes. On this block, $\tilde{Q}_N$ is the tensor product of the edge projectors. Since $\varphi_a \in \text{ran } P_{a,a}$,

$$\begin{aligned} \tilde{Q}_N \Phi_a &= P_{a,a}^{\otimes N} \Phi_a = \Phi_a, \\ \Phi_a(t_0, \dots, t_{N-1}) &= \prod_{n=0}^{N-1} \varphi_a(r_n, l_{n+1}). \end{aligned}$$

If U is the one-site spatial unitary and $\Psi_a = U^{\otimes N} \Phi_a$ is the physical loop state, unitary conjugation gives

$$Q_N \Psi_a = U^{\otimes N} \tilde{Q}_N \Phi_a = U^{\otimes N} \Phi_a = \Psi_a.$$

Theorem 12.1.89 now yields $\Psi_a \in \ker H_N$. $\square$

Lemma 12.1.93 (Common physical unitaries preserve mixed transfer maps). [Lean](#) [✓ Stmt](#) Let $A = (A_i)_i$ and $B = (B_i)_i$ be two tensors, possibly with different bond dimensions, and let u be unitary on the physical index. If

$$\tilde{A}_i = \sum_j u_{ij} A_j, \quad \tilde{B}_i = \sum_j u_{ij} B_j,$$

then

$$\mathcal{E}_{\tilde{A}, \tilde{B}} = \mathcal{E}_{A, B}.$$

Proof. [✓ Proof](#) Expand both physical sums. The coefficient of $A_j X B_k^\dagger$ is $\sum_i u_{ij} \overline{u_{ik}} = \delta_{jk}$, so only the diagonal terms remain. $\square$

Lemma 12.1.94 (Gauge-phase transport of mixed-transfer vanishing). [Lean](#) [Lean](#) [✓ Stmt](#) Let U, A and V, B be pairs of tensors with respective common bond dimensions. Suppose that $\zeta_A, \zeta_B \neq 0$ and that X_A, X_B are invertible matrices such that, for every physical index i ,

$$A^i = \zeta_A X_A U^i X_A^{-1}, \quad (12.10)$$

$$B^i = \zeta_B X_B V^i X_B^{-1}. \quad (12.11)$$

Then

$$\mathcal{E}_{A, B} = 0 \implies \mathcal{E}_{U, V} = 0. \quad (12.12)$$

Identifying equal bond dimensions on either tensor leg preserves this vanishing condition in both directions.

Proof. [✓ Proof](#) Set $D_A = \zeta_A I$ and $D_B = \zeta_B I$. For every rectangular matrix Z , covariance of the mixed transfer map gives

$$\mathcal{E}_{A, B}(X_A Z X_B^\dagger) = X_A D_A \mathcal{E}_{U, V}(Z) (X_B D_B)^\dagger. \quad (12.13)$$

If $\mathcal{E}_{A, B} = 0$, the left-hand side vanishes. The two outer factors are invertible, so cancellation gives $\mathcal{E}_{U, V}(Z) = 0$ for every Z . $\square$

Lemma 12.1.95 (Distinct normalized Beigi loops are locally orthogonal). [Lean](#) [✓ Stmt](#) If a and b are distinct positive loops in the sector graph, then the mixed transfer map of their normalized minimal loop tensors vanishes:

$$\mathcal{E}_{L_a, L_b} = 0.$$

Proof. [✓ Proof](#) In sector coordinates, every physical letter of L_a is supported in sector a , while every physical letter of L_b is supported in sector b . Hence, for every rectangular matrix Z and every physical index i ,

$$L_a^i Z (L_b^i)^\dagger = 0, \quad (12.14)$$

$$\mathcal{E}_{L_a, L_b}(Z) = \sum_i L_a^i Z (L_b^i)^\dagger = 0. \quad (12.15)$$

The common one-site physical unitary preserves the mixed transfer map by Lemma 12.1.93. $\square$

Theorem 12.1.96 (Multi-sector NNCPH ground spaces imply a fixed point for the basis direct sum). [Lean](#) [✓ Stmt](#) Let $A_1, \dots, A_g$ be the distinct basis tensors of a BNT canonical form, and set

$$B = \bigoplus_{j=1}^g A_j.$$

Suppose that, for every $N > 2$, the translated length-two parent interactions of B commute, every $V^{(N)}(A_j)$ has zero energy, and

$$\ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Then B is a renormalization fixed point. This establishes the reverse implication of [11, Theorem 3.10] for multiplicity-one unit-weight representatives.

Proof. [✓ Proof](#) Beigi’s positive-loop product states span each parent ground space. Consequently every normal basis tensor A_j agrees, up to a unit-modulus scalar and an invertible virtual change of basis, with a normalized minimal loop tensor $L_{\ell(j)}$. Distinctness of the BNT basis makes $j \mapsto \ell(j)$ injective. Thus there are $\zeta_j \neq 0$ and invertible X_j such that

$$L_{\ell(j)}^i = \zeta_j X_j A_j^i X_j^{-1}. \quad (12.16)$$

Each $L_{\ell(j)}$ has an idempotent transfer map. If $j \neq k$, then $L_{\ell(j)}$ and $L_{\ell(k)}$ have disjoint sector support, so their mixed transfer map vanishes. Lemma 12.1.94 transports this vanishing through the two virtual changes of basis:

$$\mathcal{E}_{L_{\ell(j)}, L_{\ell(k)}} = 0 \implies \mathcal{E}_{A_j, A_k} = 0. \quad (12.17)$$

Thus the diagonal mixed transfer maps of the family A_j are idempotent and the off-diagonal ones vanish. The blockwise criterion is

$$\mathcal{E}_B \circ \mathcal{E}_B = \mathcal{E}_B \iff \mathcal{E}_{A_j, A_k} \circ \mathcal{E}_{A_j, A_k} = \mathcal{E}_{A_j, A_k} \quad (12.18)$$

for every pair j, k , and proves the claim. $\square$

Theorem 12.1.97 (Fixed points and commuting parent ground spaces for the basis direct sum). [Lean](#) [✓ Stmt](#) Let $A_1, \dots, A_g$ be the distinct basis tensors of a BNT canonical form, and set $A = \bigoplus_{j=1}^g A_j$. Then

$$\mathcal{E}_A^2 = \mathcal{E}_A \iff A \text{ has all-chain commuting parent ground spaces spanned by } (A_j)_j. \quad (12.19)$$

This statement concerns the multiplicity-one, unit-weight direct sum; it does not include repeated copies or arbitrary raw weights.

Proof. [✓ Proof](#) The forward implication is Theorem 12.1.32, and the reverse implication is Theorem 12.1.96. $\square$

12.2 Decorrelation and the idempotent-product parent-commuting condition

In [11, Appendix D.2], the parent commuting Hamiltonian condition is stated for local projectors Q_{AX} and Q_{XB} . With $\widehat{Q}_{AX}$ and $\widehat{Q}_{XB}$ denoting their lifts to the tripartite space, the commutation

condition and common ground space K_{AXB} satisfy

$$\begin{aligned} [\widehat{Q}_{AX}, \widehat{Q}_{XB}] &= 0, \\ K_{AXB} &= (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB}). \end{aligned}$$

Setting $P_{AX} = 1 - Q_{AX}$ and $P_{XB} = 1 - Q_{XB}$, the corresponding product identity on the tripartite space is $\widehat{P}_{AX}\widehat{P}_{XB} = P_K$. For a positive semidefinite operator on a bipartite space, the support projector of the retained left marginal, lifted to the full space, fixes the original operator, as used in [11, Appendix D.2].

Scope restriction (abstract idempotent-product condition): The subsequent formal predicate retains only abstract commuting idempotents and their product identity; unlike [11, Appendix D.2], it omits tensor-product locality, orthogonal projectors, and the ground-space intersection, and admits the tautological choice $P_{AX} = P_{XB} = P_K$. Documented in [99].

Lemma 12.2.1 (Partial trace of the identity). Lean ✓ Stmt *For finite-dimensional spaces H_L and H_R , one has $\text{tr}_R(\mathbb{1}_{L \otimes R}) = (\dim H_R)\mathbb{1}_L$.*

Proof. ✓ Proof In product-basis coordinates,

$$(\text{tr}_R \mathbb{1}_{L \otimes R})_{ij} = \sum_k \delta_{ij} = (\dim H_R) \delta_{ij}.$$

□

For $\rho \geq 0$ on $H_L \otimes H_R$ with P_L the orthogonal projector onto the support of $\text{tr}_R \rho$, the identities $(P_L \otimes \mathbb{1}_R)\rho = \rho$ and $\rho(P_L \otimes \mathbb{1}_R) = \rho$ provide the P_{AX} support-projector identities in [11, Appendix D.2] upon setting $H_L = H_A \otimes H_X$ and $H_R = H_B$, as developed in [182].

Theorem 12.2.2 (Hermiticity of the left partial trace). Lean ✓ Stmt *If $\rho = \rho^\dagger$, then $\text{tr}_L \rho = (\text{tr}_L \rho)^\dagger$.*

Proof. ✓ Proof Entrywise Hermiticity gives

$$\overline{(\text{tr}_L \rho)_{r,s}} = \sum_k \overline{\rho_{(k,r),(k,s)}} = \sum_k \rho_{(k,s),(k,r)} = (\text{tr}_L \rho)_{s,r}.$$

□

Similarly, for the support projector P_R of $\text{tr}_L \rho$, one has $(\mathbb{1}_L \otimes P_R)\rho = \rho$ and $\rho(\mathbb{1}_L \otimes P_R) = \rho$.

We now use the explicit tripartite indexing $H_A \otimes (H_X \otimes H_B)$. Reassociation identifies it with $(H_A \otimes H_X) \otimes H_B$ whenever the AX marginal is taken.

Definition 12.2.3 (Tripartite local lifts). Lean Lean Lean Lean ✓ Stmt Writing $\widehat{M}_R$ for the lift of an operator M_R supported on $R \in \{AX, XB, A, B\}$, the four lifts to $H_A \otimes (H_X \otimes H_B)$ are

$$\widehat{M}_{AX} = M_{AX} \otimes \mathbb{1}_B, \tag{12.20}$$

$$\widehat{M}_{XB} = \mathbb{1}_A \otimes M_{XB},$$

$$\widehat{M}_A = M_A \otimes \mathbb{1}_{XB},$$

$$\widehat{M}_B = \mathbb{1}_{AX} \otimes M_B. \tag{12.21}$$

Identities (12.20) and (12.21) use the canonical identification $(H_A \otimes H_X) \otimes H_B \cong H_A \otimes (H_X \otimes H_B)$.

Lemma 12.2.4 (Products and adjoints of tripartite lifts). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) The lifts from AX and XB preserve multiplication,

$$\begin{aligned}\widehat{MN}_{AX} &= \widehat{M}_{AX}\widehat{N}_{AX}, \\ \widehat{MN}_{XB} &= \widehat{M}_{XB}\widehat{N}_{XB},\end{aligned}$$

and each of the four lifts from AX , XB , A , and B preserves adjoints:

$$(\widehat{M}_R)^\dagger = \widehat{M}_R^\dagger, \quad R \in \{AX, XB, A, B\}.$$

Proof. [✓ Proof](#) For the two-factor lifts,

$$\begin{aligned}(M_{AX} \otimes \mathbb{1}_B)(N_{AX} \otimes \mathbb{1}_B) &= (M_{AX}N_{AX}) \otimes \mathbb{1}_B, \\ (\mathbb{1}_A \otimes M_{XB})(\mathbb{1}_A \otimes N_{XB}) &= \mathbb{1}_A \otimes (M_{XB}N_{XB}).\end{aligned}$$

Moreover,

$$\begin{aligned}(M_{AX} \otimes \mathbb{1}_B)^\dagger &= M_{AX}^\dagger \otimes \mathbb{1}_B, \\ (\mathbb{1}_A \otimes M_{XB})^\dagger &= \mathbb{1}_A \otimes M_{XB}^\dagger,\end{aligned}$$

and similarly,

$$\begin{aligned}(M_A \otimes \mathbb{1}_{XB})^\dagger &= M_A^\dagger \otimes \mathbb{1}_{XB}, \\ (\mathbb{1}_{AX} \otimes M_B)^\dagger &= \mathbb{1}_{AX} \otimes M_B^\dagger,\end{aligned}$$

with the tensor factors reassocated to the tripartite space. □

Definition 12.2.5 (Tripartite observable decorrelation). [Lean](#) [✓ Stmt](#) Regions A and B are decorrelated relative to P if, for all Hermitian observables $O_A = O_A^\dagger$ and $O_B = O_B^\dagger$, one has $PO_A(\mathbb{1} - P)O_BP = 0$. This is [11, Definition D.1, lines 2187–2191].

Definition 12.2.6 (Complex-linear extension of decorrelation). [Lean](#) [✓ Stmt](#) The complex-linear extension requires $PO_A(\mathbb{1} - P)O_BP = 0$ for arbitrary complex matrices O_A and O_B .

Theorem 12.2.7 (Observable and complex-linear decorrelation are equivalent). [Lean](#) [✓ Stmt](#) Observable decorrelation holds if and only if its complex-linear extension holds.

Proof. [✓ Proof](#) Every complex matrix has the decomposition

$$\begin{aligned}O &= H + iK, \\ H &= \frac{O + O^\dagger}{2}, \\ K &= \frac{O - O^\dagger}{2i},\end{aligned}$$

where $H = H^\dagger$ and $K = K^\dagger$. For analogous decompositions $O = H + iK$ and $O' = H' + iK'$,

$$\begin{aligned}PO(\mathbb{1} - P)O'P &= PH(\mathbb{1} - P)H'P + iPH(\mathbb{1} - P)K'P \\ &\quad + iPK(\mathbb{1} - P)H'P - PK(\mathbb{1} - P)K'P.\end{aligned}$$

Each summand vanishes by observable decorrelation applied to its two Hermitian components. □

Theorem 12.2.8 (Reverse order of decorrelation). [Lean](#) [✓ Stmt](#) If $P = P^\dagger$ and A and B are decorrelated, then, for arbitrary complex matrices O_A and O_B , one has $PO_B(\mathbb{1} - P)O_AP = 0$.

Proof. [✓ Proof](#) Apply decorrelation to $O_A^\dagger$ and $O_B^\dagger$, and take adjoints. $\square$

Definition 12.2.9 (Tripartite reduced operators). [Lean](#) [Lean](#) [✓ Stmt](#) The two reduced operators are $\rho_{AX} = \text{tr}_B P$ and $\rho_{XB} = \text{tr}_A P$.

Theorem 12.2.10 (Matrix-unit expansions of the reduced operators). [Lean](#) [Lean](#) [✓ Stmt](#) For matrix units on A and B ,

$$\begin{aligned}\mathbb{1}_A \otimes \rho_{XB} &= \sum_{a,k} E_{ak}^A P E_{ka}^A, \\ \rho_{AX} \otimes \mathbb{1}_B &= \sum_{b,k} E_{bk}^B P E_{kb}^B.\end{aligned}$$

Proof. [✓ Proof](#) For the XB marginal, evaluation in the product basis gives

$$(E_{ak}^A P E_{ka}^A)_{(a_0,r),(a_1,s)} = \delta_{a,a_0} \delta_{a,a_1} P_{(k,r),(k,s)}.$$

Hence

$$\sum_{a,k} (E_{ak}^A P E_{ka}^A)_{(a_0,r),(a_1,s)} = \delta_{a_0,a_1} \sum_k P_{(k,r),(k,s)} = (\mathbb{1}_A \otimes \rho_{XB})_{(a_0,r),(a_1,s)}.$$

The AX identity follows symmetrically. $\square$

Theorem 12.2.11 (Decorrelation for the reduced operators). [Lean](#) [✓ Stmt](#) If A and B are decorrelated, then $\hat{\rho}_{XB}(\mathbb{1} - P)\hat{\rho}_{AX} = 0$.

Proof. [✓ Proof](#) Substitute the matrix-unit expansions. Every summand has the form

$$E_{ak}^A P E_{ka}^A (\mathbb{1} - P) E_{bl}^B P E_{lb}^B = E_{ak}^A [P E_{ka}^A (\mathbb{1} - P) E_{bl}^B P] E_{lb}^B = 0,$$

where the bracketed factor vanishes by decorrelation. $\square$

Theorem 12.2.12 (Reverse reduced-operator decorrelation). [Lean](#) [✓ Stmt](#) If $P = P^\dagger$ and A and B are decorrelated, then $\hat{\rho}_{AX}(\mathbb{1} - P)\hat{\rho}_{XB} = 0$.

Proof. [✓ Proof](#) Substitute the matrix-unit expansions. Every summand has the form

$$E_{bl}^B P E_{lb}^B (\mathbb{1} - P) E_{ak}^A P E_{ka}^A = E_{bl}^B [P E_{lb}^B (\mathbb{1} - P) E_{ak}^A P] E_{ka}^A = 0,$$

where the bracketed factor vanishes by reverse-order decorrelation. $\square$

Definition 12.2.13 (Tripartite marginal support projectors). [Lean](#) [Lean](#) [✓ Stmt](#) For a Hermitian operator P , let P_{AX} and P_{XB} be the support projectors of $\text{tr}_B P$ and $\text{tr}_A P$, respectively, as in [11, Appendix D.2, lines 2225–2229]. Write $\hat{P}_{AX}$ and $\hat{P}_{XB}$ for their lifts to the tripartite space.

Theorem 12.2.14 (Decorrelation for the two marginal supports). [Lean](#) [Lean](#) [✓ Stmt](#) If $P = P^\dagger$ and A and B are decorrelated, then

$$\hat{P}_{XB}(\mathbb{1} - P)\hat{P}_{AX} = 0, \tag{12.22}$$

$$\hat{P}_{AX}(\mathbb{1} - P)\hat{P}_{XB} = 0. \tag{12.23}$$

Proof. ✓ Proof For $g(0) = 0$ and $g(x) = x^{-1}$ when $x \neq 0$, functional calculus gives

$$\begin{aligned} P_{AX} &= g(\rho_{AX})\rho_{AX} = \rho_{AX}g(\rho_{AX}), \\ P_{XB} &= g(\rho_{XB})\rho_{XB} = \rho_{XB}g(\rho_{XB}). \end{aligned}$$

Hence

$$\widehat{P}_{XB}(\mathbb{1} - P)\widehat{P}_{AX} = g(\widehat{\rho_{XB}})[\widehat{\rho_{XB}}(\mathbb{1} - P)\widehat{\rho_{AX}}]g(\widehat{\rho_{AX}}) = 0,$$

and, in the reverse order,

$$\widehat{P}_{AX}(\mathbb{1} - P)\widehat{P}_{XB} = g(\widehat{\rho_{AX}})[\widehat{\rho_{AX}}(\mathbb{1} - P)\widehat{\rho_{XB}}]g(\widehat{\rho_{XB}}) = 0.$$

□

Theorem 12.2.15 (Absorption by the lifted XB support projector). Lean Lean ✓ Stmt For $P \geq 0$, $\widehat{P}_{XB}P = P = P\widehat{P}_{XB}$.

Proof. ✓ Proof For the bipartition $A|XB$, the two support-absorption identities give

$$(\mathbb{1}_A \otimes P_{XB})P = P = P(\mathbb{1}_A \otimes P_{XB}).$$

□

Theorem 12.2.16 (Absorption by the lifted AX support projector). Lean Lean ✓ Stmt For $P \geq 0$, $\widehat{P}_{AX}P = P = P\widehat{P}_{AX}$.

Proof. ✓ Proof After reassociating the tripartite space as $AX|B$, the two support-absorption identities give

$$(P_{AX} \otimes \mathbb{1}_B)P = P = P(P_{AX} \otimes \mathbb{1}_B).$$

□

Theorem 12.2.17 (Product of the two marginal support projectors). Lean ✓ Stmt If $P = P^\dagger = P^2$ and A and B are decorrelated relative to P , then

$$\widehat{P}_{XB}\widehat{P}_{AX} = P, \tag{12.24}$$

$$\widehat{P}_{AX}\widehat{P}_{XB} = P. \tag{12.25}$$

Proof. ✓ Proof Inserting $\mathbb{1} = P + (\mathbb{1} - P)$ between the two support projectors gives

$$\begin{aligned} \widehat{P}_{XB}\widehat{P}_{AX} &= \widehat{P}_{XB}P\widehat{P}_{AX} + \widehat{P}_{XB}(\mathbb{1} - P)\widehat{P}_{AX} \\ &= P + 0 = P, \end{aligned}$$

where $\widehat{P}_{XB}P\widehat{P}_{AX} = P$ by the support-absorption identities and $\widehat{P}_{XB}(\mathbb{1} - P)\widehat{P}_{AX} = 0$ by (12.22). The same calculation with the two factors reversed, using (12.23), gives (12.25). □

Definition 12.2.18 (Parent commuting Hamiltonian on the tripartite space). Lean Lean Lean ✓ Stmt There are orthogonal projectors P_{AX} and P_{XB} on the two local factors whose lifts commute and whose ranges satisfy the source ground-space intersection condition $\text{ran } P = \text{ran } \widehat{P}_{AX} \cap \text{ran } \widehat{P}_{XB}$. Thus $Q_{AX} = \mathbb{1} - P_{AX}$ and $Q_{XB} = \mathbb{1} - P_{XB}$ are exactly the commuting parent terms of [11, Definition D.2, lines 2205–2218].

Theorem 12.2.19 (Ground-space intersection and projector product). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ **Stmt** Let P , P_{AX} , and P_{XB} be orthogonal projectors, and write their lifts to the tripartite space as $\widehat{P}_{AX} = P_{AX} \otimes \mathbb{1}_B$ and $\widehat{P}_{XB} = \mathbb{1}_A \otimes P_{XB}$. Suppose that the lifted projectors commute, $\widehat{P}_{AX}\widehat{P}_{XB} = \widehat{P}_{XB}\widehat{P}_{AX}$. Then

$$\text{ran } P = \text{ran } \widehat{P}_{AX} \cap \text{ran } \widehat{P}_{XB} \iff \widehat{P}_{AX}\widehat{P}_{XB} = P. \quad (12.26)$$

This is the passage from [11, Definition D.2, lines 2205–2218] to the projector identity at lines 2279–2289.

Proof. $\checkmark$ **Proof** A vector lies in the range of $\widehat{P}_{AX}\widehat{P}_{XB}$ precisely when it is fixed by both commuting idempotents. Hence $\text{ran}(\widehat{P}_{AX}\widehat{P}_{XB}) = \text{ran } \widehat{P}_{AX} \cap \text{ran } \widehat{P}_{XB}$. The product is again an orthogonal projector. Orthogonal projectors with equal ranges are equal, giving the forward implication in (12.26); the reverse implication follows by substituting the product identity. $\square$

Theorem 12.2.20 (Parent commuting Hamiltonian–complex-linear decorrelation equivalence). [Lean](#) [Lean](#) [Lean](#) $\checkmark$ **Stmt** Let $P = P^\dagger = P^2$. Then P is the ground-space projector of a parent commuting Hamiltonian on AX and XB if and only if regions A and B satisfy complex-linear decorrelation.

Proof. $\checkmark$ **Proof** Starting from decorrelation, take the support projectors of ρ_{AX} and ρ_{XB} . The preceding product theorem proves their commutation and identifies their product with P .

Starting from commuting parent projectors, first use the ground-space intersection theorem to write $P = \widehat{P}_{AX}\widehat{P}_{XB} = \widehat{P}_{XB}\widehat{P}_{AX}$. Locality gives $[O_A, \widehat{P}_{XB}] = [O_B, \widehat{P}_{AX}] = 0$, while $\widehat{P}_{XB}(\mathbb{1} - P)\widehat{P}_{AX} = 0$. Therefore, using the two locality commutators,

$$PO_A(\mathbb{1} - P)O_BP = \widehat{P}_{AX}O_A[\widehat{P}_{XB}(\mathbb{1} - P)\widehat{P}_{AX}]O_B\widehat{P}_{XB} = 0.$$

$\square$

Theorem 12.2.21 (Parent commuting Hamiltonian–observable decorrelation equivalence). [Lean](#) $\checkmark$ **Stmt** Let $P = P^\dagger = P^2$. Then P is the ground-space projector of a parent commuting Hamiltonian on AX and XB if and only if regions A and B are decorrelated when tested on Hermitian observables. This is [11, Proposition D.3, lines 2221–2289].

Proof. $\checkmark$ **Proof** Combine the complex-linear parent equivalence with the decomposition $O = H + iK$ from Theorem 12.2.7. $\square$

Definition 12.2.22 (Decorrelation). [Lean](#) $\checkmark$ **Stmt** Given an endomorphism P_K and sets of observables $\mathcal{O}_A, \mathcal{O}_B$, regions A and B are *decorrelated* w.r.t. K when $P_K \circ O_A \circ (\mathbb{1} - P_K) \circ O_B \circ P_K = 0$ for all $O_A \in \mathcal{O}_A$ and $O_B \in \mathcal{O}_B$.

Definition 12.2.23 (Idempotent-product parent-commuting structure). [Lean](#) $\checkmark$ **Stmt** The structure used below records only the idempotent-product consequence of the parent commuting Hamiltonian condition of [11, Appendix D.2]. It consists of commuting idempotent endomorphisms P_{AX} and P_{XB} such that, for the idempotent P_K associated to K , $P_{AX} \circ P_{XB} = P_K$. In the source, this is the equation obtained from local projectors Q_{AX} and Q_{XB} satisfying $[Q_{AX}, Q_{XB}] = 0$ and $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$.

Theorem 12.2.24 (Idempotence gives tautological idempotent-product data). [Lean](#) $\checkmark$ **Stmt** If $P_K \circ P_K = P_K$, then P_K has idempotent-product data.

Proof. ✓ Proof Take $P_{AX} = P_{XB} = P_K$. Their commutativity is immediate, and the product identity is exactly the assumed idempotence of P_K . □

Lemma 12.2.25 (Cancellation for commuting idempotents). Lean ✓ Stmt *If P and Q are commuting idempotents, then $Q \circ (\mathbb{1} - P \circ Q) \circ P = 0$.*

Proof. ✓ Proof The cancellation is the computation

$$Q(\mathbb{1} - PQ)P = QP - QPQP = QP - QPPQ = QP - QP = 0,$$

using commutativity and idempotence. □

Theorem 12.2.26 (Idempotence of the product projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then $P_K \circ P_K = P_K$.*

Proof. ✓ Proof Since P_{AX} and P_{XB} are idempotent and commute, their product $P_{AX} \circ P_{XB}$ is idempotent. Substituting $P_K = P_{AX} \circ P_{XB}$ gives the claim. □

Theorem 12.2.27 (Absorption by the left projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then $P_{AX} \circ P_K = P_K$.*

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{AX} ,

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 12.2.28 (Absorption by the right projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then $P_K \circ P_{XB} = P_K$.*

Proof. ✓ Proof Using $P_K = P_{AX} \circ P_{XB}$ and the idempotence of P_{XB} ,

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 12.2.29 (Cross-absorption by the right projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then $P_{XB} \circ P_K = P_K$.*

Proof. ✓ Proof By commutativity and the idempotence of P_{XB} ,

$$P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 12.2.30 (Cross-absorption by the left projector). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then $P_K \circ P_{AX} = P_K$.*

Proof. ✓ Proof By commutativity and the idempotence of P_{AX} ,

$$P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

□

Theorem 12.2.31 (Reverse product identity). Lean ✓ Stmt *If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then $P_{XB} \circ P_{AX} = P_K$.*

Proof. ✓ Proof This follows from commutativity of P_{AX} and P_{XB} together with the identity $P_{AX} \circ P_{XB} = P_K$. $\square$

Theorem 12.2.32 (Commuting complementary projectors). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data and $Q_{AX} = \mathbb{1} - P_{AX}$ and $Q_{XB} = \mathbb{1} - P_{XB}$, then $Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}$.

Proof. ✓ Proof Expanding both products and using $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ shows that the complementary projectors commute. $\square$

Theorem 12.2.33 (Idempotent-product data imply decorrelation). Lean ✓ Stmt If P_{AX} and P_{XB} are idempotent endomorphisms satisfying $P_{AX} \circ P_{XB} = P_K$ and $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$, and A -observables commute with P_{XB} while B -observables commute with P_{AX} , then regions A and B are decorrelated w.r.t. K . This is the idempotent-product form of the backward direction of [11, Appendix D, Section D.2, Proposition D.3].

Proof. ✓ Proof The key identity is $P_{XB} \circ (\mathbb{1} - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$. Substituting $P_K = P_{AX} \circ P_{XB}$ and using $O_A \circ P_{XB} = P_{XB} \circ O_A$ and $O_B \circ P_{AX} = P_{AX} \circ O_B$ gives

$$P_K \circ O_A \circ (\mathbb{1} - P_K) \circ O_B \circ P_K = P_{AX} \circ O_A \circ [P_{XB} \circ (\mathbb{1} - P_{AX} \circ P_{XB}) \circ P_{AX}] \circ O_B \circ P_{XB} = 0.$$

$\square$

Theorem 12.2.34 (Decorrelating idempotent-product data). Lean ✓ Stmt Corollary of Theorem 12.2.33: if P_K has idempotent-product data, the A -observables commute with P_{XB} , and the B -observables commute with P_{AX} , then regions A and B are decorrelated w.r.t. K .

Proof. ✓ Proof Apply Theorem 12.2.33 to the witnessing endomorphisms P_{AX} and P_{XB} supplied by the idempotent-product data. $\square$

Lemma 12.2.35 (Frustration-free identity). Lean ✓ Stmt For any endomorphisms P and Q ,

$$(\mathbb{1} - P) + (\mathbb{1} - Q) - (\mathbb{1} - P) \circ (\mathbb{1} - Q) = \mathbb{1} - P \circ Q. \quad (12.27)$$

Proof. ✓ Proof Expanding the left-hand side of (12.27) and cancelling gives the result. $\square$

Theorem 12.2.36 (Idempotent-product Hamiltonian identity). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = \mathbb{1} - P_K, \quad (12.28)$$

where $Q_{AX} = \mathbb{1} - P_{AX}$ and $Q_{XB} = \mathbb{1} - P_{XB}$ are the complementary projectors. This is the algebraic identity obtained from the complementary projectors in [11, Appendix D, Section D.2]; it is not the full source parent commuting Hamiltonian condition.

Proof. ✓ Proof Apply Lemma 12.2.35 to P_{AX} and P_{XB} , and substitute $P_{AX} \circ P_{XB} = P_K$ to obtain (12.28). $\square$

Theorem 12.2.37 (Ground-space membership). Lean ✓ Stmt If $P_K = P_{AX} \circ P_{XB}$ is idempotent-product data, then

$$P_K v = v \iff P_{AX} v = v \text{ and } P_{XB} v = v. \quad (12.29)$$

This is the algebraic consequence corresponding to equation (D.2) from [11]: $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$. The source equation itself also carries the tensor-product local-subspace hypotheses.

Proof. [✓ Proof](#) ($\Rightarrow$) If $P_K v = v$, then

$$P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$$

by left absorption, and similarly for P_{XB} .

($\Leftarrow$) If $P_{AX} v = v$ and $P_{XB} v = v$, then

$$P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v.$$

This proves (12.29). □

12.3 Spectral gap

A frustration-free Hamiltonian is *gapped* if its smallest nonzero eigenvalue, or the spectral gap above the ground-state space, is bounded away from zero uniformly in the chain length N . For parent Hamiltonians of matrix product states, the martingale method of Fannes, Nachtergaele, and Werner [160, 191], as formulated by Cirac et al. [16, Section IV.C] and Kastoryano and Lucia [173], reduces a uniform spectral gap to an anticommutator estimate for overlapping local terms. Principal angles between local ground spaces provide a geometric interpretation of this criterion, and norm-compression bounds for cyclic-window local terms yield uniform spectral gap lower bounds on periodic chains.

Definition 12.3.1 (Parent ground space under the canonical ℓ^2 identification). [Lean](#) [✓ Stmt](#) Denote by $\mathcal{G}_{N,L}^{\text{ES}}(A)$ the parent ground space after identifying the coefficient space with the Hilbert space $\ell^2(\{0, \dots, d-1\}^N)$.

Definition 12.3.2 (Parent Hamiltonian under the canonical ℓ^2 identification). [Lean](#) [✓ Stmt](#) Denote by $H_{N,L}^{\text{ES}}(A)$ the conjugate of $H_N(A, L)$ by the canonical identification between the coefficient space and $\ell^2(\{0, \dots, d-1\}^N)$.

Theorem 12.3.3 (Kernel under the canonical ℓ^2 identification). [Lean](#) [✓ Stmt](#) *Under the canonical ℓ^2 identification of the N -site state space, the parent ground space is exactly the kernel of the corresponding parent Hamiltonian.*

Proof. [✓ Proof](#) This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and the ℓ^2 space, so the ground space is precisely the kernel after this identification. □

Lemma 12.3.4 (Membership criterion under the canonical ℓ^2 identification). [Lean](#) [✓ Stmt](#) *A vector belongs to $\mathcal{G}_{N,L}^{\text{ES}}(A)$ exactly when the parent Hamiltonian annihilates it.*

Remark 12.3.5 (Local projectors under the canonical ℓ^2 identification). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $P_L^{\text{ES}}(A)$ denote the orthogonal projector onto $(G_L^{\text{ES}}(A))^\perp$ on the canonical ℓ^2 L -site Hilbert space. For a cyclic window starting at i and an outside configuration τ , the map $R_{i,\tau}$ restricts an N -site vector to that length- L cyclic window with the outside configuration fixed by τ . The local term $h_{i,\text{ES}}$ is the corresponding N -site local projector, and $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$ is the basic positive summand in its cyclic-averaging formula.

Lemma 12.3.6 (Positivity of the Hilbert-space local summands). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The Hilbert-space local projector $P_L^{\text{ES}}(A)$ is positive. Consequently, every summand $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$ is positive, and so is the finite sum over τ .*

Proof. ✓ Proof Orthogonal projectors are positive. Conjugation by $R_{i,\tau}$ preserves positivity, and finite sums of positive operators remain positive. $\square$

Lemma 12.3.7 (Cyclic averaging for the Hilbert-space local terms). Lean Lean ✓ Stmt *The Hilbert-space parent Hamiltonian is the sum of the Hilbert-space local terms, $H_{N,\text{ES}} = \sum_i h_{i,\text{ES}}$, and each local term admits the exact cyclic-averaging formula*

$$h_{i,\text{ES}} = d^{-L} \sum_{\tau} R_{i,\tau}^{\dagger} P_L^{\text{ES}}(A) R_{i,\tau}. \quad (12.30)$$

Proof. ✓ Proof Unfold the Hilbert-space local term and compare it configuration-by-configuration with the cyclic restrictions. The d^{-L} factor compensates for the d^L choices of the window coordinates in the ambient configuration parameter τ . $\square$

Lemma 12.3.8 (Positivity of the Hilbert-space Hamiltonian). Lean Lean ✓ Stmt *Each Hilbert-space local term $h_{i,\text{ES}}$ is positive, hence the full Hilbert-space parent Hamiltonian $H_{N,\text{ES}}$ is positive.*

Proof. ✓ Proof Apply the averaging formula (12.30): $h_{i,\text{ES}}$ is a non-negative scalar multiple of a finite sum of positive conjugates of $P_L^{\text{ES}}(A)$, hence is itself positive. Summing over i yields positivity of $H_{N,\text{ES}}$. $\square$

Lemma 12.3.9 (Hilbert-space local terms are symmetric projections). Lean Lean ✓ Stmt *The Hilbert-space local projector $P_L^{\text{ES}}(A)$ is a symmetric projection, and every Hilbert-space local term $h_{i,\text{ES}}$ on the N -site chain is a symmetric projection.*

Proof. ✓ Proof The local operator $P_L^{\text{ES}}(A)$ is an orthogonal projection onto $(G_L^{\text{ES}}(A))^{\perp}$. For $L \leq N$, restricting $h_{i,\text{ES}}v$ to a cyclic window gives $P_L^{\text{ES}}(A)$ applied to the corresponding restriction of v ; applying $h_{i,\text{ES}}$ a second time therefore applies $(P_L^{\text{ES}}(A))^2 = P_L^{\text{ES}}(A)$ on that window. Lemma 12.3.8 supplies symmetry, and the case $L > N$ is the zero projection by definition. $\square$

Lemma 12.3.10 (Kernel of the Hilbert-space parent interaction). Lean ✓ Stmt *The Hilbert-space local interaction $P_L^{\text{ES}}(A)$ annihilates exactly the Hilbert-space local ground space: $P_L^{\text{ES}}(A)v = 0$ if and only if $v \in G_L^{\text{ES}}(A)$.*

Proof. ✓ Proof Since $P_L^{\text{ES}}(A)$ is the orthogonal projector onto $(G_L^{\text{ES}}(A))^{\perp}$, its kernel is the original subspace $G_L^{\text{ES}}(A)$. $\square$

Lemma 12.3.11 (Local-term kernel via cyclic restrictions). Lean Lean ✓ Stmt *For $L \leq N$, a Hilbert-space local term $h_{i,\text{ES}}$ annihilates a vector v if and only if every restriction of v to the cyclic length- L window starting at i , with the outside configuration fixed, lies in $G_L^{\text{ES}}(A)$.*

Proof. ✓ Proof Restricting $h_{i,\text{ES}}v$ to the same cyclic window gives $P_L^{\text{ES}}(A)$ applied to the corresponding restriction of v . Lemma 12.3.10 identifies the vanishing of this local projector with membership in $G_L^{\text{ES}}(A)$. $\square$

Theorem 12.3.12 (End-site and cyclic coordinate identifications). Lean Lean Lean ✓ Stmt *Let ι_L denote the canonical identification of the L -site state space with its Hilbert-space realization, and let $R_{i,\tau}^{\text{cyc}}$ denote cyclic restriction to the length- L window beginning at i , with background configuration τ . For every $L \in \mathbb{N}$, every $(L+1)$ -site Hilbert-space vector v , and every background configuration τ ,*

$$R_{\tau_L}^{\text{last}}(\iota_{L+1}^{-1}v) = \iota_L^{-1}R_{0,\tau}^{\text{cyc}}v. \quad (12.31)$$

On three sites, for every one-site prefix u and every state ψ ,

$$R_u^{\text{tail}}\psi = R_{u_0}^{\text{first}}\psi. \quad (12.32)$$

Finally, if $L > 0$, then for every $(L + 1)$ -site Hilbert-space vector v and every background configuration τ ,

$$R_{\tau_0}^{\text{first}}(\iota_{L+1}^{-1}v) = \iota_L^{-1}R_{1,\tau}^{\text{cyc}}v. \quad (12.33)$$

No hypothesis on the physical or virtual dimension is required.

Proof. ✓ Proof Expand the restriction maps on a configuration. At site 0, cyclic order is ordinary order and fixing the complement fixes the final coordinate. A one-site prefix is its unique coordinate, which gives the middle identity. If $L > 0$, cyclic restriction beginning at site 1 fixes the first coordinate and preserves the order of the remaining L coordinates. $\square$

Theorem 12.3.13 (Adjacent three-site kernels and excitation projections). Lean Lean Lean ✓ Stmt For a tensor B , let h_0 and h_1 be the two translated length-two parent interactions on the three-site Hilbert space. Then

$$\ker h_0 = \mathcal{U}_2(B), \quad \ker h_1 = \mathcal{V}_{1,2}(B). \quad (12.34)$$

Consequently, the orthogonal projectors onto the complementary excitation subspaces satisfy

$$P_{\mathcal{U}_2(B)^\perp} = h_0, \quad P_{\mathcal{V}_{1,2}(B)^\perp} = h_1. \quad (12.35)$$

These identities require neither injectivity nor a nonzero bond dimension.

Proof. ✓ Proof At site 0, fixing the final physical coordinate is the cyclic restriction defining h_0 . At site 1, fixing the one-site prefix is the first-site restriction and hence the cyclic restriction defining h_1 . The kernel characterization of a translated local parent interaction therefore gives the two displayed kernel identities. Each h_i is an orthogonal projector, so its range is the orthogonal complement of its kernel. Uniqueness of the orthogonal projector with a prescribed range gives the final two identities. $\square$

Theorem 12.3.14 (Zero energy gives the cyclic local constraints). Lean ✓ Stmt Assume $N > 0$ and $L \leq N$. If a vector ψ has zero energy for the finite periodic parent Hamiltonian, $H_N(A, L)\psi = 0$, then every cyclic length- L restriction of ψ lies in the local MPS ground space. Equivalently, $\ker H_N(A, L) \subseteq G_L^{(N)}(A)$.

Proof. ✓ Proof Let e_N be the canonical ℓ^2 identification from the Hilbert-space model to the coefficient space. Since $H_N(A, L)$ is the e_N -conjugate of $H_{N,L}^{\text{ES}}(A)$, the hypothesis gives

$$H_{N,L}^{\text{ES}}(A)(e_N^{-1}\psi) = e_N^{-1}(H_N(A, L)\psi) = 0.$$

Put $P_i = h_{i,\text{ES}}$ and $H = \sum_i P_i$. If $Hv = 0$, then

$$0 = \text{Re}\langle Hv, v \rangle = \sum_i \text{Re}\langle P_i v, v \rangle = \sum_i \|P_i v\|^2.$$

Each summand is non-negative; hence $\|P_i v\|^2 = 0$ and therefore $P_i v = 0$ for every i . Applying this to $H_{N,L}^{\text{ES}}(A) = \sum_i h_{i,\text{ES}}$ gives the vanishing of every Hilbert-space local term. Lemma 12.3.11 identifies these equations with the conditions $R_{i,\tau}(e_N^{-1}\psi) \in G_L^{\text{ES}}(A)$ for all i and τ . Transporting these conditions back through e_N gives $R_{i,\tau}\psi \in G_L(A)$ for all i and τ , which are exactly the cyclic-window defining equations of $G_L^{(N)}(A)$. $\square$

Lemma 12.3.15 (Cyclic constraints give zero energy). [Lean](#) [✓ Stmt](#) Assume $N > 0$ and $L \leq N$. Then $G_L^{(N)}(A) \subseteq \ker H_N(A, L)$.

Proof. [✓ Proof](#) If $\psi \in G_L^{(N)}(A)$, then every local parent interaction annihilates ψ , for every site i : $h_i(A, L)\psi = 0$. Summing over all sites gives $H_N(A, L)\psi = \sum_i h_i(A, L)\psi = 0$. $\square$

Definition 12.3.16 (Hilbert-space cyclic constraint space). [Lean](#) [✓ Stmt](#) The Hilbert-space cyclic constraint space is the transport of $G_L^{(N)}(A)$ by the canonical ℓ^2 identification: $G_{L, \text{ES}}^{(N)}(A) = e_N^{-1} G_L^{(N)}(A)$.

Theorem 12.3.17 (The finite-volume kernel is the cyclic constraint space). [Lean](#) [Lean](#) [✓ Stmt](#) Assume $N > 0$ and $L \leq N$. Then $\ker H_N(A, L) = G_L^{(N)}(A)$. Equivalently, after the canonical Hilbert-space identification, $\mathcal{K}_N^{\text{ES}}(A, L) = e_N^{-1} G_L^{(N)}(A)$, where $\mathcal{K}_N^{\text{ES}}(A, L)$ is the zero-energy space of $H_{N, L}^{\text{ES}}(A)$.

Proof. [✓ Proof](#) The inclusion $\ker H_N(A, L) \subseteq G_L^{(N)}(A)$ is Theorem 12.3.14. For the converse, a vector in $G_L^{(N)}(A)$ satisfies all local equations $h_i(A, L)\psi = 0$. Therefore $H_N(A, L)\psi = \sum_i h_i(A, L)\psi = 0$. The Hilbert-space statement is the same equality transported by e_N^{-1} . $\square$

Theorem 12.3.18 (Intersection property for two adjacent local kernels). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If A is injective and $L \geq 2$, then on an $(L+1)$ -site chain the kernels of the two adjacent Hilbert-space L -site local terms have common kernel equal to the Hilbert-space $(L+1)$ -site MPS ground space:

$$h_{0, \text{ES}} v = 0 \text{ and } h_{1, \text{ES}} v = 0 \iff v \in G_{L+1}^{\text{ES}}(A).$$

Proof. [✓ Proof](#) Lemma 12.3.11 translates the two local kernel assumptions into the left and right L -site ground conditions. The open-chain intersection property, Theorem 11.5.4.3, then grows back the shared $(L-1)$ -site overlap to the $(L+1)$ -site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel–restriction characterization. $\square$

Lemma 12.3.19 (Non-overlap positivity for Hilbert-space local terms). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $L \leq N$ and let $h_{i, \text{ES}}, h_{j, \text{ES}}$ be Hilbert-space local terms whose cyclic L -windows are site-disjoint: no site of the periodic chain has cyclic offset $< L$ from both i and j . Then the terms commute pointwise: for every vector v , $h_{i, \text{ES}}(h_{j, \text{ES}} v) = h_{j, \text{ES}}(h_{i, \text{ES}} v)$. The same commutation identity holds for the corresponding local terms before passing to the Hilbert-space model. Their ordered cross term is non-negative: $0 \leq \text{Re}\langle h_{i, \text{ES}} v, h_{j, \text{ES}} v \rangle$.

Proof. [✓ Proof](#) In coordinates, $h_{i, \text{ES}}$ applies $P_L^{\text{ES}}(A)$ only to the i -window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the i -window does not change the extracted j -window, and the i - and j -window replacements commute. Therefore the two embedded local operators commute. Remark 12.3.1.8 gives the non-negative ordered cross term for commuting symmetric projections. $\square$

Lemma 12.3.20 (Quadratic form to norm bound). [Lean](#) [✓ Stmt](#) Let H be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all v , $\gamma \langle v, H v \rangle \leq \langle H v, H v \rangle$, i.e., $H^2 \geq \gamma H$ as a quadratic form, for some $\gamma > 0$. Then, for every v in the orthogonal complement of $\ker H$, $\gamma \|v\| \leq \|H v\|$. In particular, every positive eigenvalue of H is at least γ . The positive-semidefiniteness hypothesis is essential: without it, $H = -1$ with $\gamma = 2$ satisfies the quadratic-form inequality vacuously but violates the norm bound.

Proof. ✓ Proof By the finite-dimensional spectral theorem, H diagonalises in an orthonormal eigenbasis with real eigenvalues λ_k . For a unit eigenvector ψ_k with eigenvalue λ_k , the hypothesis gives $\gamma\lambda_k \leq \lambda_k^2$, whence $\lambda_k \leq 0$ or $\lambda_k \geq \gamma$. Since H is positive semidefinite (its quadratic form is ≥ 0), the negative case is excluded and every nonzero eigenvalue satisfies $\lambda_k \geq \gamma$. Writing $v = \sum_k c_k \psi_k$ and projecting onto $(\ker H)^\perp$ keeps only terms with $\lambda_k \geq \gamma$, so

$$\|Hv\|^2 = \sum_k |c_k|^2 \lambda_k^2 \geq \gamma^2 \sum_k |c_k|^2 = \gamma^2 \|v\|^2.$$

□

12.3.1 Martingale condition for a finite family of projections

Lemma 12.3.1.1 (Parent-Hamiltonian quadratic-form reduction). Lean Lean ✓ Stmt Let $H_{N,L}^{\text{ES}}(A)$ be the Hilbert-space representative of the parent Hamiltonian. If a constant $\gamma > 0$ satisfies the uniform quadratic-form estimate

$$\gamma \operatorname{Re} \langle H_{N,L}^{\text{ES}}(A)v, v \rangle \leq \operatorname{Re} \langle H_{N,L}^{\text{ES}}(A)v, H_{N,L}^{\text{ES}}(A)v \rangle \quad (12.36)$$

for every $N \geq 2L$ and every vector v , then $\gamma\|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$ for every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$. In particular, with $L > 1$ and $\gamma = 1/(4L)$, the explicit gap-bound conclusion follows once this same quadratic-form estimate (12.36) is supplied.

Proof. ✓ Proof The Hilbert-space Hamiltonian is positive by Lemma 12.3.8, and its Hilbert-space ground space is its kernel by Theorem 12.3.3. Therefore the abstract spectral-theorem conversion of Lemma 12.3.20 applies directly. The explicit constant is positive because $L > 1$ implies $4L > 0$. □

Definition 12.3.1.2 (Friedrichs overlap bound). Lean ✓ Stmt Let U and V be subspaces of a finite-dimensional complex Hilbert space. Their Friedrichs overlap is at most η when

$$|\langle x, y \rangle| \leq \eta \|x\| \|y\|$$

for every $x \in U \cap (U \cap V)^\perp$ and $y \in V \cap (U \cap V)^\perp$.

Theorem 12.3.1.3 (Projector defect bounds the Friedrichs overlap). Lean ✓ Stmt Let P_U , P_V , and $P_{U \cap V}$ be the orthogonal projections onto U , V , and their intersection. If

$$\|P_U P_V - P_{U \cap V}\| \leq \eta,$$

then the Friedrichs overlap of U and V is at most η .

Proof. ✓ Proof For $y \in V \cap (U \cap V)^\perp$, the defect applied to y is $P_U y$. Hence $\|P_U y\| \leq \eta \|y\|$. If also $x \in U \cap (U \cap V)^\perp$, self-adjointness of P_U gives $\langle x, y \rangle = \langle x, P_U y \rangle$, and Cauchy–Schwarz proves the claim. □

Lemma 12.3.1.4 (Projection compression from a Friedrichs overlap bound). Lean ✓ Stmt Let P_U be the orthogonal projection onto U . If the Friedrichs overlap of U and V is at most η , where $\eta \geq 0$, then

$$\|P_U v\| \leq \eta \|v\|$$

for every $v \in V \cap (U \cap V)^\perp$.

Proof. ✓ Proof Put $x = P_U v$. Since the common intersection is contained in U , the projection preserves orthogonality to $U \cap V$. Thus $x \in U \cap (U \cap V)^\perp$, and $\langle x, v \rangle = \langle x, x \rangle = \|x\|^2$. The Friedrichs bound gives $\|x\|^2 \leq \eta \|x\| \|v\|$, and cancellation proves the result when $x \neq 0$; the case $x = 0$ is immediate. $\square$

Lemma 12.3.1.5 (Two-projection compression on the reduced space). Lean ✓ Stmt Let P_U and P_V be the orthogonal projections onto U and V . If the Friedrichs overlap of U and V is at most η , where $\eta \geq 0$, then every $v \in (U \cap V)^\perp$ satisfies

$$\|P_U P_V v\| \leq \eta \|P_V v\|.$$

Proof. ✓ Proof The vector $P_V v$ belongs to V and remains orthogonal to $U \cap V$. Apply Lemma 12.3.1.4. $\square$

Lemma 12.3.1.6 (Directional compression on a reduced projection range). Lean ✓ Stmt Under the hypotheses of Lemma 12.3.1.5, every $v \in U \cap (U \cap V)^\perp$ satisfies

$$\|P_U P_V v\| \leq \eta \|P_U v\|.$$

Proof. ✓ Proof Orthogonal projection onto V does not increase the norm, while $P_U v = v$. Hence $\|P_U P_V v\| \leq \eta \|P_V v\| \leq \eta \|v\| = \eta \|P_U v\|$. $\square$

Lemma 12.3.1.7 (Directional compression for kernel-complement projections). Lean ✓ Stmt Let P and Q be the orthogonal projections onto $K^\perp$ and $L^\perp$. If the Friedrichs overlap of these two ranges is at most $\eta \geq 0$, then every $v \in K^\perp \cap (K^\perp \cap L^\perp)^\perp$ satisfies

$$\|PQv\| \leq \eta \|Pv\|.$$

Proof. ✓ Proof Apply Lemma 12.3.1.6 to the two subspaces $K^\perp$ and $L^\perp$. $\square$

Remark 12.3.1.8 (Projection-geometry identities for row-sum arguments). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity $\text{Re}\langle Pv, Pv \rangle = \text{Re}\langle Pv, v \rangle$ for symmetric projections; nonnegativity of this quadratic form; nonnegativity of $\text{Re}\langle Pv, Qv \rangle$ for commuting symmetric projections P, Q ; the Cauchy–Schwarz conversion from $\|PQv\| \leq c\|Pv\|$ to $\text{Re}\langle Pv, Qv \rangle \geq -c \text{Re}\langle Pv, v \rangle$; the symmetric variant which, for $c \geq 0$, converts the operator-product bound $\|PQv\| \leq c\|Qv\|$ to the anticommutator estimate $\text{Re}\langle Pv, Qv \rangle + \text{Re}\langle Qv, Pv \rangle \geq -c(\text{Re}\langle Pv, v \rangle + \text{Re}\langle Qv, v \rangle)$; finite-sum expansions of $\text{Re}\langle (\sum_i P_i)v, v \rangle$ and $\text{Re}\langle (\sum_i P_i)v, (\sum_i P_i)v \rangle$; the conclusion that a vector annihilated by a finite sum of symmetric projections is annihilated by each projection; the diagonal/off-diagonal split of the ordered double sum; the aggregate off-diagonal estimate obtained from row sums $\sum_{j \neq i} c_{ij} \leq 1$; and the corresponding aggregate estimate for the anticommutator martingale condition.

Lemma 12.3.1.9 (Ordered projection row-sum quadratic-form reduction). Lean ✓ Stmt Let P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. If the ordered off-diagonal forms obey

$$\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) c_{ij} \text{Re}\langle P_i v, v \rangle \quad (i \neq j),$$

with row sums $\sum_{j \neq i} c_{ij} \leq 1$ and $\gamma \leq 1$, then $H^2 \geq \gamma H$ as a quadratic form.

Proof. ✓ Proof Expand $\text{Re}\langle Hv, Hv \rangle$ as an ordered double sum. Symmetric idempotence identifies each diagonal term with $\text{Re}\langle P_i v, v \rangle$, while the row-summable ordered cross-term bounds control the off-diagonal contribution by $-(1 - \gamma) \sum_i \text{Re}\langle P_i v, v \rangle$. $\square$

Lemma 12.3.1.10 (Projection anticommutator row-sum reduction). [Lean](#) [✓ Stmt](#) Let P_i be a finite family of symmetric projections and let $H = \sum_i P_i$. Suppose the anticommutator forms obey

$$\operatorname{Re}\langle P_i v, P_j v \rangle + \operatorname{Re}\langle P_j v, P_i v \rangle \geq -(1 - \gamma) c_{ij} (\operatorname{Re}\langle P_i v, v \rangle + \operatorname{Re}\langle P_j v, v \rangle) \quad (i \neq j),$$

with row sums $\sum_{j \neq i} c_{ij} \leq 1$, column sums $\sum_{i \neq j} c_{ij} \leq 1$, and $\gamma \leq 1$. Then $H^2 \geq \gamma H$ as a quadratic form.

Proof. [✓ Proof](#) Expand $\operatorname{Re}\langle H v, H v \rangle$ as an ordered double sum. The anticommutator bound controls the sum of each ordered term with its reversed term. The row and column estimates bound the coefficient-weighted diagonal contribution by twice $\sum_i \operatorname{Re}\langle P_i v, v \rangle$; after dividing by two, the same quadratic-form estimate follows. $\square$

Lemma 12.3.1.11 (Finite-overlap projection anticommutator reduction). [Lean](#) [✓ Stmt](#) Let P_i be a finite family of symmetric projections and let $\gamma \leq 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row and in each column, where $m > 0$. If noninteracting pairs have non-negative anticommutator form and interacting pairs satisfy

$$\operatorname{Re}\langle P_i v, P_j v \rangle + \operatorname{Re}\langle P_j v, P_i v \rangle \geq -(1 - \gamma) \frac{1}{m} (\operatorname{Re}\langle P_i v, v \rangle + \operatorname{Re}\langle P_j v, v \rangle),$$

then $H = \sum_i P_i$ satisfies $H^2 \geq \gamma H$ as a quadratic form.

Proof. [✓ Proof](#) Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row and column cardinality bounds give the corresponding coefficient sums, and the noninteracting anticommutator non-negativity supplies the zero-coefficient estimates. Apply Lemma 12.3.1.10. $\square$

Lemma 12.3.1.12 (Finite-overlap projection row-sum reduction). [Lean](#) [✓ Stmt](#) Let P_i be a finite family of symmetric projections and let $\gamma \leq 1$. Suppose an interaction predicate $i \sim j$ has at most m off-diagonal neighbours in each row for some positive integer m . If noninteracting ordered cross terms are non-negative and, on each interacting pair,

$$\operatorname{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) \frac{1}{m} \operatorname{Re}\langle P_i v, v \rangle,$$

then $H = \sum_i P_i$ satisfies $H^2 \geq \gamma H$ as a quadratic form.

Proof. [✓ Proof](#) Choose $c_{ij} = 1/m$ on interacting pairs and $c_{ij} = 0$ otherwise. The row cardinality bound gives $\sum_{j \neq i} c_{ij} \leq 1$; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 12.3.1.9. $\square$

Lemma 12.3.1.13 (Finite-overlap projection norm-compression reduction). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let P_i be a finite family of symmetric projections and let $\gamma \leq 1$. Suppose an interaction predicate has at most $m > 0$ off-diagonal neighbours in each row, noninteracting ordered cross terms are non-negative, and interacting pairs satisfy the norm-compression estimate

$$\|P_i P_j v\| \leq (1 - \gamma) \frac{1}{m} \|P_i v\|. \quad (12.37)$$

Then $H = \sum_i P_i$ satisfies $H^2 \geq \gamma H$ as a quadratic form. The same conclusion holds from a separate compression coefficient η whenever $\eta \leq (1 - \gamma)/m$. In particular, if $0 \leq \eta$ and $\eta m < 1$, then $H^2 \geq (1 - \eta m)H$ and $1 - \eta m > 0$.

Proof. ✓ Proof Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into $\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma)m^{-1} \text{Re}\langle P_i v, v \rangle$. If a separate coefficient η is used, the assumption $\eta \leq (1 - \gamma)/m$ first weakens that estimate to (12.37). The finite-overlap row-sum reduction then applies. For the strict form, take $\gamma = 1 - \eta m$. $\square$

Definition 12.3.1.14 (Cyclic window and complement indexing). Lean Lean Lean ✓ Stmt If $L \leq N$ and $i \in \{0, \dots, N-1\}$, identify the sites of the periodic chain with the disjoint union of the cyclic window $i, i+1, \dots, i+L-1$ and its ordered complement $i+L, i+L+1, \dots, i+N-1$, with all indices taken modulo N .

Theorem 12.3.1.15 (Cyclic product factorization). Lean ✓ Stmt Let M be a commutative monoid. If $L \leq N$, $i \in \{0, \dots, N-1\}$, and $f: \{0, \dots, N-1\} \rightarrow M$, then the product of f over the chain equals the product over the cyclic window beginning at i times the product over its ordered complement.

Proof. ✓ Proof Reindex the product by the cyclic window and complement identification of Definition 12.3.1.14, then split the product over the resulting disjoint union. $\square$

Theorem 12.3.1.16 (Three-site anticommutator transport to cyclic adjacent pairs). Lean ✓ Stmt Fix a tensor A with physical dimension $d > 0$ and a coefficient $c \in \mathbb{R}$. For every chain length N and site $i \in \{0, \dots, N-1\}$, set $h_i^{(N)} := h_{i, \text{ES}}^{(N, 2)}(A)$, the Euclidean range-two parent term of this same tensor A at i . Suppose that, for every three-site vector w ,

$$-c(\text{Re}\langle h_0^{(3)} w, w \rangle + \text{Re}\langle h_1^{(3)} w, w \rangle) \leq \text{Re}\langle h_0^{(3)} w, h_1^{(3)} w \rangle + \text{Re}\langle h_1^{(3)} w, h_0^{(3)} w \rangle. \quad (12.38)$$

Then, for every $N \geq 3$, every site $i \in \{0, \dots, N-1\}$, and every N -site vector v , the forward pair satisfies

$$-c(\text{Re}\langle h_i^{(N)} v, v \rangle + \text{Re}\langle h_{i+1}^{(N)} v, v \rangle) \leq \text{Re}\langle h_i^{(N)} v, h_{i+1}^{(N)} v \rangle + \text{Re}\langle h_{i+1}^{(N)} v, h_i^{(N)} v \rangle, \quad (12.39)$$

and the backward pair satisfies

$$-c(\text{Re}\langle h_i^{(N)} v, v \rangle + \text{Re}\langle h_{i-1}^{(N)} v, v \rangle) \leq \text{Re}\langle h_i^{(N)} v, h_{i-1}^{(N)} v \rangle + \text{Re}\langle h_{i-1}^{(N)} v, h_i^{(N)} v \rangle. \quad (12.40)$$

All site indices are taken modulo N , so these conclusions include pairs and three-site windows crossing the periodic cut.

Proof. ✓ Proof For a cyclic three-site window beginning at i , write $\mathcal{C}_{N-3}$ for its ordered complementary configurations and $R_{i, \tau}^{(3)}$ for restriction with complement τ fixed. Definition 12.3.1.14 identifies every chain configuration uniquely with its window and complement, while the cyclic restriction operators of Remark 11.5.5.2 give the fiber decomposition

$$\text{Re}\langle u, v \rangle = \sum_{\tau \in \mathcal{C}_{N-3}} \text{Re}\langle R_{i, \tau}^{(3)} u, R_{i, \tau}^{(3)} v \rangle. \quad (12.41)$$

The restriction identities in Theorem 12.1.5.9 intertwine the two adjacent terms:

$$R_{i, \tau}^{(3)} h_i^{(N)} = h_0^{(3)} R_{i, \tau}^{(3)}, \quad R_{i, \tau}^{(3)} h_{i+1}^{(N)} = h_1^{(3)} R_{i, \tau}^{(3)}. \quad (12.42)$$

Applying (12.38) to $w_\tau = R_{i, \tau}^{(3)} v$ and summing over the complement gives

$$\sum_{\tau \in \mathcal{C}_{N-3}} -c(\text{Re}\langle h_0^{(3)} w_\tau, w_\tau \rangle + \text{Re}\langle h_1^{(3)} w_\tau, w_\tau \rangle) \leq \sum_{\tau \in \mathcal{C}_{N-3}} (\text{Re}\langle h_0^{(3)} w_\tau, h_1^{(3)} w_\tau \rangle + \text{Re}\langle h_1^{(3)} w_\tau, h_0^{(3)} w_\tau \rangle), \quad (12.43)$$

By the fiber decomposition (12.41) and restriction intertwining (12.42), this is (12.39); in particular, summation leaves c unchanged. For the backward pair, begin the forward window at $i - 1$. Moving one step forward from that site returns i , including when $i = 0$, and yields (12.40). $\square$

Lemma 12.3.1.17 (Quadratic-form estimate from the martingale condition). [Lean](#) [✓ Stmt](#) For a fixed chain length N , suppose the Hilbert-space local terms $h_{i,\text{ES}}$ satisfy the ordered row-sum estimate

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \gamma)c_{ij} \text{Re}\langle h_{i,\text{ES}}v, v \rangle \quad (i \neq j), \quad (12.44)$$

with $\sum_{j \neq i} c_{ij} \leq 1$ and $\gamma \leq 1$. Then the Hilbert-space parent Hamiltonian satisfies

$$\gamma \text{Re}\langle H_{N,L}^{\text{ES}}v, v \rangle \leq \text{Re}\langle H_{N,L}^{\text{ES}}v, H_{N,L}^{\text{ES}}v \rangle$$

for every vector v .

Proof. [✓ Proof](#) Lemma 12.3.9 supplies the symmetric-projection hypothesis for the family $P_i = h_{i,\text{ES}}$. Apply Lemma 12.3.1.9 to this family and use $H_{N,L}^{\text{ES}} = \sum_i h_{i,\text{ES}}$. $\square$

Lemma 12.3.1.18 (Parent-Hamiltonian gap from ordered local row bounds). [Lean](#) [✓ Stmt](#) Assume uniformly for every $N \geq 2L$ that the Hilbert-space local terms satisfy the ordered row-sum estimate (12.44) with $\gamma = 1/(4L)$. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$, $\frac{1}{4L}\|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$.

Proof. [✓ Proof](#) The fixed-chain row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N . The positivity and kernel identification of the Hilbert-space Hamiltonian then allow Lemma 12.3.1.1 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. $\square$

Lemma 12.3.1.19 (Parent-Hamiltonian gap from the anticommutator martingale condition). [Lean](#) [Lean](#) [✓ Stmt](#) Assume uniformly for every $N \geq 2L$ that the Hilbert-space local terms satisfy the anticommutator martingale estimate

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle + \text{Re}\langle h_{j,\text{ES}}v, h_{i,\text{ES}}v \rangle \geq -c_{ij} \left(1 - \frac{1}{4L}\right) (\text{Re}\langle h_{i,\text{ES}}v, v \rangle + \text{Re}\langle h_{j,\text{ES}}v, v \rangle)$$

for $i \neq j$, with row and column coefficient sums bounded by one. If $L > 1$, then $1/(4L) > 0$ and, for every $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$, $\frac{1}{4L}\|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$.

Proof. [✓ Proof](#) The fixed-chain anticommutator row-sum reduction supplies the quadratic-form estimate with $\gamma = 1/(4L)$ for every admissible N . The positivity and kernel identification of the Hilbert-space Hamiltonian then convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. $\square$

Lemma 12.3.1.20 (Parent-Hamiltonian finite-overlap anticommutator reduction). [Lean](#) [Lean](#) [✓ Stmt](#) Fix $L > 1$ and set $m = 2(L - 1)$. Suppose uniformly for every $N \geq 2L$ that the overlap relation on local windows has at most m off-diagonal neighbours in each row and in each column. If non-overlapping local terms have non-negative anticommutator form, and overlapping local terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle + \text{Re}\langle h_{j,\text{ES}}v, h_{i,\text{ES}}v \rangle \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{m} (\text{Re}\langle h_{i,\text{ES}}v, v \rangle + \text{Re}\langle h_{j,\text{ES}}v, v \rangle),$$

then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$.

Proof. [✓ Proof](#) Apply the finite-overlap anticommutator projection reduction to the Hilbert-space local terms. The resulting quadratic-form estimate has $\gamma = 1/(4L)$, and Lemma 12.3.1.1 converts it to the norm lower bound. $\square$

Lemma 12.3.1.21 (Parent-Hamiltonian finite-overlap martingale quadratic reduction). [Lean](#) [✓ Stmt](#) For a fixed chain length N , let m be a positive integer and let $\gamma \leq 1$. Suppose an overlap predicate on local windows has at most m overlapping off-diagonal windows in each row. If non-overlapping local terms have non-negative ordered cross terms, and overlapping local terms satisfy the one-sided ordered cross-term estimate

$$\operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1-\gamma)\frac{1}{m} \operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle,$$

and the Hilbert-space local terms are symmetric projections, then the Hilbert-space Hamiltonian satisfies $H_{N,L}^{\text{ES},2} \geq \gamma H_{N,L}^{\text{ES}}$ as a quadratic form.

Proof. [✓ Proof](#) Apply the finite-overlap projection reduction with $P_i = h_{i,\text{ES}}$. $\square$

12.3.2 Cyclic-window overlap estimates

Definition 12.3.2.1 (Cyclic-window support). [Lean](#) [✓ Stmt](#) For a length- L cyclic window on $\{0, \dots, N-1\}$ starting at i , its support is $S_i = \{i, i+1, \dots, i+L-1\} \pmod{N}$, with repeated visits counted once.

Lemma 12.3.2.2 (Cyclic-window support offsets). [Lean](#) [✓ Stmt](#) If $L \leq N$, then a site k belongs to the cyclic support S_i exactly when its cyclic offset from i is below L : $k \in S_i \iff (k - i) \pmod{N} < L$.

Proof. [✓ Proof](#) Write $k = i + r \pmod{N}$ for a representative offset r . Membership in S_i gives an offset $r < L$, and conversely an offset below L gives the corresponding point of the support. $\square$

Definition 12.3.2.3 (Cyclic-window overlap). [Lean](#) [✓ Stmt](#) Two length- L cyclic windows overlap, written $i \sim_L j$, when their supports meet: $S_i \cap S_j \neq \emptyset$.

Lemma 12.3.2.4 (Cyclic-window overlap symmetry). [Lean](#) [✓ Stmt](#) For length- L cyclic windows, $i \sim_L j$ if and only if $j \sim_L i$.

Proof. [✓ Proof](#) Both assertions say that the two cyclic supports have a common site. $\square$

Lemma 12.3.2.5 (Concrete non-overlap positivity for cyclic supports). [Lean](#) [Lean](#) [✓ Stmt](#) If $L \leq N$ and the cyclic supports S_i and S_j do not meet, then the Hilbert-space local terms have non-negative ordered cross term: $0 \leq \operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle$.

Proof. [✓ Proof](#) The support-offset characterization converts failure of $S_i \cap S_j \neq \emptyset$ into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies. $\square$

Lemma 12.3.2.6 (Cyclic-window overlap row cardinality). [Lean](#) [✓ Stmt](#) If $2L \leq N$ and $L > 1$, then, for every $i \in \{0, \dots, N-1\}$, $\#\{j \neq i : i \sim_L j\} \leq 2(L-1)$.

Proof. [✓ Proof](#) If S_i and S_j meet, choose offsets $a, b < L$ with $i + a \equiv j + b \pmod{N}$. Since $2L \leq N$, the offset from i to j is either the clockwise distance $a - b \in \{1, \dots, L-1\}$ or the counterclockwise distance $b - a \in \{1, \dots, L-1\}$; the zero distance is excluded by $j \neq i$. Thus every overlapping off-diagonal start lies among the $L-1$ clockwise starts or the $L-1$ counterclockwise starts. $\square$

Lemma 12.3.2.7 (Parent-Hamiltonian gap from a finite-overlap martingale estimate). [Lean](#)
[✓ Stmt](#) Fix $L > 1$ and set $m = 2(L-1)$. Suppose uniformly for every $N \geq 2L$ that the hypotheses of the fixed-chain finite-overlap martingale reduction hold with $\gamma = 1/(4L)$ and this value of m . Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$.

Proof. [✓ Proof](#) The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible N . Lemma 12.3.1.1 converts that estimate to the norm lower bound. $\square$

Lemma 12.3.2.8 (Parent-Hamiltonian gap from a cyclic-window martingale estimate). [Lean](#)
[✓ Stmt](#) Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length- L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \frac{1}{4L}) \frac{1}{2(L-1)} \text{Re}\langle h_{i,\text{ES}}v, v \rangle. \quad (12.45)$$

Then $1/(4L) > 0$, and the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$.

Proof. [✓ Proof](#) Lemma 12.3.2.6 gives the row-cardinality bound $m = 2(L-1)$ for the overlap relation among length- L local windows, and Lemma 12.3.9 supplies the symmetric projection structure of each Hilbert-space local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 12.3.2.5 gives the required non-negative ordered cross term. Apply Lemma 12.3.2.7 with these facts and the overlapping-window estimate (12.45). $\square$

Lemma 12.3.2.9 (Parent-Hamiltonian gap from a cyclic anticommutator estimate). [Lean](#)
[✓ Stmt](#) Fix $L > 1$ and let $i \sim_L j$ mean that the cyclic supports of the length- L windows at i and j meet. Suppose uniformly for every $N \geq 2L$ that overlapping off-diagonal terms satisfy

$$\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle + \text{Re}\langle h_{j,\text{ES}}v, h_{i,\text{ES}}v \rangle \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} (\text{Re}\langle h_{i,\text{ES}}v, v \rangle + \text{Re}\langle h_{j,\text{ES}}v, v \rangle).$$

Then the explicit norm lower bound with constant $1/(4L)$ follows for vectors in $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$.

Proof. [✓ Proof](#) The cyclic overlap relation is symmetric, and Lemma 12.3.2.6 gives at most $2(L-1)$ overlapping off-diagonal windows in each row and column. If two cyclic supports do not meet, Lemma 12.3.2.5 gives non-negative ordered cross terms in both orders. Apply the finite-overlap anticommutator reduction. $\square$

12.3.3 The weighted virtual matrix space

Theorem 12.3.3.1 (Weighted virtual inner product). [Lean](#) [✓ Stmt](#) Let $\rho \in M_D(\mathbb{C})$ be positive definite. The ρ -weighted inner product on $M_D(\mathbb{C})$ is

$$\langle X, Y \rangle_\rho = \text{tr}(\rho X^\dagger Y). \quad (12.46)$$

This is the auxiliary inner product in [160, Equation (5.6)].

Theorem 12.3.3.2 (Weighted norm formula). [Lean](#) [✓ Stmt](#) For every $X \in M_D(\mathbb{C})$,

$$\|X\|_\rho^2 = \text{Re tr}(\rho X^\dagger X). \quad (12.47)$$

Proof. ✓ **Proof** By Theorem 12.3.3.1 with $Y = X$,

$$\|X\|_\rho^2 = \operatorname{Re}\langle X, X \rangle_\rho \quad (12.48)$$

$$= \operatorname{Re} \operatorname{tr}(\rho X^\dagger X). \quad (12.49)$$

□

Definition 12.3.3.3 (Euclidean realization of the weighted matrix space). Lean Lean Lean

✓ **Stmnt** The map

$$\Phi_\rho: M_D(\mathbb{C}) \longrightarrow \mathbb{C}^{D^2}, \quad \Phi_\rho(X) = \operatorname{vec}(X\sqrt{\rho}) \quad (12.50)$$

is a complex linear isometric isomorphism. Its inverse is

$$\Phi_\rho^{-1}(z) = \operatorname{unvec}(z) (\sqrt{\rho})^{-1}. \quad (12.51)$$

Thus the weighted matrix space has complex Hilbert dimension D^2 . This Euclidean realization is not asserted as a result of Fannes, Nachtergaele, and Werner.

12.3.4 Nested ground projections and martingale differences

Definition 12.3.4.1 (Nested ground projections and their martingale differences). Lean Lean

✓ **Stmnt** Let $\mathcal{H}$ be a complex Hilbert space. A *nested ground-projection family* is a sequence $(G_n)_{n \geq 0}$ of orthogonal projections such that

$$m \leq n \implies \operatorname{ran}(G_n) \subseteq \operatorname{ran}(G_m). \quad (12.52)$$

Its n th martingale difference is

$$E_n = G_n - G_{n+1}. \quad (12.53)$$

Theorem 12.3.4.2 (Orthogonal martingale-difference resolution). Lean Lean Lean ✓ **Stmnt**
Each E_n is an orthogonal projection, and

$$E_m E_n = \delta_{m,n} E_m. \quad (12.54)$$

With Nachtergaele's endpoint conventions $G_0 = \mathbf{1}$ and $G_{N+1} = 0$, the differences form a complete family:

$$\sum_{n=0}^N E_n = \mathbf{1}. \quad (12.55)$$

These are the assertions following the display labelled E_n in [191].

Proof. ✓ **Proof** If $m \leq n$, nestedness and orthogonality give $G_m G_n = G_n G_m = G_n$. Consequently,

$$E_n^2 = (G_n - G_{n+1})^2 = G_n - G_{n+1} = E_n, \quad (12.56)$$

$$E_m E_n = 0 \quad (m \neq n). \quad (12.57)$$

Self-adjointness follows from $E_n^\dagger = G_n^\dagger - G_{n+1}^\dagger = E_n$. The complete sum telescopes with both endpoints visible:

$$\sum_{n=0}^N E_n = \sum_{n=0}^N (G_n - G_{n+1}) = G_0 - G_{N+1} = \mathbf{1}. \quad (12.58)$$

□

Theorem 12.3.4.3 (Truncated martingale-difference reconstruction). [Lean](#) [✓ Stmt](#) Suppose $G_0 = \mathbf{1}$. For every $v \perp \text{ran}(G_N)$,

$$v = \sum_{n=0}^{N-1} E_n v. \quad (12.59)$$

This is the identity resolutionpsi in the proof of [191, Theorem 2.1(i)].

Proof. [✓ Proof](#) Since G_N is the orthogonal projection onto its range, $v \perp \text{ran}(G_N)$ implies $G_N v = 0$. Hence

$$\sum_{n=0}^{N-1} E_n v = (G_0 - G_N)v = v. \quad (12.60)$$

□

Theorem 12.3.4.4 (Pythagorean identity for martingale differences). [Lean](#) [Lean](#) [✓ Stmt](#) For every $v \in \mathcal{H}$ and finite $S \subset \mathbb{N}$,

$$\left\| \sum_{n \in S} E_n v \right\|^2 = \sum_{n \in S} \|E_n v\|^2. \quad (12.61)$$

In particular, for $S = \{0, \dots, N-1\}$,

$$\left\| \sum_{n=0}^{N-1} E_n v \right\|^2 = \sum_{n=0}^{N-1} \|E_n v\|^2. \quad (12.62)$$

Proof. [✓ Proof](#) The summands are pairwise orthogonal because $\langle E_m v, E_n v \rangle = \langle v, E_m E_n v \rangle = 0$ for $m \neq n$. The identity follows by the finite Pythagorean theorem. □

Theorem 12.3.4.5 (Norm-square decomposition on the final orthogonal complement). [Lean](#) [✓ Stmt](#) Suppose $G_0 = \mathbf{1}$. For every $v \perp \text{ran}(G_N)$,

$$\|v\|^2 = \sum_{n=0}^{N-1} \|E_n v\|^2. \quad (12.63)$$

This is the identity resolutionnormpsi in the proof of [191, Theorem 2.1(i)].

Proof. [✓ Proof](#) Substitute the truncated reconstruction $v = \sum_{n=0}^{N-1} E_n v$ into the Pythagorean identity. □

Theorem 12.3.4.6 (Outside-window martingale cancellation). [Lean](#) [Lean](#) [✓ Stmt](#) Let Q_n be a local ground projection and suppose that

$$E_m Q_n = Q_n E_m \quad \text{whenever } m < n - l \text{ or } n < m. \quad (12.64)$$

Then every index outside $[n - l, n]$ cancels:

$$E_m Q_n E_n = 0 \quad \text{whenever } m < n - l \text{ or } n < m. \quad (12.65)$$

If $n < N$, the complete resolution term therefore reduces to

$$\left\langle v, \sum_{m=0}^{N-1} E_m Q_n E_n v \right\rangle = \left\langle \sum_{m=n-l}^n E_m v, Q_n E_n v \right\rangle. \quad (12.66)$$

This is the reduction from Enpsi to the first line of Enpsi2 in [191, Theorem 2.1(i)].

Proof. ✓ Proof For an outside index, commutation and orthogonality give

$$E_m Q_n E_n = Q_n E_m E_n = 0, \quad (12.67)$$

since either inequality implies $m \neq n$. Hence only $n - l \leq m \leq n$ remains in the finite sum. Finally, symmetry of each E_m gives

$$\langle v, E_m Q_n E_n v \rangle = \langle E_m v, Q_n E_n v \rangle, \quad (12.68)$$

and summing over the active interval proves the displayed identity. $\square$

Definition 12.3.4.7 (Fixed-volume prefix and interval ground projections). Lean Lean Lean

✓ Stmt Fix an N -site Hilbert space and let h_i denote the nonwrapping interaction of length L starting at i . For every $m \geq 0$, set

$$H_m^{(N)} = \sum_{i+L \leq m} h_i, \quad G_m^{(N)} = P_{\ker H_m^{(N)}}. \quad (12.69)$$

All operators act on the same N -site space. The local interval ground projection in the proof of Theorem 2.1(i) is

$$Q_n = P_{\ker H_{[n-l, n+1]}}, \quad H_{[n-l, n+1]} = \sum_{n-l \leq i, i+L \leq n+1} h_i. \quad (12.70)$$

Lemma 12.3.4.8 (Positivity of the fixed-volume Hamiltonians). Lean Lean ✓ Stmt *The prefix and interval Hamiltonians are positive and symmetric:*

$$H_m^{(N)} = \sum_{i+L \leq m} h_i \geq 0, \quad (H_m^{(N)})^\dagger = H_m^{(N)}, \quad (12.71)$$

$$H_{[n-l, n+1]} = \sum_{n-l \leq i, i+L \leq n+1} h_i \geq 0, \quad H_{[n-l, n+1]}^\dagger = H_{[n-l, n+1]}. \quad (12.72)$$

Proof. ✓ Proof Each h_i is an orthogonal projection and is therefore positive. Finite sums of positive operators are positive, and every positive operator is symmetric. $\square$

Theorem 12.3.4.9 (Nested fixed-volume prefix ground projections). Lean Lean Lean ✓ Stmt *If $m \leq n$, then*

$$\ker H_n^{(N)} \subseteq \ker H_m^{(N)}, \quad \text{ran } G_n^{(N)} \subseteq \text{ran } G_m^{(N)}. \quad (12.73)$$

Hence $(G_m^{(N)})_{m \geq 0}$ is a nested ground-projection family.

Proof. ✓ Proof If $H_n^{(N)} v = 0$, then orthogonality of the local projections gives

$$0 = \langle v, H_n^{(N)} v \rangle = \sum_{i+L \leq n} \|h_i v\|^2. \quad (12.74)$$

Every summand is nonnegative, so $h_i v = 0$ whenever $i + L \leq n$. When $m \leq n$, every summand of $H_m^{(N)}$ occurs in $H_n^{(N)}$, and hence $H_m^{(N)} v = 0$. This proves the kernel inclusion, and the range inclusion follows from $\text{ran } G_m^{(N)} = \ker H_m^{(N)}$. $\square$

Lemma 12.3.4.12 (Right-spectator coordinates and kernel projections). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) For $n, r \in \mathbb{N}$, restriction to the first n sites and the final r sites gives a canonical equivalence

$$[d]^{n+r} \simeq [d]^n \times [d]^r \quad (12.81)$$

and the corresponding Euclidean linear isometry. A vector belongs to $\ker(T^{\oplus S})$ exactly when each spectator fiber belongs to $\ker T$. If $P_{\ker T}$ is the orthogonal projection onto $\ker T$, then

$$P_{\ker(T^{\oplus S})} = (P_{\ker T})^{\oplus S}. \quad (12.82)$$

Proof. ✓ Proof The configuration equivalence is the inverse of concatenation. The kernel characterization follows by evaluating each coordinate. Fiberwise extension preserves symmetry and idempotence, and its range consists exactly of the vectors whose fibers lie in $\ker T$. Equality of the two symmetric projections follows from equality of their ranges. $\square$

Theorem 12.3.4.13 (Nonwrapping interactions with right spectators). Lean StmT Let $i + L \leq n$. Under the canonical identification

$$\ell^2([d]^{n+r}) \simeq \ell^2([d]^n \times [d]^r), \quad (12.83)$$

the nonwrapping length- L interaction beginning at i is

$$h_{i,L}^{(n+r)} = (h_{i,L}^{(n)})^{\oplus [d]^r}. \quad (12.84)$$

Proof. ✓ Proof The nonwrapping hypothesis identifies cyclic restriction with contiguous restriction on both chain lengths. Concatenation leaves the first n coordinates unchanged and fixes the final r spectator coordinates. Hence window extraction and restriction agree on every spectator fiber, and applying the same local parent interaction gives the identity. $\square$

Theorem 12.3.4.14 (Open Hamiltonian sums with right spectators). [Lean] [Lean] [Stm] *Let $L > 0$ and $p \leq n$. Under the canonical identification*

$$\ell^2([d]^{n+r}) \simeq \ell^2([d]^n \times [d]^r), \quad (12.85)$$

both the prefix Hamiltonian on $[0, p)$ and every suffix-interval Hamiltonian ending at p are direct sums over the right-spectator configurations of their operators in the n -site volume.

Proof. ✓ **Proof** The condition $L > 0$ makes the active starts in the two finite sums exactly the starts before p . Concatenation gives a bijection between the corresponding start sets in the two volumes. Apply Theorem 12.3.4.13 to each summand and then sum over this bijection. $\square$

Theorem 12.3.4.15 (Fixed-volume C3 ground projections with right spectators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $K, l, n, r \in \mathbb{N}$. Suppose that A has nonzero bond dimension, is block injective at length $L_0 > 0$, and satisfies $L_0 + 1 \leq n$. The kernel of the range- $(L_0 + 1)$ prefix Hamiltonian on the first n sites of the $(n + 1)$ -site volume is the left open-chain ground space. When that prefix instead fills its own n -site volume, its kernel is the whole n -site ground space. Independently, the kernel of the full final length- $(l + 1)$ suffix window beginning at K is the tail open-chain ground space. Let U_{pre} and U_{tail} be the canonical right-spectator isometries for the active-volume decompositions $(n + 1 + r) = (n + 1) + r$ and $(K + l + 1 + r) = (K + l + 1) + r$, respectively. Their orthogonal projections satisfy

$$U_{\text{pre}} G_{[0,n)}^{(n+1+r)} U_{\text{pre}}^{-1} = (G_{[0,n)}^{(n+1)})^{\oplus [d]^r}, \quad (12.86)$$

$$U_{\text{tail}} G_{[K, K+l+1]}^{(K+l+1+r)} U_{\text{tail}}^{-1} = (G_{[K, K+l+1]}^{(K+l+1)})^{\oplus [d]^r}. \quad (12.87)$$

Proof. ✓ Proof The prefix kernel is read fiberwise. Block injectivity identifies every fiber kernel with the n -site MPS ground space, which is precisely the left-window membership condition after fixing the last physical index. When the active prefix fills its smaller volume, the same argument identifies its kernel projection with the whole open-chain ground projection. The full suffix interval contains one interaction; its cyclic restriction is the ordinary restriction obtained by fixing the K -site prefix, giving the tail-window membership condition. Orthogonal kernel projections are preserved by the canonical isometry, and the fiberwise kernel-projection identity gives the two displayed formulas. $\square$

Theorem 12.3.4.16 (C3 boundary indices in the fixed final volume). Lean Lean Lean Lean Lean ✓ Stmt Suppose that $0 < l$ and $l + 1 \leq N$. Take interaction length $L = l + 1$ and define $E_n = G_n^{(N)} - G_{n+1}^{(N)}$. Then the two boundary ranges in the fixed N -site space are explicit:

$$n < l \implies E_n = 0. \quad (12.88)$$

At the remaining boundary index,

$$Q_l E_l = 0. \quad (12.89)$$

At $n = l$, both Q_l and $G_{l+1}^{(N)}$ are the kernel projection of the unique interaction on $[0, l + 1)$. Thus

$$E_l = \mathbb{1} - G_{l+1}^{(N)}.$$

Hence

$$Q_l E_l = G_{l+1}^{(N)}(\mathbb{1} - G_{l+1}^{(N)}) = 0. \quad (12.90)$$

Proof. ✓ Proof If $n < l$, neither the n -site prefix nor the $(n + 1)$ -site prefix contains a full length- $(l + 1)$ interaction, so both ground projections are the identity, proving (12.88). At $n = l$, the next prefix and the interval $[0, l + 1)$ contain exactly the same single interaction. Idempotence of its kernel projection gives (12.89) via (12.90). $\square$

12.3.5 Spectator-indexed boundary maps

The tail and left boundary maps embed virtual-space data into a common $(K + L)$ -site Hilbert space, expressing overlapping ground spaces as ranges of linear maps on the same ambient space to compare their orthogonal projectors.

Definition 12.3.5.1 (Prefix restriction). Lean Lean ✓ Stmt For $K, L \in \mathbb{N}$ and a fixed suffix configuration $\tau: \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$, the *prefix restriction* is the $\mathbb{C}$ -linear map $P_\tau: \mathbb{C}^{\{0, \dots, d-1\}^{K+L}} \rightarrow \mathbb{C}^{\{0, \dots, d-1\}^K}$ defined by $(P_\tau \psi)(\sigma) = \psi(\sigma, \tau)$ for $\sigma: \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$. This is the left-sided analogue of the suffix restriction of Definition 11.5.5.5.

Lemma 12.3.5.2 (Prefix restriction of a ground-space vector). Lean ✓ Stmt For every boundary matrix $X \in M_D(\mathbb{C})$ and every suffix τ ,

$$P_\tau(\Gamma_{K+L}(X)) = \Gamma_K(A^\tau X). \quad (12.91)$$

Proof. ✓ Proof Expand $\Gamma_{K+L}(X)(\sigma, \tau) = \text{tr}(A^{\sigma, \tau} X)$, factor $A^{\sigma, \tau} = A^\sigma A^\tau$ by Lemma 2.1.4, and reassociate: $\text{tr}((A^\sigma A^\tau)X) = \text{tr}(A^\sigma(A^\tau X)) = \Gamma_K(A^\tau X)(\sigma)$. $\square$

Definition 12.3.5.3 (Tail boundary map). [Lean](#) [Lean](#) [✓ Stmt](#) For $K, L \in \mathbb{N}$, the *tail boundary map* is the $\mathbb{C}$ -linear map

$$\Gamma_{K,L}^{\text{tail}} : ((\{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}) \rightarrow M_D(\mathbb{C})) \rightarrow \mathbb{C}^{\{0, \dots, d-1\}^{K+L}} \quad (12.92)$$

defined for every family $Y : (\{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}) \rightarrow M_D(\mathbb{C})$ by

$$\Gamma_{K,L}^{\text{tail}}(Y)(u, \tau) = \text{tr}(A^\tau Y_u), \quad (12.93)$$

where $u : \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$ is the prefix and $\tau : \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$ is the suffix. When $Y_u = XA^u$ the output equals the standard $\Gamma_{K+L}(X)$.

Definition 12.3.5.4 (Left boundary map). [Lean](#) [Lean](#) [✓ Stmt](#) For $K, L \in \mathbb{N}$, the *left boundary map* is the $\mathbb{C}$ -linear map

$$\Gamma_{K,L}^{\text{left}} : ((\{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}) \rightarrow M_D(\mathbb{C})) \rightarrow \mathbb{C}^{\{0, \dots, d-1\}^{K+L}} \quad (12.94)$$

defined for every family $Z : (\{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}) \rightarrow M_D(\mathbb{C})$ by

$$\Gamma_{K,L}^{\text{left}}(Z)(u, \tau) = \text{tr}(A^u Z_\tau), \quad (12.95)$$

where $u : \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$ is the prefix and $\tau : \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$ is the suffix. When $Z_\tau = A^\tau X$ the output equals the standard $\Gamma_{K+L}(X)$.

Theorem 12.3.5.5 (Tail boundary map on a prefix-suffix pair). [Lean](#) [✓ Stmt](#) For every prefix $u : \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$ and suffix $\tau : \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$,

$$\Gamma_{K,L}^{\text{tail}}(Y)(u, \tau) = \Gamma_L(Y_u)(\tau). \quad (12.96)$$

Proof. [✓ Proof](#) Unfold the definition of $\Gamma_{K,L}^{\text{tail}}$ on the pair $(u, \tau) = \{0, \dots, -1\} \text{append } u \tau$, use the decomposition $(\{0, \dots, -1\} \text{append } u \tau) \circ \{0, \dots, -1\} \text{natAdd } K = \tau$ and $(\{0, \dots, -1\} \text{append } u \tau) \circ \{0, \dots, -1\} \text{castAdd } L = u$, and apply $\Gamma_L(Y_u)(\tau) = \text{tr}(A^\tau Y_u)$. $\square$

Theorem 12.3.5.6 (Suffix restriction of the tail boundary map). [Lean](#) [✓ Stmt](#) Restricting the tail boundary map to the suffix for a fixed prefix $u \in \{0, \dots, d-1\}^K$ recovers $\Gamma_L(Y_u)$:

$$R_u^{\text{tail}}(\Gamma_{K,L}^{\text{tail}}(Y)) = \Gamma_L(Y_u). \quad (12.97)$$

Proof. [✓ Proof](#) This is Theorem 12.3.5.5 applied pointwise: for each τ , $(R_u^{\text{tail}}\psi)(\tau) = \psi(u, \tau)$. $\square$

Theorem 12.3.5.7 (Tail boundary map factorization). [Lean](#) [✓ Stmt](#) The standard length- $(K+L)$ boundary map factors through the tail boundary map:

$$\Gamma_{K+L}(X) = \Gamma_{K,L}^{\text{tail}}(u \mapsto XA^u) \quad (12.98)$$

for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Write a configuration $\sigma \in \{0, \dots, d-1\}^{K+L}$ as $\sigma = (u, \tau)$ with $u \in \{0, \dots, d-1\}^K$ and $\tau \in \{0, \dots, d-1\}^L$. Substitute $Y_u = XA^u$ into the definition of $\Gamma_{K,L}^{\text{tail}}$, apply the three-factor trace cycle $\text{tr}(A^\tau(XA^u)) = \text{tr}(A^u(A^\tau X))$, reassociate $\text{tr}(A^u(A^\tau X)) = \text{tr}((A^u A^\tau)X)$, factor $A^u A^\tau = A^{u, \tau}$ by multiplicativity of word evaluation (Lemma 2.1.4), and identify the result with $\Gamma_{K+L}(X)(u, \tau) = \text{tr}(A^{u, \tau} X)$. $\square$

Theorem 12.3.5.8 (Tail boundary map range). [Lean](#) [✓ Stmt](#) *The range of the tail boundary map is exactly the tail ground space:*

$$\text{range}(\Gamma_{K,L}^{\text{tail}}) = \{\psi \in \mathbb{C}^{\{0,\dots,d-1\}^{K+L}} \mid \forall u \in \{0, \dots, d-1\}^K, R_u^{\text{tail}}\psi \in G_L(A)\}. \quad (12.99)$$

Proof. [✓ Proof](#) For the forward inclusion, if $\psi = \Gamma_{K,L}^{\text{tail}}(Y)$ then $R_u^{\text{tail}}\psi = \Gamma_L(Y_u)$ for every prefix u , so the suffix restriction lies in $G_L(A)$. Conversely, given ψ with $R_u^{\text{tail}}\psi \in G_L(A)$ for every u , choose for each u a matrix $Y_u \in M_D(\mathbb{C})$ with $\Gamma_L(Y_u) = R_u^{\text{tail}}\psi$. Then $\Gamma_{K,L}^{\text{tail}}(Y)(u, \tau) = \text{tr}(A^\tau Y_u) = \Gamma_L(Y_u)(\tau) = (R_u^{\text{tail}}\psi)(\tau) = \psi(u, \tau)$, so ψ lies in the range. $\square$

Theorem 12.3.5.9 (Left boundary map on a prefix-suffix pair). [Lean](#) [✓ Stmt](#) *For every prefix $u: \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$ and suffix $\tau: \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$,*

$$\Gamma_{K,L}^{\text{left}}(Z)(u, \tau) = \Gamma_K(Z_\tau)(u). \quad (12.100)$$

Proof. [✓ Proof](#) Unfold the definition of $\Gamma_{K,L}^{\text{left}}$ on the pair (u, τ) , use $(\{0, \dots, -1\} \text{append } u \tau) \circ \{0, \dots, -1\} \text{castAdd } L = u$ and $(\{0, \dots, -1\} \text{append } u \tau) \circ \{0, \dots, -1\} \text{natAdd } K = \tau$, and apply $\Gamma_K(Z_\tau)(u) = \text{tr}(A^u Z_\tau)$. $\square$

Theorem 12.3.5.10 (Prefix restriction of the left boundary map). [Lean](#) [✓ Stmt](#) *Restricting the left boundary map to the prefix for a fixed suffix $\tau \in \{0, \dots, d-1\}^L$ recovers $\Gamma_K(Z_\tau)$:*

$$P_\tau(\Gamma_{K,L}^{\text{left}}(Z)) = \Gamma_K(Z_\tau). \quad (12.101)$$

Proof. [✓ Proof](#) Pointwise application of Theorem 12.3.5.9: for each σ , $(P_\tau\psi)(\sigma) = \psi(\sigma, \tau)$. $\square$

Theorem 12.3.5.11 (Left boundary map factorization). [Lean](#) [✓ Stmt](#) *The standard length- $(K+L)$ boundary map factors through the left boundary map:*

$$\Gamma_{K+L}(X) = \Gamma_{K,L}^{\text{left}}(\tau \mapsto A^\tau X) \quad (12.102)$$

for every $X \in M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Substitute $Z_\tau = A^\tau X$ into the definition of $\Gamma_{K,L}^{\text{left}}$, giving $\text{tr}(A^u(A^\tau X)) = \text{tr}((A^u A^\tau)X)$. Factor $A^u A^\tau = A^{u,\tau}$ by Lemma 2.1.4 and identify the result with $\Gamma_{K+L}(X)(u, \tau)$. $\square$

Theorem 12.3.5.12 (Left boundary map range). [Lean](#) [✓ Stmt](#) *The range of the left boundary map is exactly the left ground space:*

$$\text{range}(\Gamma_{K,L}^{\text{left}}) = \{\psi \in \mathbb{C}^{\{0,\dots,d-1\}^{K+L}} \mid \forall \tau \in \{0, \dots, d-1\}^L, P_\tau\psi \in G_K(A)\}. \quad (12.103)$$

Proof. [✓ Proof](#) The proof is symmetric to Theorem 12.3.5.8. If $\psi = \Gamma_{K,L}^{\text{left}}(Z)$ then $P_\tau\psi = \Gamma_K(Z_\tau)$, so the prefix restriction lies in $G_K(A)$. Conversely, given ψ with $P_\tau\psi \in G_K(A)$ for every τ , choose Z_τ with $\Gamma_K(Z_\tau) = P_\tau\psi$; then $\Gamma_{K,L}^{\text{left}}(Z)(u, \tau) = \text{tr}(A^u Z_\tau) = \Gamma_K(Z_\tau)(u) = (P_\tau\psi)(u) = \psi(u, \tau)$. $\square$

Lemma 12.3.5.13 (Prefix restriction at $L = 1$). [Lean](#) [✓ Stmt](#) *For $L = 1$, the prefix restriction P_τ coincides with the last-site restriction: $P_\tau\psi = R^{\text{last}}(\psi, \tau(0))$ for every $\tau: \{0, \dots, 1-1\} \rightarrow \{0, \dots, d-1\}$.*

Proof. [✓ Proof](#) Unfolding both sides: for $\sigma \in \{0, \dots, d-1\}^K$, $(P_\tau\psi)(\sigma) = \psi(\sigma, \tau)$ while $R^{\text{last}}(\psi, j)(\sigma) = \psi(\sigma_1, \dots, \sigma_K, j)$ with $j = \tau(0)$. These are equal because $\{0, \dots, -1\} \text{append } \sigma \tau$ puts τ after σ , matching the $\{0, \dots, -1\} \text{snoc}$ pattern of R^{last} . $\square$

Theorem 12.3.5.14 (Tail boundary map range: Euclidean-space transport). [Lean](#) [✓ Stmt](#) For every $K, L \in \mathbb{N}$, transport of the tail boundary map range by the inverse canonical ℓ^2 isomorphism identifies it with the open-chain tail ground submodule:

$$\iota_{K+L}(\text{range}(\Gamma_{K,L}^{\text{tail}})) = \mathcal{V}_{K,L}(A). \quad (12.104)$$

Concretely, a vector v in the $(K+L)$ -site Euclidean space belongs to $\mathcal{V}_{K,L}(A)$ exactly when the corresponding function satisfies the tail ground condition (Definition 11.5.5.6).

Proof. [✓ Proof](#) By Theorem 12.3.5.8, the range equals the intersection over prefixes u of the ground-space comaps $(R_u^{\text{tail}})^{-1}(G_L(A))$. Transporting this intersection through the canonical ℓ^2 identification and unfolding the definition of $\mathcal{V}_{K,L}(A)$ (Definition 11.5.5.12) gives the stated equality. $\square$

Theorem 12.3.5.15 (Left boundary map range: Euclidean-space transport ($L = 1$)). [Lean](#) [✓ Stmt](#) At $L = 1$, transport of the left boundary map range by the inverse canonical ℓ^2 isomorphism identifies it with the open-chain left ground submodule:

$$\iota_{K+1}(\text{range}(\Gamma_{K,1}^{\text{left}})) = \mathcal{U}_K(A). \quad (12.105)$$

Concretely, a vector v in the $(K+1)$ -site Euclidean space belongs to $\mathcal{U}_K(A)$ exactly when the corresponding function satisfies the left ground condition (every fixed-final-site restriction lies in $G_K(A)$; Definition 11.5.1.4).

Proof. [✓ Proof](#) By Theorem 12.3.5.12 with $L = 1$, the range equals the intersection over suffixes $\tau: \{0, \dots, 1-1\} \rightarrow \{0, \dots, d-1\}$ of the comaps $P_\tau^{-1}(G_K(A))$. For $L = 1$, each suffix τ is determined by a single index $j \in \{0, \dots, d-1\}$, and Lemma 12.3.5.13 identifies P_τ with $R^{\text{last}}(\cdot, j)$, the last-site restriction of Definition 11.5.1.2. Transporting the resulting intersection $\bigcap_{j \in \{0, \dots, d-1\}} (R^{\text{last}}(\cdot, j))^{-1}(G_K(A))$ through the canonical ℓ^2 identification and unfolding Definition 11.5.5.12 gives $\mathcal{U}_K(A)$. $\square$

Definition 12.3.5.16 (Euclidean spectator boundary maps). [Lean](#) [Lean](#) [✓ Stmt](#) Transport the tail and left boundary maps through the canonical Euclidean identifications on their virtual matrix families and physical coefficient spaces. Denote the resulting continuous maps by $\Gamma_{K,L}^{\text{tail}}$ and $\Gamma_{K,L}^{\text{left}}$.

Theorem 12.3.5.17 (Euclidean spectator boundary ranges and factorizations). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Equip each virtual matrix fiber with the unweighted Hilbert–Schmidt inner product and flatten the finite family of fibers into a Euclidean space. The resulting continuous tail and left boundary maps have ranges $\mathcal{V}_{K,L}(A)$ and, at suffix length one, $\mathcal{U}_K(A)$. If $J_K^{\text{tail}}(X)_u = XA^u$ and $J_L^{\text{left}}(X)_\tau = A^\tau X$, then

$$\Gamma_{K,L}^{\text{tail}} J_K^{\text{tail}} = \Gamma_{K+L}, \quad \Gamma_{K,L}^{\text{left}} J_L^{\text{left}} = \Gamma_{K+L}. \quad (12.106)$$

Proof. [✓ Proof](#) Transport the algebraic range descriptions through the Euclidean identification. The two factorization identities follow by evaluating a virtual matrix X fiberwise and applying the corresponding algebraic tail or left factorization. $\square$

Theorem 12.3.5.18 (Injectivity of the left spectator boundary map). [Lean](#) [✓ Stmt](#) If the length- K ground-space map is injective, then the left boundary map with any spectator length L is injective.

Proof. ✓ **Proof** Recover every spectator fiber by restricting to its fixed suffix and use injectivity of the length- K ground-space map. $\square$

Theorem 12.3.5.19 (Direct-sum spectator Gram operators). Lean Lean Lean Lean ✓ **Stmt** *The tail Gram is the direct sum over prefix configurations of copies of $\mathcal{K}_L$, and the left Gram is the direct sum over suffix configurations of copies of $\mathcal{K}_K$. Whenever the corresponding local boundary map is injective, the inverse Gram is the direct sum of the local inverse Grams. These identities include empty spectator index types and assert no norm estimate.*

Proof. ✓ **Proof** Reindex the physical inner product by prefix-suffix configurations and evaluate each fiber. This gives the direct-sum Gram identities. Composing fiberwise with the local inverse Gram on either side gives the stated inverse formulas. $\square$

Theorem 12.3.5.20 (Norm of a fiberwise boundary operator). Lean ✓ **Stmt** *Let S be a finite spectator index set, let V be the Euclidean virtual matrix space, and let $G : V \rightarrow V$ be linear. If $G^{\oplus S}$ acts on a boundary family $x = (x_s)_{s \in S}$ by $(G^{\oplus S}x)_s = Gx_s$, then*

$$\|G^{\oplus S}\| \leq \|G\|. \quad (12.107)$$

This estimate also holds when S is empty.

Proof. ✓ **Proof** Square both sides of the pointwise estimate $\|Gx_s\| \leq \|G\|\|x_s\|$, sum over $s \in S$, and use the Euclidean direct-sum norm identity. $\square$

Theorem 12.3.5.21 (Exact C3 projector residual identity). Lean ✓ **Stmt** *Fix $K, L_0 \in \mathbb{N}$. Suppose that the tail boundary map for $(K, L_0 + 1)$, the left boundary map for $(K + L_0, 1)$, and Γ_{K+L_0+1} are injective. The difference between the product of the first two inverse-Gram range projectors and the third inverse-Gram range projector is exactly the common-factorization residual obtained from the tail and left virtual maps. Their ranges are, respectively, $\mathcal{V}_{K, L_0+1}(A)$, $\mathcal{U}_{K+L_0}(A)$, and $\mathcal{G}_{K+L_0+1}(A)$. This identity asserts no numerical contraction estimate.*

Proof. ✓ **Proof** Apply the common-factorization projector identity to the two spectator boundary maps, the full boundary map, and their tail and left virtual maps. These range equalities identify the three physical ranges. $\square$

Theorem 12.3.5.22 (Arbitrary-increment mixed Gram of the overlapping spectator boundary maps). Lean Lean ✓ **Stmt** *Let $q \in \mathbb{N}$, let $X = (X_u)_{u \in \{0, \dots, d-1\}^\kappa}$, and let $Z = (Z_j)_{j \in \{0, \dots, d-1\}^q}$ be virtual boundary-matrix families. After the canonical identification of $K + (l+q)$ with $(K+l) + q$, the mixed Gram for the tail window of length $l+q$ and the left window with a q -site increment is*

$$\langle X, (\Gamma_{K, l+q}^{\text{tail}})^\dagger \Gamma_{K+l, q}^{\text{left}} Z \rangle = \sum_{u \in \{0, \dots, d-1\}^\kappa} \sum_{j \in \{0, \dots, d-1\}^q} \langle U(A^j X_u), \mathcal{K}_l U(Z_j A^u) \rangle, \quad (12.108)$$

where U is Frobenius vectorization and $\mathcal{K}_l = \Gamma_l^\dagger \Gamma_l$. The multiplication order is fixed by $\text{tr}(A^{\tau, j} X_u) = \text{tr}(A^\tau (A^j X_u))$ and $\text{tr}(A^{u, \tau} Z_j) = \text{tr}(A^\tau (Z_j A^u))$. The one-site identity is the specialization $q = 1$. This identity expresses the algebraic form of condition (C3') in [191, Equation (2.5)].

Proof. ✓ **Proof** Split a physical configuration into a prefix u , an overlap word τ of length l , and an increment configuration j of length q . The adjoint relation turns the mixed Gram into the physical inner product of the two boundary vectors. For each (u, τ, j) , word multiplicativity and trace cyclicity give

$$\text{tr}(A^{\tau, j} X_u) = \text{tr}(A^\tau (A^j X_u)), \quad \text{tr}(A^{u, \tau} Z_j) = \text{tr}(A^\tau (Z_j A^u)). \quad (12.109)$$

For fixed (u, j) , summing the resulting inner products over τ is exactly the boundary-map Gram pairing $\langle U(A^j X_u), \mathcal{K}_l U(Z_j A^u) \rangle$. Summing over u and j proves the arbitrary-increment identity; setting $q = 1$ gives the original formula. $\square$

Theorem 12.3.5.23 (Centered mixed Gram at the limiting metric). [Lean](#) [✓ Stmt](#) *Specialize Theorem 12.3.5.22 to $q = 1$. Let $X = (X_u)_{u \in \{0, \dots, d-1\}^\kappa}$ and $Z = (Z_j)_{j \in \{0, \dots, d-1\}^1}$ be the resulting boundary families. If $\rho \in M_D(\mathbb{C})$ has nonzero trace and $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$, then*

$$\langle X, (\Gamma_{K,l+1}^{\text{tail}})^\dagger \Gamma_{K+l,1}^{\text{left}} Z \rangle - \sum_{u,j} \langle U(A^j X_u), \mathcal{K}_\infty U(Z_j A^u) \rangle \quad (12.110)$$

$$= \sum_{u,j} \langle U(A^j X_u), (\mathcal{K}_l - \mathcal{K}_\infty) U(Z_j A^u) \rangle. \quad (12.111)$$

Here $\mathcal{K}_\infty U(Z_j A^u) = U((\text{tr } \rho)^{-1} Z_j A^u \rho)$, so the limiting multiplication is on the right and is generally not the identity.

Proof. [✓ Proof](#) Substitute the exact mixed-Gram formula and the right-multiplication action of $\mathcal{K}_\infty$, then distribute subtraction through the two finite sums and the second argument of the inner product. $\square$

Theorem 12.3.5.24 (Limiting metric inverse and virtual-map adjoints). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For arbitrary virtual dimension, the adjoints of the tail and left virtual maps are*

$$(J_K^{\text{tail}})^\dagger (X_u)_u = U\left(\sum_u X_u (A^u)^\dagger\right), \quad (J_L^{\text{left}})^\dagger (Z_\tau)_\tau = U\left(\sum_\tau (A^\tau)^\dagger Z_\tau\right). \quad (12.112)$$

If, in addition, $D > 0$ and $\rho \in M_D(\mathbb{C})$ is positive definite, set $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$ and $S_\infty = \mathcal{K}_\infty^{-1}$. Then

$$\mathcal{K}_\infty U(X) = U((\text{tr } \rho)^{-1} X \rho), \quad S_\infty U(X) = U((\text{tr } \rho) X \rho^{-1}). \quad (12.113)$$

Proof. [✓ Proof](#) For the tail adjoint, test against an arbitrary matrix Y and use the Hilbert–Schmidt trace pairing fiberwise:

$$\sum_u \langle U(Y A^u), U(X_u) \rangle = \left\langle U(Y), U\left(\sum_u X_u (A^u)^\dagger\right) \right\rangle. \quad (12.114)$$

The left formula follows in the same way from $\langle U(A^\tau Y), U(Z_\tau) \rangle = \langle U(Y), U((A^\tau)^\dagger Z_\tau) \rangle$.

Under the additional positive-definiteness hypothesis, the forward action of $\mathcal{K}_\infty$ is the exact limiting-metric formula and the operator is invertible. Applying the forward formula to $U((\text{tr } \rho) X \rho^{-1})$ cancels both scalar factors and the two right matrix factors, which gives the displayed inverse action. $\square$

Theorem 12.3.5.25 (Exact limiting-metric telescope for the C3 residual). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\rho \in M_D(\mathbb{C})$ be positive definite and set $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$ and $S_\infty = \mathcal{K}_\infty^{-1}$. Write J_t and J_l for the tail and left virtual maps, and write $S_n = \mathcal{K}_n^{-1}$ whenever the length- n boundary map is injective. The virtual residual in the exact tail-after-left C3 projector factorization has*

the telescoping decomposition

$$M_l - \mathcal{K}_{l+1}^{\oplus K} J_t S_{K+l+1} J_1^\dagger \mathcal{K}_{K+l}^{\oplus 1} \quad (12.115)$$

$$= (M_l - \mathcal{K}_\infty^{\oplus K} J_t S_\infty J_1^\dagger \mathcal{K}_\infty^{\oplus 1}) \quad (12.116)$$

$$- (\mathcal{K}_{l+1}^{\oplus K} - \mathcal{K}_\infty^{\oplus K}) J_t S_{K+l+1} J_1^\dagger \mathcal{K}_{K+l}^{\oplus 1} \quad (12.117)$$

$$- \mathcal{K}_\infty^{\oplus K} J_t (S_{K+l+1} - S_\infty) J_1^\dagger \mathcal{K}_{K+l}^{\oplus 1} \quad (12.118)$$

$$- \mathcal{K}_\infty^{\oplus K} J_t S_\infty J_1^\dagger (\mathcal{K}_{K+l}^{\oplus 1} - \mathcal{K}_\infty^{\oplus 1}), \quad (12.119)$$

where $M_l = (\Gamma_{K,l+1}^{\text{tail}})^\dagger \Gamma_{K+l,1}^{\text{left}}$ and the superscript $\oplus$ denotes the corresponding fiberwise spectator operator. Substituting this equality into the exact inverse-Gram factorization gives the same decomposition of the physical projector residual, sandwiched between the tail boundary map, the two local fiberwise inverse Grams, and the adjoint left boundary map.

The limiting product in the first parenthesis has, without an additional hypothesis, the explicit limiting mixed pairing

$$\langle X, \mathcal{K}_\infty^{\oplus K} J_t S_\infty J_1^\dagger \mathcal{K}_\infty^{\oplus 1} Z \rangle \quad (12.120)$$

$$= \sum_{u,j} \langle U(A^j X_u), \mathcal{K}_\infty U(Z_j A^u) \rangle. \quad (12.121)$$

Consequently, pairing the first parenthesis against boundary families is exactly the same finite sum with $\mathcal{K}_l - \mathcal{K}_\infty$ in place of $\mathcal{K}_l$.

These identities are reconstructed operator algebra. They assert neither the numerical FNW/Nachtergaele contraction nor any identification of a physical ground projector with P_ρ .

Proof. ✓ Proof Add and subtract the product in which all three finite Gram factors are replaced by their limiting values, and telescope the resulting product from left to right. Substitute the direct-sum Gram and inverse-Gram formulas into the exact projector factorization.

It remains to identify the leading limiting product. In Frobenius coordinates the limiting metric and its inverse act by

$$\mathcal{K}_\infty U(X) = U((\text{tr } \rho)^{-1} X \rho), \quad S_\infty U(X) = U((\text{tr } \rho) X \rho^{-1}). \quad (12.122)$$

The adjoint left virtual map sends a family $W = (W_j)_j$ to $U(\sum_j (A^j)^\dagger W_j)$. For $W_j = (\text{tr } \rho)^{-1} Z_j \rho$, the right multiplication by ρ and the factor $(\text{tr } \rho)^{-1}$ supplied by the left limiting spectator metric cancel exactly through S_∞ . Applying the tail virtual map and the tail limiting spectator metric leaves, in the fiber indexed by u ,

$$U\left((\text{tr } \rho)^{-1} \sum_j (A^j)^\dagger Z_j A^u \rho\right). \quad (12.123)$$

Its inner product with X_u , followed by summation over u , is the displayed limiting mixed pairing by the Frobenius trace identity. No transfer fixed-point equation is needed for this zeroth-order cancellation. The centered mixed-Gram identity then turns the leading term into the pairing with $\mathcal{K}_l - \mathcal{K}_\infty$. $\square$

12.3.6 Martingale gap consequences

Lemma 12.3.6.1 (Weighted inner-product estimate). Lean ✓ Stmt Let x and y belong to a complex inner-product space. For every $c > 0$,

$$\text{Re}\langle x, y \rangle \leq \frac{1}{2c} \|x\|^2 + \frac{c}{2} \|y\|^2. \quad (12.124)$$

Proof. ✓ Proof Cauchy–Schwarz bounds the left-hand side by $\|x\|\|y\|$. Apply the weighted arithmetic–geometric mean inequality to $\|y\|$ and $\|x\|$. $\square$

Lemma 12.3.6.2 (C3 projection-composition estimate). Lean ✓ Stmt *Let P be an orthogonal projection on a complex inner-product space, and let Q be a bounded linear operator from that space to another. If*

$$\|QP\| \leq \epsilon, \quad (12.125)$$

then, for every v ,

$$\|QPv\|^2 \leq \epsilon^2 \|Pv\|^2. \quad (12.126)$$

This includes $\epsilon = 0$.

Proof. ✓ Proof Idempotency gives $QPv = QP(Pv)$. Apply the operator-norm estimate for QP to Pv and square the resulting nonnegative inequality. The hypothesis itself implies $\epsilon \geq 0$. $\square$

Theorem 12.3.6.3 (Moving-window double counting). Lean Lean Lean ✓ Stmt *Let $(a_m)_{m \geq 0}$ be a nonnegative real sequence. For all $l, N \in \mathbb{N}$,*

$$\sum_{n=0}^{N-1} \sum_{m=\max\{0, n-l\}}^n a_m \leq (l+1) \sum_{m=0}^{N-1} a_m. \quad (12.127)$$

The same bound holds after omitting the terms with $n < l$. More generally, for a threshold $n_0 \in \mathbb{N}$ only the coefficients indexed by $m \geq \max\{0, n_0 - l\}$ occur, so

$$\sum_{n=n_0}^{N-1} \sum_{m=\max\{0, n-l\}}^n a_m \leq (l+1) \sum_{m=\max\{0, n_0-l\}}^{N-1} a_m. \quad (12.128)$$

Proof. ✓ Proof Write $k = n - m$. Each pair in the double sum has $0 \leq k \leq l$. For fixed k , the map $n \mapsto n - k$ is injective and has image in $\{0, \dots, N-1\}$. Nonnegativity bounds its contribution by $\sum_{m=0}^{N-1} a_m$. Summing over the $l+1$ values of k proves (12.127). This is the counting step in the proof of Nachtergaele’s Theorem 2.1(i), between Equation (Enpsi2) and the following summed estimate: for $a_m = \|E_m \psi\|^2$, each term occurs in at most $l+1$ consecutive windows [191, Proof of Theorem 2.1(i)]. For (12.128), replace a_m by the sequence that vanishes below $\max\{0, n_0 - l\}$ and agrees with a_m above it. Every window with $n \geq n_0$ is unchanged, and the previous bound applied to the replacement sequence gives the claim. $\square$

Theorem 12.3.6.4 (Threshold C1–C3 energy estimate). Lean ✓ Stmt *Let $(G_n)_{n \geq 0}$ be nested orthogonal projections on a finite-dimensional complex Hilbert space $\mathcal{H}$, put $E_n = G_n - G_{n+1}$, and assume $G_0 = \mathbf{1}$. Fix $l, N, n_0 \in \mathbb{N}$ with $n_0 \leq N$, and let Q_n be orthogonal projections for $n_0 \leq n < N$. Suppose $E_m Q_n = Q_n E_m$ whenever $m < n - l$ or $n < m$. Let H_n be the local Hamiltonians and H the full Hamiltonian. Assume the quadratic-form version of C1 on the same range, for every $x \in \mathcal{H}$,*

$$0 \leq \sum_{n=n_0}^{N-1} \operatorname{Re} \langle H_n x, x \rangle \leq d_{l+1} \operatorname{Re} \langle H x, x \rangle, \quad (12.129)$$

the quadratic-form version of C2 for every $x \in \mathcal{H}$ and $n_0 \leq n < N$,

$$\gamma_{l+1} \|(\mathbf{1} - Q_n)x\|^2 \leq \operatorname{Re} \langle H_n x, x \rangle, \quad (12.130)$$

and the literal condition C3 for $n_0 \leq n < N$,

$$\|Q_n E_n\| \leq \epsilon_l, \quad 0 \leq \epsilon_l < \frac{1}{\sqrt{l+1}}. \quad (12.131)$$

If $\gamma_{l+1} > 0$, $d_{l+1} > 0$, and $v \perp \text{ran}(G_N)$, then

$$\begin{aligned} \frac{\gamma_{l+1}}{d_{l+1}} \left(1 - \epsilon_l \sqrt{l+1}\right)^2 \sum_{n=n_0}^{N-1} \|E_n v\|^2 &\leq \text{Re}\langle H v, v \rangle \\ &+ \frac{\gamma_{l+1}}{d_{l+1}} \left(1 - \epsilon_l \sqrt{l+1}\right) \epsilon_l \sqrt{l+1} \sum_{m=\max\{0, n_0-l\}}^{n_0-1} \|E_m v\|^2. \end{aligned} \quad (12.132)$$

Conditions C2 and C3 have the source forms, with their onset represented by n_0 . The displayed C1 inequality follows from source C1 when $l \leq n_0$ and the omitted local terms are nonnegative. Thus a source-derived application chooses n_0 above both l and the C2–C3 onset. If $n_0 < l$, the displayed C1 inequality is instead an independent stronger assumption. The coefficient of the martingale mass above the threshold is exactly the one in Theorem 2.1(i) of [191]. The correction is an explicit upper bound for the contribution of the at most l martingale differences immediately below the threshold; it uses multiplicity $l+1$ uniformly. At $n_0 = 0$ the correction is an empty sum.

Proof. ✓ Proof Write $P = \sum_{n=n_0}^{N-1} \|E_n v\|^2$ and $S = \sum_{m=\max\{0, n_0-l\}}^{n_0-1} \|E_m v\|^2$. The martingale resolution $v = \sum_{n=0}^{N-1} E_n v$ and outside-window cancellation reduce the identity labelled Enpsi at index n to the active window $m \in [n-l, n]$; this uses only $n < N$, so it is available for every $n \geq n_0$. Apply the weighted inner-product estimate with positive parameters c_1, c_2 , then C2 and C3, and sum over $n_0 \leq n < N$. The martingale differences are mutually orthogonal, so each window norm square splits into its summands, and (12.128) bounds the resulting double sum by $(l+1)(S+P)$. With C1 this gives

$$\left(2 - c_1 - \frac{\epsilon_l^2}{c_2} - (l+1)c_2\right) P \leq \frac{d_{l+1}}{c_1 \gamma_{l+1}} \text{Re}\langle H v, v \rangle + (l+1)c_2 S. \quad (12.133)$$

For $\epsilon_l > 0$, take $c_1 = 1 - \epsilon_l \sqrt{l+1}$ and $c_2 = \epsilon_l / \sqrt{l+1}$, so that $\epsilon_l^2 / c_2 = (l+1)c_2 = \epsilon_l \sqrt{l+1}$ and the left factor is $1 - \epsilon_l \sqrt{l+1}$. Multiplying by $c_1 \gamma_{l+1} / d_{l+1}$ gives (12.132). If $\epsilon_l = 0$, then $c_2 = 0$ is not an admissible weighted parameter. In that case C3 instead gives $Q_n E_n = 0$, so the second inner product in Enpsi vanishes; applying only the first weighted estimate with $c_1 = 1$ yields $P \leq (d_{l+1} / \gamma_{l+1}) \text{Re}\langle H v, v \rangle$, which is the same formula because the correction term vanishes. This repair is documented in [83]. □

Lemma 12.3.6.5 (Martingale differences below a fixed level). Lean ✓ Stmt Let $(G_n)_{n \geq 0}$ be nested orthogonal projections on a complex Hilbert space, and put $E_n = G_n - G_{n+1}$. If $G_{n_0} v = v$ for some n_0 , then $E_n v = 0$ for every $n < n_0$.

Proof. ✓ Proof For $m \leq n_0$ the ranges are nested, so $G_m G_{n_0} = G_{n_0}$ and hence $G_m v = G_m G_{n_0} v = G_{n_0} v = v$. For $n < n_0$ this applies to both n and $n+1$, so $E_n v = G_n v - G_{n+1} v = v - v = 0$. □

Theorem 12.3.6.6 (C1–C3 energy estimate with no mass below the threshold). Lean ✓ Stmt Assume the hypotheses of Theorem 12.3.6.4, except that $l \leq n_0$ is assumed and (12.129) is replaced by condition C1 on the range $l \leq n < M$ of its own window sum at the two volumes

at which the source asserts it: the full inequality $0 \leq \sum_{n=l}^{N-1} \text{Re}\langle H_n x, x \rangle \leq d_{l+1} \text{Re}\langle H x, x \rangle$ at $M = N$, and its non-negativity clause $0 \leq \sum_{n=l}^{n_0-1} \text{Re}\langle H_n x, x \rangle$ at $M = n_0$. Assume in addition that $E_n v = 0$ for every $n < n_0$. Then

$$\frac{\gamma_{l+1}}{d_{l+1}} (1 - \epsilon_l \sqrt{l+1})^2 \|v\|^2 \leq \text{Re}\langle H v, v \rangle. \quad (12.134)$$

This is the conclusion of Theorem 2.1(i) of [191] with each of C1, C2, and C3 imposed on the range at which the source imposes it. The added vanishing hypothesis is the minimal repair of that theorem: the two below-threshold quantities in (12.132) enter (12.134) with non-negative weights, so both have to vanish, and the vanishing of the omitted martingale mass forces the vanishing of the window correction. Without the added hypothesis the statement is false, by Theorem 12.3.6.7. The source defect and the repair are recorded in [82].

Proof. ✓ Proof Condition C1 on the threshold range $n_0 \leq n < N$ follows from the assumed form: its upper bound by subtracting from the sum over $l \leq n < N$ the initial sum over $l \leq n < n_0$, which is non-negative by C1 at the volume Λ_{n_0} , and its non-negativity clause termwise from C2, each of whose left-hand sides is non-negative. The window correction in (12.132) is a sum of squared norms of martingale differences with index below n_0 , hence zero. The Pythagorean decomposition of a vector orthogonal to $\text{ran}(G_N)$ splits $\|v\|^2$ into the sum over $n < n_0$, which is zero for the same reason, and the martingale mass $\sum_{n=n_0}^{N-1} \|E_n v\|^2$. Hence the latter equals $\|v\|^2$, and (12.132) becomes (12.134). □

Theorem 12.3.6.7 (The source conditions on their own ranges do not give the estimate). Lean

✓ Stmt In the notation of Theorem 12.3.6.6: for every k there are, on a complex Hilbert space of dimension $k+2$, nested orthogonal projections $(G_n)_{n \geq 0}$ with $G_0 = \mathbf{1}$, $G_{N+1} = 0$, and $G_n \neq 0$ for every $n \leq N$, nonzero orthogonal projections Q_n , local operators H_n , volume Hamiltonians H_{Λ_M} , and a vector $v \perp \text{ran}(G_N)$ such that, with $l = 0$, $n_0 = 1$, $N = k+1$, $\gamma_1 = d_1 = 1$, and $\epsilon_0 = 0$, the required commutations hold, condition C1 holds with the single constant d_1 at every volume Λ_M on the range $l \leq n < M$ carried by its own window sum, the gap inequality of condition C2 and condition C3 hold on the range $n_0 \leq n < N$ from which they are imposed, every ground space is the kernel of its own Hamiltonian, that is $\ker H_{\Lambda_M} = \text{ran}(G_M)$ for $M \leq N$ and $\ker H_n = \text{ran}(Q_n)$ for every window inside Λ_N , that is for $0 \leq n < N$, where the Q_n are also nonzero orthogonal projections commuting with the martingale differences outside their own windows, and

$$\text{Re}\langle H_{\Lambda_N} v, v \rangle = \frac{1}{2} \|v\|^2 < \frac{\gamma_1}{d_1} (1 - \epsilon_0)^2 \|v\|^2. \quad (12.135)$$

Hence condition C1 at every volume on the range of its own window sum together with conditions C2 and C3 on $n_0 \leq n < N$, the ranges at which [191] imposes them, do not imply the conclusion of its Theorem 2.1(i), even when every ground space is the kernel of its Hamiltonian and is nontrivial, as condition C2 also requires, and even under the conventions the source attaches to the extreme volumes.

Proof. ✓ Proof Fix an orthonormal basis $e_0, \dots, e_{k+1}$ and let every operator be diagonal in it. The first $N = k+1$ basis vectors carry the single excitations of a chain of N sites and the last carries its ground state. Let G_n have weight 1 at e_i when $n \leq i$ and weight 0 otherwise, so that $G_0 = \mathbf{1}$, $G_{N+1} = 0$ because no basis vector has index at least $N+1$, every G_n with $n \leq N$ is nonzero because it retains e_{k+1} , G_N is the projection onto e_{k+1} , and E_n is the projection onto e_n for $n < N$. Let $c_0 = \frac{1}{2}$, let $c_i = 1$ for $1 \leq i \leq k$, and let $c_{k+1} = 0$; let H_{Λ_M} have weight c_i at

e_i when $i < M$ and weight 0 otherwise, let H_n have weight c_n at e_n and weight 0 elsewhere, and let $Q_n = \mathbf{1} - E_n$, which is nonzero because the space has dimension at least two. All operators are diagonal, so the required commutations hold, and $Q_n E_n = 0$ gives condition C3 with $\epsilon_0 = 0$. Summing H_n over the C1 range $0 \leq n < M$ gives H_{Λ_M} exactly, at every M , and the weights c_i are nonnegative, so condition C1 holds at every volume with the single constant $d_1 = 1$ and with equality in its upper bound. Since $c_i \neq 0$ for $i \leq k$, the kernel of H_{Λ_M} is spanned by the e_i with $i \geq M$ whenever $M \leq N$, which is $\text{ran}(G_M)$, and the kernel of H_n is spanned by the e_i with $i \neq n$ whenever $0 \leq n < N$, which is $\text{ran}(Q_n)$ and is nonzero. For $1 \leq n < N$ one has $c_n = 1$, so $\|(\mathbf{1} - Q_n)x\|^2 = \|E_n x\|^2$ equals $\text{Re}\langle H_n x, x \rangle$, which is condition C2 with $\gamma_1 = 1$. Take $v = e_0$. Then $v \perp \text{ran}(G_N)$, the line through e_{k+1} , and $\text{Re}\langle H_{\Lambda_N} v, v \rangle = c_0 = \frac{1}{2}$, while $\|v\|^2 = 1$, which is (12.135). The whole martingale mass of v sits at the index $n = 0$, which C1 covers while C2 and C3 exclude it, and it is at that index that C2 fails. $\square$

Theorem 12.3.6.8 (Full-range C1–C3 energy estimate). [Lean] [Stmt] *Let $(G_n)_{n \geq 0}$ be nested orthogonal projections on a finite-dimensional complex Hilbert space $\mathcal{H}$, put $E_n = G_n - G_{n+1}$, and assume $G_0 = \mathbf{1}$. Fix $l, N \in \mathbb{N}$, and let Q_n be orthogonal projections for $0 \leq n < N$. Suppose $E_m Q_n = Q_n E_m$ whenever $m < n - l$ or $n < m$. Let H_n be the local Hamiltonians and H the full Hamiltonian. Assume the quadratic-form version of C1 for every $x \in \mathcal{H}$,*

$$0 \leq \sum_{n=0}^{N-1} \text{Re}\langle H_n x, x \rangle \leq d_{l+1} \text{Re}\langle H x, x \rangle, \quad (12.136)$$

and the quadratic-form version of C2 for every $x \in \mathcal{H}$ and $0 \leq n < N$,

$$\gamma_{l+1} \|(\mathbf{1} - Q_n)x\|^2 \leq \text{Re}\langle H_n x, x \rangle, \quad (12.137)$$

and the literal condition C3,

$$\|Q_n E_n\| \leq \epsilon_l, \quad 0 \leq \epsilon_l < \frac{1}{\sqrt{l+1}}. \quad (12.138)$$

If $\gamma_{l+1} > 0$, $d_{l+1} > 0$, and $v \perp \text{ran}(G_N)$, then

$$\text{Re}\langle H v, v \rangle \geq \frac{\gamma_{l+1}}{d_{l+1}} (1 - \epsilon_l \sqrt{l+1})^2 \|v\|^2. \quad (12.139)$$

This full-range statement is stronger than the unrestricted source theorem in general. It assumes C1, C2, and C3 for every $0 \leq n < N$, whereas the source imposes lower thresholds on these conditions. Those thresholds cannot simply be dropped: without separate control of the initial volumes the source hypotheses imply neither the assumptions above nor the conclusion, by Theorem 12.3.6.7. The coefficient is exactly the one in Theorem 2.1(i) of [191]: γ_{l+1}/d_{l+1} times $(1 - \epsilon_l \sqrt{l+1})^2$. This is Theorem 12.3.6.4 at threshold $n_0 = 0$, where nothing lies below the threshold.

Proof. [Proof] Apply Theorem 12.3.6.4 with $n_0 = 0$ and the given window length l . Its window correction is a sum over an empty range, hence zero, and its martingale mass over $0 \leq n < N$ is $\|v\|^2$ by the Pythagorean decomposition of a vector orthogonal to $\text{ran}(G_N)$. What remains is (12.139). $\square$

Theorem 12.3.6.9 (Energy form to norm bound). [Lean] [Stmt] *Let H be a linear operator on a complex Hilbert space. If*

$$\kappa \|v\|^2 \leq \text{Re}\langle H v, v \rangle \quad (12.140)$$

for every $v \in (\ker H)^\perp$, then $\kappa \|v\| \leq \|H v\|$ on the same subspace.

Proof. ✓ Proof Cauchy–Schwarz gives

$$\kappa \|v\|^2 \leq \operatorname{Re}\langle Hv, v \rangle \leq \|Hv\| \|v\|. \quad (12.141)$$

The claim is immediate for $v = 0$; otherwise cancel $\|v\|$. □

Theorem 12.3.6.10 (Norm gap from the full-range C1–C3 estimate). Lean ✓ Stmt *In the setting of Theorem 12.3.6.8, suppose that H is positive and $\operatorname{ran}(G_N) = \ker H$. Then every $v \in (\ker H)^\perp$ satisfies*

$$\frac{\gamma_{l+1}}{d_{l+1}} \left(1 - \epsilon_l \sqrt{l+1}\right)^2 \|v\| \leq \|Hv\|. \quad (12.142)$$

Proof. ✓ Proof Denote the coefficient in (12.142) by κ . For an arbitrary x , remove its orthogonal projection onto $\ker H$ and call the remainder w . Then $Hx = Hw$ and $\operatorname{Re}\langle Hx, x \rangle = \operatorname{Re}\langle Hw, w \rangle$. The energy estimate and Theorem 12.3.6.9 give $\kappa \|w\| \leq \|Hx\|$. Hence Cauchy–Schwarz gives

$$\kappa \operatorname{Re}\langle Hx, x \rangle \leq \kappa \|Hx\| \|w\| \leq \|Hx\|^2. \quad (12.143)$$

Thus the hypothesis of Lemma 12.3.20 holds globally. The finite-dimensional quadratic-form criterion then yields the stated norm bound. □

Theorem 12.3.6.11 (Martingale criterion for a finite family of projections). Lean ✓ Stmt *Let P_i be a finite family of symmetric projections on a finite-dimensional complex Hilbert space, let $H = \sum_i P_i$, and let $0 < \gamma \leq 1$. Suppose that a real coefficient matrix (c_{ij}) has row and column sums at most one away from the diagonal, and that, for $i \neq j$ and every v ,*

$$\operatorname{Re}\langle P_i v, P_j v \rangle + \operatorname{Re}\langle P_j v, P_i v \rangle \quad (12.144)$$

$$\geq -(1 - \gamma) c_{ij} (\operatorname{Re}\langle P_i v, v \rangle + \operatorname{Re}\langle P_j v, v \rangle). \quad (12.145)$$

Then

$$\gamma \operatorname{Re}\langle Hv, v \rangle \leq \operatorname{Re}\langle Hv, Hv \rangle$$

for every v . Consequently, for every $v \in (\ker H)^\perp$,

$$\gamma \|v\| \leq \|Hv\|.$$

Proof. ✓ Proof The row-and-column reduction gives the quadratic-form inequality. The sum of symmetric projections is positive, so the abstract martingale criterion converts that inequality into the norm lower bound on $(\ker H)^\perp$. □

Theorem 12.3.6.12 (Reduced ground-space overlap bounds the excitation anticommutator). Lean ✓ Stmt *Let U and V be ground-space ranges in a finite-dimensional complex Hilbert space, let $W = U \cap V$, and let q_U and q_V be the orthogonal projectors onto U and V . Suppose that $\eta \geq 0$ and*

$$|\langle x, y \rangle| \leq \eta \|x\| \|y\| \quad (12.146)$$

whenever $x \in U \cap W^\perp$ and $y \in V \cap W^\perp$. Define the complementary excitation projections by $h_U = \mathbb{1} - q_U$ and $h_V = \mathbb{1} - q_V$. Then, for every vector v ,

$$\operatorname{Re}\langle h_U v, h_V v \rangle + \operatorname{Re}\langle h_V v, h_U v \rangle \geq -\eta (\operatorname{Re}\langle h_U v, v \rangle + \operatorname{Re}\langle h_V v, v \rangle). \quad (12.147)$$

Proof. ✓ Proof Remove the component of v in the common intersection W ; both excitation projections are unchanged. If $\eta \leq 1$, apply (12.146) to the reduced ground-space components $q_U v$ and $q_V v$, and combine the resulting principal-angle estimate with the orthogonal decompositions $v = q_U v + h_U v = q_V v + h_V v$. If $\eta > 1$, the same conclusion follows from the universal overlap bound with coefficient 1 and the non-negativity of the two diagonal excitation forms. □

12.3.7 The compatible open-chain Hamiltonian

For an interval, the open-chain parent Hamiltonian contains only local terms whose supports lie entirely within the interval [191, Equation (3.12)]. The corresponding zero-energy space coincides with the open-chain ground space. Each compatible open Hamiltonian is positive as a sum of orthogonal projections. Conditions (C1) and (C2) hold across the full martingale range: before a complete interaction window fits, the suffix Hamiltonian vanishes, and on larger intervals it equals the orthogonal projection onto the local excited space. Combined with condition (C3), these identities specialize Theorem 12.3.6.10 with $d_{l+1} = \gamma_{l+1} = 1$ and overlap bound ϵ_l .

Definition 12.3.7.1 (Compatible open-chain parent Hamiltonian). [Lean](#) [Lean](#) [✓ Stmt](#) For $L, N \in \mathbb{N}$, let

$$I_{L,N} = \{i \in \{0, \dots, N-1\} \mid i + L \leq N\}. \quad (12.148)$$

The compatible length- N open-chain parent Hamiltonian of interaction length L is

$$H_{N,L}^{\text{open}}(A) = \sum_{i \in I_{L,N}} h_{i,\text{ES}}(A, L). \quad (12.149)$$

Thus no summand crosses the two endpoints of the interval. For $L = l + 1$, the starting sites are $0, \dots, N - l - 1$, in agreement with [191, Equation (3.12)].

Theorem 12.3.7.2 (Full-window compatible interaction). [Lean](#) [✓ Stmt](#) If $L > 0$, then

$$H_{L,L}^{\text{open}}(A) = P_L^{\text{ES}}(A). \quad (12.150)$$

Proof. [✓ Proof](#) The interaction has one window:

$$I_{L,L} = \{0\}. \quad (12.151)$$

The outside region is empty, so it has a unique configuration τ . Cyclic restriction to the sole window is the identity:

$$R_{0,\tau}v = v. \quad (12.152)$$

Likewise,

$$\sigma|_{[0,L-1]} = \sigma. \quad (12.153)$$

Therefore

$$H_{L,L}^{\text{open}}(A) = \sum_{i \in I_{L,L}} h_{i,\text{ES}}(A, L) = h_{0,\text{ES}}(A, L) = P_L^{\text{ES}}(A). \quad (12.154)$$

□

Theorem 12.3.7.3 (Unit gap at the full-window volume). [Lean](#) [Lean](#) [✓ Stmt](#) If $L > 0$ and $v \in (\ker H_{L,L}^{\text{open}}(A))^\perp$, then

$$\|H_{L,L}^{\text{open}}(A)v\| = \|v\|. \quad (12.155)$$

In particular,

$$\|v\| \leq \|H_{L,L}^{\text{open}}(A)v\|. \quad (12.156)$$

Thus $\gamma_{l+1} = 1$ at the full-window volume $N = L = l + 1$ in [191, Theorem 2.1(i)]. This neither asserts that the excited subspace is nonzero nor supplies a gap for any $N > L$.

Proof. ✓ Proof The full-window identity and the local kernel formula give

$$\ker H_{L,L}^{\text{open}}(A) = \ker P_L^{\text{ES}}(A) = G_L^{\text{ES}}(A). \quad (12.157)$$

Hence v belongs to $(G_L^{\text{ES}}(A))^\perp$, where the orthogonal projector $P_L^{\text{ES}}(A)$ acts as the identity. Thus

$$\|H_{L,L}^{\text{open}}(A)v\| = \|P_L^{\text{ES}}(A)v\| = \|v\|. \quad (12.158)$$

Consequently,

$$\|v\| \leq \|H_{L,L}^{\text{open}}(A)v\|. \quad (12.159)$$

□

Lemma 12.3.7.4 (Positivity of the compatible open-chain Hamiltonian). Lean ✓ Stmt For every A , L , and N , the operator $H_{N,L}^{\text{open}}(A)$ is positive.

Proof. ✓ Proof Each $h_{i,\text{ES}}(A, L)$ is an orthogonal projection and hence is positive. Positivity is preserved by finite sums. □

Definition 12.3.7.5 (Compatible suffix Hamiltonian). Lean ✓ Stmt For $L, l, n, N \in \mathbb{N}$, the Hamiltonian on the last l sites of the prefix $[0, n)$, regarded as an operator on the N -site chain, is

$$H_{N,L;l,n}^{\text{suffix}}(A) = \sum_{\substack{i \in I_{L,N} \\ n-l \leq i, \ i+L \leq n}} h_{i,\text{ES}}(A, L). \quad (12.160)$$

Its local windows lie in the nonwrapping interval $[n-l, n)$.

Theorem 12.3.7.6 (Vanishing suffix before a complete window). Lean ✓ Stmt If $n < L$, then

$$H_{N,L;l,n}^{\text{suffix}}(A) = 0. \quad (12.161)$$

Proof. ✓ Proof A summand would require a starting site i satisfying both $n-L \leq i$ and $i+L \leq n$. The latter inequality is impossible when $n < L$ and $i \geq 0$, so the defining sum is empty. □

Theorem 12.3.7.7 (Shifted full-range suffix sum). Lean ✓ Stmt If $L > 0$, then

$$\sum_{n=0}^{N-1} H_{N,L;l,n+1}^{\text{suffix}}(A) = \sum_{n=L}^N H_{N,L;l,n}^{\text{suffix}}(A). \quad (12.162)$$

Proof. ✓ Proof The substitution $m = n + 1$ identifies the left-hand side with the sum over $1 \leq m \leq N$. Every term with $m < L$ vanishes by Theorem 12.3.7.6, leaving precisely the stated interval. □

Theorem 12.3.7.8 (Finite-overlap condition C1 for the compatible open chain). Lean ✓ Stmt If $L \leq l$, then

$$0 \leq \sum_{n=l}^N H_{N,L;l,n}^{\text{suffix}}(A) \leq (l-L+1)H_{N,L}^{\text{open}}(A). \quad (12.163)$$

Thus condition C1 of [191] holds with $d_l = l - L + 1$. This is the source value $pl - r + 1$ in the present convention $p = 1$ and interaction range $r = L$.

Proof. [✓ Proof](#) Fix a starting site $i \in I_{L,N}$. The suffix Hamiltonian indexed by n contains its local term exactly when

$$i + L \leq n \leq i + l. \quad (12.164)$$

This interval contains at most $l - L + 1$ integers. Reversing the two finite sums therefore writes the left-hand side as

$$\sum_{i \in I_{L,N}} c_i h_{i,\text{ES}}(A, L), \quad (12.165)$$

where $0 \leq c_i \leq l - L + 1$. Positivity of each local term turns the coefficient bound into the upper operator inequality. Positivity of their sum gives the lower inequality. $\square$

Theorem 12.3.7.9 (Full-range compatible open-chain counting bound). [Lean](#) [✓ Stmt](#) Set $L = l + 1$. For every vector v on the N -site chain,

$$0 \leq \sum_{n=0}^{N-1} \text{Re} \langle H_{N,L;L,n+1}^{\text{suffix}}(A)v, v \rangle \leq \text{Re} \langle H_{N,L}^{\text{open}}(A)v, v \rangle. \quad (12.166)$$

Thus the full martingale range satisfies C1 with $d_{l+1} = 1$.

Proof. [✓ Proof](#) The shift identity (12.162) rewrites the sum over the interval $L \leq n \leq N$. Apply Theorem 12.3.7.8 with suffix length L . Each interaction then occurs once, since $L - L + 1 = 1$. Positivity of the suffix Hamiltonians gives the lower bound. $\square$

Theorem 12.3.7.10 (Embedded fixed-window condition C2). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let $L > 0$ and $L \leq m \leq N$. The suffix of length L ending at m contains one interaction:

$$H_{N,L;L,m}^{\text{suffix}}(A) = h_{m-L,\text{ES}}(A, L). \quad (12.167)$$

Write $G_{N,L;m}^{\text{ker}}$ for the orthogonal projection onto the kernel of this suffix Hamiltonian. Then

$$H_{N,L;L,m}^{\text{suffix}}(A) = \text{Proj}_{(\ker H_{N,L;L,m}^{\text{suffix}}(A))^\perp} = I - G_{N,L;m}^{\text{ker}}. \quad (12.168)$$

Consequently condition C2 holds in Loewner order with $\gamma_L = 1$:

$$H_{N,L;L,m}^{\text{suffix}}(A) \geq I - G_{N,L;m}^{\text{ker}}. \quad (12.169)$$

Equivalently, for every v ,

$$\text{Re} \langle (I - G_{N,L;m}^{\text{ker}})v, v \rangle \leq \text{Re} \langle H_{N,L;L,m}^{\text{suffix}}(A)v, v \rangle. \quad (12.170)$$

If $v \in (\ker H_{N,L;L,m}^{\text{suffix}}(A))^\perp$, then

$$\|H_{N,L;L,m}^{\text{suffix}}(A)v\| = \|v\|. \quad (12.171)$$

In particular,

$$\|v\| \leq \|H_{N,L;L,m}^{\text{suffix}}(A)v\|. \quad (12.172)$$

In the notation of [191], set $L = l + 1$ and $m = n + 1$. The interval is then $\Lambda_{n+1} \setminus \Lambda_{n-l} = [n-l, n+1)$, so

$$d_{l+1} = 1. \quad (12.173)$$

Condition C2 has the constant

$$\gamma_{l+1} = 1. \quad (12.174)$$

The later condition C3 uses the distinct overlap parameter ϵ_l .

Proof. ✓ Proof An admissible start i satisfies

$$m - L \leq i. \quad (12.175)$$

It also satisfies

$$i + L \leq m. \quad (12.176)$$

Hence $i = m - L$, which proves (12.167). The sole local term is an orthogonal projection. For any orthogonal projection H ,

$$\text{ran } H = (\ker H)^\perp. \quad (12.177)$$

Therefore

$$H = \text{Proj}_{(\ker H)^\perp} = I - \text{Proj}_{\ker H}. \quad (12.178)$$

This gives (12.168). The Loewner and quadratic-form inequalities are equalities. On $(\ker H)^\perp = \text{ran } H$, idempotence gives $Hv = v$, and hence exact norm preservation and the unit lower bound. Condition C1 at suffix length L gives $d_L = L - L + 1 = 1$. With $L = l + 1$, the two constants are therefore $d_{l+1} = 1$ and $\gamma_{l+1} = 1$. $\square$

Theorem 12.3.7.11 (Full-range fixed-window local gap). Lean ✓ Stmt Set $L = l + 1$, let $0 \leq n < N$, and let Q_n be the orthogonal projection onto the kernel of $H_{N,L;L,n+1}^{\text{suffix}}(A)$. Then every vector v satisfies

$$\|(I - Q_n)v\|^2 \leq \text{Re} \langle H_{N,L;L,n+1}^{\text{suffix}}(A)v, v \rangle. \quad (12.179)$$

Hence C2 holds on the full martingale range with $\gamma_{l+1} = 1$.

Proof. ✓ Proof If $n < l$, the suffix Hamiltonian is zero and $Q_n = I$, so both sides of (12.179) vanish. If $n \geq l$, the suffix contains the single length- $(l + 1)$ interaction and equals $I - Q_n$ by Theorem 12.3.7.10. Since $I - Q_n$ is an orthogonal projection,

$$\|(I - Q_n)v\|^2 = \text{Re} \langle (I - Q_n)v, v \rangle. \quad (12.180)$$

$\square$

Theorem 12.3.7.12 (Zero energy forces every compatible local constraint). Lean ✓ Stmt If $H_{N,L}^{\text{open}}(A)v = 0$, then, for every $i \in I_{L,N}$,

$$h_{i,\text{ES}}(A, L)v = 0. \quad (12.181)$$

Proof. ✓ **Proof** Taking the inner product of the sum with v gives

$$\sum_{i \in I_{L,N}} \operatorname{Re} \langle h_{i,\text{ES}}(A, L)v, v \rangle = \operatorname{Re} \langle H_{N,L}^{\text{open}}(A)v, v \rangle = 0. \quad (12.182)$$

Each summand is the squared norm of the image of v under an orthogonal projection. Hence every local term annihilates v . $\square$

Theorem 12.3.7.13 (Restriction of an open-boundary MPS vector). Lean ✓ **Stmt** Assume $s + L \leq N$. For every boundary matrix X and complementary configuration τ ,

$$\operatorname{Res}_{s,L}^{\tau} \Gamma_N^A(X) \in G_L(A). \quad (12.183)$$

Proof. ✓ **Proof** For the fixed words

$$w_L = (\tau_0, \dots, \tau_{s-1}), \quad (12.184)$$

and

$$w_R = (\tau_{s+L}, \dots, \tau_{N-1}), \quad (12.185)$$

the word of a configuration obtained by inserting the window word ω is

$$A^{w_L} \omega^{w_R} = A^{w_L} A^{\omega} A^{w_R}. \quad (12.186)$$

Cyclicity of the trace gives

$$\operatorname{tr}(A^{w_L} A^{\omega} A^{w_R} X) = \operatorname{tr}(A^{\omega} A^{w_R} X A^{w_L}), \quad (12.187)$$

and hence

$$\operatorname{Res}_{s,L}^{\tau} \Gamma_N^A(X) = \Gamma_L^A(A^{w_R} X A^{w_L}) \in G_L(A). \quad (12.188)$$

$\square$

Theorem 12.3.7.14 (Open-boundary vectors have zero compatible energy). Lean ✓ **Stmt** For every A , L , and N ,

$$G_N^{\text{ES}}(A) \subseteq \ker H_{N,L}^{\text{open}}(A). \quad (12.189)$$

Proof. ✓ **Proof** Let $v = \iota_N(\Gamma_N^A(X)) \in G_N^{\text{ES}}(A)$. For every non-wrapping window $[s, s + L - 1]$ and complementary configuration τ , Theorem 12.3.7.13 gives

$$\operatorname{Res}_{s,L}^{\tau} \Gamma_N^A(X) \in G_L(A). \quad (12.190)$$

The non-wrapping identification and Lemma 11.3.2 then give

$$R_{s,\tau} v = \iota_L(\operatorname{Res}_{s,L}^{\tau} \Gamma_N^A(X)) \in G_L^{\text{ES}}(A). \quad (12.191)$$

The local kernel criterion therefore gives

$$h_{s,\text{ES}}(A, L)v = 0. \quad (12.192)$$

Thus every compatible local term annihilates v . $\square$

Theorem 12.3.7.15 (Compatible zero energy restricts to each window). [Lean](#) [✓ Stmt](#) Let $L \leq N$. If $H_{N,L}^{\text{open}}(A)v = 0$, then for every nonwrapping start s with $s+L \leq N$ and every complementary configuration τ ,

$$\text{Res}_{s,L}^{\tau} v \in G_L(A).$$

This is the frustration-free extraction in [191, Equation (3.12)].

Proof. [✓ Proof](#) A kernel vector of a sum of orthogonal projections is a kernel vector of each summand. The local kernel criterion then places every window restriction in the local ground space. $\square$

Theorem 12.3.7.16 (Compatible zero energy has open-boundary form). [Lean](#) [✓ Stmt](#) Assume $D \geq 1$, $L_0 > 0$, and $N \geq L_0 + 1$. If A is L_0 -block-injective, then

$$\ker H_{N,L_0+1}^{\text{open}}(A) \subseteq G_N^{\text{ES}}(A). \quad (12.193)$$

Proof. [✓ Proof](#) In the following display, s ranges over all starts satisfying $s + L_0 + 1 \leq N$, and τ ranges over all complementary configurations. The window restriction and the open-chain intersection theorem give

$$H_{N,L_0+1}^{\text{open}}(A)v = 0 \implies (\forall s, \tau, \text{Res}_{s,L_0+1}^{\tau} v \in G_{L_0+1}(A)) \implies v \in G_N^{\text{ES}}(A). \quad (12.194)$$

$\square$

Theorem 12.3.7.17 (Ground-state property of the compatible open chain). [Lean](#) [✓ Stmt](#) Assume $D \geq 1$, $L_0 > 0$, and $N \geq L_0 + 1$. If A is L_0 -block-injective, then

$$\ker H_{N,L_0+1}^{\text{open}}(A) = G_N^{\text{ES}}(A). \quad (12.195)$$

This is the canonical MPS instance of the ground-state property in [191, Equation (3.13)]. The right-hand side is the open boundary-condition space and may therefore be degenerate.

Proof. [✓ Proof](#) Combine the mutual inclusions Theorem 12.3.7.14 and Theorem 12.3.7.16. $\square$

Theorem 12.3.7.18 (Open-boundary states span the local ground space). [Lean](#) [Lean](#) [✓ Stmt](#) For every tensor A and every L , the local ground space $G_L(A)$ is spanned by the open-boundary states

$$\sigma \mapsto (l|A^{\sigma}|r), \quad ((l|, |r)) \in \mathbb{C}^D \times \mathbb{C}^D.$$

Here $(l|$ and $|r)$ are the boundary vectors on both sides of [16, line 1174], written with the bilinear pairing.

Proof. [✓ Proof](#) The rank-one boundary matrix $|r)(l|$ gives $\Gamma_L(|r)(l|)(\sigma) = \text{tr}(A^{\sigma}|r)(l|) = (l|A^{\sigma}|r)$. Every boundary matrix is a linear combination of matrix units, and each matrix unit is rank one. $\square$

Theorem 12.3.7.19 (Dimension of the open-chain ground space). [Lean](#) [Lean](#) [✓ Stmt](#) If A is L -block-injective, then $\dim G_L^{\text{ES}}(A) = D^2$. If A is L_0 -block-injective with $L_0 > 0$, $D \geq 1$ and $N \geq L_0 + 1$, then

$$\dim \ker H_{N,L_0+1}^{\text{open}}(A) = D^2.$$

The local support spaces of a pure generalized valence-bond-solid state with auxiliary algebra $M_k(\mathbb{C})$ have dimension k^2 on long intervals, and they are the kernels of the open-interval Hamiltonian [191, lines 1505–1510 and 1545–1558]; the growth argument on an open chain is [193, lines 1211–1233]. The review states that the open-boundary degeneracy is at least the square of the dimension of the irreducible projective representation carried by the boundary [16, line 1174]. That dimension is at most D , so the exact count D^2 is sharper than the review’s lower bound.

Proof. ✓ Proof By Lemma 13.2.5, $\dim G_L(A) = D^2$, and the canonical ℓ^2 identification is a linear isomorphism. Block injectivity at length L_0 persists to every length $N \geq L_0$, and Theorem 12.3.7.17 identifies the kernel with $G_N^{\text{ES}}(A)$. $\square$

Definition 12.3.7.20 (Block-injective MPS realization of the open-chain martingale difference).

Lean ✓ Stmt Assume $D \geq 1$, $l > 0$, and that A is l -block-injective. Fix $K > 0$, set $n = K + l$ and $n_l = l + 1$, and work in the Hilbert space on Λ_{n+1} . The physical open-chain martingale difference is

$$E_n := G_{\Lambda_n} - G_{\Lambda_{n+1}} \quad (12.196)$$

$$= P_{\mathcal{U}_{K+l}(A)} - P_{\mathcal{G}_{K+l+1}^{\text{ES}}(A)}. \quad (12.197)$$

The first projector imposes the ground condition on the left n sites while acting on all $n + 1$ sites. Block injectivity identifies these MPS spaces with the corresponding open-parent Hamiltonian kernels. The condition $K > 0$ excludes the undersized volume, where an interaction of length $l + 1$ has no nonwrapping window and its physical ground projector is the identity.

Lemma 12.3.7.21 (Whole ground space lies in the C3 tail space). Lean ✓ Stmt For every $K, l \in \mathbb{N}$, set $n = K + l$. Then

$$\mathcal{G}_{n+1}^{\text{ES}}(A) \subseteq \mathcal{V}_{K,l+1}(A). \quad (12.198)$$

Proof. ✓ Proof Every fixed-prefix suffix restriction of a whole-chain ground-space vector is a ground-space vector on the suffix. This is exactly membership in $\mathcal{V}_{K,l+1}(A)$. $\square$

Lemma 12.3.7.22 (Reverse C3 projector composition). Lean ✓ Stmt For every $K, l \in \mathbb{N}$, the tail projector fixes the whole-ground projector:

$$G_{\Lambda_{n+1} \setminus \Lambda_{n-l}} G_{\Lambda_{n+1}} = G_{\Lambda_{n+1}}. \quad (12.199)$$

Proof. ✓ Proof The inner projection has range in $\mathcal{G}_{n+1}^{\text{ES}}(A)$, which lies in the tail ground space by Lemma 12.3.7.21. Orthogonal projection onto that tail space therefore leaves every vector in the inner range unchanged. $\square$

Theorem 12.3.7.23 (Block-injective MPS instance of the exact C3 identity). Lean ✓ Stmt Assume $D \geq 1$, $l > 0$, and that A is l -block-injective. Fix $K > 0$. With $n = K + l$ and $n_l = l + 1$, the MPS parent chain satisfies Nachtergaele’s identity following condition C3:

$$G_{\Lambda_{n+1} \setminus \Lambda_{n-l}} E_n = G_{\Lambda_{n+1} \setminus \Lambda_{n-l}} G_{\Lambda_n} - G_{\Lambda_{n+1}}. \quad (12.200)$$

Proof. ✓ Proof Distribute the tail projector over $E_n = G_{\Lambda_n} - G_{\Lambda_{n+1}}$. The second product is $G_{\Lambda_{n+1}}$ by Lemma 12.3.7.22, which gives the displayed identity. $\square$

Theorem 12.3.7.24 (Nachtergaele open-chain projector defect). [Lean](#) [✓ Stmt](#) Assume $D \geq 1$ and let A be L_0 -block-injective with $L_0 > 0$. Set $n = K + L_0$ and $l = L_0$, and write

$$S = \mathcal{U}_{K+L_0}(A), \quad T = \mathcal{V}_{K,L_0+1}(A).$$

Thus S and T are the left and tail ground spaces in the Hilbert space on Λ_{n+1} . Let $q_S = P_{S^\perp}$ and $q_T = P_{T^\perp}$ be the complementary excitation projections. If

$$\|G_{\Lambda_{n+1} \setminus \Lambda_{n-l}} G_{\Lambda_n} - G_{\Lambda_{n+1}}\| \leq \eta,$$

then every vector v in the Hilbert space on Λ_{n+1} satisfies

$$-\eta(\operatorname{Re}\langle q_T v, v \rangle + \operatorname{Re}\langle q_S v, v \rangle) \leq \operatorname{Re}\langle q_T v, q_S v \rangle + \operatorname{Re}\langle q_S v, q_T v \rangle.$$

Proof. [✓ Proof](#) The open-chain intersection theorem identifies $S \cap T$ with the ground space on Λ_{n+1} . The projector-defect adapter gives a Friedrichs overlap bound, and the reduced two-projection theorem gives the stated anticommutator inequality. $\square$

Theorem 12.3.7.25 (Uniform tail virtual-map bound). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a primitive MPS tensor with nonzero virtual dimension, and let $J_K = J_K^{\text{tail}}$ be its length- K tail virtual map. Then there is a constant $C > 0$ such that, for every $K \in \mathbb{N}$,

$$\|J_K\| \leq C. \quad (12.201)$$

No positive-definiteness hypothesis on the fixed point is required.

Proof. [✓ Proof](#) The operator norm is controlled using the adjoint identity

$$J_K^\dagger J_K U(X) = U(X \mathcal{E}_A^K(\mathbf{1})). \quad (12.202)$$

The Hilbert–Schmidt product estimate then gives the fourth-power bound

$$\|J_K\|^4 \leq \sum_{i,j} |\mathcal{E}_A^K(\mathbf{1})_{ij}|^2. \quad (12.203)$$

For $K \geq 1$, decompose the transfer power into its fixed-point projection and the K -th power of the complementary map. The complementary powers are geometrically bounded by Theorem 11.4.4, while the entrywise mass is controlled by the L^2 operator norm with the virtual dimension factor from Theorem 11.3.27. The case $K = 0$ is bounded directly. This first gives a uniform bound for $\|J_K\|^4$ and hence a strictly positive uniform bound for $\|J_K\|$. No FNW or Nachtergaele contraction estimate enters. $\square$

Theorem 12.3.7.26 (Left-canonical normalization of fixed-length words). [Lean](#) [✓ Stmt](#) If A is left-canonical, then for every $Q \in \mathbb{N}$,

$$\sum_{j \in \Omega_Q} (A^j)^\dagger A^j = \mathbf{1}, \quad (12.204)$$

where $\Omega_Q = \{0, \dots, (-1)d^Q\}$ is the set of physical words of length Q .

Proof. [✓ Proof](#) Induct on Q . Split a word of length $Q + 1$ into its first letter and its remaining suffix, apply the induction hypothesis to the suffix sum, and then use left-canonicity for the outer one-site sum. $\square$

Theorem 12.3.7.27 (Whole-increment left virtual norm). [Lean](#) [Lean](#) [✓ Stmt](#) If A is left-canonical, then the length- Q left virtual map J_Q^{left} preserves the Hilbert–Schmidt norm. In particular,

$$\|J_Q^{\text{left}} X\| = \|X\|, \quad (12.205)$$

$$\|J_Q^{\text{left}}\| \leq 1. \quad (12.206)$$

Proof. [✓ Proof](#) Expand the squared norm as a sum over $j \in \Omega_Q$. The identity from Theorem 12.3.7.26 reduces the sum to the Hilbert–Schmidt norm of X . The operator-norm bound follows immediately. $\square$

Definition 12.3.7.28 (Physical reassociation). [Lean](#) [✓ Stmt](#) For $K, L, Q \in \mathbb{N}$, let

$$\mathcal{A}_{K,L,Q} : \ell^2(\Omega_{K+(L+Q)}) \longrightarrow \ell^2(\Omega_{(K+L)+Q}) \quad (12.207)$$

be the unitary coordinate map induced by associativity of addition.

Theorem 12.3.7.29 (Reassociation of the ground-space boundary map). [Lean](#) [✓ Stmt](#) For every $K, L, Q \in \mathbb{N}$,

$$\mathcal{A}_{K,L,Q} \Gamma_{K+(L+Q)}^{\text{ES}} = \Gamma_{K+L+Q}^{\text{ES}}. \quad (12.208)$$

Proof. [✓ Proof](#) Evaluate both sides at a physical word. Reassociation changes only the parenthesization of its coordinate set, so both coefficients are the trace pairing with the same word product. $\square$

Definition 12.3.7.30 (Reassociated tail boundary map). [Lean](#) [✓ Stmt](#) Define the tail boundary map in the common $(K + L + Q)$ -site coordinates by

$$\tilde{\Gamma}_{K;L,Q}^{\text{tail}} := \mathcal{A}_{K,L,Q} \Gamma_{K,L+Q}^{\text{tail}}. \quad (12.209)$$

Theorem 12.3.7.31 (One-site reassociation). [Lean](#) [✓ Stmt](#) A one-site suffix increment needs no reassociation:

$$\tilde{\Gamma}_{K;L,1}^{\text{tail}} = \Gamma_{K,L+1}^{\text{tail}}. \quad (12.210)$$

Proof. [✓ Proof](#) Reassociating a window whose last block carries a single site is the identity, so the reassociated tail map is the ordinary tail map with $L + 1$ spectator sites. $\square$

Theorem 12.3.7.32 (Reassociated tail boundary map on three intervals). [Lean](#) [✓ Stmt](#) Let $K, L, Q \in \mathbb{N}$ and let $Y : (\{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}) \rightarrow M_D(\mathbb{C})$ be a boundary family. For all $u : \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$, $w : \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$, and $v : \{0, \dots, Q-1\} \rightarrow \{0, \dots, d-1\}$,

$$\tilde{\Gamma}_{K;L,Q}^{\text{tail}}(Y)((u, w), v) = \Gamma_{L+Q}(Y_u)(w, v). \quad (12.211)$$

Proof. [✓ Proof](#) The reassociation map only regroups the physical coordinates: for every vector f on $K + (L + Q)$ sites,

$$(\mathcal{A}_{K,L,Q} f)((u, w), v) = f(u, (w, v)). \quad (12.212)$$

Applying this to $f = \Gamma_{K,L+Q}^{\text{tail}}(Y)$ and then Theorem 12.3.5.5 with suffix (w, v) of length $L + Q$ gives

$$\tilde{\Gamma}_{K;L,Q}^{\text{tail}}(Y)((u, w), v) = \Gamma_{K,L+Q}^{\text{tail}}(Y)(u, (w, v)) = \Gamma_{L+Q}(Y_u)(w, v). \quad (12.213)$$

$\square$

Theorem 12.3.7.33 (Factorization of the reassociated tail map). [Lean](#) [✓ Stmt](#) *The reassociated tail map satisfies*

$$\tilde{\Gamma}_{K;L,Q}^{\text{tail}} J_K^{\text{tail}} = \Gamma_{K+L+Q}^{\text{ES}}. \quad (12.214)$$

Proof. [✓ Proof](#) Substitute the definition of the reassociated tail map, use the ordinary tail factorization, and then reassociate the full boundary map. $\square$

Theorem 12.3.7.34 (Whole-increment inverse-Gram projector residual). [Lean](#) [✓ Stmt](#) *Let $T = \tilde{\Gamma}_{K;L,Q}^{\text{tail}}$, $S = \Gamma_{K+L,Q}^{\text{left}}$, and $C = \Gamma_{K+L+Q}^{\text{ES}}$. Assume separately that T , S , and C are injective. Since $TJ_K^{\text{tail}} = C = SJ_Q^{\text{left}}$, the difference $P_T P_S - P_C$ equals the exact inverse-Gram residual obtained from these two factorizations.*

Proof. [✓ Proof](#) Apply the common-factorization range-projector identity to T , S , and C , using the two displayed factorizations. No canonicity or norm estimate is used. $\square$

Theorem 12.3.7.35 (Whole-increment mixed Gram after reassociation). [Lean](#) [✓ Stmt](#) *With the notation of Theorem 12.3.5.22,*

$$\langle X, (\tilde{\Gamma}_{K;L,Q}^{\text{tail}})^\dagger \Gamma_{K+L,Q}^{\text{left}} Z \rangle = \sum_{u \in \Omega_K} \sum_{j \in \Omega_Q} \langle U(A^j X_u), \mathcal{K}_L U(Z_j A^u) \rangle. \quad (12.215)$$

Proof. [✓ Proof](#) Expand the reassociated tail map. Its adjoint introduces $\mathcal{A}_{K,L,Q}^{-1}$, so the coordinate formula of Theorem 12.3.5.22 applies. $\square$

Theorem 12.3.7.36 (Centered whole-increment mixed Gram). [Lean](#) [✓ Stmt](#) *Let $\rho \in M_D(\mathbb{C})$ have nonzero trace and set $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$. Then*

$$\langle X, (\tilde{\Gamma}_{K;L,Q}^{\text{tail}})^\dagger \Gamma_{K+L,Q}^{\text{left}} Z \rangle - \sum_{u,j} \langle U(A^j X_u), \mathcal{K}_\infty U(Z_j A^u) \rangle \quad (12.216)$$

$$= \sum_{u,j} \langle U(A^j X_u), (\mathcal{K}_L - \mathcal{K}_\infty) U(Z_j A^u) \rangle. \quad (12.217)$$

Proof. [✓ Proof](#) Substitute the reassociated mixed-Gram formula and the action of the limiting metric, then distribute subtraction over the finite sums. $\square$

Definition 12.3.7.37 (Whole-increment left overlap factor). [Lean](#) [✓ Stmt](#) *For $X = (X_u)_{u \in \Omega_K}$, define*

$$(F_{K,Q} X)_{u,j} := A^j X_u. \quad (12.218)$$

Definition 12.3.7.38 (Whole-increment right overlap factor). [Lean](#) [✓ Stmt](#) *For $Z = (Z_j)_{j \in \Omega_Q}$, define*

$$(G_{K,Q} Z)_{u,j} := Z_j A^u. \quad (12.219)$$

Theorem 12.3.7.39 (Fiber formula for the whole-increment left factor). [Lean](#) [✓ Stmt](#) *The matrix fiber of $F_{K,Q} X$ at (u, j) is $A^j X_u$.*

Proof. [✓ Proof](#) Unfold the direct-sum factor and the Frobenius coordinate identification. $\square$

Theorem 12.3.7.40 (Fiber formula for the whole-increment right factor). [Lean](#) [✓ Stmt](#) *The matrix fiber of $G_{K,Q} Z$ at (u, j) is $Z_j A^u$.*

Proof. ✓ Proof Unfold the direct-sum factor and the Frobenius coordinate identification. □

Theorem 12.3.7.41 (Norm bounds for the whole-increment overlap factors). Lean Lean ✓ Stmt
The overlap factors satisfy

$$\|F_{K,Q}\| \leq \|J_Q^{\text{left}}\|, \quad (12.220)$$

$$\|G_{K,Q}\| \leq \|J_K^{\text{tail}}\|. \quad (12.221)$$

Proof. ✓ Proof Expand the squared direct-sum norms fiberwise and apply the operator-norm bound to each fiber. Summing gives the two inequalities without a factor depending on the number of spectator words. □

Definition 12.3.7.42 (Whole-increment centered residual). Lean ✓ Stmt Assume $\text{tr}(\rho) \neq 0$ and let $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$. Define

$$R_{K,L,Q} := (\tilde{\Gamma}_{K;L,Q}^{\text{tail}})^\dagger \Gamma_{K+L,Q}^{\text{left}} - F_{K,Q}^\dagger \mathcal{K}_\infty^{\oplus(\Omega_K \times \Omega_Q)} G_{K,Q}. \quad (12.222)$$

Theorem 12.3.7.43 (Whole-increment centered factorization). Lean ✓ Stmt *The centered residual factors as*

$$R_{K,L,Q} = F_{K,Q}^\dagger (\mathcal{K}_L - \mathcal{K}_\infty)^{\oplus(\Omega_K \times \Omega_Q)} G_{K,Q}. \quad (12.223)$$

Proof. ✓ Proof Test both sides against arbitrary boundary families, use the two fiber formulas, and apply the centered mixed-Gram identity. □

Theorem 12.3.7.44 (Whole-increment centered residual bound). Lean ✓ Stmt *The factorization gives*

$$\|R_{K,L,Q}\| \leq \|J_Q^{\text{left}}\| \|\mathcal{K}_L - \mathcal{K}_\infty\| \|J_K^{\text{tail}}\|. \quad (12.224)$$

Proof. ✓ Proof Apply submultiplicativity to the three factors and use the operator norm of a fiberwise direct sum. □

Theorem 12.3.7.45 (Left-canonical whole-increment centered residual bound). Lean ✓ Stmt *If A is left-canonical, then*

$$\|R_{K,L,Q}\| \leq \|\mathcal{K}_L - \mathcal{K}_\infty\| \|J_K^{\text{tail}}\|. \quad (12.225)$$

Proof. ✓ Proof Substitute $\|J_Q^{\text{left}}\| \leq 1$ from Theorem 12.3.7.27 into Theorem 12.3.7.44. □

Theorem 12.3.7.46 (Primitive whole-increment centered residual bound). Lean ✓ Stmt *Let A be a primitive MPS tensor with positive-definite fixed point ρ . Then A satisfies the left-canonical centered residual estimate of Theorem 12.3.7.45.*

Proof. ✓ Proof Use the left-canonical normalization contained in the primitive-MPS hypotheses together with positive definiteness of ρ . □

Definition 12.3.7.47 (Common overlap factors and centered residual). Lean Lean Lean ✓ Stmt
Let A be an MPS tensor, let Ω_K be the set of length- K words in the physical alphabet, and let $\mathcal{B}_K = \bigoplus_{u \in \Omega_K} M_D(\mathbb{C})$ with its Hilbert–Schmidt direct-sum norm. Define

$$F_K : \mathcal{B}_K \longrightarrow \bigoplus_{(u,j) \in \Omega_K \times \Omega_1} M_D(\mathbb{C}) \quad (12.226)$$

by applying the one-site left virtual map independently to each prefix fiber. Define

$$G_K : \mathcal{B}_1 \longrightarrow \bigoplus_{(u,j) \in \Omega_K \times \Omega_1} M_D(\mathbb{C}) \quad (12.227)$$

by applying the length- K tail virtual map independently to each final-site fiber. These definitions make sense without any nonemptiness assumption on the virtual dimension or on either configuration set.

When $D \geq 1$ and $\rho \in M_D(\mathbb{C})$ is positive definite, set $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$ and define the centered virtual residual by

$$R_{K,l} = (\Gamma_{K,l+1}^{\text{tail}})^\dagger \Gamma_{K+l,1}^{\text{left}} - \mathcal{K}_\infty^{\oplus \Omega_K} J_K^{\text{tail}} \mathcal{K}_\infty^{-1} (J_1^{\text{left}})^\dagger \mathcal{K}_\infty^{\oplus \Omega_1}. \quad (12.228)$$

Theorem 12.3.7.48 (Centered overlap factorization and geometric bound). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For every $K \in \mathbb{N}$, the common overlap factors have fiber formulas

$$(F_K X)_{u,j} = A^j X_u, \quad (G_K Z)_{u,j} = Z_j A^u, \quad (12.229)$$

and norm bounds

$$\|F_K\| \leq \|J_1^{\text{left}}\|, \quad \|G_K\| \leq \|J_K^{\text{tail}}\|. \quad (12.230)$$

The formulas and bounds remain valid for empty direct sums and incur no factor depending on the number of configurations.

Assume $D \geq 1$ and that $\rho \in M_D(\mathbb{C})$ is positive definite. The centered residual of Definition 12.3.7.47 factors exactly at the generally nonidentity limiting metric as

$$R_{K,l} = F_K^\dagger (\mathcal{K}_l - \mathcal{K}_\infty)^{\oplus (\Omega_K \times \Omega_1)} G_K, \quad (12.231)$$

and hence

$$\|R_{K,l}\| \leq \|J_1^{\text{left}}\| \|\mathcal{K}_l - \mathcal{K}_\infty\| \|J_K^{\text{tail}}\|. \quad (12.232)$$

If, in addition, A is primitive with fixed-point witness ρ , then there are constants $C > 0$ and $0 < r < 1$ such that, for every $K \in \mathbb{N}$ and every $l \geq 1$,

$$\|R_{K,l}\|^2 \leq D^3 (Cr^l)^2. \quad (12.233)$$

All formulas in this theorem are reconstructed algebra. They are not sourced formulas and assert neither a full projector-defect estimate nor the FNW–Nachtergaele rational coefficient.

Proof. [✓ Proof](#) Evaluate the two factors on each common direct-sum fiber. The direct-sum norm identity gives the two bounds without a factor counting spectator configurations. Substitute these fiber formulas into the centered residual identity from Theorem 12.3.5.25 to obtain the exact factorization. Submultiplicativity and Theorem 12.3.5.20 give the product bound. Under primitivity, combine the uniform bound for J_K^{tail} with the squared geometric Gram estimate and absorb the fixed norm of J_1^{left} into the constant. $\square$

Theorem 12.3.7.49 (Fiberwise subtraction). [Lean](#) [✓ Stmnt](#) Let S be a finite set and let $G, H : V \rightarrow V$ be continuous linear endomorphisms of the virtual Euclidean space. Then

$$(G - H)^{\oplus S} = G^{\oplus S} - H^{\oplus S}. \quad (12.234)$$

Proof. [✓ Proof](#) Evaluate both sides on each fiber indexed by $s \in S$. $\square$

Theorem 12.3.7.50 (Invertibility of the finite ground-space Gram). Lean ✓ Stmt *If the ground-space map Γ_n^{ES} is injective, then its Gram endomorphism $\mathcal{K}_n = (\Gamma_n^{\text{ES}})^\dagger \Gamma_n^{\text{ES}}$ is invertible.*

Proof. ✓ Proof The equality $\langle X, \mathcal{K}_n X \rangle = \|\Gamma_n^{\text{ES}} X\|^2$ shows that $\mathcal{K}_n$ is injective. An injective endomorphism of a finite-dimensional space is bijective and hence invertible. $\square$

Theorem 12.3.7.51 (Finite-to-limiting inverse resolvent identity). Lean ✓ Stmt *Assume $D \geq 1$ and let $\rho \in M_D(\mathbb{C})$ be positive definite. Set $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$ and $S_\infty = \mathcal{K}_\infty^{-1}$. If Γ_n^{ES} is injective and $S_n = \mathcal{K}_n^{-1}$, then*

$$S_n - S_\infty = S_\infty (\mathcal{K}_\infty - \mathcal{K}_n) S_n. \quad (12.235)$$

Proof. ✓ Proof Both Gram endomorphisms are invertible. Apply the identity $A^{-1} - B^{-1} = B^{-1}(B - A)A^{-1}$ with $A = \mathcal{K}_n$ and $B = \mathcal{K}_\infty$. $\square$

Theorem 12.3.7.52 (Left finite-Gram cancellation). Lean ✓ Stmt *Suppose that Γ_K^{ES} is injective and write $S_K = \mathcal{K}_K^{-1}$. Then*

$$(J_L^{\text{left}})^\dagger \mathcal{K}_K^{\oplus \Omega_L} S_K^{\oplus \Omega_L} (\Gamma_{K,L}^{\text{left}})^\dagger = (\Gamma_{K+L}^{\text{ES}})^\dagger. \quad (12.236)$$

Proof. ✓ Proof The direct-sum Gram and inverse Gram cancel fiberwise. The remaining composition is the adjoint of $\Gamma_{K,L}^{\text{left}} J_L^{\text{left}} = \Gamma_{K+L}^{\text{ES}}$. $\square$

Theorem 12.3.7.53 (Tail spectator pseudoinverse bound). Lean ✓ Stmt *If Γ_L^{ES} is injective, then*

$$\|\Gamma_{K,L}^{\text{tail}} S_{\Gamma_{K,L}^{\text{tail}}}\| \leq \sqrt{\|S_L\|}. \quad (12.237)$$

Proof. ✓ Proof The square-root pseudoinverse identity gives the left-hand side as the square root of the norm of the tail boundary inverse Gram. This inverse Gram is the fiberwise lift of S_L , whose norm is at most $\|S_L\|$. $\square$

Theorem 12.3.7.54 (Left spectator pseudoinverse bound). Lean ✓ Stmt *If Γ_K^{ES} is injective, then*

$$\|S_{\Gamma_{K,L}^{\text{left}}} (\Gamma_{K,L}^{\text{left}})^\dagger\| \leq \sqrt{\|S_K\|}. \quad (12.238)$$

Proof. ✓ Proof The adjoint-side square-root identity reduces the left-hand side to the norm of the left boundary inverse Gram. The direct-sum inverse-Gram formula and the fiberwise norm estimate give the result. $\square$

Definition 12.3.7.55 (Finite-Gram corrections in the C3 residual). Lean Lean Lean ✓ Stmt *Assume $D \geq 1$ and let $\rho \in M_D(\mathbb{C})$ be positive definite. Set $\mathcal{K}_\infty = \mathcal{R}(P_\rho)$ and $S_\infty = \mathcal{K}_\infty^{-1}$. Suppose that Γ_{l+1}^{ES} , Γ_{K+l}^{ES} , and $\Gamma_{K+l+1}^{\text{ES}}$ are injective, and write*

$$T = \Gamma_{K,l+1}^{\text{tail}}, \quad (12.239)$$

$$L = \Gamma_{K+l,1}^{\text{left}}, \quad (12.240)$$

$$C = \Gamma_{K+l+1}^{\text{ES}}. \quad (12.241)$$

The three correction sandwiches in Theorem 12.3.5.25 are

$$Q_1 = T S_{l+1}^{\oplus \Omega_K} (\mathcal{K}_{l+1} - \mathcal{K}_\infty)^{\oplus \Omega_K} J_K^{\text{tail}} S_{K+l+1} (J_1^{\text{left}})^\dagger \mathcal{K}_{K+l}^{\oplus \Omega_1} S_{K+l}^{\oplus \Omega_1} L^\dagger, \quad (12.242)$$

$$Q_2 = T S_{l+1}^{\oplus \Omega_K} \mathcal{K}_\infty^{\oplus \Omega_K} J_K^{\text{tail}} (S_{K+l+1} - S_\infty) (J_1^{\text{left}})^\dagger \mathcal{K}_{K+l}^{\oplus \Omega_1} S_{K+l}^{\oplus \Omega_1} L^\dagger, \quad (12.243)$$

$$Q_3 = T S_{l+1}^{\oplus \Omega_K} \mathcal{K}_\infty^{\oplus \Omega_K} J_K^{\text{tail}} S_\infty (J_1^{\text{left}})^\dagger (\mathcal{K}_{K+l} - \mathcal{K}_\infty)^{\oplus \Omega_1} S_{K+l}^{\oplus \Omega_1} L^\dagger. \quad (12.244)$$

Theorem 12.3.7.56 (Exact factorizations of the C3 corrections). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#)
Under the hypotheses of Definition 12.3.7.55, the three corrections have the exact factorizations

$$Q_1 = TS_{l+1}^{\oplus \Omega_K} (\mathcal{K}_{l+1} - \mathcal{K}_\infty)^{\oplus \Omega_K} J_K^{\text{tail}} S_{K+l+1} C^\dagger, \quad (12.245)$$

$$Q_2 = TS_{l+1}^{\oplus \Omega_K} \mathcal{K}_\infty^{\oplus \Omega_K} J_K^{\text{tail}} S_\infty (\mathcal{K}_\infty - \mathcal{K}_{K+l+1}) S_{K+l+1} C^\dagger, \quad (12.246)$$

$$Q_3 = TS_{l+1}^{\oplus \Omega_K} \mathcal{K}_\infty^{\oplus \Omega_K} J_K^{\text{tail}} S_\infty (J_1^{\text{left}})^\dagger (\mathcal{K}_{K+l} - \mathcal{K}_\infty)^{\oplus \Omega_1} S_{\Gamma_{K+l,1}^{\text{left}}} L^\dagger. \quad (12.247)$$

Proof. [Stmnt](#) The direct-sum inverse-Gram formula identifies the initial factor with the pseudoinverse factor of T . Theorem 12.3.7.52 replaces the final product in Q_1 and Q_2 by $C^\dagger$. For Q_2 , apply the resolvent identity

$$S_{K+l+1} - S_\infty = S_\infty (\mathcal{K}_\infty - \mathcal{K}_{K+l+1}) S_{K+l+1}. \quad (12.248)$$

Fiberwise subtraction gives the displayed Gram differences. The factorization of Q_3 leaves the left pseudoinverse adjacent to $L^\dagger$. $\square$

Theorem 12.3.7.57 (Raw norm bounds for the C3 corrections). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#)
Under the hypotheses of Definition 12.3.7.55,

$$\|Q_1\| \leq \sqrt{\|S_{l+1}\|} \|\mathcal{K}_{l+1} - \mathcal{K}_\infty\| \|J_K^{\text{tail}}\| \sqrt{\|S_{K+l+1}\|}, \quad (12.249)$$

$$\|Q_2\| \leq \sqrt{\|S_{l+1}\|} \|\mathcal{K}_\infty\| \|J_K^{\text{tail}}\| \|S_\infty\| \|\mathcal{K}_{K+l+1} - \mathcal{K}_\infty\| \sqrt{\|S_{K+l+1}\|}, \quad (12.250)$$

$$\|Q_3\| \leq \sqrt{\|S_{l+1}\|} \|\mathcal{K}_\infty\| \|J_K^{\text{tail}}\| \|S_\infty\| \|J_1^{\text{left}}\| \|\mathcal{K}_{K+l} - \mathcal{K}_\infty\| \sqrt{\|S_{K+l}\|}. \quad (12.251)$$

Proof. [Stmnt](#) Apply submultiplicativity to the factorizations in Theorem 12.3.7.56. The two spectator pseudoinverse estimates control the exterior factors, and Theorem 12.3.5.20 controls every fiberwise Gram factor without a term depending on $|\Omega_K|$. The square-root pseudoinverse identity controls $S_{K+l+1} C^\dagger$. $\square$

Theorem 12.3.7.58 (Uniform decay of the individual-block projector error). [Lean](#) [Stmnt](#) Let A be a normalized primitive tensor with nonzero bond dimension and a positive-definite fixed point. For every $\varepsilon > 0$, all sufficiently large L and every $K \geq 0$, $Q > 0$ satisfy

$$\|P_{\mathcal{H}_K \otimes G_{L+Q}(A)} P_{G_{K+L}(A) \otimes \mathcal{H}_Q} - P_{G_{K+L+Q}(A)}\| \leq \varepsilon \quad (12.252)$$

This is the uniform decay of the individual-block errors used in [191, Section 6].

Proof. [Stmnt](#) The geometric FNW estimate bounds the norm by $c\lambda^L(1 + c\lambda^L)/(1 - c\lambda^L)$ once L exceeds the injectivity length and $c\lambda^L < 1$. Since $0 < \lambda < 1$, this coefficient tends to zero. Neither the coefficient nor the lower bound on L depends on K or Q . $\square$

Theorem 12.3.7.59 (Uniform whole-increment defect at the coefficient 7/16). [Lean](#) [Stmnt](#) Let A be primitive with positive-definite fixed point p and nonzero bond dimension. There is an integer $p > 0$, measured in sites of A , such that A is p -block injective and, for every prefix length K ,

$$\|P_{K;p,p}^{\text{tail}} P_{K+p,p}^{\text{left}} - P_{\mathcal{G}_{K+2p}^{\text{ES}}(A)}\| \leq \frac{7}{16}. \quad (12.253)$$

Proof. [Stmnt](#) Primitivity supplies a positive injective length. Choose p above it and large enough for the uniform decay estimate with tolerance 7/16. Block injectivity propagates to p . Set the suffix length equal to p in that estimate, leaving the prefix length unrestricted. $\square$

Theorem 12.3.7.60 (Three-block whole-increment defect at the coefficient 7/16). [Lean](#) [Stmnt](#) Under the same hypotheses, there is an integer $p > 0$, measured in sites of A , such that A is p -block injective and

$$\left\| P_{p;p,p}^{\text{tail}} P_{2p,p}^{\text{left}} - P_{g_{3p}^{\text{ES}}(A)} \right\| \leq \frac{7}{16}. \quad (12.254)$$

Proof. [Stmnt](#) Apply the prefix-uniform theorem at $K = p$. The three consecutive regions then each contain exactly p sites of the input tensor. $\square$

Theorem 12.3.7.61 (Uniform open-chain projector-defect threshold). [Lean](#) [Lean](#) [Stmnt](#) Let A be a primitive MPS tensor with nonzero bond dimension and a positive-definite fixed point. There are an overlap length $l > 1$, measured in sites of A , and a number $\epsilon_l \geq 0$ such that A is l -block injective, the projector defect is bounded uniformly in the left length K by ϵ_l , and

$$\epsilon_l < \frac{1}{\sqrt{l+1}}.$$

Nachtergaele's projector-defect identity following condition C3 [191, Eq. (2.4)], together with the two-projection estimate, gives the corresponding anticommutator bound for every K . If A is already a blocked representative, one site of A is one blocked-chain filtration step.

Proof. [Stmnt](#) Apply Theorem 12.9.5 with $a = 1$ and choose one of its sufficiently large lengths l . Since $\sqrt{l+1} > 0$, the inequality $\epsilon_l \sqrt{l+1} < 1$ gives $\epsilon_l < 1/\sqrt{l+1}$. The open-chain projector-defect identity gives the anticommutator statement. $\square$

Theorem 12.3.7.62 (Literal open-chain martingale condition C3). [Lean](#) [Stmnt](#) Let A be a primitive MPS tensor with nonzero bond dimension and a positive-definite fixed point. There are $l > 1$ and $\epsilon_l \geq 0$, with A l -block injective, such that for every $K > 0$, after setting $n = K + l$ and $n_l = l + 1$,

$$\left\| G_{\Lambda_{n+1} \setminus \Lambda_{n-l}} E_n \right\| \leq \epsilon_l. \quad (12.255)$$

The coefficient satisfies

$$\epsilon_l < \frac{1}{\sqrt{l+1}}. \quad (12.256)$$

The length l and coefficient ϵ_l are exactly those chosen by Theorem 12.3.7.61.

Proof. [Stmnt](#) The uniform threshold theorem gives

$$\left\| G_{\Lambda_{n+1} \setminus \Lambda_{n-l}} G_{\Lambda_n} - G_{\Lambda_{n+1}} \right\| \leq \epsilon_l \quad (12.257)$$

for every $K > 0$. Rewrite its norm by Theorem 12.3.7.23; the existing strict threshold on ϵ_l is unchanged. $\square$

Theorem 12.3.7.63 (Literal fixed-final-volume martingale condition C3). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let A be a primitive MPS tensor with nonzero bond dimension and a positive-definite fixed point. There are $l > 1$ and $\epsilon_l \geq 0$, with A l -block injective, such that for every final volume N and every $n < N$,

$$\left\| G_{\Lambda_{n+1} \setminus \Lambda_{n-l}}^{(N)} E_n^{(N)} \right\| \leq \epsilon_l, \quad (12.258)$$

$$\epsilon_l < \frac{1}{\sqrt{l+1}}. \quad (12.259)$$

This is the same overlap length and the same coefficient as in Theorem 12.3.7.62. More generally, for any tensor of positive bond dimension that is l -block injective with $l > 1$, a uniform physical open-chain bound $\epsilon \geq 0$ gives the same bound in every final volume, without requiring $\epsilon < 1/\sqrt{l+1}$.

Proof. ✓ Proof If $n < l$, then $E_n^{(N)} = 0$. If $n = l$, the endpoint identity gives $G_{\Lambda_{l+1}}^{(N)} E_l^{(N)} = 0$. These cases use only $\epsilon_l \geq 0$.

Suppose that $n > l$, and set $K = n - l > 0$ and $r = N - (n + 1)$. The prefix and whole-prefix conjugacies identify $E_n^{(N)}$ with the direct sum of the $(n + 1)$ -site martingale difference over the r right spectators. The interval-projector conjugacy and fiberwise composition then identify the product in (12.258) with the direct sum of its open-chain counterpart. Isometric conjugacy preserves operator norm, and the fiberwise extension does not increase it. The bound therefore follows from Theorem 12.3.7.62. $\square$

Theorem 12.3.7.64 (Compatible open-chain gap from fixed-volume overlap). Lean ✓ Stmt Let $D \geq 1$, let $l > 0$, suppose that A is l -block injective, and let $N \geq l + 1$. Assume $0 \leq \epsilon_l < 1/\sqrt{l+1}$ and, for every $0 \leq n < N$,

$$\|Q_n E_n^{(N)}\| \leq \epsilon_l, \quad (12.260)$$

where Q_n is the ground-space projection of the interval $[n - l, n + 1]$ and $E_n^{(N)}$ is the fixed-volume martingale difference. Then every $v \in (G_N^{\text{ES}}(A))^\perp$ satisfies

$$(1 - \epsilon_l \sqrt{l+1})^2 \|v\| \leq \|H_{N,l+1}^{\text{open}}(A)v\|. \quad (12.261)$$

This intermediate MPS specialization assumes C3 on the full range $0 \leq n < N$, rather than only above the lower threshold in the source statement. It applies the full-range estimate with $d_{l+1} = \gamma_{l+1} = 1$ and retains the coefficient of Theorem 2.1(i) in [191]. It is not a formulation of the unrestricted source theorem. Theorem 12.3.7.65 supplies the full-range C3 hypothesis from the physical open-chain estimates.

Proof. ✓ Proof Use the fixed-volume prefix ground-space projections and their differences $E_n^{(N)}$. Theorem 12.3.4.16 gives $G_0^{(N)} = I$, while Theorem 12.3.4.15 identifies $\text{ran } G_N^{(N)}$ with the kernel of the open Hamiltonian. The outside-window commutation theorem gives the locality hypothesis in the martingale estimate. Theorems 12.3.7.9 and 12.3.7.11 give C1 and C2 with $d_{l+1} = \gamma_{l+1} = 1$, while (12.260) is C3. Positivity of the open Hamiltonian and $\ker H_{N,l+1}^{\text{open}}(A) = G_N^{\text{ES}}(A)$ identify the final orthogonal complement. Substitution into (12.142) yields (12.261). $\square$

Theorem 12.3.7.65 (Uniform gap for primitive compatible open chains). Lean ✓ Stmt Let A be a primitive MPS tensor with nonzero bond dimension and a positive-definite fixed point. There exist $l > 1$ and $\epsilon_l \geq 0$ such that A is l -block injective,

$$\epsilon_l < \frac{1}{\sqrt{l+1}}, \quad (12.262)$$

and, for every $N \geq l + 1$ and every $v \in (G_N^{\text{ES}}(A))^\perp$,

$$(1 - \epsilon_l \sqrt{l+1})^2 \|v\| \leq \|H_{N,l+1}^{\text{open}}(A)v\|. \quad (12.263)$$

Proof. [✓ Proof](#) Theorem 12.3.7.63 supplies $l > 1$, ϵ_l , l -block injectivity, and the C3 estimate for every final volume N and every $n < N$. Apply Theorem 12.3.7.64 for each $N \geq l + 1$. The value of ϵ_l is unchanged, so the coefficient is exactly the specialization

$$\frac{\gamma_{l+1}}{d_{l+1}} (1 - \epsilon_l \sqrt{l+1})^2 = (1 - \epsilon_l \sqrt{l+1})^2. \quad (12.264)$$

□

Lemma 12.3.7.66 (Blocked three-site projector and adjacent-pair transport). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with nonzero physical dimension, let p be any blocking length, and set $B = \text{block}_p(A)$. Then

$$\|P_{1,2}^{\text{tail}}(B)P_2^{\text{left}}(B) - P_{\mathcal{G}_3^{\text{ES}}(B)}\| = \|P_{p;p,p}^{\text{tail}}P_{2p,p}^{\text{left}} - P_{\mathcal{G}_{3p}^{\text{ES}}(A)}\|. \quad (12.265)$$

If B is injective, then the bound on either side

$$\|P_{1,2}^{\text{tail}}(B)P_2^{\text{left}}(B) - P_{\mathcal{G}_3^{\text{ES}}(B)}\| \leq \frac{7}{16} \quad (12.266)$$

gives the adjacent three-site anticommutator inequality

$$h_0 h_1 + h_1 h_0 \geq -\frac{7}{16}(h_0 + h_1). \quad (12.267)$$

The same quadratic-form inequality holds for every forward or backward adjacent pair of cyclic range-two terms on every chain of length at least three, including pairs crossing the periodic cut.

Theorem 12.3.7.67 (Uniform gap for a blocked primitive MPS parent Hamiltonian). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a primitive MPS tensor with nonzero physical and bond dimensions, and suppose that its fixed point is positive definite. There is a blocking length $p > 0$ such that A is p -block injective and, with $B = \text{block}_p(A)$, for every $N \geq 4$ and every $v \in (\ker H_{N,2}(B))^\perp$,

$$\frac{1}{8}\|v\| \leq \|H_{N,2}(B)v\|. \quad (12.268)$$

Proof. [✓ Proof](#) Choose p so that the three-block projector defect is at most $7/16$. Lemma 12.3.7.66 gives the same coefficient for every overlapping cyclic pair. For range two each row has at most two such pairs, so the martingale estimate gives $1 - 2(7/16) = 1/8$. □

Theorem 12.3.7.68 (Anticommutator bound for two projections). [Lean](#) [✓ Stmt](#) Let P and Q be orthogonal projections on a real or complex Hilbert space. For every vector v ,

$$\text{Re}\langle Pv, Qv \rangle + \text{Re}\langle Qv, Pv \rangle \leq \text{Re}\langle Pv, v \rangle + \text{Re}\langle Qv, v \rangle. \quad (12.269)$$

Proof. [✓ Proof](#) Positivity of the inner product gives

$$\begin{aligned} 0 &\leq \text{Re}\langle (P - Q)v, (P - Q)v \rangle \\ &= \text{Re}\langle Pv, v \rangle - \text{Re}\langle Pv, Qv \rangle - \text{Re}\langle Qv, Pv \rangle + \text{Re}\langle Qv, v \rangle, \end{aligned} \quad (12.270)$$

where the equality expands the inner product using symmetry and idempotence $P^\dagger = P = P^2$, $Q^\dagger = Q = Q^2$. Rearranging (12.270) proves the claim. □

Proof. ✓ Proof For fixed offsets $a, b \in \{0, \dots, m-1\}$, translation by a gives

$$\sum_s h_{s+a} h_{s+b} = \sum_i h_i h_{i+b-a}. \quad (12.282)$$

The case $a = b$ contributes mH , while each of the differences $b-a = q$ and $b-a = -q$ contributes $(m-q)Q_q$. Summing (12.282) over a and b gives (12.279). Taking the quadratic form of that operator identity at v turns each product $h_i h_{i\pm q}$ into the pair $\text{Re}\langle h_i v, h_{i\pm q} v \rangle$, giving (12.280). Translation invariance also gives

$$\sum_s \text{Re}\langle A_s v, v \rangle = m \text{Re}\langle H v, v \rangle. \quad (12.283)$$

□

Theorem 12.3.7.72 (Cyclic cross-term estimates). Lean Lean Lean Lean ✓ Stmt For Q_q as in Theorem 12.3.7.71,

$$\text{Re}\langle Q_q v, v \rangle \leq 2 \text{Re}\langle H v, v \rangle. \quad (12.284)$$

Suppose that h_i and h_{i+d} commute whenever $R \leq d \leq N-R$. Then the cross term is nonnegative for these offsets. If $1 \leq m$, $R \leq m$, and $2m \leq N$, it follows that

$$\text{Re}\langle H v, v \rangle + \sum_{q=1}^{m-1} \text{Re}\langle Q_q v, v \rangle \leq \text{Re}\langle H v, H v \rangle. \quad (12.285)$$

Proof. ✓ Proof Translation of the reversed leg gives

$$\sum_i \text{Re}\langle h_{i+q} v, h_i v \rangle = \sum_i \text{Re}\langle h_i v, h_{i-q} v \rangle. \quad (12.286)$$

Summing Theorem 12.3.7.68 over i and using (12.286) proves the upper bound. If h_i and h_{i+d} commute, their symmetry and idempotence give

$$0 \leq \text{Re}\langle h_i v, h_{i+d} v \rangle. \quad (12.287)$$

Partitioning the nonzero cyclic offsets into $1 \leq q < m$, their reflections $N-q$, and $m \leq k \leq N-m$ gives

$$\text{Re}\langle H v, H v \rangle = \text{Re}\langle H v, v \rangle + \sum_{q=1}^{m-1} \text{Re}\langle Q_q v, v \rangle + \sum_{k=m}^{N-m} \sum_i \text{Re}\langle h_i v, h_{i+k} v \rangle. \quad (12.288)$$

The last sum in (12.288) is nonnegative by (12.287), which proves the lower bound. □

Theorem 12.3.7.73 (Finite-range cyclic Knabe inequality). Lean ✓ Stmt Let $1 \leq R \leq m$, $2m \leq N$, and let h_i be cyclically indexed orthogonal projections such that h_i commutes with h_{i+d} whenever $R \leq d \leq N-R$. Suppose that, for every s and v ,

$$\gamma \text{Re}\langle A_s v, v \rangle \leq \text{Re}\langle A_s v, A_s v \rangle. \quad (12.289)$$

If $(R-1)^2 < m\gamma$, then, with

$$\delta = \frac{m\gamma - (R-1)^2}{m - R + 1} > 0, \quad (12.290)$$

one has

$$\delta \text{Re}\langle H v, v \rangle \leq \text{Re}\langle H v, H v \rangle. \quad (12.291)$$

The coefficient for general R is derived here and is not attributed to the cited papers.

Proof. ✓ Proof Put $c = m - R + 1$. Summing the window gaps and applying Theorem 12.3.7.71 gives

$$m\gamma \operatorname{Re}\langle Hv, v \rangle \leq \sum_s \operatorname{Re}\langle A_s v, A_s v \rangle. \quad (12.292)$$

Decompose $m - q = c + (R - 1 - q)$ and expand (12.280): the weights $q \leq R - 2$ carry the coefficient c plus the positive part $R - 1 - q$, while for $q \geq R$ the coefficient $R - 1 - q$ is nonpositive and the cross term is nonnegative, so those summands are bounded by $c C_q(v)$. For the positive parts, the anticommutator bound and the triangular identity

$$2 \sum_{q=1}^{R-1} (R - 1 - q) = (R - 1)(R - 2) \quad (12.293)$$

combine with the decomposition

$$m - c + (R - 1)(R - 2) = (R - 1)^2, \quad c = m - R + 1, \quad (12.294)$$

so that

$$\begin{aligned} \sum_s \operatorname{Re}\langle A_s v, A_s v \rangle &\leq (m + (R - 1)(R - 2)) \operatorname{Re}\langle Hv, v \rangle + c \sum_{q=1}^{m-1} C_q(v) \\ &\leq (m + (R - 1)(R - 2)) \operatorname{Re}\langle Hv, v \rangle + c(\operatorname{Re}\langle Hv, Hv \rangle - \operatorname{Re}\langle Hv, v \rangle). \end{aligned}$$

The cyclic cross-term estimate therefore gives

$$\sum_s \operatorname{Re}\langle A_s v, A_s v \rangle \leq c \operatorname{Re}\langle Hv, Hv \rangle + (R - 1)^2 \operatorname{Re}\langle Hv, v \rangle. \quad (12.295)$$

Rearranging and dividing by $c > 0$ proves the claim. □

Theorem 12.3.7.74 (Nearest-neighbor cyclic Knabe inequality). Lean ✓ Stmt Under the hypotheses of Theorem 12.3.7.73 with $R = 2$, if $1 < m\gamma$, then

$$\frac{m\gamma - 1}{m - 1} \operatorname{Re}\langle Hv, v \rangle \leq \operatorname{Re}\langle Hv, Hv \rangle. \quad (12.296)$$

Equivalently, the open-window threshold is $\gamma > 1/m$. This is Knabe's nearest-neighbor threshold [177], quoted in [193, lines 1483–1489] and [16, lines 2194–2197].

Proof. ✓ Proof Substitute $R = 2$ into the coefficient of Theorem 12.3.7.73. □

Definition 12.3.7.75 (Cyclic coordinates for periodic local interactions). Lean ✓ Stmt For an N -site periodic chain with $N > 0$, index the range- R local interactions by $\mathbb{Z}/N\mathbb{Z}$. If $s \in \mathbb{Z}/N\mathbb{Z}$ and $i \in \{0, \dots, N - 1\}$ is its standard representative, write

$$h_s^{(N, R)} = h_{i, \text{ES}}(A, R). \quad (12.297)$$

Theorem 12.3.7.76 (Total sum in cyclic coordinates). Lean ✓ Stmt The sum over the cyclic coordinates is the periodic parent Hamiltonian:

$$\sum_{s \in \mathbb{Z}/N\mathbb{Z}} h_s^{(N, R)} = H_{N, R}^{\text{ES}}(A). \quad (12.298)$$

Proof. ✓ Proof Reindex the standard sum over $\{0, \dots, N-1\}$ by its canonical bijection with $\mathbb{Z}/N\mathbb{Z}$. Each summand is unchanged. $\square$

Theorem 12.3.7.77 (Oriented separation of cyclic interactions). Lean Lean Lean ✓ Stmt Let $R \leq e$ and $e + R \leq N$. Addition by e in $\mathbb{Z}/N\mathbb{Z}$ moves the standard representative forward by e sites. The two length- R cyclic windows beginning at s and $s + e$ are disjoint. Consequently,

$$h_s^{(N,R)} h_{s+e}^{(N,R)} = h_{s+e}^{(N,R)} h_s^{(N,R)}. \quad (12.299)$$

Proof. ✓ Proof Measure every site from the first window's initial site in the forward cyclic order. Membership in both windows would give an offset smaller than R and also an offset of the form $e + b$ with $b < R$. The inequalities $R \leq e$ and $e + R \leq N$ exclude this possibility. Local interactions on disjoint cyclic windows commute, which gives (12.299). $\square$

Definition 12.3.7.78 (Cyclic active-block coordinates). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $W \leq N$ and fix a site i . Splitting the cyclically ordered sites into the first W sites and their complement gives

$$[d]^N \simeq [d]^W \times [d]^{N-W}, \quad (12.300)$$

$$U_i: \ell^2([d]^N) \longrightarrow \ell^2([d]^W \times [d]^{N-W}). \quad (12.301)$$

If (σ, τ) is an active configuration and a complementary configuration, then their inverse image has value σ_r at the site $i + r$ for $0 \leq r < W$, and value τ_r at the site $i + W + r$ for $0 \leq r < N - W$. The isometry and its inverse act by composition with this configuration bijection.

Theorem 12.3.7.79 (A local interaction inside a cyclic active block). Lean ✓ Stmt Suppose that $W \leq N$ and $q + R \leq W$. Under the isometry U_i in (12.301), the range- R interaction beginning at the cyclic site $i + q$ becomes the corresponding nonwrapping interaction on the active W -site interval, acting independently on the complementary configuration:

$$U_i h_{i+q}^{(N,R)} U_i^{-1} = (h_q^{(W,R)})^{\oplus [d]^{N-W}}. \quad (12.302)$$

Proof. ✓ Proof The condition $q + R \leq W$ places the whole interaction window inside the active block. Joining an active configuration to a complementary one therefore preserves both the extracted length- R word and every configuration obtained by replacing that word. The local projector acts only on the active coordinates, which proves (12.302) on each complementary configuration. $\square$

Theorem 12.3.7.80 (A cyclic Knabe block is an open parent Hamiltonian). Lean ✓ Stmt Let $1 \leq R \leq m$ and $2m \leq N$, and put

$$W = m + R - 1. \quad (12.303)$$

For every $s \in \mathbb{Z}/N\mathbb{Z}$, let $A_s = \sum_{q=0}^{m-1} h_{s+q}^{(N,R)}$. The m interaction windows occupy exactly the W cyclic sites beginning at s . Under the active block isometry U_s ,

$$U_s A_s U_s^{-1} = (H_{W,R}^{\text{open}}(A))^{\oplus [d]^{N-W}}. \quad (12.304)$$

Proof. ✓ Proof The starts $q = 0, \dots, m-1$ are precisely the nonwrapping range- R starts in an interval of length $W = m + R - 1$. Apply Theorem 12.3.7.79 to every summand and reindex the sum by these nonwrapping starts. Fiberwise extension commutes with finite sums, giving (12.304). $\square$

Theorem 12.3.7.81 (Periodic finite-range gap from one open active volume). [Lean](#) [Stmt](#) Let $1 \leq R \leq m$, put $W = m + R - 1$, and suppose that $H_{W,R}^{\text{open}}(A)$ has norm gap γ . Thus, for every $u \in (\ker H_{W,R}^{\text{open}}(A))^\perp$,

$$\gamma \|u\| \leq \|H_{W,R}^{\text{open}}(A)u\|. \quad (12.305)$$

If $(R - 1)^2 < m\gamma$, define

$$\delta = \frac{m\gamma - (R - 1)^2}{m - R + 1}. \quad (12.306)$$

Then $\delta > 0$, and for every $N \geq 2m$ and every $v \in (\ker H_{N,R}^{\text{ES}}(A))^\perp$,

$$\delta \|v\| \leq \|H_{N,R}^{\text{ES}}(A)v\|. \quad (12.307)$$

The coefficient in (12.306) is the *TNLean* finite-range derivation documented in `docs/paper-gaps/knabe88_finite_range_coefficient.tex`. It is not a theorem of the MPS sources cited for the gap definition and the nearest-neighbor Knabe principle.

Proof. [Proof](#) The norm gap (12.305) and positivity give $(H_{W,R}^{\text{open}})^2 \geq \gamma H_{W,R}^{\text{open}}$ as a quadratic form. Lemma 12.3.7.70 gives the same inequality after adjoining the $N - W$ complementary sites, and Theorem 12.3.7.80 gives it for every cyclic sum A_s . The oriented-separation theorem verifies the commutation hypothesis in Theorem 12.3.7.73. That theorem gives

$$\delta H_{N,R}^{\text{ES}}(A) \leq (H_{N,R}^{\text{ES}}(A))^2 \quad (12.308)$$

as a quadratic form, using the total-sum identity (12.298). The inequality $(R - 1)^2 < m\gamma$ makes δ positive, and the finite-dimensional spectral criterion yields (12.307). $\square$

Theorem 12.3.7.82 (Uniform finite-range gap for primitive MPS parent Hamiltonians). [Lean](#) [Stmt](#) Let A be a primitive MPS tensor with nonzero bond dimension and a positive-definite fixed point. There exist $l > 1$, $\epsilon_l \geq 0$, an integer $m \geq l + 1$, and $\delta > 0$ such that A is l -block injective and

$$\epsilon_l < \frac{1}{\sqrt{l+1}}. \quad (12.309)$$

Set

$$R = l + 1, \quad (12.310)$$

$$W = m + R - 1 = m + l, \quad (12.311)$$

$$\gamma_{\text{open}} = (1 - \epsilon_l \sqrt{l+1})^2. \quad (12.312)$$

The integer m may be chosen so that

$$l^2 < m\gamma_{\text{open}}, \quad (12.313)$$

and

$$\delta = \frac{m\gamma_{\text{open}} - l^2}{m - l}. \quad (12.314)$$

For every $N \geq 2m$ and every $v \in (\ker H_{N,l+1}^{\text{ES}}(A))^{\perp}$,

$$\delta \|v\| \leq \|H_{N,l+1}^{\text{ES}}(A)v\|. \quad (12.315)$$

The existence of a uniform MPS parent-Hamiltonian gap and the nearest-neighbor Knabe principle are stated in [193, lines 1460–1500] and [16, lines 2184–2188, 2194–2197]. Those passages do not state the arbitrary finite-range coefficient (12.314).

Proof. ✓ **Proof** Theorem 12.3.7.65 gives l , ϵ_l , and the norm gap γ_{open} on every open volume at least $l+1$. Since $\gamma_{\text{open}} > 0$, choose $m \geq l+1$ with $l^2 < m\gamma_{\text{open}}$. At the active volume $W = m+l$, block injectivity identifies the open ground space with the kernel of the open Hamiltonian, so (12.263) is exactly the hypothesis (12.305) with $R = l+1$. Apply Theorem 12.3.7.81. Substituting $R-1 = l$ and $m-R+1 = m-l$ gives (12.314) and (12.315). $\square$

12.4 Overlaps of distinct block ground spaces

Theorem 12.4.1 (Mixed boundary Gram identity). Lean Lean ✓ Stmt Let A and B have bond dimensions D_A and D_B , respectively, and let $\Gamma_n^{\text{ES}}(A)$ and $\Gamma_n^{\text{ES}}(B)$ be their Hilbert-space boundary maps, writing E_{ab} also for the coordinate vector of a matrix unit. Let F_{BA}^{rect} be the rectangular mixed transfer operator on $M_{D_B \times D_A}(\mathbb{C})$. For matrix units E_{ab} and E_{ce} ,

$$\langle \Gamma_n^{\text{ES}}(A)E_{ab}, \Gamma_n^{\text{ES}}(B)E_{ce} \rangle = [(F_{BA}^{\text{rect}})^n(E_{ca})]_{eb}. \quad (12.316)$$

Thus $\Gamma_n^{\text{ES}}(A)*\Gamma_n^{\text{ES}}(B)$ is the rectangular reshuffling of $(F_{BA}^{\text{rect}})^n$. This is the boundary-coordinate form of equations ip and P12 in the proof of the lemma disjoint in [191].

Proof. ✓ **Proof** Expanding the trace boundary maps gives

$$\langle \Gamma_n^{\text{ES}}(A)E_{ab}, \Gamma_n^{\text{ES}}(B)E_{ce} \rangle = \sum_{|w|=n} \overline{A_{ba}^w} B_{ec}^w = \sum_{|w|=n} [B^w E_{ca} (A^w)^*]_{eb}. \quad (12.317)$$

The word expansion of $(F_{BA}^{\text{rect}})^n$ gives the assertion. $\square$

Theorem 12.4.2 (Mixed boundary Gram decay). Lean Lean ✓ Stmt If $(F_{BA}^{\text{rect}})^n(X) \rightarrow 0$ for every boundary matrix X , then $\|\Gamma_n^{\text{ES}}(A)*\Gamma_n^{\text{ES}}(B)\| \rightarrow 0$. In particular this holds when the spectral radius of F_{BA}^{rect} is less than one.

Proof. ✓ **Proof** Theorem 12.4.1 gives convergence of every entry of the mixed Gram operator. Its domain and codomain have fixed finite dimensions, so entrywise convergence implies convergence in operator norm. A spectral radius less than one gives decay of the powers of F_{BA}^{rect} , as in equation limP12 in the proof of the lemma disjoint in [191]. $\square$

Theorem 12.4.3 (Ground-space overlap from mixed Gram bounds). Lean ✓ Stmt Suppose that $\Gamma_n^{\text{ES}}(A)$ and $\Gamma_n^{\text{ES}}(B)$ are injective, and put $\mathcal{K}_n(A) = \Gamma_n^{\text{ES}}(A)*\Gamma_n^{\text{ES}}(A)$ and $\mathcal{K}_n(B) = \Gamma_n^{\text{ES}}(B)*\Gamma_n^{\text{ES}}(B)$. For vectors $x \in \text{ran } \Gamma_n^{\text{ES}}(A)$ and $y \in \text{ran } \Gamma_n^{\text{ES}}(B)$,

$$|\langle x, y \rangle| \leq \sqrt{\|\mathcal{K}_n(A)^{-1}\|} \|\Gamma_n^{\text{ES}}(A)*\Gamma_n^{\text{ES}}(B)\| \sqrt{\|\mathcal{K}_n(B)^{-1}\|} \|x\| \|y\|. \quad (12.318)$$

Proof. ✓ **Proof** Write $x = \Gamma_n^{\text{ES}}(A)u$ and $y = \Gamma_n^{\text{ES}}(B)v$. The left inverse $\mathcal{K}_n(A)^{-1}\Gamma_n^{\text{ES}}(A)^*$ has norm $\sqrt{\|\mathcal{K}_n(A)^{-1}\|}$, so $\|u\| \leq \sqrt{\|\mathcal{K}_n(A)^{-1}\|} \|x\|$; the analogous estimate holds for v . Apply these bounds to $|\langle u, \Gamma_n^{\text{ES}}(A)*\Gamma_n^{\text{ES}}(B)v \rangle|$. This makes explicit the normalization step using equation C1C2 in [191]. $\square$

Theorem 12.4.4 (Uniform decay of block ground-space overlaps). [Lean] [✓ Stmt] *Let A and B be trace-preserving tensors with positive-definite fixed matrices ρ and σ . Suppose their complementary transfer maps have spectral radius less than one, and that $\rho_{\text{spec}}(F_{BA}^{\text{rect}}) < 1$. For every $\varepsilon > 0$, there is n_0 such that for every $n \geq n_0$, $x \in G_n(A)$, and $y \in G_n(B)$,*

$$|\langle x, y \rangle| \leq \varepsilon \|x\| \|y\|. \quad (12.319)$$

This is the bound on inner products of ground-space vectors obtained from the spectral-radius estimate in the proof of the lemma disjoint in [191].

Proof. [✓ Proof] By Theorem 11.4.23, the self-Gram operators $\mathcal{K}_n(A)$ and $\mathcal{K}_n(B)$ converge to invertible limits $\mathcal{K}_\infty(A)$ and $\mathcal{K}_\infty(B)$ determined by ρ and σ , and their inverses converge to $\mathcal{K}_\infty(A)^{-1}$ and $\mathcal{K}_\infty(B)^{-1}$. By Theorem 11.4.22, $\mathcal{K}_n(A)$ and $\mathcal{K}_n(B)$ are eventually invertible, and hence $\Gamma_n^{\text{ES}}(A)$ and $\Gamma_n^{\text{ES}}(B)$ are injective, for all sufficiently large n . By Theorem 12.4.2, the coefficient in Theorem 12.4.3 satisfies

$$c(n) = \sqrt{\|\mathcal{K}_n(A)^{-1}\|} \|\Gamma_n^{\text{ES}}(A)^* \Gamma_n^{\text{ES}}(B)\| \sqrt{\|\mathcal{K}_n(B)^{-1}\|} \longrightarrow \sqrt{\|\mathcal{K}_\infty(A)^{-1}\|} \cdot 0 \cdot \sqrt{\|\mathcal{K}_\infty(B)^{-1}\|} = 0. \quad (12.320)$$

Choose n_0 with $c(n) \leq \varepsilon$ for $n \geq n_0$; then Theorem 12.4.3 gives the bound, uniformly in x and y . □

Theorem 12.4.5 (Asymptotic orthogonality of inequivalent blocks). [Lean] [✓ Stmt] *Let A and B be trace-preserving tensors with positive-definite fixed matrices and complementary transfer spectral radii less than one. Suppose that, if their bond dimensions agree, they are not gauge-phase equivalent. Then, for every $\varepsilon > 0$ and all sufficiently large n , every $x \in G_n(A)$ and $y \in G_n(B)$ satisfy*

$$|\langle x, y \rangle| \leq \varepsilon \|x\| \|y\|. \quad (12.321)$$

This is the normalized tensor form of equation limitepsilon in [191].

Proof. [✓ Proof] The positive-definite fixed matrices and complementary gaps imply irreducibility by Lemma C.5.7. For equal bond dimensions, inequivalence and Theorem 5.6.2 give $\rho_{\text{spec}}(F_{BA}^{\text{rect}}) = \rho_{\text{spec}}(F_{BA}) < 1$. For unequal bond dimensions, a peripheral eigenvector of F_{BA}^{rect} would force equality of the bond dimensions, so Theorem 5.6.4 gives

$$\rho_{\text{spec}}(F_{BA}^{\text{rect}}) < 1. \quad (12.322)$$

Apply Theorem 12.4.4. □

12.5 Overlap with a sum of subspaces

Theorem 12.5.1 (Lower Gram estimate). [Lean] [✓ Stmt] *Let $v_1, \dots, v_m$ lie in a complex inner product space, and let $B \in M_m(\mathbb{R})$ have zero diagonal. Suppose that $|\langle v_i, v_j \rangle| \leq B_{ij} \|v_i\| \|v_j\|$ whenever $i \neq j$. With the Euclidean operator norm on B ,*

$$(1 - \|B\|) \sum_i \|v_i\|^2 \leq \left\| \sum_i v_i \right\|^2. \quad (12.323)$$

This is the lower Gram estimate in the proof of [191, Lemma 9].

Proof. ✓ Proof Put $w_i = \|v_i\|$. Expanding the squared norm gives

$$\sum_i \|v_i\|^2 - \left\| \sum_i v_i \right\|^2 \leq \sum_{i,j} B_{ij} w_i w_j = \langle w, Bw \rangle \leq \|B\| \|w\|^2. \quad (12.324)$$

Rearranging proves the assertion. □

Theorem 12.5.2 (Overlap with a sum of subspaces). Lean ✓ Stmt Let $U, V_1, \dots, V_m$ be subspaces of a complex inner product space. Let $a \in \mathbb{R}^m$ and $B \in M_m(\mathbb{R})$, where B has zero diagonal and $\|B\| < 1$. Suppose that $|\langle u, v \rangle| \leq B_{ij} \|u\| \|v\|$ for $i \neq j$, $u \in V_i$, and $v \in V_j$. Suppose also that $|\langle x, u \rangle| \leq a_i \|x\| \|u\|$ for every i , $x \in U$, and $u \in V_i$. Then every $x \in U$ and $y \in V_1 + \dots + V_m$ satisfy

$$|\langle x, y \rangle| \leq \frac{\|a\|}{\sqrt{1 - \|B\|}} \|x\| \|y\|. \quad (12.325)$$

This is [191, Lemma 9], expressed as a bound on inner products.

Proof. ✓ Proof Write $y = \sum_i v_i$ with $v_i \in V_i$. The triangle inequality and Cauchy–Schwarz give

$$|\langle x, y \rangle| \leq \|x\| \sum_i a_i \|v_i\| \leq \|x\| \|a\| \left(\sum_i \|v_i\|^2 \right)^{1/2}. \quad (12.326)$$

Apply Theorem 12.5.1 to the final factor. □

Theorem 12.5.3 (Uniform overlap bound). Lean ✓ Stmt Let $U, V_1, \dots, V_m$ be subspaces of a complex inner product space, and let $\varepsilon \geq 0$ satisfy $\varepsilon m \leq 1$. Suppose that every pair of vectors in distinct V_i satisfies $|\langle u, v \rangle| \leq \varepsilon \|u\| \|v\|$, and that the same inequality holds for vectors in U and V_i , for every i . Set

$$\eta = \frac{\varepsilon \sqrt{m}}{\sqrt{1 - \varepsilon(m-1)}}. \quad (12.327)$$

Then $\eta \leq 1$, and every $x \in U$ and $y \in V_1 + \dots + V_m$ satisfy $|\langle x, y \rangle| \leq \eta \|x\| \|y\|$. This is the uniform bound in [191, Lemma 9].

Proof. ✓ Proof For $y = \sum_i v_i$, the off-diagonal terms satisfy

$$\sum_{i \neq j} \varepsilon \|v_i\| \|v_j\| = \varepsilon \left[\left(\sum_i \|v_i\| \right)^2 - \sum_i \|v_i\|^2 \right] \leq \varepsilon(m-1) \sum_i \|v_i\|^2. \quad (12.328)$$

Take $a_i = \varepsilon$ and $B_{ij} = \varepsilon(1 - \delta_{ij})$. For $w \in \mathbb{R}^m$ write $|w|$ for the vector with entries $|w_i|$. Since B is symmetric with nonnegative entries, the same computation applied to $|w|$ gives

$$|\langle w, Bw \rangle| \leq \langle |w|, B|w| \rangle = \varepsilon \left[\left(\sum_i |w_i| \right)^2 - \sum_i w_i^2 \right] \leq \varepsilon(m-1) \|w\|^2, \quad (12.329)$$

and hence $\|B\| = \sup_{\|w\|=1} |\langle w, Bw \rangle| \leq \varepsilon(m-1) \leq 1 - \varepsilon$, which is less than one when $\varepsilon > 0$ and vanishes when $\varepsilon = 0$. With $\|a\| = \varepsilon \sqrt{m}$, Theorem 12.5.2 gives

$$|\langle x, y \rangle| \leq \frac{\varepsilon \sqrt{m}}{\sqrt{1 - \varepsilon(m-1)}} \|x\| \|y\|. \quad (12.330)$$

Finally, $\varepsilon^2 m \leq 1 - \varepsilon(m-1)$ follows from $\varepsilon m \leq 1$, and gives $\eta \leq 1$. □

12.6 Comparison of sums of orthogonal projectors

Theorem 12.6.1 (Off-diagonal pairing estimate). Lean ✓ Stmt Let (v_i) and (w_i) be finite families in a complex inner product space. Let B be a real matrix with zero diagonal and $\beta = \|B\| < 1$. Suppose that, for $i \neq j$, each of $|\langle v_i, v_j \rangle|$, $|\langle w_i, w_j \rangle|$, and $|\langle v_i, w_j \rangle|$ is bounded by B_{ij} times the product of the corresponding vector norms. Put $\delta = \beta/(1 - \beta)$. Then

$$\sum_{i \neq j} |\langle v_i, w_j \rangle| \leq \delta \left\| \sum_i v_i \right\| \left\| \sum_i w_i \right\|. \quad (12.331)$$

This is the pairing estimate in the proof of [191, Lemma 6.1].

Proof. ✓ Proof Write $a_i = \|v_i\|$ and $b_i = \|w_i\|$. The left-hand side is at most $\langle a, Bb \rangle \leq \beta \|a\| \|b\|$. The lower Gram estimate gives $\|a\| \leq \|\sum_i v_i\|/\sqrt{1 - \beta}$, and likewise for b . $\square$

Theorem 12.6.2 (Sum of block projectors). Lean Lean ✓ Stmt Let (V_i) be a finite family of subspaces of a finite-dimensional complex inner product space. Let B have zero diagonal and $\beta = \|B\| < 1/2$, and suppose that $|\langle u, v \rangle| \leq B_{ij} \|u\| \|v\|$ for $i \neq j$, $u \in V_i$, and $v \in V_j$. Write P_i for the orthogonal projector onto V_i , P for the projector onto $\sum_i V_i$, $X = \sum_i P_i$, and $\delta = \beta/(1 - \beta)$. Then

$$\|X - P\| \leq \frac{\delta}{1 - \delta}, \quad \|X\| \leq \frac{1}{1 - \delta}. \quad (12.332)$$

This is the finite-dimensional projector estimate underlying [191, Lemma 6.1(i)].

Proof. ✓ Proof For $u = \sum_j v_j$ with $v_j \in V_j$, self-adjointness gives

$$\langle z, Xu - u \rangle = \sum_{i \neq j} \langle P_i z, v_j \rangle. \quad (12.333)$$

The pairing estimate bounds its modulus by $\delta \|Xz\| \|u\|$. Since $XP = X$, it follows that $\|X - P\| \leq \delta \|X\|$. Combine this with $\|X\| \leq \|X - P\| + 1$ and $\delta < 1$. For the second bound, use $\|P\| \leq 1$ and the first bound:

$$\begin{aligned} \|X\| &\leq \|X - P\| + \|P\| \leq \frac{\delta}{1 - \delta} + 1 \\ &= \frac{1}{1 - \delta}. \end{aligned}$$

$\square$

Theorem 12.6.3 (Off-diagonal products of projectors). Lean ✓ Stmt Let (U_i) and (V_i) be finite families of subspaces of a finite-dimensional complex inner product space. Let B have zero diagonal and $\beta = \|B\| < 1$. Suppose that, for $i \neq j$, $|\langle u, v \rangle| \leq B_{ij} \|u\| \|v\|$ whenever (u, v) lies in $U_i \times U_j$, $V_i \times V_j$, or $U_i \times V_j$. Write P_i, Q_i for their orthogonal projectors and put $X_U = \sum_i P_i$, $X_V = \sum_i Q_i$, $\delta = \beta/(1 - \beta)$. Then

$$\left\| X_U X_V - \sum_i P_i Q_i \right\| \leq \delta \|X_U\| \|X_V\|. \quad (12.334)$$

Proof. ✓ Proof Pair the operator on the left with unit vectors z, x . Its matrix coefficient is $\sum_{i \neq j} \langle P_i z, Q_j x \rangle$. Apply the off-diagonal pairing estimate and take the supremum over z, x . $\square$

Theorem 12.6.4 (Comparison of joint projectors). Lean ✓ Stmt Under the hypotheses of Theorem 12.6.3, assume $\beta < 1/2$. Let $W_i \subseteq U_i$, and let R_i be the orthogonal projector onto W_i . Let P, Q, R project onto $\sum_i U_i$, $\sum_i V_i$, and $\sum_i W_i$, respectively. Then

$$\|R - PQ\| \leq \frac{4\delta}{(1-\delta)^2} + \sum_i \|R_i - P_i Q_i\|. \quad (12.335)$$

This is the abstract projector comparison used in [191, Lemma 6.1(ii)].

Proof. ✓ Proof Put $X_W = \sum_i R_i$, $Z = \sum_i P_i Q_i$, $a = \delta/(1-\delta)$, and $q = 1/(1-\delta)$. The triangle inequality, applied through X_W , Z , and $X_U X_V$, gives

$$\|R - PQ\| \leq a + \sum_i \|R_i - P_i Q_i\| + aq + (aq + a) \quad (12.336)$$

$$\leq 4aq + \sum_i \|R_i - P_i Q_i\|. \quad (12.337)$$

Here $\|X_U X_V - PQ\| \leq aq + a$ follows by expanding $(X_U - P)X_V + P(X_V - Q)$, and $a \leq aq$ since $q \geq 1$. $\square$

12.7 Overlap bounds with spectator sites

Throughout this section, local ground spaces are subspaces of the Hilbert spaces of configurations, so $G_L(A)$ stands for $G_L^{\text{ES}}(A) = \iota_L(G_L(A))$ of Definition 11.3.1.

Theorem 12.7.1 (Overlap from restrictions to a common interval). Lean ✓ Stmt Let A, B be tensors with the same physical dimension and arbitrary bond dimensions. Suppose $\varepsilon \geq 0$ and $|\langle a, b \rangle| \leq \varepsilon \|a\| \|b\|$ for every $a \in G_L(A)$ and $b \in G_L(B)$. Let x, y be vectors on three successive intervals of lengths K, L, Q . For each prefix configuration u and suffix configuration v , define $x_{u,v}(w) = x(u, w, v)$ and $y_{u,v}(w) = y(u, w, v)$. If $x_{u,v} \in G_L(A)$ and $y_{u,v} \in G_L(B)$ for all u, v , then

$$|\langle x, y \rangle| \leq \varepsilon \|x\| \|y\|. \quad (12.338)$$

No positivity assumption on K, L, Q is required. This is the finite-sum estimate used to extend the overlap bounds in [191, Lemma 6.1(ii)].

Proof. ✓ Proof Decomposing along the prefix and suffix configurations gives

$$\begin{aligned} \langle x, y \rangle &= \sum_{u,v} \langle x_{u,v}, y_{u,v} \rangle, \\ \|x\|^2 &= \sum_{u,v} \|x_{u,v}\|^2, \quad \|y\|^2 = \sum_{u,v} \|y_{u,v}\|^2. \end{aligned}$$

The triangle inequality, the overlap bound, and Cauchy–Schwarz for the sequences $(\|x_{u,v}\|)$ and $(\|y_{u,v}\|)$ give

$$|\langle x, y \rangle| \leq \sum_{u,v} |\langle x_{u,v}, y_{u,v} \rangle| \quad (12.339)$$

$$\leq \varepsilon \sum_{u,v} \|x_{u,v}\| \|y_{u,v}\| \leq \varepsilon \|x\| \|y\|. \quad (12.340)$$

$\square$

Theorem 12.7.2 (Overlap of extended boundary ranges). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Assume the ground-space overlap bound of Theorem 12.7.1. Let U_A be the range of $\Gamma_{K+L,Q}^{\text{left}}(A)$ and V_A the range of the reassociated tail boundary map $\tilde{\Gamma}_{K,L,Q}^{\text{tail}}(A)$, both in the physical space on $K+L+Q$ sites. Define U_B, V_B in the same way. Then $|\langle x, y \rangle| \leq \varepsilon \|x\| \|y\|$ for vectors in any of the pairs of ranges $U_A \times U_B, V_A \times V_B$, and $U_A \times V_B$.

Proof. [✓ Proof](#) For a left boundary vector with boundary family (Z_v) , fixing u, v leaves the middle vector $\Gamma_L(Z_v A^u)$. For a right boundary vector with boundary family (Y_u) , the corresponding middle vector is $\Gamma_L(A^v Y_u)$. Both belong to the appropriate length- L ground space, so the common-interval estimate applies to each of the three pairs. $\square$

Theorem 12.7.3 (Joint projectors on overlapping MPS intervals). [Lean](#) [✓ Stmt](#) Let (A_i) be a finite family of tensors with the same physical dimension, with bond dimensions allowed to depend on i . Let B be a nonnegative real matrix with zero diagonal and $\beta = \|B\| < 1/2$. Suppose that for $i \neq j$, $x \in G_L(A_i)$, and $y \in G_L(A_j)$, $|\langle x, y \rangle| \leq B_{ij} \|x\| \|y\|$. In the physical space on $K+L+Q$ sites, put

$$U_i = G_{K+L}(A_i) \otimes \mathcal{H}_Q, \quad V_i = \mathcal{H}_K \otimes G_{L+Q}(A_i), \quad W_i = G_{K+L+Q}(A_i). \quad (12.341)$$

Let P_i, Q_i, R_i be the orthogonal projectors onto these spaces, and let P, Q, R project onto their respective sums. With $\delta = \beta/(1-\beta)$,

$$\|R - PQ\| \leq \frac{4\delta}{(1-\delta)^2} + \sum_i \|R_i - P_i Q_i\|. \quad (12.342)$$

This is the physical interval comparison in the proof of [191, Lemma 6.1(ii)], before inserting the individual block error bounds.

Proof. [✓ Proof](#) The boundary-map factorization gives $W_i \subseteq U_i$. The extended-boundary overlap estimate supplies all three bounds for $U_i \times U_j, V_i \times V_j$, and $U_i \times V_j$ with the same matrix B . Apply the joint-projector comparison. $\square$

Theorem 12.7.4 (Uniform decay of the joint projector error). [Lean](#) [✓ Stmt](#) Let $(A_i)_{i \in I}$ be a finite family of tensors, and use the interval spaces and projectors of Theorem 12.7.3. Suppose that, for each $\varepsilon > 0$ and all sufficiently large L , the following two bounds hold. For distinct i, j , $x \in G_L(A_i)$, and $y \in G_L(A_j)$, $|\langle x, y \rangle| \leq \varepsilon \|x\| \|y\|$. For every i , $K \geq 0$, and $Q > 0$, $\|Q_i P_i - R_i\| \leq \varepsilon$. Then, for every $\eta > 0$, all sufficiently large L , and every $K \geq 0$ and $Q > 0$,

$$\|R - PQ\| \leq \eta. \quad (12.343)$$

Proof. [✓ Proof](#) Put $r = |I|$ and let C have zero diagonal and all off-diagonal entries equal to one. As $\varepsilon \downarrow 0$, the quantities $\beta = \varepsilon C$, $\delta = \beta/(1-\beta)$, and $4\delta/(1-\delta)^2$ tend to zero. Choose $\varepsilon > 0$ so that $\beta < 1/2$ and the last quantity is less than $\eta/(r+1)$. For all sufficiently large L , the overlap bound holds with $B = \varepsilon C$, and every individual projector error is at most $\eta/(r+1)$. Since P_i, Q_i , and R_i are self-adjoint, $(R_i - P_i Q_i)^* = R_i - Q_i P_i$, and hence

$$\|R_i - P_i Q_i\| = \|Q_i P_i - R_i\|. \quad (12.344)$$

The interval comparison therefore gives

$$\|R - PQ\| \leq \frac{\eta}{r+1} + r \frac{\eta}{r+1} = \eta. \quad (12.345)$$

$\square$

Theorem 12.7.5 (Uniform projector error decay for primitive blocks). Lean ✓ Stmt Let $(A_i)_{i \in I}$ be a finite family of normalized primitive tensors with positive bond dimensions and positive-definite fixed points. Suppose that, if the bond dimensions of two distinct members agree, they are not related by a gauge transformation and a phase. Use the interval projectors of Theorem 12.7.3. For every $\eta > 0$, all sufficiently large L , and every $K \geq 0$ and $Q > 0$,

$$\|R - PQ\| \leq \eta. \quad (12.346)$$

This is the uniform decay consequence of [191, Lemma 6.1(ii)] for a positive right outer interval.

Proof. ✓ Proof Inequivalence of distinct blocks gives, for every $\varepsilon > 0$, a length L_1 such that for all $L \geq L_1$, all distinct i, j , and all $x \in G_L(A_i)$, $y \in G_L(A_j)$,

$$|\langle x, y \rangle| \leq \varepsilon \|x\| \|y\|.$$

Primitivity gives, for every $\varepsilon > 0$, a length L_2 such that for all $L \geq L_2$, every block i , and every $K \geq 0$ and $Q > 0$,

$$\|Q_i P_i - R_i\| \leq \varepsilon.$$

The family is finite, so both bounds hold at the common threshold

$$L \geq L_1 \vee L_2.$$

Apply Theorem 12.7.4. □

Theorem 12.7.6 (Interval spaces of a block-diagonal tensor). Lean Lean Lean ✓ Stmt Let $B = \bigoplus_{j=1}^r \mu_j A_j$ with $\mu_j \neq 0$ for every j . For every $K, L \geq 0$, the extended interval spaces satisfy

$$G_K(B) \otimes \mathcal{H}_L = \sum_{j=1}^r (G_K(A_j) \otimes \mathcal{H}_L), \quad (12.347)$$

$$\mathcal{H}_K \otimes G_L(B) = \sum_{j=1}^r (\mathcal{H}_K \otimes G_L(A_j)). \quad (12.348)$$

For every $K, L, Q \geq 0$, with the reassociation $\mathcal{A}_{K,L,Q}$ of Definition 12.3.7.28, the right interval space of length $L + Q$ satisfies

$$\mathcal{A}_{K,L,Q}(\mathcal{H}_K \otimes G_{L+Q}(B)) = \sum_{j=1}^r \mathcal{A}_{K,L,Q}(\mathcal{H}_K \otimes G_{L+Q}(A_j)). \quad (12.349)$$

Proof. ✓ Proof Let Ω_L be the set of physical words of length L . For subspaces $S_1, \dots, S_r$ of a vector space, the finite coordinate identity

$$\prod_{\tau \in \Omega_L} \left(\sum_{j=1}^r S_j \right) = \sum_{j=1}^r \prod_{\tau \in \Omega_L} S_j \quad (12.350)$$

holds: the right side is contained in the left, and a family $(\psi_\tau)_\tau$ with $\psi_\tau = \sum_j \psi_{\tau,j}$ and $\psi_{\tau,j} \in S_j$ is the sum over j of the families $(\psi_{\tau,j})_\tau$. Let $T_{K,L}$ be the coordinate isomorphism sending $(\psi_\tau)_{\tau \in \Omega_L}$ to the vector with coefficient $\psi_\tau(\sigma)$ at $(\sigma, \tau) \in \Omega_K \times \Omega_L$. Then $G_K(C) \otimes \mathcal{H}_L =$

$T_{K,L} \prod_{\tau \in \Omega_L} G_K(C)$ for every tensor C . The local identity $G_K(B) = \sum_j G_K(A_j)$ and the finite coordinate identity give

$$G_K(B) \otimes \mathcal{H}_L = T_{K,L} \prod_{\tau \in \Omega_L} \left(\sum_{j=1}^r G_K(A_j) \right) = \sum_{j=1}^r T_{K,L} \prod_{\tau \in \Omega_L} G_K(A_j) = \sum_{j=1}^r (G_K(A_j) \otimes \mathcal{H}_L), \quad (12.351)$$

since linear maps preserve sums of subspaces. The same argument, with the free sites on the left and coordinates indexed by Ω_K , gives the second identity. Applying the linear map $\mathcal{A}_{K,L,Q}$ to the second identity at lengths K and $L+Q$ gives the third. $\square$

Theorem 12.7.7 (Projector error decay for a block-diagonal tensor). [Lean] [Stmt] *Let $B = \bigoplus_{j=1}^r \mu_j A_j$, where every μ_j is nonzero and the A_j are pairwise gauge-phase inequivalent normalized primitive tensors with positive bond dimensions and positive-definite fixed points. Write $P_I(B)$ for the orthogonal projector onto the ground space on the interval I , extended by the identity on the remaining sites. For every $\eta > 0$, all sufficiently large L , and every $K \geq 0$ and $Q > 0$,*

$$\|P_{[1,K+L+Q]}(B) - P_{[1,K+L]}(B)P_{[K+1,K+L+Q]}(B)\| \leq \eta. \quad (12.352)$$

Proof. [Proof] For each block put

$$U_j = G_{K+L}(A_j) \otimes \mathcal{H}_Q, \quad V_j = \mathcal{A}_{K,L,Q}(\mathcal{H}_K \otimes G_{L+Q}(A_j)), \quad W_j = G_{K+L+Q}(A_j), \quad (12.353)$$

and let P, S, R be the orthogonal projectors onto $\sum_j U_j$, $\sum_j V_j$, and $\sum_j W_j$. These are the projectors written P, Q, R in Theorem 12.7.3; the middle one is renamed here because Q denotes a length. The local identity at length $K+L+Q$, the first identity of Theorem 12.7.6 at lengths $K+L$ and Q , and its reassociated identity at lengths K, L, Q give

$$P_{[1,K+L+Q]}(B) = P_{G_{K+L+Q}(B)} = P_{\sum_j W_j} = R, \quad (12.354)$$

$$P_{[1,K+L]}(B) = P_{G_{K+L}(B) \otimes \mathcal{H}_Q} = P_{\sum_j U_j} = P, \quad (12.355)$$

$$P_{[K+1,K+L+Q]}(B) = P_{\mathcal{A}_{K,L,Q}(\mathcal{H}_K \otimes G_{L+Q}(B))} = P_{\sum_j V_j} = S. \quad (12.356)$$

Hence the left side of the asserted bound is $\|R - PS\|$, and Theorem 12.7.5 with the prescribed η bounds it by η for all sufficiently large L , every $K \geq 0$, and every $Q > 0$. $\square$

12.8 The normalized primitive gauge of a normal tensor

A normal tensor need not be trace-preserving or have spectral radius one. Standard treatments normalize the tensor by scaling the spectral radius of the transfer map to one and applying a gauge transformation by the square root of a positive fixed point [191, 193]. This gauge reduction extends the finite-range spectral gap to arbitrary normal tensors.

Theorem 12.8.1 (Rescaling invariance of block injectivity and normality). [Lean] [Lean] [Lean] [Stmt] *Let $\zeta \neq 0$ and let A be a tensor. For every N the length- N word spans of A and of ζA agree, so ζA is N -block injective exactly when A is, and ζA is normal exactly when A is.*

Proof. [Proof] Every length- N word of ζA is the corresponding word of A multiplied by ζ^N , and ζ^N is invertible, so each span contains the other. $\square$

Theorem 12.8.2 (A normal tensor has a nonzero matrix). [Lean](#) [✓ Stmt](#) Let $D > 0$ and let A be a normal tensor. Then $A^i \neq 0$ for at least one i .

Proof. [✓ Proof](#) If every A^i vanished, then every word of positive length would vanish, so the span of the length- N words would be $\{0\}$ for the length $N > 0$ at which A is block injective. That span is $M_D(\mathbb{C})$, which contains $1 \neq 0$ because $D > 0$. $\square$

Lemma 12.8.3 (Trace pairing of a conjugated word against a boundary matrix). [Lean](#) [✓ Stmt](#) Let X be invertible and let E and Y be $D \times D$ matrices. Then

$$\text{tr}(XEX^{-1}Y) = \text{tr}(EX^{-1}YX).$$

Proof. [✓ Proof](#) Cyclicity of the trace moves the leading factor X to the end of the product. $\square$

Theorem 12.8.4 (Gauge invariance of the local ground space). [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be gauge equivalent. For every L one has $G_L(A) = G_L(B)$.

Proof. [✓ Proof](#) If $B^i = XA^iX^{-1}$, then $B^\sigma = XA^\sigma X^{-1}$ for every length- L configuration σ by Lemma 2.4.18, so by Lemma 12.8.3 the boundary parametrization of B at Y agrees with that of A at $X^{-1}YX$. This gives $G_L(B) \subseteq G_L(A)$, and the symmetric argument applied to $A^i = X^{-1}B^iX$ gives the reverse inclusion. $\square$

Theorem 12.8.5 (Rescaling invariance of the local ground space). [Lean](#) [✓ Stmt](#) Let $\zeta \neq 0$. For every L one has $G_L(\zeta A) = G_L(A)$.

Proof. [✓ Proof](#) The boundary parametrization of ζA at length L sends X to the value of the boundary parametrization of A at $\zeta^L X$, and multiplication by ζ^L is a bijection of $M_D(\mathbb{C})$. $\square$

Theorem 12.8.6 (Local ground spaces determine the parent Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B have the same bond dimension and satisfy $G_L(A) = G_L(B)$. Then their canonical parent interactions at length L agree, hence so do all their translated local terms and their parent Hamiltonians $H_{N,L}(A) = H_{N,L}(B)$ and $H_{N,L}^{\text{ES}}(A) = H_{N,L}^{\text{ES}}(B)$ on every periodic chain.

Proof. [✓ Proof](#) The parent interaction at length L is the orthogonal projector onto $G_L(A)^\perp$, so it depends on $G_L(A)$ alone. Every translated local term is built from that single projector, and the parent Hamiltonian is their sum. $\square$

Theorem 12.8.7 (Normality alone gives an irreducible tensor). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a normal tensor. Then A is primitive in the sense of Definition 6.1.1.3, and A has no nontrivial invariant projection, so A is irreducible in the sense of Definition 7.1.1.1. No normalization is imposed on A .

Proof. [✓ Proof](#) Normality is eventual full Kraus rank (Remark 6.1.1.12), so $S_N(A) = M_D(\mathbb{C})$ for some $N > 0$. Applying the length- N words to a nonzero φ then spans $\mathbb{C}^D$, which is the spreading condition. A subspace invariant under all matrices of A and containing a nonzero vector therefore contains the span of the images of that vector, namely all of $\mathbb{C}^D$; so no invariant projection is nontrivial. $\square$

Theorem 12.8.8 (Normalized primitive gauge of a normal tensor). [Lean](#) [✓ Stmt](#) Let $D > 0$ and let A be a normal tensor. There exist a scalar $\zeta \neq 0$, a tensor B , and a matrix $\rho > 0$ such that ζA and B are gauge equivalent, the pair (B, ρ) is primitive, and

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$$

for every chain length N and every configuration σ . The local ground spaces, the parent interactions, and the periodic chain ground spaces of A and B also coincide:

$$\begin{aligned} G_L(A) &= G_L(B), \\ h_L(A) &= h_L(B), \\ \mathcal{G}_{N,L}(A) &= \mathcal{G}_{N,L}(B) \end{aligned}$$

for all L and N , where h_L is the canonical parent interaction and $\mathcal{G}_{N,L}$ the periodic chain ground space.

This is the normalization taken without loss of generality in [191, equations (3.1)–(3.2b), lines 1394–1435] and at the start of the proof of [193, Theorem 4, lines 765–770]. The gauge here is built from the Perron eigenvector of the adjoint transfer map, giving the trace-preserving orientation. That orientation is what conditions 1 and 2 of [193, Theorem 4, lines 752–758], $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ and $\sum_i (A^i)^\dagger \Lambda A^i = \Lambda$ with $\Lambda > 0$, become after the further gauge by $\Lambda^{1/2}$: condition 2 turns into $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ and condition 1 into the statement that Λ is a positive definite fixed point of the transfer map of B .

Proof. ✓ Proof Theorem 12.8.7 makes A an irreducible tensor, with no normalization imposed on A , and Theorem 12.8.2 supplies the nonzero matrix that the Perron–Frobenius eigenvector requires. Theorem 4.5.3 therefore returns a positive definite σ and an eigenvalue $r > 0$ of the adjoint transfer map. Set $\zeta = r^{-1/2}$ and $B^i = \zeta \sigma^{1/2} A^i \sigma^{-1/2}$; then $\sum_i (B^i)^\dagger B^i = \mathbb{1}$, and ζA and B are gauge equivalent. Rescaling preserves normality by Theorem 12.8.1 and gauging preserves it by Theorem 2.5.7, so B is normal. Since B is trace preserving, Theorem 6.1.1.16 makes B strongly irreducible, and Theorem 6.7.4.2 then returns a positive definite ρ with (B, ρ) primitive. Gauge equivalence leaves the matrix product vectors unchanged by Theorem 2.4.19, and multiplying every matrix by ζ multiplies every length- N word by ζ^N by Lemma 2.4.12, hence every length- N coefficient of B is ζ^N times that of A . By Theorem 12.8.5 together with gauge invariance of the local ground space, $G_L(A) = G_L(\zeta A) = G_L(B)$. The parent interaction then agrees by Theorem 12.8.6 and the periodic chain ground space by Lemma 11.6.5. $\square$

Theorem 12.8.9 (Uniform finite-range gap for normal MPS parent Hamiltonians). Lean ✓ Stmt
Let $D > 0$ and let A be a normal tensor. There exist $l > 1$, $\epsilon_l \geq 0$, an integer $m \geq l + 1$, and $\delta > 0$ such that A is l -block injective, $\epsilon_l < 1/\sqrt{l+1}$,

$$\delta = \frac{m(1 - \epsilon_l \sqrt{l+1})^2 - l^2}{m - l}, \quad (12.357)$$

and, for every $N \geq 2m$ and every $v \in (\ker H_{N,l+1}^{\text{ES}}(A))^\perp$,

$$\delta \|v\| \leq \|H_{N,l+1}^{\text{ES}}(A)v\|. \quad (12.358)$$

No normalization is imposed on A . The uniform gap of matrix product state parent Hamiltonians is asserted at [16, lines 2183–2187]; the arbitrary finite-range coefficient (12.357) is the derivation recorded in `docs/paper-gaps/knabe88_finite_range_coefficient.tex`.

Proof. ✓ Proof Replace A by the normalized primitive representative B of Theorem 12.8.8 and apply Theorem 12.3.7.82 to B . Since $G_L(A) = G_L(B)$ for every L , Theorem 12.8.6 makes the two parent Hamiltonians equal, so both the kernel and the norm bound transfer verbatim. Block injectivity of B at length l passes to ζA by Lemma 2.5.6 applied to the inverse gauge, and from ζA to A by Theorem 12.8.1. $\square$

Lemma 12.8.10 (Positive gap transfer under Loewner domination). [Lean](#) [Lean](#) [✓ Stmt](#) Let P and Q be finite-dimensional complex endomorphisms, and let $\gamma \geq 0$. Suppose that P is positive and

$$\gamma\|v\| \leq \|Pv\| \quad (12.359)$$

for every $v \in (\ker P)^\perp$. Then

$$\gamma\|v\|^2 \leq \operatorname{Re}\langle Pv, v \rangle \quad (12.360)$$

on $(\ker P)^\perp$. If also $P \leq Q$ and $\ker P = \ker Q$, then

$$\gamma\|v\| \leq \|Qv\| \quad (12.361)$$

for every $v \in (\ker Q)^\perp$.

Theorem 12.8.11 (Kernels of positively weighted sums of positive operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $(P_i)_{i \in I}$ be a finite family of positive operators on an inner product space. A positive operator P annihilates a vector v as soon as $\operatorname{Re}\langle Pv, v \rangle = 0$, and consequently, for real weights $w_i > 0$,

$$\ker\left(\sum_{i \in I} w_i P_i\right) = \ker\left(\sum_{i \in I} P_i\right) = \bigcap_{i \in I} \ker P_i, \quad (12.362)$$

equivalently $(\sum_{i \in I} w_i P_i)v = 0$ exactly when $P_i v = 0$ for every $i \in I$. For the empty family both sums are zero and all three kernels are the full space. This is the analytic kernel step of the blocked-sum comparison in Lemma `block-obc` of Rozmán–Molnár–Schuch¹, in which the blocked parent Hamiltonian is a weighted sum $\sum_i \gamma_i h_i$ of positive local terms with $\gamma_i > 0$.

Proof. [✓ Proof](#) If $(\sum_i w_i P_i)v = 0$, the real part of the inner product with v expands as

$$\operatorname{Re}\left\langle \left(\sum_i w_i P_i\right)v, v \right\rangle = \sum_i w_i \operatorname{Re}\langle P_i v, v \rangle = 0. \quad (12.363)$$

Each summand is nonnegative, so each vanishes, and $w_i > 0$ forces $\operatorname{Re}\langle P_i v, v \rangle = 0$. For a positive operator P , the quadratic form is nonnegative along the real affine line $t \mapsto v + t \cdot (Pv)$ and vanishes at $t = 0$; its linear coefficient is $2\|Pv\|^2$, so evaluating at small negative t forces $\|Pv\| = 0$, that is $Pv = 0$. The converse inclusion holds term by term, and the unweighted comparison is the case $w_i = 1$. $\square$

Theorem 12.8.12 (Loewner domination of positively weighted sums). [Lean](#) [Lean](#) [✓ Stmt](#) Let $(P_i)_{i \in I}$ be a finite family of positive operators on an inner product space, with real weights $(w_i)_{i \in I}$. If $w_i \geq 0$ for every i , then $\sum_{i \in I} w_i P_i$ is positive. If $w_i \leq 1$ for every i , then

$$\sum_{i \in I} w_i P_i \leq \sum_{i \in I} P_i \quad (12.364)$$

in Loewner order. This is the comparison step of the blocked-sum argument in the source cited in Theorem 12.8.11, where the coefficients satisfy $0 < \gamma_i \leq 1$.

¹arXiv:2607.19078v1.

Proof. ✓ **Proof** Positivity of the weighted sum follows from positivity of each scalar multiple $w_i P_i$ and of finite sums of positive operators. For the domination, the difference is again a weighted sum,

$$\sum_{i \in I} P_i - \sum_{i \in I} w_i P_i = \sum_{i \in I} (1 - w_i) P_i, \quad (12.365)$$

now with nonnegative weights $1 - w_i \geq 0$, hence positive. □

Theorem 12.8.13 (Gap transfer through a positively weighted sum). Lean ✓ **Stmnt** *Let $(P_i)_{i \in I}$ be a finite family of positive operators on a finite-dimensional complex inner product space, with real weights $0 < w_i \leq 1$, and let $\gamma \geq 0$. If*

$$\gamma \|v\| \leq \left\| \left(\sum_{i \in I} w_i P_i \right) v \right\| \quad (12.366)$$

for every $v \in (\ker \sum_{i \in I} w_i P_i)^\perp$, then

$$\gamma \|v\| \leq \left\| \left(\sum_{i \in I} P_i \right) v \right\| \quad (12.367)$$

for every $v \in (\ker \sum_{i \in I} P_i)^\perp$.

Proof. ✓ **Proof** The weighted sum is positive and, by Theorem 12.8.12, dominated by the unweighted sum; by the kernel equality (12.362) of Theorem 12.8.11, the two sums have the same kernel. Lemma 12.8.10 transfers the norm gap from the smaller weighted sum to the larger unweighted sum. □

Theorem 12.8.14 (Positive gap at a fixed length). Lean ✓ **Stmnt** *For any tensor A , interaction range L , and chain length N , there exists $\delta_N > 0$ such that $\delta_N \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$ for every $v \in (\ker H_{N,L}^{\text{ES}}(A))^\perp$.*

Proof. ✓ **Proof** The restriction of $H_{N,L}^{\text{ES}}(A)$ to its kernel complement is injective. Its inverse on its image is bounded, since the domain is finite dimensional. Thus $\|v\| \leq C_N \|H_{N,L}^{\text{ES}}(A)v\|$ for some $C_N > 0$; take $\delta_N = C_N^{-1}$. If the kernel complement is zero, any positive constant suffices. □

Theorem 12.8.15 (Extension of an eventual uniform gap). Lean ✓ **Stmnt** *Fix A and L . Suppose $\delta > 0$ and $M \in \mathbb{N}$ satisfy $\delta \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$ for every $N \geq M$ and every $v \in (\ker H_{N,L}^{\text{ES}}(A))^\perp$. Then some $\gamma > 0$ satisfies the inequality with γ in place of δ for every N .*

Proof. ✓ **Proof** Take the minimum of δ and the finitely many positive constants δ_N supplied by Theorem 12.8.14 for $N < M$. Equivalently, induct on M , replacing δ by $\min\{\delta, \delta_{M-1}\}$ at each step. □

Theorem 12.8.16 (Uniform gap for normal MPS parent Hamiltonians). Lean ✓ **Stmnt** *Let $D > 0$ and let A be a normal tensor. There exist $l > 1$ and $\gamma > 0$ such that A is l -block injective and, for every N and every $v \in (\ker H_{N,l+1}^{\text{ES}}(A))^\perp$,*

$$\gamma \|v\| \leq \|H_{N,l+1}^{\text{ES}}(A)v\|.$$

This proves the normal-tensor case of the uniform gap assertion in [16, lines 2183–2187], at a suitable interaction range.

Proof. ✓ Proof Theorem 12.8.9 supplies $l > 1$, $m \geq l + 1$, and $\delta > 0$ with the required norm inequality for $N \geq 2m$. Apply Theorem 12.8.15 with $L = l + 1$ and $M = 2m$. $\square$

Theorem 12.8.17 (Gap bound from a cyclic anticommutator estimate). ✓ Stmt Let $L > 1$. Suppose that for every $N \geq 2L$ and every overlapping off-diagonal pair of cyclic length- L windows,

$$h_{i,\text{ES}}h_{j,\text{ES}} + h_{j,\text{ES}}h_{i,\text{ES}} \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)}(h_{i,\text{ES}} + h_{j,\text{ES}})$$

in quadratic form. Then the parent Hamiltonian has the explicit gap constant $\gamma_L := 1/(4L) > 0$. Concretely, if $N \geq 2L$ and $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$, then $\gamma_L \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$.

Proof. ✓ Proof The finite-overlap martingale reduction applies with the cyclic overlap relation: at most $2(L-1)$ off-diagonal neighbours per row; for non-overlapping pairs the disjoint-window estimate $\text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq 0$ holds; and the anticommutator inequality is retained as an explicit hypothesis. $\square$

Theorem 12.8.18 (Conditional spectral gap for MPS parent Hamiltonians). Lean ✓ Stmt For a tensor A and a range $L > 1$, assume the cyclic-window anticommutator estimate of Theorem 12.8.17. Then there exists $\gamma > 0$ such that for every $N \geq 2L$ and every vector $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ one has $\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$. By Lemma 12.3.20, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

Proof. ✓ Proof This follows immediately from the corresponding explicit theorem by taking γ to be the constant produced there. $\square$

Definition 12.8.19 (Grouped prefix and interval projections). Lean Lean ✓ Stmt Fix $p \geq 1$, $M \geq 2$, and an open chain of length $N = pM$. For $0 \leq n \leq M$, let G_n be the ground projection of the range- $2p$ parent Hamiltonian on $[0, pn)$, extended by the identity on the remaining sites. For $0 \leq n < M$, let Q_n be the ground projection of the same interaction restricted to $[\max\{0, p(n-1)\}, p(n+1))$. Thus $Q_0 = I$, while for $n \geq 1$ the interval Hamiltonian is a single parent interaction.

Lemma 12.8.20 (Locality and the initial grouped differences). Lean Lean Lean ✓ Stmt Set $E_n = G_n - G_{n+1}$. For $0 \leq m, n < M$ with $m < n-1$ or $m > n$, one has

$$[E_m, Q_n] = 0. \tag{12.368}$$

The initial differences satisfy $E_0 = 0$ and $Q_1 E_1 = 0$.

Proof. ✓ Proof In the first case both prefix Hamiltonians have support disjoint from the interval defining Q_n . In the second case their kernels are contained in the local interval kernel. The corresponding orthogonal projections therefore commute in either case. The prefixes of lengths zero and p contain no interaction, so $G_0 = G_1 = I$. At the next step $Q_1 = G_2$, giving $Q_1 E_1 = G_2(I - G_2) = 0$. $\square$

Theorem 12.8.21 (The first two grouped differences need no estimate). Lean ✓ Stmt Let $p \geq 1$, let $N \geq 2p$, and let $M \geq 0$. Define G_n , Q_n , and $E_n = G_n - G_{n+1}$ on the N -site chain by the formulas of Definition 12.8.19. Let $\epsilon \geq 0$ and suppose that $\|Q_n E_n\| \leq \epsilon$ for $2 \leq n < M$. Then $\|Q_n E_n\| \leq \epsilon$ for $0 \leq n < M$.

Proof. ✓ Proof By Lemma 12.8.20,

$$Q_0 E_0 = 0, \quad Q_1 E_1 = 0, \tag{12.369}$$

so both norms are $0 \leq \epsilon$. For $2 \leq n < M$ the bound is the hypothesis. $\square$

Theorem 12.8.22 (Grouped open-chain coercivity). [Lean](#) [✓ Stmt](#) Let $p \geq 1$, $M \geq 2$, and $N = pM$. Consider the range- $2p$ open parent Hamiltonian H_N and the grouped projections above. Let $0 \leq \epsilon < 1/\sqrt{2}$ and suppose that $\|Q_n E_n\| \leq \epsilon$ for $0 \leq n < M$. Then, for every $v \perp \ker H_N$,

$$(1 - \epsilon\sqrt{2})^2 \|v\| \leq \|H_N v\|. \quad (12.370)$$

Proof. [✓ Proof](#) Apply the full-range martingale estimate to the grouped filtration, with window width one. Each local window Hamiltonian is the projection $I - Q_n$, so Theorem 12.8.24 gives condition C2 with constant one. The nonzero local window terms are distinct summands of H_N , so Theorem 12.8.23 gives condition C1 with constant one. Lemma 12.3.7.4 supplies positivity of H_N . The locality lemma supplies the commutation condition. The assumed norm bound supplies C3, and the terminal projection is the projection onto $\ker H_N$. $\square$

Theorem 12.8.23 (Energy counting for grouped open windows). [Lean](#) [✓ Stmt](#) Let $p \geq 1$, $M \geq 0$, and $N = pM$. For every vector v on the N -site chain,

$$0 \leq \sum_{n=0}^{M-1} \operatorname{Re} \langle H_{N,2p;2p,p(n+1)}^{\text{suffix}}(A) v, v \rangle \leq \operatorname{Re} \langle H_{N,2p}^{\text{open}}(A) v, v \rangle. \quad (12.371)$$

Proof. [✓ Proof](#) The indices $p(n+1) - 1$, for $0 \leq n < M$, are distinct and belong to $\{0, \dots, pM - 1\}$. Positivity bounds the sum over these indices by the full sum, which is bounded by the open-chain energy. $\square$

Theorem 12.8.24 (Unit local gap for grouped open windows). [Lean](#) [✓ Stmt](#) Let $p \geq 1$, $0 \leq n < M$, and $N = pM$. Write Q_n for the orthogonal projection onto the kernel of $H_{N,2p;2p,p(n+1)}^{\text{suffix}}(A)$. Then

$$\|(I - Q_n)v\|^2 \leq \operatorname{Re} \langle H_{N,2p;2p,p(n+1)}^{\text{suffix}}(A) v, v \rangle \quad (12.372)$$

for every vector v on the N -site chain.

Proof. [✓ Proof](#) Apply the full-range local gap bound with $l = 2p - 1$ and index $p(n+1) - 1 < N$. $\square$

Theorem 12.8.25 (Prefix ground spaces with free right sites). [Lean](#) [Lean](#) [✓ Stmt](#) Let $R > 0$ and suppose that the length- n open parent Hamiltonian has kernel $G_n(A)$. On $n + r$ sites, the kernel of the same interactions restricted to the first n sites is $G_n(A) \otimes (\mathbb{C}^d)^{\otimes r}$. A vector belongs to this space if and only if each restriction obtained by fixing the last r physical coordinates belongs to $G_n(A)$.

Proof. [✓ Proof](#) Decompose a vector v on $n + r$ sites according to the last r physical coordinates, writing v_s for the vector on n sites obtained by fixing these coordinates to $s \in \{0, \dots, d - 1\}^r$. The prefix Hamiltonian acts on each component v_s by the length- n open Hamiltonian, whose kernel is $G_n(A)$. Hence $v \in G_n(A) \otimes (\mathbb{C}^d)^{\otimes r}$ if and only if, for every $s \in \{0, \dots, d - 1\}^r$,

$$v_s \in G_n(A). \quad (12.373)$$

$\square$

Theorem 12.8.26 (Reassociated suffix ground space). [Lean](#) [Lean](#) [✓ Stmt](#) Reassociation identifies any subspace of the physical space on $K + (L + Q)$ sites with the same subspace on $(K + L) + Q$ sites. In particular, when $L + Q > 0$, the kernel of the single parent interaction on the last $L + Q$ sites is $(\mathbb{C}^d)^{\otimes K} \otimes G_{L+Q}(A)$ in these common coordinates.

Proof. ✓ Proof Write σ for the reassociation isometry from the physical space on $K + (L + Q)$ sites to that on $(K + L) + Q$ sites. It preserves the ordered physical coordinates, so for every family $(S_n)_n$ of subspaces of the n -site physical spaces,

$$\sigma(S_{K+(L+Q)}) = S_{(K+L)+Q}. \quad (12.374)$$

A parent interaction is the orthogonal projection onto the complement of the local boundary-condition space, acting trivially on the other sites. Its kernel on the last $L + Q$ sites is therefore $(\mathbb{C}^d)^{\otimes K} \otimes G_{L+Q}(A)$, and the display transports this description to the common coordinates. $\square$

Theorem 12.8.27 (Active grouped projector error). Lean ✓ Stmt *Let $R = L + Q > 0$ and suppose that the range- R open parent Hamiltonians on $K + L$ and $K + L + Q$ sites have their boundary-condition spaces as kernels. On $K + L + Q$ sites, let P project onto the ground space of the first $K + L$ sites, let T project onto the ground space of the last $L + Q$ sites, and let F project onto the whole-chain ground space. Then*

$$\|T(P - F)\| = \|F - PT\|.$$

The right-hand side is the three-interval projector error in [191, condition C3].

Proof. ✓ Proof The whole-chain boundary space lies in the suffix boundary space, so $TF = F$. Thus $T(P - F) = TP - F$. Taking adjoints, using that orthogonal projections are self-adjoint, and reversing the sign gives the identity. $\square$

Theorem 12.8.28 (Periodic gaps from divisible open intervals). Lean ✓ Stmt *Let A be a tensor, let $p, R \geq 1$ be integers, and let $\gamma > 0$. Suppose that, for every sufficiently large integer M and every $u \perp \ker H_{pM, R}^{\text{open}}(A)$, $\gamma\|u\| \leq \|H_{pM, R}^{\text{open}}(A)u\|$. Then there is $\delta > 0$ such that, for every sufficiently large N and every $v \perp \ker H_{N, R}^{\text{ES}}(A)$, $\delta\|v\| \leq \|H_{N, R}^{\text{ES}}(A)v\|$.*

Proof. ✓ Proof Choose n with $(R - 1)^2 < n\gamma$ and then choose $M \geq n + 2R$ for which the open-chain bound holds. Set $m = pM - R + 1$. Then $m \geq \max\{n, R\}$ and the active open volume is $m + R - 1 = pM$. The finite-range Knabe comparison gives the positive constant $\delta = (m\gamma - (R - 1)^2)/(m - R + 1)$ and the asserted bound for every $N \geq 2m$. $\square$

Theorem 12.8.29 (Spectator extension of a grouped projection difference). Lean ✓ Stmt *Let the interaction range satisfy $R > 0$, let $p \leq n$, and add $r \geq 0$ sites to the right of an active interval of length n . In each ambient volume $s \in \{n, n + r\}$, write $P_a^{(s)}$ for the kernel projection of the prefix Hamiltonian ending at a , and write $Q^{(s)}$ for the kernel projection of the suffix Hamiltonian of length l ending at n . Then*

$$\|Q^{(n+r)}(P_p^{(n+r)} - P_n^{(n+r)})\| \leq \|Q^{(n)}(P_p^{(n)} - P_n^{(n)})\|. \quad (12.375)$$

Proof. ✓ Proof Let Φ be the isometry $\ell^2([d]^{n+r}) \rightarrow \ell^2([d]^n \times [d]^r)$ splitting a configuration into its active and spectator coordinates, and write $T^{\oplus[d]^r}$ for the fiberwise extension of an operator T on n sites. By Theorem 12.3.4.14, Φ conjugates each of the three Hamiltonians to its fiberwise extension. Lemma 12.3.4.12 transfers this to their kernel projections: for $a \in \{p, n\}$,

$$\Phi P_a^{(n+r)} \Phi^{-1} = (P_a^{(n)})^{\oplus[d]^r}, \quad \Phi Q^{(n+r)} \Phi^{-1} = (Q^{(n)})^{\oplus[d]^r}. \quad (12.376)$$

Since fiberwise extension respects composition and subtraction (Lemma 12.3.4.11),

$$\Phi Q^{(n+r)}(P_p^{(n+r)} - P_n^{(n+r)}) \Phi^{-1} = \left(Q^{(n)}(P_p^{(n)} - P_n^{(n)}) \right)^{\oplus[d]^r}. \quad (12.377)$$

Conjugation by Φ preserves the operator norm, and $\|T^{\oplus[d]^r}\| \leq \|T\|$ by Lemma 12.3.4.11. $\square$

Theorem 12.8.30 (Grouped C3 from a three-interval estimate). [Lean](#) [✓ Stmt](#) Let $p \geq 1$. Suppose that the range- $2p$ open parent Hamiltonian of a tensor has its boundary-condition space as kernel at every length at least $2p$. For every $K \geq 0$, let P_K , T_K , and F_K be the boundary-space projections on the first $K + p$ sites, the last $2p$ sites, and all $K + 2p$ sites, respectively, with free sites elsewhere. Suppose that $\epsilon \geq 0$ and that, for every $K \geq 0$,

$$\|F_K - P_K T_K\| \leq \epsilon.$$

Then on every chain of length pM with $M \geq 2$, the grouped martingale projections satisfy

$$\|Q_n E_n\| \leq \epsilon$$

for every $0 \leq n < M$.

Proof. [✓ Proof](#) The products at indices zero and one vanish, so by Theorem 12.8.21 it suffices to treat $n \geq 2$. For such n , put $K = p(n - 1)$. The active interval has length $p(n + 1)$, and its preceding prefix has length pn . The kernel hypothesis and the active projector identity give the desired bound on this interval. Adjoining the remaining $pM - p(n + 1)$ sites does not increase the operator norm. $\square$

Theorem 12.8.31 (Open-chain gaps on divisible lengths for primitive block sums). [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{r-1}$ be normalized primitive tensors with common physical dimension $d \geq 1$, positive bond dimensions, and positive-definite invariant matrices. Assume that no two distinct blocks are related by a virtual similarity and a phase. Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$, with every $\mu_j \neq 0$. For every $R_* \in \mathbb{N}$, there are $p \geq \max\{1, R_*\}$ and $\gamma > 0$ such that

$$\gamma \|v\| \leq \|H_{pM, 2p}^{\text{open}}(B)v\|$$

for every $M \geq 2$ and every $v \perp \ker H_{pM, 2p}^{\text{open}}(B)$. This follows from the grouped martingale estimate of [191, Theorem 2.1(ii) and Section 6].

Proof. [✓ Proof](#) Choose p above R_* and the open-kernel threshold, and sufficiently large that the three-interval projector error is at most $1/2$ for every pair of outer lengths at least p . The grouped C3 estimate then holds with $\epsilon = 1/2 < 1/\sqrt{2}$. The grouped open-chain bound gives $\gamma = (1 - \sqrt{2}/2)^2 > 0$ at every length pM , $M \geq 2$. $\square$

Theorem 12.8.32 (Open-chain gaps at every length for primitive block sums). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 12.8.31, for every $R_* \in \mathbb{N}$ there exist $p \geq \max\{1, R_*\}$ and $\delta > 0$ such that

$$\delta \|v\| \leq \|H_{N, 2p}^{\text{open}}(B)v\|$$

for every $N \in \mathbb{N}$ and every $v \perp \ker H_{N, 2p}^{\text{open}}(B)$. At $N < 2p$ the kernel complement is zero. This is the all-length conclusion of [191, Theorem 1.2 and Section 6] for the sufficiently large interaction range chosen here.

Proof. [✓ Proof](#) Choose p above the kernel threshold and the projector-decay threshold. The grouped theorem gives a uniform gap $\gamma > 0$ at lengths pM , $M \geq 2$. Write $N = pM + q$, $0 \leq q < p$. If $N \geq 3p$ and $q > 0$, use a prefix of length pM and a terminal interval of length $2p + q$. Their overlap has length $2p$, so their projection defect is at most $1/2$. The finitely many terminal Hamiltonians have a common gap $\eta > 0$ on their kernel complements. Preservation under free-site extension and the two-interval estimate give the gap $\min\{\gamma, \eta\}/16$ for all $N \geq 3p$. The case $q = 0$ follows directly from the grouped bound. Taking a finite minimum with the positive gaps at $N < 3p$ completes the proof. $\square$

Theorem 12.8.33 (Uniformly gapped interaction ranges for primitive block sums). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{r-1}$ be normalized primitive tensors with common physical dimension $d \geq 1$, positive bond dimensions, and positive-definite invariant matrices. Assume that no two blocks are related by a virtual similarity and a phase. Let $\mu_j \neq 0$ for every j , and set $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$. For every prescribed lower bound R_* , there exist $R \geq \max\{1, R_*\}$, $\gamma > 0$, and N_0 such that the periodic range- R parent Hamiltonian satisfies

$$\gamma \|v\| \leq \|H_{R,N} v\|$$

for every $N \geq N_0$ and every $v \perp \ker H_{R,N}$. In particular, the block sum admits a parent interaction with a uniform positive periodic gap.

Proof. [✓ Proof](#) Choose $p \geq \max\{1, R_*\}$ by Theorem 12.8.31 and set $R = 2p$. The uniform open-chain gap on lengths pM , $M \geq 2$, gives a positive periodic gap at every sufficiently large length by Theorem 12.8.28. $\square$

Theorem 12.8.34 (Operator order after adjoining free sites). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let T be an operator on a finite-dimensional physical space and let $(e_s)_s$ be an orthonormal basis for an additional physical space. For $x = \sum_s x_s \otimes e_s$ and $y = \sum_s y_s \otimes e_s$,

$$\langle (T \otimes \mathbb{1})x, y \rangle = \sum_s \langle T x_s, y_s \rangle.$$

Consequently $T \geq 0$ implies $T \otimes \mathbb{1} \geq 0$, and $T \leq U$ implies $T \otimes \mathbb{1} \leq U \otimes \mathbb{1}$. Scalar multiplication commutes with extension: $(cT) \otimes \mathbb{1} = c(T \otimes \mathbb{1})$. If V is a unitary identification of physical spaces, then $VTV^{-1} \leq VUV^{-1}$ if and only if $T \leq U$.

Proof. [✓ Proof](#) Expand the inner product in the product orthonormal basis and sum first over the additional coordinates. Positivity follows term by term; the order assertion follows by applying positivity to $U - T$. The scalar identity follows on each coordinate. Unitary conjugation preserves and reflects positivity, hence also operator order. $\square$

Lemma 12.8.35 (Comparison on a finite interval). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor, and let R, W be interaction and interval lengths. Write $K_{R,W} = H_{W,R}^{\text{open}}(A)$ for the open parent Hamiltonian of interaction length R on W sites, and suppose that its kernel is $\mathcal{G}_W(A)$. Writing h_W for the orthogonal projection onto $\mathcal{G}_W(A)^\perp$, there are positive constants κ, C such that

$$\kappa h_W \leq K_{R,W} \leq C h_W.$$

Proof. [✓ Proof](#) The operator $K_{R,W}$ is a sum of orthogonal projections, hence positive. By hypothesis its kernel is $\mathcal{G}_W(A)$, so the orthogonal projection onto $(\ker K_{R,W})^\perp$ is h_W . Lemma 12.8.36 applied to $K_{R,W}$ gives the two bounds. $\square$

Lemma 12.8.36 (Comparison with the kernel projection). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let H be a positive operator on a finite-dimensional complex Hilbert space. If P is the orthogonal projection onto $(\ker H)^\perp$, then $aP \leq H \leq bP$ for some $a, b > 0$. More precisely, a norm gap $\gamma \geq 0$ on $(\ker H)^\perp$ implies $\gamma P \leq H$.

Proof. [✓ Proof](#) The restriction of H to $(\ker H)^\perp$ is injective and thus has a positive norm lower bound. Since H is positive, its eigenvalues on this subspace have the same lower bound. The operator norm gives the upper bound. Orthogonal projection onto $(\ker H)^\perp$ leaves the quadratic form of H unchanged. $\square$

Lemma 12.8.37 (Kernels and gaps under positive comparison). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $0 \leq P \leq Q$, then $\ker Q \subseteq \ker P$. Consequently, if P, Q are positive operators and $a, b, c > 0$ satisfy

$$aP \leq bQ \leq cP,$$

their kernels agree. In finite dimension, a norm gap $\gamma > 0$ for P on its kernel complement implies a norm gap $a\gamma/b$ for Q on the same subspace.

Proof. [✓ Proof](#) A positive quadratic form vanishes only on the kernel of its operator. Evaluating the order bounds on kernel vectors proves the inclusions. On the common kernel complement, the norm gap for P is equivalent to $P \geq \gamma I$. Hence $bQ \geq a\gamma I$, which gives the stated norm gap by Cauchy–Schwarz. $\square$

Theorem 12.8.38 (Transfer to a prescribed interaction range). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor, and let R, W be positive integers with $W \geq 2R$. Suppose that the open range- R Hamiltonian on W sites has kernel $\mathcal{G}_W(A)$. If the periodic parent Hamiltonians of range W have a uniform positive gap for all sufficiently large chain lengths, then the same holds at range R .

Proof. [✓ Proof](#) By Lemma 12.8.35, choose $\kappa, C > 0$ such that $\kappa h_W \leq K_{R,W} \leq C h_W$. Summing all cyclic translates (Theorem 12.8.40) gives

$$\kappa H_{W,N} \leq (W - R + 1) H_{R,N} \leq C H_{W,N}$$

for sufficiently large N . By Lemma 12.8.37, the periodic kernels therefore agree, and a gap γ at range W transfers to the gap $\kappa\gamma/(W - R + 1)$ at range R . $\square$

Lemma 12.8.39 (The interaction on a full interval). [Lean](#) [✓ Stmt](#) For an MPS tensor A and an interval of positive length W , the length- W parent interaction starting at the first site is the orthogonal projection h_W onto $\mathcal{G}_W(A)^\perp$.

Proof. [✓ Proof](#) Restriction to the full interval is the identity, so the local interaction is precisely the defining orthogonal projection. $\square$

Theorem 12.8.40 (Periodic comparison from an open interval). [Lean](#) [✓ Stmt](#) Let $1 \leq R \leq m$, $N \geq 2m$, and $W = m + R - 1$. Write $K_{R,W}$ for the open parent Hamiltonian of interaction length R on W sites, and $H_{R,N}$ for the corresponding periodic Hamiltonian on N sites. If real constants κ, C satisfy

$$\kappa h_W \leq K_{R,W} \leq C h_W,$$

where h_W projects onto $\mathcal{G}_W(A)^\perp$, then

$$\kappa H_{W,N} \leq m H_{R,N} \leq C H_{W,N}.$$

Proof. [✓ Proof](#) Extend each local inequality by the identity on the complementary sites and conjugate by the cyclic identification of the chosen interval. Both operations preserve operator order. Summing over all cyclic origins gives the asserted bounds: each range- R interaction occurs exactly m times, once for each of its possible positions in the open interval. $\square$

Theorem 12.8.41 (Uniform gap above an interaction threshold). [Lean](#) [✓ Stmt](#) Let $B = \bigoplus_j \mu_j A_j$, where the coefficients are nonzero and $(A_j)_j$ is a finite family of normalized primitive tensors of positive bond dimension, pairwise inequivalent under similarity and phase. Assume that each

recorded invariant density is positive definite and that the physical dimension is positive. There exists $R_0 \geq 1$ such that, for every $R \geq R_0$, there exist $\gamma_R > 0$ and N_R for which

$$\gamma_R \|v\| \leq \|H_{R,N}(B)v\|$$

for every $N \geq N_R$ and every $v \perp \ker H_{R,N}(B)$.

Proof. ✓ Proof Choose R_0 so that every open parent Hamiltonian of range at least R_0 has the expected local ground space. Fix $R \geq R_0$, and choose a uniformly gapped range $W \geq 2R$. On W sites, the range- R Hamiltonian and the orthogonal projection onto the complement of the local MPS space have the same kernel. Finite dimensionality gives two-sided positive order bounds. Summing their cyclic translates gives two-sided bounds between the periodic Hamiltonians of ranges R and W . These bounds identify their kernels and transfer the positive gap to range R . $\square$

Theorem 12.8.42 (Gap at the common block-injectivity range). Lean ✓ Stmt Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$, where $r \geq 2$, all coefficients are nonzero, and the normalized primitive tensors A_j are pairwise inequivalent under similarity and phase. Assume positive physical and bond dimensions, positive definite recorded invariant densities, and a common block-injectivity length $L_0 \geq 1$. For every $R \geq 3(r-1)(L_0+1)+1$, there exist $\gamma_R > 0$ and N_R such that

$$\gamma_R \|v\| \leq \|H_{R,N}(B)v\|$$

for every $N \geq N_R$ and every $v \perp \ker H_{R,N}(B)$. This removes the restriction to the range constructed by the overlap argument, as recorded in [74].

Proof. ✓ Proof The common block-injectivity hypothesis identifies the kernel of the open-chain range- R Hamiltonian on every interval of length at least R with the local MPS space. Choose a uniformly gapped range $W \geq 2R$. The two-sided comparison on the interval of length W and its cyclic translates transfers the gap from range W to range R . $\square$

Theorem 12.8.43 (Gap uniform in every chain length). Lean ✓ Stmt Let $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$, where $r \geq 2$, all coefficients are nonzero, and the normalized primitive tensors A_j are pairwise inequivalent under similarity and phase. Assume positive physical and bond dimensions, positive definite recorded invariant densities, and a common block-injectivity length $L_0 \geq 1$. For every $R \geq 3(r-1)(L_0+1)+1$, there exists $\gamma_R > 0$ such that, for every $N \in \mathbb{N}$ and every $v \perp \ker H_{R,N}(B)$,

$$\gamma_R \|v\| \leq \|H_{R,N}(B)v\|. \tag{12.378}$$

Proof. ✓ Proof The eventual uniform gap leaves only finitely many chain lengths untreated. At each such length the Hamiltonian, restricted to its kernel complement, has a positive lower norm bound. Taking the minimum of these bounds and the eventual uniform constant proves the assertion. $\square$

Theorem 12.8.44 (Nonzero eigenvalues of the transfer remainder). Lean ✓ Stmt Let A be primitive with nonzero bond dimension and fixed point ρ . Write $E = E_{\text{FNW},A}$ for its observable transfer map and $P = E_\infty^\rho$ for its stationary projection. Every nonzero eigenvalue of $E - P$ is an eigenvalue of E .

Proof. ✓ Proof The identities $PE = P$ and $P^2 = P$ imply $P(E - P) = 0$. If $(E - P)X = \nu X$ with $X \neq 0$ and $\nu \neq 0$, applying P gives $PX = 0$, hence $EX = \nu X$. $\square$

Theorem 12.8.45 (FNW decay rates from transfer eigenvalues). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be primitive with nonzero bond dimension and positive-definite fixed point ρ . Suppose $\lambda > 0$ and every eigenvalue $\nu \neq 1$ of its observable transfer map satisfies $|\nu| < \lambda$. Then the spectral radius of the transfer remainder in the ρ -weighted norm is less than λ . If $\text{tr } \rho = 1$, there is a constant $C_\lambda > 0$ such that, for every integer $n \geq 1$,

$$a(n) \leq C_\lambda \lambda^n. \quad (12.379)$$

This establishes the eigenvalue prescription following [191, Section 6, equation (6.1)].

Proof. [✓ Proof](#) A nonzero remainder eigenvalue is a transfer eigenvalue, and primitivity excludes the eigenvalue 1. The zero eigenvalue also has modulus less than λ . In finite dimension these strict eigenvalue bounds give a strict spectral-radius bound, to which the prescribed-rate mixing estimate applies. $\square$

Theorem 12.8.46 (Prescribed FNW prefactors after an overlap threshold). [Lean](#) [✓ Stmt](#) Let A be primitive with nonzero bond dimension and a positive-definite trace-one fixed point ρ . Suppose $\lambda > 0$ exceeds the modulus of every nonunit transfer eigenvalue. For every $c > 0$, there is an integer M such that, for every $n \geq M$,

$$a(n) \leq c \lambda^n. \quad (12.380)$$

The onset M may depend on the state, rate, and prescribed prefactor.

Proof. [✓ Proof](#) Choose an intermediate rate μ strictly between the remainder spectral radius and λ . The mixing estimate gives $a(n) \leq C_\mu \mu^n$. Since $(\mu/\lambda)^n$ tends to zero, eventually $C_\mu (\mu/\lambda)^n \leq c$. $\square$

Theorem 12.8.47 (Eventual projector bound with a prescribed prefactor). [Lean](#) [✓ Stmt](#) Under the preceding hypotheses, suppose additionally that $\lambda < 1$. For every $c > 0$, there exists an onset M such that, for every overlap $m \geq M$, every prefix length $r \geq 0$, and every suffix length $\ell \geq 1$, we have $c \lambda^m < 1$ and

$$\|P_{r;m,\ell}^{\text{tail}} P_{r+m;\ell}^{\text{left}} - G_{r+m+\ell}\| \leq \frac{c \lambda^m (1 + c \lambda^m)}{1 - c \lambda^m}. \quad (12.381)$$

The onset depends on the state, rate, and prefactor; it is not asserted to be the injectivity threshold. This is the version of [191, Section 6, equation (6.1)] with the additional onset stated explicitly [83].

Proof. [✓ Proof](#) Increase the onset until the overlap exceeds an injectivity length and $c \lambda^m < 1$. The preceding mixing estimate then satisfies the hypotheses of the fixed-overlap FNW projector estimate, uniformly in the two exterior lengths. $\square$

12.8.1 The dimension-only prefactor at a prescribed rate

Definition 12.8.1.1 (Weighted matrix-unit path vectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $t = (t_{ij})$ and $z = (z_{ij})$ be real matrices indexed by a finite set of k vertices. Set $A^{ij} = t_{ij} E_{ij}$. For endpoints a, f , define the four-site vector

$$v_{a,f}(z) = \sum_{b,c,e} z_{be} t_{ab} t_{bc} t_{ce} t_{ef} |ab, bc, ce, ef\rangle.$$

Write $\mathbf{1}$ for the mark with all entries one and, for a real function u on the vertices, $u \otimes \mathbf{1}$ and $\mathbf{1} \otimes u$ for the marks $z_{be} = u_b$ and $z_{be} = u_e$. For $X \in M_k(\mathbb{C})$, the four-site boundary vector of A has the coordinates

$$\Gamma_4(X)(i_0 j_0, i_1 j_1, i_2 j_2, i_3 j_3) = \text{tr}(A^{i_0 j_0} A^{i_1 j_1} A^{i_2 j_2} A^{i_3 j_3} X).$$

Theorem 12.8.1.2 (Path contractions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Writing $T_{abcef} = t_{ab} t_{bc} t_{ce} t_{ef}$, one has

$$\begin{aligned} \langle v_{a,f}(z), v_{a,f}(z') \rangle &= \sum_{b,c,e} z_{be} z'_{be} T_{abcef}^2, \\ \langle v_{a,f}(z), \Gamma_4(X) \rangle &= X_{fa} \sum_{b,c,e} z_{be} T_{abcef}^2. \end{aligned}$$

For every configuration $\sigma = (i_0 j_0, i_1 j_1, i_2 j_2, i_3 j_3)$ and every real function u on the vertices,

$$\begin{aligned} v_{a,f}(u \otimes \mathbf{1})(\sigma) &= u_{j_0} v_{a,f}(\mathbf{1})(\sigma), \\ v_{a,f}(\mathbf{1} \otimes u)(\sigma) &= u_{i_3} v_{a,f}(\mathbf{1})(\sigma). \end{aligned}$$

Proof. [Stmnt](#) Distinct vertex paths (b, c, e) give distinct physical basis vectors $|ab, bc, ce, ef\rangle$, which gives the first identity. For the second, multiply four consecutive letters:

$$\begin{aligned} A^{ab} A^{b'c} A^{c'e} A^{e'f} &= \delta_{bb'} \delta_{cc'} \delta_{ee'} T_{abcef} E_{af}, \\ \text{tr}(E_{af} X) &= X_{fa}. \end{aligned}$$

Hence $\Gamma_4(X)(ab, bc, ce, ef) = T_{abcef} X_{fa}$ on every vertex path. On the support of $v_{a,f}$, the first letter ab determines $b = j_0$ and the last letter ef determines $e = i_3$, which gives the two mark identities. $\square$

Theorem 12.8.1.3 (Spectator multipliers preserve interval spaces). [Lean](#) [Lean](#) [Stmnt](#) Let $K, L \in \mathbb{N}$ and $X \in M_D(\mathbb{C})$. Let φ be a complex function of the prefix configurations $u: \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$ and ψ a complex function of the suffix configurations $\tau: \{0, \dots, L-1\} \rightarrow \{0, \dots, d-1\}$. Then the vector $(u, \tau) \mapsto \varphi(u) \Gamma_{K+L}(X)(u, \tau)$ belongs to the range of $\Gamma_{K,L}^{\text{tail}}$, and the vector $(u, \tau) \mapsto \psi(\tau) \Gamma_{K+L}(X)(u, \tau)$ belongs to the range of $\Gamma_{K,L}^{\text{left}}$.

Proof. [Stmnt](#) Factor the matrix product at the boundary of the interval:

$$\begin{aligned} \varphi(u) \Gamma_{K+L}(X)(u, \tau) &= \text{tr}(A^\tau \varphi(u) X A^u) = \Gamma_{K,L}^{\text{tail}}(Y)(u, \tau), \\ \psi(\tau) \Gamma_{K+L}(X)(u, \tau) &= \text{tr}(A^u \psi(\tau) A^\tau X) = \Gamma_{K,L}^{\text{left}}(Z)(u, \tau), \end{aligned}$$

where $Y_u = \varphi(u) X A^u$ and $Z_\tau = \psi(\tau) A^\tau X$. $\square$

Theorem 12.8.1.4 (Matrix-unit boundary vectors). [Lean](#) [Stmnt](#) For the weighted matrix-unit tensor $A^{ij} = t_{ij} E_{ij}$ and all endpoints a, f , one has $v_{a,f}(\mathbf{1}) = \Gamma_4(E_{fa})$.

Proof. [Stmnt](#) For every configuration,

$$\text{tr}(A^{i_0 j_0} A^{i_1 j_1} A^{i_2 j_2} A^{i_3 j_3} E_{fa}) = \delta_{i_0 a} \delta_{j_0 i_1} \delta_{j_1 i_2} \delta_{j_2 i_3} \delta_{j_3 f} t_{i_0 j_0} t_{i_1 j_1} t_{i_2 j_2} t_{i_3 j_3}.$$

The nonzero coordinates of $\Gamma_4(E_{fa})$ are therefore those of the paths (ab, bc, ce, ef) , with value $t_{ab} t_{bc} t_{ce} t_{ef}$. These are the coordinates of $v_{a,f}(\mathbf{1})$. $\square$

Proof. ✓ Proof The four displayed sign vectors are pairwise orthogonal and have squared norm four. Multiplication gives the identities. Each entry of P belongs to $\{1/8, 1/4, 3/8\}$; its row and column sums are one. $\square$

Theorem 12.8.1.10 (Scalar path sums). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Regard P , u and w above as real. Then $P_{ij} > 0$, $\sum_j P_{ij} = \sum_j P_{ji} = 1$ and $u_i^2 = w_i^2 = 1$ for every i , and the identities $Pu = 0$, $w^\top P = 0$, $P^2 = Q + uw^\top/64$ and $P^3 = Q$ of Theorem 12.8.1.9 hold over $\mathbb{R}$. For every pair of endpoints a, f , set $p_{abcef} = P_{ab}P_{bc}P_{ce}P_{ef}$. Then*

$$\sum_{b,c,e} p_{abcef} = \frac{1}{4}, \quad \sum_{b,c,e} u_b p_{abcef} = 0, \quad \sum_{b,c,e} u_b w_e p_{abcef} = \frac{1}{64}. \quad (12.387)$$

The amplitudes $t_{ij} = \sqrt{P_{ij}}$ are strictly positive and satisfy $t_{ij}^2 = P_{ij}$.

Proof. ✓ Proof The real identities are those of Theorem 12.8.1.9, read in $\mathbb{R}$. For the path sums, apply the identities for P^2 and P^3 , and use $u_b^2 = w_e^2 = 1$. The assertions concerning square roots follow from positivity. $\square$

Theorem 12.8.1.11 (A dimension-four physical projection defect). Lean ✓ Stmt *For the transition matrix P and weighted matrix-unit tensor $A^{ij} = \sqrt{P_{ij}}E_{ij}$ defined above, let G_I denote the orthogonal projection onto the ground space on the interval I , extended by the identity on the remaining sites. Then*

$$\|G_{[1,3]}G_{[0,2]} - G_{[0,3]}\| \geq \frac{1}{16}.$$

Proof. ✓ Proof Fix external vertices a, f and write $T_{abcef} = \sqrt{P_{ab}P_{bc}P_{ce}P_{ef}}$. Consider

$$\begin{aligned} x &= 2 \sum_{b,c,e} u_b T_{abcef} |ab, bc, ce, ef\rangle, \\ y &= 2 \sum_{b,c,e} w_e T_{abcef} |ab, bc, ce, ef\rangle. \end{aligned}$$

With $t_{ij} = \sqrt{P_{ij}}$ these are $x = 2v_{a,f}(u \otimes \mathbf{1})$ and $y = 2v_{a,f}(\mathbf{1} \otimes w)$. By Theorem 12.8.1.5, x belongs to the range of $G_{[1,3]}$ and y belongs to the range of $G_{[0,2]}$. The path contraction identities give $\|x\| = \|y\| = 1$, $x \perp \text{range } G_{[0,3]}$, and $\langle y, x \rangle = 1/16$. Thus the matrix coefficient of $G_{[0,2]}G_{[1,3]} - G_{[0,3]}$ between y and x is $1/16$. Taking the adjoint gives the asserted order of the two projections. $\square$

Theorem 12.8.1.12 (Weighted matrix-unit normalization). Lean Lean ✓ Stmt *Let $A_{ij} = t_{ij}E_{ij}$, where t is a real matrix. If every entry of t is nonzero, then the letters of A span the full matrix algebra. If the squared entries in every column sum to one, then*

$$\sum_{i,j} A_{ij}^* A_{ij} = I. \quad (12.388)$$

Proof. ✓ Proof When $t_{ij} \neq 0$, the matrix unit E_{ij} is a scalar multiple of A_{ij} . The normalization follows from $A_{ij}^* A_{ij} = t_{ij}^2 E_{jj}$ and the column-sum assumption. $\square$

Theorem 12.8.1.13 (Transfer map of a weighted matrix-unit tensor). [Lean](#) [✓ Stmt](#) Let $t = (t_{ij})$ be a real matrix and define physical letters $A_{ij} = t_{ij}E_{ij}$. The associated transfer map satisfies

$$\mathcal{E}_A(X) = \text{diag} \left(\sum_j t_{ij}^2 X_{jj} \right)_i. \quad (12.389)$$

Proof. [✓ Proof](#) Multiply the matrix units in $\sum_{i,j} A_{ij} X A_{ij}^\dagger$. Only the diagonal entries of X contribute. $\square$

Theorem 12.8.1.14 (The positive Markov matrix-unit channel). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For the four-dimensional matrix P above, let $A_{ij} = \sqrt{P_{ij}}E_{ij}$. Then A is injective at length one, its transfer map $\mathcal{E}_A$ is unital and trace preserving, and every matrix X satisfies

$$\mathcal{E}_A^3(X) = \frac{\text{tr}(X)}{4} I. \quad (12.390)$$

These are the channel properties of the explicit construction used to examine [191, Section 6, equation (6.1)].

Proof. [✓ Proof](#) The positive coefficients ensure that the physical letters span all matrix units. Column stochasticity gives trace preservation, and row stochasticity gives unitality. On diagonal matrices the transfer map acts by P , while it annihilates off-diagonal matrix units. Therefore $P^3 = ee^\top/4$ gives the asserted formula. $\square$

Theorem 12.8.1.15 (A faithful primitive fixed point). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For the weighted matrix-unit tensor of the Markov example, the matrix $\rho = I/4$ is positive definite, has trace one, and is stationary. The tensor has a complementary transfer spectral gap at ρ . Every nonunit eigenvalue of its FNW observable transfer map is zero.

Proof. [✓ Proof](#) The row sums of the Markov matrix give stationarity of $I/4$. For a nonunit eigenvalue ν of the trace-preserving transfer map, trace preservation forces an associated eigenvector X to have trace zero. The depolarizing third iterate then gives $\nu^3 X = 0$, and hence $\nu = 0$. The bilinear trace adjoint has the same eigenvalues, giving the assertion for the FNW observable transfer map. One-site injectivity gives irreducibility. Together with the unique peripheral eigenvalue and the faithful stationary density, this implies the complementary spectral gap. $\square$

Theorem 12.8.1.16 (Failure of the dimension-only coefficient at a prescribed rate). [Lean](#) [Lean](#) [✓ Stmt](#) There exists an injective matrix-product tensor of physical dimension 16 and auxiliary dimension 4 with a faithful primitive stationary density matrix of trace one such that every eigenvalue other than 1 of its observable transfer map has modulus smaller than $\lambda = 1/1000$. Nevertheless, at overlap length $m = 2$,

$$16\lambda^2 < 1, \quad \left\| G_{[1,3]} G_{[0,2]} - G_{[0,3]} \right\| > \frac{16\lambda^2(1 + 16\lambda^2)}{1 - 16\lambda^2}.$$

Thus the dimension-only coefficient and the prescribed-rate assertion following Nachtergaele's bound cannot hold simultaneously at this fixed overlap length; see [83].

Proof. [✓ Proof](#) Take the weighted matrix-unit tensor associated with the transition matrix above and the stationary density $I/4$. Its transfer map becomes completely depolarizing after three steps, so every nonunit eigenvalue of the observable transfer map is zero. The rate $1/1000$ is therefore admissible. The physical projection defect is at least $1/16$, whereas the proposed coefficient is strictly smaller than $1/16$. $\square$

Theorem 12.8.1.17 (Block injectivity from a simultaneous word span). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ be tensors with bond dimensions $D_0, \dots, D_{r-1}$, and let $L \in \mathbb{N}$. Suppose that their simultaneous word evaluations at length L span $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$. Then every A^j is L -block injective.

Proof. ✓ Proof Let $M \in M_{D_j}(\mathbb{C})$ and let X be the tuple with $X_j = M$ and $X_k = 0$ for $k \neq j$. By hypothesis,

$$X = \sum_{|\omega|=L} c_\omega(A^{0,\omega}, \dots, A^{r-1,\omega})$$

for some coefficients c_ω . Its j -th component is $M = \sum_{|\omega|=L} c_\omega A^{j,\omega}$, so the length- L words of A^j span $M_{D_j}(\mathbb{C})$. □

Theorem 12.8.1.18 (Separation from simultaneous injectivity). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ be tensors with positive bond dimensions. Suppose that their simultaneous word evaluations at length S span the product of their matrix algebras. Then no two distinct blocks are gauge-phase equivalent. This is a consequence of the simultaneous block-injectivity condition in [16, Section IV.C].

Proof. ✓ Proof The local ground spaces of the blocks at length S are independent. Each block is S -block injective by Theorem 12.8.1.17, so each boundary map at that length is injective, and its value at the identity matrix, the periodic matrix product vector $V^{(S)}(A^i) = \Gamma_S^{A^i}(\mathbb{1})$, is nonzero. Suppose that $i \neq j$ and $A^{i,a} = \zeta X A^{j,a} X^{-1}$ for every physical index a . Then

$$V^{(S)}(A^i) = \zeta^S V^{(S)}(A^j) \in G_S(A^i) \cap G_S(A^j) = \{0\},$$

so $\Gamma_S^{A^i}(\mathbb{1}) = 0$, contradicting injectivity of the boundary map. □

Theorem 12.8.1.19 (Propagation of a positive-length simultaneous span). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ have positive bond dimensions and a common physical dimension. Suppose that their simultaneous length- S word tuples span the product matrix algebra, where $S > 0$. Then their word tuples span the same algebra at every length $n \geq S$. This is the propagation of block injectivity used in [16, Section IV.C].

Proof. ✓ Proof Each block is S -block injective and hence normal. Rescale and conjugate each block into normalized primitive form. Both operations preserve the simultaneous span at S . Trace preservation propagates that span to $n \geq S$. Inverse conjugation and rescaling return the full span to the original family at length n . □

Theorem 12.8.1.20 (Open ground spaces at a simultaneous injectivity length). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ have positive physical and bond dimensions. Suppose that their simultaneous length- S word tuples span the product matrix algebra, where $S > 0$. Set $B = \bigoplus_j \mu_j A^j$, with every $\mu_j \neq 0$. For every $S+1 \leq R \leq N$, $\ker H_{N,R}^{\text{open}}(B) = G_N^{\text{ES}}(B)$. This sufficient range follows from the inversion and regrowth argument in [16, Section IV.C, lines 2114–2129].

Proof. ✓ Proof Normalize the blocks while preserving their simultaneous span at S and every component local space. Trace preservation gives the span at all lengths at least S , so the open-chain intersection theorem gives the kernel identity for the normalized block sum. The original and normalized sums have equal local spaces at R and N ; hence their open Hamiltonians and their spaces G_N^{ES} agree. □

Theorem 12.8.1.21 (Open-chain gaps at a simultaneous injectivity length). [Lean](#) [✓ Stmt](#) Let $A^0, \dots, A^{r-1}$ have positive bond dimensions and common physical dimension $d \geq 1$. Suppose that their simultaneous length- S word tuples span the product matrix algebra, where $S > 0$. Let $B = \bigoplus_j \mu_j A^j$, with every $\mu_j \neq 0$. For every interaction range $R \geq S + 1$, there exists $\delta > 0$ such that

$$\delta \|v\| \leq \|H_{N,R}^{\text{open}}(B)v\|$$

for every $N \geq R$ and every $v \perp G_N(B)$. No normalization or inequivalence assumption on the original blocks is required. This is the finite-interval conclusion of [191, Theorem 1.2 and Section 6] at the sufficient interaction ranges supplied by block injectivity.

Proof. [✓ Proof](#) Independently normalize the blocks into primitive form while preserving their simultaneous span at S and all component local ground spaces. Simultaneous injectivity separates these normalized blocks. The all-length primitive theorem gives a uniform gap at some range $2p$ with $p \geq R$. At range R , the open kernel is G_N for every $N \geq R$. On the interval of length $2p$, finite-dimensional comparison gives $\kappa h_{2p} \leq H_{2p,R}^{\text{open}}$ for some $\kappa > 0$. The interval comparison transfers the long-range gap to range R at every $N \geq 2p$, with loss $2p - R + 1$. A finite minimum includes $R \leq N < 2p$. Equality of the local spaces returns the same Hamiltonians and ground spaces for the original weighted block tensor. $\square$

Theorem 12.8.1.22 (Dimension-bound open gaps for normal blocks). [Lean](#) [✓ Stmt](#) Let A^j be normal tensors with common physical dimension $d \geq 1$, pairwise inequivalent under virtual similarity and phase. Set $D_* = \max\{\sum_j D_j, 1\}$ and $B = \bigoplus_j \mu_j A^j$, with every $\mu_j \neq 0$. For every $R \geq 3D_*^5$, there exists $\delta > 0$ such that $\delta \|v\| \leq \|H_{N,R}^{\text{open}}(B)v\|$ for every $N \geq R$ and every $v \perp G_N(B)$.

Proof. [✓ Proof](#) The dimension-bound blocking theorem gives a positive simultaneous injectivity length S with $S + 1 \leq 3D_*^5$. Apply the open-chain gap theorem at $R \geq S + 1$. $\square$

Theorem 12.8.1.23 (Open parent gaps for canonical tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a canonical tensor with physical dimension $d \geq 1$, and choose its distinct normal representatives B^j . At every positive interaction range, its open parent Hamiltonian agrees with that of $\bigoplus_j B^j$, including all multiplicities and the ambient reconstruction of A . If the representatives are simultaneously injective at $S > 0$ and $R \geq S + 1$, then $\ker H_{N,R}^{\text{open}}(A) = G_N(A)$ for every $N \geq R$, and there is $\delta > 0$ such that

$$\delta \|v\| \leq \|H_{N,R}^{\text{open}}(A)v\|$$

for every $N \geq R$ and every $v \perp G_N(A)$. If the bond dimension D is positive, the conclusion holds at every $R \geq 3D^5$ without a supplied injectivity length.

Proof. [✓ Proof](#) The canonical reconstruction preserves every positive-length local ground space. This identifies the open parent Hamiltonians and their boundary-condition spaces with those of the distinct representatives. Apply the block open-chain gap and kernel theorems. For the dimension bound, the blocking estimate supplies $S > 0$ with $S + 1 \leq 3D^5$. $\square$

Definition 12.8.1.24 (Open sums of a local interaction). [Lean](#) [✓ Stmt](#) For a local operator h on R sites, let $H_N^{\text{open}}(h)$ be the sum of its translates over starts i satisfying $0 \leq i$ and $i + R \leq N$. Each translate acts as the identity on the remaining physical sites.

Lemma 12.8.1.25 (Properties of open interaction sums). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Open interaction sums preserve order and positivity, commute with scalar multiplication, and vanish for the zero interaction. When $R > 0$, the sum of the canonical parent projection is the canonical open parent Hamiltonian.

Proof. [✓ Proof](#) Extension by the identity and physical coordinate isometries preserve order, positivity, and scalar multiplication. Summation preserves these properties. Extending the local parent projection gives each canonical parent term. $\square$

Theorem 12.8.1.26 (Open gaps for fixed positive parent interactions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let $R > 0$, and let $h \geq 0$ have kernel $G_R(A)$. There are $\kappa, C > 0$, independent of N , such that

$$\kappa H_{N,R}^{\text{open}}(A) \leq H_N^{\text{open}}(h) \leq C H_{N,R}^{\text{open}}(A).$$

Hence a uniform canonical open-chain gap $\gamma > 0$ for $N \geq R$ gives the gap $\kappa\gamma$ for $H_N^{\text{open}}(h)$ on its kernel complement. In particular, this applies to every fixed positive parent interaction of a canonical tensor when its normal representatives are simultaneously injective at $S > 0$ and $R \geq S + 1$.

Proof. [✓ Proof](#) Finite-dimensional positivity and the prescribed local kernel give $\kappa h_R(A) \leq h \leq C h_R(A)$. Sum the translated comparisons. The two positive operators bound one another by positive constants, so their kernels coincide. The lower bound transfers the canonical gap to the common kernel complement. $\square$

Theorem 12.8.1.27 (Compact comparison for open parent interactions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let K be a compact parameter set. Suppose the local ground projections of A_x are continuous, and let h_x be a continuous family of positive interactions with kernel $G_R(A_x)$, where $R > 0$. If the canonical open parent Hamiltonians have one positive gap for every $x \in K$ and every $N \geq N_0$, then the interaction sums $H_N^{\text{open}}(h_x)$ also have one positive gap for those parameters and lengths.

For a fixed canonical tensor A , this conclusion requires no supplied canonical gap: simultaneous injectivity at $S > 0$ and $R \geq S + 1$ give one gap uniform over $x \in K$ and every $N \geq R$. When its bond dimension D is positive, $R \geq 3D^5$ suffices without a supplied injectivity length.

Proof. [✓ Proof](#) Compactness and continuity give a single $\kappa > 0$ with $\kappa h_R(A_x) \leq h_x$ for every $x \in K$. Each interaction also has a finite positive upper comparison constant. The preceding proof transfers the supplied gap with the common lower constant; the upper constants identify the kernels and do not affect the resulting gap. For a fixed canonical tensor, apply its all-length open gap theorem. $\square$

Theorem 12.8.1.28 (Compact gaps on bounded open intervals). [Lean](#) [Lean](#) [✓ Stmt](#)
Let K be a compact parameter set, and let A_x^j be continuous block tensors with positive physical and bond dimensions. Suppose that their length- S word tuples span the product matrix algebra for every parameter, where $S > 0$. Set $B_x = \bigoplus_j \mu_j(x) A_x^j$, with all weights nonzero; the weights need not be continuous. Fix $R \geq S + 1$ and a finite upper length M . There is $\delta > 0$ such that

$$\delta \|v\| \leq \|H_{N,R}^{\text{open}}(B_x)v\|$$

for every $x \in K$, every $R \leq N \leq M$, and every $v \perp \ker H_{N,R}^{\text{open}}(B_x)$. The constant may depend on M .

Proof. ✓ Proof At each fixed $N \geq R$, simultaneous injectivity gives continuous local ground projections at lengths R and N . The open Hamiltonian depends continuously on the first projection, and its kernel is G_N . The compact finite-dimensional norm-gap theorem supplies one positive bound at this N . Taking the minimum over $R \leq N \leq M$ gives the claim. Nonzero weights preserve every local ground space. $\square$

Theorem 12.8.1.29 (Open-chain gaps at a prescribed simultaneous injectivity range). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ have positive physical and bond dimensions. Suppose that their simultaneous length- S word tuples span the product matrix algebra, where $S > 0$. Set $B = \bigoplus_j \mu_j A^j$, with every $\mu_j \neq 0$, and fix $R \geq S + 1$. There are $p > 0$, with $R \leq 2p$, and $\delta > 0$ such that

$$\delta \|v\| \leq \|H_{pM,R}^{\text{open}}(B)v\|$$

for every $M \geq 2$ and every $v \perp G_{pM}^{\text{ES}}(B)$. This combines the block-injective local spaces of [16, Section IV.C] with the martingale estimate of [191, Theorem 2.1(ii) and Section 6].

Proof. ✓ Proof Normalize the blocks while preserving all local spaces and the span at S . The simultaneous span separates the blocks. Choose $p \geq R$ and a uniform open-chain gap $\gamma > 0$ at range $2p$ on every length pM , $M \geq 2$. The range- R and range- $2p$ open kernels both equal G_{pM}^{ES} . On the interval of length $2p$, finite-dimensional comparison gives $\kappa h_{2p} \leq H_{2p,R}^{\text{open}}$ with $\kappa > 0$. Open-range comparison transfers the gap to $\delta = \kappa\gamma/(2p - R + 1) > 0$ at range R . Equality of all local spaces transports this bound to the original block sum. $\square$

Theorem 12.8.1.30 (Strict Knabe intervals at a simultaneous injectivity range). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ have positive physical and bond dimensions. Suppose that their simultaneous length- S word tuples span the product matrix algebra, where $S > 0$. Set $B = \bigoplus_j \mu_j A^j$, with every $\mu_j \neq 0$, and fix $R \geq S + 1$. There are $m \geq R$ and $\gamma > 0$ such that $(R - 1)^2 < m\gamma$ and

$$\gamma \|v\| \leq \|H_{m+R-1,R}^{\text{open}}(B)v\|$$

for every $v \perp G_{m+R-1}^{\text{ES}}(B)$. The strict inequality is the finite-range threshold of Theorem 12.3.7.81; the coefficient is discussed in `docs/paper-gaps/knabe88_finite_range_coefficient.tex`.

Proof. ✓ Proof Choose $p > 0$ and $\delta > 0$ giving the open-chain gap on all lengths pM , $M \geq 2$. Choose $n \in \mathbb{N}$ with $(R - 1)^2 < n\delta$. Set $M = n + 2R + 2$ and $m = pM + 1 - R$. Then $m \geq R$, $n \leq m$, and $m + R - 1 = pM$. Thus $\gamma = \delta$ has both the required gap and $(R - 1)^2 < m\gamma$. $\square$

Theorem 12.8.1.31 (A uniform gap at the simultaneous injectivity range). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ have positive physical and bond dimensions. Suppose their word tuples of a supplied positive length S span the full product matrix algebra. Let $B = \bigoplus_{j=0}^{r-1} \mu_j A^j$, where every μ_j is nonzero. For each interaction range $R \geq S + 1$, there is a constant $\gamma_R > 0$ such that

$$\gamma_R \|v\| \leq \|H_{R,N} v\| \tag{12.391}$$

for every chain length $N \in \mathbb{N}$ and every $v \perp \ker H_{R,N}$, where $H_{R,N}$ is the canonical periodic parent Hamiltonian of B . This is the block-injective gap assertion of [16, Section IV.C] at the simultaneous injectivity range.

Proof. ✓ Proof Normalize the blocks while retaining simultaneous injectivity at S and all local ground spaces. Simultaneous injectivity forces distinct blocks to be inequivalent. The grouped martingale estimate and finite-range periodic comparison provide a gapped interaction range

$W \geq 2R$. Let B' be the weighted sum of the normalized blocks. Trace preservation propagates simultaneous injectivity from S to every larger length, so, since $R \geq S + 1$, the open-chain ground-space theorem gives

$$\ker H_{W,R}^{\text{open}}(B') = G_W^{\text{ES}}(B').$$

Two-sided comparison of the periodic range- R and range- W Hamiltonians therefore transfers the gap to R . Taking a minimum with the finitely many remaining finite-volume gaps makes the bound uniform over every N . Finally, normalization left the original parent Hamiltonians unchanged. $\square$

Theorem 12.8.1.32 (Canonical parent Hamiltonians and normal representatives). [Lean](#) [✓ Stmt](#) *Let A be a tensor in canonical form, with distinct normal representatives $B_0, \dots, B_{g-1}$, and set $B = \bigoplus_j B_j$. For every positive interaction range R and every chain length N , the canonical periodic parent Hamiltonians satisfy*

$$H_{R,N}(A) = H_{R,N}(B).$$

This equality includes the multiplicity copies and the ambient reconstruction of the original tensor.

Proof. [✓ Proof](#) For every positive length R , the local spaces satisfy

$$G_R(A) = \bigvee_j G_R(B_j) = G_R(B).$$

Their orthogonal complementary projections therefore agree. Translating and summing those projections gives the same Hamiltonian. $\square$

Theorem 12.8.1.33 (Canonical gap for every canonical decomposition). [Lean](#) [✓ Stmt](#) *Let A be a tensor in canonical form with positive physical dimension. Fix any canonical decomposition of A , and let $B_0, \dots, B_{g-1}$ be its distinct normal representatives. Let S be a positive integer such that the simultaneous length- S word tuples of $B_0, \dots, B_{g-1}$ span the product of the block matrix algebras. Then, for every $R \geq S + 1$, there exists $\gamma > 0$ such that, for every chain length N and every vector $v \perp \ker H_{R,N}(A)$,*

$$\gamma \|v\| \leq \|H_{R,N}(A)v\|.$$

Proof. [✓ Proof](#) Set $B = \bigoplus_j B_j$. Its blocks satisfy the hypotheses of the uniform gap at the simultaneous injectivity range, so $H_{R,N}(B)$ has a positive uniform gap. Since $H_{R,N}(A) = H_{R,N}(B)$ for every N , the same constant works for A , without changing the range or the chain length. $\square$

Theorem 12.8.1.34 (Canonical gap at a supplied simultaneous injectivity length). [Lean](#) [✓ Stmt](#) *Let A be a tensor in canonical form with positive physical dimension. One may choose a basis of normal tensors $B_0, \dots, B_{g-1}$ such that the following holds for every positive integer S : if the simultaneous length- S word tuples span the product of the block matrix algebras, then, for every $R \geq S + 1$, there exists $\gamma > 0$ such that, for every chain length N and every vector $v \perp \ker H_{R,N}(A)$,*

$$\gamma \|v\| \leq \|H_{R,N}(A)v\|.$$

The representatives are chosen once, before S and R . This includes canonical forms with repeated blocks, as in [16, Section IV.C].

Proof. ✓ Proof Fix one canonical decomposition of A and take its distinct normal representatives, which form a basis of normal tensors for A . The preceding theorem applies to this fixed choice for every S and every $R \geq S + 1$. $\square$

Theorem 12.8.1.35 (A dimension bound for canonical parent gaps). Lean ✓ Stmt *Let A be a tensor in canonical form with positive physical dimension and positive bond dimension D . For every $R \geq 3D^5$, there is $\gamma > 0$ such that*

$$\gamma \|v\| \leq \|H_{R,N}(A)v\|$$

for every $N \in \mathbb{N}$ and every $v \perp \ker H_{R,N}(A)$. The constant may depend on A and R , but not on N . This sufficient range combines the blocking estimate of [11, lines 317–345] with the gap assertion of [16, Section IV.C]. It need not be optimal; the sharper sufficient condition $R \geq S + 1$ is recorded in [74]. The unrestricted assertion at shorter ranges is false; see [68].

Proof. ✓ Proof Choose a canonical decomposition. The dimension estimate supplies a simultaneous injectivity length $S > 0$ for its distinct normal representatives, with $S + 1 \leq 3D^5 \leq R$. The canonical gap at that simultaneous injectivity length gives the assertion. $\square$

Theorem 12.8.1.36 (Comparison of positive parent interactions). Lean ✓ Stmt *Let h be a positive interaction on R sites whose kernel is the local MPS space $G_R(A)$, in the sense of [16, Section IV.C]. Write $h_R = I - P_{G_R(A)}$ for the canonical parent projection. There are constants $\kappa, C > 0$ such that*

$$\kappa h_R \leq h \leq C h_R. \quad (12.392)$$

Proof. ✓ Proof Apply the finite-dimensional comparison with the projection onto the orthogonal complement of the kernel of h . This projection is h_R . $\square$

Definition 12.8.1.37 (Periodic sum of a fixed interaction). Lean Lean ✓ Stmt *Let h act on the physical space of R consecutive sites. For $R \leq N$, let U_i be the unitary which places the cyclic interval of length R starting at i first and the remaining sites second. Define*

$$h_i = U_i^*(h \otimes I)U_i, \quad H_N(h) = \sum_{i=0}^{N-1} h_i.$$

When $R > N$, set each translated interaction, and hence their sum, to zero. This agrees with the convention for the canonical parent Hamiltonian.

Theorem 12.8.1.38 (Order and positivity of periodic interaction sums). Lean Lean Lean ✓ Stmt *Cyclic extension and its periodic sum preserve operator inequalities and commute with scalar multiplication. In particular, for every volume N ,*

$$\begin{aligned} h \leq k &\implies H_N(h) \leq H_N(k), \\ H_N(ch) &= cH_N(h), & H_N(0) &= 0, \\ h \geq 0 &\implies H_N(h) \geq 0. \end{aligned}$$

The scalar identity holds for every complex number c .

Proof. ✓ Proof Extension by the identity preserves order and scalar multiplication, as does unitary conjugation. Summing over the cyclic positions preserves these relations. Positivity is the comparison with zero. $\square$

Theorem 12.8.1.39 (Canonical parent projections give the canonical Hamiltonian). [Lean](#) [Lean](#)

[✓ Stmt](#) For a tensor A and a positive interaction range R , the periodic sum of the canonical parent projection $h_R = I - P_{G_R(A)}$ is exactly the canonical parent Hamiltonian, at every volume:

$$H_N(h_R) = H_{R,N}(A).$$

Proof. [✓ Proof](#) In cyclic active-interval coordinates, each translated canonical term is the full-interval parent projection extended by the identity on the spectator sites. Conjugating back identifies the terms individually. When $R > N$, both definitions give zero. $\square$

Theorem 12.8.1.40 (Uniform comparison of periodic parent Hamiltonians). [Lean](#) [✓ Stmt](#) Let $R > 0$ and let h be a fixed positive operator with kernel $G_R(A)$, and let $H_{R,N}(A)$ be the canonical periodic parent Hamiltonian. There are $\kappa, C > 0$, independent of N , such that

$$\kappa H_{R,N}(A) \leq H_N(h) \leq C H_{R,N}(A). \quad (12.393)$$

Proof. [✓ Proof](#) Theorem 12.8.1.35 gives $\kappa, C > 0$ with $\kappa h_R \leq h \leq C h_R$. Each translated term h_i is h extended by the identity and conjugated by the unitary U_i , so these inequalities hold term by term, $\kappa(h_R)_i \leq h_i \leq C(h_R)_i$. Summing over the cyclic translates and using the monotonicity and homogeneity of the periodic sum from Theorem 12.8.1.37 gives

$$\kappa H_N(h_R) \leq H_N(h) \leq C H_N(h_R).$$

Theorem 12.8.1.38 identifies $H_N(h_R)$ with $H_{R,N}(A)$. $\square$

Theorem 12.8.1.41 (Uniform gap for a fixed positive parent interaction). [Lean](#) [✓ Stmt](#) Let A be a matrix product tensor and let $R > 0$. Suppose that the periodic Hamiltonians formed from the orthogonal projection onto $G_R(A)^\perp$ have a positive gap uniformly in the chain length. Let h be a fixed positive operator on R sites with kernel $G_R(A)$, and let $H_N(h)$ be the sum of its cyclic translates. Then there is $\delta > 0$ such that, for every chain length N and every $v \perp \ker H_N(h)$,

$$\delta \|v\| \leq \|H_N(h)v\|. \quad (12.394)$$

For $N < R$ both periodic Hamiltonians are defined to be zero.

Proof. [✓ Proof](#) Finite-dimensional positivity gives constants $\kappa, C > 0$ such that $\kappa h_R \leq h \leq C h_R$, where $h_R = I - P_{G_R(A)}$ is the canonical parent projection. Cyclic extension and summation give the same inequalities for the periodic Hamiltonians, with constants independent of N . The two inequalities imply equality of the kernels. Restriction to their common orthogonal complement therefore transfers a canonical gap γ to the gap $\kappa\gamma$. This is the projector reduction of [16, Section IV.C]. $\square$

Theorem 12.8.1.42 (Positive parent interactions for every canonical decomposition). [Lean](#)

[✓ Stmt](#) Let A be a tensor in canonical form with positive physical dimension. Fix any canonical decomposition of A , and let $B_0, \dots, B_{g-1}$ be its distinct normal representatives. Suppose that $S > 0$ and the tuples of length- S words span the product of the block matrix algebras. For every $R \geq S + 1$ and every fixed positive operator h on R sites with kernel $G_R(A)$, there is $\gamma > 0$ such that, for every N and every $v \perp \ker H_N(h)$,

$$\gamma \|v\| \leq \|H_N(h)v\|. \quad (12.395)$$

Proof. [✓ Proof](#) Apply the canonical projector gap of Theorem 12.8.1.32 at range R , then the comparison theorem for the fixed interaction h . The constants depend on h but not on N . $\square$

Theorem 12.8.1.43 (Positive parent interactions at the simultaneous injectivity range). [Lean](#) [✓ Stmt](#) Let A be a tensor in canonical form with positive physical dimension. One may choose a basis of normal tensors $B_0, \dots, B_{g-1}$ such that the following holds. Suppose that $S > 0$ and the tuples of length- S words span the product of the block matrix algebras. For every $R \geq S + 1$ and every fixed positive operator h on R sites with kernel $G_R(A)$, there is $\gamma > 0$ such that, for every N and every $v \perp \ker H_N(h)$,

$$\gamma \|v\| \leq \|H_N(h)v\|. \quad (12.396)$$

The representatives are chosen once, independently of S , R , and h .

Proof. [✓ Proof](#) Apply the canonical projector gap at range R , then the comparison theorem for the fixed interaction h . The constants depend on h but not on N . $\square$

Theorem 12.8.1.44 (A dimension bound for fixed positive parent interactions). [Lean](#) [✓ Stmt](#) Let A be a tensor in canonical form with positive physical dimension and positive bond dimension D . Let $R \geq 3D^5$ and let h be any fixed positive operator on R sites with $\ker h = G_R(A)$. Then there is $\gamma > 0$ such that

$$\gamma \|v\| \leq \|H_N(h)v\|$$

for every $N \in \mathbb{N}$ and every $v \perp \ker H_N(h)$. The constant may depend on A , R , and h , but not on N . This is the positive-interaction comparison of [16, Section IV.C] at the sufficient dimension bound of [11, lines 317–345].

Proof. [✓ Proof](#) The canonical projector interaction has a uniform positive periodic gap at range R . Comparison with the fixed interaction h preserves the kernel and gives a positive gap independent of N . $\square$

12.9 Prescribed gaps at sufficiently long interaction ranges

Write $H_{R,N}$ for the periodic parent Hamiltonian on N sites with interaction range R . The local interactions in this section are orthogonal projectors with coefficient one. The range increases with the requested gap; no rescaling of the Hamiltonian is made. The bounds follow from the projector-decay and martingale estimates of [191, Theorem 2.1 and Section 6] and the finite-range comparison. They do not improve the gap at a prescribed shorter interaction range.

Theorem 12.9.1 (A prescribed periodic gap from an open-chain estimate). [Lean](#) [✓ Stmt](#) Let $0 < \eta < 1$, $1 \leq R \leq m$, and $2(R-1)^2 \leq m\eta$. Suppose the open parent Hamiltonian on $m+R-1$ sites has gap at least $1 - \eta/2$. Then, for every $N \geq 2m$ and every $v \perp \ker H_{R,N}$,

$$(1 - \eta)\|v\| \leq \|H_{R,N}v\|. \quad (12.397)$$

Proof. [✓ Proof](#) Put $l = R - 1$ and $\gamma = 1 - \eta/2$. The finite-range comparison gives $\delta = (m\gamma - l^2)/(m - l)$. Its denominator is positive, and

$$(m - l)(\delta - (1 - \eta)) = m\eta/2 - l^2 + (1 - \eta)l \quad (12.398)$$

$$\geq 0. \quad (12.399)$$

Thus $\delta \geq 1 - \eta > 0$, which also verifies the positive-numerator hypothesis of the comparison. $\square$

Theorem 12.9.2 (Preservation of finite-interval gaps). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $R > 0$. If the range- R parent Hamiltonian on n sites has norm gap $\kappa > 0$ on its kernel complement, its extension to the prefix of an $(n + q)$ -site chain has the same lower bound. Likewise, if $R \leq W \leq n \leq N$ and the range- R Hamiltonian on W sites has gap at least κ , its interaction sum on $[n - W, n]$, acting on the N -site chain, has gap at least κ . The physical coordinate isometry identifies this interval interaction sum with the W -site Hamiltonian tensored with the identity on the remaining sites. These are the finite-interval gaps used in condition C2 of [191].

Proof. [✓ Proof](#) For the finite-interval operator T and its ground projection P , the supplied gap gives $\kappa(I - P) \leq T$. Extension by free sites preserves this inequality. The kernel projection extends to $P \otimes I$, and the physical coordinate isometry transports both the Hamiltonian and this projection to the chosen interval. Thus the extended operator dominates κ times its excitation projection. On the kernel complement, the quadratic-form bound and Cauchy–Schwarz give the asserted norm bound. $\square$

Theorem 12.9.3 (Kernels of embedded terminal Hamiltonians). [Lean](#) [Lean](#) [✓ Stmt](#) Let $0 < R \leq W \leq n \leq N$, and suppose $\ker H_{W,R}^{\text{open}}(A) = G_W(A)$. Then the kernel of the range- R interaction sum on $[n - W, n]$, acting on N sites, is the extension of $G_W(A)$ by all free sites. It is also the kernel of the single parent interaction on the whole W -site interval. In particular, for $W = L + Q$ and $N = K + L + Q$, its kernel on the terminal interval is the reassociated tail boundary space from the three-interval estimate.

Proof. [✓ Proof](#) The physical coordinate isometry places the active interval first and the free sites second. The resulting operator acts independently on each free-site coordinate. Its kernel consists precisely of vectors whose restrictions to those coordinates belong to the active kernel $G_W(A)$. The full-window parent interaction has this same kernel. Reassociation gives the asserted terminal boundary space. $\square$

Theorem 12.9.4 (Combining gaps on two open intervals). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let H be the range- R parent Hamiltonian on an open chain of length N , and let H_L and H_T be its interaction sums on a prefix of length n and on a terminal interval of length W , respectively. Write P_L , P_T , and P for their orthogonal ground projections. Suppose $\kappa \geq 0$, and suppose both interval Hamiltonians have a norm gap at least κ on their respective kernel complements. If $\|P - P_L P_T\| \leq 1/2$, then $(\kappa/16)\|v\| \leq \|Hv\|$ for every $v \perp \ker H$. The interval interaction sums always satisfy $H_L \leq H$ and $H_T \leq H$.

Proof. [✓ Proof](#) Each interval interaction sum is a subfamily of the positive terms of H , so $H_L + H_T \leq 2H$. The interval gaps give $\kappa(P_L^\perp + P_T^\perp) \leq H_L + H_T$. For $v \perp \ker H$, the projection defect gives $\|P_L P_T v\| \leq \|v\|/2$. Decomposing $v = P_L P_T v + P_L P_T^\perp v + P_L^\perp v$ yields $\|v\| \leq 2(\|P_L^\perp v\| + \|P_T^\perp v\|)$. Squaring gives $\|v\|^2 \leq 8(\|P_L^\perp v\|^2 + \|P_T^\perp v\|^2)$. Hence $\kappa\|v\|^2 \leq 16\langle Hv, v \rangle$. Cauchy–Schwarz and cancellation of $\|v\|$ prove the claim. $\square$

Theorem 12.9.5 (Prescribed open-chain projector defects). [Lean](#) [✓ Stmt](#) Let A be a normalized primitive tensor with positive bond dimension and a positive-definite invariant matrix. For every $a > 0$, every sufficiently large integer l satisfies $l > 1$, l -block injectivity, and a number $\epsilon_l \geq 0$ with $\epsilon_l \sqrt{l+1} < a$ such that, for every prefix length $K \geq 0$,

$$\|G_{[K, K+l+1]} G_{[0, K+l]} - G_{[0, K+l+1]}\| \leq \epsilon_l. \quad (12.400)$$

Here G_I denotes the projection onto the matrix-product boundary space on the indicated interval, extended by the identity on its complement in the chain of length $K + l + 1$.

normal, pairwise inequivalent blocks. Its simultaneous injectivity length is exactly two. Its two-site local ground space is $\text{Sym}^2(\mathbb{C}^2)$, and its canonical parent interaction is the singlet projection $h_2 = (I - F)/2$, where F exchanges the two physical sites.

Proof. [✓ Proof](#) The word tuples for 00, 11 and 01 are $(1, 0, 1/2)$, $(0, 1, 1/2)$ and $(0, 0, 1/2)$, which span $\mathbb{C}^3$; two one-site words cannot span this space. Each scalar block has transfer normalization one. Simultaneous spanning separates the blocks and gives eventual independence of their periodic vectors. The two-site boundary vectors span $|00\rangle$, $|11\rangle$ and $|++\rangle$, exactly the symmetric subspace. Its orthogonal complement is the singlet line. $\square$

Theorem 12.9.10 (Failure of the unrestricted parent-Hamiltonian gap assertion). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For the preceding tensor, the periodic range-two parent Hamiltonian has a nonzero eigenvector of eigenvalue $\varepsilon_N = 1 - \cos(2\pi/N) > 0$ for every $N \geq 2$. These eigenvalues tend to zero. Consequently there are no $\delta > 0$ and N_0 such that

$$\delta\|v\| \leq \|H_{N,2}v\| \quad (N \geq N_0, v \perp \ker H_{N,2}).$$

Thus the unrestricted assertion in [16, Section IV.C, lines 2183–2187] is false. The sufficient-range theorems remain valid, since this example has $R = S = 2$. See [68].

Proof. [✓ Proof](#) On configurations with one down spin the sum of singlet projections acts as $f(j) - (f(j+1) + f(j-1))/2$. The first Fourier character of $\mathbb{Z}/N\mathbb{Z}$ is therefore an eigenvector with the stated eigenvalue. Its coefficient at the zero site is one, so it is nonzero. Self-adjointness makes it orthogonal to the kernel. A uniform positive norm bound would give $\delta \leq \varepsilon_N$ for every sufficiently large N , contradicting $\varepsilon_N \rightarrow 0$. $\square$

12.10 Continuity of the parent-Hamiltonian gap

Along a continuous path of tensors, the phase argument of [204, Appendix A, lines 2475–2580] needs a gap of the parent Hamiltonians that is uniform both in the path parameter and in the chain length. The source obtains it from Nachtergaele’s bound, whose finite-size input is the gap of the Hamiltonian restricted to a fixed number of sites; that gap has a uniform lower bound because the restricted Hamiltonian depends continuously on the parameter [204, lines 2575–2578]. This section proves the continuity step and combines it with Knabe’s nearest-neighbor criterion and compactness. The single-block results apply to normal tensors with a common injectivity length p , at every fixed interaction range $R > p$. Quantum Wielandt supplies $p = D^4$ from normality alone. The sharper one-site injective result has range two. The source’s path also covers normal forms with several blocks; the scope of the compact-gap conclusions is recorded in [152].

Theorem 12.10.1 (Gap stability under perturbation of operator and kernel projection). [Lean](#) [Lean](#) [✓ Stmt](#) Let H, H_0, P, P_0 be bounded operators on a complex normed space, and suppose $\delta\|v - P_0v\| \leq \|H_0v\|$ for all v , with $\delta \geq 0$, $\|H - H_0\| \leq \varepsilon$ and $\|P - P_0\| \leq r$. Then $(\delta(1-r) - \varepsilon)\|v\| \leq \|Hv\|$ for every v with $Pv = 0$. Consequently, if $x \mapsto H_x$ and the orthogonal projection onto a subspace K_x are continuous at x_0 , $K_{x_0} \subseteq \ker H_{x_0}$, and $\delta\|v\| \leq \|H_{x_0}v\|$ on $K_{x_0}^\perp$ with $\delta > 0$, then every $\delta' < \delta$ satisfies $\delta'\|v\| \leq \|H_xv\|$ on $K_x^\perp$ for all x near x_0 .

Proof. [✓ Proof](#) For $Pv = 0$ one has $\|P_0v\| = \|(P - P_0)v\| \leq r\|v\|$ and $\|H_0v\| \leq \|Hv\| + \varepsilon\|v\|$; combining these with the hypothesis gives the first claim. For the second, the gap on $K_{x_0}^\perp$ and $K_{x_0} \subseteq \ker H_{x_0}$ give $\delta\|v - P_{x_0}v\| \leq \|H_{x_0}v\|$ for all v , and one chooses $\varepsilon = (\delta - \delta')/2$ and $r = \varepsilon/\delta$. $\square$

Theorem 12.10.2 (Open-chain gaps at every sufficient range). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let A be injective at a length $p > 0$, and let $R > p$. There is $\delta > 0$ such that

$$\delta \|v\| \leq \|H_{N,R}^{\text{open}}(A)v\| \quad (N \geq R, v \perp \ker H_{N,R}^{\text{open}}(A)).$$

In particular, for a normal tensor of bond dimension $D > 0$, every fixed $R \geq D^4 + 1$ has this property. No normalization is required.

Proof. [✓ Proof](#) Injectivity propagates to length $R - 1$. The long-chain martingale estimate and range comparison give a positive gap at range R . The exact open kernel identity and finite dimensionality handle the finitely many shorter admissible chains. Quantum Wielandt supplies $p = D^4$ for a normal tensor. $\square$

Theorem 12.10.3 (Continuity of injective range projectors). [Lean](#) [✓ Stmt](#) Let E be a finite-dimensional Hilbert space, F a Hilbert space, and $T_x: E \rightarrow F$ a continuous family of injective bounded maps. Then the orthogonal projection $T_x(T_x^\dagger T_x)^{-1}T_x^\dagger$ onto the range of T_x depends continuously on x .

Proof. [✓ Proof](#) The Gram operator $T_x^\dagger T_x$ is invertible and continuous in x , and inversion is continuous at invertible elements of a Banach algebra. $\square$

Theorem 12.10.4 (Continuity of finite-window parent Hamiltonians). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $x \mapsto A_x$ be a continuous family of tensors. For fixed L , the boundary map $X \mapsto \sum_\sigma \text{tr}(A_{x,\sigma} X) |\sigma\rangle$ on L sites is continuous in x . If every A_x is L -block injective, the orthogonal projection onto the local ground space on L sites is continuous in x . Hence, for $L \leq N$, the open and periodic parent Hamiltonians with interaction range L on N sites are continuous in x .

Proof. [✓ Proof](#) Each matrix entry of the boundary map is a trace of a fixed product of tensor letters. Block injectivity makes the boundary map injective, and its range is the local ground space, so the previous theorem applies. The Hamiltonians are finite sums of translates of one minus that projection. $\square$

Theorem 12.10.5 (Local stability of a finite-window gap). [Lean](#) [✓ Stmt](#) Let $x \mapsto A_x$ be continuous, $R \leq N$, and suppose every A_x is R -block and N -block injective. If $\delta \|v\| \leq \|H_{N,R}^{\text{open}}(A_{x_0})v\|$ for all $v \perp G_N^{\text{ES}}(A_{x_0})$, with $\delta > 0$, then for every $\delta' < \delta$ and every x near x_0 , $\delta' \|v\| \leq \|H_{N,R}^{\text{open}}(A_x)v\|$ for all $v \perp G_N^{\text{ES}}(A_x)$. This is the continuity step of [204, lines 2575–2578].

Proof. [✓ Proof](#) The open parent Hamiltonian annihilates the ground space, and both it and the ground-space projection are continuous in x . $\square$

Lemma 12.10.6 (Comparison of open-chain ranges). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $0 < R \leq W \leq N$. If $\kappa h_W \leq H_{W,R}^{\text{open}}$ for the range- W parent interaction h_W and some $\kappa > 0$, then $\kappa H_{N,W}^{\text{open}} \leq (W - R + 1)H_{N,R}^{\text{open}}$. For a tensor for which both open kernels equal G_N^{ES} , a gap $\gamma \|v\| \leq \|H_{N,W}^{\text{open}}v\|$ on $(G_N^{\text{ES}})^\perp$ gives $\frac{\kappa\gamma}{W-R+1} \|v\| \leq \|H_{N,R}^{\text{open}}v\|$ on the same complement. The assertion may be used at one volume or uniformly over every $N \geq W$. For a one-site injective tensor with $R = 2$ and $W \geq 2$, the kernel hypotheses hold automatically.

Proof. [✓ Proof](#) Translate the local comparison to every window of W consecutive sites and sum; each range- R term lies in at most $W - R + 1$ windows. Both open Hamiltonians have kernel G_N^{ES} , so the operator inequality transfers the gap. $\square$

Theorem 12.10.7 (Strict Knabe windows for injective tensors). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a one-site injective tensor of nonzero bond dimension. Then the open-chain Hamiltonian $H_{N,2}^{\text{open}}(A)$ built from the canonical two-site parent interaction has a positive gap above $G_N^{\text{ES}}(A)$, uniform in all sufficiently large N . In particular, there are $m \geq 2$ and γ with $m\gamma > 1$ such that $\gamma\|v\| \leq \|H_{m+1,2}^{\text{open}}(A)v\|$ for all $v \perp G_{m+1}^{\text{ES}}(A)$.

Proof. [✓ Proof](#) Gauge A to a primitive tensor with a positive-definite fixed point, which has the same local ground spaces, and take its open-chain gap $\gamma_W\|v\| \leq \|H_{N,W}^{\text{open}}(A)v\|$ at some range $W \geq 2$ for all $N \geq W$. The finite-interval comparison gives $\kappa > 0$ with $\kappa h_W \leq H_{W,2}^{\text{open}}(A)$, and Lemma 12.10.6 with $R = 2$ yields, for every $N \geq W$ and every $v \perp G_N^{\text{ES}}(A)$,

$$\frac{\kappa\gamma_W}{W-1}\|v\| \leq \|H_{N,2}^{\text{open}}(A)v\|. \quad (12.404)$$

Writing $\delta = \kappa\gamma_W/(W-1)$, choose n with $n\delta > 1$ and put $m = n + W$; then $m\delta > 1$ and (12.404) holds at $N = m + 1$. $\square$

Theorem 12.10.8 (Gap in a parameter neighborhood). [Lean](#) [✓ Stmt](#) Let $x \mapsto A_x$ be a continuous family of one-site injective tensors of nonzero bond dimension, let $m \geq 2$, and suppose $\gamma\|v\| \leq \|H_{m+1,2}^{\text{open}}(A_{x_0})v\|$ for $v \perp G_{m+1}^{\text{ES}}(A_{x_0})$ with $m\gamma > 1$. Then there is $\delta > 0$ such that, for all x near x_0 and all $N \geq 2m$, the periodic parent Hamiltonian $H_{N,2}(A_x)$ satisfies $\delta\|v\| \leq \|H_{N,2}(A_x)v\|$ on the orthogonal complement of its kernel.

Proof. [✓ Proof](#) Choose $1/m < \gamma' < \gamma$. The strict window persists with γ' near x_0 , and Knabe's nearest-neighbor criterion turns it into a periodic gap for every $N \geq 2m$ with a constant depending only on m and γ' . $\square$

Theorem 12.10.9 (Spectral gap along an ordered interpolation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If every eigenvalue z of a Hermitian matrix M has $\text{Re } z = 0$ or $\text{Re } z \geq \delta$, then $\delta\|x\|^2 \leq \text{Re}\langle x, Mx \rangle$ for all $x \perp \ker M$. Conversely, if $M \geq 0$ satisfies this quadratic-form bound, then every eigenvalue z of M has $\text{Re } z \geq 0$, and $\text{Re } z = 0$ or $\text{Re } z \geq \delta$. Let $0 \leq A \leq B$ be matrices with $\ker A \subseteq \ker B$ and such a gap δ for A , and let $t \geq 0$. Then $C_t = A + t(B - A)$ is positive semidefinite, $\ker C_t = \ker A$, and every eigenvalue z of C_t satisfies $\text{Re } z = 0$ or $\text{Re } z \geq \delta$. If $\ker A \neq 0$, then 0 is an eigenvalue of every C_t .

Proof. [✓ Proof](#) Since $t(B - A) \geq 0$, a vector annihilated by C_t has $\langle x, Ax \rangle = 0$, hence $Ax = 0$, and conversely $Ax = 0$ forces $Bx = 0$. On $(\ker A)^\perp$ one has $\langle x, C_t x \rangle \geq \langle x, Ax \rangle \geq \delta\|x\|^2$. The equivalence with the spectral statement follows by expanding x in an orthonormal eigenbasis of the Hermitian matrix. $\square$

Lemma 12.10.10 (Uniform bounds from local bounds on a compact set). [Lean](#) [✓ Stmt](#) Let S be a compact subset of a topological space and $P(\delta, N, x)$ a property of a real δ , a natural number N and a point x that is preserved by decreasing δ and by increasing N . If every $x \in S$ has $\delta > 0$ and N such that $P(\delta, N, y)$ holds for all y near x , then one $\delta > 0$ and one N satisfy $P(\delta, N, x)$ for all $x \in S$.

Proof. [✓ Proof](#) Take a finite subcover of the neighborhoods, the minimum of the finitely many constants and the maximum of the finitely many thresholds. $\square$

Theorem 12.10.11 (Uniform finite-dimensional kernel gap). [Lean](#) [✓ Stmt](#) Let E be a finite-dimensional complex Hilbert space, let $x \mapsto H_x: E \rightarrow E$ be continuous, and suppose the orthogonal projections onto $\ker H_x$ are continuous in x . For every compact parameter set S , there is $\delta > 0$ such that $\delta\|v\| \leq \|H_x v\|$ for every $x \in S$ and $v \perp \ker H_x$. No positivity or self-adjointness is required.

continuous family of blocks whose length- S word tuples span the product of their matrix algebras, with $S > 0$. The projection onto the sum of the local block spaces is continuous at every length at least S . For the weighted block sum, the periodic kernel projection is continuous whenever $S + 1 \leq R \leq N$. Nonzero block weights may vary arbitrarily. Thus each such fixed volume has a positive gap uniform on any compact parameter set. The overlaps of two local block spaces are also continuous at any common injectivity length.

Proof. ✓ Proof The joint boundary map has fixed domain, is injective, and has range equal to the sum of the local block spaces. Its range projection therefore depends continuously on the parameter. Restricting the boundary matrices to scalar multiples of the identity gives the injective map onto the periodic ground space. The kernel identity then gives continuity of the periodic kernel projection. Compactness yields a common positive finite-volume gap. $\square$

Theorem 12.10.16 (Compact families of several normal blocks). Lean Lean Lean Lean ✓ Stmt
Under the preceding simultaneous spanning hypothesis, fix $R \geq S + 1$. On a compact parameter set there is $\delta > 0$ such that the canonical periodic parent Hamiltonian of the weighted block sum satisfies

$$\delta \|v\| \leq \|H_{N,R}(x)v\| \quad (N \geq R, v \perp \ker H_{N,R}(x)).$$

In particular, for pairwise inequivalent normalized normal blocks of dimensions $D_j > 0$, one may choose any $R \geq 3 \max(\sum_j D_j, 1)^5$. The simultaneous spanning length is then derived from the dimensions. This includes the two-site interaction of the initially blocked path in [204, Appendix A]; see [152].

Proof. ✓ Proof The open-chain multiblock gap gives a strict finite-range Knabe window at each parameter. Continuity preserves the strict inequality in a neighborhood. A finite subcover gives a common gap for all sufficiently long rings. The preceding fixed-volume result handles the finitely many remaining lengths, including $N = R$. For normalized normal blocks, the simultaneous word-span bound supplies a common length $S = 3 \max(\sum_j D_j, 1)^5 - 1$. $\square$

Theorem 12.10.17 (Continuous positive parent interactions). Lean Lean Lean Lean Lean ✓ Stmt
Suppose the local ground-space projection and a positive parent interaction h_x depend continuously on x , with the same local kernel. On a compact parameter set there is $\kappa > 0$ such that $h_x \geq \kappa P_{G_R(x)^\perp}$. Consequently every canonical periodic gap uniform in the parameter and chain length transfers to the chosen interactions. This applies, in particular, to compact normal families at every fixed range $R \geq D^4 + 1$.

Proof. ✓ Proof Compactness gives a common positive local gap away from the kernel. Positivity turns it into the stated order bound. Translation and summation preserve this comparison. A pointwise upper comparison identifies the periodic kernels, and the lower comparison transfers the canonical gap with factor κ . $\square$

Theorem 12.10.18 (Invertible changes of the physical space). Lean Lean Lean Lean Lean ✓ Stmt
Let $(A^j)_j$ have simultaneous spanning length $S > 0$. An invertible change M of the physical space preserves this spanning property. More generally, a rectangular change with a left inverse preserves the joint word span. If $x \mapsto M_x$ is continuous and invertible, then on a compact parameter set the weighted block sums of $(M_x A^j)_j$ have a uniform periodic parent gap at every fixed $R \geq S + 1$, for all $N \geq R$. All block weights are nonzero.

Proof. ✓ Proof Expanding each transformed word as a linear combination of original words gives one span inclusion. Applying a left inverse gives the reverse inclusion. Continuity of the transformed letters and the compact-family gap theorem give the assertion. $\square$

Chapter 13

Block-Injective Parent Hamiltonians and Degenerate Ground Spaces

For a block-injective tensor $A^i = \bigoplus_j \mu_j A_j^i$, the local ground spaces decompose according to the normal blocks. For sufficiently long chains and upon satisfying boundary-closure conditions, open-segment intersection relations and trace identities separate distinct blocks, identifying the periodic ground space with $\text{span}\{V^{(N)}(A_j)\}_j$, as in [16, Theorem IV.16] and [193, Theorem 12].

13.1 Block injectivity and degenerate ground spaces

For an MPS with several canonical blocks, the relevant local space is the linear sum of the block ranges. Simultaneous word-span separation, the one-step restriction intersection, and the global change-of-cut boundary comparison culminate in the source-normalized kernel theorem (Theorem 13.4.2).

Definition 13.1.1 (Vector space generated by a basis of normal tensors). Lean ✓ Stmt For a basis of normal tensors $A_1, \dots, A_g$, the vector space generated by their length- N matrix product vectors is $\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$.

Lemma 13.1.2 (BNT span as the sum of one-vector spaces). Lean ✓ Stmt For a family $A_0, \dots, A_{r-1}$ and a chain length N , let $M_j = \text{span}\{V^{(N)}(A_j)\}$. Then

$$\bigvee_{j=0}^{r-1} M_j = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

Proof. ✓ Proof Both sides are the span of the same family of vectors. □

Lemma 13.1.3 (Inclusion of a block ground space). Lean ✓ Stmt For each block index j with $\mu_j \neq 0$, and whenever $0 < N$ and $L \leq N$, the periodic-boundary ground space of the block A_j embeds into the corresponding ground space of the direct-sum tensor:

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right).$$

Proof. [✓ Proof](#) The definition by length- L windows on the ring reduces the inclusion to the local block-embedding statement that each ground vector of A_j on the window lifts to the direct sum by the identity

$$\Gamma_L^{\oplus_k \mu_k A_k}(\iota_j(\mu_j^{-L} X)) = \Gamma_L^{A_j}(X).$$

Here ι_j inserts a matrix in the j -th diagonal block and has zero off that block. Thus $\iota_j(\mu_j^{-L} X)$ has exactly the local vector of the j -th block on that window. $\square$

Lemma 13.1.4 (Forward inclusion from the block ground spaces). [Lean](#) [✓ Stmt](#) Assume $\mu_j \neq 0$ for every block index j . Whenever $0 < N$ and $L \leq N$, the linear sum of the blockwise ground spaces with periodic boundary conditions is contained in the direct-sum tensor's ground space with periodic boundary conditions:

$$\sum_j \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right),$$

where j ranges over all block indices.

Proof. [✓ Proof](#) Take the linear sum of the inclusions $\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}(\bigoplus_k \mu_k A_k)$ over all block indices j . $\square$

13.2 Block separation and intersection

Block-separating words imply that the simultaneous block-word tuples $\omega \mapsto (A_1^\omega, \dots, A_g^\omega)$ span $\prod_j M_{D_j}(\mathbb{C})$ at a single length. Together with the normalization $\sum_a A_a^j A_a^{j\dagger} = I$, this yields the one-step intersection identity $\bigvee_j G_{n+2}(A^j)$ used for the degenerate parent-Hamiltonian ground space.

Definition 13.2.1 (Pairwise block-separating equations). [Lean](#) [Lean](#) [✓ Stmt](#) Fix a finite target set T of block indices. A length- S block-separating equation for k against T is a choice of coefficients c_ω such that

$$\sum_{|\omega|=S} c_\omega A_\ell^\omega = \begin{cases} I, & \ell = k, \\ 0, & \ell \in T. \end{cases}$$

The pairwise version requires this equation for each ordered pair of distinct blocks, with $T = \{j\}$.

Definition 13.2.2 (Trace-dual pair product-span condition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For two blocks A and B , the length- S pair word tuple is $w \mapsto (A^w, B^w)$. We also use the subalgebra of $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$ generated by the one-letter pairs (A_i, B_i) . The pair product-span condition says that the length- S tuples span $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$. Equivalently, in the trace-dual formulation of the same product-span condition, the only pair of test matrices (Δ_A, Δ_B) whose trace pairing with every length- S tuple vanishes is $(0, 0)$. The cumulative variant replaces one fixed length by all word lengths at most S , and the all-length variant allows every finite word.

Lemma 13.2.3 (Three-block trace reduction from a nonzero middle matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be tensors with virtual bond dimensions D_A and D_B . Assume that A is L -block

injective. Let $\Delta_A \in M_{D_A}(\mathbb{C})$ with $\Delta_A \neq 0$, and let $\Delta_B \in M_{D_B}(\mathbb{C})$. Suppose that, for all words u, v, t of length L ,

$$\text{tr}(\Delta_A A_v A_t A_u) + \text{tr}(\Delta_B B_v B_t B_u) = 0.$$

Then for every $Z \in M_{D_A}(\mathbb{C})$ there is a matrix $W \in M_{D_B}(\mathbb{C})$ such that, for all words t of length L , $\text{tr}(Z A_t) + \text{tr}(W B_t) = 0$. This is the trace-dual algebraic step in the two-block case of the direct-sum lemma of [193].

Proof. ✓ Proof Since $\Delta_A \neq 0$, Lemma C.1.5 writes Z as a linear combination of matrices $A_u \Delta_A A_v$. For one such term, cyclicity of the trace rewrites the three-block relation as

$$\text{tr}(A_u \Delta_A A_v A_t) + \text{tr}(B_u \Delta_B B_v B_t) = 0.$$

Taking the same linear combination of the matrices $B_u \Delta_B B_v$ gives the required W , and linearity gives the general case. $\square$

Lemma 13.2.4 (Finite-chain image inclusion from a three-block relation). Lean Lean Lean Lean ✓ Stmt Under the hypotheses of Lemma 13.2.3, the length- L spaces $\mathcal{G}_L^A$ and $\mathcal{G}_L^B$ satisfy $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$. If the corresponding nonzero middle matrix exists on both sides, then $\mathcal{G}_L^A = \mathcal{G}_L^B$.

Proof. ✓ Proof Write a vector in $\mathcal{G}_L^A$ as $t \mapsto \text{tr}(A_t Z)$. The preceding lemma supplies W with $\text{tr}(Z A_t) + \text{tr}(W B_t) = 0$ for every word t . Cyclicity of the trace gives $\text{tr}(A_t Z) = \text{tr}(Z A_t) = -\text{tr}(W B_t) = \text{tr}(B_t(-W))$, so the vector lies in $\mathcal{G}_L^B$. The equality statement follows by applying the same argument with the two blocks exchanged. $\square$

Lemma 13.2.5 (Dimension step for two-block local spaces). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt If A and B are length- L block-injective, the map $Z \mapsto (t \mapsto \text{tr}(A_t Z))$ has image dimension D_A^2 , and similarly for B . Therefore an inclusion $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$, together with $D_B \leq D_A$, forces $D_A = D_B$ and $\mathcal{G}_L^A = \mathcal{G}_L^B$. In particular, the three-block trace relation from Lemma 13.2.4 gives this equality whenever the ordering $D_B \leq D_A$ is available. If the inclusion comes from the strictly larger block, $D_B < D_A$, then the same dimension step already contradicts the existence of the three-block trace relation. More generally, let H be a source hypothesis which excludes the simultaneous conclusion

$$\begin{aligned} D_A &= D_B, \\ \mathcal{G}_L^A &= \mathcal{G}_L^B. \end{aligned}$$

Then the three-block intersection vanishes:

$$H \implies \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\}.$$

With injectivity at the three-block length, the strict-size branch gives

$$D_B < D_A, \quad \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\} \implies \text{span}\{(A^\omega, B^\omega) : |\omega| = 3L\} = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}).$$

One such source hypothesis is non-proportionality of the periodic MPS vectors at a sufficiently long chain length: $\mathbb{C} V^{(N)}(A) \neq \mathbb{C} V^{(N)}(B)$. Indeed,

$$\mathcal{G}_L^A = \mathcal{G}_L^B \implies \mathcal{G}_{N,L}(A) = \mathcal{G}_{N,L}(B) \implies \mathbb{C} V^{(N)}(A) = \mathbb{C} V^{(N)}(B),$$

by equality of the periodic-boundary constraint spaces, followed by injective uniqueness. For same-dimensional normalized BNT representatives that are not gauge-phase equivalent, the non-equivalence condition supplies this non-proportionality at arbitrarily large chain lengths; the direct-sum injectivity hypotheses remain explicit. Combining the equal-dimension and unequal-dimension branches over all ordered block pairs gives one common positive length S_0 with

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = S_0\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

For an ordered pair of blocks, the trace-separation conclusion has the form

$$[\forall |\omega| = S, \text{tr}(\Delta_A A^\omega) + \text{tr}(\Delta_B B^\omega) = 0] \implies \Delta_A = \Delta_B = 0.$$

Equivalently, the homogeneous pair span is full:

$$\text{span}\{(A^\omega, B^\omega) : |\omega| = S\} = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}).$$

Concatenating words propagates this equality to positive multiples of S , and hence to a finite period window for every ordered pair. Combining that window with the conditional period-window homogenization theorem gives a single length S_0 such that

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = S_0\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

The same conclusion applies to the corresponding normal canonical-form hypotheses with a basis of normal tensors once one-site injectivity is supplied explicitly, since those hypotheses carry the irreducibility and normalization assumptions used in the direct-sum comparison.

Proof. ✓ Proof Block injectivity makes the boundary-to-chain map injective, so $\dim \mathcal{G}_L^A = D_A^2$ and $\dim \mathcal{G}_L^B = D_B^2$. Inclusion gives $D_A^2 \leq D_B^2$, while $D_B \leq D_A$ gives the reverse inequality. Hence $D_A = D_B$, and equality of dimensions inside the inclusion gives $\mathcal{G}_L^A = \mathcal{G}_L^B$. In the strict-size branch this contradicts $D_B < D_A$. For the conditional directness statement, suppose $\psi \in \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B$ has boundary representations $\psi = \Gamma_{3L}^A(X) = \Gamma_{3L}^B(Y)$. If $X = 0$, injectivity of the A boundary-to-chain map gives $\psi = 0$. Otherwise equality of the coefficient formulas gives $\forall |t| = 3L, \quad \text{tr}(XA^t) - \text{tr}(YB^t) = 0$, equivalently, $\forall |t| = 3L, \quad \text{tr}(XA^t) + \text{tr}((-Y)B^t) = 0$. The dimension step applied to this homogeneous three-block trace relation gives $D_A^2 \leq D_B^2$, contradicting $D_B < D_A$ in the strict-size branch. In the equal-size branch, equal local spaces impose the same length- L constraints on the N -site ring:

$$\mathcal{G}_L^A = \mathcal{G}_L^B \implies \mathcal{G}_{N,L}(A) = \mathcal{G}_{N,L}(B).$$

The injective uniqueness theorem identifies these periodic ground spaces with the one-dimensional MPS spans,

$$\begin{aligned} \mathcal{G}_{N,L}(A) &= \mathbb{C} V^{(N)}(A), \\ \mathcal{G}_{N,L}(B) &= \mathbb{C} V^{(N)}(B), \end{aligned}$$

and hence $\mathbb{C} V^{(N)}(A) = \mathbb{C} V^{(N)}(B)$, contradicting non-proportionality of the two MPS vectors. $\square$

Proof. ✓ Proof Let $M = (M_j)_j$ be a target tuple. Write

$$M_j = \sum_{|u|=L} a_u A_u^j,$$

$$I = \sum_{|v|=S} b_v A_v^j$$

simultaneously for every block j . Multiplying the two expansions gives

$$M_j = M_j I = \sum_{|u|=L} \sum_{|v|=S} a_u b_v A_u^j A_v^j = \sum_{|u|=L} \sum_{|v|=S} a_u b_v A_{uv}^j.$$

The words uv have length $L + S$, so the target tuple lies in the length- $L + S$ span. $\square$

Lemma 13.2.12 (Period-window propagation for blockwise product spans). Lean Lean ✓ Stmt Let $p > 0$. Suppose the length- p simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$, and suppose that for every $0 \leq r < p$, the length- $s + r$ simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$. Then there is a length N such that, for every $n \geq N$, the length- n simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$. More concretely, one may take $N = s$: writing $n = s + r + qp$, with $0 \leq r < p$, the span at length $s + r$ propagates to the span at length n by multiplying q times by the period- p span.

Proof. ✓ Proof The Euclidean division of $n - s$ by p gives $n = s + r + qp$ with $0 \leq r < p$. The assumed residue-window span gives the full product algebra at length $s + r$. Applying homogeneous padding with the length- p span once gives the full product algebra at length $s + r + p$; iterating gives the full product algebra at length $s + r + qp$. $\square$

Lemma 13.2.13 (Blockwise word span from block-selector equations). Lean ✓ Stmt Let J be a finite set of blocks, and let $(A_j)_{j \in J}$ be a family of tensors such that every A_j is L -block injective. Suppose that for each block k there are coefficients $c_\omega^{(k)}$, indexed by words of length S , such that

$$\sum_{|\omega|=S} c_\omega^{(k)} A_\omega^j = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

Then the simultaneous length- $(L + S)$ block-word evaluations span $\prod_{j \in J} M_{D_j}(\mathbb{C})$.

Proof. ✓ Proof For a target tuple $(M_j)_{j \in J}$, block injectivity gives coefficients $a_u^{(k)}$ such that $M_k = \sum_{|u|=L} a_u^{(k)} A_k^u$. Multiply this expansion by the selector equation for k and sum over k . For every block j this gives

$$M_j = \sum_{k \in J} \sum_{|u|=L} \sum_{|v|=S} a_u^{(k)} c_v^{(k)} A_j^u A_j^v.$$

Each product $A_j^u A_j^v$ is the evaluation of the concatenated word uv , which has length $L + S$. Hence the length- $(L + S)$ word tuples span the full product algebra. $\square$

13.2.1 Trace identities for block intersections

Once the simultaneous block-word tuples span the full product algebra at a single length, non-degeneracy of the product trace pairing turns blockwise trace equalities into blockwise matrix identities $A_b^j C_a^j = A_b^j E^j A_a^j$. The lemmas here use those identities to place left-boundary trace decompositions in the block sum $\bigvee_j G_{n+2}(A^j)$ and to obtain the one-step block-intersection identity.

Lemma 13.2.1.1 (Arbitrary interpolation on one block of a pair). Lean ✓ Stmt Let $(A^\ell)_{\ell=1}^r$ be a finite family of tensors. Suppose that the length- S pair evaluations of blocks k and j span their product matrix algebra. For every $X \in M_{D_k}(\mathbb{C})$, there is a tuple $M = (M_\ell)_\ell$ in the span of the simultaneous length- S word evaluations such that $M_k = X$ and $M_j = 0$.

Proof. ✓ Proof Full pair span gives coefficients satisfying $(X, 0) = \sum_{|w|=S} c_w(A_w^k, A_w^j)$. Define $M_\ell = \sum_{|w|=S} c_w A_w^\ell$ for every block ℓ . Then M lies in the simultaneous word-tuple span, and its k - and j -components are X and 0, respectively. $\square$

Lemma 13.2.1.2 (Blockwise matrix equality from trace pairings). Lean ✓ Stmt Let $A^1, \dots, A^g$ be block tensors whose length- m simultaneous word tuples span the product algebra. If, for every word w of length m , two blockwise matrix families X_j and Y_j satisfy $\sum_j \text{tr}(X_j A_w^j) = \sum_j \text{tr}(Y_j A_w^j)$, then $X_j = Y_j$ for every block j .

Proof. ✓ Proof For every word w of length m ,

$$\sum_j \text{tr}((X_j - Y_j) A_w^j) = \sum_j \text{tr}(X_j A_w^j) - \sum_j \text{tr}(Y_j A_w^j) = 0.$$

Nondegeneracy of the product trace pairing, together with the spanning hypothesis, gives $X_j - Y_j = 0$ for every block j . $\square$

Lemma 13.2.1.3 (Boundary identities from block-sum membership). Lean ✓ Stmt Let $A^1, \dots, A^g$ be block tensors whose length- m simultaneous word tuples span the product algebra. Let $C_a^j \in M_{D_j}(\mathbb{C})$ be matrices indexed by blocks j and physical letters a . Suppose that, for every pair of boundary letters a, b and every word w of length m , a vector ψ has the left-boundary trace form $\psi(a, w, b) = \sum_j \text{tr}(A_b^j C_a^j A_w^j)$. If $\psi \in \bigvee_j G_{m+2}(A^j)$, then there are matrices E_j such that, for every block j and all a, b , $A_b^j C_a^j = A_b^j E_j A_a^j$.

Proof. ✓ Proof Write $\psi = \sum_j \phi_j$ with $\phi_j \in G_{m+2}(A^j)$. For each j , choose E_j whose image under the length- $m+2$ ground-space map for A^j is ϕ_j . Comparing the coefficient of the word awb in the two expressions for ψ gives

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(A_b^j E_j A_a^j A_w^j).$$

Lemma 13.2.1.2 gives the displayed blockwise identities. $\square$

Lemma 13.2.1.4 (Blockwise boundary compatibility from traces). Lean ✓ Stmt Let $A^1, \dots, A^g$ be block tensors with the following common word-span property: for a fixed m , the simultaneous tuples $w \mapsto (A_w^1, \dots, A_w^g)$, $|w| = m$, span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Suppose that matrices $C_a^j, D_b^j \in M_{D_j}(\mathbb{C})$ satisfy, for all physical indices a, b and all words w of length m ,

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Then $A_b^j C_a^j = D_b^j A_a^j$ for every block j and all a, b .

Proof. ✓ Proof For fixed a, b , set $\Delta_j = A_b^j C_a^j - D_b^j A_a^j$. The assumed trace equality says that $\sum_j \text{tr}(\Delta_j A_w^j) = 0$ for every word w of length m . Since the tuples $(A_w^1, \dots, A_w^g)$ span the product algebra, the product trace pairing is nondegenerate and hence every Δ_j is zero. $\square$

Lemma 13.2.1.5 (Word-valued boundary compatibility from traces). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A^1, \dots, A^g$ be block tensors with the common word-span property at length m . Suppose that $C_\rho^j \in M_{D_j}(\mathbb{C})$ is indexed by a block j and a word ρ of some fixed length M , and that $D_\beta^j \in M_{D_j}(\mathbb{C})$ is indexed by a block j and a word β of some fixed length K . If, for every word ρ of length M , every word β of length K , and every middle word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \text{tr}(D_\beta^j A_\rho^j A_w^j),$$

then $A_\beta^j C_\rho^j = D_\beta^j A_\rho^j$ for every block j and all words β, ρ . In particular, taking $D_\beta^j = X_j A_\beta^j$ gives $A_\beta^j C_\rho^j = (X_j A_\beta^j) A_\rho^j$.

Proof. [✓ Proof](#) Fix β and ρ , and set $\Delta_j = A_\beta^j C_\rho^j - D_\beta^j A_\rho^j$. The trace equality says that $\sum_j \text{tr}(\Delta_j A_w^j) = 0$ for every word w of length m . The same nondegenerate product trace-pairing argument as in Lemma 13.2.1.4 gives $\Delta_j = 0$ for every j . The specialization $D_\beta^j = X_j A_\beta^j$ is immediate. $\square$

Lemma 13.2.1.6 (Fixed complementary-word compatibility from traces). [Lean](#) [✓ Stmt](#) Let $A^1, \dots, A^g$ be block tensors with the common word-span property at length m . Fix a complementary word ρ of length M and matrices $C_\rho^j, X_j \in M_{D_j}(\mathbb{C})$. If, for every wrapped word β of length K and every middle word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \text{tr}((X_j A_\beta^j) A_\rho^j A_w^j),$$

then $A_\beta^j C_\rho^j = (X_j A_\beta^j) A_\rho^j$ for every block j and every wrapped word β .

Proof. [✓ Proof](#) Fix β and set $\Delta_j = A_\beta^j C_\rho^j - (X_j A_\beta^j) A_\rho^j$. The trace equality gives $\sum_j \text{tr}(\Delta_j A_w^j) = 0$ for every word w of length m . The common word-span property and the product trace pairing imply $\Delta_j = 0$ for every j . $\square$

Lemma 13.2.1.7 (Normalized boundary matrix identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $(A_a)_{a \in I}$, $(C_a)_{a \in I}$, and $(D_b)_{b \in I}$ be matrices in $M_D(\mathbb{C})$, indexed by a finite alphabet. If

$$\sum_a A_a A_a^\dagger = \mathbb{1},$$

$$A_b C_a = D_b A_a$$

for all $a, b \in I$, and if $E = \sum_a C_a A_a^\dagger$, then

$$D_b = A_b E,$$

$$A_b C_a = A_b E A_a$$

for all $a, b \in I$.

In particular, the same calculation applies when the alphabet is the finite set of words of a fixed length and A_β denotes $A_{\beta_1} \cdots A_{\beta_K}$.

The same argument also applies when the normalized family is indexed by a finite set J , while $(A_b)_{b \in I}$ and $(D_b)_{b \in I}$ are indexed by a set I . If $(B_a)_{a \in J}$ and $(C_a)_{a \in J}$ are indexed by J , and if, for all $a \in J$ and $b \in I$,

$$\sum_a B_a B_a^\dagger = \mathbb{1},$$

$$A_b C_a = D_b B_a,$$

then, with $E = \sum_a C_a B_a^\dagger$, one has

$$\begin{aligned} D_b &= A_b E, \\ A_b C_a &= A_b E B_a. \end{aligned}$$

This includes the case where I and J are word sets of different lengths.

Proof. ✓ **Proof** Multiply D_b by the normalization:

$$D_b = D_b \sum_a A_a A_a^\dagger = \sum_a D_b A_a A_a^\dagger = \sum_a A_b C_a A_a^\dagger = A_b E.$$

Substituting this identity into $A_b C_a = D_b A_a$ gives $A_b C_a = A_b E A_a$. For two alphabets, the corresponding step is

$$D_b = D_b \sum_a B_a B_a^\dagger = \sum_a D_b B_a B_a^\dagger = \sum_a A_b C_a B_a^\dagger = A_b E,$$

where $E = \sum_a C_a B_a^\dagger$. Substituting this identity into $A_b C_a = D_b B_a$ gives $A_b C_a = A_b E B_a$. □

Lemma 13.2.1.8 (Explicit C^j, D^j, E^j identities from trace decompositions). Lean ✓ **Stmt** Let $A^1, \dots, A^g$ be block tensors with the common word-span property at length m . Fix matrices X_j , and put $D_\beta^j = X_j A_\beta^j$. Suppose that C_ρ^j is indexed by a block j and a complementary word ρ of length M , and assume that, for every wrapped word β of length K , every complementary word ρ , and every middle word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \text{tr}(D_\beta^j A_\rho^j A_w^j).$$

If each block satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$, then, for every block j , there is a matrix $E^j = \sum_\rho C_\rho^j (A_\rho^j)^\dagger$ such that, for every wrapped word β and complementary word ρ ,

$$\begin{aligned} D_\beta^j &= A_\beta^j E^j, \\ A_\beta^j C_\rho^j &= A_\beta^j E^j A_\rho^j. \end{aligned}$$

This is the calculation in [193, Theorem 12, proof lines 1446–1451], with the adjoint on A_ρ^j required by the displayed normalization.

Proof. ✓ **Proof** Lemma 13.2.1.5 gives $A_\beta^j C_\rho^j = D_\beta^j A_\rho^j$. The single-letter normalization gives the word normalization $\sum_\rho A_\rho^j (A_\rho^j)^\dagger = \mathbb{1}$. Lemma 13.2.1.7, applied to the two word alphabets, gives

$$\begin{aligned} E^j &= \sum_\rho C_\rho^j (A_\rho^j)^\dagger, \\ D_\beta^j &= A_\beta^j E^j, \\ A_\beta^j C_\rho^j &= A_\beta^j E^j A_\rho^j. \end{aligned}$$

□

Lemma 13.2.1.9 (Complementary-word boundary identities from a compatibility hypothesis).

[Lean](#) [✓ Stmt](#) Let A_β be the products over wrapped words of length K , let A_ρ be the products over complementary words of length M , and let $X \in M_D(\mathbb{C})$ be a matrix. Suppose that the complementary-word products satisfy $\sum_\rho A_\rho A_\rho^\dagger = \mathbb{1}$ and that there are matrices C_ρ , indexed by complementary words, such that, for every wrapped word β and complementary word ρ , $A_\beta C_\rho = (XA_\beta)A_\rho$. Then, for every complementary word ρ , with

$$E_\rho = \left(\sum_\gamma C_\gamma A_\gamma^\dagger \right) A_\rho,$$

one has, for every wrapped word β , $(XA_\beta)A_\rho = A_\beta E_\rho$.

Proof. [✓ Proof](#) Apply Lemma 13.2.1.7 with $D_\beta = XA_\beta$. It gives $A_\beta C_\rho = A_\beta E A_\rho$ for $E = \sum_\gamma C_\gamma A_\gamma^\dagger$. With $E_\rho = E A_\rho$, the compatibility hypothesis gives

$$(XA_\beta)A_\rho = A_\beta C_\rho = A_\beta E A_\rho = A_\beta E_\rho.$$

□

Lemma 13.2.1.10 (Complementary-word boundary identities from trace decompositions). [Lean](#)

[✓ Stmt](#) Let $A^1, \dots, A^g$ be block tensors with the common word-span property at length m . Fix matrices X_j . Suppose that C_ρ^j is indexed by a block j and a complementary word ρ of length M . Assume that, for every wrapped word β of length K , every complementary word ρ , and every middle word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \text{tr}((X_j A_\beta^j) A_\rho^j A_w^j).$$

If each block satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$, then, for every block j and every complementary word ρ , there is a matrix $E_{j,\rho}$ such that, for every wrapped word β , $(X_j A_\beta^j) A_\rho^j = A_\beta^j E_{j,\rho}$.

Proof. [✓ Proof](#) Lemma 13.2.1.5, applied with $D_\beta^j = X_j A_\beta^j$, gives $A_\beta^j C_\rho^j = (X_j A_\beta^j) A_\rho^j$. The hypothesis $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ extends to words of length M : $\sum_\rho A_\rho^j (A_\rho^j)^\dagger = \mathbb{1}$. Lemma 13.2.1.9 gives the matrices $E_{j,\rho}$. □

Lemma 13.2.1.11 (A left-boundary summand is a ground-space vector). [Lean](#) [✓ Stmt](#) Let A be a block tensor, and let $C_a, E \in M_D(\mathbb{C})$ satisfy $A_b C_a = A_b E A_a$ for all physical indices a, b . For a word $\sigma = (\sigma_1, \dots, \sigma_{n+2})$, define

$$\alpha(\sigma) = \text{tr}(A_{\sigma_{n+2}} C_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_{n+1}}).$$

Then $\alpha \in G_{n+2}(A)$.

Proof. [✓ Proof](#) The boundary identity gives

$$A_{\sigma_{n+2}} C_{\sigma_1} = A_{\sigma_{n+2}} E A_{\sigma_1}.$$

Hence, by cyclicity of the trace,

$$\alpha(\sigma) = \text{tr}(A_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_{n+2}} E),$$

which is the ground-space parametrization of E . □

Lemma 13.2.1.12 (Left-boundary summands are ground-space vectors). [Lean](#) [✓ Stmt](#) Let $A^1, \dots, A^g$ be block tensors, and let $C_a^j, E^j \in M_{D_j}(\mathbb{C})$ satisfy $A_b^j C_a^j = A_b^j E^j A_a^j$ for every block j and all physical indices a, b . For a word $\sigma = (\sigma_1, \dots, \sigma_{n+2})$, define

$$\alpha_j(\sigma) = \text{tr}(A_{\sigma_{n+2}}^j C_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+1}}^j).$$

Then $\sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j)$.

Proof. [✓ Proof](#) Fix j . By the assumed boundary identity and cyclicity of the trace,

$$\text{tr}(A_{\sigma_{n+2}}^j C_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+1}}^j) = \text{tr}(A_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+2}}^j E^j).$$

Thus α_j is the image of E^j under the parametrization of $G_{n+2}(A^j)$. Hence each α_j lies in $G_{n+2}(A^j)$, and the finite sum lies in the supremum of these subspaces. $\square$

Lemma 13.2.1.13 (Left-boundary trace decompositions lie in the block ground spaces). [Lean](#) [✓ Stmt](#) Let $A^1, \dots, A^g$ be block tensors with common word-span separation at length n , and suppose that $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ for every block j . Suppose a vector $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$ has the left-boundary trace decomposition

$$\psi = \sum_j \alpha_j,$$

$$\alpha_j(i_1, \dots, i_{n+2}) = \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j),$$

and suppose that the two trace decompositions agree for every middle word:

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Then $\psi \in \bigvee_j G_{n+2}(A^j)$.

Proof. [✓ Proof](#) The equality of the two boundary trace expressions and the common word-span hypothesis give $A_b^j C_a^j = D_b^j A_a^j$ for every j, a, b . By the normalization,

$$\begin{aligned} D_b^j &= D_b^j \sum_a A_a^j A_a^{j\dagger} = \sum_a A_b^j C_a^j A_a^{j\dagger} = A_b^j E^j, \\ E^j &:= \sum_a C_a^j A_a^{j\dagger}. \end{aligned}$$

Hence $A_b^j C_a^j = A_b^j E^j A_a^j$. Lemma 13.2.1.12 then gives $\psi = \sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j)$. $\square$

Lemma 13.2.1.14 (Trace decompositions from fixed-boundary restrictions). [Lean](#) [✓ Stmt](#) Let $A^1, \dots, A^g$ be block tensors. If $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$ satisfies

$$\begin{aligned} \psi(a, -) &\in \bigvee_j G_{n+1}(A^j), \\ \psi(-, b) &\in \bigvee_j G_{n+1}(A^j) \end{aligned}$$

for every physical index a, b , then there are matrices C_a^j and D_b^j such that

$$\begin{aligned} \psi &= \sum_j \alpha_j, \\ \alpha_j(i_1, \dots, i_{n+2}) &= \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j), \end{aligned}$$

and, for every middle word w of length n ,

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Proof. ✓ Proof Choose finite decompositions of the fixed-boundary vectors:

$$\begin{aligned} \psi(a, i_2, \dots, i_{n+2}) &= \sum_j \text{tr}(A_{i_2}^j \cdots A_{i_{n+2}}^j C_a^j), \\ \psi(i_1, \dots, i_{n+1}, b) &= \sum_j \text{tr}(A_{i_1}^j \cdots A_{i_{n+1}}^j D_b^j). \end{aligned}$$

For every middle word w ,

$$\psi(a, w, b) = \sum_j \text{tr}(A_w^j A_b^j C_a^j) = \sum_j \text{tr}(A_b^j C_a^j A_w^j),$$

and the second decomposition gives

$$\psi(a, w, b) = \sum_j \text{tr}(A_a^j A_w^j D_b^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Hence

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

For an arbitrary word $(i_1, \dots, i_{n+2})$, the first decomposition gives

$$\psi(i_1, \dots, i_{n+2}) = \sum_j \text{tr}(A_{i_2}^j \cdots A_{i_{n+2}}^j C_{i_1}^j) = \sum_j \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j),$$

by cyclicity of the trace. □

Lemma 13.2.1.15 (One-step block intersection from block restrictions). Lean ✓ Stmt Let $A^1, \dots, A^g$ be block tensors with common word-span separation at length n , and suppose that $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ for every block j . If $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$ satisfies

$$\begin{aligned} \psi(a, -) &\in \bigvee_j G_{n+1}(A^j), \\ \psi(-, b) &\in \bigvee_j G_{n+1}(A^j) \end{aligned}$$

for every physical index a, b , then $\psi \in \bigvee_j G_{n+2}(A^j)$.

Proof. ✓ Proof Choose finite decompositions of the fixed-boundary vectors:

$$\begin{aligned} \psi(a, i_2, \dots, i_{n+2}) &= \sum_j \text{tr}(A_{i_2}^j \cdots A_{i_{n+2}}^j C_a^j), \\ \psi(i_1, \dots, i_{n+1}, b) &= \sum_j \text{tr}(A_{i_1}^j \cdots A_{i_{n+1}}^j D_b^j). \end{aligned}$$

Evaluating these two decompositions on (a, w, b) , where w has length n , gives

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

The first decomposition also gives

$$\begin{aligned}\psi &= \sum_j \alpha_j, \\ \alpha_j(i_1, \dots, i_{n+2}) &= \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \dots A_{i_{n+1}}^j).\end{aligned}$$

Lemma 13.2.1.13 therefore gives $\psi \in \bigvee_j G_{n+2}(A^j)$. $\square$

Lemma 13.2.1.16 (One-step block intersection characterization). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Lemma 13.2.1.15, an $(n+2)$ -site vector satisfies $\psi \in \bigvee_j G_{n+2}(A^j)$ if and only if*

$$\begin{aligned}\psi(a, -) &\in \bigvee_j G_{n+1}(A^j), \\ \psi(-, b) &\in \bigvee_j G_{n+1}(A^j)\end{aligned}$$

for every physical index a, b .

Proof. [✓ Proof](#) The reverse implication is Lemma 13.2.1.15. For the forward implication, write

$$\begin{aligned}\psi &= \sum_j \gamma_j, \\ \gamma_j &\in G_{n+2}(A^j).\end{aligned}$$

Each γ_j has fixed-boundary restrictions in $G_{n+1}(A^j)$:

$$\begin{aligned}\gamma_j(a, -) &\in G_{n+1}(A^j), \\ \gamma_j(-, b) &\in G_{n+1}(A^j).\end{aligned}$$

Taking the finite sum over j gives the two restrictions of ψ in $\bigvee_j G_{n+1}(A^j)$. $\square$

Lemma 13.2.1.17 (Subspace form of the one-step block intersection). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Lemma 13.2.1.15, set $S_n := \bigvee_j G_{n+1}(A^j)$. Then*

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. [✓ Proof](#) Lemma 13.2.1.16 gives the displayed equivalence for each vector ψ . Interpreting the two sides of that equivalence as subspaces gives the stated equality. $\square$

13.2.2 Eventual block-intersection consequences

Propagating the single-length product span through a period window gives the block-intersection identity for all sufficiently large n , both from a general period window and from the normalized BNT representative hypotheses. The full product span at length n also makes the sum of the local spaces $G_n(A^j)$ direct.

Lemma 13.2.2.1 (Period-window form of the block intersection identity). [Lean](#) [✓ Stmt](#) *Let $p > 0$. Suppose the length- p simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$, and suppose that, for every $0 \leq r < p$, the length- $(s+r)$ simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$. Assume*

also the normalization $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ for every block j . Then there is a length N such that, for every $n \geq N$, if $S_n := \bigvee_j G_{n+1}(A^j)$, then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Lemma 13.2.12 gives a length N such that the simultaneous block-word tuples span $\prod_j M_{D_j}(\mathbb{C})$ at every length $n \geq N$. For each such n , Lemma 13.2.1.17 applies and gives the displayed subspace equality. $\square$

Lemma 13.2.2.2 (BNT block separation gives one block-intersection step). Lean ✓ Stmt Under the hypotheses of Lemma 8.3.3, assume in addition the normalization $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ for every block j . Let

$$\begin{aligned} n &= (L_0 + 1) + (r - 1)3(L_0 + 1), \\ S_n &= \bigvee_j G_{n+1}(A^j). \end{aligned}$$

Then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Lemma 8.3.3 gives $\text{span}\{A_w^1, \dots, A_w^r : |w| = n\} = \prod_j M_{D_j}(\mathbb{C})$. Substituting this span into Lemma 13.2.1.17, together with the normalization, gives the stated subspace equality. $\square$

Lemma 13.2.2.3 (BNT block separation with a block-injectivity window). Lean ✓ Stmt Let $A^1, \dots, A^r$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Assume $L > 1$, and assume that, for every block j ,

$$\begin{aligned} \text{span}\{A_a^j : a = 0, \dots, d - 1\} &= M_{D_j}(\mathbb{C}), \\ \text{span}\{A_u^j : |u| = L\} &= M_{D_j}(\mathbb{C}), \\ \text{span}\{A_v^j : |v| = 3L\} &= M_{D_j}(\mathbb{C}). \end{aligned}$$

Set $S = 3L$ and $q = (r - 1)S$. Suppose that $p > 0$ and let s_0 be a starting length. Assume that, for every block j , $\text{span}\{A_u^j : |u| = p\} = M_{D_j}(\mathbb{C})$, and that, for every $0 \leq s < p + q$ and every block j , $\text{span}\{A_v^j : |v| = s_0 + s\} = M_{D_j}(\mathbb{C})$. Assume also the normalization $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ for every block j . Then there is N such that, for every $n \geq N$, if $S_n := \bigvee_j G_{n+1}(A^j)$, then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof The direct-sum dimension step gives, for each ordered pair of distinct blocks $k \neq j$, coefficients $c_\omega^{k,j}$ of length S such that

$$\begin{aligned} \sum_{|\omega|=S} c_\omega^{k,j} A_\omega^k &= \mathbb{1}, \\ \sum_{|\omega|=S} c_\omega^{k,j} A_\omega^j &= 0. \end{aligned}$$

Multiplying these equations over the $r - 1$ blocks $j \neq k$ gives coefficients $c_\omega^{(k)}$ of total length q satisfying

$$\sum_{|\omega|=q} c_\omega^{(k)} A_\omega^j = \begin{cases} \mathbb{1}, & j = k, \\ 0, & j \neq k. \end{cases}$$

Combining this block-separating equation with block injectivity at length p gives $\text{span}\{(A_w^1, \dots, A_w^r) : |w| = p + q\} = \prod_j M_{D_j}(\mathbb{C})$. Combining the same separating equations with the assumed block-injectivity window gives the full product span at lengths $s_0 + q + s$, for $0 \leq s < p + q$. The period-window form of Lemma 13.2.2.1 then gives the displayed equality for all sufficiently large n . $\square$

Lemma 13.2.2.4 (Unitality propagates simultaneous product spans). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the length- q simultaneous products span the product algebra, $\text{span}\{(A_w^1, \dots, A_w^r) : |w| = q\} = \prod_j M_{D_j}(\mathbb{C})$, and that every block satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Then the simultaneous products span the product algebra at every length $n \geq q$. This is the simultaneous-tuple form of the unital propagation in [193, lines 893–898].

Proof. [✓ Proof](#) Given matrices M_j , first represent the tuple $((A_a^j)^\dagger M_j)_j$ by words of length q . Prefixing the letter a and summing over a gives $(\sum_a A_a^j A_a^{j\dagger} M_j)_j = (M_j)_j$. This proves the step from q to $q + 1$; induction gives every $n \geq q$. $\square$

Lemma 13.2.2.5 (Normalized block-separating equations give all large product spans). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A^1, \dots, A^r$ be blocks satisfying the normalization $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ and assume that every block is injective at length L : $\text{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C})$. If there are block-separating equations of length S , then, for every $n \geq L + (r - 1)S$, $\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C})$. In particular, for normalized BNT representatives satisfying the displayed unital normalization, if $L_0 > 0$ and $\Gamma_{L_0}^{A^j}$ is injective for every block j , the same conclusion holds with $S = 3(L_0 + 1)$ and $n \geq (L_0 + 1) + (r - 1)3(L_0 + 1)$.

Proof. [✓ Proof](#) By Theorem 6.8.1, the normalization propagates the length- L span equality to every larger homogeneous length:

$$m \geq L \implies \text{span}\{A_u^j : |u| = m\} = M_{D_j}(\mathbb{C}).$$

For $n \geq L + (r - 1)S$, take $m = n - (r - 1)S$. Combining the length- m block span with the $r - 1$ block-separating equations gives the displayed product span at length n . In the BNT case, the injectivity of $\Gamma_{L_0}^{A^j}$ gives the full span at length L_0 , and the normalization propagates it to the lengths $L_0 + 1$ and $3(L_0 + 1)$. Lemma 8.3.3 then supplies the block-separating equations with $S = 3(L_0 + 1)$. $\square$

Theorem 13.2.2.6 (Tensor irreducibility implies transfer-map irreducibility). [✓ Stmt](#) If an MPS tensor is irreducible, then its transfer map has no nontrivial invariant face.

Proof. [✓ Proof](#) Tensor irreducibility is the absence of a nontrivial invariant matrix projection, which is precisely the corresponding irreducibility condition for the transfer map. $\square$

Lemma 13.2.2.7 (Positive square-root gauging preserves irreducibility). [Lean](#) [✓ Stmt](#) Let A have positive bond dimension and irreducible transfer map. If $\Lambda > 0$, then the square-root-gauged tensor $\Lambda^{1/2} A \Lambda^{-1/2}$ is irreducible.

Proof. ✓ **Proof** The square-root gauge conjugates the transfer map by an invertible similarity. Invariant faces therefore correspond bijectively before and after gauging. $\square$

Lemma 13.2.2.8 (Gauge-phase equivalence is invariant under gauge replacement). Lean ✓ **Stmt** *Let A, A', B, B' have the same bond dimension. If A is gauge equivalent to A' , A' is gauge-phase equivalent to B' , and B is gauge equivalent to B' , then A is gauge-phase equivalent to B .*

Proof. ✓ **Proof** If X, Y, Z are the left gauge, gauge-phase intertwiner, and right gauge, respectively, then $Z^{-1}YX$ is the required intertwiner from A to B ; it carries the same unit-modulus phase as Y . $\square$

Lemma 13.2.2.9 (Gauge-phase equivalence transports across two gauges). Lean ✓ **Stmt** *Gauge equivalent replacements on both sides preserve gauge-phase equivalence, including after identifying equal bond dimensions.*

Proof. ✓ **Proof** Compose the left gauge, the gauge-phase intertwiner, and the inverse right gauge. After transporting across the bond-dimension identification, the composite is the required gauge-phase intertwiner. $\square$

Theorem 13.2.2.10 (Prepared-gauge simultaneous product span). Lean ✓ **Stmt** *Let $r \geq 2$, and let the bond dimensions $D_1, \dots, D_r$ be positive. Suppose that every block A^j is irreducible and that no two distinct blocks of the same bond dimension are gauge-phase equivalent. Let $L_0 > 0$, and suppose that, for every j ,*

$$\begin{aligned} \text{span}\{A_w^j : |w| = L_0\} &= M_{D_j}(\mathbb{C}), \\ \text{span}\{A_w^j : |w| = L_0 + 1\} &= M_{D_j}(\mathbb{C}), \\ \text{span}\{A_w^j : |w| = 3(L_0 + 1)\} &= M_{D_j}(\mathbb{C}). \end{aligned}$$

For every j , let Λ^j be positive definite and satisfy

$$\sum_a (A_a^j)^\dagger \Lambda^j A_a^j = \Lambda^j.$$

Then

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = 3(r-1)(L_0 + 1)\} = \prod_j M_{D_j}(\mathbb{C}).$$

Diagonality of Λ^j is not needed.

Proof. ✓ **Proof** Define the prepared blocks by

$$B_a^j = (\Lambda^j)^{1/2} A_a^j (\Lambda^j)^{-1/2}.$$

Positive definiteness makes this an invertible gauge, while the dual fixed-point equation gives

$$\sum_a (B_a^j)^\dagger B_a^j = \mathbb{1}.$$

Gauge transport preserves irreducibility, the three fixed-length injectivity statements, and pairwise inequivalence up to gauge and phase. Thus every B^j is algebraically normal and left-canonical, so

$$O_{B^j B^j}(N) \longrightarrow 1.$$

The sharp separation theorem now gives

$$\text{span}\{(B_w^1, \dots, B_w^r) : |w| = 3(r-1)(L_0+1)\} = \prod_j M_{D_j}(\mathbb{C}).$$

Theorem 8.3.2 transports this equality back to the source representatives. $\square$

Theorem 13.2.2.11 (Simultaneous product span at the source length). [Lean](#) [✓ Stmt](#) *Let $r \geq 2$, and let the bond dimensions $D_1, \dots, D_r$ be positive. Suppose that every block A^j is irreducible and that no two distinct blocks of the same bond dimension are gauge-phase equivalent. Let $L_0 > 0$, and suppose every block is injective at length L_0 :*

$$\text{span}\{A_w^j : |w| = L_0\} = M_{D_j}(\mathbb{C}).$$

For every j , let Λ^j be positive definite and assume the canonical identities from [193]:

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}, \quad \sum_a (A_a^j)^\dagger \Lambda^j A_a^j = \Lambda^j.$$

For every $n \geq 3(r-1)(L_0+1)$,

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

No identity $\sum_a (A_a^j)^\dagger A_a^j = \mathbb{1}$ is assumed for the source blocks. This is the simultaneous spanning estimate used in [193, Theorem 12, proof lines 1424–1456].

Proof. [✓ Proof](#) Theorem 6.8.1 first gives

$$\begin{aligned} \text{span}\{A_w^j : |w| = L_0\} = M_{D_j}(\mathbb{C}) &\implies \text{span}\{A_w^j : |w| = L_0 + 1\} = M_{D_j}(\mathbb{C}) \\ &\implies \text{span}\{A_w^j : |w| = 3(L_0 + 1)\} = M_{D_j}(\mathbb{C}). \end{aligned}$$

Theorem 13.2.2.10 gives the full simultaneous span at $q = 3(r-1)(L_0+1)$ on the source representatives. The source unital identity and Lemma 13.2.2.4 propagate this equality to every $n \geq q$. $\square$

Theorem 13.2.2.12 (Block-space independence at the source length). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 13.2.2.11, the spaces $G_n(A^j)$ form an internal sum for every $n \geq 3(r-1)(L_0+1)$.*

Proof. [✓ Proof](#) Write $\phi_j = \Gamma_n^{A^j}(X_j)$ and suppose that $\sum_j \phi_j = 0$. The simultaneous spanning equality gives

$$\sum_j \Gamma_n^{A^j}(X_j) = 0 \implies \left(\forall (M_j)_j, \sum_j \text{tr}(X_j M_j) = 0 \right) \implies \forall j, X_j = 0.$$

Hence every ϕ_j vanishes. $\square$

Theorem 13.2.2.13 (Direct-sum restriction intersection at the source length). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 13.2.2.11, let $m \geq 3(r-1)(L_0+1) + 1$. Then*

$$\mathbb{C}^d \otimes \left(\sum_j G_m(A^j) \right) \cap \left(\sum_j G_m(A^j) \right) \otimes \mathbb{C}^d = \sum_j G_{m+1}(A^j).$$

Theorem 13.2.2.12 also shows that the sums at lengths m and $m+1$ are internal. This is the direct-sum lemma in the proof of [193, Theorem 12, proof lines 1346–1456].

Proof. [✓ Proof](#) Let a vector belong to both restricted spaces. The two endpoint decompositions give boundary matrices C_a^j and D_b^j . For every pair of physical indices a, b and every middle word w of length $m - 1$, their coefficients obey

$$\sum_j \text{tr}((A_b^j C_a^j) A_w^j) = \sum_j \text{tr}((D_b^j A_a^j) A_w^j).$$

The simultaneous product span at length $m - 1$ and nondegeneracy of the trace pairing therefore give, block by block,

$$A_b^j C_a^j = D_b^j A_a^j. \quad (13.1)$$

Set $E^j = \sum_a C_a^j A_a^{j\dagger}$. Identity (13.1) and the right-unital normalization imply $A_b^j C_a^j = A_b^j E^j A_a^j$. Thus the component in block j lies in $G_{m+1}(A^j)$. The reverse inclusion follows by splitting a word of length $m + 1$ at either endpoint. $\square$

Theorem 13.2.2.14 (Doubly normalized simultaneous product span). [Lean](#) [✓ Stmt](#) *Let $r \geq 2$, and let the bond dimensions be positive. Suppose that the blocks are irreducible, have normalized self-overlap, are pairwise inequivalent up to gauge and phase, and are injective at a common length $L_0 > 0$. Assume both normalization identities*

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}, \quad \sum_a (A_a^j)^\dagger A_a^j = \mathbb{1}.$$

Then for every $n \geq 3(r - 1)(L_0 + 1)$,

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

Proof. [✓ Proof](#) Propagate injectivity to the two longer lengths used by the sharp block separation theorem, obtain the simultaneous span at $3(r - 1)(L_0 + 1)$, and propagate it using the right-unital identity. $\square$

Theorem 13.2.2.15 (Doubly normalized block-space independence). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 13.2.2.14, the spaces $G_n(A^j)$ form an internal sum for every $n \geq 3(r - 1)(L_0 + 1)$.*

Proof. [✓ Proof](#) Apply nondegeneracy of the trace pairing to the simultaneous product span. $\square$

Theorem 13.2.2.16 (Doubly normalized restriction intersection). [Lean](#) [✓ Stmt](#) *Under the same hypotheses, if $m \geq 3(r - 1)(L_0 + 1) + 1$, then*

$$\mathbb{C}^d \otimes \left(\sum_j G_m(A^j) \right) \cap \left(\sum_j G_m(A^j) \right) \otimes \mathbb{C}^d = \sum_j G_{m+1}(A^j).$$

Proof. [✓ Proof](#) Use the simultaneous product span at length $m - 1$ in the endpoint restriction calculation. $\square$

Lemma 13.2.2.17 (Product span makes the sum of local spaces direct). [Lean](#) [Lean](#) [✓ Stmt](#) *Suppose $\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C})$. Then the sum of the local spaces $G_n(A^j)$ is direct: if*

$$\begin{aligned} \phi_j &\in G_n(A^j), \\ \sum_j \phi_j &= 0, \end{aligned}$$

then $\phi_j = 0$ for every j . In particular, for the normalized BNT hypotheses above, this directness holds for every $n \geq (L_0 + 1) + (r - 1)3(L_0 + 1)$.

Proof. ✓ Proof Write $\phi_j(\sigma) = \text{tr}(A_\sigma^j X_j)$. Evaluating $\sum_j \phi_j = 0$ on each word w of length n gives $\sum_j \text{tr}(A_w^j X_j) = 0$. Since $\text{tr}(A_w^j X_j) = \text{tr}(X_j A_w^j)$, the product-span equation gives, for every $(M_1, \dots, M_r) \in \prod_j M_{D_j}(\mathbb{C})$, $\sum_j \text{tr}(X_j M_j) = 0$. Taking only the j -th component nonzero and using nondegeneracy of the trace pairing yields $X_j = 0$, hence $\phi_j = 0$. The BNT statement follows by substituting Lemma 13.2.2.5. $\square$

13.3 Block-diagonal boundary conditions for periodic ground spaces

For a block-diagonal tensor $B = \bigoplus_{j=1}^g \mu_j A_j$ with $\mu_j \neq 0$, simultaneous injectivity at length $S > 0$ gives the periodic ground-space identity at every $S + 1 \leq L \leq N$, including the minimal ring $N = S + 1$; see Theorem 13.3.35. The earlier separation argument starts from individual block injectivity and uses a larger interaction range. Its explicit boundary identities yield component membership $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$, establishing the periodic boundary step at that larger range. The simultaneous-span argument gives the bound of [16, Theorem IV.16] directly.

Lemma 13.3.1 (Normalized BNT representatives give the large-length block intersection). Lean
Lean Lean ✓ Stmt *Let $A^1, \dots, A^r$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose $L_0 > 0$, and, for every block j , the map*

$$\Gamma_{L_0}^{A_j} : M_{D_j}(\mathbb{C}) \longrightarrow (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Set $N = (L_0 + 1) + (r - 1)3(L_0 + 1)$. For every $n \geq N$, set $S_n := \bigvee_j G_{n+1}(A^j)$. The two consecutive sums are internal direct sums:

$$\begin{aligned} \bigvee_j G_{n+1}(A^j) &= \bigoplus_j G_{n+1}(A^j), \\ \bigvee_j G_{n+2}(A^j) &= \bigoplus_j G_{n+2}(A^j). \end{aligned}$$

With this notation,

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Lemma 13.2.2.5 gives $\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C})$ for every $n \geq N$. Substituting this product-span equation into Lemma 13.2.1.17, together with the normalization equation, gives the displayed intersection identity. Lemma 13.2.2.17 at lengths $n + 1$ and $n + 2$ gives the two directness statements. $\square$

Lemma 13.3.2 (Normalized BNT representatives give an eventual block intersection). Lean
✓ Stmt *Let $A^1, \dots, A^r$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two*

distinct same-dimensional representatives are gauge-phase equivalent. Suppose $L_0 > 0$, and, for every block j , the map

$$\Gamma_{L_0}^{A^j} : M_{D_j}(\mathbb{C}) \longrightarrow (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Then there is a length N such that, for every $n \geq N$, with $S_n := \bigvee_j G_{n+1}(A^j)$, one has

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

Proof. ✓ Proof Apply Lemma 13.3.1 and take $N = (L_0 + 1) + (r - 1)3(L_0 + 1)$. □

Lemma 13.3.3 (Block-diagonal boundary-condition formula). Lean ✓ Stmt Let $B = \bigoplus_{j=1}^g \mu_j A_j$. For block-diagonal boundary conditions X_j ,

$$\Gamma_L^B \left(\bigoplus_{j=1}^g X_j \right) = \sum_{j=1}^g \Gamma_L^{A_j}(\mu_j^L X_j).$$

Proof. ✓ Proof This is the diagonal-block trace formula specialized to a block-diagonal boundary matrix. The j -th diagonal block of $\bigoplus_k X_k$ is X_j , giving the displayed sum. □

Lemma 13.3.4 (Boundary matrices for a sum of open-boundary block spaces). Lean ✓ Stmt Let $B = \bigoplus_{j=1}^g \mu_j A_j$, where $\mu_j \neq 0$ for every j . For every length L and every $\psi \in \bigvee_{j=1}^g G_L(A_j)$, there are boundary matrices X_j such that

$$\psi = \Gamma_L^B \left(\bigoplus_{j=1}^g X_j \right).$$

For every j , one also has $\Gamma_L^{A_j}(\mu_j^L X_j) \in G_L(A_j)$.

Proof. ✓ Proof Choose $\phi_j \in G_L(A_j)$ such that

$$\psi = \sum_{j=1}^g \phi_j.$$

Choose Y_j with $\phi_j = \Gamma_L^{A_j}(Y_j)$, and set $X_j = \mu_j^{-L} Y_j$. Since $\mu_j \neq 0$, one has, for every j ,

$$\Gamma_L^{A_j}(\mu_j^L X_j) = \phi_j.$$

Lemma 13.3.3 now gives

$$\Gamma_L^B \left(\bigoplus_{j=1}^g X_j \right) = \sum_{j=1}^g \Gamma_L^{A_j}(\mu_j^L X_j) = \sum_{j=1}^g \phi_j = \psi.$$

□

Lemma 13.3.5 (One block lies in the local parent space of the direct-sum tensor). Lean ✓ Stmt Let $B = \bigoplus_{k=1}^g \mu_k A_k$. If $\mu_j \neq 0$, then, for every length L , one has $G_L(A_j) \subseteq G_L(B)$.

Proof. ✓ Proof A vector in $G_L(A_j)$ has the form $\Gamma_L^{A_j}(X)$. Put $\mu_j^{-L}X$ in the j -th diagonal block of a boundary matrix for B and put zero in every other block. If $\iota_j(Z)$ denotes this block insertion, Lemma 13.3.3 gives

$$\Gamma_L^B(\iota_j(\mu_j^{-L}X)) = \sum_{k=1}^g \Gamma_L^{A_k}(\mu_k^L(\iota_j(\mu_j^{-L}X))_{kk}) = \Gamma_L^{A_j}(X).$$

Hence $\Gamma_L^{A_j}(X) \in G_L(B)$. □

Theorem 13.3.6 (Local ground space of a block-diagonal tensor). Lean Lean ✓ Stmt Let $B = \bigoplus_{j=1}^g \mu_j A_j$, where $\mu_j \neq 0$ for $1 \leq j \leq g$. Then, for every length L ,

$$G_L(B) = \bigvee_{j=1}^g G_L(A_j),$$

$$G_L^{\text{ES}}(B) = \bigvee_{j=1}^g G_L^{\text{ES}}(A_j).$$

The second identity is the image of the first under the canonical identification ι_L , using $G_L^{\text{ES}}(A) = \iota_L(G_L(A))$.

Proof. ✓ Proof A boundary matrix X for B has diagonal blocks X_j , and block-diagonality gives $\Gamma_L^B(X) = \sum_{j=1}^g \Gamma_L^{A_j}(\mu_j^L X_j)$. This proves $G_L(B) \subseteq \bigvee_j G_L(A_j)$. The reverse inclusion is Lemma 13.3.5 for each block j . Applying ι_L , which preserves finite sums of subspaces, gives $G_L^{\text{ES}}(B) = \bigvee_j G_L^{\text{ES}}(A_j)$. □

Lemma 13.3.7 (Periodic constraints for a block-diagonal tensor). Lean ✓ Stmt Let $B = \bigoplus_{j=1}^g \mu_j A_j$, where $\mu_j \neq 0$ for $1 \leq j \leq g$, and set $S_M := \bigvee_{j=1}^g G_M(A_j)$. If, for every $M \geq L$, $\mathbb{C}^d \otimes S_M \cap S_M \otimes \mathbb{C}^d = S_{M+1}$, then $\mathcal{G}_{N,L}(B) \subseteq S_N$.

Proof. ✓ Proof Theorem 13.3.6 gives $G_L(B) = S_L$. Apply Lemma 11.5.5.18 to the sequence S_M . □

Lemma 13.3.8 (Periodic constraints lie in the block local-space sum). Lean ✓ Stmt Let $A_1, \dots, A_g$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose $L_0 > 0$, and, for every block j , the map

$$\Gamma_{L_0}^{A_j} : M_{D_j}(\mathbb{C}) \longrightarrow (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. For nonzero scalars μ_j , set

$$B = \bigoplus_{j=1}^g \mu_j A_j,$$

$$S_M := \bigvee_{j=1}^g G_M(A_j).$$

If $L \geq (L_0 + 1) + (g - 1)3(L_0 + 1) + 1$, then, for every $N \geq L$, $\mathcal{G}_{N,L}(B) \subseteq S_N$.

Proof. [✓ Proof](#) Put $T = (L_0 + 1) + (g - 1)3(L_0 + 1)$. For $M \geq L$, one has $M - 1 \geq T$. Applying Lemma 13.3.1 at $n = M - 1$ gives $\mathbb{C}^d \otimes S_M \cap S_M \otimes \mathbb{C}^d = S_{M+1}$ for every $M \geq L$. Lemma 13.3.7 then propagates the local constraint from L to N . $\square$

Lemma 13.3.9 (Periodic constraints and internality of the block local-space sum). [Lean](#) [✓ Stmt](#)
With the hypotheses and notation of Lemma 13.3.8, set $S_N := \bigvee_{j=1}^g G_N(A_j)$. Then, for every $N \geq L$, $\mathcal{G}_{N,L}(B) \subseteq S_N$, and the sum defining S_N is internal: if

$$\begin{aligned} \phi_j &\in G_N(A_j), \\ \sum_{j=1}^g \phi_j &= 0, \end{aligned}$$

then $\phi_j = 0$ for every j .

Proof. [✓ Proof](#) The inclusion is Lemma 13.3.8. Since $N \geq L$ and $L \geq (L_0 + 1) + (g - 1)3(L_0 + 1) + 1$, the product-span directness range of Lemma 13.2.2.17 applies at length N :

$$\begin{aligned} \text{span}\{(A_w^1, \dots, A_w^g) : |w| = N\} &= \prod_j M_{D_j}(\mathbb{C}) \\ \implies \ker \left(\bigoplus_{j=1}^g G_N(A_j) \longrightarrow (\mathbb{C}^d)^{\otimes N}, \quad (\phi_j)_j \mapsto \sum_j \phi_j \right) &= 0. \end{aligned}$$

$\square$

Lemma 13.3.10 (Periodic parent-space vectors decompose in the local block spaces). [Lean](#) [✓ Stmt](#)
With the hypotheses and notation of Lemma 13.3.8, every vector $\psi \in \mathcal{G}_{N,L}(\bigoplus_{j=1}^g \mu_j A_j)$ has a unique family $(\psi_j)_{j=1}^g$ such that

$$\begin{aligned} \psi &= \sum_{j=1}^g \psi_j, \\ \psi_j &\in G_N(A_j). \end{aligned}$$

Proof. [✓ Proof](#) Lemma 13.3.9 gives

$$\mathcal{G}_{N,L} \left(\bigoplus_{j=1}^g \mu_j A_j \right) \subseteq \bigvee_{j=1}^g G_N(A_j),$$

and the spaces $G_N(A_j)$ are independent: for every j and every $x_j \in G_N(A_j)$, $\sum_j x_j = 0 \implies x_j = 0$. Thus every $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ has a unique decomposition $\psi = \sum_j \psi_j$ with $\psi_j \in G_N(A_j)$. $\square$

Lemma 13.3.11 (Block-diagonal boundary conditions for periodic parent-space vectors). [Lean](#) [✓ Stmt](#)
With the hypotheses and notation of Lemma 13.3.8, every vector $\psi \in \mathcal{G}_{N,L}(\bigoplus_{j=1}^g \mu_j A_j)$ admits block-diagonal boundary conditions X_j such that

$$\begin{aligned} \psi &= \Gamma_N^B \left(\bigoplus_{j=1}^g X_j \right), \\ \Gamma_N^{A_j}(\mu_j^N X_j) &\in G_N(A_j) \end{aligned}$$

for every j .

Proof. ✓ **Proof** Lemma 13.3.8 places ψ in $\bigvee_{j=1}^g G_N(A_j)$. Applying Lemma 13.3.4 gives boundary matrices X_j for which, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j)$, and

$$\psi = \Gamma_N^B \left(\bigoplus_{j=1}^g X_j \right).$$

□

Lemma 13.3.12 (Block-diagonal boundary conditions at the source length). Lean Lean Lean ✓ **Stmt** Let $g \geq 2$. Let A_j be irreducible blocks, pairwise inequivalent up to gauge and phase, and injective at the common length $L_0 > 0$. Assume the canonical normalization of [193]:

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}, \quad \sum_a A_a^{j\dagger} \Lambda_j A_a^j = \Lambda_j, \quad \Lambda_j > 0.$$

Let every weight μ_j be nonzero. If $0 < L \leq N$ and $L \geq 3(g-1)(L_0+1)+1$, then

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) \subseteq \bigoplus_j G_N(A_j).$$

In particular, every vector ψ in the left-hand side admits matrices X_j such that

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j).$$

This is the open-boundary decomposition used in [193, Theorem 12, proof lines 1424–1456].

Proof. ✓ **Proof** Put $S_M = \sum_j G_M(A_j)$. For every M with $L \leq M < N$, Theorem 13.2.2.13 gives

$$\left(\bigcap_b \text{Res}_{-,b}^{-1} S_M \right) \cap \left(\bigcap_a \text{Res}_{a,-}^{-1} S_M \right) = S_{M+1}.$$

Induction through the cyclic local constraints therefore gives

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) \subseteq S_N.$$

Since $N \geq L \geq 3(g-1)(L_0+1)+1$, Theorem 13.2.2.12 shows that the spaces $G_N(A_j)$ form an internal sum. Lemma 13.3.4 then supplies matrices X_j such that $\Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j)$ for every j and

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right).$$

□

Theorem 13.3.13 (Doubly normalized block-diagonal boundary conditions at the source length). [Lean](#) [Lean](#) [✓ Stmt](#) Let $g \geq 2$, assume the physical alphabet is nonempty, and let A_j be irreducible blocks with positive bond dimensions, pairwise inequivalent up to gauge and phase, and injective at the common length $L_0 > 0$. Assume normalized self-overlap and

$$\sum_a (A_a^j)^\dagger A_a^j = \mathbb{1},$$

$$\sum_a A_a^j (A_a^j)^\dagger = \mathbb{1}.$$

Let every weight μ_j be nonzero. If $0 < L \leq N$ and $L \geq 3(g-1)(L_0+1)+1$, then

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) \subseteq \bigoplus_j G_N(A_j).$$

In particular, every vector ψ in the left-hand side admits matrices X_j such that

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j)$$

for every j .

Proof. [✓ Proof](#) Put $S_M = \sum_j G_M(A_j)$. For every M with $L \leq M < N$, Theorem 13.2.2.16 gives

$$\left(\bigcap_b \text{Res}_{-,b}^{-1} S_M \right) \cap \left(\bigcap_a \text{Res}_{a,-}^{-1} S_M \right) = S_{M+1}.$$

Induction through the cyclic local constraints therefore yields

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) \subseteq S_N.$$

At the stated length, the block-space independence theorem makes this sum internal. Lemma 13.3.4 then supplies the matrices X_j . $\square$

The theorem gives $\Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j)$. This is not enough for periodic-boundary membership when a cyclic window crosses the chosen cut. The boundary-cut comparison of Lemma 13.3.1.24 concerns the separate assertion $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$.

Theorem 13.3.14 (Boundary compatibility in the trace-preserving normalization). [Lean](#) [✓ Stmt](#) Suppose that $\sum_b (A^b)^\dagger A^b = \mathbb{1}$ and that $A^b C_a = D_b A^a$ for every pair of physical letters a, b . Then $C_a = E A^a$, where $E = \sum_b (A^b)^\dagger D_b$. This is the trace-preserving form of the boundary calculation in [193, Theorem 12].

Proof. [✓ Proof](#) Multiplication on the left by $(A^b)^\dagger$ and summation over b gives $C_a = \sum_b (A^b)^\dagger D_b A^a = E A^a$. $\square$

Theorem 13.3.15 (Block intersections in the trace-preserving normalization). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A^0, \dots, A^{r-1}$ satisfy $\sum_a (A^{j,a})^\dagger A^{j,a} = \mathbb{1}$ for each j . Suppose that the tuples $(A^{0,w}, \dots, A^{r-1,w})$, with $|w| = n$, span $\prod_j M_{D_j}(\mathbb{C})$. Writing $S_m = \sum_j G_m(A^j)$, one has

$$(\mathbb{C}^d \otimes S_{n+1}) \cap (S_{n+1} \otimes \mathbb{C}^d) = S_{n+2}.$$

Proof. ✓ Proof The two restrictions provide boundary matrices C_a^j, D_b^j (Lemma 13.2.1.14). The simultaneous word span gives $A^{j,b}C_a^j = D_b^jA^{j,a}$ (Lemma 13.2.1.4). Set $E_j = \sum_b (A^{j,b})^\dagger D_b^j$. Then $C_a^j = E_jA^{j,a}$ by Theorem 13.3.14, so each boundary summand lies in $G_{n+2}(A^j)$ (Lemma 13.2.1.12). The reverse inclusion follows by restricting a boundary-condition vector at its first or last physical letter (Lemmas 11.5.2.1 and 11.5.2.2):

$$\begin{aligned} G_{n+2}(A^j) &\subseteq \mathbb{C}^d \otimes G_{n+1}(A^j), \\ G_{n+2}(A^j) &\subseteq G_{n+1}(A^j) \otimes \mathbb{C}^d. \end{aligned}$$

□

Theorem 13.3.16 (Open-chain ground space of trace-preserving blocks). Lean Lean ✓ Stmt Let $d > 0$, and let $B = \bigoplus_j \mu_j A^j$, where every μ_j is nonzero and $\sum_a (A^{j,a})^\dagger A^{j,a} = \mathbb{1}$ for every j . Suppose that the simultaneous word tuples span the full product matrix algebra at every length $n \geq L_0$. For $L_0 + 1 \leq L \leq N$, the open-chain parent Hamiltonian with interaction range L satisfies

$$\ker H_{N,L}^{\text{open}}(B) = G_N(B) = \sum_j G_N(A^j).$$

Proof. ✓ Proof By Definition 12.3.7.1, the Hamiltonian is a sum of orthogonal projectors,

$$\begin{aligned} H_{N,L}^{\text{open}}(B) &= \sum_{i \in I_{L,N}} h_{i,\text{ES}}(B, L), \\ h_{i,\text{ES}}(B, L) &\geq 0. \end{aligned}$$

Hence, by Theorem 12.3.7.12,

$$\ker H_{N,L}^{\text{open}}(B) = \bigcap_{i \in I_{L,N}} \ker h_{i,\text{ES}}(B, L),$$

so, by Lemma 12.3.11, every contiguous length- L restriction of a zero-energy vector lies in $G_L(B)$ under the identification of Lemma 11.3.2. The block intersection identity of Theorem 13.3.15 extends these constraints one site at a time (Lemma 11.5.5.17), giving membership in $G_N(B)$. Conversely, $G_N(B) \subseteq \ker H_{N,L}^{\text{open}}(B)$ by Theorem 12.3.7.14. Nonzero block weights identify $G_N(B)$ with the sum of the block spaces (Theorem 13.3.6). □

Theorem 13.3.17 (Three-block pairwise separation from block injectivity). Lean ✓ Stmt Let $A^0, \dots, A^{r-1}$ be tensors with positive bond dimensions such that each A^j is irreducible, satisfies $\sum_a (A^{j,a})^\dagger A^{j,a} = \mathbb{1}$, and has self-overlap tending to 1 with the chain length. Suppose that no two distinct blocks are related by a virtual gauge and a nonzero complex scalar. Let $L_0 > 0$ and suppose that every A^j is block injective at the lengths $L_0, L_0 + 1$ and $S = 3(L_0 + 1)$. Then for all $k \neq j$ there are coefficients c_ω with

$$\begin{aligned} \sum_{|\omega|=S} c_\omega A_k^\omega &= I, \\ \sum_{|\omega|=S} c_\omega A_j^\omega &= 0. \end{aligned}$$

Proof. ✓ Proof Fix $k \neq j$. If $D_k \neq D_j$, the strict-size branch of Lemma 13.2.5 at the comparison length $L_0 + 1$ gives

$$\text{span}\{(A_k^\omega, A_j^\omega) : |\omega| = S\} = M_{D_k}(\mathbb{C}) \times M_{D_j}(\mathbb{C}).$$

If $D_k = D_j$, the blocks are not related by a virtual gauge and a nonzero scalar, so their periodic matrix product vectors are not proportional at arbitrarily long chain lengths, and the same lemma gives $\mathcal{G}_S^{A_k} \cap \mathcal{G}_S^{A_j} = \{0\}$ and the same pair span. In both cases the pair is trace separated at length S , which is Lemma E.3.1, and Lemma 13.2.9 converts the pair span into the displayed equations. $\square$

Theorem 13.3.18 (Eventual word span from block selectors). Lean ✓ Stmt *Let $A^0, \dots, A^{r-1}$ be tensors, let $p > 0$, and suppose that every A^j is block injective at length p . Suppose that for every k there are coefficients $c_\omega^{(k)}$ with*

$$\sum_{|\omega|=s} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

Then there is L_0 such that, for every $L \geq L_0$, the tuples $(A^{0,w}, \dots, A^{r-1,w})$, with $|w| = L$, span $\prod_j M_{D_j}(\mathbb{C})$.

Proof. ✓ Proof Take $L_0 = s + p$ and let $L \geq L_0$. Since $L - s \geq p$, every A^j is block injective at length $L - s$ by Theorem 6.1.1.24. Lemma 13.2.13 with the selectors of length s gives the full product span at length $(L - s) + s = L$. $\square$

Theorem 13.3.19 (Eventual simultaneous word span of primitive blocks). Lean ✓ Stmt *Let $A^0, \dots, A^{r-1}$ be trace-preserving primitive tensors with positive bond dimensions and positive definite fixed points. Suppose that no two distinct blocks are related by conjugation and a nonzero complex scalar. Then there is L_0 such that, for every $L \geq L_0$, the tuples $(A^{0,w}, \dots, A^{r-1,w})$, with $|w| = L$, span $\prod_j M_{D_j}(\mathbb{C})$.*

Proof. ✓ Proof Primitivity gives normality (Theorem 6.7.4.4), hence a common positive length p at which every block is injective (Theorem 6.5.8). Injectivity persists at every length at least p (Theorem 6.1.1.24), in particular at $p+1$ and $3(p+1)$. Each block is irreducible and has primitive transfer map, so its self-overlap satisfies $O_{A^j A^j}(N) \rightarrow 1$ (Theorem C.5.2). Theorem 13.3.17 with $L_0 = p$ gives, for every ordered pair $k \neq j$, a word polynomial of length $3(p+1)$ equal to I on block k and to 0 on block j . Lemma 13.2.10 multiplies these polynomials into selectors of the common length $s = (r-1) \cdot 3(p+1)$:

$$\sum_{|\omega|=s} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

Theorem 13.3.18 then gives the full product span at every length $L \geq s + p$. $\square$

Theorem 13.3.20 (Uniform open-chain kernel threshold for primitive blocks). Lean ✓ Stmt *Let $d > 0$ and $B = \bigoplus_j \mu_j A^j$, with nonzero coefficients μ_j . Suppose that the blocks have positive bond dimensions, are trace-preserving and primitive with positive definite fixed points, and are pairwise inequivalent under conjugation and a nonzero scalar. There is $L_0 > 0$ such that for every $L_0 \leq L \leq N$,*

$$\ker H_{N,L}^{\text{open}}(B) = G_N(B).$$

Proof. ✓ Proof Choose L_0 one larger than the eventual simultaneous word-span threshold of Theorem 13.3.19 and apply Theorem 13.3.16. $\square$

Theorem 13.3.21 (Nonzero rescalings preserve gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let A and B be tensors of the same bond dimension, and let $c, e \in \mathbb{C}$ be nonzero. If cA and eB are gauge-phase equivalent, then A and B are gauge-phase equivalent.

Proof. [✓ Proof](#) If $eB^i = \zeta X(cA^i)X^{-1}$ for every i , with $\zeta \neq 0$, then

$$B^i = \frac{\zeta c}{e} X A^i X^{-1}$$

for every i , and $\zeta c/e \neq 0$. □

Theorem 13.3.22 (Primitive representatives of inequivalent normal blocks). [Lean](#) [✓ Stmt](#) Let $(A^i)_{i \in I}$ be normal tensors with positive bond dimensions. Suppose that no two distinct blocks are related by a virtual gauge and a nonzero complex scalar. Then there are tensors $(B^i)_{i \in I}$ of the same respective bond dimensions and positive definite matrices ρ_i such that each B^i is primitive and trace preserving, ρ_i is its fixed point, and $G_L(A^i) = G_L(B^i)$ for every i and every length L . The tensors B^i remain pairwise inequivalent under a virtual gauge and a nonzero scalar.

Proof. [✓ Proof](#) Normalize each block independently by Theorem 12.8.8. The resulting nonzero rescaling and virtual gauge leave its local ground spaces unchanged. Thus $B^{i,a} = \zeta_i X_i A^{i,a} X_i^{-1}$ for every physical index a , with $\zeta_i \neq 0$ and X_i invertible. If $D_i = D_j$ and $B^{i,a} = \zeta Y B^{j,a} Y^{-1}$ for every a , with $\zeta \neq 0$ and Y invertible, then composing the gauges gives

$$A^{i,a} = \frac{\zeta \zeta_j}{\zeta_i} (X_i^{-1} Y X_j) A^{j,a} (X_i^{-1} Y X_j)^{-1}$$

for every a (Lemma 13.2.2.9 transports the gauges, and Theorem 13.3.21 removes the nonzero rescalings), contrary to the hypothesis. □

Theorem 13.3.23 (Blockwise replacement preserves parent Hamiltonians). [Lean](#) [Lean](#) [✓ Stmt](#) Let A^j and B^j have the same respective bond dimensions, and let all coefficients μ_j, ν_j be nonzero. If $G_L(A^j) = G_L(B^j)$ for every j , then the assembled tensors $A = \bigoplus_j \mu_j A^j$ and $B = \bigoplus_j \nu_j B^j$ satisfy $G_L(A) = G_L(B)$. Their canonical periodic parent Hamiltonians of range L coincide at every volume: $H_{N,L}^{\text{ES}}(A) = H_{N,L}^{\text{ES}}(B)$.

Proof. [✓ Proof](#) The assembled local ground spaces are the sums of the respective block spaces. Equality of these spaces gives equality of their orthogonal complement projections and hence of every translated parent interaction. □

Theorem 13.3.24 (Primitive block representatives with unchanged Hamiltonians). [Lean](#) [✓ Stmt](#) Let $A = \bigoplus_j \mu_j A^j$ have nonzero coefficients and normal blocks of positive bond dimensions. Suppose that no two distinct blocks are related by conjugation and a nonzero scalar. There are normalized primitive blocks B^j , with positive definite invariant matrices, which remain pairwise inequivalent and satisfy $H_{N,L}^{\text{ES}}(A) = H_{N,L}^{\text{ES}}(B)$ for every L, N , where $B = \bigoplus_j \mu_j B^j$.

Proof. [✓ Proof](#) Normalize each block while retaining its local ground spaces and the inequivalence of distinct blocks. The assembled parent Hamiltonians therefore coincide at each interaction range and volume. □

Lemma 13.3.25 (Primitive normalization at a fixed C1 length). [Lean](#) [✓ Stmt](#) Suppose that a finite family of nonzero-dimensional MPS tensors is block injective at a common positive word length L_0 , and that its members are pairwise inequivalent under virtual conjugation and nonzero

scalar multiplication. Then each tensor admits a normalized primitive representative, with positive definite invariant matrix, which remains block injective at length L_0 . The representatives remain pairwise inequivalent. For any nonzero weights, the original and normalized block sums have exactly the same periodic parent Hamiltonians at every interaction range and chain length.

Proof. [✓ Proof](#) By Theorem 12.8.8, normalize each tensor by a nonzero scalar and an invertible virtual conjugation. These operations preserve block injectivity at each fixed word length, preserve all local ground spaces, and preserve inequivalence. By Theorem 13.3.23, equal block ground spaces give equal ground spaces of the weighted block sums, hence every local parent projection, and therefore every periodic parent Hamiltonian, is unchanged. $\square$

Theorem 13.3.26 (Propagation of simultaneous spans in trace-preserving form). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that $\sum_a (A^{j,a})^\dagger A^{j,a} = \mathbb{1}$ for every block j . If the simultaneous length- L word tuples span the product matrix algebra, the same holds at every length $n \geq L$.

Proof. [✓ Proof](#) For an arbitrary tuple $(M_j)_j$, represent $(M_j(A^{j,a})^\dagger)_j$ as a linear combination of length- L word tuples. Append the letter a and sum over a . The resulting tuple is $(M_j)_j$ by trace-preserving normalization. This proves the one-site extension, and induction gives every larger length. $\square$

Theorem 13.3.27 (The printed open-chain threshold for primitive blocks). [Lean](#) [✓ Stmt](#) Let $d > 0$, $r \geq 2$, and $B = \bigoplus_{j=0}^{r-1} \mu_j A^j$, with all coefficients nonzero. Suppose that the blocks have positive bond dimensions, are trace-preserving and primitive with positive definite invariant matrices, and are pairwise inequivalent under conjugation and multiplication by a nonzero scalar. Suppose that each block is block injective at the supplied common length $L_0 > 0$. Then for

$$3(r-1)(L_0+1)+1 \leq R \leq N,$$

the open-chain parent Hamiltonian satisfies $\ker H_{N,R}^{\text{open}}(B) = G_N(B)$. This is the interaction threshold of [193, Theorem 12] in trace-preserving coordinates.

Proof. [✓ Proof](#) The simultaneous-span estimate of Lemma 8.3.3, the direct-sum lemma of [193, lines 1346–1408], gives the full product matrix algebra at length $3(r-1)(L_0+1)$. Trace-preserving propagation keeps the span full at every larger length. The block intersection theorem therefore gives the kernel identity at the stated interaction range. $\square$

Theorem 13.3.28 (A periodic gap for normal block sums). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A^0, \dots, A^{r-1}$ be normal MPS tensors with positive physical and bond dimensions, pairwise inequivalent under virtual conjugation and nonzero scalar multiplication. Let $B = \bigoplus_j \mu_j A^j$, where every μ_j is nonzero. There is a positive integer R_0 such that, for every $R \geq R_0$, there are constants $\gamma_R > 0$ and $N_R \in \mathbb{N}$ for which the canonical periodic parent Hamiltonian $H_{R,N}$ satisfies

$$\gamma_R \|v\| \leq \|H_{R,N} v\| \tag{13.2}$$

for every $N \geq N_R$ and every $v \perp \ker H_{R,N}$. In particular, at least one positive interaction range has a uniform eventual periodic gap.

Proof. [✓ Proof](#) Replace each block by a normalized primitive representative. This preserves inequivalence and leaves every parent Hamiltonian unchanged. The uniform gap theorem for primitive block sums therefore applies with the same interaction ranges and the same constants. $\square$

Theorem 13.3.29 (The gap at the prescribed interaction range). [Lean](#) [✓ Stmt](#) Let $A^0, \dots, A^{r-1}$, $r \geq 2$, be MPS tensors with positive physical and bond dimensions, block injective at a common positive word length L_0 , and pairwise inequivalent under virtual conjugation and nonzero scalar multiplication. Let $B = \bigoplus_j \mu_j A^j$, with every $\mu_j \neq 0$. For every prescribed range

$$R \geq 3(r-1)(L_0 + 1) + 1, \quad (13.3)$$

there is a constant $\gamma_R > 0$ such that the canonical periodic parent Hamiltonian $H_{R,N}$ of B satisfies

$$\gamma_R \|v\| \leq \|H_{R,N} v\| \quad (13.4)$$

for every chain length $N \in \mathbb{N}$ and every $v \perp \ker H_{R,N}$. This is the block-injective parent-Hamiltonian gap of [16, Section IV.C] at the interaction bound of [193, Theorem 12].

Proof. [✓ Proof](#) Normalize the blocks to primitive representatives without changing the specified injectivity length L_0 . Their periodic parent Hamiltonians agree with the original ones at every range and chain length. Apply the primitive-block gap theorem at the prescribed range. $\square$

Theorem 13.3.30 (Cyclic propagation of boundary intertwiners). [Lean](#) [✓ Stmt](#) Let $N \geq 1$, let $s \in \mathbb{Z}/N\mathbb{Z}$, and suppose matrices Y_i satisfy $Y_i A^a = A^a Y_{s+i}$ for every i and physical label a . Then, for every word w of length N and every i ,

$$Y_i A^w = A^w Y_i. \quad (13.5)$$

Proof. [✓ Proof](#) Propagating the intertwining identity through the word gives $Y_i A^w = A^w Y_{N+s+i}$. Since $Ns = 0$ in the cyclic index group, the two boundary matrices coincide. $\square$

Theorem 13.3.31 (Cyclic closure of block boundary matrices). [Lean](#) [✓ Stmt](#) Let $N \geq 2$. Suppose that the simultaneous words of length $N-1$ in the tensors A^j span the product of their matrix algebras. Let ψ be a vector such that, for every cyclic cut $i \in \mathbb{Z}/N\mathbb{Z}$, there are boundary matrices X_i^j with

$$T_i \psi = \sum_j \Gamma_N^{A^j}(X_i^j).$$

Then ψ belongs to the span of the periodic MPS vectors of the blocks. This is the block-diagonal cyclic closure argument in [16, Section IV.C].

Proof. [✓ Proof](#) Choose the boundary matrices X_i^j of the hypothesis at each cyclic cut i . Comparing adjacent cuts against all complementary words of length $N-1$ gives $X_i^j A^{j,a} = A^{j,a} X_{i-1}^j$. Iterating this identity once around the ring yields

$$X_i^j A^{j,a_1} \dots A^{j,a_N} = A^{j,a_1} \dots A^{j,a_N} X_i^j.$$

Each block is injective at length $N-1$, hence also at length N . Its boundary matrix therefore commutes with the full matrix algebra and is scalar. The representation at cut zero is consequently a linear combination of the periodic block vectors, that is, an element of the sum of their lines (Lemma 13.1.2). $\square$

Theorem 13.3.32 (Periodic ground space at a simultaneous injectivity length). [Lean](#) [✓ Stmt](#) Let $B = \bigoplus_j \mu_j A^j$ have nonzero block coefficients and positive physical dimension. Suppose that each block is trace preserving and that the simultaneous words of length $S > 0$ span the product matrix algebra. For every $S+1 \leq L \leq N$, the periodic parent Hamiltonian satisfies

$$\ker H_L^{(N)}(B) = \text{span}\{V^{(N)}(A^j) : j\}.$$

Proof. [✓ Proof](#) Trace-preserving normalization propagates simultaneous injectivity to every length at least S , in particular to $N - 1$. The open-chain intersection property expresses each periodic ground vector as a sum of block boundary vectors at every cut. Cyclic closure places this vector in the span of the periodic block vectors. Conversely, each periodic block vector satisfies every local parent constraint. $\square$

Theorem 13.3.33 (Independent rescaling of simultaneous word spans). [Lean](#) [✓ Stmt](#) Suppose the length- S word tuples of a finite family $A^0, \dots, A^{r-1}$ span the full product matrix algebra. If every ζ_j is nonzero, the word tuples of $\zeta_0 A^0, \dots, \zeta_{r-1} A^{r-1}$ also span that algebra at the same length S .

Proof. [✓ Proof](#) Rescaling the j th block multiplies its length- S words by ζ_j^S . For any target tuple $(M_j)_j$, first express $(\zeta_j^{-S} M_j)_j$ in the original word span. The same coefficients express $(M_j)_j$ in the rescaled word span. $\square$

Theorem 13.3.34 (Primitive normalization at a simultaneous injectivity length). [Lean](#) [Lean](#) [✓ Stmt](#) Let $S > 0$, and suppose the length- S word tuples of a finite family $A^0, \dots, A^{r-1}$ with positive bond dimensions span the full product matrix algebra. There are normalized primitive representatives $B^0, \dots, B^{r-1}$, with positive definite fixed points, whose word tuples still span the full product algebra at length S . For every $0 \leq j < r$, every length L , and every chain length N ,

$$G_L(A^j) = G_L(B^j),$$

$$\text{span}\{V^{(N)}(A^k) : 0 \leq k < r\} = \text{span}\{V^{(N)}(B^k) : 0 \leq k < r\}.$$

For any nonzero weights μ_j , the periodic parent Hamiltonians of $\bigoplus_j \mu_j A^j$ and $\bigoplus_j \mu_j B^j$ are equal at every interaction range and chain length.

Proof. [✓ Proof](#) Projection of the simultaneous span onto each matrix summand proves injectivity of every block at length S . Each block is therefore normal and admits a primitive representative obtained by a rescaling by some $\zeta_j \neq 0$ and an invertible virtual conjugation. Both operations preserve the simultaneous word span at the specified length. They preserve every component ground space, and the periodic component vectors satisfy

$$V^{(N)}(B^j) = \zeta_j^N V^{(N)}(A^j).$$

Since $\zeta_j^N \neq 0$, each block spans the same line for A^j and B^j , and the two spans, being the sums of these lines (Lemma 13.1.2), agree. Taking sums of the component ground spaces proves equality of the assembled parent Hamiltonians. $\square$

Theorem 13.3.35 (Periodic ground space at simultaneous block injectivity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A^0, \dots, A^{r-1}$ have a common positive physical dimension and positive bond dimensions. Suppose their length- S word tuples span the full product matrix algebra, where $S > 0$, and let $B = \bigoplus_j \mu_j A^j$ with all $\mu_j \neq 0$. For every interaction range and chain length satisfying

$$S + 1 \leq L \leq N,$$

the periodic local constraints have ground space

$$G_L^{(N)}(B) = \text{span}\{V^{(N)}(A^j) : 0 \leq j < r\}.$$

This is the kernel of the canonical parent Hamiltonian. It is also the kernel of the sum of any positive terms whose cyclic local kernels agree with those of the canonical parent terms. The result includes the minimal ring $N = S + 1$ and gives the block-injective closure statement of [16, Section IV.C] at the interaction range specified by the preceding intersection argument.

Proof. ✓ **Proof** Normalize the blocks while retaining simultaneous injectivity at length S . The parent Hamiltonians and the span of the periodic component vectors are unchanged. Apply the trace-preserving block closure theorem. For positive operators H_i ,

$$\ker\left(\sum_i H_i\right) = \bigcap_i \ker H_i.$$

Replacing the local projectors by positive terms with the same local kernels therefore preserves the ground space. $\square$

Theorem 13.3.36 (A bounded range including the minimal periodic ring). Lean Lean Lean ✓ **Stmt** Let A be a canonical tensor with positive physical dimension and positive bond dimension D . There are distinct normal representatives $(A_j)_j$ and a positive integer $R \leq 3D^5$ such that, for every $R \leq L \leq N$, every collection of positive operators H_i with the same kernels as the canonical range- L parent terms satisfies

$$\ker\left(\sum_i H_i\right) = \text{span}\{V^{(N)}(A_j) : j\}.$$

The equality includes $N = L = R$, and in particular holds at $N = L = 3D^5$ for the canonical parent Hamiltonian. Repeated copies of the normal representatives are allowed in A .

Proof. ✓ **Proof** The representative bond dimensions sum to at most D . The simultaneous span estimate supplies $S > 0$ with $S + 1 \leq 3D^5$. Take $R = S + 1$ and apply periodic closure for canonical tensors at every $R \leq L \leq N$. $\square$

Theorem 13.3.37 (Local spaces of canonical representatives). Lean ✓ **Stmt** Let A be in canonical form, and let $B^0, \dots, B^{g-1}$ be its distinct normal representatives. Repeated copies and their nonzero coefficients do not change the positive-length local spaces: for every $L > 0$,

$$G_L(A) = \bigvee_{j=0}^{g-1} G_L(B^j).$$

This is the local-space consequence of the canonical decomposition in [16].

Proof. ✓ **Proof** The coisometric reconstruction preserves the local space on every positive-length interval. The local space of the resulting direct sum is the sum of its block spaces. Each repeated copy differs from its representative by a virtual similarity and a nonzero scalar, both of which preserve the local space. Taking the sum once over each distinct representative therefore gives the same space. $\square$

Theorem 13.3.38 (Canonical ground spaces at a simultaneous injectivity length). Lean Lean Lean ✓ **Stmt** Let A be a canonical MPS tensor with positive physical dimension, including arbitrary multiplicities of its distinct normal blocks $B^0, \dots, B^{g-1}$. Suppose that their simultaneous words of length $S > 0$ span the product matrix algebra. For every $S + 1 \leq L \leq N$, every periodic parent Hamiltonian with positive interactions and the prescribed length- L local kernels satisfies

$$\ker H_L^{(N)}(A) = \text{span}\{V^{(N)}(B^j) : 0 \leq j < g\}.$$

The conclusion includes the minimal ring $N = L = S + 1$ and the canonical orthogonal-projection interaction. This is the block-injective closure conclusion of [16, Section IV.C].

Proof. ✓ Proof Replace the original canonical tensor by the direct sum containing one copy of each normal representative. Their local spaces, hence their periodic local constraints, coincide. Apply the simultaneous block-injectivity closure theorem at the same supplied length S . For positive operators H_i ,

$$\ker\left(\sum_i H_i\right) = \bigcap_i \ker H_i.$$

Replacing each orthogonal projection by any positive interaction with the same kernel therefore leaves the periodic ground space unchanged. □

13.3.1 Cyclic windows at the boundary cut

A cyclic length- L window either stays inside the chosen linear interval ($i + L \leq N$) or crosses the boundary cut ($N < i + L$). The non-wrapping case follows from trace cyclicity, while the crossing case requires a boundary-matrix identity $\mu_j^N X_j A_\beta^j = A_\beta^j Y$. The lemmas below establish the single-block periodic membership $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ in both cases.

Lemma 13.3.1.1 (Cyclic restrictions of block-diagonal boundary vectors). Lean ✓ Stmt Let $B = \bigoplus_{j=1}^g \mu_j A_j$, and fix block-diagonal boundary conditions X_j . For every cyclic restriction $R_{i,\tau}$ from N sites to L sites,

$$R_{i,\tau}\left(\Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right)\right) = \sum_{j=1}^g R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right).$$

Proof. ✓ Proof Apply Lemma 13.3.3 at length N , then use linearity of the cyclic restriction $R_{i,\tau}$. □

Lemma 13.3.1.2 (Local direct-sum constraint from block-diagonal boundary conditions). Lean ✓ Stmt Let $B = \bigoplus_{j=1}^g \mu_j A_j$, with $\mu_j \neq 0$, and suppose

$$\begin{aligned} \psi &= \Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right), \\ \psi &\in \mathcal{G}_{N,L}(B). \end{aligned}$$

Then every local constraint of ψ gives

$$\sum_{j=1}^g R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right) \in \bigvee_{j=1}^g G_L(A_j).$$

Proof. ✓ Proof The hypothesis $\psi \in \mathcal{G}_{N,L}(B)$ says that the left-hand side before decomposition lies in $G_L(B)$. Lemma 13.3.1.1 rewrites this local vector as the displayed sum, and Theorem 13.3.6 identifies $G_L(B)$ with $\bigvee_j G_L(A_j)$. □

Lemma 13.3.1.3 (Cyclic windows not crossing the cut). Lean ✓ Stmt Fix block-diagonal boundary conditions X_j . If the cyclic window beginning at i stays inside the chosen linear interval, $i + L \leq N$, then for every block j ,

$$R_{i,\tau}\left(\Gamma_N^{A_j}(\mu_j^N X_j)\right) \in G_L(A_j).$$

Proof. ✓ Proof When the window stays inside the chosen linear interval, the boundary matrix remains outside the window. Write $A_{\text{left}} = A_{\tau_0}^j \cdots A_{\tau_{i-1}}^j$ and $A_{\text{right}} = A_{\tau_{i+L}}^j \cdots A_{\tau_{N-1}}^j$. Trace cyclicity gives

$$R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j)) = \Gamma_L^{A_j}(A_{\text{right}} \mu_j^N X_j A_{\text{left}}).$$

Hence the restricted vector lies in $G_L(A_j)$. □

Lemma 13.3.1.4 (Boundary-crossing coefficient formula). Lean ✓ Stmt Assume $0 < N, L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Write $a = i + L - N$. For a local word σ of length L , set

$$\begin{aligned}\beta &= \sigma_{N-i} \cdots \sigma_{L-1}, \\ \rho &= \tau_a \cdots \tau_{i-1}, \\ \alpha &= \sigma_0 \cdots \sigma_{N-i-1}.\end{aligned}$$

Then the corresponding coefficient of the j -th block component is

$$R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j))(\sigma) = \text{tr}(A_\beta^j A_\rho^j A_\alpha^j \mu_j^N X_j).$$

Proof. ✓ Proof Let ω be the N -site word obtained by inserting σ in the cyclic interval beginning at i and by using τ outside that interval. Since the interval crosses the cut, $\omega = \beta \rho \alpha$. Hence

$$\Gamma_N^{A_j}(\mu_j^N X_j)(\omega) = \text{tr}(A_\beta^j A_\rho^j A_\alpha^j \mu_j^N X_j).$$

This is exactly $R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j))(\sigma)$. □

Lemma 13.3.1.5 (Boundary-crossing trace decomposition from the local constraint). Lean ✓ Stmt Assume $0 < N, L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Put $a = i + L - N$. For a local word σ of length L , write

$$\begin{aligned}\beta &= \sigma_{N-i} \cdots \sigma_{L-1}, \\ \rho &= \tau_a \cdots \tau_{i-1}, \\ \alpha &= \sigma_0 \cdots \sigma_{N-i-1}.\end{aligned}$$

These are the word coordinates obtained by opening the periodic boundary at the chosen cut, in the comparison with the source expressions involving $C_{i_1}^j$ and $D_{i_{m+1}}^j$. If

$$\sum_j R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j)) \in \bigvee_j G_L(A_j),$$

then there are matrices $C_{i,\tau}^j$ such that, for every σ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\tau}^j A_\alpha^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_\alpha^j).$$

Proof. ✓ Proof Decompose the local vector in $\bigvee_j G_L(A_j)$ as a sum of vectors $\Gamma_L^{A_j}(C_{i,\tau}^j)$. Evaluating at σ gives $\sum_j \text{tr}(A_{\alpha\beta}^j C_{i,\tau}^j)$. Trace cyclicity rewrites this as $\sum_j \text{tr}(A_\beta^j C_{i,\tau}^j A_\alpha^j)$. On the other hand, Lemma 13.3.1.4 gives the second trace expression for the same local coefficient sum. □

Lemma 13.3.1.6 (Boundary representation gives fixed-tail boundary traces). [Lean](#) [✓ Stmt](#)
 Assume that the physical dimension d is positive, $0 < N$, $L \leq N$, and $\mu_j \neq 0$ for every block j .
 Let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Let

$$\begin{aligned}\psi &\in \mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right), \\ \psi &= \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right).\end{aligned}$$

For every outside word ρ of length $N - L$, there are matrices $C_{i,\rho}^j$ such that, for every word β of length $i + L - N$ before the cut and every wrapped tail word α of length $N - i$,

$$\sum_j \text{tr} \left(A_\beta^j C_{i,\rho}^j A_\alpha^j \right) = \sum_j \text{tr} \left(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_\alpha^j \right).$$

Proof. [✓ Proof](#) Since $d > 0$, extend the outside word ρ to a full outside configuration τ whose entries between the end of the interval and the cut are the word ρ . By the nonzero-weight hypothesis, Lemma 13.3.1.2 applies and puts the sum of the corresponding block restrictions in $\bigvee_j G_L(A_j)$. Applying Lemma 13.3.1.5 gives the displayed trace equality after rewriting the opened cyclic word as $\beta\rho\alpha$. $\square$

Lemma 13.3.1.7 (Boundary representation gives all fixed-tail boundary traces). [Lean](#) [✓ Stmt](#)
 Assume that the physical dimension d is positive, $0 < N$, $L \leq N$, and $\mu_j \neq 0$ for every block j .
 Let

$$\begin{aligned}\psi &\in \mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right), \\ \psi &= \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right).\end{aligned}$$

Then there are matrices $C_{i,\rho}^j$, indexed by the boundary-crossing start i and by an outside word ρ of length $N - L$, such that whenever $N < i + L$,

$$\sum_j \text{tr} \left(A_\beta^j C_{i,\rho}^j A_\alpha^j \right) = \sum_j \text{tr} \left(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_\alpha^j \right)$$

for every word β of length $i + L - N$ before the cut and every wrapped tail word α of length $N - i$.

Proof. [✓ Proof](#) Apply Lemma 13.3.1.6 to each boundary-crossing start i and each outside word ρ , and choose the resulting matrices $C_{i,\rho}^j$. $\square$

Theorem 13.3.1.8 (Fixed boundary-crossing C^j, D^j comparison from the local block-sum constraint). [Lean](#) [✓ Stmt](#) Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Put $a = i + L - N$ and $\rho = \tau_a \cdots \tau_{i-1}$. Suppose also that the block tensors have the common word-span property at length $N - i$. If

$$\sum_j R_{i,\tau} \left(\Gamma_N^{A_j} (\mu_j^N X_j) \right) \in \bigvee_j G_L(A_j),$$

then there are matrices $C_{i,\tau}^j$ such that, for every block j and every word β of length a ,

$$A_\beta^j C_{i,\tau}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j. \quad (13.6)$$

Proof. [✓ Proof](#) Lemma 13.3.1.5 gives matrices $C_{i,\tau}^j$ with

$$\sum_j \operatorname{tr}(A_\beta^j C_{i,\tau}^j A_\alpha^j) = \sum_j \operatorname{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_\alpha^j)$$

for every word α of length $N - i$ and every word β of length a . The fixed complementary-word trace comparison, applied with X_j replaced by $\mu_j^N X_j$, gives (13.6). $\square$

Theorem 13.3.1.9 (C^j, D^j comparison for a block-diagonal boundary representation). [Lean](#)

[✓ Stmt](#) Let

$$\begin{aligned} \psi &\in \mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right), \\ \psi &= \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right). \end{aligned}$$

Assume that the physical alphabet is nonempty, that every μ_j is nonzero, that $0 < N$, and that $L \leq N$. For every boundary-crossing interval beginning at i , assume that the simultaneous products

$$\begin{aligned} w &\mapsto (A_w^1, \dots, A_w^r), \\ |w| &= N - i, \end{aligned}$$

span the product algebra $\prod_j M_{D_j}(\mathbb{C})$. Then there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, every block j , and every word β before the cut, $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$.

Proof. [✓ Proof](#) Lemma 13.3.1.7 gives matrices $C_{i,\rho}^j$ such that, for every boundary-crossing start i , outside word ρ of length $N - L$, word β of length $i + L - N$ before the cut, and wrapped tail word α of length $N - i$,

$$\sum_j \operatorname{tr}(A_\beta^j C_{i,\rho}^j A_\alpha^j) = \sum_j \operatorname{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_\alpha^j).$$

Apply Lemma 13.2.1.6 with the stated product-algebra span at length $N - i$. Equality of the traces for all α then gives $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$ for every block j . $\square$

Theorem 13.3.1.10 (Fixed boundary-crossing component in the local block space). [Lean](#)

[✓ Stmt](#) Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Suppose that the block tensors have the common word-span property at length $N - i$ and that

$$\sum_j R_{i,\tau} \left(\Gamma_N^{A_j} (\mu_j^N X_j) \right) \in \bigvee_j G_L(A_j).$$

Then each block component of the fixed boundary-crossing restriction lies in its local block space:

$$R_{i,\tau} \left(\Gamma_N^{A_j} (\mu_j^N X_j) \right) \in G_L(A_j).$$

Proof. [✓ Proof](#) Theorem 13.3.1.8 gives matrices $C_{i,\tau}^j$ satisfying (13.6). Thus $E = C_{i,\tau}^j$ satisfies the boundary matrix identity required in Lemma 13.3.1.11, and the displayed membership follows for every block j . $\square$

Lemma 13.3.1.11 (Boundary-crossing cyclic intervals from the boundary matrix identity).

[Lean](#) [✓ Stmt](#) Assume $0 < N$, $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Put $a = i + L - N$. Suppose that there is a matrix E such that for every word β of length a ,

$$\mu_j^N X_j A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j = A_\beta^j E. \quad (13.7)$$

Then

$$R_{i,\tau} \left(\Gamma_N^{A_j} (\mu_j^N X_j) \right) \in G_L(A_j).$$

Proof. [✓ Proof](#) Write a length- L word in the boundary-crossing interval as $\alpha\beta$, where α is the segment inside $\{i, \dots, N-1\}$ and β is the segment after the interval wraps past the cut, inside $\{0, \dots, a-1\}$. The full N -site word appearing in the restriction is then $\beta\tau_a \cdots \tau_{i-1}\alpha$. Trace cyclicity gives

$$\begin{aligned} \text{tr} \left(A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j A_\alpha^j \mu_j^N X_j \right) &= \text{tr} \left(A_\alpha^j \mu_j^N X_j A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j \right) \\ &= \text{tr} \left(A_\alpha^j A_\beta^j E \right), \end{aligned}$$

applying (13.7) to the inside product. The last expression is the coefficient of $\Gamma_L^{A_j}(E)$ at the word $\alpha\beta$. Hence the restricted vector lies in $G_L(A_j)$. $\square$

Lemma 13.3.1.12 (Last crossing-window component trace form). [Lean](#) [✓ Stmt](#) Write $N = M+1$ and let the local length be $m+2 \leq M+1$. Consider the cyclic interval beginning at the last site M . For an outside configuration τ , put

$$\begin{aligned} \rho &= \tau_{m+1} \cdots \tau_{M-1}, \\ C_a^j &= A_\rho^j A_a^j (\mu_j^{M+1} X_j). \end{aligned}$$

Then the j -th component of the last crossing-window restriction has the left-boundary trace form

$$\begin{aligned} R_{M,\tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) &= \Phi^j, \\ \Phi^j(a, w, b) &= \text{tr} \left(A_b^j C_a^j A_w^j \right). \end{aligned}$$

Proof. [✓ Proof](#) The cyclic interval beginning at M reads the last site first and then wraps to the sites $0, \dots, m$. Thus, for the local word awb , the full word in the trace is $wb\rho a$. Hence the j -th coefficient is

$$\text{tr} \left(A_w^j A_b^j A_\rho^j A_a^j (\mu_j^{M+1} X_j) \right) = \text{tr} \left(A_b^j A_\rho^j A_a^j (\mu_j^{M+1} X_j) A_w^j \right),$$

where the equality follows from trace cyclicity. This is the displayed left-boundary trace form. $\square$

Lemma 13.3.1.13 (Last crossing-window trace form). [Lean](#) [✓ Stmt](#) Write $N = M+1$ and let the local length be $m+2 \leq M+1$. Consider the cyclic interval beginning at the last site M . For an outside configuration τ , put

$$\begin{aligned} \rho &= \tau_{m+1} \cdots \tau_{M-1}, \\ C_a^j &= A_\rho^j A_a^j (\mu_j^{M+1} X_j). \end{aligned}$$

Then the block sum of the last crossing-window restrictions has the left-boundary trace form

$$\begin{aligned} \sum_j R_{M,\tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) &= \sum_j \Phi^j, \\ \Phi^j(a, w, b) &= \text{tr} \left(A_b^j C_a^j A_w^j \right). \end{aligned}$$

Proof. ✓ Proof Sum Lemma 13.3.1.12 over the block index. □

Lemma 13.3.1.14 (Last crossing-window coefficient comparison). Lean ✓ Stmt *In the setting of Lemma 13.3.1.13, assume that the length- m simultaneous word tuples span the product algebra and that*

$$\sum_j R_{M,\tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j).$$

Then there are matrices E_j such that, for every block j and all boundary letters a, b ,

$$A_b^j (A_\rho^j A_a^j (\mu_j^{M+1} X_j)) = A_b^j E_j A_a^j. \quad (13.8)$$

Proof. ✓ Proof Lemma 13.3.1.13 writes the local block sum in the form

$$\begin{aligned} \Phi(a, w, b) &= \sum_j \text{tr} \left(A_b^j C_a^j A_w^j \right), \\ C_a^j &= A_\rho^j A_a^j (\mu_j^{M+1} X_j). \end{aligned}$$

Applying Lemma 13.2.1.3 gives (13.8). □

Lemma 13.3.1.15 (Last crossing-window component membership). Lean ✓ Stmt *In the setting of Lemma 13.3.1.13, assume that the length- m simultaneous word tuples span the product algebra and that*

$$\sum_j R_{M,\tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j).$$

Then, for every block j ,

$$R_{M,\tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in G_{m+2}(A^j).$$

Proof. ✓ Proof Lemma 13.3.1.14 gives matrices E_j satisfying (13.8). By Lemma 13.3.1.12, the j -th cyclic restriction is the left-boundary component determined by C_a^j . The displayed identity makes that component equal to $\Gamma_{m+2}^{A_j}(E_j)$, hence it lies in $G_{m+2}(A^j)$. □

Lemma 13.3.1.16 (Last crossing-window coefficients from a block-diagonal chain vector). Lean ✓ Stmt *Let $B = \bigoplus_j \mu_j A^j$, and assume $\mu_j \neq 0$ for every block j . Suppose*

$$\begin{aligned} \psi &= \Gamma_{M+1}^B \left(\bigoplus_j X_j \right), \\ \psi &\in \mathcal{G}_{M+1, m+2}(B). \end{aligned}$$

Assume also that the length- m simultaneous word tuples span the product algebra. Then, for every outside configuration τ , with $\rho = \tau_{m+1} \cdots \tau_{M-1}$, there are matrices E_j such that $A_b^j (A_\rho^j A_a^j (\mu_j^{M+1} X_j)) = A_b^j E_j A_a^j$ for every block j and all boundary letters a, b .

Proof. ✓ Proof The local constraint of ψ at the cyclic interval beginning at M gives

$$\sum_j R_{M,\tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j)$$

by Lemma 13.3.1.2. Applying Lemma 13.3.1.14 gives the displayed equations. □

Lemma 13.3.1.17 (Last crossing-window component membership from a chain vector). [Lean](#)

[✓ Stmt](#) Let $B = \bigoplus_j \mu_j A^j$, and assume $\mu_j \neq 0$ for every block j . Suppose

$$\begin{aligned}\psi &= \Gamma_{M+1}^B \left(\bigoplus_j X_j \right), \\ \psi &\in \mathcal{G}_{M+1, m+2}(B).\end{aligned}$$

If the length- m simultaneous word tuples span the product algebra, then for every outside configuration τ and every block j ,

$$R_{M, \tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in G_{m+2}(A^j).$$

Proof. [✓ Proof](#) Lemma 13.3.1.2, applied to the window beginning at the last site, gives

$$\sum_j R_{M, \tau} \left(\Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j).$$

Lemma 13.3.1.15 then applies to each block component. □

Lemma 13.3.1.18 (Boundary-crossing intervals from a right boundary-matrix identity). [Lean](#)

[✓ Stmt](#) Assume $0 < N$ and $L \leq N$, and let the cyclic interval beginning at i cross the boundary cut, so $N < i + L$. Put $a = i + L - N$. Suppose that there is a matrix Y such that, for every word β of length a , $\mu_j^N X_j A_\beta^j = A_\beta^j Y$. Then

$$R_{i, \tau} \left(\Gamma_N^{A_j} (\mu_j^N X_j) \right) \in G_L(A_j).$$

Proof. [✓ Proof](#) Let $E := Y A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j$. Then, for every word β on the segment $0, \dots, a-1$,

$$\mu_j^N X_j A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j = A_\beta^j E.$$

Lemma 13.3.1.11 gives the stated local constraint. □

Lemma 13.3.1.19 (Boundary-crossing cyclic intervals suffice). [Lean](#) [✓ Stmt](#) Assume $0 < N$ and $L \leq N$. Fix block-diagonal boundary conditions X_j . Suppose that, for every block j , every cyclic interval beginning at i with $N < i + L$, and every configuration τ ,

$$R_{i, \tau} \left(\Gamma_N^{A_j} (\mu_j^N X_j) \right) \in G_L(A_j).$$

Then, for every block j , $\Gamma_N^{A_j} (\mu_j^N X_j) \in \mathcal{G}_{N, L}(A_j)$.

Proof. [✓ Proof](#) By definition, membership in $\mathcal{G}_{N, L}(A_j)$ is the collection of all cyclic length- L local constraints. For an interval beginning at i , either $i + L \leq N$ or $N < i + L$. In the first case, use Lemma 13.3.1.3. In the second case, use the displayed hypothesis. □

Lemma 13.3.1.20 (Right boundary-matrix identities give single-block periodic-boundary constraints). [Lean](#) [✓ Stmt](#) Assume $0 < N$ and $L \leq N$. Fix block-diagonal boundary conditions X_j . Suppose that, for every block j , every boundary-crossing interval beginning at i , and every configuration τ , there is a matrix $Y_{j, i, \tau}$ such that, for every word β of length $a = i + L - N$, $\mu_j^N X_j A_\beta^j = A_\beta^j Y_{j, i, \tau}$. Then, for every block j , $\Gamma_N^{A_j} (\mu_j^N X_j) \in \mathcal{G}_{N, L}(A_j)$.

Proof. [✓ Proof](#) Lemma 13.3.1.18 turns the displayed boundary identity into the local constraint for each boundary-crossing interval. Lemma 13.3.1.19 then yields $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$, handling both boundary-crossing intervals $N < i + L$ and non-crossing intervals $i + L \leq N$. $\square$

Lemma 13.3.1.21 (Spanning matrix identities give single-block constraints). [Lean](#) [✓ Stmt](#) Assume $0 < N$ and $L \leq N$, and assume that, for every block j , the length- $N - L$ word products of A_j span the full matrix algebra. Suppose that, for every block j , every boundary-crossing interval beginning at i , and every outside word ρ of length $N - L$, there is a matrix $E_{j,i,\rho}$ such that, for every word β before the cut of length $i + L - N$, $\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}$. Then, for every block j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$.

Proof. [✓ Proof](#) For fixed j, i , apply Theorem 11.10.2 to the family

$$\begin{aligned} Z_\beta &= \mu_j^N X_j A_\beta^j, \\ F_\beta &= A_\beta^j. \end{aligned}$$

Since the length- $N - L$ outside-word products span the full matrix algebra, there is a single matrix $Y_{j,i}$ such that, for every β , $\mu_j^N X_j A_\beta^j = A_\beta^j Y_{j,i}$. Lemma 13.3.1.20 then gives the periodic-boundary constraints for each block. $\square$

Lemma 13.3.1.22 (Matrix identities under block injectivity give single-block constraints). [Lean](#) [✓ Stmt](#) Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0 -block-injective and satisfies the normalization equation $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Suppose also that the boundary-crossing matrix identities of Lemma 13.3.1.21 hold for every outside word ρ of length $N - L$. Then, for every block j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$.

Proof. [✓ Proof](#) Since $L + L_0 \leq N$, the complementary length satisfies $L_0 \leq N - L$. Theorem 6.8.1 therefore gives, for every block j , $S_{N-L}(A_j) = M_{D_j}(\mathbb{C})$. Apply Lemma 13.3.1.21. $\square$

Lemma 13.3.1.23 (Normalization for words). [Lean](#) [✓ Stmt](#)

If $\sum_a A_a A_a^\dagger = \mathbb{1}$, then, for every length M , $\sum_\rho A_\rho A_\rho^\dagger = \mathbb{1}$, where ρ ranges over all words of length M .

Proof. [✓ Proof](#)

The case $M = 0$ is the identity matrix. For the induction step, write a word of length $M + 1$ as $a\rho$. Since $A_{a\rho} = A_a A_\rho$,

$$\sum_{a,\rho} A_a A_\rho A_\rho^\dagger A_a^\dagger = \sum_a A_a \left(\sum_\rho A_\rho A_\rho^\dagger \right) A_a^\dagger = \sum_a A_a A_a^\dagger = \mathbb{1}.$$

$\square$

Lemma 13.3.1.24 (Compatibility hypotheses under block injectivity give single-block constraints). [Lean](#) [✓ Stmt](#) Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0 -block-injective and satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Fix block-diagonal boundary conditions X_j . For every boundary-crossing interval beginning at i , suppose there are matrices $C_{i,\rho}^j$, indexed by outside words ρ of length $N - L$, such that, for every word β before the cut of length $i + L - N$,

$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. After opening the periodic boundary at the chosen cut, this is the source comparison

$$\begin{aligned} A_{i_{m+1}}^j C_{i_1}^j &= D_{i_{m+1}}^j A_{i_1}^j, \\ D_\beta^j &= (\mu_j^N X_j) A_\beta^j. \end{aligned}$$

Then, for every block j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$.

Proof. ✓ **Proof** The normalization equation gives, by induction on word length, $\sum_\rho A_\rho^j A_\rho^{j\dagger} = \mathbb{1}$ for words ρ of length $N - L$. Applying Lemma 13.2.1.9 with $X = \mu_j^N X_j$ gives, for every outside word ρ , a matrix $E_{j,i,\rho}$ such that, for every word β before the cut, $((\mu_j^N X_j) A_\beta^j) A_\rho^j = A_\beta^j E_{j,i,\rho}$. Lemma 13.3.1.22 then gives the periodic-boundary constraint for each block. $\square$

Lemma 13.3.1.25 (Boundary representation and crossing constraints give periodic components). Lean ✓ Stmt Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0 -block-injective and satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Let

$$\begin{aligned} \psi &\in \mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right), \\ \psi &= \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right). \end{aligned}$$

If every boundary-crossing interval has the corresponding tail-word spanning property, then there are block-diagonal boundary conditions X_j with the displayed representation of ψ and, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$.

Proof. ✓ **Proof** Theorem 13.3.1.9 supplies matrices $C_{i,\rho}^j$ satisfying $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. Lemma 13.3.1.24 then gives the single-block periodic constraints. $\square$

Lemma 13.3.1.26 (Boundary comparison implies trace comparison). Lean ✓ Stmt Fix block-diagonal boundary matrices X_j . Suppose the opened-boundary matrices $C_{i,\rho}^j$ satisfy, for every boundary-crossing interval i , every outside word ρ , and every word β before the cut, $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. Then, for every middle word w ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

Proof. ✓ **Proof** Substituting the displayed comparison identity in each summand gives

$$\text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

Summing over j gives the conclusion. $\square$

Lemma 13.3.1.27 (Boundary trace comparisons give single-block constraints). Lean ✓ Stmt Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0 -block-injective and satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Assume also that the simultaneous tuples $w \mapsto (A_w^1, \dots, A_w^g)$ at the middle-word length m span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Fix complex numbers μ_j and block-diagonal boundary conditions X_j . For every boundary-crossing interval beginning at i , suppose there are

matrices $C_{i,\rho}^j$, indexed by outside words ρ of length $N - L$, such that, for every word β before the cut, outside word ρ , and middle word w ,

$$\sum_j \operatorname{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \operatorname{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

Then, for every block j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$.

Proof. ✓ Proof Applying Lemma 13.2.1.8 with the boundary matrix $D_\beta^j = (\mu_j^N X_j) A_\beta^j$, for every block j and every boundary-crossing interval i , gives a matrix E_i^j such that, for every word β before the cut, $(\mu_j^N X_j) A_\beta^j = A_\beta^j E_i^j$. Hence, for every outside word ρ , taking $E_{j,i,\rho} = E_i^j A_\rho^j$ gives $((\mu_j^N X_j) A_\beta^j) A_\rho^j = A_\beta^j E_{j,i,\rho}$. These are precisely the block-diagonal boundary-condition equations of Lemma 13.3.1.22, which gives the periodic-boundary constraint for each block. $\square$

Lemma 13.3.1.28 (Boundary trace comparisons with a block-diagonal representation). Lean
✓ Stmt Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0 -block-injective and satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Assume also that the simultaneous tuples $w \mapsto (A_w^1, \dots, A_w^g)$ at the middle-word length m span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Let

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right)$$

for some block-diagonal boundary conditions X_j . Suppose that whenever ψ has such a representation, there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, every word β of length $i + L - N$ before the cut, and every word w of length m ,

$$\sum_j \operatorname{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \operatorname{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

Then there are block-diagonal boundary conditions X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right)$$

and, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$.

Proof. ✓ Proof Choose a block-diagonal boundary representation of ψ . For that representation, the boundary trace comparison supplies the matrices $C_{i,\rho}^j$. Applying Lemma 13.3.1.27 gives the desired containment $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ for every block j . $\square$

Lemma 13.3.1.29 (C^j boundary comparison with a block-diagonal representation). Lean
✓ Stmt Assume $0 < N$, $L \leq N$, and $L + L_0 \leq N$. Suppose each block A_j is L_0 -block-injective and satisfies $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$. Assume also that the simultaneous tuples $w \mapsto (A_w^1, \dots, A_w^g)$ at the middle-word length m span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Let $\psi = \Gamma_N^{\oplus_j \mu_j A_j} (\bigoplus_j X_j)$ for some block-diagonal boundary conditions X_j . Suppose that whenever ψ has such a representation, there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, and every word β of length $i + L - N$ before the cut, $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. Then

there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ **Proof** Choose a block-diagonal boundary representation of ψ . For that representation, Lemma 13.3.1.26 gives, for every boundary-crossing interval i , outside word ρ , word β before the cut, and middle word w ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

Applying Lemma 13.3.1.28 to these trace comparisons gives $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ for every j . $\square$

Lemma 13.3.1.30 (Boundary trace comparisons give periodic-boundary single-block states).

Lean ✓ **Stmnt** With the hypotheses and notation of Lemma 13.3.11, assume in addition that $L + L_0 \leq N$. Assume also that, for some middle-word length m , the simultaneous tuples $w \mapsto (A_w^1, \dots, A_w^g)$, where w ranges over words of length m in $\{0, \dots, d-1\}$, span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Let $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Suppose that whenever $\psi = \Gamma_N^{\oplus_j \mu_j A_j}(\bigoplus_j X_j)$, there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, every word β before the cut, and every word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

Then there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ **Proof** By Lemma 13.3.11, there are matrices X_j with $\psi = \Gamma_N^{\oplus_j \mu_j A_j}(\bigoplus_j X_j)$. Lemma 13.3.1.28 applies to this representation and the assumed equality of boundary trace expressions. Thus, $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ for every j . $\square$

Lemma 13.3.1.31 (Basis-of-normal-tensors product span gives boundary-trace periodic components). Lean ✓ **Stmnt** With the hypotheses and notation of Lemma 13.3.11, assume in addition that $L + L_0 \leq N$ and $L \geq (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1$. Put $m = (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1)))$. Let $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Suppose that whenever $\psi = \Gamma_N^{\oplus_j \mu_j A_j}(\bigoplus_j X_j)$, there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, every word β before the cut, and every word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

Then there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof Lemma 13.2.2.5 gives $\text{span}\{(A_w^1, \dots, A_w^g) : |w| = m\} = \prod_j M_{D_j}(\mathbb{C})$. Applying Lemma 13.3.1.30 with this middle-word span gives the stated block-diagonal boundary conditions and the periodic-boundary constraint in each block. $\square$

Lemma 13.3.1.32 (C^j boundary comparison gives periodic single-block states). Lean ✓ Stmt With the hypotheses and notation of Lemma 13.3.11, assume in addition that $L + L_0 \leq N$. Let $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Suppose that whenever $\psi = \Gamma_N^{\oplus_j \mu_j A_j}(\bigoplus_j X_j)$, there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, and every word β before the cut, $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. Then there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof By Lemma 13.3.11, there are matrices X_j with $\psi = \Gamma_N^{\oplus_j \mu_j A_j}(\bigoplus_j X_j)$. For these boundary conditions, the assumed C^j boundary comparison supplies the matrices $C_{i,\rho}^j$. Lemma 13.3.1.24 gives $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ for every j . $\square$

Theorem 13.3.1.33 (Crossing-tail spans give periodic single-block states). Lean ✓ Stmt With the hypotheses and notation of Lemma 13.3.11, assume in addition that $L + L_0 \leq N$ and that every boundary-crossing interval beginning at i has the simultaneous tail-word spanning property: the products

$$w \mapsto (A_w^1, \dots, A_w^g),$$

$$|w| = N - i,$$

span the product algebra $\prod_j M_{D_j}(\mathbb{C})$. Let $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Then there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof Lemma 13.3.11 supplies block-diagonal boundary conditions X_j , and Lemma 13.3.1.25, applied with the assumed tail-word spans, places each $\Gamma_N^{A_j}(\mu_j^N X_j)$ in $\mathcal{G}_{N,L}(A_j)$. $\square$

Theorem 13.3.1.34 (Short crossing-tail spans give periodic single-block states). [Lean](#) [✓ Stmt](#) With the hypotheses and notation of Lemma 13.3.11, assume in addition that $L + L_0 \leq N$. Assume also that, for every boundary-crossing interval beginning at i whose tail length lies below the block-separation bound, $N - i < (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1)))$, the simultaneous products

$$w \mapsto (A_w^1, \dots, A_w^g),$$

$$|w| = N - i,$$

span the product algebra $\prod_j M_{D_j}(\mathbb{C})$. Let $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Then there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

This derives the boundary-condition comparison of [193, Theorem 12, proof lines 1446–1456] at every crossing tail whose length reaches the displayed block-separation bound.

Proof. [✓ Proof](#) For a boundary-crossing interval beginning at i , either $N - i < (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1)))$ or $N - i \geq (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1)))$. In the first case, the assumed short-tail span applies; in the second case, Lemma 13.2.2.5 gives the simultaneous product span at length $N - i$. Hence every boundary-crossing interval has the tail-word spanning property, and Theorem 13.3.1.33 gives the block-diagonal boundary conditions and $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ for every j . $\square$

Definition 13.3.1.35 (Cyclic translation of configurations). [Lean](#) [✓ Stmt](#) For a periodic configuration σ on N sites and $s \in \mathbb{Z}/N\mathbb{Z}$, define $(T_s \sigma)(k) = \sigma(k + s)$.

Definition 13.3.1.36 (Cyclic translation of states). [Lean](#) [✓ Stmt](#) For a state ψ on a periodic chain, define $(T_s \psi)(\sigma) = \psi(T_s \sigma)$.

Theorem 13.3.1.37 (Cyclic translations compose). [Lean](#) [✓ Stmt](#) For offsets $s, t \in \mathbb{Z}/N\mathbb{Z}$ and a state ψ on the periodic chain,

$$T_s(T_t \psi) = T_{s+t} \psi.$$

Proof. [✓ Proof](#) For every configuration σ , the two sides evaluate ψ on the same configuration because $T_t(T_s \sigma) = T_{s+t} \sigma$. $\square$

Lemma 13.3.1.38 (Cyclic translation preserves the periodic chain space). [Lean](#) [✓ Stmt](#) If $\psi \in \mathcal{G}_{N,L}(A)$, then $T_s \psi \in \mathcal{G}_{N,L}(A)$ for every cyclic translation s .

Proof. [✓ Proof](#) A length- L interval of $T_s \psi$ beginning at i is the corresponding interval of ψ beginning at $i - s$, with the outside configuration translated by s . $\square$

Lemma 13.3.1.39 (Translation exchanges the two words at a cut). [Lean](#) [✓ Stmt](#) If β has length K and α has positive length S , then $T_K(\beta\alpha) = \alpha\beta$ as configurations on the periodic chain of length $K + S$.

Proof. ✓ Proof This is addition modulo $K + S$, separated according to whether the site lies before or after the cut. $\square$

Lemma 13.3.1.40 (Comparison across a global cut). Lean ✓ Stmt Let $X_j, Y_j \in M_{D_j}(\mathbb{C})$, and suppose

$$T_K \left(\sum_j \Gamma_{K+S}^{A_j}(X_j) \right) = \sum_j \Gamma_{K+S}^{A_j}(Y_j).$$

If the simultaneous block words of length K span $\prod_j M_{D_j}(\mathbb{C})$, then, for every block j and every word α of length S , $X_j A_\alpha^j = A_\alpha^j Y_j$.

Proof. ✓ Proof Evaluate the translated equality on the word $\beta\alpha$, where $|\beta| = K$. Cyclicity of the trace gives

$$\sum_j \text{tr}(X_j A_\alpha^j A_\beta^j) = \sum_j \text{tr}(A_\alpha^j Y_j A_\beta^j).$$

The simultaneous spanning hypothesis separates the blocks and gives the asserted matrix identities. $\square$

Theorem 13.3.1.41 (A global change of cut gives periodic single-block states). Lean ✓ Stmt With the hypotheses and notation of Lemma 13.3.12, assume in addition that $L + L_0 \leq N$. Thus

$$\begin{aligned} g &\geq 2, \\ L &\geq 3(g-1)(L_0+1)+1, \\ N &\geq L + L_0. \end{aligned}$$

Let $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Then there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:

$$\begin{aligned} \psi &= \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right), \\ \Gamma_N^{A_j}(\mu_j^N X_j) &\in \mathcal{G}_{N,L}(A_j). \end{aligned}$$

No simultaneous spanning assumption is imposed at the short crossing tails.

Proof. ✓ Proof Lemma 13.3.12 gives boundary matrices X_j . Translate the periodic state so that the cut moves past the final L_0 sites. By Lemma 13.3.1.38, the translated state remains in the periodic chain space, so applying Lemma 13.3.12 again yields boundary matrices Y_j for the translated state. For a word α of length L_0 , comparison of the two decompositions against every complementary word w of length $N - L_0$ gives

$$\sum_j \text{tr}((\mu_j^N X_j) A_\alpha^j A_w^j) = \sum_j \text{tr}(A_\alpha^j (\mu_j^N Y_j) A_w^j).$$

The length bound and Theorem 13.2.2.11 imply that the simultaneous products of length $N - L_0$ span the product algebra. Therefore, Lemma 13.3.1.40 gives, block by block, $(\mu_j^N X_j) A_\alpha^j =$

$A_\alpha^j(\mu_j^N Y_j)$. Since the words of length L_0 span each block matrix algebra, evaluation at the identity gives

$$\mu_j^N X_j = \mu_j^N Y_j. \quad (13.9)$$

Thus (13.9) says that $\mu_j^N X_j$ commutes with every word of length L_0 , and consequently with every one-site matrix. The boundary condition may then be moved across every interval crossing the cut, which places each block component in its periodic chain space. This is the change-of-cut argument of [193, Theorem 12, proof lines 1276–1289 and 1454–1456]. $\square$

Theorem 13.3.1.42 (Doubly normalized global-cut boundary closing). Lean ✓ Stmt *Let $g \geq 2$, assume the physical alphabet is nonempty, and let A_j be irreducible blocks with positive bond dimensions, pairwise inequivalent up to gauge and phase, and injective at a common length $L_0 > 0$. Assume normalized self-overlap and the two normalization identities*

$$\begin{aligned} \sum_a (A_a^j)^\dagger A_a^j &= \mathbb{1}, \\ \sum_a A_a^j (A_a^j)^\dagger &= \mathbb{1}. \end{aligned}$$

Let every μ_j be nonzero, and suppose

$$\begin{aligned} L &\geq 3(g-1)(L_0+1)+1, \\ N &\geq L+L_0. \end{aligned}$$

If $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$, then there are matrices X_j such that, for every j ,

$$\begin{aligned} \psi &= \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right), \\ \Gamma_N^{A_j}(\mu_j^N X_j) &\in \mathcal{G}_{N,L}(A_j). \end{aligned}$$

This theorem is restricted to the case in which the positive dual fixed matrices of the source normalization are all the identity, with normalized self-overlap imposed separately.

Proof. ✓ Proof Obtain block-diagonal boundary matrices at the chosen cut. Move the cut past the final L_0 sites and compare the two boundary representations. The doubly normalized simultaneous product-span theorem separates the blocks and gives

$$(\mu_j^N X_j) A_\alpha^j = A_\alpha^j (\mu_j^N Y_j).$$

for every word α of length L_0 . Injectivity identifies the two boundary matrices and permits the cut to move across every local interval, proving the periodic single-block constraints. $\square$

Theorem 13.3.1.43 (Periodic block decomposition from a global cut). Lean ✓ Stmt *Let $g \geq 2$. Let A_j be irreducible blocks, pairwise inequivalent up to gauge and phase, and injective at a common length $L_0 > 0$. Assume $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ and, for positive-definite Λ_j , $\sum_a A_a^{j\dagger} \Lambda_j A_a^j = \Lambda_j$. Let $\mu_j \neq 0$ for every j , and assume*

$$\begin{aligned} L &\geq 3(g-1)(L_0+1)+1, \\ N &\geq L+L_0. \end{aligned}$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \sum_j \mathcal{G}_{N,L}(A_j),$$

as asserted in [193, Theorem 12, proof lines 1424–1456].

Proof. ✓ Proof This is the equality component of Theorem 13.3.1.44. □

Theorem 13.3.1.44 (Periodic block decomposition and block-space independence). Lean

✓ Stmt Under the same hypotheses,

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \sum_j \mathcal{G}_{N,L}(A_j).$$

The open-boundary spaces $G_N(A_j)$ also form an internal sum.

Proof. ✓ Proof Theorem 13.3.1.41 gives the forward inclusion in the displayed equality. Every periodic single-block vector is a periodic vector for the block-diagonal tensor, which gives the reverse inclusion. For independence, write $\phi_j = \Gamma_N^{A_j}(X_j)$ and suppose that $\sum_j \phi_j = 0$. Theorem 13.2.2.12 and nondegeneracy of the trace pairing give

$$\sum_j \Gamma_N^{A_j}(X_j) = 0 \implies \left(\forall (M_j)_j, \sum_j \text{tr}(X_j M_j) = 0 \right) \implies \forall j, X_j = 0.$$

The last implication follows by choosing a tuple supported in one block. Thus every ϕ_j vanishes, and the open-boundary spaces form an internal sum. □

Theorem 13.3.1.45 (Doubly normalized periodic block decomposition and independence).

Lean ✓ Stmt Under the hypotheses of Theorem 13.3.1.42,

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \sum_j \mathcal{G}_{N,L}(A_j),$$

and the open-boundary spaces $G_N(A_j)$ form an internal sum.

Proof. ✓ Proof Apply Theorem 13.3.1.44 with $\Lambda_j = \mathbb{1}$. Left canonicity gives the required dual fixed-point identity, while the remaining hypotheses are unchanged. □

Theorem 13.3.1.46 (Doubly normalized periodic block decomposition). Lean ✓ Stmt Let $g \geq 2$, assume the physical alphabet is nonempty, and let A_j be irreducible blocks with positive bond dimensions, pairwise inequivalent up to gauge and phase, and injective at a common length $L_0 > 0$. Assume normalized self-overlap and the two normalization identities

$$\begin{aligned} \sum_a (A_a^j)^\dagger A_a^j &= \mathbb{1}, \\ \sum_a A_a^j (A_a^j)^\dagger &= \mathbb{1}. \end{aligned}$$

Let every μ_j be nonzero, and suppose

$$\begin{aligned} L &\geq 3(g-1)(L_0+1)+1, \\ N &\geq L+L_0. \end{aligned}$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \sum_j \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ **Proof** This is the equality component of Theorem 13.3.1.45. □

Theorem 13.3.1.47 (Unique decomposition into open-boundary block spaces). Lean ✓ **Stmt** *Let $g \geq 2$, assume the physical alphabet is nonempty, and let A_j be irreducible blocks with positive bond dimensions, pairwise inequivalent up to gauge and phase, and injective at a common length $L_0 > 0$. Assume the source normalization*

$$\begin{aligned} \sum_a A_a^j (A_a^j)^\dagger &= \mathbb{1}, \\ \sum_a (A_a^j)^\dagger \Lambda_j A_a^j &= \Lambda_j, \\ \Lambda_j &> 0. \end{aligned}$$

Let every μ_j be nonzero. If $0 < L \leq N$ and $L \geq 3(g-1)(L_0+1)+1$, then every $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ has a unique decomposition

$$\begin{aligned} \psi &= \sum_j \phi_j, \\ \phi_j &\in G_N(A_j) \quad \text{for every } j. \end{aligned}$$

Proof. ✓ **Proof** The source-length open-boundary theorem places ψ in $\sum_j G_N(A_j)$ and proves that the spaces $G_N(A_j)$ form an internal sum. Existence follows from membership in the sum, and internality makes the components unique. □

Corollary 13.3.1.48 (Kernel containment from the global cut). Lean ✓ **Stmt** *Under the same hypotheses, suppose additionally that every periodic single-block chain space is contained in its matrix-product-vector line. Then*

$$\ker H_L^{(N)}\left(\bigoplus_j \mu_j A_j\right) \subseteq \text{span}\{V^{(N)}(A_j) : 1 \leq j \leq g\}.$$

Proof. ✓ **Proof** The parent-Hamiltonian kernel is the periodic chain space. Apply the preceding block decomposition and then the assumed single-block containments. □

Theorem 13.3.1.49 (Doubly normalized kernel containment from a global cut). Lean ✓ **Stmt** *Under the hypotheses of Theorem 13.3.1.42, suppose in addition that, for every j ,*

$$\mathcal{G}_{N,L}(A_j) \subseteq \text{span}\{|V^{(N)}(A_j)\rangle\}.$$

Then

$$\ker H_L^{(N)}\left(\bigoplus_j \mu_j A_j\right) \subseteq \text{span}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g\}.$$

This theorem is restricted to the doubly normalized case in which every positive dual fixed matrix is the identity and normalized self-overlap is assumed.

Proof. ✓ Proof Theorem 13.3.1.46 identifies the periodic chain space of the block sum with the sum of the single-block chain spaces. Apply the assumed containment to each summand, and use the identification of the parent-Hamiltonian kernel with the periodic chain space. $\square$

Lemma 13.3.1.50 (Matrix identities give periodic single-block states). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.3.11, assume in addition that $L + L_0 \leq N$. Let $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$. Suppose that whenever $\psi = \Gamma_N^{\bigoplus_j \mu_j A_j}(\bigoplus_j X_j)$, the corresponding boundary conditions satisfy the block-diagonal boundary-condition equations: for every boundary-crossing interval i , every word β before the cut, and every outside word ρ of length $N - L$, $\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}$. Then there are block-diagonal boundary conditions X_j such that the displayed representation holds and, for every j , the displayed component lies in the corresponding single-block ground space:*

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ Proof By Lemma 13.3.11, there are matrices X_j with $\psi = \Gamma_N^{\bigoplus_j \mu_j A_j}(\bigoplus_j X_j)$. These X_j satisfy the assumed identities, so Lemma 13.3.1.22 gives $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ for every j . $\square$

13.3.2 Periodic-boundary ground space for a block-diagonal tensor

For sufficiently long periodic chains, the periodic-boundary ground space of the parent Hamiltonian is generated by a basis of normal tensors $A_1, \dots, A_g$ of the initial tensor A [16, Theorem IV.16]. When boundary-compatibility conditions across a chosen cut yield words β and ρ satisfying

$$A_\beta^j C_{i,\rho}^j = (\mu_j^N X_j A_\beta^j) A_\rho^j,$$

this supplies the boundary-comparison input $A_{i_{m+1}}^j C_{i_1}^j = D_{i_{m+1}}^j A_{i_1}^j$ with $D_\beta^j = (\mu_j^N X_j) A_\beta^j$ from [193], reducing the periodic-boundary ground space to the individual block components.

Lemma 13.3.2.1 (The block-diagonal periodic-boundary space lies between two block sums).

Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.3.8, one has*

$$\bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L} \left(\bigoplus_{j=1}^g \mu_j A_j \right) \subseteq \bigvee_{j=1}^g G_N(A_j),$$

and the sum defining the right-hand side is internal.

Proof. ✓ Proof The first inclusion is Lemma 13.1.4. The second inclusion and internal directness are Lemma 13.3.9. $\square$

Lemma 13.3.2.2 (Boundary inclusion gives the block-diagonal periodic-boundary equality).

Lean ✓ Stmt *Let $B = \bigoplus_{j=1}^g \mu_j A_j$, where $\mu_j \neq 0$ for $1 \leq j \leq g$. If the periodic-boundary comparison with block-diagonal boundary conditions gives*

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j),$$

then $\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j)$.

Proof. ✓ **Proof** The reverse inclusion is Lemma 13.1.4; combine the two inclusions. $\square$

Lemma 13.3.2.3 (Componentwise periodic-boundary decomposition gives the reverse inclusion). Lean ✓ **Stmnt** Let $B = \bigoplus_{j=1}^g \mu_j A_j$. Suppose that every vector $\psi \in \mathcal{G}_{N,L}(B)$ can be written as

$$\begin{aligned} \psi &= \sum_{j=1}^g \psi_j, \\ \psi_j &\in \mathcal{G}_{N,L}(A_j). \end{aligned}$$

Then

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ **Proof** For $\psi \in \mathcal{G}_{N,L}(B)$, choose the displayed decomposition. Each summand lies in the corresponding subspace $\mathcal{G}_{N,L}(A_j)$, hence their finite sum lies in $\bigvee_j \mathcal{G}_{N,L}(A_j)$. $\square$

Lemma 13.3.2.4 (Block-diagonal boundary vectors in the periodic-boundary block sum). Lean ✓ **Stmnt** Let $B = \bigoplus_{j=1}^g \mu_j A_j$. Fix block-diagonal boundary conditions X_j . Suppose that, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$. Then

$$\Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right) \in \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ **Proof** The block-diagonal boundary formula gives

$$\Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right) = \sum_{j=1}^g \Gamma_N^{A_j}(\mu_j^N X_j).$$

Each summand belongs to the corresponding block periodic-boundary space by assumption, so the finite sum lies in the linear span of these spaces. $\square$

Lemma 13.3.2.5 (Block-diagonal boundary representation gives the reverse inclusion). Lean ✓ **Stmnt** Let $B = \bigoplus_{j=1}^g \mu_j A_j$. Suppose every $\psi \in \mathcal{G}_{N,L}(B)$ admits block-diagonal boundary conditions X_j such that $\psi = \Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right)$, and, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$. Then

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

Proof. ✓ **Proof** Apply Lemma 13.3.2.4 to the block-diagonal boundary representation of each $\psi \in \mathcal{G}_{N,L}(B)$. $\square$

Lemma 13.3.2.6 (Block-diagonal boundary representation reduces to periodic-boundary equality). Lean ✓ **Stmnt** With the hypotheses and notation of Lemma 13.3.8, suppose every vector $\psi \in \mathcal{G}_{N,L}(B)$ has block-diagonal boundary conditions X_j such that $\psi = \Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right)$, and, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$. Then $\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j)$, and the sum defining $\bigvee_j \mathcal{G}_{N,L}(A_j)$ is internal.

Proof. ✓ Proof The block-diagonal boundary representation gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j)$$

by Lemma 13.3.2.5. The equality follows from Lemma 13.3.2.2; the internality statement is Lemma 13.3.9. □

Lemma 13.3.2.7 (Complementary-word matrix identities reduce to periodic-boundary equality). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.3.2.6, assume in addition that $L + L_0 \leq N$. Assume further that, for every $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ and every block-diagonal boundary representation*

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

the complementary-word matrix identities hold: for every block j , boundary-crossing interval i , word β before the cut, and outside word ρ of length $N - L$, there is a matrix $E_{j,i,\rho}$ such that $\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}$. Then

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N -site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. ✓ Proof For each ψ , Lemma 13.3.1.50 gives matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right)$$

and, for every j , the membership $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ holds. Lemma 13.3.2.6 then gives

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the $G_N(A_j)$. □

Lemma 13.3.2.8 (C^j, D^j boundary-condition comparisons reduce to periodic-boundary equality). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.3.2.6, assume in addition that $L + L_0 \leq N$. Assume further that, for every $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ and every block-diagonal boundary representation*

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, and every word β before the cut, $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. This is the C^j, D^j compatibility input; the subsequent normalized E^j identities follow from Lemma 13.2.1.9. Then

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N -site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. ✓ **Proof** For each ψ , Lemma 13.3.1.32 gives matrices X_j with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left(\bigoplus_j X_j \right)$$

and, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$. Lemma 13.3.2.6 then gives

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the $G_N(A_j)$. □

Lemma 13.3.2.9 (C^j, D^j comparisons give the periodic block ground-space equality). Lean

✓ **Stmt** With the hypotheses and notation of Lemma 13.3.2.8, one has

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j).$$

This records only the ground-space equality conclusion obtained after the C^j, D^j boundary-condition comparison input in [193, Theorem 12, proof lines 1446–1456]; the local normalized E^j calculation is Lemma 13.2.1.9. The independence of the N -site spaces is a separate conclusion of Lemma 13.3.2.8. The statement is still in the length- L_0 injectivity range displayed in that lemma, not the shorter range in [193, Theorem 12].

Proof. ✓ **Proof** The claim is the equality part of Lemma 13.3.2.8. □

Lemma 13.3.2.10 (C^j, D^j comparisons at boundary-crossing intervals reduce to periodic-boundary equality). Lean ✓ **Stmt** With the hypotheses and notation of Lemma 13.3.2.6, assume in addition that $L + L_0 \leq N$. Assume also that, for every boundary-crossing interval beginning at i , the simultaneous products

$$\begin{aligned} w &\mapsto (A_w^1, \dots, A_w^g), \\ |w| &= N - i, \end{aligned}$$

span the product algebra $\prod_j M_{D_j}(\mathbb{C})$. Then

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N -site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum. The present statement is the span-dependent consequence, after opening the boundary, of the C^j, D^j compatibility input in [193, Theorem 12, proof lines 1436–1451] in the length- L_0 injectivity range $L \geq (L_0 + 1) + 3(g - 1)(L_0 + 1) + 1$. The source theorem has the shorter bound $L \geq 3(g - 1)(L_0 + 1) + 1$.

Proof. ✓ **Proof** Lemma 13.3.2.6 reduces the claim to producing, for each ψ , block-diagonal boundary conditions X_j whose single-block components lie in $\mathcal{G}_{N,L}(A_j)$. The crossing comparison theorem supplies this: it first uses the boundary-comparison-free BNT representation to obtain X_j , and then uses the stated tail-word spans and Theorem 13.3.1.9 to produce matrices $C_{i,\rho}^j$ satisfying $A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j$. Lemma 13.3.1.24 then gives $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$ for every j . Applying Lemma 13.3.2.6 to these boundary conditions gives the displayed equality and internality. □

Lemma 13.3.2.11 (Boundary-crossing comparisons give the periodic block ground-space equality). [Lean](#) [Stmnt](#) With the hypotheses and notation of Lemma 13.3.2.10, one has

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j).$$

This records the ground-space equality conclusion of the conditional crossing-span version above. It is the corresponding part of [193, Theorem 12], but only under the visible tail-word span hypothesis and in the longer stated range. The independence of the N -site spaces is a separate conclusion of Lemma 13.3.2.10. This intermediate statement remains conditional on the short tail-word spans. The source-range conclusion without them is supplied by the global change-of-cut theorem Theorem 13.3.1.41 under the unrestricted source normalization, with positive definite dual fixed points satisfying

$$\sum_a A_j^{a\dagger} \Lambda_j A_j^a = \Lambda_j.$$

The doubly normalized theorem with $\Lambda_j = \mathbb{1}$ remains available as a specialization of that result.

Proof. [Proof](#) The claim is the equality part of Lemma 13.3.2.10. $\square$

Lemma 13.3.2.12 (Boundary trace comparisons reduce to periodic-boundary equality). [Lean](#)

[Stmnt](#) With the hypotheses and notation of Lemma 13.3.2.6, assume in addition that $L + L_0 \leq N$. Assume also that, for some middle-word length m , the simultaneous tuples $w \mapsto (A_w^1, \dots, A_w^g)$, where w ranges over words of length m in $\{0, \dots, d-1\}$, span the full product algebra $\prod_j M_{D_j}(\mathbb{C})$. Assume further that, for every $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, every word β before the cut, and every word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N -site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. [Proof](#) For each ψ , Lemma 13.3.1.30 gives matrices X_j with

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right)$$

and, for every j , $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$. Lemma 13.3.2.6 then gives

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the $G_N(A_j)$. $\square$

Lemma 13.3.2.13 (Basis-of-normal-tensors product span gives boundary-trace ground-space equality). [Lean](#) [✓ Stmt](#) With the hypotheses and notation of Lemma 13.3.2.6, assume in addition that $L + L_0 \leq N$ and $L \geq (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1$. Put $m = (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1)))$. Assume that, for every $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left(\bigoplus_j X_j \right),$$

there are matrices $C_{i,\rho}^j$ such that, for every boundary-crossing interval i , every outside word ρ of length $N - L$, every word β before the cut, and every word w of length m ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

Then

$$\mathcal{G}_{N,L} \left(\bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the N -site ground spaces $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$ have internal sum.

Proof. [✓ Proof](#) Lemma 13.2.2.5 gives $\text{span}\{(A_w^1, \dots, A_w^g) : |w| = m\} = \prod_j M_{D_j}(\mathbb{C})$. Applying Lemma 13.3.2.12 with this middle-word span gives the equality of periodic-boundary ground spaces and the internality of the sum of the N -site spaces. $\square$

Lemma 13.3.2.14 (Boundary decomposition reduces to periodic-boundary equality). [Lean](#) [✓ Stmt](#) Let $A_1, \dots, A_g$ be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent, and let $B = \bigoplus_{j=1}^g \mu_j A_j$, where $\mu_j \neq 0$. Assume each block is injective at length L_0 , assume $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ for every j , and suppose

$$\begin{aligned} L &\geq (L_0 + 1) + (g - 1)3(L_0 + 1) + 1, \\ N &\geq L. \end{aligned}$$

If every vector $\psi \in \mathcal{G}_{N,L}(B)$ can be written as

$$\begin{aligned} \psi &= \sum_{j=1}^g \psi_j, \\ \psi_j &\in \mathcal{G}_{N,L}(A_j), \end{aligned}$$

then $\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j)$, and the sum $\bigvee_j G_N(A_j)$ is internal.

Proof. [✓ Proof](#) Lemma 13.3.2.3 gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

The blockwise inclusion gives the reverse containment, hence equality. Internality follows from the directness conclusion of Lemma 13.3.9. $\square$

13.4 Periodic-boundary ground space generated by a basis of normal tensors

Let $A_0, \dots, A_{g-1}$ be irreducible canonical blocks with weights $\mu_j \neq 0$. Under the source normalization of [193, Theorem 12], with each block equipped with a positive definite matrix Λ_j satisfying

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}, \quad \sum_a A_a^{j\dagger} \Lambda_j A_a^j = \Lambda_j,$$

and under common block-injectivity, explicit length bounds, and gauge-phase inequivalence of distinct blocks, the parent-Hamiltonian kernel coincides with the span of the basis of normal tensors.

Scope restriction (gauge-phase inequivalence and canonical normalization): The formalized kernel equality requires that distinct blocks be inequivalent up to gauge and phase, whereas [193, Theorem 12] assumes only distinct finite-ring component vectors; documented in [50]. In addition, the doubly-normalized corollary assumes normalized self-overlap; documented in [126].

Theorem 13.4.1 (Doubly-normalized PGVWC07 source-length kernel equality). [Lean](#) [✓ Stmt](#)
Let $d \geq 1$ and $g \geq 2$. For $0 \leq j < g$, let $D_j \geq 1$ and let $A_j = (A_a^j)_{a=0}^{d-1}$ be an irreducible left-canonical block with normalized self-overlap. Assume that no two distinct blocks are equivalent up to gauge and phase. For every $0 \leq j < g$, let $\mu_j \neq 0$. Suppose there is a common $L_0 > 0$ such that the words of length L_0 in each block A_j span $M_{D_j}(\mathbb{C})$. In this doubly-normalized specialization, assume for every $0 \leq j < g$ that

$$\begin{aligned} \sum_{a=0}^{d-1} A_a^j A_a^{j\dagger} &= \mathbb{1}, \\ \sum_{a=0}^{d-1} A_a^{j\dagger} A_a^j &= \mathbb{1}. \end{aligned} \tag{13.10}$$

In the general canonical form, (13.10) is replaced by the dual fixed-point condition with a positive definite matrix Λ_j . If

$$\begin{aligned} L &\geq 3(g-1)(L_0+1)+1, \\ N &\geq L+L_0, \end{aligned}$$

then the parent-Hamiltonian kernel is exactly the span of the BNT component vectors:

$$\ker H_L^{(N)} \left(\bigoplus_j \mu_j A_j \right) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, g-1\}. \tag{13.11}$$

Proof. [✓ Proof](#) Apply Theorem 13.4.2 with $\Lambda_j = \mathbb{1}$. Positive definiteness is immediate, and the assumed left-canonical identity is the dual fixed-point equation at $\Lambda_j = \mathbb{1}$. $\square$

Theorem 13.4.2 (PGVWC07 source-length kernel equality at general canonical normalization). [Lean](#) [✓ Stmt](#) *Retain the same nonzero dimensions and weights, irreducibility, gauge-phase inequivalence, common block-injectivity length, and source-length bounds, but do not assume*

left-canonicity. For every $0 \leq j < g$, assume there is a positive definite matrix Λ_j such that

$$\sum_{a=0}^{d-1} A_a^j A_a^{j\dagger} = \mathbb{1},$$

$$\sum_{a=0}^{d-1} A_a^{j\dagger} \Lambda_j A_a^j = \Lambda_j.$$

Then

$$\ker H_L^{(N)} \left(\bigoplus_j \mu_j A_j \right) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, g-1\}.$$

Proof. ✓ Proof The source length bounds imply $L_0 < L$, $L \leq N$, and $L_0 + 1 < N$. Theorem 11.11.2.3 identifies each single-block periodic chain space with its matrix-product-vector line. Corollary 13.3.1.48 therefore gives the containment of the parent-Hamiltonian kernel in the BNT span. Conversely, Lemma 12.1.20 gives the reverse containment, and antisymmetry yields equality. $\square$

Chapter 14

Exponential Decay of Correlations

In the thermodynamic limit, spatial correlation functions of matrix product states are governed by the subleading spectrum of the transfer map $\mathcal{E}_A$. Subtracting the leading fixed-point contribution leaves a connected correlator decaying exponentially (with polynomial factors n^k for non-diagonal Jordan blocks) at a rate defining the spatial correlation length. For tensors generating parent Hamiltonians (Chapter 11), this spectral gap controls ground-state correlations.

While Chapter 5 analyzes the peripheral spectrum and overlap decay, this chapter develops the one-point and two-point functionals, their spectral decomposition, and quantitative decay bounds.

14.1 Connected correlators from the transfer map

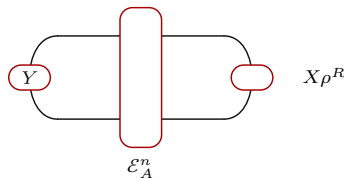
Definition 14.1.1 (Auxiliary-space one-point functional). [Lean](#) [✓ Stmt](#) Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as A and the fixed point as ρ^R . Under the hypotheses of Theorem 4.3.2, that fixed point exists and is unique up to scaling; we normalize it so that $\text{tr}(\rho^R) = 1$. For an auxiliary-space insertion $X \in M_D(\mathbb{C})$, define

$$\langle X_0 \rangle = \text{tr}(X \rho^R).$$

Definition 14.1.2 (Auxiliary-space two-point functional at distance n). [Lean](#) [✓ Stmt](#) For auxiliary-space insertions $X, Y \in M_D(\mathbb{C})$ and $n \geq 0$, define

$$\langle X_0 Y_n \rangle = \text{tr}(Y \mathcal{E}_A^n(X \rho^R)).$$

Equivalently, insert X at site 0, propagate by $\mathcal{E}_A^n$, then insert Y and take the trace.



Definition 14.1.3 (Auxiliary-space connected correlator). [Lean](#) [✓ Stmt](#) The connected two-point function is

$$C(X, Y; n) = \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

Definition 14.1.4 (Fixed-point-centered auxiliary insertion). [Lean](#) [✓ Stmt](#) For a matrix $\rho \in M_D(\mathbb{C})$ and an auxiliary insertion $X \in M_D(\mathbb{C})$, define

$$Z_\rho(X) = X\rho - \text{tr}(X\rho)\rho. \quad (14.1)$$

When $\text{tr}(\rho) = 1$, this removes the trace component of $X\rho$.

Lemma 14.1.5 (Trace of the centered insertion). [Lean](#) [✓ Stmt](#) If $\text{tr}(\rho) = 1$, then

$$\text{tr}(Z_\rho(X)) = 0. \quad (14.2)$$

Proof. [✓ Proof](#) By linearity of the trace,

$$\text{tr}(Z_\rho(X)) = \text{tr}(X\rho) - \text{tr}(X\rho)\text{tr}(\rho) = 0. \quad (14.3)$$

□

Theorem 14.1.6 (Fixed-point centering identity for connected correlators). [Lean](#) [✓ Stmt](#) Suppose that ρ is a right fixed point of the transfer map, $\mathcal{E}_A(\rho) = \rho$. Then, for every $n \geq 0$,

$$C(X, Y; n) = \text{tr}(Y \mathcal{E}_A^n(Z_\rho(X))). \quad (14.4)$$

When moreover $\text{tr}(\rho) = 1$, the preceding lemma shows that $Z_\rho(X)$ is traceless. The identity isolates the fixed-point contribution before the remaining transfer iterates are analysed.

Proof. [✓ Proof](#) Since $\mathcal{E}_A(\rho) = \rho$, one has $\mathcal{E}_A^n(\rho) = \rho$ for every n . Linearity therefore gives

$$\text{tr}(Y \mathcal{E}_A^n(Z_\rho(X))) = \text{tr}(Y \mathcal{E}_A^n(X\rho)) - \text{tr}(X\rho)\text{tr}(Y\rho), \quad (14.5)$$

which is the definition of $C(X, Y; n)$. □

14.2 Spectral expansion and exponential decay

Theorem 14.2.1 (Sum of exponentials under diagonalizability). Let $\mathcal{E}_A$ be trace-preserving, with fixed point ρ satisfying $\text{tr} \rho = 1$. Suppose its restriction to the traceless matrices is diagonalizable, with eigenvalues $\lambda_1, \dots, \lambda_{D^2-1}$, counted with multiplicity. For every X, Y , there are coefficients $c_j \in \mathbb{C}$ such that, for all $n \geq 0$,

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \lambda_j^n. \quad (14.6)$$

This is the diagonalizable case of the expansion in [16, Section II.B.3]. Without diagonalizability, Jordan blocks can contribute polynomial factors multiplying the exponential terms, so the source's unqualified pure-exponential form needs this hypothesis [113].

Proof. The matrix $Z_\rho(X) = X\rho - \text{tr}(X\rho)\rho$ is traceless. Expanding it in an eigenbasis (v_j) of the traceless subspace,

$$Z_\rho(X) = \sum_{j=1}^{D^2-1} \gamma_j(X) v_j, \quad (14.7)$$

$$\mathcal{E}_A^n(v_j) = \lambda_j^n v_j. \quad (14.8)$$

By Theorem 14.1.6,

$$C(X, Y; n) = \text{tr}(Y \mathcal{E}_A^n(Z_\rho(X))) = \sum_{j=1}^{D^2-1} \gamma_j(X) \text{tr}(Y v_j) \lambda_j^n, \quad (14.9)$$

so $c_j = \gamma_j(X) \text{tr}(Y v_j)$. $\square$

Theorem 14.2.2 (Exponential decay bound). *Let $\mathcal{E}_A$ be primitive and trace-preserving, with fixed point ρ satisfying $\text{tr} \rho = 1$, and put $P(Z) = \text{tr}(Z)\rho$. Write $\rho_{\text{spec}}(\mathcal{E}_A - P)$ for the spectral radius of $\mathcal{E}_A - P$. For every r with $\rho_{\text{spec}}(\mathcal{E}_A - P) < r < 1$ and every X, Y , there is a constant $C_{XY} > 0$ such that, for all $n \geq 0$,*

$$|C(X, Y; n)| \leq C_{XY} r^n. \quad (14.10)$$

If the complementary transfer map is diagonalizable, the rate may instead be taken to be its spectral radius, as in the pure-exponential expansion of [16, Section II.B.3]; at a defective subleading eigenvalue the source's rate $|\lambda_2|$ needs this qualification [113].

Proof. Primitivity gives $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ [182, “Complementary transfer-map gap for primitive maps”], so such rates r exist. The matrix $Z_\rho(X)$ is traceless, so $P(Z_\rho(X)) = 0$ and $\mathcal{E}_A^n(Z_\rho(X)) = (\mathcal{E}_A - P)^n(Z_\rho(X))$. The Gelfand formula yields $\|(\mathcal{E}_A - P)^n\| \leq C r^n$ for some constant C . By Theorem 14.1.6 and the Cauchy–Schwarz inequality for Hilbert–Schmidt norms,

$$|C(X, Y; n)| = |\text{tr}(Y (\mathcal{E}_A - P)^n(Z_\rho(X)))| \leq \|Y\| \|(\mathcal{E}_A - P)^n(Z_\rho(X))\| \leq \|Y\| C r^n \|Z_\rho(X)\|, \quad (14.11)$$

which is (14.10) with $C_{XY} = \|Y\| C \|Z_\rho(X)\| + 1$. In the diagonalizable case, the expansion in the proof of Theorem 14.2.1 and the triangle inequality give

$$|C(X, Y; n)| \leq \sum_{j=1}^{D^2-1} |\gamma_j(X) \text{tr}(Y v_j)| |\lambda_j|^n \leq \left(\sum_{j=1}^{D^2-1} |\gamma_j(X) \text{tr}(Y v_j)| \right) \rho_{\text{spec}}(\mathcal{E}_A - P)^n. \quad (14.12)$$

$\square$

Remark 14.2.3 (Transfer-map spectral hypothesis). The rate $r < 1$ in Theorem 14.2.2 gives exponential decay. It is a transfer-map gap condition: all non-leading eigenvalues of $\mathcal{E}_A$ lie strictly inside the unit disk. It is distinct from the Hamiltonian spectral gap of Chapter 12. The complementary transfer-map gap $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ is established for primitive channels (the corresponding theorem in the companion quantum-channel volume [182, “Complementary transfer-map gap for primitive maps”]). The mixed-transfer gap needed for separation of non-equivalent canonical blocks is supplied for irreducible trace-preserving tensors (Theorem 5.6.2), and quantitative single-tensor transfer-map bounds for injective tensors are recorded in Theorem 14.3.3.

These transfer-map gap results give exponential convergence of the complementary transfer map at every rate strictly exceeding its spectral radius.

Definition 14.2.4 (Correlation length). Lean ✓ Stmt For every $\lambda_2 \in \mathbb{C}$, define the associated correlation-length quantity by

$$\xi = -\frac{1}{\log |\lambda_2|}.$$

For a nonzero subleading eigenvalue in the physical range $0 < |\lambda_2| < 1$, this quantity is positive. If all subleading spectral values vanish and correlations therefore vanish after finitely many transfer steps, the total definition gives the limiting value $\xi = 0$.

Lemma 14.2.5 (Correlation length is positive). Lean ✓ Stmt If $|\lambda_2| < 1$ and $\lambda_2 \neq 0$, then $\xi > 0$.

Proof. ✓ Proof Since $0 < |\lambda_2| < 1$, we have $\log |\lambda_2| < 0$, so $\xi = -1/\log |\lambda_2| > 0$. □

14.3 Quantitative transfer-map gap bounds

The qualitative bound $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ for primitive channels [182, “Complementary transfer-map gap for primitive maps”] admits quantitative refinements in terms of geometric constants and decay rates for the transfer map $\mathcal{E}_A$.

Theorem 14.3.1 (Exponential convergence of injective primitive channels). Lean ✓ Stmt Fix an injective, trace-preserving MPS tensor A with positive definite fixed point ρ . Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho \tag{14.13}$$

be the corresponding fixed-point projection. Then there exist $C > 0$ and $0 < \delta \leq 1$ such that, for all $n \geq 0$ and $X \in M_D(\mathbb{C})$,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|. \tag{14.14}$$

The transfer-map gap δ exists by primitivity (the corresponding theorem in the companion quantum-channel volume [182, “Complementary transfer-map gap for primitive maps”], the complementary transfer-map gap).

Proof. ✓ Proof Injectivity implies primitivity of the transfer map. The complementary spectral-radius bound $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ then follows from the primitive transfer-map gap theorem. Choose r with $\rho_{\text{spec}}(\mathcal{E}_A - P) < r < 1$. The Gelfand formula yields $\|(\mathcal{E}_A - P)^n\| \leq Cr^n$ for a suitable constant $C > 0$. Since P is the fixed-point projection, $\mathcal{E}_A P = P \mathcal{E}_A = P$, and hence $\mathcal{E}_A^n - P = (\mathcal{E}_A - P)^n$ for $n \geq 1$. The case $n = 0$ is absorbed by increasing C if necessary. Setting $\delta = 1 - r$ gives (14.14). □

Theorem 14.3.2 (Correlation length bound). Lean ✓ Stmt For an injective trace-preserving MPS tensor, there exist $C > 0$ and $\xi > 0$ such that, for all $n \geq 0$ and all traceless $X \in M_D(\mathbb{C})$, i.e. $\text{tr}(X) = 0$,

$$\|\mathcal{E}_A^n(X)\| \leq Ce^{-n/\xi} \|X\|. \tag{14.15}$$

Traceless matrices lie in $\ker P$, where P is the fixed-point projection (14.13). Since $\mathcal{E}_A - P$ has spectral radius strictly less than 1, choose any r with $\rho_{\text{spec}}(\mathcal{E}_A - P) < r < 1$. The Gelfand formula then gives exponential decay with $\xi = -1/\log r$. Choosing a rate strictly above the spectral radius is necessary in general because a Jordan block at the spectral radius can prevent a uniform bound proportional to $\rho_{\text{spec}}(\mathcal{E}_A - P)^n$.

Proof. ✓ Proof For traceless X , we have $P(X) = 0$, so $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$. Applying the Gelfand-formula estimate with $\rho_{\text{spec}}(\mathcal{E}_A - P) < r < 1$ gives $\|\mathcal{E}_A^n(X)\| \leq Cr^n\|X\|$. With $\xi = -1/\log r > 0$, the identity $r^n = e^{-n/\xi}$ yields (14.15). $\square$

Theorem 14.3.3 (Transfer-map gap from injectivity). Lean ✓ Stmt *For an injective trace-preserving MPS tensor, there exists $\delta > 0$ such that every eigenvalue $\mu \neq 1$ of the transfer map satisfies $|\mu| \leq 1 - \delta$.*

Proof. ✓ Proof Injectivity implies eventual full Kraus rank. This yields primitivity, hence a complementary transfer-map gap. Since the transfer map has only finitely many eigenvalues, one may choose a uniform $\delta > 0$ separating 1 from the remaining spectrum. $\square$

14.4 Relation to parent Hamiltonians

Remark 14.4.1. Suppose that the MPS tensor A generates a parent Hamiltonian $H_N(A, L)$ (Definition 11.4.42) whose ground state is the matrix product vector (Lemma 11.4.46), and that physical observable insertions reduce to the auxiliary functionals defined above. If the hypotheses of Theorem 14.2.2 hold, then for every rate r with $\rho_{\text{spec}}(\mathcal{E}_A - P) < r < 1$ the corresponding ground-state correlations decay as $r^n = e^{-n/\xi}$ with finite correlation length $\xi = -1/\log r$, the quantity of Definition 14.2.4 evaluated at r . When $\mathcal{E}_A - P$ is diagonalizable, one may take $r = |\lambda_2|$, the largest modulus of a subleading eigenvalue, provided $\lambda_2 \neq 0$.

Chapter 15

Concrete Examples

Concrete matrix product state tensors illustrate injectivity, canonical forms, symmetries, and parent Hamiltonians [16, Section III]. Each example is specified by its physical dimension d , bond dimension D , and matrix family $\{A^i\}$. The cases formalized here comprise foundational models from the quantum information and many-body literature: product states, the Greenberger–Horne–Zeilinger (GHZ) state [168, 167], the Affleck–Kennedy–Lieb–Tasaki (AKLT) spin-1 valence-bond state [2, 3], the Majumdar–Ghosh dimer ground state [183, 184], the W state [158], the one-dimensional cluster state [6, 196], the Kitaev (Majorana) chain, of which only the ungraded contraction is formalized [175], and the correlation-independent non-fixed-point tensor of Cirac et al. [12], followed by their parent-Hamiltonian constructions [160, 193].

15.1 Product states

Every product state is a matrix product state with bond dimension one [16, Appendix A, “Product states”]. For site-dependent local tensors, the matrix product is closed by boundary vectors.

Definition 15.1.1 (Product vector and its tensor). [Lean](#) [Lean](#) [✓ Stmt](#) For local vectors $|\phi^s\rangle = \sum_i a^{i,[s]}|i\rangle \in \mathbb{C}^d$, $s = 1, \dots, N$, the product vector $|\psi\rangle = |\phi^1\rangle \otimes \dots \otimes |\phi^N\rangle$ has coefficient $a^{i_1,[1]} \dots a^{i_N,[N]}$ on $|i_1, \dots, i_N\rangle$. Its tensor has bond dimension $D = 1$: at site s the matrix of physical index i is the 1×1 matrix $a^{i,[s]}$.

The boundary vectors are the scalars $v_L = v_R = 1$. Theorem 15.1.4 identifies the resulting contraction with the product vector; Corollary 15.1.5 gives the translation-invariant specialization and its periodic closure.

Definition 15.1.2 (Site-dependent open-boundary contraction). [Lean](#) [✓ Stmt](#) Let $A^{[1]}, \dots, A^{[N]}$ be tensors with bond dimension D , one for each site, and let $v_L, v_R \in \mathbb{C}^D$. The open-boundary contraction of the configuration $(\sigma_1, \dots, \sigma_N)$ is

$$v_L^\top A^{[1]\sigma_1} A^{[2]\sigma_2} \dots A^{[N]\sigma_N} v_R.$$

When all sites carry the same tensor this is the open-boundary contraction of Definition 15.5.2.

Lemma 15.1.3 (Site-independent families). [Lean](#) [✓ Stmt](#) If $A^{[s]} = A$ for every site s , the site-dependent open-boundary contraction of $(\sigma_1, \dots, \sigma_N)$ is the amplitude of the open-boundary state of A at that configuration.

Proof. ✓ Proof Both sides are $v_L^T A^{\sigma_1} \dots A^{\sigma_N} v_R$. □

Theorem 15.1.4 (Product states are MPS with $D = 1$). Lean ✓ Stmt *Every product state is a matrix product state with bond dimension one [16, Appendix A, “Product states”]: for the tensor of Definition 15.1.1, with $A^{[s]i} = a^{i,[s]}$ and trivial boundary vectors $v_L = v_R = 1$, the site-dependent open-boundary contraction of every configuration $(i_1, \dots, i_N)$ is the coefficient of the product vector,*

$$v_L^T A^{[1]i_1} A^{[2]i_2} \dots A^{[N]i_N} v_R = a^{i_1,[1]} \dots a^{i_N,[N]},$$

so that $|\psi\rangle = \sum_{i_1, \dots, i_N} a^{i_1,[1]} \dots a^{i_N,[N]} |i_1, \dots, i_N\rangle$ is the matrix product state of this tensor.

Proof. ✓ Proof A product of 1×1 matrices is the 1×1 matrix of the product of their entries; induct on N . □

Corollary 15.1.5 (Translation-invariant product states). Lean Lean ✓ Stmt *If every site carries the same local vector $\sum_i a^i |i\rangle$, the one-site tensor with 1×1 matrices a^i generates the product vector $(\sum_i a^i |i\rangle)^{\otimes N}$ both as an open-boundary state with trivial boundary vectors and as a matrix product vector with periodic boundary conditions.*

Proof. ✓ Proof The open-boundary statement is Theorem 15.1.4 combined with Lemma 15.1.3. For bond dimension one the trace of a 1×1 matrix is its only entry, so the periodic closure agrees with the open one. □

15.2 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state [168, 167] is the prototypical non-injective, renormalization-fixed-point MPS [16, Section III.A]. On a d -level system it is an MPS with bond dimension d [16, Appendix A, “The GHZ state”].

Definition 15.2.1 (GHZ tensor on a d -level system). Lean Lean Lean ✓ Stmt Set $D = d$. The GHZ tensor on a d -level system is the family $\{A^i\}_{i=0}^{d-1} \subset M_d(\mathbb{C})$ with entries

$$A_{\alpha\beta}^i = \delta_{i=\alpha=\beta},$$

that is, $A^i = |i\rangle\langle i|$ [16, Appendix A, “The GHZ state”].

The virtual indices are closed by the ordinary trace. The resulting periodic vector is identified in Theorem 15.2.3. Theorems 15.2.4–15.2.6 give the idempotent transfer map, non-injectivity for $d \geq 2$, and on-site permutation symmetry. For the qubit tensor, the virtual symmetry and zero correlation length are Theorems 15.2.16 and 15.2.19; its parent ground space is Theorem 15.9.3.

Definition 15.2.2 (GHZ state). Lean ✓ Stmt The (unnormalized) GHZ state on N sites of a d -level system is

$$|\text{GHZ}\rangle = \sum_{i=0}^{d-1} |i, i, \dots, i\rangle.$$

Theorem 15.2.3 (The GHZ state is an MPS). Lean ✓ Stmt *For every N , the matrix product vector of the tensor of Definition 15.2.1 on N sites is the GHZ state $\sum_{i=0}^{d-1} |i, \dots, i\rangle$ [16, Appendix A, “The GHZ state”]. For $N = 0$ both sides equal d .*

Proof. ✓ Proof A product $A^{\sigma_1} \dots A^{\sigma_N}$ is the diagonal matrix whose x -th entry is 1 if $\sigma_1 = \dots = \sigma_N = x$ and 0 otherwise; its trace counts the x with this property. $\square$

Theorem 15.2.4 (GHZ transfer map on a d -level system). Lean Lean ✓ Stmt *The transfer map of the GHZ tensor on a d -level system extracts the diagonal, $\mathcal{E}_A(X)_{\alpha\beta} = \delta_{\alpha\beta} X_{\alpha\alpha}$. In particular $\mathcal{E}_A^2 = \mathcal{E}_A$, so the tensor is a renormalization fixed point.*

Proof. ✓ Proof Since $A^i = |i\rangle\langle i|$, the term $A^i X (A^i)^\dagger$ is $X_{ii} |i\rangle\langle i|$. Extracting the diagonal is idempotent. $\square$

Theorem 15.2.5 (GHZ tensor is not injective for $d \geq 2$). Lean ✓ Stmt *For $d \geq 2$ the GHZ tensor on a d -level system is not injective.*

Proof. ✓ Proof Every A^i has vanishing $(0, 1)$ entry, while the all-ones matrix does not. $\square$

Theorem 15.2.6 (Permutation and $\mathbb{Z}_d$ on-site symmetry). Lean Lean Lean ✓ Stmt *Let the symmetric group on $\{0, \dots, d-1\}$ act on $\mathbb{C}^d$ by permutation matrices. The GHZ tensor on a d -level system is on-site symmetric under this action. In particular, for $d \geq 1$ it is on-site symmetric under the cyclic group $\mathbb{Z}_d$ acting by the cyclic shift $|i\rangle \mapsto |i+1 \bmod d\rangle$.*

Proof. ✓ Proof The twisted tensor of a permutation π is the relabelling $i \mapsto A^{\pi^{-1}(i)}$, so its matrix product vector is the GHZ state evaluated on the relabelled configuration, which equals the GHZ state. $\square$

For $d = 2$ the tensor of Definition 15.2.1 is the two-level GHZ tensor of Definition 15.2.7, which is the representative used in the rest of this section.

Definition 15.2.7 (Two-level GHZ tensor). Lean Lean ✓ Stmt Set $d = D = 2$. The GHZ tensor is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ defined by

$$A^0 = |0\rangle\langle 0| = \text{diag}(1, 0), \quad A^1 = |1\rangle\langle 1| = \text{diag}(0, 1).$$

For a system of N sites, the resulting MPV is the GHZ state

$$|\text{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Lemma 15.2.8 (The two-level case). Lean ✓ Stmt *The GHZ tensor on a d -level system with $d = 2$ is the tensor of Definition 15.2.7.*

Proof. ✓ Proof Both tensors have $A^i = |i\rangle\langle i|$ for $i \in \{0, 1\}$. $\square$

Definition 15.2.9 (Two-site computational vector). Lean ✓ Stmt For $a, b \in \{0, 1\}$, the vector $|ab\rangle \in (\{0, 1\}^2 \rightarrow \mathbb{C})$ is the indicator of the configuration $(\sigma_1, \sigma_2) = (a, b)$.

Definition 15.2.10 (Constant product vector). Lean ✓ Stmt For $a \in \{0, 1\}$ and $N \geq 0$,

$$|a\rangle^{\otimes N}(\sigma_1, \dots, \sigma_N) = \begin{cases} 1, & (\sigma_1, \dots, \sigma_N) = (a, \dots, a), \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 15.2.11 (GHZ transfer map is idempotent). Lean ✓ Stmt *The transfer map of the GHZ tensor satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$.*

Proof. ✓ Proof The transfer map acts by $\mathcal{E}_A(X)_{ij} = \delta_{ij} X_{ij}$ (extraction of the diagonal). This operation is idempotent. $\square$

Theorem 15.2.12 (GHZ tensor is a renormalization fixed point). [Lean](#) [✓ Stmt](#) *The GHZ tensor is a renormalization fixed point.*

Proof. [✓ Proof](#) The claim follows from Theorem 15.2.11. $\square$

Theorem 15.2.13 (GHZ tensor is not injective). [Lean](#) [✓ Stmt](#) *The GHZ tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ is the two-dimensional space of diagonal matrices in $M_2(\mathbb{C})$, which does not equal $M_2(\mathbb{C})$.*

Proof. [✓ Proof](#) Every element of $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span. $\square$

Theorem 15.2.14 ($\mathbb{Z}_2$ on-site symmetry of the GHZ tensor). [Lean](#) [✓ Stmt](#) *Let $\mathbb{Z}_2 = \{1, \sigma_x\}$ act on $\mathbb{C}^2$ by the Pauli-X matrix σ_x . The GHZ tensor is on-site symmetric under this action: for each $g \in \mathbb{Z}_2$, the twisted tensor $\tilde{A}_g$ defines the same MPV family as A , i.e. $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$.*

Proof. [✓ Proof](#) The identity element is trivial. For the generator $g = \sigma_x$, the twisted tensor satisfies $\tilde{A}_g^i = A^{1-i}$, i.e. the two matrices are swapped. Conjugation by σ_x implements the same swap, giving gauge equivalence and hence the same MPV family. $\square$

Definition 15.2.15 (Virtual $\mathbb{Z}_2$ representation of the GHZ tensor). [Lean](#) [✓ Stmt](#) *The virtual $\mathbb{Z}_2$ representation on the bond space is the homomorphism $\mathbb{Z}_2 \rightarrow M_2(\mathbb{C})$ sending the identity to I and the generator to $\sigma_z = \text{diag}(1, -1)$.*

Theorem 15.2.16 (Virtual $\mathbb{Z}_2$ symmetry of the GHZ tensor). [Lean](#) [✓ Stmt](#) *The matrix σ_z commutes with every GHZ matrix A^i . A non-scalar matrix commuting with all A^i is the hallmark of a non-injective ($\mathbb{Z}_2$ -injective) tensor: the on-site symmetry of Theorem 15.2.14 swaps the two one-dimensional injective blocks $A^0 \leftrightarrow A^1$, while σ_z acts on each block by a phase.*

Proof. [✓ Proof](#) Both A^0 and A^1 are diagonal, so each commutes with the diagonal matrix σ_z . $\square$

Theorem 15.2.17 (GHZ order parameter is a transfer-map fixed point). [Lean](#) [✓ Stmt](#) *The order-parameter operator σ_z is a fixed point of the GHZ transfer map, $\mathcal{E}_A(\sigma_z) = \sigma_z$.*

Proof. [✓ Proof](#) Since $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ and both A^i are diagonal, the claim is the entrywise computation $\mathcal{E}_A(\sigma_z) = \sigma_z$. $\square$

Remark 15.2.18. A primitive (injective) tensor has a one-dimensional transfer-map fixed-point space, spanned by a positive matrix that equals the identity only in a normalized gauge. That the GHZ transfer map fixes the additional non-scalar operator σ_z is the transfer-matrix signature of the extra non-decaying mode behind its long-range ferromagnetic correlations.

Theorem 15.2.19 (GHZ zero correlation length). [Lean](#) [✓ Stmt](#) *The GHZ tensor has zero correlation length: its idempotent transfer map (Theorem 15.2.12) gives correlations independent of the separation. This contrasts with the AKLT state (Theorem 15.3.1.4), whose subleading eigenvalue $-\frac{1}{3}$ produces a finite correlation length $1/\log 3$.*

Proof. [✓ Proof](#) The claim follows from the idempotence of the GHZ transfer map and the equivalence of zero correlation length with transfer-map idempotence (Theorem 17.5.20). $\square$

15.2.1 The G -injective fixed-point tensor

For a finite group G , the review constructs a G -injective MPS that is a renormalization fixed point and generalizes the GHZ state to arbitrary groups [16, Section III.A, “Virtual symmetries”]. Write L_g for the left regular representation, $L_g|k\rangle = |gk\rangle$.

Definition 15.2.1.1 (G -injective fixed-point tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The tensor with physical index $(i, j) \in G \times G$ and bond dimension $|G|$ is

$$A^{ij} = \frac{1}{|G|} \sum_{g \in G} \sum_{\alpha, \beta} (L_g)_{\alpha i} (\bar{L}_g)_{\beta j} |\alpha\rangle\langle\beta|$$

[16, Section III.A, “Virtual symmetries”]. Only $g = \alpha i^{-1}$ contributes, so $A_{\alpha\beta}^{ij} = |G|^{-1}$ if $\beta = \alpha i^{-1}j$ and 0 otherwise.

Theorem 15.2.1.2 (The G -injective tensor is a renormalization fixed point). [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#)
 *Stmt* The transfer map of the G -injective tensor is $\mathcal{E}_A(X)_{\alpha\beta} = |G|^{-1} \sum_{k \in G} X_{\alpha k, \beta k}$. It is idempotent, so every nonzero eigenvalue of the transfer map equals 1 [16, Section III.A, “Virtual symmetries”].

Proof. ✔ **Proof** The term $A^{ij}X(A^{ij})^\dagger$ has entries $|G|^{-2}X_{\alpha i-1j, \beta i-1j}$; summing over j and then i gives the formula. Applying it twice averages over k and then over k' , and the substitution $k' \mapsto kk'$ returns the single average. $\square$

Theorem 15.2.1.3 (*G*-injectivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let G act on the two virtual legs by $L_g \otimes \bar{L}_g$. The map $\mathcal{P}(A) = \sum_{ij} \sum_{\alpha\beta} A_{\alpha\beta}^{ij} |ij\rangle \langle \alpha\beta|$ is invariant under this action and injective on its invariant subspace, so the tensor is *G*-injective [16, Section III.A, “Virtual symmetries”], [203, Section “Two dimensions: *G*-injective PEPS”].

Proof. ✓ Proof The condition $\beta = \alpha i^{-1}j$ is unchanged by $(\alpha, \beta) \mapsto (h\alpha, h\beta)$. An invariant vector satisfies $x_{\alpha\beta} = x_{1, \alpha^{-1}\beta}$, and the component of $\mathcal{P}(A)x$ at $(1, w)$ is $|G|^{-1} \sum_{\alpha} x_{\alpha, \alpha w} = x_{1, w}$, so $\mathcal{P}(A)x = 0$ forces $x = 0$. $\square$

Theorem 15.2.1.4 (Invariance under the regular representation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let G act on the physical index by $L_g \otimes \bar{L}_g$, that is $(i, j) \mapsto (gi, gj)$. Each matrix A^{ij} is unchanged by this action, so the tensor is on-site symmetric under the regular representation [16, Section III.A, “Virtual symmetries”].

Proof. ✓ Proof $(gi)^{-1}(gj) = i^{-1}j$.

Theorem 15.2.1.5 (The case $G = \mathbb{Z}_2$ is the GHZ tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#). For $G = \mathbb{Z}_2$ the tensor is $A^{ij} = \frac{1}{2}\sigma_x^{i+j}$. Let $H = (\begin{smallmatrix} 1 & \\ & -1 \end{smallmatrix})$. After the Fourier transform $\frac{1}{2}H \otimes H$ on the physical index (i, j), the resulting tensor is gauge equivalent, through H on the bond, to the GHZ tensor on two-level systems blocked over two sites [16, Section III.A, “Virtual symmetries”].

Proof. ✓ Proof The transformed tensor at (a, b) is $\frac{1}{4} \sum_{ij} (-1)^{ai+bj} \sigma_x^{i+j}$, and conjugation by H turns σ_x into σ_z , giving $\frac{1}{4} (1 + (-1)^a \sigma_z) (1 + (-1)^b \sigma_z) = |a\rangle\langle a| |b\rangle\langle b|$. $\square$

15.3 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state [2, 3] is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order [194, 204, 16].

Definition 15.3.1 (AKLT tensor from singlet bonds). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Place a spin- $\frac{1}{2}$ singlet on every bond, encoded by

$$Y = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

and project the two spin- $\frac{1}{2}$ particles at each site onto their joint spin-1 subspace. With the physical states labelled by $S_z = 0, \pm 1$, the resulting tensor with $d = 3$ and $D = 2$ is

$$A^{+1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} Y, \quad A^0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} Y, \quad A^{-1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} Y. \quad (15.1)$$

The three left factors are the spin-1 states $|\uparrow\uparrow\rangle$, $(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}$, and $|\downarrow\downarrow\rangle$ written as coefficient matrices. This is the tensor of [16, Appendix A, “The AKLT state”], for the state of [2].

On a ring the virtual indices are closed by the trace; on an open chain the two end spins remain as boundary data. The physical-basis and scalar-gauge relations to the normalized representative are Theorem 15.3.6 and Lemma 15.3.8. Theorems 15.3.31 and 15.3.32 give normality, on-site symmetry and the projective virtual action. For the normalized representative, the correlation length is $1/\log 3$ (Theorem 15.3.1.4). The periodic parent-Hamiltonian statement is Theorem 15.3.33, and the open-boundary ground space is described in Theorem 15.9.18.

Lemma 15.3.2 (Matrices of the singlet-bond AKLT tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The matrices of (15.1) are $A^0 = \sigma_z/\sqrt{2}$, $A^{+1} = -|0\rangle\langle 1|$, and $A^{-1} = |1\rangle\langle 0|$.

Theorem 15.3.3 (The site map projects onto spin 1). [Lean](#) [Lean](#) [✓ Stmt](#) The three coefficient matrices P^s with $A^s = P^s Y$ are symmetric, $(P^s)^T = P^s$, so they are states of the symmetric (spin-1) subspace of two spin- $\frac{1}{2}$ particles, and they are orthonormal: $\text{tr}((P^s)^\dagger P^t) = \delta_{st}$.

Definition 15.3.4 (The Pauli basis of the spin-1 site). [Lean](#) [Lean](#) [✓ Stmt](#) The basis of [16, Appendix A, “The AKLT state”] is

$$|+\rangle = \frac{i}{\sqrt{2}}(|-1\rangle + | +1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|-1\rangle - | +1\rangle), \quad |0\rangle. \quad (15.2)$$

The second basis replaces $|+\rangle$ with $-\frac{i}{\sqrt{2}}(|-1\rangle + | +1\rangle)$ and keeps $|-\rangle$ and $|0\rangle$.

Theorem 15.3.5 (The Pauli bases are orthonormal). [Lean](#) [Lean](#) [✓ Stmt](#) The basis (15.2) is orthonormal, and so is the basis obtained by replacing $|+\rangle$ with $-\frac{i}{\sqrt{2}}(|-1\rangle + | +1\rangle)$: the matrices whose columns are these vectors are unitary.

Proof. [✓ Proof](#) The vectors $|\pm\rangle$ are supported on $|\pm 1\rangle$ and $|0\rangle$ is orthogonal to both; the inner product of $|+\rangle$ and $|-\rangle$ is $\frac{-i}{2}(1 - 1) = 0$, and each vector has norm 1. A phase on $|+\rangle$ preserves all of these relations. $\square$

Theorem 15.3.6 (Pauli form of the AKLT tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let U be the unitary whose columns are the vectors of (15.2). The tensor (15.1) expressed in this basis, that is, the tensor $B^a = \sum_s \overline{U_{sa}} A^s$ obtained by rotating the physical index by $U^\dagger$, is

$$B^- = \frac{1}{\sqrt{2}}\sigma_x, \quad B^+ = -\frac{1}{\sqrt{2}}\sigma_y, \quad B^0 = \frac{1}{\sqrt{2}}\sigma_z.$$

This is the form $\sigma_i/\sqrt{2}$ printed in [16, Appendix A, “The AKLT state”] up to the virtual gauge σ_y and the scalar -1 : $B^a = \sigma_y(-\sigma_a/\sqrt{2})\sigma_y^{-1}$ for $a = -, +, 0$ with $\sigma_- = \sigma_x$, $\sigma_+ = \sigma_y$, $\sigma_0 = \sigma_z$. Hence on N sites the state coefficients of B are $(-1)^N$ times those of the printed form, and the two describe the same state up to a global sign. In the basis with $|+\rangle = -\frac{i}{\sqrt{2}}(|-1\rangle + |+1\rangle)$ the tensor becomes exactly $\sigma_i/\sqrt{2}$. The printed identity $A^+ = \sigma_y/\sqrt{2}$ in the basis (15.2) holds only up to this gauge and sign, as recorded in [131].

Proof. [✓ Proof](#) Since $\langle +|\pm 1\rangle = -i/\sqrt{2}$, one has $B^+ = -\frac{i}{\sqrt{2}}(A^{-1} + A^{+1}) = -\frac{i}{\sqrt{2}}Y = -\sigma_y/\sqrt{2}$, while $B^- = (A^{-1} - A^{+1})/\sqrt{2} = \sigma_x/\sqrt{2}$; conjugating the phase of $|+\rangle$ reverses the sign of B^+ . Conjugation by σ_y fixes σ_y and reverses the signs of σ_x and σ_z , which gives the gauge relation; a gauge leaves the state unchanged and the scalar -1 multiplies it by $(-1)^N$. $\square$

Definition 15.3.7 (AKLT tensor). [Lean](#) [✓ Stmt](#) Set $d = 3$ (spin-1) and $D = 2$. The AKLT tensor is the family $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$ defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}}\sigma_z, \quad A^1 = \frac{\sqrt{2}}{\sqrt{3}}|0\rangle\langle 1|, \quad A^2 = -\frac{\sqrt{2}}{\sqrt{3}}|1\rangle\langle 0|.$$

Here $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, so equivalently $A^1 = \frac{1}{\sqrt{3}}\sigma_+$ and $A^2 = -\frac{1}{\sqrt{3}}\sigma_-$ for $\sigma_+ = \sqrt{2}|0\rangle\langle 1|$ and $\sigma_- = \sqrt{2}|1\rangle\langle 0|$. Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin- $\frac{1}{2}$ to spin- $\frac{1}{2}$. This is the representative of the tensor of Definition 15.3.1 used below (Lemma 15.3.8).

Lemma 15.3.8 (The singlet-bond tensor is a rescaled gauge of the representative). [Lean](#) [✓ Stmt](#) With the physical labels 0, 1, 2 of Definition 15.3.7 read as $S_z = 0, +1, -1$, the tensor A of Definition 15.3.1 and the tensor T of Definition 15.3.7 satisfy $A^s = \sqrt{3/2}\sigma_z T^s \sigma_z^{-1}$ for every s .

Proof. [✓ Proof](#) Conjugation by σ_z fixes σ_z and reverses the sign of $|0\rangle\langle 1|$ and $|1\rangle\langle 0|$; compare with Lemma 15.3.2 entrywise. $\square$

Theorem 15.3.9 (AKLT tensor is not injective). [Lean](#) [✓ Stmt](#) The AKLT tensor is not injective at a single site: $\text{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$ is the three-dimensional space of traceless matrices $\mathfrak{sl}(2, \mathbb{C})$, which is a proper subspace of $M_2(\mathbb{C})$.

Theorem 15.3.10 (AKLT tensor is normal). [Lean](#) [✓ Stmt](#) The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.

Theorem 15.3.11 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the AKLT tensor). [Lean](#) [✓ Stmt](#) The AKLT tensor is on-site symmetric under the $\mathbb{Z}_2 \times \mathbb{Z}_2$ subgroup of $\text{SO}(3)$ generated by two commuting π -rotations about orthogonal axes, acting on the spin-1 physical index.

Theorem 15.3.12 (Anticommuting virtual gauges of the AKLT symmetry). [Lean](#) [✓ Stmt](#) The two virtual matrices that implement the symmetry of Theorem 15.3.11, namely $i\sigma_y$ and σ_z , anticommute.

Definition 15.3.13 (AKLT factor system). [Lean](#) [✓ Stmt](#) The AKLT factor system on $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $\omega(g, h) = (-1)^{(g_1+g_2)h_1}$ (the value -1 when $g_1 + g_2 = 1$ and $h_1 = 1$, and 1 otherwise). It is the factor system of the explicit projective representation with virtual gauges $i\sigma_y$ and σ_z ; the diagonal term $g_1 h_1$ relative to the cluster case reflects $(i\sigma_y)^2 = -\mathbb{1}$.

Lemma 15.3.14 (AKLT commutator phase). [Lean](#) [✓ Stmt](#) The commutator phase of the AKLT factor system on the two generators is -1 .

Theorem 15.3.15 (The AKLT factor system is a 2-cocycle). [Lean](#) [✓ Stmt](#) The AKLT factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{U}(1))$.

Theorem 15.3.16 (The AKLT $\mathbb{Z}_2 \times \mathbb{Z}_2$ phase is non-trivial). [Lean](#) [✓ Stmt](#) The AKLT factor system (Theorem 15.3.15), built from the anticommuting gauges $i\sigma_y$ and σ_z (Theorem 15.3.12), is not a coboundary: its commutator phase on the two generators is -1 . Its class in $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{U}(1)) \cong \mathbb{Z}_2$ is therefore non-trivial, so the AKLT chain realizes a non-trivial SPT phase rather than a trivial product state.

Remark 15.3.17. The subgroup $\mathbb{Z}_2 \times \mathbb{Z}_2$ is the minimal subgroup of $\text{SO}(3)$ admitting a non-trivial 2-cocycle.

Definition 15.3.18 (Blocked AKLT tensor). [Lean](#) [✓ Stmt](#) The length-2 blocked AKLT tensor groups two physical sites into one, with blocked physical dimension $9 = 3^2$ and the same bond dimension $D = 2$.

Definition 15.3.19 (Blocked $\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site action). [Lean](#) [✓ Stmt](#) The length-2 blocked on-site action of $\mathbb{Z}_2 \times \mathbb{Z}_2$ on the 9-dimensional blocked physical space is the Kronecker square $g \mapsto P_g \otimes P_g$ of the single-site spin-1 π -rotation representation P_g that implements the symmetry of Theorem 15.3.11. Identifying the blocked physical index with the pair of single-site indices, the entry of $P_g \otimes P_g$ at $(i_1 i_2, j_1 j_2)$ is the product $(P_g)_{i_1 j_1} (P_g)_{i_2 j_2}$ over the two grouped sites. This is the on-site action that makes the blocked twist agree with blocking the single-site twist.

Theorem 15.3.20 (The blocked AKLT tensor is injective). [Lean](#) [✓ Stmt](#) The length-2 blocked AKLT tensor is injective: its nine matrices span $M_2(\mathbb{C})$.

Proof. [✓ Proof](#) This is the blocked-tensor form of the normality of the single-site tensor (Theorem 15.3.10): the single-site tensor is 2-block injective, which is exactly injectivity of its length-2 blocking. $\square$

Theorem 15.3.21 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked AKLT tensor). [Lean](#) [✓ Stmt](#) The length-2 blocked AKLT tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$, with each group element acting by the Kronecker square $P_g \otimes P_g$ of the corresponding single-site spin-1 π -rotation P_g (Definition 15.3.19).

Proof. [✓ Proof](#) Writing A for the single-site tensor and $A^{[2]}$ for its length-2 blocking, twisting the blocked tensor by $P_g \otimes P_g$ equals blocking the twisted single-site tensor, $(A^{[2]})^{P_g \otimes P_g} = (A^{P_g})^{[2]}$, where the superscript denotes the twist. Since the single-site tensor is symmetric, A^{P_g} has the same matrix product vector family as A (Theorem 15.3.11), and blocking preserves the matrix product vector family, so $(A^{[2]})^{P_g \otimes P_g}$ has the same matrix product vector family as $A^{[2]}$. $\square$

Theorem 15.3.22 (The single-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ action is unitary). [Lean](#) [✓ Stmt](#) Each single-site spin-1 π -rotation matrix P_g of the $\mathbb{Z}_2 \times \mathbb{Z}_2$ action is unitary: $P_g P_g^\dagger = \mathbb{1}$.

Theorem 15.3.23 (Time-reversal and reflection symmetry of the AKLT tensor). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $R = e^{i\pi S_y}$ be the spin-1 rotation, which sends $|0\rangle \mapsto -|0\rangle$ and exchanges $|\pm 1\rangle$, and let $X = i\sigma_y$. The AKLT tensor satisfies $\sum_j R_{ij} A^j = X^\dagger A^i X$ and $(A^i)^T = -X^\dagger A^i X$ for every i , and $X\bar{X} = -\mathbb{1}$ [16, Section III.A].

Proof. [Lean](#) [Stmnt](#) Both identities are checked entry by entry on the three matrices of the AKLT tensor, which are real. $\square$

Theorem 15.3.24 (Non-trivial time-reversal and reflection indices of the AKLT state). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Every unitary gauge implementing the time-reversal or the reflection symmetry of the AKLT tensor has index $X\bar{X} = -\mathbb{1}$.

Proof. [Lean](#) [Stmnt](#) The AKLT tensor is normal, so by Theorems 10.4.5.2 and 10.4.5.3 the index does not depend on the gauge; the gauge $i\sigma_y$ of Theorem 15.3.23 has index $-\mathbb{1}$. $\square$

Theorem 15.3.25 (Doubly degenerate entanglement spectrum of the AKLT state). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let Λ be a positive definite fixed point of the adjoint transfer map of the length-2 blocked AKLT tensor. Every eigenspace of Λ has even dimension.

Proof. [Lean](#) [Stmnt](#) The blocked tensor is injective and satisfies $\mathcal{E}(\mathbb{1}) = \mathbb{1}$. The two generators of $\mathbb{Z}_2 \times \mathbb{Z}_2$ act by unitary Kronecker squares with virtual gauges $i\sigma_y$ and σ_z , which carry over from the single site by Lemma 10.1.9 and anticommute. Apply Theorem 10.4.4.5. $\square$

Theorem 15.3.26 (The AKLT state has string order under its $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry). [Lean](#) [Lean](#) [Stmnt](#) For every element g of $\mathbb{Z}_2 \times \mathbb{Z}_2$, the length-2 blocked AKLT tensor has string order in the sense of Definition 10.3.11 for the twist by g , with the maximally mixed boundary state $\Lambda = \frac{1}{2}\mathbb{1}$ as its stationary boundary state.

Proof. [Lean](#) [Stmnt](#) The blocked tensor is injective (Theorem 15.3.20) and on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$ (Theorem 15.3.21), and each group element acts by a real symmetric involutive matrix, hence unitarily. The maximally mixed state $\Lambda = \frac{1}{2}\mathbb{1}$ is positive definite and has trace 1. The single-site AKLT matrices satisfy $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. These identities propagate to the length-2 blocked tensor $A^{[2]}$, whose matrices are the products $A^{i_1} A^{i_2}$. In particular, the blocked transfer map is unital:

$$\mathcal{E}_{A^{[2]}}(\mathbb{1}) = \sum_{i_1, i_2} A^{i_1} A^{i_2} (A^{i_2})^\dagger (A^{i_1})^\dagger = \mathbb{1}, \quad (15.3)$$

$$\mathcal{E}_{A^{[2]}}^\dagger(\Lambda) = \sum_{i_1, i_2} (A^{i_2})^\dagger (A^{i_1})^\dagger \Lambda A^{i_1} A^{i_2} = \Lambda. \quad (15.4)$$

Identity (15.3) sums the right-canonical relation $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ over the inner letter and then the outer; (15.4) sums the left-canonical relation $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ over the outer letter and then the inner, using $\Lambda = \frac{1}{2}\mathbb{1}$. Applying Theorem 10.4.4 yields string order for every g . $\square$

Definition 15.3.27 (Cartesian AKLT tensor). [Lean](#) [Lean](#) [Stmnt](#) The Cartesian presentation of the AKLT tensor has $d = 3$, $D = 2$, and $A^x = \sigma_x$, $A^y = \sigma_y$, $A^z = \sigma_z$, the three Pauli matrices. This is the rotationally natural form used in [16] (around line 1159); it is the AKLT state in the Cartesian basis of the spin-1 site, the counterpart of the $|m\rangle$ -basis form of Definition 15.3.7.

Theorem 15.3.28 (The spin- $\frac{1}{2}$ cover is surjective onto $\text{SO}(3)$). [Lean](#) [Lean](#) [Stmnt](#) Every rotation $R \in \text{SO}(3)$ is realized by conjugating the Pauli vector by some $U \in \text{SU}(2)$: there is $U \in \text{SU}(2)$ with $(\frac{1}{2} \text{tr}(\sigma_i U \sigma_j U^{-1}))_{ij} = R$. Equivalently, the adjoint double cover $\text{SU}(2) \rightarrow \text{SO}(3)$ is surjective.

Proof. [✓ Proof](#) Factor $R = R_z(\alpha)R_x(\beta)R_z(\gamma)$ by its ZXZ Euler decomposition and map each coordinate rotation to an explicit $SU(2)$ generator:

$$\begin{aligned} \text{diag}(e^{-i\theta/2}, e^{i\theta/2}) &\mapsto R_z(\theta), \\ \begin{pmatrix} \cos(\beta/2) & -i\sin(\beta/2) \\ -i\sin(\beta/2) & \cos(\beta/2) \end{pmatrix} &\mapsto R_x(\beta). \end{aligned}$$

□

Theorem 15.3.29 ($SO(3)$ on-site symmetry of the AKLT tensor). [Lean](#) [✓ Stmt](#) *The Cartesian AKLT tensor is on-site symmetric under the spin-1 (defining) representation of $SO(3)$ on the physical index.*

Remark 15.3.30. The bond index carries the projective spin- $\frac{1}{2}$ representation: the gauge for a rotation is either of its two $SU(2)$ preimages, so the assignment is a representation only up to sign and factors through the double cover $SU(2) \rightarrow SO(3)$. Its class is the non-trivial element of $H^2(SO(3), U(1)) \cong \mathbb{Z}_2$, witnessing the SPT phase.

Theorem 15.3.31 (Symmetries and normality of the singlet-bond AKLT tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The tensor of Definition 15.3.1 is normal and on-site symmetric under the $\mathbb{Z}_2 \times \mathbb{Z}_2$ action of Theorem 15.3.11. Its Pauli form $\sigma_i/\sqrt{2}$ of Theorem 15.3.6 is on-site symmetric under the spin-1 representation of $SO(3)$.*

Proof. [✓ Proof](#) Normality and on-site symmetry are unchanged by a nonzero rescaling and by a virtual gauge (Lemma 15.3.8); the Pauli form is $1/\sqrt{2}$ times the Cartesian tensor of Definition 15.3.27. □

Theorem 15.3.32 (The singlet-bond AKLT tensor lies in a non-trivial SPT phase). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every $g \in \mathbb{Z}_2 \times \mathbb{Z}_2$, the twist of the tensor A of Definition 15.3.1 by g satisfies $\sum_j (P_g)_{ij} A^j X_g = X_g A^i$, where X_g is the projective representation generated by $i\sigma_y$ and σ_z . Its factor system is the non-trivial class of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, U(1))$ (Theorem 15.3.16).*

Proof. [✓ Proof](#) With the physical states ordered $S_z = 0, +1, -1$, the two generators act by

$$P_{(1,0)} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad P_{(0,1)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

and the virtual matrices are

$$X_{(1,0)} = i\sigma_y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad X_{(0,1)} = \sigma_z, \quad X_{(1,1)} = X_{(1,0)}X_{(0,1)} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

For $g = (1,0)$ the twist is $\tilde{A}_g^0 = -A^0$, $\tilde{A}_g^{+1} = A^{-1}$, $\tilde{A}_g^{-1} = A^{+1}$, and with the matrices of Lemma 15.3.2,

$$\begin{aligned} \tilde{A}_g^{+1} X_g &= |1\rangle\langle 0| \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = |1\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} (-|0\rangle\langle 1|) = X_g A^{+1}, \\ \tilde{A}_g^0 X_g &= -\frac{1}{\sqrt{2}} \sigma_z i\sigma_y = \frac{1}{\sqrt{2}} i\sigma_y \sigma_z = X_g A^0, \end{aligned}$$

since σ_y and σ_z anticommute; the letter -1 is checked in the same way. For $g = (0,1)$ the twist multiplies $A^{\pm 1}$ by -1 and fixes A^0 , which is conjugation by σ_z . The case $g = (1,1)$ is the composite of the two, and $g = 1$ is trivial. □

15.4 The Majumdar–Ghosh state

The Majumdar–Ghosh state is the exact dimer ground state of the spin- $\frac{1}{2}$ antiferromagnetic Heisenberg chain with next-nearest-neighbour coupling at the Majumdar–Ghosh point $J_2 = J_1/2$ [183, 184]: $H = \sum S_i \cdot S_{i+1} + \frac{1}{2} \sum S_i \cdot S_{i+2}$. On an even periodic chain it is the symmetric superposition of the two nearest-neighbour singlet coverings $(1, 2)(3, 4) \cdots$ and $(2, 3)(4, 5) \cdots (N, 1)$, forming the one-dimensional prototype of a resonating valence-bond state. The review [16, Appendix A, “The Majumdar–Ghosh model”] writes it as a matrix product state with a tensor built from the singlet matrix

$$Y = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad (15.5)$$

whose entry Y_{ab} is the coefficient of $|ab\rangle$ in the singlet $|10\rangle - |01\rangle$.

Definition 15.4.1 (Majumdar–Ghosh tensor of the review). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Set $d = 2$. Let E_{ab} denote the matrix units of $M_3(\mathbb{C})$, and let

$$B^0 = E_{02} + E_{20}, \quad B^1 = E_{12} + E_{21}. \quad (15.6)$$

The review prints the tensor $A = [|0\rangle[|02\rangle + |20\rangle] + |1\rangle[|12\rangle + |21\rangle]] \otimes Y$ [16, Appendix A, “The Majumdar–Ghosh model”]. Read literally, with $(ab|$ the bra E_{ab} of the pair of virtual indices and $\otimes$ the Kronecker product, it is the tensor $\tilde{A}^i = B^i \otimes Y \in M_6(\mathbb{C})$ on the bond $\mathbb{C}^3 \otimes \mathbb{C}^2 \cong \mathbb{C}^6$, with Y as in (15.5).

The periodic coefficients are traces of products of these matrices. The literal reading above and the singlet-bond reading have different periodic vectors; their relation and the source ambiguity are recorded in [135] and Theorem 15.4.10. The singlet-covering state is identified in Theorem 15.4.8, and its bond-dimension-three tensor is left-canonical, non-injective and non-normal (Theorems 15.4.11, 15.4.13 and 15.4.14). Its two-dimensional parent ground space on even rings of at least four sites is Theorem 15.4.24; the corresponding Hamiltonian comparison is Theorem 15.4.28.

Theorem 15.4.2 (Coefficients of the review tensor factor). [Lean](#) [✓ Stmt](#) For every word w , $\text{tr}(\tilde{A}^w) = \text{tr}(B^w) \text{tr}(Y^{|w|})$.

Proof. [✓ Proof](#) By induction on w and the mixed-product rule $(B^i \otimes Y)(M \otimes Y^n) = B^i M \otimes Y^{n+1}$, the word evaluation is $\tilde{A}^w = B^w \otimes Y^{|w|}$ up to the identification of the bond with $\mathbb{C}^6$, which preserves traces. The trace of a Kronecker product is the product of the traces. $\square$

Definition 15.4.3 (Pair coverings). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $f(a, b)$ be a pair amplitude on two sites. For a chain of N sites with configurations $\sigma = (\sigma_0, \dots, \sigma_{N-1})$ put

$$P_f^{\text{even}}(\sigma) = \prod_{k=0}^{\lfloor N/2 \rfloor - 1} f(\sigma_{2k}, \sigma_{2k+1}), \quad P_f^{\text{odd}}(\sigma) = \prod_{k=0}^{\lfloor N/2 \rfloor - 1} f(\sigma_{2k+1}, \sigma_{(2k+2) \bmod N}). \quad (15.7)$$

For even N these are the coverings $(1, 2)(3, 4) \cdots (N-1, N)$ and $(2, 3)(4, 5) \cdots (N, 1)$ of the periodic chain by the pair state $\sum_{ab} f(a, b)|ab\rangle$. The normalized singlet amplitude is $s(a, b) = Y_{ab}/\sqrt{2}$. For a word w , the pair product is the product of f over the consecutive pairs $(w_0, w_1), (w_2, w_3), \dots$, and zero if $|w|$ is odd.

Proof. [✓ Proof](#) Both vectors are sums of the two pair coverings, with amplitudes Y_{ab} and $Y_{ab}/\sqrt{2}$; each covering is a product of $N/2$ amplitudes. $\square$

Theorem 15.4.10 (The literal review tensor does not give the dimer superposition). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $N = 2m > 0$ and $\delta(a, b) = \delta_{ab}$. Then

$$|V^{(N)}(\tilde{A})\rangle = 2(-1)^m(P_\delta^{\text{even}} + P_\delta^{\text{odd}}), \quad (15.10)$$

the sum of the two coverings by the pair state $|00\rangle + |11\rangle$, and $|V^{(N)}(\tilde{A})\rangle$ is not a scalar multiple of $|V^{(N)}(A)\rangle$.

Proof. [✓ Proof](#) The level $z = 2$ satisfies the hypotheses of Theorem 15.4.4 for $\{B^0, B^1\}$, with $(B^a B^b)_{22} = \delta_{ab}$, so Theorem 15.4.4 gives $\text{tr}(B^\sigma) = P_\delta^{\text{even}}(\sigma) + P_\delta^{\text{odd}}(\sigma)$. Since $Y^2 = -I_2$, $\text{tr}(Y^{2m}) = 2(-1)^m$, and Theorem 15.4.2 gives (15.10). On the configuration with all spins 0 the right side of (15.10) equals $4(-1)^m \neq 0$, while Theorem 15.4.8 gives $V^{(N)}(A)_{0\dots 0} = 0$ because $s(0, 0) = 0$. $\square$

Theorem 15.4.11 (Majumdar–Ghosh tensor is in left-canonical form). [Lean](#) [✓ Stmt](#) The Majumdar–Ghosh tensor is isometric: $(A^0)^\dagger A^0 + (A^1)^\dagger A^1 = \mathbb{1}$.

Proof. [✓ Proof](#) Direct multiplication of the two displayed matrices gives $(A^0)^\dagger A^0 = \text{diag}(\frac{1}{2}, 1, 0)$ and $(A^1)^\dagger A^1 = \text{diag}(\frac{1}{2}, 0, 1)$. $\square$

Theorem 15.4.12 (Transfer map of the Majumdar–Ghosh tensor). [Lean](#) [✓ Stmt](#) The transfer map of the Majumdar–Ghosh tensor sends the identity to the diagonal matrix $\mathcal{E}_A(\mathbb{1}) = \text{diag}(2, \frac{1}{2}, \frac{1}{2})$.

Proof. [✓ Proof](#) Substitute the two matrices in $\mathcal{E}_A(\mathbb{1}) = A^0(A^0)^\dagger + A^1(A^1)^\dagger$ and multiply. $\square$

Theorem 15.4.13 (Majumdar–Ghosh tensor is not injective). [Lean](#) [✓ Stmt](#) The Majumdar–Ghosh tensor is not injective: every element of $\text{span}_{\mathbb{C}}\{A^0, A^1\}$ has vanishing $(2, 2)$ entry, so the span is a proper subspace of $M_3(\mathbb{C})$.

Proof. [✓ Proof](#) The matrix unit E_{22} has nonzero $(2, 2)$ entry, while every linear combination of A^0 and A^1 has zero $(2, 2)$ entry. $\square$

Theorem 15.4.14 (Majumdar–Ghosh tensor is not normal). [Lean](#) [✓ Stmt](#) The Majumdar–Ghosh tensor is not normal: no blocking length makes the products of that length span the full matrix algebra $M_3(\mathbb{C})$. This non-normality is the algebraic signature expected from the two-periodic dimer-covering structure.

Proof. [✓ Proof](#) The bond carries a two-periodic grading with complementary support sets

$$\begin{aligned} &\{(0, 1), (0, 2), (1, 0), (2, 0)\}, \\ &\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2)\}. \end{aligned}$$

Left multiplication with a generator exchanges these sets. Let E_{ij} denote the (i, j) matrix unit. Then, for every blocking length N ,

$$\begin{aligned} N \text{ odd} &\Rightarrow (\forall w, |w| = N \Rightarrow (A^w)_{22} = 0) \Rightarrow E_{22} \notin \text{span}_{\mathbb{C}}\{A^w : |w| = N\}, \\ N \text{ even} &\Rightarrow (\forall w, |w| = N \Rightarrow (A^w)_{10} = 0) \Rightarrow E_{10} \notin \text{span}_{\mathbb{C}}\{A^w : |w| = N\}. \end{aligned}$$

Hence no fixed-length word span is all of $M_3(\mathbb{C})$. $\square$

Remark 15.4.15. Read literally, the valence-bond formula of [16] specifies the six-dimensional tensor of Definition 15.4.1, which by Theorem 15.4.10 gives coverings by $|00\rangle + |11\rangle$ rather than by singlets. Placing the singlet on the spin levels of a three-level bond (Definition 15.4.5) gives the Majumdar–Ghosh superposition; see [135].

Definition 15.4.16 (Majumdar–Ghosh Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $S^\alpha = \sigma^\alpha/2$ for $\alpha = x, y, z$ be the spin- $\frac{1}{2}$ operators, and for two sites j, k of a chain let $\mathbf{S}_j \cdot \mathbf{S}_k = \sum_\alpha S_j^\alpha S_k^\alpha$. On a periodic chain of N sites, with indices modulo N ,

$$H_N = \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1} + \frac{1}{2} \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+2} \quad (15.11)$$

[16, Appendix A, “The Majumdar–Ghosh model”]. The three-site term is $h = \frac{1}{2}(\mathbf{S}_0 \cdot \mathbf{S}_1 + \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_0 \cdot \mathbf{S}_2)$ on the sites 0, 1, 2 of a window.

Theorem 15.4.17 (Exchange as a transposition). [Lean](#) [Lean](#) [✓ Stmt](#) For distinct sites $j \neq k$, $\mathbf{S}_j \cdot \mathbf{S}_k = \frac{1}{2}P_{jk} - \frac{1}{4}$, where P_{jk} exchanges the spins at j and k .

Proof. [✓ Proof](#) The completeness relation $\sum_\alpha S_{pa}^\alpha S_{qb}^\alpha = \frac{1}{2}\delta_{pb}\delta_{qa} - \frac{1}{4}\delta_{pa}\delta_{qb}$ is checked on the sixteen index values. $\square$

Theorem 15.4.18 (Three-site ground space of the Majumdar–Ghosh tensor). [Lean](#) [✓ Stmt](#) The three-site local ground space $\mathcal{G}_3(A)$ equals the eigenspace of h for the eigenvalue $-\frac{3}{8}$.

Proof. [✓ Proof](#) By Theorem 15.4.17, $h = \frac{1}{4}(P_{01} + P_{12} + P_{02}) - \frac{3}{8}$, so the eigenspace consists of the vectors v with $v(\sigma P_{01}) + v(\sigma P_{12}) + v(\sigma P_{02}) = 0$ for every σ , that is,

$$v_{000} = v_{111} = 0, \quad v_{001} + v_{010} + v_{100} = 0, \quad v_{011} + v_{101} + v_{110} = 0. \quad (15.12)$$

Every $\Gamma_3(X)$ satisfies (15.12) by expansion. Conversely, a vector satisfying these conditions is $\Gamma_3(X)$ for the matrix X with $X_{01} = -2v_{110}$, $X_{02} = -2v_{001}$, $X_{10} = \sqrt{2}v_{100}$, $X_{20} = -\sqrt{2}v_{011}$ and all other entries zero. $\square$

Theorem 15.4.19 (Twisted periodic vectors satisfy the window constraints). [Lean](#) [Lean](#) [✓ Stmt](#) Let $G \in M_D(\mathbb{C})$ and $c \in \mathbb{C}$ satisfy $GA^i = cA^iG$ for every i . Then for $0 < N$ and $L \leq N$ the vector $\sigma \mapsto \text{tr}(A^{\sigma_0} \cdots A^{\sigma_{N-1}}G)$ lies in $\mathcal{G}_{N,L}(A)$.

Proof. [✓ Proof](#) For the window starting at site i , split the configuration as uv with $|u| = i$, so that vu begins with the window. Then $\text{tr}(A^u A^v G) = \text{tr}(A^v G A^u) = c^i \text{tr}(A^v A^u G)$, and $A^{vu}G$ is the window word followed by the boundary matrix $c^i A^{\text{out}}G$, where A^{out} is the product over the sites outside the window. $\square$

Theorem 15.4.20 (Each dimer covering lies in the chain ground space). [Lean](#) [Lean](#) [✓ Stmt](#) For even $N > 0$ and $L \leq N$, both coverings P_s^{even} and P_s^{odd} lie in $\mathcal{G}_{N,L}(A)$.

Proof. [✓ Proof](#) The grading $G = \text{diag}(1, -1, -1)$ satisfies $GA^i = -A^iG$. For even N , $\text{tr}(A^\sigma G) = 2(A^\sigma)_{00} - \text{tr}(A^\sigma) = P_s^{\text{even}}(\sigma) - P_s^{\text{odd}}(\sigma)$ by Theorem 15.4.4 and Theorem 15.4.8. The two coverings are half the sum and half the difference of this vector and $|V^{(N)}(A)\rangle$, both of which lie in $\mathcal{G}_{N,L}(A)$ by Theorem 15.4.19. $\square$

Theorem 15.4.21 (Energy of the dimer coverings). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $N \geq 3$, every $\psi \in \mathcal{G}_{N,3}(A)$ satisfies $H_N \psi = -\frac{3N}{8}\psi$. In particular, for even $N \geq 4$ each of the two singlet coverings, and for $N \geq 3$ the periodic vector $|V^{(N)}(A)\rangle$, is an eigenvector of (15.11) with eigenvalue $-\frac{3N}{8}$.

Proof. [✓ Proof](#) Each window of three sites $i, i+1, i+2$ of ψ lies in $\mathcal{G}_3(A)$, so by Theorem 15.4.18 the three transposed copies of ψ in that window sum to zero. Summing over i and shifting the index of the middle transposition gives $2 \sum_i P_{i,i+1} \psi + \sum_i P_{i,i+2} \psi = 0$. Theorem 15.4.17 then gives $H_N \psi = \frac{1}{4} (2 \sum_i P_{i,i+1} \psi + \sum_i P_{i,i+2} \psi) - \frac{3N}{8} \psi = -\frac{3N}{8} \psi$. $\square$

Because the Majumdar–Ghosh tensor is non-normal (Theorem 15.4.14), the intersection property of Theorem 11.5.4.3 does not apply to it. The bond grading $G = \text{diag}(1, -1, -1)$, which satisfies $GA^i = -A^i G$, takes the place of normality in the following two theorems.

Theorem 15.4.22 (Graded injectivity of the boundary map). [Lean](#) [Lean](#) [✓ Stmt](#) *Every vector of $\mathcal{G}_K(A)$ equals $\Gamma_K(X)$ for a matrix X with $GX = (-1)^K XG$. For $K \geq 3$, a matrix X with $GX = (-1)^K XG$ and $\Gamma_K(X) = 0$ is zero.*

Proof. [✓ Proof](#) Since $G^2 = 1$ and $GA^i = -A^i G$, the identity $\Gamma_K(GXG) = (-1)^K \Gamma_K(X)$ holds, so X may be replaced by $\frac{1}{2}(X + (-1)^K GXG)$. For $K = 3$ the matrix X is supported on the entries $(0, 1), (0, 2), (1, 0), (2, 0)$, and the coordinates of $\Gamma_3(X)$ at $001, 100, 011, 110$ are $-\frac{1}{2}X_{02}, \frac{1}{\sqrt{2}}X_{10}, -\frac{1}{\sqrt{2}}X_{20}$ and $-\frac{1}{2}X_{01}$. For the step from K to $K+1$, fixing the last site to a gives $\Gamma_K(A^a X) = 0$, where $A^a X$ has parity $(-1)^K$; hence $A^a X = 0$ for both a , and $X = \sum_a (A^a)^\dagger A^a X = 0$ by Theorem 15.4.11. $\square$

Theorem 15.4.23 (Intersection property of the Majumdar–Ghosh tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *For $M \geq 3$, $(\mathcal{G}_M(A) \otimes \mathbb{C}^2) \cap (\mathbb{C}^2 \otimes \mathcal{G}_M(A)) = \mathcal{G}_{M+1}(A)$. Consequently, for $N \geq 3$ every vector of $\mathcal{G}_{N,3}(A)$ lies in $\mathcal{G}_N(A)$.*

Proof. [✓ Proof](#) Let ψ lie in the intersection. By Theorem 15.4.22, its restriction with first letter i is $\Gamma_M(Y_i)$ and its restriction with last letter j is $\Gamma_M(Z_j)$, with Y_i and Z_j of parity $(-1)^M$. Comparing on the middle $M-1$ sites gives $\Gamma_{M-1}(A^j Y_i) = \Gamma_{M-1}(Z_j A^i)$ (Theorem 11.5.4.1), and both matrices have parity $(-1)^{M-1}$. For $M \geq 4$ the second statement of Theorem 15.4.22 gives $A^j Y_i = Z_j A^i$; with $X = \sum_j (A^j)^\dagger Z_j$ the left-canonical equation $\sum_j (A^j)^\dagger A^j = 1$ and Theorem 11.5.4.2 give $Y_i = X A^i$, hence $\psi = \Gamma_{M+1}(X)$. For $M = 3$, the conditions (15.12) on the four three-site restrictions with a fixed first or last letter give $\psi = \Gamma_4(X)$ for the matrix X with $X_{00} = -2\psi_{0110}, X_{11} = -2\psi_{1100}, X_{12} = 2\psi_{0100}, X_{21} = 2\psi_{1011}, X_{22} = -2\psi_{0011}$ and all other entries zero, as a check of the sixteen coordinates shows. The second statement follows by enlarging a window of three sites one site at a time. $\square$

Theorem 15.4.24 (Ground space of the Majumdar–Ghosh parent Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *On an even periodic chain of length $N = 2m \geq 4$, let $H_{N,\text{MG}}^{(3)}$ be the range-three parent Hamiltonian of the Majumdar–Ghosh tensor, the periodic sum of translates of the local projector with kernel $\mathcal{G}_3(A_{\text{MG}})$. Then $H_{N,\text{MG}}^{(3)}$ has a two-dimensional ground space $\mathcal{G}_{N,3}(A_{\text{MG}})$, spanned by the two nearest-neighbour singlet coverings $(1, 2)(3, 4) \cdots (N-1, N)$ and $(2, 3)(4, 5) \cdots (N, 1)$.*

Proof. [✓ Proof](#) The two coverings lie in $\mathcal{G}_{N,3}(A)$ by Theorem 15.4.20. Conversely, let $\psi \in \mathcal{G}_{N,3}(A)$. By Theorem 15.4.23, $\psi = \Gamma_N(X)$ and its translate by one site, which lies in $\mathcal{G}_{N,3}(A)$ by Lemma 13.3.1.38, is $\Gamma_N(Y)$, with X and Y commuting with G . Comparing the two cuts on the remaining $N-1$ sites (Theorem 11.11.1.10) gives $\Gamma_{N-1}(X A^a) = \Gamma_{N-1}(A^a Y)$, so $X A^a = A^a Y$ by Theorem 15.4.22. The entries of these two equations force $X = \text{diag}(x, y, y) = \frac{x+y}{2} 1 + \frac{x-y}{2} G$, and $\Gamma_N(1) = P_s^{\text{even}} + P_s^{\text{odd}}, \Gamma_N(G) = P_s^{\text{even}} - P_s^{\text{odd}}$ as in the proof of Theorem 15.4.20. The two coverings are linearly independent because Γ_N is injective on the matrices commuting with G . The kernel of $H_{N,\text{MG}}^{(3)}$ is $\mathcal{G}_{N,3}(A)$ by Theorem 12.3.17. $\square$

Theorem 15.4.29 (The dimer energy on odd rings). [Lean](#) [✓ Stmt](#) For odd $N \geq 5$, $-\frac{3N}{8}$ is not an eigenvalue of H_N , so every eigenvalue of H_N is strictly larger than $-\frac{3N}{8}$.

Proof. [✓ Proof](#) By Theorem 15.4.28, the eigenspace for $-\frac{3N}{8}$ is the kernel of $H_{N,\text{MG}}^{(3)}$, which is $\mathcal{G}_{N,3}(A)$ by Theorem 12.3.17 and zero by Theorem 15.4.25. $\square$

Remark 15.4.30. By Theorem 15.4.28, on an even ring the ground space of H_N is two-dimensional. The one-site translation exchanges the two singlet coverings, so this degeneracy reflects the spontaneous breaking of the one-site translation symmetry of the chain.

15.5 The W state

The W state on N sites is the single-excitation symmetric state

$$|W_N\rangle = |10 \cdots 0\rangle + |010 \cdots 0\rangle + \cdots + |0 \cdots 01\rangle,$$

the equal-weight superposition of all configurations with exactly one excitation [158]. Following [16, Appendix A, “The W state”], it is presented with open boundary conditions, where the matrix product is closed by a left and a right boundary vector rather than by a trace. The tensor of Definition 15.5.1 is the one printed there, in the same basis and with the same boundary vectors, so the results below are stated for it directly.

Definition 15.5.1 (W tensor and boundary vectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Following [16, Appendix A, “The W state”], set $d = D = 2$. The W tensor is the site-independent family

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

$$A^1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

with left boundary covector $\langle l| = \langle 0|$ and right boundary vector $|r\rangle = |1\rangle$. Here A^1 is the single raising operator: it sends $|1\rangle \mapsto |0\rangle$ and annihilates $|0\rangle$, and squares to zero.

Theorem 15.5.5 proves the open-boundary state identity. Closing the same matrices by a trace instead gives the vacuum term of Theorem 15.5.7. The single-length periodic bond-dimension question is distinct: the review’s general bound $D^3 \log D = \Omega(N)$ remains open, with the conditional and restricted-length results summarized in Remark 15.5.26 and [48].

Definition 15.5.2 (Open-boundary contraction). [Lean](#) [✓ Stmt](#) For a tensor A with bond dimension D , a left boundary covector $v_L \in \mathbb{C}^D$, a right boundary vector $v_R \in \mathbb{C}^D$, and a word $w = (i_1, \dots, i_k)$, the open-boundary contraction is the bilinear pairing

$$(v_L | A^{i_1} A^{i_2} \cdots A^{i_k} | v_R) = v_L^T (A^{i_1} \cdots A^{i_k}) v_R.$$

Definition 15.5.3 (Open-boundary state). [Lean](#) [✓ Stmt](#) The open-boundary state on N sites assigns to each configuration $\sigma = (\sigma_1, \dots, \sigma_N)$ the amplitude $(v_L | A^{\sigma_1} \cdots A^{\sigma_N} | v_R)$.

Definition 15.5.4 (W-state indicator). [Lean](#) [✓ Stmt](#) For $N \geq 0$, the W-state amplitude is the indicator of the single-excitation configurations:

$$|W_N\rangle(\sigma_1, \dots, \sigma_N) = \begin{cases} 1, & \text{exactly one } \sigma_j = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Theorem 15.5.5 (The open-boundary contraction is the W state). [Lean](#) [✓ Stmt](#) For every N , the open-boundary state of the W tensor with boundary vectors $|0\rangle$ and $|1\rangle$ equals the W state:

$$|W_N\rangle = \sum_{\sigma} (0|A^{\sigma_1} \dots A^{\sigma_N}|1)|\sigma\rangle.$$

Proof. [✓ Proof](#) Because A^0 is the identity and A^1 is the raising operator with $A^1 A^1 = 0$, the ordered product $A^{\sigma_1} \dots A^{\sigma_N}$ depends only on the number of excitations of σ : it is the identity with no excitation, the raising operator with one excitation, and zero with two or more. Reading off the $(0,1)$ matrix element against the boundary vectors therefore gives 1 on the single-excitation configurations and 0 elsewhere, which is the W -state amplitude. $\square$

Theorem 15.5.6 (Three-site single excitations). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On three sites, each of the three single-excitation configurations $|100\rangle$, $|010\rangle$, $|001\rangle$ receives W -state amplitude 1, while the vacuum $|000\rangle$ and the doubly-excited $|110\rangle$ receive amplitude 0.

Proof. [✓ Proof](#) Direct instances of Theorem 15.5.5: the excitation count is one in the first three cases and zero or two in the others. $\square$

The representation of Definition 15.5.1 is not translationally invariant, because its boundary is not periodic [16, Appendix A, “The W state”]. The following computation illustrates this remark by closing the same tensor with a trace.

Theorem 15.5.7 (The periodic closure is not the W state). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Closing the tensor of Definition 15.5.1 with a trace gives

$$\text{tr}(A^{\sigma_1} \dots A^{\sigma_N}) = \begin{cases} 2, & \text{no } \sigma_j = 1, \\ 0, & \text{otherwise.} \end{cases}$$

In particular the trace contraction vanishes on every single-excitation configuration, and for $N \geq 1$ it is not $|W_N\rangle$.

Proof. [✓ Proof](#) The ordered product is the identity, the raising operator, or zero according to the number of excitations, as in the proof of Theorem 15.5.5. Their traces are 2, 0, and 0. For $N \geq 1$ the configuration $|10 \dots 0\rangle$ has W -state amplitude 1 and trace contraction 0. $\square$

Definition 15.5.8 (Periodic representation of the W state). [Lean](#) [✓ Stmt](#) A tensor B with physical dimension 2 and bond dimension D is a periodic representation of $|W_N\rangle$ when $\text{tr}(B^{\sigma_1} \dots B^{\sigma_N}) = |W_N\rangle(\sigma)$ for every configuration σ of N sites.

Theorem 15.5.9 (Bond dimension one). [Lean](#) [✓ Stmt](#) For $N \geq 2$, no tensor of bond dimension 1 is a periodic representation of $|W_N\rangle$.

Proof. [✓ Proof](#) Write $B^0 = b_0$ and $B^1 = b_1$ as scalars. The vacuum amplitude gives $b_0^N = 0$, so $b_0 = 0$. The configuration $|10 \dots 0\rangle$ then has amplitude $b_1 b_0^{N-1} = 0$, whereas its W -state amplitude is 1. $\square$

Theorem 15.5.10 (No fixed bond dimension represents every W state). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let B be a tensor with physical dimension 2 and bond dimension D . If B is a periodic representation of $|W_k\rangle$ for every $1 \leq k \leq D$, then $(B^0)^D = 0$, and B is a periodic representation of $|W_N\rangle$ for no $N > D$. Consequently there is a length $1 \leq N \leq D + 1$ for which B is not a periodic representation of $|W_N\rangle$, and no single tensor of fixed bond dimension is a periodic representation of $|W_N\rangle$ for every $N \geq 1$.

Proof. ✓ Proof The vacuum amplitudes give $\text{tr}((B^0)^k) = 0$ for $1 \leq k \leq D$. By the Newton–Girard identities the characteristic polynomial of B^0 is x^D , and the Cayley–Hamilton theorem gives $(B^0)^D = 0$. For $N > D$ the configuration $|10 \cdots 0\rangle$ then has amplitude $\text{tr}(B^1(B^0)^{N-1}) = 0$, whereas its W-state amplitude is 1. If B represented every $|W_N\rangle$ with $1 \leq N \leq D+1$, the case $N = D+1$ would contradict this. $\square$

Definition 15.5.11 (Excitation-number indicator). Lean ✓ Stmt For $N \geq 0$ and $m \geq 0$, let $u_m^{(N)}$ be the indicator of the configurations of N sites with exactly m excitations. In particular $u_1^{(N)}$ is the W amplitude of Definition 15.5.4.

Lemma 15.5.12 (Cut rank of the W state). Lean Lean Lean ✓ Stmt Split $N = R + R'$ sites into a first piece of R sites and a second piece of R' sites. For configurations σ of the first piece and τ of the second,

$$\langle \sigma \tau | W_N \rangle = u_1^{(R)}(\sigma) u_0^{(R')}(\tau) + u_0^{(R)}(\sigma) u_1^{(R')}(\tau). \quad (15.14)$$

Hence for every τ the function $\sigma \mapsto \langle \sigma \tau | W_N \rangle$ lies in the span of $u_0^{(R)}$ and $u_1^{(R)}$, and the reduced state of $|W_N\rangle$ on the first piece has rank at most two.

Proof. ✓ Proof The word $\sigma \tau$ has exactly one excitation if and only if one of the two pieces has one excitation and the other has none, which is (15.14). Fixing τ , the right-hand side is $u_0^{(R')}(\tau) u_1^{(R)}(\sigma) + u_1^{(R')}(\tau) u_0^{(R)}(\sigma)$. $\square$

Lemma 15.5.13 (Simultaneous product span of distinct blocks). Lean Lean ✓ Stmt Let A be in the canonical form of Definition 23.3.6 with blocks $A^1, \dots, A^b$ of sizes $D_1, \dots, D_b$, each L_0 -block injective for a common $L_0 > 0$, no two related by a gauge transformation and a phase. For every $n \geq \max(L_0, 3(b-1)(L_0+1))$ the tuples $(A_w^1, \dots, A_w^b)$ over words w of length n span $\bigoplus_j M_{D_j}(\mathbb{C})$. The same holds for any family of blocks $A^1, \dots, A^b$, not assumed to come from a canonical form, provided each block satisfies $\sum_i A_i^j (A_i^j)^\dagger = \mathbb{1}$, its dual map $X \mapsto \sum_i (A_i^j)^\dagger X A_i^j$ has a positive definite fixed point, and no orthogonal projection other than 0 and $\mathbb{1}$ is invariant under every A_i^j .

Proof. ✓ Proof For one block this is L_0 -block injectivity, which persists at all larger lengths. For several blocks it is the direct-sum lemma of [193] for distinct blocks with condition C1; the length L_0 in the threshold covers the single-block case. $\square$

Lemma 15.5.14 (Cut-rank lower bound). Lean ✓ Stmt Let $A^1, \dots, A^b$ be tensors of bond dimensions $D_1, \dots, D_b$ whose word tuples span $\bigoplus_j M_{D_j}(\mathbb{C})$ at lengths R and R' , and let $c_1, \dots, c_b$ be nonzero. If for every word τ of length R' the function $\sigma \mapsto \sum_j c_j \text{tr}(A_\sigma^j A_\tau^j)$ of words σ of length R lies in a subspace V , then $\sum_j D_j^2 \leq \dim V$.

Proof. ✓ Proof The linear map sending $(X_j)_j \in \bigoplus_j M_{D_j}(\mathbb{C})$ to $\sigma \mapsto \sum_j c_j \text{tr}(A_\sigma^j X_j)$ is injective, because the word tuples at length R span and the trace pairing is nondegenerate. Its image lies in V , because the word tuples at length R' span as well. $\square$

Theorem 15.5.15 (At most two product states). Lean ✓ Stmt Let $r \leq 2$, let $x_0, \dots, x_{r-1}$ and $y_0, \dots, y_{r-1}$ be complex numbers, and let $c \neq 0$. Suppose $N \geq 2$, and $N \geq 3$ when $r = 2$. Then the equalities

$$\sum_{k=0}^{r-1} y_k^m x_k^{N-m} = \begin{cases} c & m = 1, \\ 0 & m = 2, 3, \end{cases} \quad (15.15)$$

for all $1 \leq m \leq \min(3, N)$ cannot all hold. In particular a sum of at most two translation-invariant product states, with one-site amplitudes x_k and y_k for the letters 0 and 1, is not a nonzero multiple of $|W_N\rangle$: its amplitude on the configuration $1^m 0^{N-m}$ is the left-hand side of (15.15). This supplies the step that [193, lines 2208–2210] calls trivial, after the rank count has reduced the canonical form to at most two blocks of size one by one.

Proof. ✓ Proof For $r = 0$ the case $m = 1$ gives $c = 0$. For $r = 1$, combining the cases $m = 1$ and $m = 2$ gives $c y_0 = 0$, so $y_0 = 0$ and again $c = 0$. For $r = 2$ both sequences $m \mapsto y_k^m x_k^{N-m}$ are annihilated by the recurrence with characteristic polynomial $(x_0 S - y_0)(x_1 S - y_1)$; applied to the cases $m = 1, 2, 3$ it gives $c y_0 y_1 = 0$. Hence one of the y_k vanishes, and the remaining single product state is excluded as for $r = 1$. $\square$

Theorem 15.5.16 (Single-length bound under a canonical form with condition C1). Lean Lean ✓ Stmt Let A be a translation-invariant tensor of bond dimension D in the canonical form of Definition 23.3.6, with b blocks, no two of which are related by a gauge transformation and a phase. Suppose that for some N and some $c \neq 0$ the periodic vector of A on N sites is $c|W_N\rangle$.

1. If every block is L_0 -block injective for a common $L_0 > 0$, then $N < 2 \max(L_0, 3(b-1)(L_0 + 1))$.
2. If every block is L -block injective for some $L > 0$ and its matrix A_0^j for the letter 0 is invertible, then $N < 2 \max(D^2 + 1, 3(D-1)(D^2 + 2))$, so $D = \Omega(N^{1/3})$.
3. If the dual transfer channel of every block is primitive, then $N < 2 \max((D^2 + 1)^2, 3(D-1)((D^2 + 1)^2 + 1))$.

The first item holds more generally for $\sum_j c_j |\phi_{A_j}\rangle = c|W_N\rangle$ with arbitrary nonzero coefficients c_j in place of λ_j^N , for blocks satisfying the three conditions of the canonical form. The first item is the inequality of the proof in [193], conditional on the canonical form with condition C1, and with the threshold enlarged by L_0 so that it also covers a single block. The distinctness hypothesis is stricter than the source's pairwise-different block states. The reduction of an arbitrary tensor to this form holds when $N \geq 2$ has no divisor between 2 and D (Theorem 15.5.24); the review's asymptotic bound $D^3 \log D = \Omega(N)$ at every length is outside the scope of these results [48].

Proof. ✓ Proof Cut the chain into two pieces of at least $n_0 = \max(L_0, 3(b-1)(L_0 + 1))$ sites. By Lemmas 15.5.13 and 15.5.14, the reduced state of the first piece has rank at least $\sum_j D_j^2$, while Lemma 15.5.12 bounds it by two. Hence there are at most two blocks, each of size one by one, with scalar letters $A_0^j = x_j$ and $A_1^j = y_j$, so that the amplitude of $1^m 0^{N-m}$ is $\sum_j c_j y_j^m x_j^{N-m}$. Choosing z_j with $z_j^N = c_j$ absorbs each coefficient into the one-site amplitudes, $c_j y_j^m x_j^{N-m} = (z_j y_j)^m (z_j x_j)^{N-m}$. By Theorem 15.5.15, a sum of at most two translation-invariant product states is not a nonzero multiple of $|W_N\rangle$ for $N \geq 3$, a contradiction when $N \geq 2n_0$. For the second item, an invertible letter gives condition C1 at $L_0 = D^2 + 1$ by the fixed-length Wielandt theorem; for the third, the general quantum Wielandt bound gives $L_0 \leq (D^2 + 1)^2$. $\square$

Theorem 15.5.17 (Trace pairings across a cut). Lean Lean ✓ Stmt Let $B^1, \dots, B^b$ have bond dimensions $D_1, \dots, D_b$ and suppose their word tuples of length R' span $\bigoplus_j M_{D_j}(\mathbb{C})$. Let every $c_j \neq 0$. If each function $\sigma \mapsto \sum_j c_j \text{tr}(B_\sigma^j B_\tau^j)$ lies in a subspace V of functions on words of length R , then every function $\sigma \mapsto \sum_j \text{tr}(X_j B_\sigma^j)$ also lies in V . If the specified block B^j is R -block injective, then $D_j^2 \leq \dim V$.

Proof. ✓ Proof For an arbitrary tuple $(X_j)_j$, simultaneous spanning gives coefficients a_τ such that $\sum_\tau a_\tau c_j B_\tau^j = X_j$. Hence

$$\sum_\tau a_\tau \sum_j c_j \operatorname{tr}(B_\sigma^j B_\tau^j) = \sum_j \operatorname{tr} \left(B_\sigma^j \sum_\tau a_\tau c_j B_\tau^j \right) = \sum_j \operatorname{tr}(X_j B_\sigma^j). \quad (15.16)$$

To obtain the dimension bound, choose $X_k = 0$ for $k \neq j$. The map $X_j \mapsto (\operatorname{tr}(X_j B_\sigma^j))_\sigma$ is injective: if all its values vanish, R -block injectivity gives $\operatorname{tr}(X_j Y) = 0$ for every matrix Y , and matrix units then give $X_j = 0$. Its image lies in V by (15.16), so $D_j^2 \leq \dim V$. $\square$

Theorem 15.5.18 (W-state bound from unequal cuts). Lean Lean ✓ Stmt Let $B^1, \dots, B^b$ be binary tensors of positive bond dimensions, each unital and irreducible, with a positive definite fixed point of its dual transfer map. Suppose no two blocks differ by a gauge transformation and a phase, and each is L_0 -block injective for a common $L_0 > 0$. If $\sum_j c_j |V^{(N)}(B^j)\rangle = c |W_N\rangle$ with every $c_j \neq 0$ and $c \neq 0$, then, writing $F = \max\{L_0, 3(b-1)(L_0+1)\}$,

$$N < \max\{L_0 + F, 12(b-1)\}. \quad (15.17)$$

The same conclusion holds for canonical-form coefficients $c_j = \lambda_j^N$. Here $b-1$ means zero when $b = 0$. The threshold in (15.17) is at most $2F$; for $L_0 \geq 3$ it is $L_0 + F$.

Proof. ✓ Proof Suppose $N \geq \max\{L_0 + F, 12(b-1)\}$ and cut at L_0 . The word tuples on the other side, of length $N - L_0 \geq F$, span the product algebra by Lemma 15.5.13. Individual injectivity on the first side and Theorem 15.5.17 give $D_j^2 \leq 2$, since the W-state cut rank is at most two. Thus every $D_j = 1$.

For scalar letters b_i^j , unitality reads $\sum_i |b_i^j|^2 = 1$. A nonzero letter spans the one-dimensional matrix algebra, so the same blocks are injective at length one. Reapply Theorem 15.5.16 with this length to obtain

$$N < 2 \max\{1, 6(b-1)\} = \max\{2, 12(b-1)\} \leq \max\{L_0 + F, 12(b-1)\}, \quad (15.18)$$

a contradiction. Finally, $L_0 + F \leq 2F$ and $12(b-1) \leq 2F$. If $L_0 \geq 3$ and $b \geq 2$, then $3(b-1)(L_0+1) \geq 12(b-1)$; if $b \leq 1$, the latter term is zero. $\square$

Lemma 15.5.19 (Quadratic word-span bound from a nonzero eigenvalue). Lean ✓ Stmt Let B be a tensor of bond dimension $D \geq 1$ whose exact word spans equal $M_D(\mathbb{C})$ at all sufficiently large lengths. If $S_1(B)$ contains a matrix with a nonzero eigenvalue, then $S_{D^2}(B) = M_D(\mathbb{C})$. This is the nonzero-eigenvalue bound of [212, Theorem 6.9(3)], in its word-span form.

Theorem 15.5.20 (Cubic bound from a nonzero one-step eigenvalue). Lean ✓ Stmt Let A be a binary tensor of bond dimension D in canonical form, with no two blocks related by a gauge transformation and a phase. Suppose each block B^j is injective at some positive length and there are $X_j \in \operatorname{span}_{\mathbb{C}}\{B_i^j\}$, $\mu_j \neq 0$, and $\phi_j \neq 0$ satisfying $X_j \phi_j = \mu_j \phi_j$. If $V^{(N)}(A) = c W_N$ with $c \neq 0$, then, for $L = \max\{1, D^2\}$,

$$N < \max\{L + \max\{L, 3(D-1)(L+1)\}, 12(D-1)\}. \quad (15.19)$$

For positive D , $L = D^2$, so this gives $D = \Omega(N^{1/3})$. The maximum with one includes $D = N = 0$.

Proof. ✓ Proof Positive-length injectivity implies full word span at every larger length. Lemma 15.5.19 gives injectivity at D_j^2 in each block. Since $D_j \leq D$, this propagates to the common length L . Apply Theorem 15.5.18 and use $b \leq D$. $\square$

Theorem 15.5.21 (The period is at most the bond dimension). [Lean](#) [✓ Stmt](#) *If a tensor of bond dimension D is periodic with period m , then $m \leq D$.*

Proof. [✓ Proof](#) The cyclic sectors have positive dimensions D_j and support isometries $V_j : \mathbb{C}^{D_j} \rightarrow \mathbb{C}^D$, with $V_j^\dagger V_j = \mathbb{1}_{D_j}$, $V_j V_j^\dagger = P_j$, and $\sum_j P_j = \mathbb{1}_D$. Taking traces gives

$$\mathrm{tr} P_j = \mathrm{tr}(V_j V_j^\dagger) = \mathrm{tr}(V_j^\dagger V_j) = D_j, \quad (15.20)$$

$$D = \mathrm{tr} \mathbb{1}_D = \sum_{j=0}^{m-1} \mathrm{tr} P_j = \sum_{j=0}^{m-1} D_j \geq \sum_{j=0}^{m-1} 1 = m. \quad (15.21)$$

□

Lemma 15.5.22 (The blocks of the canonical form are irreducible). [Lean](#) [✓ Stmt](#) *Let $\sum_i A_i A_i^\dagger = \mathbb{1}$, let the dual map $X \mapsto \sum_i A_i^\dagger X A_i$ have a positive definite fixed point, and let every fixed point of $X \mapsto \sum_i A_i X A_i^\dagger$ be a multiple of the identity. Then no orthogonal projection other than 0 and $\mathbb{1}$ is invariant under every A_i .*

Proof. [✓ Proof](#) If $(\mathbb{1} - P)A_i P = 0$ for an orthogonal projection P , then $P - \sum_i A_i P A_i^\dagger = \sum_i B_i B_i^\dagger$ with $B_i = P A_i (\mathbb{1} - P)$ is positive, and its pairing with the positive definite dual fixed point vanishes. Hence P is a fixed point, a multiple of the identity, so P is 0 or $\mathbb{1}$. □

Lemma 15.5.23 (Periodic representative of an irreducible block). [Lean](#) [Lean](#) [✓ Stmt](#) *A nonzero irreducible tensor A , rescaled by a nonzero ζ and gauged, becomes a periodic tensor B of some period m , with $|\phi_B\rangle = \zeta^N |\phi_A\rangle$ on N sites. A tensor is periodic with period one exactly when it is irreducible, left canonical and its transfer channel is primitive.*

Proof. [✓ Proof](#) Rescale by the inverse square root of the spectral radius of the transfer map and gauge by its positive definite eigenvector; the peripheral spectrum of the resulting trace-preserving irreducible channel is the group of m -th roots of unity. Period one means that 1 is the only peripheral eigenvalue, which is primitivity. □

Theorem 15.5.24 (Single-length bound without small divisors). [Lean](#) [✓ Stmt](#) *Let A be any translation-invariant tensor of bond dimension D . Let $N \geq 2$, and suppose no integer m with $2 \leq m \leq D$ divides N . If the periodic vector of A on N sites is $c|W_N\rangle$ with $c \neq 0$, then*

$$N < 2 \max((D^2 + 1)^2, 3(D - 1)((D^2 + 1)^2 + 1)), \quad (15.22)$$

The admissible lengths include every N whose prime divisors exceed D , such as $N = p^2$ for a prime $p > D$.

Proof. [✓ Proof](#) Write the periodic vector as $\sum_j \lambda_j^N |\phi_{A^j}\rangle$ with unital irreducible blocks A^j . Each block is irreducible by Lemma 15.5.22, and, rescaled and gauged, periodic by Lemma 15.5.23, with a period $m_j \leq D_j \leq D$ by Theorem 15.5.21. If $m_j \neq 1$, then $2 \leq m_j \leq D$, so the divisor assumption gives $m_j \nmid N$. The block state vanishes by Theorem 7.3. If $m_j = 1$ the block is primitive, and the general quantum Wielandt bound of Theorem 6.4.8 gives injectivity after $L_0 = (D^2 + 1)^2$ sites. Group the blocks into the phase classes of Definition 8.6.1.7: the q -th member $A^{(j,q)}$ of class j satisfies $|\phi_{A^{(j,q)}}\rangle = \xi_{j,q}^N |\phi_{A^{(j,0)}}\rangle$, with $A^{(j,0)}$ the class representative. Then

$$\sum_k \lambda_k^N |\phi_{A^k}\rangle = \sum_j c_j |\phi_{A^{(j,0)}}\rangle, \quad c_j = \sum_q \lambda_{j,q}^N \xi_{j,q}^N, \quad (15.23)$$

and no two representatives are related by a gauge transformation and a phase. Dropping the classes with $c_j = 0$ or period other than one leaves at most D blocks. The generalized first item of Theorem 15.5.16 then gives (15.22). $\square$

Theorem 15.5.25 (Single-length bound at prime length). [Lean](#) [✓ Stmt](#) *Let A be any translation-invariant tensor of bond dimension D and let N be prime. If its periodic vector is $c|W_N\rangle$ with $c \neq 0$, then the bound (15.22) holds. In particular, $D = \Omega(N^{1/5})$ along prime lengths.*

Proof. [✓ Proof](#) If $N \leq D$, the bound is immediate. Otherwise a divisor $m > 1$ of the prime N equals $N > D$. Thus no m with $2 \leq m \leq D$ divides N , and Theorem 15.5.24 applies. $\square$

Remark 15.5.26 (Growth of the translation-invariant bond dimension). Theorem 15.5.10 concerns one tensor representing the W state on a range of lengths. The review states the stronger single-length bound: a translation-invariant representation of $|W_N\rangle$ with bond dimension D requires $D^3 \log D = \Omega(N)$, hence $D = \Omega(N^{1/(3+\delta)})$ for every $\delta > 0$, obtained by combining the representation theory of translation-invariant matrix product states with a quantitative quantum Wielandt inequality [16, Appendix A, “The W state”]. This single-length bound is outside the scope of the results here: Theorem 15.5.16 proves the conditional single-length bound of its source, and Theorem 15.5.25 establishes the bound $D = \Omega(N^{1/5})$ at prime lengths. Theorem 15.5.24 also covers lengths with no divisor between 2 and D [48].

15.6 The cluster state

The cluster state [6, 196] is a universal resource for measurement-based quantum computation; its one-dimensional chain provides another canonical example of a non-trivial SPT phase [204, 16]. We follow the presentation of [16, Appendix A, “The cluster state”]. Throughout, $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

Definition 15.6.1 (Cluster-state tensor of the review). [Lean](#) [✓ Stmt](#) Set $d = D = 2$. The cluster-state tensor of [16, Appendix A, “The cluster state”] is the family $\{A^0, A^1\} \subset M_2(\mathbb{C})$ given by

$$A^0 = |0\rangle\langle+| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad A^1 = |1\rangle\langle-| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}.$$

Equivalently, $(A^s)_{\alpha\beta} = \delta_{\alpha s} (-1)^{s\beta} / \sqrt{2}$.

Close the virtual indices periodically by the trace. Theorem 15.6.2 identifies this vector with the controlled- Z circuit state, including its short-ring convention; Lemma 15.6.5 relates the two tensor representatives. Theorem 15.6.6 gives normality, the $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry and its virtual action, string order under the stated boundary restriction, and zero correlation length after blocking two sites. The nontrivial projective class is Theorem 15.6.15; the periodic stabilizer ground state for $N \geq 3$ is Theorem 15.9.10, using the gauge-equivalent representative.

Theorem 15.6.2 (Controlled- Z construction of the cluster state). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every $N \geq 1$ and every $s \in \{0, 1\}^N$,*

$$\text{tr}(A^{s_1} A^{s_2} \dots A^{s_N}) = \left(\prod_{j=1}^N \text{CZ}_{j,j+1} |+\rangle^{\otimes N} \right)_s = 2^{-N/2} \prod_{j=1}^N (-1)^{s_j s_{j+1}},$$

with indices modulo N , where $\text{CZ}_{a,b}$ multiplies the coefficient of $|s\rangle$ by $(-1)^{s_a s_b}$. Thus the periodic matrix product vector of the tensor of Definition 15.6.1 is exactly the normalized cluster state

obtained by acting with controlled-Z gates between nearest neighbours of a ring on $|+\rangle^{\otimes N}$ [16, Appendix A, “The cluster state”], [196]. For $N \geq 3$ the N gates act on the distinct bonds of the ring. For $N = 2$ the two gates act on the same pair and cancel, giving $|+\rangle^{\otimes 2}$; for $N = 1$ the single gate CZ_{1,1} is the Pauli Z, giving $|-\rangle$.

Proof. ✓ Proof Since $A^s = |s\rangle\langle h_s|$ with $h_0 = +$ and $h_1 = -$, the trace of the product is $\prod_j \langle h_{s_j} | s_{j+1} \rangle$ with indices modulo N , and $\langle h_a | b \rangle = (-1)^{ab} / \sqrt{2}$. On the other side, every coefficient of $|+\rangle^{\otimes N}$ equals $2^{-N/2}$, and the gates multiply the coefficient of $|s\rangle$ by $\prod_j (-1)^{s_j s_{j+1}}$. $\square$

Lemma 15.6.3 (Valence-bond form of the cluster-state tensor). Lean Lean Lean ✓ Stmt In the valence-bond description of [210], neighbouring virtual qubits share the bond $|00\rangle + |01\rangle + |10\rangle - |11\rangle$ and each site maps both of its virtual qubits to the physical one by $|0\rangle\langle 00| + |1\rangle\langle 11|$, so that the coefficient of $|s\rangle$ on a ring of $N \geq 1$ sites is $\prod_{j=1}^N H_{s_j s_{j+1}}$ with $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and indices modulo N . Let $B^s = |s\rangle\langle s|H$. Then for every $N \geq 1$ and every $s \in \{0, 1\}^N$,

$$\text{tr}(B^{s_1} B^{s_2} \dots B^{s_N}) = \prod_{j=1}^N H_{s_j s_{j+1}},$$

so B is a matrix product description of the valence-bond state, and $B^s = \sqrt{2} A^s$.

Proof. ✓ Proof The entries are $(B^s)_{\alpha\beta} = \delta_{\alpha s} H_{s\beta}$, so in the expansion of the trace as a sum over cyclic index paths only the path $\alpha_j = s_j$ survives, and it contributes $\prod_j H_{s_j s_{j+1}}$. The identity $B^s = \sqrt{2} A^s$ is read off from $|s\rangle\langle s|H = \sqrt{2} |s\rangle\langle h_s|$ with $h_0 = +$ and $h_1 = -$. $\square$

Definition 15.6.4 (Transposed representative). Lean ✓ Stmt The results below are proved for the transpose of the tensor of Definition 15.6.1, the family

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

and transferred back by Lemma 15.6.5.

Lemma 15.6.5 (The two cluster-state tensors are gauge equivalent). Lean Lean Lean Lean ✓ Stmt The tensor of Definition 15.6.1 is the transpose of the tensor of Definition 15.6.4, and the two are gauge equivalent: with $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, the review tensor is $HA^s H^{-1}$, since $H|\pm\rangle = \sqrt{2}|0\rangle$ and $\sqrt{2}|1\rangle$ respectively. Hence both tensors generate the same matrix product vectors at every length, and their length-2 blockings are gauge equivalent by the same matrix.

Theorem 15.6.6 (Properties of the review cluster-state tensor). Lean Lean Lean Lean Lean Lean ✓ Stmt The tensor of Definition 15.6.1 is not injective, its length-2 blocking is injective, and so it is normal. The length-2 blocked tensor is on-site symmetric under the $\mathbb{Z}_2 \times \mathbb{Z}_2$ action of Theorem 15.6.10; the two generators are implemented on the bond space by σ_x and σ_z , the conjugates by H of the gauges of Theorem 15.6.11, and these form a projective representation with the factor system of Definition 15.6.12, whose class is non-trivial (Theorem 15.6.15). The blocked tensor has string order for every group element with the maximally mixed boundary state, and has zero correlation length.

Proof. ✓ Proof Each property is invariant under the gauge equivalence of Lemma 15.6.5 and holds for the transposed representative by Theorems 15.6.7, 15.6.8, 15.6.10, 15.6.19 and 15.6.21. The twisted blocked tensors and their virtual action are the conjugates by H of those of the transposed representative, so the intertwining relations carry over. For the string order, the blocked transfer

channel of the transposed representative is unital and its adjoint fixes the maximally mixed state; since H is real symmetric with $H^2 = 2\mathbb{1}$, conjugation by H carries both properties to the blocked tensor of Definition 15.6.1. $\square$

Theorem 15.6.7 (Cluster-state tensor is not injective). [Lean](#) [✓ Stmt](#) *The cluster-state tensor is not injective: $\text{span}_{\mathbb{C}}\{A^0, A^1\} = \text{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$ has dimension 2, not $\dim M_2(\mathbb{C}) = 4$.*

Proof. [✓ Proof](#) Both A^0 and A^1 are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than $\dim M_2(\mathbb{C}) = 4$. $\square$

Theorem 15.6.8 (Cluster-state tensor is normal). [Lean](#) [Lean](#) [✓ Stmt](#) *Although the cluster-state tensor is not injective at a single site, its length-2 blocking is injective: the four length-2 products span $M_2(\mathbb{C})$. In particular, the cluster-state tensor is normal.*

Remark 15.6.9. Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$, and $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$, the four products $A^{ij} = A^i A^j$ are

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\frac{1}{\sqrt{2}}|-\rangle\langle 1|. \quad (15.24)$$

Since $\{|+\rangle, |-\rangle\}$ and $\{|0\rangle, |1\rangle\}$ are bases, these four rank-one matrices are linearly independent and span $M_2(\mathbb{C})$.

Theorem 15.6.10 ($\mathbb{Z}_2 \times \mathbb{Z}_2$ on-site symmetry of the blocked cluster-state tensor). [Lean](#) [✓ Stmt](#) *The length-2 blocked cluster-state tensor is on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$. The two generators act on the 4-dimensional blocked physical space $(\mathbb{C}^2)^{\otimes 2}$ by $\sigma_x \otimes \mathbb{1}$ and $\mathbb{1} \otimes \sigma_x$, which commute and each square to the identity, defining a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$.*

Theorem 15.6.11 (Anticommuting virtual gauges of the cluster-state symmetry). [Lean](#) [✓ Stmt](#) *The virtual matrices implementing the two generators of Theorem 15.6.10, namely $V_1 = \sigma_z$ and $V_2 = \sigma_x$, anticommute ($V_1 V_2 = -V_2 V_1$).*

Definition 15.6.12 (Cluster-state factor system). [Lean](#) [✓ Stmt](#) *The cluster-state factor system on $\mathbb{Z}_2 \times \mathbb{Z}_2$ is $\omega(g, h) = (-1)^{g_2 h_1}$ (the value -1 when $g_2 = 1$ and $h_1 = 1$, and 1 otherwise). It is the factor system of the explicit projective representation with virtual gauges σ_z and σ_x , both squaring to the identity, so there is no diagonal term.*

Lemma 15.6.13 (Cluster-state commutator phase). [Lean](#) [✓ Stmt](#) *The commutator phase of the cluster-state factor system on the two generators is -1 .*

Theorem 15.6.14 (The cluster-state factor system is a 2-cocycle). [Lean](#) [✓ Stmt](#) *The cluster-state factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1))$.*

Theorem 15.6.15 (The cluster-state $\mathbb{Z}_2 \times \mathbb{Z}_2$ phase is non-trivial). [Lean](#) [✓ Stmt](#) *The cluster-state factor system (Theorem 15.6.14), built from the anticommuting gauges σ_z and σ_x (Theorem 15.6.11), is not a coboundary: its commutator phase on the two generators is -1 . Its class in $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$ is therefore non-trivial, so the cluster state realizes a non-trivial SPT phase.*

Theorem 15.6.16 (Virtual implementation of the cluster-state symmetry). [Lean](#) [✓ Stmt](#) Let u be the on-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ action of Theorem 15.6.10 and let X be the projective representation with factor system of Definition 15.6.12 whose generators act by σ_z and σ_x . For every g and every letter i of the length-2 blocked cluster-state tensor A ,

$$\left(\sum_j u(g)_{ij} A^j \right) X_g = X_g A^i. \quad (15.25)$$

Proof. [✓ Proof](#) For g the identity both sides equal A^i . For each of the other three elements, with X_g equal to σ_z , σ_x or $\sigma_z \sigma_x$, a direct computation on the four blocked letters gives $\sum_j u(g)_{ij} A^j = X_g A^i X_g^{-1}$; multiplying on the right by X_g gives (15.25). $\square$

Theorem 15.6.17 (The cluster state is the $\mathbb{Z}_2 \times \mathbb{Z}_2$ SPT fixed point). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let B be the SPT fixed-point tensor of the projective representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ by σ_z and σ_x . The length-2 blocked cluster-state tensor satisfies $A^i = \sum_j V_{ij} B^j$ for a unitary V on the four-dimensional physical space. This unitary intertwines the two symmetries: $u(g) V = V U(g)$ for every g , where u is the on-site $\mathbb{Z}_2 \times \mathbb{Z}_2$ action on the blocked cluster state and U is the physical action of the fixed point. Every virtual projective representation of B has a non-trivial class [16, Section III.A].

Proof. [✓ Proof](#) Take $V_{i,(a,b)} = \sqrt{2} (A^i)_{ab}$; the four blocked letters are orthogonal with $\text{tr}(A^i A^{j\dagger}) = \frac{1}{2} \delta_{ij}$, so V is unitary. Both tensors realize the symmetry g with the same virtual gauge, so $\sum_j (u(g)V)_{ij} B^j = \sum_j (VU(g))_{ij} B^j$, and the letters B^j are linearly independent. The class statement follows from Theorem 10.4.2.3 and the non-triviality of the cluster factor system. $\square$

Theorem 15.6.18 (Doubly degenerate entanglement spectrum of the cluster state). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let Λ be a positive definite fixed point of the adjoint transfer map of the length-2 blocked cluster-state tensor. Every eigenspace of Λ has even dimension.

Proof. [✓ Proof](#) The blocked tensor is injective and satisfies $\mathcal{E}(\mathbb{1}) = \mathbb{1}$; the two generators of $\mathbb{Z}_2 \times \mathbb{Z}_2$ act by unitary permutation matrices with anticommuting virtual gauges σ_z and σ_x . Apply Theorem 10.4.4.5. $\square$

Theorem 15.6.19 (The cluster state has string order under its $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry). [Lean](#) [✓ Stmt](#) For every element g of $\mathbb{Z}_2 \times \mathbb{Z}_2$, the length-2 blocked cluster-state tensor has string order in the sense of Definition 10.3.11 for the twist by g , with the maximally mixed boundary state $\Lambda = \frac{1}{2} \mathbb{1}$ as its stationary boundary state.

Proof. [✓ Proof](#) The blocked tensor is injective (Remark 15.6.9) and on-site symmetric under $\mathbb{Z}_2 \times \mathbb{Z}_2$ (Theorem 15.6.10), and each group element acts by a real symmetric involutive permutation matrix, hence unitarily. The maximally mixed state $\Lambda = \frac{1}{2} \mathbb{1}$ is positive definite and has trace 1. The four blocked matrices in (15.24) satisfy $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, so the transfer map is unital and Λ is a fixed point of the adjoint transfer map. Applying Theorem 10.4.4 yields string order for every g . $\square$

Theorem 15.6.20 (Blocked cluster transfer map is idempotent). [Lean](#) [Lean](#) [✓ Stmt](#) The length-2 blocked cluster transfer map satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$. In closed form, it sends every X to the scalar $(X_{00} + X_{11})/2$ times the identity, so its image is one-dimensional.

Proof. [✓ Proof](#) The single-site cluster transfer map is not idempotent, but the four blocked matrices give the closed form $\mathcal{E}_A(X) = \frac{1}{2}(X_{00} + X_{11})\mathbb{1}$, which is manifestly idempotent. $\square$

Theorem 15.6.21 (Cluster zero correlation length). [Lean](#) [✓ Stmt](#) *The length-2 blocked cluster tensor has zero correlation length: its idempotent transfer map (Theorem 15.6.20) gives correlations independent of the separation. Like the GHZ state, and unlike the AKLT state, the cluster state is a renormalization fixed point with zero correlation length.*

Proof. [✓ Proof](#) The claim follows from the idempotence of the blocked cluster transfer map and the equivalence of zero correlation length with transfer-map idempotence (Theorem 17.5.20). $\square$

15.7 The Kitaev chain

The Kitaev chain, or Majorana chain [175], is a chain of spinless fermions. In the limit where the Majorana modes pair up only across neighbouring sites, its ground state is the non-trivial fixed point of the topological phase, and it is described by a graded (fermionic) matrix product state with bond dimension two and a twist at the boundary [16, Appendix A, “The Kitaev chain”], [8].

Definition 15.7.1 (Kitaev-chain tensor and twist). [Lean](#) [Lean](#) [✓ Stmt](#) Set $d = D = 2$. The Kitaev-chain tensor is

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

with the twist $Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ at the boundary [16, Appendix A, “The Kitaev chain”], [8].

On a ring of N sites the state is

$$|\psi\rangle = \sum_{i_1, \dots, i_N} \text{tr}(Y A^{i_1} A^{i_2} \dots A^{i_N}) |i_1\rangle \otimes_g |i_2\rangle \otimes_g \dots \otimes_g |i_N\rangle, \quad (15.26)$$

where $\otimes_g$ is the graded tensor product; the state has odd fermion parity [16, Section II.B.4].

Scope. The results below concern the ordinary contraction of the matrices, which gives the coefficients of (15.26) in the ordered graded product basis; the graded contraction rules and the graded notion of injectivity are outside the scope of these results [77].

The boundary twist enters as $\text{tr}(Y A^{i_1} \dots A^{i_N})$. Theorem 15.7.5 and Corollary 15.7.6 compute both twisted and untwisted coefficients and their parity supports. Lemma 15.7.7 gives the open closures. Theorems 15.7.8 and 15.7.9 describe non-normality and the two injective sectors of the ordinary matrix tensor. These are not statements of graded injectivity; that distinction is maintained in [77].

Definition 15.7.2 (Twisted periodic amplitude). [Lean](#) [✓ Stmt](#) For a word $w = (i_1, \dots, i_N)$ the twisted periodic amplitude is $\text{tr}(Y A^{i_1} \dots A^{i_N})$, the coefficient of w in (15.26).

Lemma 15.7.3 (The twist squares to minus the identity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) $A^0 = \mathbb{1}$, $A^1 = Y$, $(A^1)^2 = Y^2 = -\mathbb{1}$, and Y commutes with A^0 and A^1 [8].

Proof. [✓ Proof](#) Direct multiplication of 2×2 matrices. $\square$

Lemma 15.7.4 (Products of the Kitaev-chain matrices). [Lean](#) [✓ Stmt](#) *If the word w contains the letter 1 exactly k times, then $A^{i_1} \dots A^{i_N} = Y^k$.*

Proof. [✓ Proof](#) Induction on the word, using $A^0 = \mathbb{1}$ and $A^1 = Y$. $\square$

Theorem 15.7.5 (Periodic amplitudes). [Lean](#) [Lean](#) [✓ Stmt](#) *Let the word w contain the letter 1 exactly k times. Then*

$$\mathrm{tr}(A^{i_1} \dots A^{i_N}) = \begin{cases} 2(-1)^{k/2}, & k \text{ even}, \\ 0, & k \text{ odd}, \end{cases} \quad (15.27)$$

$$\mathrm{tr}(YA^{i_1} \dots A^{i_N}) = \begin{cases} 0, & k \text{ even}, \\ 2(-1)^{(k+1)/2}, & k \text{ odd}. \end{cases} \quad (15.28)$$

Neither amplitude depends on the length N except through k .

Proof. [✓ Proof](#) By Lemma 15.7.4 the products are Y^k and Y^{k+1} . Since $Y^2 = -\mathbb{1}$, an even power Y^{2m} is $(-1)^m \mathbb{1}$, of trace $2(-1)^m$, and an odd power Y^{2m+1} is $(-1)^m Y$, of trace zero. $\square$

Corollary 15.7.6 (Complementary parity supports). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *By (15.28), $\mathrm{tr}(YA^w) \neq 0 \iff k \equiv 1 \pmod{2}$, matching the odd fermion parity of (15.26); by (15.27), $\mathrm{tr}(A^w) \neq 0 \iff k \equiv 0 \pmod{2}$. Consequently, $\mathrm{tr}(YA^w) \mathrm{tr}(A^w) = 0$ for every word w .*

Proof. [✓ Proof](#) Immediate from (15.27) and (15.28). $\square$

Lemma 15.7.7 (Open-boundary closures). [Lean](#) [✓ Stmt](#) *For every word, the product $M = A^{i_1} \dots A^{i_N}$ satisfies*

$$M_{00} = M_{11}, \quad M_{01} = -M_{10}. \quad (15.29)$$

Of the four ways to close the virtual indices at the two ends of an open chain, the two diagonal closures give the same amplitudes and the two off-diagonal closures differ by a sign [8].

Proof. [✓ Proof](#) Both $\mathbb{1}$ and Y have this form, and so does every power of Y . $\square$

Theorem 15.7.8 (The ordinary contraction is reducible and not normal). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Read as an ordinary matrix product state, the Kitaev-chain tensor is reducible: the vector $(1, i)$ is a common eigenvector of A^0 and A^1 , with eigenvalues 1 and i , and the orthogonal projection onto the line it spans is a nontrivial projection invariant under both matrices. The tensor is not injective, and it is not normal: no blocking length makes the products span the full matrix algebra [16, Section III.A.4].*

Proof. [✓ Proof](#) The line spanned by $(1, i)$ is invariant under $A^0 = \mathbb{1}$ and under $A^1 = Y$. The linear functional $M \mapsto M_{00} - M_{11}$ vanishes on every product by (15.29) (Lemma 15.7.7), while the matrix unit $E_{00} = |0\rangle\langle 0|$ satisfies $(E_{00})_{00} - (E_{00})_{11} = 1 \neq 0$, so $\mathrm{span}\{A^w : |w| = N\} \subsetneq M_2(\mathbb{C})$ for every N . $\square$

Theorem 15.7.9 (Sum of two injective matrix product states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For $c \in \mathbb{C}$ let a_c be the tensor of bond dimension one with $a_c^0 = 1$ and $a_c^1 = c$; it is injective. Write $a_{\pm} = a_{\pm i}$. For every word w ,*

$$\mathrm{tr}(YA^{i_1} \dots A^{i_N}) = i a_+^{i_1} \dots a_+^{i_N} - i a_-^{i_1} \dots a_-^{i_N}, \quad (15.30)$$

so the ordinary contraction of the twisted state is a sum of two injective matrix product states [16, Section III.A.4].

Proof. [✓ Proof](#) With k the number of letters 1 in w , the right-hand side is $i^{k+1} + (-i)^{k+1}$, which is the closed form of the twisted amplitude. $\square$

The Kitaev-chain tensor is related to the even-parity tensor $E^0 = \frac{1}{\sqrt{2}}\mathbb{1}$, $E^1 = \frac{1}{\sqrt{2}}\sigma^x$, whose periodic contraction is the equal superposition of even-parity configurations.

Lemma 15.7.10 (Relation to the even-parity tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $X = \text{diag}(1, i)$, which is invertible with inverse $\text{diag}(1, -i)$, and let $\varphi_0 = 1$, $\varphi_1 = i$. Then $A^j X = \sqrt{2} \varphi_j X E^j$ for $j = 0, 1$: up to the on-site phase $\text{diag}(1, i)$ on the physical index and the factor $\sqrt{2}$, the Kitaev-chain tensor is gauge equivalent to the even-parity tensor.*

Proof. [✓ Proof](#) $X\sigma^x X^{-1} = \sigma^y$ and $i\sigma^y = Y$. □

15.8 Correlation independence without a renormalization fixed point

Example 3.4 of [11, 12] demonstrates physical correlation independence without satisfying the renormalization fixed-point condition.

Definition 15.8.1 (The tensor of CPSV16 Example 3.4). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Set $d = D = 2$ and

$$A^0 = \text{diag}\left(1, \frac{1}{\sqrt{2}}\right), \quad A^1 = \text{diag}\left(0, \frac{1}{\sqrt{2}}\right).$$

The first and second virtual diagonal entries define the bond-dimension-one product tensors for $|0\rangle$ and $|+\rangle$, respectively.

Close the virtual indices by the trace. Theorem 15.8.2 identifies the resulting periodic vector. Physical correlation independence is Theorem 15.8.3, while Theorem 15.8.4 proves that the tensor is not a renormalization fixed point. The blocked overlap calculation is Theorem 15.8.5.

Theorem 15.8.2 (Positive-length amplitudes). [Lean](#) [✓ Stmt](#) For $N > 0$ and $i = (i_1, \dots, i_N) \in \{0, 1\}^N$,

$$\text{Tr}(A^{i_1} \dots A^{i_N}) = \mathbf{1}_{\{i=(0,\dots,0)\}} + \left(\frac{1}{\sqrt{2}}\right)^N.$$

Hence the generated MPV is $|0\rangle^{\otimes N} + |+\rangle^{\otimes N}$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Proof. [✓ Proof](#) Both matrices are diagonal. The first diagonal entry contributes 1 only to the all-zero word, while the second contributes $(1/\sqrt{2})^N$ to every word. □

Theorem 15.8.3 (Physical correlations are independent of distance). [Lean](#) [✓ Stmt](#) The tensor in Definition 15.8.1 satisfies physical correlation independence for arbitrary disjoint physical regions, including adjacent regions.

Proof. [✓ Proof](#) Every word matrix is diagonal. Consequently every observable-inserted transfer operator and the ordinary transfer operator act entrywise on bond matrices, so they commute. The two-region expectation therefore depends on the two complementary gap lengths only through their sum. □

Theorem 15.8.4 (Failure of the renormalization fixed-point equation). [Lean](#) [Lean](#) [✓ Stmt](#) The transfer map is not idempotent, and the tensor has no physical blocking isometry. Thus it does not satisfy the pure-state renormalization fixed-point equation.

Proof. ✓ Proof Let $E_{01} = |0\rangle\langle 1|$. The transfer map obeys

$$\mathcal{E}_A(E_{01}) = \frac{1}{\sqrt{2}}E_{01}, \quad \mathcal{E}_A^2(E_{01}) = \frac{1}{2}E_{01}.$$

Hence $\mathcal{E}_A^2 \neq \mathcal{E}_A$. The equivalence between transfer idempotence and a physical blocking isometry gives the second conclusion. $\square$

Theorem 15.8.5 (Blocked product-state overlap). Lean ✓ Stmt Let A_0 and A_+ be the bond-dimension-one product tensors obtained from the first and second virtual diagonal entries of A . Their formal MPV overlap at length n is

$$O_{A_0 A_+}(n) = \left(\frac{1}{\sqrt{2}} \right)^n.$$

Proof. ✓ Proof Only the all-zero configuration contributes, with amplitude $(1/\sqrt{2})^n$. $\square$

15.9 Parent Hamiltonian statements for the examples

Following the parent-Hamiltonian construction of [16, lines 1988–2000] reviewed in Chapter 11, originally introduced for finitely correlated states by Fannes, Nachtergaele, and Werner [160] and developed for matrix product states by Perez-Garcia et al. [193], the range- L local parent space is

$$\mathcal{G}_L(A) = \left\{ \sum_{i_1, \dots, i_L} \text{tr}(A^{i_1} \dots A^{i_L} X) |i_1 \dots i_L\rangle : X \in M_D(\mathbb{C}) \right\}.$$

Then $\ker h_L = \mathcal{G}_L(A)$, the range- L parent Hamiltonian is $H_N^{(L)} = \sum_i (h_L)_i$, and $\text{GS}(H) = \ker(H - E_0(H)\mathbb{1})$ denotes the ground eigenspace.

Theorem 15.9.1 (GHZ two-site parent space). Lean ✓ Stmt For the GHZ tensor A of Definition 15.2.7,

$$\mathcal{G}_2(A) = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

Proof. ✓ Proof By definition, $\Gamma_2(X)(\sigma_1, \sigma_2) = \text{tr}(A^{\sigma_1} A^{\sigma_2} X)$. For the GHZ matrices $A^0 = \text{diag}(1, 0)$ and $A^1 = \text{diag}(0, 1)$, whenever $\sigma_1 \neq \sigma_2$, one has $A^{\sigma_1} A^{\sigma_2} = 0$. Hence $\Gamma_2(X) = X_{00}|00\rangle + X_{11}|11\rangle$, and the image of Γ_2 is $\text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}$. $\square$

Theorem 15.9.2 (Constant GHZ product vectors satisfy the cyclic constraints). Lean ✓ Stmt For $N \geq 2$ and $a \in \{0, 1\}$, one has $|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$.

Proof. ✓ Proof For a cyclic nearest-neighbour window and outside assignment τ , set $R_{i,\tau} := |a\rangle^{\otimes N}|_{[i, i+1], \tau}$. For $b, c \in \{0, 1\}$ with $b \neq c$, the completed configuration is not constant, hence $R_{i,\tau}(b, c) = 0$. Therefore

$$R_{i,\tau} \in \{v : v(0, 1) = v(1, 0) = 0\} = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\} = \mathcal{G}_2(A_{\text{GHZ}}),$$

so $|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$. $\square$

Theorem 15.9.3 (GHZ periodic nearest-neighbour ground space). [Lean](#) [Stmt](#) For $N \geq 2$, the common nearest-neighbour cyclic ground space of the GHZ tensor satisfies

$$\mathcal{G}_{N,2}(A_{\text{GHZ}}) = \text{span}_{\mathbb{C}}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}.$$

This is the cyclic-window form of the two-fold degeneracy in [16, line 2205].

Proof. [Proof](#) Let $\psi \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$. For every cyclic nearest-neighbour window and outside assignment τ , one has

$$\psi|_{[i,i+1],\tau} \in \mathcal{G}_2(A_{\text{GHZ}}) = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

Taking the outside assignment from the word σ gives $\psi|_{[i,i+1],\sigma}(\sigma_i, \sigma_{i+1}) = \psi(\sigma)$. Hence a mixed nearest-neighbour word has zero coefficient: $\sigma_i \neq \sigma_{i+1} \implies \psi(\sigma) = 0$. A nonconstant binary cyclic word has a mixed edge: $\sigma \notin \{(0, \dots, 0), (1, \dots, 1)\} \implies \exists i, \sigma_i \neq \sigma_{i+1}$. Therefore

$$\psi = \psi(0, \dots, 0)|0\rangle^{\otimes N} + \psi(1, \dots, 1)|1\rangle^{\otimes N}.$$

Conversely, Theorem 15.9.2 gives $|0\rangle^{\otimes N}, |1\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$, and hence their span lies in $\mathcal{G}_{N,2}(A_{\text{GHZ}})$. $\square$

Remark 15.9.4 (GHZ parent interaction). For the GHZ tensor of Definition 15.2.7,

$$\begin{aligned} \mathcal{G}_2(A) &= \text{span}\{|00\rangle, |11\rangle\}, \\ h_{\text{GHZ}} &= |01\rangle\langle 01| + |10\rangle\langle 10| = \mathbb{1} - |00\rangle\langle 00| - |11\rangle\langle 11|. \end{aligned}$$

The periodic nearest-neighbour parent Hamiltonian therefore satisfies

$$\begin{aligned} H_{\text{GHZ},N} &= \sum_i (h_{\text{GHZ}})_i, \\ \ker H_{\text{GHZ},N} &= \text{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}, \end{aligned}$$

using the cyclic-window equality $\mathcal{G}_{N,2}(A_{\text{GHZ}}) = \text{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}$ from Theorem 15.9.3. This is the ferromagnetic Ising parent Hamiltonian [16, lines 1194–1196 and line 2205]; see also [162, Section 3] for the displayed two-site space and interaction.

Definition 15.9.5 (Cluster-state stabilizer). [Lean](#) [Stmt](#) The three-site stabilizer of [193, local TeX lines 374–387] is $K = \sigma_1^z \sigma_2^x \sigma_3^z$. In the computational basis, with $\sigma^z|s\rangle = (-1)^s|s\rangle$ and $\sigma^x|s\rangle = |s+1\rangle$, it acts on a three-site vector v by

$$(Kv)(s_1, s_2, s_3) = (-1)^{s_1+s_3} v(s_1, s_2+1, s_3).$$

Theorem 15.9.6 (Cluster three-site parent space is the stabilizer eigenspace). [Lean](#) [Stmt](#) For the cluster tensor A of Definition 15.6.4,

$$\mathcal{G}_3(A) = \{v : Kv = v\},$$

the $+1$ eigenspace of $K = \sigma_1^z \sigma_2^x \sigma_3^z$.

Proof. [Proof](#) Write $q = 1/\sqrt{2}$. The matrices $A^s = |\pm\rangle\langle s|$ have entries $A_{ij}^s = q(-1)^{is}\delta_{sj}$, so

$$\Gamma_3(X)(s_1, s_2, s_3) = \text{tr}(A^{s_1} A^{s_2} A^{s_3} X) = q^3 (-1)^{s_1 s_2 + s_2 s_3} (X_{s_3 0} + (-1)^{s_1} X_{s_3 1}). \quad (15.31)$$

The sign $(-1)^{s_1 s_2 + s_2 s_3}$ changes by $(-1)^{s_1 + s_3}$ when s_2 is replaced by $s_2 + 1$, so $K\Gamma_3(X) = \Gamma_3(X)$ by (15.31). Conversely, if $Kv = v$, then $v(s_1, 1, s_3) = (-1)^{s_1+s_3} v(s_1, 0, s_3)$, and the matrix X with rows $X_{s_0} = v(0, 0, s) + v(1, 0, s)$, $X_{s_1} = v(0, 0, s) - v(1, 0, s)$ satisfies $\Gamma_3(X) = 2q^3 v$ by (15.31); hence $v = \Gamma_3(X)/(2q^3) \in \mathcal{G}_3(A)$. $\square$

Theorem 15.9.7 (The stabilizer has eigenvalue -1 on the source's matrices). [Lean](#) [Lean](#)
[✓ Stmt](#) Let A_{src} be the tensor with the two matrices displayed in [193, local TeX lines 378–387], the source's physical labels 1, 2 written as 0, 1:

$$A_{\text{src}}^0 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \quad A_{\text{src}}^1 = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}.$$

Then

$$\mathcal{G}_3(A_{\text{src}}) = \{v : Kv = -v\},$$

the -1 eigenspace of $K = \sigma_1^z \sigma_2^x \sigma_3^z$.

Proof. [✓ Proof](#) With $A_{\text{src}}^t = |1-t\rangle\langle r_t|$, $r_0 = (1, 1)$, $r_1 = (1, -1)$, one has $\langle r_t | u \rangle = (-1)^{tu}$ and

$$\Gamma_3(X)(t_1, t_2, t_3) = (-1)^{t_1(1-t_2)+t_2(1-t_3)}(X_{0,1-t_1} + (-1)^{t_3}X_{1,1-t_1}). \quad (15.32)$$

Replacing t_2 by $t_2 + 1$ changes the exponent by $1 + t_1 + t_3$ modulo 2, so $K\Gamma_3(X) = -\Gamma_3(X)$. Conversely, if $Kv = -v$, then $v(t_1, 1, t_3) = -(-1)^{t_1+t_3}v(t_1, 0, t_3)$, and the matrix X with $X_{0,1-t} = (-1)^t(v(t, 0, 0) + v(t, 0, 1))$, $X_{1,1-t} = (-1)^t(v(t, 0, 0) - v(t, 0, 1))$ satisfies $\Gamma_3(X) = 2v$ by (15.32); hence $v = \Gamma_3(X/2) \in \mathcal{G}_3(A_{\text{src}})$. $\square$

Definition 15.9.8 (Translated cluster stabilizers). [Lean](#) [✓ Stmt](#) On a periodic chain of N sites, the stabilizer at site i is the operator

$$(K_i\psi)(\sigma) = (-1)^{\sigma_i+\sigma_{i+2}} \psi(\sigma_1, \dots, \sigma_i, \sigma_{i+1} + 1, \sigma_{i+2}, \dots, \sigma_N),$$

with site indices modulo N . For $N \geq 3$ the sites $i, i+1, i+2$ are distinct and $K_i = \sigma_i^z \sigma_{i+1}^x \sigma_{i+2}^z$; the results below assume $N \geq 3$.

Theorem 15.9.9 (Cluster stabilizer ground space is the periodic chain ground space). [Lean](#)
[✓ Stmt](#) For $N \geq 3$, the common $+1$ eigenspace of the translated stabilizers is the periodic three-site chain ground space of the cluster tensor:

$$\bigcap_i \{\psi : K_i\psi = \psi\} = \mathcal{G}_{N,3}(A_{\text{cl}}).$$

Proof. [✓ Proof](#) For a cyclic three-site window at i and an outside assignment τ , the restriction $\psi|_{[i,i+2],\tau}$ satisfies $K(\psi|_{[i,i+2],\tau}) = (K_i\psi)|_{[i,i+2],\tau}$, since K_i reads σ_i, σ_{i+2} and flips σ_{i+1} inside the window. Thus $K_i\psi = \psi$ implies that every restriction is fixed by K ; conversely, if every restriction is fixed by K , then evaluating at the window extracted from σ with outside assignment σ itself gives $(K_i\psi)(\sigma) = \psi(\sigma)$. By Theorem 15.9.6, a restriction is fixed by K exactly when it lies in $\mathcal{G}_3(A_{\text{cl}})$. $\square$

Theorem 15.9.10 (The cluster state spans the common $+1$ eigenspace of the ZXZ stabilizers). [Lean](#) [Lean](#) [✓ Stmt](#) For $N \geq 3$,

$$\bigcap_i \{\psi : K_i\psi = \psi\} = \mathbb{C} V^{(N)}(A_{\text{cl}}),$$

and this common eigenspace is one-dimensional.

Proof. [✓ Proof](#) By Theorem 15.9.9 the common eigenspace is $\mathcal{G}_{N,3}(A_{\text{cl}})$. The cluster tensor is injective after blocking $L_0 = 2$ sites (Theorem 15.6.8), so Theorem 11.11.2.3 with $L = 3 \leq N$ gives $\mathcal{G}_{N,3}(A_{\text{cl}}) = \mathbb{C} V^{(N)}(A_{\text{cl}})$, and Theorem 11.11.2.4 gives $\dim \mathcal{G}_{N,3}(A_{\text{cl}}) = 1$. $\square$

Theorem 15.9.11 (Periodic ground space for the source cluster matrices). [Lean](#) [✓ Stmt](#) For $N \geq 3$, the periodic three-site ground space of the source tensor is the common -1 eigenspace of the translated stabilizers:

$$\bigcap_i \{\psi : K_i \psi = -\psi\} = \mathcal{G}_{N,3}(A_{\text{src}}).$$

Proof. [✓ Proof](#) The three-site calculation in Theorem 15.9.7 identifies $\mathcal{G}_3(A_{\text{src}})$ with the -1 eigenspace of $K = ZXZ$. The periodic ground space imposes this local condition at every position around the ring. Each translated operator K_i acts only on its three sites, so we can apply the local calculation with the outside spins fixed.

Fix a site i and fix the spins outside $[i, i+2]$ to a configuration τ . The remaining coefficients form a three-site vector $\psi|_{[i, i+2], \tau}$. The operator K_i acts only on these three sites: it flips the middle spin and multiplies by the two endpoint signs. Since $N \geq 3$, these sites are distinct, and therefore

$$K(\psi|_{[i, i+2], \tau}) = (K_i \psi)|_{[i, i+2], \tau}.$$

It follows that $K_i \psi = -\psi$ if and only if $K(\psi|_{[i, i+2], \tau}) = -\psi|_{[i, i+2], \tau}$ for every τ . For the reverse implication, every coefficient of $K_i \psi + \psi$ occurs in one of these three-site vectors, so all its coefficients vanish.

Substituting the local ground-space identity, we obtain $K_i \psi = -\psi$ for every i exactly when every three-site restriction belongs to $\mathcal{G}_3(A_{\text{src}})$, which is the defining condition for $\psi \in \mathcal{G}_{N,3}(A_{\text{src}})$. $\square$

Theorem 15.9.12 (Source cluster ground state). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $N \geq 3$ and put $U = (\sigma^z)^{\otimes N}$ and $H = \sum_i K_i$. Then

$$|V^{(N)}(A_{\text{src}})\rangle = (\sqrt{2})^N U |V^{(N)}(A_{\text{cl}})\rangle \neq 0. \quad (15.33)$$

The common -1 eigenspace of the K_i and the ground eigenspace of H are both $\mathbb{C}|V^{(N)}(A_{\text{src}})\rangle$. More precisely, with the usual inner product $\langle v, w \rangle = \sum_s \bar{v}_s w_s$, the bound below holds for every vector v :

$$\begin{aligned} \operatorname{Re}\langle v, Hv \rangle &\geq -N\|v\|^2, \\ \ker(H + N\mathbb{1}) &= \mathbb{C}|V^{(N)}(A_{\text{src}})\rangle. \end{aligned}$$

Thus the ground energy is $-N$, and the normalized ground state is unique up to phase. This proves the periodic-chain claim for the matrices in [193], for odd as well as even $N \geq 3$.

Proof. [✓ Proof](#) Applying σ^z at every site changes every stabilizer eigenvalue from $+1$ to -1 . We first relate the vector associated with the normalized cluster tensor to the vector defined by the source matrices.

Put

$$G = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad G^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

Direct multiplication gives, for $t = 0, 1$,

$$A_{\text{src}}^t = \sqrt{2}(-1)^t G A_{\text{cl}}^t G^{-1}.$$

For a configuration $s = (s_0, \dots, s_{N-1})$, adjacent inverse factors cancel in the ordered product:

$$A_{\text{src}}^{s_0} \dots A_{\text{src}}^{s_{N-1}} = (\sqrt{2})^N (-1)^{\sum_j s_j} G(A_{\text{cl}}^{s_0} \dots A_{\text{cl}}^{s_{N-1}}) G^{-1}. \quad (15.34)$$

Taking the trace in (15.34) and using $\text{tr}(GBG^{-1}) = \text{tr}(B)$ gives

$$\begin{aligned} V^{(N)}(A_{\text{src}})_s &= \text{tr}(A_{\text{src}}^{s_0} \dots A_{\text{src}}^{s_{N-1}}) \\ &= (\sqrt{2})^N (-1)^{\sum_j s_j} \text{tr}(A_{\text{cl}}^{s_0} \dots A_{\text{cl}}^{s_{N-1}}) \\ &= (\sqrt{2})^N (-1)^{\sum_j s_j} V^{(N)}(A_{\text{cl}})_s. \end{aligned}$$

The parity factor is precisely the action of U on the coefficient at s . This proves (15.33); nonvanishing follows from the nonzero cluster vector, since $(\sqrt{2})^N \neq 0$ and $U^2 = \mathbb{1}$.

Conjugating by σ^z fixes σ^z and sends σ^x to $-\sigma^x$. The three sites in each K_i are distinct, so $UK_iU = -K_i$. If $K_i v = v$ for every i , then $K_i(Uv) = -Uv$ for every i . Conversely, if $K_i w = -w$ for every i , then $K_i(Uw) = Uw$; applying U again returns w . Thus U maps the common $+1$ eigenspace bijectively onto the common -1 eigenspace. By Theorem 15.9.10 and (15.33),

$$\bigcap_i \{v : K_i v = -v\} = \mathbb{C} |V^{(N)}(A_{\text{src}})\rangle.$$

This space is one-dimensional.

Each K_i changes coefficient signs and permutes the computational basis, so it preserves the norm. Expanding the squares gives

$$\frac{1}{2} \sum_i \|(\mathbb{1} + K_i)v\|^2 = N\|v\|^2 + \text{Re}\langle v, Hv \rangle \geq 0. \quad (15.35)$$

Equality holds exactly when every $(\mathbb{1} + K_i)v$ vanishes, that is, when $v \in \mathbb{C} |V^{(N)}(A_{\text{src}})\rangle$. Conversely, summing the equations $K_i |V^{(N)}(A_{\text{src}})\rangle = -|V^{(N)}(A_{\text{src}})\rangle$ gives $H |V^{(N)}(A_{\text{src}})\rangle = -N |V^{(N)}(A_{\text{src}})\rangle$. Together with (15.35), this identifies $-N$ as the attained ground energy and its eigenspace as $\mathbb{C} |V^{(N)}(A_{\text{src}})\rangle$. $\square$

Remark 15.9.13 (Cluster-state stabilizer convention). The cluster tensor of Definition 15.6.4 is the reflected convention of the tensor displayed in [16, lines 2364–2369]. By (15.31) and its reflected counterpart, the three-site parent space of either convention is the $+1$ eigenspace of $\sigma_1^z \sigma_2^x \sigma_3^z$. Since each K_i is a self-adjoint involution, $\frac{1}{2}(\mathbb{1} - K_i)$ is the orthogonal projector onto its -1 eigenspace, and Theorem 15.9.10 says that the cluster state is the unique ground state of $\sum_i \frac{1}{2}(\mathbb{1} - \sigma_i^z \sigma_{i+1}^x \sigma_{i+2}^z)$, with ground energy 0.

The older review states instead that cluster states are unique ground states of $\sum_i \sigma_i^z \sigma_{i+1}^x \sigma_{i+2}^z$ with the sign as printed [193, local TeX lines 374–387]. Both statements are correct for their own tensors: on the matrices A_{src} displayed there every stabilizer has eigenvalue -1 (Theorem 15.9.7), while on A_{cl} it has eigenvalue $+1$. The two tensors are related by $A_{\text{src}}^t = \sqrt{2} \sigma^x A^t$ for the tensor A of [16, lines 2364–2369], and since $\langle \pm | \sigma^x = \pm \langle \pm |$ the states differ by σ^z on every site, which conjugates each K_i to $-K_i$ and exchanges the two Hamiltonians $\sum_i K_i$ and $-\sum_i K_i = \sum_i (\mathbb{1} - K_i) - N\mathbb{1}$. Theorem 15.9.12 proves this on-site relation for the matrix product vectors and identifies the ground energy and eigenspace for the printed matrices; see [56].

Theorem 15.9.14 (AKLT two-site parent space). Lean ✓ Stmt For the AKLT tensor A of Definition 15.3.7, a two-site vector v lies in $\mathcal{G}_2(A)$ if and only if

$$v_{11} = v_{22} = 0, \quad v_{01} + v_{10} = 0, \quad v_{02} + v_{20} = 0, \quad 2v_{00} + v_{12} + v_{21} = 0. \quad (15.36)$$

Proof. ✓ Proof Write $A^0 = c\sigma^z$, $A^1 = b\sigma^+$, $A^2 = -b\sigma^-$ with $c = 1/\sqrt{3}$, $b = \sqrt{2}/\sqrt{3}$, so that $b^2 = 2c^2$. The products are $A^0A^0 = c^2\mathbb{1}$, $A^0A^1 = -A^1A^0 = cb|0\rangle\langle 1|$, $A^0A^2 = -A^2A^0 = cb|1\rangle\langle 0|$, $A^1A^2 = -b^2|0\rangle\langle 0|$, $A^2A^1 = -b^2|1\rangle\langle 1|$, and $A^1A^1 = A^2A^2 = 0$. Hence $\Gamma_2(X)(a, b) = \text{tr}(A^a A^b X)$ has coordinates

$$\Gamma_2(X) = (c^2(X_{00} + X_{11}), cbX_{10}, cbX_{01}, -cbX_{10}, 0, -b^2X_{00}, -cbX_{01}, -b^2X_{11}, 0) \quad (15.37)$$

in the order 00, 01, 02, 10, 11, 12, 20, 21, 22, which satisfies (15.36) because $2c^2 = b^2$. Conversely, given v satisfying (15.36), the matrix X with $X_{00} = -c^2v_{12}$, $X_{01} = cbv_{02}$, $X_{10} = cbv_{01}$, $X_{11} = -c^2v_{21}$ satisfies $\Gamma_2(X) = c^2b^2v$ by (15.37), so $v = \Gamma_2(X/(c^2b^2))$. $\square$

Remark 15.9.15 (AKLT two-site parent space as a span). The constraints (15.36) of Theorem 15.9.14 describe $\mathcal{G}_2(A)$ as the four-dimensional span of $|01\rangle - |10\rangle$, $|02\rangle - |20\rangle$, $|00\rangle - 2|12\rangle$, and $|00\rangle - 2|21\rangle$. This span is the spin-0 and spin-1 subspace of two spin-1 particles.

Theorem 15.9.16 (AKLT three-site intersection property). Lean Lean ✓ Stmt For the AKLT tensor A , the three-site intersection property holds:

$$(\mathcal{G}_{[1,2]} \otimes \mathcal{H}_3) \cap (\mathcal{H}_1 \otimes \mathcal{G}_{[2,3]}) = \mathcal{G}_{[1,3]},$$

where $\mathcal{G}_{[a,b]}$ denotes the local ground space $\mathcal{G}_{b-a+1}(A)$ placed on sites $a, \dots, b$. This is the identity $\mathcal{G}_{1,2} \cap \mathcal{G}_{2,3} = \mathcal{G}_{1,2,3}$ that [16, line 2095] asks to check by hand; it is not an instance of the general intersection property, since the AKLT tensor is not injective on one site.

Proof. ✓ Proof The inclusion $\supseteq$ is the restriction property of local ground spaces (Lemmas 11.5.2.1 and 11.5.2.2). For $\subseteq$, let ψ be a three-site vector whose restrictions $\psi(i, \cdot, \cdot)$ and $\psi(\cdot, \cdot, k)$ all lie in $\mathcal{G}_2(A)$, so that each satisfies (15.36). With c, b as in the proof of Theorem 15.9.14, the products $A^a A^b A^c$ give

$$\begin{aligned} \Gamma_3(X) = & c^3(X_{00} - X_{11})|000\rangle + c^2bX_{10}(|001\rangle - |010\rangle + |100\rangle) - c^2bX_{01}(|002\rangle - |020\rangle + |200\rangle) \\ & + cb^2X_{00}(|102\rangle - |012\rangle - |120\rangle) + cb^2X_{11}(|021\rangle - |201\rangle + |210\rangle) - b^3X_{10}|121\rangle + b^3X_{01}|212\rangle. \end{aligned} \quad (15.38)$$

Set $X_{00} = c\psi_{102}$, $X_{01} = -b\psi_{002}$, $X_{10} = b\psi_{001}$, $X_{11} = c\psi_{210}$. Then $\Gamma_3(X) = c^2b^2\psi$ coordinate by coordinate: the twelve coordinates absent from (15.38) vanish for ψ by the constraints $v_{11} = v_{22} = 0$ on the six restrictions, together with $\psi_{101} = -\psi_{110} = 0$ and $\psi_{202} = -\psi_{220} = 0$; the sign relations $\psi_{010} = -\psi_{001} = -\psi_{100}$, $\psi_{020} = -\psi_{002} = -\psi_{200}$, $\psi_{012} = -\psi_{102} = \psi_{120}$, $\psi_{201} = -\psi_{210}$ come from $v_{01} + v_{10} = 0$ and $v_{02} + v_{20} = 0$; and the remaining coordinates use $2v_{00} + v_{12} + v_{21} = 0$ with $b^2 = 2c^2$: $2\psi_{000} = \psi_{102} - \psi_{210}$, $\psi_{021} = \psi_{210}$, $\psi_{121} = -2\psi_{001}$, $\psi_{212} = -2\psi_{002}$. Hence $\psi = \Gamma_3(X/(c^2b^2)) \in \mathcal{G}_3(A)$. $\square$

Theorem 15.9.17 (Unique ground state of the two-site AKLT parent Hamiltonian). Lean Lean ✓ Stmt For $N \geq 3$, the periodic two-site chain ground space of the AKLT tensor is spanned by the AKLT state:

$$\mathcal{G}_{N,2}(A_{\text{AKLT}}) = \mathbb{C}V^{(N)}(A_{\text{AKLT}}).$$

Thus the two-site parent Hamiltonian has a unique ground state although the AKLT tensor becomes injective only after blocking $L_0 = 2$ sites [16, line 2095]. On two periodic sites the two-site chain ground space is $\mathcal{G}_2(A_{\text{AKLT}})$ itself, so the condition $N \geq 3$ is necessary.

Proof. [✓ Proof](#) Let $\psi \in \mathcal{G}_{N,2}(A_{\text{AKLT}})$. Every cyclic three-site window restricts on either side to a cyclic two-site window, so both restrictions of $\psi|_{[i,i+2],\tau}$ lie in $\mathcal{G}_2(A_{\text{AKLT}})$, and Theorem 15.9.16 places $\psi|_{[i,i+2],\tau}$ in $\mathcal{G}_3(A_{\text{AKLT}})$. Hence $\mathcal{G}_{N,2}(A_{\text{AKLT}}) \subseteq \mathcal{G}_{N,3}(A_{\text{AKLT}})$. The AKLT tensor is injective after blocking $L_0 = 2$ sites (Theorem 15.3.10), so Theorem 11.11.2.3 with $L = 3 \leq N$ gives $\mathcal{G}_{N,3}(A_{\text{AKLT}}) = \mathbb{C} V^{(N)}(A_{\text{AKLT}})$. The reverse inclusion holds because the periodic MPS vector satisfies every cyclic two-site constraint (Lemma 11.9.8). Finally $V^{(N)}(A_{\text{AKLT}}) \neq 0$ by Theorem 11.9.9 with $L_0 = 2 < N$, so $\dim \mathcal{G}_{N,2}(A_{\text{AKLT}}) = 1$. $\square$

Theorem 15.9.18 (Fourfold open-boundary degeneracy of the AKLT chain). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $N \geq 2$, the kernel of the open-chain two-site projector parent Hamiltonian of the AKLT tensor is the open-boundary MPS space,

$$\ker H_{N,2}^{\text{open}}(A_{\text{AKLT}}) = G_N^{\text{ES}}(A_{\text{AKLT}}) = \text{span}_{\mathbb{C}} \{ \sigma \mapsto (l|A_{\text{AKLT}}^\sigma|r) : ((l,|r)) \in \mathbb{C}^2 \times \mathbb{C}^2 \},$$

and it is four-dimensional. The count is for this open-chain projector parent Hamiltonian. The passage [16, lines 1171–1174] states a fourfold degeneracy for the AKLT chain with open boundary conditions, whose displayed Hamiltonian is the polynomial spin interaction. Summing the local identity $P^{(2)} = \frac{1}{2}(h_{\text{poly}} + \frac{2}{3}I)$ over the $N - 1$ open bonds gives $H_{\text{proj}} = \frac{1}{2}(H_{\text{poly}} + \frac{2(N-1)}{3}I)$. The projector kernel is the ground eigenspace of the polynomial Hamiltonian, equivalently the kernel of this shifted operator, and not the kernel of the polynomial Hamiltonian itself. For $N = 2$ the polynomial interaction has eigenvalues $-\frac{2}{3}$ and $\frac{4}{3}$, so its kernel is trivial, while the projector kernel is four-dimensional. The identification of the projector kernel with the open-boundary matrix-product space uses the boundary vectors of that passage together with the three-site intersection check of [16, line 2095], which the review prints for the periodic chain. As a consequence of Theorem 12.3.7.19 at $L_0 = 2$, and not as a statement of the review, the three-site open-chain parent Hamiltonian also has a four-dimensional kernel for $N \geq 3$. The spin-0 statement of the same passage is not part of this theorem; see [56].

Proof. [✓ Proof](#) For $N = 2$ the Hamiltonian is the projector onto the orthogonal complement of $\mathcal{G}_2(A_{\text{AKLT}})$. For $N \geq 3$, a kernel vector satisfies every nonwrapping two-site constraint. Each nonwrapping three-site window restricts on either side to a nonwrapping two-site window, so Theorem 15.9.16 places every three-site restriction in $\mathcal{G}_3(A_{\text{AKLT}})$; this is the local check of [16, line 2095], used here on nonwrapping windows. The AKLT tensor is injective after blocking $L_0 = 2$ sites (Theorem 15.3.10). For $N \geq 3$, with s ranging over starts satisfying $s + 3 \leq N$ and τ over complementary configurations,

$$H_{N,2}^{\text{open}}(A_{\text{AKLT}})v = 0 \implies (\forall s, \tau, \text{Res}_{s,3}^\tau v \in \mathcal{G}_3(A_{\text{AKLT}})) \implies v \in G_N^{\text{ES}}(A_{\text{AKLT}}).$$

The last step is the open-chain intersection theorem. The reverse inclusion holds for every tensor. The span description is Theorem 12.3.7.18, and Theorem 12.3.7.19 with $D = 2$ gives dimension $D^2 = 4$. $\square$

Definition 15.9.19 (AKLT polynomial Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let S^x, S^y, S^z be the spin-1 operators in the basis $|0\rangle = |m = 0\rangle$, $|1\rangle = |m = +1\rangle$, $|2\rangle = |m = -1\rangle$, with $S^z = \text{diag}(0, 1, -1)$, $S^+|0\rangle = \sqrt{2}|1\rangle$, $S^+|2\rangle = \sqrt{2}|0\rangle$, $S^x = \frac{1}{2}(S^+ + S^-)$ and $S^y = \frac{1}{2i}(S^+ - S^-)$. This is the labelling in which the three matrices of the AKLT tensor of Definition 15.3.7 are proportional to $\sigma^z, \sqrt{2}\sigma^+, -\sqrt{2}\sigma^-$ on the indices 0, 1, 2. For two sites j, k of a chain let $\vec{S}_j \cdot \vec{S}_k = \sum_\alpha S_j^\alpha S_k^\alpha$. On a periodic chain of N spin-1 sites, with indices modulo N , the AKLT Hamiltonian is

$$H_{\text{AKLT},N} = \sum_i \left(\vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2 \right) \quad (15.39)$$

[2, 3]; see also [193, local TeX lines 340–349], and, up to an additive constant, [204, lines 2126–2130]. Its two-site term is $h = \vec{S}_0 \cdot \vec{S}_1 + \frac{1}{3}(\vec{S}_0 \cdot \vec{S}_1)^2$.

Theorem 15.9.20 (The AKLT term and the spin-two projector). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On two spin-1 sites,

$$P^{(2)} = \frac{1}{2} \left(\vec{S}_0 \cdot \vec{S}_1 + \frac{1}{3}(\vec{S}_0 \cdot \vec{S}_1)^2 + \frac{2}{3}\mathbb{1} \right), \quad (15.40)$$

where $P^{(2)}$ is the orthogonal projector onto the subspace of total spin 2, the eigenspace of $\vec{S}_{\text{tot}}^2$ for $6 = 2(2+1)$, with $\vec{S}_{\text{tot}} = \vec{S}_0 + \vec{S}_1$. This projector is the two-site parent interaction of the AKLT tensor, the orthogonal projector onto $\mathcal{G}_2(A_{\text{AKLT}})^\perp$. Writing the Hamiltonian as a sum of such spin-2 projectors is the form of [2].

Proof. [✓ Proof](#) In the basis of Definition 15.9.19, $\vec{S}_0 \cdot \vec{S}_1 = S^z \otimes S^z + \frac{1}{2}(S^+ \otimes S^- + S^- \otimes S^+)$ has integer entries: it fixes $|11\rangle$ and $|22\rangle$, exchanges $|01\rangle \leftrightarrow |10\rangle$ and $|02\rangle \leftrightarrow |20\rangle$, and maps $|00\rangle \mapsto |12\rangle + |21\rangle$, $|12\rangle \mapsto |00\rangle - |12\rangle$, $|21\rangle \mapsto |00\rangle - |21\rangle$. A computation on the nine coordinates shows that the right side Q of (15.40) is $\sum_k w_k w_k^T / \|w_k\|^2$ for the five vectors $|11\rangle$, $|22\rangle$, $|01\rangle + |10\rangle$, $|02\rangle + |20\rangle$, $|12\rangle + 2|00\rangle + |21\rangle$. Hence Q is real symmetric and idempotent, and its kernel is cut out by the five constraints (15.36), which is $\mathcal{G}_2(A_{\text{AKLT}})$ by Theorem 15.9.14. A self-adjoint idempotent is the orthogonal projector onto the orthogonal complement of its kernel. The same computation gives $(\vec{S}_0 \cdot \vec{S}_1)Q = Q$, so a vector is fixed by $\vec{S}_0 \cdot \vec{S}_1$ exactly when it is fixed by Q . Since $\sum_\alpha (S^\alpha)^2 = 2$ on each site, $\vec{S}_{\text{tot}}^2 = 4 + 2\vec{S}_0 \cdot \vec{S}_1$, so the eigenspace of $\vec{S}_{\text{tot}}^2$ for 6 is the set of vectors fixed by $\vec{S}_0 \cdot \vec{S}_1$, which is the range of Q . $\square$

Theorem 15.9.21 (Ground state of the AKLT Hamiltonian). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $N \geq 2$ the Hamiltonian (15.39) satisfies

$$H_{\text{AKLT},N} + \frac{2N}{3} = 2 \sum_i P_{i,i+1}^{(2)} = 2H_{N,\text{AKLT}}^{(2)},$$

where $H_{N,\text{AKLT}}^{(2)}$ is the two-site parent Hamiltonian of the AKLT tensor. Hence every eigenvalue of $H_{\text{AKLT},N}$ is real and at least $-\frac{2N}{3}$. For $N \geq 3$ the eigenspace for $-\frac{2N}{3}$ is $\mathbb{C}V^{(N)}(A_{\text{AKLT}})$, so the AKLT state is the unique ground state of $H_{\text{AKLT},N}$ [2], [193, local TeX lines 340–349]. The ground energy $-\frac{2N}{3}$ is derived here from the identity above; the local source states only the ground-state property, for no specified boundary condition.

Proof. [✓ Proof](#) Restricting a chain vector to the window $i, i+1$ turns $\vec{S}_i \cdot \vec{S}_{i+1}$ into $\vec{S}_0 \cdot \vec{S}_1$, so the local term of the two-site parent Hamiltonian at i is $\frac{1}{2}(h_i + \frac{2}{3})$ by Theorem 15.9.20. Summing over the N bonds gives the identity. The parent Hamiltonian is positive by Lemma 12.3.8, so a nonzero v with $H_{\text{AKLT},N}v = \mu v$ satisfies $\mu + \frac{2N}{3} \geq 0$. The eigenspace for $-\frac{2N}{3}$ is the kernel of $H_{N,\text{AKLT}}^{(2)}$, which is $\mathcal{G}_{N,2}(A_{\text{AKLT}})$ by Theorem 12.3.17 and $\mathbb{C}V^{(N)}(A_{\text{AKLT}})$, a line, by Theorem 15.9.17. $\square$

Remark 15.9.22 (AKLT edge modes). For open boundary conditions and $N \geq 2$,

$$\begin{aligned} \dim \text{GS}(H_{\text{AKLT},N}^{\text{open}}) &= 4, \\ \dim(\text{GS}(H_{\text{AKLT},N}^{\text{open}}) \cap \mathcal{H}_{S=0}) &= 1 \end{aligned}$$

[16, lines 1169–1174]. Theorem 15.9.18 proves the fourfold count for the open-chain projector parent Hamiltonian. Summing the displayed local identity over the open chain gives $H_{\text{proj}} =$

$\frac{1}{2}(H_{\text{poly}} + \frac{2(N-1)}{3}I)$, so the projector kernel is the ground eigenspace of the polynomial Hamiltonian and the kernel of that shifted operator. It is not the kernel of the polynomial Hamiltonian: for $N = 2$ that interaction has eigenvalues $-\frac{2}{3}$ and $\frac{4}{3}$. The theorem says nothing about the spin-0 count in the second equality; see [\[56\]](#).

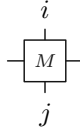
Chapter 16

Matrix Product Operators and Density Operators

Fix a physical dimension d and a bond dimension D . An MPO tensor carries two physical indices, one ket index and one bra index, together with the virtual left and right indices. Matrix product density operators (MPDOs) and matrix product operators (MPOs) describe 1D mixed states and quantum operators with bounded virtual entanglement [11, 12, 16]. The renormalization fixed-point conditions and their algebraic consequences are developed in Chapters 20 and 21.

16.1 MPO tensors

Definition 16.1.1 (MPO tensor). [Lean](#) [✓ Stmt](#) An *MPO tensor* is a family of matrices $\{M^{ij}\}_{i,j=0}^{d-1}$ with $M^{ij} \in M_D(\mathbb{C})$, indexed by a ket index i and a bra index j . Diagrammatically,



Definition 16.1.2 (Doubled-index MPS view). [Lean](#) [✓ Stmt](#) By combining the pair (i, j) into a single physical index in $\{0, \dots, d^2 - 1\}$, one obtains an MPS tensor with physical dimension d^2 and bond dimension D , with $M_{(i,j)}^{\text{MPS}} = M^{ij}$, identifying $\{0, \dots, d-1\} \times \{0, \dots, d-1\}$ with $\{0, \dots, d^2 - 1\}$ via the product encoding.

Definition 16.1.3 (Word evaluation). [Lean](#) [✓ Stmt](#) For words $u = (i_1, \dots, i_N)$ and $v = (j_1, \dots, j_N)$ in $\{0, \dots, d-1\}^N$, the *word evaluation* is

$$M^{u,v} := M^{i_1 j_1} M^{i_2 j_2} \dots M^{i_N j_N} \in M_D(\mathbb{C}). \quad (16.1)$$

The empty pair of words evaluates to the identity, $M^{\emptyset, \emptyset} = \mathbb{1}_D$, and word pairs of different lengths evaluate to 0.

Lemma 16.1.4 (Word-evaluation factorization). [Lean](#) [✓ Stmt](#) For words u_1, v_1 of equal length and arbitrary words u_2, v_2 , word evaluation factors as $M^{u_1 u_2, v_1 v_2} = M^{u_1, v_1} M^{u_2, v_2}$.

Proof. ✓ Proof The trace of a block-diagonal sum is the sum of the traces of the two blocks; apply Lemma 16.1.9 and take traces entrywise. $\square$

Definition 16.1.11 (Cyclic-shift reindexing). Lean ✓ Stmt The *cyclic shift* of configurations on N sites is the bijection that relabels each site by one position, $\sigma \mapsto \sigma \circ r$, where r is the cyclic rotation $i \mapsto i + 1$ of $\{0, \dots, N - 1\}$.

Theorem 16.1.12 (Translation invariance of the periodic MPDO). Lean ✓ Stmt The operator $\rho^{(N)}(M)$ is invariant under the simultaneous cyclic shift of its bra and ket configurations: $\rho^{(N)}(M)_{r\sigma, r\tau} = \rho^{(N)}(M)_{\sigma, \tau}$.

Proof. ✓ Proof The entry is a closed bond trace, $\rho^{(N)}(M)_{\sigma, \tau} = \text{tr}(M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N})$. Shifting both configurations by r rotates the matrix product by one factor, and the trace is invariant under that cyclic rotation:

$$\text{tr}(M^{\sigma_2 \tau_2} \dots M^{\sigma_N \tau_N} M^{\sigma_1 \tau_1}) = \text{tr}(M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N}),$$

so $\rho^{(N)}(M)_{r\sigma, r\tau} = \rho^{(N)}(M)_{\sigma, \tau}$. $\square$

Definition 16.1.13 (Hermitian MPO tensor). Lean ✓ Stmt An MPO tensor is *Hermitian* if $M^{ij} = (M^{ji})^\dagger$ for all $i, j \in \{0, \dots, d-1\}$.

Definition 16.1.14 (Adjoint MPO tensor). Lean Lean ✓ Stmt The *adjoint tensor* $M^\dagger$ of an MPO tensor M exchanges the ket and bra physical indices and conjugate-transposes each virtual matrix: $(M^\dagger)^{ij} = (M^{ji})^\dagger$. A tensor equals its adjoint tensor, $M^\dagger = M$, precisely when it is Hermitian.

Theorem 16.1.15 (Reflection identity for the adjoint tensor). Lean Lean Lean Lean ✓ Stmt The adjoint tensor generates the adjoint operator family up to spatial reflection: entrywise,

$$\rho^{(N)}(M^\dagger)_{\sigma, \tau} = \overline{\rho^{(N)}(M)_{\tau \circ \text{rev}, \sigma \circ \text{rev}}}, \quad (16.4)$$

where *rev* reverses the site order. For an MPDO and $N \geq 1$, the conjugation disappears:

$$\rho^{(N)}(M^\dagger)_{\sigma, \tau} = \rho^{(N)}(M)_{\sigma \circ \text{rev}, \tau \circ \text{rev}}, \quad (16.5)$$

and the adjoint tensor is again an MPDO.

Proof. ✓ Proof At the level of word evaluations, the conjugate transpose of a product reverses the factors, so

$$(M^\dagger)^{\sigma, \tau} = (M^{\tau \circ \text{rev}, \sigma \circ \text{rev}})^\dagger.$$

Taking traces gives the entrywise identity

$$\rho^{(N)}(M^\dagger)_{\sigma, \tau} = \text{tr}((M^{\tau \circ \text{rev}, \sigma \circ \text{rev}})^\dagger) = \overline{\rho^{(N)}(M)_{\tau \circ \text{rev}, \sigma \circ \text{rev}}}.$$

For an MPDO, $\rho^{(N)}(M)$ is Hermitian, so

$$\overline{\rho^{(N)}(M)_{\tau \circ \text{rev}, \sigma \circ \text{rev}}} = \rho^{(N)}(M)_{\sigma \circ \text{rev}, \tau \circ \text{rev}};$$

positive semidefiniteness is preserved under the simultaneous relabeling of rows and columns by the reflection. $\square$

16.2 Transfer maps

Definition 16.2.1 (MPO transfer map). [Lean](#) [✓ Stmt](#) The transfer map associated to M is the linear map $\mathcal{E}_M : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_M(X) = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger. \quad (16.6)$$

Lemma 16.2.2 (Identification with the doubled-index MPS transfer map). [Lean](#) [✓ Stmt](#) The transfer map of an MPO tensor agrees with the transfer map of its doubled-index MPS view: $\mathcal{E}_M = \mathcal{E}_{M^{\text{MPS}}}$.

Proof. [✓ Proof](#) After identifying the single physical index of the doubled tensor with a pair (i, j) , for every $X \in M_D(\mathbb{C})$,

$$\mathcal{E}_{M^{\text{MPS}}}(X) = \sum_{(i,j)} M_{(i,j)}^{\text{MPS}} X (M_{(i,j)}^{\text{MPS}})^\dagger = \sum_{i,j=0}^{d-1} M^{ij} X (M^{ij})^\dagger = \mathcal{E}_M(X).$$

□

Theorem 16.2.3 (Positivity of the MPO transfer map). [Lean](#) [✓ Stmt](#) If $X \geq 0$, then $\mathcal{E}_M(X) \geq 0$.

Proof. [✓ Proof](#) By Lemma 16.2.2, $\mathcal{E}_M$ is the transfer map of the doubled-index MPS tensor. The corresponding MPS transfer map is positive, so $\mathcal{E}_M(X) \geq 0$ whenever $X \geq 0$. □

16.3 MPDOs and LPDOs

Definition 16.3.1 (MPDO). [Lean](#) [✓ Stmt](#) An MPO tensor is an MPDO if, for every $N \geq 1$,

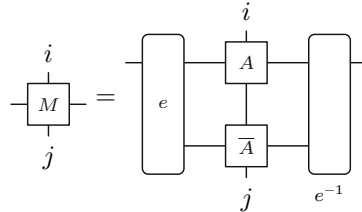
$$\rho^{(N)}(M) \geq 0. \quad (16.7)$$

This is the global positivity condition of [11, Section 4].

Definition 16.3.2 (LPDO). [Lean](#) [✓ Stmt](#) An MPO tensor is an LPDO if there exist a Kraus dimension d_K , an inner bond dimension D' , a bond-space identification $e : \{0, \dots, D-1\} \simeq \{0, \dots, D'-1\} \times \{0, \dots, D'-1\}$ (equivalently, D and $(D')^2$ have the same cardinality), and matrices $A^{(i,k)} \in M_{D'}(\mathbb{C})$ such that, for all physical indices i, j ,

$$M^{ij} = \left(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}, \quad (16.8)$$

where $\otimes$ denotes the Kronecker product, $\overline{(\cdot)}$ denotes entrywise complex conjugation, and $(\cdot)_{e,e}$ denotes reindexing along e back to $\{0, \dots, D-1\}$. See [11, Section 4.3]. The local contraction is



Theorem 16.3.3 (LPDOs are MPDOs). [Lean](#) [Stmnt](#) Every LPDO tensor is an MPDO.

Proof. [Proof](#) By the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, the N -site product decomposes as a sum of Kronecker products after transporting along the bond-space identification e :

$$M^{\sigma_1 \tau_1} \dots M^{\sigma_N \tau_N} = \left(\sum_{\kappa} P_{\kappa} \otimes \overline{Q_{\kappa}} \right) \Big|_e,$$

where $P_{\kappa} = A^{(\sigma_1, \kappa_1)} \dots A^{(\sigma_N, \kappa_N)}$ and $Q_{\kappa} = A^{(\tau_1, \kappa_1)} \dots A^{(\tau_N, \kappa_N)}$. Since the trace is invariant under the reindexing by e , we apply $\text{tr}(X \otimes Y) = \text{tr}(X) \text{tr}(Y)$ to obtain

$$\rho^{(N)}(M)_{\sigma, \tau} = \sum_{\kappa} \text{tr}(P_{\kappa}) \overline{\text{tr}(Q_{\kappa})} = \sum_{\kappa} \psi_{\kappa}(\sigma) \overline{\psi_{\kappa}(\tau)},$$

where $\psi_{\kappa}(\sigma) = \text{tr}(P_{\kappa})$. Hence $\rho^{(N)}(M) = \sum_{\kappa} |\psi_{\kappa}\rangle \langle \psi_{\kappa}| \geq 0$. $\square$

16.4 Horizontal and Vertical Canonical Forms

Definition 16.4.1 (Normalized BNT-refined horizontal form for an MPO). [Lean](#) [Stmnt](#) An MPO tensor is in *normalized BNT-refined horizontal form* if its doubled-index MPS tensor is a BNT sector decomposition with an independent gauge on every repeated copy:

$$M^{ab} = \bigoplus_{j=0}^{g-1} \bigoplus_{q=0}^{r_j-1} \mu_{j,q} X_{j,q} A_j^{ab} X_{j,q}^{-1}. \quad (16.9)$$

Here the A_j are the distinct minimal representatives, $r_j > 0$, and every weight $\mu_{j,q}$ is nonzero, $|\mu_{j,q}| \leq 1$, and at least one weight has modulus one. The BNT hypotheses include irreducibility, left-canonicity, normalized self-overlap, eventual linear independence of the representative MPV families, and the exclusion of similarity up to phase between distinct representatives. Repeated gauge-equivalent copies are retained over the same representative rather than separated as distinct blocks. The total bond dimension of the displayed direct sum is exactly D . Equivalently, with the block-diagonal matrix $X = \bigoplus_{j,q} X_{j,q}$,

$$M^{ab} = X \left[\bigoplus_{j,q} \mu_{j,q} A_j^{ab} \right] X^{-1},$$

$$X = \bigoplus_{j,q} X_{j,q}.$$

This is the strengthened, normalized BNT-refined form of the canonical decomposition in [11, Section 2.3]: unlike the literal predicate (Definition 7.6.7), the direct sum exhausts the full bond dimension D , each copy weight is nonzero and normalized, and gauge-equivalent normal blocks are grouped over a minimal BNT representative with an independent gauge for each copy.

Definition 16.4.2 (MPV representation associated with horizontal canonical form). [Lean](#) [Stmnt](#) An MPO tensor has a *horizontal-canonical-form MPV representation* if its doubled-index MPS tensor generates the same positive-length MPV family as a BNT canonical form $\bigoplus_{j=0}^{g-1} \bigoplus_{q=0}^{r_j-1} \mu_{j,q} A_j$. Its length- N matrix product vector is

$$\sum_{j=0}^{g-1} \left(\sum_{q=0}^{r_j-1} \mu_{j,q}^N \right) V^{(N)}(A_j). \quad (16.10)$$

This is the MPV-level decomposition in [11, Section 2.3, lines 298–301].

Theorem 16.4.3 (BNT-refined horizontal form gives the BNT MPV representation). [Lean](#) [✓ Stmt](#) Every MPO tensor in normalized BNT-refined horizontal form has the associated positive-length BNT matrix-product-vector representation.

Proof. [✓ Proof](#) Each copy similarity cancels inside every positive-length closed matrix-product trace. $\square$

$$K = X \left(\bigoplus_j \mathcal{K}_j \right) X^{-1}$$

Figure 16.1: Horizontal canonical form of the doubled tensor: the gauge X , the direct sum of basis tensors $\mathcal{K}_j$ carrying the ket and bra indices, and X^{-1} . In the source specialized simple-biCF/ZCL display the weights μ_j are independent of the copy index and absorbed into the basis tensors, so the basis tensors depend only on the block index j while the gauge factors $X_{j,q}$ depend on both block indices. This is [11, Appendix C.2, lines 1660–1665].

Definition 16.4.4 (Vertically viewed tensor of an MPO). [Lean](#) [Lean](#) [✓ Stmt](#) The *vertical direction* of an MPO tensor exchanges the notion of physical and virtual indices [11, line 943]: the vertically viewed tensor $\widetilde{M}$ has its letters indexed by the horizontal bond pair (a, b) , and each letter is the operator on the physical space with matrix elements $(\widetilde{M}_{ab})_{ij} = M_{ab}^{ij}$. Concatenating $\widetilde{M}$ vertically generates the matrix product vectors of the vertical direction.

Lemma 16.4.5 (Vertical reflection identity for the adjoint tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Adjunction exchanges the two oriented horizontal bond indices in the vertical view:

$$\widetilde{M}_{a,b}^\dagger = (\widetilde{M}_{b,a})^\dagger. \quad (16.11)$$

Proof. [✓ Proof](#) This follows entrywise from the exchange of the ket and bra indices and conjugate transposition of the virtual matrix. $\square$

Definition 16.4.6 (Vertical canonical form for an MPO). [Lean](#) [✓ Stmt](#) An MPO tensor is in *vertical canonical form* if there are a basis of normal tensors $\{M_\alpha\}_\alpha$ for its vertically viewed tensor $\widetilde{M}$, positive diagonal matrices $\mu_\alpha = \text{diag}(\omega_{\alpha,0}, \dots, \omega_{\alpha,r_\alpha-1})$ with $r_\alpha \geq 1$, and a matrix U from the physical space onto the retained nonzero sector space such that

$$\begin{aligned} U \widetilde{M} U^\dagger &= B, \\ \widetilde{M} &= U^\dagger B U, \\ B &= \bigoplus_\alpha \mu_\alpha \otimes M_\alpha. \end{aligned} \quad (16.12)$$

This is the conclusion of [11, Proposition 4.13, lines 1863–1870]. In this orientation $U U^\dagger = I$ on the retained space, while $U^\dagger U$ is its support projection in the original physical space. The reconstruction identity says that every letter of $\widetilde{M}$ is supported on this projection, so the discarded orthogonal complement and the off-diagonal corners vanish. The orthogonal complement may be a zero sector: the source’s general canonical-form construction explicitly permits the sum of the nonzero block dimensions to be strictly smaller than the original dimension [11, lines 214–225]. Since each μ_α is diagonal, the summand $\mu_\alpha \otimes M_\alpha$ is the direct sum

$\bigoplus_{k=0}^{r_\alpha-1} \omega_{\alpha,k} M_\alpha$ of weighted copies of M_α , so the right-hand side is a block-diagonal tensor over the pairs (α, k) in which the multiplicities r_α and the diagonal entries $\omega_{\alpha,k}$ stay explicit; the coefficients $m_\alpha = \text{tr}(\mu_\alpha) = \sum_{k=0}^{r_\alpha-1} \omega_{\alpha,k}$ appearing in the renormalization fixed-point characterization are recovered from them.

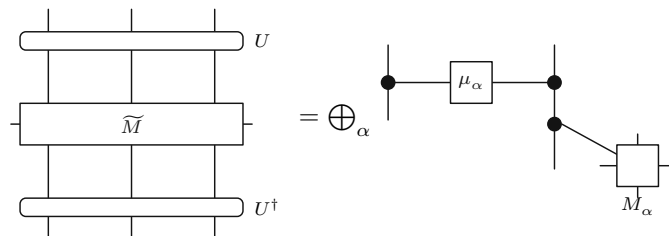


Figure 16.2: Vertical direct-sum form of the tensor: after the physical isometry U , each block factorizes as the positive diagonal weight μ_α on the multiplicity factor times the basis tensor M_α , and only M_α carries the virtual legs. This is the decomposition $U\widetilde{M}U^\dagger = \bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ of [11, Proposition 4.13, lines 1864–1870].

Remark 16.4.7 (Finite-word span and trace separation). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [✓ Stmt](#) Let $(A_k)_k$ be a finite family of block tensors of physical dimension d , and fix a length L ; write A_k^w for the length- L word evaluation of A_k along a word $w \in \{0, \dots, d-1\}^L$. The finite-length spanning hypothesis is that the tuple-valued evaluations span the full product algebra of block matrices,

$$\text{span}_{\mathbb{C}} \{ (A_k^w)_k : w \in \{0, \dots, d-1\}^L \} = \prod_k M_{D_k}(\mathbb{C}). \quad (16.13)$$

The corresponding finite-length separation property is that some length L makes the trace pairing against block words nondegenerate: there is a length L such that, for every tuple $(\Delta_k)_k$ of block matrices,

$$\left(\forall w \in \{0, \dots, d-1\}^L, \sum_k \text{tr}(\Delta_k A_k^w) = 0 \right) \implies \forall k, \Delta_k = 0. \quad (16.14)$$

The spanning and separation conditions are equivalent after existentially quantifying the length. Spanning at length L gives separation at that length by nondegeneracy of the product trace pairing. Conversely, if the span were proper, a nonzero linear functional f would annihilate it. Nondegeneracy of the matrix trace pairing represents f as

$$f((M_k)_k) = \sum_k \text{tr}(\Delta_k M_k),$$

for some tuple $(\Delta_k)_k$, contradicting separation. This finite-word property generalizes the one-letter simultaneous span in the source definition of block-injective canonical form to length- L words.

The spanning hypothesis has an equivalent entrywise form. For a block index k and matrix entry (a, b) , let $w \mapsto (A_k^w)_{a,b}$ be the corresponding scalar word family, indexed jointly by the triple (k, a, b) . Linear independence of this scalar family implies the product-algebra spanning condition, and hence trace separation.

The block-injectivity proposition of [11, lines 340–345] gives a concrete sufficient criterion for the spanning condition: a common length L with every block L -block-injective, together with a

length S at which, for every k , there is a coefficient family $(c_w)_{w \in \{0, \dots, d-1\}^S}$ with

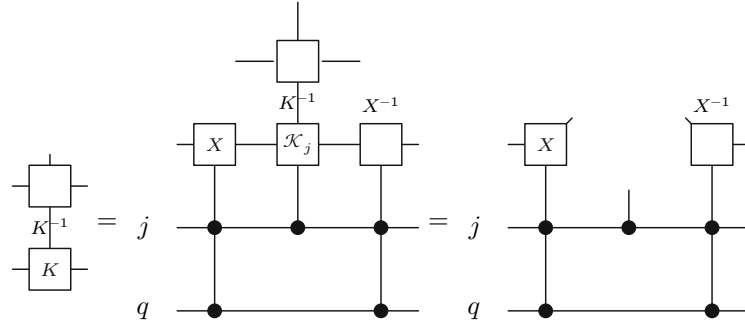
$$\begin{aligned} \sum_w c_w A_k^w &= \mathbb{1}, \\ \sum_w c_w A_j^w &= 0 \quad (j \neq k). \end{aligned}$$

Concatenating the injective prefix with this finite selector family produces the spanning condition at length $L + S$, hence the product-algebra spanning and trace-separation properties.

The inverse tensor of a block-injective canonical form satisfies

$$\sum_p (\mathcal{K}_j^p)_{\alpha, \beta} (K_{j'}^{-1, p})_{\alpha', \beta'} = \delta_{\alpha, \alpha'} \delta_{\beta, \beta'} \delta_{j, j'}. \quad (16.15)$$

Writing $K = X(\bigoplus_j \mathcal{K}_j)X^{-1}$ in this contraction places the gauges on the virtual legs of K and identifies the block index of $\mathcal{K}_j$ with the label j :



Consequently, $K^{-1}K = \bigoplus_j \mathbb{1}_j$. This is the inverse identity in [11, Appendix C.2, lines 1666–1670].

Theorem 16.4.8 (Finite-word trace-separation criterion). [Lean](#) [✓ Stmt](#) *If a block family has a common injective word length and finite words that select each block from the others, then the family has finite-word trace separation.*

Proof. [✓ Proof](#) If the common injective prefix has length L and the selector has length S , concatenation gives simultaneous words of length $L + S$. For every tuple $(X_j)_j$, the resulting coefficients satisfy $\sum_{|w|=L+S} c_w A_j^w = X_j$. Hence the simultaneous word tuples span the product algebra, and nondegeneracy of the trace pairing gives finite-length trace separation. $\square$

Lemma 16.4.9 (Single-block word-tuple span). [Lean](#) [✓ Stmt](#) *Let $(A_j)_{j=0}^{g-1}$ be a family with $g = 1$. If its unique block is L -block injective, then the simultaneous length- L word evaluations span the full one-block product algebra.*

Proof. [✓ Proof](#) For every matrix X , block injectivity gives coefficients such that $X = \sum_{|w|=L} c_w A_0^w$. Since the family has only the block A_0 , this is precisely the required simultaneous word-tuple spanning identity. $\square$

Definition 16.4.10 (Representative word-tuple separation). [Lean](#) [Lean](#) [✓ Stmt](#) Let $(A_j)_{j=0}^{g-1}$ be the basis representatives of a sector decomposition. They have *representative word-tuple separation at length L* when

$$\text{span}_{\mathbb{C}} \{(A_j^w)_{j=0}^{g-1} : w \in \{0, \dots, d-1\}^L\} = \prod_{j=0}^{g-1} M_{D_j}(\mathbb{C}). \quad (16.16)$$

They have *eventual representative word-tuple separation* when this equality holds at every sufficiently large length. Notice that the family is indexed by the minimal representatives j , and not by the repeated copies lying over each j .

Theorem 16.4.11 (Representative-grouped Lemma L). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let A_{tot} be the tensor of a sector decomposition with basis representatives $(A_j)_{j=0}^{g-1}$, multiplicities $r_j > 0$, nonzero weights $\mu_{j,q}$, and grouped coefficients $c_j^{(N)} = \sum_{q=0}^{r_j-1} \mu_{j,q}^N$. Suppose that the first-site actions of $Y, Z \in M_d(\mathbb{C})$ agree on the MPV family of A_{tot} . If representative word-tuple separation holds at length L and $c_{j_0}^{(L+1)} \neq 0$, then $YA_{j_0} = ZA_{j_0}$. Consequently, under eventual representative word-tuple separation, $YA_j = ZA_j$ for every representative j . The same conclusion holds if the first-site equality is initially given for any tensor with the same positive-length MPV family as A_{tot} .

The representative grouping is the argument of [11, Appendix C.3, lines 1835–1858].

In the first auxiliary assertion, eventual representative word-tuple separation is an explicit assumption. The later post-blocked theorem derives the needed finite representative separation from the full BNT canonical-form hypotheses. The sharp $3D^5$ representative-separation bound is Theorem 8.6.1.24; its detailed source comparison is recorded in [130].

Proof. [Proof](#) At a fixed tail length L , expand the first-site equality at system size $L+1$ and group all copies lying over the same representative. For every length- L word w this gives

$$\sum_{j=0}^{g-1} \text{tr}(c_j^{(L+1)}(YA_j - ZA_j)A_j^w) = 0.$$

Representative word-tuple separation therefore forces each matrix $c_j^{(L+1)}(YA_j - ZA_j)$ to vanish. A nonzero coefficient may then be cancelled.

For the eventual assertion, choose a threshold L_0 and put $M = L_0 + 1$. For a fixed representative j , apply Corollary 8.5.2 to the nonzero numbers $\mu_{j,q}^M$. It yields $r \geq 1$ such that $c_j^{(Mr)} = \sum_q (\mu_{j,q}^M)^r \neq 0$. Since $Mr - 1 \geq L_0$, the fixed-length argument applies with $L = Mr - 1$. Finally, equality of the positive-length MPV families gives the first-site hypothesis for the sector-decomposition tensor. $\square$

Definition 16.4.12 (Direct sum with one marked letter). [Lean](#) [Stmnt](#) Let

$$A_{\text{tot}}^i = \bigoplus_{j=0}^{g-1} \bigoplus_{q=0}^{r_j-1} \mu_{j,q} A_j^i, \quad (16.17)$$

be the tensor of a sector decomposition. Let s range over a second finite alphabet, and for each representative A_j let C_j^s be a matrix on its bond space. The corresponding direct sum with one marked letter is

$$C_{\text{tot}}^s = \bigoplus_{j=0}^{g-1} \bigoplus_{q=0}^{r_j-1} \mu_{j,q} C_j^s. \quad (16.18)$$

The marked alphabet need not coincide with the alphabet of the unmarked tensor. This is the algebraic form of the marked matrices in the two diagrams of [11, Proposition 4.13, lines 1909–1919].

Lemma 16.4.13 (Trace expansion with one marked letter). [Lean](#) [✓ Stmt](#) For every marked index s and every unmarked word w ,

$$\begin{aligned} \mathrm{tr}(C_{\mathrm{tot}}^s A_{\mathrm{tot}}^w) &= \sum_{j=0}^{g-1} c_j^{(|w|+1)} \mathrm{tr}(C_j^s A_j^w), \\ c_j^{(N)} &= \sum_{q=0}^{r_j-1} \mu_{j,q}^N. \end{aligned} \quad (16.19)$$

The exponent $|w| + 1$ includes the weight at the marked site.

Proof. [✓ Proof](#) Expand both matrices as direct sums. Their product is the direct sum of $\mu_{j,q}^{(|w|+1)} C_j^s A_j^w$. Taking the trace and summing first over q gives the stated coefficient $c_j^{(|w|+1)}$. $\square$

Theorem 16.4.14 (Algebraic marked-letter extension of representative separation). [Lean](#) [Lean](#) [✓ Stmt](#) Let (C_j^s) and (E_j^s) be two marked families over the same sector decomposition. Fix a tail length L , and suppose that, for every marked index s and every unmarked word w of length L ,

$$\mathrm{tr}(C_{\mathrm{tot}}^s A_{\mathrm{tot}}^w) = \mathrm{tr}(E_{\mathrm{tot}}^s A_{\mathrm{tot}}^w).$$

If representative word-tuple separation holds at length L and $c_{j_0}^{(L+1)} \neq 0$, then $C_{j_0} = E_{j_0}$. Under eventual representative word-tuple separation, agreement for every marked index and every positive-length unmarked word implies $C_j = E_j$ for every j .

This extends the argument of [11, Appendix C.3, lines 1848–1858] from first-site physical marks to general marked matrices in the marked slot.

Proof. [✓ Proof](#) Fix s and a tail length L , and put $\Delta_j = c_j^{(L+1)}(C_j^s - E_j^s)$. Lemma 16.4.13 gives $\sum_{j=0}^{g-1} \mathrm{tr}(\Delta_j A_j^w) = 0$ for every length- L word w . Representative word-tuple separation implies $\Delta_j = 0$ for every j . If $c_{j_0}^{(L+1)} \neq 0$, cancellation gives $C_{j_0}^s = E_{j_0}^s$. Since s was arbitrary, $C_{j_0} = E_{j_0}$.

Under eventual separation, apply Corollary 8.5.2 to the nonzero weights above each fixed representative, exactly as in Theorem 16.4.11. This selects a sufficiently large L with $c_j^{(L+1)} \neq 0$ and proves $C_j = E_j$. $\square$

Theorem 16.4.15 (Full-coordinate marked trace identity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let T_{grp} be the full grouped tensor of Theorem 8.6.1.9, and let P_{rep} be its representative sector decomposition. For a marked family $(C_j^s)_j$, define the marked tensor $\widehat{C}_{\mathrm{grp}}$ on all grouped listed coordinates as follows. The copy (j, q) carries the marked block $\lambda_{j,q} C_j^s$, where $\lambda_{j,q} = \nu_{k(j,q)} \zeta_{k(j,q)}$.

For every marked letter s and every unmarked word w , one has

$$\begin{aligned} \mathrm{tr}(\widehat{C}_{\mathrm{grp}}^s T_{\mathrm{grp}}^w) &= \mathrm{tr}(C_{\mathrm{rep}}^s P_{\mathrm{rep}}^w) \\ &= \sum_j \left(\sum_q \lambda_{j,q}^{|w|+1} \right) \mathrm{tr}(C_j^s C_j^w). \end{aligned} \quad (16.20)$$

The exponent $|w| + 1$ includes the factor at the marked site.

Proof. [✓ Proof](#) Reindex the finite sum over listed coordinates by the class-copy pairs (j, q) and use the bond-dimension identifications to replace each block by its unit-phase representative. The marked block supplies one factor $\lambda_{j,q}$ and the word supplies $|w|$ further factors. Summing first over q gives (16.20). $\square$

Theorem 16.4.16 (Retained-coordinate representative-grouped Lemma L). [Lean](#) [Lean](#) [✓ Stmt](#)
 Let $(C_j^s)_j$ and $(E_j^s)_j$ be two marked families on the representatives. Form their full-coordinate marked tensors. If

$$\text{tr}(\widehat{C}_{\text{grp}}^s T_{\text{grp}}^w) = \text{tr}(\widehat{E}_{\text{grp}}^s T_{\text{grp}}^w) \quad (16.21)$$

for every s and every word w of positive length, then $C_j = E_j$ for every representative j .

In particular, let $Y, Z \in M_d(\mathbb{C})$. If their first-site actions agree on the positive-length matrix product vectors of T_{grp} , then

$$(YC_j)^i = (ZC_j)^i \quad (16.22)$$

for every representative j and every physical letter i .

Proof. [✓ Proof](#) Theorem 16.4.15 rewrites (16.21) into marked-trace agreement for P_{rep} . Its representatives have eventual word-tuple separation, and every copy weight is nonzero. Theorem 16.4.14 therefore gives $C_j = E_j$.

First-site agreement for T_{grp} also holds for P_{rep} because their matrix product vectors agree at every positive length. Applying Theorem 16.4.11 gives (16.22), namely

$$(YC_j)^i = (ZC_j)^i. \quad (16.23)$$

□

Theorem 16.4.17 (Original-coordinate marked Lemma L). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) Let A be in literal CPSV canonical form. For canonical-form data and its BNT refinement, write

$$\begin{aligned} A^i &= V^\dagger B^i V, \\ VV^\dagger &= \mathbb{1}. \end{aligned} \quad (16.24)$$

where $B^i = GT_{\text{grp}}^i G^{-1}$ and G is the listed block gauge. If marked matrices satisfy $C = V^\dagger E V$, then for every word w , including the empty word,

$$\text{tr}(CA^w) = \text{tr}(EB^w). \quad (16.25)$$

The inserted marked matrix ensures that the closed chain is nonempty.

For a linear physical mark f , the linear marking of T_{grp} is exactly its full grouped marked tensor. Each copy (j, q) carries $\lambda_{j,q} f(C_j)$, where $\lambda_{j,q} = \nu_{k(j,q)} \zeta_{k(j,q)}$. For two linear marks f and g , put $F_A^u = \sum_i f_{ui} A^i$ and $G_A^u = \sum_i g_{ui} A^i$. If

$$\text{tr}(F_A^u A^w) = \text{tr}(G_A^u A^w) \quad (16.26)$$

for every marked letter u and every word w of positive length, then $F_A = G_A$ on the original bond space.

This positive-tail formulation corresponds to the argument in [11, Appendix C.3, lines 1835–1858]: first-site agreement at positive system sizes suffices to separate representatives at a nonzero tail length without requiring empty-word data.

Proof. [✓ Proof](#) Induction on w in (16.24) gives $CA^w = V^\dagger EB^w V$. Cyclicity of trace and $VV^\dagger = \mathbb{1}$ give (16.25). Linearity of the physical mark gives the grouped marked tensor.

Apply (16.25) to both sides of (16.26), remove the block gauge by cyclicity of trace, and use Theorem 16.4.16 on the representatives. The resulting equality of the full grouped marked tensors, followed by (16.24), gives $F_A = G_A$. □

Definition 16.4.18 (First-site action on a physical block). [Lean](#) [✓ Stmt](#) Let $Y \in M_d(\mathbb{C})$, and write physical indices of an $(L+1)$ -site block as $I = (i_0, \dots, i_L)$ and $J = (j_0, \dots, j_L)$. The induced matrix on the blocked physical space is

$$Y_{I,J}^{[L+1]} = Y_{i_0,j_0} \prod_{k=1}^L \delta_{i_k,j_k}. \quad (16.27)$$

Lemma 16.4.19 (First-site actions under physical blocking). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) This identity connects the block-injectivity argument [11, lines 318–345] with Lemma L [11, Appendix C.3, lines 1835–1858].

Let A be a tensor, and let $Y^{[L+1]}$ be the physical-block action induced by $Y \in M_d(\mathbb{C})$. For $I = (i_0, \dots, i_L)$, its insertion into the blocked tensor satisfies

$$(Y^{[L+1]}A^{[L+1]})^I = (YA)^{i_0}A^{i_1} \dots A^{i_L}. \quad (16.28)$$

Hence equality of the first-site actions of Y, Z on the MPV family of A implies equality of the first-block actions of $Y^{[L+1]}, Z^{[L+1]}$ on the MPV family of $A^{[L+1]}$.

The first-site actions of Y, Z agree at every length if and only if

$$\text{tr}((YA)^i A^w) = \text{tr}((ZA)^i A^w), \quad (16.29)$$

for every physical index i and every finite word w .

Conversely, suppose that A is L -block injective. If the two induced insertion tensors on $A^{[L+1]}$ are equal, then $YA = ZA$.

Proof. [✓ Proof](#) For the forward implication, split the first block into its first letter and its remaining L letters. The insertion identity (16.28) reduces the blocked MPV equality to the original first-site equality, with the remaining letters of the first block prepended to the flattened tail.

For the converse, put $\Delta^i = (YA)^i - (ZA)^i$. Equality after blocking gives $\Delta^i A^w = 0$ for every word w of length L . Since the length- L words span $M_D(\mathbb{C})$, the same equality holds with A^w replaced by the identity, and therefore $\Delta^i = 0$ for every physical index i . $\square$

Theorem 16.4.20 (Representative separation after injective blocking). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let P be a BNT canonical form with basis representatives $(A_j)_{j=0}^{g-1}$, and suppose that the representatives have already been blocked so that every A_j is injective. Then representative word-tuple separation holds at every length $L \geq 6(g-1) + 1$. Consequently, if the first-site actions of $Y, Z \in M_d(\mathbb{C})$ agree on the positive-length MPV family of the tensor represented by P , then $YA_j = ZA_j$ for $0 \leq j < g$. The same conclusion holds when the first-site equality is given for any tensor with the same positive-length MPV family as the tensor represented by P .

More generally, let P be a BNT canonical form before physical blocking, let $p > 0$, and suppose that every p -blocked representative $A_j^{[p]}$ is injective. Then the blocked sector decomposition $P^{[p]}$ has representative word-tuple separation at every blocked length $L \geq 6(g-1) + 1$.

Without changing the physical alphabet, a BNT canonical form also has eventual representative word-tuple separation. If $p > 0$ is a common block-injectivity length, then one may take pairwise separating words of length $6p$, form selectors of length $6p(g-1)$, and append homogeneous words of any length at least p .

Finally, the BNT canonical-form hypotheses imply that there is a common $L > 0$ for which every unblocked representative A_j is L -block injective. Indeed, irreducibility and left-canonicity give a finite peripheral period m_j . The self-overlap tends to m_j along multiples of m_j , but by normalization it also tends to 1; hence $m_j = 1$. Thus every representative is normal, and

finiteness gives a common positive block-injectivity length. If the first-site actions of Y, Z agree on the tensor represented by P , then $YA_j = ZA_j$ for $0 \leq j < g$. The same conclusion holds when the first-site equality is given for any tensor with the same positive-length MPV family. This composes the blocking conclusion of [11, lines 318–345] with Appendix C.3, Lemma L [11, lines 1835–1858].

No additional common-length hypothesis is required.

Proof. ✓ Proof Injectivity of A_j implies that its homogeneous words span $M_{D_j}(\mathbb{C})$ at every positive length. Apply the three-block separation argument with source length 1. For each ordered pair of distinct representatives it gives a pairwise block-separating equation at length 6. Multiplying the equations for the $g-1$ representatives different from A_j gives coefficients $c_w^{(j)}$ satisfying

$$\sum_{|w|=6(g-1)} c_w^{(j)} A_k^w = \begin{cases} 1, & k = j, \\ 0, & k \neq j. \end{cases} \quad (16.30)$$

If $L \geq 6(g-1) + 1$, put $m = L - 6(g-1) > 0$. The length- m words of each A_j span $M_{D_j}(\mathbb{C})$. Concatenating these words with the selectors in (16.30) shows that the simultaneous length- L word tuples span $\prod_j M_{D_j}(\mathbb{C})$. This proves eventual representative word-tuple separation. The representative-grouped Lemma L gives the two first-site conclusions.

For the unblocked canonical form, propagate p -block injectivity to the lengths $2p-1$, $2p$, and $6p$. Applying the direct-sum argument to the resulting injective tensors gives pair trace separation at original length $6p$. The bijection between blocked words of length 6 and original words of length $6p$ gives pair trace separation for the p -blocked representatives. The preceding selector argument then applies to $P^{[p]}$.

The same pairwise equations at original length $6p$ give selectors for the unblocked representatives directly. For every $L \geq 6p(g-1) + p$, the remaining prefix has length at least p and hence spans each full representative matrix algebra. Concatenating the prefix with the selectors proves eventual representative word-tuple separation before blocking.

For the final assertion, first consider one representative A_j . Irreducibility and left-canonicity give a positive peripheral period m_j . The periodic self-overlap formula and the normalized self-overlap hypothesis give, along the same subsequence,

$$\begin{aligned} \lim_{k \rightarrow \infty} O_{A_j A_j}(m_j k) &= m_j, \\ \lim_{k \rightarrow \infty} O_{A_j A_j}(m_j k) &= 1. \end{aligned}$$

Hence $m_j = 1$. Primitivity then implies normality. Taking a common multiple of the finitely many normality lengths gives a positive length L at which every representative is block injective. Block $L+1$ sites. Injectivity propagates from length L to length $L+1$, so the blocked representatives are injective. Act on the first letter of each physical block and apply the blocked representative separation followed by the representative-grouped Lemma L. This gives equality of the two induced insertions on every blocked representative. The remaining L letters of the first block form a spanning family, so Lemma 16.4.19 recovers $YA_j = ZA_j$ before blocking. $\square$

Corollary 16.4.21 (Marked-letter separation for a BNT canonical form). Lean ✓ Stmt Let (C_j^s) and (E_j^s) be two marked families over a BNT canonical form. If their closed marked chains agree for every marked index s and every unmarked word w of positive length, then $C_j^s = E_j^s$ for every representative j and every marked index s .

Proof. ✓ Proof The BNT canonical-form hypotheses give eventual representative word-tuple separation. Apply Theorem 16.4.14. $\square$

Corollary 16.4.22 (Physical-letter marked separation in BNT-refined horizontal form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let M be an MPO tensor in normalized BNT-refined horizontal form. For two coefficient families $f_{u,z}$ and $g_{u,z}$, put $C^u = \sum_z f_{u,z} M^z$ and $E^u = \sum_z g_{u,z} M^z$. If $\text{tr}(C^u M^w) = \text{tr}(E^u M^w)$ for every marked index u and every word w of positive length, then $C^u = E^u$ for every u .

Restricting the marks to the span of the physical letters isolates the physical-insertion consequence of [11, Appendix C.3, lines 1835–1858] needed for the contractions in [11, Proposition 4.13, lines 1909–1919].

Proof. [✓ Proof](#) Pass through the block-diagonal similarity in the normalized BNT-refined horizontal form. Linear combinations of physical letters are covariant under this similarity, and cyclicity of the trace removes it from each closed marked chain. On the representative-indexed sector decomposition one has

$$C_{\text{tot}}^u = \bigoplus_{j,q} \mu_{j,q} \sum_z f_{u,z} A_j^z,$$

$$E_{\text{tot}}^u = \bigoplus_{j,q} \mu_{j,q} \sum_z g_{u,z} A_j^z.$$

Corollary 16.4.21 identifies the two marks on every minimal representative. Taking their weighted direct sums and restoring the common block-diagonal similarity gives $C^u = E^u$. $\square$

Theorem 16.4.23 (Representative and literal reduction from first-site contractions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let M be in normalized BNT-refined horizontal form. For doubled-index matrices Y, Z , suppose that $Y_1 V^{(N)}(M) = Z_1 V^{(N)}(M)$ for every $N \geq 1$. Then the corresponding inserted tensors agree on the original bond space, $YM = ZM$.

Now suppose in addition that M generates an MPDO, $Q = Q^\dagger$, and that $Q\tilde{M} = Q\tilde{M}Q$ on the vertically viewed tensor. Then, on every representative, the insertions corresponding to $\tilde{M}Q$ and $Q\tilde{M}Q$ agree: $Y_R(Q)A_j = Y_C(Q)A_j$ for $0 \leq j < g$. This equality extends through every repeated weighted sector and the direct sum of the copy gauges, so that the original MPO tensor satisfies $\tilde{M}Q = Q\tilde{M}Q$. The specialized assertion follows from the two entrywise identities $Q_1 H^{(N)} = Q_1 H^{(N)} Q_1$ and $Q_1 H^{(N)} = H^{(N)} Q_1$ at every positive length.

The corresponding literal-CF assertion also holds. Let A be in literal CPSV canonical form. If the first-site actions of $Y, Z \in M_d(\mathbb{C})$ agree on every positive-length matrix product vector of A , then

$$YA = ZA \tag{16.31}$$

on the original bond space. In particular, suppose that M generates matrix product density operators, $P = P^\dagger$, and $P\tilde{M} = P\tilde{M}P$. Positivity gives agreement of the first-site actions associated with $\tilde{M}P$ and $P\tilde{M}P$ at every positive length. Hence (16.31) gives the opposite one-sided identity on the original bond space.

This is the invariant-projection step of [11, Proposition 4.13, lines 1873–1887], grouping copies over each normal representative as in Lemma L [11, Appendix C.3]; Hermiticity of Q suffices for the implication.

Proof. [✓ Proof](#) For the normalized BNT-refined assertion, positive-length MPV equality gives the first-site identity for the BNT sector decomposition. The representative-grouped Lemma L gives $YA_j = ZA_j$ for every representative. First-site insertion commutes with the weighted direct sum, and conjugating each copy by its gauge gives $YM = ZM$ on the original bond space.

For the specialized assertion, positivity and Hermiticity give $Q_1 H^{(N)} = Q_1 H^{(N)} Q_1 = H^{(N)} Q_1$. These two identities give the corresponding first-site equalities for the BNT sector decomposition. Apply the representative-grouped Lemma L to identify first $Y_L(Q)A_j$ with $Y_C(Q)A_j$ and then with $Y_R(Q)A_j$. Transitivity gives the stated equality for every representative, and the same direct-sum and copy-gauge argument gives the literal equality on $\widetilde{M}$.

For literal CPSV canonical form, first-site agreement is a marked-trace identity on the original tensor. Theorem 16.4.17 needs only the identities with a nonempty unmarked tail; it identifies the two linear marks and gives (16.31). In the invariant-projection case, positivity first yields $P_1 H^{(N)} = P_1 H^{(N)} P_1 = H^{(N)} P_1$; its matrix coefficients are exactly the required first-site agreement. $\square$

Lemma 16.4.24 (Blockwise equality from agreement on the MPV family). Lean ✓ Stmt *Let $\bigoplus_k \mu_k A_k$ be a block-diagonal tensor whose blocks are injective, left-canonical, weighted by nonzero scalars, and jointly block-injective in the sense that for some blocking length L the tuple of trace pairings against length- L block words is faithful across all blocks simultaneously. For $Y \in M_d(\mathbb{C})$, write*

$$(Y A_k)^i := \sum_{j=0}^{d-1} Y_{i,j} A_k^j$$

for the physical-index insertion of Y into block A_k , and for a configuration $\sigma = (i_1, \dots, i_N)$ write

$$(Y_1 V^{(N)}(A))_{(i_1, \dots, i_N)} := \sum_{j=0}^{d-1} Y_{i_1, j} V^{(N)}(A)_{(j, i_2, \dots, i_N)}$$

for Y acting on the first physical index of the MPV. If $Y, Z \in M_d(\mathbb{C})$ satisfy $Y_1 V^{(N)}(A) = Z_1 V^{(N)}(A)$ for every $N \geq 1$, then $Y A_k = Z A_k$ for every block k .

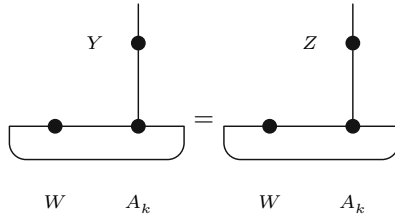
Scope restriction (block-injective specialization): The block-injective hypothesis excludes gauge-equivalent repeated blocks, specializing the source lemma [11, Lemma L]; documented in [130].

Proof. ✓ Proof This is the per-block-separated form of Lemma L of [11, Appendix C.3, lines 1835–1846], using the finite criterion from Theorem 16.4.8 applied directly to the full block family rather than to a minimal basis of representatives. Expanding the hypothesis over a word of length $L + 1$ beginning with the fixed site and folding each block contribution into a trace pairing against the corresponding first-site insertion tensor yields, upon taking the Y - Z difference, a tuple $(\Delta_k)_k$ of block matrices. For every word w of length L ,

$$\sum_k \text{tr}(\mu_k^{L+1} \Delta_k A_k^w) = 0.$$

The finite block-separation hypothesis applied to the tuple $(\mu_k^{L+1} \Delta_k)_k$ forces each weighted block difference to be zero, and the nonzero weights give the blockwise equality of the two first-site insertions. $\square$

The trace-pairing step used above is the following local identity, $\text{tr}(W Y A_k) = \text{tr}(W Z A_k)$:



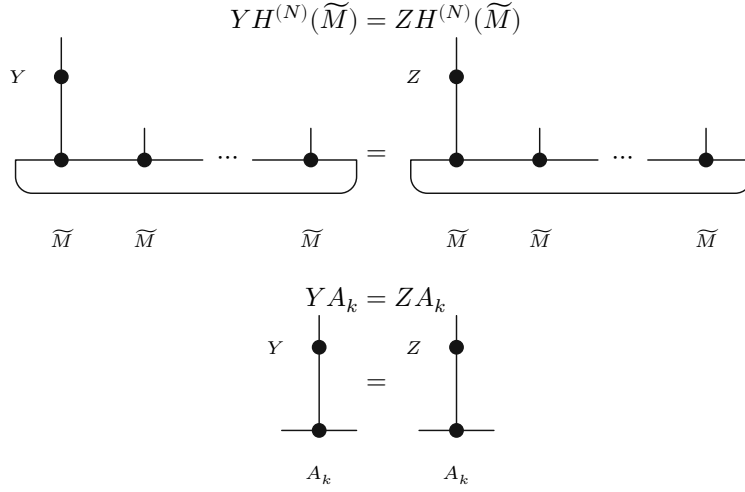


Figure 16.3: Blockwise equality from agreement on the MPV family. Top: the hypothesis $Y_1 V^{(N)}(A) = Z_1 V^{(N)}(A)$ for every positive chain length $N \geq 1$, with the operator inserted on the physical leg of the first site. Bottom: the conclusion that the two insertions agree on every block A_k of the canonical form. These are the two displays of Lemma L, [11, Appendix C.3, lines 1835–1846].

For every W in the span of the length- N word evaluations of the canonical form, it states $\text{tr}(WA^Y) = \text{tr}(WA^Z)$, where A^Y is obtained by applying Y to the physical index. Compare the trace-pairing identities in the proof of Lemma L, [11, Appendix C.3, lines 1848–1858].

Theorem 16.4.25 (One-sided invariant corners of Hermitian operators). [Lean] [Stmt] Let H and P be Hermitian matrices. If $PH = PHP$, then $(\mathbb{1} - P)HP = 0$.

Proof. [Proof] Taking adjoints gives $HP = PHP$, and the conclusion follows by subtraction. $\square$

Theorem 16.4.26 (One-sided invariant corners of positive operators). [Lean] [Stmt] Let H be a nonempty-chain density operator generated by an MPDO, and let $P = P^\dagger$. If $PH = PHP$, then $(\mathbb{1} - P)HP = 0$. This is the operator identity used in the proof of [11, Proposition 4.13].

Proof. [Proof] Since H and P are Hermitian, taking adjoints gives $HP = (PH)^\dagger = (PHP)^\dagger = PHP$. Therefore $(\mathbb{1} - P)HP = HP - PHP = 0$. $\square$

Theorem 16.4.27 (First-site contraction reduction). [Lean] [Stmt] Let $\bigoplus_{j,q} \mu_{j,q} A_j$ be a BNT canonical form and let $Y_L, Y_C, Y_R \in M_d(\mathbb{C})$, with $Y_1 V^{(N)}(A)$ denoting Y acting on the first physical index of the MPV. If, for every $N \geq 1$,

$$\begin{aligned} (Y_L)_1 V^{(N)}(A) &= (Y_C)_1 V^{(N)}(A), \\ (Y_L)_1 V^{(N)}(A) &= (Y_R)_1 V^{(N)}(A), \end{aligned}$$

then $Y_R A_j = Y_C A_j$ for every minimal representative A_j . This application of Lemma L specializes to the physical-index insertions of P , PHP , and HP in Theorem 16.4.31.

Proof. [Proof] Apply the representative-grouped Lemma L first to the actions Y_L, Y_R and then to Y_L, Y_C . The BNT canonical-form hypotheses supply the required common block-injectivity

length. Transitivity gives the stated equality on every minimal representative; the repeated copies have already been combined through $\sum_q \mu_{j,q}^N$. $\square$

Definition 16.4.28 (Doubled-index forms of the three first-site contractions). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let $P \in M_d(\mathbb{C})$ and write doubled indices as $p = (a, b)$ and $q = (c, e)$. The first-site matrices corresponding respectively to $PH^{(N)}$, $H^{(N)}P$, and $PH^{(N)}P$ are

$$\begin{aligned}(Y_L)_{p,q} &= P_{a,c} \delta_{b,e}, \\ (Y_R)_{p,q} &= \delta_{a,c} P_{e,b}, \\ (Y_C)_{p,q} &= P_{a,c} P_{e,b}.\end{aligned}$$

The horizontal operator $H^{(N)}(\widetilde{M})$ is the cyclic word $H^{(N)}(\widetilde{M}) = \text{tr}_v(\widetilde{M}_1 \cdots \widetilde{M}_N)$:

$$H^{(N)}(\widetilde{M}) = \begin{array}{c} \begin{array}{ccccccc} \sigma_1 & \sigma_2 & & \sigma_N \\ | & | & & | \\ \bullet & \bullet & \cdots & \bullet \\ | & | & & | \\ \tau_1 & \tau_2 & & \tau_N \end{array} \\ \hline \widetilde{M}_1 \quad \widetilde{M}_2 \quad \quad \quad \widetilde{M}_N \end{array}$$

Its virtual indices are contracted around the row, while every ket and bra index remains external. Reading each ket–bra pair as one doubled index identifies its entries with the matrix product vector of the doubled-index tensor. This is the contraction in [11, Proposition 4.13, lines 1876–1880].

Lemma 16.4.29 (Coefficient form of the three first-site contractions). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) For ket and bra configurations σ, τ , $V^{(N)}(\widetilde{M})_{(\sigma, \tau)} = H^{(N)}(M)_{\sigma, \tau}$. If $p_0 = (a, b)$ is the first doubled index and σ', τ' are the ket and bra configurations on the remaining N sites, contraction with Y_L , Y_R , and Y_C gives, respectively,

$$\begin{aligned} & \sum_c P_{a,c} H_{(c, \sigma'), (b, \tau')}^{(N+1)}, \\ & \sum_e H_{(a, \sigma'), (e, \tau')}^{(N+1)} P_{e,b}, \\ & \sum_{c,e} P_{a,c} H_{(c, \sigma'), (e, \tau')}^{(N+1)} P_{e,b}. \end{aligned} \tag{16.32}$$

Proof. [✓ Proof](#) Decode the doubled index at every site as a ket–bra pair. The defining trace of the resulting matrix word is exactly the corresponding entry of $H^{(N)}(M)$. Summing the first doubled index against Y_L , Y_R , or Y_C leaves the single or double sums in (16.32). $\square$

Lemma 16.4.30 (Doubled-index insertions are vertical multiplications). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let $\widetilde{M}$ be the doubled-index tensor associated with M . Then insertion of Y_L , Y_R , and Y_C gives, respectively, the doubled-index tensors associated with $P\widetilde{M}$, $\widetilde{M}P$, and $P\widetilde{M}P$.

Proof. [✓ Proof](#) Write the output and summed doubled indices as (a, b) and (c, e) . The Kronecker deltas in Y_L and Y_R remove, respectively, the sums over e and c ; Y_C leaves both multiplications. The resulting coefficients are $\sum_c P_{a,c} M^{cb}$, $\sum_e P_{e,b} M^{ae}$, and $\sum_{c,e} P_{a,c} P_{e,b} M^{ce}$. $\square$

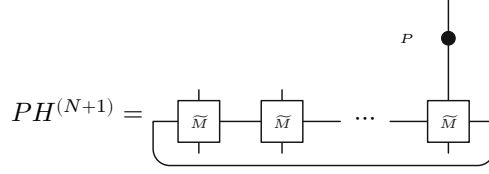


Figure 16.4: The periodic contraction representing $PH^{(N+1)}$. The analogous contractions with Y_R and Y_C place P on the bra leg or on both legs and represent $H^{(N+1)}P$ and $PH^{(N+1)}P$. These are the three first-site contractions in [11, Proposition 4.13, lines 1881–1887].

Theorem 16.4.31 (Representative reduction for the MPO contractions). [Lean](#) [✓ Stmt](#) *Let the doubled-index tensor $\widehat{M}$ generate the same positive-length MPV family as a BNT canonical form $\bigoplus_{j,q} \mu_{j,q} A_j$. If, entrywise at every positive length,*

$$\begin{aligned} PH^{(N)} &= PH^{(N)}P, \\ PH^{(N)} &= H^{(N)}P, \end{aligned}$$

then for every minimal representative A_j the insertions associated with $H^{(N)}P$ and $PH^{(N)}P$ agree. Repeated copies over A_j are grouped through the coefficient $\sum_q \mu_{j,q}^N$.

Proof. [✓ Proof](#) Lemma 16.4.29 turns the two operator identities into equality of the corresponding first-site contractions. Equality of matrix product vectors gives these equalities for the chosen canonical-form decomposition. Apply Theorem 16.4.27 with (Y_L, Y_C, Y_R) . $\square$

Definition 16.4.32 (One-site actions of a matrix on the vertically viewed tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) View the MPO tensor M vertically as the family of physical-space operators M_{ab} of Definition 16.4.4, indexed by the virtual indices, with matrix elements $(M_{ab})_{ij} = M_{ab}^{ij}$. A matrix $P \in M_d(\mathbb{C})$ then acts on the tensor by

$$\begin{aligned} (PM)^{ij} &= \sum_k P_{ik} M^{kj}, \\ (MP)^{ij} &= \sum_k P_{kj} M^{ik}, \end{aligned}$$

and on an $(N+1)$ -site chain by the first-spin operator $P \otimes \mathbb{1}^{\otimes N}$. On the letters of the vertically viewed tensor these actions are the left and right matrix products, $(PM)_{ab} = PM_{ab}$ and $(MP)_{ab} = M_{ab}P$. The first-spin operators compose site by site, $P_1Q_1 = (PQ)_1$.

Theorem 16.4.33 (One-sided invariance and density-operator commutation). [Lean](#) [Lean](#) [✓ Stmt](#) *Let M generate an MPDO and let $P = P^\dagger$ satisfy $PM = PMP$ on the vertically viewed tensor. For every total length $L \geq 1$, write $P_1 = P \otimes \mathbb{1}^{\otimes(L-1)}$. Then $P_1H^{(L)} = P_1H^{(L)}P_1 = H^{(L)}P_1$. This is the equality chain in the proof of [11, Proposition 4.13]; the proposition assumes P is an orthogonal projector, but only self-adjointness of P is used.*

Proof. [✓ Proof](#) Put $L = N+1$. Splitting off the first site of a density-operator entry gives

$$H_{(i,\sigma),(j,\tau)}^{(N+1)} = \text{tr}(M^{ij}M^{\sigma_0\tau_0} \dots M^{\sigma_{N-1}\tau_{N-1}}).$$

Summing the first ket index against P therefore pairs the operator $(PM)^{ij}$ against the remaining word; the hypothesis $PM = PMP$ replaces it by $(PMP)^{ij}$, which unfolds to the entry of

$P_1 H^{(N+1)} P_1$. This proves $P_1 H^{(N+1)} = P_1 H^{(N+1)} P_1$. Since $H^{(N+1)}$ is positive semidefinite and P is self-adjoint, Theorem 16.4.26 gives $(\mathbb{1} - P_1) H^{(N+1)} P_1 = 0$, hence $H^{(N+1)} P_1 = P_1 H^{(N+1)} P_1 = P_1 H^{(N+1)}$. $\square$

Theorem 16.4.34 (One-sided invariance on BNT representatives). [Lean](#) [✓ Stmt](#) *Let M generate an MPDO whose doubled-index tensor $\widehat{M}$ generates the same positive-length MPV family as a BNT canonical form $\bigoplus_{j,q} \mu_{j,q} A_j$, and let $P = P^\dagger$ satisfy $PM = PMP$ on the vertically viewed tensor. Then on every minimal representative A_j the insertions of MP and PMP agree; equivalently, $(\mathbb{1} - P)MP = 0$ on every representative.*

Proof. [✓ Proof](#) For every tail length N , Theorem 16.4.33 turns $PM = PMP$ into the entrywise identities $P_1 H^{(N+1)} = P_1 H^{(N+1)} P_1$ and $P_1 H^{(N+1)} = H^{(N+1)} P_1$. These are exactly the hypotheses of Theorem 16.4.31, whose conclusion is the equality of the insertions of MP and PMP on every minimal representative. $\square$

Theorem 16.4.35 (Commutation with a power of a positive semidefinite matrix). [Lean](#) [✓ Stmt](#) *Let $H \geq 0$, let $p \geq 1$, and let $QH^p = H^p Q$. Then $QH = HQ$.*

Proof. [✓ Proof](#) Diagonalize H . The function $x \mapsto x^p$ is injective on the non-negative reals, so a matrix commutes with the diagonal matrix of the p -th powers exactly when it commutes with the diagonal matrix itself. $\square$

Theorem 16.4.36 (Commutation with a power of a positive semidefinite matrix). [Lean](#) [✓ Stmt](#) *A matrix commuting with a nonzero power of a nonempty-chain density operator generated by an MPDO commutes with the density operator itself. This is the final operator implication in the periodic-sector contradiction in the proof of [11, Proposition 4.13].*

Proof. [✓ Proof](#) Diagonalize $H = U\Lambda U^\dagger$ with U unitary and Λ the diagonal matrix of eigenvalues $\lambda_i \geq 0$. Conjugating the hypothesis by U gives, entrywise for $Q' = U^\dagger Q U$,

$$Q'_{ij} \lambda_j^p = \lambda_i^p Q'_{ij}.$$

Since $x \mapsto x^p$ is injective on the non-negative reals, every entry Q'_{ij} with $\lambda_i \neq \lambda_j$ vanishes, so $Q'\Lambda = \Lambda Q'$. Conjugating back gives $QH = HQ$. $\square$

Definition 16.4.37 (Periodic-sector projector). [Lean](#) [✓ Stmt](#) *Let M be an MPO tensor. A periodic-sector projector for M with exponent $p \geq 1$ is an orthogonal projector Q on the physical space such that for every $N \geq 0$, with $Q_1 = Q \otimes \mathbb{1}^{\otimes N}$ acting on the first site,*

$$Q_1 H^{(N+1)} \neq H^{(N+1)} Q_1, \tag{16.33}$$

$$Q_1 [H^{(N+1)}]^p = [H^{(N+1)}]^p Q_1. \tag{16.34}$$

These are the commutation identities that the proof of [11, Proposition 4.13], lines 1889–1891, attaches to a nontrivial vertical period; the second family is the displayed identity at lines 1890–1891.

Theorem 16.4.38 (No periodic-sector projector for a matrix product density operator). [Lean](#) [✓ Stmt](#) *Let M generate matrix product density operators. Then M admits no periodic-sector projector.*

Proof. [✓ Proof](#) Suppose Q were such a projector. Every $H^{(N+1)}$ is positive semidefinite, so commutation of Q_1 with $[H^{(N+1)}]^p$ implies commutation with $H^{(N+1)}$ (Theorem 16.4.36), contradicting $Q_1 H^{(N+1)} \neq H^{(N+1)} Q_1$. This is the contradiction concluding the periodic-sector step in the proof of [11, Proposition 4.13], lines 1892–1893. $\square$

Definition 16.4.39 (Periodic vectors supply projectors). [Lean](#) [✓ Stmt](#) Say that *periodic vectors of M supply projectors* if for every nontrivial p -periodic vector of the vertically viewed tensor $\widetilde{M}$ — an irreducible restriction B of $\widetilde{M}$ to a minimal invariant subspace whose transfer map has a peripheral eigenvalue different from its spectral radius, as in Definition 7.7.4 — there is an exponent $q \geq 1$ for which M admits a periodic-sector projector. The proof of [11, Proposition 4.13], lines 1889–1891, asserts this implication from the general theory of the peripheral spectrum of completely positive maps, with q equal to the period p of the vector; only the existence of some exponent enters the exclusion argument. The projector, its word invariance, and its single-letter displacement are constructed from the cyclic decomposition of the peripheral spectrum (Theorem 16.4.47); the single non-commutation instance suffices for the periodic-sector contradiction below.

Theorem 16.4.40 (No periodic vectors in the vertical direction). [Lean](#) [✓ Stmt](#) Let M generate matrix product density operators and suppose that periodic vectors of M supply projectors. Then the vertically viewed tensor $\widetilde{M}$ has no nontrivial p -periodic vectors. This is the periodic-sector step in the proof of [11, Proposition 4.13], lines 1888–1893, granted the projector that the source draws from the general theory of the peripheral spectrum.

Proof. [✓ Proof](#) A nontrivial p -periodic vector of $\widetilde{M}$ supplies a periodic-sector projector for M , and Theorem 16.4.38 shows that no such projector exists. $\square$

Definition 16.4.41 (Vertically stacked layers). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Placing one MPO layer on top of another and contracting the shared physical index gives the *layer product* of two MPO tensors M and M' with the same physical dimension, $(MM')^{ij} = \sum_k M^{ik} \otimes M'^{kj}$, an MPO tensor on the tensor product of the virtual spaces. The p -fold vertical stack $M^{[p]}$ of M is the iterated layer product of p copies of M ; the empty stack $M^{[0]}$ is the trivial-bond tensor with letters δ_{ij} , which generates the identity operator at every system size.

Theorem 16.4.42 (Stacked layers generate powers of the density operator). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every system size N , the operator generated by a layer product is the product of the density operators generated by the layers, $H^{(N)}(MM') = H^{(N)}(M) H^{(N)}(M')$. Consequently the p -fold vertical stack generates $H^{(N)}(M^{[p]}) = [H^{(N)}(M)]^p$ for all N and p ; this is the reading of the powers in the displayed identity of [11, Proposition 4.13, lines 1890–1891] as density operators of stacked layers. In particular, the vertical stack of a tensor generating matrix product density operators again generates matrix product density operators on every nonempty chain.

Proof. [✓ Proof](#) The word product of layer-product letters along a ket–bra configuration expands, by induction on the length and the mixed-product property of the Kronecker product, as a sum over configurations κ of the contracted physical index of Kronecker products of the layers' word products. Taking traces, each summand factors into the product of the two traces, so the entry of $H^{(N)}(MM')$ at (σ, τ) equals $\sum_{\kappa} H^{(N)}(M)_{\sigma\kappa} H^{(N)}(M')_{\kappa\tau}$, the entry of the matrix product. The power identity follows by induction on p , and positive semidefiniteness of $[H^{(N)}]^p$ follows from that of $H^{(N)}$. $\square$

Theorem 16.4.43 (Vertical letters of the stacked tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The vertically viewed letters of a layer product are the products of the layers' vertically viewed letters, $(MM')_{(a,a'),(b,b')} = M_{ab} M'_{a'b'}$. Hence every vertically viewed letter of the p -fold stack $M^{[p]}$ is the evaluation of a length- p word of $\widetilde{M}$. In particular, if a matrix Q satisfies the one-sided invariance $QA_w = QA_w Q$ for every product A_w of p letters of $\widetilde{M}$, then $QM^{[p]} = QM^{[p]}Q$ in the sense of Definition 16.4.32.

Proof. ✓ Proof The letter identity is a direct computation from the layer product: contracting the shared physical index composes the physical-space operators. Induction on p writes each vertical letter of $M^{[p]}$ as a length- p word. Thus, for every bond index v , there is a word w of length p such that $\widetilde{M}^{[p]}_v = A_w$, and $QA_w = QA_wQ$ implies $Q\widetilde{M}^{[p]}_v = Q\widetilde{M}^{[p]}_vQ$. Applying this identity to the two one-site actions gives $QM^{[p]} = QM^{[p]}Q$. $\square$

Definition 16.4.44 (Periodic-sector projector from cyclic data). Lean ✓ Stmt Let M generate matrix product density operators, let $p \geq 1$, and let Q be an orthogonal projector on the physical space satisfying $QA_w = QA_wQ$ for every product A_w of p letters of $\widetilde{M}$ and $Q_1H^{(N)} \neq H^{(N)}Q_1$ for every length. Then Q is a periodic-sector projector for M with exponent p .

Proof. ✓ Proof By Theorem 16.4.43 the word invariance gives $QM^{[p]} = QM^{[p]}Q$ on the stacked tensor, which generates matrix product density operators by Theorem 16.4.42. Theorem 16.4.33 applied to $M^{[p]}$ yields $Q_1H^{(N)}(M^{[p]}) = H^{(N)}(M^{[p]})Q_1$ for every length, and $H^{(N)}(M^{[p]}) = [H^{(N)}]^p$ by the stacked-layers identity. Together with the assumed non-commutation family this is exactly the periodic-sector projector of Definition 16.4.37. $\square$

Definition 16.4.45 (Periodic vectors supply cyclic projectors). Lean ✓ Stmt Say that *periodic vectors of M supply cyclic projectors* if for every nontrivial p -periodic vector of the vertically viewed tensor $\widetilde{M}$ — presented as in Definition 7.7.4 — there are an exponent $q \geq 1$ and an orthogonal projector Q on the physical space satisfying the one-sided invariance $QA_w = QA_wQ$ for every product A_w of q letters of $\widetilde{M}$, together with $Q_1H^{(N)} \neq H^{(N)}Q_1$ for every length. This is the shape of the cyclic decomposition of the peripheral spectrum [212, Theorem 6.6] applied to the vertical transfer map: each cyclic projection of a period- p sector returns to itself after p applications of the map, while its nontrivial cyclic displacement under one application prevents commutation with $H^{(N)}$. Compared with Definition 16.4.39, the commutation of Q with $[H^{(N)}]^p$ is no longer assumed; it is derived through the stacked-layers identity.

Theorem 16.4.46 (Reduction of the periodic-sector step to the cyclic decomposition). Lean ✓ Stmt Let M generate matrix product density operators. If periodic vectors of M supply cyclic projectors, then they supply periodic-sector projectors. This reduces the projector-supply hypothesis of the periodic-sector step in the proof of [11, Proposition 4.13, lines 1888–1893] to the cyclic decomposition of the peripheral spectrum of the vertical transfer map.

Proof. ✓ Proof A nontrivial p -periodic vector supplies an invariant non-commuting projector, which is a periodic-sector projector by Definition 16.4.44. $\square$

Theorem 16.4.47 (The displaced invariant projector of a nontrivial periodic vector). Lean ✓ Stmt Let M be an MPO tensor and let $\widetilde{M}$ carry a nontrivial p -periodic vector, presented as in Definition 7.7.4: an irreducible restriction B of $\widetilde{M}$ to a minimal invariant subspace, reached through an isometry V , with a positive definite transfer eigenvector ρ of eigenvalue $r > 0$ and a transfer eigenvalue $\mu \neq r$ of modulus r . Then there are an exponent $q \geq 1$ and an orthogonal projector Q on the physical space such that $QA_w = QA_wQ$ for every product A_w of q letters of $\widetilde{M}$, while $Q\widetilde{M} \neq Q\widetilde{M}Q$ in the sense of Definition 16.4.32. This realizes the projector that the proof of [11, Proposition 4.13, lines 1889–1891] draws from the general theory of the peripheral spectrum.

Proof. ✓ Proof The rescaled unital gauge K of the corner satisfies the hypotheses of the cyclic decomposition of the peripheral spectrum (Theorem D.3.5), and the eigenvalue $\mu/r \neq 1$ of modulus one (Lemma D.3.6) forces the peripheral cyclic group to have order $m \geq 2$. Theorem [182, “Cyclic decomposition for irreducible finite Kraus maps”] produces orthogonal projections $P_0, \dots, P_{m-1}$

summing to the identity with $\mathcal{E}_K(P_{k+1}) = P_k$, and the letter-level shift $K_v P_{k+1} = P_k K_v$ holds (Lemma 5.8.1.1). Set $q = m$ and let Q be the complement of the support projection π of $YY^\dagger$ for $Y = V\rho^{1/2}P_0$. A word A_w of m letters of $\widetilde{M}$ satisfies $A_w Y = Y G_w$, where G_w is a scalar multiple of the corresponding word in the letters of K , which commutes with P_0 by the full-period shift; so the range of Y is invariant and Lemma D.3.3 gives $Q A_w = Q A_w Q$. If a single letter also satisfied the invariance, then π would fix $A_v Y$, whose columns lie in the sector $V\rho^{1/2}P_{m-1}$ by the cyclic shift; cancelling the injective factor $V\rho^{1/2}$ and using orthogonality of P_{m-1} and P_0 would force $P_{m-1}K_v = 0$ for every letter, hence $P_{m-1} = \mathcal{E}_K(P_0) = 0$, contradicting the nonvanishing of the cyclic projections. $\square$

Theorem 16.4.48 (A displaced idempotent fails to commute at some length). [Lean](#) [Stmt](#) *Let M be an MPO tensor whose doubled-index MPS tensor is in literal CPSV canonical form. If $Q^2 = Q$ and $Q\widetilde{M} \neq Q\widetilde{M}Q$, then there is an integer $N \geq 0$ such that, for $Q_1 = Q \otimes \mathbb{1}^{\otimes N}$,*

$$Q_1 H^{(N+1)} \neq H^{(N+1)} Q_1. \quad (16.35)$$

This existential non-commutation statement supplies the single length needed for the contradiction in [11, Proposition 4.13, lines 1888–1893].

Proof. [Proof](#) Suppose instead that $Q_1 H^{(N+1)} = H^{(N+1)} Q_1$ for every $N \geq 0$. Idempotence gives

$$Q_1 H^{(N+1)} = Q_1^2 H^{(N+1)} = Q_1 H^{(N+1)} Q_1.$$

Taking matrix coefficients yields equality of the one-sided first-site contractions. The literal form of Lemma L in Theorem 16.4.23 identifies the two inserted tensors on the original bond space. By Lemma 16.4.30, this equality is $Q\widetilde{M} = Q\widetilde{M}Q$, contradicting the displacement. $\square$

Theorem 16.4.49 (Literal canonical form excludes vertical periodic sectors). [Lean](#) [Stmt](#) *Let M generate matrix product density operators, and suppose that its doubled-index MPS tensor is in literal CPSV canonical form. Then the vertically viewed tensor $\widetilde{M}$ has no nontrivial periodic vectors.*

Proof. [Proof](#) A nontrivial periodic vector supplies an integer $p \geq 1$ and an orthogonal projector Q such that every word A_w of length p in the letters of $\widetilde{M}$ satisfies $Q A_w = Q A_w Q$, while $Q\widetilde{M} \neq Q\widetilde{M}Q$ (Theorem 16.4.47). By Theorem 16.4.48, there is a length $N + 1$ for which

$$Q_1 H^{(N+1)} \neq H^{(N+1)} Q_1.$$

The word invariance and the vertical-stack identity give

$$Q_1 [H^{(N+1)}]^p = [H^{(N+1)}]^p Q_1.$$

Since $H^{(N+1)}$ is positive semidefinite, Theorem 16.4.36 implies $Q_1 H^{(N+1)} = H^{(N+1)} Q_1$, a contradiction. This is the periodic exclusion in [11, Proposition 4.13, lines 1888–1893], with the single noncommuting length supplied by Lemma L. $\square$

Theorem 16.4.50 (BNT-refined horizontal form excludes vertical periodic sectors). [Lean](#) [Stmt](#) *Let M be in normalized BNT-refined horizontal form. If an idempotent Q satisfies $Q\widetilde{M} \neq Q\widetilde{M}Q$, then for some $N \geq 0$, $Q_1 H^{(N+1)} \neq H^{(N+1)} Q_1$. Consequently, if M generates matrix product density operators, then $\widetilde{M}$ has no nontrivial periodic vectors.*

Scope restriction (BNT-refined horizontal form): *The normalized BNT-refined horizontal hypothesis is stronger than the literal CPSV canonical-form hypothesis; documented in [153].*

Proof. [✓ Proof](#) Suppose first that Q_1 commutes with $H^{(N+1)}$ for every N . Since $Q^2 = Q$, this gives $Q_1 H^{(N+1)} = Q_1 H^{(N+1)} Q_1$ at every positive length. Theorem 16.4.23 and Lemma 16.4.30 then imply $Q\tilde{M} = Q\tilde{M}Q$, a contradiction. Thus a displaced idempotent has a noncommuting length $N + 1$.

A nontrivial periodic vector supplies the orthogonal projector Q of Theorem 16.4.47, invariant under every word of some full period $p \geq 1$. At the noncommuting length just obtained, the stacked-layer identity and the invariant-projection calculation give $Q_1[H^{(N+1)}]^p = [H^{(N+1)}]^p Q_1$. Positive semidefiniteness and Theorem 16.4.36 imply that Q_1 commutes with $H^{(N+1)}$, which yields the contradiction in [11, Proposition 4.13, lines 1888–1893] using a single noncommuting length. $\square$

Theorem 16.4.51 (Commutation at chain length two forces letter invariance, for an injective tensor). [Lean](#) [✓ Stmt](#) *Let M be an MPO tensor whose doubled-index tensor is injective, and let Q be an idempotent on the physical space (Hermiticity is not needed). If $Q_1 H^{(2)} = H^{(2)} Q_1$, then $Q\tilde{M} = Q\tilde{M}Q$.*

Proof. [✓ Proof](#) Idempotence of Q upgrades the commutation to the one-sided reduction $Q_1 H^{(2)} = Q_1 H^{(2)} Q_1$. Unfolding this identity against every trailing letter of M gives, for every pair of physical indices a, b , a trace identity pairing $(Q\tilde{M})_{ab}$ against $(Q\tilde{M}Q)_{ab}$ over every letter of the doubled-index tensor. Injectivity's spanning property and nondegeneracy of the trace pairing force the two matrices to agree. $\square$

Theorem 16.4.52 (No periodic vectors for an injective matrix product density operator). [Lean](#) [✓ Stmt](#) *Let M generate matrix product density operators and let M 's doubled-index tensor be injective. Then the vertically viewed tensor $\tilde{M}$ has no nontrivial periodic vectors, unconditionally.*

Proof. [✓ Proof](#) A nontrivial periodic vector supplies the invariant, displaced projector Q of Theorem 16.4.47. Its word invariance transfers to the stacked tensor, giving commutation with $[H^{(2)}]^p$ unconditionally through the invariant-projection step applied to the stack; positive semidefiniteness of the density operators removes the power, and Theorem 16.4.51 turns the resulting commutation with $H^{(2)}$ into the letter-level invariance that the displacement forbids.

Theorem 16.4.50 gives the corresponding result for normalized BNT-refined horizontal form. $\square$

Theorem 16.4.53 (Positive trace of nonzero positive semidefinite matrices). [Lean](#) [✓ Stmt](#) *A nonzero positive semidefinite matrix has positive trace: the trace is a positive real.*

Proof. [✓ Proof](#) The trace of a positive semidefinite matrix is a non-negative real, and it vanishes only if the matrix itself is zero. $\square$

Theorem 16.4.54 (Nonzero sector compressions have positive trace). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $H^{(N)}$, $N \geq 1$, be a density operator generated by an MPDO and let $P = P^\dagger$. Then $PH^{(N)}P \geq 0$, and if $PH^{(N)}P \neq 0$ its trace is a positive real. In particular this holds for the first-site compressions $P_1 H^{(N+1)} P_1$. Applied to the sector projectors $P_{\alpha,k}$ of the vertical decomposition, this is the positivity $\text{tr}(P_{\alpha,k} H^{(N)} P_{\alpha,k}) > 0$ from which the proof of [11, Proposition 4.13] concludes $\mu_{\alpha,k} d_\alpha > 0$; the identification of the trace with $\mu_{\alpha,k} d_\alpha$ is Theorem 16.4.62.*

Proof. [✓ Proof](#) Compression by a Hermitian matrix preserves positive semidefiniteness, since $PH^{(N)}P = PH^{(N)}P^\dagger$. By Theorem 16.4.53, a nonzero compression therefore has positive trace. The first-site case follows because $P_1 = P \otimes \mathbb{1}^{\otimes N}$ is Hermitian whenever P is. $\square$

$(E_\alpha)_{ab} = \text{tr}(M_\alpha^{(a,b)})$, independently of k . Indeed, cyclicity of trace cancels the internal similarity $X_{\alpha,k} M_\alpha^{(a,b)} X_{\alpha,k}^{-1}$. Applied to the physical isometry of a grouped normal sector, the gauge-corner identity constructs the projector and proves its loop identity.

Theorem 16.4.59 (Literal sector-compression separation). [Lean] [✓ Stmt] *Let M be an MPO tensor whose doubled-index MPS tensor is in literal CPSV canonical form, and let $P \in M_d(\mathbb{C})$. If one vertical letter has a nonzero two-sided corner, then some finite-chain first-site compression is nonzero:*

$$(\exists v, P \widetilde{M}_v P \neq 0) \implies (\exists N \geq 0, P_1 H^{(N+1)} P_1 \neq 0), \quad (16.37)$$

where $P_1 = P \otimes \mathbb{1}^{\otimes N}$. This is the separation assertion at [11, Proposition 4.13, line 1898], obtained from Lemma L in [11, Appendix C.3, lines 1835–1858]. It uses neither positivity of $H^{(N+1)}$ nor a grouping of the vertical sectors.

Proof. [✓ Proof] Suppose that $P_1 H^{(N+1)} P_1 = 0$ for every $N \geq 0$. By the coefficient formula for the doubled-index first-site contraction, the two-sided action $Y_C(P)$ and the zero action agree on every positive-length matrix product vector of the doubled-index tensor. Literal Lemma L therefore identifies their inserted tensors. For every vertical letter v , the insertion identity then gives

$$P \widetilde{M}_v P = 0. \quad (16.38)$$

This contradicts the hypothesis in (16.37). □

Theorem 16.4.60 (A gauge corner gives a nonzero range-projector corner). [Lean] [✓ Stmt] *Let M be an MPO tensor of physical dimension d and bond dimension D . Let A be a CPSV normal tensor of positive bond dimension n and physical dimension D^2 , let $V \in \mathbb{C}^{d \times n}$ be an isometry, let $X \in \mathbb{C}^{n \times n}$ be invertible, and let ω be nonzero. Suppose that, for every vertical letter v ,*

$$\omega X A_v X^{-1} = V^\dagger \widetilde{M}_v V. \quad (16.39)$$

Then the range projector $VV^\dagger$ has a nonzero vertical corner:

$$(VV^\dagger) \widetilde{M}_v (VV^\dagger) \neq 0 \quad (16.40)$$

for some v . This is the algebraic nonvanishing step used in [11, Proposition 4.13, line 1898].

Proof. [✓ Proof] Normality supplies a letter v for which $A_v \neq 0$. Since ω is nonzero and X is invertible, the left-hand side of (16.39) is nonzero. If (16.40) vanished, multiplying it by $V^\dagger$ and V and using $V^\dagger V = 1$ would contradict (16.39). □

Theorem 16.4.61 (A nonzero vertical corner has a nonzero sector compression). [Lean] [✓ Stmt] *Let M be in normalized BNT-refined horizontal form and let P be a matrix for which $P \widetilde{M}_v P$ is nonzero for some vertical letter v . Then the first-site compression $P_1 H^{(N+1)} P_1$ is nonzero for some N . This is the separation step used in [11, Proposition 4.13, line 1898].*

The normalized BNT-refined horizontal hypothesis provides the grouped vertical sectors used here; the corresponding literal separation statement is Theorem 16.4.59.

Proof. [✓ Proof] If every compression vanished, its matrix entries would say that the first-site insertion by P has the same matrix product vectors as zero. Lemma L on the normalized BNT-refined horizontal form would then force the inserted tensor itself to vanish, contrary to the assumed nonzero corner. □

Theorem 16.4.62 (The sector trace identity). [Lean](#) [✓ Stmt](#) Let P be a sector projector for the weight μ and the matrix E_α . Then for every chain length $N + 1$,

$$\text{tr}(P_1 H^{(N+1)} P_1) = \mu d_\alpha^{(N+1)}. \quad (16.41)$$

Once spectral normalization and grouping identify the reducing projectors with sector projectors $P_{\alpha,k}$ satisfying Definition 16.4.58, this specializes to the trace identification $\text{tr}(P_{\alpha,k} H^{(N)} P_{\alpha,k}) = \mu_{\alpha,k} d_\alpha$ used in the proof of [11, Proposition 4.13]; the source states it at the particular length where the compression does not vanish.

Proof. [✓ Proof](#) By Theorem 16.4.57 the trace of the compression is $\text{tr}(E_P E^N)$, and the sector-projector property replaces E_P by μE_α . $\square$

Theorem 16.4.63 (Positivity of the vertical weights). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let M generate an MPDO, let Q and P be sector projectors for the weights $\mu_{\alpha,0}$ and $\mu_{\alpha,k}$ and a common matrix E_α , and suppose $\mu_{\alpha,0} > 0$. If the compression of $H^{(N+1)}$ by P at the first site is nonzero, then

$$\begin{aligned} \mu_{\alpha,k} d_\alpha^{(N+1)} &> 0, \\ d_\alpha^{(N+1)} &> 0, \\ \mu_{\alpha,k} &> 0. \end{aligned}$$

Granted such sector projectors, this yields the conclusion of the trace argument in the proof of [11, Proposition 4.13]: since w.l.o.g. $\mu_{\alpha,0} > 0$, the positive trace forces $d_\alpha > 0$ and hence $\mu_{\alpha,k} > 0$.

Proof. [✓ Proof](#) By Theorem 16.4.54 the nonzero compression has positive trace, which Theorem 16.4.62 identifies with $\mu_{\alpha,k} d_\alpha^{(N+1)}$. In particular $d_\alpha^{(N+1)} \neq 0$, so the trace of the compression by the sector projector of $(\alpha, 0)$ at the same length equals $\mu_{\alpha,0} d_\alpha^{(N+1)} \neq 0$; that compression is therefore also nonzero, and its trace is a positive real. Dividing by $\mu_{\alpha,0} > 0$ gives $d_\alpha^{(N+1)} > 0$, and dividing the first inequality by $d_\alpha^{(N+1)}$ gives $\mu_{\alpha,k} > 0$. $\square$

Theorem 16.4.64 (Unitary normalization of a scalar Gram matrix). [Lean](#) [✓ Stmt](#) If $\omega > 0$ and $X^\dagger X = \omega \mathbb{1}$, then $\omega^{-1/2} X$ is unitary.

Proof. [✓ Proof](#) Divide the Gram identity by ω . $\square$

Theorem 16.4.65 (Gram rigidity of normal tensors). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let M_α be a normal tensor. Any matrix commuting with every matrix of M_α is a scalar multiple of the identity. In particular, if $X \neq 0$ and $X^\dagger X$ commutes with every matrix of M_α , then $X^\dagger X = \omega \mathbb{1}$ for a necessarily positive constant ω , and $\omega^{-1/2} X$ is unitary. This is the normalized form $X_{\alpha,k}^\dagger X_{\alpha,k} = \omega_{\alpha,k} \mathbb{1}$ of the Gram-matrix proportionality in the proof of [11, Proposition 4.13], with $X_{\alpha,0} = \mathbb{1}$ and $U_{\alpha,k} = \omega_{\alpha,k}^{-1/2} X_{\alpha,k}$. Deriving the commutation from self-adjointness of the sector compressions is a separate step.

Proof. [✓ Proof](#) Commutation with the matrices of M_α extends to all word evaluations, and after blocking these span the full matrix algebra, so the commuting matrix is central and hence scalar. The Gram matrix $X^\dagger X$ is positive semidefinite and nonzero, so its scalar value is a positive real ω ; dividing X by $\sqrt{\omega}$ makes the Gram matrix the identity. $\square$

Theorem 16.4.66 (Reflected marked-chain identity). Lean Lean Lean Lean ✓ Stmt Let M generate positive semidefinite operators on every nonempty chain, let A be a vertical tensor, and suppose that a matrix V selects the vertical corner $V^\dagger \widetilde{M}_v V = cA^v$, where $c > 0$. For vertical indices r, s , write $(\mathcal{R}(A)^{rs})_{ab} = (A^{(a,b)})_{rs}$, and set $(A^\sharp)^v = (A^{v^{\text{op}}})^\dagger$. Then, for every pair of equal-length tail words σ, τ ,

$$\text{tr}(\mathcal{R}(A)^{rs} M^{\sigma_1 \tau_1} \dots M^{\sigma_L \tau_L}) = \text{tr}(\mathcal{R}(A^\sharp)^{rs} (M^\dagger)^{\sigma_L \tau_L} \dots (M^\dagger)^{\sigma_1 \tau_1}). \quad (16.42)$$

Adjunction exchanges the oriented horizontal bond pair and reverses the unmarked tail, without identifying A with $A^\sharp$ letterwise.

Proof. ✓ Proof Compress the first site of the positive operator by V . The resulting matrix is Hermitian. Expanding one matrix entry gives the marked chain on the left and the complex conjugate of the entry with r, s and the ket–bra words exchanged. Taking the conjugate transpose reverses the tail product, cyclicity of trace returns the marked factor to the first position, and positivity of c permits cancellation without a phase. $\square$

Theorem 16.4.67 (Gram-dressed reflected marked-chain identity). Lean Lean ✓ Stmt Let M generate positive semidefinite operators on every nonempty chain, and suppose that one vertical corner is a positive scalar multiple of a similarity of a tensor A ,

$$V^\dagger \widetilde{M}^v V = c X A^v X^{-1}, \quad c > 0.$$

Then the marked-chain coefficient of the Gram-dressed tensor $(X^\dagger X) A^v (X^\dagger X)^{-1}$ equals the reflected-adjoint marked-chain coefficient of A in the adjoint tensor, with the two tail words reversed. This is the single-corner, open-index conjugation step from Figures 7–8 of [11, Proposition 4.13, lines 1909–1919].

Proof. ✓ Proof Apply the reflected marked-chain identity to the similar tensor $X A^v X^{-1}$. Conjugate its two open virtual indices by $X^\dagger$ and $(X^{-1})^\dagger$. On the original side this gives the Gram dressing of A ; on the reflected side the gauge factors cancel, leaving the reflected adjoint of A . $\square$

Theorem 16.4.68 (Pairwise Gram conjugation of grouped vertical corners). Lean Lean Lean ✓ Stmt Let M be in normalized BNT-refined horizontal form and generate positive semidefinite operators on every nonempty chain. Suppose two vertical corners are positive scalar multiples of similarities of the same tensor A :

$$\begin{aligned} V_X^\dagger \widetilde{M}^v V_X &= c_X X A^v X^{-1}, & c_X &> 0, \\ V_Y^\dagger \widetilde{M}^v V_Y &= c_Y Y A^v Y^{-1}, & c_Y &> 0. \end{aligned}$$

Then the two Gram conjugations agree for every vertical letter,

$$(X^\dagger X) A^v (X^\dagger X)^{-1} = (Y^\dagger Y) A^v (Y^\dagger Y)^{-1}. \quad (16.43)$$

Normality of A is not required for this equality; it enters only in the subsequent proportionality of the Gram matrices. This is the blockwise statement underlying the unnormalized pairwise conclusion of Figures 7–8 in [11, Proposition 4.13, lines 1909–1919].

Scope restriction (BNT-refined horizontal form): The normalized BNT-refined horizontal hypothesis is stronger than the literal CPSV canonical-form hypothesis; documented in [153].

Proof. ✓ Proof Conjugate the open indices in the reflected marked-chain identity for the first corner by $X^\dagger$. Its left side becomes the marked chain of $(X^\dagger X)A^v(X^\dagger X)^{-1}$, while the gauge factors on the reflected side cancel and leave the marked chain of $A^\sharp$. The same calculation with Y gives the identical reflected chain. Thus the two Gram-dressed marks have equal closed chains for every tail word. The reflected identity also holds for the empty tail, but marked separation uses only nonempty tails, as in the proof of Lemma L in [11, Appendix C.3, lines 1848–1858].

These marks lie in the physical-letter span. Indeed, with $L_X = V_X X$ and $R_X = V_X (X^{-1})^\dagger$, their coefficients are $f_{(r,s),(i,j)}^X = c_X^{-1} (L_X)_{ir} (R_X)_{js}$, and similarly for Y . The corner identities show that these linear combinations are precisely the horizontal slices of the two Gram-dressed tensors. Physical-letter marked separation therefore identifies every slice, hence every vertical letter. $\square$

Definition 16.4.69 (Pairwise Gram dressing for grouped corners). Lean ✓ Stmt An MPO tensor M has the *grouped-corner Gram-dressing property* if the following holds. Let A be a tensor, let V_X, V_Y be two physical maps, let X, Y be invertible matrices, and let $c_X, c_Y > 0$. Suppose that, for every vertical letter v , the following two corner identities hold:

$$\begin{aligned} V_X^\dagger \widetilde{M}^v V_X &= c_X X A^v X^{-1}, \\ V_Y^\dagger \widetilde{M}^v V_Y &= c_Y Y A^v Y^{-1} \end{aligned}$$

Then, for every vertical letter v ,

$$(X^\dagger X) A^v (X^\dagger X)^{-1} = (Y^\dagger Y) A^v (Y^\dagger Y)^{-1}. \quad (16.44)$$

Theorem 16.4.70 (Distinguished reference corner of a grouped phase class). Lean ✓ Stmt Suppose a phase-class grouping has positive effective coefficients $c_{\alpha,q}$, corner identities

$$V_{\alpha,q}^\dagger \widetilde{M}^v V_{\alpha,q} = c_{\alpha,q} X_{\alpha,q} A_{\alpha,q}^v X_{\alpha,q}^{-1},$$

and distinguished gauges $X_{\alpha,0} = \mathbb{1}$. Fix a class α and a copy q . After identifying the representative bond space with that of copy q , there are a physical map $W_{\alpha,q}$ and a scalar $c_{\alpha,0} > 0$ such that, for every vertical letter v ,

$$W_{\alpha,q}^\dagger \widetilde{M}^v W_{\alpha,q} = c_{\alpha,0} A_{\alpha,q}^v. \quad (16.45)$$

Proof. ✓ Proof Identify the distinguished bond space with that of copy q and set $W_{\alpha,q} = V_{\alpha,0}$. Since the distinguished gauge is the identity, its corner equation reduces to

$$W_{\alpha,q}^\dagger \widetilde{M}^v W_{\alpha,q} = c_{\alpha,0} A_{\alpha,q}^v,$$

yielding (16.45) because $c_{\alpha,0} > 0$. $\square$

Theorem 16.4.71 (Gram conjugation from grouped-corner dressing). Lean ✓ Stmt Suppose a phase-class grouping has positive effective coefficients $c_{\alpha,q}$ and corner identities

$$V_{\alpha,q}^\dagger \widetilde{M}^v V_{\alpha,q} = c_{\alpha,q} X_{\alpha,q} A_{\alpha,q}^v X_{\alpha,q}^{-1}.$$

Assume that the distinguished copy in each class has identity gauge and that M has the grouped-corner Gram-dressing property. Then for every class α , copy q , and vertical letter v ,

$$(X_{\alpha,q}^\dagger X_{\alpha,q}) A_{\alpha,q}^v (X_{\alpha,q}^\dagger X_{\alpha,q})^{-1} = A_{\alpha,q}^v. \quad (16.46)$$

Proof. [✓ Proof](#) Theorem 16.4.70 transports the distinguished identity-gauge corner to the bond space of copy q . The grouped-corner Gram-dressing property gives

$$(X_{\alpha,0}^\dagger X_{\alpha,0}) A_{\alpha,q}^v (X_{\alpha,0}^\dagger X_{\alpha,0})^{-1} = (X_{\alpha,q}^\dagger X_{\alpha,q}) A_{\alpha,q}^v (X_{\alpha,q}^\dagger X_{\alpha,q})^{-1}. \quad (16.47)$$

The left side is $A_{\alpha,q}^v$ because $X_{\alpha,0} = \mathbb{1}$. This gives (16.46). $\square$

Theorem 16.4.72 (Gram conjugation from the adjoint dressing). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let X be invertible and suppose that B and a target C satisfy

$$XBX^{-1} = (X^{-1})^\dagger CX^\dagger.$$

Then conjugation by the Gram matrix carries B to C ,

$$(X^\dagger X)B(X^\dagger X)^{-1} = C. \quad (16.48)$$

Hence any two gauges satisfying this conditional identity with the same target have equal Gram conjugations, and the ratio of their Gram matrices commutes with B . This gives the relative identity from an abstract common target in [11, Proposition 4.13, lines 1909–1919]. In [11, Proposition 4.13, lines 1909–1919], the target appears via the reflected marked-chain argument without separate identification for either sector; the specializations $C = B^\dagger$ and $B^\dagger = B$ follow as corollaries.

Proof. [✓ Proof](#) Multiply the dressed-target identity by $X^\dagger$ on the left and by $(X^{-1})^\dagger$ on the right. The left side becomes $(X^\dagger X)B(X^\dagger X)^{-1}$ and the right side collapses to C . Eliminating the common target for two gauges and multiplying by the inverse Gram matrix gives the relative commutation relation. $\square$

$$X_\alpha^\dagger X_\alpha = Y_\alpha^\dagger Y_\alpha$$

Figure 16.5: Equality of the Gram conjugations for two copies of the same vertical normal tensor: $(X_{\alpha,k}^\dagger X_{\alpha,k}) M_\alpha (X_{\alpha,k}^\dagger X_{\alpha,k})^{-1} = (X_{\alpha,0}^\dagger X_{\alpha,0}) M_\alpha (X_{\alpha,0}^\dagger X_{\alpha,0})^{-1}$. Theorem 16.4.68 gives this equality from self-adjointness of the corresponding sector compressions and horizontal marked separation. Normality is used only in the subsequent proportionality argument. This is [11, Proposition 4.13, lines 1915–1921].

Theorem 16.4.73 (Gram-matrix proportionality from relative Gram conjugations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let M_α be a normal tensor, and let X and Y be invertible gauges whose Gram matrices induce the same conjugation on each letter M_α^v . Then there is a positive real number ω such that

$$X^\dagger X = \omega Y^\dagger Y, \quad (16.49)$$

and $\omega^{-1/2}XY^{-1}$ is unitary. Theorem 16.4.68 gives the Figure 8 comparison in [11, Proposition 4.13, lines 1909–1921]. This comparison supplies precisely the relative hypothesis for a sector and the distinguished sector. In the normalization $Y = \mathbb{1}$, the resulting unitary is $\omega^{-1/2}X$. The common-target declarations attached to this entry are conditional algebraic corollaries, not the reflected marked-chain statement itself.

Proof. [✓ Proof](#) By Theorem 16.4.72, the ratio $(Y^\dagger Y)^{-1}(X^\dagger X)$ commutes with every letter of M_α . Theorem 16.4.65 makes this ratio scalar. When the matrix dimension is positive, positive definiteness of the two Gram matrices makes the scalar a positive real number ω . In dimension zero the equality is vacuous and one may choose $\omega = 1$. Conjugating the relative Gram identity by Y^{-1} shows that XY^{-1} has Gram matrix $\omega \mathbb{1}$; division by $\sqrt{\omega}$ makes it unitary. $\square$

Theorem 16.4.74 (Positive scalar Gram matrix and unitary normalization from grouped-corner dressing). [Lean](#) [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 16.4.71, suppose in addition that every representative $A_{\alpha,q}$ is normal and has positive bond dimension. Then for every class α and copy q there is a real number $\omega_{\alpha,q} > 0$ such that*

$$X_{\alpha,q}^\dagger X_{\alpha,q} = \omega_{\alpha,q} \mathbb{1}. \quad (16.50)$$

Consequently $U_{\alpha,q} = \omega_{\alpha,q}^{-1/2} X_{\alpha,q}$ is unitary. This is the gauge normalization of [11, Proposition 4.13, lines 1903–1921]. Literal CPSV canonical form and normalized BNT-refined horizontal form each supply the Gram-dressing hypothesis on their own.

Proof. [✓ Proof](#) Theorem 16.4.71 shows that conjugation by $X_{\alpha,q}^\dagger X_{\alpha,q}$ fixes every letter of the normal representative. Apply the identity-gauge specialization of Theorem 16.4.73; it gives a real number $\omega_{\alpha,q} > 0$ with $X_{\alpha,q}^\dagger X_{\alpha,q} = \omega_{\alpha,q} \mathbb{1}$. Dividing each gauge by the positive square root of its Gram scalar gives the unitary normalization. $\square$

Definition 16.4.75 (Normalized grouped-sector map). [Lean](#) [✓ Stmt](#) After identifying the bond space of a copy with that of its representative, define its normalized physical map by

$$W_{\alpha,q} = V_{\alpha,q} \omega_{\alpha,q}^{-1/2} X_{\alpha,q}.$$

Theorem 16.4.76 (Normalized grouped-sector maps from Gram scalars). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let a phase-class grouping have gauges $X_{\alpha,q}$, physical maps $V_{\alpha,q}$, and effective coefficients $c_{\alpha,q}$. Suppose that, for every class α and copy q , $\omega_{\alpha,q} > 0$ and*

$$X_{\alpha,q}^\dagger X_{\alpha,q} = \omega_{\alpha,q} \mathbb{1}.$$

Suppose also that the maps $V_{\alpha,q}$ are isometries with pairwise orthogonal ranges, they intertwine $\widetilde{M}$ with the corresponding gauged representatives, and their grouped corner sum reconstructs every letter of $\widetilde{M}$. Then there are maps $W_{\alpha,q}$ such that, for every class α and copy q ,

$$W_{\alpha,q}^\dagger W_{\alpha,q} = \mathbb{1}. \quad (16.51)$$

For distinct pairs $(\alpha, q) \neq (\beta, p)$,

$$W_{\alpha,q}^\dagger W_{\beta,p} = 0. \quad (16.52)$$

For every vertical letter v , these maps also satisfy

$$\begin{aligned} \widetilde{M}_v W_{\alpha,q} &= W_{\alpha,q} (c_{\alpha,q} M_\alpha^v), \\ \widetilde{M}_v &= \sum_\alpha \sum_q W_{\alpha,q} (c_{\alpha,q} M_\alpha^v) W_{\alpha,q}^\dagger. \end{aligned} \quad (16.53)$$

Proof. ✓ Proof After identifying the copy bond space with the representative bond space, define

$$\begin{aligned} Q_{\alpha,q} &= \omega_{\alpha,q}^{-1/2} X_{\alpha,q}, \\ W_{\alpha,q} &= V_{\alpha,q} Q_{\alpha,q}. \end{aligned}$$

Since $\omega_{\alpha,q} > 0$, the Gram identity implies

$$\begin{aligned} X_{\alpha,q}^\dagger &= \omega_{\alpha,q} X_{\alpha,q}^{-1}, \\ Q_{\alpha,q}^\dagger Q_{\alpha,q} &= \mathbb{1}. \end{aligned}$$

Hence $W_{\alpha,q}^\dagger W_{\alpha,q} = \mathbb{1}$, and inserting the two normalized gauges on either side of the original orthogonality equation preserves pairwise orthogonality. The original intertwining identity gives

$$\widetilde{M}_v W_{\alpha,q} = W_{\alpha,q} (c_{\alpha,q} M_\alpha^v).$$

The same Gram identity and the positive square-root normalization give the corner equality

$$Q_{\alpha,q} (c_{\alpha,q} M_\alpha^v) Q_{\alpha,q}^\dagger = c_{\alpha,q} X_{\alpha,q} M_\alpha^v X_{\alpha,q}^{-1}. \quad (16.54)$$

Multiplying (16.54) on the left by $V_{\alpha,q}$ and on the right by $V_{\alpha,q}^\dagger$ gives

$$W_{\alpha,q} (c_{\alpha,q} M_\alpha^v) W_{\alpha,q}^\dagger = V_{\alpha,q} (c_{\alpha,q} X_{\alpha,q} M_\alpha^v X_{\alpha,q}^{-1}) V_{\alpha,q}^\dagger. \quad (16.55)$$

Summing these physical-space identities over all copies and classes, then applying the original grouped reconstruction, proves (16.53). $\square$

Theorem 16.4.77 (Normalized grouped-sector maps from grouped-corner dressing). Lean
✓ Stmt *Let a phase-class grouping have gauges $X_{\alpha,q}$, physical maps $V_{\alpha,q}$, and positive effective coefficients $c_{\alpha,q}$, with the distinguished gauge of every class equal to the identity. Assume that each representative is a normal tensor and that M has the grouped-corner Gram-dressing property. Put*

$$\begin{aligned} Q_{\alpha,q} &= \omega_{\alpha,q}^{-1/2} X_{\alpha,q}, \\ W_{\alpha,q} &= V_{\alpha,q} Q_{\alpha,q}, \end{aligned}$$

after identifying the bond space of each copy with that of its chosen representative. Then

$$\begin{aligned} W_{\alpha,q}^\dagger W_{\alpha,q} &= \mathbb{1}, \\ W_{\alpha,q}^\dagger W_{\beta,p} &= 0 \quad ((\alpha, q) \neq (\beta, p)), \\ \widetilde{M}_v W_{\alpha,q} &= W_{\alpha,q} (c_{\alpha,q} M_\alpha^v), \end{aligned}$$

and every vertical letter is reconstructed exactly:

$$\widetilde{M}_v = \sum_\alpha \sum_q W_{\alpha,q} (c_{\alpha,q} M_\alpha^v) W_{\alpha,q}^\dagger. \quad (16.56)$$

These are the normalized sector identities in [11, Proposition 4.13, lines 1895–1921]. Literal CPSV canonical form and normalized BNT-refined horizontal form each supply the Gram-dressing hypothesis on their own.

Proof. ✓ Proof Theorem 16.4.74 supplies the positive Gram scalars. Apply Theorem 16.4.76 to the phase-class grouping. Its corner identity preserves the grouped reconstruction and gives (16.56). $\square$

Lemma 16.4.78 (Matrix product vector of the diagonal tensor). Lean ✓ Stmt *For an MPO tensor M and a configuration σ , the matrix product vector of the diagonal tensor $i \mapsto M^{ii}$ equals the matrix product vector of the doubled-index tensor $\widehat{M}$ evaluated on the diagonal-paired configuration $k \mapsto (\sigma_k, \sigma_k)$: $V^{(N)}(i \mapsto M^{ii})(\sigma) = V^{(N)}(\widehat{M})(k \mapsto (\sigma_k, \sigma_k))$.*

Proof. ✓ Proof At each site value i , the diagonal tensor agrees with the doubled-index tensor on the diagonal pair, $M^{ii} = \widehat{M}^{(i,i)}$. The matrix product vector is multiplicative over sites, so replacing each factor $M^{\sigma_k \sigma_k}$ by $\widehat{M}^{(\sigma_k, \sigma_k)}$ reindexes the configuration from σ to $k \mapsto (\sigma_k, \sigma_k)$, which is the stated equality. $\square$

Lemma 16.4.79 (Diagonal tensor as a block decomposition under horizontal form). Lean ✓ Stmt *Suppose the doubled-index tensor $\widehat{M}$ shares its positive-length matrix product vector family with a block-diagonal direct sum $\bigoplus_k \mu_k A_k$ of blocks A_k on the doubled index, as in a horizontal canonical form. Then the diagonal tensor of M generates the same positive-length matrix product vector family as the block-diagonal direct sum $\bigoplus_k \mu_k A_k^{\text{diag}}$ on the single-spin index, where $A_k^{\text{diag}}(i) = A_k(i, i)$ is the diagonal restriction of each block. For every $N \geq 1$,*

$$V^{(N)}(i \mapsto M^{ii})(\sigma) = V^{(N)}\left(\bigoplus_k \mu_k A_k^{\text{diag}}\right)(\sigma).$$

Proof. ✓ Proof By Lemma 16.4.78, the diagonal matrix product vector equals that of $\widehat{M}$ on the diagonal-paired configuration, which by hypothesis equals that of the block-diagonal direct sum there. Expanding the direct sum as the weighted sum $\sum_k \mu_k^N V^{(N)}(A_k)$ and restricting each block to the diagonal pair gives $\sum_k \mu_k^N V^{(N)}(A_k^{\text{diag}})$, the matrix product vector of the single-spin block-diagonal direct sum. $\square$

Theorem 16.4.80 (Per-copy horizontal hypotheses do not control diagonal normality). Lean ✓ Stmt *There is a left-canonical, injective doubled-index tensor with a nonzero block weight satisfying the per-copy finite-length separation hypothesis whose restriction to the diagonal physical pairs is not normal. On bond space $\mathbb{C}^2$, take*

$$A^{(a,b)} = c_a E_{ab}, \quad c_0 = \frac{3}{5}, \quad c_1 = \frac{4}{5}.$$

Then the four matrices $A^{(a,b)}$ span $M_2(\mathbb{C})$ and $\sum_{a,b} (A^{(a,b)})^ A^{(a,b)} = \mathbb{1}$. For a single block, injectivity also gives finite-length block separation. The diagonal restriction, however, consists only of*

$$A^{\text{diag}}(0) = \frac{3}{5} E_{00}, \quad A^{\text{diag}}(1) = \frac{4}{5} E_{11}.$$

Every product of these two matrices is diagonal, so no blocking length spans $M_2(\mathbb{C})$.

Proof. ✓ Proof The four nonzero scalar multiples of the matrix units span the full matrix algebra. The identity $(3/5)^2 + (4/5)^2 = 1$ gives left-canonicity. After diagonal restriction, $E_{ii} E_{jj} = \delta_{ij} E_{ii}$. Hence every word is either zero or a scalar multiple of some E_{ii} , and therefore has vanishing $(0, 1)$ entry. Since E_{01} does not, it lies outside the span at every blocking length. $\square$

Lemma 16.4.81 (Diagonal tensor word evaluation). [Lean](#) [✓ Stmt](#) For any word w , the matrix product of the diagonal tensor along w equals the operator word evaluation with equal ket and bra words, since both multiply the matrices $M^{w_k w_k}$ site by site.

Proof. [✓ Proof](#) By induction on the word. □

Lemma 16.4.82 (Diagonal tensor evaluates the density-operator diagonal). [Lean](#) [✓ Stmt](#) For an MPO tensor M and a configuration σ , the matrix product vector of the diagonal tensor equals the diagonal entry of the generated operator:

$$V^{(N)}(i \mapsto M^{ii})(\sigma) = \langle \sigma | \rho^{(N)}(M) | \sigma \rangle. \quad (16.57)$$

Both sides of (16.57) equal $\text{tr}(M^{\sigma_1 \sigma_1} \dots M^{\sigma_N \sigma_N})$.

Proof. [✓ Proof](#) Word evaluation of the diagonal tensor multiplies the matrices $M^{\sigma_k \sigma_k}$, which is exactly the operator word evaluation with equal ket and bra indices; taking the trace gives the diagonal entry. □

Lemma 16.4.83 (Nonnegativity of the diagonal matrix product vector). [Lean](#) [✓ Stmt](#) If $\rho^{(N)}(M)$ is positive semidefinite, then $V^{(N)}(i \mapsto M^{ii})(\sigma) \geq 0$ for every configuration σ .

Proof. [✓ Proof](#) By Lemma 16.4.82, the value is a diagonal entry of $\rho^{(N)}(M)$, and the diagonal entries of a positive semidefinite matrix are non-negative. □

Lemma 16.4.84 (Matrix product vector of the vertical direct-sum tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let g be a block count, $\text{dim}, \text{mult} : \{0, \dots, g-1\} \rightarrow \mathbb{N}$, weights $\omega_{\alpha j} \in \mathbb{C}$ for $0 \leq j < \text{mult}_\alpha$, and blocks A_α of bond dimension dim_α . The vertical direct-sum tensor is the block-diagonal direct sum over the flattened index (α, j) that repeats A_α with weight $\omega_{\alpha j}$, written $\bigoplus_\alpha \bigoplus_j \omega_{\alpha j} A_\alpha$. Its matrix product vector decomposes as

$$V^{(N)}\left(\bigoplus_\alpha \bigoplus_j \omega_{\alpha j} A_\alpha\right)(\sigma) = \sum_\alpha \sum_j (\omega_{\alpha j})^N V^{(N)}(A_\alpha)(\sigma). \quad (16.58)$$

Proof. [✓ Proof](#) The vertical direct-sum tensor is the block-diagonal direct sum over the single flattened index q , with flattened weights $\tilde{\omega}_q$ and blocks $\tilde{A}_q$. By the block matrix-product-vector decomposition (Theorem 2.6.9),

$$V^{(N)}\left(\bigoplus_q \tilde{\omega}_q \tilde{A}_q\right)(\sigma) = \sum_q (\tilde{\omega}_q)^N V^{(N)}(\tilde{A}_q)(\sigma).$$

The flattening identifies q with the pair (α, j) via the bijection

$$\{(\alpha, j) : 0 \leq \alpha < g, 0 \leq j < \text{mult}_\alpha\} \simeq \{0, \dots, \sum_\alpha \text{mult}_\alpha - 1\},$$

under which $\tilde{\omega}_q = \omega_{\alpha j}$ and $\tilde{A}_q = A_\alpha$. Reindexing the sum along this bijection yields the double sum over (α, j) . □

Lemma 16.4.85 (Regrouping a block family by repeated representatives). [Lean](#) [✓ Stmt](#) Let $(B_k)_{k \in I}$ be a finite family of blocks with weights ν_k , and let $(A_\alpha)_{\alpha \in J}$ be a finite family of representative blocks. Suppose there is a bijection

$$e : I \simeq \{(\alpha, j) : \alpha \in J, 0 \leq j < m_\alpha\}$$

such that, writing $e(k) = (\alpha, j)$, one has $\nu_k = \omega_{\alpha j}$ and, for every positive length N , $V^{(N)}(B_k) = V^{(N)}(A_\alpha)$. Then

$$V^{(N)}\left(\bigoplus_{k \in I} \nu_k B_k\right) = V^{(N)}\left(\bigoplus_{\alpha \in J} \bigoplus_{j=0}^{m_\alpha-1} \omega_{\alpha j} A_\alpha\right)$$

for every positive length N . This is the finite-sum reindexing after the vertical sectors have been grouped in [11, Proposition 4.13].

Proof. ✓ **Proof** For $N > 0$ and every configuration σ , the two sides expand as

$$\sum_{k \in I} \nu_k^N V^{(N)}(B_k)(\sigma),$$

$$\sum_{\alpha \in J} \sum_{j=0}^{m_\alpha-1} \omega_{\alpha j}^N V^{(N)}(A_\alpha)(\sigma).$$

The summand indexed by k in the first sum equals the summand indexed by $e(k)$ in the second, so reindexing by e proves the equality. $\square$

Theorem 16.4.86 (Invariant vertical projections in literal CPSV canonical form). Lean Lean ✓ **Stmt** Let M generate matrix product density operators, and suppose that its doubled-index MPS tensor is in literal CPSV canonical form. If an orthogonal projection P satisfies $P\widetilde{M}_v = P\widetilde{M}_vP$ for every vertical letter v , then

$$\widetilde{M}_vP = P\widetilde{M}_vP \tag{16.59}$$

for every v . Thus the vertically viewed tensor has invariant-projector closure.

Proof. ✓ **Proof** The left-invariance hypothesis gives $P_1H^{(N)} = P_1H^{(N)}P_1$ at every positive length. Since $H^{(N)}$ is positive and $P = P^\dagger$,

$$P_1H^{(N)} = P_1H^{(N)}P_1 = H^{(N)}P_1. \tag{16.60}$$

The matrix coefficients of (16.60) give agreement of the corresponding first-site actions. In particular,

$$P_1H^{(N)} = H^{(N)}P_1. \tag{16.61}$$

The original-coordinate marked Lemma L gives equality of their inserted tensors on the original bond space. Reading this equality as a vertical one-site action yields

$$\widetilde{M}_vP = P\widetilde{M}_vP. \tag{16.62}$$

$\square$

Theorem 16.4.87 (Invariant vertical projections are reducing). Lean ✓ **Stmt** Let M be in normalized BNT-refined horizontal form and generate matrix product density operators. If an orthogonal projection P satisfies $P\widetilde{M}_v = P\widetilde{M}_vP$ for every vertical letter v , then it also satisfies $\widehat{M}_vP = P\widehat{M}_vP$ for every vertical letter v . Thus the vertically viewed tensor has invariant-projector closure.

Proof. ✓ **Proof** Translate the left-invariance identity into the one-site ket and bra actions of the original tensor. The normalized BNT-refined version of Lemma L gives the opposite one-sided identity. Translating back to the vertically viewed tensor proves the claim. $\square$

Theorem 16.4.88 (Irreducible reducing sectors of the vertically viewed tensor). [Lean](#) [Stmnt](#) Under the hypotheses of Theorem 16.4.87, there are positive integers d_k , tensors B_k of bond dimension d_k , and isometries $V_k : \mathbb{C}^{d_k} \rightarrow \mathbb{C}^d$ such that

$$\begin{aligned} V_k^\dagger V_k &= \mathbb{1}, \\ \sum_k V_k V_k^\dagger &= \mathbb{1}, \\ V_k^\dagger V_l &= 0 \quad (k \neq l). \end{aligned}$$

Every B_k is irreducible, and for every vertical letter v ,

$$\begin{aligned} \widetilde{M}_v V_k &= V_k B_k^v, \\ V_k^\dagger \widetilde{M}_v &= B_k^v V_k^\dagger, \\ B_k^v &= V_k^\dagger \widetilde{M}_v V_k. \end{aligned}$$

The vertically viewed tensor $\widetilde{M}$ and the unit-weight direct sum $\bigoplus_k B_k$ have the same matrix product vectors at every positive length.

Proof. [Proof](#) Apply the irreducible-corner decomposition under invariant-projector closure to $\widetilde{M}$. $\square$

Theorem 16.4.89 (Complete vertical reducing projectors). [Lean](#) [Stmnt](#) Under the same hypotheses, there are orthogonal projections P_k that are complete and pairwise orthogonal and, for every vertical letter v , reduce the corresponding letter:

$$\begin{aligned} \sum_k P_k &= \mathbb{1}, \\ P_k P_l &= 0 \quad (k \neq l), \\ \widetilde{M}_v P_k &= P_k \widetilde{M}_v. \end{aligned}$$

Proof. [Proof](#) For the isometries of Theorem 16.4.88, set $P_k = V_k V_k^\dagger$. The isometry, orthogonality, completeness, and intertwining identities give the four assertions directly. $\square$

$$U \mathcal{K} U^\dagger = \bigoplus_\alpha M_\alpha.$$

Figure 16.6: The vertically viewed tensor decomposes through complete orthogonal physical sectors. Each V_k embeds the irreducible tensor B_k , while $P_k = V_k V_k^\dagger$ is its reducing range projection. The decomposition $U \mathcal{K} U^\dagger = \bigoplus_\alpha M_\alpha$ is equivalent, for every vertical letter v , to $\widetilde{M}_v = \sum_k V_k B_{k,v} V_k^\dagger$. This is the iterated invariant-subspace construction in [11, lines 200–219], applied to the reduction in [11, Proposition 4.13, lines 1873–1887].

Theorem 16.4.90 (Spectrally normalized vertical sectors with physical isometries). [Lean](#) [Stmnt](#) Under the same hypotheses, there are positive integers d_k , positive scalars ν_k , normal tensors B_k of bond dimension d_k , and isometries $V_k : \mathbb{C}^{d_k} \rightarrow \mathbb{C}^d$ with pairwise orthogonal ranges. For every sector k and every vertical letter v ,

$$\begin{aligned} \widetilde{M}_v V_k &= V_k (\nu_k B_k^v), \\ V_k^\dagger \widetilde{M}_v &= (\nu_k B_k^v) V_k^\dagger, \\ \nu_k B_k^v &= V_k^\dagger \widetilde{M}_v V_k. \end{aligned}$$

Every vertical letter is reconstructed by

$$\widetilde{M}_v = \sum_k V_k(\nu_k B_k^v) V_k^\dagger. \quad (16.63)$$

Proof. [✓ Proof](#) Retain exactly the irreducible corners containing a nonzero letter. The Perron–Frobenius eigenvalue of each retained corner is positive, and the absence of nontrivial periodic vectors makes the inverse-square-root rescaling normal. Put the removed square root into ν_k and reindex the retained corners. The original complete reducing decomposition expresses each vertical letter as the sum of all compressed corners; every omitted summand vanishes, which gives (16.63). Thus the retained range projections need not sum to the identity, but their orthogonal complement is a zero sector. $\square$

Theorem 16.4.91 (Normal vertical corners in literal CPSV canonical form). [Lean](#) [✓ Stmt](#) *Let M generate matrix product density operators, and suppose that its doubled-index MPS tensor is in literal CPSV canonical form. Then there are positive integers d_k , positive scalars ν_k , normal tensors B_k of bond dimension d_k , and isometries $V_k : \mathbb{C}^{d_k} \rightarrow \mathbb{C}^d$ with pairwise orthogonal ranges. For every sector k and every vertical letter v ,*

$$\begin{aligned} \widetilde{M}_v V_k &= V_k(\nu_k B_k^v), \\ V_k^\dagger \widetilde{M}_v &= (\nu_k B_k^v) V_k^\dagger, \\ \nu_k B_k^v &= V_k^\dagger \widetilde{M}_v V_k. \end{aligned}$$

Every vertical letter is reconstructed by

$$\widetilde{M}_v = \sum_k V_k(\nu_k B_k^v) V_k^\dagger. \quad (16.64)$$

The omitted orthogonal complement is an all-zero sector. This is the normal-corner intermediate between the periodic exclusion at lines 1889–1893 and the grouped vertical representation used at line 1898 of [11, Proposition 4.13].

Proof. [✓ Proof](#) Theorem 16.4.86 gives invariant-projector closure for $\widetilde{M}$, and Theorem 16.4.49 excludes its nontrivial periodic vectors. Applying Theorem 7.7.5 then gives normal nonzero corners with positive spectral weights and orthogonal isometries, together with the stated intertwining and compression formulas. Their reconstruction is

$$\widetilde{M}_v = \sum_k V_k(\nu_k B_k^v) V_k^\dagger. \quad (16.65)$$

$\square$

Definition 16.4.92 (Grouped vertical BNT decomposition with physical isometries). [Lean](#) [✓ Stmt](#) *An MPO tensor M has a grouped vertical BNT decomposition with physical isometries if there are positive bond dimensions d_k , positive weights μ_k , normal tensors B_k , and isometries $V_k : \mathbb{C}^{d_k} \rightarrow \mathbb{C}^d$ with pairwise orthogonal ranges such that*

$$\begin{aligned} \widetilde{M}_v V_k &= V_k(\mu_k B_k^v), \\ V_k^\dagger \widetilde{M}_v &= (\mu_k B_k^v) V_k^\dagger, \\ \mu_k B_k^v &= V_k^\dagger \widetilde{M}_v V_k, \\ \widetilde{M}_v &= \sum_k V_k(\mu_k B_k^v) V_k^\dagger. \end{aligned}$$

Let $k = (\alpha, q)$ be the phase-class enumeration. There are equalities of the representative and copy bond dimensions, invertible gauges $X_{\alpha,q}$, and nonzero phases $\zeta_{\alpha,q}$ of unit modulus, with $X_{\alpha,0} = \mathbb{1}$ and $\zeta_{\alpha,0} = 1$, such that

$$B_{\alpha,q}^v = \zeta_{\alpha,q} X_{\alpha,q} M_{\alpha}^v X_{\alpha,q}^{-1}.$$

The representatives M_{α} form a basis of normal tensors for the weighted block tensor and are pairwise inequivalent up to gauge and phase. Every effective coefficient $c_{\alpha,q} = \mu_{\alpha,q} \zeta_{\alpha,q}$ is nonzero and positive.

For $P_{\alpha,q} = V_{\alpha,q} V_{\alpha,q}^{\dagger}$, the triple consisting of $P_{\alpha,q}$, $c_{\alpha,q}$, and the representative loop of M_{α} satisfies the sector-projector identity of Definition 16.4.58, and for every (α, q) there is a finite chain length for which the first-site compression by $P_{\alpha,q}$ is nonzero. Finally, the grouped physical maps remain isometries with pairwise orthogonal ranges, satisfy both grouped intertwining identities and the grouped corner identity, and reconstruct each vertical letter exactly:

$$\widetilde{M}_v = \sum_{\alpha} \sum_q V_{\alpha,q} (c_{\alpha,q} X_{\alpha,q} M_{\alpha}^v X_{\alpha,q}^{-1}) V_{\alpha,q}^{\dagger}. \quad (16.66)$$

Theorem 16.4.93 (Vertical normal sectors grouped into BNT representatives). [Lean](#) [Stmt](#)
Under the same hypotheses, partition the normalized vertical sectors by equality of their matrix product vector families up to a fixed nonzero scalar per site. There are normal representatives M_{α} , an enumeration $k = (\alpha, q)$ of the original sectors, invertible matrices $X_{\alpha,q}$, and complex numbers $\zeta_{\alpha,q}$ of unit modulus, with $X_{\alpha,0} = \mathbb{1}$ and $\zeta_{\alpha,0} = 1$, such that

$$B_{\alpha,q}^v = \zeta_{\alpha,q} X_{\alpha,q} M_{\alpha}^v X_{\alpha,q}^{-1}. \quad (16.67)$$

The representatives are pairwise gauge-phase distinct and form a basis of normal tensors for the weighted block tensor $\bigoplus_k \nu_k B_k$. Every grouped coefficient $\nu_{\alpha,q} \zeta_{\alpha,q}$ is positive. With the physical isometries from Theorem 16.4.90 retained unchanged, one has both reducing identities, exact compression, and the literal reconstruction

$$\widetilde{M}_v = \sum_{\alpha} \sum_q V_{\alpha,q} (\nu_{\alpha,q} \zeta_{\alpha,q} X_{\alpha,q} M_{\alpha}^v X_{\alpha,q}^{-1}) V_{\alpha,q}^{\dagger}. \quad (16.68)$$

For each pair (α, q) , the range projector of $V_{\alpha,q}$ obeys the representative-loop identity and gives a nonzero compression at some finite length. This is the decomposition and positivity argument of [11, Proposition 4.13, lines 1898–1902]. The relative Gram identity and proportional-unitary normalization of lines 1903–1908 are not asserted here.

Proof. [✓ Proof](#) Take one representative from each phase class. Theorem 8.2.8 gives equality of bond dimensions and (16.67) for every copy. Distinct representatives are gauge-phase separated by construction, so Theorem 8.2.13 gives eventual linear independence. The phase-class regrouping of the finite sum gives the BNT expansion. The representative copy is assigned the identity gauge and phase one, while the self-overlap normalization of normal tensors shows that every phase has unit modulus. Substitution into the two intertwining identities, the compression identity, and the reconstruction (16.63) proves all remaining algebraic assertions without altering any physical isometry. Cyclicity of trace gives the sector-loop identities. The nonzero-corner separation theorem supplies a nonzero compression for every sector, and the sector trace identity and MPDO positivity then make every grouped coefficient positive. $\square$

Theorem 16.4.94 (Literal vertical normal sectors grouped into BNT representatives). [Lean](#) [Stmt](#)
Under the same hypotheses, partition the normalized vertical sectors by equality of their

matrix product vector families up to a fixed nonzero scalar per site. There are normal representatives M_α , an enumeration $k = (\alpha, q)$ of the original sectors, invertible matrices $X_{\alpha,q}$, and complex numbers $\zeta_{\alpha,q}$ of unit modulus, with $X_{\alpha,0} = \mathbb{1}$ and $\zeta_{\alpha,0} = 1$, such that

$$B_{\alpha,q}^v = \zeta_{\alpha,q} X_{\alpha,q} M_\alpha^v X_{\alpha,q}^{-1}. \quad (16.69)$$

The representatives are pairwise gauge-phase distinct and form a basis of normal tensors for the weighted block tensor $\bigoplus_k \nu_k B_k$. Every grouped coefficient $\nu_{\alpha,q} \zeta_{\alpha,q}$ is positive. With the physical isometries from Theorem 16.4.91 retained unchanged, one has both reducing identities, exact compression, and the literal reconstruction

$$\widetilde{M}_v = \sum_\alpha \sum_q V_{\alpha,q} (\nu_{\alpha,q} \zeta_{\alpha,q} X_{\alpha,q} M_\alpha^v X_{\alpha,q}^{-1}) V_{\alpha,q}^\dagger. \quad (16.70)$$

For each pair (α, q) , the range projector of $V_{\alpha,q}$ obeys the representative-loop identity and gives a nonzero compression at some finite length. This is the decomposition and positivity argument of [11, Proposition 4.13, lines 1898–1902]. The relative Gram identity and proportional-unitary normalization of lines 1903–1908 are not asserted here.

Proof. ✓ Proof Take one representative from each phase class. Theorem 8.2.8 gives equality of bond dimensions and (16.69) for every copy. Distinct representatives are gauge-phase separated by construction, so Theorem 8.2.13 gives eventual linear independence. The phase-class regrouping of the finite sum gives the BNT expansion. The representative copy is assigned the identity gauge and phase one, while the self-overlap normalization of normal tensors shows that every phase has unit modulus. Substitution into the two intertwining identities, the compression identity, and the reconstruction (16.64) proves all remaining algebraic assertions without altering any physical isometry. Cyclicity of trace gives the sector-loop identities. The nonzero-corner separation theorem supplies a nonzero compression for every sector, and the sector trace identity and MPDO positivity then make every grouped coefficient positive. $\square$

Definition 16.4.95 (Block-row map of sector isometries). Lean ✓ Stmt Let $W_k : \mathbb{C}^{D_k} \rightarrow \mathbb{C}^d$ be a finite family of rectangular matrices. Their block-row map $U : \mathbb{C}^d \rightarrow \bigoplus_k \mathbb{C}^{D_k}$ is defined by $Ux = (W_k^\dagger x)_k$. This realizes the coisometry in the conclusion of [11, Proposition 4.13, lines 1863–1870] once the sector isometries have been normalized.

Theorem 16.4.96 (Conjugation and reconstruction from orthogonal sectors). Lean Lean Lean ✓ Stmt Suppose that

$$\begin{aligned} W_k^\dagger W_k &= \mathbb{1}, \\ W_k^\dagger W_l &= 0 \quad (k \neq l), \end{aligned}$$

and, for every letter v and every sector k , $T_v W_k = W_k B_k^v$. Then the block-row map U is a coisometry and conjugates T to the direct sum of the sector tensors:

$$UU^\dagger = \mathbb{1}, \quad (16.71)$$

$$UT_v U^\dagger = \bigoplus_k B_k^v. \quad (16.72)$$

If the letters also satisfy the literal sector decomposition

$$T_v = \sum_k W_k B_k^v W_k^\dagger, \quad (16.73)$$

then

$$T_v = U^\dagger \left(\bigoplus_k B_k^v \right) U.$$

This last equality is a conditional consequence of (16.73); no additional assertion from the source is used.

Proof. ✓ Proof The (k, l) block of $UU^\dagger$ is $W_k^\dagger W_l$, so orthonormality yields (16.71). The corresponding block of $UT_v U^\dagger$ is $W_k^\dagger T_v W_l = W_k^\dagger W_l B_l^v$, which equals B_k^v for $k = l$ and zero otherwise, yielding (16.72). Finally, multiplication of the block diagonal matrix by the block row gives the sum $\sum_k W_k B_k^v W_k^\dagger$, which is T_v by hypothesis. The row indices of U form the dependent disjoint union $\coprod_k \{0, \dots, D_k - 1\}$. Its canonical order-preserving bijection with $\{0, \dots, \sum_k D_k - 1\}$ gives the coordinate reindexing required by the vertical canonical-form definition. $\square$

Theorem 16.4.97 (Vertical canonical form from orthogonal grouped sectors). Lean ✓ Stmt Let $\{A_\alpha\}_{\alpha=0}^{g-1}$ be a BNT for the vertically viewed tensor $\widetilde{M}$. For each α , let $r_\alpha > 0$, let $\omega_{\alpha,q} > 0$ for $0 \leq q < r_\alpha$, and let $W_{\alpha,q} : \mathbb{C}^{D_\alpha} \rightarrow \mathbb{C}^d$ satisfy

$$\begin{aligned} W_{\alpha,q}^\dagger W_{\alpha,q} &= \mathbb{1}, \\ W_{\alpha,q}^\dagger W_{\beta,p} &= 0 \quad ((\alpha, q) \neq (\beta, p)). \end{aligned}$$

Suppose that, for every vertical letter v and every pair (α, q) , $\widetilde{M}_v W_{\alpha,q} = W_{\alpha,q} (\omega_{\alpha,q} A_\alpha^v)$, and that the letters have the exact sector decomposition

$$\widetilde{M}_v = \sum_{\alpha=0}^{g-1} \sum_{q=0}^{r_\alpha-1} W_{\alpha,q} (\omega_{\alpha,q} A_\alpha^v) W_{\alpha,q}^\dagger.$$

Then M is in vertical canonical form. This is the final finite-dimensional implication in [11, Proposition 4.13, lines 1863–1870].

Proof. ✓ Proof Form the block-row map U from the adjoints $W_{\alpha,q}^\dagger$. Theorem 16.4.96 gives

$$\begin{aligned} UU^\dagger &= \mathbb{1}, \\ U \widetilde{M}_v U^\dagger &= \bigoplus_{\alpha,q} \omega_{\alpha,q} A_\alpha^v, \\ \widetilde{M}_v &= U^\dagger \left(\bigoplus_{\alpha,q} \omega_{\alpha,q} A_\alpha^v \right) U. \end{aligned}$$

Identifying the dependent pairs (α, q) and their bond indices with the standard coordinates of the two finite direct sums gives the required matrix dimensions. Positivity of the multiplicities and weights, together with the BNT hypothesis, supplies the remaining conditions of vertical canonical form. $\square$

Lemma 16.4.98 (A retained vertical phase class exists). Lean ✓ Stmt Under the hypotheses of Theorem 16.4.93, the retained vertical decomposition has at least one sector and hence at least one phase class.

Proof. ✓ Proof The unit-modulus weight witness in the normalized BNT-refined horizontal form, together with left canonicity of its representative, makes the doubled-index tensor nonzero. Hence its vertical reshaping $\widetilde{M}$ is nonzero. If the retained decomposition had no sector, (16.68) would make every letter of $\widetilde{M}$ zero, a contradiction. Passing from the nonempty finite set of retained sectors to its phase partition gives a retained representative. $\square$

Theorem 16.4.99 (Nonvanishing of the vertical reshaping). Lean ✓ Stmt *If an MPO tensor M is in normalized BNT-refined horizontal form, then its vertical reshaping is nonzero: $\widetilde{M} \neq 0$. This is the nonvanishing observation used in [11, Proposition 4.13, lines 1895–1902].*

Proof. ✓ Proof Write the doubled-index tensor in normalized BNT-refined horizontal form as $\widetilde{M} = GSG^{-1}$, where S is the weighted direct sum of its basis tensors. Choose a copy whose weight has unit modulus. Its basis tensor cannot have every letter equal to zero, since left canonicity gives $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. The corresponding nonzero weight and nonzero letter give $S \neq 0$, hence $\widetilde{M} \neq 0$. If $\widetilde{M} = 0$, the bijection between its vertical index and the two horizontal bond indices would give $\widetilde{M} = 0$, a contradiction. $\square$

Theorem 16.4.100 (Algebraic BNT from a grouped vertical decomposition). Lean ✓ Stmt *Let M be an MPO tensor and put $T = \widetilde{M}$. Let B_k have bond dimension d_k , let $\mu_k \in \mathbb{C}$, and let $V_k : \mathbb{C}^{d_k} \rightarrow \mathbb{C}^d$ be an isometry for $0 \leq k < r$. Assume, for every vertical letter v , that*

$$\begin{aligned} V_k^\dagger T_v &= (\mu_k B_k^v) V_k^\dagger, \\ T_v &= \sum_{k=0}^{r-1} V_k (\mu_k B_k^v) V_k^\dagger. \end{aligned}$$

Partition the tensors B_k into matrix-product-vector phase classes and choose one representative A_α from each class. If $\{A_\alpha\}_\alpha$ is a basis of normal tensors in the spectral sense for the weighted direct sum $A_{\text{ret}} = \bigoplus_{k=0}^{r-1} \mu_k B_k$, then $\{A_\alpha\}_\alpha$ is an algebraic basis of normal tensors for T .

This is the basis-of-normal-tensors conclusion in [11, Proposition 4.13, lines 1863–1921], stated for the grouped decomposition before the final coisometry is formed.

Proof. ✓ Proof Put $C_k^v = \mu_k B_k^v$. Induction on a word w gives $V_k^\dagger T^w = C_k^w V_k^\dagger$. For a nonempty word $w = (v, w')$, reconstruction of its first letter, cyclicity of trace, and $V_k^\dagger V_k = \mathbb{1}$ therefore give

$$\begin{aligned} \text{tr}(T^w) &= \sum_k \text{tr}(V_k C_k^v V_k^\dagger T^{w'}) \\ &= \sum_k \text{tr}(C_k^v C_k^{w'}) = \sum_k \mu_k^{|w|} \text{tr}(B_k^w) = V^{(|w|)}(A_{\text{ret}})_w. \end{aligned}$$

Thus T and A_{ret} have the same matrix product vectors at every positive length. The spectral BNT hypothesis gives an algebraic BNT for A_{ret} by Theorem 8.2.5; in particular, it supplies the required positive bond dimensions. Lemma E.1.5 then gives the same algebraic BNT for T . $\square$

Theorem 16.4.101 (Pairwise Figure 8 identity under literal canonical form). Lean ✓ Stmt *Let M be an MPO tensor whose doubled-index MPS tensor is in the literal CPSV canonical form, and suppose that M generates positive semidefinite operators on every nonempty chain. Let A be a tensor, let X and Y be invertible matrices, and suppose that two vertical corners satisfy*

$$\begin{aligned} V_X^\dagger \widetilde{M}^v V_X &= c_X X A^v X^{-1}, & c_X &> 0, \\ V_Y^\dagger \widetilde{M}^v V_Y &= c_Y Y A^v Y^{-1}, & c_Y &> 0. \end{aligned}$$

Then the Gram-dressed tensors agree:

$$(X^\dagger X)A^v(X^\dagger X)^{-1} = (Y^\dagger Y)A^v(Y^\dagger Y)^{-1} \quad (16.74)$$

for every vertical letter v .

Proof. ✓ Proof For $Z = X, Y$, set

$$f_{(r,s),(i,j)}^Z = c_Z^{-1} \overline{(V_Z Z)_{ir}} (V_Z (Z^{-1})^\dagger)_{js}.$$

The corner identity identifies the mark with coefficients f^Z with the horizontal slice of $(Z^\dagger Z)A^v(Z^\dagger Z)^{-1}$. Self-adjointness of the two corner compressions identifies both closed marked chains with the same reflected chain. The original-coordinate marked Lemma L for literal canonical form therefore gives equality of the two marks, hence (16.74). $\square$

Theorem 16.4.102 (Positive relative Gram scalar under literal canonical form). Lean ✓ Stmt Let M be an MPO tensor whose doubled-index MPS tensor is in the literal CPSV canonical form, and suppose that M generates positive semidefinite operators on every nonempty chain. Let A be normal, let X and Y be invertible matrices, and suppose that

$$\begin{aligned} V_X^\dagger \widetilde{M}^v V_X &= c_X X A^v X^{-1}, & c_X &> 0, \\ V_Y^\dagger \widetilde{M}^v V_Y &= c_Y Y A^v Y^{-1}, & c_Y &> 0. \end{aligned}$$

Then there is a real number $\omega > 0$ such that

$$X^\dagger X = \omega Y^\dagger Y. \quad (16.75)$$

Proof. ✓ Proof Equation (16.74) makes the ratio of the two Gram matrices commute with every letter of A . Normality makes this ratio scalar, and positive definiteness makes the scalar a positive real number. $\square$

Theorem 16.4.103 (Vertical canonical form from grouped sectors and Gram dressing). Lean ✓ Stmt If M has a grouped vertical BNT decomposition with physical isometries in the sense of Definition 16.4.92, and M has the grouped-corner Gram-dressing property of Definition 16.4.69, then M is in vertical canonical form.

Proof. ✓ Proof Theorem 16.4.74 gives $X_{\alpha,q}^\dagger X_{\alpha,q} = \omega_{\alpha,q} \mathbb{1}$ with $\omega_{\alpha,q} > 0$. Absorb $\omega_{\alpha,q}^{-1/2} X_{\alpha,q}$ into the physical corner maps. The normalized maps are isometries with orthogonal ranges, intertwine $\widetilde{M}$ with the undressed representatives, and preserve the exact reconstruction. The grouped representatives form a BNT, so Theorem 16.4.97 gives the vertical canonical form. $\square$

Theorem 16.4.104 (Vertical canonical form of a canonical MPDO). Lean ✓ Stmt Let M be an MPO tensor whose doubled-index MPS tensor is in the literal CPSV canonical form. If M generates matrix product density operators, then M is in canonical form in the vertical direction. More precisely, there is a basis of normal tensors $\{M_\alpha\}_\alpha$ for the vertically viewed tensor $\widetilde{M}$, together with positive diagonal matrices μ_α and a coisometry U from the physical space onto the retained nonzero sectors. Writing $B = \bigoplus_\alpha \mu_\alpha \otimes M_\alpha$, one has

$$UU^\dagger = \mathbb{1}. \quad (16.76)$$

$$U \widetilde{M} U^\dagger = B. \quad (16.77)$$

$$\widetilde{M} = U^\dagger B U. \quad (16.78)$$

In [11, Proposition 4.13], U is called an isometry and (16.77) is displayed. In this rectangular orientation, U maps the physical space onto the retained sector space and satisfies (16.76), making U a coisometry and $U^\dagger$ an isometric embedding. The reverse identity (16.78) supplies the exact reconstruction on the physical space.

Proof. ✓ Proof Theorem 16.4.94 supplies the grouped vertical BNT decomposition with physical isometries. The literal pairwise Figure 8 theorem supplies the grouped-corner Gram-dressing property. Apply Theorem 16.4.103. $\square$

Theorem 16.4.105 (BNT-refined horizontal form implies vertical canonical form). Lean ✓ Stmt *A tensor in normalized BNT-refined horizontal form that generates matrix product density operators is also in canonical form in the vertical direction: there exist a basis of normal tensors $\{M_\alpha\}_\alpha$ for the vertically viewed tensor $\widetilde{M}$, positive diagonal matrices μ_α , and a coisometry U from the physical space onto the retained nonzero sectors. Writing $B = \bigoplus_\alpha \mu_\alpha \otimes M_\alpha$, one has*

$$\begin{aligned} U U^\dagger &= \mathbb{1}, \\ U \widetilde{M} U^\dagger &= B, \\ \widetilde{M} &= U^\dagger B U. \end{aligned}$$

This establishes vertical canonical form under the normalized BNT-refined horizontal hypothesis of Definition 16.4.1, complementing the literal CPSV formulation of Theorem 16.4.104.

Proof. ✓ Proof Theorem 16.4.93 supplies the grouped vertical decomposition. The pairwise Figure 8 theorem supplies its grouped-corner Gram-dressing property. Apply Theorem 16.4.103. $\square$

16.5 Strong and weak symmetries

For a unitary on-site symmetry U , a density operator ρ satisfies $[U, \rho] = 0$ (weak symmetry) or $U\rho = \lambda\rho$ (strong symmetry) [156, 165]. Sun formulates both notions for families of matrix product operators [208]. The statements below consider periodic operators $\rho^{(L)} = \rho^{(L)}(M)$ with identity boundary matrix [141].

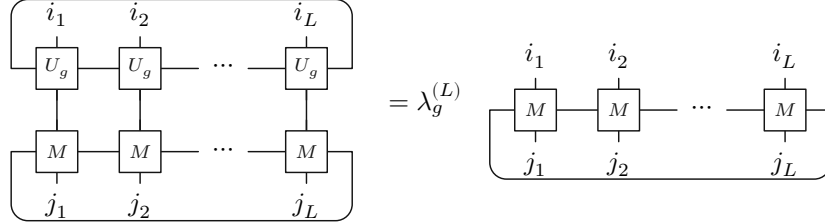
Definition 16.5.1 (Strong and weak symmetry). Lean Lean Lean Lean ✓ Stmt An operator O is a *strong symmetry* of a matrix ρ if $O\rho = \lambda\rho$ for some $\lambda \in \mathbb{C}$, and a *weak symmetry* if $[O, \rho] = 0$. A family of periodic matrix product operators $O_a^{(L)}$ is a strong symmetry of M , with eigenvalues $\lambda_a^{(L)}$, if $O_a^{(L)}\rho^{(L)} = \lambda_a^{(L)}\rho^{(L)}$ for every label a and every $L \geq 1$. It is a weak symmetry if $[O_a^{(L)}, \rho^{(L)}] = 0$ for every a and every $L \geq 1$ [208]. The eigenvalues may depend on L and need not be phases. The definition requires neither positivity of $\rho^{(L)}$ nor fusion algebra relations on O_a ; these hypotheses are stated where used.

Definition 16.5.2 (On-site symmetry). Lean Lean Lean ✓ Stmt For a representation $g \mapsto U_g$ on $\mathbb{C}^d$, the on-site operators $U_g^{\otimes L}$ are the periodic operators of the bond dimension one tensor with entries $(U_g)_{ij}$. Strong and weak on-site symmetry are the two notions of Definition 16.5.1 for this family. The eigenvalue 1 gives the strong symmetry $U\rho = \rho$ [165].

Theorem 16.5.3 (Periodic operators of an on-site tensor). Lean ✓ Stmt *The periodic operator of length L of the bond dimension one tensor with entries u_{ij} is $u^{\otimes L}$.*

Proof. ✓ Proof Each word evaluates to the scalar $\prod_l u_{\sigma_l \tau_l}$ times the one-by-one identity, whose trace is that scalar. $\square$

By Theorem 16.5.3, strong on-site symmetry with eigenvalue $\lambda_g^{(L)}$ is the following identity of periodic contractions, in which the upper row is the bond dimension one tensor with entries $(U_g)_{ij}$.



Theorem 16.5.4 (Strong symmetry implies weak symmetry). Lean Lean Lean Lean ✓ Stmt Let ρ be Hermitian. If O and $O^\dagger$ are both strong symmetries of ρ , then O is a weak symmetry of ρ . Hence a strong symmetry of an MPDO whose family contains the adjoint of each of its operators is a weak symmetry, and a strong on-site symmetry under a unitary representation is a weak on-site symmetry [208].

Proof. ✓ Proof From $O^\dagger \rho = \mu \rho$ one gets $\rho O = \bar{\mu} \rho$. Then $\rho O \rho$ equals both $\lambda \rho^2$ and $\bar{\mu} \rho^2$. If $\rho \neq 0$ then $\rho^2 \neq 0$, so $\lambda = \bar{\mu}$ and $O \rho = \rho O$. For unitary O , the relation $\rho = O^\dagger O \rho = \lambda O^\dagger \rho$ shows that $O^\dagger$ is a strong symmetry. $\square$

Theorem 16.5.5 (Eigenvalues of strong symmetries). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $\rho^{(L)} \neq 0$. A unitary strong symmetry of $\rho^{(L)}$ has an eigenvalue of modulus one. If the periodic operators obey fusion rules $O_a O_b = \sum_c N_{ab}^c O_c$, the eigenvalues of a strong symmetry obey $\lambda_a^{(L)} \lambda_b^{(L)} = \sum_c N_{ab}^c \lambda_c^{(L)}$, and they form a one-dimensional representation of the fusion ring when the eigenvalue of the unit is nonzero. For an on-site representation, $g \mapsto \lambda_g^{(L)}$ is a character of the group.

Proof. ✓ Proof For unitary O , $\rho^\dagger \rho = (O \rho)^\dagger O \rho = |\lambda|^2 \rho^\dagger \rho$ with $\rho^\dagger \rho \neq 0$. The fusion relation follows by applying $O_a O_b$ to $\rho^{(L)}$ in both forms. $\square$

Theorem 16.5.6 (Purification form of strong symmetry). Lean Lean Lean Lean Lean ✓ Stmt Let $\rho = \Phi \Phi^\dagger$, where Φ has system configurations as rows and ancilla configurations as columns, so that Φ is a purification Ψ of ρ . Then $O \rho = \lambda \rho$ if and only if $(O \otimes \mathbb{1}_{\text{anc}}) \Psi = \lambda \Psi$. For a locally purified tensor the purification is the spin-ancilla matrix product state of the purifying tensor, and M is strongly symmetric under a family O_a if and only if every $O_a^{(L)} \otimes \mathbb{1}_{\text{anc}}$ acts on that state by $\lambda_a^{(L)}$ [208].

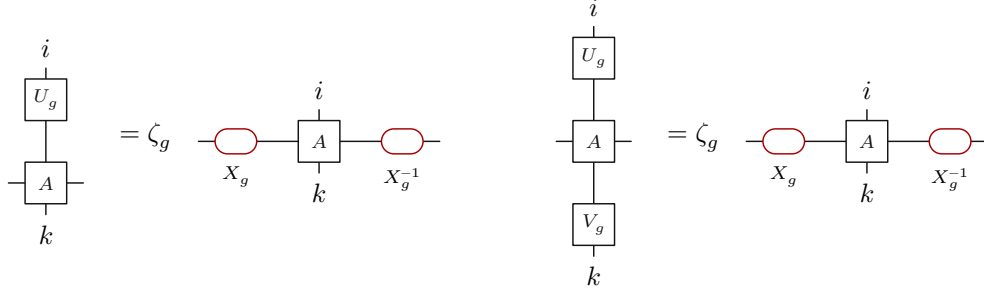
Proof. ✓ Proof If $O \rho = \lambda \rho$, the matrix $(O - \lambda) \Phi$ has Gram matrix $(O - \lambda) \rho (O - \lambda)^\dagger = 0$, so it vanishes. The converse is immediate. $\square$

Theorem 16.5.7 (Local conditions for strong and weak symmetry). Lean Lean ✓ Stmt Let M be locally purified by A . Suppose that $\sum_j (U_g)_{ij} A^{(j,k)} = \zeta_g X_g A^{(i,k)} X_g^{-1}$ for invertible X_g . Then M is strongly on-site symmetric under U with eigenvalues ζ_g^L . If instead $\sum_{j,l} (U_g)_{ij} (V_g)_{kl} A^{(j,l)} = \zeta_g X_g A^{(i,k)} X_g^{-1}$ with U_g and V_g unitary, then M is weakly on-site symmetric under U . These conditions are analogous to the criteria of de Groot, Turzillo and Schuch for channels with a

symmetric unitary dilation, with trivial and with symmetric action on the ancilla [156]; that work contains no statement about matrix product density operators.

Proof. ✓ Proof The twisted purifying tensor is a gauge transform of A up to the scalar ζ_g , so the purifying state is multiplied by ζ_g^L . For the weak condition, $U_g^{\otimes L} \rho U_g^{\otimes L \dagger} = |\zeta_g|^{2L} \rho$, and comparing traces gives $|\zeta_g|^{2L} = 1$ whenever $\rho \neq 0$. $\square$

The two local conditions of Theorem 16.5.7 read diagrammatically as follows, the first for strong and the second for weak symmetry.



Theorem 16.5.8 (Strong symmetry with a normal purification). Lean Lean Lean Lean Lean Lean ✓ Stmt Let M be locally purified by A , let the spin-ancilla tensor with entries $A^{(i,k)}$ be normal, and let U be a unitary on-site representation. Then M is strongly on-site symmetric under U if and only if for every g there are an invertible X_g and a scalar ζ_g with $|\zeta_g| = 1$ such that $\sum_j (U_g)_{ij} A^{(j,k)} = \zeta_g X_g A^{(i,k)} X_g^{-1}$. In that case the eigenvalues are $\lambda_g^{(L)} = \zeta_g^L$ at every length with $\rho^{(L)} \neq 0$. The symmetry thus acts on the purification by a virtual gauge transformation and trivially on the ancilla, in analogy with the strongly symmetric channel dilations of de Groot, Turzillo and Schuch [156].

Proof. ✓ Proof Let B_g be the twisted tensor with entries $B_g^{(i,k)} = \sum_j (U_g)_{ij} A^{(j,k)}$. Acting with a unitary on the spin leg leaves the transfer map unchanged and preserves irreducibility, so B_g is again normal. By Theorem 16.5.6 there are scalars $c_g(N)$ with

$$|V^{(N)}(B_g)\rangle = c_g(N) |V^{(N)}(A)\rangle$$

for every $N > 0$. Normality of B_g gives $O_{B_g B_g}(N) \rightarrow 1$, so $O_{B_g B_g}(N) \neq 0$ for all large N , and then $c_g(N) \neq 0$. Theorems 8.2.12 and 8.2.8 turn this proportionality into the gauge relation. The converse is Theorem 16.5.7. $\square$

Theorem 16.5.9 (Symmetries of the vectorized density operator). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Regard $\rho^{(L)}(M)$ as the matrix product vector $|\rho\rangle\rangle$ of the tensor with doubled physical index (i, j) and matrices M^{ij} . Acting with $u \otimes v$ on every doubled index sends $\rho^{(L)}$ to $u^{\otimes L} \rho^{(L)} (v^{\otimes L})^T$. For a unitary on-site representation U , the operator M is weakly on-site symmetric if and only if the doubled tensor is invariant under $g \mapsto U_g \otimes \bar{U}_g$, and strongly on-site symmetric with eigenvalues $\lambda_g^{(L)}$ if and only if $(U_g \otimes \mathbb{1})^{\otimes L} |\rho\rangle\rangle = \lambda_g^{(L)} |\rho\rangle\rangle$ at every $L \geq 1$ [165].

Proof. ✓ Proof The matrix product vector of the doubled tensor at the configuration $((\sigma_1, \tau_1), \dots, (\sigma_L, \tau_L))$ is the entry $\rho_{\sigma\tau}^{(L)}$. Expanding the trace of a product of linear combinations gives the twisted operator. For unitary u , $[u, \rho] = 0$ is equivalent to $u\rho u^\dagger = \rho$, and $(\bar{u}^{\otimes L})^T = (u^{\otimes L})^\dagger$. $\square$

Chapter 17

Pure States: Renormalization of Matrix Product States

Renormalization fixed points (RFPs) for matrix product states are characterized by physical two-site blocking, transfer map idempotence, zero correlation length (ZCL), and the square-root isometry canonical form. The presentation follows the pure-state analysis of [11, Section 3].

17.1 Physical blocking and transfer idempotence

Definition 17.1.1 (Physical blocking isometry). [Lean](#) [✓ Stmt](#) An MPS tensor A is a *renormalization fixed point* if there exists an isometry $V: \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$ with coefficients $V_{(i_1, i_2), j}$ satisfying $V^\dagger V = \mathbb{1}$ such that for all physical indices $i_1, i_2 \in \{0, \dots, d-1\}$,

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j. \quad (17.1)$$

This expresses the physical two-site blocking relation $AA = A$ of [11, Definition 3.2].

Definition 17.1.2 (Idempotent transfer map). [Lean](#) [✓ Stmt](#) The transfer map $\mathcal{E}_A$ of an MPS tensor A satisfies *transfer map idempotence* when $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$.

Theorem 17.1.3 (Eigenvalues of an idempotent transfer map). [Lean](#) [✓ Stmt](#) If $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$, then every nonzero eigenvalue of $\mathcal{E}_A$ equals 1, the form in which [16, Section III.A] states the renormalization fixed-point property.

Proof. [✓ Proof](#) If $\mathcal{E}_A(X) = \mu X$ with $X \neq 0$, then $\mu^2 X = \mathcal{E}_A^2(X) = \mathcal{E}_A(X) = \mu X$, so $\mu^2 = \mu$. $\square$

Theorem 17.1.4 (A physical blocking isometry gives transfer idempotence). [Lean](#) [✓ Stmt](#) Suppose there exists an isometry $V: \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1, i_2), j}$, such that

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j \quad (17.2)$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$. Then $\mathcal{E}_A^2 = \mathcal{E}_A$.

Proof. ✓ **Proof** Substituting the decomposition (17.2) into the double Kraus sum gives, for every bond matrix X ,

$$\begin{aligned}\mathcal{E}_A^2(X) &= \sum_{i_1, i_2} (A^{i_1} A^{i_2}) X (A^{i_1} A^{i_2})^\dagger \\ &= \sum_{j, k} \left(\sum_{i_1, i_2} V_{(i_1, i_2), j} \overline{V_{(i_1, i_2), k}} \right) A^j X (A^k)^\dagger \\ &= \sum_{j, k} (V^\dagger V)_{j, k} A^j X (A^k)^\dagger = \sum_{j, k} \delta_{j, k} A^j X (A^k)^\dagger = \sum_j A^j X (A^j)^\dagger = \mathcal{E}_A(X).\end{aligned}$$

□

Theorem 17.1.5 (Kraus-isometry criterion for transfer idempotence). Lean ✓ **Stmt** *The transfer map of A is idempotent if and only if there exists an isometry $V: \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$, written as coefficients $V_{(i_1, i_2), j}$ with $V^\dagger V = \mathbb{1}$, such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j \quad (17.3)$$

for all $i_1, i_2 \in \{0, \dots, d-1\}$.

Proof. ✓ **Proof** The backward implication is Theorem 17.1.4. For the forward implication, expanding $\mathcal{E}_A^2 = \mathcal{E}_A$ as a Kraus sum gives, for every X ,

$$\sum_{i_1, i_2} (A^{i_1} A^{i_2}) X (A^{i_1} A^{i_2})^\dagger = \sum_j A^j X (A^j)^\dagger.$$

Rectangular Kraus freedom [182, “Rectangular Kraus freedom with general finite index types”] then supplies the isometry V relating the two Kraus families. □

Theorem 17.1.6 (Physical and transfer-map fixed-point criteria). Lean ✓ **Stmt** *The transfer map $\mathcal{E}_A$ of an MPS tensor A is idempotent, $\mathcal{E}_A^2 = \mathcal{E}_A$, if and only if there exists an isometry $V: \mathbb{C}^d \rightarrow \mathbb{C}^d \otimes \mathbb{C}^d$ such that*

$$A^{i_1} A^{i_2} = \sum_j V_{(i_1, i_2), j} A^j.$$

Proof. ✓ **Proof** Follows from Theorem 17.1.5 and Definition 17.1.1. □

Definition 17.1.7 (Limit of the renormalization flow). Lean ✓ **Stmt** Let $A = (A^i)_{i=0}^{d-1}$ be an MPS tensor with bond dimension D . The physical-isometry class of A appears as a limit of the renormalization flow if there exist a physical dimension e and an MPS tensor $B = (B^a)_{a=0}^{e-1}$ of bond dimension D such that

$$\hat{\mathcal{E}}_B^{2^n} \rightarrow \hat{\mathcal{E}}_A.$$

Here $\hat{\mathcal{E}}_A$ and $\hat{\mathcal{E}}_B$ are the transfer matrices, and convergence is taken in the standard topology on the finite-dimensional matrix space $M_{D^2}(\mathbb{C})$. The physical dimension may change under blocking, while the bond dimension D is preserved. Transfer matrices represent the quotient by physical isometries [11, lines 382–394 and 1205–1209].

Theorem 17.1.8 (A dyadic power limit is idempotent). Lean ✓ **Stmt** *Let R be a Hausdorff topological monoid with continuous multiplication. If $E^{2^n} \rightarrow P$, then $P^2 = P$.*

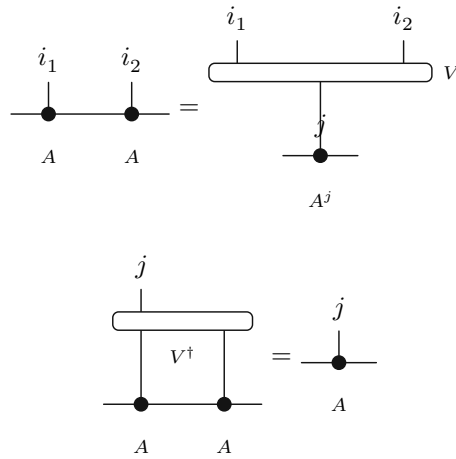
Proof. ✓ Proof Continuity of multiplication gives $E^{2^n} E^{2^n} \rightarrow P^2$. On the other hand, $E^{2^n} E^{2^n} = E^{2^{n+1}} \rightarrow P$. Uniqueness of limits gives $P^2 = P$. $\square$

Theorem 17.1.9 (Limits of the renormalization flow). Lean Lean Lean Lean ✓ Stmt *A tensor class appears as a limit of the two-site renormalization flow if and only if it has a physical blocking isometry. Equivalently, there is an isometry $V: \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$ such that*

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j$$

for every i_1, i_2 . This is [11, Theorem 3.1 and lines 1205–1209].

Proof. ✓ Proof If $\hat{\mathcal{E}}_B^{2^n} \rightarrow \hat{\mathcal{E}}_A$, then Theorem 17.1.8 gives $\hat{\mathcal{E}}_A^2 = \hat{\mathcal{E}}_A$. Since the transfer-matrix representation is injective, $\mathcal{E}_A^2 = \mathcal{E}_A$. Theorem 17.1.6 supplies the physical blocking isometry. Conversely, if A has such an isometry, its transfer map is idempotent by Theorem 17.1.6. Setting $B = A$ makes the dyadic orbit constant: $\hat{\mathcal{E}}_A^{2^n} = \hat{\mathcal{E}}_A \rightarrow \hat{\mathcal{E}}_A$. $\square$

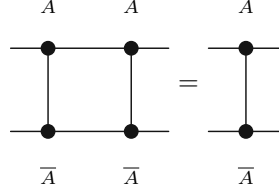


$$\sum_{i_1, i_2} (V^\dagger)_{i_1, i_2}^j A^{i_1} A^{i_2} = A^j. \quad (17.4)$$

Figure 17.1: The fusion identity of the renormalization fixed-point characterization: two contracted copies of A equal a single copy of A with the isometry V merging the summed index j into the physical pair (i_1, i_2) , as stated in (17.3); conversely, applying $V^\dagger$ to the two physical legs of the contracted pair recovers the single tensor, as stated in (17.4). The source draws the isometry as U [11, Theorem 3.1, lines 396–412].

17.2 Zero-correlation-length conditions

Definition 17.2.1 (Virtual one-block idempotence convention). Lean ✓ Stmt An MPS tensor A satisfies one-block transfer idempotence if $\mathcal{E}_A^2 = \mathcal{E}_A$. Diagrammatically, two doubled columns equal one:

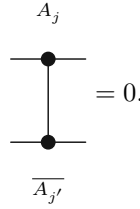


In [11, Definition 3.5], local orthogonality denotes instead the vanishing of mixed transfer maps between distinct BNT components (Definition 17.2.2).

Definition 17.2.2 (BNT local orthogonality). [Lean](#) [✓ Stmt](#) Let $(A_j)_j$ be a basis of normal tensors. The family is *locally orthogonal* if for every pair of distinct components $j \neq j'$,

$$\mathcal{E}_{j,j'} = \sum_i A_j^i \otimes \overline{A_{j'}^i} = 0.$$

Diagrammatically,



This is the mixed-sector condition of [11, Definition 3.5].

Definition 17.2.3 (Virtual-insertion auxiliary BNT condition). [Lean](#) [✓ Stmt](#) Let $(A_j)_j$ be a basis of normal tensors for A . The family $(A_j)_j$ satisfies *virtual-insertion BNT zero correlation length* for A if A satisfies virtual-insertion distance independence (Definition 17.5.11) and the components $(A_j)_j$ are locally orthogonal: $\mathcal{E}_{j,j'} = 0$ for all $j \neq j'$ (Definition 17.2.2). In contrast, [11, Definition 3.6] quantifies over physical observables on all disjoint regions (Definition 17.2.6).

Definition 17.2.4 (Positive-gap restriction of BNT zero correlation length). [Lean](#) [✓ Stmt](#) A family $(A_j)_j$ of MPS tensors has *positive-gap BNT zero correlation length* for A if $(A_j)_j$ is a basis of normal tensors for A (Definition 8.1.2), its components are locally orthogonal (Definition 17.2.2), and the physical correlations of A are independent of separation whenever both complementary gaps are positive (Definition 17.5.15). Adjacent regions are excluded.

Lemma 17.2.5 (Positive-gap restriction, unfolded). [Lean](#) [✓ Stmt](#) A family $(A_j)_j$ has *positive-gap BNT zero correlation length* for A if and only if $(A_j)_j$ is a basis of normal tensors for A (Definition 8.1.2), the physical correlations of A are independent of separation whenever both complementary gaps are positive (Definition 17.5.15), and the components $(A_j)_j$ are locally orthogonal (Definition 17.2.2).

Proof. [✓ Proof](#) This is the defining conjunction of Definition 17.2.4. □

Definition 17.2.6 (Physical BNT zero correlation length). [Lean](#) [✓ Stmt](#) A basis of normal tensors $(A_j)_j$ for A has *physical BNT zero correlation length* if the generated matrix product states have physical correlations independent of distance for all disjoint regions (Definition 17.5.14) and the components $(A_j)_j$ are locally orthogonal (Definition 17.2.2). This is [11, Definition 3.6].

Proof. ✓ Proof The unique basis block has bond dimension one. It is irreducible and left-canonical, its self-overlap is identically 1, and its singleton MPV family is linearly independent. Pairwise distinctness is vacuous. The two weights have moduli at most 1, and the first has modulus 1; hence P satisfies the auxiliary BNT sector hypotheses.

The two one-dimensional weighted copies yield $A^0 = \text{diag}(1, 1/2)$. This equality gives equality of the generated matrix-product vectors at every length, and therefore at every positive length. The sector power sum is

$$1^N + \left(\frac{1}{2}\right)^N = 1 + 2^{-N}.$$

The scalar block is a normal tensor, the displayed coefficient spans the MPV of A , and the singleton family is eventually linearly independent. Since $d = 1$, every physical observable is scalar and each two-point expectation depends only on the total gap length. Thus physical correlations are independent of distance, while local orthogonality is vacuous for the singleton BNT family.

Finally, if $\mathcal{E}_A$ were idempotent, the one-letter blocking equation would give $(A^0)^2 = vA^0$ for some scalar v . Its two diagonal entries yield $1 = v$ and $1/4 = v/2$, respectively, which is impossible. $\square$

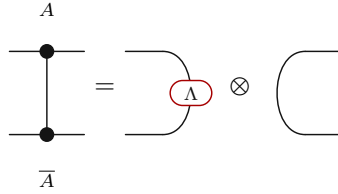
Theorem 17.2.9 (Adjacent regions obstruct the RFP–CID forward implication). Lean Lean Lean Lean Lean Lean ✓ Stmt Let A be the tensor with physical index pairs $(\alpha, \beta) \in \{0, 1\}^2$ and letters

$$A_{\gamma, \delta}^{(\alpha, \beta)} = \delta_{\gamma, \alpha} \delta_{\delta, \beta} \sqrt{\frac{1}{2}},$$

the canonical normal-tensor renormalization fixed point with maximally mixed spectrum $\Lambda = (1/2, 1/2)$: each node carries two spins, and the pair $(|00\rangle + |11\rangle)/\sqrt{2}$ is shared between the second spin of one node and the first spin of the next. The transfer map is

$$\mathcal{E}_A(X) = \text{tr}(X) \text{diag}(\Lambda) = \frac{1}{2} \text{tr}(X) \mathbb{1},$$

Diagrammatically, the transfer map is the rank-one map whose output legs carry $\text{diag}(\Lambda)$ and whose input legs are traced:



which is idempotent since $\text{tr} \text{diag}(\Lambda) = 1$, and A is a single normal tensor, so no copy weights are involved. With the Pauli matrix Z on the second spin of one node and on the first spin of the next node, the two regions are adjacent, and the two-region expectation is

$$\text{Tr}_{M_2(\mathbb{C})}(\mathcal{E}_{I \otimes Z} \circ \mathcal{E}_A^0 \circ \mathcal{E}_{Z \otimes I} \circ \mathcal{E}_A^2) = 1,$$

while after a shift by one site, which leaves one free site on each complementary arc, it is

$$\text{Tr}_{M_2(\mathbb{C})}(\mathcal{E}_{I \otimes Z} \circ \mathcal{E}_A^1 \circ \mathcal{E}_{Z \otimes I} \circ \mathcal{E}_A^1) = 0.$$

Thus A is a renormalization fixed point whose correlations are not independent of distance in the unrestricted sense of Definition 17.5.14. The block $\mathcal{E}_A^0 = 1$ between adjacent observables is not governed by idempotence.

Proof. ✓ Proof The letters are the rescaled matrix units $A^{(\alpha, \beta)} = \sqrt{1/2} |\alpha\rangle\langle\beta|$, so a conjugated letter product collapses to a diagonal matrix unit and the transfer map sums to $\mathcal{E}_A(X) = \frac{1}{2} \text{tr}(X) \mathbb{1}$. Idempotence follows from $\frac{1}{2} \text{tr}(\mathbb{1}) = 1$. The one-letter observable transfer maps are $\mathcal{E}_{Z \otimes I}(X) = \frac{1}{2} \text{tr}(X) Z$ and $\mathcal{E}_{I \otimes Z}(X) = \frac{1}{2} \text{tr}(ZX) \mathbb{1}$. With zero middle gap the composition reduces by $\text{tr}(Z^2) = 2$ to the transfer map itself, whose operator trace is 1; with one idempotent block on each side the composition is the zero map, since $\text{tr}(Z) = 0$. $\square$

Lemma 17.2.10 (Source BNT zero correlation length implies its positive-gap form). Lean
✓ Stmt If a BNT family $(A_j)_j$ for A has zero correlation length for all disjoint physical regions $(\text{ZCL}_{\text{phys}})$, then it satisfies the positive-gap zero correlation length condition (ZCL_+) :

$$\text{ZCL}_{\text{phys}}((A_j)_j, A) \implies \text{ZCL}_+((A_j)_j, A).$$

Proof. ✓ Proof Follows from Lemma 17.5.16 by restricting the physical distance-independence condition to positive complementary gaps. $\square$

Theorem 17.2.11 (Direct-sum RFP implies positive-gap BNT zero correlation length). Lean
✓ Stmt Let $(A_j)_j$ be nonzero-dimensional, irreducible, left-canonical, pairwise gauge-phase-distinct BNT components. If the direct-sum tensor $A = \bigoplus_j A_j$ satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$, then A has positive-gap physical correlations independent of distance and $\mathcal{E}_{j,j'} = 0$ for all $j \neq j'$. Thus A has positive-gap BNT zero correlation length (Definition 17.2.4). As shown in Theorem 17.2.9, this does not extend to adjacent regions.

Proof. ✓ Proof Idempotence implies positive-gap physical CID by Theorem 17.5.17. On each off-diagonal bond block, idempotence of $\mathcal{E}_A$ gives $\mathcal{E}_{j,j'} \circ \mathcal{E}_{j,j'} = \mathcal{E}_{j,j'}$. Distinct irreducible left-canonical blocks satisfy $\rho(\mathcal{E}_{j,j'}) < 1$. Since the spectrum of an idempotent is contained in $\{0, 1\}$, the conditions $\mathcal{E}_{j,j'}^2 = \mathcal{E}_{j,j'}$ and $\rho(\mathcal{E}_{j,j'}) < 1$ imply $\mathcal{E}_{j,j'} = 0$. $\square$

Theorem 17.2.12 (Basis direct sum is in literal CPSV canonical form). Lean ✓ Stmt Let P be an auxiliary BNT sector decomposition with distinct basis tensors $(A_j)_j$. The multiplicity-one unit-weight direct sum $A = \bigoplus_j A_j$ is in CPSV canonical form (Definition 7.6.7), with unit retained weights, retained blocks $(A_j)_j$, and the identity ambient coisometry.

Proof. ✓ Proof Every retained dimension is positive. Each A_j is irreducible and left-canonical, hence normal by Corollary 8.6.1.19. The unit-weight retained direct sum equals A , so the identity coisometry gives the required exact reconstruction. $\square$

Theorem 17.2.13 (Auxiliary BNT basis gives a basis for its direct sum). Lean Lean ✓ Stmt Let P be an auxiliary BNT sector decomposition with distinct basis tensors $(A_j)_j$, and let $A = \bigoplus_j A_j$ be their direct sum with one unit-weight copy of each sector. Then $(A_j)_j$ is a basis of normal tensors for A : every A_j is a normal tensor, at every positive system length N

$$V^{(N)}(A) = \sum_j V^{(N)}(A_j),$$

and the states $\{|V^{(N)}(A_j)\rangle\}_j$ are linearly independent for all sufficiently large N .

Proof. ✓ Proof Every basis block is irreducible and left-canonical with normalized self-overlap converging to 1, hence a normal tensor by Corollary 8.6.1.19. The direct sum is the unit-weight block-diagonal tensor, so its length- N matrix-product vector splits as the sum of the block vectors by Theorem 2.6.9. Eventual linear independence is part of the auxiliary BNT sector hypotheses. $\square$

Theorem 17.2.14 (Multiplicity-one fixed point implies positive-gap ZCL). [Lean](#) [✓ Stmt](#) Let P be an auxiliary BNT sector decomposition with distinct basis tensors $(A_j)_j$, and let $A = \bigoplus_j A_j$ be their multiplicity-one direct sum. If A is a renormalization fixed point ($\mathcal{E}_A^2 = \mathcal{E}_A$), then A has positive-gap BNT zero correlation length: its physical correlations are independent of separation whenever both complementary gaps are positive, and for all distinct BNT components $j \neq j'$,

$$\mathcal{E}_{j,j'} = 0.$$

This establishes the positive-gap, multiplicity-one formulation of [11, Theorem 3.8].

Proof. [✓ Proof](#) By Theorem 17.2.13, the distinct basis tensors of P are a basis of normal tensors for A . The basis blocks are irreducible, left-canonical, and pairwise gauge-phase distinct, so Theorem 17.2.11 applies to the direct sum. $\square$

Theorem 17.2.15 (Simultaneous inverse for BNT word evaluations). [Lean](#) [✓ Stmt](#) Suppose the simultaneous length- L word evaluations of $(A_j)_j$ span $\bigoplus_j M_{D_j}(\mathbb{C})$. There are coefficients $C_{(j,a,b),w}$ such that, for every sector k ,

$$\sum_{|w|=L} C_{(j,a,b),w} (A_k^w)_{x,y} = \delta_{j,k} \delta_{a,x} \delta_{b,y}.$$

Proof. [✓ Proof](#) The tuple whose j -th component is the matrix unit $|a\rangle\langle b|$ and whose other components vanish belongs to the full simultaneous word span. Reading its (k, x, y) entry gives the displayed identity. $\square$

Theorem 17.2.16 (Sector-supported physical observable realization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A = \bigoplus_k A_k$, and suppose the simultaneous length- L word evaluations span $\bigoplus_k M_{D_k}(\mathbb{C})$. For every sector j and matrices $R, l \in M_{D_j}(\mathbb{C})$, there is an observable O on L sites whose inserted transfer map vanishes on every sector pair except (j, j) and satisfies

$$\mathcal{E}_O(X)_{j,j} = R \operatorname{tr}(lX_{j,j}). \quad (17.5)$$

In particular, arbitrary virtual matrix-unit maps supported on one sector are realized by physical observables.

Proof. [✓ Proof](#) Apply Theorem 17.2.15 independently to the left and right word evaluations. The observable with coefficients

$$O_{\tau,\sigma} = C_{(j,a,b),\sigma} \overline{C_{(j,c,e),\tau}}$$

sends the (b, e) entry of $X_{j,j}$ to the (a, c) entry and annihilates every other entry and sector pair. Summing these matrix-unit observables with coefficients $R_{a,c} l_{e,b}$ gives (17.5). $\square$

Theorem 17.2.17 (Multiplicity-one BNT rank-one observables at the sharp blocking length). [Lean](#) [✓ Stmt](#) Let P be an auxiliary BNT sector decomposition of total bond dimension D , and let $A = \bigoplus_j A_j$ be the direct sum of its basis tensors. There exists a length $L \leq 3D^5$ such that, for every basis sector j and every $R, l \in M_{D_j}(\mathbb{C})$, a physical observable on L sites realizes the sector-supported insertion (17.5). This establishes the multiplicity-one case of the block-injective observable construction in [11, proof of Theorem 3.8, lines 1252–1256].

Proof. [✓ Proof](#) Theorem 8.6.1.24 gives a positive $L \leq 3D^5$ for which the simultaneous word evaluations span the product matrix algebra. Apply Theorem 17.2.16 at this length. $\square$

Lemma 17.2.18 (Sector compression calculus for the direct sum). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stm](#) Let $A = \bigoplus_k A_k$. Denote compression to the (j, j) bond block by C_j and its trace-pairing adjoint by I_j . Then

$$C_j I_j = \text{id}.$$

If $\mathcal{E}_A$ and $\mathcal{E}_j$ are the transfer maps of A and A_j , respectively, then, for every $n \geq 0$,

$$C_j \mathcal{E}_A^n = \mathcal{E}_j^n C_j.$$

The sector-supported rank-one map $\mathcal{R}_{j;R,l}(X) = I_j(R) \operatorname{tr}(lC_j(X))$ has operator trace

$$\mathrm{Tr}_{\mathrm{End}}(\mathcal{R}_{j;R,l}) = \mathrm{tr}(lR).$$

Finally, if $\mathcal{E}_j^*(l) = \lambda l$, then

$$\mathrm{tr}(l\mathcal{E}_j^n(X)) = \lambda^n \mathrm{tr}(lX).$$

Proof. ✓ **Proof** The inclusion is the trace-pairing adjoint of compression; expanding it in matrix units gives $C_j I_j = \text{id}$ and the rank-one trace formula. The block-diagonal form of A gives $C_j \mathcal{E}_A = \mathcal{E}_j C_j$, and induction gives the identity for every power. Moving one factor of $\mathcal{E}_j$ across the trace pairing and iterating proves the eigenvector formula. $\square$

Theorem 17.2.19 (Subleading spectral pair excludes positive-gap CID). [Lean] [Stmt] *Let P be an auxiliary BNT sector decomposition, let $A = \bigoplus_k A_k$, and fix a sector j . Suppose there are matrices $r, l \in M_{D_j}(\mathbb{C})$ and a scalar $\lambda \neq 0$ with $|\lambda| < 1$ such that*

$$(\mathcal{E}_j(r), \mathcal{E}_j^*(l), \text{tr}(lr)) = (\lambda r, \lambda l, 1).$$

Then A does not have positive-gap physical correlations independent of distance (Definition 17.5.15).

Proof. ✓ Proof Let ρ_j be the trace-one positive fixed point of $\mathcal{E}_j$. At the common blocking length, choose physical observables with inserted transfer maps

$$(\mathcal{E}_{O_1}(X), \mathcal{E}_{O_2}(X)) = (\rho_j \operatorname{tr}(lX_{j,j}), r \operatorname{tr}(X_{j,j})).$$

The left eigenvector equation, together with $\mathcal{E}_j(\rho_j) = \rho_j$, $\text{tr}(\rho_j) = 1$, and $\text{tr}(lr) = 1$, gives

$$\langle O_1 O_2 \rangle_{n_1, n_2} = \lambda^{n_1}.$$

Positive-gap CID compares $(n_1, n_2) = (1, 2)$ with $(2, 1)$, so $\lambda = \lambda^2$. Since $\lambda \neq 0$, this forces $\lambda = 1$, contrary to $|\lambda| < 1$. $\square$

Lemma 17.2.20 (Sector-supported probes without spectral assumptions). [Lean] [✓ Stmt] *Let P be an auxiliary BNT sector decomposition in BNT canonical form, let $A = \bigoplus_j A_j$, and fix a sector j . For every $u, v \in M_{D_j}(\mathbb{C})$, there are a positive integer L and observables O_1, O_2 on L sites such that, for all $n_1, n_2 \in \mathbb{N}$,*

$$\mathrm{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2}\mathcal{E}_A^{n_2}\mathcal{E}_{O_1}\mathcal{E}_A^{n_1}) = \mathrm{tr}(u\mathcal{E}_{A_i}^{n_1}(v)).$$

Proof. ✓ Proof Choose the trace-one stationary state ρ_j of $\mathcal{E}_{A_j}$. Simultaneous block injectivity supplies observables with inserted maps

$$\mathcal{E}_{O_1} = |\rho_j\rangle\langle u|, \quad \mathcal{E}_{O_2} = |v\rangle\langle \mathbb{1}|.$$

Compression to sector j intertwines powers of $\mathcal{E}_A$ with powers of $\mathcal{E}_{A_j}$. Hence $\mathcal{E}_A^{n_2}$ fixes the middle state ρ_j , and the composed inserted map is

$$|v\rangle\langle (\mathcal{E}_{A_j}^{n_1})^*(u)|.$$

The operator trace of this rank-one map is $\text{tr}((\mathcal{E}_{A_j}^{n_1})^*(u)v) = \text{tr}(u\mathcal{E}_{A_j}^{n_1}(v))$. □

Theorem 17.2.21 (Unconditional reverse implication at the multiplicity-one representative).

Lean ✓ Stmt Let P be an auxiliary BNT sector decomposition in BNT canonical form and $A = \bigoplus_j A_j$. If A has positive-gap BNT zero correlation length, then $\mathcal{E}_A^2 = \mathcal{E}_A$.

Proof. ✓ Proof Fix j and arbitrary $u, v \in M_{D_j}(\mathbb{C})$. Apply positive-gap CID to the observables of Lemma 17.2.20, comparing complementary gaps $(1, 2)$ and $(2, 1)$. This gives

$$\text{tr}(u\mathcal{E}_{A_j}(v)) = \text{tr}(u\mathcal{E}_{A_j}^2(v)).$$

Since u and v are arbitrary and the trace pairing is nondegenerate, $\mathcal{E}_{A_j}^2 = \mathcal{E}_{A_j}$. Local orthogonality makes every mixed map $\mathcal{E}_{A_j, A_k}$ with $j \neq k$ equal to zero. Therefore every diagonal and off-diagonal mixed map is idempotent, and the direct-sum criterion gives $\mathcal{E}_A^2 = \mathcal{E}_A$. □

Theorem 17.2.22 (Positive-gap BNT zero correlation length at the multiplicity-one representative). Lean ✓ Stmt Let P be an auxiliary BNT sector decomposition in BNT canonical form and $A = \bigoplus_j A_j$. Then

$$A \text{ has positive-gap BNT zero correlation length} \iff \mathcal{E}_A^2 = \mathcal{E}_A.$$

Proof. ✓ Proof The forward implication is Theorem 17.2.21. Conversely, transfer idempotence implies positive-gap physical correlation independence, while the BNT hypotheses give local orthogonality for the distinct sectors. □

Theorem 17.2.23 (Physical BNT zero correlation length at the multiplicity-one representative).

Lean ✓ Stmt Let P be an auxiliary BNT sector decomposition in BNT canonical form and $A = \bigoplus_j A_j$. If A has physical BNT zero correlation length, then $\mathcal{E}_A^2 = \mathcal{E}_A$.

Proof. ✓ Proof Physical correlation independence includes all positive-gap configurations. Restrict to those configurations and apply Theorem 17.2.21. □

Theorem 17.2.24 (Positive-gap BNT zero correlation length gives commuting parent ground spaces). Lean ✓ Stmt Let P be an auxiliary BNT sector decomposition in BNT canonical form and $A = \bigoplus_j A_j$. If A has positive-gap BNT zero correlation length, then for every $N > 2$ its translated two-site parent interactions satisfy

$$\begin{aligned} h_i h_j &= h_j h_i, \\ h_i V^{(N)}(A) &= 0, \\ \ker H_2^{(N)}(A) &= \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}. \end{aligned}$$

Proof. ✓ **Proof** Theorem 17.2.21 gives $\mathcal{E}_A^2 = \mathcal{E}_A$. The multiplicity-one fixed-point theorem 12.1.32 gives the stated interactions and ground spaces. $\square$

Theorem 17.2.25 (Positive-gap zero correlation length and commuting parent ground spaces for the basis direct sum). Lean ✓ **Stmt** Let $A_1, \dots, A_g$ be the distinct basis tensors of a BNT canonical form, and set $A = \bigoplus_{j=1}^g A_j$. Then

A has positive-gap BNT zero correlation length $\iff A$ has all-chain commuting parent ground spaces spanned by

This equivalence holds for the multiplicity-one direct sum of distinct BNT basis tensors with positive complementary gaps.

Proof. ✓ **Proof** By Theorem 17.2.22, positive-gap BNT zero correlation length is equivalent to transfer idempotence. The latter is equivalent to the all-chain commuting parent ground-space condition by Theorem 12.1.97. $\square$

Remark 17.2.26 (Scope and obstructions in the converse). For the multiplicity-one unit-weight direct sum, sector-supported observables require no additional hypotheses. In the general canonical form of [11, equations (1252) and (1256)], observable insertions must also cover repeated-copy sectors with their respective weights. Furthermore, the source argument infers from $\mathcal{E}_j^2 \neq \mathcal{E}_j$ the existence of an eigenvalue λ with $0 < |\lambda| < 1$, which omits non-idempotent channels whose non-trivial Jordan blocks occur at eigenvalue zero. At multiplicity one, Theorem 17.2.21 resolves this unconditionally using trace probes (Lemma 17.2.20), while Theorem 17.2.19 covers the spectral case when a nonzero subleading eigenpair exists.

17.3 Normal tensors and fixed-point isometries

17.3.1 Auxiliary block-family hypotheses

Definition 17.3.1.1 formulates an auxiliary condition for a family of injective left-canonical blocks with non-increasing nonzero weights and converging self-overlap, distinguishing it from the ambient canonical forms of Definitions 7.6.7 and 7.8.3.

Definition 17.3.1.1 (Auxiliary injective block-family hypotheses). Lean ✓ **Stmt** A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *auxiliary block-family hypotheses* if:

1. each A_k is injective,
2. each A_k satisfies the TP normalization $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$,
3. the moduli are non-increasing: $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$,
4. all $\mu_k \neq 0$,
5. every block bond dimension is positive: $D_k \geq 1$ for every k ,
6. the self-overlap converges: $O_{A_k A_k}(N) \rightarrow 1$ for each k .

Condition (6) corresponds to channel primitivity: for a primitive channel T , $T^n(\rho) \rightarrow \rho_\infty$ for every state ρ [212, Theorem 6.7(3)], which implies convergence of the self-overlap to 1 (see also [212, Theorem 6.8] and Chapter 5).

Theorem 17.3.1.2 (Auxiliary block-family hypotheses from peripheral primitivity). [Lean](#) [✓ Stmt](#) Let $(\mu_k, A_k)_{k=1}^r$ be nonzero weights and injective left-canonical blocks with positive bond dimensions, non-increasing moduli $|\mu_1| \geq \dots \geq |\mu_r| > 0$, and $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$. Then $(\mu_k, A_k)_{k=1}^r$ satisfies the auxiliary hypotheses of Definition 17.3.1.1, and $O_{A_k A_k}(N) \rightarrow 1$ for every k .

Proof. [✓ Proof](#) The injectivity, left-canonical identity, weight ordering, and nonzero-weight clauses are exactly the hypotheses. For each block, peripheral primitivity gives the complementary spectral gap $\rho_{\text{spec}}(\mathcal{E}_{A_k} - P) < 1$ by the complementary transfer-map gap for primitive maps [182, “Complementary transfer-map gap for primitive maps”]. Theorem 5.10.2 then yields $O_{A_k A_k}(N) \rightarrow 1$. $\square$

Theorem 17.3.1.3 (RFP normal tensor is injective). [Lean](#) [✓ Stmt](#) If A is a normal RFP tensor, then A is injective.

Proof. [✓ Proof](#) The RFP condition $\mathcal{E}_A^2 = \mathcal{E}_A$ gives coefficients $V_{i_1 i_2, j}$ such that

$$A^{i_1} A^{i_2} = \sum_j V_{i_1 i_2, j} A^j.$$

Induction on the word length gives $A^{i_1} \dots A^{i_n} \in \text{span}\{A^j : 0 \leq j < d\}$ for $n \geq 1$. Normality implies that products of some length N span $M_D(\mathbb{C})$, so the one-site matrices $\{A^j\}_{j=0}^{d-1}$ already span $M_D(\mathbb{C})$, establishing injectivity. $\square$

Theorem 17.3.1.4 (Left-canonical normal RFP tensor is injective). [Lean](#) [✓ Stmt](#) Assume $D > 0$. Let A be a normal renormalization fixed point such that $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then A is injective. This is the left-canonical injectivity step in [11, Appendix B].

Proof. [✓ Proof](#) Follows from Theorem 17.3.1.3. $\square$

Theorem 17.3.1.5 (Unitary diagonal fixed point for a left-canonical normal RFP). [Lean](#) [✓ Stmt](#) Assume $D > 0$. Let A be a normal renormalization fixed point such that $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that, for $B^i := U^\dagger A^i U$, $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ and $\mathcal{E}_B(\Lambda) = \Lambda$. This is the diagonal fixed-point reduction in [11, Appendix B].

Proof. [✓ Proof](#) By Theorem 17.3.1.4, A is injective. By Theorem 2.5.2, its transfer map $\mathcal{E}_A$ is irreducible. Applying canonical form II (Theorem 7.8.6) yields a unitary matrix U preserving the left-canonical condition and a diagonal positive-definite matrix Λ satisfying $\mathcal{E}_B(\Lambda) = \Lambda$ for $B^i = U^\dagger A^i U$. $\square$

Theorem 17.3.1.6 (Rank-one classification for RFP transfer maps). [Lean](#) [✓ Stmt](#) Let A be an injective left-canonical RFP tensor with positive-definite fixed point ρ of the transfer map $\mathcal{E}_A$. Then $\mathcal{E}_A = P_\rho$, where $P_\rho(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$ is the rank-one fixed-point projection.

Proof. [✓ Proof](#) Exponential convergence of primitive channels (Theorem 14.3.1) gives $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1 - \delta)^n \|X\|$. Transfer map idempotence implies $\mathcal{E}_A^n = \mathcal{E}_A$ for all $n \geq 1$, so $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1 - \delta)^n \|X\|$ for all $n \geq 1$. Taking $n \rightarrow \infty$ yields $\mathcal{E}_A = P_\rho$. $\square$

Theorem 17.3.1.7 (Diagonal Kraus form for a normal fixed point). [Lean](#) [✓ Stmt](#) A normal left-canonical renormalization fixed-point tensor A admits a decomposition $A^i = X \Lambda U^i X^{-1}$, where Λ is diagonal positive, $\sum_i (U^i)^\dagger U^i = I$, and

$$\sum_i \overline{(U^i)_{\alpha, \beta}} (U^i)_{\alpha', \beta'} = D^{-1} \delta_{\alpha, \alpha'} \delta_{\beta, \beta'}.$$

Proof. ✓ Proof After the diagonal fixed-point reduction, write the conjugated tensor as B and its positive diagonal fixed point as ρ . The rank-one classification gives $\mathcal{E}_B(Y) = P_\rho(Y) = \frac{\text{tr}(Y)}{\text{tr}(\rho)}\rho$. Set

$$\begin{aligned}\Lambda_\alpha &= \sqrt{\frac{D\rho_{\alpha,\alpha}}{\text{tr}(\rho)}}, \\ E_{\alpha,\beta} &= D^{-1/2}e_{\alpha,\beta}, \\ K_{\alpha,\beta} &= \Lambda E_{\alpha,\beta}.\end{aligned}$$

Direct multiplication of matrix units yields

$$\sum_{\alpha,\beta} K_{\alpha,\beta} Y K_{\alpha,\beta}^\dagger = P_\rho(Y).$$

Kraus freedom therefore gives an isometry V such that $B^i = \sum_{\alpha,\beta} V_{i,(\alpha,\beta)} K_{\alpha,\beta}$. With $U^i = \sum_{\alpha,\beta} V_{i,(\alpha,\beta)} E_{\alpha,\beta}$, one has $B^i = \Lambda U^i$ and

$$\begin{aligned}U_{\alpha,\beta}^i &= D^{-1/2}V_{i,(\alpha,\beta)}, \\ \sum_i \overline{U_{\alpha,\beta}^i} U_{\alpha',\beta'}^i &= D^{-1}\delta_{\alpha,\alpha'}\delta_{\beta,\beta'}.\end{aligned}$$

Conjugating back gives $A^i = X\Lambda U^i X^{-1}$. □

Theorem 17.3.1.8 (Diagonal Kraus form with square-sum normalization). Lean ✓ Stmt *In the decomposition of Theorem 17.3.1.7, the diagonal weights satisfy $\sum_\alpha \Lambda_\alpha^2 = D$.*

Proof. ✓ Proof From $\Lambda_\alpha = \sqrt{D\rho_{\alpha,\alpha}/\text{tr}\rho}$, squaring and summing over α gives $\sum_\alpha \Lambda_\alpha^2 = \frac{D}{\text{tr}\rho} \sum_\alpha \rho_{\alpha,\alpha} = D$. □

Theorem 17.3.1.9 (Unit pair-index form for a normal fixed point). Lean ✓ Stmt *A normal left-canonical renormalization fixed-point tensor A admits a decomposition $A^i = X\Lambda U^i X^{-1}$, where Λ is diagonal positive and*

$$\sum_i \overline{(U^i)_{\alpha,\beta}} (U^i)_{\alpha',\beta'} = \delta_{\alpha,\alpha'}\delta_{\beta,\beta'}.$$

This is the unit pair-index convention of [11, Section 3].

Proof. ✓ Proof Apply Theorem 17.3.1.12, which gives $A^i = X\sqrt{\Lambda}U^i X^{-1}$ with U already a unit pair-index isometry. Set $\tilde{\Lambda}_\alpha = \sqrt{\Lambda_\alpha}$, so that $A^i = X\tilde{\Lambda}U^i X^{-1}$ with $\tilde{\Lambda}$ diagonal positive; the decomposition and the pair-index condition are immediate. □

Definition 17.3.1.10 (Square-root fixed-point form). Lean ✓ Stmt Let $\lambda_\alpha > 0$ satisfy $\sum_\alpha \lambda_\alpha = 1$, and set $\rho = \text{diag}(\lambda_\alpha)$. A normal tensor A has the *square-root fixed-point form* if there exist an invertible matrix X and a tensor U satisfying the unit pair-index isometry

$$\sum_i \overline{(U^i)_{\alpha,\beta}} (U^i)_{\alpha',\beta'} = \delta_{\alpha,\alpha'}\delta_{\beta,\beta'},$$

such that

$$A^i = X\sqrt{\rho}U^i X^{-1}.$$

Local fix (square-root factor in canonical form): The definition uses the square-root diagonal $\sqrt{\rho}$ rather than the bare diagonal printed in [11, line 1278], as required by the unit pair-index isometry and reference-tensor construction; documented in [100].

Explicitly, the reference tensor is

$$\widehat{A}_{\alpha,\beta}^{(\alpha',\beta')} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'} \sqrt{\lambda_\alpha}.$$

Thus $\rho = \text{diag}(\lambda_\alpha)$ is the trace-one diagonal fixed point. This records the $j = j'$ part of the joint isometry condition [11, Appendix B, lines 1278, 1281–1283, 1300].

Lemma 17.3.1.11 (Gauge invariance of the square-root fixed-point form). [Lean] [✓ Stmt] *If $A \sim_{\text{gauge}} B$ (so $B^i = Y A^i Y^{-1}$ for invertible Y), then A has the square-root fixed-point form if and only if B does.*

Proof. [✓ Proof] Write $A^i = X \sqrt{\rho} U^i X^{-1}$ and $B^i = Y A^i Y^{-1}$. Then

$$B^i = (Y X) \sqrt{\rho} U^i (Y X)^{-1}.$$

Thus the diagonal weight and pair-index isometry are unchanged. Applying the same argument to the inverse gauge proves the converse. □

Theorem 17.3.1.12 (Trace-normalized square-root form for a normal fixed point). [Lean] [✓ Stmt] *Every normal left-canonical renormalization fixed-point tensor has the square-root fixed-point form (Definition 17.3.1.10): setting $\lambda_\alpha = \rho_{\alpha,\alpha} / \text{tr } \rho$ and $\rho_0 = \text{diag}(\lambda_\alpha)$, there exist an invertible matrix X and a unit pair-index isometry U such that*

$$A^i = X \sqrt{\rho_0} U^i X^{-1}.$$

This is the normal-tensor characterization of [11, Appendix B, lines 1278–1300], incorporating the square-root weight correction.

Proof. [✓ Proof] By Theorem 17.3.1.8, A admits a decomposition $A^i = X \widetilde{\Lambda} U_{\text{raw}}^i X^{-1}$ with $\sum_\alpha \widetilde{\Lambda}_\alpha^2 = D$. Rescaling to the unit pair-index convention via $U^i = \sqrt{D} U_{\text{raw}}^i$ and setting

$$\rho_0 = \text{diag} \left(\left(\frac{\widetilde{\Lambda}_\alpha}{\sqrt{D}} \right)^2 \right),$$

one has $\text{tr}(\rho_0) = D^{-1} \sum_\alpha \widetilde{\Lambda}_\alpha^2 = D^{-1} \cdot D = 1$, while $\sqrt{(\rho_0)_{\alpha,\alpha}} = \widetilde{\Lambda}_\alpha / \sqrt{D}$ recovers the decomposition $A^i = X \sqrt{\rho_0} U^i X^{-1}$. □

Lemma 17.3.1.13 (Reference-tensor transfer identity for a unit pair-index isometry). [Lean] [✓ Stmt] *Let U be a family of bond matrices satisfying the unit pair-index isometry condition*

$$\sum_i \overline{(U^i)_{\alpha,\beta}} (U^i)_{\alpha',\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

Then for every bond matrix Z ,

$$\sum_i U^i Z (U^i)^\dagger = \text{tr}(Z) I.$$

$$A = X\sqrt{\rho}U X^{-1}$$

Figure 17.2: The square-root form of a normal renormalization fixed point: $\rho = \text{diag}(\lambda_\alpha)$ and $A^i = X\sqrt{\rho}U^i X^{-1}$. The pair-index isometry is the one in lines 1281–1283, and the square root follows from the reference tensor at line 1300 [11, Appendix B].

Proof. ✓ Proof Reading off the (x, y) entry, the pair-index condition gives $\sum_i (U^i)_{x,\alpha} \overline{(U^i)_{y,\beta}} = \delta_{x,y} \delta_{\alpha,\beta}$, so the entry collapses to $\delta_{x,y} \sum_\alpha Z_{\alpha,\alpha} = \delta_{x,y} \text{tr}(Z)$, which is the (x, y) entry of $\text{tr}(Z)I$. $\square$

Theorem 17.3.1.14 (The square-root form implies the fixed-point property). Lean ✓ Stmt *If an MPS tensor A has the square-root fixed-point form, then its transfer map $\mathcal{E}_A$ is idempotent:*

$$\mathcal{E}_A^2 = \mathcal{E}_A.$$

In particular, A is a renormalization fixed point [11, Section 3].

Proof. ✓ Proof Write $A^i = X\sqrt{\rho}U^i X^{-1}$. Substituting into the transfer map and pulling the index-independent factors $X\sqrt{\rho}$ and $\sqrt{\rho}X^\dagger$ outside the sum, the unit pair-index isometry identity (Lemma 17.3.1.13) gives the rank-one form

$$\begin{aligned} \mathcal{E}_A(Y) &= \text{tr}(X^{-1}Y(X^{-1})^\dagger) R, \\ R &= X \rho X^\dagger. \end{aligned}$$

Idempotence then reduces to $\text{tr}(X^{-1}R(X^{-1})^\dagger) = \text{tr}(\rho) = 1$, the trace normalization, so $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$. $\square$

Theorem 17.3.1.15 (Normal-tensor fixed-point characterization). Lean ✓ Stmt *Let A be a normal tensor. Then its transfer map satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$ if and only if A has the square-root fixed-point form*

$$A^i = X\sqrt{\rho}U^i X^{-1}.$$

In particular, the bond dimension of A is positive. This establishes the corrected form of [11, Lemma B.1, lines 1274–1301].

Proof. ✓ Proof Normality first gives positive bond dimension by Theorem 8.2.1. For the forward implication, choose the pure trace-preserving Perron gauge from Theorem 8.2.3. Idempotence and algebraic normality are invariant under this gauge, so Theorem 17.3.1.12 gives the square-root form for the gauged tensor. Lemma 17.3.1.11 gives the form for the original tensor. The reverse implication is Theorem 17.3.1.14. $\square$

Across all blocks, the repaired trace-normalized decomposition has the following representation. Here j indexes the distinct normal blocks and q their repeated copies. The joint isometry condition across distinct blocks is established in Theorem 17.4.16.

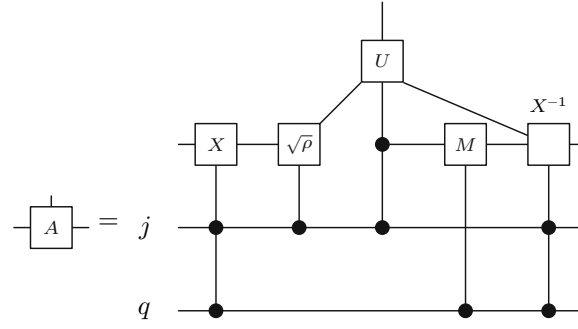


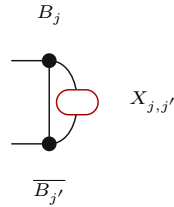
Figure 17.3: Block and multiplicity indices in the fixed-point decomposition. The independent rails j and q pass through the coefficient network without being fused: X and X^{-1} depend on both, $\sqrt{\rho}$ and U carry the block label j , and M carries the multiplicity label q . The upper leg of U is the physical index, as in the source figure [11, Section 3.4, lines 543–563].

17.4 Direct sums and the joint isometry condition

Lemma 17.4.1 (Block decomposition of the direct-sum transfer map). [Lean](#) [✓ Stmt](#) Let $(B_k)_k$ be a family of tensors and let $\bigoplus_k B_k$ be their direct sum on the total bond space. For every bond matrix X and every pair of blocks, the (j, j') bond block of the transfer sum

$$\sum_i \left(\bigoplus_k B_k^i \right) X \left(\bigoplus_k B_k^i \right)^\dagger$$

equals $\mathcal{E}_{j,j'}$ applied to the (j, j') bond block of X :



Proof. [✓ Proof](#) A block-diagonal factor connects only matching blocks, so the (j, j') block of the product is

$$\left(\sum_i \left(\bigoplus_k B_k^i \right) X \left(\bigoplus_k B_k^i \right)^\dagger \right)_{j,j'} = \sum_i B_j^i X_{j,j'} (B_{j'}^i)^\dagger = \mathcal{E}_{j,j'}(X_{j,j'}).$$

□

Theorem 17.4.2 (Reindexing the direct-sum transfer map). [Lean](#) [✓ Stmt](#) Let $(B_k)_k$ be a family of tensors, let Φ be the canonical identification of the block-labelled bond space with the total bond space, and let $\mathcal{S}_B$ be the block transfer sum. Then, for every block matrix Y ,

$$\mathcal{E}_{\bigoplus_k B_k}(\Phi(Y)) = \Phi(\mathcal{S}_B(Y)).$$

Proof. ✓ Proof Since Φ preserves products and adjoints, expanding the transfer map gives

$$\begin{aligned}\mathcal{E}_{\oplus_k B_k}(\Phi(Y)) &= \sum_i \Phi\left(\bigoplus_k B_k^i\right) \Phi(Y) \Phi\left(\bigoplus_k B_k^i\right)^\dagger \\ &= \Phi\left(\sum_i \left(\bigoplus_k B_k^i\right) Y \left(\bigoplus_k B_k^i\right)^\dagger\right) \\ &= \Phi(\mathcal{S}_B(Y)).\end{aligned}$$

□

Theorem 17.4.3 (Direct-sum quasi-idempotence on the block transfer sum). Lean ✓ Stmt Let $(B_k)_k$ be a family of tensors and let $c \in \mathbb{C}$. If

$$\mathcal{E}_{\oplus_k B_k}^2 = c\mathcal{E}_{\oplus_k B_k},$$

then the corresponding block transfer sum $\mathcal{S}_B$ satisfies, for every bond matrix Y ,

$$\mathcal{S}_B(\mathcal{S}_B(Y)) = c\mathcal{S}_B(Y).$$

Proof. ✓ Proof Reindex the total bond space by its block labels. By Theorem 17.4.2, the direct-sum transfer map becomes $\mathcal{S}_B$. Transporting the assumed equation through this identification gives the displayed identity. □

Theorem 17.4.4 (Direct-sum quasi-idempotence on every mixed corner). Lean ✓ Stmt Let $(B_k)_k$ be a family of tensors and let $c \in \mathbb{C}$. If

$$\mathcal{E}_{\oplus_k B_k}^2 = c\mathcal{E}_{\oplus_k B_k},$$

then, for every ordered pair of blocks (j, j') whose two bond dimensions are positive,

$$\mathcal{E}_{j,j'}^2 = c\mathcal{E}_{j,j'}.$$

Proof. ✓ Proof Let Z be an arbitrary rectangular matrix and realize it as the (j, j') block of a matrix Y on the total bond space. By Lemma 17.4.1, $[\mathcal{S}_B(Y)]_{j,j'} = \mathcal{E}_{j,j'}(Z)$. Theorem 17.4.3 gives $\mathcal{S}_B^2 = c\mathcal{S}_B$, and hence

$$\mathcal{E}_{j,j'}(\mathcal{E}_{j,j'}(Z)) = [\mathcal{S}_B(\mathcal{S}_B(Y))]_{j,j'} = [c\mathcal{S}_B(Y)]_{j,j'} = c\mathcal{E}_{j,j'}(Z).$$

□

Lemma 17.4.5 (Pairwise idempotence of the block transfer sum). Lean ✓ Stmt Let $(B_k)_k$ be a family of tensors. If every mixed transfer operator $\mathcal{E}_{j,j'}$ is idempotent, then the block transfer sum $\mathcal{S}_B$ is idempotent: $\mathcal{S}_B(\mathcal{S}_B(X)) = \mathcal{S}_B(X)$.

Proof. ✓ Proof By Lemma 17.4.1, the (j, j') block satisfies

$$(\mathcal{S}_B^2(X))_{j,j'} = \mathcal{E}_{j,j'}^2(X_{j,j'}) = \mathcal{E}_{j,j'}(X_{j,j'}) = (\mathcal{S}_B(X))_{j,j'}.$$

Equality on every block gives the asserted equality on the total bond space. □

Theorem 17.4.6 (Pairwise mixed-transfer criterion for a direct sum). Lean ✓ Stmt Suppose every bond dimension is positive. The transfer map of $\bigoplus_k B_k$ is idempotent if and only if, for every pair j, j' , the mixed transfer operator is idempotent:

$$\mathcal{E}_{\oplus_k B_k}^2 = \mathcal{E}_{\oplus_k B_k} \iff \mathcal{E}_{j,j'}^2 = \mathcal{E}_{j,j'}.$$

Proof. ✓ Proof By Lemma 17.4.1, the (j, j') block of the direct-sum transfer map depends only on the (j, j') block of its argument and is obtained by applying $\mathcal{E}_{j, j'}$. Thus pairwise idempotence gives

$$(\mathcal{E}_{\oplus_k B_k}^2(X))_{j, j'} = \mathcal{E}_{j, j'}^2(X_{j, j'}) = \mathcal{E}_{j, j'}(X_{j, j'}) = (\mathcal{E}_{\oplus_k B_k}(X))_{j, j'}$$

for every pair of blocks, and hence gives idempotence on the total bond space, as in Lemma 17.4.5. Conversely, every matrix M of the appropriate rectangular size occurs as the (j, j') block of some Y on the total bond space. Applying Lemma 17.4.1 and whole-tensor idempotence gives

$$\mathcal{E}_{j, j'}^2(M) = (\mathcal{E}_{\oplus_k B_k}^2(Y))_{j, j'} = (\mathcal{E}_{\oplus_k B_k}(Y))_{j, j'} = \mathcal{E}_{j, j'}(M).$$

□

Lemma 17.4.7 (Scaling of a rectangular mixed transfer operator). Lean ✓ Stmt For complex scalars c, e and tensors A, B of possibly different bond dimensions, $\mathcal{E}_{cA, eB} = c\bar{e} \mathcal{E}_{A, B}$.

Proof. ✓ Proof Conjugate transposition changes the second scalar to its complex conjugate: $\sum_i (cA^i)X(eB^i)^\dagger = c\bar{e} \sum_i A^i X(B^i)^\dagger$. □

Theorem 17.4.8 (Phase coherence for literal repeated blocks). Lean ✓ Stmt Let A have positive bond dimension and a nonzero idempotent transfer map. Let μ_q have unit modulus. If the literal bond direct sum $\bigoplus_q \mu_q A$ has an idempotent transfer map, then $\mu_q = \mu_{q'}$ for every pair q, q' .

Proof. ✓ Proof The pairwise criterion (Theorem 17.4.6) gives that $\mathcal{E}_{\mu_q A, \mu_{q'} A}$ is idempotent. Lemma 17.4.7 and $\mathcal{E}_{A, A} = \mathcal{E}_A$ give $\mathcal{E}_{\mu_q A, \mu_{q'} A} = \mu_q \overline{\mu_{q'}} \mathcal{E}_{A, A} = \mu_q \overline{\mu_{q'}} \mathcal{E}_A$. Hence $(\mu_q \overline{\mu_{q'}}) \mathcal{E}_A$ is idempotent. Since $\mathcal{E}_A$ is nonzero and $|\mu_q \overline{\mu_{q'}}| = |\mu_q| |\mu_{q'}| = 1$,

$$(\mu_q \overline{\mu_{q'}})^2 \mathcal{E}_A = \mu_q \overline{\mu_{q'}} \mathcal{E}_A \implies \mu_q \overline{\mu_{q'}} (\mu_q \overline{\mu_{q'}} - 1) = 0 \implies \mu_q \overline{\mu_{q'}} = 1.$$

Since $|\mu_{q'}| = 1$ gives $\overline{\mu_{q'}} \mu_{q'} = 1$, one has $\mu_q = \mu_q (\overline{\mu_{q'}} \mu_{q'}) = (\mu_q \overline{\mu_{q'}}) \mu_{q'} = \mu_{q'}$. □

Theorem 17.4.9 (Cross-block transfer vanishing for a direct-sum fixed point). Lean ✓ Stmt Let $(A_j)_j$ be a family of distinct irreducible left-canonical blocks, no two of which are gauge-phase equivalent, and suppose the direct sum $\bigoplus_j A_j$ is a renormalization fixed point. Then for every pair of distinct components $j \neq j'$, the mixed transfer operator vanishes:

$$\mathcal{E}_{j, j'} = \sum_i A_j^i \otimes \overline{A_{j'}^i} = 0.$$

This establishes the cross-block $(\delta_{j, j'})$ orthogonality of [11, Theorem 3.11 and Corollary 3.12]. The passage to the residual tensors U_j is given by Theorem 17.4.16.

Proof. ✓ Proof By Lemma 17.4.1 the (j, j') bond block of the direct-sum transfer map acts as $\mathcal{E}_{j, j'}$. Whole-tensor idempotence $\mathcal{E}^2 = \mathcal{E}$ therefore restricts to each block as $\mathcal{E}_{j, j'}^2 = \mathcal{E}_{j, j'}$. For distinct irreducible left-canonical blocks, Theorem 5.6.2 gives $\rho_{\text{spec}}(\mathcal{E}_{j, j'}) < 1$ when the bond dimensions agree, because the blocks are not gauge-phase equivalent; Theorem 5.6.4 gives the same inequality when the bond dimensions differ. Consequently, Lemma 5.5.3 gives

$$\mathcal{E}_{j, j'}^2 = \mathcal{E}_{j, j'}, \quad \rho_{\text{spec}}(\mathcal{E}_{j, j'}) < 1 \implies \mathcal{E}_{j, j'} = 0.$$

□

Definition 17.4.10 (Joint isometry condition). [Lean](#) [✓ Stmt](#) A family $(U_j)_j$ satisfies the *joint isometry condition* if

$$\sum_i (U_j^i)_{\alpha,\beta} \overline{(U_{j'}^i)_{\alpha',\beta'}} = \delta_{j,j'} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

Diagrammatically, contracting U_j with the conjugate of $U_{j'}$ over the physical index identifies the virtual legs pairwise:

$$\begin{array}{c} \boxed{U_j} \\ | \\ \boxed{U_{j'}} \end{array} = \delta_{j,j'} \begin{array}{c} \text{---} \text{---} \end{array} \otimes \begin{array}{c} \text{---} \text{---} \end{array}$$

Each block satisfies unit pair-index orthonormality ($j = j'$), and the cross-block sums between distinct blocks vanish ($j \neq j'$), as in [11, Theorem 3.11].

Theorem 17.4.11 (One-letter Gram matrix under the joint isometry condition). [Lean](#) [✓ Stmt](#) Let $\mathcal{V} = \bigsqcup_j \{j\} \times [D_j] \times [D_j]$ be the disjoint union of the within-sector virtual pairs. For $x = (j, \alpha, \beta)$ in $\mathcal{V}$, set $u_x(i) = (U_j^i)_{\alpha,\beta}$. Then $\sum_i u_x(i) \overline{u_y(i)} = \delta_{x,y}$.

Proof. [✓ Proof](#) If x and y belong to the same sector, the assertion is the within-sector part of the joint isometry condition. If they belong to distinct sectors, it is the cross-sector vanishing condition. $\square$

Lemma 17.4.12 (Covariance of the mixed transfer operator). [Lean](#) [✓ Stmt](#) For decompositions $A^i = X_A D_A U^i X_A^{-1}$ and $B^i = X_B D_B V^i X_B^{-1}$, the mixed transfer operator of A, B is the mixed transfer operator of the tensors U, V conjugated by the outer factors:

$$\mathcal{E}_{A,B}(Y) = X_A D_A \mathcal{E}_{U,V}(X_A^{-1} Y (X_B^{-1})^\dagger) (X_B D_B)^\dagger.$$

Proof. [✓ Proof](#) Substitute the two decompositions into $\mathcal{E}_{A,B}(Y) = \sum_i A^i Y (B^i)^\dagger$ and collect the outer factors, which are independent of the summation index, outside the sum. $\square$

Lemma 17.4.13 (Vanishing after removing the diagonal factors). [Lean](#) [✓ Stmt](#) With decompositions $A^i = X_A D_A U^i X_A^{-1}$ and $B^i = X_B D_B V^i X_B^{-1}$ whose factors X_A, D_A, X_B, D_B are invertible, if $\mathcal{E}_{A,B} = 0$ then $\mathcal{E}_{U,V} = 0$.

Proof. [✓ Proof](#) The covariance identity Lemma 17.4.12 expresses $\mathcal{E}_{A,B}$ as $\mathcal{E}_{U,V}$ conjugated by the invertible outer factors $X_A D_A$ and $(X_B D_B)^\dagger$, so $\mathcal{E}_{A,B} = 0$ forces $\mathcal{E}_{U,V} = 0$. $\square$

Lemma 17.4.14 (Entrywise cross-block isometry sum). [Lean](#) [✓ Stmt](#) If $\mathcal{E}_{U,V} = 0$, then, for all virtual indices,

$$\sum_i (U^i)_{\alpha,\beta} \overline{(V^i)_{\alpha',\beta'}} = 0.$$

Proof. [✓ Proof](#) Apply $\mathcal{E}_{U,V} = 0$ to the matrix unit supported at (β, β') and read the (α, α') entry. $\square$

Lemma 17.4.15 (Each block of a direct-sum fixed point is a fixed point). [Lean](#) [✓ Stmt](#) If the direct sum $\bigoplus_k B_k$ is a renormalization fixed point, with $\dim_k \geq 1$ for all k , then each block B_j is a renormalization fixed point.

Proof. ✓ Proof The diagonal mixed transfer operator $\mathcal{E}_{j,j}$ is the transfer map of B_j , and whole-tensor idempotence makes it idempotent, which is exactly the renormalization fixed-point condition for B_j . $\square$

Theorem 17.4.16 (Joint isometry condition for a direct-sum fixed point). Lean ✓ Stmt Let $(A_j)_j$ be a family of normal, irreducible, left-canonical blocks with bond dimensions $D_j \geq 1$, no two of which are gauge-phase equivalent. If the direct sum $\bigoplus_j A_j$ is a renormalization fixed point, then there exist invertible matrices X_j , trace-one positive diagonal matrices $\rho_j = \text{diag}(\lambda_{j,\alpha})$, and tensors U_j with

$$A_j^i = X_j \sqrt{\rho_j} U_j^i X_j^{-1},$$

that is,

such that $(U_j)_j$ satisfies the joint isometry condition (Definition 17.4.10). This formalizes the isometry equations of [11, Corollary 3.12] for distinct normal blocks.

Proof. ✓ Proof Whole-tensor idempotence makes the diagonal mixed transfer operator of each block its own transfer map, hence each block is a renormalization fixed point; the square-root fixed-point form (Theorem 17.3.1.12) supplies the decomposition $A_j^i = X_j \sqrt{\rho_j} U_j^i X_j^{-1}$ with the within-block orthonormality, which is the $j = j'$ case. For $j \neq j'$, Theorem 17.4.9 gives $\mathcal{E}_{A_j, A_{j'}} = 0$; by the covariance Lemma 17.4.12 and invertibility of the outer factors $\mathcal{E}_{U_j, U_{j'}} = 0$, and reading the entry of this operator at a matrix unit yields the cross-block sum $\sum_i (U_j^i)_{\alpha, \beta} \overline{(U_{j'}^i)_{\alpha', \beta'}} = 0$. $\square$

Theorem 17.4.17 (Direct sum of square-root fixed-point blocks). Lean ✓ Stmt Let $(B_k)_k$ be a family of tensors in square-root fixed-point form whose cross-block mixed transfer operators vanish ($\mathcal{E}_{j,j'} = 0$ for all $j \neq j'$). Then the direct sum $\bigoplus_k B_k$ is a renormalization fixed point. This establishes the converse direction of the pure-state fixed-point characterization [11, Section 3] for distinct blocks.

Proof. ✓ Proof The direct-sum transfer map decouples block by block: by Lemma 17.4.1 its (j, j') bond block acts as $\mathcal{E}_{j,j'}$ on the (j, j') bond block of the argument. Each diagonal block $\mathcal{E}_{j,j}$ is the transfer map of B_j , which is idempotent because B_j is a renormalization fixed point (Theorem 17.3.1.14); each off-diagonal block $\mathcal{E}_{j,j'}$ with $j \neq j'$ vanishes by hypothesis. Therefore, for every bond block,

$$[\mathcal{E}_{\bigoplus B}^2(Y)]_{j,j'} = \mathcal{E}_{j,j'}(\mathcal{E}_{j,j'}(Y_{j,j'})) = \mathcal{E}_{j,j'}(Y_{j,j'}) = [\mathcal{E}_{\bigoplus B}(Y)]_{j,j'},$$

so $\mathcal{E}_{\bigoplus B}^2 = \mathcal{E}_{\bigoplus B}$ and the direct sum is a renormalization fixed point. $\square$

Theorem 17.4.18 (Distinct-block fixed-point characterization). Lean ✓ Stmt Let $(B_k)_k$ be a family of normal, irreducible, left-canonical blocks with $D_k \geq 1$ for all k , no two of which are gauge-phase equivalent. Then the direct sum $\bigoplus_k B_k$ is a renormalization fixed point if and only if each block has the square-root fixed-point form and all cross-block mixed transfer operators vanish:

$$\left(\bigoplus_k B_k \text{ is an RFP} \right) \iff (\forall k, B_k \text{ in square-root form}) \wedge (\forall j \neq j', \mathcal{E}_{j,j'} = 0).$$

This is the distinct-block characterization of pure-state renormalization fixed points [11, Theorem 3.11].

Proof. ✓ Proof For the forward direction, whole-tensor idempotence makes each diagonal mixed transfer operator the transfer map of its block (Lemma 17.4.15), so each block is a renormalization fixed point and hence in the square-root fixed-point form (Theorem 17.3.1.12), while the off-diagonal operators vanish, $\mathcal{E}_{j,j'} = 0$ for $j \neq j'$, by Theorem 17.4.9. Conversely, a direct sum of square-root fixed-point blocks whose cross-block operators vanish is a renormalization fixed point (Theorem 17.4.17). $\square$

Corollary 17.4.19 (BNT basis direct-sum characterization). Lean Lean ✓ Stmt Let P satisfy the auxiliary BNT sector hypotheses, with distinct basis tensors $(B_j)_j$. Then the multiplicity-one direct sum $\bigoplus_j B_j$ is a renormalization fixed point if and only if every B_j has the square-root fixed-point form and $\mathcal{E}_{j,j'} = 0$ for all $j \neq j'$.

In this case, there exist invertible matrices X_j , trace-one positive diagonal matrices $\rho_j = \text{diag}(\lambda_{j,\alpha})$, and tensors U_j such that $B_j^i = X_j \sqrt{\rho_j} U_j^i X_j^{-1}$, where $(U_j)_j$ satisfies the joint isometry condition (Definition 17.4.10).

Proof. ✓ Proof Positive basis dimensions supply the nonzero bond spaces. For each basis tensor, irreducibility and left-canonical normalization are part of the auxiliary BNT sector hypotheses. If $m_j > 0$ is the peripheral period of B_j , the normalized self-overlap hypothesis and the periodic self-overlap formula used in Theorem 16.4.20 give, along the same subsequence,

$$\begin{aligned} \lim_{k \rightarrow \infty} O_{B_j B_j}(m_j k) &= 1, \\ \lim_{k \rightarrow \infty} O_{B_j B_j}(m_j k) &= m_j. \end{aligned}$$

Hence $m_j = 1$. The transfer map is primitive by the corresponding theorem in the companion quantum-channel volume [182, “Primitivity equals period one”], and normality follows from Theorem 7.11.1.1. Gauge-phase distinctness is also part of the auxiliary BNT sector hypotheses. The equivalence now follows from Theorem 17.4.18; the same derived hypotheses in Theorem 17.4.16 gives the joint isometry condition. $\square$

Theorem 17.4.20 (Transfer idempotence under coisometric reconstruction). Lean ✓ Stmt Let A and B be tensors with bond dimensions D and R . Suppose that $U \in \mathbb{C}^{R \times D}$ satisfies $UU^\dagger = I_R$ and, for every physical index i ,

$$A^i = U^\dagger B^i U.$$

Then A is a renormalization fixed point if and only if B is a renormalization fixed point.

Proof. ✓ Proof For every bond matrix X ,

$$\mathcal{E}_A(X) = U^\dagger \mathcal{E}_B(UXU^\dagger)U.$$

The identity $UU^\dagger = I_R$ gives $U(U^\dagger Y U)U^\dagger = Y$. Applying these two identities twice proves each implication. $\square$

Theorem 17.4.21 (A fixed nonzero scaling of a normal tensor). Lean ✓ Stmt Let B be a normal tensor and let $c \in \mathbb{C} \setminus \{0\}$. If cB is a renormalization fixed point, then

$$|c| = 1, \quad \mathcal{E}_B^2 = \mathcal{E}_B.$$

Proof. ✓ Proof Normality gives $r(\mathcal{E}_B) = 1$, while $\mathcal{E}_{cB} = |c|^2 \mathcal{E}_B$. Since $c \neq 0$, this operator is nonzero. Theorem 5.5.2 gives $r(\mathcal{E}_{cB}) = 1$, and Theorem 5.5.1 gives

$$1 = r(\mathcal{E}_{cB}) = |c|^2 r(\mathcal{E}_B) = |c|^2.$$

Thus $|c| = 1$, so $\mathcal{E}_{cB} = \mathcal{E}_B$ and the idempotence of $\mathcal{E}_B$ follows. $\square$

Theorem 17.4.22 (Blocks of a CPSV fixed point). Lean ✓ Stmt Suppose A has CPSV canonical-form data

$$A^i = U^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) U, \quad UU^\dagger = I,$$

where every A_k is normal. If A is a renormalization fixed point, then every block k satisfies

$$|\mu_k| = 1, \quad \mathcal{E}_{A_k}^2 = \mathcal{E}_{A_k}.$$

Proof. ✓ Proof By Theorem 17.4.20, the direct sum $\bigoplus_k \mu_k A_k$ is a renormalization fixed point. Lemma 17.4.15 implies that each block $\mu_k A_k$ is also a renormalization fixed point. Since $\mu_k \neq 0$, Theorem 17.4.21 gives both conclusions. $\square$

Theorem 17.4.23 (Phase-class residual isometries). Lean Lean ✓ Stmt Let A be a CPSV canonical-form renormalization fixed point. Choose one representative C_j from each matrix-product-vector phase class. Then there are invertible matrices X_j , positive numbers $\lambda_{j,\alpha}$ with $\sum_\alpha \lambda_{j,\alpha} = 1$, and tensors U_j such that

$$C_j^i = X_j \text{diag} \left(\sqrt{\lambda_{j,\alpha}} \right) U_j^i X_j^{-1},$$

$$\sum_i (U_j^i)_{\alpha,\beta} \overline{(U_{j'}^i)_{\alpha',\beta'}} = \delta_{j,j'} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

If no classes are present, the statement holds vacuously.

Local fix (square-root factor in canonical form): The formula uses the square-root weights $\sqrt{\lambda_{j,\alpha}}$ rather than the bare diagonal printed in [11, line 547], as required by the reference-tensor construction; documented in [100].

Proof. ✓ Proof The retained weighted direct sum is a renormalization fixed point. For each representative, Theorem 17.4.22 gives $|\mu_j| = 1$ and $\mathcal{E}_{C_j}^2 = \mathcal{E}_{C_j}$. By Theorem 17.3.1.15, each normal fixed point C_j admits the displayed square-root form. Putting the representatives independently in left-canonical gauge, distinct phase-class representatives are not gauge-phase equivalent. Since the direct sum is a fixed point, Theorem 17.4.9 gives

$$\mathcal{E}_{\mu_j C_j, \mu_{j'} C_{j'}} = 0 \quad (j \neq j').$$

Cancelling the unit-modulus scalars and gauge transformations yields $\mathcal{E}_{C_j, C_{j'}} = 0$. Removing the positive diagonal factors and evaluating on matrix units gives the off-diagonal residual orthogonality. Combined with the within-block orthonormality of each U_j , this yields the joint isometry condition. $\square$

Theorem 17.4.24 (Square-root characterization of fixed points). Not ready Cirac et al. [11, Theorem 3.11] assert that a tensor A in canonical form is a renormalization fixed point if and only if it can be written as

$$A^i = \bigoplus_{j=1}^g \bigoplus_{q=1}^{r_j} \mu_{j,q} X_{j,q} \sqrt{\rho_j} U_j^i X_{j,q}^{-1},$$

where $|\mu_{j,q}| = 1$, each ρ_j is positive diagonal with $\text{tr}(\rho_j) = 1$, and

$$\sum_i (U_j^i)_{\alpha,\beta} \overline{(U_{j'}^i)_{\alpha',\beta'}} = \delta_{j,j'} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

The source specifies that $\bigoplus_{j,q}$ is a simultaneous direct sum in both the physical and virtual spaces [11, lines 561–562]. However, it does not provide an explicit coordinate embedding for the physical routing of q , a dimension constraint, or a q -indexed isometry equation. Consequently, this statement remains unformalized pending clarification of the coordinate-level routing in the source.

Local fix (square-root factor in canonical form): CPSV prints a bare Λ_j in Theorem 3.11, whereas the reference tensor at line 1300 has coefficients $\sqrt{(\rho_j)_{\alpha,\alpha}}$, which forces the factor $\sqrt{\rho_j}$ used above. This repair is documented in [100].

17.5 Renormalization flow and physical correlations

One renormalization step blocks two sites of a tensor B and compresses the blocked physical index by an isometry U with $U^\dagger U = \mathbb{1}$, producing a tensor A with

$$B^{i_1} B^{i_2} = \sum_j U_{(i_1, i_2), j} A^j$$

[11, lines 389–394]. Since $U^\dagger U = \mathbb{1}$, the isometry cancels in the doubled contraction, so the step acts on transfer maps as $\mathcal{E}_A = \mathcal{E}_B^2$ [11, line 1209]; repeated steps produce the dyadic powers $\mathcal{E}_B^{2^n}$.

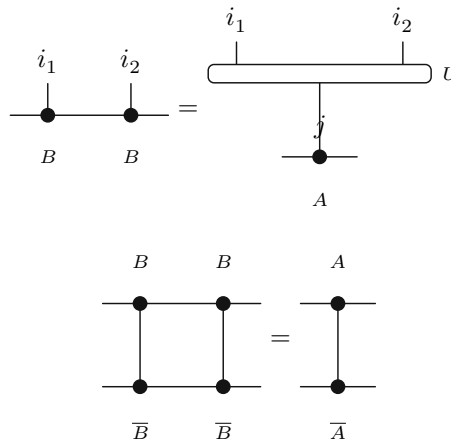


Figure 17.4: One renormalization step. Top: blocking two sites of B and compressing the pair index by the isometry U gives the tensor A . Bottom: on the doubled network the isometry cancels, so $\mathcal{E}_B^2 = \mathcal{E}_A$ [11, lines 389–394 and 1209].

Theorem 17.5.1 (Primitive-transfer convergence for each auxiliary block). [Lean](#) [Stmt](#) Fix one block A_k in a family satisfying Definition 17.3.1.1. Then the dyadic powers of its transfer map converge pointwise to an idempotent linear map: there exists $\mathcal{E}_{k,\infty}$ with $\mathcal{E}_{k,\infty}^2 = \mathcal{E}_{k,\infty}$ such that $\mathcal{E}_{A_k}^{2^n}(\rho) \rightarrow \mathcal{E}_{k,\infty}(\rho)$ as $n \rightarrow \infty$ for every virtual matrix ρ . This formalizes the per-block primitive convergence of [11, Appendix B, lines 1211–1244].

Proof. ✓ Proof Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point ρ_0 . The exponential convergence bound $\|\mathcal{E}^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|$ (Theorem 14.3.1) yields pointwise convergence $\mathcal{E}^n(X) \rightarrow P(X)$, and restricting to the dyadic subsequence $2^n \rightarrow \infty$ yields the claim with idempotent projection $\mathcal{E}_{k,\infty} = P_k$. $\square$

Theorem 17.5.2 (Phase obstruction to canonical-form flow convergence). Lean ✓ Stmt Let $\omega = e^{2\pi i/3}$ and consider the tensor with one physical component

$$A^1 = \begin{pmatrix} 1 & 0 \\ 0 & \omega \end{pmatrix}.$$

This tensor is in canonical form, with two one-dimensional normal components of weights 1 and ω . For the matrix unit e_{21} ,

$$\mathcal{E}_A^{2^n}(e_{21}) = \omega^{2^n} e_{21}.$$

The right-hand side alternates between ωe_{21} and $\omega^2 e_{21}$, so the dyadic renormalization flow does not converge. This refutes the unrestricted canonical-form convergence assertion of [11, lines 417 and 1211–1244]; it does not affect the single-component convergence of Theorem 17.5.1.

Proof. ✓ Proof The two one-dimensional components are normal and both weights have modulus one. Direct calculation gives $\mathcal{E}_A(e_{21}) = \omega e_{21}$. Since 2^n is congruent alternately to 1 and 2 modulo 3, the sequence alternates between ωe_{21} and $\omega^2 e_{21}$, precluding convergence. $\square$

Definition 17.5.3 (One-letter doubled-MPO virtual-gauge condition). Lean Lean ✓ Stmt For an MPO tensor M with one physical component, the Appendix D blocking diagram holds up to virtual gauge if $M^{[2]}$ and M are related by similarity on the virtual indices. For a pure tensor A , this condition is imposed on the doubled tensor

$$M = A \otimes \bar{A}.$$

Since the physical algebra is one-dimensional, every physical unitary channel in the diagram is trivial, recovering the relation in [11, Appendix D, lines 2100–2107].

Definition 17.5.4 (One-letter MPS blocking witness). Lean ✓ Stmt A tensor A with one physical component has a blocking witness if there exist an invertible matrix X and a scalar ζ with $|\zeta| = 1$ such that

$$A = \zeta X A^{[2]} X^{-1}.$$

Definition 17.5.5 (Doubled virtual gauge). Lean ✓ Stmt Given an invertible virtual matrix X , the induced gauge on the doubled bond space is $X \otimes \bar{X}$.

Definition 17.5.6 (Cube-phase swap gauge). Lean ✓ Stmt The permutation matrix

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

defines an invertible virtual gauge that exchanges the 1 and ω coordinates of the cube-phase tensor.

Theorem 17.5.7 (An MPS blocking witness gives the doubled-MPO condition). [Lean](#) [✓ Stmt](#)
 If $A = \zeta X A^{[2]} X^{-1}$ is a one-letter MPS blocking witness with $|\zeta| = 1$, then the doubled tensor $M = A \otimes \bar{A}$ satisfies

$$M = (X \otimes \bar{X}) M^{[2]} (X \otimes \bar{X})^{-1},$$

satisfying the doubled-MPO virtual-gauge condition (Definition 17.5.3).

Proof. [✓ Proof](#) Tensoring $A = \zeta X A^{[2]} X^{-1}$ with its complex conjugate yields $(X \otimes \bar{X}) M^{[2]} (X \otimes \bar{X})^{-1}$, with scalar factor $\zeta \bar{\zeta} = |\zeta|^2 = 1$. $\square$

Theorem 17.5.8 (The printed pure blocking-up-to-gauge equivalence is false). [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\omega = e^{2\pi i/3}$, and set

$$A = \begin{pmatrix} 1 & 0 \\ 0 & \omega \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then $\omega^3 = 1$, $\omega \neq 1$, and

$$A = \omega S A^2 S^{-1}.$$

The doubled MPO tensor generates positive semidefinite operators on every nonempty chain, is in CPSV canonical form, and has four scalar-block weights satisfying the source normalization convention. It also satisfies the literal one-letter virtual-gauge condition in the source diagram, but its transfer map is not idempotent. Hence the canonical-restricted pure-state equivalence claimed in [11, Appendix D, lines 2100–2107] is false.

Proof. [✓ Proof](#) On every nonempty chain, $M = A \otimes \bar{A}$ generates the rank-one operator $|V^{(N)}(A)\rangle\langle V^{(N)}(A)|$, which is positive semidefinite. Moreover, M is the direct sum of four bond-one scalar blocks with weights $\mu_p \bar{\mu}_q$, where $\mu_0 = 1$ and $\mu_1 = \omega$. Each block is normal and every weight has modulus one, proving both the CPSV canonical-form condition and its normalization.

Swapping the two virtual coordinates transforms $\text{diag}(1, \omega^2)$ into $\text{diag}(\omega^2, 1)$; multiplication by ω gives A . Writing $M = A \otimes \bar{A}$, doubling cancels the phase:

$$(\omega S A^{[2]} S^{-1}) \otimes (\bar{\omega} \bar{S} \bar{A}^{[2]} \bar{S}^{-1}) = (S \otimes \bar{S}) M^{[2]} (S \otimes \bar{S})^{-1}.$$

However, idempotence $\mathcal{E}_A^2 = \mathcal{E}_A$ would imply $\mathcal{E}_A^{2^n} = \mathcal{E}_A$ for all $n \geq 0$, contradicting the non-convergent alternating dyadic orbit of Theorem 17.5.2. $\square$

Remark 17.5.9. Theorem 17.5.8 refutes the pure-state equivalence assertion of [11, Appendix D]; the general mixed-state case is treated separately in MPO theory.

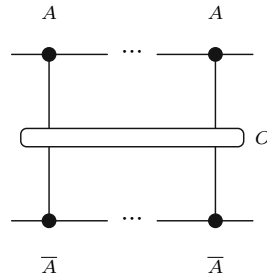
Remark 17.5.10. When the transfer map is idempotent ($\mathcal{E}^2 = \mathcal{E}$), connected correlations become separation-independent for all separations $n \geq 1$. In the rank-one (primitive) case where the peripheral spectrum consists only of the simple eigenvalue 1, all subleading spectral values vanish, yielding zero correlation length ($\xi = 0$).

Definition 17.5.11 (Virtual-insertion auxiliary condition). [Lean](#) [✓ Stmt](#) An MPS tensor satisfies virtual-insertion distance independence if for every positive-definite right fixed point $\rho^R > 0$, every pair of virtual bond matrices X, Y , and all separations $n, m \geq 1$, the connected correlator satisfies $C(X, Y; n) = C(X, Y; m)$.

Definition 17.5.12 (Physical observable transfer). [Lean](#) [✓ Stmt](#) For an observable O on a block of L physical spins, define the linear map on the virtual matrix space by

$$\mathcal{E}_O(X) = \sum_{i,j} \langle j|O|i\rangle A^i X (A^j)^\dagger.$$

Diagrammatically, O is inserted between the two layers of the doubled chain on its L sites:

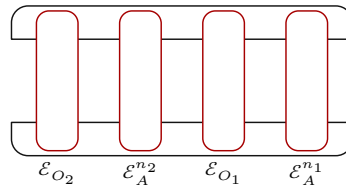


This is the observable transfer map of [11, lines 490–496].

Definition 17.5.13 (Two-observable periodic expectation). [Lean](#) [✓ Stmt](#) Let O_1, O_2 act on blocks of L_1, L_2 physical spins, separated around the periodic chain by complementary gaps of lengths n_1, n_2 . Their two-observable expectation is

$$\text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A^{n_2} \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A^{n_1}).$$

Diagrammatically, the four maps are composed around the ring:



The trace is the operator trace on the virtual matrix space, as in [11, lines 490–496].

Definition 17.5.14 (Physical correlations independent of distance). [Lean](#) [✓ Stmt](#) Let O_1, O_2 act on two disjoint contiguous regions of positive lengths $L_1, L_2 \geq 1$ in a periodic chain, with complementary gap lengths $n_1, n_2 \geq 0$. Physical correlations are independent of distance if, for all $m_1, m_2 \geq 0$ satisfying $n_1 + n_2 = m_1 + m_2$,

$$\text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A^{n_2} \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A^{n_1}) = \text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A^{m_2} \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A^{m_1}).$$

This allows either region to be translated along the chain without crossing the other, including adjacent regions ($n_i = 0$) [11, Definition 3.3].

Definition 17.5.15 (Positive-gap restriction of physical distance independence). [Lean](#) [✓ Stmt](#) Let O_1, O_2 be observables on disjoint blocks of a periodic chain, with observable transfer maps $\mathcal{E}_{O_1}, \mathcal{E}_{O_2}$ (Definition 17.5.12). The tensor has *positive-gap physical correlations independent of distance* if for all positive complementary gaps $n_1, n_2, m_1, m_2 > 0$ satisfying $n_1 + n_2 = m_1 + m_2$,

$$\text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A^{n_2} \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A^{n_1}) = \text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A^{m_2} \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A^{m_1}).$$

This restricts Definition 17.5.14 to strictly positive gaps, excluding adjacent regions [11, Definition 3.3].

Lemma 17.5.16 (Physical CID implies positive-gap physical CID). [Lean](#) [✓ Stmt](#) If an MPS tensor satisfies physical distance independence (Definition 17.5.14), then it satisfies the positive-gap condition (Definition 17.5.15).

Proof. [✓ Proof](#) Restrict the gap lengths in Definition 17.5.14 to strictly positive integers. $\square$

Theorem 17.5.17 (Idempotent transfer implies positive-gap physical CID). [Lean](#) [✓ Stmt](#) If $\mathcal{E}_A^2 = \mathcal{E}_A$, then A satisfies positive-gap physical distance independence (Definition 17.5.15).

Proof. [✓ Proof](#) Idempotence yields $\mathcal{E}_A^n = \mathcal{E}_A$ for all $n \geq 1$. Since $n_1, n_2, m_1, m_2 \geq 1$, substituting $\mathcal{E}_A$ for each power in the periodic expectation gives

$$\begin{aligned} \text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A^{n_2} \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A^{n_1}) &= \text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A) \\ &= \text{Tr}_{M_D(\mathbb{C})}(\mathcal{E}_{O_2} \circ \mathcal{E}_A^{m_2} \circ \mathcal{E}_{O_1} \circ \mathcal{E}_A^{m_1}). \end{aligned}$$

$\square$

Definition 17.5.18 (Virtual-insertion zero correlation length). [Lean](#) [✓ Stmt](#) A tensor A satisfies the virtual-insertion zero-correlation-length condition if its transfer map is idempotent ($\mathcal{E}_A^2 = \mathcal{E}_A$) and it satisfies virtual-insertion distance independence (Definition 17.5.11).

Theorem 17.5.19 (Virtual-insertion distance independence implies RFP). [Lean](#) [✓ Stmt](#) If the virtual-insertion condition (Definition 17.5.11) holds for every positive-definite right fixed point $\rho^R > 0$ of the transfer map, then $\mathcal{E}_A^2 = \mathcal{E}_A$.

Proof. [✓ Proof](#) Let $\rho > 0$ be a right fixed point. For any virtual matrix Z , set $X = Z\rho^{-1}$, so that $X\rho = Z$. Applying distance independence with virtual insertions X and Y at separations $n = 2$ and $m = 1$, the disconnected terms cancel, yielding

$$\text{tr}(Y\mathcal{E}_A^2(Z)) = \text{tr}(Y\mathcal{E}_A(Z))$$

for all Y . Nondegeneracy of the trace pairing gives $\mathcal{E}_A^2(Z) = \mathcal{E}_A(Z)$ for all Z , concluding $\mathcal{E}_A^2 = \mathcal{E}_A$. $\square$

Theorem 17.5.20 (Redundancy of the one-block auxiliary condition). [Lean](#) [✓ Stmt](#) A tensor satisfies the virtual-insertion zero-correlation-length condition (Definition 17.5.18) if and only if its transfer map is idempotent:

$$\text{IsZCL}(A) \iff \mathcal{E}_A^2 = \mathcal{E}_A.$$

Proof. [✓ Proof](#) The forward direction is immediate from Definition 17.5.18. Conversely, idempotence gives $\mathcal{E}_A^n = \mathcal{E}_A$ for all $n \geq 1$. For every positive-definite right fixed point ρ^R , all virtual matrices X, Y , and all $n \geq 1$,

$$\begin{aligned} C(X, Y; n) &= \text{tr}(Y\mathcal{E}_A^n(X\rho^R)) - \text{tr}(X\rho^R)\text{tr}(Y\rho^R) \\ &= \text{tr}(Y\mathcal{E}_A(X\rho^R)) - \text{tr}(X\rho^R)\text{tr}(Y\rho^R) \\ &= C(X, Y; 1). \end{aligned}$$

Hence connected correlators are independent of distance. $\square$

Chapter 18

Matrix Product Unitaries

Matrix product unitaries describe translation-invariant unitary operators generated by local tensor networks [15]. The nil-matrix bounds of Theorem 18.1 and Theorem 18.2 establish the blocking lengths required for structural reduction.

Theorem 18.1 (Dimension bound for a nilpotent Lie action). Lean ✓ Stmt *Let a Lie algebra L act nilpotently on a finite-dimensional vector space V over a field K . Write $C_r(L, V)$ for the r th term of the lower central series of this action. Then*

$$C_{\dim_K(V)}(L, V) = 0. \quad (18.1)$$

Proof. ✓ Proof As long as $C_r(L, V)$ is nonzero, nilpotency prevents two consecutive terms from being equal: equality would make every later term equal, in contradiction with eventual vanishing. Thus each nonzero step strictly lowers dimension, and at most $\dim_K(V)$ such steps are possible. $\square$

Theorem 18.2 (Finite-dimensional nil-matrix theorem). Lean ✓ Stmt *Let V be a finite-dimensional vector space over a field K , and let $A \subseteq \text{End}_K(V)$ be a not necessarily unital associative subalgebra. If every element of A is nilpotent, then every ordered product of $\dim_K(V)$ elements of A is zero. The empty product in dimension zero is also zero because the identity endomorphism of the zero space is zero.*

Proof. ✓ Proof Regard A as a Lie algebra under the commutator bracket. Engel's theorem makes V a nilpotent Lie module, so the dimension bound gives

$$C_{\dim_K(V)}(A, V) = 0. \quad (18.2)$$

For every nonnegative integer r , every ordered family $a_1, \dots, a_r$ in A , and every vector v in V , induction on r gives

$$(a_1 \cdots a_r)v \in C_r(A, V). \quad (18.3)$$

Taking $r = \dim_K(V)$ proves that the product annihilates every vector and is therefore zero. $\square$

Theorem 18.3 (Nil matrix subalgebras). Lean ✓ Stmt *Let A be a not necessarily unital associative subalgebra of $n \times n$ matrices over a field. If every matrix in A is nilpotent, then every ordered product of n matrices from A vanishes.*

Proof. ✓ Proof Transport the matrix algebra to the endomorphisms of the coordinate space through the standard algebra equivalence. Nilpotency, membership, and the ordered product are preserved, so the finite-dimensional nil-matrix theorem applies; injectivity of the equivalence returns the zero matrix. $\square$

The bound n is sharp for strictly upper-triangular matrices, correcting the intermediate strict inequality in [15]. Applied once to an algebra of $D^2 \times D^2$ matrices, it gives the blocking length D^2 of Theorem 18.3.24.

18.1 The matrix product unitary condition

Let $\mathcal{U} = \{U^{ij}\}_{i,j=0}^{d-1}$ be a matrix product operator tensor of bond dimension D , and write $U^{(N)}$ for the periodic operator obtained by closing a chain of N copies of $\mathcal{U}$.

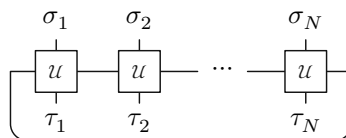
Theorem 18.1.1 (Unitarity under simultaneous reindexing). Lean ✓ Stmt Let I and J be finite sets, let $e: I \rightarrow J$ be a bijection, and let A be a unitary matrix indexed by I . The matrix indexed by J whose (j, k) entry is

$$A_{e^{-1}(j), e^{-1}(k)} \quad (18.4)$$

is unitary.

Proof. ✓ Proof Simultaneous reindexing preserves the identity and matrix multiplication, and it commutes with the conjugate transpose. Reindex the equation $AA^\dagger = \mathbb{1}$. $\square$

Definition 18.1.2 (Matrix product unitary tensor). Lean Lean ✓ Stmt The tensor $\mathcal{U}$ generates matrix product unitaries if $U^{(N)}$ is unitary for every integer $N > 1$. This is the matrix product unitary condition in [15, Section III, equation (10)].



Lemma 18.1.3 (First matrix product unitarity equation). Lean ✓ Stmt If $\mathcal{U}$ generates matrix product unitaries and $N > 1$, then

$$U^{(N)}(U^{(N)})^\dagger = \mathbb{1}. \quad (18.5)$$

Proof. ✓ Proof Expand the unitary-group condition for $U^{(N)}$. $\square$

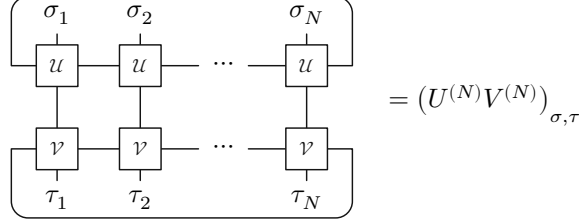
Lemma 18.1.4 (Second matrix product unitarity equation). Lean ✓ Stmt If $\mathcal{U}$ generates matrix product unitaries and $N > 1$, then

$$(U^{(N)})^\dagger U^{(N)} = \mathbb{1}. \quad (18.6)$$

This is the unitarity equation displayed in [15, Section III, equation (12)].

Proof. ✓ Proof Use the equivalent left-multiplied characterization of a unitary matrix. $\square$

Theorem 18.1.5 (Composition preserves matrix product unitarity). Lean ✓ Stmt Let $\mathcal{U}$ and $\mathcal{V}$ be matrix product unitary tensors with the same physical dimension. Contracting the output index of $\mathcal{U}$ with the input index of $\mathcal{V}$ gives a matrix product unitary tensor whose periodic operator is $U^{(N)}V^{(N)}$, in that order. This is the concatenation tensor $\mathcal{W}$ in the proof of [15, Theorem IV.6(ii)].



Proof. ✓ Proof For every $N > 1$, both $U^{(N)}$ and $V^{(N)}$ are unitary. The defining contraction of the product tensor gives the periodic operator $U^{(N)}V^{(N)}$, which is unitary because the unitary matrices form a group under multiplication. $\square$

Lemma 18.1.6 (Physical blocking preserves matrix product unitarity). Lean ✓ Stmt Suppose that $\mathcal{U}$ generates matrix product unitaries. For every positive integer L , the tensor $\mathcal{U}^{[L]}$ obtained by blocking L adjacent sites also generates matrix product unitaries. This is the blocking operation of [15, Definition II.5 and equation (9)]; preservation of the MPU property under blocking is used in the proof of [15, Proposition III.3].

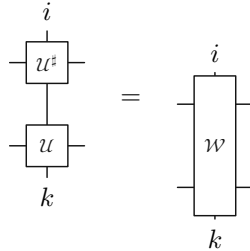
Proof. ✓ Proof At system size N , closing the blocked tensor gives $U^{(NL)}$, up to the canonical bijection between N blocked configurations and NL unblocked configurations. If $N > 1$ and $L > 0$, then $NL > 1$, so $U^{(NL)}$ is unitary. Simultaneous reindexing preserves unitarity. $\square$

18.2 Simple matrix product unitary tensors

Definition 18.2.1 (Double layer). Lean Lean ✓ Stmt For an MPO tensor $\mathcal{U}$, let $\mathcal{W}$ be the double-layer tensor obtained by placing the physical adjoint above $\mathcal{U}$ and contracting the adjacent physical legs:

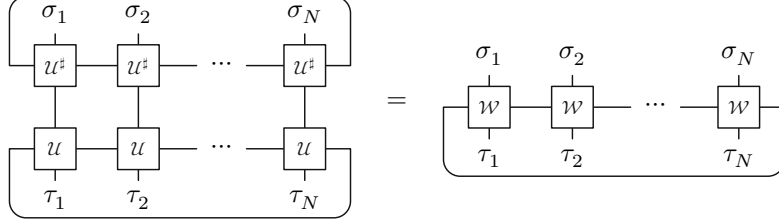
$$\mathcal{W}^{ik} = \sum_j (\mathcal{U}^\sharp)^{ij} \otimes \mathcal{U}^{jk}. \quad (18.7)$$

The bond space of $\mathcal{W}$ is the product of the two original bond spaces. The physical adjoint exchanges and conjugates the physical indices without reversing the order of the virtual chain, exactly as in the barred upper layer of Figures II_Simple1.png and II_Simple2.png of [15].



Theorem 18.2.2 (Closed double layer). [Lean](#) [✓ Stmt](#) For every chain length N , closing the double layer gives

$$W^{(N)} = (U^{(N)})^\dagger U^{(N)}. \quad (18.8)$$



Proof. [✓ Proof](#) The operator family is multiplicative under the local product tensor, and the physical adjoint generates the conjugate transpose without reflecting the chain. $\square$

Theorem 18.2.3 (Physical-adjoint identities for matrix product unitaries). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The physical adjoint is involutive. If $\mathcal{U}$ generates matrix product unitaries, then so does $\mathcal{U}^\sharp$, and for every chain length N ,

$$(\mathcal{U}^\sharp)^\sharp = \mathcal{U}, \quad (\mathcal{U}^\sharp)^{(N)} = (\mathcal{U}^{(N)})^\dagger. \quad (18.9)$$

Proof. [✓ Proof](#) Applying the physical adjoint twice conjugates every entry twice and exchanges the physical indices twice. Closing a physical-adjoint tensor exchanges the two physical configurations and conjugates the resulting scalar. Hence its periodic operator is the conjugate transpose. The first MPU unitarity equation for $U^{(N)}$ is therefore the second unitarity equation for $(U^\sharp)^{(N)}$. $\square$

Theorem 18.2.4 (Output-first double-layer entry). [Lean](#) [✓ Stmt](#) Order the left doubled bond as (α, γ) and the right doubled bond as (β, δ) . Then the double layer whose lower tensor is $\mathcal{U}^\sharp$ has entries

$$\mathcal{W}(\mathcal{U}^\sharp)_{(\alpha, \gamma), (\beta, \delta)}^{ik} = \sum_j u_{\alpha, \beta}^{ij} \overline{u_{\gamma, \delta}^{kj}}. \quad (18.10)$$

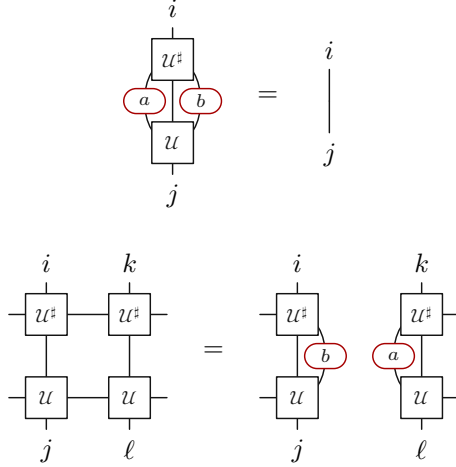
Proof. [✓ Proof](#) Expand the double layer and then apply the physical adjoint to its upper factor. The two exchanges of physical indices leave the contracted index j in the second physical position of both factors. The product-index convention keeps (α, γ) and (β, δ) in the displayed order. $\square$

Definition 18.2.5 (Simple matrix product unitary tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The tensor $\mathcal{U}$ is *simple* if there are vectors a, b in the double-layer bond space such that, for all physical indices i, j, k, ℓ ,

$$(a|W^{ij}|b) = \delta_{ij}, \quad (\text{simple1})$$

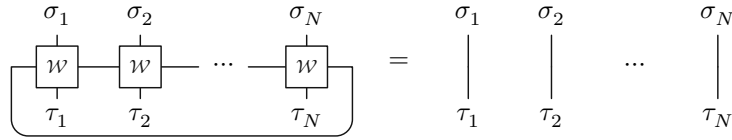
$$W^{ij}W^{k\ell} = W^{ij}|b)(a|W^{k\ell}. \quad (\text{simple2})$$

Thus (simple1) closes one double-layer letter to the physical identity, while (simple2) inserts the rank-one operator $|b)(a|$ between adjacent letters. These are precisely the two diagrams in [15, Definition III.2].



Theorem 18.2.6 (Sequential contraction of a simple double layer). [Lean](#) [✓ Stmt](#) If $\mathcal{U}$ is simple and $N > 1$, then

$$W^{(N)} = \mathbb{1}. \quad (18.11)$$



Proof. [✓ Proof](#) Apply [\(simple2\)](#) sequentially between neighboring letters. The inserted rank-one operators split the periodic contraction into the product of the one-letter contractions $(a|W^{\sigma_x \tau_x}|b)$. Identity [\(simple1\)](#) turns this product into $\prod_x \delta_{\sigma_x, \tau_x}$, which is the matrix element of the physical identity. $\square$

Theorem 18.2.7 (A simple tensor generates matrix product unitaries). [Lean](#) [✓ Stmt](#) Every simple tensor generates matrix product unitaries. This is [\[15, Proposition III.3\(i\)\]](#).

Proof. [✓ Proof](#) For every $N > 1$, the closed-double-layer identity and the sequential contraction give

$$(U^{(N)})^\dagger U^{(N)} = \mathbb{1}. \quad (18.12)$$

Hence $U^{(N)}$ is unitary. $\square$

18.3 Double-layer blocking and mixed physical traces

Theorem 18.3.1 (Physical adjoints and double layers commute with blocking). [Lean](#) [Lean](#) [✓ Stmt](#) For every blocking length L , physical adjunction commutes with blocking, and consequently

$$(\mathcal{U}^\dagger)^{[L]} = (\mathcal{U}^{[L]})^\dagger, \quad (18.13)$$

$$\mathcal{W}(\mathcal{U}^{[L]}) = \mathcal{W}(\mathcal{U})^{[L]}. \quad (18.14)$$

The virtual order is unchanged on both sides.

Proof. ✓ **Proof** Identity (18.13) holds entrywise since physical adjunction conjugates each matrix entry without reversing the virtual word; (18.14) follows because physical blocking commutes with the local operator product. $\square$

Theorem 18.3.2 (Blocked double-layer diagonal is a transfer power). Lean Lean Lean Lean
✓ **Stmt** Assume $d > 0$, and let E be the normalized transfer matrix of $\mathcal{U}$. After blocking L sites, the normalized physical diagonal of the double layer is exactly E^L . More precisely, if

$$q : \{0, \dots, D-1\} \times \{0, \dots, D-1\} \longrightarrow \{0, \dots, D^2-1\} \quad (18.15)$$

is the standard product encoding, then its entries are

$$(E^L)_{q^{-1}(a), q^{-1}(b)}. \quad (18.16)$$

Thus the row and column reindexings have the same explicit orientation. If $\mathcal{U}$ is an MPU and $D > 1$, then at the stabilization exponent J this diagonal is the same reindexing of the rank-one projector $|\rho\rangle\langle\Phi|$.

Proof. ✓ **Proof** On one site, expanding the diagonal double-layer contraction gives the transfer matrix after the stated product reindexing. The factor d^{-1} is the square of the local factor $d^{-1/2}$. Summing blocked diagonal letters over words and distributing the ordered product shows that normalized diagonals turn blocking into powers. Simultaneous reindexing preserves multiplication, giving the displayed formula. Transfer stabilization gives the final rank-one specialization. $\square$

Theorem 18.3.3 (Normalized diagonal as a diagonal-word sum). Lean ✓ **Stmt** Let $\mathcal{W}$ be an MPO tensor of physical dimension d , and let

$$R = \frac{1}{d} \sum_{i=0}^{d-1} W^{ii} \quad (18.17)$$

be its normalized physical diagonal. Here and in the two diagonal-tail identities below, scalar inverses are the totalized complex-field inverses, so $0^{-1} = 0$. For every $K \geq 0$,

$$R^K = d^{-K} \sum_{\tau \in \{0, \dots, d-1\}^K} W^{\tau_0 \tau_0} \dots W^{\tau_{K-1} \tau_{K-1}}. \quad (18.18)$$

At $K = 0$, the word product is the identity matrix. This is the algebraic diagonal-tail expansion used in the source-factor networks of [15, Theorem III.8, Section III.B, equations (31)–(32)].

Proof. ✓ **Proof** Expand the K th power of the finite sum defining R . Distributivity indexes the resulting terms by physical words τ , and each term is the ordered diagonal word displayed above. $\square$

Theorem 18.3.4 (A matrix basis adapted to the trace). Lean ✓ **Stmt** If $d > 0$, the identity matrix extends to a basis of $M_d(\mathbb{C})$ whose other elements are traceless.

Proof. ✓ **Proof** The trace functional is nonzero on the identity. Extend the identity by a basis of the kernel of the trace. $\square$

Theorem 18.3.5 (Identity plus traceless physical decomposition). [Lean](#) [Lean](#) [✓ Stmt](#) Assume $d > 0$. For every double-layer tensor $\mathcal{W}$, there is a basis $\{\sigma_0, \sigma_\alpha\}$ of the physical matrix algebra and virtual matrices E, S_α such that $\sigma_0 = \mathbb{1}$ and, for every nonzero index α , $\text{tr}(\sigma_\alpha) = 0$, while

$$\mathcal{W} = E \otimes \mathbb{1} + \sum_{\alpha \neq 0} S_\alpha \otimes \sigma_\alpha. \quad (18.19)$$

Here E is the normalized physical diagonal. Equivalently, contraction against a physical matrix X separates as

$$\text{contr}_{\text{phys}}(\mathcal{W}, X) = \text{tr}(X)E + R_{\mathcal{W}}(X), \quad (18.20)$$

$$R_{\mathcal{W}}(X) = \text{contr}_{\text{phys}}\left(\mathcal{W}, X - \frac{\text{tr}(X)}{d}\mathbb{1}\right), \quad (18.21)$$

where the second physical argument is traceless. This is equation *WIsom* in [15, Proposition III.3(ii)].

Proof. [✓ Proof](#) The trace functional is nonzero on the identity. Extend the identity to a basis adapted to the kernel of the trace; all remaining basis vectors are therefore traceless. The bilinear pairing $(X, Y) \mapsto \text{tr}(XY)$ is nondegenerate. Its dual basis gives the virtual coefficients by physical contraction. The dual vector paired with the identity is $d^{-1}\mathbb{1}$, so its virtual coefficient is precisely E . Expanding matrix units in the dual basis yields the decomposition. Subtracting the trace component gives the equivalent residual formula. $\square$

Theorem 18.3.6 (Multilinear physical contraction). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose a length- N closed tensor equals the physical identity. For arbitrary physical matrices $X_0, \dots, X_{N-1}$,

$$\text{tr}\left(\prod_{k=0}^{N-1} \text{contr}_{\text{phys}}(\mathcal{W}, X_k)\right) = \prod_{k=0}^{N-1} \text{tr}(X_k). \quad (18.22)$$

In particular this applies to the double layer of an MPU whenever $N > 1$.

Proof. [✓ Proof](#) Expand each contraction and distribute the ordered product. The closed tensor coefficient is a Kronecker delta between the two physical words, so only equal words survive. The remaining independent diagonal sums factor into the product of the physical traces. $\square$

Theorem 18.3.7 (Positive-length mixed residual traces). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU with physical dimension $d > 0$. Every closed product of positive length that contains at least one residual coefficient has zero trace. In particular, this includes a single residual coefficient. This corresponds to the mixed-trace property in [15, Proposition III.3(ii)].

Proof. [✓ Proof](#) At lengths greater than one, represent identity factors by $d^{-1}\mathbb{1}$ and residual factors by traceless physical arguments. The multilinear contraction identity then vanishes. For a single residual coefficient A , repeat the same coefficient at every length $q > 1$ to obtain $\text{tr}(A^q) = 0$. Thus the one-letter case is recovered after the longer words: the shifted zero-trace-power theorem makes A nilpotent, and therefore $\text{tr}(A) = 0$. $\square$

Theorem 18.3.8 (Controlled mixed residual traces and rank-one sandwich). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU with physical dimension $d > 0$, and let $N > 1$. Every closed product of N identity and residual coefficients that contains at least one residual coefficient has zero trace. If the normalized diagonal is rank one, then every nonempty residual word P satisfies

$$EPE = 0. \quad (18.23)$$

The same conclusion holds after blocking $J > 0$ sites whenever the J th normalized transfer power is $|\rho\rangle\langle\Phi|$, with both virtual indices reindexed by $q^{-1} : \{0, \dots, D^2-1\} \rightarrow \{0, \dots, D-1\} \times \{0, \dots, D-1\}$.

Proof. ✓ Proof Represent identity factors by $d^{-1}\mathbb{1}$ and residual factors by their traceless physical arguments. Since $N > 1$, the multilinear contraction identity applies, and one trace factor vanishes. For a nonempty residual word, adjoin E as one extra factor; its closed trace has length at least two and therefore vanishes. Cyclicity of trace and

$$|\rho\rangle\langle\Phi|P|\rho\rangle\langle\Phi| = \text{tr}(P|\rho\rangle\langle\Phi|)|\rho\rangle\langle\Phi| \quad (18.24)$$

give $EPE = 0$. The blocked statement follows from the transfer-power formula with the same product-index reindexing in both matrix indices. $\square$

Definition 18.3.9 (Residual nonunital algebra). Lean Lean Lean ✓ Stmt Let $\mathcal{U}$ have bond dimension D . Contract the double layer against arbitrary physical matrices, subtract the normalized diagonal coefficient, and let $\mathcal{S}$ be the resulting set of residual matrices in $M_{D^2}(\mathbb{C})$. The residual algebra $\mathfrak{S}$ is the not necessarily unital algebra generated by $\mathcal{S}$. This is the algebra generated by the matrices S'_α in [15, Proposition III.3(ii)].

Theorem 18.3.10 (Trace and nilpotence of the residual algebra). Lean Lean Lean Lean ✓ Stmt If $\mathcal{U}$ is an MPU with positive physical dimension, then every nonempty product of residual generators has zero trace. Consequently, for every $A \in \mathfrak{S}$ and every integer $N > 1$,

$$\text{tr}(A) = 0, \quad \text{tr}(A^N) = 0, \quad (18.25)$$

and every element of $\mathfrak{S}$ is nilpotent.

Proof. ✓ Proof A nonempty product in the multiplicative semigroup generated by $\mathcal{S}$ is a nonempty residual word, so its trace vanishes by the positive-length mixed residual identity. The generated nonunital algebra is the linear span of this semigroup. Linearity of the trace therefore gives $\text{tr}(A) = 0$ for every $A \in \mathfrak{S}$. Since every positive power of A remains in $\mathfrak{S}$, the same argument gives $\text{tr}(A^N) = 0$ for $N > 1$. The shifted zero-trace-power theorem now implies that A is nilpotent. $\square$

Theorem 18.3.11 (Length- D^2 residual products vanish). Lean Lean ✓ Stmt Let $\mathcal{U}$ be an MPU of bond dimension D and positive physical dimension. Every ordered product of exactly D^2 elements of $\mathfrak{S}$ vanishes. In particular, for arbitrary residual generators,

$$S_{\alpha_1} \cdots S_{\alpha_{D^2}} = 0. \quad (18.26)$$

This is equation **Prime=0** in [15, Proposition III.3(ii)], with the corrected exact length D^2 in place of the unsupported strict intermediate bound.

Proof. ✓ Proof The algebra $\mathfrak{S}$ acts on a space of dimension D^2 , and every element is nilpotent. The finite-dimensional nil-matrix theorem therefore annihilates every ordered product of D^2 elements. Each residual generator belongs to $\mathfrak{S}$, giving the stated specialization. $\square$

Definition 18.3.12 (Three-form residual subspace). Lean Lean Lean Lean ✓ Stmt Let E be the normalized diagonal after the first blocking and let $\mathfrak{S}$ be its residual nonunital algebra. Define

$$\mathcal{T} = \text{span}\{EA : A \in \mathfrak{S}\} + \text{span}\{AE : A \in \mathfrak{S}\} \quad (18.27)$$

$$+ \text{span}\{AEB : A, B \in \mathfrak{S}\}. \quad (18.28)$$

This is the basis-independent sum of the three forms in equation **primeforms** of [15, Proposition III.3(ii)].

Theorem 18.3.13 (Ordered second-block expansion). [Lean](#) [✓ Stmt](#) Suppose $E^2 = E$, $EAE = 0$ for every $A \in \mathfrak{S}$, and every ordered product of $N > 0$ elements of $\mathfrak{S}$ vanishes. For scalars c_k and elements $S_k \in \mathfrak{S}$,

$$\prod_{k=0}^{N-1} (c_k E + S_k) - \left(\prod_{k=0}^{N-1} c_k \right) E \in \mathcal{T}. \quad (18.29)$$

Products are taken in increasing site order and need not commute.

Proof. [✓ Proof](#) Expand the product as a sum over binary choices of E or S_k , retaining the original order. The all- E choice is the subtracted term because $E^q = E$ for $q > 0$. The all-residual choice vanishes by the length- N hypothesis. In every remaining word, contiguous copies of E collapse. If two resulting copies of E are separated by a nonempty residual word, the factor EAE makes the word zero. Thus each surviving word has one contiguous projector block and belongs to one of the three summands of $\mathcal{T}$. $\square$

Theorem 18.3.14 (Second-block residuals lie in the three-form subspace). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let the first-block tensor be an MPU of bond dimension $D > 0$. Assume its normalized double-layer diagonal E is a rank-one idempotent. After a second blocking of exactly D^2 sites, every residual coefficient belongs to $\mathcal{T}$.

The conclusion is membership in the sum of the three subspaces. It does not assign one exclusive type to a coefficient in an arbitrary physical basis.

Proof. [✓ Proof](#) First extend $EPE = 0$ from nonempty residual words P to every element of $\mathfrak{S}$ by writing the generated algebra as the linear span of the multiplicative semigroup of residual generators. For a blocked matrix unit, each local double-layer entry is

$$\delta_{ij} E + R_{ij}. \quad (18.30)$$

The ordered expansion theorem applies with $N = D^2$. Its all- E coefficient is the blocked Kronecker delta and is exactly removed by the residual slice. Hence every blocked matrix-unit residual lies in $\mathcal{T}$. Expanding an arbitrary physical matrix in matrix units and using linearity proves the result. $\square$

Theorem 18.3.15 (Three-form contraction identities). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose $E^2 = E$ and $EAE = 0$ for every $A \in \mathfrak{S}$. For all $X, Y \in \mathcal{T}$,

$$XY = XEY, \quad \text{tr}(XE) = 0. \quad (18.31)$$

Proof. [✓ Proof](#) It is enough by bilinearity and linearity of trace to consider the generators EA , AE , and AEB of the three summands. Representative products reduce as

$$(EA)(BE) = E(AB)E = 0, \quad (18.32)$$

$$(AE)(EB) = AEB = (AE)E(EB), \quad (18.33)$$

$$\text{tr}((AEB)E) = \text{tr}(A(EBE)) = 0. \quad (18.34)$$

The remaining generator pairs follow by the same use of associativity, $E^2 = E$, and $EAE = 0$ for $A \in \mathfrak{S}$. $\square$

Theorem 18.3.16 (Exact simplicity after the second blocking). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{V}$ be an MPU of positive physical and bond dimensions whose normalized double-layer diagonal is

$$E = |\rho\rangle\langle\Phi|, \quad \langle\Phi|\rho\rangle = 1. \quad (18.35)$$

Then blocking $\mathcal{V}$ by exactly D^2 sites produces a simple tensor with witnesses $a = \Phi$ and $b = \rho$.

Proof. ✓ Proof The pairing normalization makes E idempotent. Write every blocked double-layer coefficient as $\delta_{ij}E + R_{ij}$, where $R_{ij} \in \mathcal{T}$. The trace identity for $R_{ij}E$ gives the diagonal contraction, and $R_{ij}R_{kl} = R_{ij}ER_{kl}$, together with $E^2 = E$, gives the rank-one insertion identity after expanding both coefficients. $\square$

Definition 18.3.17 (Physical-basis reindexing). Lean Lean ✓ Stmt For a bijection e between two physical index sets, the reindexed tensor and its action on length- N configurations are

$$(e^*\mathcal{U})^{ij} = \mathcal{U}^{e(i)e(j)}, \quad (e_*\sigma)(n) = e(\sigma(n)). \quad (18.36)$$

Theorem 18.3.18 (Words under physical reindexing). Lean ✓ Stmt For ket and bra words i, j , physical reindexing gives

$$(e^*\mathcal{U})[i, j] = \mathcal{U}[e_*i, e_*j]. \quad (18.37)$$

Proof. ✓ Proof Induct simultaneously along the two words. The empty pair gives the identity matrix, mismatched lengths give zero, and the nonempty case applies the defining equality to the first letters and the induction hypothesis to the tails. $\square$

Theorem 18.3.19 (Finite-size operators under physical reindexing). Lean ✓ Stmt At every length N , the sitewise configuration bijection identifies

$$(e^*\mathcal{U})^{(N)} \cong \mathcal{U}^{(N)}. \quad (18.38)$$

Proof. ✓ Proof Apply the word identity to every pair of configurations and take the virtual trace. This is simultaneous row and column reindexing by e_* . $\square$

Theorem 18.3.20 (Double layer under physical reindexing). Lean ✓ Stmt For every physical-index bijection e , taking the double layer commutes with physical reindexing:

$$W(e^*\mathcal{U})^{ij} = W(\mathcal{U})^{e(i)e(j)}. \quad (18.39)$$

Proof. ✓ Proof This follows entrywise from the definitions of physical reindexing and the double layer. $\square$

Lemma 18.3.21 (Iterated physical blocking). Lean ✓ Stmt Let e be the canonical bijection which groups a word of length mn into n consecutive words of length m . Iterating blocking at lengths m and n , then applying this physical reindexing, agrees with direct blocking at length mn :

$$e^*((\mathcal{U}_m)_n) = \mathcal{U}_{mn}. \quad (18.40)$$

Proof. ✓ Proof Expand the two blocked words. The canonical regrouping concatenates their n words of length m into the direct word of length mn . $\square$

Definition 18.3.22 (Double-layer algebra). Lean Lean Lean ✓ Stmt Let $\mathcal{U}$ have bond dimension D . The double-layer algebra $\mathcal{A} \subseteq M_{D^2}(\mathbb{C})$ is the not necessarily unital algebra generated by the double-layer letters W^{ij} . It contains every W^{ij} and, as a linear combination of them, the normalized diagonal E .

Theorem 18.3.23 (The trace is a character on the double-layer algebra). Lean Lean Lean Lean ✓ Stmt Let $\mathcal{U}$ be an MPU of positive physical and bond dimensions, and let $\mathcal{A}$ be its double-layer algebra. The normalized diagonal E has trace one. Every nonempty word satisfies

$$\text{tr}(W^{i_1j_1} \dots W^{i_Nj_N}) = \prod_{q=1}^N \delta_{i_qj_q}. \quad (18.41)$$

Consequently $\text{tr}(ab) = \text{tr}(a)\text{tr}(b)$ for all $a, b \in \mathcal{A}$.

Proof. ✓ **Proof** Let $E^J = |\rho\rangle\langle\Phi| = p$ with $J > 0$, $\langle\Phi|\rho\rangle = 1$ and $E|\rho\rangle = |\rho\rangle$. Then $Ep = pE = p^2 = p$, so $(E - p)^N = E^N - p$ for $N > 0$ and $E - p$ is nilpotent. Hence $\text{tr } E = \text{tr } p = 1$. Writing $W^{ij} = \delta_{ij}E + R_{ij}$ with a traceless residual R_{ij} gives the one-letter case. For $N > 1$, the trace of the word is a matrix entry of $(\mathcal{U}^{(N)})^\dagger \mathcal{U}^{(N)} = \mathbb{1}$. The algebra $\mathcal{A}$ is the linear span of nonempty words, and the trace of a concatenated word is the product of the traces of its two parts; bilinearity gives multiplicativity on $\mathcal{A}$. $\square$

Theorem 18.3.24 (Simplicity after blocking D^2 sites). Lean Lean Lean Lean ✓ **Stmt** *Every MPU of positive physical and bond dimensions becomes simple after blocking exactly D^2 sites, and hence after blocking any $k \geq D^2$ sites. The witnesses at length D^2 are the normalized fixed vectors Φ and ρ of the transfer matrix. Simplicity is preserved under every further iterated blocking by a positive factor and under a bijective relabeling of the physical basis.*

The length D^2 is the one stated in [199, Theorem 2]; [15, Proposition III.3(ii)] gives a length below D^4 .

Proof. ✓ **Proof** Let $\mathcal{A}$ be the double-layer algebra, $p = E^J = |\rho\rangle\langle\Phi|$ the stabilized transfer power, and

$$\mathcal{N} = \{a \in \mathcal{A} : \text{tr } a = 0\}. \quad (18.42)$$

Since the trace is a character on $\mathcal{A}$, the set $\mathcal{N}$ is a subalgebra, and $\text{tr}(n^m) = \text{tr}(n)^m = 0$ for every $n \in \mathcal{N}$ and $m > 0$. Every element of $\mathcal{N}$ is therefore nilpotent, and by the nil-matrix theorem every product of D^2 elements of $\mathcal{N}$ vanishes. Moreover $p \in \mathcal{A}$ and $\text{tr } p = 1$, so

$$W^{ij} = \delta_{ij}p + n_{ij}, \quad n_{ij} \in \mathcal{N}, \quad (18.43)$$

$$pnp = \text{tr}(np)p = 0, \quad n \in \mathcal{N}. \quad (18.44)$$

The ordered expansion theorem, applied with p in place of the projector and $\mathcal{N}$ in place of the residual algebra, writes every blocked letter of length D^2 as $\delta_{IJ}p + t_{IJ}$ with t_{IJ} in the three-form subspace of p and $\mathcal{N}$. The three-form contraction identity $t t' = t p t'$ and $p^2 = p$ give

$$W^{IJ}W^{KL} = W^{IJ}|\rho\rangle\langle\Phi|W^{KL}. \quad (18.45)$$

MPU unitarity then gives the one-letter contraction with the same witnesses, and simplicity persists at every later blocking. $\square$

Theorem 18.3.25 (Matching contractions at the direct stabilized block). Lean ✓ **Stmt** *Let $\mathcal{U}$ be an MPU of positive physical and bond dimensions. Suppose that ρ has trace one and that, for some positive integer J , the normalized transfer matrix satisfies*

$$E^J = |\rho\rangle\langle\mathbb{1}|. \quad (18.46)$$

Put $K = JD^2$. After the standard product reindexing of the physical coordinates, the direct length- K block satisfies the contractions (simple1) and (simple2) with the specific witnesses ρ and $\mathbb{1}$ from this transfer-power identity.

Proof. ✓ **Proof** First block J sites. The normalized diagonal of this block is the reindexed rank-one matrix $|\rho\rangle\langle\mathbb{1}|$, and the trace-one condition is exactly $\langle\mathbb{1}|\rho\rangle = 1$. The exact second-block contraction theorem then supplies both identities after a further D^2 sites. The canonical equivalence between iterated blocking and direct length- JD^2 blocking transports each double-layer coefficient without changing either virtual witness. $\square$

Theorem 18.3.26 (Rank-one transfer power after the second blocking). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor with $d, D > 0$, let E be the transfer matrix of its normalized flattening, and suppose that for some $J \geq 0$ and some trace-one ρ ,

$$E^J = |\rho\rangle\langle 1|. \quad (18.47)$$

Then the normalized diagonal of the double layer of the direct length- JD^2 block is the same rank-one matrix, after the standard product reindexing of the doubled bond.

Proof. [✓ Proof](#) Trace normalization gives $\langle 1|\rho\rangle = 1$, so $|\rho\rangle\langle 1|$ is idempotent and every positive power of it equals itself. The blocked normalized diagonal is the corresponding power of E , and E^{JD^2} is the D^2 -th power of E^J . $\square$

Theorem 18.3.27 (Normalized input-tail isometry). [Lean](#) [✓ Stmt](#) Let $d > 0$, let $\mathcal{U}$ be an MPU, and let $K \geq 0$. For $p, q \in \{0, \dots, d-1\} \times \{0, \dots, d-1\}$, separate the first two input sites from a common input tail $\tau \in \{0, \dots, K-1\} \rightarrow \{0, \dots, d-1\}$. Then

$$d^{-K} \sum_{\tau} \sum_{\eta} \overline{U_{\eta, (p, \tau)}^{(K+2)}} U_{\eta, (q, \tau)}^{(K+2)} = \delta_{p, q}. \quad (18.48)$$

The global contraction is the input-first unitarity $(U^{(K+2)})^\dagger U^{(K+2)} = \mathbb{1}$. This is the input-oriented tail contraction supporting the network calculation in [15, Theorem III.8, Section III.B, equations (31)–(32)].

Proof. [✓ Proof](#) Apply input-first unitarity for each fixed tail. The inner output sum is $\delta_{p, q}$. Summing over the d^K tails and multiplying by d^{-K} leaves the same Kronecker delta. $\square$

Theorem 18.3.28 (Simplicity at every later direct blocking). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ generate an MPU. If specified vectors a, b satisfy the two-letter insertion identity

$$W^{ij}W^{k\ell} = W^{ij}|b\rangle\langle a|W^{k\ell}, \quad (18.49)$$

then the same vectors satisfy

$$(a|W^{ij}|b) = \delta_{ij}. \quad (18.50)$$

If the direct length- k block is simple for some $k > 0$, then its direct length- k' block is simple for every $k' \geq k$, without any divisibility condition on k' by k [15, Proposition III.3].

Proof. [✓ Proof](#) First suppose vectors a, b satisfy the rank-one insertion identity

$$W^{ij}W^{k\ell} = W^{ij}|b\rangle\langle a|W^{k\ell}. \quad (18.51)$$

Set $x_{ij} = (a|W^{ij}|b)$. Applying this identity around the constant two-site and three-site words, and then using MPU unitarity, gives

$$x_{ij}^2 = \delta_{ij}, \quad (18.52)$$

$$x_{ij}^3 = \delta_{ij}. \quad (18.53)$$

Hence $x_{ii} = 1$, while $x_{ij} = 0$ for $i \neq j$, so the same vectors obey the one-letter contraction $(a|W^{ij}|b) = \delta_{ij}$.

For the successor step, take four words I, J, K, L of length $k+1$ and write $W^{I,J} = W^{i_0 j_0} \dots W^{i_k j_k}$, with the analogous notation for $W^{K,L}$. Apply the length- k rank-one insertion identity to the suffixes $I_+ = (i_1, \dots, i_k)$, $J_+ = (j_1, \dots, j_k)$ and the prefixes $K_- = (k_0, \dots, k_{k-1})$, $L_- = (\ell_0, \dots, \ell_{k-1})$. Then

$$W^{I,J} W^{K,L} = W^{i_0 j_0} (W^{I_+, J_+} W^{K_-, L_-}) W^{k_k \ell_k} \quad (18.54)$$

$$= W^{I,J} |b\rangle \langle a| W^{K,L}. \quad (18.55)$$

Thus the length- $(k+1)$ block has the same insertion witnesses, and the two- and three-site argument gives their one-letter contraction. Induction on $k' - k$ proves the claim. $\square$

Lemma 18.3.29 (Common simple blocking for two or three MPU tensors). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}_1, \mathcal{U}_2$ be MPU tensors with positive physical and bond dimensions, which may differ between the tensors. There is a length $k > 0$ for which both direct length- k blockings are simple. The same conclusion holds for three such tensors $\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3$, with one common length k . This is a finite-family consequence of [15, Proposition III.3], not an additional hypothesis on the tensors.*

Proof. [✓ Proof](#) A tensor of bond dimension D is simple after blocking any $k \geq D^2$ sites. The sum of the squares of the bond dimensions is positive and bounds every square from above, so it is a common simple blocking length for two or for three tensors. $\square$

Theorem 18.3.30 (Common simple blocking along a continuous MPU path). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $d, D > 0$. Every bond- D MPU tensor becomes simple after blocking exactly D^4 sites. For each fixed L , the map*

$$\mathcal{U} \mapsto \mathcal{U}_L \quad (18.56)$$

is continuous. Consequently, if $x \mapsto \mathcal{W}(x)$ is a continuous family of bond- D MPU tensors, then $x \mapsto \mathcal{W}(x)_{D^4}$ is a continuous family of simple tensors. This provides the common-blocking step for Proposition 18.9.100.

Proof. [✓ Proof](#) Every tensor is simple after blocking any $k \geq D^2$ sites, in particular after blocking $D^4 \geq D^2$ sites. For fixed blocked ket and bra words, every matrix entry of the blocked tensor is a finite sum of products of entries of the original tensor. These entries are continuous, and the finite product and sum operations preserve continuity. Applying this fixed blocking map to the original path proves the path statement. $\square$

Theorem 18.3.31 (Supplied-projector simple2 contraction). [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ be simple with positive physical dimension, let W be its double layer, and let E be the normalized diagonal of W . Suppose that for some $J > 0$,*

$$E^J = |\rho\rangle \langle \Phi|. \quad (18.57)$$

Then the simple insertion identity holds with these same supplied witnesses:

$$W^{ij} W^{k\ell} = W^{ij} |\rho\rangle \langle \Phi| W^{k\ell}. \quad (18.58)$$

Proof. [✓ Proof](#) Choose the witnesses a, b from simplicity and put $R = |b\rangle \langle a|$. Expanding the normalized diagonal as the average of the diagonal letters, the two defining simple contractions give

$$W^{ij} E W^{k\ell} = W^{ij} W^{k\ell}. \quad (18.59)$$

Indeed, insert R on both sides of each diagonal letter; its middle sandwich is $RW^{qq}R = R$ by the first contraction, and the normalized sum contains d equal terms. The same identity makes $W^{ij}E^N = W^{ij}E$ for every $N > 0$. Hence

$$W^{ij}E^JW^{k\ell} = W^{ij}W^{k\ell}. \quad (18.60)$$

Substituting the supplied equality $E^J = |\rho\rangle(\Phi|$ proves the claim. No uniqueness of rank-one factorizations is used. $\square$

Theorem 18.3.32 (Fixed boundaries of a normalized rank-one power). [Lean](#) [✓ Stmt](#) *Let R be the normalized physical diagonal of the double layer of an MPO tensor. Suppose*

$$(\Phi|\rho) = 1, \quad R^J = |\rho\rangle(\Phi|. \quad (18.61)$$

Then the two boundaries of the stabilized diagonal tail are fixed:

$$R^J|\rho\rangle = |\rho\rangle, \quad (\Phi|R^J = (\Phi|. \quad (18.62)$$

This is the fixed-pair normalization from the canonical-form-II fixed-point equations in [15, equations (6a)–(6b)] and underlies the tail contractions in [15, Theorem III.8, Section III.B, equations (31)–(32)].

Proof. [✓ Proof](#) Substitute $R^J = |\rho\rangle(\Phi|$ in each expression and use $(\Phi|\rho) = 1$. $\square$

Theorem 18.3.33 (Rank-one insertion as a normalized diagonal tail). [Lean](#) [✓ Stmt](#) *Let W be the double layer of an MPO tensor, and let R be its normalized physical diagonal. If*

$$R^J = |\rho\rangle(\Phi|, \quad (18.63)$$

then, for all physical indices i, j, k, ℓ ,

$$W^{ij}|\rho\rangle(\Phi|W^{k\ell} = d^{-J} \sum_{\tau \in \{0, \dots, d-1\}^J} W^{ij}W^{\tau_0\tau_0} \dots W^{\tau_{J-1}\tau_{J-1}}W^{k\ell}. \quad (18.64)$$

This is the exact diagonal-word tail between the two retained letters in [15, Theorem III.8, Section III.B, equations (31)–(32)].

Proof. [✓ Proof](#) Replace the rank-one insertion by R^J , apply the diagonal-word expansion, and distribute the two retained letters across the normalized finite sum. $\square$

Theorem 18.3.34 (Simple two-letter product as a normalized diagonal tail). [Lean](#) [✓ Stmt](#) *In the setting of Theorem 18.3.33, suppose in addition that the same fixed pair satisfies the supplied two-letter contraction*

$$W^{ij}W^{k\ell} = W^{ij}|\rho\rangle(\Phi|W^{k\ell} \quad (18.65)$$

for all physical indices. Then

$$W^{ij}W^{k\ell} = d^{-J} \sum_{\tau \in \{0, \dots, d-1\}^J} W^{ij}W^{\tau_0\tau_0} \dots W^{\tau_{J-1}\tau_{J-1}}W^{k\ell}. \quad (18.66)$$

Thus the supplied insertion and its stabilized transfer power occur in one aligned network, as required in the calculation of [15, Theorem III.8, Section III.B, equations (31)–(32)].

Proof. ✓ Proof Apply the supplied two-letter contraction and then replace its rank-one insertion by the normalized diagonal-word tail. $\square$

Theorem 18.3.35 (The fixed-point simple1 contraction). Lean Lean ✓ Stmt *For every MPU of positive physical and bond dimensions there are normalized left and right fixed vectors Φ and ρ of the transfer matrix such that the unblocked double layer satisfies*

$$(\Phi|W^{ij}|\rho) = \delta_{ij}. \quad (18.67)$$

The unblocked tensor need not satisfy the rank-one insertion identity.

Proof. ✓ Proof Let $E^J = |\rho)(\Phi|$ be a stabilized transfer power. Decompose $W^{ij} = \delta_{ij}E + R_{ij}$. Fixedness and normalization give

$$\text{tr}(E|\rho)(\Phi|) = 1. \quad (18.68)$$

The mixed residual trace identity applied to one residual factor followed by J diagonal factors gives

$$\text{tr}(R_{ij}E^J) = 0. \quad (18.69)$$

Since $E^J = |\rho)(\Phi|$, these two trace identities are exactly the displayed contraction. $\square$

18.4 Trace-power lemma

In the transfer-matrix spectrum analysis of [15, Proposition 1], a matrix whose positive powers all have trace 1 has characteristic polynomial $X^{n-1}(X - 1)$. Assuming this trace condition only for powers with exponent strictly greater than one determines the nonzero spectrum as a set.

Definition 18.4.1 (Rank-one diagonal idempotent). Lean ✓ Stmt For an index i_0 in a finite set I of cardinality n , let M_{i_0} be the $n \times n$ diagonal matrix with a 1 at position i_0 and zeros elsewhere.

Lemma 18.4.2 (Properties of the rank-one diagonal idempotent). Lean Lean Lean Lean ✓ Stmt *The matrix M_{i_0} from definition 18.4.1 satisfies*

- $M_{i_0}^k = M_{i_0}$ for every $k \geq 1$,
- $\text{tr}(M_{i_0}^k) = 1$ for every $k \geq 1$,
- $\chi_{M_{i_0}}(X) = X^{n-1}(X - 1)$.

Proof. ✓ Proof The first two items follow from $1^k = 1$ and $0^k = 0$ for $k \geq 1$. The characteristic polynomial is computed from $\chi_{\text{diag}(d)}(X) = \prod_i (X - d_i)$. $\square$

Theorem 18.4.3 (Finite-range trace-power lemma). Lean ✓ Stmt *Let A be an $n \times n$ complex matrix with $n \geq 1$. If $\text{tr}(A^k) = 1$ for every k with $1 \leq k \leq n$, then*

$$\chi_A(X) = X^{n-1}(X - 1). \quad (18.70)$$

Proof. ✓ Proof Choose any index i_0 (the hypothesis $n \geq 1$ makes this possible) and let B be the rank-one diagonal idempotent from Lemma 18.4.2. For $1 \leq k \leq n$ we have $\text{tr}(A^k) = 1 = \text{tr}(B^k)$. The finite-range half of the Newton–Girard the corresponding theorem in the companion quantum-channel volume [182, “Newton–Girard trace recursion”] then yields $\chi_A = \chi_B = X^{n-1}(X - 1)$. $\square$

Theorem 18.4.4 (All-positive-powers trace-power lemma). [Lean](#) [✓ Stmt](#) Let A be an $n \times n$ complex matrix. If $\operatorname{tr}(A^k) = 1$ for every $k \geq 1$, then

$$\chi_A(X) = X^{n-1}(X - 1). \quad (18.71)$$

(The hypothesis already forces $n \geq 1$ because the trace of a 0×0 matrix is 0, contradicting $\operatorname{tr}(A) = 1$.)

Proof. [✓ Proof](#) If $n = 0$ the hypothesis $\operatorname{tr}(A) = 1$ contradicts the fact that the trace of an empty matrix is 0; the conclusion follows vacuously. When $n \geq 1$, the hypothesis on all positive powers implies the finite-range hypothesis of Theorem 18.4.3. $\square$

Theorem 18.4.5 (Shifted moments under matrix powers). [Lean](#) [✓ Stmt](#) Let A be an $n \times n$ complex matrix such that $\operatorname{tr}(A^k) = 1$ for every $k > 1$, and let $m > 1$. Then $\operatorname{tr}((A^m)^r) = 1$ for every $r \geq 1$.

Proof. [✓ Proof](#) For $r \geq 1$, one has $(A^m)^r = A^{mr}$ and $mr > 1$, so the hypothesis at exponent mr gives the result. $\square$

Theorem 18.4.6 (Trace powers above one determine the nonzero spectrum). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an $n \times n$ complex matrix. If $\operatorname{tr}(A^k) = 1$ for every integer $k > 1$, then

$$\sigma(A) \setminus \{0\} = \{1\}. \quad (18.72)$$

Thus $1 \in \sigma(A)$ and every nonzero spectral value equals 1 [15, Proposition III.1].

Proof. [✓ Proof](#) For every positive integer r , the shifted moments give

$$\operatorname{tr}((A^2)^r) = \operatorname{tr}(A^{2r}) = 1, \quad \operatorname{tr}((A^3)^r) = \operatorname{tr}(A^{3r}) = 1. \quad (18.73)$$

Hence

$$\chi_{A^2}(X) = \chi_{A^3}(X) = X^{n-1}(X - 1). \quad (18.74)$$

If $\mu \in \sigma(A)$ is nonzero, spectral mapping places μ^2 in $\sigma(A^2)$ and μ^3 in $\sigma(A^3)$, so

$$\mu^2 = \mu^3 = 1, \quad \mu = 1. \quad (18.75)$$

Conversely, spectral mapping from $1 \in \sigma(A^2)$ supplies a nonzero spectral value of A , which must be 1. $\square$

Theorem 18.4.7 (Generalized eigenspace inclusion under squaring). [Lean](#) [✓ Stmt](#) For a complex square matrix A , write $G_1(A)$ for its maximal generalized eigenspace at the eigenvalue 1. Then

$$G_1(A) \subseteq G_1(A^2). \quad (18.76)$$

Proof. [✓ Proof](#) The commuting factorization

$$A^2 - \mathbb{1} = (A + \mathbb{1})(A - \mathbb{1}) \quad (18.77)$$

gives, for every nonnegative integer k ,

$$(A^2 - \mathbb{1})^k = (A + \mathbb{1})^k (A - \mathbb{1})^k. \quad (18.78)$$

Therefore every vector annihilated by a power of $A - \mathbb{1}$ is annihilated by the corresponding power of $A^2 - \mathbb{1}$. $\square$

Theorem 18.4.8 (Multiplicity at one does not decrease under squaring). [Lean](#) [✓ Stmt](#) For every complex square matrix A ,

$$\text{mult}_1(\chi_A) \leq \text{mult}_1(\chi_{A^2}). \quad (18.79)$$

Proof. [✓ Proof](#) With G_1 as above, algebraic multiplicity equals the dimension of the maximal generalized eigenspace. Hence

$$\text{mult}_1(\chi_A) = \dim G_1(A) \quad (18.80)$$

$$\leq \dim G_1(A^2) = \text{mult}_1(\chi_{A^2}). \quad (18.81)$$

□

Theorem 18.4.9 (Trace powers above one determine the characteristic polynomial). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an $n \times n$ complex matrix. If $\text{tr}(A^k) = 1$ for every integer $k > 1$, then

$$\text{mult}_1(\chi_A) = 1, \quad \chi_A(X) = X^{n-1}(X - 1). \quad (18.82)$$

Thus 1 is the sole nonzero eigenvalue and has algebraic multiplicity one. This is the exact shifted-moment conclusion of [15, Proposition III.1].

Proof. [✓ Proof](#) Applying the all-positive trace-power theorem to A^2 gives

$$\chi_{A^2}(X) = X^{n-1}(X - 1), \quad \text{mult}_1(\chi_{A^2}) = 1. \quad (18.83)$$

The multiplicity inequality therefore bounds $\text{mult}_1(\chi_A)$ by one. The set-spectrum theorem shows that 1 occurs, so this multiplicity is positive and hence equals one. The same set-spectrum theorem excludes every other nonzero root. Splitting the monic characteristic polynomial now yields

$$\text{mult}_1(\chi_A) = 1, \quad \chi_A(X) = X^{n-1}(X - 1). \quad (18.84)$$

□

Theorem 18.4.10 (Semisimple rank-one Jordan–Chevalley reduction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let V be a finite-dimensional complex vector space and let S be a semisimple endomorphism of V . Suppose that $c \in \mathbb{C}$ is nonzero and

$$\text{tr}((cS)^k) = 1 \quad (18.85)$$

for every integer $k > 1$. Then

$$\dim \text{range}(S) = 1. \quad (18.86)$$

If N is nilpotent and commutes with S , then, for every positive integer k ,

$$\text{tr}((N + S)^k) = \text{tr}(S^k), \quad (18.87)$$

$$NS = SN = 0. \quad (18.88)$$

There is an integer $J > 0$ such that, for every $k \geq J$,

$$(N + S)^k = S^k. \quad (18.89)$$

This is the generic Jordan–Chevalley step in [187, Proposition 20, lines 3839–3866].

Proof. [✓ Proof](#) Choose an arbitrary basis of V and represent cS by a matrix A . The shifted trace-power theorem gives

$$\chi_A(X) = X^{\dim V-1}(X-1). \quad (18.90)$$

Since cS is semisimple, its generalized zero-eigenspace is its ordinary zero-eigenspace, namely its kernel. The dimension of the generalized zero-eigenspace is the root multiplicity of zero in the characteristic polynomial, so

$$\dim \ker(cS) = \dim V - 1. \quad (18.91)$$

Rank-nullity and $c \neq 0$ therefore give $\dim \text{range}(S) = 1$.

For $k > 0$, commutation gives the binomial expansion

$$(N + S)^k = \sum_{m=0}^k \binom{k}{m} N^m S^{k-m}. \quad (18.92)$$

If $m > 0$, write $N^m = NN^{m-1}$. The first factor commutes with the remaining factors, so nilpotence of N makes the entire summand nilpotent and its trace vanishes. Only the term $m = 0$ remains, giving $\text{tr}((N + S)^k) = \text{tr}(S^k)$. This is the trace reduction used in [187, Proposition 20, line 3865].

Commutation makes $\text{range}(S)$ invariant under N . The restriction of N to this one-dimensional space is both nilpotent and a scalar endomorphism, hence zero. Therefore $NS = 0$, and commutation also gives $SN = 0$. For every positive k , induction using these two identities yields

$$(N + S)^k = N^k + S^k. \quad (18.93)$$

Taking J at least the nilpotency index of N removes the first summand. $\square$

Theorem 18.4.11 (Vanishing positive trace powers determine the characteristic polynomial).

[Lean](#) [✓ Stmt](#) Let A be an $n \times n$ complex matrix. If $\text{tr}(A^k) = 0$ for every integer $k \geq 1$, then

$$\chi_A(X) = X^n. \quad (18.94)$$

Proof. [✓ Proof](#) The zero matrix has the same positive trace powers as A and characteristic polynomial X^n . Newton–Girard therefore identifies the two characteristic polynomials. This is the characteristic-polynomial consequence of the all-positive trace condition obtained in [15, proof of Proposition III.3]. $\square$

Theorem 18.4.12 (Shifted vanishing moments under matrix powers). [Lean](#) [✓ Stmt](#) Let A be an $n \times n$ complex matrix such that $\text{tr}(A^k) = 0$ for every $k > 1$, and let $m > 1$. Then $\text{tr}((A^m)^r) = 0$ for every $r \geq 1$. This auxiliary result supports the generalization below to a weaker hypothesis; the source itself supplies vanishing for every positive exponent.

Proof. [✓ Proof](#) For $r \geq 1$, one has $(A^m)^r = A^{mr}$ and $mr > 1$, so the hypothesis at exponent mr gives the result. $\square$

Theorem 18.4.13 (Shifted zero trace powers force nilpotence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an $n \times n$ complex matrix. If $\text{tr}(A^N) = 0$ for every integer $N > 1$, then A is nilpotent and

$$\chi_A(X) = X^n. \quad (18.95)$$

The source obtains the all-positive trace hypothesis in [15, proof of Proposition III.3]; its specialization is the elementwise nilpotence assertion used before the nil-matrix theorem. The result above is a stronger generalization because it assumes vanishing only for $N > 1$. It is useful independently and requires no positivity, normality, or diagonalizability.

Proof. [✓ Proof](#) Every positive power of A^2 has vanishing trace. Hence

$$\chi_{A^2}(X) = X^n. \quad (18.96)$$

If μ is a spectral value of A , spectral mapping shows that μ^2 is a root of this characteristic polynomial. Therefore $\mu^2 = 0$, and hence $\mu = 0$. Thus every root of the split monic polynomial χ_A is zero, so $\chi_A(X) = X^n$. Cayley–Hamilton now proves that A is nilpotent. $\square$

The standard-form construction uses the compact singular-value decomposition in which the intermediate dimension is the rank rather than the smaller ambient dimension.

Theorem 18.4.14 (Compact rectangular singular-value decomposition). [Lean](#) [✓ Stmt](#) Let $M \in M_{m,n}(\mathbb{C})$ be a complex matrix of rank r . There are matrices $V \in M_{r,m}(\mathbb{C})$ and $U \in M_{r,n}(\mathbb{C})$ and strictly positive real numbers $s_0, \dots, s_{r-1}$ such that, for $D = \text{diag}(s_0, \dots, s_{r-1})$,

$$M = V^\dagger D U, \quad V V^\dagger = \mathbb{1}_r, \quad U U^\dagger = \mathbb{1}_r. \quad (18.97)$$

Thus no zero singular values occur in the intermediate space. This is the compact decomposition used in the standard-form construction of [15, Section III].

Proof. [✓ Proof](#) Apply the spectral theorem to $MM^\dagger$. Its nonzero eigenvalues are positive, and their number is $\text{rank}(MM^\dagger) = \text{rank}(M) = r$. For an orthonormal eigenvector e_j with eigenvalue $\lambda_j > 0$, put $w_j = M^\dagger e_j$. Then $\langle w_j, w_k \rangle = \lambda_j \delta_{jk}$. Hence the vectors $\lambda_j^{-1/2} w_j$ are orthonormal. Expanding each column of M in the eigenbasis and discarding the zero-eigenvalue terms gives

$$M = \sum_{\lambda_j > 0} \sqrt{\lambda_j} e_j (\lambda_j^{-1/2} w_j)^\dagger, \quad (18.98)$$

which has the form in (18.97). $\square$

Singular-value decompositions of a matrix product unitary tensor in Section III of [15] rely on two cut matrices. Write $\mathcal{U}_{j,\alpha,\beta}^i$ for the entries of the tensor, where i, j are physical indices and α, β are auxiliary indices.

Definition 18.4.15 (First matrix associated to a Matrix Product Unitary tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The matrix M_1 is obtained by combining the upper physical index with the right auxiliary index, and the left auxiliary index with the lower physical index. Thus

$$(M_1)_{(i,\beta),(\alpha,j)} = \mathcal{U}_{j,\alpha,\beta}^i. \quad (18.99)$$

Transporting the row and column indices along the standard product enumerations gives a matrix $\widetilde{M}_1$ indexed by $\{0, \dots, dD - 1\} \times \{0, \dots, dD - 1\}$. Its entries are obtained by applying M_1 to the inverse images of the enumerated indices, and

$$\text{rank}(\widetilde{M}_1) = \text{rank}(M_1). \quad (18.100)$$

Definition 18.4.16 (Second matrix associated to a Matrix Product Unitary tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The matrix M_2 is obtained by combining the left auxiliary index with the upper physical index, and the lower physical index with the right auxiliary index. Thus

$$(M_2)_{(\alpha,i),(j,\beta)} = \mathcal{U}_{j,\alpha,\beta}^i. \quad (18.101)$$

Transporting the row and column indices along the standard product enumerations gives a matrix $\widetilde{M}_2$ indexed by $\{0, \dots, dD - 1\} \times \{0, \dots, dD - 1\}$. Its entries are obtained by applying M_2 to the inverse images of the enumerated indices, and

$$\text{rank}(\widetilde{M}_2) = \text{rank}(M_2). \quad (18.102)$$

Definition 18.4.17 (Right rank). [Lean](#) [Lean](#) [✓ Stmt](#) The right rank is

$$r = \text{rank}(M_1). \quad (18.103)$$

Definition 18.4.18 (Left rank). [Lean](#) [Lean](#) [✓ Stmt](#) The left rank is

$$\ell = \text{rank}(M_2). \quad (18.104)$$

Theorem 18.4.19 (Source cuts under physical adjunction). [Lean](#) [Lean](#) [✓ Stmt](#) *Physical adjunction exchanges the two source cuts and takes their adjoints:*

$$M_1(\mathcal{U}^\sharp) = M_2(\mathcal{U})^\dagger, \quad (18.105)$$

$$M_2(\mathcal{U}^\sharp) = M_1(\mathcal{U})^\dagger. \quad (18.106)$$

Proof. [✓ Proof](#) Physical adjunction exchanges the upper and lower physical indices and complex conjugates every entry. Under the source-cut regroupings, it also exchanges rows and columns. Hence the first cut becomes the conjugate transpose of the second, and conversely. $\square$

Definition 18.4.20 (Physical-pair swap). [Lean](#) [Lean](#) [✓ Stmt](#) On the flattened physical alphabet, define the involution

$$s(i, j) = (j, i). \quad (18.107)$$

This acts on physical coordinates and is distinct from the doubled-bond exchange used in reflected transfer calculations.

Theorem 18.4.21 (Normalized flattening under physical adjunction). [Lean](#) [✓ Stmt](#) *The normalized flattening of the physical adjoint satisfies*

$$\widetilde{\mathcal{U}}^\sharp{}^{(i,j)} = \overline{\widetilde{\mathcal{U}}^{(j,i)}}. \quad (18.108)$$

Proof. [✓ Proof](#) For every pair of physical indices (i, j) ,

$$\widetilde{\mathcal{U}}^\sharp{}^{(i,j)} = \frac{1}{\sqrt{d}} (\mathcal{U}^\sharp)^{(i,j)} = \frac{1}{\sqrt{d}} \overline{\mathcal{U}^{(j,i)}} = \overline{\widetilde{\mathcal{U}}^{(j,i)}}, \quad (18.109)$$

where the intermediate steps follow by definition of normalized flattening, physical adjunction, and the reality of $d^{-1/2}$. $\square$

Definition 18.4.22 (Canonical-form-II data under physical adjunction). [Lean](#) [✓ Stmt](#) Given canonical-form-II data for $\widetilde{\mathcal{U}}$, define the transformed tuple by

$$r' = r, \quad (18.110)$$

$$D'_k = D_k, \quad (18.111)$$

$$\mu'_k = \overline{\mu_k}, \quad (18.112)$$

$$(A'_k)^{(i,j)} = \overline{A_k^{(j,i)}}, \quad (18.113)$$

$$V' = \overline{V}. \quad (18.114)$$

These data reconstruct $\widetilde{\mathcal{U}}^\sharp$.

Theorem 18.4.23 (Canonical form II under physical adjunction). [Lean](#) [✓ Stmt](#) *If $\widetilde{\mathcal{U}}$ is in canonical form II, then $\widetilde{\mathcal{U}}^\sharp$ is in canonical form II.*

Proof. ✓ Proof Theorem 7.6.4 independently preserves normality and left-canonical normalization under entrywise conjugation, and Theorem 7.6.8 preserves these properties under the physical-pair swap. Conjugating the coisometry equation gives

$$V'V'^{\dagger} = \overline{VV^{\dagger}} = \mathbb{1}. \quad (18.115)$$

Conjugating the reconstruction at the swapped physical index gives

$$\widetilde{\mathcal{U}}^{\sharp(i,j)} = (V')^{\dagger} \left(\bigoplus_k \mu'_k(A'_k)^{(i,j)} \right) V'. \quad (18.116)$$

Finally, each diagonal positive matrix Λ_k is real, and therefore

$$\mathcal{E}_{\overline{A_k}}(\Lambda_k) = \overline{\mathcal{E}_{A_k}(\Lambda_k)} = \overline{\Lambda_k} = \Lambda_k. \quad (18.117)$$

Physical relabeling leaves this transfer map unchanged, which proves the transformed fixed-point equation. $\square$

Theorem 18.4.24 (Full support under physical adjunction). Lean ✓ Stmt *If canonical-form-II data for $\widetilde{\mathcal{U}}$ have full support, then the data from Definition 18.4.22 also have full support.*

Proof. ✓ Proof Complex conjugation of the weights leaves the block indices and their bond dimensions unchanged, so the full-support sum is unchanged. $\square$

Theorem 18.4.25 (Source ranks under physical adjunction). Lean Lean ✓ Stmt *Physical adjunction exchanges the two source ranks:*

$$r(\mathcal{U}^{\sharp}) = \ell(\mathcal{U}), \quad (18.118)$$

$$\ell(\mathcal{U}^{\sharp}) = r(\mathcal{U}). \quad (18.119)$$

Proof. ✓ Proof Apply Theorem 18.4.19 and use invariance of matrix rank under conjugate transpose. $\square$

Theorem 18.4.26 (Source cuts under physical reindexing). Lean Lean ✓ Stmt *A bijective relabeling of the physical basis simultaneously relabels the physical component of every source-cut row and column. Hence*

$$M_1(e^*\mathcal{U}) \cong M_1(\mathcal{U}), \quad M_2(e^*\mathcal{U}) \cong M_2(\mathcal{U}), \quad (18.120)$$

with the right and left orientations unchanged.

Proof. ✓ Proof Evaluate each cut after relabeling. In both cuts the upper physical index lies in the row and the lower physical index lies in the column, with the virtual index occupying the other product coordinate. Applying e to both physical coordinates gives the two displayed row-and-column equivalences. $\square$

Theorem 18.4.27 (Source ranks under physical reindexing). Lean Lean ✓ Stmt *Bijective physical relabeling preserves both source ranks:*

$$r(e^*\mathcal{U}) = r(\mathcal{U}), \quad \ell(e^*\mathcal{U}) = \ell(\mathcal{U}). \quad (18.121)$$

Proof. ✓ Proof Matrix rank is unchanged by bijective row and column relabelings. Apply this fact to the two source-cut identities. $\square$

Definition 18.4.28 (Independent tensor product of matrix product tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A = \{A^i\}_{i \in \{0, \dots, d-1\}}$ and $B = \{B^k\}_{k \in \{0, \dots, e-1\}}$ have auxiliary dimensions D and E , respectively. Write

$$\pi_{m,n} : \{0, \dots, m-1\} \times \{0, \dots, n-1\} \longrightarrow \{0, \dots, mn-1\} \quad (18.122)$$

for the standard finite-product coordinate bijection. Their independent tensor product $A \boxtimes B$ has physical dimension de , auxiliary dimension DE , and local matrices determined by

$$(A \boxtimes B)_{\pi_{D,E}(\alpha,\gamma), \pi_{D,E}(\beta,\delta)}^{\pi_{d,e}(i,k)} = A_{\alpha,\beta}^i B_{\gamma,\delta}^k. \quad (18.123)$$

For every N , the same coordinates give the pointwise splitting

$$(\{0, \dots, N-1\} \rightarrow \{0, \dots, de-1\}) \simeq (\{0, \dots, N-1\} \rightarrow \{0, \dots, d-1\}) \times (\{0, \dots, N-1\} \rightarrow \{0, \dots, e-1\}), \quad (18.124)$$

$$\sigma \mapsto (\sigma_d, \sigma_e), \quad \sigma(n) = \pi_{d,e}(\sigma_d(n), \sigma_e(n)) \quad (18.125)$$

This is infrastructure for the tensoring clause in the proof of the MPU Index Theorem IV.6(ii) [15, lines 824–845], not a separately stated theorem of that paper. The notation $A = \{A^i\}_i$ agrees with the canonical-form block notation in [11, Section 2.3, lines 214–245].

Theorem 18.4.29 (Word evaluation of an independent tensor product). [Lean](#) [✓ Stmt](#) If w has length n and

$$w = (\pi_{d,e}(i_1, k_1), \dots, \pi_{d,e}(i_n, k_n)), \quad (18.126)$$

$$w_d = (i_1, \dots, i_n), \quad w_e = (k_1, \dots, k_n), \quad (18.127)$$

then

$$(A \boxtimes B)^w = \text{reind}_{\pi_{D,E}}(A^{w_d} \otimes B^{w_e}). \quad (18.128)$$

Proof. [✓ Proof](#) Induct on the length of w . For the empty word, use $\mathbb{1}_D \otimes \mathbb{1}_E = \mathbb{1}_{DE}$ and invariance of the identity under the coordinate bijection. For a leading letter $\pi_{d,e}(i, k)$, equation (18.123) and the induction hypothesis reduce the claim to

$$(A^i \otimes B^k)(X \otimes Y) = (A^i X) \otimes (B^k Y), \quad (18.129)$$

after transporting multiplication through the same coordinate bijection. $\square$

Theorem 18.4.30 (Homogeneous word spans of an independent tensor product). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $S_N(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = N\}$. The pointwise splitting of length- N words gives

$$S_N(A \boxtimes B) = \text{span}_{\mathbb{C}}\{\text{reind}_{\pi_{D,E}}(A^u \otimes B^v) : |u| = |v| = N\}. \quad (18.130)$$

In particular, if $S_N(A) = M_D(\mathbb{C})$ and $S_N(B) = M_E(\mathbb{C})$, then $S_N(A \boxtimes B) = M_{DE}(\mathbb{C})$. If A and B are algebraically normal, then $A \boxtimes B$ is algebraically normal.

This is the homogeneous-word infrastructure for the tensoring sentence in the proof of the MPU Index Theorem IV.6(ii) [15, lines 824–845], not a separate theorem stated there. It applies to the retained blocks in the canonical-form decomposition [11, Section 2.3, lines 214–245].

Proof. ✓ Proof For every product word w , let (u, v) be its pointwise component words. The bijection between w and (u, v) and Theorem 18.4.29 give

$$(A \boxtimes B)^w = \text{reind}_{\pi_{D,E}}(A^u \otimes B^v), \quad (18.131)$$

which proves (18.130) after taking linear spans. Pure tensors span $M_D(\mathbb{C}) \otimes M_E(\mathbb{C})$. The Kronecker linear equivalence, followed by simultaneous reindexing along $\pi_{D,E}$, is an isomorphism onto $M_{DE}(\mathbb{C})$; this proves the fixed-length consequence. If $S_{N_A}(A) = M_D(\mathbb{C})$ and $S_{N_B}(B) = M_E(\mathbb{C})$ for positive N_A, N_B , then persistence at positive multiples gives

$$S_{N_A N_B}(A) = M_D(\mathbb{C}), \quad S_{N_A N_B}(B) = M_E(\mathbb{C}). \quad (18.132)$$

Applying the fixed-length consequence at $N_A N_B$ proves algebraic normality of $A \boxtimes B$. □

Theorem 18.4.31 (Left-canonical normal blocks under independent tensor products). Lean

Lean ✓ Stmt Let A and B be CPSV normal tensors, and suppose that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}_D, \quad \sum_k (B^k)^\dagger B^k = \mathbb{1}_E. \quad (18.133)$$

Then $A \boxtimes B$ is a CPSV normal tensor and is left-canonical:

$$\sum_{i,k} ((A \boxtimes B)^{\pi_{d,e}(i,k)})^\dagger (A \boxtimes B)^{\pi_{d,e}(i,k)} = \mathbb{1}_{DE}. \quad (18.134)$$

This is the retained-normal-block infrastructure for the tensoring sentence in the proof of the MPU Index Theorem IV.6(ii) [15, lines 824–845], not a separate theorem stated there. The retained-normal-block terminology is that of the canonical-form decomposition in [11, Section 2.3, lines 214–245]; the two left-canonical identities are explicit infrastructure hypotheses, not an additional assertion attributed to that passage.

Proof. ✓ Proof The adjoint and multiplication laws for Kronecker products, together with the fact that simultaneous reindexing preserves adjoints, products, and finite sums, give

$$\sum_{i,k} ((A \boxtimes B)^{\pi_{d,e}(i,k)})^\dagger (A \boxtimes B)^{\pi_{d,e}(i,k)} \quad (18.135)$$

$$= \text{reind}_{\pi_{D,E}} \left(\sum_{i,k} (A^i \otimes B^k)^\dagger (A^i \otimes B^k) \right) \quad (18.136)$$

$$= \text{reind}_{\pi_{D,E}} \left(\left(\sum_i (A^i)^\dagger A^i \right) \otimes \left(\sum_k (B^k)^\dagger B^k \right) \right) \quad (18.137)$$

$$= \text{reind}_{\pi_{D,E}} (\mathbb{1}_D \otimes \mathbb{1}_E) = \mathbb{1}_{DE}, \quad (18.138)$$

proving (18.134). Spectral normality gives positive lengths N_A, N_B at which the two homogeneous word spans are full. Theorem 18.4.30 therefore makes $A \boxtimes B$ algebraically normal at the common length $N_A N_B$. The algebraically normal, left-canonical tensor is CPSV normal by Theorem 7.6.3. □

Definition 18.4.32 (Independent tensor product of matrix product operator tensors). Lean

Lean ✓ Stmt Let $\mathcal{U}$ and $\mathcal{V}$ have physical dimensions d, e and auxiliary dimensions D, E , respectively. Their independent tensor product $\mathcal{U} \boxtimes \mathcal{V}$ has dimensions de and DE and entries

$$(\mathcal{U} \boxtimes \mathcal{V})_{(\alpha,\gamma),(\beta,\delta)}^{(i,k),(j,l)} = \mathcal{U}_{\alpha,\beta}^{i,j} \mathcal{V}_{\gamma,\delta}^{k,l}. \quad (18.139)$$

A product-valued physical configuration splits sitewise into one configuration for $\mathcal{U}$ and one for $\mathcal{V}$.

Definition 18.4.33 (Blocked product-alphabet equivalence). [Lean](#) [✓ Stmt](#) A blocked word of L product letters separates canonically into two blocked words:

$$\{0, \dots, (de)^L - 1\} \simeq \{0, \dots, d^L e^L - 1\}. \quad (18.140)$$

The map decodes the length- L word, splits every letter into its d and e components, and re-encodes the two resulting words.

Theorem 18.4.34 (Decoding the blocked product alphabet). [Lean](#) [Lean](#) [✓ Stmt](#) If q denotes the blocked product-alphabet equivalence, then each site s satisfies

$$\text{dec}_d(q(I)_1, s) = \text{dec}_{de}(I, s)_1, \quad (18.141)$$

$$\text{dec}_e(q(I)_2, s) = \text{dec}_{de}(I, s)_2. \quad (18.142)$$

Proof. [✓ Proof](#) Each equality follows by composing the inverse product encoding at every site with the corresponding projection. $\square$

Theorem 18.4.35 (Finite-size operator of an independent tensor product). [Lean](#) [✓ Stmt](#) For every system size N , the sitewise configuration bijection identifies the finite-size operator with

$$(\mathcal{U} \boxtimes \mathcal{V})^{(N)} \cong \mathcal{U}^{(N)} \otimes \mathcal{V}^{(N)}. \quad (18.143)$$

Proof. [✓ Proof](#) Grouping the two auxiliary indices makes every closed-word matrix a Kronecker product. The identity

$$\text{Tr}(A \otimes B) = \text{Tr}(A) \text{Tr}(B) \quad (18.144)$$

then gives the stated finite-size operator after the sitewise configuration bijection. $\square$

Theorem 18.4.36 (Independent tensor products preserve matrix product unitarity). [Lean](#) [✓ Stmt](#) The independent tensor product of two matrix product unitary tensors is a matrix product unitary tensor. This is the tensoring operation in the proof of [15, Theorem IV.6(ii)].

Proof. [✓ Proof](#) The multiplication rule $(A \otimes B)(C \otimes D) = AC \otimes BD$ gives

$$(\mathcal{U}^{(N)} \otimes \mathcal{V}^{(N)})(\mathcal{U}^{(N)} \otimes \mathcal{V}^{(N)})^\dagger \quad (18.145)$$

$$= \mathcal{U}^{(N)}(\mathcal{U}^{(N)})^\dagger \otimes \mathcal{V}^{(N)}(\mathcal{V}^{(N)})^\dagger = \mathbb{1} \otimes \mathbb{1} = \mathbb{1}. \quad (18.146)$$

Simultaneous reindexing preserves this unitarity equation. $\square$

Theorem 18.4.37 (Blocking an independent tensor product). [Lean](#) [✓ Stmt](#) For every blocking length L , separating each blocked product letter by the canonical equivalence q gives the exact identity

$$(\mathcal{U} \boxtimes \mathcal{V})^{[L]} = q^*(\mathcal{U}^{[L]} \boxtimes \mathcal{V}^{[L]}). \quad (18.147)$$

The ordering of the two component words agrees with the independent tensor-product convention.

Proof. [✓ Proof](#) Evaluate both sides on blocked ket and bra letters. Word multiplication of local Kronecker products gives

$$(\mathcal{U} \boxtimes \mathcal{V})[I, J] = \mathcal{U}[I_1, J_1] \boxtimes \mathcal{V}[I_2, J_2]. \quad (18.148)$$

The two decoding identities identify I_1, I_2 and J_1, J_2 with the corresponding letters of $q(I)$ and $q(J)$. $\square$

Definition 18.4.38 (Source-cut shuffle for an independent tensor product). [Lean](#) [✓ Stmt](#) For four finite index sets, the source-cut shuffle sends

$$((x, y), (x', y')) \mapsto ((x, x'), (y, y')). \quad (18.149)$$

It is applied to the row and column products with the orders appropriate to M_1 and M_2 .

Theorem 18.4.39 (Source cuts of an independent tensor product). [Lean](#) [Lean](#) [✓ Stmt](#) The source cuts satisfy

$$M_1(\mathcal{U} \boxtimes \mathcal{V}) \cong M_1(\mathcal{U}) \otimes M_1(\mathcal{V}), \quad (18.150)$$

$$M_2(\mathcal{U} \boxtimes \mathcal{V}) \cong M_2(\mathcal{U}) \otimes M_2(\mathcal{V}). \quad (18.151)$$

Thus the paper's right and left orientations are unchanged.

Proof. [✓ Proof](#) Evaluate both source cuts entry by entry after the displayed row and column shuffles. Each entry is the product of the corresponding entries of the two source cuts. $\square$

Theorem 18.4.40 (Source-cut ranks of an independent tensor product). [Lean](#) [Lean](#) [✓ Stmt](#) The right and left source-cut ranks are multiplicative:

$$r(\mathcal{U} \boxtimes \mathcal{V}) = r(\mathcal{U})r(\mathcal{V}), \quad (18.152)$$

$$\ell(\mathcal{U} \boxtimes \mathcal{V}) = \ell(\mathcal{U})\ell(\mathcal{V}). \quad (18.153)$$

This is the rank-multiplication ingredient for the tensoring case in the proof of [15, Theorem IV.6(ii)].

Proof. [✓ Proof](#) Matrix rank is unchanged by row and column bijections and satisfies $\text{rank}(A \otimes B) = \text{rank}(A) \text{rank}(B)$. $\square$

Lemma 18.4.41 (Double layer of an independent tensor product). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $W_{\mathcal{U}}$ and $W_{\mathcal{V}}$ be the double layers of $\mathcal{U}$ and $\mathcal{V}$. Identify the double-layer bond space of $\mathcal{U} \boxtimes \mathcal{V}$ with the tensor product of the two double-layer bond spaces by $((\alpha, \gamma), (\alpha', \gamma')) \mapsto ((\alpha, \alpha'), (\gamma, \gamma'))$. Under this identification,

$$W_{\mathcal{U} \boxtimes \mathcal{V}}^{(i,k)(j,\ell)} = W_{\mathcal{U}}^{ij} \otimes W_{\mathcal{V}}^{k\ell}. \quad (18.154)$$

Proof. [✓ Proof](#) The contracted physical index of the product runs over pairs (m, n) , and each summand factors as a product of an entry of $\sum_m (\mathcal{U}^{mi})^* \otimes \mathcal{U}^{mj}$ and an entry of $\sum_n (\mathcal{V}^{nk})^* \otimes \mathcal{V}^{n\ell}$. $\square$

Theorem 18.4.42 (Independent tensor products preserve simplicity). [Lean](#) [✓ Stmt](#) If $\mathcal{U}$ and $\mathcal{V}$ are simple, then $\mathcal{U} \boxtimes \mathcal{V}$ is simple. If a, b are witnesses for $\mathcal{U}$ and a', b' are witnesses for $\mathcal{V}$, then $a \otimes a'$ and $b \otimes b'$ are witnesses for the product. This is the step behind the sentence “The case of tensoring is trivial” in the proof of [15, Theorem IV.6(ii)].

Proof. [✓ Proof](#) By Lemma 18.4.41, and since $|b \otimes b' \rangle (a \otimes a' | = |b \rangle (a | \otimes |b' \rangle (a' |$,

$$(a \otimes a' | W_{\mathcal{U}}^{ij} \otimes W_{\mathcal{V}}^{k\ell} | b \otimes b') = (a | W_{\mathcal{U}}^{ij} | b) (a' | W_{\mathcal{V}}^{k\ell} | b') = \delta_{ij} \delta_{k\ell}. \quad (18.155)$$

The rule $(A \otimes B)(C \otimes D) = AC \otimes BD$ reduces (simple2) for the product to the same identity for each factor. $\square$

Theorem 18.4.43 (One-site blocking bounds for the source-cut ranks). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}^{[L]}$ denote the direct blocking of L sites of a tensor $\mathcal{U}$ with physical dimension d , and let r_L and ℓ_L be its right and left source-cut ranks. Then

$$r_{L+1} \leq d r_L, \quad (18.156)$$

$$\ell_{L+1} \leq d \ell_L. \quad (18.157)$$

These are the one-site forms of the two upper bounds in the proof of [15, Proposition IV.2].

Proof. [✓ Proof](#) For the first cut, choose a compact factorization of the source matrix of $\mathcal{U}^{[L]}$ through a space of dimension r_L . Split each physical word of length $L+1$ into its first letter and its remaining word of length L . For suitable rectangular matrices $A_{1,L}$ and $B_{1,L}$, the successor source matrix factors as

$$M_{1,L+1} = A_{1,L} B_{1,L}. \quad (18.158)$$

The intermediate space is $\mathbb{C}^d \otimes \mathbb{C}^{r_L}$, and hence

$$\text{rank}(M_{1,L+1}) \leq \dim(\mathbb{C}^d \otimes \mathbb{C}^{r_L}) = d r_L. \quad (18.159)$$

Interchanging the upper and lower physical indices gives matrices $A_{2,L}$ and $B_{2,L}$ with

$$M_{2,L+1} = A_{2,L} B_{2,L}. \quad (18.160)$$

This factorization passes through $\mathbb{C}^d \otimes \mathbb{C}^{\ell_L}$, so

$$\text{rank}(M_{2,L+1}) \leq \dim(\mathbb{C}^d \otimes \mathbb{C}^{\ell_L}) = d \ell_L. \quad (18.161)$$

□

Theorem 18.4.44 (Source-cut rank bounds between two blocking lengths). [Lean](#) [Lean](#) [✓ Stmt](#) If $L_0 \leq L$, then

$$r_L \leq d^{L-L_0} r_{L_0}, \quad (18.162)$$

$$\ell_L \leq d^{L-L_0} \ell_{L_0}. \quad (18.163)$$

These are the two blocking upper bounds used in the proof of [15, Proposition IV.2].

Proof. [✓ Proof](#) Induct on $L - L_0$. The case $L = L_0$ is equality. For the right rank, the successor bound and the induction hypothesis give

$$r_{L+1} \leq d r_L \leq d(d^{L-L_0} r_{L_0}) = d^{L+1-L_0} r_{L_0}. \quad (18.164)$$

The same multiplication by d gives

$$\ell_{L+1} \leq d \ell_L \leq d(d^{L-L_0} \ell_{L_0}) = d^{L+1-L_0} \ell_{L_0}. \quad (18.165)$$

□

Theorem 18.4.45 (Exact blocking growth from endpoint rank products). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $d > 0$ and $L_0 \leq L$. Suppose the endpoint source-cut ranks satisfy the supplied product identities

$$r_{L_0} \ell_{L_0} = d^{2L_0}, \quad r_L \ell_L = d^{2L}. \quad (18.166)$$

Then the blocking bounds are exact:

$$r_L = d^{L-L_0} r_{L_0}, \quad \ell_L = d^{L-L_0} \ell_{L_0}. \quad (18.167)$$

This is the saturation step in the proof of [15, Proposition IV.2].

Proof. ✓ Proof The blocking bounds give

$$r_L \leq d^{L-L_0} r_{L_0}, \quad \ell_L \leq d^{L-L_0} \ell_{L_0}. \quad (18.168)$$

Their upper bounds have product

$$(d^{L-L_0} r_{L_0})(d^{L-L_0} \ell_{L_0}) = d^{2(L-L_0)} d^{2L_0} = d^{2L} = r_L \ell_L. \quad (18.169)$$

Since $d > 0$, the latter product is positive, so both r_L and ℓ_L are positive. Hence neither upper bound can be strict, and both equalities follow. $\square$

Definition 18.4.46 (Product-index source weight). Lean ✓ Stmt The row space of M_1 has product order (upper physical, right auxiliary). For $\rho \in M_D(\mathbb{C})$, define

$$W_\rho = \mathbb{1}_d \otimes \rho. \quad (18.170)$$

This is the source weight in [15, equation labelled Y1Y1X1X1].

Theorem 18.4.47 (Positivity of the product-index source weight). Lean ✓ Stmt If ρ is positive definite, then W_ρ is positive definite.

Proof. ✓ Proof The Kronecker product of the positive-definite matrices $\mathbb{1}_d$ and ρ is positive definite. $\square$

Definition 18.4.48 (Source singular-value data and compressed source metric). Lean Lean ✓ Stmt Choose compact singular-value decompositions

$$M_i = V_i^\dagger D_i U_i, \quad V_i V_i^\dagger = \mathbb{1}, \quad U_i U_i^\dagger = \mathbb{1}, \quad (18.171)$$

with intermediate dimensions r and ℓ , respectively. For $\rho \in M_D(\mathbb{C})$, define the compressed positive metric

$$A = V_1 W_\rho V_1^\dagger. \quad (18.172)$$

The inverse square root $A^{-1/2}$, rather than A itself, is the normalization factor used in the first weighted source factor. This is the source-factor construction of [15, Section III].

Theorem 18.4.49 (Positivity of the first compressed source metric). Lean ✓ Stmt If ρ is positive definite, then A is positive definite.

Proof. ✓ Proof The map $V_1^\dagger$ is injective because $V_1 V_1^\dagger = \mathbb{1}_r$. Positive definiteness of W_ρ is therefore preserved by this congruence. $\square$

Definition 18.4.50 (Weighted source factors). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $\rho \in M_D(\mathbb{C})$ be positive definite. Define

$$X_1 = V_1^\dagger A^{-1/2}, \quad Y_1 = A^{1/2} D_1 U_1, \quad Z_1 = U_1^\dagger D_1^{-1} A^{-1/2}, \quad (18.173)$$

$$X_2 = V_2^\dagger, \quad Y_2 = D_2 U_2, \quad Z_2 = U_2^\dagger D_2^{-1}. \quad (18.174)$$

These are the factors in [15, Section III].

Proof. ✓ Proof Since A is positive definite, its support projection is the identity and

$$A^{-1/2}A^{1/2} = \mathbb{1}_r, \quad A^{-1/2}AA^{-1/2} = \mathbb{1}_r. \quad (18.175)$$

Therefore

$$X_1Y_1 = V_1^\dagger A^{-1/2}A^{1/2}D_1U_1 = V_1^\dagger D_1U_1 = M_1, \quad (18.176)$$

$$X_1^\dagger W_\rho X_1 = A^{-1/2}V_1W_\rho V_1^\dagger A^{-1/2} = A^{-1/2}AA^{-1/2} = \mathbb{1}_r, \quad (18.177)$$

$$Y_1Z_1 = A^{1/2}D_1U_1U_1^\dagger D_1^{-1}A^{-1/2} = A^{1/2}A^{-1/2} = \mathbb{1}_r. \quad (18.178)$$

For the second cut, $V_2V_2^\dagger = U_2U_2^\dagger = \mathbb{1}_\ell$ and $D_2D_2^{-1} = \mathbb{1}_\ell$ give

$$X_2Y_2 = V_2^\dagger D_2U_2 = M_2, \quad (18.179)$$

$$X_2^\dagger X_2 = V_2V_2^\dagger = \mathbb{1}_\ell, \quad (18.180)$$

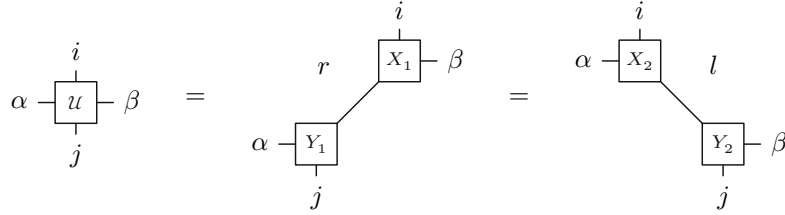
$$Y_2Z_2 = D_2U_2U_2^\dagger D_2^{-1} = \mathbb{1}_\ell. \quad (18.181)$$

□

Theorem 18.4.51 (Source-cut factorizations). Lean Lean ✓ Stmt *The two factorizations are*

$$M_1 = X_1Y_1, \quad M_2 = X_2Y_2. \quad (18.182)$$

These are the source decompositions in [15, Section III].



Proof. ✓ Proof The source-factor construction supplies $M_1 = X_1Y_1$. The second compact singular-value decomposition gives $M_2 = X_2Y_2$ directly. □

Theorem 18.4.52 (Entry forms of the source-cut factorizations). Lean Lean Lean Lean ✓ Stmt *The entries of the two products satisfy*

$$(X_1Y_1)_{(i,\beta),(\alpha,j)} = u_{j,\alpha,\beta}^i, \quad (18.183)$$

$$(X_2Y_2)_{(\alpha,i),(j,\beta)} = u_{j,\alpha,\beta}^i. \quad (18.184)$$

These are the two graphical entry identifications in [15, Section III].

Proof. ✓ Proof Evaluate the two matrix factorizations at $((i, \beta), (\alpha, j))$ and $((\alpha, i), (j, \beta))$, respectively. □

Theorem 18.4.53 (Source-factor normalizations). Lean Lean Lean Lean ✓ Stmt *The left factors satisfy*

$$X_1^\dagger W_\rho X_1 = \mathbb{1}_r, \quad X_2^\dagger X_2 = \mathbb{1}_\ell, \quad (18.185)$$

$$\sum_{x,y} \overline{(X_1)_{x,r}} (W_\rho)_{x,y} (X_1)_{y,r'} = \delta_{r,r'}, \quad (18.186)$$

$$\sum_{\beta,i} \overline{(X_2)_{(\beta,i),l}} (X_2)_{(\beta,i),l'} = \delta_{l,l'}. \quad (18.187)$$

These are [15, Section III].



Theorem 18.4.58 (Paper-gate two-site factorization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Identify a two-site physical configuration with an ordered pair in $\{0, \dots, d-1\} \times \{0, \dots, d-1\}$. Then

$$\text{reindex}(U^{(2)}) = v \text{ swap}_{r,\ell}(u), \quad (18.194)$$

where $\text{swap}_{r,\ell}$ reorders the source index of u from $\ell \times r$ to the $r \times \ell$ order carried by the columns of v and reverses the two physical input coordinates. This is the exact two-site factorization through the paper gates u and v of CPSV17 SVDforms2- $uuvv$.

Proof. [✓ Proof](#) Expand the periodic trace through $M_1 = X_1 Y_1$ and $M_2 = X_2 Y_2$. Reordering the four scalar finite sums gives

$$\sum_{l,r} v_{(i_1, i_2), (r, l)} u_{(l, r), (j_2, j_1)}. \quad (18.195)$$

Reindexing both physical configuration spaces and swapping (l, r) to (r, l) gives the matrix identity. $\square$

Theorem 18.4.59 (Open two-site factorization through v). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $q = (q_1, q_2)$ and $j = (j_1, j_2)$, the untraced two-site product is

$$(\mathcal{U}^{q_1 j_1} \mathcal{U}^{q_2 j_2})_{\alpha, \gamma} = \sum_{r, l} v_{q, (r, l)} (Y_1)_{r, (\alpha, j_1)} (Y_2)_{l, (j_2, \gamma)}. \quad (18.196)$$

The intermediate contraction is

$$\sum_r v_{(q_1, q_2), (r, l)} (Y_1)_{r, (\alpha, j_1)} = \sum_{\beta} \mathcal{U}_{j_1, \alpha, \beta}^{q_1} (X_2)_{(\beta, q_2), l}. \quad (18.197)$$

Proof. [✓ Proof](#) Expand v as the contraction of X_1 with X_2 . Contracting X_1 with Y_1 gives the first tensor letter, while contracting X_2 with Y_2 gives the second. Reordering the finite sums proves the open product identity. The concrete blocked formula follows by encoding both physical pairs through $\{0, \dots, d-1\} \times \{0, \dots, d-1\} \simeq \{0, \dots, (-1)d^2\}$. $\square$

Theorem 18.4.60 (Open two-site factorization through u). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The same untraced two-site product also satisfies

$$(\mathcal{U}^{i_1 j_1} \mathcal{U}^{i_2 j_2})_{\alpha, \gamma} = \sum_{l, r} (X_2)_{(\alpha, i_1), l} u_{(l, r), (j_1, j_2)} (X_1)_{(i_2, \gamma), r}. \quad (18.198)$$

The product indices of u are (l, r) on the source side and (j_1, j_2) on the physical side.

Proof. [✓ Proof](#) Expand the first tensor letter through $M_2 = X_2 Y_2$ and the second through $M_1 = X_1 Y_1$. The common auxiliary index contracts Y_2 with Y_1 to give u . Summing the two remaining source indices gives the formula. The concrete blocked identity follows by decoding its physical indices into ordered pairs. $\square$

Theorem 18.4.61 (Closed double-layer trace for the normalized input tail). [Lean](#) [Stmt](#) Assume $d > 0$. For any MPO tensor $\mathcal{U}$, tail length K , and $p, q \in \{0, \dots, d-1\} \times \{0, \dots, d-1\}$, let W be its double layer and let $E = d^{-1} \sum_i W^{ii}$ be the normalized physical diagonal. Then

$$d^{-K} \sum_{\tau, \eta} \overline{U_{\eta, (p, \tau)}^{(K+2)}} U_{\eta, (q, \tau)}^{(K+2)} = \text{tr}[W^{p_1 q_1} W^{p_2 q_2} E^K]. \quad (18.199)$$

Thus p labels the starred input and q the unstarred input. This is the closed input-tail expression occurring on the right-hand side of [15, equation `uUnitary` and lines 550–556].

Proof. [Proof](#) Closing the common output word gives the double-layer identity

$$\sum_{\eta} \overline{U_{\eta, \sigma}^{(K+2)}} U_{\eta, \sigma'}^{(K+2)} = W_{\sigma, \sigma'}^{(K+2)}. \quad (18.200)$$

Set $\sigma = (p, \tau)$ and $\sigma' = (q, \tau)$. Splitting off the two retained sites rewrites the right-hand side as

$$\text{tr}[W^{p_1 q_1} W^{p_2 q_2} W^{\tau_1 \tau_1} \dots W^{\tau_K \tau_K}]. \quad (18.201)$$

Finally,

$$E^K = d^{-K} \sum_{\tau} W^{\tau_1 \tau_1} \dots W^{\tau_K \tau_K}. \quad (18.202)$$

Substitute this expansion and use linearity of the trace. □

Definition 18.4.62 (Tensorized source dressing). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Put

$$Y = Y_1 \otimes Y_2, \quad Z = Z_1 \otimes Z_2, \quad (18.203)$$

with source-bond order $r \times \ell$. Thus Y maps $(\mathbb{C}^D \otimes \mathbb{C}^d) \otimes (\mathbb{C}^d \otimes \mathbb{C}^D)$ to $\mathbb{C}^r \otimes \mathbb{C}^\ell$, and Z maps in the opposite direction. The regrouping of the open indices is

$$R((a, i), (j, b)) = ((i, j), \text{flat}(a, b)), \quad (18.204)$$

where $\text{flat} : \{0, \dots, D-1\} \times \{0, \dots, D-1\} \simeq \{0, \dots, (-1)^D\}$ is the canonical product enumeration. The dressed source- v Gram equation is

$$Y^\dagger (v^\dagger v) Y = Y^\dagger Y. \quad (18.205)$$

These are the tensorized factors surrounding $v^\dagger v$ in [15, Theorem III.8, Section III.B, equations (31)–(32)].

Theorem 18.4.63 (Four-letter source contractions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let $p = (p_1, p_2)$ be the starred physical pair, $q = (q_1, q_2)$ the unstarred pair, and let $a = (a_1, a_2)$ and $b = (b_1, b_2)$ be virtual paths. The weighted first-cut contraction is

$$\sum_r \overline{(Y_1)_{r, (a_1, p_1)}} (Y_1)_{r, (b_1, q_1)} = \sum_{i_1, \beta, \beta'} \overline{U_{a_1 \beta}^{i_1 p_1}} \rho_{\beta \beta'} U_{b_1 \beta'}^{i_1 q_1}. \quad (18.206)$$

The entry of two ordinary double-layer letters from row (a_1, b_1) to column (a_2, b_2) is

$$\sum_{\alpha, \beta, j_1, j_2} \overline{U_{a_1 \alpha}^{j_1 p_1}} U_{b_1 \beta}^{j_1 q_1} \overline{U_{\alpha a_2}^{j_2 p_2}} U_{\beta b_2}^{j_2 q_2}. \quad (18.207)$$

Set

$$\rho'_x = \text{vec}(\rho)_{\text{flat}^{-1}(x)}, \quad \Phi'_x = \text{vec}(\mathbb{1}_D)_{\text{flat}^{-1}(x)}. \quad (18.208)$$

Inserting $|\rho'\rangle\langle\Phi'|$ between the two letters gives

$$(W^{p_1 q_1} |\rho'\rangle\langle\Phi'| W^{p_2 q_2})_{(a_1, b_1), (a_2, b_2)} \quad (18.209)$$

$$= \sum_{\alpha, \alpha', \beta, \beta_1, \beta_2} \overline{U_{a_1 \alpha}^{j_1 p_1}} U_{b_1 \alpha'}^{j_1 q_1} \rho_{\alpha' \alpha} \overline{U_{\beta a_2}^{j_2 p_2}} U_{\beta b_2}^{j_2 q_2}. \quad (18.210)$$

Finally, the Gram entry of $Y_1 \otimes Y_2$ expands as

$$\sum_{i_1, \beta, \beta', \delta, i_2} \overline{U_{a_1 \beta}^{i_1 p_1}} \rho_{\beta \beta'} U_{b_1 \beta'}^{i_1 q_1} \overline{U_{\delta a_2}^{i_2 p_2}} U_{\delta b_2}^{i_2 q_2}. \quad (18.211)$$

Proof. [✓ Proof](#) Expand the ordinary and rank-one-inserted matrix products entrywise. For the tensorized Gram entry, use $X_1^\dagger(\mathbb{1}_d \otimes \rho)X_1 = \mathbb{1}_r$ on the first cut and $X_2^\dagger X_2 = \mathbb{1}_\ell$ on the second cut, then reorder the finite sums. No ambient identity $X_2 X_2^\dagger = \mathbb{1}$ is used. $\square$

Theorem 18.4.64 (Paper- v open-leg contraction). [Lean](#) [✓ Stmt](#) Let $\rho > 0$. For $z = (z_1, z_2)$, $p = (p_1, p_2)$, and $a = (a_1, a_2)$,

$$(v(Y_1 \otimes Y_2))_{z, ((a_1, p_1), (p_2, a_2))} = \sum_{\gamma} U_{a_1 \gamma}^{z_1 p_1} U_{\gamma a_2}^{z_2 p_2}. \quad (18.212)$$

This is the local open-leg identity for the literal gate $v = X_1 - X_2$.

Proof. [✓ Proof](#) Expand the matrix product and tensor product, reorder the finite sums, and contract $X_1 Y_1$ and $X_2 Y_2$ by the two source-cut factorizations. $\square$

Theorem 18.4.65 (Tensorized source right inverse). [Lean](#) [✓ Stmt](#) The tensorized source factors satisfy

$$YZ = \mathbb{1}_{r\ell}. \quad (18.213)$$

Proof. [✓ Proof](#) Multiplicativity of the Kronecker product gives

$$YZ = (Y_1 Z_1) \otimes (Y_2 Z_2) = \mathbb{1}_r \otimes \mathbb{1}_\ell = \mathbb{1}_{r\ell}. \quad (18.214)$$

$\square$

Theorem 18.4.66 (Regrouped entries of the dressed Gram equation). [Lean](#) [✓ Stmt](#) For every positive definite source weight ρ , and for $x = R^{-1}(p, a)$ and $y = R^{-1}(q, b)$, the dressed Gram equation is equivalent to

$$\sum_t \left(\sum_s \overline{Y_{s,x}} \sum_z \overline{v_{z,s}} v_{z,t} \right) Y_{t,y} = \sum_t \overline{Y_{t,x}} Y_{t,y}. \quad (18.215)$$

Here $p, q \in \{0, \dots, d-1\} \times \{0, \dots, d-1\}$, $a, b \in \{0, \dots, (-1\}D^2\}$, the indices s, t range over $\{0, \dots, r-1\} \times \{0, \dots, \ell-1\}$, and z ranges over $\{0, \dots, d-1\} \times \{0, \dots, d-1\}$.

Proof. [✓ Proof](#) Evaluate both sides of $Y^\dagger(v^\dagger v)Y = Y^\dagger Y$ at (x, y) and expand the three matrix products. Conversely, apply the displayed equality after regrouping each pair of open source-cut indices by R . $\square$

Theorem 18.4.67 (Right-inverse cancellation of a dressed Gram equation). [Lean](#) [✓ Stmt](#) If Y has a right inverse Z , so that $YZ = \mathbb{1}$, then for every square matrix G on the row space of Y ,

$$Y^\dagger G Y = Y^\dagger Y \iff G = \mathbb{1}. \quad (18.216)$$

Proof. [✓ Proof](#) Multiplying $Y^\dagger G Y = Y^\dagger Y$ by $Z^\dagger$ on the left and Z on the right, and using $YZ = \mathbb{1}$, gives

$$G = (YZ)^\dagger G (YZ) = Z^\dagger (Y^\dagger G Y) Z = Z^\dagger (Y^\dagger Y) Z = (YZ)^\dagger (YZ) = \mathbb{1}. \quad (18.217)$$

The converse is immediate. $\square$

Theorem 18.4.68 (Cancellation of the source- v dressing). [Lean](#) [✓ Stmt](#) For every positive definite source weight ρ ,

$$Y^\dagger (v^\dagger v) Y = Y^\dagger Y \iff v^\dagger v = \mathbb{1}_{r_\ell}. \quad (18.218)$$

Proof. [✓ Proof](#) Apply right-inverse cancellation with $G = v^\dagger v$ and $Z = Z_1 \otimes Z_2$. $\square$

Theorem 18.4.69 (Right-rank bound). [Lean](#) [✓ Stmt](#) If the physical and auxiliary dimensions are d and D , respectively, then

$$r \leq Dd. \quad (18.219)$$

Proof. [✓ Proof](#) The matrix M_1 has dD columns. Its rank is at most its number of columns, hence at most $dD = Dd$. $\square$

Theorem 18.4.70 (Left-rank bound). [Lean](#) [✓ Stmt](#) If the physical and auxiliary dimensions are d and D , respectively, then

$$\ell \leq Dd. \quad (18.220)$$

Proof. [✓ Proof](#) The matrix M_2 has dD columns. Its rank is at most its number of columns, hence at most $dD = Dd$. $\square$

18.5 Full-support canonical representatives

Definition 18.5.1 (Full support). [Lean](#) [✓ Stmt](#) Let

$$A^i = V^* \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) V \quad (18.221)$$

be CPSV canonical-form data in ambient bond dimension D , where every A_k has positive bond dimension D_k . The data have *full support* when

$$\sum_{k=1}^r D_k = D. \quad (18.222)$$

Theorem 18.5.2 (Reduced canonical representative of the normalized tensor). [Lean](#) [✓ Stmt](#) Let $d, D > 0$, let U be an MPU tensor of physical dimension d and bond dimension D , and let $\tilde{U} = d^{-1/2} U$ be its normalized doubled-index tensor. There are an integer $D_{\text{red}} > 0$, a normal

tensor A_{red} of bond dimension D_{red} , canonical-form-II data for A_{red} , and a matrix ρ on the ambient bond space of A_{red} such that, for every integer $N > 0$ and every word σ of length N ,

$$D_{\text{red}} \leq D, \quad (18.223)$$

$$V^{(N)}(A_{\text{red}})_{\sigma} = V^{(N)}(\widetilde{U})_{\sigma}, \quad (18.224)$$

$$\sum_{k=1}^r D_k = D_{\text{red}}, \quad (18.225)$$

and ρ is positive definite and diagonal with

$$\text{tr}(\rho) = 1, \quad \mathcal{E}_{A_{\text{red}}}(\rho) = \rho. \quad (18.226)$$

This organizes the dimension-bounded canonical replacement used in [15, lines 257–294 and 319–326]. Its irreducible reduction and canonical-form-II gauge are those of [11, lines 195–255 and 1058–1077]. The shifted transfer argument is the one used in [15, lines 344–356], and the diagonal positive trace-one ambient right fixed matrix is the one of [15, lines 269–281].

Proof. ✓ Proof Remove the zero irreducible blocks of $\widetilde{U}$ and write

$$A_0^i = \bigoplus_{k=1}^r B_k^i, \quad \sum_{k=1}^r D_k \leq D, \quad (18.227)$$

$$V^{(N)}(A_0)_{\sigma} = V^{(N)}(\widetilde{U})_{\sigma}, \quad (18.228)$$

for every $N > 0$ and every word σ of length N . At $N = 2$ the common self-overlap is one, so the retained sum is nonempty and $D_0 = \sum_k D_k > 0$. For every integer $N > 1$, overlap invariance gives

$$\text{tr}(E_{A_0}^N) = \sum_{\sigma} |V^{(N)}(A_0)_{\sigma}|^2 = \sum_{\sigma} |V^{(N)}(\widetilde{U})_{\sigma}|^2 = 1. \quad (18.229)$$

Let $J_k : \mathbb{C}^{D_k} \rightarrow \mathbb{C}^{D_0}$ be the inclusion of the k th block. Then

$$J_k^{\dagger} J_k = \mathbb{1}, \quad A_0^i J_k = J_k B_k^i, \quad (18.230)$$

$$\mathcal{E}_{A_0}(J_k X J_k^{\dagger}) = J_k \mathcal{E}_{B_k}(X) J_k^{\dagger}. \quad (18.231)$$

Thus every nonzero transfer eigenvalue of B_k is a nonzero eigenvalue of E_{A_0} and hence equals one by (18.229). Perron–Frobenius theory supplies $\rho_k > 0$ and $s_k > 0$ with $\mathcal{E}_{B_k}(\rho_k) = s_k \rho_k$. Therefore $s_k = 1$, and every peripheral eigenvalue of B_k is also one. Each B_k is consequently normal.

The direct sum now has literal canonical-form data with full support. Apply the nonsingular blockwise canonical-form-II gauge. It preserves the block dimensions and all periodic vectors, so the resulting tensor A_1 still satisfies

$$\sum_{k=1}^r D_k = D_{\text{red}}, \quad \text{tr}(E_{A_1}^N) = 1, \quad (18.232)$$

for every integer $N > 1$. The shifted traces force $r = 1$ and $|\mu_1| = 1$, and full support identifies the unique retained block with the ambient bond space, so its inclusion J_1 is a unitary of $\mathbb{C}^{D_{\text{red}}}$:

$$J_1^{\dagger} J_1 = \mathbb{1}, \quad J_1 J_1^{\dagger} = \mathbb{1}, \quad A_1^i J_1 = J_1 \mu_1 B_1^i. \quad (18.233)$$

The retained block carries a diagonal positive definite fixed matrix Λ_0 of its own transfer map; rescale it to $\Lambda = \text{tr}(\Lambda_0)^{-1}\Lambda_0$, so that

$$\Lambda > 0, \quad \text{tr}(\Lambda) = 1, \quad (18.234)$$

$$\mathcal{E}_{\mu_1 B_1}(\Lambda) = |\mu_1|^2 \mathcal{E}_{B_1}(\Lambda) = \Lambda, \quad (18.235)$$

the last equality because $|\mu_1| = 1$. Now conjugate the ambient tensor by J_1 and read the block matrix in the resulting coordinates:

$$A_{\text{red}}^i = J_1^\dagger A_1^i J_1 = \mu_1 B_1^i, \quad \rho = \Lambda. \quad (18.236)$$

Since J_1 is unitary this is a change of ambient coordinates: $A_1^i = J_1 A_{\text{red}}^i J_1^\dagger$, so A_1 and A_{red} have the same matrix product vectors at every positive length. The canonical-form-II data travel along the same factor: if Φ is the ambient coisometry reconstructing A_1 from the blocks B_k and the weights μ_k , then ΦJ_1 is again a coisometry and reconstructs A_{red} from those same blocks and weights,

$$(\Phi J_1)(\Phi J_1)^\dagger = \Phi J_1 J_1^\dagger \Phi^\dagger = \Phi \Phi^\dagger = \mathbb{1}, \quad (18.237)$$

$$(\Phi J_1)^\dagger \left(\bigoplus_{k=1}^r \mu_k B_k^i \right) (\Phi J_1) = J_1^\dagger A_1^i J_1 = A_{\text{red}}^i. \quad (18.238)$$

The fixed matrix transports along the same conjugation:

$$\mathcal{E}_{A_1}(J_1 \rho J_1^\dagger) = J_1 \mathcal{E}_{\mu_1 B_1}(\Lambda) J_1^\dagger = J_1 \rho J_1^\dagger, \quad (18.239)$$

$$\mathcal{E}_{A_{\text{red}}}(\rho) = J_1^\dagger \mathcal{E}_{A_1}(J_1 \rho J_1^\dagger) J_1 = \rho, \quad (18.240)$$

and ρ is positive definite, diagonal, and of trace one because Λ is. This is (18.226).

It remains to see that the conjugated tensor is normal. Transport the trace identity to it: for every integer $N > 1$,

$$\text{tr}(E_{A_{\text{red}}}^N) = \sum_{\sigma} |V^{(N)}(A_{\text{red}})_{\sigma}|^2 = \sum_{\sigma} |V^{(N)}(\tilde{U})_{\sigma}|^2 = 1. \quad (18.241)$$

These shifted traces force the spectral radius of $E_{A_{\text{red}}}$ to be one and $E_{A_{\text{red}}}$ to be primitive. Together with $r = 1$ and full support, both preserved by the conjugation, they make A_{red} normal. $\square$

Theorem 18.5.3 (Reduced canonical representative of an MPU). [Lean](#) [✓ Stmt](#) *Let $d, D > 0$ and let U be an MPU tensor. There are $D_{\text{red}} > 0$ and an MPO tensor U_{red} of bond dimension D_{red} carrying a canonical-form-II presentation in the sense of Definition 18.8.8 such that, for every integer $N > 0$,*

$$D_{\text{red}} \leq D, \quad (18.242)$$

$$U_{\text{red}}^{(N)} = U^{(N)}. \quad (18.243)$$

In particular U_{red} is an MPU, the retained block dimensions of its normalized doubled-index tensor sum to D_{red} , and the ambient right fixed matrix of that presentation is positive definite, diagonal, and of trace one. No equality is asserted at length zero, and no virtual gauge is asserted between the two potentially unequal-dimensional bond spaces.

Proof. [✓ Proof](#) Apply Theorem 18.5.2 and write its representative as A_{red} . Define

$$U_{\text{red}}^{ij} = \sqrt{d} A_{\text{red}}^{(i,j)}. \quad (18.244)$$

Since $d > 0$, normalization recovers A_{red} exactly. For every $N > 0$ and every ket–bra pair (σ, τ) , the doubled-index coefficient identity gives

$$(U_{\text{red}}^{(N)})_{\sigma, \tau} = d^{N/2} V^{(N)}(A_{\text{red}})_{(\sigma, \tau)} \quad (18.245)$$

$$= d^{N/2} V^{(N)}(\tilde{U})_{(\sigma, \tau)} \quad (18.246)$$

$$= (U^{(N)})_{\sigma, \tau}. \quad (18.247)$$

Hence the periodic operators agree at every positive length. For $N > 1$ the right side is unitary, so U_{red} is an MPU. Its normalized doubled-index tensor is A_{red} , so the canonical-form-II data, the full-support identity, and the diagonal positive trace-one ambient right fixed matrix supplied by Theorem 18.5.2 assemble into a canonical-form-II presentation of U_{red} . $\square$

Theorem 18.5.4 (Full support under physical blocking). [Lean](#) [✓ Stmt](#) *Suppose that canonical-form data have full support:*

$$\sum_{k=1}^r D_k = D. \quad (18.248)$$

Then every positive physical blocking of these data also has full support.

Proof. [✓ Proof](#) Positive blocking supplies canonical-form data with the same block dimensions, so the full-support sum is unchanged. $\square$

Theorem 18.5.5 (Full support under physical relabeling). [Lean](#) [✓ Stmt](#) *A bijective relabeling of the physical alphabet preserves full support of canonical-form data.*

Proof. [✓ Proof](#) Physical relabeling leaves the blocks and their bond dimensions unchanged, so the full-support sum is unchanged. $\square$

Theorem 18.5.6 (An ambient-dimensional block is irreducible). [Lean](#) [✓ Stmt](#) *If a block has bond dimension D , then the ambient tensor A is irreducible.*

Proof. [✓ Proof](#) If k is the block with $D_k = D$, the total-dimension bound gives

$$D = D_k \leq \sum_{j=1}^r D_j \leq D. \quad (18.249)$$

Since every retained block dimension is positive, these equalities force k to be the only retained block. After reindexing its bond coordinates, the ambient coisometry U is square, hence unitary, and reconstruction reads

$$A^i = U^*(\mu_k B^i)U, \quad (18.250)$$

where B is the reindexed normal block and $\mu_k \neq 0$. Thus Theorem D.3.4 transports irreducibility of B through the nonzero rescaling and unitary conjugation. $\square$

Theorem 18.5.7 (Unique block with full support). [Lean](#) [✓ Stmt](#) *Canonical-form data with one block and full support reconstruct an irreducible ambient tensor.*

Proof. ✓ Proof Let k be the unique block. The singleton sum and full-support identity give

$$\sum_{j=1}^r D_j = D_k = D. \quad (18.251)$$

Thus k has ambient bond dimension, so Theorem 18.5.6 applies. $\square$

Theorem 18.5.8 (Normality of a unique full-support block). Lean ✓ Stmt *Suppose that canonical-form data have one block and full support. If the transfer map has spectral radius one and is primitive, then the ambient tensor is a normal tensor.*

Proof. ✓ Proof Combine Theorem 18.5.7 with the supplied spectral-radius and primitivity properties. $\square$

18.6 Transfer-matrix identities

Let D be a positive integer and let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map. The transfer matrix $\widehat{T}$ of T under the column-stacking identification $M_D(\mathbb{C}) \cong \mathbb{C}^{D^2}$ is developed in the companion quantum-channel volume [182, “Transfer matrix of a matrix endomorphism”, “Transfer matrix in one family”, “Functoriality of transfer matrices”, “Trace preservation fixes the vectorized identity on the left”, “Powers and traces of transfer matrices”].

Theorem 18.6.1 (Eigenvalues of a transfer matrix). Lean ✓ Stmt *A complex number μ is an eigenvalue of T if and only if it is an eigenvalue of $\widehat{T}$.*

Proof. ✓ Proof

Column stacking is a linear bijection and intertwines T with multiplication by $\widehat{T}$. It therefore carries nonzero eigenmatrices to nonzero eigenvectors and conversely. $\square$

For a matrix product tensor A , write $V^{(N)}(A)_\sigma$ for the coefficient of its length- N periodic vector at the configuration σ .

Theorem 18.6.2 (Transfer-matrix powers as periodic overlaps). Lean Lean ✓ Stmt *Let A and B be matrix product tensors with the same positive bond dimension, and let $F_{A,B}(X) = \sum_i A^i X (B^i)^\dagger$ be their mixed transfer map. Then*

$$\mathrm{tr} \left(\widehat{F_{A,B}}^N \right) = \sum_{\sigma} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}. \quad (18.252)$$

In particular, taking $B = A$ identifies the trace of the N th power of the ordinary transfer matrix with the length- N self-overlap.

Proof. ✓ Proof Replace the matrix power by the transfer matrix of the corresponding operator power, identify matrix trace with operator trace, and expand the latter over periodic words. For the ordinary self-transfer statement, specialize to $B = A$ and use the identity $F_{A,A} = \mathcal{E}_A$ from Theorem B.1.1. $\square$

Definition 18.6.3 (Normalized flattening of an MPO tensor). Lean ✓ Stmt For an MPO tensor U of physical dimension $d > 0$, its normalized flattening is the doubled-physical-index MPS tensor

$$\widetilde{U}^{ij} = \frac{1}{\sqrt{d}} U^{ij}. \quad (18.253)$$

This is the normalization used in the normal-tensor argument of [15, Section III, equation (13)].

Definition 18.6.4 (Doubled product-coordinate equivalence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Write $\pi_{m,n} : \{0, \dots, m-1\} \times \{0, \dots, n-1\} \rightarrow \{0, \dots, mn-1\}$ for the standard product coordinate bijection. For natural numbers d, e , the canonical doubled product-coordinate equivalence $s_{d,e} : \{0, \dots, (de)^2-1\} \rightarrow \{0, \dots, d^2e^2-1\}$ and its inverse satisfy

$$\begin{aligned} s_{d,e}(\pi_{de,de}(\pi_{d,e}(i,k), \pi_{d,e}(j,\ell))) &= \pi_{d^2,e^2}(\pi_{d,d}(i,j), \pi_{e,e}(k,\ell)), \\ s_{d,e}^{-1}(\pi_{d^2,e^2}(\pi_{d,d}(i,j), \pi_{e,e}(k,\ell))) &= \pi_{de,de}(\pi_{d,e}(i,k), \pi_{d,e}(j,\ell)). \end{aligned} \quad (18.254)$$

Theorem 18.6.5 (Normalized flattening of an independent tensor product). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ and $\mathcal{V}$ have physical dimensions d, e and auxiliary dimensions D, E , respectively. With the coordinate equivalence $s_{d,e}$ of Definition 18.6.4, normalized flattening obeys

$$\begin{aligned} \widetilde{\mathcal{U} \boxtimes \mathcal{V}} &= s_{d,e}^*(\widetilde{\mathcal{U}} \boxtimes \widetilde{\mathcal{V}}), \\ (\widetilde{\mathcal{U} \boxtimes \mathcal{V}})_{\pi_{D,E}(\alpha,\gamma), \pi_{D,E}(\beta,\delta)}^{\pi_{d,e}(i,k), \pi_{d,e}(j,\ell)} &= \widetilde{\mathcal{U}}_{\alpha,\beta}^{i,j} \widetilde{\mathcal{V}}_{\gamma,\delta}^{k,\ell}. \end{aligned} \quad (18.255)$$

For positive physical dimensions this uses the normalization of [15, Section III, equation (13)]. With the totalized inverse convention $0^{-1} = 0$, the algebraic identity also holds when $d = 0$ or $e = 0$. This identity expresses the normalized-flattening compatibility for the tensor product in [15, Theorem IV.6(ii), lines 824–845].

Proof. [✓ Proof](#) Apply the two product-coordinate entry formulas after the regrouping in (18.254). The scalar normalizations agree because

$$(\sqrt{de})^{-1} = (\sqrt{d}\sqrt{e})^{-1} = (\sqrt{d})^{-1}(\sqrt{e})^{-1}. \quad (18.256)$$

Multiplying this identity by $\mathcal{U}_{\alpha,\beta}^{i,j} \mathcal{V}_{\gamma,\delta}^{k,\ell}$ gives (18.255). $\square$

Theorem 18.6.6 (Canonical-form-II data under independent tensor products). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B have chosen canonical-form-II data with retained blocks A_a and B_b , dimensions D_a and E_b , weights μ_a and ν_b , diagonal positive-definite fixed points Λ_a and Γ_b , and ambient coisometries V_A and V_B . Write $\pi_{m,n}$ for the standard product-coordinate equivalence and let χ be the retained-coordinate equivalence

$$\chi((a, \alpha), (b, \beta)) = (\pi_{r_A, r_B}(a, b), \pi_{D_a, E_b}(\alpha, \beta)). \quad (18.257)$$

The product ambient coisometry is

$$W = \text{reind}_{\chi, \pi_{D,E}}(V_A \otimes V_B). \quad (18.258)$$

For each retained pair (a, b) , the chosen product data are

$$D_{(a,b)} = D_a E_b, \quad (18.259)$$

$$\omega_{(a,b)} = \mu_a \nu_b, \quad (18.260)$$

$$C_{(a,b)}^{\pi_{d,e}(i,k)} = \text{reind}_{\pi_{D_a, E_b}, \pi_{D_a, E_b}}(A_a^i \otimes B_b^k), \quad (18.261)$$

and

$$\Lambda_{(a,b)} = \text{reind}_{\pi_{D_a, E_b}, \pi_{D_a, E_b}}(\Lambda_a \otimes \Gamma_b). \quad (18.262)$$

The exact reconstruction is

$$A \boxtimes B = W^\dagger \left(\bigoplus_{a,b} \mu_a \nu_b (A_a \boxtimes B_b) \right) W. \quad (18.263)$$

Thus the retained direct sum is conjugated by the reindexed Kronecker ambient coisometry, rather than identified directly with the ambient tensor. For MPO tensors $\mathcal{U}$ and $\mathcal{V}$, the doubled physical equivalence of Theorem 18.6.5 identifies these data with canonical-form-II data for $\widetilde{\mathcal{U} \boxtimes \mathcal{V}}$.

Full support is a supplied reduced-representative restriction. If the two chosen input data satisfy

$$\sum_a D_a = D, \quad (18.264)$$

and

$$\sum_b E_b = E, \quad (18.265)$$

then the chosen product data satisfy

$$\sum_{a,b} D_a E_b = \left(\sum_a D_a \right) \left(\sum_b E_b \right) = DE. \quad (18.266)$$

Identifying canonical-form-II data along an equality of ambient tensors leaves this retained-dimension sum unchanged. This full-support assertion is not stated by the tensoring sentence in [15, Theorem IV.6(ii)]; it is the supplied restriction documented in [66].

For the independent tensor product, the raw source ranks satisfy $r(\mathcal{U} \boxtimes \mathcal{V}) = r(\mathcal{U})r(\mathcal{V})$ and $\ell(\mathcal{U} \boxtimes \mathcal{V}) = \ell(\mathcal{U})\ell(\mathcal{V})$ by Theorem 18.4.40.

Proof. ✓ Proof For physical labels i and k , set

$$\mathbf{B}_A^i = \bigoplus_a \mu_a A_a^i, \quad (18.267)$$

and

$$\mathbf{B}_B^k = \bigoplus_b \nu_b B_b^k. \quad (18.268)$$

Regrouping retained coordinates by χ gives the weighted-block-diagonal identity

$$\bigoplus_{a,b} \mu_a \nu_b C_{(a,b)}^{\pi_{d,e}(i,k)} = \text{reind}_{\chi, \chi}(\mathbf{B}_A^i \otimes \mathbf{B}_B^k). \quad (18.269)$$

Substitute $A^i = V_A^\dagger \mathbf{B}_A^i V_A$ and $B^k = V_B^\dagger \mathbf{B}_B^k V_B$. The mixed-product law for Kronecker products and the definition of W then give (18.263). The relation $WW^\dagger = \mathbb{1}$ follows from $V_A V_A^\dagger = \mathbb{1}$ and $V_B V_B^\dagger = \mathbb{1}$. Left-canonicity and normality follow from Theorem 18.4.31.

For arbitrary matrices X and Y , direct expansion of the finite Kraus sums gives

$$\mathcal{E}_{A \boxtimes B}(\text{reind}_{\pi_{D,E}, \pi_{D,E}}(X \otimes Y)) = \text{reind}_{\pi_{D,E}, \pi_{D,E}}(\mathcal{E}_A(X) \otimes \mathcal{E}_B(Y)). \quad (18.270)$$

Taking $X = \Lambda_a$ and $Y = \Gamma_b$ proves the fixed-point equation; Kronecker products and bijective reindexing preserve positive definiteness and diagonality. The finite double-sum identity proves the supplied full-support conclusion. Finally apply the normalized-flattening identity and physical relabeling to obtain the MPO statement. $\square$

Theorem 18.6.7 (Canonical form II under independent tensor products). [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ and $\mathcal{V}$ be MPUs in canonical form II, with recorded ambient right fixed matrices ρ_U and ρ_V . Their independent tensor product is in canonical form II in the product bond coordinates. Its recorded matrix is*

$$\rho_{U \boxtimes V} = \text{reind}_{\pi_{D,E}, \pi_{D,E}}(\rho_U \otimes \rho_V). \quad (18.271)$$

In particular, the product presentation has full support. This is the canonical-form-II consequence of the independent-product construction used in [15, Theorem IV.6(ii)]; the full-support convention is recorded in [66].

Proof. [✓ Proof](#) Tensor the chosen canonical blocks and their diagonal positive fixed matrices, then apply the doubled physical-coordinate relabeling. The retained dimensions multiply and their sum is the product bond dimension. Independent tensor products preserve the MPU property. The product of the two ambient fixed matrices is positive definite, diagonal, and of trace one. In the product bond coordinates, the transfer identity gives

$$\mathcal{E}_{\widetilde{U \boxtimes V}}(\rho_{U \boxtimes V}) = \text{reind}_{\pi_{D,E}, \pi_{D,E}}(\mathcal{E}_{\widetilde{U}}(\rho_U) \otimes \mathcal{E}_{\widetilde{V}}(\rho_V)) \quad (18.272)$$

$$= \text{reind}_{\pi_{D,E}, \pi_{D,E}}(\rho_U \otimes \rho_V) = \rho_{U \boxtimes V}. \quad (18.273)$$

$\square$

Lemma 18.6.8 (Normalized flattening of a composition tensor). [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ and $\mathcal{V}$ be MPO tensors with the same positive physical dimension d , and let*

$$W^{ik} = \sum_{j=0}^{d-1} U^{ij} \otimes V^{jk} \quad (18.274)$$

be their composition tensor. Under the canonical identification of its auxiliary space with $\mathbb{C}^{D_U} \otimes \mathbb{C}^{D_V}$, the normalized flattenings satisfy

$$\widetilde{W}^{ik} = \sqrt{d} \sum_{j=0}^{d-1} \widetilde{U}^{ij} \otimes \widetilde{V}^{jk}. \quad (18.275)$$

This identity combines the raw composition tensor used in the proof of [15, Theorem IV.6(ii)] with the normalization in [15, Section III, equation (13)]; it is not a separately stated result of that proof.

Proof. [✓ Proof](#) Substitute $U^{ij} = \sqrt{d} \widetilde{U}^{ij}$ and $V^{jk} = \sqrt{d} \widetilde{V}^{jk}$ into the defining contraction for W^{ik} , and then multiply the result by $d^{-1/2}$. $\square$

Theorem 18.6.9 (Normalized flattening under blocking). Lean ✓ Stmt Let $\mathcal{U}$ be an MPO tensor of physical dimension d , and let $L \geq 0$. For the normalized-tensor interpretation assume $d > 0$, as in Definition 18.6.3; with the convention $0^{-1} = 0$, the algebraic identity below also holds for $d = 0$. Pairing the ket and bra letters at each site gives a canonical bijection

$$\vartheta_L : \text{Fin}(d^L)^2 \longrightarrow \text{Fin}(d^2)^L. \quad (18.276)$$

If $\widetilde{\mathcal{U}} = d^{-1/2}\mathcal{U}$ denotes normalized flattening, then

$$\widetilde{\mathcal{U}^{[L]}} = \vartheta_L^*(\widetilde{\mathcal{U}}^{[L]}). \quad (18.277)$$

Thus normalized flattening commutes with blocking, up to the canonical doubled-index relabeling. This is the normalized form of the blocking operation in [15, Definition II.5 and equation (9)].

Proof. ✓ Proof A blocked letter is a product of L original letters. The identity

$$(\sqrt{d^L})^{-1} = ((\sqrt{d})^{-1})^L \quad (18.278)$$

identifies its normalization with the product of the L one-site normalizations. Theorem 20.15.2 identifies the two physical indexings for every $L \geq 0$; at $L = 0$ both blocked words are empty and both local matrix products are the identity. $\square$

Theorem 18.6.10 (Left-canonical normalization under blocking and relabeling). ✓ Stmt Suppose that an MPS tensor A is left-canonical:

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}. \quad (18.279)$$

Then for every $L \geq 0$ its blocked tensor is left-canonical:

$$\sum_I ((A^{[L]})^I)^\dagger (A^{[L]})^I = \mathbb{1}. \quad (18.280)$$

If e is a bijection of physical alphabets, then e^*A is left-canonical if and only if A is:

$$\sum_j ((e^*A)^j)^\dagger (e^*A)^j = \mathbb{1} \iff \sum_i (A^i)^\dagger A^i = \mathbb{1}. \quad (18.281)$$

Proof. ✓ Proof Expand (18.280) as a sum over length- L words and sum one letter at a time, repeatedly applying the one-site left-canonical identity. For $L = 0$, the unique empty word evaluates to the identity. The equivalence (18.281) follows by reindexing the finite sum along e . $\square$

Definition 18.6.11 (Canonical-form-II data under blocking and relabeling). Lean Lean Lean ✓ Stmt Let A have canonical-form-II data with retained weights μ_k , blocks A_k , and positive definite diagonal fixed matrices Λ_k . For every positive integer L , the blocked tensor $A^{[L]}$ has canonical-form-II data with weights μ_k^L , blocks $A_k^{[L]}$, and the same matrices Λ_k . The retained blocks remain left-canonical:

$$\sum_I ((A_k^{[L]})^I)^\dagger (A_k^{[L]})^I = \mathbb{1}. \quad (18.282)$$

A bijective relabeling of the physical alphabet likewise preserves canonical-form-II data, including both the matrices Λ_k and the left-canonical identities.

Consequently, if the normalized flattening $\widetilde{\mathcal{U}}$ of an MPO tensor has canonical-form-II data, then for every $L > 0$ the normalized flattening of $\mathcal{U}^{[L]}$ has the data obtained by first blocking and then relabeling by ϑ_L from Theorem 18.6.9. Theorem 18.6.10 preserves the left-canonical identity under both operations. For each retained block, Lemma D.4.9 gives

$$\begin{aligned}\mathcal{E}_{A_k^{[L]}} &= \mathcal{E}_{A_k}^L, \\ \mathcal{E}_{A_k}^L(\Lambda_k) &= \Lambda_k,\end{aligned}\tag{18.283}$$

where (18.283) follows by iterating $\mathcal{E}_{A_k}(\Lambda_k) = \Lambda_k$. Thus the same positive definite diagonal matrix is fixed after blocking. A physical bijection leaves the transfer map unchanged. The final construction applies these two operations to (18.277).

Theorem 18.6.12 (Canonical form II under positive physical blocking). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ be an MPU in canonical form II, with recorded ambient right fixed matrix ρ . For every positive physical blocking length p , the blocked tensor $\mathcal{U}^{[p]}$ is in canonical form II in the same bond coordinates, with full support and the same matrix ρ . This is the positive-blocking assertion following [15, Proposition III.1].*

Proof. [✓ Proof](#) Block each canonical tensor and relabel the blocked physical alphabet. Blocking leaves the retained bond dimensions unchanged, so their sum is still the ambient bond dimension. The transfer identities in Definition 18.6.11, together with invariance under physical relabeling, give

$$\mathcal{E}_{\widetilde{\mathcal{U}^{[p]}}}(\rho) = \mathcal{E}_{\mathcal{U}}^p(\rho) = \rho.\tag{18.284}$$

Positive definiteness, diagonality, and trace normalization of ρ are unchanged, and blocking preserves the MPU property. $\square$

Theorem 18.6.13 (Self-overlap scaling). [Lean](#) [Lean](#) [✓ Stmt](#) *For a matrix product tensor A , a complex number c , and a nonnegative integer N , scaling every local matrix by c gives*

$$\sum_{\sigma} V^{(N)}(cA)_{\sigma} \overline{V^{(N)}(cA)_{\sigma}} = (c\bar{c})^N \sum_{\sigma} V^{(N)}(A)_{\sigma} \overline{V^{(N)}(A)_{\sigma}},\tag{18.285}$$

Consequently, if $d > 0$ and U is an MPO tensor of physical dimension d , then

$$\sum_{\sigma} V^{(N)}(\widetilde{U})_{\sigma} \overline{V^{(N)}(\widetilde{U})_{\sigma}} = d^{-N} \sum_{\sigma} V^{(N)}(U^{\text{MPS}})_{\sigma} \overline{V^{(N)}(U^{\text{MPS}})_{\sigma}}.\tag{18.286}$$

Proof. [✓ Proof](#) Each length- N coefficient satisfies $V^{(N)}(cA)_{\sigma} = c^N V^{(N)}(A)_{\sigma}$, yielding (18.285) upon multiplying by its complex conjugate; setting $c = 1/\sqrt{d}$ gives (18.286). $\square$

Theorem 18.6.14 (Normalized MPU transfer trace). [Lean](#) [✓ Stmt](#) *Let U be an MPU tensor with positive physical and bond dimensions, and let E be the transfer matrix of its normalized flattening. For every integer $N > 1$,*

$$\text{tr}(E^N) = 1.\tag{18.287}$$

This is the trace identity in the proof of the normal-tensor result [15, Proposition III.1].

Proof. ✓ Proof Scaling both tensor copies by $1/\sqrt{d}$ multiplies the length- N self-overlap by d^{-N} . The doubled-index overlap identity and MPU unitarity give the complete chain

$$\begin{aligned}
\mathrm{tr}(E^N) &= \sum_{\sigma} V^{(N)}(\widetilde{U})_{\sigma} \overline{V^{(N)}(\widetilde{U})_{\sigma}} \\
&= d^{-N} \sum_{\sigma} V^{(N)}(U^{\mathrm{MPS}})_{\sigma} \overline{V^{(N)}(U^{\mathrm{MPS}})_{\sigma}} \\
&= d^{-N} \mathrm{tr}(U^{(N)}(U^{(N)})^{\dagger}) \\
&= d^{-N} \mathrm{tr}(\mathbb{1}) \\
&= d^{-N} d^N \\
&= 1.
\end{aligned} \tag{18.288}$$

Here (18.288) uses $N > 1$, so that the MPU equation applies. □

18.7 Block transfer multiplicity

Definition 18.7.1 (Flattened dependent-block inclusion). Lean ✓ Stmt Let the blocks be indexed by $k \in \{0, \dots, r-1\}$, let n_k be their nonnegative dimensions, and put $n = \sum_{k \in \{0, \dots, r-1\}} n_k$. The standard block-order bijection is

$$\phi: \bigsqcup_{k \in \{0, \dots, r-1\}} \{0, \dots, n_k-1\} \longrightarrow \{0, \dots, n-1\}, \tag{18.289}$$

$$\phi(k, a) = \sum_{j=0}^{k-1} n_j + a. \tag{18.290}$$

For each $k \in \{0, \dots, r-1\}$, define $J_k: \mathbb{C}^{n_k} \rightarrow \mathbb{C}^n$ by

$$(J_k)_{x,a} = \begin{cases} 1, & x = \phi(k, a), \\ 0, & x \neq \phi(k, a). \end{cases} \tag{18.291}$$

Theorem 18.7.2 (Dependent-block inclusion identities). Lean Lean Lean Lean ✓ Stmt The inclusions satisfy the following identities, with the second holding for distinct k and ℓ :

$$J_k^{\dagger} J_k = \mathbb{1}, \tag{18.292}$$

$$J_k^{\dagger} J_{\ell} = 0. \tag{18.293}$$

If $B^i = \bigoplus_k \mu_k A_k^i$ is a weighted block-diagonal tensor, then

$$B^i J_k = J_k (\mu_k A_k^i). \tag{18.294}$$

Proof. ✓ Proof Reindex the canonical dependent-summand inclusion along the equivalence between dependent pairs and flattened coordinates. Transporting its isometry, cross-block orthogonality, and block-diagonal intertwining identities gives the three displayed formulas. □

Definition 18.7.3 (Ambient inclusion of a canonical block). Lean ✓ Stmt If U is the reconstruction coisometry and J_k is the inclusion of the k th retained block, define

$$V_k = U^{\dagger} J_k. \tag{18.295}$$

Theorem 18.7.4 (Ambient block isometry and intertwining). [Lean](#) [Lean](#) [✓ Stmt](#) For every retained block and physical index i ,

$$V_k^\dagger V_k = \mathbb{1}, \quad (18.296)$$

$$A^i V_k = V_k(\mu_k A_k^i). \quad (18.297)$$

Proof. [✓ Proof](#) The coisometry $UU^\dagger = \mathbb{1}$ and $J_k^\dagger J_k = \mathbb{1}$ give

$$V_k^\dagger V_k = J_k^\dagger U U^\dagger J_k = J_k^\dagger J_k = \mathbb{1}. \quad (18.298)$$

The canonical reconstruction and block-diagonal intertwining give

$$A^i V_k = U^\dagger \left(\bigoplus_j \mu_j A_j^i \right) J_k = U^\dagger J_k (\mu_k A_k^i) = V_k(\mu_k A_k^i). \quad (18.299)$$

□

Definition 18.7.5 (Weighted-block transfer eigenvalue). [Lean](#) [✓ Stmt](#) For a block k , define its weighted transfer eigenvalue by

$$q_k = \mu_k \overline{\mu_k} = |\mu_k|^2. \quad (18.300)$$

Theorem 18.7.6 (Shifted traces normalize every weight). [Lean](#) [✓ Stmt](#) If the ambient transfer matrix satisfies $\text{tr}(E^N) = 1$ for every integer $N > 1$, then every block satisfies

$$q_k = 1. \quad (18.301)$$

Proof. [✓ Proof](#) Choose a nonzero fixed matrix X_k for the unweighted normal block and set $Y_k = V_k X_k V_k^\dagger$. The isometry and intertwining identities give

$$Y_k \neq 0, \quad E(Y_k) = q_k Y_k. \quad (18.302)$$

Thus q_k lies in the ambient transfer spectrum; it is nonzero because $\mu_k \neq 0$. The shifted trace-power spectrum is $\{0, 1\}$, hence $q_k = 1$. □

Theorem 18.7.7 (Unique full-support block geometry). [Lean](#) [Lean](#) [✓ Stmt](#) If there is exactly one block k and the blocks fill the ambient bond space, then

$$D_k = D, \quad V_k V_k^\dagger = \mathbb{1}. \quad (18.303)$$

Proof. [✓ Proof](#) The full-support sum reduces to the unique summand:

$$D = \sum_{j=1}^r D_j = D_k. \quad (18.304)$$

Therefore V_k is a square matrix. Since $V_k^\dagger V_k = \mathbb{1}$, finite equal-dimensional inverse symmetry gives $V_k V_k^\dagger = \mathbb{1}$. □

Theorem 18.7.8 (Shifted transfer traces determine one block). [Lean](#) [✓ Stmt](#) Let A have literal CPSV canonical-form data with r blocks. If its transfer matrix E satisfies $\text{tr}(E^N) = 1$ for every integer $N > 1$, then

$$r = 1. \quad (18.305)$$

This is the block multiplicity consequence used in [15, Proposition III.1].

Proof. ✓ Proof Each normal block A_k has a nonzero transfer fixed vector X_k . Let U be the reconstruction coisometry and let J_k be the flattened inclusion of the k th block. Define the ambient inclusion and transported matrix by

$$V_k = U^\dagger J_k, \quad Y_k = V_k X_k V_k^\dagger. \quad (18.306)$$

The matrix Y_k is nonzero and is an ambient transfer eigenmatrix with eigenvalue $q_k = \mu_k \overline{\mu_k} \neq 0$. Theorem 18.4.6 gives $q_k = 1$.

For distinct k and ℓ , the inclusion identities give

$$V_k^\dagger Y_k V_k = X_k, \quad (18.307)$$

$$V_k^\dagger Y_\ell V_k = 0. \quad (18.308)$$

Hence the matrices Y_k , for $1 \leq k \leq r$, are linearly independent. Vectorizing gives

$$\text{vec}(Y_k) \in \ker(E^2 - \mathbb{1}). \quad (18.309)$$

For every $r \geq 1$, the shifted traces and Theorem 18.4.4 give

$$\text{tr}((E^2)^r) = 1, \quad (18.310)$$

$$\chi_{E^2}(X) = X^{D^2-1}(X - 1). \quad (18.311)$$

The eigenspace dimension is bounded by the algebraic multiplicity, so

$$r \leq \dim \ker(E^2 - \mathbb{1}) \leq \text{mult}_1(\chi_{E^2}) = 1. \quad (18.312)$$

Finally, if r were zero, the retained direct sum and therefore A would vanish, contradicting $\text{tr}(E^2) = 1$. Consequently $r = 1$. $\square$

18.8 Normalized transfer spectrum and conditional normality

Theorem 18.8.1 (Shifted transfer traces imply radius one and primitivity). Lean ✓ Stmt Let $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map with $D > 0$. If $\text{tr}(\widehat{T}^N) = 1$ for every integer $N > 1$, then

$$r(T) = 1, \quad \text{peripheral}(T) = \{1\}. \quad (18.313)$$

This is the radius-and-primitivity part of [15, Proposition III.1].

Proof. ✓ Proof The shifted traces give

$$\sigma(\widehat{T}) \setminus \{0\} = \{1\}. \quad (18.314)$$

Column stacking identifies the eigenvalues of T and $\widehat{T}$. Consequently every spectral value of T is zero or one, and one occurs, so

$$r(T) = \max_{\lambda \in \sigma(T)} |\lambda| = 1. \quad (18.315)$$

Restricting to eigenvalues of modulus one gives

$$\text{peripheral}(T) = \{1\}, \quad (18.316)$$

which is primitivity. $\square$

Theorem 18.8.2 (Exact spectrum of the normalized MPU transfer matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let U be an MPU tensor with positive physical and bond dimensions, and let E be the transfer matrix of its normalized flattening. Then

$$\sigma(E) \setminus \{0\} = \{1\}, \quad \text{mult}_1(\chi_E) = 1, \quad \chi_E(X) = X^{D^2-1}(X-1). \quad (18.317)$$

This is the transfer-matrix conclusion of [15, Proposition III.1].

Proof. [✓ Proof](#) For every integer $N > 1$, the normalized transfer identity gives

$$\text{tr}(E^N) = 1. \quad (18.318)$$

Applying the exact shifted trace-power theorem in dimension D^2 yields

$$\chi_E(X) = X^{D^2-1}(X-1), \quad \text{mult}_1(\chi_E) = 1, \quad \sigma(E) \setminus \{0\} = \{1\}. \quad (18.319)$$

□

Theorem 18.8.3 (Finite blocking eliminates the zero-primary transfer component). [Lean](#) [✓ Stmt](#) Let U be an MPU tensor with physical dimension $d > 0$ and bond dimension $D > 1$, and let E be the transfer matrix of its normalized flattening. There is an integer J satisfying

$$0 < J \leq D^2 - 1 \quad (18.320)$$

and vectors ρ, Φ satisfying

$$(\Phi \mid \rho) = 1, \quad E\rho = \rho, \quad (\Phi \mid E = (\Phi \mid, \quad (18.321)$$

such that

$$E^J = |\rho\rangle\langle\Phi|. \quad (18.322)$$

For every integer $k \geq J$,

$$E^k = E^J. \quad (18.323)$$

Thus blocking J sites removes the entire generalized zero-eigenspace. The one-dimensional case is stated separately below.

This is the Jordan-elimination step in [15, proof of Proposition III.2(ii)].

Proof. [✓ Proof](#) The exact characteristic polynomial is

$$\chi_E(X) = X^{D^2-1}(X-1). \quad (18.324)$$

Cayley–Hamilton gives $E^{D^2} = E^{D^2-1}$. Hence every power from $J = D^2 - 1$ onward is equal, and $P = E^J$ is idempotent. Since $J > 1$, the normalized transfer trace identity gives $\text{tr}(P) = 1$. The range of an idempotent has dimension equal to its trace, so P has rank one and factors as $P = |\rho\rangle\langle\Phi|$. Taking the factors with $(\Phi \mid \rho) = 1$ gives the displayed fixed-point identities from $EP = PE = P$. □

Theorem 18.8.4 (Uniqueness of the normalized fixed-pair factorization). [Lean](#) [✓ Stmt](#) Let $P = E^J$ be a stabilized rank-one transfer power with normalized left and right factors. If ρ' and Φ' are any other fixed vectors with $(\Phi' \mid \rho') = 1$, then

$$P = |\rho'\rangle\langle\Phi'|. \quad (18.325)$$

Proof. ✓ Proof Applying P to ρ' and applying Φ' to P shows that the two pairs of rank-one factors differ by reciprocal scalars. Their normalized pairings force the product of those scalars to be one. $\square$

Theorem 18.8.5 (Identification with chosen reduced-CFII fixed points). Lean ✓ Stmt Suppose $d > 0$, $D > 1$, and $\mathcal{U}$ is an MPU whose normalized flattening is presented by chosen canonical-form-II data with blocks filling the bond space. There is a unique block k . Normalize its diagonal positive definite fixed matrix Λ_k to have trace one, and let V_k be its inclusion into the ambient bond space. Then

$$\rho = V_k \Lambda_k V_k^\dagger, \quad \rho > 0, \quad \text{tr}(\rho) = 1, \quad (18.326)$$

$$E|\rho\rangle = |\rho\rangle, \quad (\Phi|E = (\Phi|, \quad (\Phi|\rho) = 1, \quad (18.327)$$

where $(\Phi|$ is the vectorized identity. Consequently,

$$E^{D^2-1} = |\rho\rangle(\Phi|. \quad (18.328)$$

The diagonal matrix is the block matrix Λ_k in the canonical-form-II fixed-point equations in [15, equations (6a)–(6b)]; its ambient transport ρ need not be diagonal, corresponding to the reduced representative in [15, proof of Proposition III.2(ii)].

Proof. ✓ Proof The shifted trace identities satisfy $\text{tr}(E^N) = 1$ for every integer $N > 1$ and give

$$\text{tr}(E^N) = 1 \implies r = 1. \quad (18.329)$$

Let k be this block. Canonical form II supplies a diagonal $\Lambda_0 > 0$ fixed by the unweighted block transfer map. Normalize it by

$$\Lambda = \text{tr}(\Lambda_0)^{-1} \Lambda_0, \quad \text{tr}(\Lambda) = 1. \quad (18.330)$$

If ω_k is its weight, then

$$|\omega_k|^2 = 1, \quad E_k(\Lambda) = \Lambda, \quad (18.331)$$

where E_k is the transfer map of the weighted block $\widetilde{B}_k = \omega_k B_k$. Full support makes the block inclusion $V = V_k$ both isometric and onto:

$$V^\dagger V = I, \quad VV^\dagger = I. \quad (18.332)$$

Hence $\rho = V\Lambda V^\dagger$ is positive definite with trace one, and the intertwining equations $A_i V = V\widetilde{B}_k(i)$ give the fixedness transport

$$E(V\Lambda V^\dagger) = VE_k(\Lambda)V^\dagger = V\Lambda V^\dagger = \rho. \quad (18.333)$$

The blockwise left-canonical identity transports similarly:

$$\sum_i \widetilde{B}_k(i)^\dagger \widetilde{B}_k(i) = I \implies \sum_i A_i^\dagger A_i = I \implies (\Phi|E = (\Phi|. \quad (18.334)$$

Finally,

$$(\Phi|\rho) = \text{tr}(\rho) = 1, \quad E^{D^2-1} = |\rho\rangle(\Phi|, \quad (18.335)$$

where the last equality is the uniqueness of the normalized fixed-pair factorization for the stabilized power. $\square$

Theorem 18.8.6 (Forced simple block in canonical form II). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU in canonical form II, with ambient fixed matrix ρ . Put $J = \max(D^2 - 1, 1)$ and $W = \mathcal{U}_{JD^2}$. With the vectorized pair ρ', Φ' of ρ and of the identity, the blocked tensor W satisfies the two aligned simple contractions

$$\Phi' \cdot W^{ij} \rho' = \delta_{ij}, \quad (18.336)$$

$$W^{ij} W^{k\ell} = W^{ij} | \rho' \rangle \langle \Phi' | W^{k\ell}. \quad (18.337)$$

Proof. [✓ Proof](#) The convention records a positive trace-one ambient fixed matrix, and Theorem 18.8.14 identifies the stabilized power with $|\rho\rangle\langle\Phi|$ at the positive exponent J . Applying Theorem 18.3.25 with this pair gives both contractions on the direct block $W = \mathcal{U}_{JD^2}$. $\square$

Theorem 18.8.7 (One-dimensional normalized transfer matrix). [Lean](#) [✓ Stmt](#) If the physical dimension satisfies $d > 0$ and the bond dimension is $D = 1$, then the normalized transfer matrix satisfies

$$E = 1. \quad (18.338)$$

Consequently its powers stabilize at the positive exponent one.

Proof. [✓ Proof](#) In dimension one the characteristic polynomial identity reduces to $\chi_E(X) = X - 1$, and Cayley–Hamilton gives $E = 1$. $\square$

Definition 18.8.8 (Canonical form II for a matrix product unitary). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor, and write $\tilde{\mathcal{U}} = d^{-1/2} \mathcal{U}$ and $E = \mathcal{E}_{\tilde{\mathcal{U}}}$. A canonical-form-II presentation of $\mathcal{U}$ consists of the MPU condition, a literal CPSV canonical-form-II representation

$$\begin{aligned} \tilde{\mathcal{U}}^a &= V^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^a \right) V, \\ VV^\dagger &= \mathbb{1}, \\ \sum_{k=1}^r D_k &= D. \end{aligned} \quad (18.339)$$

The ambient fixed pair has $\Phi = \mathbb{1}_D$ and positive coefficients $\rho_0, \dots, \rho_{D-1}$. It satisfies

$$\begin{aligned} \rho &= \sum_{n=0}^{D-1} \rho_n |n\rangle\langle n|, \\ \text{tr}(\rho) &= 1, \\ E(\rho) &= \rho. \end{aligned} \quad (18.340)$$

Thus the retained blocks have literal full support, and ρ is a positive diagonal trace-one right fixed point in the ambient bond coordinates. This is the standing convention of [15, lines 269–281, 344–356, and 488–502].

Theorem 18.8.9 (Canonical representative from a positive left fixed point). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ generate an MPU, suppose its normalized flattening $\tilde{\mathcal{U}}$ is normal, and let L be positive definite with $\mathcal{E}_{\tilde{\mathcal{U}}}^*(L) = L$. There is a unitary V on the same bond space such that

$$\mathcal{B}^{ij} = V^\dagger L^{1/2} \mathcal{U}^{ij} L^{-1/2} V \quad (18.341)$$

admits a full-support canonical-form-II presentation. For every $N \geq 0$, the periodic operators satisfy $O_N(\mathcal{B}) = O_N(\mathcal{U})$. Both source ranks are preserved: $r[\mathcal{B}] = r[\mathcal{U}]$ and $\ell[\mathcal{B}] = \ell[\mathcal{U}]$. This is the canonical-gauge construction used in the proof of [15, Proposition IV.5, lines 798–804].

Proof. ✓ Proof Put $S = L^{1/2}$ and $\mathcal{G}^{ij} = S\mathcal{U}^{ij}S^{-1}$. The adjoint fixed-point equation gives

$$\sum_{i,j} (\tilde{\mathcal{G}}^{ij})^\dagger \tilde{\mathcal{G}}^{ij} = S^{-1} \mathcal{E}_{\mathcal{U}}^*(L) S^{-1} = \mathbb{1}.$$

This is the positive-fixed-point gauge construction of the companion quantum-channel volume [182, Trace-preserving gauge from an adjoint fixed point, Kraus form]. Normality is preserved by similarity. Theorem 7.10.1 therefore supplies a unitary V and a positive diagonal fixed matrix Λ after conjugating $\tilde{\mathcal{G}}$ by V . Replace Λ by $\rho = \Lambda/\text{tr}(\Lambda)$. Positivity gives $\text{tr}(\Lambda) > 0$, and linearity preserves the fixed-point equation. The resulting normal, left-canonical tensor has a one-block canonical-form-II presentation with weight one and no omitted bond coordinates. Finally, put $T = S^{-1}V$, so that $\mathcal{B}^{ij} = T^{-1}\mathcal{U}^{ij}T$. The interior factors TT^{-1} cancel, and cyclicity of the trace removes the outer pair:

$$\text{tr}(\mathcal{B}^{i_1 j_1} \dots \mathcal{B}^{i_N j_N}) = \text{tr}(T^{-1}\mathcal{U}^{i_1 j_1} \dots \mathcal{U}^{i_N j_N} T) = \text{tr}(\mathcal{U}^{i_1 j_1} \dots \mathcal{U}^{i_N j_N}).$$

Since these traces are the coefficients of the periodic operators, $O_N(\mathcal{B}) = O_N(\mathcal{U})$ for every N . Since S , S^{-1} , V , and $V^\dagger$ are invertible, Theorem 18.9.94 applied to the two successive bond transformations proves equality of both source ranks. $\square$

Theorem 18.8.10 (Canonical form II forces positive dimensions). Lean Lean ✓ Stmt *If an MPO tensor $\mathcal{U}$ admits a canonical-form-II presentation, then its bond space is nonempty and its physical dimension is positive:*

$$D > 0, \quad d > 0. \quad (18.342)$$

Proof. ✓ Proof Both dimensions are forced by the normalization $\text{tr}(\rho) = 1$ of (18.340). If $D = 0$, the ambient bond index set is empty and every trace over it vanishes,

$$\text{tr}(\rho) = \sum_{n \in \emptyset} \rho_{nn} = 0 \neq 1. \quad (18.343)$$

If $d = 0$, the physical index set is empty, so the normalized transfer map is an empty sum,

$$E(\rho) = \sum_{i \in \emptyset} \tilde{\mathcal{U}}^i \rho (\tilde{\mathcal{U}}^i)^\dagger = 0. \quad (18.344)$$

The fixed-point equation $E(\rho) = \rho$ of (18.340) then gives $\rho = 0$, hence $\text{tr}(\rho) = 0 \neq 1$. $\square$

Definition 18.8.11 (MPU canonical form). Lean Lean Lean ✓ Stmt A tensor A of physical dimension d and bond dimension D is in MPU canonical form if there are positive block dimensions D_k , nonzero weights $\mu_k \in \mathbb{C}$, tensors A_k of bond dimension D_k , and a coisometry

$$C: \mathbb{C}^D \longrightarrow \bigoplus_{k=1}^r \mathbb{C}^{D_k}, \quad CC^\dagger = \mathbb{1}, \quad (18.345)$$

such that $\sum_k D_k = D$ and

$$A^i = C^\dagger \left(\bigoplus_{k=1}^r \mu_k A_k^i \right) C. \quad (18.346)$$

Every A_k is irreducible and the spectral radius of its transfer map is one. The blocks may be periodic, so no uniqueness condition is imposed on their peripheral eigenvalues. This is the canonical form of [15, lines 259–267]; normal tensors form a strictly stronger class. The nonzero-weight and full-support conventions are recorded in [93, 66].

Theorem 18.8.12 (Normality of an MPU in canonical form). [Lean](#) [✓ Stmt](#) *Let U be an MPU tensor with positive physical and bond dimensions. Suppose that the original local tensor is in canonical form, with nonzero weights and blocks occupying the full ambient bond space. Then the normalized flattening is a normal tensor.*

This is the normality assertion of [15, Proposition III.1] for the canonical representative adopted before that proposition. The nonzero-weight and full-support conventions are recorded in [93, 66]. The spectral assertion of the proposition is Theorem 18.8.2.

Proof. [✓ Proof](#) Normalizing the local tensor multiplies each weight μ_k by $d^{-1/2} \neq 0$ and leaves each block unchanged. Write $\tilde{U}$ for the normalized flattening and E for its transfer matrix. For every integer $N > 1$, unitarity gives

$$\mathrm{tr}(E^N) = 1. \quad (18.347)$$

Every nonzero eigenvalue of a weighted block transfer map is an eigenvalue of the ambient transfer map, hence equals one. Irreducibility and the positive Perron eigenvector therefore make each weighted block normal. Absorbing its weight into the block gives a decomposition into normal blocks. The block-multiplicity and transfer-spectrum results then give $r = 1$, $r(\mathcal{E}_{\tilde{U}}) = 1$, and $\mathrm{peripheral}(\mathcal{E}_{\tilde{U}}) = \{1\}$. Since the canonical blocks occupy the whole bond space,

$$\sum_{k=1}^r D_k = D. \quad (18.348)$$

Thus the unique normal block has ambient dimension D . Its irreducibility, together with the spectral identities, makes $\tilde{U}$ normal. $\square$

Theorem 18.8.13 (Normality in canonical form II). [Lean](#) [✓ Stmt](#) *Let $d, D > 0$. If an MPO tensor $\mathcal{U}$ has a canonical-form-II presentation in the sense of Definition 18.8.8, then its normalized flattening $\tilde{\mathcal{U}} = d^{-1/2}\mathcal{U}$ is a normal tensor. This is the normality assertion of [15, Proposition III.1] in the ambient convention (18.340).*

Proof. [✓ Proof](#) The canonical-form-II presentation has normal blocks and full support. The shifted transfer traces give one block, transfer spectral radius one, and peripheral spectrum $\{1\}$. The unique block has dimension D and makes $\tilde{\mathcal{U}}$ irreducible. $\square$

Theorem 18.8.14 (Ambient fixed pair in canonical form II). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ be an MPU in canonical form II with ambient fixed matrix ρ , and put $J = \max(D^2 - 1, 1)$. Then*

$$(\Phi|E = (\Phi|, \quad E^J = |\rho)(\Phi|. \quad (18.349)$$

Let $q : \{0, \dots, D-1\} \times \{0, \dots, D-1\} \rightarrow \{0, \dots, D^2-1\}$ be the standard product encoding, let R be the normalized physical diagonal of the double layer $\mathcal{W}(\mathcal{U})$, and let ρ', Φ' be the vectorizations of ρ and of $\Phi = \mathbb{1}_D$ in the doubled bond coordinates $\{0, \dots, D^2-1\}$, that is $\rho'_{q(m,n)} = \rho_{mn}$ and $\Phi'_{q(m,n)} = (\mathbb{1}_D)_{mn}$. Then also

$$R^J = |\rho')(\Phi'|. \quad (18.350)$$

The exponent is positive in every bond dimension, so no separate one-dimensional branch is needed.

Proof. ✓ Proof If $D = 1$, the normalized transfer matrix is the identity, $\rho = 1$ by $\text{tr}(\rho) = 1$, and both identities hold at $J = 1$. If $D > 1$, then $J = D^2 - 1$, and Theorem 18.8.5 applied to the canonical-form-II data of the presentation produces a positive trace-one matrix σ with $(\Phi|E = (\Phi|$ and $E^J = |\sigma)(\Phi|$. Since ρ is a fixed point of E ,

$$|\rho) = E^J |\rho) \quad (18.351)$$

$$= |\sigma)(\Phi| \rho) \quad (18.352)$$

$$= \text{tr}(\rho) |\sigma) \quad (18.353)$$

$$= |\sigma), \quad (18.354)$$

so $\sigma = \rho$. For the double-layer form, Theorem 18.3.2 identifies the normalized diagonal of the J -fold block with the J -th power of the normalized transfer matrix after the encoding q , so (18.349) gives, for all $a, b \in \{0, \dots, D^2 - 1\}$,

$$(R^J)_{ab} = (E^J)_{q^{-1}(a), q^{-1}(b)} = (|\rho)(\Phi|)_{q^{-1}(a), q^{-1}(b)} = (|\rho')(\Phi')_{ab}, \quad (18.355)$$

which is (18.350). □

Theorem 18.8.15 (Canonical form from a positive stabilized transfer factor). Lean ✓ Stmt Let $\mathcal{U}$ be an MPU with physical dimension $d > 0$, and let E be the transfer matrix of its normalized flattening. Suppose that $\rho \in M_D(\mathbb{C})$ is positive definite and diagonal and that, for some integer $K \geq 0$,

$$E^K = |\rho)(\Phi|, \quad (\Phi|X) = \text{tr}(X). \quad (18.356)$$

Then $\text{tr}(\rho) = 1$, and $\mathcal{U}$ is in canonical form II in its original bond coordinates, with ambient right fixed matrix ρ . This is the converse to the stabilized form of [15, Proposition III.1].

Proof. ✓ Proof Write $t = \text{tr}(\rho)$ and $P = |\rho)(\Phi|$. Then $P^2 = tP$ and $\text{tr}(P) = t$. If $K \geq 2$, the normalized transfer trace at length K gives $t = 1$. If $K = 1$, lengths two and three give $t^2 = t^3 = 1$, hence $t = 1$. If $K = 0$, then $P = \mathbb{1}_{D^2}$; its rank is at most one, so $D = 1$. In that dimension the normalized transfer matrix is the identity, and the first power also equals P .

Thus there is a positive integer J with $E^J = P$ and $t = 1$. Since E commutes with P and $\text{tr}(E^{J+1}) = 1$, the rank-one identity gives $PEP = \text{tr}(EP)P = P$, hence $EP = PE = P$. Therefore ρ is fixed by the transfer map and its trace functional is fixed by the adjoint map, so the normalized flattening is left-canonical. Every fixed matrix X satisfies $X = E^J X = \text{tr}(X)\rho$. Positivity of ρ and this uniqueness imply irreducibility. The normalized transfer traces also give spectral radius one and primitivity. Taking the normalized flattening itself as the sole normal block gives the asserted canonical-form-II presentation. □

18.9 Standard form, index, and symmetry classification

Definition 18.9.1 (Reduced full-support source datum). Not ready Let $\tilde{\mathcal{U}} = d^{-1/2}\mathcal{U}$ be the normalized flattening, and let E be its transfer matrix. A reduced full-support source datum for $\mathcal{U}$, with physical dimension $d > 0$ and positive bond dimension, consists of an integer $J > 0$, a diagonal positive definite matrix ρ with $\text{tr}(\rho) = 1$, and $(\Phi| = \text{vec}(\mathbb{1}_D)^\top$ such that

$$E^J = |\rho)(\Phi|. \quad (18.357)$$

The one-dimensional case is supplied directly: if $D = 1$, take $J = 1$ and $\rho = \mathbb{1}_1$ using Theorem 18.8.7. If $D > 1$, chosen canonical-form-II data with full support give a positive trace-one

ambient fixed point with $J = D^2 - 1$ by Theorem 18.8.5. Diagonality of this ambient fixed point is an additional hypothesis: the theorem gives a diagonal matrix in the unique block coordinates, but conjugation into the ambient coordinates need not preserve diagonality. For $D > 1$, this states the fixed-pair convention in [15, equations labelled **Erightleft** and **II_UTransfer**, and proof of Proposition III.2(ii)]. The $D = 1$ branch is the direct one-dimensional specialization. When $\mathcal{U}$ is an MPU, Theorem 18.8.15 gives a canonical-form-II presentation of this same tensor whose ambient fixed matrix is the supplied ρ ; it makes no assertion about continuous choices along a path.

Theorem 18.9.2 (Staggered Gram expansion of the source tensor u). [Lean](#) [✓ Stmt](#) Let $M_i = X_i Y_i$ be supplied source factorizations for $\mathcal{U}$ and the weight ρ . For retained physical pairs $p = (p_1, p_2)$ and $q = (q_1, q_2)$, one has

$$\sum_{l,r} u_{(l,r),q} \overline{u_{(l,r),p}} = \sum_{\alpha, \alpha', \beta, \beta', \gamma, j_1, j_2} \overline{\mathcal{U}_{p_2, \alpha, \beta}^{j_1} \rho_{\beta, \beta'} \mathcal{U}_{q_2, \alpha', \beta'}^{j_1}} \cdot \overline{\mathcal{U}_{p_1, \gamma, \alpha}^{j_2} \mathcal{U}_{q_1, \gamma, \alpha'}^{j_2}}. \quad (18.358)$$

The p entries are starred and the q entries are unstarred. This is precisely the staggered reflected-direct four-tensor expression obtained from $u = Y_2 - Y_1$ and the two identities in [15, equations **X1X2b** and **uUnitary**].

Proof. [✓ Proof](#) Since the first cut has rows (i, β) and columns (α, j) , its weighted normalization gives

$$\sum_r \overline{(Y_1)_{r, (\alpha, p_2)}} (Y_1)_{r, (\alpha', q_2)} = \sum_{\beta, \beta', j_1} \overline{\mathcal{U}_{p_2, \alpha, \beta}^{j_1} \rho_{\beta, \beta'} \mathcal{U}_{q_2, \alpha', \beta'}^{j_1}}. \quad (18.359)$$

The unweighted normalization of the second cut gives

$$\sum_l \overline{(Y_2)_{l, (p_1, \alpha)}} (Y_2)_{l, (q_1, \alpha')} = \sum_{\gamma, j_2} \overline{\mathcal{U}_{p_1, \gamma, \alpha}^{j_2} \mathcal{U}_{q_1, \gamma, \alpha'}^{j_2}}. \quad (18.360)$$

Expand both copies of $u_{(l,r), (i_1, i_2)} = \sum_{\alpha} (Y_2)_{l, (i_1, \alpha)} (Y_1)_{r, (\alpha, i_2)}$ and substitute these two Gram identities. Reordering the finite sums yields the stated formula. $\square$

Theorem 18.9.3 (Transpose fixed-pair trace closure). [Lean](#) [✓ Stmt](#) [Discussion: #7632](#) Let $\mathcal{U}$ be an MPO tensor, let ρ be a bond matrix, and let $M_i = X_i Y_i$ be supplied source factorizations for $\mathcal{U}$ and ρ . Construct the source tensor $u = Y_2 - Y_1$. For retained physical pairs $p = (p_1, p_2)$ and $q = (q_1, q_2)$, let W^{ij} denote the (i, j) double-layer letter, let $|\rho^\top\rangle$ be the column-stacking vector of $\rho^\top$, and set $(\Phi| = \text{vec}(\mathbb{1}_D)^\top$. Then

$$\sum_{l,r} u_{(l,r),q} \overline{u_{(l,r),p}} = \text{tr}(W^{p_2 q_2} |\rho^\top\rangle (\Phi| W^{p_1 q_1}). \quad (18.361)$$

Thus the fixed-pair weight is genuinely transposed at this stage. In the diagonal canonical-form-II coordinates fixed above, $\rho^\top = \rho$; this is the identification used in the complete contraction below.

Proof. [✓ Proof](#) Enumerate each doubled-bond coordinate as $\text{flat}(\alpha, \alpha')$. Expanding a diagonal entry of the matrix under the trace gives

$$\sum_{\beta, \beta', \gamma, j_1, j_2} \overline{\mathcal{U}_{p_2, \alpha, \beta}^{j_1} \mathcal{U}_{q_2, \alpha', \beta'}^{j_1} (\rho^\top)_{\beta', \beta}} \overline{\mathcal{U}_{p_1, \gamma, \alpha}^{j_2} \mathcal{U}_{q_1, \gamma, \alpha'}^{j_2}}. \quad (18.362)$$

Since $(\rho^\top)_{\beta', \beta} = \rho_{\beta, \beta'}$, summing over α, α' gives the identity in Theorem 18.9.2. $\square$

Theorem 18.9.4 (Retained-interior source- u closure). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7633](#) Let $\mathcal{U}$ be an MPO tensor with physical dimension $d > 0$, let W be its double layer, and put $E = d^{-1} \sum_i W^{ii}$. Let ρ be symmetric, set $(\Phi| = \text{vec}(\mathbb{1}_D)^\top$, and suppose that for some $K \geq 0$,

$$E^K = |\rho)(\Phi|. \quad (18.363)$$

For supplied source factors built from the same weight ρ , the paper gate $u = Y_2 - Y_1$ satisfies

$$\sum_{l,r} u_{(l,r),q} \overline{u_{(l,r),p}} = \text{tr}(W^{p_1 q_1} W^{p_2 q_2} E^K). \quad (18.364)$$

If ρ is diagonal, the same quantity is

$$\sum_{l,r} u_{(l,r),q} \overline{u_{(l,r),p}} = d^{-K} \sum_{\tau,\eta} \overline{U_{\eta,(p,\tau)}^{(K+2)}} U_{\eta,(q,\tau)}^{(K+2)}. \quad (18.365)$$

Thus only the K -site interior is blocked; the two endpoint letters remain one-site tensors. The p entries are conjugated and the q entries are not. This is the supplied-fixed-pair contraction in [15, Lemma III.7].

Proof. [✓ Proof](#) Symmetry changes the inserted vector $|\rho^\top)$ to $|\rho)$ in Theorem 18.9.3. Substitute the fixed pair and cycle the first endpoint letter to obtain (18.364). The normalized input-tail identity Theorem 18.4.61 expands that trace into (18.365). $\square$

Theorem 18.9.5 (Complete source- u network contraction). [Lean](#) [✓ Stmt](#) [Discussion: #5982](#) Let $\mathcal{U}$ be an MPU tensor with physical dimension $d > 0$ and bond dimension $D > 0$. Let $\rho > 0$ be diagonal with $\text{tr}(\rho) = 1$, and put $(\Phi| = \text{vec}(\mathbb{1}_D)^\top$. Suppose that for some $J > 0$ the normalized transfer matrix satisfies

$$E^J = |\rho)(\Phi|. \quad (18.366)$$

Construct the source factors and u from $\mathcal{U}$ using ρ , and set $K = JD^2$. For retained physical pairs $p, q \in \{0, \dots, d-1\}^2$, the complete source- u contraction satisfies

$$\sum_{l,r} u_{(l,r),q} \overline{u_{(l,r),p}} = d^{-K} \sum_{\tau,\eta} \overline{U_{\eta,(p,\tau)}^{(K+2)}} U_{\eta,(q,\tau)}^{(K+2)}. \quad (18.367)$$

Here τ ranges over input words of length K , and η ranges over output words of length $K+2$. This is the supplied-fixed-pair form of the complete-network equality in [15, Lemma III.7].

Proof. [✓ Proof](#) Set $K = JD^2$. Trace normalization makes the supplied rank-one transfer matrix idempotent, and the normalized blocked-transfer identity therefore gives

$$E^K = |\rho)(\Phi|. \quad (18.368)$$

Diagonality gives $\rho^\top = \rho$, so Theorem 18.9.4 applies at this K and yields the displayed normalized input-tail sum. $\square$

Theorem 18.9.6 (Complete source- v network). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\rho > 0$, and compute both source cuts, the source operator v , and the right source factors Y_1, Y_2 from the same tensor $\mathcal{U}$ using ρ . Let W be the double layer of $\mathcal{U}$, let $|\rho^\top)$ be the column-stacking vector of $\rho^\top$, and let $(\Phi| = \text{vec}(\mathbb{1}_D)^\top$. For the starred physical pair $p = (p_1, p_2)$, the unstarred pair $q = (q_1, q_2)$, and

virtual paths $a = (a_1, a_2)$ and $b = (b_1, b_2)$, put $x = ((a_1, p_1), (p_2, a_2))$ and $y = ((b_1, q_1), (q_2, b_2))$. Then

$$((Y_1 \otimes Y_2)^\dagger v^\dagger v(Y_1 \otimes Y_2))_{x,y} = (W^{p_1 q_1} W^{p_2 q_2})_{(a_1, b_1), (a_2, b_2)}, \quad (18.369)$$

$$((Y_1 \otimes Y_2)^\dagger (Y_1 \otimes Y_2))_{x,y} = (W^{p_1 q_1} |\rho^\top|)(\Phi | W^{p_2 q_2})_{(a_1, b_1), (a_2, b_2)}. \quad (18.370)$$

Consequently the dressed Gram equation

$$(Y_1 \otimes Y_2)^\dagger v^\dagger v(Y_1 \otimes Y_2) = (Y_1 \otimes Y_2)^\dagger (Y_1 \otimes Y_2) \quad (18.371)$$

holds if and only if

$$W^{ij} W^{k\ell} = W^{ij} |\rho^\top| (\Phi | W^{k\ell}) \quad (18.372)$$

for all i, j, k, ℓ . The source gate and source factors use the same weight ρ throughout. The transpose occurs only in the inserted column-stacking vector. In the source, ρ is diagonal, so this vector equals $|\rho\rangle$ and the displayed equivalence is exactly [15, equation *vUnitary* and proof of Theorem III.8].

Proof. ✓ Proof Regrouping gives $(Y_1 \otimes Y_2)^\dagger v^\dagger v(Y_1 \otimes Y_2) = (v(Y_1 \otimes Y_2))^\dagger v(Y_1 \otimes Y_2)$, and Theorem 18.4.64 closes each factor by the two source-cut factorizations:

$$(v(Y_1 \otimes Y_2))_{z,x} = \sum_{\gamma} U_{a_1 \gamma}^{z_1 p_1} U_{\gamma a_2}^{z_2 p_2}. \quad (18.373)$$

Summing the conjugate of this expression against the corresponding expression for y over z and reordering the finite sums gives the ordinary two-letter entry of Theorem 18.4.63, which proves (18.369). For (18.370), the same theorem expands the Gram entry of $Y_1 \otimes Y_2$ with the weight $\rho_{\beta\beta'}$ attached to the starred index β , and expands the inserted entry with $(\rho^\top)_{\alpha'\alpha} = \rho_{\alpha\alpha'}$ attached to the starred index α ; the two finite sums agree after reordering. Both directions of the equivalence compare the two entry identities at every x and y . The network is kept combined, and neither source cut is contracted separately. $\square$

Theorem 18.9.7 (Source- v isometry of a simple tensor). Lean Lean ✓ Stmt Let $\mathcal{U}$ be simple and in canonical form II, with ambient fixed matrix ρ . Let v be the source operator of $\mathcal{U}$ built from that same weight ρ . Then $v^\dagger v = \mathbb{1}_{r\ell}$, and consequently $r\ell \leq d^2$. This is the implication $1 \rightarrow 2$ in [15, proof of Theorem III.8], before the standing isometry of u is used.

Proof. ✓ Proof Theorem 19.5.6 gives the insertion identity for the recorded canonical boundaries: $W^{ij} W^{k\ell} = W^{ij} |\rho| (\Phi | W^{k\ell})$. Since ρ is diagonal, $\rho^\top = \rho$, so Theorem 18.9.6 gives the dressed Gram equation for the factors built from the same weight ρ . Then Theorem 18.4.68 cancels the dressing. Since $v^\dagger v = \mathbb{1}$ on $\mathbb{C}^r \otimes \mathbb{C}^\ell$, the map v is injective, so $r\ell \leq d^2$. $\square$

Theorem 18.9.8 (Source- v isometry after positive blocking). Lean Lean ✓ Stmt Let $\mathcal{U}$ be an MPU in canonical form II, with recorded weight ρ . If $p > 0$ and $\mathcal{U}^{[p]}$ is simple, then the source operator v computed from the source cuts of $\mathcal{U}^{[p]}$, using this same ρ , satisfies $v^\dagger v = \mathbb{1}$. In particular, some integer $1 \leq p \leq D^2$ gives a simple blocked tensor and this source- v isometry. The statement includes $D = 1$.

Proof. ✓ Proof The blocked tensor retains canonical form II with the same recorded matrix ρ . The source- v contraction for a simple tensor therefore applies to its own source cuts and gives the isometry. The blocking length D^2 is simple for every MPU, including in bond dimension one. $\square$

Theorem 18.9.9 (Simplicity from the source- v isometry). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU tensor, let $\rho > 0$, and let v be its source operator built from ρ . If $v^\dagger v = \mathbb{1}_{r\ell}$, then $\mathcal{U}$ is simple, with witnesses $(\Phi|$ and $|\rho^\top)$. This is the implication $4 \rightarrow 1$ in [15, proof of Theorem III.8].

Proof. [✓ Proof](#) By Theorem 18.4.68, $v^\dagger v = \mathbb{1}$ gives the dressed Gram equation, and Theorem 18.9.6 turns it into $W^{ij}W^{k\ell} = W^{ij}|\rho^\top)(\Phi|W^{k\ell}$. The same-witness implication in Theorem 18.3.28 supplies $(\Phi|W^{ij}|\rho^\top) = \delta_{ij}$ from unitarity, so both simple contractions hold. $\square$

Lemma 18.9.10 (The source operator u is an isometry). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #5708](#) For any MPU tensor $\mathcal{U}$ under the source's standing canonical-form-II convention, let r and ℓ be the two source-cut ranks and let u be the source operator. Then

$$u^\dagger u = \mathbb{1}. \quad (18.374)$$

Consequently u is an isometry and $r\ell \geq d^2$. This is the unrestricted statement of [15, Lemma III.7].

Proof. [✓ Proof](#) The convention records a positive diagonal ambient weight ρ and, by Theorem 18.8.14, the stabilized power $E^J = |\rho)(\Phi|$ at the positive exponent $J = \max(D^2 - 1, 1)$. Lemma 18.9.11, applied to the source factors built from that same weight, gives $u^\dagger u = \mathbb{1}$ and $d^2 \leq r\ell$. $\square$

Lemma 18.9.11 (Supplied-fixed-pair source- u isometry). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #5708](#) Let $\mathcal{U}$ be an MPU tensor with physical dimension $d > 0$. Let W be its double layer, put $E = d^{-1} \sum_i W^{ii}$, and let $\rho > 0$ be diagonal. Put $(\Phi| = \text{vec}(\mathbb{1}_D)^\top$ and suppose that for some $K \geq 0$,

$$E^K = |\rho)(\Phi|. \quad (18.375)$$

Let r and ℓ be the source-cut ranks, and construct the compact-SVD source factors and u from $\mathcal{U}$ using ρ . Then

$$u^\dagger u = \mathbb{1}, \quad (18.376)$$

$$d^2 \leq r\ell. \quad (18.377)$$

Thus the compact-SVD paper gate is an isometry. This is the supplied-fixed-pair version of [15, Lemma III.7].

Proof. [✓ Proof](#) The retained-interior identity and input-first MPU unitarity give, for all physical pairs p, q ,

$$(u^\dagger u)_{p,q} = d^{-K} \sum_{\tau, \eta} \overline{U_{\eta, (p, \tau)}^{(K+2)}} U_{\eta, (q, \tau)}^{(K+2)} = \delta_{p,q}. \quad (18.378)$$

Hence $u^\dagger u = \mathbb{1}$. Since u maps a space of dimension d^2 isometrically into one of dimension $r\ell$, one also has $d^2 \leq r\ell$. $\square$

Theorem 18.9.12 (Simple-tensor equivalence theorem). [Lean](#) [✓ Stmt](#) [Discussion: #5708](#) Let $\mathcal{U}$ be an MPU tensor under the source's standing canonical-form-II convention. Let r and ℓ be its source-cut ranks, and let u and v be its source operators, both built from the ambient weight ρ recorded by the convention. Then the following conditions are equivalent:

1. $\mathcal{U}$ is simple;
2. $r\ell = d^2$;

3. u is unitary;

4. v is unitary.

This is the unrestricted statement of [15, Theorem III.8].

Proof. ✓ Proof The convention gives $u^\dagger u = \mathbb{1}$ and $d^2 \leq r\ell$ before any of the four conditions is assumed. Follow the cycle $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$.

If $\mathcal{U}$ is simple, the source- v contraction for the same weight ρ gives $v^\dagger v = \mathbb{1}$, hence $r\ell \leq d^2$ and therefore $r\ell = d^2$. Conversely, this equality makes the standing isometry u square, hence unitary. The two-site factorization gives

$$\text{reindex}(U^{(2)}) = v \text{ swap}_{r,\ell}(u). \quad (18.379)$$

Reindexing preserves unitarity. Since $U^{(2)}$ and u are unitary, cancellation makes v unitary. Finally, $v^\dagger v = \mathbb{1}$ gives the second simple contraction, while the MPU identity gives the first, so $\mathcal{U}$ is simple. $\square$

Definition 18.9.13 (Standard form). Not ready Discussion: #6019 Let $\mathcal{U}$ be a simple tensor under the source's standing canonical-form-II convention. Its standard form is the depth-two form of the two-site block $\mathcal{U}_2$ determined by the source unitaries

$$u: \mathbb{C}^d \otimes \mathbb{C}^d \longrightarrow \mathbb{C}^\ell \otimes \mathbb{C}^r, \quad (18.380)$$

$$v: \mathbb{C}^r \otimes \mathbb{C}^\ell \longrightarrow \mathbb{C}^d \otimes \mathbb{C}^d. \quad (18.381)$$

On four consecutive blocked legs its unitary is

$$U = v_{23} v_{41} u_{12} u_{34}. \quad (18.382)$$

This is the unrestricted definition of [15, Definition III.9].

Definition 18.9.14 (Two-site standard form with supplied gates). Lean ✓ Stmt Let d, D, ℓ, r be positive integers and let W be a two-site tensor of physical dimension d^2 and bond dimension D . A standard form with supplied gates consists of unitary maps

$$u: \mathbb{C}^d \otimes \mathbb{C}^d \longrightarrow \mathbb{C}^\ell \otimes \mathbb{C}^r, \quad (18.383)$$

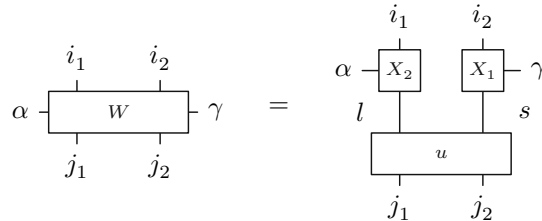
$$v: \mathbb{C}^r \otimes \mathbb{C}^\ell \longrightarrow \mathbb{C}^d \otimes \mathbb{C}^d, \quad (18.384)$$

and half factors X_1, X_2 such that

$$v_{s,l}^{(i_1, i_2)} = \sum_{\beta=0}^{D-1} (X_1)_{(i_1, \beta), s} (X_2)_{(\beta, i_2), l}, \quad (18.385)$$

$$W_{\alpha, \gamma}^{(i_1, i_2), (j_1, j_2)} = \sum_{l=0}^{\ell-1} \sum_{s=0}^{r-1} (X_2)_{(\alpha, i_1), l} u_{(l, s), (j_1, j_2)} (X_1)_{(i_2, \gamma), s}. \quad (18.386)$$

This is the open contraction underlying [15, Definition III.9]. No unitary periodic operator is assumed for W .



Definition 18.9.15 (The two layers of a supplied standard form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#)
Let $N > 0$. Write each physical output and input label as an ordered pair $i_x = (i_{x,1}, i_{x,2})$ and $j_x = (j_{x,1}, j_{x,2})$, and each intermediate label as $h_x = (h_{x,1}, h_{x,2}) \in \{0, \dots, \ell - 1\} \times \{0, \dots, r - 1\}$. Indices x are taken modulo N . Define the unshifted and shifted layer matrices by

$$(C_{u,N})_{h,j} = \prod_{x \in \mathbb{Z}/N\mathbb{Z}} u_{(h_{x,1}, h_{x,2}), (j_{x,1}, j_{x,2})}, \quad (18.387)$$

$$(C_{v,N})_{i,h} = \prod_{x \in \mathbb{Z}/N\mathbb{Z}} v_{(i_{x,2}, i_{x+1,1}), (h_{x,2}, h_{x+1,1})}. \quad (18.388)$$

The cyclic relabeling of pairs is $h \mapsto ((h_{x,2}, h_{x+1,1}))_x$, with inverse $q \mapsto ((q_{x-1,2}, q_{x,1}))_x$. The alphabet of the two-site tensor is identified with ordered pairs $(i_{x,1}, i_{x,2})$. This is the ordering in [\[15, Equation StandardForm\]](#).

Theorem 18.9.16 (Unitarity of the two circuit layers). [Lean](#) [Lean](#) [Stmnt](#) *If the supplied gates u and v are unitary between their stated coordinate spaces, then for every $N > 0$ both $C_{u,N}$ and $C_{v,N}$ are unitary between their corresponding configuration spaces.*

Proof. [Stmnt](#) The two Gram matrices of the product of a rectangular gate factor sitewise: for example,

$$(C_{u,N}^\dagger C_{u,N})_{j,j'} = \prod_{x \in \mathbb{Z}/N\mathbb{Z}} (u^\dagger u)_{j_x, j'_x} = \delta_{j,j'}. \quad (18.389)$$

The reverse Gram identity is analogous. The shifted layer is the product of v on relabeled configurations, and the cyclic relabeling preserves both Gram identities. $\square$

Theorem 18.9.17 (Periodic circuit of a supplied two-site standard form). [Lean](#) [Lean](#) [Stmnt](#)
Let W have supplied two-site standard-form data (u, v, X_1, X_2) . For every $N > 0$ and pair configurations i, j ,

$$U^{(N)}(W)_{i,j} = \sum_h \prod_{x \in \mathbb{Z}/N\mathbb{Z}} v_{(i_{x,2}, i_{x+1,1}), (h_{x,2}, h_{x+1,1})} \prod_{x \in \mathbb{Z}/N\mathbb{Z}} u_{(h_{x,1}, h_{x,2}), (j_{x,1}, j_{x,2})} = (C_{v,N} C_{u,N})_{i,j}. \quad (18.390)$$

Thus, after the stated identification of pair configurations with the two-site alphabet, $U^{(N)}(W) = C_{v,N} C_{u,N}$. This includes $N = 1$. The formula is the positive-length closed contraction of the supplied open form in [\[15, Equations uvuv and StandardForm\]](#).

Proof. [Stmnt](#) Expand the periodic trace as a sum over closed bond configurations. Insert the supplied X_2 - u - X_1 expression at each site. For each x , the bond label between sites x and $x+1$ contracts $(X_1)_{(i_{x,2}, \beta), h_{x,2}}$ with $(X_2)_{(\beta, i_{x+1,1}), h_{x+1,1}}$; their sum is precisely $v_{(i_{x,2}, i_{x+1,1}), (h_{x,2}, h_{x+1,1})}$. The remaining factors form $C_{u,N}$, giving the displayed matrix product in its stated order. $\square$

Theorem 18.9.18 (Positive-length unitarity of the supplied standard form). [Lean](#) [Lean](#) [Stmnt](#)
If W has supplied two-site standard-form data, then

$$U^{(N)}(W)^\dagger U^{(N)}(W) = U^{(N)}(W) U^{(N)}(W)^\dagger = \mathbb{1} \quad (18.391)$$

for every $N > 0$. In particular, W is an MPU under the source's $N > 1$ length convention.

Proof. [Stmnt](#) The periodic operator is the product $C_{v,N} C_{u,N}$ of two unitary layers, and the physical-pair identification preserves unitarity. $\square$

Proof. ✓ Proof The two internal contractions $X_1 X_2$ give one v gate on each neighboring output pair. Thus, for suitable exterior factors L, R ,

$$F_{\alpha, \gamma}(a, b; j) = \sum_{s_0, t_1, s_1, t_2} L_{\alpha}(s_0) v_{a, (s_0, t_1)} u_{(t_1, s_1), j} v_{b, (s_1, t_2)} R_{\gamma}(t_2), \quad (18.399)$$

where L and R retain the exterior half factors and input gates. Since u and v are unitary between their stated spaces, $uA = (uAu^{\dagger})u$ and $(v \otimes v)\widetilde{C} = B(A)(v \otimes v)$. Inserting these two identities into the displayed contraction proves the matrix equality without a virtual trace. $\square$

Lemma 18.9.22 (Virtual-end and word-buffer locality). Lean Lean ✓ Stmt Replace the open word in $F_{\alpha, \gamma}$ by its virtual trace with arbitrary matrices P, Q at the left and right ends. The resulting matrix $F_{P, Q}$ obeys

$$F_{P, Q}A = B(A)F_{P, Q}. \quad (18.400)$$

The same conclusion holds when P and Q are the matrix products of arbitrary complete-block prefix and suffix words, respectively. Their physical row and column words remain explicit and may have different lengths on the two sides of the three-block window.

Proof. ✓ Proof If E is the open three-letter word, cyclicity of the virtual trace gives

$$\text{tr}(PEQ) = \sum_{\gamma, \alpha} (QP)_{\gamma, \alpha} E_{\alpha, \gamma}. \quad (18.401)$$

The open identity therefore survives this linear combination. For complete-block prefix and suffix words, apply word concatenation twice to identify their product with PEQ , leaving both factors intact. $\square$

Theorem 18.9.23 (Left-canonical normalization of a canonical MPU). Lean ✓ Stmt Let $\mathcal{U}$ be a matrix product unitary in full-support canonical form II , with bond dimension D and physical dimension d . Its normalized flattening $A^{ij} = d^{-1/2}U^{ij}$ is left-canonical:

$$\sum_{i, j=0}^{d-1} A^{ij\dagger} A^{ij} = \mathbb{1}_D. \quad (18.402)$$

This is the ambient normalization implicit in [15, lines 269–281 and 344–356].

Proof. ✓ Proof The normalized transfer traces satisfy $\text{tr}(E^N) = 1$ for $N > 1$. Hence the canonical form has one retained block B , with coefficient μ satisfying $|\mu|^2 = 1$. Let W be its inclusion into the ambient bond space. Full support gives $W^{\dagger}W = \mathbb{1}$ and $WW^{\dagger} = \mathbb{1}$, while the reconstruction gives $A^{ij}W = W(\mu B^{ij})$. Therefore

$$\sum_{i, j} A^{ij\dagger} A^{ij} = W \left(\sum_{i, j} (\mu B^{ij})^{\dagger} (\mu B^{ij}) \right) W^{\dagger} = WW^{\dagger} = \mathbb{1}_D. \quad (18.403)$$

$\square$

Theorem 18.9.24 (Virtual unitary gauge from equal periodic operators). Lean ✓ Stmt Let $\mathcal{U}$ and $\mathcal{V}$ be matrix product unitaries on the same physical space, each in full-support canonical form II , with bond dimensions $D_{\mathcal{U}}$ and $D_{\mathcal{V}}$. Suppose that their periodic operators agree at every length $N > 1$:

$$U^{(N)} = V^{(N)}. \quad (18.404)$$

Then $D_{\mathcal{U}} = D_{\mathcal{V}}$, and, after identifying their bond spaces by this equality, there is a unitary $z \in \mathbf{U}(D_{\mathcal{V}})$ such that, for every i, j ,

$$V^{ij} = zU^{ij}z^\dagger. \quad (18.405)$$

This is the virtual-bond implication underlying the local relation in [15, Equations (SFuu) and (SFvv), lines 624–652]. The full-support convention is recorded in [66].

Proof. ✓ Proof Write $A_{\mathcal{U}}^{ij} = d^{-1/2}U^{ij}$ and $A_{\mathcal{V}}^{ij} = d^{-1/2}V^{ij}$. Both are normal tensors. Equality of periodic operators for $N > 1$ makes their matrix product vectors equal at these lengths. The normal-tensor phase alternative gives a unit scalar a such that

$$V^{(N)}(A_{\mathcal{U}})_\sigma = a^N V^{(N)}(A_{\mathcal{V}})_\sigma \quad (18.406)$$

at every positive length. At $N = 2, 3$ the normalized states are nonzero, so the given equality yields $a^2 = a^3 = 1$, hence $a = 1$. Gauge recovery for normal tensors now gives equal bond dimensions and an invertible X and nonzero scalar ζ satisfying $A_{\mathcal{V}}^{ij} = \zeta X A_{\mathcal{U}}^{ij} X^{-1}$. Applying this relation to lengths two and three, where the common states are nonzero, gives $\zeta^2 = \zeta^3 = 1$, so $\zeta = 1$. Theorem 18.9.23 makes both tensors left-canonical, so the irreducible unitary-gauge theorem replaces X by a unitary z . Cancelling the common factor $d^{-1/2}$ gives the asserted relation for U^{ij} and V^{ij} . $\square$

Definition 18.9.25 (Open standard form from source cuts). Lean ✓ Stmt Let $\mathcal{U}$ be simple and have a reduced full-support canonical-form-II presentation with positive diagonal fixed matrix ρ . The two-site tensor $\mathcal{U}_2$ has the supplied standard form with the source gates u, v and the source half factors X_1, X_2 . In particular,

$$(\mathcal{U}_2)_{\alpha, \gamma}^{(i_1, i_2), (j_1, j_2)} = \sum_{l, s} (X_2)_{(\alpha, i_1), l} u_{(l, s), (j_1, j_2)} (X_1)_{(i_2, \gamma), s}, \quad (18.407)$$

$$v_{s, l}^{(i_1, i_2)} = \sum_{\beta} (X_1)_{(i_1, \beta), s} (X_2)_{(\beta, i_2), l}. \quad (18.408)$$

The full-support restriction is recorded in [66].

Proof. ✓ Proof The source-cut factorization gives the open contraction of $\mathcal{U}_2$ through X_2, u, X_1 , while the definition of v gives its X_1, X_2 contraction. The simple-tensor equivalence makes both source gates unitary. Its rank identity $r\ell = d^2$ also makes the two intermediate dimensions positive. $\square$

Theorem 18.9.26 (Source circuit on the original even chain). Lean Lean ✓ Stmt Let $\mathcal{U}$ be simple and have a reduced full-support canonical-form-II presentation with positive diagonal fixed matrix ρ . Let u and v be its source gates. For $N > 0$, flatten each pair configuration by the bijection Φ_N given by

$$(\Phi_N i)_{2x} = i_{x, 1}, \quad (\Phi_N i)_{2x+1} = i_{x, 2}. \quad (18.409)$$

Then the original chain of $2N$ sites satisfies

$$U^{(2N)}(\mathcal{U})_{\Phi_N i, \Phi_N j} = (C_{v, N} C_{u, N})_{i, j}. \quad (18.410)$$

In particular, numbering four original sites from 1 to 4 gives

$$U^{(4)}(\mathcal{U}) = v_{23} v_{41} u_{12} u_{34}. \quad (18.411)$$

Proof. [✓ Proof](#) Apply the supplied circuit identity to the two-site block of $\mathcal{U}$. The two physical-blocking identities identify its length- N periodic operator with $U^{(2N)}(\mathcal{U})$ under Φ_N . For $N = 2$, the two shifted pairs are $(2, 3)$ and $(4, 1)$. $\square$

Theorem 18.9.27 (Bounded block with an open source standard form). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ have a reduced full-support canonical-form-II presentation. There is an integer k with $0 < k \leq D^2$ such that $\mathcal{U}^{[k]}$ is simple and its two-site block has the open source standard form of Definition 18.9.25. If $\eta_{d,k}$ identifies a word of length $2k$ with two consecutive words of length k , the corresponding physical relabeling satisfies

$$\text{reindex}_{\eta_{d,k}}((\mathcal{U}^{[k]})_2) = \mathcal{U}^{[2k]}. \quad (18.412)$$

The source cuts and both gates are formed on the same tensor $\mathcal{U}^{[k]}$, using the fixed matrix of the given presentation.

Proof. [✓ Proof](#) Choose the simple blocking length $k = D^2$. Canonical form II persists under this positive blocking, with the same fixed matrix, so Definition 18.9.25 applies to $\mathcal{U}^{[k]}$. Two-site blocking is direct blocking by $2k$ after the displayed relabeling: first identify the ordinary two-site block with length-two blocking, then group the direct word into consecutive length- k subwords and use $k \cdot 2 = 2k$. $\square$

Lemma 18.9.28 (Original-site residual words around two-site blocks). [Lean](#) [✓ Stmt](#) Let U be any MPO tensor, let $k, p, m, s \in \mathbb{N}$, and write $W = (U^{[k]})_2$. For a word σ of m blocks of length $2k$, let $\eta\sigma$ relabel each block as two consecutive words of length k ; use the same convention for τ . If i^-, j^- are words of length p and i^+, j^+ are words of length s , then

$$\rho^{(p+2km+s)}(U)_{i^-\hat{\sigma}i^+, j^-\hat{\tau}j^+} = \text{tr}(U^{i^-, j^-} \cdot W^{\eta\sigma, \eta\tau} \cdot U^{i^+, j^+}). \quad (18.413)$$

Both residual products remain present, including when $m = 0$; no unitarity, canonical form, or divisibility of the total length is assumed.

Proof. [✓ Proof](#) Residual-site blocking gives the trace with middle factor $(U^{[2k]})^{\sigma, \tau}$. The two-site relabeling identifies this factor, letter by letter and in the same order, with $W^{\eta\sigma, \eta\tau}$; the prefix and suffix products are unchanged. $\square$

Theorem 18.9.29 (Original-chain coefficient at a three-block window). [Lean](#) [✓ Stmt](#) Suppose $W = (U^{[k]})_2$ has supplied two-site standard-form data. Let three consecutive blocks of length $2k$ lie between an arbitrary original-site prefix of length p and suffix of length s . Under the two-site relabeling, suppose their output labels are $(e_-, a_0), (a_1, b_0), (b_1, e_+)$ and their input labels are j_0, j, j_2 . Let $P = U^{i^-, j^-}$ and $Q = U^{i^+, j^+}$ be the residual products. The corresponding entry of the original length- $p + 6k + s$ MPO is

$$\rho^{(p+6k+s)}(U)_{i^-\hat{\sigma}i^+, j^-\hat{\tau}j^+} = F_{P,Q}(a, b; j). \quad (18.414)$$

The equality holds for every choice of the residual words and window labels; no unitarity or MPU hypothesis on U is required.

Proof. [✓ Proof](#) Apply the residual-word identity to the three middle blocks. Their relabeled words are precisely the three displayed letters, so the middle product is the one defining $F_{P,Q}$. $\square$

Theorem 18.9.30 (Buffered original-chain three-block locality). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose $W = (U^{[k]})_2$ has supplied two-site standard-form data. Let i^-, j^- be original-site words of length p and i^+, j^+ original-site words of length s . Let (σ^L, τ^L) and (σ^R, τ^R) be complete-block

word pairs, with $|\sigma^L| = |\tau^L|$. For each choice of the window labels a, b, j , let $\sigma(a, b)$ and $\tau(j)$ be words of m blocks of length $2k$ whose relabelings in the sense of the residual-word identity are

$$\eta\sigma(a, b) = \sigma^L((e_-, a_0), (a_1, b_0), (b_1, e_+)) \sigma^R, \quad (18.415)$$

$$\eta\tau(j) = \tau^L(j_0, j, j_2) \tau^R. \quad (18.416)$$

Put

$$P = U^{i^-, j^-} W^{\sigma^L, \tau^L}, \quad Q = W^{\sigma^R, \tau^R} U^{i^+, j^+}, \quad (18.417)$$

$$F^U(a, b; j) = \rho^{(p+2km+s)}(U)_{i^-, \widehat{\sigma(a,b)} i^+, j^-, \widehat{\tau(j)} j^+}. \quad (18.418)$$

Then the original-chain coefficient matrix F^U satisfies

$$F^U(a, b; j) = F_{P,Q}(a, b; j), \quad (18.419)$$

$$F^U A = B(A) F^U. \quad (18.420)$$

Both equalities are finite-chain identities with every exterior physical label fixed; the virtual boundary is not discarded.

Proof. ✓ Proof The residual-word identity first leaves both original-site products in the trace. Two word concatenations split the relabeled middle product into the left complete-block word, the three window letters, and the right complete-block word, yielding (18.419) entrywise with P and Q the displayed end products. Applying the arbitrary-virtual-end three-block identity to these two matrices yields (18.420). □

Lemma 18.9.31 (Three-block locality with original-site residuals). Lean ✓ Stmt Let $W = (U^{[k]})_2$ have supplied two-site standard-form data. For arbitrary original-site prefix and suffix word pairs, put $P = U^{i^-, j^-}$ and $Q = U^{i^+, j^+}$. The three-block coefficient contracted against these end matrices satisfies

$$F_{P,Q} A = B(A) F_{P,Q}. \quad (18.421)$$

Thus the local observable on the two neighboring output pairs is independent of both residual words.

Proof. ✓ Proof Apply the virtual-end identity with the stated word products as its two end matrices. □

Theorem 18.9.32 (Fundamental Theorem of MPU). Not ready Discussion: #6020 Let $\mathcal{U}$ and $\mathcal{V}$ be simple tensors written in standard form, with source unitaries $(u_{\mathcal{U}}, v_{\mathcal{U}})$ and $(u_{\mathcal{V}}, v_{\mathcal{V}})$. They generate the same MPU for every N ,

$$U^{(N)} = V^{(N)}, \quad (18.422)$$

if and only if their standard forms are related by the local unitary gauges of [15, Eqns. (SFuu) and (SFvu)]: there are unitaries x, y , and z on the three corresponding source and canonical-form-II legs. In particular, the source unitaries obey the local relation

$$u_{\mathcal{V}} = (x \otimes y) u_{\mathcal{U}}, \quad (18.423)$$

while the two canonical-form-II tensors are related on their virtual leg by z and on their source legs by x and y . This records the source's unblocked all-length wording. Its converse has the odd-length parity obstruction of Theorem 18.9.35; the blocked-family interpretation is distinguished in [150].

Theorem 18.9.33 (Scalar invariance of the double layer). [Lean](#) [Lean](#) [✓ Stmt](#) For any tensor $\mathcal{U}$ and scalar $c \in \mathbb{C}$, $(c\mathcal{U})^\sharp = \bar{c}\mathcal{U}^\sharp$. If $\bar{c}c = 1$, the double layers of $c\mathcal{U}$ and $\mathcal{U}$ agree.

Proof. [✓ Proof](#) Conjugation sends c to $\bar{c}$, and contracting the two layers multiplies their scalars to $\bar{c}c = 1$. $\square$

Theorem 18.9.34 (Simplicity of the identity tensor). [Lean](#) [✓ Stmt](#) For every physical dimension d , the bond-one identity tensor $\mathcal{I}^{ij} = \delta_{ij}$ is simple.

Proof. [✓ Proof](#) Its double-layer letters are δ_{ij} , so both simplicity identities hold with boundary vectors $a = b = 1$. $\square$

Theorem 18.9.35 (Parity obstruction to an unblocked standard-form converse). [Lean](#) [✓ Stmt](#) There are simple matrix product unitaries $\mathcal{U}, \mathcal{V}$ of physical dimension two and bond dimension one whose two-site blocked tensors agree, but whose periodic operators differ at length three:

$$\mathcal{U}_2 = \mathcal{V}_2, \quad U^{(3)} \neq V^{(3)}. \quad (18.424)$$

This is the parity obstruction to reading the converse of [15, Theorem FundamentalMPU, lines 624–648] as an assertion about the original tensors at every length. Definition SF of that source concerns the two-site blocks; the explicit source factors and local unitary relations for this example are recorded in [150].

Proof. [✓ Proof](#) Let $\mathcal{U}^{ij} = \delta_{ij}$ on its one-dimensional bond space and set $\mathcal{V}^{ij} = -\mathcal{U}^{ij}$. Their double-layer letters are both δ_{ij} , so the simplicity contractions hold with unit boundary vectors. Two-site blocking removes the pair of minus signs, whereas closing a chain of length three leaves one minus sign:

$$\mathcal{V}_2 = \mathcal{U}_2, \quad U^{(3)} = I_8, \quad V^{(3)} = -I_8. \quad (18.425)$$

$\square$

Remark 18.9.36. The source-labelled Fundamental Theorem above remains unformalized. The source’s displayed standard forms concern $\mathcal{U}_2, \mathcal{V}_2$; their local relations can control the blocked families, but do not distinguish the original families at odd lengths. The explicit local-gauge calculation and the possible corrected scopes are given in [150]. No odd-length premise has been inserted into the source-labelled statement.

Theorem 18.9.37 (Rank of a Kronecker product). [Lean](#) [✓ Stmt](#) Let A and B be finite matrices over a field. Then

$$\text{rank}(A \otimes B) = \text{rank}(A) \text{rank}(B). \quad (18.426)$$

Proof. [✓ Proof](#) Identify the image of the Kronecker product with the tensor product of the two images and multiply their dimensions. $\square$

Definition 18.9.38 (Specified-tensor source-index value). [Lean](#) [✓ Stmt](#) For a specified tensor $\mathcal{U}$ whose right and left source-cut ranks r and ℓ are positive, define

$$\text{ind}_{\text{src}}(\mathcal{U}) = \frac{1}{2}(\log_2 r - \log_2 \ell). \quad (18.427)$$

This definition makes no choice of a simple blocking.

Theorem 18.9.39 (Auxiliary source-index identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A specified tensor with equal positive source ranks $r_{\mathcal{U}} = \ell_{\mathcal{U}}$ has $\text{ind}_{\text{src}}(\mathcal{U}) = 0$. Let $\mathcal{U}$ and $\mathcal{V}$ be specified tensors. Suppose that their source ranks obey

$$r_{\mathcal{V}} = cr_{\mathcal{U}}, \quad \ell_{\mathcal{V}} = c\ell_{\mathcal{U}}, \quad (18.428)$$

where c , $r_{\mathcal{U}}$, and $\ell_{\mathcal{U}}$ are positive. Then

$$\text{ind}_{\text{src}}(\mathcal{V}) = \text{ind}_{\text{src}}(\mathcal{U}). \quad (18.429)$$

Let $\mathcal{U}$, $\mathcal{V}$, and $\mathcal{W}$ instead have positive source ranks and suppose that

$$r_{\mathcal{W}} = cr_{\mathcal{U}}r_{\mathcal{V}}, \quad (18.430)$$

$$\ell_{\mathcal{W}} = c\ell_{\mathcal{U}}\ell_{\mathcal{V}}, \quad (18.431)$$

where c is a positive integer. Then

$$\text{ind}_{\text{src}}(\mathcal{W}) = \text{ind}_{\text{src}}(\mathcal{U}) + \text{ind}_{\text{src}}(\mathcal{V}). \quad (18.432)$$

If d , $r_{\mathcal{U}}$, and $\ell_{\mathcal{U}}$ are positive and a specified tensor has $r_{\mathcal{U}}\ell_{\mathcal{U}} = d^2$, then

$$\text{ind}_{\text{src}}(\mathcal{U}) = \log_2 \frac{r_{\mathcal{U}}}{d} = -\log_2 \frac{\ell_{\mathcal{U}}}{d}. \quad (18.433)$$

Proof. [✓ Proof](#) Expand the definition. Equal ranks give a vanishing difference of logarithms. For common scaling, apply $\log_2(cx) = \log_2 c + \log_2 x$ to both ranks and cancel the two copies of $\log_2 c$. For the three-tensor identity, apply the product formula twice to each rank and cancel the common term. For the equivalent formulas, apply the same product identity to $r_{\mathcal{U}}\ell_{\mathcal{U}} = d^2$ and use $\log_2(x/y) = \log_2 x - \log_2 y$. $\square$

Theorem 18.9.40 (Source-index value of an independent tensor product). [Lean](#) [✓ Stmt](#) If $\mathcal{U}$ and $\mathcal{V}$ have positive right and left source ranks, then

$$\text{ind}_{\text{src}}(\mathcal{U} \boxtimes \mathcal{V}) = \text{ind}_{\text{src}}(\mathcal{U}) + \text{ind}_{\text{src}}(\mathcal{V}). \quad (18.434)$$

This is the specified-tensor form of the tensoring case of [15, Theorem IV.6(ii)]; it makes no choice of a simple blocking.

Proof. [✓ Proof](#) Both source ranks are multiplicative, so the logarithms add. $\square$

Theorem 18.9.41 (Conditional blocking invariance of the specified source-index value). [Lean](#) [✓ Stmt](#) Let $d > 0$ and $k_0 \leq k$. Suppose that the endpoint source ranks satisfy

$$r_{k_0}\ell_{k_0} = d^{2k_0}, \quad (18.435)$$

$$r_k\ell_k = d^{2k}. \quad (18.436)$$

Then the specified source-index values agree:

$$\text{ind}_{\text{src}}(\mathcal{U}_k) = \text{ind}_{\text{src}}(\mathcal{U}_{k_0}). \quad (18.437)$$

This statement assumes the endpoint products; it neither selects a simple blocking nor defines the public index.

Proof. ✓ **Proof** Exact blocking growth gives the common positive factor d^{k-k_0} :

$$r_k = d^{k-k_0} r_{k_0}, \quad (18.438)$$

$$\ell_k = d^{k-k_0} \ell_{k_0}. \quad (18.439)$$

Its logarithm occurs once in each rank term and therefore cancels from their difference. □

Definition 18.9.42 (Index). Not ready Discussion: #6014 Let k be any blocking length for which $\mathcal{U}_k$ is simple, and let r and ℓ be its right and left source-cut ranks. The source index is

$$\text{ind}(\mathcal{U}) = \frac{1}{2}(\log_2 r - \log_2 \ell). \quad (18.440)$$

Since $r\ell = d^{2k}$, this is equivalently

$$\text{ind}(\mathcal{U}) = \log_2 \frac{r}{d^k} = -\log_2 \frac{\ell}{d^k}. \quad (18.441)$$

This is the unrestricted definition of [15, Definition IV.1].

Definition 18.9.43 (Index of a canonical-form-II tensor). Lean Lean Lean ✓ Stmt Let $\mathcal{U}$ be an MPU in canonical form II. Choose any positive blocking length k for which $\mathcal{U}_k$ is simple, and let r_k and ℓ_k be its right and left source-cut ranks. Such a blocking exists, and both ranks are positive. Define

$$\text{ind}(\mathcal{U}) = \frac{1}{2}(\log_2 r_k - \log_2 \ell_k). \quad (18.442)$$

Since $r_k \ell_k = d^{2k}$, one also has

$$\text{ind}(\mathcal{U}) = \log_2 \frac{r_k}{d^k} = -\log_2 \frac{\ell_k}{d^k}. \quad (18.443)$$

This is the canonical-form-II representative case of [15, Definition IV.1].

Theorem 18.9.44 (Comparison of canonical source indices). Lean Lean ✓ Stmt Let U and V be full-support canonical-form-II tensors on the same physical space, possibly with different bond dimensions. If $U^{(N)} = V^{(N)}$ for every $N > 1$, then their right and left source-cut ranks agree, and $\text{ind}(U) = \text{ind}(V)$. This is a comparison of canonical representatives in the setting of [15, Definition IV.1 and Proposition IV.2].

Proof. ✓ **Proof** Equality of the periodic operators gives a unitary virtual similarity, which preserves both source-cut ranks. Choose a common positive simple blocking length k for U and V . The periodic operators of the k -site blocks agree, so the ranks r_k and ℓ_k of these blocks satisfy $r_k(U) = r_k(V)$ and $\ell_k(U) = \ell_k(V)$. Hence

$$\text{ind}(U) = \frac{1}{2}(\log_2 r_k(U) - \log_2 \ell_k(U)) = \frac{1}{2}(\log_2 r_k(V) - \log_2 \ell_k(V)) = \text{ind}(V). \quad (18.444)$$

□

Definition 18.9.45 (Index through a reduced canonical representative). Lean ✓ Stmt Let $d, D > 0$, and let U be an MPU tensor. Choose a reduced canonical-form-II representative U_{red} with $U_{\text{red}}^{(N)} = U^{(N)}$ for every $N > 0$, and set

$$\text{ind}(U) = \text{ind}(U_{\text{red}}). \quad (18.445)$$

The definition does not identify the source-cut ranks of U itself with those of U_{red} .

Theorem 18.9.46 (Independence of the reduced representative). [Lean](#) [Lean](#) [✓ Stmt](#) Every canonical-form-II representative with the same positive-length periodic operators as U has index $\text{ind}(U)$. In particular, positive-dimensional MPU tensors U, V with $U^{(N)} = V^{(N)}$ for all $N > 1$ have equal indices, even when their bond dimensions differ.

Proof. [✓ Proof](#) Choose reduced canonical-form-II representatives of U and V . Their periodic operators agree for $N > 1$, so their canonical source indices agree by Theorem 18.9.44. $\square$

Theorem 18.9.47 (Blocking and adjunction of the representative index). [Lean](#) [Lean](#) [✓ Stmt](#) Let U be a positive-dimensional MPU tensor. For every positive integer p , physical blocking preserves its representative index, while physical adjunction reverses its sign:

$$\text{ind}(U_p) = \text{ind}(U), \quad (18.446)$$

$$\text{ind}(U^\sharp) = -\text{ind}(U). \quad (18.447)$$

Proof. [✓ Proof](#) Choose a reduced canonical-form-II representative U_{red} . Under the canonical identification of blocked configurations with concatenated configurations, the periodic operators satisfy

$$(U_p)^{(N)} = U^{(Np)}, \quad (U^\sharp)^{(N)} = (U^{(N)})^\dagger. \quad (18.448)$$

Hence blocking and adjunction preserve equality of the positive-length periodic operators. The respective canonical-form-II index identities give $\text{ind}((U_{\text{red}})_p) = \text{ind}(U_{\text{red}})$ and $\text{ind}(U_{\text{red}}^\sharp) = -\text{ind}(U_{\text{red}})$. Representative independence then gives both displayed identities. $\square$

Proposition 18.9.48 (Well-definedness of the index). [Not ready](#) [Discussion: #6014](#) The source index is independent of the chosen simple blocking. More precisely, if $k \geq k_0$ and both $\mathcal{U}_{k_0}$ and $\mathcal{U}_k$ are simple, then

$$r_k = d^{k-k_0} r_{k_0}, \quad \ell_k = d^{k-k_0} \ell_{k_0}. \quad (18.449)$$

This is the unrestricted statement of [15, Proposition IV.2].

Theorem 18.9.49 (Well-definedness on a canonical-form-II representative). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU in canonical form II. If k_0 and k are positive blocking lengths for which $\mathcal{U}_{k_0}$ and $\mathcal{U}_k$ are simple, then both blocks give the same index. When $k \geq k_0$, their ranks satisfy

$$r_k = d^{k-k_0} r_{k_0}, \quad \ell_k = d^{k-k_0} \ell_{k_0}. \quad (18.450)$$

The value is unchanged by positive physical blocking or by choosing a different canonical-form-II presentation of the same tensor. This is the canonical-form-II representative case of [15, Proposition IV.2].

Proof. [✓ Proof](#) Each simple block retains canonical form II. The source-gate criterion gives $r_j \ell_j = d^{2j}$ at $j = k_0, k$, and therefore both ranks are positive. The source-rank blocking bounds, together with these two products, force the displayed equalities when $k \geq k_0$; the common factor d^{k-k_0} cancels from the logarithmic difference. The same calculation compares either block with the one used to define the index. A further physical block is the corresponding direct block after the canonical reindexing, so its index is unchanged. Two canonical-form-II presentations of the same tensor are compared at a common simple block. $\square$

Theorem 18.9.50 (Index under physical adjunction in canonical form II). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU in canonical form II. Its physical adjoint is again in canonical form II, and

$$\text{ind}(\mathcal{U}^\sharp) = -\text{ind}(\mathcal{U}). \quad (18.451)$$

This is the canonical-form-II representative case of the index-sign statement in the time-reversal discussion of [15]. It does not assert that $\mathcal{U}$ and $\mathcal{U}^\sharp$ generate the same MPU.

Proof. [✓ Proof](#) Choose a positive simple block $\mathcal{U}_k$. Physical adjunction exchanges its two source-cut ranks, so the rank-product criterion makes $(\mathcal{U}_k)^\sharp$ simple as well. At this block the right-rank formula for $\mathcal{U}^\sharp$ is the negative of the left-rank formula for $\mathcal{U}$:

$$\log_2 \frac{r((\mathcal{U}_k)^\sharp)}{d^k} = \log_2 \frac{\ell(\mathcal{U}_k)}{d^k} = -\text{ind}(\mathcal{U}). \quad (18.452)$$

Both values are independent of the positive simple block chosen. $\square$

Lemma 18.9.51 (Tensor-product additivity). [Not ready](#) [Discussion: #6015](#) For MPU tensors under the source's standing canonical-form convention,

$$\text{ind}(\mathcal{U} \otimes \mathcal{V}) = \text{ind}(\mathcal{U}) + \text{ind}(\mathcal{V}). \quad (18.453)$$

This is the unrestricted tensor-product statement in [15, Theorem IV.6(ii)].

Theorem 18.9.52 (Tensor-product additivity in canonical form II). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ and $\mathcal{V}$ be MPUs in canonical form II. Their independent tensor product is again in canonical form II, and

$$\text{ind}(\mathcal{U} \boxtimes \mathcal{V}) = \text{ind}(\mathcal{U}) + \text{ind}(\mathcal{V}). \quad (18.454)$$

In particular, if $n > 0$ and $\mathbb{1}_n$ is the bond-one tensor generating the identity on a physical space of dimension n , then

$$\text{ind}(\mathcal{U} \boxtimes \mathbb{1}_n) = \text{ind}(\mathcal{U}). \quad (18.455)$$

The bridge to physical identity-ancilla attachment is Theorem 18.9.54. This is the canonical-form-II representative case of the tensoring claim in [15, Theorem IV.6(ii)].

Proof. [✓ Proof](#) Choose a common positive simple blocking length k for $\mathcal{U}$, $\mathcal{V}$, and their tensor product. Blocking commutes with the independent tensor product up to a physical relabeling, so their source ranks satisfy

$$r((\mathcal{U} \boxtimes \mathcal{V})_k) = r(\mathcal{U}_k)r(\mathcal{V}_k), \quad (18.456)$$

$$\ell((\mathcal{U} \boxtimes \mathcal{V})_k) = \ell(\mathcal{U}_k)\ell(\mathcal{V}_k). \quad (18.457)$$

Taking the half-difference of the base-two logarithms gives additivity; independence of k identifies these values with the three indices. The bond-one identity has index zero by Theorem 18.9.3.22, giving the last assertion. $\square$

Theorem 18.9.53 (Tensor-product additivity of the representative index). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let U and V be positive-dimensional MPU tensors, possibly with different physical and bond dimensions. Their independent tensor product satisfies

$$\text{ind}(U \boxtimes V) = \text{ind}(U) + \text{ind}(V). \quad (18.458)$$

The bond-one identity tensor $\mathbb{1}_n$ has index zero for every $n > 0$; hence $\text{ind}(U \boxtimes \mathbb{1}_n) = \text{ind}(U)$. This is the reduced-representative form of the tensoring assertion in [15, Theorem IV.6(ii)].

Proof. [✓ Proof](#) Choose reduced canonical-form-II representatives U_{red} and V_{red} . By Theorem [18.4.35](#), for every $N > 0$,

$$(U_{\text{red}} \boxtimes V_{\text{red}})^{(N)} \cong U_{\text{red}}^{(N)} \otimes V_{\text{red}}^{(N)} = U^{(N)} \otimes V^{(N)} \cong (U \boxtimes V)^{(N)}. \quad (18.459)$$

Representative independence and canonical tensor-product additivity give

$$\text{ind}(U \boxtimes V) = \text{ind}(U_{\text{red}} \boxtimes V_{\text{red}}) \quad (18.460)$$

$$= \text{ind}(U_{\text{red}}) + \text{ind}(V_{\text{red}}). \quad (18.461)$$

The identity tensor itself is a canonical representative of index zero, and the ancilla assertion follows from additivity. $\square$

Theorem 18.9.54 (Physical identity ancillas and the representative index). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let U be a tensor and let $U^{(n)} = U \otimes \mathbb{1}_n$ denote attachment of an identity ancilla of dimension n at each physical site, keeping the bond dimension fixed. For every chain length N , its periodic operator equals that of the independent tensor product with the bond-one identity tensor:

$$(U^{(n)})^{(N)} = (U \boxtimes \mathbb{1}_n)^{(N)} \cong U^{(N)} \otimes I_{n^N}. \quad (18.462)$$

Here the identification groups the original physical coordinates and the ancilla coordinates separately. If U is a positive-dimensional MPU and $n > 0$, then

$$\text{ind}(U^{(n)}) = \text{ind}(U). \quad (18.463)$$

Proof. [✓ Proof](#) Both constructions give the displayed Kronecker product in the same sitewise coordinates. Equality of the periodic operators identifies their representative indices, and the bond-one identity factor has index zero. $\square$

Lemma 18.9.55 (Source ranks under rectangular reduction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a matrix product operator tensor of physical dimension d and bond dimension D . For arbitrary matrices $A \in \mathbb{C}^{E \times D}$ and $B \in \mathbb{C}^{D \times E}$, put $\mathcal{V}^{ij} = AU^{ij}B$. Then

$$M_1(\mathcal{V}) = (I_d \otimes B^\top) M_1(\mathcal{U}) (A^\top \otimes I_d), \quad (18.464)$$

$$M_2(\mathcal{V}) = (A \otimes I_d) M_2(\mathcal{U}) (I_d \otimes B), \quad (18.465)$$

$$r[\mathcal{V}] \leq r[\mathcal{U}], \quad (18.466)$$

$$\ell[\mathcal{V}] \leq \ell[\mathcal{U}]. \quad (18.467)$$

In particular, a rectangular reduction cannot increase either source rank. No equality of ranks is asserted, and no unitarity, simplicity, or canonical-form assumption is needed. If $\mathcal{V}$ is a rectangular reduction of $\mathcal{U}$, then for every common blocking length L ,

$$r[\mathcal{V}_L] \leq r[\mathcal{U}_L], \quad (18.468)$$

$$\ell[\mathcal{V}_L] \leq \ell[\mathcal{U}_L]. \quad (18.469)$$

These identities follow directly from the source cuts of [\[15, Section III, definition of \$r, \ell\$ \]](#) and permit upper bounds for composition to pass to a reduced representative.

Proof. ✓ Proof Expand the matrix products entrywise in the two source cuts. The first cut places the right virtual coordinate on its row side, which accounts for the two transposes. The second cut keeps the ordinary order of the virtual coordinates. Multiplication on either side cannot increase matrix rank, proving both inequalities. For a rectangular reduction, its single-letter word identity gives $\mathcal{V}^{ij} = A\mathcal{U}^{ij}B$. Its all-word identity gives the same relation for the blocked tensors, with the same matrices A, B , so the inequalities apply at every length. $\square$

Lemma 18.9.56 (Four-site composition source-rank bounds). Lean Lean ✓ Stmt Let $\mathcal{U}$ and $\mathcal{V}$ be arbitrary matrix product operator tensors of common physical dimension d . For their product tensor, four-site blocking satisfies

$$r[(\mathcal{U} \cdot \mathcal{V})_4] \leq d^2 r[\mathcal{U}] r[\mathcal{V}], \quad (18.470)$$

$$\ell[(\mathcal{U} \cdot \mathcal{V})_4] \leq d^2 \ell[\mathcal{U}] \ell[\mathcal{V}]. \quad (18.471)$$

No unitarity or simplicity assumption is required.

Proof. ✓ Proof Use the canonical physical reindexing to identify iterated two-site blocking with direct four-site blocking. For the right cut, write each two-site block in the v - Y_1 - Y_2 form. Group v with Y_2 on one side of the cut and retain Y_1 on the other. The cut therefore carries one physical leg and one right source leg from each tensor. Thus, for suitable rectangular matrices $A_{1,4}$ and $B_{1,4}$,

$$M_1((\mathcal{U} \cdot \mathcal{V})_4) = A_{1,4} B_{1,4}, \quad (18.472)$$

where the columns of $A_{1,4}$ and the rows of $B_{1,4}$ are indexed by

$$\mathcal{K}_1 = (\mathbb{C}^d \otimes \mathbb{C}^{r[\mathcal{U}]}) \otimes (\mathbb{C}^d \otimes \mathbb{C}^{r[\mathcal{V}]}). \quad (18.473)$$

Consequently,

$$r[(\mathcal{U} \cdot \mathcal{V})_4] = \text{rank}(A_{1,4} B_{1,4}) \leq \dim(\mathcal{K}_1) = d^2 r[\mathcal{U}] r[\mathcal{V}]. \quad (18.474)$$

For the left cut, write each two-site block in the X_2 - u - X_1 form. Retain X_2 on one side and group u with X_1 on the other. The cut carries one physical leg and one left source leg from each tensor. For suitable rectangular matrices $A_{2,4}$ and $B_{2,4}$,

$$M_2((\mathcal{U} \cdot \mathcal{V})_4) = A_{2,4} B_{2,4}, \quad (18.475)$$

where the intermediate space is

$$\mathcal{K}_2 = (\mathbb{C}^d \otimes \mathbb{C}^{\ell[\mathcal{U}]}) \otimes (\mathbb{C}^d \otimes \mathbb{C}^{\ell[\mathcal{V}]}). \quad (18.476)$$

Hence

$$\ell[(\mathcal{U} \cdot \mathcal{V})_4] = \text{rank}(A_{2,4} B_{2,4}) \leq \dim(\mathcal{K}_2) = d^2 \ell[\mathcal{U}] \ell[\mathcal{V}]. \quad (18.477)$$

Both source-cut ranks are unchanged by the canonical physical relabeling. $\square$

Lemma 18.9.57 (Composition additivity). Not ready Discussion: #6016 Let $\mathcal{W}$ generate the composition of the MPUs generated by $\mathcal{U}$ and $\mathcal{U}'$. Under the source's standing canonical-form convention,

$$\text{ind}(\mathcal{W}) = \text{ind}(\mathcal{U}) + \text{ind}(\mathcal{U}'). \quad (18.478)$$

This is the unrestricted composition statement in [15, Theorem IV.6(ii)].

Theorem 18.9.58 (Reduction to a canonical representative). Lean ✓ Stmt Let $\mathcal{U}$ and $\mathcal{V}$ have the same physical dimension and bond dimensions D and E , respectively. Suppose $\mathcal{V}$ is in canonical form II and their periodic operators agree at every positive length. Then there are matrices $A \in \mathbb{C}^{E \times D}$ and $B \in \mathbb{C}^{D \times E}$ such that, for every word w in the paired physical alphabet,

$$AU^wB = \mathcal{V}^w. \quad (18.479)$$

The empty word is included, so $AB = I_E$.

Proof. ✓ Proof The normalized flattening of $\mathcal{V}$ is normal. Multiplication by the nonzero scalar $\sqrt{d}$ preserves normality, so its unnormalized flattening is normal as well. Equality of the periodic operators gives equality of all positive-length word traces. The equality-case rectangular reduction theorem now supplies A and B . $\square$

Theorem 18.9.59 (Saturation of the composition rank bounds). Lean ✓ Stmt Let $\mathcal{U}, \mathcal{V}, \mathcal{R}$ be canonical-form-II tensors of physical dimension d . Suppose $\mathcal{U}$ and $\mathcal{V}$ are simple, $\mathcal{R}_4$ is simple, and $\mathcal{R}$ is a rectangular reduction of $\mathcal{U} \cdot \mathcal{V}$. Then

$$r[\mathcal{R}_4] = d^2 r[\mathcal{U}] r[\mathcal{V}], \quad (18.480)$$

$$\ell[\mathcal{R}_4] = d^2 \ell[\mathcal{U}] \ell[\mathcal{V}]. \quad (18.481)$$

Proof. ✓ Proof Rectangular reduction and the four-site composition bounds give the corresponding upper bounds. Simplicity gives

$$r[\mathcal{U}] \ell[\mathcal{U}] = r[\mathcal{V}] \ell[\mathcal{V}] = d^2, \quad (18.482)$$

$$r[\mathcal{R}_4] \ell[\mathcal{R}_4] = d^8. \quad (18.483)$$

The product of the two upper bounds is therefore d^8 . Since both ranks of $\mathcal{R}_4$ are positive, neither inequality can be strict. $\square$

Theorem 18.9.60 (Composition additivity of the representative index). Lean Lean ✓ Stmt Let $\mathcal{U}$ and $\mathcal{V}$ be MPUs with a common positive physical dimension and positive bond dimensions. For the index defined through a reduced canonical-form-II representative,

$$\text{ind}(\mathcal{U} \cdot \mathcal{V}) = \text{ind}(\mathcal{U}) + \text{ind}(\mathcal{V}). \quad (18.484)$$

In particular, if $\mathcal{U}, \mathcal{V}, \mathcal{R}$ are in canonical form II and the positive-length periodic operators of $\mathcal{R}$ equal the products of those of $\mathcal{U}$ and $\mathcal{V}$, then their canonical indices satisfy the same identity. The argument is the rank-saturation proof of [15, Theorem IV.6(ii)].

Proof. ✓ Proof Choose reduced canonical representatives for the two factors and their composition. Their positive-length periodic operators still satisfy the composition identity. Theorem 18.9.58 gives a rectangular reduction from the product tensor to its representative $\mathcal{R}$. Choose a common positive blocking length k at which the three representatives are simple, and put $d_k = d^k$. The rectangular reduction persists under blocking. Rank saturation gives

$$r[\mathcal{R}_{4k}] = d_k^2 r[\mathcal{U}_k] r[\mathcal{V}_k], \quad (18.485)$$

$$\ell[\mathcal{R}_{4k}] = d_k^2 \ell[\mathcal{U}_k] \ell[\mathcal{V}_k]. \quad (18.486)$$

Subtracting the logarithms cancels d_k^2 and adds the two indices. Independence of positive blocking and of the chosen canonical representatives gives the asserted identity. $\square$

Definition 18.9.61 (Strict equivalence). [Lean](#) [✓ Stmt](#) Fix a virtual bond dimension D . Let $\mathcal{U}$ and $\mathcal{V}$ be canonical-form MPU tensors with physical dimensions d_a and d_b . Given an equality $h: d_a = d_b$, reindex $\mathcal{U}$ from d_a to d_b along h^{-1} . The tensors are strictly equivalent with respect to h if there is a continuous path $\mathcal{W}(p)$, $p \in [0, 1]$, in the fixed-bond MPU locus such that

$$\mathcal{W}(0) = \text{reind}_{h^{-1}}(\mathcal{U}), \quad (18.487)$$

$$\mathcal{W}(1) = \mathcal{V}. \quad (18.488)$$

Only the canonical-form requirement is relaxed along the path, matching the fixed-bond interpretation of [15, lines 708–714].

Theorem 18.9.62 (Normalized transfer data under identity ancillas). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $x > 0$. After adjoining an identity ancilla of dimension x , the normalized flattening is the physical reindexing of the original flattening with a normalized diagonal ancilla alphabet. Its transfer map is exactly the original transfer map. Consequently, chosen canonical-form-II data with full support determine canonical-form-II data with full support for $\mathcal{U}^{(x)} = \mathcal{U} \otimes \mathbb{1}_x$.

The product-index equivalence has the orientation

$$((i, a), (j, b)) \mapsto ((i, j), (a, b)). \quad (18.489)$$

This result preserves the normalized transfer data.

Proof. [✓ Proof](#) Only the x diagonal ancilla letters are nonzero, each with coefficient $x^{-1/2}$. Hence their total transfer contribution is

$$x (x^{-1/2})^2 = 1. \quad (18.490)$$

Thus the transfer map and each diagonal positive fixed matrix are unchanged. The block dimensions, weights, and ambient coisometry are unchanged as well, so full support follows from the same block-dimension sum. The displayed physical reindexing gives the normalized flattening of the enlarged MPO. $\square$

Definition 18.9.63 (Equivalence after blocking and ancillas). [Lean](#) [✓ Stmt](#) Fix a virtual bond dimension D . Let $\mathcal{U}$ and $\mathcal{V}$ be MPU tensors with physical dimensions d_a and d_b . They are equivalent if there are a positive blocking length k and coprime positive integers p_a, p_b with

$$p_a d_a = p_b d_b \quad (18.491)$$

such that $(\mathcal{U}^{(p_a)})_k$ and $(\mathcal{V}^{(p_b)})_k$ are strictly equivalent. Here the positive-dimensional identity ancillas are attached before the common blocking, and

$$\mathcal{U}^{(p)} = \mathcal{U} \otimes \mathbb{1}_p. \quad (18.492)$$

This provides the fixed-bond formulation of [15, lines 716–724].

Definition 18.9.64 (Admissible equivalence and path datum). [Not ready](#) An admissible strict-equivalence datum for a path $[0, 1] \ni p \mapsto \mathcal{W}(p)$ consists of a common positive blocking length m such that the actual blocked path tensor $\mathcal{W}(p)_m$ is simple and carries a reduced full-support source datum for every p . The blocked tensors and their source data vary continuously with p , after fixed identifications of the finite-dimensional spaces involved. No equivalent or reduced representative may replace $\mathcal{W}(p)_m$. In particular, the actual blocked endpoint tensors carry their own source data.

When the path is used under a specified symmetry $\mathcal{S}$, the datum also includes continuously varying reduced full-support source data for the fixed family of comparison paths

$$\overline{\mathcal{W}(p)_m}, \quad \mathcal{W}(p)_m^\sharp, \quad \mathcal{W}(p)_m^\top, \quad \mathcal{S}[\mathcal{W}(p)_m]. \quad (18.493)$$

Every tensor in this family is simple. This requirement is part of the datum independently of which comparisons a later proof uses; no preservation under conjugation, physical adjoint, transposition, or $\mathcal{S}$ is assumed.

Two tensors are admissibly strictly equivalent if their strict equivalence is witnessed by such a path. They are admissibly equivalent if the blocked, ancilla-enlarged endpoints in Definition 18.9.63 are joined by an admissible strict-equivalence path. The source arguments motivating these additional path hypotheses are Proposition IV.5, Theorem IV.6, Corollary IV.7, and the lemma following the symmetry-equivalence definitions in [15]; the datum itself is introduced here and has no source-labelled counterpart.

Theorem 18.9.65 (Generic two-projection angle block). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

Let P and Q be orthogonal projections on a complex inner-product space. The compression of Q to $\text{ran } P$ is self-adjoint. If a unit vector $x \in \text{ran } P$ satisfies

$$PQx = \mu x, \quad 0 < \mu < 1, \quad (18.494)$$

then there is a unit vector $w \perp x$ such that

$$Px = x, \quad Pw = 0, \quad (18.495)$$

$$Qx = \mu x + \sqrt{\mu(1-\mu)} w, \quad Qw = \sqrt{\mu(1-\mu)} x + (1-\mu)w. \quad (18.496)$$

Thus $\text{span}\{x, w\}$ is the two-dimensional angle block associated with the eigenvalue $\mu \in (0, 1)$.

Proof. [✓ Proof](#) Set $d = Qx - \mu x$. Symmetry and idempotence give

$$Pd = 0, \quad \langle x, d \rangle = 0, \quad \|d\|^2 = \mu(1-\mu), \quad (18.497)$$

$$Qd = \mu(1-\mu)x + (1-\mu)d. \quad (18.498)$$

Since $0 < \mu < 1$, the vector $w = d/\sqrt{\mu(1-\mu)}$ is well defined and has the stated properties. $\square$

Definition 18.9.66 (Compression spectral data). [Lean](#) [Lean](#) [✓ Stmt](#) For orthogonal projections P and Q on a finite-dimensional complex inner-product space, let $(x_i)_i$ be an orthonormal eigenbasis of the compression of Q to $\text{ran } P$, with ordered real eigenvalues $(\mu_i)_i$.

Theorem 18.9.67 (Compression-spectrum endpoint and angle-block trichotomy). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let P and Q be orthogonal projections on a finite-dimensional complex inner-product space, with compression spectral data (x_i, μ_i) . Then, for every i , $\mu_i \in [0, 1]$ and exactly one of the following alternatives holds:

1. $\mu_i = 0$ and $Qx_i = 0$;
2. $\mu_i = 1$ and $Qx_i = x_i$;
3. $0 < \mu_i < 1$, and x_i belongs to the two-dimensional angle block described in Theorem 18.9.65.

This theorem classifies the compression eigenvectors.

Proof. [✓ Proof](#) Apply the finite-dimensional spectral theorem to the self-adjoint compression of Q to $\text{ran } P$. For a unit compression eigenvector x with eigenvalue μ , the angle-defect identity gives

$$\|Qx - \mu x\|^2 = \mu(1 - \mu), \quad (18.499)$$

hence $0 \leq \mu \leq 1$. At $\mu = 0$ or 1 the defect vanishes, giving the two common one-dimensional blocks. For $0 < \mu < 1$, apply Theorem [18.9.65](#). $\square$

Definition 18.9.68 (Compression defect block). [Lean](#) [Lean](#) [✓ Stmt](#) For compression spectral data (x_i, μ_i) , define

$$d_i = Qx_i - \mu_i x_i, \quad (18.500)$$

$$B_i = \text{span}\{x_i, d_i\}. \quad (18.501)$$

Theorem 18.9.69 (Orthogonality of compression angle blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) For all indices i, j ,

$$\langle x_i, d_j \rangle = 0, \quad (18.502)$$

$$\langle d_i, x_j \rangle = 0, \quad (18.503)$$

$$\langle d_i, d_j \rangle = \mu_j(1 - \mu_j)\langle x_i, x_j \rangle. \quad (18.504)$$

In particular, $\langle d_i, d_j \rangle = 0$ when $i \neq j$, including when $\mu_i = \mu_j$. The block B_i has dimension at most two, is invariant under both P and Q , and satisfies $B_i \perp B_j$ for distinct indices.

Proof. [✓ Proof](#) Since $Px_i = x_i$, symmetry of P and the compression eigenvalue equation give

$$\langle x_i, d_j \rangle = \langle x_i, Qx_j \rangle - \mu_j \langle x_i, x_j \rangle \quad (18.505)$$

$$= \langle Px_i, Qx_j \rangle - \mu_j \langle x_i, x_j \rangle \quad (18.506)$$

$$= \langle x_i, PQx_j \rangle - \mu_j \langle x_i, x_j \rangle = 0. \quad (18.507)$$

Conjugate symmetry gives $\langle d_i, x_j \rangle = 0$. For the exact defect pairing, symmetry of Q and the local action formula give

$$Qd_j = \mu_j(1 - \mu_j)x_j + (1 - \mu_j)d_j, \quad (18.508)$$

$$\langle d_i, d_j \rangle = \langle Qx_i, d_j \rangle - \mu_i \langle x_i, d_j \rangle, \quad (18.509)$$

$$\langle Qx_i, d_j \rangle - \mu_i \langle x_i, d_j \rangle = \langle x_i, Qd_j \rangle, \quad (18.510)$$

$$\langle x_i, Qd_j \rangle = \mu_j(1 - \mu_j)\langle x_i, x_j \rangle + (1 - \mu_j)\langle x_i, d_j \rangle, \quad (18.511)$$

$$\langle d_i, d_j \rangle = \mu_j(1 - \mu_j)\langle x_i, x_j \rangle. \quad (18.512)$$

Orthogonality of the generated spans follows by linearity. The action formulas

$$Px_i = x_i, \quad (18.513)$$

$$Pd_i = 0, \quad (18.514)$$

$$Qx_i = \mu_i x_i + d_i, \quad (18.515)$$

$$Qd_i = \mu_i(1 - \mu_i)x_i + (1 - \mu_i)d_i \quad (18.516)$$

show that both projections preserve $B_i = \text{span}\{x_i, d_i\}$. Its dimension bound follows from its two generators. $\square$

By definition, $\text{ran}(\tilde{P}) = \text{ran}(PQ)$, which also gives $P\tilde{P} = \tilde{P}$. For all x and z ,

$$\langle Qx - PQx, PQz \rangle = 0. \quad (18.527)$$

Hence $Qx - PQx \in \text{ran}(PQ)^\perp$, so orthogonal projection onto this range yields $\tilde{P}Q = PQ$. Taking adjoints gives $Q\tilde{P} = QP$. Finally,

$$\text{ran}(\tilde{P}Q) = \text{ran}(PQ) = \text{ran}(\tilde{P}). \quad (18.528)$$

□

Theorem 18.9.74 (Transversality of the reduced range). [Lean](#) [Lean](#) [✓ Stmt](#) For orthogonal projections P and Q , the reduced range contains no nonzero vector annihilated by Q :

$$\ker Q \cap \text{ran}(\tilde{P}) = \{0\}. \quad (18.529)$$

Proof. [✓ Proof](#) If $x = PQy$ and $Qx = 0$, then $Px = x$ and symmetry of the two projections gives

$$\langle x, x \rangle = \langle PQy, PQy \rangle = \langle Qy, PQy \rangle = \langle y, QPQy \rangle = 0. \quad (18.530)$$

Hence $x = 0$. □

Theorem 18.9.75 (Invertible right factor for the reduced projection). [Lean](#) [✓ Stmt](#) There is an invertible linear map Y such that

$$\tilde{P}QY = \tilde{P}. \quad (18.531)$$

Proof. [✓ Proof](#) The equality

$$\text{ran}(\tilde{P}Q) = \text{ran}(\tilde{P}) \quad (18.532)$$

and the equal-ranges right-factor theorem give an invertible Y satisfying $\tilde{P}QY = \tilde{P}$. □

Definition 18.9.76 (Matrix reduced projection). [Lean](#) [✓ Stmt](#) In standard Euclidean coordinates, let $\tilde{P}$ be the matrix of the reduced projection associated with two square matrices P and Q .

Theorem 18.9.77 (Coordinate identities for the reduced projection). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If P and Q are orthogonal projections, then $\tilde{P}$ is an orthogonal projection and

$$P\tilde{P} = \tilde{P}, \quad \tilde{P}P = \tilde{P}, \quad \tilde{P}Q = PQ, \quad Q\tilde{P} = QP. \quad (18.533)$$

There is also an invertible matrix Y satisfying $\tilde{P}QY = \tilde{P}$.

Lemma 18.9.78 (Transfer map under virtual sandwiching). [Lean](#) [✓ Stmt](#) Let $U = (U^{ij})$ be an MPO tensor, and let $\widehat{U}^{ij} = AU^{ij}B$. Then

$$E_{\widehat{U}}(X) = AE_U(BXB^\dagger)A^\dagger. \quad (18.534)$$

Proof. [✓ Proof](#) Expand the transfer map and collect the factors A , B , $B^\dagger$, and $A^\dagger$ outside the finite sum. □

Theorem 18.9.79 (Normalized transfer under virtual sandwiching). [Lean](#) [Lean](#) [✓ Stmt](#) Let $W = (W^{ij})$ be an MPO tensor, let $\widehat{E}_W$ denote the transfer map of its normalized flattening, and set $\widehat{W}^{ij} = AW^{ij}B$. For every bond matrix X ,

$$\widehat{E}_{\widehat{W}}(X) = A\widehat{E}_W(BXB^\dagger)A^\dagger. \quad (18.535)$$

In particular, suppose that $\widehat{E}_W(X) = \text{tr}(LX)R$ for every X . For any bond matrices P, Q , let T be their reduced projection and set $\widetilde{W}^{ij} = TW^{ij}T$. Then

$$\widehat{E}_{\widetilde{W}}(X) = \text{tr}((TLT)X)TRT. \quad (18.536)$$

This is the normalized compression identity in the proof of [15, Proposition IV.5].

Proof. [✓ Proof](#) Scalar normalization commutes with left and right multiplication. Expanding the normalized transfer map therefore gives the first identity by collecting the constant factors outside the finite sum. For the reduced projection, $T^\dagger = T$, and cyclicity of the trace gives $\text{tr}(LTX) = \text{tr}(TLTX)$. $\square$

Lemma 18.9.80 (Positive definite support compression). [Lean](#) [✓ Stmt](#)

Let $\rho \succeq 0$. If

$$\text{supp}(\rho)(Vx) \neq 0 \quad \text{for every } x \neq 0, \quad (18.537)$$

then

$$V^\dagger \rho V \succ 0. \quad (18.538)$$

Proof. [✓ Proof](#) If the quadratic form of $V^\dagger \rho V$ vanished at a nonzero vector x , positive semidefiniteness would give $\rho Vx = 0$. The support projection would then annihilate Vx , contrary to the hypothesis. $\square$

Theorem 18.9.81 (Reduced representative under a supplied positive fixed pair). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let $W = (W^{ij})$ be an MPO tensor. Let $L, R \succeq 0$ satisfy

$$E_W(X) = \text{tr}(LX)R, \quad \text{tr}(LR) = 1, \quad (18.539)$$

and let P and Q be their support projections. Assume, letter by letter,

$$W^{ij}P = W^{ij}, \quad QW^{ij} = W^{ij}. \quad (18.540)$$

Set $T = \widetilde{P}$ and $\widetilde{W}^{ij} = TW^{ij}T$. Then, for every $N \geq 0$,

$$\text{MPO}_N(\widetilde{W}) = \text{MPO}_N(W), \quad (18.541)$$

$$E_{\widetilde{W}}(X) = \text{tr}((TLT)X)TRT, \quad (18.542)$$

$$\text{tr}((TLT)(TRT)) = 1. \quad (18.543)$$

If $V^\dagger V = I$ and $VV^\dagger = T$, then

$$V^\dagger LV \succ 0, \quad V^\dagger RV \succ 0. \quad (18.544)$$

Proof. ✓ Proof For every nonempty word w , idempotence of T and (18.533) give

$$\widetilde{W}_w = TW_w T. \quad (18.545)$$

Cyclicity of the trace, together with $QW_w = W_w = W_w P$, therefore gives $\text{tr}(TW_w T) = \text{tr}(W_w)$. Length zero is immediate. Lemma 18.9.78, with $A = B = T$ and $T^\dagger = T$, gives

$$E_{\widetilde{W}}(X) = \text{tr}(LTXT)TRT = \text{tr}((TLT)X)TRT. \quad (18.546)$$

Since P and Q are the support projections of L and R , respectively,

$$PL = LP = L, \quad QR = RQ = R. \quad (18.547)$$

Combining these absorptions with (18.533), $T^2 = T$, and cyclicity of the trace gives

$$\text{tr}((TLT)(TRT)) = \text{tr}(LTRT) = \text{tr}(LRT) = \text{tr}(RTL) = \text{tr}(RL) = \text{tr}(LR) = 1. \quad (18.548)$$

Let $x \neq 0$ and set $u = Vx$. Since $V^\dagger V = I$, one has $u \neq 0$; since $VV^\dagger = T$, one has $Tu = u$. Equation (18.533) gives $Pu = u$, and (18.529) gives $Qu \neq 0$. Thus

$$\text{supp}(L)u = Pu = u \neq 0, \quad \text{supp}(R)u = Qu \neq 0. \quad (18.549)$$

Applying Lemma 18.9.80 to L and R yields

$$x^\dagger(V^\dagger LV)x = u^\dagger Lu > 0, \quad x^\dagger(V^\dagger RV)x = u^\dagger Ru > 0. \quad (18.550)$$

□

Theorem 18.9.82 (Simplicity under a rectangular bond inclusion). Lean Lean Lean ✓ Stmt Let W have bond dimension D , and let $A \in \mathbb{C}^{E \times D}$ and $B \in \mathbb{C}^{D \times E}$ satisfy $BA = I_D$. The tensor with letters $AW^{ij}B$ is simple if and only if W is simple. In particular, if $V^\dagger V = I_k$ and $VV^\dagger = T$, simplicity of $(TW^{ij}T)$ implies simplicity of $(V^\dagger W^{ij}V)$.

Proof. ✓ Proof Put $\mathcal{A} = \overline{A} \otimes A$ and $\mathcal{B} = \overline{B} \otimes B$, in the doubled bond coordinates. The double-layer letters F^{ij} of W transform to $\mathcal{A}F^{ij}\mathcal{B}$, and $\mathcal{B}\mathcal{A} = I$. If $(a|, |b)$ are simplicity witnesses for W , then $(a|\mathcal{B}, \mathcal{A}|b)$ are witnesses for the included tensor. Conversely, witnesses $(a|, |b)$ for the included tensor give $(a|\mathcal{A}, \mathcal{B}|b)$ for W . Substitution in (simple1) and (simple2) proves both claims. For the assertion about support coordinates, use $V(V^\dagger W^{ij}V)V^\dagger = TW^{ij}T$. □

Theorem 18.9.83 (Simplicity under reduced support compression). Lean ✓ Stmt Let W be simple with $d > 0$. Let P be an orthogonal projection and Q a square matrix on its bond space, with $W^{ij}P = W^{ij} = QW^{ij}$. Suppose the normalized diagonal E of its double layer has a rank-one factorization $E = |\rho\rangle\langle\Phi|$. If T is the reduced projection associated with P, Q , then $(TW^{ij}T)$ is simple.

Proof. ✓ Proof Write F^{ij} for the double-layer letters and put $\mathcal{P} = \overline{P} \otimes P$, $\mathcal{Q} = \overline{Q} \otimes Q$, and $H = \overline{T} \otimes T$. The letterwise support identities also hold for their normalized diagonal E . Since $TQ = PQ$, they give

$$XHY = XHQY = X\mathcal{P}QY = XY \quad (18.551)$$

whenever X, Y are double-layer letters or E . The supplied-projector identity is $F^{ij}F^{k\ell} = F^{ij}EF^{k\ell}$. Using (18.551) and $H^2 = H$ therefore yields

$$(HF^{ij}H)(HF^{k\ell}H) = (HF^{ij}H)(HEH)(HF^{k\ell}H). \quad (18.552)$$

These are the double-layer letters of the compressed tensor, and $HEH = |H\rho)(\Phi H|$ supplies its two simplicity witnesses. Support compression preserves the periodic operators, so the compressed tensor still generates an MPU. Its two-letter insertion identity then implies the first simplicity contraction with the same witnesses. $\square$

Theorem 18.9.84 (Invertible factor for a reduced projection). [Lean](#) [✓ Stmt](#) *Let $L, R \succeq 0$, let P and Q be their support projections, and let $\tilde{P}$ be the reduced projection of P and Q . There are invertible support-inverse extensions of L and R , together with an invertible matrix Y , such that $PQY = \tilde{P}$.*

Proof. [✓ Proof](#) Extend the inverses on the supports by the identity on their orthogonal complements, and choose an invertible right factor for the reduced projection. $\square$

Theorem 18.9.85 (Source-rank comparison of the hatted and reduced tensors). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $W = (W^{ij})$ be an MPO tensor, let L, R, X, Z, Y be square matrices of the same size, let P be an orthogonal projection, let Q be a square matrix of the same size, and set $\tilde{P}$ equal to their reduced projection. Assume, letter by letter, $W^{ij}P = W^{ij} = QW^{ij}$. Set $\widehat{W}^{ij} = LW^{ij}R$ and $\widetilde{W}^{ij} = \tilde{P}W^{ij}\tilde{P}$. If X, Z, Y satisfy $XL = P$, $RZ = Q$, and $PQY = \tilde{P}$, then*

$$X\widehat{W}ZY = PWQY \quad (18.553)$$

$$PWQY = PQWPQY \quad (18.554)$$

$$PQWPQY = \tilde{P}W\tilde{P} \quad (18.555)$$

$$\tilde{P}W\tilde{P} = \widetilde{W}. \quad (18.556)$$

If in addition X, Z , and Y are invertible, then

$$r[\widehat{W}] = r[\widetilde{W}], \quad (18.557)$$

$$\ell[\widehat{W}] = \ell[\widetilde{W}]. \quad (18.558)$$

If $L, R \succeq 0$ and $P = \text{supp}(L)$, $Q = \text{supp}(R)$, such invertible matrices may be chosen as $X = L^+ + (\mathbb{1} - P)$, $Z = R^+ + (\mathbb{1} - Q)$, together with an invertible right factor Y of the reduced projection.

Proof. [✓ Proof](#) Expanding each virtual sandwich and using $XL = P$ and $RZ = Q$ gives (18.553). The absorptions $QW^{ij} = W^{ij} = W^{ij}P$ give, letter by letter,

$$PW^{ij}QY = P(QW^{ij})QY = PQ(W^{ij}P)QY = PQW^{ij}PQY,$$

which is (18.554). Since $\tilde{P}Q = PQ$ and $PQY = \tilde{P}$,

$$PQW^{ij}PQY = (\tilde{P}Q)W^{ij}\tilde{P} = \tilde{P}W^{ij}\tilde{P}. \quad (18.559)$$

This proves (18.555) and (18.556). Thus $\widetilde{W}$ is obtained from $\widehat{W}$ by left multiplication by X and right multiplication by ZY . Invertibility of these factors and the source ranks of a virtual sandwich (Theorem 18.9.94) give

$$\begin{aligned} r[\widehat{W}] &= r[X\widehat{W}(ZY)] = r[\widetilde{W}], \\ \ell[\widehat{W}] &= \ell[X\widehat{W}(ZY)] = \ell[\widetilde{W}], \end{aligned}$$

by invertibility of X and ZY together with the relation $X\widehat{W}(ZY) = \widetilde{W}$ from (18.553)–(18.556), yielding (18.557) and (18.558). For positive semidefinite L and R , their ambient support-inverse extensions supply X and Z ; the invertible right-factor identity for $\tilde{P}Q$ supplies Y . $\square$

Theorem 18.9.86 (Lower semicontinuity of finite matrix rank). [Lean](#) [Lean](#) [✓ Stmt](#) For finite index sets M, N , matrix rank defines a lower-semicontinuous map

$$\text{rank}: \text{Mat}_{M \times N}(\mathbb{C}) \longrightarrow \mathbb{N}. \quad (18.560)$$

Consequently, if $A: X \rightarrow \text{Mat}_{M \times N}(\mathbb{C})$ is continuous, then $x \mapsto \text{rank } A(x)$ is lower semicontinuous.

Proof. [✓ Proof](#) For every $k \in \mathbb{N}$, the sublevel set is the closed determinantal set

$$\{A : \text{rank } A \leq k\}. \quad (18.561)$$

These closed sublevel sets characterize lower semicontinuity. Composition with a continuous map preserves lower semicontinuity. $\square$

Theorem 18.9.87 (Fixed positive product gives local constancy). [Lean](#) [✓ Stmt](#) Let $r, \ell: X \rightarrow \mathbb{N}$ be lower semicontinuous. Suppose that $c > 0$ and

$$r(x)\ell(x) = c \quad (18.562)$$

for every $x \in X$. Then both r and ℓ are locally constant.

Proof. [✓ Proof](#) Fix $x \in X$. Since $c > 0$, both $r(x)$ and $\ell(x)$ are positive. Lower semicontinuity gives a neighborhood U of x on which

$$r(y) \geq r(x), \quad \ell(y) \geq \ell(x). \quad (18.563)$$

For $y \in U$, monotonicity of multiplication and the fixed-product identity give

$$r(x)\ell(x) \leq r(y)\ell(x) \leq r(y)\ell(y) = r(x)\ell(x). \quad (18.564)$$

Positivity permits cancellation, first of $\ell(x)$ and then of $r(x)$, so $r(y) = r(x)$ and $\ell(y) = \ell(x)$. $\square$

Theorem 18.9.88 (Fixed positive product on a preconnected set). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 18.9.87, if $S \subseteq X$ is preconnected, then for all $x, y \in S$,

$$r(x) = r(y), \quad \ell(x) = \ell(y). \quad (18.565)$$

In particular, both functions are constant when X is preconnected and when $X = [0, 1]$.

Proof. [✓ Proof](#) A locally constant function has open fibers. A preconnected set cannot meet two disjoint nonempty open fibers, so each of r and ℓ has one value on S . The unit interval is connected, hence preconnected. $\square$

Definition 18.9.89 (Virtual sandwich). [Lean](#) [Lean](#) [✓ Stmt](#) For an MPO tensor $\mathcal{U} = (U^{ij})_{0 \leq i, j < d}$ of bond dimension D and matrices $A, B \in \mathbb{C}^{D \times D}$, define

$$\widehat{\mathcal{U}}^{ij} = AU^{ij}B. \quad (18.566)$$

Theorem 18.9.90 (Continuity of the virtual sandwich). [Lean](#) [✓ Stmt](#) If $x \mapsto A(x)$, $x \mapsto \mathcal{U}(x)$, and $x \mapsto B(x)$ are continuous families, then

$$x \mapsto A(x)\mathcal{U}(x)B(x) \quad (18.567)$$

is continuous.

Proof. ✓ **Proof** Every matrix entry is a finite sum of products of entries of $A(x)$, $\mathcal{U}(x)$, and $B(x)$, hence is continuous. $\square$

Definition 18.9.91 (Doubled-virtual contraction and factor-free sandwich). Lean Lean Lean Lean ✓ **Stmt** Let $q : \{0, \dots, D-1\} \times \{0, \dots, D-1\} \rightarrow \{0, \dots, D^2-1\}$ be the standard enumeration of a pair. For an MPO tensor $\mathcal{W}$ and a doubled-virtual matrix $E \in \mathbb{C}^{D^2 \times D^2}$, define

$$C(\mathcal{W}, E)_{ab}^{ij} = \sum_{c,e=0}^{D-1} E_{q(b,e),q(c,a)} W_{ce}^{ij}. \quad (18.568)$$

If $E(\mathcal{W})$ denotes the normalized physical diagonal of the physical-adjoint double layer of $\mathcal{W}$, let $E'(\mathcal{W})$ be the matrix obtained by exchanging the two entries of its column pair, $E'(\mathcal{W})_{p,q(c,a)} = E(\mathcal{W})_{p,q(a,c)}$, and define

$$\widehat{\mathcal{W}}_{\text{ff}} = C(\mathcal{W}, E'(\mathcal{W})), \quad (18.569)$$

$$(\widehat{\mathcal{W}}_{\text{ff}}^{ij})_{ab} = \sum_{c,e=0}^{D-1} E(\mathcal{W})_{q(b,e),q(a,c)} W_{ce}^{ij}. \quad (18.570)$$

This is the contraction shown in Figure IV_index4.png of [15, Proposition IV.5].

Theorem 18.9.92 (Rank-one identity and continuity of the factor-free sandwich). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ **Stmt** Let $L, R \in \mathbb{C}^{D \times D}$ and let vec stack matrix columns, so

$$\text{vec}(L)_{q(c,a)} = L_{ac}, \quad (18.571)$$

$$\text{vec}(R)_{q(b,e)} = R_{eb}. \quad (18.572)$$

Then the rank-one doubled-virtual matrix satisfies

$$C(\mathcal{W}, \text{vec}(R) \text{vec}(L)^T)^{ij} = L W^{ij} R. \quad (18.573)$$

No adjoint or normalization factor occurs in the contraction result. The binary map $(\mathcal{W}, E) \mapsto C(\mathcal{W}, E)$, the map $\mathcal{W} \mapsto E(\mathcal{W})$, and consequently the map $\mathcal{W} \mapsto \widehat{\mathcal{W}}_{\text{ff}}$ are continuous. If $d > 0$ and the normalized transfer map satisfies $\mathcal{E}_{\mathcal{W}}(X) = \text{tr}(LX)R$ for every X , then $\widehat{\mathcal{W}}_{\text{ff}}^{ij} = L W^{ij} R$. Consequently, for a continuous family $\mathcal{W}(x)$ with pointwise trace factors $L(x), R(x)$, the tensors $L(x)W^{ij}(x)R(x)$ are continuous, whether or not the factors themselves are chosen continuously. These assertions neither choose L and R continuously nor infer their positivity. They make no claim about reduced representatives or constancy of ranks.

Proof. ✓ **Proof** Expanding the left-hand side of (18.573) and using column stacking gives

$$C(\mathcal{W}, \text{vec}(R) \text{vec}(L)^T)_{ab}^{ij} = \sum_{c,e} R_{eb} L_{ac} W_{ce}^{ij} \quad (18.574)$$

$$= \sum_{c,e} L_{ac} W_{ce}^{ij} R_{eb} = (L W^{ij} R)_{ab}. \quad (18.575)$$

This proves (18.573). Each entry in (18.568) is a finite sum of products, so the binary contraction is continuous. The double-layer entries are finite sums of products of entries of $\mathcal{W}$ and their complex conjugates, and the normalized physical diagonal is another finite sum. Hence $\mathcal{W} \mapsto E(\mathcal{W})$ is continuous. Substitution into (18.569) proves continuity of the factor-free sandwich. For the trace-factor identity, column vectorization gives $E(\mathcal{W}) = \text{vec}(R) \text{vec}(L^T)^T$. Therefore $E'(\mathcal{W}) = \text{vec}(R) \text{vec}(L)^T$, and (18.573) yields $\widehat{\mathcal{W}}_{\text{ff}}^{ij} = L W^{ij} R$. Applied pointwise, this equality proves the asserted continuity without any regularity of the factors. $\square$

Theorem 18.9.93 (Source cuts of a virtual sandwich). [Lean](#) [Lean](#) [✓ Stmt](#) *The two raw source cuts satisfy the literal formulas*

$$\begin{aligned} M_1(\widehat{\mathcal{U}}) &= (I_d \otimes B^\top) M_1(\mathcal{U}) (A^\top \otimes I_d), \\ M_2(\widehat{\mathcal{U}}) &= (A \otimes I_d) M_2(\mathcal{U}) (I_d \otimes B). \end{aligned} \quad (18.576)$$

The transposes in (18.576) result from the corrected orientation of M_1 ; no complex conjugation occurs.

Proof. [✓ Proof](#) Expand an entry of each right-hand side. The identity factors collapse the two physical sums. For the first cut, the entries of $B^\top$ and $A^\top$ put $B_{\delta\beta}$ on the left and $A_{\alpha\gamma}$ on the right; commutativity of scalar multiplication gives

$$\sum_{\gamma,\delta} A_{\alpha\gamma} U_{\gamma\delta}^{ij} B_{\delta\beta} = (AU^{ij}B)_{\alpha\beta}. \quad (18.577)$$

The second cut gives the same entry directly. $\square$

Theorem 18.9.94 (Source ranks of a virtual sandwich). [Lean](#) [Lean](#) [✓ Stmt](#) *If A and B are invertible, then*

$$r[\widehat{\mathcal{U}}] = r[\mathcal{U}], \quad \ell[\widehat{\mathcal{U}}] = \ell[\mathcal{U}]. \quad (18.578)$$

This assertion concerns the raw source ranks only.

Proof. [✓ Proof](#) The determinant identities for Kronecker products show that $A \otimes I_d$ and $I_d \otimes B$ are invertible. Left and right multiplication by these matrices preserves rank. Apply this observation to each source-cut formula in Theorem 18.9.93. $\square$

Theorem 18.9.95 (Source factors under a unitary virtual conjugation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ and $\mathcal{V}$ be simple tensors in canonical form II with $V^{ij} = zU^{ij}z^\dagger$ for a unitary z on the virtual space. Their recorded positive matrices need not be related. Write $X_a(\mathcal{U}), Y_a(\mathcal{U})$ and $X_a(\mathcal{V}), Y_a(\mathcal{V})$ for their selected source factors, $a = 1, 2$. After identifying the equal source ranks, there are unitaries W_1 and W_2 on the respective source-rank spaces such that*

$$X_1(\mathcal{V}) = (I_d \otimes (z^\dagger)^T) X_1(\mathcal{U}) W_1, \quad Y_1(\mathcal{V}) = W_1^\dagger Y_1(\mathcal{U}) (z^T \otimes I_d), \quad (18.579)$$

$$X_2(\mathcal{V}) = (z \otimes I_d) X_2(\mathcal{U}) W_2, \quad Y_2(\mathcal{V}) = W_2^\dagger Y_2(\mathcal{U}) (I_d \otimes z^\dagger). \quad (18.580)$$

This is the forward factor comparison associated with the diagrams in [15, lines 624–648]; it does not assert a converse from the local relations to equality at every chain length. The scalar-free step is the isometric specialization of [163, lines 1052–1066].

Proof. [✓ Proof](#) The two cut identities of Theorem 18.9.93, applied to the selected factorizations of Theorem 18.4.51, give transported factorizations with the four exterior matrices displayed in the statement. The right exterior matrices are unitary, so the transported right factors retain the right inverses of Theorem 18.4.56. The left exterior matrix of the second cut is unitary, so its transported left factor remains an isometry. The contraction defining u is unchanged: its interior virtual product is $z^\dagger z = I_D$. Rank preservation, Theorem 18.9.94, identifies the transported factors with the selected source-rank spaces. Simplicity and the source-gate equivalence of Theorem 18.9.12 give unitarity of both u contractions. The selected factors have left inverses from their weighted first-cut and ordinary second-cut normalizations in Theorem 18.4.53. The normalized source-decomposition comparison of Corollary 19.5.10.3 now gives the two unitary changes W_1, W_2 and the four displayed equalities. $\square$

Remark 18.9.96. In the setting of Theorem 18.9.95, write $u_{\mathcal{U}}, u_{\mathcal{V}}$ for the source gates of Definition 18.4.57 built from the selected factors. Substituting the two $Y_a(\mathcal{V})$ identities into that contraction, the interior virtual product is $\sum_{\beta} (z^{\dagger})_{\beta' \beta} z_{\beta \beta''} = (z^{\dagger} z)_{\beta' \beta''} = \delta_{\beta' \beta''}$, so

$$u_{\mathcal{V}} = (W_2^{\dagger} \otimes W_1^{\dagger}) u_{\mathcal{U}}. \quad (18.581)$$

Thus the unitaries $x = W_2^{\dagger}$ and $y = W_1^{\dagger}$ realize the local relation $u_{\mathcal{V}} = (x \otimes y) u_{\mathcal{U}}$ of Theorem 18.9.32. This identification of the gauges is a consequence of the displayed factor identities and is not asserted by Theorem 18.9.95. It is stated, together with the corresponding relation for v , in Theorem 18.9.97.

Theorem 18.9.97 (Source gates under a unitary virtual conjugation). Lean ✓ Stmt *Let $\mathcal{U}$ and $\mathcal{V}$ be simple tensors in canonical form II with $V^{ij} = z U^{ij} z^{\dagger}$ for a unitary z . Let $e_r : r[\mathcal{U}] \simeq r[\mathcal{V}]$ and $e_{\ell} : \ell[\mathcal{U}] \simeq \ell[\mathcal{V}]$ be the source-rank bijections. Reindex the source gates of $\mathcal{U}$ by*

$$\tilde{u}_{(\ell, r), (p, q)} = u_{\mathcal{U}, (e_{\ell}^{-1} \ell, e_r^{-1} r), (p, q)}, \quad \tilde{v}_{(p, q), (r, \ell)} = v_{\mathcal{U}, (p, q), (e_r^{-1} r, e_{\ell}^{-1} \ell)}. \quad (18.582)$$

There are unitaries x on $\ell[\mathcal{V}]$ and y on $r[\mathcal{V}]$ such that

$$u_{\mathcal{V}} = (x \otimes y) \tilde{u}, \quad (18.583)$$

$$v_{\mathcal{V}} = \tilde{v} (y^{\dagger} \otimes x^{\dagger}). \quad (18.584)$$

Relation (18.583) is the forward u relation of [15, Equation (SFuu)]. The source's [15, Equation (SFvv)] displays two relations between the open half-factors involving z ; contraction of those half-factors yields (18.584). No converse for the original periodic operators at odd length is asserted.

Proof. ✓ Proof Use the four selected-factor identities of Theorem 18.9.95. Their source-rank changes are $x = W_2^{\dagger}$ and $y = W_1^{\dagger}$. The two source-gate contractions have entries

$$u_{(\ell, r), (p, q)} = \sum_{\beta} (Y_2)_{\ell, (p, \beta)} (Y_1)_{r, (\beta, q)}, \quad (18.585)$$

$$v_{(p, q), (r, \ell)} = \sum_{\alpha} (X_1)_{(p, \alpha), r} (X_2)_{(\alpha, q), \ell}. \quad (18.586)$$

In the first contraction, the two virtual matrices cancel as $z^{\dagger} z = I_D$, leaving $(x \otimes y) \tilde{u}$. In the second, the first-cut transpose and second-cut left action likewise contract to $z^{\dagger} z = I_D$, leaving $\tilde{v} (y^{\dagger} \otimes x^{\dagger})$. $\square$

Definition 18.9.98 (Ambient support-inverse extension). Lean Lean ✓ Stmt

Let $A \succeq 0$ have support projection P , and let A^+ denote its inverse on the support. Define

$$X_A = A^+ + (\mathbb{1} - P). \quad (18.587)$$

Define also

$$Z_A = A + (\mathbb{1} - P). \quad (18.588)$$

This construction gives the invertible matrices for Proposition IV.5 of [15].

Theorem 18.9.99 (Invertibility and support identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) The matrices X_A and Z_A are mutually inverse, and

$$X_A A = A X_A = P. \quad (18.589)$$

Consequently, X_A is a unit in the matrix algebra, with inverse Z_A , and $\det(X_A)$ is a unit in $\mathbb{C}$.

In the notation of Proposition IV.5 of [15], let L and R be the positive semidefinite outer factors with support projections P and Q , respectively. The required witnesses are

$$X = X_L = L^+ + (\mathbb{1} - P), \quad (18.590)$$

$$Z = X_R = R^+ + (\mathbb{1} - Q). \quad (18.591)$$

Thus

$$X_L L = P, \quad (18.592)$$

$$R X_R = Q, \quad (18.593)$$

or equivalently $XL = P$ and $RZ = Q$. Here the paper's matrix Z is X_R , not the inverse-extension matrix Z_R from Definition 18.9.98.

Proof. [✓ Proof](#) The support identities give

$$A^+ A = A A^+ = P. \quad (18.594)$$

Their complementary forms are

$$A^+ (\mathbb{1} - P) = (\mathbb{1} - P) A^+ = 0, \quad (18.595)$$

$$A (\mathbb{1} - P) = (\mathbb{1} - P) A = 0, \quad (18.596)$$

and idempotence of P gives

$$(\mathbb{1} - P)^2 = \mathbb{1} - P. \quad (18.597)$$

Therefore

$$X_A Z_A = P + (\mathbb{1} - P) = \mathbb{1}. \quad (18.598)$$

Likewise,

$$Z_A X_A = P + (\mathbb{1} - P) = \mathbb{1}, \quad (18.599)$$

while multiplication by A gives $X_A A = A X_A = P$. □

Proposition 18.9.100 (Constancy of the source ranks). [Not ready](#) [Discussion: #6022](#) Let $[0, 1] \ni x \mapsto \mathcal{W}(x)$ be a continuous path of tensors generating MPUs, without requiring the path to be in canonical form. For each x , take the canonical form and the two source-cut ranks $r(x)$ and $\ell(x)$ of its associated simple tensor. Then $r(x)$ and $\ell(x)$ are constant on $[0, 1]$.

Proposition 18.9.101 (Admissible constancy of the source ranks). [Not ready](#) [Discussion: #6022](#) Let $[0, 1] \ni x \mapsto \mathcal{W}(x)$ be a continuous path of tensors generating MPUs, without requiring the path to be in canonical form. Suppose that the path carries an admissible equivalence datum with blocking length m . Then the ranks $r(x)$ and $\ell(x)$ of the actual blocked path tensors $\mathcal{W}(x)_m$ are constant on $[0, 1]$.

Theorem 18.9.102 (Source ranks along a canonical-form-II path). [Lean](#) [Lean](#) [✓ Stmt](#) Let $[0, 1] \ni x \mapsto \mathcal{U}(x)$ be a continuous path of tensors of fixed physical dimension d and bond dimension D . Suppose that each $\mathcal{U}(x)$ is in canonical form II. Then for all $x, y \in [0, 1]$,

$$r[\mathcal{U}(x)_{D^4}] = r[\mathcal{U}(y)_{D^4}], \quad \ell[\mathcal{U}(x)_{D^4}] = \ell[\mathcal{U}(y)_{D^4}]. \quad (18.600)$$

This is the canonical-form-II path case of [15, Proposition IV.5]. Its hypotheses are not comparable with those of Proposition 18.9.101: here each tensor of the path is in canonical form II, but no continuously varying source data are assumed; there the path may leave canonical form, but continuously varying source data at a common blocking length are assumed.

Proof. [✓ Proof](#) Blocking is continuous, and the two source cuts are coordinate rearrangements of the blocked tensor. Thus their ranks are lower semicontinuous functions of x . The common block of length D^4 is simple at every point. Its ranks obey

$$r[\mathcal{U}(x)_{D^4}] \ell[\mathcal{U}(x)_{D^4}] = d^{2D^4} > 0. \quad (18.601)$$

Lower semicontinuity and this fixed positive product make both ranks constant on the connected interval. $\square$

Theorem 18.9.103 (Index along a canonical-form-II path). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 18.9.102, the index of $\mathcal{U}(x)$ is independent of $x \in [0, 1]$.

Proof. [✓ Proof](#) Every D^4 -site block is simple, so for every $x \in [0, 1]$,

$$\text{ind}(\mathcal{U}(x)) = \frac{1}{2} (\log_2 r[\mathcal{U}(x)_{D^4}] - \log_2 \ell[\mathcal{U}(x)_{D^4}]). \quad (18.602)$$

Both ranks on the right-hand side are independent of x by Theorem 18.9.102. $\square$

Theorem 18.9.104 (Index Theorem). [Not ready](#) [Discussion: #6023](#) For MPU tensors under the source's standing canonical-form convention, take the index defined from any simple blocking. This index has the following properties.

1. It is invariant under blocking.
2. It is additive under tensor products and composition.
3. It is constant under continuous deformations through MPU tensors.
4. Two MPU tensors have the same index if and only if they are equivalent.

This is the unrestricted statement of [15, Theorem IV.6].

Theorem 18.9.105 (Admissible Index Theorem). [Not ready](#) [Discussion: #6023](#) Assume the following operation-specific admissibility hypotheses. Every pair of simple blockings compared for blocking independence carries reduced full-support source data. Tensor products need no hypothesis, by Theorem 18.9.53. For a composition tensor $\mathcal{W}$ of $\mathcal{U}$ and $\mathcal{U}'$, there is one positive length k such that

$$\mathcal{U}_k, \quad \mathcal{U}'_k, \quad \mathcal{W}_k, \quad \mathcal{U}_{2k}, \quad \mathcal{U}'_{2k}, \quad \mathcal{W}_{4k} \quad (18.603)$$

are simple and carry their own data. Continuous-deformation and equivalence claims use the path data of Definition 18.9.64. Under these hypotheses, the MPU index has the following properties.

1. It is invariant under blocking.
2. It is additive under tensor products and composition.
3. It is constant under continuous deformations through MPU tensors.
4. Admissibly equivalent MPU tensors have the same index. Conversely, tensors with the same index are admissibly equivalent whenever the standard-form interpolation is supplied with an admissible path datum.

These are the conclusions of [15, Theorem IV.6], restricted to the explicit admissibility hypotheses above.

Corollary 18.9.106 (Continuous choice of standard form). Not ready Discussion: #6024 For every continuous path $p \mapsto \mathcal{U}(p)$ of MPU tensors, not necessarily in canonical form, there are an integer $k_0 \leq D^4$ and continuous paths of unitaries $u(p)$ and $v(p)$ such that $U^{(2k_0N)}(p)$ has standard form given by $u(p)$ and $v(p)$.

Corollary 18.9.107 (Admissible continuous choice of standard form). Not ready Discussion: #6024 Let k_0 be the common blocking length in an admissible equivalence datum for a continuous path $\mathcal{U}(p)$ of MPU tensors, and suppose that this supplied length satisfies $k_0 \leq D^4$. Then there are continuous paths of unitaries $u(p)$ and $v(p)$ such that the MPU $U^{(2k_0N)}(p)$ has standard form given by $u(p)$ and $v(p)$ for every $p \in [0, 1]$. This is the reduced-source version of [15, Corollary IV.7].

18.9.1 Local characterization of symmetries

Theorem 18.9.1.1 (Common transpose sign of the conjugation factors). Lean Lean Lean Lean
✓ Stmt Let x and y be unitary matrices over $\mathbb{C}$, indexed by nonempty finite sets, whose entrywise conjugates satisfy

$$\bar{x}x \otimes \bar{y}y = \mathbb{1}. \quad (18.604)$$

Then there is a sign $\sigma \in \{1, -1\}$ with

$$x = \sigma x^T, \quad y = \sigma y^T. \quad (18.605)$$

Equivalently, either $x = x^T$ and $y = y^T$, or $x = -x^T$ and $y = -y^T$. This is the algebraic conclusion in [15, lines 1033–1038].

Proof. ✓ Proof A Kronecker product equal to the identity has reciprocal scalar factors, so $\bar{x}x = e^{i\phi}\mathbb{1}$ and $\bar{y}y = e^{-i\phi}\mathbb{1}$. Since x is unitary, $x^T\bar{x} = \mathbb{1}$; multiplying $\bar{x}x = e^{i\phi}\mathbb{1}$ on the left by x^T gives $x = e^{i\phi}x^T$, and similarly $y = e^{-i\phi}y^T$. Transposing $x = e^{i\phi}x^T$ gives $x^T = e^{i\phi}x$, hence $x = e^{2i\phi}x$; since x is invertible, $e^{2i\phi} = 1$, so $e^{i\phi} = \pm 1$ and $e^{-i\phi} = e^{i\phi}$. The two matrices therefore carry one transpose sign. □

Proposition 18.9.1.2 (Conjugation symmetry). Not ready Discussion: #5713 An MPU in standard form is invariant under complex conjugation if and only if there are unitaries x and y such that either $x = x^T$ and $y = y^T$, or $x = -x^T$ and $y = -y^T$, and

$$\bar{u} = (x \otimes y)u, \quad \bar{v} = v(y^\dagger \otimes x^\dagger). \quad (18.606)$$

This is the unrestricted conjugation characterization in [15, Proposition V.1].

Proposition 18.9.1.3 (Admissible conjugation symmetry). Not ready Discussion: #5713 Let a tensor and its conjugate comparison tensor both be equipped with reduced full-support source data and written in the standard forms determined by those data. The generated MPU is invariant under complex conjugation if and only if there are unitaries x and y such that either $x = x^T$ and $y = y^T$, or $x = -x^T$ and $y = -y^T$, and

$$\bar{u} = (x \otimes y)u, \quad \bar{v} = v(y^\dagger \otimes x^\dagger). \quad (18.607)$$

This is the reduced-source version of the conjugation characterization in [15, Proposition V.1].

Lemma 18.9.1.4 (Symmetric branch of the unitary congruence normal form). Lean ✓ Stmt Let x be a symmetric unitary matrix. There is a symmetric unitary matrix S such that

$$x = S^T S. \quad (18.608)$$

This is the symmetric branch of [15, Lemma lemma:conjclass-normalform-continuous].

Proof. ✓ Proof Write the finite spectral decomposition of x as

$$x = \sum_{\lambda \in \sigma(x)} \lambda P_\lambda \quad (18.609)$$

and set

$$S = \sum_{\lambda \in \sigma(x)} \sqrt{\lambda} P_\lambda. \quad (18.610)$$

Each spectral value has modulus one, so its square root has modulus one, and hence S is unitary and $S^2 = x$. On the finite spectrum, interpolate the square-root function by a polynomial q . Then $S = q(x)$, and $x^T = x$ implies $S^T = q(x^T) = q(x) = S$. Therefore $x = S^2 = S^T S$. $\square$

Lemma 18.9.1.5 (Skew congruence from paired spectral projectors). Lean ✓ Stmt Let K and I be finite sets, let $E_k \in \mathbb{R}$ for $k \in K$, and let P_k be Hermitian projectors on $\mathbb{C}^I$. Suppose that the family $\{P_k, P_k^T\}_{k \in K}$ consists of mutually orthogonal projectors and sums to the identity. Define

$$x = \sum_{k \in K} e^{-iE_k} (P_k - P_k^T). \quad (18.611)$$

Then the matrices

$$S = e^{-i\pi/4} \sum_{k \in K} e^{-iE_k/2} (P_k + P_k^T), \quad (18.612)$$

and

$$\tilde{\Lambda} = i \sum_{k \in K} (P_k - P_k^T) \quad (18.613)$$

are unitary, $S^T = S$, $\tilde{\Lambda} = \tilde{\Lambda}$, $\tilde{\Lambda}^T = -\tilde{\Lambda}$, and

$$x = S^T \tilde{\Lambda} S. \quad (18.614)$$

This is the paired-projector construction in the skew branch of [15, Lemma lemma:conjclass-normalform-continuous].

Proof. [✓ Proof](#) Write the two projectors in each pair as separate members of one complete orthogonal family. A weighted sum of this family is unitary whenever every weight has modulus one. This applies to S , whose two weights at k both equal $e^{-i\pi/4}e^{-iE_k/2}$, and to $\tilde{\Lambda}$, whose weights are i and $-i$. Transposition exchanges the two members of each pair, proving the symmetry of S and the skew symmetry of $\tilde{\Lambda}$. Hermiticity gives $\overline{P_k} = P_k^T$, hence $\tilde{\tilde{\Lambda}} = \tilde{\Lambda}$. Finally, orthogonality reduces the product to its weights, and

$$(e^{-i\pi/4}e^{-iE_k/2})^2 i = e^{-iE_k}. \quad (18.615)$$

The transposed member has the negative weight, which gives $x = S^T \tilde{\Lambda} S$. $\square$

Lemma 18.9.1.6 (Normal form of symmetric and skew-symmetric unitaries). [Not ready](#) [Discussion: #5713](#)

Let x be unitary and satisfy $x = x^T$ or $x = -x^T$. There are a symmetric unitary S and a matrix $\tilde{\Lambda}$ such that

$$x = S^T \tilde{\Lambda} S, \quad S = S^T. \quad (18.616)$$

In the symmetric case, $\tilde{\Lambda}$ is real and symmetric. In the skew-symmetric case, it is real and skew-symmetric. In both cases, $\det(\tilde{\Lambda}) = 1$.

Lemma 18.9.1.7 (Odd-dimensional skew-symmetric determinant). [Lean](#) [✓ Stmt](#) Let I be a finite set of odd cardinality and let $A \in \text{Mat}_I(\mathbb{C})$ satisfy $A^T = -A$. Then

$$\det A = 0. \quad (18.617)$$

Proof. [✓ Proof](#) Transpose invariance, skew symmetry, and the determinant of a negated matrix give

$$\det A = \det(A^T) = \det(-A) = (-1)^{|I|} \det A = -\det A. \quad (18.618)$$

Since the complex numbers have characteristic zero, $\det A = 0$. $\square$

Lemma 18.9.1.8 (Parity of a nonsingular skew-symmetric matrix). [Lean](#) [✓ Stmt](#) Let I be finite and let $A \in \text{Mat}_I(\mathbb{C})$ satisfy $A^T = -A$ and $\det A \neq 0$. Then $|I|$ is even.

Proof. [✓ Proof](#) If $|I|$ were odd, Lemma 18.9.1.7 would give $\det A = 0$. Hence $|I|$ is not odd and is therefore even. $\square$

Theorem 18.9.1.9 (Even dimension of a skew-symmetric unitary). [Lean](#) [✓ Stmt](#) Let $U \in \text{Mat}_d(\mathbb{C})$ be unitary and satisfy $U^T = -U$. Then d is even. This is the parity obstruction in [15, Proposition V.2, Case II].

Proof. [✓ Proof](#) The determinant of a unitary matrix is a unit, so it is nonzero. Apply Lemma 18.9.1.8 to U . $\square$

Proposition 18.9.1.10 (Orthogonal and symplectic standard forms). [Not ready](#) [Discussion: #5713](#)
Let an MPU in standard form be invariant under complex conjugation. Its standard-form unitaries can be chosen in exactly one of the following two cases. In Case I,

$$\overline{u'} = u', \quad \overline{v'} = v', \quad (18.619)$$

and the choice is unique up to local orthogonal transformations. In Case II,

$$\overline{u'} = (\Sigma_\ell \otimes \Sigma_r) u', \quad \overline{v'} = v' (\Sigma_r \otimes \Sigma_\ell), \quad (18.620)$$

where

$$\Sigma_x = \mathbb{1}_{x/2} \otimes Y, \quad Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (18.621)$$

The choice is unique up to local symplectic transformations S_x satisfying $S_x^T \Sigma_x S_x = \Sigma_x$. Case II can occur only when r and ℓ are even.

Proposition 18.9.1.11 (Admissible orthogonal and symplectic standard forms). Not ready

Discussion: #5713 Let an MPU in the standard form determined by its supplied source datum be invariant under complex conjugation, and suppose every standard-form tensor compared below carries reduced full-support source data. Its standard-form unitaries can be chosen in exactly one of the following two cases. In Case I,

$$\overline{u'} = u', \quad \overline{v'} = v', \quad (18.622)$$

and the choice is unique up to local orthogonal transformations. In Case II,

$$\overline{u'} = (\Sigma_\ell \otimes \Sigma_r) u', \quad \overline{v'} = v' (\Sigma_r \otimes \Sigma_\ell), \quad (18.623)$$

where

$$\Sigma_x = \mathbb{1}_{x/2} \otimes Y, \quad Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (18.624)$$

The choice is unique up to local transformations S_x satisfying $S_x^T \Sigma_x S_x = \Sigma_x$. Case II can occur only when r and ℓ are even.

Lemma 18.9.1.12 (The four-dimensional symplectic congruence). Lean Lean Lean ✓ Stmt Let

$Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $R = \frac{1-i}{2}[\mathbb{1}_4 + i(Y \otimes Y)]$, with the two tensor factors in that order. Then

$$R^\dagger R = \mathbb{1}_4, \quad (18.625)$$

$$R^T(Y \otimes Y)R = \mathbb{1}_4. \quad (18.626)$$

Proof. ✓ Proof Direct multiplication of the four entries of Y and the corresponding entries of $Y \otimes Y$ gives both identities. □

Corollary 18.9.1.13 (Reduction of the symplectic case to orthogonal unitaries). Not ready

Discussion: #5713 In Case II of Proposition 18.9.1.10, one can write

$$u' = (\mathbb{1}_{r\ell/4} \otimes R)\tilde{u}, \quad v' = \tilde{v}(\mathbb{1}_{r\ell/4} \otimes R), \quad (18.627)$$

where $\tilde{u}$ and $\tilde{v}$ are orthogonal and

$$R = \frac{1-i}{2}[\mathbb{1}_4 + i(Y \otimes Y)]. \quad (18.628)$$

Corollary 18.9.1.14 (Admissible reduction of the symplectic case to orthogonal unitaries).

Not ready Discussion: #5713 In the admissible Case II of Proposition 18.9.1.11, one can write

$$u' = (\mathbb{1}_{r\ell/4} \otimes R)\tilde{u}, \quad v' = \tilde{v}(\mathbb{1}_{r\ell/4} \otimes R), \quad (18.629)$$

where $\tilde{u}$ and $\tilde{v}$ are orthogonal and

$$R = \frac{1-i}{2}[\mathbb{1}_4 + i(Y \otimes Y)]. \quad (18.630)$$

Proposition 18.9.1.15 (Local characterization of time reversal). Not ready Discussion: #5714 *Let an MPU be in standard form, with one-site physical space $\mathbb{C}^{d_0} \otimes \mathbb{C}^{d_0}$. Time-reversal symmetry forces $\text{ind}(\mathcal{U}) = 0$. The MPU satisfies $U^{(N)} = U^{(N)\dagger}$ for every even N if and only if there is a unitary x such that*

$$(\mathbb{1}_1 \otimes v_{23} \otimes \mathbb{1}_4)(u_{12} \otimes u_{34}) = \pm(x_{12} \otimes x_{34}^\dagger)(\mathbb{1}_1 \otimes v_{23}^\dagger \otimes \mathbb{1}_4). \quad (18.631)$$

Proof. Under time reversal, the two-site block $\mathcal{U}_2$ and its physical-adjoint comparison tensor $\mathcal{U}_2^\sharp$ generate the same MPU. The fundamental theorem identifies their source spaces, so their corresponding source-cut ranks agree. Physical adjunction exchanges the two cuts by Theorem 18.4.25. Hence

$$r(\mathcal{U}_2) = r(\mathcal{U}_2^\sharp) = \ell(\mathcal{U}_2). \quad (18.632)$$

Definition 18.9.42 therefore gives $\text{ind}(\mathcal{U}_2) = 0$, and blocking independence gives $\text{ind}(\mathcal{U}) = 0$.

Apply the two gauge equations of the fundamental theorem to the standard forms of $\mathcal{U}_2$ and $\mathcal{U}_2^\sharp$. Substituting the second equation into the first gives

$$x_{12}y_{12} \otimes y_{34}x_{34} = \mathbb{1}. \quad (18.633)$$

Therefore $y = \pm x^\dagger$, which yields the displayed local relation. Conversely, contract that relation and its adjoint alternately around an even periodic chain. $\square$

Proposition 18.9.1.16 (Admissible local characterization of time reversal). Not ready Discussion: #5714 *Let an MPU be in the standard form determined by its supplied source datum. Suppose its one-site tensor $\mathcal{U}$, its actual two-site block $\mathcal{U}_2$, and the physical-adjoint comparison tensor $\mathcal{U}_2^\sharp$ are simple and each carries its own reduced full-support source datum. Write the one-site physical space as $\mathbb{C}^{d_0} \otimes \mathbb{C}^{d_0}$. Time-reversal symmetry then forces $\text{ind}(\mathcal{U}) = 0$. Under these hypotheses, $U^{(N)} = U^{(N)\dagger}$ for every even N if and only if there is a unitary x such that*

$$(\mathbb{1}_1 \otimes v_{23} \otimes \mathbb{1}_4)(u_{12} \otimes u_{34}) = \pm(x_{12} \otimes x_{34}^\dagger)(\mathbb{1}_1 \otimes v_{23}^\dagger \otimes \mathbb{1}_4). \quad (18.634)$$

Proof. Under time reversal, $\mathcal{U}_2$ and $\mathcal{U}_2^\sharp$ generate the same MPU. Apply the same-tensor converse to each supplied pair; the resulting presentations record the original fixed matrices. Theorem 18.9.32 then identifies their source spaces, so their corresponding source-cut ranks agree. Physical adjunction exchanges the two cuts. Hence

$$r(\mathcal{U}_2) = r(\mathcal{U}_2^\sharp) = \ell(\mathcal{U}_2). \quad (18.635)$$

Definition 18.9.43 therefore gives $\text{ind}(\mathcal{U}_2) = 0$, and blocking independence gives $\text{ind}(\mathcal{U}) = 0$. Apply Theorem 18.9.32 to the standard forms of $\mathcal{U}_2$ and $\mathcal{U}_2^\sharp$ and follow the source proof. The converse follows by contracting the displayed local relation around every even periodic chain. $\square$

Proposition 18.9.1.17 (Local characterization of transposition). Not ready Discussion: #5714 *Let an MPU be in standard form, with one-site physical space $\mathbb{C}^{d_0} \otimes \mathbb{C}^{d_0}$. Transposition symmetry forces $\text{ind}(\mathcal{U}) = 0$. The MPU satisfies $U^{(N)} = U^{(N)T}$ for every even N if and only if there are a unitary x and a phase $e^{i\phi}$ such that*

$$(\mathbb{1}_1 \otimes v_{23} \otimes \mathbb{1}_4)(u_{12} \otimes u_{34}) = e^{i\phi}(x_{12} \otimes x_{34}^T)(\mathbb{1}_1 \otimes v_{23}^T \otimes \mathbb{1}_4). \quad (18.636)$$

Proof. Under transposition, the two-site block $\mathcal{U}_2$ and its transposed comparison tensor $\mathcal{U}_2^\top$ generate the same MPU. The fundamental theorem identifies their source spaces, so their corresponding source-cut ranks agree. Swapping the physical input and output indices transposes the first source matrix into the second and therefore exchanges the two ranks. Hence

$$r(\mathcal{U}_2) = r(\mathcal{U}_2^\top) = \ell(\mathcal{U}_2). \quad (18.637)$$

Thus $\text{ind}(\mathcal{U}_2) = 0$ by Definition 18.9.42, and blocking independence gives $\text{ind}(\mathcal{U}) = 0$.

Apply the two gauge equations of the fundamental theorem to the standard forms of $\mathcal{U}_2$ and $\mathcal{U}_2^\top$. Their compatibility gives $y = e^{i\phi}x^T$ for some phase $e^{i\phi}$, which yields the displayed local relation. Conversely, contract that relation around an even periodic chain; the phases cancel pairwise. $\square$

Proposition 18.9.1.18 (Admissible local characterization of transposition). Not ready Discussion: #5714

Let an MPU be in the standard form determined by its supplied source datum. Suppose its one-site tensor $\mathcal{U}$, its actual two-site block $\mathcal{U}_2$, and the transposed comparison tensor $\mathcal{U}_2^\top$ are simple and each carries its own reduced full-support source datum. Write the one-site physical space as $\mathbb{C}^{d_0} \otimes \mathbb{C}^{d_0}$. Transposition symmetry then forces $\text{ind}(\mathcal{U}) = 0$. Under these hypotheses, $U^{(N)} = U^{(N)T}$ for every even N if and only if there are a unitary x and a phase $e^{i\phi}$ such that

$$(\mathbb{1}_1 \otimes v_{23} \otimes \mathbb{1}_4)(u_{12} \otimes u_{34}) = e^{i\phi}(x_{12} \otimes x_{34}^T)(\mathbb{1}_1 \otimes v_{23}^T \otimes \mathbb{1}_4). \quad (18.638)$$

Proof. Under transposition, $\mathcal{U}_2$ and $\mathcal{U}_2^\top$ generate the same MPU. Apply the same-tensor converse to each supplied pair; the resulting presentations record the original fixed matrices. Theorem 18.9.32 then identifies their source spaces, so their corresponding source-cut ranks agree. Transposition exchanges the two cuts. Hence

$$r(\mathcal{U}_2) = r(\mathcal{U}_2^\top) = \ell(\mathcal{U}_2). \quad (18.639)$$

Thus $\text{ind}(\mathcal{U}_2) = 0$ by Definition 18.9.43, and blocking independence gives $\text{ind}(\mathcal{U}) = 0$. Apply Theorem 18.9.32 to the standard forms of $\mathcal{U}_2$ and $\mathcal{U}_2^\top$ and repeat the time-reversal argument, retaining the unrestricted phase $e^{i\phi}$. The converse follows by contracting the local relation around every even periodic chain. $\square$

18.9.2 Equivalence under symmetries

Definition 18.9.2.1 (Finite-chain operator symmetry). Lean Lean ✓ Stmt At physical dimension d , a finite-chain operator symmetry $\mathcal{S}$ specifies a set A of applicable chain lengths and, for every length N , a map

$$\mathcal{S}_{d,N}: \text{Mat}_{dN}(\mathbb{C}) \longrightarrow \text{Mat}_{dN}(\mathbb{C}). \quad (18.640)$$

An MPO tensor $\mathcal{W}$ is invariant under $\mathcal{S}$ if

$$\mathcal{S}_{d,N}[\mathcal{W}^{(N)}] = \mathcal{W}^{(N)} \quad (18.641)$$

for every $N \in A$.

Definition 18.9.2.2 (Coherent symmetry family under identity ancillas). Lean Lean Lean ✓ Stmt A coherent symmetry family supplies an action $\mathcal{S}_{d,N}$ for every physical dimension d , with one common set A of applicable lengths. For an ancilla dimension x , let

$$\iota_x(X) = R_{d,x,N}(X \otimes I_{x^N})R_{d,x,N}^{-1}, \quad (18.642)$$

where $R_{d,x,N}$ is the canonical sitewise identification of the d -dimensional and x -dimensional configurations. Coherence means

$$\iota_x(\mathcal{S}_{d,N}(X)) = \mathcal{S}_{dx,N}(\iota_x(X)). \quad (18.643)$$

Thus the action on the full enlarged operator algebra is supplied data; it is not obtained by extending its values on operators $X \otimes I$.

Theorem 18.9.2.3 (Identity-ancilla transport of symmetry invariance). [Lean](#) [✓ Stmt](#) *If $\mathcal{U}$ is invariant under the d -dimensional component of a coherent symmetry family, then $\mathcal{U}^{(x)}$ is invariant under its dx -dimensional component.*

Proof. [✓ Proof](#) The finite-chain operator of $\mathcal{U}^{(x)}$ is $\iota_x(U^{(N)})$. Semiconjugacy and invariance of $U^{(N)}$ give

$$\mathcal{S}_{dx,N}(\iota_x(U^{(N)})) = \iota_x(\mathcal{S}_{d,N}(U^{(N)})) = \iota_x(U^{(N)}). \quad (18.644)$$

□

Definition 18.9.2.4 (Blocked finite-chain symmetry). [Lean](#) [Lean](#) [✓ Stmt](#) For a blocking length k , the blocked symmetry has applicable lengths

$$A_k = \{N \mid Nk \in A\}. \quad (18.645)$$

Its action is obtained by conjugating $\mathcal{S}_{d,Nk}$ along the canonical identification $(\mathbb{C}^{d^k})^{\otimes N} \cong (\mathbb{C}^d)^{\otimes Nk}$.

Theorem 18.9.2.5 (Blocking transport of symmetry invariance). [Lean](#) [✓ Stmt](#) *If $\mathcal{U}$ is invariant under $\mathcal{S}$, then its k -site blocking is invariant under the blocked symmetry.*

Proof. [✓ Proof](#) Let $R_{d,N,k}$ be the canonical identification from blocked length- N configurations to unblocked length- Nk configurations. For $N \in A_k$, one has $Nk \in A$ and

$$\mathcal{S}_{k,N}[U_k^{(N)}] = R_{d,N,k}^{-1}(\mathcal{S}_{d,Nk}[R_{d,N,k}(U_k^{(N)})]) \quad (18.646)$$

$$= R_{d,N,k}^{-1}(\mathcal{S}_{d,Nk}[U^{(Nk)}]) \quad (18.647)$$

$$= R_{d,N,k}^{-1}(U^{(Nk)}) \quad (18.648)$$

$$= U_k^{(N)}. \quad (18.649)$$

The second and final equalities are the inverse reindexing identities, and the third is invariance of $U^{(Nk)}$ because $Nk \in A$. □

Definition 18.9.2.6 (Strict equivalence under a symmetry). [Lean](#) [✓ Stmt](#) [Discussion: #6948](#) Fix a virtual bond dimension D . Two canonical-form tensors $\mathcal{U}$ and $\mathcal{V}$ are strictly equivalent under a symmetry $\mathcal{S}$ if they have the same physical dimension and there is a continuous path $\mathcal{W}(p)$ of fixed-bond MPU tensors from $\mathcal{U}$ to $\mathcal{V}$, not necessarily in canonical form, such that

$$\mathcal{S}[W^{(N)}(p)] = W^{(N)}(p) \quad (18.650)$$

for every $p \in [0, 1]$ and every applicable chain length N .

Lemma 18.9.2.7 (Forgetting symmetry from strict equivalence). [Lean](#) [✓ Stmt](#) *Strict equivalence under a symmetry implies strict equivalence.*

Proof. ✓ Proof The same path lies in the larger locus obtained by forgetting invariance under $\mathcal{S}$. □

Definition 18.9.2.8 (Admissible strict equivalence under a symmetry). Not ready Discussion: #6017 Two tensors $\mathcal{U}$ and $\mathcal{V}$ are strictly equivalent under a symmetry $\mathcal{S}$ if they have the same physical dimension and there is a continuous path $\mathcal{W}(p)$ of MPU tensors from $\mathcal{U}$ to $\mathcal{V}$ such that

$$\mathcal{S}[W^{(N)}(p)] = W^{(N)}(p) \quad (18.651)$$

for every $p \in [0, 1]$ and every chain length N under consideration. The path need not be in canonical form. The equivalence is admissible if its witnessing path carries an admissible equivalence datum for the actual blocked path tensor and the fixed conjugate, physical-adjoint, transposed, and $\mathcal{S}$ -image comparison paths, and if the common blocking length m of that datum satisfies $m \leq D^4$.

Definition 18.9.2.9 (Equivalence under a symmetry). Not ready Discussion: #7422 Let $\mathcal{U}$ and $\mathcal{V}$ have physical dimensions d_a and d_b . They are equivalent under a symmetry $\mathcal{S}$ if there are a positive blocking length k and coprime positive ancilla dimensions p_a, p_b with $p_a d_a = p_b d_b$ such that $(\mathcal{U}^{(p_a)})_k$ and $(\mathcal{V}^{(p_b)})_k$ are strictly equivalent under $\mathcal{S}$.

Definition 18.9.2.10 (Fixed-bond coherent-family equivalence under a symmetry). Lean ✓ Stmt Discussion: #7015 Fix a virtual bond dimension D , and let $\mathcal{U}$ and $\mathcal{V}$ have physical dimensions d_a and d_b . They are coherently equivalent under a dimension-indexed symmetry family $\mathcal{S}$ if they are MPUs and there are a positive blocking length k and coprime positive ancilla dimensions p_a, p_b with $p_a d_a = p_b d_b$ such that $(\mathcal{U}^{(p_a)})_k$ and $(\mathcal{V}^{(p_b)})_k$ are strictly equivalent under the k -site blocking of the supplied enlarged-dimension symmetry. The family obeys the identity-ancilla semiconjugacy law, and its applicable blocked lengths are precisely the lengths N for which $Nk \in A$. The ancillas are attached before the common blocking.

Theorem 18.9.2.11 (Forgetting symmetry from coherent stabilized equivalence). Lean ✓ Stmt *Fixed-bond equivalence under a coherent symmetry family implies ordinary equivalence after the same ancillas and blocking.*

Proof. ✓ Proof Retain the MPU hypotheses, positive coprime ancilla dimensions, common positive blocking length, and physical-dimension equality from the symmetry-preserving witness. Forgetting invariance from its final strict path gives the required ordinary stabilized equivalence. □

Definition 18.9.2.12 (Admissible equivalence under a symmetry). Not ready Discussion: #6017 Let $\mathcal{U}$ and $\mathcal{V}$ have physical dimensions d_a and d_b . They are equivalent under a symmetry $\mathcal{S}$ if there are a positive blocking length k and coprime positive ancilla dimensions p_a, p_b with $p_a d_a = p_b d_b$ such that $(\mathcal{U}^{(p_a)})_k$ and $(\mathcal{V}^{(p_b)})_k$ are strictly equivalent under $\mathcal{S}$. Thus the ancillas are attached before the common blocking. The equivalence is admissible if this strict-equivalence witness carries an admissible equivalence datum for the actual blocked path and the fixed conjugate, physical-adjoint, transposed, and $\mathcal{S}$ -image comparison paths, with common blocking length at most D^4 .

Lemma 18.9.2.13 (Standard-form path criterion). Not ready Discussion: #6017 *Two MPUs described by simple tensors are strictly equivalent under $\mathcal{S}$ if and only if their standard-form unitaries are joined by continuous families $\tilde{u}(p)$ and $\tilde{v}(p)$ whose generated MPU is invariant under $\mathcal{S}$ for every $p \in [0, 1]$.*

Lemma 18.9.2.14 (Admissible standard-form path criterion). Not ready Discussion: #6017 Fix a positive integer $m \leq D^4$. The actual m -site blocked endpoint MPUs $U_1^{(mN)}$ and $U_2^{(mN)}$ are admissibly strictly equivalent under $\mathcal{S}$ if and only if there are continuous families of unitaries $\tilde{u}(p)$ and $\tilde{v}(p)$ whose endpoints generate $U_1^{(mN)}$ and $U_2^{(mN)}$, whose generated MPU is invariant under $\mathcal{S}$ for every $p \in [0, 1]$, and whose tensor path and fixed conjugate, physical-adjoint, transposed, and $\mathcal{S}$ -image comparison paths carry continuously varying reduced full-support source data. These families arise from the actual blocked path $\mathcal{W}(p)_m$ in an admissible equivalence datum of common length m ; the blocked path itself is the length-one admissible witness for the displayed endpoint families, corresponding to [15, Lemma following the symmetry-equivalence definitions].

Theorem 18.9.2.15 (Conjugation symmetry when one source rank is odd). Not ready Discussion: #5713 Suppose that either r or ℓ is odd. Two conjugation-invariant MPUs $\mathcal{U}_1$ and $\mathcal{U}_2$ are strictly equivalent under conjugation if and only if

$$\det(u_1 v_1) = \det(u_2 v_2). \quad (18.652)$$

Theorem 18.9.2.16 (Admissible conjugation symmetry when one source rank is odd). Not ready Discussion: #5713 For $i = 1, 2$, suppose the original one-site endpoint tensor $\mathcal{U}_i$ and its conjugate comparison tensor are simple and each carries reduced full-support source data. Suppose that either r or ℓ is odd and that the standard-form interpolation determined below carries an admissible path datum of blocking length one. Then $\mathcal{U}_1$ and $\mathcal{U}_2$ are admissibly strictly equivalent under conjugation symmetry if and only if

$$\det(u_1 v_1) = \det(u_2 v_2). \quad (18.653)$$

Corollary 18.9.2.17 (Equivalence of conjugation-symmetric MPUs after blocking). Not ready If either r or ℓ is odd, all conjugation-invariant MPUs with these source ranks are equivalent under conjugation symmetry.

Corollary 18.9.2.18 (Admissible equivalence of conjugation-symmetric MPUs after blocking). Not ready Discussion: #5713 For $i = 1, 2$, suppose the endpoint tensor $\mathcal{U}_i$, its conjugate comparison $\overline{\mathcal{U}_i}$, the actual two-site block $\mathcal{U}_{i,2}$, and its conjugate comparison $\overline{\mathcal{U}_{i,2}}$ are simple and each carries reduced full-support source data. Suppose that either r or ℓ is odd and that the resulting standard-form interpolation carries an admissible path datum of blocking length one. Then $\mathcal{U}_1$ and $\mathcal{U}_2$ are admissibly equivalent under conjugation symmetry.

Proof. If either r or ℓ is odd, both endpoints lie in Case I of Proposition 18.9.1.11. The full admissible classification in Corollary 18.9.2.24 handles both possible parities of the source ranks after blocking and gives the claimed admissible equivalence. $\square$

Theorem 18.9.2.19 (Determinants of orthogonal Kronecker products). Lean Lean Lean ✓ Stmt Let O and O' be orthogonal matrices over a commutative ring, indexed by finite sets of cardinalities m and n . For every even integer $k \geq 0$, $\det(O)^k = 1$. Over an integral domain one also has $\det(O) \in \{1, -1\}$. If m and n are even, then

$$\det(O \otimes O') = \det(O)^n \det(O')^m = 1. \quad (18.654)$$

These identities give the orthogonal determinant calculation in [15].

Proof. ✓ Proof Taking determinants in $OO^T = \mathbb{1}$ gives $\det(O)^2 = 1$, and the same holds for O' . The determinant of a Kronecker product is $\det(O)^n \det(O')^m$; both powers are one when m and n are even. $\square$

Theorem 18.9.2.20 (Determinant of a symplectic Kronecker product). [Lean](#) [✓ Stmt](#) Let S and S' preserve the standard symplectic forms on spaces of dimensions $2p$ and $2q$ over a commutative ring. Then

$$\det(S \otimes S') = \det(S)^{2q} \det(S')^{2p} = 1. \quad (18.655)$$

No positivity assumption on either dimension is needed. This theorem uses the standard symplectic form; the interleaved form in [15] requires a separate change of coordinates before this calculation applies to its matrices.

Proof. [✓ Proof](#) Each symplectic determinant equals one. Apply the Kronecker determinant formula. $\square$

Theorem 18.9.2.21 (Strict conjugation equivalence for even source ranks). [Not ready](#) [Discussion: #5713](#) Suppose r and ℓ are even. Two conjugation-invariant MPUs in the standard forms of Proposition 18.9.1.10 are strictly equivalent under conjugation if and only if they belong to the same case and

$$\det(u_1) = \det(u_2), \quad \det(v_1) = \det(v_2). \quad (18.656)$$

Theorem 18.9.2.22 (Admissible strict conjugation equivalence for even source ranks). [Not ready](#) [Discussion: #5713](#) Suppose that r and ℓ are both even and that the standard-form interpolation determined below carries an admissible path datum of blocking length one. Choose standard forms from the supplied source data of two conjugation-invariant MPUs as in Proposition 18.9.1.11. They are admissibly strictly equivalent under conjugation symmetry if and only if they belong to the same case of Proposition 18.9.1.11 and

$$\det(u_1) = \det(u_2), \quad \det(v_1) = \det(v_2). \quad (18.657)$$

Corollary 18.9.2.23 (Equivalence under conjugation). [Not ready](#) [Discussion: #5713](#) Two conjugation-invariant MPUs are equivalent under conjugation symmetry if and only if they belong to the same case of Proposition 18.9.1.10.

Corollary 18.9.2.24 (Admissible equivalence under conjugation). [Not ready](#) [Discussion: #5713](#) For $i = 1, 2$, suppose the original one-site endpoint tensor $\mathcal{U}_i$, its conjugate comparison $\overline{\mathcal{U}}_i$, the actual two-site block $\mathcal{U}_{i,2}$, and its conjugate comparison $\overline{\mathcal{U}}_{i,2}$ are simple and each carries reduced full-support source data. Suppose also that the resulting standard-form interpolation carries an admissible path datum of blocking length one. Then $\mathcal{U}_1$ and $\mathcal{U}_2$ are admissibly equivalent under conjugation symmetry if and only if they belong to the same case of Proposition 18.9.1.11.

Proof. Suppose first that either r or ℓ is odd. Then both original endpoints lie in Case I. After one blocking, write $r^{(2)}$ and $\ell^{(2)}$ for the blocked source ranks. Admissible blocking independence, applied to the supplied one-site and two-site endpoint data, gives

$$r^{(2)} = dr, \quad (18.658)$$

and

$$\ell^{(2)} = d\ell. \quad (18.659)$$

For each endpoint,

$$\det(u_i^{(2)} v_i^{(2)}) = \det(u_i v_i)^{2r\ell} = 1. \quad (18.660)$$

Blocking preserves Case I. If either $r^{(2)}$ or $\ell^{(2)}$ is odd, the odd-rank theorem applied to the blocked endpoints gives admissible strict equivalence. If both blocked ranks are even, then d is even because one original source rank is odd. Hence $r\ell = d^2$ is even. Consequently,

$$\det(u_i^{(2)}) = (\det(u_i)^2 \det(v_i))^{r\ell} = 1, \quad (18.661)$$

and

$$\det(v_i^{(2)}) = \det(v_i)^{r\ell} = 1. \quad (18.662)$$

The even-rank theorem now applies to the preserved Case I endpoints. Thus the odd-source-rank branch has a single admissible equivalence class without imposing a parity condition on the blocked ranks.

If r and ℓ are both even, admissible blocking independence shows that the blocked ranks remain even. Since $r\ell$ is even, the same calculations give

$$\det(u_i^{(2)}) = (\det(u_i)^2 \det(v_i))^{r\ell} = 1, \quad (18.663)$$

and

$$\det(v_i^{(2)}) = \det(v_i)^{r\ell} = 1. \quad (18.664)$$

Blocking preserves Case I and Case II. The even-rank theorem therefore shows that the blocked endpoints are admissibly strictly equivalent exactly when they belong to the same case. This is precisely admissible equivalence of the original endpoints. $\square$

Lemma 18.9.2.25 (SWAP trace contraction). [Lean](#) [Lean](#) [✓ Stmt](#) For matrices $A, B \in \mathbb{C}^{n \times n}$ and the exchange matrix $\mathbb{S}$ on $\mathbb{C}^n \otimes \mathbb{C}^n$,

$$\mathrm{tr}[\mathbb{S}(A \otimes B)] = \mathrm{tr}(AB). \quad (18.665)$$

In particular, if U is unitary, then

$$\mathrm{tr}[\mathbb{S}(U \otimes U^\dagger)] = n. \quad (18.666)$$

Proof. [✓ Proof](#) In product coordinates, the exchange matrix has entries $\mathbb{S}_{(i,j),(k,l)} = \delta_{i,l} \delta_{j,k}$. Therefore

$$\mathrm{tr}[\mathbb{S}(A \otimes B)] = \sum_{i,j} A_{j,i} B_{i,j} = \sum_{i,j} A_{i,j} B_{j,i} = \mathrm{tr}(AB). \quad (18.667)$$

Taking $A = U$ and $B = U^\dagger$ gives $\mathrm{tr}(UU^\dagger) = \mathrm{tr}(\mathbb{1}) = n$. $\square$

Theorem 18.9.2.26 (The exchange intertwines the Kronecker factors). [Lean](#) [Lean](#) [✓ Stmt](#) For matrices $A, B \in \mathbb{C}^{n \times n}$ and the exchange matrix $\mathbb{S}$ on $\mathbb{C}^n \otimes \mathbb{C}^n$,

$$\mathbb{S}(A \otimes B) = (B \otimes A)\mathbb{S}, \quad (18.668)$$

$$(A \otimes B)\mathbb{S} = \mathbb{S}(B \otimes A). \quad (18.669)$$

The equivalent two-sided exchange identity is stated in [15, lines 1316–1319]; the gauge-invariance argument at [15, lines 1624–1630] uses it without a printed derivation.

Proof. ✓ Proof In product coordinates the exchange matrix has entries $\mathbb{S}_{(i,j),(k,l)} = \delta_{i,l}\delta_{j,k}$, so both sides of (18.668) have entry

$$[\mathbb{S}(A \otimes B)]_{(i_1, i_2), (j_1, j_2)} = A_{i_2, j_1} B_{i_1, j_2} = [(B \otimes A)\mathbb{S}]_{(i_1, i_2), (j_1, j_2)}. \quad (18.670)$$

Swapping factors A and B gives (18.669). □

Theorem 18.9.2.27 (Exchange contraction of a scaled unitary Kronecker product). Lean

✓ Stmt Let $x \in \mathbb{C}^{n \times n}$ be unitary, let $\sigma \in \mathbb{C}$, and let $M = \sigma x \otimes x^\dagger$. Then

$$\text{tr}[\mathbb{S}M] = \sigma n. \quad (18.671)$$

This is the exchange-trace evaluation in [15, lines 1608–1610]; there $n = d^2$.

Proof. ✓ Proof The trace is linear, so $\text{tr}[\mathbb{S}M] = \sigma \text{tr}[\mathbb{S}(x \otimes x^\dagger)]$, and Lemma 18.9.2.25 evaluates the remaining contraction as n . □

Lemma 18.9.2.28 (Trace formula for the time-reversal sign). Not ready Discussion: #5714 For a time-reversal-invariant MPU in standard form, the sign σ of Proposition 18.9.1.15 is

$$\sigma = \frac{1}{d^2} \text{tr}[\mathbb{S}_{12,34} v_{23} u_{12} u_{34} v_{23}], \quad (18.672)$$

where $\mathbb{S}_{12,34}$ exchanges the pairs of sites 12 and 34.

Lemma 18.9.2.29 (Admissible trace formula for the time-reversal sign). Not ready Discussion: #5714 For a time-reversal-invariant MPU in the standard form determined by its supplied source datum, suppose its one-site tensor $\mathcal{U}$, its actual two-site block $\mathcal{U}_2$, and the physical-adjoint comparison tensor $\mathcal{U}_2^\sharp$ are simple and each carries its own reduced full-support source datum. Then the sign σ in Proposition 18.9.1.16 is

$$\sigma = \frac{1}{d^2} \text{tr}[\mathbb{S}_{12,34} v_{23} u_{12} u_{34} v_{23}], \quad (18.673)$$

where $\mathbb{S}_{12,34}$ exchanges the pairs of sites 12 and 34.

Lemma 18.9.2.30 (Gauge invariance of the time-reversal sign). Not ready Discussion: #5714 Under the hypotheses of Lemma 18.9.2.28, let x and y be unitaries and change the standard-form gauge by

$$u' = (x \otimes y)u, \quad v' = v(y^\dagger \otimes x^\dagger). \quad (18.674)$$

Then the trace formula has the same value for (u', v') as for (u, v) . Consequently the sign σ is independent of the standard-form gauge.

Proof. Write x_i and y_i for the unitaries acting on site i . Then $u'_{12} = x_1 y_2 u_{12}$, $u'_{34} = x_3 y_4 u_{34}$, and $v'_{23} = v_{23} y_2^\dagger x_3^\dagger$. Substitution into Lemma 18.9.2.28, cancellation on the middle legs, and $\mathbb{S}_{12,34} x_1 y_4 = x_3 y_2 \mathbb{S}_{12,34}$ give

$$\text{tr}[\mathbb{S}_{12,34} v'_{23} u'_{12} u'_{34} v'_{23}] = \text{tr}[x_3 y_2 \mathbb{S}_{12,34} v_{23} u_{12} u_{34} v_{23} y_2^\dagger x_3^\dagger] \quad (18.675)$$

$$= \text{tr}[\mathbb{S}_{12,34} v_{23} u_{12} u_{34} v_{23}]. \quad (18.676)$$

The last equality follows from cyclicity of the trace and unitarity of x and y . □

Lemma 18.9.2.31 (Admissible gauge invariance of the time-reversal sign). Not ready Discussion: #5714
Under the hypotheses of Lemma 18.9.2.29, let x and y be unitaries and change the standard-form gauge by

$$u' = (x \otimes y)u, \quad v' = v(y^\dagger \otimes x^\dagger). \quad (18.677)$$

Then the trace formula has the same value for (u', v') as for (u, v) . Consequently the sign σ is independent of the standard-form gauge.

Proof. Write x_i and y_i for the unitaries acting on site i . Then $u'_{12} = x_1 y_2 u_{12}$, $u'_{34} = x_3 y_4 u_{34}$, and $v'_{23} = v_{23} y_2^\dagger x_3^\dagger$. Substitution into Lemma 18.9.2.29, cancellation on the middle legs, and $\mathbb{S}_{12,34} x_1 y_4 = x_3 y_2 \mathbb{S}_{12,34}$ give

$$\text{tr}[\mathbb{S}_{12,34} v'_{23} u'_{12} u'_{34} v'_{23}] = \text{tr}[x_3 y_2 \mathbb{S}_{12,34} v_{23} u_{12} u_{34} v_{23} y_2^\dagger x_3^\dagger] \quad (18.678)$$

$$= \text{tr}[\mathbb{S}_{12,34} v_{23} u_{12} u_{34} v_{23}]. \quad (18.679)$$

The last equality follows from cyclicity of the trace and unitarity of x and y . $\square$

Proposition 18.9.2.32 (Well-definedness of the time-reversal sign). Not ready Discussion: #5714
For a time-reversal-invariant MPU in standard form, the sign $\sigma = \pm 1$ is independent of the gauge and is constant under continuous deformations that preserve time-reversal symmetry.

Proposition 18.9.2.33 (Admissible well-definedness of the time-reversal sign). Not ready Discussion: #5714
Suppose a time-reversal-preserving path carries an admissible equivalence datum of blocking length one. At every path point, require the one-site tensor, its actual two-site block, and the physical-adjoint comparison of that two-site block to be simple and to carry continuously varying reduced full-support source data. Then the sign $\sigma = \pm 1$ is independent of the admissible standard-form gauge and is constant along the path.

Theorem 18.9.2.34 (Scalar parity law for the blocked sign). Lean Lean Lean Stmt Let $\sigma^{(1)}, \sigma^{(2)} \in \{1, -1\}$ and put $\zeta = \sigma^{(1)} \sigma^{(2)}$. For every $k \geq 1$,

$$\sigma^{(1)} \zeta^{k-1} = \begin{cases} \sigma^{(1)}, & k \text{ odd}, \\ \sigma^{(2)}, & k \text{ even}. \end{cases} \quad (18.680)$$

This is the scalar conclusion of [15, lines 1687 and 1730–1732].

Proof. Proof Both signs square to one, hence $\zeta^2 = 1$. For odd k the exponent $k-1$ is even, so $\zeta^{k-1} = 1$ and the product is $\sigma^{(1)}$. For even $k \geq 2$ the exponent $k-1$ is odd, so $\zeta^{k-1} = \zeta$ and the product is $(\sigma^{(1)})^2 \sigma^{(2)} = \sigma^{(2)}$. $\square$

Proposition 18.9.2.35 (Blocking parity of the time-reversal sign). Not ready Discussion: #5714 Let $\mathcal{U}$ be a time-reversal-invariant MPU in standard form, and let $\sigma^{(k)}$ be the sign obtained from a standard form of its k -site block. Then

$$\sigma^{(k)} = \begin{cases} \sigma^{(1)}, & k \text{ odd}, \\ \sigma^{(2)}, & k \text{ even}. \end{cases} \quad (18.681)$$

Proof. Gauge invariance permits the standard blocked representatives used in the source proof. Choose

$$u^{(2)} = v_{23} u_{12} u_{34}, \quad v^{(2)} = v_{45}, \quad (18.682)$$

and put $\zeta = \sigma^{(2)}/\sigma^{(1)}$. The one- and two-site local time-reversal relations give the unwinding identity

$$\zeta[(1 \otimes v \otimes 1)(u \otimes x)]^\dagger = (1 \otimes v \otimes 1)(x^\dagger \otimes u). \quad (18.683)$$

Applying this identity successively through a block of length k gives

$$\sigma^{(k)} = \sigma^{(1)} \zeta^{k-1}. \quad (18.684)$$

Since each sign is ± 1 , this equals $\sigma^{(1)}$ for odd k and $\sigma^{(2)}$ for even k . $\square$

Proposition 18.9.2.36 (Admissible blocking parity of the time-reversal sign). Not ready Discussion: #5714

For every blocking length k under consideration, suppose the endpoint tensor $\mathcal{U}_k$, its actual two-site block $\mathcal{U}_{2k}$, and the physical-adjoint comparison tensor $\mathcal{U}_{2k}^\sharp$ are simple and each carries its own reduced full-support source datum. Let $\sigma^{(k)}$ be the sign of the resulting standard form. Then

$$\sigma^{(k)} = \begin{cases} \sigma^{(1)}, & k \text{ odd}, \\ \sigma^{(2)}, & k \text{ even}. \end{cases} \quad (18.685)$$

Proof. Gauge invariance permits the standard blocked representatives used in the source proof. Choose

$$u^{(2)} = v_{23}u_{12}u_{34}, \quad v^{(2)} = v_{45}, \quad (18.686)$$

and put $\zeta = \sigma^{(2)}/\sigma^{(1)}$. The one- and two-site local time-reversal relations give the unwinding identity

$$\zeta[(1 \otimes v \otimes 1)(u \otimes x)]^\dagger = (1 \otimes v \otimes 1)(x^\dagger \otimes u). \quad (18.687)$$

Applying this identity successively through a block of length k gives

$$\sigma^{(k)} = \sigma^{(1)} \zeta^{k-1}. \quad (18.688)$$

Since each sign is ± 1 , this equals $\sigma^{(1)}$ for odd k and $\sigma^{(2)}$ for even k . $\square$

Corollary 18.9.2.37 (Necessary time-reversal invariants). Not ready Discussion: #5714 Strictly equivalent time-reversal-invariant MPUs have the same $\sigma^{(1)}$ and $\sigma^{(2)}$. Equivalent time-reversal-invariant MPUs have the same $\sigma^{(2)}$.

Corollary 18.9.2.38 (Admissible time-reversal invariants of supplied witnesses). Not ready

Discussion: #5714 Let two time-reversal-invariant MPUs be compared by an admissible symmetry path $p \mapsto \mathcal{W}(p)$ whose path datum has blocking length one. Suppose that the five path families $\mathcal{W}(p)$, $\mathcal{W}(p)_2$, $\mathcal{W}(p)_4$, $\mathcal{W}(p)_2^\sharp$, and $\mathcal{W}(p)_4^\sharp$ are simple and carry continuously varying reduced full-support source data. Then admissible strict equivalence requires equality of both $\sigma^{(1)}$ and $\sigma^{(2)}$.

For admissible equivalence after adjoining ancillas and applying the source blocking, let $p \mapsto \mathcal{W}_{\text{anc}}(p)$ be the actual ancilla-enlarged blocked witness path, not a reduced or equivalent representative, and regard it as a length-one admissible path. Require continuous reduced full-support source data along $\mathcal{W}_{\text{anc}}(p)$, its actual two- and four-site blocks, and the physical-adjoint comparisons of those two blocks. The conclusion is only

$$\sigma^{(2)}(\mathcal{W}_{\text{anc}}(0)) = \sigma^{(2)}(\mathcal{W}_{\text{anc}}(1)). \quad (18.689)$$

No equality between $\sigma^{(2)}$ of the original two MPUs is asserted. Such a conclusion requires preservation under adjoining identity ancillas, which is not established here.

Proposition 18.9.2.39 (Conjugation class and the MPS sign). Not ready Discussion: #5714 For a conjugation-invariant MPU, the MPS sign $G\bar{G} = \pm 1$ agrees with Case I and Case II, respectively, of Proposition 18.9.1.10.

Proposition 18.9.2.40 (Admissible conjugation class and the MPS sign). Not ready Discussion: #5714 Let $\mathcal{U}$ be conjugation invariant. Suppose $\mathcal{U}$ and its conjugate comparison tensor are simple and carry their own reduced full-support source data. Let $p \mapsto \mathcal{W}(p)$ be the actual standard-form interpolation from $\mathcal{U}$ to its Case I or Case II representative. Suppose this interpolation carries an admissible conjugation-symmetry path datum of blocking length one: the path $\mathcal{W}(p)$ and its fixed conjugate, physical-adjoint, transposed, and conjugation-image comparison paths are simple and carry continuously varying source data. Then the sign $G\bar{G} = \pm 1$ in the MPS classification agrees with Case I and Case II, respectively, of Proposition 18.9.1.11.

Proposition 18.9.2.41 (Time-reversal class and the MPS sign). Not ready Discussion: #5714 For a time-reversal-invariant MPU in standard form, the MPS sign $G\bar{G} = \pm 1$ equals the sign $\sigma^{(2)}$ obtained after blocking two sites.

Proposition 18.9.2.42 (Admissible time-reversal class and the MPS sign). Not ready Discussion: #5714 Let an MPU in the standard form determined by its supplied source datum be invariant under time reversal. Suppose $\mathcal{U}$, $\mathcal{U}_2$, and $\mathcal{U}_4$, together with the physical-adjoint comparison tensors $\mathcal{U}_2^\sharp$ and $\mathcal{U}_4^\sharp$, are simple and each carries its own reduced full-support source datum. Then the MPS sign $G\bar{G} = \pm 1$ equals the sign $\sigma^{(2)}$.

18.9.3 Symmetry examples

Proposition 18.9.3.1 (All phases under conjugation). Not ready Discussion: #5715 Let $u_+ = v_+ = \mathbb{1}_{r\ell}$ and $u_- = v_- = \text{diag}(-1, 1, \dots, 1)$. If at least one of r and ℓ is odd, (u_+, v_+) and (u_+, v_-) realize the two classes of Theorem 18.9.2.15. If both are even, the four pairs $(u_\pm, v_\pm)$ realize the four Case I classes. Multiplying by the matrix R of Corollary 18.9.1.13 realizes the four Case II classes.

Proposition 18.9.3.2 (Admissible phases under conjugation). Not ready Discussion: #5715 Assume the standard-form tensors displayed below carry reduced full-support source data and that the comparison paths used to classify them carry admissible path data of blocking length one. Let

$$u_+ = v_+ = \mathbb{1}_{r\ell}, \quad u_- = v_- = \text{diag}(-1, 1, \dots, 1). \quad (18.690)$$

If at least one of r and ℓ is odd, the pairs (u_+, v_+) and (u_+, v_-) represent the two classes of Theorem 18.9.2.16. If r and ℓ are both even, the four pairs $(u_\pm, v_\pm)$ represent the four Case I classes of Theorem 18.9.2.22. With

$$u'_\pm = (\mathbb{1}_{r\ell/4} \otimes R)u_\pm, \quad v'_\pm = v_\pm(\mathbb{1}_{r\ell/4} \otimes R), \quad (18.691)$$

the four pairs $(u'_\pm, v'_\pm)$ represent the four Case II classes.

Theorem 18.9.3.3 (Normalization obstruction in the printed CZX tensor). Lean Lean Lean ✓ Stmt Discussion: #7738 Let $u, d, \ell, r \in \{0, 1\}$ and set $P_{\ell r}^{ud} = \delta_{u, 1-d} \delta_{u, \ell} 2^{-1/2} (-1)^{\ell r}$, the tensor printed in the CZX example of [15]. In the ordinary periodic trace contraction, the two-site operator generated by P has all-zero output row

$$\text{MPO}_2(P)_{00,b} = \frac{1}{2} \delta_{b,11}. \quad (18.692)$$

In particular, P does not generate a matrix product unitary. The normalization discrepancy and comparison with the Z-decorated tensor are detailed in [42].

Proof. ✓ Proof The copy deltas fix both virtual indices to zero. The physical bit flips then force the input to be 11, and the two Hadamard entries contribute $1/2$. If the operator were unitary, its all-zero diagonal entry after multiplication by its adjoint would give

$$1 = \sum_b |\text{MPO}_2(P)_{00,b}|^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}, \quad (18.693)$$

a contradiction. □

Remark 18.9.3.4. The normalized Hadamard in the following printed CZX formulas has the obstruction above. In contrast, the blue tensors in the CZX tensor diagram of [163] are $\sqrt{2}H$, and this diagram also has a Z gate on each output qubit. Let $A = \sqrt{2}P$, let A_2 denote its two-site block, and let B denote that displayed Z -decorated tensor. The output-leg relation is

$$B = (Z \otimes Z)A_2. \quad (18.694)$$

The coordinate derivation of (18.694) is given in [42].

Proposition 18.9.3.5 (CZX). Not ready Discussion: #5715 *The literal normalized tensor printed below is not an MPU, by Theorem 18.9.3.3. The following is the source classification claim, not an assertion of unitarity for that literal tensor; its normalization and comparison with the Z -decorated tensor are discussed in [42]. The CZX MPU of Chen, Liu and Wen [10] (see also [15, lines 1906–1943]) is defined by*

$$u = CZ(\sigma_x \otimes \sigma_x), \quad v = CZ, \quad (18.695)$$

where $CZ = \text{diag}(1, 1, 1, -1)$. Let

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (18.696)$$

be the Hadamard matrix. The corresponding tensor is

$$\mathcal{U}_{\ell r}^{ud} = (\sigma_x)_{ud} \delta_{u,\ell} H_{\ell,r}. \quad (18.697)$$

It has $\sigma^{(1)} = \sigma^{(2)} = -1$.

Proposition 18.9.3.6 (Admissible CZX data). Not ready Discussion: #5715 *The literal normalized tensor printed below is not an MPU, by Theorem 18.9.3.3. The classification below evaluates the supplied tensor data under [42]. Suppose the CZX tensors $\mathcal{U}$, $\mathcal{U}_2$, and $\mathcal{U}_4$, together with the physical-adjoint comparison tensors $\mathcal{U}_2^\sharp$ and $\mathcal{U}_4^\sharp$, are simple and each carries its own reduced full-support source datum. The CZX MPU of Chen, Liu and Wen [10] (see also [15, lines 1906–1943]) is defined by*

$$u = CZ(\sigma_x \otimes \sigma_x), \quad v = CZ, \quad (18.698)$$

$$CZ = \text{diag}(1, 1, 1, -1). \quad (18.699)$$

Its tensor is

$$\mathcal{U}_{\ell r}^{ud} = (\sigma_x)_{ud} \delta_{u,\ell} H_{\ell,r}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (18.700)$$

It has $\sigma^{(1)} = \sigma^{(2)} = -1$.

For the left-shift tensor $A^\sharp$ they are

$$M_1(A^\sharp) = zz^\top, \quad (18.713)$$

$$M_2(A^\sharp) = \mathbb{1}. \quad (18.714)$$

In every physical dimension, including $d = 0$,

$$r(A) = d^2, \quad (18.715)$$

$$\ell(A^\sharp) = d^2. \quad (18.716)$$

If $d > 0$, the remaining two ranks are

$$\ell(A) = 1, \quad (18.717)$$

$$r(A^\sharp) = 1. \quad (18.718)$$

These are the raw source-cut quantities from [15, Section III]; no choice of source factors is involved.

Proof. ✓ Proof The entry formula $A_{\alpha\beta}^{ij} = \delta_{i\alpha}\delta_{j\beta}$ gives $M_1(A) = \mathbb{1}$ and $M_2(A) = zz^\top$. Hence the first rank is d^2 . For $d > 0$, the vector z has a nonzero diagonal coordinate, so its outer product has rank one. Physical adjunction exchanges the two cuts and their ranks, which gives all four left-shift formulas without a second rank calculation. $\square$

Theorem 18.9.3.12 (Unitarity of the swap operator). Lean ✓ Stmt Let $d \in \mathbb{N}$. On $\mathbb{C}^d \otimes \mathbb{C}^d$, let $\mathbb{S}$ be the swap operator used in [15, Section VII], defined on the standard basis by $\mathbb{S}(e_k \otimes e_\ell) = e_\ell \otimes e_k$. With columns indexed by source coordinates (k, ℓ) and rows indexed by target coordinates (i, j) , its entries are

$$\mathbb{S}_{(i,j),(k,\ell)} = \delta_{i,\ell}\delta_{j,k}. \quad (18.719)$$

Then $\mathbb{S}$ is unitary between the two product-coordinate orders:

$$\mathbb{S}^\dagger \mathbb{S} = \mathbb{1}, \quad (18.720)$$

$$\mathbb{S} \mathbb{S}^\dagger = \mathbb{1}. \quad (18.721)$$

Proof. ✓ Proof The swap operator is self-adjoint and involutive, so $\mathbb{S}^\dagger = \mathbb{S}$ and $\mathbb{S}^2 = \mathbb{1}$. Substitution gives both identities. $\square$

Definition 18.9.3.13 (Three topological-insulator shift MPUs). Lean Lean Lean Lean ✓ Stmt On two d -dimensional spins per site, define

$$U_1^{(N)} = (\mathbb{1} \otimes \mathbb{1})^{\otimes N}, \quad (18.722)$$

$$U_2^{(N)} = T^{(N)\dagger} \otimes T^{(N)}, \quad (18.723)$$

$$U_3^{(N)} = T^{(N)} \otimes T^{(N)\dagger}. \quad (18.724)$$

Definition 18.9.3.14 (Trace-normalized shift weight). Lean Lean Lean ✓ Stmt Canonical form II requires the positive right fixed point ρ to have trace one. If the virtual weight on the bond space of a single shift is chosen to be a scalar identity, this normalization forces

$$\rho_d = d^{-1} \mathbb{1}_d \quad (18.725)$$

Proof. [✓ Proof](#) For the identity-weight factors, direct substitution in $u = Y_2 - Y_1$ gives

$$u_{(0,(a,b)),(i_1,i_2)}^{(0)} = d^{1/2} \delta_{a,i_1} \delta_{b,i_2}. \quad (18.733)$$

The trace-normalized first cut replaces Y_1 by $d^{-1/2}Y_1$ and leaves Y_2 unchanged. Therefore

$$u_{(0,(a,b)),(i_1,i_2)} = d^{-1/2} u_{(0,(a,b)),(i_1,i_2)}^{(0)} = \delta_{a,i_1} \delta_{b,i_2}. \quad (18.734)$$

Summing the product of two such entries over (a,b) gives $\delta_{p,q}$, which is the entrywise identity $u^\dagger u = \mathbb{1}$. $\square$

Theorem 18.9.3.18 (Local physical-species swap for the shift MPUs). [Lean](#) [✓ Stmt](#) Write $[U_a]^{(i_1,i_2),(j_1,j_2)}$ for the virtual matrix of the one-site tensor U_a with physical output coordinate (i_1,i_2) and input coordinate (j_1,j_2) . Let $\mathbb{S}$ be the SWAP matrix on the two virtual factors, defined by

$$\mathbb{S}|\alpha_1, \alpha_2\rangle = |\alpha_2, \alpha_1\rangle. \quad (18.735)$$

Then

$$[U_3]^{(i_2,i_1),(j_2,j_1)} = \mathbb{S}[U_2]^{(i_1,i_2),(j_1,j_2)}\mathbb{S}. \quad (18.736)$$

Thus exchanging the two physical species exchanges the local tensors U_2 and U_3 , with the same exchange on their virtual factors.

Proof. [✓ Proof](#) In virtual coordinates (α_1, α_2) and (β_1, β_2) , multiplication by $\mathbb{S}$ on both sides selects the entry with row (α_2, α_1) and column (β_2, β_1) . Consequently,

$$[U_3]_{(\alpha_1, \alpha_2), (\beta_1, \beta_2)}^{(i_2, i_1), (j_2, j_1)} = T_{\alpha_1 \beta_1}^{i_2 j_2} (T^\dagger)_{\alpha_2 \beta_2}^{i_1 j_1} \quad (18.737)$$

$$= (T^\dagger)_{\alpha_2 \beta_2}^{i_1 j_1} T_{\alpha_1 \beta_1}^{i_2 j_2} \quad (18.738)$$

$$= [U_2]_{(\alpha_2, \alpha_1), (\beta_2, \beta_1)}^{(i_1, i_2), (j_1, j_2)}. \quad (18.739)$$

This is exactly the entrywise form of conjugation by $\mathbb{S}$. $\square$

Theorem 18.9.3.19 (Finite-chain formulas for the three shift MPUs). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The three tensors generate the operators displayed in Definition 18.9.3.13 on every nonempty chain, and each tensor is a matrix product unitary.

Proof. [✓ Proof](#) The bond-one identity tensor generates the identity at every length. The finite-chain operator of an independent tensor product is the Kronecker product of the two finite-chain operators, after the sitewise identification $(\mathbb{C}^d \otimes \mathbb{C}^d)^{\otimes N} \cong (\mathbb{C}^d)^{\otimes N} \otimes (\mathbb{C}^d)^{\otimes N}$. Applying this identity to the right and left shifts gives the three displayed formulas. The identity tensor is unitary, and tensor products preserve unitarity, so all three tensors are matrix product unitaries. $\square$

Theorem 18.9.3.20 (Specified-tensor source-index values of the shifts). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Assume $d > 0$. For the displayed right- and left-shift tensors T and $T^\sharp$, the specified-tensor values are

$$\text{ind}_{\text{src}}(T) = \log_2 d. \quad (18.740)$$

Likewise,

$$\text{ind}_{\text{src}}(T^\sharp) = -\log_2 d. \quad (18.741)$$

The bond-one identity tensor and the three tensors U_1, U_2, U_3 satisfy

$$\text{ind}_{\text{src}}(\mathbb{1}) = \text{ind}_{\text{src}}(U_1) = \text{ind}_{\text{src}}(U_2) = \text{ind}_{\text{src}}(U_3) = 0. \quad (18.742)$$

Here ind_{src} is the specified-tensor quantity from Definition 18.9.38, evaluated on the displayed unblocked tensors. It is not the public MPU index ind of Definition 18.9.42, whose definition uses a simple blocking and requires independence of that choice. The public index values follow in Theorem 18.9.3.21.

Proof. ✓ Proof The right shift has source ranks $(d^2, 1)$, while the left shift has $(1, d^2)$. Substitution into the source-index expression gives the first two formulas. The identity tensor has equal source ranks (d, d) . Source ranks multiply under the independent tensor product, so each of U_1, U_2, U_3 has equal right and left rank d^2 . Their logarithmic differences therefore vanish. $\square$

Theorem 18.9.3.21 (Public indices of the shift examples). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

✓ **Stmt** Let $d > 0$. The right shift is in canonical form II in its original bond coordinates, with ambient right fixed matrix $\rho = I/d$. The right and left shifts and the three two-spin families satisfy

$$\text{ind}(T) = \log_2 d, \quad \text{ind}(T^\sharp) = -\log_2 d, \quad (18.743)$$

$$\text{ind}(U_1) = \text{ind}(U_2) = \text{ind}(U_3) = 0. \quad (18.744)$$

These are the blocking-independent indices of the families in Definition 18.9.3.13, as asserted in [15].

Proof. ✓ **Proof** The normalized transfer map of the right shift sends X to $\text{tr}(X)I/d$. Its positive diagonal fixed matrix I/d gives a canonical-form-II presentation. The source-rank computation therefore yields the public index $\log_2 d$. Physical adjunction negates this index, giving the left-shift value. The identity has index zero, and additivity under the tensor product gives all three remaining values. $\square$

Theorem 18.9.3.22 (Index of the bond-one identity tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

Lemma 3.10. *Let $d > 0$. The bond-one tensor generating the identity operator has a canonical-form-II presentation with fixed matrix $\rho = 1$. For every canonical-form-II presentation of this tensor,*

$$\text{ind}(\mathbb{1}) = 0. \quad (18.745)$$

This follows from the index definition of [15, Definition IV.1]; it is distinct from the source's later examples of three two-spin tensors with vanishing index.

Proof. ✓ Proof In bond dimension one the normalized transfer matrix is the identity; the positive fixed matrix $\rho = 1$ therefore gives a canonical-form-II presentation. Both source ranks of the identity tensor equal d , so the source-gate criterion makes it simple. Blocking once only relabels the physical alphabet. The defining logarithmic difference for any canonical-form-II presentation is consequently $\frac{1}{2}(\log_{\gamma} d - \log_{\gamma} d) = 0$. $\square$

Theorem 18.9.3.23 (Source ranks and index values of the shifts at every blocking). [Lean](#) [Lean](#)

Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean

Lean Lean Lean Lean Lean Lean Lean Lean **Stmnt** Assume $d > 0$ and let $k > 0$. The right-shift

tensor T and the left-shift tensor $T^\sharp$ are simple, and so are their blockings T_{k+1} and $T_{k+1}^\sharp$ over $k+1$ sites. The blocked letters are

$$T_{k+1}^{\sigma\tau} = \left(\prod_{m < k} \delta_{\tau_m, \sigma_{m+1}} \right) |\sigma_0\rangle \langle \tau_k|, \quad (18.746)$$

and the source ranks are

$$(r, \ell)(T_{k+1}) = (d^{k+2}, d^k), \quad (r, \ell)(T_{k+1}^\sharp) = (d^k, d^{k+2}). \quad (18.747)$$

Both source ranks of the blockings of U_2 and U_3 equal d^{2k+2} . Consequently

$$\text{ind}_{\text{src}}(T_{k+1}) = \log_2 d, \quad \text{ind}_{\text{src}}(T_{k+1}^\sharp) = -\log_2 d, \quad (18.748)$$

and the blockings of U_2 and U_3 have value zero. If $d > 1$, then $r(T_{k+1}) \neq \ell(T_{k+1})$ and $\text{ind}_{\text{src}}(T_{k+1}) \neq 0$.

The tensors U_1 , U_2 , and U_3 are simple, and so are the blockings of U_2 and U_3 over $k+1$ sites, because independent tensor products of simple tensors are simple. Thus all four values are taken at simple blockings, which is the simplicity requirement of Definition 18.9.42. The canonical-form construction and representative invariance in Theorem 18.9.3.21 identify these values with the public indices and transport them to any positive-dimensional MPU generating the same periodic operators. Unequal ranks do not by themselves exclude a circuit: after blocking, every matrix product unitary, the shift included, is a depth-two circuit of nearest-neighbour unitaries in its standard form. The statement that the shift is not a finite-depth circuit of local gates needs parts (ii)–(iv) of Theorem 18.9.104 together with index zero for one layer of nearest-neighbour gates, and is not asserted here.

Proof. ✓ Proof With $\Phi = \sum_j |jj\rangle$, the double-layer letters of T are $|\Phi\rangle\langle ik|$, so the boundary vectors $|00\rangle$ and Φ satisfy both simplicity identities; for $T^\sharp$ the letters are $|ik\rangle\langle\Phi|$ and the boundary vectors are exchanged. The identity tensor has bond dimension one and double-layer letters δ_{ij} , so $a = b = 1$ are boundary vectors. Simplicity survives independent tensor products and blocking, which gives simplicity of U_1 , U_2 , U_3 and of the blockings of U_2 and U_3 . A product of matrix units $|\sigma_m\rangle\langle\tau_m|$ is $|\sigma_0\rangle\langle\tau_k|$ when adjacent indices match and zero otherwise, which gives the blocked letters. The first source cut of T_{k+1} sends the row (σ, β) to the single column $(\sigma_0, \sigma_1 \cdots \sigma_k \beta)$; this is a bijection, so the cut is a permutation matrix of rank d^{k+2} . The second source cut factors through the d^k middle words $\sigma_1 \cdots \sigma_k$, and its rows $(0, 0y)$ and columns $(y0, 0)$ form an identity minor of size d^k . Physical adjunction exchanges the two ranks, and ranks multiply under tensor products and are unchanged by physical reindexing. Substituting into the source-index expression gives $\frac{1}{2}((k+2) - k) \log_2 d = \log_2 d$ and the remaining values. For $d > 1$ the powers d^{k+2} and d^k differ. □

Definition 18.9.3.24 (Supplied source factors for the three shift MPUs). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Assume $d > 0$. Equip U_1 with the tensor product of the two bond-one identity source factorizations. Equip U_2 with the tensor product of the left-shift and right-shift source factorizations, in that order, and equip U_3 with the tensor product in the reverse order. In all three cases the virtual weight is the identity. These are explicit supplied source-factor witnesses; no identification with an arbitrary compact singular-value decomposition is asserted.

Theorem 18.9.3.25 (Unitarity of the four-spin shift matrices). Lean Lean Lean Lean ✓ Stmt Let $\mathbb{S}$ exchange two d -dimensional spins. The two swap factors appearing in the blocked forms of U_2 and U_3 , and both ordered products, are unitary:

$$1 \otimes \mathbb{S} \otimes 1, \quad \mathbb{S} \otimes \mathbb{S}, \quad (\mathbb{S} \otimes \mathbb{S})(1 \otimes \mathbb{S} \otimes 1), \quad (1 \otimes \mathbb{S} \otimes 1)(\mathbb{S} \otimes \mathbb{S}). \quad (18.749)$$

Proof. [✓ Proof](#) The swap is unitary by Theorem 18.9.3.12. Kronecker products preserve isometries, and a square isometry is unitary. Thus both factors are unitary. Closure of the unitary matrices under multiplication gives the two ordered products. $\square$

Theorem 18.9.3.26 (Supplied paper gates for U_1). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *In the four-spin coordinates of the supplied source factorization,*

$$u_1 = \mathbb{1} \otimes \mathbb{1}, \quad v_1 = \mathbb{1} \otimes \mathbb{1}. \quad (18.750)$$

Proof. [✓ Proof](#) The tensor-product formulas for $u = Y_2 - Y_1$ and $v = X_1 - X_2$ reduce each entry to the product of two primitive identity entries. $\square$

Theorem 18.9.3.27 (Paper gates in the two-site factorizations of U_2). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Assume $d > 0$. In the four-spin coordinates of the paper, the supplied source gates satisfy*

$$u_2^{(2)} = d (\mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}), \quad (18.751)$$

$$d v_2^{(2)} = (\mathbb{S} \otimes \mathbb{S})(\mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}). \quad (18.752)$$

The blocked tensor factors through $X_2, u_2^{(2)}, X_1$ and, equivalently, through $v_2^{(2)}, Y_1, Y_2$. The scalar factors record the normalization of the supplied source-factor witnesses.

Proof. [✓ Proof](#) The tensor-product formulas for the paper gates reduce their entries to products of the left- and right-shift source gates. The Kronecker deltas give the two displayed swap matrices, with the factors d and d^{-1} fixed by the identity-weight normalization. Substitute these evaluations into the two exact open source-gate factorizations. $\square$

Theorem 18.9.3.28 (Paper gates in the two-site factorizations of U_3). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Assume $d > 0$. In the four-spin coordinates of the paper, the supplied source gates satisfy*

$$u_3^{(2)} = d (\mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1})(\mathbb{S} \otimes \mathbb{S}), \quad (18.753)$$

$$d v_3^{(2)} = \mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}. \quad (18.754)$$

The blocked tensor factors through $X_2, u_3^{(2)}, X_1$ and, equivalently, through $v_3^{(2)}, Y_1, Y_2$. The scalar factors record the normalization of the supplied source-factor witnesses.

Proof. [✓ Proof](#) Reverse the two primitive shift factors in the tensor-product formulas for u and v . Their Kronecker deltas give the displayed ordered swap products, with the factors d and d^{-1} fixed by the identity-weight normalization. Substitute these evaluations into the two exact open source-gate factorizations. $\square$

Theorem 18.9.3.29 (Trace-normalized source gates for the counterpropagating shifts). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For $d > 0$, the two counterpropagating shift tensors have source factors with trace-one weight $\rho = d^{-2} \mathbb{1}_{d^2}$. In the four-spin coordinates used above, their source gates are*

$$u_2^{(2)} = \mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}, \quad v_2^{(2)} = (\mathbb{S} \otimes \mathbb{S})(\mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}), \quad (18.755)$$

$$u_3^{(2)} = (\mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1})(\mathbb{S} \otimes \mathbb{S}), \quad v_3^{(2)} = \mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}. \quad (18.756)$$

Proof. ✓ Proof Starting with the identity-weight source factors, replace X_1, Y_1, Z_1 by $dX_1, d^{-1}Y_1, dZ_1$, respectively, and leave the second cut unchanged. The cut factorizations and right inverses are preserved, while the first weighted isometry now uses $d^{-2}\mathbb{1}_{d^2}$. Writing $\hat{u}_k, \hat{v}_k$ for the identity-weight gates of Theorem 18.9.3.27 and Theorem 18.9.3.28, the source gates rescale as

$$u_k^{(2)} = d^{-1}\hat{u}_k, \quad v_k^{(2)} = d\hat{v}_k, \quad k = 2, 3, \quad (18.757)$$

which cancels the scalar factors in those formulas. □

Theorem 18.9.3.30 (Two-site decompositions of the three shift MPUs). Lean Lean Lean Lean ✓ Stmt Assume $d > 0$. The three shift families have two-site decompositions with supplied gates whose unitary gates satisfy

$$u_1 = \mathbb{1} \otimes \mathbb{1}, \quad v_1 = \mathbb{1} \otimes \mathbb{1}, \quad (18.758)$$

$$u_3 = \mathbb{S}, \quad v_3 = \mathbb{S}, \quad (18.759)$$

$$u_2^{(2)} = \mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}, \quad v_2^{(2)} = (\mathbb{S} \otimes \mathbb{S})(\mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}), \quad (18.760)$$

$$u_3^{(2)} = (\mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1})(\mathbb{S} \otimes \mathbb{S}), \quad v_3^{(2)} = \mathbb{1} \otimes \mathbb{S} \otimes \mathbb{1}. \quad (18.761)$$

The two-site blockings in the last two lines use the physical order $((1, 2), (3, 4))$. These are unitary gates with normalized half factors satisfying the open standard-form equation. For the two blocked families, the half factors and gates are those of the source factors of Theorem 18.9.3.29. For U_1 and the unblocked U_3 , the one-site tensor is read as a pair of spins and decomposed directly; these decompositions are not identified with the standard form of Definition 18.9.13 [142].

Proof. ✓ Proof For the identity family, take $X_1(i, \gamma; r) = \delta_{i,r}$ and $X_2(\alpha, i; \ell) = \delta_{i,\ell}$. For the unblocked third family, take

$$X_1(i, (a, b); r) = \sqrt{d} \delta_{a,r} \delta_{b,i}, \quad X_2((a, b), i; \ell) = d^{-1/2} \delta_{a,i} \delta_{b,\ell}. \quad (18.762)$$

Its gates are the swap, which is unitary by Theorem 18.9.3.12. The reciprocal scalars cancel in both contractions. Summing the Gram entries proves $X_1^*(\mathbb{1} \otimes \rho)X_1 = \mathbb{1}$ and $X_2^*X_2 = \mathbb{1}$, with $\rho = d^{-2}\mathbb{1}_{d^2}$ for the third family and $\rho = 1$ for the identity. For the two blocked families, use the trace-normalized source factors of Theorem 18.9.3.29, relabeling the intermediate coordinates by pairs of spin indices. Write $e_L : \{0, \dots, d^2 - 1\} \rightarrow L$ and $e_R : \{0, \dots, d^2 - 1\} \rightarrow R$ for these bijections, obtained by $\ell \leftrightarrow (a, b)$ and $r \leftrightarrow (c, e)$ and the source-factor rank identifications. The open contraction becomes

$$\sum_{p \in L} \sum_{q \in R} X_2(\alpha, i_1; p) u(p, q; j_1, j_2) X_1(i_2, \gamma; q) = \sum_{\ell=0}^{d^2-1} \sum_{r=0}^{d^2-1} X_2(\alpha, i_1; e_L(\ell)) u(e_L(\ell), e_R(r); j_1, j_2) X_1(i_2, \gamma; e_R(r)).$$

Thus the source-factor contraction theorem gives the open equation in the displayed coordinates. The four-spin permutation matrices are unitary. The relabeled half factors are $\widehat{X}_1(i, \gamma; r) = X_1(i, \gamma; e_R(r))$ and $\widehat{X}_2(\alpha, i; \ell) = X_2(\alpha, i; e_L(\ell))$. Since e_R and e_L are injective, the source normalization identities give

$$(\widehat{X}_1^*(\mathbb{1} \otimes \rho) \widehat{X}_1)_{r,t} = (X_1^*(\mathbb{1} \otimes \rho) X_1)_{e_R(r), e_R(t)} = \delta_{e_R(r), e_R(t)} = \delta_{r,t}, \quad (18.763)$$

$$(\widehat{X}_2^* \widehat{X}_2)_{\ell,m} = (X_2^* X_2)_{e_L(\ell), e_L(m)} = \delta_{e_L(\ell), e_L(m)} = \delta_{\ell,m}. \quad (18.764)$$

□

Proof. ✓ Proof The physical swap exchanges the two output indices of each tensor. For the first family, retain the identity family's half factors and replace its unshifted gate by the swap. For the second family, choose $X_1(i, (a, b); r) = \sqrt{d} \delta_{a,i} \delta_{b,r}$ and $X_2((a, b), i; \ell) = d^{-1/2} \delta_{a,\ell} \delta_{b,i}$. Contracting over the bond index $\beta = (a, b)$ gives the shifted swap gate,

$$\sum_{a,b} X_1(i_1, (a, b); s) X_2((a, b), i_2; t) = \delta_{i_1,t} \delta_{i_2,s} = \mathbb{S}_{(i_1, i_2), (s, t)}. \quad (18.776)$$

Inserting the unshifted identity gate $\mathbb{1} \otimes \mathbb{1}$ between the half factors gives, for $\alpha = (a, b)$ and $\gamma = (a', b')$,

$$\sum_{\ell, r} X_2(\alpha, i_1; \ell) (\mathbb{1} \otimes \mathbb{1})_{(\ell, r), (j_1, j_2)} X_1(i_2, \gamma; r) = X_2(\alpha, i_1; j_1) X_1(i_2, \gamma; j_2) = \delta_{a, j_1} \delta_{b, i_1} \delta_{a', i_2} \delta_{b', j_2}, \quad (18.777)$$

which is the entry of $\tilde{\mathcal{U}}_2$ with output (i_1, i_2) , input (j_1, j_2) , and bond indices α, γ . The swap is unitary by Theorem 18.9.3.12, so both gates are unitary. With $\rho = d^{-2} \mathbb{1}_{d^2}$, the Gram sums are

$$(X_1^* (\mathbb{1} \otimes \rho) X_1)_{r, r'} = d^{-2} \sum_{i, a, b} d \delta_{a,i} \delta_{b,r} \delta_{b, r'} = \delta_{r, r'}, \quad (18.778)$$

$$(X_2^* X_2)_{\ell, \ell'} = d^{-1} \sum_{a, b, i} \delta_{a,\ell} \delta_{a, \ell'} \delta_{b,i} = \delta_{\ell, \ell'}, \quad (18.779)$$

since d pairs (i, a) satisfy $a = i$ and d pairs (b, i) satisfy $b = i$. □

Lemma 18.9.3.35 (Transformation with swaps and symmetry equivalence). Not ready Discussion: #5715

Two MPUs $U_i^{(N)}$ and $U_j^{(N)}$ are in the same phase under time reversal combined with the local swap, transposition combined with the local swap, or conjugation if and only if the corresponding $\tilde{U}_i^{(N)}$ and $\tilde{U}_j^{(N)}$ are in the same phase under time reversal, transposition, or conjugation, respectively.

Lemma 18.9.3.36 (Admissible transformation with swaps and symmetry equivalence). Not ready

Discussion: #5715 Fix one of the three symmetry comparisons and let $p \mapsto \mathcal{W}(p)$ be the actual source-constructed interpolation used in that comparison. Suppose it carries an admissible symmetry-path datum of blocking length one. Thus the path $\mathcal{W}(p)$ and the four fixed comparison paths $\bar{\mathcal{W}}(p)$, $\mathcal{W}(p)^\sharp$, $\mathcal{W}(p)^\top$, and $\mathcal{S}[\mathcal{W}(p)]$ are simple and carry continuously varying reduced full-support source data. Suppose also that every one- and two-site endpoint tensor used in the local comparison carries its own datum. Then $U_i^{(N)}$ and $U_j^{(N)}$ are in the same admissible phase under time reversal combined with the local swap, transposition combined with the local swap, or conjugation if and only if the corresponding $\tilde{U}_i^{(N)}$ and $\tilde{U}_j^{(N)}$ are in the same admissible phase under time reversal, transposition, or conjugation, respectively.

Proposition 18.9.3.37 (Conjugation phases of U_2 and U_3). Not ready Discussion: #5715 After blocking two sites, $U_2^{(N)}$ and $U_3^{(N)}$ are both in the trivial phase under conjugation.

Proposition 18.9.3.38 (Admissible conjugation phases of U_2 and U_3). Not ready Discussion: #5715

Suppose the two-site blocked tensors carry reduced full-support source data and the standard-form comparison paths carry admissible path data of blocking length one. Then the actual two-site blocked tensors of U_2 and U_3 are both in the trivial admissible strict phase under conjugation. Consequently U_2 and U_3 lie in the same admissible equivalence class.

Theorem 18.9.3.39 (Continuous unitary path to the factor swap). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ **Stmt** Let $\mathbb{S}$ denote the swap operator on $\mathbb{C}^d \otimes \mathbb{C}^d$. There is a continuous map $S: \mathbb{R} \rightarrow \mathrm{U}(d^2)$ such that

$$(S(0), S(1)) = (\mathbb{1}, \mathbb{S}). \quad (18.780)$$

One concrete choice is

$$S(x) = P_{\mathrm{sym}} + e^{i\pi x} P_{\mathrm{anti}}, \quad (18.781)$$

where

$$P_{\mathrm{sym}} = \frac{\mathbb{1} + \mathbb{S}}{2}, \quad (18.782)$$

and

$$P_{\mathrm{anti}} = \frac{\mathbb{1} - \mathbb{S}}{2}. \quad (18.783)$$

Proof. $\checkmark$ **Proof** The swap is a self-adjoint involution:

$$\mathbb{S}^2 = \mathbb{1}, \quad (18.784)$$

and

$$\mathbb{S}^\dagger = \mathbb{S}. \quad (18.785)$$

Hence the two projections satisfy

$$(P_{\mathrm{sym}}^2, P_{\mathrm{anti}}^2) = (P_{\mathrm{sym}}, P_{\mathrm{anti}}), \quad (18.786)$$

$$(P_{\mathrm{sym}} P_{\mathrm{anti}}, P_{\mathrm{anti}} P_{\mathrm{sym}}) = (0, 0), \quad (18.787)$$

$$(P_{\mathrm{sym}}^\dagger, P_{\mathrm{anti}}^\dagger) = (P_{\mathrm{sym}}, P_{\mathrm{anti}}), \quad (18.788)$$

and

$$P_{\mathrm{sym}} + P_{\mathrm{anti}} = \mathbb{1}. \quad (18.789)$$

Since $|e^{i\pi x}| = 1$, these identities imply

$$S(x)S(x)^\dagger = P_{\mathrm{sym}} + P_{\mathrm{anti}} = \mathbb{1}. \quad (18.790)$$

Continuity follows from continuity of the exponential, while $e^0 = 1$ and $e^{i\pi} = -1$ give the endpoint identities. $\square$

Proposition 18.9.3.40 (Time-reversal and transposition phases of U_2 and U_3). [Not ready](#)

[Discussion: #5715](#) The MPUs $\widetilde{U}_2^{(N)}$ and $\widetilde{U}_3^{(N)}$ are in the same phase under both time reversal and transposition.

Proposition 18.9.3.41 (Admissible time-reversal and transposition phases of U_2 and U_3).

Not ready Discussion: #5715 For each of time reversal and transposition, let $p \mapsto \mathcal{W}_S(p)$ be the actual source-constructed interpolation from $\widetilde{U}_2$ to $\widetilde{U}_3$. Suppose it carries an admissible symmetry-path datum of blocking length one. Thus the path $\mathcal{W}_S(p)$ and its fixed conjugate, physical-adjoint, transposed, and $\mathcal{S}$ -image comparison paths are simple and carry continuously varying reduced full-support source data. Suppose also that the one- and two-site endpoint tensors used in the local comparison carry their own data. Then $\widetilde{U}_2^{(N)}$ and $\widetilde{U}_3^{(N)}$ are in the same admissible phase under both time reversal and transposition.

Proposition 18.9.3.42 (Conjugation phases of U_1 and U_2). Not ready Discussion: #5715 The MPUs $U_1^{(N)}$ and $U_2^{(N)}$ lie in different strict conjugation phases exactly when $d = 4k + 2$ for a nonnegative integer k . Otherwise they lie in the same strict phase. All these phases are in Case I, so the two MPUs are equivalent after blocking and adjoining ancillas.

Proposition 18.9.3.43 (Admissible conjugation phases of U_1 and U_2). Not ready Discussion: #5715 For each endpoint tensor $\mathcal{A}$ among U_1 , U_2 , $\widetilde{U}_1$, and $\widetilde{U}_2$, suppose that $\mathcal{A}$, its conjugate comparison $\overline{\mathcal{A}}$, its actual two-site block $\mathcal{A}_2$, and the conjugate comparison $\overline{\mathcal{A}_2}$ are simple and carry their own reduced full-support source data. For every strict comparison used below, let $p \mapsto \mathcal{W}(p)$ be the actual source-constructed interpolation between the corresponding endpoint tensors. Suppose it carries an admissible conjugation-symmetry path datum of blocking length one: the path and its fixed conjugate, physical-adjoint, transposed, and conjugation-image comparison paths are simple and carry continuously varying source data. Require a length-one admissible path datum also for the actual blocked interpolation used in the final equivalence claim. Under these hypotheses, $U_1^{(N)}$ and $U_2^{(N)}$ are in different admissible strict phases under conjugation exactly when k is a nonnegative integer and

$$d = 4k + 2. \quad (18.791)$$

Otherwise they are in the same admissible strict phase. All these standard forms lie in Case I, so $U_1^{(N)}$ and $U_2^{(N)}$ are admissibly equivalent after blocking and adding ancillas.

Definition 18.9.3.44 (Two-species ancilla configurations). Lean ✓ Stmt Discussion: #7470 For a periodic chain of length N , the configuration space consists of triples $((\sigma_1, \sigma_2), a)$ of two physical configurations and one d -dimensional ancilla configuration.

Definition 18.9.3.45 (Sitewise physical swap). Lean ✓ Stmt Discussion: #7470 The sitewise physical swap is

$$S^{1,2}((\sigma_1, \sigma_2), a) = ((\sigma_2, \sigma_1), a). \quad (18.792)$$

Definition 18.9.3.46 (Sitewise second-species-ancilla swap). Lean ✓ Stmt Discussion: #7470 The sitewise swap of the second species with the ancilla is

$$S^{2,a}((\sigma_1, \sigma_2), a) = ((\sigma_1, a), \sigma_2). \quad (18.793)$$

Definition 18.9.3.47 (Crossed first-species-ancilla swap). Lean Lean ✓ Stmt Discussion: #7470 Let R cyclically rotate configurations on the chain. The crossed swap is

$$S_+^{1,a}((\sigma_1, \sigma_2), a) = ((Ra, \sigma_2), R^{-1}\sigma_1). \quad (18.794)$$

Definition 18.9.3.48 (Literal three-swap circuit). Lean ✓ Stmt Discussion: #7470 Let C be the permutation obtained by applying $S_+^{1,a}$ first, then $S^{1,2}$, and finally $S^{2,a}$.

Definition 18.9.3.49 (Counter-shift permutation). [Lean](#) [✓ Stmt](#) [Discussion: #7470](#) Define

$$K((\sigma_1, \sigma_2), a) = ((R\sigma_2, R^{-1}\sigma_1), a). \quad (18.795)$$

Theorem 18.9.3.50 (Literal three-swap permutation identity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

[Discussion: #7470](#) The literal three-swap circuit satisfies

$$C^{-1}S^{1,2}C = K. \quad (18.796)$$

Writing P_f for the permutation matrix of f , one also has

$$P_C P_{S^{1,2}} P_{C^{-1}} = P_K. \quad (18.797)$$

At the coordinate level, K is the configuration map of $\widetilde{U}_3$, whereas $\widetilde{U}_2$ has configuration map $((\sigma_1, \sigma_2), a) \mapsto ((R^{-1}\sigma_2, R\sigma_1), a)$. The literal source labels produce a discrepancy at the $\widetilde{U}_2$ boundary on a nondegenerate periodic chain; see [15, Figure 16].

Proof. [✓ Proof](#) Evaluating the three swaps gives

$$C^{-1}S^{1,2}C((\sigma_1, \sigma_2), a) = ((R\sigma_2, R^{-1}\sigma_1), a). \quad (18.798)$$

Extensionality gives the permutation equality. The permutation-matrix identity follows by functoriality. $\square$

Proposition 18.9.3.51 (Ancilla equivalence of U_1 and U_2). [Not ready](#) [Discussion: #5715](#) After adding one ancilla of dimension d per site, $\widetilde{U}_1^{(N)}$ and $\widetilde{U}_2^{(N)}$ are strictly equivalent under both time reversal and transposition.

Proposition 18.9.3.52 (Admissible ancilla equivalence of U_1 and U_2). [Not ready](#) [Discussion: #5715](#) After adding one ancilla of dimension d per site and performing the source blocking, denote the actual enlarged blocked endpoints by $\widehat{U}_1$ and $\widehat{U}_2$. For either time reversal or transposition, let $p \mapsto \mathcal{W}_S(p)$ be the source-constructed interpolation from $\widehat{U}_1$ to $\widehat{U}_2$. Suppose this interpolation carries an admissible symmetry-path datum of blocking length one. Thus the path $\mathcal{W}_S(p)$ and its fixed conjugate, physical-adjoint, transposed, and $\mathcal{S}$ -image comparison paths are simple and carry continuously varying reduced full-support source data. Suppose also that the one- and two-site enlarged endpoint tensors used in the local comparison carry their own data. Then $\widehat{U}_1$ and $\widehat{U}_2$ are admissibly strictly equivalent under both time reversal and transposition. No equivalence of the original ancilla-free tensors is asserted, because the equal ancilla dimensions used here need not satisfy the coprimality condition in Definition 18.9.2.12.

18.10 Finite-region observable algebras

Definition 18.10.1 (Finite-region local algebra). [Lean](#) [Lean](#) [✓ Stmt](#) Fix a nonnegative integer d . For a finite region $\Lambda \subset \mathbb{Z}$, the configuration space and local observable algebra are

$$\text{Config}_d(\Lambda) = \{0, \dots, d-1\}^\Lambda, \quad (18.799)$$

$$\mathcal{A}_\Lambda = M_{\text{Config}_d(\Lambda)}(\mathbb{C}) \cong M_d^{\otimes \Lambda}. \quad (18.800)$$

The algebra $\mathcal{A}_\Lambda$ carries its operator norm. This is the finite-region construction used in the QCA appendix of [15]; no completion is taken here.

Definition 18.10.2 (Canonical inclusion of finite-region observables). [Lean](#) [✓ Stmt](#) If $\Lambda \subseteq \Gamma$ are finite regions, define

$$\iota_{\Lambda, \Gamma} : \mathcal{A}_\Lambda \longrightarrow \mathcal{A}_\Gamma, \quad (18.801)$$

$$\iota_{\Lambda, \Gamma}(A) = A \otimes \mathbf{1}_{\Gamma \setminus \Lambda}. \quad (18.802)$$

Here product configurations are identified with configurations on Γ . This is a unital star-algebra homomorphism.

Lemma 18.10.3 (Injectivity of the finite-region inclusion). [Lean](#) [✓ Stmt](#) For $d > 0$, the map $\iota_{\Lambda, \Gamma}$ is injective.

Proof. [✓ Proof](#) Since $d > 0$, choose a configuration c on $\Gamma \setminus \Lambda$. For configurations x, y on Λ ,

$$\iota_{\Lambda, \Gamma}(A)((x, c), (y, c)) = A(x, y) \cdot \mathbf{1}_{\Gamma \setminus \Lambda}(c, c) = A(x, y). \quad (18.803)$$

Hence $\iota_{\Lambda, \Gamma}(A) = \iota_{\Lambda, \Gamma}(B)$ implies $A = B$. $\square$

Lemma 18.10.4 (Operator-norm preservation). [Lean](#) [✓ Stmt](#) If $d > 0$, then for every $A \in \mathcal{A}_\Lambda$,

$$\|\iota_{\Lambda, \Gamma}(A)\| = \|A\|. \quad (18.804)$$

Proof. [✓ Proof](#) An injective star-algebra homomorphism between complex C^* -algebras is isometric. $\square$

Lemma 18.10.5 (Isometry of the finite-region inclusion). [Lean](#) [✓ Stmt](#) For $d > 0$, the map $\iota_{\Lambda, \Gamma}$ is an operator-norm isometry.

Proof. [✓ Proof](#) Use injectivity and the general theorem that an injective star-algebra homomorphism between complex C^* -algebras is an isometry. $\square$

Lemma 18.10.6 (Identity inclusion). [Lean](#) [✓ Stmt](#) For every finite region Λ , $\iota_{\Lambda, \Lambda}$ is the identity on $\mathcal{A}_\Lambda$.

Proof. [✓ Proof](#) When the two regions agree, their complement is empty. Hence

$$\iota_{\Lambda, \Lambda}(A) = A \otimes \mathbf{1}_\emptyset = A \otimes 1 = A. \quad (18.805)$$

$\square$

Lemma 18.10.7 (Composition of finite-region inclusions). [Lean](#) [✓ Stmt](#) If $\Lambda \subseteq \Gamma \subseteq \Delta$, then

$$\iota_{\Gamma, \Delta} \circ \iota_{\Lambda, \Gamma} = \iota_{\Lambda, \Delta}. \quad (18.806)$$

Proof. [✓ Proof](#) The complement decomposes as the disjoint union

$$\Delta \setminus \Lambda = (\Gamma \setminus \Lambda) \sqcup (\Delta \setminus \Gamma), \quad (18.807)$$

$$\mathbf{1}_{\Delta \setminus \Lambda} = \mathbf{1}_{\Gamma \setminus \Lambda} \otimes \mathbf{1}_{\Delta \setminus \Gamma}. \quad (18.808)$$

Therefore

$$\iota_{\Gamma, \Delta}(\iota_{\Lambda, \Gamma}(A)) = (A \otimes \mathbf{1}_{\Gamma \setminus \Lambda}) \otimes \mathbf{1}_{\Delta \setminus \Gamma} \quad (18.809)$$

$$= A \otimes \mathbf{1}_{\Delta \setminus \Lambda} \quad (18.810)$$

$$= \iota_{\Lambda, \Delta}(A). \quad (18.811)$$

$\square$

Theorem 18.10.8 (Disjoint finite-region observables commute). Lean ✓ Stmt If $\Lambda \cap \Gamma = \emptyset$, $A \in \mathcal{A}_\Lambda$, and $B \in \mathcal{A}_\Gamma$, then their images in the algebra of the union satisfy

$$\iota_{\Lambda, \Lambda \cup \Gamma}(A) \iota_{\Gamma, \Lambda \cup \Gamma}(B) = \iota_{\Gamma, \Lambda \cup \Gamma}(B) \iota_{\Lambda, \Lambda \cup \Gamma}(A). \quad (18.812)$$

This identity provides the local commutativity of disjoint regions in the quasi-local algebra [15, lines 2292–2300].

Proof. ✓ Proof For configurations x, y on $\Lambda \cup \Gamma$, split each intermediate configuration z into $(z|_\Lambda, z|_\Gamma)$. The matrix entry of the left-hand side is

$$\sum_{z_\Lambda, z_\Gamma} A(x|_\Lambda, z_\Lambda) \delta_{x|_\Gamma, z_\Gamma} B(z_\Gamma, y|_\Gamma) \delta_{z_\Lambda, y|_\Lambda} = A(x|_\Lambda, y|_\Lambda) B(x|_\Gamma, y|_\Gamma). \quad (18.813)$$

Reversing the two factors gives the same scalar product. □

Theorem 18.10.9 (Finite relative commutant). Lean ✓ Stmt Let $\Lambda \subseteq \Gamma$ be finite regions and $A \in \mathcal{A}_\Gamma$. Then A acts trivially outside Λ if and only if it commutes with the full observable algebra on the complement:

$$A \in \iota_{\Lambda, \Gamma}(\mathcal{A}_\Lambda) \iff \forall B \in \mathcal{A}_{\Gamma \setminus \Lambda}, \quad A \iota_{\Gamma \setminus \Lambda, \Gamma}(B) = \iota_{\Gamma \setminus \Lambda, \Gamma}(B) A. \quad (18.814)$$

This is a finite-dimensional tensor-factor identity. It is not a statement of Schumacher–Werner or the QCA appendix of [15].

Proof. ✓ Proof Splitting configurations gives $\mathcal{A}_\Gamma \cong \mathcal{A}_\Lambda \otimes \mathcal{A}_{\Gamma \setminus \Lambda}$. Under this identification, the two inclusions are $C \mapsto C \otimes \mathbf{1}$ and $B \mapsto \mathbf{1} \otimes B$. Write an arbitrary matrix in blocks indexed by configurations on Λ . If it commutes with every $\mathbf{1} \otimes B$, each block commutes with the full matrix algebra on $\Gamma \setminus \Lambda$ and is therefore scalar. Thus the matrix is $C \otimes \mathbf{1}$ for some $C \in \mathcal{A}_\Lambda$. The converse follows from

$$(C \otimes \mathbf{1})(\mathbf{1} \otimes B) = C \otimes B = (\mathbf{1} \otimes B)(C \otimes \mathbf{1}). \quad (18.815)$$

□

18.10.1 Bipartite support algebras

Let I and J be finite sets. For $X \in M_{I \times J}(\mathbb{C})$, the existing left coefficient slice and right operator block are the matrices

$$X_{rs}^L(i, j) = X((i, r), (j, s)), \quad X_{ij}^R(r, s) = X((i, r), (j, s)). \quad (18.816)$$

For a star-subalgebra $\mathcal{D} \subseteq M_K(\mathbb{C})$, write $\mathcal{D}' = \{Y \in M_K(\mathbb{C}) \mid [Y, Z] = 0 \text{ for every } Z \in \mathcal{D}\}$ for its commutant. The construction below is the full complex matrix-algebra specialization of the coefficient construction in [205, Section 4.3] and of the leastness and commutant statements in [169, Lemma 7].

The commutant characterization of [169, Lemma 7] requires star-closure [205, lines 1161–1171], as documented in [140]. For the MPU spin chain, full matrix factors $\mathcal{B}_1 = M_I(\mathbb{C})$ and $\mathcal{B}_2 = M_J(\mathbb{C})$ suffice [15, lines 2320–2334]; see [65].

Theorem 18.10.1.1 (Adjoint of a right coefficient block). Lean ✓ Stmt For every $X \in M_{I \times J}(\mathbb{C})$ and $i, j \in I$,

$$(X^\dagger)_{ij}^R = (X_{ji}^R)^\dagger. \quad (18.817)$$

Proof. ✓ Proof Compare the (r, s) entries. Both sides equal $\overline{X((j, s), (i, r))}$. □

Definition 18.10.1.2 (Left coefficient set). Lean ✓ Stmt Let $\mathcal{A} \subseteq M_{I \times J}(\mathbb{C})$ be a star-subalgebra. Its set of left coefficients is

$$\text{Coeff}_L(\mathcal{A}) = \{X_{rs}^L \mid X \in \mathcal{A}, r, s \in J\} \subseteq M_I(\mathbb{C}). \quad (18.818)$$

Definition 18.10.1.3 (Right coefficient set). Lean ✓ Stmt Let $\mathcal{A} \subseteq M_{I \times J}(\mathbb{C})$ be a star-subalgebra. Its set of right coefficients is

$$\text{Coeff}_R(\mathcal{A}) = \{X_{ij}^R \mid X \in \mathcal{A}, i, j \in I\} \subseteq M_J(\mathbb{C}). \quad (18.819)$$

Definition 18.10.1.4 (Left support algebra). Lean ✓ Stmt The left support algebra of $\mathcal{A}$ is the star-subalgebra

$$\text{Spp}_L(\mathcal{A}) = \text{alg}^*(\text{Coeff}_L(\mathcal{A})) \subseteq M_I(\mathbb{C}) \quad (18.820)$$

generated by all left coefficients.

Definition 18.10.1.5 (Right support algebra). Lean ✓ Stmt The right support algebra of $\mathcal{A}$ is the star-subalgebra

$$\text{Spp}_R(\mathcal{A}) = \text{alg}^*(\text{Coeff}_R(\mathcal{A})) \subseteq M_J(\mathbb{C}) \quad (18.821)$$

generated by all right coefficients.

Theorem 18.10.1.6 (Left coefficient generators belong to the support algebra). Lean ✓ Stmt If $X \in \mathcal{A}$ and $r, s \in J$, then

$$X_{rs}^L \in \text{Spp}_L(\mathcal{A}). \quad (18.822)$$

Proof. ✓ Proof The matrix X_{rs}^L is one of the generators of the defining star-algebra closure. □

Theorem 18.10.1.7 (Right coefficient generators belong to the support algebra). Lean ✓ Stmt If $X \in \mathcal{A}$ and $i, j \in I$, then

$$X_{ij}^R \in \text{Spp}_R(\mathcal{A}). \quad (18.823)$$

Proof. ✓ Proof The matrix X_{ij}^R is one of the generators of the defining star-algebra closure. □

Theorem 18.10.1.8 (Reconstruction from left coefficients). Lean ✓ Stmt For every $X \in M_{I \times J}(\mathbb{C})$,

$$X = \sum_{r, s \in J} X_{rs}^L \otimes E_{rs}. \quad (18.824)$$

Proof. ✓ Proof At the entry $((i, r), (j, s))$, only the matrix unit E_{rs} contributes, and its coefficient is $X_{rs}^L(i, j) = X((i, r), (j, s))$. □

Theorem 18.10.1.9 (Reconstruction from right coefficients). Lean ✓ Stmt For every $X \in M_{I \times J}(\mathbb{C})$,

$$X = \sum_{i, j \in I} E_{ij} \otimes X_{ij}^R. \quad (18.825)$$

Proof. ✓ Proof At the entry $((i, r), (j, s))$, only the matrix unit E_{ij} contributes, and its coefficient is $X_{ij}^R(r, s) = X((i, r), (j, s))$. □

Theorem 18.10.1.10 (Minimality of the left support algebra). [Lean](#) [✓ Stmt](#) For every star-subalgebra $\mathcal{B} \subseteq M_I(\mathbb{C})$,

$$\text{Spp}_L(\mathcal{A}) \subseteq \mathcal{B} \iff X_{rs}^L \in \mathcal{B} \text{ for every } X \in \mathcal{A} \text{ and } r, s \in I. \quad (18.826)$$

Proof. [✓ Proof](#) Use the universal property of the star-subalgebra generated by $\text{Coeff}_L(\mathcal{A})$. $\square$

Theorem 18.10.1.11 (Minimality of the right support algebra). [Lean](#) [✓ Stmt](#) For every star-subalgebra $\mathcal{C} \subseteq M_J(\mathbb{C})$,

$$\text{Spp}_R(\mathcal{A}) \subseteq \mathcal{C} \iff X_{ij}^R \in \mathcal{C} \text{ for every } X \in \mathcal{A} \text{ and } i, j \in I. \quad (18.827)$$

Proof. [✓ Proof](#) Use the universal property of the star-subalgebra generated by $\text{Coeff}_R(\mathcal{A})$. $\square$

Definition 18.10.1.12 (Kronecker-product submodule). [Lean](#) [✓ Stmt](#) For submodules $S \subseteq M_I(\mathbb{C})$ and $T \subseteq M_J(\mathbb{C})$, define

$$S \boxtimes T = \text{span}_{\mathbb{C}}\{A \otimes B \mid A \in S, B \in T\} \subseteq M_{I \times J}(\mathbb{C}). \quad (18.828)$$

This is the tensor-product subspace used below, rather than a conjunction of two one-sided containment conditions.

Theorem 18.10.1.13 (Coefficient criterion for the Kronecker-product submodule). [Lean](#) [✓ Stmt](#) For $X \in M_{I \times J}(\mathbb{C})$,

$$X \in S \boxtimes T \iff X_{rs}^L \in S \text{ for every } r, s \in J, \quad (18.829)$$

$$\text{and } X_{ij}^R \in T \text{ for every } i, j \in I. \quad (18.830)$$

Proof. [✓ Proof](#) For $A \in S$ and $B \in T$, direct evaluation gives

$$(A \otimes B)_{rs}^L = B_{rs} A, \quad (A \otimes B)_{ij}^R = A_{ij} B. \quad (18.831)$$

Thus the forward implication extends by linearity. Conversely, expand the right blocks in a basis of T . Applying the dual coordinate functionals to those blocks produces matrices in S from the left-slice hypothesis, and the resulting finite sum reconstructs X as a sum of Kronecker products from S and T . $\square$

Theorem 18.10.1.14 (Kronecker-submodule criterion for left support). [Lean](#) [✓ Stmt](#) For every star-subalgebra $\mathcal{B} \subseteq M_I(\mathbb{C})$,

$$\text{Spp}_L(\mathcal{A}) \subseteq \mathcal{B} \iff \mathcal{A} \subseteq \mathcal{B} \boxtimes M_J(\mathbb{C}). \quad (18.832)$$

Proof. [✓ Proof](#) By Theorem 18.10.1.13 with the right submodule the full matrix space, membership in the Kronecker span is exactly membership of every left coefficient in $\mathcal{B}$. Apply Theorem 18.10.1.10. $\square$

Theorem 18.10.1.15 (Kronecker-submodule criterion for right support). [Lean](#) [✓ Stmt](#) For every star-subalgebra $\mathcal{C} \subseteq M_J(\mathbb{C})$,

$$\text{Spp}_R(\mathcal{A}) \subseteq \mathcal{C} \iff \mathcal{A} \subseteq M_I(\mathbb{C}) \boxtimes \mathcal{C}. \quad (18.833)$$

Proof. [✓ Proof](#) The coefficient criterion reduces the right-hand containment to membership of every right block in $\mathcal{C}$. This is equivalent to the left-hand inclusion by Theorem 18.10.1.11. $\square$

Theorem 18.10.1.16 (Two-sided support-algebra containment). [Lean](#) [✓ Stmt](#) Every bipartite star-subalgebra satisfies the literal two-sided containment

$$\mathcal{A} \subseteq \text{Spp}_L(\mathcal{A}) \boxtimes \text{Spp}_R(\mathcal{A}). \quad (18.834)$$

This is the full complex matrix-algebra specialization of the support-algebra containment in Schumacher–Werner Section 4.3 and of Gross–Nesme–Vogts–Werner equation (22). The latter follows after applying Lemma 7 to both tensor factors. The displayed assertion is an inclusion in an actual tensor-product span, not merely the conjunction of the two one-sided inclusions. The MPU appendix applies this construction to $\mathcal{R}_{2x}$ in [15, lines 2320–2334].

Proof. [✓ Proof](#) If $X \in \mathcal{A}$, every left coefficient of X lies in the left support algebra and every right coefficient lies in the right support algebra. Theorem 18.10.1.13 places X in their Kronecker-product submodule. $\square$

Theorem 18.10.1.17 (Left Kronecker commutation is coefficientwise). [Lean](#) [✓ Stmt](#) For $B \in M_I(\mathbb{C})$ and $X \in M_{I \times J}(\mathbb{C})$,

$$[B \otimes \mathbf{1}_J, X] = 0 \iff [B, X_{rs}^L] = 0 \text{ for every } r, s \in J. \quad (18.835)$$

Proof. [✓ Proof](#) For $i, j \in I$ and $r, s \in J$, the two product entries are

$$[(B \otimes \mathbf{1}_J)X]_{(i,r),(j,s)} = \sum_{k \in I} B_{ik} X_{(k,r),(j,s)} = (BX_{rs}^L)_{ij}, \quad (18.836)$$

$$[X(B \otimes \mathbf{1}_J)]_{(i,r),(j,s)} = \sum_{k \in I} X_{(i,r),(k,s)} B_{kj} = (X_{rs}^L B)_{ij}. \quad (18.837)$$

Equality of the two products is therefore equivalent to $[B, X_{rs}^L] = 0$ for every $r, s \in J$. $\square$

Theorem 18.10.1.18 (Right Kronecker commutation is coefficientwise). [Lean](#) [✓ Stmt](#) For $C \in M_J(\mathbb{C})$ and $X \in M_{I \times J}(\mathbb{C})$,

$$[\mathbf{1}_I \otimes C, X] = 0 \iff [C, X_{ij}^R] = 0 \text{ for every } i, j \in I. \quad (18.838)$$

Proof. [✓ Proof](#) For $i, j \in I$ and $r, s \in J$, the two product entries are

$$[(\mathbf{1}_I \otimes C)X]_{(i,r),(j,s)} = \sum_{t \in J} C_{rt} X_{(i,t),(j,s)} = (CX_{ij}^R)_{rs}, \quad (18.839)$$

$$[X(\mathbf{1}_I \otimes C)]_{(i,r),(j,s)} = \sum_{t \in J} X_{(i,r),(j,t)} C_{ts} = (X_{ij}^R C)_{rs}. \quad (18.840)$$

Equality of the two products is therefore equivalent to $[C, X_{ij}^R] = 0$ for every $i, j \in I$. $\square$

Theorem 18.10.1.19 (Left support-algebra centralizer). [Lean](#) [✓ Stmt](#) For $B \in M_I(\mathbb{C})$,

$$B \otimes \mathbf{1}_J \in \mathcal{A}' \iff B \in \text{Spp}_L(\mathcal{A})'. \quad (18.841)$$

Equivalently,

$$\text{Spp}_L(\mathcal{A})' = \{B \in M_I(\mathbb{C}) \mid B \otimes \mathbf{1}_J \in \mathcal{A}'\}. \quad (18.842)$$

Proof. [✓ Proof](#) By Theorem 18.10.1.17, commutation of $B \otimes \mathbf{1}_J$ with every $X \in \mathcal{A}$ is equivalent to commutation of B with every left coefficient. The coefficient set is closed under adjoints because $\mathcal{A}$ is a star-subalgebra. The universal property of the generated star-subalgebra then extends this commutation relation to all of $\text{Spp}_L(\mathcal{A})$; the converse follows from generator membership. $\square$

Theorem 18.10.1.20 (Right support-algebra centralizer). [Lean](#) [Stmt](#) For $C \in M_J(\mathbb{C})$,

$$\mathbf{1}_I \otimes C \in \mathcal{A}' \iff C \in \text{Spp}_R(\mathcal{A})'. \quad (18.843)$$

Equivalently,

$$\text{Spp}_R(\mathcal{A})' = \{C \in M_J(\mathbb{C}) \mid \mathbf{1}_I \otimes C \in \mathcal{A}'\}. \quad (18.844)$$

Proof. [Proof](#) By Theorem 18.10.1.18, commutation of $\mathbf{1}_I \otimes C$ with every $X \in \mathcal{A}$ is equivalent to commutation of C with every right coefficient. Theorem 18.10.1.1 shows that the right coefficient set is closed under adjoints. The universal property of the generated star-subalgebra extends commutation to all of $\text{Spp}_R(\mathcal{A})$; the converse follows from generator membership. $\square$

Theorem 18.10.1.21 (Commutation of overlapping support algebras). [Lean](#) [Stmt](#) Let I, J , and K be finite sets, and let

$$\mathcal{A}_1 \subseteq M_{I \times J}(\mathbb{C}), \quad \mathcal{A}_2 \subseteq M_{J \times K}(\mathbb{C}) \quad (18.845)$$

be star-subalgebras. Suppose that, for every $X \in \mathcal{A}_1$ and $Y \in \mathcal{A}_2$,

$$(X \otimes \mathbf{1}_K)(\mathbf{1}_I \otimes Y) = (\mathbf{1}_I \otimes Y)(X \otimes \mathbf{1}_K). \quad (18.846)$$

Then the two support algebras on the common factor commute:

$$RL = LR \quad \text{for every } R \in \text{Spp}_R(\mathcal{A}_1), \quad L \in \text{Spp}_L(\mathcal{A}_2). \quad (18.847)$$

This is the full complex matrix-algebra specialization of Schumacher–Werner Lemma *sppcomm* [205, lines 1174–1194] and GNVW Lemma *sppcomm* [169, Section 7.1, lines 1221–1246].

The input star-subalgebra must be closed under involution, and the basis element belongs to the third factor; see the notes [140] and [98]. The statement treats full complex matrix algebras, which suffices for the MPU application [15, lines 2300–2306, 2313–2334]; see [65].

Proof. [Proof](#) For $X \in \mathcal{A}_1$, $Y \in \mathcal{A}_2$, $i, i' \in I$, and $k, k' \in K$, applying Theorem [182, “Middle-factor coefficients of commuting overlapping operators”] to (18.846) gives

$$X_{ii'}^R Y_{kk'}^L = Y_{kk'}^L X_{ii'}^R. \quad (18.848)$$

Applying the same identity to $Y^\dagger \in \mathcal{A}_2$, with k and k' exchanged, also gives commutation with $(Y_{kk'}^L)^\dagger$. Hence every generator of $\text{Spp}_R(\mathcal{A}_1)$ lies in the star-subalgebra commuting with all left coefficients of $\mathcal{A}_2$. Passing to the generated star-subalgebra yields

$$\begin{aligned} LR &= RL, \\ L^\dagger R &= RL^\dagger \end{aligned} \quad (18.849)$$

for every $R \in \text{Spp}_R(\mathcal{A}_1)$ and every left coefficient L of $\mathcal{A}_2$.

Since $\text{Spp}_R(\mathcal{A}_1)$ is closed under adjoints, applying (18.849) to $R^\dagger$ shows that each left coefficient commutes with both R and $R^\dagger$. Passing to the star-subalgebra generated by the left coefficients gives (18.847). $\square$

18.10.2 Products of commuting matrix factors

Let $\mathcal{C}$ be a nonzero complex star-algebra and let $\mathcal{A}, \mathcal{B} \subseteq \mathcal{C}$ be unital star-subalgebras such that $ab = ba$ for every $a \in \mathcal{A}$ and $b \in \mathcal{B}$.

Definition 18.10.2.1 (Multiplication of commuting factors). [Lean](#) [Lean](#) [✓ Stmt](#) Multiplication defines a unital star-algebra homomorphism

$$\begin{aligned}\mu_{\mathcal{A}, \mathcal{B}}: \mathcal{A} \otimes_{\mathbb{C}} \mathcal{B} &\longrightarrow \mathcal{C}, \\ \mu_{\mathcal{A}, \mathcal{B}}(a \otimes b) &= ab.\end{aligned}\tag{18.850}$$

Theorem 18.10.2.2 (Algebraic product of commuting factors). [Lean](#) [Lean](#) [✓ Stmt](#) The image of $\mu_{\mathcal{A}, \mathcal{B}}$ is the algebraic join $\mathcal{A} \vee \mathcal{B}$. Its underlying complex vector space is

$$\mathcal{AB} = \text{span}_{\mathbb{C}}\{ab \mid a \in \mathcal{A}, b \in \mathcal{B}\}.\tag{18.851}$$

Thus every element of the join is a finite complex-linear sum of products ab .

Proof. [✓ Proof](#) The involution reverses products. Since $a^* \in \mathcal{A}$ and $b^* \in \mathcal{B}$ commute,

$$\mu_{\mathcal{A}, \mathcal{B}}((a \otimes b)^*) = a^*b^* = b^*a^* = (ab)^*.\tag{18.852}$$

The image contains both factors because $a = a\mathbf{1}$ and $b = \mathbf{1}b$. Conversely, every pure tensor maps to an element of the join, and linearity gives the reverse inclusion. The description of the underlying vector space is the range formula for the linear multiplication map. $\square$

Theorem 18.10.2.3 (Product of two full matrix factors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that $\mathcal{C} = M_I(\mathbb{C})$ for a finite nonempty set I , that $p, q > 0$, and that

$$\begin{aligned}\mathcal{A} &\cong M_p(\mathbb{C}), \\ \mathcal{B} &\cong M_q(\mathbb{C})\end{aligned}\tag{18.853}$$

as star-algebras. Then $\mu_{\mathcal{A}, \mathcal{B}}$ is injective and

$$\begin{aligned}\mathcal{A} \vee \mathcal{B} &\cong M_{pq}(\mathbb{C}), \\ \dim_{\mathbb{C}}(\mathcal{A} \vee \mathcal{B}) &= (pq)^2.\end{aligned}\tag{18.854}$$

This is the unital single-block specialization of Schumacher–Werner Proposition *Cscom* [205, lines 2119–2138], used for the adjacent support algebras in Gross et al. [169, lines 1270–1308]; see [133].

Proof. [✓ Proof](#) The source of the multiplication homomorphism is

$$M_p(\mathbb{C}) \otimes_{\mathbb{C}} M_q(\mathbb{C}) \cong M_{pq}(\mathbb{C}).\tag{18.855}$$

This is a simple ring. The kernel of a unital homomorphism from a simple ring to a nonzero ring is not the whole ring, and is therefore zero. Hence multiplication is injective. Its image is the join by Theorem 18.10.2.2; the matrix presentation and dimension formula in Equation (18.854) follow. $\square$

Theorem 18.10.2.4 (Canonical Kronecker coordinates). [Lean](#) [✓ Stmt](#) Let I, J be finite sets, and let $\mathcal{S} \subseteq M_I(\mathbb{C})$ and $\mathcal{T} \subseteq M_J(\mathbb{C})$ be unital star-subalgebras. In the canonical bipartite coordinates, the underlying complex vector space of the join of $\mathcal{S} \otimes \mathbf{1}_J$ and $\mathbf{1}_I \otimes \mathcal{T}$ is exactly

$$\mathcal{S} \boxtimes \mathcal{T} = \text{span}_{\mathbb{C}}\{S \otimes T \mid S \in \mathcal{S}, T \in \mathcal{T}\}.\tag{18.856}$$

Proof. ✓ Proof If I or J is empty, the ambient matrix space is zero, so both subspaces in the claimed equality are zero. Assume that both sets are nonempty. The two canonical factors commute, and their products satisfy

$$(S \otimes \mathbf{1}_J)(\mathbf{1}_I \otimes T) = S \otimes T. \quad (18.857)$$

Equation (18.851) therefore becomes exactly (18.856). □

Definition 18.10.2.5 (Translation of a finite region). Lean Lean Lean Lean Lean Lean ✓ Stmt
 For a finite region $\Lambda \subset \mathbb{Z}$ and $a \in \mathbb{Z}$, define

$$\Lambda + a = \{i + a \mid i \in \Lambda\}. \quad (18.858)$$

Thus

$$j \in \Lambda + a \iff j - a \in \Lambda, \quad (18.859)$$

$$\Lambda + 0 = \Lambda, \quad (18.860)$$

$$(\Lambda + a) + b = \Lambda + (a + b). \quad (18.861)$$

If $\Lambda \subseteq \Gamma$, then $\Lambda + a \subseteq \Gamma + a$.

Definition 18.10.2.6 (Sumset of finite regions). Lean Lean Lean Lean Lean Lean Lean Lean
Lean ✓ Stmt For finite regions $\Lambda, \mathcal{N} \subset \mathbb{Z}$, define their sumset by

$$\Lambda + \mathcal{N} = \{i + a \mid i \in \Lambda, a \in \mathcal{N}\}. \quad (18.862)$$

Hence $j \in \Lambda + \mathcal{N}$ exactly when $j = i + a$ for some $i \in \Lambda$ and $a \in \mathcal{N}$. The sumset is monotone in both arguments, and the singleton $\{0\}$ is an identity. A singleton displacement recovers translation,

$$\Lambda + \{a\} = \Lambda + a. \quad (18.863)$$

Sumsets are associative:

$$(\Lambda + \mathcal{N}) + \mathcal{M} = \Lambda + (\mathcal{N} + \mathcal{M}). \quad (18.864)$$

These identities support composition of propagation bounds. This is the finite region $\Lambda + \mathcal{N}$ in the propagation condition of the QCA appendix of [15].

Definition 18.10.2.7 (Translation of sites and configurations). Lean Lean Lean Lean Lean
✓ Stmt Translation gives bijections

$$t_{a,\Lambda} : \Lambda \longrightarrow \Lambda + a, \quad t_{a,\Lambda}(i) = i + a, \quad (18.865)$$

$$T_{a,\Lambda} : \text{Config}_d(\Lambda) \longrightarrow \text{Config}_d(\Lambda + a), \quad (T_{a,\Lambda}x)(j) = x(j - a). \quad (18.866)$$

Their inverses translate sites and configurations by $-a$.

Definition 18.10.2.8 (Finite-region translation of local observables). Lean Lean ✓ Stmt Relabelling configurations defines a star-algebra equivalence

$$\tau_{a,\Lambda} : \mathcal{A}_\Lambda \longrightarrow \mathcal{A}_{\Lambda+a}, \quad (18.867)$$

$$\tau_{a,\Lambda}(A)(x, y) = A(T_{a,\Lambda}^{-1}x, T_{a,\Lambda}^{-1}y). \quad (18.868)$$

This is the finite-region translation associated with the lattice translation $i \mapsto i + a$.

Lemma 18.10.2.9 (Norm preservation of finite-region translation). Lean Lean ✓ Stmt Let d be a nonnegative integer, let $\Lambda \subset \mathbb{Z}$ be finite, and let $a \in \mathbb{Z}$. For every $A \in \mathcal{A}_\Lambda$,

$$\|\tau_{a,\Lambda}(A)\| = \|A\|. \quad (18.869)$$

Consequently, $\tau_{a,\Lambda}$ is an isometry from $\mathcal{A}_\Lambda$ onto $\mathcal{A}_{\Lambda+a}$.

Proof. ✓ Proof Translation bijectively relabels the configuration basis. The resulting star-algebra equivalence between the two finite matrix algebras preserves the operator norm. Applying this identity to $A - B$ and using linearity gives

$$\|\tau_{a,\Lambda}(A) - \tau_{a,\Lambda}(B)\| = \|\tau_{a,\Lambda}(A - B)\| = \|A - B\|, \quad (18.870)$$

so distances are preserved. □

Lemma 18.10.2.10 (Finite-region translation action laws). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Translation by zero acts trivially on sites, configurations, and finite-region observables. Successive translations by a and b agree with translation by $a + b$. Pointwise, for $i \in \Lambda$, $x, y \in \text{Config}_d(\Lambda)$, and $A \in \mathcal{A}_\Lambda$,

$$t_{0,\Lambda}(i) = i, \quad (18.871)$$

$$t_{b,\Lambda+a}(t_{a,\Lambda}(i)) = t_{a+b,\Lambda}(i), \quad (18.872)$$

$$(T_{a,\Lambda}x)(t_{a,\Lambda}(i)) = x(i), \quad (18.873)$$

$$(T_{b,\Lambda+a}T_{a,\Lambda}x)(t_{b,\Lambda+a}(t_{a,\Lambda}(i))) = (T_{a+b,\Lambda}x)(t_{a+b,\Lambda}(i)), \quad (18.874)$$

$$(\tau_{a,\Lambda}(A))(T_{a,\Lambda}x, T_{a,\Lambda}y) = A(x, y), \quad (18.875)$$

$$(\tau_{b,\Lambda+a}\tau_{a,\Lambda}(A))(T_{b,\Lambda+a}T_{a,\Lambda}x, T_{b,\Lambda+a}T_{a,\Lambda}y) = (\tau_{a+b,\Lambda}(A))(T_{a+b,\Lambda}x, T_{a+b,\Lambda}y). \quad (18.876)$$

These identities express the additive action without identifying the distinct configuration spaces before applying their canonical translation bijections.

Proof. ✓ Proof The site identities follow from

$$(i + a) + b = i + (a + b). \quad (18.877)$$

Relabelling a configuration is precomposition by the inverse site translation, so evaluation at the corresponding translated site gives

$$(T_{a,\Lambda}x)(t_{a,\Lambda}(i)) = x(i). \quad (18.878)$$

Applying this identity successively proves the configuration action laws. Finally, the local observable translation relabels both matrix indices by $T_{a,\Lambda}^{-1}$; evaluating on translated configurations therefore gives $A(x, y)$, and applying this entrywise identity successively proves the local observable action laws. □

Lemma 18.10.2.11 (Naturality of finite-region translation). Lean ✓ Stmt If $\Lambda \subseteq \Gamma$, then for every $A \in \mathcal{A}_\Lambda$,

$$\tau_{a,\Gamma}(\iota_{\Lambda,\Gamma}(A)) = \iota_{\Lambda+a,\Gamma+a}(\tau_{a,\Lambda}(A)). \quad (18.879)$$

Proof. ✓ Proof The lattice translation identifies $\Gamma \setminus \Lambda$ with $(\Gamma + a) \setminus (\Lambda + a)$. Hence relabelling configurations carries the identity on the first complement to the identity on the second complement, and

$$\tau_{a,\Gamma}(A \otimes \mathbf{1}_{\Gamma \setminus \Lambda}) = \tau_{a,\Lambda}(A) \otimes \mathbf{1}_{(\Gamma+a) \setminus (\Lambda+a)}. \quad (18.880)$$

□

Definition 18.10.2.12 (Algebraic local algebra). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The finite-region local algebras form a directed system under the maps $\iota_{\Lambda, \Gamma}$. Its algebraic direct limit is the algebraic local algebra

$$\mathcal{A}_{\text{loc}} = \lim_{\substack{\longrightarrow \\ \Lambda \in \mathbb{Z}}} \mathcal{A}_{\Lambda}. \quad (18.881)$$

For every finite region Λ , write $\iota_{\Lambda} : \mathcal{A}_{\Lambda} \rightarrow \mathcal{A}_{\text{loc}}$ for the canonical inclusion. Whenever $\Lambda \subseteq \Gamma$, these inclusions satisfy

$$\iota_{\Gamma} \circ \iota_{\Lambda, \Gamma} = \iota_{\Lambda}. \quad (18.882)$$

This is the union $\bigcup_{\Lambda \in \mathbb{Z}} \mathcal{A}_{\Lambda}$ in the Appendix of [15]. No norm completion is taken.

Definition 18.10.2.13 (Finite-chain realization of a periodic MPU). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $a \in \mathbb{Z}$ and $N \in \mathbb{N}$, let $I_{a, N} = [a, a + N) \cap \mathbb{Z}$. The increasing bijection $\{0, \dots, N-1\} \simeq I_{a, N}$ identifies the periodic configuration basis with $\text{Config}_d(I_{a, N})$. In this basis, write $U_a^{(N)}$ for the matrix obtained by simultaneously reindexing the rows and columns of $U^{(N)}$.

Definition 18.10.2.14 (Residual words on an anchored interval). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix a positive block length L and an integer anchor a . The number of sites from a to the next globally aligned block boundary is $q(a, L) = (-a) \bmod L$. If an interval of length N has the decomposition $N = p + mL + s$, its configuration may be specified by a prefix word of length p , a word of m blocked letters, and a suffix word of length s . Their ordered concatenation defines a configuration on $I_{a, N}$, including when $m = 0$. This concatenation is a bijection from triples of such words to configurations on the interval.

Theorem 18.10.2.15 (Decomposition of an arbitrary anchored interval). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $a \in \mathbb{Z}$, $N \in \mathbb{N}$, and $L > 0$, there are $p, m, s \in \mathbb{N}$ such that

$$p, s < L, \quad N = p + mL + s. \quad (18.883)$$

If $m > 0$, then $a + p$ is divisible by L ; the site at residue r of the t th complete middle block has integer coordinate $a + p + tL + r$. If $N < q(a, L)$, then $p = N$ and $m = s = 0$. The anchored configuration reads the prefix, middle blocks, and suffix in exactly this increasing site order.

Proof. [✓ Proof](#) Set $p = \min(N, q(a, L))$ and divide $N - p$ by L , taking its quotient as m and its remainder as s . Euclidean division gives $s < L$ and the length identity. If $m > 0$, then $p = q(a, L)$ and $a + p$ is aligned. The order-preserving site identification sends position k to $a + k$; applying it at $k = p + tL + r$ gives the middle-block coordinate. When $N < q(a, L)$, one has $p = N$ and both other lengths vanish. $\square$

Theorem 18.10.2.16 (Finite-chain MPO entries with residual sites). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let U be any MPO tensor and let $N = p + mL + s$. For two anchored configurations with residual prefix words i^-, j^- , blocked middle words σ, τ , and residual suffix words i^+, j^+ , the entry of the finite-chain matrix is

$$(U_a^{(N)})_{i^-, j^-} = \text{tr}(U^{i^-, j^-} (U^{[L]})^{\sigma, \tau} U^{i^+, j^+}). \quad (18.884)$$

This holds at every integer anchor and for arbitrary L, p, m, s , including an empty middle word. Neither unitarity nor divisibility of N by L is assumed.

Proof. [✓ Proof](#) Undo the increasing reindexing of the finite-chain MPO matrix. The resulting periodic MPO entry is the trace of the prefix, blocked middle, and suffix products by residual-site blocking. $\square$

Definition 18.10.2.17 (Finite-chain MPU conjugation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $\mathcal{U}$ is an MPU and $N > 1$, then $U_a^{(N)}$ is unitary and defines the star-algebra automorphism

$$\text{Ad}_{U_a^{(N)}}(B) = U_a^{(N)} B U_a^{(N)\dagger}. \quad (18.885)$$

The factor order is the one in [15, lines 2300–2306].

Proof. [✓ Proof](#) Simultaneous reindexing preserves both unitarity equations for $U^{(N)}$. Conjugation by the resulting unitary preserves multiplication, the unit, complex scalars, and the adjoint. $\square$

Theorem 18.10.2.18 (Conjugation coefficients in residual-word coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) Let U be an MPU, $N = p + mL + s > 1$, and $A \in \mathcal{A}_\Lambda$ for $\Lambda \subseteq I_{a,N}$. Write $\hat{x}$ for the anchored configuration represented by a triple $x = (x^-, x^0, x^+)$ of residual prefix, complete-block, and residual suffix words, and put

$$T(x, y) = \text{tr}(U^{x^-, y^-} (U^{[L]})^{x^0, y^0} U^{x^+, y^+}). \quad (18.886)$$

For all such triples x, y , the matrix coefficient of finite-chain conjugation is

$$(U_a^{(N)} \iota_{\Lambda, I_{a,N}}(A) U_a^{(N)\dagger})_{\hat{x}, \hat{y}} = \sum_{z, w} T(x, z) A_{z|_\Lambda, \hat{w}|_\Lambda} \cdot \delta_{z|_{I_{a,N} \setminus \Lambda}, \hat{w}|_{I_{a,N} \setminus \Lambda}} \overline{T(y, w)}. \quad (18.887)$$

Both MPO factors retain their residual prefix and suffix products. This finite-chain identity does not assert independence of the interval or support of the conjugated observable.

Proof. [✓ Proof](#) Expand conjugation as a double sum over intermediate configurations. The local inclusion contributes the matrix entry of A on Λ and the equality condition on its complement. Reindex both sums by the residual-word bijection and apply the finite-chain residual-trace identity to each MPO coefficient. $\square$

Definition 18.10.2.19 (Three-block coordinates on an anchored finite chain). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix $k, p, s \geq 0$ and an integer anchor a . On the interval of length $N = p + 3(2k) + s$, distinguish three consecutive complete blocks of length $2k$ between an arbitrary prefix and suffix. Let $H = d^k$. On the output side, the four neighboring half-blocks form a factor X of dimension H^4 ; the prefix, suffix, and two exposed outer half-blocks form the exterior factor E_{out} . On the input side, the central complete block forms a factor J of dimension H^2 ; the prefix, suffix, and the two outer complete blocks form E_{in} . Ordered concatenation gives bijections from $E_{\text{out}} \times X$ and $E_{\text{in}} \times J$ to the respective configuration bases of the same finite interval. Write $M_{a;p,k,s}$ for the finite-chain MPO matrix in these two bases.

Theorem 18.10.2.20 (Anchored three-block operator identity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the two-site blocking of U at length k is supplied with standard-form factors u, v . For every observable A on the central input block, let $B(A)$ be the corresponding observable on the two neighboring output links. Then

$$M_{a;p,k,s}(I_{E_{\text{in}}} \otimes A) = (I_{E_{\text{out}}} \otimes B(A)) M_{a;p,k,s}. \quad (18.888)$$

This identity holds for arbitrary original-site prefix and suffix words; U need not be an MPU. If U is an MPU and $N > 1$, then

$$M_{a;p,k,s}(I_{E_{\text{in}}} \otimes A)M_{a;p,k,s}^\dagger = I_{E_{\text{out}}} \otimes B(A). \quad (18.889)$$

Thus the conjugated operator is the identity on every exterior output coordinate. This statement assumes the displayed supplied standard form and does not yet identify the right-hand side with a site-indexed local inclusion or prove independence of the enclosing interval.

Proof. ✓ Proof Fixing all exterior input and output coordinates makes a slice of $M_{a;p,k,s}$ equal to the original-site three-block window. The open-word identity intertwines each slice with the same $B(A)$. Summing over exterior coordinates yields (18.888). When U is an MPU, $M_{a;p,k,s}$ is unitary; postmultiplying (18.888) by $M_{a;p,k,s}^\dagger$ yields (18.889). $\square$

Definition 18.10.2.21 (Finite-chain term on a local observable). Lean Lean ✓ Stmt Let $\Lambda \subseteq I_{a,N}$ and $A \in \mathcal{A}_\Lambda$. Set $A_L = \iota_{\Lambda, I_{a,N}}(A)$. For $N > 1$, the finite-chain term in [15, lines 2300–2306] is the star-algebra map

$$\mathcal{A}_\Lambda \longrightarrow \mathcal{A}_{\text{loc}}, \quad (18.890)$$

$$A \mapsto \iota_{I_{a,N}}(U_a^{(N)} A_L U_a^{(N)\dagger}). \quad (18.891)$$

No stabilization in N is asserted.

Definition 18.10.2.22 (Operator norm on the local algebra). Lean Lean Lean Lean ✓ Stmt Assume $d > 0$. If a local observable $A \in \mathcal{A}_{\text{loc}}$ is represented by $A_\Lambda \in \mathcal{A}_\Lambda$ on a finite region Λ , define its operator norm by

$$\|A\| = \|A_\Lambda\|. \quad (18.892)$$

This definition is independent of the finite region and chosen representative, giving the norm completed in the Appendix of [15] to obtain the quasi-local algebra.

Proof. ✓ Proof If A_Λ and A_Γ represent the same local observable, then the defining relation for the inductive limit gives a finite region $\Delta \supseteq \Lambda \cup \Gamma$ such that

$$\iota_{\Lambda, \Delta}(A_\Lambda) = \iota_{\Gamma, \Delta}(A_\Gamma). \quad (18.893)$$

Both inclusions preserve the operator norm, so $\|A_\Lambda\| = \|A_\Gamma\|$. $\square$

Definition 18.10.2.23 (Normed star algebra of local observables). Lean Lean Lean ✓ Stmt For $d > 0$, the local algebra with the operator norm is a normed complex star algebra. No completeness assertion is made.

Proof. ✓ Proof Choose finite-region representatives of the local observables and enlarge their regions to a common finite region. The norm axioms, the product inequality, and compatibility with complex scalar multiplication then follow from the corresponding properties of the finite-region matrix algebra. $\square$

Lemma 18.10.2.24 (C-star identity for local observables). Lean ✓ Stmt For $d > 0$ and every local observable A ,

$$\|A^*A\| = \|A\|^2. \quad (18.894)$$

Proof. ✓ Proof Choose a finite region Λ and a representative $A_\Lambda \in \mathcal{A}_\Lambda$. Then

$$\|A^*A\| = \|A_\Lambda^*A_\Lambda\| = \|A_\Lambda\|^2 = \|A\|^2. \quad (18.895)$$

□

Definition 18.10.2.25 (C-star axiom structure on local observables). Lean ✓ Stmt The normed star algebra of local observables satisfies the C^* -norm identity.

Proof. ✓ Proof The C-star identity gives the required norm inequality in the reverse direction; submultiplicativity gives the other direction. □

Lemma 18.10.2.26 (Local observables embed isometrically). Lean ✓ Stmt For $d > 0$ and every finite region Λ , the canonical map

$$\iota_\Lambda : \mathcal{A}_\Lambda \longrightarrow \mathcal{A}_{\text{loc}} \quad (18.896)$$

is an operator-norm isometry.

Proof. ✓ Proof The norm of $\iota_\Lambda(A)$ is defined to be the operator norm of its representative $A \in \mathcal{A}_\Lambda$. Hence $\|\iota_\Lambda(A)\| = \|A\|$; applying this equality to differences proves preservation of distance. □

Theorem 18.10.2.27 (Star-algebra universal property). Lean Lean Lean Lean ✓ Stmt Let B be a complex star algebra. Suppose that for every finite region Λ there is a star-algebra homomorphism $g_\Lambda : \mathcal{A}_\Lambda \rightarrow B$. Whenever $\Lambda \subseteq \Gamma$, suppose that

$$g_\Gamma \circ \iota_{\Lambda, \Gamma} = g_\Lambda. \quad (18.897)$$

Then there is a unique star-algebra homomorphism $g : \mathcal{A}_{\text{loc}} \rightarrow B$ satisfying $g \circ \iota_\Lambda = g_\Lambda$ for every finite region Λ .

Proof. ✓ Proof Define g on a local observable represented by $A \in \mathcal{A}_\Lambda$ as $g_\Lambda(A)$. Compatibility makes this independent of the representing region. Since each g_Λ preserves addition, multiplication, the unit, scalar multiplication, and the star operation, so does g . Every element of the direct limit has a finite-region representative, which also proves uniqueness. □

Definition 18.10.2.28 (Support of a local observable). Lean ✓ Stmt A local observable $A \in \mathcal{A}_{\text{loc}}$ is supported in the finite region Λ if it lies in the image of $\iota_\Lambda : \mathcal{A}_\Lambda \rightarrow \mathcal{A}_{\text{loc}}$.

Theorem 18.10.2.29 (Finite supports of local observables). Lean Lean Lean Lean Lean Lean ✓ Stmt Every local observable is supported in some finite region. If A is supported in Λ and $\Lambda \subseteq \Gamma$, then A is supported in Γ . If A and B are supported in Λ and Γ , respectively, then $A + B$ and AB are supported in $\Lambda \cup \Gamma$. The adjoint A^* is supported in Λ , and both 0 and 1 are supported in every finite region. For positive on-site dimension, each map ι_Λ is injective.

Proof. ✓ Proof Enlarge representatives by tensoring with the identity on the added sites. Two representatives can therefore be moved to the finite union of their regions, where sums and products are formed. The canonical inclusions are unital star-algebra homomorphisms, so they preserve the adjoint, zero, and the unit. For $d > 0$, every transition map between finite-region algebras is injective; the canonical map from each member of such a directed system to its direct limit is therefore injective. Finally, every element of an algebraic direct limit has a representative in one member of the directed system. □

Theorem 18.10.2.30 (Disjointly supported local observables commute). [Lean](#) [Lean](#) [✓ Stmt](#) If $X, Y \in \mathcal{A}_{\text{loc}}$ are supported in finite regions Λ and Γ with $\Lambda \cap \Gamma = \emptyset$, then

$$XY = YX. \quad (18.898)$$

In particular, for $A \in \mathcal{A}_\Lambda$ and $B \in \mathcal{A}_\Gamma$,

$$\iota_\Lambda(A)\iota_\Gamma(B) = \iota_\Gamma(B)\iota_\Lambda(A). \quad (18.899)$$

This is the algebraic-local form of the tensor-factor identity underlying the construction in [15, lines 2292–2300], not a named theorem of the appendix.

Proof. [✓ Proof](#) Choose representatives $X = \iota_\Lambda(A)$ and $Y = \iota_\Gamma(B)$. Embed both into $\mathcal{A}_{\Lambda \cup \Gamma}$. Theorem 18.10.8 gives

$$XY = \iota_{\Lambda \cup \Gamma}(\iota_{\Lambda, \Lambda \cup \Gamma}(A)\iota_{\Gamma, \Lambda \cup \Gamma}(B)) \quad (18.900)$$

$$= \iota_{\Lambda \cup \Gamma}(\iota_{\Gamma, \Lambda \cup \Gamma}(B)\iota_{\Lambda, \Lambda \cup \Gamma}(A)) = YX. \quad (18.901)$$

□

Theorem 18.10.2.31 (Algebraic-local relative commutant). [Lean](#) [✓ Stmt](#) Assume $d > 0$. A local observable $X \in \mathcal{A}_{\text{loc}}$ is supported in a finite region Λ if and only if it commutes with every local observable having a finite support disjoint from Λ :

$$X \in \iota_\Lambda(\mathcal{A}_\Lambda) \iff \forall \Gamma, \forall Y, (\Gamma \cap \Lambda = \emptyset \wedge Y \in \iota_\Gamma(\mathcal{A}_\Gamma)) \implies XY = YX. \quad (18.902)$$

This characterizes locally supported observables on the inductive limit algebra.

Proof. [✓ Proof](#) The forward implication is commutation of disjointly supported observables. Conversely, choose a finite support Δ of X and represent X by $A \in \mathcal{A}_{\Lambda \cup \Delta}$. The hypothesis says that A commutes with every observable on $(\Lambda \cup \Delta) \setminus \Lambda$. Since $d > 0$, the canonical map from $\mathcal{A}_{\Lambda \cup \Delta}$ into $\mathcal{A}_{\text{loc}}$ is injective, so this commutation already holds in the finite-region algebra. The finite relative-commutant theorem gives $A = \iota_{\Lambda, \Lambda \cup \Delta}(A_\Lambda)$ for some $A_\Lambda \in \mathcal{A}_\Lambda$. Hence $X = \iota_\Lambda(A_\Lambda)$. □

Definition 18.10.2.32 (Algebraic-local translation homomorphism). [Lean](#) [Lean](#) [✓ Stmt](#) For $a \in \mathbb{Z}$, finite-region translations induce a star-algebra homomorphism

$$\tau_a : \mathcal{A}_{\text{loc}} \longrightarrow \mathcal{A}_{\text{loc}} \quad (18.903)$$

determined on every finite-region representative by

$$\tau_a(\iota_\Lambda(A)) = \iota_{\Lambda+a}(\tau_{a,\Lambda}(A)). \quad (18.904)$$

Proof. [✓ Proof](#) Naturality gives, for $\Lambda \subseteq \Gamma$,

$$\iota_{\Gamma+a}(\tau_{a,\Gamma}(\iota_{\Lambda,\Gamma}(A))) = \iota_{\Gamma+a}(\iota_{\Lambda+a,\Gamma+a}(\tau_{a,\Lambda}(A))) \quad (18.905)$$

$$= \iota_{\Lambda+a}(\tau_{a,\Lambda}(A)). \quad (18.906)$$

Hence the finite-region maps form a compatible family and induce the stated homomorphism on the algebraic direct limit. □

Lemma 18.10.2.33 (Action laws for algebraic-local translation homomorphisms). [Lean](#) [Lean](#) [✓ Stmt](#) The homomorphisms satisfy

$$\tau_0 = \text{id}, \quad (18.907)$$

$$\tau_b \circ \tau_a = \tau_{a+b}. \quad (18.908)$$

Proof. ✓ Proof On a representative $A \in \mathcal{A}_\Lambda$, the two sides of the composition law are represented on the equal finite regions $(\Lambda + a) + b$ and $\Lambda + (a + b)$. The finite-region action identity therefore gives

$$\tau_b(\tau_a(\iota_\Lambda(A))) = \iota_{\Lambda+(a+b)}(\tau_{a+b,\Lambda}(A)) = \tau_{a+b}(\iota_\Lambda(A)). \quad (18.909)$$

The zero law is proved in the same way. Equality on all finite-region representatives determines equality on the algebraic direct limit. $\square$

Definition 18.10.2.34 (Translation of algebraic-local observables). Lean Lean Lean ✓ Stmt For every $a \in \mathbb{Z}$, the homomorphism τ_a is a star-algebra automorphism of $\mathcal{A}_{\text{loc}}$, and on finite-region representatives

$$\tau_a(\iota_\Lambda(A)) = \iota_{\Lambda+a}(\tau_{a,\Lambda}(A)). \quad (18.910)$$

Proof. ✓ Proof The homomorphism action laws give

$$\tau_{-a} \circ \tau_a = \tau_0 = \text{id}, \quad (18.911)$$

$$\tau_a \circ \tau_{-a} = \tau_0 = \text{id}. \quad (18.912)$$

Thus τ_a is bijective with inverse τ_{-a} . The formula on finite-region representatives is the defining formula for the homomorphism. $\square$

Lemma 18.10.2.35 (Algebraic-local translation action laws). Lean Lean Lean ✓ Stmt The automorphisms form an additive action of $\mathbb{Z}$:

$$\tau_0 = \text{id}, \quad (18.913)$$

$$\tau_b \circ \tau_a = \tau_{a+b}, \quad (18.914)$$

$$\tau_a^{-1} = \tau_{-a}. \quad (18.915)$$

Proof. ✓ Proof The first two identities are the homomorphism action laws expressed as identities of star-algebra automorphisms. Taking $b = -a$ in the composition law gives

$$\tau_{-a} \circ \tau_a = \tau_0 = \text{id}, \quad (18.916)$$

$$\tau_a \circ \tau_{-a} = \tau_0 = \text{id}, \quad (18.917)$$

which is the inverse identity. $\square$

Lemma 18.10.2.36 (Exact translation of algebraic-local supports). Lean Lean ✓ Stmt A local observable A is supported in a finite region Λ if and only if $\tau_a(A)$ is supported in $\Lambda + a$.

Proof. ✓ Proof If $A = \iota_\Lambda(A_\Lambda)$, then

$$\tau_a(A) = \iota_{\Lambda+a}(\tau_{a,\Lambda}(A_\Lambda)), \quad (18.918)$$

which proves the forward implication. Apply the same argument to $\tau_a(A)$ with displacement $-a$, and use $\tau_{-a}\tau_a = \text{id}$ and $(\Lambda + a) - a = \Lambda$, for the reverse implication. $\square$

Lemma 18.10.2.37 (Norm isometry of algebraic-local translation). Lean Lean ✓ Stmt If $d > 0$, then every translation preserves the operator norm and distance:

$$\|\tau_a(A)\| = \|A\|, \quad (18.919)$$

$$\|\tau_a(A) - \tau_a(B)\| = \|A - B\|. \quad (18.920)$$

Proof. ✓ Proof Choose $A_\Lambda \in \mathcal{A}_\Lambda$ with $A = \iota_\Lambda(A_\Lambda)$. Finite-region translation is a star-algebra equivalence of matrix algebras, hence preserves the operator norm. Therefore

$$\|\tau_a(A)\| = \|\tau_{a,\Lambda}(A_\Lambda)\| = \|A_\Lambda\| = \|A\|. \quad (18.921)$$

Applying this equality to $A - B$ gives distance preservation. □

Definition 18.10.2.38 (C-star structure on a completion). Lean Lean Lean Lean Lean Lean ✓ Stmt The norm completion of a complex normed star algebra satisfying the C-star identity carries the continuously extended involution and a C-star algebra structure. The canonical map into the completion is a star-algebra homomorphism.

Proof. ✓ Proof Extend the isometric involution continuously. The algebraic identities and the C-star identity hold on the dense canonical image and hence on the completion. □

Definition 18.10.2.39 (Quasi-local observable algebra). Lean Lean ✓ Stmt For $d > 0$, the quasi-local observable algebra is the operator-norm completion

$$\mathcal{A} = \overline{\mathcal{A}_{\text{loc}}}^{\|\cdot\|}. \quad (18.922)$$

The involution on $\mathcal{A}$ is the unique continuous extension of the involution on $\mathcal{A}_{\text{loc}}$. It makes $\mathcal{A}$ a complex C-star algebra, and the canonical star-algebra homomorphism

$$j: \mathcal{A}_{\text{loc}} \longrightarrow \mathcal{A} \quad (18.923)$$

is the completion map.

Proof. ✓ Proof The involution is isometric on $\mathcal{A}_{\text{loc}}$, hence extends continuously to the completion. The identities $(A + B)^* = A^* + B^*$, $(AB)^* = B^*A^*$, and $(A^*)^* = A$ hold on the dense subalgebra $j(\mathcal{A}_{\text{loc}})$ and therefore on $\mathcal{A}$. The conjugate-linear identity $(cA)^* = \bar{c}A^*$ extends by the same argument. Density also extends

$$\|A^*A\| = \|A\|^2, \quad (18.924)$$

so the completed algebra satisfies the C-star identity. □

Definition 18.10.2.40 (Finite-region quasi-local observable). Lean ✓ Stmt For every finite region Λ , define

$$j_\Lambda: \mathcal{A}_\Lambda \longrightarrow \mathcal{A}, \quad j_\Lambda = j \circ \iota_\Lambda. \quad (18.925)$$

Lemma 18.10.2.41 (Finite-region observables in the quasi-local algebra). Lean Lean Lean Lean ✓ Stmt For every finite region Λ , the map j_Λ is an injective isometry. If $\Lambda \subseteq \Gamma$, then

$$j_\Gamma \circ \iota_{\Lambda,\Gamma} = j_\Lambda. \quad (18.926)$$

Proof. ✓ Proof The algebraic inclusion ι_Λ and the completion map j both preserve the norm. Their composite is therefore an isometry and hence injective. Compatibility follows from $\iota_\Gamma \circ \iota_{\Lambda,\Gamma} = \iota_\Lambda$. □

Theorem 18.10.2.42 (Density of the canonical completion map). Lean ✓ Stmt The canonical star-algebra homomorphism from a normed star algebra into its completion has dense range.

Proof. ✓ Proof This is the defining density property of the completion. □

Theorem 18.10.2.43 (Density of finite-region observables). [Lean](#) [Lean](#) [✓ Stmt](#) Algebraic local observables are dense in the quasi-local algebra. Equivalently,

$$\overline{j(\mathcal{A}_{\text{loc}})} = \overline{\bigcup_{\Lambda \in \mathbb{Z}} j_{\Lambda}(\mathcal{A}_{\Lambda})} = \mathcal{A}. \quad (18.927)$$

Proof. [✓ Proof](#) Density of $j(\mathcal{A}_{\text{loc}})$ is the defining density property of the completion. Every algebraic local observable has a representative in some finite region, so

$$j(\mathcal{A}_{\text{loc}}) = \bigcup_{\Lambda \in \mathbb{Z}} j_{\Lambda}(\mathcal{A}_{\Lambda}), \quad (18.928)$$

and taking closures yields

$$\overline{\bigcup_{\Lambda \in \mathbb{Z}} j_{\Lambda}(\mathcal{A}_{\Lambda})} = \overline{j(\mathcal{A}_{\text{loc}})} = \mathcal{A}. \quad (18.929)$$

□

Lattice translation acts covariantly on the quasi-local algebra in the setting of [15, lines 2292–2306].

Definition 18.10.2.44 (Extension of star-algebra equivalences to completions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be complex normed star algebras, and let $f: A \rightarrow B$ be a star-algebra equivalence. If f and f^{-1} are continuous, then f extends to a star-algebra equivalence

$$\widehat{f}: \widehat{A} \longrightarrow \widehat{B} \quad (18.930)$$

between their norm completions. Its underlying map is the continuous extension of f . Writing j_A and j_B for the canonical maps into the completions, the extension satisfies

$$\widehat{f}(j_A(x)) = j_B(f(x)). \quad (18.931)$$

Proof. [✓ Proof](#) Apply the completion functor to f and f^{-1} . On the dense canonical images their composites satisfy

$$\widehat{f^{-1}}(\widehat{f}(j_A(x))) = j_A(x), \quad \widehat{f}(\widehat{f^{-1}}(j_B(y))) = j_B(y). \quad (18.932)$$

Continuity gives the two inverse identities on the completions. Multiplication, addition, scalar multiplication, and the star operation commute with the extension because the corresponding identities hold on the canonical images and both sides are continuous. □

Definition 18.10.2.45 (Translation of quasi-local observables). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $d > 0$ and every $a \in \mathbb{Z}$, algebraic-local translation extends to a star-algebra automorphism

$$\tau_a: \mathcal{A} \longrightarrow \mathcal{A} \quad (18.933)$$

of the quasi-local algebra. On the dense algebraic-local subalgebra and on finite-region observables it satisfies

$$\tau_a(j(A)) = j(\tau_a(A)), \quad (18.934)$$

$$\tau_a(j_{\Lambda}(A_{\Lambda})) = j_{\Lambda+a}(\tau_{a,\Lambda}(A_{\Lambda})). \quad (18.935)$$

Proof. [✓ Proof](#) Algebraic-local translation and its inverse translation by $-a$ are operator-norm isometries, hence continuous. Extend this equivalence to the norm completion to obtain (18.934). Substituting $A = \iota_{\Lambda}(A_{\Lambda})$ and applying the finite-region translation formula yields (18.935). □

Lemma 18.10.2.46 (Quasi-local translation action laws). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $d > 0$, the quasi-local automorphisms form an additive action of $\mathbb{Z}$:

$$\tau_0 = \text{id}, \quad (18.936)$$

$$\tau_b \circ \tau_a = \tau_{a+b}, \quad (18.937)$$

$$\tau_a^{-1} = \tau_{-a}. \quad (18.938)$$

Proof. [✓ Proof](#) On the canonical image of the algebraic-local algebra,

$$\tau_b(\tau_a(j(A))) = j(\tau_b(\tau_a(A))) = j(\tau_{a+b}(A)), \quad (18.939)$$

$$\tau_0(j(A)) = j(A). \quad (18.940)$$

Continuity extends these identities to the completion. Taking $(a, b) = (a, -a)$ and $(-a, a)$ identifies the inverse of τ_a with τ_{-a} . $\square$

Definition 18.10.2.47 (Translation covariance). [Lean](#) [✓ Stmt](#) For $d > 0$, a star-algebra automorphism ω of the quasi-local algebra $\mathcal{A}$ is translation covariant if

$$\omega \circ \tau_a = \tau_a \circ \omega \quad (18.941)$$

for every $a \in \mathbb{Z}$. This is the covariance condition in the definition of a one-dimensional QCA [15, line 2298].

Theorem 18.10.2.48 (Pointwise characterization of translation covariance). [Lean](#) [✓ Stmt](#) For $d > 0$, a star-algebra automorphism ω of $\mathcal{A}$ is translation covariant if and only if

$$\omega(\tau_a(A)) = \tau_a(\omega(A)) \quad (18.942)$$

for every $a \in \mathbb{Z}$ and every $A \in \mathcal{A}$.

Proof. [✓ Proof](#) Two automorphisms are equal if and only if they agree on every observable. Apply this observation to the two composites in the definition. $\square$

Theorem 18.10.2.49 (Translation covariance from unit translation). [Lean](#) [✓ Stmt](#) For $d > 0$, a star-algebra automorphism ω of $\mathcal{A}$ commutes with every integer translation if and only if it commutes with unit translation:

$$\omega \circ \tau_1 = \tau_1 \circ \omega. \quad (18.943)$$

Proof. [✓ Proof](#) The forward implication follows by taking the translation step to be 1. Conversely, use two-sided integer induction. The zero case follows from $\tau_0 = \text{id}$. For the successor step, the additive action law gives $\tau_{i+1} = \tau_1 \circ \tau_i$, and

$$\omega \circ \tau_{i+1} = \omega \circ \tau_1 \circ \tau_i = \tau_1 \circ \omega \circ \tau_i \quad (18.944)$$

$$= \tau_1 \circ \tau_i \circ \omega = \tau_{i+1} \circ \omega. \quad (18.945)$$

For the predecessor step, $\tau_{-1} = \tau_1^{-1}$ transfers commutation to τ_{-1} , while $\tau_{i-1} = \tau_{-1} \circ \tau_i$, so

$$\omega \circ \tau_{i-1} = \omega \circ \tau_{-1} \circ \tau_i = \tau_{-1} \circ \omega \circ \tau_i \quad (18.946)$$

$$= \tau_{-1} \circ \tau_i \circ \omega = \tau_{i-1} \circ \omega. \quad (18.947)$$

$\square$

Theorem 18.10.2.50 (Identity and composition of translation-covariant automorphisms). [Lean](#) [Lean](#) [✓ Stmt](#) For $d > 0$, the identity automorphism of $\mathcal{A}$ is translation covariant. If ω and η are translation covariant, then so is the composite

$$A \mapsto \eta(\omega(A)), \quad (18.948)$$

which first applies ω and then η .

Proof. [✓ Proof](#) The identity commutes pointwise with every translation. For the composite, translation covariance gives

$$\eta(\omega(\tau_a(A))) = \eta(\tau_a(\omega(A))) = \tau_a(\eta(\omega(A))). \quad (18.949)$$

□

Lemma 18.10.2.51 (Norm isometry of quasi-local translation). [Lean](#) [Lean](#) [✓ Stmt](#) For $d > 0$, every quasi-local translation preserves the operator norm and distance:

$$\|\tau_a(A)\| = \|A\|, \quad (18.950)$$

$$\|\tau_a(A) - \tau_a(B)\| = \|A - B\|. \quad (18.951)$$

Proof. [✓ Proof](#) A star-algebra equivalence between complex C-star algebras preserves the C-star norm. Since τ_a is linear,

$$\|\tau_a(A) - \tau_a(B)\| = \|\tau_a(A - B)\| = \|A - B\|. \quad (18.952)$$

□

Definition 18.10.2.52 (Finite-region support in the quasi-local algebra). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $d > 0$, a quasi-local observable $x \in \mathcal{A}$ is supported in a finite region $\Lambda \subset \mathbb{Z}$ if it belongs to the image of the canonical embedding $j_\Lambda: \mathcal{A}_\Lambda \rightarrow \mathcal{A}$. Equivalently,

$$x \text{ is supported in } \Lambda \iff x \in j_\Lambda(\mathcal{A}_\Lambda) \quad (18.953)$$

$$\iff \text{there exists } A \in \mathcal{A}_\Lambda \text{ such that } j_\Lambda(A) = x. \quad (18.954)$$

In particular, $j_\Lambda(A)$ is supported in Λ for every $A \in \mathcal{A}_\Lambda$.

Proof. [✓ Proof](#) These assertions follow directly from membership in the range of j_Λ . □

Lemma 18.10.2.53 (Monotonicity of quasi-local support). [Lean](#) [✓ Stmt](#) Let $d > 0$. If $x \in \mathcal{A}$ is supported in Λ and $\Lambda \subseteq \Gamma$, then x is supported in Γ .

Proof. [✓ Proof](#) Write $x = j_\Lambda(A)$. Enlarge A to $\mathcal{A}_\Gamma$ by tensoring with the identity on $\Gamma \setminus \Lambda$. Compatibility of the canonical embeddings identifies the image of this enlargement with x . □

Theorem 18.10.2.54 (Disjointly supported quasi-local observables commute). [Lean](#) [Lean](#) [✓ Stmt](#) Let $d > 0$. If $x, y \in \mathcal{A}$ are supported in finite regions Λ and Γ with $\Lambda \cap \Gamma = \emptyset$, then

$$xy = yx. \quad (18.955)$$

Equivalently, for $A \in \mathcal{A}_\Lambda$ and $B \in \mathcal{A}_\Gamma$,

$$j_\Lambda(A)j_\Gamma(B) = j_\Gamma(B)j_\Lambda(A). \quad (18.956)$$

This is finite-support infrastructure for the quasi-local algebra in [15, lines 2292–2300]; it is not stated there as a separate theorem.

Proof. [✓ Proof](#) The completion map is a star-algebra homomorphism. Applying it to the algebraic-local identity gives

$$j(XY) = j(YX), \quad (18.957)$$

and finite-region support provides representatives $x = j_\Lambda(A)$ and $y = j_\Gamma(B)$. $\square$

Commutation with observables on the complementary tensor factor characterizes exact local support via the Weyl twirl.

Definition 18.10.2.55 (Complementary-factor Weyl twirl). [Lean](#) [Lean](#) [✓ Stmt](#) Let S and C be finite sets, let $n = |C| > 0$, and choose an equivalence $e: C \simeq \mathbb{Z}/n\mathbb{Z}$ and a primitive n -th root of unity ζ . If $W_{a,b}$ is the corresponding reindexed Weyl operator on $\mathbb{C}^C$, define

$$\mathcal{E}_C(M) = \frac{1}{n^2} \sum_{a,b \in \mathbb{Z}/n\mathbb{Z}} (\mathbf{1}_S \otimes W_{a,b}) M (\mathbf{1}_S \otimes W_{a,b})^\dagger. \quad (18.958)$$

Lemma 18.10.2.56 (Complementary-factor Weyl average). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $M \in \text{Mat}_{S \times C}(\mathbb{C})$,

$$\mathcal{E}_C(M) = \text{tr}_C(M) \otimes \frac{\mathbf{1}_C}{n} \quad (18.959)$$

$$= \left(\frac{1}{n} \text{tr}_C(M) \right) \otimes \mathbf{1}_C. \quad (18.960)$$

Proof. [✓ Proof](#) Split M into $C \times C$ blocks indexed by S . The Weyl average sends each block B to $(\text{tr } B/n) \mathbf{1}_C$, so its block trace is the corresponding entry of $\text{tr}_C(M)$. $\square$

Lemma 18.10.2.57 (Norm estimate for the complementary-factor twirl). [Lean](#) [Lean](#) [✓ Stmt](#) Let S be a nonempty finite set. The average is contractive in the operator norm; if every complementary Weyl conjugation fixes X , then

$$\|\mathcal{E}_C(M) - X\| \leq \|M - X\|. \quad (18.961)$$

Proof. [✓ Proof](#) Unitary conjugation preserves the operator norm, and the coefficients of the finite average sum to one. Apply this contraction to $M - X$. $\square$

Theorem 18.10.2.58 (Exact support from outside commutation). [Lean](#) [✓ Stmt](#) Let $d > 0$, let $\Lambda \subset \mathbb{Z}$ be finite, and let $x \in \mathcal{A}$. Then

$$x \in \mathcal{A}_\Lambda \iff [x, y] = 0 \text{ for every } y \in \mathcal{A}_\Gamma \text{ and every finite } \Gamma \cap \Lambda = \emptyset. \quad (18.962)$$

This characterizes local support through commutants on complementary regions in the UHF algebra [15, 205].

Proof. [✓ Proof](#) The forward implication follows from commutation of disjoint tensor factors. Conversely, approximate x by $A \in \mathcal{A}_\Delta$, where $\Lambda \subseteq \Delta$. Split $\Delta = \Lambda \sqcup (\Delta \setminus \Lambda)$ and average A over Weyl unitaries on the second factor. Every such conjugation fixes x by the commutation hypothesis, so unitary invariance and convexity give

$$\|\mathcal{E}_{\Delta \setminus \Lambda}(A) - x\| \leq \|A - x\|. \quad (18.963)$$

The Weyl identity writes the average as $B \otimes \mathbf{1}$ for some $B \in \mathcal{A}_\Lambda$. Thus elements of $\mathcal{A}_\Lambda$ approximate x arbitrarily well. The embedding of $\mathcal{A}_\Lambda$ is isometric, hence has closed range, and therefore $x \in \mathcal{A}_\Lambda$. $\square$

The homogeneous quasi-local UHF algebra has a trivial center consisting of scalar multiples of the identity [15, 169, 205].

Theorem 18.10.2.59 (Scalar center of the homogeneous quasi-local algebra). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $d > 0$ and let $z \in \mathcal{A}$. If

$$\forall \Gamma \subset \mathbb{Z} \text{ finite}, \quad \forall A \in \mathcal{A}_\Gamma, \quad z j_\Gamma(A) = j_\Gamma(A) z, \quad (18.964)$$

then there is a unique $c \in \mathbb{C}$ such that

$$z = c \mathbf{1}. \quad (18.965)$$

Consequently,

$$\{z \in \mathcal{A} \mid za = az \text{ for every } a \in \mathcal{A}\} = \mathbb{C} \mathbf{1}. \quad (18.966)$$

Proof. [Stmnt](#) Apply Theorem 18.10.2.58 with $\Lambda = \emptyset$. The commutation hypothesis gives $z \in j_\emptyset(\mathcal{A}_\emptyset)$, so there is an $A \in \mathcal{A}_\emptyset$ with $z = j_\emptyset(A)$. Let $e: \mathbb{C} \xrightarrow{\sim} \mathcal{A}_\emptyset$ be the canonical isomorphism and write $A = e(c)$. If also $z = j_\emptyset(e(c'))$, then injectivity of $j_\emptyset$ and of e give

$$j_\emptyset(e(c)) = j_\emptyset(e(c')) \implies e(c) = e(c') \implies c = c'. \quad (18.967)$$

Thus c is the unique scalar for which $z = c \mathbf{1}$.

Write $Z(\mathcal{A})$ for the center. Since every $j_\Gamma(B)$ belongs to $\mathcal{A}$, the two inclusions are

$$z \in Z(\mathcal{A}) \implies (\forall \Gamma \subset \mathbb{Z} \text{ finite}, \forall B \in \mathcal{A}_\Gamma, z j_\Gamma(B) = j_\Gamma(B) z) \implies z \in \mathbb{C} \mathbf{1}, \quad (18.968)$$

$$z = c \mathbf{1} \quad (c \in \mathbb{C}) \implies (\forall a \in \mathcal{A}, (c \mathbf{1})a = a(c \mathbf{1})) \implies z \in Z(\mathcal{A}). \quad (18.969)$$

Hence $Z(\mathcal{A}) = \mathbb{C} \mathbf{1}$. □

The appendix requires that “there exists a finite subset $\mathcal{N} \subset \mathbb{Z}$ ” such that, for every finite $\Lambda \subset \mathbb{Z}$,

$$\omega(\mathcal{A}_\Lambda) \subset \mathcal{A}_{\Lambda + \mathcal{N}}. \quad (18.970)$$

This is the forward propagation condition of [15, line 2298].

Definition 18.10.2.60 (Propagation within a fixed neighborhood). [Lean](#) [Stmnt](#) Let $d > 0$, let $\omega: \mathcal{A} \rightarrow \mathcal{A}$ be a star-algebra automorphism, and let $\mathcal{N} \subset \mathbb{Z}$ be finite. The automorphism ω propagates within $\mathcal{N}$ if, for every finite region Λ and every $x \in \mathcal{A}$,

$$x \text{ supported in } \Lambda \implies \omega(x) \text{ supported in } \Lambda + \mathcal{N}. \quad (18.971)$$

Definition 18.10.2.61 (Finite forward propagation). [Lean](#) [Stmnt](#) Let $d > 0$ and let $\omega: \mathcal{A} \rightarrow \mathcal{A}$ be a star-algebra automorphism. The automorphism ω has finite forward propagation if there exists a finite set $\mathcal{N} \subset \mathbb{Z}$ such that ω propagates within $\mathcal{N}$.

Lemma 18.10.2.62 (Equivalent formulations of forward propagation). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $d > 0$, let $\omega: \mathcal{A} \rightarrow \mathcal{A}$ be a star-algebra automorphism, and let $\mathcal{N} \subset \mathbb{Z}$ be finite. The following conditions are equivalent:

$$\omega \text{ propagates within } \mathcal{N}; \quad (a)$$

$$\forall \Lambda \in \mathbb{Z}, \forall A \in \mathcal{A}_\Lambda, \quad \omega(j_\Lambda(A)) \text{ is supported in } \Lambda + \mathcal{N}; \quad (b)$$

$$\forall \Lambda \in \mathbb{Z}, \quad \omega(j_\Lambda(\mathcal{A}_\Lambda)) \subseteq j_{\Lambda + \mathcal{N}}(\mathcal{A}_{\Lambda + \mathcal{N}}); \quad (c)$$

$$\forall \Lambda \in \mathbb{Z}, \forall A \in \mathcal{A}_\Lambda, \exists B \in \mathcal{A}_{\Lambda + \mathcal{N}}, \quad \omega(j_\Lambda(A)) = j_{\Lambda + \mathcal{N}}(B). \quad (d)$$

Thus condition (c) is precisely the range-inclusion formula from line 2298 of [15], expressed inside the quasi-local algebra, and condition (d) is its finite-region witness form.

Proof. ✓ **Proof** Every observable supported in Λ has the form $j_\Lambda(A)$, so (a) and (b) are equivalent. Replacing support by membership in the range of the canonical embedding gives (c). Finally, membership in $j_{\Lambda+\mathcal{N}}(\mathcal{A}_{\Lambda+\mathcal{N}})$ is equivalent to the existence of a representative B in that finite-region algebra, which gives (d). $\square$

Lemma 18.10.2.63 (Enlargement of propagation neighborhoods). Lean Lean ✓ Stmt Let $d > 0$ and let $\omega: \mathcal{A} \rightarrow \mathcal{A}$ be a star-algebra automorphism. If $\mathcal{N} \subseteq \mathcal{M}$ and ω propagates within $\mathcal{N}$, then ω propagates within $\mathcal{M}$. More explicitly, for every finite region Λ ,

$$\Lambda + \mathcal{N} \subseteq \Lambda + \mathcal{M}, \quad (18.972)$$

$$x \text{ supported in } \Lambda \implies \omega(x) \text{ supported in } \Lambda + \mathcal{N} \implies \omega(x) \text{ supported in } \Lambda + \mathcal{M}. \quad (18.973)$$

Consequently, if ω has finite forward propagation, then for every prescribed finite set $\mathcal{N} \subset \mathbb{Z}$ there is a finite set $\mathcal{M} \subset \mathbb{Z}$ such that

$$\mathcal{N} \subseteq \mathcal{M}, \quad (18.974)$$

$$\forall \Lambda \in \mathbb{Z}, \quad \omega(\mathcal{A}_\Lambda) \subseteq \mathcal{A}_{\Lambda+\mathcal{M}}. \quad (18.975)$$

This is the neighborhood-enlargement consequence of the propagation condition in Appendix line 2298 of [15].

Proof. ✓ **Proof** Monotonicity of the sumset in its second argument gives $\Lambda + \mathcal{N} \subseteq \Lambda + \mathcal{M}$. Monotonicity of quasi-local support then enlarges the support of $\omega(x)$ from $\Lambda + \mathcal{N}$ to $\Lambda + \mathcal{M}$. For the final assertion, choose a finite propagation neighborhood $\mathcal{K}$ and take $\mathcal{M} = \mathcal{K} \cup \mathcal{N}$. $\square$

Lemma 18.10.2.64 (Composition of forward propagation bounds). Lean Lean ✓ Stmt Let $d > 0$, let $\omega, \eta: \mathcal{A} \rightarrow \mathcal{A}$ be star-algebra automorphisms, and let $\mathcal{N}, \mathcal{M} \subset \mathbb{Z}$ be finite. Suppose that ω propagates within $\mathcal{N}$ and η propagates within $\mathcal{M}$. The composite $\eta \circ \omega$ first applies ω and then η , so

$$(\eta \circ \omega)(x) = \eta(\omega(x)), \quad (18.976)$$

$$x \text{ supported in } \Lambda \implies \omega(x) \text{ supported in } \Lambda + \mathcal{N} \quad (18.977)$$

$$\implies \eta(\omega(x)) \text{ supported in } (\Lambda + \mathcal{N}) + \mathcal{M} = \Lambda + (\mathcal{N} + \mathcal{M}). \quad (18.978)$$

Hence $\eta \circ \omega$ propagates within $\mathcal{N} + \mathcal{M}$. In particular, if ω and η have finite forward propagation, then $\eta \circ \omega$ has finite forward propagation. This is the composition closure of the finite-neighborhood condition in Appendix line 2298 of [15].

Proof. ✓ **Proof** Apply the two propagation bounds successively and use associativity of finite region sumsets. For the existential assertion, choose one finite propagation neighborhood for each automorphism and use their sumset. $\square$

Lemma 18.10.2.65 (Symmetric-interval normalization of finite propagation). Lean Lean Lean ✓ Stmt Every finite set $\mathcal{N} \subset \mathbb{Z}$ is contained in $[-R, R] \cap \mathbb{Z}$ for some $R \in \mathbb{N}$. This includes $\mathcal{N} = \emptyset$, for which one may take $R = 0$. If ω propagates within a fixed set $\mathcal{N}$, then R may be chosen so that

$$\mathcal{N} \subseteq [-R, R] \cap \mathbb{Z}, \quad (18.979)$$

$$\omega \text{ propagates within } [-R, R] \cap \mathbb{Z}. \quad (18.980)$$

Consequently, finite forward propagation admits a symmetric-interval witness:

$$\exists R \in \mathbb{N}, \quad \omega \text{ propagates within } [-R, R] \cap \mathbb{Z}. \quad (18.981)$$

Proof. ✓ Proof Let R be the maximum of $|n|$ over $n \in \mathcal{N}$, with value 0 when $\mathcal{N}$ is empty. Then $-R \leq n \leq R$ for every $n \in \mathcal{N}$, which gives $\mathcal{N} \subseteq [-R, R] \cap \mathbb{Z}$. Enlargement of propagation neighborhoods gives the fixed-neighborhood assertion. Finally, choose a finite propagation neighborhood and apply this enlargement to its symmetric interval. $\square$

Lemma 18.10.2.66 (Disjointness for reflected neighborhoods). Lean ✓ Stmt Let $\Gamma, \Lambda, \mathcal{N} \subset \mathbb{Z}$ be finite. Then

$$\Gamma \cap (\Lambda + (-\mathcal{N})) = \emptyset \iff (\Gamma + \mathcal{N}) \cap \Lambda = \emptyset. \quad (18.982)$$

Proof. ✓ Proof If $g + n = \lambda$ for $g \in \Gamma$, $n \in \mathcal{N}$, and $\lambda \in \Lambda$, then $g = \lambda + (-n) \in \Lambda + (-\mathcal{N})$, proving the forward implication. Conversely, if $g \in \Gamma$ and $g = \lambda + (-n)$ with $\lambda \in \Lambda$ and $n \in \mathcal{N}$, then $\lambda = g + n \in \Gamma + \mathcal{N}$. $\square$

Theorem 18.10.2.67 (Inverse propagation and inverse covariance). Lean Lean Lean ✓ Stmt Let $d > 0$, let ω be a star-algebra automorphism of $\mathcal{A}$, and let $\mathcal{N} \subset \mathbb{Z}$ be finite. If ω propagates within $\mathcal{N}$, then ω^{-1} propagates within $-\mathcal{N}$. Therefore finite forward propagation of ω implies finite forward propagation of ω^{-1} . If ω is translation covariant, then ω^{-1} is also translation covariant.

Schumacher and Werner [205, Theorem 6, Corollary 7] derive inverse locality through generalized Margolus partitioning; the proof here establishes it via commutants on complementary regions.

Proof. ✓ Proof Let $x \in \mathcal{A}_\Lambda$ and let $y \in \mathcal{A}_\Gamma$, where $\Gamma \cap (\Lambda + (-\mathcal{N})) = \emptyset$. Forward locality gives $\omega(y) \in \mathcal{A}_{\Gamma + \mathcal{N}}$. The regions $\Gamma + \mathcal{N}$ and Λ are disjoint, so x commutes with $\omega(y)$. Applying ω^{-1} shows that $\omega^{-1}(x)$ commutes with y . The exact outside-commutant characterization now gives

$$\omega^{-1}(x) \in \mathcal{A}_{\Lambda + (-\mathcal{N})}. \quad (18.983)$$

Choosing the reflected finite neighborhood proves the existential statement. For each lattice translation τ_a , multiplying the covariance identity on the left and right by ω^{-1} gives

$$\omega \circ \tau_a = \tau_a \circ \omega \implies \omega^{-1} \circ \tau_a = \tau_a \circ \omega^{-1}. \quad (18.984)$$

$\square$

Definition 18.10.2.68 (One-dimensional quantum cellular automaton). Lean ✓ Stmt Let $d > 0$ and let $\omega: \mathcal{A} \rightarrow \mathcal{A}$ be a star-algebra automorphism. The automorphism ω is a one-dimensional quantum cellular automaton if it is translation covariant and has finite forward propagation. Thus there exists a finite set $\mathcal{N} \subset \mathbb{Z}$ such that, for every finite $\Lambda \subset \mathbb{Z}$,

$$\omega(\mathcal{A}_\Lambda) \subseteq \mathcal{A}_{\Lambda + \mathcal{N}}. \quad (18.985)$$

This is the definition at line 2298 of [15]. It imposes only the displayed forward propagation condition and makes no locality claim for ω^{-1} .

Theorem 18.10.2.69 (Exact interface for quantum cellular automata). Lean Lean Lean Lean ✓ Stmt For $d > 0$, a star-algebra automorphism ω of $\mathcal{A}$ is a quantum cellular automaton if and only if it is translation covariant and has finite forward propagation. These two properties therefore construct an automaton, and every quantum cellular automaton satisfies each of them.

Proof. ✓ Proof This is the conjunction defining a quantum cellular automaton, together with its introduction and elimination rules. $\square$

Theorem 18.10.2.70 (Identity quantum cellular automaton). [Lean](#) [✓ Stmt](#) For $d > 0$, the identity automorphism of $\mathcal{A}$ is a quantum cellular automaton.

Proof. [✓ Proof](#) The identity is translation covariant. For finite forward propagation, choose the neighborhood $\{0\}$. An observable supported in Λ remains unchanged, and $\Lambda + \{0\} = \Lambda$. $\square$

Theorem 18.10.2.71 (Composition of quantum cellular automata). [Lean](#) [✓ Stmt](#) Let $d > 0$. If ω and η are quantum cellular automata on $\mathcal{A}$, then $\eta \circ \omega$ is a quantum cellular automaton. This composite first applies ω and then η . No propagation condition for either inverse is required.

Proof. [✓ Proof](#) Both defining properties are closed under composition in the stated order. Translation covariance of ω and η gives translation covariance of $\eta \circ \omega$. If $\mathcal{N}$ and $\mathcal{M}$ are forward propagation neighborhoods for ω and η , respectively, then $\mathcal{N} + \mathcal{M}$ is a forward propagation neighborhood for $\eta \circ \omega$. Hence the composite is a quantum cellular automaton. $\square$

Theorem 18.10.2.72 (Inverse quantum cellular automaton). [Lean](#) [✓ Stmt](#) Let $d > 0$. If ω is a quantum cellular automaton on $\mathcal{A}$, then ω^{-1} is also a quantum cellular automaton.

Proof. [✓ Proof](#) Translation covariance and finite forward propagation are both preserved by inversion. These are the two defining properties of a quantum cellular automaton. $\square$

Definition 18.10.2.73 (Expansion of a blocked region). [Lean](#) [Lean](#) [✓ Stmt](#) Let $L > 0$ be a block length and let $\Lambda \subset \mathbb{Z}$ be finite. The expansion of Λ is

$$\widehat{\Lambda}_L = \{Lx + r : x \in \Lambda, 0 \leq r < L\}. \quad (18.986)$$

Every $z \in \widehat{\Lambda}_L$ has unique block coordinates (x, r) with $x \in \Lambda$ and $0 \leq r < L$. Increasing r gives the consecutive-site order within the block, matching the finite-region site grouping in [15, lines 2308 and 2313–2320].

Theorem 18.10.2.74 (Coordinates and monotonicity of expanded regions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) An integer z belongs to $\widehat{\Lambda}_L$ exactly when its quotient by L belongs to Λ . The inverse block-coordinate map sends (x, r) to $Lx + r$, and $\Lambda \subseteq \Gamma$ implies $\widehat{\Lambda}_L \subseteq \widehat{\Gamma}_L$.

Proof. [✓ Proof](#) Apply Euclidean division by the positive integer L . Its quotient and residue are unique, the residue lies in $\{0, \dots, L-1\}$, and taking the inverse coordinates gives $Lx + r$. Monotonicity follows by retaining the same quotient and residue. $\square$

Definition 18.10.2.75 (Blocking equivalence for finite configurations). [Lean](#) [✓ Stmt](#) For on-site dimension d , configurations of d^L -level blocked spins on Λ are equivalent to configurations of d -level spins on $\widehat{\Lambda}_L$. The blocked label at x is decoded into its ordered length- L word, and its r th letter is placed at $Lx + r$.

Theorem 18.10.2.76 (Evaluation and restriction of blocked configurations). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Evaluation at $Lx + r$ reads the r th letter of the blocked spin at x . Conversely, encoding collects these letters in increasing residue order. At the configuration level, decoding commutes with restriction from Γ to $\Lambda \subseteq \Gamma$.

Proof. [✓ Proof](#) The two evaluation formulas are the inverse laws for the ordered word encoding. Restriction preserves the quotient block coordinate and residue, so both sides have the same value at every expanded site. $\square$

Definition 18.10.2.77 (Finite local-algebra blocking equivalence). [Lean](#) [✓ Stmt](#) Reindexing matrix rows and columns by the configuration blocking equivalence gives a star-algebra equivalence

$$\mathcal{A}_\Lambda^{(d^L)} \simeq \mathcal{A}_{\widehat{\Lambda}_L}^{(d)}. \quad (18.987)$$

This is the finite-region equivalence underlying the algebraic-local blocking construction below. No quasi-local completion, compatibility with translations, propagation statement, circuit representation, or transport of quantum cellular automata is asserted here.

Theorem 18.10.2.78 (Finite local-algebra blocking formulas). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Matrix entries are transported by applying the inverse configuration equivalence to both indices. The blocking equivalence preserves the operator norm and is an isometry.*

Proof. [✓ Proof](#) The entry formula is matrix reindexing. A star-algebra equivalence between finite matrix algebras preserves the C-star norm and hence distances. $\square$

Definition 18.10.2.79 (Finite block hull). [Lean](#) [✓ Stmt](#) For $L > 0$ and a finite original region $\Delta \subset \mathbb{Z}$, define its block hull by

$$\text{Hull}_L(\Delta) = \{\lfloor z/L \rfloor : z \in \Delta\}. \quad (18.988)$$

This is the finite set of length- L blocks that meet Δ .

Lemma 18.10.2.80 (Block-hull containment and monotonicity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $L > 0$. Every finite original region is contained in the expansion of its block hull,*

$$\Delta \subseteq \widehat{\text{Hull}_L(\Delta)}_L. \quad (18.989)$$

If $\Delta \subseteq \Theta$, then $\text{Hull}_L(\Delta) \subseteq \text{Hull}_L(\Theta)$. Expansion and block hull are exact in the blocked direction:

$$\text{Hull}_L(\widehat{\Lambda}_L) = \Lambda. \quad (18.990)$$

Proof. [✓ Proof](#) The quotient coordinate of each $z \in \Delta$ belongs to the image defining the hull, which proves containment after expansion. Monotonicity follows by retaining the same witness z in the larger region. Finally, every expanded site has quotient in Λ , while for each $x \in \Lambda$ the site Lx belongs to $\widehat{\Lambda}_L$ and has quotient x . Hence $\text{Hull}_L(\widehat{\Lambda}_L) = \Lambda$. $\square$

Lemma 18.10.2.81 (Blocking commutes with finite local inclusions). [Lean](#) [✓ Stmt](#) *Let $L > 0$. For finite blocked regions $\Lambda \subseteq \Gamma$, the square formed by finite blocking and the two canonical local inclusions commutes:*

$$\mathcal{B}_{L,\Gamma} \iota_{\Lambda,\Gamma} = \iota_{\widehat{\Lambda}_L, \widehat{\Gamma}_L} \mathcal{B}_{L,\Lambda}. \quad (18.991)$$

Proof. [✓ Proof](#) For configurations x, y on $\widehat{\Gamma}_L$, let $\tilde{x}$ and $\tilde{y}$ be their block encodings on Γ . The two matrix entries in the commuting square are

$$[\mathcal{B}_{L,\Gamma}(\iota_{\Lambda,\Gamma}(A))]_{x,y} = A_{\tilde{x}|_\Lambda, \tilde{y}|_\Lambda} \mathbf{1}_{\{\tilde{x}|_{\Gamma \setminus \Lambda} = \tilde{y}|_{\Gamma \setminus \Lambda}\}}, \quad (18.992)$$

$$[\iota_{\widehat{\Lambda}_L, \widehat{\Gamma}_L}(\mathcal{B}_{L,\Lambda}(A))]_{x,y} = A_{\tilde{x}|_\Lambda, \tilde{y}|_\Lambda} \mathbf{1}_{\{x|_{\widehat{\Gamma}_L \setminus \widehat{\Lambda}_L} = y|_{\widehat{\Gamma}_L \setminus \widehat{\Lambda}_L}\}}. \quad (18.993)$$

Restriction commutes with block encoding, and decoding identifies $\Gamma \setminus \Lambda$ with the original sites in $\widehat{\Gamma}_L \setminus \widehat{\Lambda}_L$. Hence the two indicator conditions are equivalent and the entries agree. $\square$

Definition 18.10.2.82 (Forward and reverse algebraic-local blocking maps). [Lean](#) [Lean](#) [✓ Stmt](#)
Let $L > 0$. The compatible finite blocking maps induce a star-algebra homomorphism

$$\mathcal{B}_L : \mathcal{A}_{\text{loc}}^{(d^L)} \longrightarrow \mathcal{A}_{\text{loc}}^{(d)}. \quad (18.994)$$

In the reverse direction there is a star-algebra homomorphism

$$\mathcal{U}_L : \mathcal{A}_{\text{loc}}^{(d)} \longrightarrow \mathcal{A}_{\text{loc}}^{(d^L)}. \quad (18.995)$$

A representative on an arbitrary original region Δ is included into $\widehat{\text{Hull}_L(\Delta)}_L$ before inverting the finite blocking.

Proof. [✓ Proof](#) For $\Lambda \subseteq \Gamma$, the finite commuting square gives

$$\iota_{\widehat{\Gamma}_L} \mathcal{B}_{L,\Gamma}(\iota_{\Lambda,\Gamma}(A)) = \iota_{\widehat{\Lambda}_L} \mathcal{B}_{L,\Lambda}(A), \quad (18.996)$$

so the forward finite-region maps form a compatible family. For the reverse family, block-hull monotonicity sends $\Delta \subseteq \Theta$ to $\text{Hull}_L(\Delta) \subseteq \text{Hull}_L(\Theta)$. Transitivity of the canonical inclusions and the inverse commuting square then give

$$\iota_{\text{Hull}_L(\Theta)} \mathcal{U}_{L,\Theta}(\iota_{\Delta,\Theta}(A)) = \iota_{\text{Hull}_L(\Delta)} \mathcal{U}_{L,\Delta}(A). \quad (18.997)$$

The universal property of the algebraic direct limit therefore induces both star-algebra homomorphisms. $\square$

Lemma 18.10.2.83 (Evaluation of the algebraic-local blocking maps). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

Let $L > 0$. The forward map applies finite blocking to a representative on Λ and regards the result as supported on $\widehat{\Lambda}_L$. The reverse map enlarges a representative on Δ to the expansion of its block hull, applies the inverse finite blocking equivalence, and regards the result as supported on $\text{Hull}_L(\Delta)$. The forward composite is the identity,

$$\mathcal{B}_L \mathcal{U}_L = \text{id}_{\mathcal{A}_{\text{loc}}^{(d)}}. \quad (18.998)$$

This holds for every on-site dimension d .

Proof. [✓ Proof](#) These are the evaluation formulas for the two compatible-family lifts. Forward compatibility is the commuting square. Reverse compatibility follows by monotonicity of block hulls, composition of local inclusions, and the same square applied in the inverse direction. Forward blocking after the reverse map reduces to the canonical inclusion of Δ into its expanded block hull. $\square$

Definition 18.10.2.84 (Algebraic-local site-blocking equivalence). [Lean](#) [✓ Stmt](#) For $d > 0$ and $L > 0$, forward blocking gives a star-algebra equivalence

$$\mathcal{A}_{\text{loc}}^{(d^L)} \simeq \mathcal{A}_{\text{loc}}^{(d)}. \quad (18.999)$$

This corresponds to the algebraic-local change of site grouping in [15, lines 2308 and 2313–2320].

Proof. [✓ Proof](#) The identity $\mathcal{B}_L \mathcal{U}_L = \text{id}$ makes $\mathcal{B}_L$ surjective. For injectivity, represent two blocked observables on finite regions Λ and Γ , enlarge both to $\Omega = \Lambda \cup \Gamma$, and suppose their images under $\mathcal{B}_L$ agree. Injectivity of the canonical map from the finite algebra on $\widehat{\Omega}_L$, followed by injectivity of $\mathcal{B}_{L,\Omega}$ and the finite commuting square, gives equality of the two representatives in the algebra on Ω . Thus $\mathcal{B}_L$ is bijective, with inverse $\mathcal{U}_L$, and defines the stated star-algebra equivalence. $\square$

Finally, for all quasi-local observables X and Y on the blocked chain,

$$\|\overline{\mathcal{B}}_L(X)\| = \|X\|, \quad (18.1009)$$

$$\|\overline{\mathcal{B}}_L(X) - \overline{\mathcal{B}}_L(Y)\| = \|X - Y\|. \quad (18.1010)$$

Proof. ✓ Proof Equalities (18.1004) and (18.1005) are the defining continuous extensions of the algebraic equivalence and its inverse. Composing them with the finite-region canonical maps gives (18.1006) and (18.1007). The exact formula (18.1008) uses $\text{Hull}_L(\widehat{\Lambda}_L) = \Lambda$. Because a star-algebra equivalence of C-star algebras preserves the norm, (18.1009) holds, and applying it to $X - Y$ gives (18.1010). $\square$

Lemma 18.10.2.88 (Support transport under site blocking). Lean Lean Lean ✓ Stmt Let $d > 0$ and $L > 0$. If a blocked-chain observable X is supported in Λ , then $\overline{\mathcal{B}}_L(X)$ is supported in $\widehat{\Lambda}_L$. If an original-chain observable Y is supported in Δ , then $\overline{\mathcal{B}}_L^{-1}(Y)$ is supported in $\text{Hull}_L(\Delta)$. In particular, support transport is exact on expanded regions:

$$\overline{\mathcal{B}}_L(X) \text{ is supported in } \widehat{\Lambda}_L \iff X \text{ is supported in } \Lambda. \quad (18.1011)$$

Proof. ✓ Proof Choose a finite-region representative and apply the corresponding forward or inverse finite evaluation formula. For the reverse implication in the exact statement, apply inverse blocking and simplify the inverse composite and the block hull of the expanded region. $\square$

Lemma 18.10.2.89 (Expansion of a translated blocked region). Lean ✓ Stmt Let $L > 0$, let $a \in \mathbb{Z}$, and let $\Lambda \subset \mathbb{Z}$ be a finite blocked region. Then

$$\widehat{\Lambda + aL} = \widehat{\Lambda}_L + aL. \quad (18.1012)$$

This is the translation rule for the site grouping used in [15, lines 2308 and 2313–2320].

Proof. ✓ Proof Write each original site uniquely as $xL + r$, with $0 \leq r < L$. Translation of the blocked coordinate by a sends this site to $(x+a)L + r = xL + r + aL$. The argument is unchanged when x or a is negative. $\square$

Definition 18.10.2.90 (Carry-aware blocked neighborhood). Lean ✓ Stmt Let $L > 0$ and let $\mathcal{N} \subset \mathbb{Z}$ be a finite original-lattice neighborhood. Write $\widehat{\{0\}}_L = \{0, \dots, L-1\}$ for the sites in the reference block. Define the induced blocked-lattice neighborhood by

$$\mathcal{C}_L(\mathcal{N}) = \text{Hull}_L(\widehat{\{0\}}_L + \mathcal{N}). \quad (18.1013)$$

The full reference block is included because the block reached after a displacement can depend on the starting residue. This is the finite-region geometry implicit in the grouping of sites in [15, lines 2308 and 2313–2320].

Lemma 18.10.2.91 (Residue description of the blocked neighborhood). Lean Lean ✓ Stmt Let $L > 0$, let $\mathcal{N} \subset \mathbb{Z}$ be finite, and let $b \in \mathbb{Z}$. Then

$$b \in \mathcal{C}_L(\mathcal{N}) \iff \exists r \in \{0, \dots, L-1\}, \exists n \in \mathcal{N}, \left\lfloor \frac{r+n}{L} \right\rfloor = b, \quad (18.1014)$$

$$\iff \exists r, s \in \{0, \dots, L-1\}, \exists n \in \mathcal{N}, r + n = bL + s. \quad (18.1015)$$

The quotient is the Euclidean quotient by the positive integer L . Consequently these formulas include negative displacements and both positive and negative block carries. In general, $\text{Hull}_L(\mathcal{N})$ alone omits some carries caused by the starting residue. This makes explicit the boundary geometry implicit in [15, lines 2313–2320].

Proof. ✓ **Proof** An element of $\widehat{\{0\}}_L + \mathcal{N}$ has the form $r + n$ with $0 \leq r < L$ and $n \in \mathcal{N}$. Taking its block coordinate gives the first equivalence. Euclidean division writes $r + n = bL + s$ with the unique remainder $0 \leq s < L$, which gives the second equivalence without any sign restriction on n . $\square$

Lemma 18.10.2.92 (Exact block hull after adding a neighborhood). Lean ✓ Stmt Let $L > 0$. For every finite blocked region Λ and every finite original-lattice neighborhood $\mathcal{N}$,

$$\text{Hull}_L(\widehat{\Lambda}_L + \mathcal{N}) = \Lambda + \mathcal{C}_L(\mathcal{N}). \quad (18.1016)$$

This is the exact finite-region carry identity associated with the site grouping in [15, lines 2313–2320].

Proof. ✓ **Proof** Write an expanded site as $xL + r$, where $x \in \Lambda$ and $0 \leq r < L$. For $n \in \mathcal{N}$, Euclidean division gives

$$\left\lfloor \frac{xL + r + n}{L} \right\rfloor = x + \left\lfloor \frac{r + n}{L} \right\rfloor. \quad (18.1017)$$

The second summand ranges over exactly $\mathcal{C}_L(\mathcal{N})$ by Lemma 18.10.2.91. This proves both inclusions, including for negative x and n . $\square$

Lemma 18.10.2.93 (Carry-aware expansion containment). Lean ✓ Stmt Let $L > 0$. For every finite blocked region Λ and every finite original-lattice neighborhood $\mathcal{N}$,

$$\widehat{\Lambda}_L + \mathcal{N} \subseteq \Lambda + \widehat{\mathcal{C}_L(\mathcal{N})}_L. \quad (18.1018)$$

Proof. ✓ **Proof** Every finite original region is contained in the expansion of its block hull. Apply this containment to $\widehat{\Lambda}_L + \mathcal{N}$ and use Lemma 18.10.2.92 to identify that hull. $\square$

Lemma 18.10.2.94 (Algebraic-local blocking intertwines scaled translations). Lean ✓ Stmt Let $d, L > 0$. If $\mathcal{B}_L$ denotes the algebraic-local blocking equivalence, then for every $a \in \mathbb{Z}$,

$$\mathcal{B}_L \tau_a^{(dL)} = \tau_{aL}^{(d)} \mathcal{B}_L. \quad (18.1019)$$

This is the translation rule induced by the blocking of sites used in [15, lines 2308 and 2313–2320].

Proof. ✓ **Proof** On a representative supported in a finite blocked region Λ , both sides apply the same matrix reindexing. Their supports agree because

$$\widehat{\Lambda + aL} = \widehat{\Lambda}_L + aL. \quad (18.1020)$$

Equality on all finite-region representatives proves the identity on the algebraic direct limit. $\square$

Lemma 18.10.2.95 (Quasi-local blocking intertwines scaled translations). Lean ✓ Stmt Let $d, L > 0$. If $\overline{\mathcal{B}}_L$ denotes the quasi-local blocking equivalence, then for every $a \in \mathbb{Z}$,

$$\overline{\mathcal{B}}_L \overline{\tau}_a^{(dL)} = \overline{\tau}_{aL}^{(d)} \overline{\mathcal{B}}_L. \quad (18.1021)$$

This is the completion of the translation rule induced by the site grouping in [15, lines 2308 and 2313–2320].

Proof. ✓ **Proof** The two sides are continuous maps on the quasi-local algebra. On the dense algebraic-local subalgebra their values agree by Lemma 18.10.2.94. They therefore agree on the completion. $\square$

Under finite blocking, every one-dimensional quantum cellular automaton remains a QCA, providing the blocked grouping used in the representation theorem of [15, line 2308].

Definition 18.10.2.96 (Blocked automorphism). [Lean](#) [✓ Stmt](#) Let $d, L > 0$, let $\overline{\mathcal{B}}_L: \mathcal{A}^{(d^L)} \simeq \mathcal{A}^{(d)}$ be the quasi-local blocking equivalence, and let ω be a star-algebra automorphism of $\mathcal{A}^{(d)}$. The induced automorphism of the blocked chain is

$$\omega^{[L]} = \overline{\mathcal{B}}_L^{-1} \circ \omega \circ \overline{\mathcal{B}}_L. \quad (18.1022)$$

Lemma 18.10.2.97 (Evaluation of the blocked automorphism). [Lean](#) [✓ Stmt](#) For every $X \in \mathcal{A}^{(d^L)}$,

$$\omega^{[L]}(X) = \overline{\mathcal{B}}_L^{-1}(\omega(\overline{\mathcal{B}}_L(X))). \quad (18.1023)$$

Proof. [✓ Proof](#) This is the evaluation formula for the defining conjugation. □

Lemma 18.10.2.98 (Inverse of the blocked automorphism). [Lean](#) [✓ Stmt](#) The inverse of the blocked automorphism is the blocked inverse:

$$(\omega^{[L]})^{-1} = (\omega^{-1})^{[L]}. \quad (18.1024)$$

Proof. [✓ Proof](#) Evaluate both sides on a blocked-chain observable. The blocking equivalence and its inverse cancel on the two sides of ω^{-1} . □

Lemma 18.10.2.99 (Propagation neighborhood after site blocking). [Lean](#) [✓ Stmt](#) Let $\mathcal{N} \subset \mathbb{Z}$ be finite. If ω propagates within $\mathcal{N}$, then $\omega^{[L]}$ propagates within the carry-aware blocked neighborhood $\mathcal{C}_L(\mathcal{N})$. Explicitly, for every finite blocked region Λ and every $X \in \mathcal{A}^{(d^L)}$ supported in Λ , the observable $\omega^{[L]}(X)$ is supported in $\Lambda + \mathcal{C}_L(\mathcal{N})$.

Proof. [✓ Proof](#) Forward blocking sends support in Λ to support in $\widehat{\Lambda}_L$. Apply the propagation bound for ω , then apply inverse blocking. The resulting support is

$$\text{Hull}_L(\widehat{\Lambda}_L + \mathcal{N}) = \Lambda + \mathcal{C}_L(\mathcal{N}) \quad (18.1025)$$

by the exact carry-aware block-hull identity. □

Lemma 18.10.2.100 (Finite propagation after site blocking). [Lean](#) [✓ Stmt](#) If ω has finite forward propagation, then $\omega^{[L]}$ has finite forward propagation.

Proof. [✓ Proof](#) Choose a finite propagation neighborhood $\mathcal{N}$ for ω . Lemma 18.10.2.99 gives the finite neighborhood $\mathcal{C}_L(\mathcal{N})$ for $\omega^{[L]}$. □

Theorem 18.10.2.101 (Translation covariance of the blocked automorphism). [Lean](#) [✓ Stmt](#) The blocked automorphism is translation covariant if and only if the original automorphism commutes with translation by one whole block:

$$\omega^{[L]} \text{ is translation covariant} \iff \omega \circ \overline{\tau}_L^{(d)} = \overline{\tau}_L^{(d)} \circ \omega. \quad (18.1026)$$

Thus blocked unit-translation covariance records commutation of ω with translation by L , not necessarily commutation with original unit translation. For $L \neq 1$, the latter condition can be strictly stronger.

Proof. ✓ Proof By Theorem 18.10.2.49, translation covariance on the blocked chain is equivalent to commutation with blocked unit translation. Under $\overline{\mathcal{B}}_L$, blocked unit translation corresponds to original translation by L . Conjugating the commutation identity by the blocking equivalence proves both implications. $\square$

Lemma 18.10.2.102 (Translation covariance after site blocking). Lean ✓ Stmt *If ω is translation covariant, then $\omega^{[L]}$ is translation covariant.*

Proof. ✓ Proof Translation covariance of ω gives commutation with $\overline{\tau}_L^{(d)}$. Apply Theorem 18.10.2.101. $\square$

Lemma 18.10.2.103 (QCA transport through site blocking). Lean ✓ Stmt *Let $d, L > 0$. If ω is a one-dimensional QCA on $\mathcal{A}^{(d)}$, then*

$$\omega^{[L]} = \overline{\mathcal{B}}_L^{-1} \circ \omega \circ \overline{\mathcal{B}}_L \quad (18.1027)$$

is a one-dimensional QCA on $\mathcal{A}^{(d^L)}$.

Proof. ✓ Proof The two defining properties of a one-dimensional QCA are finite forward propagation and translation covariance. They are preserved by Lemma 18.10.2.100 and Lemma 18.10.2.102, respectively. $\square$

Regrouping sites normalizes the finite propagation neighborhood of a QCA to $[-1, 1] \cap \mathbb{Z}$, setting up the depth-two circuit representation in [15, line 2308] and [205].

Lemma 18.10.2.104 (Nearest-neighbor carries from a strict displacement bound). Lean ✓ Stmt *Let $L > 0$ and let $\mathcal{N} \subset \mathbb{Z}$ be finite. If $|n| < L$ for every $n \in \mathcal{N}$, then*

$$\mathcal{C}_L(\mathcal{N}) \subseteq [-1, 1] \cap \mathbb{Z}. \quad (18.1028)$$

Proof. ✓ Proof If $\mathcal{N} = \emptyset$, membership in $\mathcal{C}_L(\mathcal{N})$ is impossible. In general, a point $b \in \mathcal{C}_L(\mathcal{N})$ supplies residues $0 \leq r, s < L$ and $n \in \mathcal{N}$ such that

$$r + n = bL + s. \quad (18.1029)$$

Since $-L < n < L$, the equality is incompatible with $b \leq -2$ and with $b \geq 2$. Thus $-1 \leq b \leq 1$, including for negative displacements. $\square$

Lemma 18.10.2.105 (Supremum criterion for nearest-neighbor carries). Lean ✓ Stmt *Let $L > 0$ and let $\mathcal{N} \subset \mathbb{Z}$ be finite. If*

$$\sup_{n \in \mathcal{N}} |n| < L, \quad (18.1030)$$

then

$$\mathcal{C}_L(\mathcal{N}) \subseteq [-1, 1] \cap \mathbb{Z}. \quad (18.1031)$$

Proof. ✓ Proof Every $|n|$ is bounded above by the finite supremum, so the strict supremum bound supplies the pointwise hypothesis. When $\mathcal{N} = \emptyset$, the supremum is 0 and the conclusion is vacuous. $\square$

Lemma 18.10.2.106 (Canonical nearest-neighbor block length). [Lean](#) [✓ Stmt](#) For every finite $\mathcal{N} \subset \mathbb{Z}$, define

$$L_{\mathcal{N}} = 1 + \sup_{n \in \mathcal{N}} |n|. \quad (18.1032)$$

Then $L_{\mathcal{N}} > 0$ and

$$\mathcal{C}_{L_{\mathcal{N}}}(\mathcal{N}) \subseteq [-1, 1] \cap \mathbb{Z}. \quad (18.1033)$$

With the finite-supremum convention, $L_{\emptyset} = 1$.

Proof. [✓ Proof](#) The finite supremum is strictly smaller than its successor, which is positive even when $\mathcal{N}$ is empty. $\square$

Lemma 18.10.2.107 (Nearest-neighbor propagation after a bounded blocking). [Lean](#) [✓ Stmt](#) Let $d, L > 0$. Suppose that ω propagates within the finite neighborhood $\mathcal{N}$ and that $|n| < L$ for every $n \in \mathcal{N}$. For every finite blocked region Λ and every observable $X \in \mathcal{A}^{(d^L)}$ supported in Λ , the observable $\omega^{[L]}(X)$ is supported in

$$\Lambda + ([-1, 1] \cap \mathbb{Z}). \quad (18.1034)$$

Proof. [✓ Proof](#) Blocking first gives propagation within $\mathcal{C}_L(\mathcal{N})$. The carry bound places this neighborhood inside $[-1, 1] \cap \mathbb{Z}$, and enlargement of propagation neighborhoods gives the claim. $\square$

Theorem 18.10.2.108 (Finite propagation becomes nearest-neighbor). [Lean](#) [✓ Stmt](#) Let $d > 0$. If ω has finite forward propagation, then there exists a positive integer L such that, for every finite blocked region Λ and every observable $X \in \mathcal{A}^{(d^L)}$ supported in Λ , the observable $\omega^{[L]}(X)$ is supported in

$$\Lambda + ([-1, 1] \cap \mathbb{Z}). \quad (18.1035)$$

Proof. [✓ Proof](#) Choose a finite propagation neighborhood $\mathcal{N}$ and take $L = 1 + \sup_{n \in \mathcal{N}} |n|$. This L is positive, including when $\mathcal{N}$ is empty. Blocking gives propagation within $\mathcal{C}_L(\mathcal{N})$; the canonical carry inclusion and propagation monotonicity then give the nearest-neighbor bound. $\square$

Theorem 18.10.2.109 (A QCA becomes nearest-neighbor after blocking). [Lean](#) [✓ Stmt](#) Let $d > 0$ and let ω be a one-dimensional QCA on $\mathcal{A}^{(d)}$. There exists a positive integer L such that $\omega^{[L]}$ is a QCA on $\mathcal{A}^{(d^L)}$ and, for every finite blocked region Λ and every observable $X \in \mathcal{A}^{(d^L)}$ supported in Λ , the observable $\omega^{[L]}(X)$ is supported in

$$\Lambda + ([-1, 1] \cap \mathbb{Z}). \quad (18.1036)$$

Proof. [✓ Proof](#) Apply Theorem 18.10.2.108 to the finite propagation of ω , using the canonical positive block length. At the same length, site blocking preserves both finite propagation and translation covariance, hence the QCA property. $\square$

Lines 2292–2298 of [15] specify the finite local algebras, their canonical inclusions, translation covariance, and the locality inclusion

$$\omega(\mathcal{A}_{\Lambda}) \subseteq \mathcal{A}_{\Lambda + \mathcal{N}}. \quad (18.1037)$$

The finite-dimensional maps determined by this locality inclusion follow the adjacent-pair ordering convention of [169].

Definition 18.10.2.110 (Finite-region restriction). [Lean](#) [✓ Stmt](#) Let $d > 0$, let ω be a star-algebra automorphism of $\mathcal{A}^{(d)}$, and suppose that ω propagates within the finite neighborhood $\mathcal{N}$. For every finite region Λ , define the finite-region restriction

$$\omega_{\Lambda, \mathcal{N}}: \mathcal{A}_{\Lambda} \longrightarrow \mathcal{A}_{\Lambda + \mathcal{N}} \quad (18.1038)$$

to be the unique star-algebra homomorphism satisfying

$$\iota_{\Lambda + \mathcal{N}}(\omega_{\Lambda, \mathcal{N}}(A)) = \omega(\iota_{\Lambda}(A)) \quad (A \in \mathcal{A}_{\Lambda}). \quad (18.1039)$$

Theorem 18.10.2.111 (Finite-region restriction is an embedding). [Lean](#) [Lean](#) [✓ Stmt](#) *The finite-region restriction is injective, and for every $A \in \mathcal{A}_{\Lambda}$ one has*

$$\iota_{\Lambda + \mathcal{N}}(\omega_{\Lambda, \mathcal{N}}(A)) = \omega(\iota_{\Lambda}(A)). \quad (18.1040)$$

Proof. [✓ Proof](#) The defining equality follows from the locality inclusion and the injectivity of $\iota_{\Lambda + \mathcal{N}}$. If two local observables have the same image under $\omega_{\Lambda, \mathcal{N}}$, their canonical images become equal after applying ω . Injectivity of ω and of ι_{Λ} then gives equality of the two local observables. $\square$

Theorem 18.10.2.112 (Enlargement of finite-region restrictions). [Lean](#) [✓ Stmt](#) *If $\Lambda \subseteq \Gamma$, then for every $A \in \mathcal{A}_{\Lambda}$,*

$$\omega_{\Gamma, \mathcal{N}}(\iota_{\Lambda, \Gamma}(A)) = \iota_{\Lambda + \mathcal{N}, \Gamma + \mathcal{N}}(\omega_{\Lambda, \mathcal{N}}(A)). \quad (18.1041)$$

Proof. [✓ Proof](#) Apply the canonical inclusion into the quasi-local algebra to both sides. Both expressions become $\omega(\iota_{\Lambda}(A))$; injectivity of the target inclusion gives the equality. $\square$

Theorem 18.10.2.113 (Translation of finite-region restrictions). [Lean](#) [✓ Stmt](#) *Suppose in addition that ω is translation covariant. For $a \in \mathbb{Z}$ and $A \in \mathcal{A}_{\Lambda}$,*

$$\iota_{(\Lambda + a) + \mathcal{N}}(\omega_{\Lambda + a, \mathcal{N}}(\tau_{a, \Lambda}(A))) = \iota_{(\Lambda + \mathcal{N}) + a}(\tau_{a, \Lambda + \mathcal{N}}(\omega_{\Lambda, \mathcal{N}}(A))). \quad (18.1042)$$

Here $(\Lambda + a) + \mathcal{N} = (\Lambda + \mathcal{N}) + a$.

Proof. [✓ Proof](#) The left-hand side is $\omega(\tau_a(\iota_{\Lambda}(A)))$. Translation covariance moves τ_a past ω , and the finite-region translation formula gives the right-hand side. $\square$

Definition 18.10.2.114 (Bipartite coordinates for disjoint local algebras). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\Gamma, \Delta \subset \mathbb{Z}$ be disjoint finite regions. Writing $\mathcal{C}_{\Gamma}$ and $\mathcal{C}_{\Delta}$ for their configuration spaces, restriction to the two regions gives

$$\mathcal{C}_{\Gamma \cup \Delta} \simeq \mathcal{C}_{\Gamma} \times \mathcal{C}_{\Delta}, \quad (18.1043)$$

with the Γ -configuration first. Reindexing matrix rows and columns gives a star-algebra equivalence

$$\beta_{\Gamma, \Delta}: \mathcal{A}_{\Gamma \cup \Delta} \simeq M_{\mathcal{C}_{\Gamma} \times \mathcal{C}_{\Delta}}(\mathbb{C}). \quad (18.1044)$$

Theorem 18.10.2.115 (Canonical inclusions in bipartite coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) *For $A \in \mathcal{A}_{\Gamma}$ and $B \in \mathcal{A}_{\Delta}$,*

$$\beta_{\Gamma, \Delta}(\iota_{\Gamma, \Gamma \cup \Delta}(A)) = A \otimes \mathbf{1}, \quad (18.1045)$$

$$\beta_{\Gamma, \Delta}(\iota_{\Delta, \Gamma \cup \Delta}(B)) = \mathbf{1} \otimes B. \quad (18.1046)$$

Proof. ✓ Proof Evaluate both matrices on pairs of configurations. For the first local inclusion, restriction to Γ gives the matrix coefficient of A , while equality of the complementary Δ -configurations gives the matrix coefficient of the identity. This is $A \otimes \mathbf{1}$. Interchanging the two regions gives $\mathbf{1} \otimes B$. $\square$

Definition 18.10.2.116 (Finite local image in bipartite coordinates). Lean Lean Lean Lean ✓ Stmt Define the finite local image by

$$\mathcal{R}_{\Lambda, \mathcal{N}} = \omega_{\Lambda, \mathcal{N}}(\mathcal{A}_\Lambda) \subseteq \mathcal{A}_{\Lambda + \mathcal{N}}. \quad (18.1047)$$

It is a star-subalgebra star-isomorphic to $\mathcal{A}_\Lambda$. If $\Lambda + \mathcal{N} = \Gamma \cup \Delta$ with $\Gamma \cap \Delta = \emptyset$, then

$$\beta_{\Gamma, \Delta}(\mathcal{R}_{\Lambda, \mathcal{N}}) \subseteq M_{\mathcal{C}_\Gamma \times \mathcal{C}_\Delta}(\mathbb{C}) \quad (18.1048)$$

is the bipartite matrix star-subalgebra to which the left and right support-algebra constructions apply. No support algebra is defined in this statement.

Definition 18.10.2.117 (Nearest-neighbour pair regions). Lean Lean Lean ✓ Stmt For $x \in \mathbb{Z}$, set

$$E_x = \{2x, 2x + 1\}, \quad L_x = \{2x - 1, 2x\}, \quad P_x = \{2x + 1, 2x + 2\}. \quad (18.1049)$$

The order L_x, P_x is the left-right factor order in equation **RR2x** of [169, lines 1251–1266]. The same four-site localization occurs in [15, lines 2313–2318].

Lemma 18.10.2.118 (Geometry of the nearest-neighbour pair image). Lean Lean Lean ✓ Stmt For the neighbourhood $\mathcal{N} = \{-1, 0, 1\}$,

$$E_x + \mathcal{N} = L_x \cup P_x, \quad L_x \cap P_x = \emptyset, \quad P_x = L_{x+1}. \quad (18.1050)$$

Proof. ✓ Proof Substitute (18.1049). Adding $-1, 0, 1$ to the two points of E_x gives

$$E_x + \{-1, 0, 1\} = \{2x - 1, 2x, 2x + 1, 2x + 2\} = L_x \cup P_x. \quad (18.1051)$$

The largest point of L_x is $2x$, while the smallest point of P_x is $2x + 1$, so the two sets are disjoint. Finally, $L_{x+1} = \{2x + 1, 2x + 2\} = P_x$. $\square$

Definition 18.10.2.119 (Site-indexed support algebras). Lean Lean Lean Lean Lean Lean ✓ Stmt Let ω be a star-algebra automorphism of the homogeneous quasi-local algebra which propagates within $\{-1, 0, 1\}$. Write the finite image of $\mathcal{A}_{E_x}$ in the ordered $L_x \times P_x$ coordinates as

$$S_x = \beta_{L_x, P_x}(\omega(\mathcal{A}_{E_x})) \subseteq M_{\mathcal{C}_{L_x} \times \mathcal{C}_{P_x}}(\mathbb{C}). \quad (18.1052)$$

Define

$$\mathcal{R}_{2x} = \text{Spp}_L(S_x) \subseteq M_{\mathcal{C}_{L_x}}(\mathbb{C}), \quad (18.1053)$$

$$\mathcal{R}_{2x+1} = \text{Spp}_R(S_x) \subseteq M_{\mathcal{C}_{P_x}}(\mathbb{C}). \quad (18.1054)$$

After the canonical finite-region embeddings, these form one family of star-subalgebras of the quasi-local algebra. Here the matrix embedding is the composition

$$M_{\mathcal{C}_\Lambda}(\mathbb{C}) \cong \mathcal{A}_\Lambda \longrightarrow \mathcal{A}. \quad (18.1055)$$

Under this identification, the canonical local inclusion in [15, lines 2292–2298] is exactly the displayed composition. Formula (18.1054) is equation RR2x in [169, lines 1261–1266]. Cirac et al. use the homogeneous even factor in [15, lines 2322–2328]; the odd factor and unified family are supplied by the GNVW construction.

While GNVW allow the cell size $d(y)$ to depend on y , the present definition treats a homogeneous chain with fixed size d ; see [71].

Theorem 18.10.2.120 (Adjacent-pair support containment). [Lean](#) [✓ Stmt](#) For every $x \in \mathbb{Z}$, the finite image satisfies

$$S_x \subseteq \mathcal{R}_{2x} \boxtimes \mathcal{R}_{2x+1} \subseteq M_{\mathcal{C}_{L_x} \times \mathcal{C}_{P_x}}(\mathbb{C}). \quad (18.1056)$$

This is the homogeneous, full-matrix instance used in the two-sided containment at [169, lines 1276–1278]; see [71]. Cirac et al. state only the weaker inclusion in the left support algebra tensored with the unrestricted right physical factor at [15, lines 2322–2328].

Proof. [✓ Proof](#) Apply Theorem 18.10.1.16 to S_x in the ordered coordinates of (18.1052). Its left support is $\mathcal{R}_{2x}$ and its right support is $\mathcal{R}_{2x+1}$ by (18.1054), which gives (18.1056). $\square$

Theorem 18.10.2.121 (Physical support of the site-indexed support algebras). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $\widehat{\mathcal{R}}_y$ denotes the canonical quasi-local copy of $\mathcal{R}_y$, then for every $x \in \mathbb{Z}$,

$$\widehat{\mathcal{R}}_{2x} \subseteq \mathcal{A}_{L_x}, \quad \widehat{\mathcal{R}}_{2x+1} \subseteq \mathcal{A}_{P_x}. \quad (18.1057)$$

More generally, the canonical image of any matrix on a finite region Λ is supported in Λ . This is the homogeneous specialization of the localization accompanying equation RR2x in [169, lines 1261–1274]; see [71].

Proof. [✓ Proof](#) An element of $\widehat{\mathcal{R}}_{2x}$ is the canonical image of a matrix in $\mathcal{R}_{2x} \subseteq \mathcal{A}_{L_x}$ and is therefore supported in L_x . The same argument places $\widehat{\mathcal{R}}_{2x+1}$ in $\mathcal{A}_{P_x}$. $\square$

Theorem 18.10.2.122 (Generation by two disjoint finite regions). [Lean](#) [Lean](#) [✓ Stmt](#) If $\Lambda \subseteq \Gamma$ are finite regions, then $\iota_\Lambda(\mathcal{A}_\Lambda) \subseteq \iota_\Gamma(\mathcal{A}_\Gamma)$. If the finite regions Λ and Γ are disjoint, then the two canonical images generate the image of their union,

$$\iota_{\Lambda \cup \Gamma}(\mathcal{A}_{\Lambda \cup \Gamma}) = \iota_\Lambda(\mathcal{A}_\Lambda) \vee \iota_\Gamma(\mathcal{A}_\Gamma). \quad (18.1058)$$

This is the expansion of an observable on several cells into a product of observables on disjoint groups of cells used in [169, lines 592–596] and proved in [205, lines 341–364].

Proof. [✓ Proof](#) Monotonicity follows from $\iota_\Gamma \circ \iota_{\Lambda, \Gamma} = \iota_\Lambda$, and it gives the inclusion $\supseteq$ in (18.1058). For the remaining inclusion, let $\mathcal{V} \subseteq \mathcal{A}_{\Lambda \cup \Gamma}$ be the star-subalgebra of those A whose canonical image lies in the join on the right-hand side. In the bipartite coordinates $\beta_{\Lambda, \Gamma}$, the canonical inclusion of $\mathcal{A}_\Lambda$ is $X \mapsto X \otimes \mathbf{1}$ and that of $\mathcal{A}_\Gamma$ is $Y \mapsto \mathbf{1} \otimes Y$, so $\beta_{\Lambda, \Gamma}(\mathcal{V})$ contains $M_{\mathcal{C}_\Lambda}(\mathbb{C}) \otimes \mathbf{1}$ and $\mathbf{1} \otimes M_{\mathcal{C}_\Gamma}(\mathbb{C})$. The join of these two subalgebras has underlying vector space $M_{\mathcal{C}_\Lambda}(\mathbb{C}) \boxtimes M_{\mathcal{C}_\Gamma}(\mathbb{C})$, and every matrix on $\mathcal{C}_\Lambda \times \mathcal{C}_\Gamma$ lies in that span. Hence $\beta_{\Lambda, \Gamma}(\mathcal{V})$ is everything, and $\beta_{\Lambda, \Gamma}$ is bijective, so $\mathcal{V} = \mathcal{A}_{\Lambda \cup \Gamma}$. $\square$

Lemma 18.10.2.123 (Even pairs cover the chain). [Lean](#) [Lean](#) [✓ Stmt](#) For $x \neq y$ the pairs E_x and E_y are disjoint, and every finite region Λ satisfies

$$\Lambda \subseteq \bigcup_{x \in X_\Lambda} E_x, \quad (18.1059)$$

where $X_\Lambda = \{\lfloor i/2 \rfloor \mid i \in \Lambda\}$.

Proof. [✓ Proof](#) Both statements are integer arithmetic: the four sites $2x, 2x+1, 2y, 2y+1$ are distinct for $x \neq y$, and every i satisfies $i = 2\lfloor i/2 \rfloor$ or $i = 2\lfloor i/2 \rfloor + 1$. $\square$

Theorem 18.10.2.124 (Generation of the algebraic local algebra by the pair algebras). [Lean](#) [Lean](#) [✓ Stmt](#) *The pair algebras generate the algebraic local algebra,*

$$\bigvee_{x \in \mathbb{Z}} \iota_{E_x}(\mathcal{A}_{E_x}) = \mathcal{A}_{\text{loc}}. \quad (18.1060)$$

The completion is not involved: this is an equality of algebras of local observables.

Proof. [✓ Proof](#) For a finite set X of pair labels, induction on X gives

$$\iota_{\bigcup_{x \in X} E_x}(\mathcal{A}_{\bigcup_{x \in X} E_x}) \subseteq \bigvee_{x \in \mathbb{Z}} \iota_{E_x}(\mathcal{A}_{E_x}). \quad (18.1061)$$

For $X = \emptyset$ the region is empty and contained in E_0 , so monotonicity applies. For $X \cup \{x_0\}$ with $x_0 \notin X$, the pair E_{x_0} is disjoint from $\bigcup_{x \in X} E_x$, so (18.1058) splits the left-hand side into the join of $\iota_{E_{x_0}}(\mathcal{A}_{E_{x_0}})$ with the algebra handled by the induction hypothesis. Every local observable is supported in some finite region Λ , which is covered by (18.1059); monotonicity and (18.1061) place it in the join. $\square$

Definition 18.10.2.125 (Join of the site-indexed support algebras). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a star-algebra automorphism ω of the quasi-local algebra propagating within $\{-1, 0, 1\}$, define

$$G = \bigvee_{y \in \mathbb{Z}} \widehat{\mathcal{R}}_y \subseteq \mathcal{A}. \quad (18.1062)$$

Each $\widehat{\mathcal{R}}_y$ is contained in G . This is the algebraic join associated with the algebra generated by all $\mathcal{R}_y$ in [169, lines 1276–1279]. No claim that G is proper or that it fails to be norm closed is made here.

Theorem 18.10.2.126 (Images of the pair algebras lie in the join). [Lean](#) [Lean](#) [✓ Stmt](#) *For every $x \in \mathbb{Z}$,*

$$\omega(j_{E_x}(\mathcal{A}_{E_x})) \subseteq G, \quad (18.1063)$$

and consequently $\omega(j(\mathcal{A}_{\text{loc}})) \subseteq G$. Inclusion (18.1063) is the containment $\alpha(\mathcal{A}_{2x} \otimes \mathcal{A}_{2x+1}) \subseteq \mathcal{R}_{2x} \otimes \mathcal{R}_{2x+1}$ at [169, lines 1276–1278], in the homogeneous fixed- d setting; see [71].

Proof. [✓ Proof](#) Write S_x for the finite image of $\mathcal{A}_{E_x}$ in the ordered $L_x \times P_x$ coordinates. By (18.1056), S_x is contained in $\mathcal{R}_{2x} \boxtimes \mathcal{R}_{2x+1}$, which is the underlying vector space of the join of $\mathcal{R}_{2x} \otimes \mathbf{1}$ and $\mathbf{1} \otimes \mathcal{R}_{2x+1}$. The canonical quasi-local map on $L_x \cup P_x$ sends $X \otimes \mathbf{1}$ to the canonical embedding of $X \in \mathcal{R}_{2x}$ and $\mathbf{1} \otimes Y$ to the canonical embedding of $Y \in \mathcal{R}_{2x+1}$, so it maps that join into G . It sends the image of $A \in \mathcal{A}_{E_x}$ to $\omega(j_{E_x}(A))$, which proves (18.1063). By (18.1060), every local observable lies in the algebra generated by the $\iota_{E_x}(\mathcal{A}_{E_x})$, and $\omega \circ j$ is a star-algebra homomorphism, so its image lies in G . $\square$

Theorem 18.10.2.127 (Norm-closed generation by the support algebras). [Lean](#) [✓ Stmt](#) *The site-indexed support algebras generate the quasi-local algebra in the norm-closed sense,*

$$\overline{G}^{\|\cdot\|} = \mathcal{A}. \quad (18.1064)$$

In [169, lines 1276–1279] the algebras $\mathcal{R}_y$ are said to generate an algebra containing $\alpha(\mathcal{A}(\mathbb{Z}))$, which equals $\mathcal{A}(\mathbb{Z})$ because α is an automorphism. Here this generation statement is expressed by the norm-closure equality (18.1064) in the homogeneous fixed- d setting; see [71].

Proof. [✓ Proof](#) An injective star-algebra homomorphism of complex C-star algebras is isometric, so ω is continuous and $\omega^{-1}(\overline{G})$ is closed. By (18.1063) it contains $j(\mathcal{A}_{\text{loc}})$, which is dense, so $\omega^{-1}(\overline{G}) = \mathcal{A}$. Since ω is surjective, every $z \in \mathcal{A}$ equals $\omega(\omega^{-1}(z)) \in \overline{G}$. $\square$

Corollary 18.10.2.128 (Commutant of the site-indexed support algebras). [Lean](#) [✓ Stmt](#) *If $z \in \mathcal{A}$ satisfies $[z, R] = 0$ for every $y \in \mathbb{Z}$ and every $R \in \widehat{\mathcal{R}}_y$, then $[z, A] = 0$ for every $A \in \mathcal{A}$. This is the step from commutation with all $\mathcal{R}_y$ to commutation in the quasi-local algebra at [169, lines 1279–1282]; see [71].*

Proof. [✓ Proof](#) The elements commuting with z and with z^* form a norm-closed star-subalgebra. Each $\widehat{\mathcal{R}}_y$ is star closed, so the hypothesis applied to R^* gives $[z^*, R] = 0$ as well; hence that star-subalgebra contains every $\widehat{\mathcal{R}}_y$, therefore contains G , and being closed it contains $\overline{G} = \mathcal{A}$ by (18.1064). $\square$

Definition 18.10.2.129 (Tripartite finite-region coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) For pairwise disjoint finite regions Λ, Γ, Δ , successive restriction in left–middle–right order gives

$$\mathcal{C}_{(\Lambda \cup \Gamma) \cup \Delta} \cong (\mathcal{C}_\Lambda \times \mathcal{C}_\Gamma) \times \mathcal{C}_\Delta, \quad (18.1065)$$

$$\mathcal{A}_{(\Lambda \cup \Gamma) \cup \Delta} \cong M_{(\mathcal{C}_\Lambda \times \mathcal{C}_\Gamma) \times \mathcal{C}_\Delta}(\mathbb{C}). \quad (18.1066)$$

Lemma 18.10.2.130 (Geometry of consecutive pair images). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $T_x = (L_x \cup P_x) \cup P_{x+1}$. The three regions L_x, P_x, P_{x+1} are pairwise disjoint, and*

$$T_x = \{2x - 1, 2x, 2x + 1, 2x + 2, 2x + 3, 2x + 4\}. \quad (18.1067)$$

Proof. [✓ Proof](#) Substitute the three pair definitions and compare their six sites. $\square$

Theorem 18.10.2.131 (Canonical inclusions in tripartite coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) *In the coordinates of Definition 18.10.2.129, the canonical inclusion from $\Lambda \cup \Gamma$ is $X \mapsto X \otimes \mathbf{1}$, and the canonical inclusion from $\Gamma \cup \Delta$ is $Y \mapsto \mathbf{1} \otimes Y$, with the ordered factors Λ, Γ, Δ .*

Proof. [✓ Proof](#) Restrict each reconstructed configuration to the retained region and its complement. In the first inclusion the complementary coordinates are equal precisely on Δ ; in the second they are equal precisely on Λ . The two matrix entries are therefore the left and right overlapping lifts, respectively. $\square$

Theorem 18.10.2.132 (Commutation across the common physical pair). [Lean](#) [✓ Stmt](#) *For every $x \in \mathbb{Z}$, the canonically embedded support algebras $\widehat{\mathcal{R}}_{2x+1}$ and $\widehat{\mathcal{R}}_{2x+2}$ commute elementwise.*

Proof. [✓ Proof](#) The source regions E_x and E_{x+1} are disjoint, so their canonical quasi-local observables commute. Multiplicativity of ω preserves this equality. Embed the two finite images into $\mathcal{A}_{T_x}$ and use the tripartite coordinates above to obtain

$$(X \otimes \mathbf{1})(\mathbf{1} \otimes Y) = (\mathbf{1} \otimes Y)(X \otimes \mathbf{1}). \quad (18.1068)$$

The overlapping-support theorem, in the order $\text{Spp}_R(S_x)$ followed by $\text{Spp}_L(S_{x+1})$, gives the claim. $\square$

Definition 18.10.2.133 (Physical region of a site-indexed support algebra). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Define $Q_{2x} = L_x$ and $Q_{2x+1} = P_x$.

Theorem 18.10.2.134 (Unified physical localization). [Lean](#) [✓ Stmt](#) *Every element of $\widehat{\mathcal{R}}_y$ is supported in Q_y .*

Proof. ✓ Proof Split according to the parity of y and apply the corresponding localization of $\widehat{\mathcal{R}}_{2x}$ or $\widehat{\mathcal{R}}_{2x+1}$. □

Theorem 18.10.2.135 (Classification of overlapping support regions). Lean ✓ Stmt If $y \neq z$, then Q_y and Q_z are disjoint unless

$$(y, z) = (2x + 1, 2x + 2) \quad (18.1069)$$

for some x , or unless this ordered pair is reversed.

Proof. ✓ Proof Write each index in its even or odd form. Two regions of equal parity are disjoint. In the mixed-parity cases, the defining two-site sets intersect exactly for the displayed consecutive odd–even pair. □

Theorem 18.10.2.136 (Pairwise commutation of distinct support algebras). Lean ✓ Stmt Distinct members of the canonically embedded family commute elementwise:

$$[X, Y] = 0, \quad X \in \widehat{\mathcal{R}}_y, \quad Y \in \widehat{\mathcal{R}}_z, \quad y \neq z. \quad (18.1070)$$

This is the homogeneous fixed- d specialization of the consecutive/disjoint argument in [169, lines 1270–1274], using Lemma `sppcomm` from lines 1221–1246; compare [205, lines 1174–1194]. The four-site localization is established in [15, lines 2313–2329]; see [71] for the homogeneous scope of the adjacent-generation theorem.

Proof. ✓ Proof In the exceptional odd–even order apply Theorem 18.10.2.132; the reversed order follows by symmetry. Every remaining ordered pair has disjoint physical supports, so disjoint quasi-locality applies. □

Lemma 18.10.2.137 (Two-site translation of the pair regions). Lean Lean Lean ✓ Stmt Translation by two sites advances each of the three two-site regions by one index:

$$E_x + 2 = E_{x+1}, \quad L_x + 2 = L_{x+1}, \quad P_x + 2 = P_{x+1}. \quad (18.1071)$$

Proof. ✓ Proof Add 2 to each of the two points of (18.1049) and compare with the same set at the index $x + 1$. □

Lemma 18.10.2.138 (Matrix coefficients as compressions by matrix units). Lean Lean ✓ Stmt Let $X \in M_{I \times J}(\mathbb{C})$ and write e_{rs} for the matrix units of either factor. Then

$$X_{rs}^L \otimes \mathbf{1} = \sum_{t \in J} (\mathbf{1} \otimes e_{tr}) X (\mathbf{1} \otimes e_{st}), \quad (18.1072)$$

$$\mathbf{1} \otimes X_{ij}^R = \sum_{k \in I} (e_{ki} \otimes \mathbf{1}) X (e_{jk} \otimes \mathbf{1}). \quad (18.1073)$$

Every coefficient of the expansion used at [169, lines 1188–1191] and [205, lines 1137–1169] is therefore a sum of products formed inside the ambient algebra.

Proof. ✓ Proof Compare entries. On the left of (18.1072), the entry at $((i, t), (j, u))$ equals $X((i, r), (j, s))$ when $t = u$ and vanishes otherwise. On the right, the summand indexed by t contributes $X((i, r), (j, s))$ exactly when both second coordinates equal t , so the sum over t reproduces the same entry. Exchanging the roles of the two factors yields (18.1073). □

Lemma 18.10.2.139 (Quasi-local form of the coefficient compressions). [Lean](#) [Lean](#) [✓ Stmt](#) Let Γ and Δ be disjoint finite regions and let $A \in \mathcal{A}_{\Gamma \cup \Delta}$, written in the ordered $\Gamma \times \Delta$ coordinates. For configurations $r, s \in \mathcal{C}_\Delta$ and $i, j \in \mathcal{C}_\Gamma$, the canonical quasi-local copies of the two coefficient families are

$$\widehat{A}_{rs}^L = \sum_{t \in \mathcal{C}_\Delta} \widehat{e}_{tr}^\Delta \widehat{A} \widehat{e}_{st}^\Delta, \quad (18.1074)$$

$$\widehat{A}_{ij}^R = \sum_{k \in \mathcal{C}_\Gamma} \widehat{e}_{ki}^\Gamma \widehat{A} \widehat{e}_{jk}^\Gamma, \quad (18.1075)$$

where a hat denotes the canonical image in the quasi-local algebra. Both right-hand sides are formed inside the quasi-local algebra alone.

Proof. [✓ Proof](#) In the bipartite coordinates the inclusion of $\mathcal{A}_\Gamma$ is $X \mapsto X \otimes \mathbf{1}$ and the inclusion of $\mathcal{A}_\Delta$ is $Y \mapsto \mathbf{1} \otimes Y$. Transport (18.1072) and (18.1073) through the inverse of the bipartite coordinate equivalence, then apply the canonical map of $\mathcal{A}_{\Gamma \cup \Delta}$ into the quasi-local algebra, under which a finite-region inclusion leaves the quasi-local observable unchanged. $\square$

Theorem 18.10.2.140 (Two-site translation covariance of the support algebras). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ω be a star-algebra automorphism of the homogeneous quasi-local algebra which propagates within $\{-1, 0, 1\}$ and commutes with every lattice translation. Then, for every $y \in \mathbb{Z}$,

$$\tau_2(\widehat{\mathcal{R}}_y) = \widehat{\mathcal{R}}_{y+2}, \quad (18.1076)$$

and in the two parities

$$\tau_2(\widehat{\mathcal{R}}_{2x}) = \widehat{\mathcal{R}}_{2x+2}, \quad \tau_2(\widehat{\mathcal{R}}_{2x+1}) = \widehat{\mathcal{R}}_{2x+3}. \quad (18.1077)$$

The displacement is two sites of the original chain (one blocked cell); the shift automorphism in [169, lines 1267–1268] shows that single-site translation exchanges the inequivalent even and odd support algebras.

Covariance of the support algebras under translation is the identity displayed at [205, lines 1218–1226] for the algebras defined at [205, lines 1199–1206], and the ordered even and odd factors are equation RR2x at [169, lines 1251–1266]. The translation-covariant fixed- d setting is that of [15, lines 2292–2298].

The statement treats the one-dimensional translation-invariant setting [95], with homogeneous cell size d [71].

Proof. [✓ Proof](#) Fix x and put $\Lambda = E_x$, $\Gamma = L_x$, $\Delta = P_x$. A finite-region observable belongs to the finite local image exactly when its quasi-local copy lies in the image of $\mathcal{A}_\Lambda$ under ω . Hence, by (18.1074), the canonical quasi-local copy of $\mathcal{R}_{2x}$ is generated by the elements

$$\sum_{t \in \mathcal{C}_\Delta} \widehat{e}_{tr}^\Delta \omega(\widehat{B}) \widehat{e}_{st}^\Delta, \quad B \in \mathcal{A}_\Lambda, \quad r, s \in \mathcal{C}_\Delta, \quad (18.1078)$$

and the copy of $\mathcal{R}_{2x+1}$ is generated by the analogous elements built from (18.1075) with Γ in place of Δ . Since a star-algebra automorphism carries the algebra generated by a set to the algebra generated by the image of that set, it suffices to translate the two generating families.

Translation carries a matrix unit of a region to the matrix unit of the translated region at the translated configurations, and covariance gives $\tau_2(\omega(\widehat{B})) = \omega(\tau_2(\widehat{B}))$, whose right-hand side is ω applied to the quasi-local copy of the translated observable. Reindexing the summation over $\mathcal{C}_\Delta$

along the configuration translation therefore turns (18.1078) into the same family for the regions $\Lambda + 2$ and $\Delta + 2$, and the translation is a bijection between the two families. By (18.1071) those regions are E_{x+1} and P_{x+1} , so the translated family generates $\widehat{\mathcal{R}}_{2x+2}$. The odd case is identical with Γ in place of Δ and L_{x+1} in place of P_{x+1} .

Finally, splitting y by parity and using $2x + 2 = 2(x + 1)$ and $2x + 3 = 2(x + 1) + 1$ gives (18.1076) and (18.1077). $\square$

In [15, lines 2300–2306], eventual constancy of finite-chain conjugation defines a norm-preserving star-algebra automorphism of $\mathcal{A}_{\text{loc}}$ satisfying the support bound $\mathcal{N} = [-4D^4, 4D^4] \cap \mathbb{Z}$ and translation covariance, extending uniquely to $\mathcal{A}$. The following statements develop this direct-limit completion from compatible forward and inverse finite-local actions.

Definition 18.10.2.141 (Compatible finite-local automorphism data). [Lean](#) [✓ Stmt](#) For each finite region Λ , let

$$f_\Lambda, g_\Lambda: \mathcal{A}_\Lambda \longrightarrow \mathcal{A}_{\text{loc}} \quad (18.1079)$$

be star-algebra homomorphisms such that, whenever $\Lambda \subseteq \Gamma$,

$$f_\Gamma \circ \iota_{\Lambda, \Gamma} = f_\Lambda, \quad g_\Gamma \circ \iota_{\Lambda, \Gamma} = g_\Lambda. \quad (18.1080)$$

Let $f, g: \mathcal{A}_{\text{loc}} \rightarrow \mathcal{A}_{\text{loc}}$ be the maps induced by these compatible families. Compatible finite-local automorphism data require

$$g(f_\Lambda(A)) = \iota_\Lambda(A), \quad f(g_\Lambda(A)) = \iota_\Lambda(A) \quad (18.1081)$$

for every $A \in \mathcal{A}_\Lambda$.

Definition 18.10.2.142 (Induced algebraic-local homomorphisms). [Lean](#) [Lean](#) [✓ Stmt](#) The compatible families induce star-algebra homomorphisms

$$f, g: \mathcal{A}_{\text{loc}} \longrightarrow \mathcal{A}_{\text{loc}}. \quad (18.1082)$$

Proof. [✓ Proof](#) Apply the star-algebra universal property to each compatible family. $\square$

Lemma 18.10.2.143 (Evaluation of the induced homomorphisms). [Lean](#) [Lean](#) [✓ Stmt](#) For every finite region Λ and $A \in \mathcal{A}_\Lambda$,

$$f(\iota_\Lambda(A)) = f_\Lambda(A), \quad g(\iota_\Lambda(A)) = g_\Lambda(A). \quad (18.1083)$$

Proof. [✓ Proof](#) These are the evaluation identities supplied by the universal property. $\square$

Definition 18.10.2.144 (Induced algebraic-local equivalence). [Lean](#) [✓ Stmt](#) Compatible finite-local automorphism data determine a star-algebra equivalence

$$F: \mathcal{A}_{\text{loc}} \simeq \mathcal{A}_{\text{loc}} \quad (18.1084)$$

whose forward and inverse maps are f and g , respectively.

Proof. [✓ Proof](#) Induct on finite-region representatives in the algebraic direct limit. The two inverse assumptions give

$$g(f(\iota_\Lambda(A))) = g(f_\Lambda(A)) = \iota_\Lambda(A), \quad (18.1085)$$

$$f(g(\iota_\Lambda(A))) = f(g_\Lambda(A)) = \iota_\Lambda(A). \quad (18.1086)$$

Thus f and g are mutually inverse. $\square$

Lemma 18.10.2.145 (Evaluation of the algebraic equivalence). [Lean](#) [Lean](#) [✓ Stmt](#) For every $A \in \mathcal{A}_{\text{loc}}$,

$$F(A) = f(A), \quad F^{-1}(A) = g(A). \quad (18.1087)$$

Proof. [✓ Proof](#) Both identities follow from the defining forward and inverse maps of F . $\square$

Definition 18.10.2.146 (Forward local norm hypothesis). [Lean](#) [✓ Stmt](#) Assume $d > 0$. The forward local norm hypothesis is

$$\|f_{\Lambda}(A)\| = \|A\| \quad (18.1088)$$

for every finite region Λ and every $A \in \mathcal{A}_{\Lambda}$.

Lemma 18.10.2.147 (Forward algebraic norm formula). [Lean](#) [✓ Stmt](#) Under the forward local norm hypothesis, every $A \in \mathcal{A}_{\text{loc}}$ satisfies

$$\|F(A)\| = \|A\|. \quad (18.1089)$$

Proof. [✓ Proof](#) Induct on finite-region representatives. For $A = \iota_{\Lambda}(A_{\Lambda})$,

$$\|F(A)\| = \|f_{\Lambda}(A_{\Lambda})\| = \|A_{\Lambda}\| = \|A\|. \quad (18.1090)$$

$\square$

Lemma 18.10.2.148 (Forward and inverse algebraic isometries). [Lean](#) [Lean](#) [✓ Stmt](#) Under the forward local norm hypothesis, for all $A, B \in \mathcal{A}_{\text{loc}}$,

$$\|F(A) - F(B)\| = \|A - B\|, \quad \|F^{-1}(A) - F^{-1}(B)\| = \|A - B\|. \quad (18.1091)$$

Proof. [✓ Proof](#) Apply the forward norm identity to $A - B$ to prove that F is an isometry. Since $F \circ F^{-1}$ is the identity, the right-inverse law then makes F^{-1} an isometry as well. $\square$

Lemma 18.10.2.149 (Inverse algebraic norm formula). [Lean](#) [✓ Stmt](#) Under the forward local norm hypothesis, every $A \in \mathcal{A}_{\text{loc}}$ satisfies

$$\|F^{-1}(A)\| = \|A\|. \quad (18.1092)$$

Proof. [✓ Proof](#) Apply the inverse isometry to the distance from A to 0 and use $F^{-1}(0) = 0$. $\square$

Definition 18.10.2.150 (Completed compatible local automorphism). [Lean](#) [✓ Stmt](#) Under the forward local norm hypothesis, the algebraic equivalence extends to a star-algebra equivalence

$$\overline{F}: \mathcal{A} \simeq \mathcal{A} \quad (18.1093)$$

of the quasi-local completion.

Proof. [✓ Proof](#) The forward norm formula makes F an isometry, and the inverse law makes F^{-1} an isometry. Apply the completion extension for star-algebra equivalences to these two continuous maps. $\square$

Lemma 18.10.2.151 (Dense evaluation of the completed automorphism). [Lean](#) [Lean](#) [✓ Stmt](#) Let $j: \mathcal{A}_{\text{loc}} \rightarrow \mathcal{A}$ be the canonical completion map. For every $A \in \mathcal{A}_{\text{loc}}$,

$$\overline{F}(j(A)) = j(F(A)), \quad \overline{F}^{-1}(j(A)) = j(F^{-1}(A)). \quad (18.1094)$$

Proof. ✓ Proof These are the canonical-image formulas for the continuous extension of a star-algebra equivalence and for its inverse. $\square$

Lemma 18.10.2.152 (Finite evaluation of the completed automorphism). Lean Lean ✓ Stmt
For every finite region Λ and $A \in \mathcal{A}_\Lambda$,

$$\overline{F}(j_\Lambda(A)) = j(f_\Lambda(A)), \quad \overline{F}^{-1}(j_\Lambda(A)) = j(g_\Lambda(A)). \quad (18.1095)$$

Proof. ✓ Proof Substitute $j_\Lambda = j \circ \iota_\Lambda$ into the dense evaluation formulas and use $F(\iota_\Lambda(A)) = f_\Lambda(A)$ and $F^{-1}(\iota_\Lambda(A)) = g_\Lambda(A)$. $\square$

Definition 18.10.2.153 (Finite-local support bound). Lean ✓ Stmt For a finite neighborhood $\mathcal{N} \subset \mathbb{Z}$, the forward family has support bound $\mathcal{N}$ if

$$f_\Lambda(A) \in \iota_{\Lambda+\mathcal{N}}(\mathcal{A}_{\Lambda+\mathcal{N}}) \quad (18.1096)$$

for every finite region Λ and every $A \in \mathcal{A}_\Lambda$.

Theorem 18.10.2.154 (Propagation from a finite-local support bound). Lean ✓ Stmt *If the forward family has support bound $\mathcal{N}$, then the completed automorphism $\overline{F}$ propagates within $\mathcal{N}$.*

Proof. ✓ Proof Let $X = j_\Lambda(A)$ be supported in Λ . Choose $B \in \mathcal{A}_{\Lambda+\mathcal{N}}$ such that $f_\Lambda(A) = \iota_{\Lambda+\mathcal{N}}(B)$. Then

$$\overline{F}(X) = j(f_\Lambda(A)) = j_{\Lambda+\mathcal{N}}(B), \quad (18.1097)$$

so the image is supported in $\Lambda + \mathcal{N}$. $\square$

Definition 18.10.2.155 (Finite-local unit-translation formula). Lean ✓ Stmt The forward family commutes with unit translation if, for every finite region Λ and every $A \in \mathcal{A}_\Lambda$,

$$f_{\Lambda+1}(\tau_{1,\Lambda}(A)) = \tau_1(f_\Lambda(A)). \quad (18.1098)$$

Lemma 18.10.2.156 (Algebraic commutation with unit translation). Lean ✓ Stmt *If the finite-local unit-translation formula holds, then every $A \in \mathcal{A}_{\text{loc}}$ satisfies*

$$F(\tau_1(A)) = \tau_1(F(A)). \quad (18.1099)$$

Proof. ✓ Proof Induct on finite-region representatives and write $A = \iota_\Lambda(A_\Lambda)$. Then

$$F(\tau_1(A)) = f_{\Lambda+1}(\tau_{1,\Lambda}(A_\Lambda)) = \tau_1(f_\Lambda(A_\Lambda)) = \tau_1(F(A)). \quad (18.1100)$$

$\square$

Theorem 18.10.2.157 (Translation covariance from unit translation). Lean ✓ Stmt *If the finite-local unit-translation formula holds, then the completed automorphism $\overline{F}$ is translation covariant.*

Proof. ✓ Proof By Theorem 18.10.2.49, it suffices to prove that $\overline{F}$ commutes with translation by one site. Induct on the completion. On every $A \in \mathcal{A}_{\text{loc}}$,

$$\overline{F}(\overline{\tau}_1(j(A))) = j(F(\tau_1(A))) = j(\tau_1(F(A))) = \overline{\tau}_1(\overline{F}(j(A))). \quad (18.1101)$$

The equality set is closed because both composites are continuous, so the identity extends from the canonical dense subalgebra to the completion. $\square$

Theorem 18.10.2.158 (QCA assembled from compatible finite-local actions). [Lean](#) [✓ Stmt](#)
 Let $d > 0$. Suppose the compatible forward family preserves the local operator norm, has a finite support bound, and satisfies the finite-local unit-translation formula. Then the completed automorphism $\bar{F}$ is a one-dimensional QCA.

Proof. [✓ Proof](#) The support hypothesis supplies a finite forward propagation neighborhood, while the unit-translation formula supplies translation covariance. These are exactly the two defining properties of a one-dimensional QCA. $\square$

18.10.3 Full-matrix presentation of a factor with scalar centre

Theorem 18.10.3.1 (Scalar centre forces a full matrix algebra of a definite size). [Lean](#) [✓ Stmt](#)
 Let I be a finite nonempty set and let $S \subseteq M_I(\mathbb{C})$ be a unital star-subalgebra whose centre consists of the multiples of its unit,

$$\{z \in S \mid zs = sz \text{ for every } s \in S\} = \mathbb{C}\mathbf{1}. \quad (18.1102)$$

Then there is an $r \geq 1$ with

$$\begin{aligned} S &\cong M_r(\mathbb{C}), \\ \dim_{\mathbb{C}}(S) &= r^2, \end{aligned} \quad (18.1103)$$

the first isomorphism being one of star-algebras.

This is the single-block case of the structure theorem for finite-dimensional C^* -algebras [205, Proposition Csform, lines 2082–2098] for an algebra presented inside a matrix algebra; see [23].

Proof. [✓ Proof](#) A star-subalgebra of a complex matrix algebra is semisimple, so it is isomorphic to a finite product $\prod_{k=1}^m M_{d_k}(\mathbb{C})$ with every $d_k \geq 1$. The element that is the unit in the k -th coordinate and zero in the others is central, hence by (18.1102) a multiple $\lambda\mathbf{1}$; comparing its k -th coordinate with another one gives $\lambda = 1$ and $\lambda = 0$, so $m \leq 1$. The unit of $M_I(\mathbb{C})$ is nonzero because I is nonempty, so $m = 1$ and S is a simple ring. The Wedderburn theorem over the algebraically closed field $\mathbb{C}$ then produces an $r \geq 1$ and an isomorphism of $\mathbb{C}$ -algebras $S \cong M_r(\mathbb{C})$, and conjugating it by a positive definite matrix makes it preserve adjoints [182, “A $*$ -subalgebra with scalar centre is a full matrix algebra”]. A star-isomorphism is in particular a $\mathbb{C}$ -linear isomorphism, and $\dim_{\mathbb{C}} M_r(\mathbb{C}) = r^2$ [182, “Dimension of a subalgebra presented as a full matrix algebra”]. $\square$

Gross, Nesme, Vogts and Werner apply this conclusion to the site-indexed support algebras [169, lines 1276–1282]: an element of $\mathcal{R}_x$ commuting with all of $\mathcal{R}_x$ commutes with every $\mathcal{R}_y$, hence with the algebra they generate, which is the whole quasi-local algebra; its centre is scalar, so (18.1102) holds for $\mathcal{R}_x$ and $\mathcal{R}_x \cong M_{r(x)}(\mathbb{C})$.

18.11 Depth estimates for a merging tree

The merging construction of [207] gives numerical recurrences for circuit depth. The results below concern those recurrences; they do not assert that an arbitrary tensor admits the required circuit construction or conditioning bounds.

Theorem 18.11.1 (Product of the conditioning costs). [Lean](#) [✓ Stmt](#) Let T_j, q_j be real sequences, let $a \in \mathbb{R}$ and $C \geq 0$, and suppose $q_j \geq 1$ for every nonnegative integer j . If

$$T_0 \leq a, \quad T_{j+1} \leq q_j(2T_j + C2^{j+1}), \quad (18.1104)$$

then for every nonnegative integer L ,

$$T_L \leq 2^L(a + CL) \prod_{j=0}^{L-1} q_j. \quad (18.1105)$$

Proof. ✓ Proof Induct on L . The initial case is $T_0 \leq a$. At the induction step, use $\prod_{j < L} q_j \geq 1$ to bound the additive overhead by $C2^{L+1} \prod_{j < L} q_j$. Multiplication by $q_L \geq 0$ gives

$$T_{L+1} \leq q_L \left(2^{L+1}(a + CL) \prod_{j < L} q_j + C2^{L+1} \prod_{j < L} q_j \right) \quad (18.1106)$$

$$= 2^{L+1}(a + C(L+1)) \prod_{j < L+1} q_j. \quad (18.1107)$$

□

Theorem 18.11.2 (Polynomial product bound). Lean ✓ Stmt Under the hypotheses of Theorem 18.11.1, suppose also $a \geq 0$. Let L, c be nonnegative integers. If $\prod_{j < L} q_j \leq (2^L)^c$, then

$$T_L \leq (a + CL)(2^L)^{c+1}. \quad (18.1108)$$

Proof. ✓ Proof Substitute the assumed product bound into the preceding estimate. Its coefficient $2^L(a + CL)$ is nonnegative. □

Theorem 18.11.3 (Constant conditioning). Lean ✓ Stmt Let T_j be a real sequence, $a \in \mathbb{R}$, $C \geq 0$, and $q \geq 1$. Suppose $T_0 \leq a$ and $T_{j+1} \leq q(2T_j + C2^{j+1})$ for every nonnegative integer j . Then

$$T_L \leq (a + CL)(2q)^L. \quad (18.1109)$$

Proof. ✓ Proof Apply the product estimate to $q_j = q$ and use $2^L q^L = (2q)^L$. □

Theorem 18.11.4 (Linear overhead at unit conditioning). Lean ✓ Stmt For every real C and every nonnegative integer L ,

$$C(L+1)2^{L+1} = 2(CL2^L) + C2^{L+1}. \quad (18.1110)$$

Proof. ✓ Proof Expand $2^{L+1} = 2 \cdot 2^L$ and distribute multiplication. □

For $N = 2^L$, the sequence $T_L = CL2^L$ has order $N \log N$ when $C > 0$. Thus a recurrence with conditioning equal to one need not yield a linear estimate. This numerical observation is not a lower bound for implementing an MPU.

18.11.1 Boundary Gram matrices of a parity rotation

The factorization $I_A \otimes cI_B + P_A \otimes isP_B$, with Hermitian involutions P_A, P_B , gives the following two Gram matrices. Their algebraic inverse-trace identity is distinct from the identification of all admissible boundary states and the infimum defining the conditioning cost.

Definition 18.11.1.1 (Parity boundary Gram matrices). [Lean](#) [Lean](#) [✓ Stmt](#) For complex numbers c, s, x, y , define

$$G_A(x) = \begin{pmatrix} 1 & x \\ x & 1 \end{pmatrix}, \quad G_B(c, s, y) = \begin{pmatrix} c^2 & icsy \\ -icsy & s^2 \end{pmatrix}. \quad (18.1111)$$

When the parameters are real, x, y represent the expectations of the two involutions in exterior density operators.

Theorem 18.11.1.2 (Inverse trace of the parity Gram matrices). [Lean](#) [✓ Stmt](#) For the matrices of Definition 18.11.1.1,

$$\text{tr}[G_B(c, s, y)^{-1}(G_A(x)^{-1})^T] = \frac{c^2 + s^2}{c^2 s^2 (1 - x^2)(1 - y^2)}. \quad (18.1112)$$

The identity holds for arbitrary complex parameters, with the convention that the inverse of a singular matrix and the reciprocal of zero are zero.

Proof. [✓ Proof](#) Express each inverse as its adjugate divided by its determinant. The determinants are $1 - x^2$ and $c^2 s^2 (1 - y^2)$. The mixed off-diagonal contributions to the trace cancel, leaving numerator $c^2 + s^2$. $\square$

18.11.2 Interval isometries for product-projection phase gates

Let P_j be orthogonal projections on the individual sites and $|z| = 1$. The unitary $I + (z - 1) \otimes_j P_j$ is a product-projection instance of an MPO-projector unitary from [207]. The following boundary frames supply interval isometries directly, without requiring realization by exterior density operators.

Definition 18.11.2.1 (Projection interval maps). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a square complex matrix P and a complex number z , define

$$V_{\text{pre}}(P) = \begin{pmatrix} I - P \\ P \end{pmatrix}, \quad (18.1113)$$

$$V_{\text{int}}(P) = \begin{pmatrix} I - P/2 \\ P/2 \\ -P/2 \\ P/2 \end{pmatrix}, \quad (18.1114)$$

$$V_{\text{suf}}(P, z) = \begin{pmatrix} I + (z - 1)P/2 \\ (z - 1)P/2 \end{pmatrix}. \quad (18.1115)$$

The displayed components are stacked vertically. For the interior map, their order is the lexicographic order of the two bond labels.

Theorem 18.11.2.2 (Normalization of projection interval maps). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If P is an orthogonal projection, the prefix and interior maps of Definition 18.11.2.1 are isometries. If also $|z| = 1$, the suffix map is an isometry.

Proof. [✓ Proof](#) Sum the Gram products of the displayed components, using $P^\dagger = P$ and $P^2 = P$. For the prefix and interior maps this gives I . For the suffix map, the coefficient of P is

$$\frac{z - 1}{2} + \frac{\bar{z} - 1}{2} + \frac{|z - 1|^2}{2} = \frac{|z|^2 - 1}{2} = 0. \quad (18.1116)$$

$\square$

Definition 18.11.2.3 (Projection boundary frames and merging kernel). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 Define

$$C_L = \begin{pmatrix} 1 & 1/2 \\ 0 & 1/2 \end{pmatrix}, \quad C_R = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad K = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}. \quad (18.1117)$$

The columns of C_L, C_R are the left and right bond vectors, respectively. The matrix K gives the coefficients of their bilinear contraction.

Theorem 18.11.2.4 (Projection bond contraction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The frames in Definition 18.11.2.3 are invertible and obey*

$$C_R^T K C_L = I, \quad (18.1118)$$

$$\text{tr}(K^\dagger K) = 4, \quad (18.1119)$$

$$\text{tr}[(C_R^\dagger C_R)^{-1}((C_L^\dagger C_L)^{-1})^T] = 4. \quad (18.1120)$$

Proof. [✓ Proof](#) Multiply the two-by-two matrices. Their determinants are $\det C_L = 1/2$ and $\det C_R = 1$. Evaluate the inverse Gram matrices by their adjugates and determinants. $\square$

The bond contraction in Theorem 18.11.2.4 identifies matching tensor labels. For adjacent intervals it multiplies P_{jk} and $P_{k+1,l}$ by their tensor product, giving P_{jl} . Its norm is two and is independent of the phase z .

18.11.3 Finite coefficient alphabets at a cut

A coefficient matrix at a cut factors through the corresponding MPO bond. When its coefficients belong to a finite alphabet, the number of distinct residual coefficient functions can be bounded without a smallest-Schmidt-value assumption.

Theorem 18.11.3.1 (Columns distinguishing the rows). [Lean](#) [✓ Stmt](#) *Let M be a matrix over a field K , indexed by an arbitrary row set I and a finite column set J . There is a subset $J_0 \subseteq J$ of cardinality $\text{rank } M$ such that two rows of M are equal whenever they agree at every column in J_0 .*

Proof. [✓ Proof](#) Choose a basis of the column space among the columns of M . Agreement at these columns is preserved under linear combinations, so it implies agreement at every column. $\square$

Theorem 18.11.3.2 (Number of finite-alphabet rows). [Lean](#) [Lean](#) [✓ Stmt](#) *Let M have an arbitrary row set and a finite column set over a field. Suppose every entry belongs to a finite set S . The set of distinct rows is finite and has cardinality at most $|S|^{\text{rank } M}$. If also S is nonempty and $\text{rank } M \leq D$, this number is at most $|S|^D$.*

Proof. [✓ Proof](#) Restrict each row to the chosen columns. This injects the set of distinct rows into a set of $|S|^{\text{rank } M}$ functions. The rank-bound consequence uses $|S| \geq 1$. $\square$

Theorem 18.11.3.3 (Finite-alphabet rows from a bond factorization). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A and B be matrices over a field with compatible middle index set of cardinality D and a finite column set for B . Suppose every entry of AB belongs to a nonempty finite set S . The set of distinct rows of AB is finite and has cardinality at most $|S|^D$. In particular, zero-one entries give at most 2^D distinct rows.*

Proof. [✓ Proof](#) Use $\text{rank}(AB) \leq \text{rank } A \leq D$. For the zero-one specialization, take $S = \{0, 1\}$. $\square$

18.11.4 Unit-circle geometry and diagonal circuits

The following elementary geometry is the rank-two step in the diagonal circuit argument of [209].

Theorem 18.11.4.1 (Three points on a vertical line). [Lean](#) [✓ Stmt](#) *Let $u, v, w \in \mathbb{C}$ have the same real part and the same squared modulus. Then $u = v$, $u = w$, or $v = w$.*

Proof. [✓ Proof](#) Equality of squared moduli gives equal squares of the imaginary parts. Three real numbers with the same square have at most two values. $\square$

Theorem 18.11.4.2 (Constant real functional on the unit circle). [Lean](#) [✓ Stmt](#) *Let u, v, w be three distinct complex numbers of modulus one. If $\operatorname{Re}(zu) = \operatorname{Re}(zv) = \operatorname{Re}(zw)$, then $z = 0$.*

Proof. [✓ Proof](#) The three products have the same real part and the same modulus. Two are equal; their factors are distinct, so $z = 0$. $\square$

Theorem 18.11.4.3 (Three affine unit-circle values). [Lean](#) [✓ Stmt](#) *Let u, v, w be distinct complex numbers of modulus one. If $|a + bu| = |a + bv| = |a + bw| = 1$, then $a = 0$ or $b = 0$.*

Proof. [✓ Proof](#) Expand the squared moduli. The real parts of $a\bar{b}\bar{u}$, $a\bar{b}\bar{v}$ and $a\bar{b}\bar{w}$ agree. Thus $a\bar{b} = 0$. $\square$

Theorem 18.11.4.4 (A unit-modulus factorization dichotomy). [Lean](#) [✓ Stmt](#) *Let I, J be sets with J nonempty. Suppose $a, b : I \rightarrow \mathbb{C}$ and $t : J \rightarrow \mathbb{C}$ satisfy $|t_j| = 1$ and $|a_i + b_i t_j| = 1$ for all i, j . Then either $a_i = 0$ or $b_i = 0$ for every i , or there are $j_0, j_1 \in J$ such that $t_j \in \{t_{j_0}, t_{j_1}\}$ for every $j \in J$.*

Proof. [✓ Proof](#) If t has three distinct values, apply the preceding theorem to each row. Otherwise choose one value, and, if it exists, a second distinct value. There can be no third. $\square$

The matrix-basis argument and reversible circuit constructions in [209] are mathematical proofs and are not yet formalized. They give an exact circuit with $O(N^{\log_2 3})$ gates for any diagonal MPU with consecutive operator Schmidt ranks at most two. A sequential traversal uses $O(N)$ auxiliary qubits; a parallel version uses $O(N^{\log_2 3})$ auxiliary qubits and logarithmic depth. For diagonal MPUs of arbitrary maximal bond dimension D , reversible phase evaluation gives operator-norm error at most ε with resources polynomial in $N, D, \log(1/\varepsilon)$. Neither conclusion is asserted here for arbitrary, nondiagonal MPUs.

Theorem 18.11.4.5 (An affine product of bond factorizations). [Lean](#) [✓ Stmt](#) *Let K be a field, P, Q finite sets, and let $A : I \times P \rightarrow K$, $B : P \times J \rightarrow K$, $C : I \times Q \rightarrow K$, and $E : Q \times J \rightarrow K$ be matrices. For $s, t \in K$, the matrix $H_{ij} = s(AB)_{ij}(CE)_{ij} + t$ factors through the index set $(P \times Q) \sqcup \{*\}$.*

Proof. [✓ Proof](#) At (p, q) use left entry $sA_{ip}C_{iq}$ and right entry $B_{pj}E_{qj}$. At $*$ use left entry t and right entry 1. Distributivity gives their product equal to H . $\square$

Theorem 18.11.4.6 (Boolean selectors from two bond factorizations). [Lean](#) [✓ Stmt](#) *Under the preceding hypotheses suppose J is finite and every entry of H is zero or one. The set of distinct rows of H is finite and has cardinality at most $2^{|P||Q|+1}$. In particular, $|P| = |Q| = 2$ gives at most 32 rows.*

Proof. [✓ Proof](#) Apply the finite-alphabet row bound to the displayed factorization. Its middle index has cardinality $|P||Q| + 1$. $\square$

Chapter 19

Finite-Group MPU Representations and Scalar Gauge Data

19.1 Finite-group unitary averages

Definition 19.1.1 (Finite-group unitary average). ✓ ML stmt Discussion: #7458 Let G be a finite group and let $\rho: G \rightarrow \mathbf{U}(n)$ be a unitary representation. Define

$$P_\rho = \frac{1}{|G|} \sum_{g \in G} \rho(g). \quad (19.1)$$

For a fixed lattice site j , the specialization $\rho(g) = \mathcal{G}_g^j$ gives the local Gauss-law average P_j in [163, lines 459–461].

Theorem 19.1.2 (Finite-group unitary average is an orthogonal projection). Lean Lean ✓ Stmt Discussion: #7458 For every finite-dimensional unitary representation ρ of a finite group,

$$P_\rho^\dagger = P_\rho, \quad (19.2)$$

$$P_\rho^2 = P_\rho. \quad (19.3)$$

Hence P_ρ is an orthogonal projection.

Proof. ✓ Proof Regard each matrix $\rho(g)$ as the corresponding linear map and let $e_G = |G|^{-1} \sum_{g \in G} g$ be the normalized group-algebra average. The finite-group representation-averaging theorem used here is the projection identity whose concrete group-algebra calculation is

$$he_G = e_G \quad (h \in G), \quad e_G^2 = \frac{1}{|G|} \sum_{h \in G} he_G = e_G. \quad (19.4)$$

Applying the linear representation and transporting back to matrices gives $P_\rho^2 = P_\rho$. Unitarity gives $\rho(g)^\dagger = \rho(g^{-1})$. Since inversion permutes G ,

$$P_\rho^\dagger = \frac{1}{|G|} \sum_{g \in G} \rho(g)^\dagger = \frac{1}{|G|} \sum_{g \in G} \rho(g^{-1}) = P_\rho. \quad (19.5)$$

□

19.2 Finite-group representations by matrix product unitaries

Let G be a group, let $d \in \mathbb{N}$, and let $D: G \rightarrow \mathbb{N}_{>0}$ assign a virtual bond dimension to each group element. For every $g \in G$, let $\mathcal{U}_g$ be an MPO tensor with physical input and output dimension d and virtual bond dimension D_g . Write $U_g^{(N)}$ for the periodic operator obtained by closing a chain of N copies of $\mathcal{U}_g$. Throughout this section the chain length is positive: $N \geq 1$.

Definition 19.2.1 (Matrix product unitarity on nonempty chains). [Lean](#) [✓ Stmt](#) An MPO tensor $\mathcal{U}$ is a matrix product unitary on nonempty chains when $U^{(N)}$ is unitary for every $N \geq 1$. This is the system-size convention in [163, line 207].

Theorem 19.2.2 (Comparison with the $N > 1$ MPU convention). [Lean](#) [✓ Stmt](#) Every matrix product unitary on nonempty chains is a matrix product unitary in the sense of Definition 18.1.2.

Proof. [✓ Proof](#) The nonempty-chain condition applies in particular to every $N > 1$. $\square$

Theorem 19.2.3 (Physical blocking preserves nonempty-chain unitarity). [Lean](#) [✓ Stmt](#) If $\mathcal{U}$ is a matrix product unitary on nonempty chains and $L > 0$, then $\mathcal{U}^{[L]}$ is also a matrix product unitary on nonempty chains.

Proof. [✓ Proof](#) At chain length $N \geq 1$, the canonical reindexing of configurations gives

$$O_N(\mathcal{U}^{[L]}) \cong U^{(NL)}. \quad (19.6)$$

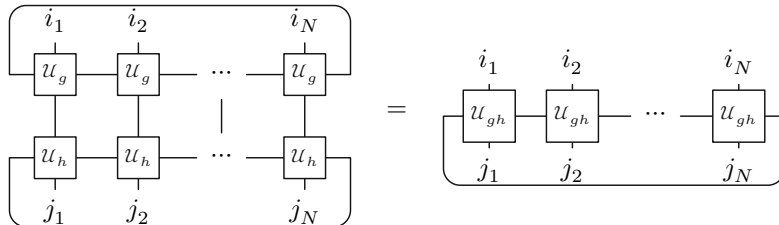
Since $N, L > 0$, the length NL is positive. Thus $U^{(NL)}$ is unitary, and simultaneous reindexing preserves unitarity. $\square$

Definition 19.2.4 (Group representation by simple injective MPUs). [Lean](#) [Lean](#) [✓ Stmt](#) A group representation by simple injective MPUs consists of the dependent-bond family $\{\mathcal{U}_g\}_{g \in G}$ above such that every $\mathcal{U}_g$ is injective and simple, every $\mathcal{U}_g$ is a matrix product unitary on nonempty chains, and

$$U_e^{(N)} = \mathbb{1}, \quad (19.7)$$

$$U_g^{(N)} U_h^{(N)} = U_{gh}^{(N)} \quad (19.8)$$

for all $g, h \in G$ and all $N \geq 1$. The multiplicative law is the representation-level input in [163, Subsection “MPU group representations”]. The identity law is its standard consequence and is recorded explicitly for positive lengths. It is also compatible with the later, stronger convention that $\mathcal{U}_e = \mathbb{1}$ has bond dimension one in [163, Subsection “Assumptions for the rest of the paper”]. The dimensions $D_g D_h$ and D_{gh} need not agree, so (19.8) is an identity of periodic operators, not an equality of local tensors.



Definition 19.2.5 (The operator representation on one chain). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{\mathcal{U}_g\}_{g \in G}$ be a dependent-bond family in which every $\mathcal{U}_g$ is a matrix product unitary on nonempty chains, and which satisfies (19.7) and (19.8) for all $N \geq 1$. For a fixed $N \geq 1$, the map $g \mapsto U_g^{(N)}$ is then a unitary representation of G on $(\mathbb{C}^d)^{\otimes N}$. Injectivity and simplicity of the tensors are not required.

Theorem 19.2.6 (Group fusion rules). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let G be a finite group and $\{\mathcal{U}_g\}_{g \in G}$ matrix product operator tensors whose periodic operators satisfy $U_g^{(N)} U_h^{(N)} = U_{gh}^{(N)}$ for all $g, h \in G$ and $N \geq 1$. Then $\{\mathcal{U}_g\}$ is a fusion algebra with structure constants

$$N_{gh}^k = \delta_{k,gh}, \quad (19.9)$$

and every label g is invertible with respect to the unit e , with inverse g^{-1} [164, Section 4]. In particular this holds for a representation as in Definition 19.2.4; only the multiplicative law (19.8) is used.

Proof. [✓ Proof](#) By the multiplicative law, $U_g^{(N)} U_h^{(N)} = U_{gh}^{(N)} = \sum_k \delta_{k,gh} U_k^{(N)}$. For the inverse, $N_{g,g^{-1}}^k = \delta_{k,gg^{-1}} = \delta_{k,e}$ and likewise $N_{g^{-1},g}^k = \delta_{k,e}$. $\square$

Theorem 19.2.7 (Powers of represented operators). [Lean](#) [✓ Stmt](#) [Discussion: #7445](#) For every $g \in G$, every $n \in \mathbb{N}$, and every $N \geq 1$,

$$(U_g^{(N)})^n = U_{g^n}^{(N)}. \quad (19.10)$$

Proof. [✓ Proof](#) Induct on n . For $n = 0$, the identity law gives $U_e^{(N)} = \mathbb{1}$. For the induction step, multiply the induction hypothesis by $U_g^{(N)}$ and apply the representation multiplication law:

$$(U_g^{(N)})^{n+1} = U_{g^n}^{(N)} U_g^{(N)} = U_{g^{n+1}}^{(N)}. \quad (19.11)$$

$\square$

Theorem 19.2.8 (Finite order of represented operators). [Lean](#) [✓ Stmt](#) [Discussion: #7445](#) Suppose that G is finite. For every $g \in G$ and every $N \geq 1$, the represented operator has finite order dividing the order of g :

$$(U_g^{(N)})^{\text{ord}(g)} = \mathbb{1}. \quad (19.12)$$

This is the finite-order assertion in [163, line 1547].

Proof. [✓ Proof](#) Apply (19.10) with $n = \text{ord}(g)$. The finite-order identity $g^{\text{ord}(g)} = e$ and the representation identity law then give $(U_g^{(N)})^{\text{ord}(g)} = U_e^{(N)} = \mathbb{1}$. $\square$

Theorem 19.2.9 (Product-tensor operator identity). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}_g \cdot \mathcal{U}_h$ denote the product MPO tensor, whose virtual bond dimension is $D_g D_h$. For every $N \geq 1$,

$$O_N(\mathcal{U}_g \cdot \mathcal{U}_h) = U_g^{(N)} U_h^{(N)} = U_{gh}^{(N)}. \quad (19.13)$$

Thus the product tensor and the selected tensor $\mathcal{U}_{gh}$ generate the same nonempty-chain periodic operators, although their virtual bond dimensions may differ.

Proof. [✓ Proof](#) Multiplicativity of the periodic operator under product-tensor contraction gives $O_N(\mathcal{U}_g \cdot \mathcal{U}_h) = U_g^{(N)} U_h^{(N)}$, which equals $U_{gh}^{(N)}$ by (19.8). $\square$

Theorem 19.2.10 (The physical adjoint represents the inverse). [Lean](#) [✓ Stmt](#) For every $g \in G$ and every $N \geq 1$, the physical adjoint of $\mathcal{U}_g$ generates the operator representing g^{-1} :

$$O_N(\mathcal{U}_g^\#) = U_{g^{-1}}^{(N)}. \quad (19.14)$$

This is the first assertion in [163, line 1552]. It requires no finiteness assumption on G .

Proof. [✓ Proof](#) The representation law and unitarity give

$$U_{g^{-1}}^{(N)} U_g^{(N)} = U_e^{(N)} = \mathbb{1}, \quad (19.15)$$

$$U_g^{(N)} (U_g^{(N)})^\dagger = \mathbb{1}. \quad (19.16)$$

Hence

$$U_{g^{-1}}^{(N)} = U_{g^{-1}}^{(N)} (U_g^{(N)} (U_g^{(N)})^\dagger) = (U_{g^{-1}}^{(N)} U_g^{(N)}) (U_g^{(N)})^\dagger = (U_g^{(N)})^\dagger. \quad (19.17)$$

Closing the physical-adjoint tensor gives the conjugate transpose, which proves (19.14). $\square$

Theorem 19.2.11 (Positive-length matrix product vector consequence). [Lean](#) [✓ Stmt](#) View $\mathcal{U}_g^\#$ and $\mathcal{U}_{g^{-1}}$ as MPS tensors by pairing their two physical indices. They generate the same matrix product vectors at every positive length:

$$\mathcal{V}^+((\mathcal{U}_g^\#)^{\text{MPS}}) = \mathcal{V}^+(\mathcal{U}_{g^{-1}}^{\text{MPS}}). \quad (19.18)$$

Proof. [✓ Proof](#) For a doubled physical configuration ρ , decode each site as a pair of ket and bra indices (σ_n, τ_n) . The corresponding matrix product vector coefficient is the (σ, τ) matrix entry of the periodic operator. Equality (19.14) therefore gives equality of every coefficient in (19.18). $\square$

Definition 19.2.12 (Exchange of product bond factors). [Lean](#) [✓ Stmt](#) For bond dimensions D_M and D_N , identify a product bond index with its pair of factor indices. The exchange of product bond factors is the equivalence

$$\{0, \dots, D_M - 1\} \times \{0, \dots, D_N - 1\} \longrightarrow \{0, \dots, D_N - 1\} \times \{0, \dots, D_M - 1\}, \quad (\alpha, \gamma) \longmapsto (\gamma, \alpha). \quad (19.19)$$

Theorem 19.2.13 (Physical adjunction reverses the operator product). [Lean](#) [✓ Stmt](#) Let M and N be MPO tensors with bond dimensions D_M and D_N . Write a left bond index of the product tensor as a pair (α, γ) , with α in the bond space of M and γ in the bond space of N , and a right bond index as (β, δ) likewise. Then, for every pair of physical indices i, k ,

$$((M \cdot N)^\#)_{(\alpha, \gamma), (\beta, \delta)}^{ik} = (N^\# \cdot M^\#)_{(\gamma, \alpha), (\delta, \beta)}^{ik}. \quad (19.20)$$

The two bond spaces need not have the same dimension, so the exchange of the two factors of a product bond space is recorded here for unequal dimensions. This is the inverse-product step used in the proof of the inversion identities for ζ and ω in [163, lines 1662–1677], for the dagger operation described there, which conjugates every entry and exchanges the two legs lying on the same side of a tensor.

Proof. [✓ Proof](#) Expand both sides. The left side is the complex conjugate of the $((\alpha, \gamma), (\beta, \delta))$ entry of $\sum_j M^{kj} \otimes N^{ji}$, namely $\sum_j \overline{M_{\alpha\beta}^{kj}} \overline{N_{\gamma\delta}^{ji}}$. The right side is the $((\gamma, \alpha), (\delta, \beta))$ entry of $\sum_j (N^\#)^{ij} \otimes (M^\#)^{jk}$, namely $\sum_j \overline{N_{\gamma\delta}^{ji}} \overline{M_{\alpha\beta}^{kj}}$. The two sums agree term by term. $\square$

Suppose now that the normalized flattening of every $\mathcal{U}_g$ is left-canonical. This is the canonical-form hypothesis used in [163, lines 1552–1557].

Theorem 19.2.14 (Unitary gauges for the dagger and the group inverse). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [StmT](#) Let G be a group, and let $\{\mathcal{U}_g\}_{g \in G}$ be a representation by simple injective MPU tensors whose normalized flattenings are left-canonical. Then

$$D_q = D_{q^{-1}}. \quad (19.21)$$

After identifying these bond spaces, for every $g \in G$ there is a unitary matrix $T_g \in \text{Mat}_{D_g}(\mathbb{C})$ such that

$$\mathcal{U}_g^\dagger = T_g^\dagger \mathcal{U}_{g^{-1}} T_g. \quad (19.22)$$

Here $\mathcal{U}_g^\dagger$ is the physical-adjoint tensor. This is the dagger-inverse gauge relation in [163, lines 1552–1557].

$$\begin{array}{c} i \\ \boxed{u_g^\dagger} \\ j \end{array} = \begin{array}{c} i \\ \text{red oval } T_g^\dagger \end{array} \begin{array}{c} \boxed{u_{g^{-1}}} \\ j \end{array} \begin{array}{c} \text{red oval } T_g \end{array}$$

Proof. ✓ Proof Apply the equality-case reduction theorem first to $\mathcal{U}_g^\dagger$ and $\mathcal{U}_{g^{-1}}$, and then with g and g^{-1} exchanged. The two reductions give $D_{g^{-1}} \leq D_g$ and $D_g \leq D_{g^{-1}}$, proving (19.21). Their positive-length matrix product vectors agree; at length zero they agree because the bond dimensions have just been identified. The single-block Fundamental Theorem therefore gives an invertible gauge. The normalized flattenings of both tensors are left-canonical, while injectivity implies irreducibility, so the canonical-form unitary-gauge theorem replaces this gauge by a unitary one. Reversing its orientation gives (19.22). $\square$

For a complex matrix A , write $A^* = (\overline{A_{ij}})_{i,j}$ for its entrywise complex conjugate. This is not the Hermitian adjoint $A^\dagger = (A^*)^\top$.

Theorem 19.2.15 (Scalar relation for the dagger-inverse gauges). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The gauges in Theorem 19.2.14 may be chosen together with scalars $\sigma_a \in \mathbb{C}$ satisfying*

$$T_g T_g^* = \sigma_g \mathbf{1}. \quad (19.23)$$

Every such scalar lies in $U(1)$, and

$$|\sigma_a| = 1, \quad (19.24)$$

$$\sigma_q \sigma_{q^{-1}} = 1. \quad (19.25)$$

If $g^{-1} = g$, the chosen gauge at g^{-1} , transported to the bond coordinates at g , equals T_g . Thus the same chosen scalar satisfies $T_g T_g^* = \sigma_g \mathbf{1}$, and $\sigma_g \in \{1, -1\}$. These are the dagger-inverse scalar relation and its immediate consequences in [163, lines 1557–1567].

Proof. ✔ Proof Apply physical adjunction to (19.22) for g^{-1} . Its involutivity and unitarity give a second gauge from $\mathcal{U}_{g^{-1}}$ to $\mathcal{U}_g^\dagger$, whose matrix is $T_{g^{-1}}^*$. Injective gauge uniqueness says that

$T_{g^{-1}}^*$ is a nonzero scalar multiple of $T_g^\dagger$. Multiplication on the left by T_g gives (19.23) with the displayed orientation. The unit-modulus and inverse-pair conclusions follow from Theorems 19.2.4.26 and 19.2.4.27. If $g^{-1} = g$, transporting the chosen gauge along this equality gives T_g itself in the current bond coordinates. Substitution in (19.23) gives $T_g T_g^* = \sigma_g \mathbf{1}$ with the same chosen scalar. Apply Theorem 19.2.4.28 to obtain $\sigma_g \in \{1, -1\}$. $\square$

The first-cut factorization in [163, lines 5390–5432] factors through the second-cut space with explicit one-sided inverses. This construction does not identify the two cut ranks, establish weighted normalizations, or construct the comparison matrix K and full compatibility properties of [163, lines 5342–5350].

Definition 19.2.16 (Gauge matrices for the first cut). Lean Lean ✓ Stmt For a unitary matrix T on $\mathbb{C}^D$, define matrices on $\mathbb{C}^d \otimes \mathbb{C}^D$ and $\mathbb{C}^D \otimes \mathbb{C}^d$, respectively, by

$$A = I_d \otimes \bar{T}, \quad (19.26)$$

$$B = T^\top \otimes I_d. \quad (19.27)$$

These are the gauge matrices in the first construction of [163, lines 5390–5432]. The order in B follows the virtual–physical column order of the first cut.

Definition 19.2.17 (Candidate first-cut factors). Lean Lean ✓ Stmt Let $\mathcal{U}$ be a tensor of physical dimension d and bond dimension D , and let T be unitary on $\mathbb{C}^D$. Let $\ell = \text{rank } M_2(\mathcal{U})$. For the chosen compact singular-value decomposition $M_2(\mathcal{U}) = V_2^\dagger D_2 U_2$, set

$$X_2 = V_2^\dagger, \quad (19.28)$$

$$Y_2 = D_2 U_2, \quad (19.29)$$

$$Z_2 = U_2^\dagger D_2^{-1}. \quad (19.30)$$

With A, B from Definition 19.2.16, define

$$X = A Y_2^\dagger: \mathbb{C}^\ell \longrightarrow \mathbb{C}^d \otimes \mathbb{C}^D, \quad (19.31)$$

$$Y = X_2^\dagger B: \mathbb{C}^D \otimes \mathbb{C}^d \longrightarrow \mathbb{C}^\ell. \quad (19.32)$$

These are the candidate factors of [163, line 5432]. Their intermediate coordinate space is $\mathbb{C}^\ell$, indexed by $\{0, \dots, \ell - 1\}$. The second-cut factors require no choice of a positive weight.

Theorem 19.2.18 (Reversing a unitary adjoint gauge). Lean ✓ Stmt Let $\mathcal{U}$ be a tensor of physical dimension d and bond dimension D . Suppose that T is unitary on $\mathbb{C}^D$ and that, for every physical pair i, j , $(\mathcal{U}^\sharp)^{ij} = T^\dagger \mathcal{U}^{ij} T$. Then, for every physical pair i, j , $\mathcal{U}^{ij} = T(\mathcal{U}^\sharp)^{ij} T^\dagger$. This reverses the gauge relation used in [163, lines 5391–5410].

Proof. ✓ Proof Unitarity gives $T(\mathcal{U}^\sharp)^{ij} T^\dagger = (T T^\dagger) \mathcal{U}^{ij} (T T^\dagger) = \mathcal{U}^{ij}$. $\square$

Theorem 19.2.19 (The first cut as a dressed second cut). Lean ✓ Stmt Let $\mathcal{U}$ be a tensor of physical dimension d and bond dimension D . Suppose that T is unitary and that, for every physical pair i, j , $(\mathcal{U}^\sharp)^{ij} = T^\dagger \mathcal{U}^{ij} T$. With $A = I_d \otimes \bar{T}$ and $B = T^\top \otimes I_d$, the source cuts satisfy $M_1(\mathcal{U}) = A M_2(\mathcal{U})^\dagger B$. This is the cut identity underlying [163, lines 5391–5431].

Proof. ✓ Proof Physical adjunction gives $M_2(\mathcal{U})^\dagger = M_1(\mathcal{U}^\sharp)$. For physical indices i, j and virtual indices a, b , expand the matrix product and apply Theorem 19.2.18:

$$(A M_1(\mathcal{U}^\sharp) B)_{(i,b),(a,j)} = \sum_{x,y=0}^{D-1} \overline{T_{bx}} (\mathcal{U}^\sharp)_{yx}^{ij} T_{ay} \quad (19.33)$$

$$= (T(\mathcal{U}^\sharp)^{ij} T^\dagger)_{ab} = \mathcal{U}_{ab}^{ij}. \quad (19.34)$$

□

Theorem 19.2.20 (Factorization through the second-cut space). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a tensor and T a unitary virtual matrix such that, for every physical pair i, j , $(\mathcal{U}^\sharp)^{ij} = T^\dagger \mathcal{U}^{ij} T$. The candidates X, Y of Definition 19.2.17 satisfy $M_1(\mathcal{U}) = XY$. This is the candidate decomposition of [163, line 5432].

Proof. [✓ Proof](#) Substitute the second-cut factorization to obtain

$$M_1(\mathcal{U}) = A(X_2 Y_2)^\dagger B = (A Y_2^\dagger)(X_2^\dagger B) = XY. \quad (19.35)$$

□

Theorem 19.2.21 (A left inverse for the candidate first factor). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be any tensor and T any unitary virtual matrix. For the candidate $X = A Y_2^\dagger$ on the second-cut space $\mathbb{C}^\ell$, the matrix $Z_2^\dagger A^\dagger$ is a left inverse: $(Z_2^\dagger A^\dagger)X = I_\ell$. No adjoint gauge relation is required. This verifies the left-invertibility used in [163, line 5432].

Proof. [✓ Proof](#) Unitarity of T gives $A^\dagger A = I_d \otimes (T^\top \bar{T}) = I_{dD}$. Hence

$$(Z_2^\dagger A^\dagger)X = Z_2^\dagger (A^\dagger A) Y_2^\dagger = (Y_2 Z_2)^\dagger = I_\ell. \quad (19.36)$$

□

Theorem 19.2.22 (A right inverse for the candidate second factor). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be any tensor and T any unitary virtual matrix. For the candidate $Y = X_2^\dagger B$ on the second-cut space $\mathbb{C}^\ell$, the matrix $B^\dagger X_2$ is a right inverse: $Y(B^\dagger X_2) = I_\ell$. No adjoint gauge relation is required. This verifies the right-invertibility used in [163, line 5432].

Proof. [✓ Proof](#) Unitarity of T gives $BB^\dagger = (T^\top \bar{T}) \otimes I_d = I_{Dd}$. Hence $Y(B^\dagger X_2) = X_2^\dagger (BB^\dagger) X_2 = X_2^\dagger X_2 = I_\ell$. □

Theorem 19.2.23 (The chosen adjoint gauge at an involutive element). [Lean](#) [✓ Stmt](#) Let G be a group and $\{\mathcal{U}_a\}_{a \in G}$ a representation by simple injective MPUs. Assume that the normalized flattening of $\mathcal{U}_a$ is left-canonical for every $a \in G$. If $g^{-1} = g$, the chosen unitary gauge T_g of Theorem 19.2.14 satisfies, for every physical pair i, j , $(\mathcal{U}_g^\sharp)^{ij} = T_g^\dagger \mathcal{U}_g^{ij} T_g$. This specializes the gauge relation of [163, lines 1552–1557] to the involutive case discussed in [163, lines 1933–1955].

Proof. [✓ Proof](#) The equality $g^{-1} = g$ identifies both the tensors and their bond spaces. Thus the chosen gauge satisfies $(\mathcal{U}_g^\sharp)^{ij} = T_g^\dagger \mathcal{U}_{g^{-1}}^{ij} T_g = T_g^\dagger \mathcal{U}_g^{ij} T_g$. □

Theorem 19.2.24 (The first-cut construction at an involutive element). [Lean](#) [✓ Stmt](#) Let G be a group and $\{\mathcal{U}_a\}_{a \in G}$ a representation by simple injective MPUs whose normalized flattenings are left-canonical for every $a \in G$. Let $g^{-1} = g$, and take the chosen gauge T_g of Theorem 19.2.14. For $\mathcal{U}_g$, put $\ell = \text{rank } M_2(\mathcal{U}_g)$, $A = I_d \otimes \bar{T}_g$, $B = T_g^\top \otimes I_d$, $X = A Y_2^\dagger$, and $Y = X_2^\dagger B$, using its second-cut factors X_2, Y_2, Z_2 . Then

$$M_1(\mathcal{U}_g) = XY, \quad (19.37)$$

$$(Z_2^\dagger A^\dagger)X = I_\ell, \quad (19.38)$$

$$Y(B^\dagger X_2) = I_\ell. \quad (19.39)$$

This is the first algebraic construction of [163, lines 5390–5432], with the gauge chosen from the representation rather than supplied separately.

Proof. [✓ Proof](#) Theorem 19.2.23 gives $(\mathcal{U}_g^\#)^{ij} = T_g^\dagger \mathcal{U}_g^{ij} T_g$. Theorem 19.2.20 therefore gives $M_1(\mathcal{U}_g) = XY$. Theorems 19.2.21 and 19.2.22 give $(Z_2^\dagger A^\dagger)X = I_\ell$ and $Y(B^\dagger X_2) = I_\ell$. $\square$

19.2.1 Rank equality and the invertible first-cut comparison

The invertible-comparison step of [163, lines 5432–5443] relates the source factors via their one-sided inverses, with inverse comparison matrix J . No unitarity of K , equality $J = K^\dagger$, or full inverse-compatible decomposition is asserted here.

Theorem 19.2.1.1 (Equality of the two source-cut ranks). [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ be an MPO tensor of physical dimension d and bond dimension D . Suppose that T is unitary on $\mathbb{C}^D$ and $(\mathcal{U}^\#)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every $0 \leq i, j < d$. Then its source-cut ranks satisfy $r = \text{rank } M_1(\mathcal{U}) = \text{rank } M_2(\mathcal{U}) = \ell$. No positivity, simplicity, or positive-dimension assumption is required.*

Proof. [✓ Proof](#) Put $X = AY_2^\dagger$, $Y = X_2^\dagger B$, $L = Z_2^\dagger A^\dagger$, and $R = B^\dagger X_2$, with the gauge matrices and second-cut factors of Definition 19.2.17. The factorization $M_1 = XY$ through $\mathbb{C}^\ell$ gives $r \leq \ell$. The identities $LX = I_\ell$ and $YR = I_\ell$ give $LM_1R = I_\ell$. Hence $\ell = \text{rank}(LM_1R) \leq \text{rank } M_1 = r$. $\square$

Theorem 19.2.1.2 (Both source-cut ranks equal the physical dimension). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ be a simple MPU of physical dimension d and bond dimension D in canonical form II. Suppose that T is unitary on $\mathbb{C}^D$ and that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every $0 \leq i, j < d$. Then $\ell = r = d$, and the resulting bijection of $\{0, \dots, \ell - 1\}$ with $\{0, \dots, d - 1\}$ carries the raw source-cut index to the physical index. Neither the phase relation $T\bar{T} = \sigma I_D$, nor simplicity of $\mathcal{U}^\dagger$, nor a further virtual gauge enters. This is the self-inverse case of [163, lines 1552–1563], in which the general finite-group index argument of line 1547 is not needed.*

Proof. [✓ Proof](#) Theorem 19.2.1.1 gives $r = \ell$ from the adjoint gauge relation alone. Simplicity of a canonical-form-II tensor is equivalent to $r\ell = d^2$ by Theorem 18.9.12. Substituting $r = \ell$ gives $\ell^2 = d^2$, hence $\ell = d$ and $r = d$. $\square$

Definition 19.2.1.3 (The weighted comparison matrix). [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ be an MPO tensor of dimensions d, D , let T be unitary on $\mathbb{C}^D$, and let ρ be a positive-definite matrix on $\mathbb{C}^D$. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. Let $\tilde{X}, \tilde{Y}, \tilde{Z}$ be the first source factors constructed with weight ρ , and let $X = AY_2^\dagger$ be the candidate first factor. Define*

$$\tilde{L} = \tilde{X}^\dagger(I_d \otimes \rho), \quad (19.40)$$

$$K = \tilde{L}X: \mathbb{C}^\ell \longrightarrow \mathbb{C}^r. \quad (19.41)$$

Here $\tilde{L}$ is the weighted left inverse supplied by $\tilde{X}^\dagger(I_d \otimes \rho)\tilde{X} = I_r$. The definition retains the weight ρ and requires no adjoint gauge relation.

Definition 19.2.1.4 (The reverse comparison matrix). [Lean](#) [✓ Stmt](#) *Let $\mathcal{U}$ be an MPO tensor of dimensions d, D , let T be unitary on $\mathbb{C}^D$, and let ρ be positive definite on $\mathbb{C}^D$. Let $\tilde{X}$ be its first source factor with weight ρ , and let Y_2, Z_2 be its second-cut factors with $Y_2 Z_2 = I_\ell$, where $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. For $A = I_d \otimes \bar{T}$, define*

$$L = Z_2^\dagger A^\dagger, \quad (19.42)$$

$$J = L\tilde{X}: \mathbb{C}^r \longrightarrow \mathbb{C}^\ell. \quad (19.43)$$

No adjoint gauge relation is needed to define J .

Theorem 19.2.1.5 (Mutually inverse rectangular comparisons). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor of dimensions d, D , let T be unitary on $\mathbb{C}^D$, and suppose that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every $0 \leq i, j < d$. Let ρ be positive definite on $\mathbb{C}^D$. Write $\widetilde{X}, \widetilde{Y}, \widetilde{Z}$ for the first source factors with weight ρ , and X_2, Y_2, Z_2 for the second-cut factors. Set

$$A = I_d \otimes \overline{T}, \quad (19.44)$$

$$B = T^\top \otimes I_d, \quad (19.45)$$

$$X = AY_2^\dagger, \quad (19.46)$$

$$Y = X_2^\dagger B, \quad (19.47)$$

$$L = Z_2^\dagger A^\dagger, \quad (19.48)$$

$$R = B^\dagger X_2, \quad (19.49)$$

$$\widetilde{L} = \widetilde{X}^\dagger (I_d \otimes \rho), \quad (19.50)$$

$$K = \widetilde{L}X, \quad (19.51)$$

$$J = L\widetilde{X}. \quad (19.52)$$

With $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$, the matrices $K: \mathbb{C}^\ell \rightarrow \mathbb{C}^r$ and $J: \mathbb{C}^r \rightarrow \mathbb{C}^\ell$ satisfy

$$\begin{aligned} K &= \widetilde{Y}R, \\ X &= \widetilde{X}K, \\ \widetilde{Y} &= KY, \\ J &= Y\widetilde{Z}, \\ \widetilde{X} &= XJ, \\ Y &= J\widetilde{Y}, \\ KJ &= I_r, \\ JK &= I_\ell. \end{aligned} \quad (19.53)$$

The two source-coordinate spaces are retained separately, without identifying their bases along the equality $r = \ell$. These are inverse identities, not adjoint or unitarity identities.

Proof. [✓ Proof](#) The two first-cut factorizations give $XY = \widetilde{X}\widetilde{Y}$. Their actual one-sided inverses satisfy $\widetilde{L}\widetilde{X} = I_r$, $LX = I_\ell$, $YR = I_\ell$, and $\widetilde{Y}\widetilde{Z} = I_r$. Apply Theorem 19.5.9.4 first with $\widetilde{L}, R$, then with the factorizations exchanged and inverses $L, \widetilde{Z}$. This proves the factor relations in (19.53). Finally,

$$KJ = KY\widetilde{Z} = \widetilde{Y}\widetilde{Z} = I_r, \quad (19.54)$$

$$JK = J\widetilde{Y}R = YR = I_\ell. \quad (19.55)$$

□

Theorem 19.2.1.6 (The second source factor from the adjoint comparison). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor of physical dimension d and bond dimension D , let ρ be positive definite on $\mathbb{C}^D$, and let T be unitary on $\mathbb{C}^D$. Suppose that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j . Write $\widetilde{X}_1, \widetilde{Y}_1$ for the first source factors with weight ρ , and X_2, Y_2 for the second source factors.

Put $A = I_d \otimes \bar{T}$ and take the weighted comparison $K = \widetilde{X}_1^\dagger (I_d \otimes \rho) A Y_2^\dagger$ of Definition 19.2.1.3. Then

$$Y_2 = K^\dagger \widetilde{X}_1^\dagger A. \quad (19.56)$$

This is the adjointed comparison used in [163, lines 5440–5443], in the maintained physical–virtual column order.

Proof. ✓ Proof Theorem 19.2.1.5 gives $A Y_2^\dagger = \widetilde{X}_1 K$. Since $A^\dagger A = I$,

$$K^\dagger \widetilde{X}_1^\dagger A = (A Y_2^\dagger)^\dagger A = Y_2 A^\dagger A = Y_2. \quad (19.57)$$

No identification of the reverse comparison with $K^\dagger$ is used. □

Theorem 19.2.1.7 (The inverse-compatible source-gate identity). Lean ✓ Stmt Let $\mathcal{U}$ be an MPO tensor of physical dimension d and bond dimension D , let ρ be positive definite on $\mathbb{C}^D$, and let T be unitary on $\mathbb{C}^D$. Suppose that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j , and that $T\bar{T} = \sigma I_D$ for a scalar $\sigma \in \mathbb{C}$. Write $\widetilde{X}_1, \widetilde{Y}_1$ for the first source factors constructed with this ρ , and X_2, Y_2 for the second source factors. Set $r = \text{rank } M_1(\mathcal{U})$, $\ell = \text{rank } M_2(\mathcal{U})$, and $K = \widetilde{X}_1^\dagger (I_d \otimes \rho) (I_d \otimes \bar{T}) Y_2^\dagger: \mathbb{C}^\ell \rightarrow \mathbb{C}^r$. For the source gates u, v of Definition 18.4.57 formed from $\widetilde{X}_1, \widetilde{Y}_1, X_2, Y_2$,

$$u = \sigma(K^\dagger \otimes K)v^\dagger. \quad (19.58)$$

The row orders of u and $v^\dagger$ are respectively (ℓ, r) and (r, ℓ) ; the Kronecker product maps between these orders without an additional swap or identification of the rank coordinates. This is the contraction in [163, lines 5444–5487], retaining the diagram’s $\widetilde{Y}_1$ [88]. Neither canonical form, simplicity, gate unitarity, nor $|\sigma| = 1$ is a hypothesis. The conclusion is the algebraic identity, not unitarity of K or the gates.

Proof. ✓ Proof Conjugating the phase relation by T gives $\bar{T}T = T^\dagger(T\bar{T})T = \sigma I_D$. Theorem 19.2.1.6 and Theorem 19.2.1.5 give

$$Y_2 = K^\dagger \widetilde{X}_1^\dagger (I_d \otimes \bar{T}), \quad (19.59)$$

$$\widetilde{Y}_1 = K X_2^\dagger (T^\dagger \otimes I_d). \quad (19.60)$$

Fix physical indices i, j . Define slices $C \in \text{Mat}_{r \times D}(\mathbb{C})$ and $E \in \text{Mat}_{D \times \ell}(\mathbb{C})$ by $C_{a\gamma} = \overline{(\widetilde{X}_1)_{(i,\gamma),a}}$ and $E_{\delta b} = \overline{(X_2)_{(\delta,j),b}}$. For every virtual index β and source indices l, s ,

$$(Y_2)_{l,(i,\beta)} = (K^\dagger C \bar{T})_{l\beta}, \quad (19.61)$$

$$(\widetilde{Y}_1)_{s,(\beta,j)} = (T E K^\dagger)_{\beta s}. \quad (19.62)$$

Contract the virtual index in the defining expression for u :

$$u_{(l,s),(i,j)} = \sum_{\beta=0}^{D-1} (Y_2)_{l,(i,\beta)} (\widetilde{Y}_1)_{s,(\beta,j)} \quad (19.63)$$

$$= (K^\dagger C \bar{T} T E K^\dagger)_{ls} = \sigma(K^\dagger C E K^\dagger)_{ls}. \quad (19.64)$$

Finally expand the matrix products and regroup the finite sums:

$$(K^\dagger C E K^\dagger)_{ls} = \sum_{a=0}^{r-1} \sum_{b=0}^{\ell-1} (K^\dagger)_{la} K_{sb} \sum_{\gamma=0}^{D-1} \overline{(\widetilde{X}_1)_{(i,\gamma),a}} (X_2)_{(\gamma,j),b} \quad (19.65)$$

$$= ((K^\dagger \otimes K)v^\dagger)_{(l,s),(i,j)}. \quad (19.66)$$

□

19.2.2 Unitarity of the first-cut comparison

Theorem 19.2.2.1 (Unitarity of the first-cut comparison). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU of physical dimension d and bond dimension D , with a canonical-form-II presentation and its recorded positive-definite matrix ρ . Let T be unitary on $\mathbb{C}^D$ and let $\sigma \in \mathbb{C}$. Suppose that, for every physical pair i, j , $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$, and that $T\bar{T} = \sigma I_D$. Construct the first source factor $\tilde{X}$ with this same ρ and the second source factor Y_2 . Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. The comparison

$$K = \tilde{X}^\dagger (I_d \otimes \rho) (I_d \otimes \bar{T}) Y_2^\dagger : \mathbb{C}^\ell \longrightarrow \mathbb{C}^r \quad (19.67)$$

is unitary between these spaces: $K^\dagger K = I_\ell$ and $KK^\dagger = I_r$. This is the comparison-unitarity step of [163, lines 5444–5487]. The comparison carries the weight of the chosen normalization [15, lines 487–494], which the printed definition of K at [163, lines 5436–5439] omits; the omission and its repair are recorded in [88].

Proof. [✓ Proof](#) Canonical form gives $d, D > 0$, and Lemma 18.9.10 gives $d^2 \leq r\ell$, so both rank spaces are nonempty. By Theorem 19.2.4.26, $\sigma\bar{\sigma} = 1$. Theorem 18.9.12 gives unitary source gates u, v , constructed with the recorded ρ . Put $H = K^\dagger \otimes K : \mathbb{C}^r \otimes \mathbb{C}^\ell \rightarrow \mathbb{C}^\ell \otimes \mathbb{C}^r$. Theorem 19.2.1.7 gives $u = \sigma H v^\dagger$. Cancelling the unitary right factor makes σH unitary. Its two Gram identities give

$$I_{r\ell} = (\sigma H)^\dagger (\sigma H) = \bar{\sigma} \sigma H^\dagger H = H^\dagger H, \quad (19.68)$$

$$I_{\ell r} = (\sigma H)(\sigma H)^\dagger = \sigma \bar{\sigma} H H^\dagger = H H^\dagger. \quad (19.69)$$

Thus H is unitary. Theorem 19.5.2 implies that K is unitary without identifying the two rank bases. $\square$

Theorem 19.2.2.2 (Adjoint inverse and weighted normalization). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU of dimensions d, D , with a canonical-form-II presentation and recorded positive-definite matrix ρ . Suppose that T is unitary on $\mathbb{C}^D$, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j , and that $T\bar{T} = \sigma I_D$ for a scalar $\sigma \in \mathbb{C}$. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. Form the first source factors with this same ρ , the candidate $X = (I_d \otimes \bar{T}) Y_2^\dagger$, and the comparisons $K : \mathbb{C}^\ell \rightarrow \mathbb{C}^r$ and $J : \mathbb{C}^r \rightarrow \mathbb{C}^\ell$ of Definitions 19.2.1.3 and 19.2.1.4. Then

$$J = K^\dagger, \quad (19.70)$$

$$X^\dagger (I_d \otimes \rho) X = I_\ell. \quad (19.71)$$

These give the adjoint identification and first weighted normalization in [163, lines 5432–5487], retaining the source normalization in [15, lines 487–494].

Proof. [✓ Proof](#) Theorems 19.2.2.1 and 19.2.1.5 give $K^\dagger K = I_\ell$ and $KJ = I_r$. Hence $J = (K^\dagger K)J = K^\dagger(KJ) = K^\dagger$. For the normalization, let $\tilde{X}$ be the first source factor constructed with ρ . Substituting $X = \tilde{X}K$ gives

$$X^\dagger (I_d \otimes \rho) X = K^\dagger [\tilde{X}^\dagger (I_d \otimes \rho) \tilde{X}] K = K^\dagger K = I_\ell. \quad (19.72)$$

$\square$

Theorem 19.2.2.3 (The unitary comparison at an involutive element). [Lean](#) [✓ Stmt](#) Let G be a group and let $\{\mathcal{U}_a\}_{a \in G}$ be a representation by simple injective MPUs of physical dimension

d. Suppose that the normalized flattening of $\mathcal{U}_a$ is left-canonical for every $a \in G$. Fix $g \in G$ with $g^{-1} = g$, and give $\mathcal{U}_g$ a canonical-form-II presentation with recorded positive-definite matrix ρ_g . Take the chosen gauge T_g of Theorem 19.2.14. For $\mathcal{U}_g$, set $r = \text{rank } M_1(\mathcal{U}_g)$ and $\ell = \text{rank } M_2(\mathcal{U}_g)$. Construct K_g, J_g with this T_g and this same ρ_g , and put $X_g = (I_d \otimes \overline{T}_g)Y_2^\dagger$, using its second-cut factor Y_2 . Then

$$K_g^\dagger K_g = I_\ell, \quad (19.73)$$

$$K_g K_g^\dagger = I_r, \quad (19.74)$$

$$J_g = K_g^\dagger, \quad (19.75)$$

$$X_g^\dagger (I_d \otimes \rho_g) X_g = I_\ell. \quad (19.76)$$

Canonical form II is required only at g ; left-canonicity of the normalized flattenings is required throughout the family.

Proof. ✓ Proof The representation makes $\mathcal{U}_g$ simple. Theorem 19.2.23 supplies $(\mathcal{U}_g^\dagger)^{ij} = T_g^\dagger \mathcal{U}_g^{ij} T_g$ for every physical pair. For the same chosen scalar σ_g , Theorem 19.2.15 gives $T_g \overline{T}_g = \sigma_g I_{D_g}$. Apply Theorems 19.2.2.1 and 19.2.2.2 with the recorded ρ_g . No new gauge, scalar, or simplicity assumption is needed. $\square$

19.2.3 The compatible decomposition and its literal gates

The candidate decomposition in [163, lines 5390–5487] yields source factors and unitary gates satisfying the factor and phase identities. This construction establishes the algebraic factor identities and the phase relation between the new gates, but does not establish the full collection of pleasant properties or the truncated-chain identity asserted in that passage.

Definition 19.2.3.1 (The identification of the first-cut coordinates). Lean ✓ Stmt Let $\mathcal{U}$ be an MPO tensor of dimensions d, D and let T be unitary on $\mathbb{C}^D$. Suppose that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j . Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. The equality $\ell = r$ of Theorem 19.2.1.1 defines the order-preserving bijection $e: \{0, \dots, \ell - 1\} \rightarrow \{0, \dots, r - 1\}$, with $e(l) = l$. Write $P_e: \mathbb{C}^\ell \rightarrow \mathbb{C}^r$ for its permutation matrix. This explicitly identifies the intermediate coordinates in the comparison of [163, lines 5432–5443].

Definition 19.2.3.2 (The inverse-compatible source factors). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $\mathcal{U}$ be a simple MPU in canonical form II, with recorded positive-definite matrix ρ . Suppose that T is unitary on $\mathbb{C}^D$, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j , and that $T \overline{T} = \sigma I_D$ for a scalar $\sigma \in \mathbb{C}$. Take the chosen second-cut factors X_2, Y_2, Z_2 and put $X = (I_d \otimes \overline{T})Y_2^\dagger$, $Y = X_2^\dagger (T^\top \otimes I_d)$. With e, P_e as in Definition 19.2.3.1, define the source factors $\widehat{X}_1, \widehat{Y}_1, \widehat{Z}_1, \widehat{X}_2, \widehat{Y}_2, \widehat{Z}_2$ by

$$\widehat{X}_1 = X P_e^\dagger, \quad (19.77)$$

$$\widehat{Y}_1 = P_e Y, \quad (19.78)$$

$$\widehat{Z}_1 = Y^\dagger P_e^\dagger, \quad (19.79)$$

$$\widehat{X}_2 = X_2, \quad (19.80)$$

$$\widehat{Y}_2 = Y_2, \quad (19.81)$$

$$\widehat{Z}_2 = Z_2. \quad (19.82)$$

Thus only the first intermediate coordinate is transported. For physical–virtual indices a , virtual–physical indices b , and $0 \leq l < \ell$, their coordinates are

$$(\widehat{X}_1)_{a,e(l)} = X_{a,l}, \quad (19.83)$$

$$(\widehat{Y}_1)_{e(l),b} = Y_{l,b}, \quad (19.84)$$

$$(\widehat{Z}_1)_{b,e(l)} = \overline{Y_{l,b}}. \quad (19.85)$$

These factors satisfy the two source-cut factorizations, the weighted first isometry, the ordinary second isometry, and the two right-inverse identities at the same weight ρ . This is the candidate construction of [163, lines 5390–5432], with its normalization supplied by the comparison argument at lines 5444–5487.

Proof. ✓ Proof The second factors and their identities are unchanged. Put $W = I_d \otimes \rho$. The candidate identities are $XY = M_1(\mathcal{U})$, $X^\dagger W X = I_\ell$, and $YY^\dagger = I_\ell$; the last follows because the right inverse $B^\dagger X_2$ of Y is $Y^\dagger$, where $B = T^\top \otimes I_d$. Transporting along P_e gives

$$\widehat{X}_1 \widehat{Y}_1 = X P_e^\dagger P_e Y = XY = M_1(\mathcal{U}), \quad (19.86)$$

$$\widehat{X}_1^\dagger W \widehat{X}_1 = P_e (X^\dagger W X) P_e^\dagger = I_r, \quad (19.87)$$

$$\widehat{Y}_1 \widehat{Z}_1 = P_e Y Y^\dagger P_e^\dagger = I_r. \quad (19.88)$$

Evaluating the transported matrices at $e(l)$ gives the coordinate formulas, since $e^{-1}(e(l)) = l$. $\square$

Theorem 19.2.3.3 (Unitary transport of the first source factors). Lean Lean ✓ Stmt Let $\mathcal{U}$ be a simple MPU of dimensions d, D in canonical form Π , with recorded positive-definite matrix ρ . Suppose that T is unitary on $\mathbb{C}^D$, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j , and that $T\bar{T} = \sigma I_D$ for a scalar $\sigma \in \mathbb{C}$. Let $\widetilde{X}_1, \widetilde{Y}_1$ be the chosen first source factors at this same weight, and let $\widehat{X}_1, \widehat{Y}_1$ be the inverse-compatible first factors of Definition 19.2.3.2. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. Take the weighted comparison $K: \mathbb{C}^\ell \rightarrow \mathbb{C}^r$ of Definition 19.2.1.3 and the permutation matrix $P_e: \mathbb{C}^\ell \rightarrow \mathbb{C}^r$ of Definition 19.2.3.1. The square matrix $C = K P_e^\dagger$ is unitary on $\mathbb{C}^r$, and

$$\widehat{X}_1 = \widetilde{X}_1 C, \quad (19.89)$$

$$\widehat{Y}_1 = C^\dagger \widetilde{Y}_1. \quad (19.90)$$

Consequently, the Gram matrices on the external first-cut spaces are unchanged:

$$\widehat{X}_1 \widehat{X}_1^\dagger = \widetilde{X}_1 \widetilde{X}_1^\dagger, \quad (19.91)$$

$$\widehat{Y}_1^\dagger \widehat{Y}_1 = \widetilde{Y}_1^\dagger \widetilde{Y}_1. \quad (19.92)$$

This is the transport of the comparison in [163, lines 5432–5443], after its unitarity is established at lines 5444–5487. It asserts no transport formula for the chosen first right inverse and no further pleasant properties.

Proof. ✓ Proof The comparison results give $K^\dagger K = I_\ell$, $KK^\dagger = I_r$, $X = \widetilde{X}_1 K$, and $Y = K^\dagger \widetilde{Y}_1$ for the untransported candidates X, Y . Reindexing the columns of K along e gives $C = K P_e^\dagger$. Since P_e identifies the two coordinate bases,

$$C^\dagger C = P_e K^\dagger K P_e^\dagger = I_r, \quad (19.93)$$

$$CC^\dagger = K P_e^\dagger P_e K^\dagger = I_r. \quad (19.94)$$

Reindexing the two candidate factors gives

$$\widehat{X}_1 = X P_e^\dagger = \widetilde{X}_1 K P_e^\dagger = \widetilde{X}_1 C, \quad (19.95)$$

$$\widehat{Y}_1 = P_e Y = P_e K^\dagger \widetilde{Y}_1 = C^\dagger \widetilde{Y}_1. \quad (19.96)$$

Cancel $CC^\dagger$ in the external Gram matrices:

$$\widehat{X}_1 \widehat{X}_1^\dagger = \widetilde{X}_1 (CC^\dagger) \widetilde{X}_1^\dagger = \widetilde{X}_1 \widetilde{X}_1^\dagger, \quad (19.97)$$

$$\widehat{Y}_1^\dagger \widehat{Y}_1 = \widetilde{Y}_1^\dagger (CC^\dagger) \widetilde{Y}_1 = \widetilde{Y}_1^\dagger \widetilde{Y}_1. \quad (19.98)$$

□

Theorem 19.2.3.4 (Physical closures of the compatible right factors). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU of dimensions d, D in canonical form II, with recorded positive-definite matrix ρ . Suppose that T is unitary, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and that $T\bar{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. Let $\widehat{Y}_1, \widehat{Y}_2$ be the inverse-compatible source factors at this same weight. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. For physical indices p, q , one has

$$\sum_{s=0}^{r-1} \sum_{\alpha=0}^{D-1} \overline{(\widehat{Y}_1)_{s,(\alpha,p)}} (\widehat{Y}_1)_{s,(\alpha,q)} = \delta_{p,q}, \quad (19.99)$$

$$\sum_{t=0}^{\ell-1} \sum_{\alpha,\beta=0}^{D-1} \overline{(\widehat{Y}_2)_{t,(p,\alpha)}} (\widehat{Y}_2)_{t,(q,\beta)} \rho_{\beta\alpha} = \delta_{p,q}. \quad (19.100)$$

These are the physical closures of [163, lines 869–917], transported as in lines 5486–5487. Simplicity of $\mathcal{U}^\dagger$ is not required.

Proof. [✓ Proof](#) Write $\widetilde{Y}_1, Y_2$ for the original source factors at ρ . The external Gram equality gives

$$\sum_{\alpha=0}^{D-1} (\widehat{Y}_1^\dagger \widehat{Y}_1)_{(\alpha,p),(\alpha,q)} = \sum_{\alpha=0}^{D-1} (\widetilde{Y}_1^\dagger \widetilde{Y}_1)_{(\alpha,p),(\alpha,q)} = \delta_{p,q}. \quad (19.101)$$

Expanding each Gram entry gives the first closure. The second follows from $\widehat{Y}_2 = Y_2$ and Theorem 19.4.4.2, with the same weight entries $\rho_{\beta\alpha}$. □

Theorem 19.2.3.5 (Rank normalizations of the compatible right factors). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU of dimensions d, D in canonical form II, with recorded positive-definite matrix ρ . Assume that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Suppose that T is unitary, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and that $T\bar{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. Let $\widehat{Y}_1, \widehat{Y}_2$ be its inverse-compatible source factors at ρ , with cut ranks r, ℓ . Then

$$\widehat{Y}_1 \widehat{Y}_1^\dagger = \frac{d}{r} I_r, \quad (19.102)$$

$$\widehat{Y}_2 (I_d \otimes \rho) \widehat{Y}_2^\dagger = \frac{d}{\ell} I_\ell. \quad (19.103)$$

These are the rank normalizations of [163, lines 1064–1119], transported at lines 5486–5487. The adjoint-simplicity hypothesis is retained.

Proof. [✓ Proof](#) Let $\widetilde{Y}_1, Y_2$ be the original factors. Unitary transport gives $\widehat{Y}_1 = C^\dagger \widetilde{Y}_1$ with $C^\dagger C = I_r$. Conjugating the original scalar Gram matrix yields

$$\widehat{Y}_1 \widehat{Y}_1^\dagger = C^\dagger (\widetilde{Y}_1 \widetilde{Y}_1^\dagger) C = C^\dagger \left(\frac{d}{r} I_r \right) C = \frac{d}{r} I_r. \quad (19.104)$$

The second normalization is unchanged because $\widehat{Y}_2 = Y_2$. □

Theorem 19.2.3.6 (Physical closures of the compatible left factors). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU of dimensions d, D in canonical form II, with recorded positive-definite matrix ρ . Assume that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Suppose that T is unitary, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and that $T\bar{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. Let $\widehat{X}_1, \widehat{X}_2$ be its inverse-compatible source factors at ρ , with cut ranks r, ℓ . For physical indices i, j ,

$$\sum_{s=0}^{r-1} \sum_{\alpha, \beta=0}^{D-1} (\widehat{X}_1)_{(i, \alpha), s} \overline{(\widehat{X}_1)_{(j, \beta), s}} = \frac{r}{d} \delta_{i, j}, \quad (19.105)$$

$$\sum_{t=0}^{\ell-1} \sum_{x=0}^{D-1} (\widehat{X}_2)_{(x, i), t} \overline{(\widehat{X}_2)_{(x, j), t}} = \frac{\ell}{d} \delta_{i, j}. \quad (19.106)$$

These are the physical left-factor closures of [163, lines 1120–1162], transported at lines 5486–5487. The canonical weight is not replaced by the identity.

Proof. [✓ Proof](#) Write $\widetilde{X}_1, X_2$ for the original factors. Multiply the $((i, \alpha), (j, \beta))$ entries of the external Gram equality by $\rho_{\beta\alpha}$ and sum over the virtual indices:

$$\sum_{\alpha, \beta=0}^{D-1} (\widehat{X}_1 \widehat{X}_1^\dagger)_{(i, \alpha), (j, \beta)} \rho_{\beta\alpha} = \sum_{\alpha, \beta=0}^{D-1} (\widetilde{X}_1 \widetilde{X}_1^\dagger)_{(i, \alpha), (j, \beta)} \rho_{\beta\alpha} \quad (19.107)$$

$$= \frac{r}{d} \delta_{i, j}. \quad (19.108)$$

Expanding each Gram entry and regrouping the finite sums gives the first closure. The second is the original physical closure because $\widehat{X}_2 = X_2$. $\square$

These six identities concern only the stated closures and normalizations. They do not assert movement identities, truncated-gate identities, transport of the chosen first right inverse, or the complete set of pleasant properties.

19.2.4 Movement gates of the inverse-compatible source factors

Theorem 19.2.4.1 (Transport of the movement gates). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU of dimensions d, D with a canonical-form-II presentation and recorded positive-definite matrix ρ . Let T be unitary on $\mathbb{C}^D$ and let $\sigma \in \mathbb{C}$ satisfy $T\bar{T} = \sigma I_D$. Suppose that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j . Let S_0 be the chosen source factors at this ρ , and let S be the inverse-compatible source factors of Definition 19.2.3.2 for the same data. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. Let $e: \{0, \dots, \ell-1\} \rightarrow \{0, \dots, r-1\}$ be the rank identification of Definition 19.2.3.1. Reindex the columns of the comparison K by e to obtain $C \in \text{Mat}_r(\mathbb{C})$, with $C_{a,b} = K_{a, e^{-1}(b)}$. Then the crossed movement gates satisfy

$$w_L(S) = w_L(S_0), \quad (19.109)$$

$$w_R(S) = (I_d \otimes C^\dagger) w_R(S_0) (C \otimes I_d). \quad (19.110)$$

The right gate has row order (physical, first rank) and column order (first rank, physical); no leg permutation occurs. These are the crossed contractions of [163, lines 811–869], transported by the unitary comparison as in [163, lines 5486–5487].

Proof. [✓ Proof](#) The second factors are unchanged, giving the left-gate equality. For the first factors, Theorem 19.2.3.3 gives $X_1(S) = X_1(S_0)C$ and $Y_1(S) = C^\dagger Y_1(S_0)$. For physical indices

i, j and first-rank indices t, t' , substitute these into the crossed contraction:

$$(w_R(S))_{(i,t),(t',j)} = \sum_{a,b=0}^{r-1} \overline{C_{b,t}} \left(\sum_{\beta=0}^{D-1} (X_1(S_0))_{(i,\beta),a} (Y_1(S_0))_{b,(\beta,j)} \right) C_{a,t'} \quad (19.111)$$

$$= \sum_{a,b=0}^{r-1} \overline{C_{b,t}} (w_R(S_0))_{(i,b),(a,j)} C_{a,t'}. \quad (19.112)$$

Regrouping the finite sums gives the right-gate formula in (19.110). $\square$

Theorem 19.2.4.2 (Unitarity of the transported movement gates). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU of dimensions d, D with a canonical-form-II presentation and recorded positive-definite matrix ρ . Suppose that T is unitary on $\mathbb{C}^D$, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j , and that $T\bar{T} = \sigma I_D$ for a scalar $\sigma \in \mathbb{C}$. Let S be its inverse-compatible source factors constructed with this same ρ . With source-cut ranks r, ℓ , both movement gates are unitary maps between their indicated spaces:

$$w_L(S): \mathbb{C}^d \otimes \mathbb{C}^\ell \longrightarrow \mathbb{C}^\ell \otimes \mathbb{C}^d, \quad (19.113)$$

$$w_R(S): \mathbb{C}^r \otimes \mathbb{C}^d \longrightarrow \mathbb{C}^d \otimes \mathbb{C}^r. \quad (19.114)$$

No simplicity hypothesis on the physical adjoint is required.

Proof. [✓ Proof](#) Theorem 19.2.3.3 makes C unitary, and Theorem 19.4.1.5 makes the chosen gates $w_L(S_0), w_R(S_0)$ unitary. The transport theorem leaves the left gate unchanged. For the right gate, use (19.110). The two outer factors of $(I_d \otimes C^\dagger)w_R(S_0)(C \otimes I_d)$ are unitary by Theorem 19.4.2.5; the middle factor $w_R(S_0)$ is already unitary. Their composition is the right gate. This is the movement-gate consequence of the comparison transport in [163, lines 5486–5487]. $\square$

Definition 19.2.4.3 (Literal gates on the common second-cut space). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be any MPO tensor of dimensions d, D and T any unitary virtual matrix. Put $\ell = \text{rank } M_2(\mathcal{U})$. From the candidates $X = (I_d \otimes \bar{T})Y_2^\dagger$ and $Y = X_2^\dagger(T^\dagger \otimes I_d)$ and the second factors X_2, Y_2 , define $u_{\text{new}}: \mathbb{C}^d \otimes \mathbb{C}^d \rightarrow \mathbb{C}^\ell \otimes \mathbb{C}^\ell$ and $v_{\text{new}}: \mathbb{C}^\ell \otimes \mathbb{C}^\ell \rightarrow \mathbb{C}^d \otimes \mathbb{C}^d$. For physical indices i, j and second-cut indices l, s , set

$$(u_{\text{new}})_{(l,s),(i,j)} = \sum_{\beta=0}^{D-1} (Y_2)_{l,(i,\beta)} Y_{s,(\beta,j)}, \quad (19.115)$$

$$(v_{\text{new}})_{(i,j),(l,s)} = \sum_{\beta=0}^{D-1} X_{(i,\beta),l} (X_2)_{(\beta,j),s}. \quad (19.116)$$

These are the literal contractions of the candidate factors, before transporting a rank coordinate. They are the new gates in [163, lines 5486–5487].

Theorem 19.2.4.4 (Comparison transport of the new left gate). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor, ρ a positive-definite virtual matrix, and T a unitary virtual matrix satisfying $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair. Let u be its source gate at weight ρ and J its reverse comparison at that weight. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. Then $u_{\text{new}} = (I_\ell \otimes J)u$. This is the comparison transport in [163, lines 5432–5443 and 5486–5487]; it does not require $J = K^\dagger$ or gate unitarity.

Proof. ✓ **Proof** Write $\widetilde{Y}_1$ for the old first factor. Substitute $Y = J\widetilde{Y}_1$ and interchange the finite sums:

$$(u_{\text{new}})_{(l,s),(i,j)} = \sum_{a=0}^{r-1} J_{sa} \sum_{\beta=0}^{D-1} (Y_2)_{l,(i,\beta)} (\widetilde{Y}_1)_{a,(\beta,j)} \quad (19.117)$$

$$= ((I_\ell \otimes J)u)_{(l,s),(i,j)}. \quad (19.118)$$

□

Theorem 19.2.4.5 (Comparison transport of the new right gate). Lean ✓ **Stmt** Let $\mathcal{U}$ be an MPO tensor, ρ a positive-definite virtual matrix, and T a unitary virtual matrix satisfying $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair. Let v be its source gate at weight ρ and K its weighted comparison. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$. Then $v_{\text{new}} = v(K \otimes I_\ell)$. This is the comparison transport used in [163, lines 5432–5443 and 5486–5487].

Proof. ✓ **Proof** Write $\widetilde{X}_1$ for the old first factor. Substitute $X = \widetilde{X}_1 K$ and interchange the finite sums:

$$(v_{\text{new}})_{(i,j),(l,s)} = \sum_{a=0}^{r-1} \left(\sum_{\beta=0}^{D-1} (\widetilde{X}_1)_{(i,\beta),a} (X_2)_{(\beta,j),s} \right) K_{al} \quad (19.119)$$

$$= (v(K \otimes I_\ell))_{(i,j),(l,s)}. \quad (19.120)$$

□

Theorem 19.2.4.6 (Unitarity and the phase relation of the new gates). Lean ✓ **Stmt** Let $\mathcal{U}$ be a simple MPU in canonical form II, with recorded positive-definite matrix ρ . Suppose that T is unitary, $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and $TT = \sigma I_D$ for $\sigma \in \mathbb{C}$. The literal gates of Definition 19.2.4.3 are unitary between $\mathbb{C}^d \otimes \mathbb{C}^d$ and $\mathbb{C}^\ell \otimes \mathbb{C}^\ell$, and satisfy $u_{\text{new}} = \sigma v_{\text{new}}^\dagger$. This is the new-gate conclusion of [163, lines 5440–5487]. The equality uses the common second-cut coordinate space, not the differently ordered rank spaces of the gates formed after transporting the first factors.

Proof. ✓ **Proof** Use the old source gates at the recorded weight ρ . The comparison results give $J = K^\dagger$ and $K^\dagger K = I_\ell$. Thus

$$u_{\text{new}} = (I_\ell \otimes K^\dagger)u, \quad (19.121)$$

$$v_{\text{new}} = v(K \otimes I_\ell). \quad (19.122)$$

By Theorem 18.9.12, u, v are unitary. Theorem 19.4.2.5 makes both Kronecker factors unitary, so the two compositions are unitary. Substitute the old gate identity and cancel $K^\dagger K$:

$$u_{\text{new}} = \sigma(I_\ell \otimes K^\dagger)(K^\dagger \otimes K)v^\dagger \quad (19.123)$$

$$= \sigma(K^\dagger \otimes I_\ell)v^\dagger = \sigma(v(K \otimes I_\ell))^\dagger = \sigma v_{\text{new}}^\dagger. \quad (19.124)$$

□

Theorem 19.2.4.7 (The new gates at an involutive group element). Lean ✓ **Stmt** Let G be a group and $\{\mathcal{U}_a\}_{a \in G}$ a representation by simple injective MPUs. Assume that the normalized flattening of $\mathcal{U}_a$ is left-canonical for every $a \in G$. Fix g with $g^{-1} = g$ and give $\mathcal{U}_g$ a canonical-form-II presentation. Use its chosen gauge T_g and scalar σ_g from Theorems 19.2.14 and 19.2.15.

The literal new gates of $\mathcal{U}_g, T_g$ are unitary and satisfy $u_{\text{new},g} = \sigma_g v_{\text{new},g}^\dagger$ on the common second-cut rank-pair space. Canonical form II is needed only at g ; left-canonicity of normalized flattenings is required for the whole family. This specializes the construction of [163, lines 5440–5487] using its dagger-inverse gauges at lines 1552–1562.

Proof. ✓ Proof The representation supplies simplicity at g . Its chosen gauge and scalar satisfy

$$(\mathcal{U}_g^\dagger)^{ij} = T_g^\dagger \mathcal{U}_g^{ij} T_g, \quad (19.125)$$

$$T_g \overline{T_g} = \sigma_g I_{D_g}. \quad (19.126)$$

Apply Theorem 19.2.4.6 with the canonical-form-II presentation at g . □

Theorem 19.2.4.8 (Source properties at an involutive group element). Lean ✓ Stmt Let G be a group and $\{\mathcal{U}_a\}_{a \in G}$ a representation by simple injective MPUs whose normalized flattenings are all left-canonical. Fix g with $g^{-1} = g$, give $\mathcal{U}_g$ a canonical-form-II presentation with recorded positive-definite matrix ρ , and build the inverse-compatible source factors $\widehat{X}_1, \widehat{Y}_1, \widehat{X}_2, \widehat{Y}_2$ from the chosen gauge T_g and scalar σ_g of Theorems 19.2.14 and 19.2.15. Write r, ℓ for the two source-cut ranks of $\mathcal{U}_g$ and D_g for its bond dimension. Then all three pairs of source properties hold: for physical indices p, q ,

$$\sum_{s=0}^{r-1} \sum_{\alpha=0}^{D_g-1} \overline{(\widehat{Y}_1)_{s,(\alpha,p)}} (\widehat{Y}_1)_{s,(\alpha,q)} = \delta_{p,q}, \quad (19.127)$$

$$\sum_{t=0}^{\ell-1} \sum_{\alpha, \beta=0}^{D_g-1} \overline{(\widehat{Y}_2)_{t,(p,\alpha)}} (\widehat{Y}_2)_{t,(q,\beta)} \rho_{\beta\alpha} = \delta_{p,q}; \quad (19.128)$$

the two rank normalizations

$$\widehat{Y}_1 \widehat{Y}_1^\dagger = \frac{d}{r} I_r, \quad (19.129)$$

$$\widehat{Y}_2 (I_d \otimes \rho) \widehat{Y}_2^\dagger = \frac{d}{\ell} I_\ell; \quad (19.130)$$

and, for physical indices i, j ,

$$\sum_{s=0}^{r-1} \sum_{\alpha, \beta=0}^{D_g-1} (\widehat{X}_1)_{(i,\alpha),s} \rho_{\beta\alpha} \overline{(\widehat{X}_1)_{(j,\beta),s}} = \frac{r}{d} \delta_{i,j}, \quad (19.131)$$

$$\sum_{t=0}^{\ell-1} \sum_{x=0}^{D_g-1} (\widehat{X}_2)_{(x,i),t} \overline{(\widehat{X}_2)_{(x,j),t}} = \frac{\ell}{d} \delta_{i,j}. \quad (19.132)$$

Simplicity of $\mathcal{U}_g^\dagger$, which [163, line 5344] assumes, is here a consequence of the canonical-form-II presentation at g , establishing the properties labelled eq:MPUnice2–eq:MPUnice4 in [163, lines 1064–1162], transported as at lines 5486–5487.

Proof. ✓ Proof The representation supplies simplicity of $\mathcal{U}_g$, and its chosen gauge and scalar satisfy $(\mathcal{U}_g^\dagger)^{ij} = T_g^\dagger \mathcal{U}_g^{ij} T_g$ and $T_g \overline{T_g} = \sigma_g I_{D_g}$ by Theorems 19.2.23 and 19.2.15. For the adjoint premise, Theorem 18.9.12 turns simplicity of the canonical-form-II tensor $\mathcal{U}_g$ into $r\ell = d^2$. The adjoint carries the canonical boundaries of Definition 19.5.5, and Theorem 18.4.25 exchanges the two source-cut ranks, so $\mathcal{U}_g^\dagger$ has rank product $\ell r = d^2$ and Theorem 18.9.12 makes it simple as well. The three pairs are then Theorems 19.2.3.4, 19.2.3.5, and 19.2.3.6 at $\mathcal{U}_g, T_g$. □

Theorem 19.2.4.9 (The adjoint gauge along a word). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor of dimensions d, D and T a unitary matrix on $\mathbb{C}^D$. Suppose that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair i, j . For equal-length physical words p, q , including the empty words, their ordered matrix products satisfy

$$(\mathcal{U}^\dagger)[p, q] = T^\dagger \mathcal{U}[p, q] T. \quad (19.133)$$

This iterates the local gauge relation of [163, lines 1552–1557].

Proof. [✓ Proof](#) Induct on the first word, using equality of lengths to split the second word at the same position. The empty case is $I_D = T^\dagger I_D T$. For $p = ip'$ and $q = jq'$, cancellation gives

$$(\mathcal{U}^\dagger)[ip', jq'] = T^\dagger \mathcal{U}^{ij} T T^\dagger \mathcal{U}[p', q'] T = T^\dagger \mathcal{U}[ip', jq'] T. \quad (19.134)$$

□

Theorem 19.2.4.10 (Recovering the second endpoint from the candidate). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be any MPO tensor and T any unitary virtual matrix. For $A = I_d \otimes \bar{T}$ and the candidate $X = AY_2^\dagger$, one has $Y_2 = X^\dagger A$. This is the endpoint recovery from the candidate in [163, line 5432].

Proof. [✓ Proof](#) Unitarity gives $A^\dagger A = I$, so $X^\dagger A = Y_2 A^\dagger A = Y_2$. □

Definition 19.2.4.11 (The two endpoint matrices along a word). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor of dimensions d, D and T a unitary virtual matrix. Take the second source factors X_2, Y_2 and the raw candidates $X = (I_d \otimes \bar{T})Y_2^\dagger$, $Y = X_2^\dagger(T^\top \otimes I_d)$. Put $\ell = \text{rank } M_2(\mathcal{U})$ and $\Omega_N = \{0, \dots, d-1\}^{\{0, \dots, N-1\}}$ for $N \geq 0$. Define

$$\mathcal{H}_N = \mathbb{C}^d \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^d), \quad (19.135)$$

$$\mathcal{R}_N = \mathbb{C}^\ell \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^\ell). \quad (19.136)$$

The matrices $A_N: \mathcal{H}_N \rightarrow \mathcal{R}_N$ and $B_N: \mathcal{R}_N \rightarrow \mathcal{H}_N$ have the following entries for $p, q \in \Omega_N$, physical indices a, b , and source indices l, s :

$$(A_N)_{(l, (p, s)), (a, (q, b))} = \sum_{\alpha, \beta=0}^{D-1} (Y_2)_{l, (a, \alpha)} \mathcal{U}[p, q]_{\alpha\beta} Y_{s, (\beta, b)}, \quad (19.137)$$

$$(B_N)_{(a, (p, b)), (l, (q, s))} = \sum_{\alpha, \beta=0}^{D-1} X_{(a, \alpha), l} \mathcal{U}[p, q]_{\alpha\beta} (X_2)_{(\beta, b), s}. \quad (19.138)$$

The words are read in their given order. These are the all-word endpoint contractions extending [163, lines 5444–5487]. With $\emptyset$ the unique empty configuration, the zero-length entries recover the literal new gates:

$$(A_0)_{(l, (\emptyset, s)), (a, (\emptyset, b))} = (u_{\text{new}})_{(l, s), (a, b)}, \quad (19.139)$$

$$(B_0)_{(a, (\emptyset, b)), (l, (\emptyset, s))} = (v_{\text{new}})_{(a, b), (l, s)}. \quad (19.140)$$

Proof. [✓ Proof](#) At length zero, $\mathcal{U}[\emptyset, \emptyset] = I_D$. The entry $\delta_{\alpha\beta}$ contracts the two sums, respectively, to

$$\sum_{\alpha} (Y_2)_{l, (a, \alpha)} Y_{s, (\alpha, b)}, \quad (19.141)$$

$$\sum_{\alpha} X_{(a, \alpha), l} (X_2)_{(\alpha, b), s}. \quad (19.142)$$

□

Theorem 19.2.4.12 (All-word adjunction of the endpoints). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor and T a unitary virtual matrix. Suppose that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair and that $T\bar{T} = \sigma I_D$ for a scalar $\sigma \in \mathbb{C}$. For every $N \geq 0$, the endpoint matrices of Definition 19.2.4.11 satisfy $A_N = \sigma B_N^\dagger$. The scalar occurs once, independently of N . Adjoining exchanges the physical configurations without reversing either word or inserting an additional swap. No canonical form, simplicity, comparison matrix, or rank transport is required. This is an all-word local extension of [163, lines 5444–5487], not the final truncated-chain sewing identity.

Proof. [✓ Proof](#) Fix $p, q \in \Omega_N$ and physical indices a, b . Put $M = \mathcal{U}[p, q]$ and $W = \overline{\mathcal{U}[q, p]}$, and define slices $C_{t\gamma} = \overline{X_{(a, \gamma), t}}$ and $E_{\delta t} = \overline{X_{2(\delta, b), t}}$. Physical adjunction conjugates each letter and exchanges its two physical indices, so $W = (\mathcal{U}^\dagger)[p, q] = T^\dagger M T$ by Theorem 19.2.4.9. The phase relation gives $\bar{T} = \sigma T^\dagger$. Endpoint recovery and the definition of Y give, for virtual indices α, β ,

$$(Y_2)_{l, (a, \alpha)} = (C\bar{T})_{l\alpha}, \quad (19.143)$$

$$Y_{s, (\beta, b)} = (TE)_{\beta s}. \quad (19.144)$$

The two virtual sums therefore contract as

$$(A_N)_{(l, (p, s)), (a, (q, b))} = (C\bar{T}MTE)_{ls} = \sigma(C(T^\dagger M T)E)_{ls} = \sigma(CWE)_{ls}. \quad (19.145)$$

Expanding the remaining product gives

$$(CWE)_{ls} = \overline{\sum_{\alpha, \beta=0}^{D-1} X_{(a, \alpha), t} \mathcal{U}[q, p]_{\alpha\beta} X_{2(\beta, b), s}} \quad (19.146)$$

$$= \overline{(B_N)_{(a, (q, b)), (l, (p, s))}}. \quad (19.147)$$

This is precisely the entry of $B_N^\dagger$ at the original row and column, with the spatial order unchanged. $\square$

Definition 19.2.4.13 (Common positive physical blocking). [Lean](#) [✓ Stmt](#) For a positive integer L , the common length- L physical blocking is the family

$$\tilde{\mathcal{U}}_g = \mathcal{U}_g^{[L]}. \quad (19.148)$$

Each blocked tensor has physical dimension d^L and the same virtual bond dimension D_g as $\mathcal{U}_g$. The physical alphabet is replaced simultaneously for every group element by words of length L .

Theorem 19.2.4.14 (Common blocking preserves the representation laws). [Lean](#) [Lean](#) [✓ Stmt](#) Let $L > 0$. The commonly blocked family $\{\mathcal{U}_g^{[L]}\}_{g \in G}$ is again a group representation by simple injective MPUs. Under the canonical identification of blocked configurations with words of length NL ,

$$O_N(\mathcal{U}_g^{[L]}) \cong U_g^{(NL)}, \quad (19.149)$$

$$O_N(\mathcal{U}_e^{[L]}) = \mathbb{1}, \quad (19.150)$$

$$O_N(\mathcal{U}_g^{[L]})O_N(\mathcal{U}_h^{[L]}) = O_N(\mathcal{U}_{gh}^{[L]}) \quad (19.151)$$

for every $N \geq 1$. Physical blocking also commutes with tensor products:

$$(\mathcal{U}_g \cdot \mathcal{U}_h)^{[L]} = \mathcal{U}_g^{[L]} \cdot \mathcal{U}_h^{[L]}. \quad (19.152)$$

Proof. ✓ Proof Physical blocking preserves nonempty-chain unitarity, injectivity, and simplicity. By (19.149), the blocked periodic operator is the original operator at length NL up to the canonical simultaneous reindexing. Since $N, L > 0$, the length NL is positive, so (19.7) and (19.8) apply. Simultaneous reindexing preserves identities and products, which gives (19.150) and (19.151). Expanding one blocked letter gives (19.152). $\square$

Assume from now until the end of this section that G is finite and $d > 0$.

Definition 19.2.4.15 (Raw MPU representation). Lean ✓ Stmt A raw MPU representation is a dependent-bond family $\{\mathcal{U}_g\}_{g \in G}$ of injective matrix product unitaries on nonempty chains satisfying (19.7) and (19.8) for all $g, h \in G$ and all $N \geq 1$. No simplicity hypothesis is imposed.

Definition 19.2.4.16 (Common simplicity length). Lean ✓ Stmt For a finite dependent-bond family, define

$$L_* = \sum_{g \in G} D_g^4. \quad (19.153)$$

Theorem 19.2.4.17 (A finite MPU family has a common simple blocking). Lean Lean Lean ✓ Stmt Let $\{\mathcal{U}_g\}_{g \in G}$ be a raw MPU representation. Then $L_* > 0$, and every $\mathcal{U}_g^{[L_*]}$ is simple. For every $K > 0$, each further block $(\mathcal{U}_g^{[L_*]})^{[K]}$ is simple. Consequently the blocked family at length L_* satisfies Definition 19.2.4, and every further common positive blocking does as well.

Proof. ✓ Proof Positivity of the bond dimensions gives $L_* > 0$. For each g , the direct block of length D_g^4 is simple, and $D_g^4 \leq L_*$ by (19.153). Simplicity holds at every later direct blocking, hence at the common length L_* . Nonempty-chain unitarity and injectivity are preserved by physical blocking. Reindexing the closed chain at length NL_* carries the identity and multiplication laws of the raw representation to the blocked family. The length- L_* family is therefore a group representation by simple injective MPUs. Applying Theorem 19.2.4.14 to this established representation proves the assertion for every further positive blocking. $\square$

Definition 19.2.4.18 (Selected reductions for the pairs of a representation). Lean Lean Lean Lean ✓ Stmt For $g, h \in G$, write

$$B_{g,h} = \mathcal{U}_g \cdot \mathcal{U}_h, \quad A_{g,h} = \mathcal{U}_{gh} \quad (19.154)$$

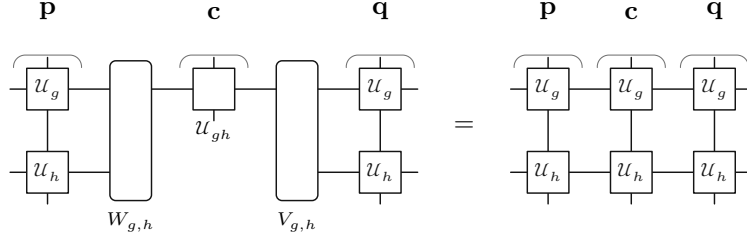
for the product tensor and the target tensor, both regarded as MPS tensors whose physical letters are the ket–bra pairs (i, j) , of bond dimensions $D_g D_h$ and D_{gh} . A *family of selected reductions* consists of matrices

$$V_{g,h} \in \mathbb{C}^{D_{gh} \times D_g D_h}, \quad W_{g,h} \in \mathbb{C}^{D_g D_h \times D_{gh}} \quad (19.155)$$

such that $(V_{g,h}, W_{g,h})$ is a reduction from $B_{g,h}$ to $A_{g,h}$ for every pair: $V_{g,h} W_{g,h} = 1$ and $V_{g,h} B_{g,h}^{\mathbf{a}} W_{g,h} = A_{g,h}^{\mathbf{a}}$ for every word $\mathbf{a}$. The identity on words is the interior fusion equation in [163, equation labelled *eq:fusion_2*, lines 1459–1497], with $F_{g,h}^< = V_{g,h}$ and $F_{g,h}^> = W_{g,h}$. Write $N_0(g, h) = \ell(B_{g,h}, A_{g,h}; V_{g,h}, W_{g,h})$ for the residual nilpotency length of the selected reduction of (g, h) . The family has *exterior buffer length* m if m is an exterior buffer length for every selected reduction:

$$B_{g,h}^{\mathbf{p}} W_{g,h} A_{g,h}^{\mathbf{c}} V_{g,h} B_{g,h}^{\mathbf{q}} = B_{g,h}^{\mathbf{pcq}} \quad (19.156)$$

for all $g, h \in G$, every nonempty word $\mathbf{c}$, and all words with $|\mathbf{p}|, |\mathbf{q}| \geq m$. This is the exterior fusion equation in [163, equation labelled *eq:fusion_1*, lines 1406–1457, with the quantifiers on line 1498], with $m \geq \ell$.



Theorem 19.2.4.19 (Existence of selected reductions). [Lean](#) [✓ Stmt](#) Every MPU representation by simple injective tensors admits a family of selected reductions.

Proof. [✓ Proof](#) By (19.13), $B_{g,h}$ and $A_{g,h}$ have the same positive-length matrix product vectors, and $A_{g,h}$ is injective with $D_{gh} > 0$. Theorem 25.11.5 gives a reduction for each pair; choose one for every pair. $\square$

Lemma 19.2.4.20 (Exterior buffers from the nilpotency lengths). [Lean](#) [Lean](#) [✓ Stmt](#) For a family of selected reductions of an MPU representation by simple injective tensors, $N_0(g, h) - 1$ is an exterior buffer length for the selected reduction of (g, h) . Consequently, if $N_0(g, h) \leq m + 1$ for every pair, the family has exterior buffer length m .

Proof. [✓ Proof](#) Equality of the positive-length matrix product vectors of $B_{g,h}$ and $A_{g,h}$ makes $N_0(g, h)$ a residual-word nilpotency bound, so Theorem 25.11.25 applies; enlarging the buffer preserves (19.156). $\square$

Definition 19.2.4.21 (Blocked selected reductions and the common buffer length). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $L > 0$. Under the canonical pairing of blocked ket and bra words,

$$\mathcal{U}_g^{[L]} \cdot \mathcal{U}_h^{[L]} \cong B_{g,h}^{[L]}, \quad \mathcal{U}_{gh}^{[L]} \cong A_{g,h}^{[L]}, \quad (19.157)$$

so a family of selected reductions $(V_{g,h}, W_{g,h})$ of $\{\mathcal{U}_g\}_{g \in G}$ is, with the same matrices, a family of selected reductions of the blocked representation $\{\mathcal{U}_g^{[L]}\}_{g \in G}$. The *common buffer length* of the family is the common buffer length (25.40) of the finite family $(N_0(g, h))_{(g,h) \in G \times G}$; it is positive and satisfies $N_0(g, h) \leq L + 1$ for every pair.

Theorem 19.2.4.22 (Common blocking for the one-site fusion identities). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7396](#) Let $\{\mathcal{U}_g\}_{g \in G}$ be an MPU representation by simple injective tensors with a family of selected reductions, and let $L > 0$ satisfy $N_0(g, h) \leq L + 1$ for every pair. Then the blocked family of selected reductions of $\{\mathcal{U}_g^{[L]}\}_{g \in G}$ has exterior buffer length 1: for all $g, h \in G$ and all nonempty blocked words $\mathbf{p}, \mathbf{c}, \mathbf{q}$,

$$(B_{g,h}^{[L]})^{\mathbf{p}} W_{g,h} (A_{g,h}^{[L]})^{\mathbf{c}} V_{g,h} (B_{g,h}^{[L]})^{\mathbf{q}} = (B_{g,h}^{[L]})^{\mathbf{pcq}}. \quad (19.158)$$

The common buffer length is such an L . Consequently, every MPU representation of a finite group by simple injective tensors admits a positive L such that $\{\mathcal{U}_g^{[L]}\}_{g \in G}$ is again an MPU representation by simple injective tensors and has a family of selected reductions with exterior buffer length 1.

This is the assertion of [163], *main.tex* line 1498, and of the footnote to Theorem 1 of [166], that the nilpotency length can be reduced to 1 by blocking, read as the one-site exterior identity (19.158). Blocking need not make a blocked residual letter vanish, and the blocked nilpotency length $\ell(B_{g,h}^{[L]}, A_{g,h}^{[L]}; V_{g,h}, W_{g,h})$ of Definition 8 in [187] need not be 1 [90].

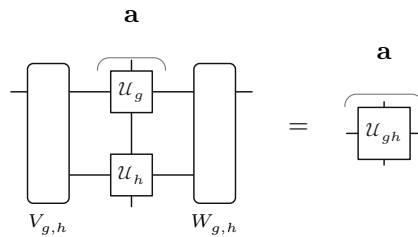
Proof. ✓ Proof By Lemma 19.2.4.20, $N_0(g, h) - 1$ is an exterior buffer length for the pair (g, h) , and $N_0(g, h) - 1 \leq L$. Theorem 25.11.25 with $k = 1$ makes 1 an exterior buffer length for $(V_{g,h}, W_{g,h})$ as a reduction from $B_{g,h}^{[L]}$ to $A_{g,h}^{[L]}$, and the pairing (19.157), obtained from the commutation of blocking with the doubled-index view and with the product tensor, transports it to the blocked representation. For the last assertion, choose a family of selected reductions by Theorem 19.2.4.19, take L to be its common buffer length, and apply Theorem 19.2.4.14. $\square$

Definition 19.2.4.23 (Fusion tensors of a pair). Lean Lean ✓ Stmt Let $\{\mathcal{U}_g\}_{g \in G}$ be an MPU representation by simple injective tensors with a family of selected reductions. The *fusion tensors* of the pair (g, h) are the two matrices of the selected reduction of that pair,

$$F_{g,h}^< = V_{g,h}, \quad F_{g,h}^> = W_{g,h}. \quad (19.159)$$

Under this identification the reduction identities $V_{g,h} W_{g,h} = \mathbb{1}$ and $V_{g,h} B_{g,h}^{\mathbf{a}} W_{g,h} = A_{g,h}^{\mathbf{a}}$ are the interior fusion equation of [163, lines 1459–1497], and (19.156) is its exterior fusion equation.

Two features of the source are recorded here. First, the existence of the fusion tensors is not proved in [163]: at `main.tex` line 1405 it is attributed to [187], whose Proposition 20 and Theorem 22 carry the existence and the uniqueness; the existence statement used here is Theorem 19.2.4.19. Second, the reduction of the nilpotency length to 1 by blocking is asserted at `main.tex` line 1498 without proof; the content used here is the one-site exterior identity (19.158). Blocking need not make a blocked residual letter vanish, and the blocked nilpotency length of Definition 8 in [187] need not be 1 [90].



Theorem 19.2.4.24 (Reciprocal scalar rescaling of the fusion tensors). Lean Lean Lean Lean ✓ Stmt Discussion: #7397 Let (V, W) be a reduction from B to A and let $\beta \in \mathbb{C}^\times$. Then $(\beta^{-1}V, \beta W)$ is again a reduction from B to A ; its residual letters $N^i = B^i - W A^i V$ are unchanged, hence so are its residual-word nilpotency bounds and its residual nilpotency length $\ell(B, A; V, W)$; and a natural number is an exterior buffer length for $(\beta^{-1}V, \beta W)$ if and only if it is one for (V, W) .

Consequently, let $\beta: G^2 \rightarrow \mathbb{C}^\times$ be a scalar 2-cochain and let $\{\mathcal{U}_g\}_{g \in G}$ have a family of selected reductions. The rescaled matrices of (19.257), namely $W_{g,h} \mapsto \beta_{g,h} W_{g,h}$ and $V_{g,h} \mapsto \beta_{g,h}^{-1} V_{g,h}$, again form a family of selected reductions of $\{\mathcal{U}_g\}_{g \in G}$. It has exterior buffer length m exactly when the original family does, and rescaling commutes with blocking by any length L .

This is the scalar gauge freedom of the fusion tensors in [163, lines 1500–1504] under the identification (19.159); uniqueness up to boundary dressing is given by Theorem 25.11.28.

Proof. ✓ Proof In each identity the two scalars occur in one of the combinations $(\beta^{-1}V)(\beta W)$, $(\beta W)A^c(\beta^{-1}V)$, and $(\beta^{-1}V)B^a(\beta W)$; in every case they multiply to $\beta\beta^{-1} = 1$ and disappear. The residual letters and the exterior identity are therefore unchanged, and equality of the two sets of residual-word nilpotency bounds gives equality of their infima. The statements for a family follow pair by pair, and blocking changes only the physical alphabet of the tensors, leaving $V_{g,h}$ and $W_{g,h}$ untouched. $\square$

Theorem 19.2.4.25 (Zero specified source-index value determines the source ranks). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a specified MPO tensor of positive physical dimension d , with positive right and left source-cut ranks r and ℓ . If

$$r\ell = d^2 \quad (19.160)$$

and $\text{ind}_{\text{src}}(\mathcal{U}) = 0$, then

$$r = d, \quad (19.161)$$

$$\ell = d. \quad (19.162)$$

This is the arithmetic conclusion in [163, line 1547], under the displayed hypotheses.

Proof. [✓ Proof](#) The source-index identity gives $0 = \log_2(r/d)$. Since $r/d > 0$, this implies $r/d = 1$, hence $r = d$. Substitution into $r\ell = d^2$ and cancellation of the positive factor d give $\ell = d$. $\square$

Theorem 19.2.4.26 (Scalar phase from a unitary relation). [Lean](#) [✓ Stmt](#) Let I be a nonempty finite set, let $T, S \in \text{Mat}_I(\mathbb{C})$ be unitary, and let $\sigma \in \mathbb{C}$. If

$$TS^* = \sigma \mathbf{1}, \quad (19.163)$$

then

$$\sigma \bar{\sigma} = 1. \quad (19.164)$$

This is the scalar phase conclusion following [163, lines 1557–1562] once the matrix relation has been supplied.

Proof. [✓ Proof](#) The matrix S^* is unitary, so $TS^* = \sigma \mathbf{1}$ is unitary. Hence

$$(\sigma \mathbf{1})(\sigma \mathbf{1})^\dagger = \sigma \bar{\sigma} \mathbf{1} = \mathbf{1}. \quad (19.165)$$

Evaluating a diagonal entry gives (19.164). $\square$

Theorem 19.2.4.27 (Reciprocal scalars in the paired relations). [Lean](#) [✓ Stmt](#) Let I be a nonempty finite set, let $T, S \in \text{Mat}_I(\mathbb{C})$ be unitary, and let $\sigma, \tau \in \mathbb{C}$. If

$$TS^* = \sigma \mathbf{1}, \quad (19.166)$$

$$ST^* = \tau \mathbf{1}, \quad (19.167)$$

then

$$\sigma \tau = 1. \quad (19.168)$$

This is the conditional matrix-algebra content of $\sigma_g \sigma_{g^{-1}} = 1$ in [163, lines 1559–1563].

Proof. [✓ Proof](#) Multiplying (19.166) by T^{-1} on the left and T on the right gives $S^*T = \sigma \mathbf{1}$. Entrywise conjugation of (19.167) gives $S^*T = \bar{\tau} \mathbf{1}$. A diagonal entry therefore gives $\sigma = \bar{\tau}$. Applying Theorem 19.2.4.26 to (19.167) yields $\tau \bar{\tau} = 1$, and hence $\sigma \tau = 1$. $\square$

Theorem 19.2.4.28 (Sign dichotomy in the self relation). [Lean](#) [✓ Stmt](#) Let I be a nonempty finite set, let $T \in \text{Mat}_I(\mathbb{C})$ be unitary, and let $\sigma \in \mathbb{C}$. If

$$TT^* = \sigma \mathbf{1}, \quad (19.169)$$

then

$$\sigma \in \{1, -1\}. \quad (19.170)$$

This is the sign step $\mathbb{1} = xx^\dagger = e^{2i\phi}\mathbb{1}$ in [15, lines 1036–1038], and the involution conclusion in [163, lines 1563–1567], conditional on the self relation.

Proof. ✓ Proof Apply Theorem 19.2.4.27 with $S = T$ and $\tau = \sigma$. Then $\sigma^2 = 1$, so $\sigma = 1$ or $\sigma = -1$. □

19.3 A nondegenerate alternating-network comparison

Definition 19.3.1 (Connected open alternating network). Lean ✓ Stmt Let $n \geq 0$, and let $D_0, \dots, D_n$ be horizontal bond dimensions. At each $0 \leq i < n$, let A_i have square bond dimension D_i , and let B_i have left and right bond dimensions D_i and D_{i+1} . For fixed physical indices $\sigma = (\sigma_i)_i$ and $\tau = (\tau_i)_i$, define the connected network with its horizontal ends open by

$$\mathcal{N}_{A,B}(\sigma, \tau)_{a,b} := \sum_{\substack{x_0, \dots, x_n \\ x_0=a, x_n=b}} \sum_{y_0, \dots, y_{n-1}} \prod_{i=0}^{n-1} (A_i^{\sigma_i})_{x_i, y_i} (B_i^{\tau_i})_{y_i, x_{i+1}}.$$

The indices y_i join A_i to B_i , while x_{i+1} joins B_i to A_{i+1} when $i+1 < n$; x_0 and x_n are the two open end indices. This is the coordinate form of the connected open diagram in [163, lines 2299–2361].

Lemma 19.3.2 (One-sided-inverse contraction of an alternating network). Lean ✓ Stmt Suppose that every A_i is injective. For the matrix unit E_{z_i, q_i} , let $c_i^{\sigma_i}(z_i, q_i)$ be its coefficients under a one-sided inverse of A_i , so that

$$\sum_{\sigma_i} c_i^{\sigma_i}(z_i, q_i) A_i^{\sigma_i} = E_{z_i, q_i}.$$

Then, for every horizontal path $z = (z_i)_{i=0}^n$, every intermediate path $q = (q_i)_{i=0}^{n-1}$, and every τ , one has

$$\sum_{\sigma_0, \dots, \sigma_{n-1}} \left(\prod_{i=0}^{n-1} c_i^{\sigma_i}(z_i, q_i) \right) \mathcal{N}_{A,B}(\sigma, \tau)_{z_0, z_n} = \prod_{i=0}^{n-1} (B_i^{\tau_i})_{q_i, z_{i+1}}.$$

This is the simultaneous inverse-tensor contraction in [163, lines 2365–2392].

Proof. ✓ Proof Expand the connected network and interchange the finite sums. At site i , the sum over σ_i is the (x_i, y_i) entry of E_{z_i, q_i} . Its two Kronecker deltas force $x_i = z_i$ and $y_i = q_i$. Thus every internal sum collapses, leaving precisely the displayed product of entries of the B_i . □

Theorem 19.3.3 (Nondegenerate corrected alternating-network proportionality). Lean ✓ Stmt Let $n \geq 0$, and let $D_0, \dots, D_n$ be horizontal bond dimensions. For each $0 \leq i < n$, let A_i be an injective MPS tensor with square bond dimension D_i , and let B_i, B'_i be tensors whose left and right horizontal bond dimensions are D_i and D_{i+1} . Suppose that every B_i is nonzero and that the connected open networks agree:

$$A_0 B_0 A_1 B_1 \cdots A_{n-1} B_{n-1} = A_0 B'_0 A_1 B'_1 \cdots A_{n-1} B'_{n-1}. \quad (19.171)$$

Here the equality retains the two horizontal end legs and all physical legs. Then there are scalars $\beta_i \in \mathbb{C}$, for $0 \leq i < n$, such that

$$B_i = \beta_i B'_i, \quad (19.172)$$

$$\prod_{i=0}^{n-1} \beta_i = 1. \quad (19.173)$$

This is the nondegenerate corrected form of Lemma [lemma:1](#) in [163, lines 2296–2395]. The printed wording omits nonvanishing. The paper’s truncated symmetry U_g^L is unitary at line 2099; lines 2444–2466 identify the side used at line 2567 with $(U_{g^{-1}}^L)^\dagger$, while lines 3033–3053 identify the side used at line 3157 with $\sigma_g(U_{g^{-1}}^L)^\dagger$ for $\sigma_g \in \{1, -1\}$. Both displayed networks are therefore nonzero. A vanishing intervening factor would make its entire network zero, so taking the factors on these sides as the B_i supplies $B_i \neq 0$ at both applications. This nondegeneracy condition is recorded in [92].

Proof. [✓ Proof](#) Apply Lemma 19.3.2 to both sides of (19.171), choosing the matrix units to prescribe arbitrary entries of all B_i . The resulting connected horizontal contractions retain precisely those entries. Since the entries were arbitrary, this gives the pure-tensor identity

$$\bigotimes_{i=0}^{n-1} B_i = \bigotimes_{i=0}^{n-1} B'_i. \quad (19.174)$$

Apply Theorem 22.2.3.1 to (19.174), with global scalar 1 and the nonzero reference entries of the B_i . It gives the factorwise proportionalities and (19.173); reversing the resulting nonzero scalars gives the orientation (19.172). $\square$

19.4 Staircase gates and the first source network

Definition 19.4.1 (Staircase gates). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7314](#) Let $\mathcal{U}$ be an MPO tensor of physical dimension d and bond dimension D . Put $d_l = \ell$ and $d_r = r$ for its source-cut ranks. Let $X_1, Y_1, Z_1, X_2, Y_2, Z_2$ be source factors with $M_1 = X_1 Y_1$, $M_2 = X_2 Y_2$, $X_1^\dagger (I_d \otimes \rho) X_1 = I_r$, $X_2^\dagger X_2 = I_\ell$, and $Y_i Z_i = I$ for $i = 1, 2$, where ρ is a matrix on the virtual space. Define

$$(w_L)_{(l,i),(j,l')} = \sum_{\beta=0}^{D-1} (Y_2)_{l,(j,\beta)} (X_2)_{(\beta,i),l'}, \quad (19.175)$$

$$(w_R)_{(i,r),(r',j)} = \sum_{\beta=0}^{D-1} (X_1)_{(i,\beta),r'} (Y_1)_{r,(\beta,j)}. \quad (19.176)$$

Thus $w_L: \mathbb{C}^d \otimes \mathbb{C}^{d_l} \rightarrow \mathbb{C}^{d_l} \otimes \mathbb{C}^d$ and $w_R: \mathbb{C}^{d_r} \otimes \mathbb{C}^d \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{d_r}$. Rows are the upper legs and columns the lower legs, each in spatial left-to-right order. These are the crossed contractions of [163, lines 811–869].

$$\begin{array}{c} \begin{array}{|c|c|} \hline l & i \\ \hline \boxed{w_L} \\ \hline j & l' \\ \hline \end{array} & = & \begin{array}{c} \begin{array}{|c|c|} \hline l & i \\ \hline \bullet & \bullet \\ \hline j & l' \\ \hline \end{array} \\ Y_2 \quad X_2 \end{array} \end{array} \quad \begin{array}{c} \begin{array}{|c|c|} \hline i & r \\ \hline \boxed{w_R} \\ \hline r' & j \\ \hline \end{array} & = & \begin{array}{c} \begin{array}{|c|c|} \hline i & r \\ \hline \bullet & \bullet \\ \hline r' & j \\ \hline \end{array} \\ X_1 \quad Y_1 \end{array} \end{array}$$

Definition 19.4.2 (An operator on the middle two legs). [Lean](#) [✓ Stmt](#) Let A, B, C, B', C', E be index sets and let M be a complex matrix with rows indexed by $B \times C$ and columns by $B' \times C'$. Its action on the middle two legs is the matrix M_{23} defined by

$$(M_{23})_{((a,b),(c,e)),((a',b'),(c',e'))} = \delta_{a,a'} M_{(b,c),(b',c')} \delta_{e,e'}. \quad (19.177)$$

This is $(I_A \otimes M) \otimes I_E$ after regrouping $((a,b),(c,e))$ as $((a,(b,c)),e)$, as in the first network in the proof of the staircase-gate corollary of [163, lines 918–974].

Theorem 19.4.3 (Composition on the middle two legs). [Lean](#) [Lean](#) [✓ Stmt](#) Let M and N be composable complex matrices between pairs of index sets. If their common index sets and the two outer index sets are finite, then $(MN)_{23} = M_{23}N_{23}$. The identity matrix satisfies $I_{23} = I$.

Proof. [✓ Proof](#) In the matrix product, the two outer Kronecker deltas fix the outer indices. The remaining sum is the entry formula for MN . For the identity, the four Kronecker deltas give equality of the two four-leg indices. $\square$

Theorem 19.4.4 (The first staircase network before cancellation). [Lean](#) [✓ Stmt](#) [Discussion: #7314](#) Let $\mathcal{U}$ have source factors as in Definition 19.4.1, and form u, v, w_L, w_R from those same factors. Then

$$(w_L \otimes w_R)u_{23} = v_{23}(u \otimes u). \quad (19.178)$$

No simplicity or unitarity assumption on $\mathcal{U}$ is required. This is the first diagram in the proof of the staircase-gate corollary of [163, lines 918–968].

Proof. [✓ Proof](#) Expand the crossed contractions on both sides. For output index $((l, i_2), (i_3, r))$ and input index $((j_1, j_2), (j_3, j_4))$, the two open factorizations identify both entries with

$$\sum_{\alpha, \gamma=0}^{D-1} (Y_2)_{l, (j_1, \alpha)} (\mathcal{U}_{j_2}^{i_2} \mathcal{U}_{j_3}^{i_3})_{\alpha, \gamma} (Y_1)_{r, (\gamma, j_4)}. \quad (19.179)$$

$\square$

Theorem 19.4.5 (The first staircase network after cancellation). [Lean](#) [✓ Stmt](#) [Discussion: #7314](#) Let $\mathcal{U}$ have source factors as in Definition 19.4.1, and form u, v, w_L, w_R from those same factors. If u is a coisometry, so that $uu^\dagger = I_{d_l d_r}$, then

$$w_L \otimes w_R = v_{23}(u \otimes u)(u^\dagger)_{23} \quad (19.180)$$

$$= (I \otimes v \otimes I)(u \otimes u)(I \otimes u^\dagger \otimes I). \quad (19.181)$$

The second expression uses the regrouping of Definition 19.4.2. This is the displayed network identity in [163, lines 969–974].

Proof. [✓ Proof](#) Multiply (19.178) on the right by $(u^\dagger)_{23}$. The middle layers compose to $(uu^\dagger)_{23} = I$, giving (19.181). $\square$

19.4.1 The second network and unitarity of the staircase gates

Theorem 19.4.1.1 (Isometries on the middle two legs). [Lean](#) [✓ Stmt](#) Let A, B, C, E be finite index sets, let B', C' be index sets, and let M be a complex matrix with rows indexed by $B \times C$ and columns by $B' \times C'$. If $M^\dagger M = I$, then $(M_{23})^\dagger M_{23} = I$, with the middle-leg placement of Definition 19.4.2.

Proof. [✓ Proof](#) Tensoring an isometry with identity matrices preserves its Gram matrix. Regrouping the four indices preserves this identity. $\square$

Theorem 19.4.1.2 (The second staircase network). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7314](#) Let $\mathcal{U}$ have source factors as in Definition 19.4.1, and form u, v, w_L from those same factors. Write d_l, d_r for the source-cut ranks. Under the natural associativity identifications of the three tensor factors,

$$(w_L \otimes I_d)(I_d \otimes w_L) = (I_{d_l} \otimes v)(u \otimes I_{d_l}). \quad (19.182)$$

Both sides map $\mathbb{C}^d \otimes \mathbb{C}^d \otimes \mathbb{C}^{d_l}$ to $\mathbb{C}^{d_l} \otimes \mathbb{C}^d \otimes \mathbb{C}^d$. Explicitly, for $0 \leq l, k < d_l$ and $0 \leq i, j, a, b < d$,

$$\sum_{t=0}^{d_l-1} (w_L)_{(l,i),(a,t)} (w_L)_{(t,j),(b,k)} = \sum_{r=0}^{d_r-1} u_{(l,r),(a,b)} v_{(i,j),(r,k)}. \quad (19.183)$$

This is the second network in [163, lines 977–1011]. It uses two copies of w_L and requires no simplicity or unitarity hypothesis.

Proof. [✓ Proof](#) Expand the crossed gates and use $M_2 = X_2 Y_2$. The left-hand entry is

$$\sum_{\alpha, \beta=0}^{D-1} (Y_2)_{l,(a,\alpha)} \mathcal{U}_{\alpha\beta}^{i,b} (X_2)_{(\beta,j),k}. \quad (19.184)$$

Expanding u, v and using $M_1 = X_1 Y_1$ gives the same expression on the right. The displayed matrix equality is this entry identity after regrouping the three external indices. $\square$

Theorem 19.4.1.3 (Isometry of the first staircase network). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ have source factors as in Definition 19.4.1. If the associated operators u and v are unitary between their respective spaces, then

$$(w_L \otimes w_R)^\dagger (w_L \otimes w_R) = I. \quad (19.185)$$

Proof. [✓ Proof](#) Theorem 19.4.5 expresses $w_L \otimes w_R$ as $v_{23}(u \otimes u)(u^\dagger)_{23}$. Each factor is an isometry, so their product is an isometry. $\square$

Theorem 19.4.1.4 (Isometry of the second staircase network). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ have source factors as in Definition 19.4.1, with left source-cut rank d_l . If $u^\dagger u = I$ and $v^\dagger v = I$, then $K = (w_L \otimes I_d)(I_d \otimes w_L)$ satisfies $K^\dagger K = I$, with the three-leg associativity identifications understood.

Proof. [✓ Proof](#) By Theorem 19.4.1.2, $K = (I_{d_l} \otimes v)(u \otimes I_{d_l})$. Tensoring with an identity, regrouping indices, and composition preserve isometries. $\square$

Theorem 19.4.1.5 (Unitary staircase gates). [Lean](#) [✓ Stmt](#) [Discussion: #7314](#) Let $\mathcal{U}$ be a simple MPU in canonical form II, with recorded positive diagonal fixed matrix ρ . Construct the source factors from $\mathcal{U}$ and this same ρ , and let d_l, d_r be their source-cut ranks. Then the staircase gates are unitary maps

$$w_L : \mathbb{C}^d \otimes \mathbb{C}^{d_l} \longrightarrow \mathbb{C}^{d_l} \otimes \mathbb{C}^d, \quad (19.186)$$

$$w_R : \mathbb{C}^{d_r} \otimes \mathbb{C}^d \longrightarrow \mathbb{C}^d \otimes \mathbb{C}^{d_r}. \quad (19.187)$$

This is part (a) of the staircase-gate corollary in [163, lines 811–1012].

Proof. ✓ Proof Theorem 18.9.12 gives unitary u, v and $d_r d_l = d^2 > 0$. Theorem 19.4.1.3 gives

$$(w_L^\dagger w_L) \otimes (w_R^\dagger w_R) = I. \quad (19.188)$$

By Theorem 19.5.1, the two Gram matrices are δI and $\delta^{-1} I$ for a nonzero scalar δ . Positivity of the first Gram matrix makes $\delta \geq 0$. Each of the two left-gate layers in $K = (w_L \otimes I_d)(I_d \otimes w_L)$ has Gram matrix δI , so $K^\dagger K = \delta^2 I$. By Theorem 19.4.1.4, this equals I . Thus $\delta^2 = 1$, equivalently $\delta^{-2} = 1$, and positivity gives $\delta = 1$. Both gates are isometries between spaces of equal dimension, hence unitary. $\square$

19.4.2 Endpoint contraction and growth of truncated symmetries

Definition 19.4.2.1 (Truncated symmetry). Lean Lean ✓ Stmt Discussion: #7319 Let $\mathcal{U}$ be an MPO tensor of physical dimension d and bond dimension D , and let $d_l = \text{rank } M_2$ and $d_r = \text{rank } M_1$. Choose arbitrary complex endpoint matrices $Y_2: \mathbb{C}^d \otimes \mathbb{C}^D \rightarrow \mathbb{C}^{d_l}$ and $Y_1: \mathbb{C}^D \otimes \mathbb{C}^d \rightarrow \mathbb{C}^{d_r}$. For $N \geq 0$ and physical words σ, τ of length N , put

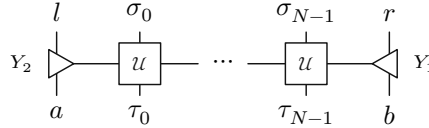
$$\mathcal{U}^{\sigma, \tau} = \mathcal{U}^{\sigma_0, \tau_0} \dots \mathcal{U}^{\sigma_{N-1}, \tau_{N-1}}, \quad (19.189)$$

$$(U^{N+2})_{(l, \sigma, r), (a, \tau, b)} = \sum_{\alpha, \beta=0}^{D-1} (Y_2)_{l, (a, \alpha)} (\mathcal{U}^{\sigma, \tau})_{\alpha, \beta} (Y_1)_{r, (\beta, b)}. \quad (19.190)$$

Here $0 \leq l < d_l$, $0 \leq r < d_r$, and a, b are physical indices. The empty ordered matrix product is I_D . Thus

$$U^{N+2}: \mathbb{C}^d \otimes (\mathbb{C}^D)^{\otimes N} \otimes \mathbb{C}^d \longrightarrow \mathbb{C}^{d_l} \otimes (\mathbb{C}^D)^{\otimes N} \otimes \mathbb{C}^{d_r}. \quad (19.191)$$

This is the endpoint contraction of [163, lines 2062–2100], with the group label suppressed. The source superscript is the total length $L = N + 2 \geq 2$, including the two endpoint triangles; N counts only the bulk tensors. No cut factorization is needed. For the normalized source factors of Definition 18.4.50, choosing their matrices Y_2, Y_1 recovers the source-factor endpoint contraction. No group representation or unitarity assertion is included in this definition; arbitrary endpoint matrices need not give a unitary map.



Theorem 19.4.2.2 (The zero-bulk contraction). Lean Lean ✓ Stmt Let $\mathcal{U}$ and the arbitrary endpoint matrices Y_2, Y_1 have the dimensions in Definition 19.4.2.1. For endpoint indices l, r, a, b and the empty words $\emptyset$,

$$(U^2)_{(l, \emptyset, r), (a, \emptyset, b)} = \sum_{\alpha=0}^{D-1} (Y_2)_{l, (a, \alpha)} (Y_1)_{r, (\alpha, b)}. \quad (19.192)$$

This is total length two and requires no cut factorization. For normalized source factors, the right-hand side is $u_{(l, r), (a, b)}$, with u as in Definition 18.4.57.

Proof. ✓ Proof The bulk product $\mathcal{U}^{\emptyset, \emptyset}$ is the empty ordered product I_D , so $(\mathcal{U}^{\emptyset, \emptyset})_{\alpha\beta} = \delta_{\alpha\beta}$. Instantiating (19.190) at $N = 0$ gives

$$(U^2)_{(l, \emptyset, r), (a, \emptyset, b)} = \sum_{\alpha, \beta=0}^{D-1} (Y_2)_{l, (a, \alpha)} \delta_{\alpha\beta} (Y_1)_{r, (\beta, b)} = \sum_{\alpha=0}^{D-1} (Y_2)_{l, (a, \alpha)} (Y_1)_{r, (\alpha, b)}. \quad (19.193)$$

For normalized source factors this is the defining contraction of u . $\square$

Theorem 19.4.2.3 (Left growth from the second cut). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7319](#) Let $\mathcal{U}$ be an MPO tensor with physical dimension d and bond dimension D , and put $d_l = \text{rank } M_2$ and $d_r = \text{rank } M_1$. Let $Y_1: \mathbb{C}^D \otimes \mathbb{C}^d \rightarrow \mathbb{C}^{d_r}$, $Y_2: \mathbb{C}^d \otimes \mathbb{C}^D \rightarrow \mathbb{C}^{d_l}$, and $X_2: \mathbb{C}^{d_l} \rightarrow \mathbb{C}^D \otimes \mathbb{C}^d$ be complex matrices satisfying only $M_2 = X_2 Y_2$, that is,

$$\mathcal{U}_{\gamma,\alpha}^{i,b} = \sum_{t=0}^{d_l-1} (X_2)_{(\gamma,i),t} (Y_2)_{t,(b,\alpha)}. \quad (19.194)$$

Define the crossed contraction

$$(w_L)_{(l,i),(a,t)} = \sum_{\gamma=0}^{D-1} (Y_2)_{l,(a,\gamma)} (X_2)_{(\gamma,i),t}. \quad (19.195)$$

For $N \geq 0$, physical words σ, τ of length N , source indices l, r , and physical indices i, a, b, c ,

$$(U^{N+3})_{(l,i\sigma,r),(a,b\tau,c)} = \sum_{t=0}^{d_l-1} (w_L)_{(l,i),(a,t)} (U^{N+2})_{(t,\sigma,r),(b,\tau,c)}. \quad (19.196)$$

The right endpoint Y_1 is arbitrary and unchanged; no first-cut factorization is assumed. For normalized source factors, w_L is the movement operator of Definition 19.4.1. The identity is the left-end factorization step of [163, lines 2101–2174], using the second decomposition in [15, lines 479–530].

Proof. [✓ Proof](#) The second cut factorization gives, for each virtual index α ,

$$\sum_{\gamma=0}^{D-1} (Y_2)_{l,(a,\gamma)} \mathcal{U}_{\gamma,\alpha}^{i,b} = \sum_{\gamma=0}^{D-1} \sum_{t=0}^{d_l-1} (Y_2)_{l,(a,\gamma)} (X_2)_{(\gamma,i),t} (Y_2)_{t,(b,\alpha)} \quad (19.197)$$

$$= \sum_{t=0}^{d_l-1} (w_L)_{(l,i),(a,t)} (Y_2)_{t,(b,\alpha)}. \quad (19.198)$$

Expand $\mathcal{U}^{i\sigma,b\tau} = \mathcal{U}^{i,b} \mathcal{U}^{\sigma,\tau}$ and substitute this equality into the endpoint contraction. Regrouping the finite sums yields

$$(U^{N+3})_{(l,i\sigma,r),(a,b\tau,c)} = \sum_{t=0}^{d_l-1} (w_L)_{(l,i),(a,t)} \sum_{\alpha,\beta=0}^{D-1} (Y_2)_{t,(b,\alpha)} (\mathcal{U}^{\sigma,\tau})_{\alpha,\beta} (Y_1)_{r,(b,\beta,c)} \quad (19.199)$$

$$= \sum_{t=0}^{d_l-1} (w_L)_{(l,i),(a,t)} (U^{N+2})_{(t,\sigma,r),(b,\tau,c)}. \quad (19.200)$$

□

Theorem 19.4.2.4 (Two-sided movement-operator growth). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7319](#) Let $\mathcal{U}$ be an MPO tensor of physical dimension d and bond dimension D , with $d_l = \text{rank } M_2$ and $d_r = \text{rank } M_1$. Let $X_1: \mathbb{C}^{d_r} \rightarrow \mathbb{C}^d \otimes \mathbb{C}^D$, $Y_1: \mathbb{C}^D \otimes \mathbb{C}^d \rightarrow \mathbb{C}^{d_r}$, $X_2: \mathbb{C}^{d_l} \rightarrow \mathbb{C}^D \otimes \mathbb{C}^d$, and $Y_2: \mathbb{C}^d \otimes \mathbb{C}^D \rightarrow \mathbb{C}^{d_l}$ be complex matrices satisfying only $M_1 = X_1 Y_1$ and $M_2 = X_2 Y_2$. In coordinates,

$$\mathcal{U}_{\beta,\delta}^{j,c} = \sum_{s=0}^{d_r-1} (X_1)_{(j,\delta),s} (Y_1)_{s,(\beta,c)}, \quad (19.201)$$

$$\mathcal{U}_{\gamma,\alpha}^{i,b} = \sum_{t=0}^{d_l-1} (X_2)_{(\gamma,i),t} (Y_2)_{t,(b,\alpha)}. \quad (19.202)$$

Form U^{N+2} using the endpoints Y_2, Y_1 , and define the crossed contractions

$$(w_L)_{(l,i),(a,t)} = \sum_{\gamma=0}^{D-1} (Y_2)_{l,(a,\gamma)} (X_2)_{(\gamma,i),t}, \quad (19.203)$$

$$(w_R)_{(j,r),(s,e)} = \sum_{\delta=0}^{D-1} (X_1)_{(j,\delta),s} (Y_1)_{r,(\delta,e)}. \quad (19.204)$$

For $N \geq 0$, physical words σ, τ of length N , source indices l, r , and physical indices i, j, a, b, c, e , one has

$$(U^{N+4})_{(l,i\sigma j,r),(a,b\tau c,e)} = \sum_{t=0}^{d_l-1} \sum_{s=0}^{d_r-1} (w_L)_{(l,i),(a,t)} (U^{N+2})_{(t,\sigma,s),(b,\tau,c)} (w_R)_{(j,r),(s,e)}. \quad (19.205)$$

Concatenation $i\sigma j$ inserts i before the word and j after it. This is the source-factorization identity in the growth diagram of [163, lines 2101–2174]. It adds one bulk site at each end, using only the two decompositions in [15, lines 479–530]. For normalized source factors, the crossed contractions yield the movement operators of Definition 19.4.1, extending the identity to the source-factor endpoint contraction. No simplicity or unitarity assumption on $\mathcal{U}$ is needed. These algebraic identities do not assert unitarity for arbitrary factors, the finite-group physical specialization, or the complementary finite-chain movement relation.

Proof. ✓ Proof Expand the enlarged bulk product as $\mathcal{U}^{i\sigma j, b\tau c} = \mathcal{U}^{i,b} \mathcal{U}^{\sigma, \tau} \mathcal{U}^{j,c}$. The second source-cut factorization gives

$$\sum_{\gamma} (Y_2)_{l,(a,\gamma)} \mathcal{U}_{\gamma, \alpha}^{i,b} = \sum_t (w_L)_{(l,i),(a,t)} (Y_2)_{t,(b,\alpha)}. \quad (19.206)$$

The first source-cut factorization similarly gives

$$\sum_{\delta} \mathcal{U}_{\beta, \delta}^{j,c} (Y_1)_{r,(\delta,e)} = \sum_s (Y_1)_{s,(\beta,c)} (w_R)_{(j,r),(s,e)}. \quad (19.207)$$

Substitute both equalities into the endpoint contraction and regroup the finite sums. The remaining interior contraction is U^{N+2} . □

Theorem 19.4.2.5 (Kronecker product of unitary maps). Lean ✓ Stmt Let A, B, C, E be finite sets, possibly empty, and let $M: \mathbb{C}^B \rightarrow \mathbb{C}^A$ and $P: \mathbb{C}^E \rightarrow \mathbb{C}^C$ be unitary maps: $M^\dagger M = I_B$, $MM^\dagger = I_A$, $P^\dagger P = I_E$, and $PP^\dagger = I_C$. Then $M \otimes P$ is a unitary map from $\mathbb{C}^{B \times E}$ to $\mathbb{C}^{A \times C}$.

Proof. ✓ Proof The adjoint and multiplication identities for Kronecker products give

$$(M \otimes P)^\dagger (M \otimes P) = (M^\dagger M) \otimes (P^\dagger P) = I_B \otimes I_E = I_{B \times E}, \quad (19.208)$$

$$(M \otimes P)(M \otimes P)^\dagger = (MM^\dagger) \otimes (PP^\dagger) = I_A \otimes I_C = I_{A \times C}. \quad (19.209)$$

□

Theorem 19.4.2.6 (Unitarity at every bulk length). Lean ✓ Stmt Discussion: #7319 Let $\mathcal{U}$ be a simple MPU with a canonical-form-II presentation and recorded positive diagonal fixed matrix ρ . Construct the source factors from this tensor and this same ρ . For every $N \geq 0$, their endpoint contraction U^{N+2} is a unitary map

$$U^{N+2}: \mathcal{H}_{\text{in}, N} \longrightarrow \mathcal{H}_{\text{out}, N}, \quad (19.210)$$

$$\mathcal{H}_{\text{in}, N} = \mathbb{C}^d \otimes (\mathbb{C}^d)^{\otimes N} \otimes \mathbb{C}^d, \quad (19.211)$$

$$\mathcal{H}_{\text{out}, N} = \mathbb{C}^{d_l} \otimes (\mathbb{C}^d)^{\otimes N} \otimes \mathbb{C}^{d_r}. \quad (19.212)$$

The subscripts of $\mathcal{H}_{\text{in},N}$ and $\mathcal{H}_{\text{out},N}$ record the bulk length N . Equivalently,

$$(U^{N+2})^\dagger U^{N+2} = I_{\mathcal{H}_{\text{in},N}}, \quad U^{N+2} (U^{N+2})^\dagger = I_{\mathcal{H}_{\text{out},N}}. \quad (19.213)$$

This is the endpoint-coordinate unitarity part of the assertion following the truncated-symmetry diagram in [163, lines 2062–2099], between the displayed input and output spaces. The scope restriction is recorded in [55]. Neither simplicity of the physical adjoint nor unitarity of supplied gates is assumed. The assertion does not identify each endpoint source space with a physical site, specialize to a finite-group family, prove agreement with a global symmetry on interior observables, or establish the complementary finite-chain relation following the movement display at [163, lines 2101–2174].

Proof. ✓ Proof Theorem 18.9.12 gives unitary u , and Theorem 19.4.1.5 gives unitary w_L , both for the constructed source factors. Induct on the bulk length N . For $N = 0$, the total length is two. In U^2 , remove the unique empty word from both index sets; by Theorem 19.4.2.2, the resulting matrix is u .

For the successor step put $\mathcal{R}_N = (\mathbb{C}^d)^{\otimes N} \otimes \mathbb{C}^{d_r}$. After the natural associativity identifications, the left-end growth identity expresses U^{N+3} as

$$(w_L \otimes I_{\mathcal{R}_N})(I_d \otimes U^{N+2}). \quad (19.214)$$

The output reindexing sends (l, i, σ, r) to $(l, i\sigma, r)$, and the input reindexing sends (a, b, τ, c) to $(a, b\tau, c)$. These are bijections given by splitting off the first bulk letter; they leave both endpoint coordinates unchanged. Theorem 19.4.2.5 makes both $w_L \otimes I_{\mathcal{R}_N}$ and $I_d \otimes U^{N+2}$ unitary. Composition and the two coordinate bijections preserve both Gram identities. The induction therefore covers every N , not just one parity of bulk lengths. $\square$

Theorem 19.4.2.7 (Unitary coordinate transport of the endpoint contraction). Lean Lean
✓ Stmt Let $\mathcal{U}$ be a simple MPU of dimensions d, D in canonical form II , with recorded positive-definite matrix ρ . Suppose that T is unitary, that $(U^\dagger)^{ij} = T^\dagger U^{ij} T$ for every physical pair, and that $T\bar{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. Let S_0 denote its chosen source factors at p and S its inverse-compatible source factors. Put $r = \text{rank } M_1(\mathcal{U})$ and $\ell = \text{rank } M_2(\mathcal{U})$, and take the square unitary comparison $C = KP_e^\dagger$ of Theorem 19.2.3.3. For $N \geq 0$, let Ω_N be the set of physical configurations on N sites. Write U_0^{N+2} and $\widehat{U}^{N+2}$ for the endpoint contractions of Definition 19.4.2.1 formed from S_0 and S , respectively. Their common domain and codomain, with the indicated association, are

$$\mathcal{H}_N = \mathbb{C}^d \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^d), \quad (19.215)$$

$$\mathcal{R}_N = \mathbb{C}^\ell \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^r). \quad (19.216)$$

Then

$$\widehat{U}^{N+2} = [I_\ell \otimes (I_{\Omega_N} \otimes C^\dagger)] U_0^{N+2}. \quad (19.217)$$

Explicitly, for source indices l, s , configurations $p, q \in \Omega_N$, and physical indices a, b ,

$$(\widehat{U}^{N+2})_{(l,(p,s)),(a,(q,b))} = \sum_{t=0}^{r-1} (C^\dagger)_{st} (U_0^{N+2})_{(l,(p,t)),(a,(q,b))}. \quad (19.218)$$

Only the right source coordinate changes. Here N counts bulk sites and the total source length is $N + 2$, including both endpoints. This applies the comparison transport of [163, lines 5432–5487] to the endpoint contraction at lines 2062–2099. No simplicity assumption on the physical adjoint is needed.

Proof. ✓ Proof Write $\widetilde{Y}_1, Y_2$ for the chosen endpoints and $\widehat{Y}_1, \widehat{Y}_2$ for the compatible ones. Substitute $\widehat{Y}_1 = C^\dagger \widetilde{Y}_1$ and $\widehat{Y}_2 = Y_2$ into the endpoint contraction. Regrouping the finite sums gives

$$(\widehat{U}^{N+2})_{(l,(p,s)),(a,(q,b))} = \sum_{t=0}^{r-1} (C^\dagger)_{st} \sum_{\alpha,\beta=0}^{D-1} (Y_2)_{l,(a,\alpha)} \mathcal{U}_{\alpha\beta}[p,q](\widetilde{Y}_1)_{t,(\beta,b)} \quad (19.219)$$

$$= \sum_{t=0}^{r-1} (C^\dagger)_{st} (U_0^{N+2})_{(l,(p,t)),(a,(q,b))}. \quad (19.220)$$

Expanding the two identity factors in $I_\ell \otimes (I_{\Omega_N} \otimes C^\dagger)$ fixes the left source index and the bulk configuration. Its matrix product with U_0^{N+2} therefore has exactly these entries. $\square$

Theorem 19.4.2.8 (Unitarity of the compatible endpoint contractions). Lean ✓ Stmt Let $\mathcal{U}$ be a simple MPU of dimensions d, D in canonical form II, with recorded positive-definite matrix ρ . Suppose that T is unitary, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and that $T\bar{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. For every bulk length $N \geq 0$, the endpoint contraction $\widehat{U}^{N+2}$ formed from the inverse-compatible source factors is unitary from $\mathcal{H}_N = \mathbb{C}^d \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^d)$ to $\mathcal{R}_N = \mathbb{C}^\ell \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^r)$, where Ω_N is the set of length- N physical configurations and r, ℓ are the first and second cut ranks. The case $N = 0$ has total length two. No simplicity assumption on the physical adjoint is needed. This transports the unitarity assertion following the endpoint diagram of [163, lines 2062–2099], using its comparison argument at lines 5486–5487.

Proof. ✓ Proof The comparison C is unitary. By Theorem 19.4.2.5, $Q_N = I_\ell \otimes (I_{\Omega_N} \otimes C^\dagger)$ is unitary. The chosen-source contraction U_0^{N+2} is unitary by Theorem 19.4.2.6, and the transport theorem gives $\widehat{U}^{N+2} = Q_N U_0^{N+2}$. Hence

$$(\widehat{U}^{N+2})^\dagger \widehat{U}^{N+2} = (U_0^{N+2})^\dagger Q_N^\dagger Q_N U_0^{N+2} = I_{\mathcal{H}_N}, \quad (19.221)$$

$$\widehat{U}^{N+2} (\widehat{U}^{N+2})^\dagger = Q_N U_0^{N+2} (U_0^{N+2})^\dagger Q_N^\dagger = I_{\mathcal{R}_N}. \quad (19.222)$$

$\square$

These are coordinate transport and unitarity statements, not the final truncated-chain sewing identity or a finite-group physical-action assertion.

Theorem 19.4.2.9 (Raw endpoint unitarity by rank-coordinate transport). Lean Lean ✓ Stmt Let $\mathcal{U}$ be a simple MPU of dimensions d, D in canonical form II. Suppose that T is unitary, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and that $T\bar{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. Construct the inverse-compatible source factors at the recorded canonical weight. Put $r = \text{rank } M_1(\mathcal{U})$, $\ell = \text{rank } M_2(\mathcal{U})$, and take the rank bijection $e: \{0, \dots, \ell - 1\} \rightarrow \{0, \dots, r - 1\}$ of Definition 19.2.3.1. For every $N \geq 0$, let Ω_N be the length- N physical configurations. The raw and packaged source spaces and the physical space are, with right association,

$$\mathcal{R}_N^{\text{raw}} = \mathbb{C}^\ell \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^\ell), \quad (19.223)$$

$$\mathcal{R}_N^{\text{pkg}} = \mathbb{C}^\ell \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^r), \quad (19.224)$$

$$\mathcal{H}_N = \mathbb{C}^d \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^d). \quad (19.225)$$

Let $E_N(l, (p, s)) = (l, (p, e(s)))$, and let $P_{E_N}: \mathcal{R}_N^{\text{raw}} \rightarrow \mathcal{R}_N^{\text{pkg}}$ be its permutation matrix. If $\widehat{U}^{N+2}$ is the endpoint contraction of the inverse-compatible source factors, then the raw matrix A_N of Definition 19.2.4.11 satisfies

$$A_N = P_{E_N}^\dagger \widehat{U}^{N+2}. \quad (19.226)$$

Equivalently, for raw source indices l, s , configurations p, q , and physical indices a, b ,

$$(A_N)_{(l,(p,s)),(a,(q,b))} = (\widehat{U}^{N+2})_{(l,(p,e(s))), (a,(q,b))}. \quad (19.227)$$

Thus $A_N: \mathcal{H}_N \rightarrow \mathcal{R}_N^{\text{raw}}$ is unitary. This transports only the right endpoint back along e^{-1} ; no spatial permutation is introduced. The total source length is $N+2$, so $N=0$ has total length two. This is coordinate transport of the endpoint construction in [163, lines 2062–2099 and 5390–5487].

Proof. ✓ **Proof** The compatible endpoints obey $\widehat{Y}_2 = Y_2$ and $(\widehat{Y}_1)_{e(s),(\beta,b)} = Y_{s,(\beta,b)}$. Substitution gives

$$(\widehat{U}^{N+2})_{(l,(p,e(s))), (a,(q,b))} = \sum_{\alpha,\beta=0}^{D-1} (Y_2)_{l,(a,\alpha)} \mathcal{U}[p,q]_{\alpha\beta} Y_{s,(\beta,b)} = (A_N)_{(l,(p,s)), (a,(q,b))}. \quad (19.228)$$

Hence $A_N = P_{E_N}^\dagger \widehat{U}^{N+2}$. Theorem 19.4.2.8 makes $\widehat{U}^{N+2}$ unitary. Since E_N is a bijection, its permutation matrix satisfies $P_{E_N}^\dagger P_{E_N} = I_{\mathcal{R}_N^{\text{raw}}}$ and $P_{E_N} P_{E_N}^\dagger = I_{\mathcal{R}_N^{\text{pkg}}}$, so both Gram products collapse:

$$A_N^\dagger A_N = (\widehat{U}^{N+2})^\dagger P_{E_N} P_{E_N}^\dagger \widehat{U}^{N+2} = (\widehat{U}^{N+2})^\dagger \widehat{U}^{N+2} = I_{\mathcal{H}_N}, \quad (19.229)$$

$$A_N A_N^\dagger = P_{E_N}^\dagger \widehat{U}^{N+2} (\widehat{U}^{N+2})^\dagger P_{E_N} = P_{E_N}^\dagger P_{E_N} = I_{\mathcal{R}_N^{\text{raw}}}. \quad (19.230)$$

□

Theorem 19.4.2.10 (Local cancellation with one phase). Lean ✓ **Stmt** Let $\mathcal{U}$ be a simple MPU of dimensions d, D in canonical form II. Suppose that T is unitary, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and that $T\bar{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. For every bulk length $N \geq 0$, take the raw endpoint matrices $A_N: \mathcal{H}_N \rightarrow \mathcal{R}_N^{\text{raw}}$ and $B_N: \mathcal{R}_N^{\text{raw}} \rightarrow \mathcal{H}_N$ of Definition 19.2.4.11, where $\mathcal{H}_N = \mathbb{C}^d \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^d)$ and $\mathcal{R}_N^{\text{raw}} = \mathbb{C}^\ell \otimes (\mathbb{C}^{\Omega_N} \otimes \mathbb{C}^\ell)$. Here Ω_N is the set of length- N physical configurations and ℓ is the second-cut rank. Then

$$A_N B_N = \sigma I_{\mathcal{R}_N^{\text{raw}}}, \quad (19.231)$$

$$B_N A_N = \sigma I_{\mathcal{H}_N}. \quad (19.232)$$

There is a single phase, independent of N ; the total source length is $N+2$. These are local endpoint cancellations extending [163, lines 5444–5487], with the phase relation at lines 1559–1562.

Proof. ✓ **Proof** Canonical form II gives $D > 0$. Theorem 19.2.4.26 applied to $T\bar{T} = \sigma I_D$ therefore gives $\sigma\bar{\sigma} = 1$, without any further positivity assumption. Adjoining $A_N = \sigma B_N^\dagger$ yields the reverse relation

$$\sigma A_N^\dagger = \sigma\bar{\sigma} B_N = B_N. \quad (19.233)$$

Unitarity of A_N now gives

$$A_N B_N = \sigma A_N A_N^\dagger = \sigma I_{\mathcal{R}_N^{\text{raw}}}, \quad (19.234)$$

$$B_N A_N = \sigma A_N^\dagger A_N = \sigma I_{\mathcal{H}_N}. \quad (19.235)$$

□

19.4.3 Periodic sewing of the two source arcs

The periodic chain carries two marked sites. Cutting the marked tensors by the two source cuts produces two open arcs that share both marked sites, and the shared rank indices sew them back to the periodic trace. The arcs have lengths $n + 2$ and $m + 2$ on a ring of $n + m + 2$ sites; the second is not the disjoint complement of the first.

Definition 19.4.3.1 (The two sewn source arcs). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor of dimensions d, D , and let X_1, Y_1 and X_2, Y_2 be any factorizations of its two source cuts through $\mathbb{C}^r$ and $\mathbb{C}^\ell$. For $N \geq 0$ and configurations $p, q \in \Omega_N$, define matrices A_N^{sew} and B_N^{sew} by

$$(A_N^{\text{sew}})_{(s,(p,t)),(a,(q,b))} = \sum_{\alpha,\beta=0}^{D-1} (Y_2)_{s,(a,\alpha)} \mathcal{U}[p, q]_{\alpha\beta} (Y_1)_{t,(\beta,b)}, \quad (19.236)$$

$$(B_N^{\text{sew}})_{(a,(p,b)),(t,(q,s))} = \sum_{\alpha,\beta=0}^{D-1} (X_1)_{(a,\alpha),t} \mathcal{U}[p, q]_{\alpha\beta} (X_2)_{(\beta,b),s}, \quad (19.237)$$

with $0 \leq s < \ell$, $0 \leq t < r$, and physical indices a, b . Both words are read in their given order; neither is reversed, and the row order (ℓ, r) of the first arc is the order in which its endpoints are traversed. This is the local algebra of the sewing referred to in [163, lines 5486–5487], using its source decompositions at lines 5390–5432.

Theorem 19.4.3.2 (Sewing the periodic trace at two marked sites). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor with $M_1(\mathcal{U}) = X_1 Y_1$ and $M_2(\mathcal{U}) = X_2 Y_2$. Fix bulk lengths n, m , marked physical indices a, a', b, b' , and configurations $p, p' \in \Omega_n$, $q, q' \in \Omega_m$. Then

$$\text{tr}(\mathcal{U}^{aa'} \mathcal{U}[p, p'] \mathcal{U}^{bb'} \mathcal{U}[q, q']) = \sum_{t=0}^{r-1} \sum_{s=0}^{\ell-1} (B_n^{\text{sew}})_{(a,(p,b)),(t,(p',s))} (A_m^{\text{sew}})_{(s,(q,t)),(b',(q',a'))}. \quad (19.238)$$

The same sum is the entry of the periodic operator $\mathcal{U}^{(n+m+2)}$ at the concatenated configurations (a, p, b, q) and (a', p', b', q') . The ring has $n+m+2$ sites, including two distinct marked sites even when $n = m = 0$, and the two endpoint-inclusive arcs overlap at both marked sites. This identity requires only the two cut factorizations, without assuming canonicity, simplicity, normalization, or unitarity.

Proof. [✓ Proof](#) Write $P = \mathcal{U}[p, p']$ and $Q = \mathcal{U}[q, q']$, and define slices

$$C_{t\alpha} = (X_1)_{(a,\alpha),t}, \quad (19.239)$$

$$H_{\delta t} = (Y_1)_{t,(\delta,a')}, \quad (19.240)$$

$$E_{\beta s} = (X_2)_{(\beta,b),s}, \quad (19.241)$$

$$F_{s\gamma} = (Y_2)_{s,(b',\gamma)}. \quad (19.242)$$

Reading the two cut factorizations entrywise gives $\mathcal{U}^{aa'} = HC$ and $\mathcal{U}^{bb'} = EF$. Cyclicity of the trace and associativity then give

$$\text{tr}(HCPEFQ) = \text{tr}((CPE)(FQH)). \quad (19.243)$$

The entries of the two products are $(CPE)_{ts} = (B_n^{\text{sew}})_{(a,(p,b)),(t,(p',s))}$ and $(FQH)_{st} = (A_m^{\text{sew}})_{(s,(q,t)),(b',(q',a'))}$, so expanding the trace of the product as a double sum over t and s gives (19.238). For the operator entry, expand word evaluation along the concatenation (a, p, b, q) : splitting off the two marked letters and the two bulk words returns the same trace. $\square$

Definition 19.4.3.3 (Grouped periodic coordinates and the complementary arc). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix bulk lengths n, m and group a periodic configuration as $((a, p, b), q)$ with $p \in \Omega_n$ and $q \in \Omega_m$, placed on the ring in the order (a, p, b, q) . Let $M_{n,m}$ be the periodic operator $\mathcal{U}^{(n+m+2)}$ pulled back along this grouping in both its row and its column index. With A_N, B_N the endpoint matrices of Definition 19.2.4.11, put

$$\widehat{A}_{n,m} = A_n \otimes I_{\Omega_m}, \quad (19.244)$$

which acts as the first endpoint operator on (a, p, b) and as the identity on the second bulk. Define the complementary arc $C_{n,m}$ independently by

$$(C_{n,m})_{((l,(p,s)),q),((a',(p',b')),q')} = \delta_{p,p'} (A_m)_{(s,(q,l)),(b',(q',a'))}. \quad (19.245)$$

It retains the first bulk unchanged and traverses the other arc from the second marked site back to the first, with rank order (s, q, l) . These are the shared-boundary coordinates of the sewing referred to in [163, lines 5486–5487]; no interval hierarchy, spatial reflection, or identification of rank with physical coordinates is introduced.

Theorem 19.4.3.4 (Factorization of the grouped periodic operator). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor of dimensions d, D and T a unitary virtual matrix with $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair. For all bulk lengths n, m ,

$$M_{n,m} = (B_n \otimes I_{\Omega_m}) C_{n,m}. \quad (19.246)$$

Neither simplicity, canonical form, nor a phase relation is used; the complementary arc is the independently defined $C_{n,m}$, not a rearrangement of $M_{n,m}$.

Proof. [✓ Proof](#) Apply Theorem 19.4.3.2 to the factorization $M_1 = \widehat{X}_1 \widehat{Y}_1$ of Theorem 19.2.20 and the factorization $M_2 = X_2 Y_2$ of Theorem 18.4.51. The two sewn arcs of Definition 19.4.3.1 built from these factors are exactly B_n and A_m . On the right-hand side of (19.246) the identity factor contributes $\delta_{q,q'}$ on the second bulk and the Kronecker delta of $C_{n,m}$ contributes $\delta_{p,p'}$, so the matrix product collapses to the double sum of (19.238) at the grouped configurations. $\square$

Theorem 19.4.3.5 (One phase in the periodic composition). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU of dimensions d, D in canonical form II. Suppose that T is unitary, that $(\mathcal{U}^\dagger)^{ij} = T^\dagger \mathcal{U}^{ij} T$ for every physical pair, and that $T\overline{T} = \sigma I_D$ for $\sigma \in \mathbb{C}$. For all bulk lengths n, m ,

$$\widehat{A}_{n,m} M_{n,m} = \sigma C_{n,m}. \quad (19.247)$$

Exactly one phase occurs, independently of n and m . The two endpoint-inclusive arcs share both marked sites, so this is a shared-boundary composition and not the disjoint-complement identity of [163, lines 5486–5487].

Proof. [✓ Proof](#) Substituting (19.246) and using $(A_n \otimes I_{\Omega_m})(B_n \otimes I_{\Omega_m}) = (A_n B_n) \otimes I_{\Omega_m}$ gives

$$\widehat{A}_{n,m} M_{n,m} = ((A_n B_n) \otimes I_{\Omega_m}) C_{n,m}. \quad (19.248)$$

Theorem 19.4.2.10 gives $A_n B_n = \sigma I$ on the raw source space, and the scalar passes through the Kronecker product. The cancellation used is the one on the source space, not the reverse product $B_n A_n$. $\square$

19.4.4 Physical closures of the source factors

Theorem 19.4.4.1 (The canonical one-letter physical contraction). [Lean](#) [✓ Stmt](#) [Discussion: #7314](#)
 Let $\mathcal{U}$ be a simple MPU in canonical form II, with physical dimension d , bond dimension D , and recorded fixed matrix ρ . For $0 \leq p, q < d$,

$$\sum_{x, \alpha, \beta=0}^{D-1} \sum_{z=0}^{d-1} \overline{\mathcal{U}_{x\alpha}^{z,p}} \mathcal{U}_{x\beta}^{z,q} \rho_{\beta\alpha} = \delta_{p,q}. \quad (19.249)$$

This is the one-letter diagram in the proof of part (b) of the staircase-gate corollary in [163, lines 1013–1030].

Proof. [✓ Proof](#) Theorem 19.5.6 supplies the two-letter insertion identity with the recorded canonical boundaries $(\Phi|_{(a,b)} = \delta_{a,b}$ and $|\rho)_{(a,b)} = \rho_{ba}$. Applying Theorem 18.3.28 to this fixed pair gives the one-letter contraction. Expanding its coefficient yields

$$W^{pq} W^{kl} = W^{pq} |\rho) (\Phi| W^{kl}, \quad (19.250)$$

$$(\Phi| W^{pq} |\rho) = \sum_{x, \alpha, \beta=0}^{D-1} \sum_{z=0}^{d-1} \overline{\mathcal{U}_{x\alpha}^{z,p}} \mathcal{U}_{x\beta}^{z,q} \rho_{\beta\alpha} = \delta_{p,q}. \quad (19.251)$$

The left boundary identifies the two bond indices x , while the right boundary contributes $\rho_{\beta\alpha}$. $\square$

Theorem 19.4.4.2 (Physical closures of the two source factors). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7314](#)
 Let $\mathcal{U}$ be a simple MPU in canonical form II, with physical dimension d , bond dimension D , and recorded positive diagonal fixed matrix ρ . Let Y_1, Y_2 be the source factors constructed from this tensor and this same ρ , with source-cut ranks d_r, d_l . For $0 \leq p, q < d$,

$$\sum_{l=0}^{d_l-1} \sum_{\alpha, \beta=0}^{D-1} \overline{(Y_2)_{l,(p,\alpha)}} (Y_2)_{l,(q,\beta)} \rho_{\beta\alpha} = \delta_{p,q}, \quad (19.252)$$

$$\sum_{r=0}^{d_r-1} \sum_{\alpha=0}^{D-1} \overline{(Y_1)_{r,(\alpha,p)}} (Y_1)_{r,(\alpha,q)} = \delta_{p,q}. \quad (19.253)$$

The open indices are physical; these are identities on $\mathbb{C}^d$, not on the source-cut spaces. They are the two diagrams of part (b) of the staircase-gate corollary in [163, lines 869–917].

Proof. [✓ Proof](#) Theorem 18.4.54 at $a = \alpha, b = \beta$ gives

$$\sum_{l=0}^{d_l-1} \overline{(Y_2)_{l,(p,\alpha)}} (Y_2)_{l,(q,\beta)} = \sum_{x=0}^{D-1} \sum_{z=0}^{d-1} \overline{\mathcal{U}_{x\alpha}^{z,p}} \mathcal{U}_{x\beta}^{z,q}. \quad (19.254)$$

Multiplying by $\rho_{\beta\alpha}$ and summing over α, β rewrites the first sum as the one-letter contraction of Theorem 19.4.4.1. Theorem 18.4.63 at the diagonal case $a_1 = b_1 = \alpha, p_1 = p, q_1 = q$ gives

$$\sum_{r=0}^{d_r-1} \overline{(Y_1)_{r,(\alpha,p)}} (Y_1)_{r,(\alpha,q)} = \sum_{\beta, \gamma=0}^{D-1} \sum_{z=0}^{d-1} \overline{\mathcal{U}_{\alpha\beta}^{z,p}} \rho_{\beta\gamma} \mathcal{U}_{\alpha\gamma}^{z,q}. \quad (19.255)$$

Summing over α rewrites the second sum in the same way, after renaming (α, β, γ) to (x, α, β) ; diagonality of ρ gives $\rho_{\alpha\beta} = \rho_{\beta\alpha}$. Both sums therefore equal $\delta_{p,q}$. These Gram identities cancel the X_2 and X_1 factors using their respective normalizations, as in [163, lines 1030–1050]. $\square$

19.5 Scalar three-cocycles and fusion-tensor gauge freedom

Let G be a group and write $\mathbb{C}^\times$ for the multiplicative group of nonzero complex numbers. The fusion tensors have the scalar gauge freedom

$$F_{g,h}^> \mapsto \beta_{g,h} F_{g,h}^>, \quad (19.256)$$

$$F_{g,h}^< \mapsto \beta_{g,h}^{-1} F_{g,h}^<. \quad (19.257)$$

This specifies the fusion-tensor gauge freedom of [163] for the scalar functions β , ω , γ , L , σ , and Ξ , with representation-level data as in Definition 19.2.4.

Theorem 19.5.1 (Scalar factors of an identity Kronecker product). Lean ✓ Stmt *Let M and N be nonempty sets with decidable equality, and let $A \in \text{Mat}_M(\mathbb{C})$ and $B \in \text{Mat}_N(\mathbb{C})$. If $A \otimes B = \mathbf{1}$, then there is a nonzero $c \in \mathbb{C}$ such that*

$$A = c\mathbf{1}, \quad B = c^{-1}\mathbf{1}. \quad (19.258)$$

Proof. ✓ Proof Choose $i_0 \in M$ and $j_0 \in N$. The $((i_0, j_0), (i_0, j_0))$ entry gives $A_{i_0 i_0} B_{j_0 j_0} = 1$. Thus $c = A_{i_0 i_0}$ is nonzero and $B_{j_0 j_0} = c^{-1}$. Comparing every $((i, j_0), (i', j_0))$ entry gives $A_{ii'} = c\delta_{ii'}$; comparing every $((i_0, j), (i_0, j'))$ entry gives $B_{jj'} = c^{-1}\delta_{jj'}$. $\square$

Theorem 19.5.2 (Unitarity from an adjoint Kronecker product). Lean ✓ Stmt *Let M, N be nonempty finite sets, and let $K: \mathbb{C}^N \rightarrow \mathbb{C}^M$ be a complex matrix. If $K^\dagger \otimes K: \mathbb{C}^{M \times N} \rightarrow \mathbb{C}^{N \times M}$ is unitary, then K is unitary between its coordinate spaces:*

$$K^\dagger K = I_N, \quad (19.259)$$

$$K K^\dagger = I_M. \quad (19.260)$$

No equality of $|M|$ and $|N|$ is assumed. This is the matrix implication used in [163, lines 5444–5487]; the tensor-network identity to which it is applied is a separate result.

Proof. ✓ Proof The isometry identity for $K^\dagger \otimes K$ gives

$$(K K^\dagger) \otimes (K^\dagger K) = I_{M \times N}. \quad (19.261)$$

By Theorem 19.5.1, there is a nonzero $c \in \mathbb{C}$ with $K K^\dagger = c I_M$ and $K^\dagger K = c^{-1} I_N$. Consequently,

$$(K K^\dagger)^2 = K (K^\dagger K) K^\dagger = c^{-1} K K^\dagger = I_M. \quad (19.262)$$

Choose $i \in M$. The (i, i) entry gives $c^2 = 1$, while positivity of the Gram matrix gives $c = (K K^\dagger)_{ii} \geq 0$. Thus $c = 1$, and the two scalar Gram identities become $K^\dagger K = I_N$ and $K K^\dagger = I_M$. $\square$

Theorem 19.5.3 (Unitary factors of a unitary Kronecker product). Lean Lean ✓ Stmt *Let M and N be nonempty finite sets. If $K_1 \in \text{Mat}_M(\mathbb{C})$ and $K_2 \in \text{Mat}_N(\mathbb{C})$ satisfy that $K_1 \otimes K_2$ is unitary, then there are a positive real number δ and unitary matrices W_1, W_2 such that*

$$K_1 = \delta W_1, \quad K_2 = \delta^{-1} W_2. \quad (19.263)$$

Equivalently, $\delta^{-1} K_1$ and δK_2 are unitary. This is only the tensor-factor step in the decomposition result of [163]; it does not construct or compare the three-legged decompositions occurring there.

Proof. ✓ Proof Unitarity gives

$$(K_1^\dagger K_1) \otimes (K_2^\dagger K_2) = \mathbf{1}. \quad (19.264)$$

Theorem 19.5.1 supplies a nonzero $c \in \mathbb{C}$ such that $K_1^\dagger K_1 = c\mathbf{1}$ and $K_2^\dagger K_2 = c^{-1}\mathbf{1}$. Positivity of the first Gram matrix makes c a positive real number. Taking $\delta = \sqrt{c}$ shows directly that $\delta^{-1}K_1$ and δK_2 have identity Gram matrices; these are W_1 and W_2 in (19.263). $\square$

Theorem 19.5.4 (Cross-product rigidity). Lean Lean ✓ Stmt Let A_i and B_i be two families of square matrices of the same size, indexed by the same finite alphabet, and suppose that, for every i, j ,

$$B_i A_j = A_i B_j. \quad (19.265)$$

If A is injective, or more generally normal, then there is a scalar $c \in \mathbb{C}$ with $B_i = cA_i$ for every i . This is the commuting-tensor step in the proof of the source-factor contraction identities in [163, lines 1255–1325], where the tensor commuting with a normal matrix product unitary tensor is shown to be a multiple of it.

Proof. ✓ Proof Let $N \geq 1$ be a blocking length at which the length- N words of A span the full matrix algebra, and decompose the identity as $\mathbf{1} = \sum_{\sigma} c_{\sigma} A^{\sigma_1} \dots A^{\sigma_N}$. The cross relation moves the marked letter successively through the word,

$$B_i A^{\sigma_1} \dots A^{\sigma_N} = A_i A^{\sigma_1} \dots A^{\sigma_{N-1}} B^{\sigma_N}.$$

When $N = 1$, the middle word on the right is absent. Thus $B_i = A_i C$ with $C = \sum_{\sigma} c_{\sigma} A^{\sigma_1} \dots A^{\sigma_{N-1}} B^{\sigma_N}$. Substituting this identity into (19.265) gives $A_i (C A_j - A_j C) = 0$ for all i and j . Multiplying on the left by a length- N word, whose last letter absorbs one of the A_i , and recombining the words into the identity gives $C A_j = A_j C$. A matrix commuting with every letter of a normal family is a multiple of the identity, so $C = c\mathbf{1}$ and $B_i = cA_i$. The injective case is the case of blocking length one. $\square$

Definition 19.5.5 (Canonical boundaries under physical adjunction). Lean ✓ Stmt Let $\mathcal{U}$ be an MPU with a canonical-form-II presentation and recorded positive diagonal fixed matrix ρ , with $\text{tr } \rho = 1$. Its physical adjoint has letters $(\mathcal{U}^\dagger)^{ij} = \overline{\mathcal{U}^{ji}}$; the virtual indices are not transposed. Give its normalized flattening the conjugated canonical blocks with the physical indices exchanged, and retain the same full support and the same fixed matrix ρ . This defines canonical-form-II data for $\mathcal{U}^\dagger$ with the same left and right boundaries I, ρ . These are the reflected boundaries used in [163, lines 1164–1254].

Proof. ✓ Proof Physical adjunction preserves periodic unitarity and transports the canonical blocks without changing their dimensions. Writing $\mathcal{E}_{\mathcal{U}}$ for the transfer map of $\mathcal{U}$, the transported fixed-point equation is

$$\mathcal{E}_{\mathcal{U}^\dagger}(\rho) = \overline{\mathcal{E}_{\mathcal{U}}(\overline{\rho})} = \overline{\mathcal{E}_{\mathcal{U}}(\rho)} = \overline{\rho} = \rho, \quad (19.266)$$

using $\overline{\rho} = \rho$ and $\mathcal{E}_{\mathcal{U}}(\rho) = \rho$. The transfer map is invariant under the physical-index permutation, so the transported data share the same fixed matrix. $\square$

Theorem 19.5.6 (Canonical boundary contractions). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7317](#) Let $\mathcal{U}$ be a simple MPU with a canonical-form-II presentation and recorded fixed matrix ρ . Let W be its raw double layer, put $(\Phi|_{(a,b)} = \delta_{a,b}$ and $|\rho)_{(a,b)} = \rho_{ba}$. For all physical indices i, j, k, l ,

$$W^{ij}W^{kl} = W^{ij}|\rho)(\Phi|W^{kl}, \quad (19.267)$$

$$(\Phi|W^{ij}W^{kl} = \delta_{ij}(\Phi|W^{kl}, \quad (19.268)$$

$$W^{ij}W^{kl}|\rho) = \delta_{kl}W^{ij}|\rho). \quad (19.269)$$

These are the boundary removals used in the six-tensor argument of [163, lines 1164–1254].

Proof. [✓ Proof](#) Aligning the simplicity witnesses with the canonical fixed pair using the stabilized normalized transfer power gives (19.267) and $(\Phi|W^{ij}|\rho) = \delta_{ij}$. Contracting (19.267) on the left with $(\Phi|$ yields (19.268); contracting on the right with $|\rho)$ yields (19.269). $\square$

Theorem 19.5.7 (Canonical boundary contractions for a simple physical adjoint). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU with a canonical-form-II presentation and recorded fixed matrix ρ . Assume that $\mathcal{U}^\dagger$ is simple. Let $W_\dagger$ be the raw double layer of $\mathcal{U}^\dagger$, put $(\Phi|_{(a,b)} = \delta_{a,b}$ and $|\rho)_{(a,b)} = \rho_{ba}$. For all physical indices i, j, k, l ,

$$(\Phi|W_\dagger^{ij}W_\dagger^{kl} = \delta_{ij}(\Phi|W_\dagger^{kl}, \quad (19.270)$$

$$W_\dagger^{ij}W_\dagger^{kl}|\rho) = \delta_{kl}W_\dagger^{ij}|\rho). \quad (19.271)$$

The simplicity assumption is precisely the additional hypothesis used for the reflected reduction in [163, lines 1221–1254].

Proof. [✓ Proof](#) Transport the canonical presentation to $\mathcal{U}^\dagger$ with the same ρ , then apply Theorem 19.5.6 using the assumed simplicity of $\mathcal{U}^\dagger$. $\square$

Definition 19.5.8 (The three-tensor contraction s). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPO tensor of physical dimension d and bond dimension D , and let ρ be a complex $D \times D$ matrix. Define an MPO tensor s of the same dimensions by

$$s_{ab}^{ij} = \sum_{p,q=0}^{d-1} \sum_{x,r,t=0}^{D-1} \mathcal{U}_{xb}^{ip} \overline{\mathcal{U}_{xr}^{qp}} \mathcal{U}_{at}^{qj} \rho_{tr}. \quad (19.272)$$

The left open bond is on the bottom tensor and the right open bond is on the top tensor. For diagonal ρ only $t = r$ contributes. This is the tensor marked s in [163, lines 1190–1218]. No factor $1/d$ occurs: the letters in this contraction are unnormalized.

Theorem 19.5.9 (The two reductions of the six-tensor network). [Lean](#) [✓ Stmt](#) [Discussion: #7317](#) Let $\mathcal{U}$ have a canonical-form-II presentation with recorded fixed matrix ρ , and assume that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Construct s from this same ρ by (19.272). For all physical indices i, j, k, l ,

$$s^{ij}\mathcal{U}^{kl} = \mathcal{U}^{ij}s^{kl}. \quad (19.273)$$

This is the equality obtained from the two reductions in [163, lines 1164–1254].

Proof. [✓ Proof](#) For fixed open indices i, j, k, l, a, b , expand the six-tensor network as

$$\mathcal{N}_{ab}^{ij,kl} = \sum_{p,q,v,w} \sum_{x,m,n,z,r,t} \mathcal{U}_{xm}^{ip} \mathcal{U}_{mb}^{kq} \overline{\mathcal{U}_{xn}^{vp}} \overline{\mathcal{U}_{nr}^{wq}} \mathcal{U}_{az}^{vj} \mathcal{U}_{zt}^{wl} \rho_{tr}. \quad (19.274)$$

The first sum ranges over physical indices and the second over virtual indices. Applying the right-boundary contraction to the lower two layers gives $\mathcal{N}_{ab}^{ij,kl} = (s^{ij}\mathcal{U}^{kl})_{ab}$. Regrouping the same finite sum and applying the left-boundary contraction for the physical adjoint to the upper two layers gives $\mathcal{N}_{ab}^{ij,kl} = (\mathcal{U}^{ij}s^{kl})_{ab}$. Both expressions are therefore equal. $\square$

Theorem 19.5.10 (The contraction s is a scalar multiple of the MPU tensor). [Lean](#) [✓ Stmt](#)
[Discussion: #7317](#) Let $\mathcal{U}$ have a canonical-form-II presentation with recorded fixed matrix ρ , and suppose that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Then the tensor s constructed from this ρ satisfies

$$s^{ij} = \delta \mathcal{U}^{ij} \quad (19.275)$$

for a single scalar $\delta \in \mathbb{C}$ and every physical pair i, j . This is the proportionality conclusion in [163, lines 1255–1325].

Proof. [✓ Proof](#) Put $\tilde{\mathcal{U}}^{ij} = d^{-1/2} \mathcal{U}^{ij}$. Canonical form II gives $d, D > 0$ and normality of $\tilde{\mathcal{U}}$. Scaling (19.273) gives $s^{ij} \tilde{\mathcal{U}}^{kl} = \tilde{\mathcal{U}}^{ij} s^{kl}$. Cross-product rigidity, using blocking to an injective tensor, gives $s^{ij} = c \tilde{\mathcal{U}}^{ij}$ for one scalar c . Set $\delta = c/\sqrt{d}$. $\square$

19.5.1 The anomaly three-cocycle from fusion tensors

For matrix product operator tensors M and N of bond dimensions D_M and D_N , write MN for the stacked product tensor, whose periodic operators are the products of those of M and N , and write $\mathbf{a}$ for the bond associator of Definition 21.3.6, the permutation from $\mathbb{C}^{D_1} \otimes (\mathbb{C}^{D_2} \otimes \mathbb{C}^{D_3})$, the bond space of $M(NP)$, to $(\mathbb{C}^{D_1} \otimes \mathbb{C}^{D_2}) \otimes \mathbb{C}^{D_3}$, the bond space of $(MN)P$.

Lemma 19.5.1.1 (Composition of reductions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Reductions compose: if (V_1, W_1) reduces C onto B and (V_2, W_2) reduces B onto A , then $(V_2 V_1, W_1 W_2)$ reduces C onto A . If (V, W) reduces the doubled-index tensor of X onto that of A , then $(V \otimes \mathbf{1}, W \otimes \mathbf{1})$ reduces XP onto AP and $(\mathbf{1} \otimes V, \mathbf{1} \otimes W)$ reduces PX onto PA . A reduction (V, W) of $M(NP)$ gives the reduction $(V \mathbf{a}^{-1}, \mathbf{a} W)$ of $(MN)P$, and conversely.*

Proof. [✓ Proof](#) The first claim is immediate from the intertwining equations. For the second, Lemma 21.3.8 expands a word of XP along a configuration pair (σ, τ) as a sum over the contracted middle configurations ρ , and $V \otimes \mathbf{1}$ and $W \otimes \mathbf{1}$ act on the first factor only:

$$(V \otimes \mathbf{1}) (XP)^{\sigma\tau} (W \otimes \mathbf{1}) = \sum_{\rho} V X^{\sigma\rho} W \otimes P^{\rho\tau} = \sum_{\rho} A^{\sigma\rho} \otimes P^{\rho\tau} = (AP)^{\sigma\tau}.$$

The case PX is the same with the factors exchanged. For the third, the letters satisfy $((MN)P)^i \mathbf{a} = \mathbf{a} (M(NP))^i$, so the words do too. $\square$

Lemma 19.5.1.2 (Scalar comparison of two reductions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be normal with positive bond dimension, and let (V, W) and (V', W') reduce B onto A , where B and A generate the same matrix product vectors on nonempty chains. Then there is a unique $z \in \mathbb{C}^\times$ such that $VB^{\mathbf{w}} = z V'B^{\mathbf{w}}$ for all sufficiently long words $\mathbf{w}$. If $XA^{\mathbf{w}} = z YA^{\mathbf{w}}$ for all long words, then $XVB^{\mathbf{w}} = z YVB^{\mathbf{w}}$ for all long words.*

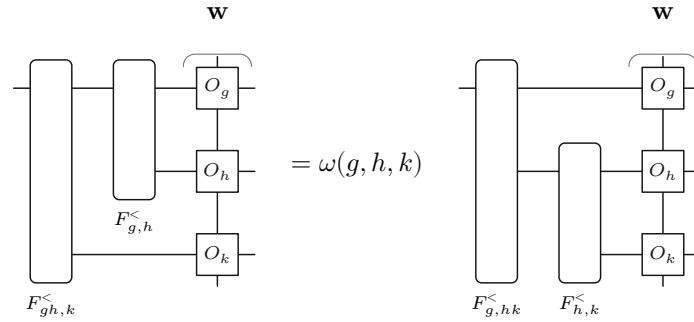
Proof. [✓ Proof](#) Existence is Theorem 25.11.28. Uniqueness holds because $VB^{\mathbf{w}}W = A^{\mathbf{w}}$ and a normal tensor has nonzero words of every multiple of its injectivity length. For the last claim, once both exterior parts of a long word $\mathbf{w} = \mathbf{pcq}$ exceed the residual nilpotency length, $VB^{\mathbf{w}} = A^{\mathbf{pc}}VB^{\mathbf{q}}$. $\square$

Definition 19.5.1.3 (The anomaly three-cochain). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{O_g\}_{g \in G}$ be matrix product operator tensors of positive bond dimension whose doubled-index tensors are normal and whose periodic operators satisfy $O_g O_h = O_{gh}$ on every nonempty chain.

A choice of fusion tensors is, for every pair g, h , a reduction $(F_{g,h}^<, F_{g,h}^>)$ of $O_g O_h$ onto O_{gh} . The triple product $T = (O_g O_h) O_k$ reduces onto O_{ghk} along the two fusion trees with left boundaries

$$V^L = F_{gh,k}^< (F_{g,h}^< \otimes \mathbf{1}), \quad V^R = F_{g,hk}^< (\mathbf{1} \otimes F_{h,k}^<) \mathbf{a}^{-1}, \quad (19.276)$$

where the target bond space of $F_{g,hk}^<$, that of $O_{g(hk)}$, is identified with that of $O_{(gh)k}$ through the equality $g(hk) = (gh)k$ in G ; the same identification enters the right boundaries $W^L = (F_{g,h}^> \otimes \mathbf{1}) F_{gh,k}^>$ and $W^R = \mathbf{a}(\mathbf{1} \otimes F_{h,k}^>) F_{g,hk}^>$. The anomaly three-cochain value $\omega(g, h, k) \in \mathbb{C}^\times$ is the scalar with $V^L T^{\mathbf{w}} = \omega(g, h, k) V^R T^{\mathbf{w}}$ for all long words $\mathbf{w}$. Equivalently $T^{\mathbf{w}} W^R = \omega(g, h, k) T^{\mathbf{w}} W^L$ for the right boundaries, which is the defining display of [163] before its three-cocycle equation. No normalization of ω is imposed.



Lemma 19.5.1.4 (The two fusion trees are reductions). [Lean](#) [Lean](#) [✓ Stmt](#) The pairs (V^L, W^L) and (V^R, W^R) of Definition 19.5.1.3 are reductions of the triple product $(O_g O_h) O_k$ onto O_{ghk} .

Proof. [✓ Proof](#) For (V^L, W^L) , tensor the reduction $(F_{g,h}^<, F_{g,h}^>)$ with the identity on the third factor and compose with $(F_{gh,k}^<, F_{gh,k}^>)$. For (V^R, W^R) , tensor $(F_{h,k}^<, F_{h,k}^>)$ with the identity on the first factor, compose with $(F_{g,hk}^<, F_{g,hk}^>)$, and transport along the associator $\mathbf{a}$; all three steps are Lemma 19.5.1.1. $\square$

Theorem 19.5.1.5 (Characterization of the anomaly three-cochain). [Lean](#) [Lean](#) [✓ Stmt](#) The scalar $\omega(g, h, k)$ exists, is nonzero, and is the only scalar with the dressed identity of Definition 19.5.1.3.

Proof. [✓ Proof](#) By Lemma 19.5.1.4 both fusion trees are reductions of T onto the normal tensor O_{ghk} , and T and O_{ghk} have the same periodic operators on nonempty chains. Apply Lemma 19.5.1.2. $\square$

Theorem 19.5.1.6 (The anomaly three-cocycle). [Lean](#) [Lean](#) [✓ Stmt](#) For every $g, h, k, l \in G$,

$$\omega(gh, k, l) \omega(g, h, kl) = \omega(g, h, k) \omega(g, hk, l) \omega(h, k, l). \quad (19.277)$$

Proof. [✓ Proof](#) The five fusion trees of the fourfold product $((O_g O_h) O_k) O_l$ are reductions onto O_{ghkl} . Two trees that differ by one change of bracketing are related against long words by one value of ω : either directly, through Lemma 19.5.1.1 with an identity factor, or after pulling back along a fusion tensor by Lemma 19.5.1.2. Write $\mathbf{a}_{p,q,r}$ for $\mathbf{a}$ on $\mathbb{C}^p \otimes \mathbb{C}^q \otimes \mathbb{C}^r$. Identifying the boundaries uses the naturality of $\mathbf{a}$, for $X: \mathbb{C}^n \rightarrow \mathbb{C}^m$,

$$\begin{aligned} \mathbf{a}_{m,q,r}^{-1} (X \otimes \mathbf{1}_q \otimes \mathbf{1}_r) &= (X \otimes \mathbf{1}_{qr}) \mathbf{a}_{n,q,r}^{-1}, \\ \mathbf{a}_{p,m,r}^{-1} (\mathbf{1}_p \otimes X \otimes \mathbf{1}_r) &= (\mathbf{1}_p \otimes (X \otimes \mathbf{1}_r)) \mathbf{a}_{p,n,r}^{-1}, \\ \mathbf{a}_{p,q,m}^{-1} (\mathbf{1}_{pq} \otimes X) &= (\mathbf{1}_p \otimes (\mathbf{1}_q \otimes X)) \mathbf{a}_{p,q,n}^{-1}, \end{aligned}$$

and the pentagon identity for the reassociation permutations,

$$(\mathbf{1}_p \otimes a_{q,r,s}^{-1}) a_{p,q,r,s}^{-1} (a_{p,q,r}^{-1} \otimes \mathbf{1}_s) = a_{p,q,rs}^{-1} a_{pq,r,s}^{-1}.$$

The two paths around the pentagon compare the same pair of trees, and uniqueness of the scalar on a nonvanishing word gives (19.277). $\square$

Theorem 19.5.1.7 (Independence of the fusion tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let ω and ω' come from two choices of fusion tensors, and let $\beta_{g,h} \in \mathbb{C}^\times$ be the scalar with $F_{g,h}^{<} B^{\mathbf{w}} = \beta_{g,h} F_{g,h}'^{<} B^{\mathbf{w}}$ for all long words of $B = O_g O_h$. Then*

$$\omega'(g, h, k) = \frac{\beta_{g,hk} \beta_{h,k}}{\beta_{g,h} \beta_{gh,k}} \omega(g, h, k), \quad (19.278)$$

so the cohomology class of ω does not depend on the fusion tensors, as in (19.257).

Proof. [✓ Proof](#) The left boundaries of the two choices of the tree fusing g with h first differ by $\beta_{g,h} \beta_{gh,k}$, and those of the tree fusing h with k first by $\beta_{h,k} \beta_{g,hk}$, by the two parts of Lemma 19.5.1.2. Compare with Theorem 19.5.1.5. $\square$

Definition 19.5.1.8 (The anomaly class). [Lean](#) [✓ Stmt](#) Let the doubled-index tensors be normal. The anomaly class in $H^3(G, \mathbb{C}_{\text{triv}}^\times)$ is the class of the anomaly three-cocycle ω , for any choice of fusion tensors.

Corollary 19.5.1.9 (The anomaly class is independent of the fusion tensors). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The class $[\omega] \in H^3(G, \mathbb{C}_{\text{triv}}^\times)$ of the anomaly three-cocycle is the same for every choice of fusion tensors. It vanishes exactly when ω has trivial gauge class.*

Proof. [✓ Proof](#) Two choices of fusion tensors give cohomologous three-cocycles by Theorem 19.5.1.7, hence equal classes by (19.405). The second statement is (19.406). $\square$

19.5.2 Reductions through action tensors

Definition 19.5.2.1 (The action tensor). [Lean](#) [✓ Stmt](#) For a matrix product operator tensor T of bond dimension D_1 and a matrix product state tensor B of bond dimension D_2 , the action tensor $T \cdot B$ has bond dimension $D_1 D_2$ and letters

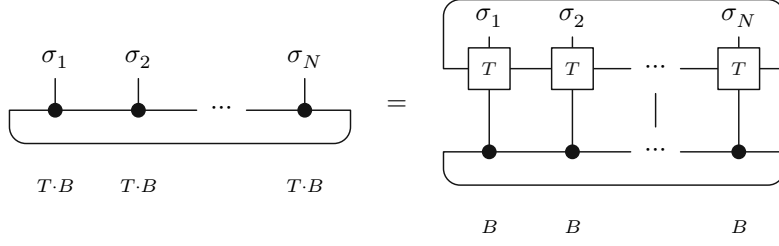
$$(T \cdot B)^i = \sum_j T^{ij} \otimes B^j. \quad (19.279)$$

It is the tensor whose local decomposition onto an image block gives the action tensors of the periodic-boundary case of [164, “Periodic boundary condition case”].

Theorem 19.5.2.2 (The action tensor generates the acted state). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every word σ of length N ,*

$$(T \cdot B)^\sigma = \sum_\tau T^{\sigma\tau} \otimes B^\tau, \quad (19.280)$$

the sum running over words τ of length N . Consequently $|V^{(N)}(T \cdot B)\rangle = \rho^{(N)}(T) |V^{(N)}(B)\rangle$ for every N .



Proof. ✓ Proof Induction on N : the empty word gives $\mathbf{1} \otimes \mathbf{1}$, and one more letter multiplies both sides by $(T \cdot B)^i$, since $(T^{\sigma\tau} \otimes B^\tau)(T^{ij} \otimes B^j) = T^{\sigma\tau} T^{ij} \otimes B^\tau B^j$. Taking traces and using $\text{tr}(X \otimes Y) = \text{tr } X \text{ tr } Y$ gives $V^{(N)}(T \cdot B)_\sigma = \sum_\tau \text{tr}(T^{\sigma\tau}) \text{tr}(B^\tau) = (\rho^{(N)}(T) V^{(N)}(B))_\sigma$. $\square$

Lemma 19.5.2.3 (Reductions through action tensors).

Lean Lean Lean Lean Lean Lean Lean ✓ Stmt If (V, W) reduces B onto A , then $(\mathbf{1} \otimes V, \mathbf{1} \otimes W)$ reduces $T \cdot B$ onto $T \cdot A$. If (V, W) reduces the doubled-index tensor of X onto that of Y , then $(V \otimes \mathbf{1}, W \otimes \mathbf{1})$ reduces $X \cdot B$ onto $Y \cdot B$. The letters satisfy $((MN) \cdot A)^i \mathbf{a} = \mathbf{a} (M \cdot (N \cdot A))^i$, so a reduction (V, W) of $M \cdot (N \cdot A)$ onto Z gives the reduction $(V \mathbf{a}^{-1}, \mathbf{a} W)$ of $(MN) \cdot A$ onto Z , and a reduction (V, W) of $(MN) \cdot A$ onto Z gives the reduction $(V \mathbf{a}, \mathbf{a}^{-1} W)$ of $M \cdot (N \cdot A)$ onto Z , as in Lemma 19.5.1.1. If P intertwines the letters of two operator tensors, $X'^{ij} P = P X^{ij}$ for all i, j , then $(X' \cdot B)^i (P \otimes \mathbf{1}) = (P \otimes \mathbf{1}) (X \cdot B)^i$. Finally, a relation $V B^\mathbf{w} = z V' B^\mathbf{w}$ for all long words lifts to $(\mathbf{1} \otimes V)(T \cdot B)^\mathbf{w} = z (\mathbf{1} \otimes V')(T \cdot B)^\mathbf{w}$, and the analogous relation for the doubled-index tensor of X lifts through $V \otimes \mathbf{1}$ to $X \cdot B$.

Proof. ✓ Proof By Theorem 19.5.2.2, every word of $T \cdot B$ is the sum (19.280) of Kronecker products of a word of T and a word of B of the same length. A boundary of the form $\mathbf{1} \otimes V$ passes through the first factor of each summand:

$$(\mathbf{1} \otimes V)(T \cdot B)^\sigma (\mathbf{1} \otimes W) = \sum_\tau T^{\sigma\tau} \otimes V B^\tau W = \sum_\tau T^{\sigma\tau} \otimes A^\tau = (T \cdot A)^\sigma,$$

which is the reduction of the state. For the operator, the word $X^{\sigma\tau}$ is the word of the doubled-index tensor of X on the pair configuration (σ, τ) , so $(V \otimes \mathbf{1})(X \cdot B)^\sigma (W \otimes \mathbf{1}) = \sum_\tau V X^{\sigma\tau} W \otimes B^\tau = (Y \cdot B)^\sigma$. For a dressed proportionality, if $V B^\tau = z V' B^\tau$ for all words τ of length at least N_0 , then for every word σ of such a length

$$(\mathbf{1} \otimes V)(T \cdot B)^\sigma = \sum_\tau T^{\sigma\tau} \otimes V B^\tau = z \sum_\tau T^{\sigma\tau} \otimes V' B^\tau = z (\mathbf{1} \otimes V')(T \cdot B)^\sigma,$$

because every τ in the sum has the length of σ ; the same substitution on the pair words of X gives the lift through $V \otimes \mathbf{1}$. For the intertwiner, $(X'^{ij} \otimes B^j)(P \otimes \mathbf{1}) = X'^{ij} P \otimes B^j = P X^{ij} \otimes B^j = (P \otimes \mathbf{1})(X^{ij} \otimes B^j)$. The letter identity is the reassociation of Kronecker products:

$$((MN) \cdot A)^i = \sum_{j,k} (M^{ij} \otimes N^{jk}) \otimes A^k = \mathbf{a} \left(\sum_{j,k} M^{ij} \otimes (N^{jk} \otimes A^k) \right) \mathbf{a}^{-1} = \mathbf{a} (M \cdot (N \cdot A))^i \mathbf{a}^{-1}.$$

$\square$

19.5.3 An anomaly obstructs invariant normal states

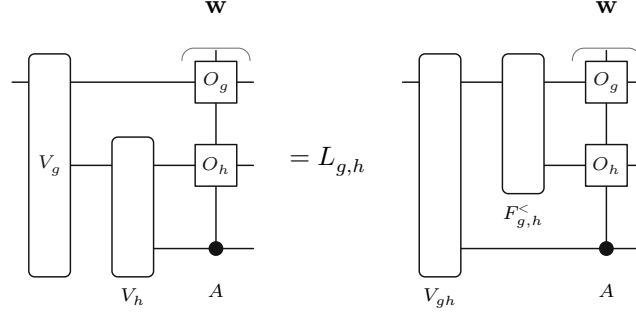
Definition 19.5.3.1 (L-symbol of an invariant normal state).

Lean Lean Lean Lean Lean ✓ Stmt Let $\{O_g\}$ be as in Definition 19.5.1.3, and let A be a matrix product state tensor with $O_g |V^{(N)}(A)\rangle = |V^{(N)}(A)\rangle$ for every g and every $N \geq 1$. Action tensors for A are, for every g ,

a reduction (V_g, W_g) of $O_g \cdot A$ onto A . The stacked action $(O_g O_h) \cdot A$ then has the two left boundaries

$$\alpha_{g,h} = V_g(\mathbf{1} \otimes V_h) \mathbf{a}^{-1}, \quad \varphi_{g,h} = V_{gh}(F_{g,h}^< \otimes \mathbf{1}),$$

and $L_{g,h}$ is a nonzero scalar z with $\alpha_{g,h} B^{\mathbf{w}} = z \varphi_{g,h} B^{\mathbf{w}}$ for all long words $\mathbf{w}$ of $B = (O_g O_h) \cdot A$, or 1 if there is none. This is the L -symbol of a single invariant block [163].



Theorem 19.5.3.2 (Characterization of the single-block L -symbol). [Lean](#) [Lean](#) [✓ Stmt](#) If A in Definition 19.5.3.1 is normal of positive bond dimension, then action tensors for A exist [164, lines 1064–1090], and for every choice of them $\alpha_{g,h} B^{\mathbf{w}} = L_{g,h} \varphi_{g,h} B^{\mathbf{w}}$ for all long words $\mathbf{w}$ of $B = (O_g O_h) \cdot A$.

Proof. [✓ Proof](#) Existence is Corollary 25.6.4. By Lemmas 19.5.2.3 and 19.5.1.1, both $\alpha_{g,h}$ and $\varphi_{g,h}$ are left boundaries of reductions of B onto A , and B generates the same vectors as A , so Lemma 19.5.1.2 gives a nonzero scalar relating them against long words. $\square$

Theorem 19.5.3.3 (Compatibility of the L -symbol with the anomaly). [Lean](#) [✓ Stmt](#) In the setting of Theorem 19.5.3.2, for all $g, h, k \in G$,

$$L_{g,hk} L_{h,k} = \omega(g, h, k) L_{g,h} L_{gh,k}. \quad (19.281)$$

This is (19.412) for a one-point set of blocks.

Proof. [✓ Proof](#) Consider the triple action $B = ((O_g O_h) O_k) \cdot A$, and write V^L, V^R for the two fusion trees of (g, h, k) as in Definition 19.5.1.3. The naturality and pentagon identities of $\mathbf{a}$ give five identities between left boundaries of reductions of B onto A :

$$\begin{aligned} \alpha_{g,h}(\mathbf{1} \otimes V_k) \mathbf{a}^{-1} &= V_g(\mathbf{1} \otimes \alpha_{h,k}) \mathbf{a}^{-1}(\mathbf{a}^{-1} \otimes \mathbf{1}), \\ V_g(\mathbf{1} \otimes \varphi_{h,k}) \mathbf{a}^{-1}(\mathbf{a}^{-1} \otimes \mathbf{1}) &= \alpha_{g,hk}((\mathbf{1} \otimes F_{h,k}^<) \mathbf{a}^{-1} \otimes \mathbf{1}), \\ \varphi_{g,h}(\mathbf{1} \otimes V_k) \mathbf{a}^{-1} &= \alpha_{gh,k}((F_{g,h}^< \otimes \mathbf{1}) \otimes \mathbf{1}), \\ \varphi_{gh,k}((F_{g,h}^< \otimes \mathbf{1}) \otimes \mathbf{1}) &= V_{ghk}(V^L \otimes \mathbf{1}), \\ \varphi_{g,hk}((\mathbf{1} \otimes F_{h,k}^<) \mathbf{a}^{-1} \otimes \mathbf{1}) &= V_{ghk}(V^R \otimes \mathbf{1}). \end{aligned} \quad (19.282)$$

Starting from $\alpha_{g,h}(\mathbf{1} \otimes V_k) \mathbf{a}^{-1} B^{\mathbf{w}}$, replace $\alpha_{g,h}$ by $L_{g,h} \varphi_{g,h}$, then $\alpha_{gh,k}$ by $L_{gh,k} \varphi_{gh,k}$, then V^L by $\omega(g, h, k) V^R$; or apply (19.282), replace $\alpha_{h,k}$ by $L_{h,k} \varphi_{h,k}$ inside the action of g , and then $\alpha_{g,hk}$ by $L_{g,hk} \varphi_{g,hk}$. Each replacement holds against long words by Theorem 19.5.3.2, lifted through the action tensors by Lemma 19.5.2.3 and pulled back along the reductions by Lemma 19.5.1.2. For all long words,

$$\alpha_{g,h}(\mathbf{1} \otimes V_k) \mathbf{a}^{-1} B^{\mathbf{w}} = L_{g,h} L_{gh,k} \omega(g, h, k) V_{ghk}(V^R \otimes \mathbf{1}) B^{\mathbf{w}} = L_{h,k} L_{g,hk} V_{ghk}(V^R \otimes \mathbf{1}) B^{\mathbf{w}}.$$

The boundary $V_{ghk}(V^R \otimes \mathbf{1})$ belongs to a reduction onto the normal tensor A , so it is nonzero against arbitrarily long words, and the two scalars agree. $\square$

Theorem 19.5.3.4 (An anomaly obstructs invariant normal states). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\{O_g\}$ be matrix product operator tensors with normal doubled-index tensors and $O_g O_h = O_{gh}$ on every nonempty chain. If a normal matrix product state tensor of positive bond dimension satisfies $O_g |V^{(N)}(A)\rangle = |V^{(N)}(A)\rangle$ for every g and every $N \geq 1$, then the anomaly three-cocycle has trivial gauge class. Equivalently, a group with nontrivial anomaly has no invariant normal matrix product state. This is the statement that an anomalous representation cannot have an invariant injective matrix product state [163, line 1545], [164, line 738], with normal in place of injective and without unitarity or simplicity of the operators.*

Proof. [✓ Proof](#) Action tensors exist by Theorem 19.5.3.2, so Theorem 19.5.3.3 applies, and Theorem 19.5.10.60 for the single block gives $\omega = dL$. $\square$

19.5.4 Blocks permuted by the group

Let $\{O_g\}$ be as in Definition 19.5.1.3, and let G act on a set X . Consider normal matrix product state tensors A_x , $x \in X$, of positive bond dimensions, possibly different for different x , with $O_g |V^{(N)}(A_x)\rangle = |V^{(N)}(A_{gx})\rangle$ for every g , x and $N \geq 1$ [164, line 1064].

Definition 19.5.4.1 (L-symbols of permuted blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Action tensors for the family (A_x) are, for every g and x , a reduction $(V_{g,x}, W_{g,x})$ of $O_g \cdot A_x$ onto A_{gx} . The stacked action $B = (O_g O_h) \cdot A_x$ then reduces onto $A_{g(hx)} = A_{(gh)x}$ along two trees, with left boundaries

$$V_{g,hx}(\mathbf{1} \otimes V_{h,x}) \mathbf{a}^{-1}, \quad V_{gh,x}(F_{g,h}^< \otimes \mathbf{1}),$$

and $L_{g,h}^x$ is a nonzero scalar relating them against all long words of B , or 1 if there is none, as in Definition 19.5.3.1.

Theorem 19.5.4.2 (Characterization of the L-symbols of permuted blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Action tensors for the family (A_x) of Definition 19.5.4.1 exist; for the bond-one identity tensor they are the identification of the bond spaces. For every choice of them, $L_{g,h}^x$ relates the two left boundaries against all long words of $(O_g O_h) \cdot A_x$.*

Proof. [✓ Proof](#) As in Theorem 19.5.3.2: existence is Corollary 25.6.4 applied to $O_g \cdot A_x$ and A_{gx} , which generate the same vectors. By Lemmas 19.5.2.3 and 19.5.1.1 both boundaries are left boundaries of reductions of $(O_g O_h) \cdot A_x$ onto the normal tensor $A_{(gh)x}$, so Lemma 19.5.1.2 gives a nonzero scalar relating them against long words. $\square$

Theorem 19.5.4.3 (Compatibility for permuted blocks). [Lean](#) [Lean](#) [✓ Stmt](#) *For all $x \in X$ and $g, h, k \in G$,*

$$L_{g,hk}^x L_{h,k}^x = \omega(g, h, k) L_{g,h}^{kx} L_{gh,k}^x, \quad (19.283)$$

which is (19.412). Consequently the restriction of ω to any subgroup fixing one block has trivial gauge class [164, lines 740–745].

Proof. [✓ Proof](#) Write

$$\alpha_{g,h}^x = V_{g,hx}(\mathbf{1} \otimes V_{h,x}) \mathbf{a}^{-1}, \quad \varphi_{g,h}^x = V_{gh,x}(F_{g,h}^< \otimes \mathbf{1})$$

for the two left boundaries of Definition 19.5.4.1, so that $\alpha_{g,h}^x = L_{g,h}^x \varphi_{g,h}^x$ against long words of $(O_g O_h) \cdot A_x$ by Theorem 19.5.4.2. The block labels $h(kx)$ and $(hk)x$ are equal, and so are $g(h(kx))$ and $(ghk)x$, and the bond spaces of blocks with equal labels are identified, so that $V_{g,h(kx)} = V_{g,(hk)x}$. With this identification, the naturality and pentagon identities of $\mathbf{a}$ give, for $B = ((O_g O_h) O_k) \cdot A_x$ and the fusion trees V^L, V^R of (g, h, k) , the five identities

$$\begin{aligned} \alpha_{g,h}^{kx} (\mathbf{1} \otimes V_{k,x}) \mathbf{a}^{-1} &= V_{g,(hk)x} (\mathbf{1} \otimes \alpha_{h,k}^x) \mathbf{a}^{-1} (\mathbf{a}^{-1} \otimes \mathbf{1}), \\ V_{g,(hk)x} (\mathbf{1} \otimes \varphi_{h,k}^x) \mathbf{a}^{-1} (\mathbf{a}^{-1} \otimes \mathbf{1}) &= \alpha_{g,hk}^x ((\mathbf{1} \otimes F_{h,k}^<) \mathbf{a}^{-1} \otimes \mathbf{1}), \\ \varphi_{g,h}^{kx} (\mathbf{1} \otimes V_{k,x}) \mathbf{a}^{-1} &= \alpha_{gh,k}^x ((F_{g,h}^< \otimes \mathbf{1}) \otimes \mathbf{1}), \\ \varphi_{gh,k}^x ((F_{g,h}^< \otimes \mathbf{1}) \otimes \mathbf{1}) &= V_{ghk,x} (V^L \otimes \mathbf{1}), \\ \varphi_{g,hk}^x ((\mathbf{1} \otimes F_{h,k}^<) \mathbf{a}^{-1} \otimes \mathbf{1}) &= V_{ghk,x} (V^R \otimes \mathbf{1}). \end{aligned} \tag{19.284}$$

The first factor $\alpha_{g,h}^{kx}$ acts on the block kx reached by O_k , which is why $L_{g,h}^{kx}$ appears. Replacing, as in the proof of Theorem 19.5.3.3, $\alpha_{g,h}^{kx}$, $\alpha_{gh,k}^x$ and V^L along one route, and (19.284), $\alpha_{h,k}^x$ and $\alpha_{g,hk}^x$ along the other, gives for all long words

$$\begin{aligned} \alpha_{g,h}^{kx} (\mathbf{1} \otimes V_{k,x}) \mathbf{a}^{-1} B^{\mathbf{w}} &= L_{g,h}^{kx} L_{g,h,k}^x \omega(g, h, k) V_{ghk,x} (V^R \otimes \mathbf{1}) B^{\mathbf{w}} \\ &= L_{h,k}^x L_{g,hk}^x V_{ghk,x} (V^R \otimes \mathbf{1}) B^{\mathbf{w}}. \end{aligned}$$

The boundary $V_{ghk,x} (V^R \otimes \mathbf{1})$ belongs to a reduction onto the normal tensor $A_{(ghk)x}$, so it is nonzero against arbitrarily long words, and the two scalars agree. The second statement is Theorem 19.5.10.60. $\square$

Theorem 19.5.4.4 (The anomaly at an involution). [Lean](#) [✓ Stmt](#) *If L is compatible with ω and $g^2 = e$, then for every $x \in \mathbf{X}$*

$$\omega(g, g, g) = \frac{L_{g,g}^x L_{g,e}^x}{L_{g,g}^{gx} L_{e,g}^x}. \tag{19.285}$$

This is the formula of [163, line 5121].

Proof. [✓ Proof](#) Put $h = k = g$ in (19.412) and use $gg = e$. $\square$

Theorem 19.5.4.5 (The involution ratio from the self-pair and identity L-symbols). [Lean](#) [✓ Stmt](#) *If L is compatible with ω and $g^2 = e$, then for every $x \in \mathbf{X}$*

$$\frac{L_{g,g}^x L_{e,e}^x}{L_{g,g}^{gx} L_{e,e}^{gx}} = \omega(g, g, g) \omega(g, e, g).$$

For L-symbols and cocycles that are trivial when an argument is the identity, this is $L_{g,g}^x / L_{g,g}^{gx} = \omega(g, g, g)$, the relation $L_A / L_B = \omega$ of [166, Section III.C].

Proof. [✓ Proof](#) Compatibility at (g, e, g) gives $L_{e,g}^x = \omega(g, e, g) L_{g,e}^{gx}$, and at (g, e, e) it gives $L_{e,e}^z = \omega(g, e, e) L_{g,e}^z$ for $z = x$ and $z = gx$. Substituting into the formula of Theorem 19.5.4.4 gives the claim. $\square$

19.5.5 Domain walls between permuted blocks

Keep the setting of the previous subsection. A domain wall between the blocks x and y is a family $e = (e^i)_i$ of $D_x \times D_y$ matrices, placed between a region of A_x and a region of A_y [166, Section III.B]. For an operator tensor T write $(T \cdot e)^i = \sum_j T^{ij} \otimes e^j$.

Definition 19.5.5.1 (Local action on domain walls). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let e be a domain wall between x and y , and e' a domain wall between gx and gy . The element g carries e to e' with phase c if $e \neq 0$, $e' \neq 0$, $c \neq 0$, and there is N such that

$$V_{g,x}(O_g \cdot A_x)^{\mathbf{u}}(O_g \cdot e)^i(O_g \cdot A_y)^{\mathbf{v}}W_{g,y} = c A_{gx}^{\mathbf{u}} e'^i A_{gy}^{\mathbf{v}}$$

for every letter i and all words $\mathbf{u}, \mathbf{v}$ of length at least N . This is the local relation of [166, Section III.B], drawn there at the renormalization fixed point with one site on each side of the wall; the words $\mathbf{u}, \mathbf{v}$ are the blocked sites of the discussion away from the fixed point [52].

Theorem 19.5.5.2 (Fractionalization of the symmetry on domain walls). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose h carries the domain wall e_1 between x and y to e_2 with phase $B_{x,y}^h$ and g carries e_2 to e_3 with phase $B_{hx,hy}^g$. Then gh carries e_1 to e_3 with phase $(L_{g,h}^y/L_{g,h}^x)B_{hx,hy}^gB_{x,y}^h$. Consequently, if gh carries e_1 to e_3 with phase $B_{x,y}^{gh}$, then

$$B_{hx,hy}^g B_{x,y}^h = \frac{L_{g,h}^x}{L_{g,h}^y} B_{x,y}^{gh}.$$

This is [166, Section IV.B].

Proof. [✓ Proof](#) Act with O_g on the local relation of h , which holds for every configuration of a chain with long regions: the boundary matrices $V_{h,x}, W_{h,y}$ pass through as $\mathbf{1} \otimes V_{h,x}$ and $\mathbf{1} \otimes W_{h,y}$, and the action of O_g on the action of O_h is the action of $O_g O_h$ up to the bond associator. The local relation of g then gives

$$V_{g,hx}(\mathbf{1} \otimes V_{h,x})\mathbf{a}^{-1}(O_g O_h \cdot \mathbf{C})\mathbf{a}(\mathbf{1} \otimes W_{h,y})W_{g,hy} = B_{hx,hy}^g B_{x,y}^h \mathbf{C}_3$$

for the chain $\mathbf{C} = A_x^{\mathbf{u}} e_1^i A_y^{\mathbf{v}}$ and the chain $\mathbf{C}_3$ with e_3 . Against long words the left boundary equals $L_{g,h}^x$ times the left boundary through the fusion tensor $F_{g,h}^<$; by the two-sided comparison of reductions onto a normal tensor, the right boundary equals $(L_{g,h}^y)^{-1}$ times the right boundary through $F_{g,h}^>$. The fusion tensors reduce $O_g O_h \cdot \mathbf{C}$ to $O_{gh} \cdot \mathbf{C}$, which gives the local action of gh with the stated phase. Since A_{ghx} and A_{ghy} are normal and $e_3 \neq 0$, some chain $\mathbf{C}_3$ with long regions is nonzero, so the phase of a local action is unique, and the two phases of gh agree. $\square$

Theorem 19.5.5.3 (The symmetry on states with two domain walls). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $|\psi(x-y-x)\rangle$ be the periodic state with an A_x region, a domain wall e between x and y , an A_y region, a domain wall f between y and x , and a second A_x region. If g carries e to e' with phase c and f to f' with phase c' , then

$$O_g |\psi(x-y-x)\rangle = c c' |\psi(gx-gy-gx)\rangle$$

whenever the first A_x region and the A_y region are long enough. For $\mathbb{Z}_2$ this is $U|\psi(A-B-A)\rangle = c_{AB} c_{BA} |\psi(B-A-B)\rangle$ [166, Section III.B].

Proof. [✓ Proof](#) The coefficient of $O_g |\psi\rangle$ is the trace of the acted chain. In each of the two acted regions insert the reduced block $W_{g,z} A_{gz}^{\mathbf{c}} V_{g,z}$, which leaves long words unchanged by the residual nilpotency of a reduction [187]. By cyclicity of the trace the chain splits into the two local relations, with phases c and c' , and the remaining trace is the coefficient of $|\psi(gx-gy-gx)\rangle$. $\square$

Proof. [✓ Proof](#) Both sides are multiples of the state with the domain walls $e_{AB}, e_{BA}, e_{AB}, e_{BA}$ at i_2, i_1, j_1, j_2 . If the inner string acts first, the outer string passes over its two domain walls, and Theorem 19.5.5.4 gives the phase $c_{AB}c_{BA}$. If the outer string acts first, the inner string acts on the B region it created, and Theorem 19.5.5.6 gives no phase. $\square$

Lemma 19.5.5.8 (The square of the exchange phase). [Lean](#) [Lean](#) [✓ Stmt](#) Let g carry e_{AB} to e_{BA} with phase c_{AB} and e_{BA} to e_{AB} with phase c_{BA} , where A and B are normal, and suppose that $U = O_g$ satisfies $U^2 = \mathbf{1}$ on the periodic chain of every length. Then $(c_{AB}c_{BA})^2 = 1$ [166, Section III.B].

Proof. [✓ Proof](#) By Theorem 19.5.5.3, $U|\psi(A-B-A)\rangle = c_{AB}c_{BA}|\psi(B-A-B)\rangle$ and $U|\psi(B-A-B)\rangle = c_{BA}c_{AB}|\psi(A-B-A)\rangle$ whenever the regions are long, so $U^2 = \mathbf{1}$ gives $|\psi(A-B-A)\rangle = (c_{AB}c_{BA})^2|\psi(A-B-A)\rangle$. The state is nonzero for suitable long regions: the words of a fixed long length span the full matrix algebras of A and B , so if every coefficient vanished, then $\text{tr}(Xe_{AB}^i Y e_{BA}^j) = 0$ for all matrices X and Y , and choosing matrix units for X and Y gives $e_{AB}^i = 0$ or $e_{BA}^j = 0$ for every pair of letters. $\square$

Corollary 19.5.5.9 (Exchange of two domain-wall strings, printed orientation). [Lean](#) [Lean](#) [✓ Stmt](#) In the setting of Theorem 19.5.5.7, suppose that $U = O_g$ satisfies $U^2 = \mathbf{1}$ on the periodic chain of every length. Then

$$O^{[i_1, j_1]} O^{[i_2, j_2]} |\psi_A\rangle = c_{AB}c_{BA} O^{[i_2, j_2]} O^{[i_1, j_1]} |\psi_A\rangle,$$

the relation of [166, Section III.F].

Proof. [✓ Proof](#) The tensors A and B have left inverses, so their matrices span the full matrix algebras and they are normal. Lemma 19.5.5.8 gives $(c_{AB}c_{BA})^2 = 1$. Multiply the identity of Theorem 19.5.5.7 by $c_{AB}c_{BA}$. $\square$

Theorem 19.5.5.10 (Exchange of two left endpoints of domain-wall strings). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) In the setting of Theorem 19.5.5.7, let $j < i$ and let $r_1 \neq r_2$ be two sites to the right of i . If the regions between the four sites j, i, r_1, r_2 are longer than a fixed buffer, then

$$O^{[j, r_1]} O^{[i, r_2]} |\psi_A\rangle = c_{AB} O^{[i, r_1]} O^{[j, r_2]} |\psi_A\rangle.$$

This is the exchange of two left endpoints $O_x^{[j]} O_y^{[i]} |\psi_A\rangle = c_{AB} O_x^{[i]} O_y^{[j]} |\psi_A\rangle$ of [166, Section III.F], with the half-chain operators glued with the right endpoints of their strings at r_1 and r_2 ; the identity of half-chain objects with open legs is not formalized [57].

Proof. [✓ Proof](#) Every side is a multiple of the state with the domain walls $e_{AB}, e_{BA}, e_{AB}, e_{BA}$ at the four sites in increasing order. For $r_2 < r_1$ the left side is a nested pair whose outer string acts second, with phase $c_{AB}c_{BA}$ by Theorem 19.5.5.7; on the right side the string on $[i, r_1]$ starts in the B region, passes over the domain wall e_{BA} at r_2 and ends in the A region, so the local action gives c_{BA} . For $r_1 < r_2$ the string on $[j, r_1]$ on the left side passes over the domain wall e_{AB} at i , with phase c_{AB} , and the right side is a nested pair whose inner string acts second, with no phase. $\square$

Theorem 19.5.5.11 (Rescaling domain walls). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If g carries e to e' with phase c and $a \neq 0$, then it carries $a e$ to e' with phase ac , and e to $a e'$ with phase c/a . In particular, if an involution g exchanges e_{AB} and e_{BA} with phases c_{AB} and c_{BA} , then for every s with $s^2 = c_{AB}c_{BA}$ the walls e_{AB} and $(c_{AB}/s)e_{BA}$ are exchanged with both phases equal to s , as in [166, Section III.F].

Proof. [✓ Proof](#) The rectangular action is linear in the domain wall; for the last claim rescale e_{BA} by c_{AB}/s . $\square$

Definition 19.5.5.12 (A family of domain walls permuted by the group). [Lean](#) [Lean](#) [✓ Stmt](#) A family of domain walls permuted by the group is a nonzero domain wall e_{yz} between y and z for every ordered pair of blocks, together with nonzero phases $B_{y,z}^g$ such that every g carries e_{yz} to $e_{gy,gz}$ with phase $B_{y,z}^g$ [166, Section IV.B]. The walls and phases are nonzero by convention [52].

Theorem 19.5.5.13 (The interchange phase of a family of domain walls). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a family of domain walls permuted by the group, every g with $g^n = e$ and every block x ,

$$\prod_{k=0}^{n-1} B_{g^k(gx), g^kx}^g = \prod_{k=0}^{n-1} \frac{L_{g,g^k}^{gx}}{L_{g,g^k}^x} = \left(\prod_{k=0}^{n-1} \omega(g, g^k, g) \right)^{-1}.$$

This is [166, Section IV.C]; the interchange phase depends only on the anomaly. The middle product includes the factor $k=0$, $L_{g,e}^{gx}/L_{g,e}^x$, which is one for normalized L -symbols; the source's middle product runs over $1 \leq k < n$, and for normalized L -symbols that product is again $(\prod_{k=0}^{n-1} \omega(g, g^k, g))^{-1}$.

Proof. [✓ Proof](#) Write $\beta_k = B_{gx,x}^{g^k}$. By Theorem 19.5.5.2 for (g, g^k) , $B_{g^k(gx), g^kx}^g \beta_k = (L_{g,g^k}^{gx}/L_{g,g^k}^x) \beta_{k+1}$. Multiplying over $k < n$ and using $\beta_n = \beta_0$ gives the first equality. Compatibility at (g, g^k, g) gives $L_{g,g^k}^{gx}/L_{g,g^k}^x = \omega(g, g^k, g)^{-1} (L_{g,g^{k+1}}^x/L_{g,g^k}^x) (L_{g^{k+1},g}^x/L_{g^k,g}^x)^{-1}$, and both quotients telescope to one over $k < n$. $\square$

Theorem 19.5.5.14 (The stabilizer of two blocks acts projectively). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $H_{y,z}$ be the subgroup of elements fixing the blocks y and z . For $a, b \in H_{y,z}$,

$$B_{y,z}^a B_{y,z}^b = \alpha(a, b) B_{y,z}^{ab}, \quad \alpha(a, b) = \frac{L_{a,b}^y}{L_{a,b}^z},$$

and α is a two-cocycle on $H_{y,z}$; for $y = z$ the phases form a linear representation. The source states this for the subgroup fixing every block [166, Section IV.C], which is contained in $H_{y,z}$; see [151].

Proof. [✓ Proof](#) The first identity is Theorem 19.5.5.2 with $by = y$ and $bz = z$. Dividing the compatibility relations at y and at z for (a, b, c) with $c \in H_{y,z}$ cancels $\omega(a, b, c)$ and gives the two-cocycle equation. $\square$

Theorem 19.5.5.15 (Domain walls of an involution detect the anomaly). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $g^2 = e$ exchange the blocks x and y , let g carry the domain wall e_{AB} between x and y to e_{BA} with phase c_{AB} and e_{BA} to e_{AB} with phase c_{BA} . Then

$$c_{AB} c_{BA} = \frac{L_{g,g}^x L_{e,e}^x}{L_{g,g}^y L_{e,e}^y} = \omega(g, g, g) \omega(g, e, g),$$

so $c_{AB} c_{BA}$ does not depend on the action tensors, the fusion tensors or the domain walls, and $U|\psi(A-B-A)\rangle = \omega(g, g, g) \omega(g, e, g) |\psi(B-A-B)\rangle$. When the tensors of the identity are trivial, as in the source, $L_{e,e} = 1$ and $\omega(g, e, g) = 1$, and this is $c_{AB} c_{BA} = L_A/L_B = \omega$ [166, Section III.C].

Proof. ✓ Proof Applying g twice, Theorem 19.5.5.2 shows that $g^2 = e$ carries e_{AB} to itself with phase $b = (L_{g,g}^y/L_{g,g}^x) c_{BA} c_{AB} \neq 0$. The same theorem for (e, e) gives $b^2 = (L_{e,e}^x/L_{e,e}^y) b$, so $b = L_{e,e}^x/L_{e,e}^y$, which is the first equality. The second equality is Theorem 19.5.4.5 for the compatible L-symbols of Theorem 19.5.4.3. Its right-hand side involves only the fusion tensors, and it is invariant under their gauge action, so the product is independent of all choices. Finally, by Theorem 19.5.5.3, $U|\psi(A-B-A)\rangle = c_{AB} c_{BA} |\psi(B-A-B)\rangle = \omega(g, g, g) \omega(g, e, g) |\psi(B-A-B)\rangle$. $\square$

19.5.6 Detecting the anomaly with truncated symmetries

Lemma 19.5.6.1 (The commutator of two unitaries on a vector). Lean Lean Lean ✓ Stmt Let U_1 and U_2 be unitary, let ψ be a vector and c a scalar with $U_2 U_1 \psi = c U_1 U_2 \psi$. Then

$$U_2^\dagger U_1^\dagger U_2 U_1 \psi = c \psi, \quad \langle \psi | U_2^\dagger U_1^\dagger U_2 U_1 | \psi \rangle = c \langle \psi | \psi \rangle.$$

For two truncations $U_1 = U^{[i_1, j_1]}$ and $U_2 = U^{[i_2, j_2]}$ of a symmetry whose exchange relation on a ground state has the phase $c = \omega$, this is the last step of the detection of the anomaly in [166, Section V]: the expectation value of the commutator measures the exchange phase.

Proof. ✓ Proof Substitute the exchange relation, then cancel $U_1^\dagger U_1$ and $U_2^\dagger U_2$. $\square$

Theorem 19.5.6.2 (Truncations of an on-site symmetry). Lean Lean Lean Lean Lean ✓ Stmt Let u satisfy $u^\dagger u = \mathbf{1}$ and, for a set S of sites of a chain, let $U^S = \bigotimes_{i \in S} u^{[i]}$ act by u on the sites of S and by the identity elsewhere. For any two sets S_1 and S_2 ,

$$(U^{S_2})^\dagger (U^{S_1})^\dagger U^{S_2} U^{S_1} = \mathbf{1},$$

so the expectation value of the commutator on every normalized state is 1, as for the trivial anomaly of on-site symmetries [166, Section V].

Proof. ✓ Proof The four operators are tensor products over the sites. On a site in both sets the product of the four factors is $u^\dagger u^\dagger u u = \mathbf{1}$, on a site in one set it is $u^\dagger u = \mathbf{1}$, and elsewhere all four factors are the identity. $\square$

19.5.7 Normalization of the first source factor

Theorem 19.5.7.1 (The raw first-cut contraction). Lean ✓ Stmt For an arbitrary tensor A of physical dimension d and bond dimension D , and an arbitrary $D \times D$ matrix ρ , put $M = M_1(A)$ and $W = I_d \otimes \rho$. Then

$$M_1(s(A, \rho)) = M M^\dagger W^\top M. \quad (19.286)$$

No positivity, canonical-form, or simplicity hypothesis is needed.

Proof. ✓ Proof In the first-cut coordinates, multiplication and the physical Kronecker delta in $W^\top$ give

$$(M M^\dagger W^\top M)_{(i,b),(a,j)} = \sum_{p,q=0}^{d-1} \sum_{x,r,t=0}^{D-1} A_{xb}^{i,p} \overline{A_{xr}^{q,p}} A_{at}^{q,j} \rho_{tr} \quad (19.287)$$

$$= s(A, \rho)_{ab}^{i,j}. \quad (19.288)$$

The transpose preserves the boundary entry ρ_{tr} in the defining contraction. $\square$

Theorem 19.5.7.2 (The first cut of the adjoint-simple contraction). [Lean](#) [✓ Stmt](#) [Discussion: #7317](#)
Let $\mathcal{U}$ be an MPU in canonical form II with recorded fixed matrix ρ , and let X_1, Y_1 be its chosen source factors for this same ρ . Construct s from $\mathcal{U}, \rho$ as in Definition 19.5.8. With the first cut convention $(M_1(A))_{(i,b),(a,j)} = A_{ab}^{i,j}$,

$$M_1(s) = X_1(Y_1 Y_1^\dagger)Y_1. \quad (19.289)$$

This factorization requires no simplicity assumption. It is the source-factor expansion in [163, lines 1326–1374].

Proof. [✓ Proof](#) Put $W = I_d \otimes \rho$. Diagonality gives $W^\top = W$. The cut factorization and weighted isometry then yield

$$M_1(s) = (X_1 Y_1)(X_1 Y_1)^\dagger W(X_1 Y_1) \quad (19.290)$$

$$= X_1(Y_1 Y_1^\dagger)(X_1^\dagger W X_1)Y_1 \quad (19.291)$$

$$= X_1(Y_1 Y_1^\dagger)Y_1. \quad (19.292)$$

□

Theorem 19.5.7.3 (The first source Gram matrix is scalar). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU in canonical form II, and suppose that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Construct the source factors from $\mathcal{U}$ and its recorded fixed matrix ρ . If d_r is the first source-cut rank, then there is a scalar $\delta \in \mathbb{C}$ such that

$$Y_1 Y_1^\dagger = \delta I_{d_r}. \quad (19.293)$$

This is the scalar Gram identity in [163, lines 1375–1397].

Proof. [✓ Proof](#) By Theorem 19.5.10, $s = \delta \mathcal{U}$. Applying the first cut and Theorem 19.5.7.2 gives

$$X_1(Y_1 Y_1^\dagger)Y_1 = \delta X_1 Y_1. \quad (19.294)$$

Multiply on the left by $X_1^\dagger(I_d \otimes \rho)$ and on the right by Z_1 . The identities $X_1^\dagger(I_d \otimes \rho)X_1 = I_{d_r}$ and $Y_1 Z_1 = I_{d_r}$ cancel the two outer factors. □

Theorem 19.5.7.4 (Trace of the first source Gram matrix). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU in canonical form II, with physical dimension d , bond dimension D , first source-cut rank d_r , and recorded fixed matrix ρ . Its chosen first source factor satisfies

$$\text{tr}(Y_1 Y_1^\dagger) = d. \quad (19.295)$$

This is the physical-trace calculation in [163, line 1398]; it does not require simplicity of $\mathcal{U}^\dagger$.

Proof. [✓ Proof](#) Cyclic invariance gives $\text{tr}(Y_1 Y_1^\dagger) = \text{tr}(Y_1^\dagger Y_1)$. Grouping the latter sum by its physical index gives

$$\text{tr}(Y_1^\dagger Y_1) = \sum_{p=0}^{d-1} \sum_{r=0}^{d_r-1} \sum_{\alpha=0}^{D-1} \overline{(Y_1)_{r,(\alpha,p)}} (Y_1)_{r,(\alpha,p)} = \sum_{p=0}^{d-1} 1 = d \quad (19.296)$$

by Theorem 19.4.4.2. □

Theorem 19.5.7.5 (Normalization of the first source Gram matrix). [Lean](#) [✓ Stmt](#) [Discussion: #7317](#) Let $\mathcal{U}$ be an MPU in canonical form II of physical dimension d , and suppose that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Let Y_1 be the first source factor constructed from $\mathcal{U}$ and its recorded fixed matrix ρ , and let d_r be the first source-cut rank. Then

$$Y_1 Y_1^\dagger = \frac{d}{d_r} I_{d_r}. \quad (19.297)$$

This establishes the normalization $Y_1 Y_1^\dagger = (d/d_r) I_{d_r}$ from the source-factor normalization proposition in [163, lines 1064–1119].

Proof. [✓ Proof](#) Write $Y_1 Y_1^\dagger = \delta I_{d_r}$ using Theorem 19.5.7.3. Taking traces and using Theorem 19.5.7.4 gives $\delta d_r = d$. Canonical form II gives $d > 0$, so $d_r \neq 0$. Dividing gives $\delta = d/d_r$, as in [163, line 1398]. $\square$

19.5.8 Normalization of the weighted second source factor

Theorem 19.5.8.1 (The first cut of the reflected contraction). [Lean](#) [✓ Stmt](#) [Discussion: #7317](#) Let $\mathcal{U}$ be an MPU in canonical form II with recorded fixed matrix ρ , and let X_2, Y_2 be its chosen second source factors. Set $W = I_d \otimes \rho$ and let $s_\dagger$ be the raw contraction of Definition 19.5.8, applied to $\mathcal{U}^\dagger$ and this same ρ . Then

$$M_1(s_\dagger) = Y_2^\dagger (Y_2 W Y_2^\dagger) X_2^\dagger. \quad (19.298)$$

This is the reflected factorization for $Y_2(I_d \otimes \rho)Y_2^\dagger$ in the normalization proposition of [163, lines 1064–1119 and 1398]. No simplicity hypothesis is needed for this factorization.

Proof. [✓ Proof](#) Theorem 19.5.7.1 gives

$$M_1(s(A, \rho)) = M_1(A) M_1(A)^\dagger W^\top M_1(A). \quad (19.299)$$

The transpose records the boundary coefficient ρ_{tr} . Here ρ is diagonal, so $W^\top = W$. Substitute $M_1(\mathcal{U}^\dagger) = M_2(\mathcal{U})^\dagger = Y_2^\dagger X_2^\dagger$ and cancel $X_2^\dagger X_2 = I$. All factors are those of $\mathcal{U}$, not newly chosen factors of its adjoint. $\square$

Theorem 19.5.8.2 (The weighted second source Gram matrix is scalar). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be an MPU in canonical form II with recorded fixed matrix ρ , and assume that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Let Y_2 be its chosen second source factor and d_l its second source-cut rank. There is a scalar $\varepsilon \in \mathbb{C}$ such that

$$Y_2(I_d \otimes \rho)Y_2^\dagger = \varepsilon I_{d_l}. \quad (19.300)$$

Proof. [✓ Proof](#) Transport canonical form II to $\mathcal{U}^\dagger$ with the same ρ . Theorem 19.5.10 gives $s_\dagger = \varepsilon \mathcal{U}^\dagger$, since physical adjunction is involutive and both required simplicity hypotheses hold. Taking the first cut and applying Theorem 19.5.8.1 gives

$$Y_2^\dagger (Y_2(I_d \otimes \rho)Y_2^\dagger) X_2^\dagger = \varepsilon Y_2^\dagger X_2^\dagger. \quad (19.301)$$

Multiply on the left by $Z_2^\dagger$ and on the right by X_2 . Use $Z_2^\dagger Y_2^\dagger = I$ and $X_2^\dagger X_2 = I$. The scalar comes from the adjoint contraction; it has not been identified with the scalar for the original tensor. $\square$

Theorem 19.5.8.3 (Trace of the weighted second source Gram matrix). [Lean](#) [✓ Stmt](#) Let $\mathcal{U}$ be a simple MPU in canonical form Π of physical dimension d , with recorded fixed matrix ρ and chosen second source factor Y_2 . Then

$$\text{tr}(Y_2(I_d \otimes \rho)Y_2^\dagger) = d. \quad (19.302)$$

This is the physical-trace calculation for Y_2 in [163, line 1398]. Simplicity of $\mathcal{U}^\dagger$ is not required.

Proof. [✓ Proof](#) Cyclic invariance and expanding $I_d \otimes \rho$ give

$$\text{tr}(Y_2(I_d \otimes \rho)Y_2^\dagger) = \sum_{p=0}^{d-1} \sum_{x,y=0}^{D-1} \sum_{l=0}^{d_l-1} \overline{(Y_2)_{l,(p,y)}} (Y_2)_{l,(p,x)} \rho_{xy} = \sum_{p=0}^{d-1} 1 = d \quad (19.303)$$

by Theorem 19.4.4.2 at $q = p$. □

Theorem 19.5.8.4 (Normalization of the weighted second source Gram matrix). [Lean](#) [✓ Stmt](#) [Discussion: #7317](#) Let $\mathcal{U}$ be an MPU in canonical form Π of physical dimension d , and suppose that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Let Y_2 be the second source factor constructed from $\mathcal{U}$ and its recorded fixed matrix ρ , and let d_l be the second source-cut rank. Then

$$Y_2(I_d \otimes \rho)Y_2^\dagger = \frac{d}{d_l} I_{d_l}. \quad (19.304)$$

This establishes the normalization $Y_2(I_d \otimes \rho)Y_2^\dagger = (d/d_l)I_{d_l}$ from the source-factor normalization proposition in [163, lines 1064–1119].

Proof. [✓ Proof](#) Write the Gram matrix as εI_{d_l} using Theorem 19.5.8.2. Its trace is d by Theorem 19.5.8.3, so $\varepsilon d_l = d$. As $d > 0$, the rank d_l is nonzero, giving $\varepsilon = d/d_l$. This is the trace argument indicated in [163, line 1398]. □

19.5.9 Physical normalizations of the two source X factors

Theorem 19.5.9.1 (The canonical one-letter output contraction). [Lean](#) [✓ Stmt](#) [Discussion: #7317](#) Let $\mathcal{U}$ be an MPU in canonical form Π of physical dimension d and bond dimension D , with recorded fixed matrix ρ . Suppose that $\mathcal{U}^\dagger$ is simple. For $0 \leq i, j < d$,

$$\sum_{x,\alpha,\beta=0}^{D-1} \sum_{p=0}^{d-1} U_{x\alpha}^{i,p} \overline{U_{x\beta}^{j,p}} \rho_{\beta\alpha} = \delta_{i,j}. \quad (19.305)$$

The open indices are the physical output indices. This is the reflected one-letter contraction used for the physical source-factor normalizations in [163, lines 1120–1162 and 1398]. Simplicity of $\mathcal{U}$ is not required for this contraction.

Proof. [✓ Proof](#) Apply Theorem 19.4.4.1 to $\mathcal{U}^\dagger$ with its canonical presentation carrying the same ρ . Its letters are $(\mathcal{U}^\dagger)_{x\alpha}^{p,i} = \overline{U_{x\alpha}^{i,p}}$. Substitution and double conjugation give the displayed sum. □

Theorem 19.5.9.2 (Physical normalization of the first source X factor). [Lean](#) [✓ Stmt](#) [Discussion: #7317](#) Let $\mathcal{U}$ be an MPU in canonical form Π of physical dimension d and bond dimension D . Assume that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Let X_1 be the first source factor constructed from $\mathcal{U}$ and its recorded fixed matrix ρ , and let d_r be the first source-cut rank. For $0 \leq i, j < d$,

$$\sum_{r=0}^{d_r-1} \sum_{\alpha,\beta=0}^{D-1} (X_1)_{(i,\alpha),r} \rho_{\beta\alpha} \overline{(X_1)_{(j,\beta),r}} = \frac{d_r}{d} \delta_{i,j}. \quad (19.306)$$

This is the first physical source-factor normalization in [163, lines 1120–1162]. The matrix ρ acts on the virtual legs; the identity acts on the physical space.

Proof. ✓ **Proof** Factor $M_1(\mathcal{U}) = X_1 Y_1$ and apply Theorem 19.5.7.5 to obtain

$$M_1(\mathcal{U})M_1(\mathcal{U})^\dagger = \frac{d}{d_r} X_1 X_1^\dagger. \quad (19.307)$$

Multiply the $((i, \alpha), (j, \beta))$ entry by $\rho_{\beta\alpha}$ and sum over α, β . Theorem 19.5.9.1 makes the left-hand side $\delta_{i,j}$. The trace identity in Theorem 19.5.7.4 gives $d_r \neq 0$, and canonical form II gives $d > 0$. Multiplication by d_r/d gives the required physical contraction. $\square$

Theorem 19.5.9.3 (Physical normalization of the second source X factor). Lean ✓ Stmt

Discussion: #7317 Let $\mathcal{U}$ be an MPU in canonical form II of physical dimension d and bond dimension D . Assume that both $\mathcal{U}$ and $\mathcal{U}^\dagger$ are simple. Let X_2 be its chosen second source factor and let d_l be the second source-cut rank. For $0 \leq i, j < d$,

$$\sum_{l=0}^{d_l-1} \sum_{x=0}^{D-1} (X_2)_{(x,i),l} \overline{(X_2)_{(x,j),l}} = \frac{d_l}{d} \delta_{i,j}. \quad (19.308)$$

This is the second physical source-factor normalization in [163, lines 1120–1162].

Proof. ✓ **Proof** Put $W = I_d \otimes \rho$. The factorization $M_2(\mathcal{U}) = X_2 Y_2$ and Theorem 19.5.8.4 give

$$M_2(\mathcal{U})W M_2(\mathcal{U})^\dagger = \frac{d}{d_l} X_2 X_2^\dagger. \quad (19.309)$$

Sum the $((x, i), (x, j))$ entry over x . Expanding the left-hand side, using $\rho_{\alpha\beta} = \rho_{\beta\alpha}$, gives $\delta_{i,j}$ by Theorem 19.5.9.1. The trace identity in Theorem 19.5.8.3 gives $d_l \neq 0$, and $d > 0$. Multiply by d_l/d to conclude. $\square$

Theorems 19.5.7.5, 19.5.8.4, 19.5.9.2, and 19.5.9.3 establish all four identities of the source-factor normalization proposition in [163, lines 1064–1162] for the compact source factors under canonical form II and simplicity of $\mathcal{U}$ and $\mathcal{U}^\dagger$; no additional boundary or Gram identities are assumed.

Theorem 19.5.9.4 (Comparison of factorizations from one-sided inverses). Lean ✓ Stmt Let $A, B, \widetilde{B}, C$ be finite index sets, possibly empty, and let $\mathbb{S}$ be a semiring. Suppose that

$$X \in \text{Mat}_{A \times B}(\mathbb{S}), \quad (19.310)$$

$$Y \in \text{Mat}_{B \times C}(\mathbb{S}), \quad (19.311)$$

$$\widetilde{X} \in \text{Mat}_{A \times \widetilde{B}}(\mathbb{S}), \quad (19.312)$$

$$\widetilde{Y} \in \text{Mat}_{\widetilde{B} \times C}(\mathbb{S}) \quad (19.313)$$

satisfy $XY = \widetilde{X}\widetilde{Y}$. If $L \in \text{Mat}_{\widetilde{B} \times A}(\mathbb{S})$ and $R \in \text{Mat}_{C \times B}(\mathbb{S})$ satisfy $L\widetilde{X} = I_{\widetilde{B}}$ and $YR = I_B$, then the rectangular comparison matrix $K = LX \in \text{Mat}_{\widetilde{B} \times B}(\mathbb{S})$ obeys

$$K = \widetilde{Y}R, \quad (19.314)$$

$$X = \widetilde{X}K, \quad (19.315)$$

$$\widetilde{Y} = KY. \quad (19.316)$$

The intermediate index sets B and $\widetilde{B}$ are independent; no equality of their cardinalities is assumed or concluded. These are the one-sided-inverse identities in the proof of [163, lines 1052–1066].

Proof. ✓ **Proof** Associativity and the two inverse identities give

$$LX = (LX)(YR) = L(XY)R = L(\widetilde{X}\widetilde{Y})R = \widetilde{Y}R. \quad (19.317)$$

Multiplying $XY = \widetilde{X}\widetilde{Y}$ on the right by R gives $X = \widetilde{X}K$. Multiplying it on the left by L gives $\widetilde{Y} = KY$. $\square$

Theorem 19.5.9.5 (Reciprocal unitary gauges from a Kronecker comparison). Lean ✓ **Stmt**
For $i = 1, 2$, suppose that two compatible matrix factorizations are related by square comparison matrices K_i through

$$X_i = \widetilde{X}_i K_i, \quad \widetilde{Y}_i = K_i Y_i. \quad (19.318)$$

If $K_1 \otimes K_2$ is unitary, then there are a positive real number δ and unitary matrices W_1, W_2 such that

$$\widetilde{X}_1 = \delta^{-1} X_1 W_1, \quad \widetilde{Y}_1 = \delta W_1^\dagger Y_1, \quad (19.319)$$

$$\widetilde{X}_2 = \delta X_2 W_2, \quad \widetilde{Y}_2 = \delta^{-1} W_2^\dagger Y_2. \quad (19.320)$$

Thus the positive scalar factors in the orientation of [163, Lemma at lines 1052–1066] are $\delta_1 = \delta^{-1}$ and $\delta_2 = \delta$, and hence $\delta_1 \delta_2 = 1$.

Proof. ✓ **Proof** Apply Theorem 19.5.3 to $K_1 \otimes K_2$. It gives $\delta > 0$ and unitary matrices U_1, U_2 with $K_1 = \delta U_1$ and $K_2 = \delta^{-1} U_2$. Set $W_i = U_i^\dagger$. Substitution into (19.318) gives (19.320); positivity of δ gives $\delta^{-1} \delta = 1$. $\square$

19.5.10 Uniqueness of the source decompositions

Theorem 19.5.10.1 (Transport of the source contraction). Lean ✓ **Stmt** Discussion: #7315 Let U have physical dimension d and bond dimension D , and write $r = \text{rank } M_1(U)$ and $\ell = \text{rank } M_2(U)$. Let $Y_1, \widetilde{Y}_1 \in \text{Mat}_{r \times (D \times d)}(\mathbb{C})$ and $Y_2, \widetilde{Y}_2 \in \text{Mat}_{\ell \times (d \times D)}(\mathbb{C})$. For square matrices $K_1 \in \text{Mat}_r(\mathbb{C})$ and $K_2 \in \text{Mat}_\ell(\mathbb{C})$, suppose that $\widetilde{Y}_i = K_i Y_i$ for $i = 1, 2$. Define $u, \tilde{u}: \mathbb{C}^d \otimes \mathbb{C}^d \rightarrow \mathbb{C}^\ell \otimes \mathbb{C}^r$ by

$$u_{(l, r'), (p, q)} = \sum_{\beta=0}^{D-1} (Y_2)_{l, (p, \beta)} (Y_1)_{r', (\beta, q)}, \quad (19.321)$$

$$\tilde{u}_{(l, r'), (p, q)} = \sum_{\beta=0}^{D-1} (\widetilde{Y}_2)_{l, (p, \beta)} (\widetilde{Y}_1)_{r', (\beta, q)}. \quad (19.322)$$

Here $0 \leq l < \ell$, $0 \leq r' < r$, and $0 \leq p, q < d$. Then $\tilde{u} = (K_2 \otimes K_1)u$.

Proof. ✓ **Proof** Substitute $\widetilde{Y}_i = K_i Y_i$ into (19.322) and interchange the three finite sums. The coefficient of $u_{(l', r''), (p, q)}$ is $(K_2)_{l, l'} (K_1)_{r', r''}$, the corresponding entry of $K_2 \otimes K_1$. $\square$

Theorem 19.5.10.2 (Reciprocal unitary gauges for the source decompositions). Lean ✓ **Stmt**
Discussion: #7315 Let U have physical dimension d and bond dimension D . Put $r = \text{rank } M_1(U)$ and $\ell = \text{rank } M_2(U)$. Suppose that

$$X_1, \widetilde{X}_1 \in \text{Mat}_{(d \times D) \times r}(\mathbb{C}), \quad Y_1, \widetilde{Y}_1 \in \text{Mat}_{r \times (D \times d)}(\mathbb{C}), \quad (19.323)$$

$$X_2, \widetilde{X}_2 \in \text{Mat}_{(D \times d) \times \ell}(\mathbb{C}), \quad Y_2, \widetilde{Y}_2 \in \text{Mat}_{\ell \times (d \times D)}(\mathbb{C}) \quad (19.324)$$

satisfy $X_i Y_i = \widetilde{X}_i \widetilde{Y}_i = M_i(U)$ for $i = 1, 2$. Suppose there are matrices L_i, R_i with $L_i \widetilde{X}_i = I$ and $Y_i R_i = I$. Assume that the maps $u, \tilde{u}: \mathbb{C}^d \otimes \mathbb{C}^d \rightarrow \mathbb{C}^\ell \otimes \mathbb{C}^r$ with coordinates

$$u_{(l,r'),(p,q)} = \sum_{\beta=0}^{D-1} (Y_2)_{l,(p,\beta)} (Y_1)_{r',(\beta,q)}, \quad (19.325)$$

$$\tilde{u}_{(l,r'),(p,q)} = \sum_{\beta=0}^{D-1} (\widetilde{Y}_2)_{l,(p,\beta)} (\widetilde{Y}_1)_{r',(\beta,q)} \quad (19.326)$$

are unitary. The indices range over $0 \leq l < \ell$, $0 \leq r' < r$, and $0 \leq p, q < d$. Then there exist $\delta > 0$ and unitaries $W_1 \in \text{Mat}_r(\mathbb{C})$, $W_2 \in \text{Mat}_\ell(\mathbb{C})$ such that

$$\widetilde{X}_1 = \delta^{-1} X_1 W_1, \quad \widetilde{Y}_1 = \delta W_1^\dagger Y_1, \quad (19.327)$$

$$\widetilde{X}_2 = \delta X_2 W_2, \quad \widetilde{Y}_2 = \delta^{-1} W_2^\dagger Y_2. \quad (19.328)$$

Thus $\delta_1 = \delta^{-1}$ and $\delta_2 = \delta$ satisfy $\delta_1 \delta_2 = 1$. This proves the decomposition lemma of [163, lines 1052–1066]: its cut factorizations, one-sided inverses, and unitarity of $u, \tilde{u}$ suffice. Unitarity of the second contraction v is not needed.

Proof. ✓ Proof If $d = 0$, every factor equality has an empty physical row or column index. Thus $\delta = 1$ and identity gauges satisfy all four equalities, and $\delta^{-1} \delta = 1$. Suppose now that $d > 0$. Since u is an isometry, $\ell r \geq d^2 > 0$; hence both source ranks are positive. Theorem 19.5.9.4 gives $K_i = L_i X_i = \widetilde{Y}_i R_i$ with $X_i = \widetilde{X}_i K_i$ and $\widetilde{Y}_i = K_i Y_i$. Theorem 19.5.10.1 yields $\tilde{u} = (K_2 \otimes K_1)u$. Therefore $K_2 \otimes K_1 = \tilde{u} u^\dagger$ is unitary. Interchanging the two tensor factors makes $K_1 \otimes K_2$ unitary. Apply Theorem 19.5.9.5 to obtain (19.328). In particular, if $K_1 = \delta V_1$ and $K_2 = \delta^{-1} V_2$ with V_i unitary, the required orientation is $W_i = V_i^\dagger$. $\square$

Corollary 19.5.10.3 (Unitary gauges for normalized source decompositions). Lean ✓ Stmt Under the hypotheses of Theorem 19.5.10.2, suppose in addition that

$$X_2^\dagger X_2 = I_\ell, \quad \widetilde{X}_2^\dagger \widetilde{X}_2 = I_\ell. \quad (19.329)$$

Then there exist unitaries $W_1 \in \text{Mat}_r(\mathbb{C})$ and $W_2 \in \text{Mat}_\ell(\mathbb{C})$ such that, for $i = 1, 2$,

$$\widetilde{X}_i = X_i W_i, \quad \widetilde{Y}_i = W_i^\dagger Y_i. \quad (19.330)$$

This is the isometrically normalized specialization of [163, lines 1052–1066]; the additional isometry hypotheses remove the scalar freedom.

Proof. ✓ Proof If $d = 0$, all four equalities have an empty physical index and hold with identity gauges. If $d > 0$, unitarity of u gives $r\ell = d^2 > 0$, so $\ell > 0$. Theorem 19.5.10.2 gives $\widetilde{X}_2 = \delta X_2 W_2$ with $\delta > 0$. The two isometry identities and unitarity of W_2 imply $I_\ell = \delta^2 I_\ell$. Since $\ell > 0$, a diagonal entry gives $\delta^2 = 1$, hence $\delta = 1$. Substitution into the four comparison identities proves the claim. $\square$

Definition 19.5.10.4 (Unitary factorization comparison). Lean Lean ✓ Stmt Let I be a set and let $R_{a,b}$ be a family of unitary matrices of a fixed size, indexed by $(a,b) \in I^2$. The unitary comparison from (a,b) to the reference pair (c,d) is

$$\lambda_{a,b}^{c,d} = (R_{c,d})^{-1} R_{a,b} = R_{c,d}^\dagger R_{a,b}. \quad (19.331)$$

This is the orientation in the corollary after Proposition 3 of [163, lines 2782–2821]. In that corollary, $I = G$ and comparisons are used when $ab = cd$; this equality selects the factorizations and is not needed in the unitary calculation.

Theorem 19.5.10.9 (Gauss operators form a unitary representation). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7503](#) The local Gauss operators satisfy

$$\mathcal{G}_e = \text{id}. \quad (19.342)$$

Their product law is

$$\mathcal{G}_g \mathcal{G}_h = \mathcal{G}_{gh}. \quad (19.343)$$

Finally,

$$\mathcal{G}_g^\dagger = \mathcal{G}_{g^{-1}}. \quad (19.344)$$

Hence $g \mapsto \mathcal{G}_g$ is a unitary representation. This is the representation part of the local Gauss-law proposition in [163, lines 3380–3387]; invariance of the gauged tensor is a separate assertion and is not included here.

Proof. [✓ Proof](#) In a column labelled by (a, b) , multiplication has only the intermediate gauge label $T_h(a, b)$. The surviving matter block is

$$\lambda_{T_h(a,b)}^{T_g(T_h(a,b))} \lambda_{a,b}^{T_h(a,b)} = \lambda_{a,b}^{T_{gh}(a,b)}, \quad (19.345)$$

by the comparison composition law and $T_g T_h = T_{gh}$. At e , every surviving comparison is the identity. Reversing a comparison gives $(\lambda_{a,b}^{c,d})^\dagger = \lambda_{c,d}^{a,b}$, so taking the adjoint gives $\mathcal{G}_g^\dagger = \mathcal{G}_{g^{-1}}$. The product with this adjoint is $\mathcal{G}_e = \text{id}$, which proves unitarity. $\square$

Definition 19.5.10.10 (Unplaced local Gauss projector kernel). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7578](#)
Let G be finite and let $\mathcal{G}_g$ act on $\mathcal{H} = \mathbb{C}^n \otimes \mathbb{C}[G]^{\otimes 2}$. Define the unplaced local projector kernel

$$P_G(R) = \frac{1}{|G|} \sum_{g \in G} \mathcal{G}_g. \quad (19.346)$$

This is the local average introduced in [163, lines 3494–3499], before a position on the chain is chosen. This definition neither places the kernel on $(j, j+1)$ nor asserts invariance of the gauged matrix product state.

Theorem 19.5.10.11 (The local Gauss average is a projection). [Lean](#) [✓ Stmt](#) [Discussion: #7578](#)
The kernel in (19.346) satisfies

$$P_G(R)^2 = P_G(R), \quad (19.347)$$

$$P_G(R)^\dagger = P_G(R). \quad (19.348)$$

Proof. [✓ Proof](#) Identify the matrix average with the averaging operator of the unitary representation $g \mapsto \mathcal{G}_g$. The invariant-space averaging theorem makes this operator idempotent. For self-adjointness, use $\mathcal{G}_g^\dagger = \mathcal{G}_{g^{-1}}$ and reindex the finite sum by the bijection $g \mapsto g^{-1}$. $\square$

Definition 19.5.10.12 (Gauss projector on a periodic chain). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) [Discussion: #7605](#) Let the matter degree of freedom at one site have dimension d . The local kernel has row and column coordinates

$$((i_0, i_1), (a, b)) \in (\{0, \dots, d-1\}^{\{0,1\}}) \times G^2. \quad (19.349)$$

Identify these with two site coordinates by the bijection

$$\kappa((i_0, i_1), (a, b)) = ((i_0, a), (i_1, b)), \quad (19.350)$$

followed at each site by a fixed enumeration of $\{0, \dots, d-1\} \times G$. If $\widetilde{P}_G(R)$ denotes the matrix obtained from $P_G(R)$ by this simultaneous row and column relabelling, define its placement on a periodic chain of length $N \geq 2$ by

$$P_j(R) = \text{Embed}_{j,j+1}(\widetilde{P}_G(R)). \quad (19.351)$$

Here the second site is $j+1$ modulo N . This is the position-dependent projector P_j in [163, lines 3494–3499].

Theorem 19.5.10.13 (The placed Gauss projector is a projection). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7605](#)
For every $N \geq 2$ and every position j , the operator in (19.351) satisfies

$$P_j(R)^2 = P_j(R), \quad (19.352)$$

$$P_j(R)^\dagger = P_j(R). \quad (19.353)$$

Proof. [✓ Proof](#) Let U_κ be the permutation unitary determined by the coordinate bijection in Definition 19.5.10.12, with the orientation

$$U_\kappa|((i_0, i_1), (a, b))\rangle = |\kappa((i_0, i_1), (a, b))\rangle.$$

Simultaneous row and column reindexing is therefore the conjugation

$$\widetilde{P}_G(R) = U_\kappa P_G(R) U_\kappa^{-1}. \quad (19.354)$$

Since $U_\kappa^{-1} = U_\kappa^\dagger$, Theorem 19.5.10.11 and (19.354) give

$$\widetilde{P}_G(R)^2 = U_\kappa P_G(R)^2 U_\kappa^{-1} = \widetilde{P}_G(R), \quad (19.355)$$

$$\widetilde{P}_G(R)^\dagger = U_\kappa P_G(R)^\dagger U_\kappa^{-1} = \widetilde{P}_G(R). \quad (19.356)$$

Let V_j be the permutation unitary that sends a chain configuration to its restriction to the cyclic window $(j, j+1)$ followed by its restriction to the complement. By Theorem 20.13.2,

$$V_j P_j(R) V_j^{-1} = \widetilde{P}_G(R) \otimes \mathbb{1}_{\text{comp}}. \quad (19.357)$$

Using $V_j^{-1} = V_j^\dagger$ and (19.355)–(19.356) in (19.357), we obtain

$$V_j P_j(R)^2 V_j^{-1} = (\widetilde{P}_G(R) \otimes \mathbb{1}_{\text{comp}})^2 = \widetilde{P}_G(R)^2 \otimes \mathbb{1}_{\text{comp}} = V_j P_j(R) V_j^{-1}, \quad (19.358)$$

$$V_j P_j(R)^\dagger V_j^{-1} = (\widetilde{P}_G(R) \otimes \mathbb{1}_{\text{comp}})^\dagger = \widetilde{P}_G(R)^\dagger \otimes \mathbb{1}_{\text{comp}} = V_j P_j(R) V_j^{-1}. \quad (19.359)$$

Conjugating by V_j^{-1} gives $P_j(R)^2 = P_j(R)$ and $P_j(R)^\dagger = P_j(R)$. $\square$

Definition 19.5.10.14 (Common fixed subspace). [Lean](#) [✓ Stmt](#) Let $\mathcal{H}$ be a complex vector space and let $(P_j)_{j \in J}$ be any family of endomorphisms of $\mathcal{H}$. Define their common fixed subspace by

$$\mathcal{V}(P) = \bigcap_{j \in J} \ker(P_j - \mathbb{1}_{\mathcal{H}}). \quad (19.360)$$

No commutativity of the operators is required. For the averaged Gauss projectors, this is the common $+1$ eigenspace called the gauge-invariant subspace in [163, lines 4327–4335 and line 5198]. Here P_j is an averaged projector, not an individual Gauss operator $\mathcal{G}_g$.

Theorem 19.5.10.15 (Membership in the common fixed subspace). [Lean](#) [✓ Stmt](#) A vector $\psi \in \mathcal{H}$ belongs to $\mathcal{V}(P)$ if and only if $P_j\psi = \psi$ for every $j \in J$.

Proof. [✓ Proof](#) Membership in an indexed intersection is equivalent to membership in every factor, and membership in $\ker(P_j - \mathbb{1}_{\mathcal{H}})$ is equivalent to $P_j\psi = \psi$. $\square$

Theorem 19.5.10.16 (Nonzero-vector lower bound). [Lean](#) [✓ Stmt](#) If $\mathcal{H}$ is finite-dimensional, $\psi \in \mathcal{V}(P)$, and $\psi \neq 0$, then $1 \leq \dim_{\mathbb{C}} \mathcal{V}(P)$.

Proof. [✓ Proof](#) The common fixed subspace is nonzero because it contains ψ , so its dimension is at least one. $\square$

Definition 19.5.10.17 (Prescribed defect maps). [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) Let $\mathcal{H}_m$ be the matter space of two sites. Prescribed defect maps assign to every ordered pair $(a, b) \in G^2$ a defect subspace

$$\mathcal{K}_{a,b} \subseteq \mathcal{H}_m \quad (19.361)$$

together with a map $D_{a,b}$ of $\mathcal{H}_m$ whose restriction to that subspace is an isometry: for every $|\xi\rangle \in \mathcal{K}_{a,b}$,

$$D_{a,b}^\dagger D_{a,b} |\xi\rangle = |\xi\rangle. \quad (19.362)$$

When the fusion scalars are not block independent, the state-level construction of [163, lines 4215–4326] prescribes the modified fusion operators only on such subspaces.

Definition 19.5.10.18 (Full unitary completion class). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) The full completion class of prescribed defect maps $D = (D_{a,b})_{a,b \in G}$ is

$$\mathfrak{C}(D) = \left\{ R = (U_{a,b})_{a,b \in G} : U_{a,b} \in \mathcal{U}(\mathcal{H}_m), U_{a,b}|_{\mathcal{K}_{a,b}} = D_{a,b}|_{\mathcal{K}_{a,b}} \right\}. \quad (19.363)$$

When $\mathcal{K}_{a,b}$ is a proper subspace of the matter space, the unitary extension of $D_{a,b}$ is non-unique [163, lines 4215–4326].

Theorem 19.5.10.19 (Existence of full unitary completions). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7702](#) Let $\mathcal{H}_m$ be finite-dimensional. Every family of prescribed defect maps satisfying (19.362) admits a simultaneous unitary extension:

$$\mathfrak{C}(D) \neq \emptyset. \quad (19.364)$$

Equivalently, there is a family $R = (U_{a,b})_{(a,b) \in G^2}$ of unitary operators such that, for every $a, b \in G$ and every $|\xi\rangle \in \mathcal{K}_{a,b}$,

$$U_{a,b} |\xi\rangle = D_{a,b} |\xi\rangle. \quad (19.365)$$

This is the finite-dimensional extension step implicit in the freedom of the modified fusion operators at [163, lines 4215–4326]. The subspaces $\mathcal{K}_{a,b}$ and their prescribed isometries are hypotheses here.

Proof. [✓ Proof](#) For each pair (a, b) , regard the restriction of $D_{a,b}$ to $\mathcal{K}_{a,b}$ as a linear isometry. Extend it to a unitary operator on the whole finite-dimensional matter space. Choosing one such extension for every pair gives a family R satisfying (19.365), hence a member of $\mathfrak{C}(D)$. $\square$

Theorem 19.5.10.20 (Transport identity for a completion). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#)
[Discussion: #7569](#) Let $R = (U_{a,b}) \in \mathfrak{C}(D)$, let $|\xi\rangle \in \mathcal{K}_{a,b}$ and $|\eta\rangle \in \mathcal{K}_{c,d}$, and assume the exact covariance

$$D_{a,b}|\xi\rangle = D_{c,d}|\eta\rangle. \quad (19.366)$$

Then

$$U_{a,b}|\xi\rangle = D_{a,b}|\xi\rangle, \quad (19.367)$$

$$U_{a,b}|\xi\rangle = U_{c,d}|\eta\rangle, \quad (19.368)$$

$$\lambda_{a,b}^{c,d}|\xi\rangle = |\eta\rangle. \quad (19.369)$$

Taking $(c,d) = T_g(a,b)$ gives the transport identity used for the modified Gauss laws of [163, lines 4325–4335].

Proof. [Stmnt](#) Identity (19.367) is the defining property of a completion on the defect subspace. Applying it at (a,b) and at (c,d) and inserting (19.366) gives (19.368), an equality of vectors in the whole matter space. Multiplying that equality by $U_{c,d}^\dagger$ and using $U_{c,d}^\dagger U_{c,d} = \mathbb{1}$ gives

$$\lambda_{a,b}^{c,d}|\xi\rangle = U_{c,d}^\dagger U_{a,b}|\xi\rangle = U_{c,d}^\dagger U_{c,d}|\eta\rangle = |\eta\rangle. \quad (19.370)$$

The adjoint is applied only after the equality in the whole matter space is available; no step asserts that $U_{c,d}^\dagger$ restricts to $D_{c,d}^\dagger$ on a defect subspace. $\square$

Definition 19.5.10.21 (Label action at one window). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) [Discussion: #7569](#)
Let $\alpha = (a_1, \dots, a_N)$ be a configuration of gauge labels on a periodic chain of length $N \geq 2$. For a window $(j, j+1)$ and $g \in G$, the configuration $T_{j,g}\alpha$ carries the labels

$$(a_j g^{-1}, g a_{j+1}) \quad (19.371)$$

on that window and agrees with α at every other site.

Theorem 19.5.10.22 (Composition of the label action). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) [Discussion: #7569](#)
For all $g, h \in G$,

$$T_{j,e} = \text{id}, \quad (19.372)$$

$$T_{j,g} T_{j,h} = T_{j,gh}. \quad (19.373)$$

In particular $T_{j,g}$ is a bijection of gauge-label configurations with inverse $T_{j,g^{-1}}$.

Proof. [Stmnt](#) Outside the window both sides leave α unchanged. On the window, replacing the labels twice keeps only the second replacement, and the two labels are

$$(a_j h^{-1} g^{-1}, g h a_{j+1}) = (a_j (gh)^{-1}, (gh) a_{j+1}). \quad (19.374)$$

At $g = e$ the labels are unchanged, so $T_{j,e} = \text{id}$, and the composition law with $h = g^{-1}$ gives the inverse. $\square$

Definition 19.5.10.23 (Placed Gauss operator of a completion). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#)
[Discussion: #7569](#) Let $R \in \mathfrak{C}(D)$ and let a chain site carry one matter index and one gauge label. Relabelling the row and column coordinates of $\mathcal{G}_g$ by the bijection κ of Definition 19.5.10.12 and embedding the result on the periodic window $(j, j+1)$ defines

$$\tilde{\mathcal{G}}_g^{j,j+1}(R) = \text{Embed}_{j,j+1}(\tilde{\mathcal{G}}_g(R)), \quad (19.375)$$

the modified Gauss law of [163, lines 4325–4335], whose block from (a,b) to $T_g(a,b)$ is $U_{T_g(a,b)}^\dagger U_{a,b}$.

Theorem 19.5.10.24 (The placed projector is the average of the placed Gauss operators). [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) For every window $(j, j+1)$,

$$\tilde{P}_j(R) = \frac{1}{|G|} \sum_{g \in G} \tilde{\mathcal{G}}_g^{j,j+1}(R). \quad (19.376)$$

Proof. [✓ Proof](#) Comparing matrix entries, the relabelling is entrywise and the embedding multiplies each entry by the indicator that the two chain configurations agree outside the window. Both operations commute with the normalized sum over G defining the local projector kernel in (19.346). $\square$

Definition 19.5.10.25 (Gauged vector of matter defect states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) Every chain configuration splits into its matter part and its gauge-label part. Given matter defect states $|\Psi_\alpha\rangle$ indexed by the gauge-label configurations α of a chain of length N , the gauged vector is

$$|\Psi_{\text{gauged}}\rangle = \sum_{\alpha} |\Psi_\alpha\rangle \otimes |\alpha\rangle, \quad (19.377)$$

whose component at a chain configuration is the component of $|\Psi_\alpha\rangle$ at its matter part, α being its gauge-label part. This is the gauged state of [163, lines 4325–4335].

Definition 19.5.10.26 (Local defect support and exact defect covariance). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) The matter defect states have local defect support when, for every α and every window $(j, j+1)$ whose adjacent labels are (a_j, a_{j+1}) , the matter support of $|\Psi_\alpha\rangle$ lies in

$$\mathcal{K}_{a_j, a_{j+1}} \otimes \mathcal{H}_{\text{rest}}. \quad (19.378)$$

They satisfy exact defect covariance when

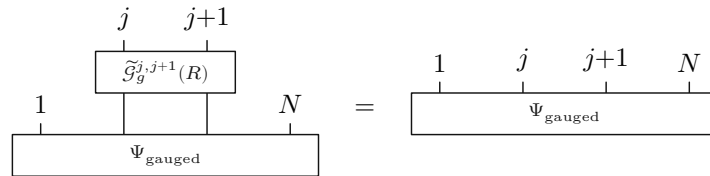
$$(D_{a_j, a_{j+1}} \otimes \text{id}_{\text{rest}}) |\Psi_\alpha\rangle = (D_{T_g(a_j, a_{j+1})} \otimes \text{id}_{\text{rest}}) |\Psi_{T_{j,g}\alpha}\rangle \quad (19.379)$$

for every α , every window $(j, j+1)$, and every $g \in G$.

Theorem 19.5.10.27 (Universal invariant vector). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) Assume local defect support and exact defect covariance (19.379). Here these properties are hypotheses rather than consequences of locally orthogonal MPS blocks [47]. For every $R \in \mathfrak{C}(D)$, every window $(j, j+1)$, and every $g \in G$,

$$\tilde{\mathcal{G}}_g^{j,j+1}(R) |\Psi_{\text{gauged}}\rangle = |\Psi_{\text{gauged}}\rangle, \quad (19.380)$$

$$\tilde{P}_j(R) |\Psi_{\text{gauged}}\rangle = |\Psi_{\text{gauged}}\rangle. \quad (19.381)$$



Proof. [✓ Proof](#) Fix an output chain configuration and let α be its gauge-label part, with $(a, b) = (a_j, a_{j+1})$ the two labels adjacent to the window. Among the columns contributing to this fixed

output row of $\tilde{\mathcal{G}}_g^{j,j+1}(R)$, only the gauge labels $T_{j,g^{-1}}\alpha$ survive. Thus, with $\beta = T_{j,g^{-1}}\alpha$, the component of $\tilde{\mathcal{G}}_g^{j,j+1}(R)|\Psi_{\text{gauged}}\rangle$ equals the component of

$$(U_{a,b}^\dagger U_{ag,g^{-1}b} \otimes \text{id}_{\text{rest}}) |\Psi_\beta\rangle, \quad (19.382)$$

since the labels of β adjacent to the window are $(ag, g^{-1}b)$ and $T_g(ag, g^{-1}b) = (a, b)$, whence $T_{j,g}\beta = \alpha$ by the composition law (19.373). Local defect support puts the matter slices of $|\Psi_\beta\rangle$ and $|\Psi_\alpha\rangle$ over that window in $\mathcal{K}_{ag,g^{-1}b}$ and $\mathcal{K}_{a,b}$, and exact covariance (19.379) relates them, so the transport identity (19.369) turns that vector into $|\Psi_\alpha\rangle$. Identifying the label configuration β with α through the bijection $T_{j,g}$ is the reindexing of the sum over gauge-label configurations in (19.377). Averaging (19.380) over g via (19.376) yields (19.381). $\square$

Definition 19.5.10.28 (Gauge-invariant subspace of a completion). [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) For $R \in \mathfrak{C}(D)$ and a chain of length $N \geq 2$, the gauge-invariant subspace is the common fixed subspace

$$\mathcal{V}_N(R) = \{|\phi\rangle : \tilde{P}_j(R)|\phi\rangle = |\phi\rangle \text{ for every } j\}. \quad (19.383)$$

The modified projectors on neighbouring windows need not commute [163, lines 4325–4335].

Theorem 19.5.10.29 (The gauged vector spans at least one dimension). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7569](#) Under local defect support and exact defect covariance (19.379),

$$|\Psi_{\text{gauged}}\rangle \in \mathcal{V}_N(R) \quad (19.384)$$

for every $R \in \mathfrak{C}(D)$. If $|\Psi_{\text{gauged}}\rangle \neq 0$, then

$$\dim_{\mathbb{C}} \mathcal{V}_N(R) \geq 1. \quad (19.385)$$

Neither the minimum nor the maximum of $\dim_{\mathbb{C}} \mathcal{V}_N(R)$ over $\mathfrak{C}(D)$ is asserted; the growth of this dimension with N is the question raised in [163, line 5198].

Proof. [✓ Proof](#) Membership is the projector identity in Theorem 19.5.10.27 at every window. A nonzero member of a subspace forces its dimension to be at least one. $\square$

Definition 19.5.10.30 (Scalar cochains and normalization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Recall from Definition 10.1.18 that a scalar 2-cochain is a function $\beta: G^2 \rightarrow \mathbb{C}^\times$. A scalar 3-cochain is a function $\omega: G^3 \rightarrow \mathbb{C}^\times$. Following the paper’s later standing convention, it is normalized when, for all $g, h, k \in G$,

$$\omega(e, h, k) = 1, \quad (19.386)$$

$$\omega(g, e, k) = 1, \quad (19.387)$$

$$\omega(g, h, e) = 1. \quad (19.388)$$

The standing convention of [163] makes every fusion tensor involving the identity trivial, which is what forces all three identity axes. Its printed display repeats the third axis in place of the first two; the three-axis form is the local correction recorded in [116]. The scalar 2-cochain β is normalized when $\beta(e, g) = \beta(g, e) = 1$ for every $g \in G$.

Definition 19.5.10.31 (Scalar three-cocycle). [Lean](#) [✓ Stmt](#) A scalar 3-cochain is a 3-cocycle when, for all $g, h, k, l \in G$,

$$\omega(gh, k, l)\omega(g, h, kl) = \omega(g, h, k)\omega(g, hk, l)\omega(h, k, l). \quad (19.389)$$

This is the scalar 3-cocycle relation of [163].

Definition 19.5.10.32 (Three-coboundary and fusion-tensor gauge action). [Lean](#) [Lean](#) [✓ Stmt](#) For a scalar 2-cochain β , define

$$(d\beta)(g, h, k) = \frac{\beta(g, hk)\beta(h, k)}{\beta(g, h)\beta(gh, k)}, \quad (19.390)$$

$$(\beta \triangleright \omega)(g, h, k) = (d\beta)(g, h, k)\omega(g, h, k). \quad (19.391)$$

Formula (19.391) is the change of ω induced by (19.257), as in [163].

Theorem 19.5.10.33 (Identity and composition of fusion gauges). [Lean](#) [Lean](#) [✓ Stmt](#) Let 1 denote the constant scalar 2-cochain and let $(\beta\gamma)(g, h) = \beta(g, h)\gamma(g, h)$. Then

$$1 \triangleright \omega = \omega, \quad (19.392)$$

$$\gamma \triangleright (\beta \triangleright \omega) = (\beta\gamma) \triangleright \omega. \quad (19.393)$$

Proof. [✓ Proof](#) The constant cochain has $d1 = 1$. Commutativity of $\mathbb{C}^\times$ gives $d(\beta\gamma) = (d\beta)(d\gamma)$, and substitution into (19.391) gives (19.392) and (19.393). $\square$

Theorem 19.5.10.34 (A three-coboundary is a three-cocycle). [Lean](#) [✓ Stmt](#) For every scalar 2-cochain β , the cochain $d\beta$ satisfies (19.389).

Proof. [✓ Proof](#) Expanding the two sides gives the same product after cancellation:

$$(d\beta)(gh, k, l)(d\beta)(g, h, kl) \quad (19.394)$$

$$= \frac{\beta(k, l)\beta(h, kl)\beta(g, hkl)}{\beta(ghk, l)\beta(gh, k)\beta(g, h)} \quad (19.395)$$

$$= (d\beta)(g, h, k)(d\beta)(g, hk, l)(d\beta)(h, k, l). \quad (19.396)$$

$\square$

Theorem 19.5.10.35 (Products and fusion gauges preserve three-cocycles). [Lean](#) [Lean](#) [✓ Stmt](#) The pointwise product of two scalar 3-cocycles is a scalar 3-cocycle. Consequently, if ω satisfies (19.389), then so does $\beta \triangleright \omega$.

Proof. [✓ Proof](#) Multiply the two instances of (19.389) and reorder their factors in $\mathbb{C}^\times$. Apply this product identity to $(d\beta)\omega$, using Theorem 19.5.10.34 for $d\beta$. $\square$

Theorem 19.5.10.36 (Normalization under a fusion gauge). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The 3-coboundary $d\beta$ is normalized if and only if there is a constant $c \in \mathbb{C}^\times$ such that, for every $g \in G$,

$$\beta(e, g) = c = \beta(g, e). \quad (19.397)$$

In particular, a normalized scalar 2-cochain has normalized 3-coboundary. Hence $\beta \triangleright \omega$ is normalized whenever both β and ω are normalized.

Proof. [✓ Proof](#) Set one argument at a time equal to e in (19.390). If $d\beta$ is normalized, then (19.388) for $d\beta$ at (e, e, g) and at (g, e, e) gives $\beta(e, g) = \beta(e, e) = \beta(g, e)$. Conversely, (19.397) gives

$$(d\beta)(e, h, k) = 1, \quad (19.398)$$

$$(d\beta)(g, e, k) = 1, \quad (19.399)$$

$$(d\beta)(g, h, e) = 1. \quad (19.400)$$

Definition 19.5.10.42 (Cyclic invariant of a scalar three-cochain). [Lean](#) [✓ Stmt](#) For a scalar 3-cochain ω , an element $g \in G$, and a natural number n , set

$$I_{g,n}(\omega) = \prod_{k=0}^{n-1} \omega(g, g^k, g). \quad (19.407)$$

Theorem 19.5.10.43 (Gauge invariance of the cyclic invariant). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $g^n = e$. For every scalar 2-cochain β ,

$$I_{g,n}(d\beta) = 1, \quad (19.408)$$

$$I_{g,n}(\beta \triangleright \omega) = I_{g,n}(\omega). \quad (19.409)$$

Consequently cohomologous 3-cochains have the same cyclic invariant, every 3-cochain with trivial gauge class has $I_{g,n}(\omega) = 1$, and a cyclic invariant different from 1 shows that the gauge class of ω is nontrivial. No normalization of β or ω is assumed.

Proof. [✓ Proof](#) By (19.390),

$$(d\beta)(g, g^k, g) = \frac{\beta(g, g^{k+1})}{\beta(g, g^k)} \cdot \frac{\beta(g^k, g)}{\beta(g^{k+1}, g)}. \quad (19.410)$$

Both quotients telescope over $k = 0, \dots, n-1$, giving $\beta(g, g^n)/\beta(g, e)$ and $\beta(e, g)/\beta(g^n, g)$, which equal 1 because $g^n = e$. Multiplying by ω pointwise proves the second identity, and the remaining statements follow from Definitions 19.5.10.37 and 19.5.10.39. $\square$

Definition 19.5.10.44 (Restriction of scalar three-cochains). [Lean](#) [Lean](#) [✓ Stmt](#) For a group homomorphism $f: H \rightarrow G$, the restriction of a scalar 3-cochain ω on G is $(f^*\omega)(a, b, c) = \omega(f(a), f(b), f(c))$.

Theorem 19.5.10.45 (Restriction preserves cocycles and gauge classes). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Restriction along f sends 3-cocycles to 3-cocycles and satisfies $f^*(\beta \triangleright \omega) = (f^*\beta) \triangleright f^*\omega$, where $(f^*\beta)(a, b) = \beta(f(a), f(b))$. Hence it preserves the relation of being cohomologous and sends cochains with trivial gauge class to cochains with trivial gauge class.

Proof. [✓ Proof](#) Since $f(ab) = f(a)f(b)$, for all $a, b, c, d \in H$

$$\begin{aligned} (f^*\omega)(ab, c, d) (f^*\omega)(a, b, cd) &= \omega(f(a)f(b), f(c), f(d)) \omega(f(a), f(b), f(c)f(d)), \\ (df^*\beta)(a, b, c) &= \frac{\beta(f(a), f(b)f(c))\beta(f(b), f(c))}{\beta(f(a), f(b))\beta(f(a)f(b), f(c))} = (d\beta)(f(a), f(b), f(c)). \end{aligned}$$

The first line and the corresponding expansion of the right-hand side of (19.389) turn the cocycle relation for $f^*\omega$ into (19.389) at $f(a), f(b), f(c), f(d)$. The second line, which is (19.390) evaluated at $f(a), f(b), f(c)$, gives $d(f^*\beta) = f^*(d\beta)$, hence $f^*(\beta \triangleright \omega) = (f^*\beta) \triangleright f^*\omega$. The last two claims follow from Definitions 19.5.10.37 and 19.5.10.39. $\square$

Definition 19.5.10.46 (Restriction of a group of matrix product operators). [Lean](#) [Lean](#) [✓ Stmt](#) For a group homomorphism $f: H \rightarrow G$ and a family $\{O_g\}_{g \in G}$ as in Definition 19.5.1.3, the restricted family is $\{O_{f(a)}\}_{a \in H}$. A choice of fusion tensors $(F_{g,h}^<, F_{g,h}^>)$ of the family restricts to the fusion tensors $(F_{f(a),f(b)}^<, F_{f(a),f(b)}^>)$ of the restricted family, where the bond space of $O_{f(a)f(b)}$ is identified with that of $O_{f(ab)}$ through the equality $f(a)f(b) = f(ab)$.

Theorem 19.5.10.51 (Identity and composition of joint scalar gauges). [Lean](#) [Lean](#) [✓ Stmt](#)
The constant fusion and action gauges leave every L -symbol unchanged. Successive joint gauges compose by pointwise multiplication:

$$(\delta, \varepsilon) \triangleright ((\beta, \gamma) \triangleright L) = (\beta\delta, \gamma\varepsilon) \triangleright L. \quad (19.414)$$

Proof. [✓ Proof](#) Substitute (19.413) twice and cancel the intermediate factors. The identity case follows by taking all four cochains equal to 1. $\square$

Definition 19.5.10.52 (Normalized L -symbols and action-tensor gauges). [Lean](#) [Lean](#) [✓ Stmt](#)
 The L -symbol is normalized when, for every $x \in X$ and $g \in G$,

$$L_{g,e}^x = 1. \quad (19.415)$$

$$L_{e,g}^x = 1. \quad (19.416)$$

Equations (19.415) and (19.416) are the normalization of the L -symbols in [163]. The action-tensor gauge is normalized when

$$\gamma_e^x = 1 \quad (19.417)$$

for every $x \in X$. This last condition is not a displayed equation of the source: it is the scalar shadow of the standing convention that every action tensor involving the identity is trivial.

Definition 19.5.10.53 (Restricted scalar gauges). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7700](#) A fusion scalar $\beta: G^2 \rightarrow \mathbb{C}^\times$ is *inverse-normalized* when, for every $g \in G$,

$$\beta_{g,e} = \beta_{e,g} = \beta_{g,g^{-1}} = 1. \quad (19.418)$$

A restricted scalar gauge is a pair (β, γ) in which β is inverse-normalized and, for every $x \in X$,

$$\gamma_e^x = 1. \quad (19.419)$$

The conditions are the standing restriction imposed at [163, lines 2050–2054] after fixing the fusion and action gauges of Definition 19.5.10.50.

Theorem 19.5.10.54 (Joint scalar gauges preserve L -symbol compatibility). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If L is compatible with ω , then $(\beta, \gamma) \triangleright L$ is compatible with $\beta \triangleright \omega$. On the identity axes, the gauge formula is exactly*

$$((\beta, \gamma) \triangleright L)_{g,e}^x = \frac{\beta_{g,e}}{\gamma_e^x} L_{g,e}^x. \quad (19.420)$$

$$((\beta, \gamma) \triangleright L)_{e,g}^x = \frac{\beta_{e,g}}{\gamma_e^{gx}} L_{e,g}^x. \quad (19.421)$$

Thus the gauged L -symbol is normalized exactly when $\beta_{g,e} L_{g,e}^x = \gamma_e^x$ and $\beta_{e,g} L_{e,g}^x = \gamma_e^{gx}$ for all x and g . In particular, normalized β , γ , and L give a normalized gauged L -symbol.

Proof. [✓ Proof](#) Write $L' = (\beta, \gamma) \triangleright L$. Substituting (19.413) into the left side of (19.412) for L' cancels the factor γ_{hk}^x and leaves

$$L'_{g,hk}{}^x L'_{h,k}{}^x = \frac{\gamma_{ghk}^x}{\gamma_g^{(hk)x} \gamma_h^{kx} \gamma_k^x} \beta_{g,hk} \beta_{h,k} L_{g,hk}^x L_{h,k}^x. \quad (19.422)$$

The same substitution on the right side cancels γ_{gh}^{kx} and leaves

$$L'_{g,h}{}^{kx} L'_{gh,k}{}^x = \frac{\gamma_{ghk}^x}{\gamma_g^{h(kx)} \gamma_h^{kx} \gamma_k^x} \beta_{g,h} \beta_{gh,k} L_{g,h}^{kx} L_{gh,k}^x. \quad (19.423)$$

The action law $h(kx) = (hk)x$ makes the two action-tensor prefactors of (19.422) and (19.423) equal. Applying (19.412) for L and ω to the right of (19.422) therefore gives

$$L'_{g,hk}{}^x L'_{h,k}{}^x = \frac{\beta_{g,hk} \beta_{h,k}}{\beta_{g,h} \beta_{gh,k}} \omega(g, h, k) L'_{g,h}{}^{kx} L'_{gh,k}{}^x, \quad (19.424)$$

whose scalar factor is $(\beta \triangleright \omega)(g, h, k)$ by (19.390) and (19.391). Setting either group argument equal to e in (19.413) gives (19.420) and (19.421), from which the normalization statements follow. $\square$

Definition 19.5.10.55 (Unimodular square root and modulus of a phase). [Lean](#) [Lean](#) [✓ Stmt](#) For $u \in \mathbb{C}^\times$ put $s_u = e^{i \arg(u)/2}$, the unimodular square root of the phase of u , and regard $|u|$ as an element of $\mathbb{C}^\times$.

Theorem 19.5.10.56 (Unimodular square roots of phases). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $u \in \mathbb{C}^\times$, $\overline{s_u} = s_u^{-1}$ and $u = s_u^2 |u|$. If $w \in \mathbb{C}^\times$ and $\overline{w} = pw$ for a positive real number p , then $p = 1$ and w is real.

Proof. [✓ Proof](#) The first two claims are $\overline{e^{i\theta}} = e^{-i\theta}$ and the polar form $u = |u|e^{i \arg u}$. For the third, taking absolute values in $\overline{w} = pw$ gives $p = 1$. $\square$

Theorem 19.5.10.57 (Time reversal makes the anomaly real). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ω be a scalar 3-cochain and β a scalar 2-cochain with $\beta \triangleright \omega = \overline{\omega}$. Then $s \triangleright \omega$ is real-valued, where $s(g, h) = s_{\beta(g, h)}$ is the unimodular square root of the phase of $\beta(g, h)$ from Definition 19.5.10.55. Consequently, a scalar 3-cochain cohomologous to its complex conjugate is cohomologous to a real-valued one. This is the conclusion of [164, Section 4.3] that, when time reversal commutes with a group matrix product operator symmetry, the three-cocycle can be gauge transformed to a real number; no finiteness of G is needed.

Proof. [✓ Proof](#) The coboundary is multiplicative in β and commutes with complex conjugation. Write $\beta = s^2 N$ with $N(g, h) = |\beta(g, h)|$. Since $\overline{s} = s^{-1}$,

$$\overline{s \triangleright \omega} = s^{-1} \triangleright \overline{\omega} = s^{-1} \triangleright (s^2 N \triangleright \omega) = N \triangleright (s \triangleright \omega). \quad (19.425)$$

The values of dN are positive real numbers, so Theorem 19.5.10.56 gives $dN = 1$ and $\overline{s \triangleright \omega} = s \triangleright \omega$. If $\omega = \beta \triangleright \overline{\omega}$, conjugating gives $\overline{\beta} \triangleright \omega = \overline{\omega}$, and the first part applies. $\square$

Theorem 19.5.10.58 (Time reversal makes the anomaly and the L-symbols jointly real). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ω be a scalar 3-cochain, L an L-symbol, β a scalar 2-cochain and γ an action-tensor scalar gauge with $\beta \triangleright \omega = \overline{\omega}$ and $(\beta, \gamma) \triangleright L = \overline{L}$. Let s and t be the unimodular square roots of the phases of β and γ . Then $s \triangleright \omega$ and $(s, t) \triangleright L$ are both real-valued. This result is not stated in [164, Section 4.3]. The source derives $\beta \triangleright \omega = \overline{\omega}$ and, with the fusion-tensor phase omitted, $(1, \gamma) \triangleright L = \overline{L}$; the hypothesis here, with the same β in both relations, is the gauge-consistent pair and is assumed rather than derived from time reversal.

are again normalized, ω' is a scalar 3-cocycle, and L' is compatible with ω' . The identity gauge acts trivially, successive restricted gauges compose by pointwise multiplication, and applying κ^{-1} after κ recovers (ω, L) . These are scalar statements only.

Proof. ✓ Proof The three specified values of β and the identity value of γ remain equal to 1 under pointwise products and inverses. Apply the general preservation laws to (19.430). The group laws and the composition formulas for the two gauge actions give the identity, composition, and inverse assertions. □

Theorem 19.5.10.62 (Cocycles and coboundaries for trivial coefficients). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) *Let G act trivially on $\mathbb{C}^\times$, and let $f(g, h) = \omega(g, h)$. The multiplicative cocycle equation*

$$f(qh, k)f(q, h) = f(h, k)f(q, hk) \quad (19.431)$$

is equivalent to Definition 10.2.4. Likewise, the multiplicative coboundary equation

$$\frac{\varphi(h)}{\varphi(qh)}\varphi(g) = f(g, h) \quad (19.432)$$

is equivalent to Definition 10.2.1.

If π denotes the canonical map from multiplicative 2-cocycles to degree-two group cohomology, then for any two cocycles ω_1 and ω_2 ,

$$\omega_1 \sim \omega_2 \iff \pi(\omega_1) = \pi(\omega_2). \quad (19.433)$$

Proof. ✓ **Proof** Since $\mathbb{C}^\times$ is commutative, (19.431) is obtained from the cocycle equation by interchanging the two factors on each side, and (19.432) is obtained from $\varphi(g)\varphi(h)\varphi(gh)^{-1}$ by reordering its factors. Equivalence $\omega_1 \sim \omega_2$ holds when $\omega_1\omega_2^{-1}$ is a coboundary. In additive notation this ratio is the difference of the corresponding cochains, so the quotient criterion for π gives (19.433). □

Definition 19.5.10.63 (Comparison with low-degree group cohomology). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Give $\mathbb{C}^\times$ the trivial G -action. Currying and uncurrying identify the cocycle representatives in Definition 10.2.6 with the multiplicative representatives in the inhomogeneous cochain complex. By (19.433), this correspondence induces a canonical equivalence

$$Z^2(G, \mathbb{C}^\times)/\sim \simeq H^2(G, \mathbb{C}_{\text{triv}}^\times). \quad (19.434)$$

The left-hand side is the quotient already defined in Definition 10.2.6; no additional quotient is introduced. This is the $H^2(H, \mathbb{C}^\times)$ appearing in the pair (H, ψ) in [163, lines 1927–1931].

Definition 19.5.10.64 (Pullback of scalar cohomology). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $f : K \rightarrow G$ be a group homomorphism. For a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$, define $(f^*\omega)(a, b) = \omega(f(a), f(b))$. The induced pullback map $f^* : H^2(G, \mathbb{C}^\times) \rightarrow H^2(K, \mathbb{C}^\times)$ is defined by $f^*[\omega] = [f^*\omega]$.

Theorem 19.5.10.65 (Pullback preserves cocycles and cohomology). [Lean](#) [Lean](#) [✓ Stmt](#) Let $f : K \rightarrow G$ be a group homomorphism. If ω is a scalar 2-cocycle on G , then $f^*\omega$ is a scalar 2-cocycle on K . If ω_1 and ω_2 are cohomologous scalar 2-cochains on G , then $f^*\omega_1$ and $f^*\omega_2$ are cohomologous on K .

Proof. ✓ Proof For $a, b, c \in K$, substitute $f(a), f(b), f(c)$ in the cocycle equation for ω and use $f(ab) = f(a)f(b)$ and $f(bc) = f(b)f(c)$. If $\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)$, set $\eta = \varphi \circ f$. Substituting $g = f(a)$ and $h = f(b)$ gives $(f^*\omega_1)(a, b) = \eta(a)\eta(b)\eta(ab)^{-1}(f^*\omega_2)(a, b)$, so η witnesses the required cohomology relation. $\square$

Theorem 19.5.10.66 (Exact extension of a subgroup cocycle). Lean Lean ✓ Stmt Let $H \leq G$ be any subgroup, let $\iota : H \hookrightarrow G$ be its inclusion, and let $\psi : H \times H \rightarrow \mathbb{C}^\times$ be a 2-cocycle. Then $[\psi]$ belongs to the image of ι^* if and only if there is a 2-cocycle $\Psi : G \times G \rightarrow \mathbb{C}^\times$ with $\Psi(a, b) = \psi(a, b)$ for every $a, b \in H$. No normalization of either cocycle is required. This is the representative adjustment implicit in the extension condition of [163, lines 7042–7064], here stated with $\mathbb{C}^\times$ coefficients.

Proof. ✓ Proof Suppose $[\psi] = \iota^*[\omega]$. Choose a cochain $\varphi : H \rightarrow \mathbb{C}^\times$ such that $\psi(a, b) = \varphi(a)\varphi(b)\varphi(ab)^{-1}\omega(a, b)$. Extend φ to a function $\Phi : G \rightarrow \mathbb{C}^\times$, taking value 1 outside H , and set $\Psi(g, h) = \Phi(g)\Phi(h)\Phi(gh)^{-1}\omega(g, h)$. In the cocycle equation for Ψ , cancellation leaves the common factor $\Phi(g)\Phi(h)\Phi(k)\Phi(ghk)^{-1}$ multiplied by the two sides of the cocycle equation for ω . Hence Ψ is a cocycle, and $\Phi|_H = \varphi$ gives $\Psi|_{H \times H} = \psi$ exactly. Conversely, any exact extension Ψ satisfies $\iota^*[\Psi] = [\psi]$. $\square$

Theorem 19.5.10.67 (Circle and nonzero-complex coefficients in degree two). Lean Lean Lean Lean Lean Lean ✓ Stmt Let G be finite, and give $U(1)$ and $\mathbb{C}^\times$ the trivial G -actions. Inclusion of the unit circle induces an equivalence

$$H^2(G, U(1)) \simeq H^2(G, \mathbb{C}^\times). \quad (19.435)$$

More concretely, for a $\mathbb{C}^\times$ -valued cocycle ω , define its pointwise phase and radial residual by

$$u(g, h) = \frac{\omega(g, h)}{|\omega(g, h)|} \in U(1), \quad r(g, h) = \frac{\omega(g, h)}{u(g, h)} = |\omega(g, h)|. \quad (19.436)$$

Then u is a cocycle, r is a positive coboundary, and hence ω is cohomologous to the inclusion of u . The inclusion map is injective for every group because pointwise phase is its left inverse; finiteness is used only to prove surjectivity.

This is an algebraic group-cohomology statement for a finite discrete group. The footnote at [163, line 1931] instead invokes a comparison of sheaf cohomology with topological coefficient groups; the distinction and the direct finite-group proof are recorded in [34].

Proof. ✓ Proof The phase map $q(z) = z/|z|$ is multiplicative and satisfies $q(u) = u$ for $u \in U(1)$, so it preserves the cocycle equation and retracts inclusion. The polar identity in (19.436) identifies the residual r with a positive cocycle. Apply Theorem 19.5.10.97 to the one-point block space to write $r = d\chi$. Therefore $\omega \sim u$, which proves surjectivity of coefficient inclusion on degree-two classes. Its left inverse induced by q proves injectivity. $\square$

Definition 19.5.10.68 (Cocycle-twisted regular matrices). Lean Lean Lean ✓ Stmt Let G be a group and let ω be a $\mathbb{C}^\times$ -valued scalar cochain. Write

$$\phi(g, h) = \frac{\omega(g, h)}{|\omega(g, h)|} \in U(1). \quad (19.437)$$

On the standard basis of $\mathbb{C}^G$, define

$$L_g^\phi = \sum_{h \in G} \phi(g, h)^{-1} |gh\rangle \langle h|, \quad R_g^\phi = \sum_{h \in G} \phi(h, g^{-1})^{-1} |hg^{-1}\rangle \langle h|. \quad (19.438)$$

The source uses a $U(1)$ -valued cocycle Ω ; taking $\Omega = \phi$ gives exactly the leg and inverse conventions in [163, lines 412–418].

Lemma 19.5.10.69 (Cocycle equation for the circle phase). [Lean](#) [✓ Stmt](#) *If ω is a cocycle, then its circle phase satisfies*

$$\phi(g, h)\phi(gh, k) = \phi(g, hk)\phi(h, k) \quad (19.439)$$

for all $g, h, k \in G$.

Proof. [✓ Proof](#) The pointwise phase map $z \mapsto z/|z|$ is multiplicative, so it preserves the cocycle equation. $\square$

Theorem 19.5.10.70 (Action of the twisted regular matrices). [Lean](#) [Lean](#) [✓ Stmt](#) *If G is finite, then for $v \in \mathbb{C}^G$ and $g, h \in G$,*

$$(L_g^\phi v)(gh) = \phi(g, h)^{-1}v(h), \quad (R_g^\phi v)(hg^{-1}) = \phi(h, g^{-1})^{-1}v(h). \quad (19.440)$$

Proof. [✓ Proof](#) In each matrix-vector product, exactly one summand is nonzero: the column indexed by h . $\square$

Theorem 19.5.10.71 (Multiplication laws for twisted regular matrices). [Lean](#) [Lean](#) [✓ Stmt](#) *If G is finite and ω is a cocycle, then*

$$L_g^\phi L_h^\phi = \phi(g, h)^{-1} L_{gh}^\phi, \quad (19.441)$$

$$R_g^\phi R_h^\phi = \phi(h^{-1}, g^{-1})^{-1} R_{gh}^\phi. \quad (19.442)$$

Proof. [✓ Proof](#) Expand the matrix products entrywise. The two scalar identities are the cocycle equation at (g, h, k) and (k, h^{-1}, g^{-1}) , respectively. $\square$

Theorem 19.5.10.72 (Commutation of twisted left and right actions). [Lean](#) [✓ Stmt](#) *If G is finite and ω is a cocycle, then*

$$L_g^\phi R_h^\phi = R_h^\phi L_g^\phi. \quad (19.443)$$

Proof. [✓ Proof](#) Expand both matrix products entrywise and use the cocycle equation at (g, k, h^{-1}) . $\square$

Theorem 19.5.10.73 (Unitarity of twisted regular matrices). [Lean](#) [Lean](#) [✓ Stmt](#) *If G is finite, the matrices L_g^ϕ and R_g^ϕ are unitary for every $g \in G$.*

Proof. [✓ Proof](#) Each matrix is a permutation matrix whose nonzero entries have modulus one. Distinct columns therefore have disjoint support, while every column has squared norm one. $\square$

Theorem 19.5.10.74 (Onsite Gauss coefficient identity). [Lean](#) [✓ Stmt](#) *Suppose that ω is a cocycle and impose the additional normalization*

$$\phi(t^{-1}, t) = 1 \quad (19.444)$$

for every $t \in G$. Then

$$\frac{\phi(ag^{-1}, gb)}{\phi(a, b)} = \frac{1}{\phi(a, g^{-1})\phi(g, b)}. \quad (19.445)$$

The scalar convention $S_{t^{-1}} = S_t^{-1}$ appears in [163, line 238], and the coefficient rewrite is at lines 3763–3766. The displayed normalization remains an explicit assumption here; it is not a consequence of cocyclicity.

Proof. [✓ Proof](#) Use the cocycle equation first at (a, g^{-1}, gb) and then at (g^{-1}, g, b) . The normalization at $t = e$ also gives $\phi(e, b) = 1$. $\square$

Definition 19.5.10.75 (Block scalar and block independence). [Lean](#) [Lean](#) [✓ Stmt](#) For an L -symbol, define

$$\ell_{a,b;r}^x = \frac{L_{ar,r^{-1}b}^x}{L_{a,b}^x}. \quad (19.446)$$

The scalar ℓ is block-independent when

$$\ell_{a,b;r}^x = \ell_{a,b;r}^y \quad (19.447)$$

for all $a, b, r \in G$ and $x, y \in \mathsf{X}$.

Definition 19.5.10.76 (Four scalar gauge formulations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) These are the four scalar L -symbol properties used in the proof of the source proposition. Write $L' = (\beta, \gamma) \triangleright L$ for a joint scalar gauge of a supplied L -symbol. The four properties are, respectively, the existence of β and γ such that

$$(L')_{g,h}^x = 1 \quad \text{for all } x, g, h, \quad (19.448)$$

$$(L')_{a,b}^x = (L')_{a,b}^y \quad \text{for all } x, y, a, b, \quad (19.449)$$

$$\ell[L']_{a,b;r}^x = 1 \quad \text{for all } x, a, b, r, \quad (19.450)$$

$$\ell[L']_{a,b;r}^x = \ell[L']_{a,b;r}^y \quad \text{for all } x, y, a, b, r. \quad (19.451)$$

Theorem 19.5.10.77 (Corrected block factorization). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that L is compatible with ω and that ℓ is block-independent. Fix $x_0 \in \mathsf{X}$ and set

$$A(g, h) = (\omega(gh, e, e) \ell_{g,h;h}^{x_0})^{-1}. \quad (19.452)$$

Then, for every $x \in \mathsf{X}$ and $g, h \in G$,

$$L_{g,h}^x = A(g, h) L_{e,e}^x. \quad (19.453)$$

The reciprocal factor in this identity is recorded in [115].

Proof. [✓ Proof](#) Definition 19.5.10.75 gives

$$L_{g,h}^x = \frac{L_{gh,e}^x}{\ell_{g,h;h}^x}. \quad (19.454)$$

Block independence replaces x by x_0 in the denominator. Compatibility at (gh, e, e) gives $L_{gh,e}^x = L_{e,e}^x / \omega(gh, e, e)$, which proves (19.453). $\square$

Theorem 19.5.10.78 (Relative block character). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 19.5.10.77, define

$$\rho(g) = (\ell_{e,g;g}^{x_0} \omega(g, e, e) \omega(e, e, g))^{-1}. \quad (19.455)$$

The reciprocal factor $(\ell_{e,g;g}^{x_0})^{-1}$ corrects the corresponding printed factor [115]. Then

$$L_{e,e}^{gx} = \rho(g) L_{e,e}^x, \quad (19.456)$$

$$\rho(gh) = \rho(g) \rho(h). \quad (19.457)$$

Thus $\rho: G \rightarrow \mathbb{C}^\times$ is a one-dimensional character. No transitivity or finiteness assumption on the action is required.

Proof. [✓ Proof](#) Compatibility at (e, e, g) gives $L_{e,g}^x = \omega(e, e, g)L_{e,e}^{gx}$. Compare this with (19.453) at (e, g) to obtain (19.456). Applying that identity first to h and then to g , and comparing with its application to gh , proves (19.457) by cancellation. $\square$

Theorem 19.5.10.79 (Explicit inverse gauge). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) With A and ρ as above, define the forward gauges by

$$\beta_{g,h}^+ = \frac{A(g, h)}{\rho(h)}, \quad (19.458)$$

$$\gamma_g^{+,x} = (L_{e,e}^{gx})^{-1}. \quad (19.459)$$

Their joint gauge multiplier equals L :

$$(\beta^+, \gamma^+) \triangleright 1 = L. \quad (19.460)$$

Consequently the pointwise inverse gauges trivialize L :

$$((\beta^+)^{-1}, (\gamma^+)^{-1}) \triangleright L = 1. \quad (19.461)$$

The distinction between the forward multiplier and its pointwise inverse corrects the orientation of the displayed source calculation [115]. If $\mathbf{X}$ is nonempty, a choice of x_0 therefore gives a trivializing joint scalar gauge for every compatible block-independent L -symbol.

Proof. [✓ Proof](#) Substitute (19.458) and (19.459) into (19.413). Equation (19.456) reduces the multiplier to $A(g, h)L_{e,e}^x$, which equals $L_{g,h}^x$ by (19.453). Joint gauges compose by pointwise multiplication, so the inverse gauges give (19.461). $\square$

Theorem 19.5.10.80 (Trivial and block-constant gauges). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let L be a supplied L -symbol compatible with a scalar 3-cochain, and suppose that $\mathbf{X}$ is nonempty. A joint scalar gauge trivializes L if and only if a joint scalar gauge makes $L_{g,h}^x$ independent of x .

Proof. [✓ Proof](#) A trivial L -symbol is block-constant. Conversely, a block-constant representative has block-independent ℓ . Apply Theorem 19.5.10.79 to that representative and compose the two joint gauges. $\square$

Theorem 19.5.10.81 (Trivial gauges and trivial block scalars). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let L be a supplied L -symbol compatible with a scalar 3-cochain, and suppose that $\mathbf{X}$ is nonempty. A joint scalar gauge trivializes L if and only if a joint scalar gauge makes every $\ell_{a,b;r}^x$ equal to 1.

Proof. [✓ Proof](#) A trivial L -symbol has $\ell = 1$. Conversely, the identity $\ell = 1$ implies block independence. Apply Theorem 19.5.10.79 and compose the gauges. $\square$

Theorem 19.5.10.82 (Scalar gauge triviality and block independence). [Lean](#) [Lean](#) [✓ Stmt](#) Let L be a supplied L -symbol compatible with a scalar 3-cochain, and suppose that $\mathbf{X}$ is nonempty. A joint scalar gauge trivializes L if and only if a joint scalar gauge makes $\ell_{a,b;r}^x$ independent of x .

Proof. [✓ Proof](#) Gauge triviality implies block constancy and hence block independence. For the converse, apply Theorem 19.5.10.79 to the block-independent representative and compose its trivializing gauge with the original gauge. $\square$

Theorem 19.5.10.83 (Four scalar gauge formulations of block independence). [Lean](#) [✓ Stmt](#) Let L be a supplied L -symbol compatible with a scalar 3-cochain, and suppose that X is nonempty. The four scalar gauge properties (19.448), (19.449), (19.450), and (19.451) of L are equivalent. This is the scalar core of the block-independence proposition of [163], whose fourth condition is stated for an MPS–MPU pair.

Proof. [✓ Proof](#) Theorems 19.5.10.80, 19.5.10.81, and 19.5.10.82 identify each of the last three properties with the first. $\square$

For a supplied compatible scalar L -symbol, Theorem 19.5.10.83 establishes the algebraic equivalence among its four scalar gauge properties; the corresponding tensor-level bridge to an underlying MPS–MPU pair is not asserted here.

Theorem 19.5.10.84 (Stabilizer transition elements). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that G acts transitively on X , and fix $x_0 \in \mathsf{X}$. There are representatives $k_x \in G$ satisfying $k_x x_0 = x$ and $k_{x_0} = e$. With $H = \text{Stab}_G(x_0)$, define

$$t_k(g, x) = k_{gx}^{-1} g k_x. \quad (19.462)$$

Then $t_k(g, x) \in H$ and

$$t_k(g_1 g_2, x) = t_k(g_1, g_2 x) t_k(g_2, x). \quad (19.463)$$

Consequently, the source factors are

$$h_2 = t_k(g_2, x) = k_{g_2 x}^{-1} g_2 k_x, \quad (19.464)$$

$$h_1 = t_k(g_1, g_2 x) = k_{g_1 g_2 x}^{-1} g_1 k_{g_2 x}. \quad (19.465)$$

The last representative in h_1 corrects the printed $k_{g_1 x}$ [119]. No finiteness or normality assumption is required.

Proof. [✓ Proof](#) Transitivity supplies one element carrying x_0 to each x ; choose the identity at $x = x_0$. Acting with (19.462) on x_0 gives $k_{gx}^{-1} g(k_x x_0) = k_{gx}^{-1}(gx) = x_0$. For (19.463), substitute (19.462) and cancel $k_{g_2 x} k_{g_2 x}^{-1}$. The two formulas for h_1 and h_2 are the corresponding specializations. $\square$

Theorem 19.5.10.85 (L -symbol induced by a stabilizer cocycle). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $H = \text{Stab}_G(x_0)$ and let $\psi: H \times H \rightarrow \mathbb{C}^\times$ be a scalar 2-cocycle. Define

$$L_{g,h}^x = \psi(t_k(g, hx), t_k(h, x)). \quad (19.466)$$

Formula (19.466) is the $\omega = 1$ scalar specialization of [164, Eq. (20)]. The L -symbol satisfies (19.412) with $\omega = 1$, and its base-point restriction recovers ψ :

$$L_{a,b}^{x_0} = \psi(a, b) \quad (19.467)$$

for all $a, b \in H$.

More precisely, two stabilizer cocycles are cohomologous exactly when their induced L -symbols differ by an action-tensor gauge and the trivial fusion-tensor gauge:

$$\psi_1 \sim \psi_2 \iff \exists \gamma, \quad L[\psi_1] = (1, \gamma) \triangleright L[\psi_2]. \quad (19.468)$$

If the cohomology relation is witnessed by $\varphi: H \rightarrow \mathbb{C}^\times$, with

$$\psi_1(a, b) = \varphi(a)\varphi(b)\varphi(ab)^{-1}\psi_2(a, b), \quad (19.469)$$

then the forward action-tensor gauge is

$$\gamma_{g,x} = \varphi(t_k(g, x))^{-1}. \quad (19.470)$$

Thus the induced action-gauge class depends only on the cohomology class of ψ .

This establishes the scalar part of the L -symbol reduction in [163, lines 7028–7042]; the representation-theoretic claim for a tensor-derived L -symbol is not asserted here.

Proof. ✓ Proof Put $a = t_k(g, h k x)$, $b = t_k(h, k x)$, and $c = t_k(k, x)$. The product law (19.463) gives $t_k(gh, kx) = ab$ and $t_k(hk, x) = bc$. Compatibility is therefore exactly

$$\psi(a, b)\psi(ab, c) = \psi(a, bc)\psi(b, c), \quad (19.471)$$

which is the cocycle equation. If $p, q \in H$, then $t_k(p, x_0) = p$ and $t_k(q, x_0) = q$ because $k_{x_0} = e$, proving (19.467). For the forward gauge calculation, set $a = t_k(g, h x)$ and $b = t_k(h, x)$, so that $t_k(gh, x) = ab$. Substituting (19.470) into (19.413) gives the multiplier

$$\frac{\gamma_{gh,x}}{\gamma_{g,hx}\gamma_{h,x}} = \varphi(ab)^{-1}(\varphi(a)^{-1}\varphi(b)^{-1})^{-1} = \varphi(a)\varphi(b)\varphi(ab)^{-1}. \quad (19.472)$$

Formula (19.469) now identifies the two induced L -symbols. Conversely, suppose $L[\psi_1] = (1, \gamma) \triangleright L[\psi_2]$ and define $\varphi(a) = \gamma_{a,x_0}^{-1}$ for $a \in H$. Evaluating the gauge equality at x_0 and $a, b \in H$ gives

$$\psi_1(a, b) = \frac{\gamma_{ab,x_0}}{\gamma_{a,x_0}\gamma_{b,x_0}}\psi_2(a, b) = \varphi(a)\varphi(b)\varphi(ab)^{-1}\psi_2(a, b). \quad (19.473)$$

Hence $\psi_1 \sim \psi_2$, which proves the reverse implication in (19.468). □

Theorem 19.5.10.86 (Explicit gauge from a stabilizer cocycle extension). Lean Lean Lean

✓ Stmt Let G act on X , with basepoint x_0 , stabilizer $H = \text{Stab}_G(x_0)$, and representatives k_x satisfying $k_x x_0 = x$ and $k_{x_0} = e$. Let $\psi: H \times H \rightarrow \mathbb{C}^\times$ and let $\Psi: G \times G \rightarrow \mathbb{C}^\times$ be a two-cocycle whose restriction to $H \times H$ is ψ . Define

$$\gamma_g^x = \frac{\Psi(k_{g_x}^{-1}, g k_x) \Psi(g, k_x)}{\Psi(e, e) \Psi(k_x, k_x^{-1})}. \quad (19.474)$$

With $h_1 = t_k(g_1, g_2 x)$ and $h_2 = t_k(g_2, x)$, the induced symbols satisfy

$$L_{g_1, g_2}^x = \psi(h_1, h_2) = \Psi(g_1, g_2) \frac{\gamma_{g_1 g_2}^x}{\gamma_{g_1}^x \gamma_{g_2}^x}. \quad (19.475)$$

Consequently the inverse fusion and action gauges (Ψ^{-1}, γ^{-1}) make L identically one. This is the scalar representative calculation in the forward proof of the block-independence extension proposition of [163]; here extension means the equality $\Psi|_{H \times H} = \psi$ of chosen cocycle representatives, rather than only equality of their classes. See [134] for the relationship between cocycle representatives, cohomology classes, and tensor data.

Proof. ✓ Proof Put $a = k_{g_1 g_2 x}$, $b = k_{g_2 x}$, $c = k_x$, $u = g_1$, and $v = g_2$. The cocycle equation gives $\Psi(z, e) = \Psi(e, z) = \Psi(e, e)$ and $\Psi(b^{-1}, b) = \Psi(b, b^{-1})$. Apply it successively to the triples

$$(a^{-1}ub, b^{-1}, vc), \quad (a^{-1}ub, b^{-1}, b), \quad (a^{-1}, u, vc), \quad (a^{-1}, u, b), \quad (u, v, c). \quad (19.476)$$

Solving these equalities for the corresponding factors of $\Psi(a^{-1}ub, b^{-1}vc)$ and cancelling yields (19.475) with (19.474). The restriction hypothesis identifies its left side with $\psi(h_1, h_2)$. Substitution into (19.413) cancels the multiplier under the inverse gauges. $\square$

Theorem 19.5.10.87 (Reconstruction of stabilizer L-symbols). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let G act on X with basepoint x_0 , stabilizer $H = \text{Stab}_G(x_0)$, and representatives k_x as in Theorem 19.5.10.84. Let ω be a scalar 3-cocycle on G with $\omega(a, b, c) = 1$ for all $a, b, c \in H$, and let ψ be a scalar 2-cocycle on H . With $h_2 = t_k(g_2, x)$ and $h_1 = t_k(g_1, g_2 x)$, define

$$L_{g_1, g_2}^x = \frac{\omega(g_1 g_2 k_x, h_2^{-1}, h_1^{-1}) \psi(h_1, h_2)}{\omega(g_1, g_2 k_x, h_2^{-1}) \omega(g_1, g_2, k_x)}. \quad (19.477)$$

Then L satisfies (19.412), and $L_{a, b}^{x_0} = \psi(a, b)$ for $a, b \in H$. Formula (19.477) is [164, Eq. (20)].

If only $\omega|_H$ is a coboundary, some fusion gauge β on G makes $\beta \triangleright \omega$ equal to one on $H \times H \times H$, and applying the inverse gauge $(\beta^{-1}, 1)$ to (19.477) built from $\beta \triangleright \omega$ gives an L -symbol compatible with ω . The hypothesis that ω is one on the stabilizer is the fusion gauge in which the source's formula holds; for a general ω with $\omega|_H = d\beta$ the printed formula can fail [67]. No normalization of ω is required.

Proof. ✓ Proof The formula factors as its value at $\psi = 1$ times the induced L -symbol (19.466), and pointwise products of compatible L -symbols are compatible with the product of the 3-cochains. By Theorem 19.5.10.85 it suffices to treat $\psi = 1$. Fix x and g, h, k , put $p = k_x$, $a = t_k(k, x)$, $b = t_k(h, kx)$, and $c = t_k(g, h k x)$, so that $k_{kx} = kpa^{-1}$ and, by (19.463), $t_k(hk, x) = ba$ and $t_k(gh, kx) = cb$. The compatibility equation then follows from the 3-cocycle equation at the four quadruples

$$(g, h, k, p), \quad (g, h, kp, a^{-1}), \quad (g, hkp, a^{-1}, b^{-1}), \quad (ghkp, a^{-1}, b^{-1}, c^{-1}), \quad (19.478)$$

together with $\omega(a^{-1}, b^{-1}, c^{-1}) = 1$. At x_0 and $a, b \in H$ every argument of ω in (19.477) lies in H , which gives $L_{a, b}^{x_0} = \psi(a, b)$. If $\omega|_H = d\beta_0$, extend β_0^{-1} by one outside $H \times H$; the coboundary of the extension cancels ω on $H \times H \times H$. Finally, $(\beta^{-1}, 1)$ changes $\beta \triangleright \omega$ back to ω . $\square$

Theorem 19.5.10.88 (Classification and torsor of stabilizer L-symbols). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Keep the setting of Theorem 19.5.10.87, and call two L -symbols action-gauge equivalent when they differ by $(1, \gamma) \triangleright$ for some action-tensor gauge γ .

1. If R is compatible with the trivial 3-cochain, then $R = (1, \gamma) \triangleright L[\psi]$ for the induced L -symbol (19.466) of the 2-cocycle $\psi(a, b) = R_{a, b}^{x_0}$, with $\gamma_g^x = R_{k_{gx}, t_k(g, x)}^{x_0} / R_{g, k_x}^{x_0}$.
2. If ω is a 3-cocycle equal to one on $H \times H \times H$, every L -symbol compatible with ω is action-gauge equivalent to (19.477) for some 2-cocycle ψ on H . For a general fusion gauge β making $\beta \triangleright \omega$ trivial on H , the same holds for $(\beta, 1) \triangleright L$ and the formula built from $\beta \triangleright \omega$.

3. Let L_0 be compatible with an arbitrary scalar 3-cochain ω . An L -symbol L is compatible with ω if and only if L is action-gauge equivalent to $L_0 L[\psi]$ for some 2-cocycle ψ on H , and $L_0 L[\psi_1]$ and $L_0 L[\psi_2]$ are action-gauge equivalent if and only if $\psi_1 \sim \psi_2$. Hence sending the class of ψ to the class of $L_0 L[\psi]$ is a bijection from $H^2(H, \mathbb{C}^\times)$ onto the action-gauge classes of L -symbols compatible with ω .
4. Sending the class of ψ and the class of L to the class of $L L[\psi]$ is a well-defined action of the group cohomology $H^2(H, \mathbb{C}^\times)$, for the trivial action of H on $\mathbb{C}^\times$, on these classes, and for any two classes exactly one cohomology class carries the first to the second.

The last two items are the torsor statement of [164] for fixed (G, ω, H) , with the fusion gauge held fixed as in the source. In all four items G , and hence H , is a small group: its underlying set lies in a fixed set-theoretic universe containing $\mathbb{Z}$, in which the group cohomology $H^2(H, \mathbb{C}^\times)$ is formed. In the last two items the set $\mathbf{X}$ lies in the same universe.

Proof. ✓ Proof For the first item, compatibility at x_0 with arguments (g, h, k_x) expresses $R_{g,h}^x$ through basepoint values. Rewrite $hk_x = k_{hx} t_k(h, x)$ and $gk_{hx} = k_{ghx} t_k(g, hx)$ and use compatibility at x_0 with arguments $(g, k_{hx}, t_k(h, x))$ and $(k_{ghx}, t_k(g, hx), t_k(h, x))$. The three equations combine into $R_{g,h}^x = \gamma_{gh}^x (\gamma_g^{hx} \gamma_h^x)^{-1} \psi(t_k(g, hx), t_k(h, x))$. Compatibility at x_0 with arguments in H is the cocycle equation for ψ .

For the third item, if L and L_0 are compatible with ω , then LL_0^{-1} is compatible with the trivial 3-cochain, and the first item applies; conversely $L_0 L[\psi]$ is compatible with ω , and an action gauge with trivial fusion part preserves compatibility and commutes with multiplication by L_0 . Cancelling L_0 reduces the freeness statement to (19.468). The second item is the third with L_0 the formula at $\psi = 1$, which is compatible by Theorem 19.5.10.87. For the last item, cocycles modulo cohomology form the second group cohomology of H with coefficients in $\mathbb{C}^\times$, and

$$\begin{aligned} [\psi_1] + [\psi_2] &= [\psi_1 \psi_2], \\ L[\psi_1 \psi_2] &= L[\psi_1] L[\psi_2]. \end{aligned}$$

Cohomologous cocycles give induced L -symbols related by an action gauge, so the rule is well defined, and by the two identities acting by $[\psi_1] + [\psi_2]$ is acting by $[\psi_2]$ and then by $[\psi_1]$. With base point L the action is the bijection of the third item. □

Definition 19.5.10.89 (Generalized coboundary and cocycle). Lean Lean Lean Lean Lean Lean

✓ Stmt Let $\chi: G \times \mathbf{X} \rightarrow \mathbb{C}^\times$ be block-dependent. Its generalized coboundary is

$$(d\chi)_{g,h}^x = \frac{\chi_g^{hx} \chi_h^x}{\chi_{gh}^x}. \quad (19.479)$$

For a block-dependent scalar 2-cochain $f: \mathbf{X} \times G^2 \rightarrow \mathbb{C}^\times$, set

$$(df)_{g,h,k}^x = \frac{f_{g,hk}^x f_{h,k}^x}{f_{g,h}^{kx} f_{gh,k}^x}. \quad (19.480)$$

The final group index in the denominator is k ; this local correction of the printed formula is recorded in [62]. For a block-dependent scalar 3-cochain $\omega: \mathbf{X} \times G^3 \rightarrow \mathbb{C}^\times$, the corresponding next coboundary is

$$(d\omega)_{g,h,k,l}^x = \frac{\omega_{g,h,k}^{lx} \omega_{g,hk,l}^x \omega_{h,k,l}^x}{\omega_{gh,k,l}^x \omega_{g,h,kl}^x}. \quad (19.481)$$

A function $\Omega: \mathsf{X} \times G^2 \rightarrow \mathbb{C}^\times$ is a generalized 2-cocycle when

$$\Omega_{g,hk}^x \Omega_{h,k}^x = \Omega_{g,h}^{kx} \Omega_{gh,k}^x. \quad (19.482)$$

The differential identities $d\Omega = 1$ and $d(df) = 1$ follow the group cohomology conventions of [163, lines 5914–5922], with the three-to-four-cochain coboundary given explicitly by (19.481). No transitivity or finiteness assumption is imposed on X .

Theorem 19.5.10.90 (Generalized coboundaries square to one). [Lean](#) [Lean](#) [✓ Stmt](#) *The generalized coboundary of every block-dependent scalar 1-cochain is a generalized 2-cocycle. For every block-dependent scalar 2-cochain f ,*

$$d(df) = 1. \quad (19.483)$$

This is the identity stated in [163, line 5922], with the index correction in [62].

Proof. [✓ Proof](#) Substitute the coboundary formulas. The action law $k(lx) = (kl)x$ and associativity identify the shifted block labels and group products. Every value of the original cochain then occurs once in the numerator and once in the denominator, so all factors cancel. $\square$

Definition 19.5.10.91 (Positive block-dependent cochains). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\iota: \mathbb{R}_{>0} \rightarrow \mathbb{C}^\times$ be the natural multiplicative embedding. A block-dependent scalar 1-cochain χ is positive when

$$\chi_g^x \in \iota(\mathbb{R}_{>0}) \quad (19.484)$$

for every $x \in \mathsf{X}$ and $g \in G$. A block-dependent scalar 2-cochain Ω is positive when

$$\Omega_{g,h}^x \in \iota(\mathbb{R}_{>0}) \quad (19.485)$$

for every $x \in \mathsf{X}$ and $g, h \in G$. Thus positivity means membership in the embedded subgroup of strictly positive real numbers, not positivity of the real part.

Definition 19.5.10.92 (Hat operation on block-dependent cochains). [Lean](#) [Lean](#) [✓ Stmt](#) For block-dependent scalar cochains, define

$$\hat{\chi}_g^x = \chi_{g^{-1}}^{gx}, \quad (19.486)$$

and

$$\hat{\Omega}_{g,h}^x = \Omega_{h^{-1},g^{-1}}^{(gh)x}. \quad (19.487)$$

These are the one-index and two-index cases of the hat operation in [163, line 5912].

Theorem 19.5.10.93 (Hat involution and generalized coboundaries). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The block-dependent hat operation is an involution:*

$$\hat{\hat{\chi}} = \chi, \quad (19.488)$$

and

$$\hat{\hat{\Omega}} = \Omega. \quad (19.489)$$

It commutes with the generalized coboundary:

$$\widehat{d\chi} = d\hat{\chi}. \quad (19.490)$$

Proof. ✓ Proof Applying (19.486) twice and cancelling each group element with its inverse gives (19.488), and applying (19.487) twice gives (19.489). For (19.490), the action law $g(hx) = (gh)x$ and $g^{-1}((gh)x) = hx$ give

$$(d\hat{\chi})_{g,h}^x = \frac{\hat{\chi}_g^{hx} \hat{\chi}_h^x}{\hat{\chi}_{gh}^x} = \frac{\chi_{g^{-1}}^{(gh)x} \chi_{h^{-1}}^{hx}}{\chi_{(gh)^{-1}}^{(gh)x}} = (d\chi)_{h^{-1},g^{-1}}^{(gh)x} = \widehat{d\chi}_{g,h}^x. \quad (19.491)$$

The two sides therefore carry the same three values of χ , reordered in $\mathbb{C}^\times$. □

Definition 19.5.10.94 (Regular-action matrices). Lean ✓ Stmt Let $\Omega: \mathbf{X} \times G^2 \rightarrow \mathbb{C}^\times$ be an L -symbol. On the group algebra $\mathbb{C}G$, define X_g^x by

$$X_g^x |k\rangle = \Omega_{g,k}^{k^{-1}x} |gk\rangle. \quad (19.492)$$

This is the regular-action matrix in [163, Lemma lemma:G_cocycle].

Theorem 19.5.10.95 (Generalized projective multiplication law). Lean ✓ Stmt Let G be finite. If Ω satisfies (19.482), then

$$X_g^{hx} X_h^x = \Omega_{g,h}^x X_{gh}^x. \quad (19.493)$$

Proof. ✓ Proof Apply the left side to $|k\rangle$. Its coefficient is

$$\Omega_{g,hk}^{(hk)^{-1}hx} \Omega_{h,k}^{k^{-1}x} = \Omega_{g,hk}^{k^{-1}x} \Omega_{h,k}^{k^{-1}x} \quad (19.494)$$

$$= \Omega_{g,h}^x \Omega_{gh,k}^{k^{-1}x}, \quad (19.495)$$

where the last equality is (19.482) at the block $k^{-1}x$. Formula (19.492) now gives (19.493). □

Theorem 19.5.10.96 (Finite-group power trivialization). Lean Lean Lean Lean Lean ✓ Stmt Let G be finite and let Ω be a generalized 2-cocycle. Then $\Omega^{|G|}$ is a generalized coboundary. More precisely, with

$$\chi_g^x = \det X_g^x, \quad (19.496)$$

one has

$$\left(\Omega_{g,h}^x\right)^{|G|} = \frac{\det X_g^{hx} \det X_h^x}{\det X_{gh}^x} = (d\chi)_{g,h}^x. \quad (19.497)$$

This is [163, Lemma lemma:G_cocycle]. The second numerator determinant in (19.497) carries the index h ; this local correction of the printed formula is recorded in [117].

Proof. ✓ Proof Let P_g be the regular-action permutation matrix defined by $P_g |k\rangle = |gk\rangle$, and set $D_g^x = \text{diag}(\Omega_{g,k}^{k^{-1}x})_{k \in G}$. Then

$$X_g^x = P_g D_g^x. \quad (19.498)$$

Since $\det P_g \in \{1, -1\}$ and every diagonal entry of D_g^x is nonzero, $\det X_g^x \neq 0$. Thus (19.496) takes values in $\mathbb{C}^\times$. Taking determinants in (19.493), using multiplicativity of the determinant on the left and $\det(cX) = c^{|G|} \det X$ for the $|G| \times |G|$ matrix X_{gh}^x on the right, gives

$$\det X_g^{hx} \det X_h^x = \left(\Omega_{g,h}^x\right)^{|G|} \det X_{gh}^x. \quad (19.499)$$

Dividing (19.499) by $\det X_{gh}^x$ gives (19.497). □

Definition 19.5.10.99 (Inversion cochains). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) These conventions and inversion cochains follow [163, lines 5910–5916 and 5967–5973]. For scalar cochains with one, two, and three arguments, set

$$(d\sigma)(g, h) = \frac{\sigma(g)\sigma(h)}{\sigma(gh)}. \quad (19.510)$$

The hat reverses the arguments and replaces each group element by its inverse:

$$\hat{\beta}(g, h) = \beta(h^{-1}, g^{-1}). \quad (19.511)$$

$$\hat{\omega}(g, h, k) = \omega(k^{-1}, h^{-1}, g^{-1}). \quad (19.512)$$

Define the inversion cochains by

$$\Xi(g, h) = \frac{\omega(h^{-1}, g^{-1}, g)}{\omega(h^{-1}g^{-1}, g, h)}. \quad (19.513)$$

$$\sigma(g) = \omega(g, g^{-1}, g). \quad (19.514)$$

Theorem 19.5.10.100 (The hat operation is an involution). [Lean](#) [Lean](#) [✓ Stmt](#) Let G be an arbitrary group. Every scalar 2-cochain β and every scalar 3-cochain ω satisfy

$$\hat{\hat{\beta}} = \beta, \quad (19.515)$$

$$\hat{\hat{\omega}} = \omega. \quad (19.516)$$

Proof. [✓ Proof](#) Reversing the arguments of (19.511) and inverting each of them twice restores the original arguments, and likewise for (19.512). $\square$

Theorem 19.5.10.101 (Two inversion-cochain formulas). [Lean](#) [Lean](#) [✓ Stmt](#) These formulas are part of the cocycle calculation in [163, lines 5967–5979]. Let G be an arbitrary group and let ω be a normalized scalar 3-cocycle. Then

$$\Xi(g, h) = \frac{\omega(h^{-1}, g^{-1}, g)}{\omega(h^{-1}g^{-1}, g, h)} = \frac{\omega(h^{-1}, g^{-1}, gh)}{\omega(g^{-1}, g, h)}. \quad (19.517)$$

The inversion one-cochain satisfies

$$\sigma(g^{-1}) = \sigma(g)^{-1}. \quad (19.518)$$

Proof. [✓ Proof](#) Apply (19.389) to (h^{-1}, g^{-1}, g, h) and use (19.388); this gives the second quotient in (19.517). Applying (19.389) to (g, g^{-1}, g, g^{-1}) gives $1 = \sigma(g)\sigma(g^{-1})$, hence (19.518). $\square$

Theorem 19.5.10.102 (Inversion identities for a normalized three-cocycle). [Lean](#) [Lean](#) [✓ Stmt](#) These inversion identities and their cocycle substitutions are given in [163, lines 5967–5979]. Let ω be a normalized scalar 3-cocycle. Then

$$\hat{\omega} = \omega d\Xi. \quad (19.519)$$

The inversion two-cochain also satisfies

$$\Xi = \hat{\Xi} d\sigma. \quad (19.520)$$

Proof. ✓ Proof For (19.519), multiply the cocycle identities at

$$(k^{-1}, h^{-1}, g^{-1}, g), \quad (k^{-1}, h^{-1}g^{-1}, g, h), \quad (k^{-1}h^{-1}g^{-1}, g, h, k), \quad (19.521)$$

and cancel the normalized factors. For (19.520), use the second expression in (19.517) for $\widehat{\Xi}$, then combine the identities at

$$(h^{-1}g^{-1}, g, h, h^{-1}g^{-1}), \quad (h^{-1}, h, h^{-1}, g^{-1}), \quad (h^{-1}, g^{-1}, g, g^{-1}), \quad (g, g^{-1}, g, g^{-1}). \quad (19.522)$$

The last identity at g , h , and gh supplies (19.518) in the three required places. $\square$

Let $\mathbb{Z}_2 = \{e, s\}$ with $s^2 = e$.

Theorem 19.5.10.103 (Normalized L -symbol ratio). Lean ✓ Stmt *Let G act on a set X . Suppose that a normalized L -symbol L is compatible with a scalar 3-cochain ω . Then, for every $x \in X$ and $g \in G$,*

$$\sigma(g) = \frac{L_{g^{-1}, g}^x}{L_{g, g^{-1}}^{gx}}. \quad (19.523)$$

This is the scalar identity in [163, lines 5327–5329].

Proof. ✓ Proof Apply the compatibility relation (19.412) to (g, g^{-1}, g) . The two factors with an identity argument are 1 by normalization, leaving (19.523). $\square$

Theorem 19.5.10.104 (Trivial scalar gauge class for $\mathbb{Z}_2$). Lean ✓ Stmt *Let ω be a normalized scalar 3-cochain on $\mathbb{Z}_2$. Its gauge class is trivial if and only if*

$$\sigma(s) = 1. \quad (19.524)$$

This strengthens the scalar step in [163, lines 5339–5341]: only normalization, rather than the cocycle identity, is needed for this equivalence.

Proof. ✓ Proof Suppose first that $\omega = d\beta$ for an arbitrary scalar 2-cochain β . Although β need not be normalized, the normalization of $d\beta$ implies that there is a constant $c \in \mathbb{C}^\times$ such that $\beta(e, h) = \beta(h, e) = c$ for every $h \in \mathbb{Z}_2$. Since $s^2 = e$,

$$(d\beta)(s, s, s) = \frac{\beta(s, e)\beta(s, s)}{\beta(s, s)\beta(e, s)} = \frac{c}{c} = 1. \quad (19.525)$$

Thus every scalar 2-cochain whose coboundary is normalized contributes the factor 1 at (s, s, s) . In particular, an arbitrary gauge between normalized scalar 3-cochains cannot change this value, and $\sigma(s) = 1$. Conversely, normalization determines every value of ω except $\omega(s, s, s) = \sigma(s)$. Hence (19.524) makes ω the constant cochain 1. $\square$

Theorem 19.5.10.105 (Sign dichotomy for the $\mathbb{Z}_2$ inversion scalar). Lean ✓ Stmt *Let ω be a normalized scalar 3-cocycle on $\mathbb{Z}_2$. Then*

$$\sigma(s) \in \{1, -1\}. \quad (19.526)$$

Proof. ✓ Proof Since $s^{-1} = s$, formula (19.518) gives $\sigma(s) = \sigma(s)^{-1}$. Hence $\sigma(s)^2 = 1$, whose two solutions in $\mathbb{C}^\times$ are 1 and -1 . $\square$

For an element g whose square is the identity, the source condition $\xi_g(x)\xi_{g^{-1}}(g \cdot x) = -1$ reduces to $\xi_g(x)\xi_g(g \cdot x) = -1$ because $g^{-1} = g$.

Definition 19.5.10.106 (Anomalous sign for an involution). [Lean](#) [Lean](#) [✓ Stmt](#) Let $f: X \rightarrow X$. An anomalous sign for f is a function $\xi: X \rightarrow \mathbb{C}^\times$ such that

$$\xi(x) \in \{1, -1\}, \quad (19.527)$$

and

$$\xi(x)\xi(f(x)) = -1 \quad (19.528)$$

for every $x \in X$.

Theorem 19.5.10.107 (Anomalous signs and fixed points). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $f: X \rightarrow X$ be an involution. An anomalous sign exists if and only if f has no fixed point. In particular, let a group act on X from the left, and suppose that $g^2 = e$. The action $x \mapsto g \cdot x$ admits an anomalous sign if and only if $g \cdot x \neq x$ for every $x \in X$. This is the sign condition used for anomalous defect tensors in [163, lines 2979–2981].

Proof. [✓ Proof](#) Choose a well-order on X . In each two-element orbit $\{x, f(x)\}$, assign 1 to the smaller element and -1 to the larger. Involutivity exchanges the two elements, so their signs multiply to -1 . Conversely, at a fixed point the product is the square of either 1 or -1 , hence equals 1 rather than -1 . For the action statement, $g \cdot (g \cdot x) = (g^2) \cdot x = x$. $\square$

Theorem 19.5.10.108 (Nontrivial scalar gauge class for $\mathbb{Z}_2$). [Lean](#) [✓ Stmt](#) Let ω be a normalized scalar 3-cocycle on $\mathbb{Z}_2$. Its gauge class is nontrivial if and only if

$$\sigma(s) = -1. \quad (19.529)$$

This is the scalar cohomology classification used in the proof at [163, lines 5339–5341].

Proof. [✓ Proof](#) By Theorem 19.5.10.104, a nontrivial class has $\sigma(s) \neq 1$. The dichotomy (19.526) then forces $\sigma(s) = -1$, and the converse follows from the same two statements. $\square$

This completes the classification of the normalized scalar 3-cocycle by $\sigma(s)$ in the setting of [163, lines 5207–5338].

19.6 The GHZ–blocked-cluster tensor

Definition 19.6.1 (GHZ–blocked-cluster tensor and its two blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7699](#) Let $C = \{C^q\}_{q=0}^3$ be the once-blocked cluster-state tensor. The tensor obtained by taking its independent tensor product with the GHZ tensor has physical dimension 8 and bond dimension 4:

$$A = A^{\text{GHZ}} \boxtimes C. \quad (19.530)$$

For $x \in \{0, 1\}$, let E_x be the one-dimensional GHZ sector supported at the physical value x . The two bond-dimension-2 tensors

$$A_x = E_x \boxtimes C \quad (19.531)$$

satisfy, for $p, x \in \{0, 1\}$ and $q \in \{0, 1, 2, 3\}$,

$$A_x^{(p,q)} = \delta_{p,x} C^q. \quad (19.532)$$

Their canonical direct sum is denoted $A_0 \oplus A_1$. This is the tensor and block formula in [163, lines 4427–4479].

Theorem 19.6.2 (Injective and locally orthogonal GHZ–cluster blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) [Discussion: #7699](#) Each tensor A_x is injective. For $x \neq y$ and every $X \in \text{Mat}_2(\mathbb{C})$, their one-site mixed transfer map satisfies

$$\mathcal{E}_{x,y}(X) = \sum_{p=0}^1 \sum_{q=0}^3 A_x^{(p,q)} X(A_y^{(p,q)})^\dagger = 0. \quad (19.533)$$

Thus the two blocks are locally orthogonal, and

$$A_0 \oplus A_1 = A \quad (19.534)$$

after the canonical identification of the direct-sum and product bond coordinates. The local orthogonality is the observation at [163, line 4481].

Proof. [Proof](#) The one-dimensional sector E_x is injective, and injectivity is preserved by independent tensor products, so $A_x = E_x \boxtimes C$ is injective. If $x \neq y$, formula (19.532) makes every summand in (19.533) vanish: a physical value p cannot equal both x and y . For $p, x, y \in \{0, 1\}$, $q \in \{0, 1, 2, 3\}$, and $a, b \in \{0, 1\}$, the direct-sum and product tensors have the same entry in their canonical bond coordinates:

$$(A_0 \oplus A_1)_{(x,a),(y,b)}^{(p,q)} = A_{(x,a),(y,b)}^{(p,q)} = \delta_{x,y} \delta_{p,x} C_{a,b}^q. \quad (19.535)$$

Equality of all entries proves (19.534). $\square$

Theorem 19.6.3 (GHZ–cluster physical representation and virtual action table). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) [Discussion: #7747](#) Write the physical basis as $|p, s, r\rangle$, with integer index $4p + 2s + r$ for $p, s, r \in \{0, 1\}$. Thus the blocked cluster index is $q = 2s + r$. The operators $U_{(1,0)} = (X_1 \otimes X_2) \text{CNOT}_{1 \rightarrow 2}$ and $U_{(0,1)} = X_3$ generate a unitary representation of $G = \mathbb{Z}_4 \times \mathbb{Z}_2$, with $U_{(2,0)} = X_2$. Put $g \cdot x = x + a \pmod{2}$ for $g = (a, b)$ and $x \in \{0, 1\}$. Let $V_{g,x}$ be given by the following table, where $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$:

g	(0, 0)	(2, 0)	(0, 1)	(2, 1)	(1, 0)	(3, 0)	(1, 1)	(3, 1)
$x = 0$	$\mathbb{1}$	X	Z	Y	X	$\mathbb{1}$	Y	Z
$x = 1$	$\mathbb{1}$	X	Z	Y	$\mathbb{1}$	X	Z	Y

Every $V_{g,x}$ is Hermitian and squares to the identity. For every physical index i , the blocks satisfy

$$\sum_j (U_g)_{i,j} A_x^j = V_{g,x} A_{g \cdot x}^i V_{g,x}^\dagger. \quad (19.536)$$

On the subgroup $H = \{(2a, b) : a, b \in \{0, 1\}\}$, the comparison with the cluster virtual representatives $W_{(b,a)} = Z^b X^a$ is $V_{(2a,b),x} = (-i)^{ab} W_{(b,a)}$. These are the physical representation and matrix action identities of [163, lines 4481–4489]. No claim is made about defect normalization, L -symbols, cocycle nonextension, block independence, or gauging.

Proof. ✓ **Proof** On basis vectors the generators send (p, s, r) to $(p+1, s+p+1, r)$ and $(p, s, r+1)$, with coordinates added modulo two. Their permutations have orders four and two and commute. The corresponding permutation matrices are unitary and preserve multiplication. Substitute $A_x^{(p,q)} = \delta_{p,x} C^q$ into (19.536). For each of the eight group elements and two block labels, matrix multiplication gives the displayed table. Hermiticity and $V_{g,x}^2 = \mathbb{1}$ turn forward transport of the physical index into the stated conjugation identity. Finally, $ZX = iY$, so the only phase needed to compare the restriction with $Z^b X^a$ is $-i$ when $a = b = 1$. $\square$

19.7 The gauge-invariant subspace of the CZX circuit tuple

The CZX model of [163, lines 4503–5183] is the anomalous $\mathbb{Z}_2 = \{e, g\}$ matrix product unitary acting on the GHZ state. After blocking, one site carries two matter qubits, and the fusion operators on two neighboring blocked sites are $\lambda_{e,e}^R = \lambda_{e,g}^R = \mathbb{1}$, $\lambda_{g,e}^R = w_R$, together with the modified fusion operator $\tilde{\lambda}_{g,g}^R = \lambda_{g,g}^R Z_1$ introduced at the end of that passage. For the displayed circuit tuple $(\mathbb{1}, \mathbb{1}, w, \tilde{\lambda})$, the common $+1$ eigenspace of the resulting Gauss projectors has dimension 2^N on $N \geq 3$ blocked sites, answering the scaling question of [163, lines 5198–5204] for this tuple only. The displayed tuple is not identified with a member of the full physical completion class of the CZX defect data; that identification requires the three defect domains and maps, which the displayed matrices do not supply.

19.7.1 Monomial operators and their common fixed space

Definition 19.7.1.1 (Monomial matrix). Lean Lean Lean Lean ✓ **Stmt** Let σ be a permutation of a finite index set S and let $\varphi: S \rightarrow \mathbb{C}$. The monomial matrix $M(\sigma, \varphi)$ is the operator on $\mathbb{C}^S$ with

$$M(\sigma, \varphi)|s\rangle = \varphi(s)|\sigma(s)\rangle. \quad (19.537)$$

Its entry in row $\sigma(s)$ and column s is $\varphi(s)$, and every other entry vanishes.

Theorem 19.7.1.2 (Calculus of monomial matrices). Lean Lean Lean Lean Lean Lean Lean Lean ✓ **Stmt** A vector $v \in \mathbb{C}^S$ is fixed by $M(\sigma, \varphi)$ if and only if $v_{\sigma(s)} = \varphi(s)v_s$ for every $s \in S$. Products, adjoints, scalar multiples, negatives, and simultaneous relabellings of monomial matrices are monomial:

$$M(\sigma, \varphi)M(\tau, \psi) = M(\sigma\tau, s \mapsto \varphi(\tau(s))\psi(s)), \quad (19.538)$$

$$M(\sigma, \varphi)^\dagger = M(\sigma^{-1}, t \mapsto \overline{\varphi(\sigma^{-1}(t))}). \quad (19.539)$$

The identity is $M(\text{id}, 1)$, and $M(\sigma, \varphi)$ is unitary when $|\varphi(s)| = 1$ for every s . The trace is $\text{tr } M(\sigma, \varphi) = \sum_{s: \sigma(s)=s} \varphi(s)$, which vanishes when σ has no fixed point.

Proof. ✓ **Proof** The action of $M(\sigma, \varphi)$ on a vector is $(M(\sigma, \varphi)v)_t = \varphi(\sigma^{-1}(t))v_{\sigma^{-1}(t)}$, which gives the fixed-vector criterion after the substitution $t = \sigma(s)$. The remaining identities are entrywise: in a product only the column $\tau(s)$ of the first factor meets the row $\tau(s)$ of the second, the adjoint transposes the single nonzero entry of each column, and a relabelling conjugates the permutation. For unitarity, (19.538) and (19.539) give $M(\sigma, \varphi)^\dagger M(\sigma, \varphi) = M(\text{id}, |\varphi|^2) = \mathbb{1}$. The diagonal entry at s is $\varphi(s)$ if $\sigma(s) = s$ and zero otherwise, which gives the trace. $\square$

Theorem 19.7.1.3 (Flat phases on the hypercube have a potential). [Lean](#) [Lean](#) [Stmnt](#) Let M be a commutative group and let $\varphi_j: \mathbb{F}_2^r \rightarrow M$ for $0 \leq j < r$ be phases attached to the edges $(\gamma, \gamma + e_j)$ of the hypercube. Assume that

$$\varphi_j(\gamma)\varphi_j(\gamma + e_j) = 1, \quad (19.540)$$

$$\varphi_j(\gamma)\varphi_k(\gamma + e_j) = \varphi_k(\gamma)\varphi_j(\gamma + e_k) \quad (19.541)$$

for all j, k and γ . Then there is $\Phi: \mathbb{F}_2^r \rightarrow M$ with $\Phi(0) = 1$ and $\Phi(\gamma + e_j) = \varphi_j(\gamma)\Phi(\gamma)$ for every j and γ . In addition, a property of vectors of $\mathbb{F}_2^r$ that holds at 0 and is preserved by every coordinate flip holds everywhere.

Proof. [Stmnt](#) [Proof](#) Connectivity: every γ is the sum of the e_j over its support, so induction on the support carries the property from 0 to γ .

The potential is constructed by induction on r . Write $\gamma = (a, \gamma')$ with $a \in \mathbb{F}_2$ and $\gamma' \in \mathbb{F}_2^{r-1}$. The phases $\gamma' \mapsto \varphi_{j+1}(0, \gamma')$ for $0 \leq j < r - 1$ satisfy (19.540) and (19.541) on $\mathbb{F}_2^{r-1}$, so the induction hypothesis gives a potential Φ' for them. Define

$$\Phi(0, \gamma') = \Phi'(\gamma'), \quad \Phi(1, \gamma') = \varphi_0(0, \gamma')\Phi'(\gamma'). \quad (19.542)$$

The edge relation in direction 0 at $(0, \gamma')$ is the definition, and at $(1, \gamma')$ it follows from (19.540). The edge relation in a direction $j + 1$ at $(0, \gamma')$ is the relation for Φ' , and at $(1, \gamma')$ it is the relation for Φ' combined with (19.541) for the square spanned by the directions 0 and $j + 1$ at $(0, \gamma')$. $\square$

Definition 19.7.1.4 (Trivial holonomy on a fiber). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let X be a finite set and consider the basis $X \times \mathbb{F}_2^r$. For $0 \leq j < r$ let σ_j be the coordinate flip $(x, \gamma) \mapsto (x, \gamma + e_j)$ and let $\varphi_j: X \times \mathbb{F}_2^r \rightarrow \mathbb{C}$. The phases have trivial holonomy on the fiber over $x \in X$ when

$$\varphi_j(x, \gamma)\varphi_k(x, \gamma + e_j) = \varphi_k(x, \gamma)\varphi_j(x, \gamma + e_k) \quad (19.543)$$

for all j, k and $\gamma \in \mathbb{F}_2^r$.

Theorem 19.7.1.5 (Monomial fixed-space criterion). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) In the setting of Definition 19.7.1.4, let $T_j = M(\sigma_j, \varphi_j)$ and assume that every T_j is an involution, that is, $\varphi_j(s)\varphi_j(\sigma_j(s)) = 1$ for every basis element s . Then

$$\dim \mathcal{V}(T) = \#\{x \in X : \text{the phases have trivial holonomy on the fiber over } x\}. \quad (19.544)$$

The dimension of a common fixed subspace is also unchanged by a simultaneous relabelling of the basis.

Proof. [Stmnt](#) [Proof](#) By Theorem 19.7.1.2, a vector v lies in $\mathcal{V}(T)$ exactly when $v_{(x, \gamma + e_j)} = \varphi_j(x, \gamma)v_{(x, \gamma)}$ for all j, x , and γ . Let X_0 be the set of fibers with trivial holonomy and consider the linear map $E: \mathcal{V}(T) \rightarrow \mathbb{C}^{X_0}$, $v \mapsto (v_{(x, 0)})_{x \in X_0}$.

E is injective. If $v \in \mathcal{V}(T)$ vanishes at one point of a fiber, the transport relation and the connectivity statement of Theorem 19.7.1.3 make it vanish on the whole fiber. On a fiber with trivial holonomy, v therefore vanishes as soon as $v_{(x, 0)} = 0$. On a fiber where (19.543) fails at some (j, k, γ_0) , the two paths from (x, γ_0) to $(x, \gamma_0 + e_j + e_k)$ give

$$v_{(x, \gamma_0 + e_j + e_k)} = \varphi_j(x, \gamma_0)\varphi_k(x, \gamma_0 + e_j)v_{(x, \gamma_0)}, \quad (19.545)$$

$$v_{(x, \gamma_0 + e_j + e_k)} = \varphi_k(x, \gamma_0)\varphi_j(x, \gamma_0 + e_k)v_{(x, \gamma_0)}. \quad (19.546)$$

The two phase products differ, so $v_{(x,\gamma_0)} = 0$ and again v vanishes on the fiber.

E is surjective. Given $w \in \mathbb{C}^{X_0}$, take on each fiber with trivial holonomy the potential Φ_x of Theorem 19.7.1.3 for the invertible phases $\gamma \mapsto \varphi_j(x, \gamma)$, and set $v_{(x,\gamma)} = w_x \Phi_x(\gamma)$ there and $v = 0$ on the other fibers. The potential relation is the transport relation, so $v \in \mathcal{V}(T)$, and $\Phi_x(0) = 1$ gives $Ev = w$.

Hence $\dim \mathcal{V}(T) = \dim \mathbb{C}^{X_0} = \#X_0$. For the relabelling statement, a relabelled matrix acts by relabelling the vector, acting, and relabelling back, so the common fixed subspace is the image of the original one under the coordinate relabelling. $\square$

Theorem 19.7.1.6 (Local Gauss operator of monomial matter unitaries). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Suppose that every matter unitary of the tuple R is monomial, $R_{a,b} = M(\sigma_{a,b}, \varphi_{a,b})$. Then the local Gauss operator of Definition 19.5.10.8 is the monomial operator

$$\mathcal{G}_g|i, (a, b)\rangle = \overline{\varphi_{T_g(a,b)}(\sigma_{T_g(a,b)}^{-1}\sigma_{a,b}(i))} \varphi_{a,b}(i) |\sigma_{T_g(a,b)}^{-1}\sigma_{a,b}(i), T_g(a, b)\rangle. \quad (19.547)$$

Proof. [Stmnt](#) The block of $\mathcal{G}_g$ from (a, b) to $T_g(a, b)$ is $R_{T_g(a,b)}^\dagger R_{a,b}$, which is monomial by (19.538) and (19.539); all other blocks vanish. $\square$

Theorem 19.7.1.7 (Periodic placement of monomial operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $M(\sigma, \varphi)$ be a monomial operator on L consecutive sites. Its placement on the periodic window beginning at site i of an N -site chain is the monomial operator whose permutation replaces the window of a configuration s by σ applied to that window and whose phase is φ evaluated on the window of s . For $L = 2$ and $N \geq 2$ the window at j is (s_j, s_{j+1}) , and replacing it changes exactly the sites j and $j + 1$.

Proof. [Stmnt](#) The placed operator has entry (t, s) equal to the window entry $M(\sigma, \varphi)_{t_W, s_W}$ when t and s agree outside the window and 0 otherwise. Both conditions together say that t is s with its window replaced by $\sigma(s_W)$, and the entry is then $\varphi(s_W)$. The two-site statements are the cyclic offset computations $(j + N - j) \bmod N = 0$ and $(j + 1 + N - j) \bmod N = 1$. $\square$

19.7.2 The exact CZX matrix product unitary

Definition 19.7.2.1 (The once-blocked CZX tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Write a physical index as $2a + b$, with $a, b \in \{0, 1\}$, and let $(a, b) = (1 - a, 1 - b)$. The tensor in [163, lines 4503–4547] is the two-site block $B^{ab,cd} = C^{a,c}C^{b,d}$ of the bond-two tensor

$$C_{\ell r}^{u,d} = \delta_{u,1-d} \delta_{\ell,u} (-1)^{u+\ell r}. \quad (19.548)$$

Here the constituent indices u, d, ℓ, r are binary, the weight is $W_{\ell r} = (-1)^{\ell r} = \sqrt{2}H_{\ell r}$, where H is the normalized Hadamard matrix, and the factor $(-1)^u$ is the output Pauli Z . Thus, if P denotes the normalized tensor printed in [15, lines 1906–1943], then $B = (Z \otimes Z) \text{block}_2(\sqrt{2}P)$, not $\text{block}_2(P)$. Define the local integer exponent $e((a, b), r) = a + b + ab + br$. For a periodic configuration $s_j = (a_j, b_j)$ of positive length N , let $\kappa_N(s)_j = \overline{s_j}$ and set

$$E(s) = \sum_{j=0}^{N-1} e(s_j, a_{j+1}) = \sum_{j=0}^{N-1} (a_j + b_j + a_j b_j + b_j a_{j+1}), \quad (19.549)$$

where $j + 1$ is interpreted modulo N and the bit values in the sum are integers.

Theorem 19.7.2.2 (CZX tensor as general length-two blocking). [Lean](#) [✓ Stmt](#) Let $\varphi : \{0, 1, 2, 3\} \rightarrow \{0, 1\}^2$ be the pair-index bijection $\varphi(2a + b) = (a, b)$, regarded as a word of length two. Write $C^{[L]}$ for the general physical blocking of C from Definition 20.15.1. For physical indices i, j ,

$$B^{i,j} = (C^{[L]})^{\varphi(i), \varphi(j)} \big|_{L=2}. \quad (19.550)$$

Proof. [✓ Proof](#) Apply Theorem 20.15.8 to C . For $i = 2a + b$ and $j = 2c + d$, both sides are the same ordered product:

$$(C^{[L]})^{\varphi(i), \varphi(j)} \big|_{L=2} = C^{a,c} C^{b,d} = B^{ab,cd}. \quad (19.551)$$

□

Theorem 19.7.2.3 (Coordinates and unitarity of the exact CZX tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The tensor B has bond dimension two and coordinates

$$B_{\ell r}^{ab,cd} = \delta_{a,1-c} \delta_{b,1-d} \delta_{\ell,a} (-1)^{a+b+ab+br}. \quad (19.552)$$

Its periodic operator is unitary for every $N > 0$, with entries and monomial form

$$O_N(B)_{s,t} = \delta_{s, \kappa_N(t)} (-1)^{E(s)}, \quad (19.553)$$

$$O_N(B) = M(\kappa_N, t \mapsto (-1)^{E(\kappa_N(t))}). \quad (19.554)$$

In particular, B is a matrix product unitary. This assertion does not include its phase classification, GHZ action, or defect data.

Proof. [✓ Proof](#) The two-site block is the matrix product contracted over a binary intermediate bond m . The second factor vanishes unless $m = b$, so

$$B_{\ell r}^{ab,cd} = \sum_{m \in \{0,1\}} C_{\ell m}^{a,c} C_{mr}^{b,d} \quad (19.555)$$

$$= C_{\ell b}^{a,c} C_{br}^{b,d} \quad (19.556)$$

$$= \delta_{a,1-c} \delta_{b,1-d} \delta_{\ell,a} (-1)^{a+\ell b+b+br} \quad (19.557)$$

$$= \delta_{a,1-c} \delta_{b,1-d} \delta_{\ell,a} (-1)^{a+b+ab+br}. \quad (19.558)$$

For the periodic contraction, write $s_j = (a_j, b_j)$ and $t_j = (c_j, d_j)$. The cyclic trace expansion is

$$O_N(B)_{s,t} = \sum_{g: \mathbb{Z}/N\mathbb{Z} \rightarrow \{0,1\}} \prod_{j=0}^{N-1} B_{g_j, g_{j+1}}^{s_j, t_j}. \quad (19.559)$$

The factor at j vanishes unless $g_j = a_j$, so only the path $g_j = a_j$ can contribute. Substituting it gives

$$O_N(B)_{s,t} = \prod_{j=0}^{N-1} B_{a_j, a_{j+1}}^{s_j, t_j} \quad (19.560)$$

$$= \prod_{j=0}^{N-1} \delta_{a_j, 1-c_j} \delta_{b_j, 1-d_j} (-1)^{a_j+b_j+a_j b_j+b_j a_{j+1}} \quad (19.561)$$

$$= \delta_{s, \kappa_N(t)} (-1)^{E(s)}. \quad (19.562)$$

Thus the operator is the monomial matrix for complementation, with column phase $(-1)^{E(\kappa_N(t))}$. Each phase has absolute value one, so Theorem 19.7.1.2 proves unitarity. □

Theorem 19.7.2.4 (Injectivity of the once-blocked CZX tensor). [Lean](#) [✓ Stmt](#) [Discussion: #7354](#)
The once-blocked CZX tensor B of Definition 19.7.2.1, with the unnormalized Hadamard weights and both output Z gates, is injective:

$$\text{span}_{\mathbb{C}}\{B^{i,j} : 0 \leq i, j < 4\} = \text{Mat}_2(\mathbb{C}). \quad (19.563)$$

Here $B^{2a+b, 2c+d} = B^{ab, cd}$ for binary a, b, c, d . This corresponds to the injectivity assertion in [163, lines 4505–4547].

Proof. [✓ Proof](#) Let E_{ab} be the bond matrix units, with $a, b \in \{0, 1\}$. The coordinate formula in Theorem 19.7.2.3 gives

$$E_{00} = \frac{1}{2}(B^{0,3} - B^{1,2}), \quad (19.564)$$

$$E_{01} = \frac{1}{2}(B^{0,3} + B^{1,2}), \quad (19.565)$$

$$E_{10} = -\frac{1}{2}(B^{2,1} + B^{3,0}), \quad (19.566)$$

$$E_{11} = \frac{1}{2}(B^{3,0} - B^{2,1}). \quad (19.567)$$

Thus every bond matrix unit lies in the span of the physical letters. Since every 2×2 matrix is a linear combination of these units, the letters span the full bond matrix algebra. $\square$

19.7.3 The explicit CZX adjoint gauge

Definition 19.7.3.1 (The Pauli adjoint gauge). [Lean](#) [✓ Stmt](#) [Discussion: #7354](#) On the bond space $\mathbb{C}^2$, with basis indexed by $0, 1$, define the Pauli matrix

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \quad (19.568)$$

This is the matrix on both virtual legs in the adjoint diagram of [163, lines 4547–4559].

Theorem 19.7.3.2 (Elementary identities for the Pauli adjoint gauge). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
The matrix Y of Definition 19.7.3.1 satisfies

$$Y^\dagger = Y, \quad (19.569)$$

$$Y^2 = I_2. \quad (19.570)$$

In particular, $Y \in \text{U}(2)$. These identities give the unitary gauge used in [163, lines 4547–4559].

Proof. [✓ Proof](#) Conjugate-transpose the displayed matrix and use the Pauli multiplication rule $Y^2 = I_2$. Thus $YY^\dagger = I_2$. $\square$

Theorem 19.7.3.3 (The physical adjoint of the exact CZX tensor). [Lean](#) [✓ Stmt](#) [Discussion: #7354](#)
Let B be the exact once-blocked CZX tensor of Definition 19.7.2.1. For $0 \leq u, v < 4$,

$$(B^\dagger)^{u,v} = Y B^{u,v} Y. \quad (19.571)$$

Here $(B^\dagger)_{\ell_r}^{u,v} = \overline{B_{\ell_r}^{v,u}}$; the virtual indices are not transposed. This is the adjoint diagram in [163, lines 4547–4559], with no scalar prefactor.

Proof. [✓ Proof](#) Insert the coordinate formula from Theorem 19.7.2.3 on both sides and expand the two matrix products over the binary bond indices. Substituting the four physical values and the two bond values verifies each entry. $\square$

Theorem 19.7.3.4 (An explicit unitary adjoint gauge for CZX). [Lean](#) [Lean](#) [✓ Stmt](#) For the exact once-blocked CZX tensor B , the matrix $T = Y$, regarded as an element of $U(2)$, satisfies, for $0 \leq u, v < 4$,

$$(B^\dagger)^{u,v} = T^\dagger B^{u,v} T. \quad (19.572)$$

This is the unitary gauge convention of [163, lines 1552–1557 and 4547–4559].

Proof. [✓ Proof](#) Theorem 19.7.3.2 gives $Y \in U(2)$ and $Y^\dagger = Y$. Substitute into Theorem 19.7.3.3. $\square$

Theorem 19.7.3.5 (The CZX adjoint-gauge sign). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7354](#) For the Pauli adjoint gauge Y , entrywise complex conjugation gives

$$\bar{Y} = -Y, \quad (19.573)$$

$$Y\bar{Y} = -I_2. \quad (19.574)$$

Hence the sign in $Y\bar{Y} = \sigma_g I_2$ is $\sigma_g = -1$, as stated in [163, lines 4547–4559].

Proof. [✓ Proof](#) Conjugating each entry of the displayed Pauli matrix gives $\bar{Y} = -Y$. Then $Y\bar{Y} = -Y^2 = -I_2$ by Theorem 19.7.3.2. $\square$

19.7.4 The displayed CZX source factors

Definition 19.7.4.1 (Four explicit CZX source factors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7354](#) For $0 \leq i < 4$, write $i = 2a(i) + b(i)$ with $a(i), b(i) \in \{0, 1\}$, and put $\bar{i} = 3 - i$. Let Y be the Pauli matrix of Definition 19.7.3.1. Define matrices $X_2 \in \text{Mat}_{(2 \times 4) \times 4}(\mathbb{C})$ and $Y_2 \in \text{Mat}_{4 \times (4 \times 2)}(\mathbb{C})$ by

$$(X_2)_{(\alpha, i), k} = \delta_{i, k} \delta_{\alpha, a(i)} (-1)^{a(i) + b(i)}, \quad (19.575)$$

$$(Y_2)_{k, (j, \beta)} = \delta_{k, \bar{j}} (-1)^{a(k)b(k) + b(k)\beta}, \quad (19.576)$$

where $0 \leq i, j, k < 4$ and $\alpha, \beta \in \{0, 1\}$. Define $X_1 \in \text{Mat}_{(4 \times 2) \times 4}(\mathbb{C})$ and $Y_1 \in \text{Mat}_{4 \times (2 \times 4)}(\mathbb{C})$ by

$$(X_1)_{(i, \beta), k} = \sum_{t=0}^1 Y_{\beta t} \overline{(Y_2)_{k, (i, t)}}, \quad (19.577)$$

$$(Y_1)_{k, (\alpha, j)} = \sum_{t=0}^1 \overline{(X_2)_{(t, j), k}} Y_{t\alpha}. \quad (19.578)$$

Product indices in these matrix spaces retain their displayed order. These are the filled and empty factors in [163, lines 4559–4659]. The weights $(-1)^{ab}$ are unnormalized Hadamard weights, and $(-1)^{a(i) + b(i)}$ includes both output Z gates. In the formula for X_1 , the Pauli entry is $Y_{\beta t}$, not $Y_{t\beta}$.

Theorem 19.7.4.2 (The displayed factors recover both CZX cuts). [Lean](#) [✓ Stmt](#) For the exact once-blocked CZX tensor B , the factors of Definition 19.7.4.1 satisfy

$$X_1 Y_1 = M_1(B), \quad (19.579)$$

$$X_2 Y_2 = M_2(B). \quad (19.580)$$

Here $(M_1(B))_{(i, \beta), (\alpha, j)} = B_{\alpha\beta}^{i,j}$ and $(M_2(B))_{(\alpha, i), (j, \beta)} = B_{\alpha\beta}^{i,j}$. These are the two decompositions in [163, lines 4559–4659].

Proof. ✓ Proof Expand each matrix product over its four-dimensional factor index. Substitute the displayed formulas and the coordinate formula from Theorem 19.7.2.3. Evaluating the binary bond indices and the four physical values proves both entry identities. $\square$

Theorem 19.7.4.3 (Four Gram matrices of the displayed CZX factors). Lean ✓ Stmt *The four explicit factors of Definition 19.7.4.1 satisfy*

$$X_1^\dagger X_1 = 2I_4, \quad (19.581)$$

$$X_2^\dagger X_2 = I_4, \quad (19.582)$$

$$Y_1 Y_1^\dagger = I_4, \quad (19.583)$$

$$Y_2 Y_2^\dagger = 2I_4. \quad (19.584)$$

The factors of two retain the unnormalized Hadamard weights of [163, lines 4559–4659].

Proof. ✓ Proof The empty factors are Pauli-gauge contractions of the filled factors, $X_1 = (I_4 \otimes Y)Y_2^\dagger$ and $Y_1 = X_2^\dagger(Y \otimes I_4)$. By Theorem 19.7.3.2, Y is unitary, hence so are $I_4 \otimes Y$ and $Y \otimes I_4$; composing a unitary matrix with $Y_2^\dagger$, respectively with $X_2^\dagger$, leaves the Gram matrix unchanged, so $X_1^\dagger X_1 = Y_2 Y_2^\dagger$ and $Y_1 Y_1^\dagger = X_2^\dagger X_2$. It remains to expand the two filled Gram matrices $X_2^\dagger X_2$ and $Y_2 Y_2^\dagger$ over their product-index spaces and substitute the factor coordinates; evaluation at each pair of factor indices gives the two displayed scalar identities. $\square$

Definition 19.7.4.4 (Right inverses of the displayed CZX factors). Lean Lean ✓ Stmt For the factors of Definition 19.7.4.1, set

$$Z_1 = Y_1^\dagger, \quad (19.585)$$

$$Z_2 = \frac{1}{2}Y_2^\dagger. \quad (19.586)$$

These are the right inverses associated with the explicit decompositions in [163, lines 4559–4659].

Theorem 19.7.4.5 (Right inverses and weighted normalization). Lean ✓ Stmt *Set $\rho = I_2/2$. The displayed factors and right inverses satisfy*

$$Y_1 Z_1 = I_4, \quad (19.587)$$

$$Y_2 Z_2 = I_4, \quad (19.588)$$

$$X_1^\dagger(I_4 \otimes \rho)X_1 = I_4. \quad (19.589)$$

Proof. ✓ Proof The two inverse identities follow directly from Theorem 19.7.4.3. Since $I_4 \otimes \rho = I_8/2$, the weighted identity is $\frac{1}{2}X_1^\dagger X_1 = I_4$. $\square$

Theorem 19.7.4.6 (Both CZX source-cut ranks are four). Lean ✓ Stmt *For the exact once-blocked CZX tensor B ,*

$$d_r = \text{rank } M_1(B) = 4, \quad (19.590)$$

$$d_l = \text{rank } M_2(B) = 4. \quad (19.591)$$

Thus the four-dimensional legs in [163, lines 4559–4659] have the actual source-cut dimensions.

Proof. ✓ Proof Each cut factors through a four-dimensional space, so its rank is at most four. For the first cut take $L_1 = X_1^\dagger/2$, and for the second take $L_2 = X_2^\dagger$. By (19.581) and (19.582), $L_i X_i = I_4$; by (19.587) and (19.588), $Y_i Z_i = I_4$, hence $L_i(X_i Y_i)Z_i = I_4$. Multiplication cannot increase rank, so each cut has rank at least four. $\square$

Definition 19.7.4.7 (The displayed factors in source-rank coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Stmnt](#) Let d_r, d_l be the two source-cut ranks of the exact CZX tensor B . By Theorem 19.7.4.6, both are four. Let

$$e_r : \{0, \dots, d_r - 1\} \longrightarrow \{0, 1, 2, 3\}, \quad (19.592)$$

$$e_l : \{0, \dots, d_l - 1\} \longrightarrow \{0, 1, 2, 3\} \quad (19.593)$$

be the bijections that preserve each integer index. Pull back the four-dimensional factor index of X_1, Y_1, Z_1 along e_r , and that of X_2, Y_2, Z_2 along e_l . Denote the resulting matrices by $\widehat{X}_i, \widehat{Y}_i, \widehat{Z}_i$. Together with $\rho = I_2/2$, these give source factors for B in the sense of Definition 18.4.50, satisfying both cut factorizations, the two left-factor normalizations, and both right-inverse identities.

Explicitly, for $0 \leq k, i < 4$ and $\alpha, \beta \in \{0, 1\}$, evaluating the pullbacks gives

$$(\widehat{X}_1)_{(i, \alpha), e_r^{-1}(k)} = (X_1)_{(i, \alpha), k}, \quad (19.594)$$

$$(\widehat{Y}_1)_{e_r^{-1}(k), (\alpha, i)} = (Y_1)_{k, (\alpha, i)}, \quad (19.595)$$

$$(\widehat{Z}_1)_{(\alpha, i), e_r^{-1}(k)} = (Z_1)_{(\alpha, i), k}, \quad (19.596)$$

$$(\widehat{X}_2)_{(\alpha, i), e_l^{-1}(k)} = (X_2)_{(\alpha, i), k}, \quad (19.597)$$

$$(\widehat{Y}_2)_{e_l^{-1}(k), (i, \alpha)} = (Y_2)_{k, (i, \alpha)}, \quad (19.598)$$

$$(\widehat{Z}_2)_{(i, \alpha), e_l^{-1}(k)} = (Z_2)_{(i, \alpha), k}, \quad (19.599)$$

and substituting the definitions of X_1, Y_1, Z_1, Z_2 gives the Pauli and inverse coordinates

$$(\widehat{X}_1)_{(i, \beta), e_r^{-1}(k)} = \sum_{t=0}^1 Y_{\beta t} \overline{(\widehat{Y}_2)_{e_l^{-1}(k), (i, t)}}, \quad (19.600)$$

$$(\widehat{Y}_1)_{e_r^{-1}(k), (\beta, i)} = \sum_{t=0}^1 \overline{(\widehat{X}_2)_{(t, i), e_l^{-1}(k)}} Y_{t\beta}, \quad (19.601)$$

$$(\widehat{Z}_1)_{(\beta, i), e_r^{-1}(k)} = \overline{(\widehat{Y}_1)_{e_r^{-1}(k), (\beta, i)}}, \quad (19.602)$$

$$(\widehat{Z}_2)_{(i, \beta), e_l^{-1}(k)} = \frac{1}{2} \overline{(\widehat{Y}_2)_{e_l^{-1}(k), (i, \beta)}}. \quad (19.603)$$

In particular, transport preserves the Pauli orientation $Y_{\beta t}$ in the first factor of [163, lines 4611–4659].

Proof. [Stmnt](#) [Proof](#) Transport the cut factorizations and inverse identities along the two index bijections. Products, conjugate transposes, and identities are preserved. The weighted first normalization and the ordinary second normalization therefore hold in the source-rank coordinates. Evaluating the pullbacks, using $e_r(e_r^{-1}(k)) = k$ and $e_l(e_l^{-1}(k)) = k$, gives the coordinate identities; substituting the definitions of X_1, Y_1, Z_1, Z_2 and cancelling the inverse bijections gives the Pauli and inverse coordinates. $\square$

19.7.5 Normalization of the displayed CZX comparison

Definition 19.7.5.1 (Reflected candidates and the two displayed comparisons). [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Stmnt](#) Let Y be the Pauli adjoint gauge of Definition 19.7.3.1 and let X_1, Y_1, X_2, Y_2 be the displayed factors of Definition 19.7.4.1, with $d = 4$ and $D = 2$. Put $A = I_4 \otimes \overline{Y}$ and $B = Y^\top \otimes I_4$, and define the reflected candidates

$$\widehat{X}_1 = AY_2^\dagger, \quad (19.604)$$

$$\widehat{Y}_1 = X_2^\dagger B. \quad (19.605)$$

With $\rho = I_2/2$, define the raw and the density-weighted comparisons

$$K_{\text{raw}} = X_1^\dagger \widehat{X}_1, \quad (19.606)$$

$$K_\rho = X_1^\dagger (I_4 \otimes \rho) \widehat{X}_1. \quad (19.607)$$

These are the candidates and the comparison of [163, lines 5432–5441] evaluated on the explicit factors of lines 4559–4659.

Theorem 19.7.5.2 (The reflected candidates are the negatives of the displayed factors). [Lean](#)

[✓ Stmt](#) For the maintained Pauli gauge,

$$\widehat{X}_1 = -X_1, \quad (19.608)$$

$$\widehat{Y}_1 = -Y_1. \quad (19.609)$$

Proof. [✓ Proof](#) Theorem 19.7.3.5 gives $\overline{Y} = -Y$, and $Y^\top = \overline{Y}^\dagger = \overline{Y} = -Y$ by Theorem 19.7.3.2. Expanding both entries of the displayed factors against these two gauges gives the sign. $\square$

Theorem 19.7.5.3 (Coordinates of the two comparisons and the right overlap). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) On the displayed factors,

$$K_{\text{raw}} = -2I_4, \quad (19.610)$$

$$K_\rho = -I_4, \quad (19.611)$$

$$Y_1 \widehat{Y}_1^\dagger = -I_4. \quad (19.612)$$

Consequently $K_{\text{raw}}^\dagger K_{\text{raw}} = 4I_4$, so K_{raw} is not unitary, and $K_{\text{raw}} \neq Y_1 \widehat{Y}_1^\dagger$: the unweighted left overlap and the right overlap of [163, lines 5438–5441] disagree on these factors.

Proof. [✓ Proof](#) Substituting Theorem 19.7.5.2 turns each overlap into the negative of a Gram matrix of the displayed factors. The unweighted Gram $X_1^\dagger X_1 = 2I_4$ of Theorem 19.7.4.3 gives (19.610); the weighted normalization $X_1^\dagger (I_4 \otimes \rho) X_1 = I_4$ of Theorem 19.7.4.5 gives (19.611); and $Y_1 Y_1^\dagger = I_4$ gives (19.612). Squaring (19.610) gives the Gram matrix $4I_4$, whose diagonal entry 4 differs from 1, and comparing the diagonal entries of (19.610) and (19.612) separates the two overlaps. $\square$

Theorem 19.7.5.4 (The weighted comparison is unitary and recovers both factors). [Lean](#)

[✓ Stmt](#) The weighted comparison K_ρ is unitary, and

$$X_1 = \widehat{X}_1 K_\rho^\dagger, \quad (19.613)$$

$$Y_1 = K_\rho \widehat{Y}_1. \quad (19.614)$$

The weight is the one carried by the normalization of the first left factor. This calculation concerns the explicit displayed witness of Definition 19.7.4.1, not choice-selected factors, so the unequal overlaps of Theorem 19.7.5.3 diagnose the intermediate formula of [163, lines 5438–5441] on these factors rather than contradicting the proposition being proved there. The weighted comparison is the one retained in [88]; no gauge phase is changed.

Proof. [✓ Proof](#) By (19.611), $K_\rho = -I_4$, which is unitary. Theorem 19.7.5.2 then gives $\widehat{X}_1 K_\rho^\dagger = (-X_1)(-I_4) = X_1$ and $K_\rho \widehat{Y}_1 = (-I_4)(-Y_1) = Y_1$. $\square$

19.7.6 The blocked GHZ state and its CZX invariance

Definition 19.7.6.1 (The GHZ tensor in the blocked CZX alphabet). [Lean](#) [✓ Stmt](#) [Discussion: #7354](#)
 Let $A^0 = |0\rangle\langle 0|$ and $A^1 = |1\rangle\langle 1|$ be the GHZ tensor. Identify the blocked physical alphabet with $\{0, 1, 2, 3\}$ by $\varphi(2a + b) = (a, b)$, for $a, b \in \{0, 1\}$. Define

$$\widehat{A}^{2a+b} = A^a A^b = (A^{[2]})^{\varphi(2a+b)}. \quad (19.615)$$

This is the once-blocked GHZ tensor used in [163, lines 4660–4670], with the same physical-index order as the CZX tensor B .

Theorem 19.7.6.2 (The two blocked GHZ sectors). [Lean](#) [✓ Stmt](#) For $x \in \{0, 1\}$, let $A_x^a = \delta_{a,x}$ be the bond-one GHZ sector. Under $\varphi(2a + b) = (a, b)$ its two-site block is

$$(A_x^{[2]})^{\varphi(i)} = \delta_{i,3x}, \quad i \in \{0, 1, 2, 3\}. \quad (19.616)$$

Thus the blocked sector vectors are $|00\rangle$ and $|11\rangle$, as in [163, lines 4660–4670].

Proof. [✓ Proof](#) For $i = 2a + b$, the block coefficient is $A_x^a A_x^b = \delta_{a,x} \delta_{b,x}$, which equals one exactly when $a = b = x$, or equivalently $i = 3x$. $\square$

Theorem 19.7.6.3 (The matrix product vector of the blocked GHZ tensor). [Lean](#) [✓ Stmt](#) For every $N \geq 0$, the periodic matrix product vector of $\widehat{A}$ is

$$|\psi_N\rangle = |0\rangle^{\otimes N} + |3\rangle^{\otimes N}. \quad (19.617)$$

In the original two-qubit alphabet the two summands are $|00\rangle^{\otimes N}$ and $|11\rangle^{\otimes N}$. At $N = 0$ they are the same empty configuration, with total coefficient two.

Proof. [✓ Proof](#) The blocked letters are diagonal, with $\widehat{A}^i = \text{diag}(\delta_{i,0}, \delta_{i,3})$. Hence, for $s \in \{0, 1, 2, 3\}^N$,

$$\text{tr}(\widehat{A}^{s_1} \dots \widehat{A}^{s_N}) = \prod_{j=1}^N \delta_{s_j,0} + \prod_{j=1}^N \delta_{s_j,3}. \quad (19.618)$$

These are the coefficients of the two constant configurations. $\square$

Theorem 19.7.6.4 (CZX invariance of the blocked GHZ state). [Lean](#) [✓ Stmt](#) [Discussion: #7354](#)
 Let B be the once-blocked CZX tensor, with both output Z gates and the unnormalized Hadamard weights, and let $N > 0$. Transport the matrix product vector of the two-site-blocked GHZ tensor $A^{[2]}$ by the alphabet bijection $\varphi(2a + b) = (a, b)$:

$$\psi_N(s) = V^{(N)}(A^{[2]})_{(\varphi(s_1), \dots, \varphi(s_N))}, \quad s \in \{0, 1, 2, 3\}^N. \quad (19.619)$$

Then the periodic CZX operator satisfies

$$O_N(B)|\psi_N\rangle = |\psi_N\rangle. \quad (19.620)$$

This is the GHZ-invariance assertion in [163, lines 4660–4670].

Proof. [✓ Proof](#) Reindexing the alphabet gives the vector in (19.617). The CZX operator complements every matter bit. Its phase on either constant configuration is one: the cyclic exponent is zero on 0^N and $4N$ on 3^N . Consequently

$$O_N(B)|0\rangle^{\otimes N} = |3\rangle^{\otimes N}, \quad O_N(B)|3\rangle^{\otimes N} = |0\rangle^{\otimes N}. \quad (19.621)$$

Their sum is invariant. $\square$

19.7.7 The printed CZX action pair and its empty-word boundary

The action tensors displayed in [163, lines 4690–4694] satisfy the interior identity for nonempty words and the exterior identity with nonempty buffers. Their empty contraction is one half, whereas the general interior identity in [163, lines 1841–1870] includes length zero. The distinction and a local replacement are recorded in [53]. The replacement below is a different pair, not a scalar gauge of the printed pair or a substitution in the unbuffered defect formulas.

Definition 19.7.7.1 (The acted GHZ sectors). Lean Lean Lean Lean Lean ✓ Stmt For $x \in \{0, 1\}$, let A_x be the actual two-site block of the bond-one GHZ sector, with physical indices identified by $\varphi(2a + b) = (a, b)$. Thus $A_x^i = \delta_{i, 3x}$. Write $B^{i,j}$ for the CZX tensor and define

$$B_x^i = \sum_{j=0}^3 B^{i,j} A_x^j, \quad (19.622)$$

$$k_x = 3(1 - x), \quad (19.623)$$

$$C_x = B^{k_x, 3x}. \quad (19.624)$$

The target sector is A_{1-x} . These are the physical contraction and exchanged GHZ blocks of [163, lines 4660–4694].

Theorem 19.7.7.2 (Coordinates of the acted sectors). Lean Lean Lean ✓ Stmt For $x \in \{0, 1\}$ and $i \in \{0, 1, 2, 3\}$,

$$B_x^i = \delta_{i, k_x} C_x, \quad (19.625)$$

$$A_{1-x}^i = \delta_{i, k_x}, \quad (19.626)$$

$$C_0 = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}, \quad (19.627)$$

$$C_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}. \quad (19.628)$$

Proof. ✓ Proof Since $A_x^j = \delta_{j, 3x}$, the defining sum of Definition 19.7.7.1 collapses to $B_x^i = \sum_{j=0}^3 B^{i,j} A_x^j = B^{i, 3x}$. The exact CZX tensor of Definition 19.7.2.1 vanishes off its support, giving

$$B^{i, 3x} = \delta_{i, 3(1-x)} B^{3(1-x), 3x} = \delta_{i, k_x} C_x, \quad (19.629)$$

so $B_x^i = \delta_{i, k_x} C_x$; likewise $A_x^j = \delta_{j, 3x}$ at $x \mapsto 1 - x$ gives $A_{1-x}^i = \delta_{i, k_x}$. Expanding the defining CZX matrices at $B^{k_x, 3x}$ evaluates the two displayed matrices C_0, C_1 . $\square$

Definition 19.7.7.3 (The printed action pair). Lean Lean ✓ Stmt For $x \in \{0, 1\}$, denote the printed left and right action tensors by p_x and q_x . With $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$, the formulas of [163, lines 4690–4694] read

$$p_0 = (0, -i), \quad (19.630)$$

$$p_1 = (1, 0), \quad (19.631)$$

$$q_0 = \frac{1}{2} \begin{pmatrix} -i \\ i \end{pmatrix}, \quad (19.632)$$

$$q_1 = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (19.633)$$

The normalization of the kets is not specified explicitly in that passage. These coordinates agree with the later defect evaluations; see [53].

Theorem 19.7.7.4 (Finite contractions of the printed action pair). Lean ✓ Stmt For each $x \in \{0, 1\}$,

$$C_x^2 = C_x, \quad (19.634)$$

$$p_x q_x = \frac{1}{2}, \quad (19.635)$$

$$p_x C_x q_x = 1, \quad (19.636)$$

$$C_x q_x p_x C_x = C_x. \quad (19.637)$$

Proof. ✓ Proof Multiply the two-by-two matrices and the printed rows and columns. □

Theorem 19.7.7.5 (Nonempty acted and target words). Lean Lean ✓ Stmt For a word $\mathbf{a}$ in $\{0, 1, 2, 3\}$, write $B_x^{\mathbf{a}}$ and $A_{1-x}^{\mathbf{a}}$ for ordered products of letters. If $|\mathbf{a}| > 0$, then

$$B_x^{\mathbf{a}} = \begin{cases} C_x & \text{if every letter is } k_x, \\ 0 & \text{otherwise,} \end{cases} \quad (19.638)$$

$$A_{1-x}^{\mathbf{a}} = \begin{cases} 1 & \text{if every letter is } k_x, \\ 0 & \text{otherwise.} \end{cases} \quad (19.639)$$

Proof. ✓ Proof An unsupported letter makes the product zero. Otherwise every acted letter is C_x , whose positive powers equal C_x by idempotency. □

Theorem 19.7.7.6 (Printed interior and buffered exterior identities). Lean Lean ✓ Stmt Let $x \in \{0, 1\}$. For every nonempty word $\mathbf{c}$,

$$p_x B_x^{\mathbf{c}} q_x = A_{1-x}^{\mathbf{c}}. \quad (19.640)$$

If $\mathbf{a}, \mathbf{b}$ are also nonempty, then

$$B_x^{\mathbf{a}} q_x A_{1-x}^{\mathbf{c}} p_x B_x^{\mathbf{b}} = B_x^{\mathbf{acb}}. \quad (19.641)$$

No equality of buffer lengths is needed. The equal-length case is the exterior diagram of [163, lines 1788–1838]. The interior equation covers only the nonempty part of that source assertion; see [53].

Proof. ✓ Proof If an unsupported letter occurs, both sides vanish. Otherwise the interior is $p_x C_x q_x = 1$ and the exterior is $C_x q_x p_x C_x = C_x$. □

Theorem 19.7.7.7 (The printed pair is not an all-word reduction). Lean Lean ✓ Stmt For each $x \in \{0, 1\}$, the printed pair (p_x, q_x) is not a reduction from B_x to A_{1-x} . Indeed, if ε is the empty word, then

$$p_x B_x^{\varepsilon} q_x = \frac{1}{2} \neq 1 = A_{1-x}^{\varepsilon}. \quad (19.642)$$

This is the empty-word obstruction of [53].

Proof. ✓ Proof Empty products are identities, whereas $p_x q_x = 1/2$. □

Definition 19.7.7.8 (A separately dressed left action tensor). [Lean](#) [✓ Stmt](#) For $x \in \{0, 1\}$, define $V_x = p_x C_x$ and retain $W_x = q_x$. This is the local replacement of [53], not the literal printed pair. In coordinates $V_0 = (i, -i)$ and $V_1 = (1, 1)$, so neither is a scalar multiple of the corresponding printed row. No replacement in an unbuffered defect formula is asserted.

Theorem 19.7.7.9 (Dressing is absorbed by a nonempty buffer). [Lean](#) [✓ Stmt](#) For $x \in \{0, 1\}$ and every nonempty word $\mathbf{b}$,

$$V_x B_x^{\mathbf{b}} = p_x B_x^{\mathbf{b}}. \quad (19.643)$$

Proof. [✓ Proof](#) The word is either zero or C_x ; use $C_x^2 = C_x$ in the latter case. $\square$

Theorem 19.7.7.10 (An all-word reduction with exterior buffer length one). [Lean](#) [Lean](#) [✓ Stmt](#) For each $x \in \{0, 1\}$, (V_x, W_x) is a reduction from B_x to A_{1-x} with exterior buffer length one. The reduction identity holds for every word $\mathbf{c}$, including the empty word; the exterior identity holds for nonempty words $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$. Thus

$$V_x W_x = 1, \quad (19.644)$$

$$V_x B_x^{\mathbf{c}} W_x = A_{1-x}^{\mathbf{c}}, \quad (19.645)$$

$$B_x^{\mathbf{a}} W_x A_{1-x}^{\mathbf{c}} V_x B_x^{\mathbf{b}} = B_x^{\mathbf{a}\mathbf{c}\mathbf{b}}. \quad (19.646)$$

This proves the all-word relation for the separately dressed pair, not the empty-word claim for the printed pair.

Proof. [✓ Proof](#) The empty contraction is $V_x W_x = p_x C_x q_x = 1$. For a nonempty interior word, absorb C_x and apply the printed interior identity. In the exterior identity, absorb C_x into the right buffer and apply (19.641). $\square$

19.7.8 The CZX g, g fusion caps

Definition 19.7.8.1 (The CZX g, g fusion caps). [Lean](#) [Lean](#) [✓ Stmt](#) In the ordered bond coordinates $0, 1, 2, 3$, define the row vector $F_{g,g}^<: \mathbb{C}^4 \rightarrow \mathbb{C}$ and the column vector $F_{g,g}^>: \mathbb{C} \rightarrow \mathbb{C}^4$ by

$$F_{g,g}^< = (0, -i, i, 0), \quad (19.647)$$

$$F_{g,g}^> = \frac{1}{2}(0, i, -i, 0)^T. \quad (19.648)$$

These are the concrete caps in [163, lines 4671–4689], the $g = h = g^{-1}$ specialization of the general fusion-tensor display [163, lines 2000–2037].

Theorem 19.7.8.2 (Product coordinates of the CZX fusion caps). [Lean](#) [✓ Stmt](#) Let $F_{g,g}^<, F_{g,g}^>$ be the caps of Definition 19.7.8.1, and let Y be the Pauli matrix of Definition 19.7.3.1. Identify the product-bond coordinate (a, b) with $2a + b$. For all bond bits $a, b \in \{0, 1\}$,

$$F_{g,g}^<(a, b) = Y_{a,b}, \quad (19.649)$$

$$F_{g,g}^>(a, b) = \frac{1}{2}Y_{b,a}. \quad (19.650)$$

Thus both caps vanish at the diagonal bond coordinates and carry Y in opposite orientations, as in the printed diagram of [163, lines 4671–4689].

Proof. [✓ Proof](#) Evaluate both caps at $2a + b$ for the four pairs (a, b) and compare with the entries of Y to obtain (19.649) and (19.650). $\square$

Theorem 19.7.8.3 (The CZX fusion reduction). [Lean](#) [Lean](#) [✓ Stmt](#) Let B be the doubled-index tensor of the actual product $\mathcal{U}_g \cdot \mathcal{U}_g$ (Definition 21.3.5 applied to Definition 19.7.2.1 with itself, viewed through Definition 16.1.2), and let I be the doubled-index bond-one identity tensor of Definition 18.9.3.13. Then $(F_{g,g}^<, F_{g,g}^>)$ is a reduction from B to I with exterior buffer length one. For every word $\mathbf{c}$, including the empty word,

$$F_{g,g}^< F_{g,g}^> = 1, \quad (19.651)$$

$$F_{g,g}^< B^{\mathbf{c}} F_{g,g}^> = I^{\mathbf{c}}. \quad (19.652)$$

For nonempty words $\mathbf{p}, \mathbf{q}, \mathbf{c}$,

$$B^{\mathbf{p}} F_{g,g}^> I^{\mathbf{c}} F_{g,g}^< B^{\mathbf{q}} = B^{\mathbf{p}\mathbf{c}\mathbf{q}}. \quad (19.653)$$

This is the concrete g, g instance of the fusion reduction [163, lines 1405–1498], with the fusion tensors of [163, lines 2000–2037], specialized to the CZX display [163, lines 4671–4689].

Proof. [✓ Proof](#) Put $V = F_{g,g}^<$ and $W = F_{g,g}^>$. Substituting the cap and tensor coordinates gives the three local identities

$$VW = 1, \quad (19.654)$$

$$VB^i W = I^i, \quad (19.655)$$

$$B^i W V B^j = B^i B^j. \quad (19.656)$$

The all-word identity $VB^{\mathbf{c}}W = I^{\mathbf{c}}$ follows by induction on $\mathbf{c}$, removing its first letter. The empty word is (19.654) and a single letter is (19.655). For $\mathbf{c} = ij\mathbf{c}'$, insert WV after the first letter by (19.656), split the caps off both factors, and apply (19.655) together with the induction hypothesis for $j\mathbf{c}'$:

$$VB^{ij\mathbf{c}'}W = (VB^i W)(VB^{j\mathbf{c}'}W) = I^i I^{j\mathbf{c}'} = I^{ij\mathbf{c}'}. \quad (19.657)$$

For nonempty words $\mathbf{p}, \mathbf{q}$ a second induction gives

$$B^{\mathbf{p}} W V B^{\mathbf{q}} = B^{\mathbf{p}\mathbf{q}}. \quad (19.658)$$

Induct on the left word $\mathbf{p}$, again removing its first letter. For the singleton case $\mathbf{p} = i$ and $\mathbf{q} = j\mathbf{q}'$, (19.656) gives

$$B^i W V B^{j\mathbf{q}'} = (B^i W V B^j) B^{\mathbf{q}'} = B^i B^j B^{\mathbf{q}'} = B^{ij\mathbf{q}'}. \quad (19.659)$$

If $\mathbf{p} = i\mathbf{p}'$ with $\mathbf{p}'$ nonempty, apply the induction hypothesis to the tail:

$$B^{i\mathbf{p}'} W V B^{\mathbf{q}} = B^i (B^{\mathbf{p}'} W V B^{\mathbf{q}}) = B^i B^{\mathbf{p}'} B^{\mathbf{q}} = B^{i\mathbf{p}'\mathbf{q}}. \quad (19.660)$$

For nonempty buffers and middle word, substitute $I^{\mathbf{c}} = VB^{\mathbf{c}}W$ and apply (19.658) twice:

$$B^{\mathbf{p}} W I^{\mathbf{c}} V B^{\mathbf{q}} = B^{\mathbf{p}} W V B^{\mathbf{c}} W V B^{\mathbf{q}} = B^{\mathbf{p}\mathbf{c}\mathbf{q}}. \quad (19.661)$$

□

19.7.9 Literal fusion–action coordinate contractions

The contractions below evaluate the printed action pair and right fusion cap of [163, lines 4671–4694] in fixed product-bond order. They are compared with the fusion–action relation of [163, lines 1875–1913], but do not identify its L -symbol or its retained physical scalar. Compatibility of the oriented legs, density insertion, and fusion gauge remains open [39]. No cap, action tensor, block label, or physical gate is changed.

Definition 19.7.9.1 (Sequential action covector). [Lean](#) [✓ Stmt](#) For $x \in \{0, 1\}$, let p_x be the printed action row of Definition 19.7.7.3. Define the row $P_x: \mathbb{C}^4 \rightarrow \mathbb{C}$ by the following coordinates, for $\alpha, \gamma \in \{0, 1\}$:

$$P_x = p_{1-x} \otimes p_x, \quad (P_x)_{2\alpha+\gamma} = p_{1-x}(\alpha)p_x(\gamma). \quad (19.662)$$

The first action factor occupies the first product-bond coordinate.

Definition 19.7.9.2 (Sequential action vector). [Lean](#) [✓ Stmt](#) For $x \in \{0, 1\}$, let q_x be the printed action column of Definition 19.7.7.3. Define the column $Q_x: \mathbb{C} \rightarrow \mathbb{C}^4$ by the following coordinates, for $\alpha, \gamma \in \{0, 1\}$:

$$Q_x = q_{1-x} \otimes q_x, \quad (Q_x)_{2\alpha+\gamma} = q_{1-x}(\alpha)q_x(\gamma). \quad (19.663)$$

Rows contract with columns by ordinary matrix multiplication, with no additional complex conjugation.

Definition 19.7.9.3 (Supported product letter). [Lean](#) [✓ Stmt](#) Let T be the exact blocked CZX tensor of Definition 19.7.2.1. For $x \in \{0, 1\}$, define $B_x \in \text{Mat}_4(\mathbb{C})$ to be the physical $(3x, 3x)$ letter of $T \cdot T$. Explicitly, for $\alpha, \gamma, \beta, \delta \in \{0, 1\}$,

$$(B_x)_{2\alpha+\gamma, 2\beta+\delta} = \sum_{j=0}^3 T_{\alpha\beta}^{3x,j} T_{\gamma\delta}^{j,3x}. \quad (19.664)$$

The physical index $3x$ encodes the two bits (x, x) .

Theorem 19.7.9.4 (Coordinates of the supported product letters). [Lean](#) [✓ Stmt](#) *The two supported product letters have coordinates*

$$B_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (19.665)$$

$$B_1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (19.666)$$

Proof. [✓ Proof](#) The letter of $T \cdot T$ at a physical pair (i, i') vanishes unless $i = i'$, because the blocked tensor complements the physical index it carries. At $i = i' = 3x$ the summation index of Definition 19.7.9.3 retains the single term $j = 3(1-x)$, whose two factors force the left virtual pair $(\alpha, \gamma) = (x, 1-x)$, the row of index $x+1$. On that row the surviving sign is $(-1)^{1+(1-x)\delta+x\beta}$, which is the row displayed above. $\square$

Definition 19.7.9.5 (Literal fusion–action coefficient). [Lean](#) [✓ Stmt](#) For $x \in \{0, 1\}$, let P_x and B_x be the sequential action row and supported product letter, and put $W = F_{g,g}^> = \frac{1}{2}(0, i, -i, 0)^\top$, in the product-bond order of Definition 19.7.8.1. Define

$$c_x = P_x B_x W \in \mathbb{C}. \quad (19.667)$$

The one-by-one matrix on the right is identified with its sole entry.

Theorem 19.7.9.6 (Values of the literal coefficient). [Lean](#) [✓ Stmt](#) The literal coefficients satisfy $c_0 = 1$ and $c_1 = -1$.

Proof. [✓ Proof](#) The printed rows are $P_0 = (0, -i, 0, 0)$ and $P_1 = (0, 0, -i, 0)$, and the column is $W = \frac{1}{2}(0, i, -i, 0)^\top$. With $e_0, \dots, e_3$ the standard basis of the product bond, the single nonzero row of each supported letter of (19.666) contracts with W as

$$B_0 W = \frac{1}{2}(i + i)e_1 = ie_1, \quad (19.668)$$

$$B_1 W = \frac{1}{2}(-i - i)e_2 = -ie_2, \quad (19.669)$$

so the two finite matrix products are

$$c_0 = P_0 B_0 W = (0, -i, 0, 0) ie_1 = 1, \quad (19.670)$$

$$c_1 = P_1 B_1 W = (0, 0, -i, 0) (-ie_2) = -1. \quad (19.671)$$

□

Theorem 19.7.9.7 (Normalization, proportionality, and idempotence). [Lean](#) [✓ Stmt](#) For each $x \in \{0, 1\}$, the literal rows, columns, and supported product letter satisfy

$$P_x B_x Q_x = 1, \quad B_x W = c_x B_x Q_x, \quad B_x^2 = B_x. \quad (19.672)$$

Proof. [✓ Proof](#) Substituting the printed action columns gives $Q_0 = \frac{1}{4}(-i, i, -i, i)^\top$ and $Q_1 = \frac{1}{4}(-i, -i, i, i)^\top$. With $e_0, \dots, e_3$ the standard basis of the product bond, the two supported letters of (19.666) contract with these columns and with the cap W as

$$B_0 Q_0 = ie_1, \quad (19.673)$$

$$B_0 W = ie_1, \quad (19.674)$$

$$B_1 Q_1 = ie_2, \quad (19.675)$$

$$B_1 W = -ie_2. \quad (19.676)$$

The printed rows are $P_0 = (0, -i, 0, 0)$ and $P_1 = (0, 0, -i, 0)$, so applying P_x to (19.676) gives $P_x B_x Q_x = 1$ and $P_x B_x W = c_x$, with $c_0 = 1$ and $c_1 = -1$. Comparing the two right-hand sides for each x gives $B_x W = c_x B_x Q_x$. Each matrix of (19.666) has a single nonzero row, whose entry in the supporting column is 1, so $B_x^2 = B_x$. □

Theorem 19.7.9.8 (Positive repetitions of the supported letter). [Lean](#) [✓ Stmt](#) For every $x \in \{0, 1\}$ and every integer $n \geq 0$,

$$B_x^{n+1} = B_x, \quad P_x B_x^{n+1} W = c_x, \quad (19.677)$$

$$P_x B_x^{n+1} Q_x = 1, \quad B_x^{n+1} W = c_x B_x^{n+1} Q_x. \quad (19.678)$$

These statements concern repetitions of this single supported letter, not arbitrary words or an identification with the source L -symbol.

Proof. [✓ Proof](#) Induct on n . The first power is B_x ; multiplication by B_x and idempotence prove the successor step. Substitute $B_x^{n+1} = B_x$ into the remaining three identities, and use the definition of c_x and the normalization and proportionality of Theorem 19.7.9.7. □

19.7.10 The displayed CZX circuit tuple

Definition 19.7.10.1 (The displayed CZX circuit tuple). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Number the four matter qubits of two neighboring blocked sites by 0, 1, 2, 3, so that (x_0, x_1) are the qubits of the first site and (x_2, x_3) those of the second, and write $|x\rangle = |x_0\rangle \otimes |x_1\rangle \otimes |x_2\rangle \otimes |x_3\rangle$ for $x \in \mathbb{F}_2^4$. The elementary gates act by

$$X_r|x\rangle = |x + e_r\rangle, \quad (19.679)$$

$$Z_r|x\rangle = (-1)^{x_r}|x\rangle, \quad (19.680)$$

$$\text{CZ}_{r_s}|x\rangle = (-1)^{x_r x_s}|x\rangle, \quad (19.681)$$

and operator products act from right to left. Define

$$w = (X_0 X_1) CZ_{01} CZ_{12} (Z_2 Z_3), \quad (19.682)$$

$$\lambda = -i(X_0 X_1 X_2 Z_3) \text{CZ}_{01} \text{CZ}_{12} X_2, \quad (19.683)$$

$$\tilde{\lambda} = \lambda Z_0. \quad (19.684)$$

These are the fusion operators $\lambda_{g,e}^R = w_R$ and $\lambda_{g,g}^R$ and the modified fusion operator $\tilde{\lambda}_{g,g}^R = \lambda_{g,g}^R Z_1$ of [163, lines 4503–5183] in this qubit numbering. The displayed circuit tuple, indexed by the ordered pair of gauge labels in $\mathbb{Z}_2$ with $e \leftrightarrow 0$ and $g \leftrightarrow 1$, is

$$R^{\text{CZX}} = (U_{0,0}, U_{0,1}, U_{1,0}, U_{1,1}) = (\mathbb{1}, \mathbb{1}, w, \tilde{\lambda}). \quad (19.685)$$

Each site index in $\{0, 1, 2, 3\}$ is read as the pair of matter bits $(m^{(1)}, m^{(2)})$ with $m^{(1)}$ the leading binary digit, and the four-qubit operators are transported to the two-site matter space by this identification.

Theorem 19.7.10.2 (Phase tables of the CZX fusion operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Write
 $\bar{x} = (x_0 + 1, x_1 + 1, x_2, x_3)$ and

$$e(x) = x_0x_1 + x_1x_2 + x_2 + x_3, \quad (19.686)$$

$$h(x) = x_0x_1 + x_1x_2 + x_1 + x_3, \quad (19.687)$$

$$f(x) = x_0x_1 + x_1x_2 + x_0 + x_1 + x_3. \quad (19.688)$$

Then

$$w|x\rangle = (-1)^{e(x)}|\overline{x}\rangle, \quad (19.689)$$

$$\lambda|x\rangle = -i(-1)^{h(x)}|\overline{x}\rangle, \quad (19.690)$$

$$\tilde{\lambda}|x\rangle = -i(-1)^{f(x)}|\bar{x}\rangle, \quad (19.691)$$

the exponents satisfy $f(\bar{x}) = e(x) + 1$ and $e(\bar{x}) = f(x) + 1$. After transport to the two-site matter coordinates, w and $\tilde{\lambda}$ are unitary.

Proof. ✓ Proof Each gate is monomial, so (19.538) expresses each product as a monomial operator. Reading (19.682) from right to left, the diagonal gates contribute the phase $(-1)^{x_0 x_1 + x_1 x_2 + x_2 + x_3}$ and the two bit flips send x to $\bar{x}$. In (19.683) the first X_2 replaces x_2 by $x_2 + 1$ in the phase of CZ_{12} , giving $x_1(x_2 + 1) = x_1 x_2 + x_1$, and the second X_2 restores x_2 , so the permutation is again τ . The factor Z_0 in (19.684) adds x_0 to the exponent. The identities between e and f are checked on the sixteen bit strings. After relabelling the computational basis as two four-level matter sites, unitarity follows from Theorem 19.7.1.2 because all phases are unimodular. $\square$

Proof. ✓ Proof By Theorem 19.7.1.6 applied to the monomial unitaries of Theorem 19.7.10.2, the operator $\mathcal{G}_g$ is monomial. The permutation part is τ on the matter bits in all four gauge sectors, since τ is an involution, and T_g flips both gauge bits. The phases are read off (19.689)–(19.691) and the adjoint formula (19.539) together with $f(\bar{x}) = e(x) + 1$ and $e(\bar{x}) = f(x) + 1$: for example the first row is $-i(-1)^{e(x)} = i^{3+2e(x)} = i^{3+u(x)+2x_2}$, and the other three rows follow in the same way. The four identities between the phase table and e, f are checked on the sixteen bit strings. $\square$

19.7.11 Orbits, holonomy, and the exact dimension

Definition 19.7.11.1 (Chain configurations, bond flips, and labels). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt A computational-basis configuration of the periodic chain of N blocked sites is

$$s = (m_k^{(1)}, m_k^{(2)}, g_k)_{k \in \mathbb{Z}/N\mathbb{Z}} \in S_N := (\mathbb{F}_2^3)^N, \quad (19.700)$$

read from the enumerated site indices of the placed Gauss projector. The bond- j flip σ_j adds 1 to $m_j^{(1)}, m_j^{(2)}, g_j$, and g_{j+1} . The bond- j phase exponent $q_j(s)$ is the table (19.699) evaluated at $x = (m_j^{(1)}, m_j^{(2)}, m_{j+1}^{(1)}, m_{j+1}^{(2)})$ and $(a, a') = (g_j, g_{j+1})$. The affine labels are

$$p_k(s) = m_k^{(1)} + m_k^{(2)}, \quad (19.701)$$

$$b_k(s) = g_k + m_k^{(1)} + m_{k-1}^{(1)}. \quad (19.702)$$

Theorem 19.7.11.2 (Permutation-orbit classification). Lean Lean Lean Lean Lean ✓ Stmt For $N \geq 2$, the map

$$((p, b), \gamma) \mapsto (m_k^{(1)}, m_k^{(2)}, g_k)_k = (\gamma_k, p_k + \gamma_k, b_k + \gamma_k + \gamma_{k-1})_k \quad (19.703)$$

is a bijection $\mathbb{F}_2^N \times \mathbb{F}_2^N \times \mathbb{F}_2^N \rightarrow S_N$ whose inverse reads off $p(s)$, $b(s)$, and $\gamma = m^{(1)}$. In these coordinates σ_j is $\gamma \mapsto \gamma + e_j$ and leaves (p, b) unchanged.

Proof. ✓ Proof Both compositions of (19.703) with the label map are the identity, by the identity $a + a = 0$ in $\mathbb{F}_2$. The flip σ_j adds $((1, 1), 1)$ at site j and $((0, 0), 1)$ at site $j + 1$; in fiber coordinates this is $\gamma_j \mapsto \gamma_j + 1$ because $k - 1 = j$ exactly when $k = j + 1$. $\square$

Theorem 19.7.11.3 (Involution and neighboring holonomy). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For every $N \geq 3$, every bond j , and every configuration s ,

$$q_j(s) + q_j(\sigma_j s) = 0, \quad (19.704)$$

$$q_j(s) + q_{j+1}(\sigma_j s) = q_{j+1}(s) + q_j(\sigma_{j+1} s) + 2b_{j+1}(s) \quad (19.705)$$

modulo 4, and $q_j(\sigma_k s) = q_j(s)$ whenever $k \notin \{j - 1, j, j + 1\}$. For the phases $\varphi_j(s) = i^{q_j(s)}$, (19.704) gives $\varphi_j(s)\varphi_j(\sigma_j s) = 1$, while (19.705) gives the neighboring square relation with factor $(-1)^{b_{j+1}(s)}$.

Proof. ✓ Proof The exponent q_j depends only on the six bits at the sites j and $j + 1$, and σ_k changes only the sites k and $k + 1$; this gives the disjoint-bond statement. For (19.704) the flip σ_j changes x_0, x_1, a , and a' , and the identity is verified on the 64 values of the six bits. For (19.705) the sites $j, j + 1, j + 2$ are distinct because $N \geq 3$; σ_j changes the left gauge bit seen by the bond $(j + 1, j + 2)$, and σ_{j+1} changes x_2, x_3 , and a' seen by the bond $(j, j + 1)$, so the identity is verified on the 512 values of the nine bits at the three sites. $\square$

Proof. [✓ Proof](#) By Theorem 19.7.12.5, $\mathcal{V}_N(R^*)$ is the common fixed subspace of monomial operators which, in fiber coordinates, have permutations $\gamma \mapsto \gamma + e_j$ on the fibers over (p, b) , and by (19.715) these operators are involutions. The monomial fixed-space criterion (19.544) counts the fibers with trivial holonomy, which by Theorem 19.7.12.4 are exactly the fibers whose labels satisfy the N equations $c_j(p, b) = 0$. $\square$

Theorem 19.7.12.7 (Dimensions for the second displayed tuple at three and four sites). [Lean](#) [Lean](#) [✓ Stmt](#) On periodic chains of three and four blocked sites,

$$\dim \mathcal{V}_3(R^*) = 14, \quad (19.718)$$

$$\dim \mathcal{V}_4(R^*) = 36. \quad (19.719)$$

These two values are certificates at two system sizes. They carry no asymptotic content, and no growth law for $\dim \mathcal{V}_N(R^)$ follows from them. Since R^* extends the four prescribed maps of Definition 19.8.10 by Theorem 19.9.3, these are two values of the dimension on that completion class, recorded as lower bounds on its supremum in Theorem 19.9.4; they give no value for the minimum or the maximum of the dimension over completions. Compared with (19.707), which gives 8 and 16 at the same two sizes, they exhibit dependence of the dimension on the displayed circuit data at those two sizes.*

Proof. [✓ Proof](#) Evaluate the right-hand side of (19.717) by exhaustive enumeration: at $N = 3$ the 64 pairs (p, b) yield 14 solutions of the three equations $c_j(p, b) = 0$, and at $N = 4$ the 256 pairs yield 36 solutions of the four equations. $\square$

19.8 The CZX defect domains and prescribed defect maps

The modified-fusion lemma of [163, lines 4215–4254] attaches to each pair (a, b) of gauge labels and each injective block x a two-site defect vector $v(a, b, x)$ mapped unitarily to $v(e, ab, x)$. For the CZX model, the defect subspaces and prescribed maps are evaluated by finite contraction in [41].

Definition 19.8.1 (Single-site CZX defects). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7739](#) One blocked site of the CZX model carries two matter qubits, and the two injective blocks of the once-blocked GHZ state are the product vectors $|\circ\rangle = |00\rangle$ and $|\bullet\rangle = |11\rangle$ [163, lines 4660–4670], both of bond dimension one. Writing $e \leftrightarrow 0$ and $g \leftrightarrow 1$ for the labels of $\mathbb{Z}_2 = \{e, g\}$, the identity defect and the domain-wall defect of [163, lines 2831–2845] at the block $x \in \mathbb{F}_2$ are

$$e[x] = |xx\rangle, \quad (19.720)$$

$$g[0] = |00\rangle, \quad (19.721)$$

$$g[1] = -i|11\rangle. \quad (19.722)$$

Contracting the action tensors $A_{g,0}^\Gamma = -i\langle 1|$ and $A_{g,1}^\Gamma = \langle 0|$ of [163, lines 4690–4694] against the defect diagram gives (19.721) and (19.722), which are the evaluations displayed in [163, lines 4800–4859]; the derivation is carried out in [41].

Definition 19.8.2 (Two-site CZX defect vectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The two-site defect vector of the label pair (a, b) and the block x is the left defect $a[bx]$ tensored with the right defect $b[x]$, in that order [163, lines 4234–4245],

$$v(a, b, x) = a[bx] \otimes b[x]. \quad (19.723)$$

The four qubits of two neighboring blocked sites are numbered 0, 1 on the left site and 2, 3 on the right, and the computational basis of the sixteen-dimensional two-site matter space is written $|x_0x_1x_2x_3\rangle$.

Theorem 19.8.3 (The eight CZX defect vectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) *The eight two-site defect vectors of the CZX model are*

$$v(e, e, 0) = |0000\rangle, \quad v(e, e, 1) = |1111\rangle, \quad (19.724)$$

$$v(e, g, 0) = |1100\rangle, \quad v(e, g, 1) = -i|0011\rangle, \quad (19.725)$$

$$v(g, e, 0) = |0000\rangle, \quad v(g, e, 1) = -i|1111\rangle, \quad (19.726)$$

$$v(g, g, 0) = -i|1100\rangle, \quad v(g, g, 1) = -i|0011\rangle. \quad (19.727)$$

For every label pair the two vectors $v(a, b, 0)$ and $v(a, b, 1)$ are orthonormal.

Proof. [✓ Proof](#) Substituting (19.720), (19.721), and (19.722) into (19.723), with $bx = x + b$ in $\mathbb{F}_2$, gives the eight values. The two vectors of a label pair differ in the last two bits, hence are orthogonal, and each carries a scalar of modulus one, hence is a unit vector. $\square$

Lemma 19.8.4 (Transported monomial operators on the computational basis). [Lean](#) [Lean](#)
[✓ Stmt](#) *Write $|x\rangle$ for the computational basis vector of the two-site matter space attached to $x \in \mathbb{F}_2^4$ by the qubit numbering of Definition 19.8.2. After transport to that space a monomial four-qubit operator still acts on the computational basis by its phase and its permutation, so that for every $x \in \mathbb{F}_2^4$*

$$M(\sigma, \varphi)|x\rangle = \varphi(x)|\sigma(x)\rangle. \quad (19.728)$$

Proof. [✓ Proof](#) Transport is the simultaneous relabelling of rows and columns along the qubit numbering. It carries $M(\sigma, \varphi)$ to the monomial matrix of the relabelled permutation and phase, and carries the basis vector of x to $|x\rangle$, so reading (19.537) in the new labels gives (19.728). $\square$

Theorem 19.8.5 (Unmodified CZX fusion contractions). [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7355](#) *Let $\mathbb{Z}_2 = \{e, g\}$, and let $v(a, b, x)$ be the two-site defect vectors of Definition 19.8.2, where $x \in \{0, 1\}$. On the four matter qubits, numbered 0, 1, 2, 3, the unmodified operator*

$$\lambda = \lambda_{g,g}^R = -i (X_0 X_1 X_2 Z_3) \text{CZ}_{01} \text{CZ}_{12} X_2 \quad (19.729)$$

satisfies

$$\lambda v(g, g, 0) = -v(e, e, 0), \quad (19.730)$$

$$\lambda v(g, g, 1) = v(e, e, 1). \quad (19.731)$$

These are the two contractions displayed in [163, lines 4989–5067]. Here λ is the unmodified operator, not $\tilde{\lambda} = \lambda Z_0$ used in the displayed circuit tuple.

Proof. [✓ Proof](#) The phase formula (19.690), read on the two-site matter space through (19.728), gives

$$\lambda|1100\rangle = -i|0000\rangle, \quad (19.732)$$

$$\lambda|0011\rangle = i|1111\rangle, \quad (19.733)$$

since $h(1100) = 0$ and $h(0011) = 1$. By Theorem 19.8.3, the input defect vectors are $-i|1100\rangle$ and $-i|0011\rangle$, while the target vectors are $|0000\rangle$ and $|1111\rangle$. Multiplying (19.732) and (19.733) by $-i$ gives the claimed signs. $\square$

Definition 19.8.10 (Prescribed defect maps of the CZX model). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The four domains (19.737) and (19.738) together with the four maps (19.739) are prescribed defect maps in the sense of Definition 19.5.10.17 for the group $\mathbb{Z}_2$ and the two-site matter space of the CZX model, the isometry condition (19.362) being Theorem 19.8.9. That the circuit tuple R^{CZX} lies in the resulting completion class (19.363) is proved unconditionally in Theorem 19.9.1 below.

19.9 The displayed CZX tuple as a completion

The four defect domains and four prescribed maps of Section 19.8 constrain each fusion operator on one plane of the two-site matter space. The operators $(\mathbb{1}, \mathbb{1}, w, \tilde{\lambda})$ of Definition 19.7.10.1 satisfy these constraints, so the displayed tuple belongs to the completion class of the CZX prescribed defect maps D^{CZX} , and (19.707) lower-bounds the supremum of the gauge-invariant dimension over the full class.

Theorem 19.9.1 (The displayed CZX tuple is a completion). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every ordered pair (a, b) of labels of $\mathbb{Z}_2$ and every injective block x , the displayed operator of the pair carries the defect vector of the pair to the defect vector of the product label,

$$U_{a,b}^{\text{CZX}} v(a, b, x) = v(e, ab, x), \quad (19.745)$$

that is,

$$\mathbb{1} v(e, b, x) = v(e, b, x), \quad (19.746)$$

$$w v(g, e, x) = v(e, g, x), \quad (19.747)$$

$$\tilde{\lambda} v(g, g, x) = v(e, e, x). \quad (19.748)$$

Consequently the displayed operator of every pair agrees on $\mathcal{K}_{a,b}$ with the prescribed map $D_{a,b}$, and the displayed tuple lies in the full completion class,

$$R^{\text{CZX}} \in \mathfrak{C}(D^{\text{CZX}}). \quad (19.749)$$

Proof. [✓ Proof](#) The phase tables (19.689) and (19.691), evaluated at the four computational basis states carrying the defect vectors of Theorem 19.8.3, give

$$w|0000\rangle = |1100\rangle, \quad w|1111\rangle = |0011\rangle, \quad (19.750)$$

$$\tilde{\lambda}|1100\rangle = i|0000\rangle, \quad \tilde{\lambda}|0011\rangle = i|1111\rangle, \quad (19.751)$$

because $e(0000) = e(1111) = 0$ and $f(1100) = f(0011) = 1$. Substituting (19.724)–(19.727) and cancelling the factors $-i$ gives (19.747) and (19.748); the pairs with left label e carry the identity operator, so (19.746) is immediate. The two defect vectors of a pair span $\mathcal{K}_{a,b}$ by (19.734), and the prescribed map obeys the same identity (19.740), so the displayed operator and the prescribed map agree on the whole plane, which is (19.749). $\square$

Theorem 19.9.2 (Lower bound on the supremum over the CZX completion class). [Lean](#) [Lean](#) [✓ Stmt](#) On a periodic chain of $N \geq 3$ blocked sites the gauge-invariant dimensions of the members of the CZX completion class are bounded above by the dimension of the chain space, so their supremum is finite, and

$$\sup_{R \in \mathfrak{C}(D^{\text{CZX}})} \dim \mathcal{V}_N(R) \geq 2^N. \quad (19.752)$$

Nothing is asserted about the minimum of $\dim \mathcal{V}_N(R)$ over the class, about the exact value of the supremum, or about which completion attains it. Those extrema are questions about this completion class, not questions the source raises; [163, lines 5198–5204] asks only whether the gauge-invariant subspace of the state-level gauging stays bounded or grows with the system size, and nothing is asserted about that growth either. The domains and the maps entering the class are derived in [41].

Proof. ✓ Proof Every gauge-invariant subspace (19.383) is a subspace of the chain space, so the dimensions of the members of the class are bounded above by the dimension of that space and their supremum is finite. By Theorem 19.9.1 the displayed tuple lies in the class, and by Theorem 19.7.11.7 its gauge-invariant subspace has dimension 2^N , which is therefore at most the supremum. $\square$

Theorem 19.9.3 (The second displayed CZX tuple is a completion). Lean Lean Lean Lean ✓ Stmt *The second displayed tuple obeys the same prescription (19.745) as the first,*

$$U_{a,b}^* v(a, b, x) = v(e, ab, x), \quad (19.753)$$

and therefore lies in the full completion class,

$$R^* \in \mathfrak{C}(D^{\text{CZX}}). \quad (19.754)$$

Proof. ✓ Proof The pairs (e, e) , (e, g) , and (g, e) carry the same operators in R^* as in R^{CZX} , so Theorem 19.9.1 gives (19.753) there. For the pair (g, g) the operator is $\tilde{\lambda}R_*$, and the defect domain is $\mathcal{K}_{g,g} = \mathcal{L}_1 = \text{span}_{\mathbb{C}}\{|1100\rangle, |0011\rangle\}$ by (19.738). Since $r_*(1100) = r_*(0011) = 0$, the factor R_* fixes both spanning vectors, so $\tilde{\lambda}R_*$ has the same values there as $\tilde{\lambda}$, namely $i|0000\rangle$ and $i|1111\rangle$. The two defect vectors of the pair span its domain, which gives (19.754). $\square$

Theorem 19.9.4 (Two further values of the dimension on the CZX completion class). Lean Lean Lean ✓ Stmt *The dimensions 14 at three sites and 36 at four sites are values of the gauge-invariant dimension on the CZX completion class, so*

$$\sup_{R \in \mathfrak{C}(D^{\text{CZX}})} \dim \mathcal{V}_3(R) \geq 14, \quad (19.755)$$

$$\sup_{R \in \mathfrak{C}(D^{\text{CZX}})} \dim \mathcal{V}_4(R) \geq 36. \quad (19.756)$$

Beside the values 8 and 16 of the first displayed tuple at the same two chain lengths, these exhibit two different values of the dimension on the class at each of those lengths. Nothing follows about the minimum, the exact supremum, which completion attains it, or the behaviour of any of these as N grows.

Proof. ✓ Proof By Theorem 19.9.3 the second displayed tuple lies in the class, and by Theorem 19.7.12.7 its gauge-invariant subspace has dimensions 14 and 36 at three and four sites. Each value is therefore at most the supremum over the class at that chain length, by the same boundedness argument as in Theorem 19.9.2. $\square$

Chapter 20

Mixed States: Renormalization of Matrix Product Operators

Building on the matrix product density operators of Chapter 16, renormalization fixed points and zero-correlation-length conditions characterize mixed-state chains [11, Section 4]. Local channels govern the fixed-point equation, the saturation of the area law for mutual information, the structure of simple and commuting fixed points, and the induced boundary theory for two-dimensional PEPS. The general algebraic classification continues in Chapter 21.

20.1 Preliminaries for physical renormalization

The physical renormalization of matrix product density operators rests on channel extensions, projector-controlled positive sums, and physical closures of virtual contractions.

Theorem 20.1.1 (Multiplicativity from reflected short contractions). Lean ✓ Stmt *Let A, B be complex algebras with involutions, and let $V : B \rightarrow A$ be complex linear and preserve the involutions. Let $\tau_A : A \rightarrow \mathbb{C}$ and $\tau_B : B \rightarrow \mathbb{C}$ be complex linear, with $\tau_A(a^*a) = 0$ implying $a = 0$. Suppose, for every $x, y, z \in B$, that*

$$\begin{aligned}\tau_A(VxVy) &= \tau_B(xy), \\ \tau_A(VxVyVz) &= \tau_B(xyz), \\ \tau_A((Vy)^*(Vx)^*VxVy) &= \tau_B(y^*x^*xy).\end{aligned}$$

Then $V(xy) = VxVy$ for every $x, y \in B$. No finite-dimensionality, positivity, traciality, or unitality of V is required beyond the stated quadratic faithfulness.

Proof. ✓ Proof Put $d = VxVy - V(xy)$. Expanding $\tau_A(d^*d)$ gives four terms. The reflected four-point identity evaluates the first, the three-point identity evaluates the two cross terms, and the two-point identity evaluates the last. After preserving involutions and associating products, their values are all $\tau_B(y^*x^*xy)$, with coefficients 1, -1 , -1 , 1. Hence $\tau_A(d^*d) = 0$, and quadratic faithfulness gives $d = 0$. □

This is a finite-moment criterion for an already specified involution on the compared algebras. In a boundary reconstruction, preservation of the oriented involution must be proved separately before applying the criterion; preservation of the physical adjoint alone does not supply it.

Theorem 20.1.2 (Tensor extension preserves trace-preserving complete positivity). [Lean](#) [Lean](#)

[✓ Stmt](#)

Let $\mathcal{S} : \text{End}_{\mathbb{C}}(H) \rightarrow \text{End}_{\mathbb{C}}(K)$ be trace-preserving and completely positive, and let R be another finite-dimensional space. Then

$$\begin{aligned}\mathcal{S} \otimes \text{id}_R &: \text{End}_{\mathbb{C}}(H \otimes R) \longrightarrow \text{End}_{\mathbb{C}}(K \otimes R), \\ \text{id}_R \otimes \mathcal{S} &: \text{End}_{\mathbb{C}}(R \otimes H) \longrightarrow \text{End}_{\mathbb{C}}(R \otimes K)\end{aligned}$$

are trace-preserving and completely positive.

Proof. [✓ Proof](#) The first assertion is the tensor-extension closure theorem. For the second, let $\tau_H : R \otimes H \simeq H \otimes R$ and $\tau_K : K \otimes R \simeq R \otimes K$ be the canonical factor swaps. Then

$$\text{id}_R \otimes \mathcal{S} = \mathcal{R}_{\tau_K} \circ (\mathcal{S} \otimes \text{id}_R) \circ \mathcal{R}_{\tau_H},$$

where $\mathcal{R}_{\tau}$ denotes reindexing both matrix coordinates along τ . Both reindexing maps are trace-preserving and completely positive, so closure under composition proves the claim. $\square$

Lemma 20.1.3 (Compressed sums of completely positive maps). [Lean](#) [✓ Stmt](#)

Let $(P_i)_i$ be finitely many operators on H and let each $\Phi_i : \text{End}_{\mathbb{C}}(H) \rightarrow \text{End}_{\mathbb{C}}(K)$ be completely positive. Then

$$X \mapsto \sum_i \Phi_i(P_i X P_i^{\dagger})$$

is completely positive. No completeness or orthogonality condition on the P_i is required.

Proof. [✓ Proof](#)

Composing each Kraus family of Φ_i with the corresponding P_i gives one combined rectangular Kraus family representing the sum. $\square$

Lemma 20.1.4 (Trace selection by projector-controlled sums). [Lean](#) [✓ Stmt](#)

Let $(P_i)_i$ be finitely many orthogonal projections on H , not necessarily resolving the identity, and let each $\Phi_i : \text{End}_{\mathbb{C}}(H) \rightarrow \text{End}_{\mathbb{C}}(K)$ be trace-preserving and completely positive. Then, for every X ,

$$\text{tr}\left(\sum_i \Phi_i(P_i X P_i)\right) = \text{tr}\left(\left(\sum_i P_i\right)X\right).$$

Proof. [✓ Proof](#) Trace preservation of each Φ_i reduces the left side to $\sum_i \text{tr}(P_i X P_i)$; cyclicity of the trace and idempotence of each projection give $\text{tr}(P_i X P_i) = \text{tr}(P_i X)$. $\square$

The state-preparation map $\mathcal{P}_{\rho}(X) = X \otimes \rho$ from operators on A to operators on $A \otimes B$, used in the maps $\mathcal{T}_1$ and $\mathcal{S}_1$ of [11, Appendix C.2, lines 1527–1533 and 1551–1555], is stated in [182, “State-preparation map”].

Definition 20.1.5 (Physical closure maps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For an MPO tensor M , a length N , and a virtual operator X , define the physical operator $M_N(X)$ by

$$M_N(X)_{i,j} = \text{tr}(M^{i_0 j_0} M^{i_1 j_1} \dots M^{i_{N-1} j_{N-1}} X).$$

This is linear in X . The displayed order chooses the cut of the cyclic virtual contraction immediately before the first site, so that X follows the last site. The one- and two-site operators are constructed in [11, lines 638–654 and Definition 4.1]. When $M = \mathcal{K}$, write these operators as $\mathcal{K}_N(X)$; the cases $N = 2, 3$ occur in [11, Proposition C.7, lines 1510–1516].

Lemma 20.1.6 (Periodic closure). [Lean](#) [✓ Stmt](#) Closing the virtual legs with the identity gives the periodic MPO operator: $M_N(1) = \rho^{(N)}(M)$.

Proof. [✓ Proof](#) For every pair of physical words σ, τ ,

$$M_N(1)_{\sigma, \tau} = \text{tr}(M^{\sigma, \tau} \cdot 1) = \text{tr}(M^{\sigma, \tau}) = \rho^{(N)}(M)_{\sigma, \tau}.$$

□

Theorem 20.1.7 (One- and two-site physical closures). [Lean](#) [Lean](#) [✓ Stmt](#) Under the canonical identifications of one-site configurations with physical indices and two-site configurations with pairs of physical indices,

$$\begin{aligned} M_1(X)_{i,j} &= \text{tr}(M^{ij} X), \\ M_2(X)_{(i_0, i_1), (j_0, j_1)} &= \text{tr}(M^{i_0 j_0} M^{i_1 j_1} X). \end{aligned}$$

Proof. [✓ Proof](#) Under these identifications, the length-one and length-two word evaluations are M^{ij} and $M^{i_0 j_0} M^{i_1 j_1}$, respectively. □

Lemma 20.1.8 (Three-site physical closure). [Lean](#) [Lean](#) [✓ Stmt](#) The general three-site physical closure has coefficients

$$M_3(X)_{(i_0, (i_1, i_2)), (j_0, (j_1, j_2))} = \text{tr}(M^{i_0 j_0} M^{i_1 j_1} M^{i_2 j_2} X).$$

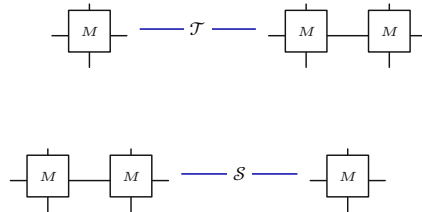
For $M = \mathcal{K}$, this is the operator $\mathcal{K}_3(X)$ in [11, Proposition C.7, lines 1510–1516].

Proof. [✓ Proof](#) The length-three word evaluation is $M^{i_0 j_0} M^{i_1 j_1} M^{i_2 j_2}$. □

20.2 Renormalization fixed points

In the mixed-state renormalization fixed point of [11, Definition 4.1], two local channels extend to periodic chains of arbitrary length by acting on the first one or two sites and leaving the remaining sites unchanged.

Definition 20.2.1 (Renormalization fixed point via trace-preserving maps). [Lean](#) [✓ Stmt](#) An MPO tensor M is a *renormalization fixed point* (RFP) if there exist trace-preserving completely positive maps $\mathcal{S}, \mathcal{T}$ on the physical indices such that $\mathcal{S}[M_2(X)] = M_1(X)$ and $\mathcal{T}[M_1(X)] = M_2(X)$ for every virtual operator X , where $M_1(X)_{ij} = \text{tr}(M^{ij} X)$ and $M_2(X)_{(i_1 i_2)(j_1 j_2)} = \text{tr}(M^{i_1 j_1} M^{i_2 j_2} X)$ are the one- and two-site physical operators obtained by closing one or two tensors with X . In [11, Definition 4.1], this condition is formulated for tensors in canonical form; the zero-correlation-length consequence concerns the physical-trace transfer (Definition 20.6.1) rather than idempotence of the doubled-index transfer map. On the open coefficient tensors the two maps have types:



Theorem 20.2.2 (Pure-state specialization of the physical-map fixed point). [Lean](#) [Lean](#) [✓ Stmt](#)
Let $A = (A^i)_{i=0}^{d-1}$ be an MPS tensor, and let $M(A)^{ij} = A^i \otimes \overline{A^j}$ be its doubled MPO tensor. Then

$$\begin{aligned} M(A) \text{ satisfies Definition 20.2.1} &\iff \mathcal{E}_A^2 = \mathcal{E}_A \\ &\iff A \text{ has a physical blocking isometry.} \end{aligned}$$

More explicitly, if $V: \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$ is the blocking isometry, the refinement channel is given by

$$\mathcal{T}(X) = VXV^\dagger.$$

If $d > 0$, put $P = VV^\dagger$ and choose a density matrix ρ_0 on the one-site space. A trace-preserving coarse-graining channel is

$$\mathcal{S}(Y) = V^\dagger Y V + \text{tr}((\mathbb{1} - P)Y(\mathbb{1} - P))\rho_0.$$

On the range of V , the second term vanishes and this agrees with the source construction $\mathcal{S}(Y) = \text{tr}_2(U^\dagger Y U)$ after extending V to a unitary with a fixed pure ancilla. If $d = 0$, both matrix spaces are trivial and the unique zero maps give $\mathcal{S}$ and $\mathcal{T}$. This is the pure-state specialization stated in [11, lines 668, 675–687].

Proof. [✓ Proof](#) The blocking relation $A^{i_1} A^{i_2} = \sum_j V_{(i_1, i_2), j} A^j$ implies

$$M_2(X) = V M_1(X) V^\dagger.$$

For $d > 0$, the refinement identity follows directly:

$$\mathcal{T}(M_1(X)) = V M_1(X) V^\dagger = M_2(X).$$

For the coarse-graining identity, put $P = VV^\dagger$. Since $V^\dagger V = \mathbb{1}$ and $(\mathbb{1} - P)V = 0$, substitution gives

$$\begin{aligned} \mathcal{S}(M_2(X)) &= V^\dagger (V M_1(X) V^\dagger) V + \text{tr}((\mathbb{1} - P)(V M_1(X) V^\dagger)(\mathbb{1} - P))\rho_0 \\ &= M_1(X) + 0 = M_1(X). \end{aligned}$$

If $d = 0$, the unique zero channels satisfy both identities. Conversely, trace preservation of $\mathcal{T}$ gives idempotence of the physical-trace transfer of $M(A)$, as in [11, Appendix C, lines 1333–1340]. After exchanging the two tensor factors, this matrix is the transfer matrix of $\mathcal{E}_A$. Faithfulness of the transfer-matrix representation therefore gives $\mathcal{E}_A^2 = \mathcal{E}_A$. The physical-isometry formulation follows from Theorem 17.1.6. $\square$

Definition 20.2.3 (Fibonacci vacuum-sector support tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Fibonacci vacuum-sector support tensor [11, Appendix D, lines 2116–2121] has physical labels given by triples $i, j, k \in \{0, 1\}$ and virtual labels $\alpha, \beta \in \{0, 1\}$. The fusion weights satisfy

$$N_{ijk} = 0 \iff (i, j, k) \in \{(0, 0, 1), (0, 1, 0), (1, 0, 0)\}. \quad (20.1)$$

Every weight outside this zero set is positive, and

$$A_{\alpha\beta}^{ijk} = \delta_{i,\alpha} \delta_{k,\beta} N_{ijk}.$$

The canonical support tensor takes every nonzero N_{ijk} to be one.

Definition 20.2.4 (Fibonacci transition matrix). [Lean](#) [✓ Stmt](#) The transition matrix from [11, Appendix D, lines 2146–2149] is

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Definition 20.2.5 (Fibonacci boundary transition counts). [Lean](#) [✓ Stmt](#) For $n \in \mathbb{N}$ and $\alpha, \beta \in \{0, 1\}$, define

$$x_{\alpha\beta}^n = (B^n)_{\alpha\beta}. \quad (20.2)$$

These are the transition counts governed by the recurrence in [11, Appendix D, lines 2131–2149].

Theorem 20.2.6 (Fibonacci boundary transition recurrence). [Lean](#) [Lean](#) [✓ Stmt](#) In the order $(00, 01, 10, 11)$, one has $x^1 = (1, 1, 1, 2)$ and

$$x_{\alpha 0}^{n+1} = x_{\alpha 0}^n + x_{\alpha 1}^n, \quad x_{\alpha 1}^{n+1} = x_{\alpha 0}^n + 2x_{\alpha 1}^n.$$

This is the recurrence in [11, Appendix D, lines 2131–2145].

Proof. [✓ Proof](#) Multiply the row indexed by α in B^n by the two columns of B . □

Definition 20.2.7 (Periodic Fibonacci transition count). [Lean](#) [✓ Stmt](#) Closing the two boundary labels gives the transition count

$$a_N = x_{00}^N + x_{11}^N = \text{tr}(B^N).$$

This is the boundary-label closure used in [11, Appendix D, line 2157].

Theorem 20.2.8 (Characteristic recurrence for periodic Fibonacci transition counts). [Lean](#) [✓ Stmt](#) For every $N \in \mathbb{N}$,

$$a_{N+2} + a_N = 3a_{N+1}.$$

Proof. [✓ Proof](#) Expand $B^{N+2} = B^{N+1}B$ and $B^{N+1} = B^N B$. Substitution of $B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ into the diagonal sums gives $a_{N+2} + a_N = 3a_{N+1}$. □

Theorem 20.2.9 (First periodic Fibonacci transition counts). [Lean](#) [Lean](#) [✓ Stmt](#) The first two periodic transition counts are $a_1 = 3$ and $a_2 = 7$.

Proof. [✓ Proof](#) Substitute $x^1 = (1, 1, 1, 2)$ into the recurrence once and close the two diagonal boundary labels. □

Theorem 20.2.10 (Golden-ratio formula for periodic Fibonacci transition counts). [Lean](#) [✓ Stmt](#) Set $\tau_+ = (1 + \sqrt{5})/2$ and $\tau_- = (1 - \sqrt{5})/2$. For every $N \in \mathbb{N}$,

$$a_N = \tau_+^{2N} + \tau_-^{2N}.$$

This is the transition-count formula in [11, Appendix D, lines 2146–2176].

Proof. [✓ Proof](#) Use two-step induction on N . The cases $N = 0$ and $N = 1$ follow by direct calculation. Since $\tau_{\pm}^2 = \tau_{\pm} + 1$, each sequence $\tau_{\pm}^{2N}$ satisfies $\tau_{\pm}^{2(N+2)} + \tau_{\pm}^{2N} = 3\tau_{\pm}^{2(N+1)}$. The induction step now follows from the same recurrence for a_N . □

Theorem 20.2.11 (Periodic Fibonacci transition counts are not geometric). [Lean](#) [✓ Stmt](#) *There are no $r, s \in \mathbb{N}$ such that $a_N = rs^{N-1}$ for every $N > 0$. This is the non-geometricity assertion at [11, Appendix D, line 2125] for the transition-count sequence.*

Proof. [✓ Proof](#) Such a formula would give $r = a_1 = 3$ and $7 = a_2 = 3s$, which is impossible in $\mathbb{N}$. $\square$

Definition 20.2.12 (Open-boundary Fibonacci operator). [Lean](#) [Lean](#) [✓ Stmt](#) For $n > 0$ and $\alpha, \beta \in \{0, 1\}$, define the matrix $\rho_{\alpha\beta}^{(n)}(A)$, indexed by physical words σ and τ , by

$$(\rho_{\alpha\beta}^{(n)}(A))_{\sigma,\tau} = \begin{cases} (\alpha|A^{\sigma_1} \dots A^{\sigma_n}|\beta), & \sigma = \tau, \\ 0, & \sigma \neq \tau. \end{cases} \quad (20.3)$$

Theorem 20.2.13 (Open-boundary Fibonacci operator ranks). [Lean](#) [✓ Stmt](#) For $n > 0$ and $\alpha, \beta \in \{0, 1\}$,

$$\text{rank } \rho_{\alpha\beta}^{(n)}(A) = x_{\alpha\beta}^n. \quad (20.4)$$

This is [11, Appendix D, lines 2127–2130].

Proof. [✓ Proof](#) A diagonal entry is nonzero exactly when its physical labels form an allowed path from α to β . Splitting a path at its first physical label shows that the numbers of supported words obey the transition rule determined by B , with the same length-zero initial values as $x_{\alpha\beta}^n$. Induction therefore identifies the number of nonzero diagonal entries with $x_{\alpha\beta}^n$. The rank of a diagonal matrix is the number of its nonzero diagonal entries. $\square$

Theorem 20.2.14 (Fibonacci periodic operator-rank formula). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be obtained from any positive choice of fusion weights with the support prescribed in Definition 20.2.3. Set $\tau_{\pm} = (1 \pm \sqrt{5})/2$, as in [11, Appendix D, line 2125]. For every $N > 0$,

$$\text{rank } \rho^{(N)}(A) = a_N = x_{00}^N + x_{11}^N = \tau_+^{2N} + \tau_-^{2N}. \quad (20.5)$$

This is [11, Appendix D, lines 2122–2125 and 2157–2176].

Proof. [✓ Proof](#) The diagonal MPO rank counts physical words with nonzero periodic amplitude. For $N > 0$, the supports obtained by closing the virtual endpoint at 0 and at 1 are disjoint, since the first physical bit determines the initial endpoint. Their cardinalities are x_{00}^N and x_{11}^N by the open-boundary formula. The golden-ratio identity for a_N gives the final equality. $\square$

Theorem 20.2.15 (Fibonacci periodic operator ranks are not geometric). [Lean](#) [✓ Stmt](#) Let A be obtained from any positive choice of fusion weights with the support prescribed in Definition 20.2.3. There are no $r, s \in \mathbb{N}$ such that $\text{rank } \rho^{(N)}(A) = rs^{N-1}$ for every $N > 0$. This is the operator-level conclusion in [11, Appendix D, line 2125].

Proof. [✓ Proof](#) The operator rank equals a_N at every positive length, while $a_1 = 3$ and $a_2 = 7$. A geometric formula would therefore give $r = 3$ and $7 = 3s$, which is impossible in $\mathbb{N}$. $\square$

Theorem 20.2.16 (The Fibonacci boundary tensor is not a strong fixed point). [Lean](#) [✓ Stmt](#) Let A be obtained from any positive choice of fusion weights with the support prescribed in Definition 20.2.3. Then its canonical diagonal MPO tensor is not a strong renormalization fixed point.

Proof. ✓ Proof Suppose otherwise, and let P be the positive witness for the strong fixed-point relation. Then, for every $N > 0$, the periodic ranks satisfy

$$\text{rank } \rho^{(N)}(A) = \text{rank } \rho^{(1)}(A)(\text{rank } P)^{N-1}. \quad (20.6)$$

(20.6) contradicts the non-geometricity of the Fibonacci periodic ranks. □

Definition 20.2.17 (Initial-coordinate regroupings). Lean Lean ✓ Stmt Let R_1 identify an $(N+1)$ -tuple with its first coordinate and the remaining N -tuple. Let R_2 identify an $(N+2)$ -tuple with its first two coordinates and the remaining N -tuple.

Definition 20.2.18 (Renormalization maps on a longer ring). Lean Lean ✓ Stmt Let N count the sites that are left unchanged. For physical maps $\mathcal{T} : M_d \rightarrow M_{d^2}$ and $\mathcal{S} : M_{d^2} \rightarrow M_d$, define

$$\begin{aligned} \widehat{\mathcal{T}}_N &= \mathcal{T} \otimes \text{id}_{d^N}, \\ \widehat{\mathcal{S}}_N &= \mathcal{S} \otimes \text{id}_{d^N}, \end{aligned}$$

using R_1 and R_2 . Thus $N = 0$ leaves no spectator site, but the maps still act on a one-site or two-site operator; no empty ring occurs.

Lemma 20.2.19 (Regrouping the first physical sites). Lean Lean Lean Lean ✓ Stmt For a physical word σ of length $N + 1$,

$$\begin{aligned} R_1(\sigma) &= (\sigma_0, (\sigma_1, \dots, \sigma_N)), \\ R_1^{-1}(i, (a_1, \dots, a_N)) &= (i, a_1, \dots, a_N). \end{aligned}$$

For a physical word σ of length $N + 2$,

$$\begin{aligned} R_2(\sigma) &= ((\sigma_0, \sigma_1), (\sigma_2, \dots, \sigma_{N+1})), \\ R_2^{-1}((i, j), (a_1, \dots, a_N)) &= (i, j, a_1, \dots, a_N). \end{aligned}$$

Proof. ✓ Proof All four identities hold by definition of the product-index equivalences. □

Lemma 20.2.20 (Localized renormalization maps are channels). Lean Lean ✓ Stmt If $\mathcal{S}$ and $\mathcal{T}$ are trace-preserving completely positive, then so are $\widehat{\mathcal{S}}_N$ and $\widehat{\mathcal{T}}_N$ for every N .

Proof. ✓ Proof If $(K_j)_j$ are Kraus operators for $\mathcal{T}$ with $\sum_j K_j^\dagger K_j = I_d$, then the localized map has Kraus operators $K_j \otimes I_{d^N}$, and

$$\sum_j (K_j \otimes I_{d^N})^\dagger (K_j \otimes I_{d^N}) = \left(\sum_j K_j^\dagger K_j \right) \otimes I_{d^N} = I_d \otimes I_{d^N}.$$

The two regroupings are unitary conjugations and preserve complete positivity and the trace. The argument for $\mathcal{S}$ is identical. □

Theorem 20.2.21 (Global action of the renormalization maps). Lean Lean Lean Lean Lean ✓ Stmt Suppose that $\mathcal{S}[M_2(X)] = M_1(X)$ and $\mathcal{T}[M_1(X)] = M_2(X)$ for every virtual operator X . Then, for every N and X ,

$$\begin{aligned} \widehat{\mathcal{S}}_N[M_{N+2}(X)] &= M_{N+1}(X), \\ \widehat{\mathcal{T}}_N[M_{N+1}(X)] &= M_{N+2}(X). \end{aligned}$$

In particular, these identities hold for every periodic MPO operator. Thus the renormalization channels act on periodic rings of arbitrary length [11, Appendix C, lines 1333–1341].

Proof. ✓ Proof For words u, v on the N spectator sites, write $M^{u,v}$ for their virtual word evaluation and write $[Y]_{u,v}^{(r)}$ for the block of an operator Y on the first r sites. Regrouping the first site gives

$$[M_{N+1}(X)]_{u,v}^{(1)} = M_1(M^{u,v}X).$$

Therefore

$$\mathcal{T}([M_{N+1}(X)]_{u,v}^{(1)}) = M_2(M^{u,v}X) = [M_{N+2}(X)]_{u,v}^{(2)}.$$

Equality of all blocks proves the refinement identity. Interchanging M_1, M_2 and using $\mathcal{S}[M_2(Y)] = M_1(Y)$ proves the coarsening identity. Setting $X = 1$ and using $M_N(1) = \rho^{(N)}(M)$ (Lemma 20.1.6) gives the periodic-ring identities

$$\begin{aligned}\widehat{\mathcal{T}}_N[\rho^{(N+1)}(M)] &= \rho^{(N+2)}(M), \\ \widehat{\mathcal{S}}_N[\rho^{(N+2)}(M)] &= \rho^{(N+1)}(M).\end{aligned}$$

□

Theorem 20.2.22 (One-site positivity at a renormalization fixed point). Lean Lean ✓ Stmt Suppose that the one-site and two-site closures of M are related by the two channels of Definition 20.2.1. Then $\rho^{(N)}(M) \succeq 0$ at every positive length if and only if $\rho^{(1)}(M) \succeq 0$.

Proof. ✓ Proof One direction follows by taking $N = 1$. For the other, apply the localized refinement channel repeatedly. By Theorem 20.2.21, it sends $\rho^{(N)}(M)$ to $\rho^{(N+1)}(M)$, and complete positivity preserves each matrix inequality. □

20.3 Pure-state recovery inside the MPO formalism

Definition 20.3.1 (Diagonal pure-state embedding as an MPO). Lean ✓ Stmt An MPS tensor $A = \{A^i\}_{i=0}^{d-1}$ determines an MPO tensor by placing A^i on the diagonal in the bra and ket indices:

$$M^{ij} = \delta_{ij} A^i.$$

Equivalently, $M^{ii} = A^i$ and $M^{ij} = 0$ for $i \neq j$.

Theorem 20.3.2 (Operator and rank of the diagonal pure-state embedding). Lean Lean Lean ✓ Stmt For every N , the embedded MPO is diagonal with diagonal $V^{(N)}(A)$, and hence

$$\rho^{(N)}(A^{\text{MPO}}) = \text{diag } V^{(N)}(A), \quad \text{rank } \rho^{(N)}(A^{\text{MPO}}) = \#\{\sigma : V^{(N)}(A)_\sigma \neq 0\}.$$

Proof. ✓ Proof A ket–bra word evaluation vanishes at the first unequal pair of physical labels. Equal words recover the original MPS word evaluation. The rank formula follows by counting the nonzero entries of a diagonal matrix. □

Theorem 20.3.3 (Transfer map of the diagonal pure-state embedding). Lean ✓ Stmt The transfer map of the diagonal MPO associated to A agrees with the original MPS transfer map: $\mathcal{E}_{A^{\text{MPO}}} = \mathcal{E}_A$.

Proof. ✓ Proof In the double sum defining $\mathcal{E}_{A^{\text{MPO}}}$, all off-diagonal terms vanish because $M^{ij} = 0$ for $i \neq j$, while the diagonal terms are exactly $A^i X (A^i)^\dagger$. Thus only the sum over i remains, which is the defining formula for $\mathcal{E}_A$. □

Theorem 20.3.4 (Pure-state recovery of transfer-map idempotence). [Lean](#) [✓ Stmt](#) For an MPS tensor A , the diagonal MPO embedding has idempotent transfer map if and only if the MPS transfer map is idempotent:

$$\mathcal{E}_{A^{\text{MPO}}} \circ \mathcal{E}_{A^{\text{MPO}}} = \mathcal{E}_{A^{\text{MPO}}} \iff \mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A.$$

Proof. [✓ Proof](#) Both conditions assert idempotence of the corresponding transfer map. Theorem 20.3.3 identifies those transfer maps, so the two predicates are equivalent. $\square$

Theorem 20.3.5 (Pure-state recovery of zero correlation length). [Lean](#) [✓ Stmt](#) For an MPS tensor A , the diagonal MPO embedding has idempotent transfer map if and only if A has zero correlation length.

Proof. [✓ Proof](#) Combine Theorem 20.3.4 with the characterization of zero correlation length by idempotence of the MPS transfer map (Theorem 17.5.20). $\square$

20.4 PEPS boundary tensors

For physical dimension d , the local contraction admits separate horizontal virtual dimension D_h and vertical virtual dimension D_v .

Definition 20.4.1 (Local PEPS boundary tensor). [Lean](#) [✓ Stmt](#) Let $A_{l,r,u,b}^s$ be a rank-five component function, with physical index $0 \leq s < d$, horizontal virtual indices $0 \leq l, r < D_h$, and vertical virtual indices $0 \leq u, b < D_v$. The same component function is used at every lattice site. Contracting both its physical leg and its inward vertical leg between a ket copy and a bra copy defines an MPO tensor E_A with physical dimension D_v and bond dimension D_h^2 .

This local ket–bra contraction corresponds to the diagram in [11, lines 721–724]. Under the PEPS renormalization hypothesis [11, lines 716–724], this contraction represents the boundary tensor, with the common contracted inward leg denoted by b .

Theorem 20.4.2 (Component formula for the local boundary tensor). [Lean](#) [Lean](#) [✓ Stmt](#) With b denoting the contracted inward vertical leg,

$$(E_A)^{u,u'}_{(l,l'),(r,r')} = \sum_{s=0}^{d-1} \sum_{b=0}^{D_v-1} A_{l,r,u,b}^s \overline{A_{l',r',u',b}^s}. \quad (20.7)$$

The physical coordinates are the remaining ket and bra indices u, u' , and the bond coordinates are (l, l') and (r, r') . Legs are ordered as left, right, up, down, physical, with horizontal ket–bra indices grouped into a single bond coordinate.

Proof. [✓ Proof](#) Expand the contractions over the common physical and inward vertical indices, and decode the two packed bond coordinates. This gives (20.7). $\square$

Theorem 20.4.3 (The local PEPS boundary tensor is an LPDO). [Lean](#) [✓ Stmt](#) The local boundary tensor E_A is a locally purifiable density operator. One may take ancillary dimension dD_v with coordinate (s, b) , purifying bond dimension D_h , and purifying matrices

$$B_{l,r}^{(u;(s,b))} = A_{l,r,u,b}^s.$$

Under the PEPS renormalization hypothesis [11, lines 716–724], this contraction represents the boundary tensor. The Gram factorization exhibits E_A as an LPDO for every rank-five tensor A , independently of any transfer fixed point or renormalization channel.

Proof. ✓ Proof Identify each MPO bond coordinate with a pair of horizontal PEPS virtual coordinates. Substituting the displayed purifying matrices into the LPDO contraction and summing over the ancillary pair (s, b) gives exactly (20.7). $\square$

Corollary 20.4.4 (The local PEPS boundary tensor is an MPDO). Lean ✓ Stmt *For every nonempty periodic chain, the matrix product operator generated by E_A is positive semidefinite.*

Proof. ✓ Proof Apply the general implication from local purification to matrix product density operators. $\square$

Definition 20.4.5 (Boundary fixed point). Not ready Discussion: #7371 The row-to-row transfer operator of a homogeneous PEPS is represented by a strip of local ket-bra transfer tensors. A boundary fixed point is a left or right boundary operator invariant under the corresponding transfer action. The transfer-operator diagram at source lines 709–712 of [11] depicts the operator from which the fixed point is selected; it does not depict the fixed point itself. On a finite cylinder, [13, Section II.2, Equations (10)–(13)] defines the positive boundary operators σ_L, σ_R and the support isometry U , with $\rho_\ell = U \sqrt{\sigma_L^T \sigma_R} \sqrt{\sigma_L^T} U^\dagger$. For a long cylinder, [202, Equation (1) and the following paragraph] identifies σ_L, σ_R with eigenvectors for the largest eigenvalue of the transfer operator and notes that topological degeneracy can make the fixed point non-unique. These constructions specify the boundary operators in their respective cylinder settings, but not their passage to the local boundary tensor and channels.

Definition 20.4.6 (Algebraic PEPS renormalization fixed-point relation). Not ready Discussion: #7371 The PEPS renormalization diagram at source lines 716–720 of [11] identifies two neighboring tensors with one effective local tensor, up to an isometry on their physical indices. An algebraic definition must specify the domain and codomain of that isometry and the identifications of the external virtual spaces.

Definition 20.4.7 (Boundary coarse-graining channel). Not ready Discussion: #7371 A coarse-graining channel $\mathcal{S}$ acts from two boundary sites to one and satisfies $\mathcal{S}[M_2(X)] = M_1(X)$ for every virtual operator X . The coarse-graining map in the diagram at source lines 725–729 of [11] must be defined on the full two-site boundary algebra and proved completely positive and trace-preserving, not merely on the image selected by the PEPS isometry.

Definition 20.4.8 (Boundary refinement channel). Not ready Discussion: #7371 A refinement channel $\mathcal{T}$ acts from one boundary site to two and satisfies $\mathcal{T}[M_1(X)] = M_2(X)$ for every virtual operator X . It is obtained in the diagram at source lines 725–729 of [11] by inserting the displayed local factor. The factor and its normalization must be specified so that the map on the full one-site boundary algebra is completely positive and trace-preserving.

Theorem 20.4.9 (PEPS renormalization induces a boundary MPO fixed point). Not ready Discussion: #7371 *Suppose a homogeneous PEPS satisfies the neighboring blocking relation at source lines 716–720 of [11], and let its transfer fixed points determine the boundary MPO displayed at source lines 721–724. Then that boundary MPO is a renormalization fixed point in the sense of Definition 20.2.1; the maps $\mathcal{S}$ and $\mathcal{T}$ are those displayed at source lines 725–729.*

Proof. Not ready The domain and codomain of the physical isometry, the virtual-space identifications, the transition from the finite-cylinder operators σ_L, σ_R to the local tensor, and the normalization of the refinement channel are unspecified in [11]; complete positivity, trace preservation, and the intertwining identities $\mathcal{S}[M_2(X)] = M_1(X)$ and $\mathcal{T}[M_1(X)] = M_2(X)$ await complete algebraic formulation [87]. $\square$

20.5 The unnormalized strong fixed-point relation

In the Strong-RFP condition [11, Appendix D, lines 2109–2116], the two-site tensor is obtained by coupling a positive physical matrix to one site followed by a two-site unitary change of basis, without requiring unit trace.

Definition 20.5.1 (Strong renormalization fixed-point relation). Lean ✓ Stmt An MPO tensor M satisfies the strong renormalization fixed-point relation if it admits a matrix $P \geq 0$ on the physical space and a unitary U on the two-site physical space such that, for all two-site physical indices $i = (i_1, i_2)$ and $j = (j_1, j_2)$,

$$M^{i_1 j_1} M^{i_2 j_2} = \sum_{a,b} U_{i,a} P_{a_2 b_2} \overline{U_{j,b}} M^{a_1 b_1}. \quad (20.8)$$

This identity expresses $M_2 = U(M_1 \otimes P)U^\dagger$ with M on the left site and P on the right, with no virtual change of basis or scalar proportionality, labelled *Strong-RFP* in [11, Appendix D, lines 2109–2113].

Theorem 20.5.2 (Physical closures of the strong fixed-point relation). Lean Lean ✓ Stmt For fixed matrices P and U , the tensor identity (20.8) holds if and only if every virtual matrix X satisfies

$$M_2(X) = U(M_1(X) \otimes P)U^\dagger. \quad (20.9)$$

Hence the strong renormalization fixed-point relation is equivalent to the existence of $P \geq 0$ and a two-site unitary U satisfying (20.9) for every X .

Proof. ✓ Proof Multiplying (20.8) on the right by X and taking the virtual trace gives, entrywise,

$$\mathrm{tr}(M^{i_1 j_1} M^{i_2 j_2} X) = \sum_{a,b} U_{i,a} P_{a_2 b_2} \overline{U_{j,b}} \mathrm{tr}(M^{a_1 b_1} X),$$

which is (20.9). Conversely, subtract the right-hand side of (20.8). By (20.9), the trace of the product with every X vanishes, so nondegeneracy of the trace pairing makes the difference zero. $\square$

Definition 20.5.3 (First-site preparation shuffle). Lean ✓ Stmt After R_1 separates the first physical coordinate from the N spectator coordinates, define

$$\Sigma_N : ((i, u), p) \mapsto ((i, p), u), \quad (20.10)$$

where $i, p \in \{0, \dots, d-1\}$ and $u \in \{0, \dots, d-1\}^N$. Thus, for a matrix Z indexed by $((i, u), p)$,

$$(\Sigma_N[Z])_{((i,p),u),((j,q),v)} = Z_{((i,u),p),((j,v),q)}. \quad (20.11)$$

Lemma 20.5.4 (Localized first-site preparation). Lean ✓ Stmt Localizing the preparation $Y \mapsto Y \otimes P$ at the first site gives

$$(\mathrm{prep}_P \otimes \mathrm{id})[Y] = \Sigma_N[Y \otimes P]. \quad (20.12)$$

Proof. ✓ Proof At row $((i, p), u)$ and column $((j, q), v)$, both sides equal $Y_{(i,u),(j,v)} P_{p,q}$. $\square$

Theorem 20.5.5 (Global operator identity for a fixed choice of the defining matrices). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix matrices P and U satisfying (20.9). For every $N \geq 0$ and every virtual matrix X , write R_1 and R_2 for the initial-coordinate regroupings of lengths $N + 1$ and $N + 2$. Then

$$R_2[M_{N+2}(X)] = (U \otimes \mathbb{1}) \Sigma_N(R_1[M_{N+1}(X)] \otimes P) (U^\dagger \otimes \mathbb{1}). \quad (20.13)$$

Closing the virtual indices with the identity yields the corresponding formula for the periodic operators $\rho^{(k)}(M)$.

Proof. [✓ Proof](#) Apply Theorem 20.2.21 to the local map $Y \mapsto U(Y \otimes P)U^\dagger$ and the identity (20.9). This gives the localized closure equality. Expanding the regroupings R_1 and R_2 , localized unitary conjugation is conjugation by $U \otimes \mathbb{1}$, while Lemma 20.5.4 gives the middle factor $\Sigma_N(R_1[M_{N+1}(X)] \otimes P)$. The identity closure from Lemma 20.1.6 gives the periodic formula. $\square$

Theorem 20.5.6 (Rank invariance under unitary conjugation). [Lean](#) [✓ Stmt](#) If V is unitary, then

$$\text{rank}(VAV^\dagger) = \text{rank } A. \quad (20.14)$$

Proof. [✓ Proof](#) Left and right multiplication by invertible matrices preserve rank. $\square$

Theorem 20.5.7 (Geometric periodic rank growth of a strong fixed point). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix witnesses $P \geq 0$ and U for the strong fixed-point relation. For every $N \geq 0$, the ordinary full periodic MPO matrix ranks satisfy

$$\text{rank } \rho^{(N+2)}(M) = \text{rank } \rho^{(N+1)}(M) \text{ rank } P, \quad (20.15)$$

$$\text{rank } \rho^{(N+1)}(M) = \text{rank } \rho^{(1)}(M) (\text{rank } P)^N. \quad (20.16)$$

Consequently every strong fixed point admits a single positive witness P whose matrix rank is the factor in both formulas, without assuming normalization of P or positivity of the periodic operators.

Proof. [✓ Proof](#) Unitary conjugation and relabelling of matrix coordinates preserve rank, while rank is multiplicative under Kronecker products. Applying these facts to (20.13) gives the recurrence. Induction from chain length one gives the geometric formula. Finally, Theorem 20.5.2 exposes the same positive witness P for any tensor satisfying the existential strong fixed-point relation. $\square$

20.6 Zero correlation length

Definition 20.6.1 (Physical-trace transfer). [Lean](#) [Lean](#) [✓ Stmt](#) The physical-trace transfer of an MPO tensor M is the virtual matrix obtained by contracting the ket and bra physical legs of one tensor:

$$\mathcal{T}_M = \sum_i M^{ii}.$$

This is the transfer object appearing in the zero-correlation-length condition of [11, Definition 4.2, lines 735–739].

Definition 20.6.2 (Literal physical-trace idempotence). [Lean](#) [✓ Stmt](#) An MPO tensor M has zero correlation length when its physical-trace transfer is idempotent:

$$\mathcal{T}_M^2 = \mathcal{T}_M.$$

This is the literal fixed-tensor condition of [11, Definition 4.2, lines 735–739].

Theorem 20.6.3 (Equation for literal physical-trace idempotence). [Lean](#) [✓ Stmt](#) An MPO tensor satisfies literal physical-trace idempotence if and only if $\mathcal{T}_M^2 = \mathcal{T}_M$.

Proof. [✓ Proof](#) This is the defining equation. □

Definition 20.6.4 (Periodic two-point MPDO correlation). [Lean](#) [✓ Stmt](#) For one-site observables $O_1, O_2 \in M_d(\mathbb{C})$ and complementary gap lengths $g_{\text{mid}}, g_{\text{wrap}} \geq 0$, set

$$C_M(O_1, O_2; g_{\text{mid}}, g_{\text{wrap}}) = \text{tr}(\mathbb{E}_{O_1} \mathcal{T}_M^{g_{\text{mid}}} \mathbb{E}_{O_2} \mathcal{T}_M^{g_{\text{wrap}}}), \quad (20.17)$$

where $\mathbb{E}_O = \sum_{i,j} O_{ji} M^{ij}$. In the notation of the two-observable formula in [11, lines 490–496], $g_{\text{mid}} = n_2 - n_1 - 1$, $g_{\text{wrap}} = N + n_1 - n_2 - 1$, and $N = g_{\text{mid}} + g_{\text{wrap}} + 2$. This unnormalized contraction matches [11, lines 490–496].

Theorem 20.6.5 (Zero correlation length removes positive wrapping-gap dependence). [Lean](#) [✓ Stmt](#) Let M be an MPO tensor satisfying the literal zero-correlation-length condition. Fix O_1, O_2 and a middle gap $g_{\text{mid}} \geq 0$. For two positive wrapping gaps $g_{\text{wrap}}, g'_{\text{wrap}} > 0$,

$$C_M(O_1, O_2; g_{\text{mid}}, g_{\text{wrap}}) = C_M(O_1, O_2; g_{\text{mid}}, g'_{\text{wrap}}). \quad (20.18)$$

This establishes independence of the periodic correlation under variations of the ring length at fixed observable separation, including adjacent marked sites ($g_{\text{mid}} = 0$), in the setting of [11, Definition 4.2, lines 735–742]. The restriction to strictly positive wrapping gaps is documented in [105].

Proof. [✓ Proof](#) Idempotence gives $\mathcal{T}_M^r = \mathcal{T}_M$ for every $r > 0$. Apply this identity to the wrapping power in each side of (20.18); the middle power is unchanged. □

Corollary 20.6.6 (Zero correlation length gives positive-gap distance independence). [Lean](#) [✓ Stmt](#) Let M be an MPO tensor satisfying the literal zero-correlation-length condition. Fix O_1, O_2 . For two pairs of positive complementary gaps $g_{\text{mid}}, g_{\text{wrap}} > 0$ and $g'_{\text{mid}}, g'_{\text{wrap}} > 0$,

$$C_M(O_1, O_2; g_{\text{mid}}, g_{\text{wrap}}) = C_M(O_1, O_2; g'_{\text{mid}}, g'_{\text{wrap}}). \quad (20.19)$$

For equal gap sums, this establishes correlation independence under varying observable distance on a fixed ring when the observables are nonadjacent on both complementary arcs [11, lines 433–500]; the positive-gap regime is documented in [105].

Proof. [✓ Proof](#) Idempotence gives $\mathcal{T}_M^r = \mathcal{T}_M$ for every $r > 0$. Applying this identity to both powers in each instance of (20.17) proves (20.19). □

Definition 20.6.7 (Nonzero up-to-scalar physical-trace relation). [Lean](#) [✓ Stmt](#) An MPO tensor satisfies the nonzero up-to-scalar physical-trace relation when, for some real number $\lambda > 0$, the following conditions hold:

$$\begin{aligned} \mathcal{T}_M &\neq 0, \\ \mathcal{T}_M^2 &= \lambda \mathcal{T}_M. \end{aligned}$$

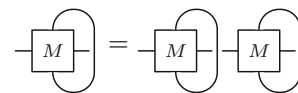
This scale-invariant condition excludes the zero transfer while remaining invariant under positive rescaling of M , extending the literal $\lambda = 1$ diagram of [11, Definition 4.2] [24].

Theorem 20.6.8 (The nonzero physical-trace relation requires a nonzero bond space). [Lean](#)
[✓ Stmt](#) If an MPO tensor satisfies the nonzero up-to-scalar physical-trace relation, then its bond dimension is positive.

Proof. [✓ Proof](#) On a zero-dimensional bond space every endomorphism vanishes, contrary to the nonzero physical-trace transfer in Definition 20.6.7. $\square$

Theorem 20.6.9 (Literal idempotence gives the up-to-scalar physical-trace relation). [Lean](#)
[✓ Stmt](#) If $\mathcal{T}_M \neq 0$ and M has literal physical-trace idempotence, then M satisfies the nonzero up-to-scalar physical-trace relation.

Proof. [✓ Proof](#) This is Definition 20.6.7 with $\lambda = 1$. $\square$

$$\mathcal{T}_M = \mathcal{T}_M^2.$$


This is the literal identity in [11, Definition 4.2, lines 735–739]; Definition 20.6.7 permits the positive factor λ .

Lemma 20.6.10 (Physical-trace normalization gives an idempotent transfer). [Lean](#) [✓ Stmt](#) If M satisfies the nonzero up-to-scalar physical-trace relation, then there exists $\lambda > 0$ such that $\lambda^{-1}\mathcal{T}_M$ is idempotent.

Proof. [✓ Proof](#) Choose $\lambda > 0$ from the up-to-scalar relation, so that $\mathcal{T}_M^2 = \lambda\mathcal{T}_M$. Then

$$(\lambda^{-1}\mathcal{T}_M)^2 = \lambda^{-2}\mathcal{T}_M^2 = \lambda^{-1}\mathcal{T}_M.$$

$\square$

Definition 20.6.11 (Doubled-index transfer idempotence for MPO tensors). [Lean](#) [✓ Stmt](#) An MPO tensor M satisfies the doubled-index transfer condition when its completely positive transfer map is idempotent:

$$\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M.$$

This is the condition obtained from the doubled-index MPS view. It is not the physical-trace condition of Definition 20.6.2.

Theorem 20.6.12 (Doubled-index transfer idempotence iff MPS RFP). [Lean](#) [✓ Stmt](#) An MPO tensor satisfies the doubled-index transfer condition if and only if its doubled-index MPS view is a renormalization fixed point.

Proof. [✓ Proof](#) By Lemma 16.2.2, the transfer map of M agrees with the transfer map of its doubled-index MPS view. Thus $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ holds exactly when the transfer map of that doubled-index MPS tensor is idempotent, which is the RFP condition. $\square$

Theorem 20.6.13 (Renormalization fixed points have idempotent physical-trace transfer). [Lean](#) [✓ Stmt](#) If M is a renormalization fixed point via trace-preserving maps, then $\mathcal{T}_M^2 = \mathcal{T}_M$. This proves the zero-correlation-length component of implication (i) $\Rightarrow$ (ii) in [11, Theorem 4.9, lines 851–893] and [11, Appendix C, lines 1333–1340].

Proof. ✓ Proof For every virtual matrix X , the one-site and two-site closures satisfy

$$\begin{aligned}\mathrm{tr} M_1(X) &= \mathrm{tr}(\mathcal{T}_M X), \\ \mathrm{tr} M_2(X) &= \mathrm{tr}(\mathcal{T}_M^2 X).\end{aligned}$$

Write Φ for the one-to-two trace-preserving map $\mathcal{T}$ of Definition 20.2.1; this is distinct from the physical-trace transfer $\mathcal{T}_M$. Then $\Phi[M_1(X)] = M_2(X)$, and hence

$$\mathrm{tr}(\mathcal{T}_M^2 X) = \mathrm{tr} M_2(X) = \mathrm{tr} \Phi[M_1(X)] = \mathrm{tr} M_1(X) = \mathrm{tr}(\mathcal{T}_M X).$$

Nondegeneracy of the trace pairing now gives $\mathcal{T}_M^2 = \mathcal{T}_M$. □

Theorem 20.6.14 (An idempotent physical-trace transfer without fixed-tensor renormalization maps). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt
Let $Z = \mathrm{diag}(1, -1)$ and let K be the bond-two MPO tensor whose nonzero physical letters are

$$K^{00} = \mathrm{diag}(1/2, 1/4), \quad K^{11} = \mathrm{diag}(1/2, -1/4).$$

For every $N \geq 1$, its closed operator is

$$\rho_N(K) = 2^{-N} I + 4^{-N} Z^{\otimes N} \geq 0, \quad \mathrm{tr} \rho_N(K) = 1.$$

Its physical-trace transfer is

$$\mathcal{T}_K = \mathrm{diag}(1, 0), \quad \mathcal{T}_K^2 = \mathcal{T}_K.$$

For every virtual matrix X , the physical closures satisfy

$$\begin{aligned}K_1(X) &= \frac{X_{00}}{2} I + \frac{X_{11}}{4} Z, \\ K_2(X) &= \frac{X_{00}}{4} I \otimes I + \frac{X_{11}}{16} Z \otimes Z.\end{aligned}$$

Nevertheless, K is not a renormalization fixed point in the sense of Definition 20.2.1.

This tensor corresponds to the $p = 1/2$ model of [174, lines 709–838] conjugated by Hadamard matrices on each physical site. While its physical-trace transfer is idempotent, the one- and two-site physical closures prevent any trace-preserving completely positive coarse-graining channel from satisfying $\mathcal{S}[K_2(X)] = K_1(X)$. Canonical form and area-law saturation for K are established in Theorem 20.6.15.

Proof. ✓ Proof Since the two virtual sectors are diagonal, set $z_0 = 1$ and $z_1 = -1$. A physical configuration σ has weight

$$2^{-N} + 4^{-N} \prod_{k=0}^{N-1} z_{\sigma_k}.$$

This proves the closed-operator formula. Each diagonal entry is either $2^{-N} + 4^{-N}$ or $2^{-N} - 4^{-N}$ and is nonnegative. Contracting the physical indices gives $\mathcal{T}_K = \mathrm{diag}(1, 0)$, whence $\mathrm{tr} \rho_N(K) = \mathrm{tr}(\mathcal{T}_K^N) = 1$ for $N \geq 1$ and $\mathcal{T}_K^2 = \mathcal{T}_K$.

Direct contraction with a virtual matrix X gives the two closure formulas. Set $X_* = \mathrm{diag}(2, 8)$. Then

$$\begin{aligned}K_2(X_*) &= \frac{1}{2}(I \otimes I + Z \otimes Z) \geq 0, \\ K_1(X_*) &= I + 2Z = \mathrm{diag}(3, -1) \not\geq 0.\end{aligned}$$

If the coarse-graining channel $\mathcal{S}$ of Definition 20.2.1 existed, complete positivity would give $\mathcal{S}[K_2(X_*)] \geq 0$, whereas its intertwining identity would give $\mathcal{S}[K_2(X_*)] = K_1(X_*)$. This is a contradiction. $\square$

Theorem 20.6.15 (A canonical-form, area-law-saturating MPDO without fixed-tensor renormalization maps). Lean Lean Lean ✓ Stmt *Let K be the bond-two MPO tensor of Theorem 20.6.14, and let A_K be its doubled-index MPS tensor, an MPS tensor with four physical symbols. Then A_K is in literal CPSV canonical form: every letter is the weighted direct sum $A_K^{(i,j)} = \mu_0 A_0^{(i,j)} \oplus \mu_1 A_1^{(i,j)}$ for $i, j \in \{0, 1\}$, with weights and bond-one normal tensors*

$$\begin{aligned} \mu_0 &= \frac{1}{\sqrt{2}}, & \mu_1 &= \frac{1}{2\sqrt{2}}, \\ A_0^{(i,j)} &= \frac{\delta_{ij}}{\sqrt{2}}, & A_1^{(i,j)} &= (-1)^i \frac{\delta_{ij}}{\sqrt{2}}. \end{aligned}$$

Moreover, K saturates the area law: for every $N \geq 2$ and every $1 \leq L < N$, the reduced state of the first L sites is maximally mixed,

$$\rho_L^{(N)}(K) = 2^{-L} I, \quad S_L^{(N)}(K) = L \log 2.$$

Hence the mutual information $I_L^{(N)} = S_L^{(N)}(K) + S_{N-L}^{(N)}(K) - S_N^{(N)}(K) = N \log 2 - S_N^{(N)}(K)$ is constant in L , giving $I_1 = I_2 = \dots = I_{\lfloor N/2 \rfloor}$. Consequently, there is a bond-two MPO tensor whose doubled-index MPS tensor is in literal CPSV canonical form, which generates an MPDO, whose physical-trace transfer is idempotent, which saturates the area law, and which is not a renormalization fixed point in the sense of Definition 20.2.1.

The canonical-form decomposition follows [11, Section 2.3, lines 214–245].

Proof. ✓ Proof Every letter of A_K is diagonal and the block letters are scalars, so the weighted direct sum is the entrywise check

$$\mu_0 A_0^{(i,i)} = \frac{1}{2}, \quad \mu_1 A_1^{(i,i)} = \frac{(-1)^i}{4},$$

whose right-hand sides are the two virtual diagonal entries of K^{ii} ; the letters with $i \neq j$ vanish on both sides. Each block satisfies $\sum_{i,j} A_k^{(i,j)} \overline{A_k^{(i,j)}} = 1$, hence has identity transfer map and is a normal tensor by Lemma E.1.3. The weighted direct sum of positive-dimensional normal blocks is in literal CPSV canonical form by Definition 7.6.7.

Since $\text{tr } \rho_N(K) = 1$, the normalized state is $\rho_N(K)$ itself. For $1 \leq L < N$, the partial trace over the complementary $N - L$ sites annihilates the summand $4^{-N} Z^{\otimes N}$ of $\rho_N(K) = 2^{-N} I + 4^{-N} Z^{\otimes N}$, since $\text{tr } Z = 0$, and contracts the remaining summand to $2^{-L} I$. Hence $\rho_L^{(N)}(K) = 2^{-L} I$ and $S_L^{(N)}(K) = L \log 2$. Substitution in $I_L^{(N)} = S_L^{(N)}(K) + S_{N-L}^{(N)}(K) - S_N^{(N)}(K)$ gives $I_L^{(N)} = N \log 2 - S_N^{(N)}(K)$, which is independent of L ; in particular $I_L = I_{L+1}$ whenever $1 \leq L < \lfloor N/2 \rfloor$, so K saturates the area law by Definition 20.8.4.

Theorem 20.6.14 supplies the remaining three properties: K generates an MPDO, its physical-trace transfer is idempotent, and it is not a renormalization fixed point. Together with canonical form and area-law saturation, this proves the claim. $\square$

Theorem 20.6.16 (Renormalization fixed points satisfy the up-to-scalar physical-trace relation). Lean ✓ Stmt *If M is a renormalization fixed point via trace-preserving maps and $\mathcal{T}_M \neq 0$, then M satisfies the nonzero up-to-scalar physical-trace relation.*

Proof. ✓ Proof Apply Theorem 20.6.13 and take the positive scalar in Definition 20.6.7 to be 1. $\square$

20.7 Purification fixed points

Definition 20.7.1 (Purifying spin–ancilla tensor). [Lean](#) [✓ Stmt](#) Given matrices $A^{(i,k)}$ with a spin index i and an ancillary index k , the associated MPS tensor on the product physical space is $\widehat{A}^{(i,k)} = A^{(i,k)}$. This is the tensor whose finite-chain vector appears in the purification formula of [11, lines 744–751].

Definition 20.7.2 (Ancillary trace of a purifying MPS). [Lean](#) [✓ Stmt](#) For a purifying spin–ancilla tensor A , the reduced spin density matrix is defined by

$$[\rho_p^{(N)}(A)]_{\sigma,\tau} = \sum_{\kappa} \psi_A(\sigma, \kappa) \overline{\psi_A(\tau, \kappa)},$$

$$\psi_A(\sigma, \kappa) = \text{tr}(A^{(\sigma_0, \kappa_0)} A^{(\sigma_1, \kappa_1)} \dots A^{(\sigma_{N-1}, \kappa_{N-1})}).$$

This is the coefficient form of the ancillary trace in [11, eq. (4.7), line 751].

Definition 20.7.3 (Single-site ancillary trace map). [Lean](#) [✓ Stmt](#)

For a matrix X on the spin–ancilla space, the ancillary trace map is

$$[\text{tr}_a(X)]_{ij} = \sum_k X_{(i,k),(j,k)}.$$

This is the one-site trace over ancillary degrees of freedom appearing in [11, lines 751 and 761–764].

Theorem 20.7.4 (The ancillary trace is trace-preserving completely positive). [Lean](#) [✓ Stmt](#) For every ancillary dimension, the one-site ancillary trace map is trace-preserving and completely positive.

Proof. [✓ Proof](#)

This is Lemma [182, “The right partial trace is trace-preserving completely positive”] with A the spin space and B the ancillary space. $\square$

Definition 20.7.5 (Global purification equation). [Lean](#) [✓ Stmt](#) An MPO tensor M satisfies the global purification equation with a purifying tensor A if $\rho^{(N)}(M) = \rho_p^{(N)}(A)$ for every positive length N . Equivalently, for every positive length N ,

$$\rho^{(N)}(M) = \text{tr}_a(|\Psi^{(N)}(A)\rangle\langle\Psi^{(N)}(A)|),$$

as in [11, line 751].

Definition 20.7.6 (Existence of a purifying fixed-point tensor). [Lean](#) [✓ Stmt](#) For an MPO tensor M , require the existence of a spin–ancilla tensor A such that $\rho^{(N)}(M) = \rho_p^{(N)}(A)$ for every $N > 0$ and $\widehat{A}$ is a pure-state renormalization fixed point. This condition records the density-operator equation of [11, line 751].

Definition 20.7.7 (Trace-preserving spin reduction). [Lean](#) [✓ Stmt](#) The spin reduction obtained by tracing the ancillary index is trace-preserving completely positive [182, “Trace-preserving completely positive map”]. This formalizes the structure described immediately after Definition 4.3 in [11, lines 761–764].

Corollary 20.7.8 (Trace-preserving spin reduction of the ancillary trace). [Lean](#) [✓ Stmt](#) For every ancillary dimension, the spin reduction obtained by tracing the ancillary index is trace-preserving and completely positive.

Proof. ✓ Proof This is exactly Theorem 20.7.4, unpacked through Definition 20.7.7. □

Definition 20.7.9 (Purification renormalization fixed point). Lean ✓ Stmt An MPO tensor M is a purification renormalization fixed point if there is a spin-ancilla tensor A and an identification of the doubled purifying bond space with the bond space of M such that

$$M^{ij} = \left(\sum_k A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e},$$

and $\widehat{A}$ is a pure-state renormalization fixed point. This defines a purification renormalization fixed point (PRFP) in the one-site presentation of [11, lines 744–763].

Definition 20.7.10 (Nondegenerate purification RFP). Lean ✓ Stmt An MPO tensor is a nondegenerate purification renormalization fixed point if it is a purification renormalization fixed point and $\mathcal{T}_M = \sum_i M^{ii} \neq 0$.

Theorem 20.7.11 (A purification RFP is an LPDO). Lean ✓ Stmt *Every purification renormalization fixed point is a local purification density operator.*

Proof. ✓ Proof Forget the renormalization-fixed-point condition on the purifying tensor. □

Theorem 20.7.12 (A purification RFP generates MPDOs). Lean ✓ Stmt *Every purification renormalization fixed point generates matrix product density operators.*

Proof. ✓ Proof Apply Theorem 16.3.3 to the local purification supplied by Theorem 20.7.11. □

Theorem 20.7.13 (Purification RFP does not imply doubled-index idempotence). Lean ✓ Stmt *There is a purification renormalization fixed point for which the doubled-index transfer-map equation $\mathcal{E}_M \circ \mathcal{E}_M = \mathcal{E}_M$ fails.*

Proof. ✓ Proof Take the diagonal purification at $d = d_K = 2$ and $D = D' = 1$ with amplitudes $A = [\frac{1}{\sqrt{2}}, 0, 0, \frac{1}{\sqrt{2}}]$. The purifying tensor is a pure-state renormalization fixed point, since $\sum |A|^2 = 1$ makes its transfer map the identity. The ancilla contraction $M^{ij} = \sum_k A^{(i,k)} \overline{A^{(j,k)}}$ gives the scalar entries $M^{00} = M^{11} = \frac{1}{2}$ (off-diagonal 0), so the induced transfer map is $\frac{1}{2} \cdot \text{id}$ and $\mathcal{E}_M \circ \mathcal{E}_M = \frac{1}{4} \cdot \text{id} \neq \mathcal{E}_M$. The trace contraction has dropped the leading eigenvalue from 1 to $\frac{1}{2}$. By contrast, the physical-trace transfer is 1 and hence is literally idempotent. Thus this example separates doubled-index idempotence from the paper’s physical-trace condition. □

Lemma 20.7.14 (Physical-trace transfer of the maximally mixed tensor). Lean ✓ Stmt *For the maximally mixed tensor M_w , closing the ket and bra physical legs gives $\mathcal{T}_{M_w} = M_w^{00} + M_w^{11} = \frac{1}{2} + \frac{1}{2} = 1$.*

Proof. ✓ Proof This is the direct computation of the two nonzero diagonal entries of the tensor. □

Theorem 20.7.15 (The maximally mixed tensor satisfies the nonzero physical-trace relation). Lean ✓ Stmt *The maximally mixed tensor M_w satisfies the nonzero up-to-scalar physical-trace relation.*

Proof. ✓ Proof Lemma 20.7.14 gives $\mathcal{T}_{M_w} = 1$. The identity is nonzero and idempotent, so the relation holds with $\lambda = 1$. □

Theorem 20.7.16 (Fixed local purification: pure RFP iff normalized physical-trace idempotence). [Lean](#) [✓ Stmt](#) Suppose that M is the ancillary contraction of a fixed spin-ancilla tensor A through a bond identification e :

$$M^{ij} = \left(\sum_k A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}.$$

Then the purification tensor $\hat{A}$ is a pure-state renormalization fixed point if and only if $\mathcal{T}_M^2 = \mathcal{T}_M$. This is the tensor-level equivalence between the first two conditions of the purification theorem in [11, lines 744–786].

Proof. [✓ Proof](#) Let K' be the transfer matrix of $\hat{A}$, let s interchange the two Kronecker factors, and let K denote the same Kronecker sum in the physical ordering. The local contraction gives

$$\begin{aligned} \mathcal{T}_M &= K_e, \\ K &= (K')_s. \end{aligned}$$

Since both reindexings are bijective and commute with matrix multiplication,

$$(K')^2 = K' \iff \mathcal{T}_M^2 = \mathcal{T}_M.$$

Injectivity of the transfer-matrix representation gives

$$(K')^2 = K' \iff \mathcal{E}_{\hat{A}} \circ \mathcal{E}_{\hat{A}} = \mathcal{E}_{\hat{A}}.$$

□

Theorem 20.7.17 (A purification RFP has idempotent physical-trace transfer). [Lean](#) [Lean](#) [✓ Stmt](#) If M is a purification renormalization fixed point, then

$$\mathcal{T}_M^2 = \mathcal{T}_M.$$

This establishes the forward implication from PRFP to literal zero correlation length [11, Theorem 4.4].

Proof. [✓ Proof](#) Apply the forward implication of Theorem 20.7.16 to the purifying tensor and bond identification in the PRFP presentation. □

Theorem 20.7.18 (Purification form and physical-trace idempotence characterize PRFP). [Lean](#) [✓ Stmt](#) An MPO tensor is a purification renormalization fixed point if and only if it has a local purification and its physical-trace transfer is idempotent:

$$\text{PRFP}(M) \iff \text{LPDO}(M) \text{ and } \mathcal{T}_M^2 = \mathcal{T}_M.$$

Proof. [✓ Proof](#) The forward direction combines Theorems 20.7.11 and 20.7.17. Conversely, choose the local purifying tensor and bond identification and apply Theorem 20.7.16 to that fixed purification. □

Theorem 20.7.19 (Nondegenerate PRFP equivalence). [Lean](#) [✓ Stmt](#) An MPO tensor is a nondegenerate purification renormalization fixed point if and only if it is a local purification density operator and

$$\begin{aligned} \mathcal{T}_M &\neq 0, \\ \mathcal{T}_M^2 &= \mathcal{T}_M. \end{aligned}$$

Proof. ✓ Proof Combine Theorem 20.7.18 with the nonzero clause in Definition 20.7.10. □

Theorem 20.7.20 (A nonzero purification RFP satisfies the up-to-scalar relation). Lean ✓ Stmt *If M is a purification renormalization fixed point and $\mathcal{T}_M \neq 0$, then M satisfies the nonzero up-to-scalar physical-trace relation.*

Proof. ✓ Proof Theorem 20.7.17 gives literal idempotence, and the nonzero hypothesis gives the scalar-one case of Definition 20.6.7. □

Theorem 20.7.21 (A nondegenerate purification RFP satisfies the nonzero physical-trace relation). Lean ✓ Stmt *Every nondegenerate purification renormalization fixed point satisfies the nonzero up-to-scalar physical-trace relation.*

Proof. ✓ Proof Apply Theorem 20.7.20 to the two clauses of Definition 20.7.10. □

Theorem 20.7.22 (The tensor purification identity yields the global purification equation). Lean ✓ Stmt *Let M be the ancilla contraction of a purifying spin-ancilla tensor A through a bond identification e , so that*

$$M^{ij} = \left(\sum_k A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e}.$$

Then at every length N the matrix product operator equals the ancillary trace of the pure spin-ancilla matrix product state, $\rho^{(N)}(M) = \rho_p^{(N)}(A)$. This is the coefficient form of the purification equation of [11, line 751].

Proof. ✓ Proof Applying the mixed-product property $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ repeatedly, the length- N product of the contracted tensor entries distributes over the ancillary index sum:

$$\prod_{l=0}^{N-1} \left(\sum_k A^{(\sigma_l, k)} \otimes \overline{A^{(\tau_l, k)}} \right) = \sum_{\kappa: [N] \rightarrow [d_K]} \left(\prod_l A^{(\sigma_l, \kappa_l)} \right) \otimes \overline{\left(\prod_l A^{(\tau_l, \kappa_l)} \right)}.$$

Tracing this sum and using $\text{tr}(X \otimes \overline{Y}) = \text{tr}(X) \overline{\text{tr}(Y)}$ gives the coefficient of the ancillary trace at every σ, τ . The equation holds at every length, so no positive-length restriction is needed. □

Theorem 20.7.23 (A local purification gives the global purification equation). Lean ✓ Stmt *For every purification renormalization fixed point M , its purifying tensor A satisfies $\rho^{(N)}(M) = \rho_p^{(N)}(A)$ for every $N > 0$, and $\hat{A}$ is a pure-state renormalization fixed point.*

Proof. ✓ Proof The local ancillary-contraction identity implies equality of the two generated density operators at every length by Theorem 20.7.22. □

Theorem 20.7.24 (Purification fixed-point equivalence). Discussion: #7298 Not ready *For an MPO tensor with the one-site purification presentation of [11, lines 744–763], Theorem 4.4 asserts the equivalence of three conditions: the purifying tensor is a pure-state renormalization fixed point, the physical-trace transfer is idempotent, and the generated density operators have the repeated-copy form of [11, lines 765–773].*

The equivalence between the first two conditions is Theorem 20.7.18, and Theorem 20.7.23 yields the global purification equation. The repeated-copy formula involves phases indexed by (j, q) , while the physical isometry of [11, lines 546–563] depends only on j ; the physical embedding of the repeated copies is documented in [100].

20.8 Saturation of the area law

Definition 20.8.1 (Block entropy of an MPDO). [Lean](#) [✓ Stmt](#) For a fixed system size N , a block length $L \leq N$, and an MPO tensor M such that $\rho^{(N)}(M)$ is positive semidefinite, let

$$\sigma^{(N)}(M) = \frac{\rho^{(N)}(M)}{\text{tr}[\rho^{(N)}(M)]},$$

$$\rho_L^{(N)}(M) = \text{tr}_{N \setminus L}[\sigma^{(N)}(M)]$$

denote the normalized state and the reduced state of the first L of N spins. The L -block entropy is $S_L^{(N)}(M) = S(\rho_L^{(N)}(M))$.

Theorem 20.8.2 (Block entropy from characteristic roots). [Lean](#) [✓ Stmt](#) If the characteristic roots of $\rho_L^{(N)}(M)$ are the real multiset $\{r_j\}_j$, then

$$S_L^{(N)}(M) = \sum_j -r_j \log r_j.$$

Proof. [✓ Proof](#) The spectral decomposition of the Hermitian reduced state identifies the characteristic roots with its eigenvalues. The von Neumann entropy is the sum of $-r \log r$ over these eigenvalues, counted with multiplicity. $\square$

Definition 20.8.3 (Mutual information of an MPDO). [Lean](#) [Lean](#) [✓ Stmt](#) For the same fixed system size N , block length $L \leq N$, and positive semidefiniteness of $\rho^{(N)}(M)$, the *mutual information* between a block of L spins and the rest is $I_L = S_L^{(N)}(M) + S_{N-L}^{(N)}(M) - S_N^{(N)}(M)$.

Definition 20.8.4 (Saturation of the area law). [Lean](#) [✓ Stmt](#) An MPO tensor M that generates an MPDO and whose positive-length finite-chain operators $\rho^{(N)}(M)$ all have nonzero trace (so the normalized states are well-defined) *saturates the area law* if $I_L = I_{L+1}$ for all $1 \leq L < \lfloor N/2 \rfloor$ and all N ; that is, the mutual information is constant, $I_1 = I_2 = \dots = I_{\lfloor N/2 \rfloor}$. This is [11, Definition 4.6, line 811].

Lemma 20.8.5 (Constancy on a finite interval). [Lean](#) [✓ Stmt](#) Let $a, n \in \mathbb{N}$, and let f be defined on the integer interval $[0, n]$. If $f(m) = f(m+1)$ for every $a \leq m < n$, then $f(i) = f(j)$ for all $i, j \in [a, n]$.

Proof. [✓ Proof](#) Compose the successive equalities from the smaller index to the larger. $\square$

Lemma 20.8.6 (Saturation telescopes to a constant mutual information). [Lean](#) [✓ Stmt](#) If an MPO tensor M saturates the area law, then the mutual informations coincide, $I_L = I_{L'}$, for all $1 \leq L, L' \leq \lfloor N/2 \rfloor$.

Proof. [✓ Proof](#) The defining condition supplies $I_m = I_{m+1}$ throughout the interval $1 \leq m < \lfloor N/2 \rfloor$. Apply Lemma 20.8.5. $\square$

Theorem 20.8.7 (Tripartite marginals as block entropies). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let the N -site operator generated by M be positive semidefinite, and let three consecutive regions have lengths a , b , and c , with $a + b + c \leq N$. After reindexing the reduced state of the first $a + b + c$ sites as a tripartite state on $A \otimes B \otimes C$, its four entropies in strong subadditivity are

$$\begin{aligned} S(ABC) &= S_{a+b+c}, & S(AB) &= S_{a+b}, \\ S(B) &= S_b, & S(BC) &= S_{b+c}. \end{aligned}$$

Proof. ✓ Proof Write $\rho^{(k)}$ for the first- k -site reduced state, with $\cong$ denoting equality up to a permutation of basis indices. Reindexing a matrix does not change its von Neumann entropy, giving $S(ABC) = S_{a+b+c}$ immediately. Tracing out C leaves the first $a + b$ sites, $\text{tr}_C(\rho_{ABC}) \cong \rho^{(a+b)}$, so $S(AB) = S_{a+b}$. Tracing out A and C leaves the middle b sites, while tracing out A leaves the final $b + c$ sites; translation invariance gives

$$\begin{aligned}\text{tr}_{A,C}(\rho_{ABC}) &\cong \rho^{(b)}, \\ \text{tr}_A(\rho_{ABC}) &\cong \rho^{(b+c)}.\end{aligned}$$

Hence $S(B) = S_b$ and $S(BC) = S_{b+c}$. □

Theorem 20.8.8 (Printed mutual-information proposition). Not ready Proposition 4.5 of [11] asserts that $I_L \leq I_{L+1}$ for $L < \lfloor N/2 \rfloor$, that mutual information is bounded by $4 \log D$, and that an iterated thermodynamic limit exists.

While monotonicity and the finite-chain bound hold, the thermodynamic-limit clause fails without additional convergence hypotheses: for the parity-sensitive MPDO in Theorem 20.8.13, $I_L^{(N)}$ alternates between 0 (odd N) and $h_2(1/3) > 0$ (even N), so the limit as $N \rightarrow \infty$ does not exist.

Proposition 20.8.9 (Monotonicity of the mutual information). Lean ✓ Stmt Let M generate an MPDO whose finite-chain operator $\rho^{(N)}(M)$ is positive semidefinite with nonzero trace. For every block length L with $2L + 1 \leq N$, $I_L \leq I_{L+1}$. Since $1 \leq L < \lfloor N/2 \rfloor$ implies $2L + 1 \leq N$, this contains the range in [11, Proposition 4.5, line 801].

Proof. ✓ Proof Apply strong subadditivity to the reduced state of the first $N - L$ spins, split into three contiguous segments of lengths 1, L , and $N - 2L - 1$. Tracing the appropriate segments yields the four block entropies S_{N-L} , S_L , S_{L+1} , and S_{N-L-1} , where the cyclic invariance of $\rho^{(N)}(M)$ identifies the entropy of a contiguous block with that of any block of the same length. Strong subadditivity then gives $S_{N-L} + S_L \leq S_{L+1} + S_{N-L-1}$, which rearranges to $I_L \leq I_{L+1}$ after subtracting S_N . □

Definition 20.8.10 (Parity-sensitive classical MPDO). Lean ✓ Stmt Let M have physical dimension two and bond dimension three, with

$$\begin{aligned}M^{00} &= \text{diag}(1, 0, 0), & M^{11} &= \text{diag}(0, 1, -1), \\ M^{01} &= 0, & M^{10} &= 0.\end{aligned}$$

Theorem 20.8.11 (A periodic MPDO without a thermodynamic limit). Lean Lean Lean Lean ✓ Stmt The tensor in Definition 20.8.10 generates the positive operators

$$\rho^{(N)} = |0^N\rangle\langle 0^N| + (1 + (-1)^N)|1^N\rangle\langle 1^N|.$$

Their traces are $2 + (-1)^N$. The all-zero diagonal entry of the normalized one-site reduced state is therefore one for odd N and $1/3$ for positive even N . In particular, the normalized periodic one-site reduced states do not converge as $N \rightarrow \infty$.

Thus normalized periodic marginals need not converge under the hypotheses of [11, Proposition 4.5, lines 801–806]. The mutual information for this family is analyzed in [19].

Proof. ✓ Proof A contraction around the virtual loop vanishes unless the bra and ket configurations agree and are both constant. The all-zero configuration receives weight one. The all-one configuration receives the sum of the two virtual-sector weights, namely $1 + (-1)^N$. This gives

the displayed diagonal form. Its coefficients are non-negative: the second coefficient is zero for odd N and two for even N . Hence every positive-length operator is positive semidefinite, and summing its two possible diagonal entries gives the trace $2 + (-1)^N$.

Theorem 20.8.12 (Mutual information without a thermodynamic limit). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) ✓ *Stmt* Fix $L \geq 1$. For every $N > L$, the tensor in Definition 20.8.10 satisfies

Consequently the sequence $K \mapsto I_L^{(L+K+1)}$ does not converge, giving the counterexample described in [11, Proposition 4.5, lines 801–806].

Theorem 20.8.13 (Counterexample to the Proposition 4.5 limit clause). Lean Stm *The parity-sensitive tensor generates an MPDO for which the sequence $K \mapsto I_L^{(L+K+1)}$ has no limit for any fixed $L \geq 1$, refuting the unrestricted thermodynamic-limit clause of [11, Proposition 4.5].*

The corresponding generic result is proved in the companion quantum-channel volume [182, “Map on the second tensor factor”].

Definition 20.8.14 (Normalized state across a consecutive bipartition). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For $N = L + K$, split a physical configuration into consecutive intervals of lengths L and K , flatten each interval to a finite index, and denote the resulting bipartite form of the normalized periodic MPO by $\sigma_{L:K}^{(N)}$. The flattened local maps on the two factors are denoted by $\widehat{\mathcal{T}}_L$ and $\widehat{\mathcal{S}}_K$.

Lemma 20.8.15 (Chain mutual information across a consecutive cut). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Stmnt](#) Across the consecutive cut after the first L sites, the two marginals of $\sigma_{L:N-L}^{(N)}$ are the reduced states on L and $N - L$ sites, up to the standard finite-index identifications. Consequently, $I_L(M)$ is the ordinary bipartite mutual information of $\sigma_{L:N-L}^{(N)}$.

Proof. [Proof](#) The first marginal is the defining partial trace over the last $N - L$ sites. Periodic translation invariance identifies the second marginal with the reduced state on the first $N - L$ sites. Entropy is invariant under the finite-index identifications, and retaining the full chain leaves the normalized state unchanged. $\square$

Lemma 20.8.16 (Renormalization maps transfer one site across a cut). [Lean](#) [Lean](#) [Stmnt](#)

Suppose that $N = (a + 1) + (b + 2) = (a + 2) + (b + 1)$. The local closure equations imply

$$\begin{aligned} (\widehat{\mathcal{T}}_a \otimes \widehat{\mathcal{S}}_b)(\sigma_{a+1:b+2}^{(N)}) &= \sigma_{a+2:b+1}^{(N)}, \\ (\widehat{\mathcal{S}}_a \otimes \widehat{\mathcal{T}}_b)(\sigma_{a+2:b+1}^{(N)}) &= \sigma_{a+1:b+2}^{(N)}. \end{aligned}$$

Proof. [Proof](#) Fixing the word on the second interval leaves a physical closure on the first interval, whose virtual boundary is the fixed word evaluation. Apply the appropriate global closure identity to this slice. Fixing the resulting word on the first interval gives the closure on the second interval. Cyclicity of the virtual trace identifies the two cuts of the periodic contraction. The reverse identity is the same argument with $\mathcal{S}$ and $\mathcal{T}$ interchanged. $\square$

Theorem 20.8.17 (Conditional RFP mutual-information equality). [Lean](#) [Lean](#) [Stmnt](#) Let M be a renormalization fixed point. If its N -site operator is positive semidefinite and has nonzero trace, then $I(\sigma_{a+1:b+2}^{(N)}) = I(\sigma_{a+2:b+1}^{(N)})$. This formalizes the channel calculation in [11, Appendix C, lines 1333–1341].

Proof. [Proof](#) Apply mutual-information data processing to the forward pair of local channels and to the reverse pair. The two transfer identities turn the resulting inequalities into opposite inequalities between the displayed mutual informations. $\square$

Lemma 20.8.18 (Nonvanishing of positive-length BNT-refined RFP rings). [Lean](#) [Lean](#) [Stmnt](#)

Let M be in normalized BNT-refined horizontal form, generate positive semidefinite ring operators, and satisfy the local renormalization fixed-point equations. This horizontal hypothesis is stronger than literal CPSV canonical form. Then $\text{tr} \rho^{(N)}(M) \neq 0$ for every $N \geq 1$.

Proof. [Proof](#) Normalized BNT-refined horizontal form gives a nonzero sector compression for one positive length. Positivity makes its trace nonzero. The fixed-point equations make the physical-trace transfer idempotent, so the trace of its positive powers is independent of the positive exponent. $\square$

Theorem 20.8.19 (Trace-normalized renormalization fixed points saturate the area law). [Lean](#)

[Stmnt](#) Let M generate positive semidefinite ring operators with $\text{tr}[\rho^{(N)}(M)] \neq 0$ for every $N \geq 1$. If M satisfies the local renormalization fixed-point equations of Definition 4.1, then M saturates the area law: $I_L(M) = I_{L+1}(M)$ whenever $1 \leq L < \lfloor N/2 \rfloor$. The nonvanishing trace condition ensures normalized states are well-defined.

Proof. [✓ Proof](#) Write the cut as $(a+1) : (b+2)$. The two channel identities and data processing give $I(\sigma_{a+2:b+1}^{(N)}) \leq I(\sigma_{a+1:b+2}^{(N)})$ and $I(\sigma_{a+1:b+2}^{(N)}) \leq I(\sigma_{a+2:b+1}^{(N)})$. Hence the two quantities are equal. The consecutive-cut identification gives $I_L(M) = I_{L+1}(M)$. $\square$

Corollary 20.8.20 (Horizontal renormalization fixed points saturate the area law). [Lean](#) [✓ Stmt](#) *Let M be in normalized BNT-refined horizontal form, generate positive semidefinite ring operators, and satisfy the local renormalization fixed-point equations of Definition 4.1. Then M saturates the area law.*

Proof. [✓ Proof](#) Lemma 20.8.18 gives nonzero trace at every positive length. Apply Theorem 20.8.19. $\square$

Theorem 20.8.21 (Trace-normalized renormalization fixed points have literal ZCL and SAL). [Lean](#) [✓ Stmt](#) *Let M generate positive semidefinite ring operators with $\text{tr}[\rho^{(N)}(M)] \neq 0$ for every $N \geq 1$. If M satisfies the local renormalization fixed-point equations of Definition 4.1, then $\mathcal{T}_M^2 = \mathcal{T}_M$ and M saturates the area law. This proves the ZCL and SAL forward implications of [11, Appendix C, lines 1333–1341].*

Proof. [✓ Proof](#) Trace preservation in the local renormalization equations gives $\mathcal{T}_M^2 = \mathcal{T}_M$ for the physical-trace transfer. If $\mathcal{T}_M = 0$, then $\text{tr}[\rho^{(1)}(M)] = \text{tr}(\mathcal{T}_M) = 0$, contradicting nonvanishing at length one. Thus the literal idempotence is nontrivial, and Theorem 20.8.19 gives saturation of the area law. $\square$

Corollary 20.8.22 (Horizontal renormalization fixed points have literal ZCL and SAL). [✓ Stmt](#) *Let M be in normalized BNT-refined horizontal form, generate positive semidefinite ring operators, and satisfy the local renormalization fixed-point equations of Definition 4.1. Then $\mathcal{T}_M^2 = \mathcal{T}_M$ and M saturates the area law.*

Proof. [✓ Proof](#) Lemma 20.8.18 gives nonzero trace at every positive length. Apply Theorem 20.8.21. $\square$

Lemma 20.8.23 (Nonnegativity of block entropies). [Lean](#) [✓ Stmt](#) *For a normalized positive semidefinite finite-chain operator, every block entropy is non-negative: $0 \leq S_m$.*

Proof. [✓ Proof](#) The reduced state of a block is a density matrix, and the von Neumann entropy of a density matrix is non-negative. $\square$

Lemma 20.8.24 (Nonnegativity of the mutual information). [Lean](#) [✓ Stmt](#) *Let M generate an MPDO whose finite-chain operator $\rho^{(N)}(M)$ is positive semidefinite with nonzero trace. For every block length $L \leq N$, one has $0 \leq I_L$.*

Proof. [✓ Proof](#) Apply strong subadditivity with a trivial middle subsystem (equivalently, subadditivity) to the bipartition of the chain into the first L and last $N-L$ spins: $S_N + S_0 \leq S_L + S_{N-L}$. Since $S_0 \geq 0$ (Lemma 20.8.23), rearranging gives $I_L = S_L + S_{N-L} - S_N \geq S_0 \geq 0$. $\square$

Lemma 20.8.25 (Symmetry of the mutual information across the cut). [Lean](#) [✓ Stmt](#) *For every block length $L \leq N$, the mutual information of the first L spins equals that of their complement: $I_L = I_{N-L}$.*

Proof. [✓ Proof](#) Both sides equal $S_L + S_{N-L} - S_N$ by the symmetric definition $I_L = S_L + S_{N-L} - S_N$, using $N - (N - L) = L$. $\square$

Lemma 20.8.26 (Positive rank of a normalized state). [Lean](#) [✓ Stmt](#) *A positive semidefinite matrix of trace one has positive rank.*

Proof. ✓ Proof If its rank were zero, every eigenvalue would vanish, contradicting that their sum is the trace, which equals one. $\square$

Lemma 20.8.27 (Zero entropy in rank one). Lean ✓ Stmt *A positive semidefinite matrix of trace one and rank at most one has zero von Neumann entropy.*

Proof. ✓ Proof Its rank is positive by Lemma 20.8.26, hence it is exactly one. The rank bound gives $S \leq \log 1 = 0$, while entropy is non-negative. $\square$

Lemma 20.8.28 (Entropy of the empty block). Lean ✓ Stmt *For a normalized positive semidefinite finite-chain operator, the block entropy of the empty block vanishes: $S_0 = 0$.*

Proof. ✓ Proof The reduced state of zero spins is one-dimensional, so its rank is at most one. Lemma 20.8.27 gives $S_0 = 0$. $\square$

Lemma 20.8.29 (Empty-block mutual information vanishes). Lean ✓ Stmt *For a normalized positive semidefinite finite-chain operator, the mutual information of the empty block vanishes: $I_0 = 0$.*

Proof. ✓ Proof By definition, $I_0 = S_0 + S_N - S_N = S_0$, which is zero by Lemma 20.8.28. $\square$

Definition 20.8.30 (Ancillary trace on a block). Lean Lean Lean Lean ✓ Stmt *For a block of L spin-ancilla pairs, regroup*

$$(\mathbb{C}^d \otimes \mathbb{C}^{d_K})^{\otimes L} \cong (\mathbb{C}^d)^{\otimes L} \otimes (\mathbb{C}^{d_K})^{\otimes L}$$

and trace the ancillary factor. Denote this map by $\text{Tr}_{\text{anc}, L}$. It is the blockwise analog of the one-site ancillary trace in Definition 20.7.3.

Theorem 20.8.31 (Block ancillary trace is a channel). Lean ✓ Stmt *The map $\text{Tr}_{\text{anc}, L}$ is trace-preserving and completely positive.*

Proof. ✓ Proof The reindexing equivalence is a channel, as is the partial trace over the ancillary factor. Their composition is therefore trace-preserving and completely positive. $\square$

Theorem 20.8.32 (Coefficients of the ancillary traces). Lean Lean ✓ Stmt *If X is an operator on a spin-ancilla block, then for $i, j \in [d]^L$,*

$$\text{Tr}_{\text{anc}, L}(X)_{i,j} = \sum_{\kappa \in [d_K]^L} X_{(i, \kappa), (j, \kappa)}.$$

For $i, j \in [d]^L$ and $x, y \in [d]^K$, applying the ancillary traces to both blocks across the cut $L \mid K$ gives

$$[(\text{Tr}_{\text{anc}, L} \otimes \text{Tr}_{\text{anc}, K})(X)]_{(i, x), (j, y)} = \sum_{\kappa \in [d_K]^L} \sum_{\lambda \in [d_K]^K} X_{((i, \kappa), (x, \lambda)), ((j, \kappa), (y, \lambda))}.$$

Proof. ✓ Proof After regrouping the spin and ancillary factors, the definition of the partial trace gives the first coefficient identity. Applying this identity to both tensor factors gives the displayed iterated sum; the two finite sums may be interchanged. $\square$

Theorem 20.8.33 (Local purification as a block-channel image). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix a local-purification representation

$$M^{ij} = \left(\sum_{k=0}^{d_K-1} A^{(i,k)} \otimes \overline{A^{(j,k)}} \right)_{e,e},$$

with purifying MPS tensor $\tilde{A}$ of physical dimension dd_K and bond dimension D' . For every $N = L + K$, tracing all ancillary degrees of freedom gives

$$\begin{aligned} \rho^{(N)}(M) &= \text{Tr}_{\text{anc},N} (|V^{(N)}(\tilde{A})\rangle\langle V^{(N)}(\tilde{A})|), \\ \text{tr} \rho^{(N)}(M) &= \|V^{(N)}(\tilde{A})\|^2. \end{aligned}$$

Define the normalized-form purifying operator by

$$\pi_{\tilde{A}}^{(N)} := \|V^{(N)}(\tilde{A})\|^{-2} |V^{(N)}(\tilde{A})\rangle\langle V^{(N)}(\tilde{A})|,$$

where the inverse scalar is taken with the convention $0^{-1} = 0$. After reindexing across the cut $L \mid K$,

$$\sigma_{L:K}^{(N)}(M) = (\text{Tr}_{\text{anc},L} \otimes \text{Tr}_{\text{anc},K}) \left(\left(\pi_{\tilde{A}}^{(N)} \right)_{L:K} \right).$$

This identity also holds when the common trace vanishes, in which case both normalized-form operators are zero.

This formalizes the finite-chain channel identity from the purification in [213, equation (4)] for the local purifications of [11, Section 4.3].

Proof. [✓ Proof](#) The coefficient formula for one block gives

$$\begin{aligned} & \text{Tr}_{\text{anc},N} (|V^{(N)}(\tilde{A})\rangle\langle V^{(N)}(\tilde{A})|)_{\sigma,\tau} \\ &= \sum_{\kappa \in [d_K]^N} V^{(N)}(\tilde{A})_{(\sigma,\kappa)} \overline{V^{(N)}(\tilde{A})_{(\tau,\kappa)}} = \rho^{(N)}(M)_{\sigma,\tau}. \end{aligned}$$

Trace preservation gives equality of the two normalizing traces. Splitting κ into its left and right restrictions changes the last sum into the two independent sums in Theorem 20.8.32, which proves the normalized identity across $L \mid K$. $\square$

Theorem 20.8.34 (LPDOs admit a cut-uniform block-channel representation). [Lean](#) [✓ Stmt](#) Let M be an LPDO. There exist an ancillary dimension d_K , a purifying bond dimension D' , and a local-purification family A , all independent of the cut, such that for every decomposition $N = L + K$,

$$\sigma_{L:K}^{(N)}(M) = (\text{Tr}_{\text{anc},L} \otimes \text{Tr}_{\text{anc},K}) \left(\left(\pi_{\tilde{A}}^{(N)} \right)_{L:K} \right),$$

where $\tilde{A}$ is the purifying MPS tensor associated with A .

Proof. [✓ Proof](#) Choose one local-purification representation of M . For an arbitrary decomposition $N = L + K$, apply Theorem 20.8.33 to the same purifying tensor. $\square$

Theorem 20.8.35 (Mutual-information bound for a local-channel image). [Lean](#) [✓ Stmt](#) Let M be an MPO tensor whose generated N -site operator is positive semidefinite. Let A be an MPS tensor of bond dimension $D' \geq 1$ whose N -site vector is nonzero. Suppose that, across the cut $L | (N - L)$, the normalized state generated by M is obtained from the normalized pure state of A by trace-preserving completely positive maps Φ_L and Φ_R on the two blocks:

$$\sigma_{L:N-L}^{(N)}(M) = (\Phi_L \otimes \Phi_R) \left(\frac{|V^{(N)}(A)\rangle\langle V^{(N)}(A)|}{\|V^{(N)}(A)\|^2} \right)_{L:N-L}.$$

Then $I_L(M) \leq 4 \log D'$.

Proof. [✓ Proof](#) Write $\sigma_A^{(N)}$ for the normalized pure state of A in the $L | (N - L)$ bipartition. Then

$$I_L(M) = I((\Phi_L \otimes \Phi_R)(\sigma_A^{(N)})) \leq I(\sigma_A^{(N)}) \leq 4 \log D'.$$

The equality is the channel-image hypothesis, the first inequality is data processing, and the last inequality is Proposition 20.10.23. $\square$

Theorem 20.8.36 (Mutual-information bound for a locally purified tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix a local-purification representation with purifying MPS bond dimension D' . If the N -site operator is positive semidefinite and its purifying pure state is nonzero, then $I_L(M) \leq 4 \log D'$ for every $L \leq N$.

Consequently, if M is an LPDO, $N \geq 1$, and $\text{tr} \rho^{(N)}(M) \neq 0$, then its local-purification representation supplies a bond dimension D' satisfying this estimate. The quantity D' is the bond dimension of the purifying MPS, not the MPO bond dimension D .

Proof. [✓ Proof](#) By Theorem 20.8.33,

$$\begin{aligned} I_L(M) &= I\left((\text{Tr}_{\text{anc},L} \otimes \text{Tr}_{\text{anc},N-L})\left(\left(\pi_A^{(N)}\right)_{L:N-L}\right)\right) \\ &\leq I\left(\left(\pi_{\tilde{A}}^{(N)}\right)_{L:N-L}\right) \leq 4 \log D'. \end{aligned}$$

The first inequality is data processing and the second is the pure-state boundary estimate. Nonvanishing of the purifying pure state implies $D' > 0$. For an LPDO, positivity follows from Theorem 16.3.3, while the trace identity in Theorem 20.8.33 transfers the nonzero-trace hypothesis to the purifying pure state. $\square$

Theorem 20.8.37 (Rank of diagonal coefficients across a periodic cut). [Lean](#) [Lean](#) [✓ Stmt](#) Split a periodic chain into consecutive blocks of lengths L and R . For words $x \in \{0, \dots, d-1\}^L$ and $y \in \{0, \dots, d-1\}^R$, set

$$P_{x,y} = \text{tr}(M^{x_0 x_0} \dots M^{x_{L-1} x_{L-1}} M^{y_0 y_0} \dots M^{y_{R-1} y_{R-1}}).$$

Then the ordinary complex rank of P satisfies $\text{rank}_{\mathbb{C}} P \leq D^2$.

Proof. [✓ Proof](#) For each word, write $A_x = M^{x_0 x_0} \dots M^{x_{L-1} x_{L-1}}$ and $B_y = M^{y_0 y_0} \dots M^{y_{R-1} y_{R-1}}$. The identity

$$P_{x,y} = \text{tr}(A_x B_y) = \sum_{a,b=0}^{D-1} (A_x)_{ab} (B_y)_{ba}$$

factors P through the D^2 -dimensional space indexed by pairs (a, b) . $\square$

Theorem 20.8.38 (Rank under faithful scalar extension). [Lean](#) [✓ Stmt](#) Let $K \subseteq L$ be a faithful field extension, let $\iota : K \hookrightarrow L$ be the inclusion, and let A be a finite matrix over K . Then $\text{rank}_L((\iota(A_{ij}))_{ij}) = \text{rank}_K A$.

Proof. [✓ Proof](#) For every finite selection of columns, the selected columns are linearly independent over K if and only if their scalar extensions are linearly independent over L . Selecting a basis from each column space proves the two rank inequalities. $\square$

The corresponding generic result is proved in the companion quantum-channel volume [182, “Joint probability distribution and marginals”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Entropy and classical mutual information”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Entropy is bounded by the logarithm of the support cardinality”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Classical mutual information is bounded by ordinary rank”].

Theorem 20.8.39 (Classical estimate for normalized diagonal cut coefficients). [Lean](#) [✓ Stmt](#) Let $D \geq 1$, and let W be a real diagonal coefficient matrix across a periodic cut of an MPO with bond dimension D . Suppose its scalar extension to $\mathbb{C}$ is the cut matrix $P^{\mathbb{C}}$ from Theorem 20.8.37. If $Z > 0$ and $P = Z^{-1}W$ is a joint probability distribution, then $I(X : Y)_P \leq 2 \log D$.

Proof. [✓ Proof](#) Normalization by the nonzero scalar Z^{-1} does not change real rank. Faithful scalar extension does not change rank by Theorem 20.8.38; hence Theorem 20.8.37 gives $\text{rank}_{\mathbb{R}} P \leq D^2$. Applying [182, “Classical mutual information is bounded by ordinary rank”] gives $I(X : Y)_P \leq \log \text{rank}_{\mathbb{R}} P$, and monotonicity of the logarithm gives the result. $\square$

Definition 20.8.40 (Normalized diagonal distribution at fixed length). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a chain of length $N = L + R$, call $\rho^{(N)}(M)$ diagonal if $\rho^{(N)}(M)_{u,v} = 0$ whenever $u \neq v$. Call M globally diagonal if $\rho^{(N)}(M)$ is diagonal for every $N > 0$. For configurations x and y on the two consecutive blocks, define

$$\begin{aligned} W_{x,y} &= \text{Re } \rho^{(L+R)}(M)_{(x,y),(x,y)}, \\ Z &= \sum_{x,y} W_{x,y}, \\ \widehat{P} &= Z^{-1}W. \end{aligned}$$

Lemma 20.8.41 (Sum of diagonal coefficients and trace). [Lean](#) [✓ Stmt](#) For every MPO tensor M and cut lengths L, R , the sum of the real parts of the diagonal coefficients satisfies $Z = \text{Re tr } \rho^{(L+R)}(M)$.

Proof. [✓ Proof](#) Pairing each configuration σ on $L + R$ sites with its unique block split (x, y) on L and R sites gives

$$Z = \sum_{x,y} W_{x,y} = \sum_{\sigma} \text{Re } \rho^{(L+R)}(M)_{\sigma,\sigma} = \text{Re tr } \rho^{(L+R)}(M).$$

$\square$

Theorem 20.8.42 (Classical estimate for a positive diagonal finite-chain operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $D \geq 1$. Suppose that $\rho^{(L+R)}(M)$ is diagonal and positive semidefinite, and that $Z > 0$. Then $W_{x,y} \geq 0$, the scalar extension of W to $\mathbb{C}$ is the diagonal cut matrix of Theorem 20.8.37, and $\widehat{P}$ is a joint probability distribution. Its classical mutual information satisfies $I(X : Y)_{\widehat{P}} \leq 2 \log D$.

Proof. [✓ Proof](#) Positive semidefiniteness makes every diagonal entry real and non-negative, so $W_{x,y} \geq 0$ and its scalar extension recovers the complex diagonal cut matrix. Since Z is the sum of the entries of W , $\sum_{x,y} \widehat{P}_{x,y} = Z^{-1} \sum_{x,y} W_{x,y} = 1$. Thus $\widehat{P}$ is a joint probability distribution. Diagonality says that these coefficients constitute the full classical finite-chain state, and Theorem 20.8.39 gives the bound. $\square$

Corollary 20.8.43 (Classical estimate for a diagonal matrix product density operator). [Lean](#)
[✓ Stmt](#) Let $D \geq 1$, and let M generate diagonal positive semidefinite operators at every positive chain length. If $L + R > 0$ and the trace at length $L + R$ is positive, then the classical mutual information of its normalized diagonal coefficients satisfies $I(X : Y) \leq 2 \log D$.

Proof. [✓ Proof](#) Global diagonality and positivity give the fixed-length hypotheses of Theorem 20.8.42. Since the chain operator is diagonal,

$$Z = \sum_{x,y} W_{x,y} = \text{Re tr } \rho^{(L+R)}(M),$$

so the trace positivity hypothesis supplies the required normalization. $\square$

The corresponding generic result is proved in the companion quantum-channel volume [182, “Positivity of the operator-Schmidt rank”].

Theorem 20.8.44 (Two virtual boundaries bound the operator-Schmidt rank). [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [✓ Stmt](#) Let M be an MPO tensor of operator bond dimension D , and let $N = L + K$. Opening the two virtual bonds at the consecutive cut gives

$$\sigma_{L:K}^{(N)} = \sum_{a,b=0}^{D-1} X_{ab} \otimes Y_{ab},$$

and

$$\text{OSR}(\sigma_{L:K}^{(N)}) \leq D^2.$$

For left-block words x, x' and right-block words y, y' , the factors are

$$(X_{ab})_{x,x'} = \text{tr}[\rho^{(N)}(M)]^{-1} (M^{x_0 x'_0} \dots M^{x_{L-1} x'_{L-1}})_{ab},$$

and

$$(Y_{ab})_{y,y'} = (M^{y_0 y'_0} \dots M^{y_{K-1} y'_{K-1}})_{ba}.$$

This decomposes $\sigma_{L:K}^{(N)}$ via the trace identity $\text{tr}(AB) = \sum_{a,b} A_{ab} B_{ba}$ with the normalization scalar absorbed into the left factor, formalizing the two-virtual-boundary factorization in [11, Proposition 4.5].

Proof. [✓ Proof](#) Split each closed virtual word into its left and right products. Expanding the trace over the two exposed virtual indices gives the displayed sum of D^2 product matrices. The minimality property of the operator-Schmidt rank then bounds it by the number of displayed terms. $\square$

The corresponding generic result is proved in the companion quantum-channel volume [182, “Coefficient-space characterization of operator-Schmidt rank”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Operator-Schmidt rank under local isometries”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Support of a bipartite positive operator”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Compression to both marginal supports”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Partial traces under product isometries”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Faithful marginals after simultaneous support compression”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Channel of a faithful bipartite marginal”].

20.9 Support compression for entropy functionals

The corresponding generic result is proved in the companion quantum-channel volume [182, “Isometric conjugation into a larger matrix algebra”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Functional calculus under rectangular isometric conjugation”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Trace preservation under isometric inclusion”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Compression cancels isometric expansion”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Reconstruction from a reference support”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Relative entropy under support compression”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Functional calculus under transpose”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Logarithm of a positive square root”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Logarithm of a real power”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Support-aware vector-state Jensen inequality for the logarithm”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Order-two sandwiched trace functional”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Nonnegativity of the order-two functional”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Faithful cyclic form of the order-two functional”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Relative entropy bounded by the faithful order-two functional”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Order-two sandwiched functional under support compression”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Reduction of the singular order-two comparison to the faithful case”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Relative entropy bounded by the support-restricted order-two functional”].

20.10 Rectangular Choi matrices and a range-dimension estimate

The rectangular Choi correspondence [212, Proposition 2.1] bounds the order-two Choi functional of a Hilbert–Schmidt contraction by its range dimension, bounding mutual information in terms of operator-Schmidt rank.

The corresponding generic result is proved in the companion quantum-channel volume [182, “Matrix of a linear map in matrix-unit bases”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Unnormalized rectangular Choi matrix”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Frobenius norm under rectangular Choi reshaping”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Matrix-unit Parseval identity for a rectangular Choi matrix”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Hilbert–Schmidt contraction”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Rectangular Choi norm bounded by the range dimension”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Full-support eigenbasis whitening as a Choi matrix”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Full-support eigenbasis whitened Choi estimate”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Real powers under unitary conjugation”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Real powers of positive tensor products”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Inverse quarter-power from iterated square roots”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Unitary invariance of the order-two functional”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Reindexing invariance of the order-two functional”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Faithful marginal whitening bound”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Marginal-support whitening bound”].

For trace-one ρ and $\alpha > 1$, the total sandwiched Rényi trace is the quantity inside the logarithm of the sandwiched Rényi divergence [5, Equation (3)] [189, Definition 2] for $\rho, \omega \geq 0$ with $\ker \omega \subseteq \ker \rho$. For singular ω , negative powers denote the generalized inverse on $\text{supp } \omega$. The regime $1 < \alpha \leq 2$ connects the Umegaki relative entropy at $\alpha \rightarrow 1$ to the order-two functional.

The corresponding generic result is proved in the companion quantum-channel volume [182, “Total sandwiched Rényi trace”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Nonnegativity on the faithful domain”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Order-one sandwiched trace”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Order-two sandwiched trace”].

The corresponding generic result is proved in the companion quantum-channel volume [182, “Mutual information from operator-Schmidt rank”].

Theorem 20.10.1 (Finite-chain mutual information from the operator bond). Lean ✓ Stmt *Let M be an MPO tensor of operator bond dimension D . Fix a chain of length N and a cut with $0 \leq L \leq N$. If the corresponding finite-chain operator is positive semidefinite and has nonzero trace, then*

$$I_L^{(N)} \leq 2 \log D.$$

The bound holds for all cuts $0 \leq L \leq N$ without assuming local purification or canonical form.

Proof. ✓ Proof Reindex the normalized chain operator across the consecutive cut. Its trace is one, and its positivity is preserved by this reindexing. Hence

$$I_L^{(N)} \leq \log \text{OSR}(\sigma_{L:N-L}^{(N)}) \leq \log D^2 = 2 \log D.$$

Nonzero trace makes the normalized operator nonzero, so its operator-Schmidt rank is positive. If $D = 0$, then every finite-chain trace vanishes. Hence $D > 0$, which justifies the last logarithmic identity. $\square$

Theorem 20.10.2 (Finite-chain mutual-information estimate for matrix product density operators). Lean ✓ Stmt *Let M be a matrix product density operator of operator bond dimension D . For every positive chain length N , every cut $0 \leq L \leq N$, and every nonzero finite-chain trace,*

$$I_L^{(N)} \leq 4 \log D.$$

This establishes the finite-chain mutual-information bound invoked in [11, Proposition 4.5].

Proof. ✓ Proof Global positivity supplies positivity at the chosen positive chain length. Apply Theorem 20.10.1. The same nonzero-trace argument gives $D > 0$, hence $0 \leq \log D$ and $2 \log D \leq 4 \log D$. $\square$

Remark 20.10.3 (Boundedness of mutual information). The proof of [11, Proposition 4.5] invokes the uniform estimate $I_L \leq 4 \log D$ for every MPDO, referencing the purification of local completely positive maps [213]. For such maps, data processing and the pure-state boundary estimate yield

$$I(A : B)_\rho \leq I(A : B)_{|\Psi\rangle\langle\Psi|} = 2S(A)_{|\Psi\rangle\langle\Psi|} \leq 2|\partial A| \log D = 4 \log D.$$

General positive MPOs need not admit such a local purification.

The two virtual bonds bound the operator-Schmidt rank by D^2 [182, “Coefficient-space characterization of operator-Schmidt rank”], even though the ordinary rank of a marginal can grow exponentially (for instance, a product of full-rank single-site density matrices has $\text{rank}(\rho_L) = (\text{rank } q)^L$ while $I_L = 0$).

Theorem 20.10.1 provides the uniform bound $I_L^{(N)} \leq 2 \log D$ whenever the finite-chain operator is positive semidefinite with positive trace. For diagonal states, Theorem 20.8.42 gives an independent verification.

While the finite-chain bound $I_L^{(N)} \leq 2 \log D$ holds uniformly, the iterated thermodynamic limit in [11, Proposition 4.5] can fail, as demonstrated by Theorem 20.8.11 and documented in [19]. The channel-image case is treated in Theorem 20.8.35, pure states in Proposition 20.10.23, and locally purified states in Theorem 20.8.36.

Definition 20.10.4 (Block entropy and area-law saturation for pure states). [Lean](#) [Lean](#) [✓ Stmt](#) For an MPS tensor A , a system size N , and a block length $L \leq N$, let

$$\sigma^{(N)}(A) = \frac{|V^{(N)}(A)\rangle\langle V^{(N)}(A)|}{\|V^{(N)}(A)\|^2},$$

$$\rho_L^{(N)}(A) = \text{tr}_{N \setminus L}[\sigma^{(N)}(A)]$$

denote the normalized state and the reduced state of the first L spins. The L -block entropy is $S_L^{(N)}(A) = S(\rho_L^{(N)}(A))$. The tensor saturates the area law if $S_L^{(N)}(A) = S_{L+1}^{(N)}(A)$ for all $1 \leq L < \lfloor N/2 \rfloor$ and all N . This is [11, Definition 3.13, line 600].

Definition 20.10.5 (Purification tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor A on bond space $\mathbb{C}^D$, define the associated MPO tensor M on bond space $\mathbb{C}^{D^2}$ by $M^{ij} = A^i \otimes \bar{A}^j$. This is the single-ancilla purification tensor of [11].

Lemma 20.10.6 (Pure state as a purification MPDO). [Lean](#) [✓ Stmt](#) For an MPS tensor A , the pure-state operator $|V^{(N)}(A)\rangle\langle V^{(N)}(A)|$ equals the MPDO generated by the tensor $M^{ij} = A^i \otimes \bar{A}^j$, acting on a bond space of dimension D^2 :

$$\rho^{(N)}(A \otimes \bar{A}) = |V^{(N)}(A)\rangle\langle V^{(N)}(A)|.$$

This is the single-ancilla case of the purification picture of [11]; it lets the MPDO block-entropy theory apply to pure-state block entropies.

Proof. [✓ Proof](#) The word evaluation of the doubled tensor is the Kronecker product of the word evaluation of A and its conjugate, $M^{\sigma, \tau} = A^\sigma \otimes \bar{A}^\tau$, so its trace factors as $\text{tr}(M^{\sigma, \tau}) = \text{tr}(A^\sigma) \text{tr}(\bar{A}^\tau) = \overline{V^{(N)}(A)}_\tau V^{(N)}(A)_\sigma$, the corresponding matrix element of $|V^{(N)}(A)\rangle\langle V^{(N)}(A)|$. $\square$

Lemma 20.10.7 (Block entropy of the purification tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor A , system size N , and block length $L \leq N$, the MPDO block entropy of the tensor $M^{ij} = A^i \otimes \bar{A}^j$ agrees with the pure-state block entropy: $S_L^{(N)}(M) = S_L^{(N)}(A)$.

Proof. [✓ Proof](#) Lemma 20.10.6 gives $\rho^{(N)}(M) = |V^{(N)}(A)\rangle\langle V^{(N)}(A)|$, hence

$$\sigma^{(N)}(M) = \frac{\rho^{(N)}(M)}{\text{tr}[\rho^{(N)}(M)]} = \frac{|V^{(N)}(A)\rangle\langle V^{(N)}(A)|}{\|V^{(N)}(A)\|^2} = \sigma^{(N)}(A).$$

Therefore the reduced block states agree:

$$\rho_L^{(N)}(M) = \text{tr}_{N \setminus L}[\sigma^{(N)}(M)] = \text{tr}_{N \setminus L}[\sigma^{(N)}(A)] = \rho_L^{(N)}(A).$$

Thus $S_L^{(N)}(M) = S(\rho_L^{(N)}(M)) = S(\rho_L^{(N)}(A)) = S_L^{(N)}(A)$. $\square$

Lemma 20.10.8 (Schmidt symmetry of the block entropy). [Lean](#) [✓ Stmt](#) For an MPS tensor A and a block length $L \leq N$, the block entropies of a block and its complement coincide: $S_L^{(N)}(A) = S_{N-L}^{(N)}(A)$.

Proof. [✓ Proof](#) The reduced state of the first L spins is $\rho_L \propto WW^\dagger$, where W is the matrix of amplitudes $\langle u w | V^{(N)}(A) \rangle$ with u the first L spins and w the remaining $N - L$. By cyclic invariance of the amplitudes, the reduced state of the complement is $\propto \overline{W^\dagger W}$. Cyclic invariance of the entropy, $S(WW^\dagger) = S(W^\dagger W)$, together with invariance of the entropy under entrywise conjugation of a Hermitian matrix, $S(\overline{W^\dagger W}) = S(W^\dagger W)$, gives $S_L^{(N)}(A) = S_{N-L}^{(N)}(A)$. $\square$

Proposition 20.10.9 (Pure-state area-law monotonicity). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with $V^{(N)}(A) \neq 0$. For every block length L with $2L + 1 \leq N$, covering $1 \leq L < \lfloor N/2 \rfloor$, $S_L^{(N)}(A) \leq S_{L+1}^{(N)}(A)$. This is the pure-state area law of [11, Section 3, line 599].

Proof. [✓ Proof](#) Apply the block-entropy form of strong subadditivity, the inequality underlying Proposition 20.8.9, to the purification MPDO of A , giving $S_{N-L} + S_L \leq S_{L+1} + S_{N-L-1}$. By the Schmidt symmetry $S_{N-L} = S_L$ and $S_{N-L-1} = S_{L+1}$, this reduces to $S_L \leq S_{L+1}$. $\square$

Lemma 20.10.10 (Operator-Schmidt Gram is a transfer-map power). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an MPS tensor of bond dimension D . For a length- ℓ configuration σ and a length- m configuration τ , write the operator-Schmidt factors as

$$\begin{aligned} (L_\ell)_{\sigma, (a_1, a_2)} &= (A^\sigma)_{a_1 a_2}, \\ (R_m)_{(p_1, p_2), \tau} &= (A^\tau)_{p_2 p_1}. \end{aligned}$$

The $D^2 \times D^2$ Gram matrix of the left factor equals the ℓ -fold transfer map evaluated on a matrix unit:

$$\begin{aligned} (L_\ell^\dagger L_\ell)_{(a_1, a_2), (b_1, b_2)} &= \sum_{\sigma} \overline{(A^\sigma)_{a_1 a_2}} (A^\sigma)_{b_1 b_2} \\ &= (\mathcal{E}_A^\ell(|b_2\rangle\langle a_2|))_{b_1, a_1}, \end{aligned}$$

where the sum runs over length- ℓ configurations σ . The Gram of the right factor R_m of an m -spin complement satisfies the bond-swapped identity

$$\begin{aligned} (R_m R_m^\dagger)_{(p_1, p_2), (q_1, q_2)} &= \sum_{\tau} (A^\tau)_{p_2 p_1} \overline{(A^\tau)_{q_2 q_1}} \\ &= (\mathcal{E}_A^m(|p_1\rangle\langle q_1|))_{p_2, q_2}, \end{aligned}$$

where the sum runs over length- m configurations τ .

Proof. [✓ Proof](#) The left Gram entry is $\sum_{\sigma} \overline{(A^\sigma)_{a_1 a_2}} (A^\sigma)_{b_1 b_2}$ by definition of L_ℓ . Expanding

$$\mathcal{E}_A^\ell(|b_2\rangle\langle a_2|) = \sum_{\sigma} A^\sigma |b_2\rangle\langle a_2| (A^\sigma)^\dagger$$

and reading off the (b_1, a_1) entry gives the same sum, since

$$(A^\sigma |b_2\rangle\langle a_2| (A^\sigma)^\dagger)_{b_1, a_1} = (A^\sigma)_{b_1 b_2} \overline{(A^\sigma)_{a_1 a_2}}.$$

The right-factor formula is the same expansion with the two boundary bonds interchanged: the (p, q) entry of $R_m R_m^\dagger$ is $\sum_{\tau} (A^\tau)_{p_2 p_1} \overline{(A^\tau)_{q_2 q_1}}$, which is the (p_2, q_2) entry of $\mathcal{E}_A^m(|p_1\rangle\langle q_1|)$. $\square$

Lemma 20.10.11 (RFP collapse of the operator-Schmidt Grams). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be a renormalization fixed point. For every $L \geq 1$ and $m \geq 1$, the operator-Schmidt Grams of the L -block and of the m -site complement are the one-site Grams:*

$$\begin{aligned} L_L^\dagger L_L &= L_1^\dagger L_1, \\ R_m R_m^\dagger &= R_1 R_1^\dagger. \end{aligned}$$

Proof. [✓ Proof](#) By definition, a renormalization fixed point satisfies $\mathcal{E}_A^2 = \mathcal{E}_A$, so $\mathcal{E}_A^k = \mathcal{E}_A$ for every $k \geq 1$. Applying Lemma 20.10.10 to the left and right Gram entries gives the two displayed identities. $\square$

Lemma 20.10.12 (Pure block entropy as a bond-environment charpoly sum). [Lean](#) [✓ Stmt](#) *Let A be an MPS tensor, let $L \leq N$, put $m = N - L$, and write $c = (\text{tr } \rho^{(N)})^{-1}$. Then the pure block entropy is the charpoly-root entropy sum of the $D^2 \times D^2$ bond environment:*

$$S_L^{(N)}(A) = \sum_{\lambda \in \text{roots } \chi_{c(R_m R_m^\dagger)(L_L^\dagger L_L)}} -\text{Re}(\lambda) \log \text{Re}(\lambda).$$

The roots are counted with algebraic multiplicity.

Proof. [✓ Proof](#) The reduced block state has the exact factorization

$$\rho_L^{(N)}(A) = c L_L (R_m R_m^\dagger) L_L^\dagger.$$

Cyclic invariance of the charpoly-root entropy sum gives

$$S(c L_L (R_m R_m^\dagger) L_L^\dagger) = S(c (R_m R_m^\dagger) (L_L^\dagger L_L)),$$

which is the displayed formula. $\square$

Theorem 20.10.13 (A renormalization fixed point saturates the area law). [Lean](#) [✓ Stmt](#) *If an MPS tensor A is a renormalization fixed point, then it saturates the area law: for every chain length N and every block length $1 \leq L < \lfloor N/2 \rfloor$, the pure-state block entropy is constant in the block size, $S_L^{(N)}(A) = S_{L+1}^{(N)}(A)$. This strengthens [11, Proposition 3.14, lines 606–608] by deducing area-law saturation from transfer-map idempotence alone, without assuming canonical form.*

Proof. [✓ Proof](#) Write $c = (\text{tr } \rho^{(N)})^{-1}$. Lemma 20.10.12 computes the entropy from the bond environment:

$$S_L^{(N)}(A) = \sum_{\lambda \in \text{roots } \chi_{c(R_{N-L} R_{N-L}^\dagger)(L_L^\dagger L_L)}} -\text{Re}(\lambda) \log \text{Re}(\lambda).$$

Since $1 \leq L < \lfloor N/2 \rfloor$, the four lengths L , $L + 1$, $N - L$, and $N - (L + 1)$ are positive. Lemma 20.10.11 gives

$$\begin{aligned} L_L^\dagger L_L &= L_1^\dagger L_1, \\ L_{L+1}^\dagger L_{L+1} &= L_1^\dagger L_1, \\ R_{N-L} R_{N-L}^\dagger &= R_1 R_1^\dagger, \\ R_{N-(L+1)} R_{N-(L+1)}^\dagger &= R_1 R_1^\dagger. \end{aligned}$$

Therefore the two bond environments agree:

$$\begin{aligned} (R_{N-L} R_{N-L}^\dagger)(L_L^\dagger L_L) &= (R_1 R_1^\dagger)(L_1^\dagger L_1) \\ &= (R_{N-(L+1)} R_{N-(L+1)}^\dagger)(L_{L+1}^\dagger L_{L+1}). \end{aligned}$$

The scalar c depends only on N , so the two charpoly-root entropy sums are equal. Hence $S_L^{(N)}(A) = S_{L+1}^{(N)}(A)$. $\square$

Theorem 20.10.14 (A renormalization fixed point has a constant block-entropy chain). [Lean](#)

[✓ Stmt](#) If an MPS tensor A is a renormalization fixed point, then all block entropies in the range $1 \leq L \leq \lfloor N/2 \rfloor$ coincide:

$$S_1^{(N)}(A) = S_2^{(N)}(A) = \dots = S_{\lfloor N/2 \rfloor}^{(N)}(A).$$

This formalizes the constant-entropy consequence of area-law saturation [11, Definition 3.13, line 600] for fixed points.

Proof. [✓ Proof](#) The fixed point saturates the area law (Theorem 20.10.13), so by the telescoping of saturation (Lemma 20.10.21) the block entropies at any two lengths $1 \leq L, L' \leq \lfloor N/2 \rfloor$ agree: $S_L^{(N)}(A) = S_{L'}^{(N)}(A)$. $\square$

Lemma 20.10.15 (Operator-Schmidt rank bound). [Lean](#) [✓ Stmt](#) For an MPS tensor A of bond dimension D and a block length $L \leq N$, the reduced state $\rho_L^{(N)}(A)$ of the first L spins has rank at most D^2 .

Proof. [✓ Proof](#) The reduced state is $\rho_L \propto WW^\dagger$, where the amplitude matrix

$$W(u, w) = \langle u | w \rangle V^{(N)}(A) = \text{tr}(A^{u_1} \dots A^{u_L} A^{w_1} \dots A^{w_{N-L}})$$

factors through the two bond pairs cut by the block boundary. Writing the trace as a sum over the bond index pair $(a, b) \in \{1, \dots, D\}^2$ exhibits $W = L'R'$ with L' having D^2 columns, so $\text{rank } \rho_L \leq \text{rank } W \leq D^2$. $\square$

Lemma 20.10.16 (The reduced pure block state is a density matrix). [Lean](#) [Lean](#) [✓ Stmt](#) For an MPS tensor A with $V^{(N)}(A) \neq 0$, the reduced state $\rho_L^{(N)}(A)$ of the first L spins is a density matrix: it is positive semidefinite and has unit trace.

Proof. [✓ Proof](#) Positive semidefiniteness is preserved by the partial trace of the positive semidefinite normalized pure state, and the partial trace preserves the trace, which the normalization fixes to one. $\square$

Proposition 20.10.17 (Pure-state area-law upper bound). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor of bond dimension $D \geq 1$ with $V^{(N)}(A) \neq 0$. For every block length $L \leq N$, $S_L^{(N)}(A) \leq 2 \log D$, uniformly in L and N . This is the area law of [11, Section 3, line 599]: the block entropy is bounded by a constant independent of the block size.

Proof. [✓ Proof](#) The reduced state $\rho_L^{(N)}(A)$ is a density matrix (Lemma 20.10.16), so its rank is positive by Lemma 20.8.26. Its rank is at most D^2 by Lemma 20.10.15. The rank bound on the entropy gives $S_L^{(N)}(A) \leq \log(D^2) = 2 \log D$. $\square$

Lemma 20.10.18 (Nonnegativity of the pure-state block entropy). [Lean](#) [✓ Stmt](#) For an MPS tensor A with $V^{(N)}(A) \neq 0$ and every block length $L \leq N$, the pure-state block entropy is non-negative: $0 \leq S_L^{(N)}(A)$.

Proof. [✓ Proof](#) The reduced state $\rho_L^{(N)}(A)$ is a density matrix (Lemma 20.10.16), and the von Neumann entropy of a density matrix is non-negative. $\square$

Lemma 20.10.19 (Full-block pure-state entropy vanishes). [Lean](#) [✓ Stmt](#) For an MPS tensor A with $V^{(N)}(A) \neq 0$, the entropy of the whole chain vanishes: $S_N^{(N)}(A) = 0$.

Proof. [✓ Proof](#) After reindexing the full block, the normalized operator is the rank-one pure state

$$\frac{|V^{(N)}(A)\rangle\langle V^{(N)}(A)|}{\langle V^{(N)}(A)|V^{(N)}(A)\rangle}.$$

A pure state has von Neumann entropy zero. $\square$

Lemma 20.10.20 (Empty-block pure-state entropy vanishes). [Lean](#) [✓ Stmt](#) For an MPS tensor A with $V^{(N)}(A) \neq 0$, the pure-state block entropy of the empty block vanishes: $S_0^{(N)}(A) = 0$.

Proof. [✓ Proof](#) By the Schmidt symmetry $S_0^{(N)}(A) = S_N^{(N)}(A)$, and the entropy of the full system vanishes for a pure state (Lemma 20.10.19). $\square$

Lemma 20.10.21 (Saturation telescopes to a constant block entropy). [Lean](#) [✓ Stmt](#) If an MPS tensor A saturates the area law, then the block entropies coincide for all $1 \leq L, L' \leq \lfloor N/2 \rfloor$: $S_L^{(N)}(A) = S_{L'}^{(N)}(A)$.

Proof. [✓ Proof](#) The defining condition supplies $S_m^{(N)}(A) = S_{m+1}^{(N)}(A)$ throughout the interval $1 \leq m < \lfloor N/2 \rfloor$. Apply Lemma 20.8.5. $\square$

Lemma 20.10.22 (Pure-state mutual information is twice the block entropy). [Lean](#) [✓ Stmt](#) For an MPS tensor A with $V^{(N)}(A) \neq 0$, the mutual information of the purification MPDO equals twice the pure-state block entropy: $I_L = 2S_L^{(N)}(A)$.

Proof. [✓ Proof](#) By definition, $I_L = S_L + S_{N-L} - S_N$ for the generated state. The global entropy of the pure state vanishes (Lemma 20.10.19), and the Schmidt symmetry (Lemma 20.10.8) gives $S_{N-L} = S_L$, so $I_L = 2S_L$. $\square$

Proposition 20.10.23 (Pure-state mutual-information bound). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor of bond dimension $D \geq 1$ with $V^{(N)}(A) \neq 0$. For the purification MPDO of A , whose generated operator is the pure state $|V^{(N)}(A)\rangle\langle V^{(N)}(A)|$, the mutual information between a block of L spins and the rest satisfies $I_L \leq 4 \log D$, uniformly in L and N .

Proof. [✓ Proof](#) For a pure state, $I_L = 2S_L^{(N)}(A)$, since the global entropy S_N vanishes and a block and its complement share the same Schmidt spectrum. The block-entropy bound $S_L^{(N)}(A) \leq 2 \log D$ (Proposition 20.10.17) then gives $I_L \leq 4 \log D$. $\square$

20.11 Isometric enlargements of the physical space

Definition 20.11.1 (Doubled physical matrix). [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ be a linear map. Its action on the doubled ket–bra index is the matrix $W : \mathbb{C}^{d^2} \rightarrow \mathbb{C}^{e^2}$ with

$$W_{(i,j),(p,q)} = V_{i,p} \overline{V_{j,q}}. \quad (20.20)$$

Lemma 20.12.2 (Identification with the tensor-level transfer). [Lean](#) [✓ Stmt](#) *The physical-trace transfer of the doubled-index view of an MPO tensor is the physical-trace transfer of the tensor itself.*

Proof. [✓ Proof](#) If M^{dbl} is the doubled-index view of M , then $(M^{\text{dbl}})^{(i,j)} = M^{ij}$, and hence $\mathcal{T}_{M^{\text{dbl}}} = \sum_i (M^{\text{dbl}})^{ii} = \sum_i M^{ii} = \mathcal{T}_M$. $\square$

Lemma 20.12.3 (Blockwise form of the physical-trace transfer). [Lean](#) [✓ Stmt](#) *For a sector decomposition with basis elements A_j and weights $\mu_{j,q}$, the physical-trace transfer of the direct-sum tensor is the block direct sum of the weighted transfers of the copies:*

$$\mathcal{T} = \bigoplus_j \bigoplus_q \mu_{j,q} \mathcal{T}_{A_j}. \quad (20.25)$$

Proof. [✓ Proof](#) The diagonal physical sum distributes over the block direct sum:

$$\begin{aligned} \mathcal{T} &= \sum_i S^{ii} = \sum_i \bigoplus_{j,q} \mu_{j,q} A_j^{ii} \\ &= \bigoplus_{j,q} \mu_{j,q} \sum_i A_j^{ii} = \bigoplus_{j,q} \mu_{j,q} \mathcal{T}_{A_j}. \end{aligned}$$

$\square$

Theorem 20.12.4 (Gauge-phase invariance of nilpotent physical trace). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If two doubled-index tensors are related by an invertible gauge and a nonzero phase, then the physical-trace transfer of one is nilpotent if and only if that of the other is nilpotent. Identifying equal bond dimensions does not alter this condition.*

Proof. [✓ Proof](#) The physical-trace transfer transforms by the same nonzero phase and simultaneous similarity. Powers therefore transform by the corresponding nonzero scalar power and similarity, which preserves whether a power vanishes. $\square$

Theorem 20.12.5 (Presentation independence of BNT nonnilpotency). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT sector presentations of the same doubled-index tensor. Every representative of P has non-nilpotent physical-trace transfer if and only if every representative of Q does.*

Proof. [✓ Proof](#) Match the representatives by Theorem 8.6.1.11, then apply Theorem 20.12.4 to each matched pair. $\square$

Definition 20.12.6 (Simple tensor). [Lean](#) [✓ Stmt](#) A tensor generating MPDOs is *simple* if there is a positive physical blocking length L such that the doubled-index tensor of the L -site blocking has a BNT sector presentation P for which every representative B_j has non-nilpotent physical-trace transfer $\mathcal{T}_{B_j}$. This records the standing canonical-block convention of [11, lines 217–246] together with the simplicity condition of [11, lines 815–822].

Every representative in P has a positive number of copies, and every copy has nonzero weight. The blocking length is existential, and presentation independence follows from Theorem 20.12.5. The definition omits the global unit-weight normalization of [11, line 246] and does not require a normalized horizontal canonical-form witness.

Theorem 20.12.7 (Simplicity implies positive-length nontriviality). [Lean](#) [✓ Stmt](#) *If M is simple, then $\rho^{(N)}(M) \neq 0$ for some $N > 0$.*

Proof. ✓ Proof Choose a representative and one of its copies. Every copy weight over that representative is nonzero, so its power-sum coefficient cannot vanish at every sufficiently large length. At a large length where this coefficient is nonzero, eventual linear independence of the BNT representatives makes the blocked matrix product vector nonzero. The closed-chain blocking identity then gives a nonzero unblocked MPO at a positive length. $\square$

Lemma 20.12.8 (Scaling law for MPO words). Lean ✓ Stmt *Let M be an MPO tensor and let $c \in \mathbb{C}$. If two physical words u, v have the same length, then*

$$(cM)^{u,v} = c^{|u|} M^{u,v}.$$

Proof. ✓ Proof Induct on the common word length and collect one factor of c from each tensor letter. $\square$

Theorem 20.12.9 (Scaling a BNT presentation). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let P be a BNT sector presentation of A and let $c \in \mathbb{C} \setminus \{0\}$. Multiplying every copy weight of P by c gives a BNT sector presentation of cA with the same representatives and multiplicities.*

Proof. ✓ Proof The new copy weights remain nonzero. Normality and eventual linear independence of the representatives are unchanged, and both total matrix product vectors acquire the common factor c^N at length N . $\square$

Theorem 20.12.10 (Transport of a normal basis along equal MPV families). Lean ✓ Stmt *If A and A' generate the same positive-length MPV family, then every basis of normal tensors for A is also a basis of normal tensors for A' .*

Proof. ✓ Proof The basis tensors, their normality, and their eventual linear independence are unchanged. For every $N > 0$,

$$|V^{(N)}(A')\rangle = |V^{(N)}(A)\rangle \tag{20.26}$$

$$= \sum_{j=1}^g c_j^{(N)} |V^{(N)}(B_j)\rangle, \tag{20.27}$$

where (20.26) is Definition 2.4.4 and (20.27) is the spanning identity in Definition 8.1.2. Thus the unchanged coefficients $c_j^{(N)}$ give the required expansion for A' . $\square$

Theorem 20.12.11 (Scaling law for closed matrix-product operators). Lean ✓ Stmt *Let M be an MPO tensor and let $c \in \mathbb{C}$. For every chain length N , the closed operator generated by cM is*

$$\rho^{(N)}(cM) = c^N \rho^{(N)}(M).$$

Proof. ✓ Proof Apply Lemma 20.12.8 to the two length- N physical words, unfold the closed MPO family, and use linearity of the virtual trace. $\square$

Theorem 20.12.12 (Normalization is invariant under nonzero rescaling). Lean ✓ Stmt *Let M be an MPO tensor and let $c \in \mathbb{C}$ be nonzero. For every chain length N , the normalized density operators agree:*

$$\text{tr}[\rho^{(N)}(cM)]^{-1} \rho^{(N)}(cM) = \text{tr}[\rho^{(N)}(M)]^{-1} \rho^{(N)}(M),$$

with the inverse of zero understood to be zero.

Proof. ✓ **Proof** By Theorem 20.12.11, the closed operator and its trace are both multiplied by c^N , and the two nonzero factors cancel in the normalization. When the trace vanishes, both sides are zero by the convention on the inverse of zero. □

Definition 20.12.13 (Word encoded by a physical-block index). Lean ✓ **Stmt** A physical index for an L -site block determines a word by decoding the index into its L original physical letters and listing them in site order.

Lemma 20.12.14 (Length of an encoded physical-block word). Lean ✓ **Stmt** The word encoded by an L -site physical-block index has length L .

Proof. ✓ **Proof** This follows from the fixed-length block-word encoding. □

Theorem 20.12.15 (Scaling law for physically blocked MPO tensors). Lean ✓ **Stmt** Let M be an MPO tensor and let $c \in \mathbb{C}$. For every blocking length L , one has

$$(cM)^{[L]} = c^L M^{[L]}.$$

Proof. ✓ **Proof** Unfold physical blocking and apply Lemma 20.12.8 to the two encoded words; Lemma 20.12.14 gives their length L . □

Theorem 20.12.16 (Nonnegative rescaling preserves the MPDO condition). Lean ✓ **Stmt** Let M generate positive semidefinite closed operators at every positive length, and let $r \geq 0$ be real. Then rM has the same property.

Proof. ✓ **Proof** The length- N operator is multiplied by the nonnegative scalar r^N . □

Theorem 20.12.17 (Positive rescaling invariance of the MPDO condition). Lean ✓ **Stmt** Let M be an MPO tensor and let $r > 0$ be real. Then rM generates positive semidefinite closed operators at every positive chain length if and only if M does.

Proof. ✓ **Proof** Apply Theorem 20.12.16 first with r and then with r^{-1} . □

Theorem 20.12.18 (Positive rescaling preserves simplicity). Lean ✓ **Stmt** Let M be simple and let $r > 0$ be real. Then rM is simple.

Proof. ✓ **Proof** Positive rescaling preserves the MPDO condition. For a blocking length L , it multiplies the blocked tensor by the nonzero scalar r^L . Multiply every copy weight by r^L and apply Theorem 20.12.9; the representatives and their non-nilpotent physical-trace transfers are unchanged. □

Theorem 20.12.19 (Positive rescaling invariance of simplicity). Lean ✓ **Stmt** Let M be an MPO tensor and let $r > 0$ be real. Then rM is simple if and only if M is simple.

Proof. ✓ **Proof** Apply the preceding implication first with r and then with r^{-1} . □

Definition 20.12.20 (Normalized simplicity in blocked canonical form). Lean ✓ **Stmt** A tensor generating MPDOs satisfies this predicate if it has a normalized horizontal basis-of-normal-tensors decomposition whose basis elements all have non-nilpotent physical-trace transfer. This records the horizontal-canonical-form data of the already-blocked tensor K fixed in [11, Appendix C.2, line 1628].

This definition fixes the canonical representative and incorporates the global unit-weight convention of [11, line 246]. It instantiates the Appendix C.2 hypothesis on the already-blocked tensor, in contrast to Definition 20.12.6 which quantifies over blocking lengths and presentations.

Lemma 20.12.21 (Normalized simple canonical forms are horizontal canonical). [Lean](#) [✓ Stmt](#) *A tensor in normalized simple canonical form is in horizontal canonical form.*

Proof. [✓ Proof](#) The sector decomposition, its basis-of-normal-tensors property, and the block-diagonal gauge directly satisfy the requirements of a horizontal canonical form. $\square$

Theorem 20.12.22 (Normalized simple canonical forms are simple). [Lean](#) [✓ Stmt](#) *Every tensor in normalized simple canonical form is simple in the sense of Definition 20.12.6.*

Proof. [✓ Proof](#) Take the blocking length to be one. The canonical bijection between the one-site blocked ket–bra alphabet and the original doubled alphabet relabels the displayed sector decomposition into a BNT presentation of the blocked tensor. Normality and eventual linear independence are preserved by this relabeling, and the global block gauge gives equality of all positive-length closed matrix-product coefficients. Write e_1 for the one-site physical bijection, $\hat{e} = e_1 \times e_1$ for its induced bijection of ket–bra pairs, and $\hat{e}^* A_j$ for the corresponding relabeling of the j -th representative. Since $\hat{e}(i, i) = (e_1(i), e_1(i))$, reindexing the diagonal sum gives

$$\mathcal{T}_{\hat{e}^* A_j} = \sum_i (\hat{e}^* A_j)^{ii} = \sum_i A_j^{e_1(i), e_1(i)} = \sum_k A_j^{kk} = \mathcal{T}_{A_j}.$$

The assumed non-nilpotency certificates therefore prove simplicity. $\square$

20.13 Cyclic local-embedding coordinates

Definition 20.13.1 (Cyclic window-complement coordinates). [Lean](#) [✓ Stmt](#) A periodic configuration is identified with the ordered pair consisting of its restriction to the cyclic window W_i^L and its restriction to the complementary cyclic interval.

Theorem 20.13.2 (Local embeddings in window-complement coordinates). [Lean](#) [✓ Stmt](#) *In cyclic window-complement coordinates, every local embedding has the form $E_i^L(B) = B \otimes \mathbb{1}_{\text{comp}}$.*

Proof. [✓ Proof](#) The matrix entry between (x, u) and (y, v) vanishes unless the complementary configurations agree, $u = v$. In that case it equals $B_{x,y}$, which is the corresponding entry of $B \otimes \mathbb{1}_{\text{comp}}$. $\square$

Theorem 20.13.3 (Two-site local embeddings in pair coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) *Under the canonical identification of two-site configurations with ordered pairs, embedding an arbitrary two-site operator B beginning at site zero gives B , while embedding it beginning at site one gives SBS^{-1} , where $S(x \otimes y) = y \otimes x$ exchanges the two factors.*

Proof. [✓ Proof](#) The cyclic windows beginning at sites zero and one read the physical sites in the orders $(0, 1)$ and $(1, 0)$, respectively. Reindexing by ordered pairs therefore preserves B in the first case and conjugates it by S in the second. $\square$

20.14 Gibbs states of nearest-neighbor commuting Hamiltonians

Definition 20.14.1 (GSNNCH). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A density operator on

a periodic chain is a Gibbs state of a nearest-neighbor commuting Hamiltonian if

$$\rho^{(N)} \propto \bigoplus_x n_x \exp\left(-\sum_{j=1}^N \tau_j(h^{(x)})\right),$$

$$n_x \in \mathbb{N},$$

where the one-site spaces belonging to distinct labels x are orthogonal and $[h^{(x)}, \tau_1(h^{(x)})] = 0$. With the projector-limit convention of [11, Definition 4.8, lines 829–850], this is equivalently

$$\rho^{(N)} \propto \bigoplus_x n_x \prod_{j=1}^N \tau_j(B^{(x)}),$$

$$B^{(x)} \geq 0,$$

with commuting neighboring translates. The orthogonal direct sum decomposes the global Hilbert space into mutually orthogonal uniform product sectors [11, Definition 4.8, lines 829–850].

When restricted to a single sector with trivial multiplicity, the two-site bond in Definition 20.14.1 acts on the full two-site space without sector projections.

Definition 20.14.2 (Single-bond commuting-form data for an MPDO). [Lean](#) [✓ Stmt](#) For a fixed chain length $N \geq 2$, this consists of one positive semidefinite two-site operator $B \geq 0$ whose translated copies commute pairwise on the periodic N -site chain: $[B_{i,i+1}, B_{j,j+1}] = 0$.

Definition 20.14.3 (Single-bond commuting form). [Lean](#) [✓ Stmt](#) An MPO tensor M has the single-bond commuting-form property if for every $N \geq 2$ there is a two-site operator B as in Definition 20.14.2 and a constant $c > 0$ such that

$$\rho^{(N)}(M) = c \prod_{i=0}^{N-1} B_{i,i+1}.$$

Allowing the bond B to depend on N formalizes a chainwise version of the single translation-invariant bond in [11, Appendix C.2, lines 1597–1619]; see [37].

Theorem 20.14.4 (A commuting-form realization gives one GSNNCH sector). [Lean](#) [Lean](#) [✓ Stmt](#) A positive commuting-form bond realizing a finite-chain operator gives the one-sector instance of the explicit GSNNCH decomposition, with sector projection equal to the identity and multiplicity one.

Proof. [✓ Proof](#) Take one sector, its one-site projection to be the identity, its multiplicity to be one, and its bond to be the given bond. $\square$

Definition 20.14.5 (Restriction to an orthogonal physical sector). [Lean](#) [✓ Stmt](#) Let P be an orthogonal one-site projection and let K be an MPO tensor fixed by compression with P on its two physical indices. A physical support restriction consists of a dimension e and an isometry $V : \mathbb{C}^e \rightarrow \mathbb{C}^d$ such that

$$V^*V = \mathbb{1},$$

$$VV^* = P,$$

for which the restricted tensor $K_P = V^*KV$ is injective and its inclusion into the ambient physical space recovers K exactly.

Theorem 20.14.6 (Injective restriction to a BNT projector range). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 Suppose that the source BNT projectors have been constructed after the common physical blocking, and let $\mathcal{K}_s$ be the normal representative with its common copy weight absorbed. Then $\mathcal{K}_s$ admits a physical support restriction to the range of P_s . In particular, for some isometry V_s ,

$$\begin{aligned} V_s V_s^* &= P_s, \\ \mathcal{K}_s &= V_s (V_s^* \mathcal{K}_s V_s) V_s^*, \end{aligned}$$

and the restricted tensor $V_s^* \mathcal{K}_s V_s$ is injective.

Proof. [✓ Proof](#) The projector identity $P_s \mathcal{K}_s P_s = \mathcal{K}_s$ gives an isometric inclusion of its range. The simultaneous one-site spanning condition makes each normal representative injective, and multiplication by its nonzero common weight preserves injectivity. Write $K_{P_s} = V_s^* \mathcal{K}_s V_s$. Each ambient matrix slice satisfies

$$(\mathcal{K}_s)_{ij} = \sum_{p,q} (V_s)_{ip} \overline{(V_s)_{jq}} (K_{P_s})_{pq}.$$

Hence

$$\text{span}_{\mathbb{C}} \{(\mathcal{K}_s)_{ij}\}_{i,j} \subseteq \text{span}_{\mathbb{C}} \{(K_{P_s})_{pq}\}_{p,q}.$$

The left-hand side is the full virtual matrix algebra, so K_{P_s} is injective. The two isometry identities give the exact reconstruction. $\square$

Theorem 20.14.7 (Normalized operators under an isometric physical inclusion). [Lean](#) [✓ Stmt](#)
 Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ satisfy $V^* V = I$, and let K^V be obtained from an MPO tensor K by conjugating each physical matrix by V . The normalized periodic operators satisfy

$$\sigma^{(N)}(K^V) = V^{\otimes N} \sigma^{(N)}(K) (V^{\otimes N})^*.$$

Proof. [✓ Proof](#) Since $(V^{\otimes N})^* V^{\otimes N} = I$, cyclicity of the trace gives

$$\text{tr}[V^{\otimes N} \rho^{(N)}(K) (V^{\otimes N})^*] = \text{tr}[\rho^{(N)}(K)].$$

Therefore the scalar normalization commutes with the congruence:

$$\sigma^{(N)}(K^V) = \frac{V^{\otimes N} \rho^{(N)}(K) (V^{\otimes N})^*}{\text{tr}[\rho^{(N)}(K)]} = V^{\otimes N} \sigma^{(N)}(K) (V^{\otimes N})^*.$$

$\square$

Theorem 20.14.8 (Prefix marginal of a sitewise isometric image). [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ satisfy $V^* V = I$. If an operator on $L + K$ sites is conjugated by $V^{\otimes(L+K)}$, then its first- L -site marginal is conjugated by $V^{\otimes L}$.

Proof. [✓ Proof](#) [Lean](#) Let X denote the operator before conjugation. Reindex the chain into its first L and final K sites. The sitewise matrix becomes $V^{\otimes L} \otimes V^{\otimes K}$. Since $(V^{\otimes K})^* V^{\otimes K} = I$, the partial trace satisfies

$$\begin{aligned} \text{tr}_K[(V^{\otimes L} \otimes V^{\otimes K}) X (V^{\otimes L} \otimes V^{\otimes K})^*] \\ = V^{\otimes L} \text{tr}_K[X] (V^{\otimes L})^*. \end{aligned}$$

$\square$

Lemma 20.14.9 (Rectangular characteristic-polynomial cyclicity). [✓ ML stmt](#) Let A be an $e \times d$ matrix and B a $d \times e$ matrix, with $d \leq e$. Then

$$\chi_{AB}(X) = X^{e-d} \chi_{BA}(X).$$

Theorem 20.14.10 (Characteristic polynomial under an isometric inclusion). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ satisfy $V^*V = I_d$, and let A be a $d \times d$ matrix. Then $d \leq e$ and

$$\chi_{VAV^*}(X) = X^{e-d} \chi_A(X).$$

Equivalently, the roots of χ_{VAV^*} are the roots of χ_A together with zero of multiplicity $e - d$.

Proof. [✓ Proof](#) Write $VAV^* = V(AV^*)$. Characteristic-polynomial cyclicity for rectangular factors gives

$$\chi_{V(AV^*)}(X) = X^{e-d} \chi_{(AV^*)V}(X) = X^{e-d} \chi_A(X),$$

where the last equality uses $V^*V = I_d$. Taking roots of the product gives

$$\text{roots}(\chi_{VAV^*}) = \text{roots}(X^{e-d} \chi_A) \tag{20.28}$$

$$= \text{roots}(X^{e-d}) + \text{roots}(\chi_A) \tag{20.29}$$

$$= (e - d) \cdot \{0\} + \text{roots}(\chi_A).$$

The middle equality applies because X^{e-d} is nonzero and the monic polynomial χ_A is nonzero. $\square$

Theorem 20.14.11 (Reduced states under an isometric physical inclusion). [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ be an isometry, and let K^V be obtained from an MPO tensor K by conjugating each physical matrix by V . For every $0 \leq L \leq N$, the contiguous reduced states satisfy

$$\rho_L^{(N)}(K^V) = V^{\otimes L} \rho_L^{(N)}(K) (V^{\otimes L})^*.$$

Proof. [✓ Proof](#) The normalized periodic operators are related by the sitewise isometric congruence. Splitting the chain into the retained interval and its complement gives

$$\begin{aligned} \text{tr}_{N \setminus L} \left[(V^{\otimes L} \otimes V^{\otimes (N-L)}) \sigma^{(N)}(K) (V^{\otimes L} \otimes V^{\otimes (N-L)})^* \right] \\ = V^{\otimes L} \rho_L^{(N)}(K) (V^{\otimes L})^*. \end{aligned} \tag{20.30}$$

Here $(V^{\otimes (N-L)})^* V^{\otimes (N-L)} = I$, so the isometry on the traced factor disappears under the partial trace. $\square$

Theorem 20.14.12 (SAL descends through an isometric physical inclusion). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ be an isometry, and let K^V be obtained from an MPO tensor K by conjugating each physical matrix by V . For every $N \geq 1$,

$$\text{tr}(\sigma^{(N)}(K^V)) = \text{tr}(\sigma^{(N)}(K)).$$

If the periodic operators of K^V are positive semidefinite, then those of K are positive semidefinite. Their mutual informations agree at every contiguous cut. Consequently, if K^V satisfies saturation of the area law, then so does K .

Proof. ✓ Proof At length N , Lemma 21.3.48 identifies the changed periodic operator with conjugation by $V^{\otimes N}$. Since $(V^{\otimes N})^* V^{\otimes N} = \mathbb{1}$, cyclicity of the trace gives the displayed equality, and compression by $(V^{\otimes N})^*$ preserves positivity.

The same trace equality shows that normalized periodic operators obey the same conjugation identity. After a chain is divided into a retained interval and its complement, the isometry on the traced-out factor disappears under the partial trace. The retained marginal is therefore conjugated by the corresponding tensor power of V , so its nonzero spectrum and entropy are unchanged. Thus the mutual informations, and hence the consecutive equalities defining SAL, agree. $\square$

Theorem 20.14.13 (SAL passes to the physical support restriction). Lean ✓ Stmt *If K satisfies saturation of the area law and $K = VK_P V^*$ is a physical support restriction, then the injective restricted tensor K_P also satisfies saturation of the area law.*

Proof. ✓ Proof On a chain of every positive length, the sitewise inclusion $V^{\otimes N}$ is an isometry and

$$\rho^{(N)}(K) = V^{\otimes N} \rho^{(N)}(K_P) (V^{\otimes N})^*.$$

The trace is therefore unchanged. After a contiguous marginal is taken, the isometry on the traced-out sites disappears, while the isometry on the retained sites remains. Thus every block marginal of K is an isometric conjugate of the corresponding marginal of K_P . Their nonzero spectra, and hence their entropies and mutual informations, coincide. The consecutive equalities defining SAL for K consequently give those for K_P . $\square$

Theorem 20.14.14 (The restricted commuting product has a supported ambient realization). Lean ✓ Stmt *Let $K = VK_P V^*$ be a physical support restriction of an MPO tensor satisfying SAL. Applying [11, Proposition C.8] to the injective tensor K_P gives a positive two-site bond B_P whose cyclic translates commute pairwise on every periodic chain of length $N \geq 2$. Its lift $B = V^{\otimes 2} B_P (V^{\otimes 2})^*$ is positive, its cyclic translates commute pairwise on every such ambient chain, and $(P \otimes P)B(P \otimes P) = B$. Moreover, for every $N \geq 2$ one has the exact identity*

$$\rho^{(N)}(K) = \prod_{i=0}^{N-1} B_{i,i+1}.$$

Proof. ✓ Proof Theorem 20.14.13 gives SAL for the injective restricted tensor, so [11, Proposition C.8] supplies its positive bond. Since $V^* V = \mathbb{1}$, positivity is preserved under conjugation by $V^{\otimes 2}$. Also, $V^{\otimes 2} (V^{\otimes 2})^* = P \otimes P$; multiplying the lifted bond by this projection on either side therefore leaves it unchanged.

It remains to prove commutativity after the lift. On a three-site chain, isometric conjugation carries the product of two adjacent restricted bonds to the product of their lifted bonds. The restricted product is Hermitian because its positive factors commute. Its conjugate is therefore Hermitian; since the lifted factors are Hermitian, their product equals the product in the reverse order. The two-site periodic chain is handled by the same argument with the two oppositely oriented bonds. Locality and cyclic translation then give pairwise commutativity for every $N \geq 2$.

Finally, isometric conjugation preserves every nonempty product of restricted bonds. The image of each restricted bond is the corresponding lifted bond multiplied by the projection onto the range of $V^{\otimes N}$. The support projections contributed by the first $N - 1$ bonds cover all sites and leave the full commuting product fixed. Hence the range projection disappears, and the restricted finite-chain product identity becomes the asserted ambient identity, still with normalization scalar one. $\square$

Definition 20.14.15 (Proportionally realized orthogonal commuting sector products). [Lean](#)

[✓ Stmt](#) Let $\{K_x\}_{x=0}^{g-1}$ be MPO tensors on a common one-site space. A proportionally realized orthogonal commuting sector family consists of pairwise orthogonal one-site projections P_x and positive two-site operators $B^{(x)}$ such that

$$(P_x \otimes P_x)B^{(x)}(P_x \otimes P_x) = B^{(x)},$$

$$[\tau_i(B^{(x)}), \tau_j(B^{(x)})] = 0.$$

For every sector x and every $N \geq 2$, there is a number $a_{x,N} > 0$ such that

$$\rho^{(N)}(K_x) = a_{x,N} \prod_{i=0}^{N-1} \tau_i(B^{(x)}).$$

Theorem 20.14.16 (One-site positivity completes proportional sector positivity). [Lean](#) [Lean](#)

[✓ Stmt](#) Every sector operator of length $N \geq 2$ in a proportionally realized orthogonal commuting sector family is positive semidefinite. Consequently, if the one-site operator $\rho^{(1)}(K_x)$ is positive semidefinite for every sector x , then every K_x generates matrix product density operators at all positive lengths.

Proof. [✓ Proof](#) For $N \geq 2$, the sector operator is a positive scalar multiple of a product of commuting positive bonds, hence is positive semidefinite. The one-site hypothesis supplies the only remaining positive chain length. $\square$

Definition 20.14.17 (Exactly realized orthogonal commuting sector products). [Lean](#) [Lean](#)

[✓ Stmt](#) Let $\{K_x\}_{x=0}^{g-1}$ be MPO tensors on a common one-site space. An orthogonally supported family of commuting sector products consists of pairwise orthogonal one-site projections P_x and positive two-site operators $B^{(x)}$ such that

$$(P_x \otimes P_x)B^{(x)}(P_x \otimes P_x) = B^{(x)},$$

$$[\tau_i(B^{(x)}), \tau_j(B^{(x)})] = 0,$$

and, for every $N \geq 2$,

$$\rho^{(N)}(K_x) = \prod_{i=0}^{N-1} \tau_i(B^{(x)}).$$

Together with natural multiplicities n_x , these objects determine the corresponding finite-chain sector decomposition in Definition 20.14.1.

Definition 20.14.18 (Exact sector products as proportionally realized products). [Lean](#) [✓ Stmt](#)

Every exactly realized orthogonal commuting sector family determines a proportionally realized orthogonal commuting sector family with the same one-site projections and two-site bonds.

Proof. [✓ Proof](#) Take $a_{x,N} = 1$ for every sector x and every $N \geq 2$. $\square$

Theorem 20.14.19 (Positive factorizations on orthogonal physical restrictions). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Let $\{\mathcal{K}_x\}_{x=0}^{g-1}$ be injective MPO tensors and let $\{P_x\}_{x=0}^{g-1}$ be pairwise orthogonal projections. Suppose that every physical slice is absorbed on both sides:

$$P_x \mathcal{K}_x[\beta, \alpha] = \mathcal{K}_x[\beta, \alpha] = \mathcal{K}_x[\beta, \alpha] P_x.$$

Write $\mathcal{K}_x = V_x \mathcal{K}_{x, P_x} V_x^*$ for the restriction to the range of P_x . If every $\mathcal{K}_{x, P_x}$ admits a physical-sector factorization whose neighboring operators are positive semidefinite, then there are positive bonds $B^{(x)}$ such that

$$(P_x \otimes P_x) B^{(x)} (P_x \otimes P_x) = B^{(x)}$$

and all translates of each $B^{(x)}$ commute pairwise. For every $N \geq 2$,

$$\rho^{(N)}(\mathcal{K}_x) = \prod_{i=0}^{N-1} \tau_i(B^{(x)}).$$

Thus these bonds form an orthogonally supported family of commuting sector products.

Proof. ✓ Proof The two-sided equations identify each tensor with its compression to $\text{ran}(P_x)$ and give an injective restricted tensor. The positive neighboring operators determine the positive commuting bond on this restricted space. Conjugating by $V_x^{\otimes 2}$ preserves positivity and commutativity, gives the displayed support equation, and transports every nonempty periodic product exactly. □

Theorem 20.14.20 (SAL supplies orthogonally supported BNT sector products). Lean ✓ Stmt Suppose that the BNT physical projectors P_x have been selected and that every common-weight-absorbed representative $\mathcal{K}_x$ satisfies SAL. Then there are positive two-site bonds $B^{(x)}$ such that the projections P_x are pairwise orthogonal,

$$(P_x \otimes P_x) B^{(x)} (P_x \otimes P_x) = B^{(x)},$$

all cyclic translates of each $B^{(x)}$ commute pairwise, and, for every $N \geq 2$,

$$\rho^{(N)}(\mathcal{K}_x) = \prod_{i=0}^{N-1} \tau_i(B^{(x)}).$$

Thus the absorbed BNT representatives have an orthogonally supported family of commuting sector products with normalization scalar one.

Proof. ✓ Proof Restrict $\mathcal{K}_x$ isometrically to the range of P_x . The restricted tensor is injective and satisfies SAL, so [11, Proposition C.8] gives its positive commuting bond product. Apply Theorem 20.14.14 in every sector. The lifted bond is supported on $P_x \otimes P_x$ and realizes $\mathcal{K}_x$ exactly. The BNT projector construction gives $P_x P_y = 0$ for $x \neq y$, which completes the sector family. □

Theorem 20.14.21 (Orthogonal commuting sector products give the GSNNCH form). Lean

Lean ✓ Stmt Suppose that $\{K_x\}_{x=0}^{g-1}$ has an orthogonally supported family of commuting sector products and that, for every $N \geq 2$,

$$\rho^{(N)}(M) = \sum_{x=0}^{g-1} n_x \rho^{(N)}(K_x),$$

$$n_x \in \mathbb{N}.$$

Then M has the explicit GSNNCH sector form with outer sectors x and natural multiplicities n_x .

Proof. ✓ Proof For each $N \geq 2$, the represented unnormalized sector sum is

$$\sum_{x=0}^{g-1} n_x \prod_{i=0}^{N-1} \tau_i(B^{(x)}) = \sum_{x=0}^{g-1} n_x \rho^{(N)}(K_x) = \rho^{(N)}(M).$$

Thus the positive proportionality constant in Definition 20.14.1 is one. □

Theorem 20.14.22 (Nonnegative sector coefficients give the GSNNCH form). Lean Lean Lean
Lean Lean ✓ Stmt Suppose that $\{K_x\}_{x=0}^{g-1}$ has an orthogonally supported family of commuting sector products, and let $n_x \geq 1$ be natural numbers. For every $N \geq 2$, let $c_x(N) \geq 0$ and suppose that

$$\rho^{(N)}(M) = \sum_{x=0}^{g-1} c_x(N) \rho^{(N)}(K_x).$$

Then M has the explicit GSNNCH sector form.

Proof. ✓ Proof Fix $N \geq 2$. Replace the bond in sector x by

$$\widetilde{B}_N^{(x)} = \left(\frac{c_x(N)}{n_x} \right)^{1/N} B^{(x)}.$$

This bond is positive, has the same support, and its translates commute. Its periodic product is

$$n_x \prod_{i=0}^{N-1} \tau_i(\widetilde{B}_N^{(x)}) = c_x(N) \prod_{i=0}^{N-1} \tau_i(B^{(x)}) = c_x(N) \rho^{(N)}(K_x).$$

Taking n_x as the sector multiplicity gives the required decomposition, with positive proportionality constant one. □

Theorem 20.14.23 (Proportional local orthogonal sums preserve SAL). Lean ✓ Stmt Let $\{K_x\}_{x=0}^{g-1}$ be a nonempty finite family of MPO tensors, let n_x be positive natural numbers, and let P_x be pairwise orthogonal one-site projections. Suppose that, for every $N > 0$, there is a number $c_N > 0$ such that

$$\rho^{(N)}(M) = c_N \sum_{x=0}^{g-1} n_x \rho^{(N)}(K_x).$$

For $N \geq 2$ and $1 \leq L \leq N$, write $\hat{\rho}_{x,N,L}$ for the normalized marginal of $\rho^{(N)}(K_x)$ on the first L sites. Assume that

$$(P_x \otimes \mathbb{1}^{\otimes(L-1)}) \hat{\rho}_{x,N,L} = \hat{\rho}_{x,N,L}.$$

If every K_x satisfies SAL, then M satisfies SAL.

Proof. ✓ Proof For fixed N , put

$$S_N = \sum_x n_x \rho^{(N)}(K_x),$$

$$p_x = \frac{c_N n_x \operatorname{tr}(\rho^{(N)}(K_x))}{\operatorname{tr}(\rho^{(N)}(M))}.$$

Trace linearity and $\rho^{(N)}(M) = c_N S_N$ give

$$\mathrm{tr}(\rho^{(N)}(M))^{-1} \rho^{(N)}(M) = \mathrm{tr}(S_N)^{-1} S_N = \sum_x p_x \hat{\rho}_{x,N,N}.$$

Thus the common positive scalar cancels from the normalized chain and from every marginal.

Put $Q_x = P_x \otimes \mathbb{1}^{\otimes(L-1)}$. The support identity gives $Q_x \hat{\rho}_{x,N,L} = \hat{\rho}_{x,N,L}$, while $P_x P_y = 0$ for $x \neq y$ gives $Q_x Q_y = 0$. The entropy of a nonempty marginal is therefore

$$S_L(M) = \sum_x (-p_x \log p_x + p_x S_L(K_x)).$$

Substitution in $I_L = S_L + S_{N-L} - S_N$ gives

$$I_L(M) = - \sum_x p_x \log p_x + \sum_x p_x I_L(K_x).$$

The probabilities depend on N but not on L . Sectorwise SAL therefore gives $I_L(M) = I_{L+1}(M)$ throughout the required range. $\square$

Theorem 20.14.24 (Exact local orthogonal sums preserve SAL). Lean ✓ Stmt Let $\{K_x\}_{x=0}^{g-1}$ be a nonempty finite family of MPO tensors, let n_x be positive natural numbers, and let P_x be pairwise orthogonal one-site projections. Suppose that, for every $N > 0$,

$$\rho^{(N)}(M) = \sum_{x=0}^{g-1} n_x \rho^{(N)}(K_x).$$

For $N \geq 2$ and $1 \leq L \leq N$, write $\hat{\rho}_{x,N,L}$ for the normalized marginal of $\rho^{(N)}(K_x)$ on the first L sites. Assume that the sector label is visible on the first retained site:

$$(P_x \otimes \mathbb{1}^{\otimes(L-1)}) \hat{\rho}_{x,N,L} = \hat{\rho}_{x,N,L}.$$

If every K_x satisfies SAL, then M satisfies SAL.

This formalizes the conditional final inference in [11, Appendix C.2, lines 1801–1808], where sectorwise SAL is assumed explicitly; see [37].

Proof. ✓ Proof Take $c_N = 1$ in Theorem 20.14.23. $\square$

Theorem 20.14.25 (Proportional commuting sector products have normalized-chain support). Lean ✓ Stmt Let $\{K_x\}_{x=0}^{g-1}$ have a proportionally realized orthogonal commuting sector family with one-site projections P_x and two-site bonds $B^{(x)}$. Then, for every sector x and every $N \geq 2$,

$$(P_x \otimes \mathbb{1}^{\otimes(N-1)}) \hat{\rho}_{x,N,N} = \hat{\rho}_{x,N,N}.$$

Proof. ✓ Proof Since P_x is a projection and $B^{(x)}$ is supported on $P_x \otimes P_x$,

$$(P_x \otimes \mathbb{1}) B^{(x)} = B^{(x)}.$$

Thus the first translated bond, and hence the full product, satisfy

$$\begin{aligned} (P_x \otimes \mathbb{1}^{\otimes(N-1)}) \tau_0(B^{(x)}) &= \tau_0(B^{(x)}), \\ (P_x \otimes \mathbb{1}^{\otimes(N-1)}) \prod_{i=0}^{N-1} \tau_i(B^{(x)}) &= \prod_{i=0}^{N-1} \tau_i(B^{(x)}). \end{aligned} \tag{20.31}$$

Multiplying (20.31) by $a_{x,N}$ gives

$$(P_x \otimes \mathbb{1}^{\otimes(N-1)})\rho^{(N)}(K_x) = \rho^{(N)}(K_x). \quad (20.32)$$

Multiplying (20.32) by the scalar $(\text{tr } \rho^{(N)}(K_x))^{-1}$ gives

$$\begin{aligned} & (P_x \otimes \mathbb{1}^{\otimes(N-1)}) \left[(\text{tr } \rho^{(N)}(K_x))^{-1} \rho^{(N)}(K_x) \right] \\ &= (\text{tr } \rho^{(N)}(K_x))^{-1} \rho^{(N)}(K_x). \end{aligned}$$

The scalar multiples on both sides are the normalized chain states. This remains valid when the trace is zero, with the convention $0^{-1} = 0$. $\square$

Theorem 20.14.26 (Proportional commuting sector products have prefix-marginal support).

Lean **✓ Stmt** Let $\{K_x\}_{x=0}^{g-1}$ have a proportionally realized orthogonal commuting sector family with one-site projections P_x and two-site bonds $B^{(x)}$. Then, for every sector x , every $N \geq 2$, and every $1 \leq L \leq N$,

$$(P_x \otimes \mathbb{1}^{\otimes(L-1)})\hat{\rho}_{x,N,L} = \hat{\rho}_{x,N,L}.$$

Proof. **✓ Proof** Write $\text{Tr}_{L+1,\dots,N}$ for the partial trace over the final $N - L$ sites. By definition,

$$\hat{\rho}_{x,N,L} = \text{Tr}_{L+1,\dots,N}(\hat{\rho}_{x,N,N}).$$

The first-site action commutes with this partial trace. Therefore Theorem 20.14.25 yields

$$\begin{aligned} & (P_x \otimes \mathbb{1}^{\otimes(L-1)})\hat{\rho}_{x,N,L} \\ &= \text{Tr}_{L+1,\dots,N}((P_x \otimes \mathbb{1}^{\otimes(N-1)})\hat{\rho}_{x,N,N}) \\ &= \text{Tr}_{L+1,\dots,N}(\hat{\rho}_{x,N,N}) = \hat{\rho}_{x,N,L}. \end{aligned}$$

$\square$

Lemma 20.14.27 (Commuting sector products have local marginal support). **Lean** **✓ Stmt**

Suppose that K_x is the periodic product of translates of a positive two-site operator B_x , and that B_x is supported on $P_x \otimes P_x$. If $N \geq 2$ and $1 \leq L \leq N$, then the normalized marginal of K_x on the first L sites satisfies

$$(P_x \otimes \mathbb{1}^{\otimes(L-1)})\hat{\rho}_{x,N,L} = \hat{\rho}_{x,N,L}.$$

Proof. **✓ Proof** Apply Theorem 20.14.26 with $a_{x,N} = 1$. $\square$

Theorem 20.14.28 (Proportional orthogonal commuting sector sums preserve SAL). **Lean**

✓ Stmt Let K_x be a nonempty finite family with a proportionally realized orthogonal commuting sector family, and let n_x be positive natural numbers. Suppose that, for every $N > 0$, there is a number $c_N > 0$ such that

$$\rho^{(N)}(M) = c_N \sum_x n_x \rho^{(N)}(K_x).$$

If every K_x satisfies SAL, then M satisfies SAL.

Proof. **✓ Proof** Theorem 20.14.26 supplies the support of every nonempty marginal. Applying Theorem 20.14.23, its sector weights contain the outer scalar c_N , while the inner scalars $a_{x,N}$ have already disappeared under sector normalization. $\square$

Corollary 20.14.29 (Orthogonal commuting sector sums preserve SAL). [Lean](#) [✓ Stmt](#) Let K_x be a nonempty finite family, let an orthogonally supported commuting sector family realize the tensors K_x , and let n_x be positive natural numbers. Suppose that, for every $N > 0$,

$$\rho^{(N)}(M) = \sum_x n_x \rho^{(N)}(K_x).$$

If every K_x satisfies SAL, then M satisfies SAL.

Proof. [✓ Proof](#) Take $a_{x,N} = c_N = 1$ in Theorem 20.14.28. □

Theorem 20.14.30 (Proportional orthogonal commuting-bond sectors with ZCL satisfy SAL).

[Lean](#) [✓ Stmt](#) Let K_x be a nonempty finite family of injective tensors generating MPDOs. Suppose that there are pairwise orthogonal one-site projections P_x and fixed translation-invariant positive commuting bonds $B^{(x)}$, supported on $P_x \otimes P_x$. For every sector x and every $N \geq 2$, suppose that there is a number $a_{x,N} > 0$ such that

$$\rho^{(N)}(K_x) = a_{x,N} \prod_{i=0}^{N-1} \tau_i(B^{(x)}).$$

Suppose also that every K_x has source zero correlation length. Let n_x be positive natural numbers and suppose that, for every $N > 0$, there is a number $c_N > 0$ such that

$$\rho^{(N)}(M) = c_N \sum_x n_x \rho^{(N)}(K_x).$$

Then M satisfies SAL.

Proof. [✓ Proof](#) For each x , the fixed bond $B^{(x)}$ and the positive proportional product identities form the single-sector commuting-bond hypothesis. Theorem 20.17.56 gives SAL for every K_x . Theorem 20.14.28 then gives SAL for M . □

Theorem 20.14.31 (A compatible sector decomposition with ambient ZCL satisfies SAL).

[Lean](#) [✓ Stmt](#) Let S be a nonempty sector decomposition with representatives A_x , positive copy numbers n_x , and nonzero copy weights. Suppose its total bond dimension equals that of an MPO tensor M , and a block-diagonal gauge reconstructs M from the tensor associated with S . Suppose also that this tensor has the same matrix product vectors as M at every positive length, that all copy weights over A_x have a common value μ_x , and that the physical-trace transfer of every A_x is nonnilpotent. Assume that the one-letter tuples $(A_x^i)_x$ span the full product of the representative matrix algebras, and put $K_x = \mu_x A_x$.

Suppose that the tensors K_x form a proportionally realized orthogonal commuting sector family and that each one-site operator $\rho^{(1)}(K_x)$ is positive semidefinite. If M has source zero correlation length, then M satisfies SAL.

This establishes [11, Proposition C.14] with the ambient zero-correlation-length and one-site positivity hypotheses made explicit; see [37].

Proof. [✓ Proof](#) The one-site hypothesis and Theorem 20.14.16 show that every K_x generates matrix product density operators. The simultaneous one-letter span and the nonzero absorbed weights give one-site injectivity by Theorem 20.14.6. Ambient source zero correlation length descends to every K_x by Theorem 20.16.2.31. Finally, the supplied positive-length sector-decomposition identity gives $\rho^{(N)}(M) = \sum_x n_x \rho^{(N)}(K_x)$ for every $N > 0$, so Theorem 20.14.30 applies. □

Corollary 20.14.32 (Orthogonal BNT sector sums preserve SAL). [Lean](#) [✓ Stmt](#) *Let M have a nonempty basis-of-normal-tensors decomposition with positive copy numbers n_x and copy-independent weights μ_x . Put $K_x = \mu_x A_x$, and suppose that the decomposition generates the same positive-length matrix product vectors as M . If an independently supplied orthogonally supported commuting sector family realizes the tensors K_x , and every K_x satisfies SAL, then M satisfies SAL.*

Proof. [✓ Proof](#) The positive-length canonical-form identity gives $\rho^{(N)}(M) = \sum_x n_x \rho^{(N)}(K_x)$. Apply Corollary 20.14.29. $\square$

Corollary 20.14.33 (Orthogonal BNT sector products give the GSNNCH form). [Lean](#) [✓ Stmt](#) *Let M be an MPO tensor and let its BNT canonical form have normal representatives A_x , copy numbers n_x , and copy-independent weights μ_x . Put $\mathcal{K}_x = \mu_x A_x$, and assume that this canonical form generates the same matrix product vectors as M at every positive length. If the representatives $\mathcal{K}_x$ have orthogonally supported commuting sector products, then*

$$\rho^{(N)}(M) = \sum_x n_x \rho^{(N)}(\mathcal{K}_x).$$

Hence the BNT copy numbers n_x are the natural multiplicities of the explicit GSNNCH form.

Proof. [✓ Proof](#) Apply Theorem 20.14.21 to the positive-length BNT decomposition. Its coefficients are exactly the copy numbers n_x after the common weights have been absorbed. $\square$

Corollary 20.14.34 (Sectorwise SAL gives the BNT GSNNCH form). [Lean](#) [✓ Stmt](#) *Let M be an MPO tensor and let its BNT canonical form have normal representatives A_x , copy numbers n_x , and copy-independent weights μ_x . Put $\mathcal{K}_x = \mu_x A_x$, and assume that this canonical form generates the same matrix product vectors as M at every positive length. Equivalently, for every $N > 0$,*

$$\rho^{(N)}(M) = \sum_x n_x \rho^{(N)}(\mathcal{K}_x).$$

Under the BNT projector-selection hypotheses, if every $\mathcal{K}_x$ satisfies SAL, then M has the source GSNNCH form. Its outer sectors are the BNT sectors and its natural multiplicities are the copy numbers n_x .

Proof. [✓ Proof](#) Theorem 20.14.20 supplies the orthogonally supported commuting sector products. Apply Corollary 20.14.33. $\square$

Lemma 20.14.35 (Cyclic-shift transport of an embedded local operator). [Lean](#) [✓ Stmt](#) *Let B act on L consecutive spins of the periodic N -site chain, embedded at base site i , and let r denote the cyclic shift of sites. Reindexing both chain configurations by r carries the embedded operator to its translate at the shifted base site:*

$$\tau_i(B)[\sigma \circ r, \eta \circ r] = \tau_{r(i)}(B)[\sigma, \eta].$$

Proof. [✓ Proof](#) Entrywise from the definitions: the shift carries the window at i to the window at $r(i)$, preserves agreement of two configurations outside the window, and preserves the extracted window values. $\square$

Lemma 20.14.36 (Translates of a sector bond commute pairwise). [Lean](#) [✓ Stmt](#) *For every sector x of an explicit GSNNCH decomposition and every finite chain length $N \geq 2$, all cyclic translates of the sector bond commute: $[\tau_i(B^{(x)}), \tau_j(B^{(x)})] = 0$.*

Proof. ✓ Proof Translates with disjoint windows commute because they act on disjoint sites, and two length-two windows overlap only when their starting sites coincide or are cyclic neighbors. For the neighboring pairs, applying the transport identity of Lemma 20.14.35 p times to the neighboring commutation $[\tau_0(B^{(x)}), \tau_1(B^{(x)})] = 0$ of Definition 20.14.1 gives $[\tau_p(B^{(x)}), \tau_{p+1}(B^{(x)})] = 0$ for every p , which covers all overlapping pairs. $\square$

Lemma 20.14.37 (Positivity of the sector-bond translates). Lean Lean ✓ Stmt *An operator embedded into the periodic chain from a positive semidefinite local factor is positive semidefinite; in particular every translate $\tau_i(B^{(x)})$ of a sector bond is positive semidefinite.*

Proof. ✓ Proof In the coordinates given by the window-plus-complement splitting of the chain, the embedded operator decomposes as

$$\tau_i(B) \cong B \otimes \mathbb{1}_{\text{comp}},$$

where the identity acts on the cyclic complement of the window. A tensor product of positive semidefinite operators is positive semidefinite, and $B^{(x)} \geq 0$ by Definition 20.14.1. $\square$

Lemma 20.14.38 (Positivity of the sector sum). Lean Lean Lean ✓ Stmt *Each sector product $\prod_{j=1}^N \tau_j(B^{(x)})$ of an explicit GSNNCH decomposition is positive semidefinite; hence so are the represented operator*

$$\bigoplus_x n_x \prod_{j=1}^N \tau_j(B^{(x)})$$

and every chain operator equal to a positive multiple of it.

Proof. ✓ Proof A product of pairwise commuting positive semidefinite matrices is positive semidefinite: for two commuting factors A and B , the square root of A commutes with B , so $AB = \sqrt{A}B\sqrt{A} \geq 0$, as congruence by $\sqrt{A}$ preserves positive semidefiniteness; iterate along the product. Each sector product is such a product by Lemmas 20.14.37 and 20.14.36, and the sum with natural coefficients preserves positivity. $\square$

Lemma 20.14.39 (Translation invariance of the sector sum). Lean Lean Lean ✓ Stmt *Each sector product of an explicit GSNNCH decomposition is invariant under the cyclic shift of the periodic chain, where spin $N+1$ is identified with the first; hence so are the represented operator and every chain operator equal to a positive multiple of it [11, lines 838–842].*

Proof. ✓ Proof By the transport identity of Lemma 20.14.35, the cyclic shift carries each translate $\tau_j(B^{(x)})$ to $\tau_{j+1}(B^{(x)})$, so it permutes the factors of each sector product cyclically. All factors commute by Lemma 20.14.36, so the product is unchanged, and so is the sum over sectors. $\square$

Lemma 20.14.40 (The sector form passes to the trace normalization). Lean ✓ Stmt *If a finite-chain operator of an MPO tensor has the explicit sector form of Definition 20.14.1 and nonzero trace, then its trace normalization has the sector form as well.*

Proof. ✓ Proof By Lemma 20.14.38 the operator is positive semidefinite, so its nonzero trace is positive and the normalization multiplies the sector sum by a positive constant. $\square$

Theorem 20.14.41 (Sector form with nonvanishing traces gives GSNNCH). Lean ✓ Stmt *If every finite-chain operator of an MPO tensor has the explicit sector form of Definition 20.14.1 and a nonzero trace, then the normalized finite-chain states are density operators with that sector form, so the tensor generates Gibbs states of a nearest-neighbor commuting Hamiltonian.*

Proof. ✓ Proof Positivity of each chain operator follows from Lemma 20.14.38, and the normalized state has unit trace and keeps the sector form by Lemma 20.14.40. $\square$

20.15 Two-site and positive-length physical blocking

The two-site block is the virtual contraction of two copies of K :

$$\begin{array}{c} (j_1, j_2) \\ \downarrow \\ \boxed{M} \\ \uparrow \\ (j_1, j_2) \end{array} = \begin{array}{cc} i_1 & i_2 \\ \downarrow & \downarrow \\ \boxed{K} & \text{---} & \boxed{K} \\ \uparrow & \uparrow \\ j_1 & j_2 \end{array}$$

The two ket indices and the two bra indices are grouped into the blocked physical indices, as in [11, Theorem 4.9, lines 851–856].

Definition 20.15.1 (Physical blocking of an MPO tensor). [Lean](#) [Lean](#) [✓ Stmt](#) For an integer $L \geq 0$, group L adjacent physical sites and denote the resulting local tensor by $M^{[L]}$. It has a length- L ket word $I = (i_1, \dots, i_L)$ and a length- L bra word $J = (j_1, \dots, j_L)$ as its physical indices, and

$$(M^{[L]})^{I,J} = M^{i_1 j_1} \dots M^{i_L j_L}.$$

Thus the ket and bra words are grouped separately. For $L = 0$, both words are empty and the unique local matrix is the identity. For $L > 0$, this is the MPO version of the physical blocking used for the basis of normal tensors in [11, lines 317–345]. It is a local tensor, not the closed-chain operator $O_L(M)$ of [11, lines 962–967].

Theorem 20.15.2 (Doubled-index form of MPO blocking). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every integer $L \geq 0$, pairing the ket and bra letters site by site gives the canonical identification

$$\text{Fin}(d^L) \times \text{Fin}(d^L) \simeq (\text{Fin}(d) \times \text{Fin}(d))^{\text{Fin}(L)} \simeq \text{Fin}((d^2)^L). \quad (20.33)$$

Under this identification, the doubled-index MPS tensor of $M^{[L]}$ is the physical reindexing of the L -blocked doubled-index MPS tensor of M . In particular, one is injective if and only if the other is.

Proof. [✓ Proof](#) Decode the blocked ket and bra words, pair their letters at each site, and compare the two ordered matrix products. For $L = 0$, both words are empty and both matrix products are the identity. Physical reindexing by a bijection preserves the span of the tensor matrices. $\square$

Theorem 20.15.3 (Blocking commutes with the MPO product). [Lean](#) [✓ Stmt](#) For every integer $L \geq 0$ and two MPO tensors M, N with the same physical dimension, let juxtaposition denote the MPO tensor product obtained by contracting the intermediate physical index. Then $(MN)^{[L]} = M^{[L]}N^{[L]}$. For $L = 0$, each blocked tensor is the unique empty-word local matrix, namely the identity, so the equality reduces to the product of two identity local tensors.

Proof. [✓ Proof](#) Expand the product tensor and sum over its length- L intermediate physical word. Reindexing this word by one blocked physical index gives the product of the two blocked tensors. When $L = 0$, the intermediate word is empty and all three blocked tensors are identities. $\square$

Theorem 20.15.4 (Blocked word evaluation of an MPO tensor). [Lean](#) [✓ Stmt](#) Let $L \geq 0$, and let $\sigma = (\sigma_0, \dots, \sigma_{m-1})$ and $\tau = (\tau_0, \dots, \tau_{m-1})$ be words of the same length m in the blocked alphabet, so that each σ_k and each τ_k is a word of length L in $\{0, \dots, d-1\}$. Write $\tilde{\sigma}$ and $\tilde{\tau}$ for the concatenations $\sigma_0\sigma_1 \dots \sigma_{m-1}$ and $\tau_0\tau_1 \dots \tau_{m-1}$, words of length mL . Then

$$(M^{[L]})^{\sigma, \tau} = M^{\tilde{\sigma}, \tilde{\tau}}.$$

This is the counterpart for MPO tensors of Lemma 2.7.2, with the ket and bra words blocked separately.

Proof. ✓ Proof Induct on m . For $m = 0$ both sides are the identity. For $m \geq 1$, split off the first blocked letters σ_0, τ_0 : by Definition 20.15.1, $(M^{[L]})^{\sigma_0, \tau_0} = M^{\sigma_0, \tau_0}$, and since σ_0 and τ_0 have the same length L , Lemma 16.1.4 gives $M^{\sigma_0 \tilde{\sigma}', \tau_0 \tilde{\tau}'} = M^{\sigma_0, \tau_0} M^{\tilde{\sigma}', \tilde{\tau}'}$ for the remaining words σ', τ' . $\square$

Theorem 20.15.5 (Closed MPO chains under physical blocking). Lean Lean ✓ Stmt Let $L \in \mathbb{N}$, and let

$$\Phi_{N,L}: \{0, \dots, d^L - 1\}^N \longrightarrow \{0, \dots, d - 1\}^{NL}$$

concatenate the N blocked words. For blocked configurations σ and τ ,

$$\rho^{(N)}(M^{[L]})_{\sigma, \tau} = \rho^{(NL)}(M)_{\Phi_{N,L}(\sigma), \Phi_{N,L}(\tau)}.$$

The algebraic identity holds for every $L \in \mathbb{N}$. For $L \geq 1$, positivity of the original nonempty chain of length NL shows that blocking L adjacent physical sites of an MPDO again gives an MPDO. This is the closed-chain identification implicit when the one-site tensor and its two-site blocking are placed in vertical canonical form in [11, Appendix C.4, lines 1952–2017].

Proof. ✓ Proof The word of N blocked letters $\sigma = (\sigma_0, \dots, \sigma_{N-1})$ has concatenation $\Phi_{N,L}(\sigma)$, and similarly for τ ; this concatenation is the blocked-configuration bijection of Lemma 2.7.2. By Theorem 20.15.4,

$$(M^{[L]})^{\sigma, \tau} = M^{\Phi_{N,L}(\sigma), \Phi_{N,L}(\tau)}.$$

Taking the virtual trace gives the equality in the theorem. The map $\Phi_{N,L}$ is a bijection, and simultaneous reindexing of the rows and columns preserves positive semidefiniteness. $\square$

Lemma 20.15.6 (Closed MPO entries with residual sites). Lean ✓ Stmt Let a chain consist of a prefix of p sites, m complete blocks of length L , and a suffix of s sites. Write $M^{\alpha, \beta}$ for the ordered product of matrices indexed by two equally long physical words, and write $\tilde{\sigma}, \tilde{\tau}$ for the concatenations of blocked words σ, τ . Then, for matching row and column configurations,

$$\rho^{(p+mL+s)}(M)_{i^- \tilde{\sigma} i^+, j^- \tilde{\tau} j^+} = \text{tr}(M^{i^-, j^-} \cdot (M^{[L]})^{\sigma, \tau} \cdot M^{i^+, j^+}). \quad (20.34)$$

This identity retains both residual products, including when the chain length is not divisible by L . It extends the closed-chain blocking identity used in [11, Appendix C.4, lines 1952–2017].

Proof. ✓ Proof The concatenations $\tilde{\sigma}, \tilde{\tau}$ are identified with the flattened blocked configurations by Lemma 2.7.2. Applying word concatenation at the two boundaries (Lemma 16.1.4) and then evaluating the middle segment as a word of blocked matrices (Theorem 20.15.4) gives

$$\begin{aligned} M^{i^- \tilde{\sigma} i^+, j^- \tilde{\tau} j^+} &= M^{i^-, j^-} \cdot M^{\tilde{\sigma}, \tilde{\tau}} \cdot M^{i^+, j^+} \\ &= M^{i^-, j^-} \cdot (M^{[L]})^{\sigma, \tau} \cdot M^{i^+, j^+}. \end{aligned}$$

Taking the virtual trace yields the stated entry identity. $\square$

Definition 20.15.7 (Two-site blocking of an MPO tensor). Lean ✓ Stmt For an MPO tensor K , its two-site blocking $K^{[2]}$ has physical indices (i_0, i_1) and (j_0, j_1) and matrices

$$(K^{[2]})^{(i_0, i_1), (j_0, j_1)} = K^{i_0 j_0} K^{i_1 j_1}.$$

This is the blocking used in [11, Theorem 4.9, lines 851–856].

Theorem 20.15.8 (Concrete two-site blocking as length-two blocking). [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Let $\varphi : \{0, \dots, d^2 - 1\} \rightarrow \{0, \dots, d - 1\}^2$ be the canonical bijection. Then $(K^{[2]})^{i,j} = (K^{[L]})^{\varphi(i), \varphi(j)} \Big|_{L=2}$. Consequently, for every nonempty chain length N , the closed MPO of $K^{[2]}$ is obtained from the closed MPO of the general length-two block by applying φ independently at every ket and bra site. In particular, if K generates matrix product density operators, then so does $K^{[2]}$.

Proof. [✓ Proof](#) Decode a blocked index as an ordered pair. The two local tensors are then the same product $K^{i_0 j_0} K^{i_1 j_1}$. Applying this equality at every site gives the closed-chain reindexing, and simultaneous row and column reindexing preserves positive semidefiniteness. $\square$

Theorem 20.15.9 (MPO blocking preserves literal CPSV doubled-index form). [Lean](#) [Lean](#)
[✓ Stmt](#) Let A_M be the doubled-index MPS tensor associated to an MPO tensor M . If A_M is in literal CPSV canonical form and $L \geq 1$, then the doubled-index MPS tensor associated to $M^{[L]}$ is again in literal CPSV canonical form. The same conclusion holds for the concrete two-site blocking $M^{[2]}$.

Proof. [✓ Proof](#) The doubled-index tensor of $M^{[L]}$ is, under the canonical pairing of ket and bra words, a physical relabeling of $A_M^{[L]}$:

$$A_{M^{[L]}} = \psi_L^*(A_M^{[L]}). \quad (20.35)$$

Theorem 7.6.15 gives literal CPSV canonical form for $A_M^{[L]}$, and Theorem 7.6.8 carries it across ψ_L . For two-site blocking one takes $L = 2$ and uses the concrete identification of Theorem 20.15.8. $\square$

Theorem 20.15.10 (Vertical decomposition from grouped sectors). [Lean](#) [✓ Stmt](#) Suppose that M admits the phase-class grouping with physical isometries of Definition 16.4.92 and the grouped-corner Gram comparison of Definition 16.4.69. Then there are normal tensors A_α , positive multiplicities r_α , positive numbers $c_{\alpha,q}$, and a coisometry U such that the A_α form a CPSV basis of normal tensors for $\widetilde{M}$ and, writing

$$B_v = \bigoplus_{\alpha} \bigoplus_{q=0}^{r_\alpha-1} c_{\alpha,q} A_\alpha^v, \quad (20.36)$$

one has

$$\begin{aligned} UU^\dagger &= \mathbb{1}, \\ U\widetilde{M}_v U^\dagger &= B_v, \\ \widetilde{M}_v &= U^\dagger B_v U. \end{aligned} \quad (20.37)$$

Proof. [✓ Proof](#) Group the normal vertical corners into phase classes and normalize their physical sector maps. The exact original-sector reconstruction gives positive-length matrix-product-vector equality with the corresponding weighted direct sum. Hence the spectral representatives retain the CPSV basis-of-normal-tensors property for $\widetilde{M}$. Flatten the pair (α, q) into one finite index. Orthogonality of the normalized sector maps gives $UU^\dagger = \mathbb{1}$; their intertwining identities give the middle equality in (20.37), and their exact reconstruction gives the last equality. $\square$

For a matrix product density operator, either literal CPSV canonical form or normalized BNT-refined horizontal form supplies both hypotheses of Theorem 20.15.10. Each canonical form and positivity are preserved by two-site blocking. Applying the theorem to M and $M^{[2]}$ therefore gives their vertical decompositions simultaneously.

Lemma 20.15.11 (Two-site blocking preserves normalized BNT-refined horizontal form). [Lean](#)

[✓ Stmt](#) If an MPO tensor K is in normalized BNT-refined horizontal form, then its two-site blocking $K^{[2]}$ is again in that form.

Proof. [✓ Proof](#) Block every normal representative in the horizontal sector decomposition by two sites and square each copy weight. Positive blocking preserves irreducibility, left-canonical normalization, eventual linear independence of the representative states, and inequivalence of distinct representatives. Normalized self-overlap is preserved by $O_{A_j^{[2]} A_j^{[2]}}(N) = O_{A_j A_j}(2N) \rightarrow 1$. Hence the blocked sector decomposition is again a BNT canonical form. If φ is the canonical bijection between a pair of ket–bra indices and a length-two word in the doubled alphabet, then $(K^{[2]})^{\text{MPS}} = \varphi^*(\text{block}_2(K^{\text{MPS}}))$. Thus φ only relabels the physical basis. Finally, if X is the original block-diagonal virtual gauge and S its sector tensor, word evaluation gives $((K^{[2]})^{\text{MPS}})_i = X(\varphi^*(S^{[2]}))_i X^{-1}$. Therefore this blocked decomposition and the same copy gauges witness normalized BNT-refined horizontal form for $K^{[2]}$. $\square$

Theorem 20.15.12 (One-site and two-site vertical canonical forms). [Lean](#) [✓ Stmt](#) Let K be an MPO tensor in normalized BNT-refined horizontal form that generates matrix product density operators. Then both K and $K^{[2]}$ satisfy the vertical canonical-form conditions of Definition 16.4.6. This is the initial canonical-form step in [11, Appendix C.4, lines 1951–1956].

Proof. [✓ Proof](#) By Theorem 20.15.8 and Lemma 20.15.11, $K^{[2]}$ is in normalized BNT-refined horizontal form and generates matrix product density operators. Apply Theorem 16.4.105 first to K and then to $K^{[2]}$. $\square$

Definition 20.15.13 (Normalized vertical-sector maps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Let the vertical sectors be labelled by α , with simple matrix algebra M_{d_α} and positive diagonal multiplicity matrix

$$\mu_\alpha = \text{diag}(\mu_{\alpha,0}, \dots, \mu_{\alpha,r_\alpha-1}).$$

Write $m_\alpha = \text{tr}(\mu_\alpha)$. The normalized embedding and the left partial trace are

$$\begin{aligned} R_\mu \left(\bigoplus_\alpha X_\alpha \right) &= \bigoplus_\alpha \frac{\mu_\alpha}{m_\alpha} \otimes X_\alpha, \\ \widetilde{R}_\mu \left(\bigoplus_\alpha Y_\alpha \right) &= \bigoplus_\alpha \text{tr}_{\text{mult}}(Y_\alpha). \end{aligned}$$

On a weighted sector, $\widetilde{R}_\mu(\lambda_\alpha \mu_\alpha \otimes X_\alpha) = \lambda_\alpha m_\alpha X_\alpha$. Thus the partial trace is the canonical extension to the full block matrix space of the inverse map displayed in [11, Appendix C.4, lines 1957–1971].

Theorem 20.15.14 (Complete positivity of the normalized vertical-sector maps). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Let $(d_\alpha)_\alpha$ and $(r_\alpha)_\alpha$ be finite families of dimensions with $r_\alpha > 0$, and suppose that every diagonal entry of μ_α is positive. Then the normalized embedding

$$\begin{aligned} R_\mu: \bigoplus_\alpha M_{d_\alpha}(\mathbb{C}) &\longrightarrow \bigoplus_\alpha M_{r_\alpha d_\alpha}(\mathbb{C}), \\ R_\mu((X_\alpha)_\alpha) &= \left(\frac{\mu_\alpha}{m_\alpha} \otimes X_\alpha \right)_\alpha \end{aligned}$$

is a direct-sum Kraus map. Without any condition on the multiplicities, the map

$$\begin{aligned}\widetilde{R}: \bigoplus_{\alpha} M_{r_{\alpha} d_{\alpha}}(\mathbb{C}) &\longrightarrow \bigoplus_{\alpha} M_{d_{\alpha}}(\mathbb{C}), \\ \widetilde{R}((Y_{\alpha})_{\alpha}) &= (\mathrm{tr}_{\mathrm{mult}}(Y_{\alpha}))_{\alpha}\end{aligned}$$

is a direct-sum Kraus map. Its restriction to the weighted sectors is the retraction in [11, Appendix C.4, lines 1961–1970].

Proof. ✓ Proof

Exchange the two coordinates within each sector by $w(\alpha, q, i) = (\alpha, i, q)$. Let R_w be the corresponding matrix reindexing. For the normalized embedding, set $\rho_{\alpha} = m_{\alpha}^{-1} \mu_{\alpha}$. The hypotheses give $\rho_{\alpha} \geq 0$ and $\mathrm{tr}(\rho_{\alpha}) = 1$. Let $\mathcal{P}$ be the orthogonally controlled preparation of these density matrices. Thus

$$[\mathcal{P}(Z)]_{\langle \alpha, (i, q) \rangle, \langle \beta, (j, s) \rangle} = \begin{cases} Z_{\langle \alpha, i \rangle, \langle \alpha, j \rangle} \rho_{\alpha}(q, s), & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

The preparation Kraus operators resolve the identity in each sector, so $\mathcal{P}$ is trace-preserving and completely positive. Reindexing by w^{-1} changes $X_{\alpha} \otimes \rho_{\alpha}$ into $\rho_{\alpha} \otimes X_{\alpha}$. Consequently, the full-matrix extension satisfies

$$\begin{aligned}[\widehat{R}_{\mu}(Z)]_{\langle \alpha, (q, i) \rangle, \langle \beta, (s, j) \rangle} &= \begin{cases} \rho_{\alpha}(q, s) Z_{\langle \alpha, i \rangle, \langle \alpha, j \rangle}, & \alpha = \beta, \\ 0, & \alpha \neq \beta, \end{cases} \\ \widehat{R}_{\mu} &= R_{w^{-1}} \circ \mathcal{P}.\end{aligned}$$

This proves complete positivity of the normalized embedding.

Now let $\mathcal{C}_{\mathrm{tr}}$ be the controlled partial trace over the $\mathbb{C}^{r_{\alpha}}$ factors. For a matrix Y on $\bigoplus_{\alpha} (\mathbb{C}^{r_{\alpha}} \otimes \mathbb{C}^{d_{\alpha}})$, the full-matrix extension satisfies

$$[\widehat{R}(Y)]_{\langle \alpha, i \rangle, \langle \beta, j \rangle} = \begin{cases} \sum_{q=0}^{r_{\alpha}-1} Y_{\langle \alpha, (q, i) \rangle, \langle \alpha, (q, j) \rangle}, & \alpha = \beta, \\ 0, & \alpha \neq \beta. \end{cases}$$

The diagonal case is the ordinary right partial trace after applying R_w , and the off-diagonal case vanishes under the orthogonal sector control. Hence $\widehat{R} = \mathcal{C}_{\mathrm{tr}} \circ R_w$. Equivalence reindexing and the controlled dependent partial trace are trace-preserving completely positive maps, so their composition has a Kraus representation. $\square$

Lemma 20.15.15 (Retraction of the normalized vertical-sector embedding). Lean Lean ✓ Stmt
Suppose that every multiplicity space is nonzero and every diagonal entry of μ_{α} is positive. Then $m_{\alpha} \neq 0$ for every α , and $\widetilde{R}_{\mu} \circ R_{\mu} = \mathrm{id}$.

Proof. ✓ Proof Positivity of the diagonal entries and nonemptiness of the multiplicity space give $m_{\alpha} = \sum_q \mu_{\alpha, q} > 0$. The Kronecker-product identity for the left partial trace then gives

$$\mathrm{tr}_{\mathrm{mult}}(m_{\alpha}^{-1} \mu_{\alpha} \otimes X_{\alpha}) = m_{\alpha}^{-1} \mathrm{tr}(\mu_{\alpha}) X_{\alpha} = X_{\alpha}.$$

$\square$

Definition 20.15.16 (Products of weighted vertical contractions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 For vertical BNT tensors M_α with simple matrix algebras M_{d_α} and a family of complex scalars $(c_\alpha)_\alpha$, define

$$V_c(X) = \bigoplus_{\alpha} c_\alpha M_\alpha(X),$$

$$M_\alpha(X) = \sum_{a,b} X_{ba} M_{\alpha,ab}.$$

For each $L \geq 0$, let

$$C_L(V_c) = \text{span}_{\mathbb{C}} \{V_c(X_1) \cdots V_c(X_L) : X_1, \dots, X_L \in M_D\} \subseteq \prod_{\alpha} M_{d_\alpha}.$$

Write V_m for the specialization obtained by taking $c_\alpha = m_\alpha = \text{tr}(\mu_\alpha)$.

Lemma 20.15.17 (Contraction against a matrix unit). [Lean](#) [✓ Stmt](#) *Let $M = (M_{ab})_{a,b=0}^{D-1}$ be a matrix-product tensor whose physical index is identified with the ordered pair (a, b) . Then $M(E_{ba}) = M_{ab}$.*

Lemma 20.15.18 (Contraction in the encoded product coordinate). [Lean](#) [✓ Stmt](#) *If $r \in \{0, \dots, D^2 - 1\}$ encodes the ordered pair $(\lfloor r/D \rfloor, r \bmod D)$, then $M(E_{(r \bmod D), \lfloor r/D \rfloor}) = M_r$.*

Definition 20.15.19 (Sectorwise scaling equivalence). [Lean](#) [✓ Stmt](#) For scalars $c_\alpha \neq 0$, the map

$$\varphi_c : \prod_{\alpha} M_{d_\alpha} \longrightarrow \prod_{\alpha} M_{d_\alpha},$$

$$(X_\alpha)_\alpha \longmapsto (c_\alpha X_\alpha)_\alpha$$

is a linear equivalence.

Lemma 20.15.20 (Weighted contraction of matrix-unit words). [Lean](#) [✓ Stmt](#) *For a word $w = ((a_1, b_1), \dots, (a_L, b_L))$,*

$$V_c(E_{b_1 a_1}) \cdots V_c(E_{b_L a_L}) = \bigoplus_{\alpha} c_\alpha^L M_\alpha^w.$$

Lemma 20.15.21 (Transfer of simultaneous word spanning). [Lean](#) [✓ Stmt](#) *Fix $L \geq 0$ and suppose that every c_α is nonzero. If the simultaneous word tuples $(M_\alpha^w)_\alpha$ of length L span $\prod_{\alpha} M_{d_\alpha}$, then $C_L(V_c) = \prod_{\alpha} M_{d_\alpha}$.*

Theorem 20.15.22 (Nonzero-weight vertical bond contractions generate the sector algebra). [Lean](#) [✓ Stmt](#) *Suppose that $(M_\alpha)_\alpha$ is a basis of normal tensors and that $(c_\alpha)_\alpha$ is any family of nonzero complex scalars. Then there is a positive integer L such that $C_L(V_c) = \prod_{\alpha} M_{d_\alpha}$.*

Proof. [✓ Proof](#) Choose $L > 0$ for which the simultaneous word tuples $(M_\alpha^w)_\alpha$ span $\prod_{\alpha} M_{d_\alpha}$. Since every c_α is nonzero, so is c_α^L . Apply Lemma 20.15.21. $\square$

Theorem 20.15.23 (Vertical bond contractions generate the sector algebra). [Lean](#) [✓ Stmt](#) *Suppose that $(M_\alpha)_\alpha$ is a basis of normal tensors, every multiplicity space is nonzero, and every diagonal entry of μ_α is positive. Then there is a positive integer L such that $C_L(V_m) = \prod_{\alpha} M_{d_\alpha} =: \mathcal{A}_1$. This is the spanning assertion used in [11, Appendix C.4, lines 1980–1990].*

Proof. [✓ Proof](#) Since every multiplicity space is nonzero and every diagonal entry is positive, $m_\alpha = \sum_{q=0}^{\text{mult}(\alpha)-1} \mu_{\alpha,q} > 0$. Apply Theorem 20.15.22 to the nonzero family $c_\alpha = m_\alpha = \text{tr}(\mu_\alpha)$. $\square$

Theorem 20.15.24 (Fixed spanning contraction products determine an endomorphism). [Lean](#) [✓ Stmt](#) Let F be a linear endomorphism of $\prod_{\alpha} M_{d_{\alpha}}$. Suppose that $C_L(V_c)$ is the whole sector algebra and, for every choice of bond matrices $X_1, \dots, X_L$,

$$F(V_c(X_1) \cdots V_c(X_L)) = V_c(X_1) \cdots V_c(X_L).$$

Then F is the identity.

Proof. [✓ Proof](#) Since the displayed products span the domain, every $Y \in \prod_{\alpha} M_{d_{\alpha}}$ has the form $Y = \sum_i \lambda_i V_c(X_1^{(i)}) \cdots V_c(X_L^{(i)})$. Hence

$$F(Y) = \sum_i \lambda_i F(V_c(X_1^{(i)}) \cdots V_c(X_L^{(i)})) = \sum_i \lambda_i V_c(X_1^{(i)}) \cdots V_c(X_L^{(i)}) = Y.$$

Thus F is the identity. □

Definition 20.15.25 (Products of fixed points). [Lean](#) [✓ Stmt](#) Let F be a linear endomorphism of $\mathcal{A} = \prod_{\alpha} M_{d_{\alpha}}$, and write $\mathcal{F} = \{X \in \mathcal{A} : F(X) = X\}$. For $L \geq 0$, define

$$C_L(\mathcal{F}) = \text{span}_{\mathbb{C}}\{X_1 \cdots X_L : X_1, \dots, X_L \in \mathcal{F}\}.$$

This is the product space used in the dimension argument of [11, Appendix C.4, lines 1980–1993].

Theorem 20.15.26 (Density blocks of a faithful fixed family bound products). [Lean](#) [✓ Stmt](#) Let $\mathcal{A} = \prod_{\alpha} M_{d_{\alpha}}$, and let $F : \mathcal{A} \rightarrow \mathcal{A}$ be positive and trace preserving. Suppose that the trace adjoint F^* satisfies the Schwarz inequality and that F fixes a family $\rho = (\rho_{\alpha})_{\alpha}$ for which every ρ_{α} is positive definite. If $\mathcal{F} = \text{Fix}(F)$, then, for every $L > 0$, $\dim C_L(\mathcal{F}) \leq \dim \mathcal{F}$.

Proof. [✓ Proof](#)

Extend F to the full matrix algebra by diagonal compression followed by block-diagonal embedding, and choose a numbered basis of the ambient space. The extension is positive and trace preserving, its trace adjoint satisfies the Schwarz inequality, and the embedded family ρ is a positive definite fixed point. The fixed points therefore have the form

$$U \left(\bigoplus_k \sigma_k \otimes X_k \right) U^*, \quad X_k \in M_{h_k}(\mathbb{C}),$$

where $\text{tr}(\sigma_k) = 1$. The parametrization $(X_k)_k \mapsto \bigoplus_k \sigma_k \otimes X_k$ is injective, since its left partial trace is $(X_k)_k$. Every product of L fixed points lies in the range of

$$(Y_k)_k \mapsto U \left(\bigoplus_k \sigma_k^L \otimes Y_k \right) U^*.$$

Thus the span of these products has dimension at most $\sum_k h_k^2$, which is the dimension of the fixed-point space of the extension. Writing $\mathcal{E}$ for the fixed-point space of the full-matrix extension, the injective block-diagonal embedding and diagonal compression give the dimension chain

$$\dim C_L(\mathcal{F}) \leq \dim C_L(\mathcal{E}) \leq \dim \mathcal{E} \leq \dim \mathcal{F}.$$

□

Theorem 20.15.27 (Fixed contractions and the dimension of their fixed-point space). [Lean](#)

[✓ Stmt](#) Let F be a linear endomorphism of $\mathcal{A} = \prod_{\alpha} M_{d_{\alpha}}$. Suppose that the following three conditions hold, with the second required for every bond matrix X :

$$\begin{aligned} C_L(V_c) &= \mathcal{A}, \\ F(V_c(X)) &= V_c(X), \\ \dim C_L(\mathcal{F}) &\leq \dim \mathcal{F}, \quad \mathcal{F} = \{Y \in \mathcal{A} : F(Y) = Y\}. \end{aligned}$$

Then $F = \text{id}_{\mathcal{A}}$.

This is the finite-dimensional conclusion in [11, Appendix C.4, lines 1980–1993]. It does not assume that the products $V_c(X_1) \cdots V_c(X_L)$ are fixed.

Proof. [✓ Proof](#) Since every $V_c(X)$ belongs to $\mathcal{F}$, every product occurring in $C_L(V_c)$ belongs to $C_L(\mathcal{F})$. Hence

$$\mathcal{A} = C_L(V_c) \subseteq C_L(\mathcal{F}) \subseteq \mathcal{A},$$

so $C_L(\mathcal{F}) = \mathcal{A}$. Therefore

$$\dim \mathcal{A} = \dim C_L(\mathcal{F}) \leq \dim \mathcal{F} \leq \dim \mathcal{A}.$$

Thus $\mathcal{F} = \mathcal{A}$, and F is the identity. □

Theorem 20.15.28 (Faithful fixed family in a finite product). [Lean](#) [✓ Stmt](#) Let F be a positive trace-preserving endomorphism of $\mathcal{A} = \prod_{\alpha} M_{d_{\alpha}}$. Suppose that each V_j is fixed by F and, for some $L > 0$, the identity belongs to

$$\text{span}_{\mathbb{C}} \{V_{j_1} \cdots V_{j_L} : j_1, \dots, j_L \in J\}.$$

Then there is a fixed family $\rho = (\rho_{\alpha})_{\alpha}$ such that every ρ_{α} is positive definite.

This provides the intermediate step for the fixed-point argument in [11, Appendix C.4, lines 1980–1993].

Proof. [✓ Proof](#) Extend F to the full matrix algebra by diagonal compression followed by block-diagonal embedding. This extension is positive and trace preserving. Block-diagonal embedding preserves the identity and every product $V_{j_1} \cdots V_{j_L}$, so Theorem 20.15.66 gives a positive-definite fixed point $\hat{\rho}$ of the extension. Fixed points of the extension are block diagonal. Thus

$$\begin{aligned} \hat{\rho} &= \bigoplus_{\alpha} \rho_{\alpha}, \\ F(\rho) &= \rho. \end{aligned}$$

Every diagonal block ρ_{α} of the positive-definite matrix $\hat{\rho}$ is positive definite. □

Theorem 20.15.29 (Identity criterion from fixed contractions). [Lean](#) [✓ Stmt](#) Let F be a positive trace-preserving endomorphism of $\mathcal{A} = \prod_{\alpha} M_{d_{\alpha}}$ whose trace adjoint satisfies the Schwarz inequality. Suppose that, for some $L > 0$, $C_L(V_c) = \mathcal{A}$ and $F(V_c(X)) = V_c(X)$ for every bond matrix X . Then $F = \text{id}_{\mathcal{A}}$. No product $V_c(X_1) \cdots V_c(X_L)$ is assumed to be fixed.

This is the implication used for each transported square composite in [11, Appendix C.4, lines 1980–1995].

Proof. [✓ Proof](#) The fixed contractions and their positive-length product span give a positive-definite fixed family $\rho = (\rho_{\alpha})_{\alpha}$. Hence Theorem 20.15.26 gives $\dim C_L(\mathcal{F}) \leq \dim \mathcal{F}$, where $\mathcal{F} = \text{Fix}(F)$. Since $C_L(V_c) = \mathcal{A}$ and $V_c(X) \in \mathcal{F}$, Theorem 20.15.27 gives $F = \text{id}_{\mathcal{A}}$. □

Proof. ✓ Proof Distributivity of matrix multiplication over finite sums gives

$$\sum_{a,b} X_{ba} (LA_{ab}K) = L \left(\sum_{a,b} X_{ba} A_{ab} \right) K.$$

□

Lemma 20.15.39 (Contraction of a weighted block sum). Lean ✓ Stmt For a weighted block-diagonal tensor $A = \bigoplus_{\alpha} \mu_{\alpha} A_{\alpha}$, one has $A(X) = \bigoplus_{\alpha} \mu_{\alpha} A_{\alpha}(X)$.

Proof. ✓ Proof Since each tensor letter is block diagonal,

$$\sum_{a,b} X_{ba} \left(\bigoplus_{\alpha} \mu_{\alpha} A_{\alpha} \right)_{ab} = \bigoplus_{\alpha} \mu_{\alpha} \sum_{a,b} X_{ba} (A_{\alpha})_{ab} = \bigoplus_{\alpha} \mu_{\alpha} A_{\alpha}(X).$$

□

Lemma 20.15.40 (Vertical contraction and physical closure). Lean ✓ Stmt For the vertically viewed tensor $\widetilde{M}_{ab,ij} = M_{ab}^{ij}$, bond-matrix contraction is the one-site physical closure: $\widetilde{M}(X) = M(X)$.

Proof. ✓ Proof For every pair of physical indices,

$$\widetilde{M}(X)_{ij} = \sum_{a,b} X_{ba} M_{ab}^{ij} = \text{tr}(M^{ij}X) = M(X)_{ij}.$$

□

Lemma 20.15.41 (Contraction of the assembled vertical tensor). Lean ✓ Stmt The contraction of the assembled vertical tensor is the weighted block diagonal of the sector contractions:

$$\left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right) (X) = \bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}(X).$$

Proof. ✓ Proof Applying Lemma 20.15.39 to the block form $\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}$ of the assembled tensor gives

$$\left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right) (X) = \bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}(X).$$

□

Lemma 20.15.42 (Contraction of a letterwise forward identity). Lean ✓ Stmt Let B_{ab} be an arbitrary vertical tensor. If $U \widetilde{M}_{ab} U^{\dagger} = B_{ab}$ for every pair (a, b) , then, for every bond matrix X , $UM(X)U^{\dagger} = B(X)$.

Proof. ✓ Proof By the definition of bond-matrix contraction,

$$UM(X)U^{\dagger} = U \left(\sum_{a,b} X_{ba} \widetilde{M}_{ab} \right) U^{\dagger} = \sum_{a,b} X_{ba} (U \widetilde{M}_{ab} U^{\dagger}) = \sum_{a,b} X_{ba} B_{ab} = B(X).$$

□

Lemma 20.15.43 (Contraction of a letterwise reconstruction identity). [Lean](#) [✓ Stmt](#) Let B_{ab} be an arbitrary vertical tensor. If $\widetilde{M}_{ab} = U^\dagger B_{ab} U$ for every pair (a, b) , then, for every bond matrix X , $M(X) = U^\dagger B(X) U$.

Proof. [✓ Proof](#) Contracting the letterwise identity gives

$$M(X) = \sum_{a,b} X_{ba} \widetilde{M}_{ab} = \sum_{a,b} X_{ba} (U^\dagger B_{ab} U) = U^\dagger \left(\sum_{a,b} X_{ba} B_{ab} \right) U = U^\dagger B(X) U.$$

□

Lemma 20.15.44 (Contraction of the one-site vertical canonical form). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that

$$U \widetilde{M}_{ab} U^\dagger = \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right)_{ab}$$

is a one-site vertical canonical-form identity. Contracting against an arbitrary bond matrix X gives

$$U M(X) U^\dagger = \bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}(X).$$

If the reverse letterwise identity is also given, then

$$M(X) = U^\dagger \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}(X) \right) U.$$

These are the contracted canonical-form identities used in [11, Appendix C.4, lines 1955–1979].

Proof. [✓ Proof](#) Since $A(X)_{ij} = \sum_{a,b} X_{ba} A_{ab,ij}$ and $\widetilde{M}_{ab,ij} = M_{ab}^{ij}$, one has $M(X) = \sum_{a,b} X_{ba} \widetilde{M}_{ab}$. Distributivity of fixed left and right multiplication over finite sums gives

$$U \left(\sum_{a,b} X_{ba} \widetilde{M}_{ab} \right) U^\dagger = \sum_{a,b} X_{ba} (U \widetilde{M}_{ab} U^\dagger).$$

Substituting the forward hypothesis into this equality gives

$$U M(X) U^\dagger = \sum_{a,b} X_{ba} \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right)_{ab} = \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right)(X).$$

Applied blockwise, the same calculation gives

$$\left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right)(X) = \bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}(X).$$

If the reverse letterwise identity is given, the same calculation yields

$$\begin{aligned} M(X) &= \sum_{a,b} X_{ba} \widetilde{M}_{ab} \\ &= U^\dagger \left(\sum_{a,b} X_{ba} \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right)_{ab} \right) U \\ &= U^\dagger \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}(X) \right) U. \end{aligned}$$

□

Definition 20.15.45 (Four-site physical closure). [Lean](#) [Lean](#) [✓ Stmt](#) For a virtual matrix X , the four-site physical closure of K is the operator $K_4(X)$ whose matrix coefficients are

$$K_4(X)_{(i_0, (i_1, (i_2, i_3))), (j_0, (j_1, (j_2, j_3)))} = \text{tr}(K^{i_0 j_0} K^{i_1 j_1} K^{i_2 j_2} K^{i_3 j_3} X).$$

Lemma 20.15.46 (Coordinates of the four-site regrouping). [Lean](#) [Lean](#) [✓ Stmt](#) The canonical four-coordinate regrouping and its inverse satisfy

$$R_4(\sigma) = (\sigma_0, (\sigma_1, (\sigma_2, \sigma_3))), \\ R_4^{-1}(i_0, (i_1, (i_2, i_3))) = (i_0, i_1, i_2, i_3).$$

Proof. [✓ Proof](#) Both identities follow from the definition of the canonical right-associated regrouping. $\square$

Theorem 20.15.47 (General and right-associated four-site closures). [Lean](#) [✓ Stmt](#) Under the canonical right-associated identification of four-site configurations with quadruples of physical indices,

$$K_4(X)_{(i_0, (i_1, (i_2, i_3))), (j_0, (j_1, (j_2, j_3)))} = \text{tr}(K^{i_0 j_0} K^{i_1 j_1} K^{i_2 j_2} K^{i_3 j_3} X).$$

Proof. [✓ Proof](#) Expand the general four-site closure in the right-associated coordinates. $\square$

Theorem 20.15.48 (One-site closure of the two-site blocking). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) After identifying one blocked physical index with two original indices, $(K^{[2]})_1(X) = K_2(X)$.

Proof. [✓ Proof](#) Expand one blocked tensor matrix and decode its two physical indices. $\square$

Definition 20.15.49 (Physical maps in vertical canonical coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) Let U_1 and U_2 be the physical coordinate changes in vertical canonical forms of M and its two-site blocking, respectively. Let J be the canonical relabelling from matrices indexed by pairs of one-site physical indices to matrices indexed by one blocked physical index. For the refinement and coarse-graining maps T and S , define

$$T_v(Z) = U_2 J(T(U_1^\dagger Z U_1)) U_2^\dagger, \\ S_v(W) = U_1 S(J^{-1}(U_2^\dagger W U_2)) U_1^\dagger.$$

These are the physical maps after absorbing the two vertical coordinate changes, as in [11, Appendix C.4, lines 1955–1979]. They are intermediate maps on the retained physical coordinate spaces; the normalized sector maps $R_1, R_2, \widetilde{R}_1, \widetilde{R}_2$ have not yet been composed with them.

Theorem 20.15.50 (Action of the transported physical maps). [Lean](#) [Lean](#) [✓ Stmt](#) For a bond matrix X , write $\mathcal{M}^{(1)}(X)$ and $\mathcal{M}^{(2)}(X)$ for the one-site and two-site physical closures, respectively, and write

$$B_1(X) = \bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha}(X), \\ B_2(X) = \bigoplus_{\beta} \nu_{\beta} \otimes A_{\beta}(X)$$

for the contracted one-site and two-site vertical canonical forms. If

$$T(\mathcal{M}^{(1)}(X)) = \mathcal{M}^{(2)}(X), \\ S(\mathcal{M}^{(2)}(X)) = \mathcal{M}^{(1)}(X),$$

then

$$\begin{aligned} T_v(B_1(X)) &= B_2(X), \\ S_v(B_2(X)) &= B_1(X). \end{aligned}$$

These are the physical-coordinate identities which, after composition with $R_1, R_2, \widetilde{R}_1, \widetilde{R}_2$, give the two displayed identities for $\widetilde{T}$ and $\widetilde{S}$ in [11, Appendix C.4, lines 1955–1979].

Proof. ✓ Proof The reverse one-site canonical-form identity gives $U_1^\dagger B_1(X) U_1 = \mathcal{M}^{(1)}(X)$. Applying T , relabelling its two-site output by J , and using the forward blocked canonical-form identity gives

$$T_v(B_1(X)) = U_2 J(T(\mathcal{M}^{(1)}(X))) U_2^\dagger = U_2 J(\mathcal{M}^{(2)}(X)) U_2^\dagger = B_2(X).$$

The reverse blocked canonical-form identity and the equation for S give, in the other direction,

$$S_v(B_2(X)) = U_1 S(\mathcal{M}^{(2)}(X)) U_1^\dagger = U_1 \mathcal{M}^{(1)}(X) U_1^\dagger = B_1(X).$$

□

Definition 20.15.51 (Refinement and coarse-graining on vertical sectors). Lean Lean ✓ Stmt Define maps between the one-site and two-site simple-sector algebras by

$$\begin{aligned} \widetilde{T} &= \widetilde{R}_2 T_v R_1, \\ \widetilde{S} &= \widetilde{R}_1 S_v R_2. \end{aligned}$$

The partial trace and diagonal-summand extraction define these maps on the full matrix algebras; on the weighted block-diagonal image they match the normalized maps in [11, Appendix C.4, lines 1972–1979].

Definition 20.15.52 (Positive vertical-sector families and total trace). Lean Lean ✓ Stmt A vertical-sector family $X = (X_\alpha)_\alpha$ is positive when every X_α is positive semidefinite. Its total sector trace is

$$\mathrm{Tr}_{\mathrm{sec}}(X) = \sum_\alpha \mathrm{tr}(X_\alpha).$$

Lemma 20.15.53 (Positivity and trace in normalized sector coordinates). Lean Lean Lean

Lean Lean Lean Lean Lean Lean ✓ Stmt Suppose that every multiplicity space is nonzero and every diagonal entry of μ_α is positive. The normalized embedding R_μ sends a positive sector family to a positive retained matrix and satisfies $\mathrm{tr}(R_\mu(X)) = \mathrm{Tr}_{\mathrm{sec}}(X)$. Conversely, the sectorwise partial trace sends a positive retained matrix to a positive sector family and satisfies $\mathrm{Tr}_{\mathrm{sec}}(\widetilde{R}_\mu(Z)) = \mathrm{tr}(Z)$. These trace identities follow by summing the traces of the diagonal sectors; deleting the off-diagonal sector entries does not change the trace.

Proof. ✓ Proof A positive diagonal weight matrix has positive trace, and its Kronecker product with each positive X_α is positive. Partial trace preserves positivity. The two trace formulas follow from $\mathrm{tr}(A \otimes B) = \mathrm{tr}(A) \mathrm{tr}(B)$ and the normalization by $\mathrm{tr}(\mu_\alpha)$. □

Lemma 20.15.54 (Normalized embedding of multiplicity-trace-scaled sectors). Lean ✓ Stmt Suppose that every multiplicity space is nonzero and every diagonal entry of μ_α is positive. For $m_\alpha = \mathrm{tr}(\mu_\alpha)$,

$$R_\mu \left(\bigoplus_\alpha m_\alpha X_\alpha \right) = \bigoplus_\alpha \mu_\alpha \otimes X_\alpha.$$

Proof. ✓ Proof In each sector,

$$R_\mu(m_\alpha X_\alpha) = m_\alpha^{-1} m_\alpha (\mu_\alpha \otimes X_\alpha) = \mu_\alpha \otimes X_\alpha.$$

□

Lemma 20.15.55 (Partial trace of weighted vertical sectors). Lean ✓ Stmt For $m_\alpha = \text{tr}(\mu_\alpha)$,

$$\widetilde{R}_\mu \left(\bigoplus_\alpha \mu_\alpha \otimes X_\alpha \right) = \bigoplus_\alpha m_\alpha X_\alpha.$$

Proof. ✓ Proof Taking the partial trace over the multiplicity factor gives

$$\widetilde{R}_\mu(\mu_\alpha \otimes X_\alpha) = \text{tr}(\mu_\alpha) X_\alpha = m_\alpha X_\alpha.$$

□

Lemma 20.15.56 (Trace under coisometric compression). Lean Lean Lean Lean ✓ Stmt Let $U : \mathbb{C}^n \rightarrow \mathbb{C}^r$ satisfy $UU^\dagger = 1$. If $Z \geq 0$, then

$$\begin{aligned} \text{tr}(UZU^\dagger) &\leq \text{tr}(Z), \\ \text{tr}(UZU^\dagger) &= \text{tr}(Z) \iff U^\dagger UZ = Z. \end{aligned}$$

For every $W \in \mathbb{C}^{r \times r}$, one also has $\text{tr}(U^\dagger WU) = \text{tr}(W)$.

Proof. ✓ Proof Put $P = U^\dagger U$. The matrices P and $1 - P$ are orthogonal projections, and

$$\begin{aligned} \text{tr}(UZU^\dagger) &= \text{tr}(PZ), \\ \text{tr}(Z) - \text{tr}(PZ) &= \text{tr}((1 - P)Z) \geq 0. \end{aligned}$$

The last trace vanishes precisely when $(1 - P)Z = 0$. The formula for W follows by cyclicity of trace and $UU^\dagger = 1$. □

Definition 20.15.57 (Matrices before active-sector compression). Lean Lean ✓ Stmt Let $Z_T(X)$ be the matrix obtained by applying the normalized one-site embedding, expanding by $U_1^\dagger$, applying T , and relabelling the two-site indices, before compression by U_2 . Define $Z_S(Y)$ in the reverse direction, before compression by U_1 .

Lemma 20.15.58 (Positivity and trace before active-sector compression). Lean Lean Lean ✓ Stmt Suppose that the physical maps are completely positive and trace preserving and $U_i U_i^\dagger = 1$. For positive sector families $X \in \mathcal{A}_1$ and $Y \in \mathcal{A}_2$, the matrices $Z_T(X)$ and $Z_S(Y)$ are positive and

$$\begin{aligned} \text{tr}(Z_T(X)) &= \text{Tr}_{\text{sec}}(X), \\ \text{tr}(Z_S(Y)) &= \text{Tr}_{\text{sec}}(Y). \end{aligned}$$

After the final compression and sectorwise partial trace,

$$\begin{aligned} \text{Tr}_{\text{sec}}(\widetilde{T}(X)) &= \text{tr}(U_2 Z_T(X) U_2^\dagger), \\ \text{Tr}_{\text{sec}}(\widetilde{S}(Y)) &= \text{tr}(U_1 Z_S(Y) U_1^\dagger). \end{aligned}$$

Proof. ✓ Proof Positivity follows successively from the normalized embedding, expansion by an adjoint, the physical map, relabelling, and partial trace. The trace identities follow from Lemma 20.15.56, Lemma 20.15.58, and trace preservation of the physical maps. □

Theorem 20.15.59 (Positivity and trace loss of the transported sector maps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the multiplicity spaces are nonzero, their diagonal weights are positive, the physical maps T and S are completely positive and trace preserving, and the vertical coordinate maps satisfy $U_i U_i^\dagger = 1$. Then $\tilde{T}$ and $\tilde{S}$ preserve pointwise positive semidefiniteness. For pointwise positive families $X \in \mathcal{A}_1$ and $Y \in \mathcal{A}_2$, their total sector traces obey

$$\begin{aligned}\mathrm{Tr}_{\mathrm{sec}}(\tilde{T}(X)) &\leq \mathrm{Tr}_{\mathrm{sec}}(X), \\ \mathrm{Tr}_{\mathrm{sec}}(\tilde{S}(Y)) &\leq \mathrm{Tr}_{\mathrm{sec}}(Y).\end{aligned}$$

The trace loss vanishes precisely when

$$\begin{aligned}U_2^\dagger U_2 Z_T(X) &= Z_T(X), \\ U_1^\dagger U_1 Z_S(Y) &= Z_S(Y),\end{aligned}$$

respectively.

The condition on vanishing trace loss characterizes when the transported maps preserve the total sector trace.

Proof. [✓ Proof](#) The normalized diagonal embedding and the sectorwise partial trace preserve positivity. Before the final compression, trace preservation of the physical map and $U_i U_i^\dagger = 1$ give the original sector trace. For a positive matrix Z and a coisometry U , put $P = U^\dagger U$. Then

$$\mathrm{tr}(UZU^\dagger) = \mathrm{tr}(PZ) \leq \mathrm{tr}(Z).$$

Since $1 - P$ is an orthogonal projection, equality is equivalent to $\mathrm{tr}((1 - P)Z) = 0$, hence to $(1 - P)Z = 0$. $\square$

Lemma 20.15.60 (Two-site support after refinement). [Lean](#) [✓ Stmt](#) For a bond matrix X , set $V_1(X) = \bigoplus_\alpha m_\alpha M_\alpha(X)$. Suppose that both vertical canonical forms satisfy their exact reconstruction identities, the one-site multiplicity spaces are nonzero, their diagonal weights are positive, $T(\mathcal{M}^{(1)}(X)) = \mathcal{M}^{(2)}(X)$, and $U_2 U_2^\dagger = 1$. Then $U_2^\dagger U_2 Z_T(V_1(X)) = Z_T(V_1(X))$. This is the refinement support identity used in [11, Appendix C.4, lines 1955–1980].

Proof. [✓ Proof](#) Let $\mathcal{R}^{-1}$ relabel a matrix indexed by two one-site indices as a matrix indexed by the blocked two-site index. Then

$$\begin{aligned}Z_T(V_1(X)) &= \mathcal{R}^{-1} [T(U_1^\dagger B_1(X) U_1)] \\ &= \mathcal{R}^{-1} [T(\mathcal{M}^{(1)}(X))] \\ &= \mathcal{R}^{-1} [\mathcal{M}^{(2)}(X)] \\ &= U_2^\dagger B_2(X) U_2,\end{aligned}$$

where $Z_T(V_1(X)) = \mathcal{R}^{-1} [T(U_1^\dagger B_1(X) U_1)]$ is the normalized embedding of $V_1(X)$, $U_1^\dagger B_1(X) U_1 = \mathcal{M}^{(1)}(X)$ and $\mathcal{R}^{-1} [\mathcal{M}^{(2)}(X)] = U_2^\dagger B_2(X) U_2$ are the two reconstruction identities, and $T(\mathcal{M}^{(1)}(X)) = \mathcal{M}^{(2)}(X)$ is the refinement identity. Therefore

$$U_2^\dagger U_2 Z_T(V_1(X)) = U_2^\dagger (U_2 U_2^\dagger) B_2(X) U_2 = Z_T(V_1(X)).$$

$\square$

Lemma 20.15.61 (One-site support after coarse-graining). [Lean](#) [Stmt](#) For a bond matrix X , set $V_2(X) = \bigoplus_{\beta} n_{\beta} A_{\beta}(X)$. Suppose that both vertical canonical forms satisfy their exact reconstruction identities, the two-site multiplicity spaces are nonzero, their diagonal weights are positive, $S(\mathcal{M}^{(2)}(X)) = \mathcal{M}^{(1)}(X)$, and $U_1 U_1^{\dagger} = 1$. Then $U_1^{\dagger} U_1 Z_S(V_2(X)) = Z_S(V_2(X))$. This is the coarse-graining support identity used in [11, Appendix C.4, lines 1955–1980].

Proof. [Proof](#) Let $\mathcal{R}$ decode a matrix indexed by the blocked two-site index as a matrix indexed by two one-site indices. Then

$$\begin{aligned} Z_S(V_2(X)) &= S(\mathcal{R}[U_2^{\dagger} B_2(X) U_2]) \\ &= S(\mathcal{M}^{(2)}(X)) \\ &= \mathcal{M}^{(1)}(X) \\ &= U_1^{\dagger} B_1(X) U_1, \end{aligned}$$

where $Z_S(V_2(X)) = S(\mathcal{R}[U_2^{\dagger} B_2(X) U_2])$ is the normalized embedding of $V_2(X)$, $\mathcal{R}[U_2^{\dagger} B_2(X) U_2] = \mathcal{M}^{(2)}(X)$ is the blocked-index relabelling identity, $S(\mathcal{M}^{(2)}(X)) = \mathcal{M}^{(1)}(X)$ is the coarse-graining identity, and $\mathcal{M}^{(1)}(X) = U_1^{\dagger} B_1(X) U_1$ is the one-site reconstruction. Therefore

$$U_1^{\dagger} U_1 Z_S(V_2(X)) = U_1^{\dagger} (U_1 U_1^{\dagger}) B_1(X) U_1 = Z_S(V_2(X)).$$

□

Lemma 20.15.62 (Trace preservation on one-site canonical-form contractions). [Lean](#) [Stmt](#) Under the hypotheses of Lemma 20.15.60, suppose in addition that $U_1 U_1^{\dagger} = 1$ and that T is completely positive and trace preserving. Then $\text{Tr}_{\text{sec}}(\tilde{T}(V_1(X))) = \text{Tr}_{\text{sec}}(V_1(X))$.

Proof. [Proof](#) Cyclicity of trace and the support identity of Lemma 20.15.60 give

$$\text{Tr}_{\text{sec}}(\tilde{T}(V_1(X))) = \text{tr}(U_2^{\dagger} U_2 Z_T(V_1(X))) = \text{tr}(Z_T(V_1(X))) = \text{Tr}_{\text{sec}}(V_1(X)).$$

□

Lemma 20.15.63 (Trace preservation on two-site canonical-form contractions). [Lean](#) [Stmt](#) Under the hypotheses of Lemma 20.15.61, suppose in addition that $U_2 U_2^{\dagger} = 1$ and that S is completely positive and trace preserving. Then $\text{Tr}_{\text{sec}}(\tilde{S}(V_2(X))) = \text{Tr}_{\text{sec}}(V_2(X))$.

Proof. [Proof](#) Cyclicity of trace and the support identity of Lemma 20.15.61 give

$$\text{Tr}_{\text{sec}}(\tilde{S}(V_2(X))) = \text{tr}(U_1^{\dagger} U_1 Z_S(V_2(X))) = \text{tr}(Z_S(V_2(X))) = \text{Tr}_{\text{sec}}(V_2(X)).$$

□

Theorem 20.15.64 (Action of the normalized vertical-sector maps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Suppose that every multiplicity space for μ_{α} and ν_{β} is nonzero, every diagonal entry of these two matrices is positive, and

$$\begin{aligned} T_v \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes X_{\alpha} \right) &= \bigoplus_{\beta} \nu_{\beta} \otimes Y_{\beta}, \\ S_v \left(\bigoplus_{\beta} \nu_{\beta} \otimes Y_{\beta} \right) &= \bigoplus_{\alpha} \mu_{\alpha} \otimes X_{\alpha}. \end{aligned}$$

Writing $m_\alpha = \text{tr}(\mu_\alpha)$ and $n_\beta = \text{tr}(\nu_\beta)$, one has

$$\begin{aligned}\tilde{T}\left(\bigoplus_{\alpha} m_{\alpha} X_{\alpha}\right) &= \bigoplus_{\beta} n_{\beta} Y_{\beta}, \\ \tilde{S}\left(\bigoplus_{\beta} n_{\beta} Y_{\beta}\right) &= \bigoplus_{\alpha} m_{\alpha} X_{\alpha}.\end{aligned}$$

In particular, take $X_{\alpha} = M_{\alpha}(X)$ and $Y_{\beta} = A_{\beta}(X)$ from the contractions of the one-site and two-site vertical canonical forms. The physical closure identities for T and S then imply the two hypotheses above, so the same conclusions hold for these source-generated sector operators.

Proof. ✓ **Proof** Since $\tilde{T} = \tilde{R}_2 T_{\vee} R_1$, one has

$$\tilde{T}\left(\bigoplus_{\alpha} m_{\alpha} X_{\alpha}\right) = \tilde{R}_2\left(T_{\vee}\left(\bigoplus_{\alpha} \mu_{\alpha} \otimes X_{\alpha}\right)\right) = \tilde{R}_2\left(\bigoplus_{\beta} \nu_{\beta} \otimes Y_{\beta}\right) = \bigoplus_{\beta} n_{\beta} Y_{\beta}.$$

Likewise, $\tilde{S} = \tilde{R}_1 S_{\vee} R_2$ gives

$$\tilde{S}\left(\bigoplus_{\beta} n_{\beta} Y_{\beta}\right) = \tilde{R}_1\left(S_{\vee}\left(\bigoplus_{\beta} \nu_{\beta} \otimes Y_{\beta}\right)\right) = \tilde{R}_1\left(\bigoplus_{\alpha} \mu_{\alpha} \otimes X_{\alpha}\right) = \bigoplus_{\alpha} m_{\alpha} X_{\alpha}.$$

□

Lemma 20.15.65 (Fixed contraction families of the transported composites). Lean Lean

✓ **Stmt** For every bond matrix X , set

$$\begin{aligned}V_1(X) &= \bigoplus_{\alpha} m_{\alpha} M_{\alpha}(X), \\ V_2(X) &= \bigoplus_{\beta} n_{\beta} A_{\beta}(X).\end{aligned}$$

Under the hypotheses of Theorem 20.15.64, the transported composites satisfy

$$\begin{aligned}(\tilde{S} \circ \tilde{T})(V_1(X)) &= V_1(X), \\ (\tilde{T} \circ \tilde{S})(V_2(X)) &= V_2(X).\end{aligned}$$

Thus the one-site and two-site contraction families lie in the respective fixed-point spaces [11, Appendix C.4, lines 1974–1980].

Proof. ✓ **Proof** Theorem 20.15.64 gives $\tilde{T}(V_1(X)) = V_2(X)$ and $\tilde{S}(V_2(X)) = V_1(X)$. Hence

$$\begin{aligned}(\tilde{S} \circ \tilde{T})(V_1(X)) &= \tilde{S}(V_2(X)) = V_1(X), \\ (\tilde{T} \circ \tilde{S})(V_2(X)) &= \tilde{T}(V_1(X)) = V_2(X).\end{aligned}$$

□

Theorem 20.15.66 (Faithful fixed point from fixed product generators). Lean ✓ **Stmt** Let $\mathcal{M}$ be a finite full matrix algebra over $\mathbb{C}$, and let $F : \mathcal{M} \rightarrow \mathcal{M}$ be positive and trace nonincreasing. Suppose that a family $(V_j)_{j \in J}$ consists of fixed points of F and that, for some $L > 0$, the identity belongs to the linear span of the products $V_{j_1} \cdots V_{j_L}$ in $\mathcal{M}$. Then F has a positive-definite fixed point. This is the local full-matrix intermediate implication used in [11, Appendix C.4, lines 1980–1995]; it is not stated there as a separate theorem.

Proof. ✓ Proof Boundedness of the forward orbits gives the mean-ergodic projection P . Set $\rho = P(\mathbf{1})$ and let Q be the support projection of ρ . For every j , the maximal-support property of ρ gives $QV_jQ = V_j$. Since Q is a projection, this implies $QV_j = V_j = V_jQ$. By induction, for every $k \geq 1$,

$$Q \cdot V_{j_1} \cdots V_{j_k} \cdot Q = V_{j_1} \cdots V_{j_k}.$$

The product-span hypothesis and linearity then give $Q\mathbf{1}Q = \mathbf{1}$. Since $Q^2 = Q$, one has $Q = Q^2 = Q\mathbf{1}Q = \mathbf{1}$. Thus ρ is positive definite, and the construction of P gives $F(\rho) = \rho$. $\square$

Theorem 20.15.67 (Trace preservation from fixed product generators). Lean ✓ Stmt Let $\mathcal{M}$ be a finite full matrix algebra over $\mathbb{C}$, and let $F : \mathcal{M} \rightarrow \mathcal{M}$ be positive and trace nonincreasing. Suppose that $F(V_j) = V_j$ for every $j \in J$. Put $\mathcal{V} = \text{span}_{\mathbb{C}}\{V_j : j \in J\}$ and, in the notation of [11, Appendix C.4], define

$$C_L(\mathcal{V}) = \text{span}_{\mathbb{C}}\{v_1 \cdots v_L : v_1, \dots, v_L \in \mathcal{V}\}.$$

If $\mathbf{1} \in C_L(\mathcal{V})$ for some $L > 0$, then F is trace preserving.

Proof. ✓ Proof Multilinearity gives

$$C_L(\mathcal{V}) = \text{span}_{\mathbb{C}}\{V_{j_1} \cdots V_{j_L} : j_1, \dots, j_L \in J\}.$$

By Theorem 20.15.66, there is a positive-definite matrix ρ such that $F(\rho) = \rho$. Let F^* be the trace-pairing adjoint and set $\Delta = \mathbf{1} - F^*(\mathbf{1})$. For every positive semidefinite X , trace nonincrease gives $\text{tr}(\Delta X) = \text{tr}(X) - \text{tr}(F(X)) \geq 0$. Positivity of F^* and self-duality of the positive semidefinite cone therefore give $\Delta \geq 0$. The fixed-point equation gives $\text{tr}(\rho\Delta) = 0$, so positive definiteness of ρ forces $\Delta = 0$. Thus $F^*(\mathbf{1}) = \mathbf{1}$, which is equivalent to trace preservation. $\square$

Theorem 20.15.68 (Trace transfer through the canonical full-matrix extension). Lean Lean

✓ Stmt

Let $\mathcal{A} = \bigoplus_{\alpha \in I} \mathcal{M}_{d_\alpha \times d_\alpha}$, let $\iota : \mathcal{A} \rightarrow \text{End}(\bigoplus_{\alpha} \mathbb{C}^{d_\alpha})$ be the block-diagonal embedding, and let π be diagonal compression. For a linear map $F : \mathcal{A} \rightarrow \mathcal{A}$, set $\widehat{F} = \iota \circ F \circ \pi$. Then:

1. if F is trace nonincreasing on positive elements of $\mathcal{A}$, then $\widehat{F}$ is trace nonincreasing;
2. if $\widehat{F}$ is trace preserving, then F preserves the total trace on $\mathcal{A}$.

Proof. ✓ Proof If $Y \geq 0$, then $\pi(Y) \geq 0$, and the trace identities for diagonal compression and block-diagonal embedding give

$$\text{tr}(\widehat{F}(Y)) = \text{tr}_{\mathcal{A}}(F(\pi(Y))) \leq \text{tr}_{\mathcal{A}}(\pi(Y)) = \text{tr}(Y).$$

This proves the first assertion. For $X \in \mathcal{A}$, the identity $\pi \circ \iota = \text{id}_{\mathcal{A}}$ gives, whenever $\widehat{F}$ is trace preserving,

$$\text{tr}_{\mathcal{A}}(F(X)) = \text{tr}(\widehat{F}(\iota(X))) = \text{tr}(\iota(X)) = \text{tr}_{\mathcal{A}}(X).$$

$\square$

Theorem 20.15.69 (Trace preservation on a finite sum from fixed products). Lean ✓ Stmt

Let $\mathcal{A} = \bigoplus_{\alpha \in I} \mathcal{M}_{d_\alpha \times d_\alpha}$ be a finite sum of full matrix algebras, with total trace $\text{tr}_{\mathcal{A}}(X) = \sum_{\alpha \in I} \text{tr}(X_\alpha)$. Let $F : \mathcal{A} \rightarrow \mathcal{A}$ be positive and trace nonincreasing. Suppose that $F(V_j) = V_j$ for every $j \in J$. With $\mathcal{V} = \text{span}_{\mathbb{C}}\{V_j : j \in J\}$, put

$$C_L(\mathcal{V}) = \text{span}_{\mathbb{C}}\{v_1 \cdots v_L : v_1, \dots, v_L \in \mathcal{V}\}.$$

If $\mathbf{1}_{\mathcal{A}} \in C_L(\mathcal{V})$ for some $L > 0$, then F preserves the total trace on $\mathcal{A}$.

Proof. ✓ **Proof** Multilinearity first gives

$$C_L(\mathcal{V}) = \text{span}_{\mathbb{C}}\{V_{j_1} \cdots V_{j_L} : j_1, \dots, j_L \in J\}.$$

Let $\iota : \mathcal{A} \rightarrow \text{End}(\bigoplus_{\alpha} \mathbb{C}^{d_{\alpha}})$ be the block-diagonal embedding and let $\widehat{F}$ be the canonical full-matrix extension. These maps satisfy

$$\begin{aligned}\iota(XY) &= \iota(X)\iota(Y), \\ \iota(\mathbf{1}_{\mathcal{A}}) &= \mathbf{1}, \\ \widehat{F}(\iota(X)) &= \iota(F(X)).\end{aligned}$$

The extension is positive and trace nonincreasing. The matrices $\iota(V_j)$ are fixed by $\widehat{F}$, and the displayed identities carry the product-span hypothesis to the full matrix algebra. Theorem 20.15.67 makes $\widehat{F}$ trace preserving. Restricting to block-diagonal matrices proves preservation of $\text{tr}_{\mathcal{A}}$ by F . □

Theorem 20.15.70 (Trace sandwich on finite sums of matrix algebras). Lean ✓ **Stmt** *Let $\mathcal{A}$ and $\mathcal{B}$ be finite sums of full matrix algebras, and let $T : \mathcal{A} \rightarrow \mathcal{B}$ and $S : \mathcal{B} \rightarrow \mathcal{A}$ be linear maps. Suppose that T is positive, both maps are trace nonincreasing, and $S \circ T$ is trace preserving. Then T is trace preserving.*

Proof. ✓ **Proof** For every positive $X \in \mathcal{A}$, positivity of T and the two trace-nonincreasing inequalities give

$$\text{tr}_{\mathcal{A}}(X) = \text{tr}_{\mathcal{A}}(S(T(X))) \leq \text{tr}_{\mathcal{B}}(T(X)) \leq \text{tr}_{\mathcal{A}}(X).$$

Hence the middle term equals the endpoints. Equality extends from the positive cone to Hermitian matrices by positive and negative parts. For arbitrary X , put

$$\begin{aligned}H_1 &= X + X^{\dagger}, \\ H_2 &= i(X - X^{\dagger}), \\ X &= \frac{1}{2}(H_1 - iH_2).\end{aligned}$$

Both H_1 and H_2 are Hermitian, so linearity proves trace preservation for every X . □

Theorem 20.15.71 (Trace preservation of the transported vertical-sector maps). Lean ✓ **Stmt** *Suppose that every one-site and two-site multiplicity space is nonzero and every corresponding diagonal weight is positive. Let $\widetilde{M}^{(1)}$ and $\widetilde{M}^{(2)}$ be the vertical tensors of M and its two-site blocking, and let B_1 and B_2 be their assembled one-site and two-site canonical forms. Assume that both canonical forms are bases of normal tensors and that, for every vertical letter ℓ ,*

$$\begin{aligned}U_i \widetilde{M}_{\ell}^{(i)} U_i^{\dagger} &= (B_i)_{\ell}, \\ \widetilde{M}_{\ell}^{(i)} &= U_i^{\dagger} (B_i)_{\ell} U_i, \\ U_i U_i^{\dagger} &= \mathbf{1} \quad (i = 1, 2).\end{aligned}$$

Assume in addition that the physical maps T and S are completely positive and trace preserving and that, for every horizontal bond matrix Z , the two physical-coordinate identities hold:

$$\begin{aligned}T(\mathcal{M}^{(1)}(Z)) &= \mathcal{M}^{(2)}(Z), \\ S(\mathcal{M}^{(2)}(Z)) &= \mathcal{M}^{(1)}(Z).\end{aligned}$$

Then each transported map and both square composites preserve the appropriate total sector trace. For the individual maps,

$$\begin{aligned}\sum_{\beta} \operatorname{tr}(\tilde{T}(X)_{\beta}) &= \sum_{\alpha} \operatorname{tr}(X_{\alpha}), \\ \sum_{\alpha} \operatorname{tr}(\tilde{S}(Y)_{\alpha}) &= \sum_{\beta} \operatorname{tr}(Y_{\beta}).\end{aligned}$$

For the square composites,

$$\begin{aligned}\sum_{\alpha} \operatorname{tr}((\tilde{S}\tilde{T})(X)_{\alpha}) &= \sum_{\alpha} \operatorname{tr}(X_{\alpha}), \\ \sum_{\beta} \operatorname{tr}((\tilde{T}\tilde{S})(Y)_{\beta}) &= \sum_{\beta} \operatorname{tr}(Y_{\beta}).\end{aligned}$$

Neither of the stronger range inclusions is assumed:

$$\begin{aligned}T(R_1(\mathcal{A}_1)) &\subseteq R_2(\mathcal{A}_2), \\ S(R_2(\mathcal{A}_2)) &\subseteq R_1(\mathcal{A}_1).\end{aligned}$$

Proof. ✓ Proof Each transported map is positive and trace nonincreasing, and hence so are the square composites $\tilde{S}\tilde{T}$ and $\tilde{T}\tilde{S}$. The fixed-generator lemma and the product-spanning theorem give the hypotheses of Theorem 20.15.69 for both composites. Thus both preserve the corresponding total trace.

For positive $X \in \mathcal{A}_1$, one has

$$\operatorname{tr}_{\mathcal{A}_1}(X) = \operatorname{tr}_{\mathcal{A}_1}((\tilde{S}\tilde{T})(X)) \leq \operatorname{tr}_{\mathcal{A}_2}(\tilde{T}(X)) \leq \operatorname{tr}_{\mathcal{A}_1}(X).$$

Theorem 20.15.70 therefore makes $\tilde{T}$ trace preserving. Applying the same theorem to $\tilde{S}$ and $\tilde{T}\tilde{S}$ proves the assertion for $\tilde{S}$. □

Lemma 20.15.72 (Complete positivity of refinement in vertical coordinates). Lean ✓ Stmt *If the physical refinement map T is completely positive, then the transported map T_v in vertical canonical coordinates is completely positive.*

Proof. ✓ Proof By Definition 20.15.49,

$$T_v(Z) = U_2 J(T(U_1^\dagger Z U_1)) U_2^\dagger.$$

The two single-operator conjugations and the reindexing J are completely positive, so the assertion follows by composition. □

Lemma 20.15.73 (Complete positivity of coarse-graining in vertical coordinates). Lean ✓ Stmt *If the physical coarse-graining map S is completely positive, then the transported map S_v in vertical canonical coordinates is completely positive.*

Proof. ✓ Proof By Definition 20.15.49,

$$S_v(W) = U_1 S(J^{-1}(U_2^\dagger W U_2)) U_1^\dagger.$$

Again, single-operator conjugation, reindexing, and composition preserve complete positivity. □

Lemma 20.15.74 (Retained-coordinate block-diagonal inclusion). [Lean](#) [✓ Stmt](#) Let ι be block-diagonal inclusion of the weighted vertical-sector block family into its full matrix algebra, and let W be the canonical reindexing to retained physical coordinates. Then the retained-coordinate inclusion is $\iota_v = W \circ \iota$.

Proof. [✓ Proof](#) This is the defining relation between the direct-sum and retained coordinates. $\square$

Lemma 20.15.75 (Retained-coordinate diagonal-block extraction). [Lean](#) [✓ Stmt](#) Let π be diagonal-block extraction to the weighted vertical-sector block family, and let W be the canonical reindexing to retained physical coordinates. Then the retained-coordinate extraction is $\pi_v = \pi \circ W^{-1}$.

Proof. [✓ Proof](#) This is the inverse defining relation between the retained and direct-sum coordinates. $\square$

Lemma 20.15.76 (Full-matrix factorization of transported refinement). [Lean](#) [✓ Stmt](#) Write $\widehat{R}_1$ for normalized state preparation, $\widehat{R}_2$ for controlled partial trace, and W_i for retained-coordinate reindexing. The full-matrix extension of the transported refinement map is

$$\widehat{T} = \widehat{R}_2 \circ W_2^{-1} \circ T_v \circ W_1 \circ \widehat{R}_1. \quad (20.38)$$

Proof. [✓ Proof](#) Substitute the inclusion and extraction identities of Lemma 20.15.74 and Lemma 20.15.75 into $\widehat{T} = \widehat{R}_2 T_v R_1$. $\square$

Lemma 20.15.77 (Full-matrix factorization of transported coarse-graining). [Lean](#) [✓ Stmt](#) Write $\widehat{R}_2$ for normalized state preparation, $\widehat{R}_1$ for controlled partial trace, and W_i for retained-coordinate reindexing. The full-matrix extension of the transported coarse-graining map is

$$\widehat{S} = \widehat{R}_1 \circ W_1^{-1} \circ S_v \circ W_2 \circ \widehat{R}_2. \quad (20.39)$$

Proof. [✓ Proof](#) Substitute the inclusion and extraction identities of Lemma 20.15.74 and Lemma 20.15.75 into $\widehat{S} = \widehat{R}_1 S_v R_2$. $\square$

Theorem 20.15.78 (Complete positivity and adjoint Schwarz for the transported maps). [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 20.15.71, the maps

$$\begin{aligned} \widehat{T} &: \mathcal{A}_1 \longrightarrow \mathcal{A}_2, \\ \widehat{S} &: \mathcal{A}_2 \longrightarrow \mathcal{A}_1 \end{aligned}$$

are completely positive. The trace adjoints of the two square composites satisfy the Schwarz inequality in every simple summand.

Proof. [✓ Proof](#) Let ι_i and π_i be block-diagonal inclusion and diagonal-block extraction, respectively, and let W_i denote the canonical reindexing from the weighted vertical-sector coordinates to the retained physical coordinates. The full-matrix extension of the transported refinement map has the factorization (20.38). Here $\widehat{R}_1$ is normalized state preparation, $\widehat{R}_2$ is controlled partial trace, and

$$T_v(Z) = U_2 J \left(T(U_1^\dagger Z U_1) \right) U_2^\dagger.$$

Every factor is completely positive. The analogous identities

$$S_v(W) = U_1 S \left(J^{-1}(U_2^\dagger W U_2) \right) U_1^\dagger$$

and (20.39) prove complete positivity of $\tilde{S}$. These factorizations are exact on the full matrix algebras: $\pi_i W_i^{-1}$ is precisely extraction of the retained diagonal sector blocks. Thus no range inclusion is used.

Composition gives complete positivity of $\tilde{S}\tilde{T}$ and $\tilde{T}\tilde{S}$. Theorem 20.15.71 gives trace preservation of these two composites. Their trace adjoints are therefore unital completely positive maps, and the Kadison–Schwarz inequality applies in each simple summand. $\square$

Theorem 20.15.79 (Transported vertical-sector composites are identities). Lean Stmt *Under the hypotheses of Theorem 20.15.71, the transported maps satisfy*

$$\tilde{S} \circ \tilde{T} = \text{id}_{\mathcal{A}_1}, \quad (20.40)$$

$$\tilde{T} \circ \tilde{S} = \text{id}_{\mathcal{A}_2}. \quad (20.41)$$

Thus $\tilde{T}$ and $\tilde{S}$ are mutually inverse. This is the identity-composition conclusion of [11, Appendix C.4, lines 1974–1995].

Proof. Proof For each square composite F , positivity follows from positivity of the two transported maps. Theorem 20.15.71 and Theorem 20.15.78 give trace preservation and the Schwarz inequality for F^* , respectively. Lemma 20.15.65 gives $F(V_i(X)) = V_i(X)$ for every bond matrix X . Theorem 20.15.23 gives a positive integer L_i such that $C_{L_i}(V_i) = \mathcal{A}_i$. Applying Theorem 20.15.29 first to $F = \tilde{S} \circ \tilde{T}$ and then to $F = \tilde{T} \circ \tilde{S}$ proves (20.40) and (20.41). $\square$

Theorem 20.15.80 (Unitary relabelling of transported vertical sectors). Lean Stmt *Under the hypotheses of Theorem 20.15.79, write*

$$\mathcal{A}_1 = \prod_{\alpha} M_{D_{\alpha}^{(1)}}(\mathbb{C}), \quad \mathcal{A}_2 = \prod_{\beta} M_{D_{\beta}^{(2)}}(\mathbb{C}).$$

There are an equivalence σ of the two sector index sets, equalities $D_{\alpha}^{(1)} = D_{\sigma(\alpha)}^{(2)}$, and unitaries $V_{\alpha} \in M_{D_{\sigma(\alpha)}^{(2)}}(\mathbb{C})$ such that

$$\tilde{T}(0, \dots, 0, X, 0, \dots, 0) = (0, \dots, 0, V_{\alpha} \iota_{\alpha}(X) V_{\alpha}^*, 0, \dots, 0),$$

where the nonzero entry on the right is in position $\sigma(\alpha)$, and

$$\tilde{S}(0, \dots, 0, Y, 0, \dots, 0) = (0, \dots, 0, \iota_{\alpha}^{-1}(V_{\alpha}^* Y V_{\alpha}), 0, \dots, 0),$$

where the entries on the left and right are in positions $\sigma(\alpha)$ and α , respectively. These identities hold for every $X \in M_{D_{\alpha}^{(1)}}(\mathbb{C})$ and $Y \in M_{D_{\sigma(\alpha)}^{(2)}}(\mathbb{C})$. Here ι_{α} is the reindexing induced by the displayed dimension equality. These are the mutually inverse unitary formulas of [11, Appendix C.4, line 1997]. They make no assertion about the multiplicities, weights, or coefficient identity at lines 2001–2008.

Proof. Proof The two basis-of-normal-tensors decompositions have positive matrix dimensions. The transported maps are completely positive and preserve the total block trace, and Theorem 20.15.79 gives both inverse laws. Apply Theorem [182, “Unitary action of an invertible CPTP map”] to this pair. Its equivalence of summands, dimension equalities, and unitary matrices give the two displayed formulas without changing the chosen relabelling. $\square$

Theorem 20.15.81 (Coefficient comparison for transported vertical sectors). [Lean](#) [✓ Stmt](#) Under the hypotheses and with the notation of Theorem 20.15.80, let μ_α and ν_β be the positive diagonal multiplicity matrices in the one-site and two-site vertical canonical forms, and set $m_\alpha = \text{tr}(\mu_\alpha)$ and $n_\beta = \text{tr}(\nu_\beta)$. For every horizontal bond matrix X , every sector α , and every tensor letter a ,

$$A_{\sigma(\alpha)}^{(2)}(X) = \frac{m_\alpha}{n_{\sigma(\alpha)}} V_\alpha \iota_\alpha(A_\alpha^{(1)}(X)) V_\alpha^*, \quad (20.42)$$

$$A_{\sigma(\alpha),a}^{(2)} = \frac{m_\alpha}{n_{\sigma(\alpha)}} V_\alpha \iota_\alpha(A_{\alpha,a}^{(1)}) V_\alpha^*. \quad (20.43)$$

This is the coefficient identity in [11, Appendix C.4, lines 2001–2008]. It compares the traces of the multiplicity matrices; it does not assert equality of their dimensions or of their individual diagonal entries.

Proof. [✓ Proof](#) Decompose the weighted contraction family into its individual matrix summands. At the summand paired with α , the unitary formula gives

$$m_\alpha V_\alpha \iota_\alpha(A_\alpha^{(1)}(X)) V_\alpha^* = n_{\sigma(\alpha)} A_{\sigma(\alpha)}^{(2)}(X).$$

Positivity of $\nu_{\sigma(\alpha)}$ makes $n_{\sigma(\alpha)}$ nonzero, so division gives (20.42). Taking X to be a matrix unit gives (20.43). $\square$

Theorem 20.15.82 (Simultaneous representations of the blocked vertical operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses and with the sector correspondence of Theorem 20.15.81, let B be the vertical reading of the two-site blocking, and write $O_L(A) = \text{Tr}_{\text{bond}}(A_{i_1} \cdots A_{i_L})$ for the length- L closed-chain operator. If $L > 0$, then the same equivalence σ , dimension identifications, and unitaries appearing in the tensor-letter identity satisfy

$$O_L(B) = \sum_\alpha \left(\frac{m_\alpha}{n_{\sigma(\alpha)}} \right)^L \text{tr}(\nu_{\sigma(\alpha)}^L) O_L(A_\alpha^{(1)}), \quad (20.44)$$

$$O_L(B) = \sum_{\alpha,\beta} \text{tr}(\mu_\alpha^L) \text{tr}(\mu_\beta^L) O_L(A_\alpha^{(1)}) O_L(A_\beta^{(1)}). \quad (20.45)$$

These are the two representations in [11, Appendix C.4, lines 2011–2018].

In [11, Appendix C.4, line 2013], the exponent is printed as m_γ^L/n_γ ; the tensor scaling requires $(m_\gamma/n_\gamma)^L$ (see [145]).

Proof. [✓ Proof](#) The two-site vertical canonical form first gives $O_L(B) = \sum_\delta \text{tr}(\nu_\delta^L) O_L(A_\delta^{(2)})$. Reindex this sum by $\delta = \sigma(\alpha)$. The tensor-letter identity from Theorem 20.15.81 multiplies every letter of $A_\alpha^{(1)}$ by $m_\alpha/n_{\sigma(\alpha)}$. Along a chain of length L this scalar therefore occurs to the power L ; unitary conjugation and the bond-index identification leave the closing trace unchanged. This proves (20.44).

For (20.45), the vertical reading of a two-site block is the product tensor of two copies of the one-site vertical reading. Closed chains turn this tensor product into multiplication of operators. Expand each copy by the one-site vertical canonical form, $O_L(\tilde{M}) = \sum_\alpha \text{tr}(\mu_\alpha^L) O_L(A_\alpha^{(1)})$, and distribute the product of the two finite sums. $\square$

Theorem 20.15.83 (Canonical hypotheses for the retained vertical product). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the vertical reading of M is reconstructed from a retained tensor

A by a coisometry U . The two-site blocking is then reconstructed from the product of two copies of A by the Kronecker square of U . Moreover, if M is in normalized BNT-refined horizontal form and generates positive operators, this retained product tensor has invariant-projector closure and has no nontrivial periodic vectors. No full-support assumption is imposed on the retained bond space.

Proof. [✓ Proof](#) The Kronecker square of a coisometry is again a coisometry. Expanding a blocked vertical letter gives the product of the two one-site reconstructions and hence the asserted exact compression formula. Normalized BNT-refined horizontal form and positivity are preserved by two-site blocking. Invariant-projector closure descends to an exact reducing compression, while a periodic vector in the compression would embed isometrically as a periodic vector of the blocked vertical tensor. $\square$

Theorem 20.15.84 (Canonical hypotheses for every retained copy pair). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *In the retained coordinates, the product tensor is the orthogonal direct sum over ordered pairs of BNT copies. Its summand indexed by $((\alpha, q), (\beta, r))$ is $\mu_{\alpha, q} \mu_{\beta, r} (M_\alpha M_\beta)$. Every such summand has invariant-projector closure and has no nontrivial periodic vectors. This includes zero-dimensional summands and requires no nonvanishing assumption on the displayed scalar.*

Proof. [✓ Proof](#) The canonical inclusion of a dependent direct-sum block is an isometry. After transporting this inclusion through the retained-coordinate equivalence, it intertwines the chosen summand with the full retained product tensor on both sides. Its range projection therefore commutes with every tensor letter. Invariant-projector closure descends by exact compression, and any periodic vector of the summand would embed as a periodic vector of the full retained product. $\square$

Theorem 20.15.85 (Normal-corner decomposition of a retained copy pair). [Lean](#) [✓ Stmt](#) *For every ordered pair of retained BNT copies, the weighted product tensor has an exact orthogonal decomposition into positive multiples of normal tensors. The corresponding corner isometries are retained, intertwine in both directions, and reconstruct every tensor letter. The active family is allowed to be empty; in particular, no normal block is introduced for a zero-dimensional or identically zero copy-pair tensor.*

Proof. [✓ Proof](#) Apply the canonical-form sufficient condition to the invariant-projector closure and periodic-exclusion properties of Theorem 20.15.84. Its spectral construction discards precisely the corners on which every letter vanishes and retains the isometry of every nonzero corner. $\square$

Definition 20.15.86 (Simultaneous spectral data for retained copy pairs). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *A retained-product spectral family records, simultaneously for every ordered pair of retained BNT copies, all nonzero normal corners, their positive spectral coefficients, and their local isometries. Its active labels form the dependent union of the local corner families; this union is enumerated by a single finite type. The local isometries and their inclusions remain part of the data, and their range projections can be formed when needed.*

Definition 20.15.87 (Grouped vertical product hypotheses). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $M^{(2)}$ be the two-site blocking of M and write $A = \widehat{M^{(2)}}$. The grouped vertical product hypotheses require the phase-class grouping with physical reducing isometries and the pairwise grouped-corner Gram comparison for both M and $M^{(2)}$. They require in addition the following three properties of A :*

- Both normalized BNT-refined horizontal form and literal CPSV canonical form, together with positivity of the generated operators, supply these hypotheses independently.

Proof. ✓ Proof In copy coordinates the assembled tensor is block diagonal. The canonical dependent-sum inclusion is an isometry, and the two block-diagonal intertwining identities give its forward, adjoint, and compression formulas. Choose the exact normal-corner decomposition of every copy-pair tensor. The canonical copy-pair inclusions are isometries and select the weighted diagonal blocks of the retained product tensor. Composition with each local isometry gives the retained inclusion and both intertwining identities. Finally, write the blocked vertical tensor as $T = W^\dagger C W$ and the ambient inclusion as $J = W^\dagger R$. The coisometry identity $W W^\dagger = 1$ transports both retained intertwining identities from C to T ; the compression identity then follows from $J^\dagger J = 1$. For any blocked BNT label, choose its first multiplicity copy. Composing its canonical inclusion with the adjoint reconstruction coisometry gives an isometric reference corner of the blocked vertical tensor. Transport along an equality of bond dimensions preserves its isometry and exact compression formula. □

$$C^i = \sum_j W_j (\lambda_j A_j^i) W_j^\dagger,$$

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Each compression $W_j^\dagger C^i W_j$ is $\lambda_j A_j^i$, and C has the same positive-length closed chains as the finite direct sum $\bigoplus_j \lambda_j A_j$.

Proof. ✓ Proof First reconstruct the retained product as the sum of its canonical copy-pair blocks. Substitute the exact local spectral reconstruction in every pair, distribute the two inclusion maps across the finite sums, and identify the iterated sum with the dependent active-label sum. Finally transport this sum through the active-label enumeration. Isometry, orthogonality, intertwining, and compression are inherited by composition; the closed-chain equality follows from the exact isometric reconstruction. This is the active-corner decomposition in [11, Appendix C.4, lines 2020–2029]. $\square$

Theorem 20.15.90 (Coverage of active product corners by the blocked BNT). Lean Lean Lean

✓ Stmt Suppose that the one-site vertical tensor has the stated exact coisometric reconstruction, and let $(A_\gamma^{(2)})_\gamma$ be a basis of normal tensors for the blocked vertical tensor. For every active product corner A_j there are a label $\gamma(j)$, an equality of bond dimensions, an invertible matrix X_j , and a scalar ζ_j of modulus one such that, after transporting the blocked tensor along the dimension equality,

$$A_j^i = \zeta_j X_j A_{\gamma(j)}^{(2),i} X_j^{-1}.$$

The choices may be made simultaneously, giving one-way coverage of the active product corners [20]. Here fixed-pair support follows from exact squared reconstruction and the canonical direct-sum corner, with a zero pair contributing an empty active family [101].

Proof. ✓ Proof The squared coisometric reconstruction gives equality of all positive-length closed chains between the blocked vertical tensor and the retained product. Transport the blocked basis of normal tensors along this equality. Proposition 2.7, applied to the flattened positive normal decomposition, then covers each active corner by a blocked BNT tensor up to an invertible gauge and a unit-modulus phase. Classical choice makes these comparisons simultaneous. $\square$

Theorem 20.15.91 (Distinguished blocked reference corners). Lean Lean Lean Lean Lean

Lean Lean Lean ✓ Stmt Assume that every blocked BNT label has positive multiplicity and that the blocked vertical tensor has the exact coisometric reconstruction $T = U_2^\dagger C_2 U_2$. Let $E_{\gamma,0}$ be the canonical inclusion of the first copy of label γ , and put $F_\gamma = U_2^\dagger E_{\gamma,0}$. Then F_γ is an isometry, intertwines T with the distinguished weighted BNT copy in both directions, and compresses T to that copy exactly. For an active product corner covered by γ , transport F_γ along the stated equality of bond dimensions. The transported map remains an isometry and obeys the corresponding transported compression identity.

Proof. ✓ Proof The copy inclusion is isometric and selects one diagonal block of the assembled tensor. Inserting $T = U_2^\dagger C_2 U_2$ and $U_2 U_2^\dagger = 1$ gives both intertwining identities and compression. Transport along equality of bond dimensions preserves these equations. $\square$

Theorem 20.15.92 (Active product positivity from sector separation). Lean ✓ Stmt

Assume that M generates matrix product density operators and that the one-site and blocked vertical tensors admit exact coisometric reconstructions. Assume positive multiplicities and weights for the blocked BNT and normality of its representatives. Suppose moreover that every projection with a nonzero two-sided corner of the blocked vertical tensor has a nonzero finite-chain sector compression. Then every active product corner j satisfies $0 < \lambda_j \zeta_j$.

Proof. ✓ Proof Compare the active corner with the distinguished positive-weight copy of its blocked BNT representative. The active coefficient is nonzero because $\lambda_j > 0$ and $|\zeta_j| = 1$. The assumed separation property supplies a nonzero finite-chain compression, so the sector-weight comparison gives $\lambda_j \zeta_j > 0$. $\square$

Theorem 20.15.93 (Positivity of active product coefficients). Lean ✓ Stmt *Suppose that M generates positive operators and has the grouped vertical product hypotheses. Assume exact coisometric reconstructions of its one-site and blocked vertical tensors, positive multiplicities and weights for the blocked BNT, and normality of its BNT representatives. For every active product corner j , one has $0 < \lambda_j \zeta_j$.*

Proof. ✓ Proof The sector-compression separation hypothesis supplies a nonzero finite-chain compression for each active corner. Apply Theorem 20.15.92 with this separation property. $\square$

Theorem 20.15.94 (Unitary normalization of active product gauges). Lean ✓ Stmt *Under the hypotheses of Theorem 20.15.93, for every active product corner j there is a real number $\omega_j > 0$ such that*

$$\begin{aligned} X_j^\dagger X_j &= \omega_j \mathbb{1}, \\ \omega_j^{-1/2} X_j &\in \mathbf{U}(D_j). \end{aligned}$$

This is the gauge normalization used in [11, Appendix C.4, lines 2025–2029].

Proof. ✓ Proof Apply the Figure 8 comparison to the active corner and the distinguished blocked reference corner. Both coefficients are positive, and the two-site Gram comparison identifies their Gram-dressed normal tensors. Normal-tensor rigidity then makes $X_j^\dagger X_j$ a positive real multiple of the identity. Division by the positive square root gives a unitary matrix. $\square$

Definition 20.15.95 (Original-label corner family). Lean ✓ Stmt *An original-label corner family for the retained product records, for every active corner of each copy pair, an original one-site BNT label, a positive coefficient, and an isometric inclusion into the raw product bond space. Distinct corners belonging to one copy pair have orthogonal ranges. Each inclusion intertwines the raw product tensor with the stated positive multiple of its original BNT representative, and the resulting corners reconstruct every tensor letter exactly.*

A copy pair with no active corner is represented by an empty family [101].

Theorem 20.15.96 (Positive original-label corner decomposition). Lean ✓ Stmt *Let the active product corners be covered by blocked BNT representatives, let the blocked representatives be identified with the original BNT by the trace-ratio unitary conjugacies, and suppose that the active gauge-phase coefficients are positive. If $X_j^\dagger X_j = \omega_j \mathbb{1}$ with $\omega_j > 0$, then every retained copy pair has an original-label corner family. Its coefficient at an active corner is*

$$\frac{\lambda_j \zeta_j}{w_{\alpha,q} w_{\beta,r}} \frac{\mathrm{tr} D_{\gamma(j)}}{\mathrm{tr} D_{\sigma(\gamma(j))}^{(2)}},$$

which is positive. The normalized inclusions are isometries with pairwise orthogonal ranges, intertwine the raw product tensor with these positive multiples of the original BNT representatives, and reconstruct every raw product letter exactly.

For copy pairs with empty active support, exact weighted reconstruction and positivity of the outer copy weight imply that the corresponding raw product vanishes [101].

Proof. ✓ Proof Transport each local inclusion first by the normalized active comparison gauge and then by the unitary identifying the blocked representative with its original one-site label. The first normalization preserves isometry and orthogonality; the second unitary transport does likewise. The two conjugacy equations give the stated coefficient and intertwining identity. Positivity follows from the active gauge–phase coefficient, the two positive multiplicity traces, and the positive outer copy weight. Conjugating each corner term through the two transports recovers its local spectral term. Summing these identities gives the weighted reconstruction, and cancellation of the nonzero outer weight gives the raw reconstruction. $\square$

Theorem 20.15.97 (Fusion coisometries from original-label corners). Lean ✓ Stmt *Given an original-label corner family, choose the distinguished retained copy of every BNT label. For fixed labels α, β , let the multiplicity of γ be the number of active corners of the distinguished copy pair whose original label is γ , and put the corresponding positive corner coefficients on the diagonal of $\chi_{\alpha, \beta, \gamma}$. The block row of the adjoints of the corner inclusions, regrouped by these label fibers, defines a fusion coisometry. It satisfies*

$$\begin{aligned}\chi_{\alpha, \beta, \gamma; r} &> 0, \\ U_{\alpha, \beta} U_{\alpha, \beta}^\dagger &= 1, \\ U_{\alpha, \beta} (M_\alpha M_\beta)^{ij} U_{\alpha, \beta}^\dagger &= \bigoplus_\gamma \chi_{\alpha, \beta, \gamma} \otimes M_\gamma^{ij},\end{aligned}$$

together with the exact reconstruction

$$(M_\alpha M_\beta)^{ij} = U_{\alpha, \beta}^\dagger \left(\bigoplus_\gamma \chi_{\alpha, \beta, \gamma} \otimes M_\gamma^{ij} \right) U_{\alpha, \beta}.$$

Empty label fibers contribute zero-dimensional diagonal blocks, yielding the decomposition of [11, Appendix C.4, lines 2020–2029]. Here only active product corners are retained [20], with fixed-pair support handled by empty families [101] and the retained-row map acting as a coisometry onto the active direct sum [61].

Proof. ✓ Proof For each distinguished copy pair, form the block row from the adjoints of its pairwise orthogonal isometric corner inclusions. The block-row identity makes this map a coisometry. Its forward conjugation is block diagonal over the active corners, while the corner reconstruction gives the reverse identity. Regrouping the active-corner sum by the original label turns each label fiber into the diagonal matrix whose entries are precisely the positive corner coefficients. These identities are valid without separate cases when a fiber or the whole active family is empty. $\square$

Theorem 20.15.98 (Fusion coisometries from normalized product corners). Lean ✓ Stmt *Suppose that every active product corner has positive gauge–phase coefficient and that its gauge Gram matrix is a positive scalar multiple of the identity. Given the trace-ratio unitary correspondence between the one-site and two-site vertical sectors, there are positive diagonal matrices $\chi_{\alpha, \beta, \gamma}$ and coisometries $U_{\alpha, \beta}$ satisfying the forward fusion identity and exact reconstruction for every product tensor $M_\alpha M_\beta$. This yields the active-corner construction of [11, Appendix C.4, lines 2020–2029], retaining active product corners [20], handling fixed-pair support [101], and using coisometric fusion onto the active subspace [61].*

Proof. ✓ Proof The positive coefficients and normalized gauges give an original-label corner family. Apply the block-row construction of Theorem 20.15.97 to its orthogonal isometric inclusions. $\square$

Theorem 20.15.99 (Positive fusion from transported vertical sectors). [Lean](#) [✓ Stmt](#) Under the one-site and two-site vertical basis-of-normal-tensors decompositions and the two completely positive, trace-preserving physical-closure transport identities, assume that the one-site tensor has grouped vertical product inputs and defines a matrix product density operator. Then there are positive diagonal multiplicity matrices $\chi_{\alpha,\beta,\gamma}$ and matrices $U_{\alpha,\beta}$ such that every diagonal entry is positive and

$$\begin{aligned} U_{\alpha,\beta} U_{\alpha,\beta}^\dagger &= 1, \\ U_{\alpha,\beta} (M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger &= \bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij}, \end{aligned}$$

while the exact reconstruction is

$$(M_\alpha M_\beta)^{ij} = U_{\alpha,\beta}^\dagger \left(\bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij} \right) U_{\alpha,\beta}.$$

This establishes the positive fusion decomposition of [11, Appendix C.4, lines 2020–2029], accounting for active product corners [20], fixed-pair support [101], and coisometric fusion [61].

Proof. [✓ Proof](#) First decompose every retained copy-pair tensor into its active normal corners and compare these corners with the two-site vertical basis. Positivity of matrix product density operators makes every resulting scalar positive, while the Gram comparison normalizes each gauge to a unitary. The inverse vertical-sector transport then returns the corners to the original one-site labels. Their orthogonal isometric inclusions form the rows of $U_{\alpha,\beta}$. Grouping these rows by the original label gives the positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$; corner intertwining and corner reconstruction give the two displayed identities. $\square$

Theorem 20.15.100 (Positive fusion from grouped vertical product inputs). [Lean](#) [✓ Stmt](#) Let M generate matrix product density operators and have grouped vertical product inputs. Suppose that its one-site and two-site vertical tensors have exact coisometric weighted reconstructions with positive multiplicities and weights, and that the two-site representatives form a basis of normal tensors. Suppose also that a sector bijection σ identifies the bond dimensions and that, after this identification, the unitary matrices V_α satisfy $A_{2,\sigma(\alpha)}^v = \frac{\text{tr}(\mu_{1,\alpha})}{\text{tr}(\mu_{2,\sigma(\alpha)})} V_\alpha A_{1,\alpha}^v V_\alpha^\dagger$ for every vertical letter v . Then there are positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ and matrices $U_{\alpha,\beta}$ such that

$$\begin{aligned} U_{\alpha,\beta} U_{\alpha,\beta}^\dagger &= 1, \\ U_{\alpha,\beta} (M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger &= \bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij}, \\ (M_\alpha M_\beta)^{ij} &= U_{\alpha,\beta}^\dagger \left(\bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij} \right) U_{\alpha,\beta}. \end{aligned}$$

Every diagonal entry of $\chi_{\alpha,\beta,\gamma}$ is positive. This establishes the positive fusion decomposition of [11, Appendix C.4, lines 2020–2029], with active corners and coisometric fusion following [20, 101, 61].

Proof. [✓ Proof](#) The grouped vertical product hypotheses give simultaneous spectra for all retained product corners. Compare each active corner with the blocked vertical BNT. Sector separation makes its coefficient positive, and the blocked Gram comparison normalizes its gauge to a unitary. Transporting these corners through the one-site–two-site sector correspondence gives positive original-label corners; their adjoints form the rows of the coisometries $U_{\alpha,\beta}$ and yield both displayed identities. $\square$

Theorem 20.15.101 (Idempotent trace scalars from blocked representations). [Lean](#) [✓ Stmt](#) Fix one-site and two-site vertical canonical decompositions whose sectors are identified by the transported unitary relabelling. Suppose that the product tensors have positive diagonal fusion matrices $\chi_{\alpha,\beta,\gamma}$ and exact forward and reverse coisometric fusion identities. If the blocked operator has the two simultaneous representations of Theorem 20.15.82, then

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

This is the idempotent coefficient identity in [11, Appendix C.4, lines 2030–2042].

Proof. [✓ Proof](#) For every sufficiently large L , expand the product of one-site sector operators with the fusion identity. Eventual linear independence of the BNT operators and the two blocked representations then identify, for every γ , the two positive power sums

$$\text{tr} \left[\left(\frac{m_\gamma}{n_{\sigma(\gamma)}} \nu_{\sigma(\gamma)} \right)^L \right] = \sum_{\alpha,\beta} \text{tr} \left[(\mu_\alpha \otimes \mu_\beta \otimes \chi_{\alpha,\beta,\gamma})^L \right].$$

The eventual power-sum comparison identifies the two multisets of positive entries. Summing this multiset equality and cancelling $n_{\sigma(\gamma)}$ gives the displayed length-one identity. $\square$

Theorem 20.15.102 (Positive fusion decomposition of vertical products). [Lean](#) [✓ Stmt](#) Let M be in normalized BNT-refined horizontal form, generate matrix product density operators, and satisfy the renormalization fixed-point condition of Definition 20.2.1. Then one may choose a vertical basis of normal tensors $\{M_\alpha\}_\alpha$ for which there are positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ and matrices $U_{\alpha,\beta}$. Every diagonal entry of $\chi_{\alpha,\beta,\gamma}$ is positive, and

$$\begin{aligned} U_{\alpha,\beta} U_{\alpha,\beta}^\dagger &= 1, \\ U_{\alpha,\beta} (M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger &= \bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij}, \\ (M_\alpha M_\beta)^{ij} &= U_{\alpha,\beta}^\dagger \left(\bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij} \right) U_{\alpha,\beta}. \end{aligned}$$

Thus the renormalization fixed-point and matrix-product-density-operator assumptions, together with normalized BNT-refined horizontal form, imply the positive fusion decomposition of [11, Appendix C.4, lines 2020–2029] on the active corners [20, 101, 61].

Proof. [✓ Proof](#) Apply the proved normalized BNT-refined specialization of Proposition 4.13 to M and to its two-site blocking. In each case, retain the CPSV basis-of-normal-tensors assertion, the positive multiplicity matrices, and both coisometric decomposition identities. Unpack the two completely positive, trace-preserving maps in the renormalization fixed-point condition. Theorem 20.15.99 then supplies the positive diagonal family, the row coisometries, the forward fusion identity, and exact reconstruction for the chosen one-site vertical basis. $\square$

Theorem 20.15.103 (Two-site closure of the two-site blocking). [Lean](#) [✓ Stmt](#) After identifying two blocked physical indices with four original indices, one has $(K^{[2]})_2(X) = K_4(X)$.

Proof. [✓ Proof](#) Expand the two blocked tensor matrices and reassociate their product. $\square$

20.16 Simple local structure from SAL and ZCL

20.16.1 Simple canonical-form weights

Lemma 20.16.1.1 (Blockwise zero-correlation-length equation). [Lean](#) [✓ Stmt](#) Suppose a tensor satisfying the literal zero-correlation-length equation is gauge equivalent, through block-diagonal copy gauges, to a sector decomposition. Every copy satisfies

$$(\mu_{j,q} \mathcal{T}_{A_j})^2 = \mu_{j,q} \mathcal{T}_{A_j}. \quad (20.46)$$

This is the block equation in [11, Appendix C.2, lines 1652–1657].

Proof. [✓ Proof](#) Write $\mathcal{T}_M = G \mathcal{T}_S G^{-1}$, where G is the induced block-diagonal gauge. The equation $\mathcal{T}_M^2 = \mathcal{T}_M$ therefore becomes $\mathcal{T}_S^2 = \mathcal{T}_S$ after conjugation by G^{-1} . Restriction to the (j, q) -th diagonal block gives (20.46). $\square$

Theorem 20.16.1.2 (Copy independence for a non-nilpotent sector decomposition). [Lean](#) [✓ Stmt](#) Let a tensor satisfying the literal zero-correlation-length equation be gauge equivalent, through block-diagonal copy gauges, to a sector decomposition whose basis elements all have non-nilpotent physical-trace transfer. Then the weights do not depend on the copy index: $\mu_{j,q} = \mu_j$ [11, Appendix C.2, lines 1646–1661].

Proof. [✓ Proof](#) Literal zero correlation length states $\mathcal{T}_M^2 = \mathcal{T}_M$. If $\mathcal{T}_M = X \mathcal{T}_S X^{-1}$ is the block-diagonal gauge to the transfer of the sector decomposition, then conjugation gives $\mathcal{T}_S^2 = \mathcal{T}_S$. By Lemmas 20.12.2 and 20.12.3, the identity descends to every copy:

$$\begin{aligned} \mu_{j,q}^2 \mathcal{T}_{A_j}^2 &= \mu_{j,q} \mathcal{T}_{A_j}, \\ \mathcal{T}_{A_j}^2 &= \frac{1}{\mu_{j,q}} \mathcal{T}_{A_j}. \end{aligned}$$

The left side does not depend on q , and $\mathcal{T}_{A_j} \neq 0$ because it is not nilpotent, so $1/\mu_{j,q}$ is independent of q , which gives $\mu_{j,q} = \mu_{j,q'}$. $\square$

Corollary 20.16.1.3 (Simple canonical-form tensors with zero correlation length). [Lean](#) [✓ Stmt](#) A simple canonical-form tensor (Definition 20.12.20) with zero correlation length has non-nilpotent basis transfers and weights independent of the copy index.

Proof. [✓ Proof](#) Apply Theorem 20.16.1.2 to the witness supplied by the simple canonical-form predicate. $\square$

Lemma 20.16.1.4 (Scale-invariant blockwise zero-correlation-length equation). [Lean](#) [✓ Stmt](#) Suppose a tensor with scale-invariant source zero correlation length is gauge equivalent, through block-diagonal copy gauges, to a sector decomposition. There is a scalar $\lambda > 0$ such that every copy satisfies

$$(\mu_{j,q} \mathcal{T}_{A_j})^2 = \lambda \mu_{j,q} \mathcal{T}_{A_j}.$$

Proof. [✓ Proof](#) Conjugate the scale-invariant physical-trace equation through the induced block-diagonal gauge and restrict it to the (j, q) -th diagonal block. $\square$

Theorem 20.16.1.5 (Scale-invariant copy-independence generalization). [Lean](#) [✓ Stmt](#) *More generally, suppose that, for some $\lambda > 0$, the nonzero physical-trace transfer satisfies*

$$\mathcal{T}_M^2 = \lambda \mathcal{T}_M.$$

Under the same canonical-form and non-nilpotency conditions, the weights remain independent of the copy index for every positive scaling factor λ .

Proof. [✓ Proof](#) The weighted block equation gives $\mathcal{T}_{A_j}^2 = (\lambda/\mu_{j,q})\mathcal{T}_{A_j}$. Comparing two copy indices and using $\lambda \neq 0$ and $\mathcal{T}_{A_j} \neq 0$ gives $\mu_{j,q} = \mu_{j,q'}$. $\square$

Corollary 20.16.1.6 (Simple canonical-form tensors with scale-invariant zero correlation length). [Lean](#) [✓ Stmt](#) *A simple canonical-form tensor with scale-invariant source zero correlation length has non-nilpotent basis transfers and weights independent of the copy index.*

Proof. [✓ Proof](#) Apply Theorem 20.16.1.5 to the witness supplied by the simple canonical-form predicate. $\square$

Definition 20.16.1.7 (Common sector weight and absorbed representative). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the weights over each representative do not depend on the copy index. Choose the common weight μ_j from any copy over j and absorb it into the representative:

$$\mathcal{K}_j = \mu_j A_j.$$

This is the convention of [11, Appendix C.2, equation CFK, lines 1660–1665].

Lemma 20.16.1.8 (Every copy has the common weight). [Lean](#) [✓ Stmt](#) *For every copy q over j , one has $\mu_{j,q} = \mu_j$.*

Proof. [✓ Proof](#) Compare the q -th copy with the distinguished copy used to define μ_j . $\square$

Theorem 20.16.1.9 (Coefficient after common-weight absorption). [Lean](#) [✓ Stmt](#) *Let r_j be the number of horizontal copies of representative j . Then its coefficient on a chain of length N is*

$$\sum_q \mu_{j,q}^N = r_j \mu_j^N.$$

Thus the natural multiplicity is the horizontal copy count r_j .

Proof. [✓ Proof](#) Every term in the copy sum equals μ_j^N , and there are r_j terms. $\square$

Theorem 20.16.1.10 (Nonvanishing of the common-weight coefficient). [Lean](#) [Lean](#) [✓ Stmt](#) *The common weight is nonzero, and hence for every chain length N ,*

$$q_j = \sum_q \mu_{j,q}^N = r_j \mu_j^N \neq 0.$$

This establishes the coefficient nonvanishing used for the factorization $m_j = q_j \tilde{m}_j$ in [11, Appendix C.2, lines 1714–1718].

Proof. [✓ Proof](#) The distinguished copy weight is nonzero by the sector data. Since $r_j > 0$, both factors in $r_j \mu_j^N$ are nonzero. $\square$

Theorem 20.16.1.11 (Common nonzero BNT tail selection). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the weights over each representative are independent of the copy index, and none of the physical-trace transfers $\mathcal{B}_j = \sum_i A_j^{ii}$ is nilpotent. There is one positive tail length L such that, for every representative A_j , some virtual indices α, β satisfy $\widetilde{m}_j = [\mathcal{B}_j^L]_{\alpha, \beta} \neq 0$. At total chain length $L + 3$, the corresponding copy coefficient is $q_j = \sum_q \mu_{j,q}^{L+3} \neq 0$. Thus the three-site closing matrix satisfies

$$R_j = q_j \mathcal{B}_j^L, \\ (R_j)_{\alpha, \beta} = q_j \widetilde{m}_j \neq 0.$$

This is the traced-tail selection made in [11, Appendix C.2, lines 1714–1718].

Proof. [✓ Proof](#) Since $\mathcal{B}_j$ is not nilpotent, every positive power of $\mathcal{B}_j$ is nonzero; in particular, one common positive power has a nonzero entry for every j . The preceding theorem supplies the nonzero coefficient at length $L + 3$. Therefore its product with the selected entry of $\mathcal{B}_j^L$ is nonzero. $\square$

Theorem 20.16.1.12 (Canonical form with natural multiplicities). [Lean](#) [✓ Stmt](#) After absorbing the common weight, the matrix product vector of the direct-sum tensor is

$$\mathcal{V}_N(S) = \sum_j r_j \mathcal{V}_N(\mathcal{K}_j).$$

This is equation CFK of [11, Appendix C.2, lines 1660–1665]; the same multiplicities occur in equation sigmaNKj of [11, Appendix C.2, lines 1756–1759].

Proof. [✓ Proof](#) Expand the direct-sum tensor by representatives, use Theorem 20.16.1.9, and absorb μ_j^N into the length- N matrix product vector of A_j . $\square$

Definition 20.16.1.13 (Three-site family closure and outer inverse contraction). [Lean](#) [Lean](#) [✓ Stmt](#) Let $(\mathcal{K}_s)_s$ be a finite family of MPO tensors, with $\mathcal{K}_s^{ij} \in M_{D_s}(\mathbb{C})$, and let $R_s \in M_{D_s}(\mathbb{C})$. A three-site operator is a family closure when

$$\rho_{i_1 i_2 i_3, j_1 j_2 j_3} = \sum_s \text{tr}(\mathcal{K}_s^{i_1 j_1} \mathcal{K}_s^{i_2 j_2} \mathcal{K}_s^{i_3 j_3} R_s).$$

Given a simultaneous left inverse C as in Theorem 21.3.84, contracting C against the first and third physical sites of ρ defines the outer inverse contraction

$$(\Gamma_{x,y})_{i_2 j_2} = \sum_{i_1, j_1, i_3, j_3} C_{x, (i_1, j_1)} C_{y, (i_3, j_3)} \rho_{(i_1, i_2, i_3), (j_1, j_2, j_3)}.$$

The closure is the sector-sum form of the displayed equation for $\sigma_3^{(N)}$ in [11, Appendix C.2, lines 1343–1348], used with the block-canonical tensor in the Case II calculation at [11, Appendix C.2, lines 1726–1732]. The outer contraction is the sector-family form of the Case I contraction in [11, Appendix C.2, lines 1415–1438]; the same construction is used with sector projectors at lines 1719–1725.

Theorem 20.16.1.14 (Normalized three-site BNT family closure). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the doubled-index tensor has the horizontal BNT representation S , the copy weights over each representative agree, the physical-trace transfers $\mathcal{B}_j = \sum_i A_j^{ii}$ are not nilpotent,

and the original tensor satisfies SAL. After identifying length-three configurations with ordered triples, the normalized three-site marginal of the four-site state is the family closure

$$(\sigma_3^{(4)})_{i_1 i_2 i_3, j_1 j_2 j_3} = \sum_j \text{tr}(A_j^{i_1 j_1} A_j^{i_2 j_2} A_j^{i_3 j_3} \widehat{R}_j),$$

where

$$\begin{aligned} \widehat{R}_j &= \text{tr}(\rho^{(4)})^{-1} q_j \mathcal{B}_j, \\ q_j &= \sum_q \mu_{j,q}^4. \end{aligned}$$

Every $\widehat{R}_j$ is nonzero. More generally, a tail of length L gives $\widehat{R}_j = \text{tr}(\rho^{(L+3)})^{-1} q_j \mathcal{B}_j^L$. This is the normalized traced-tail expansion used in [11, Appendix C.2, lines 1714–1732].

Proof. ✓ Proof Equality of the positive-length matrix-product-vector families expands every matrix entry of the original operator as the sum of the BNT entries with coefficients q_j . Summing the diagonal physical configurations of the tail gives $\mathcal{B}_j^L$. The global normalization multiplies every closing matrix by $\text{tr}(\rho^{(L+3)})^{-1}$. At $L = 1$, SAL makes this scalar nonzero, copy independence makes q_j nonzero, and nonnilpotence makes $\mathcal{B}_j$ nonzero. $\square$

Theorem 20.16.1.15 (Collapse of the outer inverse contraction). Lean ✓ Stmt Let $x = (s, \alpha_1, \beta_1)$ and $y = (t, \alpha_3, \beta_3)$. For a three-site family closure and a simultaneous left inverse, the outer inverse contraction vanishes when $s \neq t$. When $s = t$, it is $(\mathcal{K}_s^{i_2 j_2})_{\beta_1, \alpha_3} (R_s)_{\beta_3, \alpha_1}$. In particular, the contraction retains precisely one entry of the middle physical slice and one entry of the closing matrix, with the orientation shown above. This is the sector-family form of the Case I contraction in [11, Appendix C.2, lines 1415–1438]. The basis-of-normal-tensors inverse is constructed at lines 1666–1676 and the resulting contraction is used in the Case II calculation at lines 1719–1732 of the same appendix. The closing entry $R_s(\beta_3, \alpha_1)$ is the corrected tail-index orientation of the display at lines 1422–1438, recorded in [123].

Proof. ✓ Proof Expand the family closure and interchange the finite sector and physical sums. For $x = (x_s, x_\alpha, x_\beta)$, the inverse identity is

$$\sum_{i,j} C_{x,(i,j)} \mathcal{K}_s^{ij} = \begin{cases} E_{x_\alpha, x_\beta}, & x_s = s, \\ 0, & x_s \neq s. \end{cases}$$

Hence distinct labels give zero. For equal labels, the remaining trace is

$$\text{tr}(E_{\alpha_1, \beta_1} \mathcal{K}_s^{i_2 j_2} E_{\alpha_3, \beta_3} R_s) = (\mathcal{K}_s^{i_2 j_2})_{\beta_1, \alpha_3} (R_s)_{\beta_3, \alpha_1}.$$

$\square$

Corollary 20.16.1.16 (Three-site collapse from the one-letter span). Lean ✓ Stmt Under the one-letter product-algebra spanning hypothesis, there is a simultaneous left inverse for which every three-site family closure has the preceding diagonal-sector contraction and vanishing off-diagonal contractions.

Proof. ✓ Proof Choose the simultaneous left inverse from Theorem 21.3.84 and apply Theorem 20.16.1.15. $\square$

20.16.2 Markov decomposition and inverse-map sector factorization

For the simple MPDO case in Appendix C.2 of [11], the hypothesis of saturation of the area law is used locally through the equality case of strong subadditivity for a normalized three-site reduced state.

Definition 20.16.2.1 (Local η -structure for a three-site reduced state). [Lean](#) [✓ Stmt](#) For a three-site density operator ρ_{ABC} , the local η -structure is the quantum Markov decomposition on the middle subsystem B produced by equality in strong subadditivity. Concretely, after a unitary change of basis on B , one has a finite direct-sum decomposition

$$B = \bigoplus_k (b_1^{(k)} \otimes b_2^{(k)}),$$

$$\rho_{ABC} = \bigoplus_k \rho_{Ab_1}^{(k)} \otimes \rho_{b_2C}^{(k)}.$$

Definition 20.16.2.2 (Canonical projections of a Hayashi decomposition). [Lean](#) [Lean](#) [✓ Stmt](#)

Let $B = \bigoplus_k (b_1^{(k)} \otimes b_2^{(k)})$ be the middle-site decomposition, and let E_k be the coordinate projection onto its k -th summand. In the original physical basis, define $Q_k = U_B^* E_k U_B$. These are the projectors introduced in [11, Appendix C.2, lines 1364–1368]; the unitary implements the basis change described in [11, Appendix C.2, lines 1437–1440].

Theorem 20.16.2.3 (Resolution of the identity by Hayashi-sector projections). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Each Q_k is an orthogonal projection, distinct summands are mutually orthogonal, and the family resolves the identity:

$$Q_k Q_\ell = 0 \quad \text{if } k \neq \ell,$$

$$\sum_k Q_k = \mathbb{1}.$$

The first two assertions follow from the orthogonal direct-sum decomposition in [11, Appendix C.2, lines 1364–1368]; the resolution of the identity is stated in [11, Appendix C.2, lines 1693–1697].

Proof. [✓ Proof](#) The coordinate projections satisfy

$$E_k^* = E_k,$$

$$E_k^2 = E_k,$$

$$\sum_k E_k = \mathbb{1}.$$

Reindexing the direct sum and conjugating by U_B preserve these identities. Hence each $Q_k = U_B^* E_k U_B$ is a star projection and therefore an orthogonal projection, while $\sum_k Q_k = U_B^* (\sum_k E_k) U_B = \mathbb{1}$. Finally, multiplying the resolution on both sides by Q_k gives

$$\sum_\ell Q_k Q_\ell Q_k = Q_k,$$

$$Q_k Q_\ell Q_k = (Q_\ell Q_k)^* (Q_\ell Q_k) \geq 0.$$

The $\ell = k$ term already equals Q_k , so every term with $\ell \neq k$ vanishes. Thus $(Q_\ell Q_k)^* (Q_\ell Q_k) = 0$, hence $Q_\ell Q_k = 0$. Taking adjoints gives $Q_k Q_\ell = 0$ for $\ell \neq k$. $\square$

Definition 20.16.2.4 (Markov factors of a simultaneous inverse). [Lean](#) [Lean](#) [✓ Stmt](#) Let C_x and C_y be two rows of a simultaneous left inverse, and choose a Hayashi decomposition of the three-site operator. In its k -th Markov sector, set

$$A_x^{(k)}(l, l') = \sum_{i_1, j_1} C_{x, (i_1, j_1)} \rho_{Ab_1; (i_1, l), (j_1, l')}^{(k)},$$

$$B_y^{(k)}(r, r') = \sum_{i_3, j_3} C_{y, (i_3, j_3)} \rho_{b_2 C; (r, i_3), (r', j_3)}^{(k)}.$$

These are the two factors obtained in the projected contraction of [11, Appendix C.2, lines 1719–1725].

Theorem 20.16.2.5 (Outer inverse contraction in Markov coordinates). [Lean](#) [✓ Stmt](#) Let $\Gamma_{x,y}$ denote the outer inverse contraction. In the basis of the chosen Hayashi decomposition,

$$[U_B \Gamma_{x,y} U_B^\dagger]_{(k,l,r), (k',l',r')} = \begin{cases} p_k A_x^{(k)}(l, l') B_y^{(k)}(r, r'), & k = k', \\ 0, & k \neq k'. \end{cases}$$

In particular, the probability p_k remains explicit. This identity requires neither a nonvanishing condition on p_k nor assumptions of saturated area law or zero correlation length.

Proof. [✓ Proof](#) Expand the two matrix products and the outer inverse contraction. The Hayashi block decomposition makes the middle-site contraction zero for $k \neq k'$. For $k = k'$, the remaining four outer sums separate into the product defining $A_x^{(k)}$ and $B_y^{(k)}$, with the common factor p_k . $\square$

Theorem 20.16.2.6 (Entrywise BNT–Markov identity). [Lean](#) [✓ Stmt](#) Let $x = (s, \alpha_1, \beta_1)$ and $y = (t, \alpha_3, \beta_3)$. Comparing the two expressions for the same outer inverse contraction gives

$$\begin{aligned} & \begin{cases} p_k A_x^{(k)}(l, l') B_y^{(k)}(r, r'), & k = k', \\ 0, & k \neq k' \end{cases} \\ &= \begin{cases} (R_s)_{\beta_3, \alpha_1} [U_B \kappa_{\beta_1, \alpha_3}^{(s)} U_B^\dagger]_{(k,l,r), (k',l',r')}, & s = t, \\ 0, & s \neq t. \end{cases} \end{aligned}$$

Here $\kappa_{\beta_1, \alpha_3}^{(s)}(i, j) = (\mathcal{K}_s^{ij})_{\beta_1, \alpha_3}$. This is the entrywise identity used in [11, Appendix C.2, lines 1719–1732], retaining both the normal-sector and Markov-sector off-diagonal entries.

The closing entry $(R_s)_{\beta_3, \alpha_1}$ is the corrected tail-index orientation of the display at lines 1422–1438, recorded in [123].

Proof. [✓ Proof](#) The preceding theorem identifies the left side with the transformed outer inverse contraction. The collapse theorem identifies the contraction with zero when $s \neq t$ and with $(R_s)_{\beta_3, \alpha_1} \kappa_{\beta_1, \alpha_3}^{(s)}$ when $s = t$. Conjugating this matrix by U_B gives the right side. $\square$

Theorem 20.16.2.7 (Off-diagonal Markov blocks of a BNT sector). [Lean](#) [✓ Stmt](#) Fix a normal sector s and virtual indices α_1, β_3 for which $(R_s)_{\beta_3, \alpha_1} \neq 0$. Then, for every β_1, α_3 and every pair of distinct Markov sectors $k \neq k'$, one has

$$[U_B \kappa_{\beta_1, \alpha_3}^{(s)} U_B^\dagger]_{(k,l,r), (k',l',r')} = 0.$$

Thus each physical slice of the s -th normal tensor is block diagonal in the Markov decomposition. This is equation Qks in [11, Appendix C.2, lines 1682–1688 and 1730–1735].

The nonzero closing entry uses the corrected orientation $(R_s)_{\beta_3, \alpha_1}$ recorded in [123].

Proof. [✓ Proof](#) In the entrywise BNT–Markov identity, take equal normal-sector labels and distinct Markov-sector labels. The left side vanishes, while the right side is the selected closing entry times the displayed block. Cancel the nonzero closing entry. $\square$

Definition 20.16.2.8 (Diagonal BNT block in a Markov sector). [Lean](#) [Lean](#) [✓ Stmt](#) For a normal sector s and a Markov sector k , let

$$O_s^{(k)}(\beta, \alpha) = [U_B \kappa_{\beta, \alpha}^{(s)} U_B^\dagger]_{k, k}.$$

The assertion $O_s^{(k)} \neq 0$ means that this block is nonzero for at least one pair of virtual indices β, α .

Theorem 20.16.2.9 (Existence of a normal-sector label). [Lean](#) [✓ Stmt](#) If $p_k \neq 0$, then there is a normal sector s for which $O_s^{(k)} \neq 0$.

This is the existence part of equation QkKjs in [11, Appendix C.2, lines 1733–1737]. Restricting to $p_k \neq 0$ is the literal version of the source’s preceding removal of summands for which $Q_k \sigma_3^{(N)} Q_k = 0$.

Proof. [✓ Proof](#) Choose nonzero diagonal entries of the trace-one left and right states in sector k . The corresponding diagonal entry of the Hayashi decomposition is nonzero because $p_k \neq 0$. Expanding the same entry by the three-site normal-sector family therefore leaves at least one nonzero summand, whose middle tensor has $O_s^{(k)} \neq 0$. $\square$

Theorem 20.16.2.10 (Uniqueness from nonzero closing matrices). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose, in addition, that every closing matrix R_s is nonzero. If $p_k \neq 0$, there is a unique normal sector j_k for which $O_{j_k}^{(k)} \neq 0$.

This is the cancellation argument in equation QkKjs. The stated nonzero-closing-matrix condition isolates the step obtained in the source from simplicity and the common-weight choice at [11, Appendix C.2, lines 1714–1718]. Its derivation from those preceding hypotheses is Theorem 20.16.1.14.

Proof. [✓ Proof](#) Assume that sectors $s \neq t$ both have nonzero k -blocks. A diagonal instance of the BNT–Markov identity for s gives a nonzero left factor. A mixed instance with normal-sector labels s, t forces the right factor selected from t to vanish. A diagonal instance for t then has zero left-hand side and a nonzero right-hand side, a contradiction. $\square$

Definition 20.16.2.11 (Normal-sector projections on the active support). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 20.16.2.10, define

$$P_s = \sum_{\substack{k: p_k \neq 0 \\ j_k = s}} Q_k,$$

$$P_{\text{act}} = \sum_{k: p_k \neq 0} Q_k.$$

This is equation Pis, restricted to the positive-weight Markov support retained here.

Theorem 20.16.2.12 (Orthogonality and resolution on the active support). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Every P_s is an orthogonal projection, distinct P_s are mutually orthogonal, and $\sum_s P_s = P_{\text{act}}$.

Proof. ✓ Proof Each P_s is a sum of mutually orthogonal Markov-sector projections. Distinct labels have disjoint sums. Finally, summing over s counts each positive-weight Markov sector exactly once. $\square$

Theorem 20.16.2.13 (SAL and arbitrary-cut equality in strong subadditivity). Lean Lean

Lean Lean Lean ✓ Stmt Let $1 \leq m \leq \lfloor N/2 \rfloor$, and divide the chain into four consecutive regions A, B, C, D of lengths $m-1, 1, N-m-1, 1$. If the tensor saturates the area law, then its marginal on ABC satisfies $S(ABC) + S(B) = S(AB) + S(BC)$. Indeed, this is the entropy form of $I_1 = I_m$. Taking $m = 2$ gives the regions of lengths $1, 1, N-3$, and then taking $N = 4$ shows that the three-site reduced state of the four-site chain satisfies equality in strong subadditivity.

Conversely, suppose the tensor generates positive semidefinite periodic operators and their traces are nonzero at every positive chain length. If the displayed equality holds for every N and every $1 \leq m \leq \lfloor N/2 \rfloor$, then the tensor saturates the area law. This is the choice of regions used in [11, Appendix C.2, lines 1760–1780]. The endpoint corrects the strict inequality printed at source line 1771: Definition 4.6 and the concluding comparison both require $m = \lfloor N/2 \rfloor$. This local correction is recorded in [123].

Proof. ✓ Proof By saturation, $I_1 = I_m$. Expanding the two mutual informations and cancelling the entropy of the full chain gives $S_{N-1} + S_1 = S_m + S_{N-m}$, which is the stated equality for the regions A, B, C . Conversely, this equality gives $I_1 = I_m$. If it holds for every admissible m , comparing the cases $m = L$ and $m = L + 1$ gives $I_L = I_{L+1}$ and hence saturation of the area law. $\square$

Theorem 20.16.2.14 (All-cut quantum Markov structure implies SAL). Lean ✓ Stmt Suppose that the periodic operators are positive semidefinite and have nonzero trace at every positive chain length. For every N and every $1 \leq m \leq \lfloor N/2 \rfloor$, divide the chain into four consecutive regions of lengths $m-1, 1, N-m-1, 1$. If the marginal on the first three regions admits a quantum Markov decomposition on its one-site middle system, then the tensor saturates the area law.

Proof. ✓ Proof The quantum Markov decomposition gives equality in strong subadditivity at every admissible cut: $S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$. These equalities at every admissible cut imply saturation of the area law by Theorem 20.16.2.13. $\square$

Theorem 20.16.2.15 (SAL implies local η -structure). Lean ✓ Stmt If a normalized three-site reduced state satisfies the equality case of strong subadditivity — the local entropy consequence of saturation of the area law for a simple MPDO — then it admits the local η -structure of Definition 20.16.2.1.

Proof. ✓ Proof This follows from the characterization of states saturating strong subadditivity [171]. Petz recovery with a product reference and the Koashi–Imoto decomposition yield the direct-sum decomposition into tensor products on the middle system, giving the local η -structure of Definition 20.16.2.1. $\square$

Lemma 20.16.2.16 (Partial trace along a right tensor factor). Lean Lean Lean Lean ✓ Stmt Suppose that $C \simeq C' \otimes E$. The partial trace over E is given entrywise by

$$(\mathrm{tr}_E X)_{(a,c'),(a',d')} = \sum_e X_{(a,c',e),(a',d',e)}.$$

It preserves positivity and trace.

Proof. ✓ Proof Reindex C as $C' \otimes E$ and apply the ordinary right partial trace. Its entry formula gives the displayed sum. Positivity follows by tracing a positive matrix, and invariance of the full trace under reindexing gives trace preservation. $\square$

Theorem 20.16.2.17 (Right marginals preserve a quantum Markov decomposition). [Lean](#) [Lean](#)
[✓ Stmt](#) Suppose that $C \simeq C' \otimes E$ and

$$\rho_{ABC} = \bigoplus_k p_k \rho_{Ab_1}^{(k)} \otimes \rho_{b_2C}^{(k)}$$

is a quantum Markov decomposition on B . Then the marginal $\rho_{ABC'} = \text{tr}_E(\rho_{ABC})$ has the decomposition

$$\rho_{ABC'} = \bigoplus_k p_k \rho_{Ab_1}^{(k)} \otimes \text{tr}_E(\rho_{b_2C}^{(k)}).$$

Proof. [✓ Proof](#) The middle-system decomposition, its unitary change of basis, and the probabilities p_k are unchanged. Partial trace preserves positivity and trace, so each $\text{tr}_E(\rho_{b_2C}^{(k)})$ is a density operator. Applying tr_E to the block-diagonal identity gives the displayed formula. $\square$

Lemma 20.16.2.18 (Tracing a terminal block). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $a, b, c, N \in \mathbb{N}$ satisfy $a + b + c \leq N$. For configurations u, v on the first $a + b$ sites,

$$\sum_y \sigma_{a+b+c}^{(N)}(K)_{u\|y, v\|y} = \sigma_{a+b}^{(N)}(K)_{u,v},$$

where the sum ranges over configurations on the last c sites. The same identity holds after replacing an arithmetically equal block length by its canonical identification.

Proof. [✓ Proof](#) Expand both reduced states as partial traces of the normalized periodic state. The left-hand side first traces the complement of the $(a + b + c)$ -site block and then its last c sites; combining the two sums gives the partial trace onto the first $a + b$ sites. $\square$

Theorem 20.16.2.19 (SAL three-site marginals have Hayashi decompositions). [Lean](#) [✓ Stmt](#)
Let K satisfy saturation of the area law and let $N \geq 4$. If $\sigma_3^{(N)}(K)$ is obtained from the normalized N -site periodic state by tracing out all but its first three sites, then $\sigma_3^{(N)}(K)$ admits a quantum Markov decomposition on its middle site. This is the three-site marginal result proved in [11, Appendix C.2, lines 1351–1371]. The decomposition may depend on N .

Proof. [✓ Proof](#) Divide the first $N - 1$ sites into consecutive regions of lengths 1, 1, and $N - 3$. Let ρ_{ABC}^{flat} denote this marginal in the corresponding index set $\{0, \dots, d^1 - 1\} \times \{0, \dots, d^1 - 1\} \times \{0, \dots, d^{N-3} - 1\}$. By Theorem 20.16.2.13,

$$S(\rho_{ABC}^{\text{flat}}) + S(\rho_B^{\text{flat}}) = S(\rho_{AB}^{\text{flat}}) + S(\rho_{BC}^{\text{flat}}).$$

The first two singleton blocks have the canonical identification $e_{\text{site}} : \{0, \dots, d - 1\} \simeq \{0, \dots, d^1 - 1\}$, and reindexing these factors preserves the displayed equality. Positivity and unit trace follow from the normalized reduced state, so the Hayashi characterization gives a quantum Markov decomposition of the $(N - 1)$ -site marginal. Finally split the third region as $C \simeq c \otimes E$, where c is its first site and E contains the remaining $N - 4$ sites. Theorem 20.16.2.17 gives the same middle-site direct sum after tracing out E . The contiguous-block reduction gives

$$\sigma_3^{(N)}(K) = \text{tr}_{4, \dots, N-1} \left[\sigma_{N-1}^{(N)}(K) \right],$$

which identifies the resulting operator with $\sigma_3^{(N)}(K)$. $\square$

Theorem 20.16.2.20 (Four-site SAL marginals have Hayashi decompositions). [Lean](#) [✓ Stmt](#) Let K satisfy saturation of the area law. If $\sigma_3^{(4)}(K)$ is obtained from its normalized four-site periodic state by tracing out the fourth site, then $\sigma_3^{(4)}(K)$ admits a quantum Markov decomposition on its middle site.

Proof. [✓ Proof](#) This is the case $N = 4$ of Theorem 20.16.2.19. □

Theorem 20.16.2.21 (BNT sector projectors for the four-site SAL marginal). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the common blocking has been performed, so that the BNT representatives have a simultaneous one-letter span. For the normalized three-site marginal $\sigma_3^{(4)}$, there are a simultaneous inverse C and a Hayashi decomposition such that every Markov sector k with $p_k \neq 0$ has a unique BNT label j_k . Consequently,

$$P_s = \sum_{\substack{k: p_k \neq 0 \\ j_k = s}} Q_k.$$

Each P_s is an orthogonal projection, $P_s P_t = 0$ for $s \neq t$, and $\sum_s P_s = P_{\text{act}}$. No independent nonzero-closing-matrix hypothesis is required. This is the four-site specialization of equations QkKjs and Pis in [11, Appendix C.2, lines 1714–1737]. The Hayashi decomposition uses the forward equality characterization of strong subadditivity recorded in the corresponding theorem in the companion quantum-channel volume [182, “Hayashi SSA-equality characterization”]; no additional analytic assumption is introduced here.

Proof. [✓ Proof](#) The canonical ordered-triple reindexing is the submatrix convention of the SAL-to-Hayashi theorem. Choose the simultaneous inverse from the one-letter span and the Hayashi decomposition from SAL. By Theorem 20.16.1.14, the same marginal is a BNT family closure whose closing matrices are all nonzero. The generic uniqueness and projector theorems now give the unique labels, orthogonality, mutual disjointness, and resolution of the active support. □

Definition 20.16.2.22 (Sitewise physical action and BNT coefficients). [Lean](#) [Lean](#) [✓ Stmt](#) For a one-site matrix V , write $V^{\otimes N}$ in the configuration basis as

$$(V^{\otimes N})_{s,t} = \prod_{n=1}^N V_{s_n, t_n}.$$

If the copies of the j -th normal tensor have common coefficient μ_j , set $\mathcal{K}_j = \mu_j A_j$.

Theorem 20.16.2.23 (Local selection by the BNT projectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 20.16.2.21, the resulting projectors satisfy $P_s A_i P_t = 0$ whenever $s \neq t$ or $i \neq s$. For the matching label, they satisfy $P_i A_i P_i = A_i$. Consequently, $\sum_s P_s A_i P_s = A_i$, which is the local diagonal decomposition required for the sector entropy argument. Equivalently, compressing the physical legs of A_i by P_s retains A_i for $s = i$ and gives the zero local tensor otherwise. This is the local projection relation in [11, Appendix C.2, lines 1680–1755].

The Markov summands of zero Hayashi weight are retained in the ambient decomposition. They annihilate every normal-sector physical slice, so the displayed local identities do not require the stronger assertion $\sum_s P_s = \mathbb{1}$ outside the active support.

Proof. [✓ Proof](#) Distinct Hayashi sectors give zero off-diagonal corners. On an active diagonal sector, uniqueness of the BNT label kills every normal tensor except the matching one. A zero-weight diagonal sector also kills every physical slice: otherwise the nonzero closing matrix would make the corresponding three-site Markov block nonzero. Summing these statements over the Hayashi sectors gives the asserted $P_s A_i P_t$ identities. □

Definition 20.16.2.24 (Completion on the zero-weight Markov sectors). [Lean](#) [Lean](#) [✓ Stmt](#)
 Fix a BNT label s_0 . Each positive-weight Markov sector retains its unique BNT label, while every zero-weight Markov sector is assigned the label s_0 . If $\widehat{S}_s$ is the set of Markov sectors with completed label s , define

$$\widehat{P}_s = \sum_{k \in \widehat{S}_s} Q_k.$$

Theorem 20.16.2.25 (Separating projectors with copy-independent weights). [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let K be simple, in biCF, and satisfy SAL. Assume in addition that the common blocking has been performed, the copy weights over each BNT representative agree, and the representatives have a simultaneous one-site span. There is a distinguished BNT label s_0 such that the completed projectors are orthogonal and, for distinct labels s and t ,

$$\begin{aligned} \widehat{P}_s \widehat{P}_t &= 0, \\ \sum_s \widehat{P}_s &= \mathbb{1}. \end{aligned}$$

For every physical slice of the i -th BNT representative, whenever $s \neq t$ or $i \neq s$,

$$\widehat{P}_s K_i^{\beta\alpha} \widehat{P}_t = 0.$$

This is the copy-independent-weight specialization of the separating-projector lemma in [11, Appendix C.2, lines 1680–1737].

Proof. [✓ Proof](#) A unit-modulus BNT weight supplies the label s_0 . The Markov-sector projections Q_k are mutually orthogonal and sum to the identity, so partitioning all labels by $\widehat{S}_s$ proves orthogonality and the identity resolution. If $p_k = 0$, then $Q_k K_i^{\beta\alpha} Q_k = 0$ for every i, β, α . If $p_k \neq 0$, then $Q_k K_i^{\beta\alpha} Q_k = 0$ whenever the BNT label of sector k is not i . For distinct Markov sectors, $Q_k K_i^{\beta\alpha} Q_l = 0$ whenever $k \neq l$. Hence, whenever $s \neq t$ or $i \neq s$,

$$\widehat{P}_s K_i^{\beta\alpha} \widehat{P}_t = \sum_{k \in \widehat{S}_s} \sum_{l \in \widehat{S}_t} Q_k K_i^{\beta\alpha} Q_l = 0,$$

since every summand vanishes by one of the three preceding cases. $\square$

Theorem 20.16.2.26 (Separating projectors for a simple tensor satisfying ZCL and SAL). [Lean](#) [✓ Stmt](#)
 Let K be a simple tensor in block-injective canonical form which satisfies ZCL and SAL. For the representatives $\mathcal{K}_i$ of its BNT, there are orthogonal projectors P_s such that

$$\begin{aligned} \sum_s P_s &= \mathbb{1}, \\ P_s K_i^{\beta\alpha} P_t &= 0 \quad \text{whenever } s \neq t \text{ or } i \neq s. \end{aligned}$$

This is the separating-projector lemma in [11, Appendix C.2, lines 1626–1691].

Proof. [✓ Proof](#) Write the horizontal canonical form as $K = G(\bigoplus_{j,q} \mu_{j,q} \mathcal{K}_j) G^{-1}$. Restricting the ZCL equation to the (j, q) -th block and using the non-nilpotence supplied by simplicity gives $\mu_{j,q} = \mu_{j,q'}$ for all j, q, q' . The block-diagonal gauge preserves every positive-length periodic state, so the canonical-form tensor and K generate the same such states. The preceding copy-independent-weight theorem now supplies the projectors. $\square$

Theorem 20.16.2.27 (Positive-length BNT-sector compression). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) For every nonempty periodic chain and every normal-sector label s ,

$$P_s^{\otimes N} \sigma^{(N)}(K) P_s^{\otimes N} = n_s \sigma^{(N)}(\mathcal{K}_s), \quad N \geq 1. \quad (20.47)$$

This is equation sigmaNKj in [11, Appendix C.2, lines 1756–1759].

Here and throughout the MPDO discussion, a periodic chain has positive length.

Proof. [Proof](#) Sitewise multiplication by P_s is the periodic MPO obtained by changing every local physical matrix by P_s . The preceding theorem replaces each normal representative by itself when its label is s and by the zero tensor otherwise. For $N \geq 1$, the latter has zero periodic MPO. The surviving coefficient is $n_s \mu_s^N$, which is precisely n_s times the MPO of the representative with μ_s absorbed locally. $\square$

Corollary 20.16.2.28 (Each absorbed BNT element generates MPDOs). [Lean](#) [Lean](#) [Stmt](#) Under the hypotheses of Theorem 20.16.2.27, every absorbed BNT element $\mathcal{K}_s$ generates positive semidefinite operators on all nonempty periodic chains. Indeed, $P_s^{\otimes N} \sigma^{(N)}(K) P_s^{\otimes N} \geq 0$ and $n_s > 0$, so (20.47) gives $\sigma^{(N)}(\mathcal{K}_s) \geq 0$. This is the first positivity conclusion in [11, Appendix C.2, lines 1753–1759].

Proof. [Proof](#) Congruence by the Hermitian projection $P_s^{\otimes N}$ preserves positive semidefiniteness. Multiplication by the inverse of the positive integer n_s then gives positivity of the sector operator. $\square$

Theorem 20.16.2.29 (Physical-trace transfer after common-weight absorption). [Lean](#) [Lean](#) [Stmt](#) Suppose the weights over each BNT representative are copy-independent. For $\mathcal{K}_s = \mu_s A_s$ and $\mathcal{B}_s = \mathcal{T}_{A_s}$, one has

$$\mathcal{T}_{\mathcal{K}_s} = \mu_s \mathcal{B}_s.$$

Proof. [Proof](#) Substitute $\mathcal{K}_s = \mu_s A_s$ into the defining diagonal sum for the physical-trace transfer and factor out the scalar μ_s . $\square$

Theorem 20.16.2.30 (Literal ZCL descends to absorbed BNT representatives). [Lean](#) [Stmt](#) Let K be gauge equivalent to the weighted direct sum of BNT representatives A_s , with copy weights $\mu_{s,q}$. If its physical-trace transfer is literally idempotent, then every weighted copy satisfies

$$\mathcal{T}_{\mu_{s,q} A_s}^2 = \mathcal{T}_{\mu_{s,q} A_s}.$$

If the copy weights over A_s agree and $\mathcal{K}_s = \mu_s A_s$, then

$$\mathcal{T}_{\mathcal{K}_s}^2 = \mathcal{T}_{\mathcal{K}_s}.$$

Proof. [Proof](#) Conjugating $\mathcal{T}_K^2 = \mathcal{T}_K$ by the block-diagonal gauge gives the same identity for the weighted direct sum. Its diagonal block indexed by (s, q) is $\mathcal{T}_{\mu_{s,q} A_s}^2 = \mathcal{T}_{\mu_{s,q} A_s}$. When the weights are copy-independent, $\mathcal{T}_{\mathcal{K}_s} = \mu_s \mathcal{T}_{A_s}$ is that block for any copy over s . $\square$

Theorem 20.16.2.31 (Scale-invariant source ZCL descends to absorbed BNT representatives). [Lean](#) [Lean](#) [Stmt](#) Let K be a simple tensor in its chosen canonical form and suppose that K has scale-invariant source zero correlation length. Then the copy weights over each normal representative A_s agree. For every A_s with non-nilpotent physical-trace transfer $\mathcal{B}_s$, absorb the

common weight and write $\mathcal{K}_s = \mu_s A_s$. Each absorbed BNT representative $\mathcal{K}_s$ has scale-invariant source zero correlation length. More precisely, $\mathcal{T}_{\mathcal{K}_s} = \mu_s \mathcal{B}_s \neq 0$ and, for some $\lambda > 0$,

$$\mathcal{T}_{\mathcal{K}_s}^2 = \lambda \mathcal{T}_{\mathcal{K}_s}.$$

This is the final zero-correlation-length conclusion in [11, Appendix C.2, line 1781].

Proof. ✓ Proof Corollary 20.16.1.6 gives the common weight on every copy. Lemma 20.16.1.4 then gives the displayed quadratic identity. The common weight is nonzero, and nonnilpotence of $\mathcal{B}_s$ implies $\mathcal{B}_s \neq 0$, so the absorbed transfer is nonzero. $\square$

Remark 20.16.2.32 (Sectorwise hypotheses in the canonical BNT application). In the canonical absorbed BNT presentation, ambient source zero correlation length supplies source zero correlation length for every absorbed representative by Theorem 20.16.2.31. The standing biCF one-letter span and the nonzero common weights give one-site injectivity by Theorem 20.14.6.

The positive commuting-bond realization begins at length two. It therefore proves positivity of $\rho^{(N)}(K_x)$ for all $N \geq 2$, while positivity of $\rho^{(1)}(K_x)$ remains the only separate sector-positivity hypothesis. Theorem 20.14.31 combines these facts and proves SAL at every original cut, without a normality hypothesis on the absorbed sectors. The printed GSNCH-only proposition remains incomplete because it omits ambient source zero correlation length; an arbitrary GSNCH presentation need not inherit this tensor-level property.

Theorem 20.16.2.33 (Strict normalization at positive length). Lean Lean Lean ✓ Stmt Let M generate matrix product density operators and have source zero correlation length. Then, for every $N \geq 1$, $\text{tr}(\sigma^{(N)}(M)) > 0$. In particular, after a common blocking has made the simultaneous BNT word span full at one letter, the absorbed BNT elements selected in Theorem 20.16.2.31 have a well-defined normalized state at every physical chain length. This is the biCF hypothesis imposed at the start of Case II in the source; its relation with finite physical blocking is recorded in [130]. This supplies the strict form of the sector normalization used in [11, Appendix C.2, lines 1760–1780].

Proof. ✓ Proof Write the physical-trace transfer as $\mathcal{T}_M = \lambda E$, where $\lambda > 0$ and E is a nonzero idempotent. A nonzero idempotent on a finite-dimensional complex vector space has nonzero trace, while $E^N = E$ for $N \geq 1$. Hence

$$\text{tr}(\sigma^{(N)}(M)) = \text{tr}(\mathcal{T}_M^N) = \lambda^N \text{tr}(E) \neq 0.$$

Positivity of the density operator then makes this trace strictly positive. $\square$

Theorem 20.16.2.34 (Saturated area law for the absorbed BNT elements). Lean Lean Lean Lean Lean ✓ Stmt Let K saturate the area law and have zero correlation length, and let its horizontal BNT canonical form, up to block-diagonal gauge, have normal representatives A_s , common copy weights μ_s , and copy numbers n_s . Assume that the physical-trace transfer of every A_s is nonnilpotent. After a common blocking, assume explicitly that the simultaneous one-letter word tuples of the A_s span the direct sum of their full matrix algebras. Put $\mathcal{K}_s = \mu_s A_s$.

For every nonempty chain, define

$$p_s^{(N)} = \frac{n_s \text{tr}(\sigma^{(N)}(\mathcal{K}_s))}{\text{tr}(\sigma^{(N)}(K))}.$$

Then $p_s^{(N)} > 0$, and every nonempty prefix marginal has the same sector decomposition with these probabilities. Translation invariance identifies the other contiguous blocks of the same length. Consequently, every absorbed representative $\mathcal{K}_s$ saturates the area law.

The one-letter simultaneous span is the block-injective canonical-form hypothesis imposed at the start of Case II in [11, line 1628]. Its relation with finite physical blocking is recorded in [130]. The entropy conclusion is the argument of [11, Appendix C.2, lines 1748–1781].

Proof. [✓ Proof](#) The BNT projections carry every normalized marginal on mutually annihilating supports, so the preceding support form of entropy additivity gives

$$S_L(K) = H(p^{(N)}) + \sum_s p_s^{(N)} S_L(\mathcal{K}_s), \quad 1 \leq L \leq N.$$

The four marginals entering strong subadditivity have the same probabilities. Substitution in the equality for K cancels all four Shannon terms. Strong subadditivity gives the corresponding inequality in each sector, and strict positivity of every $p_s^{(N)}$ forces every such inequality to be an equality. Comparing the equalities for block lengths L and $L + 1$ proves saturation for $\mathcal{K}_s$. $\square$

Theorem 20.16.2.35 (Analytic properties of absorbed BNT representatives). [Lean](#) [✓ Stmt](#)
Let K be an MPDO tensor in biCF, written up to a block-diagonal gauge as a weighted BNT direct sum whose BNT representatives have nonnilpotent physical-trace transfers. Suppose that the weights over each BNT label are copy-independent, $\mu_{s,q} = \mu_{s,q'}$, and that K saturates the area law and satisfies literal zero correlation length, $\mathcal{T}_K^2 = \mathcal{T}_K$. For every BNT label s , set $\mathcal{K}_s = \mu_s A_s$. Then $\mathcal{K}_s$ is one-site injective, generates positive semidefinite operators on every nonempty periodic chain, saturates the area law, and satisfies

$$\mathcal{T}_{\mathcal{K}_s}^2 = \mathcal{T}_{\mathcal{K}_s}.$$

Proof. [✓ Proof](#) The block-diagonal gauge identifies the MPV families of K and its weighted BNT direct sum. Saturation gives $\text{tr}(\sigma^{(1)}(K)) > 0$. Since $\text{tr}(\sigma^{(1)}(K)) = \text{tr}(\mathcal{T}_K)$, one has $\mathcal{T}_K \neq 0$. Together with $\mathcal{T}_K^2 = \mathcal{T}_K$, this gives scale-invariant source zero correlation length with scalar 1. The one-letter BNT span gives injectivity, the sector compression gives positivity, and the sector entropy decomposition gives saturation of the area law. Theorem 20.16.2.30 supplies the final literal idempotence identity. $\square$

Definition 20.16.2.36 (Injective inverse map for a simple MPO tensor). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) A simple MPO tensor K is called injective if its doubled-index MPS tensor is injective. For such a tensor, the chosen decomposition map of the doubled-index MPS furnishes a concrete inverse tensor K^{-1} together with right- and left-physical realization maps that turn virtual bond insertions back into local physical operators.

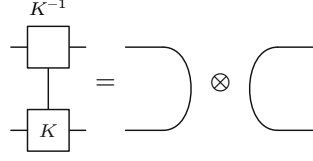
Theorem 20.16.2.37 (Inverse-map identities for an injective simple MPO tensor). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) For an injective simple MPO tensor, contracting K^{-1} with K recovers the matrix units on the virtual bond space. Likewise, every right virtual insertion admits a physical realization, this realization is multiplicative, and the analogous left-insertion statement also holds.

For an injective simple tensor, contracting K^{-1} with K through the physical index identifies the corresponding virtual legs. In components, this is the matrix-unit identity

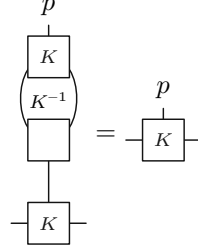
$$\sum_p (K^{-1})_p^{\alpha,\beta} (K^p)_{\alpha',\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}. \quad (20.48)$$

Its two Kronecker factors are the two bare cups:



This is the identity in [11, Appendix C.2, lines 1376–1380].

Applying this inverse map to the doubled physical index and closing its virtual legs against a further copy of K recovers K :



This is the single-block identity in [11, Appendix C.2, lines 1671–1676].

Definition 20.16.2.38 (Normalized fourth-site virtual tail). [Lean](#) [✓ Stmt](#) Let $\mathcal{K}$ be an MPO tensor and set $Z_4 = \text{tr}[\rho^{(4)}(\mathcal{K})]$. Its normalized fourth-site virtual tail is

$$R_4 := Z_4^{-1} \mathcal{J}_{\mathcal{K}} = \text{tr}[\rho^{(4)}(\mathcal{K})]^{-1} \sum_{i_4=0}^{d-1} \mathcal{K}^{i_4 i_4}. \quad (20.49)$$

Thus the contraction of the fourth ket and bra indices is incorporated into the virtual closing matrix. This is the $N = 4$ specialization of the normalization convention and three-site marginal in [11, lines 792–793 and 1343–1348].

Theorem 20.16.2.39 (Three-site marginal closed by the normalized fourth-site tail). [Lean](#) [✓ Stmt](#) Let $u = (i_1, i_2, i_3)$ and $v = (j_1, j_2, j_3)$ be physical words of length three. The corresponding entry of the first-three-site marginal of the normalized four-site periodic operator is

$$\sigma_3^{(4)}(\mathcal{K})_{u,v} = \text{tr}(\mathcal{K}^{u,v} R_4) = \text{tr}(\mathcal{K}^{i_1 j_1} \mathcal{K}^{i_2 j_2} \mathcal{K}^{i_3 j_3} R_4). \quad (20.50)$$

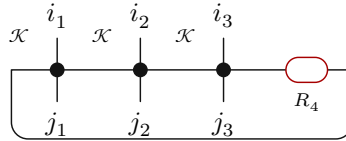


Figure 20.1: Tracing the fourth site of the normalized four-site periodic MPO leaves the physical legs of sites 1, 2, 3 open and replaces the fourth site by the virtual closing operator $R_4 = \text{tr}[\rho^{(4)}(\mathcal{K})]^{-1} \sum_{i_4} \mathcal{K}^{i_4 i_4}$; compare the normalization convention and three-site marginal in [11, lines 792–793 and 1343–1348].

Proof. ✓ Proof Expanding the partial trace over the fourth site and factoring each length-four word at its last letter gives

$$\sigma_3^{(4)}(\mathcal{K})_{u,v} = Z_4^{-1} \sum_{i_4=0}^{d-1} \text{tr}(\mathcal{K}^{u,v} \mathcal{K}^{i_4 i_4}).$$

Linearity of the trace and (20.49) now yield

$$Z_4^{-1} \sum_{i_4=0}^{d-1} \text{tr}(\mathcal{K}^{u,v} \mathcal{K}^{i_4 i_4}) = \text{tr} \left(\mathcal{K}^{u,v} \left(Z_4^{-1} \sum_{i_4=0}^{d-1} \mathcal{K}^{i_4 i_4} \right) \right) = \text{tr}(\mathcal{K}^{u,v} R_4).$$

□

Theorem 20.16.2.40 (Non-vanishing of the normalized fourth-site tail). Lean ✓ Stmt If $Z_4 = \text{tr}[\rho^{(4)}(\mathcal{K})] \neq 0$, then $R_4 \neq 0$.

Proof. ✓ Proof If $R_4 = 0$, then the tail formula makes every entry of $\sigma_3^{(4)}(\mathcal{K})$ vanish, so $\text{tr}[\sigma_3^{(4)}(\mathcal{K})] = 0$. On the other hand, $Z_4 \neq 0$ implies $\text{tr}[\sigma_3^{(4)}(\mathcal{K})] = 1$, a contradiction. □

Theorem 20.16.2.41 (A nonzero entry of the normalized fourth-site tail). Lean ✓ Stmt If $Z_4 \neq 0$, there exist virtual indices β and α such that $(R_4)_{\beta,\alpha} \neq 0$. These are the indices selected before the sector factorization in [11, Appendix C.2, lines 1431–1437].

Proof. ✓ Proof If every matrix entry of R_4 vanished, then R_4 would be the zero matrix, contrary to the preceding theorem. □

Definition 20.16.2.42 (Three-site inverse-map contraction). Lean ✓ Stmt Let $\mathcal{K}$ be an injective simple MPO tensor, and let $R \in M_D(\mathbb{C})$ be the virtual contraction of the sites following a three-site word. For virtual indices $\alpha_1, \beta_1, \alpha_3, \beta_3$, define

$$C_{\alpha_1, \beta_1, \alpha_3, \beta_3}(p_2; R) := \sum_{p_1, p_3} (\mathcal{K}^{-1})_{p_1}^{\alpha_1, \beta_1} (\mathcal{K}^{-1})_{p_3}^{\alpha_3, \beta_3} \text{tr}(\mathcal{K}^{p_1} \mathcal{K}^{p_2} \mathcal{K}^{p_3} R). \quad (20.51)$$

This is the contraction used in the proof of [11, Appendix C.2, lines 1415–1438].

Theorem 20.16.2.43 (Evaluation of the three-site inverse-map contraction). Lean ✓ Stmt For every middle physical index p_2 ,

$$C_{\alpha_1, \beta_1, \alpha_3, \beta_3}(p_2; R) = \mathcal{K}_{\beta_1, \alpha_3}^{p_2} R_{\beta_3, \alpha_1}. \quad (20.52)$$

Proof. ✓ Proof Move the two finite sums through the trace. By the inverse-map identity

$$\sum_p (\mathcal{K}^{-1})_p^{\alpha, \beta} \mathcal{K}^p = E_{\alpha, \beta},$$

which is the matrix form of (20.48), the double sum collapses to

$$\text{tr}(E_{\alpha_1, \beta_1} \mathcal{K}^{p_2} E_{\alpha_3, \beta_3} R) = \mathcal{K}_{\beta_1, \alpha_3}^{p_2} R_{\beta_3, \alpha_1}.$$

□

$$\left(\alpha_1 \text{---} \overset{x^{-1}}{\bullet} \beta_1 \otimes \mathbb{1} \otimes \alpha_3 \text{---} \overset{x^{-1}}{\bullet} \beta_3 \right) \begin{array}{c} p_1 \quad p_2 \quad p_3 \\ \downarrow \quad \downarrow \quad \downarrow \\ \text{---} \bullet \text{---} \bullet \text{---} \bullet \text{---} \textcircled{R} \text{---} \end{array} = \beta_1 \text{---} \overset{p_2}{\bullet} \alpha_3 \otimes \begin{array}{c} \beta_3 \text{---} \textcircled{R} \text{---} \alpha_1 \\ x \quad x \quad x \end{array}$$

Figure 20.2: Applying $\mathcal{K}^{-1}$ at the first and third sites collapses the three-site contraction to $\mathcal{K}_{\beta_1, \alpha_3}^{p_2} R_{\beta_3, \alpha_1}$, where $p_2 = (i_2, j_2)$ is the doubled physical index; compare [11, Appendix C.2, lines 1415–1438].

Definition 20.16.2.44 (Three-site closure against a virtual tail). [Lean](#) [StmT](#) Let $\mathcal{K}$ be an MPO tensor and let $R \in M_D(\mathbb{C})$. A tripartite operator ρ is the three-site closure of $\mathcal{K}$ against R if

$$\rho_{i_1 i_2 i_3, j_1 j_2 j_3} = \text{tr}(\mathcal{K}^{i_1 j_1} \mathcal{K}^{i_2 j_2} \mathcal{K}^{i_3 j_3} R). \quad (20.53)$$

This is the algebraic form of the three-site reduced operator in [11, Appendix C.2, lines 1422–1433].

Definition 20.16.2.45 (Physical slice and Hayashi inverse factors). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 Fix virtual indices β, α and set $\kappa_{\beta, \alpha}(i, j) = \mathcal{K}_{\beta, \alpha}^{ij}$. For a Hayashi sector k , define the two inverse factors

$$\begin{aligned} A_{\alpha_1, \beta_1}^{(k)}(l, l') &:= \sum_{i_1, j_1} (\mathcal{K}^{-1})_{(i_1, j_1)}^{\alpha_1, \beta_1} (\rho_{Ab_1}^{(k)})_{(i_1, l), (j_1, l')}, \\ B_{\alpha_3, \beta_3}^{(k)}(r, r') &:= \sum_{i_3, j_3} (\mathcal{K}^{-1})_{(i_3, j_3)}^{\alpha_3, \beta_3} (\rho_{b_2C}^{(k)})_{(r, i_3), (r', j_3)}. \end{aligned}$$

These are the two factors in the sectorwise inverse-map calculation of [11, Appendix C.2, lines 1407–1428].

Theorem 20.16.2.46 (Outer inverse-map contraction of a closure). [Lean](#) [Lean](#) [✓ Stmt](#) *Let $\mathcal{K}$ be injective and let ρ be its three-site closure against R . Applying the inverse tensor at the two outer sites collapses the closure to one entry of the middle physical slice and one entry of the tail:*

$$\begin{aligned} & \sum_{i_1, j_1, i_3, j_3} (\mathcal{K}^{-1})_{(i_1, j_1)}^{\alpha_1, \beta_1} (\mathcal{K}^{-1})_{(i_3, j_3)}^{\alpha_3, \beta_3} \rho_{i_1 i_2 i_3, j_1 j_2 j_3} \\ &= \mathcal{K}_{\beta_1, \alpha_3}^{i_2 j_2} R_{\beta_3, \alpha_1}. \end{aligned} \quad (20.54)$$

Consequently, for any matrix U on the middle site,

$$R_{\beta_3, \alpha_1}(U \kappa_{\beta_1, \alpha_3} U^\dagger)_{b, b'} = \sum_{i_1, j_1, i_3, j_3} (\mathcal{K}^{-1})_{(i_1, j_1)}^{\alpha_1, \beta_1} (\mathcal{K}^{-1})_{(i_3, j_3)}^{\alpha_3, \beta_3} \sum_{i_2, j_2} U_{b, i_2} \rho_{i_1 i_2 i_3, j_1 j_2 j_3} \overline{U_{b', j_2}}. \quad (20.55)$$

Proof. Substituting the three-site closure relation (20.53) converts the left-hand side of (20.54) to $C_{\alpha_1, \beta_1, \alpha_3, \beta_3}((i_2, j_2); R)$. Theorem 20.16.2.43 then gives

$$C_{\alpha_1, \beta_1, \alpha_3, \beta_3}((i_2, j_2); R) = \mathcal{K}_{\beta_1, \alpha_3}^{i_2 j_2} R_{\beta_3, \alpha_1},$$

which is (20.54). Identity (20.55) follows by expanding the matrix product entrywise,

$$(U\kappa_{\beta_1,\alpha_3}U^\dagger)_{b,b'} = \sum_{i_2,j_2} U_{b,i_2}\kappa_{\beta_1,\alpha_3}(i_2,j_2)\overline{U_{b',j_2}},$$

multiplying by R_{β_3,α_1} , substituting

$$R_{\beta_3,\alpha_1}\kappa_{\beta_1,\alpha_3}(i_2,j_2) = \sum_{i_1,j_1,i_3,j_3} (\mathcal{K}^{-1})_{(i_1,j_1)}^{\alpha_1,\beta_1} (\mathcal{K}^{-1})_{(i_3,j_3)}^{\alpha_3,\beta_3} \rho_{i_1i_2i_3,j_1j_2j_3},$$

and reordering the finite sums. $\square$

Theorem 20.16.2.47 (Sectorwise inverse-map comparison). Lean ✓ Stmt Suppose that $\mathcal{K}$ is injective, ρ is its three-site closure against R , and a Hayashi decomposition of ρ has been chosen. Write

$$\tilde{\kappa}_{\beta_1,\alpha_3}^{(k)}((l,r),(l',r')) := [U_B\kappa_{\beta_1,\alpha_3}U_B^\dagger]_{(k,l,r),(k,l',r')}.$$

Then every diagonal sector satisfies

$$R_{\beta_3,\alpha_1}\tilde{\kappa}_{\beta_1,\alpha_3}^{(k)}((l,r),(l',r')) = p_k A_{\alpha_1,\beta_1}^{(k)}(l,l') B_{\alpha_3,\beta_3}^{(k)}(r,r'). \quad (20.56)$$

The probability p_k remains explicit because both sector density operators are normalized. No zero-correlation-length or non-vanishing hypothesis is used in this comparison.

For fixed k, α_1, β_3 , the identity (20.56) reads coefficientwise as follows.

$$R_{\beta_3,\alpha_1} \begin{array}{c} (l,r) \\ \text{---} \square \text{---} \\ (l',r') \end{array} \alpha_3 = p_k A_{\alpha_1,\beta_1}^{(k)}(l,l') B_{\alpha_3,\beta_3}^{(k)}(r,r').$$

Proof. ✓ Proof At fixed indices (k,l,r) and (k,l',r') , the Hayashi block equality is

$$\begin{aligned} & \sum_{i_2,j_2} [U_B]_{(k,l,r),i_2} \rho_{i_1i_2i_3,j_1j_2j_3} \overline{[U_B]_{(k,l',r'),j_2}} \\ &= p_k (\rho_{Ab_1}^{(k)})_{(i_1,l),(j_1,l')} (\rho_{b_2C}^{(k)})_{(r,i_3),(r',j_3)}. \end{aligned}$$

For each i_2, j_2 , the preceding theorem gives

$$\begin{aligned} & \sum_{i_1,j_1,i_3,j_3} (\mathcal{K}^{-1})_{(i_1,j_1)}^{\alpha_1,\beta_1} (\mathcal{K}^{-1})_{(i_3,j_3)}^{\alpha_3,\beta_3} \rho_{i_1i_2i_3,j_1j_2j_3} \\ &= \mathcal{K}_{\beta_1,\alpha_3}^{i_2j_2} R_{\beta_3,\alpha_1}. \end{aligned} \quad (20.57)$$

Apply

$$\sum_{i_1,j_1,i_3,j_3} (\mathcal{K}^{-1})_{(i_1,j_1)}^{\alpha_1,\beta_1} (\mathcal{K}^{-1})_{(i_3,j_3)}^{\alpha_3,\beta_3} (-)$$

to the Hayashi block equality. Substituting (20.57) and interchanging the finite sums gives

$$\begin{aligned} & \sum_{i_1, j_1, i_3, j_3} (\mathcal{K}^{-1})_{(i_1, j_1)}^{\alpha_1, \beta_1} (\mathcal{K}^{-1})_{(i_3, j_3)}^{\alpha_3, \beta_3} \\ & \times \left(\sum_{i_2, j_2} [U_B]_{(k, l, r), i_2} \rho_{i_1 i_2 i_3, j_1 j_2 j_3} \overline{[U_B]_{(k, l', r'), j_2}} \right) \\ & = \sum_{i_2, j_2} [U_B]_{(k, l, r), i_2} \mathcal{K}_{\beta_1, \alpha_3}^{i_2 j_2} R_{\beta_3, \alpha_1} \overline{[U_B]_{(k, l', r'), j_2}}. \end{aligned}$$

The right-hand side is

$$\begin{aligned} & R_{\beta_3, \alpha_1} \sum_{i_2, j_2} [U_B]_{(k, l, r), i_2} \kappa_{\beta_1, \alpha_3}(i_2, j_2) \overline{[U_B]_{(k, l', r'), j_2}} \\ & = R_{\beta_3, \alpha_1} \tilde{\kappa}_{\beta_1, \alpha_3}^{(k)}((l, r), (l', r')). \end{aligned}$$

The contraction of the right-hand side of the Hayashi block equality is $p_k A_{\alpha_1, \beta_1}^{(k)}(l, l') B_{\alpha_3, \beta_3}^{(k)}(r, r')$. Since the Hayashi block equality holds, equating the two contractions gives (20.56). $\square$

Theorem 20.16.2.48 (The four-site marginal is a three-site closure). [Lean](#) [✓ Stmt](#) *Carried onto the tripartite site index, the three-site marginal of the normalized four-site chain is the three-site closure of $\mathcal{K}$ against the normalized fourth-site tail R_4 .*

Proof. [✓ Proof](#) The entry formula (20.50) for $\sigma_3^{(4)}(\mathcal{K})$, carried onto the tripartite site index (i_1, i_2, i_3) , gives

$$\rho_{i_1 i_2 i_3, j_1 j_2 j_3} = \text{tr}(\mathcal{K}^{i_1 j_1} \mathcal{K}^{i_2 j_2} \mathcal{K}^{i_3 j_3} R_4),$$

which is the three-site closure condition (20.53) against R_4 . $\square$

Theorem 20.16.2.49 (Vanishing of the off-diagonal Hayashi sectors). [Lean](#) [✓ Stmt](#) *In the setting of Theorem 20.16.2.47, for all $k \neq k'$, every off-diagonal sector satisfies*

$$R_{\beta_3, \alpha_1} [U_B \kappa_{\beta_1, \alpha_3} U_B^\dagger]_{(k, l, r), (k', l', r')} = 0. \quad (20.58)$$

Together with the diagonal comparison this gives the direct-sum structure of the factorized tensor: the direct sum over sectors is inherited from the splitting of the middle site.

Proof. [✓ Proof](#) For b in sector k and b' in sector k' with $k \neq k'$, the Hayashi block equality at off-diagonal indices gives

$$\sum_{i_2, j_2} (U_B)_{b, i_2} \rho_{i_1 i_2 i_3, j_1 j_2 j_3} \overline{(U_B)_{b', j_2}} = 0.$$

Substituting this vanishing into the expansion of Theorem 20.16.2.46 gives (20.58). $\square$

Definition 20.16.2.50 (Sector tensors). [Lean](#) [Lean](#) [✓ Stmt](#) Fix outer virtual indices α_1, β_3 and a Hayashi sector k . The sector tensors are

$$\begin{aligned} (l_k)_{\beta_1} &:= R_{\beta_3, \alpha_1}^{-1} p_k A_{\alpha_1, \beta_1}^{(k)}, \\ (r_k)_{\alpha_3} &:= B_{\alpha_3, \beta_3}^{(k)}. \end{aligned}$$

These are the tensors of the factorization displayed in [11, Appendix C.2, lines 1435–1448].

Lemma 20.16.2.51 (Zero-weight left sector tensor). [Lean](#) [Lean](#) [✓ Stmt](#) If $p_k = 0$, then $(l_k)_\beta = 0$ for every virtual index β . Consequently, $\eta_{h,k} = 0$ for every sector h .

Proof. [✓ Proof](#) The scalar weight p_k is a factor of every matrix entry of $(l_k)_\beta$. Each neighboring operator entering sector k contains $(l_k)_\beta$ as its target factor. $\square$

Theorem 20.16.2.52 (Sector factorization of an injective simple tensor). [Lean](#) [✓ Stmt](#) Suppose that $\mathcal{K}$ is injective, ρ is its three-site closure against R , a Hayashi decomposition of ρ has been chosen, and the outer indices satisfy $R_{\beta_3, \alpha_1} \neq 0$. Then for all virtual indices β_1, α_3 ,

$$U_B \kappa_{\beta_1, \alpha_3} U_B^\dagger = \bigoplus_k (l_k)_{\beta_1} \otimes (r_k)_{\alpha_3}. \quad (20.59)$$

The direct sum refers to the physical indices and is inherited from the splitting of the middle site. This is the factorization displayed at [11, Appendix C.2, lines 1435–1448].

In the middle-site basis selected by U_B , the same factorization (20.59) is displayed as

Proof. [✓ Proof](#) On a diagonal sector, the sectorwise comparison gives $R_{\beta_3, \alpha_1} \tilde{\kappa}^{(k)} = p_k A^{(k)} B^{(k)}$, and dividing by the nonzero entry R_{β_3, α_1} produces $(l_k)_{\beta_1} \otimes (r_k)_{\alpha_3}$. On an off-diagonal sector, the product with R_{β_3, α_1} vanishes, and since $R_{\beta_3, \alpha_1} \neq 0$ the sector itself vanishes. $\square$

Theorem 20.16.2.53 (Existence of the sector tensors). [Lean](#) [✓ Stmt](#) Let $\mathcal{K}$ be injective with nonzero four-site trace, let ρ be a three-site closure of $\mathcal{K}$ against the normalized fourth-site tail R_4 , and choose a Hayashi decomposition of ρ . Then there exist sector tensors l_k and r_k such that, for all virtual indices β_1, α_3 ,

$$U_B \kappa_{\beta_1, \alpha_3} U_B^\dagger = \bigoplus_k (l_k)_{\beta_1} \otimes (r_k)_{\alpha_3}.$$

The neighboring operators $\eta_{k,h}$ are formed from these tensors in Definition 20.16.3.1; the remaining steps of [11, Appendix C.2, Lemma C.4] prove the all-length identity and primitivity.

Proof. [✓ Proof](#) A nonzero four-site trace supplies outer indices with $(R_4)_{\beta_3, \alpha_1} \neq 0$, and the sector factorization applies. $\square$

20.16.3 Neighboring operators and commuting bond products

Definition 20.16.3.1 (Neighboring operators). [Lean](#) [✓ Stmt](#) Fix outer virtual indices α_1, β_3 with the chosen Hayashi decomposition, and let l_k, r_k be the sector tensors of Definition 20.16.2.50. For each pair of sectors (k, h) , the neighboring operator contracts the right sector tensor of one site with the left sector tensor of the following site over the virtual bond they share:

$$\eta_{k,h} = \sum_\gamma (r_k)_\gamma \otimes (l_h)_\gamma.$$

Definition 20.16.3.6 (Virtual-matrix transport of a physical-sector factorization). [Lean](#) [✓ Stmt](#) Let K and L have virtual dimensions D and E , respectively. Suppose that matrices $X \in M_{E \times D}(\mathbb{C})$ and $Y \in M_{D \times E}(\mathbb{C})$ satisfy the following identity for every physical index i :

$$L^i = XK^iY.$$

Every physical-sector factorization of K induces one of L with the same physical decomposition. Its virtual tensor families are

$$(l_k^L)_\beta = \sum_\gamma X_{\beta,\gamma} (l_k^K)_\gamma, \quad (r_k^L)_\alpha = \sum_\delta Y_{\delta,\alpha} (r_k^K)_\delta.$$

This construction abstracts the virtual corner in [11, Section 2.3, lines 195–219] and the physical-sector factorization in [11, Appendix C.2, lines 1381–1439].

Theorem 20.16.3.7 (Neighboring operators under virtual-matrix transport). [Lean](#) [✓ Stmt](#) *The neighboring operators of the transported factorization satisfy*

$$\eta_{k,h}^L = \eta_{k,h}^K(YX).$$

This identity combines the virtual corner in [11, Section 2.3, lines 195–219] with the neighboring contraction in [11, Appendix C.2, lines 1441–1450].

Proof. [✓ Proof](#) Replace each left factor by $\sum_\gamma X_{\beta,\gamma} (l_k)_\gamma$ and each right factor by $\sum_\delta Y_{\delta,\alpha} (r_k)_\delta$. Contracting the common virtual index gives

$$\sum_\beta Y_{\delta,\beta} X_{\beta,\gamma} = (YX)_{\delta,\gamma},$$

which proves the neighboring-operator identity. $\square$

Definition 20.16.3.8 (Virtual compression of a physical-sector factorization). [Lean](#) [✓ Stmt](#) Let K and L have virtual dimensions D and E , respectively, and suppose that a rectangular matrix $V \in M_{D \times E}(\mathbb{C})$ satisfies the following identity for every physical index i :

$$L^i = V^\dagger K^i V.$$

Every physical-sector factorization of K induces one of L with the same physical decomposition. This construction combines the virtual corner in [11, Section 2.3, lines 195–219] with the physical-sector factorization in [11, Appendix C.2, lines 1381–1439].

Theorem 20.16.3.9 (Range projection of a matrix-block inclusion). [Lean](#) [✓ Stmt](#) *For the canonical inclusion V_s of a matrix block, $V_s V_s^\dagger$ is the projection onto that block.*

Proof. [✓ Proof](#) Compute the entries of the inclusion and its conjugate transpose. $\square$

Theorem 20.16.3.10 (Flattened sector as an isometric virtual corner). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $A = \bigoplus_t \mu_t A_t$ be a weighted block-diagonal tensor in flattened virtual coordinates. For every block s , the canonical inclusion V_s of that block satisfies*

$$\begin{aligned} V_s^\dagger V_s &= 1, \\ V_s V_s^\dagger &= P_s, \\ V_s^\dagger A^i V_s &= \mu_s A_s^i. \end{aligned}$$

Here P_s is the corresponding block projection, transported to the flattened coordinates. Moreover,

$$\begin{aligned} A^i V_s &= V_s (\mu_s A_s^i), \\ V_s^\dagger A^i &= (\mu_s A_s^i) V_s^\dagger. \end{aligned}$$

This is the exact blockwise virtual-corner identity from [11, Section 2.3, equation II_Aiplusk1, lines 195–219], providing the virtual corner structure used in Appendix C.2.

Proof. ✓ Proof Let E_s be the canonical inclusion of the s -th summand in the dependent direct sum, let Q_s be the corresponding unflattened block projection, and let e flatten the direct-sum coordinates. Thus V_s is the reindexing of E_s , while P_s is the reindexing of Q_s . Multiplicativity of reindexing transports

$$\begin{aligned} E_s^\dagger E_s &= 1, \\ E_s E_s^\dagger &= Q_s \end{aligned}$$

to the two corresponding identities for V_s . If $B_t^i = \mu_t A_t^i$, block diagonality gives

$$\begin{aligned} A^i V_s &= V_s B_s^i, \\ V_s^\dagger A^i &= B_s^i V_s^\dagger. \end{aligned}$$

Therefore

$$V_s^\dagger A^i V_s = V_s^\dagger V_s B_s^i = B_s^i = \mu_s A_s^i.$$

□

Theorem 20.16.3.11 (Neighboring operators under virtual compression). Lean ✓ Stmt The neighboring operators of the factorization obtained by virtual compression satisfy

$$\eta_{k,h}^L = \eta_{k,h}^K (VV^\dagger).$$

This identity combines the virtual corner in [11, Section 2.3, lines 195–219] with the neighboring contraction in [11, Appendix C.2, lines 1441–1450].

Proof. ✓ Proof Apply the virtual-matrix transport identity with $X = V^\dagger$ and $Y = V$. □

Theorem 20.16.3.12 (Virtual-gauge invariance of physical-sector factorizations). Lean Lean ✓ Stmt Suppose that two MPO tensors of the same virtual dimension differ by an invertible virtual gauge X . The induced factorization is precisely the virtual-matrix transport by X and X^{-1} . Thus every physical-sector factorization of the first tensor induces one of the second tensor with the same physical decomposition and the same neighboring operators. In particular, positive semidefiniteness of all neighboring operators is invariant under the gauge.

Proof. ✓ Proof Specialize the virtual-matrix neighboring identity to a gauge and its inverse. Their product is the identity matrix, and contraction through the identity matrix is the ordinary neighboring operator. □

Definition 20.16.3.13 (One-site matrix spaces of sector sets). Lean Lean ✓ Stmt For a sector k and physical matrix indices x, y in its summand, let $A_{k;x,y} \in M_D(\mathbb{C})$ be the corresponding entry of the sector-coordinate tensor. For a set S of sectors, define

$$\mathcal{A}_S = \text{span}\{A_{k;x,y} : k \in S\}.$$

These are the matrix spaces associated with the physical projections in [11, Appendix C.2, lines 1463–1467].

Theorem 20.16.3.14 (Directed-cut annihilation of one-site matrix spaces). [Lean](#) [Lean](#) [✓ Stmt](#) Let S and C be sets of sectors. If $\eta_{k,h} = 0$ for every $k \in S$ and $h \in C$, then $XY = 0$ for every $X \in \mathcal{A}_S$ and $Y \in \mathcal{A}_C$. This is the directed local-orthogonality implication used in [11, Appendix C.2, lines 1463–1470].

Proof. [✓ Proof](#) Expand two generating matrices in sector coordinates. Their product has entries

$$\sum_{\gamma} (l_k)_{\beta} (r_k)_{\gamma} (l_h)_{\gamma} (r_h)_{\alpha}.$$

The middle contraction is a matrix entry of $\eta_{k,h}$ and therefore vanishes. Bilinearity extends the conclusion from the generators to the two linear spans. $\square$

Definition 20.16.3.15 (Vertex rephasing of the sector tensors). [Lean](#) [✓ Stmt](#) Let $z_k \in \mathbb{C}$ satisfy $|z_k| = 1$ for every sector. Replacing

$$\begin{aligned} (l_k)_a &\mapsto z_k (l_k)_a, \\ (r_k)_a &\mapsto z_k^{-1} (r_k)_a \end{aligned}$$

leaves every physical-slice factorization unchanged.

Theorem 20.16.3.16 (Rephasing identities for the sector data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Vertex rephasing transforms the neighboring operators by $\eta_{k,h} \mapsto z_k^{-1} z_h \eta_{k,h}$. It leaves every sector virtual matrix unchanged, preserves the nonzero neighboring support, and leaves every complete cyclic product of neighboring operators unchanged. These are the rephasing identities used in [11, Appendix C.2, lines 1434–1450].

Definition 20.16.3.17 (Vanishing right tensors on inactive sectors). [Lean](#) [Lean](#) [✓ Stmt](#) Let S be a set of active sectors in a physical-sector factorization, and suppose that $(l_k)_{\beta} = 0$ for every $k \notin S$ and every β . Retain every physical summand and define

$$\begin{aligned} \hat{l}_k &= l_k, \\ \hat{r}_k &= \begin{cases} r_k, & k \in S, \\ 0, & k \notin S. \end{cases} \end{aligned}$$

Then $\hat{l}_k \otimes \hat{r}_k = l_k \otimes r_k$ for every k , so the physical-slice factorization is unchanged. The neighboring operators become

$$\hat{\eta}_{k,h} = \begin{cases} \eta_{k,h}, & k \in S, \\ 0, & k \notin S. \end{cases}$$

This is the reparameterization freedom in the factorization and neighboring contraction of [11, Appendix C.2, lines 1434–1445].

Definition 20.16.3.18 (Inverse-map physical-sector factorization). [Lean](#) [✓ Stmt](#) Let $\mathcal{K}$ be injective, let ρ be its three-site closure against a virtual tail R , and choose a Hayashi decomposition of ρ . Fix outer indices such that $R_{\beta_3, \alpha_1} \neq 0$. The inverse-map tensors l_k, r_k , together with the Hayashi physical decomposition and unitary, determine a physical-sector factorization of $\mathcal{K}$.

The left and right factor spaces are nonzero because the corresponding Hayashi density matrices have trace one; their nonzero dimensions are conclusions rather than hypotheses. This gives the factorization in [11, Appendix C.2, equations AppUkU=r1 and formK, lines 1383–1387 and 1434–1439].

Theorem 20.16.3.19 (Identification of the neighboring operators). [Lean](#) [✓ Stmt](#) For all sectors k, h , the neighboring contraction of the resulting physical-sector factorization is the inverse-map operator

$$\eta_{k,h} = \sum_{\gamma} (r_k)_{\gamma} \otimes (l_h)_{\gamma}.$$

This is equation `etar1` in [11, Appendix C.2, lines 1441–1445].

Definition 20.16.3.20 (Zero-weight inverse-map reparameterization). [Lean](#) [✓ Stmt](#) In the inverse-map factorization, retain every physical sector and its factor spaces, and set

$$\begin{aligned} \hat{l}_k &= l_k, \\ \hat{r}_k &= \begin{cases} r_k, & p_k \neq 0, \\ 0, & p_k = 0. \end{cases} \end{aligned}$$

The physical-slice factorization is unchanged. This treats the zero-weight freedom left by the Hayashi decomposition used in [11, Appendix C.2, lines 1415–1445].

Theorem 20.16.3.21 (Neighboring operators after zero-weight reparameterization). [Lean](#)

[Lean](#) [✓ Stmt](#) The neighboring operators of the reparameterized factorization are

$$\hat{\eta}_{k,h} = \begin{cases} \eta_{k,h}, & p_k \neq 0, \\ 0, & p_k = 0. \end{cases}$$

Thus a zero-weight sector has no outgoing support edge. Since $p_h = 0$ also gives $l_h = 0$, it has no incoming support edge either.

Proof. [✓ Proof](#) The neighboring contraction uses r_k at its source and l_h at its target. The first assertion follows from the definition of $\hat{r}_k$, and the last assertion follows from Lemma 20.16.2.51. $\square$

Corollary 20.16.3.22 (SAL gives a raw physical-sector factorization). [Lean](#) [✓ Stmt](#) Every injective MPO tensor satisfying SAL admits a physical-sector factorization. This is only the raw algebraic factorization from Lemma C.4 and equations `formK` and `etar1`; coherent positivity of the neighboring operators, recurrence, and primitivity are not part of this conclusion.

Definition 20.16.3.23 (Finite-chain coordinates of a physical-sector factorization). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let $N \geq 1$. For a cyclic sector configuration $k = (k_0, \dots, k_{N-1})$, write

$$I_k = \prod_{n=0}^{N-1} (B_{k_n}^L \times B_{k_n}^R).$$

Applying the one-site sector decomposition at every site gives a bijection

$$\Phi_N : \{0, \dots, d-1\}^N \longrightarrow \prod_{k_0, \dots, k_{N-1}} I_k.$$

On I_k , define the cyclic neighboring product by

$$\Omega_k(x, y) = \prod_{n=0}^{N-1} \eta_{k_n, k_{n+1}}((x_n^R, x_{n+1}^L), (y_n^R, y_{n+1}^L)),$$

where site labels are read modulo N .

Theorem 20.16.3.24 (Finite-chain decomposition of a physical-sector factorization). [Lean](#) [Lean](#) [Stmt](#) Suppose that the physical-sector factorization is given. For every $N \geq 1$, the physically transformed MPO is block diagonal in the cyclic sector configurations, and

$$\text{reindex}_{\Phi_N}(M_N(\widetilde{\mathcal{K}})) = \bigoplus_{k_0, \dots, k_{N-1}} \Omega_k.$$

This establishes the all-length algebraic decomposition of [11, Appendix C.2, lines 1435–1450] under the displayed physical-sector factorization.

Proof. [Proof](#) Fix a sector configuration k and entries $x, y \in I_k$. Expanding the closed horizontal contraction gives

$$\sum_g \prod_n (l_{k_n})_{g_n} (x_n^L, y_n^L) (r_{k_n})_{g_{n+1}} (x_n^R, y_n^R).$$

Cyclically shifting the summation variables and collecting the factors with the same horizontal index turns this expression into $\Omega_k(x, y)$. If the row and column sector configurations differ, one transformed physical slice lies between distinct direct summands and vanishes. Thus all off-diagonal sector blocks vanish. $\square$

Corollary 20.16.3.25 (SAL gives positive cyclic sector products). [Lean](#) [Stmt](#) Every injective MPO tensor satisfying SAL admits a physical-sector factorization such that, for every $N \geq 1$ and every cyclic sector configuration $k = (k_0, \dots, k_{N-1})$ with $k_N = k_0$,

$$\bigotimes_{n=0}^{N-1} \eta_{k_n, k_{n+1}} \geq 0.$$

This establishes the projected-chain inequality of [11, Appendix C.2, Lemma C.4, lines 1446–1450].

Proof. [Proof](#) Choose the raw physical-sector factorization supplied by SAL. The physical basis change sends the positive N -site MPO to a positive congruence. Each cyclic neighboring product is a diagonal sector compression of this congruence and is therefore positive semidefinite. $\square$

Definition 20.16.3.26 (Boundary contraction against a virtual matrix). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) For an arbitrary virtual matrix X , define the outer boundary operator on $B_k^L \otimes B_h^R$ by

$$B_{k,h}(X) = \sum_{a,b} X_{b,a} (l_k)_a \otimes (r_h)_b.$$

The map $X \mapsto B_{k,h}(X)$ is complex-linear. No positivity of X or $B_{k,h}(X)$ is assumed.

Definition 20.16.3.27 (Regrouped two-site sector closure). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) First pass to the physical basis selected by U , so that each slice is written in the sector coordinates of Definition 20.16.3.3. Form the two-site closure of this sector-coordinate tensor, restrict it to the physical sectors (k, h) , and reindex its physical factors by

$$((L_k, R_k), (L_h, R_h)) \mapsto ((L_k, R_h), (R_k, L_h)).$$

Denote the resulting matrix by $\mathcal{K}_{2;k,h}(X)$.

Theorem 20.16.3.28 (Two-site closure factorization). [Lean](#) [Stmt](#) For every virtual matrix X and every sector pair (k, h) ,

$$\mathcal{K}_{2;k,h}(X) = B_{k,h}(X) \otimes \eta_{k,h}.$$

Proof. ✓ Proof At the regrouped matrix entry indexed by $((\lambda_k, \rho_h), (\rho_k, \lambda_h))$ and $((\lambda'_k, \rho'_h), (\rho'_k, \lambda'_h))$, the virtual trace expands as the scalar

$$\sum_{a,b,c} X_{b,a} [(l_k)_a]_{\lambda_k, \lambda'_k} [(r_k)_c]_{\rho_k, \rho'_k} [(l_h)_c]_{\lambda_h, \lambda'_h} [(r_h)_b]_{\rho_h, \rho'_h}.$$

Commuting scalar factors and collecting the sums over (a, b) and c gives respectively $B_{k,h}(X)$ and $\eta_{k,h}$. $\square$

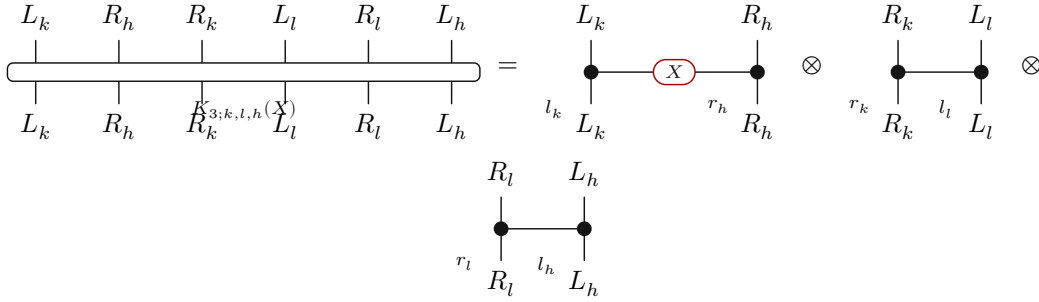
Definition 20.16.3.29 (Regrouped three-site sector closure). Lean Lean Lean Lean ✓ Stmt First pass to the physical basis selected by U and form the three-site closure of the resulting sector-coordinate tensor. Restrict this closure to the sectors (k, l, h) and reindex its physical factors by

$$((L_k, R_k), ((L_l, R_l), (L_h, R_h))) \mapsto ((L_k, R_h), ((R_k, L_l), (R_l, L_h))).$$

Denote the resulting matrix by $\mathcal{K}_{3;k,l,h}(X)$.

Theorem 20.16.3.30 (Three-site closure factorization). Lean ✓ Stmt For every virtual matrix X and fixed sectors (k, l, h) ,

$$\mathcal{K}_{3;k,l,h}(X) = B_{k,h}(X) \otimes (\eta_{k,l} \otimes \eta_{l,h}).$$



Proof. ✓ Proof At the regrouped matrix entry indexed by $((\lambda_k, \rho_h), ((\rho_k, \lambda_l), (\rho_l, \lambda_h)))$ and its primed counterpart, the left-hand side is the scalar

$$\sum_{a,b,c,e} X_{b,a} [(l_k)_a]_{\lambda_k, \lambda'_k} [(r_k)_c]_{\rho_k, \rho'_k} [(l_l)_c]_{\lambda_l, \lambda'_l} [(r_l)_e]_{\rho_l, \rho'_l} [(l_h)_e]_{\lambda_h, \lambda'_h} [(r_h)_b]_{\rho_h, \rho'_h}.$$

Reordering the four finite sums separates the (a, b) boundary contraction from the neighboring contractions indexed by c and e , which gives the three factors on the right-hand side. $\square$

Definition 20.16.3.31 (Three-site neighboring operator). Lean ✓ Stmt For every outer-sector pair (k, h) , define

$$\Omega_{k,h} = \bigoplus_l (\eta_{k,l} \otimes \eta_{l,h}).$$

This is the direct sum of the two neighboring contractions appearing in the three-site closure factorization [11, Appendix C.2, lines 1510–1516].

$$\mathfrak{R}_2(\mathcal{K}_2(X))_{k,h} = a_k b_h B_{k,h}(X).$$

The figure consists of two parts. The top part shows a cylinder with two vertical lines, the left one labeled L_k and the right one labeled R_h . A horizontal line segment connects the two vertical lines, with the label $a_k b_h$ placed below it. The bottom part shows a similar cylinder with vertical lines L_k and R_h . A horizontal line segment connects the two vertical lines, with a red oval labeled X in the middle. Below the left vertical line is the label l_k , and below the right vertical line is the label r_h .

$$\mathfrak{R}_3(\mathcal{K}_3(X))_{k,l,h} = B_{k,h}(X) \otimes \eta_{k,l} \otimes \eta_{l,h}.$$

Figure 20.3: First each slice is conjugated by U and expressed in the corresponding sector coordinates. After restriction to fixed sectors and the displayed regroupings, the arbitrary- X two- and three-site closures separate into the boundary factor and respectively one or two neighboring contractions, following [11, lines 638–655 and Appendix C.2, lines 1435–1448].

Definition 20.16.3.32 (Neighboring-operator trace factorization). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

 **Stmt** Suppose that every neighboring operator is positive semidefinite and that there are real numbers a_k, b_h such that

$$\begin{aligned} \mathrm{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_k a_k b_k &= 1. \end{aligned}$$

These are precisely the remaining hypotheses used in the construction of Proposition C.7 after the physical-sector factorization has been fixed [11, Appendix C.2, lines 1389–1403].

Theorem 20.16.3.33 (Unitary-between physical-coordinate invariance of neighboring trace factorizations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ satisfy $V^\dagger V = \mathbb{1}_d$ and $VV^\dagger = \mathbb{1}_e$, and define $K_{\beta,\alpha}^V = VK_{\beta,\alpha}V^\dagger$. Then K admits a physical-sector factorization with

$$\begin{aligned}\eta_{k,h} &\geq 0, \\ \text{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_k a_k b_k &= 1.\end{aligned}$$

if and only if K^V admits one. The same sector spaces, tensors l_k, r_k , neighboring operators $\eta_{k,h}$, and numbers a_k, b_h may be used on both sides. This invariance is project-derived; the data being transported are those of [11, Theorem 4.9(iv) and Appendix C.2, lines 1381–1450], but the equivalence itself is not stated there.

Proof. ✓ Proof The two one-sided unitary identities give $d \leq e$ and $e \leq d$, respectively, so $d = e$. If the physical isometry for K is U , use $UV^\dagger$ for K^V ; then $(UV^\dagger)K_{\beta,\alpha}^V(UV^\dagger)^\dagger = UK_{\beta,\alpha}U^\dagger$. Conversely, if the isometry for K^V is U' , use $U'V$ for K . In both directions the tensors l_k, r_k are unchanged, so $\eta_{k,h} = \sum_\gamma (r_k)_\gamma \otimes (l_h)_\gamma$ is unchanged. The positivity, trace, and normalization identities therefore remain the same identities. $\square$

Theorem 20.16.3.34 (Gauge and support-coordinate independence of neighboring trace factorizations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that two MPO tensors of the same physical and virtual dimensions differ by an invertible virtual gauge. One admits a physical-sector factorization with neighboring data $\eta_{k,h}$, a_k , and b_h if and only if the other does. The same neighboring operators and the same numbers a_k, b_h may be used on both sides.

Let $V_F : \mathbb{C}^{d_F} \rightarrow \mathbb{C}^d$ and $V_G : \mathbb{C}^{d_G} \rightarrow \mathbb{C}^d$ be two isometric inclusions with the same range projection P :

$$\begin{aligned} V_F^\dagger V_F &= \mathbb{1}_{d_F}, & V_G^\dagger V_G &= \mathbb{1}_{d_G}, \\ V_F V_F^\dagger &= P, & V_G V_G^\dagger &= P. \end{aligned} \quad (20.60)$$

Set $Q = V_F^\dagger V_G$. Then

$$\begin{aligned} Q^\dagger Q &= \mathbb{1}_{d_G}, & Q Q^\dagger &= \mathbb{1}_{d_F}, \\ Q(V_G^\dagger K V_G) Q^\dagger &= V_F^\dagger K V_F. \end{aligned} \quad (20.61)$$

Consequently, the restrictions of K along V_F and V_G admit neighboring trace factorizations simultaneously.

Both equivalences are project-derived and are not stated in [11]. The gauge transformation is the one discussed in Section 2.3, lines 187–195; the support-restriction setting is that of Appendix C.2, lines 1680–1691 and 1733–1770.

Proof. [✓ Proof](#) For the virtual gauge $L^i = X K^i X^{-1}$, write $\eta_{k,h}^K(M)$ for the neighboring contraction with M inserted between the transported virtual factors. Their composition is $X^{-1}X$, and hence

$$\eta_{k,h}^L = \eta_{k,h}^K(X^{-1}X) = \eta_{k,h}^K(\mathbb{1}) = \eta_{k,h}^K.$$

The families a_k and b_h are retained literally, together with their positivity, trace, and normalization identities. The reverse implication uses the inverse gauge.

For the support coordinates, (20.60) gives

$$\begin{aligned} Q^\dagger Q &= V_G^\dagger (V_F V_F^\dagger) V_G = V_G^\dagger P V_G = \mathbb{1}_{d_G}, \\ Q Q^\dagger &= V_F^\dagger (V_G V_G^\dagger) V_F = V_F^\dagger P V_F = \mathbb{1}_{d_F}, \\ Q V_G^\dagger &= V_F^\dagger (V_G V_G^\dagger) = V_F^\dagger P = V_F^\dagger. \end{aligned}$$

The last identity gives (20.61) by composition of physical coordinate changes. The unitarity-between invariance above, with the resulting equivalence reversed, proves the claim. $\square$

Theorem 20.16.3.35 (Trace of the three-site neighboring operator). [Lean](#) [Lean](#) [✓ Stmt](#) Assume the neighboring-operator trace factorization of Definition 20.16.3.32. Then, for every outer-sector pair (k, h) ,

$$\mathrm{tr}(\Omega_{k,h}) = \sum_l \mathrm{tr}(\eta_{k,l}) \mathrm{tr}(\eta_{l,h}) = a_k b_h.$$

This is the normalization used in the coarse-graining channel of [11, Appendix C.2], lines 1547–1555.

Proof. [✓ Proof](#) The trace of a direct sum is the sum of the traces of its blocks, and the trace of a tensor product is the product of the traces. Hence

$$\mathrm{tr}(\Omega_{k,h}) = \sum_l (a_k b_l) (a_l b_h) = a_k b_h \sum_l a_l b_l = a_k b_h.$$

$\square$

Lemma 20.16.3.36 (Restricting finite sums to nonzero weights). [Lean](#) [✓ Stmt](#) Let I be finite, let p_i be weights in a type with zero, and let f_i take values in a commutative additive monoid. If $f_i = 0$ whenever $p_i = 0$, then

$$\sum_{i \in I} f_i = \sum_{\substack{i \in I \\ p_i \neq 0}} f_i.$$

Proof. [✓ Proof](#) Split the full sum according to whether p_i vanishes. Every term in the zero-weight part vanishes by hypothesis. $\square$

The corresponding generic result is proved in the companion quantum-channel volume [182, “Cyclic trace identity for rectangular powers”].

Theorem 20.16.3.37 (Unit closure traces from a neighboring trace factorization). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let M be an MPO tensor admitting a physical-sector factorization whose neighboring operators satisfy the trace factorization of Definition 20.16.3.32. Closing the physical legs of the factorized slices writes the physical-trace transfer as a rectangular product

$$\mathcal{T}_M = LQ, \quad L_{\beta,k} = \text{tr}((l_k)_\beta), \quad Q_{k,\alpha} = \text{tr}((r_k)_\alpha),$$

whose opposite product is the rank-one coefficient matrix

$$(QL)_{k,h} = \text{tr}(\eta_{k,h}) = a_k b_h.$$

Consequently every positive power of the physical-trace transfer and every nonempty closed chain has unit trace:

$$\text{tr}(\mathcal{T}_M^N) = 1, \quad \text{tr} \rho^{(N)}(M) = 1, \quad N \geq 1.$$

In particular, a tensor whose closed chains do not all have unit trace admits no physical-sector factorization with positive neighboring operators and normalized rank-one trace coefficients. The normalization $\text{tr}(T) = 1$ invoked in [11, Appendix C.2, line 1498] is therefore a constraint on the tensor rather than a convention, and the ambient normalization at [11, Appendix C.2, lines 1755–1759] constrains only the weighted sum over the representatives of a basis of normal tensors, not each representative separately.

Proof. [✓ Proof](#) Closing the physical legs of each factorized slice and using unitary invariance of the trace expresses each entry of the physical-trace transfer as $\sum_k \text{tr}((l_k)_\beta) \text{tr}((r_k)_\alpha)$, which is the rectangular product LQ . The entries of the opposite product are the contractions

$$(QL)_{k,h} = \sum_{\alpha} \text{tr}((r_k)_\alpha) \text{tr}((l_h)_\alpha) = \text{tr}(\eta_{k,h}) = a_k b_h.$$

Cyclic invariance of the trace gives $\text{tr}((LQ)^N) = \text{tr}((QL)^N)$ for every $N \geq 1$, and every positive power of the rank-one matrix $(a_k b_h)_{k,h}$ with $\sum_k a_k b_k = 1$ equals the matrix itself, of trace one. The closed chain of length N has trace $\text{tr}(\mathcal{T}_M^N)$. $\square$

Theorem 20.16.3.38 (Partial traces of the sector closures). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every virtual matrix X ,

$$\text{tr}_{R_k \otimes L_h}(\mathcal{K}_{2;k,h}(X)) = \text{tr}(\eta_{k,h}) B_{k,h}(X) = a_k b_h B_{k,h}(X).$$

Similarly,

$$\text{tr}_{(R_k \otimes L_l) \otimes (R_l \otimes L_h)}(\mathcal{K}_{3;k,l,h}(X)) = \text{tr}(\eta_{k,l}) \text{tr}(\eta_{l,h}) B_{k,h}(X),$$

and summing the scalar coefficient over l gives $a_k b_h$.

Proof. [✓ Proof](#) The partial trace of $A \otimes B$ is $\text{tr}(B)A$. The two-site identity follows immediately. For three sites, the coefficient is

$$\sum_l (a_k b_l)(a_l b_h) = a_k \left(\sum_l a_l b_l \right) b_h = a_k b_h.$$

□

Definition 20.16.3.39 (Global two- and three-site sector coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The two- and three-site physical spaces are reindexed by their canonical direct sums over sector pairs and triples, with each summand further regrouped into its boundary and neighboring factors. Write $\mathfrak{G}_2$ and $\mathfrak{G}_3$ for these reindexings of matrices.

Theorem 20.16.3.40 (Global direct-sum form of the sector closures). [Lean](#) [Lean](#) [✓ Stmt](#) After the canonical regrouping of all physical-sector indices, the full closures are

$$\begin{aligned} \mathfrak{G}_2(\mathcal{K}_2(X)) &= \bigoplus_{k,h} B_{k,h}(X) \otimes \eta_{k,h}, \\ \mathfrak{G}_3(\mathcal{K}_3(X)) &= \bigoplus_{k,l,h} B_{k,h}(X) \otimes (\eta_{k,l} \otimes \eta_{l,h}). \end{aligned}$$

Every matrix entry between distinct sector tuples vanishes.

Proof. [✓ Proof](#) On a diagonal sector tuple these are the fixed-sector factorizations. For distinct sector labels $k \neq p$, the block structure of the sector-coordinate tensor gives $[U\kappa_{\beta,\alpha}U^\dagger]_{k,p} = 0$. Consequently, if $(k,h) \neq (p,q)$ or $(k,l,h) \neq (p,m,q)$, respectively, then

$$\begin{aligned} [\mathfrak{G}_2(\mathcal{K}_2(X))]_{(k,h),(p,q)} &= 0, \\ [\mathfrak{G}_3(\mathcal{K}_3(X))]_{(k,l,h),(p,m,q)} &= 0. \end{aligned}$$

□

Definition 20.16.3.41 (Two-site sector bond). [Lean](#) [✓ Stmt](#) In the two-site sector coordinates, define

$$B_{\text{sector}} = \bigoplus_{k,h} \left(\mathbb{1}_{B_k^L \otimes B_h^R} \otimes \eta_{k,h} \right).$$

Thus the outer factors carry the identity, while the neighboring factors carry the operator $\eta_{k,h}$. This is the local bond in [11, Appendix C.2, Proposition C.8, lines 1581–1593].

Theorem 20.16.3.42 (Positivity of the two-site sector bond). [Lean](#) [✓ Stmt](#) Suppose that $\eta_{k,h}$ is positive semidefinite for every pair of sectors. Then B_{sector} is positive semidefinite. The positivity premise is the conclusion at [11, Appendix C.2, lines 1446–1450].

Proof. [✓ Proof](#) Each summand $\mathbb{1}_{B_k^L \otimes B_h^R} \otimes \eta_{k,h}$ is positive semidefinite. Positivity is preserved by finite block diagonals and by the reindexing from the regrouped direct sum to the two-site sector coordinates. □

Definition 20.16.3.43 (Two-site bond in physical coordinates). [Lean](#) [Lean](#) [✓ Stmt](#) Let V_2 be the two-site tensor product of the project's coordinate map, so that $V_2 X V_2^\dagger$ expresses a physical

operator X in sector coordinates. Thus $V_2 = U_{\text{paper}}^\dagger$ in the convention of [11, Appendix C.2, lines 1581–1583]. Define

$$B_{\text{physical}} = V_2^\dagger B_{\text{sector}} V_2,$$

and identify pairs of physical indices with functions $\{0, 1\} \rightarrow \{0, \dots, d-1\}$. This is the physical two-site bond in [11, Appendix C.2, Proposition C.8, lines 1581–1593].

Theorem 20.16.3.44 (Two-site translation identities). Lean Lean ✓ Stmt *Under the canonical identification of two-site configurations with ordered pairs, the translates beginning at sites zero and one are respectively $B_{01} = B_{\text{physical}}$ and $B_{10} = S B_{\text{physical}} S^{-1}$, where $S(x \otimes y) = y \otimes x$ exchanges the two physical factors.*

Proof. ✓ Proof Apply Theorem 20.13.3 to B_{physical} . Its pair-coordinate matrix is the operator in Definition 20.16.3.43, so the two generic identities give the stated formulas. $\square$

Theorem 20.16.3.45 (Positivity of the physical two-site bond). Lean Lean ✓ Stmt *Suppose that $\eta_{k,h}$ is positive semidefinite for every pair of sectors. Then B_{physical} is positive semidefinite, both with pair indices and with two-site configuration indices.*

Proof. ✓ Proof Apply positivity of B_{sector} to the congruence $V_2^\dagger B_{\text{sector}} V_2$. Reindexing the resulting matrix by two-site configurations preserves positivity. $\square$

Definition 20.16.3.46 (Three-site lifts of a two-site operator). Lean Lean ✓ Stmt Let B be an operator on two copies of a finite-dimensional space H . On the right-associated space $H \otimes (H \otimes H)$, define

$$B_{01} = \alpha(B \otimes \mathbb{1}_H) \alpha^{-1}, \quad B_{12} = \mathbb{1}_H \otimes B,$$

where $\alpha : (H \otimes H) \otimes H \rightarrow H \otimes (H \otimes H)$ is the canonical associator.

Theorem 20.16.3.47 (Three-site translation identities). Lean Lean ✓ Stmt *Under the canonical identification of three-site configurations with $H \otimes (H \otimes H)$, the translates of the physical bond beginning at sites 0 and 1 are respectively $(B_{\text{physical}})_{01} = B_{01}$ and $(B_{\text{physical}})_{12} = B_{12}$.*

Proof. ✓ Proof An entry of the first translate vanishes unless the third indices agree; its remaining entry is that of B_{physical} . Similarly, an entry of the second translate vanishes unless the first indices agree. These are precisely the matrix entries of B_{01} and B_{12} . $\square$

Theorem 20.16.3.48 (Commutativity on a fixed sector triple). Lean ✓ Stmt *Fix sectors (k, l, h) . On the regrouped sector space, set*

$$\begin{aligned} C_{01}^{k,l,h} &= \mathbb{1}_{B_k^L \otimes B_h^R} \otimes (\eta_{k,l} \otimes \mathbb{1}_{B_l^R \otimes B_h^L}), \\ C_{12}^{k,l,h} &= \mathbb{1}_{B_k^L \otimes B_h^R} \otimes (\mathbb{1}_{B_k^R \otimes B_l^L} \otimes \eta_{l,h}). \end{aligned}$$

Then $C_{01}^{k,l,h} C_{12}^{k,l,h} = C_{12}^{k,l,h} C_{01}^{k,l,h}$. This is the fixed-sector calculation in [11, Appendix C.2, Proposition C.8, lines 1589–1593].

On the sector triple (k, l, h) , the two operators act on distinct neighboring factors, while the outer factors B_k^L and B_h^R remain unchanged. Their commutativity is therefore the local identity $\eta_{k,l}^{(12)} \eta_{l,h}^{(23)} = \eta_{l,h}^{(23)} \eta_{k,l}^{(12)}$.

Proof. ✓ Proof The operators $\eta_{k,l}$ and $\eta_{l,h}$ act on the distinct factors $B_k^R \otimes B_l^L$ and $B_l^R \otimes B_h^L$, respectively. The claim follows from the mixed-product identity for tensor products. $\square$

Theorem 20.16.3.49 (Commutativity of adjacent physical bonds). [Lean](#) [✓ Stmt](#) *The two translates of the physical bond on three sites commute:*

$$(B_{\text{physical}})_{01}(B_{\text{physical}})_{12} = (B_{\text{physical}})_{12}(B_{\text{physical}})_{01}.$$

This is the adjacent-bond calculation in [11, Appendix C.2, Proposition C.8, lines 1589–1593].

Proof. [✓ Proof](#) Regroup the three-site sector coordinates into the direct sum over triples (k, l, h) . The two bonds become block diagonal, with blocks $C_{01}^{k,l,h}$ and $C_{12}^{k,l,h}$, which commute by Theorem 20.16.3.48. Conjugating by the three-fold physical coordinate unitary preserves the equality. The three-site translation identities then give the displayed physical commutator. $\square$

Theorem 20.16.3.50 (Agreement outside a cyclic window). [Lean](#) [✓ Stmt](#) *Let $W_i^L = \{i, i+1, \dots, i+L-1\}$ be a cyclic window of length $L \leq N$. Two configurations agree outside W_i^L precisely when their values agree at every site not contained in W_i^L .*

Proof. [✓ Proof](#) Replacing the entries in W_i^L does not affect any complementary site. Conversely, if the two configurations agree on the complement, replacing the window of the first by the window of the second reconstructs the second configuration. $\square$

Theorem 20.16.3.51 (Multiplicativity of cyclic local embeddings). [Lean](#) [✓ Stmt](#) *Let E_i^L embed an operator on L consecutive sites into an N -site periodic chain at the cyclic window beginning at i . Then $E_i^L(BC) = E_i^L(B)E_i^L(C)$.*

Proof. [✓ Proof](#) Identify a periodic configuration with its restriction to W_i^L and its restriction to the cyclic complement. In these coordinates, $E_i^L(B) = B \otimes \mathbb{1}$, so the assertion is the mixed-product identity. $\square$

Theorem 20.16.3.52 (Commutativity on disjoint cyclic windows). [Lean](#) [✓ Stmt](#) *Let $L \leq N$, and let W_i^L and W_j^L be disjoint cyclic windows in an N -site periodic chain. For arbitrary operators B and C on L sites, $E_i^L(B)E_j^L(C) = E_j^L(C)E_i^L(B)$.*

Proof. [✓ Proof](#) Apply both sides to a chain amplitude. The first order of application is a double sum over replacements on W_i^L and W_j^L . Since the windows are disjoint, each replacement leaves the other restricted configuration unchanged, and the two replacements commute. Interchanging the two finite sums gives the reverse order. $\square$

Theorem 20.16.3.53 (Adjacent physical bonds on periodic chains). [Lean](#) [✓ Stmt](#) *Let $N \geq 3$. For every site $i \in \mathbb{Z}/N\mathbb{Z}$, including $i = N-1$, the adjacent translates of the physical bond commute:*

$$(B_{\text{physical}})_{i,i+1}(B_{\text{physical}})_{i+1,i+2} = (B_{\text{physical}})_{i+1,i+2}(B_{\text{physical}})_{i,i+1}.$$

This is the periodic-chain form of the adjacent-bond calculation in [11, Appendix C.2, Proposition C.8, lines 1571–1593]. The case $N = 2$ is treated separately below, because its two translated bonds have the same support with opposite cyclic order.

Proof. [✓ Proof](#) Enclose the two bonds in the cyclic three-site window beginning at i . In these coordinates they are the bonds beginning at sites 0 and 1, respectively. Embedding an operator into a fixed cyclic window preserves products, so the assertion follows from Theorem 20.16.3.49. $\square$

Theorem 20.16.3.54 (Commutativity of the crossed two-site translates). [Lean](#) [✓ Stmt](#) *On the periodic chain of length two, the translates beginning at sites zero and one read the physical sites in the opposite cyclic orders $(0, 1)$ and $(1, 0)$. Nevertheless they commute: $B_{01}B_{10} = B_{10}B_{01}$. This establishes the translated-bond commutativity of [11, Appendix C.2, Proposition C.8] at length two.*

Proof. [✓ Proof](#) Fix sectors (k, h) and regroup the two physical sites as $(L_k \otimes R_h) \otimes (R_k \otimes L_h)$. The translate in the order $(0, 1)$ is $\mathbb{1}_{L_k \otimes R_h} \otimes \eta_{k,h}$, while the translate in the order $(1, 0)$ is $\eta_{h,k}^{\text{op}} \otimes \mathbb{1}_{R_k \otimes L_h}$, where $\eta_{h,k}^{\text{op}}$ denotes the same matrix after exchanging the factors R_h and L_k . These operators act on complementary factors and commute. Taking the direct sum over (k, h) and conjugating by the two-fold physical coordinate unitary proves the claim. $\square$

Theorem 20.16.3.55 (Disjoint physical bonds on periodic chains). [Lean](#) [✓ Stmt](#) *Let $N \geq 2$, and suppose that the cyclic bonds beginning at i and j have disjoint two-site supports. Then their physical bond operators commute: $(B_{\text{physical}})_{i,i+1}(B_{\text{physical}})_{j,j+1} = (B_{\text{physical}})_{j,j+1}(B_{\text{physical}})_{i,i+1}$. This is the locality part of the pairwise commutation assertion in [11, Appendix C.2, Proposition C.8, lines 1571–1593].*

Proof. [✓ Proof](#) This is Theorem 20.16.3.52 applied to the same physical bond operator in both windows. No positivity or additional property of the sector factorization is needed for this locality step. $\square$

Theorem 20.16.3.56 (Pairwise commutativity of the physical bonds). [Lean](#) [✓ Stmt](#) *Suppose that the physical-sector factorization is given. For every periodic chain of length $N \geq 2$ and every pair of sites $i, j \in \mathbb{Z}/N\mathbb{Z}$, the corresponding translates of the physical bond commute:*

$$(B_{\text{physical}})_{i,i+1}(B_{\text{physical}})_{j,j+1} = (B_{\text{physical}})_{j,j+1}(B_{\text{physical}})_{i,i+1}.$$

This is the pairwise commutation assertion in [11, Appendix C.2, Proposition C.8, lines 1571–1593], conditional here on the physical-sector factorization constructed in the preceding results.

Proof. [✓ Proof](#) For $N = 2$, equal translates commute trivially, while the two distinct translates commute by Theorem 20.16.3.54. Let $N \geq 3$. If the two cyclic supports are disjoint, apply Theorem 20.16.3.55. Otherwise, Lemma 12.1.12 shows that the two starting sites either coincide or are cyclic neighbors. The coincident case is immediate, and the two neighboring orientations follow from Theorem 20.16.3.53 and its symmetric equality. $\square$

Theorem 20.16.3.57 (Exact product of physical-sector bonds). [Lean](#) [✓ Stmt](#) *Suppose that K is equipped with a physical-sector factorization F . Let B_i denote the cyclic translate beginning at site i of the physical two-site bond determined by F . For every $N \geq 2$, the N -site periodic operator is*

$$\rho_N = B_0 B_1 \cdots B_{N-1}.$$

No positivity, zero-correlation-length, injectivity, or normalization hypothesis is required. This is the product identity in [11, Appendix C.2, Proposition C.8, lines 1581–1593], conditional here on the given physical-sector factorization.

Proof. [✓ Proof](#) For a fixed cyclic sector sequence, contraction of the virtual index between the right tensor at site i and the left tensor at site $i + 1$ produces $\eta_{k_i, k_{i+1}}$. Taking the virtual trace therefore gives the cyclic product of these neighboring operators. The translated sector bonds give the same ordered product. Conjugating by the tensor power of the one-site coordinate unitary gives the stated physical identity. For $N = 2$, the second translate has the opposite cyclic order; the two factors are the ordinary and crossed bonds. $\square$

Theorem 20.16.3.58 (Unit scalar in the physical-sector product). [Lean](#) [✓ Stmt](#) Under the physical-sector factorization F of the preceding theorem, for every $N \geq 2$ there is a real number $c > 0$ such that $\rho_N = c B_0 B_1 \cdots B_{N-1}$. Here the bonds are the canonical bonds determined by F , and one may take $c = 1$. No positivity hypothesis on the neighboring operators $\eta_{k,h}$ is required.

Proof. [✓ Proof](#) Take $c = 1$ in Theorem [20.16.3.57](#). □

Theorem 20.16.3.59 (Positive translation-invariant bond datum). [Lean](#) [✓ Stmt](#) Suppose that every neighboring operator $\eta_{k,h}$ is positive semidefinite. Then the physical bond is positive semidefinite and all of its cyclic translates commute on every periodic chain of length at least two. Thus it determines a single translation-invariant positive bond datum.

Proof. [✓ Proof](#) Positivity is Theorem [20.16.3.45](#); pairwise commutativity is Theorem [20.16.3.56](#). □

Theorem 20.16.3.60 (η -local structure from a positive physical-sector factorization). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that a physical-sector factorization of K is given and that every neighboring operator $\eta_{k,h}$ is positive semidefinite. Then the physical two-site bond determines an η -local structure for K . Its translated copies commute pairwise, and for every $N \geq 2$, $\rho_N = B_0 B_1 \cdots B_{N-1}$. If the virtual dimension of K is positive, then K itself is a fixed tensor representation of this product family, with the same virtual dimension. Thus the positive normalization scalar may be chosen to be 1. This is Proposition C.8 conditional on the coherently positive physical-sector factorization that the source derives from SAL.

Proof. [✓ Proof](#) Theorem [20.16.3.59](#) gives the positive translation-invariant bond and its pairwise commuting translates. Theorem [20.16.3.57](#) realizes every finite-chain operator with normalization scalar 1. □

Corollary 20.16.3.61 (η -local structure from SAL). [Lean](#) [✓ Stmt](#) Every injective MPO tensor satisfying SAL admits an η -local structure. Thus there is a single positive two-site bond whose cyclic translates commute pairwise and whose ordered product is the finite-chain density operator at every length $N \geq 2$, with normalization scalar one. This is the conclusion of Proposition C.8.

Proof. [✓ Proof](#) Theorem [20.16.4.41](#) supplies the positive physical-sector factorization. Apply Theorem [20.16.3.60](#). □

Corollary 20.16.3.62 (Normal representative for the SAL-selected bond). [Lean](#) [✓ Stmt](#) Let K be injective and satisfy SAL, and suppose that its doubled-index matrix-product tensor is normal. There is a positive physical-sector factorization whose selected two-site bond B has K itself as a normal fixed tensor representation. Its virtual dimension is the virtual dimension of K , and for every $N \geq 2$, $\rho_N(K) = B_0 B_1 \cdots B_{N-1}$. In particular, the scalar is one, providing the normal fixed tensor representation selected in Proposition C.8 of [\[11\]](#).

Proof. [✓ Proof](#) SAL implies that the virtual dimension is positive. The positive physical-sector factorization is supplied by Theorem [20.16.4.41](#). For its selected bond, use K as the fixed tensor. The exact product identity is Theorem [20.16.3.57](#), and normality is the given hypothesis on the doubled-index tensor. □

20.16.4 Closed-sector contractions and coherent rephasing

Definition 20.16.4.1 (Closed sector tensors). [Lean](#) [Lean](#) [✓ Stmt](#) Tracing the two physical legs of a sector tensor leaves a vector over the virtual bond:

$$\begin{aligned} |l_k)_\gamma &:= \text{tr} [(l_k)_\gamma], \\ (r_k|_\gamma &:= \text{tr} [(r_k)_\gamma]. \end{aligned}$$

These are the closed tensors of [11, Appendix C.2, lines 1473–1477].

Definition 20.16.4.2 (Closed-sector pairing operator). [Lean](#) [✓ Stmt](#) For sector tensors $(l_k)_\beta$ and $(r_k)_\alpha$, close their physical legs and set

$$S := \sum_k |l_k)(r_k|. \quad (20.62)$$

This is the pairing operator displayed in [11, Appendix C.2, lines 1473–1493].

Theorem 20.16.4.3 (Sector trace pairing). [Lean](#) [✓ Stmt](#) For all sectors k, h , the trace of the neighboring operator is the pairing of the closed sector tensors: $T_{k,h} = \text{tr}(\eta_{k,h}) = (r_k|l_h)$. This is [11, Appendix C.2, lines 1452–1455 and 1478–1481].

Proof. [✓ Proof](#) Expanding the definition of the neighboring operator and applying $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$ gives

$$\text{tr}(\eta_{k,h}) = \sum_\gamma \text{tr}((r_k)_\gamma \otimes (l_h)_\gamma) = \sum_\gamma \text{tr}[(r_k)_\gamma] \text{tr}[(l_h)_\gamma] = (r_k|l_h).$$

□

Definition 20.16.4.4 (Concrete closed-sector trace matrix). [Lean](#) [Lean](#) [✓ Stmt](#) For the inverse-map sector tensors, define the complex matrix $T_{k,h} := (r_k|l_h)$. Equivalently, $T_{k,h} = \text{tr}(\eta_{k,h})$. This is the trace matrix in [11, Appendix C.2, lines 1478–1481].

Theorem 20.16.4.5 (Closed-sector form of the physical-trace transfer). [Lean](#) [✓ Stmt](#) More generally, any sector factorization $U_B \kappa_{\beta,\alpha} U_B^\dagger = \bigoplus_k (l_k)_\beta \otimes (r_k)_\alpha$ gives the closed-sector pairing operator

$$\mathcal{T}_K = \sum_k |l_k)(r_k|. \quad (20.63)$$

This identifies the transfer matrix of Definition 20.6.1 with the operator on the left-hand side of the zero-correlation-length identity in [11, Appendix C.2, lines 1489–1493].

Proof. [✓ Proof](#) Trace the sector factorization over the physical index. Unitary invariance gives $\text{tr}(U_B \kappa_{\beta,\alpha} U_B^\dagger) = \text{tr}(\kappa_{\beta,\alpha})$. The trace of the block diagonal is the sum of the block traces. In each sector,

$$\text{tr}((l_k)_\beta \otimes (r_k)_\alpha) = \text{tr}[(l_k)_\beta] \text{tr}[(r_k)_\alpha] = [|l_k)(r_k|]_{\beta,\alpha}.$$

Summing over k gives (20.63). □

Corollary 20.16.4.6 (Physical-trace transfer from the inverse-map sectors). [Lean](#) [✓ Stmt](#) Let K be injective. Let R and ρ give a three-site closure of K , choose a Hayashi decomposition of ρ , and fix virtual indices α_1, β_3 such that $R_{\beta_3, \alpha_1} \neq 0$. For the corresponding closed sector tensors, $\mathcal{T}_K = \sum_k |l_k)(r_k|$.

Proof. ✓ Proof The inverse-map sector tensors satisfy the sector factorization of Theorem 20.16.2.52. Apply Theorem 20.16.4.5. □

Theorem 20.16.4.7 (Normalized closed-sector pairing operator). Lean Lean Lean ✓ Stmt Let K be injective. Let R and ρ give a three-site closure of K , choose a Hayashi decomposition of ρ , and fix virtual indices α_1, β_3 such that $R_{\beta_3, \alpha_1} \neq 0$. Form the corresponding closed sector tensors $|l_k\rangle$ and $(r_k|$. If K satisfies the positive-real up-to-scalar physical-trace relation, set $S := \sum_k |l_k\rangle(r_k|$. Then there is a real number $\lambda > 0$ such that

$$\begin{aligned} S^2 &= \lambda S, \\ (\lambda^{-1}S)^2 &= \lambda^{-1}S. \end{aligned} \tag{20.64}$$

If K satisfies the paper's literal source zero-correlation-length identity $\mathcal{T}_K^2 = \mathcal{T}_K$, then the raw pairing operator itself satisfies $S^2 = S$. This is the display preceding [11, Appendix C.2, Lemma C.5]. Identity (20.64) records the broader relation invariant under positive real rescaling; the paper's identity is its $\lambda = 1$ specialization.

Proof. ✓ Proof Substitute $S = \mathcal{T}_K$ from Corollary 20.16.4.6 into the defining relation $\mathcal{T}_K^2 = \lambda \mathcal{T}_K$, and then apply Lemma 20.6.10 to obtain (20.64) and its normalized form. Under literal idempotence, the same substitution gives $S^2 = S$ directly. □

Theorem 20.16.4.8 (Normalized closed-sector trace relations). Lean ✓ Stmt Under the hypotheses of Theorem 20.16.4.7, suppose that K satisfies the positive-real up-to-scalar physical-trace relation. There is a real number $\lambda > 0$ such that, for $\hat{T} := \lambda^{-1}T$, one has

$$\hat{T}^2 = \hat{T}^3. \tag{20.65}$$

For every positive integer N , one also has

$$\text{tr}(\hat{T}^N) = \text{tr}(\hat{T}). \tag{20.66}$$

For $\lambda = 1$, this gives the trace-power display in [11, Appendix C.2, lines 1490–1497].

Proof. ✓ Proof Let L be the matrix whose h th column is $|l_h\rangle$, and let Q be the matrix whose k th row is $(r_k|$. Then $S = LQ$ and $T = QL$. Place the normalization scalar on L . Since $\lambda^{-1}LQ$ is idempotent, associativity gives

$$(Q\lambda^{-1}L)^2 = Q(\lambda^{-1}LQ)\lambda^{-1}L = Q(\lambda^{-1}LQ)^2\lambda^{-1}L = (Q\lambda^{-1}L)^3.$$

Cyclicity of trace and the idempotence of $\lambda^{-1}LQ$ give

$$\text{tr}(Q\lambda^{-1}L) = \text{tr}(\lambda^{-1}LQ) = \text{tr}((\lambda^{-1}LQ)^2) = \text{tr}((Q\lambda^{-1}L)^2).$$

By (20.65), $\hat{T}^N = \hat{T}^2$ for every $N \geq 2$, which proves (20.66). □

Definition 20.16.4.9 (Explicit neighboring η -operator data). Lean Lean ✓ Stmt Fix a Hayashi decomposition h_η . An explicit neighboring η -family is a dependent family $(\eta_{k,h})$ indexed by sector pairs, where $\eta_{k,h}$ acts on the neighboring bond space $B_k^R \otimes B_h^L$. Requiring positivity for each pair gives the corresponding positive η -data.

Definition 20.16.4.10 (Neighboring operators from sector tensors). Lean ✓ Stmt Let $(l_k)_a$ and $(r_k)_a$ be sector tensors indexed by the horizontal virtual index a . For neighboring sectors k, h , define an operator on $B_k^R \otimes B_h^L$ by

$$\eta_{k,h}((x_R, x_L), (y_R, y_L)) = \sum_a (r_k)_a(x_R, y_R) (l_h)_a(x_L, y_L). \quad (20.67)$$

Thus $\eta_{k,h} = r_k l_h$ is the contraction over the shared horizontal index in [11, Appendix C.2, equation (etarl)]. Positivity is not part of this definition.

Lemma 20.16.4.11 (Entries of the neighboring operators). Lean ✓ Stmt For every pair of sectors and every pair of matrix indices, the entry of $\eta_{k,h}$ is the contraction displayed in (20.67).

Proof. ✓ Proof This is immediate from the definition of $\eta_{k,h}$. □

Neighboring right and left sector tensors meet in the bond operator $\eta_{k,h} = r_k l_h$:

This is the contraction in [11, Appendix C.2, lines 1441–1445].

Theorem 20.16.4.12 (Cyclic contraction of the sector tensors). Lean ✓ Stmt Let $N \geq 1$ and let $k_0, \dots, k_{N-1}$ be a cyclic sequence of sectors. Fix matrix indices in the left and right factor of every sector, and abbreviate

$$\begin{aligned} L_n(a) &= (l_{k_n})_a(x_n^L, y_n^L), \\ R_n(a) &= (r_{k_n})_a(x_n^R, y_n^R). \end{aligned}$$

With all site labels read modulo N ,

$$\sum_{g_0, \dots, g_{N-1}} \prod_{n=0}^{N-1} L_n(g_n) R_n(g_{n+1}) = \prod_{n=0}^{N-1} \eta_{k_n, k_{n+1}}((x_n^R, x_{n+1}^L), (y_n^R, y_{n+1}^L)). \quad (20.68)$$

This is the entrywise cyclic contraction at [11, Appendix C.2, lines 1446–1450].

Proof. ✓ Proof Expand the product of the sums defining the neighboring operators and set $g'_n = g_{n+1}$. Cyclic invariance of the finite product and commutativity of scalar multiplication give

$$\begin{aligned} \sum_{g_0, \dots, g_{N-1}} \prod_{n=0}^{N-1} L_n(g_n) R_n(g_{n+1}) &= \sum_{g'_0, \dots, g'_{N-1}} \prod_{n=0}^{N-1} R_n(g'_n) L_{n+1}(g'_n) \\ &= \prod_{n=0}^{N-1} \sum_a R_n(a) L_{n+1}(a). \end{aligned}$$

The last expression is the product of the corresponding entries of $\eta_{k_n, k_{n+1}}$, as in (20.68). □

Theorem 20.16.4.13 (Closed sector word as neighboring contractions). Lean ✓ Stmt Under the hypotheses of the preceding theorem, define $M_n(a, b) = L_n(a) R_n(b)$. Then

$$\text{tr}(M_0 M_1 \cdots M_{N-1}) = \prod_{n=0}^{N-1} \eta_{k_n, k_{n+1}}((x_n^R, x_{n+1}^L), (y_n^R, y_{n+1}^L)). \quad (20.69)$$

$$\text{tr}(M_{k_0} \cdots M_{k_{N-1}}) = \text{tr}(\eta_{k_0, k_1} \otimes \cdots \otimes \eta_{k_{N-1}, k_0}).$$

Figure 20.4: Closing the horizontal indices of the factorized sector tensors pairs each right factor with the left factor at the next site. The cycle therefore becomes the product of the neighboring operators $\eta_{k_n, k_{n+1}}$; compare [11, Appendix C.2, lines 1446–1450].

Proof. ✓ Proof Expanding the trace of the matrix word gives

$$\text{tr}(M_0 M_1 \cdots M_{N-1}) = \sum_{g_0, \dots, g_{N-1}} \prod_{n=0}^{N-1} M_n(g_n, g_{n+1}).$$

Substituting $M_n(a, b) = L_n(a)R_n(b)$ gives the cyclic sum in (20.68); applying that identity yields (20.69). □

Definition 20.16.4.14 (Sector-adapted chain coordinates). Lean Lean Lean ✓ Stmt For a sector configuration $k = (k_0, \dots, k_{N-1})$, let

$$I_k = \prod_{n=0}^{N-1} (\{0, \dots, d_L(k_n) - 1\} \times \{0, \dots, d_R(k_n) - 1\}).$$

Applying the one-site Hayashi decomposition at every site gives a bijection

$$\Phi_N : \{0, \dots, d-1\}^N \longrightarrow \prod_{k_0, \dots, k_{N-1}} I_k.$$

We write $\text{reindex}_{\Phi_N}(M)$ for the matrix obtained by transporting both indices of M along this bijection.

Definition 20.16.4.15 (Physical conjugation of an MPO tensor). Lean ✓ Stmt Let U be a matrix on the physical space. The physical conjugation of K by U is the tensor $\tilde{K}$ defined by

$$\tilde{\kappa}_{\beta, \alpha}^{ij} = (U \kappa_{\beta, \alpha} U^\dagger)_{ij}.$$

Its horizontal bond indices are unchanged. When U is unitary, this is the corresponding physical basis change.

Definition 20.16.4.16 (Cyclic tensor product of neighboring operators). Lean ✓ Stmt Let $N \geq 1$ and let k be a sector configuration. Define the matrix Ω_k on I_k by

$$\Omega_k(x, y) = \prod_{n=0}^{N-1} \eta_{k_n, k_{n+1}}((x_n^R, x_{n+1}^L), (y_n^R, y_{n+1}^L)). \quad (20.70)$$

Thus $\Omega_k = \eta_{k_0, k_1} \otimes \cdots \otimes \eta_{k_{N-1}, k_0}$ after the neighboring tensor factors are ordered cyclically.

Theorem 20.16.4.17 (A fixed sector block is the cyclic neighboring product). Lean ✓ Stmt Suppose that the physical slices in the Hayashi basis satisfy

$$U_B \kappa_{\beta, \alpha} U_B^\dagger = \bigoplus_q (l_q)_\beta \otimes (r_q)_\alpha.$$

Let $\eta_{q, h} = r_q l_h$. For every $N \geq 1$ and every sector configuration k , the corresponding diagonal block of $\text{reindex}_{\Phi_N}(M_N(\tilde{K}))$ is Ω_k .

Proof. [✓ Proof](#) Fix $x, y \in I_k$ and expand the closed horizontal trace of the MPO word. The sector factorization writes each local entry as a product of entries of l_{k_n} and r_{k_n} . Set

$$M_n(\beta, \alpha) = (l_{k_n})_\beta(x_n^L, y_n^L)(r_{k_n})_\alpha(x_n^R, y_n^R).$$

Then

$$\begin{aligned} [\text{reindex}_{\Phi_N}(M_N(\widetilde{K}))]_{(k,x),(k,y)} &= \text{tr}(M_0 M_1 \cdots M_{N-1}) \\ &= \prod_{n=0}^{N-1} \eta_{k_n, k_{n+1}}((x_n^R, x_{n+1}^L), (y_n^R, y_{n+1}^L)) \\ &= \Omega_k(x, y). \end{aligned} \quad (20.71)$$

where (20.71) is Theorem 20.16.4.13 and the final identification is (20.70). $\square$

Theorem 20.16.4.18 (Sector-adapted MPO decomposition). [Lean](#) [✓ Stmt](#) Under the same sector factorization hypothesis, for every $N \geq 1$,

$$\text{reindex}_{\Phi_N}(M_N(\widetilde{K})) = \bigoplus_{k_0, \dots, k_{N-1}} \eta_{k_0, k_1} \otimes \eta_{k_1, k_2} \otimes \cdots \otimes \eta_{k_{N-1}, k_0}. \quad (20.72)$$

This is the algebraic sector decomposition in [11, Appendix C.2, lines 1446–1450]. It does not assert that the individual neighboring operators are positive.

Proof. [✓ Proof](#) If the row and column sector configurations coincide, the corresponding block is Theorem 20.16.4.17. If they differ, choose a site n at which their labels k_n and h_n are unequal. Block diagonality gives $[U_B \kappa_{\beta, \alpha} U_B^\dagger]_{(k_n, \cdot), (h_n, \cdot)} = 0$ for every pair of horizontal indices. This local factor occurs in every cyclic horizontal-index product, so every summand in the closed trace is zero, proving (20.72). $\square$

Theorem 20.16.4.19 (Positive rescaling of Kronecker factors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be nonzero complex square matrices whose Kronecker product $A \otimes B$ is positive semidefinite. Then there is a scalar $c \neq 0$ with $cA \geq 0$ and $c^{-1}B \geq 0$. Since $(cA) \otimes (c^{-1}B) = A \otimes B$, the rescaled factors represent the same product.

Proof. [✓ Proof](#) On a product vector $x \otimes y$ the quadratic form of $A \otimes B$ factors, so $(x^\dagger A x)(y^\dagger B y) \geq 0$ for all vectors x and y . By polarization a complex matrix is determined by its quadratic form; since A and B are nonzero there are vectors with $a = v_0^\dagger A v_0 \neq 0$ and $c = w_0^\dagger B w_0 \neq 0$. Every value $x^\dagger (cA) x = (x^\dagger A x) c$ is non-negative, and over the complex numbers a non-negative quadratic form already forces the matrix to be Hermitian, hence positive semidefinite. For the second factor, $ac > 0$ and $a(y^\dagger B y) \geq 0$ give $c^{-1}(y^\dagger B y) = (a(y^\dagger B y))(ac)^{-1} \geq 0$. $\square$

Lemma 20.16.4.20 (A matrix and its negative cannot both be positive). [Lean](#) [✓ Stmt](#) If X and $-X$ are positive semidefinite complex matrices, then $X = 0$.

Proof. [✓ Proof](#) The quadratic form of X is both non-negative and nonpositive, hence vanishes identically. Polarization gives $X = 0$. $\square$

Definition 20.16.4.21 (Finite Kronecker product). [Lean](#) [✓ Stmt](#) For square matrices A_n indexed by $n = 0, \dots, N-1$, possibly with different index types, their finite Kronecker product is the matrix with entries

$$\left(\bigotimes_{n=0}^{N-1} A_n \right)_{x,y} = \prod_{n=0}^{N-1} (A_n)_{x_n, y_n}.$$

Lemma 20.16.4.22 (Elementary finite Kronecker identities). [Lean](#) [Lean](#) [✓ Stmt](#) *A finite Kronecker product of nonzero matrices is nonzero. Replacing one factor A_i by cA_i multiplies the whole finite product by c .*

Proof. [✓ Proof](#) Choose a nonzero entry in each factor and evaluate the product at the resulting row and column. The rescaling identity follows immediately from the entry formula in Definition 20.16.4.21. $\square$

Theorem 20.16.4.23 (Uniqueness of the positive rescaling phase). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $M \neq 0$. If $c_1, c_2 \neq 0$ and both c_1M and c_2M are positive semidefinite, then $c_1 = tc_2$ for some positive real number t . Every complex scalar of unit modulus is nonzero. In particular, every nonzero scalar c has a unit-modulus part $u = c/|c|$, positivity of cM implies positivity of uM , and there is at most one such unit scalar when $M \neq 0$.*

Proof. [✓ Proof](#) Hermiticity of c_1M and c_2M gives $c_2\bar{c}_1 = c_1\bar{c}_2$, so c_1/c_2 is real. It cannot be negative: otherwise c_1M and its negative would both be positive semidefinite, forcing $c_1M = 0$. Thus the quotient is positive. Multiplication by $|c|^{-1} > 0$ replaces c by its unit-modulus part without changing positivity. Two admissible unit scalars differ by a positive real number of modulus one, and are therefore equal. $\square$

Theorem 20.16.4.24 (Positive rescaling of a finite Kronecker product). [Lean](#) [✓ Stmt](#) *Let $N \geq 1$, and let $A_0, \dots, A_{N-1}$ be nonzero complex square matrices, with possibly different dimensions. If their finite Kronecker product is positive semidefinite, then there are nonzero scalars $c_0, \dots, c_{N-1}$ such that*

$$\prod_{n=0}^{N-1} c_n = 1, \\ c_n A_n \geq 0 \quad (0 \leq n < N).$$

Proof. [✓ Proof](#) The assertion is proved by induction on N . Separate the first factor from the remaining Kronecker product and apply Theorem 20.16.4.19. Absorb the inverse of the first rescaling coefficient into one factor of the tail, then apply the induction hypothesis. The resulting coefficients are nonzero, their product is one, and every rescaled factor is positive semidefinite. $\square$

Lemma 20.16.4.25 (One-sided product obstruction in a full matrix algebra). [Lean](#) [✓ Stmt](#) *Let $\mathcal{A}_1, \mathcal{A}_2 \subseteq M_D(\mathbb{C})$ be nonzero linear subspaces. If $XY = 0$ for every $X \in \mathcal{A}_1$ and $Y \in \mathcal{A}_2$, then $\mathcal{A}_1 + \mathcal{A}_2 \neq M_D(\mathbb{C})$. No vanishing condition on products in the reverse order is required. This is the matrix-algebra obstruction used in [11, Appendix C.2, lines 1465–1470].*

Proof. [✓ Proof](#) Choose nonzero matrices $X \in \mathcal{A}_1$ and $Y \in \mathcal{A}_2$, together with nonzero entries X_{ip} and Y_{qj} . If the two subspaces spanned the full matrix algebra, write the matrix unit $E_{pq} = E_1 + E_2$ with $E_1 \in \mathcal{A}_1$ and $E_2 \in \mathcal{A}_2$. The one-sided product hypothesis gives

$$XE_{pq}Y = X(E_1Y) + (XE_2)Y = 0.$$

Its (i, j) entry is $X_{ip}Y_{qj} \neq 0$, a contradiction. $\square$

Definition 20.16.4.26 (Nonzero sector support and recurrence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The nonzero support graph of a neighboring-operator family has a directed edge $k \rightarrow h$ precisely when $\eta_{k,h} \neq 0$. Its support is recurrent when every such edge belongs to a directed cycle, equivalently when h can reach k whenever $k \rightarrow h$ is an edge.*

Definition 20.16.4.27 (Sector-coordinate virtual matrix families). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) In sector coordinates, let $A_{k;x,y}$ be the virtual matrix obtained from the (x, y) entry within sector k . The sector-coordinate family contains all transformed physical matrices, while the virtual-matrix family consists of all matrices $A_{k;x,y}$.

Theorem 20.16.4.28 (Sector spanning from injectivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let K be injective and let

$$U\kappa_{\beta,\alpha}U^\dagger = \bigoplus_k (l_k)_\beta \otimes (r_k)_\alpha$$

be a physical-sector factorization. Then the virtual matrices $A_{k;x,y}$ obtained from all within-sector physical entries span the full virtual matrix algebra $M_D(\mathbb{C})$.

Proof. [✓ Proof](#) Since U is unitary, every original physical matrix of K is a linear combination of the matrices in the transformed physical coordinates. The off-diagonal sector entries in those coordinates vanish, while each diagonal sector entry is one of the matrices $A_{k;x,y}$. Injectivity says that the original physical matrices span $M_D(\mathbb{C})$, and the conclusion follows. $\square$

Theorem 20.16.4.29 (Triangle closure from sector spanning). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a physical-sector factorization, let $A_{k;x,y} \in M_D(\mathbb{C})$ be the virtual matrix obtained from the physical matrix entry (x, y) within sector k . Suppose that all such matrices span $M_D(\mathbb{C})$ and that both endpoint sectors of every nonzero neighboring operator contain a nonzero matrix of this form. Then every nonzero $\eta_{k,h}$ admits a two-edge return: there is a sector j for which

$$\begin{aligned}\eta_{h,j} &\neq 0, \\ \eta_{j,k} &\neq 0.\end{aligned}$$

In particular, the nonzero neighboring support is recurrent.

Proof. [✓ Proof](#) A nonzero entry of $\eta_{k,h}$, together with nonzero sector matrices at k and h , gives matrices A from sector k and B from sector h with $AB \neq 0$. By nondegeneracy of the trace pairing, choose Y with $\text{tr}(ABY) \neq 0$. Expanding Y in the spanning family, some sector matrix C from a sector j satisfies $\text{tr}(ABC) \neq 0$. Cyclicity of the trace gives $BC \neq 0$ and $CA \neq 0$. The sector multiplication identity then forces $\eta_{h,j} \neq 0$ and $\eta_{j,k} \neq 0$. $\square$

Lemma 20.16.4.30 (Equivalent forms of sector reachability). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A vertex h is reachable from k in the nonzero sector support if and only if there is a finite directed walk from k to h . Consequently, the support is recurrent if and only if every directed edge admits a directed return walk.

Proof. [✓ Proof](#) A walk gives an element of the reflexive transitive closure by induction on its edges. Conversely, append the final edge at each induction step in the reflexive transitive closure. $\square$

Lemma 20.16.4.31 (Finite cyclic form of a closed directed walk). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Every nonempty closed directed walk of length N determines a cyclic family of vertices $v_0, \dots, v_{N-1}$. Each $v_n \rightarrow v_{n+1}$ is an edge, where $v_N = v_0$, and for any edge weights in a commutative monoid, $\text{wt}(w) = \prod_{n=0}^{N-1} \kappa_{v_n, v_{n+1}}$.

Proof. [✓ Proof](#) Enumerate the sources of the successive edges. The last edge ends at the initial vertex because the walk is closed. The product identity follows by induction on the number of edges. $\square$

Theorem 20.16.4.32 (Closed-walk criterion for vertex phases). [Lean](#) [✓ Stmt](#) Let E be a directed relation on a set V , and suppose that every edge $a \rightarrow b$ admits a directed return walk from b to a . Let G be a group and assign a weight $\kappa_{a,b} \in G$ to each ordered pair of vertices. If the product of the edge weights along every closed directed walk is one, then there are vertex weights $z_a \in G$ such that $\kappa_{a,b} = z_a^{-1} z_b$ whenever $a \rightarrow b$ is an edge. This is the graph-theoretic coboundary step needed to make the phase choices in [11, Appendix C.2, lines 1446–1450] coherent across a recurrent support component.

Proof. [✓ Proof](#) Reachability is an equivalence relation because every edge has a return walk. Choose one vertex in each reachability class and, for every vertex a , choose a walk from the class representative to a . Define z_a as the product of the edge weights along this walk. If two such walks have the same endpoints, append a common return walk; the two resulting closed walks have weight one, so the original walk weights agree. Comparing the chosen walk to b with the chosen walk to a followed by the edge $a \rightarrow b$ gives $z_b = z_a \kappa_{a,b}$. $\square$

Definition 20.16.4.33 (Horizontal bond fiber of a sector cycle). [Lean](#) [Lean](#) [✓ Stmt](#) For a cyclic sector assignment $(k_n)_{n \in \mathbb{Z}/N\mathbb{Z}}$, the horizontal bond fiber at n is

$$H_n = B_{k_n}^R \times B_{k_{n+1}}^L.$$

Shifting the left component by one site identifies $\prod_n H_n$ with the vertex-indexed sector fiber used in the cyclic neighboring-operator product.

Lemma 20.16.4.34 (A cyclic block is a finite Kronecker product). [Lean](#) [✓ Stmt](#) Reindexing a cyclic neighboring-operator block along its horizontal bond fiber gives the finite Kronecker product $\bigotimes_{n=0}^{N-1} \eta_{k_n, k_{n+1}}$.

Proof. [✓ Proof](#) Expand both sides entrywise. The shifted left component of the bond fiber at site $n+1$ is precisely the left index paired with the right component at site n . $\square$

Theorem 20.16.4.35 (Positive rescaling around one sector cycle). [Lean](#) [✓ Stmt](#) Let $N \geq 1$, and fix a cyclic sector assignment $k_0, \dots, k_{N-1}$ such that every operator $\eta_{k_n, k_{n+1}}$ is nonzero. If the corresponding cyclic tensor product is positive semidefinite, then there are nonzero scalars c_n satisfying

$$\prod_{n=0}^{N-1} c_n = 1, \\ c_n \eta_{k_n, k_{n+1}} \geq 0 \quad (0 \leq n < N).$$

This establishes a coherent positive rescaling along a single fixed cycle.

Proof. [✓ Proof](#) Reindex the cyclic sector block by the shared horizontal-bond indices. Its matrix entries then become the finite Kronecker product of the consecutive neighboring operators. Apply Theorem 20.16.4.24. $\square$

Corollary 20.16.4.36 (Positive rescaling of the concrete sector operators around one cycle). [Lean](#) [✓ Stmt](#) Let K be an injective MPDO, let R give a three-site closure, and fix an η -structure and a nonzero tail entry. Let $N \geq 1$. For a cyclic sector assignment $k_0, \dots, k_{N-1}$, suppose that every concrete operator $\eta_{k_n, k_{n+1}}$ obtained from these data is nonzero. Then there are nonzero scalars c_n such that

$$\prod_{n=0}^{N-1} c_n = 1, \\ c_n \eta_{k_n, k_{n+1}} \geq 0 \quad (0 \leq n < N).$$

This is the specialization of Theorem 20.16.4.35 to the inverse-map sector operators; positivity of the cyclic product follows from the MPDO property.

Proof. ✓ Proof The fixed-sector cyclic product is positive semidefinite by Theorem 20.16.4.38. Apply Theorem 20.16.4.35. □

Theorem 20.16.4.37 (The conjugated chain is a congruence). Lean Lean ✓ Stmt For every matrix U on the physical space and every $N \geq 1$,

$$\rho^{(N)}(\tilde{K}) = U^{\otimes N} \rho^{(N)}(K) (U^{\otimes N})^\dagger.$$

In particular $\rho^{(N)}(\tilde{K}) \geq 0$ whenever $\rho^{(N)}(K) \geq 0$.

Proof. ✓ Proof Expand the closed trace over bond configurations g (with $g_{N+1} = g_1$), and each conjugated local entry over its two physical indices:

$$\begin{aligned} [\rho^{(N)}(\tilde{K})]_{\sigma, \tau} &= \sum_g \prod_{n=1}^N \tilde{\kappa}_{g_n, g_{n+1}}^{\sigma_n \tau_n} \\ &= \sum_g \prod_{n=1}^N \sum_{p_n, q_n} U_{\sigma_n p_n} \kappa_{g_n, g_{n+1}}^{p_n q_n} \overline{U_{\tau_n q_n}}. \end{aligned}$$

Distributing the site product over the physical sums factors the two U -products out of the bond sum, which leaves the matrix entry of $U^{\otimes N} \rho^{(N)}(K) (U^{\otimes N})^\dagger$. Positivity is preserved by any congruence. □

Theorem 20.16.4.38 (Positivity of the projected chain blocks). Lean ✓ Stmt Let K be an MPDO whose transformed physical slices satisfy the sector factorization hypothesis. Then for every $N \geq 1$ and every cyclic sector assignment $k_1, \dots, k_N$ with $k_{N+1} = k_1$,

$$\begin{aligned} 0 &\leq [Q_{k_1} \otimes \dots \otimes Q_{k_N}] \tilde{\sigma} [Q_{k_1} \otimes \dots \otimes Q_{k_N}] \\ &= \bigotimes_{n=1}^N \eta_{k_n, k_{n+1}}. \end{aligned}$$

This is the positivity half of the display at [11, Appendix C.2, lines 1446–1450].

Proof. ✓ Proof The cyclic tensor product Ω_k of Definition 20.16.4.16 is a diagonal sector block of the basis-conjugated chain by Theorem 20.16.4.17. The conjugated chain is positive semidefinite by Theorem 20.16.4.37 and the MPDO property, and a principal submatrix of a positive semidefinite matrix is positive semidefinite. □

Theorem 20.16.4.39 (Positivity of the diagonal and paired neighboring operators). Lean Lean Lean ✓ Stmt Let K be an injective MPDO with the chosen Hayashi decomposition and a nonzero tail entry. Then for all sectors k, h :

1. $\eta_{k,k} \geq 0$;
2. $\eta_{k,h} \otimes \eta_{h,k} \geq 0$;
3. if $\eta_{k,h} \neq 0$ and $\eta_{h,k} \neq 0$, there is a scalar $c \neq 0$ with $c \eta_{k,h} \geq 0$, $c^{-1} \eta_{h,k} \geq 0$, and $(c \eta_{k,h}) \otimes (c^{-1} \eta_{h,k}) = \eta_{k,h} \otimes \eta_{h,k}$.

Proof. [✓ Proof](#) The chain of length one projected to the sector k is $\eta_{k,k}$ up to the exchange of its two tensor factors, and the chain of length two projected to (k, h) is $\eta_{k,h} \otimes \eta_{h,k}$, both positive by Theorem 20.16.4.38. The third item, including the Kronecker-product-preserving equality, is Theorem 20.16.4.19. $\square$

Theorem 20.16.4.40 (Coherent positive rephasing under recurrent support). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $(\eta_{k,h})$ be a neighboring-operator family for which every nonempty cyclic tensor product is positive semidefinite. Suppose, in addition, that every nonzero edge $k \rightarrow h$ admits a directed return path from h to k . Then there are unit complex numbers z_k such that $z_k^{-1} z_h \eta_{k,h} \geq 0$ for all k and h . Consequently, if the neighboring operators arise from the inverse-map physical-sector factorization of an injective MPDO and its nonzero support is recurrent, the sector tensors admit a reciprocal rephasing for which all neighboring operators are positive semidefinite. This factorization determines the positive commuting nearest-neighbor product structure of Proposition C.8.

This is a scope-restricted result. Recurrence is not a hypothesis of Lemma C.4 in [11, Appendix C.2, lines 1406–1450]. Its derivation for the zero-weight reparameterized inverse-map factorization is treated in Theorem 20.16.4.41 and recorded in [123].

Proof. [✓ Proof](#) For a nonzero edge, choose a directed return path and close it with the edge. Positive rescaling of the corresponding cyclic tensor product supplies a scalar that makes the edge operator positive; retain only its unit-modulus part. On any closed directed walk, positive rescaling of the associated cyclic tensor product gives phases whose product is one. Uniqueness of the unit phase identifies them edge by edge with the chosen phases. Hence the chosen edge phases have product one around every closed walk. The closed-walk criterion gives $\kappa_{k,h} = z_k^{-1} z_h$ on every nonzero edge. Zero edge operators are already positive semidefinite. Finally, reciprocal rephasing of the left and right sector tensors multiplies each neighboring operator by this phase and preserves the physical-slice factorization. The conditional η -local construction then gives the positive commuting product data. $\square$

Theorem 20.16.4.41 (Coherent positive choice of the neighboring operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let K be an injective MPO tensor satisfying SAL. Then it admits a physical-sector factorization for which every neighboring operator $\eta_{k,h} = \sum_a (r_k)_a \otimes (l_h)_a$ is positive semidefinite. Zero-weight Hayashi sectors remain in the physical direct sum, but their right tensors may be set to zero without changing the physical slices.

Proof. [✓ Proof](#) First set the right tensor to zero in every zero-weight sector. The left tensor already vanishes there, so this does not alter the factorization and removes every support edge incident to such a sector. Every remaining endpoint has positive weight. If all virtual matrices in a positive-weight sector vanished, then the corresponding middle-site block of the three-site closure would vanish; this contradicts its Hayashi expression as a nonzero weight times two trace-one sector states. Thus every endpoint of a nonzero edge contains a nonzero virtual matrix. Injectivity supplies spanning by all sector virtual matrices, and triangle closure gives recurrence. The projected chain identity makes every cyclic Kronecker product positive semidefinite. Apply Theorem 20.16.4.40 to obtain the coherent vertex rephasing. $\square$

Theorem 20.16.4.42 (Positivity choice with symmetric support). [Lean](#) [✓ Stmt](#) In the setting of Theorem 20.16.4.39, assume in addition that the nonzero neighboring operators occur in symmetric pairs: $\eta_{k,h} = 0$ forces $\eta_{h,k} = 0$. Then there is a positive semidefinite family $\eta'_{k,h}$ with $\eta'_{k,k} = \eta_{k,k}$ and, for every pair of sectors, scalars $c_{k,h} \neq 0$ with $\eta'_{k,h} = c_{k,h} \eta_{k,h}$ and $\eta'_{h,k} = c_{k,h}^{-1} \eta_{h,k}$. In particular every two-site block $\eta'_{k,h} \otimes \eta'_{h,k} = \eta_{k,h} \otimes \eta_{h,k}$ of the sector-adapted decomposition is unchanged.

The source asserts at [11, Appendix C.2, lines 1446–1450] that the η ’s can be chosen positive semidefinite, without the symmetric-support hypothesis. That hypothesis excludes a scalar obstruction: a nonzero neighboring operator whose reverse pair vanishes is constrained only by cycles through at least three sectors, and a coherent choice for the whole family can then fail.

Proof. ✓ Proof The diagonal operators are positive by Theorem 20.16.4.39, and their scalar is taken to be one. For a pair of distinct sectors, either both neighboring operators vanish and any nonzero scalar works, or both are nonzero by symmetric support and Theorem 20.16.4.39 supplies the reciprocal pair of scalars. Choosing the scalar once per unordered pair makes the two assignments reciprocal, and the block-preserving equality of Theorem 20.16.4.19 applied to that scalar gives $\eta'_{k,h} \otimes \eta'_{h,k} = \eta_{k,h} \otimes \eta_{h,k}$. □

20.16.5 Normalized preparations and controlled partial traces

Definition 20.16.5.1 (Trace matrices of explicit η -data). Lean Lean Lean Lean ✓ Stmt Every explicit neighboring η -family determines a sector-by-sector trace matrix. We give both its complex-valued form and the real-part version, which is the direct data for the real Perron–Frobenius matrix T used in the rank-one step.

Definition 20.16.5.2 (Rank-one factorization of neighboring-operator traces). Lean ✓ Stmt Let $(\eta_{k,h})$ be a positive neighboring-operator family. A rank-one trace factorization consists of real families (a_k) and (b_h) such that

$$\begin{aligned} \text{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_l a_l b_l &= 1. \end{aligned}$$

These identities occur in [11, Appendix C.2, lines 1395–1402].

Definition 20.16.5.3 (Direct-sum neighboring operator). Lean Lean ✓ Stmt For sectors k, h , define the positive-operator candidate

$$\Omega_{k,h} = \bigoplus_l (\eta_{k,l} \otimes \eta_{l,h}).$$

The summand indexed by l acts on $(B_k^R \otimes B_l^L) \otimes (B_l^R \otimes B_h^L)$. This is the operator adjointed by $\mathcal{T}_{k,h}$ in [11, Appendix C.2, lines 1527–1535].

Theorem 20.16.5.4 (Positivity of finite dependent block diagonals). Lean ✓ Stmt Let $(A_i)_{i \in I}$ be a finite family of positive semidefinite matrices, where the size of A_i may depend on i . Then the block-diagonal matrix $\bigoplus_{i \in I} A_i$ is positive semidefinite.

Proof. ✓ Proof Write each $A_i = B_i^* B_i$ using its positive square root. The dependent block diagonal then satisfies $\bigoplus_i A_i = (\bigoplus_i B_i)^* (\bigoplus_i B_i)$. □

Theorem 20.16.5.5 (Positivity of the direct-sum neighboring operator). Lean ✓ Stmt Every operator $\Omega_{k,h}$ is positive semidefinite.

Proof. ✓ Proof For every l , positivity is preserved by the Kronecker product: $\eta_{k,l} \succeq 0$ and $\eta_{l,h} \succeq 0$ imply $\eta_{k,l} \otimes \eta_{l,h} \succeq 0$. Apply Theorem 20.16.5.4 to the family $(\eta_{k,l} \otimes \eta_{l,h})_l$. □

Theorem 20.16.5.6 (Trace of the direct-sum neighboring operator). [Lean](#) [Lean](#) [✓ Stmt](#) The trace is

$$\text{tr}(\Omega_{k,h}) = \sum_l \text{tr}(\eta_{k,l}) \text{tr}(\eta_{l,h}).$$

Under the rank-one trace factorization, this becomes

$$\text{tr}(\Omega_{k,h}) = a_k b_h \sum_l a_l b_l = a_k b_h.$$

This is the trace calculation in [11, Appendix C.2, lines 1531–1535].

Proof. [✓ Proof](#) The direct-sum and Kronecker trace identities give

$$\text{tr}(\Omega_{k,h}) = \sum_l \text{tr}(\eta_{k,l} \otimes \eta_{l,h}) = \sum_l \text{tr}(\eta_{k,l}) \text{tr}(\eta_{l,h}).$$

Substitute $\text{tr}(\eta_{k,h}) = a_k b_h$ and $\sum_l a_l b_l = 1$. □

Theorem 20.16.5.7 (Vanishing of zero-weight neighboring operators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Every product $a_k b_h$ is non-negative. If $a_k b_h = 0$, then $\eta_{k,h} = 0$ and $\Omega_{k,h} = 0$.

Proof. [✓ Proof](#) Since $\eta_{k,h} \succeq 0$, its trace is non-negative, and hence $a_k b_h = \text{tr}(\eta_{k,h}) \geq 0$. If $a_k b_h = 0$, then $\eta_{k,h} \succeq 0$ and $\text{tr}(\eta_{k,h}) = a_k b_h = 0$, which imply $\eta_{k,h} = 0$. The same implication gives $\Omega_{k,h} = 0$ because $\Omega_{k,h} \succeq 0$ and $\text{tr}(\Omega_{k,h}) = a_k b_h = 0$. □

Definition 20.16.5.8 (Normalized active neighboring operators). [Lean](#) [Lean](#) [✓ Stmt](#) With the convention $0^{-1} = 0$, define for every pair (k, h)

$$\begin{aligned}\hat{\eta}_{k,h} &= (a_k b_h)^{-1} \eta_{k,h}, \\ \hat{\Omega}_{k,h} &= (a_k b_h)^{-1} \Omega_{k,h}.\end{aligned}$$

These operators are normalized when the pair is active, namely when $a_k b_h \neq 0$.

Theorem 20.16.5.9 (Positivity of normalized neighboring operators). [Lean](#) [Lean](#) [✓ Stmt](#) For every pair (k, h) , the operators $\hat{\eta}_{k,h}$ and $\hat{\Omega}_{k,h}$ are positive semidefinite.

Proof. [✓ Proof](#) By Theorem 20.16.5.7, $a_k b_h \geq 0$, and hence $(a_k b_h)^{-1} \geq 0$. Since $\eta_{k,h} \succeq 0$ and $\Omega_{k,h} \succeq 0$, multiplication by $(a_k b_h)^{-1}$ gives $\hat{\eta}_{k,h} \succeq 0$ and $\hat{\Omega}_{k,h} \succeq 0$. □

Theorem 20.16.5.10 (Normalized active preparations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $a_k b_h \neq 0$, then $\hat{\eta}_{k,h}$ and $\hat{\Omega}_{k,h}$ have trace one. Consequently the maps

$$\begin{aligned}X &\mapsto X \otimes \hat{\eta}_{k,h}, \\ X &\mapsto X \otimes \hat{\Omega}_{k,h}\end{aligned}$$

are trace-preserving and completely positive. These are the individual preparations $\mathcal{S}_{k,h}$ and $\mathcal{T}_{k,h}$ in [11, Appendix C.2, lines 1527–1535 and 1551–1555].

Proof. [✓ Proof](#) On an active pair, $\text{tr}(\hat{\eta}_{k,h}) = (a_k b_h)^{-1} \text{tr}(\eta_{k,h}) = (a_k b_h)^{-1} a_k b_h = 1$. Likewise, $\text{tr}(\hat{\Omega}_{k,h}) = 1$ follows from $\text{tr}(\Omega_{k,h}) = a_k b_h$. Apply Lemma [182, “State preparation is trace-preserving completely positive”] to each operator. □

Lemma 20.16.5.11 (Trace-one matrices have nonempty index spaces). [Lean](#) [✓ Stmt](#) *If a matrix indexed by a finite type has trace one, then its index type is nonempty.*

Proof. [✓ Proof](#) A matrix on an empty index space has trace zero, contrary to the hypothesis. $\square$

Lemma 20.16.5.12 (Nonemptiness of the Hayashi sector factors). [Lean](#) [Lean](#) [✓ Stmt](#) *Every left and right factor in a Hayashi sector is nonzero-dimensional.*

Proof. [✓ Proof](#) A matrix on an empty index space has trace zero. Hence $\text{tr}(\rho_k^R) = 1$ implies $\dim(B_k^R \otimes C) > 0$, which implies $\dim B_k^R > 0$, and the same argument applied to ρ_k^L gives $d_k^L > 0$. $\square$

Lemma 20.16.5.13 (Existence of a Hayashi sector). [Lean](#) [✓ Stmt](#) *The set of Hayashi sectors is nonempty.*

Proof. [✓ Proof](#) The sector-weight identity $\sum_k p_k = 1$ implies $m > 0$. $\square$

Lemma 20.16.5.14 (Nonemptiness of the neighboring auxiliary spaces). [Lean](#) [Lean](#) [✓ Stmt](#) *For every pair of sectors (k, h) , the spaces carrying $\eta_{k,h}$ and $\Omega_{k,h}$ are nonzero-dimensional.*

Proof. [✓ Proof](#) The tensor product carrying $\eta_{k,h}$ is nonzero-dimensional by Lemma 20.16.5.12. Choosing an intermediate sector l by Lemma 20.16.5.13 then supplies an element of the direct-sum space carrying $\Omega_{k,h}$. $\square$

Definition 20.16.5.15 (Faithful uniform density matrix). [Lean](#) [✓ Stmt](#) On a nonzero finite-dimensional space H , define

$$\tau_H = \frac{1}{\dim H} I_H.$$

Theorem 20.16.5.16 (The uniform density matrix is faithful and normalized). [Lean](#) [Lean](#) [✓ Stmt](#) *The matrix τ_H is positive definite and has trace one.*

Proof. [✓ Proof](#) Since $\dim H > 0$, the scalar $(\dim H)^{-1}$ is positive. Thus τ_H is positive definite, and $\text{tr}(\tau_H) = (\dim H)^{-1} \dim H = 1$. $\square$

Definition 20.16.5.17 (Completed neighboring-state preparation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every sector pair define

$$\bar{\eta}_{k,h} = \begin{cases} \hat{\eta}_{k,h}, & a_k b_h \neq 0, \\ \tau_{B_k^R \otimes B_h^L}, & a_k b_h = 0, \end{cases}$$

$$\bar{\mathcal{S}}_{k,h}(X) = X \otimes \bar{\eta}_{k,h}.$$

On an active pair this is precisely the preparation $\mathcal{S}_{k,h}$ of [11, Appendix C.2, lines 1551–1555]; the second branch gives a normalized extension on a pair where the displayed source quotient is undefined.

Theorem 20.16.5.18 (Completed neighboring-state preparations are trace-preserving completely positive). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every pair (k, h) , the operator $\bar{\eta}_{k,h}$ is positive semidefinite with trace one. Hence $\bar{\mathcal{S}}_{k,h}$ is trace-preserving and completely positive. If $a_k b_h \neq 0$, then both the density matrix and the preparation equal their normalized active forms.*

Proof. ✓ Proof The two branches give

$$\begin{aligned} a_k b_h \neq 0 &\implies \bar{\eta}_{k,h} = \hat{\eta}_{k,h} \succeq 0, \quad \text{tr}(\bar{\eta}_{k,h}) = 1, \\ a_k b_h = 0 &\implies \bar{\eta}_{k,h} = \tau_{B_k^R \otimes B_h^L} \succ 0, \quad \text{tr}(\bar{\eta}_{k,h}) = 1. \end{aligned}$$

The state-preparation result applies in either case. □

Definition 20.16.5.19 (Completed direct-sum preparation). Lean Lean Lean ✓ Stmt For every sector pair define

$$\begin{aligned} \mathcal{H}_{k,h}^\Omega &= \bigoplus_l (B_k^R \otimes B_l^L) \otimes (B_l^R \otimes B_h^L), \\ \bar{\Omega}_{k,h} &= \begin{cases} \hat{\Omega}_{k,h}, & a_k b_h \neq 0, \\ \tau_{\mathcal{H}_{k,h}^\Omega}, & a_k b_h = 0, \end{cases} \\ \bar{\mathcal{T}}_{k,h}(X) &= X \otimes \bar{\Omega}_{k,h}. \end{aligned}$$

Thus $\mathcal{H}_{k,h}^\Omega$ is the carrier space, whereas $\Omega_{k,h}$, $\hat{\Omega}_{k,h}$, and $\bar{\Omega}_{k,h}$ are operators on it: respectively the unnormalized neighboring operator, its normalization on an active pair, and its completed density on an arbitrary pair. On an active pair this is precisely the preparation $\mathcal{T}_{k,h}$ of [11, Appendix C.2, lines 1527–1535]; the inactive branch is a normalized extension.

Theorem 20.16.5.20 (Completed direct-sum preparations are trace-preserving completely positive). Lean Lean Lean Lean Lean ✓ Stmt For every pair (k, h) , the operator $\bar{\Omega}_{k,h}$ is positive semidefinite with trace one. Hence $\bar{\mathcal{T}}_{k,h}$ is trace-preserving and completely positive. On every active pair it equals the normalized active preparation.

Proof. ✓ Proof The two branches give

$$\begin{aligned} a_k b_h \neq 0 &\implies \bar{\Omega}_{k,h} = \hat{\Omega}_{k,h} \succeq 0, \quad \text{tr}(\bar{\Omega}_{k,h}) = 1, \\ a_k b_h = 0 &\implies \bar{\Omega}_{k,h} = \tau_{\mathcal{H}_{k,h}^\Omega} \succ 0, \quad \text{tr}(\bar{\Omega}_{k,h}) = 1. \end{aligned}$$

The state-preparation result therefore applies to $\bar{\mathcal{T}}_{k,h}$. □

Definition 20.16.5.21 (Globally controlled neighboring-state preparation). Lean ✓ Stmt Let $H = \bigoplus_{k,h} H_{k,h}$. Orthogonally controlling the completed preparations over the sector pairs gives a map

$$\bar{\mathcal{S}}_1(X) = \sum_{k,h,a} \tilde{A}_{k,h,a} X \tilde{A}_{k,h,a}^\dagger,$$

where $(A_{k,h,a})_a$ is a Kraus family for $\bar{\mathcal{S}}_{k,h}$ and the tilde denotes its extension by zero outside $H_{k,h}$. Thus the map discards inter-sector coherences and applies the completed preparation in each diagonal sector. On active sectors this is the sector-control formula for $\mathcal{S}_1$ in [11, Appendix C.2, lines 1548–1555].

Theorem 20.16.5.22 (Diagonal-sector action of the completed neighboring preparation). Lean ✓ Stmt If $X_{k,h}$ is the (k, h) diagonal block of X , then

$$[\bar{\mathcal{S}}_1(X)]_{k,h} = \bar{\mathcal{S}}_{k,h}(X_{k,h}) = X_{k,h} \otimes \bar{\eta}_{k,h}.$$

Proof. ✓ Proof

The diagonal-block formula and invariance under Kraus-index relabeling give

$$\begin{aligned} [\bar{\mathcal{S}}_1(X)]_{k,h} &= \Phi_{\tilde{A}_{k,h}}(X_{k,h}) = \Phi_{A_{k,h}}(X_{k,h}), \\ \Phi_{A_{k,h}}(X_{k,h}) &= X_{k,h} \otimes \bar{\eta}_{k,h}. \end{aligned}$$

□

Theorem 20.16.5.23 (The globally controlled neighboring-state preparation is trace-preserving completely positive). Lean ✓ Stmt *The map $\bar{\mathcal{S}}_1$ is trace-preserving and completely positive.*

Proof. ✓ Proof If $(A_{k,h,a})_a$ is the sectorwise Kraus family, then

$$\sum_{k,h,a} \tilde{A}_{k,h,a}^\dagger \tilde{A}_{k,h,a} = \bigoplus_{k,h} \left(\sum_a A_{k,h,a}^\dagger A_{k,h,a} \right) = \bigoplus_{k,h} I_{H_{k,h}} = I_H.$$

Thus the controlled Kraus family is trace-preserving. □

Definition 20.16.5.24 (Globally controlled direct-sum preparation). Lean ✓ Stmt Orthogonally controlling the completed direct-sum preparations gives

$$\bar{\mathcal{T}}_1(X) = \sum_{k,h,a} \tilde{B}_{k,h,a} X \tilde{B}_{k,h,a}^\dagger,$$

where $(B_{k,h,a})_a$ is a Kraus family for $\bar{\mathcal{T}}_{k,h}$. This map discards inter-sector coherences and, on active sectors, is the sector-control formula for $\mathcal{T}_1$ in [11, Appendix C.2, lines 1523–1535].

Theorem 20.16.5.25 (Diagonal-sector action of the completed direct-sum preparation). Lean ✓ Stmt *If $X_{k,h}$ is the (k,h) diagonal block of X , then*

$$[\bar{\mathcal{T}}_1(X)]_{k,h} = \bar{\mathcal{T}}_{k,h}(X_{k,h}) = X_{k,h} \otimes \bar{\Omega}_{k,h}.$$

Proof. ✓ Proof

The diagonal-block formula and invariance under Kraus-index relabeling give

$$\begin{aligned} [\bar{\mathcal{T}}_1(X)]_{k,h} &= \Phi_{\tilde{B}_{k,h}}(X_{k,h}) = \Phi_{B_{k,h}}(X_{k,h}), \\ \Phi_{B_{k,h}}(X_{k,h}) &= X_{k,h} \otimes \bar{\Omega}_{k,h}. \end{aligned}$$

□

Theorem 20.16.5.26 (The globally controlled direct-sum preparation is trace-preserving completely positive). Lean ✓ Stmt *The map $\bar{\mathcal{T}}_1$ is trace-preserving and completely positive.*

Proof. ✓ Proof For the sectorwise Kraus family $(B_{k,h,a})_a$,

$$\sum_{k,h,a} \tilde{B}_{k,h,a}^\dagger \tilde{B}_{k,h,a} = \bigoplus_{k,h} \left(\sum_a B_{k,h,a}^\dagger B_{k,h,a} \right) = \bigoplus_{k,h} I_{H_{k,h}} = I_H.$$

The orthogonal-control theorem now gives the claim. □

Proof. ✓ Proof

If Φ_2 is the two-site regrouping, then $\mathcal{S}_2 = R_{\Phi_2^{-1}}$. It is therefore trace-preserving and completely positive by Theorem [182, “Equivalence reindexing is trace-preserving completely positive”]. $\square$

Definition 20.16.5.33 (Three-site subspin regrouping). Lean Lean Lean Lean Lean Lean
Lean Lean Lean ✓ Stmt Regroup three sites by

$$((l_k, r_k), ((l_l, r_l), (l_h, r_h))) \mapsto ((l_k, r_h), ((r_k, l_l), (r_l, l_h))).$$

Equivalently,

$$\left(\bigoplus_k L_k \otimes R_k \right)^{\otimes 3} \simeq \bigoplus_{k,h} (L_k \otimes R_h) \otimes \left[\bigoplus_l (R_k \otimes L_l) \otimes (R_l \otimes L_h) \right].$$

The forward equivalence prepares the four middle subspins for $\mathcal{S}_0$; its inverse is the shift $\mathcal{T}_2$ in [11, Appendix C.2, lines 1535–1540 and 1547].

Definition 20.16.5.34 (The three-site partial-trace map $\mathcal{S}_0$). Lean Lean Lean ✓ Stmt Define $\mathcal{S}_0$ by the three-site regrouping followed, for every outer pair (k, h) , by the partial trace over

$$\bigoplus_l (R_k \otimes L_l) \otimes (R_l \otimes L_h).$$

Thus, writing R_{Φ_3} for the regrouping reindexing,

$$\mathcal{S}_0 = \text{CTr}_{\left(\bigoplus_l (R_k \otimes L_l) \otimes (R_l \otimes L_h) \right)_{k,h}} \circ R_{\Phi_3}. \quad (20.74)$$

Theorem 20.16.5.35 ($\mathcal{S}_0$ is trace-preserving completely positive). Lean Lean ✓ Stmt The map $\mathcal{S}_0$ is trace-preserving and completely positive.

Proof. ✓ Proof

By (20.74), $\mathcal{S}_0$ is the composition of the controlled dependent partial trace with reindexing along the three-site equivalence. The controlled trace and equivalence reindexing are trace-preserving and completely positive by Theorems [182, “The controlled dependent partial trace is trace-preserving completely positive”] and [182, “Equivalence reindexing is trace-preserving completely positive”], respectively. Their composition has the same property. $\square$

Definition 20.16.5.36 (The three-site shift $\mathcal{T}_2$). Lean Lean ✓ Stmt The map $\mathcal{T}_2$ is matrix reindexing along the inverse three-site equivalence. On each summand it sends

$$((l_k, r_h), ((r_k, l_l), (r_l, l_h))) \mapsto ((l_k, r_k), ((l_l, r_l), (l_h, r_h))).$$

Theorem 20.16.5.37 ($\mathcal{T}_2$ is trace-preserving completely positive). Lean Lean ✓ Stmt The shift $\mathcal{T}_2$ is trace-preserving and completely positive.

Proof. ✓ Proof

If Φ_3 is the three-site regrouping, then $\mathcal{T}_2 = R_{\Phi_3^{-1}}$. It is therefore trace-preserving and completely positive by Theorem [182, “Equivalence reindexing is trace-preserving completely positive”]. $\square$

Definition 20.16.5.38 (Fiberwise partial-trace maps). Lean Lean ✓ Stmt For fixed sectors (k, h) , define the fiberwise form of $\mathcal{T}_0$ by regrouping two sites and tracing $R_k \otimes L_h$. For fixed (k, l, h) , define the fiberwise form of $\mathcal{S}_0$ by regrouping three sites and tracing $(R_k \otimes L_l) \otimes (R_l \otimes L_h)$.

Theorem 20.16.5.39 (The fiberwise partial traces are trace-preserving completely positive).
Lean Lean ✓ Stmt *Every fiberwise form of $\mathcal{T}_0$ and $\mathcal{S}_0$ is trace-preserving and completely positive.*

Proof. ✓ Proof

The fiberwise forms factor as

$$\begin{aligned}\mathcal{T}_0^{(k,h)} &= \text{tr}_{R_k \otimes L_h} \circ R_{\phi_{k,h}}, \\ \mathcal{S}_0^{(k,l,h)} &= \text{tr}_{(R_k \otimes L_l) \otimes (R_l \otimes L_h)} \circ R_{\phi_{k,l,h}},\end{aligned}$$

where $\phi_{k,h}$ is the two-site equivalence of Definition 20.16.5.28 on the (k, h) sector, $\phi_{k,l,h}$ is the three-site equivalence of Definition 20.16.5.33 on the (k, l, h) sector, and R_ϕ denotes equivalence reindexing. Both factors in each composition are trace-preserving and completely positive, so the composition is as well. $\square$

Definition 20.16.5.40 (Fiberwise shifts). Lean Lean ✓ Stmt On fixed sector pairs, define $\mathcal{S}_2$ and $\mathcal{T}_2$ by the same inverse regroupings as the corresponding global shifts.

Theorem 20.16.5.41 (The fiberwise shifts are trace-preserving completely positive). Lean
Lean ✓ Stmt *Every fiberwise shift is trace-preserving and completely positive.*

Proof. ✓ Proof

The shifts $\mathcal{S}_2^{(k,h)}$ and $\mathcal{T}_2^{(k,h)}$ are reindexings along the inverse two-site and three-site equivalences of Definitions 20.16.5.28 and 20.16.5.33, respectively. They are therefore trace-preserving and completely positive by Theorem [182, “Equivalence reindexing is trace-preserving completely positive”]. $\square$

To record the domains and codomains, set

$$\mathcal{H}_{k,h}^\Omega = \bigoplus_l (B_k^R \otimes B_l^L) \otimes (B_l^R \otimes B_h^L),$$

and introduce the matrix algebras

$$\begin{aligned}\mathcal{A}_2 &= \text{End} \left(\left[\bigoplus_j (B_j^L \otimes B_j^R) \right]^{\otimes 2} \right), & \mathcal{A}_\partial &= \text{End} \left(\bigoplus_{k,h} B_k^L \otimes B_h^R \right), \\ \mathcal{A}_\eta &= \text{End} \left(\bigoplus_{k,h} [(B_k^L \otimes B_h^R) \otimes (B_k^R \otimes B_h^L)] \right), & \mathcal{A}_\Omega &= \text{End} \left(\bigoplus_{k,h} [(B_k^L \otimes B_h^R) \otimes \mathcal{H}_{k,h}^\Omega] \right), \\ \mathcal{A}_3 &= \text{End} \left(\left[\bigoplus_j (B_j^L \otimes B_j^R) \right]^{\otimes 3} \right).\end{aligned}$$

Thus the global maps have types

$$\begin{aligned}\mathcal{T}_0 &: \mathcal{A}_2 \rightarrow \mathcal{A}_\partial, \\ \mathcal{S}_2 &: \mathcal{A}_\eta \rightarrow \mathcal{A}_2, \\ \mathcal{S}_0 &: \mathcal{A}_3 \rightarrow \mathcal{A}_\partial, \\ \mathcal{T}_2 &: \mathcal{A}_\Omega \rightarrow \mathcal{A}_3.\end{aligned}$$

The controlled maps $\mathcal{T}_0$ and $\mathcal{S}_0$ discard coherences between distinct outer-sector pairs. For fixed (k, h) , the global map $\mathcal{S}_0$ traces the entire carrier $\mathcal{H}_{k,h}^\Omega$, including its direct sum over l . The

shifts $\mathcal{S}_2$ and $\mathcal{T}_2$ are global reindexings, with all displayed tensor factors present at their inputs [11, Appendix C.2, lines 1521–1559].

For the fiberwise maps, write

$$\begin{aligned}\mathcal{A}_2^{k,h} &= \text{End}((B_k^L \otimes B_k^R) \otimes (B_h^L \otimes B_h^R)), & \mathcal{A}_\partial^{k,h} &= \text{End}(B_k^L \otimes B_h^R), \\ \mathcal{A}_\eta^{k,h} &= \text{End}((B_k^L \otimes B_h^R) \otimes (B_k^R \otimes B_h^L)), & \mathcal{A}_\Omega^{k,h} &= \text{End}((B_k^L \otimes B_h^R) \otimes \mathcal{H}_{k,h}^\Omega), \\ \mathcal{A}_3^{k,l,h} &= \text{End}((B_k^L \otimes B_k^R) \otimes (B_l^L \otimes B_l^R) \otimes (B_h^L \otimes B_h^R)), \\ \mathcal{A}_3^{k,\bullet,h} &= \text{End}\left(\bigoplus_l [(B_k^L \otimes B_k^R) \otimes (B_l^L \otimes B_l^R) \otimes (B_h^L \otimes B_h^R)]\right).\end{aligned}$$

The map $\mathcal{S}_0^{(k,l,h)}$ fixes the middle sector l before taking the partial trace; it is therefore distinct from the fixed- (k,h) restriction of the global controlled map $\mathcal{S}_0$. The four fiberwise types are

$$\mathcal{A}_2^{k,h} \xrightarrow{\mathcal{T}_0^{(k,h)}} \mathcal{A}_\partial^{k,h}$$

$$\mathcal{A}_\eta^{k,h} \xrightarrow{\mathcal{S}_2^{(k,h)}} \mathcal{A}_2^{k,h}$$

$$\mathcal{A}_3^{k,l,h} \xrightarrow{\mathcal{S}_0^{(k,l,h)}} \mathcal{A}_\partial^{k,h}$$

$$\mathcal{A}_\Omega^{k,h} \xrightarrow{\mathcal{T}_2^{(k,h)}} \mathcal{A}_3^{k,\bullet,h}$$

20.16.6 Primitivity and rank-one trace matrices

Definition 20.16.6.1 (Explicit η -data from the sector tensors). [Lean](#) [✓ Stmt](#) If every neighboring operator $\eta_{k,h}$ of Definition 20.16.3.1 is positive semidefinite, the family constitutes explicit neighboring η -data. The source obtains the positivity from the projected chain identity: writing $\tilde{\sigma} = U^{\dagger \otimes N} \sigma^{(N)}(K) U^{\otimes N}$ for the N -site operator in the sector basis and Q_k for the projector onto the k -th sector of the physical splitting, every cyclic sector assignment $k_1, \dots, k_N$ (with $k_{N+1} = k_1$) satisfies

$$0 \leq [Q_{k_1} \otimes \dots \otimes Q_{k_N}] \tilde{\sigma} [Q_{k_1} \otimes \dots \otimes Q_{k_N}] = \bigotimes_{n=1}^N \eta_{k_n, k_{n+1}},$$

and the sector tensors can be chosen so that each factor is positive semidefinite [11, Appendix C.2, lines 1446–1450]. The all-length sector identity is Theorem 20.16.4.18, with positive semidefiniteness assumed for the chosen representatives here. Theorem 20.16.4.39 provides the diagonal and paired positivity, and Theorem 20.16.4.42 a rescaled positive family under symmetric support.

Theorem 20.16.6.2 (Trace matrix of the sector-tensor η -data). [Lean](#) [✓ Stmt](#) Under the hypotheses of Definition 20.16.6.1, the trace matrix of the explicit η -data is the pairing of the closed sector tensors:

$$T_{k,h} = \text{tr}(\eta_{k,h}) = (r_k | l_h).$$

Proof. [✓ Proof](#) Immediate from Theorem 20.16.4.3 and the definition of the trace matrix. $\square$

Theorem 20.16.6.3 (Primitivity from returns of lengths two and three). [Lean](#) [✓ Stmt](#) Let T be a non-negative irreducible matrix on a finite index set. If $(T^2)_{k,k} > 0$ and $(T^3)_{k,k} > 0$ for every index k , then T is primitive.

Proof. [✓ Proof](#) Choose a positive path between every ordered pair of indices, and let L be the sum of the chosen path lengths. Every chosen length is at most L . For a chosen path of length m , write the difference $2L + 2 - m$ as $2a + 3b$ with $a, b \geq 0$. Appending these returns at the terminal index gives a positive path of common length $2L + 2$ between every ordered pair. $\square$

Definition 20.16.6.4 (Nonzero-weight sector trace matrix). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For sector weights p_k , let $I_* = \{k : p_k \neq 0\}$. The one-site matrices and their spaces on I_* are

$$A_{k;x,y}, \\ A_S = \text{span}\{A_{k;x,y} : k \in S\}, \quad S \subseteq I_*.$$

The real trace matrix on this index set is

$$T_{k,h}^* = \text{Re tr}(\eta_{k,h}), \quad k, h \in I_*.$$

Only the trace matrix is restricted: the physical direct sum still contains every zero-weight summand.

Theorem 20.16.6.5 (Identification and spanning of active one-site matrices). [Lean](#) [Lean](#) [✓ Stmt](#) For every sector entry, the one-site matrix used in the directed-cut argument equals the virtual matrix used in the spanning argument. If all sector virtual matrices span $M_D(\mathbb{C})$ and the matrices in every zero-weight sector vanish, then $\text{span}\{A_{k;x,y} : k \in I_*\} = M_D(\mathbb{C})$.

Theorem 20.16.6.6 (Complementary active-sector spaces). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the active one-site matrices span $M_D(\mathbb{C})$ and each active sector contains a nonzero such matrix. For every $S \subseteq I_*$, $A_S + A_{I_* \setminus S} = M_D(\mathbb{C})$, and both spaces are nonzero when the corresponding sector sets are nonempty. If $\eta_{k,h} = 0$ for $k \in S$ and $h \in C$, then $A_S A_C = 0$.

Theorem 20.16.6.7 (Strong connectivity of the active neighboring support). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the active one-site matrices span $M_D(\mathbb{C})$ and each active sector contains a nonzero such matrix. Then for every $k, h \in I_*$ there is a directed walk

$$k = k_0 \longrightarrow k_1 \longrightarrow \cdots \longrightarrow k_m = h,$$

such that $\eta_{k_j, k_{j+1}} \neq 0$. In particular, every active sector has an outgoing nonzero neighboring operator.

Theorem 20.16.6.8 (Positive entries of the active trace matrix). [Lean](#) [Lean](#) [✓ Stmt](#) If every neighboring operator is positive semidefinite, then T^* is entrywise non-negative and $T_{k,h}^* > 0 \iff \eta_{k,h} \neq 0$.

Theorem 20.16.6.9 (Two-step return through every active sector). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the active one-site matrices span $M_D(\mathbb{C})$ and each active sector contains a nonzero such matrix. For every $k \in I_*$ there is an $h \in I_*$ such that

$$\eta_{k,h} \neq 0, \\ \eta_{h,k} \neq 0.$$

If the neighboring operators are positive semidefinite, then $(T^{*2})_{k,k} > 0$.

$$P_j \mathcal{K}_i P_{j'} = 0 \quad \text{unless } i = j = j'.$$

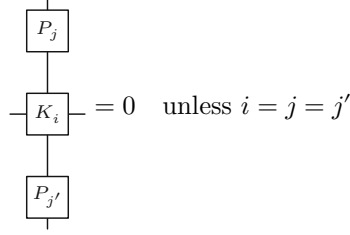


Figure 20.5: The local-orthogonality contraction of [11, Appendix C.2, lines 1457–1470]. After decomposing the sector trace matrix into primitive components, the source assumes that two components occur, obtains two locally orthogonal summands, and chooses orthogonal projectors P_1 and P_2 . The displayed identity gives the cross contractions $P_1 K P_2 = P_2 K P_1 = 0$, and the corresponding two-site contraction also vanishes. For $A_i = \text{span}\{\text{tr}(X P_i K P_i) : X\}$, these vanishings and injectivity imply $A_1 + A_2 = M_D(\mathbb{C})$ and $A_1 A_2 = 0$, a contradiction. Theorem 20.16.6.7 does not construct P_1 or P_2 : an arbitrary reachability cut $I_* = S \sqcup C$ supplies only $A_S A_C = 0$. The common final step is the same one-sided obstruction: two nonzero matrix spaces cannot sum to $M_D(\mathbb{C})$ while their ordered products vanish.

Theorem 20.16.6.10 (Irreducibility of the active trace matrix). [Lean](#) [✓ Stmt](#) Suppose that the neighboring operators are positive semidefinite, the active one-site matrices span $M_D(\mathbb{C})$, and each active sector contains a nonzero such matrix. Then the non-negative matrix T^* is irreducible.

Theorem 20.16.6.11 (Three-step returns and a common positive power). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the neighboring operators are positive semidefinite, the active one-site matrices span $M_D(\mathbb{C})$, each active sector contains a nonzero such matrix, and every nonzero edge closes through a third active sector. Then $(T^{*3})_{k,k} > 0$ for every $k \in I_*$. Together with the length-two returns and irreducibility, this implies that T^* is primitive.

Proof. [✓ Proof](#) Triangle closure supplies a positive closed walk of length three through each active sector. Combine these walks with the length-two returns and apply Theorem 20.16.6.3. $\square$

Theorem 20.16.6.12 (Existence of a positive factorization with full active span). [Lean](#) [Lean](#) [✓ Stmt](#) Let K be an injective tensor generating MPDOs, let R close a three-site state ρ , and choose a Hayashi decomposition with weights p_k . Fix outer indices for which the corresponding entry of R is nonzero, and let F be the zero-weight reparameterized inverse-map factorization. There are coherent sector phases for which all neighboring operators are positive semidefinite and the active one-site matrices span $M_D(\mathbb{C})$. Every active sector contains a nonzero virtual matrix, every virtual matrix in a zero-weight sector vanishes, every nonzero neighboring edge closes through a third active sector, every inactive left tensor vanishes, and I_* is nonempty.

Proof. [✓ Proof](#) Choose the coherent positive phases. Injectivity gives the full virtual matrix span, while zero-weight reparameterization removes the inactive summands from the active span. A nonzero virtual matrix exists in every active sector, and the neighboring-support recurrence gives the three-edge return. Rephasing preserves the nonzero neighboring edges, so this return persists after the coherent rephasing. Inactive left tensors vanish by construction. Finally, $\sum_k p_k = 1$ gives a nonempty active sector. $\square$

Theorem 20.16.6.13 (Primitivity of the active sector trace matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let K be an injective tensor that generates MPDOs and satisfies SAL, and choose the positive semidefinite neighboring operators of Theorem 20.16.6.12. Restrict the sector index to the positive-weight summands of the Hayashi decomposition. Then positive semidefiniteness makes every neighboring trace real and non-negative, and the matrix

$$T_{k,h}^* = \text{Re tr}(\eta_{k,h})$$

is primitive. Zero-weight summands remain in the physical direct sum but are not indices of T^* .

Proof. [✓ Proof](#) Theorem 20.16.2.20 supplies the four-site Hayashi decomposition, and Theorem 20.16.2.41 supplies nonzero outer indices for the normalized four-site tail. Its reduced block state gives the required three-site closure; form the zero-weight reparameterized inverse-map factorization. Then Theorem 20.16.6.12 supplies one rephased factorization with positive neighboring operators, full active span, nonzero active virtual matrices, and the three-edge return condition. Apply Theorem 20.16.6.11 to conclude that the active trace matrix is primitive. $\square$

Theorem 20.16.6.14 (Zero-correlation-length relations on the primitive active trace matrix). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let K be injective and suppose that it satisfies SAL and source zero correlation length. Then there are a physical-sector factorization, weights $p_k \geq 0$ with $\sum_k p_k = 1$, and positive semidefinite neighboring operators $\eta_{k,h}$ such that the active trace matrix

$$T_{k,h}^* = \text{Re tr}(\eta_{k,h}), \quad k, h \in I_* := \{j : p_j \neq 0\},$$

is primitive. There is also a real number $\lambda > 0$ for which, upon setting $\widehat{T}^* := \lambda^{-1}T^*$, one has $(\widehat{T}^*)^2 = (\widehat{T}^*)^3$. For every positive integer N , $\text{tr}((\widehat{T}^*)^N) = \text{tr}(\widehat{T}^*)$. Thus the primitivity and the two normalized identities concern the same active, positively rephased trace matrix.

Proof. [✓ Proof](#) Theorem 20.16.6.12 gives one rephased factorization with positive neighboring operators and vanishing left tensors on the zero-weight sectors. Apply Theorem 20.16.6.11 to obtain primitivity of its active trace matrix. Lemma 20.6.10 supplies normalized idempotence of the physical-trace transfer, from which the same active trace matrix satisfies the square-cube identity and the positive-power trace identities. $\square$

Lemma 20.16.6.15 (SAL makes the physical-trace transfer nonzero). [Lean](#) [✓ Stmt](#) If an MPO tensor K satisfies SAL, then its physical-trace transfer is nonzero:

$$\mathcal{T}_K \neq 0.$$

Proof. [✓ Proof](#) The SAL normalization at one site gives a nonzero trace for the one-site periodic operator. Lemma 16.4.56 identifies this trace with that of $\mathcal{T}_K$, which therefore cannot vanish. $\square$

Theorem 20.16.6.16 (SAL and literal idempotence imply source zero correlation length). [Lean](#) [✓ Stmt](#) If K satisfies SAL and

$$\mathcal{T}_K^2 = \mathcal{T}_K,$$

then K has source zero correlation length.

Proof. [✓ Proof](#) The preceding lemma gives $\mathcal{T}_K \neq 0$. Apply Theorem 20.6.9. $\square$

Theorem 20.16.6.17 (Literal-ZCL specialization of the active trace relations). [Lean](#) [✓ Stmt](#)
Let K be injective and satisfy SAL and the literal identity

$$\mathcal{T}_K^2 = \mathcal{T}_K.$$

Then K admits a physical-sector factorization, nonnegative weights of sum one, and positive semidefinite neighboring operators for which the active trace matrix T^* is primitive. For some $\lambda > 0$, the normalized matrix $\widehat{T}^* = \lambda^{-1}T^*$ satisfies $(\widehat{T}^*)^2 = (\widehat{T}^*)^3$ and, for every positive integer N ,

$$\mathrm{tr}((\widehat{T}^*)^N) = \mathrm{tr}(\widehat{T}^*).$$

Proof. [✓ Proof](#) SAL makes the one-site periodic trace nonzero, and hence $\mathcal{T}_K \neq 0$. Theorem 20.16.6.16 therefore gives source zero correlation length. Apply Theorem 20.16.6.14. $\square$

Theorem 20.16.6.18 (Virtual spanning does not force active-sector idempotence). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) There is a tensor of bond dimension two with four one-dimensional physical sectors and weights p_k such that the following statements hold simultaneously: the tensor is injective and has source zero correlation length; its physical-trace transfer is idempotent; for every $k \in \{0, 1, 2, 3\}$, the weights satisfy

$$\begin{aligned} p_k &= \frac{1}{4}, \\ 0 &\leq p_k, \\ \sum_{k=0}^3 p_k &= 1; \end{aligned}$$

its sector virtual matrices span $M_2(\mathbb{C})$; every neighboring operator is positive semidefinite; and its active trace matrix is primitive and has trace one. Nevertheless, if $S = LQ$ is the physical-trace transfer and $T = QL$ is the active trace matrix, then $Q(1 - S)L = T - T^2 \neq 0$. In particular, full virtual-matrix spanning does not by itself imply that the active trace matrix is idempotent.

Proof. [✓ Proof](#) Take

$$L = \begin{pmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 1 & -1 & 1 & -1 \end{pmatrix},$$

$$Q = \begin{pmatrix} 1 & \frac{1}{8} \\ 1 & \frac{1}{8} \\ 1 & -\frac{1}{8} \\ 1 & -\frac{1}{8} \end{pmatrix}.$$

For each k , the k th sector virtual matrix is the outer product of the k th column of L with the k th row of Q . Their four coordinate functions are proportional to the Walsh functions $1, x, y, xy$ on four points. More explicitly, the matrix whose columns are their coordinates in the order $(00, 01, 10, 11)$ has determinant $-1/64$. Hence they form a basis of $M_2(\mathbb{C})$. Placing them on the diagonal physical entries gives an injective tensor and a physical-sector factorization with scalar sectors.

Direct multiplication gives

$$LQ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

$$QL = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Thus LQ is idempotent, whereas every entry of QL is strictly positive, so QL is primitive. Its trace is one. Since each sector is one-dimensional, the neighboring operators are the scalar matrices $\eta_{k,h} = ((QL)_{k,h}) \geq 0$.

The $(0, 1)$ entry of $(QL)^2$ is $1/4$, while the corresponding entry of QL is $1/8$. Hence QL is not idempotent, and associativity gives $Q(1 - LQ)L = QL - (QL)^2 \neq 0$. $\square$

Theorem 20.16.6.19 (A one-sector factorization of the four-sector tensor). [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) The tensor in Theorem 20.16.6.18 also admits a physical-sector factorization with one sector. Write each physical label as a pair $(x, y) \in \{0, 1\}^2$ and put $Z = \text{diag}(1, -1)$. The left and right factors are

$$\begin{aligned} l_0 &= \frac{1}{4}I, & l_1 &= Z, \\ r_0 &= I, & r_1 &= \frac{1}{8}Z. \end{aligned}$$

Its sole neighboring operator is

$$\eta = I \otimes \frac{1}{4}I + \frac{1}{8}Z \otimes Z = \text{diag}\left(\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8}\right).$$

In particular, η is positive semidefinite and $\text{tr}(\eta) = 1$. Taking $a = b = 1$ gives the normalized rank-one trace factorization. Thus the four sectors selected by the inverse-map construction are nonminimal; their non-rank-one trace matrix is not an obstruction to the existence of another physical-sector factorization for this tensor.

Proof. ✓ Proof Reindex the four physical basis vectors by parity and quotient. The diagonal entries of the tensor then factor as $(l_\beta)_{x,x}(r_\alpha)_{y,y}$, and all off-diagonal entries vanish. Contracting the common virtual label gives the displayed expression for η . Its four diagonal entries prove positivity and trace one. $\square$

Theorem 20.16.6.20 (Pair-preserving trace obstruction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✔ Stmt](#) Let $\eta_{k,h}$ be the neighboring operators of a physical-sector factorization, and define

$$\begin{aligned} C_{k,h} &= \text{tr}(\eta_{k,h}), \\ \Theta_{k,h} &= \bigoplus_{l,m} \eta_{k,l} \otimes \eta_{l,m} \otimes \eta_{m,h}. \end{aligned}$$

Then

$$\mathrm{tr}(\Theta_{k,h}) = (C^3)_{k,h}.$$

Suppose that, for every fixed pair (k, h) , there is either a trace-preserving linear map from the four-site middle algebra to the two-site middle algebra which sends $\Theta_{k,h}$ to $\eta_{k,h}$, or a trace-preserving linear map in the reverse direction which sends $\eta_{k,h}$ to $\Theta_{k,h}$. If the same direction is chosen for all pairs, then

$$C = C^3.$$

Proof. [✓ Proof](#) Multiplicativity of the trace on tensor products and additivity on direct sums give

$$\text{tr}(\Theta_{k,h}) = \sum_{l,m} C_{k,l} C_{l,m} C_{m,h} = (C^3)_{k,h}.$$

Trace preservation identifies this quantity with $\text{tr}(\eta_{k,h}) = C_{k,h}$ in either direction. The entrywise identities give $C = C^3$. $\square$

Theorem 20.16.6.21 (Doubled-index self-overlap is the Hilbert–Schmidt trace). [Lean](#) [Lean](#) [✓ Stmt](#) Let K be an MPO tensor. For every chain length N , the self-overlap of the doubled-index MPS family equals the Hilbert–Schmidt trace of the closed operator:

$$O_{K^{\text{MPS}} K^{\text{MPS}}}(N) = \text{tr}(\rho^{(N)}(K) \rho^{(N)}(K)^\dagger).$$

This is a project derivation preparing the Lemma C.5 route of [\[11, Appendix C.2, lines 1473–1499\]](#); the pure-state analogue is the normalized-MPV self-overlap lemma of [\[11, lines 1080–1100\]](#). It is not a statement of that source. Until a trace normalization is established, the right-hand side is the raw squared Hilbert–Schmidt norm of the MPO family, not a normalized-state purity.

Proof. [✓ Proof](#) Both sides are sums over physical configurations. Expanding the trace of the product gives the Frobenius inner product

$$\text{tr}(\rho^{(N)}(K) \rho^{(N)}(K)^\dagger) = \sum_{\sigma, \tau} \rho^{(N)}(K)_{\sigma, \tau} \overline{\rho^{(N)}(K)_{\sigma, \tau}}$$

over pairs (σ, τ) of ket and bra configurations, while the overlap sums $V^{(N)}(K^{\text{MPS}})_\rho \overline{V^{(N)}(K^{\text{MPS}})_\rho}$ over doubled-index words ρ . The product encoding of Definition 16.1.2 reindexes each configuration pair (σ, τ) as the doubled-index word $n \mapsto (\sigma_n, \tau_n)$, and decoding each doubled index as a ket–bra pair identifies the summands: $V^{(N)}(K^{\text{MPS}})_{(\sigma, \tau)} = \rho^{(N)}(K)_{\sigma, \tau}$. $\square$

Corollary 20.16.6.22 (Self-overlap as the trace of the square for an MPDO). [Lean](#) [✓ Stmt](#) If K generates MPDOs and $N \geq 1$, then

$$O_{K^{\text{MPS}} K^{\text{MPS}}}(N) = \text{tr}(\rho^{(N)}(K) \rho^{(N)}(K)).$$

Proof. [✓ Proof](#) Positivity on the nonempty chain makes $\rho^{(N)}(K)$ positive semidefinite, hence Hermitian. Substituting $\rho^{(N)}(K)^\dagger = \rho^{(N)}(K)$ into Theorem 20.16.6.21 gives the identity. $\square$

Corollary 20.16.6.23 (Real part of the self-overlap is the squared Frobenius norm). [Lean](#) [✓ Stmt](#) For every MPO tensor K and every chain length N ,

$$\text{Re } O_{K^{\text{MPS}} K^{\text{MPS}}}(N) = \|\rho^{(N)}(K)\|_F^2 = \sum_{\sigma, \tau} |\rho^{(N)}(K)_{\sigma, \tau}|^2.$$

Proof. [✓ Proof](#) Cyclicity of the trace rewrites the Hilbert–Schmidt trace as $\text{tr}(\rho^{(N)}(K)^\dagger \rho^{(N)}(K))$, whose real part is the sum of the squared absolute entries. $\square$

Theorem 20.16.6.24 (Trace factorization of the cyclic neighboring product). [Lean](#) [Lean](#) [✓ Stmt](#)
 Let $N \geq 1$ and let $k = (k_0, \dots, k_{N-1})$ be a cyclic sector configuration of a physical-sector factorization. The trace of the cyclic neighboring product factors into the product of the neighboring traces, and so does the trace of its square:

$$\begin{aligned}\mathrm{tr}(\Omega_k) &= \prod_{n=0}^{N-1} \mathrm{tr}(\eta_{k_n, k_{n+1}}), \\ \mathrm{tr}(\Omega_k^2) &= \prod_{n=0}^{N-1} \mathrm{tr}(\eta_{k_n, k_{n+1}}^2),\end{aligned}$$

with site labels read modulo N . These are the coefficient-extraction identities for the cyclic direct-sum form of [11, Appendix C.2, lines 1580–1593].

Proof. [✓ Proof](#) The fiber bijection of Definition 20.16.3.23 identifies the diagonal entries of Ω_k with tuples of diagonal entries of the neighboring factors. Hence

$$\mathrm{tr}(\Omega_k) = \sum_x \prod_{n=0}^{N-1} \eta_{k_n, k_{n+1}}(x_n, x_n) = \prod_{n=0}^{N-1} \sum_{x_n} \eta_{k_n, k_{n+1}}(x_n, x_n),$$

expanding the product of sums. For the square, write $\mathrm{tr}(\Omega_k^2) = \sum_{x,y} \Omega_k(x, y) \Omega_k(y, x)$; the same reindexing and expansion turn the inner sums into the diagonal entries of $\eta_{k_n, k_{n+1}}^2$. $\square$

Lemma 20.16.6.25 (Trace-squared matrix of the neighboring operators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let K be an MPO tensor with a physical-sector factorization and sector weights p_k , and suppose that every neighboring operator $\eta_{k,h}$ is positive semidefinite. On the active sector set I_* , define the trace-squared matrix

$$S_{k,h}^* := \mathrm{Re} \, \mathrm{tr}(\eta_{k,h}^2), \quad k, h \in I_*.$$

Then, entrywise,

$$0 \leq S_{k,h}^* \leq (T_{k,h}^*)^2.$$

This matrix is a project derivation preparing the Case-I route toward [11, Appendix C.2, Lemma C.5, lines 1473–1499]; it is not a matrix defined in that source.

Proof. [✓ Proof](#) Hermiticity of $\eta_{k,h}$ gives $\eta_{k,h}^2 = \eta_{k,h} \eta_{k,h}^\dagger$, which is positive semidefinite, so its trace is a non-negative real number. The upper bound is the corresponding theorem in the companion quantum-channel volume [182, “Forward Lorentz-cone trace inequality”] applied to $\eta_{k,h}$. $\square$

Theorem 20.16.6.26 (Literal zero correlation length stabilizes the active trace matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let K be an MPO tensor with a physical-sector factorization and sector weights p_k . Suppose that every neighboring operator is positive semidefinite, that the physical-trace transfer is literally idempotent,

$$\mathcal{T}_K^2 = \mathcal{T}_K,$$

as in [11, Definition 4.2, lines 735–741], and that the left tensor of every zero-weight sector vanishes. Then the active trace matrix stabilizes after the second power:

$$(T^*)^2 = (T^*)^3.$$

If, in addition, the active one-site matrices span $M_D(\mathbb{C})$, each active sector contains a nonzero such matrix, every nonzero neighboring operator closes through a third active sector, and there is at least one active sector, then $(T^*)^2$ is strictly positive entrywise.

These are matrix-algebra components of the Case-I argument under the assumptions of [11, Appendix C.2, lines 1374–1381], for the factorization supplied by [11, Appendix C.2, Lemma C.4, lines 1406–1471] and the zero-correlation-length step of [11, Appendix C.2, Lemma C.5, lines 1473–1499]. They are project derivations, not statements of that source.

Proof. ✓ Proof On the active sectors, collect the left and right sector-tensor traces into rectangular matrices L and Q with

$$L_{\beta,h} = \text{tr}((l_h)_\beta), \quad Q_{k,\alpha} = \text{tr}((r_k)_\alpha).$$

The left tensor vanishes on every zero-weight sector, so the rectangular product LQ is the full physical-trace transfer, and literal zero correlation length makes LQ idempotent. Associativity then gives

$$(QL)^2 = Q(LQ)L = Q(LQ)(LQ)L = (QL)^3.$$

Positivity of the neighboring operators makes their traces real, and expanding the products identifies QL with the complexification of T^* ; injectivity of the real-to-complex inclusion gives $(T^*)^2 = (T^*)^3$.

Under the additional hypotheses, Theorem 20.16.6.11 makes T^* primitive: some power $m \geq 1$ has strictly positive entries. Stabilization gives $(T^*)^{2m} = (T^*)^2$, so

$$((T^*)^2)_{k,h} = \sum_j ((T^*)^m)_{k,j} ((T^*)^m)_{j,h} > 0,$$

which is the positivity argument of Theorem 20.16.6.55. □

Lemma 20.16.6.27 (Indicator-path expansion of a matrix power). Lean ✓ Stmt Let S be a square matrix with entries in a commutative semiring, indexed by a finite set I , and let $N \geq 0$. Then, for $a, b \in I$,

$$(S^N)_{a,b} = \sum_{\substack{v: \{0, \dots, N\} \rightarrow I \\ v_0 = a, v_N = b}} \prod_{i=0}^{N-1} S_{v_i, v_{i+1}},$$

the sum, over length- N walks v in I from a to b , of the product of matrix entries along the walk. This is an elementary combinatorial identity used to derive the cyclic-product form of $\text{tr}(S^N)$ in the Case-I route.

Proof. ✓ Proof Induction on N : the base case is the identity-matrix entry $\delta_{a,b}$, indexed by the unique length-0 walk; the inductive step expands $(S^{N+1})_{a,b} = \sum_c (S^N)_{a,c} S_{c,b}$ and reindexes the walk-and-endpoint pair (v, c) as the length- $(N+1)$ walk obtained by appending c to v . □

Theorem 20.16.6.28 (Trace of a matrix power as a sum over cyclic products). Lean ✓ Stmt Let S be a square matrix with entries in a commutative semiring, indexed by a finite set I , and let $N \geq 1$. Then

$$\text{tr}(S^N) = \sum_{k: \{0, \dots, N-1\} \rightarrow I} \prod_{i=0}^{N-1} S_{k_i, k_{i+1}},$$

with indices read modulo N , the sum over labelings k of the N cyclic positions of the product of matrix entries along the cycle. Applied to the trace-squared matrix S^* of Lemma 20.16.6.25, this is the combinatorial identity used to express the self-overlap $O_{K^{\text{MPS}} K^{\text{MPS}}}(N)$ in terms of $\text{tr}((S^*)^N)$.

Proof. ✓ Proof Expand $\text{tr}(S^N) = \sum_a (S^N)_{a,a}$ by Lemma 20.16.6.27, collapse the sum over the shared endpoint $a = v_0 = v_N$, and reindex the resulting closed length- N walks v (with $v_0 = v_N$) as cyclic labelings $k : \{0, \dots, N-1\} \rightarrow I$ by dropping the repeated final coordinate. $\square$

Lemma 20.16.6.29 (Trace invariance under the physical basis change). Lean Lean ✓ Stmt Let K be an MPO tensor and U an isometry on the physical space, $U^\dagger U = 1$. Then

$$\text{tr}(\rho^{(N)}(\tilde{K}) \rho^{(N)}(\tilde{K})) = \text{tr}(\rho^{(N)}(K) \rho^{(N)}(K)),$$

where $\tilde{K}$ is K conjugated by U at every physical site. This is a project derivation preparing the Case-I route of [11, Appendix C.2, lines 1434–1448] (the basis change defining σ_{tilde}); it is not a claim from that source.

Proof. ✓ Proof The sitewise product matrix $W_{\sigma, \sigma'} = \prod_n U(\sigma_n, \sigma'_n)$ satisfies $W^\dagger W = 1$ because U does: expanding $W^\dagger W$ as a sum over intermediate configurations and using $\sum_x \overline{U(x, a)} U(x, b) = \delta_{a,b}$ site by site gives the identity. Since $\rho^{(N)}(\tilde{K}) = W \rho^{(N)}(K) W^\dagger$, cyclicity of the trace and $W^\dagger W = 1$ cancel the W 's from $\text{tr}(W \rho^{(N)}(K) \rho^{(N)}(K) W^\dagger)$. $\square$

Lemma 20.16.6.30 (Reduction to the active sector). Lean Lean Lean ✓ Stmt Let K be an MPO tensor with a physical-sector factorization and sector weights p_k , such that the left tensor of every zero-weight sector vanishes. Then, for $N \geq 1$, the sum over all sector labelings of the cyclic trace-square product reduces to the sum over active-sector labelings:

$$\sum_{k: \{0, \dots, N-1\} \rightarrow I} \prod_{i=0}^{N-1} \text{tr}(\eta_{k_i, k_{i+1}}^2) = \sum_{k: \{0, \dots, N-1\} \rightarrow I_*} \prod_{i=0}^{N-1} \text{tr}(\eta_{k_i, k_{i+1}}^2).$$

This reconciles the full sector index set I with the active sector set I_* .

Proof. ✓ Proof A neighboring operator with an inactive target sector vanishes, because the left tensor of that sector vanishes identically. Hence any cyclic labeling touching an inactive sector has a vanishing factor (that sector is the target of some edge in the cycle), so its product is zero. The active-sector sum is therefore exactly the full-sector sum, restricted to a labeling-by-labeling bijection with the inactive terms discarded. $\square$

Theorem 20.16.6.31 (The overlap formula). Lean ✓ Stmt Let K be an MPO tensor with a physical-sector factorization and sector weights p_k , such that every neighboring operator is positive semidefinite and the left tensor of every zero-weight sector vanishes. Then, for $N \geq 1$,

$$O_{K^{\text{MPS}} K^{\text{MPS}}}(N) = \text{tr}((S^*)^N).$$

This is a project derivation completing the first step of the Case-I route toward [11, Appendix C.2, Lemma C.5, lines 1473–1499]; it is not a statement of that source.

Proof. ✓ Proof Write $\rho = \rho^{(N)}(K)$, $\tilde{\rho} = \rho^{(N)}(\tilde{K})$ for the physically conjugated MPO, and Ω_k for the cyclic neighboring product of Theorem 20.16.6.24. The overlap unwinds into a chain of trace

identities:

$$\begin{aligned}
O_{K^{\text{MPS}} K^{\text{MPS}}}(N) &= \text{tr}(\rho \rho) && \text{Theorem 20.16.6.21} \\
&= \text{tr}(\tilde{\rho} \tilde{\rho}) && \text{Lemma 20.16.6.29} \\
&= \text{tr}\left(\left(\bigoplus_k \Omega_k\right)^2\right) && \text{block-diagonal product realization} \\
&= \sum_{k:\{0,\dots,N-1\} \rightarrow I} \text{tr}(\Omega_k^2) && \text{trace of a block-diagonal matrix} \\
&= \sum_{k:\{0,\dots,N-1\} \rightarrow I} \prod_{i=0}^{N-1} \text{tr}(\eta_{k_i, k_{i+1}}^2) && \text{Theorem 20.16.6.24} \\
&= \sum_{k:\{0,\dots,N-1\} \rightarrow I_*} \prod_{i=0}^{N-1} \text{tr}(\eta_{k_i, k_{i+1}}^2) && \text{Lemma 20.16.6.30} \\
&= \sum_{k:\{0,\dots,N-1\} \rightarrow I_*} \prod_{i=0}^{N-1} S_{k_i, k_{i+1}}^* && \text{Hermiticity, } \text{tr}(A) = \text{Re tr}(A) \text{ for } A = A^\dagger \\
&= \text{tr}((S^*)^N) && \text{Theorem 20.16.6.28}
\end{aligned}$$

The third step combines the block-diagonal product realization of the sector-coordinate MPO with the reindexing invariance of trace-of-square. The sixth step uses Hermiticity of $\eta_{k,h}$ to identify its complex trace-square with the real entry $S_{k,h}^* = \text{Re tr}(\eta_{k,h}^2)$. $\square$

Theorem 20.16.6.32 (Trace invariance under squared diagonal similarity). [Lean](#) [✓ Stmt](#) Let I be a finite index set, let $S \in \mathbb{R}^{I \times I}$, and let $v \in \mathbb{R}^I$ satisfy $v_i > 0$ for every $i \in I$. Put

$$D = \text{diag}(v_i^2).$$

Then, for every $N \geq 0$,

$$\text{tr}((D^{-1}SD)^N) = \text{tr}(S^N).$$

This project lemma isolates the diagonal-conjugation calculation used in the Case-I argument. It is not a separately stated result of [11].

Proof. [✓ Proof](#) Every diagonal entry of D is nonzero, so D is invertible. Induction on N gives

$$(D^{-1}SD)^N = D^{-1}S^N D.$$

Cyclicity of trace and $DD^{-1} = 1$ then give the claimed equality. $\square$

Theorem 20.16.6.33 (Trace-power decay below a stochastic Hadamard square). [Lean](#) [✓ Stmt](#) Let I be a finite set with at least two elements, and let $P, S \in \mathbb{R}^{I \times I}$. Suppose that P is primitive and row-stochastic, and that, for all $i, j \in I$,

$$0 \leq S_{i,j} \leq (P \odot P)_{i,j} = P_{i,j}^2.$$

Then, as $N \rightarrow \infty$,

$$\text{tr}(S^N) \rightarrow 0.$$

This project matrix-theoretic lemma is used in the Case-I argument. It is not a separately stated result of [11].

Proof. ✓ Proof Entrywise non-negativity gives $\text{tr}(S^N) \geq 0$. Entrywise domination is preserved by matrix powers, and hence

$$\text{tr}(S^N) \leq \text{tr}((P \odot P)^N).$$

Every power of $P \odot P$ has L^∞ operator norm at most one. Primitivity gives a positive power whose L^∞ operator norm is strictly below one; the division algorithm then shows that $(P \odot P)^N \rightarrow 0$. Thus its trace tends to zero, and the squeeze theorem proves the result. $\square$

Theorem 20.16.6.34 (The active sector is unique). Lean ✓ Stmt *Let K be an MPO tensor of positive bond dimension D with a physical-sector factorization and sector weights p_k , and let $I_* = \{k : p_k \neq 0\}$. Suppose that every neighboring operator $\eta_{k,h}$ is positive semidefinite,*

$$\text{span}_{\mathbb{C}}\{A_{k;x,y} : k \in I_*\} = M_D(\mathbb{C}),$$

and, for every $k \in I_$, some virtual matrix $A_{k;x,y}$ is nonzero. Suppose also that every nonzero $\eta_{k,h}$ with $k, h \in I_*$ admits $j \in I_*$ such that $\eta_{h,j} \neq 0$ and $\eta_{j,k} \neq 0$, that the physical-trace transfer is literally idempotent,*

$$\mathcal{T}_K^2 = \mathcal{T}_K,$$

that $p_k = 0$ implies that the left tensor of sector k vanishes, that the underlying tensor is normal in the sense of Definition 7.6.1, and that I_ is nonempty. Then*

$$|I_*| = 1.$$

This is a project derivation completing the Case-I route toward [11, Appendix C.2, Lemma C.5, lines 1473–1499]; it is not a statement of that source.

Proof. ✓ Proof By contradiction, suppose $|I_*| \geq 2$. The overlap formula and the normal self-overlap limit ([11, lines 1080–1100]) give $\text{tr}((S^*)^N) \rightarrow 1$. The relation $(T^*)^2 = (T^*)^3$ and primitivity supply a positive Perron eigenvector with eigenvalue 1. Conjugating by its diagonal gives a row-stochastic primitive matrix P . The matching squared diagonal conjugate $\hat{S}$ of S^* is entrywise non-negative and bounded by $P \odot P$. On the nontrivial active-sector set, Theorem 20.16.6.33 gives $\text{tr}(\hat{S}^N) \rightarrow 0$. Theorem 20.16.6.32 then gives $\text{tr}((S^*)^N) \rightarrow 0$, contradicting the limit 1. $\square$

Theorem 20.16.6.35 (The Case-I relation $(T^*)^2 = T^*$). Lean ✓ Stmt *Let K be an MPO tensor of positive bond dimension D with a physical-sector factorization and sector weights p_k , and let $I_* = \{k : p_k \neq 0\}$. Suppose that every neighboring operator $\eta_{k,h}$ is positive semidefinite,*

$$\text{span}_{\mathbb{C}}\{A_{k;x,y} : k \in I_*\} = M_D(\mathbb{C}),$$

and, for every $k \in I_$, some virtual matrix $A_{k;x,y}$ is nonzero. Suppose also that every nonzero $\eta_{k,h}$ with $k, h \in I_*$ admits $j \in I_*$ such that $\eta_{h,j} \neq 0$ and $\eta_{j,k} \neq 0$, that the physical-trace transfer is literally idempotent,*

$$\mathcal{T}_K^2 = \mathcal{T}_K,$$

that $p_k = 0$ implies that the left tensor of sector k vanishes, that the underlying tensor is normal in the sense of Definition 7.6.1, and that I_ is nonempty. Then*

$$(T^*)^2 = T^*.$$

This is the Case-I input to the proof of Lemma C.5 ([11, Appendix C.2, lines 1473–1499]), not the lemma’s headline statement.

Proof. ✓ Proof On the singleton active sector set, T^* is a scalar t . Stabilization gives $t^2 = t^3$, and positivity of $(T^*)^2$ gives $t^2 > 0$, hence $t = 1$ and $t^2 = t$. $\square$

Theorem 20.16.6.36 (Normalized Case-I rank-one coefficients). Lean ✓ Stmt *Under the complete normal Case-I hypotheses of Theorem 20.16.6.35, there are real functions $a, b : I_* \rightarrow \mathbb{R}$ such that, for all $k, h \in I_*$,*

$$T_{k,h}^* = a_k b_h,$$

and

$$\sum_{k \in I_*} a_k b_k = 1.$$

This establishes the normal Case-I rank-one and normalization equations on the active sectors from [11, Appendix C.2, lines 1394–1401 and 1484–1499].

Proof. ✓ Proof Write $I_* = \{k_0\}$. Strict positivity of $(T^*)^2$ and $(T^*)^2 = T^*$ give

$$0 < (T_{k_0,k_0}^*)^2 = T_{k_0,k_0}^*.$$

Thus the sole entry is $T_{k_0,k_0}^* = 1$. Taking $a_k = b_k = 1$ gives $T_{k,h}^* = a_k b_h$ and $\sum_{k \in I_*} a_k b_k = 1$. $\square$

Theorem 20.16.6.37 (Saturation-of-the-area-law form of the normalized Case-I coefficients). Lean ✓ Stmt *Let K be an injective MPO tensor. Suppose that K satisfies saturation of the area law, that its physical-trace transfer is literally idempotent,*

$$\mathcal{T}_K^2 = \mathcal{T}_K,$$

and that the underlying tensor is normal in the sense of Definition 7.6.1. Then there is a physical-sector factorization with sector weights $p_k \geq 0$ satisfying $\sum_k p_k = 1$, for which every neighboring operator is positive semidefinite. There are real functions a, b on $I_ = \{k : p_k \neq 0\}$. For all $k, h \in I_*$,*

$$T_{k,h}^* = a_k b_h,$$

and

$$\sum_{k \in I_*} a_k b_k = 1.$$

This is the active-sector part of the normal Case-I conclusion of [11, Appendix C.2, Case I, lines 1374–1381]. The active Hayashi factorization comes from [11, Appendix C.2, Lemma C.4, lines 1406–1471]; Theorem 20.16.6.40 extends these coefficients to every sector label, as in Lemma C.5.

Proof. ✓ Proof Apply Theorem 20.16.6.39 and retain its active coefficient functions and equations. $\square$

Theorem 20.16.6.38 (Vanishing at a zero-weight incident sector). Lean ✓ Stmt *Let $(\eta_{k,h})_{k,h}$ be the neighboring operators of the zero-weight-reparameterized inverse-map factorization, after a coherent sector rephasing, and let $(p_k)_k$ be its Hayashi weights. If $p_k = 0$ or $p_h = 0$, then*

$$\eta_{k,h} = 0.$$

The reparameterization makes the right tensor of an inactive source sector zero, while its left tensor is already zero. Thus this is a statement about the chosen reparameterized and rephased factorization, not about the raw inverse-map factorization.

Proof. ✓ **Proof** Rephasing changes a neighboring operator by

$$(\eta_z)_{k,h} = z_k^{-1} z_h \eta_{k,h}.$$

Hence $\eta_{k,h} = 0$ implies $(\eta_z)_{k,h} = 0$. If $p_k = 0$, the reparameterized right tensor vanishes, so $\eta_{k,h} = 0$. If $p_h = 0$, the left tensor vanishes and again $\eta_{k,h} = 0$. $\square$

Theorem 20.16.6.39 (Existence of normalized Case-I rank-one coefficients). Lean ✓ **Stmt** *Let K be an injective MPO tensor satisfying saturation of the area law. Suppose that its physical-trace transfer is literally idempotent and that its underlying tensor is normal. Then one can choose a single coherently rephased, zero-weight-reparameterized inverse-map factorization with nonnegative weights $(p_k)_k$ satisfying $\sum_k p_k = 1$, positive semidefinite neighboring operators $\eta_{k,h}$, and real functions a, b on $I_* = \{k : p_k \neq 0\}$. For all $k, h \in I_*$,*

$$T_{k,h}^* = a_k b_h,$$

and

$$\sum_{k \in I_*} a_k b_k = 1.$$

For all ambient sector labels k, h ,

$$p_k = 0 \text{ or } p_h = 0 \implies \eta_{k,h} = 0.$$

The incident-vanishing clause belongs to this chosen reparameterized and rephased factorization, not to the raw inverse-map factors. This is the normal Case-I construction from [11, Appendix C.2, Case I, Lemmas C.4–C.5, lines 1374–1505]; the incident-vanishing clause is the chosen zero-weight convention for inactive summands.

Proof. ✓ **Proof** Normality makes the bond dimension positive. Saturation of the area law supplies the four-site Hayashi decomposition, while the normalized four-site tail gives nonzero outer indices and the required three-site closure. Choose the coherently rephased zero-weight-reparameterized inverse-map factorization. Its active trace matrix admits the normalized rank-one factorization, and Theorem 20.16.6.38 gives the final incident-vanishing clause for the same factorization. $\square$

Theorem 20.16.6.40 (All-sector normalized Case-I rank-one coefficients). Lean ✓ **Stmt** *Let K be an injective normal MPO tensor satisfying saturation of the area law and literal physical-trace idempotence,*

$$\mathcal{T}_K^2 = \mathcal{T}_K.$$

Then there is a physical-sector factorization with nonnegative weights $(p_k)_k$ satisfying $\sum_k p_k = 1$, positive semidefinite neighboring operators $\eta_{k,h}$, and real families $(a_k)_k, (b_k)_k$ indexed by all sector labels. For all k, h ,

$$\begin{aligned} \operatorname{Re} \operatorname{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_k a_k b_k &= 1. \end{aligned}$$

On the nonzero-weight sectors these are the coefficient families of Theorem 20.16.6.39; if $p_k = 0$, then $a_k = b_k = 0$. This realizes the all-index coefficient statement of [11, Appendix C.2, Lemma C.5, lines 1473–1499] for literal zero correlation length by extending the active-sector coefficients to inactive sectors by zero.

Proof. [✓ Proof](#) Extend the active coefficient families of this factorization by zero. In the chosen zero-weight-reparameterized and rephased factorization, $p_k = 0$ makes every neighboring operator incident to k vanish. Therefore the trace factorization is zero whenever either index is inactive, while it is the active factorization otherwise. Filtering the full diagonal sum to the nonzero-weight sectors gives

$$\sum_k a_k b_k = \sum_{k:p_k \neq 0} a_k b_k = 1.$$

□

Theorem 20.16.6.41 (Rectangular form of the Case-I relation). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 20.16.6.35, define the rectangular trace-pairing matrices

$$L_{\beta,h} = \text{tr}((l_h)_\beta), \quad Q_{k,\alpha} = \text{tr}((r_k)_\alpha).$$

Then

$$Q(1 - LQ)L = 0.$$

This expresses the operator identity corresponding to the l_h, r_k factors in [11, Appendix C.2, Lemma C.5, lines 1473–1499].

Proof. [✓ Proof](#) The trace pairing gives $QL = T^*$. Therefore

$$Q(1 - LQ)L = QL - (QL)^2 = T^* - (T^*)^2 = 0,$$

where the last equality is Theorem 20.16.6.35. □

Theorem 20.16.6.42 (Positivity of bipartite partial traces). [Lean](#) [Lean](#) [✓ Stmt](#) If ρ is positive semidefinite on a bipartite space $A \otimes B$, then both reduced operators $\text{tr}_A \rho$ and $\text{tr}_B \rho$ are positive semidefinite.

Proof. [✓ Proof](#) Write each partial trace as the finite sum, over the traced index, of a principal submatrix of ρ . Principal submatrices of a positive semidefinite matrix are positive semidefinite, and finite sums of positive semidefinite matrices remain positive semidefinite. □

Definition 20.16.6.43 (Hayashi–Markov extraction of explicit η -operators). [Lean](#) [✓ Stmt](#) Every Hayashi decomposition h_η canonically produces an explicit neighboring η -family by Kronecker-multiplying the sector-reduced states on the neighboring bond spaces:

$$\eta_{k,h} := \text{tr}_C(\rho_{b_2 C}^{(k)}) \otimes \text{tr}_A(\rho_{A b_1}^{(h)}).$$

Positive semidefiniteness of each $\eta_{k,h}$ follows from the fact that partial traces preserve positivity and that positive semidefiniteness is closed under Kronecker products. This is the sector-reduced extraction; the remaining connection to [11, Appendix C.2] is the identification of this family with the inverse-map construction in the original simple-MPDO tensor coordinates.

Theorem 20.16.6.44 (Trace matrix of the Hayashi–Markov η -operators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The trace matrix induced by the extracted family is entrywise $T_{k,h} = 1$ in both its complex-valued form and its real-valued form. If there are m sectors, then $T = \mathbf{1}\mathbf{1}^\top$ and $\text{tr}(T) = m$.

Proof. [✓ Proof](#) Each $\rho_{b_2C}^{(k)}$ is a density matrix, hence has unit trace; the partial trace tr_C preserves the trace. The analogous statement holds for $\text{tr}_A \rho_{Ab_1}^{(h)}$. Multiplicativity of the trace under Kronecker products gives $T_{k,h} = 1 \cdot 1 = 1$, and taking real parts gives the real-valued statement. Summing the m diagonal entries gives $\text{tr}(T) = m$. $\square$

Theorem 20.16.6.45 (Positive semidefiniteness of the Hayashi–Markov trace matrix). [Lean](#) [✓ Stmt](#) *The real trace matrix of the sector-reduced family is positive semidefinite.*

Proof. [✓ Proof](#) This trace matrix is the all-ones matrix $\mathbf{1}\mathbf{1}^\top$, hence is positive semidefinite. $\square$

Theorem 20.16.6.46 (Entrywise nonnegativity of the real trace matrix). [Lean](#) [✓ Stmt](#) *Positivity of the neighboring operators gives entrywise nonnegativity of the real trace matrix. Matrix-level positive semidefiniteness of T is a separate structural condition.*

Proof. [✓ Proof](#) The trace of a positive semidefinite neighboring operator is non-negative in the complex order. Taking real parts gives the stated entrywise nonnegativity. $\square$

Definition 20.16.6.47 (Auxiliary matrix predicates for the rank-one step). [Lean](#) [Lean](#) [✓ Stmt](#) The auxiliary predicates state two matrix-theoretic ingredients used in the rank-one step of [11, Appendix C.2, Lemma C.5]: constant traces of all positive powers and existence of a rank-one factorization $T = ab^\top$.

The proposed implication from primitivity and constant trace powers to a rank-one factorization is not a valid general theorem. A concrete 3×3 counterexample appears in the archive. The corrected criterion proved here adds positive semidefiniteness and trace normalization, which provide the diagonalizability needed to turn the trace-power condition into a rank-one factorization.

Theorem 20.16.6.48 (Trace square of a Hermitian matrix). [Lean](#) [✓ Stmt](#) *Let T be a finite Hermitian matrix over a real or complex scalar field, with Hermitian eigenvalues λ_i . Then $\text{tr}(T^2) = \sum_i \lambda_i^2$.*

Proof. [✓ Proof](#) The spectral theorem writes T as a unitary conjugate of the diagonal matrix of its Hermitian eigenvalues. Squaring commutes with this conjugation, and cyclicity of trace reduces the identity to the trace of the squared diagonal matrix. $\square$

Theorem 20.16.6.49 (Rank-one eigenvector resolution of a Hermitian matrix). [Lean](#) [✓ Stmt](#) *Let A be a finite Hermitian matrix. Choose an orthonormal eigenbasis $(u_k)_k$ with eigenvalues $(\lambda_k)_k$, and let $E_k = |u_k\rangle\langle u_k|$ be the rank-one projection onto u_k . Then $A = \sum_k \lambda_k E_k$.*

Proof. [✓ Proof](#) The finite-dimensional spectral theorem gives $A = U \text{diag}(\lambda_k) U^\dagger$, where the columns of U are the vectors u_k . Expanding the diagonal matrix as the sum of its rank-one coordinate projections gives the stated identity. $\square$

Theorem 20.16.6.50 (PSD trace-power rank-one criterion). [Lean](#) [✓ Stmt](#) *Let T be a finite real square matrix. If T is positive semidefinite, normalized by $\text{tr}(T) = 1$, and has constant traces on all positive powers, then T has a rank-one factorization.*

Proof. [✓ Proof](#) The spectral theorem diagonalizes the positive semidefinite matrix with non-negative real eigenvalues. The trace gives that their sum is 1, while the second trace moment gives that the sum of their squares is also 1. Hence exactly one eigenvalue is equal to 1 and all others vanish, so the diagonal form is rank one. Unitary conjugation preserves the existence of an outer-product factorization. $\square$

Theorem 20.16.6.51 (Idempotent trace-one criterion). [Lean](#) [✓ Stmt](#) Let T be a finite real square matrix. If $T^2 = T$ and $\text{tr}(T) = 1$, then there are real vectors a, b such that $T = ab^\top$.

Proof. [✓ Proof](#) For an idempotent endomorphism f , one has $\text{tr}(f) = \dim(\text{range}(f))$. Hence the range of T has dimension one. Choosing a nonzero vector a spanning it expresses every column of T as $b_j a$ for a scalar b_j , and therefore $T = ab^\top$. $\square$

Theorem 20.16.6.52 (Constant trace powers of an idempotent matrix). [Lean](#) [✓ Stmt](#) Let T be a finite real square matrix with $T^2 = T$. Then, for every positive integer N , one has $\text{tr}(T^N) = \text{tr}(T)$. This recovers the trace identity displayed in the proof of [11, Appendix C.2, Lemma C.5] from idempotence.

Proof. [✓ Proof](#) Induction on N : for $N \geq 1$, one has $T^{N+1} = T^N T = T T = T$, so every positive power of T equals T . $\square$

Theorem 20.16.6.53 (Positive entries in every row and column of a primitive matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be a primitive non-negative matrix. Then every row of T contains a strictly positive entry, and so does every column.

Proof. [✓ Proof](#) If row k of T vanishes, then, for every $m \geq 1$,

$$(T^m)_{k,k} = \sum_j T_{k,j} (T^{m-1})_{j,k} = 0,$$

contradicting the strict entrywise positivity of some power of T . Likewise, if column h of T vanishes, then, for every $m \geq 1$,

$$(T^m)_{h,h} = \sum_j (T^{m-1})_{h,j} T_{j,h} = 0,$$

again contradicting the strict positivity of some power of T . $\square$

Theorem 20.16.6.54 (Rank of a strictly positive idempotent). [Lean](#) [✓ Stmt](#) Let I be a nonempty finite index set, and let P be a real matrix indexed by $I \times I$. Suppose that $P^2 = P$ and $P_{i,j} > 0$ for every $i, j \in I$. Then $\text{rank}(P) = 1$.

Proof. [✓ Proof](#) Put $u = P\mathbf{1}$, so every coordinate of u is positive and $Pu = u$. For a fixed vector x satisfying $Px = x$, choose an index k at which x_i/u_i is maximal and put $c = x_k/u_k$. Then $y = cu - x$ is non-negative, $y_k = 0$, and $P y = y$. Hence

$$0 = y_k = (P y)_k = \sum_j P_{k,j} y_j.$$

Since every $P_{k,j}$ is strictly positive, all coordinates of y vanish. Thus every fixed vector is proportional to u . The range of an idempotent is its fixed space, so it is one-dimensional. $\square$

Theorem 20.16.6.55 (Rank and factorization of a stationary primitive square). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be a primitive real $N \times N$ matrix, where $N \geq 1$. Multiplication by a positive scalar preserves primitivity. If $T^2 = T^3$, then $\text{rank}(T^2) = 1$. There are vectors $a, b \in \mathbb{R}^N$ such that

$$T^2 = ab^\top, \\ \sum_i a_i b_i = 1.$$

This finite-dimensional matrix lemma supports the argument sought at [11, Appendix C.2, line 1613]; it is not a theorem asserted there.

Proof. ✓ Proof Primitivity gives $m \geq 1$ such that $(T^m)_{i,j} > 0$ for every i, j . Stabilization gives $T^{2m} = T^2$. Thus, for every i, j ,

$$(T^2)_{i,j} = (T^{2m})_{i,j} = \sum_k (T^m)_{i,k} (T^m)_{k,j} > 0.$$

The same stabilization identity gives $(T^2)^2 = T^2$. The preceding theorem applied to $P = T^2$ now gives the conclusion. Since an idempotent has trace equal to its rank, $\text{tr}(T^2) = 1$. Factoring its one-dimensional range gives $T^2 = ab^\top$, and taking the trace yields $\sum_i a_i b_i = 1$. $\square$

Theorem 20.16.6.56 (Idempotence of the sector trace matrix). Lean Lean ✓ Stmt *Let V be a module over a commutative ring, let $l_1, \dots, l_n \in V$, let $r_1, \dots, r_n$ be linear functionals on V , and let $T_{k,h} = (r_k | l_h)$ be the pairing matrix; for the tensors of [11, Appendix C.2] this is the sector trace matrix. Assume the identity*

$$\sum_k |l_k)(r_k| = \sum_{k,h} T_{k,h} |l_k)(r_h|,$$

whose right-hand side is the square of its left-hand side, so that the operator $M = \sum_k |l_k)(r_k|$ satisfies $M^2 = M$. If the vectors $l_1, \dots, l_n$ are linearly independent, then $T^2 = T$. The same conclusion holds if instead the functionals $r_1, \dots, r_n$ are linearly independent.

Proof. ✓ Proof Evaluating $M^2 v = Mv$ at an arbitrary $v \in V$ gives

$$\sum_k ((r_k | Mv) - (r_k | v)) l_k = 0,$$

so linear independence of the l_k yields $(r_k | Mv) = (r_k | v)$ for all k and v . Taking $v = l_h$ and expanding $Ml_h = \sum_j T_{j,h} l_j$ turns the left-hand side into $\sum_j T_{k,j} T_{j,h} = (T^2)_{k,h}$ and the right-hand side into $T_{k,h}$. For linearly independent functionals, apply the same argument in the dual space, with the r_k as vectors and evaluation at l_k as functionals. The dual pairing matrix therefore satisfies $(T^\top)^\top = T^\top$. Taking transposes gives $T^2 = T$. $\square$

Equivalently, the zero-correlation-length identity is the algebraic equality

$$\sum_k |l_k)(r_k| = \sum_{k,h} T_{k,h} |l_k)(r_h|, \quad T_{k,h} = (r_k | l_h),$$

which is the displayed identity of [11, Appendix C.2, lines 1490–1493].

Theorem 20.16.6.57 (Unconditional constant trace powers of the pairing matrix). Lean Lean Lean ✓ Stmt *Let V be a real vector space, let $l_1, \dots, l_n \in V$, let $r_1, \dots, r_n$ be real linear functionals on V , and let $T_{k,h} = (r_k | l_h)$ be the pairing matrix, as in Theorem 20.16.6.56. Assuming only the identity $M^2 = M$ for the operator $M = \sum_k |l_k)(r_k|$, without assuming linear independence of $l_1, \dots, l_n$ or of $r_1, \dots, r_n$, one has the following identities: the first unconditionally, the second for every $N \geq 2$, and the third for every positive integer N :*

$$\begin{aligned} T^2 &= T^3, \\ T^N &= T^2, \\ \text{tr}(T^N) &= \text{tr}(T). \end{aligned}$$

Proof. [✓ Proof](#) Multiplying the identity $M^2 = M$ on the left by the functional map and on the right by the coefficient map turns the two sides directly into T^2 and T^3 , with no independence hypothesis; induction then gives $T^N = T^2$ for every $N \geq 2$. For the trace identity, restrict the operator M to the finite-dimensional span of $l_1, \dots, l_n$: there it is an idempotent endomorphism of a finite free module, and cyclicity of the trace of a composition between that span and the coefficient space identifies $\text{tr}(T)$ with the trace of the restriction and $\text{tr}(T^2)$ with the trace of its square, which agree since the restriction is idempotent. $\square$

Theorem 20.16.6.58 (Rank-one factorization of the sector trace matrix). [Lean](#) [✓ Stmt](#) *In the setting of Theorem 20.16.6.56 with real scalars and linearly independent vectors $l_1, \dots, l_n$, if in addition $\text{tr}(T) = 1$, then there are real vectors a, b such that $T = ab^\top$. This is the factorization $T_{k,h} = a_k b_h$ obtained in the proof of [11, Appendix C.2, Lemma C.5] for the sector trace matrix, derived here from the displayed zero-correlation-length identity together with the linear independence of the l_k , instead of the Perron–Frobenius trace-power argument of the source.*

Proof. [✓ Proof](#) Theorem 20.16.6.56 gives $T^2 = T$, and the idempotent trace-one criterion of Theorem 20.16.6.51 gives the factorization $T = ab^\top$. $\square$

Theorem 20.16.6.59 (PSD-corrected rank-one T criterion). [Lean](#) [✓ Stmt](#) *Let T be a finite real square matrix. Assume that T is primitive, positive semidefinite, normalized by $\text{tr}(T) = 1$, and has constant traces on all positive powers. Then $T = ab^\top$ for some vectors a and b , with $a \cdot b = 1$.*

Proof. [✓ Proof](#) The positive-semidefinite matrix criterion diagonalizes T , uses the first two trace moments to show that exactly one eigenvalue is nonzero, and hence gives the factorization. Taking traces and using $\text{tr}(ab^\top) = a \cdot b$ gives the normalization $a \cdot b = 1$. $\square$

Definition 20.16.6.60 (Sector tensors paired into the sector trace matrix). [Lean](#) [✓ Stmt](#) *For a real $n \times n$ matrix T and a real vector space V , this structure consists of vectors $|l_0), \dots, |l_{n-1}) \in V$ and linear functionals $(r_0|, \dots, (r_{n-1}|$ on V satisfying the pairing identity $T_{k,h} = (r_k|l_h)$ of [11, Appendix C.2] together with the zero-correlation-length identity*

$$\sum_k |l_k)(r_k| = \sum_{k,h} T_{k,h} |l_k)(r_h|,$$

displayed in the proof of [11, Lemma C.5]. The vectors are the closed sector tensors obtained there from the sector splitting of an injective simple tensor.

Theorem 20.16.6.61 (Idempotence of the sector trace matrix from the zero-correlation-length identity). [Lean](#) [Lean](#) [✓ Stmt](#) *For sector tensors paired into T as in Definition 20.16.6.60, the operator $M = \sum_k |l_k)(r_k|$ satisfies $M^2 = M$. If in addition the vectors $|l_0), \dots, |l_{n-1})$ are linearly independent, then $T^2 = T$.*

Proof. [✓ Proof](#) Substituting $T_{k,h} = (r_k|l_h)$ into the right-hand side of the zero-correlation-length identity gives, for every $v \in V$,

$$\sum_{k,h} T_{k,h} (r_h|v) |l_k) = \sum_{k,h} (r_k|l_h) (r_h|v) |l_k) = \sum_k (r_k|Mv) |l_k) = M^2 v,$$

so $M^2 = M$. Linear independence of the $|l_k)$ then gives $T^2 = T$ by Theorem 20.16.6.56. $\square$

Theorem 20.16.6.62 (Unconditional constant trace powers of the sector trace matrix). [Lean](#) [Lean](#) [✓ Stmt](#) *For sector tensors paired into T as in Definition 20.16.6.60, without any independence hypothesis on the closed sector tensors or the functionals, one has $T^2 = T^3$, and, for every positive integer N , $\text{tr}(T^N) = \text{tr}(T)$.*

Proof. [✓ Proof](#) The operator identity $M^2 = M$ established in the proof of Theorem 20.16.6.61 gives $T^2 = T^3$ and the constant trace powers by Theorem 20.16.6.57, with no independence hypothesis. $\square$

Theorem 20.16.6.63 (Nonvanishing and independence of the sector tensors). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the sector trace matrix T of Definition 20.16.6.60 is primitive. Then every closed sector tensor satisfies $|l_k) \neq 0$. If, in addition, each $|l_k)$ lies in its own member of an independent family of subspaces of V , then the vectors $|l_0), \dots, |l_{n-1})$ are linearly independent.

Proof. [✓ Proof](#) By Theorem 20.16.6.53, some entry of the k -th column of T is positive, and that entry is the pairing $(r_j|l_k)$, so $|l_k) \neq 0$. Nonzero vectors chosen from an independent family of subspaces, one from each member, are linearly independent. $\square$

Theorem 20.16.6.64 (Rank one from sector-pairing idempotence under linear independence). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be a sector trace matrix with $\text{tr}(T) = 1$, paired from sector tensors satisfying the zero-correlation-length identity of Definition 20.16.6.60. If the closed sector tensors $|l_0), \dots, |l_{n-1})$ are linearly independent, then there are real numbers a_k, b_h such that

$$T_{k,h} = a_k b_h, \\ \sum_k a_k b_k = 1.$$

The same conclusion holds if, instead of the independence, T is primitive and the closed sector tensors lie in distinct members of an independent family of subspaces. In contrast with the positive-semidefinite criterion of Theorem 20.16.6.59, the constant trace powers are derived rather than assumed.

Proof. [✓ Proof](#) Theorem 20.16.6.61 gives $T^2 = T$, and Theorem 20.16.6.62 gives the constant trace powers unconditionally. The idempotent trace-one criterion of Theorem 20.16.6.51 yields the factorization $T = ab^\top$, and taking traces gives $\sum_k a_k b_k = \text{tr}(T) = 1$. When the independence is not assumed directly, Theorem 20.16.6.63 derives it from primitivity once each sector tensor lies in its own member of an independent family of subspaces. $\square$

20.16.7 Refinement and coarse-graining channels

Theorem 20.16.7.1 (Positivity and zero-weight neighboring operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The sector set and every middle-subspin space are nonempty. Each $\Omega_{k,h}$ is positive semidefinite, and $a_k b_h = 0$ implies $\Omega_{k,h} = 0$.

Definition 20.16.7.2 (The three-site neighboring-state preparation $\mathcal{T}_1$). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On a sector pair with $a_k b_h \neq 0$, set $\widehat{\Omega}_{k,h} = (a_k b_h)^{-1} \Omega_{k,h}$. If $a_k b_h = 0$, positivity and vanishing trace imply $\Omega_{k,h} = 0$; choose any density operator on that summand and denote the resulting completed density by $\bar{\Omega}_{k,h}$. The sectorwise preparation is $X \mapsto X \otimes \bar{\Omega}_{k,h}$. Taking the orthogonal direct sum over the outer sectors (k, h) gives the preparation stage $\mathcal{T}_1$ of [11, Appendix C.2, lines 1523–1535]. The zero-weight choice makes the source map total without imposing a nonzero-weight hypothesis.

Theorem 20.16.7.3 (Properties of the three-site neighboring densities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The normalized density is positive semidefinite and has trace one on every active pair. The density chosen on an inactive pair is positive definite and has trace one. Hence $\Omega_{k,h}$ is positive semidefinite with trace one for every pair, and agrees with $\widehat{\Omega}_{k,h}$ on active pairs. The corresponding sectorwise and controlled maps prepare these densities on their diagonal summands.

Definition 20.16.7.9 (The coarse-graining channel $\mathcal{S}$). [Lean](#) [✓ Stmt](#) Define the channel from three physical sites to two physical sites by $\mathcal{S} = \mathcal{S}_2 \mathcal{S}_1 \mathcal{S}_0$. This is the coarse-graining map in Proposition C.7 [11, Appendix C.2, lines 1547–1563].

Theorem 20.16.7.10 ($\mathcal{S}$ is trace-preserving completely positive). [Lean](#) [✓ Stmt](#) The channel $\mathcal{S}$ is trace-preserving and completely positive.

Proof. [✓ Proof](#)

Each of the three factors is trace-preserving and completely positive, and this property is preserved under composition. $\square$

Definition 20.16.7.11 (Two- and three-site sector coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The coordinate equivalences used by the coarse-graining channel expose the direct sums over site-sector pairs and triples. Write $\mathfrak{R}_2$ and $\mathfrak{R}_3$ for the corresponding matrix reindexings. Their inverses send every sector block to the corresponding physical matrix entries. These reindexings preserve the left-right order within every physical site. They differ from the regrouped coordinates $\mathfrak{G}_2$ and $\mathfrak{G}_3$ of Definition 20.16.3.39, which place the two outer subspins before the neighboring subspins. If R_{Φ_2} and R_{Φ_3} denote the two regrouping maps, then

$$\begin{aligned}\mathfrak{G}_2 &= R_{\Phi_2} \mathfrak{R}_2, \\ \mathfrak{G}_3 &= R_{\Phi_3} \mathfrak{R}_3.\end{aligned}$$

Theorem 20.16.7.12 (Diagonal outer-sector block of the three-site closure). [Lean](#) [✓ Stmt](#) For every virtual matrix X and outer-sector pair (k, h) ,

$$\left[R_{\Phi_3} \mathfrak{R}_3(\mathcal{K}_3(X)) \right]_{(k,h),(k,h)} = B_{k,h}(X) \otimes \Omega_{k,h}.$$

Proof. [✓ Proof](#) On the summand labelled by the middle sector l , this is the fixed-sector factorization $B_{k,h}(X) \otimes (\eta_{k,l} \otimes \eta_{l,h})$. Taking the direct sum over l gives the stated block. $\square$

Theorem 20.16.7.13 (Off-diagonal outer-sector entries of the three-site closure). [Lean](#) [✓ Stmt](#) Every matrix entry of $R_{\Phi_3} \mathfrak{R}_3(\mathcal{K}_3(X))$ between distinct outer-sector pairs $(k, h) \neq (p, q)$ vanishes.

Proof. [✓ Proof](#) Distinct outer-sector pairs differ either in their first sector or in their third sector. In the corresponding physical slice, the sector-coordinate tensor is block diagonal, so that factor vanishes. $\square$

Theorem 20.16.7.14 (Regrouped two-site closure block). [Lean](#) [✓ Stmt](#) For every virtual matrix X and outer-sector pair (k, h) ,

$$\left[R_{\Phi_2} \mathfrak{R}_2(\mathcal{K}_2(X)) \right]_{(k,h),(k,h)} = B_{k,h}(X) \otimes \eta_{k,h}.$$

Proof. [✓ Proof](#) This is the fixed-sector two-site factorization after identifying the sitewise sector coordinates with the regrouped boundary and neighboring coordinates. $\square$

Theorem 20.16.7.15 (Diagonal-block action of $\mathcal{T}_0$). [Lean](#) [✓ Stmt](#) For every two-site matrix Y , outer-sector pair (k, h) , and boundary indices a, b ,

$$[\mathcal{T}_0(Y)]_{(k,h,a),(k,h,b)} = \left[\text{tr}_{R_k \otimes L_h} \left([R_{\Phi_2} Y]_{(k,h),(k,h)} \right) \right]_{a,b}.$$

Proof. [✓ Proof](#) The controlled partial trace acts on a diagonal outer-sector block by the ordinary partial trace over its neighboring factor. $\square$

Theorem 20.16.7.16 (Partial trace of a factorized two-site block). [Lean](#) [✓ Stmt](#) If the (k, h) diagonal block of $R_{\Phi_2} Y$ is $B \otimes \eta_{k,h}$, then $[\mathcal{T}_0(Y)]_{(k,h),(k,h)} = a_k b_h B$.

Proof. [✓ Proof](#) The partial trace of $B \otimes \eta_{k,h}$ is $\text{tr}(\eta_{k,h})B = a_k b_h B$. $\square$

Theorem 20.16.7.17 (Preparation of a scaled boundary block). [Lean](#) [✓ Stmt](#) For every outer-sector pair (k, h) and boundary matrix B , the completed sectorwise preparation satisfies $\bar{\mathcal{T}}_{k,h}(a_k b_h B) = B \otimes \Omega_{k,h}$.

Proof. [✓ Proof](#) If $a_k b_h \neq 0$, the scalar cancels the normalization in $\hat{\Omega}_{k,h} = (a_k b_h)^{-1} \Omega_{k,h}$. If $a_k b_h = 0$, both sides vanish because $\Omega_{k,h} = 0$. $\square$

Theorem 20.16.7.18 (Diagonal-block action of $\mathcal{T}_1$). [Lean](#) [✓ Stmt](#) If the (k, h) diagonal block of a boundary matrix Y is $a_k b_h B$, then $[\mathcal{T}_1(Y)]_{(k,h),(k,h)} = B \otimes \Omega_{k,h}$.

Proof. [✓ Proof](#) On a diagonal outer-sector block, the controlled map is the completed sectorwise preparation, to which the preceding theorem applies. $\square$

Theorem 20.16.7.19 (Off-diagonal action of $\mathcal{T}_1$). [Lean](#) [✓ Stmt](#) The map $\mathcal{T}_1$ annihilates every matrix entry between distinct outer-sector pairs.

Proof. [✓ Proof](#) This is the orthogonal control in the definition of $\mathcal{T}_1$. $\square$

Theorem 20.16.7.20 (Reindexing action of $\mathcal{T}_2$). [Lean](#) [✓ Stmt](#) Let Φ_3 be the three-site regrouping. For every prepared matrix Y and three-site indices p, q , $[\mathcal{T}_2(Y)]_{p,q} = Y_{\Phi_3(p), \Phi_3(q)}$.

Proof. [✓ Proof](#) This is matrix reindexing along the inverse regrouping Φ_3^{-1} . $\square$

Theorem 20.16.7.21 (Refinement of the two-site closure). [Lean](#) [✓ Stmt](#) Assume the neighboring-operator trace factorization of Definition 20.16.3.32. For every virtual matrix X , in the canonical physical-sector coordinates selected by the isometry U , the refinement channel satisfies

$$\mathcal{T}(\mathfrak{R}_2(\mathcal{K}_2(X))) = \mathfrak{R}_3(\mathcal{K}_3(X)). \quad (20.75)$$

This is the refinement half of Proposition C.7 [11, Appendix C.2, lines 1510–1516 and 1522–1545].

Proof. [✓ Proof](#) The regrouping contained in $\mathcal{T}_0$ changes $\mathfrak{R}_2$ into $\mathfrak{G}_2$. On the (k, h) diagonal block, the partial trace then gives

$$\mathcal{T}_0(\mathfrak{R}_2(\mathcal{K}_2(X)))_{k,h} = a_k b_h B_{k,h}(X).$$

On an active pair, $\mathcal{T}_1$ adjoins $(a_k b_h)^{-1} \Omega_{k,h}$, and hence

$$\mathcal{T}_1 \mathcal{T}_0(\mathfrak{R}_2(\mathcal{K}_2(X)))_{k,h} = B_{k,h}(X) \otimes \Omega_{k,h}.$$

If $a_k b_h = 0$, both sides vanish because $\Omega_{k,h} = 0$. After the canonical reassociation of the intermediate sector sum, these are precisely the diagonal blocks of $\mathfrak{G}_3(\mathcal{K}_3(X))$; all off-diagonal sector blocks vanish. Finally, the inverse regrouping $\mathcal{T}_2$ changes $\mathfrak{G}_3$ into $\mathfrak{R}_3$, which gives (20.75). $\square$

Theorem 20.16.7.22 (Coarse-graining of the three-site closure). [Lean](#) [Lean](#) [✓ Stmt](#) Assume the neighboring-operator trace factorization of Definition 20.16.3.32. For every virtual matrix X , in the canonical physical-sector coordinates selected by the isometry U , the coarse-graining channel satisfies

$$\mathcal{S}(\mathfrak{R}_3(\mathcal{K}_3(X))) = \mathfrak{R}_2(\mathcal{K}_2(X)). \quad (20.76)$$

Proof. ✓ Proof On the (k, h) diagonal block, $\mathcal{S}_0$ traces $\Omega_{k,h}$ and gives $a_k b_h B_{k,h}(X)$. On an active sector pair, $\mathcal{S}_1((a_k b_h) B_{k,h}(X)) = B_{k,h}(X) \otimes \eta_{k,h}$. If $a_k b_h = 0$, positivity and zero trace give $\eta_{k,h} = 0$, so both sides of this identity vanish. Finally, $\mathcal{S}_2$ regroups the four factors into two physical sites. For distinct sector pairs $(k, h) \neq (p, q)$,

$$\begin{aligned} (\mathcal{S}_1 Y)_{(k,h),(p,q)} &= 0, \\ [\mathfrak{R}_2(\mathcal{K}_2(X))]_{(k,h),(p,q)} &= 0. \end{aligned}$$

These diagonal and off-diagonal identities give (20.76). □

For the following construction, through Theorem 20.16.7.42, assume the neighboring-operator trace factorization of Definition 20.16.3.32. Write $q = \dim(\bigoplus_k L_k \otimes R_k)$ for the physical dimension of the sector-coordinate tensor $\widehat{\mathcal{K}}$.

Definition 20.16.7.23 (Refinement in ordinary physical coordinates). Lean ✓ Stmt Transporting $\mathcal{T}$ through the canonical two-site and three-site coordinate identifications gives a channel $\widehat{\mathcal{T}} : M_{q^2}(\mathbb{C}) \rightarrow M_{q^3}(\mathbb{C})$, where all multiple physical indices are associated to the right.

Theorem 20.16.7.24 (The physical refinement map is a channel). Lean ✓ Stmt The map $\widehat{\mathcal{T}}$ is trace-preserving and completely positive.

Proof. ✓ Proof The coordinate identifications are unitary permutation channels. Composing them with a trace-preserving completely positive map preserves both properties. □

Theorem 20.16.7.25 (Refinement identity in ordinary physical coordinates). Lean ✓ Stmt For every virtual matrix X , $\widehat{\mathcal{T}}(\widehat{\mathcal{K}}_2(X)) = \widehat{\mathcal{K}}_3(X)$.

Proof. ✓ Proof Transport the sector-coordinate identity through the inverse three-site coordinate identification. □

Definition 20.16.7.26 (Physical coarse-graining). Lean ✓ Stmt Transporting $\mathcal{S}$ through the canonical three-site and two-site coordinate identifications gives a channel $\widehat{\mathcal{S}} : M_{q^3}(\mathbb{C}) \rightarrow M_{q^2}(\mathbb{C})$.

Theorem 20.16.7.27 (The physical coarse-graining map is a channel). Lean ✓ Stmt The map $\widehat{\mathcal{S}}$ is trace-preserving and completely positive.

Proof. ✓ Proof This follows by composition with the two coordinate-permutation channels. □

Theorem 20.16.7.28 (Coarse-graining identity in ordinary physical coordinates). Lean ✓ Stmt For every virtual matrix X , $\widehat{\mathcal{S}}(\widehat{\mathcal{K}}_3(X)) = \widehat{\mathcal{K}}_2(X)$.

Proof. ✓ Proof Transport the sector-coordinate identity through the inverse two-site coordinate identification. □

Definition 20.16.7.29 (Localized physical refinement). Lean ✓ Stmt Reassociating three right-associated sites as $(12)3$, applying $\widehat{\mathcal{T}} \otimes \text{id}$ to the first two, and reassociating the result back to $1(2(34))$ defines the localized refinement channel $\widehat{\mathcal{T}}_{(12)3} : M_{q^3}(\mathbb{C}) \rightarrow M_{q^4}(\mathbb{C})$ on right-associated physical coordinates.

Theorem 20.16.7.30 (The localized physical refinement is a channel). Lean ✓ Stmt The localized refinement map is trace-preserving and completely positive.

Proof. ✓ Proof Tensoring a channel with the identity preserves complete positivity and trace, as do the two associativity permutations. $\square$

Theorem 20.16.7.31 (Localized refinement of physical closures). Lean ✓ Stmt *For every virtual matrix X , $(\widehat{\mathcal{T}} \otimes \text{id})(\widehat{\mathcal{K}}_3(X)) = \widehat{\mathcal{K}}_4(X)$, with the map understood in right-associated coordinates.*

Proof. ✓ Proof Fixing the last ket and bra indices turns the three-site closure into a two-site closure with the last tensor matrix absorbed into X . Apply the two-to-three-site refinement identity to this virtual matrix. $\square$

Definition 20.16.7.32 (Localized physical coarse-graining). Lean ✓ Stmt Reassociating four right-associated sites as $(123)4$, applying $\widehat{\mathcal{S}} \otimes \text{id}$ to the first three, and reassociating the result back to $1(23)$ defines the localized coarse-graining channel $\widehat{\mathcal{S}}_{(123)4} : M_{q^4}(\mathbb{C}) \rightarrow M_{q^3}(\mathbb{C})$ on right-associated physical coordinates.

Theorem 20.16.7.33 (The localized physical coarse-graining is a channel). Lean ✓ Stmt *The localized coarse-graining map is trace-preserving and completely positive.*

Proof. ✓ Proof Tensor the physical coarse-graining channel with the identity and compose with the two associativity permutations. $\square$

Theorem 20.16.7.34 (Localized coarse-graining of physical closures). Lean ✓ Stmt *For every virtual matrix X , $(\widehat{\mathcal{S}} \otimes \text{id})(\widehat{\mathcal{K}}_4(X)) = \widehat{\mathcal{K}}_3(X)$ in right-associated coordinates.*

Proof. ✓ Proof Fixing the last ket and bra indices identifies the four-site closure with a three-site closure having the last tensor matrix absorbed into X . Apply the three-to-two-site coarse-graining identity. $\square$

Definition 20.16.7.35 (Two-step physical refinement). Lean ✓ Stmt Let F be a physical-sector factorization whose neighboring operators satisfy Definition 20.16.3.32, and let $\widehat{\mathcal{K}}$ be its sector-coordinate tensor. Define

$$\begin{aligned}\widehat{\mathcal{T}}^2 &= \widehat{\mathcal{T}}_{(12)3} \circ \widehat{\mathcal{T}}, \\ \widehat{\mathcal{T}}^2 &: M_{q^2}(\mathbb{C}) \rightarrow M_{q^4}(\mathbb{C}).\end{aligned}$$

Here the second application acts on the first two sites and leaves the third site unchanged. Thus the superscript denotes two successive overlapping applications, as in [11, Appendix C.2, lines 1821–1825].

Theorem 20.16.7.36 (The two-step physical refinement is a channel). Lean ✓ Stmt *The map $\widehat{\mathcal{T}}^2$ is trace-preserving and completely positive.*

Proof. ✓ Proof It is the composition of the physical refinement channel and its localization on the first two sites. $\square$

Theorem 20.16.7.37 (Two-step refinement of the sector-coordinate tensor). Lean ✓ Stmt *For every virtual matrix X , the sector-coordinate tensor satisfies $\widehat{\mathcal{T}}^2(\widehat{\mathcal{K}}_2(X)) = \widehat{\mathcal{K}}_4(X)$.*

Proof. ✓ Proof The first refinement gives $\widehat{\mathcal{K}}_3(X)$; applying the localized refinement to its first two sites gives $\widehat{\mathcal{K}}_4(X)$. $\square$

Definition 20.16.7.38 (Two-step physical coarse-graining). [Lean](#) [✓ Stmt](#) Define

$$\begin{aligned}\hat{\mathcal{S}}^2 &= \hat{\mathcal{S}} \circ \hat{\mathcal{S}}_{(123)4}, \\ \hat{\mathcal{S}}^2 : M_{q^4}(\mathbb{C}) &\longrightarrow M_{q^2}(\mathbb{C}).\end{aligned}$$

Here the first application acts on the first three sites and leaves the fourth site unchanged. The superscript again denotes successive overlapping applications.

Theorem 20.16.7.39 (The two-step physical coarse-graining is a channel). [Lean](#) [✓ Stmt](#) The map $\hat{\mathcal{S}}^2$ is trace-preserving and completely positive.

Proof. [✓ Proof](#) It is the composition of the localized coarse-graining channel and the physical coarse-graining channel. $\square$

Theorem 20.16.7.40 (Two-step coarse-graining of the sector-coordinate tensor). [Lean](#) [✓ Stmt](#) For every virtual matrix X , the sector-coordinate tensor satisfies $\hat{\mathcal{S}}^2(\hat{\mathcal{K}}_4(X)) = \hat{\mathcal{K}}_2(X)$.

Proof. [✓ Proof](#) The localized coarse-graining first gives $\hat{\mathcal{K}}_3(X)$; the second coarse-graining gives $\hat{\mathcal{K}}_2(X)$. $\square$

Theorem 20.16.7.41 (Blocked sector-coordinate tensor is a renormalization fixed point). [Lean](#) [✓ Stmt](#) Let F be a physical-sector factorization whose neighboring operators satisfy Definition 20.16.3.32. Let $\hat{\mathcal{K}}^{[2]}$ be the tensor obtained by blocking two adjacent sites of its sector-coordinate tensor $\hat{\mathcal{K}}$. Then $\hat{\mathcal{K}}^{[2]}$ is a renormalization fixed point in the sense of Definition 20.2.1. More precisely, after the canonical identifications of one blocked site with two original sites and two blocked sites with four original sites, the required channels are $\hat{\mathcal{S}}^2$ and $\hat{\mathcal{T}}^2$.

Proof. [✓ Proof](#) Decode one blocked index as a pair of physical indices and two blocked indices as four physical indices. The one-site and two-site closures of $\hat{\mathcal{K}}^{[2]}$ thereby become respectively $\hat{\mathcal{K}}_2(X)$ and $\hat{\mathcal{K}}_4(X)$. Transporting the two channels through these permutation identifications preserves complete positivity and trace, and the two closure identities give the required equations in Definition 20.2.1. $\square$

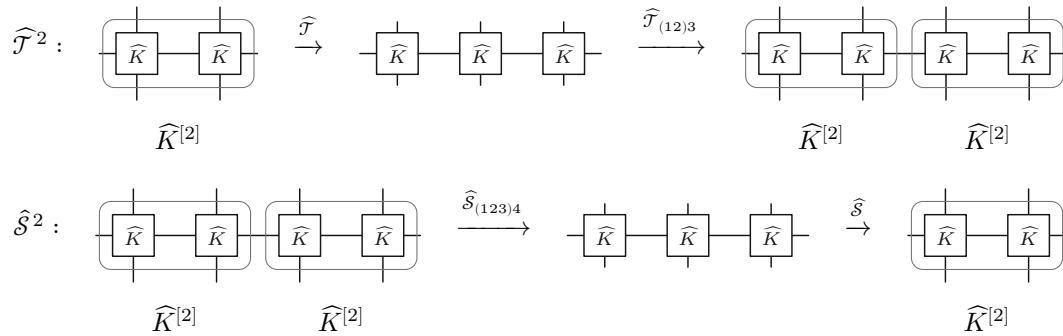


Figure 20.6: The superscripts in $\hat{\mathcal{T}}^2$ and $\hat{\mathcal{S}}^2$ denote successive overlapping applications. The enclosures identify the two-site and four-site endpoints with one and two sites of the blocked tensor. This is the channel construction in [11, Appendix C.2, lines 1821–1825].

Theorem 20.16.7.42 (Conditional two-site blocking theorem from neighboring trace factorization). [Lean](#) [✓ Stmt](#) Let F be a physical-sector factorization of $\mathcal{K}$ whose neighboring operators satisfy Definition 20.16.3.32. Thus they are positive semidefinite and there are real families $(a_k)_k, (b_k)_k$ with $\text{tr}(\eta_{k,h}) = a_k b_h$ and $\sum_k a_k b_k = 1$. Let $\mathcal{K}^{[2]}$ be the tensor obtained by blocking two adjacent sites of $\mathcal{K}$. Then $\mathcal{K}^{[2]}$ is a renormalization fixed point in the sense of Definition 20.2.1. This is the conditional channel construction of [11, Proposition C.7] for one supplied factorization. It does not extend to the printed all-BNT implication [11, Theorem 4.9, (iv) $\Rightarrow$ (v)]: condition (iv) constrains each BNT representative but not its raw repeated-copy weights, and that implication is refuted by Theorem 20.16.8.24. After condition (ii) has forced equal copy weights and their common values have been absorbed, these one-factorization channels can be inputs to the projector-controlled channel construction when one factorization is supplied for each representative. The printed route obtains those factorizations from the false Proposition C.12, so this conditional theorem does not prove the bare implication (ii) $\Rightarrow$ (v). It also does not exclude different channel maps; see [22].

Proof. [✓ Proof](#) Transport the refinement and coarse-graining channels for $\widehat{\mathcal{K}}^{[2]}$ through the two-site and four-site tensor powers of the physical isometry, as in (20.81). These unitary congruences preserve complete positivity and trace, while the corresponding closure identities carry the two fixed-point equations back to $\mathcal{K}^{[2]}$. $\square$

Theorem 20.16.7.43 (Arbitrary virtual boundaries of canonical block sums). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let a tensor be gauge equivalent, by an invertible virtual matrix G , to a finite block sum with component tensors A_s and weights μ_s . For an arbitrary virtual boundary X and an arbitrary word w , its closure is the sum of the component closures with boundaries given by the diagonal blocks of $G^{-1}XG$:

$$\text{tr}(B^w X) = \sum_s \mu_s^{|w|} \text{tr}(A_s^w (G^{-1}XG)_{ss}).$$

The conjugated boundary $G^{-1}XG$ is independent of the word length. This is the arbitrary-boundary form of the block calculation used in [11, lines 186–195 and Appendix C.2, lines 1646–1665 and 1810–1825].

Proof. [✓ Proof](#) A word of length $|w|$ in the block-sum tensor is block diagonal. Its s -th diagonal block is $\mu_s^{|w|} A_s^w$. On multiplying by an arbitrary boundary matrix, only the diagonal boundary blocks contribute to the trace, which gives the asserted sum before the gauge change.

Under the gauge G , the word A^w is conjugated to GA^wG^{-1} . Cyclicity of the trace moves the two gauge factors to the boundary and replaces X by $G^{-1}XG$. Combining these two identities proves the formula. Since this conjugation depends only on G and X , the transformed boundary is the same at every word length. $\square$

Theorem 20.16.7.44 (Absorbed-representative boundary decomposition). [Lean](#) [Lean](#) [✓ Stmt](#) Let the complete tensor B be exactly gauge equivalent, by an invertible virtual matrix G , to a sector decomposition: an arbitrary finite family of basis blocks A_j , $j = 1, \dots, g$, each carrying a copy multiplicity r_j and nonzero weights $\mu_{j,q}$. Suppose that the repeated-copy weights over each basis block are equal, with no equality required between different basis blocks, and absorb this common weight into its representative $\mathcal{K}_j$ as in Definition 20.16.1.7. Then, for every length N and every virtual boundary X , the physical closure of the complete tensor is the sum of the absorbed-representative closures, with the boundary of a representative given by the sum of the

diagonal copy blocks of $G^{-1}XG$ lying over it:

$$B_N(X) = \sum_{j=1}^g (\mathcal{K}_j)_N \left(\sum_{q=1}^{r_j} (G^{-1}XG)_{(j,q),(j,q)} \right).$$

Neither the conjugation $X \mapsto G^{-1}XG$ nor the subsequent sum of diagonal copy blocks depends on N . In particular, the same representative boundary map applies at the two lengths entering the two channel equations of Definition 20.2.1. This is the arbitrary-boundary form of the block calculation used in [11, lines 186–195 and Appendix C.2, lines 1646–1665 and 1810–1825].

Proof. ✓ Proof Apply Theorem 20.16.7.43 to the block sum indexed by the individual copies. Reindex this finite sum by a representative j and a copy label q . For fixed j , all copy weights are equal, so their common N -th power may be absorbed into the length N word of the representative tensor. The remaining dependence on q is linear in the diagonal boundary block. Summing over q therefore replaces these blocks by their sum, which is the stated representative boundary. Neither the conjugation $X \mapsto G^{-1}XG$ nor the subsequent sum of diagonal copy blocks depends on N . □

Theorem 20.16.7.45 (Conditional composition of absorbed BNT channels). Lean ✓ Stmt Let K be exactly gauge equivalent, by an invertible virtual matrix G , to a canonical BNT sum. Suppose that the repeated-copy weights agree within each fixed representative and have been absorbed into tensors $(\mathcal{K}_s)_s$. Let $(P_s)_s$ be orthogonal projections on the one-site physical space such that

$$\sum_s P_s = I, \quad P_s P_t = 0 \quad (s \neq t).$$

Suppose also that, for every s and every pair of virtual indices β, α , the corresponding physical slice satisfies

$$P_s \mathcal{K}_s[\beta, \alpha] = \mathcal{K}_s[\beta, \alpha] P_s.$$

If every blocked representative $\mathcal{K}_s^{[2]}$ is a renormalization fixed point via trace-preserving completely positive maps, then $K^{[2]}$ is also such a renormalization fixed point.

The global maps first measure the outer sector on the first original site of the blocked input. On one blocked site the corresponding projection is $P_s \otimes I$; on two blocked sites it acts on the first original site and as the identity on the remaining three. For every virtual boundary X , the boundary supplied to sector s is the sum of the diagonal copy blocks of $G^{-1}XG$ over that representative. The same boundary is used in the length-two and length-four identities. This is the conditional composition underlying [11, Appendix C.2, lines 1646–1665 and 1810–1825]; it is not the printed implication (iv) $\Rightarrow$ (v) of Theorem 4.9.

Proof. ✓ Proof Write $\mathcal{C}_N(A, X)$ for the length- N closure of A with boundary X , and write $B_s(X)$ for the boundary supplied to representative s . At both relevant lengths the boundary decomposition gives

$$\mathcal{C}_N(K, X) = \sum_t \mathcal{C}_N(\mathcal{K}_t, B_t(X)), \quad N \in \{2, 4\}.$$

Let $\Phi_{s,N}$ be compression by P_s on the first site and by the identity on the remaining N sites. Two-sided physical support and $P_s P_t = 0$ for $s \neq t$ give the selection identity

$$\Phi_{s,N}(\mathcal{C}_{N+1}(\mathcal{K}_t, B_t(X))) = \begin{cases} \mathcal{C}_{N+1}(\mathcal{K}_t, B_t(X)), & s = t, \\ 0, & s \neq t. \end{cases}$$

Apply the representative channel to the surviving summand and sum over s . The projective resolution makes both resulting maps trace-preserving and completely positive. The boundary decomposition then identifies their outputs with the length-two and length-four closures of K carrying the original boundary. $\square$

Theorem 20.16.7.46 (Completion outside the absorbed BNT supports). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) *Retain the hypotheses of Theorem 20.16.7.45, except that the pairwise orthogonal outer support projections are not required to sum to the identity. If every blocked absorbed representative is a renormalization fixed point via trace-preserving completely positive maps, then the blocked complete tensor is also such a fixed point.*

Proof. [✓ Proof](#) If the one-site physical space has dimension zero, all of its endomorphisms are equal. Hence $\sum_s P_s = I$, and Theorem 20.16.7.45 applies directly. Assume from now on that the physical dimension is positive. Fixed basis vectors witness that the two output spaces are nonempty; normalizing their identity matrices gives maximally mixed density matrices $\rho_2 \in M_{q^2}(\mathbb{C})$ and $\rho_4 \in M_{q^4}(\mathbb{C})$.

Let $Q = \sum_s P_s$, an orthogonal projection by pairwise orthogonality. For $N \in \{2, 4\}$ write $Q_N = Q \otimes \mathbb{1}^{\otimes(N-1)}$ for the lift of Q to the first original site of an N -site chain, and likewise $P_{s,N} = P_s \otimes \mathbb{1}^{\otimes(N-1)}$, so that $Q_N = \sum_s P_{s,N}$. Two-sided support of each absorbed representative carries over to the length-two and length-four closures of the complete tensor K ,

$$Q_2 K_2(X) Q_2 = K_2(X), \quad Q_4 K_4(X) Q_4 = K_4(X), \quad (20.77)$$

for every virtual boundary X .

By hypothesis every $\mathcal{K}_s^{[2]}$ carries trace-preserving completely positive maps $\mathcal{S}_s, \mathcal{T}_s$ witnessing its fixed-point property. Sum these channels after compression by the matching first-site projection, $\mathcal{S}_{\text{act}}(Y) = \sum_s \mathcal{S}_s(P_{s,4} Y P_{s,4})$ and $\mathcal{T}_{\text{act}}(Z) = \sum_s \mathcal{T}_s(P_{s,2} Z P_{s,2})$. Each sum is completely positive, and pairwise orthogonality of the P_s makes it preserve exactly the trace selected by Q ,

$$\text{tr}(\mathcal{S}_{\text{act}}(Y)) = \text{tr}(Q_4 Y), \quad \text{tr}(\mathcal{T}_{\text{act}}(Z)) = \text{tr}(Q_2 Z).$$

Since Q_N is Hermitian and idempotent, applying Theorem 21.2.6.1 to each active sum with $P = Q_N$ completes it on the orthogonal input complement,

$$\mathcal{S}_{\text{glob}}(Y) = \mathcal{S}_{\text{act}}(Y) + \text{tr}((I - Q_4)Y)\rho_2, \quad \mathcal{T}_{\text{glob}}(Z) = \mathcal{T}_{\text{act}}(Z) + \text{tr}((I - Q_2)Z)\rho_4,$$

and both global maps are trace-preserving and completely positive. By (21.53) and (20.77), $\mathcal{S}_{\text{glob}}(K_4(X)) = \mathcal{S}_{\text{act}}(K_4(X))$ and $\mathcal{T}_{\text{glob}}(K_2(X)) = \mathcal{T}_{\text{act}}(K_2(X))$ for every virtual boundary X , so the two channel equations for $\mathcal{S}_{\text{glob}}, \mathcal{T}_{\text{glob}}$ reduce to those of $\mathcal{S}_{\text{act}}, \mathcal{T}_{\text{act}}$ and hence, by construction of the active sums, to the representative equations. $\square$

Theorem 20.16.7.47 (Blocking channels from representative trace factorizations). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) *Let M be exactly gauge equivalent, by an invertible virtual matrix G , to a canonical BNT sum whose repeated-copy weights agree within each fixed representative and have been absorbed into tensors $(\mathcal{K}_s)_s$. Suppose that the simultaneous one-site tuples of the decomposition span the product of the virtual matrix algebras and that distinct absorbed representatives satisfy the unstarred layer orthogonality identities. If every $\mathcal{K}_s$ carries a physical-sector factorization with positive semidefinite neighboring operators $\eta_{k,h}^s$ and real families $(a_k^s)_k, (b_k^s)_k$ with*

$$\text{tr}(\eta_{k,h}^s) = a_k^s b_h^s, \quad \sum_k a_k^s b_k^s = 1,$$

then $M^{[2]}$ is a renormalization fixed point via trace-preserving completely positive maps. In particular, this conclusion holds whenever every representative carries a single-sector factorization whose sole neighboring operator is positive semidefinite of unit trace, since the coefficient families may then be taken identically one. The factorizations are assumed here. They cannot in general be derived from saturation of the area law and zero correlation length: Theorem 20.16.8.71 refutes that representativewise assertion under the complete standing hypotheses; see [22]. Thus this conditional result neither proves Theorem 20.16.7.50 nor itself excludes a different construction of its two channel maps. The independent trace-norm argument in that theorem refutes the source conclusion.

Proof. ✓ Proof Positivity of the neighboring operators makes every absorbed representative generate positive semidefinite periodic operators, and one-site spanning makes it injective. With the unstarred layer identities, Corollary 20.16.8.18 gives pairwise orthogonal two-sided outer physical supports, realizing the projection onto each local subspace in the closing proof of the source. Each supplied trace factorization gives a blocked channel pair for its representative by Theorem 20.16.7.42. The projector-controlled combination of these pairs across the outer supports, completed trace-preservingly outside their sum, is Theorem 20.16.7.46. In the single-sector case the sole trace condition $\text{tr}(\eta^s) = 1$ realizes the required rank-one normalization with unit coefficient families. □

Definition 20.16.7.48 (Sitewise support completion of an adjoint map). Lean ✓ Stmt Let $d > 0$, let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ be a linear map, put $W_N = V^{\otimes N}$ and $P_N = W_N W_N^\dagger$, and let $\tau_N = d^{-N} I \in M_{d^N}(\mathbb{C})$ be the uniform faithful density matrix. The sitewise support-completion map is

$$\mathcal{R}_N(Y) = W_N^\dagger Y W_N + \text{tr}((I - P_N)Y(I - P_N))\tau_N. \quad (20.78)$$

Lemma 20.16.7.49 (The sitewise recovery is a channel). Lean Lean Lean ✓ Stmt If V is an isometry, then $\mathcal{R}_N : M_{e^N}(\mathbb{C}) \rightarrow M_{d^N}(\mathbb{C})$ is trace-preserving and completely positive. It is a left inverse to the physical embedding:

$$\mathcal{R}_N(W_N Z W_N^\dagger) = Z. \quad (20.79)$$

In particular, if $\tilde{A}$ is obtained from an MPO tensor A by the one-site physical isometry V , then

$$\mathcal{R}_N(\mathcal{K}_N^{\tilde{A}}(X)) = \mathcal{K}_N^A(X)$$

for every virtual boundary X . This is a general conditional transport construction: it applies after an isometric physical embedding has been supplied. It neither constructs nor implies the refinement channel asserted in Proposition C.15.

Proof. ✓ Proof Since $W_N^\dagger W_N = I$, the first term of (20.78) has trace $\text{tr}(P_N Y)$. The second is completely positive and has trace $\text{tr}((I - P_N)Y)$. Their sum is therefore a channel. On matrices supported by P_N , the second term vanishes and conjugation by $W_N^\dagger$ proves (20.79). The closure identity follows because a sitewise physical change sends every N -site closure to $W_N \mathcal{K}_N^A(X) W_N^\dagger$. □

Theorem 20.16.7.50 (The SAL–ZCL blocking-channel conclusion is false). Lean ✓ Stmt Discussion: #7479

There is a five-letter MPO tensor K of virtual bond dimension three which generates matrix product density operators, has normalized simple canonical form, is simple in the sense of Definition 4.7, and is in biCF. Its two normal representatives occur once with coefficients μ and 1,

where $0 < |\mu| < 1$, so the paper's global normalization convention holds. The tensor saturates the area law and has literal zero correlation length,

$$\mathcal{T}_K^2 = \mathcal{T}_K.$$

Nevertheless, no trace-preserving completely positive map sends every two-site closure of K to the corresponding three-site closure with the same virtual boundary. Nor does such a map exist from two sites to four sites. Consequently the two-site blocking $K^{[2]}$ is not a renormalization fixed point in the sense of Definition 4.1. Thus Proposition C.15 and its twice-applied contribution to the condition-(ii) route to Theorem 4.9(v) are false as printed [122].

Proof. ✓ Proof Let A be the four-letter rank-one tensor underlying the ambient construction, and choose the virtual boundary X displayed in the paper-gap note [122]. The resulting physical closures are real diagonal Hermitian matrices, and direct finite summation gives

$$\|\mathcal{K}_2^A(X)\|_1 = \frac{337}{250}, \quad \|\mathcal{K}_3^A(X)\|_1 = \|\mathcal{K}_4^A(X)\|_1 = \frac{27}{20}.$$

A trace-preserving positive map contracts the trace norm on Hermitian matrices, but the latter value exceeds the former by $1/500$. Hence the two-to-three and two-to-four refinement maps do not exist for this tensor.

Let V embed its four physical letters into the first four letters of the ambient tensor, let $\tilde{A}$ be the resulting embedded tensor, and let Y_X be the ambient virtual boundary supported on its obstruction block. Moving the virtual gauge to the boundary, discarding the zero terminal block, and applying the physical embedding give, for every N ,

$$\begin{aligned} \mathcal{K}_N^K(Y_X) &= \mathcal{K}_N^{\tilde{A}}(X), \\ \mathcal{K}_N^{\tilde{A}}(X) &= V^{\otimes N} \mathcal{K}_N^A(X) (V^{\otimes N})^\dagger. \end{aligned} \tag{20.80}$$

Lemma 20.16.7.49 supplies a trace-preserving completely positive left inverse to (20.80). Therefore a refinement map for the ambient tensor would descend to one of the two forbidden maps above. Finally, the refinement map of an RFP structure for $K^{[2]}$, transported from blocked coordinates, would give the forbidden two-to-four map for K . $\square$

$$\begin{aligned} U^{\otimes 4} \widehat{\mathcal{T}}^2(U^\dagger)^{\otimes 2} &= \mathcal{T}, \\ U^{\otimes 2} \widehat{\mathcal{S}}^2(U^\dagger)^{\otimes 4} &= \mathcal{S}. \end{aligned} \tag{20.81}$$

Conjugation by $U^{\otimes n}$ and $(U^\dagger)^{\otimes n}$ carries the sector-coordinate channels to the ordinary-coordinate channels. This combines Proposition C.7 with [11, Appendix C.2, lines 1510–1563 and 1821–1825].

20.16.8 Local simple-MPDO structure

Remark 20.16.8.1 (Normalization boundary in Lemma C.5). Lemma C.5 of [11, Appendix C.2, lines 1484–1499] claims that an injective matrix product density operator tensor saturating the area law and having zero correlation length has real families $(a_k)_k, (b_k)_k$ with

$$\begin{aligned} T_{k,h} &= \text{tr}(\eta_{k,h}) = a_k b_h, \\ \sum_k a_k b_k &= 1. \end{aligned}$$

The lemma is proved inside Case I, where the tensor is also assumed to be an injective normal tensor. The raw four-sector tensor below has the source inverse-map factorization and no such families a, b , but it is not the normalized tensor of Case I: its doubled-index transfer satisfies $\mathcal{E}_{\mathcal{K}}^2 = (5/16)\mathcal{E}_{\mathcal{K}}$. Its normalized representative $A = (4/\sqrt{5})\mathcal{K}$ satisfies $\mathcal{B}_A^2 = (4/\sqrt{5})\mathcal{B}_A \neq \mathcal{B}_A$ and therefore fails the paper's literal zero-correlation-length diagram. Thus it does not obstruct the all-sector coefficient theorem proved under the complete normal Case-I hypotheses. The remaining open step is the structural statement used by the following corollary.

Theorem 20.16.8.2 (SAL–ZCL rank-one normalization obstruction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) *There is an injective matrix product density operator tensor satisfying saturation of the area law and source zero correlation length whose source-selected inverse-map factorization has positive semidefinite neighboring operators and a primitive trace-one active trace matrix T , but for which there are no families a, b satisfying $T_{k,h} = a_k b_h$ for all k, h . Equivalently, for the corresponding closed sector tensors L, Q , one has $Q(1 - LQ)L = T - T^2 \neq 0$. Thus the rank-one conclusion in Lemma C.5 fails for this raw representative. The normal-tensor hypothesis imposed in Case I at source lines 1374–1381 is not included in this statement.*

Proof. [Stmt](#) The four physical sectors are one-dimensional. Their cyclic transition law is doubly stochastic. After tracing one site, conditioning on the middle physical label gives, at every admissible cut,

$$\rho_{ABC} = \bigoplus_{k=0}^3 \frac{1}{4} \rho_{A,k}^L \otimes \rho_{k,C}^R.$$

Thus the reduced state has a four-sector quantum-Markov decomposition and the tensor satisfies saturation of the area law. Its physical-trace transfer is the idempotent matrix LQ , so it has source zero correlation length. The explicit Hayashi decomposition and inverse map recover the displayed tensors L, Q . The active trace matrix is primitive and has trace one, but direct multiplication gives $T^2 \neq T$. If $T = ab^\top$, then

$$T^2 = a(b^\top a)b^\top = \left(\sum_k a_k b_k \right) T = T,$$

where the last equality uses $\text{tr}(T) = 1$. Hence T has no rank-one factorization. $\square$

Theorem 20.16.8.3 (Blocked renormalization maps from two-to-four maps). [Lean](#) [Stmt](#) *Let K be an MPO tensor. Suppose that trace-preserving completely positive maps S_{24} and T_{24} satisfy the following identities for every virtual matrix X :*

$$S_{24}[\mathcal{K}_4(X)] = \mathcal{K}_2(X), \quad T_{24}[\mathcal{K}_2(X)] = \mathcal{K}_4(X).$$

Then the two-site physical blocking $K^{[2]}$ is a renormalization fixed point.

Proof. [Stmt](#) Identify the two-site physical space of K with the one-site physical space of $K^{[2]}$, and identify the four-site physical space of K with the two-site physical space of $K^{[2]}$. These identifications merely permute orthonormal coordinates, so conjugating S_{24} and T_{24} by them preserves complete positivity and trace preservation. The two assumed closure identities then become precisely the two defining renormalization identities for $K^{[2]}$. $\square$

Theorem 20.16.8.4 (Two-site blocking preserves renormalization maps). [Lean](#) [Stmt](#) *Suppose that a tensor K has trace-preserving completely positive maps S, T satisfying*

$$S[\mathcal{K}_2(X)] = \mathcal{K}_1(X), \quad T[\mathcal{K}_1(X)] = \mathcal{K}_2(X)$$

for every virtual matrix X . Then the two-site physical blocking $K^{[2]}$ has maps satisfying the same two identities and is therefore a renormalization fixed point.

Proof. ✓ Proof Apply T twice to the first physical site, first passing from two sites to three and then from three sites to four. This gives a trace-preserving completely positive map from the two-site closure to the four-site closure. Applying S twice to the first two physical sites in the reverse direction gives the coarse-graining map. The canonical identifications of two neighboring sites with one blocked site are permutations of orthonormal coordinates and preserve both complete positivity and trace preservation. $\square$

Theorem 20.16.8.5 (Renormalization maps and normalization obstruction for the four-sector tensor). Lean Lean Lean Lean ✓ Stmt The four-sector tensor in Theorem 20.16.8.2 has trace-preserving completely positive maps S, T such that, for every virtual matrix X ,

$$S[\mathcal{K}_2(X)] = \mathcal{K}_1(X), \quad T[\mathcal{K}_1(X)] = \mathcal{K}_2(X).$$

Its two-site physical blocking has maps satisfying the same two identities and is therefore also a renormalization fixed point. On the other hand, for the source-normalized scalar representative

$$A = \frac{4}{\sqrt{5}}\mathcal{K},$$

the two-site blocking $A^{[2]}$ is not a renormalization fixed point.

The last assertion concerns literal trace-preserving diagrammatic renormalization. The rescaled tensor obeys an idempotence relation up to a positive scalar, but it does not satisfy the paper's literal zero-correlation-length diagram. Consequently it is not, by itself, a counterexample to Theorem 4.9 of [11]. The simple-biCF-BNT hypotheses of that theorem are not part of this theorem's statement.

Proof. ✓ Proof Let $\tau = QL$ be the neighboring transition matrix and set

$$f(i, j) = (i \bmod 2) + 2 \left\lfloor \frac{j}{2} \right\rfloor.$$

Direct multiplication of the sector matrices gives

$$S_i S_j = \tau_{ij} S_{f(i, j)}.$$

For each k , the transition weights on the fibre of f satisfy

$$\sum_{(i, j): f(i, j)=k} \tau_{ij} = 1.$$

The refinement channel has Kraus operators $\sqrt{\tau_{ij}}|i, j\rangle\langle f(i, j)|$; the coarse-graining channel has Kraus operators $|f(i, j)\rangle\langle i, j|$. Their identity resolutions give trace preservation, and the two displayed sector identities give the two required physical-closure identities. Applying each channel twice and regrouping two neighboring physical sites gives the maps for the blocked tensor, as in Theorem 20.16.8.4.

Let $P = \text{diag}(1, 0)$. The physical-trace transfer of the normalized representative is $(4/\sqrt{5})P$, while that of its two-site blocking is

$$\mathcal{B}_{A^{[2]}} = \frac{16}{5}P.$$

Hence the neighboring operators are those obtained directly from L and Q , and direct multiplication gives $Q(1 - LQ)L \neq 0$. $\square$

Corollary 20.16.8.8 (Local sector structure from SAL and ZCL). [Lean](#) [✓ Stmt](#) *Let $\mathcal{K}$ be an injective normal matrix product density operator tensor satisfying saturation of the area law and literal zero correlation length,*

$$\mathcal{T}_{\mathcal{K}}^2 = \mathcal{T}_{\mathcal{K}}.$$

Then there is a physical-sector factorization

$$\mathbb{C}^d \simeq \bigoplus_k (B_k^L \otimes B_k^R),$$

with

$$U\kappa_{\beta,\alpha}U^\dagger = \bigoplus_k (l_k)_\beta \otimes (r_k)_\alpha,$$

positive semidefinite neighboring operators $\eta_{k,h}$, and real families $(a_k)_k, (b_k)_k$ such that

$$\text{tr}(\eta_{k,h}) = a_k b_h,$$

and

$$\sum_k a_k b_k = 1.$$

The orientation of U is the physical-sector-factorization convention displayed above. No additional closure property of the sector tensors is asserted. This is the structural corollary at [11, Appendix C.2, lines 1503–1506].

Proof. [✓ Proof](#) Apply Theorem 20.16.6.40. This gives the physical isometry, the direct-sum physical-slice formula, and positive neighboring operators. Theorem 20.16.6.40 gives the real-part identity $\text{Re tr}(\eta_{k,h}) = a_k b_h$. Positivity makes each neighboring operator Hermitian, so its trace is real and the displayed complex trace identity follows. $\square$

Theorem 20.16.8.9 (Positive neighboring sectors generate positive periodic operators). [Lean](#) [✓ Stmt](#) *Suppose that an MPO tensor K admits a physical-sector factorization through an isometry U , with neighboring operators $\eta_{q,h}$. If $\eta_{q,h} \geq 0$ for every pair (q, h) , then for every $N \geq 1$,*

$$\rho^{(N)}(K) \geq 0.$$

Proof. [✓ Proof](#) In sector coordinates, the periodic operator is block diagonal, and each block is a tensor product of positive neighboring operators. Hence every block, and therefore the full operator, is positive semidefinite. Conjugation by $U^{\otimes N}$ preserves positivity and returns $\rho^{(N)}(K)$. $\square$

Definition 20.16.8.10 (Unstarred orthogonality of BNT layers). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For a family $\{\mathcal{K}_x\}_{x=0}^{g-1}$, write $\mathcal{K}_x[\beta, \alpha] \in M_d(\mathbb{C})$ for the physical matrix obtained by fixing the two virtual indices. The layers are orthogonal when, for $x \neq y$,*

$$\mathcal{K}_y[\beta_y, \alpha_y] \mathcal{K}_x[\beta_x, \alpha_x] = 0 \tag{20.82}$$

$$\sum_{k=0}^{d-1} (\mathcal{K}_y)_{\beta_y \alpha_y}^{ik} (\mathcal{K}_x)_{\beta_x \alpha_x}^{kj} = 0.$$

The factor order is $\mathcal{K}_y \mathcal{K}_x$ without adjoints. This is the orthogonality relation in [11, Appendix C.2, lines 1634–1639].

$$\sum_{k=0}^{d-1} (\mathcal{K}_y)_{\beta_y \alpha_y}^{ik} (\mathcal{K}_x)_{\beta_x \alpha_x}^{kj} = 0.$$

The factor order is $\mathcal{K}_y \mathcal{K}_x$ without adjoints. This is the orthogonality relation in [11, Appendix C.2, lines 1634–1639].

$$(K^\sharp)^{ij}_{\beta\alpha} = \overline{K^{ji}_{\beta\alpha}}.$$
$$\rho^{(N)}(K^\sharp)_{\sigma, \tau} = \overline{\rho^{(N)}(K)_{\tau, \sigma}}.$$
$$K[\beta, \alpha]^* = \sum_{\nu, u=0}^{D-1} X_{\beta\mu} (X^{-1})_{\nu\alpha} K[\mu, \nu]$$

Proof. ✓ Proof Positivity makes every periodic operator Hermitian, so K and $K^\sharp$ generate the same matrix-product vectors. The single-block fundamental theorem gives the inner gauge X . Taking its virtual matrix entries yields the displayed expansion. $\square$

$$UK[\beta, \alpha]U^* = \bigoplus_k l_{k, \beta} \otimes r_{k, \alpha}$$
$$\begin{aligned} P_k^L l_{k,\beta} &= l_{k,\beta} = l_{k,\beta} P_k^L, & P_k^R r_{k,\alpha} &= r_{k,\alpha} = r_{k,\alpha} P_k^R, \\ \text{supp}\{l_{k,\beta} \otimes r_{k,\alpha}\}_{\beta,\alpha} &= P_k^L \otimes P_k^R. \end{aligned}$$

Moreover, there are positive integers p_k, q_k and isometries

$$V_k^L : \mathbb{C}^{p_k} \longrightarrow \mathbb{C}^{d_k^L}, \quad V_k^R : \mathbb{C}^{q_k} \longrightarrow \mathbb{C}^{d_k^R}$$

whose range projections are P_k^L and P_k^R . If k is inactive, every product $l_{k,\beta} \otimes r_{k,\alpha}$ vanishes; in particular, there is no active sector when the virtual dimension is zero.

Proof. ✓ Proof The sector-coordinate tensor remains injective. Positivity of the neighboring operators makes it a matrix product density operator, so Theorem 20.16.8.12 closes its physical slices under conjugate transpose. Restriction to the k -th diagonal sector gives coefficients $c_{k,\beta,\alpha}(\mu, \nu)$ such that

$$(l_{k,\beta} \otimes r_{k,\alpha})^* = \sum_{\mu, \nu} c_{k,\beta,\alpha}(\mu, \nu) l_{k,\mu} \otimes r_{k,\nu}. \quad (20.83)$$

Since k is active, choose α, u, v for which $r_{k,\alpha}(u, v) \neq 0$, and put $z = \overline{r_{k,\alpha}(u, v)}$. Evaluating (20.83) at the corresponding matrix entry and dividing by z yields

$$l_{k,\beta}^* = \sum_{\mu} \left(z^{-1} \sum_{\nu} c_{k,\beta,\alpha}(\mu, \nu) r_{k,\nu}(v, u) \right) l_{k,\mu}. \quad (20.84)$$

Interchanging the two factors gives the corresponding identity for $r_{k,\alpha}^*$. Thus both factor spans are closed under conjugate transpose. The joint column-support projections then absorb the factors on both sides. Their tensor product is the support projection of the product family, and finite-dimensional range decompositions give the two isometries. $\square$

Theorem 20.16.8.14 (Canonical restriction constructed from the active sector factors). Lean

Lean Lean Lean Lean Lean Lean ✓ Stmt Let K be injective and let

$$UK[\beta, \alpha]U^* = \bigoplus_k l_{k,\beta} \otimes r_{k,\alpha}$$

be a physical-sector factorization whose neighboring operators are positive semidefinite. In every active sector choose isometries onto the joint column supports of the left and right factor families, and retain no coordinates in an inactive sector. Their direct sum, transported back by U^* , is an isometry

$$V : \mathbb{C}^e \longrightarrow \mathbb{C}^d$$

whose range projection is the canonical joint column support P_K of the physical slices of K . The restricted tensor $K_{\text{act}} = V^*KV$ is injective, satisfies

$$VK_{\text{act}}V^* = K,$$

and, in the dependent sector coordinates, has physical slices

$$K_{\text{act}}[\beta, \alpha] = \bigoplus_k ((V_k^L)^* l_{k,\beta} V_k^L) \otimes ((V_k^R)^* r_{k,\alpha} V_k^R).$$

This construction also applies when every sector is inactive: then $e = 0$, with no complementary summand added.

Proof. ✓ Proof Take the tensor product of the two support isometries in each active sector and their dependent direct sum over all sectors, using an empty coordinate space in every inactive sector. Block multiplication gives orthonormal columns and the displayed compressed factorization. The range projection absorbs every physical slice. Conversely, every projection that absorbs all physical slices also absorbs the direct sum of their joint column supports. Hence this range projection is P_K . Two-sided absorption then gives exact re-embedding, and injectivity descends through the physical isometry. $\square$

Theorem 20.16.8.15 (Neighboring positivity and trace factorization on the canonical active restriction). Lean Lean Lean Lean Lean Lean ✓ Stmt *In the setting of Theorem 20.16.8.14, the restricted tensor K_{act} has a physical-sector factorization indexed precisely by the active ambient sectors. For active sectors k, h , put*

$$W_{k,h} = V_k^R \otimes V_h^L.$$

Its neighboring operator is

$$\eta_{k,h}^{\text{act}} = W_{k,h}^* \eta_{k,h} W_{k,h}.$$

In particular, every $\eta_{k,h}^{\text{act}}$ is positive semidefinite. If the active family is empty, the restricted factorization has no sectors and the assertion is vacuous.

Suppose moreover that the ambient neighboring operators admit real coefficients a_k, b_h such that

$$\begin{aligned} \text{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_k a_k b_k &= 1. \end{aligned}$$

For the active-sector correspondence $j \mapsto k(j)$, set $\tilde{a}_j = a_{k(j)}$ and $\tilde{b}_j = b_{k(j)}$. Then

$$\begin{aligned} \text{tr}(\eta_{j,l}^{\text{act}}) &= \tilde{a}_j \tilde{b}_l, \\ \sum_j \tilde{a}_j \tilde{b}_j &= 1. \end{aligned}$$

Scope restriction (active physical support compression): *The identities being preserved correspond to [11, Theorem 4.9(iv), lines 864–889]. Their transport through active physical support compression is an auxiliary construction rather than an assertion of that source theorem; documented in [25].*

Proof. ✓ Proof Remove the empty inactive fibers from the dependent direct sum and enumerate the remaining sector subtype by a finite ordinal. The compressed right factor of k and compressed left factor of h give

$$\sum_{\alpha} ((V_k^R)^* r_{k,\alpha} V_k^R) \otimes ((V_h^L)^* l_{h,\alpha} V_h^L) = W_{k,h}^* \eta_{k,h} W_{k,h}.$$

Positive semidefiniteness is preserved by this matrix congruence. The range projections of V_k^R and V_h^L satisfy the two-sided absorption identity

$$(P_k^R \otimes P_h^L) \eta_{k,h} = \eta_{k,h} = \eta_{k,h} (P_k^R \otimes P_h^L).$$

Cyclicity of trace therefore gives $\text{tr}(\eta_{k,h}^{\text{act}}) = \text{tr}(\eta_{k,h})$. For an inactive sector, $\eta_{k,k} = 0$, hence $a_k b_k = 0$. Consequently, splitting the normalized sum and reindexing its active part by $j \mapsto k(j)$ gives

$$\begin{aligned} 1 &= \sum_k a_k b_k \\ &= \sum_{\substack{k \\ k \text{ active}}} a_k b_k + \sum_{\substack{k \\ k \text{ inactive}}} a_k b_k \\ &= \sum_j \tilde{a}_j \tilde{b}_j + 0 \\ &= \sum_j \tilde{a}_j \tilde{b}_j. \end{aligned}$$

□

Theorem 20.16.8.16 (Neighboring-trace factorization descends through a full-support physical embedding). [Lean](#) [✓ Stmt](#) *Let $\mathcal{K}$ be injective, and suppose that the joint column support of its physical slices is the whole one-site space, $\Pi_{\mathcal{K}} = \mathbb{1}_d$. Let $V : \mathbb{C}^d \rightarrow \mathbb{C}^e$ be an isometry and put*

$$\mathcal{K}_V^{ij} = \sum_{p,q} V_{i,p} \overline{V_{j,q}} \mathcal{K}^{pq}.$$

If $\mathcal{K}_V$ admits a physical-sector factorization for which there exist families $(a_k)_k$ and $(b_h)_h$ of real numbers such that the neighboring operators satisfy

$$\begin{aligned} \eta_{k,h} &\geq 0, \\ \text{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_k a_k b_k &= 1, \end{aligned}$$

then $\mathcal{K}$ admits such a factorization as well.

This implication is project-derived and is not stated in [11]. It transports the neighboring data of Theorem 4.9(iv), lines 864–889, through the physical-support restriction used in Appendix C.2, lines 1680–1691 and 1733–1770.

Proof. [✓ Proof](#) The isometric embedding theorem gives

$$\Pi_{\mathcal{K}_V} = V \Pi_{\mathcal{K}} V^* = V V^*.$$

It also preserves injectivity. Restrict the assumed factorization to its canonical active physical support. The preceding theorem preserves positivity, the identities $\text{tr}(\eta_{k,h}) = a_k b_h$, and the normalized sum $\sum_k a_k b_k = 1$ on that restriction. Its isometric inclusion W satisfies $W W^* = \Pi_{\mathcal{K}_V} = V V^*$.

The support-coordinate comparison therefore carries the factorization from $W^* \mathcal{K}_V W$ to $V^* \mathcal{K}_V V$. Finally,

$$V^* \mathcal{K}_V V = (V^* V) \mathcal{K} (V^* V) = \mathcal{K},$$

which proves the assertion. □

Theorem 20.16.8.17 (Orthogonal two-sided supports from unstarred layer orthogonality).

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{\mathcal{K}_x\}_{x=0}^{g-1}$ be injective MPO tensors, each of which generates positive semidefinite periodic operators at every positive length. If, for every pair $x \neq y$, the unstarred identities

$$\mathcal{K}_y[\beta_y, \alpha_y] \mathcal{K}_x[\beta_x, \alpha_x] = 0$$

hold for all virtual indices, then there are pairwise orthogonal projections $P_x \in M_d(\mathbb{C})$ such that

$$P_x \mathcal{K}_x[\beta, \alpha] = \mathcal{K}_x[\beta, \alpha] = \mathcal{K}_x[\beta, \alpha] P_x$$

for every x, β, α . Conversely, any pairwise annihilating family with these two-sided support identities satisfies the unstarred layer identities.

Proof. [✓ Proof](#) Let C_x be the matrix obtained by stacking the columns of all slices of $\mathcal{K}_x$, and let P_x be the orthogonal projection onto its column space. The left support identity is immediate. Theorem 20.16.8.12 gives the right support identity. For $x \neq y$, (20.82) and adjoint closure give

$$C_x^* C_y = 0, \quad P_x P_y = 0.$$

□

Corollary 20.16.8.18 (Supports of common-weight BNT representatives). [Lean](#) [✓ Stmt](#) Let a BNT decomposition have a common nonzero weight within each sector, and suppose that its one-site words span the product of the virtual matrix algebras. For every sector s , assume a physical-sector factorization with positive neighboring operators. If distinct absorbed representatives obey the unstarred layer identities, then there are pairwise orthogonal projections P_s such that

$$P_s \mathcal{K}_s[\beta, \alpha] = \mathcal{K}_s[\beta, \alpha] = \mathcal{K}_s[\beta, \alpha] P_s$$

for every s, β, α .

Proof. [✓ Proof](#) One-site spanning makes every absorbed representative injective. The positive neighboring operators make every representative an MPDO. Apply Theorem 20.16.8.17. □

Theorem 20.16.8.19 (Positive sector bonds on the printed physical supports). [Lean](#) [Lean](#) [✓ Stmt](#) Let K be injective and admit a physical-sector factorization with positive semidefinite neighboring operators. If an orthogonal projection P satisfies

$$PK[\beta, \alpha] = K[\beta, \alpha] = K[\beta, \alpha]P$$

for every β, α , then there is a positive two-site operator B whose periodic translates commute and such that

$$(P \otimes P)B(P \otimes P) = B, \\ \text{mpo}(K, N) = \prod_{j=0}^{N-1} \tau_j(B) \quad (N \geq 2).$$

Applied to pairwise orthogonal projections P_s , this construction gives an orthogonal family of positive commuting sector bonds for the tensors K_s .

Proof. [✓ Proof](#) Restrict K to its canonical joint physical support Q by Theorems [20.16.8.14](#) and [20.16.8.15](#). The positive neighboring operators on the restriction give a positive commuting bond supported on $Q \otimes Q$, with the displayed periodic product. Minimality of the joint column support and the two-sided absorption by P give

$$PQ = Q = QP.$$

Hence support of the bond on $Q \otimes Q$ implies support on $P \otimes P$. Performing this construction in every orthogonal sector gives the final assertion. $\square$

Theorem 20.16.8.20 (The five BNT identities give the GSNNCH form). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) *Let M generate positive semidefinite periodic operators and have a BNT decomposition with representatives A_x of positive bond dimension, copy numbers $n_x \geq 1$, and arbitrary copy weights $\mu_{x,q}$. Suppose that the decomposition generates the same matrix product vectors as M at every positive length. Assume that the simultaneous one-site tuples $(A_x^i)_x$, as i ranges over the physical labels, span the product of the virtual matrix algebras. Assume also that, for distinct x, y ,*

$$A_x[\beta, \alpha]A_y[\beta', \alpha'] = 0$$

for all virtual indices. For every x , suppose that A_x has the physical-sector factorization

$$U_x A_x U_x^* = \bigoplus_k l_{x,k} \otimes r_{x,k},$$

with positive neighboring operators $\eta_{k,h}^x$ and real numbers a_k^x, b_k^x satisfying

$$\text{tr}(\eta_{k,h}^x) = a_k^x b_h^x, \quad \sum_k a_k^x b_k^x = 1.$$

Then the matrix product density operators generated by M have the GSNNCH form, with the BNT copy numbers n_x as their outer multiplicities. No equality among the copy weights is assumed.

Proof. [✓ Proof](#) The physical-sector identities make every A_x an MPDO. The unstarred layer identities and one-site spanning therefore give pairwise orthogonal projections P_x supporting the raw representatives on both sides. Fix $N \geq 2$ and write

$$q_x(N) = \sum_{q=0}^{n_x-1} \mu_{x,q}^N, \quad \rho^{(N)}(M) = \sum_x q_x(N) \rho^{(N)}(A_x).$$

Sitewise compression by $P_x^{\otimes N}$ isolates $q_x(N) \rho^{(N)}(A_x)$. It is positive because $\rho^{(N)}(M)$ is positive. The operator $\rho^{(N)}(A_x)$ is positive and nonzero: the latter follows from positive bond dimension and one-site injectivity. Hence uniqueness of the positive scalar phase shows that $q_x(N)$ is a nonnegative real number.

Theorem [20.16.8.19](#) gives a supported positive commuting bond B_x with periodic product $\rho^{(N)}(A_x)$. Replace it at this chain length by

$$\widetilde{B}_{x,N} = \left(\frac{q_x(N)}{n_x} \right)^{1/N} B_x.$$

Then its multiplicity-weighted periodic product is exactly $q_x(N) \rho^{(N)}(A_x)$. The GSNNCH data may depend on N , so this construction proves the claimed form without copy independence. $\square$

Theorem 20.16.8.21 (Square-zero obstruction for arbitrary MPO tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *There is a two-label family of bond-dimension-one MPO tensors for which the unstarred layer identities hold, but no pairwise orthogonal one-site projections support both tensors on both sides. One may take both labels to have the single physical slice*

$$A = |0\rangle\langle 1|, \quad A \neq 0, A^2 = 0.$$

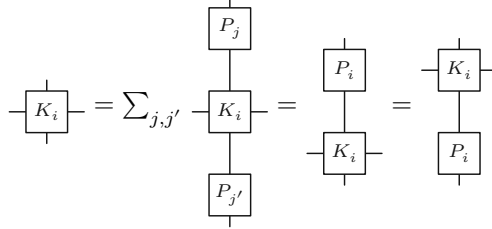
Proof. [✓ Proof](#) The equation $A^2 = 0$ gives both ordered layer products. If $P_0 P_1 = 0$ and $P_x A = A$ for both labels, then

$$A = P_0 A = P_0 P_1 A = 0,$$

contradicting $A \neq 0$. □

If the projections also satisfy $\sum_j P_j = \mathbb{1}$, then expanding this identity on both physical indices and using the sector projector relations leaves only the diagonal term:

$$\mathcal{K}_i = \sum_{j,j'} P_j \mathcal{K}_i P_{j'} = P_i \mathcal{K}_i = \mathcal{K}_i P_i.$$



Equivalently, each local tensor is supported on its own sector: $\mathcal{K}_i = P_i \mathcal{K}_i = \mathcal{K}_i P_i$. This is the calculation in [11, Appendix C.2, lines 1745–1751].

Theorem 20.16.8.22 (Normal BNT representatives, biCF, exact direct sum, and nonnilpotence in the Case-II example). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let A_0, A_1 be the bond-one tensors whose only nonzero doubled-physical entries are*

$$A_0^{00} = A_0^{11} = \frac{1}{\sqrt{2}}, \quad A_1^{22} = 1.$$

With $(\mu_0, \mu_1) = (1/\sqrt{2}, 1)$, they are normal and form a normalized BNT canonical decomposition. Their simultaneous word-tuple span is the full two-block direct sum already at length one, so the representative family is in biCF. The ambient tensor is

$$M^{ab} = \text{diag}(\mu_0 A_0^{ab}, \mu_1 A_1^{ab}),$$

both entrywise and as a doubled-index MPS tensor, and is in normalized BNT-refined horizontal form. Moreover, the doubled physical-trace transfer of neither representative is nilpotent.

Each representative occurs in exactly one copy. Thus independence of the weight from the copy label is vacuous within each representative and does not compare the two representatives or their displayed weights.

Proof. [✓ Proof](#) The two ordinary transfer maps satisfy

$$\mathcal{E}_{A_0} = \mathcal{E}_{A_1} = \text{id}_{\mathbb{C}}.$$

Their disjoint physical supports exclude equivalence by a virtual gauge and a phase. Write $\mathbf{A}(i) = (A_0^i, A_1^i)$. For every $Y = (Y_0, Y_1) \in \mathbb{C} \oplus \mathbb{C}$, the two letters 0 and 8 give

$$Y = \frac{Y_0}{1/\sqrt{2}} \mathbf{A}(0) + Y_1 \mathbf{A}(8),$$

which proves the simultaneous one-letter span. Direct evaluation gives

$$M^{ab} = \text{diag}((1/\sqrt{2})A_0^{ab}, A_1^{ab}),$$

and hence the identity block gauge. Finally,

$$\begin{aligned} \mathcal{T}_{\text{phys}}(A_0) &= \frac{2}{\sqrt{2}}, \\ \mathcal{T}_{\text{phys}}(A_1) &= 1, \end{aligned}$$

so neither physical-trace transfer is nilpotent. $\square$

Theorem 20.16.8.23 (Coefficient absorption need not preserve normality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A_0 have scalar entries $1/\sqrt{2}$ on doubled symbols $(0, 0)$ and $(1, 1)$, let A_1 have scalar entry 1 only on $(2, 2)$, and take $(\mu_0, \mu_1) = (1/\sqrt{2}, 1)$. Their weighted direct sum has

$$K^{00} = K^{11} = \text{diag}(1/2, 0), \quad K^{22} = \text{diag}(0, 1). \quad (20.85)$$

For every $N > 0$ and physical strings $\sigma, \tau \in \{0, 1, 2\}^N$,

$$\langle \sigma | \rho^{(N)}(K) | \tau \rangle = \delta_{\sigma, \tau} \left(2^{-N} \mathbf{1}_{\{\sigma_k \in \{0, 1\} \ \forall k\}} + \mathbf{1}_{\{\sigma_k = 2 \ \forall k\}} \right), \quad (20.86)$$

$$\text{tr } \rho^{(N)}(K) = 2 > 0. \quad (20.87)$$

Hence K generates positive semidefinite operators. It satisfies the saturation of the area law and the literal zero-correlation-length equation

$$\mathcal{T}_K = I_2, \quad \mathcal{T}_K^2 = \mathcal{T}_K. \quad (20.88)$$

Its displayed BNT decomposition has the global unit-weight convention, its representatives form a biCF family, and K has simple canonical form at this normalized fixed representative. Nevertheless,

$$\rho(\mathcal{E}_{\mu_0 A_0}) = \frac{1}{2}, \quad \mu_0 A_0 \text{ is not normal.} \quad (20.89)$$

Copy-weight independence is only within the single copy of each representative; it does not equate μ_0 and μ_1 . Thus all conditions used in the Case-II passage are present, but the printed absorbed-normality inference is false [11, lines 1628–1665 and 1740–1782]. This does not refute the asserted all-sector conclusion or Theorem 4.9.

Proof. ✓ Proof Set

$$\begin{aligned} P &= \text{diag}(1, 1, 0), & Q &= I - P = \text{diag}(0, 0, 1), \\ \omega_P &= \frac{1}{4}(P \otimes P), & \omega_Q &= Q \otimes Q. \end{aligned} \quad (20.90)$$

Equations (20.85) and (20.86) follow by multiplying the diagonal slices. There are 2^N binary strings and one terminal string, which gives (20.87). For a virtual matrix X , the one-site and two-site physical closures are

$$M_1(X) = \frac{X_{00}}{2}P + X_{11}Q, \quad M_2(X) = X_{00}\omega_P + X_{11}\omega_Q. \quad (20.91)$$

Define

$$\begin{aligned} \mathcal{S}(Y) &= \text{tr}_2(Y), \\ \mathcal{R}(Y) &= \text{tr}(PY P)\omega_P + \text{tr}(QY Q)\omega_Q. \end{aligned} \quad (20.92)$$

Both maps are trace-preserving and completely positive, and

$$\mathcal{S}[M_2(X)] = M_1(X), \quad \mathcal{R}[M_1(X)] = M_2(X). \quad (20.93)$$

These are the channel equations of Definition 4.1 [11, lines 638–660]; with (20.87), they imply saturation of the area law. Summing the diagonal slices gives (20.88), the literal diagram of Definition 4.2 [11, lines 735–739].

The transfer maps of the two unabsorbed representatives obey

$$\mathcal{E}_{A_0}(x) = \left(\frac{1}{2} + \frac{1}{2}\right)x = x, \quad \mathcal{E}_{A_1}(x) = x.$$

Hence both are normal. The letters 0 and 8 separately select the two scalar blocks, proving the simultaneous one-letter span and biCF. The weights satisfy $|\mu_s| \leq 1$ and $|\mu_1| = 1$, as required by the global canonical normalization at lines 224–246. Absorption nevertheless gives

$$\mathcal{E}_{\mu_0 A_0} = |\mu_0|^2 \mathcal{E}_{A_0} = \frac{1}{2} \text{id}, \quad \rho(\mathcal{E}_{\mu_0 A_0}) = \frac{1}{2}.$$

A normal tensor has transfer spectral radius one, proving (20.89) and the failure of the printed inference. □

Theorem 20.16.8.24 (Counterexample to the printed implication (iv) $\Rightarrow$ (v)). Lean ✓ Stmt
There is a one-letter matrix product density operator K whose BNT canonical presentation has one scalar normal representative $A = 1$ and two copies with raw weights

$$\mu_{0,0} = 1, \quad \mu_{0,1} = \frac{1}{2}.$$

The presentation satisfies the global unit-weight normalization, K generates positive semidefinite operators at every positive length, and the physical-trace transfer of A is nonnilpotent. Thus K satisfies the simplicity clause. The singleton BNT family has simultaneous one-letter span, so the presentation is in biCF. Moreover, A itself generates matrix product density operators, the distinct-layer equation in condition (iv) holds, and A has a physical-sector factorization with positive neighboring operator and normalized rank-one trace coefficients. Its two-site blocking also generates matrix product density operators. Nevertheless,

$$\neg \text{RFP}(K^{[2]}).$$

Consequently the literal implication (iv) $\Rightarrow$ (v) printed in [11, Theorem 4.9] is false.

Proof. [✓ Proof](#) The sole local matrix of K is the two-copy block diagonal matrix with eigenvalues 1 and $1/2$. Hence every periodic contraction equals $1 + 2^{-N} > 0$. All assertions about the scalar representative reduce to the one-by-one identity matrix; orthogonality is vacuous because there is only one BNT representative. If $K^{[2]}$ satisfied Definition 20.2.1, trace preservation would make its physical-trace transfer idempotent. Taking traces would then give

$$1 + 2^{-4} = 1 + 2^{-2},$$

a contradiction. □

Remark 20.16.8.25 (Printed simple-MPDO implication chain). For a simple tensor K in block-injective canonical form with a basis of normal tensors and $M = K^{[2]}$, [11, Theorem 4.9] prints the chain (i)⇒(ii), (ii)⇔(iii), (iii)⇒(iv), (iv)⇒(v) between the fixed-point condition, SAL with zero correlation length, the commuting Gibbs form with zero correlation length, the blockwise local structure, and the fixed-point condition for M . Theorem 20.16.8.71 refutes the representativewise assertion used in (ii)⇒(iv). Its ambient tensor has a two-representative BNT presentation in biCF. Independently, Theorem 20.16.8.24 refutes the literal implication (iv)⇒(v). The representative-channel and outer-sector constructions relevant to Proposition C.15 are proved conditionally for supplied absorbed-sector factorizations, but the printed route obtains those factorizations from the false representativewise assertion. More strongly, Theorem 20.16.7.50 gives a trace-norm counterexample to Proposition C.15 itself and shows that the same ambient tensor has a two-site blocking which is not an RFP. These false-source and proof-path dispositions are documented in [22, 122].

Remark 20.16.8.26 (Printed mixed-state separation examples). The separations asserted in [11, Examples 4.10–4.12, lines 897–942] are decided as follows. Example 4.10, read with one left and one right bit flip and $p = 1/4$ (Definition 20.16.8.27), has source zero correlation length and does not saturate the area law (Theorems 20.16.8.28 and 20.16.8.35); no claim is made about the printed universal statement in p . The literal finite periodic family of Example 4.11 fails both saturation and source zero correlation length (Theorems 20.16.8.49 and 20.16.8.43). The literal tensor of Example 4.12 is unnormalized, so its fixed-point and zero-correlation-length assertions fail for that representative (Theorem 20.16.8.60); the halved tensor saturates the area law, has literal zero correlation length, and satisfies the two channel equations without supplying the unit-weight convention (Theorems 20.16.8.61 and 20.16.8.64), and both representatives lie outside the full GSNNCH class (Theorems 20.16.8.59 and 20.16.8.62).

Definition 20.16.8.27 (Corrected tensor for Example 4.10). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Identify the local four-dimensional spin with two qubits and write $U = \sigma_x \otimes \sigma_x$. Starting from one Bell pair on every bond, apply independently at each site the channel

$$\mathcal{E}(X) = \frac{1}{4}UXU^* + \frac{3}{4}X.$$

Equivalently, use the local purification with Kraus amplitudes $\sqrt{3}/2$ for the identity and $1/2$ for the correlated flip. This is distinct from the repeated-left expression printed in the source.

Theorem 20.16.8.28 (Source zero correlation length of the corrected tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The corrected tensor is locally purified and hence positive at every chain length. It has source zero correlation length.

Proof. [✓ Proof](#) Its physical-trace transfer is the nonzero projector

$$P = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Since $P^2 = P$, the tensor has source zero correlation length.

Theorem 20.16.8.29 (One- through four-site spectra of the corrected tensor). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) *The normalized reduced states on one, two, three, and four consecutive sites have characteristic-polynomial root multisets*

$$\begin{aligned} L = 1 : & \quad \{0^{[0]}, (1/4)^{[4]}\}, \\ L = 2 : & \quad \{0^{[8]}, (5/32)^{[4]}, (3/32)^{[4]}\}, \\ L = 3 : & \quad \{0^{[48]}, (7/64)^{[4]}, (3/64)^{[12]}\}, \\ L = 4 : & \quad \{0^{[248]}, (41/128)^{[1]}, (15/128)^{[4]}, (9/128)^{[3]}\}. \end{aligned}$$

These spectra are consequences of the corrected $p = 1/4$ tensor and are not formulas stated in the source.

Proof. ✓ Proof For $1 \leq L \leq 4$, the proof constructs a bond-window weight family w_L and an isometry V_L such that

$$\rho_L = V_L \text{diag}(w_L) V_L^\dagger, \quad V_L^\dagger V_L = I.$$

The diagonal weights give the listed nonzero roots. The zero-root counts from the orthogonal complements are

$$4^1 - 2^2 = 0, \quad 4^2 - 2^3 = 8, \quad 4^3 - 2^4 = 48, \quad 4^4 - 2^4 = 240.$$

At four sites the eight odd-parity bond patterns have zero weight and supply the remaining eight zero roots. $\square$

Definition 20.16.8.30 (Correlated-flip weight families for Example 4.10). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Place one maximally entangled qubit pair on each bond of the four-spin ring and apply, independently at every spin, the corrected correlated flip with flip probability one quarter. A joint flip configuration $s \in \{0, 1\}^4$ has probability

$$P(s) = \prod_{n=1}^4 \left(\frac{1}{4}\right)^{s_n} \left(\frac{3}{4}\right)^{1-s_n},$$

and its bond pattern $t \in \{0, 1\}^4$ is $t_n = s_n \oplus s_{n+1}$, with spin five identified with spin one. Write q for the pushforward of P to bond patterns. The weight family of a window of k consecutive spins, $1 \leq k \leq 3$, assigns to each configuration of its $k - 1$ interior bond labels and its two boundary qubits one quarter of the marginal probability of those interior labels; the four-spin window weight family is q itself.

Theorem 20.16.8.31 (Exact bond-pattern distribution for Example 4.10). The flip-configuration distribution is a probability distribution on $\{0,1\}^4$. The bond-pattern distribution takes the value $41/128$ on the unflipped pattern, $15/128$ on each of the four patterns flipping two cyclically adjacent bonds, $9/128$ on the two patterns flipping two opposite bonds and on the fully flipped pattern, and zero on the eight patterns with an odd number of flipped bonds. Its one-bond marginal is $(5/8, 3/8)$, and its two-adjacent-bond marginal assigns $7/16$ to the doubly unflipped pair and $3/16$ to each of the three remaining pairs. All the window weight families are probability distributions.

Proof. [✓ Proof](#) Each value is a finite sum of products of the factors $1/4$ and $3/4$ over the sixteen flip configurations, evaluated exactly. $\square$

Theorem 20.16.8.32 (Exact window entropies for Example 4.10). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\tilde{S}_k$ denote the entropy $\sum_x h(w(x))$ of the window weight family for k consecutive spins, with $h(x) = -x \log x$ and $\log$ the natural logarithm. Then

$$\begin{aligned}\tilde{S}_1 &= 2 \log 2, \\ \tilde{S}_2 &= 5 \log 2 - \frac{5}{8} \log 5 - \frac{3}{8} \log 3, \\ \tilde{S}_3 &= 6 \log 2 - \frac{7}{16} \log 7 - \frac{9}{16} \log 3, \\ \tilde{S}_4 &= 7 \log 2 - \frac{41}{128} \log 41 - \frac{57}{64} \log 3 - \frac{15}{32} \log 5.\end{aligned}$$

Divided by $\log 2$ these are the printed decimals 2, 2.9544, 3.8802, and 2.7839.

Proof. [✓ Proof](#) The window weight families list the values $(1/4)^{[4]}$; $(5/32)^{[4]}$, $(3/32)^{[4]}$; $(7/64)^{[4]}$, $(3/64)^{[12]}$; and $(41/128)^{[1]}$, $(15/128)^{[4]}$, $(9/128)^{[3]}$. Summing h over each family and expanding the logarithms of the rational values gives the displayed closed forms. $\square$

Theorem 20.16.8.33 (Positivity of the Example 4.10 logarithmic ratio). [Lean](#) [Lean](#) [✓ Stmt](#)
The exact scalar ratio from the four-site calculation satisfies

$$\log \left(\frac{2^{32} 7^7}{3^3 5^{20}} \right) > 0.$$

Proof. [✓ Proof](#) The numerator and denominator are the integers $2^{32} 7^7 = 3537090251849728$ and $3^3 5^{20} = 2574920654296875$, so the ratio is greater than one and its logarithm is strictly positive. $\square$

Theorem 20.16.8.34 (Exact mutual-information defect for Example 4.10). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) For $\tilde{I}_1 = \tilde{S}_1 + \tilde{S}_3 - \tilde{S}_4$ and $\tilde{I}_2 = 2\tilde{S}_2 - \tilde{S}_4$,

$$\tilde{I}_2 - \tilde{I}_1 = \frac{1}{16} \log \left(\frac{2^{32} 7^7}{3^3 5^{20}} \right) > 0.$$

Proof. [✓ Proof](#) Since $\tilde{I}_2 - \tilde{I}_1 = 2\tilde{S}_2 - \tilde{S}_1 - \tilde{S}_3$, substituting the closed forms and collecting logarithms gives the displayed identity; positivity follows from the certified integer comparison. $\square$

Theorem 20.16.8.35 (Entropy obstruction for the corrected Example 4.10 tensor). [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For the corrected tensor at $p = 1/4$ on the four-site ring, let S_L be the entropy of L consecutive spins, with natural logarithms. Then

$$\begin{aligned}S_1 &= 2 \log 2, \\ S_2 &= 5 \log 2 - \frac{5}{8} \log 5 - \frac{3}{8} \log 3, \\ S_3 &= 6 \log 2 - \frac{7}{16} \log 7 - \frac{9}{16} \log 3, \\ S_4 &= 7 \log 2 - \frac{41}{128} \log 41 - \frac{57}{64} \log 3 - \frac{15}{32} \log 5.\end{aligned}$$

Consequently,

$$I_2 - I_1 = \frac{1}{16} \log \left(\frac{2^{32} 7^7}{3^3 5^{20}} \right) > 0.$$

The tensor has source zero correlation length and does not saturate the area law. These identities concern only the corrected fixed value $p = 1/4$.

Proof. [✓ Proof](#) The four root multisets give the displayed block entropies. Substitution into $I_L = S_L + S_{4-L} - S_4$ gives the exact logarithmic ratio, which is positive by Theorem [20.16.8.33](#). If the area law saturated, Lemma [20.8.6](#) would give $I_1 = I_2$, a contradiction. The source zero-correlation-length statement is Theorem [20.16.8.28](#). $\square$

Definition 20.16.8.36 (Effective binary-support model for Example 4.11). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The coefficient matrix extracted from the four diagonal neighboring operators printed in [\[11, Example 4.11, lines 907–924\]](#) is

$$W = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}.$$

In the effective binary encoding, the supported label $k \in \{0, 1\}$ selects the ambient two-qubit basis vector $|k, k\rangle$. Let M be the cyclic-edge tensor for the ket–bra diagonal edge coefficient induced by W . For a positive cycle length N , define

$$c_N(\sigma) = \prod_{n \in \mathbb{Z}/N\mathbb{Z}} W_{\sigma_n, \sigma_{n+1}}, \quad Z_N = \sum_{\sigma} c_N(\sigma).$$

This is a physical-dimension-2 support model. The ambient physical-dimension-4 realization is obtained by the support inclusion defined below; the two physical spaces are not identified by an equivalence.

Theorem 20.16.8.37 (Diagonal formula and normalization of the binary-support model). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every positive N and binary words σ, τ ,

$$\text{mpo}(M, N)_{\sigma, \tau} = \delta_{\sigma, \tau} c_N(\sigma), \quad Z_N = \text{tr}(W^N) = \frac{3^N + 1}{2^N}.$$

These identities first concern the effective binary support model. The ambient definition below transports the operator and trace identities, but no spectrum, entropy, saturation of the area law, or zero correlation length assertion is made here.

Proof. [✓ Proof](#) The closed cyclic-edge identity reduces each matrix entry to a product of local edge coefficients. Ket–bra diagonality makes the product vanish unless $\sigma = \tau$, and in the diagonal case the product is $c_N(\sigma)$. Summing these cycle weights and applying the cyclic trace-power identity gives $Z_N = \text{tr}(W^N)$. Finally, the Hadamard basis diagonalizes W with eigenvalues $3/2$ and $1/2$, so cyclicity of the trace gives $\text{tr}(W^N) = (3/2)^N + (1/2)^N = (3^N + 1)/2^N$. $\square$

Definition 20.16.8.38 (Internal transition count). [Lean](#) [✓ Stmt](#) For a binary word $x = (x_1, \dots, x_L)$, let $t(x)$ be the number of indices $n \in \{1, \dots, L-1\}$ for which $x_n \neq x_{n+1}$.

Theorem 20.16.8.39 (Exact effective-binary finite-ring block probabilities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $1 \leq L \leq N$ and let $x \in \{0, 1\}^L$. The effective-binary block state is diagonal, and its entry at x is

$$\mu_{N,L}(x) = \frac{2^L(3^{N-L+1} + (-1)^{t(x)})}{2^{t(x)+2}(3^N + 1)}.$$

This includes $L = 1$ and $L = N$. It is a project-derived consequence of the finite periodic model, not a formula stated in CPSV16.

Proof. ✓ Proof Substitution in the exact finite-ring block formula gives the two displayed values. Source zero correlation length would identify the normalized two-site marginal obtained from length four with that obtained from length three, contradicting $14/41 \neq 5/14$. Isometric support inclusion preserves these entries, so the same obstruction holds in the ambient realization. At length one, the trace-loop identity and the exact effective and ambient partition functions give

$$\begin{aligned}\mathrm{tr}(\mathrm{mpo}(M, 1)) &= \mathrm{tr}(\mathcal{T}_M^1) = \mathrm{tr}(\mathcal{T}_M) = 2, \\ \mathrm{tr}(\mathrm{mpo}(\widetilde{M}, 1)) &= \mathrm{tr}(\mathcal{T}_{\widetilde{M}}^1) = \mathrm{tr}(\mathcal{T}_{\widetilde{M}}) = 2.\end{aligned}$$

Hence both physical-trace transfers are nonzero. For either $X \in \{M, \widetilde{M}\}$, literal idempotence would give

$$\mathcal{T}_X \neq 0, \quad \mathcal{T}_X^2 = \mathcal{T}_X \implies \mathcal{T}_X^2 = \lambda \mathcal{T}_X \quad \text{with } \lambda = 1 > 0,$$

which the marginal instability $14/41 \neq 5/14$ already refutes. □

Theorem 20.16.8.44 (Exact length-four effective spectra). Lean Lean Lean Lean Lean Lean
Lean Lean Lean ✓ Stmt *On the periodic ring of length four, let $\rho_L := \rho_L^{(4)}(M)$ be the effective-binary reduced state, and let χ_A denote the characteristic polynomial of a matrix A . Then*

$$\mathrm{roots}(\chi_{\rho_1}) = \{(1/2)^{[2]}\}, \tag{20.94}$$

$$\mathrm{roots}(\chi_{\rho_2}) = \{(14/41)^{[2]}, (13/82)^{[2]}\}, \tag{20.95}$$

$$\mathrm{roots}(\chi_{\rho_3}) = \{(10/41)^{[2]}, (4/41)^{[4]}, (5/82)^{[2]}\}, \tag{20.96}$$

$$\mathrm{roots}(\chi_{\rho_4}) = \{(8/41)^{[2]}, (2/41)^{[12]}, (1/82)^{[2]}\}.$$

Here $a^{[m]}$ denotes a root a of multiplicity m . These are the effective nonzero spectra, not the full ambient root multisets.

Proof. ✓ Proof The reduced-state formula is diagonal. For a binary word with t internal transitions, its diagonal probability is

$$p_{L,t} = \frac{2^L(3^{5-L} + (-1)^t)}{2^{t+2} \cdot 82}.$$

For example, at $L = 2$ and $t = 0$,

$$p_{2,0} = \frac{2^2(3^3 + 1)}{2^2 \cdot 82} = \frac{112}{328} = \frac{14}{41}.$$

Enumerating words of lengths one through four gives transition-count multiplicities (2), (2, 2), (2, 4, 2), and (2, 6, 6, 2), respectively. The same substitution for every transition count gives the displayed roots and multiplicities. □

Theorem 20.16.8.45 (Exact length-four ambient spectra). Lean Lean Lean Lean Lean Lean
✓ Stmt *For every $L \leq 4$, let $\rho_L := \rho_L^{(4)}(M)$ be the effective reduced state and $\tilde{\rho}_L := \rho_L^{(4)}(\widetilde{M})$ the ambient reduced state. Let χ_A denote the characteristic polynomial of A . Then*

$$\chi_{\tilde{\rho}_L}(X) = X^{4^L - 2^L} \chi_{\rho_L}(X).$$

For the positive block lengths $L = 1, 2, 3$, and 4, the full ambient root multiset is the effective multiset together with zero of multiplicity 2, 12, 56, and 240, respectively.

Proof. ✓ Proof The reduced ambient state is $\tilde{\rho}_L = V^{\otimes L} \rho_L (V^{\otimes L})^*$. Applying characteristic-polynomial cyclicity to the rectangular factors gives

$$\begin{aligned} \chi_{V^{\otimes L} \rho_L (V^{\otimes L})^*}(X) &= X^{4^L - 2^L} \chi_{\rho_L (V^{\otimes L})^* V^{\otimes L}}(X) \\ &= X^{4^L - 2^L} \chi_{\rho_L}(X), \end{aligned} \quad (20.97)$$

where $(V^{\otimes L})^* V^{\otimes L} = I$. The ambient and effective dimensions are 4^L and 2^L , so their difference gives the stated zero multiplicity. $\square$

Theorem 20.16.8.46 (Exact length-four effective entropies). Lean Lean Lean Lean Lean ✓ Stmt *The effective binary tensor generates positive semidefinite periodic operators. On the ring of length four, write $S_L = S(\rho_L)$. With $h(x) = -x \log x$, where $\log$ is the natural logarithm,*

$$\begin{aligned} S_1 &= \log 2, \\ S_2 &= 2h(14/41) + 2h(13/82), \\ S_3 &= 2h(10/41) + 4h(4/41) + 2h(5/82), \\ S_4 &= 2h(8/41) + 12h(2/41) + 2h(1/82). \end{aligned}$$

Proof. ✓ Proof Every positive-length periodic operator is diagonal, and each diagonal entry is a product of entries of $W = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$, hence is non-negative. For each L , summing $h(\lambda)$ over the roots and their multiplicities in Theorem 20.16.8.44 gives the displayed entropy. $\square$

Theorem 20.16.8.47 (Positivity of the Example 4.11 logarithmic ratio). Lean ✓ Stmt *The exact scalar ratio from the four-site calculation satisfies*

$$\log \left(\frac{41^{82} 5^{50}}{2^{48} 13^{52} 7^{112}} \right) > 0.$$

Proof. ✓ Proof The numerator is strictly larger than the denominator, as follows by an exact integer calculation, so the ratio is greater than one and its logarithm is strictly positive. $\square$

Theorem 20.16.8.48 (Exact length-four mutual-information obstruction). Lean Lean Lean ✓ Stmt *For $I_L = S_L + S_{4-L} - S_4$,*

$$I_2 - I_1 = \frac{1}{82} \log \left(\frac{41^{82} 5^{50}}{2^{48} 13^{52} 7^{112}} \right) > 0.$$

Consequently, the effective binary tensor does not satisfy saturation of the area law.

Proof. ✓ Proof Since $I_2 - I_1 = 2S_2 - S_1 - S_3$, expanding $h(x) = -x \log x$ and using the logarithm rules gives the displayed identity. The certified comparison $41^{82} 5^{50} > 2^{48} 13^{52} 7^{112}$ makes its logarithm strictly positive. Saturation would instead require $I_1 = I_2$ on the four-site ring. $\square$

Theorem 20.16.8.49 (Ambient length-four saturation obstruction). Lean ✓ Stmt *The isometric inclusion of the effective binary tensor into the two-qubit one-site space does not satisfy saturation of the area law.*

Proof. ✓ Proof If the ambient tensor satisfied saturation, descent through the physical isometry would imply saturation for the effective binary tensor, contrary to the preceding theorem. $\square$

Definition 20.16.8.50 (Literal tensor in Example 4.12). [Lean](#) [Lean](#) [✓ Stmt](#) Let M be the tensor with $d = D = 2$ whose only nonzero components are

$$1 = M_{00}^{00} = M_{00}^{11} = M_{11}^{00} = -M_{11}^{11},$$

and let $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$.

Definition 20.16.8.51 (One-site Pauli sign). [Lean](#) [✓ Stmt](#) For a physical letter $a \in \{0, 1\}$, define its Pauli sign by

$$z_a = (\sigma_z)_{aa}, \quad z_0 = 1, \quad z_1 = -1.$$

Definition 20.16.8.52 (Configuration sign). [Lean](#) [✓ Stmt](#) For a binary configuration $\sigma = (\sigma_0, \dots, \sigma_{N-1})$, define

$$z(\sigma) = \prod_{k=0}^{N-1} z_{\sigma_k}.$$

This is the diagonal entry of $\sigma_z^{\otimes N}$ at σ .

Theorem 20.16.8.53 (Exact positive-length operator and positivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $N > 0$,

$$\rho^{(N)}(M) = I^{\otimes N} + \sigma_z^{\otimes N}, \quad \text{tr } \rho^{(N)}(M) = 2^N,$$

and $\rho^{(N)}(M)$ is positive semidefinite.

Proof. [✓ Proof](#) Both tensor powers are diagonal. On a binary configuration their sum has eigenvalue $1 + \prod_k z_k$, where each z_k is 1 or -1 ; hence every eigenvalue is either 0 or 2. Summing the signed term over all binary configurations gives zero when $N > 0$, so the trace is 2^N . $\square$

Theorem 20.16.8.54 (One-site loss and saturation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Tracing one site removes the $\sigma_z^{\otimes N}$ term. More generally, every nonempty proper normalized L -site reduced state is $2^{-L} I^{\otimes L}$. Therefore its entropy is $L \log 2$, the mutual information is independent of L , and M satisfies saturation of the area law.

Proof. [✓ Proof](#) The identity term contributes the dimension of the traced complement, while the signed term contains the vanishing one-site sum $1 + (-1)$. The entropy of the resulting maximally mixed state is $L \log 2$; the two block terms in the mutual information then add to $N \log 2$ independently of the cut. $\square$

Definition 20.16.8.55 (Four-site tripartition in Example 4.12). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For the state in [11, Example 4.12, lines 932–938], at $N = 4$ regard the sites as $A = \{0\}$, $B = \{1, 3\}$, and $C = \{2\}$. Encode the two binary labels (b_1, b_3) as one four-dimensional label b , and write $z_a \in \{1, -1\}$ for the eigenvalue of σ_z at a . In the basis $(a, b, c) \leftrightarrow (a, b_1, c, b_3)$, let ρ_{ABC} be the normalized four-site state and set

$$w(a, b, c) = \frac{1 + z_a z_{b_1} z_c z_{b_3}}{16}.$$

Theorem 20.16.8.56 (Four-site failure of strong-subadditivity equality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The normalized four-site state is diagonal with probability

1/8 on the eight even-parity configurations and zero on the eight odd-parity configurations. Its relevant marginals are

$$\rho_{AB} = \frac{I_8}{8}, \quad \rho_{BC} = \frac{I_8}{8}, \quad \rho_B = \frac{I_4}{4}.$$

Consequently,

$$S(ABC) = 3 \log 2, \quad S(B) = 2 \log 2, \quad S(AB) = S(BC) = 3 \log 2,$$

so $S(ABC) + S(B) = 5 \log 2 < 6 \log 2 = S(AB) + S(BC)$. Thus equality in strong subadditivity fails for this tripartition.

Proof. [✓ Proof](#) The diagonal coefficient is $(1 + z_a z_{b_1} z_c z_{b_3})/16$. Summing over either a or c cancels the signed term, giving $I_8/8$; summing over both gives $I_4/4$. There are eight nonzero coefficients, all equal to $1/8$, so the full entropy is $8[-(1/8) \log(1/8)] = 3 \log 2$. The three maximally mixed marginals have the displayed entropies, and $\log 2 > 0$ gives the strict inequality. $\square$

Theorem 20.16.8.57 (Positive commuting overlapping products are quantum Markov). [Lean](#)

[✓ Stmt](#) Let ρ_{ABC} be a trace-one operator, and let X_{AB} and Y_{BC} be positive semidefinite operators. Suppose that their natural lifts to ABC commute and that, after the natural reassociation of the three tensor factors,

$$\rho_{ABC} = (X_{AB} \otimes I_C)(I_A \otimes Y_{BC}).$$

Then ρ_{ABC} admits a quantum Markov decomposition on B .

Proof. [✓ Proof](#)

The spatial decomposition [4, Lemma 2.1] gives a unitary decomposition $B = \bigoplus_q B_q^L \otimes B_q^R$ in which the two overlapping actions have the forms $R_q \otimes I_{B_q^R}$ and $I_{B_q^L} \otimes S_q$. Compression of the original positive operators shows that every R_q and S_q is positive semidefinite. Normalize the two factors in each summand and take $p_q = \text{tr}(R_q) \text{tr}(S_q)$. The factorization and the trace-one hypothesis imply $\sum_q p_q = 1$; zero factors give zero-weight summands. Thus the transformed state is the direct sum $\bigoplus_q p_q \widehat{R}_q \otimes \widehat{S}_q$, which is the quantum Markov form of [171, Theorem 6]. $\square$

Theorem 20.16.8.58 (Four-cycle GSNNCH states are quantum Markov). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let ρ be a positive semidefinite trace-one operator on four d -dimensional sites, and suppose that ρ has the sector decomposition of Definition 4.8. Regroup the sites as $A = \{0\}$, $B = \{1, 3\}$, and $C = \{2\}$. Then the resulting tripartite state admits a quantum Markov decomposition on B . Consequently,

$$S(ABC) + S(B) = S(AB) + S(BC).$$

Proof. [✓ Proof](#) Write $B_j^{(x)}$ for the four translated bonds in sector x , n_x for its multiplicity, and $c > 0$ for the common normalization. Set

$$L = c \sum_x n_x B_3^{(x)} B_0^{(x)}, \quad R = \sum_x B_1^{(x)} B_2^{(x)}.$$

Orthogonality of the one-site sector projections makes each cross-sector term in LR vanish. Within one sector the translated bonds commute, so $(B_3^{(x)} B_0^{(x)})(B_1^{(x)} B_2^{(x)})$ is the sector product in Definition 4.8. Hence $LR = \rho$. Both L and R are positive semidefinite; since $LR = \rho$ is Hermitian, also $RL = LR$.

In the displayed tripartition, $L = X_{AB} \otimes I_C$ and $R = I_A \otimes Y_{BC}$ for positive semidefinite X_{AB} and Y_{BC} . The preceding theorem therefore gives a quantum Markov decomposition of ρ on B . Its direct-sum product form gives the asserted entropy equality. $\square$

Theorem 20.16.8.62 (The normalized representative of Example 4.12 is not GSNNCH). [Lean](#)

[Lean](#) [✓ Stmt](#) *The condition of Definition 4.8 constrains only the normalized density-operator family, and that family is unchanged by any nonzero complex rescaling of the tensor. The condition is therefore invariant under nonzero rescaling. In particular the normalized representative $\widehat{M} = \frac{1}{2}M$ of Example 4.12 is not a Gibbs state of a nearest-neighbor commuting Hamiltonian.*

Proof. [✓ Proof](#) For nonzero c and every chain length, the normalized density operator of cM equals that of M , so every clause of Definition 4.8 holds for cM exactly when it holds for M . Taking $c = \frac{1}{2}$ transports the obstruction proved for the printed tensor to $\widehat{M}$. $\square$

Lemma 20.16.8.63 (Rescaling invariance of the area-law quantities). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) *Let M be an MPO tensor, let c be a nonzero complex number, and let $L \leq N$ be natural numbers. Then*

$$\rho_L^{(N)}(cM) = \rho_L^{(N)}(M).$$

If $\rho^{(N)}(cM)$ and $\rho^{(N)}(M)$ are positive semidefinite, then

$$S_L^{(N)}(cM) = S_L^{(N)}(M), \quad I_L(cM) = I_L(M).$$

For every real $r > 0$, the tensor rM saturates the area law if and only if M does.

Proof. [✓ Proof](#) By Theorem 20.12.12, the normalized operator $\sigma^{(N)}(cM)$ equals $\sigma^{(N)}(M)$, so their partial traces agree; the block entropies and mutual informations are functions of these reduced states. For $r > 0$, Theorem 20.12.17 shows that rM generates positive semidefinite operators exactly when M does, and Theorem 20.12.11 gives $\rho^{(N)}(rM) = r^N \rho^{(N)}(M)$, so the traces of the two operators vanish together. The mutual-information clause of Definition 20.8.4 transfers by the displayed identity. $\square$

Theorem 20.16.8.64 (Four properties of the normalized representative in Example 4.12).

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The normalized representative $\widehat{M} = \frac{1}{2}M$ of Example 4.12 saturates the area law, its physical-trace transfer is idempotent, it is not a Gibbs state of a nearest-neighbor commuting Hamiltonian, and it satisfies the two channel equations of Definition 4.1.*

This is a scope-restricted form of [11, Example 4.12]: the tensor is the halved representative rather than the printed one, and the fixed-point assertion is read as the two channel equations, without the global unit-weight canonical convention of the source [21, 18].

Proof. [✓ Proof](#) Lemma 20.16.8.63 with $r = \frac{1}{2}$ transfers saturation from the printed tensor (Theorem 20.16.8.54) to $\widehat{M}$. By Theorem 20.6.13, the channel equations of Theorem 20.16.8.61 imply that the physical-trace transfer is idempotent. The commuting-form obstruction is Theorem 20.16.8.62. $\square$

Theorem 20.16.8.65 (Bond-one physical-sector boundary). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) *Let K be an MPO tensor of virtual bond dimension one. If K satisfies saturation of the area law and literal zero correlation length, then there is a physical-sector factorization of K whose neighboring operator is positive semidefinite and whose trace has the normalized rank-one form*

$$\text{tr}(\eta_{k,h}) = a_k b_h.$$

The factors are normalized by

$$\sum_k a_k b_k = 1.$$

Proof. ✓ Proof Use one physical sector, with the full physical space as its left factor and a one-dimensional right factor. Its neighboring operator is the one-site periodic operator, so positivity follows from the MPDO condition included in SAL. The physical-trace transfer is a nonzero scalar. Literal idempotence therefore makes it equal to one, and the neighboring trace is normalized by $a_0 = b_0 = 1$. $\square$

Theorem 20.16.8.66 (Counterexample to the low-level absorbed-sector implication). Lean
Lean Lean Lean ✓ Stmt *There are tensors K, A of physical dimension four and virtual dimension two, and a scalar μ with $0 < |\mu| < 1$, such that $K = \mu A$, the tensor A is normal, and K is injective, saturates the area law, and obeys*

$$\mathcal{T}_K^2 = \mathcal{T}_K.$$

Nevertheless, no physical-sector factorization of K has positive neighboring operators whose traces are of the form $a_k b_h$ with $\sum_k a_k b_k = 1$. Hence these local analytic hypotheses do not by themselves imply the asserted factorization. This does not refute the all-sector assertion in [11, Theorem 4.9 and Appendix C.2, lines 1647–1782], whose hypotheses include an ambient simple tensor in biCF and its canonical reconstruction.

Proof. ✓ Proof Take

$$L = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & -7 & 4 \end{pmatrix}, \quad R = \begin{pmatrix} 1/4 & -3/100 \\ 1/4 & -1/100 \\ 1/4 & 1/100 \\ 1/4 & 3/100 \end{pmatrix}, \quad (20.98)$$

and let the four diagonal physical slices be the outer products of the corresponding columns of L and rows of R . Then $LR = \text{diag}(1, 0)$, whereas RL has rank two. The scalar slice and two simple-spectrum slices force every summand in any physical-sector factorization to have one-dimensional left and right factors. The resulting neighboring trace matrix has a nonzero two-by-two minor, so it cannot equal $(a_k b_h)_{k,h}$. The four slices span the full two-by-two matrix algebra, and their displayed scalar neighboring operators are positive. For the positive test matrix $H = \text{diag}(20, 1)$, direct calculation gives

$$H - \mathcal{E}_K^*(H) = \begin{pmatrix} 85/8 & -9/40 \\ -9/40 & 4697/5000 \end{pmatrix} > 0.$$

Pair this positive-definite gap with a positive-definite Perron eigenmatrix ρ of $\mathcal{E}_K$. The trace-adjoint identity gives

$$0 < \text{tr}((H - \mathcal{E}_K^*(H))\rho) = (1 - r) \text{tr}(H\rho).$$

Since $\text{tr}(H\rho) > 0$, the Perron value satisfies $r < 1$. Perron normalization and a canonical gauge therefore express the tensor as $K = \mu A$ with A normal and $0 < |\mu| = \sqrt{r} < 1$. $\square$

Theorem 20.16.8.67 (Prepared BNT decomposition for the rank-one neighboring-trace obstruction). Lean Lean ✓ Stmt *There are a scalar μ with $0 < |\mu| < 1$ and two normal left-canonical blocks B_{obs} and B_{term} , of bond dimensions two and one respectively, such that the embedded obstruction tensor is gauge equivalent to μB_{obs} . The two-block decomposition with one copy of each block and coefficients $(\mu, 1)$ is in normalized BNT canonical form. Its one-site physical supports P_{obs} and P_{term} are disjoint and resolve the ambient physical space:*

$$\begin{aligned} P_{\text{obs}} P_{\text{term}} &= P_{\text{term}} P_{\text{obs}} = 0, \\ P_{\text{obs}} + P_{\text{term}} &= 1. \end{aligned}$$

This is a project-derived preparation of the counterexample, not a construction asserted in [11]. It supplies the two normal blocks used in the ambient construction below.

Proof. [✓ Proof](#) Embed the four-letter obstruction tensor into four coordinates of a five-dimensional physical space and place a bond-one product block on the remaining coordinate. The coordinate projections are complementary. The terminal block has coefficient one, while the first block retains the coefficient μ . Their unequal bond dimensions separate the two BNT representatives. $\square$

Lemma 20.16.8.68 (The ambient neighboring-trace obstruction is simple). [Lean](#) [✓ Stmt](#) Let $\mu \neq 0$ and let B be a five-letter tensor such that the embedded obstruction is virtually gauge equivalent to μB . If the associated obstruction–terminal sector decomposition is in normalized BNT canonical form, then its ambient weighted direct sum is simple in the sense of Definition 4.7.

Proof. [✓ Proof](#) The displayed block-diagonal gauge and the nonnilpotent physical-trace transfers of the two representatives give a normalized simple canonical form. Theorem 20.12.22 then gives simplicity in the sense of Definition 4.7. $\square$

Theorem 20.16.8.69 (Ambient neighboring-trace counterexample properties). [Lean](#) [✓ Stmt](#) There is a five-letter MPO tensor K of virtual bond dimension three with a two-representative horizontal canonical presentation. The representatives A_0, A_1 are normal and left-canonical, have bond dimensions two and one, and each occurs once. The bond-two representative is one-site injective. Their coefficients are μ and 1, where $0 < |\mu| < 1$. In particular, every coefficient has modulus at most one and the second coefficient has modulus one. The embedded four-letter obstruction is virtually gauge equivalent to μA_0 .

The representatives have simultaneous one-letter span

$$\text{span}\{(A_0^i, A_1^i) : 0 \leq i < 25\} = M_2(\mathbb{C}) \times \mathbb{C}.$$

Thus the presentation is in biCF. The tensor K is the exact weighted direct sum $\mu A_0 \oplus A_1$ with the identity global virtual gauge. It has normalized simple canonical form and is simple in the sense of Definition 4.7. It generates positive semidefinite periodic operators with strictly positive trace at every positive length, saturates the area law, and has literal zero correlation length,

$$\mathcal{T}_K^2 = \mathcal{T}_K.$$

The obstruction and terminal sectors have one-site support projections P_0, P_1 satisfying

$$\begin{aligned} P_0 P_1 &= P_1 P_0 = 0, \\ P_0 + P_1 &= 1. \end{aligned}$$

Proof. [✓ Proof](#) Use the prepared obstruction and terminal blocks. The obstruction slices span $M_2(\mathbb{C})$ and vanish on the terminal coordinate, whereas the terminal block equals one on that coordinate. Hence the pairs of one-site slices span $M_2(\mathbb{C}) \times \mathbb{C}$ in one letter. Both representatives are normal, and the coefficients give the stated global normalization.

At every positive length, the ambient periodic operator is the sum of the two positive sector operators. Their complementary supports give the local orthogonal-sum entropy identity, so saturation of the area law passes from the sectors to K . The physical-trace transfer is block diagonal. Its two diagonal blocks are idempotent, hence so is the ambient transfer. The exact weighted direct sum and nonnilpotency of both representatives give the asserted normalized simple canonical presentation. Lemma 20.16.8.68 gives simplicity in the sense of Definition 4.7. SAL makes the physical-trace transfer nonzero; together with literal idempotence this gives source zero correlation length, hence every nonempty periodic trace is nonzero. The MPDO positivity condition then upgrades each such trace to strict positivity. $\square$

Theorem 20.16.8.70 (Universal obstruction for the absorbed representative). [Lean](#) [✓ Stmt](#) Let $\mu \neq 0$ and let B be a five-letter tensor such that μB is virtually gauge equivalent to the embedded four-letter obstruction. Then the absorbed tensor μB admits no physical-sector factorization with positive neighboring operators $\eta_{k,h}$ and real families $(a_k)_k, (b_h)_h$ satisfying

$$\begin{aligned} \text{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_k a_k b_k &= 1. \end{aligned}$$

The negation is universal over all physical-sector factorizations of μB .

Proof. [✓ Proof](#) Suppose such a factorization exists. Virtual gauge invariance gives the same neighboring-trace factorization for the embedded four-letter obstruction. The embedding is an isometry, and the original obstruction is injective with full one-site physical support. Isometric physical descent therefore gives a normalized rank-one neighboring-trace factorization of the original four-letter tensor. This contradicts Theorem 20.16.8.66. $\square$

Theorem 20.16.8.71 (The printed absorbed-representative structure assertion is false). [Lean](#) [✓ Stmt](#) There is a five-letter MPO tensor K of virtual bond dimension three with two normal left-canonical representatives A_0, A_1 whose one-site matrices simultaneously span $M_2(\mathbb{C}) \times \mathbb{C}$. The first representative is one-site injective, and the embedded four-letter obstruction is virtually gauge equivalent to μA_0 . Each representative occurs once, with coefficients μ and 1, where $0 < |\mu| < 1$; hence every coefficient has modulus at most one and the second has modulus one. The tensor has normalized simple canonical form, is simple in the sense of Definition 4.7, and is in biCF. It generates positive semidefinite periodic operators with strictly positive trace at every positive length, satisfies saturation of the area law, and obeys

$$\mathcal{T}_K^2 = \mathcal{T}_K.$$

Its two one-site sector supports are complementary. Nevertheless, the first absorbed representative $\mathcal{K}_0 = \mu A_0$ has no physical-sector factorization with positive neighboring operators and real families $(a_k)_k, (b_h)_h$ satisfying

$$\begin{aligned} \text{tr}(\eta_{k,h}) &= a_k b_h, \\ \sum_k a_k b_k &= 1. \end{aligned}$$

Hence Proposition C.12 and the representativewise assertion used in implication (ii) $\Rightarrow$ (iv) of [11, Theorem 4.9 and Appendix C.2, lines 1628–1782] are false under the stated hypotheses. This false-source result is documented in [22].

Proof. [✓ Proof](#) Choose the tensor and its two representatives from Theorem 20.16.8.69. The first representative satisfies the gauge relation required by Theorem 20.16.8.70, which excludes every normalized rank-one neighboring-trace factorization of its absorbed tensor. All remaining hypotheses are the properties listed in the ambient theorem. $\square$

20.17 Single-bond commuting form from the local structure

Definition 20.17.1 (Translation-invariant two-site bond). [Lean](#) [✓ Stmt](#) A positive semidefinite two-site bond $B \geq 0$ whose translated copies commute pairwise on the periodic N -site chain at every chain length $N \geq 2$: $[B_{i,i+1}, B_{j,j+1}] = 0$.

Proof. [✓ Proof](#) The same unitary block decomposition at both ends of B gives identity action on $H_{q,l}$ through one block identity and on $H_{h,r}$ through the other. For fixed basis indices $\ell_q^0 \in H_{q,l}$ and $r_h^0 \in H_{h,r}$, let $B' = (U^* \otimes U^*)B(U \otimes U)$. The neighboring operator has entries

$$(\eta_{q,h})_{(r,\ell),(r',\ell')} = B'_{(q,r,\ell_q^0,h,r_h^0,\ell),(q,r',\ell_q^0,h,r_h^0,\ell')}.$$

Thus $\eta_{q,h}$ is a principal compression of the positive operator B' , and is therefore positive semidefinite. $\square$

Theorem 20.17.9 (Cyclic product of a locally decomposed bond). [Lean](#) [✓ Stmt](#) Suppose that, in one fixed coordinate decomposition $H \cong \bigoplus_q H_{q,r} \otimes H_{q,l}$, after regrouping the two-site coordinates according to this decomposition, a two-site operator has the form

$$B = \bigoplus_{q,h} \mathbb{1}_{H_{q,l}} \otimes \eta_{q,h} \otimes \mathbb{1}_{H_{h,r}}.$$

For every $N \geq 2$, regroup the cyclic chain as

$$H^{\otimes N} \cong \bigoplus_{k: \mathbb{Z}/N\mathbb{Z} \rightarrow \{0, \dots, K-1\}} \bigotimes_{n \in \mathbb{Z}/N\mathbb{Z}} (H_{k_n, r} \otimes H_{k_{n+1}, l}).$$

Then, in these coordinates and with $k_N = k_0$,

$$\prod_{n=0}^{N-1} B_{n,n+1} = \bigoplus_k \bigotimes_{n=0}^{N-1} \eta_{k_n, k_{n+1}}.$$

At $N = 2$, the two translated windows have the opposite orders $(0, 1)$ and $(1, 0)$ and give the two oriented factors η_{k_0, k_1} and η_{k_1, k_0} .

Proof. [✓ Proof](#) In a fixed sector configuration, the cyclic regrouping sends the factors (r_n, l_n) to the edge factors (r_n, l_{n+1}) . The translate beginning at n acts by $\eta_{k_n, k_{n+1}}$ on the n th edge factor and by the identity on every other edge factor. Different sector configurations do not mix. Multiplying the translated operators therefore gives the stated direct sum of tensor products. $\square$

Theorem 20.17.10 (Scaled finite-chain cyclic neighboring-operator form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\sigma^{(N)}(\mathcal{K})$ be realized, for every $N \geq 2$, by translates of one positive commuting bond. There are one unitary U , positive sector dimensions, and positive semidefinite operators $\eta_{q,h}$, all independent of N , such that for every $N \geq 2$ there is a constant $c_N > 0$ satisfying, with $k_N = k_0$,

$$E_N^*(U^*)^{\otimes N} \sigma^{(N)}(\mathcal{K}) U^{\otimes N} E_N = c_N \bigoplus_{k_0, \dots, k_{N-1}} \bigotimes_{n=0}^{N-1} \eta_{k_n, k_{n+1}}.$$

Here E_N is the cyclic regrouping from the edge factors $H_{k_n, r} \otimes H_{k_{n+1}, l}$ to the site factors $H_{k_n, r} \otimes H_{k_n, l}$. The factor c_N is retained from the proportionality relation in [11, Appendix C.2, lines 1571–1576], whereas the subsequent passage at [11, Appendix C.2, lines 1603–1605] suppresses this factor when invoking equation sigmaNK2.

Proof. [✓ Proof](#) Choose the unitary and the neighboring operators from the decomposition of the fixed bond. Conjugation by $(U^*)^{\otimes N}$ carries the product of its translates to the product of the conjugated two-site bonds. The cyclic regrouping theorem changes this product into the displayed direct sum. Applying both transformations to $\sigma^{(N)}(\mathcal{K}) = c_N \prod_n B_{n,n+1}$ preserves the same positive scalar c_N . $\square$

Remark 20.17.11 (An arbitrary proportional presentation need not have unit scalar). The proportional commuting-bond form alone does not permit one common local normalization. Indeed, take physical dimension $d = 1$ and bond dimension $D = 2$, with sole matrix-product letter $M^{00} = I_2$, and take the two-site bond to be the scalar $B = 1$. Then

$$\sigma^{(N)} = \text{tr}(I_2^N) = 2 = c_N \prod_{n=0}^{N-1} B_{n,n+1}, \quad c_N = 2.$$

Any positive neighboring decomposition of the one-dimensional physical space has one sector and one scalar $a \geq 0$. A coefficient-free identity at lengths two and four would give $a^2 = 2$ and $a^4 = 2$, a contradiction. This example is not a normal injective block. More generally, an arbitrary proportional commuting-bond presentation need not be made coefficient-free by a single local rescaling. In the SAL-to-commuting-form direction, the proof chooses the neighboring operators from the source tensor and obtains scalar one directly, yielding the source-selected normal representative in Corollary 20.16.3.62.

Definition 20.17.12 (Cyclic edge-weight tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A scalar weight $w(i, j; i', j')$ on two consecutive ket–bra pairs determines a translation-invariant matrix-product tensor with virtual dimension d^2 . Its incoming virtual pair is constrained to equal the current physical ket–bra pair, while its outgoing pair records the following physical pair.

Theorem 20.17.13 (Closed cyclic edge-weight identity). [Lean](#) [✓ Stmt](#) For every positive length N , and with $\sigma_N = \sigma_0$ and $\tau_N = \tau_0$, the closed matrix-product operator of the cyclic edge-weight tensor is

$$\rho_{\sigma, \tau}^{(N)} = \prod_{n=0}^{N-1} w(\sigma_n, \tau_n; \sigma_{n+1}, \tau_{n+1}).$$

In the virtual trace there is precisely one potentially nonzero cyclic virtual configuration.

Definition 20.17.14 (Fixed tensor for a bond-product family). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A fixed tensor for a bond-product family consists of a positive virtual dimension and one matrix-product tensor C , independent of N , such that, for every $N \geq 2$,

$$\rho^{(N)}(C) = \prod_{n=0}^{N-1} B_{n,n+1}.$$

If $d > 0$, a cyclic scalar edge-weight expression for the entries of the product gives such a tensor with positive virtual dimension d^2 .

Theorem 20.17.15 (Fixed tensor representation of a commuting bond). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $B \geq 0$ be a two-site bond whose periodic translates commute. There is a positive integer R and a matrix-product tensor C of virtual dimension R , both independent of the chain length, such that for every $N \geq 2$,

$$\rho^{(N)}(C) = \prod_{n=0}^{N-1} B_{n,n+1}.$$

The construction gives a positive R independent of N . No normality, zero-correlation-length, or scalar normalization hypothesis is required.

Proof. ✓ Proof Choose Beigi's chain-independent one-site unitary and the positive neighboring operators $\eta_{q,h}$. For a common matrix-unit index set $\mathcal{A}$, write

$$\eta_{q,h}((r,l), (r',l')) = \sum_{a \in \mathcal{A}} R_a(q)_{r,r'} L_a(h)_{l,l'}.$$

Thus the contraction of the right factor at one site with the left factor at the next satisfies $\sum_{a \in \mathcal{A}} R_a(q) \otimes L_a(h) = \eta_{q,h}$. If V is the resulting unitary change to the two-site sector coordinates, this contraction gives $\tilde{B} = V B V^\dagger$ and $B_{\text{phys}} = V^\dagger \tilde{B} V = B$. For a nonzero physical space, $\mathcal{A}$ is nonempty and gives a positive virtual dimension. For the zero-dimensional physical space, take one virtual state and no physical sectors. The cyclic contraction formula then gives the product of the translates of B at every length $N \geq 2$. $\square$

Theorem 20.17.16 (Positive physical sectors of the selected fixed tensor). Lean Lean Lean Lean Lean Lean ✓ Stmt Let C be the selected fixed tensor of Theorem 20.17.15. There are physical sectors $\mathbb{C}^d \cong \bigoplus_{q=0}^{K-1} H_q^L \otimes H_q^R$, a unitary U , and matrices $L_\beta(q)$ and $R_\alpha(q)$ such that, writing $C_{\beta,\alpha}$ for the physical matrix at fixed virtual indices,

$$U C_{\beta,\alpha} U^\dagger = \bigoplus_{q=0}^{K-1} L_\beta(q) \otimes R_\alpha(q).$$

The neighboring operators $\eta_{q,h} = \sum_a R_a(q) \otimes L_a(h)$ are positive semidefinite. If U_2 denotes the induced two-site change of coordinates, then

$$B = U_2^\dagger \left[\bigoplus_{q,h} \mathbb{1}_{H_q^L \otimes H_h^R} \otimes \eta_{q,h} \right] U_2.$$

Thus the physical-sector factorization belongs to the same tensor C whose closed operators are the fixed products of B .

Proof. ✓ Proof Take the physical-sector factorization retained with C . Its defining one-site identity is

$$U C_{\beta,\alpha} U^\dagger = \bigoplus_q L_\beta(q) \otimes R_\alpha(q).$$

Contracting the adjacent right and left factors gives $\eta_{q,h} = \sum_a R_a(q) \otimes L_a(h) \succeq 0$, where positivity is the retained neighboring-operator property. Finally, the retained bond identity is

$$B = U_2^\dagger \left[\bigoplus_{q,h} \mathbb{1}_{H_q^L \otimes H_h^R} \otimes \eta_{q,h} \right] U_2.$$

$\square$

Remark 20.17.17 (The scalar edge formula depends on the one-site coordinates). A cyclic scalar edge formula need not hold in the original physical basis. For $d = 2$, let X be the Pauli flip and set $B = \mathbb{1} + X \otimes X$. This bond is positive semidefinite and its periodic translates commute. For every $N \geq 2$,

$$\langle 0^N | \left(\prod_n B_{n,n+1} \right) | 0^N \rangle = 2.$$

At $N = 2$ the periodic product is $B^2 = 2B$; for $N \geq 3$, the empty edge set and the full cycle are the two terms with zero boundary. An original-coordinate scalar edge weight would make this entry equal to a^N . The cases $N = 2$ and $N = 3$ would give $a^2 = 2$ and $a^3 = 2$, a contradiction. Equation *sigmaNK2* in [11, Appendix C.2, lines 1581–1589] is stated only after the one-site unitary coordinate change, as used in the theorem.

Theorem 20.17.18 (Eventual proportionality from a fixed product tensor). [Lean](#) [✓ Stmt](#) *Suppose one fixed tensor C generates the products of the bond carried by an eta-local structure. Then the doubled-index tensors of M and C are eventually nonzero proportional. The proof uses only lengths $N \geq 2$ and therefore assumes no relation at length one.*

Proof. [✓ Proof](#) At every $N \geq 2$, substitute the exact tensor representation into the positive realization $\rho^{(N)}(M) = c_N \prod_n B_{n,n+1}$, where $c_N > 0$, and read ket–bra pairs as doubled physical indices. $\square$

Theorem 20.17.19 (A normal rescaling gives a geometric phase). [Lean](#) [✓ Stmt](#) *Let A be the normal doubled-index source tensor of positive bond dimension D , and let C be a fixed tensor for its commuting-bond product. Suppose that sC is normal for some $s > 0$. Then there is a complex number ζ , with $|\zeta| = 1$, such that, for every $N \geq 2$,*

$$\rho^{(N)}(A) = (\zeta s)^N \prod_{n=0}^{N-1} B_{n,n+1}.$$

Proof. [✓ Proof](#) Scaling C by s gives $V^{(N)}(sC)_\sigma = s^N V^{(N)}(C)_\sigma$. Thus, for all sufficiently large N , the proportionality $V^{(N)}(A)_\sigma = c_N V^{(N)}(C)_\sigma$ becomes

$$V^{(N)}(A)_\sigma = \frac{c_N}{s^N} V^{(N)}(sC)_\sigma, \quad \frac{c_N}{s^N} \neq 0.$$

The normal-tensor overlap dichotomy gives one $\zeta \in \mathbb{C}$, with $|\zeta| = 1$, such that, at every positive length,

$$V^{(N)}(A)_\sigma = \zeta^N V^{(N)}(sC)_\sigma = (\zeta s)^N V^{(N)}(C)_\sigma.$$

$\square$

Theorem 20.17.20 (A normal rescaling forces the positive geometric law). [Lean](#) [✓ Stmt](#) *Let A be the normal doubled-index source tensor of positive bond dimension D , let C be a fixed tensor for its commuting-bond product, and suppose that sC is normal for some $s > 0$. Then the unit phase is trivial. For every $N \geq 2$,*

$$\rho^{(N)}(A) = s^N \prod_{n=0}^{N-1} B_{n,n+1}.$$

Thus a positive normal rescaling of the fixed product tensor already supplies the positive geometric realization law.

Proof. [✓ Proof](#) Normality gives $\lim_{N \rightarrow \infty} \langle V^{(N)}(A) | V^{(N)}(A) \rangle = 1$. Hence at two sufficiently large consecutive lengths, say N and $N + 1$, both the source vector and the fixed bond product are nonzero. Positivity of the realization gives positive numbers c_N, c_{N+1} satisfying $(\zeta s)^N = c_N$ and $(\zeta s)^{N+1} = c_{N+1}$. Consequently $\zeta s = c_{N+1}/c_N > 0$. Since $|\zeta| = 1$ and $s > 0$, one has $\zeta = 1$. $\square$

Theorem 20.17.21 (Finite-chain normalization under nonzero proportionality). [Lean](#) [✓ Stmt](#)
Let M and C be MPO tensors, possibly of different bond dimensions, and fix a chain length N .
Write

$$\hat{\rho}^{(N)}(M) = \text{tr}[\rho^{(N)}(M)]^{-1} \rho^{(N)}(M),$$

with the inverse of zero understood to be zero. Suppose that, for some $c \in \mathbb{C} \setminus \{0\}$, the doubled-index matrix-product vectors satisfy $|V^{(N)}(M)\rangle = c|V^{(N)}(C)\rangle$. Then

$$\hat{\rho}^{(N)}(M) = \hat{\rho}^{(N)}(C).$$

The proportionality hypothesis is the finite-length form of the hypothesis in [11, Theorem 2.10]. The conclusion is an auxiliary cancellation used in the comparison preceding [11, Appendix C.2, lines 1597–1619].

Proof. [✓ Proof](#) Reading a doubled physical index as a ket–bra pair gives $\rho^{(N)}(M) = c\rho^{(N)}(C)$, and hence $\text{tr}[\rho^{(N)}(M)] = c \text{tr}[\rho^{(N)}(C)]$. Since $c \neq 0$,

$$(c \text{tr}[\rho^{(N)}(C)])^{-1} c\rho^{(N)}(C) = \text{tr}[\rho^{(N)}(C)]^{-1} \rho^{(N)}(C).$$

This identity also holds when the two proportional traces vanish. □

Theorem 20.17.22 (Finite-chain reduced marginals under nonzero proportionality). [Lean](#) [✓ Stmt](#)
Under the hypotheses of Theorem 20.17.21, for every $L \leq N$ the reduced operators on the first L sites agree:

$$\text{tr}_{N-L}[\hat{\rho}^{(N)}(M)] = \text{tr}_{N-L}[\hat{\rho}^{(N)}(C)].$$

Proof. [✓ Proof](#) Apply the same finite-index identification and partial trace to both sides of Theorem 20.17.21. □

Theorem 20.17.23 (Eventual equality of normalized finite-chain operators). [Lean](#) [✓ Stmt](#) If the doubled-index tensors of M and C are eventually nonzero proportional, then, for every sufficiently large N ,

$$\hat{\rho}^{(N)}(M) = \hat{\rho}^{(N)}(C).$$

Proof. [✓ Proof](#) At each sufficiently large length, apply Theorem 20.17.21 to the nonzero proportionality scalar at that length. □

Theorem 20.17.24 (Eventual equality of fixed-size reduced marginals). [Lean](#) [✓ Stmt](#) If the doubled-index tensors of M and C are eventually nonzero proportional, then for each fixed L and every sufficiently large $N \geq L$,

$$\text{tr}_{N-L}[\hat{\rho}^{(N)}(M)] = \text{tr}_{N-L}[\hat{\rho}^{(N)}(C)].$$

Proof. [✓ Proof](#) At each sufficiently large length, apply Theorem 20.17.22. □

Lemma 20.17.25 (Marginal stability under source zero correlation length). [Lean](#) [Lean](#) [✓ Stmt](#)
Let M have source zero correlation length. For every $N \geq 0$, the normalized N -site marginal obtained by tracing two sites from $\sigma^{(N+2)}(M)$ equals the normalized N -site marginal obtained by tracing one site from $\sigma^{(N+1)}(M)$.

This establishes the first replacement of [11, Appendix C.2, line 1613].

Proof. ✓ Proof Write $E = \mathcal{T}_M$ and choose $\lambda > 0$ such that $E^2 = \lambda E$. For physical configurations u, v of length N , the two marginal matrix elements are $\text{tr}(M^{u,v} E^2) / \text{tr}(E^{N+2})$ and $\text{tr}(M^{u,v} E) / \text{tr}(E^{N+1})$. The numerator and denominator on the left are respectively λ times those on the right, and source zero correlation length makes the denominator on the right nonzero. $\square$

Theorem 20.17.26 (Iterated marginal stability under zero correlation length). Lean ✓ Stmt *Let M have source zero correlation length. For every $L \geq 0$, the normalized L -site marginal obtained by tracing three sites from $\sigma^{(L+3)}(M)$ equals the normalized L -site marginal obtained by tracing one site from $\sigma^{(L+1)}(M)$.*

Proof. ✓ Proof First retain $L+1$ sites and apply Lemma 20.17.25. Trace the last of these retained sites, and then apply the same lemma while retaining L sites. $\square$

Definition 20.17.27 (Fixed-sector suffix contraction). Lean Lean ✓ Stmt Let S be the finite set of physical sectors and let $k = (k_0, \dots, k_{L-1}) \in S^L$ be a retained sector word. For $R \geq 1$ and $x, y \in I_k$, define

$$\Gamma_k^{(R)}(x, y) = \sum_{t \in S^R} \sum_{z \in I_t} \Omega_{kt}(xz, yz),$$

where kt and xz denote concatenation of sector words and their corresponding fiber coordinates. The cases $R = 1, 2, 3$ are denoted by $\Gamma_k^{(1)}$, $\Gamma_k^{(2)}$, and $\Gamma_k^{(3)}$.

Theorem 20.17.28 (Contraction of a sector-chain fiber). Lean ✓ Stmt *Let $t = (t_0, \dots, t_m)$ be a nonempty sector word. For functions a on the left coordinate of t_0 and b on the right coordinate of t_m ,*

$$\begin{aligned} \sum_{z \in I_t} a(z_0^L) \prod_{i=0}^{m-1} \eta_{t_i, t_{i+1}}((z_i^R, z_{i+1}^L), (z_i^R, z_{i+1}^L)) b(z_m^R) \\ = \left(\sum_u a(u) \right) \left(\prod_{i=0}^{m-1} \text{tr}(\eta_{t_i, t_{i+1}}) \right) \left(\sum_v b(v) \right). \end{aligned} \quad (20.99)$$

Proof. ✓ Proof Write $z_i = (\ell_i, r_i)$. Finite distributivity separates the left and right coordinates:

$$\begin{aligned} \sum_{\ell, r} a(\ell_0) \prod_{i=0}^{m-1} \eta_{t_i, t_{i+1}}((r_i, \ell_{i+1}), (r_i, \ell_{i+1})) b(r_m) \\ = \left(\sum_{\ell_0} a(\ell_0) \right) \prod_{i=0}^{m-1} \left(\sum_{r_i, \ell_{i+1}} \eta_{t_i, t_{i+1}}((r_i, \ell_{i+1}), (r_i, \ell_{i+1})) \right) \left(\sum_{r_m} b(r_m) \right). \end{aligned} \quad (20.100)$$

The internal double sum is $\text{tr}(\eta_{t_i, t_{i+1}})$ for each i , which gives the asserted factorization. $\square$

Theorem 20.17.29 (Contraction of a nonempty suffix fiber). Lean ✓ Stmt *Let $k = (k_0, \dots, k_n)$ be a nonempty retained sector word and $t = (t_0, \dots, t_m)$ a nonempty suffix sector word. In open-edge coordinates $x = (x_{\text{int}}, x_{\partial})$ and $y = (y_{\text{int}}, y_{\partial})$,*

$$\sum_{z \in I_t} \Omega_{kt}(xz, yz) = \left(\prod_{j=0}^{n-1} \eta_{k_j, k_{j+1}} \right) (x_{\text{int}}, y_{\text{int}}) \left(\text{tr}_R(\eta_{k_n, t_0}) \otimes \text{tr}_L(\eta_{t_m, k_0}) \right) (x_{\partial}, y_{\partial}) \prod_{i=0}^{m-1} \text{tr}(\eta_{t_i, t_{i+1}}). \quad (20.101)$$

Proof. ✓ Proof Split the cyclic product into the retained internal edges, the two boundary edges, and the internal suffix edges. Apply Theorem 20.17.28 to the suffix coordinates. The two boundary sums are the indicated partial traces. $\square$

Theorem 20.17.30 (Sector decomposition of a suffix marginal). Lean Lean ✓ Stmt Let $\widetilde{\mathcal{K}}$ be the tensor in physical-sector coordinates. For $L \geq 0$ and $R \geq 1$, the normalized L -site marginal obtained from a chain of length $L + R$ satisfies

$$\text{reindex}_{\Phi_L} \left(\text{tr}_R \left[\hat{\rho}^{(L+R)}(\widetilde{\mathcal{K}}) \right] \right) = \text{tr} \left[\rho^{(L+R)}(\widetilde{\mathcal{K}}) \right]^{-1} \bigoplus_{k \in S^L} \Gamma_k^{(R)}.$$

In particular, this identity holds for $R = 1$, $R = 2$, and $R = 3$.

Proof. ✓ Proof Decompose each discarded physical configuration uniquely as a suffix sector word $t \in S^R$ and a fiber coordinate $z \in I_t$. By Theorem 20.16.3.24, matrix elements with distinct retained sector words vanish. For a fixed retained word k , summing the diagonal entries of the discarded fiber gives

$$\sum_{t \in S^R} \sum_{z \in I_t} \Omega_{kt}(xz, yz) = \Gamma_k^{(R)}(x, y).$$

Multiplication by the inverse trace of the length- $L + R$ operator gives the stated normalization. $\square$

Definition 20.17.31 (Rank-one trace factorization for neighboring operators). Lean ✓ Stmt A rank-one trace factorization of neighboring matrices $\eta_{q,h}$ consists of real numbers a_q, b_h such that

$$\begin{aligned} \text{tr}(\eta_{q,h}) &= a_q b_h, \\ \sum_q a_q b_q &= 1. \end{aligned}$$

The factorization is an explicit assumption; it is not deduced from zero correlation length. It is the factorization invoked at [11, Appendix C.2, line 1613].

Lemma 20.17.32 (Separation of the two-boundary edge contraction). Lean ✓ Stmt Suppose $\eta_{q,h}$ has a rank-one trace factorization. For fixed boundary sectors u, v , set

$$\begin{aligned} R_{u,q} &= \text{tr}_{H_{q,l}}(\eta_{u,q}), \\ L_{h,v} &= \text{tr}_{H_{h,r}}(\eta_{h,v}). \end{aligned}$$

Then

$$\sum_{q,h} \text{tr}(\eta_{q,h})(R_{u,q} \otimes L_{h,v}) = \left(\sum_q a_q R_{u,q} \right) \otimes \left(\sum_h b_h L_{h,v} \right).$$

This is the corrected edge contraction at [11, Appendix C.2, lines 1613–1617].

Proof. ✓ Proof Substitute $\text{tr}(\eta_{q,h}) = a_q b_h$ and distribute the two finite sums. $\square$

Theorem 20.17.33 (Conditional fourth-region boundary trace). [Lean](#) [✓ Stmt](#) Let $\eta_{q,h}$ be arbitrary matrices on $H_{q,r} \otimes H_{h,l}$ with a rank-one trace factorization. For a nonempty retained sector word $k = (k_0, \dots, k_{L-1})$, write

$$\begin{aligned} P_k &= \bigotimes_{n=0}^{L-2} \eta_{k_n, k_{n+1}}, \\ R_{u,q} &= \text{tr}_{H_{q,l}}(\eta_{u,q}), \\ L_{h,v} &= \text{tr}_{H_{h,r}}(\eta_{h,v}). \end{aligned}$$

The edge contraction corresponding to tracing the two boundary sites of the longer cyclic chain is

$$\begin{aligned} &\bigoplus_k P_k \otimes \sum_{q,h} \text{tr}(\eta_{q,h})(R_{k_{L-1},q} \otimes L_{h,k_0}) \\ &= \bigoplus_k P_k \otimes \left(\sum_q a_q R_{k_{L-1},q} \right) \otimes \left(\sum_h b_h L_{h,k_0} \right). \end{aligned}$$

This is the edge-coordinate interpretation of the displayed fourth-region trace in [11, Appendix C.2, lines 1606 and 1613–1617].

Proof. [✓ Proof](#) Apply Lemma 20.17.32 for each retained sector word. Tensor with the unchanged product P_k and take the direct sum over k . $\square$

Remark 20.17.34. The theorem applies, in particular, to the positive chain-independent neighboring family supplied by Theorem 20.17.10, given the rank-one trace factorization.

Theorem 20.17.35 (A nonminimal sector factorization need not be primitive). [Lean](#) [✓ Stmt](#) There is an injective normal tensor $\mathcal{K}$, of physical dimension two and bond dimension one, which generates MPDOs and has source zero correlation length, together with one positive two-site bond B whose translates commute. For every $N \geq 2$,

$$\sigma^{(N)}(\mathcal{K}) = \prod_{i=0}^{N-1} B_{i,i+1},$$

while the trace matrix of a chosen two-sector Beigi factorization is

$$T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

In particular, T is not primitive.

This demonstrates that primitivity depends on the choice of sector factorization in [11, Appendix C.2, line 1613]; the same tensor admits a one-sector factorization whose trace matrix is primitive.

Proof. [✓ Proof](#) Let $\mathcal{K}^{0,0} = 1$ and let all other physical entries vanish. The matrices $\{\mathcal{K}^{i,j}\}_{i,j=0}^1$ span $M_1(\mathbb{C})$. For every $X \in M_1(\mathbb{C})$, the doubled-index transfer $\mathcal{E}_{\mathcal{K}}$ and the physical-trace transfer $P_{\mathcal{K}}$ satisfy

$$\begin{aligned} \mathcal{E}_{\mathcal{K}}(X) &= X, \\ P_{\mathcal{K}} &= I_1, \\ P_{\mathcal{K}}^2 &= P_{\mathcal{K}}. \end{aligned}$$

Hence $\mathcal{K}$ is injective and normal and has source zero correlation length. Decompose the physical space nonminimally as $\mathbb{C} \oplus \mathbb{C}$. With $w_0 = 1$ and $w_1 = 0$, take both sector factors to be the scalar w_k in sector k . The neighboring operators are

$$\eta_{k,h} = w_k w_h I_1 = \delta_{k,0} \delta_{h,0} I_1 \succeq 0.$$

In the sector coordinates the corresponding physical bond is

$$\begin{aligned} B &= \sum_{k,h} \mathbb{1}_{H_L^{(k)}} \otimes \eta_{k,h} \otimes \mathbb{1}_{H_R^{(h)}} \\ &= \mathbb{1}_{H_L^{(0)}} \otimes I_1 \otimes \mathbb{1}_{H_R^{(0)}}. \end{aligned}$$

Since only the all-zero physical-index word contributes, the factorization identity is

$$\sigma^{(N)}(\mathcal{K}) = \prod_{i=0}^{N-1} B_{i,i+1}.$$

Consequently,

$$\begin{aligned} T_{k,h} &= \text{tr}(\eta_{k,h}) = \delta_{k,0} \delta_{h,0}, \\ T &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

For every $n \geq 1$, $T^n = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Thus no positive power has all entries strictly positive, so T is not primitive. For the alternative factorization, take one sector $B^L = \mathbb{C}^2$ and $B^R = \mathbb{C}$, with

$$\begin{aligned} l &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \\ r &= (1). \end{aligned}$$

Its sole neighboring operator is $\hat{\eta}_{0,0} = \text{diag}(1,0) \succeq 0$, and hence its trace matrix is $\hat{T} = (\text{tr}(\hat{\eta}_{0,0})) = (1)$. Every positive power of $\hat{T}$ equals (1) , so this one-sector trace matrix is primitive. $\square$

Definition 20.17.36 (Cyclic-active sectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) For a physical-sector factorization with neighboring operators $\eta_{k,h}$, retain a sector k when there is a sector h such that $\eta_{k,h} \neq 0$ and $h \rightsquigarrow k$, where $\rightsquigarrow$ denotes reachability by nonzero neighboring operators. The first edge $k \rightarrow h$ is explicit, so the resulting closed walk has positive length even when the return path is empty. The set C of retained sectors is therefore exactly the set of vertices occurring in positive-length directed cycles, in the sense of [4, Section III]. Define, for $k, h \in C$,

$$T_{k,h} = \text{Re tr}(\eta_{k,h}).$$

For positive semidefinite neighboring operators the trace is real, so $T_{k,h} = \text{tr}(\eta_{k,h})$, and $T_{k,h} > 0$ is equivalent to $\eta_{k,h} \neq 0$.

Theorem 20.17.37 (Deletion of sectors absent from cyclic products). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let $N \geq 1$ and let $k = (k_0, \dots, k_{N-1})$ be a cyclic sector configuration. If $k_n \notin C$ for some n , then $\Omega_k = 0$. Thus deleting the sectors outside C leaves every finite cyclic neighboring product unchanged. For every linear contraction Φ , including a composition of reindexings and partial traces, $\Omega_k = 0$ implies $\Phi(\Omega_k) = 0$.

Proof. ✓ Proof If $\Omega_k \neq 0$, each factor $\eta_{k_n, k_{n+1}}$ is nonzero; otherwise every matrix entry of the product vanishes. Starting with the edge $k_n \rightarrow k_{n+1}$ and following the remaining cyclic edges gives a return path to k_n . Hence every k_n belongs to C . The last assertion is the linear identity $\Phi(0) = 0$. $\square$

Theorem 20.17.38 (Trace-visible recurrence of selected fixed-product sectors). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let K be an injective tensor with an eta-local structure. Let C be the exact fixed-product tensor selected from its commuting bond, and let F be the associated positive physical-sector factorization. Then any two sectors in the positive-length cyclic support of F are joined by a directed path of nonzero neighboring operators.*

Proof. ✓ Proof At lengths two and three, express the closed operators of K in the physical coordinates of F . They are nonzero scalar multiples of the corresponding closed operators of C . The length-two identity and nondegeneracy of the trace pairing show that cross-sector matrices of K vanish, as do diagonal matrices belonging to sectors outside cyclic support. Hence the remaining cyclic-sector matrices span the full auxiliary matrix algebra.

Positivity supplies a nonzero diagonal coefficient on a closed sector word through every cyclic sector. Its proportional coefficient for K shows that each cyclic sector contributes a nonzero one-site matrix of K . The length-three identity transfers a zero neighboring operator of F to a zero ordered product of the corresponding two sector spaces of K .

Fix a cyclic sector and let S be the sectors reachable from it. If a cyclic sector lay outside S , the matrix spaces generated by S and by its complement would both be nonzero, their sum would be the full matrix algebra, and every product from the first space to the second would vanish. This contradicts the one-sided matrix-space obstruction. $\square$

Theorem 20.17.39 (Selected fixed-product reduced states). Lean ✓ Stmt *Let K have an eta-local structure, let F be the positive physical-sector factorization of its selected fixed-product tensor, and express K in the physical coordinates of F . For every $N \geq 2$ and $L \leq N$, the selected sector-coordinate tensor and the transformed source tensor have the same normalized L -site reduced state.*

Proof. ✓ Proof At length N , the unnormalized periodic operators satisfy

$$\rho^{(N)}(K_F) = c_N \rho^{(N)}(C_F), \quad c_N > 0.$$

The scalar cancels under trace normalization, and taking the prescribed partial trace preserves the equality. $\square$

Theorem 20.17.40 (Adjacent selected fixed-product marginals). Lean ✓ Stmt *Let K have an eta-local structure and source zero correlation length, and let C be the selected fixed-product tensor in its physical-sector coordinates. For every $L > 0$, the normalized L -site reduced states of C obtained from chain lengths $L + 2$ and $L + 1$ are equal.*

Proof. ✓ Proof In the selected physical coordinates, the periodic operators of K at both lengths are positive nonzero scalar multiples of those of C . Cancel these scalars by Theorem 20.17.39, transport through the physical isometry, and apply source marginal stability. $\square$

Theorem 20.17.41 (Iterated selected fixed-product marginals). Lean ✓ Stmt *Under the hypotheses of Theorem 20.17.40, for every $L > 0$, the normalized L -site reduced states of the selected fixed-product tensor obtained from chain lengths $L + 3$ and $L + 1$ are equal.*

Proof. ✓ **Proof** Apply the two-step source marginal identity to K , transport both sides through the selected physical isometry, and cancel the two positive realization scalars. $\square$

Theorem 20.17.42 (Positive traces of the selected fixed product). Lean ✓ **Stmt** *Let K have an eta-local structure and source zero correlation length. For every $N \geq 2$, the trace of the selected fixed-product periodic operator is a strictly positive real number.*

Proof. ✓ **Proof** The selected neighboring operators are positive semidefinite, so their cyclic product is positive semidefinite. Its trace cannot vanish: positive proportionality would then force the trace of the source periodic operator to vanish, contrary to source zero correlation length. A nonzero positive semidefinite matrix has strictly positive real trace. $\square$

Theorem 20.17.43 (Stabilization of the support of an irreducible square). Lean ✓ **Stmt** *Let T be a nonnegative irreducible real matrix on a finite index set. Suppose that*

$$(T^2)_{i,j} > 0 \iff (T^3)_{i,j} > 0$$

for every pair i, j . Then $(T^2)_{i,j} > 0$ for every pair i, j .

Proof. ✓ **Proof** Right multiplication by T propagates equality of the positive supports of two consecutive powers. Hence all powers from the second onward have the same positive support. Irreducibility supplies a positive path from i to j . If its length is one, append a positive return path at j . The resulting path has length at least two, so support stabilization reduces it to a positive path of length two from i to j . $\square$

Theorem 20.17.44 (Strict positivity from adjacent cyclic-sector marginals). Lean ✓ **Stmt** *Let F be a physical-sector factorization whose neighboring operators are positive semidefinite, and let T_C be the trace matrix on its cyclic-active sectors. Assume that every ordered pair of cyclic-active sectors is joined by a path of nonzero neighboring operators. Assume also that, for every $L > 0$, the normalized L -site marginals obtained from chain lengths $L + 2$ and $L + 1$ are equal, and that the closed operator has a positive real trace at every length $N \geq 2$. Then $(T_C^2)_{q,h} > 0$ for every pair of cyclic-active sectors q, h .*

Proof. ✓ **Proof** Fix q, h and choose a retained sector path from h to q . Positivity makes its retained trace product $P_{h,q}$ nonzero. Tracing one suffix site and two suffix sites gives the respective block-trace coefficients $P_{h,q}(T_C^2)_{q,h}$ and $P_{h,q}(T_C^3)_{q,h}$. Equality of the normalized marginals and positivity of their normalizations show that $(T_C^2)_{q,h}$ vanishes exactly when $(T_C^3)_{q,h}$ vanishes. Recurrence makes T_C irreducible, so Theorem 20.17.43 applies. $\square$

Theorem 20.17.45 (Normalization from adjacent cyclic-sector marginals). Lean ✓ **Stmt** *Let F be a physical-sector factorization whose neighboring operators are positive semidefinite, and let T_C be the trace matrix on its cyclic-active sectors. Assume that every ordered pair of cyclic-active sectors is joined by a path of nonzero neighboring operators. Assume also that, for every $L > 0$, the normalized L -site marginals obtained from chain lengths $L + 2$ and $L + 1$ are equal, and that the closed operator has a positive real trace Z_N at every length $N \geq 2$. Then there is a number $\lambda > 0$ such that*

$$(\lambda^{-1}T_C)^2 = (\lambda^{-1}T_C)^3.$$

Proof. ✓ **Proof** By Theorem 20.17.44, every entry of T_C^2 is positive. For fixed q, h , choose a positive summand $(T_C)_{h,r}(T_C)_{r,q}$ of $(T_C^2)_{h,q}$. The retained three-sector word h, r, q has nonzero

trace product. Comparing the two adjacent marginals at retained length three gives

$$(T_C^3)_{q,h} = \frac{Z_5}{Z_4}(T_C^2)_{q,h}.$$

The ratio is independent of q, h . Taking $\lambda = Z_5/Z_4$ proves the assertion.

Theorem 20.17.46 (Selected fixed-product square-cube identity). [Lean] [✓ Stmt] *Let K be an injective tensor with an eta-local structure and source zero correlation length. Let F be the positive physical-sector factorization of the selected fixed-product tensor, and let T_C be its trace matrix on cyclic-active sectors. There is a number $\lambda > 0$ such that*

$$(\lambda^{-1}T_C)^2 = (\lambda^{-1}T_C)^3.$$

Proof. ✓ Proof Express the original tensor in the physical coordinates of F . At every length $N \geq 2$, its closed operator is a positive scalar multiple of the selected closed operator. Source zero correlation length therefore gives the adjacent marginal equality for the selected tensor, and positivity gives positive selected traces. Theorem 20.17.38 supplies recurrence. Theorem 20.17.45 now gives the required normalization. $\square$

Theorem 20.17.47 (Markov cuts from the selected fixed product). Lean ✓ Stmt *Let K be an injective tensor with an eta-local structure and source zero correlation length. Express K in the one-site physical coordinates of the positive fixed-product factorization selected by the eta-local structure. For arbitrary $A, C \geq 0$, the normalized marginal on consecutive regions of lengths A , 1 , and C , obtained from the periodic chain of length $A + C + 2$, admits a Hayashi Markov decomposition.*

Proof.  **Proof** The selected trace matrix is recurrent, and its square is strictly positive. After the positive normalization of Theorem 20.17.46, its square and cube agree. Thus the normalized matrix is primitive, with its second power as a positive power, and the latter factors as ab^T for strictly positive vectors a, b .

Transport the three-suffix/one-suffix marginal identity from the source tensor to the selected fixed product. In cyclic-active coordinates, its retained blocks are the unchanged interior neighboring products tensored with the boundary factors determined by a and b . Regrouping at the distinguished middle site gives the direct sum of normalized positive left and right path states. Their trace products are nonnegative and sum to one. Finally, positive proportionality identifies this selected normalized marginal with the marginal of K in the same physical coordinates. $\square$

Theorem 20.17.48 (Primitivity on cyclic-active sectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Suppose that the virtual dimension is positive, the MPO tensor is injective, and every $\eta_{k,h}$ is positive semidefinite. Then C is nonempty and the matrix T is primitive. More precisely, T is irreducible and every $k \in C$ satisfies

$$\begin{aligned}(T^2)_{k,k} &> 0, \\ (T^3)_{k,k} &> 0.\end{aligned}$$

Proof. ✓ Proof The virtual matrices of all physical sectors span the full matrix algebra. If $A = A_k^{xy}$ is nonzero, nondegeneracy of the trace pairing gives a virtual matrix $B = A_q^{uv}$ such that $\text{tr}(AB) \neq 0$. Cyclicity gives $\text{tr}(BA) = \text{tr}(AB) \neq 0$, and therefore $\eta_{k,q} \neq 0$ and $\eta_{q,k} \neq 0$. Thus a sector containing a nonzero virtual matrix lies on a directed two-cycle. Hence every virtual matrix outside C vanishes, and the virtual matrices belonging to C still span the full algebra.

Theorem 20.17.51 (Fourth-region blocks from an explicit factorization). [Lean](#) [✓ Stmt](#) Let F be a physical-sector factorization with positive semidefinite neighboring operators. Suppose that $\lambda > 0$ and that functions a, b on the cyclic-active sectors satisfy

$$(\lambda^{-1}T_C)^2 = ab^\top.$$

For every $n \geq 0$, reindex the $(n+1)$ -site reduced state obtained from the periodic chain of length $n+4$ by its retained open-edge coordinates. The result is the inverse trace of the length- $(n+4)$ periodic operator times the direct sum over retained sector words. A word outside cyclic support contributes zero; every cyclic-active word contributes its unchanged interior neighboring product tensored with the separated boundary factors determined by λ, a, b .

Proof. [✓ Proof](#) Expand the reduced state as a direct sum of three-suffix contractions. Terms outside positive-length cyclic support vanish. For a retained cyclic-active word, the three traced suffix sites leave the interior product and the two-step boundary contraction. Substitute $(\lambda^{-1}T_C)^2 = ab^\top$ to separate its two boundary factors, retaining the normalization by the length- $(n+4)$ trace. $\square$

Theorem 20.17.52 (Cyclic-active fourth-region factorization). [Lean](#) [✓ Stmt](#) Let $\mathcal{K}$ be injective and have source zero correlation length, and suppose its neighboring operators are positive semidefinite. There are $\lambda > 0$ and strictly positive functions a, b on the cyclic-active sectors, normalized by $a \cdot b = 1$, such that every fourth-region marginal is a direct sum over retained sector words of the unchanged interior neighboring product tensored with the separated left and right boundary factors determined by a and b .

Proof. [✓ Proof](#) The additional marginal replacement leaves the retained interior product unchanged and produces the two-step coefficient on the cyclic-active sectors. Its positive rank-one factorization separates the two surviving boundary sums. All sector words leaving the cyclic-active support vanish before the two outer partial traces are evaluated. $\square$

Theorem 20.17.53 (Markov decomposition from an explicit fourth-region formula). [Lean](#) [✓ Stmt](#) Let F be a positive-dimensional physical-sector factorization whose neighboring operators are positive semidefinite. Fix $A, C \geq 0$, $\lambda \in \mathbb{R}$, and strictly positive functions a, b on the cyclic-active sectors. Suppose that the length- $(A+C+4)$ periodic operator has positive trace, the length- $(A+C+2)$ periodic operator has nonzero trace, and the normalized $(A+C+1)$ -site reduced state at length $A+C+2$ has the explicit fourth-region block formula determined by λ, a, b . Then its tripartite form on regions of lengths A , 1 , and C admits a Hayashi Markov decomposition.

Proof. [✓ Proof](#) Regroup every retained fourth-region block at the distinguished middle sector. Its positive left and right path factors are normalized by their traces, with fixed density matrices used only when a factor vanishes. The positive long-chain trace makes all block weights nonnegative. The nonzero short-chain trace makes the reduced state trace one, so these weights sum to one. The resulting direct sum is exactly the Hayashi decomposition in the sector-coordinate middle-site splitting. $\square$

Theorem 20.17.54 (Cyclic-active Markov decomposition at every cut). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 20.17.52, for arbitrary $A, C \geq 0$, the fourth-region marginal on consecutive regions of lengths A , 1 , and C admits a Hayashi Markov decomposition. The middle site decomposes as the direct sum of its left and right sector factors. Each block is the tensor product of normalized positive left and right path states, and its probability is the product of their unnormalized traces divided by the positive periodic normalization.

Proof. ✓ Proof Separate the retained chain at the distinguished middle site. The retained neighboring factors split into the left and right path products, while the two cyclic endpoint factors become their respective boundary matrices. Normalize each positive block by its trace, using an arbitrary trace-one positive matrix when the block vanishes. The trace-one identity for the full marginal shows that the resulting non-negative block weights sum to one. $\square$

Theorem 20.17.55 (Physical-sector factorization and ZCL imply SAL). Lean ✓ Stmt *Let $\mathcal{K}$ have positive bond dimension and be injective, and let F be a physical-sector factorization of $\mathcal{K}$. Suppose every neighboring operator $\eta_{q,h}^F$ is positive semidefinite. If the physical-trace transfer of $\mathcal{K}$ has source zero correlation length, then $\mathcal{K}$ saturates the area law.*

Proof. ✓ Proof Positivity of the neighboring operators makes every nonempty sector-coordinate chain operator positive. Source zero correlation length makes its trace nonzero. Apply Theorem 20.17.54 with $A = m - 1$ and $C = N - m - 1$ at every admissible cut, then use the all-cut Markov criterion. Finally transport saturation through the physical isometry of F . $\square$

Theorem 20.17.56 (A translation-invariant commuting bond and ZCL imply SAL). Lean ✓ Stmt *Let $\mathcal{K}$ be an injective tensor generating MPDOs. Suppose that there is one positive two-site bond B whose translates commute and, for every $N \geq 2$, a constant $c_N > 0$ such that*

$$\sigma^{(N)}(\mathcal{K}) = c_N \prod_{i=0}^{N-1} B_{i,i+1}.$$

If the physical-trace transfer of $\mathcal{K}$ has zero correlation length, then $\mathcal{K}$ saturates the area law. This is the proposition at [11, Appendix C.2, lines 1597–1619].

Proof. ✓ Proof Express $\mathcal{K}$ in the one-site physical coordinates selected by the fixed bond. Positivity of every periodic operator is preserved by this isometry, and its trace is unchanged, hence nonzero at every positive length by source zero correlation length.

For every $1 \leq m \leq \lfloor N/2 \rfloor$, apply Theorem 20.17.47 with $A = m - 1$ and $C = N - m - 1$. Since $A + C + 2 = N$, the resulting decomposition is the quantum Markov decomposition required at the m -th cut. The all-cut criterion gives saturation of the area law in the selected physical coordinates. Finally, isometric invariance of the chain and marginal entropies transports saturation back to $\mathcal{K}$. $\square$

Theorem 20.17.57 (η -local structure gives the one-sector GSNNCH form). Lean Lean ✓ Stmt *The single positive bond determined by the η -local structure gives the one-sector instance of the explicit GSNNCH decomposition.*

Proof. ✓ Proof At each chain length, apply the one-sector construction directly to the commuting-form data carried by the η -local structure. $\square$

Theorem 20.17.58 (Injective SAL tensors generate GSNNCH states). Lean ✓ Stmt *Every injective MPO tensor satisfying SAL generates Gibbs states of a nearest-neighbor commuting Hamiltonian: the normalized finite-chain states are density operators with the explicit sector form of Definition 20.14.1, realized through one sector. The one-sector form is the commuting product form in which [11, Proposition C.8] states its conclusion for an injective tensor, combined here with the density normalization of [11, Definition 4.8].*

Proof. ✓ Proof Corollary 20.16.3.61 gives the η -local structure, whose single positive bond gives the one-sector form by Theorem 20.17.57. SAL includes the nonvanishing of every positive-length trace, so Theorem 20.14.41 applies. $\square$

Remark 20.17.59 (Positive physical-sector factorization). Suppose that K satisfies $\text{SAL}(K)$ and admits a physical-sector factorization whose neighboring operators $\eta_{k,h}$ are simultaneously positive semidefinite. The associated two-site bond is

$$\begin{aligned} B &= \sum_{k,h} \mathbb{1}_{H_L^{(k)}} \otimes \eta_{k,h} \otimes \mathbb{1}_{H_R^{(h)}}, \\ B &\geq 0, \end{aligned} \tag{20.105}$$

and, for every $N \geq 2$,

$$\begin{aligned} \sigma^{(N)}(K) &= c_N \prod_{n=1}^N B_{n,n+1}, \\ c_N &> 0, \\ [B_{i,i+1}, B_{j,j+1}] &= 0. \end{aligned} \tag{20.106}$$

Under this hypothesis, Theorem 20.16.3.60 gives the η -local structure and proves (20.105)–(20.106) with $c_N = 1$. Theorem 20.17.6 then gives the commuting-form conclusion. Adjacent translates commute on every periodic chain of length at least three by Theorem 20.16.3.53, and the crossed two-site case is Theorem 20.16.3.54. Disjoint translates commute by Theorem 20.16.3.55. These cases give pairwise commutativity at every length $N \geq 2$ by Theorem 20.16.3.56. Given the physical-sector factorization, the all-length product identity is Theorem 20.16.3.57. The doubled-index transfer condition enters only in the subsequent renormalization fixed point and single-bond branch conclusions.

Chapter 21

Mixed States: General Case and Algebraic Structure

In the general mixed-state renormalization analysis of [11, Section 4], an MPDO tensor in vertical canonical form generates closed-chain operators whose algebraic structure is governed by diagonal χ -matrices and fusion isometries $U_{\alpha,\beta}$.

21.1 Closed-chain operators

The vertical canonical form of an MPDO tensor is $\widetilde{M} = U^\dagger B U$ with $B = \bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ (Theorem 16.4.105), as in [11, Section 4].

The closed-chain operators $O_L(M_\alpha)$ contract cyclically the BNT bond indices of L copies in the vertical reading. Decomposing each vertical physical index into a pair (a, b) of horizontal bond indices and writing $\widehat{M}_\alpha^{ab} = M_\alpha^{(a,b)}$, the matrix entries for index strings $\mathbf{a} = (a_1, \dots, a_L)$ and $\mathbf{b} = (b_1, \dots, b_L)$ are

$$\begin{aligned} [O_L(M_\alpha)]_{\mathbf{a}, \mathbf{b}} &= \text{tr}(\widehat{M}_\alpha^{a_1 b_1} \dots \widehat{M}_\alpha^{a_L b_L}) \\ &= [\rho^{(L)}(\widehat{M}_\alpha)]_{\mathbf{a}, \mathbf{b}}, \end{aligned}$$

where the trace runs over the BNT bond space. These closed-chain operators $O_L(M_\alpha) = \rho^{(L)}(\widehat{M}_\alpha)$ act on the horizontal bond strings $\mathbf{a}$ and $\mathbf{b}$ and enter the operator algebra of [11, Theorem 4.14].

21.2 Algebra structure

Definition 21.2.1 (Support algebras and blocking maps). Lean ✓ Stmt Let $\{\mathcal{A}_n\}_{n \geq 0}$ be a family of $*$ -subalgebras of $M_D(\mathbb{C})$, one at every blocking size, equipped with bilinear maps $m_n : \mathcal{A}_n \times \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$ and linear inclusion maps $\iota_n : \mathcal{A}_n \rightarrow \mathcal{A}_{n+1}$. These maps are realized in the ambient bond-space matrix algebra by

$$\begin{aligned} m_n(x, y) &= xy, \\ \iota_n(x) &= x. \end{aligned}$$

Definition 21.2.2 (Algebras of blocked adjoint fixed points). [Lean](#) [✓ Stmt](#) An MPO tensor M has a family of algebras of blocked adjoint fixed points if there exists a family of support algebras with blocking and inclusion maps as in Definition 21.2.1 such that, for every positive blocking size n ,

$$\mathcal{A}_n = \text{Fix}(E_n^\dagger),$$

where E_n is the blocked transfer map. This equality is termed compatibility with M . This condition concerns adjoint fixed points; it is not the algebra condition of [11, Theorem 4.14(ii)], which requires a common BNT $\{M_\alpha\}_\alpha$, the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, and a vector m , with $m_\alpha = \text{tr}(\mu_\alpha)$, that is an idempotent for the multiplication induced by $c^{(1)}$.

Theorem 21.2.3 (Idempotent transfer maps and stationary support algebras). [Lean](#) [✓ Stmt](#) Assume the transfer map of the doubled tensor is trace-preserving and admits a positive-definite fixed point. If $E_1^2 = E_1$, then there is a stationary family of support algebras

$$\mathcal{A}_n = \text{Fix}(E_1^\dagger),$$

with multiplication $m_n(x, y) = xy$ and inclusion $\iota_n(x) = x$.

Proof. [✓ Proof](#) Under the trace-preserving normalization and the faithful fixed point, Wolf's fixed-point-algebra theorem gives the $*$ -subalgebra $\text{Fix}(E_1^\dagger) \subseteq M_D(\mathbb{C})$. Since E_1 is idempotent, $E_n = E_1^n = E_1$ for every $n \geq 1$, so $\text{Fix}(E_n^\dagger) = \text{Fix}(E_1^\dagger)$ at every positive blocking size, giving the stationary family $\mathcal{A}_n = \text{Fix}(E_1^\dagger)$. $\square$

For the one-site and two-site vertical canonical forms, let $\{M_\alpha\}_\alpha$ and $\{A_\alpha\}_\alpha$ denote their respective BNTs. Their boundary contractions have the same local form:

$$M_\alpha(X) = \begin{array}{c} M_\alpha \\ \text{---} \square \text{---} \\ | \quad | \\ \text{---} X \text{---} \square \text{---} \\ | \quad | \end{array} \quad A_\alpha(X) = \begin{array}{c} \text{---} \square \text{---} \\ | \quad | \\ \text{---} X \text{---} \square \text{---} \\ | \quad | \end{array}$$

In each cell, X is inserted on the virtual bond and the virtual indices are closed by tr ; the physical legs remain open. In [11, Appendix C.4, lines 1974–1980], the weighted direct sums of these contractions are fixed by compositions of the reduced tpCPM. Products of these fixed operators are then used to prove that the compositions are identities on the corresponding direct sums of matrix algebras. This argument concerns the vertical canonical forms, not the fixed-point algebras $\text{Fix}(E_n^\dagger)$ of Definition 21.2.2.

Theorem 21.2.4 (Stabilized blocked adjoint fixed points yield support algebras). [Lean](#) [✓ Stmt](#) Assume the transfer map of the doubled tensor is trace-preserving and admits a positive-definite fixed point. If, for every positive blocking size n ,

$$\text{Fix}(E_n^\dagger) = \text{Fix}(E_1^\dagger),$$

then there is a compatible family of support algebras.

Proof. [✓ Proof](#) The trace-preserving map and its positive-definite fixed point give a $*$ -subalgebra $\text{Fix}(E_1^\dagger)$. Take $\mathcal{A}_n := \text{Fix}(E_1^\dagger)$ for every n , with ambient multiplication and inclusion. The stabilization hypothesis $\text{Fix}(E_n^\dagger) = \text{Fix}(E_1^\dagger)$ then gives $\mathcal{A}_n = \text{Fix}(E_n^\dagger)$ at every positive blocking size n , which is compatibility with M . $\square$

Stabilization of adjoint fixed points does not establish the converse direction of [11, Theorem 4.14]. In particular, [11, Theorem 4.14(ii)] requires positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ for the multiplication of the vertical operators $O_L(M_\alpha)$ and a vector $m_\alpha = \text{tr}(\mu_\alpha)$ that is an idempotent for the multiplication induced by $c^{(1)}$. By contrast, choosing a basis in each $\mathcal{A}_n$ gives coordinates for multiplication and inclusion between different blocking sizes. These coordinates alone do not identify the vertical BNT or its structure coefficients.

Definition 21.2.5 (Coordinates in the support algebras). [Lean](#) [✓ Stmt](#) For each blocking size n , fix a basis $\{b_i^{(n)}\}_{i \in I_n}$ of $\mathcal{A}_n$ over a finite index set I_n . The coordinates of $x \in \mathcal{A}_n$ are the unique scalars $(x_i)_{i \in I_n} \in \mathbb{C}^{I_n}$ such that

$$x = \sum_{i \in I_n} x_i b_i^{(n)}.$$

Theorem 21.2.6 (Reconstruction from coordinates). [Lean](#) [✓ Stmt](#) Reconstructing a coordinate family $(a_i)_{i \in I_n}$ gives $\sum_i a_i b_i^{(n)}$ in $\mathcal{A}_n$.

Proof. [✓ Proof](#) This is the finite-basis reconstruction formula for the chosen basis of $\mathcal{A}_n$. $\square$

Definition 21.2.7 (Multiplication coefficients between blocking sizes). [Lean](#) [✓ Stmt](#) For $i, j \in I_n$, define the multiplication coefficients $(c_{i,j,k}^{(n)})_{k \in I_{2n}}$ by

$$m_n(b_i^{(n)}, b_j^{(n)}) = \sum_{k \in I_{2n}} c_{i,j,k}^{(n)} b_k^{(2n)}.$$

These are coordinates in $\mathcal{A}_{2n}$, not the same-length structure coefficients of the vertical operators $O_L(M_\alpha)$.

Theorem 21.2.8 (Reconstruction of multiplication from its coefficients). [Lean](#) [✓ Stmt](#) In the ambient bond-space matrix algebra, $b_i^{(n)} b_j^{(n)} = \sum_{k \in I_{2n}} c_{i,j,k}^{(n)} b_k^{(2n)}$.

Proof. [✓ Proof](#) Apply the reconstruction formula of Theorem 21.2.6 to the element $m_n(b_i^{(n)}, b_j^{(n)}) \in \mathcal{A}_{2n}$. $\square$

Definition 21.2.9 (Inclusion coefficients between blocking sizes). [Lean](#) [✓ Stmt](#) For $i \in I_n$, define the inclusion coefficients $(d_{ij}^{(n)})_{j \in I_{n+1}}$ by

$$\iota_n(b_i^{(n)}) = \sum_{j \in I_{n+1}} d_{ij}^{(n)} b_j^{(n+1)}.$$

Theorem 21.2.10 (Reconstruction of inclusion from its coefficients). [Lean](#) [✓ Stmt](#) Reconstructing the inclusion coefficients recovers the image of $b_i^{(n)}$ under ι_n : $\sum_{j \in I_{n+1}} d_{ij}^{(n)} b_j^{(n+1)} = \iota_n(b_i^{(n)})$.

Proof. [✓ Proof](#) Coefficient extraction and reconstruction are mutually inverse coordinate maps for the chosen basis of $\mathcal{A}_{n+1}$. Applying them successively to $\iota_n(b_i^{(n)})$ therefore returns the same element. $\square$

Theorem 21.2.11 (Reconstruction of compatible adjoint fixed points). [Lean](#) [✓ Stmt](#) If the support algebras are compatible with an MPO tensor M and $E_n^\dagger X = X$ for a positive blocking size n , then $X \in \mathcal{A}_n$. Writing $(X_i)_{i \in I_n}$ for its coordinates gives $X = \sum_{i \in I_n} X_i b_i^{(n)}$.

Proof. [✓ Proof](#) Compatibility identifies $\mathcal{A}_n$ with the adjoint fixed-point algebra of E_n , so X belongs to $\mathcal{A}_n$. Theorem 21.2.6 then reconstructs it. $\square$

Theorem 21.2.12 (Descent of blocked adjoint fixed points). [Lean](#) [✓ Stmt](#) *If an MPO tensor has a compatible family of support algebras, then for every positive blocking size n and every X , $E_n^\dagger X = X$ implies $E_1^\dagger X = X$.*

Proof. [✓ Proof](#) Compatibility at size n places X in $\mathcal{A}_n$. The inclusion ι_n places the same ambient matrix X in $\mathcal{A}_{n+1}$. Compatibility at size $n+1$ therefore gives $E_{n+1}^\dagger X = X$. Since $E_{n+1} = E_n \circ E_1$, it follows that $X = E_{n+1}^\dagger X = E_1^\dagger E_n^\dagger X = E_1^\dagger X$. $\square$

Lemma 21.2.13 (Reverse inclusion of adjoint fixed points). [Lean](#) [✓ Stmt](#) *For every n , $E_1^\dagger X = X$ implies $E_n^\dagger X = X$.*

Proof. [✓ Proof](#) Induct on n , using $E_{n+1} = E_n \circ E_1$ and $(A \circ B)^\dagger = B^\dagger \circ A^\dagger$. The base case is $E_0 = \text{id}$. This argument does not require a compatible family of support algebras. $\square$

Lemma 21.2.14 (Blocked powers of adjoint eigenvectors). [Lean](#) [✓ Stmt](#) *If $E = E_1$ and $E^\dagger X = \lambda X$, then $E_n^\dagger X = \lambda^n X$ for every n .*

Proof. [✓ Proof](#) The case $n = 0$ is $E_0 = \text{id}$. For the induction step, $E_{n+1} = E_n \circ E_1$, hence

$$E_{n+1}^\dagger X = E_1^\dagger E_n^\dagger X = E_1^\dagger (\lambda^n X) = \lambda^n E_1^\dagger X = \lambda^{n+1} X.$$

$\square$

Theorem 21.2.15 (Finite-order adjoint eigenvalues under compatibility). [Lean](#) [✓ Stmt](#) *Assume there is a compatible family of support algebras. If $X \neq 0$, $E_1^\dagger X = \lambda X$, and $\lambda^n = 1$ for some $n > 0$, then $\lambda = 1$.*

Proof. [✓ Proof](#) By Lemma 21.2.14, $E_n^\dagger X = \lambda^n X = X$. Theorem 21.2.12 gives $E_1^\dagger X = X$. Thus $\lambda X = X$, so $(\lambda - 1)X = 0$. Since $X \neq 0$, one obtains $\lambda = 1$. $\square$

Theorem 21.2.16 (Equality of compatible adjoint fixed-point algebras). [Lean](#) [✓ Stmt](#) *If an MPO tensor has a compatible family of support algebras, then $\text{Fix}(E_n^\dagger) = \text{Fix}(E_1^\dagger)$ for every positive blocking size n .*

Proof. [✓ Proof](#) Combine Theorem 21.2.12 with Lemma 21.2.13. $\square$

Theorem 21.2.17 (Compatibility of the stationary family of support algebras). [Lean](#) [✓ Stmt](#) *Under the trace-preserving normalization and a positive-definite fixed point, if M has a compatible family of support algebras, then the stationary family $\mathcal{A}_n = \text{Fix}(E_1^\dagger)$ is itself compatible with M : for every positive blocking size n , $\text{Fix}(E_1^\dagger) = \text{Fix}(E_n^\dagger)$.*

Theorem 21.2.18 (Compatible support algebras do not imply an idempotent transfer map). [Lean](#) [✓ Stmt](#) *There exists an MPO tensor of physical and bond dimension two whose doubled-tensor transfer map is trace-preserving and has a positive-definite fixed point, and which has a compatible family of support algebras, although its doubled-tensor transfer map is not idempotent.*

Proof. ✓ **Proof** Take the phase-flip channel with Kraus operators $K_0 = \frac{3}{5}\mathbb{1}$ and $K_1 = \frac{4}{5}Z$, where $Z = \text{diag}(1, -1)$. Its transfer map satisfies

$$E(X)_{ij} = \begin{cases} X_{ij}, & i = j, \\ -\frac{7}{25}X_{ij}, & i \neq j. \end{cases}$$

The identity is a positive-definite fixed point, and $K_0^\dagger K_0 + K_1^\dagger K_1 = \mathbb{1}$ proves trace preservation. For every $n > 0$ and $i \neq j$, $E^n(X)_{ij} = (-7/25)^n X_{ij}$. Since $(-7/25)^n \neq 1$ for every $n > 0$ and E is self-adjoint, $\text{Fix}((E^n)^\dagger) = \text{Fix}(E^\dagger)$. Thus a compatible family of support algebras exists. On the other hand, for $X_{01} \neq 0$, $E^2(X)_{01} = \frac{49}{625}X_{01} \neq -\frac{7}{25}X_{01} = E(X)_{01}$. Therefore $E^2 \neq E$, so the doubled-tensor ZCL condition fails. $\square$

Remark 21.2.19 (Stabilized fixed points and the vertical algebra condition). Write $E = E_1$ for the one-site transfer map. Compatibility gives $\text{Fix}((E^n)^\dagger) = \text{Fix}(E^\dagger)$ for every $n > 0$. Hence a nonzero matrix X with $E^\dagger X = \lambda X$ and $\lambda^n = 1$ lies in $\text{Fix}(E^\dagger)$, so $(\lambda - 1)X = 0$ and $\lambda = 1$ by Theorem 21.2.15. Compatibility does not imply $E^2 = E$, as Theorem 21.2.18 shows. More generally, on $M_D(\mathbb{C})$ with $D \geq 2$, let Π_{diag} be the projection onto diagonal matrices. For $0 < \varepsilon < 1$, set $E(X) = \varepsilon X + (1 - \varepsilon)\Pi_{\text{diag}}(X)$, so that

$$\begin{aligned} E^n(X) &= \Pi_{\text{diag}}(X) + \varepsilon^n(X - \Pi_{\text{diag}}(X)), \\ E^2 - E &= (\varepsilon^2 - \varepsilon)(\text{id} - \Pi_{\text{diag}}). \end{aligned}$$

The fixed-point algebra is diagonal for every $n > 0$, but $E^2 \neq E$.

The coefficient argument in [11, Appendix C.4] instead uses the vertical canonical form of an MPDO tensor M in CF. Write its positive diagonal multiplicity matrices as μ_α and its BNT as $\{M_\alpha\}_\alpha$. The operators $O_L(M_\alpha)$ are obtained by contracting L tensors in the vertical direction, with the original physical and virtual roles exchanged. The required same-length product law is

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \text{tr}(\chi_{\alpha,\beta,\gamma}^L)O_L(M_\gamma), \quad (21.1)$$

where the $\chi_{\alpha,\beta,\gamma}$ are positive diagonal matrices. With $m_\alpha = \text{tr}(\mu_\alpha)$, the vector m must be an idempotent for the multiplication induced by $c^{(1)}$, namely

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma})m_\alpha m_\beta. \quad (21.2)$$

The proof of [11, Theorem 4.14] uses (21.1) and (21.2), positivity, and the simple-matrix lemma [11, lines 1155–1163] to identify the diagonal entries of the two-site multiplicity matrix ν_γ as a multiset:

$$\begin{aligned} \{\nu_{\gamma,k}\}_k &= \{\mu_{\alpha,k_1}\mu_{\beta,k_2}\chi_{\alpha,\beta,\gamma,k_3}\}_{\alpha,\beta,k_1,k_2,k_3}, \\ \text{tr}(\nu_\gamma) &= m_\gamma. \end{aligned} \quad (21.3)$$

The reconstruction maps in [11, lines 2065–2085] then use

$$\begin{aligned} \widetilde{R}_2 \left(\bigoplus_\gamma X_\gamma \right) &= \bigoplus_\gamma \frac{\nu_\gamma}{m_\gamma} \otimes X_\gamma, \\ \widetilde{R}_1 \left(\bigoplus_\gamma X_\gamma \right) &= \bigoplus_\gamma \frac{\mu_\gamma}{m_\gamma} \otimes X_\gamma. \end{aligned} \quad (21.4)$$

Thus $\text{tr}(\nu_\gamma) = m_\gamma$ and $m_\gamma = \text{tr}(\mu_\gamma)$ normalize the factors ν_γ/m_γ and μ_γ/m_γ , respectively, in $T = \tilde{R}_2 \tilde{T} R_1$ and $S = \tilde{R}_1 \tilde{S} R_2$. Here R_1 and R_2 compress to the canonical sectors and trace out their multiplicity factors, while $\tilde{T}$ and $\tilde{S}$ conjugate by the block unitaries relating the BNTs. These compositions are tpCPM on the retained spaces. When discarded zero-sector complements are present, extend the compressions by trace-and-prepare branches on the discarded input supports to obtain tpCPM on the full physical spaces. These additional branches vanish on the tensor letters, so the extended maps still satisfy $T[M_1(X)] = M_2(X)$ and $S[M_2(X)] = M_1(X)$, as required for an RFP.

The scalar Newton–Girard identity relates traces of matrix powers to characteristic-polynomial coefficients; see the corresponding theorem in the companion quantum-channel volume [182, “Newton–Girard trace recursion”]. Such scalar identities do not supply the vertical canonical decomposition. The trace-power conclusion of [11, Theorem 4.14(ii)] concerns a common BNT and positive diagonal matrices satisfying (21.1) and (21.2). Relating these objects to the one-site and two-site vertical canonical forms is required for the maps in (21.4); it does not follow from a choice of bases in the support algebras. In particular, Theorem 21.2.18 rules out deriving idempotence of the doubled-tensor transfer map from compatibility alone.

21.2.1 Diagonal χ -matrices and the trace-power formula

In [11, Theorem 4.14(ii)], the structure coefficients of the length- L operator algebra are traces of powers of positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$. In Definition 21.2.1.1, arbitrary complex diagonal entries are permitted to separate the scalar trace-power identity from positivity and operator multiplication.

Definition 21.2.1.1 (Diagonal χ matrices). Lean ✓ Stmt Let I be an index set. A family of diagonal complex matrices indexed by $(\alpha, \beta, \gamma) \in I^3$ consists of a size $r_{\alpha,\beta,\gamma} \in \mathbb{N}$ and entries $\chi_{\alpha,\beta,\gamma,k} \in \mathbb{C}$ for $0 \leq k < r_{\alpha,\beta,\gamma}$, defining

$$\chi_{\alpha,\beta,\gamma} = \text{diag}((\chi_{\alpha,\beta,\gamma,k})_{0 \leq k < r_{\alpha,\beta,\gamma}}).$$

A size of zero gives the empty diagonal matrix. The matrices $\chi_{\alpha,\beta,\gamma}$ in [11, Theorem 4.14(ii)] additionally have positive diagonal entries.

Theorem 21.2.1.2 (Trace of a power of a diagonal matrix). Lean ✓ Stmt For every triple (α, β, γ) and every $L \geq 0$,

$$\text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_{0 \leq k < r_{\alpha,\beta,\gamma}} \chi_{\alpha,\beta,\gamma,k}^L,$$

where $r_{\alpha,\beta,\gamma}$ is the size of $\chi_{\alpha,\beta,\gamma}$ and $\chi_{\alpha,\beta,\gamma,k}$ are its diagonal entries.

Proof. ✓ Proof Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. □

Definition 21.2.1.3 (Trace-power form of structure coefficients). Lean ✓ Stmt Let $c_{\alpha,\beta,\gamma}^{(L)}$ be a family of complex coefficients and let $\chi_{\alpha,\beta,\gamma}$ be diagonal matrices on the same index set. The coefficients have trace-power form with respect to χ if, for every $L \geq 0$ and every triple (α, β, γ) ,

$$c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_{0 \leq k < r_{\alpha,\beta,\gamma}} \chi_{\alpha,\beta,\gamma,k}^L.$$

Here $r_{\alpha,\beta,\gamma}$ is the size of $\chi_{\alpha,\beta,\gamma}$. This is the scalar identity appearing in [11, Theorem 4.14(ii)]. By itself, the definition neither requires positive diagonal entries nor asserts that the coefficients describe multiplication of vertical operators.

21.2.2 Structure coefficients and the idempotent condition

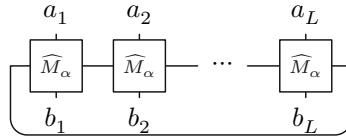
Definition 21.2.2.1 (Structure coefficients indexed by a BNT). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#)

Let Λ be a fixed set of BNT labels. Consider complex coefficients $c_{\alpha,\beta,\gamma}^{(L)}$ for $L \in \mathbb{N}$ and $\alpha, \beta, \gamma \in \Lambda$. In [11, Theorem 4.14(ii)], these are the structure coefficients for the products $O_L(M_\alpha)O_L(M_\beta)$ at the same length L , rather than for the blocked-basis multiplication $\mathcal{A}_n \times \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$. A family of diagonal matrices $\chi_{\alpha,\beta,\gamma}$ determines coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L) = \sum_r \chi_{\alpha,\beta,\gamma,r}^L$. The positive diagonal matrices constructed in [11, Appendix C.4] give coefficients of this form; the definition for a general diagonal family does not require positivity.

Definition 21.2.2.2 (Operators indexed by a BNT). [Lean](#) [Stmt](#) For each length L , let $O_L(M_\alpha)$ be operators indexed by the fixed BNT labels α . Their ambient complex vector space, equipped with multiplication, may depend on L and is not identified with a chosen blocked support algebra. Write $\widehat{M}_\alpha^{ab} = M_\alpha^{(a,b)}$ for the vertical reading. The vertical indices of the L copies are contracted cyclically, while the horizontal legs remain open, as in [11, lines 962–967]. For $\mathbf{a} = (a_1, \dots, a_L)$ and $\mathbf{b} = (b_1, \dots, b_L)$,

$$[O_L(M_\alpha)]_{\mathbf{a},\mathbf{b}} = \text{tr}(\widehat{M}_\alpha^{a_1 b_1} \dots \widehat{M}_\alpha^{a_L b_L}).$$

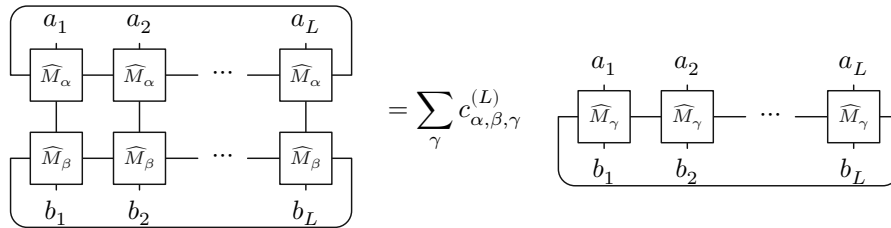
Diagrammatically, with the cyclically contracted matrix index of each copy of $\widehat{M}_\alpha$ drawn horizontally,



Definition 21.2.2.3 (Product of BNT operators at the same length). [Lean](#) [Lean](#) [Stmt](#) For a finite set of BNT labels, the operators $O_L(M_\alpha)$ satisfy the product identity with coefficients $c_{\alpha,\beta,\gamma}^{(L)}$ when, for every positive length L and all labels α, β ,

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} c_{\alpha,\beta,\gamma}^{(L)} O_L(M_\gamma).$$

In the diagram of Definition 21.2.2.2, the operator product joins the south legs of the first ring to the north legs of the second, and the identity reads



This is the product identity appearing in [11, Theorem 4.14(ii)], stated independently of the blocked-basis multiplication $\mathcal{A}_n \times \mathcal{A}_n \rightarrow \mathcal{A}_{2n}$.

Definition 21.2.2.4 (Scalars indexed by a BNT). [Lean](#) [Stmt](#) Let $m_\alpha \in \mathbb{C}$ be scalars indexed by the fixed BNT labels α . For the positive diagonal matrices μ_α in the vertical decomposition, these scalars are $m_\alpha = \text{tr}(\mu_\alpha)$, as in the idempotent condition of [11, Theorem 4.14(ii)].

Definition 21.2.2.5 (Idempotent for the multiplication induced by the coefficients). [Lean](#) [Lean](#) [✓ Stmt](#) For a finite set of BNT labels, the vector $m = (m_\alpha)_\alpha$ is an idempotent for the multiplication induced by $c^{(1)}$ when, for every label γ ,

$$m_\gamma = \sum_{\alpha, \beta} c_{\alpha, \beta, \gamma}^{(1)} m_\alpha m_\beta.$$

This is the idempotent condition in [11, Theorem 4.14(ii)], stated separately from the trace-power formula.

Definition 21.2.2.6 (Assignment of BNT labels to blocked bases). [Lean](#) [Lean](#) [✓ Stmt](#) An assignment of BNT labels consists, for every positive blocking length n , of a map σ_n from the chosen basis indices of $\mathcal{A}_n$ and a map τ_n from the chosen basis indices of $\mathcal{A}_{2n}$ to the fixed set of BNT labels. The combined map $\sigma_n \sqcup \tau_n$ is defined on the disjoint union of these two index sets.

Definition 21.2.2.7 (Comparison of BNT structure coefficients with blocked coefficients). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A comparison between the BNT structure coefficients and the chosen blocked-basis coefficients consists of an assignment of BNT labels together with the equality

$$c_{i, j, k}^{(n)} = c_{\sigma_n(i), \sigma_n(j), \tau_n(k)}^{(n)}$$

for every $n > 0$ and all basis indices i, j of $\mathcal{A}_n$ and k of $\mathcal{A}_{2n}$. On the left, $c_{i, j, k}^{(n)}$ is the coefficient of the k -th basis element of $\mathcal{A}_{2n}$ in the blocked product. This comparison is an additional hypothesis, not the product identity for BNT operators at the same length.

Definition 21.2.2.8 (Trace-power formula at positive lengths). [Lean](#) [✓ Stmt](#) The coefficients $c_{\alpha, \beta, \gamma}^{(L)}$ satisfy the trace-power formula for a family of diagonal matrices $\chi_{\alpha, \beta, \gamma}$ when, for every positive length L and all labels,

$$c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L).$$

The same matrix $\chi_{\alpha, \beta, \gamma}$ is used for all positive L . Its diagonal entries may be complex; positivity is not required here.

Lemma 21.2.2.9 (Structure coefficient as a trace power at positive length). [Lean](#) [✓ Stmt](#) Under the trace-power condition at positive lengths, for every $L > 0$, $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$.

Proof. [✓ Proof](#) Combine the defining positive-length trace-power identity with the trace formula for powers of a diagonal matrix. $\square$

Lemma 21.2.2.10 (Coefficients defined by traces of diagonal matrix powers). [Lean](#) [✓ Stmt](#) The coefficients defined by $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$ for a family of diagonal matrices satisfy the trace-power condition at positive lengths for those matrices.

Proof. [✓ Proof](#) This follows from the definition of the coefficients. $\square$

Definition 21.2.2.11 (Positive diagonal matrices giving the structure coefficients). [Lean](#) [Lean](#) [✓ Stmt](#) For given coefficients $c_{\alpha, \beta, \gamma}^{(L)}$, consider diagonal matrices $\chi_{\alpha, \beta, \gamma}$, independent of L , with strictly positive diagonal entries, such that $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$ for every $L > 0$ and all labels. This is the coefficient condition in [11, Theorem 4.14(ii)]. If the coefficients are defined by these traces, positivity of the diagonal entries suffices.

Lemma 21.2.2.12 (Trace powers of the positive diagonal matrices). [Lean](#) [✓ Stmt](#) *If the coefficients have a trace-power representation by positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ independent of L , then for every $L > 0$, $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$.*

Proof. [✓ Proof](#) Apply the trace reformulation of the trace-power identity at positive lengths. $\square$

Theorem 21.2.2.13 (Product of BNT operators with trace-power coefficients). [Lean](#) [✓ Stmt](#) *Suppose the product identity for BNT operators at the same length holds and its coefficients have a trace-power representation by positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ independent of L . Then, for every $L > 0$,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} \text{tr}(\chi_{\alpha,\beta,\gamma}^L)O_L(M_\gamma).$$

Proof. [✓ Proof](#) Substitute the positive-length trace-power formula into the same-length BNT product expansion. $\square$

Theorem 21.2.2.14 (Product of BNT operators for coefficients defined by traces). [Lean](#) [✓ Stmt](#) *If the product identity for BNT operators at the same length holds with coefficients defined by traces of powers of diagonal matrices $\chi_{\alpha,\beta,\gamma}$, then for every $L > 0$,*

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} \text{tr}(\chi_{\alpha,\beta,\gamma}^L)O_L(M_\gamma).$$

Proof. [✓ Proof](#) Substitute the defining trace-power formula for the coefficients into the same-length BNT product expansion. $\square$

Theorem 21.2.2.15 (Idempotent condition in terms of traces). [Lean](#) [✓ Stmt](#) *If m is an idempotent for the multiplication induced by $c^{(1)}$ and the coefficients have a trace-power representation by positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ independent of L , then*

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma})m_\alpha m_\beta.$$

Proof. [✓ Proof](#) Substitute the length-one trace-power formula into the idempotent coefficient identity. $\square$

Theorem 21.2.2.16 (Idempotent condition for coefficients defined by traces). [Lean](#) [✓ Stmt](#) *If m is an idempotent for the multiplication induced by $c^{(1)}$, where the coefficients are defined by traces of powers of diagonal matrices $\chi_{\alpha,\beta,\gamma}$, then*

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma})m_\alpha m_\beta.$$

Proof. [✓ Proof](#) Substitute the defining length-one trace formula into the idempotent coefficient identity. $\square$

Lemma 21.2.2.17 (Blocked coefficients inherit BNT trace powers). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Suppose the chosen blocked-basis coefficients satisfy the comparison equality with the BNT structure coefficients, and the latter have a trace-power representation by positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ independent of length. Then, for every $n > 0$ and every multiplication triple (i, j, k) ,*

$$c_{i,j,k}^{(n)} = \text{tr}(\chi_{\sigma_n(i),\sigma_n(j),\tau_n(k)}^n).$$

Proof. [✓ Proof](#) Use the comparison equality to replace the blocked-basis coefficient by the corresponding BNT structure coefficient, and then apply the trace-power formula. In the special case where the compared BNT coefficients are defined by traces of powers of χ , the same conclusion follows directly from their definition. $\square$

Definition 21.2.2.18 (Diagonal matrices for blocked multiplication coefficients). [Lean](#) [✓ Stmt](#) For each positive blocking length n , choose a family of diagonal matrices indexed by triples in the disjoint union of the chosen basis indices of $\mathcal{A}_n$ and $\mathcal{A}_{2n}$. For a multiplication triple (i, j, k) , where i, j label basis elements of $\mathcal{A}_n$ and k labels a basis element of $\mathcal{A}_{2n}$, write its size as $r_{n,i,j,k} \in \mathbb{N}$ and its complex diagonal entries as $\chi_{n,i,j,k,s}$, with $0 \leq s < r_{n,i,j,k}$. The corresponding matrix is

$$\chi_{n,i,j,k} = \text{diag}(\chi_{n,i,j,k,s})_{s=0}^{r_{n,i,j,k}-1}.$$

Unlike the matrices indexed by fixed BNT labels in [11, Theorem 4.14(ii)], the size and diagonal entries may depend on the blocking length and on the chosen basis indices.

Lemma 21.2.2.19 (Trace of a power of a blocked diagonal matrix). [Lean](#) [✓ Stmt](#) For each positive blocking length n , each multiplication triple (i, j, k) , and every exponent $L \in \mathbb{N}$, $\text{tr}(\chi_{n,i,j,k}^L) = \sum_{s=0}^{r_{n,i,j,k}-1} \chi_{n,i,j,k,s}^L$.

Proof. [✓ Proof](#) Diagonal matrices raise to powers entrywise and the trace of a diagonal matrix is the sum of its entries. $\square$

Definition 21.2.2.20 (Trace-power form for blocked multiplication coefficients). [Lean](#) [✓ Stmt](#) The blocked multiplication coefficients are in trace-power form when, for every positive blocking length n and every multiplication triple (i, j, k) , the coefficient $c_{i,j,k}^{(n)}$ equals $\sum_{s=0}^{r_{n,i,j,k}-1} \chi_{n,i,j,k,s}^n$. Equivalently, by Lemma 21.2.2.19, it is the trace of the n -th power of the corresponding diagonal matrix. The exponent n is the blocking length of the two factors in $\mathcal{A}_n$; the product is expanded in the basis of $\mathcal{A}_{2n}$.

Lemma 21.2.2.21 (Blocked coefficient as a trace power). [Lean](#) [Lean](#) [✓ Stmt](#) Under trace-power form for the blocked multiplication coefficients, for every positive blocking length n , $c_{i,j,k}^{(n)} = \text{tr}(\chi_{n,i,j,k}^n)$.

Proof. [✓ Proof](#) Combine the defining trace-power identity with the trace formula for powers of a diagonal matrix. $\square$

Definition 21.2.2.22 (Positive diagonal matrices giving the blocked coefficients). [Lean](#) [✓ Stmt](#) Consider a family of blocked diagonal matrices with strictly positive diagonal entries satisfying $c_{i,j,k}^{(n)} = \text{tr}(\chi_{n,i,j,k}^n)$ for every $n > 0$ and every multiplication triple (i, j, k) . This is a blocked-basis analogue of the positive diagonal matrices in [11, Theorem 4.14(ii)], without requiring a single family indexed only by the BNT labels.

Theorem 21.2.2.23 (Positive BNT diagonal matrices give positive blocked diagonal matrices). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the chosen blocked-basis coefficients satisfy the comparison equality with the BNT structure coefficients, and the latter have a trace-power representation by positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ independent of length. Then there are positive blocked diagonal matrices giving the blocked coefficients by the trace-power formula. They are obtained by reindexing the BNT diagonal matrices along $\sigma_n \sqcup \tau_n$; on multiplication triples this gives $\chi_{n,i,j,k} = \chi_{\sigma_n(i),\sigma_n(j),\tau_n(k)}$.

Proof. [✓ Proof](#) Reindex the positive diagonal matrices along the blocked-basis label map. Positivity is preserved by reindexing, and the trace-power identity follows by substituting the blocked-basis comparison into the BNT trace-power formula. $\square$

Lemma 21.2.2.24 (Constant power sums have entries zero or one). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let $x_0, \dots, x_{n-1}$ be non-negative real numbers whose power sums $\sum_k x_k^L$ take the same value at every integer exponent $L \geq 1$. Then every x_k equals 0 or 1. The same holds for a finite family of complex numbers whose imaginary parts vanish and whose real parts are non-negative.

Proof. [✓ Proof](#) If some entry exceeded one, its powers would grow without bound while all other summands stay non-negative, contradicting constancy of the power sums; hence every entry lies in $[0, 1]$. Comparing the exponents one and two gives $\sum_k x_k(1 - x_k) = 0$ with non-negative summands, so each summand vanishes. $\square$

Definition 21.2.2.25 (Length-independent coefficients). [Lean](#) [Lean](#) [✓ Stmt](#) The structure coefficients $c_{\alpha, \beta, \gamma}^{(L)}$ are *length independent* when $c_{\alpha, \beta, \gamma}^{(L)} = c_{\alpha, \beta, \gamma}^{(1)}$ for every positive length L and all labels; equivalently, the coefficients agree at any two positive lengths. This is the case singled out in the discussion following [11, Theorem 4.14].

Theorem 21.2.2.26 (Length-independent coefficients are natural numbers). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the BNT structure coefficients have the trace-power form $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$ with positive diagonal matrices $\chi_{\alpha, \beta, \gamma}$, and are length independent. Then every diagonal entry $\chi_{\alpha, \beta, \gamma, r}$ lies in $\{0, 1\}$, as stated in [11, Section 4.5]; by positivity every entry equals one. Consequently, at every positive length, $c_{\alpha, \beta, \gamma}^{(L)} = r_{\alpha, \beta, \gamma} \in \mathbb{N}$, where $r_{\alpha, \beta, \gamma}$ is the size of the diagonal matrix. The structure coefficient is this multiplicity, not necessarily one.

Proof. [✓ Proof](#) For fixed labels, the trace-power form turns length independence into constancy of the power sums of the diagonal entries of $\chi_{\alpha, \beta, \gamma}$, so by Lemma 21.2.2.24 every entry equals 0 or 1, and positivity excludes 0. The trace of $\chi_{\alpha, \beta, \gamma}^L$ is then the number of diagonal entries. $\square$

Theorem 21.2.2.27 (Unit entries give length independence). [Lean](#) [Lean](#) [✓ Stmt](#) If the BNT structure coefficients have a trace-power representation by diagonal matrices all of whose diagonal entries equal one, then for every $L > 0$, $c_{\alpha, \beta, \gamma}^{(L)} = r_{\alpha, \beta, \gamma}$, so the coefficients are length independent.

Proof. [✓ Proof](#) The power sums of a family of ones do not depend on the exponent. $\square$

Theorem 21.2.2.28 (Three consecutive positive power sums). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let $x_0, \dots, x_{r-1} > 0$ and $p_n = \sum_k x_k^n$. For every integer $n \geq 0$,

$$p_{n+2} - 2p_{n+1} + p_n = \sum_k x_k^n (x_k - 1)^2. \quad (21.5)$$

The second difference vanishes if and only if every x_k equals one. In particular, equality of three consecutive trace powers of a positive diagonal matrix forces that matrix to be the identity.

Proof. [✓ Proof](#) Expanding the squares gives (21.5). Every summand is nonnegative, and every factor x_k^n is strictly positive. Thus the sum vanishes exactly when all $x_k - 1$ vanish. $\square$

Theorem 21.2.2.29 (Finite criterion for length independence). [Lean](#) [✓ Stmt](#) Suppose the structure coefficients admit a trace-power presentation by positive diagonal matrices. They are length independent if and only if, for every triple of labels, $c_{\alpha, \beta, \gamma}^{(2)} = c_{\alpha, \beta, \gamma}^{(1)}$ and $c_{\alpha, \beta, \gamma}^{(3)} = c_{\alpha, \beta, \gamma}^{(1)}$.

Proof. [✓ Proof](#) Length independence implies both equalities. Conversely, apply Theorem 21.2.2.28 with $n = 1$ to each diagonal matrix. All its entries equal one, so Theorem 21.2.2.27 gives length independence. $\square$

Remark 21.2.2.30 (Topological order and length dependence). Starting from the same-length product law and the condition that the vector m is an idempotent for the multiplication induced by $c^{(1)}$, in the particular case of natural-number structure coefficients independent of the length, one recovers all known non-chiral topologically ordered phases together with a description of their excitations [7]. The RFP density operators correspond to the boundary theories of the RFP representatives of those phases, which yields a large family of non-trivial examples. The source asks whether there exist RFPs whose structure coefficients depend on the length [11, Section 4.5]. The example below gives such coefficients for its displayed BNT presentation.

Theorem 21.2.2.31 (An RFP with length-dependent BNT structure coefficients). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) *There is a tensor M in canonical form that generates matrix product density operators, is a renormalization fixed point, and has a vertical basis-of-normal-tensors presentation with one element and structure coefficient $c^{(L)} = (\sqrt{1/2})^L$. In particular, $c^{(1)} \neq c^{(2)}$, so these structure coefficients for the exhibited presentation depend on the length.*

Proof. [✓ Proof](#) The bond-dimension-two tensor with a one-element vertical basis of normal tensors generates positive operators on every nonempty chain, is in canonical form, and satisfies the displayed multiplication law and the corresponding tensor decomposition. The reverse implication of Theorem 21.2.6.3 supplies the two trace-preserving completely positive maps required for a renormalization fixed point. Finally, $\sqrt{1/2} \neq 1/2 = (\sqrt{1/2})^2$, which proves the length dependence. $\square$

Theorem 21.2.2.32 (Length dependence under every uniform positive rescaling). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Theorem 21.2.2.31 gives length-dependent structure coefficients for a specified BNT presentation, as asked in [11, lines 995–999]. Its diagonal matrix χ has the single entry $\sqrt{1/2}$, and rescaling that entry to 1 makes the coefficient constant. The diagonal matrix constructed here has two distinct positive entries, and its trace powers remain length dependent under every uniform positive rescaling. Define four matrices in $M_2(\mathbb{C})$ by*

$$A^0 = \frac{4}{5}E_{00}, \quad A^1 = \frac{4}{5}E_{11}, \quad A^2 = \frac{3}{5}E_{01}, \quad A^3 = \frac{3}{5}E_{10}.$$

Identify the physical index set $\{0, 1, 2, 3\}$ with $\{0, 1\} \times \{0, 1\}$ and let R be the bond-dimension-four MPO tensor obtained by exchanging physical and virtual indices in the matrices $B^{ab} = A^a \otimes \overline{A^b}$ and applying the indicated normalization:

$$R_{ab}^{pq} = \frac{25}{32} \left(A^a \otimes \overline{A^b} \right)_{p,q}.$$

Here (p, q) are the physical indices and (a, b) are the bond indices. Its physical-trace transfer is the rank-one projection

$$\mathcal{T}_R = \frac{25}{32} |t\rangle\langle t|, \quad t = \left(\frac{4}{5}, \frac{4}{5}, 0, 0 \right),$$

where $t_a = \text{tr}(A^a)$. It satisfies $\mathcal{T}_R^2 = \mathcal{T}_R$, the zero-correlation-length condition of [11, Definition 4.2, lines 736–741]. Define the positive diagonal matrix $\chi = \text{diag}(1, \frac{7}{25})$ and its trace powers

by

$$c^{(L)} = \text{tr}(\chi^L) = 1 + \left(\frac{7}{25}\right)^L.$$

Hence $c^{(1)} \neq c^{(2)}$, so these coefficients are not length independent. For every real number $s > 0$, the rescaled matrix $s\chi = \text{diag}(s, \frac{7s}{25})$ has trace powers

$$c_s^{(L)} = \text{tr}((s\chi)^L) = s^L \left(1 + \left(\frac{7}{25}\right)^L\right),$$

which are again not length independent. Unlike the single-entry matrix of Theorem 21.2.2.31, no uniform positive rescaling makes both diagonal entries equal to one.

The unscaled coefficients are the structure coefficients of an explicit one-element vertical basis-of-normal-tensors presentation of R , motivated by the discussion in [11, lines 995–1010].

Proof. ✓ Proof The (a, b) entry of $\mathcal{T}_R$ is $\frac{25}{32} \text{tr}(A^a) \text{tr}(A^b)$, and $\sum_a \text{tr}(A^a)^2 = 2 \cdot (\frac{4}{5})^2 = \frac{32}{25}$, so $\mathcal{T}_R^2 = \mathcal{T}_R$. The formulas for $c^{(L)}$ and $c_s^{(L)}$ follow by summing the powers of the diagonal entries of χ and $s\chi$. Write $\lambda = \frac{7}{25}$. If c_s were length independent, comparing lengths one and two and lengths two and three would give

$$\begin{aligned} s &= \frac{1 + \lambda}{1 + \lambda^2}, \\ s &= \frac{1 + \lambda^2}{1 + \lambda^3}. \end{aligned}$$

Hence $(1 + \lambda)(1 + \lambda^3) = (1 + \lambda^2)^2$, that is, $\lambda(1 - \lambda)^2 = 0$. This fails at $\lambda = \frac{7}{25}$. □

Definition 21.2.2.33 (The $\mathbb{Z}_2$ -twisted quantum dimer). Lean Lean Lean Lean Lean Lean

Lean ✓ Stmt Each site carries two qubits L, R and a flag qubit f , so the physical index set is $\{0, 1\}^3$, read as (l, r, f) . The bond index set is also $\{0, 1\}^3$, read as (p, p', k) , where k labels one of two horizontal blocks. Let X and Z denote the Pauli matrices. With $x = \frac{7}{8}$, $y = \frac{1}{8}$, $P_{\pm} = \frac{1}{2}(\mathbb{1} \pm X)$, and the 2×2 matrices $C_0 = xP_+ + yP_-$ and $C_1 = xP_+ - yP_-$, the bond matrix for the pair of physical indices $((l, r, f), (l', r', f'))$ is

$$M^{(l, r, f), (l', r', f')} = \sum_{k \in \{0, 1\}} \frac{1}{2} C_k[l, l'] (\tau_k)_{ff} \delta_{ff'} E_{(l, l', k), (r, r', k)}.$$

Here $\tau_0 = \mathbb{1}$ and $\tau_1 = Z$. Thus each bond matrix is a sum of two matrix units, one in each block. The tensor is the direct sum along k of the two bond-dimension-four blocks obtained by keeping one summand. This example is motivated by the question of [11, lines 995–1010]; it is not stated there.

Theorem 21.2.2.34 (Closures and physical-trace transfer of the twisted dimer). Lean Lean

Lean Lean Lean ✓ Stmt For a bond matrix X , the one-site physical closure $M_1(X)$ of the twisted dimer has entries

$$[M_1(X)]_{(l, r, f), (l', r', f')} = \sum_k \frac{1}{2} C_k[l, l'] (\tau_k)_{ff} \delta_{ff'} X_{(r, r', k), (l, l', k)}.$$

The two-site closure $M_2(X)$ vanishes unless the consecutive physical indices satisfy $r_1 = l_2$ and $r'_1 = l'_2$. When these conditions hold, its entry is the sum over a common block label k of the

product of the two coefficients times $X_{(r_2, r'_2, k), (l_1, l'_1, k)}$. The physical-trace transfer of the block $k = 0$ is $\sum_{l, r} C_0[l, l] E_{(l, l), (r, r)}$, and that of the block $k = 1$ vanishes because $\text{tr } \tau_1 = 0$. Thus the latter block has nilpotent physical-trace transfer in the sense used by [11, Definition 4.7, lines 815–822].

Proof. ✓ Proof The trace of a matrix unit against X picks one entry of X , and the product of two matrix units is a matrix unit or zero according to whether the inner indices agree. Since the block label is part of the bond index, only common labels survive. Summing the diagonal physical indices over the flag value gives $\sum_f (\tau_k)_{ff}$, which is 2 for $k = 0$ and 0 for $k = 1$. $\square$

Theorem 21.2.2.35 (Positivity of the twisted quantum dimer). Lean Lean Lean ✓ Stmt The closed operator $\rho^{(N)}(M)$ of the twisted quantum dimer is positive semidefinite for every positive chain length N , so M generates MPDOs. Writing l_n, r_n, f_n for the three bits at site n of a physical string σ , and using primes for those of τ , the matrix element is

$$\rho^{(N)}(M)_{\sigma\tau} = \frac{1}{2^N} \mathbb{1}[\text{cyclic bond match}](\sigma, \tau) \delta_{ff'} \left(\prod_n C_0[l_n, l'_n] + (-1)^{|f|} \prod_n C_1[l_n, l'_n] \right).$$

The indicator is one exactly when both σ and τ match their bond bits around the ring, $\delta_{ff'}$ is one exactly when the two flag strings agree, and $|f|$ is the number of sites carrying flag one. Thus $\rho^{(N)}(M)$ is 2^{-N} times the Hadamard product of the rank-one bond-matching indicator with the pullback, along the flag and left bits, of the sum of the N -fold Kronecker powers of the two local four-by-four factors $g_k[(f, l), (f', l')] = \delta_{ff'} C_k[l, l'] (\tau_k)_{ff}$. The Schur product theorem then gives positivity.

Proof. ✓ Proof Each bond matrix is a sum of two matrix units, one per block label, so an ordered product keeps a common block label and vanishes unless consecutive physical indices match on their bond bits. The surviving product is a single matrix unit whose row is determined by the left bits at the first site and whose column is determined by the right bits at the last site. Taking the trace imposes the wraparound match and leaves the sum over the block label of the product of the one-site coefficients. Each coefficient is half of g_k evaluated at the flag and left bits, so the product along the ring is 2^{-N} times an entry of the N -fold Kronecker power of g_k .

Neither the sum nor the difference of the two Kronecker powers is a Kronecker power, so both are proved positive semidefinite together by induction on the chain length. Separating the last tensor factor gives

$$g_0^{\otimes(N+1)} \pm g_1^{\otimes(N+1)} = \frac{1}{2} \left[(g_0^{\otimes N} + g_1^{\otimes N}) \otimes (g_0 \pm g_1) + (g_0^{\otimes N} - g_1^{\otimes N}) \otimes (g_0 \mp g_1) \right].$$

Both local combinations are positive semidefinite: each is block diagonal in the flag bit, with $C_0 + C_1 = \frac{7}{4}P_+$ in one flag sector and $C_0 - C_1 = \frac{1}{4}P_-$ in the other, the two sectors being exchanged between the sum and the difference. A Kronecker product of positive semidefinite matrices is positive semidefinite, and so are sums, nonnegative multiples, and pullbacks along an arbitrary reindexing. $\square$

Definition 21.2.2.36 (Controlled Bell-sign flip). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt On two bond qubits let

$$|\Phi_{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}},$$

$$\Pi_{\pm} = |\Phi_{\pm}\rangle\langle\Phi_{\pm}|.$$

Define $\sigma = \frac{7}{8}\Pi_+ + \frac{1}{8}\Pi_-$ and $\sigma' = \frac{7}{8}\Pi_+ - \frac{1}{8}\Pi_-$. On a flag qubit let $X|f\rangle = |f+1\rangle$, with addition modulo two, and $Z|f\rangle = (-1)^f|f\rangle$. The controlled Bell-sign flip is

$$V = (\mathbb{1} - \Pi_-) \otimes \mathbb{1} + \Pi_- \otimes X.$$

Lemma 21.2.2.37 (Bond-state matrix entries). [Lean](#) [✓ Stmt](#) For $k \in \{0, 1\}$, put $\sigma_k = \frac{7}{8}\Pi_+ + (-1)^k \frac{1}{8}\Pi_-$, where $\Pi_\pm$ are the Bell projections of Definition 21.2.2.36. Let C_k be the matrix of Definition 21.2.2.33. For $r, l, r', l' \in \{0, 1\}$, in right-then-left bond order,

$$\langle r, l | \sigma_k | r', l' \rangle = \begin{cases} C_k[l, l'] & \text{if } r = l \text{ and } r' = l', \\ 0 & \text{otherwise.} \end{cases}$$

Proof. [✓ Proof](#) Expand the two Bell projections in the computational basis. $\square$

Lemma 21.2.2.38 (Local unitary identities). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The operator V of Definition 21.2.2.36 is self-adjoint and satisfies $V^2 = \mathbb{1}$ on the full bond-flag space. It obeys

$$\begin{aligned} V(\sigma \otimes \mathbb{1})V^\dagger &= \sigma \otimes \mathbb{1}, \\ V(\sigma \otimes Z)V^\dagger &= \sigma' \otimes Z. \end{aligned}$$

Proof. [✓ Proof](#) Evaluate the 8×8 matrices entry by entry in the computational basis. This gives $V^\dagger = V$, $V^2 = \mathbb{1}$, $V(\sigma \otimes \mathbb{1}) = (\sigma \otimes \mathbb{1})V$, and $V(\sigma \otimes Z) = (\sigma' \otimes Z)V$. Multiplying each intertwining identity on the right by $V^\dagger$ gives the stated conjugation identities. $\square$

Definition 21.2.2.39 (Bond-flag regrouping). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $N \geq 0$, write each physical configuration as $s = ((l_m, r_m, f_m))_m$, with cyclic indices when $N > 0$. At $N = 0$ the configurations are empty. The incoming cells are $e_N(s)_m = ((r_{m-1}, l_m), f_m)$. For cells $a_m = (b_m, g_m)$, with $b_m \in \{0, 1\}^2$, the inverse reconstructs site m as $((b_m)_2, (b_{m+1})_1, g_m)$. Let $E_N|s\rangle = |e_N(s)\rangle$, and let S_N separate a cell configuration into its bond string and flag string. Define

$$\begin{aligned} \omega_N &= 2^{-N}(\mathbb{1}^{\otimes N} + Z^{\otimes N}), \\ D_N &= E_N^\dagger S_N^\dagger (\sigma^{\otimes N} \otimes \omega_N) S_N E_N, \\ U_N &= E_N^\dagger V^{\otimes N} E_N. \end{aligned}$$

For $N > 0$, ω_N is the uniform state on even-parity flag strings, and D_N is the product of independent bond states and that flag state, expressed in the physical site coordinates. At length zero, the empty-product convention gives $\omega_0 = D_0 = 2\mathbb{1}$ and $U_0 = \mathbb{1}$.

Theorem 21.2.2.40 (Unitary bond-flag factorization). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #7776](#) For every positive length N , the operator U_N of Definition 21.2.2.39 is self-adjoint and obeys $U_N U_N^\dagger = U_N^\dagger U_N = \mathbb{1}$. The closed operator of the twisted quantum dimer satisfies $\rho^{(N)}(M) = U_N D_N U_N^\dagger$.

Proof. [✓ Proof](#) Take tensor products of the identities in Lemma 21.2.2.38. After separating the bond and flag strings, we obtain

$$S_N E_N U_N D_N U_N^\dagger E_N^\dagger S_N^\dagger = 2^{-N} (\sigma^{\otimes N} \otimes \mathbb{1}^{\otimes N} + (\sigma')^{\otimes N} \otimes Z^{\otimes N}).$$

By Lemma 21.2.2.37, both bond operators vanish outside the span of $|00\rangle$ and $|11\rangle$; on this span their matrices are C_0 and C_1 , respectively. In physical coordinates the entries therefore equal the cyclic bond-matching formula of Theorem 21.2.2.35. $\square$

The gates in this factorization act on an incoming bond and its receiving flag across neighboring physical sites. After the unitary conjugation by U_N , the bond and flag factors decouple, yielding the state-level product form.

Theorem 21.2.2.41 (Identification of the flag factor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let $\widehat{M}$ be the density-normalized tensor of Theorem 20.16.8.61. For every $N \geq 0$, the flag operator satisfies $\omega_N = \rho^{(N)}(\widehat{M})$. In particular, the flag family has an MPDO representation satisfying both RFP channel equations. For $N > 0$, ω_N is positive semidefinite and has trace one. At $N = 0$, both ω_0 and $\rho^{(0)}(\widehat{M})$ are the scalar 2.

Proof. [✓ Proof](#) Substitute the identity and Pauli Z matrices in the definition of ω_N . Its expression $2^{-N}(\mathbb{1}^{\otimes N} + Z^{\otimes N})$ is the closed operator of $\widehat{M}$ by Theorem 20.16.8.53 and the factor 2^{-N} ; the empty case is evaluated directly. Positivity, normalization, and the two channel equations follow from Theorem 20.16.8.61. $\square$

Theorem 21.2.2.42 (Normalization of the bond–flag factorization). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The bond operator σ is positive semidefinite and has trace one. Every tensor power $\sigma^{\otimes N}$ has trace one, including $N = 0$. For every $N > 0$, D_N is positive semidefinite and $\text{tr } D_N = \text{tr } \rho^{(N)}(M) = 1$.

Proof. [✓ Proof](#) The Bell projectors are positive rank-one operators. By Lemma 21.2.2.37, $\text{tr } \sigma = C_0[0,0] + C_0[1,1] = 1$. Tensor products preserve positivity, and their traces multiply. Unitary changes of coordinates preserve both positivity and trace. Apply these facts to D_N and then use Theorem 21.2.2.40. $\square$

The flag factor is identified using the density-normalized tensor $\widehat{M}$; the constant multiplication coefficients for the unnormalized matrices $\mathbb{1}, Z$ differ by scaling from those for the spectrally normalized representatives $\mathbb{1}/\sqrt{2}, Z/\sqrt{2}$.

Definition 21.2.2.43 (The mixed-Bell bond dimer). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let

$$C = \begin{pmatrix} 1/2 & 3/8 \\ 3/8 & 1/2 \end{pmatrix}.$$

A physical site consists of qubits $(l, r) \in \{0, 1\}^2$. With virtual indices in $\{0, 1\}^2$, define the tensor Q by

$$Q^{(l,r),(l',r')} = C_{ll'} E_{(l,l'),(r,r')},$$

where $E_{a,b}$ is the matrix unit with row a and column b . Both physical and virtual dimensions are four. For $N \geq 0$, let B_N be the permutation of basis configurations given by

$$B_N |((l_m, r_m))_m\rangle = |((r_{m-1}, l_m))_m\rangle,$$

with cyclic indices when $N > 0$ and the unique empty configuration when $N = 0$. Its inverse sends a bond configuration $(b_m)_m$ to the site configuration $((b_m)_2, (b_{m+1})_1)_m$.

Theorem 21.2.2.44 (Closed operators of the mixed-Bell dimer). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $N > 0$, the tensor Q of Definition 21.2.2.43 satisfies

$$\rho^{(N)}(Q) = B_N^\dagger \sigma^{\otimes N} B_N.$$

Diagrammatically, the bond state σ joins the right qubit of each site to the left qubit of the next one:

$$\begin{array}{c}
\begin{array}{c} l_1 \quad r_1 \quad l_2 \quad r_2 \quad \dots \quad l_N \quad r_N \\
\boxed{Q} \text{---} \boxed{Q} \text{---} \dots \text{---} \boxed{Q} \\
\text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \end{array} \\
= \begin{array}{c} r_N \quad l_1 \quad r_1 \quad l_2 \quad \dots \quad r_{N-1} \quad l_N \\
\boxed{\sigma} \text{---} \boxed{\sigma} \text{---} \dots \text{---} \boxed{\sigma} \\
\text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \text{---} \end{array}
\end{array}$$

This operator is positive semidefinite and has trace one. In particular, Q generates MPDOs. The identity excludes $N = 0$, where the empty closed operator is 4 and the empty bond product is 1.

Proof. ✓ Proof Fix independent configurations $s_m = (l_m, r_m)$ and $t_m = (l'_m, r'_m)$, with $m = 0, \dots, N-1$ and cyclic successors. The cyclic trace expansion of Lemma 21.2.2.56 has only one possibly nonzero virtual path, namely $(l_m, l'_m)_m$. Hence

$$\text{tr} \left(\prod_{m=0}^{N-1} Q^{(l_m, r_m), (l'_m, r'_m)} \right) = \begin{cases} \prod_{m=0}^{N-1} C_{l_m l'_m} & \text{if } r_m = l_{m+1} \text{ and } r'_m = l'_{m+1} \text{ for every } m, \\ 0 & \text{otherwise.} \end{cases} \quad (21.6)$$

By Lemma 21.2.2.37, the (s, t) entry of $B_N^\dagger \sigma^{\otimes N} B_N$ is $\prod_m \sigma_{(r_{m-1}, l_m), (r'_{m-1}, l'_m)}$. Its matching conditions and value agree with (21.6) after cyclic relabeling. Positivity and trace one follow from Theorem 21.2.2.42. $\square$

Definition 21.2.2.45 (The normalized mixed-Bell representative). Lean Lean Lean ✓ Stmt For the mixed-Bell tensor Q , put $c = 4\sqrt{2}/5$ and $A = cQ$, viewed as a tensor over the sixteen-letter ket-bra alphabet.

Theorem 21.2.2.46 (One-site injectivity of the mixed-Bell factor). Lean Lean ✓ Stmt The sixteen matrices Q^{ij} span $M_4(\mathbb{C})$ and satisfy $\sum_{i,j} (Q^{ij})^\dagger Q^{ij} = (25/32)\mathbb{1}$.

Proof. ✓ Proof Each matrix unit $E_{(l, l'), (r, r')}$ is a nonzero scalar multiple of $Q^{(l, r), (l', r')}$, since every entry of C is nonzero. Summing the adjoint squares gives the stated scalar identity: the coefficient at each diagonal matrix unit is $2(1/2)^2 + 2(3/8)^2 = 25/32$. $\square$

Lemma 21.2.2.47 (Normality and trace of the mixed-Bell representative). Lean Lean Lean ✓ Stmt The representative A is left-canonical and normal. Its physical-trace transfer $\mathcal{B}_A = \sum_i A^{ii}$ satisfies $\text{tr } \mathcal{B}_A = c \neq 0$.

Proof. ✓ Proof The adjoint-square sum of A is $c^2(25/32)\mathbb{1} = \mathbb{1}$. Nonzero scalar multiplication preserves injectivity, so Theorem 7.6.3 gives normality. The one-site density operator of Q has trace one; hence $\text{tr } \mathcal{B}_A = c \text{tr } \mathcal{B}_Q = c$. $\square$

Theorem 21.2.2.48 (Simplicity of the exact mixed-Bell factor). Lean ✓ Stmt The tensor Q is simple in the sense of [11, Definition 4.7].

Proof. ✓ Proof After one-site blocking, relabel the ket-bra alphabet and use the single representative A , with one copy of weight c^{-1} . The scalar word law recovers the positive-length matrix product vectors of Q . The singleton normal family is eventually linearly independent by Theorem 8.2.13. A nilpotent complex matrix has zero trace, whereas $\text{tr } \mathcal{B}_A = c \neq 0$. Thus the only representative has nonnilpotent physical-trace transfer, as required. $\square$

Definition 21.2.2.49 (Bond insertion and removal). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let J be the basis permutation from an exterior site and an inserted bond to two sites:

$$J|(l, r), (u, v)\rangle = |(l, u), (v, r)\rangle.$$

Define the refinement and coarse-graining maps by

$$\begin{aligned}\mathcal{T}_Q(Y) &= J(Y \otimes \sigma)J^\dagger, \\ \mathcal{S}_Q(W) &= \text{tr}_{\text{bond}}(J^\dagger W J).\end{aligned}$$

The partial trace is over the inserted bond, leaving the two exterior qubits as a single physical site.

Theorem 21.2.2.50 (Channel equations for the mixed-Bell dimer). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The maps $\mathcal{T}_Q$ and $\mathcal{S}_Q$ of Definition 21.2.2.49 are trace-preserving completely positive maps. For every one-site operator Y , $\mathcal{S}_Q(\mathcal{T}_Q(Y)) = Y$. For every virtual boundary matrix $X \in M_4(\mathbb{C})$, the one-site and two-site physical closures satisfy

$$\begin{aligned}\mathcal{T}_Q(Q_1(X)) &= Q_2(X), \\ \mathcal{S}_Q(Q_2(X)) &= Q_1(X).\end{aligned}$$

Thus Q is a renormalization fixed point in the sense of Definition 20.2.1.

Proof. [✓ Proof](#) Preparation of the positive trace-one state σ , permutation of basis coordinates, and partial trace are trace-preserving completely positive maps. Tracing out the prepared state proves $\mathcal{S}_Q(\mathcal{T}_Q(Y)) = Y$. For the refinement equation, fix independent ket and bra indices $l, r, l', r', u, v, u', v' \in \{0, 1\}$. Multiplication of matrix units gives

$$Q^{(l,r),(l',r')} Q^{(u,v),(u',v')} = \begin{cases} C_{ll'} C_{uu'} E_{(l,l'),(v,v')} & \text{if } r = u \text{ and } r' = u', \\ 0 & \text{otherwise.} \end{cases}$$

For every boundary matrix $X \in M_4(\mathbb{C})$, the one-site closure has entries $[Q_1(X)]_{(l,r),(l',r')} = C_{ll'} X_{(r,r'),(l,l')}$. Put $a = ((l, r), (u, v))$ and $b = ((l', r'), (u', v'))$. The definition of the insertion map and Lemma 21.2.2.37 give

$$\begin{aligned}[\mathcal{T}_Q(Q_1(X))]_{a,b} &= [Q_1(X)]_{(l,v),(l',v')} \sigma_{(r,u),(r',u')} \\ &= \begin{cases} C_{ll'} C_{uu'} X_{(v,v'),(l,l')} & \text{if } r = u \text{ and } r' = u', \\ 0 & \text{otherwise} \end{cases} \\ &= [Q_2(X)]_{a,b}.\end{aligned}$$

Thus the refinement equation holds for arbitrary boundary matrices. Applying $\mathcal{S}_Q \mathcal{T}_Q = \text{id}$ gives the coarse-graining equation. $\square$

These channel equations establish that the mixed-Bell bond dimer is a renormalization fixed point, with nonzero bond-state eigenvalues $7/8$ and $1/8$.

Definition 21.2.2.51 (On-site separation of bond qubits and flags). [Lean](#) [✓ Stmt](#) For N sites, let H_N be the basis permutation separating each site's two bond qubits from its flag:

$$H_N |((l_m, r_m, f_m))_m\rangle = |((l_m, r_m))_m\rangle \otimes |(f_m)_m\rangle.$$

Unlike the incoming-bond regrouping, this separation does not move any qubit between physical sites.

Theorem 21.2.2.52 (Factorization into two RFP families). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let Q be the mixed-Bell bond tensor and let $\widehat{M}$ be the density-normalized tensor identified in Theorem 21.2.2.41. For every $N > 0$,

$$D_N = H_N^\dagger(\rho^{(N)}(Q) \otimes \rho^{(N)}(\widehat{M}))H_N,$$

$$\rho^{(N)}(M) = U_N H_N^\dagger(\rho^{(N)}(Q) \otimes \rho^{(N)}(\widehat{M}))H_N U_N^\dagger.$$

Both factor tensors generate MPDOs and are RFPs, by Theorems 21.2.2.44, 21.2.2.50, and 21.2.2.41.

Proof. [Stmnt](#) The coordinate permutations satisfy $(B_N \otimes \mathbb{1})H_N = S_N E_N$: both read the incoming bonds (r_{m-1}, l_m) and the unchanged flags f_m . Substitute the bond-product formula and the flag identification into the definition of D_N . The formula for $\rho^{(N)}(M)$ then follows from Theorem 21.2.2.40. The RFP channel equations for the two factors were proved independently in the cited theorems. $\square$

The separation H_N is on-site, while the conjugating operator U_N couples neighboring sites; at positive chain length $N \geq 1$, this decouples the bond qubits from the site flags.

Theorem 21.2.2.53 (The twisted quantum dimer is a renormalization fixed point). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) The twisted quantum dimer of Definition 21.2.2.33 is a renormalization fixed point in the sense of Definition 20.2.1. The refinement map $\mathcal{T}$ is completely positive and trace preserving, with eight Kraus operators, labelled by the two prepared site flags f_1, f_2 and a bond label ε . It prepares those two flags on the output sites together with the bond state $(|00\rangle + (-1)^\varepsilon |11\rangle)/\sqrt{2}$ on their shared bond, with weight x for $\varepsilon = 0$ and y for $\varepsilon = 1$, and is supported on the incoming flag $f = f_1 + f_2 + \varepsilon$. Flag addition is modulo two. The coarse-graining map $\mathcal{S}$ is completely positive and trace preserving, with sixteen Kraus operators, labelled by a member of the orthonormal bond basis $(|00\rangle \pm |11\rangle)/\sqrt{2}, |01\rangle, |10\rangle$ together with the two site flags. It measures the shared bond pair in that basis, reads the two flags, and returns the outer site qubits with flag $f_1 + f_2 + s$, where s is one for the antisymmetric bond outcome and zero otherwise. These maps satisfy $\mathcal{T}[M_1(X)] = M_2(X)$ and $\mathcal{S}[M_2(X)] = M_1(X)$ for every bond matrix X . This example is motivated by the question of [11, lines 995–1010].

Proof. [Stmnt](#) Fix a one-site basis index (l, r, f) . The two-site indices in the support of a refinement Kraus operator carry the outer bits l and r , the two prepared flags, and a free shared bond bit. Each of the four labels with $f_1 + f_2 + \varepsilon = f$ therefore contributes twice the squared amplitude $w_\varepsilon/4$, where $w_0 = x$ and $w_1 = y$. Two of the four labels carry each value of ε , and the total is $x + y = 1$. Supports belonging to distinct one-site basis indices are disjoint, so the off-diagonal entries vanish and the Kraus operators resolve the identity. For the intertwining relation, the flag signs multiply as $(\tau_k)_{f_1 f_1} (\tau_k)_{f_2 f_2} = (\tau_k)_{gg}$ with $g = f_1 + f_2$. For shared bond bits b, b' , the two prepared bond states give

$$\frac{x}{4} + (-1)^{b+b'} \frac{y}{4} = \frac{1}{2} C_0[b, b'],$$

$$\frac{x}{4} - (-1)^{b+b'} \frac{y}{4} = \frac{1}{2} C_1[b, b'].$$

These are exactly the coefficients in the two-site closure of Theorem 21.2.2.34.

For the coarse-graining map, write $B_v(b_1, b_2)$ for the amplitude of the bond-basis vector v at (b_1, b_2) . Its orthonormality gives

$$\sum_v B_v(b_1, b_2) B_v(c_1, c_2) = \delta_{b_1 c_1} \delta_{b_2 c_2}.$$

The sum over the two site flags is

$$\sum_{f_1, f_2} \delta_{f_1 F_1} \delta_{f_2 F_2} \delta_{f_1 F'_1} \delta_{f_2 F'_2} = \delta_{F_1 F'_1} \delta_{F_2 F'_2}.$$

Orthonormality of the bond basis and the flag Kronecker deltas identify all six bits of the two-site indices and give the resolution of the identity. On the support of the two-site closure, the two bond qubits of each argument agree, so the outcomes $|01\rangle$ and $|10\rangle$ contribute nothing. The symmetric and antisymmetric Bell outcomes contribute $\frac{1}{2}(\pm 1)^{b+b'}$. Summing over the two flag pairs compatible with the output flag leaves, for each block label k , the identity

$$\frac{1}{2} \left(\sum_{b, b'} C_k[b, b'] + (\tau_k)_{11} \sum_{b, b'} (-1)^{b+b'} C_k[b, b'] \right) = 1.$$

The left side is $\frac{1}{2}(\frac{7}{4} + \frac{1}{4}) = 1$ in both cases: for $k = 0$ the alternating sum is $\frac{1}{4}$ and the flag sign is 1, and for $k = 1$ it is $-\frac{1}{4}$ and the flag sign is -1 . The remaining factor is the one-site coefficient $\frac{1}{2} C_k[l, l'] (\tau_k)_{ff} \delta_{ff'}$ of Theorem 21.2.2.34. $\square$

Theorem 21.2.2.54 (Two-label structure coefficients whose length dependence survives positive rescaling). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let the labels be $\{0, 1\}$, with addition modulo two, and define positive diagonal matrices of size one, identified with their entries, by

$$\begin{aligned}\chi_{ff', f+f'} &= \alpha = \frac{7}{10}, \\ \chi_{ff', f+f'+1} &= \beta = \frac{1}{10}.\end{aligned}$$

Then $c_{ff'g}^{(L)} = \text{tr}(\chi_{ff'g}^L)$ equals α^L when $g = f + f'$ and β^L otherwise. These coefficients are not length independent. For every pair of positive rescalings (s_0, s_1) , replacing $\chi_{ff'g}$ by $(s_f s_{f'} / s_g) \chi_{ff'g}$ still gives coefficients that are not length independent. Unlike the one-label families of Theorems 21.2.2.31 and 21.2.2.32, the obstruction here is not a diagonal matrix with two distinct entries but the inconsistency of the equations $s_f s_{f'} \chi_{ff'g} = s_g$ over the eight triples of labels. This statement concerns the coefficients alone. Theorem 21.2.2.58 identifies them as the structure coefficients for the closed operators of the flag sectors of the twisted quantum dimer.

Proof. ✔ Proof If the powers of a positive real number t agree at lengths one and two, then $t = t^2$, hence $t = 1$. The triples $(0, 0, 0)$ and $(0, 1, 0)$ give $s_0\alpha = 1$ and $s_1\beta = 1$; the triple $(0, 0, 1)$ then gives $s_0^2\beta/s_1 = \beta^2/\alpha^2 = 1$, contradicting $\alpha \neq \beta$. $\square$

Definition 21.2.2.55 (Flag sectors of the twisted quantum dimer). The matrices of the twisted quantum dimer of Definition 21.2.2.33 are diagonal in the flag qubit. Viewed vertically, the tensor therefore splits along the flag value $f \in \{0, 1\}$ into two tensors M_f acting on the two-qubit space of L and R , with matrices indexed by pairs of horizontal bond indices. Write $\mu = \frac{5}{8}$, so that $\mu^2 = (x^2 + y^2)/2$ at $x = \frac{7}{8}$, $y = \frac{1}{8}$. The rescaled sector $\widehat{M}_f = M_f/\mu$ has, at the bond pair $((p, p'), k), (q, q', k')$, the matrix

$$\widehat{M}_f^{(p,p'),(q,q',k')} = [k = k'] \frac{1}{2\mu} C_k[p, p'] (\tau_k)_{ff} E_{pp'} \otimes E_{qq'},$$

where $[k = k']$ is one when $k = k'$ and zero otherwise. The matrix unit has row (p, q) and column (p', q') . The entry of this matrix between the qubit configurations (l, r) and (l', r') is μ^{-1} times

the bond-matrix entry of the twisted dimer at the physical indices $((l, r, f), (l', r', f))$ carrying the common flag value f . The horizontal blocks of Definition 21.2.2.33 and the two vertical flag sectors are all matrix-unit tensors of this form, differing only in their one-site coefficients. The closed operators $O_L(\widehat{M}_f)$ on a ring of L sites give the operators indexed by these two labels in Definition 21.2.2.2. Their basis-of-normal-tensors properties are proved below, rather than assumed in this definition. This tensor is motivated by the question of [11, lines 995–1010].

Lemma 21.2.2.56 (Cyclic path expansion of a trace). Lean Lean Lean Stmt *Let $M_0, \dots, M_{L-1}$ be square matrices of the same finite size over a commutative semiring, with $L \geq 1$. Applying the ordered product $M_0 \cdots M_{L-1}$ to a vector v and reading the entry at a sums, over all paths $(t_0, \dots, t_L)$ with $t_0 = a$, the product of the entries $(M_n)_{t_n t_{n+1}}$ along the path times v_{t_L} . Consequently,*

$$\text{tr}(M_0 \cdots M_{L-1}) = \sum_{t_0, \dots, t_{L-1}} \prod_{n=0}^{L-1} (M_n)_{t_n t_{n+1}},$$

with the cyclic convention $t_L = t_0$.

Proof. Proof The open expansion is an induction on L : splitting a path into its first step and the remaining path gives $(M_0(M_1 \cdots M_{L-1})v)_a = \sum_c (M_0)_{ac}((M_1 \cdots M_{L-1})v)_c$. For the trace, separate the last factor M_{L-1} and apply the open expansion to the vector $c \mapsto (M_{L-1})_{ca}$, using the empty product when $L = 1$. Splitting a cyclic assignment into its first entry a and the remaining entries identifies the two sides term by term, with the last factor closing the cycle back to a . □

Lemma 21.2.2.57 (Closed operators of the flag sectors). Lean Lean Lean Lean Lean Stmt *For a matrix-unit tensor with one-site coefficients c , the ordered product along a nonempty pair of words vanishes unless both words satisfy the open bond-matching condition. When they do, the product is the matrix unit whose row is given by the left bits of the first indices and whose column is given by the right bits of the last indices, multiplied by the product of the coefficients. On a ring of positive length the closed operator therefore has entries $\mathbb{1}[\text{cyclic bond match}](\sigma, \tau) \prod_n c(\sigma_n, \tau_n)$, where the first factor is the indicator that both words satisfy the cyclic bond-matching condition. In particular, for a positive length L and words $\sigma_n = (p_n, p'_n, k_n)$, $\tau_n = (q_n, q'_n, k'_n)$ of horizontal bond indices,*

$$O_L(\widehat{M}_f)_{\sigma\tau} = \mathbb{1}[\text{cyclic bond match}](\sigma, \tau) \prod_n [k_n = k'_n] \prod_n \frac{C_{k_n}[p_n, p'_n](\tau_{k_n})_{ff}}{2\mu}.$$

Proof. Proof A matrix unit multiplied on the left by a tensor matrix is a matrix unit or zero according to whether the right bits of that matrix match the row of the unit. The open-chain formula follows by induction on the length of the words, as for the twisted dimer itself in Theorem 21.2.2.35. Taking the trace imposes the cyclic bond match. For a flag sector, the coefficient at a bond pair is the one-site weight of the first index when the block labels agree and zero otherwise. Thus the product of coefficients along the ring is the product of the weights of the first word if the two label strings agree, and zero otherwise. □

Theorem 21.2.2.58 (Product law for the closed operators of the flag sectors). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Stmt *For every positive length L and all flag values $f, f' \in \{0, 1\}$, the closed operators of the two rescaled flag sectors satisfy*

$$O_L(\widehat{M}_f) O_L(\widehat{M}_{f'}) = \alpha^L O_L(\widehat{M}_{f+f'}) + \beta^L O_L(\widehat{M}_{f+f'+1}),$$

Drawing each sector with its matrix index horizontal, as in the diagram for Definition 21.2.2.3, the product law reads, with $g = f + f'$ and $\bar{g} = f + f' + 1$,

$$\begin{array}{c}
\begin{array}{c}
\begin{array}{c} a_1 \quad a_2 \quad \dots \quad a_L \\ \boxed{\widehat{M}_f} \quad \boxed{\widehat{M}_f} \quad \dots \quad \boxed{\widehat{M}_f} \\ \downarrow \quad \downarrow \quad \dots \quad \downarrow \\ \boxed{\widehat{M}_{f'}} \quad \boxed{\widehat{M}_{f'}} \quad \dots \quad \boxed{\widehat{M}_{f'}} \\ b_1 \quad b_2 \quad \dots \quad b_L \end{array}
\end{array} \\
= \alpha^L \begin{array}{c} a_1 \quad a_2 \quad \dots \quad a_L \\ \boxed{\widehat{M}_g} \quad \boxed{\widehat{M}_g} \quad \dots \quad \boxed{\widehat{M}_g} \\ b_1 \quad b_2 \quad \dots \quad b_L \end{array} + \beta^L \begin{array}{c} a_1 \quad a_2 \quad \dots \quad a_L \\ \boxed{\widehat{M}_{\bar{g}}} \quad \boxed{\widehat{M}_{\bar{g}}} \quad \dots \quad \boxed{\widehat{M}_{\bar{g}}} \\ b_1 \quad b_2 \quad \dots \quad b_L \end{array}
\end{array}$$

where $\alpha = \frac{7}{10}$, $\beta = \frac{1}{10}$, and the flag values are added modulo two. Thus the operators of Definition 21.2.2.55 satisfy the product law of Definition 21.2.2.3, in the form of [11, Theorem 4.14(ii), lines 972–985], with the structure coefficients of Theorem 21.2.2.54. These coefficients depend on L , and no positive rescaling of the two sectors makes them length independent. This product law alone does not identify the sectors as a basis of normal tensors; that property is established below.

Proof.  **Proof** Fix cyclically bond-matched words σ, τ with the common block labels k_n and left bits p_n of σ ; all other entries of both sides vanish by Lemma 21.2.2.57. The entry of the product is a sum over a middle word ρ , which must be cyclically bond matched with the same labels k_n , hence of the form $\rho_n = (t_n, t_{n+1}, k_n)$ for a bit string t . Its contribution factors as the weight of σ in the sector f times $(2\mu)^{-L} \prod_n (\tau_{k_n})_{f'f'} \prod_n C_{k_n}[t_n, t_{n+1}]$. By Lemma 21.2.2.56, the sum over t of the last product is $\text{tr}(C_{k_0} \cdots C_{k_{L-1}})$. Since $C_k = xP_+ + (-1)^k yP_-$ with orthogonal rank-one idempotents $P_\pm$, the ordered product is $x^L P_+ + (-1)^{\sum_n k_n} y^L P_-$, with trace $x^L + (-1)^{\sum_n k_n} y^L$; this is verified entrywise through the sign expansion $C_k[p, q] = \frac{1}{2}(x + (-1)^{k+p+q}y)$. The flag signs multiply as $(\tau_k)_{ff}(\tau_k)_{f'f'} = (\tau_k)_{gg}$ with $g = f + f'$, and the sign $(-1)^{\sum_n k_n} = \prod_n (\tau_{k_n})_{11}$ of the second summand shifts the label to $g + 1$. With $x/(2\mu) = \frac{7}{10}$ and $y/(2\mu) = \frac{1}{10}$, the two summands are α^L and β^L times the entries of $O_L(\widehat{M}_{f+f'})$ and $O_L(\widehat{M}_{f+f'+1})$. These are precisely the coefficients for $g = f + f'$ and $g = f + f' + 1$ in Theorem 21.2.2.54. $\square$

Theorem 21.2.2.59 (Horizontal canonical form of the twisted quantum dimer). [\[Lean\]](#) [\[Lean\]](#)
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Let B_0, B_1 denote the two horizontal blocks of the twisted quantum dimer M of Definition 21.2.2.33, read as tensors with the doubled physical index $((l, r, f), (l', r', f'))$ and bond dimension four, and let $\widehat{B}_k = \frac{8}{5} B_k$. The matrices of each $\widehat{B}_k$ span the full algebra of 4×4 matrices, and $\sum_{i,j} (\widehat{B}_k^{ij})^\dagger \widehat{B}_k^{ij} = \mathbb{1}$. Hence each $\widehat{B}_k$ is a normal tensor with spectral radius one. The doubled-index tensor is in canonical form in the sense of Definition 7.6.7: with the permutation U of the bond index (p, p', k) into the block label k followed by the block bond index (p, p') ,

$$M^{ij} = U^\dagger \left(\frac{5}{8} \widehat{B}_0^{ij} \oplus \frac{5}{8} \widehat{B}_1^{ij} \right) U.$$

The physical-trace transfer of $\widehat{B}_1$ vanishes.

Proof. ✓ Proof The matrix of B_k at the physical pair $((p, q, 0), (p', q', 0))$ is the matrix unit at the bond pair $((p, p'), (q, q'))$ with the nonzero coefficient $\frac{1}{2}C_k[p, p']$, so these matrices span the bond algebra. The product $(\widehat{B}_k^{ij})^\dagger \widehat{B}_k^{ij}$ is the diagonal matrix unit at the right bits (r, r') of i and j , with coefficient $\frac{64}{25}$ times the squared coefficient of the tensor matrix. Grouping by the right bits, the coefficients at fixed (r, r') sum to $\frac{64}{25} \cdot 2 \cdot \frac{1}{4}(2C_k[0, 0]^2 + 2C_k[0, 1]^2) = 1$, so the sum of the products is the identity. Theorem 7.6.3 gives normality with spectral radius one. The reconstruction is the block decomposition of Definition 21.2.2.33 entry by entry, with $\frac{5}{8} \cdot \frac{8}{5} = 1$ restoring the original blocks. The matrix U is a permutation matrix, hence a coisometry. The physical-trace transfer of $\widehat{B}_1$ is $\frac{8}{5}$ times that of B_1 , which vanishes by Theorem 21.2.2.34. $\square$

Lemma 21.2.2.60 (Blocking raises the physical-trace transfer to a power). [Lean] [Stm] For every tensor M and every blocking length L , the physical-trace transfer of the L -site blocking of M is the L -th power of the physical-trace transfer of M .

Proof. ✓ Proof The physical-trace transfer of the blocking is the sum over all length- L configurations of the diagonal word evaluations, which is the L -th power of the one-site loop by Lemma 16.4.56. $\square$

Theorem 21.2.2.61 (The twisted quantum dimer is not simple). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
The twisted quantum dimer of Definition 21.2.2.33 is not simple in the sense of Definition 20.12.6: for every positive blocking length L and every basis-of-normal-tensors sector presentation of the doubled-index tensor of the L -site blocking, some representative has nilpotent physical-trace transfer.

Proof.  **Proof** Fix $L > 0$ and a presentation P . Blocking the canonical form of Theorem 21.2.2.59 by L sites (Theorem 7.6.15) and relabeling the doubled alphabet along the pairing of the blocked ket and bra words (Theorems 7.6.8 and 20.15.2) gives a canonical form of the blocked doubled-index tensor whose retained blocks are the L -site blockings of $\widehat{B}_0$ and $\widehat{B}_1$. Its basis-of-normal-tensors refinement (Theorem 8.6.1.9) is a basis-of-normal-tensors sector presentation Q (Theorem 8.6.1.12) in which every retained block is related to the representative of its phase class by a similarity transformation and a phase. The blocked $\widehat{B}_1$ has vanishing physical-trace transfer by Lemma 21.2.2.60 together with Lemma 20.12.2, so by Theorem 20.12.4 the representative of its phase class has nilpotent physical-trace transfer. Theorem 20.12.5 transfers this to P . $\square$

Theorem 21.2.2.62 (Vertical canonical form of the twisted quantum dimer). [Lean](#) [Lean](#) [Lean](#)

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The twisted quantum dimer M of Definition 21.2.2.33 has a vertical decomposition in the sense of Definition 16.4.6, with the two rescaled flag sectors $\widehat{M}_0, \widehat{M}_1$ of Definition 21.2.2.55, multiplicity one each, and weight $\mu = \frac{5}{8}$. Let U be the permutation matrix reordering the original physical index $(l, r, f) \mapsto (f, (l, r))$, so that U maps $\mathbb{C}^8$ onto the direct sum of the two flag subspaces, each isomorphic to $\mathbb{C}^4$. Writing $\widetilde{M}_{ab}$ for the matrices in the vertical direction, we have $UU^\dagger = U^\dagger U = \mathbb{1}$ and, for every horizontal bond pair (a, b) ,

$$\begin{aligned} U \widetilde{M}_{ab} U^\dagger &= \mu \widehat{M}_0^{ab} \oplus \mu \widehat{M}_1^{ab}, \\ \widetilde{M}_{ab} &= U^\dagger (\mu \widehat{M}_0^{ab} \oplus \mu \widehat{M}_1^{ab}) U. \end{aligned}$$

The first identity is the vertical decomposition drawn in [11, Proposition 4.13, lines 945–959]:

$$\begin{array}{c}
\boxed{U} \\
| \\
a - \boxed{\widetilde{M}} - b \\
| \\
\boxed{U^\dagger}
\end{array}
= \mu \begin{array}{c} a - \boxed{\widehat{M}_0} - b \end{array} \oplus \mu \begin{array}{c} a - \boxed{\widehat{M}_1} - b$$

Each flag sector is injective at one site and therefore proportional to a normal tensor with spectral radius one. At every positive length N , the matrix product vector of the vertically viewed tensor is μ^N times the sum of those of the two sectors, and the matrix product vectors of the two sectors are linearly independent. The fact that these sectors already have the normalization required of normal tensors in the CPSV sense is verified below.

Proof. ✓ Proof The matrix of the twisted dimer between two physical indices with different flag values vanishes, because its coefficient contains $\delta_{ff'}$. Between two indices with common flag value f , it is μ times the matrix of $\widehat{M}_f$ by Definition 21.2.2.55. Conjugating by the permutation matrix therefore places the two sectors on the diagonal blocks, and the reconstruction follows from $U^\dagger U = \mathbb{1}$. Traces of products are unchanged by conjugation, so $|V^{(N)}(\widetilde{M})\rangle = \mu^N (|V^{(N)}(\widehat{M}_0)\rangle + |V^{(N)}(\widehat{M}_1)\rangle)$.

For every matrix unit $E_{(p,q),(p',q')}$ of the two-qubit space, the matrix of $\widehat{M}_f$ at the bond pair $((p, p', 0), (q, q', 0))$ is $C_0[p, p']/(2\mu)$ times that matrix unit, with $C_0[p, p'] \in \{\frac{1}{2}, \frac{3}{8}\}$ nonzero. Hence the matrices of each sector span the full matrix algebra, giving one-site injectivity. For linear independence, evaluate the closed sector operators of Lemma 21.2.2.57 on two cyclically bond-matched words: on the constant word of bond index $(0, 0, 0)$ both sectors give $(2/5)^N$, since the one-site weight is $C_0[0, 0]/(2\mu) = \frac{2}{5}$; on the word with bond index $(0, 0, 1)$ at the first site and $(0, 0, 0)$ elsewhere, the sector f gives $(\tau_1)_{ff} \frac{3}{10} (2/5)^{N-1}$, since $C_1[0, 0]/(2\mu) = \frac{3}{10}$, and the two flag signs are opposite. A vanishing linear combination $g_0 |V^{(N)}(\widehat{M}_0)\rangle + g_1 |V^{(N)}(\widehat{M}_1)\rangle = 0$ thus forces $g_0 + g_1 = 0$ and $g_0 - g_1 = 0$. $\square$

Theorem 21.2.2.63 (Vertical BNT algebra of the twisted quantum dimer). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The vertical sectors of the twisted quantum dimer of Definition 21.2.2.33 satisfy the algebra and idempotent conditions of Definition 21.2.2.98, using the decomposition of Theorem 21.2.2.62 and the structure coefficients of Theorem 21.2.2.54. The two flag sectors are normal tensors in the sense of [11, lines 231–235]: each is injective at one site and left-canonical,

$$\sum_{a,b} (\widehat{M}_f^{ab})^\dagger \widehat{M}_f^{ab} = \mathbb{1},$$

so its transfer map has spectral radius one. Together with the spanning and linear independence already proved, this gives a basis of normal tensors of the vertically viewed tensor, as in [11, Proposition 4.13, lines 945–959]. The positive diagonal matrices $\chi_{ff', f+f'} = \alpha$, $\chi_{ff', f+f'+1} = \beta$ give $c_{ff'g}^{(L)} = \text{tr}(\chi_{ff'g}^L)$, and the closed operators satisfy the product law of Theorem 21.2.2.58. The multiplicity matrices are $\mu_0 = \mu_1 = [\mu]$, so $m_0 = m_1 = \text{tr}([\mu]) = \frac{5}{8}$. The vector m is an idempotent for the multiplication induced by $c^{(1)}$, since

$$m_g = \sum_{f,f'} c_{ff'g}^{(1)} m_f m_{f'} = 2(\alpha + \beta) \mu^2 = \frac{8}{5} \cdot \frac{25}{64} = \frac{5}{8}.$$

No positive rescaling of these two sectors makes the structure coefficients length independent.

Proof. [✓ Proof](#) The adjoint of a sector matrix times that matrix is the diagonal matrix unit at the right bits (p', q') of the bond pair with the squared modulus of the coefficient, and the flag sign squares to one. Summing over the bond pairs $((p, p', k), (q, q', k))$ with the right bits fixed gives, at each diagonal entry,

$$\frac{1}{(2\mu)^2} \sum_k \sum_{p,q} C_k[p, p']^2 = \frac{2 \cdot 2 \cdot (\frac{1}{4} + \frac{9}{64})}{4 \cdot \frac{25}{64}} = 1,$$

and the off-diagonal entries vanish. Together with one-site injectivity from Theorem 21.2.2.62, this makes each sector a normal tensor with spectral radius one by Theorem 7.6.3. The spanning and linear independence are those of Theorem 21.2.2.62. The product law is Theorem 21.2.2.58. For each output label g , two pairs (f, f') contribute α and two contribute β , so $\sum_{f, f'} c_{ff'g}^{(1)} m_f m_{f'} = 2(\alpha + \beta)\mu^2 = \mu = m_g$. $\square$

Lemma 21.2.2.64 (Doubled-index matrices and their rank-one factorization). [Lean](#) [Lean](#) [✓ Stmt](#) Identify each physical index $p \in \{0, 1, 2, 3\}$ with a pair $(p_1, p_2) \in \{0, 1\}^2$, and similarly write $q = (q_1, q_2)$. The doubled-index matrix at the paired index (p, q) is the MPO matrix R^{pq} . This bond matrix has the outer-product factorization

$$R_{ab}^{pq} = \frac{25}{32} A_{p_1 q_1}^a A_{p_2 q_2}^b,$$

or equivalently $R^{pq} = \frac{25}{32} u_{pq} v_{pq}^\top$, where $u_{pq}(a) = A_{p_1 q_1}^a$ and $v_{pq}(b) = A_{p_2 q_2}^b$.

Proof. [✓ Proof](#) The doubled-index reading gives R^{pq} by definition. Expanding the Kronecker-product definition of R and using that the entries of the four matrices A^a are real gives $R^{pq} = \frac{25}{32} u_{pq} v_{pq}^\top$. $\square$

Lemma 21.2.2.65 (Rank-one form of the dimer transfer map). [Lean](#) [Lean](#) [✓ Stmt](#) Set $s = (\frac{4}{5}, \frac{4}{5}, \frac{3}{5}, \frac{3}{5})$, $F = \text{diag}(s_0^2, s_1^2, s_2^2, s_3^2)$, and

$$\lambda(X) = \left(\frac{25}{32}\right)^2 \sum_{b=0}^3 s_b^2 X_{bb}.$$

Then the transfer map of R has rank-one form

$$\begin{aligned} \mathcal{E}_R(X) &= \lambda(X)F, \\ \lambda(F) &= \frac{337}{512}. \end{aligned}$$

Proof. [✓ Proof](#) By Lemma 21.2.2.64, the doubled-index matrices are $R^{pq} = \frac{25}{32} u_{pq} v_{pq}^\top$. Substituting these factors into the transfer-map sum and summing over the physical indices gives $\mathcal{E}_R(X) = \lambda(X)F$. Substituting $X = F$ and evaluating the four diagonal entries gives $\lambda(F) = 337/512$. $\square$

Lemma 21.2.2.66 (One-site transfer quasi-idempotence of the dimer tensor). [Lean](#) [✓ Stmt](#) The transfer map of the dimer tensor R satisfies $\mathcal{E}_R^2 = \frac{337}{512} \mathcal{E}_R$.

Proof. [✓ Proof](#) By Lemma 21.2.2.65,

$$\mathcal{E}_R(\mathcal{E}_R(X)) = \lambda(X)\lambda(F)F = \frac{337}{512} \mathcal{E}_R(X).$$

$\square$

Theorem 21.2.2.67 (Positivity of the rescaling-stable MPDO). [Lean](#) [Lean](#) [✓ Stmt](#) *The operator $\rho^{(N)}(R)$ is positive semidefinite for every positive chain length N , so R generates MPDOs. Writing each bond matrix R^{p_q} as a rank-one product scaled by $25/32$, the closed-chain trace of a length- N configuration (σ, τ) gives*

$$\rho^{(N)}(R)_{\sigma\tau} = \left(\frac{25}{32}\right)^N \mathbb{1}[\text{cyclic bond match}](\sigma, \tau) \prod_n W(\sigma_n, \tau_n),$$

where the indicator is one exactly when both σ and τ satisfy the cyclic bond-matching condition: consecutive letters share their bond bit around the ring. In the product, W is evaluated on the corresponding bond bits. It is the local factor from Theorem 21.2.2.32, with eigenvalues $\{1, 7/25\}$. Thus $\rho^{(N)}(R)$ is $(25/32)^N$ times the Hadamard product of the rank-one bond-matching indicator with the pullback of the N -fold Kronecker power of W .

Proof. [✓ Proof](#) The entrywise factorization follows by induction on the chain length, using the rank-one contraction identity

$$\text{tr}((u_1 v_1^\top) \cdots (u_N v_N^\top)) = (v_N^\top u_1) \prod_{n=1}^{N-1} (v_n^\top u_{n+1}),$$

where u_n, v_n are the column vectors in the rank-one factorization of the n -th bond matrix. Closing the trace produces the cyclic bond-matching condition. The bond-matching indicator is positive semidefinite because it is an outer product. The pullback of the Kronecker power of W is positive semidefinite by congruence. The Schur product theorem and the positive scalar $(25/32)^N$ therefore give positivity. $\square$

Theorem 21.2.2.68 (Unitality of the transfer map). [Lean](#) [✓ Stmt](#) *For the explicit four-letter tensor A , the transfer map $\mathcal{E}_A(Y) = \sum_a A^a Y (A^a)^\dagger$ satisfies $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$.*

Proof. [✓ Proof](#) Evaluate the four matrix entries of $\mathcal{E}_A(\mathbb{1})$ from the matrices A^a . The diagonal entries are one and the off-diagonal entries vanish. $\square$

Theorem 21.2.2.69 (Traceless diagonal eigenvector of the transfer map). [Lean](#) [✓ Stmt](#) *For $Z = \text{diag}(1, -1)$, the transfer map satisfies $\mathcal{E}_A(Z) = \frac{7}{25}Z$.*

Proof. [✓ Proof](#) Evaluate $\sum_a A^a Z (A^a)^\dagger$ entrywise. Its off-diagonal entries vanish, while its diagonal entries are $7/25$ and $-7/25$. $\square$

Theorem 21.2.2.70 (Annihilation of the first off-diagonal matrix unit). [Lean](#) [✓ Stmt](#) *For the matrix unit E_{01} , the transfer map satisfies $\mathcal{E}_A(E_{01}) = 0$.*

Proof. [✓ Proof](#) Substitution of the four matrices A^a shows that every entry of $\sum_a A^a E_{01} (A^a)^\dagger$ is zero. $\square$

Theorem 21.2.2.71 (Annihilation of the second off-diagonal matrix unit). [Lean](#) [✓ Stmt](#) *For the matrix unit E_{10} , the transfer map satisfies $\mathcal{E}_A(E_{10}) = 0$.*

Proof. [✓ Proof](#) Substitution of the four matrices A^a shows that every entry of $\sum_a A^a E_{10} (A^a)^\dagger$ is zero. $\square$

Theorem 21.2.2.72 (Spectra of the local factor and the transfer map). [Lean](#) [Lean](#) [Stmnt](#) The two diagonal entries of χ are eigenvalues of the local factor W : the uniform vector has eigenvalue 1, and the alternating vector has eigenvalue $7/25$. The transfer map $\mathcal{E}_A(Y) = \sum_a A^a Y (A^a)^\dagger$ has eigenvalues $1, 7/25, 0, 0$, counted with multiplicity. The identity is fixed, the traceless diagonal matrix $\text{diag}(1, -1)$ has eigenvalue $7/25$, and the off-diagonal matrix units are annihilated. Thus the entries of χ are also the nonzero eigenvalues of $\mathcal{E}_A$.

Proof. [Proof](#) The uniform and alternating vectors are eigenvectors of the explicit 2×2 matrix W :

$$\begin{aligned} W \begin{pmatrix} 1 \\ 1 \end{pmatrix} &= \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \\ W \begin{pmatrix} 1 \\ -1 \end{pmatrix} &= \frac{7}{25} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \end{aligned}$$

Direct expansion of the letter matrices gives the action on a basis of $M_2(\mathbb{C})$:

$$\begin{aligned} \mathcal{E}_A(\mathbb{1}) &= \mathbb{1}, \\ \mathcal{E}_A(\text{diag}(1, -1)) &= \frac{7}{25} \text{diag}(1, -1), \\ \mathcal{E}_A(E_{01}) &= 0, \\ \mathcal{E}_A(E_{10}) &= 0. \end{aligned}$$

□

Lemma 21.2.2.73 (Injectivity of $R^{\bullet\bullet}$). [Lean](#) [Stmnt](#) The 16 letters R^{pq} of the doubled-index tensor $R^{\bullet\bullet}$ span the full bond matrix algebra $M_4(\mathbb{C})$.

Proof. [Proof](#) Each letter R^{pq} is, up to a nonzero scalar, one of the 16 matrix units of $M_4(\mathbb{C})$. Indeed, the outer-product structure of R^{pq} gives a single matrix unit because each factor A^a , A^b of the vertical letters $B^{ab} = A^a \otimes A^b$ is a scaled matrix unit of $M_2(\mathbb{C})$. Hence the standard matrix-unit basis lies in the span of the letters. □

Theorem 21.2.2.74 (Canonical form of R). [Lean](#) [Stmnt](#) The doubled-index tensor $R^{\bullet\bullet}$, with bond dimension 4 and 16 letters R^{pq} for $p, q \in \{0, \dots, 3\}$, is in canonical form as in Definition 7.6.7 (CF). It consists of a single normal block occupying the full bond space ($r = 1$, $D_1 = 4$, $U = \mathbb{1}$), with weight $\mu = \sqrt{337/512}$. This is the unique positive weight for which the normalized block $\mu^{-1}R^{\bullet\bullet}$ has an idempotent transfer map, and hence spectral radius one.

Proof. [Proof](#) By Lemma 21.2.2.73, the 16 letters span the full bond matrix algebra. Thus $R^{\bullet\bullet}$ is injective, and no nontrivial invariant orthogonal projection exists.

The transfer map $\mathcal{E}_{R^{\bullet\bullet}}(X) = \sum_{p,q} R^{pq} X (R^{pq})^\dagger$ has rank one: every image is a scalar multiple of a fixed diagonal matrix, and the scalar evaluated at that matrix is $337/512$. Rescaling the tensor by $\mu^{-1} = \sqrt{512/337}$ makes its transfer map idempotent. Together with injectivity, this gives a primitive transfer map with spectral radius one; all its other eigenvalues are zero. Thus $\mu^{-1}R^{\bullet\bullet}$ is a normal tensor with the required normalization. The equality $R^{\bullet\bullet} = \mu(\mu^{-1}R^{\bullet\bullet})$ gives exact reconstruction with the identity coisometry. □

Theorem 21.2.2.75 (Non-nilpotency of the retained horizontal block). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $\widehat{R} = \mu^{-1}R^{\bullet\bullet}$ be the single normal block in this canonical form, where $\mu = \sqrt{337/512}$.

Then

$$\begin{aligned}\mathcal{T}_{\widehat{R}} &= \mu^{-1}\mathcal{T}_R, \\ \mathcal{T}_{\widehat{R}}^2 &= \mathcal{T}_R, \\ (\mathcal{T}_R)_{00} &= \frac{1}{2}.\end{aligned}$$

Consequently neither $\mathcal{T}_R$ nor $\mathcal{T}_{\widehat{R}}$ is nilpotent. Thus the retained basis element satisfies the non-nilpotency condition in the definition of simplicity.

Proof. ✓ Proof A nilpotent idempotent is zero, but the $(0,0)$ entry of $\mathcal{T}_R$ is

$$\frac{25}{32} \text{tr}(A^0)^2 = \frac{25}{32} \left(\frac{4}{5}\right)^2 = \frac{1}{2},$$

so $\mathcal{T}_R$ is not nilpotent. The diagonal physical contraction is linear in the tensor, which gives $\mathcal{T}_{\widehat{R}} = \mu^{-1}\mathcal{T}_R$. Since $\mu > 0$, a nilpotent $\mathcal{T}_{\widehat{R}}$ would remain nilpotent after multiplication by μ , forcing $\mathcal{T}_R$ to be nilpotent. $\square$

Theorem 21.2.2.76 (Simplicity of the rescaling-stable dimer). Lean Lean Lean Lean ✓ Stmt
The dimer tensor R is simple, with blocking length $L = 1$. After the canonical relabeling of the doubled physical alphabet, the basis of normal tensors has one member, $\widehat{R}$, and its physical-trace transfer is non-nilpotent.

Proof. ✓ Proof Theorem 21.2.2.67 gives the MPDO condition. Choose the one-site blocking. Its doubled-index tensor is the canonical physical relabeling of $R^{\bullet\bullet}$. Refine the canonical form from Theorem 21.2.2.74 to its basis-of-normal-tensors sector presentation. The only representative is $\widehat{R}$, whose physical-trace transfer is non-nilpotent by Theorem 21.2.2.75. $\square$

Theorem 21.2.2.77 (The rescaling-stable tensor is a renormalization fixed point). Lean ✓ Stmt
The tensor R of Theorem 21.2.2.32 is a renormalization fixed point in the sense of Definition 20.2.1. The refinement map T from one site to two sites is completely positive and trace preserving, with two Kraus operators built from the eigenvectors of W , with amplitude $5/8$ on the uniform eigenvector and $\sqrt{7}/8$ on the alternating eigenvector. The coarse-graining map S from two sites to one site is completely positive and trace preserving, with four Kraus operators. Two use the amplitude $1/\sqrt{2}$ and the eigenvectors of W on physical-index pairs satisfying the bond-matching condition. The other two assign the pairs that fail this condition deterministically, so that the four Kraus operators resolve the identity. These maps satisfy $T[M_1(X)] = M_2(X)$ and $S[M_2(X)] = M_1(X)$ for every boundary matrix X , with $M = R$.

Proof. ✓ Proof Identify each physical index $p \in \{0, 1, 2, 3\}$ with a pair of bits $(p', p'') \in \{0, 1\}^2$, and write $i_1 * i_2 := (i_1', i_2'')$ for the physical index obtained from the first bit of i_1 and the second bit of i_2 . For $M = R$, the two-site physical operator satisfies

$$M_2(X)_{(i_1, i_2)(j_1, j_2)} = \frac{25}{32} \mathbb{1}[i_1'' = i_2'] \mathbb{1}[j_1'' = j_2'] W(i_1'', j_1'') M_1(X)_{i_1 * i_2, j_1 * j_2},$$

and therefore vanishes whenever either bond match fails. Write $e_0 = (1, 1)$ and $e_1 = (1, -1)$ for the eigenvectors of W , with eigenvalues $\lambda_0 = 1$ and $\lambda_1 = 7/25$ (Theorem 21.2.2.72). The two Kraus operators of T have amplitudes $\sqrt{25\lambda_s}/64$ on e_s . For $b, b' \in \{0, 1\}$, the eigendecomposition of W gives

$$\sum_{s \in \{0, 1\}} \frac{25}{64} \lambda_s e_s(b) e_s(b') = \frac{25}{32} W(b, b').$$

This reproduces the coefficient in $M_2(X)$, so $T[M_1(X)] = M_2(X)$.

The two Kraus operators of S supported on bond-matching pairs have amplitude $1/\sqrt{2}$ on e_0, e_1 . For $b, b' \in \{0, 1\}$,

$$\frac{1}{2} \sum_{s \in \{0,1\}} e_s(b) e_s(b') = \begin{cases} 1, & b = b', \\ 0, & b \neq b'. \end{cases}$$

The diagonal case gives the resolution of the identity on the bond-matching subspace, and the off-diagonal case gives cancellation between these two Kraus operators. The other two Kraus operators assign the bond-mismatched pairs deterministically. They do not contribute to $S[M_2(X)]$, because $M_2(X)$ vanishes on these pairs. Together the four Kraus operators resolve the identity on the full two-site domain and give $S[M_2(X)] = M_1(X)$. $\square$

Theorem 21.2.2.78 (Vertical BNT and algebra structure for R). Lean ✓ Stmt *The tensor R has a vertical basis of normal tensors with one element M_0 . This component has 16 physical letters, bond dimension 4, and*

$$M_0^{(a,b)} = A^a \otimes \overline{A}^b.$$

Each letter of M_0 is a ket copy of A over a bra copy of $\overline{A}$ that share no index:

$$\begin{array}{c} a \\ | \\ \boxed{M_0} \\ | \\ b \end{array} = \begin{array}{c} a \\ | \\ \boxed{A} \\ | \\ \boxed{\overline{A}} \\ | \\ b \end{array}$$

The positive diagonal multiplicity and coefficient matrices are

$$\mu_0 = \left(\frac{25}{32} \right),$$

$$\chi_{0,0,0} = \text{diag} \left(1, \frac{7}{25} \right).$$

The vertical tensor is reconstructed exactly as

$$\text{vertical}(R) = \frac{25}{32} M_0.$$

For every $L > 0$, the vertical operator satisfies

$$O_L(M_0)^2 = \left(1 + \left(\frac{7}{25} \right)^L \right) O_L(M_0) = \text{tr}(\chi_{0,0,0}^L) O_L(M_0).$$

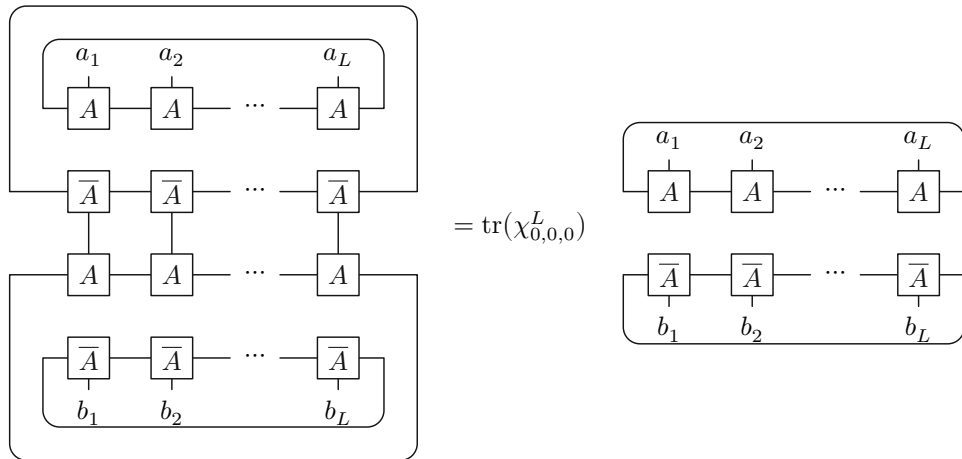
At length one, $m_0 = \text{tr}(\mu_0) = 25/32$ and $c_{0,0,0}^{(1)} = 32/25$. Thus the vector $m = (m_0)$ is an idempotent for the multiplication induced by $c^{(1)}$, since $m_0 = c_{0,0,0}^{(1)} m_0^2$:

$$\frac{25}{32} = \frac{32}{25} \left(\frac{25}{32} \right)^2.$$

Proof. [✓ Proof](#) Each letter of M_0 is a nonzero multiple of a matrix unit, so the letters span $M_4(\mathbb{C})$. The tensor A is left-canonical; hence M_0 is left-canonical and algebraically normal, and therefore is a normal tensor. The identity coisometry gives both the vertical decomposition and its exact reconstruction.

The vertical operator of $M_0 = A \otimes \bar{A}$ is $|V^{(L)}(A)\rangle\langle V^{(L)}(A)|$. Its square is its self-overlap times itself. A direct transfer-map computation gives $\langle V^{(L)}(A)|V^{(L)}(A)\rangle = \text{tr}(W^L)$, while Walsh–Hadamard diagonalization gives $\text{tr}(W^L) = \text{tr}(\chi_{0,0,0}^L) = 1 + (7/25)^L$. Finally, $c_{0,0,0}^{(1)} m_0^2 = (32/25)(25/32)^2 = m_0$. $\square$

The square computed in this proof is the following contraction. The two middle rings form the self-overlap of the periodic vector of A , a closed network equal to $\text{tr}(W^L) = \text{tr}(\chi_{0,0,0}^L)$:



Remark 21.2.2.79 (Vertical and horizontal contractions). Let $\text{rot}(\text{vertical}(R))$ denote the MPO obtained by separating the 16-valued physical index of the vertical tensor into two four-valued indices. Then

$$\rho^{(L)}(\text{rot}(\text{vertical}(R))) = \left(\frac{25}{32}\right)^L O_L(M_0).$$

This is a rotated vertical contraction, not the original horizontal tensor R , even though both physical and bond dimensions equal 4. At length one, $\rho^{(1)}(R)$ has rank two, whereas $O_1(M_0)$ has rank one, so these operators are not proportional. In particular, no identity $\rho^{(L)}(R) = (25/32)^L O_L(M_0)$ is asserted.

Theorem 21.2.2.80 (Closed-operator factorization and local diagonalization for R). [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every positive chain length N , the horizontal operator factors as

$$\rho^{(N)}(R) = \left(\frac{25}{32}\right)^N B W_N B^\top,$$

where B sends each bit-string basis vector to the basis vector of the unique cyclic-bond-matching physical-index string with those first bits, and $B^\top B = \mathbb{1}$. Here $W_N(a, b) = \prod_n W(a_n, b_n)$ is the N -fold Kronecker power of W . For the Walsh–Hadamard matrix $H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $\chi = \text{diag}(1, \frac{7}{25})$, one has $H\chi H = 2W$. Thus the entries of χ are the eigenvalues of the local factor W . Theorem 21.2.2.78 identifies this matrix as the coefficient matrix $\chi_{0,0,0}$ of a vertical basis of normal tensors with one element.

Proof. [✓ Proof](#) Let ϕ read the first bits of a cyclic-bond-matching string. Given a bit string a , each adjacent pair of entries of a determines one physical index around the ring. This gives the unique matching string whose image under ϕ is a , and defines B . Injectivity gives $B^\top B = \mathbf{1}$. Rewriting the entrywise formula of Theorem 21.2.2.67 through this embedding gives $\rho^{(N)}(R) = (25/32)^N B W_N B^\top$. The identity $H\chi H = 2W$ follows by direct 2×2 computation. $\square$

Theorem 21.2.2.81 (Trace-power formula for the structure coefficients). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The coefficients $c_{0,0,0}^{(L)}$ have a trace-power representation with the fixed positive diagonal matrix $\chi = \text{diag}(1, \frac{7}{25})$, as in Definition 21.2.2.11. For every $L > 0$,

$$c_{0,0,0}^{(L)} = \text{tr}(\chi^L) = \text{tr}(W^L) = 1 + (7/25)^L.$$

The equality of traces follows from the diagonalization in Theorem 21.2.2.80. Here W^L is the ordinary matrix power of the fixed 2×2 matrix W , not the Kronecker power W_L in $\rho^{(L)}(R)$. Theorem 21.2.2.78 identifies these coefficients with the structure coefficients of the vertical basis of normal tensors $\{M_0\}$.

Proof. [✓ Proof](#) The coefficients are defined by the diagonal entries of χ . Since these entries are positive, Lemma 21.2.2.12 gives $c_{0,0,0}^{(L)} = \text{tr}(\chi^L)$ for every $L > 0$. The Walsh–Hadamard identity $H\chi H = 2W$ and $HH = 2\mathbf{1}$ give $H\chi = WH$. Induction then gives $H\chi^L = W^L H$. Conjugating by H , whose inverse is $H/2$, and using cyclicity of the trace yields $\text{tr}(W^L) = \text{tr}(\chi^L)$ for every L . $\square$

Definition 21.2.2.82 (Linearly independent operator family). [Lean](#) [Lean](#) [✓ Stmt](#) The family $\{O_L(M_\alpha)\}_\alpha$ is *linearly independent at length L* when its operators are linearly independent, and *eventually linearly independent* when this holds for every length beyond some threshold. Eventual linear independence is the operator form of the third defining property of a basis of normal tensors in [11, Section 2.3], invoked for the vertical operators in [11, Appendix C.4].

Theorem 21.2.2.83 (Uniqueness of the structure coefficients). [Lean](#) [✓ Stmt](#) Two systems of structure coefficients satisfying the product law for the same operator family agree at every positive length at which the operators are linearly independent.

Proof. [✓ Proof](#) Applying the product law to both systems of structure coefficients gives

$$\sum_\delta c_{\alpha,\beta,\delta}^{(L)} O_L(M_\delta) = O_L(M_\alpha) O_L(M_\beta) = \sum_\delta c'_{\alpha,\beta,\delta}{}^{(L)} O_L(M_\delta).$$

Subtracting yields $\sum_\delta (c_{\alpha,\beta,\delta}^{(L)} - c'_{\alpha,\beta,\delta}{}^{(L)}) O_L(M_\delta) = 0$, and linear independence forces every coefficient of the difference to vanish. $\square$

Definition 21.2.2.84 (Change of vertical representatives). [Lean](#) [✓ Stmt](#) Let $\{O_L(M_\alpha)\}_{\alpha \in \Lambda}$ and $\{O_L(M'_\alpha)\}_{\alpha \in \Lambda'}$ be two operator families. A change of vertical representatives consists of a bijection $e : \Lambda' \rightarrow \Lambda$, nonzero complex scalars s_α , and complex algebra isomorphisms Φ_L between the ambient operator algebras. For every positive length L and every $\alpha \in \Lambda'$, these satisfy

$$O_L(M'_\alpha) = s_\alpha^L \Phi_L(O_L(M_{e(\alpha)})).$$

Definition 21.2.2.85 (Structure coefficients under a change of normalization). [Lean](#) [✓ Stmt](#) Given a label bijection $e : \Lambda' \rightarrow \Lambda$ and complex scalars s_α , define

$$\tilde{c}_{\alpha,\beta,\gamma}^{(L)} = \left(\frac{s_\alpha s_\beta}{s_\gamma} \right)^L c_{e(\alpha),e(\beta),e(\gamma)}^{(L)}.$$

For this definition, use the convention $z/0 = 0$.

Theorem 21.2.2.86 (Product law under a change of vertical representatives). [Lean](#) [✓ Stmt](#)
A change of vertical representatives carries a product law with structure coefficients $c_{\alpha,\beta,\gamma}^{(L)}$ to a product law with structure coefficients $\tilde{c}_{\alpha,\beta,\gamma}^{(L)}$.

Proof. [✓ Proof](#) For every positive length L , multiplication of the two new representatives and the original product law give

$$\begin{aligned} O_L(M'_\alpha)O_L(M'_\beta) &= (s_\alpha s_\beta)^L \Phi_L(O_L(M_{e(\alpha)})O_L(M_{e(\beta)})) \\ &= (s_\alpha s_\beta)^L \sum_{\delta} c_{e(\alpha),e(\beta),\delta}^{(L)} \Phi_L(O_L(M_\delta)). \end{aligned}$$

Reindexing by $\delta = e(\gamma)$ and using $s_\gamma \neq 0$ yields

$$\begin{aligned} O_L(M'_\alpha)O_L(M'_\beta) &= \sum_{\gamma} \left(\frac{s_\alpha s_\beta}{s_\gamma} \right)^L c_{e(\alpha),e(\beta),e(\gamma)}^{(L)} s_\gamma^L \Phi_L(O_L(M_{e(\gamma)})) \\ &= \sum_{\gamma} \tilde{c}_{\alpha,\beta,\gamma}^{(L)} O_L(M'_\gamma). \end{aligned}$$

□

Theorem 21.2.2.87 (Covariance of structure coefficients for operator families). [Lean](#) [✓ Stmt](#)
Suppose that two operator families satisfy product laws and are related by a change of vertical representatives. At a positive length where the operators of the second family are linearly independent, their structure coefficients equal the transformed coefficients $\tilde{c}_{\alpha,\beta,\gamma}^{(L)}$ of the first family.

Proof. [✓ Proof](#) The product law for the second family and Theorem 21.2.2.86 give

$$\sum_{\gamma} c'_{\alpha,\beta,\gamma}^{(L)} O_L(M'_\gamma) = O_L(M'_\alpha)O_L(M'_\beta) = \sum_{\gamma} \tilde{c}_{\alpha,\beta,\gamma}^{(L)} O_L(M'_\gamma).$$

Linear independence of these operators identifies the two systems of structure coefficients term by term. □

Theorem 21.2.2.88 (Covariance of structure coefficients for vertical BNT decompositions). [Lean](#) [✓ Stmt](#)
Suppose that two vertical basis-of-normal-tensors decompositions of tensors satisfy the algebra condition and are related by a change of vertical representatives. At every positive length L at which the operators of the second decomposition are linearly independent, their structure coefficients satisfy

$$c'_{\alpha,\beta,\gamma}^{(L)} = \left(\frac{s_\alpha s_\beta}{s_\gamma} \right)^L c_{e(\alpha),e(\beta),e(\gamma)}^{(L)}.$$

Theorem 21.2.2.92 shows why equality of the horizontal periodic operator families alone cannot replace the specified relation between vertical representatives.

Proof. [✓ Proof](#) The algebra conditions for the two vertical basis-of-normal-tensors decompositions give the product laws required by the covariance theorem for operator families. Hence

$$c'_{\alpha,\beta,\gamma}^{(L)} = \tilde{c}_{\alpha,\beta,\gamma}^{(L)} = \left(\frac{s_\alpha s_\beta}{s_\gamma} \right)^L c_{e(\alpha),e(\beta),e(\gamma)}^{(L)}.$$

□

Theorem 21.2.2.89 (Equality of structure coefficients under compatible normalization). [Lean](#)

[✓ Stmt](#) Under the hypotheses of Theorem 21.2.2.87, if $s_\alpha s_\beta = s_\gamma$, then $c'_{\alpha,\beta,\gamma}^{(L)} = c_{e(\alpha),e(\beta),e(\gamma)}^{(L)}$.

Proof. [✓ Proof](#) The compatible normalization and $s_\gamma \neq 0$ give $(s_\alpha s_\beta / s_\gamma)^L = (s_\gamma / s_\gamma)^L = 1$. Substitution in the covariance formula gives the result. $\square$

Theorem 21.2.2.90 (Equality of vertical BNT structure coefficients under compatible normalization). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 21.2.2.88, if $s_\alpha s_\beta = s_\gamma$, then $c'_{\alpha,\beta,\gamma}^{(L)} = c_{e(\alpha),e(\beta),e(\gamma)}^{(L)}$.

Proof. [✓ Proof](#) For the vertical basis-of-normal-tensors decompositions, the covariance formula gives

$$c'_{\alpha,\beta,\gamma}^{(L)} = \left(\frac{s_\alpha s_\beta}{s_\gamma} \right)^L c_{e(\alpha),e(\beta),e(\gamma)}^{(L)} = c_{e(\alpha),e(\beta),e(\gamma)}^{(L)},$$

where the last equality uses $s_\alpha s_\beta = s_\gamma$ and $s_\gamma \neq 0$. $\square$

Definition 21.2.2.91 (Horizontally gauge-equivalent tensors with distinct vertical normalizations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let

$$\begin{aligned} P &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \\ X &= \begin{pmatrix} 1 & -3/4 \\ 0 & 1 \end{pmatrix}, \\ Q &= X P X^{-1} = \begin{pmatrix} 1 & 3/4 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

Define MPO tensors with one-dimensional physical space by $M_P^{00} = P$ and $M_Q^{00} = Q$. Thus M_Q is the gauge transform of M_P by X :

$$\boxed{M_Q} = \boxed{X} \boxed{M_P} \boxed{X^{-1}}$$

Equip them with single-element vertical basis-of-normal-tensors decompositions whose normalized representatives and positive multiplicity weights are

$$\begin{aligned} A_P &= P, \\ m_P &= 1, \\ A_Q &= \frac{4}{5}Q, \\ m_Q &= \frac{5}{4}. \end{aligned}$$

The corresponding positive diagonal matrices χ in the algebra condition are $[1]$ and $[4/5]$. No horizontal canonical-form assumption is imposed on the sheared tensor M_Q .

Theorem 21.2.2.92 (Horizontal periodic equality alone does not determine vertical BNT coefficients). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The tensors M_P and M_Q are related by the nonunitary gauge X . For every positive length N , their unnormalized periodic operators

satisfy $\rho^{(N)}(M_P) = \rho^{(N)}(M_Q)$. In particular, both tensors generate matrix product density operators. Their chosen vertical basis-of-normal-tensors decompositions nevertheless have structure coefficients

$$\begin{aligned} c_P^{(L)} &= 1, \\ c_Q^{(L)} &= \left(\frac{4}{5}\right)^L. \end{aligned}$$

Hence there is no relabelling of the singleton label sets for which, at every positive length,

$$c_Q^{(L)}(\alpha, \beta, \gamma) = c_P^{(L)}(e(\alpha), e(\beta), e(\gamma)).$$

Indeed, the two sides at length one are $4/5$ and 1 .

While Theorem 21.2.2.83 guarantees uniqueness of structure coefficients for a fixed operator family, this counterexample shows that periodic operator equality alone does not constrain the structure coefficients of independent vertical decompositions across nonunitary gauge changes.

Proof. ✓ Proof The identities $P^2 = P$, $Q^2 = Q$, and $\text{tr}(P) = \text{tr}(Q) = 1$ prove the periodic equality. The Frobenius norms of P and Q are 1 and $5/4$, respectively, which gives the normalized representatives and multiplicity weights. Their products satisfy

$$\begin{aligned} A_P^2 &= A_P, \\ A_Q^2 &= \frac{4}{5}A_Q. \end{aligned}$$

Taking the L -fold vertical operators gives $c_P^{(L)} = 1$ and $c_Q^{(L)} = (4/5)^L$. Their values at $L = 1$ exclude the only possible relabelling. □

Theorem 21.2.2.93 (Length independence is not preserved by horizontal periodic equality alone). Lean Lean Lean ✓ Stmt The structure coefficients in the vertical decomposition of M_P are length independent, whereas those in the chosen vertical decomposition of M_Q are not. More precisely,

$$\begin{aligned} c_P^{(L)} &= 1, \\ c_Q^{(1)} &= \frac{4}{5}, \\ c_Q^{(2)} &= \frac{16}{25}. \end{aligned}$$

Proof. ✓ Proof The first family is constant, whereas $c_Q^{(1)} \neq c_Q^{(2)}$. □

Theorem 21.2.2.94 (Associativity constraint on the structure coefficients). Lean ✓ Stmt At every positive length L at which the operators are linearly independent, the structure coefficients of the product law satisfy

$$\sum_{\delta} c_{\alpha, \beta, \delta}^{(L)} c_{\delta, \gamma, \epsilon}^{(L)} = \sum_{\delta} c_{\beta, \gamma, \delta}^{(L)} c_{\alpha, \delta, \epsilon}^{(L)}.$$

These are the restrictions that associativity of multiplication imposes on the structure coefficients, noted in [11, Section 4.5].

Proof. [✓ Proof](#) Applying the product law twice in each order gives

$$\begin{aligned} (O_L(M_\alpha)O_L(M_\beta))O_L(M_\gamma) &= \sum_{\varepsilon'} \left(\sum_{\delta} c_{\alpha,\beta,\delta}^{(L)} c_{\delta,\gamma,\varepsilon'}^{(L)} \right) O_L(M_{\varepsilon'}), \\ O_L(M_\alpha)(O_L(M_\beta)O_L(M_\gamma)) &= \sum_{\varepsilon'} \left(\sum_{\delta} c_{\beta,\gamma,\delta}^{(L)} c_{\alpha,\delta,\varepsilon'}^{(L)} \right) O_L(M_{\varepsilon'}). \end{aligned}$$

Associativity of multiplication equates the two left-hand sides, so the right-hand sides agree; linear independence of the operators then forces the corresponding coefficient sums to agree for each ε' . $\square$

Theorem 21.2.2.95 (Associativity constraint in trace-power form). [Lean](#) [✓ Stmt](#) Suppose that the operators $\{O_L(M_\alpha)\}_\alpha$ satisfy the product law with structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ for diagonal matrices $\chi_{\alpha,\beta,\gamma}$. At every positive length L at which the operators are linearly independent, the associativity constraint becomes

$$\sum_{\delta} \text{tr}(\chi_{\alpha,\beta,\delta}^L) \text{tr}(\chi_{\delta,\gamma,\epsilon}^L) = \sum_{\delta} \text{tr}(\chi_{\beta,\gamma,\delta}^L) \text{tr}(\chi_{\alpha,\delta,\epsilon}^L).$$

In [11, Section 4.5], these restrictions in the positive diagonal setting, together with the isometry statement, lead to a pentagon-like equation for the fusion isometries. Solutions constructed from unitary fusion categories give the family of renormalization fixed point MPDOs described there.

Proof. [✓ Proof](#) Substitute the trace-power form of the coefficients into the associativity constraint. $\square$

Theorem 21.2.2.96 (The dimer RFP has length-dependent coefficients in its chosen vertical BNT decomposition). [Lean](#) [✓ Stmt](#) There is a tensor M generating MPDOs that has the direct-sum form $M^i = \bigoplus_k \mu_k M_k^i$ with normalized normal tensor blocks M_k , is a renormalization fixed point, and has a vertical basis-of-normal-tensors decomposition satisfying the algebra condition whose structure coefficients depend on the positive length L . One may take $M = R$ and the single-element basis-of-normal-tensors decomposition already constructed, for which

$$c_{0,0,0}^{(L)} = 1 + \left(\frac{7}{25} \right)^L.$$

This exhibits length-dependent structure coefficients for the chosen vertical decomposition of R , addressing the question following Theorem 4.14 of [11] without requiring the unit-weight normalization of [18].

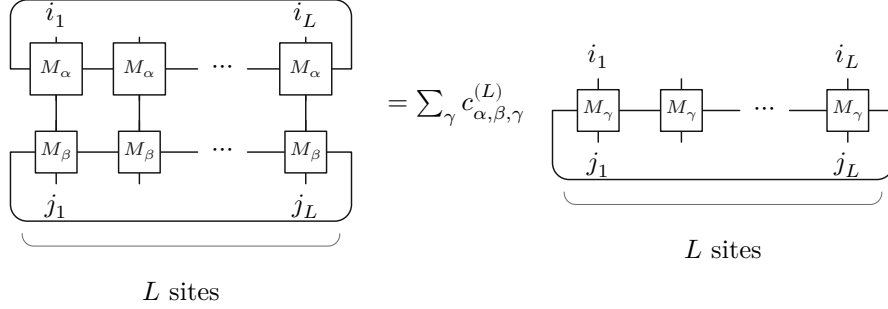
Proof. [✓ Proof](#) Theorem 21.2.2.67 and Theorem 21.2.2.74 show that R generates MPDOs and has the direct-sum form with normalized normal tensor blocks. Theorem 21.2.2.77 constructs the completely positive trace-preserving maps S, T satisfying $S[R_2(X)] = R_1(X)$ and $T[R_1(X)] = R_2(X)$ for all X , while Theorem 21.2.2.78 provides its single-element vertical basis-of-normal-tensors decomposition satisfying the algebra condition. Its structure coefficients are $c_{0,0,0}^{(L)} = 1 + (7/25)^L$. The values at lengths one and two differ, so these coefficients are not length independent. $\square$

Definition 21.2.2.97 (Algebra conditions for a basis of normal tensors). [Lean](#) [✓ Stmt](#) Fix structure coefficients $c_{\alpha,\beta,\gamma}^{(L)}$, operators $O_L(M_\alpha)$, and scalars $m_\alpha = \text{tr}(\mu_\alpha)$. The algebra conditions

require a length-independent family of positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ such that, for every $L > 0$, $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ and

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} c_{\alpha,\beta,\gamma}^{(L)} O_L(M_\gamma),$$

which reads, with each $O_L(M_\alpha)$ drawn as a closed MPO chain,



together with

$$m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta.$$

Thus the vector $m = (m_\alpha)_\alpha$ is an idempotent for the multiplication induced by $c^{(1)}$. These are the algebra conditions in [11, Theorem 4.14(ii), labelled IV.13 in the source, lines 972–985], without an additional comparison with a chosen blocked basis.

Definition 21.2.2.98 (Algebra conditions for a vertical canonical decomposition). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Choose a vertical canonical decomposition

$$U\widetilde{M}U^\dagger = \bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha},$$

$$\mu_{\alpha} = \text{diag}(\omega_{\alpha,0}, \dots, \omega_{\alpha,r_{\alpha}-1}).$$

Here the μ_{α} are positive diagonal matrices with nonempty multiplicity spaces, and U is a coisometry onto the retained sectors, so $UU^\dagger = \mathbb{1}$. The decomposition also requires exact reconstruction:

$$\widetilde{M} = U^\dagger \left(\bigoplus_{\alpha} \mu_{\alpha} \otimes M_{\alpha} \right) U.$$

These tensor identities hold for every vertical physical index. In particular, any complementary sector is zero. Separating the two factors of the doubled physical index of M_{α} gives an MPO tensor $\widehat{M}_{\alpha}$. Define

$$O_L(M_{\alpha}) = \text{mpo}(\widehat{M}_{\alpha}, L),$$

$$m_{\alpha} = \sum_q \omega_{\alpha,q} = \text{tr}(\mu_{\alpha}).$$

The chosen decomposition satisfies the algebra conditions when the tensors M_{α} form a basis of normal tensors of the vertical tensor and the conditions of Definition 21.2.2.97 hold for these operators and scalars. These are the objects and equations in [11, Theorem 4.14(ii), lines 972–993, and Appendix C.4, lines 2046–2064]; a comparison with a two-site vertical canonical decomposition is an additional requirement.

Theorem 21.2.2.99 (Positivity of the physical trace in a retained vertical sector). [Lean](#) [✓ Stmt](#) Let M generate a family of MPDOs with physical dimension d and bond dimension D , and choose a vertical canonical decomposition satisfying the algebra conditions of Definition 21.2.2.98. Let M_γ be any retained normal tensor. Separating its doubled physical index gives two physical legs of dimension D , corresponding to the original auxiliary indices of M . Closing these legs gives the bond matrix

$$T_\gamma = \sum_{a=0}^{D-1} M_\gamma^{(a,a)}. \quad (21.7)$$

Then $T_\gamma \succeq 0$. This follows from the chosen decomposition and positivity of the one-site operator $\rho^{(1)}(M)$; it does not require fusion maps or a renormalization fixed point hypothesis.

Proof. [✓ Proof](#) Apply the retained physical coisometry to the positive one-site operator. For a positive diagonal coordinate q of the multiplicity matrix μ_γ , the corresponding principal submatrix is $(\mu_\gamma)_{q,q} T_\gamma$. Principal submatrices remain positive semidefinite, and $(\mu_\gamma)_{q,q} > 0$; division by this scalar proves $T_\gamma \succeq 0$. $\square$

Lemma 21.2.2.100 (Physical traces of a nonunitary similarity pair). [Lean](#) [Lean](#) [✓ Stmt](#) Let A have letters $\mathbb{1}, E_{00}, E_{01}, E_{10}$, let $X = \text{diag}(2, 1)$, and set $B^v = X A^v X^{-1}$. After separating the doubled physical index, closing the physical legs gives

$$T_A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

$$T_B = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{pmatrix}.$$

Proof. [✓ Proof](#) Sum the two diagonal physical letters $A^{(0,0)} + A^{(1,1)}$ and then use $B^v = X A^v X^{-1}$ with $X = \text{diag}(2, 1)$. $\square$

Lemma 21.2.2.101 (The physical traces of the nonunitary similarity pair are not positive semidefinite). [Lean](#) [Lean](#) [✓ Stmt](#) Neither T_A nor T_B in Lemma 21.2.2.100 is positive semidefinite.

Proof. [✓ Proof](#) A positive semidefinite matrix is Hermitian. In both matrices the $(0, 1)$ entry is zero, whereas the $(1, 0)$ entry is respectively 1 and $1/2$. Hence neither matrix is Hermitian. $\square$

Theorem 21.2.2.102 (The nonunitary similarity pair is excluded from retained positive sectors). [Lean](#) [Lean](#) [✓ Stmt](#) Neither A nor B can occur, up to the bond-dimension identification, as a retained sector of bond dimension two in a vertical canonical decomposition of an MPDO tensor satisfying Definition 21.2.2.98.

Proof. [✓ Proof](#) For any such retained sector, closing the physical legs gives a positive semidefinite matrix by Theorem 21.2.2.99. Lemma 21.2.2.101 gives the contradiction for both tensors. $\square$

Lemma 21.2.2.103 (The nonunitary similarity pair has no positive isometric realization). [Lean](#) [✓ Stmt](#) Let A have letters $\mathbb{1}, E_{00}, E_{01}, E_{10}$, let $X = \text{diag}(2, 1)$, and set $B^i = X A^i X^{-1}$. There are no $c > 0$ and $V \in M_2(\mathbb{C})$ satisfying $V^\dagger V = \mathbb{1}$ and $V^\dagger B^i V = c A^i$ for every letter i .

Proof. [✓ Proof](#) The identity letter forces $c = 1$. The matrix $S = V^\dagger X$ then commutes with all four letters of A , which span $M_2(\mathbb{C})$, and hence S is scalar. Together with $V^\dagger V = \mathbb{1}$, this would make $X^\dagger X = \text{diag}(4, 1)$ scalar, a contradiction. $\square$

Definition 21.2.2.104 (Linear extension on each virtual block). [✓ Stmt](#) For injective families A_k and B_k with the same closed matrix product vectors, the linear extension on block k is the unique linear map $T_k : M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for every letter i .

Theorem 21.2.2.105 (Multiplicativity of the linear extension on each virtual block). [✓ Stmt](#) The linear extension on block k satisfies $T_k(MN) = T_k(M)T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.

Proof. [✓ Proof](#) The letters of A_k span $M_{D_k}(\mathbb{C})$, and equality of the closed matrix product vectors gives the identity on products of spanning letters. $\square$

Theorem 21.2.2.106 (Bijectivity of the linear extension on each virtual block). [✓ Stmt](#) The linear extension T_k on block k is bijective.

Proof. [✓ Proof](#) Multiplicativity makes the kernel an ideal of the simple matrix algebra. The map is nonzero: otherwise every letter of B_k would vanish, contradicting the nonvanishing trace pairing for the injective family. Thus the kernel is zero, and finite dimensionality gives surjectivity. $\square$

Theorem 21.2.2.107 (Positive linear extensions on virtual blocks are unitarily implemented).

[✓ Stmt](#) If T_k is positive, then there is a unitary matrix U_k such that $T_k(M) = U_k M U_k^\dagger$ for every $M \in M_{D_k}(\mathbb{C})$.

Proof. [✓ Proof](#) The map on each block is a bijective algebra endomorphism. Positivity makes it preserve adjoints, so unitary Skolem–Noether implements it by conjugation with a unitary matrix. $\square$

Theorem 21.2.2.108 (Closed matrix product vectors do not force a positive virtual extension).

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #5284](#) Regard the tensors A and B of Lemma 21.2.2.103 as one-block families. They are injective and have equal closed matrix product vectors at every length, but the linear extension on their virtual block is not positive.

Proof. [✓ Proof](#) If the linear extension were positive, its unitary-implementation theorem would give a unitary U satisfying $B^i = U A^i U^\dagger$ for every letter i . Hence $U^\dagger B^i U = A^i$, giving the forbidden realization with $c = 1$ in Lemma 21.2.2.103. $\square$

Definition 21.2.2.109 (Bond-two vertical tensor and Bell matrix). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $I = \{0, 1\}$ and let $E_{a,b}$ be the matrix units of $M_2(\mathbb{C})$. Define a tensor A by

$$A^{((a,b),(c,d))} = \begin{cases} E_{a,b}, & (a,b) = (c,d), \\ 0, & (a,b) \neq (c,d). \end{cases}$$

Its physical trace is $J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. Define $|\Phi\rangle = |0,0\rangle + |1,1\rangle$ and $P = |\Phi\rangle\langle\Phi|$. Thus P is an unnormalized rank-one positive operator, not an orthogonal projection. The tensor A is the vertical tensor of the physical model in Definition 21.2.3.1. Theorem 21.2.3.4 identifies P with its two-site operator after reindexing. For $X = \frac{1}{2} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, set $B^v = X A^v X^{-1}$ and define

$$\begin{aligned} J_X &= X J X^{-1}, \\ P_X &= (X \otimes X) P (X^{-1} \otimes X^{-1}). \end{aligned}$$

Proof. ✓ Proof The cyclic equality weights force the ket and bra words to be independently constant. The result is an outer product and is therefore positive semidefinite. $\square$

Theorem 21.2.3.4 (Two-site Bell identity under reindexing). Lean ✓ Stmt Let ε identify a binary word of length two with its ordered pair of letters, and put $|\Phi\rangle = |0, 0\rangle + |1, 1\rangle$. Then

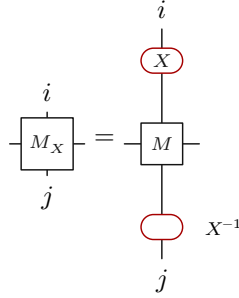
$$\text{reindex}_{\varepsilon, \varepsilon}(\rho^{(2)}(M)) = |\Phi\rangle\langle\Phi|.$$

Thus P is exactly the two-site physical operator in ordered-pair coordinates.

Proof. ✓ Proof Specialize Theorem 21.2.3.3 to $N = 2$ and evaluate the coordinate identification. $\square$

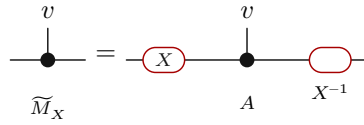
Definition 21.2.3.5 (Physical similarity of the MPO tensor). Lean ✓ Stmt For $X \in \text{GL}(2, \mathbb{C})$, define M_X by applying X to the ket index and X^{-1} to the bra index of M :

$$M_X^{i,j} = \sum_{a,b \in I} X_{i,a} (X^{-1})_{b,j} M^{a,b}.$$



This family is motivated by the sector comparison in [11, Appendix C.4, lines 2048–2057] and the physical-sector argument of [11, Proposition 4.13, lines 1898–1921]. Neither passage states this deformation or the results below.

Theorem 21.2.3.6 (Vertical tensor under physical similarity). Lean ✓ Stmt For every vertical letter v , the vertical tensor of M_X satisfies $\widetilde{M}_X^v = X A^v X^{-1}$. Drawing the vertical letter v as the upper leg, the physical similarity becomes a gauge on the virtual legs of A :



Proof. ✓ Proof Left and right multiplication on the physical indices become matrix multiplication on the vertically viewed tensor. Now use Theorem 21.2.3.2. $\square$

Theorem 21.2.3.7 (One-site physical-similarity identity). Lean ✓ Stmt After identifying one-site words with their single letters in I , the one-site operator of M_X is

$$\text{reindex}(\rho^{(1)}(M_X)) = X J X^{-1}.$$

Proof. ✓ Proof Expand the one-site contraction. The four diagonal physical letters sum to J , and the two physical multiplications give conjugation by X . $\square$

Theorem 21.2.3.8 (Two-site physical-similarity identity). [Lean](#) [✓ Stmt](#) Under the ordered-pair identification ε from Theorem 21.2.3.4,

$$\text{reindex}_{\varepsilon, \varepsilon}(\rho^{(2)}(M_X)) = (X \otimes X)P(X^{-1} \otimes X^{-1}).$$

Proof. [✓ Proof](#) Evaluate the two-site contraction in ordered-pair coordinates. Each ket index contributes X , each bra index contributes X^{-1} , and the undeformed contraction is P . $\square$

Theorem 21.2.3.9 (MPDO positivity forces a scalar physical Gram matrix). [Lean](#) [✓ Stmt](#) If M_X generates MPDOs, then there is a real number $\omega > 0$ such that $X^\dagger X = \omega \mathbb{1}$. Thus a physical similarity with nonscalar $X^\dagger X$ does not preserve MPDO positivity in this family.

Proof. [✓ Proof](#) Positivity of the one- and two-site operators implies Hermiticity. Apply the two exact reindexing identities and then Theorem 21.2.2.111. $\square$

Lemma 21.2.3.10 (The concrete physical Gram matrix is nonscalar). [Lean](#) [✓ Stmt](#) For $X = \frac{1}{2} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ and every $\omega \in \mathbb{R}$, we have $X^\dagger X \neq \omega \mathbb{1}$.

Proof. [✓ Proof](#) The $(0, 1)$ entry of $X^\dagger X$ equals $3/2$, whereas every scalar matrix has vanishing $(0, 1)$ entry. $\square$

Theorem 21.2.3.11 (The concrete physical similarity does not generate MPDOs). [Lean](#) [✓ Stmt](#) For $X = \frac{1}{2} \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$, the physical deformation M_X does not generate MPDOs.

Proof. [✓ Proof](#) Otherwise Theorem 21.2.3.9 would make $X^\dagger X$ scalar, contrary to Lemma 21.2.3.10. $\square$

Theorem 21.2.3.12 (Canonical form in the original physical direction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) As an MPS with four doubled physical symbols, M is the direct sum of four bond-one normal tensors, one supported on each symbol, all with weight one. Hence M itself is in canonical form, without a change of basis or blocking.

Proof. [✓ Proof](#) Each block has identity transfer map, and their direct sum is the diagonal matrix-unit tensor M . $\square$

Theorem 21.2.3.13 (Cyclic-edge projection in the vertical direction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Encode the operator indices of the vertical tensor A as directed edges on I . For $L \geq 1$, let $q_L(w)$ indicate that consecutive edges in w meet cyclically. The closed operator in this direction is

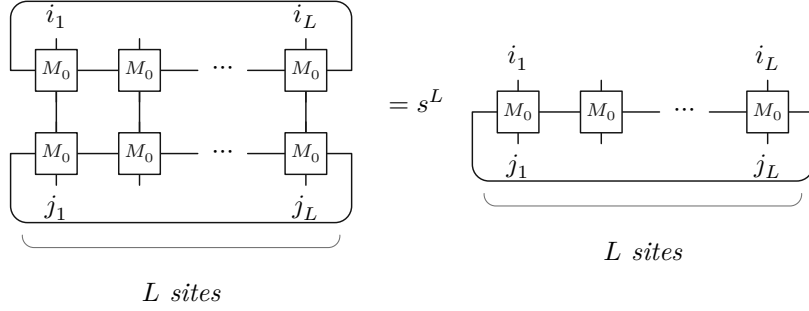
$$\begin{aligned} O_L(A) &= \text{diag}(q_L), \\ O_L(A)^\dagger &= O_L(A) = O_L(A)^2. \end{aligned}$$

Thus $O_L(A)$ is an orthogonal projection and satisfies an algebra law with one label and structure coefficient one. It is distinct from the unnormalized GHZ rank-one operator in the original physical direction in Theorem 21.2.3.3. At $L = 0$, the vertical contraction is $O_0(A) = 2\mathbb{1}$, whereas the empty cyclic indicator gives $\mathbb{1}$. In particular, this contraction is not idempotent at length zero, so neither the projection identity nor the coefficient-one algebra law extends to $L = 0$.

Proof. [✓ Proof](#) The cyclic indicator takes only the values zero and one, so its diagonal matrix is Hermitian and idempotent. $\square$

Theorem 21.2.3.14 (Normalized one-element vertical BNT and its algebra). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Put $s = 2^{-1/2}$ and $\hat{A} = sA$. The tensor $\hat{A}$ is left-canonical and is a normal tensor with transfer spectral radius one. A vertical basis of normal tensors consists of the single tensor $M_0 = \hat{A}$. With one copy, take the positive diagonal matrices $\mu_0 = (s^{-1})$ and $\chi_{0,0,0} = (s)$. The structure coefficient is $c_{0,0,0}^{(L)} = \text{tr}(\chi_{0,0,0}^L) = s^L$. For $L \geq 1$,

$$\begin{aligned} O_L(M_0) &= s^L O_L(A), \\ O_L(M_0)^2 &= s^L O_L(M_0). \end{aligned}$$



Each site carries M_0 with its doubled physical index separated into an upper and a lower leg, so each periodic row is $O_L(M_0)$. The weighted tensor $\mu_0 \otimes M_0$ reconstructs A exactly. For $m_0 = \text{tr}(\mu_0) = s^{-1}$, the identity $m_0 = c_{0,0,0}^{(1)} m_0^2$ says that the vector m is an idempotent for the multiplication induced by $c^{(1)}$. At $L = 0$, however, $O_0(M_0) = 2\mathbb{1}$, so $O_0(M_0)^2 = 4\mathbb{1} \neq 2\mathbb{1}$; the algebra identity therefore requires positive length.

Proof. [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) The identity $s^2 = 1/2$ gives left-canonical normalization. Scaling every local tensor contributes s^L to a length- L contraction, and s^{-1} cancels the one-site scaling in the weighted reconstruction. $\square$

Theorem 21.2.3.15 (Coisometry identifying the vertical sector coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) The unique label and copy identify the vertical sector coordinate with I . The corresponding permutation matrix W satisfies $WW^\dagger = \mathbb{1}$. For every $Z \in M_2(\mathbb{C})$, conjugation by W and reconstruction by its adjoint give

$$\begin{aligned} WZW^\dagger &= \text{reindex}(Z), \\ W^\dagger \text{reindex}(Z)W &= Z. \end{aligned}$$

Proof. [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Each row and column of W contains one nonzero entry, equal to one. $\square$

Theorem 21.2.3.16 (One-element vertical BNT algebra for the physical tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) The physical tensor M has a vertical basis of normal tensors with one label, one copy, and bond dimension two. Its normalized representative is $M_0 = \hat{A}$, its multiplicity matrix is $\mu_0 = (s^{-1})$, and $\chi_{0,0,0} = (s)$ gives the structure coefficient $c_{0,0,0}^{(L)} = \text{tr}(\chi_{0,0,0}^L) = s^L$ at every positive length. The coisometry W identifies the vertical sector coordinates with the original physical coordinates. Both the decomposition and its reconstruction hold exactly for every vertical letter.

Definition 21.2.3.17 (BNT multiplicity-spectrum comparison). [Lean](#) [✓ Stmt](#) Let μ_α and ν_γ be diagonal matrices with complex entries, representing one-site and two-site multiplicities, respectively. A multiplicity-spectrum comparison requires $\text{tr}(\mu_\alpha) = m_\alpha$ and, for every γ , equality with multiplicity between the entries of ν_γ and the products $\mu_{\alpha,i}\mu_{\beta,j}\chi_{\alpha,\beta,\gamma,k}$. For the positive diagonal matrices in the vertical canonical decompositions, this is the conclusion of the BNT comparison and positive power-sum argument in [11, Appendix C.4, lines 2048–2058]. The definition itself does not require positivity or a realization by an MPDO tensor.

Definition 21.2.3.18 (Two-site multiplicity spectrum). [Lean](#) [✓ Stmt](#) Let M have a vertical decomposition $\widetilde{M} = \bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ with a basis of normal tensors $\{M_\alpha\}_\alpha$ and positive diagonal matrices μ_α , satisfying the BNT algebra condition. A two-site multiplicity spectrum consists of a vertical canonical decomposition

$$\widetilde{M^{(2)}} = \bigoplus_\delta \nu_\delta \otimes A_\delta,$$

a bijection σ from the one-site sectors to the two-site sectors such that M_γ and $A_{\sigma(\gamma)}$ generate the same matrix product vectors at every positive length, and, for every γ , the multiset equality

$$\{\nu_{\sigma(\gamma),r} : r\} = \{\mu_{\alpha,i}\mu_{\beta,j}\chi_{\alpha,\beta,\gamma,k} : \alpha, \beta, i, j, k\}.$$

This is the decomposition and comparison used in [11, Appendix C.4, lines 2046–2058].

Definition 21.2.3.19 (Exact invertible gauges for the two-site sectors). [Lean](#) [✓ Stmt](#) Exact invertible gauges for a two-site multiplicity spectrum require, for every one-site sector γ , equality of the paired bond dimensions $d_\gamma = d_{\sigma(\gamma)}$ and an invertible matrix $Z_\gamma \in \text{GL}(d_\gamma, \mathbb{C})$ such that, for every physical index i of the vertical tensor,

$$A_{\sigma(\gamma)}^i = Z_\gamma M_\gamma^i Z_\gamma^{-1}.$$

In the vertical direction the gauge acts on the virtual legs of the matched normal tensor:

Thus the matched normal tensors have no residual scalar phase. This is the exact invertible conjugacy in [11, Appendix C.4, lines 2053–2057]. The definition does not require Z_γ to be unitary.

Theorem 21.2.3.20 (Two-site multiplicity spectrum from the BNT algebra condition). [Lean](#) [✓ Stmt](#) Let M generate MPDOs, and suppose that its MPS tensor obtained by pairing the physical indices is in canonical form. Suppose also that M has a vertical decomposition $\widetilde{M} = \bigoplus_\alpha \mu_\alpha \otimes M_\alpha$ satisfying the BNT algebra condition. There is a vertical canonical decomposition of the two-site blocking,

$$\widetilde{M^{(2)}} = \bigoplus_\delta \nu_\delta \otimes A_\delta.$$

Its normal-tensor sectors are in bijection, via σ , with the sectors of the one-site product expansion, in which every sector occurs. The tensors M_γ and $A_{\sigma(\gamma)}$ generate the same matrix product vectors at every positive length and, for every γ ,

$$\{\nu_{\sigma(\gamma),r} : r\} = \{\mu_{\alpha,i}\mu_{\beta,j}\chi_{\alpha,\beta,\gamma,k} : \alpha, \beta, i, j, k\} \quad (21.8)$$

as multisets. This is the multiplicity-spectrum comparison of Appendix C.4, lines 2046–2058 of [11]. The paired bond dimensions agree, and there are invertible matrices Z_γ satisfying, for every physical index i of the vertical tensor,

$$A_{\sigma(\gamma)}^i = Z_\gamma M_\gamma^i Z_\gamma^{-1}. \quad (21.9)$$

This statement asserts exact invertible conjugacy, without requiring the matrices Z_γ to be unitary.

Proof. ✓ Proof Multiplying the two one-site decompositions and using the BNT algebra law expresses the two-site closed chain in terms of $O_L(M_\gamma)$, with coefficient

$$\sum_{\alpha, \beta, i, j, k} (\mu_{\alpha, i} \mu_{\beta, j} \chi_{\alpha, \beta, \gamma, k})^L.$$

All these coefficients are positive. Hence every basis sector occurs, and the characterization of bases of normal tensors matches these sectors bijectively with those of the two-site vertical canonical decomposition. Normalization makes each matching scalar have modulus one, while positivity of two consecutive coefficients makes it positive; it is therefore one. The normal-tensor matching theorem also supplies equality of the paired bond dimensions and an invertible gauge; after the scalar is one, its gauge-phase relation is precisely the exact conjugacy in (21.9). Eventual linear independence of the basis vectors gives the corresponding equality of power sums. The finite positive power-sum identity then gives the multiset equality (21.8). $\square$

Theorem 21.2.3.21 (Physical compression onto a conjugate two-site sector). Lean Lean ✓ Stmt
 Suppose that the one-site and two-site sectors are related by the exact invertible gauges above. Let γ be a one-site sector matched with a two-site sector by the bijection σ , and identify their bond dimensions. There are an isometry V_γ from the sector bond space into the blocked physical space and a scalar $c_\gamma > 0$ such that, for every vertical letter v ,

$$V_\gamma^\dagger \widetilde{M^{(2)}}^v V_\gamma = c_\gamma Z_\gamma M_\gamma^v Z_\gamma^{-1}. \quad (21.10)$$

If, in addition, M generates matrix product density operators, then for every pair of open bond indices and every pair of two-site tail words, the marked-chain coefficient of

$$(Z_\gamma^\dagger Z_\gamma) M_\gamma^v (Z_\gamma^\dagger Z_\gamma)^{-1}$$

equals the corresponding reflected-adjoint marked-chain coefficient in the adjoint two-site blocking. This is one of the two marked comparisons in Figures 7–8 of [11, Proposition 4.13, lines 1909–1919], applied to the sector matching of Appendix C.4, lines 2048–2057.

The compression in (21.10) selects a conjugate of M_γ , not M_γ itself. It does not provide an isometric compression onto a positive multiple of M_γ or establish the Gram identity

$$(Z_\gamma^\dagger Z_\gamma) M_\gamma^v (Z_\gamma^\dagger Z_\gamma)^{-1} = M_\gamma^v.$$

This distinction is documented in [89]. The comparison with a common unblocked tail below proves the Gram identity without requiring such a compression of the two-site blocking.

Proof. [✓ Proof](#) Choose the first positive-weight copy of the matched two-site sector. Its inclusion is isometric and its compression selects the corresponding weighted normal tensor. Substituting the exact sector conjugacy gives (21.10), with the selected weight as c_γ . The reflected marked-chain identity for this compression, conjugated on its open indices by $Z_\gamma^\dagger$ and $(Z_\gamma^{-1})^\dagger$, gives the stated equality of marked-chain coefficients; the gauge factors cancel on the reflected side. $\square$

Theorem 21.2.3.22 (Physical compression onto a matched one-site sector). [Lean](#) [✓ Stmt](#) *In the setting above, each sector γ admits an isometry $W_\gamma: \mathbb{C}^{d_\gamma} \rightarrow \mathbb{C}^d$ and a scalar $a_\gamma > 0$ such that, for every vertical letter v ,*

$$W_\gamma^\dagger \widetilde{M}^v W_\gamma = a_\gamma M_\gamma^v. \quad (21.11)$$

Here the bond space of M_γ is identified with that of its matched two-site sector. This compression selects a positive-weight copy in the one-site vertical decomposition of [11, Proposition 4.13, statement lines 943–951, and Appendix C.4, lines 2046–2057].

Proof. [✓ Proof](#) Select the first copy of M_γ in the one-site vertical direct sum. Its inclusion is isometric. Compressing the reconstruction identity by this inclusion gives (21.11), with a_γ equal to the positive multiplicity weight of the selected copy. Identify the paired bond spaces using equality of their dimensions. $\square$

Theorem 21.2.3.23 (Reflected marked chains with a two-site initial block). [Lean](#) [✓ Stmt](#) *Let M generate matrix product density operators, let A be a vertical tensor, and suppose that an isometry V selects from the two-site blocking a positive multiple of an invertible conjugate of A . For some scalar $b > 0$, this means that, for every vertical letter v ,*

$$V^\dagger \widetilde{M}^{(2)}{}^v V = b X A^v X^{-1}. \quad (21.12)$$

For every pair of open bond indices and every pair of equal-length words in the original physical alphabet, the marked-chain coefficient of

$$(X^\dagger X) A^v (X^\dagger X)^{-1}$$

followed by the corresponding unblocked tail of M equals the reflected-adjoint marked-chain coefficient of A followed by the reversed tail of $M^\dagger$. Thus the first two sites are compressed jointly, while all later sites remain unblocked.

Keeping the tail unblocked supplies the common comparison needed when [11, Proposition 4.13, lines 1909–1919] is invoked in [11, Appendix C.4, lines 2048–2057]. This correction to the direct two-site comparison is recorded in [89].

Proof. [✓ Proof](#) Compress the first two physical sites of the density operator of length $N + 2$ by V . Hermiticity exchanges the open bond indices and the two physical words. Taking adjoints reverses only the remaining N one-site factors. Finally conjugate the open indices by $X^\dagger$ and $(X^{-1})^\dagger$. The left side becomes conjugation by $X^\dagger X$, while the gauge factors cancel on the reflected side. $\square$

Theorem 21.2.3.24 (Two-sided compression onto a matched two-site sector). [Lean](#) [Lean](#) [✓ Stmt](#) *For a matched two-site sector, choose V_γ and $c_\gamma > 0$ as in (21.10), and set*

$$\begin{aligned} L_\gamma &= V_\gamma (Z_\gamma^{-1})^\dagger, \\ R_\gamma &= V_\gamma Z_\gamma. \end{aligned} \quad (21.13)$$

Then, for every vertical letter v ,

$$L_\gamma^\dagger \widetilde{M^{(2)}}^v R_\gamma = c_\gamma M_\gamma^v, \quad (21.14)$$

$$R_\gamma^\dagger \widetilde{M^{(2)}}^v L_\gamma = c_\gamma (Z_\gamma^\dagger Z_\gamma) M_\gamma^v (Z_\gamma^\dagger Z_\gamma)^{-1}. \quad (21.15)$$

The first identity replaces the isometry on one side by the oblique pair (L_γ, R_γ) :

Write $\mathcal{R}(M_\gamma)^{rs}$ for the horizontal slice $v \mapsto (M_\gamma^v)_{rs}$. For open bond indices r, s and blocked physical indices i, j , define

$$f_{(r,s),(i,j)}^\gamma = c_\gamma^{-1} \overline{(L_\gamma)_{ir}} (R_\gamma)_{js}. \quad (21.16)$$

These coefficients express every horizontal slice of M_γ as a linear combination of the physical letters of the two-site blocking:

$$\sum_{i,j} f_{(r,s),(i,j)}^\gamma (M^{[2]})^{ij} = \mathcal{R}(M_\gamma)^{rs}. \quad (21.17)$$

This gives the inclusion in the physical-letter span used to compare the two vertical forms in [11, Appendix C.4, lines 2048–2057].

In general $L_\gamma \neq R_\gamma$, so (21.14) is not an isometric compression onto a positive multiple of M_γ . At the marked-chain level, adjoint reflection exchanges the two maps together with the bond-pair and tail reversal. The reflected orientation is therefore governed by (21.15), rather than by a second copy of M_γ . The linear-span identity alone gives neither a common reflected tensor nor the Gram identity

$$(Z_\gamma^\dagger Z_\gamma) M_\gamma^v (Z_\gamma^\dagger Z_\gamma)^{-1} = M_\gamma^v.$$

This prevents the direct two-site comparison from proving the Gram identity. The theorem below instead compares the one-site compression onto M_γ with the two-site compression onto $Z_\gamma M_\gamma Z_\gamma^{-1}$, keeping the same unblocked tail.

Proof. ✓ Proof Taking adjoints in the left factor of (21.13) and substituting (21.10) gives

$$L_\gamma^\dagger \widetilde{M^{(2)}}^v R_\gamma = Z_\gamma^{-1} (V_\gamma^\dagger \widetilde{M^{(2)}}^v V_\gamma) Z_\gamma = c_\gamma M_\gamma^v.$$

Reversing the two maps instead gives (21.15). Finally, taking the (r, s) matrix entry of a general two-sided compression yields

$$[L^\dagger \widetilde{M^{(2)}}^v R]_{rs} = \sum_{i,j} \overline{L_{ir}} R_{js} (M^{[2]})_v^{ij}.$$

Divide by c_γ and use (21.14) to obtain (21.17). □

Theorem 21.2.3.25 (Normalization of conjugation by a scalar Gram matrix). Lean ✓ Stmt If an invertible matrix Z satisfies $Z^\dagger Z = \omega \mathbb{1}$ with $\omega > 0$, then conjugation by $\omega^{-1/2} Z$ equals conjugation by Z .

Proof. [✓ Proof](#) The Gram identity identifies Z^{-1} with $\omega^{-1}Z^\dagger$; hence, for every matrix B of the same size, $(\omega^{-1/2}Z)B(\omega^{-1/2}Z)^\dagger = ZBZ^{-1}$. $\square$

Theorem 21.2.3.26 (Unitary sector conjugacy from a positive scalar Gram matrix). [Lean](#)

[✓ Stmt](#) Let Z_γ be an exact invertible sector gauge and suppose $Z_\gamma^\dagger Z_\gamma = \omega_\gamma \mathbb{1}$ with $\omega_\gamma > 0$. Define $U_\gamma = \omega_\gamma^{-1/2} Z_\gamma$. Then U_γ is unitary and, for every vertical letter v ,

$$A_{\sigma(\gamma)}^v = U_\gamma M_\gamma^v U_\gamma^\dagger. \quad (21.18)$$

This is the inverse-square-root normalization used in [11, Proposition 4.13, lines 1903–1908], applied at Appendix C.4, lines 2053–2057.

Proof. [✓ Proof](#) The positive scalar Gram identity gives $U_\gamma^\dagger U_\gamma = \mathbb{1}$ and $Z_\gamma^{-1} = \omega_\gamma^{-1} Z_\gamma^\dagger$. Consequently $U_\gamma M_\gamma^v U_\gamma^\dagger = Z_\gamma M_\gamma^v Z_\gamma^{-1}$. The exact sector conjugacy then gives (21.18). $\square$

Theorem 21.2.3.27 (Gram identity and unitary sector conjugacy). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let M generate MPDOs, and suppose that its MPS tensor obtained by pairing the physical indices is in canonical form. Let the one-site and two-site sectors be paired by the exact gauges Z_γ above. Then, for every sector γ and every vertical letter v ,

$$(Z_\gamma^\dagger Z_\gamma) M_\gamma^v (Z_\gamma^\dagger Z_\gamma)^{-1} = M_\gamma^v. \quad (21.19)$$

Consequently there is a real number $\omega_\gamma > 0$ such that

$$Z_\gamma^\dagger Z_\gamma = \omega_\gamma \mathbb{1}, \quad (21.20)$$

and there is a unitary U_γ satisfying

$$A_{\sigma(\gamma)}^v = U_\gamma M_\gamma^v U_\gamma^\dagger. \quad (21.21)$$

The proof compares a one-site compression onto M_γ with a two-site compression onto $Z_\gamma M_\gamma Z_\gamma^{-1}$, keeping the same unblocked tail. This supplies the common comparison missing from the direct application of [11, Proposition 4.13, lines 1909–1919] in [11, Appendix C.4, lines 2048–2057]. The correction to the direct two-site argument is recorded in [89].

Proof. [✓ Proof](#) Put $G_\gamma = Z_\gamma^\dagger Z_\gamma$. The two-site initial-block identity applied to the compression onto $Z_\gamma M_\gamma Z_\gamma^{-1}$ identifies the marked chains of $G_\gamma M_\gamma G_\gamma^{-1}$ with the reflected-adjoint chains of M_γ , followed by an arbitrary unblocked tail of $M^\dagger$. The ordinary one-site reflected identity applied to the compression onto M_γ gives the same reflected-adjoint chains.

Both marked tensors are linear combinations of the one-site physical letters. Indeed, the mark M_γ is obtained from the pair (W_γ, W_γ) , whereas the mark $G_\gamma M_\gamma G_\gamma^{-1}$ is obtained from the pair $(W_\gamma G_\gamma^\dagger, W_\gamma G_\gamma^{-1})$. Their marked traces agree against every positive-length tail. The original-coordinate form of Lemma L therefore identifies the two marked tensors and proves (21.19). Normality of M_γ gives (21.20), and normalizing Z_γ by $\omega_\gamma^{-1/2}$ gives the unitary in (21.21). $\square$

Lemma 21.2.3.28 (Idempotence in terms of the diagonal matrices). [Lean](#) [✓ Stmt](#) Suppose that the BNT algebra conditions hold. The vector m is then an idempotent for the multiplication induced by the length-one coefficients $\text{tr}(\chi_{\alpha,\beta,\gamma})$: for every γ ,

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

Proof. [✓ Proof](#) The trace-power representation at length one gives $c_{\alpha,\beta,\gamma}^{(1)} = \text{tr}(\chi_{\alpha,\beta,\gamma})$. Substitution into the idempotent law $m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta$ gives the stated identity. $\square$

Definition 21.2.3.29 (Multiplicity spectra from eventual equality of power sums). [Lean](#) [✓ Stmt](#) Let μ_α and ν_γ be the diagonal multiplicity matrices in the one-site and two-site vertical decompositions, respectively. Suppose that their diagonal entries and those of $\chi_{\alpha,\beta,\gamma}$ are strictly positive, and that $\sum_i \mu_{\alpha,i} = m_\alpha$ for every α . If, for every γ and all sufficiently large L ,

$$\sum_r \nu_{\gamma,r}^L = \sum_{\alpha,\beta,i,j,k} (\mu_{\alpha,i} \mu_{\beta,j} \chi_{\alpha,\beta,\gamma,k})^L,$$

then the multiset of entries $\nu_{\gamma,r}$ equals the multiset of products $\mu_{\alpha,i} \mu_{\beta,j} \chi_{\alpha,\beta,\gamma,k}$. These multiplicity matrices, together with this multiset equality and the one-site trace identity, give a comparison of the one-site and two-site multiplicity spectra.

Positivity makes all entries nonzero. Eventual equality of two finite sums of geometric sequences gives, for every $L \geq 1$,

$$\sum_r \nu_{\gamma,r}^L = \sum_{\alpha,\beta,i,j,k} (\mu_{\alpha,i} \mu_{\beta,j} \chi_{\alpha,\beta,\gamma,k})^L.$$

The finite power-sum identity determines both the cardinality and the multiset of entries.

Theorem 21.2.3.30 (Multiplicity trace normalization). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the BNT algebra conditions hold and that the one-site and two-site vertical multiplicity spectra satisfy the multiset comparison just defined. Then every two-site multiplicity matrix has trace m_γ :

$$\text{tr}(\nu_\gamma) = \sum_{\alpha,\beta,i,j,k} \mu_{\alpha,i} \mu_{\beta,j} \chi_{\alpha,\beta,\gamma,k} = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta = m_\gamma. \quad (21.22)$$

This is the normalization used for the preparation matrix ν_γ/m_γ in the implication from the algebra condition to the renormalization fixed point property in [11, Appendix C.4, lines 2058–2064].

Proof. [✓ Proof](#) Sum the multiset equality of the two spectra. For each α, β, γ , the resulting product sum satisfies

$$\sum_{i,j,k} \mu_{\alpha,i} \mu_{\beta,j} \chi_{\alpha,\beta,\gamma,k} = \left(\sum_i \mu_{\alpha,i} \right) \left(\sum_j \mu_{\beta,j} \right) \left(\sum_k \chi_{\alpha,\beta,\gamma,k} \right).$$

The identities

$$\begin{aligned} \sum_i \mu_{\alpha,i} &= m_\alpha, \\ \sum_j \mu_{\beta,j} &= m_\beta, \\ \sum_k \chi_{\alpha,\beta,\gamma,k} &= c_{\alpha,\beta,\gamma}^{(1)} \end{aligned}$$

therefore give

$$\text{tr}(\nu_\gamma) = \sum_{\alpha, \beta} m_\alpha m_\beta c_{\alpha, \beta, \gamma}^{(1)} = m_\gamma,$$

where the final equality is the length-one idempotent law. $\square$

Definition 21.2.3.31 (BNT algebra with a blocked-basis comparison). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Consider operators $O_L(M_\alpha)$, structure coefficients $c_{\alpha, \beta, \gamma}^{(L)}$, and scalars m_α indexed by a finite set of BNT labels. In the vertical MPDO decomposition, $m_\alpha = \text{tr}(\mu_\alpha)$. Require the BNT algebra conditions and a comparison with chosen blocked bases. The algebra conditions require positive diagonal matrices $\chi_{\alpha, \beta, \gamma}$ such that $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$ for every $L > 0$, and, at every such length, the product law

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} c_{\alpha, \beta, \gamma}^{(L)} O_L(M_\gamma).$$

The vector m is an idempotent for the multiplication induced by $c^{(1)}$, meaning that, for every γ ,

$$m_\gamma = \sum_{\alpha, \beta} c_{\alpha, \beta, \gamma}^{(1)} m_\alpha m_\beta.$$

The blocked-basis comparison is an additional condition. For every positive blocking length n , let σ_n assign BNT labels to the basis indices at length n , and let τ_n assign BNT labels to the basis indices at length $2n$. For indices i, j at length n and k at length $2n$, require

$$c_{i, j, k}^{(n)} = c_{\sigma_n(i), \sigma_n(j), \tau_n(k)}^{(n)}.$$

Here the left-hand side denotes the structure coefficient in the chosen blocked bases, while the right-hand side uses BNT labels. The same maps assign the diagonal matrix $\chi_{\sigma_n(i), \sigma_n(j), \tau_n(k)}$ to each triple of blocked-basis indices.

One may start with positive diagonal matrices $\chi_{\alpha, \beta, \gamma}$ and define the coefficients by their traces of powers. In that case, the product law, idempotent law, and blocked-basis comparison are required for those coefficients. Conversely, replacing coefficients satisfying the trace-power identity by $\text{tr}(\chi_{\alpha, \beta, \gamma}^L)$ preserves these conditions at every positive length.

Theorem 21.2.3.32 (BNT structure coefficients and product formulas). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Suppose that the BNT algebra conditions and blocked-basis comparison hold. Let $\chi_{\alpha, \beta, \gamma}$ be the associated positive diagonal matrices. For every $L > 0$,

$$c_{\alpha, \beta, \gamma}^{(L)} = c_{\alpha, \beta, \gamma}^{\chi, (L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L),$$

where $c_{\alpha, \beta, \gamma}^{\chi, (L)}$ denotes the trace of the L th power of $\chi_{\alpha, \beta, \gamma}$. Drawn as a closed chain of L copies of $\chi_{\alpha, \beta, \gamma}$, the coefficient is

$$c_{\alpha, \beta, \gamma}^{(L)} = \underbrace{\begin{array}{c} \chi_{\alpha, \beta, \gamma} \quad \chi_{\alpha, \beta, \gamma} \quad \dots \quad \chi_{\alpha, \beta, \gamma} \\ \square \quad \square \quad \dots \quad \square \end{array}}_{L \text{ factors}}$$

Every diagonal entry of every $\chi_{\alpha, \beta, \gamma}$ is strictly positive. Consequently, for every $L > 0$, the product identity reads

$$O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \text{tr}(\chi_{\alpha,\beta,\gamma}^L) O_L(M_\gamma),$$

and the idempotent identity reads

$$m_\gamma = \sum_{\alpha, \beta} \text{tr}(\chi_{\alpha, \beta, \gamma}) m_\alpha m_\beta.$$

Proof. ✓ Proof The assumed positive diagonal matrices satisfy $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$ for $L > 0$. Substitution into the product law, and into the idempotent law at length one, gives the stated formulas. $\square$

Theorem 21.2.3.33 (Positive diagonal matrices for the blocked coefficients). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the BNT algebra conditions and blocked-basis comparison hold. Positive diagonal matrices representing the blocked coefficients as traces of powers are obtained by pulling back the length-independent matrices $\chi_{\alpha,\beta,\gamma}$ along the comparison maps. At every positive blocking length n , define $\chi_{n,i,j,k} = \chi_{\sigma_n(i),\sigma_n(j),\tau_n(k)}$. Thus the two matrices have equal size, diagonal entries, and traces of all powers.

Proof. ✓ Proof Apply the construction of positive diagonal matrices for blocked coefficients to the comparison maps and the positive diagonal matrices $\chi_{\alpha, \beta, \gamma}$ representing the BNT structure coefficients. $\square$

Corollary 21.2.3.34 (Blocked coefficient trace-power formulas). [Lean](#) [Lean](#) [✓ Stmt](#) *Under the BNT algebra conditions and blocked-basis comparison, for every positive blocking length n , the blocked-basis coefficients are traces of powers of the matrices $\chi_{\alpha,\beta,\gamma}$ selected by the source and target label maps. Equivalently, they are traces of powers of the pulled-back matrices $\chi_{n,i,j,k}$:*

$$c_{i,j,k}^{(n)} = \text{tr}(\chi_{\sigma_n(i), \sigma_n(j), \tau_n(k)}^n) = \text{tr}(\chi_{n,i,j,k}^n).$$

Proof. ✓ Proof The blocked-basis comparison and the trace-power representation identify the coefficient with the trace of the n th power of the matrix selected by the source and target maps. By definition, the pulled-back matrix $\chi_{n,i,j,k}$ is this same matrix. $\square$

Definition 21.2.3.35 (BNT operators and structure coefficients). [Learn](#) [Learn](#) [Learn](#) [Learn](#)

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[Learn](#) [Learn](#) [Learn](#) [Learn](#) ✓ **Stmnt** Fix the algebra-structure data. Consider a finite set of labels α , complex vector spaces equipped with a product at each length L , and operators $O_L(M_\alpha)$ in those spaces, together with scalars m_α , structure coefficients $c_{\alpha,\beta,\gamma}^{(L)}$, and length-independent diagonal matrices $\chi_{\alpha,\beta,\gamma}$ with positive diagonal entries. Require the product, idempotent, trace-power, and blocked-basis comparison conditions of Definition [21.2.3.31](#).

In a vertical basis-of-normal-tensors decomposition, $O_L(M_\alpha)$ denotes the operator obtained by contracting L copies of M_α , and $m_\alpha = \text{tr}(\mu_\alpha)$ for the positive diagonal multiplicity matrix μ_α . The abstract conditions here do not themselves construct such a tensor decomposition. Existence means that these objects can be chosen simultaneously. The blocked-basis comparisons are additional conditions relating them to the fixed algebra-structure data, not clauses of the basis-of-normal-tensors algebra characterization.

One may instead start with the positive diagonal matrices and define $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, provided the product, idempotent, and blocked-basis comparison conditions hold. For any choice

Theorem 21.2.3.36 (Product and idempotent equations for BNT operators). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✔ Stmt](#) Suppose the objects in Definition 21.2.3.35 exist. For any such choice, the product law holds for every positive length L and all labels α, β , and the scalar identity holds for every label γ :

$$m_\gamma = \sum_{\alpha, \beta} c_{\alpha, \beta, \gamma}^{(1)} m_\alpha m_\beta. \quad (21.24)$$

$$m_\gamma = \sum_{\alpha, \beta} \text{tr}(\chi_{\alpha, \beta, \gamma}) m_\alpha m_\beta. \quad (21.26)$$

Let σ_n and τ_n be the source and target label maps in the additional blocked-basis comparison at a positive length n . For source indices i, j at length n and a target index k at length $2n$, the blocked-basis coefficients satisfy

$$c_{i,j,k}^{(n)} = c_{\sigma_n(i), \sigma_n(j), \tau_n(k)}^{(n)} = \text{tr}(\chi_{\sigma_n(i), \sigma_n(j), \tau_n(k)}^n).$$

Theorem 21.2.3.37 (Positive diagonal matrices for blocked coefficients). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the objects in Definition 21.2.3.35 exist for the chosen algebra-structure data. Then there are diagonal matrices $\chi_{n,i,j,k}$ with positive diagonal entries such that, for every positive blocked length n , source indices i, j at length n , and target index k at length $2n$.

$$\begin{aligned}\chi_{n,i,j,k} &= \chi_{\sigma_n(i),\sigma_n(j),\tau_n(k)}, \\ c_{i,j,k}^{(n)} &= \text{tr}(\chi_{n,i,j,k}^n).\end{aligned}$$

Proof. ✓ Proof Apply the blocked-basis trace-power consequence to the chosen operators, scalars, coefficients, and diagonal matrices. □

Theorem 21.2.5.3 (Trace of the ambient sector maps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For arbitrary ambient matrices $X \in M_d(\mathbb{C})$ and $Y \in M_d(\mathbb{C}) \otimes M_d(\mathbb{C})$,

$$\mathrm{Tr}_{\mathrm{sec}}(R_1(X)) = \mathrm{tr}(W_1 X W_1^\dagger), \quad (21.40)$$

$$\mathrm{Tr}_{\mathrm{sec}}(R_2(Y)) = \mathrm{tr}(W_2 \kappa_*^{-1}(Y) W_2^\dagger). \quad (21.41)$$

If X and Y are positive semidefinite, then

$$\mathrm{Tr}_{\mathrm{sec}}(R_1(X)) \leq \mathrm{tr}(X), \quad \mathrm{Tr}_{\mathrm{sec}}(R_2(Y)) \leq \mathrm{tr}(Y). \quad (21.42)$$

In the other direction, for $Z \in \mathcal{A}_1$ and $Z' \in \mathcal{A}_2^\sigma$,

$$\mathrm{tr}(\widetilde{R}_1(Z)) = \mathrm{Tr}_{\mathrm{sec}}(Z), \quad \mathrm{tr}(\widetilde{R}_2(Z')) = \mathrm{Tr}_{\mathrm{sec}}(Z'). \quad (21.43)$$

Hence the canonical full-matrix representatives of $\widetilde{R}_1$ and $\widetilde{R}_2$ are trace-preserving completely positive maps.

The inequalities in (21.42) may be strict: $W_i^\dagger W_i$ projects onto the retained nonzero sectors, so positive elements supported on the discarded orthogonal complement lose trace under the compression.

Proof. ✓ Proof Sectorwise partial trace and block-diagonal inclusion preserve the total trace, which gives (21.40) and (21.41). For $P_i = W_i^\dagger W_i$, cyclicity of trace gives

$$\mathrm{tr}(W_i Z W_i^\dagger) = \mathrm{tr}(P_i Z) \leq \mathrm{tr}(Z)$$

when $Z \succeq 0$. The relabelling κ_* preserves positivity and trace, proving (21.42). Finally, each normalized multiplicity matrix has trace one, and $W_i W_i^\dagger = I$ gives $\text{tr}(W_i^\dagger B W_i) = \text{tr}(B)$. This proves (21.43); complete positivity is supplied by Theorem 21.2.5.2. $\square$

Theorem 21.2.5.4 (Retraction identities for the ambient sector maps). [Lean](#) [Lean](#) [Lean](#)
 Stmt The raw retractions are left inverses of their normalized embeddings:

$$R_1 \widetilde{R}_1 = \text{id}_{\mathcal{A}_1}, \quad R_2 \widetilde{R}_2 = \text{id}_{\mathcal{A}_2^\sigma}. \quad (21.44)$$

Proof. ✔ Proof Compressing an expansion by $W_i^\dagger$ gives the original retained block-diagonal matrix because $W_i W_i^\dagger = I$. On every sector,

$$\begin{aligned} \mathrm{tr}_{\mathrm{mult}} \left(\frac{\mu_\gamma}{m_\gamma} \otimes X_\gamma \right) &= X_\gamma, \\ \mathrm{tr}_{\mathrm{mult}} \left(\frac{\nu_{\sigma(\gamma)}}{m_\gamma} \otimes Y_\gamma \right) &= Y_\gamma, \end{aligned} \quad (21.45)$$

where (21.45) follows from $\text{tr}(\nu_{\sigma(\gamma)}) = m_\gamma$. The two coordinate relabellings in the two-site composite cancel. These identities prove (21.44). $\square$

Theorem 21.2.5.5 (Action on the weighted tensor letters). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✔ Stmt](#) For every vertical letter v , the one-site maps satisfy

$$R_1(\widetilde{M}_v) = (m_\gamma M_\gamma^v)_{\gamma}, \quad (21.46)$$

$$\widetilde{R}_1((m_\gamma M_\gamma^v)_\gamma) = \widetilde{M}_v. \quad (21.47)$$

The relabelled two-site maps satisfy

$$R_2\left(\kappa_*(\widetilde{M^{[2]}_v})\right) = (m_\gamma A_{\sigma(\gamma)}^v)_\gamma, \quad (21.48)$$

$$\widetilde{R}_2((m_\gamma A_{\sigma(\gamma)}^v)_\gamma) = \kappa_*(\widetilde{M^{[2]}_v}). \quad (21.49)$$

Identities (21.46) and (21.47) combine the one-site weighted sector form of [11, Appendix C.4, lines 2048–2051] with, respectively, [11, Appendix C.4, lines 2067–2068] and [11, Appendix C.4, lines 2081–2083]. Identity (21.48) combines the matched weighted sector form of [11, Appendix C.4, lines 2048–2064] with the map R_2 of [11, Appendix C.4, lines 2078–2079]. Identity (21.49) combines the same sector form with the normalized map $\widetilde{R}_2$ of [11, Appendix C.4, lines 2070–2071].

Proof. ✓ Proof The vertical reconstruction identities give

$$\begin{aligned} \widetilde{M}_v &= W_1^\dagger \left(\bigoplus_\gamma \mu_\gamma \otimes M_\gamma^v \right) W_1, \\ \widetilde{M^{[2]}_v} &= W_2^\dagger \left(\bigoplus_\gamma \nu_{\sigma(\gamma)} \otimes A_{\sigma(\gamma)}^v \right) W_2. \end{aligned}$$

Compression removes the two outer coisometries, and partial trace over the multiplicity coordinate multiplies the simple-sector letter by $\text{tr}(\mu_\gamma) = m_\gamma$ or $\text{tr}(\nu_{\sigma(\gamma)}) = m_\gamma$. This proves (21.46) and (21.48). Preparing the normalized multiplicity matrix cancels the factor m_γ , yielding (21.47) and (21.49). $\square$

21.2.6 Trace-preserving completions and physical channels

Theorem 21.2.6.1 (Trace-preserving completion outside a projection). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt

Let $P \in M_m(\mathbb{C})$ be an orthogonal projection, let $\rho \in M_n(\mathbb{C})$ be positive semidefinite with $\text{tr}(\rho) = 1$, and let $\mathcal{E}: M_m(\mathbb{C}) \rightarrow M_n(\mathbb{C})$ be completely positive. Define the trace-and-prepare map and its complementary restriction by

$$\mathcal{D}_\rho(X) = \text{tr}(X)\rho, \quad \mathcal{C}_{P,\rho}(X) = \mathcal{D}_\rho((I - P)X(I - P)^\dagger). \quad (21.50)$$

Both maps are completely positive, and $\mathcal{D}_\rho$ is trace-preserving. If

$$\text{tr}(\mathcal{E}(X)) = \text{tr}(PX) \quad (21.51)$$

for every $X \in M_m(\mathbb{C})$, then the completion

$$\hat{\mathcal{E}}(X) = \mathcal{E}(X) + \mathcal{C}_{P,\rho}(X) \quad (21.52)$$

is trace-preserving and completely positive. It agrees with $\mathcal{E}$ on every matrix supported by P :

$$PXP = X \implies \hat{\mathcal{E}}(X) = \mathcal{E}(X). \quad (21.53)$$

Proof. ✓ Proof

The map $\mathcal{D}_\rho$ factors as state preparation, interchange of the two tensor factors, and partial trace. The map $X \mapsto (I - P)X(I - P)^\dagger$ has a single Kraus operator. Thus the two maps in (21.50) are completely positive. Since $P = P^\dagger = P^2$,

$$\text{tr}(\mathcal{C}_{P,\rho}(X)) = \text{tr}((I - P)X).$$

Theorem 21.2.6.3 (BNT algebra clause implies a renormalization fixed point). [Lean](#) [Lean](#)
[✓ Stmt](#) Let M be an MPDO whose doubled-index tensor is in literal CPSV canonical form. If M has a tensor-attached basis-of-normal-tensors algebra clause, then there are trace-preserving completely positive maps

$$\mathcal{T} : M_d(\mathbb{C}) \longrightarrow M_{d^2}(\mathbb{C}), \quad \mathcal{S} : M_{d^2}(\mathbb{C}) \longrightarrow M_d(\mathbb{C})$$

such that, for every virtual matrix X ,

$$\mathcal{T}[M_1(X)] = M_2(X), \quad \mathcal{S}[M_2(X)] = M_1(X). \quad (21.61)$$

Thus M is a renormalization fixed point in the sense of Definition 20.2.1. This is implication (ii) $\Rightarrow$ (i) of [11, Theorem 4.14, lines 972–993].

Here the common marked comparison needed to normalize the exact sector gauges is supplied by Theorem 21.2.3.27 [89], with the physical maps completed on the discarded complements as in [154].

Proof. [✓ Proof](#) The tensor-attached algebra clause determines the two-site vertical decomposition, the sector relabelling, and exact invertible gauges. Theorem 21.2.3.27 normalizes these gauges simultaneously to unitaries. Apply Theorem 21.2.6.2 to obtain the two trace-preserving completely positive maps and (21.61). $\square$

21.3 Transfer retracts and fusion coisometries

A linear factorization of the doubled-index blocked transfer map E_n through a subspace with the retract identities of (21.62) exists precisely when E_n is idempotent. Requiring such a factorization at every positive blocked size is equivalent to idempotence of E_1 . This transfer-retract condition is distinct from the physical renormalization fixed-point condition in Definition 20.2.1 and [11, Definition 4.1, lines 638–660]. For an MPDO tensor M in CF, the latter requires tpCPMs S, T on the physical indices satisfying $\mathcal{S}[M_2(X)] = M_1(X)$ and $\mathcal{T}[M_1(X)] = M_2(X)$ for every boundary matrix X , acting on physical indices as in [11, lines 1974–2085]. By contrast, the fusion maps in [11, Theorem 4.14(ii)–(iii), lines 972–991] act on the vertically read tensors, distinguishing support restrictions and exact reconstruction from full-support isometries.

Definition 21.3.1 (Transfer-retract datum). [Lean](#) [✓ Stmt](#) A *transfer-retract datum* at blocked size n for an MPO tensor M consists of a subspace $\mathcal{A}_n \subseteq M_D(\mathbb{C})$ together with linear maps

$$\begin{aligned} T_n : M_D(\mathbb{C}) &\longrightarrow \mathcal{A}_n, & S_n : \mathcal{A}_n &\longrightarrow M_D(\mathbb{C}), \\ T_n \circ S_n &= \text{id}_{\mathcal{A}_n}, & S_n \circ T_n &= E_n, \end{aligned} \quad (21.62)$$

where E_n denotes the blocked transfer map. No complete positivity or trace preservation is required of these linear maps.

Theorem 21.3.2 (Transfer-retract data imply transfer-map idempotence). [Lean](#) [✓ Stmt](#) A *one-site transfer-retract datum* implies doubled-index transfer-map idempotence.

Proof. [✓ Proof](#) By (21.62),

$$E_1^2 = S_1 T_1 S_1 T_1 = S_1 (T_1 S_1) T_1 = S_1 T_1 = E_1.$$

$\square$

Theorem 21.3.3 (One-site transfer-retract criterion). [Lean](#) [✓ Stmt](#) *A one-site transfer-retract datum exists if and only if the doubled-index MPO transfer map is idempotent.*

Proof. [✓ Proof](#) The forward implication is the retract calculation $E_1^2 = S_1 T_1 S_1 T_1 = S_1 T_1$. Conversely, if E_1 is idempotent, then it factors through its range, giving the required one-site datum. $\square$

Theorem 21.3.4 (Transfer retracts at positive blocked sizes). [Lean](#) [✓ Stmt](#) *If the doubled-index transfer map is idempotent, then transfer-retract data exist at every positive blocked size.*

Proof. [✓ Proof](#) Every positive blocked transfer map equals a positive power of the idempotent one-site map and is therefore idempotent. Factoring it through its range gives the transfer-retract datum. $\square$

Definition 21.3.5 (Operator product of MPO tensors). [Lean](#) [✓ Stmt](#) The *product tensor* of two MPO tensors contracts the bra index of the first factor with the ket index of the second and takes the tensor product of the bond spaces:

$$(M \cdot N)^{ik} = \sum_j M^{ij} \otimes N^{jk}. \quad (21.63)$$

In the vertical reading of [11, Section 4.5], this joins two vertically read tensors along an original horizontal bond. It is the tensor written $M_\alpha M_\beta$ in [11, Theorem 4.14(iii)].

Definition 21.3.6 (Canonical reassociation of three bond spaces). [Lean](#) [Lean](#) [✓ Stmt](#) For three bond spaces, let

$$A_{D_1, D_2, D_3} : \mathbb{C}^{D_1} \otimes (\mathbb{C}^{D_2} \otimes \mathbb{C}^{D_3}) \longrightarrow (\mathbb{C}^{D_1} \otimes \mathbb{C}^{D_2}) \otimes \mathbb{C}^{D_3}$$

be the coordinate pullback induced by canonical reassociation, defined on elementary tensors by $A_{D_1, D_2, D_3}(x_1 \otimes (x_2 \otimes x_3)) = (x_1 \otimes x_2) \otimes x_3$.

Theorem 21.3.7 (Associativity of the product tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *The letters of the two parenthesizations of a triple product are related by the canonical bond reassociation. Equivalently, for every pair of physical indices i, k ,*

$$((M \cdot N) \cdot P)^{ik} A_{D_1, D_2, D_3} = A_{D_1, D_2, D_3}(M \cdot (N \cdot P))^{ik}.$$

Proof. [✓ Proof](#) Expand both product tensors. Reassociation preserves every elementary tensor factor, while exchanging the order of the two finite sums gives the same contraction over the two intermediate physical indices. $\square$

Lemma 21.3.8 (Word evaluation of the product tensor). [Lean](#) [✓ Stmt](#) *The word evaluation of the product tensor along a configuration pair (σ, τ) expands as a sum over the contracted middle configurations ρ of tensor products of word evaluations:*

$$(M \cdot N)^{\sigma\tau} = \sum_\rho M^{\sigma\rho} \otimes N^{\rho\tau}. \quad (21.64)$$

Proof. [✓ Proof](#) Induction on the length, using the mixed-product property of the tensor product to merge one letter at a time. $\square$

Theorem 21.3.9 (The operator family is multiplicative). [Lean](#) [✓ Stmt](#) *For every length L , the operator generated by the product tensor is the product of the generated operators: $O_L(M \cdot N) = O_L(M)O_L(N)$.*

Proof. ✓ Proof Take traces in (21.64): the trace of a tensor product is the product of the traces, and the sum over middle configurations is the entrywise form of the operator product. $\square$

Lemma 21.3.10 (Word products of isometrically conjugated letters). Lean ✓ Stmt For an isometry U (a rectangular matrix with $U^\dagger U = 1$) and a nonempty word of square matrices $F_1, \dots, F_L$,

$$(UF_1U^\dagger)(UF_2U^\dagger) \cdots (UF_LU^\dagger) = UF_1F_2 \cdots F_LU^\dagger.$$

Proof. ✓ Proof The inner factors $U^\dagger U$ telescope to the identity. $\square$

Lemma 21.3.11 (Word products of block-diagonal tensor-product letters). Lean ✓ Stmt For block-diagonal letters whose block of index γ is the tensor product of a fixed matrix X_γ with a varying letter $G_{\gamma,l}$,

$$\prod_{l=1}^L \bigoplus_{\gamma} X_\gamma \otimes G_{\gamma,l} = \bigoplus_{\gamma} X_\gamma^L \otimes \prod_{l=1}^L G_{\gamma,l}.$$

Proof. ✓ Proof Blocks multiply blockwise and tensor products multiply factorwise, so the fixed factors accumulate into the matrix power X_γ^L . $\square$

Lemma 21.3.12 (Blockwise conjugation and restriction). Lean Lean ✓ Stmt Let $F = \bigoplus_a F_a$ and $G = \bigoplus_a G_a$. Then

$$FGF^\dagger = \bigoplus_a F_a G_a F_a^\dagger. \quad (21.65)$$

If r is a bijective reindexing of the row space of a matrix A , then

$$A_r X A_r^\dagger = (A X A^\dagger)_{r,r}, \quad (21.66)$$

where A_r denotes the corresponding row reindexing.

Proof. ✓ Proof Identity (21.65) follows because block-diagonal matrices multiply blockwise. Identity (21.66) follows by commuting bijective row reindexing with multiplication and adjunction. $\square$

Lemma 21.3.13 (Cancellation of a nonempty identity factor). Lean Lean Lean Lean Lean ✓ Stmt Let $D > 0$. The map $A \mapsto A \otimes I_D$ is injective and preserves matrix products. Consequently, $(F \otimes I_D)(G \otimes I_D) = I$ implies $FG = I$. In particular, orthonormal rows or columns of $F \otimes I_D$ imply the corresponding orthonormality of F .

Proof. ✓ Proof Evaluate the amplified identity on one basis vector of the nonzero factor. Multiplicativity of the Kronecker product gives the product and adjoint specializations. $\square$

Definition 21.3.14 (Fusion isometries on the full product bond space). Lean ✓ Stmt Let $\{M_\gamma\}$ be a finite family of tensors with a common physical dimension and bond dimensions depending on γ . Fusion isometries on the full product bond space consist of diagonal matrices $\chi_{\alpha,\beta,\gamma}$ with positive entries and maps $U_{\alpha,\beta}$ satisfying $U_{\alpha,\beta}^\dagger U_{\alpha,\beta} = 1$ and, site by site,

$$U_{\alpha,\beta}(M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger = \bigoplus_{\gamma} \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij}. \quad (21.67)$$

This is the full-support specialization of the fusion statement in [11, Theorem 4.14(iii)]. The definition does not require $\{M_\gamma\}$ to be a BNT. In applications to an MPDO tensor, these tensors come from the vertical BNT of [11, Proposition 4.13]; their physical dimension in the vertical reading is the bond dimension of the original tensor. Statement (iii) of the source theorem also requires the vector m , with $m_\alpha = \text{tr}(\mu_\alpha)$ for the positive diagonal multiplicity matrices μ_α of the vertical decomposition, to be an idempotent for the multiplication induced by $c_{\alpha,\beta,\gamma}^{(1)} = \text{tr}(\chi_{\alpha,\beta,\gamma})$. This condition is recorded separately in Definition 21.2.2.5.

The condition $U_{\alpha,\beta}^\dagger U_{\alpha,\beta} = 1$ on the whole product bond space is an additional restriction. The unrestricted source statement permits a common zero corner. Its support convention must therefore be made explicit: the map is a coisometry onto the retained direct sum, and exact reconstruction is required. This correction is documented in [61].

Definition 21.3.15 (Fusion coisometries onto the retained direct sum). [Lean](#) [Lean](#) [✓ Stmt](#) For the same labelled tensors and positive diagonal matrices, put $G_{\alpha,\beta}^{ij} = \bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij}$. Fusion coisometries onto the retained direct sum are maps $U_{\alpha,\beta}$ satisfying $U_{\alpha,\beta} U_{\alpha,\beta}^\dagger = 1$, the forward fusion identity

$$U_{\alpha,\beta}(M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger = G_{\alpha,\beta}^{ij},$$

and exact reconstruction

$$(M_\alpha M_\beta)^{ij} = U_{\alpha,\beta}^\dagger G_{\alpha,\beta}^{ij} U_{\alpha,\beta}. \quad (21.68)$$

Zero-dimensional retained sums are permitted. If in addition $U_{\alpha,\beta}^\dagger U_{\alpha,\beta} = 1$ for every pair, these maps are fusion isometries on the full product bond space.

The coisometry condition makes explicit the retained-row orientation of the source's vertical CF decomposition. Exact reconstruction states that the discarded common corner is zero. This correction to the full-support interpretation is documented in [61].

With $G_{\alpha,\beta}^{ij}$ denoting the retained direct-sum letter, [11, Theorem 4.14(iii), equation (89), lines 986–991] gives the following fusion identity:

Lemma 21.3.16 (Weighted zipper identities on the active support). [Lean](#) [Lean](#) [✓ Stmt](#) A fusion coisometry onto the retained direct sum satisfies

$$\begin{aligned} U_{\alpha,\beta}(M_\alpha M_\beta)^{ij} &= G_{\alpha,\beta}^{ij} U_{\alpha,\beta}, \\ (M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger &= U_{\alpha,\beta}^\dagger G_{\alpha,\beta}^{ij}. \end{aligned}$$

Proof. [✓ Proof](#) Substitute exact reconstruction (21.68) into the left-hand sides and use $U_{\alpha,\beta} U_{\alpha,\beta}^\dagger = 1$. $\square$

Theorem 21.3.17 (Weighted zipper identities). [Lean](#) [Lean](#) [✓ Stmt](#) For every pair of labels and physical indices, write $G_{\alpha,\beta}^{ij} = \bigoplus_{\gamma} \chi_{\alpha,\beta,\gamma} \otimes M_{\gamma}^{ij}$. The fusion isometry satisfies the two weighted zipper identities

$$\begin{aligned} U_{\alpha,\beta}(M_{\alpha}M_{\beta})^{ij} &= G_{\alpha,\beta}^{ij}U_{\alpha,\beta}, \\ (M_{\alpha}M_{\beta})^{ij}U_{\alpha,\beta}^{\dagger} &= U_{\alpha,\beta}^{\dagger}G_{\alpha,\beta}^{ij}. \end{aligned}$$

The corresponding identities in [7, Equation (21)] have identity matrices on the multiplicity spaces in place of the positive diagonal matrices χ . Their unweighted form requires the length-independent integer specialization and is not a consequence of the present hypotheses. Even after that specialization, comparison of the two triple-fusion orders and separation of the final labels are still needed to construct an invertible F -move.

Proof. [✓ Proof](#) Multiply the fusion identity (21.67) on the right by $U_{\alpha,\beta}$ and on the left by $U_{\alpha,\beta}^{\dagger}$, respectively, and use $U_{\alpha,\beta}^{\dagger}U_{\alpha,\beta} = 1$. $\square$

Theorem 21.3.18 (Length-independent zipper identities and reconstruction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of $L > 0$. Then $\chi_{\alpha,\beta,\gamma} = 1_{r_{\alpha,\beta,\gamma}}$ for every triple of labels, where $r_{\alpha,\beta,\gamma}$ is the dimension of the corresponding multiplicity space. If

$$H_{\alpha,\beta}^{ij} = \bigoplus_{\gamma} 1_{r_{\alpha,\beta,\gamma}} \otimes M_{\gamma}^{ij},$$

the fusion isometries satisfy

$$\begin{aligned} U_{\alpha,\beta}(M_{\alpha}M_{\beta})^{ij} &= H_{\alpha,\beta}^{ij}U_{\alpha,\beta}, \\ (M_{\alpha}M_{\beta})^{ij}U_{\alpha,\beta}^{\dagger} &= U_{\alpha,\beta}^{\dagger}H_{\alpha,\beta}^{ij}, \\ (M_{\alpha}M_{\beta})^{ij} &= U_{\alpha,\beta}^{\dagger}H_{\alpha,\beta}^{ij}U_{\alpha,\beta}. \end{aligned} \tag{21.69}$$

These are the identity-weight specialization of the fusion and zipper relations in [7, Section 3]. The MPDO fusion identity appears in [11, Theorem 4.14(iii), lines 986–991]; its identity-weight specialization follows from the length-independent case discussed in [11, line 1010].

Proof. [✓ Proof](#) Length independence makes every positive diagonal entry of $\chi_{\alpha,\beta,\gamma}$ equal to one. Substitute the resulting identity matrices into Theorem 21.3.17. For reconstruction, insert $U_{\alpha,\beta}^{\dagger}U_{\alpha,\beta} = 1$ on both sides of the product letter and apply the fusion identity between the two inserted factors, yielding (21.69). $\square$

Theorem 21.3.19 (Common positive blocking of BNT fusion isometries). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the labelled doubled-index tensors form a BNT and admit fusion isometries on the full product bond spaces. Suppose also that $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, is independent of $L > 0$. Then there is one integer $L > 0$ such that the tuples of letters of the blocked tensors $M_{\gamma}^{[L]}$ span the full product of their matrix algebras. Hence every $M_{\gamma}^{[L]}$ is injective.

For this same L , write

$$\begin{aligned} P_{\alpha,\beta}^{IJ} &= (M_{\alpha}^{[L]}M_{\beta}^{[L]})^{IJ}, \\ H_{\alpha,\beta}^{IJ} &= \bigoplus_{\gamma} 1_{r_{\alpha,\beta,\gamma}} \otimes (M_{\gamma}^{[L]})^{IJ}, \end{aligned}$$

where $r_{\alpha,\beta,\gamma}$ is the dimension of the multiplicity space on which $\chi_{\alpha,\beta,\gamma}$ acts. The original fusion isometry satisfies

$$U_{\alpha,\beta} P_{\alpha,\beta}^{IJ} U_{\alpha,\beta}^\dagger = H_{\alpha,\beta}^{IJ}, \quad (21.70)$$

the two blocked zipper identities

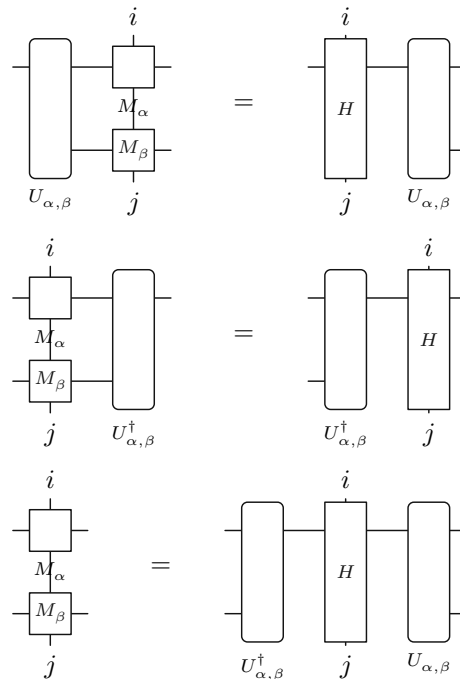
$$\begin{aligned} U_{\alpha,\beta} P_{\alpha,\beta}^{IJ} &= H_{\alpha,\beta}^{IJ} U_{\alpha,\beta}, \\ P_{\alpha,\beta}^{IJ} U_{\alpha,\beta}^\dagger &= U_{\alpha,\beta}^\dagger H_{\alpha,\beta}^{IJ}, \end{aligned}$$

and exact reconstruction

$$P_{\alpha,\beta}^{IJ} = U_{\alpha,\beta}^\dagger H_{\alpha,\beta}^{IJ} U_{\alpha,\beta}.$$

Proof. ✓ Proof Choose the positive block length supplied by simultaneous BNT blocking. The canonical pairing of the blocked ket and bra words transfers the one-letter product-algebra span to the blocked MPO tensors, and projection to each labelled factor gives injectivity. Along a nonempty block, the products of the conjugated fusion letters telescope using only $U_{\alpha,\beta}^\dagger U_{\alpha,\beta} = 1$. Length independence replaces every $\chi_{\alpha,\beta,\gamma}$ by the identity on its multiplicity space. The two zipper identities and reconstruction then follow by multiplying the blocked fusion identity (21.70) by $U_{\alpha,\beta}$ or $U_{\alpha,\beta}^\dagger$ on the appropriate side. $\square$

The three displayed diagrams represent, respectively, the two unweighted zipper identities and their local reconstruction consequence. The symbol H denotes $H_{\alpha,\beta}^{ij}$, and the open physical legs carry the indices i, j . The reconstruction relies directly on the isometry condition $U_{\alpha,\beta}^\dagger U_{\alpha,\beta} = 1$. The zipper orientation is that of [7, Section 3, Equations (19)–(21)]; the MPDO identity is given in [11, Theorem 4.14(iii), lines 986–991], with identity weights in the length-independent case discussed in [11, line 1010].



Definition 21.3.20 (Operators generated by tensors with fusion isometries). [Lean](#) [Lean](#) [✓ Stmt](#)
The tensors M_γ generate the periodic operators $O_L(M_\gamma)$. The positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ define the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$.

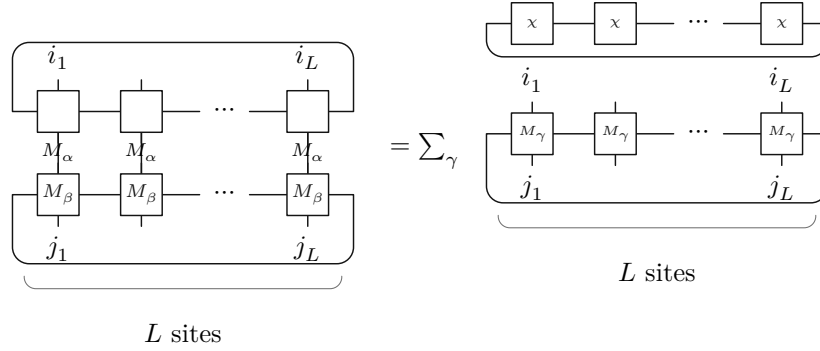


Figure 21.1: Closing an L -site chain of the local fusion identity. On the left, the bra legs of the M_α operator are contracted with the ket legs of the M_β operator. On the right, the multiplicity loop and the M_γ loop are disjoint; the former contributes $\text{tr}(\chi_{\alpha,\beta,\gamma}^L)$. Thus the diagram is the trace-power product law of [11, Theorem 4.14(ii)–(iii), equations (87) and (89), lines 962–991].

Theorem 21.3.21 (Fusion identities give the operator product law). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Fusion coisometries onto the active sectors with exact reconstruction, and also fusion isometries on the full product bond space, give the operator product law with structure coefficients determined by the diagonal matrices $\chi_{\alpha,\beta,\gamma}$. For every chain length $L \geq 1$,

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma} \text{tr}(\chi_{\alpha,\beta,\gamma}^L) O_L(M_\gamma). \quad (21.71)$$

This derives the product law of [11, Theorem 4.14(ii)] from the isometry statement of [11, Theorem 4.14(iii)]. The active-support formulation is the one used in [11, Appendix C.4, lines 2020–2029].

Proof. [✓ Proof](#) The product of the operators is generated by the product tensor. Write P^w for a nonempty word in that tensor and D^w for the corresponding word in the weighted direct sum. For a coisometry onto the active sectors, exact reconstruction and $U_{\alpha,\beta}U_{\alpha,\beta}^\dagger = 1$ give $P^w = U_{\alpha,\beta}^\dagger D^w U_{\alpha,\beta}$. Cyclicity of trace then gives $\text{tr}(P^w) = \text{tr}(D^w)$. For an isometry on the full product bond space, $D^w = U_{\alpha,\beta} P^w U_{\alpha,\beta}^\dagger$ and $\text{tr}(D^w) = \text{tr}(P^w U_{\alpha,\beta}^\dagger U_{\alpha,\beta}) = \text{tr}(P^w)$. By the fusion identity, the letters of D are block-diagonal tensor products. Their word product splits block by block with the diagonal factors accumulating into $\chi_{\alpha,\beta,\gamma}^L$; taking the trace block by block and factor by factor yields the coefficients $\text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ in (21.71). $\square$

Definition 21.3.22 (Algebra relations from fusion isometries). [Lean](#) [✓ Stmt](#) Suppose the fusion-isometry identity holds with positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$. Set $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, and suppose that the vector $m = (m_\alpha)_\alpha$ is an idempotent for the multiplication induced by $c^{(1)}$. For $L > 0$, the operators $O_L(M_\alpha)$ linearly span an algebra. The product law and the assumed

idempotent condition are, respectively,

$$\begin{aligned} O_L(M_\alpha)O_L(M_\beta) &= \sum_{\gamma} c_{\alpha,\beta,\gamma}^{(L)} O_L(M_\gamma), \\ m_\gamma &= \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta. \end{aligned}$$

This construction realizes implication (iii) $\Rightarrow$ (ii) of [11, Theorem 4.14].

Definition 21.3.23 (Active-sector fusion for a vertical canonical decomposition). [Lean](#) [Lean](#) [✓ Stmt](#) Choose a vertical canonical decomposition, with nonempty positive diagonal matrices μ_α and a BNT $\{M_\alpha\}_\alpha$:

$$U\widetilde{M}U^\dagger = \bigoplus_{\alpha} \mu_\alpha \otimes M_\alpha.$$

Here U is a coisometry onto the retained sectors, and its adjoint reconstructs $\widetilde{M}$, including any common zero-sector complement. Active-sector fusion for this decomposition consists of positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ and coisometries $U_{\alpha,\beta}$ onto the active product sectors, satisfying

$$U_{\alpha,\beta}(M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger = \bigoplus_{\gamma} \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij}. \quad (21.72)$$

They obey $U_{\alpha,\beta}U_{\alpha,\beta}^\dagger = 1$, and the active direct sum reconstructs the product tensor:

$$(M_\alpha M_\beta)^{ij} = U_{\alpha,\beta}^\dagger \left(\bigoplus_{\gamma} \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij} \right) U_{\alpha,\beta}.$$

The vector with entries $m_\alpha = \text{tr}(\mu_\alpha)$ is an idempotent for the multiplication induced by $c_{\alpha,\beta,\gamma}^{(1)} = \text{tr}(\chi_{\alpha,\beta,\gamma})$. This is the active-support formulation of statement (iii) of [11, Theorem 4.14, lines 986–993], for the chosen decomposition of [11, Proposition 4.13, lines 943–951].

Theorem 21.3.24 (Idempotent relation from paired decompositions and given fusion coisometries). [Lean](#) [✓ Stmt](#) Let $\mathcal{D}_1$ and $\mathcal{D}_2$ be positive vertical canonical decompositions of an MPO tensor and its two-site blocking. Suppose their normal sectors are identified by a bijection σ and unitary conjugacies with the corresponding multiplicity-trace ratios. Suppose also that positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ and coisometries $U_{\alpha,\beta}$ are given, with $U_{\alpha,\beta}U_{\alpha,\beta}^\dagger = 1$, and satisfy both

$$\begin{aligned} U_{\alpha,\beta}(M_\alpha M_\beta)^{ij} U_{\alpha,\beta}^\dagger &= \bigoplus_{\gamma} \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij}, \\ (M_\alpha M_\beta)^{ij} &= U_{\alpha,\beta}^\dagger \left(\bigoplus_{\gamma} \chi_{\alpha,\beta,\gamma} \otimes M_\gamma^{ij} \right) U_{\alpha,\beta}. \end{aligned}$$

Then the simultaneous blocked representations imply that the vector with entries $m_\alpha = \text{tr}(\mu_\alpha)$ is an idempotent for the multiplication induced by $c_{\alpha,\beta,\gamma}^{(1)} = \text{tr}(\chi_{\alpha,\beta,\gamma})$. Thus the chosen decomposition satisfies the active-sector fusion and idempotent relations. With the stated comparison and fusion hypotheses, this is the final step of the construction in [11, Appendix C.4, lines 2001–2046].

Proof. [✓ Proof](#) The sector correspondence expresses the blocked closed-chain operator in the one-site normal-tensor basis. Multiplication of the one-site vertical decomposition gives a second expression in the same basis. At every sufficiently large length, linear independence equates their coefficients. Equality of the resulting positive power sums identifies the multiplicity multisets, and summing their entries gives $m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta$. $\square$

Theorem 21.3.25 (Grouped product inputs give active-sector fusion). [Lean](#) [✓ Stmt](#) Let M generate matrix product density operators, have the grouped vertical product hypotheses, and satisfy the renormalization fixed-point condition. Then M has an active-sector BNT fusion tensor clause. In particular, its positive diagonal fusion matrices obey the length-one multiplicity relation $m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta$.

Proof. [✓ Proof](#) Obtain the one-site and two-site vertical decompositions from the grouped hypotheses. Transport the renormalization maps to compare their normal sectors. The positive product decomposition supplies the fusion matrices and coisometries; the paired-decomposition argument gives the length-one relation. $\square$

Theorem 21.3.26 (Renormalization fixed points in CF have active-sector fusion). [Lean](#) [✓ Stmt](#) Let M generate MPDOs $\rho^{(N)}(M)$, and suppose that its doubled-physical-index MPS tensor is in CF, with normalized NT blocks. Suppose there are trace-preserving completely positive maps (tpCPM) S, T on the original physical indices such that $S[M_2(X)] = M_1(X)$ and $T[M_1(X)] = M_2(X)$ for every X . Then a vertical canonical decomposition of M has active-sector fusion. Its positive diagonal fusion matrices satisfy

$$m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma}) m_\alpha m_\beta.$$

This is the corrected active-support formulation of implication (i) $\Rightarrow$ (iii) in [11, Theorem 4.14, lines 972–993], rather than the unqualified printed statement.

For each fixed product pair, only its active normal corners occur; an absent label has zero fusion multiplicity. See [20].

A fixed product pair may have an empty active family, so no unsupported normal sector is added. See [101].

Each fusion map is a coisometry onto the active direct sum, and its adjoint reconstructs the whole product tensor, including any discarded common zero corner. This corrects the orientation of the printed map. See [61].

Proof. [✓ Proof](#) Literal canonical form supplies the grouped vertical product hypotheses; apply the grouped active-sector fusion theorem. $\square$

Theorem 21.3.27 (Horizontal-form specialization of active-sector fusion). [Lean](#) [✓ Stmt](#) Let M generate MPDOs and satisfy the normalized BNT-refined horizontal-form hypothesis. Suppose there are tpCPM S, T on the original physical indices with $S[M_2(X)] = M_1(X)$ and $T[M_1(X)] = M_2(X)$ for every X . Then a vertical canonical decomposition of M has active-sector fusion. Its positive diagonal fusion matrices make the vector $m = (\text{tr}(\mu_\alpha))_\alpha$ an idempotent for the multiplication induced by $c^{(1)}$.

The normalized BNT-refined horizontal hypothesis is stronger than CF with normalized NT blocks. See [153].

Proof. [✓ Proof](#) Normalized BNT-refined horizontal form supplies the grouped vertical product hypotheses; apply the grouped active-sector fusion theorem. $\square$

Theorem 21.3.28 (Active-sector fusion implies the algebra relations). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose a chosen vertical canonical decomposition satisfies the active-sector fusion and idempotent relations. Then its operators $O_L(M_\alpha)$ satisfy the algebra relations. For every positive chain length L , the structure coefficients are $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, determined by the same diagonal matrices. Thus statement (iii) implies statement (ii) of [11, Theorem 4.14, lines 972–993], with the active-support interpretation of (iii).

Proof. ✓ Proof Iterating the fusion identity along L sites and taking the closed trace gives $O_L(M_\alpha)O_L(M_\beta) = \sum_\gamma \text{tr}(\chi_{\alpha,\beta,\gamma}^L)O_L(M_\gamma)$. Writing $m_\alpha = \text{tr}(\mu_\alpha)$, the assumed idempotent identity is $m_\gamma = \sum_{\alpha,\beta} \text{tr}(\chi_{\alpha,\beta,\gamma})m_\alpha m_\beta$. These are the product and idempotent relations for the chosen vertical decomposition, with the same diagonal matrices. $\square$

Theorem 21.3.29 (Printed algebra and fusion characterization). Not ready For a tensor in CF generating MPDOs, [11, Theorem 4.14] states that the RFP condition, the algebra and idempotent relations for a vertical BNT, and the fusion-isometry and idempotent relations are equivalent.

The RFP condition is equivalent to the algebra and idempotent relations. For the fusion statement, an absent product sector must have multiplicity zero, and the map onto the retained sectors is a coisometry with exact reconstruction. Under this active-support interpretation, the three conditions are equivalent by Theorem 21.3.30. Without these qualifications, the displayed fusion statement is not well-defined on empty product sectors and has the wrong isometry orientation.

Theorem 21.3.30 (BNT characterizations of MPDO renormalization fixed points). Lean Lean

✓ Stmt Let M generate MPDOs, and suppose its doubled-physical-index tensor is in CF with normalized NT blocks. Consider the following conditions:

- (i) M is an RFP: there are $\text{tpCPM } S, T$ with $S[M_2(X)] = M_1(X)$ and $T[M_1(X)] = M_2(X)$ for every X .
- (ii) A vertical canonical decomposition of M has positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$ such that the $O_L(M_\alpha)$ satisfy the product law with $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ for every $L > 0$, and the vector with entries $m_\alpha = \text{tr}(\mu_\alpha)$ is an idempotent for the multiplication induced by $c^{(1)}$.
- (iii_{act}) A vertical canonical decomposition of M has active-sector fusion coisometries with exact reconstruction and positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$, and the vector with entries $m_\alpha = \text{tr}(\mu_\alpha)$ is an idempotent for the multiplication induced by $c^{(1)}$.

Then

$$\begin{aligned} (i) &\iff (ii), \\ (i) &\iff (iii_{\text{act}}). \end{aligned} \tag{21.73}$$

The first equivalence matches $(i) \iff (ii)$ of [11, Theorem 4.14], while the second gives the active-support form (iii_{act}) .

Only nonzero product corners occur in (iii_{act}) ; a sector absent from a fixed product pair has zero fusion multiplicity. This restriction is recorded in [20].

A fixed pair may have empty active support, and the fusion map onto the retained sectors is a coisometry. These corrections are recorded in [101] and [61].

The implication $(ii) \Rightarrow (i)$ uses the comparison of mixed one-site and two-site prefixes recorded in [89].

The physical maps in $(ii) \Rightarrow (i)$ are completed on the discarded complements as recorded in [154].

Proof. ✓ Proof An RFP has active-sector fusion and the idempotent relation by Theorem 21.3.26. These imply the algebra and idempotent relations by Theorem 21.3.28. Finally, the algebra and idempotent relations imply that M is an RFP by Theorem 21.2.6.3. These three implications prove both equivalences in (21.73). $\square$

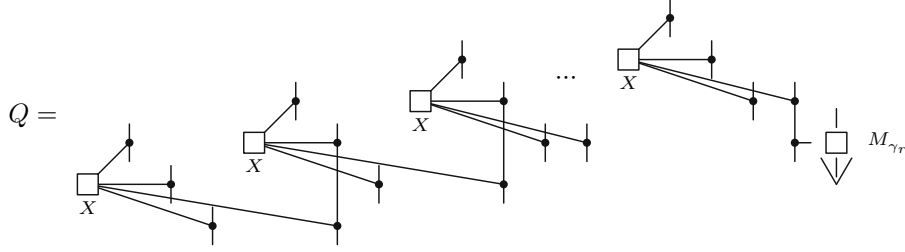


Figure 21.2: The recursive contraction defining the structure operator Q , with terminal factor $\text{tr}(M_{\gamma_r})$ [11, lines 999–1008].

Associativity of the algebra product in [11, Equation (87)] constrains the diagonal matrices $\chi_{\alpha,\beta,\gamma}$ and induces compatibility between the fusion isometries of the two bracketings $(M_\alpha M_\beta)M_\gamma$ and $M_\alpha(M_\beta M_\gamma)$ [11, lines 995–999]. Solutions constructed from the F -symbols of unitary fusion categories connect to the boundary theories of [7]. On the full product bond spaces, the tensor identities for the two bracketings determine a comparison between the full direct sums, which restricts to a fixed final label in Theorem 21.3.80.

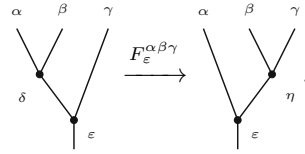


Figure 21.3: The two ways of fusing three labels to a fixed final label ϵ , using the printed- F row/column convention of [7, Section 3.4]. The solid fusion trees represent the left- and right-bracketed fusion maps U^L and U^R of Definitions 21.3.31 and 21.3.34. The arrow is the change of basis $F_{\epsilon}^{\alpha\beta\gamma}$ between left- and right-bracketed multiplicity coordinates, whose two-sided unitarity is established in Theorem 21.3.82.

Definition 21.3.31 (Iterated fusion isometry of a triple product, left bracketing). [Lean](#) [✓ Stmt](#) For labels α, β, γ , the *left iterated fusion isometry* $U_{\alpha,\beta,\gamma}^L$ is obtained by applying $U_{\alpha,\beta}$ (tensored with the identity on the bond space of M_γ) to fuse the first two tensors, then applying $U_{\delta,\gamma}$, for each label δ appearing in the resulting direct sum, to fuse the outcome with the third tensor. Its codomain is indexed by pairs of labels (δ, ϵ) , the multiplicity indices of $\chi_{\alpha,\beta,\delta}$ and $\chi_{\delta,\gamma,\epsilon}$, and the bond index of M_ϵ .

Theorem 21.3.32 (The left iterated fusion isometry is an isometry). [Lean](#) [✓ Stmt](#) The left iterated fusion isometry satisfies $(U_{\alpha,\beta,\gamma}^L)^\dagger U_{\alpha,\beta,\gamma}^L = 1$.

Proof. [✓ Proof](#) The composite of two isometries is an isometry. □

Theorem 21.3.33 (The left iterated fusion isometry conjugates a triple product onto a doubly weighted direct sum). [Lean](#) [✓ Stmt](#) Site by site,

$$\begin{aligned} & U_{\alpha,\beta,\gamma}^L ((M_\alpha M_\beta) M_\gamma)^{ij} (U_{\alpha,\beta,\gamma}^L)^\dagger \\ &= \bigoplus_{\delta} \bigoplus_{\epsilon} \chi_{\alpha,\beta,\delta} \otimes (\chi_{\delta,\gamma,\epsilon} \otimes M_\epsilon^{ij}). \end{aligned}$$

Proof. [✓ Proof](#) Apply the fusion identity for $U_{\alpha,\beta}$ to the first two tensors, summed against the physical index contracting with the third tensor. Regroup the result over δ and use blockwise conjugation to obtain $\chi_{\alpha,\beta,\delta}$ tensored with a letter of $M_\delta M_\gamma$. The fusion identity for $U_{\delta,\gamma}$ gives the second direct sum. $\square$

Definition 21.3.34 (Iterated fusion isometry of a triple product, right bracketing). [Lean](#) [✓ Stmt](#) For labels α, β, γ , the *right iterated fusion isometry* $U_{\alpha,\beta,\gamma}^R$ is obtained by applying $U_{\beta,\gamma}$, tensored with the identity on the bond space of M_α , and then applying $U_{\alpha,\delta}$ for every intermediate label δ . Its codomain is indexed by the intermediate and final labels (δ, ε) , the multiplicity indices of $\chi_{\beta,\gamma,\delta}$ and $\chi_{\alpha,\delta,\varepsilon}$, and the bond index of M_ε . This is the right-bracketed composite in the associativity equation of [7, Section “Associativity and the pentagon equation”], before the change of basis by an F -matrix.

Theorem 21.3.35 (The right iterated fusion isometry is an isometry). [Lean](#) [✓ Stmt](#) *The right iterated fusion isometry satisfies $(U_{\alpha,\beta,\gamma}^R)^\dagger U_{\alpha,\beta,\gamma}^R = 1$.*

Proof. [✓ Proof](#) The two successive fusion maps are isometries on each intermediate sector, so their block-diagonal composite is an isometry. $\square$

Theorem 21.3.36 (The right iterated fusion isometry conjugates a triple product onto a doubly weighted direct sum). [Lean](#) [✓ Stmt](#) *Site by site,*

$$\begin{aligned} U_{\alpha,\beta,\gamma}^R (M_\alpha (M_\beta M_\gamma))^{ij} (U_{\alpha,\beta,\gamma}^R)^\dagger \\ = \bigoplus_{\delta} \bigoplus_{\varepsilon} \chi_{\beta,\gamma,\delta} \otimes (\chi_{\alpha,\delta,\varepsilon} \otimes M_\varepsilon^{ij}). \end{aligned}$$

Proof. [✓ Proof](#) Apply the fusion identity for $U_{\beta,\gamma}$ to $M_\beta M_\gamma$. In the summand indexed by δ , exchange the first multiplicity factor with the bond space of M_α . Blockwise conjugation regroups the contraction as $\chi_{\beta,\gamma,\delta} \otimes (M_\alpha M_\delta)^{ij}$. The fusion identity for $U_{\alpha,\delta}$ gives the sum over ε . $\square$

Definition 21.3.37 (Full comparison of the two triple-fusion orders). [Lean](#) [✓ Stmt](#) The canonical comparison from the right-associated full direct sum to the left-associated full direct sum is

$$C_{\alpha,\beta,\gamma} = U_{\alpha,\beta,\gamma}^L A_{D_\alpha, D_\beta, D_\gamma} (U_{\alpha,\beta,\gamma}^R)^\dagger. \quad (21.74)$$

Here $A_{D_\alpha, D_\beta, D_\gamma}$ is the permutation matrix identifying the two bracketings of the product bond space. Both intermediate labels and the final label remain in the direct sums. The comparison is oriented from the right-associated sum to the left-associated sum, matching the adjoint orientation of the F -matrix convention in [7, Section 3.4].

Lemma 21.3.38 (Left range projection of the full triple-fusion comparison). [Lean](#) [✓ Stmt](#) *The product of the full comparison with its adjoint is the range projection of the left iterated fusion isometry:*

$$C_{\alpha,\beta,\gamma} C_{\alpha,\beta,\gamma}^\dagger = U_{\alpha,\beta,\gamma}^L (U_{\alpha,\beta,\gamma}^L)^\dagger.$$

Surjectivity of the left iterated fusion map corresponds to completeness of the fusion decomposition in [7, lines 181–191].

Proof. [✓ Proof](#) Substitute (21.74). The isometry identity $(U^R)^\dagger U^R = 1$ and the two-sided inverse identities for the permutation matrix A leave $U^L (U^L)^\dagger$. $\square$

Lemma 21.3.39 (Right range projection of the full triple-fusion comparison). [Lean](#) [✓ Stmt](#) The product in the opposite order is the range projection of the right iterated fusion isometry:

$$C_{\alpha,\beta,\gamma}^\dagger C_{\alpha,\beta,\gamma} = U_{\alpha,\beta,\gamma}^R (U_{\alpha,\beta,\gamma}^R)^\dagger.$$

Surjectivity of the right iterated fusion map likewise corresponds to completeness of the fusion decomposition.

Proof. [✓ Proof](#) Substitute (21.74). The isometry identity $(U^L)^\dagger U^L = 1$ and the two-sided inverse identities for A leave $U^R (U^R)^\dagger$. $\square$

Theorem 21.3.40 (The full triple-fusion comparison intertwines the unweighted direct sums).

[Lean](#) [✓ Stmt](#) Suppose the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of the positive chain length. Write $r_{\alpha,\beta}^\gamma$ for the dimension of $\chi_{\alpha,\beta,\gamma}$. For every pair of physical indices i, k , let

$$D_{ik}^L = \bigoplus_{\delta,\varepsilon} 1_{r_{\alpha,\beta}^\delta} \otimes (1_{r_{\delta,\gamma}^\varepsilon} \otimes M_\varepsilon^{ik}),$$

and define D_{ik}^R by the right-associated multiplicities $r_{\beta,\gamma}^\delta$ and $r_{\alpha,\delta}^\varepsilon$. Then

$$D_{ik}^L C_{\alpha,\beta,\gamma} = C_{\alpha,\beta,\gamma} D_{ik}^R.$$

This is an identity on the full direct sums. It neither extracts an F -transformation for a fixed final label nor proves invertibility or the pentagon identity.

Proof. [✓ Proof](#) Length independence makes every positive diagonal factor equal to the identity. Substitute the left and right triple-fusion identities into (21.74), cancel the two isometries, and apply Theorem 21.3.7. $\square$

Definition 21.3.41 (Common word selectors for the final labels). [Lean](#) [✓ Stmt](#) Fix a common length S . The tensors admit common word selectors for the final labels at length S if, for each final label ε , there are coefficients $c_\varepsilon(w)$, indexed by the words w of length S in the doubled physical alphabet, such that

$$\begin{aligned} \sum_{|w|=S} c_\varepsilon(w) M_\varepsilon^w &= 1, \\ \sum_{|w|=S} c_\varepsilon(w) M_{\varepsilon'}^w &= 0 \end{aligned}$$

for every $\varepsilon' \neq \varepsilon$. This is the finite-word form of the simultaneous inverse relation $B_d^+ B_{d'} = \delta_{d,d'} 1$ used in the fixed-channel argument of [7].

Theorem 21.3.42 (Pair fusion completeness from positive-length selectors). [Lean](#) [✓ Stmt](#) Suppose the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of the positive chain length, and suppose that common selectors exist at a positive length. Then every pair fusion isometry is surjective: $U_{\alpha,\beta} U_{\alpha,\beta}^\dagger = 1$. The common selectors are the finite-word form of the simultaneous inverse used in [7, line 269].

Proof. [✓ Proof](#) For a word w of the positive selector length, the identity $U_{\alpha,\beta}^\dagger U_{\alpha,\beta} = 1$ gives $D^w = U_{\alpha,\beta} (M_\alpha M_\beta)^w U_{\alpha,\beta}^\dagger$, where $D = \bigoplus_\gamma 1_{r_{\alpha,\beta}^\gamma} \otimes M_\gamma$. Hence the range projection $P = U_{\alpha,\beta} U_{\alpha,\beta}^\dagger$ satisfies $P D^w = D^w$. Summing the selector polynomials over the final label gives the identity on the full direct sum, and therefore $P = 1$. $\square$

Theorem 21.3.43 (Pair fusion completeness from a BNT). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the tensors $\{M_\alpha\}_\alpha$ with the given fusion isometries form a BNT and that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of the positive chain length. Then every pair fusion isometry is surjective: $U_{\alpha,\beta} U_{\alpha,\beta}^\dagger = 1$. In particular, common final-label selectors and the coisometry identity follow from these hypotheses. This is a consequence of the common BNT word span in [11, lines 317–345], the fusion identity at lines 986–993, and length independence at lines 995–1010.

Proof. [✓ Proof](#) The basis-of-normal-tensors argument gives a positive common word length at which the labelled word evaluations span the full product matrix algebra. Choosing the tuple which is the identity in one labelled block and zero in every other block gives common final-label selectors. Apply Theorem 21.3.42. $\square$

Definition 21.3.44 (One fusion layer of the topological projector). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let P_γ be operators on the bond space of the vertically read tensor M_γ . Define one fusion layer of the recursive operator Q by

$$Q_{\alpha,\beta} = \bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes P_\gamma. \quad (21.75)$$

In the source construction, the terminal matrices are $P_\gamma = \text{tr}(M_\gamma)$, with spectral refinement used to obtain projections. Here the trace closes the horizontal operator leg and leaves the bond indices of the vertically read tensor open. Before spectral refinement, the fusion identity implies

$$U_{\alpha,\beta} \text{tr}(M_\alpha M_\beta) U_{\alpha,\beta}^\dagger = \bigoplus_\gamma \chi_{\alpha,\beta,\gamma} \otimes \text{tr}(M_\gamma). \quad (21.76)$$

Summing the forward fusion identity over equal horizontal operator indices and distributing matrix multiplication gives (21.76). This is one pairwise step in the recursion of [11, lines 999–1010].

This definition describes one fusion layer. The arbitrary-chain recursion, the density decomposition, and the physical commuting Gibbs decomposition for $N \geq 2$ are given in Theorems 21.3.47, 21.3.52, and 21.3.63, respectively. The source boundary at $N = 1$ is recorded in Remark 21.3.66.

Theorem 21.3.45 (Projection property of one fusion layer). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let P_γ be self-adjoint idempotents on the terminal bond space of M_γ . Suppose that the matrices $\chi_{\alpha,\beta,\gamma}$ are diagonal with strictly positive entries, and that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ are independent of L for every positive length L . Then

$$Q_{\alpha,\beta} = \bigoplus_\gamma \mathbb{1}_{r_{\alpha,\beta}^\gamma} \otimes P_\gamma$$

is self-adjoint and idempotent, where $r_{\alpha,\beta}^\gamma$ is the dimension of $\chi_{\alpha,\beta,\gamma}$. This is the single recursive fusion step in [11, lines 999–1010]. Since the fusion map onto the nonzero sectors satisfies $U_{\alpha,\beta} U_{\alpha,\beta}^\dagger = \mathbb{1}$, the operator $U_{\alpha,\beta}^\dagger Q_{\alpha,\beta} U_{\alpha,\beta}$ is also an orthogonal projection on the product bond space.

Proof. [✓ Proof](#) The trace-power hypothesis and length independence force every diagonal entry of $\chi_{\alpha,\beta,\gamma}$ to equal one. Thus $\chi_{\alpha,\beta,\gamma} = \mathbb{1}$, while $c_{\alpha,\beta,\gamma}^{(L)} = r_{\alpha,\beta}^\gamma$ need not equal one. Hence

$$\begin{aligned} Q_{\alpha,\beta}^\dagger &= \bigoplus_\gamma (\mathbb{1} \otimes P_\gamma^\dagger) = Q_{\alpha,\beta}, \\ Q_{\alpha,\beta}^2 &= \bigoplus_\gamma (\mathbb{1} \otimes P_\gamma^2) = Q_{\alpha,\beta}. \end{aligned}$$

Finally, the coisometry identity $U_{\alpha,\beta}U_{\alpha,\beta}^\dagger = \mathbb{1}$ and Theorem [182, “Coisometric transport of a star projection”] show that $U_{\alpha,\beta}^\dagger Q_{\alpha,\beta} U_{\alpha,\beta}$ is an orthogonal projection. $\square$

Definition 21.3.46 (Recursive fusion histories and their sequential circuit). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Stmnt](#) Fix an initial label α and append labels successively along a finite chain. For total chain length $N \geq 1$, the appended-label list has length $N - 1$; the operators W_N and Q_N are indexed by this total chain length. A fusion history records, at every step, the preceding final label, the new final label, and an index in the corresponding multiplicity space. If h ends at $\gamma(h)$, let

$$w(h) = \prod_r (\chi_{\gamma_{r-1}, \beta_r, \gamma_r})_{k_r, k_r}. \quad (21.77)$$

For terminal bond operators P_γ , define

$$Q_N(P) = \bigoplus_h w(h) P_{\gamma(h)}. \quad (21.78)$$

The sequential map W_N is obtained by tensoring the preceding map with the identity on the newly appended bond and then applying the pairwise fusion coisometry onto the nonzero sectors in each preceding-history block. This gives the recursive operator and sequential circuit in [11, lines 999–1010].

Theorem 21.3.47 (Fixed-label recursive fusion identity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Stmnt](#) For every initial label and every finite list of appended labels, let $N \geq 1$ be one plus the length of that list, and let $M_{\alpha,\beta}$ denote the left-associated product tensor. The sequential fusion map W_N is a coisometry onto the retained direct sum, so $W_N W_N^\dagger = \mathbb{1}$. For every pair of horizontal operator indices i, j , it carries the product tensor to the recursive direct sum:

$$W_N M_{\alpha,\beta}^{i,j} W_N^\dagger = Q_N((M_\gamma^{i,j})_\gamma). \quad (21.79)$$

Consequently, after closing the horizontal operator index,

$$W_N \text{tr}(M_{\alpha,\beta}) W_N^\dagger = Q_N((\text{tr}(M_\gamma))_\gamma). \quad (21.80)$$

Exact reconstruction gives the reverse identities

$$M_{\alpha,\beta}^{i,j} = W_N^\dagger Q_N((M_\gamma^{i,j})_\gamma) W_N, \quad (21.81)$$

$$\text{tr}(M_{\alpha,\beta}) = W_N^\dagger Q_N((\text{tr}(M_\gamma))_\gamma) W_N. \quad (21.82)$$

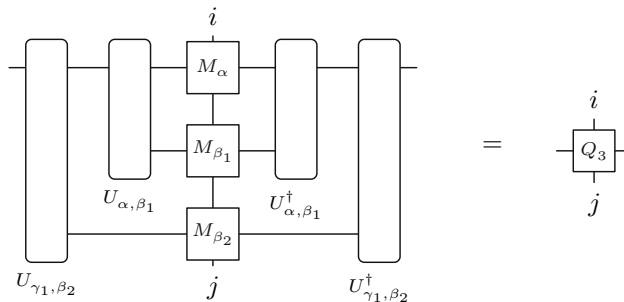
If the diagonal matrices $\chi_{\alpha,\beta,\gamma}$ have strictly positive entries and give structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ independent of every positive length L , then $Q_N(P)$ is self-adjoint and idempotent whenever every terminal matrix P_γ is self-adjoint and idempotent.

The source embedding $\widetilde{U}_N$ is $W_N^\dagger$. The correction to the source fusion map is to use a coisometry onto the nonzero sectors; exact reverse fusion also requires reconstruction from those sectors, not merely the coisometry identity. See [61].

This theorem fixes the initial label and the list of appended labels. In Theorem 21.3.52, the direct sum is taken over all labels and multiplicity coordinates. The commutator is a separate assertion; see [28].

Proof. ✓ Proof The base case has no appended labels and represents the one-site chain; its fusion history is unique. For the induction step, tensor the preceding coisometry with the identity on the new bond, apply the induction hypothesis, and then apply the pairwise fusion identity in each history block. Reindexing the nested direct sum adjoins the new final label and multiplicity coordinate to the preceding history. Summing (21.79) over equal horizontal operator indices gives (21.80). Under length independence every accumulated coefficient $w(h)$ is one, so self-adjointness and idempotence hold block by block. $\square$

For $N = 3$, with appended labels β_1, β_2 , the forward identity (21.79) is displayed below on the history block whose intermediate label is γ_1 . The inner fusion map fuses the first two bonds, and the outer one fuses the result with the third bond; the right-hand side is the corresponding block of Q_3 .



Lemma 21.3.48 (Sitewise change of physical coordinates). Lean ✓ Stmt Let V be a one-site matrix, not necessarily square. The operator $V^{\otimes N} \rho^{(N)}(M) (V^{\otimes N})^\dagger$ is generated by the tensor obtained by applying $X \mapsto V X V^\dagger$ to the one-site physical matrices of M .

Proof. ✓ Proof Expand a matrix entry as a sum over the original ket and bra configurations. The coefficient factors site by site, and the remaining virtual product is the word evaluation of the original tensor. $\square$

Theorem 21.3.49 (Positivity of the terminal transfer matrices). Lean ✓ Stmt Let M generate MPDOs, and choose a vertical canonical decomposition with BNT $\{M_\gamma\}_\gamma$ and positive diagonal multiplicity matrices μ_γ . Then every terminal transfer matrix $T_\gamma = \text{tr}(M_\gamma)$ is positive semidefinite. This supplies the positivity needed for the spectral decomposition invoked in [11, lines 1010–1016].

The trace closes the horizontal operator leg and leaves the two bond indices of the vertically read tensor M_γ open. See [28].

Proof. ✓ Proof The fusion identities imply the algebra identities for the same vertical decomposition. Apply Theorem 21.2.2.99. $\square$

Definition 21.3.50 (Recursive density operator summed over all labels). Lean Lean Lean Lean ✓ Stmt For a chain of positive length N , choose at every site a BNT label α_j and a diagonal coordinate q_j of μ_{α_j} . There are $N - 1$ appended labels after the initial site. Define

$$m(\alpha, q) = \prod_{j=0}^{N-1} (\mu_{\alpha_j})_{q_j, q_j}. \quad (21.83)$$

The first label is the initial label of the recursive fusion, while the remaining labels are appended in the physical order $\alpha_1, \dots, \alpha_{N-1}$. The recursive specification records this sequence in the reverse

order $(\alpha_{N-1}, \dots, \alpha_1)$, so removing its first entry removes the final physical site. This reversal affects only the recursive notation: the tensor product and the sequential fusion coisometry remain in the physical order $\alpha_0, \dots, \alpha_{N-1}$. Let $c = (\alpha, q)$ denote the configuration, let $W_{N,c}$ be its sequential coisometry onto the retained direct sum, and set $Q_{N,c} = Q_N((\text{tr}(M_\gamma))_\gamma)$. The corresponding density block is $m(\alpha, q)W_{N,c}^\dagger Q_{N,c} W_{N,c}$. The embedding $\tilde{U}$ in [11, line 999] corresponds to $W_{N,c}^\dagger$ on this block. The recursive density operator R_N is the direct sum of these blocks over all sitewise labels and diagonal coordinates. On the same direct-sum coordinates, define

$$\begin{aligned} A_N &= \bigoplus_c m(c) \mathbb{1}_c, \\ T_N &= \bigoplus_c W_{N,c}^\dagger Q_{N,c} W_{N,c}, \\ R_N &= A_N T_N. \end{aligned} \tag{21.84}$$

Here $\mathbb{1}_c$ is the identity on the product bond space for configuration c . The scalar coefficients of A_N are the selected diagonal coefficients of the factor $\mu^{\otimes N}$ in [11, line 999]; the identities mentioned there act on the complementary tensor factors of each block.

Theorem 21.3.51 (Density decomposition from fusion on the nonzero sectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix a vertical canonical decomposition of M with fusion coisometries onto the nonzero product sectors and exact reconstruction. For every $N > 0$, its vertical coisometry V and recursive density operator R_N satisfy

$$\begin{aligned} V^{\otimes N} \rho^{(N)}(M) (V^{\otimes N})^\dagger &= R_N, \\ ((V^\dagger)^{\otimes N}) R_N V^{\otimes N} &= \rho^{(N)}(M). \end{aligned}$$

Only nonzero product sectors enter the fusion decomposition. A normal label absent from a fixed product pair has zero fusion multiplicity. See [20].

The source fixed-pair decomposition requires a support correction: a fixed product pair may have no nonzero sectors. See [101].

The fusion map is correspondingly a coisometry onto the direct sum of nonzero sectors, and its adjoint gives exact reconstruction. See [61].

The vertical map may also be rectangular. The reverse equality uses exact reconstruction from the retained sectors rather than a square-unitary identity. See [154].

Proof. [✓ Proof](#) In retained coordinates, the one-site entries are block diagonal in the label and multiplicity coordinates. Within each configuration, multiply the scalar weights and apply the forward and adjoint recursive fusion identities. Summing over all configurations gives the forward equality; exact sitewise reconstruction from the retained sectors gives the reverse equality. $\square$

Theorem 21.3.52 (Density decomposition of an RFP MPDO). [Lean](#) [✓ Stmt](#) Let M generate MPDOs and be in normalized BNT-refined horizontal form. Suppose that M is an RFP. There is one vertical canonical decomposition and one compatible family of fusion coisometries such that, for every $N > 0$, the rectangular vertical coisometry V satisfies

$$V^{\otimes N} \rho^{(N)}(M) (V^{\otimes N})^\dagger = R_N, \tag{21.85}$$

$$((V^\dagger)^{\otimes N}) R_N V^{\otimes N} = \rho^{(N)}(M). \tag{21.86}$$

The statement applies to positive lengths under the normalized BNT-refined horizontal hypothesis.

For a rectangular vertical map, (21.86) uses exact reconstruction from the retained nonzero sectors rather than a square-unitary identity. See [154].

Proof. ✓ Proof Under the normalized BNT-refined horizontal hypothesis, the vertical canonical-form theorem gives a rectangular coisometry onto the retained nonzero vertical sectors, together with exact reconstruction from those sectors. In the retained coordinates, one-site matrix entries are block diagonal in the label and multiplicity coordinates. Hence a closed chain vanishes between different sitewise configurations. Within one configuration, the scalar entries multiply to $m(\alpha, q)$, while the remaining horizontal matrices form the fixed-label product tensor. Apply (21.79) and (21.81), then sum over all configurations. Local reconstruction applied at every site gives (21.86). $\square$

Corollary 21.3.53 (Commuting density factors from fusion on the nonzero sectors). Lean
Lean Lean Lean ✓ Stmt For any vertical canonical decomposition with fusion coisometries onto the nonzero product sectors and exact reconstruction, the recursive density operator satisfies, for every $N > 0$,

$$\begin{aligned} R_N &= A_N T_N, \\ A_N T_N &= T_N A_N. \end{aligned}$$

Thus the multiplicity factor commutes with the reconstructed recursive factor. No length-independence or spectral hypothesis is used.

Proof. ✓ Proof In every sitewise label and multiplicity configuration, A_N is a scalar multiple of the identity on the corresponding product bond space, so it commutes with the reconstructed recursive factor. Taking the direct sum gives both identities. $\square$

Corollary 21.3.54 (Commuting density factors of an RFP MPDO). Lean ✓ Stmt Let M generate MPDOs and be in normalized BNT-refined horizontal form. Suppose that M is an RFP. One choice of the vertical canonical decomposition and fusion coisometries gives the density decomposition and, for every $N > 0$, satisfies

$$R_N = A_N T_N, \tag{21.87}$$

$$A_N T_N = T_N A_N. \tag{21.88}$$

Equivalently, $[A_N, T_N] = 0$, as stated in [11, lines 1000–1002].

Proof. ✓ Proof In every sitewise label and multiplicity configuration, the factor A_N is the scalar $m(c)$ times the identity on the corresponding product bond space. It therefore commutes with $W_{N,c}^\dagger Q_{N,c} W_{N,c}$. Multiplication of the two direct sums proves (21.87) and (21.88). The renormalization fixed-point condition supplies the same fusion family used in the density decomposition. $\square$

Definition 21.3.55 (Terminal spectral refinement of a fusion decomposition). Lean Lean Lean
Lean Lean Lean Lean Lean ✓ Stmt Let M generate MPDOs, and fix a fusion decomposition of its vertical BNT $\{M_\alpha\}_\alpha$ with positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$. For every terminal label γ , close the horizontal operator leg of M_γ to obtain a positive semidefinite bond matrix K_γ . Choose an orthonormal eigenbasis of each K_γ . Its spectral indices are pairs $s = (\gamma, k)$; write λ_s for the corresponding eigenvalue and E_s for the rank-one orthogonal projection onto the chosen eigenvector. Recursively transport this family through every fusion history and the adjoint sequential coisometry. Taking the direct sum over all labels and multiplicities at positive length N defines the operator $P_s^{(N)}$.

Only nonzero product sectors occur in the fusion decomposition. A normal label absent from a fixed product pair has zero fusion multiplicity. See [20].

The fixed-pair decomposition in Figure 11 requires a support correction: a fixed product pair may have no nonzero sectors. See [101].

The fusion map restricted to the retained rows is a coisometry onto the direct sum of nonzero sectors, and its adjoint gives exact reconstruction. This corrects the fusion-map convention in Figure 11. See [61].

Closing the horizontal operator leg of M_γ produces the terminal bond matrix, and the chosen orthonormal eigenbasis expands it as a sum over the spectral indices $s = (\gamma, k)$ with first entry γ .

$$\begin{array}{c} \text{---} \\ | \\ \boxed{M_\gamma} \\ | \\ \text{---} \end{array} = \begin{array}{c} \text{---} \\ | \\ \boxed{K_\gamma} \\ | \\ \text{---} \end{array} = \sum_{s=(\gamma,k)} \lambda_s \begin{array}{c} \text{---} \\ | \\ \boxed{E_s} \\ | \\ \text{---} \end{array}$$

Definition 21.3.56 (Terminal spectral refinement for an RFP MPDO). [Lean](#) [✓ Stmt](#) For an MPDO in normalized BNT-refined horizontal form satisfying the RFP condition, fix the fusion decomposition supplied by that construction. For every terminal label γ , close the horizontal operator leg of M_γ to obtain a matrix K_γ on its bond space. Choose an orthonormal eigenbasis of each K_γ , with spectral indices $s = (\gamma, k)$ satisfying $0 \leq k < D_\gamma$. Write λ_s for the corresponding eigenvalue and E_s for the rank-one orthogonal projection onto the chosen eigenvector, extended by zero on terminal labels different from γ . Recursively transport this terminal family through every fusion history, then through the adjoint sequential fusion coisometry, and take the direct sum over all sitewise label and multiplicity configurations. Denote the resulting operator at positive chain length N by $P_s^{(N)}$.

The normalized BNT-refined horizontal hypothesis is stronger than CF as defined by CPSV.

Theorem 21.3.57 (Terminal spectral projectors from a fusion decomposition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let M generate MPDOs, fix a fusion decomposition on the nonzero product sectors, and suppose that its structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ are independent of L for all positive lengths, with $\chi_{\alpha,\beta,\gamma}$ positive diagonal. Then every terminal eigenvalue λ_s is non-negative. For every $N > 0$, the operators $P_s^{(N)}$ are pairwise orthogonal projections and

$$\begin{aligned} T_N &= \sum_s \lambda_s P_s^{(N)}, \\ A_N P_s^{(N)} &= P_s^{(N)} A_N, \\ R_N &= \sum_s \lambda_s A_N P_s^{(N)}. \end{aligned}$$

This establishes the terminal spectral decomposition of [11, lines 1003–1012].

Only nonzero product sectors occur in the fusion decomposition. A normal label absent from a fixed product pair has zero fusion multiplicity. See [20].

The fixed-pair decomposition in Figure 11 requires a support correction: a fixed product pair may have no nonzero sectors. See [101].

The fusion map restricted to the retained rows is a coisometry onto the direct sum of nonzero sectors, and its adjoint gives exact reconstruction. This corrects the fusion-map convention in Figure 11. See [61].

Proof. [✓ Proof](#) Positivity of the physical density operator makes each terminal bond matrix positive semidefinite. Its spectral theorem supplies the non-negative eigenvalues and orthogonal rank-one projections. Length independence makes every retained diagonal entry of $\chi_{\alpha,\beta,\gamma}$ equal to one, so recursive coisometric transport preserves orthogonality and the spectral sum. Corollary 21.3.53 gives the final two identities. $\square$

Theorem 21.3.58 (Terminal spectral projectors for an RFP MPDO with length-independent structure coefficients). Lean ✓ Stmt *Let M generate MPDOs, be in normalized BNT-refined horizontal form, and satisfy the RFP condition. Fix the fusion decomposition supplied by that construction, and suppose that its structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of L for all positive lengths. Then K_γ is positive semidefinite for every terminal label, so $\lambda_s \geq 0$. Write the positive chain length as $N = n + 1$, where $n \in \mathbb{N}$. For every $N > 0$,*

$$\begin{aligned}(P_s^{(N)})^\dagger &= P_s^{(N)}, \\ (P_s^{(N)})^2 &= P_s^{(N)}.\end{aligned}$$

For distinct spectral indices s and t , $P_s^{(N)} P_t^{(N)} = 0$. The recursive factor and multiplicity weights satisfy

$$T_N = \sum_s \lambda_s P_s^{(N)}, \quad (21.89)$$

$$A_N P_s^{(N)} = P_s^{(N)} A_N. \quad (21.90)$$

Consequently,

$$R_N = \sum_s \lambda_s A_N P_s^{(N)}. \quad (21.91)$$

Length independence is assumed for the coefficients of the fixed fusion decomposition supplied by the RFP construction, not for an arbitrary fusion decomposition.

The normalized BNT-refined horizontal hypothesis is stronger than CF as defined by CPSV.

This establishes the spectral decomposition of [11, lines 1003–1012].

Proof. ✓ Proof Positivity of the one-site density operator passes to every fixed BNT block; division by its strictly positive multiplicity weight shows that each K_γ is positive semidefinite. Apply the finite-dimensional spectral theorem to obtain the non-negative eigenvalues and rank-one orthogonal projections. Length independence makes every retained diagonal entry of $\chi_{\alpha,\beta,\gamma}$ equal to one, so the recursive direct sum preserves idempotence and pairwise orthogonality. Write $Q_{N,c,s}$ for the recursive spectral block in copy configuration c . The coisometry identity $W_{N,c} W_{N,c}^\dagger = \mathbb{1}$ gives

$$\begin{aligned}(W_{N,c}^\dagger Q_{N,c,s} W_{N,c})^\dagger &= W_{N,c}^\dagger Q_{N,c,s} W_{N,c}, \\ (W_{N,c}^\dagger Q_{N,c,s} W_{N,c})^2 &= W_{N,c}^\dagger Q_{N,c,s} W_{N,c}.\end{aligned}$$

Thus transport through the adjoint sequential coisometry preserves the projector identities. Linearity gives (21.89). Finally, A_N is scalar on every copy-configuration block, hence it commutes with each $P_s^{(N)}$ and gives (21.91). □

Definition 21.3.59 (Commuting Gibbs form on the retained vertical space for chains of at least two sites). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *Let M generate MPDOs, fix a fusion decomposition on the nonzero product sectors, and let μ be its strictly positive diagonal multiplicity matrix on the retained one-site space. Define $h = (-\log \mu) \otimes \mathbb{1}$ on two adjacent sites, where $\log \mu$ is the diagonal matrix obtained by taking the logarithm of each diagonal entry. For a chain of length $L = N + 2$, with $N \in \mathbb{N}$, let $H_L = \sum_{i=0}^{L-1} h_{i,i+1}$, with periodic indices.*

Theorem 21.3.61 (Commuting Gibbs decomposition on the retained vertical space for chains of at least two sites). Lean ✓ Stmt Let M generate MPDOs, be in normalized BNT-refined horizontal form, and satisfy the RFP condition. Select the fusion decomposition on the nonzero product sectors supplied by that condition, and suppose that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ of this same decomposition are independent of L for all positive lengths. Then the commuting Gibbs decomposition on the retained vertical space in Theorem 21.3.60 holds for every $L = N + 2$, with $N \in \mathbb{N}$.

The Hamiltonian and projectors act on the retained nonzero vertical sectors, with the physical-space extension treated in [104]. The conclusion holds for $L = N + 2$ (see [51] regarding length one).

Proof. ✓ Proof Select the fusion decomposition on the nonzero product sectors supplied by the RFP condition and apply Theorem 21.3.60 with the assumed length independence of its structure coefficients. $\square$

Theorem 21.3.62 (Fusion decomposition gives a physical commuting Gibbs decomposition for chains of at least two sites). Lean ✓ Stmt Let M generate MPDOs, and fix a fusion decomposition on the nonzero product sectors of a vertical canonical decomposition of M . Suppose that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of L for all positive lengths. There are non-negative numbers $\lambda_0, \dots, \lambda_{d-1}$ and one fixed Hermitian two-site matrix $\hat{h}$, all independent of the chain length. For every $L \geq 2$, let $\hat{H}_L = \sum_{j=0}^{L-1} \hat{h}_{j,j+1}$, with periodic indices. The translates of $\hat{h}$ commute pairwise, and there are orthogonal projections $\hat{P}_0^{(L)}, \dots, \hat{P}_{d-1}^{(L)}$ on the physical chain such that

$$\begin{aligned} [\hat{h}_{j-1,j}, \hat{h}_{j,j+1}] &= 0, \\ [\hat{P}_i^{(L)}, e^{-\hat{H}_L}] &= 0, \\ \rho^{(L)}(M) &= \sum_{i=0}^{d-1} \lambda_i \hat{P}_i^{(L)} e^{-\hat{H}_L}. \end{aligned} \tag{21.96}$$

This is the implication from the fusion decomposition in the proof of [11, lines 999–1016].

Only nonzero product sectors occur in the fusion decomposition. A normal label absent from a fixed product pair has zero fusion multiplicity. See [20].

The fixed-pair decomposition in Figure 11 requires a support correction: a fixed product pair may have no nonzero sectors, and no unsupported normal sector is inserted. See [101].

The fusion map restricted to the retained rows is a coisometry onto the direct sum of nonzero sectors, and its adjoint gives exact reconstruction of the product tensor. This corrects the fusion-map convention in Figure 11. See [61].

To define the operators on the full physical space, transport the retained one-site energy through the adjoint coisometry and extend it by zero energy on the orthogonal complement. Transport the retained projectors through the adjoint sitewise coisometry. This supplies the physical complement correction described in [104].

The conclusion holds for $L = N + 2$, with $N \in \mathbb{N}$. The definition of the local interaction in [11] uses two spins but gives no length-one convention. See [51].

Proof. ✓ Proof Let V be the coisometry restricted to the retained rows and let k be the retained logarithmic multiplicity energy. The physical energy $K = V^\dagger k V$ is Hermitian and

$$e^{-K} = V^\dagger e^{-k} V + (\mathbb{1} - V^\dagger V). \tag{21.97}$$

Taking the fixed two-site term $K \otimes \mathbb{1}$, its periodic translates act on separate one-site factors, hence commute, and their exponential is the tensor power of (21.97).

For the sitewise coisometry $W = V^{\otimes L}$, transport each retained projector by $\widehat{P}_i^{(L)} = W^\dagger P_i^{(L)} W$. The identity $WW^\dagger = \mathbb{1}$ preserves self-adjointness, idempotence, and pairwise orthogonality. The one-site intertwining identities transport the retained commutator to the physical Gibbs factor. Applying the same identities term by term to (21.95), and then using exact adjoint reconstruction, gives (21.96). $\square$

Theorem 21.3.63 (Physical commuting Gibbs decomposition for chains of at least two sites).

[Lean](#) [✓ Stmt](#) *Let M generate MPDOs, be in normalized BNT-refined horizontal form, and satisfy the RFP condition. Select the fusion decomposition on the nonzero product sectors supplied by that condition, and suppose that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ of this same decomposition are independent of L for all positive lengths. Then the physical commuting Gibbs decomposition of Theorem 21.3.62 holds for every $L = N + 2$, with $N \in \mathbb{N}$.*

The retained one-site energy is transported through the adjoint coisometry and extended by zero energy on the orthogonal complement. The retained projectors are transported through the adjoint sitewise coisometry, supplying the physical complement correction described in [104]. The conclusion holds for $L = N + 2$ (see [51] regarding length one).

Proof. [✓ Proof](#) Select the fusion decomposition on the nonzero product sectors supplied by the RFP condition and apply Theorem 21.3.62 with the assumed length independence of its structure coefficients. $\square$

Definition 21.3.64 (Fusion decomposition for an RFP tensor in CPSV canonical form). [Lean](#)

[✓ Stmt](#) *Let M generate MPDOs, suppose that its MPS tensor with paired physical indices is in CF as defined by CPSV, and suppose that M satisfies the RFP condition. Choose one fusion decomposition of the vertical BNT on the nonzero product sectors furnished by these hypotheses and denote it by $\mathcal{F}_{\text{CPSV}}(M)$.*

Only nonzero product sectors occur in the selected fusion decomposition. A normal label absent from a fixed product pair has zero fusion multiplicity. See [20].

The fixed-pair decomposition in Figure 11 requires a support correction: a fixed product pair may have no nonzero sectors, and no unsupported normal sector is inserted. See [101].

The fusion map restricted to the retained rows is a coisometry onto the direct sum of nonzero sectors, and its adjoint gives exact reconstruction of the product tensor. This corrects the fusion-map convention in Figure 11. See [61].

Theorem 21.3.65 (Physical commuting Gibbs decomposition in CPSV canonical form for chains of at least two sites).

[Lean](#) [✓ Stmt](#) *Let M generate MPDOs, suppose that its MPS tensor with paired physical indices is in CF as defined by CPSV, and suppose that M satisfies the RFP condition. Select the fusion decomposition on the nonzero product sectors supplied by these three hypotheses, and assume that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ of this same decomposition are independent of L for all positive lengths.*

There are non-negative numbers $\lambda_0, \dots, \lambda_{d-1}$ and one fixed Hermitian two-site matrix $\hat{h}$, all independent of the chain length. For every $L = N + 2$, with $N \in \mathbb{N}$, let $\widehat{H}_L = \sum_{j=0}^{L-1} \hat{h}_{j,j+1}$, with periodic indices. The translates of $\hat{h}$ commute pairwise, and there are pairwise orthogonal

projections $\widehat{P}_0^{(L)}, \dots, \widehat{P}_{d-1}^{(L)}$ on the physical chain such that

$$\begin{aligned} [\widehat{h}_{j-1,j}, \widehat{h}_{j,j+1}] &= 0, \\ [\widehat{P}_i^{(L)}, e^{-\widehat{H}_L}] &= 0, \\ \rho^{(L)}(M) &= \sum_{i=0}^{d-1} \lambda_i \widehat{P}_i^{(L)} e^{-\widehat{H}_L}. \end{aligned}$$

This is the physical-space conclusion of [11, lines 1013–1016], under the CF and RFP hypotheses of [11, Theorem 4.14, lines 972–993].

Only nonzero product sectors occur in the selected fusion decomposition. A normal label absent from a fixed product pair has zero fusion multiplicity. See [20].

The fixed-pair decomposition in Figure 11 requires a support correction: a fixed product pair may have no nonzero sectors, and no unsupported normal sector is inserted. See [101].

The fusion map restricted to the retained rows is a coisometry onto the direct sum of nonzero sectors, and its adjoint gives exact reconstruction of the product tensor. This corrects the fusion-map convention in Figure 11. See [61].

The retained one-site energy is transported through the adjoint coisometry and extended by zero energy on the orthogonal complement. The retained projectors are transported through the adjoint sitewise coisometry. This supplies the physical complement correction described in [104].

The conclusion holds for $L = N + 2$. The definition of the local interaction in [11] uses two spins but gives no length-one convention. See [51].

Proof. ✓ Proof Choose the fusion decomposition on the nonzero product sectors supplied by Theorem 21.3.26. Its structure coefficients are length independent by hypothesis. Apply Theorem 21.3.62. □

Remark 21.3.66 (Length-one boundary of the two-site construction). The local-interaction definition in [11] lets a local operator act on the first two spins and translates it around the periodic chain. Thus its construction has domain $L \geq 2$. Periodic identification of site labels supplies neither a map from two-site operators to one-site operators nor a multiplicity for the coincident edge.

The source states its Gibbs-decomposition theorem without an explicit lower bound on L , whereas the two-site interaction convention requires $L \geq 2$ (see [51]).

Definition 21.3.67 (One fusion layer with full support). Lean Lean ✓ Stmt Suppose that the fusion isometries have full support, and let P_γ act on the bond space of M_γ . Define

$$\begin{aligned} Q_{\alpha,\beta}^{\text{fs}} &= \bigoplus_{\gamma} \chi_{\alpha,\beta,\gamma} \otimes P_\gamma, \\ \widehat{Q}_{\alpha,\beta}^{\text{fs}} &= U_{\alpha,\beta}^\dagger Q_{\alpha,\beta}^{\text{fs}} U_{\alpha,\beta}. \end{aligned}$$

The terminal contraction is unchanged. Full support means that the active support is the whole product bond space.

Full support is an additional restriction here, not a hypothesis of the source statement for arbitrary chain lengths. This distinction is documented in [28].

Theorem 21.3.68 (Projection property with full support). Lean Lean Lean ✓ Stmt Suppose that $\{M_\alpha\}_\alpha$ is a BNT and that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of $L > 0$. If each P_γ is a self-adjoint idempotent on the bond space, then $Q_{\alpha,\beta}^{\text{fs}}$ is a self-adjoint idempotent and $\widehat{Q}_{\alpha,\beta}^{\text{fs}}$ is an orthogonal projection.

Proof. [✓ Proof](#) Length independence gives $\chi_{\alpha,\beta,\gamma} = 1$, hence $(Q_{\alpha,\beta}^{\text{fs}})^2 = Q_{\alpha,\beta}^{\text{fs}}$ and $(Q_{\alpha,\beta}^{\text{fs}})^\dagger = Q_{\alpha,\beta}^{\text{fs}}$. Completeness for a BNT gives $U_{\alpha,\beta} U_{\alpha,\beta}^\dagger = 1$, so

$$(\widehat{Q}_{\alpha,\beta}^{\text{fs}})^2 = U_{\alpha,\beta}^\dagger Q_{\alpha,\beta}^{\text{fs}} (U_{\alpha,\beta} U_{\alpha,\beta}^\dagger) Q_{\alpha,\beta}^{\text{fs}} U_{\alpha,\beta} = \widehat{Q}_{\alpha,\beta}^{\text{fs}}.$$

Conjugation also preserves self-adjointness. $\square$

Theorem 21.3.69 (Left iterated fusion completeness). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 21.3.42, $U_{\alpha,\beta,\gamma}^L (U_{\alpha,\beta,\gamma}^L)^\dagger = 1$.

Proof. [✓ Proof](#) Pair fusion completeness gives $\sum_\delta r_{a,b}^\delta D_\delta = D_a D_b$. Applying this equality first to (δ, γ) and then to (α, β) shows that the source and target dimensions of $U_{\alpha,\beta,\gamma}^L$ coincide. A square isometry is surjective. $\square$

Theorem 21.3.70 (Right iterated fusion completeness). [Lean](#) [✓ Stmt](#) Under the same hypotheses, $U_{\alpha,\beta,\gamma}^R (U_{\alpha,\beta,\gamma}^R)^\dagger = 1$.

Proof. [✓ Proof](#) The right-associated dimension calculation is

$$\sum_{\delta,\varepsilon} r_{\beta,\gamma}^\delta r_{\alpha,\delta}^\varepsilon D_\varepsilon = D_\alpha (D_\beta D_\gamma).$$

The right iterated fusion map is therefore a square isometry. $\square$

Theorem 21.3.71 (Right adjoint inverse for the full triple-fusion comparison). [Lean](#) [✓ Stmt](#) Under the same hypotheses, the comparison of the full direct sums satisfies $C_{\alpha,\beta,\gamma} C_{\alpha,\beta,\gamma}^\dagger = 1$.

Proof. [✓ Proof](#) The left range identity reduces the product to $U^L (U^L)^\dagger$, which is the identity by Theorem 21.3.69. $\square$

Theorem 21.3.72 (Left adjoint inverse for the full triple-fusion comparison). [Lean](#) [✓ Stmt](#) Under the same hypotheses, $C_{\alpha,\beta,\gamma}^\dagger C_{\alpha,\beta,\gamma} = 1$. This proves invertibility on the full direct sums, but not yet invertibility of the multiplicity matrix for a fixed final label.

Proof. [✓ Proof](#) Apply the right range identity and Theorem 21.3.70. $\square$

Theorem 21.3.73 (Vanishing between distinct final sectors). [Lean](#) [✓ Stmt](#) Suppose that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of $L > 0$ and that common word selectors exist. If $\varepsilon \neq \varepsilon'$, then the block of the full comparison from the right final sector ε' to the left final sector ε vanishes: $P_\varepsilon^L C_{\alpha,\beta,\gamma} P_{\varepsilon'}^R = 0$.

Proof. [✓ Proof](#) Fix one left multiplicity index and one right multiplicity index. Taking the corresponding bond-space block of the full letterwise identity gives $M_\varepsilon^{ij} X = X M_{\varepsilon'}^{ij}$. Induction on the word w gives $M_\varepsilon^w X = X M_{\varepsilon'}^w$. Multiply these identities by $c_\varepsilon(w)$ and sum over all words of length S . The two selector identities give $X = 0$. Since the multiplicity indices were arbitrary, the whole off-diagonal final-sector block vanishes. $\square$

Definition 21.3.74 (Multiplicity spaces for a fixed final label). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix a final label ε . The multiplicity spaces of the two bracketings are

$$\begin{aligned} \mathcal{H}_\varepsilon^L &= \bigoplus_\delta \mathbb{C}^{\dim \chi_{\alpha,\beta,\delta}} \otimes \mathbb{C}^{\dim \chi_{\delta,\gamma,\varepsilon}}, \\ \mathcal{H}_\varepsilon^R &= \bigoplus_\delta \mathbb{C}^{\dim \chi_{\beta,\gamma,\delta}} \otimes \mathbb{C}^{\dim \chi_{\alpha,\delta,\varepsilon}}. \end{aligned}$$

These spaces have the bracketing pattern of the (e, μ, ν) and (f, λ, σ) index spaces in the row and column convention for F in [7, Section 3.4]; no equality with the fusion multiplicities of that source is asserted. The corresponding spaces for a fixed final label are $\mathcal{H}_\varepsilon^L \otimes \mathbb{C}^{D_\varepsilon}$ and $\mathcal{H}_\varepsilon^R \otimes \mathbb{C}^{D_\varepsilon}$.

Theorem 21.3.75 (Equality of multiplicity dimensions for a fixed final label). Lean ✓ Stmt Suppose that the structure coefficients $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$, with positive diagonal $\chi_{\alpha, \beta, \gamma}$, are independent of $L > 0$ and that the operators $O_L(M_\alpha)$ are linearly independent for all sufficiently large L . Then, for each final label ε ,

$$\dim \mathcal{H}_\varepsilon^L = \sum_{\delta} \dim \chi_{\alpha, \beta, \delta} \dim \chi_{\delta, \gamma, \varepsilon} = \sum_{\delta} \dim \chi_{\beta, \gamma, \delta} \dim \chi_{\alpha, \delta, \varepsilon} = \dim \mathcal{H}_\varepsilon^R.$$

Consequently a matrix between these two multiplicity spaces is square, establishing the numerical associativity statement of [7, lines 237–241].

Proof. ✓ Proof Choose a positive length beyond the linear-independence threshold. Associativity of the product of the operators $O_L(M_\alpha)$ gives equality of the two sums of products of structure coefficients. Length independence and positivity identify every coefficient with the size of its corresponding diagonal χ -matrix. The two sums are precisely the dimensions of the two multiplicity spaces. $\square$

Definition 21.3.76 (Triple-fusion maps with fixed final label). Lean Lean ✓ Stmt The maps $U_{\alpha, \beta, \gamma; \varepsilon}^L$ and $U_{\alpha, \beta, \gamma; \varepsilon}^R$ are obtained by restricting the rows of the two iterated fusion isometries to the summands with final label ε . After the canonical reassociation of the triple bond space, their adjoints have the orientation and bracketing pattern of weighted versions of the two fusion maps in [7, Section 3.4].

Theorem 21.3.77 (Left triple fusion at a fixed final label). Lean ✓ Stmt For every final label ε , the left-associated restriction satisfies

$$U_{\alpha, \beta, \gamma; \varepsilon}^L ((M_\alpha M_\beta) M_\gamma)^{ij} (U_{\alpha, \beta, \gamma; \varepsilon}^L)^\dagger = \bigoplus_{\delta} \chi_{\alpha, \beta, \delta} \otimes (\chi_{\delta, \gamma, \varepsilon} \otimes M_\varepsilon^{ij}).$$

Proof. ✓ Proof Restrict Theorem 21.3.33 to the rows and columns whose final label is ε . Reindexing the conjugating matrix commutes with conjugation, and the outer block-diagonal matrix retains precisely the summands with final label ε . $\square$

Theorem 21.3.78 (Right triple fusion at a fixed final label). Lean ✓ Stmt For every final label ε , the right-associated restriction satisfies

$$U_{\alpha, \beta, \gamma; \varepsilon}^R (M_\alpha (M_\beta M_\gamma))^{ij} (U_{\alpha, \beta, \gamma; \varepsilon}^R)^\dagger = \bigoplus_{\delta} \chi_{\beta, \gamma, \delta} \otimes (\chi_{\alpha, \delta, \varepsilon} \otimes M_\varepsilon^{ij}).$$

Proof. ✓ Proof Restrict Theorem 21.3.36 to the rows and columns whose final label is ε . Reindexing the conjugating matrix commutes with conjugation, and the outer block-diagonal matrix retains precisely the summands with final label ε . $\square$

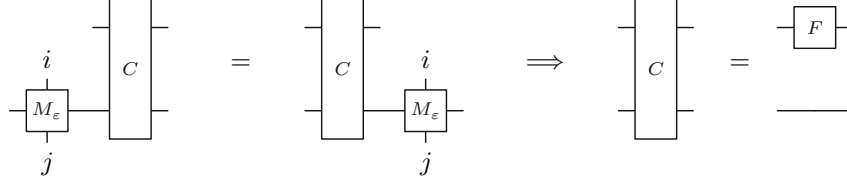
Theorem 21.3.79 (Intertwiners at an injective final sector). Lean ✓ Stmt Suppose that M_ε is injective. Let C be a matrix from the right final-sector space to the left final-sector space. Under the canonical identifications with $\mathcal{H}_\varepsilon^L \otimes \mathbb{C}^{D_\varepsilon}$ and $\mathcal{H}_\varepsilon^R \otimes \mathbb{C}^{D_\varepsilon}$, assume that, for every pair of physical indices i, j ,

$$(1_{\mathcal{H}_\varepsilon^L} \otimes M_\varepsilon^{ij}) C = C (1_{\mathcal{H}_\varepsilon^R} \otimes M_\varepsilon^{ij}).$$

Then there is a rectangular matrix $F : \mathcal{H}_\varepsilon^R \rightarrow \mathcal{H}_\varepsilon^L$ such that $C = F \otimes 1_{D_\varepsilon}$. Here the Kronecker decomposition holds for any given intertwiner C between the fixed-final-label representations.

Proof. ✓ Proof Apply Lemma 2.1.6 to the family $\{M_\varepsilon^{ij}\}_{i,j}$. This family spans the full matrix algebra by injectivity, and the assumed relation is precisely the amplified intertwining identity of that lemma. $\square$

In the diagrams below the upper strand carries the multiplicity space and the lower strand the bond space $\mathbb{C}^{D_\varepsilon}$; the hypothesis on the left forces the factorization on the right.



Theorem 21.3.80 (Kronecker form and separation at a fixed final label). Lean ✓ Stmt Suppose that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of $L > 0$ and that common word selectors separate the final labels. Fix labels $\alpha, \beta, \gamma, \varepsilon$, and suppose that M_ε is injective at the present blocking. Write $C_{\alpha,\beta,\gamma}^{\varepsilon,\varepsilon'} = P_\varepsilon^L C_{\alpha,\beta,\gamma} P_{\varepsilon'}^R$ for the block from the right final sector ε' to the left final sector ε . There is a rectangular matrix $F_\varepsilon : \mathcal{H}_\varepsilon^R \rightarrow \mathcal{H}_\varepsilon^L$ such that, after the canonical index identifications, $C_{\alpha,\beta,\gamma}^{\varepsilon,\varepsilon} = F_\varepsilon \otimes 1_{D_\varepsilon}$. For every $\varepsilon' \neq \varepsilon$, $C_{\alpha,\beta,\gamma}^{\varepsilon,\varepsilon'} = 0$.

Proof. ✓ Proof Taking the ε - ε block of the full comparison identity gives, for every pair i, j ,

$$(1_{\mathcal{H}_\varepsilon^L} \otimes M_\varepsilon^{ij}) C^{\varepsilon,\varepsilon} = C^{\varepsilon,\varepsilon} (1_{\mathcal{H}_\varepsilon^R} \otimes M_\varepsilon^{ij}).$$

Injectivity and the amplified commutant result give $C^{\varepsilon,\varepsilon} = F_\varepsilon \otimes 1_{D_\varepsilon}$. The common word selectors give $C^{\varepsilon,\varepsilon'} = 0$ whenever $\varepsilon' \neq \varepsilon$ by Theorem 21.3.73. $\square$

Theorem 21.3.81 (Unitarity of the comparison at each final sector). Lean Lean ✓ Stmt Suppose that the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$, with positive diagonal $\chi_{\alpha,\beta,\gamma}$, are independent of $L > 0$ and that common selectors exist at a positive length. For every final label ε , write $C_\varepsilon = P_\varepsilon^L C_{\alpha,\beta,\gamma} P_\varepsilon^R$. Then

$$C_\varepsilon C_\varepsilon^\dagger = 1, \tag{21.98}$$

$$C_\varepsilon^\dagger C_\varepsilon = 1. \tag{21.99}$$

The simultaneous selector identities are additional hypotheses. The conclusion gives invertibility at a fixed final label, as obtained from the simultaneous inverse in [7, lines 247–277].

Proof. ✓ Proof Order each full direct sum first by its final label. In the product $CC^\dagger$, the ε -diagonal block is $\sum_{\varepsilon'} C^{\varepsilon,\varepsilon'} (C^{\varepsilon,\varepsilon'})^\dagger$. Every term with $\varepsilon' \neq \varepsilon$ vanishes, while $CC^\dagger = 1$. Hence $C_\varepsilon C_\varepsilon^\dagger = 1$. Taking instead the ε -diagonal block of $C^\dagger C = 1$ gives (21.99). $\square$

Theorem 21.3.82 (Unitarity of the multiplicity matrix at a fixed final label). Lean ✓ Stmt Assume the hypotheses of Theorem 21.3.81. Suppose in addition that M_ε is injective at the present blocking and that $D_\varepsilon > 0$. There is a matrix $F_\varepsilon : \mathcal{H}_\varepsilon^R \rightarrow \mathcal{H}_\varepsilon^L$ such that

$$C_\varepsilon = F_\varepsilon \otimes 1_{D_\varepsilon},$$

$$F_\varepsilon F_\varepsilon^\dagger = 1,$$

$$F_\varepsilon^\dagger F_\varepsilon = 1.$$

Thus F_ε is invertible (with opposite orientation to the row and column convention for F in [7, Section 3.4]).

Proof. ✓ Proof Injectivity gives $C_\varepsilon = F_\varepsilon \otimes 1_{D_\varepsilon}$. Substituting this expression into (21.98) and (21.99) gives

$$\begin{aligned}(F_\varepsilon F_\varepsilon^\dagger) \otimes 1_{D_\varepsilon} &= 1, \\ (F_\varepsilon^\dagger F_\varepsilon) \otimes 1_{D_\varepsilon} &= 1.\end{aligned}$$

Since $D_\varepsilon > 0$, Lemma 21.3.13 removes the identity factor and gives the two asserted identities. $\square$

Definition 21.3.83 (Simultaneous MPO block left inverse). Lean ✓ Stmt Let $M_c^{il} \in M_{D_c}(\mathbb{C})$ be a finite family of MPO tensors. A simultaneous block left inverse consists of coefficients $C_{(c,x,y),(i,l)}$ such that

$$\sum_{i,l} C_{(c,x,y),(i,l)} (M_d^{il})_{x'y'} = \delta_{cd} \delta_{xx'} \delta_{yy'}.$$

This is the common block inverse used in [11, Appendix C.2, lines 1666–1676] and [7, lines 269–277].

Theorem 21.3.84 (Simultaneous inverse from a one-letter product-algebra span). Lean ✓ Stmt Let $A_c^q \in M_{D_c}(\mathbb{C})$ be a finite family of tensors. If the one-letter tuples $(A_c^q)_c$, indexed by q , span the full product algebra $\prod_c M_{D_c}(\mathbb{C})$, then there are coefficients $C_{(c,x,y),q}$ such that

$$\sum_q C_{(c,x,y),q} (A_d^q)_{x'y'} = \delta_{cd} \delta_{xx'} \delta_{yy'}.$$

For an MPO tensor the letter is the ket–bra pair $q = (i, k)$, so the same coefficients form the simultaneous block left inverse used in [7, lines 269–277]. For a BNT, the full product-algebra span is obtained after the common blocking described in [7, lines 427–431].

Proof. ✓ Proof Fix (c, x, y) and consider the product-algebra element which is the matrix unit E_{xy} in block c and zero in every other block. By the spanning hypothesis it is a linear combination of the one-letter tuples. Reading its (x', y') entry in block d gives $\sum_q C_{(c,x,y),q} (A_d^q)_{x'y'} = \delta_{cd} \delta_{xx'} \delta_{yy'}$. Reindexing the doubled physical letter by the ket–bra pair gives the MPO form. $\square$

Definition 21.3.85 (Complete zipper fusion family). Lean ✓ Stmt Let Λ be a finite set. A complete zipper fusion family consists of positive bond dimensions D_a , MPO blocks $B_a^{ij} \in M_{D_a}(\mathbb{C})$, fusion multiplicities N_{ab}^c , and matrices

$$\begin{aligned}X_{ab,\mu}^c &: \mathbb{C}^{D_c} \longrightarrow \mathbb{C}^{D_a} \otimes \mathbb{C}^{D_b}, \\ X_{ab,\mu}^{c+} &: \mathbb{C}^{D_a} \otimes \mathbb{C}^{D_b} \longrightarrow \mathbb{C}^{D_c}.\end{aligned}$$

The matrices $X_{ab,\mu}^{c+}$ are independently chosen left inverses, not adjoints. For all labels and multiplicity indices,

$$X_{ab,\nu}^{d+} X_{ab,\mu}^c = \delta_{cd} \delta_{\mu\nu} \mathbb{1}_{D_c}. \quad (21.100)$$

The opposite product $P_{ab} = \sum_{c,\mu} X_{ab,\mu}^c X_{ab,\mu}^{c+}$ is the projector onto the product-MPO support; it need not be the identity on the ambient bond space or an orthogonal projector. Write $B_{ab}^{ik} = \sum_j B_a^{ij} \otimes B_b^{jk}$. Every product-MPO matrix is reconstructed on this support:

$$B_{ab}^{ik} = \sum_{c,\mu} X_{ab,\mu}^c B_c^{ik} X_{ab,\mu}^{c+}. \quad (21.101)$$

sector cut, $\varepsilon \neq \varepsilon'$:

$$U_{\alpha,\beta,\gamma;\varepsilon}^{\text{L}} \xleftarrow{C_{\alpha,\beta,\gamma}^{\varepsilon,\varepsilon'}} U_{\alpha,\beta,\gamma;\varepsilon'}^{\text{R}} = 0$$

Figure 21.4: The fixed-final comparison in the graphical order of [7, equation (pentagon3) and lines 269–277]. At a positive common length, the simultaneous word selector is the finite-word counterpart of the inverse B_d^+ in the source: it retains the chosen final label and annihilates every other one. Thus the comparison has no off-diagonal final-label corners. On the surviving corner, injectivity gives $C_\varepsilon = F_\varepsilon \otimes 1_{D_\varepsilon}$; two-sided unitarity of C_ε and $D_\varepsilon > 0$ give two-sided unitarity of F_ε . This is the associativity mechanism described in [11, lines 995–1010].

$$B_{ab}^{ik} X_{ab,\mu}^c = X_{ab,\mu}^c B_c^{ik}, \quad (21.102)$$

Each block is injective, and the matrix of coefficients of all labelled blocks has a simultaneous left inverse. Contraction of its row indexed by (c, x, y) with B_d^{ij} gives E_{xy} when $d = c$ and zero otherwise. These are the fusion-tensor, reconstruction, zipper, and simultaneous inverse relations of [7, lines 161, 181–200, and 269–277].

Theorem 21.3.86 (Complete zipper family from a BNT). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose that the labelled tensors of a fusion-isometry family with length-independent structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ form a BNT. There is a common positive blocking length at which they determine a complete zipper fusion family. Its bond dimensions are the original labelled bond dimensions, its fusion multiplicities are $N_{\alpha\beta}^\gamma = \dim \chi_{\alpha,\beta,\gamma}$, its synthesis map is the adjoint of $U_{\alpha,\beta}$, and its analysis map is $U_{\alpha,\beta}$, after the canonical identification $\mathbb{C}^{D_\alpha D_\beta} \cong \mathbb{C}^{D_\alpha} \otimes \mathbb{C}^{D_\beta}$. The two zipper identities, exact reconstruction, injectivity, and the simultaneous left inverse of the block coefficients follow as conclusions rather than additional hypotheses.

Proof. [✓ Proof](#) Length independence makes each positive diagonal matrix $\chi_{\alpha,\beta,\gamma}$ the identity. Use a common positive blocking length for the BNT and carry the fusion equation through the blocking. Completeness of the pair decomposition gives $U_{\alpha,\beta} U_{\alpha,\beta}^\dagger = \mathbb{1}$, so the synthesis map followed by the analysis map is the identity. The other three matrix identities are the blocked zipper identities (21.102) and (21.103) and the reconstruction identity (21.101). At this blocking length, the labelled matrices span the direct sum of the block matrix algebras. This supplies both block injectivity and the simultaneous left inverse of the block coefficients. These maps satisfy Definition 21.3.85. $\square$

Theorem 21.3.87 (Complete zipper family from split compressions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let B_a^{ij} , $a \in \Lambda$, be injective MPO blocks of positive bond dimension whose labelled letters have a simultaneous left inverse as in Definition 21.3.85. Suppose that for every pair a, b the product tensor B_{ab} admits a multi-block compression, as in Theorem 25.5.1, onto N_{ab}^c copies of each block B_c , with vanishing remainder. Then the compression maps $V_{(c,\mu)}$ and $W_{(c,\mu)}$ are the fusion tensors $X_{ab,\mu}^c$ and their left inverses $X_{ab,\mu}^{c+}$ of a complete zipper fusion family. The vanishing remainder is the absence of nonzero blocks above the diagonal assumed in [7, lines 181–191]. In particular, if for every nonempty word the word trace of every product B_{ab} is the sum of the word traces of N_{ab}^c copies of each B_c , and $(B_{ab}^{ik})^\dagger$ lies in the unital algebra generated by the B_{ab}^{jl} for every i, k , then such compressions exist and the conclusion holds.

Proof. [✓ Proof](#) Biorthogonality (21.100) is $W_s V_t = \delta_{st} \mathbb{1}$. Since the remainder $B_{ab}^{ik} - \sum_s V_s B_{c(s)}^{ik} W_s$ vanishes, this sum is the reconstruction (21.101). Multiplying the reconstruction by $X_{ab,\mu}^c$ on the right, or by $X_{ab,\mu}^{c+}$ on the left, and using biorthogonality gives the two zipper identities. In the star-closed case the word-algebra module of each product is semisimple, so its multi-block compression can be chosen with vanishing remainder. $\square$

Proof. ✓ Proof Apply the simultaneous left inverse entrywise and sum over the diagonal matrix units of block d . If $e = d$, their sum is $\mathbb{1}_{D_d}$; if $e \neq d$, every term vanishes. $\square$

Theorem 21.3.91 (F-move and inverse). Lean Lean Lean ✓ Stmt Fix $a, b, c, d \in \Lambda$. There is an invertible matrix

$$F_d^{abc} : \bigoplus_e \mathbb{C}^{N_{ab}^e} \otimes \mathbb{C}^{N_{ec}^d} \longrightarrow \bigoplus_f \mathbb{C}^{N_{bc}^f} \otimes \mathbb{C}^{N_{af}^d}$$

whose rows are indexed by (f, λ, σ) and whose columns are indexed by (e, μ, ν) . Its entries satisfy

$$(X_{ab, \mu}^e \otimes \mathbb{1}) X_{ec, \nu}^d = \sum_{f, \lambda, \sigma} (F_d^{abc})_{e\mu\nu}^{f\lambda\sigma} (\mathbb{1} \otimes X_{bc, \lambda}^f) X_{af, \sigma}^d. \quad (21.104)$$

The matrix in the opposite orientation is its two-sided inverse. This is the F-move of [7, Section 3.4].

Proof. ✓ Proof Form the full left- and right-associated synthesis matrices L and R and their analysis matrices L^+ and R^+ . Applying (21.101) twice to the two associative decompositions gives

$$B_{abc}^{il} = L D_L^{il} L^+ = R D_R^{il} R^+. \quad (21.105)$$

Contract (21.105) with the coefficients s_d from Definition 21.3.89 and sum over the final labels d . The direct sums D_L and D_R both contract to the identity. Consequently,

$$\begin{aligned} P &:= LL^+ = RR^+, \\ L^+ L &= R^+ R = \mathbb{1}. \end{aligned} \quad (21.106)$$

Hence $R^+ L$ changes left-tree coordinates into right-tree coordinates and

$$R(R^+ L) = (RR^+) L = PL = (LL^+) L = L.$$

Its opposite-oriented comparison is $L^+ R$. Using (21.106),

$$\begin{aligned} (R^+ L)(L^+ R) &= R^+ P R = \mathbb{1}, \\ (L^+ R)(R^+ L) &= L^+ P L = \mathbb{1}. \end{aligned}$$

The zipper identities (21.102) and (21.103) show that $R^+ L$ intertwines the direct sums of the final-block matrices. The simultaneous left inverse annihilates its blocks with different final labels. On the d -diagonal block, injectivity gives $(R^+ L)_d = F_d^{abc} \otimes \mathbb{1}_{D_d}$. Restricting $R(R^+ L) = L$ to this block gives (21.104). The corresponding fixed-final block of $L^+ R$, together with $(R^+ L)(L^+ R) = \mathbb{1} = (L^+ R)(R^+ L)$, gives the inverse identities tensored with $\mathbb{1}_{D_d}$. Lemma 21.3.13 removes the nonempty bond-space identity factor and yields the two inverse identities for F_d^{abc} . $\square$

Theorem 21.3.92 (Gauge covariance of the F-matrices). Lean Lean Lean Lean Lean ✓ Stmt For every triple of labels let Y_{ab}^c be an invertible $N_{ab}^c \times N_{ab}^c$ matrix, write $\Pi_{ab} = \bigoplus_c (Y_{ab}^c)^\top \otimes \mathbb{1}_{D_c}$, and let X_{ab} and X_{ab}^+ be the synthesis and analysis matrices whose blocks are the $X_{ab, \mu}^c$ and the $X_{ab, \mu}^{c+}$. Replace them by $X_{ab} \Pi_{ab}$ and $\Pi_{ab}^{-1} X_{ab}^+$. The transposes occur because the fusion tensors are columns of X_{ab} . The new fusion tensors are

$$X'_{ab, \mu}{}^c = \sum_\nu (Y_{ab}^c)_{\mu\nu} X_{ab, \nu}^c.$$

Figure 21.5: The source-oriented F -move. The left-associated fusion tensor is expanded in the right-associated fusion basis. The upper indices (f, λ, σ) label the rows of F_d^{abc} , and the lower indices (e, μ, ν) label its columns. This is the printed- F row/column convention used here for [7, Section 3.4].

The new matrices, with the same blocks, multiplicities and simultaneous left inverse, again form a complete zipper fusion family. The matrix F_d^{abc} is the only matrix satisfying (21.104), and the matrix $F_d'^{abc}$ of the new family is

$$F_d'^{abc} = G_R^{-1} F_d^{abc} G_L, \quad G_L = \bigoplus_e (Y_{ab}^e)^\top \otimes (Y_{ec}^d)^\top, \quad G_R = \bigoplus_f (Y_{bc}^f)^\top \otimes (Y_{af}^d)^\top. \quad (21.107)$$

Its inverse transforms as $(F_d'^{abc})^{-1} = G_L^{-1} (F_d^{abc})^{-1} G_R$. This is the gauge freedom of the fusion tensors in [7, after eq. gauge] and the induced transformation of the F -symbols in [164, lines 415–422].

Proof. ✓ Proof Write $\mathcal{B}_{ab}^{ik} = \bigoplus_c \mathbb{1}_{N_{ab}^c} \otimes B_c^{ik}$, so that (21.101) reads $B_{ab}^{ik} = X_{ab} \mathcal{B}_{ab}^{ik} X_{ab}^+$. Since Π_{ab} acts only on the multiplicity factors, it commutes with $\mathcal{B}_{ab}^{ik}$. The biorthogonality (21.100) and the reconstruction (21.101) of the new family are

$$\begin{aligned} (\Pi_{ab}^{-1} X_{ab}^+) (X_{ab} \Pi_{ab}) &= \Pi_{ab}^{-1} (X_{ab}^+ X_{ab}) \Pi_{ab} = \mathbb{1}, \\ (X_{ab} \Pi_{ab}) \mathcal{B}_{ab}^{ik} (\Pi_{ab}^{-1} X_{ab}^+) &= X_{ab} \mathcal{B}_{ab}^{ik} \Pi_{ab} \Pi_{ab}^{-1} X_{ab}^+ = X_{ab} \mathcal{B}_{ab}^{ik} X_{ab}^+ = B_{ab}^{ik}. \end{aligned}$$

The zipper identities (21.102) and (21.103) of the new family are

$$\begin{aligned} B_{ab}^{ik} X_{ab} \Pi_{ab} &= X_{ab} \mathcal{B}_{ab}^{ik} \Pi_{ab} = (X_{ab} \Pi_{ab}) \mathcal{B}_{ab}^{ik}, \\ \Pi_{ab}^{-1} X_{ab}^+ B_{ab}^{ik} &= \Pi_{ab}^{-1} \mathcal{B}_{ab}^{ik} X_{ab}^+ = \mathcal{B}_{ab}^{ik} (\Pi_{ab}^{-1} X_{ab}^+). \end{aligned}$$

If $R(G \otimes \mathbb{1}) = L$ for the fixed-final synthesis matrices L and R , multiplying on the left by R^+ and using $R^+ R = \mathbb{1}$ gives $G \otimes \mathbb{1} = F_d^{abc} \otimes \mathbb{1}$, hence $G = F_d^{abc}$. Expanding the composites shows that the new synthesis matrices are $L' = L(G_L \otimes \mathbb{1})$ and $R' = R(G_R \otimes \mathbb{1})$. For $G = G_R^{-1} F_d^{abc} G_L$, the F -move (21.104) of the original family gives

$$R'(G \otimes \mathbb{1}) = R(G_R G_R^{-1} F_d^{abc} G_L \otimes \mathbb{1}) = R(F_d^{abc} \otimes \mathbb{1})(G_L \otimes \mathbb{1}) = L(G_L \otimes \mathbb{1}) = L'.$$

By the uniqueness just shown, applied to the new family, $F_d'^{abc} = G_R^{-1} F_d^{abc} G_L$, which is (21.107); multiplying on the left by G_R gives the intertwining form $G_R F_d'^{abc} = F_d^{abc} G_L$. For the inverse, let $H = G_L^{-1} (F_d^{abc})^{-1} G_R$. By (21.107),

$$\begin{aligned} F_d'^{abc} H &= G_R^{-1} F_d^{abc} G_L G_L^{-1} (F_d^{abc})^{-1} G_R = \mathbb{1}, \\ (F_d'^{abc})^{-1} &= (F_d^{abc})^{-1} (F_d'^{abc} H) = ((F_d^{abc})^{-1} F_d'^{abc}) H = H. \end{aligned}$$

□

Proof. ✓ Proof Apply the elementary three-block F -move along each of the five edges. Both paths give the same expansion in the right-associated fourfold fusion map. Its analysis map is a left inverse, by three successive applications of fusion-tensor biorthogonality (21.100). This gives equality after tensoring with the identity on the nonzero final bond space, and Lemma 21.3.13 gives equality of the forward composites. The reverse composites are two-sided inverses of this common matrix, hence they coincide. Taking a matrix entry gives (21.108). $\square$

Theorem 21.3.95 (Coherence of four bond-coordinate reassociations). Lean Lean ✓ Stmt Fix four labels $\alpha, \beta, \gamma, \delta$ in a BNT fusion-isometry family, and write $A = \{0, \dots, D_\alpha - 1\}$, $B = \{0, \dots, D_\beta - 1\}$, $C = \{0, \dots, D_\gamma - 1\}$, and $D = \{0, \dots, D_\delta - 1\}$ for their bond-coordinate sets. Let

$$r_{X,Y,Z} : ((X \times Y) \times Z) \longrightarrow X \times (Y \times Z)$$

denote the canonical reassociation. Define

$$\begin{aligned} r_1 &= r_{A,B,C} \times \mathbb{1}_D, \\ r_2 &= r_{A,B \times C, D}, \\ r_3 &= \mathbb{1}_A \times r_{B,C,D}, \\ s_1 &= r_{A \times B, C, D}, \\ s_2 &= r_{A,B,C \times D}. \end{aligned}$$

The three-edge and two-edge reassociations from $((A \times B) \times C) \times D$ to $A \times (B \times (C \times D))$ coincide:

$$r_3 \circ r_2 \circ r_1 = s_2 \circ s_1. \quad (21.109)$$

If $P(r)_{x,y} = 1$ when $r(x) = y$ and is zero otherwise, then the corresponding coordinate-permutation matrices satisfy

$$P(r_1)P(r_2)P(r_3) = P(s_1)P(s_2). \quad (21.110)$$

These are equalities of Cartesian bond-coordinate identifications only; they are distinct from the fusion-multiplicity identity in Theorem 21.3.94.

Proof. ✓ Proof Every element $((a, b), c), d$ is sent by either composite to $(a, (b, (c, d)))$. This proves (21.109). The identity $P(v \circ u) = P(u)P(v)$ then turns (21.109) into (21.110). $\square$

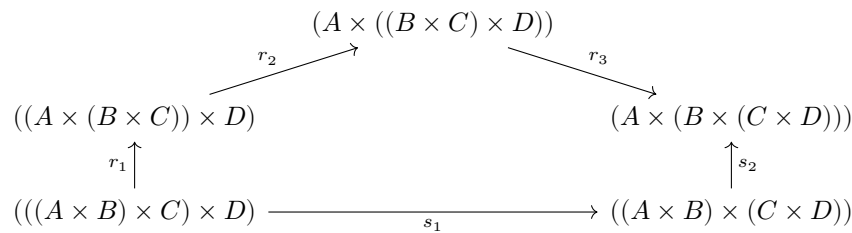


Figure 21.7: The five parenthesizations of four bond-coordinate factors. The upper path is the three-edge composite $r_3 \circ r_2 \circ r_1$, and the lower path is the two-edge composite $s_2 \circ s_1$, depicting canonical Cartesian reassociation of the bond coordinates.

Chapter 22

Periodic MPS: irreducible form and the periodic Fundamental Theorem

The periodic equal-case Fundamental Theorem of matrix product states ([157]) characterizes the gauge equivalence of periodic MPS tensors generating identical state families without prior blocking. Its principal consequences are the lifting of physical on-site symmetries of a tensor in irreducible form II to virtual Z -gauge symmetries, and the equivalence between p -refinability of a matrix-product-state family and p -divisibility of its transfer map.

22.1 Periodic irreducible form and physical-index rotation

Definition 22.1.1 (Spectrally periodic irreducible tensor). [Lean](#) [✓ Stmt](#) An irreducible MPS tensor A is *spectrally m -periodic* if $m \geq 1$, the spectral radius of its transfer map is one, and

$$\text{Spec}_{\text{per}}(\mathcal{E}_A) = \{\mu \in \mathbb{C} : \mu^m = 1\}.$$

No trace-preserving normalization is imposed. This is the form of the periodic blocks before the similarity transformation in [157, lines 248–261 and 313–332].

Theorem 22.1.2 (The period of an irreducible-form block). [Lean](#) [✓ Stmt](#) *Let A be an MPS tensor whose transfer map is irreducible with spectral radius one ([157, lines 248–261]). Then there exists a positive integer m such that A is spectrally m -periodic, so that the peripheral eigenvalues of $\mathcal{E}_A$ are precisely the m -th roots of unity ([157, lines 257–258]).*

Proof. [✓ Proof](#) Let σ be the positive definite fixed point of $\mathcal{E}_A^*$ and pass to the similar tensor $B = \sigma^{1/2} A \sigma^{-1/2}$ of [157, eq. unital, line 316], whose map is trace preserving and still irreducible of spectral radius one. Its adjoint map is unital, and the positive definite Perron fixed point of the map itself is the positive definite fixed point required of the adjoint of that unital map. Write $B^\dagger$ for the family with matrices $(B^i)^\dagger$, so that $\mathcal{E}_{B^\dagger} = \mathcal{E}_B^*$. By [212, Theorem 6.6], there are a positive integer m and a primitive m -th root of unity γ such that

$$\text{Spec}_{\text{per}}(\mathcal{E}_{B^\dagger}) = \{\gamma^k : 0 \leq k < m\} = \{\mu \in \mathbb{C} : \mu^m = 1\}.$$

Passing back from the adjoint conjugates the peripheral spectrum. Since complex conjugation preserves the set of m -th roots of unity,

$$\text{Spec}_{\text{per}}(\mathcal{E}_B) = \overline{\{\mu \in \mathbb{C} : \mu^m = 1\}} = \{\mu \in \mathbb{C} : \mu^m = 1\}.$$

Finally, similarity gives $\text{Spec}_{\text{per}}(\mathcal{E}_A) = \text{Spec}_{\text{per}}(\mathcal{E}_B)$. $\square$

Theorem 22.1.3 (A spectrally periodic tensor has positive bond dimension). [Lean](#) [✓ Stmt](#) *If A is spectrally periodic, then its bond dimension satisfies $D \geq 1$.*

Proof. [✓ Proof](#) In bond dimension zero the matrix space and its endomorphism algebra are zero-dimensional, so the transfer map has spectral radius zero. This contradicts the spectral-radius-one hypothesis. $\square$

Lemma 22.1.4 (Peripheral spectrum is invariant under bond-space similarity). [Lean](#) [✓ Stmt](#) *Let $C \in M_D(\mathbb{C})$ be invertible and let E be a linear map on $M_D(\mathbb{C})$. If*

$$E_C(X) = C^{-1}E(CXC^\dagger)(C^\dagger)^{-1},$$

then E_C and E have the same peripheral spectrum.

Proof. [✓ Proof](#) Conjugation by $X \mapsto CXC^\dagger$ gives a linear equivalence of the matrix space. Transporting eigenvectors through this equivalence preserves every eigenvalue and hence the unit-circle eigenvalue set. $\square$

Definition 22.1.5 (Left-canonical periodic tensor). [Lean](#) [✓ Stmt](#) A left-canonical irreducible MPS tensor A of bond dimension D is m -periodic (for some $m \geq 1$) if the peripheral spectrum of its transfer map $\mathcal{E}_A$ is exactly the m -th roots of unity $\{e^{2\pi i k/m} : 0 \leq k < m\}$. This is the normalized form-II convention used later in the source. A 1-periodic tensor is exactly an irreducible tensor with primitive transfer map (the corresponding definition in the companion quantum-channel volume [182, “Primitive map”]).

Theorem 22.1.6 (Irreducible left-canonical tensors are periodic). [Lean](#) [✓ Stmt](#) *Every irreducible left-canonical MPS tensor A of positive bond dimension is m -periodic for some $m \geq 1$ [157, lines 248–258 and eq. unital].*

Proof. [✓ Proof](#) The transfer map $\mathcal{E}_A$ is positive and trace-preserving on a nonzero matrix space, so $\rho_{\text{spec}}(\mathcal{E}_A) = 1$. By Theorem 22.1.2, $\text{Spec}_{\text{per}}(\mathcal{E}_A) = \{\mu \in \mathbb{C} : \mu^m = 1\}$ for some $m \geq 1$, which together with irreducibility and left-canonicity is the definition of an m -periodic tensor. $\square$

Theorem 22.1.7 (Perron normalization of spectrally periodic tensors). [Lean](#) [✓ Stmt](#) *Let A be spectrally m -periodic. There is a positive definite matrix σ such that*

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2}$$

is a left-canonical m -periodic tensor. The similarity is pure, so it leaves every matrix-product vector unchanged and introduces no scalar weight into the normalization.

Proof. [✓ Proof](#) Apply Theorem 8.2.2. The transfer map of B is similar to that of A , so Lemma 22.1.4 preserves the prescribed set of m -th roots of unity. $\square$

Definition 22.1.8 (Replacement of the representative family). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let

$$A^i = \bigoplus_{j \in J} (R_j \otimes A_j^i)$$

be a multiplicity-bearing sector decomposition. For tensors B_j with the same bond dimension as A_j , its representative replacement is

$$A_B^i = \bigoplus_{j \in J} (R_j \otimes B_j^i).$$

Thus the index set, bond dimensions, multiplicities, and every entry of the diagonal matrices R_j remain unchanged. In particular, for every length N and every representative j , the grouped coefficient remains

$$\text{tr}(R_j^N) = \sum_q \mu_{j,q}^N.$$

Theorem 22.1.9 (Similarity of sector-decomposed tensors). [Lean](#) [✓ Stmt](#) Suppose that, for every $j \in J$, there is an invertible matrix X_j such that

$$B_j^i = X_j A_j^i X_j^{-1}.$$

Then the two multiplicity-bearing tensors are related by the pure similarity

$$A_B^i = X A^i X^{-1}, \quad X = \bigoplus_{j \in J} (I_{r_j} \otimes X_j).$$

Proof. [✓ Proof](#) Repeat X_j on every copy indexed by a diagonal entry of R_j , and take their block-diagonal direct sum. On the (j, q) block, conjugation by this matrix sends $\mu_{j,q} A_j^i$ to $\mu_{j,q} B_j^i$. $\square$

Theorem 22.1.10 (Sectorwise Perron normalization). [Lean](#) [✓ Stmt](#) Suppose every representative A_j is spectrally m_j -periodic. There are left-canonical m_j -periodic representatives B_j , each purely similar to A_j , such that the two full multiplicity-bearing tensors are purely similar. Every multiplicity matrix R_j is unchanged.

Proof. [✓ Proof](#) Apply the positive Perron gauge to each representative and combine them via Theorem 22.1.9. Since every spectral radius is already one, no representative is rescaled. $\square$

Theorem 22.1.11 (A periodic tensor has positive bond dimension). [Lean](#) [✓ Stmt](#) If A is periodic, then its bond dimension satisfies $D \geq 1$.

Proof. [✓ Proof](#) The prescribed peripheral spectrum contains the eigenvalue 1. In bond dimension zero the matrix space is zero-dimensional and admits no eigenvector, which is a contradiction. $\square$

Definition 22.1.12 (Irreducible form). [Lean](#) [✓ Stmt](#) A tensor A admits an *irreducible-form representation* if it generates the same matrix-product-vector family as the block-diagonal tensor

$$\widehat{A}^i = \bigoplus_{j \in J} \mu_j A_j^i,$$

where each block A_j is a left-canonical m_j -periodic tensor. The defining relation is $|V^{(N)}(A)\rangle = |V^{(N)}(\widehat{A})\rangle$ for every $N \geq 0$, rather than literal equality of the tensors A and $\widehat{A}$. In particular, it includes the empty word and hence equality of the bond-dimension traces. The source comparison

is at positive physical lengths and permits the irreducible-form representative to have smaller bond dimension [157, lines 266–271]. Following [157, Section II], the phases are fixed by the convention $\mu_j > 0$, so the weights are positive real scalars. This is a normalized matrix-product-vector representation of the source irreducible form, whose displayed decomposition is a literal tensor equality and whose periodic blocks are only required to have irreducible transfer maps of spectral radius one. It extends the normal canonical form (Definition 7.9.2), which requires the stronger condition that every block is 1-periodic.

Remark 22.1.13 (Normal canonical form gives irreducible form). Lean Lean ✓ Stmt Let $A^i = \bigoplus_j \mu_j A_j^i$ be a normal canonical block family with nonzero complex weights. Replacing μ_j by $|\mu_j|$ and A_j by $(\mu_j/|\mu_j|)A_j$ gives the positive-weight irreducible form convention and does not change the MPV family. The corresponding irreducible-form witness has all periods equal to 1. This is the period-one part of the statement in [157, Section II] that normal tensors are periodic tensors with period 1.

Remark 22.1.14 (Irreducible form allows higher periods). Not ready Irreducible forms permit periodic blocks with period $m > 1$; for example, the antiferromagnetic state has period 2.

Physical-index rotations act on periodic tensors and their transfer maps as follows.

Theorem 22.1.15 (Transfer map preserved by physical rotation). Lean ✓ Stmt If $uu^\dagger = \mathbb{1}$, then the transfer map is invariant under physical-index rotation: $\mathcal{E}_{A_u} = \mathcal{E}_A$.

Proof. ✓ Proof This is the unitary freedom of Kraus representations from [212, Theorem 2.1(4)]. □

Theorem 22.1.16 (Left-canonical preserved by physical rotation). Lean ✓ Stmt If $uu^\dagger = \mathbb{1}$ and A is left-canonical, then A_u is left-canonical.

Proof. ✓ Proof Expand $\sum_i (B^i)^\dagger B^i$, where $B^i = \sum_j u_{ij} A_j^i$, swap sums, apply orthogonality $u^\dagger u = \mathbb{1}$, and collapse to $\sum_j (A_j)^\dagger A_j = \mathbb{1}$. □

Theorem 22.1.17 (Irreducibility preserved by physical rotation). Lean ✓ Stmt If $uu^\dagger = \mathbb{1}$ and A is irreducible, then A_u is irreducible.

Proof. ✓ Proof Contrapositive: any nontrivial invariant projection P for the rotated tensor yields $(\mathbb{1} - P)A^k P = 0$ for all k via the orthogonality of u , contradicting irreducibility of A . □

Theorem 22.1.18 (Periodicity preserved by physical rotation). Lean ✓ Stmt If $uu^\dagger = \mathbb{1}$ and A is periodic with period m , then A_u is periodic with the same period.

Proof. ✓ Proof This combines transfer-map invariance (Theorem 22.1.15), left-canonical preservation (Theorem 22.1.16), and irreducibility (Theorem 22.1.17). □

Theorem 22.1.19 (Equality of MPV families preserved by physical rotation). Lean ✓ Stmt If $\mathcal{V}(A) = \mathcal{V}(B)$, then $\mathcal{V}(A_u) = \mathcal{V}(B_u)$.

Proof. ✓ Proof Induction on the word length with a strengthened hypothesis carrying an arbitrary prefix matrix, reducing each step to the hypothesis via word concatenation. □

Theorem 22.1.20 (Physical rotation distributes over block-diagonal composition). Lean ✓ Stmt The rotated block-diagonal tensor satisfies

$$\left(\bigoplus_k \mu_k A_k \right)_u = \bigoplus_k \mu_k (A_k)_u.$$

Proof. [✓ Proof](#) Pointwise matrix-entry computation using linearity of the block-diagonal embedding and reindexing. $\square$

Theorem 22.1.21 (Irreducible form preserved by physical rotation). [Lean](#) [✓ Stmt](#) If $uu^\dagger = \mathbb{1}$ and A is in irreducible form, then A_u is in irreducible form with the same block structure and rotated blocks.

Proof. [✓ Proof](#) Each block remains periodic by Theorem 22.1.18, equality of MPV families transfers via Theorem 22.1.19, and block-diagonal compatibility follows from Theorem 22.1.20. $\square$

Corollary 22.1.22 (Periodic symmetry yields finite-order Z -gauge equivalence). [Lean](#) [✓ Stmt](#) Assume the periodic equal-case theorem: tensors in irreducible form II with equal MPV families are Z -gauge equivalent. Let A be in irreducible form II , and suppose $B := A_M$ is also in irreducible form II with $\mathcal{V}(A) = \mathcal{V}(B)$. Then there are $m > 0$, an invertible matrix Y , and a matrix $Z \in M_D(\mathbb{C})$ such that, for every physical index i ,

$$[Z, A^i] = 0, \quad Z^m = \mathbb{1}_D, \quad ZA^i = YB^iY^{-1}.$$

Hence A and B are related by a $\mathbb{Z}_m$ virtual gauge.

Proof. [✓ Proof](#) Apply the assumed periodic equal-case theorem to A and B . Its Z -gauge conclusion supplies m , Y , and Z with the three displayed identities by Definition 22.1.1.7. $\square$

22.1.1 The Z -gauge degree of freedom

Definition 22.1.1.1 (Entrywise periodic Z -gauge ratio). [Lean](#) [✓ Stmt](#) Given complex multiplicity entries μ, ν , define the entrywise Z -gauge ratio by $z(\mu, \nu) := \mu/\nu$.

Lemma 22.1.1.2 (Matched powers imply root-of-unity ratio). [Lean](#) [✓ Stmt](#) If $\mu^m = \nu^m$ and $\nu \neq 0$, then $z(\mu, \nu)^m = 1$.

Lemma 22.1.1.3 (Ratio rescales denominator back to numerator). [Lean](#) [✓ Stmt](#) If $\nu \neq 0$, then $z(\mu, \nu)\nu = \mu$.

Definition 22.1.1.4 (Diagonal periodic Z -gauge matrix). [Lean](#) [✓ Stmt](#) For matched multiplicity entries $(\mu_i)_i$ and $(\nu_i)_i$ on a finite index type, define the diagonal matrix $Z := \text{diag}(z(\mu_i, \nu_i))_i$.

Lemma 22.1.1.5 (Diagonal Z -gauge has finite order under matched powers). [Lean](#) [✓ Stmt](#) If $\mu_i^m = \nu_i^m$ and $\nu_i \neq 0$ for all i , then $Z^m = \mathbb{1}$.

Lemma 22.1.1.6 (Diagonal Z -gauge transports diagonal multiplicities). [Lean](#) [✓ Stmt](#) Under $\nu_i \neq 0$ for all i , one has $Z \text{diag}(\nu) = \text{diag}(\mu)$.

Definition 22.1.1.7 (Z -gauge equivalence). [Lean](#) [✓ Stmt](#) Let A be an m -periodic irreducible tensor of bond dimension D . A global finite-order intertwining relation is given by a matrix $Z \in M_D(\mathbb{C})$ satisfying:

1. Z commutes with every tensor matrix: $[A^i, Z] = 0$ for all physical indices i , and
2. $Z^m = \mathbb{1}_D$.

Two tensors A, B are Z -gauge equivalent if there exists an invertible Y and a Z -gauge transformation Z for A such that $ZA^i = YB^iY^{-1}$ for all i . Theorem 22.2.2.1 refines this global relation to block-multiplicity diagonal gauges Z_j .

Remark 22.1.1.8 (Z -gauge is trivial for period-1 blocks). [Lean](#) [✓ Stmt](#) When $m = 1$, the condition $Z^1 = \mathbb{1}$ forces $Z = \mathbb{1}$, so Z -gauge equivalence reduces to ordinary gauge equivalence (Definition 2.4.1). Nontrivial Z -gauges can only appear when $m > 1$.

Theorem 22.1.1.9 (Blocking a $\mathbb{Z}_m$ -gauge gives an ordinary gauge). [Lean](#) [✓ Stmt](#) If A and B are $\mathbb{Z}_m$ -gauge equivalent, then their m -blocked tensors $A^{[m]}$ and $B^{[m]}$ are gauge equivalent in the ordinary sense.

Proof. [✓ Proof](#) Write the $\mathbb{Z}_m$ -gauge relation as $ZA^i = YB^iY^{-1}$ with $Z^m = \mathbb{1}$ and $[A^i, Z] = 0$. For a blocked letter, commute one factor of Z past each physical letter in the associated length- m word. The total prefactor becomes $Z^m = \mathbb{1}$, so the entire blocked word is related by the same ordinary gauge Y . $\square$

Corollary 22.1.1.10 (Blocking a $\mathbb{Z}_m$ -gauge preserves the MPV family). [Lean](#) [✓ Stmt](#) If A and B are $\mathbb{Z}_m$ -gauge equivalent, then the blocked tensors $A^{[m]}$ and $B^{[m]}$ have the same MPV family.

Proof. [✓ Proof](#) Apply Theorem 22.1.1.9 and then the gauge invariance of MPV families. $\square$

22.2 Periodic Fundamental Theorem

While the fundamental theorems in Chapter 9 require primitive transfer maps obtained by blocking across a common period, the framework of [157] applies directly to periodic tensors.

22.2.1 Common-period blocking lemmas

Definition 22.2.1.1 (LCM period for a finite block family). [Lean](#) [✓ Stmt](#) For a finite family of positive integer periods $(p_i)_{i \in I}$, their least common multiple $\text{lcm}_{i \in I}(p_i)$ defines the shared blocking period.

Definition 22.2.1.2 (Common-period blocked family). [Lean](#) [✓ Stmt](#) For a finite family of tensors $(B_i)_{i \in I}$ with periods $(p_i)_{i \in I}$, the *common-period blocked family* consists of the blocked tensors $B_i^{[L]}$ with $L = \text{lcm}_{j \in I}(p_j)$.

Theorem 22.2.1.3 (Common-period blocking preserves transfer-map primitivity). [Lean](#) [✓ Stmt](#) If each tensor B_i has a primitive transfer map after blocking by its period $p_i \geq 1$, then each blocked tensor $B_i^{[L]}$ with $L = \text{lcm}_j(p_j)$ also has a primitive transfer map.

Proof. [✓ Proof](#) Because each p_i divides $L = \text{lcm}_j(p_j)$, Theorem D.4.5 implies that the transfer map $\mathcal{E}_{B_i^{[L]}}$ is primitive for every $i \in I$. $\square$

22.2.2 Statement of the periodic Fundamental Theorem

Theorem 22.2.2.1 formalizes the equal-case Fundamental Theorem of [157, lines 643–656] for tensors in irreducible form. Theorem 22.2.2.14 handles the case where each block’s transfer map is trace-preserving; the reduction to this normalization via blockwise similarity ([157, eq. *unital*, line 316]) is carried out in the proof.

Theorem 22.2.2.1 (Periodic Fundamental Theorem). [Lean](#) [✓ Stmt](#) Let A and B be MPS tensors in irreducible form, written

$$A^i = \bigoplus_{j \in J} (R_j \otimes A_j^i),$$

$$B^i = \bigoplus_{k \in K} (S_k \otimes B_k^i),$$

where $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ are pairwise non-repeated bases of periodic tensors, R_j and S_k are diagonal multiplicity matrices of sizes $r_j \times r_j$ and $s_k \times s_k$ with nonzero diagonal entries, and A_j^i and B_k^i act on bond spaces of dimensions D_j and D_k . The blocks need only have irreducible transfer maps of spectral radius one ([157, lines 248–261]); no trace-preserving normalization is imposed, and the periodicity m_j of each block is derived from the peripheral spectrum ([157, lines 257–258]). Suppose that $\mathcal{V}^+(A) = \mathcal{V}^+(B)$.

Then $|J| = |K|$ and there is a bijection $\pi : J \rightarrow K$ such that A_j and $B_{\pi(j)}$ are m_j -periodic for the same period m_j . Moreover, for each $j \in J$ there exist:

1. a unit-modulus scalar ξ_j and an invertible matrix Y_j with $A_j^i = \xi_j Y_j B_{\pi(j)}^i Y_j^{-1}$ for every physical index i (Definition 22.2.2.8), and
2. a diagonal matrix Z_j of size $r_j \times r_j$ with $Z_j^{m_j} = \mathbb{1}_{r_j}$, so that all its diagonal entries are m_j -th roots of unity, and a reordering T_j of the diagonal entries of $S_{\pi(j)}$ with $Z_j (\xi_j R_j) = T_j S_{\pi(j)} T_j^\dagger$.

Setting $Z = \bigoplus_j Z_j \otimes \mathbb{1}_{D_j}$ one has $Z^{\text{lcm}_j m_j} = \mathbb{1}$, $[A^i, Z] = 0$ for every i , $\mathcal{V}^+(A) = \mathcal{V}^+(ZA)$, and there is an invertible matrix Y with $ZA^i = Y B^i Y^{-1}$ for all i .

Proof. ✓ Proof Theorem 22.1.2 yields a positive period m_j for each block A_j and n_k for each block B_k . Applying Theorem 22.2.2.3 with these derived periods gives every conclusion. $\square$

Remark 22.2.2.2 (Conventions on length range and scalars). The matrix-product-vector identities are taken at positive lengths, following the convention that the empty word is not a physical chain; this extends to length zero since the empty-word coefficient equals the bond dimension. The scalars ξ_j are made explicit rather than absorbed into $S_{\pi(j)}$ as in [157, lines 667–671], allowing the statement to speak directly about the given multiplicity matrices.

Theorem 22.2.2.3 (Periodic Fundamental Theorem, periods supplied). Lean ✓ Stmt Under the hypotheses of Theorem 22.2.2.1, suppose in addition that positive integers m_j and n_k are given such that each A_j is spectrally m_j -periodic and each B_k is spectrally n_k -periodic (Definition 22.1.1). Then the conclusions of Theorem 22.2.2.1 hold for these specified periods.

Proof. ✓ Proof Normalize each representative by its positive-definite adjoint Perron fixed point, which is the source construction at [157, lines 313–332]. The normalized representatives are periodic with trace-preserving block maps and are blockwise similar to the originals, so the multiplicities and their power sums are unchanged, the non-repetition of the two bases is preserved, and the two direct-sum tensors generate the same matrix-product vectors as before. Theorem 22.2.2.14 applies to the normalized pair.

Transport the conclusion back. The bijection, the scalars, the periods and the blockwise multiplicity gauges concern only the multiplicity data, which the normalization leaves untouched, and the repeated-block relations compose with the two blockwise similarities. The multiplicity gauge is unchanged: writing $X = \bigoplus_j X_j$ for the normalizing similarity, Lemma 22.2.2.11 gives

$$ZX = \bigoplus_j (Z_j \otimes \mathbb{1}_{D_j}) X_j = XZ, \quad \text{hence} \quad X^{-1}ZX = Z.$$

Only the similarity between the two tensors is conjugated, by the two normalizing similarities. $\square$

Theorem 22.2.2.4 (Diagonal multiplicity gauge from equal powers). [Lean] [✓ Stmt] Let $(\mu_i)_{i \in I}$ and $(\nu_i)_{i \in I}$ be two finite lists of nonzero multiplicity entries. If $\mu_i^m = \nu_i^m$ for every $i \in I$, then there exists a diagonal matrix Z such that

$$Z^m = \mathbb{1}, \\ Z \operatorname{diag}(\nu) = \operatorname{diag}(\mu).$$

Proof. [✓ Proof] Take $Z = \operatorname{diag}(\mu_i/\nu_i)_{i \in I}$ and apply the two diagonal identities from Lemmas 22.1.1.5 and 22.1.1.6. □

Theorem 22.2.2.5 (Scalar multiplicity gauge from power sums). [Lean] [✓ Stmt] Let $(\mu_i)_{i \in I}$ and $(\nu_i)_{i \in I}$ be two finite lists of multiplicity entries. Suppose that the ν_i are nonzero, that $\mu_i^m = \nu_i^m$ for every $i \in I$, and that $\sum_{i \in I} \mu_i^k = \sum_{i \in I} \nu_i^k$ for every positive integer k . Then the two lists determine the same multiset, and there is a diagonal Z satisfying $Z^m = \mathbb{1}$ and $Z \operatorname{diag}(\nu) = \operatorname{diag}(\mu)$.

Proof. [✓ Proof] Apply Theorem 22.2.2.4 to obtain Z , and apply the Newton–Girard power-sum identity to recover the multiplicity multiset. □

Lemma 22.2.2.6 (Power weights of a sector decomposition). [Lean] [Lean] [✓ Stmt] Raising every multiplicity entry of a sector decomposition to a fixed power m leaves the basis blocks and the multiplicities unchanged, and the coefficient of the resulting decomposition at length n is the original coefficient at length mn :

$$\sum_q (\mu_{j,q}^m)^n = \sum_q \mu_{j,q}^{mn}.$$

Proof. [✓ Proof] The entries $\mu_{j,q}^m$ are again nonzero, so they form a sector weight family over the same basis, and the displayed identity is the exponent rule. □

Lemma 22.2.2.7 (Matched multiplicity entries along a period progression). [Lean] [✓ Stmt] Let two sector decompositions have distinguished blocks j_0 and k_0 , with multiplicity entries $(\mu_{j_0,q})_q$ and $(\nu_{k_0,q})_q$, and let ζ be a nonzero scalar. Suppose that

$$\sum_q \mu_{j_0,q}^{mn} = \zeta^{mn} \sum_q \nu_{k_0,q}^{mn}$$

for every n beyond some threshold. Then the two blocks carry the same multiplicity, and there is a bijection τ between their copy indices with

$$\nu_{k_0,\tau(q)}^m = (\zeta^{-1} \mu_{j_0,q})^m$$

for every q . This is the form in which the power-sum comparison enters [157, lines 681–684], where the comparison is available only along the multiples of a period; the recovery of the entries from their power sums is [157, lines 537–569].

Proof. [✓ Proof] Raise all multiplicity entries of both decompositions to the power m . By Lemma 22.2.2.6 the hypothesis becomes a comparison of the coefficients of the two resulting decompositions at every exponent beyond the threshold, with scalar ζ^m . Applying Lemma 8.7.1.2 to those decompositions gives the multiplicity equality and the bijection, and its conclusion is the displayed identity. □

Definition 22.2.2.8 (Similarity carrying a prescribed scalar). [Lean](#) [✓ Stmt](#) For a scalar ζ , two tensors A and B are *similar carrying* ζ if they have the same bond dimension and there is an invertible matrix Y with

$$A^i = \zeta Y B^i Y^{-1}$$

for every physical index i . This is the repeated-block relation of [157, def. *repeated*, lines 276–284] with the scalar named instead of quantified away, so that one statement can use the same scalar in the similarity and in the multiplicity relation.

Theorem 22.2.2.9 (Equal-case coefficient extraction). [Lean](#) [✓ Stmt](#) Let

$$A^i = \bigoplus_{j \in J} (R_j \otimes A_j^i),$$

$$B^i = \bigoplus_{k \in K} (S_k \otimes B_k^i),$$

be two multiplicity-bearing decompositions whose bases of periodic tensors are matched by a bijection $\pi: J \rightarrow K$ and nonzero scalars ζ_j with

$$|V^{(N)}(A_j)\rangle = \zeta_j^N |V^{(N)}(B_{\pi(j)})\rangle$$

for every length N , with $\{A_j\}_{j \in J}$ pairwise non-repeated and A_j periodic of period m_j . If $\mathcal{V}^+(A) = \mathcal{V}^+(B)$, then there is an N_0 such that

$$\text{tr}(R_j^N) = \zeta_j^{-N} \text{tr}(S_{\pi(j)}^N)$$

for every $N \geq N_0$ with $m_j \mid N$ and every $j \in J$. This is [157, lines 672–680]. The scalar ζ_j is displayed rather than suppressed: the identity compares the power sums of R_j with the rescaled power sums of $S_{\pi(j)}$, which is what the source means at [157, lines 667–671] by absorbing the phase into S_j .

Proof. [✓ Proof](#) Expand both vectors along the basis $\{A_j\}_{j \in J}$ by Lemma 8.6.1.4, rewriting each $|V^{(N)}(B_{\pi(j)})\rangle$ as $\zeta_j^{-N} |V^{(N)}(A_j)\rangle$. Equality of the two vectors makes the combination of the $|V^{(N)}(A_j)\rangle$ with coefficients $\text{tr}(R_j^N) - \zeta_j^{-N} \text{tr}(S_{\pi(j)}^N)$ vanish. By Theorem 7.3 the term for an index with $m_j \nmid N$ contributes the zero vector, so the combination runs over the indices with $m_j \mid N$, and by Theorem 22.2.3.44 those vectors are independent for every large N . Each coefficient therefore vanishes. $\square$

Theorem 22.2.2.10 (Blockwise multiplicity gauge in the equal case). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 22.2.2.9, for every $j \in J$ the multiplicity matrices R_j and $S_{\pi(j)}$ have the same size r_j , and there are a reordering τ of the diagonal entries of $S_{\pi(j)}$ and a diagonal matrix Z_j with

$$Z_j^{m_j} = \mathbb{1}_{r_j}, \quad Z_j (\zeta_j R_j) = T_j S_{\pi(j)} T_j^\dagger,$$

where T_j implements τ . This is [157, lines 681–688]. The scalar ζ_j stays visible: the diagonal entries of Z_j are m_j -th roots of unity against the rescaled multiplicity matrix $\zeta_j R_j$, and the rescaling is in general not trivial even for period one, as the pair of blocks $B_j = e^{i\theta} A_j$ shows.

Proof. [✓ Proof](#) Fix j . Theorem 22.2.2.9 compares the power sums of R_j and $S_{\pi(j)}$ at every multiple of m_j beyond N_0 , with scalar ζ_j^{-1} . Feeding that progression to Lemma 22.2.2.7 gives the equality of sizes, the reordering τ , and the entry identity $\nu_{\pi(j), \tau(q)}^{m_j} = (\zeta_j \mu_{j,q})^{m_j}$. Since the entries $\zeta_j \mu_{j,q}$ are nonzero, Theorem 22.2.2.4 produces the diagonal Z_j with the two displayed properties. $\square$

Lemma 22.2.2.11 (The multiplicity gauge commutes with block-diagonal gauges). [Lean](#) [Stmnt](#) Let $Z = \bigoplus_k z_k \mathbb{1}_{D_k}$ act by the scalar z_k on the bond space of the k -th block, and let $X = \bigoplus_k X_k$ be block diagonal for the same block decomposition. Then $ZX = XZ$.

This is why the multiplicity gauge of [157, line 653] survives the blockwise normalizing similarity of [157, lines 313–332] unchanged.

Proof. [Proof](#) On the k -th block the two factors are $z_k \mathbb{1}_{D_k}$ and X_k , and a scalar matrix is central. $\square$

Theorem 22.2.2.12 (Global multiplicity gauge and similarity). [Lean](#) [Stmnt](#) Let

$$A^i = \bigoplus_{j \in J} (R_j \otimes A_j^i),$$

$$B^i = \bigoplus_{k \in K} (S_k \otimes B_k^i),$$

be two multiplicity-bearing decompositions, let $\pi: J \rightarrow K$ be a bijection with $D_j = D_{\pi(j)}$, let τ_j reorder the multiplicity copies of the matched blocks, let ξ_j be scalars and Y_j invertible matrices with

$$A_j^i = \xi_j Y_j B_{\pi(j)}^i Y_j^{-1},$$

and let $z_{j,l}$ be scalars with

$$z_{j,l} \xi_j \mu_{j,l} = \nu_{\pi(j), \tau_j(l)}, \quad z_{j,l}^{m_j} = 1,$$

where A_j is m_j -periodic and every m_j divides L . Writing Z_j for the diagonal matrix of the $z_{j,l}$ and $Z = \bigoplus_{j \in J} Z_j \otimes \mathbb{1}_{D_j}$, one has $Z^L = \mathbb{1}$, $[A^i, Z] = 0$ for every i , $\mathcal{V}^+(A) = \mathcal{V}^+(ZA)$, and there is an invertible Y with $ZA^i = YB^iY^{-1}$ for every i . If every Y_j is unitary, then Y is unitary as well.

This establishes the global intertwining relation of [157, lines 689–690], with Y the direct sum $\bigoplus_{j \in J} T_j \otimes Y_j$ of the copy reorderings and the blockwise similarities.

Proof. [Proof](#) The multiplicity gauge acts by the scalar $z_{j,l}$ on the bond space of the copy (j, l) , so it commutes with every block-diagonal matrix and its L -th power is the identity. Multiplying A^i by it rescales the multiplicity entries by the $z_{j,l}$, and the hypothesis on the $z_{j,l}$ makes the rescaled entries equal to the matched entries of B . The matched blocks are therefore related copy by copy, so the two direct-sum tensors are related by the permutation of the matched copies followed by the direct sum of the blockwise similarities. If every Y_j is unitary, then, since each T_j is a permutation matrix,

$$YY^\dagger = \bigoplus_{j \in J} (T_j T_j^\dagger) \otimes (Y_j Y_j^\dagger) = \mathbb{1},$$

and likewise $Y^\dagger Y = \mathbb{1}$. For the matrix-product vectors, a copy whose period divides the length contributes the factor $z_{j,l}^N = 1$, and a copy whose period does not divide the length contributes the zero vector by Theorem 7.3. $\square$

Theorem 22.2.2.13 (Matched periods for trace-preserving repeated blocks). [Lean](#) [Stmnt](#) Let A and B be periodic tensors of periods m and n whose block maps are trace preserving. If A and B are repeated blocks, then $m = n$.

Proof. ✓ Proof Repeated blocks have the same peripheral transfer spectrum,

$$\text{spec}_{\text{per}}(\mathcal{E}_A) = \text{spec}_{\text{per}}(\mathcal{E}_B),$$

and for a periodic tensor that set is the set of m -th roots of unity, so the two periods agree. $\square$

Theorem 22.2.2.14 (Fundamental theorem for MPS with trace-preserving blocks, equal case).

Lean ✓ Stmt *Let A and B carry the multiplicity-bearing forms*

$$\begin{aligned} A^i &= \bigoplus_{j \in J} (R_j \otimes A_j^i), \\ B^i &= \bigoplus_{k \in K} (S_k \otimes B_k^i), \end{aligned}$$

where R_j and S_k are diagonal with nonzero entries and the two displayed families are pairwise non-repeated bases of periodic tensors whose blocks are trace preserving, of periods m_j and n_k . If $\mathcal{V}^+(A) = \mathcal{V}^+(B)$, then there is a bijection $\pi: J \rightarrow K$ and unit-modulus scalars ξ_j such that, for every j , one has $m_j = n_{\pi(j)}$, there is an invertible Y_j with

$$A_j^i = \xi_j Y_j B_{\pi(j)}^i Y_j^{-1}$$

for every physical index i , the multiplicity matrices R_j and $S_{\pi(j)}$ have the same size, and there are a reordering T_j of the diagonal entries of $S_{\pi(j)}$ and a diagonal matrix Z_j with

$$Z_j^{m_j} = \mathbb{1}, \quad Z_j (\xi_j R_j) = T_j S_{\pi(j)} T_j^\dagger.$$

Writing $Z = \bigoplus_{j \in J} Z_j \otimes \mathbb{1}_{D_j}$, one has $Z^{\text{lcm}_j m_j} = \mathbb{1}$, $[A^i, Z] = 0$ for every i , $\mathcal{V}^+(A) = \mathcal{V}^+(ZA)$, and there is a unitary Y with $ZA^i = Y B^i Y^\dagger$ for every i .

The scalar ξ_j is the same in the two displayed relations: it is the scalar of the similarity between the matched blocks, and the entries of Z_j are m_j -th roots of unity against the rescaled multiplicity matrix $\xi_j R_j$. That is what the source means at lines 667–671 by absorbing the phase into S_j ; the scalar is present, as the period-one pair $B_j = e^{i\theta} A_j$ shows.

This establishes [157, Theorem 3.8, lines 643–693] for tensors whose block maps are trace preserving ([157, eq. unital, line 316]). The periodic overlap dichotomy and the vanishing of off-period blocks apply directly in this trace-preserving setting, from which Theorem 22.2.2.1 is deduced.

Proof. ✓ Proof Equality of the generated matrix-product vectors is in particular proportionality, so Theorem 22.2.3.57 supplies non-decaying partners and Theorem 22.2.3.60 matches the two bases, each matched pair carrying a similarity with a unit-modulus scalar ξ_j . Since the blocks are left canonical and irreducible, Lemma 9.4.1.2 replaces each similarity by a unitary U_j with the same scalar:

$$\begin{aligned} A_j^i &= \xi_j U_j B_{\pi(j)}^i U_j^\dagger, \\ U_j U_j^\dagger &= \mathbb{1}_{D_j}. \end{aligned}$$

The matched periods agree by Theorem 22.2.2.13. That same ξ_j relates the matrix-product vectors of the matched pair at every length, so Theorem 22.2.2.9 compares the multiplicity power sums at every large multiple of m_j , and Lemma 22.2.2.7 turns that comparison into the reordering T_j and the entry identities $\nu_{\pi(j), \tau_j(l)}^{m_j} = (\xi_j \mu_{j,l})^{m_j}$. The ratios $z_{j,l} = \nu_{\pi(j), \tau_j(l)} / (\xi_j \mu_{j,l})$ are therefore m_j -th roots of unity; they are collected into the diagonal Z_j , and Theorem 22.2.2.12 constructs the global Z and the unitary similarity Y . $\square$

Theorem 22.2.2.15 (Unit-modulus multiplicities under equal matrix-product vectors). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 22.2.2.14, if every multiplicity of B has modulus one, then every multiplicity of A has modulus one. This is the multiplicity normalization used in the corrected forward implication of [157, Theorem 4.1, lines 743–756].

Proof. [✓ Proof](#) The matched multiplicities satisfy $z_{j,l}\xi_j\mu_{j,l} = \nu_{\pi(j),\tau_j(l)}$. Since $z_{j,l}^{m_j} = 1$ with $m_j > 0$ and $|\xi_j| = 1$, taking absolute values gives $|\mu_{j,l}| = |\nu_{\pi(j),\tau_j(l)}|$, and the hypothesis $|\nu_{\pi(j),\tau_j(l)}| = 1$ gives $|\mu_{j,l}| = 1$. $\square$

22.2.3 Periodic overlap dichotomy

Theorem 22.2.3.1 (Sector scalars from a product tensor identity). [Lean](#) [✓ Stmt](#) Let $z \in \mathbb{C}$ be nonzero. For each sector $v = 0, \dots, m-1$, let $\{A_v^i\}_i$ and $\{B_v^i\}_i$ be families of complex matrices with the same rectangular size. Suppose that for every choice of letters $\sigma = (\sigma_v)_v$,

$$\bigotimes_{v=0}^{m-1} A_v^{\sigma_v} = z \bigotimes_{v=0}^{m-1} B_v^{\sigma_v}.$$

Suppose also that for each v there is a reference letter i_v and a matrix entry (r_v, c_v) with $(B_v^{i_v})_{r_v, c_v} \neq 0$. Then there are scalars $\kappa_v \in \mathbb{C}$ such that

$$\prod_{v=0}^{m-1} \kappa_v = z, \\ A_v^i = \kappa_v B_v^i$$

for every sector v and every letter i .

This is the scalar extraction used in [157, Appendix A, lines 1072–1080], after the contraction with the maps F_u and Ω_u has produced the displayed product-tensor identity.

Proof. [✓ Proof](#) For each sector choose the coordinate functional at the nonzero reference entry and normalize it so that it takes the value 1 on $B_v^{i_v}$. Applying the tensor product of these functionals to the reference tensor identity gives $\prod_v \kappa_v = z$. To prove the proportionality in sector v , replace only the v -th functional by the coordinate functional at an arbitrary entry (r, c) and apply the tensor identity with $\sigma_v = i$ and all other letters fixed at their references. The product over the other sectors is nonzero because its product with κ_v is z , so cancellation gives $(A_v^i)_{r,c} = (\kappa_v B_v^i)_{r,c}$ for every entry. $\square$

Theorem 22.2.3.2 (Product-one sector scalars after choosing a root). [Lean](#) [✓ Stmt](#) Let $m > 0$, let $z \in \mathbb{C}$ be nonzero, and choose $\xi \in \mathbb{C}$ with $\xi^m = z$. For each sector $v = 0, \dots, m-1$, let $\{A_v^i\}_i$ and $\{B_v^i\}_i$ be families of complex matrices with the same rectangular size. Suppose that for every choice of letters $\sigma = (\sigma_v)_v$,

$$\bigotimes_{v=0}^{m-1} A_v^{\sigma_v} = z \bigotimes_{v=0}^{m-1} B_v^{\sigma_v}.$$

Suppose also that each sector has a reference letter i_v and a matrix entry (r_v, c_v) with $(B_v^{i_v})_{r_v, c_v} \neq 0$. Then there are scalars $\kappa_v \in \mathbb{C}$ such that

$$\prod_{v=0}^{m-1} \kappa_v = 1, \\ A_v^i = \kappa_v \xi B_v^i$$

for every sector v and every letter i .

This is the normalization in [157, Appendix A, lines 1072–1080]: after Theorem 22.2.3.1 has extracted scalars independent of the sector coordinates, choosing a common m -th root of their product leaves a product-one family of sector scalars.

Proof. ✓ Proof Apply Theorem 22.2.3.1 to obtain scalars μ_v with $\prod_v \mu_v = z$ and $A_v^i = \mu_v B_v^i$. Since $m > 0$, the identities $\xi^m = z$ and $z \neq 0$ imply $\xi \neq 0$. Put $\kappa_v = \mu_v/\xi$. Then $\prod_v \kappa_v = (\prod_v \mu_v)/\xi^m = z/z = 1$, and $A_v^i = (\kappa_v \xi) B_v^i$. $\square$

Theorem 22.2.3.3 (Left-canonical scalar normalization). Lean ✓ Stmt Let A have positive bond dimension, and let $c \in \mathbb{C}$. If both A and the scalar multiple cA are left-canonical, then $|c| = 1$.

Proof. ✓ Proof Left-canonicity of cA gives

$$\mathbb{1} = \sum_i (cA^i)^\dagger (cA^i) = |c|^2 \sum_i (A^i)^\dagger A^i = |c|^2 \mathbb{1}, \quad (22.1)$$

where the last equality uses left-canonicity of A . Since the bond dimension is positive, comparison of any diagonal entry gives $|c|^2 = 1$, hence $|c| = 1$. $\square$

Theorem 22.2.3.4 (Left-canonical sector scalar normalization). Lean ✓ Stmt Let A and B be left-canonical tensors with the same positive bond dimension. Suppose that $|\xi| = 1$ and that $A^i = (\kappa \xi) B^i$ for every physical index i . Then $|\kappa| = 1$.

This is the scalar normalization in [157, Appendix A, lines 1082–1084], after the sectorwise proportionality relation has been obtained.

Proof. ✓ Proof By Theorem 22.2.3.3, applied to B and the scalar $\kappa \xi$, one has $|\kappa \xi| = 1$. Since $|\xi| = 1$, the multiplicativity of the complex norm gives $|\kappa| = 1$. $\square$

Definition 22.2.3.5 (Off-diagonal sector decomposition). Lean Lean ✓ Stmt A family of compressed sector tensors $(C_k)_{k=0}^{m-1}$ on corner bond spaces forms an *off-diagonal decomposition* of the blocked tensor $A^{(m)}$ if there exist orthogonal projections $P_0, \dots, P_{m-1}$ on the ambient bond space such that $\sum_k P_k = \mathbb{1}$, the projections satisfy the cyclic-shift relation $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$ with the index $k+1$ taken cyclically modulo m , each P_k commutes with every blocked letter $P_k A_i^{(m)} = A_i^{(m)} P_k$, and $V^{(N)}(C_k)_\sigma = \text{tr}(P_k (A^{(m)})^\sigma)$ for every word σ , and each block carries a $*$ -algebra isomorphism $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k \cdot M_D(\mathbb{C}) \cdot P_k$ (multiplicative and adjoint-preserving) intertwining the compressed adjoint transfer map with the sector adjoint transfer map on the corner of P_k . Retaining the projections and corner isomorphisms as data gives the witness-bearing formulation. This is the inverse cyclic indexing of the off-diagonal convention in [157, Appendix A]. There the same adjoint transfer map is written E^* , and the source projectors satisfy $E^*(P_u) = P_{u+1}$ together with $A^i = \sum_u P_u A^i P_{u+1}$. Here P_k corresponds to the source sector $u = -k \pmod{m}$, which turns the source shift into the displayed adjoint-transfer relation.

Theorem 22.2.3.6 (Single-site cyclic grading). Lean ✓ Stmt Let A be left-canonical, and let $P_0, \dots, P_{m-1}$ be orthogonal projections on the bond space. If the adjoint transfer map satisfies $\mathcal{E}_A^\dagger(P_{k+1}) = P_k$ for all k , with indices taken modulo m , then $P_{k+1} A^i = A^i P_k$ for every physical letter i . This is the single-site grading behind [157, eq. Aoffdiag]; the compressed blocks of [157, eq. Auprop] use the inverse cyclic indexing convention of Definition 22.2.3.5.

Proof. ✓ Proof Put $K_i = (A^i)^\dagger$ and let Φ_K be the Kraus map of the family K . Since A is left-canonical, Φ_K is unital. The projection identities and the shift relation give

$$\Phi_K(P_{k+1}^\dagger P_{k+1}) = \Phi_K(P_{k+1})^\dagger \Phi_K(P_{k+1}),$$

because both sides are equal to P_k . The equality case in the Kadison–Schwarz inequality therefore gives $P_{k+1}K_i^\dagger = K_i^\dagger\Phi_K(P_{k+1})$. Substituting $K_i^\dagger = A^i$ and $\Phi_K(P_{k+1}) = P_k$ gives $P_{k+1}A^i = A^iP_k$. $\square$

Theorem 22.2.3.7 (Off-diagonal reconstruction). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 22.2.3.6, assume in addition that $\sum_{u=0}^{m-1} P_u = 1$. Then every physical letter decomposes as $A^i = \sum_{u=0}^{m-1} P_{u+1}A^iP_u$. This is [157, eq. Aoffdiag], written in the same inverse cyclic indexing convention.*

Proof. [✓ Proof](#) Insert $\sum_u P_u = 1$ on the right of A^i , then use Theorem 22.2.3.6 and $P_u^2 = P_u$ term by term. $\square$

Theorem 22.2.3.8 (Word cyclic grading). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 22.2.3.6, a word $w = i_1 \dots i_\ell$ satisfies $P_{k+\ell}A^w = A^wP_k$. Thus the single-site grading accumulates one cyclic step for each letter of the word.*

Proof. [✓ Proof](#) Induct on the length of w , using Theorem 22.2.3.6 for the first letter and the induction hypothesis for the remaining word. $\square$

Definition 22.2.3.9 (Corner-transition letters). [Lean](#) [✓ Stmt](#) Let $P_0, \dots, P_{m-1}$ be cyclic sector projections for a tensor A . The one-site corner-transition letters of [157, Appendix A] are $A_u^i = P_uA^iP_{u+1}$, with the index $u+1$ taken cyclically modulo m .

Definition 22.2.3.10 (Repeated corner products). [Lean](#) [✓ Stmt](#) For a word $\mathbf{i} = (i_1, \dots, i_\ell)$, define

$$F_u^{\mathbf{i}} = A_u^{i_1}A_{u+1}^{i_2} \dots A_{u+\ell-1}^{i_\ell}.$$

The empty word gives $F_u^\emptyset = P_u$, and a one-letter word gives $F_u^{(i)} = A_u^i$. This is the notation of [157, Appendix A]; in the Case 3 contraction the length is $\ell = mN_0$.

Lemma 22.2.3.11 (Concatenation of corner products). [Lean](#) [✓ Stmt](#) *If the sector maps P_u are orthogonal projections, then for any two words $\mathbf{i}$ and $\mathbf{j}$, $F_u^{\mathbf{ij}} = F_u^{\mathbf{i}}F_{u+|\mathbf{i}|}^{\mathbf{j}}$, where indices are taken cyclically modulo m .*

Proof. [✓ Proof](#) Induct on the word $\mathbf{i}$. For the empty word the identity is $P_uF_u^{\mathbf{j}} = F_u^{\mathbf{j}}$, which follows from the idempotent part of the orthogonal-projection hypothesis. For a nonempty word, remove the first letter and apply the induction hypothesis at the next cyclic sector. $\square$

Lemma 22.2.3.12 (Repeated-block matrix identity). [Lean](#) [✓ Stmt](#) *Let $P_0, \dots, P_{m-1}$ and $Q_0, \dots, Q_{m-1}$ be orthogonal projections, and suppose the m -block product identity*

$$A_u^{i_1} \dots A_{u+m-1}^{i_m} = c_u B_{u+q}^{i_1} \dots B_{u+q+m-1}^{i_m}$$

holds for every base sector u and every length- m word. Then for every word W of length km with $k \geq 1$, $F_u^W = c_u^k \widehat{F}_{u+q}^W$, where $\widehat{F}$ is the repeated corner product built from B . Each of the k successive length- m blocks is anchored at the same base sector u , because a full cycle of unit shifts is trivial, $m \cdot 1 = 0$ in $\mathbb{Z}_m$; hence every block contributes the same scalar c_u .

Proof. [✓ Proof](#) Induct on k . The base case $k = 1$ is the hypothesis. For the inductive step split W into its first length- m block and the remaining length- km word, apply the concatenation law, and use that the junction shift $m \cdot 1 = 0$ returns to the base sector before applying the block identity and the induction hypothesis. $\square$

Lemma 22.2.3.13 (Right-inverse contraction). [Lean](#) [✓ Stmt](#) Suppose the coefficients Ω_{k+1} recover each inserted matrix X_k from the repeated corner product F_{k+1} over length- L words: $\sum_j (\Omega_{k+1}(X_k))_j F_{k+1}^j = X_k$ ([157, eq. Omegauprop] at X_k). Then for any sector indices σ ,

$$\sum_{\rho} \left(\prod_k (\Omega_{k+1}(X_k))_{\rho_k} \right) \prod_k A_k^{\sigma_k} F_{k+1}^{\rho_k} = \prod_k A_k^{\sigma_k} X_k.$$

Summing the concatenated chain against these coefficients collapses each repeated product F_{k+1} to its inserted matrix X_k . The recovery is needed only at the inserted matrices X_k , which in the Case-3 contraction are the corner-supported tensors that Ω_{k+1} inverts.

Proof. [✓ Proof](#) Expand each $A_k^{\sigma_k} X_k = \sum_j (\Omega_{k+1}(X_k))_j A_k^{\sigma_k} F_{k+1}^j$ using the right inverse, apply the noncommutative distributivity identity

$$\prod_k \left(\sum_j f_{k,j} \right) = \sum_{\rho} \prod_k f_{k,\rho_k},$$

and pull the scalar coefficients $\prod_k (\Omega_{k+1}(X_k))_{\rho_k}$ out of each ordered product. $\square$

Lemma 22.2.3.14 (Inverse cyclic reindexing to the off-diagonal convention). [Lean](#) [✓ Stmt](#) Let A be left-canonical, $\sum_i A_i^\dagger A_i = 1$, with orthogonal sector projectors P_k satisfying the adjoint shift $\mathcal{E}^*(P_{k+1}) = P_k$, which gives the inverse-indexed grading $P_{k+1} A^i = A^i P_k$. After the inverse reindexing $P_k \mapsto P_{-k}$ the projectors satisfy the off-diagonal convention $P_{-k} A^i = A^i P_{-(k+1)}$ of [157, eq. Auprop], under which the corner-transition products are nonzero.

Proof. [✓ Proof](#) Apply the grading at sector $-(k+1)$ and simplify the cyclic index. $\square$

Lemma 22.2.3.15 (Telescoping of a corner product). [Lean](#) [✓ Stmt](#) Under the off-diagonal convention $P_k A^i = A^i P_{k+1}$, with the P_k orthogonal projections, the repeated corner-transition product collapses through its intermediate projectors:

$$F_u^i = P_u A^i P_{u+|i|},$$

where A^i is the ordinary word product and the right projector records the accumulated sector offset.

Proof. [✓ Proof](#) Induct on the word. The head junction $P_u A^i P_{u+1} = P_u A^i$ collapses by the shift and idempotency; the induction hypothesis handles the tail, and the accumulated offset realigns. $\square$

Lemma 22.2.3.16 (Blocked letter as a diagonal corner). [Lean](#) [✓ Stmt](#) Under the off-diagonal convention $P_k A^i = A^i P_{k+1}$, with the P_k orthogonal projections, the corner-transition product of a single blocked letter is the diagonal blocked corner of [157, eq. Cu]: $F_u^{(I)} = P_u (A^{[m]})^I P_u$. Combined with the corner-isomorphism letter identity, this exhibits the compressed sector letter as a corner-transition product.

Proof. [✓ Proof](#) A blocked letter is a word of length exactly m ; the accumulated offset $m \cdot 1 = 0$ vanishes, so the telescoping gives the diagonal corner. $\square$

Definition 22.2.3.17 (Flattening bijection on word indices). [Lean](#) [✓ Stmt](#) The bijection between blocked words of length L over the alphabet of d^m blocked letters and single-site words of length mL over the alphabet of d letters, $[d^m]^L \simeq [d]^{mL}$, is obtained from the iterated blocking bijection. Here $[q]^r$ denotes the set of words of length r over a q -letter alphabet.

Lemma 22.2.3.18 (Flattening bijection agrees with word flattening). [Lean](#) [✓ Stmt](#) *If $W \in [d^m]^L$ is a blocked word and $\widehat{W} \in [d]^{mL}$ is its image under the flattening bijection, then $\widehat{W}$ is exactly the concatenation of the L decoded length- m blocks of W .*

Proof. [✓ Proof](#) The decoding bijection identifies a word-indexing function with its ordered list of letters, and the iterated blocking bijection concatenates the decoded length- m blocks in order. $\square$

Lemma 22.2.3.19 (Corner product of a flattened word). [Lean](#) [✓ Stmt](#) *Let P_k be orthogonal projections satisfying the off-diagonal convention $P_k A^i = A^i P_{k+1}$, and let φ_k be the multiplicative corner isomorphisms with $\varphi_k(\bar{A}_k^I) = P_k(A^{[m]})^I P_k$. Then for a nonempty blocked word W , with flattening $\widehat{W}$, the corner-transition product is the corner image of the word evaluation,*

$$F_k^{\widehat{W}} = \varphi_k(\bar{A}_k^W),$$

the content of [157, eq. Fu] read through φ .

Proof. [✓ Proof](#) Induct on W . The head blocked letter collapses by the blocked-letter diagonal corner to $\varphi_k(\bar{A}_k^I)$, and multiplicativity of φ combines the letters:

$$F_k^{\widehat{W}} = \prod_{I \in W} \varphi_k(\bar{A}_k^I) = \varphi_k \left(\prod_{I \in W} \bar{A}_k^I \right) = \varphi_k(\bar{A}_k^W).$$

The empty word is excluded so the induction stays inside the corner. $\square$

Lemma 22.2.3.20 (Ambient corner right inverse from the sector right inverse). [Lean](#) [✓ Stmt](#) *From a sector right inverse Ω recovering every sector word product $\bar{A}_k^W$ over blocked words $W \in [d^m]^L$ with $L > 0$ ([157, eq. Omegauprop]) one obtains an ambient corner right inverse $\hat{\Omega}$ over single-site words of length mL . For every corner-supported Y , that is, $P_k Y P_k = Y$,*

$$\sum_{\mathbf{j} \in [d]^{mL}} (\hat{\Omega}_k(Y))_{\mathbf{j}} F_k^{\mathbf{j}} = Y.$$

Proof. [✓ Proof](#) Set $\hat{\Omega}_k(Y) = \Omega_k(\varphi_k^{-1}(Y))$ for corner-supported Y . Applying φ_k to the sector recovery and using the corner product of a flattened word,

$$\sum_W (\hat{\Omega}_k(Y))_{\widehat{W}} F_k^{\widehat{W}} = \varphi_k \left(\sum_W (\Omega_k(\varphi_k^{-1}(Y)))_W \bar{A}_k^W \right) = \varphi_k(\varphi_k^{-1}(Y)) = Y.$$

$\square$

Lemma 22.2.3.21 (Sector contraction). [Lean](#) [✓ Stmt](#) *Feeding the ambient corner right inverse into the right-inverse contraction yields the contracted product-tensor identity [157, eq. result-prop]. For corner-supported inserted matrices X_k ,*

$$\sum_{\rho} \left(\prod_k (\hat{\Omega}_{k+1}(X_k))_{\rho_k} \right) \prod_k A_k^{\sigma_k} F_{k+1}^{\rho_k} = \prod_k A_k^{\sigma_k} X_k.$$

Proof. [✓ Proof](#) Apply the right-inverse contraction with the ambient corner right inverse in place of the abstract right inverse. $\square$

Theorem 22.2.3.22 (Cyclic-sector single-site grading). [Lean](#) [✓ Stmt](#) *If a periodic tensor admits a cyclic sector decomposition, then for every k and every physical letter i there exists a family $P_0, \dots, P_{m-1}$ such that each P_u is an orthogonal projection,*

$$\begin{aligned} P_u^\dagger &= P_u, & P_u^2 &= P_u, \\ \sum_{u=0}^{m-1} P_u &= \mathbb{1}, \end{aligned}$$

and $P_{k+1} A^i = A^i P_k$.

Proof. [✓ Proof](#) Choose the projection family contained in the cyclic sector decomposition. It satisfies

$$\begin{aligned} P_u^\dagger &= P_u, & P_u^2 &= P_u, \\ \sum_u P_u &= \mathbb{1}, \\ \mathcal{E}_A^\dagger(P_{u+1}) &= P_u. \end{aligned}$$

Apply Theorem 22.2.3.6. □

Theorem 22.2.3.23 (Cyclic-sector off-diagonal reconstruction). [Lean](#) [✓ Stmt](#) *If a periodic tensor admits a cyclic sector decomposition, then for every physical letter i there exists a family $P_0, \dots, P_{m-1}$ such that each P_u is an orthogonal projection,*

$$\begin{aligned} P_u^\dagger &= P_u, & P_u^2 &= P_u, \\ \sum_{u=0}^{m-1} P_u &= \mathbb{1}, \end{aligned}$$

and

$$A^i = \sum_{u=0}^{m-1} P_{u+1} A^i P_u.$$

Proof. [✓ Proof](#) Choose the projection family contained in the cyclic sector decomposition. It satisfies

$$\begin{aligned} P_u^\dagger &= P_u, & P_u^2 &= P_u, \\ \sum_u P_u &= \mathbb{1}, \\ \mathcal{E}_A^\dagger(P_{u+1}) &= P_u. \end{aligned}$$

Apply Theorem 22.2.3.7. □

Theorem 22.2.3.24 (Cyclic sector decomposition of a periodic tensor). [Lean](#) [✓ Stmt](#) *If A is periodic with period m , then the blocked tensor $A^{(m)}$ admits a cyclic sector decomposition. Thus there are projectors P_u and corner tensors $C_u = P_u A^{(m)}$ which reproduce the blocked matrix-product vectors, are left-canonical, have nonzero bond dimensions, and realize the cyclic adjoint-transfer relation of Lemma bdcf in [157, lines 404–423]. The normality and non-repetition conclusions of Lemma bdcf are recorded separately in the component lemmas below.*

Proof. ✓ **Proof** This is the cyclic-sector existence part of Lemma *bdcf*, translated into the inverse indexing convention of Definition 22.2.3.5. □

Theorem 22.2.3.25 (Cyclic-sector decomposition with compressed corner letters). Lean ✓ Stmt *Under the hypotheses of the cyclic-sector construction for the blocked tensor $A^{(m)}$, there are sector tensors C_k , projections P_k , and linear corner isomorphisms*

$$\varphi_k: M_{\dim C_k}(\mathbb{C}) \longrightarrow P_k M_D(\mathbb{C}) P_k$$

whose sector tensors are left-canonical, reproduce the blocked matrix-product vectors, have nonzero bond dimensions, and carry the cyclic adjoint-transfer decomposition of Theorem 22.2.3.24. In addition, for every blocked physical letter i ,

$$\varphi_k(C_k^i) = P_k (A^{(m)})^i P_k.$$

This is the corner-letter identity used in [157, Lemma bdcf, eq. Cu].

Proof. ✓ **Proof** Let U diagonalize P_k , so that

$$U^\dagger P_k U = \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix}.$$

Write $B^i = U^\dagger (A^{(m)})^i U$, and let B_{kk}^i be the corner block on the support of P_k . The sector tensor is $C_k^i = V_k^\dagger B_{kk}^i V_k$, where V_k identifies the support with $\mathbb{C}^{\dim C_k}$. The linear corner isomorphism is the inverse transport

$$X \mapsto U \begin{pmatrix} V_k X V_k^\dagger & 0 \\ 0 & 0 \end{pmatrix} U^\dagger.$$

Hence

$$\varphi_k(C_k^i) = U \begin{pmatrix} B_{kk}^i & 0 \\ 0 & 0 \end{pmatrix} U^\dagger = P_k (A^{(m)})^i P_k.$$

□

Theorem 22.2.3.26 (Periodic cyclic sectors with support isometries). Lean ✓ Stmt *If A is periodic with period m , then the cyclic-sector decomposition of $A^{(m)}$ may be chosen with rectangular support isometries V_k such that*

$$\begin{aligned} V_k^\dagger V_k &= \mathbb{1}, \\ V_k V_k^\dagger &= P_k, \\ \varphi_k(X) &= V_k X V_k^\dagger. \end{aligned}$$

In particular, the same isometries realize the corner-letter identity $\varphi_k(C_k^i) = P_k (A^{(m)})^i P_k$.

Proof. ✓ **Proof** Apply Theorem 7.11.1 to the original tensor A with its period- m peripheral-spectrum hypotheses. The resulting decomposition is the decomposition of $A^{(m)}$, and the isometries V_k in its conclusion are the required support isometries. □

Lemma 22.2.3.27 (One-site transport of sector overlaps). Lean ✓ Stmt *Let A and B be left-canonical tensors with cyclic sector decompositions $\{A_u\}_{u \in \mathbb{Z}_m}$ and $\{B_v\}_{v \in \mathbb{Z}_m}$. For every $u, v \in \mathbb{Z}_m$ and every $N > 0$, $O_{A_{u+1}, B_{v+1}}(N) = O_{A_u, B_v}(N)$.*

Proof. ✓ Proof Choose the projection families P_u and Q_v from the two cyclic sector decompositions. The single-site relations $P_{u+1}A^i = A^iP_u$ and $Q_{v+1}B^i = B^iQ_v$ give the following trace-sum identity, where T is the one-site rotation of physical words:

$$\begin{aligned} O_{A_{u+1}, B_{v+1}}(N) &= \sum_{\tau \in [d]^{Nm}} \text{tr}(P_{u+1}A^\tau) \overline{\text{tr}(Q_{v+1}B^\tau)} \\ &= \sum_{\tau \in [d]^{Nm}} \text{tr}(P_{u+1}A^{T\tau}) \overline{\text{tr}(Q_{v+1}B^{T\tau})} \\ &= \sum_{\tau \in [d]^{Nm}} \text{tr}(P_uA^\tau) \overline{\text{tr}(Q_vB^\tau)} = O_{A_u, B_v}(N). \end{aligned} \quad (22.2)$$

The first and last equalities in (22.2) are the cyclic-sector trace formula after reindexing blocked words as physical words of length Nm . $\square$

Lemma 22.2.3.28 (Off-diagonal eigenvectors of the period transfer vanish). Lean ✓ Stmt *Let A be periodic of period m . Let $P_0, \dots, P_{m-1}$ be orthogonal projections satisfying*

$$\begin{aligned} \sum_{k=0}^{m-1} P_k &= \mathbb{1}, \\ \mathcal{E}_A^\dagger(P_{k+1}) &= P_k. \end{aligned}$$

If

$$\begin{aligned} U &= P_u U P_v, \quad P_u P_v = 0, \\ \mathcal{E}_A^m(U) &= \zeta U, \quad |\zeta| = 1, \end{aligned}$$

then $U = 0$.

Proof. ✓ Proof Since the peripheral spectrum of $\mathcal{E}_A$ consists of m -th roots of unity, the corresponding lemma in the companion quantum-channel volume [182, “Powering collapses peripheral eigenvalues”] gives $\text{spec}_{\text{per}}(\mathcal{E}_A^m) = \{1\}$, hence $\zeta = 1$. The cyclic trace gives $\text{tr} U = \text{tr}(P_u U P_v) = \text{tr}(P_v P_u U) = 0$.

Put $W = \sum_{t=0}^{m-1} \mathcal{E}_A^t(U)$. The equation $\mathcal{E}_A^m(U) = U$ gives $\mathcal{E}_A(W) = W$, and trace preservation gives $\text{tr} W = 0$. Irreducibility of A therefore gives $W = 0$. The shift relation $P_{k+1}A^i = A^iP_k$ moves $\mathcal{E}_A^t(U)$ into the $(u+t, v+t)$ corner, so only the $t = 0$ term contributes after multiplying by P_u and P_v . Thus $U = P_u W P_v = 0$. $\square$

Theorem 22.2.3.29 (Different periods imply asymptotic orthogonality). Lean ✓ Stmt *If A and B have periods $m_A \neq m_B$, then $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$.*

Proof. ✓ Proof If $D_A \neq D_B$, use Theorem 22.2.3.40. Assume $D_A = D_B$ and $A \sim_{\text{gp}} B$, say $B^i = \zeta X A^i X^{-1}$ with $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \neq 0$. The transfer maps satisfy

$$\mathcal{E}_B(Y) = \zeta \bar{\zeta} X \mathcal{E}_A(X^{-1} Y X^{-*}) X^*.$$

With $\mathcal{E}_A(\rho_A) = \rho_A$, put $\sigma = X \rho_A X^*$. Then $\mathcal{E}_B(\sigma) = |\zeta|^2 \sigma$, while $\mathcal{E}_B(\rho_B) = \rho_B$ for a nonzero positive fixed point ρ_B . Perron–Frobenius eigenvalue uniqueness [212, Theorem 6.3] gives $|\zeta|^2 = 1$. Hence $\mathcal{E}_B$ is conjugate to $\mathcal{E}_A$, so $\text{spec}_{\text{per}}(\mathcal{E}_A) = \text{spec}_{\text{per}}(\mathcal{E}_B)$. Since

$$\begin{aligned} \text{spec}_{\text{per}}(\mathcal{E}_A) &= \{e^{2\pi i r/m_A} : 0 \leq r < m_A\}, \\ \text{spec}_{\text{per}}(\mathcal{E}_B) &= \{e^{2\pi i r/m_B} : 0 \leq r < m_B\}, \end{aligned} \quad (22.3)$$

equality of the two finite sets in (22.3) gives $m_A = m_B$, a contradiction. Thus $A \sim_{\text{gp}} B$, and the irreducible trace-preserving overlap dichotomy gives $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$. $\square$

Lemma 22.2.3.30 (Sector-block normality for periodic tensors). Lean ✓ Stmt *For a periodic tensor equipped with a cyclic sector decomposition, each nonzero blocked sector tensor is normal.*

Theorem 22.2.3.31 (No sector match implies asymptotic orthogonality). Lean ✓ Stmt *Suppose A and B have the same total bond dimension and the same period m . Let A_u and B_v be period-blocked sector tensors of virtual dimensions d_u and e_v , indexed by $u, v \in \mathbb{Z}_m$, such that*

$$\begin{aligned}\bar{A} &\sim \bigoplus_{u \in \mathbb{Z}_m} A_u, \\ \bar{B} &\sim \bigoplus_{v \in \mathbb{Z}_m} B_v,\end{aligned}$$

with unit weights, and the two decompositions are cyclic sector decompositions. Assume each sector is left-canonical:

$$\begin{aligned}\sum_{\alpha} (A_u^{\alpha})^{\dagger} A_u^{\alpha} &= I_{d_u}, \\ \sum_{\alpha} (B_v^{\alpha})^{\dagger} B_v^{\alpha} &= I_{e_v}.\end{aligned}$$

Assume $d_u \neq 0$ for every u and $e_v \neq 0$ for every v , and whenever $d_u = e_v$, the corresponding sector tensors are not gauge-phase equivalent after identifying their virtual spaces:

$$d_u = e_v \implies A_u \not\sim_{\text{gp}} B_v.$$

Then $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$.

Proof. ✓ Proof Write $\bar{A} \sim \bigoplus_{u \in \mathbb{Z}_m} A_u$ and $\bar{B} \sim \bigoplus_{v \in \mathbb{Z}_m} B_v$. For every N ,

$$O_{\bar{A}\bar{B}}(N) = \sum_{u \in \mathbb{Z}_m} \sum_{v \in \mathbb{Z}_m} O_{A_u B_v}(N). \quad (22.4)$$

For each sector pair (u, v) ,

$$\begin{aligned}d_u = e_v, \quad A_u \sim_{\text{gp}} B_v &\implies \lim_{N \rightarrow \infty} O_{A_u B_v}(N) = 0, \\ d_u \neq e_v &\implies \lim_{N \rightarrow \infty} O_{A_u B_v}(N) = 0.\end{aligned}$$

The finite double sum (22.4) therefore gives

$$\lim_{N \rightarrow \infty} O_{\bar{A}\bar{B}}(N) = \sum_{u \in \mathbb{Z}_m} \sum_{v \in \mathbb{Z}_m} \lim_{N \rightarrow \infty} O_{A_u B_v}(N) = 0.$$

If $A \sim_{\text{gp}} B$, then $\bar{A} \sim_{\text{gp}} \bar{B}$, so for some $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \neq 0$, $(\bar{B})^{\alpha} = \zeta X (\bar{A})^{\alpha} X^{-1}$. Thus $V^{(N)}(\bar{B})_{\sigma} = \zeta^N V^{(N)}(\bar{A})_{\sigma}$, and

$$\begin{aligned}O_{\bar{A}\bar{B}}(N) &= \bar{\zeta}^N O_{\bar{A}\bar{A}}(N), \\ O_{\bar{B}\bar{B}}(N) &= |\zeta|^{2N} O_{\bar{A}\bar{A}}(N).\end{aligned}$$

Since $\lim_{N \rightarrow \infty} O_{\bar{A}\bar{A}}(N) = \lim_{N \rightarrow \infty} O_{\bar{B}\bar{B}}(N) = m > 0$,

$$1 = \lim_{N \rightarrow \infty} \frac{O_{\bar{B}\bar{B}}(N)}{O_{\bar{A}\bar{A}}(N)} = \lim_{N \rightarrow \infty} |\zeta|^{2N},$$

hence $|\zeta| = 1$. Therefore

$$\lim_{N \rightarrow \infty} |O_{\bar{A}\bar{B}}(N)| = \lim_{N \rightarrow \infty} O_{\bar{A}\bar{A}}(N) = m \neq 0,$$

contradicting $\lim_{N \rightarrow \infty} O_{\bar{A}\bar{B}}(N) = 0$. Thus $A \approx_{\text{gp}} B$, and the irreducible trace-preserving overlap dichotomy gives $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$. $\square$

Lemma 22.2.3.32 (Sector-match propagation under translation). [Lean](#) [✓ Stmt](#) *Let A and B be left-canonical tensors with period m , and let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical. Suppose $d_{u_0} = e_{v_0}$, $d_{u_0} \neq 0$, and, after identifying the virtual spaces, $A_{u_0} \sim_{\text{gp}} B_{v_0}$. Then for every $l \in \mathbb{Z}_m$ there is an equality $d_{u_0+l} = e_{v_0+l}$ and, after this identification, $A_{u_0+l} \sim_{\text{gp}} B_{v_0+l}$.*

Lemma 22.2.3.33 (Finite-cycle coboundary). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

Let G be a commutative group, and let $\kappa_0, \dots, \kappa_n \in G$. If $\prod_{k=0}^n \kappa_k = 1$, then there are $\varphi_0, \dots, \varphi_n \in G$ such that $\kappa_k = \varphi_k \varphi_{k+1}^{-1}$ for every $k < n$, and $\kappa_n = \varphi_n \varphi_0^{-1}$. Equivalently, on a nonempty finite cyclic index set, a product-one family satisfies $\kappa_k = \varphi_k \varphi_{k+1}^{-1}$ for every cyclic edge. In particular, a product-one family of complex unit phases admits such a choice with all φ_k again of unit modulus.

Proof. [✓ Proof](#)

Set $\varphi_k = (\kappa_0 \cdots \kappa_{k-1})^{-1}$, with the empty product equal to 1. For $k < n$,

$$\varphi_k \varphi_{k+1}^{-1} = (\kappa_0 \cdots \kappa_{k-1})^{-1} (\kappa_0 \cdots \kappa_k) = \kappa_k.$$

Separating the final factor in the product-one hypothesis gives $(\kappa_0 \cdots \kappa_{n-1}) \kappa_n = 1$, which gives

$$\varphi_n \varphi_0^{-1} = (\kappa_0 \cdots \kappa_{n-1})^{-1} = \kappa_n.$$

$\square$

Definition 22.2.3.34 (Global gauge). [Lean](#) [✓ Stmt](#) Given a positive period m , cyclic projectors P_u and Q_v , a cyclic offset q , and corner operators U_v , the global gauge is

$$U = \sum_u P_u U_{u+q} Q_{u+q}.$$

Lemma 22.2.3.35 (Global gauge as a sector sum). [Lean](#) [✓ Stmt](#) *Let $m \geq 1$. If each corner operator satisfies $U_v = P_{v-q} U_v Q_v$, then $P_u U_{u+q} Q_{u+q} = U_{u+q}$, so the global gauge is the plain sector sum $U = \sum_v U_v$.*

Proof. [✓ Proof](#) The corner relation $U_v = P_{v-q} U_v Q_v$ gives $P_u U_{u+q} Q_{u+q} = U_{u+q}$; reindexing the sum by $v = u + q$ turns $\sum_u P_u U_{u+q} Q_{u+q}$ into $\sum_v U_v$. $\square$

Lemma 22.2.3.36 (Unit-phase gauge relation gives repeated blocks). [Lean](#) [✓ Stmt](#) *Suppose $B^i = \zeta X A^i X^{-1}$ and $|\zeta| = 1$, with X invertible. Then A and B satisfy the repeated-block relation $A^i = \zeta^{-1} X^{-1} B^i X$.*

Proof. ✓ Proof Since $|\zeta| = 1$, the scalar is nonzero. Multiplication by X^{-1} on the left and by X on the right gives $X^{-1}B^iX = \zeta A^i$; multiplication by ζ^{-1} proves the claim. $\square$

Lemma 22.2.3.37 (Global-gauge construction). Lean ✓ Stmt Let $m \geq 1$, and let A and B carry the cyclic off-diagonal structure

$$\begin{aligned} A^i &= \sum_u P_u A^i P_{u+1}, \\ B^i &= \sum_v Q_v B^i Q_{v+1}, \end{aligned}$$

with P_u and Q_v families of orthogonal projections summing to the identity. Let q be the cyclic offset between the matched sector labels, and let each U_v be a corner partial isometry from the Q_v corner onto the P_{v-q} corner, that is,

$$\begin{aligned} U_v &= P_{v-q} U_v Q_v, \\ U_v^\dagger U_v &= Q_v, \\ U_v U_v^\dagger &= P_{v-q}. \end{aligned}$$

Assume the per-site proportionality of [157, eq. result],

$$P_u A^i P_{u+1} = \zeta U_{u+q} Q_{u+q} B^i Q_{u+q+1} U_{u+q+1}^\dagger, \quad |\zeta| = 1.$$

Then the global gauge $U = \sum_u P_u U_{u+q} Q_{u+q}$ is unitary and satisfies $A^i = \zeta U B^i U^\dagger$ for every i , so A and B are repeated blocks.

Proof. ✓ Proof The corner relation $U_v = P_{v-q} U_v Q_v$ gives $P_u U_{u+q} Q_{u+q} = U_{u+q}$, so $U = \sum_v U_v$ after reindexing. Distinct projections are orthogonal, $Q_v Q_w = 0$ and $P_{v-q} P_{w-q} = 0$ for $v \neq w$, which eliminates the cross terms in $UU^\dagger$ and $U^\dagger U$; the diagonal terms are $U_v U_v^\dagger = P_{v-q}$ and $U_v^\dagger U_v = Q_v$, so

$$\begin{aligned} UU^\dagger &= \sum_u P_u = \mathbb{1}, \\ U^\dagger U &= \sum_v Q_v = \mathbb{1}. \end{aligned}$$

The off-diagonal structure of B makes $Q_v B^i Q_w$ vanish unless $w = v+1$, so $U B^i U^\dagger = \sum_v U_v Q_v B^i Q_{v+1} U_{v+1}^\dagger$. Summing the per-site proportionality over u and reindexing by $v = u + q$ gives

$$\zeta U B^i U^\dagger = \sum_v \zeta U_v Q_v B^i Q_{v+1} U_{v+1}^\dagger = \sum_u P_u A^i P_{u+1} = A^i.$$

The last equality is the off-diagonal structure of A . The unit-modulus scalar and the unitary U exhibit A and B as repeated blocks. $\square$

Lemma 22.2.3.38 (Cyclic Ω_u -contraction for blocked sector gauges). Lean ✓ Stmt Let A and B be left-canonical tensors with period m , and let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical. Let the specified cyclic projectors be P_u and Q_v , and let the specified corner identifications realize the blocked sector tensors by $A_u^\alpha = P_u \bar{A}^\alpha P_u$ and $B_v^\alpha = Q_v \bar{B}^\alpha Q_v$. Assume $d_u \neq 0$ and A_u is normal for every u . Fix $q \in \mathbb{Z}_m$ and assume that for every u there are $d_u = e_{u+q}$, $\zeta_u \neq 0$, and $X_u \in \text{GL}_{d_u}(\mathbb{C})$ such that $B_{u+q}^\alpha = \zeta_u X_u A_u^\alpha X_u^{-1}$ for every blocked physical index α . The source obtains a phase-only

product identity after absorbing these sector gauges into the B -sectors: with Q_v the cyclic sector projectors of B , it writes $\widetilde{B}_v^i = U_v Q_v B^i Q_{v+1} U_{v+1}^\dagger$. The contraction input is then, for every physical block $(i_1, \dots, i_m)$,

$$A_u^{i_1} A_{u+1}^{i_2} \dots A_{u+m-1}^{i_m} = e^{i\lambda_{u+q}} \widetilde{B}_{u+q}^{i_1} \widetilde{B}_{u+q+1}^{i_2} \dots \widetilde{B}_{u+q+m-1}^{i_m}.$$

Choose N_0 so that the N_0 -fold repetition of each cyclic m -site sector product is injective. For $\mathbf{i} = (i_1, \dots, i_{mN_0})$, form the repeated product

$$F_u^{\mathbf{i}} = (A_u^{i_1} \dots A_{u+m-1}^{i_m}) \dots (A_u^{i_{m(N_0-1)+1}} \dots A_{u+m-1}^{i_{mN_0}}),$$

and choose right inverses Ω_u satisfying

$$\sum_{\mathbf{i}} (\Omega_u^{\mathbf{i}})_{\alpha, \beta} (F_u^{\mathbf{i}})_{\alpha', \beta'} = \delta_{\alpha, \alpha'} \delta_{\beta, \beta'}.$$

Let $\widetilde{F}_{u+q+r}^{\mathbf{i}}$ denote the same cyclic repeated product formed from $\widetilde{B}_{u+q+r}, \dots, \widetilde{B}_{u+q+r+m-1}$. For every $r \in \mathbb{Z}_m$, the block identity gives

$$\begin{aligned} F_{u+r}^{\mathbf{i}} &= e^{iN_0\lambda_{u+q+r}} \widetilde{F}_{u+q+r}^{\mathbf{i}}, \\ \widetilde{F}_{u+q+r}^{\mathbf{i}} &= e^{-iN_0\lambda_{u+q+r}} F_{u+r}^{\mathbf{i}}. \end{aligned}$$

Let $\mathbf{i}_1, \dots, \mathbf{i}_m$ be independent words of length mN_0 , one for each inserted repeated product. Sum each $\mathbf{i}_r$ in the contraction by the right-inverse relation for Ω_{u+r} :

$$\frac{A_u^{i_1} F_{u+1}^{\mathbf{i}_1} A_{u+1}^{i_2} F_{u+2}^{\mathbf{i}_2} \dots A_{u+m-1}^{i_m} F_u^{\mathbf{i}_m}}{\Omega_{u+1} \otimes \dots \otimes \Omega_{u+m-1} \otimes \Omega_u} \rightarrow A_u^{i_1} \otimes A_{u+1}^{i_2} \otimes \dots \otimes A_{u+m-1}^{i_m}.$$

Expand the words $\mathbf{i}_1, \dots, \mathbf{i}_m$, and write $j_{s,t}$ for the physical index in the t th sector of the s th consecutive m -site block of the expanded word, where $0 \leq s \leq mN_0$ and $1 \leq t \leq m$. Since each inserted product $F_{u+r}^{\mathbf{i}_r}$ has length mN_0 , insertion shifts the cyclic sector position by $mN_0 \equiv 0 \pmod{m}$. Hence every consecutive m -site block begins at the sector u , and the interleaved word is

$$\begin{aligned} A_u^{i_1} F_{u+1}^{\mathbf{i}_1} A_{u+1}^{i_2} F_{u+2}^{\mathbf{i}_2} \dots A_{u+m-1}^{i_m} F_u^{\mathbf{i}_m} \\ = (A_u^{j_{0,1}} \dots A_{u+m-1}^{j_{0,m}}) \dots (A_u^{j_{mN_0,1}} \dots A_{u+m-1}^{j_{mN_0,m}}). \end{aligned}$$

Applying the block identity at the sector u to each of these m -site blocks gives

$$\begin{aligned} A_u^{i_1} F_{u+1}^{\mathbf{i}_1} A_{u+1}^{i_2} F_{u+2}^{\mathbf{i}_2} \dots A_{u+m-1}^{i_m} F_u^{\mathbf{i}_m} \\ = e^{i\lambda_{u+q}(mN_0+1)} \widetilde{B}_{u+q}^{i_1} \widetilde{F}_{u+q+1}^{\mathbf{i}_1} \widetilde{B}_{u+q+1}^{i_2} \widetilde{F}_{u+q+2}^{\mathbf{i}_2} \dots \widetilde{B}_{u+q+m-1}^{i_m} \widetilde{F}_{u+q}^{\mathbf{i}_m}. \end{aligned} \quad (22.5)$$

For each $r \in \mathbb{Z}_m$, substituting $\mathbf{i} = \mathbf{i}_r$ in (22.5) gives $\widetilde{F}_{u+q+r}^{\mathbf{i}_r} = e^{-iN_0\lambda_{u+q+r}} F_{u+r}^{\mathbf{i}_r}$. Therefore the right-inverse relation gives the per-sector contraction identity

$$\sum_{\mathbf{i}_r} (\Omega_{u+r}^{\mathbf{i}_r})_{\alpha, \beta} (\widetilde{F}_{u+q+r}^{\mathbf{i}_r})_{\alpha', \beta'} = e^{-iN_0\lambda_{u+q+r}} \delta_{\alpha, \alpha'} \delta_{\beta, \beta'}.$$

Since $r \mapsto u + q + r$ is a permutation of $\mathbb{Z}_m$, $\sum_{r \in \mathbb{Z}_m} \lambda_{u+q+r} = \sum_{r \in \mathbb{Z}_m} \lambda_r$. Applying $\Omega_{u+1} \otimes \cdots \otimes \Omega_{u+m-1} \otimes \Omega_u$ to both sides of (22.5) and collecting the sector phases gives

$$\begin{aligned} A_u^{i_1} \otimes A_{u+1}^{i_2} \otimes \cdots \otimes A_{u+m-1}^{i_m} &= e^{i\lambda_{u+q}(mN_0+1) - iN_0 \sum_{r \in \mathbb{Z}_m} \lambda_r} \widetilde{B}_{u+q}^{i_1} \otimes \widetilde{B}_{u+q+1}^{i_2} \otimes \cdots \otimes \widetilde{B}_{u+q+m-1}^{i_m} \\ &= e^{i\eta_{u+q}} \widetilde{B}_{u+q}^{i_1} \otimes \widetilde{B}_{u+q+1}^{i_2} \otimes \cdots \otimes \widetilde{B}_{u+q+m-1}^{i_m}. \end{aligned}$$

Here $\eta_{u+q} = \lambda_{u+q}(mN_0+1) - N_0 \sum_{r \in \mathbb{Z}_m} \lambda_r$. Then there are a scalar ξ with $|\xi| = 1$ and a unitary matrix U such that $A^i = \xi U B^i U^\dagger$ for every physical index i .

Theorem 22.2.3.39 (Sector match yields repeated-block equivalence). [Lean](#) [Stmt](#) Let A and B be periodic tensors with the same period m . Let $\bar{A} \sim \bigoplus_u A_u$ and $\bar{B} \sim \bigoplus_v B_v$ be unit-weight cyclic sector decompositions whose sectors are left-canonical, with specified cyclic projectors P_u, Q_v and corner identifications satisfying $A_u^\alpha = P_u \bar{A}^\alpha P_u$ and $B_v^\alpha = Q_v \bar{B}^\alpha Q_v$. Assume every sector A_u has nonzero virtual dimension. If there are sectors u_0, v_0 with $d_{u_0} = e_{v_0}$ and $A_{u_0} \sim_{\text{gp}} B_{v_0}$ after identifying their virtual spaces, then A and B are repeated blocks.

Theorem 22.2.3.40 (Different bond dimensions imply asymptotic orthogonality). [Lean](#) [Stmt](#) If two periodic tensors have different bond dimensions, then $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$.

Theorem 22.2.3.41 (Periodic overlap dichotomy). [Lean](#) [Stmt](#) Let A and B be periodic tensors. Then at least one of the following alternatives holds:

1. $\langle V^{(N)}(A) | V^{(N)}(B) \rangle \rightarrow 0$ as $N \rightarrow \infty$;
2. A and B are related by the repeated-block relation, i.e. they differ by a unit-modulus phase and an invertible gauge.

Proof. [Proof](#) If $m_A \neq m_B$, then $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$ by Theorem 22.2.3.29. Assume $m_A = m_B = m$. If $D_A \neq D_B$, then $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$ by Theorem 22.2.3.40. Assume $D_A = D_B$. Choose period-blocked sector decompositions

$$\begin{aligned} \bar{A} &\sim \bigoplus_{u \in \mathbb{Z}_m} A_u, \\ \bar{B} &\sim \bigoplus_{v \in \mathbb{Z}_m} B_v. \end{aligned}$$

If no pair (u, v) satisfies $d_u = e_v$ and $A_u \sim_{\text{gp}} B_v$, then Theorem 22.2.3.31 gives $\lim_{N \rightarrow \infty} O_{AB}(N) = 0$.

Otherwise there are $u_0, v_0 \in \mathbb{Z}_m$ such that $d_{u_0} = e_{v_0}$ and $A_{u_0} \sim_{\text{gp}} B_{v_0}$. Theorem 22.2.3.39 then gives repeated-block equivalence between A and B . $\square$

Theorem 22.2.3.42 (Gram criterion for a periodic family). [Lean](#) [Stmt](#) Let $\{A_j\}_{j \in J}$ be a finite family of periodic tensors whose periods divide a common positive integer p . If

$$\langle V^{(N)}(A_i) | V^{(N)}(A_j) \rangle \rightarrow 0 \quad (i \neq j),$$

then the vectors $\{|V^{(pN)}(A_j)\rangle\}_{j \in J}$ are linearly independent for every sufficiently large N .

Proof. [Proof](#) The Gram matrix converges to the diagonal matrix whose j -th diagonal entry is the positive period of A_j . Its determinant therefore has a nonzero limit, so the Gram matrix is invertible for every sufficiently large N . $\square$

Theorem 22.2.3.43 (Eventual linear independence of periodic vectors). [Lean](#) [✓ Stmt](#) For a finite family $\{A_j\}_{j \in J}$ of pairwise non-repeated periodic tensors, in the sense that no two are related by a gauge transformation and a unit-modulus scalar, whose periods divide a common positive integer p , there is N_0 such that for every $N \geq N_0$, the map

$$\begin{aligned} \Phi_N: \mathbb{C}^J &\longrightarrow (\mathbb{C}^d)^{\otimes pN}, \\ (c_j)_{j \in J} &\longmapsto \sum_{j \in J} c_j |V^{(pN)}(A_j)\rangle \end{aligned}$$

Following [157, lines 589–609], this establishes linear independence of the periodic vectors ([111]).

Proof. [✓ Proof](#) Periodicity makes every bond dimension positive. For distinct i, j , the repeated-block alternative in Theorem 22.2.3.41 contradicts pairwise non-repetition, so the cross overlap tends to zero. Theorem 22.2.3.42 now gives the claimed independence along the common-period subsequence. $\square$

Theorem 22.2.3.44 (Independence of the non-zero periodic vectors). [Lean](#) [✓ Stmt](#) Let $\{A_j\}_{j \in J}$ be a finite family of pairwise non-repeated periodic tensors, of periods m_j . There is an N_0 such that for every single length $N \geq N_0$ the vectors

$$\{|V^{(N)}(A_j)\rangle : j \in J, m_j \mid N\}$$

are linearly independent. Since $|V^{(N)}(A_j)\rangle$ vanishes unless $m_j \mid N$, this is the assertion of [157, lines 604–608] that beyond some length the non-zero members of the family are independent, in the form used in the proof of the equal case at [157, lines 677–679].

Proof. [✓ Proof](#) There are finitely many subfamilies. For each subfamily $S \subseteq J$ apply Theorem 22.2.3.42 at the least common multiple p_S of the periods occurring in S , the cross-overlap decay coming as before from Theorem 22.2.3.41 and pairwise non-repetition; this gives a threshold M_S . Let L be the least common multiple of all the periods and take N_0 to be the largest of the numbers $L M_S$. Given $N \geq N_0$, let S be the set of indices whose period divides N . Every period occurring in S divides N , so $p_S \mid N$; writing $N = p_S n$ and using $p_S \leq L$ gives $L M_S \leq N \leq L n$, hence $n \geq M_S$ and the subfamily is independent at the length $N = p_S n$. $\square$

Theorem 22.2.3.45 (Spanning by the nonzero periodic basis vectors). [Lean](#) [✓ Stmt](#) Let A be the tensor of a sector decomposition, with representatives A_j and multiplicity entries $\mu_{j,q}$. At every length N , the nonzero vectors among $\{|V_N(A_j)\rangle\}_j$ span $|V_N(A)\rangle$. This proves the spanning assertion in [157, Proposition 3.3, lines 604–608]; this assertion does not require periodicity or normalization of the blocks.

Proof. [✓ Proof](#) The sector decomposition gives

$$|V_N(A)\rangle = \sum_j \left(\sum_q \mu_{j,q}^N \right) |V_N(A_j)\rangle. \quad (22.6)$$

Terms whose state vector vanishes contribute zero. Every remaining term belongs to the span of the nonzero state vectors, as does their sum. $\square$

Remark 22.2.3.46 (Comparison with the blocking approach). [Not ready](#) Theorem 22.2.2.1 and the equal-MPV corollary of Section 9.4 both characterize the gauge freedom between two tensors with equal MPV families, but they do so at different levels:

1. *Blocking approach* (Section 9.4, following [11]): for a common multiple p of the periods,

$$A^{[p]} \sim \bigoplus_a C_a,$$

$$B^{[p]} \sim \bigoplus_b D_b,$$

with all C_a, D_b normal. The Fundamental Theorem gives, after a permutation of the normal summands,

$$D_{\pi(a)}^\alpha = X_a C_a^\alpha X_a^{-1}.$$

This is a gauge statement for $A^{[p]}$ and $B^{[p]}$; the cyclic phases have been absorbed into the period- p words.

2. *Periodic approach* (Theorem 22.2.2.1, following [157]): keep the cyclic sectors

$$A^i = \sum_{u \in \mathbb{Z}_m} P_u A^i P_{u+1},$$

$$B^i = \sum_{v \in \mathbb{Z}_m} Q_v B^i Q_{v+1}.$$

Sector matching gives phases ϕ_v , a shift q , and unitaries U_v such that

$$P_u A^i P_{u+1} = e^{i\xi} e^{i\phi_{u+q}} U_{u+q} Q_{u+q} B^i Q_{u+q+1} U_{u+q+1}^\dagger e^{-i\phi_{u+q+1}}.$$

In the equal-MPV theorem the proportional phase is absorbed, so one obtains $ZA^i = UB^i U^{-1}$ with $Z^m = \mathbb{1}$; the trace only sees powers of the corresponding m -th roots of unity.

Chapters 7 and 9 follow only the blocking argument: remove periodicity by blocking, then apply the Fundamental Theorem. The periodic extension treats periodic blocks directly.

Theorem 22.2.3.47 (Peripheral proportional case from positive-length MPV equality). [Lean](#)

[✓ Stmt](#) *Let A and B be periodic tensors with the same matrix product vectors at every positive length. Then their bond dimensions agree, and after identifying the bond spaces, A and B are repeated blocks.*

Proof. [✓ Proof](#) By Theorem 22.2.3.41, either the overlap $\langle V^{(N)}(A) | V^{(N)}(B) \rangle$ tends to 0 or the bond dimensions agree and A and B are repeated blocks. In the decay case, equality of the MPV families gives, for every $N > 0$,

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle = \langle V^{(N)}(A) | V^{(N)}(A) \rangle. \quad (22.7)$$

Along multiples of the period m_a , (22.7) would imply

$$0 = \lim_{N \rightarrow \infty} \langle V^{(m_a N)}(A) | V^{(m_a N)}(B) \rangle = \lim_{N \rightarrow \infty} \langle V^{(m_a N)}(A) | V^{(m_a N)}(A) \rangle = m_a,$$

contradicting $m_a > 0$. □

Lemma 22.2.3.48 (A pure similarity is a repeated-block relation). [Lean](#) [✓ Stmt](#) *If $B^i = X A^i X^{-1}$ for an invertible matrix X , then A and B are repeated blocks with equal bond dimension and unit phase.*

Proof. [✓ Proof](#) Invert the pure gauge and take the repeated-block phase to be one. □

Theorem 22.2.3.49 (Repeated blocks scale their vectors by a phase power). [Lean](#) [✓ Stmt](#) Let A and B be repeated blocks. Then there is a scalar ζ with $|\zeta| = 1$ such that, for every length N ,

$$|V^{(N)}(A)\rangle = \zeta^N |V^{(N)}(B)\rangle.$$

The proportionality scalar therefore depends on the length. Already the one-dimensional blocks with a single physical index, $B^0 = 1$ and $A^0 = \zeta$, satisfy $|V^{(N)}(A)\rangle = \zeta^N |V^{(N)}(B)\rangle$, so no scalar independent of the length works unless $\zeta = 1$.

Proof. [✓ Proof](#) Write $A^i = \zeta Y B^i Y^{-1}$ with $|\zeta| = 1$. A closed chain of length N is the trace of a product of N such factors. The neighbouring similarities cancel and the trace is invariant under the outer conjugation, while each site contributes one factor ζ . $\square$

Theorem 22.2.3.50 (Absorbing the phase of a repeated-block relation). [Lean](#) [✓ Stmt](#) Let A and B be repeated blocks. Then there is a scalar ξ with $|\xi| = 1$ such that ξA and B generate the same vector at every positive length,

$$|V^{(N)}(\xi A)\rangle = |V^{(N)}(B)\rangle.$$

The phase belongs to the matched pair A, B ; distinct matched pairs of two bases of periodic tensors may require distinct phases, so no single phase is asserted for the direct-sum tensors ([157, lines 286–305, 625–627]).

Proof. [✓ Proof](#) Take ξ to be the reciprocal of the phase from Theorem 22.2.3.49, which is nonzero because it has modulus one. Rescaling every matrix of a block by ξ multiplies its length- N vector by ξ^N , and the two powers cancel. $\square$

Theorem 22.2.3.51 (Unit-norm scaling preserves the transfer map). [Lean](#) [✓ Stmt](#) If $|c| = 1$, then the tensors A and cA have the same transfer map.

Proof. [✓ Proof](#) Scaling every tensor letter by c multiplies the transfer map by $c\bar{c} = |c|^2 = 1$. $\square$

Theorem 22.2.3.52 (Repeated blocks share a peripheral spectrum). [Lean](#) [✓ Stmt](#) If A and B are repeated blocks, then their transfer maps have the same unit-circle eigenvalues.

Proof. [✓ Proof](#) Write $A^i = \xi Y B^i Y^{-1}$ with $|\xi| = 1$. The transfer map of ξC is $|\xi|^2$ times the transfer map of C , hence equals it, and the transfer map of $Y B^i Y^{-1}$ is the congruence conjugate $Y \mathcal{E}_B(Y^{-1} \cdot Y^{-\dagger}) Y^\dagger$ of the transfer map of B . Conjugation by an invertible linear map leaves the spectrum unchanged. $\square$

Theorem 22.2.3.53 (A root-of-unity set determines its positive exponent). [Lean](#) [✓ Stmt](#) Let $m, n > 0$. If $\{z \in \mathbb{C} : z^m = 1\} = \{z \in \mathbb{C} : z^n = 1\}$, then $m = n$.

Proof. [✓ Proof](#) A primitive m -th root belongs to the second set, so m divides n . Interchanging m and n gives the reverse divisibility. $\square$

Theorem 22.2.3.54 (A common peripheral spectrum determines the period). [Lean](#) [✓ Stmt](#) Let A and B be spectrally periodic tensors of periods m and n . If their transfer maps have the same peripheral spectrum, then $m = n$.

Proof. [✓ Proof](#) The two spectra are respectively the sets of m -th and n -th roots of unity, so Theorem 22.2.3.53 applies. $\square$

Theorem 22.2.3.55 (Repeated periodic blocks have the same period). [Lean](#) [✓ Stmt](#) Let A and B be spectrally periodic tensors of periods m and n . If A and B are repeated blocks, then $m = n$.

Proof. ✓ Proof Repeated blocks have the same peripheral spectrum by Theorem 22.2.3.52. Theorem 22.2.3.54 then identifies their periods. $\square$

Lemma 22.2.3.56 (Pure similarities preserve periodic overlap hypotheses). Lean ✓ Stmt *Let $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ be finite tensor families, and suppose that A'_j is purely similar to A_j and B'_k is purely similar to B_k for every j, k . Assume that every tensor in each primed family has a non-decaying mixed-overlap partner in the other primed family, and that every non-decaying primed pair is a repeated pair. Then the same three assertions hold for the two original families.*

Proof. ✓ Proof Cyclicity of the trace makes every closed-chain coefficient, and hence every mixed overlap, invariant under the two similarities. Thus decay and non-decay are unchanged. For a non-decaying pair, compose its primed repeated-block relation with the two pure similarities. $\square$

Theorem 22.2.3.57 (Non-decaying mixed overlaps for normalized bases of periodic tensors).

Lean ✓ Stmt *Let*

$$A^i = \bigoplus_{j \in J} (R_j \otimes A_j^i),$$

$$B^i = \bigoplus_{k \in K} (S_k \otimes B_k^i),$$

where $R_j = \text{diag}(\mu_{j,1}, \dots, \mu_{j,r_j})$ and $S_k = \text{diag}(\nu_{k,1}, \dots, \nu_{k,s_k})$ have nonzero complex diagonal entries. Suppose that the finite families $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ are pairwise non-repeated families of left-canonical periodic tensors. If, for every positive integer N , there is a scalar $c_N \neq 0$ such that

$$|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle.$$

Then every A_j has an index $k \in K$ for which

$$\langle V^{(N)}(A_j) | V^{(N)}(B_k) \rangle \not\rightarrow 0,$$

and, conversely, every B_k has such a partner in the first family. Moreover, whenever the overlap between A_j and B_k does not tend to zero, the two blocks have equal bond dimension and there are an invertible matrix Y and a scalar ζ with $|\zeta| = 1$ such that, for every physical index i ,

$$A_j^i = \zeta Y B_k^i Y^{-1}.$$

Normalization. In [157, lines 313–332], the trace-preserving normalization is obtained via a block-diagonal similarity from irreducible transfer maps of spectral radius one. Here, the basis tensors are taken to be normalized in advance; Theorem 22.2.3.58 transfers the result to unnormalized spectrally periodic blocks.

Proof. ✓ Proof Fix $j_0 \in J$ and suppose that its overlap with every B_k tends to zero. Let $T \subseteq J$ be the set of all indices with this property. For every $j \notin T$, choose $k(j)$ for which the mixed overlap does not decay. The periodic overlap dichotomy identifies A_j with $B_{k(j)}$ up to a gauge transformation and a phase; consequently their length- N vectors are scalar multiples of one another.

Choose a common positive multiple p of all periods. Along the lengths pN , define the coproduct-indexed family $w_N: T \amalg K \rightarrow (\mathbb{C}^d)^{\otimes pN}$ by $w_N(\text{inl}(j)) = |V^{(pN)}(A_j)\rangle$ and $w_N(\text{inr}(k)) = |V^{(pN)}(B_k)\rangle$. This family is eventually linearly independent. Indeed, distinct vectors within either summand have decaying overlaps by non-repetition and the overlap dichotomy, while the mixed overlaps vanish by the definition of T .

Expand the proportionality relation at length pN using the grouped coefficients

$$\mathrm{tr}(R_j^{pN}) = \sum_q (\mu_{j,q}^p)^N, \quad \mathrm{tr}(S_k^{pN}) = \sum_r (\nu_{k,r}^p)^N.$$

Replace each term indexed by $j \notin T$ by its scalar multiple of the corresponding $B_{k(j)}$ term, and collect terms with the same value of $k(j)$. Linear independence forces $\mathrm{tr}(R_{j_0}^{pN}) = 0$ for every sufficiently large N . Since each $\mu_{j_0,q}^p$ is nonzero, geometric extrapolation gives the same identity at exponent zero. This says $r_{j_0} = 0$, contradicting the positive size of R_{j_0} . Hence j_0 has a non-decaying partner. Interchanging A and B , inverting the nonzero scalars c_N , and using conjugate symmetry of the mixed overlap gives the converse assertion. For any non-decaying pair, the periodic overlap dichotomy excludes its decay alternative and supplies the repeated-block relation. $\square$

Theorem 22.2.3.58 (Non-decaying mixed overlaps for spectrally periodic bases). [Lean](#) [Stmt](#)
Let

$$A^i = \bigoplus_{j \in J} (R_j \otimes A_j^i), \quad B^i = \bigoplus_{k \in K} (S_k \otimes B_k^i),$$

where the diagonal entries of every R_j and S_k are nonzero. Suppose that the two representative families are pairwise non-repeated. Assign positive integers m_j and n_k to their representatives, and suppose that every transfer map is irreducible, has spectral radius one, and has peripheral spectrum respectively $\{z \in \mathbb{C} : z^{m_j} = 1\}$ or $\{z \in \mathbb{C} : z^{n_k} = 1\}$. If, for every positive N , there is a scalar $c_N \neq 0$ such that

$$|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle,$$

then every representative in either family has a representative in the other family whose mixed overlap does not tend to zero. Every such pair has equal bond dimension and differs by an invertible similarity and a unit-modulus scalar.

Proof. [Proof](#) Normalize all representatives by their pure Perron gauges and combine the gauges block diagonally. The multiplicity matrices and their grouped coefficients remain unchanged. Pure similarity preserves the total positive-length proportionality and pairwise non-repetition, so Theorem 22.2.3.57 applies to the normalized families. Finally apply Lemma 22.2.3.56. $\square$

Theorem 22.2.3.59 (Scalar-weight specialization of the mixed-overlap theorem). [Lean](#) [Stmt](#)
Let A and B each have the same length- N matrix-product vector, for every $N \geq 0$, as a scalar-weight direct sum formed from a finite, pairwise non-repeated family of left-canonical periodic tensors $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$, with a positive real scalar weight attached to each block. Suppose that, for every positive integer N , there is a scalar $c_N \neq 0$ such that

$$|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle.$$

Then every block in either family has a block in the other family whose mixed overlap does not tend to zero. Every such non-decaying pair has equal bond dimension and differs by an invertible similarity and a unit-modulus scalar.

This specializes Theorem 22.2.3.57 to block-diagonal tensors with a single scalar weight per block.

Proof. [Proof](#) Repeat the argument of Theorem 22.2.3.57 with the singleton coefficients μ_j^N and ν_k^N . Their nonvanishing at every length replaces the geometric-extrapolation step. Equality with the two scalar-weight direct sums transports the positive-length proportionality relation to the corresponding block-diagonal tensors. $\square$

Theorem 22.2.3.60 (Finite matching from the periodic overlap dichotomy). [Lean](#) [✓ Stmt](#) Let $\{A_j\}_{j \in J}$ and $\{B_k\}_{k \in K}$ be finite pairwise non-repeated families. Suppose every tensor in either family has a tensor in the other family whose mixed overlap does not tend to zero, and every such non-decaying pair is repeated. Then there is a bijection $\pi: J \rightarrow K$ such that A_j and $B_{\pi(j)}$ are repeated for every $j \in J$.

This is the finite matching step of [157, Theorem 3.4].

Proof. [✓ Proof](#) Choose a non-decaying partner $f(j) \in K$ for every $j \in J$. If $f(j_1) = f(j_2)$, the two repeated-block relations compose to show that A_{j_1} and A_{j_2} are repeated, so pairwise non-repetition makes f injective. The symmetric construction gives an injection $K \rightarrow J$. Therefore $|J| = |K|$, and the first injection is the required bijection. $\square$

Theorem 22.2.3.61 (Proportional Fundamental Theorem for irreducible-form MPS). [Lean](#) [✓ Stmt](#) Let A and B have multiplicity-bearing irreducible forms

$$A^i = \bigoplus_{j \in J} (R_j \otimes A_j^i),$$

$$B^i = \bigoplus_{k \in K} (S_k \otimes B_k^i),$$

where R_j and S_k are diagonal with nonzero diagonal entries and the two displayed block families are pairwise non-repeated bases of periodic tensors, each block being required only to have an irreducible transfer map of spectral radius one, which is the source's irreducible form at [157, lines 248–261]. Suppose that, for every positive N , the vectors $|V^{(N)}(A)\rangle$ and $|V^{(N)}(B)\rangle$ are proportional by a nonzero scalar. Then the blocks carry periods m_j and n_k for which they are spectrally periodic, and there is a bijection $\pi: J \rightarrow K$ such that, for every $j \in J$, one has $m_j = n_{\pi(j)}$ and there are a unit-modulus scalar ξ_j and an invertible matrix Y_j such that, for every physical index i ,

$$A_j^i = \xi_j Y_j B_{\pi(j)}^i Y_j^{-1}.$$

Each B_k occurs for exactly one j . This is [157, Theorem 3.4, lines 613–623].

This theorem matches the bases of periodic tensors, leaving the multiplicity matrices unconstrained. The periods m_j and n_k are derived from the transfer maps ([157, lines 257–258]).

Proof. [✓ Proof](#) Theorem 22.1.2 produces a positive period for each block of either family, and Theorem 22.2.3.62 applied to those periods gives the matching. $\square$

Theorem 22.2.3.62 (Proportional Fundamental Theorem, periods supplied). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 22.2.3.61, suppose in addition that positive integers m_j and n_k are given such that each A_j is spectrally m_j -periodic and each B_k is spectrally n_k -periodic (Definition 22.1.1). Then the conclusions of Theorem 22.2.3.61 hold for these specified periods.

Proof. [✓ Proof](#) Theorem 22.2.3.58 supplies non-decaying partners in both directions and classifies every such pair as repeated. Applying Theorem 22.2.3.60 gives a bijection $\pi: J \rightarrow K$ of repeated pairs. Finally, Theorem 22.2.3.55 gives $m_j = n_{\pi(j)}$ for every matched pair. $\square$

Theorem 22.2.3.63 (Proportional periodic blocks). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be spectrally periodic blocks. If their matrix product vectors are proportional by a nonzero scalar at every positive length, then A and B are repeated blocks. In particular, this holds when $V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma$ for a fixed $\zeta \neq 0$. Left-canonical periodic blocks satisfy the spectral-radius-one hypothesis automatically. This is the single-block case of [157, Theorem 3.4].

Proof. [✓ Proof](#) Regard each tensor as a one-block decomposition with one copy of weight one. Theorem 22.2.3.62 matches the two blocks. A left-canonical tensor defines a trace-preserving completely positive map, whose spectral radius is one. $\square$

Theorem 22.2.3.64 (Unitary matching of periodic phase classes). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A_k be a finite family of left-canonical periodic tensors, with bond dimensions D_k and periods m_k . Partition the family by the relation $V^{(N)}(A_l)_\sigma = \zeta^N V^{(N)}(A_k)_\sigma$ at every positive length for some $\zeta \neq 0$. Choose one representative ρ_j in each class and enumerate its members as $e(j, q)$. Distinct representatives are not repeated. Each member has $D_{e(j, q)} = D_{\rho_j}$ and $m_{e(j, q)} = m_{\rho_j}$, and admits a unitary matrix $U_{j, q}$ and a scalar $\xi_{j, q}$ of modulus one such that

$$A_{e(j, q)}^i = \xi_{j, q} U_{j, q} A_{\rho_j}^i U_{j, q}^\dagger. \quad (22.8)$$

More generally, any two repeated left-canonical periodic blocks admit such a unitary similarity. These are the blockwise identifications in [157, Section 2.1 and the paragraph following Theorem 3.4].

Proof. [✓ Proof](#) Theorem 22.2.3.63 makes each member repeated with its representative, giving equal dimensions and periods. Lemma 9.4.1.2 replaces the similarity by a unitary matrix while keeping its phase. Repeated representatives would have the same matrix-product-vector phase class, contrary to their choice. $\square$

Theorem 22.2.3.65 (Global similarity between matched block sums). [Lean](#) [✓ Stmt](#) Let e be a bijection between the blocks of $\bigoplus_k \mu_k A_k$ and of $\bigoplus_t \mu'_t B_t$ with $D_k = D'_{e(k)}$, and suppose $\mu_k A_k^i = \mu'_{e(k)} X_{e(k)} B_{e(k)}^i X_{e(k)}^{-1}$ for invertible X_t . Then there are matrices Y and Y' with $YY' = \mathbb{1}$, $Y'Y = \mathbb{1}$, and

$$\bigoplus_k \mu_k A_k^i = Y \left(\bigoplus_t \mu'_t B_t^i \right) Y'.$$

If every X_t is unitary, then $Y' = Y^\dagger$. This is the matrix $Y = \bigoplus_j T_j \otimes Y_j$ of [157, line 690].

Proof. [✓ Proof](#) Let Π be the permutation matrix carrying the summand $t = e(k)$ of $\bigoplus_t$ to the summand k of $\bigoplus_k$, and set $Y = \Pi(\bigoplus_t X_t)$ and $Y' = (\bigoplus_t X_t^{-1})\Pi^\dagger$. Then $YY' = Y'Y = \mathbb{1}$, and

$$Y \left(\bigoplus_t \mu'_t B_t^i \right) Y' = \Pi \left(\bigoplus_t \mu'_t X_t B_t^i X_t^{-1} \right) \Pi^\dagger = \bigoplus_k \mu_k A_k^i.$$

For unitary X_t , $X_t^{-1} = X_t^\dagger$, so $Y' = Y^\dagger$. $\square$

Definition 22.2.3.66 (Multiplicities grouped by phase class). [Lean](#) [✓ Stmt](#) Write ρ_j and $e(j, q)$ for the representatives and the enumeration of the phase-class data of Definition 8.6.1.7, as in Theorem 22.2.3.64. Given nonzero weights μ_k and unit-modulus phases $\xi_{j, q}$, let n_j be the number of copies in class j and define

$$\mu'_{j, q} = \mu_{e(j, q)} \xi_{j, q}, \quad R_j = \text{diag}(\mu'_{j, 0}, \dots, \mu'_{j, n_j-1}), \quad P^i = \bigoplus_j R_j \otimes A_{\rho_j}^i. \quad (22.9)$$

This is the multiplicity form of [157, Section 2.1].

Theorem 22.2.3.67 (Unitary grouping of periodic blocks). [Lean](#) [Stmnt](#) Let A_k be left-canonical periodic blocks and let every μ_k be nonzero. Set $T^i = \bigoplus_k \mu_k A_k^i$. There are phases $\xi_{j,q}$ in Definition 22.2.3.66 such that P has the same total bond dimension as T , its representative blocks are periodic and pairwise non-repeated, and $|\mu'_{j,q}| = |\mu_{e(j,q)}|$ for every original copy. There is a matrix Y from the grouped bond space to the original bond space with $YY^\dagger = I$ and $Y^\dagger Y = I$, satisfying $T^i = YP^iY^\dagger$ for every physical index i . In particular, $V^{(N)}(T)_\sigma = V^{(N)}(P)_\sigma$ at every positive length. This gives the grouping of [157, Section 2.1 and the paragraph after Theorem 3.4] in a unitary gauge.

Proof. [Proof](#) Choose the phases and unitaries of Theorem 22.2.3.64. Multiplying (22.8) by $\mu_{e(j,q)}$ and using Definition 22.2.3.66 gives

$$\mu_{e(j,q)} A_{e(j,q)}^i = \mu_{e(j,q)} \xi_{j,q} U_{j,q} A_{\rho_j}^i U_{j,q}^\dagger = \mu'_{j,q} U_{j,q} A_{\rho_j}^i U_{j,q}^\dagger.$$

The enumeration is a bijection between grouped copies and original blocks. Let Π be the permutation matrix sending the copy (j, q) to the block $e(j, q)$, and set $Y = \Pi(\bigoplus_{j,q} U_{j,q})$. Then

$$\begin{aligned} YY^\dagger &= \Pi \left(\bigoplus_{j,q} U_{j,q} U_{j,q}^\dagger \right) \Pi^\dagger = \mathbb{1}, \\ YP^iY^\dagger &= \Pi \left(\bigoplus_{j,q} \mu'_{j,q} U_{j,q} A_{\rho_j}^i U_{j,q}^\dagger \right) \Pi^\dagger = \bigoplus_k \mu_k A_k^i = T^i, \end{aligned}$$

and likewise $Y^\dagger Y = \mathbb{1}$. The same bijection and the equality of matched block dimensions give equality of total dimensions. The phases have modulus one, so the individual weight moduli are unchanged. For $N > 0$, the factors $Y^\dagger Y = I$ between adjacent letters cancel, and cyclicity of the trace removes the outer Y and $Y^\dagger$:

$$V^{(N)}(T)_\sigma = \text{tr}(YP^{\sigma_0}Y^\dagger \dots YP^{\sigma_{N-1}}Y^\dagger) = \text{tr}(P^{\sigma_0} \dots P^{\sigma_{N-1}}) = V^{(N)}(P)_\sigma.$$

□

Theorem 22.2.3.68 (Unit weight moduli before grouping). [Lean](#) [Stmnt](#) Let $T^i = \bigoplus_k \mu_k A_k^i$, where the A_k are left-canonical periodic blocks and every μ_k is nonzero. Let Q be a decomposition with pairwise non-repeated left-canonical periodic basis blocks and unit-modulus multiplicities. If $V^{(N)}(T) = V^{(N)}(Q)$ for every positive N , then $|\mu_k| = 1$ for every k . This is the normalization consequence needed in the corrected forward direction of [157, Theorem 4.1]; the trace-preservation correction is recorded in [144].

Proof. [Proof](#) Group the blocks of T as in Theorem 22.2.3.67. For every $N > 0$, the resulting decomposition P satisfies $V^{(N)}(P) = V^{(N)}(T) = V^{(N)}(Q)$. Theorem 22.2.2.15 matches the multiplicities of P and Q up to unit-modulus factors. Hence, for every grouped copy,

$$|\mu_{e(j,q)}| = |\mu'_{j,q}| = 1.$$

The first equality is the modulus statement of Theorem 22.2.3.67. Since e enumerates every original block, $|\mu_k| = 1$ for every k . □

22.3 Physical symmetries on periodic MPS

Theorem 22.3.1 (Conditional group-action Z -gauge reformulation of the symmetry corollary). [Lean](#) [Lean](#) [Stmnt](#) Let A be an MPS tensor in irreducible form II and let $U : G \rightarrow \text{GL}_d(\mathbb{C})$ be a

representation of a group G on the physical leg, acting unitarily ($U(g)U(g)^\dagger = \mathbb{1}$). Assume the periodic equal-case Fundamental Theorem (Theorem 22.2.2.1) for the dimensions (d, D) . If A is on-site symmetric under U — that is, the twisted tensor $\tilde{A}_g$ has the same MPV family as A for every $g \in G$ — then for each $g \in G$ there exists a positive period m_g and a $\mathbb{Z}_{m_g}$ -gauge equivalence between A and $\tilde{A}_g$.

Concretely there are matrices Z_g and Y_g , with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$, satisfying $Z_g A^i = Y_g \tilde{A}_g^i Y_g^{-1}$ for every physical index i .

Proof. ✓ Proof The twisted tensor $\tilde{A}_g$ coincides with the physical-index rotation $A_{U(g)}$ of Definition G.1.1.1. By Theorem 22.1.21 the rotated tensor remains in irreducible form II, and the on-site symmetry hypothesis supplies the equality $\mathcal{V}(A) = \mathcal{V}(A_{U(g)})$. Apply the periodic equal-case Fundamental Theorem (Theorem 22.2.2.1) to obtain the asserted $\mathbb{Z}_{m_g}$ -gauge equivalence. $\square$

Definition 22.3.2 (Conditional periodic projective-rigidity hypothesis). Lean ✓ Stmt Let A be an MPS tensor with physical dimension d and bond dimension D , and assume it is on-site symmetric under a group action $U : G \rightarrow \text{GL}_d(\mathbb{C})$. The tensor A satisfies the *periodic projective-rigidity hypothesis* if one may choose a family of virtual gauges $Y : G \rightarrow \text{GL}_D(\mathbb{C})$ and a scalar 2-cochain $u : G \times G \rightarrow \mathbb{C}^\times$ such that

1. every Y_g realizes the pointwise Z -gauge intertwining modeled on the symmetry corollary with the twisted tensor $\tilde{A}_g$ for some accompanying Z_g (with $Z_g^{m_g} = \mathbb{1}$ and $[A^i, Z_g] = 0$); and
2. the family obeys the projective multiplication law on the bond space: $Y_h Y_g = u(g, h) Y_{gh}$.

The source supplies the corresponding intertwining for a single local unitary. The group-indexed coherent choice and the projective multiplication law are additional assumptions: after proving the symmetry corollary, the source paper says that its consequences will be explored elsewhere. The injective case reduces this coherence to scalar uniqueness of gauges (cf. §10.1); in the periodic setting there is a non-trivial $\mathbb{Z}_m$ -gauge ambiguity, so coherence of the factor system u is an independent analytic assumption.

Theorem 22.3.3 (Conditional projective representation from symmetry gauges). Lean ✓ Stmt Assume the periodic projective-rigidity hypothesis (Definition 22.3.2) for the tensor A and the physical action U . Then the virtual gauge family supplied by that hypothesis defines a projective representation of G on the bond space: there exist a scalar 2-cochain $\omega : G \times G \rightarrow \mathbb{C}^\times$ and virtual gauges $X : G \rightarrow \text{GL}_D(\mathbb{C})$ satisfying $X(g)X(h) = \omega(g, h)X(gh)$, while for each g the matrix $X(g^{-1})$ still realizes the pointwise $\mathbb{Z}_{m_g}$ -intertwining, encoded in the rigidity hypothesis, between A and the twisted tensor $\tilde{A}_g$.

Proof. ✓ Proof By Definition 22.3.2, let (Y, u) be the virtual gauge family and factor system. Set $X(g) := Y(g^{-1})$ and $\omega(g, h) := u(h^{-1}, g^{-1})$. The projective multiplication law $X(g)X(h) = \omega(g, h)X(gh)$ is then exactly the (h^{-1}, g^{-1}) -instance of the factor-system identity, using $(gh)^{-1} = h^{-1}g^{-1}$. The intertwining relation between A and $\tilde{A}_g$ transports unchanged. $\square$

Theorem 22.3.4 (Cocycle identity for the upgraded factor system). Lean ✓ Stmt If the bond dimension D is positive, the factor system ω produced by Theorem 22.3.3 satisfies the multiplicative 2-cocycle identity $\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k)$.

Proof. [✓ Proof](#) Associativity of matrix multiplication applied to $X(g)X(h)X(k)$, combined with the projective multiplication law, gives

$$\omega(g, h)\omega(gh, k)X((gh)k) = \omega(g, hk)\omega(h, k)X(g(hk)). \quad (22.10)$$

Since $(gh)k = g(hk)$, both sides of (22.10) are scalar multiples of the same matrix $X((gh)k)$. Since $X((gh)k) \in \text{GL}_D(\mathbb{C})$ and $D > 0$, the matrix $X((gh)k)$ is nonzero. Cancelling $X((gh)k)$ from both sides of (22.10) yields $\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k)$ in $\mathbb{C}^\times$. $\square$

Remark 22.3.5 (External cohomological interpretation). Theorem 22.3.4 establishes a $\mathbb{C}^\times$ -valued 2-cocycle under the conditional periodic projective-rigidity hypothesis of Definition 22.3.2. In the broader SPT literature [194, 204], this factor system is given an external physical interpretation as defining a cohomology class in $H^2(G, \text{U}(1))$; no such $\text{U}(1)$ reduction is asserted here.

Theorem 22.3.6 (Conditional projective representation and cocycle from symmetry). [Lean](#) [✓ Stmt](#) *Under the periodic projective-rigidity hypothesis (Definition 22.3.2), if $D > 0$, then the virtual gauge family defines a projective representation of G on the bond space whose factor system is a 2-cocycle.*

22.4 Refinement and divisibility

Following [157, Theorem 4.1], refinement of a matrix-product-state family corresponds to divisibility of its transfer map.

Theorem 22.4.1 (Positive-weight periodic decomposition). [Lean](#) [✓ Stmt](#) *Every tensor A admits $r \in \mathbb{N}$ and, for $k = 1, \dots, r$, left-canonical periodic tensors B_k of positive bond dimensions D_k and weights $\mu_k > 0$, with $\sum_{k=1}^r D_k \leq D$, such that A has the same matrix product vectors at all positive lengths as $\bigoplus_{k=1}^r \mu_k B_k$. No physical sites are blocked [157, lines 238–271 and 313–320].*

Proof. [✓ Proof](#) Theorem 7.2.1.1 gives A the same positive-length matrix product vectors as $\bigoplus_{k=1}^r \mu_k B_k$ with $\mu_k > 0$, $\sum_k D_k \leq D$, and each B_k irreducible and left-canonical, so $\rho_{\text{spec}}(\mathcal{E}_{B_k}) = 1$. By Theorem 22.1.6, each B_k is m_k -periodic, that is, $\text{Spec}_{\text{per}}(\mathcal{E}_{B_k}) = \{\mu \in \mathbb{C} : \mu^{m_k} = 1\}$ for some $m_k \geq 1$. $\square$

Definition 22.4.2 (p -divisible CP map). [Lean](#) [✓ Stmt](#) A linear endomorphism $\mathcal{E}$ of $M_D(\mathbb{C})$ is p -divisible if there exists a CPTP map $\mathcal{E}'$ of $M_D(\mathbb{C})$ such that $\mathcal{E} = (\mathcal{E}')^p$.

Definition 22.4.3 (p -refinable MPS tensor). [Lean](#) [✓ Stmt](#) Given a tensor B , the family $\mathcal{V}(B)$ can be p -refined if there exist another tensor A and an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$ such that, for every N ,

$$|V^{(pN)}(A)\rangle = W^{\otimes N} |V^{(N)}(B)\rangle.$$

Equivalently, after identifying a p -letter with an element of $\{0, \dots, d^p - 1\}$, the p -blocked tensor $A^{[p]}$ satisfies

$$V^{(N)}(A^{[p]})_\tau = \sum_{\sigma \in \{0, \dots, d-1\}^N} \left(\prod_k W_{\tau(k), \sigma(k)} \right) V^{(N)}(B)_\sigma,$$

for every $N \in \mathbb{N}$ and every $\tau \in \{0, \dots, d^p - 1\}^N$.

Theorem 22.4.4 (The one-site boundary of refinement and divisibility). [Lean](#) [Lean](#) [✓ Stmt](#) Every tensor is 1-refinable. A linear map is 1-divisible if and only if it is a channel. Thus, at $p = 1$, the forward implication of the printed Theorem 4.1 requires trace preservation of $\mathcal{E}_B$; see [144].

Proof. [✓ Proof](#) One-site blocking only permutes the physical labels. Use the associated permutation matrix for the refinement isometry. The channel assertion follows because the first power of a linear map is the map itself. $\square$

Theorem 22.4.5 (Coordinates for a cyclic step). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $m > 0$, $p \geq 0$, and $r = \gcd(m, p)$. Every residue u modulo m has a unique expression $u = \alpha + pk \pmod{m}$, with $0 \leq \alpha < r$ and $0 \leq k < m/r$. Adding p preserves α and adds one to k modulo m/r . These are the coordinates used in [157, proof of Theorem 4.1].

Proof. [✓ Proof](#) If two such expressions agree modulo m , reduction modulo r identifies their representatives α . Cancelling that common representative gives $pk = pk' \pmod{m}$, hence $k = k' \pmod{m/r}$. Their bounds imply $k = k'$. There are $r(m/r) = m$ pairs of coordinates, so this injective map to the m residues is also surjective. Since m divides $p(m/r)$, reducing $k+1$ modulo m/r does not change $\alpha + p(k+1)$ modulo m . $\square$

Definition 22.4.6 (Projections for a prescribed blocking length). [Lean](#) [✓ Stmt](#) Let $m > 0$, $p \geq 0$, $r = \gcd(m, p)$, and let P_u be projections indexed modulo m . For $0 \leq \alpha < r$, set

$$Q_\alpha = \sum_{k=0}^{m/r-1} P_{[\alpha+pk]_m}.$$

These are the grouped projections of [157, Lemma 6].

Theorem 22.4.7 (Compression to the step orbits). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose P_u are nonzero orthogonal projections with $\sum_u P_u = \mathbb{1}_D$, A is left canonical, and $P_u A^i = A^i P_{u+1}$. There are positive integers D_α with $\sum_\alpha D_\alpha = D$, isometries $V_\alpha : \mathbb{C}^{D_\alpha} \rightarrow \mathbb{C}^D$, and left-canonical tensors C_α such that

$$\begin{aligned} V_\alpha^\dagger V_\alpha &= \mathbb{1}_{D_\alpha}, \\ V_\alpha V_\alpha^\dagger &= Q_\alpha, \\ C_\alpha^I &= V_\alpha^\dagger (A^{[p]})^I V_\alpha. \end{aligned}$$

The tensors $A^{[p]}$ and $\bigoplus_\alpha C_\alpha$ have the same MPV at every length, including zero. The adjoint transfer map satisfies $\mathcal{E}_{A^{[p]}}^\dagger(P_u) = P_{u+p}$, with indices taken modulo m .

Proof. [✓ Proof](#) The step coordinates partition the original projections, so the Q_α are mutually orthogonal nonzero projections summing to $\mathbb{1}_D$. Iterating $P_u A^i = A^i P_{u+1}$ over a word of length p shifts the projector index by p . Multiplying by the adjoint word and summing, left canonicity gives $\mathcal{E}_{A^{[p]}}^\dagger(P_u) = P_{u+p}$. Iterating the shift relation gives $P_u (A^{[p]})^I = (A^{[p]})^I P_{u+p}$. Since $u \mapsto u+p$ permutes the orbit of α , summing over that orbit gives $Q_\alpha (A^{[p]})^I = (A^{[p]})^I Q_\alpha$. Compress to the ranges of these projections. Their dimensions sum to D , and the commuting projection resolution preserves both left canonicity and traces of every word. The empty-word trace gives the same dimension sum. $\square$

Theorem 22.4.8 (Step-orbit compression of a periodic tensor). [Lean](#) [✓ Stmt](#) If A is periodic of period m and $p \geq 0$, there is a family of nonzero cyclic projections P_u with $\sum_u P_u = \mathbb{1}_D$ and $P_u A^i = A^i P_{u+1}$ for which the dimensions D_α , isometries V_α , and left-canonical tensors C_α of Theorem 22.4.7 exist, with the same identities and the same MPV at every length, including zero. If $p > 0$, every resulting block satisfies $\text{spec}_{\text{per}}(\mathcal{E}_{C_\alpha}) = \{z \in \mathbb{C} : z^{m/\gcd(m,p)} = 1\}$ and is periodic of period $m/\gcd(m,p)$. This is the prescribed-blocking conclusion of [157, Lemma 6].

Proof. [✓ Proof](#) Take the cyclic spectral projections, with their indices negated to give $P_u A^i = A^i P_{u+1}$. No projection vanishes: left canonicity and the shift relation give $\mathcal{E}_A^\dagger(P_u) = P_{u+1}$, and $\mathcal{E}_A^\dagger(0) = 0$, so $P_u = 0$ would give $P_{u+1} = 0$ and hence, around the cycle, $\mathbb{1}_D = \sum_v P_v = 0$. Apply Theorem 22.4.7. For $p > 0$, Theorem 22.4.11 identifies each block's peripheral spectrum, and Theorem 22.4.14 gives irreducibility. Together with left canonicity, these prove periodicity. $\square$

Theorem 22.4.9 (Peripheral roots after positive blocking). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $p > 0$. For any tensor A , the peripheral eigenvalues of $\mathcal{E}_{A^{[p]}}$ are the p th powers of those of $\mathcal{E}_A$. If A is periodic of period m , put $q = m/\gcd(m,p)$. Then

$$\text{spec}_{\text{per}}(\mathcal{E}_{A^{[p]}}) = \{z \in \mathbb{C} : z^q = 1\}. \quad (22.11)$$

Moreover, if $V^\dagger V = \mathbb{1}_n$, $VV^\dagger$ commutes with every letter of $A^{[p]}$, and $C^I = V^\dagger(A^{[p]})^I V$, every peripheral eigenvalue of $\mathcal{E}_C$ satisfies $z^q = 1$. This is the ambient spectral calculation in [157, Lemma 6], with the missing power in the printed root set corrected as in [102].

Proof. [✓ Proof](#) Spectral mapping for $\mathcal{E}_A^p$ gives the first assertion: a complex number has unit modulus if and only if its positive p th power does. A primitive m th root raised to the p th power has order q , giving (22.11). For compression, $(A^{[p]})^I V = V C^I$, so $\mathcal{E}_{A^{[p]}}(V X V^\dagger) = V \mathcal{E}_C(X) V^\dagger$. The isometry makes $X \mapsto V X V^\dagger$ injective; hence every eigenvalue of $\mathcal{E}_C$ is an eigenvalue of $\mathcal{E}_{A^{[p]}}$. $\square$

Theorem 22.4.10 (Cyclic motion of compressed projections). [Lean](#) [✓ Stmt](#) Suppose P_u are orthogonal projections summing to $\mathbb{1}_D$, A is left canonical, and $P_u A^i = A^i P_{u+1}$. Let $p \geq 0$, fix an orbit α , and suppose $V V^\dagger = Q_\alpha$ and $C^I = V^\dagger(A^{[p]})^I V$. For $u = [\alpha + pk]_m$,

$$\mathcal{E}_C^\dagger(V^\dagger P_u V) = V^\dagger P_{u+p} V. \quad (22.12)$$

This is the compressed form of the final display in the proof of [157, Lemma 6].

Proof. [✓ Proof](#) Orthogonality gives $Q_\alpha P_u = P_u Q_\alpha = P_u$. Therefore $V(V^\dagger P_u V) V^\dagger = P_u$. Expanding the compressed adjoint map yields $\mathcal{E}_C^\dagger(Y) = V^\dagger \mathcal{E}_{A^{[p]}}^\dagger(V Y V^\dagger) V$. Apply this identity to $Y = V^\dagger P_u V$ and use $\mathcal{E}_{A^{[p]}}^\dagger(P_u) = P_{u+p}$. $\square$

Theorem 22.4.11 (Exact peripheral spectrum of each orbit block). [Lean](#) [✓ Stmt](#) Let A be periodic of period m , let $p > 0$, and put $r = \gcd(m,p)$ and $q = m/r$. Suppose P_u are nonzero orthogonal projections summing to $\mathbb{1}_D$ with $P_u A^i = A^i P_{u+1}$. Fix $\alpha < r$, an isometry V with $V^\dagger V = \mathbb{1}$ and $V V^\dagger = Q_\alpha$, and set $C^I = V^\dagger(A^{[p]})^I V$. Then

$$\text{spec}_{\text{per}}(\mathcal{E}_C) = \{\lambda \in \mathbb{C} : \lambda^q = 1\}. \quad (22.13)$$

This is the spectral-set assertion in [157, Lemma 6], with the root notation corrected as in [102].

Proof. ✓ Proof The ambient spectral bound gives one inclusion. For the other, put $R_k = V^\dagger P_{[\alpha+pk]_m} V$. The compressed adjoint map satisfies $\mathcal{E}_C^\dagger(R_k) = R_{k+1}$. If $z^q = 1$, the sum $X_z = \sum_{k=0}^{q-1} z^k R_k$ therefore satisfies $\mathcal{E}_C^\dagger(X_z) = z^{-1} X_z$. It is nonzero: since $P_{[\alpha+pk]_m} \leq Q_\alpha = VV^\dagger$ and distinct orbit projections are orthogonal,

$$VX_zV^\dagger P_\alpha = \sum_{k=0}^{q-1} z^k P_{[\alpha+pk]_m} P_\alpha = P_\alpha \neq 0.$$

Given $\lambda^q = 1$, choose $z = \bar{\lambda}^{-1}$. The adjoint has eigenvalue $\bar{\lambda}$, so $\mathcal{E}_C$ has eigenvalue λ . Its modulus is one. $\square$

Theorem 22.4.12 (Fixed spaces of positive adjoint powers). Lean Lean ✓ Stmt *Let A be periodic of period m , let $p > 0$, and write $F = \mathcal{E}_A^\dagger$ and $r = \gcd(m, p)$. If $F^p(X) = X$, then $F^m(X) = X$. Moreover, $F^p(X) = X$ if and only if $F^r(X) = X$.*

Proof. ✓ Proof On $S = \ker(F^p - 1)$, the restriction of F is annihilated by the separable polynomial $t^p - 1$, and hence is diagonalizable. Each eigenvalue has modulus one and therefore is an m th root of unity. Thus F^m is the identity on S . A point of periods m and p also has period r ; conversely, $r \mid p$ gives the other implication. $\square$

Theorem 22.4.13 (Fixed-point rigidity on one orbit of sectors). Lean Lean ✓ Stmt *Let A be periodic of period m , let $p > 0$, and put $r = \gcd(m, p)$. Suppose P_u are orthogonal projections summing to $\mathbb{1}_D$ with $P_u A^i = A^i P_{u+1}$. Write $F = \mathcal{E}_A^\dagger$ and fix $\alpha < r$. Then $\sum_{j=0}^{r-1} F^j(Q_\alpha) = \mathbb{1}_D$. If $F^p(X) = X$ and $Q_\alpha X = X$, there is $c \in \mathbb{C}$ such that $X = cQ_\alpha$.*

Proof. ✓ Proof The projections $R_j = F^j(Q_\alpha)$, for $0 \leq j < r$, are pairwise orthogonal and sum to $\mathbb{1}_D$, since their cyclic-sector indices partition $\mathbb{Z}_m$. The latter relation gives $F(P_u Y) = P_{u+1} F(Y)$, so $R_j F^j(X) = F^j(X)$. By Theorem 22.4.12, $F^r(X) = X$. Thus $S = \sum_{j=0}^{r-1} F^j(X)$ is fixed by the irreducible unital map F , and $S = c\mathbb{1}_D$. Orthogonality gives $Q_\alpha S = X = cQ_\alpha$. $\square$

Theorem 22.4.14 (Irreducibility of each compressed orbit block). Lean ✓ Stmt *Let A be periodic of period m , let $p > 0$, and let P_u be orthogonal projections summing to $\mathbb{1}_D$ with $P_u A^i = A^i P_{u+1}$. Fix $\alpha < \gcd(m, p)$ and an isometry V with $VV^\dagger = Q_\alpha$. The compressed tensor $C^I = V^\dagger(A^{[p]})^I V$ is irreducible. This supplies the irreducibility assertion in [157, Lemma 6].*

Proof. ✓ Proof Let R be an invariant orthogonal projection for $\mathcal{E}_C^\dagger$. The projection $L = VRV^\dagger$ has invariant corner under F^p , where $F = \mathcal{E}_A^\dagger$. The faithful invariant state of $\mathcal{E}_A$ implies $F^p(L) = L$. Since $Q_\alpha L = L$, fixed-point rigidity gives $L = cQ_\alpha$, and compression gives $R = c\mathbb{1}$. Idempotence forces $R = 0$ or $R = \mathbb{1}$. Since $\mathcal{E}_C$ and $\mathcal{E}_C^\dagger$ have the same invariant orthogonal projections, C is irreducible. $\square$

Theorem 22.4.15 (Orbit roots as consecutive products). Lean ✓ Stmt *Let G be an abelian group, $m > 0$, $p \geq 0$, $r = \gcd(m, p)$, and $q = m/r$. Suppose $c_\alpha \in G$ satisfies $c_\alpha^q = 1$ for every $0 \leq \alpha < r$. There are $b_u \in G$, indexed modulo m , such that, writing $u = \alpha_u + pk_u$ for the coordinates of Theorem 22.4.5,*

$$\prod_{j=0}^{p-1} b_{u+j} = c_{\alpha_u}. \quad (22.14)$$

For $G = U(1)$, these are the local phases in [157, proof of Theorem 4.1], with the wraparound of k_u handled as in [155].

Proof. [✓ Proof](#) Put $f_u = c_{\alpha_u}^{k_u}$ and $b_u = f_{u+1}/f_u$. The coordinate shift and $c_\alpha^q = 1$ give $f_{u+p} = c_{\alpha_u} f_u$, including when $k_u = q-1$. The product telescopes to $f_{u+p}/f_u = c_{\alpha_u}$, proving (22.14). $\square$

Definition 22.4.16 (Sector rephasing). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor, let $P_0, \dots, P_{m-1} \in M_D(\mathbb{C})$, and let $c_0, \dots, c_{m-1} \in \mathbb{C}$. Define $A'^i = \sum_u c_u P_u A^i$, as in [157, before eq. Aprime-is-cPA].

Theorem 22.4.17 (Products of sector-rephased letters). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose $m \geq 1$, $P_u P_v = \delta_{uv} P_u$, $\sum_u P_u = \mathbb{1}$, and $P_u A^i = A^i P_{u+1}$, with sector indices read modulo m . For a word $w = (i_1, \dots, i_n)$, write $A^w = A^{i_1} \dots A^{i_n}$. Then

$$\begin{aligned} P_u A'^w &= \left(\prod_{k=0}^{n-1} c_{u+k} \right) P_u A^w, \\ A'^w &= \sum_u \left(\prod_{k=0}^{n-1} c_{u+k} \right) P_u A^w. \end{aligned} \quad (22.15)$$

The empty word has product $\mathbb{1}$. The case $n = p$ is the first equality of [157, eq. Aprime-is-cPA].

Proof. [✓ Proof](#) Orthogonality gives $P_u A'^i = c_u P_u A^i = c_u A^i P_{u+1}$. Induction on the word length gives the first identity in (22.15); summing it over u gives the second. $\square$

Theorem 22.4.18 (Left canonicity under sector rephasing). [Lean](#) [✓ Stmt](#) Suppose $P_u^\dagger = P_u$, $P_u P_v = \delta_{uv} P_u$, $\sum_u P_u = \mathbb{1}$, and $|c_u| = 1$. If A is left canonical, then $A'^i = \sum_u c_u P_u A^i$ is left canonical as well. Thus the rephasing in the proof of [157, Theorem 4.1] keeps a left-canonical block trace preserving.

Proof. [✓ Proof](#) Put $U = \sum_u c_u P_u$. Orthogonality gives $U^\dagger U = \sum_u |c_u|^2 P_u = \mathbb{1}$. Since $A'^i = U A^i$, $\sum_i (A'^i)^\dagger A'^i = \sum_i (A^i)^\dagger U^\dagger U A^i = \mathbb{1}$. $\square$

Theorem 22.4.19 (A left-canonical tensor with prescribed orbit phases). [Lean](#) [✓ Stmt](#) Let A be left canonical and $m > 0$. Suppose the Hermitian projections P_u satisfy $P_u P_v = \delta_{uv} P_u$, $\sum_u P_u = \mathbb{1}$, and $P_u A^i = A^i P_{u+1}$. Let $p \geq 0$, $r = \gcd(m, p)$, and $c_\alpha^{m/r} = 1$. There is a left-canonical tensor A' such that, for every word w of length p ,

$$A'^w = \sum_u c_{\alpha_u} P_u A^w. \quad (22.16)$$

This is the rephasing identity in [157, proof of Theorem 4.1].

Proof. [✓ Proof](#) Since $m/r > 0$, each root c_α has modulus one. Apply Theorem 22.4.15 in $U(1)$ to obtain unit phases b_u , and set $A'^i = \sum_u b_u P_u A^i$. Theorem 22.4.18 gives left canonicity. Substituting (22.14) into (22.15) gives (22.16). $\square$

Theorem 22.4.20 (The tensor C associated to a refinement witness). [Lean](#) [✓ Stmt](#) Let B be an MPS tensor in irreducible form II. If $\mathcal{V}(B)$ can be p -refined, then there exist a tensor A , an isometry $W : \mathbb{C}^d \rightarrow (\mathbb{C}^d)^{\otimes p}$, and a tensor C defined by $C^\alpha := \sum_{i=0}^{d-1} W_{\alpha,i} B^i$ for $\alpha \in \{0, \dots, d^p - 1\}$, such that C is again in irreducible form II, $\mathcal{E}_C = \mathcal{E}_B$, and C has the same MPV family as $A^{[p]}$.

Proof. [✓ Proof](#) Choose (A, W) from the refinement data. The definition of C gives $\mathcal{E}_C = \mathcal{E}_B$ and equality of MPV families between C and $A^{[p]}$. To see that C is still in irreducible form II, transport each periodic block B_j of B through the same physical isometry W . The left-canonical normalization is preserved because $W^\dagger W = \mathbb{1}$, while irreducibility and the peripheral spectrum

are preserved because the transfer map of each transported block agrees with that of B_j . Combining these transported blocks with the same weights gives an irreducible-form decomposition of C . $\square$

Definition 22.4.21 (*Z-gauge after applying the blocked equal-case theorem*). [Lean](#) [Stmnt](#) Fix (d, D, p) . This hypothesis says that whenever a blocked tensor C in irreducible form II has the same MPV family as the p -blocked tensor $A^{[p]}$, there exists an integer $m \geq 1$ and a $\mathbb{Z}_m$ -gauge equivalence between C and $A^{[p]}$.

Theorem 22.4.22 (*Z-gauge equivalence from the periodic theorem*). [Lean](#) [Stmnt](#) Assume the periodic equal-case Fundamental Theorem at the blocked physical dimension. Let C and $A^{[p]}$ be MPS tensors of the same bond dimension, both in irreducible form II, and suppose that they have the same MPV family. Then there exists an integer $m \geq 1$ and a $\mathbb{Z}_m$ -gauge equivalence between C and $A^{[p]}$.

Definition 22.4.23 (*Construction of $\tilde{A}$ from the Z-gauge*). [Lean](#) [Stmnt](#) Fix (d, D, p) . This hypothesis says that if $m \geq 1$ and a blocked tensor C in irreducible form II is $\mathbb{Z}_m$ -gauge equivalent to $A^{[p]}$, then one can construct a tensor $\tilde{A}$ with $\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$ and $\mathcal{E}_C = \mathcal{E}_{\tilde{A}^{[p]}}$. This channel-level condition is a strictly weaker consequence of the construction in [157, lines 765–810], which additionally supplies block scalars, intermediate matrices, and the full blocked-tensor identity $\tilde{A}^{[p]} = C$.

Definition 22.4.24 (*Construction of $\tilde{A}$ after the equal-case theorem*). [Lean](#) [Stmnt](#) Fix (d, D, p) . This states the combined construction: whenever a blocked tensor C in irreducible form II has the same MPV family as a p -blocked tensor $A^{[p]}$, one can replace A by a left-canonical tensor $\tilde{A}$ with $\mathcal{E}_C = \mathcal{E}_{\tilde{A}^{[p]}}$.

Theorem 22.4.25 (*Construction of $\tilde{A}$ from Z-gauge extraction*). [Lean](#) [Stmnt](#) If the blocked equal-case Z-gauge extraction hypothesis and the construction of $\tilde{A}$ from that Z-gauge both hold, then the combined equal-case construction holds.

Proof. [Proof](#) Apply the Z-gauge extraction hypothesis to obtain a blocked $\mathbb{Z}_m$ -gauge witness between C and $A^{[p]}$, and then apply the construction of $\tilde{A}$ to that witness. $\square$

Theorem 22.4.26 (*Reduction to the blocked equal-case construction*). [Lean](#) [Stmnt](#) Assume the blocked equal-case construction. Then every p -refinable tensor B in irreducible form II admits a left-canonical tensor $\tilde{A}$ with $\mathcal{E}_B = \mathcal{E}_{\tilde{A}^{[p]}}$.

Proof. [Proof](#) Apply Theorem 22.4.20 to the refinement witness of B . This produces a blocked tensor C in irreducible form II with $\mathcal{E}_C = \mathcal{E}_B$ and the same MPV family as $A^{[p]}$. The blocked equal-case construction then gives a left-canonical tensor $\tilde{A}$ with $\mathcal{E}_C = \mathcal{E}_{\tilde{A}^{[p]}}$. Combining the two transfer-map identities yields $\mathcal{E}_B = \mathcal{E}_{\tilde{A}^{[p]}}$. $\square$

Theorem 22.4.27 (*Forward direction from the blocked equal-case construction*). [Lean](#) [Stmnt](#) Assume the blocked equal-case construction. Then for every tensor B in irreducible form II, p -refinability of B implies p -divisibility of $\mathcal{E}_B$.

Proof. [Proof](#) Combine Theorem 22.4.26 with the witness-level forward implication: once $\mathcal{E}_B$ is identified with $\mathcal{E}_{\tilde{A}^{[p]}}$ for a left-canonical $\tilde{A}$, the blocking identity gives $\mathcal{E}_B = (\mathcal{E}_{\tilde{A}})^p$. $\square$

The refinability–divisibility equivalence of [157, Theorem 4.1] decomposes into the two one-way implications of Theorem 22.4.29 and Theorem 22.4.35.

Theorem 22.4.28 (p -refinability iff transfer-map p -divisibility, conditional). [Lean](#) [Stmnt](#) Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the forward construction of $\tilde{A}$ for refinability $\Rightarrow$ divisibility and the reverse transfer-map construction for divisibility $\Rightarrow$ refinability. Then B is p -refinable iff its transfer map $\mathcal{E}_B$ is p -divisible.

Proof. [Proof](#) Combine the forward implication (Theorem 22.4.29) with the reverse implication (Theorem 22.4.35) into a single biconditional. The forward direction assumes the construction of $\tilde{A}$; the reverse direction assumes the transfer-map construction. Together they give the full equivalence under those hypotheses. $\square$

Theorem 22.4.29 (p -refinability implies transfer-map p -divisibility). [Lean](#) [Stmnt](#) Let B be an MPS tensor in irreducible form II. Assume that every refinement witness gives a tensor $\tilde{A}$ with $\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$ and $\mathcal{E}_B = \mathcal{E}_{\tilde{A}^{[p]}}$. Then p -refinability of B implies p -divisibility of $\mathcal{E}_B$.

Proof. [Proof](#) The construction supplies $\tilde{A}$ with $\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$ and $\mathcal{E}_B = \mathcal{E}_{\tilde{A}^{[p]}}$. The left-canonical property makes $\mathcal{E}_{\tilde{A}}$ a CPTP channel, and the blocking identity $\mathcal{E}_{\tilde{A}^{[p]}} = (\mathcal{E}_{\tilde{A}})^p$ then identifies $\mathcal{E}_B$ with the p -th power of the channel $\mathcal{E}_{\tilde{A}}$. $\square$

Remark 22.4.30 (Trace preservation in the forward direction). In [157, Theorem 4.1], p -divisibility presupposes trace preservation of $\mathcal{E}_B$, which is explicit in the hypotheses here. Without trace preservation, let $p \geq 1$, let B_0 be p -refinable with witness (A_0, W) and in irreducible form II with every multiplicity equal to one, and let $\lambda > 1$. Set

$$B = \lambda B_0, \quad A = \lambda^{1/p} A_0.$$

Then B is in irreducible form II with the blocks of B_0 and the multiplicities λ , and it is p -refinable with witness (A, W) :

$$|V^{(pN)}(A)\rangle = \lambda^N |V^{(pN)}(A_0)\rangle = \lambda^N W^{\otimes N} |V^{(N)}(B_0)\rangle = W^{\otimes N} |V^{(N)}(B)\rangle.$$

Its transfer map satisfies

$$\mathcal{E}_B = \lambda^2 \mathcal{E}_{B_0}, \quad \text{tr } \mathcal{E}_B(\mathbb{1}) = \lambda^2 \text{tr } \mathbb{1} \neq \text{tr } \mathbb{1},$$

so $\mathcal{E}_B$ is not trace preserving and is not the p -th power of a CPTP map. At physical and bond dimension one, with $B_0 = (1)$, the same data give $C = A^{[p]} = (\lambda)$ related by the trivial $\mathbb{Z}_1$ -gauge, and no left-canonical $\tilde{A}$ or CPTP root has p -th power $x \mapsto \lambda^2 x$; so the constructions assumed in Definitions 22.4.23, 22.4.24, and 22.4.32 fail as well. The corrected statement takes B literally in the block form of irreducible form II from [157], as the source does, and adds trace preservation of $\mathcal{E}_B$; for such a block form this is equivalent to every multiplicity being one, since $\sum_i (B^i)^\dagger B^i = \bigoplus_j \mu_j^2 \mathbb{1}$. Under this hypothesis the printed argument closes [144]. The equivalence does not extend to the generated-family reading of Definition 22.1.12: a non-unitary similarity of a left-canonical normal tensor generates the same family with unit weights, while its transfer map is not trace preserving.

Theorem 22.4.31 (A refinable positive-weight tensor without a channel square root). [Lean](#) [Stmnt](#) Let the physical and bond dimensions both be one, and take $B = (4)$. The tensor B is literally the period-one normalized block (1) with positive weight 4. It is 2-refinable with root $A = (2)$, but its transfer map $\mathcal{E}_B(x) = 16x$ is not 2-divisible as a channel. This contradicts the forward implication of the printed [157, Theorem 4.1]; see [144].

Proof. [✓ Proof](#) The block (1) has identity transfer map and period one. Since $A^{[2]} = B$, equality of the blocked and target transfer maps gives a refinement isometry. If $\mathcal{E}_B = \mathcal{F}^2$ for a channel $\mathcal{F}$, then $\mathcal{E}_B$ preserves trace. This contradicts $\text{tr } \mathcal{E}_B(1) = 16 \neq 1$. $\square$

Definition 22.4.32 (Blocked-to-root channel-root hypothesis from Z -gauge). [Lean](#) [✓ Stmt](#) Fix (d, D, p) . This sharpened reconstruction hypothesis asks for a CPTP map $\mathcal{E}'$ such that $\mathcal{E}_C = (\mathcal{E}')^p$ and the Choi/Kraus rank of $\mathcal{E}'$ is at most d , whenever C is related to $A^{[p]}$ by a blocked $\mathbb{Z}_m$ -gauge witness.

Theorem 22.4.33 (Channel-root criterion for blocked-to-root reconstruction). [Lean](#) [✓ Stmt](#) *If the blocked-to-root channel-root hypothesis holds, then the tensor-level blocked-to-root reconstruction hypothesis holds as well.*

Proof. [✓ Proof](#) Choose a tensor A' whose matrix family realizes the root channel with at most d nonzero terms. Since this channel is trace-preserving, the matrices of A' satisfy $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$. The identity $\mathcal{E}_C = (\mathcal{E}_{A'})^p$ is then equivalent to $\mathcal{E}_C = \mathcal{E}_{A'^{[p]}}$ by the blocking formula. $\square$

Theorem 22.4.34 (Refinement from a blocked transfer identity). [Lean](#) [✓ Stmt](#) *Let A and B have the same one-site physical and bond dimensions, and let $p > 0$. If $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$, then B is p -refinable.*

Proof. [✓ Proof](#) The two tensors give Kraus representations of the same completely positive map, of respective sizes d^p and d . Since $d \leq d^p$, isometric freedom of Kraus representations gives $A^{[p],\alpha} = \sum_j W_{\alpha,j} B^j$ with $W^*W = I$. Expanding products and taking traces gives the refinement identity. $\square$

Theorem 22.4.35 (Transfer-map p -divisibility implies p -refinability). [Lean](#) [✓ Stmt](#) *Let B be an MPS tensor in irreducible form II and let $p \geq 1$. Assume the reverse transfer-map root hypothesis: there is a tensor A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$ whenever $\mathcal{E}_B$ is p -divisible. Then p -divisibility of $\mathcal{E}_B$ implies p -refinability of B .*

Proof. [✓ Proof](#) From the reverse transfer-map root hypothesis, extract A with $\mathcal{E}_B = \mathcal{E}_{A^{[p]}}$. This matches two finite Kraus families of the same completely positive map: the blocked family $A^{[p]}$ with d^p operators and the original family B with d operators. The cardinality comparison $d \leq d^p$ holds whenever $p \geq 1$, so [212, Theorem 2.1(4)] supplies an isometry $V \in M_{d^p \times d}(\mathbb{C})$ with $V^\dagger V = \mathbb{1}$ and $(A^{[p]})^\alpha = \sum_j V_{\alpha,j} B^j$. Expanding the coefficient of $A^{[p]}$ at the word τ by linearity of matrix multiplication and of the trace, the contributions reorganise into the V -weighted sum over length- N words of B required by Definition 22.4.3, exhibiting V as the isometry W in the statement of p -refinability. $\square$

Theorem 22.4.36 (Irreducibility and period constraints on a refinement root). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be a tensor and let $p \geq 0$. If its blocked tensor $A^{(p)}$ is irreducible, then A is irreducible. In particular, this holds if $\mathcal{E}_A^p = \mathcal{E}_B$ for an irreducible tensor B , even when A and B have different physical dimensions. If moreover A is periodic of period m , then $\gcd(m, p) = 1$. These are necessary conditions for the converse construction in [157, Theorem 4.1].*

Proof. [✓ Proof](#) A subspace invariant under every A^i is invariant under every product of p letters. Equality of transfer maps preserves irreducibility. For the period assertion, blocking gives $\gcd(m, p)$ nonzero orthogonal orbit projections, each commuting with every blocked letter. Irreducibility forces each projection to be the identity. Orthogonality then permits only one orbit. $\square$

Theorem 22.4.43 (Refinement for an invertible transfer map). [Lean](#) [✓ Stmt](#) *Let B have physical dimension d and an invertible transfer map, and let $p > 0$. If $\mathcal{E}_B$ is p -divisible as a channel, then B is p -refinable.*

Proof. [✓ Proof](#) Choose a channel root $\mathcal{F}$ with $\mathcal{F}^p = \mathcal{E}_B$. It is invertible, so the preceding theorem gives $\text{rank } C_{\mathcal{F}} \leq \text{rank } C_{\mathcal{E}_B} \leq d$. Apply Theorem 22.4.39. $\square$

Remark 22.4.44. This proves the converse of [157, Theorem 4.1] for invertible transfer maps without a canonical-form assumption. The corresponding assertion for singular transfer maps remains unresolved; see [120].

Theorem 22.4.45 (Word-span dimension under invertible multiplication). [Lean](#) [Lean](#) [✓ Stmt](#) *Let S be the span of a finite family of square matrices, and let S^q denote the span of products of exactly q members, with $S^0 = \mathbb{C}I$. If $p > 0$ and S^{p-1} contains an invertible matrix, then $\dim S \leq \dim S^p$. In particular, the inequality holds whenever S contains an invertible matrix.*

Proof. [✓ Proof](#) If $K \in S^{p-1}$ is invertible, the injective linear map $X \mapsto KX$ sends S into S^p . For the final assertion, take the $(p-1)$ st power of an invertible member of S . $\square$

Theorem 22.4.46 (Refinement from an invertible root word span). [Lean](#) [Lean](#) [✓ Stmt](#) *Let B have physical dimension d , and let A have the same bond dimension but any finite physical dimension. Suppose $p > 0$ and $\mathcal{E}_B = (\mathcal{E}_A)^p$. If the span of products of $p-1$ letters of A contains an invertible matrix, then $\text{rank}_{\text{Choi}} \mathcal{E}_A \leq \text{rank}_{\text{Choi}} \mathcal{E}_B \leq d$, and B is p -refinable.*

Proof. [✓ Proof](#) If S is the Kraus span of A , the preceding two theorems give

$$\text{rank}_{\text{Choi}} \mathcal{E}_A = \dim S \leq \dim S^p = \text{rank}_{\text{Choi}} \mathcal{E}_B \leq d.$$

A completely positive map has a Kraus representation with as many operators as its Choi rank. Apply Theorem 22.4.39. $\square$

Remark 22.4.47 (Continuum-limit application). [Not ready](#) Refinability along arbitrarily large p corresponds to infinite divisibility of the transfer map, which characterizes continuum limits of translationally invariant MPS ([157]).

Theorem 22.4.48 (Absorbing the orbitwise gauge into the root). [Lean](#) [✓ Stmt](#) *Let A be left canonical with orthogonal cyclic projections P_u summing to the identity and satisfying $P_u A^i = A^i P_{u+1}$. Let $m > 0$, $p \geq 0$, $r = \gcd(m, p)$, and $c_\alpha^{m/r} = 1$. If a matrix Z satisfies*

$$Z Q_\alpha (A^{[p]})^I = c_\alpha Q_\alpha (A^{[p]})^I, \quad (22.17)$$

then there is a left-canonical tensor A' of the same dimensions with $(A'^{[p]})^I = Z(A^{[p]})^I$ for every I . This is the comparison of the two phase formulas in the proof of [157, Theorem 4.1].

Proof. [✓ Proof](#) Construct A' using Theorem 22.4.19. Regroup its cyclic-sector sum by step orbits to obtain $\sum_\alpha c_\alpha Q_\alpha (A^{[p]})^I$. Substitute (22.17) and use $\sum_\alpha Q_\alpha = \mathbb{1}$. $\square$

Theorem 22.4.49 (Unitary conjugation of a blocked root). [Lean](#) [✓ Stmt](#) *If A is left canonical and $(A^{[p]})^I = UC^I U^\dagger$ for a unitary U , then C has a left-canonical root B at the original physical dimension: $B^{[p]} = C$. This is the final conjugation in the proof of [157, Theorem 4.1].*

Proof. [✓ Proof](#) Set $B^i = U^\dagger A^i U$. Then $\sum_i (B^i)^\dagger B^i = U^\dagger (\sum_i (A^i)^\dagger A^i) U = \mathbb{1}$, and $UU^\dagger = \mathbb{1}$ cancels the intermediate factors of every word:

$$B^{i_1} \dots B^{i_p} = U^\dagger A^{i_1} U U^\dagger A^{i_2} U \dots U^\dagger A^{i_p} U = U^\dagger A^{i_1} \dots A^{i_p} U.$$

Hence $B^{[p]} = U^\dagger A^{[p]} U = C$. □

Theorem 22.4.50 (Refining a weighted block sum). [Lean](#) [✓ Stmt](#) If A_k and $\bigoplus_{a=0}^{n_k-1} B_{k,a}$ have equal matrix product vectors at every length, then $\bigoplus_k \mu_k A_k$ and $\bigoplus_{k,a} \mu_k B_{k,a}$ do as well, for arbitrary complex weights μ_k .

Proof. [✓ Proof](#) Expand the trace of each block-diagonal product and substitute the refined vectors of each A_k . At every length N ,

$$V^{(N)}\left(\bigoplus_k \mu_k A_k\right)_\sigma = \sum_k \mu_k^N V^{(N)}(A_k)_\sigma = \sum_k \sum_{a=0}^{n_k-1} \mu_k^N V^{(N)}(B_{k,a})_\sigma = V^{(N)}\left(\bigoplus_{k,a} \mu_k B_{k,a}\right)_\sigma.$$

The last equality reindexes the double sum by the pairs (k, a) . □

Definition 22.4.51 (Positive blocking of an irreducible-form presentation). [Lean](#) [✓ Stmt](#) Suppose A has the same matrix product vectors at every length, including zero, as $\bigoplus_k \mu_k A_k$, where $\mu_k > 0$ and A_k is left-canonical periodic of period m_k , and let $p > 0$. The blocked presentation of $A^{[p]}$ is indexed by the pairs (k, a) with $0 \leq a < \gcd(m_k, p)$; the block $B_{k,a}$ is the a th prescribed orbit block of $A_k^{[p]}$, with weight μ_k^p and period $m_k / \gcd(m_k, p)$. It is an irreducible-form presentation of $A^{[p]}$. This is the blockwise application of [157, Lemma 6, proof of Theorem 4.1].

Proof. [✓ Proof](#) Apply the prescribed-blocking decomposition separately to each A_k , giving blocks $B_{k,a}$ with $0 \leq a < \gcd(m_k, p)$. Blocking raises μ_k to the power p : if σ' is the unblocked configuration of length pN underlying a blocked configuration σ of length N , then

$$V^{(N)}(A^{[p]})_\sigma = V^{(pN)}(A)_{\sigma'} = \sum_k \mu_k^{pN} V^{(pN)}(A_k)_{\sigma'} = \sum_k (\mu_k^p)^N V^{(N)}(A_k^{[p]})_\sigma.$$

Refinement preserves the weight μ_k^p on every orbit block. The preceding theorem assembles the resulting vector identities. All new weights remain positive. □

Theorem 22.4.52 (Total dimension of an irreducible-form presentation). [Lean](#) [✓ Stmt](#) If A of bond dimension D has an irreducible-form presentation $\bigoplus_k \mu_k A_k$ with blocks of bond dimensions D_k , then $\sum_k D_k = D$. In particular this holds for the blocked presentation of Definition 22.4.51.

Proof. [✓ Proof](#) At length zero every matrix product vector is the trace of the identity, and the presentation has the same vectors as A at length zero, so

$$D = V^{(0)}(A) = V^{(0)}\left(\bigoplus_k \mu_k A_k\right) = \sum_k D_k.$$

□

Theorem 22.4.53 (Multiplication of an orthogonal resolution). [Lean](#) [✓ Stmt](#) If orthogonal projections P_u satisfy $\sum_u P_u = \mathbb{1}$, then $P_u P_v = \delta_{uv} P_u$.

Proof. [✓ Proof](#) Each projection is idempotent, $P_u^2 = P_u$, and distinct projections in an orthogonal resolution have orthogonal ranges, $P_u P_v = 0$ for $u \neq v$. □

Theorem 22.4.54 (Unitary assembly of a resolution). [Lean](#) [✓ Stmt](#) Let $V_a : \mathbb{C}^{D_a} \rightarrow \mathbb{C}^D$ satisfy $\sum_a V_a V_a^\dagger = \mathbb{1}$ and $\sum_a D_a = D$. If $A^i V_a = V_a B_a^i$ for every a, i , then there is a unitary $U : \bigoplus_a \mathbb{C}^{D_a} \rightarrow \mathbb{C}^D$ with $U J_a = V_a$ and $A^i = U(\bigoplus_a B_a^i) U^\dagger$, where J_a is the block inclusion.

Proof. [✓ Proof](#) Concatenate the columns of the V_a to define U . The identity resolution gives $U U^\dagger = \mathbb{1}$, and equality of dimensions makes U unitary. On every summand,

$$A^i U J_a = A^i V_a = V_a B_a^i = U \left(\bigoplus_b B_b^i \right) J_a,$$

so $A^i U = U(\bigoplus_a B_a^i)$, and multiplying by $U^\dagger$ gives the asserted conjugation. $\square$

Theorem 22.4.55 (Unitary orbit coordinates and phase-weighted roots). [Lean](#) [✓ Stmt](#) Let A be left-canonical periodic of period m and let $p > 0$. Put $r = \gcd(m, p)$. The prescribed orbit blocks C_a , of period m/r , admit a unitary U such that, with Q_a the step-orbit projections of Definition 22.4.6 and J_a the coordinate inclusions,

$$(A^{[p]})^I = U \left(\bigoplus_{a=0}^{r-1} C_a^I \right) U^\dagger, \\ (U J_a)(U J_a)^\dagger = Q_a.$$

For every choice of c_a with $c_a^{m/r} = 1$, there is a left-canonical tensor A' with the same physical and bond dimensions as A such that

$$(A'^{[p]})^I = U \left(\bigoplus_{a=0}^{r-1} c_a C_a^I \right) U^\dagger.$$

The unitary and orbit blocks are chosen independently of the phases.

Proof. [✓ Proof](#) Apply Theorem 22.4.54 to the orbit isometries and set $Z = U(\bigoplus_a c_a \mathbb{1}_{D_a}) U^\dagger$. Since $U J_a$ is the original orbit isometry, $Z Q_a (A^{[p]})^I = c_a Q_a (A^{[p]})^I$. Absorbing these phases into the original cyclic sectors gives

$$(A'^{[p]})^I = Z (A^{[p]})^I = U \left(\bigoplus_a c_a C_a^I \right) U^\dagger.$$

$\square$

Theorem 22.4.56 (Weight and dimension recovery from a blocked comparison). [Lean](#) [✓ Stmt](#) Let $T = \bigoplus_k \mu_k B_k$, where $\mu_k \neq 0$ and the B_k are left-canonical periodic tensors of bond dimension D_k . Suppose that, for $p > 0$, $T^{[p]}$ has the same positive-length matrix product vectors as a periodic sector decomposition Q with a pairwise non-repeated basis and unit-modulus multiplicities. Then $|\mu_k| = 1$ for every k , and $\sum_k D_k$ equals the total bond dimension of Q . This is the weight normalization in the corrected forward direction of [157, Theorem 4.1, lines 750–756]; the unit-modulus target is the correction of [144].

Proof. [✓ Proof](#) Decompose each blocked B_k into its prescribed orbits. Every resulting block inherits weight μ_k^p . Group repeated blocks and apply the multiplicity-bearing equal-case theorem. Matching to the unit-modulus multiplicities of Q gives $|\mu_k|^p = 1$, hence $|\mu_k| = 1$. The equal-case theorem also gives matrices Y and Y' between the bond spaces of the grouped decomposition P

and of Q with $YY' = \mathbb{1}$ and $Y'Y = \mathbb{1}$, so these square matrices have the same size. Since orbit decomposition and grouping preserve the total dimension,

$$\sum_k D_k = \dim P = \dim Q.$$

□

Theorem 22.4.57 (Unit-weight presentation from a blocked comparison). Lean ✓ Stmt *Let A be any tensor of bond dimension D . Suppose that, for $p > 0$, $A^{[p]}$ has the same positive-length matrix product vectors as a periodic sector decomposition Q with a pairwise non-repeated basis and unit-modulus multiplicities. Then A has the same positive-length matrix product vectors as a unit-weight sum $\bigoplus_k C_k$ of left-canonical periodic tensors. Their total bond dimension equals that of Q and is at most D . This is the normalization step of [157, Theorem 4.1, lines 750–756] for an arbitrary input tensor.*

Proof. ✓ Proof Take the positive-weight periodic decomposition of A from Theorem 22.4.1 and block it by p . The preceding theorem gives unit modulus for each original weight and the claimed dimension. Set $C_k^I = \mu_k B_k^I$; then $\bigoplus_k C_k = \bigoplus_k \mu_k B_k$, and since $|\mu_k| = 1$ each C_k is left-canonical periodic with the period of B_k . □

Chapter 23

The Algebraic Fundamental Theorem

This chapter presents the algebraic Fundamental Theorem for injective and normal matrix product states on a finite periodic chain of fixed length, following Molnár et al. [186]. The argument works directly in the matrix algebra $M_D(\mathbb{C})$. The fixed-length development rests on two lemmas: the virtual bond gauge (Lemma 23.5.2.1) and the proportionality lemma (Lemma 23.5.3.1). These yield the main chain theorem and its corollaries for translation-invariant chains, site-dependent descriptions of translation-invariant states, and blocked chains. Block-diagonal tensors are treated through product-algebra automorphisms and their block-permutation decompositions, yielding gauge equivalence for direct sums and block matching in the equal-vector case with differing bond dimensions. The same foundational lemmas also underpin the injective PEPS development in Chapter 24.

Throughout, D denotes the bond dimension and d the physical dimension with $d \geq D^2$.

23.1 Site-dependent chains at fixed length

Molnár et al. [186] formulate the algebraic proof for periodic chains whose tensors vary from site to site, whereas Pérez-García et al. [193, Theorem 3] begin with a varying-bond open chain. Both fixed-length constructions are defined below.

Definition 23.1.1 (Periodic non-TI chain data). [Lean] [Lean] [✓ Stmt] Fix a chain length n . A *non-translation-invariant periodic MPS chain* consists of sitewise tensors $A_k = \{A_k^i\}_{i=0}^{d-1}$, where $k = 0, \dots, n-1$ and each $A_k^i \in M_D(\mathbb{C})$. For a basis configuration $\sigma = (\sigma_0, \dots, \sigma_{n-1})$, its coefficient is

$$c_{A,\sigma} := \text{tr}(A_0^{\sigma_0} A_1^{\sigma_1} \cdots A_{n-1}^{\sigma_{n-1}}).$$

Two such chains generate the *same chain state* if these coefficients agree for every σ . The chain is *injective* if each local tensor A_k is injective in the usual single-site sense. Finally, two chains are *cyclically gauge equivalent* if there are invertible bond matrices $Z_0, \dots, Z_{n-1}$ such that, for every site k and physical index i ,

$$B_k^i = Z_k A_k^i Z_{k+1}^{-1},$$

with indices understood modulo n , so that $Z_n = Z_0$. Locally, this gauge relation is represented by

Definition 23.1.2 (Varying-bond open chain). [Lean](#) [✓ Stmt](#) Fix a chain length N and dimensions $D_0, \dots, D_N \leq D$ with $D_0 = D_N = 1$. A varying-bond open chain consists of matrices $A_k^i \in \mathbb{C}^{D_k \times D_{k+1}}$ for $k = 0, \dots, N-1$ and $i = 0, \dots, d-1$. Intermediate dimensions may vanish. The number D is a common upper bound and need not be the least such bound.

This is the open-boundary representation of [193, Section II.B]. The source takes $D = \max_k D_k$; allowing an arbitrary common upper bound does not change the rectangular chain.

Definition 23.1.3 (Open-chain coefficient). [Lean](#) [✓ Stmt](#) For a physical configuration $\sigma = (\sigma_0, \dots, \sigma_{N-1})$, define

$$c_A(\sigma) := \sum_{\alpha_0=0}^{D_0-1} \cdots \sum_{\alpha_N=0}^{D_N-1} \prod_{k=0}^{N-1} (A_k^{\sigma_k})_{\alpha_k, \alpha_{k+1}}. \quad (23.1)$$

Since $D_0 = D_N = 1$, this is the scalar matrix product in [193, Section II.B]. If an intermediate dimension vanishes, the virtual-path set is empty. For $N = 0$, there is one empty path and its empty product is 1.

Definition 23.1.4 (Cyclic translation of an open-chain configuration). [Lean](#) [✓ Stmt](#) Let $N \geq 1$. For $s \in \mathbb{Z}/N\mathbb{Z}$, the cyclic translation of a physical configuration is

$$(T_s \sigma)_k = \sigma_{k+s}. \quad (23.2)$$

Only the physical configuration is translated. The dimensions $D_0, \dots, D_N$, the site matrices, and the distinguished open cut remain fixed. At $N = 0$, there is no translation parameter.

Definition 23.1.5 (Translation invariance at fixed length). [Lean](#) [✓ Stmt](#) For $N \geq 1$, a varying-bond open chain represents a translation-invariant state at length N if, for every $s \in \mathbb{Z}/N\mathbb{Z}$ and every physical configuration σ ,

$$c_A(T_s \sigma) = c_A(\sigma). \quad (23.3)$$

This condition compares coefficients at one fixed length. It does not rotate the bond dimensions or identify open cuts. At $N = 0$, the condition is vacuous.

Theorem 23.1.6 (Empty open-chain coefficient). [Lean](#) [✓ Stmt](#) At $N = 0$, the open-chain coefficient is 1.

Proof. [✓ Proof](#) The left endpoint space is one-dimensional, so there is one virtual path. Its product contains no factors and therefore equals 1. $\square$

Theorem 23.1.7 (Vanishing from a zero virtual dimension). [Lean](#) [✓ Stmt](#) If $D_k = 0$ for some $k \in \{0, \dots, N\}$, then $c_A(\sigma) = 0$ for every physical configuration σ .

Proof. [✓ Proof](#) No virtual path can have a coordinate in the empty k th bond space, so the sum in (23.1) is empty. $\square$

Definition 23.1.8 (Square padding of a varying-bond open chain). [Lean](#) [Stmnt](#) For each site k , embed A_k^i into the upper-left $D_k \times D_{k+1}$ corner of a $D \times D$ matrix $\tilde{A}_k^i$ and set every other entry to zero. The matrices $\tilde{A}_k^i$ form a square site-dependent closed chain.

This square padding embeds the rectangular open chain into square matrices, whereas [193, Theorem 3] places the rectangular matrices into the off-diagonal blocks of an $ND \times ND$ cyclic matrix.

Theorem 23.1.9 (Empty padded-chain coefficient). [Lean](#) [Stmnt](#) At $N = 0$, the coefficient of the padded square chain is D .

Proof. [Proof](#) The empty matrix product is the $D \times D$ identity, whose trace is D . $\square$

Theorem 23.1.10 (Zero-extension of a dependent finite sum). [Lean](#) [Stmnt](#) Let I be finite, let $r_i \leq s_i$ be nonnegative integers for $i \in I$, and let f take values in an additive commutative monoid. Then the sum of f over $\prod_{i \in I} \{0, \dots, r_i - 1\}$ equals its sum over $\prod_{i \in I} \{0, \dots, s_i - 1\}$ after extending f by zero outside the smaller Cartesian product.

Proof. [Proof](#) The coordinatewise inclusions identify the smaller Cartesian product with the subset of the larger product on which every coordinate is below its original bound. Split the larger sum into this subset and its complement. $\square$

Theorem 23.1.11 (Cyclic path expansion for a square chain). [Lean](#) [Stmnt](#) Let $N \geq 1$ and let $B_0, \dots, B_{N-1}$ be a square site-dependent chain of bond dimension D . For every physical configuration σ ,

$$c_{B,\sigma} = \sum_{g: \mathbb{Z}/N\mathbb{Z} \rightarrow \{0, \dots, D-1\}} \prod_{k \in \mathbb{Z}/N\mathbb{Z}} (B_k^{\sigma_k})_{g(k), g(k+1)}. \quad (23.4)$$

Proof. [Proof](#) Encode the site and physical indices as one enlarged physical index and apply the cyclic trace expansion for a closed matrix product. $\square$

Theorem 23.1.12 (Positive-length coefficient preservation). [Lean](#) [Stmnt](#) Let $N \geq 1$. For every physical configuration σ ,

$$\text{tr}(\tilde{A}_0^{\sigma_0} \dots \tilde{A}_{N-1}^{\sigma_{N-1}}) = c_A(\sigma). \quad (23.5)$$

No positivity assumption is imposed on the intermediate dimensions. The endpoint conditions already imply $D \geq 1$ whenever such a chain exists. The equality concerns only this fixed length.

Proof. [Proof](#) Expand the left-hand side by (23.4). Extend the dependent sum in (23.1) to the common dimension D by Theorem 23.1.10. Every extended path outside the rectangular bond ranges has a zero matrix entry. For the remaining paths, the one-dimensional endpoint spaces identify the last virtual coordinate with the first, and the products agree term by term. $\square$

Theorem 23.1.13 (Padded coefficient with a zero virtual dimension). [Lean](#) [Stmnt](#) Let $N \geq 1$. If $D_k = 0$ for some $k \in \{0, \dots, N\}$, then the padded square-chain coefficient vanishes for every physical configuration.

Proof. [Proof](#) Apply positive-length coefficient preservation and then the vanishing of the open-chain contraction. $\square$

Definition 23.1.14 (Cyclic block tensor at fixed length). [Lean](#) [Lean](#) [✓ Stmt](#) Let $N \geq 1$ and let $A_0, \dots, A_{N-1}$ be a periodic chain of $D \times D$ matrices. For $k \in \mathbb{Z}/N\mathbb{Z}$, write $(A^{(k)})_j = A_{j+k}$ for the cyclic rotation by k sites. Let P be the permutation matrix on the block coordinate satisfying $P(e_a \otimes e_k) = e_a \otimes e_{k-1}$. For each physical index i , set

$$B^i = N^{-1/N} \text{diag}(A_0^i, \dots, A_{N-1}^i) P \in M_{ND}(\mathbb{C}). \quad (23.6)$$

Thus the $(k, k+1)$ block of B^i is $N^{-1/N} A_k^i$, with block indices taken modulo N , and every other block vanishes. The tensor B depends on the fixed length N .

Theorem 23.1.15 (Fixed-length cyclic block average). [Lean](#) [✓ Stmt](#) For every basis configuration $\sigma \in \{0, \dots, d-1\}^N$, the length- N coefficient of the tensor in Definition 23.1.14 is

$$V^{(N)}(B)_\sigma = \frac{1}{N} \sum_{k=0}^{N-1} c_{A^{(k)}, \sigma}. \quad (23.7)$$

Reindexing the finite sum and cyclically permuting factors under the trace identifies this with the physical-configuration rotation in [193, Theorem 3]. This identity concerns only the fixed length N .

Proof. [✓ Proof](#) Write $D_i = \text{diag}(A_0^i, \dots, A_{N-1}^i)$ and omit the scalar normalization temporarily. Commuting the cyclic permutation through a block diagonal gives

$$P \text{diag}(M_0, \dots, M_{N-1}) = \text{diag}(M_1, \dots, M_0) P. \quad (23.8)$$

Repeated use of (23.8), followed by $P^N = I$, yields

$$D_{\sigma_0} P \dots D_{\sigma_{N-1}} P = \text{diag} \left(\prod_{j=0}^{N-1} A_{j+k}^{\sigma_j} \right)_{k=0}^{N-1}. \quad (23.9)$$

Taking the trace of (23.9) sums the traces of its diagonal blocks. Finally, $(N^{-1/N})^N = N^{-1}$ gives (23.7). $\square$

Theorem 23.1.16 (Conditional site-independent matrices at fixed length). [Lean](#) [Lean](#) [✓ Stmt](#) This is the cyclic-block construction in [193, Theorem 3], conditional on its open-chain input. Let $N \geq 1$. Suppose a translation-invariant state on the N -site ring has a varying-bond open-chain representation whose virtual dimensions are at most D . Then it has a site-independent trace-MPS representation of bond dimension ND with the same length- N coefficients. The constructed tensor depends on N , and the equality is asserted only at this ring length.

Proof. [✓ Proof](#) The cyclic rotation of the padded square chain satisfies

$$c_{(\tilde{A})^{(k)}, \sigma} = c_A(\tau_{-k}\sigma), \quad (23.10)$$

where $(\tau_s\sigma)_j = \sigma_{j+s}$. To prove (23.10), expand both square-chain traces as cyclic virtual-path sums and reindex the virtual path by the same rotation. The inverse shift appears when the rotated matrix product is restored to the distinguished open cut. Translation invariance makes every term in the cyclic average equal to $c_A(\sigma)$, so the factor $1/N$ cancels the sum of N identical terms. $\square$

Lemma 23.1.17 (Cyclic gauge equivalence is an equivalence relation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On fixed-length periodic chains, cyclic gauge equivalence is reflexive, symmetric, and transitive.

Proof. [✓ Proof](#) Reflexivity uses the identity gauge on every bond. Symmetry inverts each bond gauge, and transitivity multiplies the gauges pointwise along the chain. $\square$

23.2 Quadratic reconstruction of the translation-invariant canonical form

A translation-invariant state whose successive reduced density operators have rank at most D^2 can be stored as an open-boundary matrix product state with $D^2 \times D^2$ matrices. The reconstruction theorem of [193, Theorem 8] recovers a translation-invariant tensor of bond dimension D from a window of D^4 consecutive matrices of the open-boundary canonical form by solving a system of quadratic equations whose size does not depend on the chain length. The system and its uniqueness criteria follow [193, Theorems 7 and 8].

Definition 23.2.1 (Solution of the quadratic reconstruction system). Lean ✓ Stmt Let a window of matrix families on the doubled bond space be given, $A_i^{[j]} \in M_{D^2}(\mathbb{C})$ for $i \in \{0, \dots, d-1\}$ and $0 \leq j \leq m-1$. A solution of the quadratic reconstruction system over this window consists of matrices $Y_0, \dots, Y_{m-1}$ and $Z_0, \dots, Z_{m-1}$ in $M_{D^2}(\mathbb{C})$ together with matrices $B_i \in M_D(\mathbb{C})$ satisfying

$$B_i \otimes \mathbb{1} = Y_j A_i^{[j]} Z_j \quad \forall i, j, \quad (23.11)$$

$$Y_{j+1} Z_j = \mathbb{1} \quad \forall j \leq m-2, \quad (23.12)$$

$$\sum_i B_i B_i^\dagger = \mathbb{1}. \quad (23.13)$$

In the system (S) of [193] the window consists of the $m = D^4$ matrices of the unique open-boundary canonical form at the sites $L_0 + 1, \dots, L_0 + D^4$; the unknown Y_j above is the source unknown Y_{L_0+1+j} , while Z_j is the source unknown Z_{L_0+2+j} attached to the right of that site.

Lemma 23.2.2 (Invertibility of the paired gauges). Lean Lean ✓ Stmt In any solution of the quadratic reconstruction system, every matrix Y_{j+1} and every matrix Z_j appearing in (23.12) is invertible.

Proof. ✓ Proof A square matrix with a one-sided inverse is invertible. □

Lemma 23.2.3 (Nonzero combination along a shifted chain). Lean Lean Lean ✓ Stmt Let T, S be linear maps defined on the same vector spaces, and let $Y_0, \dots, Y_n$ be vectors such that $T(Y_k) = S(Y_{k+1})$ for every $0 \leq k \leq n-1$, the vectors $Y_0, \dots, Y_{n-1}$ are linearly independent, and

$$Y_n = \sum_{k=0}^{n-1} \lambda_k Y_k. \quad (23.14)$$

Let $x \neq 0$ be a solution of the polynomial equation

$$\sum_{k=0}^{n-1} \lambda_k x^{n-k} = 1, \quad (23.15)$$

and define $\mu_j = \sum_{k=0}^j \lambda_k x^{j+1-k}$. Then the combination $Y = \sum_{k=0}^{n-1} \mu_k Y_k$ is nonzero and satisfies $T(Y) = \frac{1}{x} S(Y)$. This is the first auxiliary result before the proof of [193, Theorem 7], written with indices starting at zero.

Proof. ✓ Proof The coefficients satisfy $\mu_0 = \lambda_0 x$ and the recurrence $\mu_{j+1} = x(\lambda_{j+1} + \mu_j)$, and (23.15) states $\mu_{n-1} = 1$. Applying T termwise and using the chain relation,

$$T(Y) = \sum_{k=0}^{n-1} \mu_k S(Y_{k+1}) = S\left(\sum_{k=0}^{n-2} \mu_k Y_{k+1} + Y_n\right) = S\left(\sum_{j=0}^{n-1} (\lambda_j + \mu_{j-1}) Y_j\right), \quad (23.16)$$

with the convention $\mu_{-1} = 0$, where the last equality substitutes (23.14). By the recurrence each coefficient equals μ_j/x , so the right-hand side is $\frac{1}{x}S(Y)$. Since $\mu_{n-1} = 1$ and $Y_0, \dots, Y_{n-1}$ are linearly independent, $Y \neq 0$. $\square$

Lemma 23.2.4 (Amplified intertwiners factor through the multiplicity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let $B, C \in M_n(\mathbb{C})$ and let W act on the doubled space. Then

$$W(C \otimes \mathbb{1}) = (B \otimes \mathbb{1})W \quad (23.17)$$

holds if and only if every multiplicity slice X_{ab} of W , with entries $(X_{ab})_{pq} = W_{(p,a),(q,b)}$, satisfies $X_{ab}C = BX_{ab}$. Consequently a nonzero simultaneous solution W of the amplified equations for a family of pairs (B_i, C_i) yields a nonzero matrix X with $XC_i = B_iX$ for every i . This realizes the second auxiliary result before the proof of [193, Theorem 7], which phrases the solution space of (23.17) as $\mathcal{S} \otimes M_n$ for $\mathcal{S}$ the solution space of $XC = BX$.

Proof. [✓ Proof](#) Evaluating both sides of (23.17) at a pair of doubled indices reduces the identity factor to a Kronecker delta, which collapses the double sum to the slice products $X_{ab}C$ and BX_{ab} . For the consequence, a nonzero W has a nonzero slice. $\square$

Lemma 23.2.5 (Chain of intertwiners between two solutions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let two solutions of the quadratic reconstruction system over the same window be given, with row unknowns Y_j, Y'_j , column unknowns Z_j, Z'_j , and tensors B_i, C_i . For every window site j carrying an inversion equation, the matrix

$$W_{j+1} = Y'_{j+1}Z_j = Y'_{j+1}Y_{j+1}^{-1} \quad (23.18)$$

is invertible, and whenever the neighbouring site also carries an inversion equation,

$$W_{j+1}(B_i \otimes \mathbb{1}) = (C_i \otimes \mathbb{1})W_{j+2} \quad \forall i. \quad (23.19)$$

Thus a window of m sites supplies exactly $m-1$ internal intertwiners and $m-2$ relations between consecutive internal intertwiners. This yields the internal part of the chain in [193, Theorem 8]; the full linear dependence required by Lemma 23.2.3 incorporates the boundary cuts.

Proof. [✓ Proof](#) Both solutions express $B_i \otimes \mathbb{1}$ and $C_i \otimes \mathbb{1}$ through the same window matrix at site $j+1$; substituting one expression into the other and cancelling the paired gauges with (23.12) gives (23.19). Invertibility follows from Lemma 23.2.2. $\square$

Theorem 23.2.6 (Obtaining the translation-invariant canonical form). [Not ready](#) Consider a translation-invariant state on an N -site ring with unique open-boundary canonical form such that the rank of each reduced density operator of successive spins is bounded by D^2 , and form the quadratic system of Definition 23.2.1 over the window of D^4 open-boundary canonical matrices starting after the injectivity onset. Write $A = (A_i)_i$ for its canonical translation-invariant tensor.

1. If the state admits a translation-invariant matrix product representation satisfying injectivity (condition C1), then the system has a solution.
2. Let $B = (B_i)_i$ be the tensor of a solution and suppose in addition that its length- N trace-MPS representation is the given state. Then there are a unitary U and a scalar $\xi \in \mathbb{C}$ such that $|\xi| = 1$, $\xi^N = 1$, and

$$A_i = \xi U B_i U^\dagger \quad \text{for every } i. \quad (23.20)$$

Consequently the explicitly rephased tensor $\widehat{B}_i := \xi B_i$ still represents the same length- N state and is exactly unitarily conjugate to A .

The unrestricted representation claim and exact conjugacy in [193, Theorem 8] carry a residual finite-ring phase defect [70], which is resolved by the explicit phase ξ in (23.20).

Proof. The canonical representation solves the system by the freedom in the open-boundary representation. For the converse, assume that the solution tensor B represents the given length- N state. Comparing its open-boundary representation with the canonical one supplies intertwiners at all $D^4 + 1$ cuts around the D^4 -site window, including both endpoint cuts, together with the D^4 adjacent relations [75]. Linear dependence in $M_{D^2}(\mathbb{C})$ then provides the hypothesis of Lemma 23.2.3. The internal construction in Lemma 23.2.5 gives only $D^4 - 1$ intertwiners and $D^4 - 2$ relations, so it does not force this dependence for an arbitrary algebraic solution.

Once the full cut chain is supplied, the combination lemma and the slice extraction of Lemma 23.2.4 give a nonzero matrix R and a scalar $x \neq 0$ with $RB_i = \frac{1}{x}A_iR$. The two normalizations (23.13), the invariance of the positive diagonal fixed point, and the uniqueness of the transfer-map fixed point under injectivity give $|x| = 1$ and $RR^\dagger = c\mathbb{1}$ for some $c > 0$. Thus $U = c^{-1/2}R$ is unitary and $A_i = xUB_iU^\dagger$ for every i . Since A and B represent the same nonzero length- N state, their coefficients satisfy $V^{(N)}(A) = x^N V^{(N)}(B) = V^{(N)}(B)$, and hence $x^N = 1$. The tensor $\widehat{B}_i = xB_i$ therefore represents the same state and satisfies $A_i = U\widehat{B}_iU^\dagger$. $\square$

23.3 Uniqueness of the translation-invariant canonical form at a fixed ring length

The uniqueness theorem [193, Theorem 7] compares two translation-invariant canonical representations of one state on a ring of N sites. Its printed proof rewrites both representations as open-boundary representations on the doubled bond space, compares them with the open-boundary canonical form, and obtains invertible intertwiners at the cuts of a window of D^4 sites; the two auxiliary results of Section 23.2 then produce a nonzero intertwiner between the two local tensors. The cut-vector argument supplies these intertwiners directly from injectivity, yielding the fixed-length intertwiner and the uniqueness theorem. The printed conclusion is adjusted on a finite ring by an N th-root phase [70, 125].

Definition 23.3.1 (Cut vectors). Lean Lean Lean Lean ✓ Stmt Let $A = (A_i)_i$ be a tensor of bond dimension D and let $L \geq 0$. For $\alpha, \beta \in \{0, \dots, D-1\}$ the *left cut vector* and the *right cut vector* of length L are the vectors

$$|\Phi_{\alpha,\beta}\rangle = \sum_{i_1, \dots, i_L} \langle \alpha | A_{i_1} \cdots A_{i_L} | \beta \rangle |i_1 \cdots i_L\rangle, \quad (23.21)$$

$$|\Psi_{\alpha,\beta}\rangle = \sum_{i_1, \dots, i_L} \langle \beta | A_{i_1} \cdots A_{i_L} | \alpha \rangle |i_1 \cdots i_L\rangle. \quad (23.22)$$

For a word split into a left part of length k and a right part of length m , the coefficient of the ring state factors across the cut as

$$\text{tr}(A_{i_1} \cdots A_{i_k} A_{j_1} \cdots A_{j_m}) = \sum_{\alpha, \beta} \Phi_{\alpha,\beta}(i_1 \cdots i_k) \Psi_{\alpha,\beta}(j_1 \cdots j_m), \quad (23.23)$$

and extending the left word by one site multiplies the left cut vectors by the matrix of that site, $\Phi_{\alpha,\beta}(i_1 \cdots i_k i) = \sum_{\gamma} \Phi_{\alpha,\gamma}(i_1 \cdots i_k) (A_i)_{\gamma\beta}$. These are the vectors introduced in the proof of the proposition on condition C1 in [193].

Lemma 23.3.2 (Linear independence of the cut vectors). Lean Lean Lean ✓ Stmt If A is L -block injective, then the D^2 left cut vectors of length L are linearly independent, and so are the D^2

right cut vectors of length L . If $\sum_i A_i A_i^\dagger = \mathbb{1}$ as well, then A is L' -block injective for every $L' \geq L$, so both statements hold at every length at least L .

Proof. ✓ Proof A vanishing combination $\sum_{\alpha, \beta} c_{\alpha, \beta} |\Phi_{\alpha, \beta}\rangle = 0$ says that the linear functional $M \mapsto \sum_{\alpha, \beta} c_{\alpha, \beta} M_{\alpha\beta}$ annihilates every product of L matrices of A . These products span $M_D(\mathbb{C})$, so the functional vanishes on the matrix units and $c = 0$. The right cut vectors are treated in the same way with the transposed functional. For the propagation, every matrix M equals $\sum_i A_i (A_i^\dagger M)$, and each $A_i^\dagger M$ is a combination of products of L' matrices, so M is a combination of products of $L' + 1$ matrices. $\square$

Lemma 23.3.3 (Intertwiners at the cuts). Lean Lean ✓ Stmt Let B and C be tensors of bond dimension D whose coefficients agree on every word of length $k + m$, and suppose that the left cut vectors of B of length k and the right cut vectors of B of length m are linearly independent. Then there is an invertible matrix W on the doubled bond space with

$$\Phi_a^C = \sum_b W_{b,a} \Phi_b^B \quad \text{for every index pair } a, \quad (23.24)$$

where the index pairs are written column first. If W and W' realize (23.24) at the cuts after k and after $k + 1$ sites, and the left cut vectors of B of length k are linearly independent, then

$$W(C_i \otimes \mathbb{1}) = (B_i \otimes \mathbb{1})W' \quad \text{for every } i. \quad (23.25)$$

This is the square case of the freedom theorem for open-boundary representations [193, Theorem 2] invoked in the proof of [193, Theorem 7], obtained here by a direct comparison of the two bipartite decompositions (23.23).

Proof. ✓ Proof Both decompositions (23.23) of the same coefficients have D^2 terms. The dual functionals of the linearly independent right cut vectors of B read off a matrix G with $\Phi_\mu^B = \sum_\nu G_{\mu\nu} \Phi_\nu^C$; linear independence of the left cut vectors of B shows that no nonzero row vector annihilates G , so G is invertible, and $W = (G^{-1})^\top$ satisfies (23.24). For the chain relation, extend the left words by one site: the two expressions of Φ^C at length $k + 1$ through (23.24) at the two cuts are combinations of the independent family Φ^B of length k , and comparing coefficients gives (23.25) entrywise. $\square$

Theorem 23.3.4 (Fixed-length intertwiner). Lean Lean ✓ Stmt Let B be a tensor of bond dimension $D \geq 1$ with $\sum_i B_i B_i^\dagger = \mathbb{1}$ that is L_0 -block injective, let $N \geq 2L_0 + D^4$, and let C be a tensor of bond dimension D with the same length- N coefficients as B . Then there are a nonzero matrix $R \in M_D(\mathbb{C})$ and a scalar $x \neq 0$ such that

$$RC_i = \frac{1}{x} B_i R \quad \text{for every } i. \quad (23.26)$$

This is the intertwiner obtained in the proof of [193, Theorem 7] before the unitarity argument, with the intertwiner count corrected to the $D^4 + 1$ cuts of the window [75].

Proof. ✓ Proof Consider the $D^4 + 1$ cuts after $L_0, \dots, L_0 + D^4$ sites; each leaves at least L_0 sites on both sides, so by Lemma 23.3.2 the cut vectors of B are linearly independent there, and Lemma 23.3.3 supplies invertible matrices $W_0, \dots, W_{D^4}$ on the doubled bond space with $W_k(C_i \otimes \mathbb{1}) = (B_i \otimes \mathbb{1})W_{k+1}$. Since $M_{D^2}(\mathbb{C})$ has dimension D^4 , there is a first index $n \geq 1$ such that $W_0, \dots, W_{n-1}$ are linearly independent and $W_n = \sum_{k < n} \lambda_k W_k$; some λ_k is nonzero because W_n is invertible, so the polynomial equation (23.15) has positive degree and a nonzero root x . Lemma 23.2.3, applied to right multiplication by $C_i \otimes \mathbb{1}$ and left multiplication by $B_i \otimes \mathbb{1}$, gives a nonzero combination W of $W_0, \dots, W_{n-1}$ with $W(C_i \otimes \mathbb{1}) = (\frac{1}{x} B_i \otimes \mathbb{1})W$ for every i , and Lemma 23.2.4 extracts a nonzero multiplicity slice R satisfying (23.26). $\square$

Lemma 23.3.5 (Unitarity of the intertwiner). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let B and C satisfy $\sum_i B_i B_i^\dagger = \mathbb{1}$ and $\sum_i C_i C_i^\dagger = \mathbb{1}$, let B be L_0 -block injective, and let $R \neq 0$ and $x \neq 0$ satisfy (23.26). Then R is invertible, $|x| = 1$, and there is a unitary U with

$$B_i = x U C_i U^\dagger \quad \text{for every } i. \quad (23.27)$$

Consequently the length- N coefficients of B and C satisfy $V^{(N)}(B) = x^N V^{(N)}(C)$ for every N .

Proof. [✓ Proof](#) The range of R is invariant under every B_i , hence under every product of L_0 matrices of B ; block injectivity leaves no nontrivial invariant subspace, so the range is everything and R is invertible. Block injectivity also makes the transfer maps of B and of C irreducible, the latter because $C_i = \frac{1}{x} R^{-1} B_i R$. The conjugate-transposed families are therefore left-canonical irreducible tensors related by the gauge $(R^\dagger)^{-1}$ and the scalar $\bar{x}$, and Lemma 9.4.1.2 replaces the gauge by a unitary and gives $|x| = 1$. Taking adjoints returns (23.27). This is the fixed-point argument of [193, Theorem 7], which derives $|x| = 1$ from the dual fixed point of B and $R R^\dagger = \mathbb{1}$ from the single-block property; both follow from condition C1 through the Perron–Frobenius theory of irreducible maps. The coefficient relation follows from (23.27) because conjugation by U does not change traces. $\square$

Definition 23.3.6 (Translation-invariant canonical representation). [Lean](#) [Lean](#) [✓ Stmt](#) A tensor A of bond dimension D is a *translation-invariant canonical representation* when its bond indices split into finitely many blocks of positive size, and there are weights $1 \geq \lambda_j > 0$ and tensors A_i^j on the blocks with

$$A_i = \bigoplus_j \lambda_j A_i^j \quad \text{for every } i, \quad (23.28)$$

such that every block satisfies the three conditions of Theorem 7.5.6: $\sum_i A_i^j (A_i^j)^\dagger = \mathbb{1}$, there is a diagonal positive definite Λ^j with $\sum_i (A_i^j)^\dagger \Lambda^j A_i^j = \Lambda^j$, and the identity is the only fixed point of $X \mapsto \sum_i A_i^j X (A_i^j)^\dagger$ up to scalars. This is the block form of [193, Theorem 4], which lists the blocks consecutively.

Lemma 23.3.7 (Canonical representations from the existence theorem). [Lean](#) [✓ Stmt](#) The weighted direct sum $\bigoplus_k \mu_k A_k$ of positive-dimensional blocks produced by Theorem 7.5.6, with positive weights $\mu_k \leq 1$, is a translation-invariant canonical representation with the blocks in consecutive order.

Proof. [✓ Proof](#) The weights are positive reals of modulus at most one, and the block conditions are those of Definition 23.3.6. $\square$

Lemma 23.3.8 (A block-injective canonical representation has one block). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a translation-invariant canonical representation with the block form (23.28). Every product $A_{i_1} \cdots A_{i_L}$ is block diagonal, with $\lambda_j^L A_{i_1}^j \cdots A_{i_L}^j$ on block j , and

$$\sum_i A_i A_i^\dagger = \bigoplus_j \lambda_j^2 \mathbb{1}. \quad (23.29)$$

If A has two distinct blocks, then A is not L -block injective for any L . Consequently, if $D \geq 1$ and A is L -block injective for some L , then A has a single block, and $\sum_i A_i A_i^\dagger = \lambda^2 \mathbb{1}$ with $0 < \lambda \leq 1$. This is the first assertion of the proposition on condition C1 in [193].

Proof. ✓ Proof The block-diagonal form of the products follows from (23.28) by induction on the length, and (23.29) from the first block condition. If α lies in one block and β in another, the entry (α, β) of every product vanishes while the matrix unit $|\alpha\rangle\langle\beta|$ has entry one there, so the products of any length do not span $M_D(\mathbb{C})$. When $D \geq 1$ there is at least one block, and block injectivity leaves exactly one; then (23.29) is the scalar λ^2 of that block. $\square$

Lemma 23.3.9 (Uniqueness for a single block). Lean Lean Lean Lean ✓ Stmt Let B and C be tensors of bond dimension D with $\sum_i B_i B_i^\dagger = \lambda_B^2 \mathbb{1}$ and $\sum_i C_i C_i^\dagger = \lambda_C^2 \mathbb{1}$ for some $\lambda_B, \lambda_C > 0$, let B be L_0 -block injective, let $N > 2L_0 + D^4$, and let B and C represent the same nonzero state on the ring of N sites. Then $\lambda_B = \lambda_C$, and there are a unitary U and a scalar $\xi \in \mathbb{C}$ with

$$|\xi| = 1, \quad \xi^N = 1, \quad B_i = \xi U C_i U^\dagger \quad \text{for every } i. \quad (23.30)$$

Proof. ✓ Proof For $\lambda_B = \lambda_C = 1$, Theorem 23.3.4 gives $R \neq 0$ and $x \neq 0$ with $R C_i = \frac{1}{x} B_i R$, and Lemma 23.3.5 turns R into a unitary U with $|x| = 1$ and $B_i = x U C_i U^\dagger$. The coefficients then satisfy $V^{(N)}(B) = x^N V^{(N)}(C) = x^N V^{(N)}(B)$, and since the state is nonzero, $x^N = 1$; take $\xi = x$. For general weights, $\lambda_B^{-1} B$ and $\lambda_C^{-1} C$ represent the same state and the first is unital, so Theorem 23.3.4 gives R and x for this pair. The same R satisfies $R(\lambda_C^{-1} C_i) = y^{-1}(\lambda_B^{-1} B_i)R$ with $y = x \lambda_C / \lambda_B$, and Lemma 23.3.5 applied to the two unital families gives $|y| = 1$ and $B_i = x U C_i U^\dagger$. As before $x^N = 1$, so $|x| = 1$, and $|y| = 1$ forces $\lambda_B = \lambda_C$. $\square$

Theorem 23.3.10 (Uniqueness of the canonical form, as printed). Not ready Let $|\psi\rangle = \sum \text{tr}(B_{i_1} \cdots B_{i_N}) |i_1 \cdots i_N\rangle$ be a translation-invariant canonical D -MPS such that condition C1 holds at some length L_0 , the open-boundary canonical representation of $|\psi\rangle$ is unique, and $N > 2L_0 + D^4$. Then, if $|\psi\rangle$ admits another translation-invariant canonical D -MPS representation with matrices C_i , there is a unitary U with $B_i = U C_i U^\dagger$ for every i . This is [193, Theorem 7] as printed. Its conclusion is false: for $d = D = 1$, $B_0 = 1$ and $C_0 = \eta$ with $\eta \neq 1$ an N th root of unity are two canonical representations of the same nonzero state on $N > 3$ sites, and unitary conjugation acts trivially in bond dimension one [70]. The corrected statement is Theorem 23.3.11.

Theorem 23.3.11 (Uniqueness of the canonical form, corrected). Lean Lean Lean ✓ Stmt Let $B = (B_i)_i$ and $C = (C_i)_i$ be translation-invariant canonical representations of bond dimension D in the sense of Definition 23.3.6. Assume that B is L_0 -block injective, that $N > 2L_0 + D^4$, and that B and C represent the same nonzero state on the ring of N sites. Then there are a unitary U and a scalar $\xi \in \mathbb{C}$ satisfying (23.30). The rephased representation $\widehat{C}_i := \xi C_i$ represents the same length- N state and satisfies $B_i = U \widehat{C}_i U^\dagger$ exactly; equivalently, C and B are gauge-phase equivalent with the unitary gauge U and the unit-modulus phase ξ .

Uniqueness of the open-boundary canonical representation is not needed; Theorem 23.3.4 obtains the intertwiners directly from injectivity, yielding the conclusion recorded in [125].

Proof. ✓ Proof The state is nonzero, so $D \geq 1$, and Lemma 23.3.8 leaves B with a single block: $\sum_i B_i B_i^\dagger = \lambda_B^2 \mathbb{1}$ with $0 < \lambda_B \leq 1$. The tensors $\lambda_B^{-1} B$ and $\lambda_C^{-1} C$ represent the same state and the first is unital and L_0 -block injective, so Theorem 23.3.4 gives $R \neq 0$ and $x \neq 0$ with $R(\lambda_B^{-1} C_i) = \frac{1}{x}(\lambda_B^{-1} B_i)R$, and R is invertible by Lemma 23.3.5. Hence $C_i = \frac{1}{x} R^{-1} B_i R$ is conjugate to a nonzero multiple of B_i , so C is L_0 -block injective, and Lemma 23.3.8 leaves C with a single block as well: $\sum_i C_i C_i^\dagger = \lambda_C^2 \mathbb{1}$ with $0 < \lambda_C \leq 1$. Lemma 23.3.9 now gives U and ξ . Rephasing C by ξ multiplies every length- N coefficient by $\xi^N = 1$, and $\xi \bar{\xi} = |\xi|^2 = 1$ makes (U, ξ) a unit-modulus gauge-phase equivalence. $\square$

23.4 One-sided inverse and physical realisation of virtual insertions

If A is injective, the linear combination map $\Phi_A(c) := \sum_i c_i A^i: \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ is surjective and admits a right inverse. The resulting decomposition map expresses any matrix as a linear combination of the spanning family $\{A^i\}$, and its multiplicative refinement re-expresses a virtual insertion on the bond as a physical operation on the local index.

Lemma 23.4.1 (Existence of a right inverse). Lean ✓ Stmt *If A is injective, then there exists a $\mathbb{C}$ -linear map $\Psi: M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ satisfying $\Phi_A \circ \Psi = \text{id}_{M_D(\mathbb{C})}$.*

Proof. ✓ Proof Surjectivity of Φ_A yields a linear section by choice. □

Definition 23.4.2 (Decomposition map). Lean ✓ Stmt Let A be an injective MPS tensor. The *decomposition map* $\Psi_A: M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ is a fixed choice of right inverse of Φ_A .

Lemma 23.4.3 (Decomposition property). Lean ✓ Stmt *If A is injective, then for every $X \in M_D(\mathbb{C})$ there exist coefficients $c \in \mathbb{C}^d$ such that $X = \sum_{i=0}^{d-1} c_i A^i$.*

Proof. ✓ Proof Take $c = \Psi_A(X)$. Then $\Phi_A(c) = \Phi_A(\Psi_A(X)) = X$. □

Remark 23.4.4 (Non-uniqueness of the decomposition). When $d > D^2$, the map Φ_A has a non-trivial kernel, so the section Ψ_A is not unique. The statements that follow depend only on $\Phi_A \circ \Psi_A = \text{id}_{M_D(\mathbb{C})}$, so the choice does not affect the conclusions.

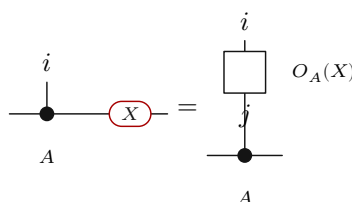
23.4.1 Physical realisation of virtual insertions

For an injective tensor A , the physical realisation map encodes the effect of inserting a virtual matrix X on the bond as a $d \times d$ operation on the physical indices. Its multiplicativity (Theorem 23.4.1.3) turns the bond algebra into an algebra acting on the physical leg, and is the source of the algebra homomorphism used in the virtual bond gauge.

Definition 23.4.1.1 (Physical realisation map). Lean ✓ Stmt Let A be an injective tensor. For $X \in M_D(\mathbb{C})$, define the $d \times d$ matrix $O_A(X) \in M_d(\mathbb{C})$ by

$$O_A(X)_{ij} := (\Psi_A(A^i X))_j,$$

so that row i of $O_A(X)$ contains the decomposition coefficients of $A^i X$ in the spanning set $\{A^j\}$. Diagrammatically, the inserted virtual matrix is re-expressed as a physical operation on the same local tensor:



The map $X \mapsto O_A(X)$ is $\mathbb{C}$ -linear.

Lemma 23.4.1.2 (Defining property of the physical realisation map). [Lean] [✓ Stmt] For every $X \in M_D(\mathbb{C})$ and every physical index i ,

$$A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j.$$

Proof. [✓ Proof] By definition, $O_A(X)_{ij} = (\Psi_A(A^i X))_j$. Applying Φ_A gives $\sum_j O_A(X)_{ij} A^j = \Phi_A(\Psi_A(A^i X)) = A^i X$, where the last step uses $\Phi_A \circ \Psi_A = \text{id}_{M_D(\mathbb{C})}$. $\square$

Theorem 23.4.1.3 (Physical realisation is multiplicative). [Lean] [✓ Stmt] Let A be an injective MPS tensor. For all $X, Y \in M_D(\mathbb{C})$, $O_A(XY) = O_A(X)O_A(Y)$.

Proof. [✓ Proof] Applying Lemma 23.4.1.2 to X yields $A^i X = \sum_{j=0}^{d-1} O_A(X)_{ij} A^j$. Multiplying by Y on the right and applying the linear map Ψ_A gives

$$\Psi_A(A^i XY) = \sum_{j=0}^{d-1} O_A(X)_{ij} \Psi_A(A^j Y).$$

Taking the k th component of both sides gives

$$O_A(XY)_{ik} = \sum_{j=0}^{d-1} O_A(X)_{ij} O_A(Y)_{jk} = (O_A(X)O_A(Y))_{ik}.$$

Since this holds for all i and k , the matrices are equal. $\square$

Remark 23.4.1.4 (General injective case). When $d > D^2$, the decomposition map Ψ_A need not be unique. Nevertheless, Theorem 23.4.1.3 is already the general injective statement: for the fixed choice of Ψ_A in Definition 23.4.2, the associated physical realisation map O_A is multiplicative. This supersedes the earlier basis-only formulation.

23.5 The virtual bond gauge and the proportionality lemma

The two structural lemmas of [186] carry the entire fixed-length argument. The virtual bond gauge (Lemma 23.5.2.1) turns equality of the combined MPV families into the virtual-bond conjugacy $X \mapsto Z^{-1}XZ$ by a fixed invertible matrix Z ; the proportionality lemma (Lemma 23.5.3.1) shows that two tensor pairs agreeing under all virtual insertions are proportional. Both rest on two elementary facts: injectivity is preserved by blocking, and the centre of $M_D(\mathbb{C})$ consists of scalars.

23.5.1 Two preliminary facts

Lemma 23.5.1.1 (Blocking preserves injectivity). [Lean] [✓ Stmt] Let $A_0, \dots, A_{m-1}$ be injective MPS tensors with compatible bond dimensions. Then the blocked tensor

$$(A_0 \otimes \dots \otimes A_{m-1})^{(i_0, \dots, i_{m-1})} := A_0^{i_0} A_1^{i_1} \dots A_{m-1}^{i_{m-1}}$$

is again injective.

Proof. [✓ Proof] The span of the blocked matrices contains all products $A_0^{i_0} \dots A_{m-1}^{i_{m-1}}$. Since each $\{A_k^i\}_i$ spans $M_D(\mathbb{C})$, these products span all of $M_D(\mathbb{C})$. $\square$

23.5.2 Same state forces a virtual bond gauge

The 3-site virtual-bond statement is used later in the proof of the PEPS fundamental theorem. Its hypothesis is the all-length MPV equality of the combined tensors, and its conclusion is the conjugacy of the middle-bond insertion data.

Lemma 23.5.2.1 (Virtual bond gauge – 3-site case). [Lean](#) [✓ Stmt](#) Let $A = (A_0, A_1, A_2)$ and $B = (B_0, B_1, B_2)$ be two 3-site injective periodic chains with common bond dimension $D \geq 1$ and $\mathcal{V}(A) = \mathcal{V}(B)$ (same MPV family). Then at the middle bond there exists an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ such that, for all $X \in M_D(\mathbb{C})$ and all physical indices i_0, i_1, i_2 , the virtual-insertion trace coefficients agree after conjugation:

$$\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Z^{-1} X Z B_1^{i_1} B_2^{i_2}).$$

In [186, Lemma 1], the statement is given for arbitrary chain length $n \geq 3$ and any bond position, with uniqueness of Z up to scalar; the three-site formulation here isolates the local algebraic content.

Proof. [✓ Proof](#) The $\mathcal{V}(A) = \mathcal{V}(B)$ hypothesis gives, for every $X \in M_D(\mathbb{C})$, equality of the virtual-insertion trace coefficients

$$\text{tr}(A_0^{i_0} X A_1^{i_1} A_2^{i_2}) = \text{tr}(B_0^{i_0} Y B_1^{i_1} B_2^{i_2})$$

for some Y depending on X . Since all three site tensors on each side are injective, the physical realisation maps O_A, O_B are algebra homomorphisms by Theorem 23.4.1.3. Setting $T(X) = Y$ defines a $\mathbb{C}$ -linear multiplicative map, hence a nonzero $\mathbb{C}$ -algebra endomorphism of $M_D(\mathbb{C})$, which is bijective by Theorem 3.2.1. Skolem–Noether (Theorem [182, “Skolem–Noether”]) gives $Z \in \text{GL}_D(\mathbb{C})$ with $T(X) = Z^{-1} X Z$. $\square$

23.5.3 Aligned gauges force proportionality

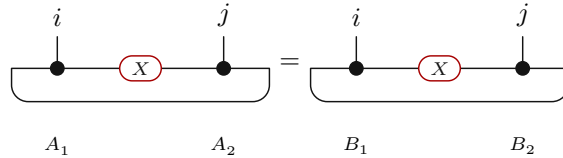
After applying Lemma 23.5.2.1 to each bond and absorbing the gauge matrices, one obtains two chains whose virtual-insertion operations agree on every bond. Lemma 2 of [186] shows that this agreement forces the local tensors to be proportional.

Lemma 23.5.3.1 (Aligned gauges force proportionality – Lemma 2 of [186]). [Lean](#) [✓ Stmt](#) Let A_1, A_2 and B_1, B_2 be injective MPS tensors with the same positive bond dimension D . Suppose that, for all $X \in M_D(\mathbb{C})$ on the internal bond, all $Y \in M_D(\mathbb{C})$ on the external bond, and all physical indices i, j ,

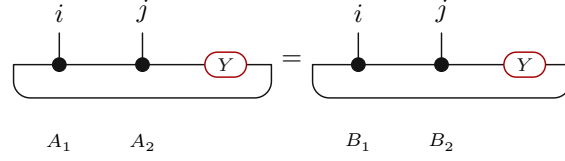
$$\text{tr}(A_1^i X A_2^j) = \text{tr}(B_1^i X B_2^j), \quad (23.31)$$

$$\text{tr}(A_2^j Y A_1^i) = \text{tr}(B_2^j Y B_1^i). \quad (23.32)$$

Diagrammatically, inserting X on the bond between the two sites, as in (23.31), gives



while the insertion Y on the outer trace bond, as in (23.32), is



so the two hypotheses distinguish the internal and external virtual loops of the two-site periodic chain.

Then there exists a nonzero scalar $\lambda \in \mathbb{C}^\times$ such that, for all physical indices i, j ,

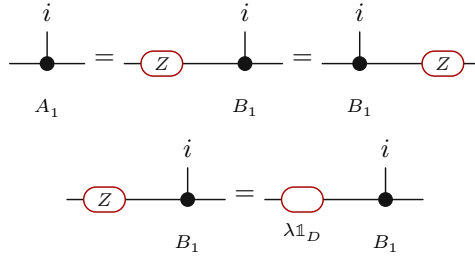
$$\begin{aligned} A_1^i &= \lambda B_1^i, \\ A_2^j &= \lambda^{-1} B_2^j. \end{aligned}$$

Proof. ✓ Proof By cyclicity and non-degeneracy of the trace pairing, the conditions in (23.31) and (23.32) give, for all physical indices i, j ,

$$A_2^j A_1^i = B_2^j B_1^i, \quad (23.33)$$

$$A_1^i A_2^j = B_1^i B_2^j. \quad (23.34)$$

Write $\mathbb{1}_D = \sum_j c_j A_2^j$ using the decomposition map Ψ_{A_2} , and define $Z := \sum_j c_j B_2^j \in M_D(\mathbb{C})$. This is the point where the two-site comparison collapses to a single boundary matrix:



The first product identity (23.33) gives $A_1^i = Z B_1^i$, and the second (23.34) gives $A_1^i = B_1^i Z$. Hence Z commutes with every B_1^i ; since $\{B_1^i\}$ spans $M_D(\mathbb{C})$, Z is central. By Lemma 2.1.5, $Z = \lambda \mathbb{1}_D$ for some $\lambda \in \mathbb{C}^\times$ (nonzero since the tensors are nonzero). Thus $A_1^i = \lambda B_1^i$, and substituting into (23.33) forces $A_2^j = \lambda^{-1} B_2^j$. $\square$

23.6 The Fundamental Theorem for injective chains

The chain theorem combines the site and physical indices into a single physical index and applies the single-block Fundamental Theorem to the resulting tensor. Its hypothesis is all-length MPV equality of the combined tensors. This hypothesis is strictly stronger than equality of one fixed-length closed-chain state: the combined-tensor MPV also tests words that repeat one site and omit the others. The corresponding fixed-length theorem is Corollary 24.9.4.10.

Theorem 23.6.1 (Fundamental Theorem for injective chains). Lean ✓ Stmt Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two chains with $n > 0$ sites and common positive bond dimension D , and assume that $\mathbf{A}$ is injective. Suppose that, after combining the site index and physical index into a single MPS tensor, the resulting two tensors generate the same MPV family. Then $\mathbf{A}$ and $\mathbf{B}$ are cyclically gauge equivalent.

Proof. ✓ Proof Choose one site of $\mathbf{A}$; its injectivity implies injectivity of the combined tensor by Lemma 23.5.1.1. Apply the single-block Fundamental Theorem (Theorem 3.2.4) to the two combined tensors. This yields one invertible matrix X such that every combined matrix of $\mathbf{B}$ equals the corresponding combined matrix of $\mathbf{A}$ conjugated by X . Reading this identity back at each site gives $B_k^i = X A_k^i X^{-1}$ for all k and i , so the constant choice $Z_k = X$ is a cyclic gauge. $\square$

Lemma 23.6.2 (Uniform site rescaling rescales the combined tensor). Lean ✓ Stmt Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ be an n -site chain and let $\zeta \in \mathbb{C}$. If $\zeta \mathbf{A}$ denotes the chain obtained by replacing each site tensor A_k with ζA_k , and if $\widetilde{\mathbf{A}}$ denotes the combined tensor of $\mathbf{A}$, then $\widetilde{\zeta \mathbf{A}} = \zeta \widetilde{\mathbf{A}}$.

Proof. ✓ Proof Combining the site and physical indices does not change which local matrix is read at the component (k, i) , so both sides evaluate to ζA_k^i there. $\square$

Theorem 23.6.3 (Injective-chain Fundamental Theorem up to a global phase). Lean ✓ Stmt Let $\mathbf{A} = (A_0, \dots, A_{n-1})$ and $\mathbf{B} = (B_0, \dots, B_{n-1})$ be two chains with $n > 0$ sites and common positive bond dimension D , and assume that $\mathbf{A}$ is injective. Suppose that the combined tensors $\widetilde{\mathbf{A}}$ and $\widetilde{\mathbf{B}}$ are gauge-phase equivalent. Then there exist invertible matrices $Z_0, \dots, Z_{n-1} \in \text{GL}_D(\mathbb{C})$ and a nonzero scalar $\zeta \in \mathbb{C}^\times$ such that

$$B_k^i = \zeta Z_k A_k^i Z_{k+1}^{-1} \quad (23.35)$$

for all sites k and physical indices i , where $k+1$ is interpreted cyclically modulo n .

Proof. ✓ Proof If the combined tensor of $\mathbf{B}$ differs from that of $\mathbf{A}$ by a gauge and a nonzero scalar ζ , rescale each site of $\mathbf{B}$ by ζ^{-1} . Lemma 23.6.2 removes this common scalar from the combined tensor, so Theorem 23.6.1 applies to the rescaled chain. Multiplying the resulting sitewise relations by ζ gives (23.35). $\square$

Remark 23.6.4 (Relationship to the translation-invariant all-sizes theorem). Theorem 23.6.1 treats a fixed chain length and assumes equality of the combined-tensor MPV families. The single-block Fundamental Theorem (Theorem 3.2.4) instead studies one tensor at all system sizes and concludes exact gauge equivalence $B^i = X A^i X^{-1}$. The chain theorem is recovered by converting the chain into one combined tensor; the price is that the hypothesis is all-length MPV equality for that combined tensor rather than equality of the chain state at one fixed length.

23.6.1 Finite-length MPV agreement

The single-block theorem can be strengthened by weakening its hypothesis from all-length to finite-length MPV agreement: for an injective tensor, agreement from any threshold N_0 onward already implies full agreement.

Definition 23.6.1.1 (Finite-length MPV agreement). Lean ✓ Stmt Two tensors A and B of the same bond dimension satisfy *finite-length MPV agreement from N_0 onward* if they generate the same MPV family for all system sizes $N \geq N_0$, meaning that $V^{(N)}(A)_\sigma = V^{(N)}(B)_\sigma$ for every $N \geq N_0$ and $\sigma \in [d]^N$.

Theorem 23.6.1.2 (Finite-length agreement implies full agreement). Lean ✓ Stmt Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B generate the same MPV family for every system size $N \geq N_0$, then they generate the same MPV family for every system size.

Proof. ✓ Proof Write $1 = \sum_i c_i A^i$ (by injectivity) and set $S = \sum_i c_i B^i$.

1. Show that the length- N_0 word span of B is all of $M_D(\mathbb{C})$ via the linear relation between the length- N_0 word-evaluation maps of A and B induced by trace agreement at length $N_0 + 1$, together with a finrank transfer argument.
2. Derive $S = 1$ by trace nondegeneracy: for every $M \in M_D(\mathbb{C})$, $\text{tr}(M(S - 1)) = 0$.
3. Downward induction on $k = N_0 - |w|$: decompose $\text{tr}(w_A)$ and $\text{tr}(w_B)$ using $1 = \sum c_i A^i$ and $S = 1$ respectively, then apply the inductive hypothesis at length $|w| + 1$.

□

Theorem 23.6.1.3 (Strengthened single-block FT (finite-length)). [Lean](#) [✓ Stmt](#) *Let A be an injective tensor of bond dimension $D \geq 1$, and let B be a tensor of the same bond dimension. If A and B agree on all MPV coefficients for system sizes $N \geq N_0$ (for any threshold N_0), then A and B are gauge equivalent.*

Proof. [✓ Proof](#) Apply Theorem 23.6.1.2 to obtain equality of the MPV families for all system sizes, then invoke the single-block Fundamental Theorem (Theorem 3.2.4). □

23.7 Corollaries: translation-invariant, non-uniform, and blocked chains

The chain theorem specializes to three settings: constant translation-invariant chains, non-uniform chains whose state happens to be translation-invariant, and constant chains of blocked tensors.

23.7.1 Translation-invariant chains

Theorem 23.7.1.1 (Single gauge for translation-invariant chains). [Lean](#) [✓ Stmt](#) *Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that $B^i = X A^i X^{-1}$ for every physical index i .*

Proof. [✓ Proof](#) Bundle the constant chains into one tensor. Injectivity of A implies injectivity of the combined tensor by Lemma 23.5.1.1, and the combined-tensor MPV equality is exactly the hypothesis needed for the single-block Fundamental Theorem. The resulting gauge matrix is independent of the site because every site carries the same local tensor. □

Corollary 23.7.1.2 (Translation-invariant collapse to one gauge). [Lean](#) [✓ Stmt](#) *Let A and B be MPS tensors of bond dimension D , and fix a chain length $n \geq 1$. Assume that A is injective and that the combined tensors of the two constant chains $(A, \dots, A)$ and $(B, \dots, B)$ generate the same MPV family. Then there exist an invertible matrix $Z \in \text{GL}_D(\mathbb{C})$ and a scalar $\lambda \in \mathbb{C}^\times$ with $\lambda^n = 1$ such that*

$$B^i = \lambda Z^{-1} A^i Z \tag{23.36}$$

for every physical index i .

Proof. [✓ Proof](#) Apply Theorem 23.7.1.1 to obtain $B^i = X A^i X^{-1}$. Set $Z = X^{-1}$ and $\lambda = 1$. Then $\lambda^n = 1$, and (23.36) is exactly the same gauge relation rewritten in the form used by the corollary. □

Symmetry-theoretic consequences of the Fundamental Theorem, including virtual representations and string order, are developed in Chapter 10.

23.7.2 Blocked chains

Definition 23.7.2.1 (Constant blocked chain). [Lean](#) [✓ Stmt](#) For an MPS tensor A , a blocking length L , and a chain length n , the *blocked chain* is the constant n -site chain whose local tensor is the blocked tensor $A^{[L]}$.

Lemma 23.7.2.2 (Blocked chains are injective). [Lean](#) [✓ Stmt](#) If A is L -block injective, then the constant blocked chain built from $A^{[L]}$ is injective at every site.

Proof. [✓ Proof](#) By Lemma 6.5.3, the local blocked tensor $A^{[L]}$ is injective, and therefore every site of the constant blocked chain is injective. $\square$

Theorem 23.7.2.3 (Fundamental Theorem for blocked chains). [Lean](#) [✓ Stmt](#) Let A and B be MPS tensors of positive bond dimension D , let L be a blocking length, and let $n > 0$ be a chain length. Assume that A is L -block injective and that the combined tensors of the two constant blocked chains built from $A^{[L]}$ and $B^{[L]}$ generate the same MPV family. Then the two blocked chains are gauge equivalent.

Proof. [✓ Proof](#) By Lemma 23.7.2.2, the blocked chain built from $A^{[L]}$ is injective. Apply Theorem 23.6.1 to the two constant blocked chains. $\square$

Remark 23.7.2.4 (Relation to normal-MPS corollary). Theorem 23.7.2.3 establishes gauge equivalence only for the blocked chains; recovering sitewise gauges for the original unblocked tensors is a separate unblocking step yielding the normal-MPS corollary of [186], outside this blocked-chain statement.

23.8 Product algebra construction and per-block gauge

The chain Fundamental Theorem (Theorem 23.6.1) handles non-translation-invariant periodic chains. A parallel development addresses block-diagonal tensors: given a family of injective block tensors $\{A_k^i\}_{k=0}^{r-1}$ with block dimensions D_k and a second family $\{B_k^i\}$ with per-block $\mathcal{V}(A_k) = \mathcal{V}(B_k)$, one seeks per-block gauge equivalence $B_k^i = X_k A_k^i X_k^{-1}$ for each k independently, and a global block-permutation plus inner-automorphism decomposition. Throughout this section, assume $D_k \geq 1$ for every block k .

23.8.1 Block permutation algebra

The k -th block ideal consists of the tuples that vanish outside the k -th coordinate, as in Definition [182, “Block ideal”].

Theorem 23.8.1.1 (Each block ideal is an atom). [Lean](#) [✓ Stmt](#)

Each block ideal I_k is an atom in the lattice of two-sided ideals of $\prod_{j=0}^{r-1} R_j$.

Proof. [✓ Proof](#) Under the order isomorphism between two-sided ideals of $\prod_{j=0}^{r-1} R_j$ and products of two-sided ideal lattices, I_k corresponds to the tuple with $\top$ in the k -th coordinate and $\perp$ elsewhere. This tuple is an atom, hence so is I_k . $\square$

Theorem 23.8.1.2 (Ring isomorphisms match block ideals). [Lean](#) [Lean](#) [✓ Stmt](#)

Let $\{R_i\}_{i \in I}$ and $\{S_j\}_{j \in J}$ be two finite families of simple rings. Every ring isomorphism $T : \prod_{i \in I} R_i \rightarrow \prod_{j \in J} S_j$ determines an equivalence $\sigma : I \simeq J$ such that T maps the i -th block ideal onto the $\sigma(i)$ -th block ideal. In particular, an automorphism of one product permutes its block ideals.

Proof. [✓ Proof](#) By Theorem 23.8.1.1, the atoms in each two-sided ideal lattice are exactly its block ideals. The induced order isomorphism therefore assigns a unique target block ideal to every source block ideal. Applying its inverse shows that this assignment is bijective and gives σ . $\square$

Theorem 23.8.1.3 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$). [Lean](#) [✓ Stmt](#) Assume $D_k \geq 1$ for all k . Every $\mathbb{C}$ -algebra automorphism T of $\prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$ decomposes as per-block inner automorphisms followed by a block permutation π : there exist $\pi \in S_r$ with $D_{\pi(k)} = D_k$ and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that, for every tuple $M = (M_0, \dots, M_{r-1})$ and every block index k , $T(M)_{\pi(k)} = X_k M_k X_k^{-1}$, where $T(M)_j$ denotes the j -th block component of $T(M)$.

Proof. [✓ Proof](#) Theorem 23.8.1.2 gives the permutation π . Theorem [182, “Summand matching and dimension preservation”] identifies the matrix sizes $D_{\pi(k)} = D_k$. For each k , Theorem [182, “Paired component maps are bijections”] gives a bijective multiplicative map from $M_{D_k}(\mathbb{C})$ to $M_{D_{\pi(k)}}(\mathbb{C})$ extracted from the elements supported on one factor of T ; after reindexing by the dimension equality, this is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$. Skolem–Noether (Theorem [182, “Skolem–Noether”]) makes each such automorphism inner.

This is a standard result in the theory of semisimple algebras; see [16, Section IV.A] for the MPS context. $\square$

Remark 23.8.1.4 (Ring level versus algebra level). Theorem 23.8.1.2 is a ring-level statement: it only uses the ideal structure of $\prod_k R_k$. Theorem [182, “Paired component maps are bijections”] stays at this ring level: once the block ideals are permuted, one can extract the corresponding single-block maps without using scalar multiplication. Theorems [182, “Summand matching and dimension preservation”] and 23.8.1.3 require an upgrade to $\mathbb{C}$ -algebra automorphisms. The reason is threefold: dimension preservation compares the blocks as $\mathbb{C}$ -vector spaces; Skolem–Noether applies to the resulting central simple $\mathbb{C}$ -algebras; and the final per-block maps must commute with scalar multiplication. Thus the argument separates the ring-level permutation step from the later algebra-level decomposition.

23.8.2 Per-block linear extension and the product automorphism

Per-block equality of MPV families produces a unique linear map on each matrix block; assembling these maps componentwise yields an algebra automorphism of the product, to which the block-permutation decomposition applies.

Definition 23.8.2.1 (Per-block linear extension). [Lean](#) [Lean](#) [✓ Stmt](#) Let A_k, B_k be MPS tensors of bond dimension D_k with A_k injective and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for each block $k \in \{0, \dots, r-1\}$. The *per-block linear extension* $T_k : M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ is the unique $\mathbb{C}$ -linear map satisfying $T_k(A_k^i) = B_k^i$ for every physical index i .

Lemma 23.8.2.2 (Per-block multiplicativity). [Lean](#) [✓ Stmt](#) For each block k , the per-block linear extension T_k is multiplicative: $T_k(MN) = T_k(M)T_k(N)$ for all $M, N \in M_{D_k}(\mathbb{C})$.

Proof. [✓ Proof](#) Since $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$, it suffices to check on spanning elements, where the result follows from the MPV identity. $\square$

Lemma 23.8.2.3 (Per-block bijectivity). [Lean](#) [✓ Stmt](#) Each T_k is bijective.

Proof. [✓ Proof](#) A nonzero multiplicative linear endomorphism of $M_{D_k}(\mathbb{C})$ is automatically bijective: injectivity follows from the simplicity of $M_{D_k}(\mathbb{C})$ (the kernel is a proper ideal, hence zero), and surjectivity from finite dimension. Non-vanishing of T_k is established via the trace pairing: if $T_k = 0$, then all $B_k^i = 0$, contradicting the non-vanishing of the trace pairing for injective tensors. $\square$

Theorem 23.8.2.4 (Unitary implementation of a positive per-block extension). [Lean](#) [✓ Stmt](#) If the per-block linear extension T_k is positive, then there is a unitary matrix U_k such that $T_k(M) = U_k M U_k^\dagger$ for every $M \in M_{D_k}(\mathbb{C})$.

Proof. [✓ Proof](#) By Lemmas 23.8.2.2 and 23.8.2.3, T_k is a bijective multiplicative linear map. Positivity makes this algebra automorphism preserve adjoints. Unitary Skolem–Noether therefore implements it by conjugation with a unitary matrix. $\square$

Definition 23.8.2.5 (Product algebra automorphism). [Lean](#) [✓ Stmt](#) The product algebra automorphism is the $\mathbb{C}$ -algebra equivalence

$$T : \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_{k=0}^{r-1} M_{D_k}(\mathbb{C})$$

defined by $T(M)_k = T_k(M_k)$, where each T_k is the per-block linear extension. It is an algebra automorphism since each T_k is a multiplicative, unital, bijective linear map. Diagrammatically, this is the componentwise map on the product algebra obtained by applying the corresponding extension to each summand.

Theorem 23.8.2.6 (Block-permutation decomposition). [Lean](#) [✓ Stmt](#) The product algebra automorphism T decomposes as a block permutation composed with per-block inner automorphisms: there exist a permutation $\sigma \in S_r$ (with $D_{\sigma(k)} = D_k$ for all k) and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that

$$(T(M))_{\sigma(k)} = X_k M X_k^{-1}$$

for all $M \in M_{D_k}(\mathbb{C})$ (viewed as supported in block k) and all k . Equivalently, the product-algebra automorphism first permutes the simple matrix blocks and then acts inside each block by an inner conjugation.

Proof. [✓ Proof](#) Apply Theorem 23.8.1.3 to the product-algebra automorphism of Definition 23.8.2.5. The theorem writes the nonzero input block as $\pi(k)$ in output component k ; for the statement above we set $\sigma := \pi^{-1}$, so an element supported in block k lands in block $\sigma(k)$. $\square$

23.8.3 Nondegeneracy of the product trace pairing

Lemma 23.8.3.1 (Nondegeneracy of the product trace pairing). [Lean](#) [✓ Stmt](#) Let $M = (M_0, \dots, M_{r-1}) \in \prod_k M_{D_k}(\mathbb{C})$. If $\sum_k \text{tr}(M_k N_k) = 0$ for all $N = (N_0, \dots, N_{r-1})$, then $M = 0$.

Proof. [✓ Proof](#) Choose N supported on a single block k (all other components zero). The hypothesis reduces to $\text{tr}(M_k N_k) = 0$ for all N_k , so $M_k = 0$ by nondegeneracy of the matrix trace pairing on $M_{D_k}(\mathbb{C})$. Since k was arbitrary, $M = 0$. $\square$

Lemma 23.8.3.2 (Injectivity of the product Gram map). [Lean](#) [✓ Stmt](#) Let $\{A_k\}$ be a family of injective block tensors. The product Gram map $M \mapsto (k, i \mapsto \text{tr}(M_k \cdot A_k^i))$ is injective.

Proof. ✓ Proof Reduces to per-block injectivity of the Gram map $M_k \mapsto (i \mapsto \text{tr}(M_k \cdot A_k^i))$, which holds by injectivity of A_k (the matrices $\{A_k^i\}$ span $M_{D_k}(\mathbb{C})$, so the trace pairing against them separates points). $\square$

23.9 Gauge equivalence for direct sums

The first lemmas in this section are elementary facts about block-diagonal tensors: they combine already-known block gauges, and they record the easy consequences of applying the injective single-block theorem to each block separately. They do not derive the block matching appearing in [11, Theorem 2.10]. In the source proof, that matching is obtained from BNT overlap, independence, and coefficient comparison; the corresponding global block-diagonal gauge is given by Theorem 9.3.4.

Lemma 23.9.1 (Gauge equivalence of direct sums with phase-weight absorption). Lean ✓ Stmt *Let A_k, B_k be block families, and let $X_k \in \text{GL}_{D_k}(\mathbb{C})$ and scalars $\zeta_k \in \mathbb{C}$ satisfy $B_k^i = \zeta_k \cdot X_k A_k^i X_k^{-1}$ for every block k and physical index i . If the weights satisfy $\mu_A(k) = \mu_B(k)\zeta_k$, then the weighted block-diagonal tensors $\bigoplus_k \mu_A(k)A_k$ and $\bigoplus_k \mu_B(k)B_k$ are gauge equivalent.*

Proof. ✓ Proof Per-block, the weight identity $\mu_A(k) = \mu_B(k)\zeta_k$ rewrites the scaled relation as $\mu_B(k)B_k^i = X_k(\mu_A(k)A_k^i)X_k^{-1}$. Each weighted block then satisfies an ordinary conjugation, and the block-diagonal conjugation formula assembles the per-block X_k into the flattened global gauge $\bigoplus_k X_k$ relating $\bigoplus_k \mu_A(k)A_k$ to $\bigoplus_k \mu_B(k)B_k$. $\square$

23.9.1 Same-index direct-sum corollaries from the single-block theorem

Remark 23.9.1.1. See Definition 2.6.7 for the block-diagonal tensor construction. This subsection records the direct-sum corollaries where the pairing between blocks is already fixed; the derivation of the block permutation from the global same-MPV condition appears in [11, Theorem 2.10].

Lemma 23.9.1.2 (Per-block application of the single-block theorem). Lean ✓ Stmt *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. ✓ Proof Apply the single-block Fundamental Theorem (Theorem 3.2.4) to each block independently. $\square$

Lemma 23.9.1.3 (Per-block same MPV implies global same MPV). Lean ✓ Stmt *If A_k and B_k generate the same MPV family for each block k , then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. ✓ Proof By the MPV decomposition (Theorem 2.6.9), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. $\square$

Remark 23.9.1.4 (Direct-sum congruence). This is a direct-sum congruence statement: per-block MPV equality lifts to global MPV equality via the block-diagonal decomposition formula.

Lemma 23.9.1.5 (Gauge equivalence for same-index direct sums). Lean ✓ Stmt *If each block A_k is injective and each pair (A_k, B_k) generates the same MPV family, then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.*

Proof. [✓ Proof](#) Apply the per-block single-block Fundamental Theorem (Lemma 23.9.1.2) to obtain blockwise gauge equivalences from injectivity and equality of MPV families, then combine them into a global gauge for the block-diagonal tensors. $\square$

Remark 23.9.1.6 (Block-diagonal gauge from matched blocks). This lemma constructs a block-diagonal gauge from pre-matched blockwise gauge equivalences, whereas the full Fundamental Theorem for direct sums derives the block correspondence and permutation from the global state.

Lemma 23.9.1.7 (Gauge equivalence of direct sums from per-block MPV equality). [Lean](#) [✓ Stmt](#) Let $\{A_k\}_{k=0}^{r-1}$ and $\{B_k\}_{k=0}^{r-1}$ be two families of MPS tensors with block dimensions D_k , each A_k injective, and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k . Then:

1. Per-block gauge equivalence: $B_k^i = X_k A_k^i X_k^{-1}$ for some $X_k \in \text{GL}_{D_k}(\mathbb{C})$, for every k and i .
2. Global gauge equivalence: the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are gauge equivalent.

Explicitly, for $X = \bigoplus_k X_k$,

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1}. \quad (23.37)$$

Proof. [✓ Proof](#) Per-block gauge equivalence follows from applying the single-block Fundamental Theorem to each block, using injectivity of A_k and $\mathcal{V}(A_k) = \mathcal{V}(B_k)$. With $X = \bigoplus_k X_k$, (23.37) follows by multiplying block-diagonal matrices block by block. $\square$

Lemma 23.9.1.8 (Explicit per-block gauges). [Lean](#) [✓ Stmt](#) Under the hypotheses of Lemma 23.9.1.7, there exist invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that $B_k^i = X_k A_k^i X_k^{-1}$ for every block k and every physical index i .

Proof. [✓ Proof](#) Extract the gauge matrix from the per-block part of Lemma 23.9.1.7 for each block. $\square$

Lemma 23.9.1.9 (Product-algebra decomposition of per-block MPV equality). [Lean](#) [✓ Stmt](#) Assume that every block dimension is nonzero. Under the hypotheses of Lemma 23.9.1.7, the product-algebra automorphism determined by the per-block linear extensions decomposes into a block permutation together with inner conjugation on each block: there exist a permutation σ of the blocks, compatible block-dimension equalities, and invertible matrices X_k such that, after reindexing the target block $\sigma(k)$ back to block k , the action on the k th matrix block is $M \mapsto X_k M X_k^{-1}$.

Proof. [✓ Proof](#) Apply Theorem 23.8.2.6 to the product-algebra automorphism built from the per-block linear extensions. $\square$

23.9.2 MPV invariance under reindexing

Lemma 23.9.2.1 (MPV invariance under block permutation). [Lean](#) [✓ Stmt](#) The SameMPV₂ family of a weighted block-diagonal tensor is invariant under permuting the block index set and transporting the weights correspondingly.

Proof. [✓ Proof](#) The sum of per-block MPV contributions is invariant under rearrangement of the summands. $\square$

Lemma 23.9.2.2 (MPV invariance under dimension cast). [Lean](#) [✓ Stmt](#) *If two block-dimension families agree pointwise, the corresponding weighted block-diagonal tensors generate the same SameMPV_2 family after transporting the blocks across the dimension equalities.*

Proof. [✓ Proof](#) Substituting the equality of dimensions makes the two tensor families identical. $\square$

Lemma 23.9.2.3 (MPV invariance under block reindexing). [Lean](#) [✓ Stmt](#) *Let $e : \{0, \dots, r' - 1\} \simeq \{0, \dots, r - 1\}$ be a bijection of block indices. For weights μ_k , block tensors A_k , word length N , and configuration σ , the coefficient of the block-diagonal MPV is unchanged by reindexing the blocks along e :*

$$V^{(N)}\left(\bigoplus_k \mu_k A_k\right)_\sigma = V^{(N)}\left(\bigoplus_k \mu_{e(k)} A_{e(k)}\right)_\sigma.$$

Proof. [✓ Proof](#) Expanding both sides by Theorem 2.6.9 gives $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$ and $\sum_k \mu_{e(k)}^N V^{(N)}(A_{e(k)})_\sigma$. These sums are equal by reindexing the finite sum along the bijection e . $\square$

23.9.3 Converse: gauge equivalence implies same MPV

The general gauge-invariance theorem, Theorem 2.4.19, already states that gauge-equivalent tensors generate the same MPV family. Applying it to the two weighted block-diagonal tensors gives the direct-sum specialization needed here; no additional formal declaration is required.

23.9.4 Canonical-form tensor as a weighted direct sum

Lemma 23.9.4.1 (Canonical-form tensor agrees with block sum). [Lean](#) [✓ Stmt](#) *For a canonical-form record C , the assembled tensor T_C is definitionally equal to the weighted block-diagonal tensor $\bigoplus_k \mu_k^C A_k^C$, where μ_k^C are the block weights and A_k^C are the block tensors stored in C .*

Proof. [✓ Proof](#) The equality holds by definition; T_C is defined as the weighted block-diagonal tensor formed from the weights and block tensors of C . $\square$

Theorem 23.9.4.2 (Fundamental theorem for canonical forms with the same block structure). [Lean](#) [✓ Stmt](#) *Let C be a canonical-form record with block weights μ_k^C and block tensors A_k^C . Let $\{B_k\}$ be a family of MPS tensors on the same block dimensions, and assume that each A_k^C is injective and generates the same MPV family as B_k . Then T_C and $\bigoplus_k \mu_k^C B_k$ are gauge equivalent.*

Proof. [✓ Proof](#) Rewrite T_C as a weighted block-diagonal tensor via Lemma 23.9.4.1, then apply the multi-block global gauge-equivalence lemma (Lemma 23.9.1.5). $\square$

Lemma 23.9.4.3 (Per-block equality of MPV families and per-block gauge equivalence). [Lean](#) [✓ Stmt](#) *Let $\{A_k\}$ be a family of injective block tensors. Then $\mathcal{V}(A_k) = \mathcal{V}(B_k)$ for all k if and only if A_k and B_k are gauge equivalent for all k .*

Proof. [✓ Proof](#) The forward direction applies the single-block Fundamental Theorem to each block. The reverse direction holds because gauge equivalence preserves all traces of word evaluations. $\square$

23.9.5 Equal-case block matching with differing bond dimensions

The same-index lemmas above assume a fixed block correspondence. The equal-case sector matching with initially different sector counts and bond dimensions is Theorem 9.2.5 of Chapter 9; its global-gauge consequence is Theorem 9.3.4.

23.10 Translation-invariant reduction and global symmetry

The single-gauge corollary for translation-invariant chains specializes to a reduction statement: equal states force one tensor to be a gauge of the other and inherit injectivity. Composing two gauges of the same tensor then yields the scalar law underlying the global-symmetry consequences.

Theorem 23.10.1 (Translation-invariant reduction corollary). [Lean](#) [✓ Stmt](#) *Let A and B be MPS tensors of bond dimension D with A injective. Consider the two constant translation-invariant chains $(A, \dots, A)$ and $(B, \dots, B)$ of length $n \geq 1$. If the combined tensors of the two chains generate the same MPV family, then:*

1. *B is gauge equivalent to A : there exists $X \in \text{GL}_D(\mathbb{C})$ with $B^i = X A^i X^{-1}$ for all i .*
2. *B is injective.*

Proof. [✓ Proof](#) Theorem 23.7.1.1 gives a single gauge matrix X with $B^i = X A^i X^{-1}$ for all i . Since gauge equivalence preserves injectivity, B is injective as well. $\square$

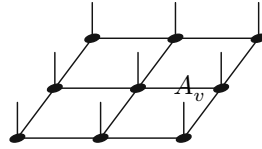
23.10.1 Global symmetry via gauge composition

The scalar uniqueness and gauge-composition results used by the virtual representation theorem are proved in Section G.2.

Chapter 24

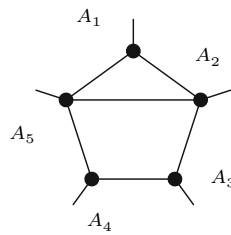
PEPS Fundamental Theorem

A *projected entangled pair state* (PEPS) is built from a finite simple graph $G = (V, E)$ by placing a local tensor A_v at every vertex v and contracting the tensors along the edges. Each vertex carries one physical index $\sigma(v) \in \{0, \dots, d-1\}$, and each edge e carries a virtual index of bond dimension D_e shared by its two endpoints. On a square lattice this is the picture



in which every black dot is a local tensor, the thin in-plane bonds are the virtual edges, and the vertical leg at each vertex is its physical index.

The state generated by A is the tensor whose coefficient at a physical configuration $\sigma: V \rightarrow \{0, \dots, d-1\}$ is obtained by summing the product of the local tensors over all virtual configurations, $c_A(\sigma) = \sum_{\eta} \prod_{v \in V} A_v(\eta_v, \sigma(v))$, where η ranges over the assignments of a bond index $\eta(e) \in \{0, \dots, D_e-1\}$ to every edge and η_v is the family of indices that η places on the edges incident to v . Closing every virtual bond while keeping the physical legs open gives the full state tensor; fixing the legs to a configuration σ evaluates the scalar $c_A(\sigma)$. On the five-vertex example graph this is



A PEPS is *injective* when at every vertex the physical vectors $A_v(\eta_v, \cdot) \in \mathbb{C}^d$, indexed by the virtual configurations η_v on the incident edges, are linearly independent; equivalently, the linear map they define from the coefficient space on virtual configurations into $\mathbb{C}^d$ has trivial kernel, $\ker A_v = \{0\}$:

$$\bigotimes_{e \ni v} \mathbb{C}^{D_e} \xrightarrow{A_v} \mathbb{C}^d$$

It is *normal* when blocking the tensors over suitable regions makes the blocked tensors injective, following [186, Section 4, lines 1316–1318].

Two PEPS that generate the same state need not have equal tensors. The virtual indices are internal, so on any edge e one may insert an invertible matrix X_e on one endpoint and its inverse on the other without changing any coefficient. Such a *gauge* on the bonds is the only freedom, and the two main results of the chapter make this precise.

- For injective PEPS on a connected graph with positive bond dimensions, two tensors generate the same state if and only if they are related by a gauge on the edges, with the gauges unique modulo the balanced edge scalars of Section 24.11. A gauge always gives the same state (Theorem 24.1.2.9); the converse is the positive-bond case of [186, Theorem 2] (Theorem 24.4.1.23).
- For a normal PEPS on a connected graph the same conclusion holds once the tensors become injective after blocking suitable regions (Theorem 24.8.6; the general normal-PEPS theorem in [186, Section 4], following its Theorem 3).

In the citations below, [186, Theorem 3] denotes only the translationally invariant square-lattice theorem. The connected-graph normal theorem and the normal-MPS corollaries are cited as results of [186, Section 4].

We begin in Section 24.1 with the core setting of PEPS on finite graphs, fixing conventions for contractions, the generated state, bond gauges, and vertex injectivity. Section 24.2 reduces a single edge to an effective three-site chain by blocking its complementary middle region. Section 24.3 turns virtual insertions into physical operations at an endpoint and extracts the per-edge gauge. Section 24.4 develops region transfer operators and proves the Fundamental Theorem for injective PEPS. Section 24.5 establishes that injective regions are closed under finite unions when bond dimensions are positive. Sections 24.6 and 24.7 supply the square-lattice and torus geometries, respectively. Section 24.8 presents the Fundamental Theorem for normal PEPS, Section 24.9 specializes it to matrix product states on closed chains, and Section 24.11 characterizes the residual gauge freedom modulo balanced edge scalars. Section 24.12 computes the states of three example tensors on the torus.

24.1 Finite Graph Tensor Networks

We establish the tensor-network setting on finite graphs: the local tensors, the state they generate, the gauge freedom on the bonds, and the vertex injectivity condition together with the local inverse it provides.

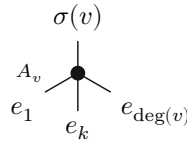
24.1.1 PEPS tensors and the states they generate

Definition 24.1.1.1 (Graph edge for PEPS). Lean ✓ Stmt For a finite simple graph G , an *edge* is an ordered pair (u, v) with $u < v$ and u adjacent to v . The ordering serves to avoid double counting.

Definition 24.1.1.2 (Incident edges at a vertex). Lean ✓ Stmt For a vertex v , an *incident edge* is an edge of G having v as one of its endpoints.

Definition 24.1.1.3 (PEPS tensor on a graph). Lean ✓ Stmt A PEPS tensor on G with physical dimension d consists of a bond dimension D_e for every edge e , together with a local tensor at each vertex v whose virtual indices are indexed by the edges incident to v and whose physical

index lies in $\{0, \dots, d-1\}$. The local tensor at v has one physical leg and one virtual leg for each incident edge:



Definition 24.1.1.4 (Virtual configuration). [Lean](#) [StmT](#) A *virtual configuration* assigns to every edge e one basis index in the corresponding bond space of dimension D_e .

Definition 24.1.1.5 (Splitting and reindexing virtual configurations). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 **Stmnt** A predicate on the edges splits a virtual configuration canonically into its restrictions to the selected and unselected edges. An equality between two bond-dimension families also induces a canonical reindexing between their virtual configurations. For a vertex v , selecting the incident edges splits a virtual configuration into its local labels at v and the labels on all nonincident edges.

Theorem 24.1.1.6 (Evaluation laws for splitting and reindexing). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The two restrictions recover the original edge labels on their respective domains. Reindexing acts edgewise by the corresponding identification of bond indices, commutes with piecewise merging, and splitting a piecewise merge recovers its selected and unselected inputs.*

Proof. ✓ Proof Each identity follows by evaluating at an edge and distinguishing whether that edge satisfies the selection predicate. □

Definition 24.1.1.7 (Piecewise and regionwise merge of virtual configurations). [Lean](#) [Lean](#)

Stmt Given two virtual configurations and a predicate on edges, their *piecewise merge* reads the first configuration on selected edges and the second configuration elsewhere. For a vertex region, the *regionwise merge* selects precisely the edges incident to that region.

Theorem 24.1.1.8 (Readback laws for merged configurations). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

Lean **✓ Stmt** *A piecewise merge reads the first input on a selected edge and the second input on an unselected edge; merging a configuration with itself leaves it unchanged. In particular, a regionwise merge reads the first input on region-incident edges and the second input on all remaining edges.*

Proof. ✓ Proof Evaluate the merge at an edge and use the defining alternative according to whether the edge is selected. □

Theorem 24.1.1.9 (Vertex products through a regionwise merge). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

The vertex product on a region reads the first input of its regionwise merge. A product on a second region reads the second input whenever the two configurations agree on edges incident to both regions. Consequently, agreement on the boundary of a region lets the complementary product read the second input through the same merge.

Proof. ✔ Proof Compare the local tensor entry at every vertex. Each incident edge is read from the prescribed input, while an edge incident to both regions is handled by the assumed agreement; a boundary edge is exactly the corresponding special case for a region and its complement. $\square$

Theorem 24.1.1.10 (Constant-cardinality fiber sum). Lean ✓ Stmt *Let a finite set map to the virtual configurations so that every fiber has cardinality n . For every function g into an additive commutative monoid, $\sum_x g(m(x)) = n \cdot \sum_\eta g(\eta)$, where the first sum ranges over the finite set and the second over all virtual configurations.*

Proof. [✓ Proof](#) Group the first sum by the value of m . Each grouped summand is repeated exactly n times by the common fiber-cardinality hypothesis. $\square$

Definition 24.1.1.11 (PEPS state coefficient). [Lean](#) [✓ Stmt](#) For a physical configuration σ on the vertices, the PEPS state coefficient is obtained by summing over all virtual configurations and, for each such configuration, multiplying the local tensor entries over all vertices.

Definition 24.1.1.12 (Equality of PEPS states). [Lean](#) [✓ Stmt](#) Two PEPS tensors represent the same state if all their state coefficients agree for every physical configuration.

24.1.2 Gauge transformations on the bonds

An invertible matrix on an edge acts on its two endpoint tensors by opposite oriented insertions. Absorbing such a matrix on every edge gives a gauge transformation that leaves the generated state unchanged.

Definition 24.1.2.1 (Edge gauge at a vertex). [Lean](#) [✓ Stmt](#) For an edge $e = (u, w)$ with $u < w$ and gauge $X_e \in \text{GL}(D_e)$, the gauge matrix applied at vertex v is X_e if $v = u$ and $(X_e^{-1})^\top$ if $v = w$.

$$X_{v,e} = \begin{cases} X_e, & v = u, \\ (X_e^{-1})^\top, & v = w. \end{cases}$$

Definition 24.1.2.2 (Gauge vertex action). [Lean](#) [✓ Stmt](#) The gauged tensor at vertex v is A_v^X . For each incident edge ie , write M_{ie} for the matrix specified by Definition 24.1.2.1, applied at the vertex v to the gauge on the underlying edge of ie . Then

$$(A^X)_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{ie \ni v} M_{ie}(\eta(ie), \eta'(ie)) \right) A_v(\eta', \sigma).$$

Definition 24.1.2.3 (Apply gauge). [Lean](#) [✓ Stmt](#) The PEPS tensor obtained by applying edge-gauge matrices to A , with vertex tensors A_v^X .

Definition 24.1.2.4 (Absorb edge gauges). [Lean](#) [✓ Stmt](#) Given a tensor family B and an oriented gauge matrix X_e on every edge, the absorbed tensor family is $\tilde{B} = B^X$; at each vertex, its local tensor is obtained by inserting the oriented endpoint matrices on the incident virtual legs.

Theorem 24.1.2.5 (Absorbed gauges keep the bond spaces). [Lean](#) [✓ Stmt](#) The absorbed tensor family has $D_{\tilde{B}}(e) = D_B(e)$ for every edge e .

Proof. [✓ Proof](#) Since $\tilde{B} = B^X$ is defined with the same bond-dimension function as B , one has $D_{\tilde{B}}(e) = D_{B^X}(e) = D_B(e)$ for every edge e . $\square$

Theorem 24.1.2.6 (Local formula for absorbed gauges). [Lean](#) [✓ Stmt](#) If M_{ie} is the oriented endpoint matrix attached to an incident edge ie at v , then $\tilde{B}_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$.

Proof. [✓ Proof](#) Expanding $\tilde{B} = B^X$ at a vertex gives $\tilde{B}_v(\eta, \sigma) = (B^X)_v(\eta, \sigma) = \sum_{\eta'} (\prod_{ie} M_{ie}(\eta(ie), \eta'(ie))) B_v(\eta', \sigma)$. $\square$

Definition 24.1.2.7 (Gauge equivalence for PEPS). [Lean](#) [✓ Stmt](#) Two PEPS tensors A, B are gauge-equivalent if they have the same bond dimensions and B is obtained from A by applying invertible gauge matrices on each edge.

Theorem 24.1.2.8 (Gauge invariance of PEPS state). Lean ✓ Stmt Applying a gauge to a PEPS tensor preserves the state coefficients. Along each internal edge, the two endpoint insertions cancel:

$$\sum_j X_e(j, i) ((X_e^{-1})^\top)(j, k) = \delta_{ik}.$$

Theorem 24.1.2.9 (Gauge equivalence implies same state). Lean ✓ Stmt If $A \sim B$ are gauge-equivalent PEPS tensors, then they represent the same state: for every physical configuration σ ,

$$\text{Coeff}_A(\sigma) = \text{Coeff}_B(\sigma).$$

24.1.3 Vertex injectivity and the local inverse

When the linear map from the coefficient space on virtual configurations to $\mathbb{C}^d$, sending the basis vector at η_v to $A_v(\eta_v, \cdot)$, is injective, the family is linearly independent and the map has a left inverse. This inverse turns a matrix inserted on an incident bond into an operation on the physical leg, the construction that drives the gauge extraction in later sections.

Definition 24.1.3.1 (Vertex injectivity). Lean ✓ Stmt A PEPS tensor A is *vertex-injective* if, at every vertex v , the family of physical vectors $(A_v(\eta, \cdot) \in \mathbb{C}^d)_\eta$ indexed by virtual configurations η on the edges incident to v is linearly independent in $\mathbb{C}^d$. Equivalently, extending A_v by linearity to a map from the free vector space on virtual configurations into the physical space $\mathbb{C}^d$ gives a linear map with trivial kernel. This matches the injectivity formulation of [186, Section 3], where the local tensor is interpreted as a map from virtual configurations to the physical space.

$$\bigotimes_{e \ni v} \mathbb{C}^{D_e} \xrightarrow{A_v} \mathbb{C}^d \quad \ker A_v = \{0\}.$$

Definition 24.1.3.2 (Local tensor map). Lean ✓ Stmt For a fixed vertex v , the family of local physical vectors $\eta \mapsto A_v(\eta, \cdot)$ extends by linearity to a map from the coefficient space on local virtual configurations at v into $\mathbb{C}^d$.

Lemma 24.1.3.3 (Trivial kernel in explicit coordinates). Lean ✓ Stmt If A is vertex-injective and R is a coefficient function on the local virtual configurations at v satisfying $\sum_\eta R(\eta) A_v(\eta, \tau) = 0$ for every physical index τ , then $R = 0$.

Proof. ✓ Proof The hypothesis says that the local tensor map sends R to the zero vector. Vertex injectivity makes this map injective, hence $R = 0$. □

Definition 24.1.3.4 (Local left inverse). Lean ✓ Stmt If the family of local physical vectors $(A_v(\eta, \cdot))_\eta$ indexed by the virtual configurations η at v is linearly independent, one may choose a linear map Ψ_v from $\mathbb{C}^d$ to the coefficient space on local virtual configurations at v such that $\Psi_v \circ \Phi_v = \mathbb{1}$.

Definition 24.1.3.5 (Physical realization of a local virtual operator). Lean ✓ Stmt For an endomorphism T of the coefficient space on local virtual configurations at v , define the local physical operator $O_T = \Phi_v \circ T \circ \Psi_v$.

Theorem 24.1.3.6 (Local virtual operations on the image). Lean ✓ Stmt For every coefficient function c on the local virtual configurations at v , one has $O_T(\Phi_v(c)) = \Phi_v(Tc)$.

Proof. ✓ Proof This follows from $\Psi_v \circ \Phi_v = \mathbb{1}$. □

Theorem 24.1.3.7 (Injectivity of local physical realization). [Lean](#) [✓ Stmt](#) *Suppose the family of local physical vectors at v is linearly independent. If two virtual operations at v have the same canonical physical realization, then they agree.*

Proof. [✓ Proof](#) Evaluate both physical realizations on vectors in the image of the local tensor map, where linear independence makes that map injective. $\square$

Definition 24.1.3.8 (Local projector). [Lean](#) [✓ Stmt](#) The local projector is the physical realization of the identity virtual operator, $P_v = \Phi_v \circ \Psi_v$.

Definition 24.1.3.9 (Virtual pullback of a local physical operator). [Lean](#) [✓ Stmt](#) For a physical operator O on the local physical space at v , define its virtual pullback by $\Psi_v \circ O \circ \Phi_v$. This is the local injectivity step used in the physical-to-virtual direction of the insertion-algebra argument in [186, Section 3].

Theorem 24.1.3.10 (Virtual pullback realizes the projected physical action). [Lean](#) [✓ Stmt](#) *For every coefficient function c , applying the local tensor map after the virtual pullback gives the projection of $O(\Phi_v(c))$ onto the image of Φ_v .*

Proof. [✓ Proof](#) Expand the definitions. The chosen left inverse gives $\Psi_v \Phi_v = \mathbb{1}$, and the remaining map is the local projector. $\square$

Theorem 24.1.3.11 (Virtual pullback for image-preserving physical actions). [Lean](#) [✓ Stmt](#) *If O sends every vector in the image of Φ_v back into that image, then its virtual pullback realizes exactly the action of O on $\Phi_v(c)$.*

Proof. [✓ Proof](#) Use the preceding theorem and the image-preservation hypothesis. $\square$

Theorem 24.1.3.12 (Recovery of a virtual operation from a physical realization). [Lean](#) [✓ Stmt](#) *Suppose the family of local physical vectors at v is linearly independent. If a physical operator O realizes a virtual operation T on every vector in the image of Φ_v , then pulling back O recovers T .*

Proof. [✓ Proof](#) After applying the local tensor map, both virtual operations have the same image. Linear independence makes the local tensor map injective, so the virtual operations agree. $\square$

Theorem 24.1.3.13 (Extensionality of physical-to-virtual pullback). [Lean](#) [✓ Stmt](#) *Suppose the family of local physical vectors at v is linearly independent. Two physical endpoint operations then have the same virtual pullback exactly when their projected actions agree on every vector in the image of the local tensor map. For every coefficient function c ,*

$$T_O = T_{O'} \iff P_v O \Phi_v(c) = P_v O' \Phi_v(c).$$

Proof. [✓ Proof](#) Apply the local tensor map to the two virtual pullbacks and use the pullback specification. The converse follows from injectivity of the local tensor map; the forward direction is obtained by rewriting the two virtual pullbacks. $\square$

Theorem 24.1.3.14 (Pullback of the canonical physical realization). [Lean](#) [✓ Stmt](#) *Pulling back the canonical physical realization of a virtual operation recovers the original virtual operation.*

Proof. [✓ Proof](#) The canonical physical realization agrees with the virtual operation on the image of the local tensor map. Apply the preceding recovery theorem. $\square$

Theorem 24.1.3.15 (Physical realization of a pulled-back physical operator). [Lean](#) [✓ Stmt](#) Let $P_v = \Phi_v \circ \Psi_v$ be the local projector, and let $T_O = \Psi_v \circ O \circ \Phi_v$ be the virtual pullback of a physical operator O . Then $\Phi_v \circ T_O \circ \Psi_v = P_v \circ O \circ P_v$.

Proof. [✓ Proof](#) Expand the definitions of the physical realization, the virtual pullback, and the local projector. $\square$

Theorem 24.1.3.16 (Projected local recovery). [Lean](#) [✓ Stmt](#) Suppose that the compressed physical action $P_v O P_v$ agrees with the canonical physical realization of a virtual operation T . Then the virtual pullback of O is T .

Proof. [✓ Proof](#) The physical realization of the virtual pullback of O is $P_v O P_v$. By hypothesis this is the physical realization of T . Injectivity of the canonical physical realization gives equality of the virtual operations. $\square$

Theorem 24.1.3.17 (Projected local recovery equivalence). [Lean](#) [✓ Stmt](#) The compressed physical action $P_v O P_v$ is the canonical physical realization of a virtual operation T if and only if the virtual pullback of O is T .

Proof. [✓ Proof](#) One implication follows by realizing both virtual operations physically and using the formula for the physical realization of a pulled-back operator. The converse is the preceding recovery theorem. $\square$

Definition 24.1.3.18 (Matrix action on one incident virtual edge). [Lean](#) [✓ Stmt](#) A matrix on one edge incident to v acts on the coefficient space of local virtual configurations by applying the matrix to that edge coordinate and summing over the input edge index while keeping the residual virtual configuration fixed.

Theorem 24.1.3.19 (One-edge matrix action on a basis configuration). [Lean](#) [✓ Stmt](#) If the coefficient function is the unit basis function at one local virtual configuration, and the matrix acts on the distinguished incident edge, the result is the sum over the new distinguished edge index with the residual local boundary held fixed.

Proof. [✓ Proof](#) Evaluate both sides on a local virtual configuration and split into the two cases according to whether the residual boundary agrees with the fixed residual boundary. $\square$

Theorem 24.1.3.20 (Local tensor form of a one-edge basis action). [Lean](#) [✓ Stmt](#) Applying the local tensor map after a one-edge matrix action on a basis virtual configuration gives the corresponding linear combination of local tensor components.

Proof. [✓ Proof](#) Use the preceding coefficient identity and linearity of the local tensor map. $\square$

Theorem 24.1.3.21 (Physical realization of one-edge virtual matrix action). [Lean](#) [✓ Stmt](#) If the family of local physical vectors at v is linearly independent, then a matrix acting on one edge incident to v is realized by a physical operator on the physical space at v .

Proof. [✓ Proof](#) Apply the physical realization of local virtual operations to the one-edge matrix action. $\square$

Definition 24.1.3.22 (Split at one incident edge). [Lean](#) [✓ Stmt](#) Choosing one incident edge at v identifies the set of local virtual configurations with the product of the bond index on that edge and the residual indices on the remaining incident edges.

24.1.4 Edge-centred decomposition of a contraction

Fixing an edge $e = (u, v)$ splits the vertex set into the two endpoints and the middle region $V \setminus \{u, v\}$. The state coefficient factors accordingly, which is the starting point for the edge-middle reduction of the next section.

Definition 24.1.4.1 (Middle region of an edge). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, the middle region is the complement $V \setminus \{u, v\}$.

Theorem 24.1.4.2 (Edge-centered vertex partition). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, the three regions $\{u\}$, $V \setminus \{u, v\}$, and $\{v\}$ cover the whole vertex set.

Proof. [✓ Proof](#) Every vertex is either u , v , or different from both. □

Theorem 24.1.4.3 (Nonempty middle region from a cardinal bound). [Lean](#) [✓ Stmt](#) If $2 < |V|$, then for every edge $e = (u, v)$ the middle region $V \setminus \{u, v\}$ is nonempty.

Proof. [✓ Proof](#) The middle region has cardinality $|V| - 2$. Hence $2 < |V|$ gives $|V| - 2 > 0$. □

Theorem 24.1.4.4 (Product splitting at an edge). [Lean](#) [✓ Stmt](#) For any edge $e = (u, v)$ and any function f on the vertex set with values in a commutative monoid M ,

$$\prod_{x \in V} f(x) = f(u) \left(\prod_{x \in V \setminus \{u, v\}} f(x) \right) f(v).$$

Proof. [✓ Proof](#) Isolate the two distinguished endpoints in the product over V and group the remaining factors into the middle region. □

Theorem 24.1.4.5 (PEPS coefficient splitting at an edge). [Lean](#) [✓ Stmt](#) For any edge $e = (u, v)$, each summand in the PEPS coefficient factors into the tensor contribution at u , the product over the middle region $V \setminus \{u, v\}$, and the tensor contribution at v .

Proof. [✓ Proof](#) Expand the state coefficient and apply the product splitting at the chosen edge to the per-vertex product inside each summand. □

Definition 24.1.4.6 (Edge-centered boundary data). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, the boundary data consist of the virtual index on e , the remaining virtual indices incident to u , and the remaining virtual indices incident to v . These are the indices over which the endpoint tensors couple to the blocked middle region.

Definition 24.1.4.7 (Boundary data read from a virtual configuration). [Lean](#) [✓ Stmt](#) A global virtual configuration determines edge-centered boundary data by reading the index on the distinguished edge and the residual endpoint indices.

Definition 24.1.4.8 (Boundary compatibility at an edge). [Lean](#) [✓ Stmt](#) A global virtual configuration is compatible with fixed edge-centered boundary data if it has the prescribed index on the distinguished edge and the prescribed residual indices at both endpoint stars.

Definition 24.1.4.9 (Middle configurations over fixed boundary data). [Lean](#) [✓ Stmt](#) For fixed boundary data β , the middle configurations are the subtype of global virtual configurations satisfying the compatibility condition with β .

Definition 24.1.4.10 (Virtual configurations fibred by edge boundary data). [Lean](#) [✓ Stmt](#) A global virtual configuration is equivalently a choice of edge-centered boundary data together with a middle configuration in the corresponding fibre.

Definition 24.1.4.11 (Left endpoint local configuration from boundary data). [Lean](#) [✓ Stmt](#) The left endpoint local virtual configuration is reconstructed from the edge index and the residual data on the left endpoint star.

Definition 24.1.4.12 (Right endpoint local configuration from boundary data). [Lean](#) [✓ Stmt](#) The right endpoint local virtual configuration is reconstructed from the edge index and the residual data on the right endpoint star.

Theorem 24.1.4.13 (Left endpoint reconstruction for compatible boundary data). [Lean](#) [✓ Stmt](#) *If a global virtual configuration is compatible with boundary data, then its restriction to the left endpoint star is the local configuration reconstructed from those data.*

Theorem 24.1.4.14 (Right endpoint reconstruction for compatible boundary data). [Lean](#) [✓ Stmt](#) *If a global virtual configuration is compatible with boundary data, then its restriction to the right endpoint star is the local configuration reconstructed from those data.*

Definition 24.1.4.15 (Boundary data with an inserted edge matrix). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, inserted-edge boundary data consist of two independent indices on the bond e , together with the residual virtual indices incident to u and to v . These are the boundary indices used when a matrix is inserted on the distinguished bond.

Definition 24.1.4.16 (Diagonal inserted-edge boundary data). [Lean](#) [✓ Stmt](#) Ordinary edge-centered boundary data determine inserted-edge boundary data by using the same bond index at both endpoints. This is the diagonal boundary datum selected by the identity matrix in the edge-insertion coefficient.

Definition 24.1.4.17 (Right-endpoint inserted-boundary reindexing). [Lean](#) [✓ Stmt](#) Ordinary edge-boundary data together with an independent right endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the left bond index and the two residual endpoint boundaries.

Definition 24.1.4.18 (Left-endpoint inserted-boundary reindexing). [Lean](#) [✓ Stmt](#) Ordinary edge-boundary data together with an independent left endpoint bond index are equivalent to inserted-edge boundary data. The ordinary boundary supplies the right bond index and the two residual endpoint boundaries.

Definition 24.1.4.19 (Diagonal inserted-boundary configurations). [Lean](#) [✓ Stmt](#) The diagonal inserted-boundary configurations are those inserted-edge boundary data for which the two distinguished bond indices agree. This is the support of the identity matrix inside the inserted-boundary sum.

Definition 24.1.4.20 (Diagonal boundary reindexing). [Lean](#) [✓ Stmt](#) Ordinary edge-boundary data are equivalent to diagonal inserted-boundary data. This is the finite reindexing map used after the identity matrix has removed all off-diagonal inserted-boundary summands.

24.2 Edge-Middle Reduction

Fixing an edge $e = (u, v)$ and blocking the whole complementary middle region $V \setminus \{u, v\}$ presents the state coefficient as a three-site contraction: the endpoint tensors at u and v joined through one blocked middle tensor. Inserting the identity on the bond e recovers the original coefficient, so a matrix inserted on e records the difference between two tensors that generate the same state. Injectivity of this middle tensor provides the foundation used by the edge-kernel argument of Section 24.3.

Definition 24.2.1 (Open middle tensor at an edge). [Lean](#) [✓ Stmt](#) Fixing the residual boundary configurations at the endpoint vertices u and v but leaving the bond e out of the middle region, the open middle tensor is the contraction over all virtual indices away from e that are compatible with those boundary configurations.

Definition 24.2.2 (Middle physical configurations at an edge). [Lean](#) [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$, the middle physical configurations are the physical indices on the vertices in $V \setminus \{u, v\}$. A global physical configuration restricts to such a middle configuration.

Definition 24.2.3 (Middle tensor with middle physical index). [Lean](#) [Lean](#) [✓ Stmt](#) The ordinary and open middle contractions may be written using only the middle physical configuration, rather than a physical configuration on all vertices. This is the middle tensor in the three-site chain obtained after the edge blocking in arXiv:1804.04964, Section 3.

Theorem 24.2.4 (Middle restriction of the blocked tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *The middle contractions written with a global physical configuration are the corresponding middle-indexed contractions evaluated on the restricted middle physical configuration.*

Proof. [✓ Proof](#) Rewrite the product over $V \setminus \{u, v\}$ as the product over the corresponding finite set of middle vertices. $\square$

Definition 24.2.5 (Restricting middle configurations away from the edge). [Lean](#) [✓ Stmt](#) An ordinary middle configuration for the edge-blocked contraction restricts to a virtual configuration on all edges except the distinguished edge. This is the first reindexing map needed to compare the identity insertion with the ordinary edge-blocked middle tensor.

Definition 24.2.6 (Restoring the distinguished edge index). [Lean](#) [✓ Stmt](#) Conversely, a compatible open middle configuration becomes an ordinary middle configuration after the fixed edge-boundary index is restored on the distinguished edge. Together with the restriction map, this gives the finite reindexing underlying the identity-insertion specialization.

Definition 24.2.7 (Open-middle reindexing equivalence). [Lean](#) [✓ Stmt](#) For fixed edge-boundary data, ordinary middle configurations are equivalent to compatible open-middle configurations. This is the finite reindexing used when the inserted matrix is the identity.

Theorem 24.2.8 (Middle-indexed open tensor reindexing). [Lean](#) [✓ Stmt](#) *The ordinary middle tensor and the open middle tensor agree after restoring the fixed distinguished-edge index and reindexing the finite sum, with the physical index already restricted to the middle block.*

Proof. [✓ Proof](#) Reindex the ordinary middle configurations by the equivalence with compatible open-middle configurations, then compare each local tensor entry. $\square$

Definition 24.2.9 (Edge insertion coefficient). [Lean](#) [✓ Stmt](#) For an edge $e = (u, v)$ and a matrix X acting on the bond space of e , the edge insertion coefficient is the contraction obtained by inserting X between the endpoint tensors and the open middle tensor:

$$C_A(e, X; \sigma) = \sum_{\beta} A_u(\beta_u, \sigma_u) X_{\beta_{e,u}, \beta_{e,v}} C_{A, M_e}^{\text{open}}(\beta_{\hat{e}, u}, \beta_{\hat{e}, v}; \sigma|_{M_e}) A_v(\beta_v, \sigma_v).$$

Theorem 24.2.10 (Middle tensor reindexing for an opened edge). [Lean](#) [✓ Stmt](#) *Restoring the fixed distinguished-edge index identifies the ordinary blocked middle tensor with the open middle tensor. This is the finite reindexing used before the identity matrix collapses the two inserted bond indices to the diagonal.*

Proof. [✓ Proof](#) Reindex the ordinary middle configurations by the finite equivalence with open-middle configurations and compare the local tensor factors pointwise. $\square$

Theorem 24.2.11 (Diagonal summand of the identity insertion). [Lean](#) [✓ Stmt](#) *On the diagonal boundary datum selected by the identity matrix, the inserted-edge summand is the ordinary edge-blocked summand. Thus the remaining identity-insertion step is purely a finite summation statement: the off-diagonal inserted boundary data vanish and the diagonal data reindex the ordinary edge-boundary sum.*

Proof. [✓ Proof](#) Substitute the identity matrix entry $\mathbb{1}_{a,a} = 1$ and apply the reindexing $C_{A,M_e}(\beta) = C_{A,M_e}^{\text{open}}(\beta)$ from Theorem 24.2.10. $\square$

Theorem 24.2.12 (Off-diagonal summands of the identity insertion). [Lean](#) [✓ Stmt](#) *If the two inserted bond indices are distinct, then the corresponding summand in the identity insertion is zero. This is the pointwise off-diagonal vanishing used before the inserted-boundary sum is restricted to its diagonal part.*

Proof. [✓ Proof](#) For $a \neq b$, the identity matrix entry $\mathbb{1}_{a,b} = 0$ makes the inserted summand vanish. $\square$

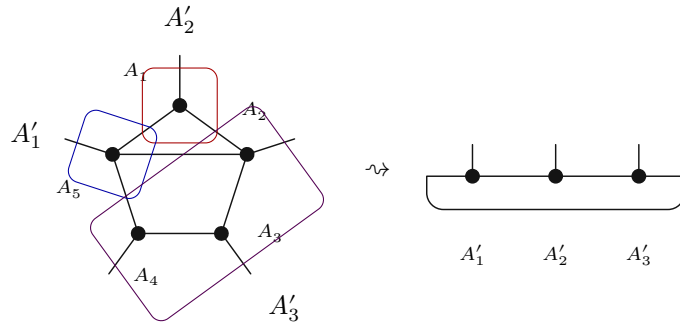
Theorem 24.2.13 (Identity insertion recovers the edge-blocked coefficient). [Lean](#) [✓ Stmt](#) *If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the ordinary edge-blocked coefficient.*

Proof. [✓ Proof](#) Split the inserted-boundary sum according to whether the two bond indices a, b agree:

$$C_A(e, \mathbb{1}; \sigma) = \sum_{\beta: a=b} A_u(\beta_u, \sigma_u) C_{A,M_e}^{\text{open}}(\beta, \sigma|_{M_e}) A_v(\beta_v, \sigma_v) + \sum_{\beta: a \neq b} 0.$$

The off-diagonal sum vanishes because $\mathbb{1}_{a,b} = 0$ for $a \neq b$, and the diagonal sum reindexes to $C_A(e; \sigma)$ through the equivalence between diagonal inserted-boundary configurations and ordinary edge-boundary configurations. $\square$

Definition 24.2.14 (Blocked middle tensor at an edge). [Lean](#) [✓ Stmt](#) Fixing the edge-centered boundary data, the blocked middle tensor is the sum over global virtual configurations compatible with those data of the product of local tensor entries over the middle region $V \setminus \{u, v\}$. The graph-local contraction is read as a three-site chain:



Definition 24.2.15 (Edge-blocked PEPS coefficient). [Lean](#) [✓ Stmt](#) The edge-blocked coefficient is the three-block contraction obtained from the left endpoint tensor, the blocked middle tensor, and the right endpoint tensor at the chosen edge.

Theorem 24.2.16 (Edge-blocked coefficient equals the PEPS coefficient). [Lean](#) [✓ Stmt](#) For every edge e , the edge-blocked three-block coefficient is exactly the original PEPS state coefficient.

Proof. [✓ Proof](#) Write $\text{Coeff}_A(\sigma)$ for the PEPS state coefficient and $C_A(e; \sigma)$ for the edge-blocked coefficient, and put $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. The equivalence $\omega \leftrightarrow (\beta, \omega_{M_e})$ between global virtual configurations and an edge boundary configuration with a compatible middle configuration rewrites the PEPS coefficient as

$$\sum_{\omega} A_u(\omega|_u, \sigma_u) \left(\prod_{x \in M_e} A_x(\omega|_x, \sigma_x) \right) A_v(\omega|_v, \sigma_v) = \sum_{\beta} A_u(\beta_u, \sigma_u) C_{A, M_e}(\beta, \sigma|_{M_e}) A_v(\beta_v, \sigma_v).$$

Under the equivalence, $\omega|_u = \beta_u$ and $\omega|_v = \beta_v$; the inner sum over ω_{M_e} is exactly $C_{A, M_e}(\beta, \sigma|_{M_e})$. Thus $C_A(e; \sigma) = \text{Coeff}_A(\sigma)$. $\square$

Theorem 24.2.17 (Identity insertion equals the PEPS coefficient). [Lean](#) [✓ Stmt](#) If the inserted matrix on the distinguished edge is the identity, then the edge insertion coefficient is the original PEPS state coefficient.

Proof. [✓ Proof](#) Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. With $\mathbb{1}$ inserted on e , the inserted-boundary sum is

$$C_A(e, \mathbb{1}; \sigma) = \sum_{a, b, \lambda, \rho} \delta_{a, b} A_u(a, \lambda; \sigma_u) C_{A, M_e}^{\text{open}}(\lambda, \rho; \sigma|_{M_e}) A_v(b, \rho; \sigma_v).$$

Here a, b are the two copies of the distinguished bond index, while λ, ρ are the remaining virtual configurations at u and v . The factor $\delta_{a, b}$ leaves only the diagonal $a = b$, and the open-middle reindexing gives $C_{A, M_e}^{\text{open}}(\lambda, \rho; \sigma|_{M_e}) = C_{A, M_e}(a, \lambda, \rho; \sigma|_{M_e})$. Hence $C_A(e, \mathbb{1}; \sigma) = C_A(e; \sigma) = \text{Coeff}_A(\sigma)$. $\square$

Theorem 24.2.18 (Same state gives the same edge-blocked coefficients). [Lean](#) [✓ Stmt](#) If two PEPS tensors A and B represent the same state, then their edge-blocked coefficients agree at every edge e and every physical configuration σ :

$$C_A(e; \sigma) = C_B(e; \sigma).$$

Proof. [✓ Proof](#) For every physical configuration σ , $C_A(e; \sigma) = \text{Coeff}_A(\sigma) = \text{Coeff}_B(\sigma) = C_B(e; \sigma)$. $\square$

Theorem 24.2.19 (Same state gives the same identity insertions). [Lean](#) [✓ Stmt](#) If two PEPS tensors A and B represent the same state, then their edge-centered identity insertions agree at every edge e and every physical configuration σ :

$$C_A(e, \mathbb{1}; \sigma) = C_B(e, \mathbb{1}; \sigma).$$

Proof. [✓ Proof](#) For every physical configuration σ , $C_A(e, \mathbb{1}; \sigma) = \text{Coeff}_A(\sigma) = \text{Coeff}_B(\sigma) = C_B(e, \mathbb{1}; \sigma)$. $\square$

Definition 24.2.20 (Middle tensor family at an edge). [Lean](#) [Lean](#) [✓ Stmt](#) For the edge-blocked three-site chain at $e = (u, v)$, the middle tensor is the family obtained by fixing the two residual endpoint boundaries and evaluating the open middle contraction on the middle physical index. Its injectivity is the middle-block part of the assertion following the edge-blocking diagram.

Lemma 24.2.21 (Middle tensor vanishes for an empty complement configuration). [Lean](#) [✓ Stmt](#) *If the virtual configuration space on the complement of the distinguished edge is empty, then for every residual boundary label ρ the vector T_{A, M_e}^ρ in the edge-middle tensor family is zero.*

Proof. [✓ Proof](#) The finite sum defining the open middle tensor is taken over the empty complement configuration space, and is therefore zero. $\square$

Lemma 24.2.22 (Failure of middle injectivity for an empty complement configuration). [Lean](#) [✓ Stmt](#) *If the complement configuration space is empty and the residual boundary-label set is nonempty, then the edge-middle tensor family is not injective.*

Proof. [✓ Proof](#) Choose a residual boundary label. By Lemma 24.2.21 its middle-tensor vector is the zero vector. A linearly independent family cannot contain a zero vector. $\square$

Definition 24.2.23 (Edge-blocked three-site injectivity). [Lean](#) [✓ Stmt](#) For a chosen edge $e = (u, v)$, the edge-blocked three-site chain is injective when the left endpoint tensor map, the middle tensor family, and the right endpoint tensor map are all injective.

Definition 24.2.24 (Singleton blocked region). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a singleton block $\{v\}$, the blocked physical index is the original physical index and the blocked tensor is $T_{A, \{v\}}(\eta, \sigma) = A_v(\eta, \sigma)$. Its injectivity is $\text{inj}(T_{A, \{v\}})$. The comparison with a region-injectivity predicate κ records the implication $\text{inj}(T_{A, \{v\}}) \implies \kappa(\{v\})$.

Theorem 24.2.25 (Singleton comparison for a vertex-injective tensor). [Lean](#) [✓ Stmt](#) *Let A be vertex-injective with every bond dimension positive. Then the concrete region-injectivity predicate of A satisfies the singleton comparison: injectivity of a singleton blocked tensor implies injectivity of the corresponding singleton region.*

Proof. [✓ Proof](#) Under vertex injectivity and positive bond dimensions every finite region is injective. In particular the singleton region $\{v\}$ is injective, so the comparison holds independently of the singleton-tensor hypothesis. $\square$

Theorem 24.2.26 (Vertex injectivity gives singleton-block injectivity). [Lean](#) [✓ Stmt](#) *If A is vertex-injective, then every singleton blocked tensor is injective. For every vertex v , $\text{inj}(T_{A, \{v\}})$.*

Proof. [✓ Proof](#) For $R = \{v\}$, the contraction defining $T_{A, R}$ has one factor. Hence $T_{A, \{v\}}(\eta, \sigma) = A_v(\eta, \sigma)$. Thus $\text{inj}(A_v)$ is exactly $\text{inj}(T_{A, \{v\}})$. Vertex injectivity gives $\forall v, \text{inj}(A_v)$, and therefore $\forall v, \text{inj}(T_{A, \{v\}})$. For a fixed v , this is the displayed assertion. $\square$

Theorem 24.2.27 (Singleton comparison gives the region base case). [Lean](#) [Lean](#) [✓ Stmt](#) *Assume that a region-injectivity predicate κ satisfies, for every vertex v , $\text{inj}(T_{A, \{v\}}) \implies \kappa(\{v\})$. Then, for every vertex v , A vertex-injective $\implies \kappa(\{v\})$.*

Proof. [✓ Proof](#) For each v , Theorem 24.2.26 gives A vertex-injective $\implies \text{inj}(T_{A, \{v\}})$. The comparison hypothesis gives $\text{inj}(T_{A, \{v\}}) \implies \kappa(\{v\})$. Their composition is the claimed implication. $\square$

Theorem 24.2.28 (Finite regions from singleton regions and union closure). [Lean](#) [✓ Stmt](#) *Assume that κ satisfies singleton comparison and union closure: $\text{inj}(T_{A, \{v\}}) \implies \kappa(\{v\})$ for every vertex v , and $\kappa(R) \wedge \kappa(S) \implies \kappa(R \cup S)$. If A is vertex-injective, then every nonempty finite region R satisfies $\kappa(R)$.*

Proof. [✓ Proof](#) For $R \neq \emptyset$, write $R = \bigcup_{v \in R} \{v\}$. Theorem 24.2.27 turns singleton comparison and vertex injectivity into $\kappa(\{v\})$ for every $v \in R$. Applying the finite-union consequence of union closure to these singleton regions gives $\kappa(R)$. $\square$

Definition 24.2.29 (Region injectivity supplies the edge-middle tensor). [Lean](#) [✓ Stmt](#) A comparison from finite-region injectivity to the edge-middle tensor records the following assertion: if the middle vertex region $V \setminus \{u, v\}$ is injective after blocking, then the middle tensor in the edge-blocked three-site chain is injective. This isolates the finite-region contraction claim in [186, Section 3].

Theorem 24.2.30 (Endpoint injectivity in the edge-blocked chain). [Lean](#) [✓ Stmt](#) *If the original PEPS tensor A is vertex-injective, then for each edge $e = (u, v)$, the two endpoint tensor maps in the edge-blocked three-site chain are injective:*

$$\text{inj}(T_{A,u}) \quad \text{and} \quad \text{inj}(T_{A,v}).$$

Proof. [✓ Proof](#) If $e = (u, v)$, vertex injectivity gives $\text{inj}(T_{A,u})$ and $\text{inj}(T_{A,v})$, which are the left and right endpoint tensor maps of the edge-blocked chain. $\square$

Theorem 24.2.31 (Endpoint injectivity at all edges). [Lean](#) [✓ Stmt](#) *If the original PEPS tensor A is vertex-injective, then the two endpoint tensor maps in the edge-blocked three-site chain are injective at every edge $e = (u, v)$:*

$$\forall e = (u, v), \quad \text{inj}(T_{A,u}) \quad \text{and} \quad \text{inj}(T_{A,v}).$$

Proof. [✓ Proof](#) For each $e = (u, v)$, Theorem 24.2.30 gives $\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,v})$. $\square$

Theorem 24.2.32 (Reducing edge-blocked injectivity to the middle block). [Lean](#) [✓ Stmt](#) *For a vertex-injective PEPS tensor A , injectivity of the edge-blocked middle tensor gives the full injectivity statement for the edge-blocked three-site chain at $e = (u, v)$:*

$$\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v}),$$

where $M_e = V \setminus \{u, v\}$.

Proof. [✓ Proof](#) Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. The endpoint theorem supplies $\text{inj}(T_{A,u})$ and $\text{inj}(T_{A,v})$, while the hypothesis is $\text{inj}(T_{A,M_e})$. Hence $\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v})$. $\square$

Theorem 24.2.33 (All edge-blocked chains from middle-tensor injectivity). [Lean](#) [✓ Stmt](#) *If every edge-middle tensor is injective, then a vertex-injective PEPS tensor A gives an injective edge-blocked three-site chain at every edge $e = (u, v)$:*

$$\forall e = (u, v), \quad \text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v}).$$

Proof. [✓ Proof](#) For each $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. Vertex injectivity gives $\text{inj}(T_{A,u})$ and $\text{inj}(T_{A,v})$, while the hypothesis gives $\text{inj}(T_{A,M_e})$. Hence $\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v})$ is the injectivity assertion for the edge-blocked three-site chain at e . $\square$

Theorem 24.2.34 (Region injectivity gives edge-blocked injectivity). [Lean](#) [✓ Stmt](#) *Suppose the middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose this region-injectivity assertion has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at the edge $e = (u, v)$.*

Proof. [✓ Proof](#) Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. The comparison gives $\kappa(M_e) \Rightarrow \text{inj}(T_{A,M_e})$. Vertex injectivity supplies $\text{inj}(T_{A,u})$ and $\text{inj}(T_{A,v})$. Theorem 24.2.32 then gives $\text{inj}(T_{A,u}) \wedge \text{inj}(T_{A,M_e}) \wedge \text{inj}(T_{A,v})$. $\square$

Theorem 24.2.35 (Middle tensors from all middle regions). [Lean](#) [✓ Stmt](#) Suppose every edge-middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then every edge-middle tensor family is injective.

Proof. [✓ Proof](#) For each edge $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. The hypotheses give $\kappa(M_e)$. The comparison implication $\kappa(M_e) \Rightarrow \text{inj}(T_{A, M_e})$ yields $\text{inj}(T_{A, M_e})$ for every edge. $\square$

Theorem 24.2.36 (All edge-blocked chains from middle-region injectivity). [Lean](#) [✓ Stmt](#) Suppose every edge-middle region $V \setminus \{u, v\}$ is injective after blocking, and suppose finite-region injectivity has been identified with injectivity of the corresponding edge-middle tensor family. Then a vertex-injective PEPS tensor gives an injective edge-blocked three-site chain at every edge.

Proof. [✓ Proof](#) For every edge $e = (u, v)$, put $M_e = V \setminus \{u, v\}$. The region hypothesis gives $\kappa(M_e)$, and the comparison gives $\kappa(M_e) \Rightarrow \text{inj}(T_{A, M_e})$. Vertex injectivity gives $\text{inj}(T_{A, u})$ and $\text{inj}(T_{A, v})$. Thus $\text{inj}(T_{A, u}) \wedge \text{inj}(T_{A, M_e}) \wedge \text{inj}(T_{A, v})$. $\square$

Theorem 24.2.37 (Edge-middle tensor from singleton regions, nonempty middle case). [Lean](#) [✓ Stmt](#) Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. Assume $M_e \neq \emptyset$. Assume also singleton comparison, union closure for κ , and the comparison $\kappa(M_e) \Rightarrow \text{inj}(T_{A, M_e})$. If A is vertex-injective, then the edge-middle tensor family T_{A, M_e} is injective.

Proof. [✓ Proof](#) The finite-region theorem applied to $M_e = \bigcup_{w \in M_e} \{w\}$ gives $\kappa(M_e)$. The edge-middle comparison sends this to $\text{inj}(T_{A, M_e})$. $\square$

Theorem 24.2.38 (One edge-blocked chain from singleton regions, nonempty middle case). [Lean](#) [✓ Stmt](#) Let $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. Under the hypotheses of Theorem 24.2.37, vertex injectivity of A gives injectivity of the edge-blocked three-site chain at e .

Proof. [✓ Proof](#) Theorem 24.2.37 gives $\text{inj}(T_{A, M_e})$. The endpoint reduction combines this middle-tensor injectivity with vertex injectivity at u and v . $\square$

Theorem 24.2.39 (All edge-blocked chains from singleton regions, nonempty middle case). [Lean](#) [✓ Stmt](#) If $M_e = V \setminus \{u, v\}$ is nonempty for every edge $e = (u, v)$, then Theorem 24.2.38 applies at every edge. Thus vertex injectivity of A gives injectivity of every edge-blocked three-site chain.

Proof. [✓ Proof](#) Apply the one-edge statement separately to each edge e . $\square$

Theorem 24.2.40 (All edge-blocked chains from singleton regions, cardinal form). [Lean](#) [Lean](#) [✓ Stmt](#) Assume singleton comparison, union closure for κ , the comparison $\kappa(M_e) \Rightarrow \text{inj}(T_{A, M_e})$, and vertex injectivity of A . If $2 < |V|$, then $V \setminus \{u, v\} \neq \emptyset$ for every edge $e = (u, v)$, and these hypotheses give $\text{inj}(T_{A, V \setminus \{u, v\}})$ and edge-blocked three-site injectivity at every edge.

Proof. [✓ Proof](#) For each $e = (u, v)$, Theorem 24.1.4.3 gives $V \setminus \{u, v\} \neq \emptyset$. Apply Theorem 24.2.39. $\square$

24.3 Edge Kernels and Gauge Transport

When the endpoint tensors are injective, a matrix inserted on the bond e is realized by a physical operation on a neighbouring physical leg, and conversely. This sets up an isomorphism between the bond- e matrix algebra of A and that of B , which by the Skolem–Noether theorem is conjugation by an invertible matrix X_e , the edge gauge: for every physical configuration σ and bond matrix M , $C_A(e, \sigma, M) = C_B(e, \sigma, X_e M X_e^{-1})$. Collecting the X_e over all edges and reconciling them on the shared bonds produces the global gauge family that relates the two tensors.

Definition 24.3.1 (Finite kernel descent datum). Lean ✓ Stmt Let V be a vertex type with decidable equality. A finite kernel descent datum is a predicate K on finite subsets $S \subseteq V$ together with implications $j \in S \implies K(S) \implies K(S \setminus \{j\})$. In the edge-blocking proof, $K(S)$ is the assertion that the boundary coefficients c_ρ , with the virtual indices exposed at that stage, give zero after the tensors in S have been contracted.

Lemma 24.3.2 (Finite descent of kernel conditions). Lean Lean ✓ Stmt For a finite kernel descent datum and any finite $R \subseteq V$, $K(R) \implies K(\emptyset)$. More generally, if $K(\emptyset) \implies Q$, then $K(R) \implies Q$.

Proof. ✓ Proof Induct on the finite set R . If $R = \emptyset$, there is nothing to prove. Otherwise write $R = S \cup \{j\}$ with $j \notin S$. The deletion implication gives $K(R) \implies K(S)$, and the induction hypothesis gives $K(S) \implies K(\emptyset)$. □

Definition 24.3.3 (Edge-middle kernel descent data). Lean Lean ✓ Stmt Let $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. An edge-middle kernel descent datum is indexed by boundary labels ρ consisting of the two residual endpoint configurations. It assigns to every finitely supported coefficient family $c = (c_\rho)_\rho$ a finite kernel descent datum $K_c(S)$ such that

$$\left(\forall \tau, \sum_\rho c_\rho T_{A, M_e}^\rho(\tau) = 0 \right) \implies K_c(M_e),$$

$$K_c(\emptyset) \implies \forall \rho, c_\rho = 0.$$

Theorem 24.3.4 (Kernel descent gives edge-middle injectivity). Lean Lean ✓ Stmt Let $e = (u, v)$ and $M_e = V \setminus \{u, v\}$. If an edge-middle kernel descent datum is given, then T_{A, M_e} is injective. Consequently, if A is vertex-injective, the edge-blocked three-site chain at e is injective.

Proof. ✓ Proof Let $c = (c_\rho)_\rho$ be a finite linear relation among the middle tensor vectors: $\sum_\rho c_\rho T_{A, M_e}^\rho(\tau) = 0$ for every τ . The initial part of the descent datum gives $K_c(M_e)$. The finite descent lemma gives $K_c(M_e) \implies K_c(\emptyset)$, and the terminal part gives $\forall \rho, c_\rho = 0$. Thus every finite relation is trivial, which is $\text{inj}(T_{A, M_e})$. Vertex injectivity at u and v gives $\text{inj}(T_{A, u})$ and $\text{inj}(T_{A, v})$; the edge-blocked three-site injectivity at e follows. □

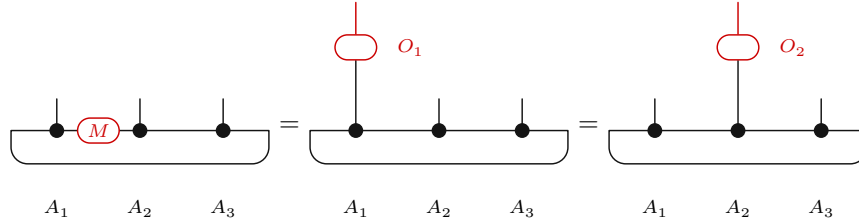
Theorem 24.3.5 (Edge blocking preserves injectivity). Lean ✓ Stmt For a tensor map or blocked tensor family T , write $\text{inj}(T)$ for the assertion that T is injective. If A is vertex-injective and every bond dimension is positive, then for every edge $e = (u, v)$, $\text{inj}(T_{A, u})$, $\text{inj}(T_{A, V \setminus \{u, v\}})$, and $\text{inj}(T_{A, v})$. Here $T_{A, u}$ and $T_{A, v}$ are the two endpoint tensor maps, and $T_{A, V \setminus \{u, v\}}$ is the blocked middle tensor family. This is the precise assertion after the edge-blocking diagram in [186, Section 3]. The positive-bond hypothesis is the source’s standing assumption that injective PEPS have nonzero virtual bond spaces. Without it, an empty complement configuration space can make the middle tensor vanish identically; Lemmas 24.2.21 and 24.2.22 record this obstruction.

Proof. [✓ Proof](#) Fix $e = (u, v)$ and put $M_e = V \setminus \{u, v\}$. From vertex injectivity, construct the edge-middle descent datum required by Theorem 24.3.4: for each finite relation $\sum_{\rho} c_{\rho} T_{A, M_e}^{\rho}(\tau) = 0$ for every τ , the associated conditions $K_c(S)$ satisfy $K_c(M_e)$, $j \in S \implies K_c(S) \implies K_c(S \setminus \{j\})$, and $K_c(\emptyset) \implies \forall \rho, c_{\rho} = 0$. The descent implication is the local contraction identity: for $j \in S$, vertex injectivity at j gives a left inverse L_j , and applying L_j to the relation contracts away that tensor, $j \in S$, $K_c(S) \xRightarrow{L_j} K_c(S \setminus \{j\})$. The contraction at j uses the factorization

$$\mathcal{C}_e \cong \mathcal{V}_j \times \prod_{\substack{f \neq e \\ f \neq j}} \{0, \dots, \dim A_f - 1\},$$

where $\mathcal{C}_e$ is the complement virtual-configuration space for the distinguished edge e and $\mathcal{V}_j$ is the local virtual configuration space at j . The terminal step $K_c(\emptyset) \implies \forall \rho, c_{\rho} = 0$ isolates each boundary coefficient by surjectivity of the boundary labelling, which is where the positive bond dimensions are used to fill the interior virtual indices of a witness configuration. The kernel descent theorem gives $\text{inj}(T_{A, M_e})$. The same theorem, together with vertex injectivity at u and v , gives $\text{inj}(T_{A, u})$, $\text{inj}(T_{A, M_e})$, and $\text{inj}(T_{A, v})$ as the edge-blocked three-site injectivity assertion. $\square$

Theorem 24.3.6 (Local virtual insertion realized physically on either endpoint). [Lean](#) [✓ Stmt](#) *Let A be vertex-injective and let $e = (u, v)$ be an edge. For every matrix M on the bond space of e , the one-leg virtual action induced by M on the local tensor at u , and likewise at v , is realized by a physical operator on that endpoint's physical space. On a closed three-site chain, the corresponding state equality is*



Proof. [✓ Proof](#) Apply the one-edge realization theorem to the two endpoint incidences of the distinguished edge. $\square$

Theorem 24.3.7 (Left-endpoint projected recovery). [Lean](#) [✓ Stmt](#) *Let $e = (u, v)$ and suppose the family of local physical vectors at the left endpoint u is linearly independent. A physical operator at that endpoint then pulls back to the one-edge virtual action of M^T if and only if its compressed physical action is the canonical realization of that one-edge virtual action.*

Proof. [✓ Proof](#) This is the projected local recovery equivalence applied to the left incident edge of e , with the transpose convention used by the left-endpoint coefficient identity. $\square$

Theorem 24.3.8 (Right-endpoint projected recovery). [Lean](#) [✓ Stmt](#) *Let $e = (u, v)$ and suppose the family of local physical vectors at the right endpoint v is linearly independent. A physical operator at that endpoint then pulls back to the one-edge virtual action of M if and only if its compressed physical action is the canonical realization of that one-edge virtual action.*

Proof. [✓ Proof](#) This is the projected local recovery equivalence applied to the right incident edge of e . $\square$

Theorem 24.3.9 (Endpoint recovery of a common virtual insertion). [Lean](#) [✓ Stmt](#) Let $e = (u, v)$. If $P_u O_1 P_u = \Phi_u T_{u,e}(M^\top) \Psi_u$ and $P_v O_2 P_v = \Phi_v T_{v,e}(M) \Psi_v$, then $\Psi_u O_1 \Phi_u = T_{u,e}(M^\top)$ and $\Psi_v O_2 \Phi_v = T_{v,e}(M)$. Here $T_{w,e}(N)$ denotes the virtual operator on the legs incident to w which acts by N on the leg belonging to e and by the identity on the other incident legs.

Proof. [✓ Proof](#) The endpoint equivalences give

$$\begin{aligned} P_u O_1 P_u = \Phi_u T_{u,e}(M^\top) \Psi_u &\iff \Psi_u O_1 \Phi_u = T_{u,e}(M^\top), \\ P_v O_2 P_v = \Phi_v T_{v,e}(M) \Psi_v &\iff \Psi_v O_2 \Phi_v = T_{v,e}(M). \end{aligned}$$

The two hypotheses are the left sides of these equivalences. $\square$

Theorem 24.3.10 (Endpoint physical-to-virtual recovery under image preservation). [Lean](#) [✓ Stmt](#) Let $e = (u, v)$. Suppose that $P_u O_1 P_u = \Phi_u T_{u,e}(M^\top) \Psi_u$ and $P_v O_2 P_v = \Phi_v T_{v,e}(M) \Psi_v$. Suppose also that O_1 maps the image of Φ_u into itself and O_2 maps the image of Φ_v into itself. Then

$$\begin{aligned} O_1 \Phi_u &= \Phi_u T_{u,e}(M^\top), \\ O_2 \Phi_v &= \Phi_v T_{v,e}(M). \end{aligned}$$

Proof. [✓ Proof](#) The preceding endpoint theorem identifies the two virtual pullbacks with the one-edge actions of $M^\top$ and M . The pullback specification gives the projected physical actions, and the image-preservation hypotheses remove the endpoint projectors. $\square$

Theorem 24.3.11 (Right-endpoint physical realization of the inserted coefficient). [Lean](#) [✓ Stmt](#) Let $e = (u, v)$ and let σ be a physical configuration. If a physical operator on v realizes the one-edge virtual matrix action on the distinguished incident edge, then the inserted-edge coefficient at σ is obtained by leaving the left endpoint and open middle contraction explicit and applying that physical operator to the right endpoint tensor.

Proof. [✓ Proof](#) Expand the physical realization on a basis local configuration at the right endpoint. The resulting sum over the right bond index is then reindexed together with the ordinary edge-boundary data. $\square$

Theorem 24.3.12 (Left-endpoint physical realization of the inserted coefficient). [Lean](#) [✓ Stmt](#) Let $e = (u, v)$ and let σ be a physical configuration. If a physical operator on u realizes the one-edge virtual action of $M^\top$ on the distinguished incident edge, then the inserted-edge coefficient for M at σ is obtained by applying that physical operator to the left endpoint tensor and leaving the open middle contraction and the right endpoint explicit.

Proof. [✓ Proof](#) Expand the physical realization on a basis local configuration at the left endpoint. The transpose converts the ordinary right bond index and the new left bond index into the coefficient M_{ij} ; the resulting finite sum is then reindexed with the ordinary edge-boundary data. $\square$

Theorem 24.3.13 (Endpoint physical realizations of the inserted coefficient). [Lean](#) [✓ Stmt](#) If the PEPS tensor is vertex-injective, then for every physical configuration σ , each inserted edge matrix is realized by physical operators at the endpoint tensors: the left endpoint realizes $M^\top$, the right endpoint realizes M , and both realizations reproduce the same inserted-edge coefficient at σ in the edge-centered three-site decomposition.

Proof. [✓ Proof](#) Apply the local physical-realization theorem at the two endpoints and then use the two coefficient identities above. $\square$

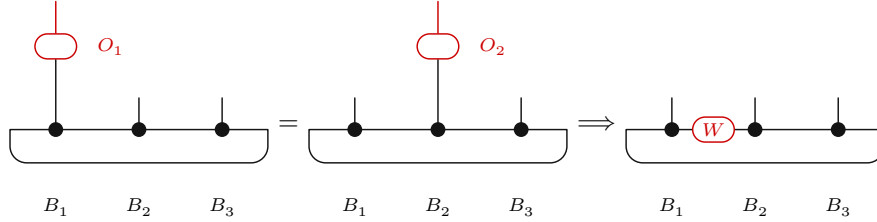
Theorem 24.3.14 (Equal neighboring physical actions recover a virtual insertion). [Lean](#) [✓ Stmt](#) Let $e = (u, v)$. Assume that the edge-blocked three-site chain at e is injective and that every virtual bond has positive dimension. Let O_1, O_2 be physical operators at the endpoint tensors, and write Φ_u, Φ_v for the two local tensor maps. Suppose that, for every physical configuration σ ,

$$\begin{aligned} \sum_{\beta} O_1(A_u(\beta_u))_{\sigma_u} C_{\text{mid}}(\sigma; \beta_L, \beta_R) A_v(\beta_v)_{\sigma_v} \\ = \sum_{\beta} A_u(\beta_u)_{\sigma_u} C_{\text{mid}}(\sigma; \beta_L, \beta_R) O_2(A_v(\beta_v))_{\sigma_v}. \end{aligned}$$

Then

$$\begin{aligned} \exists M \in M_{D(e)}(\mathbb{C}), \quad O_1 \Phi_u &= \Phi_u T_{u,e}(M^T), \\ O_2 \Phi_v &= \Phi_v T_{v,e}(M). \end{aligned}$$

The same implication is represented diagrammatically by the $O_1, O_2 \mapsto W$ step:



This is the converse part of the insertion-algebra lemma in [186, Section 3], where the recovered matrix is denoted W .

Proof. [✓ Proof](#) Contracting the middle block out of the hypothesis leaves a bond-contracted identity between the two endpoint tensors, holding for every residual boundary configuration. Inverting the right endpoint of this identity writes the action of O_1 on a left tensor vector as a bond-indexed combination of left tensor vectors with the same residual; the combining coefficients are read at a fixed reference right residual and define M . Inverting the left endpoint writes the action of O_2 on a right tensor vector as a bond-indexed combination of right tensor vectors. The left inverse identifies the two coefficient families, which is the source step $V = W$, so the right-endpoint combination is also governed by M . The reference residuals exist because every virtual bond has positive dimension. $\square$

Theorem 24.3.15 (Injectivity of an inserted edge coefficient). [Lean](#) [✓ Stmt](#) Let the edge-blocked three-site chain of B at e be injective, and assume every virtual bond of B has positive dimension. If two matrices $M, M' \in M_{D_B(e)}(\mathbb{C})$ give the same inserted coefficient at every physical configuration, $C_B(e, \sigma, M) = C_B(e, \sigma, M')$ for every σ , then $M = M'$.

Proof. [✓ Proof](#) Realize M on the right endpoint and M'^T on the left endpoint. The endpoint realizations and the equality $C_B(e, \sigma, M) = C_B(e, \sigma, M')$ give, for every physical configuration σ ,

$$\begin{aligned} \sum_{\beta} O_1(B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) (B_v(\beta_v))_{\sigma_v} \\ = \sum_{\beta} (B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) O_2(B_v(\beta_v))_{\sigma_v}. \end{aligned}$$

Theorem 24.3.14 gives a single virtual matrix realizing both endpoint operators. Endpoint uniqueness then identifies this matrix with M and with M' . $\square$

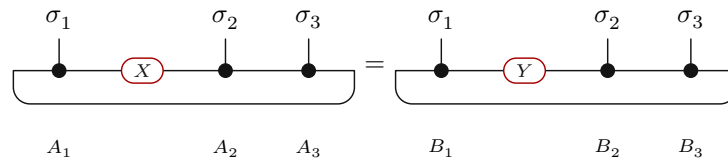
Definition 24.3.16 (Edge-blocked insertion algebra correspondence). [Lean](#) [✓ Stmt](#) For two PEPS tensors A and B and an edge e , this property is

$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\text{alg}} M_{D_B(e)}(\mathbb{C}), \quad \forall \sigma, \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

Theorem 24.3.17 (Edge-blocked insertion algebra isomorphism). [Lean](#) [✓ Stmt](#) Suppose the two edge-blocked three-site chains associated to A and B at the edge e are injective, suppose every virtual bond of A and B has positive dimension, and suppose the two PEPS tensors define the same state. Then

$$\exists \Phi_e : M_{D_A(e)}(\mathbb{C}) \simeq_{\text{alg}} M_{D_B(e)}(\mathbb{C}), \quad \forall \sigma, \forall X \in M_{D_A(e)}(\mathbb{C}), \quad C_A(e, \sigma, X) = C_B(e, \sigma, \Phi_e(X)).$$

This is the edge-blocked form of the algebra isomorphism $X \mapsto Y$ in [186, Section 3]:



Proof. [✓ Proof](#) For a matrix $X \in M_{D_A(e)}(\mathbb{C})$, realize the insertion of X at the two endpoints of the first edge-blocked chain by physical operators O_1, O_2 . Write C_{mid}^A and C_{mid}^B for the open middle weights of the two edge-blocked chains. For every physical configuration σ , endpoint realization of X and equality of the PEPS states give

$$\begin{aligned} C_A(e, \sigma, X) &= \sum_{\beta} O_1(A_u(\beta_u))_{\sigma_u} C_{\text{mid}}^A(\sigma; \beta_L, \beta_R)(A_v(\beta_v))_{\sigma_v} \\ &= \sum_{\beta} O_1(B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R)(B_v(\beta_v))_{\sigma_v}, \end{aligned}$$

and

$$\begin{aligned} C_A(e, \sigma, X) &= \sum_{\beta} (A_u(\beta_u))_{\sigma_u} C_{\text{mid}}^A(\sigma; \beta_L, \beta_R) O_2(A_v(\beta_v))_{\sigma_v} \\ &= \sum_{\beta} (B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) O_2(B_v(\beta_v))_{\sigma_v}. \end{aligned}$$

Therefore, for every physical configuration σ ,

$$\sum_{\beta} O_1(B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R)(B_v(\beta_v))_{\sigma_v} = \sum_{\beta} (B_u(\beta_u))_{\sigma_u} C_{\text{mid}}^B(\sigma; \beta_L, \beta_R) O_2(B_v(\beta_v))_{\sigma_v}.$$

The positive-bond physical-to-virtual recovery applied to the second edge-blocked chain gives a matrix $Y \in M_{D_B(e)}(\mathbb{C})$ such that $C_A(e, \sigma, X) = C_B(e, \sigma, Y)$ for every physical configuration σ . Thus $X \mapsto Y$ gives the required coefficient correspondence. Denote this correspondence by T . For every physical configuration σ ,

$$\begin{aligned} C_B(e, \sigma, T(X + X')) &= C_A(e, \sigma, X + X') \\ &= C_A(e, \sigma, X) + C_A(e, \sigma, X') \\ &= C_B(e, \sigma, T(X) + T(X')). \end{aligned}$$

Similarly, $T(cX) = cT(X)$. The multiplicative identity is read from the inserted two-endpoint realization:

$$C_B(e, \sigma, T(XX')) = C_A(e, \sigma, XX') = C_B(e, \sigma, T(X)T(X')).$$

The unit satisfies $C_B(e, \sigma, T(1)) = C_A(e, \sigma, 1) = C_B(e, \sigma, 1)$. Theorem 24.3.15 identifies the inserted matrices in these coefficient identities. Exchanging A and B gives the inverse correspondence, hence a $\mathbb{C}$ -algebra isomorphism. $\square$

Theorem 24.3.18 (Matrix-algebra isomorphisms preserve size). Lean ✓ Stmt *If the full matrix algebras $M_D(\mathbb{C})$ and $M_E(\mathbb{C})$ are isomorphic as $\mathbb{C}$ -algebras, then $D = E$.*

Proof. ✓ Proof The isomorphism preserves complex vector-space dimension, and $\dim_{\mathbb{C}} M_D(\mathbb{C}) = D^2$ and $\dim_{\mathbb{C}} M_E(\mathbb{C}) = E^2$, so $D = E$. $\square$

Theorem 24.3.19 (Matrix-algebra isomorphisms are inner after identifying size). Lean ✓ Stmt *Let $\varphi : M_D(\mathbb{C}) \rightarrow M_E(\mathbb{C})$ be a $\mathbb{C}$ -algebra isomorphism. Then $D = E$; let h denote this equality and $\iota_h : M_D(\mathbb{C}) \rightarrow M_E(\mathbb{C})$ the canonical index-relabelling isomorphism. There is $X \in \text{GL}_E(\mathbb{C})$ with $\varphi(M) = X\iota_h(M)X^{-1}$ for all $M \in M_D(\mathbb{C})$.*

Proof. ✓ Proof By Theorem 24.3.18, $D = E$, so ι_h is defined. Then $\varphi \circ \iota_h^{-1}$ is a $\mathbb{C}$ -algebra automorphism of $M_E(\mathbb{C})$, and Skolem–Noether (Theorem [182, “Skolem–Noether”]) gives $X \in \text{GL}_E(\mathbb{C})$ with $(\varphi \circ \iota_h^{-1})(N) = XNX^{-1}$. Evaluating at $N = \iota_h(M)$ gives $\varphi(M) = X\iota_h(M)X^{-1}$. $\square$

Theorem 24.3.20 (Edge gauge from the insertion algebra isomorphism). Lean ✓ Stmt *Suppose the edge insertion correspondence is a $\mathbb{C}$ -algebra isomorphism between the two full matrix algebras on the chosen bond. Then the two bond dimensions on that edge are equal. If $h_e : D_A(e) = D_B(e)$ is this equality, and if $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$ is the induced algebra isomorphism, then, for an invertible edge matrix Z_e , the insertion correspondence is $Y = Z_e\phi_{h_e}(X)Z_e^{-1}$. This is the finite-dimensional matrix-algebra step in [186, Section 3].*

Proof. ✓ Proof Apply Theorem 24.3.19 to the full matrix algebras determined by the two edge bond spaces. $\square$

Theorem 24.3.21 (Per-edge gauge family from insertion algebra isomorphisms). Lean ✓ Stmt *Let A and B be vertex-injective PEPS tensors with the same PEPS state. Suppose the corresponding virtual bond dimensions are equal and every virtual bond of A has positive dimension. Then there is a family $X_e \in \text{GL}(D_A(e))$ such that, for every edge e , there is a $\mathbb{C}$ -algebra isomorphism $\Phi_e : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$ satisfying $C_A(e, \sigma, M) = C_B(e, \sigma, \Phi_e(M))$ for every physical configuration σ and matrix M . Writing $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$ for the algebra isomorphism induced by the dimension equality on e , one has $\Phi_e(M) = \phi_{h_e}(X_e M X_e^{-1})$.*

Proof. ✓ Proof Vertex injectivity and positivity give edge-blocked three-site injectivity for A and B at each edge. Theorem 24.3.17 gives the algebra isomorphism Φ_e preserving inserted coefficients. Theorem 24.3.20 realizes Φ_e by conjugation, and the bond-dimension equality transports the conjugating matrix to the A -side bond. $\square$

Theorem 24.3.22 (Per-edge gauge at a translated horizontal interior edge). Lean ✓ Stmt *Let A and B be normal PEPS on the open $W \times H$ square lattice that satisfy the rectangular-injectivity hypotheses with union closure, have positive physical dimension $d > 0$, have equal and everywhere-positive virtual bond dimensions, and define the same PEPS state. Fix a translated horizontal interior frame with two rows and two columns of margin, that is $2 \leq x_0$, $2 \leq y_0$, $x_0 + 8 \leq W$, and*

$y_0 + 8 \leq H$, and let e be the distinguished horizontal edge of that frame. Then the bond dimensions agree on e , $D_A(e) = D_B(e)$, and there are an invertible edge matrix $Z \in \text{GL}(D_B(e))$ and a forward map fwd given by the conjugation $\text{fwd}(M) = Z\phi_{h_e}(M)Z^{-1}$, where $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$ is the reindexing induced by the bond-dimension equality on e . No single-vertex injectivity is used; the only injectivity hypotheses are the blocked-region injectivities of the interior blocking data and the red and host injectivities of B .

Proof. ✓ Proof The translated horizontal frame supplies, for both A and B , the one-edge interior blocking datum. Because the red, blue, and complementary blocks are coordinate regions, independent of the tensor, the two data share them: $R_A = R_B$, $B_A = B_B$, and $C_A = C_B$. The distinguished edge e is the unique crossing edge of the red and blue blocks. Applying the per-edge gauge construction from a one-edge blocking datum to the two blocking data yields the bond-dimension equality on e and the conjugating gauge Z . $\square$

Theorem 24.3.23 (Per-edge gauge at a translated vertical interior edge). Lean ✓ Stmt *The coordinate-swap counterpart of Theorem 24.3.22. Under the same hypotheses on A and B , for a translated vertical interior frame with two rows and two columns of margin, that is $2 \leq x_0$, $2 \leq y_0$, $x_0 + 8 \leq W$, and $y_0 + 8 \leq H$, with distinguished vertical edge e , the bond dimensions agree on e , $D_A(e) = D_B(e)$, and there are an invertible edge matrix $Z \in \text{GL}(D_B(e))$ and a forward map fwd given by the conjugation $\text{fwd}(M) = Z\phi_{h_e}(M)Z^{-1}$, with $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$ the reindexing induced by the bond-dimension equality on e .*

Proof. Lean Lean ✓ Proof Interchange the two lattice coordinates. This graph isomorphism sends the vertical translated frame to the horizontal translated frame, transposes the rectangular injectivity hypotheses, and sends the distinguished vertical edge to the distinguished horizontal edge. Apply Theorem 24.3.22 to the transported tensors. Transporting its bond-dimension equality and conjugating gauge back along the inverse coordinate swap gives the required equality on e and the matrix Z . $\square$

Definition 24.3.24 (Factorized local gauge datum). Lean ✓ Stmt Fix A, B , a bond-dimension equality, and a vertex v . Let $\mathcal{T}_v^A$ be the local tensor map of A , let L_v^A be the chosen left inverse of $\mathcal{T}_v^A$, and let $\Pi_v^A = \mathcal{T}_v^A L_v^A$. The factorized local gauge datum consists of the following two identities for local virtual configurations η, η' :

$$\begin{aligned}\Pi_v^A B_v(\eta, -) &= B_v(\eta, -), \\ L_v^A B_v(\eta, -)(\eta') &= \prod_{e \ni v} X_e(\eta_e, \eta'_e),\end{aligned}$$

with $X_e \in \text{GL}(D_A(e))$ on each incident edge. This is the local output required from the blocked-middle reduction in [186, Section 3].

Definition 24.3.25 (Blocked-middle local gauge formula). Lean ✓ Stmt The blocked-middle local gauge formula at v is the assertion that there are matrices $X_e \in \text{GL}(D_A(e))$, one on each incident edge, such that for all local virtual configurations η and physical indices σ ,

$$B_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Theorem 24.3.26 (Blocked-middle formula implies factorized local gauge). Lean ✓ Stmt *If the blocked-middle local gauge formula holds at v with matrices X_e , then the factorized local gauge datum holds at v with the same matrices.*

Proof. ✓ Proof For fixed η , set $c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$. The displayed blocked-middle formula is precisely $B_v(\eta, -) = \mathcal{T}_v^A c_\eta$. Hence

$$\begin{aligned} \Pi_v^A B_v(\eta, -) &= \Pi_v^A \mathcal{T}_v^A c_\eta = \mathcal{T}_v^A c_\eta = B_v(\eta, -), \\ L_v^A B_v(\eta, -)(\eta') &= L_v^A \mathcal{T}_v^A c_\eta(\eta') = c_\eta(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e). \end{aligned}$$

□

Theorem 24.3.27 (Blocked-middle formula gives a local gauge formula). Lean ✓ Stmt Suppose that the edge-centred three-site reduction supplies invertible matrices X_e on the incident edges of v such that, for all local virtual configurations η and physical indices σ ,

$$B_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Then the same family X_e is a local gauge at v :

Proof. ✓ Proof By Theorem 24.3.26, $L_v^A B_v(\eta, -)(\eta') = \prod_{e \ni v} X_e(\eta_e, \eta'_e)$. Since $\Pi_v^A B_v(\eta, -) = B_v(\eta, -)$ and $\Pi_v^A = \mathcal{T}_v^A L_v^A$,

$$\begin{aligned} B_v(\eta, \sigma) &= \sum_{\eta'} L_v^A B_v(\eta, -)(\eta') A_v(\eta', \sigma) \\ &= \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma). \end{aligned}$$

□

Theorem 24.3.28 (Absorbing edge gauges into the second PEPS). Lean ✓ Stmt Given edge gauges X_e , set $\tilde{B} = B^X$. Then $D_{\tilde{B}} = D_B$ and, for every vertex v ,

$$\tilde{B}_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{ie} M_{ie}(\eta(ie), \eta'(ie)) \right) B_v(\eta', \sigma),$$

where $M_{ie} = X_e$ or $M_{ie} = (X_e^{-1})^t$ according to the orientation of the endpoint ie . After this absorption, arbitrary insertions of the same matrix on the same edge are compared in the first PEPS and the modified second PEPS.

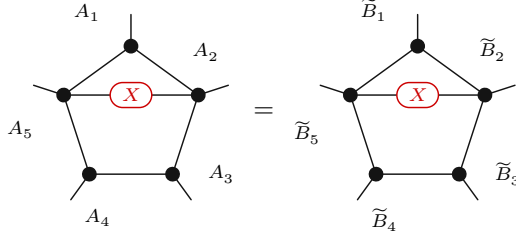
Proof. ✓ Proof With $\tilde{B} = B^X$, Theorem 24.1.2.5 gives $D_{\tilde{B}} = D_B$, and Theorem 24.1.2.6 gives

$$\tilde{B}_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{ie} M_{ie}(\eta(ie), \eta'(ie)) \right) B_v(\eta', \sigma).$$

□

Definition 24.3.29 (Post-absorption inserted-edge comparison). Lean ✓ Stmt For tensor families $A, \tilde{B}$, the post-absorption comparison holds when $h : D_A = D_{\tilde{B}}$ and the following equality holds. For each edge e , write $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_{\tilde{B}}(e)}(\mathbb{C})$ for the matrix-algebra transport induced by $h_e : D_A(e) = D_{\tilde{B}}(e)$. Then, for every matrix $X \in M_{D_A(e)}(\mathbb{C})$ and every physical configuration σ , $C_A(e, X; \sigma) = C_{\tilde{B}}(e, \phi_{h_e}(X); \sigma)$.

Theorem 24.3.30 (Post-absorption edge insertion equality). [Lean](#) [✓ Stmt](#) Let A and B be vertex-injective PEPS with the same state, assume every bond dimension is positive, and assume the bond-dimension equality $h : D_A = D_B$. After the edge gauges have been absorbed into the second tensor family, there are gauges Z_e and $\tilde{B} = B^Z$ such that $D_{\tilde{B}} = D_A$ and, for every edge e , write $\phi_{h_e} : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_{\tilde{B}}(e)}(\mathbb{C})$ for the matrix-algebra transport induced by the corresponding equality $h_e : D_A(e) = D_{\tilde{B}}(e)$. Then, for every matrix $X \in M_{D_A(e)}(\mathbb{C})$ and every physical configuration σ , $C_A(e, X; \sigma) = C_{\tilde{B}}(e, \phi_{h_e}(X); \sigma)$. On the five-site graph used in the proof, this equality reads



as in [186, Section 3, lines 1037–1065]. The positive-bond hypothesis is the source’s standing assumption that injective PEPS have nonzero virtual bond spaces; it is needed because the edge gauges come from blocking the PEPS around each edge into a three-site injective chain.

Proof. [✓ Proof](#) Block the PEPS around each edge into a three-site injective chain and compare the matched matrix insertions: this assigns to every edge an invertible bond matrix X_e realizing the insertion-algebra transport as conjugation. Absorb the transpose of X_e into the second tensor family on that edge. The open-edge gauge action then conjugates each inserted matrix by the transpose of the absorbed gauge, while the insertion-algebra transport conjugates by X_e ; the two conjugations agree because the absorbed gauge is the transpose of X_e , giving the post-absorption insertion equality. $\square$

Theorem 24.3.31 (Local gauge extraction). [Lean](#) [Lean](#) [✓ Stmt](#) Under the factorized local gauge hypothesis, the local gauge formula follows. What remains open is to upgrade equality of the edge-blocked coefficients to the blocked-middle formula using the corresponding three-site injective-chain argument. The physical-realization and physical-to-virtual steps in that argument are recorded separately as Theorems 24.3.6 and 24.3.14. After Theorem 24.3.30 supplies the factorized local gauge hypothesis, injectivity on the star neighborhood provides a left inverse that isolates the local edge gauges. Thus there are incident-edge matrices X_e such that

$$B_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_e(\eta_e, \eta'_e) \right) A_v(\eta', \sigma).$$

Proof. [✓ Proof](#) The factorized local gauge datum already supplies the incident-edge invertible matrices and the factorized local coefficient identity. $\square$

Theorem 24.3.32 (Gauge consistency). [Lean](#) [✓ Stmt](#) Suppose A and B are vertex-injective PEPS tensors with the same PEPS state, that the corresponding virtual bond dimensions are equal, that every virtual bond of A has positive dimension, and that the graph is connected. The connectivity assumption is necessary: on a disconnected graph the per-vertex scalars produced by the reduction below cannot be absorbed into edge gauges, since on the empty graph on two vertices the products of the vertex scalars can agree while no edge gauge relates the two tensors. Choose an edge $e = (u, v)$ and block all other vertices into one middle tensor. The three resulting tensors

form an injective chain, so the one-dimensional injective comparison assigns an invertible matrix to the edge e . Repeating this for every edge and absorbing these matrices into the second tensor family gives the post-absorption insertion equality of [186, Section 3]: for every edge and every matrix, inserting that matrix on the edge in the first PEPS gives the same state as inserting it on the same edge in the modified second PEPS. Applying the local extraction at each vertex then gives one edge gauge family whose endpoint actions transform the first tensor family into the second:

$$B_v(\eta, \sigma) = (A^X)_v(\eta, \sigma) = \sum_{\eta'} \left(\prod_{e \ni v} X_{v,e}(\eta_e, \eta'_e) \right) A_v(\eta', \sigma). \quad (24.1)$$

Proof. ✓ Proof Absorbing the edge gauges into the second tensor family gives the post-absorption insertion equality. On each vertex with an incident edge, the one-vertex-versus-complement comparison supplies a nonzero scalar relating the first tensor to the absorbed second tensor. Because the state is nonzero, these per-vertex scalars multiply to one, so on the connected graph a spanning-tree construction realizes their reciprocals as the oriented incidence product of one edge-scalar family. Folding those scalars into the inverse of the absorbed edge gauges produces the global gauge. On a one-vertex graph there are no edges and the gauge is trivial. $\square$

Remark 24.3.33. The gauge transformation (24.1) expresses the algebraic content of the edge-gauge construction in [186, Section 3]. On an ordered edge $e = (u, w)$, the actions X_e at u and $(X_e^{-1})^\top$ at w cancel upon contraction. Absorbing these matrices into B gives $\widetilde{B}$, for which every inserted-edge coefficient agrees with the corresponding coefficient of A . Comparing one vertex with its injective complement then yields the local gauge equation. Repeating the argument on every edge produces a single compatible gauge family; the two-injective comparison below shows that the remaining three-leg operator is scalar.

Definition 24.3.34 (Left-identity three-leg residual form). Lean ✓ Stmt For three virtual leg spaces α, β, γ and an endomorphism Z of $\beta \otimes \gamma$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as the identity on the α leg and as Z on the remaining two legs. This is the $\mathbb{1}_\alpha \otimes Z$ form appearing after the first tensor is inverted in [186, Section 3].

Definition 24.3.35 (Right-identity three-leg residual form). Lean ✓ Stmt For an endomorphism U of $\alpha \otimes \beta$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as U on the first two legs and as the identity on the γ leg. This is the $U \otimes \mathbb{1}_\gamma$ form in [186, Section 3].

Definition 24.3.36 (Middle-identity three-leg residual form). Lean ✓ Stmt For an endomorphism W of $\alpha \otimes \gamma$, define the induced endomorphism of $\alpha \otimes \beta \otimes \gamma$ by acting as W on the outer two legs and as the identity on the middle β leg, with matrix entries $\delta_{bb'} W_{ac, a'c'}$ for $(a, b, c), (a', b', c') \in \alpha \times \beta \times \gamma$. This is the third residual form in [186, Section 3].

Theorem 24.3.37 (Scalar reduction of the residual two-tensor gauges). Lean ✓ Stmt For nonzero virtual leg spaces α, β, γ , let $Z \in \text{End}(\beta \otimes \gamma)$, $U \in \text{End}(\alpha \otimes \beta)$, and $W \in \text{End}(\alpha \otimes \gamma)$ have equal induced three-leg residual forms (Definitions 24.3.34–24.3.36) on $\alpha \otimes \beta \otimes \gamma$, so that

$$\delta_{aa'} Z_{bc, b'c'} = U_{ab, a'b'} \delta_{cc'} = W_{ac, a'c'} \delta_{bb'}$$

for all $a, a' \in \alpha$, $b, b' \in \beta$, $c, c' \in \gamma$. Then there exists $\lambda \in \mathbb{C}$ such that all three residual matrices are scalar:

$$Z = \lambda \mathbb{1}_{\beta \otimes \gamma}, \quad U = \lambda \mathbb{1}_{\alpha \otimes \beta}, \quad W = \lambda \mathbb{1}_{\alpha \otimes \gamma}.$$

Local fix (nonzero virtual legs): The scalar reduction requires that each virtual leg space α, β, γ be nonzero; if any leg space is empty, the compatibility conditions hold vacuously and admit non-scalar solutions (such as $Z = \text{diag}(1, 2)$ when $\alpha = \emptyset$, $|\beta| = 1$, $|\gamma| = 2$). This models the nonzero-dimensional virtual bond spaces of [186, Section 3]; documented in [73].

Proof. ✓ Proof Write the common operator as T . The equality $\delta_{aa'} Z_{bc, b'c'} = U_{ab, a'b'} \delta_{cc'}$ gives $Z_{bc, b'c'} = 0$ whenever $c \neq c'$ (by taking $a = a'$). Hence Z is diagonal in the γ -index. Similarly, the equality $\delta_{aa'} Z_{bc, b'c'} = W_{ac, a'c'} \delta_{bb'}$ gives $Z_{bc, b'c'} = 0$ whenever $b \neq b'$. Thus $Z_{bc, b'c'} = 0$ unless $b = b'$ and $c = c'$.

It remains to compare the surviving diagonal entries. From $\delta_{aa'} Z_{bc, bc} = U_{ab, a'b}$, the value $Z_{bc, bc}$ is independent of c . From the middle-identity equality it is also independent of b . Therefore $Z = \lambda \mathbb{1}_{\beta \otimes \gamma}$. Substitution into the compatibility equalities gives $U = \lambda \mathbb{1}_{\alpha \otimes \beta}$ and $W = \lambda \mathbb{1}_{\alpha \otimes \gamma}$. $\square$

Theorem 24.3.38 (Many-leg scalar extraction). Lean ✓ Stmt Let a finite family of legs be given, each a nonzero finite index set, with invertible matrices M_i and N_i on each leg. If $\prod_i (M_i)_{p_i q_i} = \prod_i (N_i)_{p_i q_i}$ for every configuration pair (p, q) , then there are nonzero scalars c_i with $M_i = c_i N_i$ for each leg and $\prod_i c_i = 1$.

Proof. ✓ Proof Each N_i is invertible, so each of its rows has a nonzero entry; pick a reference configuration (p^0, q^0) on which every $(N_i)_{p_i^0 q_i^0}$ is nonzero. The product identity at (p^0, q^0) then makes every $(M_i)_{p_i^0 q_i^0}$ nonzero as well. Fix a leg i and a pair (a, b) , and apply the identity to the configuration that agrees with the reference except at leg i , where it takes (a, b) . Splitting off leg i gives $(M_i)_{ab} K_i = (N_i)_{ab} L_i$, where K_i, L_i are the products of the other legs at the reference, both nonzero. The reference identity at leg i relates K_i and L_i , and eliminating them yields $(M_i)_{ab} = c_i (N_i)_{ab}$ with $c_i = (M_i)_{p_i^0 q_i^0} / (N_i)_{p_i^0 q_i^0}$, independent of (a, b) . Taking the product of the scalars over all legs and using the reference identity gives $\prod_i c_i = 1$. $\square$

Theorem 24.3.39 (Pointwise product cancellation for two injective blocks). Lean ✓ Stmt Let A_1, A_2, B_1, B_2 be tensors with matching external, shared-boundary, and physical index sets. Assume that the two external boundary spaces and the shared-boundary configuration space are nonempty, and that A_1 and A_2 are injective. Suppose that, for every pair of external indices, every pair of shared-boundary configurations, and every pair of physical indices,

$$A_1(\eta_1, \mu, \sigma_1) A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1, \mu, \sigma_1) B_2(\eta_2, \nu, \sigma_2).$$

Then there exists $\lambda \in \mathbb{C}^\times$ such that $A_1 = \lambda B_1$ and $A_2 = \lambda^{-1} B_2$.

Proof. ✓ Proof Choose external and shared-boundary indices, and use injectivity of A_1 and A_2 to choose physical indices for which the two chosen coefficients are nonzero. The product equality then shows that the corresponding coefficients of B_1 and B_2 are also nonzero. Define $\lambda = B_2(\eta_2^0, \nu^0, \sigma_2^0) / A_2(\eta_2^0, \nu^0, \sigma_2^0)$. For every (η_1, μ, σ_1) , the product identity with $(\eta_2^0, \nu^0, \sigma_2^0)$ fixed gives

$$A_1(\eta_1, \mu, \sigma_1) A_2(\eta_2^0, \nu^0, \sigma_2^0) = B_1(\eta_1, \mu, \sigma_1) B_2(\eta_2^0, \nu^0, \sigma_2^0).$$

Dividing by $A_2(\eta_2^0, \nu^0, \sigma_2^0) \neq 0$ gives $A_1 = \lambda B_1$. Fixing $(\eta_1^0, \mu^0, \sigma_1^0)$ and substituting $A_1 = \lambda B_1$ into the product identity gives

$$\lambda B_1(\eta_1^0, \mu^0, \sigma_1^0) A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1^0, \mu^0, \sigma_1^0) B_2(\eta_2, \nu, \sigma_2).$$

Dividing by $B_1(\eta_1^0, \mu^0, \sigma_1^0) \neq 0$ gives $A_2 = \lambda^{-1} B_2$. $\square$

Definition 24.3.40 (Matrix unit for coefficient extraction). ✓ ML stmt For indices i and j , let E_{ij} denote the matrix whose only nonzero entry is the (i, j) entry, where it is equal to 1.

Theorem 24.3.41 (Coefficient extraction for one shared bond). Lean ✓ Stmt Suppose that the two blocks share a single virtual bond. Then inserting the matrix unit $E_{\mu, \nu}$ in that bond extracts the corresponding product of tensor coefficients:

$$C_{A_1, A_2}(b, E_{\mu, \nu}; \eta_1, \eta_2, \sigma_1, \sigma_2) = A_1(\eta_1, \mu, \sigma_1) A_2(\eta_2, \nu, \sigma_2).$$

Proof. ✓ Proof Since there is only one shared bond, the condition that the two shared configurations agree away from the distinguished bond is vacuous, and

$$C_{A_1, A_2}(b, E_{\mu, \nu}; \eta_1, \eta_2, \sigma_1, \sigma_2) = \sum_{\mu', \nu'} (E_{\mu, \nu})_{\mu'(b), \nu'(b)} A_1(\eta_1, \mu', \sigma_1) A_2(\eta_2, \nu', \sigma_2).$$

The entries of the matrix unit satisfy $(E_{\mu, \nu})_{i, j} = \delta_{i, \mu(b)} \delta_{j, \nu(b)}$. Thus the only nonzero summand is the one with $(\mu', \nu') = (\mu, \nu)$. □

Theorem 24.3.42 (One-bond two-injective comparison). Lean ✓ Stmt Suppose that the two injective blocks have exactly one shared virtual bond and nonempty external boundary spaces. If all matrix insertions on that bond give equal two-block coefficients for the pairs (A_1, A_2) and (B_1, B_2) , then there exists $\lambda \in \mathbb{C}^\times$ such that $A_1 = \lambda B_1$ and $A_2 = \lambda^{-1} B_2$.

Proof. ✓ Proof Insert the matrix unit $E_{\mu, \nu}$ and use the coefficient-extraction theorem to obtain the pointwise product equality

$$A_1(\eta_1, \mu, \sigma_1) A_2(\eta_2, \nu, \sigma_2) = B_1(\eta_1, \mu, \sigma_1) B_2(\eta_2, \nu, \sigma_2)$$

for all indices. The pointwise product cancellation theorem gives the reciprocal scalar proportionality. □

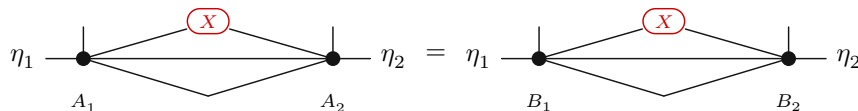
Theorem 24.3.43 (Generalized two-injective-tensor comparison). Lean Lean Lean Lean Lean Lean ✓ Stmt Let A_1, A_2, B_1, B_2 be injective tensors with the same shared virtual bonds and nonempty external boundary spaces, and assume that there is at least one shared virtual bond. For the set $\mathcal{B}$ of shared bonds, a bond $b \in \mathcal{B}$, and $X \in M_{D_b}(\mathbb{C})$, set

$$C_{A_1, A_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2) = \sum_{\substack{\mu, \nu \\ \mu|_{\mathcal{B} \setminus \{b\}} = \nu|_{\mathcal{B} \setminus \{b\}}}} X_{\mu_b, \nu_b} A_1(\eta_1, \mu, \sigma_1) A_2(\eta_2, \nu, \sigma_2).$$

Assume, for every $b, X, \eta_1, \eta_2, \sigma_1, \sigma_2$, that

$$C_{A_1, A_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2) = C_{B_1, B_2}(b, X; \eta_1, \eta_2, \sigma_1, \sigma_2).$$

The following diagram records the three-shared-bond proof subcase used in the source, with the external boundary indices restored:



Then there exists $\lambda \in \mathbb{C}^\times$ such that $A_1 = \lambda B_1$ and $A_2 = \lambda^{-1} B_2$. This is the blueprint form of the two-injective tensor comparison in [186, Section 3].

Proof. ✓ Proof Putting $X = \mathbb{1}_{D_b}$ on every shared bond contracts all bonds diagonally and, by the insertion equality, equates the two fully contracted states. Writing each contracted state as a bipartite operator with the linearly independent Schmidt families supplied by injectivity, the operator-Schmidt uniqueness of [186, Section 3] produces an invertible gauge g on the shared-bond configurations with

$$\begin{aligned} A_1(\eta_1, \mu, \sigma_1) &= \sum_{\nu} g_{\mu, \nu} B_1(\eta_1, \nu, \sigma_1), \\ B_2(\eta_2, \nu, \sigma_2) &= \sum_{\mu} g_{\mu, \nu} A_2(\eta_2, \mu, \sigma_2). \end{aligned}$$

Substitute both gauge equations into the insertion equality for the matrix unit $X = E_{p,q}$ on a single bond b , which frees that bond and contracts the others. Separating the linearly independent families B_1 and A_2 gives, for every bond b , every pair of endpoints p, q , and all configurations ν, ρ ,

$$[\rho_b = q] g_{\rho^{b \rightarrow p}, \nu} = [\nu_b = p] g_{\rho, \nu^{b \rightarrow q}}. \quad (24.2)$$

Here $\rho^{b \rightarrow p}$ replaces the value of ρ at b by p . This is the bond-wise form of the residual identity of [186, Section 3], where it is formulated as the compatibility of the three residual gauges Z, U, W (Theorem 24.3.37). Reading (24.2) at the two endpoint choices shows that g vanishes whenever two configurations differ at a bond and that its diagonal value is invariant under changing any single bond. Hence $g = \lambda \mathbb{1}$ for a scalar λ , and invertibility forces $\lambda \neq 0$.

Reinserting $g = \lambda \mathbb{1}$ into the two gauge equations gives $A_1 = \lambda B_1$ and $B_2 = \lambda A_2$, that is $A_2 = \lambda^{-1} B_2$. $\square$

24.4 Fundamental Theorem for Injective PEPS

The edge construction is transported from single edges to arbitrary regions. Reading a region R and its complement $V \setminus R$ as two-block tensors, when the blocked tensors of B on R and $V \setminus R$ are injective and the bonds are positive, the region-insertion identities determine the boundary conjugation uniquely: two invertible matrices Z_1, Z_2 realizing the same insertion identity satisfy $Z_1 M Z_1^{-1} = Z_2 M Z_2^{-1}$ for every boundary matrix M . Comparing one vertex v against its complement then upgrades this to $A_v = \lambda_v \widetilde{B}_v$ at each vertex, once the edge gauges have been absorbed into $\widetilde{B}$, and the per-edge gauges combine into one gauge relating the two tensors. This establishes the Fundamental Theorem for injective PEPS (Theorem 24.4.1.23), following [186, Theorem 2].

Definition 24.4.1 (Region two-block tensor). Lean ✓ Stmt For a finite region $R \subseteq V$, the blocked tensor of R may be read as a two-block tensor with a one-point external boundary and shared bonds the edges crossing the boundary of R : $B_{A,R}(*, \eta, \sigma) = T_{A,R}^{\eta}(\sigma)$. This is the region block used in the one-block-against-complement step of [186, Section 3].

Definition 24.4.2 (Complement-boundary identification). Lean Lean Lean ✓ Stmt The edges crossing the boundary of R are the same edges crossing the boundary of $V \setminus R$. Thus a boundary configuration η for R determines the boundary configuration η^c for the complement by reading the same edge labels through this identification.

Definition 24.4.3 (Complement-region two-block tensor). Lean ✓ Stmt For a finite region $R \subseteq V$, the complement block $V \setminus R$ is a two-block tensor over the same crossing bonds: $B_{A,R^c}(*, \eta, \tau) = T_{A,V \setminus R}^{\eta^c}(\tau)$. This is the complementary block in the region/complement comparison of [186, Section 3].

Theorem 24.4.4 (Injectivity of the complement-region two-block tensor). [Lean](#) [✓ Stmt](#) *If the blocked tensor family of $V \setminus R$ is injective, then the complement-region two-block tensor is injective: $\text{inj}(T_{A,V \setminus R}) \implies \text{inj}(B_{A,R^c})$.*

Proof. [✓ Proof](#) The one-point external boundary does not change the coefficient family, and the complement-boundary identification only reindexes the crossing bonds. More explicitly, if η_1, η_2 are boundary configurations on ∂R , then $\eta_1^c = \eta_2^c \implies \eta_1 = \eta_2$. Thus the coefficients of $T_{A,V \setminus R}$ and B_{A,R^c} differ only by an injective relabelling of boundary configurations. $\square$

Definition 24.4.5 (Region insertion coefficient). [Lean](#) [✓ Stmt](#) Let f be an edge crossing the boundary of R , and let $M \in M_{D_f}(\mathbb{C})$. The region insertion coefficient is

$$C_A(R, f, M; \sigma, \tau) = \sum_{\substack{\mu, \nu \\ \mu|_{\partial R \setminus \{f\}} = \nu|_{\partial R \setminus \{f\}}}} M_{\mu_f, \nu_f} T_{A,R}^\mu(\sigma) T_{A,V \setminus R}^{\nu^c}(\tau).$$

It is the coefficient obtained by inserting M on the boundary bond f between the region R and its complement.

Theorem 24.4.6 (Region insertion coefficients determine the inserted matrix). [Lean](#) [✓ Stmt](#) *Suppose that the blocked tensor maps of A on R and on $V \setminus R$ are injective, and that every bond dimension of A is positive. If $f \in \partial R$ and $M, M' \in M_{D_A(f)}(\mathbb{C})$ satisfy $C_A(R, f, M; \sigma, \tau) = C_A(R, f, M'; \sigma, \tau)$ for every physical configuration σ on R and τ on $V \setminus R$, then $M = M'$.*

Proof. [✓ Proof](#) Subtracting the two coefficients gives $C_A(R, f, M - M'; \sigma, \tau) = 0$ for every σ, τ . Injectivity of the region and complement blocked tensor maps reads off every entry of $M - M'$ from configurations agreeing away from f , so $M - M' = 0$ and hence $M = M'$. $\square$

Definition 24.4.7 (Bond-inserted region insert). [Lean](#) [✓ Stmt](#) Let f be an edge crossing the boundary of R , and let $M \in M_{D_f}(\mathbb{C})$. The bond-inserted region insert $I_{A,R,f,M}$ is the insert on R defined by

$$(I_{A,R,f,M})_{\nu, \sigma} = \sum_{\substack{\mu \in \partial R \\ \mu|_{\partial R \setminus \{f\}} = \nu|_{\partial R \setminus \{f\}}}} M_{\mu_f, \nu_f} T_{A,R}^\mu(\sigma).$$

Thus the complement boundary value ν is fixed away from f , while the matrix M couples the two f -labels. This is the one-bond insertion in the region/complement coefficient of [186, Section 3].

Theorem 24.4.8 (Bond insertion as a deformed region state). [Lean](#) [✓ Stmt](#) *For every physical configuration σ on R and τ on $V \setminus R$, the deformed state of the bond-inserted region insert is the region insertion coefficient: $\Psi_{A,R,I_{A,R,f,M}}(\sigma, \tau) = C_A(R, f, M; \sigma, \tau)$.*

Proof. [✓ Proof](#) Expanding the deformed state gives

$$\Psi_{A,R,I_{A,R,f,M}}(\sigma, \tau) = \sum_{\nu \in \partial R} I_{A,R,f,M}(\nu, \sigma) T_{A,V \setminus R}^{\nu^c}(\tau).$$

Substituting the definition of $I_{A,R,f,M}$ and exchanging the two finite sums gives

$$\Psi_{A,R,I_{A,R,f,M}}(\sigma, \tau) = \sum_{\nu \in \partial R} \left(\sum_{\substack{\mu \in \partial R \\ \mu|_{\partial R \setminus \{f\}} = \nu|_{\partial R \setminus \{f\}}}} M_{\mu_f, \nu_f} T_{A,R}^\mu(\sigma) \right) T_{A,V \setminus R}^{\nu^c}(\tau) = C_A(R, f, M; \sigma, \tau).$$

$\square$

For a region R , a physical configuration σ on R , and a physical configuration τ on $V \setminus R$, write $\omega_{R,\sigma,\tau}$ for the physical configuration on V with

$$\omega_{R,\sigma,\tau}(v) = \begin{cases} \sigma(v), & v \in R, \\ \tau(v), & v \notin R. \end{cases}$$

Theorem 24.4.9 (Region insertion as an edge insertion). [Lean] [✓ Stmt] *Let f be a boundary edge of R , let $N \in M_{D_B(f)}(\mathbb{C})$, let σ be a physical configuration on R , and let τ be a physical configuration on $V \setminus R$. Put $p_B(R) = \prod_{g \notin \partial R} D_B(g)$. Then the region insertion coefficient factors through the edge-inserted coefficient:*

$$C_B(R, f, N; \sigma, \tau) = p_B(R) E_B(f, \omega_{R,\sigma,\tau}, \mathcal{O}_{R,f}(N)).$$

Here $\mathcal{O}_{R,f}$ is the boundary-edge orientation, equal to the identity if the left endpoint of f lies in R and to transpose otherwise.

Proof. [✓ Proof] Let $\mathcal{C}_{R,f}(\sigma, \tau)$ denote the open-edge configurations compatible with the two physical configurations and with the two boundary readings agreeing away from f . For $\xi \in \mathcal{C}_{R,f}(\sigma, \tau)$, let $\text{Fib}_{R,f}(\xi)$ be the corresponding fibre of agreeing boundary readings. Grouping the expansion of $C_B(R, f, N; \sigma, \tau)$ by such an open-edge configuration ξ gives

$$\begin{aligned} C_B(R, f, N; \sigma, \tau) &= \sum_{\xi \in \mathcal{C}_{R,f}(\sigma, \tau)} \# \text{Fib}_{R,f}(\xi) N_{\xi_{R,f}^{(1)}, \xi_{R,f}^{(2)}} \prod_{v \in V} B_v(\xi_v, \omega_{R,\sigma,\tau}(v)), \\ \# \text{Fib}_{R,f}(\xi) &= \prod_{g \notin \partial R} D_B(g) = p_B(R). \end{aligned}$$

Substituting this fibre cardinality gives

$$\begin{aligned} C_B(R, f, N; \sigma, \tau) &= p_B(R) \sum_{\xi \in \mathcal{C}_{R,f}(\sigma, \tau)} N_{\xi_{R,f}^{(1)}, \xi_{R,f}^{(2)}} \prod_{v \in V} B_v(\xi_v, \omega_{R,\sigma,\tau}(v)) \\ &= p_B(R) E_B(f, \omega_{R,\sigma,\tau}, \mathcal{O}_{R,f}(N)). \end{aligned}$$

The final step is Definition 24.2.9 of the edge-inserted coefficient, with the inserted matrix read in the boundary-edge orientation. □

Theorem 24.4.10 (Bond insertion through a corner extension). [Lean] [✓ Stmt] *Let $R \subseteq S$, and suppose that all bond dimensions are positive. For every global physical configuration σ ,*

$$C_A(R, f, M; \sigma|_R, \sigma|_{V \setminus R}) = \Psi_{A,S, \text{Ext}_{R \subseteq S}(I_{A,R,f,M})}(\sigma).$$

In particular, the one-bond insertion on R may be read after extending the corresponding insert to any larger region S .

Proof. [✓ Proof] The preceding theorem and state preservation for corner extension give

$$C_A(R, f, M; \sigma|_R, \sigma|_{V \setminus R}) = \Psi_{A,R, I_{A,R,f,M}}(\sigma|_R, \sigma|_{V \setminus R}) = \Psi_{A,S, \text{Ext}_{R \subseteq S}(I_{A,R,f,M})}(\sigma).$$

□

Theorem 24.4.11 (Region insertion as a two-block insertion). [Lean] [✓ Stmt] *The abstract two-block insertion coefficient of the pair $(B_{A,R}, B_{A,R^c})$ is the region insertion coefficient:*

$$C_{B_{A,R}, B_{A,R^c}}(f, M; *, *, \sigma, \tau) = C_A(R, f, M; \sigma, \tau).$$

Proof. ✓ Proof Expanding the abstract two-block coefficient gives exactly the two boundary sums in the displayed formula for $C_A(R, f, M; \sigma, \tau)$. $\square$

Theorem 24.4.12 (Region insertion under bond-dimension transport). Lean ✓ Stmt *If a tensor B is transported along an equality of bond dimensions, then the region insertion coefficient is transported by the induced matrix-algebra identification on the inserted bond: $C_{B^h}(R, f, N; \sigma, \tau) = C_B(R, f, \phi_f(N); \sigma, \tau)$.*

Proof. ✓ Proof Reindex both boundary-configuration sums by the bond-dimension equivalences. The constraint away from f and the two blocked-region weights are preserved, while the inserted matrix entry becomes the transported entry $\phi_f(N)$. $\square$

Theorem 24.4.13 (Same two-block insertions from region insertions). Lean ✓ Stmt *Suppose A and B have identified bond dimensions. Write B^h for the transported tensor, and write ϕ_f for the induced transport on the boundary bond f . If $C_A(R, f, N; \sigma, \tau) = C_B(R, f, \phi_f(N); \sigma, \tau)$ for every boundary edge f , matrix N , and physical configurations σ, τ , then the two pairs $(B_{A,R}, B_{A,R^c})$ and $(B_{B^h,R}, B_{B^h,R^c})$ have the same one-bond insertion coefficients after the same bond-dimension transport.*

Proof. ✓ Proof Rewrite the two abstract insertion coefficients as region insertion coefficients and use the bond-dimension transport formula:

$$\begin{aligned} C_{B_{A,R}, B_{A,R^c}}(f, N; *, *, \sigma, \tau) &= C_A(R, f, N; \sigma, \tau) = C_B(R, f, \phi_f(N); \sigma, \tau) \\ &= C_{B_{B^h,R}, B_{B^h,R^c}}(f, N; *, *, \sigma, \tau). \end{aligned}$$

$\square$

Theorem 24.4.14 (Swapping the two blocks in a one-bond insertion). Lean ✓ Stmt *Let B_1, B_2 be two blocks joined along a shared bond f . Then inserting a matrix X while contracting B_1 against B_2 is the same as contracting B_2 against B_1 with the transposed insertion:*

$$C_{B_1, B_2}(f, X; \eta_1, \eta_2, \sigma_1, \sigma_2) = C_{B_2, B_1}(f, X^T; \eta_2, \eta_1, \sigma_2, \sigma_1).$$

Proof. ✓ Proof Exchange the two boundary-configuration sums. The condition that the two configurations agree away from f is symmetric, and $(X^T)_{\nu_f, \mu_f} = X_{\mu_f, \nu_f}$. $\square$

Theorem 24.4.15 (Complement-side reading of a region insertion). Lean ✓ Stmt *For a boundary edge $f \in \partial R$, the region-inserted coefficient can be read by contracting the complement block first:*

$$C_A(R, f, X; \sigma, \tau) = C_{A, R^c; R}(f, X^T; \tau, \sigma).$$

Here $C_{A, R^c; R}$ denotes the same one-bond insertion coefficient with the two blocks R^c and R interchanged.

Proof. ✓ Proof Identify $C_A(R, f, X; \sigma, \tau)$ with the two-block insertion of $(B_{A,R}, B_{A,R^c})$, then apply Theorem 24.4.14. $\square$

Theorem 24.4.16 (Incident form gives the transferred endpoint form). Lean ✓ Stmt *Assume that A 's and B 's components at the in-region endpoint of f are linearly independent, and that $D_B(e) > 0$ for every edge e . Let W be the virtual pullback, at the in-region endpoint of f , of the physical operator obtained by inserting X^T on f . If $W = J_f(P)$ is in incident-matrix form on the boundary leg f , then the transfer matrix read off from W reinserts to W .*

Proof. ✓ Proof The transfer matrix is defined by reading the incident matrix out of W . Reading $\mathcal{J}_f(P)$ gives P , and reinserting P returns W . $\square$

Theorem 24.4.17 (Virtual pullback realizes the endpoint operator). Lean ✓ Stmt Assume that A and B have the same PEPS state, are vertex-injective, and have positive bond dimensions. Let v be the in-region endpoint of f , and let W be the virtual pullback, for B at v , of the physical operator obtained by inserting X^T on f . Then for every local virtual coefficient c , $\mathcal{T}_{B,v}(Wc) = O_{A,R,f}(X^T)\mathcal{T}_{B,v}(c)$.

Proof. ✓ Proof The physical endpoint operator preserves the image of $\mathcal{T}_{B,v}$. The chosen left inverse defining the virtual pullback therefore realizes the same action after applying $\mathcal{T}_{B,v}$. $\square$

Definition 24.4.18 (Out-of-region endpoint and blocked-region inverse). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The out-of-region endpoint of $f \in \partial R$ is the endpoint outside R . It is the in-region endpoint of the same edge viewed as an edge of $\partial(V \setminus R)$. The blocked-region map is $c \mapsto \sum_{\mu \in \partial R} c_\mu T_{A,R}^\mu$, and injectivity of the blocked tensor gives a left inverse. In particular, $L_{A,R}(T_{A,R}^\mu) = e_\mu$.

Theorem 24.4.19 (Complement-side cast and endpoint realization). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt With f' the boundary edge f read on $V \setminus R$, one has

$$C_A(R, f, X; \sigma, \tau) = C_A(V \setminus R, f', X^T; \tau, \sigma^{cc}),$$

where σ^{cc} is σ transported across $V \setminus (V \setminus R) = R$. Consequently the same coefficient is realized through the endpoint of f lying outside R .

Proof. ✓ Proof Reindex the two boundary-configuration sums by the boundary-edge equivalence between R and its complement. The double complement carries the blocked weight of $V \setminus (V \setminus R)$ back to the blocked weight of R . Applying the usual endpoint realization to $V \setminus R$ gives the displayed out-of-region endpoint realization. $\square$

Theorem 24.4.20 (Region resonate identity and blocked read-offs). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Assume that A 's components at the in-region endpoint of f and at the in-complement-region endpoint of the complementary boundary edge are linearly independent, and that A and B define the same PEPS state. Let P_R and P_{R^c} be the interior bond products of R and R^c . Let $O_f^{\text{in}}(X)$ and $O_f^{\text{out}}(X)$ be the physical endpoint operators obtained from the two virtual insertions, and let $\Psi_B(R; \sigma, \tau)$ and $\Psi_B(R^c; \tau, \sigma^{cc})$ be the corresponding closed B -state vectors. Then the two endpoint readings of the same inserted coefficient give

$$P_R O_f^{\text{in}}(X) \Psi_B(R; \sigma, \tau) = P_{R^c} O_f^{\text{out}}(X) \Psi_B(R^c; \tau, \sigma^{cc}).$$

With $\rho_{X,\sigma}^{R^c}$ and $\rho_{X,\tau}^R$ denoting the two row functions, one has

$$\begin{aligned} C_A(R, f, X; \sigma, -) &= T_{A,R^c}(\rho_{X,\sigma}^{R^c}), \\ C_A(R, f, X; -, \tau) &= T_{A,R}(\rho_{X,\tau}^R). \end{aligned}$$

Hence the blocked left inverses read off the rows:

$$\begin{aligned} L_{A,R^c}(C_A(R, f, X; \sigma, -)) &= \rho_{X,\sigma}^{R^c}, \\ L_{A,R}(C_A(R, f, X; -, \tau)) &= \rho_{X,\tau}^R. \end{aligned}$$

Proof. ✓ Proof The in-region and out-of-region endpoint pins both evaluate to $C_A(R, f, X; \sigma, \tau)$, hence they agree. Expanding $C_A(R, f, X; \sigma, \tau)$ as a sum over boundary configurations isolates, respectively, the R^c -blocked and R -blocked tensor maps; the left inverses then read off the row functions. $\square$

Definition 24.4.21 (Incident form of the region resonance). Lean ✓ Stmt The incident-form condition associated with the region resonance asserts that, for every matrix X inserted on f , the virtual pullback W_X of the in-region endpoint operator is in incident-matrix form, namely $\exists Y: W_X = \mathcal{J}_f(Y)$.

Definition 24.4.22 (Region insertion transfer). Lean ✓ Stmt A region insertion transfer along a boundary edge $f \in \partial R$ consists of maps $T_f: M_{D_A(f)}(\mathbb{C}) \rightarrow M_{D_B(f)}(\mathbb{C})$ and $S_f: M_{D_B(f)}(\mathbb{C}) \rightarrow M_{D_A(f)}(\mathbb{C})$. For every inserted matrix and every pair of physical configurations, these maps satisfy

$$\begin{aligned} C_A(R, f, X; \sigma, \tau) &= C_B(R, f, T_f(X); \sigma, \tau), \\ C_B(R, f, Y; \sigma, \tau) &= C_A(R, f, S_f(Y); \sigma, \tau). \end{aligned}$$

The forward map is required to satisfy $T_f(XY) = T_f(X)T_f(Y)$ and $T_f(\mathbb{1}) = \mathbb{1}$.

Definition 24.4.23 (Region physical-to-virtual realization). Lean ✓ Stmt Let v be the endpoint of f lying in R . A matrix transfer T_f is realized on the region if, for every matrix X inserted on A 's boundary bond and every local virtual coefficient c for B at v ,

$$O_{A,R,f}(X^T) \mathcal{T}_{B,v}(c) = \mathcal{T}_{B,v}(\mathcal{J}_f(T_f(X)^T)c).$$

This is the region version of the physical-to-virtual insertion relation in [186, Section 3, lines 254–582].

Theorem 24.4.24 (Realized region transfers are multiplicative and unital). Lean Lean ✓ Stmt If a region matrix transfer is realized in the sense of Definition 24.4.23, then its forward map is multiplicative: $T_f(XY) = T_f(X)T_f(Y)$. If in addition the two PEPS tensors have the same state and the same bond dimensions, all bond dimensions of B are positive, and the blocked tensors of B on R and its complement are injective, then the identity insertion gives $T_f(\mathbb{1}) = \mathbb{1}$.

Proof. ✓ Proof For multiplicativity, the physical insertion for $(XY)^T$ is the composite of the physical insertions for Y^T and X^T . Realizing these insertions on the B -side gives two incident-matrix actions on the same local tensor image, $\mathcal{J}_f(T_f(XY)^T) = \mathcal{J}_f((T_f(X)T_f(Y))^T)$. Reading the incident matrix on the boundary leg gives $T_f(XY) = T_f(X)T_f(Y)$. For the identity, the coefficient $C_A(R, f, \mathbb{1}; \sigma, \tau)$ is the interior bond product times the state coefficient of A , hence equals $C_B(R, f, \mathbb{1}; \sigma, \tau)$ by equality of states and bond dimensions. The two blocked-injectivity hypotheses and positivity give injectivity of the B -side region-inserted coefficient, whence $T_f(\mathbb{1}) = \mathbb{1}$. $\square$

Definition 24.4.25 (Region insertion transfer from realized maps). Lean ✓ Stmt Suppose the forward and reverse boundary matrix transfers are realized on the two tensors, the two tensors have the same state and the same bond dimensions, and the region and complement-region blocked maps of B are injective. Then the maps T_f and S_f form a region insertion transfer:

$$\begin{aligned} C_A(R, f, X; \sigma, \tau) &= C_B(R, f, T_f(X); \sigma, \tau), \\ C_B(R, f, Y; \sigma, \tau) &= C_A(R, f, S_f(Y); \sigma, \tau), \end{aligned}$$

together with $T_f(XY) = T_f(X)T_f(Y)$ and $T_f(\mathbb{1}) = \mathbb{1}$.

Theorem 24.4.26 (Incident form gives the region insertion transfer and gauge). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Assume that A and B have the same PEPS state, are vertex-injective, have positive bond dimensions, and have injective blocked maps for R and R^c . If the incident-form condition holds in both directions, then the matrix reading gives maps $T_f : M_{D_A(f)}(\mathbb{C}) \rightarrow M_{D_B(f)}(\mathbb{C})$ and $S_f : M_{D_B(f)}(\mathbb{C}) \rightarrow M_{D_A(f)}(\mathbb{C})$ realizing all region insertion coefficients. If, in addition, $D_A(e) = D_B(e)$ for every edge e , then there is an invertible gauge $Z_f \in \text{GL}(D_B(f))$ such that $T_f(X) = Z_f \iota_f(X) Z_f^{-1}$, where $\iota_f : M_{D_A(f)}(\mathbb{C}) \simeq M_{D_B(f)}(\mathbb{C})$ is the identification induced by the equality of bond dimensions.

Proof. [Proof](#) The incident-form condition supplies incident-matrix form for every inserted matrix. Theorem 24.4.16 gives the realization identities. The forward and reverse realization identities yield the transfer datum, and the one-bond algebra theorem reads it as conjugation by an invertible matrix. $\square$

Theorem 24.4.27 (Coefficient transfer gives incident form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Assume that A and B define the same PEPS state, that B is vertex-injective, that $D_B(e) > 0$ for every edge e , and that $D_A(e) = D_B(e)$ for every edge e . Assume also that A 's and B 's components at the in-region endpoint of f are linearly independent. Suppose that for every matrix X on A 's boundary bond there is a matrix Y on B 's boundary bond such that $C_A(R, f, X; \sigma, \tau) = C_B(R, f, Y; \sigma, \tau)$ for all σ, τ . Then the virtual pullback of X is in incident-matrix form on f .

Proof. [Proof](#) Equality of the coefficients gives equality on the closed state vectors, one physical leg at a time. These state-vector coefficients span the local virtual coefficient space, so the equality extends to all coefficients. The virtual pullback is therefore the incident insertion determined by Y . $\square$

Theorem 24.4.28 (First coefficient through the second complement block). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Assume that B 's complement blocked tensor map is injective, that A and B define the same PEPS state, that $D_A(e) = D_B(e)$ for every edge e , and that A 's component at the in-region endpoint of f is linearly independent. The coefficient $C_A(R, f, X; \sigma, \tau)$, as a function of the complement physical configuration τ , lies in the image of B 's complement blocked map. Applying the complement blocked left inverse of B recovers the row obtained by applying A 's endpoint operator to B 's region weight vectors.

Proof. [Proof](#) The identity insertion expresses the closed state vector as a contraction of the region and complement blocked weights. Substituting the endpoint operator into this expression gives the complement-blocked factorization, and the left inverse reads off the corresponding row. $\square$

Theorem 24.4.29 (Transfer coefficient and double factorization). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Assume that B 's region and complement blocked tensor maps are injective, that A 's components at the in-region endpoint of f and at the in-complement-region endpoint of the complementary boundary edge are linearly independent, that A and B define the same PEPS state, and that $D_A(e) = D_B(e)$ for every edge e . For fixed X and σ , define the row

$$v_X^\sigma(\nu) = [O_{A,R,f}(X^T) T_{B,R}^{\tilde{\nu}}(\sigma)]_{\sigma(v_f)},$$

where $\tilde{\nu}$ is the R -boundary configuration corresponding to the R^c -boundary configuration ν , and v_f is the endpoint of f in R . The first tensor's inserted coefficient can be written as a double

contraction of the second tensor's region and complement blocked weights:

$$C_A(R, f, X; \sigma, \tau) = \sum_{\mu, \nu} K_X(\mu, \nu) T_{B,R}^\mu(\sigma) T_{B,R^c}^\nu(\tau).$$

If K_X has the incident form

$$K_X(\mu, \nu) = \mathbf{1}_{\mu|_{\partial R \setminus \{f\}} = \nu|_{\partial R \setminus \{f\}}} Y_{\mu_f, \nu_f}, \quad (24.3)$$

then $C_A(R, f, X; \sigma, \tau) = C_B(R, f, Y; \sigma, \tau)$.

Proof. ✓ Proof First factor C_A through B 's complement block, then through B 's region block. The two left inverses define the coefficient K_X . Substituting (24.3) into the double contraction gives the defining double sum for $C_B(R, f, Y; \sigma, \tau)$. □

Theorem 24.4.30 (Incident form of the transfer coefficient). Lean Lean Lean Lean ✓ Stmt Under the hypotheses of Theorem 24.4.29, assume in addition that A and B are vertex-injective, that $D_A(e) > 0$ and $D_B(e) > 0$ for every edge e , and that B 's component at the in-region endpoint of f is linearly independent. If the virtual pullback W_X is in incident-matrix form $W_X = \mathcal{I}_f(Y^T)$, then the transfer coefficient K_X has the incident form determined by Y :

$$K_X(\mu, \nu) = \mathbf{1}_{\mu|_{\partial R \setminus \{f\}} = \nu|_{\partial R \setminus \{f\}}} Y_{\mu_f, \nu_f}.$$

Proof. ✓ Proof The column of K_X at fixed ν is the region blocked left inverse of the row v_X^σ . If $W_X = \mathcal{I}_f(Y^T)$, this row is precisely the incident-matrix sum of Y against the region blocked weights. The left inverse sends each blocked weight to the corresponding standard boundary configuration, giving the stated formula. □

Definition 24.4.31 (Coefficient transfer map). Lean ✓ Stmt Suppose that every matrix $X \in M_{D_A(f)}(\mathbb{C})$ has a matrix $Y \in M_{D_B(f)}(\mathbb{C})$ with matching region-inserted coefficients, $C_A(R, f, X; \sigma, \tau) = C_B(R, f, Y; \sigma, \tau)$. Choosing one such Y for each X defines a forward coefficient transfer $T_f : M_{D_A(f)}(\mathbb{C}) \rightarrow M_{D_B(f)}(\mathbb{C})$.

Theorem 24.4.32 (Coefficient transfer matches coefficients and preserves the identity). Lean ✓ Stmt For every inserted matrix X and every pair of physical configurations, the chosen coefficient transfer satisfies $C_A(R, f, X; \sigma, \tau) = C_B(R, f, T_f(X); \sigma, \tau)$. If the tensors have the same state and the same bond dimensions, all B -bond dimensions are positive, and the region and complement-region blocked maps of B are injective, then $T_f(\mathbb{1}) = \mathbb{1}$.

Proof. ✓ Proof The coefficient equality is the defining property of the chosen transfer. For $X = \mathbb{1}$, both region-inserted coefficients are the corresponding interior bond products times the common PEPS state coefficient, so $C_A(R, f, \mathbb{1}; \sigma, \tau) = C_B(R, f, \mathbb{1}; \sigma, \tau)$. The two blocked-map injectivity hypotheses and positivity of the B -bond dimensions make the B -side region-inserted coefficient injective. Hence the chosen matrix $T_f(\mathbb{1})$ is equal to $\mathbb{1}$. □

Definition 24.4.33 (Region insertion transfer from coefficient transfers). Lean ✓ Stmt Suppose coefficient transfers exist in both directions:

$$\begin{aligned} C_A(R, f, X; \sigma, \tau) &= C_B(R, f, T_f(X); \sigma, \tau), \\ C_B(R, f, Y; \sigma, \tau) &= C_A(R, f, S_f(Y); \sigma, \tau). \end{aligned}$$

Assume also that the two tensors have the same state and the same bond dimensions, all B -bond dimensions are positive, and the region and complement-region blocked maps of B are injective. If the forward choice is multiplicative, $T_f(XY) = T_f(X)T_f(Y)$, then T_f and S_f form a region insertion transfer.

Theorem 24.4.34 (Per-edge gauge from coefficient transfers). [Lean](#) [✓ Stmt](#) Assume coefficient transfers exist in both directions, the forward transfer is multiplicative, the two tensors have the same state and the same bond dimensions, all bond dimensions are positive, and the region and complement-region blocked maps for both tensors are injective. Then there is an invertible boundary gauge Z_f such that, for every $X \in M_{D_A(f)}(\mathbb{C})$, one has $T_f(X) = Z_f \iota_f(X) Z_f^{-1}$, where $\iota_f : M_{D_A(f)}(\mathbb{C}) \simeq M_{D_B(f)}(\mathbb{C})$ is the identification induced by the equality of bond dimensions.

Proof. [✓ Proof](#) The coefficient transfers form a region insertion transfer by Definition 24.4.33. The insertion-transfer algebra isomorphism gives an algebra equivalence between the two boundary matrix algebras, and the Skolem–Noether theorem for matrix algebras writes this equivalence as conjugation by an invertible matrix. $\square$

Theorem 24.4.35 (Determinacy of a region insertion transfer). [Lean](#) [✓ Stmt](#) Assume that the blocked tensors of B on R and R^c are injective and that all virtual bonds of B have positive dimension. Let $T_1, T_2 : M_{D_A(f)}(\mathbb{C}) \rightarrow M_{D_B(f)}(\mathbb{C})$ be two maps. For $i = 1, 2$, suppose T_i satisfies $C_A(R, f, M; \sigma, \tau) = C_B(R, f, T_i(M); \sigma, \tau)$ for every inserted matrix and every pair of physical configurations. Then $T_1(M) = T_2(M)$ for every M .

Proof. [✓ Proof](#) For every pair of physical configurations, the two coefficient identities give the dense equality chain

$$C_B(R, f, T_1(M); \sigma, \tau) = C_A(R, f, M; \sigma, \tau) = C_B(R, f, T_2(M); \sigma, \tau).$$

Injectivity of the region and complement blocked tensors, together with positivity of all virtual bonds of B , gives

$$(\forall \sigma, \tau, C_B(R, f, T_1(M); \sigma, \tau) = C_B(R, f, T_2(M); \sigma, \tau)) \implies T_1(M) = T_2(M).$$

$\square$

Theorem 24.4.36 (Coefficient identities determine the boundary conjugation). [Lean](#) [✓ Stmt](#) Assume that the blocked tensors of B on R and R^c are injective and that all virtual bonds of B have positive dimension. Suppose that the equality $D_A(f) = D_B(f)$ fixes the canonical identification $\iota : M_{D_A(f)}(\mathbb{C}) \simeq M_{D_B(f)}(\mathbb{C})$. For $i = 1, 2$, suppose that the invertible matrix $Z_i \in \text{GL}(D_B(f))$ realizes the region-insertion identity

$$C_A(R, f, M; \sigma, \tau) = C_B(R, f, Z_i \iota(M) Z_i^{-1}; \sigma, \tau).$$

Then the two conjugations agree on every boundary matrix: $Z_1 \iota(M) Z_1^{-1} = Z_2 \iota(M) Z_2^{-1}$.

Proof. [✓ Proof](#) For $i = 1, 2$, define the transfer map $T_i(M) := Z_i \iota(M) Z_i^{-1}$. Its coefficient identity is $C_A(R, f, M; \sigma, \tau) = C_B(R, f, T_i(M); \sigma, \tau)$. Theorem 24.4.35 gives $T_1(M) = T_2(M)$. Substituting the definitions of T_1 and T_2 yields $Z_1 \iota(M) Z_1^{-1} = Z_2 \iota(M) Z_2^{-1}$. $\square$

Theorem 24.4.37 (Region vertex product across an inserted site). [Lean](#) [✓ Stmt](#) Let $v \notin R$. At a fixed global virtual configuration ζ and a physical configuration σ on $R \cup \{v\}$, the product of the local tensors over the vertices of $R \cup \{v\}$ factors as the local tensor at the inserted site v times the product over the vertices of R :

$$\prod_{w \in R \cup \{v\}} A_w(\zeta|_{\text{inc}(w)}, \sigma_w) = A_v(\zeta|_{\text{inc}(v)}, \sigma_v) \prod_{w \in R} A_w(\zeta|_{\text{inc}(w)}, \sigma|_{Rw}).$$

Proof. [✓ Proof](#) The inserted site v is isolated from the product over $R \cup \{v\}$ by splitting off a single factor, and the residual product over the remaining vertices is reindexed through the bijection between the vertices of R and the non- v vertices of $R \cup \{v\}$. $\square$

Theorem 24.4.38 (One-site split of the blocked-region weight). [Lean](#) [✓ Stmt](#) Let $v \notin R$. The blocked-region weight of $R \cup \{v\}$ is the constrained sum, over global virtual configurations ζ whose crossing-edge label on $R \cup \{v\}$ is μ , of the inserted-site tensor times the vertex product over R :

$$T_{A, R \cup \{v\}}^\mu(\sigma) = \sum_{\zeta: \zeta|_{\partial(R \cup \{v\})} = \mu} A_v(\zeta|_{\text{inc}(v)}, \sigma_v) \prod_{w \in R} A_w(\zeta|_{\text{inc}(w)}, \sigma|_{Rw}).$$

Proof. [✓ Proof](#) Each summand of the blocked-region weight is the vertex product over $R \cup \{v\}$, which factors by Theorem 24.4.37. $\square$

Theorem 24.4.39 (A non-incident crossing edge survives the insertion). [Lean](#) [✓ Stmt](#) Let $v \notin R$. A crossing edge of R whose two endpoints both differ from v is also a crossing edge of $R \cup \{v\}$.

Proof. [✓ Proof](#) A crossing edge of R has exactly one endpoint inside R . Adjoining v to R leaves both endpoints on the same side as before, since neither endpoint is v , so the edge still crosses the boundary of $R \cup \{v\}$. $\square$

Definition 24.4.40 (Boundary label of the smaller region across an insertion). [Lean](#) [✓ Stmt](#) Let $v \notin R$. Given a crossing-edge label μ for $R \cup \{v\}$ and a local virtual configuration η on the edges incident to v , the crossing-edge label $\beta(\mu, \eta)$ of R is read off these two configurations: a crossing edge of R incident to v has become internal to $R \cup \{v\}$, so its label is taken from η ; a crossing edge of R not incident to v is also a crossing edge of $R \cup \{v\}$, so its label is taken from μ .

Theorem 24.4.41 (Boundary-label fiber identity across an inserted site). [Lean](#) [✓ Stmt](#) Let $v \notin R$. If a global virtual configuration ζ restricts to μ on the crossing edges of $R \cup \{v\}$ and to η on the edges incident to v , then the crossing-edge label of R read from ζ is exactly $\beta(\mu, \eta)$ of Definition 24.4.40.

Proof. [✓ Proof](#) The two labels are compared edge by edge. On a crossing edge of R incident to v both sides read ζ on an edge incident to v , which is η . On a crossing edge of R not incident to v both sides read ζ on a crossing edge of $R \cup \{v\}$, which is μ . $\square$

Theorem 24.4.42 (Region gauge absorption). [Lean](#) [✓ Stmt](#) Let X be an edge-gauge family, write B^X for the gauged tensor, and let f be a crossing edge of R . Inserting M on f between the region and its complement in B^X equals inserting, in B , the matrix obtained by orienting M to the boundary-edge convention, conjugating by the transpose of the endpoint gauge X_f , and orienting back:

$$C_{B^X}(R, f, M; \sigma, \tau) = C_B(R, f, \mathcal{O}_{R,f}(X_f^\top \mathcal{O}_{R,f}(M) (X_f^{-1})^\top); \sigma, \tau),$$

where $\mathcal{O}_{R,f}$ is the boundary-edge orientation. This is the region-granularity analog of Theorem 24.3.28.

Proof. [✓ Proof](#) By Theorem 24.4.9 the region/complement contraction is the single-bond cut at f up to the interior-bond multiplicity $p_B(R) = \prod_{g \notin \partial R} D_B(g)$, which is unchanged by a gauge. At the edge granularity every interior edge and every crossing edge other than f cancels its gauge pairwise, while on f the endpoint gauges conjugate the oriented matrix $N = \mathcal{O}_{R,f}(M)$ by the transpose of X_f and its inverse:

$$E_{B^X}(f, \omega_{R,\sigma,\tau}, N) = E_B(f, \omega_{R,\sigma,\tau}, X_f^\top N (X_f^{-1})^\top).$$

Transporting this back through Theorem 24.4.9 on B , and using that the boundary-edge orientation is an involution ($\mathcal{O}_{R,f} \circ \mathcal{O}_{R,f} = \text{id}$), gives the displayed equality. $\square$

24.4.1 From the conjugation identity to the absorbed equality

At a region R with boundary edge f , the per-edge gauge identity takes the *conjugation* form

$$C_A(R, f, M; \sigma, \tau) = C_B(R, f, Z \phi_{h_f}(M) Z^{-1}; \sigma, \tau),$$

where Z is the invertible bond matrix on f and ϕ_{h_f} is the matrix-algebra transport induced by $h_f : D_A(f) = D_B(f)$. The region comparison of [186, Section 3] instead consumes the *absorbed* form: the gauge is absorbed into the second tensor, and the *same* matrix $\phi_{h_f}(M)$ is inserted on both sides.

Definition 24.4.1.1 (Orientation-adapted absorbing gauge). [Lean](#) [✓ Stmt](#) Let f be a boundary edge of R and let Z be an invertible matrix on the bond space of f . The orientation-adapted absorbing gauge is

$$X_f = \begin{cases} Z^\top & \text{if the left endpoint of } f \text{ lies in } R, \\ Z^{-1} & \text{otherwise.} \end{cases}$$

It is the absorbing matrix on f for which the gauge action on the region coefficient reproduces the conjugation $Z(\cdot)Z^{-1}$.

Theorem 24.4.1.2 (Matrix reconciliation for the absorbing gauge). [Lean](#) [✓ Stmt](#) With X_f the orientation-adapted absorbing gauge of Z and $\mathcal{O}_{R,f}$ the boundary-edge orientation, for every matrix N on the bond space of f ,

$$\mathcal{O}_{R,f}(X_f^\top \mathcal{O}_{R,f}(N)(X_f^{-1})^\top) = Z N Z^{-1}.$$

The conjugation by $X_f^\top \cdot (\cdot) \cdot (X_f^{-1})^\top$ carried by the region gauge action thus equals $Z(\cdot)Z^{-1}$.

Proof. [✓ Proof](#) When the left endpoint of f lies in R , the orientation is the identity and $X_f = Z^\top$, so $X_f^\top = Z$ and $(X_f^{-1})^\top = Z^{-1}$. Otherwise the orientation is transpose and $X_f = Z^{-1}$, so $X_f^\top = (Z^{-1})^\top$ and $(X_f^{-1})^\top = Z^\top$, giving

$$X_f^\top \mathcal{O}_{R,f}(N)(X_f^{-1})^\top = (Z^{-1})^\top N^\top Z^\top = (Z N Z^{-1})^\top.$$

Applying the outer transpose orientation gives $\mathcal{O}_{R,f}((Z N Z^{-1})^\top) = Z N Z^{-1}$. Both cases give $Z N Z^{-1}$. $\square$

Theorem 24.4.1.3 (Region absorbed equality from the conjugation identity). [Lean](#) [✓ Stmt](#) Let R be a region with boundary edge f and assume the bond-dimension equality $h : D_A = D_B$. Suppose the per-edge gauge Z realizes the conjugation-form region-insertion identity at f : for every matrix M on the bond space of f and all physical configurations σ on R and τ on $V \setminus R$,

$$C_A(R, f, M; \sigma, \tau) = C_B(R, f, Z \phi_{h_f}(M) Z^{-1}; \sigma, \tau).$$

Let X be an edge-gauge family whose value on f is the orientation-adapted absorbing gauge X_f of Z . Then A inserting M matches the absorbed tensor B^X inserting the same transported matrix:

$$C_A(R, f, M; \sigma, \tau) = C_{B^X}(R, f, \phi_{h_f}(M); \sigma, \tau).$$

This is the region analog of Theorem 24.3.30: the conjugation-form coefficient identity becomes the absorbed plain equality.

Proof. ✓ Proof By Theorem 24.4.42 the region gauge action conjugates the inserted matrix by $X_f^\top \cdot \mathcal{O}_{R,f}(\cdot) \cdot (X_f^{-1})^\top$ and cancels every interior gauge of the region and of its complement, so the right-hand coefficient over B^X equals the coefficient over B with the transported matrix conjugated by that gauge. Substituting the conjugation hypothesis on the left and the matrix reconciliation Theorem 24.4.1.2 matches the two conjugation forms. $\square$

Theorem 24.4.1.4 (Edge absorbed equality from the region absorbed equality). Lean ✓ Stmt
Let R be a region with boundary edge f , assume the bond-dimension equality $h : D_A = D_B$ and that every bond dimension of A is positive, and suppose the absorbed region equality holds: for every matrix M on the bond space of f and all physical configurations σ on R and τ on $V \setminus R$,

$$C_A(R, f, M; \sigma, \tau) = C_{B^X}(R, f, \phi_{h_f}(M); \sigma, \tau).$$

Then the edge-level absorbed equality holds at f : for every global physical configuration σ and every matrix N on the bond space of f ,

$$E_A(f, \sigma, N) = E_{B^X}(f, \sigma, \phi_{h_f}(N)).$$

The edge-level identity no longer mentions the region.

Proof. ✓ Proof The interior bond multiplicity $p_A(R) = \prod_{g \notin \partial R} D_A(g)$ is a positive integer, and it is shared by A and B^X because the gauge preserves bond dimensions and $D_A = D_B$. Apply Theorem 24.4.9 with the inserted matrix $\mathcal{O}_{R,f}(N)$ and with configurations $(\sigma|_R, \sigma|_{V \setminus R})$. On the left-hand side,

$$C_A(R, f, \mathcal{O}_{R,f}(N); \sigma|_R, \sigma|_{V \setminus R}) = p_A(R) E_A(f, \sigma, \mathcal{O}_{R,f}(\mathcal{O}_{R,f}(N))) = p_A(R) E_A(f, \sigma, N).$$

On the right-hand side of the assumed region equality,

$$\begin{aligned} C_{B^X}(R, f, \phi_{h_f}(\mathcal{O}_{R,f}(N)); \sigma|_R, \sigma|_{V \setminus R}) &= p_A(R) E_{B^X}(f, \sigma, \mathcal{O}_{R,f}(\phi_{h_f}(\mathcal{O}_{R,f}(N)))) \\ &= p_A(R) E_{B^X}(f, \sigma, \phi_{h_f}(N)). \end{aligned}$$

The displayed equalities also use $\mathcal{O}_{R,f} \circ \mathcal{O}_{R,f} = \text{id}$, the compatibility of $\mathcal{O}_{R,f}$ with transport, and $\omega_{R, \sigma|_R, \sigma|_{V \setminus R}} = \sigma$. Substituting these two identities into the assumed region equality and cancelling the shared positive multiplicity gives the edge-level identity. $\square$

Theorem 24.4.1.5 (Region absorbed equality from the edge absorbed equality). Lean ✓ Stmt
Assume the bond-dimension equality $h : D_A = D_B$ and suppose the edge-level absorbed equality holds at an edge e : for every global physical configuration σ and every matrix N on the bond space of e ,

$$E_A(e, \sigma, N) = E_{B^X}(e, \sigma, \phi_{h_e}(N)).$$

Then the region absorbed equality holds at every region R having e as a boundary edge: for every matrix M on the bond space of e and all physical configurations σ on R and τ on $V \setminus R$,

$$C_A(R, e, M; \sigma, \tau) = C_{B^X}(R, e, \phi_{h_e}(M); \sigma, \tau).$$

Together with the preceding theorem this expresses region-independence: the absorbed plain equality over any comparison region follows from the single edge-level identity at the boundary edge.

Proof. ✓ Proof Read both region coefficients through the region-to-edge identity Theorem 24.4.9. The interior multiplicities agree because the gauge preserves bond dimensions and $D_A = D_B$, and the orientation $\mathcal{O}_{R,e}$ commutes with the transport. Hence the two sides are $p_A(R)$ times the edge coefficients

$$\begin{aligned} E_A(e, \omega_{R,\sigma,\tau}, \mathcal{O}_{R,e}(M)), \\ E_{B^x}(e, \omega_{R,\sigma,\tau}, \phi_{h_e}(\mathcal{O}_{R,e}(M))). \end{aligned}$$

The edge-level hypothesis at $\omega_{R,\sigma,\tau}$ equates these coefficients; multiplying by the common factor $p_A(R)$ gives the region equality. $\square$

Theorem 24.4.1.6 (Transport of the complement-side blocked weight). Lean ✓ Stmt *Let $\phi : G \simeq G'$ be a graph isomorphism. The blocked weight of the transported tensor A^ϕ over the complement of the image region, read at the image complement boundary configuration and the image complement physical configuration, equals the blocked weight of A over $V \setminus R$:*

$$T_{A^\phi, V' \setminus \phi(R)}^{(\phi \cdot \eta)^c}(\phi \cdot \tau) = T_{A, V \setminus R}^{\eta^c}(\tau).$$

Proof. ✓ Proof The image of the complement of R is the complement of the image of R , and after this identification the blocked weight transports along ϕ by the change of summation variables on virtual configurations. $\square$

Theorem 24.4.1.7 (Transport preserves agreement away from a bond). Lean ✓ Stmt *Let $\phi : G \simeq G'$ be a graph isomorphism and let f be a crossing edge of R . Two crossing-edge labels of R agree away from f if and only if their images under the boundary reindex induced by ϕ agree away from the image of f .*

Proof. ✓ Proof The boundary reindex is a bijection on crossing edges that carries f to its image, so it carries the crossing edges other than f bijectively to the crossing edges other than the image of f . Agreement on the former family is therefore equivalent to agreement on the latter. $\square$

Theorem 24.4.1.8 (Covariance of the region insertion coefficient). Lean ✓ Stmt *Let $\phi : G \simeq G'$ be a graph isomorphism and let f be a crossing edge of R . The region insertion coefficient of the transported tensor A^ϕ over the image region, with M inserted on the image of f and the physical configurations relabelled by ϕ , equals the region insertion coefficient of A over R , with M carried back to the bond of A at f :*

$$C_{A^\phi}(\phi(R), \phi(f), M; \phi \cdot \sigma, \phi \cdot \tau) = C_A(R, f, \phi^* M; \sigma, \tau).$$

No endpoint orientation enters: the matrix is inserted between the two single boundary values on the edge.

Proof. ✓ Proof The two crossing-label sums reindex by the boundary action of ϕ , the agreement-away-from- f constraint is preserved by Theorem 24.4.1.7, the region weight transports along ϕ , and the complement weight transports by Theorem 24.4.1.6. The inserted matrix entry is read at the two boundary values, which the bond-dimension identification carries to the corresponding entry of $\phi^* M$. $\square$

Theorem 24.4.1.9 (Region insertion coefficient under an equality of tensors). Lean ✓ Stmt *If two tensors are equal, then their region insertion coefficients are equal, with the inserted matrix carried across the induced equality of bond dimensions:*

$$A_1 = A_2 \implies C_{A_1}(R, f, M; \sigma, \tau) = C_{A_2}(R, f, \psi M; \sigma, \tau).$$

This applies to any tensor equality $A_1 = A_2$, including the equality between a translation-invariant tensor and its transported copy.

Proof. [✓ Proof](#) Substituting the equality of tensors identifies the two coefficients, and the induced equality of bond dimensions is the identity, so the carried matrix is the original one. $\square$

Theorem 24.4.1.10 (Bond dimension at a translated crossing edge). [Lean](#) [✓ Stmt](#) *Let A be translation invariant on the discrete torus, let f be a crossing edge of R , and let $T_{a,b}$ be a translation. The bond dimension of A at the translated crossing edge equals its bond dimension at the reference crossing edge: $D_A(T_{a,b}(f)) = D_A(f)$.*

Proof. [✓ Proof](#) Translation carries the reference edge to its image, and translation invariance fixes the tensor under transport, so the bond dimension is unchanged. $\square$

Theorem 24.4.1.11 (Translation covariance of the coefficient identity). [Lean](#) [✓ Stmt](#) *Let A and B be translation invariant on the discrete torus, and let f be a crossing edge of R . Suppose a map g realizes the coefficient identity at the reference edge: $C_A(R, f, M; \sigma, \tau) = C_B(R, f, g(M); \sigma, \tau)$. Then, for every translation $T_{a,b}$, the same identity holds at the translated crossing edge of the translated region, with the inserted matrix carried between the translated and reference bonds:*

$$C_A(T_{a,b}(R), T_{a,b}(f), M'; T_{a,b} \cdot \sigma, T_{a,b} \cdot \tau) = C_B(T_{a,b}(R), T_{a,b}(f), \hat{g}(M'); T_{a,b} \cdot \sigma, T_{a,b} \cdot \tau), \quad (24.4)$$

where $\hat{g}$ applies g to the reference-bond reindex of M' and carries the result back to the translated bond.

Proof. [✓ Proof](#) Translation invariance identifies the transported tensor with the original, so Theorem 24.4.1.8 carries the reference coefficient of A to the translated edge with no change of tensor; Theorem 24.4.1.9 mediates the literal and transported tensors. Applying the reference identity for g and transporting back for B in the same way gives (24.4), the inserted matrix being carried across the equal bond dimensions of Theorem 24.4.1.10. $\square$

Definition 24.4.1.12 (Single-vertex two-block tensor). [Lean](#) [✓ Stmt](#) Fix a vertex v . The single-vertex tensor A_v is regarded as a two-block tensor whose shared bonds are the edges incident to v , whose external boundary is a one-point space, and whose physical index is the physical index at v : $T_v(*, \eta, \sigma) = A_v(\eta, \sigma)$.

Theorem 24.4.1.13 (Injectivity of the single-vertex two-block tensor). [Lean](#) [✓ Stmt](#) *If A is vertex-injective, then the single-vertex two-block tensor at every vertex is injective.*

Proof. [✓ Proof](#) Vertex injectivity is linear independence of the coefficient family $\eta \mapsto A_v(\eta, -)$. The auxiliary one-point boundary only reindexes this same family, so the two-block injectivity condition is the same linear independence statement. $\square$

Definition 24.4.1.14 (Complement two-block tensor). [Lean](#) [✓ Stmt](#) For a vertex v , the complementary block $V \setminus \{v\}$ defines a two-block tensor on the same incident shared bonds. Its coefficient is the contraction of all tensors away from v , with the incident bond indices left open at the complement endpoints: $T_{v^c}(*, \eta, \tau) = A_{v^c}(\eta, \tau)$.

Definition 24.4.1.15 (Complement-region index identifications). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a vertex v , the edges crossing the boundary of $V \setminus \{v\}$ are precisely the edges incident to v . Hence the boundary configurations of $V \setminus \{v\}$ identify with the local virtual configurations at v . Similarly, the vertices in $V \setminus \{v\}$ identify with the vertices distinct from v , giving the corresponding identification of physical configurations.

Theorem 24.4.1.16 (Vertex-complement contraction as a blocked region). [Lean](#) [Lean](#) [✓ Stmt](#) Under the preceding identifications, the blocked tensor of $V \setminus \{v\}$ is the vertex-complement tensor:

$$T_{A, V \setminus \{v\}}^\rho(\tau) = A_{v^c}(\Psi_v \rho, \Phi_v \tau).$$

Proof. [✓ Proof](#) Expanding the blocked contraction as a sum over the virtual configurations of the complement,

$$T_{A, V \setminus \{v\}}^\rho(\tau) = \sum_{\zeta: \zeta|_{\partial} = \rho} \prod_{w \neq v} A_w(\zeta|_w, \tau_w) = A_{v^c}(\Psi_v \rho, \Phi_v \tau),$$

where ζ runs over the virtual configurations of the complement whose boundary restriction $\zeta|_{\partial}$ is the prescribed boundary configuration ρ . The boundary identification Ψ_v turns the crossing-edge constraint $\zeta|_{\partial} = \rho$ into the prescribed incident-edge labels, and the vertex identification Φ_v reindexes the product over the vertices distinct from v ; both bijections match the summation sets and their factors termwise. $\square$

Theorem 24.4.1.17 (Injectivity of the vertex-complement contraction). [Lean](#) [✓ Stmt](#) If A is vertex-injective and every virtual bond space has positive dimension, then the coefficient family $\eta \mapsto (\tau \mapsto A_{v^c}(\eta, \tau))$ of the vertex-complement contraction is linearly independent.

Proof. [✓ Proof](#) Apply the injectivity theorem for an arbitrary finite region to $R = V \setminus \{v\}$. Theorem 24.4.1.16 identifies the resulting blocked tensor with the vertex-complement tensor. Reindexing the boundary family and the physical coordinates by bijections preserves linear independence. $\square$

Theorem 24.4.1.18 (Injectivity of the complement two-block tensor). [Lean](#) [✓ Stmt](#) If A is vertex-injective and every virtual bond space has positive dimension, then the complement two-block tensor at every vertex is injective.

Proof. [✓ Proof](#) The previous theorem gives linear independence of the coefficient family $\eta \mapsto A_{v^c}(\eta, -)$. The two-block tensor only adds a one-point external boundary, so this is the same injectivity condition after the harmless reindexing $(*, \eta) \leftrightarrow \eta$. $\square$

Theorem 24.4.1.19 (One-vertex versus complement comparison). [Lean](#) [✓ Stmt](#) Let the shared bond type and the two external boundary spaces be nonempty. After the edge gauges have been absorbed into the second tensor family $\tilde{B}$, block one vertex v against its complement. Assume that $A_v, A_{v^c}, \tilde{B}_v, \tilde{B}_{v^c}$ are injective as two-block tensors. The post-absorption insertion equality gives, for every shared bond b , every matrix $X \in M_{D_b}(\mathbb{C})$, and all external and physical indices,

$$C_{A_v, A_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{\tilde{B}_v, \tilde{B}_{v^c}}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Then $\exists \lambda_v \in \mathbb{C}^\times$ such that $A_v = \lambda_v \tilde{B}_v$.

Proof. [✓ Proof](#) With $A_1 = A_v$, $A_2 = A_{v^c}$, $B_1 = \tilde{B}_v$, and $B_2 = \tilde{B}_{v^c}$, the post-absorption equality is precisely, for every b , X , η_v , η_{v^c} , σ_v , and σ_{v^c} ,

$$C_{A_1, A_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}) = C_{B_1, B_2}(b, X; \eta_v, \eta_{v^c}, \sigma_v, \sigma_{v^c}).$$

Applying Theorem 24.3.43 yields $\lambda_v \in \mathbb{C}^\times$ such that

$$A_v = \lambda_v \tilde{B}_v,$$

giving the desired local proportionality. $\square$

Remark 24.4.1.20 (Normal-region form). The normal-region application in [186, Section 4] uses the same scalar comparison after comparing two injective regions $R \subset S = R \cup \{v\}$. If the region comparisons give $A_R = \lambda_R \widetilde{B}_R$ and $A_S = \lambda_S \widetilde{B}_S$, the source-paper comparison is

$$\begin{array}{c} | \\ \bullet \\ A_R \end{array} \begin{array}{c} | \\ \bullet \\ A \end{array} = \begin{array}{c} | \\ \bullet \\ A_S \end{array} \propto \begin{array}{c} | \\ \bullet \\ \widetilde{B}_S \end{array} = \begin{array}{c} | \\ \bullet \\ \widetilde{B}_R \end{array} \begin{array}{c} | \\ \bullet \\ \widetilde{B} \end{array}.$$

Applying the left inverse supplied by injectivity of the block $A_R = \lambda_R \widetilde{B}_R$ gives $A_v = (\lambda_S/\lambda_R) \widetilde{B}_v$.

Theorem 24.4.1.21 (Fundamental Theorem for injective PEPS, conditional on bond-dimension equality). [Lean](#) [✓ Stmt](#) Suppose that the corresponding virtual edge spaces of A and B are already identified, every virtual bond of A has positive dimension, and the graph is connected. If A and B are vertex-injective and give the same PEPS state, then there are invertible matrices X_e , one for each edge, such that at every vertex the tensor B_v is obtained from A_v by the oriented endpoint action of the incident matrices.

Proof. [✓ Proof](#) Gauge consistency supplies the edge gauges directly, which combine with the identified bond dimensions to yield the gauge equivalence. $\square$

Theorem 24.4.1.22 (Equal bond dimensions for injective PEPS). [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be vertex-injective PEPS tensors on the same finite simple graph, with positive virtual bond dimensions, and suppose that they represent the same state. Then

$$D_A(e) = D_B(e)$$

for every edge e . This conclusion does not require the graph to be connected. It is the bond-dimension consequence of the virtual-operation algebra isomorphism in [186, lines 279–282 and 571–582].

Proof. [✓ Proof](#) Fix an edge. Blocking around it gives two injective three-site chains. Their matched insertions on the distinguished bond form an algebra equivalence between $M_{D_A(e)}(\mathbb{C})$ and $M_{D_B(e)}(\mathbb{C})$. Equality of the linear dimensions of these full matrix algebras gives $D_A(e)^2 = D_B(e)^2$, hence $D_A(e) = D_B(e)$. $\square$

Theorem 24.4.1.23 (Fundamental Theorem for injective PEPS). [Lean](#) [✓ Stmt](#) Let A and B be vertex-injective PEPS tensors on a finite simple graph, and suppose every virtual bond of A and B has positive dimension and that the graph is connected. If they give the same PEPS state, then their corresponding virtual edge spaces have the same dimensions and, after identifying these spaces, there are invertible matrices X_e , one for each edge, such that B_v is obtained from A_v by the oriented endpoint action of the matrices incident to v .

Proof. [✓ Proof](#) By Theorem 24.4.1.22, the bond dimensions agree edgewise. With the bond dimensions identified, the conditional form of the theorem applies. $\square$

24.5 Union Closure for Normal PEPS

A normal PEPS is injective only after blocking certain regions, so the injective theorem applies once those regions are available. The closure property that makes them available is that, for positive bond dimensions, the union of two injective regions is again injective, $\text{inj}(A) \wedge \text{inj}(B) \implies \text{inj}(A \cup B)$. Writing $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$, and $P_3 = (A \cup B)^c$, a boundary

operator X attached to P_3 that is annihilated by the contraction over $A \cup B$ descends through the contraction chain

$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \implies \mathcal{C}_{\{2\}}(X) = 0 \implies \mathcal{C}_{\{1,2\}}(X) = 0 \implies X = 0,$$

so the union contraction has trivial kernel. Iterating gives injectivity of every finite union of injective regions.

Definition 24.5.1 (Four-region decomposition for a union of regions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stm](#) Let V be a finite vertex set. For two regions $A, B \subseteq V$, define the four regions $A \setminus B$, $A \cap B$, $B \setminus A$, and $V \setminus (A \cup B)$. These are the four regions used in the proof of Lemma 24.5.7. They also form an indexed family over $\{0, 1, 2, 3\}$ in the order $A \setminus B$, $A \cap B$, $B \setminus A$, $V \setminus (A \cup B)$.

Lemma 24.5.2 (Identities for the four-region decomposition). Let V be a finite vertex set and let $A, B \subseteq V$. The left-only and overlap regions reconstruct A , the overlap and right-only regions reconstruct B , the three inside regions reconstruct $A \cup B$, and all four regions reconstruct V . The same identities also hold in the indexed four-region family: pieces 0 and 1 reconstruct A , pieces 1 and 2 reconstruct B , and pieces 0, 1, 2 reconstruct $A \cup B$. These same reconstructions are available as finite unions over the corresponding subfamilies of the indexed four-region family. Piece 3 is the complement of $A \cup B$, equivalently the complement of the three inside indexed pieces, and also $A^c \cap B^c$. The singleton subfamily $\{3\}$ gives the outside block. Pieces 2, 3 reconstruct A^c , equivalently the complement of pieces 0, 1, and pieces 0, 3 reconstruct B^c , equivalently the complement of pieces 1, 2, again both as explicit unions and as finite unions over the indexed subfamilies. Conversely, pieces 0, 1 are the complement of pieces 2, 3, pieces 1, 2 are the complement of pieces 0, 3, and adding all four pieces reconstructs V . Equivalently, each vertex lies in one of the four regions, and the indexed four-region family has union V . In fact, every vertex belongs to exactly one indexed piece.

Proof. ✓ Proof Check membership of an arbitrary vertex in A and in B .

Lemma 24.5.3 (Disjointness in the four-region decomposition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) In the four-region decomposition, the left-only and right-only parts are disjoint from the overlap and from each other. Each inside part, and hence the union of the three inside regions, is disjoint from the outside region. In indexed form, the pieces forming A are disjoint from the pieces forming A^c , and the pieces forming B are disjoint from the pieces forming B^c , both as explicit binary unions and as finite unions over the corresponding selected indexed subfamilies. The indexed four-region family is pairwise disjoint.

Proof. ✓ Proof Use the standard disjointness identities for set difference and intersection, together with the reconstruction of $A \cup B$. □

Definition 24.5.4 (Contraction maps for the union injectivity proof). Lean ✓ Stmt Let K , E_{012} , E_2 , and E_{12} be complex vector spaces. A contraction-map family for the union-injectivity lemma in [186, Section 4] consists of three linear maps $\mathcal{C}_{\{0,1,2\}} : K \rightarrow E_{012}$, $\mathcal{C}_{\{2\}} : K \rightarrow E_2$, and $\mathcal{C}_{\{1,2\}} : K \rightarrow E_{12}$.

Definition 24.5.5 (Contraction chain for the union injectivity proof). [Lean] [✓ Stmt] A contraction chain consists of the three contraction maps of Definition 24.5.4, together with the three implications used in Lemma 24.5.6.

Lemma 24.5.6 (Kernel criterion for the union contraction chain). [Lean](#) [Lean](#) [✓ Stmt](#) Let $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$, and $P_3 = (A \cup B)^c$. Let K be the boundary vector space attached to P_3 , and let $\mathcal{C}_{\{0,1,2\}} : K \rightarrow E_{012}$, $\mathcal{C}_{\{2\}} : K \rightarrow E_2$, and $\mathcal{C}_{\{1,2\}} : K \rightarrow E_{12}$ be the three contraction maps in the proof of the union-injectivity lemma in [186, Section 4]. If

$$\begin{aligned}\mathcal{C}_{\{0,1,2\}}(X) = 0 &\implies \mathcal{C}_{\{2\}}(X) = 0, \\ \mathcal{C}_{\{2\}}(X) = 0 &\implies \mathcal{C}_{\{1,2\}}(X) = 0,\end{aligned}$$

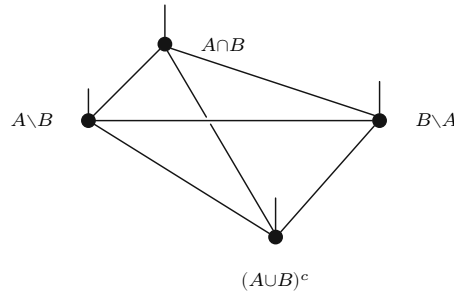
and $\mathcal{C}_{\{1,2\}}(X) = 0 \implies X = 0$, then $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$.

Proof. [✓ Proof](#) For $X \in K$ with $\mathcal{C}_{\{0,1,2\}}(X) = 0$, apply the three implications in sequence:

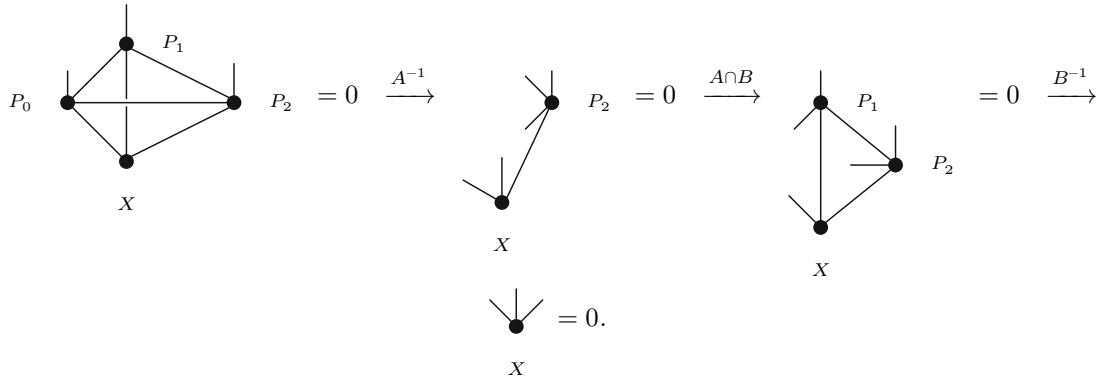
$$\mathcal{C}_{\{0,1,2\}}(X) = 0 \implies \mathcal{C}_{\{2\}}(X) = 0 \implies \mathcal{C}_{\{1,2\}}(X) = 0 \implies X = 0.$$

□

Lemma 24.5.7 (Union of injective regions). [Not ready](#) Let V be a finite vertex set. Write $\text{inj}(R)$ for the assertion that the PEPS tensor blocked to the region $R \subseteq V$ is injective. For two possibly overlapping regions $A, B \subseteq V$, $\text{inj}(A) \wedge \text{inj}(B) \implies \text{inj}(A \cup B)$. The union-injectivity proof in [186, Section 4] blocks the network into four parts $A \setminus B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$, and applies the two available left inverses in sequence:



Taking a boundary tensor X on $(A \cup B)^c$ with $\mathcal{C}_{\{0,1,2\}}(X) = 0$, injectivity of A gives $\mathcal{C}_{\{2\}}(X) = 0$; reinserting $A \cap B$ and applying injectivity of B then gives $X = 0$:



Proof. Put $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$, and $P_3 = (A \cup B)^c$. Then $P_0 \cup P_1 = A$, $P_1 \cup P_2 = B$, and $P_0 \cup P_1 \cup P_2 = A \cup B$. Let $\mathcal{C}_I$ denote contraction of the blocked tensor over the region $\bigcup_{i \in I} P_i$. To prove injectivity of $A \cup B$, take a boundary tensor X on P_3 such that $\mathcal{C}_{\{0,1,2\}}(X) = 0$. Since

$P_0 \cup P_1 = A$ is injective, its left inverse gives $\mathcal{C}_{\{0,1,2\}}(X) = 0 \implies \mathcal{C}_{\{2\}}(X) = 0$. The remaining two steps are $\mathcal{C}_{\{2\}}(X) = 0 \implies \mathcal{C}_{\{1,2\}}(X) = 0$ and $\mathcal{C}_{\{1,2\}}(X) = 0 \implies X = 0$. The first implication is obtained by contracting with the tensor on P_1 , and the second is the left inverse for the injective region $P_1 \cup P_2 = B$. Thus $\ker \mathcal{C}_{\{0,1,2\}} = \{0\}$, and therefore $P_0 \cup P_1 \cup P_2 = A \cup B$ is injective. $\square$

Lemma 24.5.8 (Union of the edge-centred blocks). [Lean](#) [✓ Stmt](#) *Fix an edge e and a partition of the vertex set into the three edge-centred blocks R, B, C (red, blue, complementary) of the normal PEPS proof, with the red block carrying one endpoint of e and the blue block the other. The injectivity of the blue and complementary blocks is part of the edge-centred blocking data; if every virtual bond dimension is positive, then their union $B \cup C = V \setminus R$ is injective.*

Proof. [✓ Proof](#) This is Lemma 24.5.7 for the edge-centred triple, where the two regions are the blue and complementary blocks and their union is the set complement of the red block. A coefficient family annihilating the blocked weights of $V \setminus R$ is stripped of the blue block by its left inverse, leaving a complement-coupling combination that vanishes for every blue boundary configuration. The complementary block then carries each coupling coefficient, read as a function of the complementary physical leg, into a combination of its blocked weights whose coefficient at a boundary configuration is the host residual coefficient reconstructed from the blue and complementary boundary data; injectivity of the complementary block forces that coefficient to vanish wherever a global configuration realizes the two boundary labels. Since every host residual is realized once the bond dimensions are positive, the coefficient family is zero. $\square$

Lemma 24.5.9 (Union of two disjoint injective regions). [Lean](#) [✓ Stmt](#) *Let $S, T \subseteq V$ be two disjoint regions. If the blocked tensors of S and of T are each injective and every virtual bond dimension is positive, then the blocked tensor of their union $S \cup T$ is injective.*

Proof. [✓ Proof](#) This is Lemma 24.5.7 in the case where the two regions are disjoint, so that the overlap $S \cap T$ is empty and the four-region picture collapses to the three blocks S, T , and the set complement $(S \cup T)^c$. The argument places S and T as the blue and complementary blocks of a three-block decomposition whose third block is the set complement of their union, and applies the two-step inverse argument of the edge-centred Lemma 24.5.8 to that decomposition. No injectivity hypothesis on the set complement and no distinguished edge are needed. $\square$

Lemma 24.5.10 (Union of two injective regions with positive bonds). [Lean](#) [✓ Stmt](#) *Let $A, B \subseteq V$ be two possibly overlapping regions. If the blocked tensors of A and of B are each injective and every virtual bond dimension is positive, then the blocked tensor of their union $A \cup B$ is injective. This extends Lemma 24.5.7 to arbitrary overlapping regions when all virtual bond dimensions are positive.*

Proof. [✓ Proof](#) Write the four parts $P_0 = A \setminus B$, $P_1 = A \cap B$, $P_2 = B \setminus A$. A coefficient family annihilating the blocked weights of $A \cup B$ is stripped of A by its left inverse, leaving the coupling through P_2 vanishing for every A -boundary configuration. The source then reinserts the overlap P_1 and inverts B . The legs running from P_0 to the outside are the open legs of the final tensor; they stay uncontracted through the reinsertion. Carrying those legs as a separate label, fix it to a reference value and restrict the coefficient family to the matching fiber. The fiber-restricted first strip still vanishes, so reinserting P_1 and inverting B forces the fiber row to vanish. The union boundary edges partition into the B -boundary edges and the P_0 -outer edges, so fixing both the reference label and the B -boundary label of a configuration realizing a host label selects that single host term, forcing the coefficient to vanish at every realizable host label. Surjectivity of the union host boundary, available once the bond dimensions are positive, realizes every label, so the coefficient family is zero. $\square$

Lemma 24.5.11 (Finite disjoint union of injective regions). Lean ✓ Stmt Let $(R_i)_{i \in s}$ be a nonempty finite family of pairwise-disjoint regions. If the blocked tensor of each R_i is injective and every virtual bond dimension is positive, then the blocked tensor of their union $\bigcup_{i \in s} R_i$ is injective.

Proof. ✓ Proof Induct on the nonempty family. The singleton case is the injectivity of the one block. For the inductive step, the next block is disjoint from the union of the remaining blocks, so Lemma 24.5.9 combines the two injective pieces. $\square$

Lemma 24.5.12 (Finite union of injective regions). Lean ✓ Stmt Let $(R_i)_{i \in s}$ be a nonempty finite family of possibly overlapping regions. If the blocked tensor of each R_i is injective and every virtual bond dimension is positive, then the blocked tensor of their union $\bigcup_{i \in s} R_i$ is injective. This is the finite iteration of the overlapping union lemma, with no disjointness assumption.

Proof. ✓ Proof Induct on the nonempty family. The singleton case is the injectivity of the one block. For the inductive step, combine the next block with the union of the remaining blocks by Lemma 24.5.10; no disjointness is needed. $\square$

Definition 24.5.13 (Injectivity predicate for blocked regions). Lean Lean ✓ Stmt A region-injectivity hypothesis assigns to each finite vertex region the assertion that the tensor obtained by blocking that region is injective. This predicate is used for regions such as R , S , T , and their complements in the normal PEPS proof. The union-closure hypothesis records the conclusion supplied by the source union-of-injective-regions lemma: the union of two injective regions is injective.

Definition 24.5.14 (Concrete region-injectivity predicate of a tensor). Lean ✓ Stmt The region-injectivity predicate of a fixed PEPS tensor A sends a finite region R to the assertion that the blocked-region tensor of R is injective.

Theorem 24.5.15 (Gauged blocked-region weight as a global sum). Lean ✓ Stmt Let $\tilde{B} = \text{Gauge}_X(B)$ be obtained from B by a gauge family X . For a region R , a boundary configuration μ , and a physical configuration τ on R , the blocked weight of $\tilde{B}$ is

$$T_{\tilde{B}, R}^\mu(\tau) = \sum_{\zeta} \left(\prod_{f \in \partial R} G_f^X(\mu_f, \zeta_f) \right) \prod_{v \in R} B_v(\zeta|_{\text{inc}(v)}, \tau_v), \quad (24.5)$$

where the sum ranges over global virtual configurations ζ , and G_f^X is the surviving gauge on the boundary edge f .

Proof. ✓ Proof In the contraction defining $T_{\tilde{B}, R}^\mu(\tau)$, the two half-edge gauge factors on every edge internal to R combine as

$$\sum_a (X_e)_{\alpha, a} (X_e^{-1})_{a, \beta} = (X_e X_e^{-1})_{\alpha, \beta} = \delta_{\alpha, \beta}.$$

Thus every interior edge contributes the identity and its gauge cancels. The only remaining factors are the boundary gauges $G_f^X(\mu_f, \zeta_f)$, giving (24.5). $\square$

Theorem 24.5.16 (Boundary-coupling form of the gauged blocked weight). Lean ✓ Stmt With the same notation, set $K_R^X(\mu, \nu) := \prod_{f \in \partial R} G_f^X(\mu_f, \nu_f)$. Then the gauged blocked weight is the original blocked weight multiplied by the boundary-coupling matrix:

$$T_{\tilde{B}, R}^\mu(\tau) = \sum_{\nu} K_R^X(\mu, \nu) T_{B, R}^\nu(\tau). \quad (24.6)$$

This is the gauge absorption of the blocked tensor in [186, Section 4, proof of Theorem 3].

Proof. ✓ Proof Start from Theorem 24.5.15 and group the global virtual configurations by their boundary label ν :

$$\sum_{\zeta} K_R^X(\mu, \zeta|_{\partial R}) \prod_{v \in R} B_v(\zeta|_{\text{inc}(v)}, \tau_v) = \sum_{\nu} K_R^X(\mu, \nu) T_{B,R}^{\nu}(\tau).$$

□

Theorem 24.5.17 (The boundary coupling has a right inverse). Lean ✓ Stmt Let $K_R^X(\mu, \nu) = \prod_{f \in \partial R} G_f^X(\mu_f, \nu_f)$ and $(K_R^X)^{-1}(\nu, \rho) = \prod_{f \in \partial R} (G_f^X)^{-1}(\nu_f, \rho_f)$. Then, for boundary configurations μ and ρ ,

$$\sum_{\nu} K_R^X(\mu, \nu) (K_R^X)^{-1}(\nu, \rho) = \begin{cases} 1, & \mu = \rho, \\ 0, & \mu \neq \rho. \end{cases} \quad (24.7)$$

Proof. ✓ Proof Exchange the finite sum over ν with the product over boundary edges:

$$\sum_{\nu} K_R^X(\mu, \nu) (K_R^X)^{-1}(\nu, \rho) = \prod_{f \in \partial R} \sum_a G_f^X(\mu_f, a) (G_f^X)^{-1}(a, \rho_f).$$

Since

$$\sum_a G_f^X(\mu_f, a) (G_f^X)^{-1}(a, \rho_f) = (G_f^X (G_f^X)^{-1})_{\mu_f, \rho_f} = \delta_{\mu_f, \rho_f},$$

each factor reduces to the Kronecker symbol at (μ_f, ρ_f) . The product of these one-edge Kronecker symbols is $\delta_{\mu, \rho}$. □

Theorem 24.5.18 (Gauge preservation of blocked-region injectivity). Lean ✓ Stmt Let $\widetilde{B} = \text{Gauge}_X(B)$. If the blocked tensor family $\{T_{B,R}^{\nu}\}_{\nu}$ is linearly independent, then the blocked tensor family $\{T_{\widetilde{B},R}^{\mu}\}_{\mu}$ is linearly independent. Equivalently, $\text{inj}(T_{B,R}) \implies \text{inj}(T_{\widetilde{B},R})$.

Proof. ✓ Proof By (24.6), $T_{\widetilde{B},R}^{\mu} = \sum_{\nu} K_R^X(\mu, \nu) T_{B,R}^{\nu}$. Suppose $\sum_{\mu} c_{\mu} T_{\widetilde{B},R}^{\mu} = 0$. Substituting (24.6) and using the linear independence of $\{T_{B,R}^{\nu}\}_{\nu}$ gives $\sum_{\mu} c_{\mu} K_R^X(\mu, \nu) = 0$ for every ν . Contracting this equality with the right inverse of K_R^X yields, for every ρ ,

$$c_{\rho} = \sum_{\nu} \left(\sum_{\mu} c_{\mu} K_R^X(\mu, \nu) \right) (K_R^X)^{-1}(\nu, \rho) = 0.$$

Thus all coefficients c_{ρ} vanish, proving the linear independence of $\{T_{\widetilde{B},R}^{\mu}\}_{\mu}$. □

Theorem 24.5.19 (Kernel descent for blocked regions). Lean ✓ Stmt Let A be vertex-injective, and assume that every virtual bond dimension is positive. Then every finite region R has an injective blocked tensor: $\text{inj}(R)$.

Proof. ✓ Proof Let $(c_{\rho})_{\rho}$ be a finitely supported coefficient family on the boundary configurations of R , and assume that, for every physical configuration τ , $\sum_{\rho} c_{\rho} T_{A,R}^{\rho}(\tau) = 0$. The kernel descent removes the vertices of R one at a time and preserves the vanishing contraction. After all vertices have been removed, the terminal contraction gives $c_{\rho} = 0$ for every boundary configuration ρ . Hence the family $\{T_{A,R}^{\rho}\}_{\rho}$ is linearly independent, which is $\text{inj}(R)$. □

Figure 1 shows three grid diagrams labeled R , S , and T . Each diagram consists of a grid of points with a shaded region.

- R is a 5×5 grid. The shaded region is a 3×3 square with the bottom-right cell removed, consisting of 7 cells.
- S is a 5×5 grid. The shaded region is a 3×3 square, consisting of 8 cells.
- T is a 6×5 grid. It contains two shaded regions: a 2×2 square in the top-left corner (4 cells) and a 3×3 square in the bottom-right corner (8 cells).

The word "and" is placed between the diagrams for R and S .

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nate rectangle is coordinatewise membership, the contiguous coordinate interval has the expected coordinate inequalities, membership in a contiguous coordinate rectangle is the conjunction of the four bounding inequalities, and membership in R is the disjunction of membership in the 2×3 lower-left piece or in the shifted 3×2 upper piece. Membership in S is the disjunction of membership in the left 2×3 piece or in the horizontally shifted 2×3 piece. Membership in the local-window T region is membership in the local 5×6 window together with nonmembership in the union of the two displayed edge blocks. Bounded contiguous coordinate rectangles of sizes 2×3 and 3×2 satisfy, respectively, the predicates for contiguous 2×3 and contiguous 3×2 square-lattice rectangles. The cardinality of a coordinate rectangle is the product of the two coordinate cardinalities, and the full coordinate rectangle is the whole finite vertex set.

Proof. [✓ Proof](#) The coordinate-rectangle statements follow from membership and cardinality identities for Cartesian products of finite coordinate sets. The contiguous-rectangle statement then unfolds the two coordinate intervals. The membership statements for R , S , and the two edge blocks removed from T additionally unfold membership in a union of two such rectangles, while membership in T also unfolds the set difference from the local 5×6 window. The shape predicates follow by using the displayed lower-left corners as witnesses. $\square$

Lemma 24.6.4 (The normal regions R and S are unions of contiguous rectangles). [Lean](#) [Lean](#)
[✓ Stmt](#) *If the displayed 3×3 window containing R and S lies in the finite square lattice, then R is the union of a contiguous 2×3 rectangle and a contiguous 3×2 rectangle, while S is the union of two contiguous 2×3 rectangles.*

Proof. [✓ Proof](#) This is the defining description of R and S , together with the coordinate bounds which make the displayed rectangles contiguous 2×3 and 3×2 rectangles. $\square$

Lemma 24.6.5 (The edge blocks removed from T are contiguous rectangles). [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) *If the two edge blocks cut out of the displayed T -region lie in the finite square lattice, then they are respectively a contiguous 2×3 rectangle and a contiguous 3×2 rectangle.*

Proof. [✓ Proof](#) This is the defining description of the two removed edge blocks, together with the coordinate bounds which make them valid contiguous rectangles. $\square$

Lemma 24.6.6 (The local T -region and its removed edge blocks fill the window). [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The two edge blocks removed from the displayed local T -region lie inside the local 5×6 coordinate window. They are disjoint from each other and from this local T region, and their union with it is exactly that local window.*

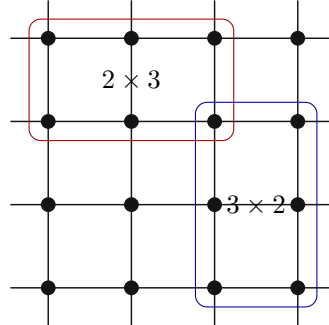
Proof. [✓ Proof](#) This follows by unfolding the definition of T as the set-theoretic complement of the two displayed edge blocks inside the local window. $\square$

Lemma 24.6.7 (The edge-complementary block fills the finite lattice). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) *The finite-lattice complementary block around the edge is disjoint from the two displayed edge blocks, and its union with them is the full finite square lattice. The local-window T region is contained in this finite-lattice complementary block.*

Proof. [✓ Proof](#) This is the set-theoretic complement of the two displayed edge blocks in the full finite square lattice. The containment of the local-window T region follows because its points also avoid the two removed edge blocks. $\square$

Definition 24.6.8 (Rectangular covers of square-lattice regions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) *A rectangular cover of a square-lattice region is a nonempty finite family of regions whose union is exactly the given region, and each member is a contiguous 2×3 or 3×2 rectangle. The*

two cases used in the proof would be a cover of the local displayed T -region and a cover of the finite-lattice complementary block A_3 . The panel below illustrates the two permitted shapes of a cover member.



Lemma 24.6.9 (The present local T -window and A_3 cover obstructions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The present origin-based local-window model for the displayed T -region admits no nonempty finite cover by contained contiguous 2×3 and 3×2 rectangles whenever the ambient lattice contains the 5×6 window. The origin-based rotated local-window model used for vertical edges has the analogous obstruction whenever the ambient lattice contains the 6×5 window. The normalized 5×6 and 6×5 cases are recorded as corollaries. The present origin-based horizontal edge-complement model has the analogous obstruction whenever the ambient lattice contains the 5×7 frame; the normalized 5×7 case is again recorded as a corollary. The present origin-based vertical edge-complement model has the corresponding obstruction whenever the ambient lattice contains the 7×5 frame; the normalized 7×5 case is recorded as a corollary.*

Proof. [✓ Proof](#) The lower-left point of the local window belongs to the present T model. Any contiguous 2×3 rectangle containing it also contains a point of the removed vertical edge block, while any contiguous 3×2 rectangle containing it also contains a point of the removed horizontal edge block. The rotated statement is the same coordinate argument with the two shapes interchanged. The horizontal edge-complement statement uses the same lower-left point, and so does the normalized vertical edge-complement statement. These cases are all instances of the same point-forcing obstruction to an exact rectangular cover. $\square$

Lemma 24.6.10 (Injectivity from a rectangular cover). [Lean](#) [✓ Stmt](#) *If a square-lattice region has a nonempty finite cover by contiguous 2×3 and 3×2 rectangles, then rectangular injectivity and finite union closure imply that the region is injective.*

Proof. [✓ Proof](#) Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. $\square$

Lemma 24.6.11 (The normal regions R and S differ in one site). [Lean](#) [Lean](#) [✓ Stmt](#) *For the displayed regions R and S with the same lower-left corner, R is contained in S . The difference $S \setminus R$ is exactly the lower-right site of the 3×3 square: in coordinates, the site one obtains by shifting two steps horizontally and zero steps vertically from the lower-left corner.*

Proof. [✓ Proof](#) Expand the two displayed unions. The only point of the second 2×3 piece of S which is not already in the union defining R is its lower-right point. $\square$

Definition 24.6.12 (Normal rectangular injectivity hypotheses). [Lean](#) [Lean](#) [✓ Stmt](#) *In the square-lattice normal theorem, every 2×3 and 3×2 rectangular region is assumed injective. The*

corresponding abstract hypotheses specify which finite regions have these two shapes and assert injectivity for all of them. In the coordinate square-lattice specialization, these two families are the contiguous coordinate 2×3 and 3×2 rectangles.

Lemma 24.6.13 (Coordinate rectangular injectivity gives abstract rectangular injectivity).

[Lean](#) [✓ Stmt](#) *The coordinate square-lattice rectangular injectivity hypotheses determine abstract rectangular injectivity hypotheses, with the two abstract rectangular families taken to be the contiguous coordinate 2×3 and 3×2 rectangles.*

Proof. [✓ Proof](#) This is the direct specialization of the abstract rectangular families to the two contiguous coordinate-rectangle predicates. $\square$

Lemma 24.6.14 (Coordinate rectangular injectivity gives injective rectangles). [Lean](#) [Lean](#)

[✓ Stmt](#) *Under the coordinate square-lattice rectangular injectivity hypotheses, every bounded contiguous 2×3 rectangle and every bounded contiguous 3×2 rectangle is injective.*

Proof. [✓ Proof](#) This is the defining rectangular injectivity hypothesis applied to the coordinate witnesses for the two rectangle shapes. $\square$

Lemma 24.6.15 (Rectangular injectivity gives injective T -edge blocks). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) *For the displayed local T -region inside its 5×6 coordinate window, if the smaller edge-block bounds $x_0 + 5 \leq n$ and $y_0 + 5 \leq m$ hold in the finite $n \times m$ square lattice, then the coordinate square-lattice rectangular injectivity hypotheses imply that the vertical 2×3 block and horizontal 3×2 block removed from the window are injective. If, in addition, region injectivity is closed under unions, then the union of these two removed edge blocks is injective.*

Proof. [✓ Proof](#) Apply the rectangular injectivity hypotheses to the already identified contiguous rectangle shapes of the two edge blocks. The final assertion is one application of union closure to these two injective blocks. $\square$

Lemma 24.6.16 (Injectivity from a rectangular cover of the local T -region). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) *If the displayed local T -region has a nonempty finite cover by contiguous 2×3 and 3×2 rectangles, then rectangular injectivity and finite union closure imply that T is injective. The rotated vertical local T -region has the same conditional injectivity criterion.*

Proof. [✓ Proof](#) Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. $\square$

Lemma 24.6.17 (Injectivity from a rectangular cover of the edge complement). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) *If the finite-lattice complementary block A_3 has a nonempty finite cover by contiguous 2×3 and 3×2 rectangles, then rectangular injectivity and finite union closure imply that A_3 is injective. The same criterion applies to every vertical edge-complementary block: a nonempty finite cover of that block by contiguous 2×3 and 3×2 rectangles implies injectivity of the vertical edge-complementary block. In the normalized horizontal-edge 5×7 frame, it is enough to give such a cover of the local T -region, since adjoining the two top-collar 3×2 rectangles gives a cover of A_3 .*

Proof. [✓ Proof](#) Apply rectangular injectivity to every rectangle in the cover, then apply finite union closure to the nonempty family. For the normalized 5×7 edge complement, first extend the local T -cover by the two top-collar rectangles. $\square$

Lemma 24.6.18 (A cover of T extends to a cover of the edge complement). [Lean](#) [Lean](#)
[✓ Stmt](#) In the normalized horizontal-edge 5×7 frame with lower-left corner $(0,0)$, any rectangular cover of the local T -region extends to a rectangular cover of the edge-complementary block A_3 by adjoining the two top-collar contiguous 3×2 rectangles. In the normalized vertical-edge 7×5 frame with lower-left corner $(0,0)$, any rectangular cover of the rotated local T -region extends to a rectangular cover of the vertical edge-complementary block by adjoining the two right-collar contiguous 2×3 rectangles.

Proof. [✓ Proof](#) Index the new cover by the old cover indices together with two extra indices for the collar rectangles. For the horizontal block, the decomposition is

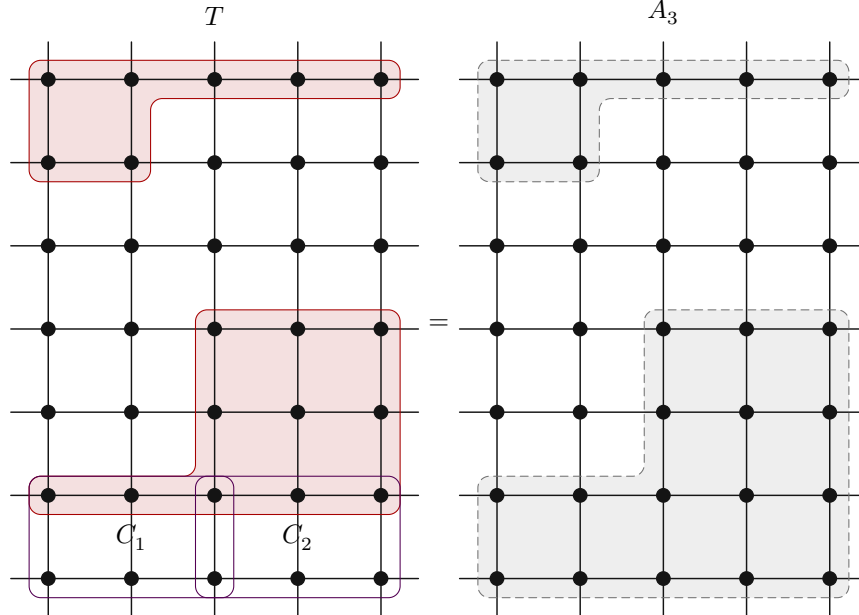
$$A_3 = T \cup (\{0, 1, 2\} \times \{5, 6\}) \cup (\{2, 3, 4\} \times \{5, 6\}).$$

For the vertical block, the decomposition is

$$A_3^v = T^v \cup (\{5, 6\} \times \{0, 1, 2\}) \cup (\{5, 6\} \times \{2, 3, 4\}).$$

□

Lemma 24.6.19 (The normalized edge complements have collars). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) In the normalized horizontal-edge 5×7 frame, the finite-lattice complementary block A_3 is the local-window T -region together with two top-collar contiguous 3×2 rectangles. Consequently, if the local T -region is injective, then rectangular injectivity and union closure imply that A_3 is injective. There is also a transposed 7×5 horizontal-frame collar identity with two contiguous 2×3 right-collar rectangles. For the rotated vertical edge-blocking regions, the corresponding complement is the rotated local T -region together with the same two right-collar rectangles; hence rotated T -injectivity, and hence a rectangular cover of rotated T , implies injectivity of the vertical complementary block.



Proof. [✓ Proof](#) The two set identities follow by expanding the coordinate inequalities. The injectivity statements apply union closure first to the local T -region and the first collar rectangle, and then to the second collar rectangle. □

Proof. ✓ Proof The endpoint memberships are coordinate checks. Rectangular injectivity gives the rectangular red and blue blocks. For the horizontal edge, the collar lemma gives the complement, and the rectangular cover criterion removes the direct T -injectivity input when such a cover is supplied. For the vertical edge, complement injectivity is an explicit input or is obtained from the rotated T -cover. The disjointness and cover statements are finite-set complement identities. $\square$

Lemma 24.6.24 (Translated horizontal edge-blocking data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *For any coordinate origin whose horizontal blocking window lies in the finite square lattice, there is a distinguished horizontal edge with the same relative position as in the normalized picture. The translated red region is the 2×3 block, the translated blue region is the 3×2 block, and the complementary region is the finite-lattice complement of their union. If this complementary region is injective, these three regions form the one-edge blocking datum; if it has a rectangular cover, rectangular injectivity and union closure supply that complementary injectivity; and two rows and columns of margin below and to the left together with eight rows and columns of total room around the frame supply the same datum directly. In each form the datum supplies the three injective regions used in the edge-blocked chain, and the distinguished edge is the only edge crossing between the red and blue regions. The distinguished edge is the corresponding coordinate right edge. At coordinate origin $0, 0$, the translated edge and the three translated regions specialize to the normalized horizontal picture.*

Proof. ✓ Proof The endpoint memberships and horizontal orientation are coordinate checks. Rectangular injectivity gives the red and blue blocks. The complementary block is supplied either as an input or from a rectangular cover, and the disjointness and covering identities follow from the finite-set complement formulas. $\square$

Lemma 24.6.25 (Translated vertical edge-blocking data). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *For any coordinate origin whose vertical blocking window lies in the finite square lattice, there is a distinguished vertical edge with the same relative position as in the normalized vertical picture. The translated red region is the 3×2 block, the translated blue region is the 2×3 block, and the complementary region is the finite-lattice complement of their union. If this complementary region is injective, these three regions form the one-edge blocking datum; if it has a rectangular cover, rectangular injectivity and union closure supply that complementary injectivity; and two rows and columns of margin below and to the left together with eight rows and columns of total room around the frame supply the same datum directly. In each form the datum supplies the three injective regions used in the edge-blocked chain, and the distinguished edge is the only edge crossing between the red and blue regions. The distinguished edge is the corresponding coordinate upward edge. At coordinate origin $0, 0$, the translated edge and the three translated regions specialize to the normalized vertical picture.*

Proof. ✓ Proof The endpoint memberships and vertical orientation are coordinate checks. Rectangular injectivity gives the red and blue blocks. The complementary block is supplied either as an input or from a rectangular cover, and the disjointness and covering identities follow from the finite-set complement formulas. $\square$

Theorem 24.6.26 (Bond-dimension equality at interior edges from rectangular injectivity). Lean ✓ Stmt *Let A and B be PEPS tensors on the $n \times m$ square lattice with positive physical dimension $d > 0$ and positive bond dimensions at every edge, generating the same state. Suppose both tensors satisfy the coordinate rectangular injectivity hypotheses (every contiguous 2×3 and*

datum recovered from the resulting hypotheses is the datum supplied by the corresponding translated window, and the resulting hypotheses give the three injective regions at every edge, together with $u_e \in R_e$, $v_e \in B_e$, $R_e \cap B_e = R_e \cap C_e = B_e \cap C_e = \emptyset$, and $R_e \cup B_e \cup C_e = V$. Equivalently, it is enough to supply the oriented margin-and-cover data at every edge; the resulting hypotheses recover the corresponding datum, give the same three injectivity conclusions, and give the same endpoint, disjointness, and covering conclusions. The obstruction theorem records that this interior margin-cover criterion does not by itself supply data at every edge of the open 7×7 lattice.

Proof. [✓ Proof](#) Apply the general construction for one-edge blocking data to the datum supplied by the chosen translated window at each edge. $\square$

Lemma 24.6.29 (Conditional injectivity of the normal R and S regions). [Lean](#) [Lean](#) [✓ Stmt](#) *If region injectivity is closed under unions, then the coordinate square-lattice rectangular injectivity hypotheses imply that the displayed regions R and S are injective.*

Proof. [✓ Proof](#) Region R is the union of one injective 2×3 rectangle and one injective 3×2 rectangle. Region S is the union of two injective 2×3 rectangles. Apply union closure in each case. $\square$

Lemma 24.6.30 (Injectivity of the normal R , S , and T regions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Fix a square-lattice PEPS tensor with every bond dimension positive whose blocked tensor is injective on every contiguous 2×3 and 3×2 rectangle. Then the displayed regions R and S are injective. If moreover the displayed region T has a cover by such rectangles, then T is injective too. The union closure used here is supplied by the overlapping union lemma, so no single-vertex injectivity is assumed.*

Proof. [✓ Proof](#) Apply the conditional injectivity of R and S (Lemma 24.6.29) and the conditional cover criterion for T (Lemma 24.6.16), with the union closure discharged by Theorem 24.5.21. $\square$

Definition 24.6.31 (Comparison square, comparison region, and interior window). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *At a coordinate offset (a, b) the comparison square is the 3×3 contiguous coordinate rectangle with lower-left corner (a, b) . The comparison region is this square with its lower-left corner (a, b) removed, presented as the union of the 2×3 contiguous rectangle at $(a + 1, b)$ and the 3×2 contiguous rectangle at $(a, b + 1)$. This is an auxiliary one-site comparison pair carrying its omitted site at the lower-left corner; it is the reflected counterpart of the displayed regions R and S , whose omitted site $S \setminus R$ sits at the lower-right corner, placed at an arbitrary offset rather than at a fixed origin. The interior window at (a, b) for a finite $n \times m$ lattice is the placement margin $4 \leq a$, $a + 9 \leq n$, $4 \leq b$, $b + 9 \leq m$, under which every boundary edge of the comparison square and of the comparison region carries the interior margins of the open edge-blocking geometry.*

Lemma 24.6.32 (Membership in the comparison region). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For any offset (a, b) a vertex with coordinates (x, y) lies in the comparison region at (a, b) if and only if*

$$(a + 1 \leq x < a + 3 \wedge b \leq y < b + 3) \vee (a \leq x < a + 3 \wedge b + 1 \leq y < b + 3). \quad (24.8)$$

Let v be a lattice vertex and place the comparison region at v 's own coordinates. Then v lies outside that region, and inserting v back in recovers the full 3×3 comparison square placed at v 's coordinates.

Proof. ✓ Proof Membership unfolds the union of the two contiguous rectangles into (24.8). Place the region at v 's own coordinates, so $(a, b) = (x, y) = (v_1, v_2)$. The corner offset (v_1, v_2) fails the first disjunct because $v_1 + 1 \leq v_1$ is false, and fails the second because $v_2 + 1 \leq v_2$ is false:

$$\neg((v_1 + 1 \leq v_1 < v_1 + 3) \wedge (v_2 \leq v_2 < v_2 + 3)) \\ \wedge \neg((v_1 \leq v_1 < v_1 + 3) \wedge (v_2 + 1 \leq v_2 < v_2 + 3)).$$

Thus v lies outside the region. Inserting v therefore yields the region together with $\{(v_1, v_2)\}$, and a vertex lies in this set exactly when it equals v or already lies in the region. That is the membership condition of the 3×3 square, the corner being the one cell of the square that both pieces omit. $\square$

Lemma 24.6.33 (Band decomposition of the comparison square complement). Lean ✓ Stmt *On a finite $n \times m$ lattice, if $a + 3 \leq n$ and $b + 3 \leq m$ then the complement of the comparison square at (a, b) is the union of the four coordinate bands*

$$\{(x, y) : 0 \leq x < a, 0 \leq y < m\} \cup \{(x, y) : a + 3 \leq x < n, 0 \leq y < m\} \\ \cup \{(x, y) : a \leq x < a + 3, 0 \leq y < b\} \cup \{(x, y) : a \leq x < a + 3, b + 3 \leq y < m\}. \quad (24.9)$$

Proof. ✓ Proof A vertex (x, y) lies in the comparison square exactly when $a \leq x < a + 3$ and $b \leq y < b + 3$. With $0 \leq x < n$ and $0 \leq y < m$ on the lattice, the negation of that conjunction is membership in (24.9):

$$\neg(a \leq x < a + 3 \wedge b \leq y < b + 3) \\ \iff (0 \leq x < a \wedge 0 \leq y < m) \vee (a + 3 \leq x < n \wedge 0 \leq y < m) \\ \vee (a \leq x < a + 3 \wedge 0 \leq y < b) \vee (a \leq x < a + 3 \wedge b + 3 \leq y < m).$$

$\square$

Lemma 24.6.34 (Band decomposition of the comparison region complement). Lean ✓ Stmt *On a finite $n \times m$ lattice, if $2 \leq a$, $1 \leq b$, $a + 3 \leq n$, and $b + 3 \leq m$ then the complement of the comparison region at (a, b) is the union of the same four bands of Lemma 24.6.33 together with the 3×2 corner rectangle $\{(x, y) : a - 2 \leq x < a + 1, b - 1 \leq y < b + 1\}$, which contains the omitted corner (a, b) .*

Proof. ✓ Proof By (24.8), a vertex (x, y) lies in the complement exactly when both disjuncts fail:

$$\neg(a + 1 \leq x < a + 3 \wedge b \leq y < b + 3) \\ \wedge \neg(a \leq x < a + 3 \wedge b + 1 \leq y < b + 3).$$

Together with $0 \leq x < n$ and $0 \leq y < m$, this places the vertex either in one of the four bands of (24.9) or in the 3×2 corner rectangle. Writing R for the comparison region and S for the comparison square,

$$V \setminus R = \{(x, y) : a - 2 \leq x < a + 1, b - 1 \leq y < b + 1\} \cup (V \setminus S),$$

since the comparison region is the comparison square with the corner (a, b) removed, and that corner lies in the displayed 3×2 rectangle. The hypotheses $2 \leq a$ and $1 \leq b$ keep the corner rectangle's lower-left coordinates $(a - 2, b - 1)$ within the lattice. $\square$

Lemma 24.6.35 (Injectivity of the comparison region and the comparison square). [Lean](#) [Lean](#)

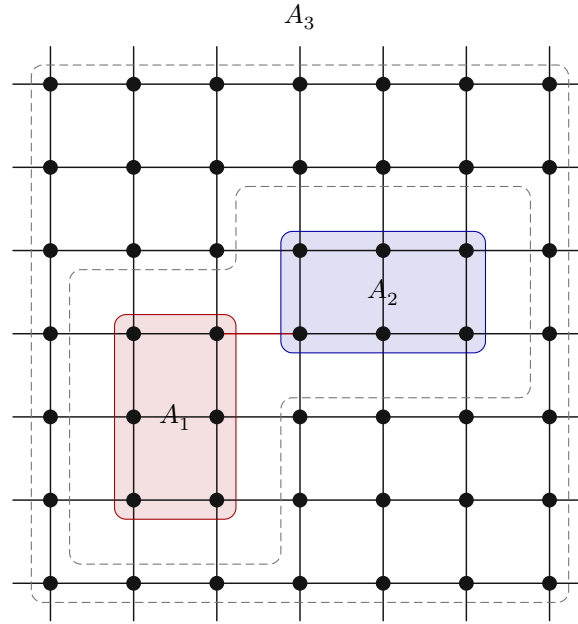
[✓ Stmt](#) Assume rectangular injectivity and union closure. On a finite $n \times m$ lattice, if $a + 3 \leq n$ and $b + 3 \leq m$ then the comparison region at (a, b) is injective, being the union of a 2×3 and a 3×2 rectangle, and the comparison square at (a, b) is injective, being a 3×3 rectangle.

Proof. [✓ Proof](#) Apply union closure to the rectangular injectivity of the contained 2×3 and 3×2 rectangles for the region, the square being a single large rectangle. $\square$

Lemma 24.6.36 (Injectivity of the comparison complements). [Lean](#) [Lean](#) [✓ Stmt](#) Assume rectangular injectivity and union closure. On a finite $n \times m$ lattice, if $2 \leq a$, $2 \leq b$, $a + 5 \leq n$, and $b + 5 \leq m$ then the complement of the comparison square at (a, b) is injective, being the union of its four surrounding bands, and the complement of the comparison region at (a, b) is injective, being that union together with one further corner rectangle.

Proof. [✓ Proof](#) Rewrite each complement by its band decomposition and apply union closure to the bands, each a large contiguous rectangle, and to the corner rectangle. $\square$

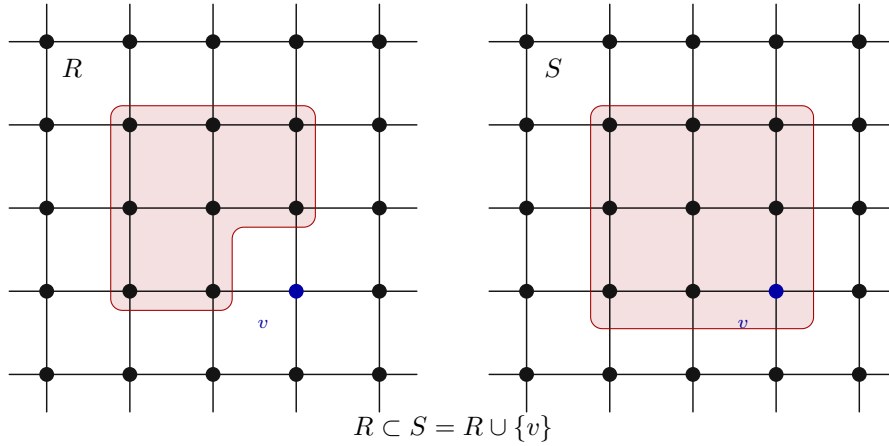
Definition 24.6.37 (Normal edge-blocking hypotheses). [Lean](#) [Lean](#) [✓ Stmt](#) Around one edge, the normal proof uses a blocking into three injective regions: a red region containing one endpoint, a blue region containing the other endpoint, and an injective complementary region. The three regions are pairwise disjoint and cover the whole vertex set. The edge-blocking hypotheses are precisely this one-edge datum at every edge; the red, blue, and complementary regions are read off from it. The local blocking picture used in the proof of [186, Theorem 3]:



Definition 24.6.38 (Left-boundary injectivity countermodel). [Lean](#) [✓ Stmt](#) In the open 7×7 rectangle, let $u = (0, 2)$ and $v = (1, 2)$ be the endpoints of a left-boundary horizontal edge. Define a region relation by

$$\text{inj}_{\partial}(R) \iff u \in R \implies v \in R.$$

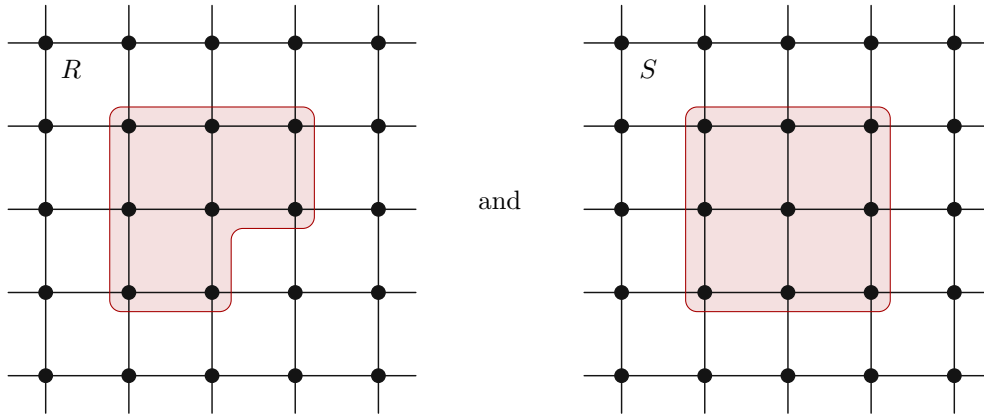
Theorem 24.6.39 (Union closure of the boundary countermodel). [Lean](#) [✓ Stmt](#) The relation inj_{∂} is closed under unions.

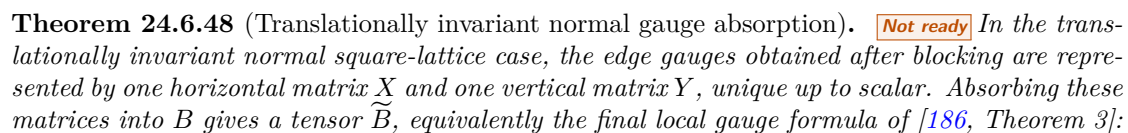
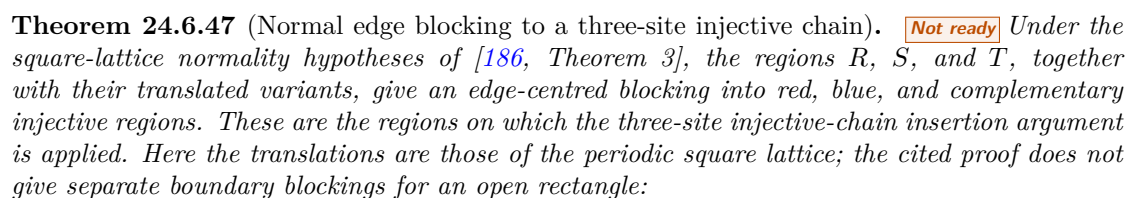


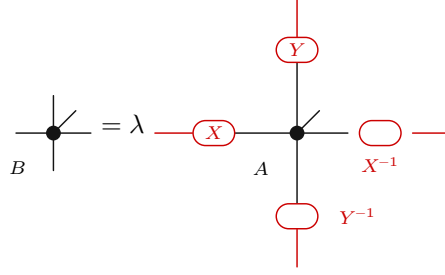
Theorem 24.6.45 (Membership in a one-site separated region). [Lean](#) [✓ Stmt](#) For the two regions associated to a site v , a vertex w belongs to the region containing v if and only if either $w = v$, or w belongs to the comparison region omitting v .

Proof. [✓ Proof](#) If $w = v$, use the distinguished-site membership and nonmembership assumptions. If $w \neq v$, use the equality of the two regions away from the distinguished site. $\square$

Definition 24.6.46 (Normal square-lattice blocking regions). [Lean](#) [Lean](#) [✓ Stmt](#) In the translationally invariant square-lattice proof, the source paper assumes that every 2×3 and 3×2 region is injective. The statement here specifies the corresponding abstract hypothesis: it records the rectangular injectivity assumptions, the size bound $n, m \geq 7$ with $|V| = nm$, and three finite regions R , S , and T whose injectivity is assumed directly. It does not by itself assert that these regions are the displayed square-lattice unions, nor that their translated variants give the edge-centred blocking around every edge. Conversely, for the coordinate square lattice with $n, m \geq 7$, the rectangular hypotheses, union closure, and a rectangular cover of the displayed T -region yield this abstract R, S, T structure, with R and S obtained from their rectangular decompositions and T obtained from the supplied cover. The source regions are:







The torus form of this absorption, with the gauge family carried onto every edge by the translations from one matrix per orientation class, is part of Theorem 24.7.1.9. This is the geometry used in the source. An analogous assertion for an open rectangle would require additional boundary data and is not an unproved case of the cited theorem.

Theorem 24.6.49 (Fundamental Theorem for normal square-lattice PEPS). Not ready Let A and B be translationally invariant normal square-lattice PEPS tensors such that every 2×3 and 3×2 region is injective. If they generate the same state on an $n \times m$ region with $n, m \geq 7$, then they are related by horizontal and vertical gauges X, Y , up to a scalar λ satisfying $\lambda^{nm} = 1$. This is [186, Theorem 3]. Theorem 24.6.50 proves a related interior-window theorem on an open rectangle, without translation invariance and with per-vertex scalars in place of the single λ ; the gauge-equivalence conclusion on a general connected graph, from region injectivity alone, is Theorem 24.8.6. The periodic statement of the source is established in Theorem 24.7.1.9. Extending the open-rectangle theorem to its boundary is a separate question.

Theorem 24.6.50 (Fundamental Theorem for normal square-lattice PEPS, interior-window form). Lean ✓ Stmt Let A and B be PEPS tensors on the open $n \times m$ square lattice with the same positive physical dimension, equal and positive bond dimensions, generating the same state, and such that every contiguous 2×3 and 3×2 coordinate rectangle of either tensor is injective. Then there is a family of invertible matrices, one on each edge, such that:

1. at every edge with interior margins (for a horizontal edge with left endpoint (x, y) : $3 \leq x$, $x+7 \leq n$, $4 \leq y$, $y+6 \leq m$; for a vertical edge the mirrored margins), inserting any matrix on the edge of A gives the same coefficient as inserting it on the corresponding edge of the gauge-absorbed B , for every physical configuration; and
2. every vertex $v = (x, y)$ of the interior comparison window $4 \leq x \leq n-9$, $4 \leq y \leq m-9$ carries a nonzero scalar λ_v such that A_v equals λ_v times the gauge action of the family on B at v , on every virtual configuration of the incident edges and every physical index.

This is the open-lattice form of [186, Theorem 3], restricted in two ways. First, the conclusion reaches only the interior: the absorbed equality holds at the edges with interior margins and the per-vertex relation at the window vertices, while vertices near the lattice boundary are not reached, the blocking frames of the open lattice requiring interior margins. Second, the scalars of distinct vertices are not asserted equal, and no condition $\lambda^{nm} = 1$ is asserted: the single λ of the source comes from translation invariance, which is not assumed here, and without it a single scalar is false — rescaling the components of A at two interior vertices by μ and μ^{-1} preserves the state and the rectangular injectivity but separates the two scalars. No lower bound on n or m is assumed; both conclusions are empty when the lattice is too small to contain interior edges or window vertices.

Proof. ✓ Proof The union closure of injective regions follows from the positive bond dimensions and Lemma 24.5.10. The translated blocking frames at the edges with interior margins produce

any graph isomorphism $\phi : G \simeq G'$ have equal state coefficients for every physical configuration on G' .

Proof. ✓ Proof By Theorem 24.7.3,

$$\text{Coeff}_{A^\phi}(\sigma) = \text{Coeff}_A(\sigma \circ \phi) = \text{Coeff}_B(\sigma \circ \phi) = \text{Coeff}_{B^\phi}(\sigma),$$

where the middle equality is the assumed equality of state coefficients. □

Definition 24.7.5 (Translation-invariant torus tensor). Lean ✓ Stmt A PEPS tensor A on the discrete torus is translation invariant if $A^{\tau_{a,b}} = A$ for every translation $\tau_{a,b}$. This is the setting of [186, Theorem 3], where one tensor is repeated at every site of a periodic lattice.

Theorem 24.7.6 (State coefficient of a translation-invariant torus tensor). Lean ✓ Stmt If A is translation invariant on the discrete torus, then $\text{Coeff}_A(\sigma \circ \tau_{a,b}) = \text{Coeff}_A(\sigma)$ for every translation $\tau_{a,b}$ and every physical configuration σ .

Proof. ✓ Proof Theorem 24.7.3 gives

$$\text{Coeff}_A(\sigma \circ \tau_{a,b}) = \text{Coeff}_{A^{\tau_{a,b}}}(\sigma) = \text{Coeff}_A(\sigma),$$

using $A^{\tau_{a,b}} = A$ from translation invariance. □

Definition 24.7.7 (Translation-invariant torus state). Lean ✓ Stmt A PEPS tensor A on the discrete torus represents a translation-invariant state if

$$\text{Coeff}_A(\sigma \circ \tau_{a,b}) = \text{Coeff}_A(\sigma)$$

for every lattice translation $\tau_{a,b}$ and every physical configuration σ . This condition concerns the contracted state; the local tensor family itself may depend on the lattice site.

Theorem 24.7.8 (Tensor translation invariance implies state translation invariance). Lean ✓ Stmt A translation-invariant torus tensor A represents a translation-invariant state: for every lattice translation $\tau_{a,b}$ and physical configuration σ ,

$$\text{Coeff}_A(\sigma \circ \tau_{a,b}) = \text{Coeff}_A(\sigma).$$

Proof. ✓ Proof This is Theorem 24.7.6 for every translation and physical configuration. □

Theorem 24.7.9 (State equality with every translated tensor). Lean ✓ Stmt If A represents a translation-invariant state, then A and $A^{\tau_{a,b}}$ represent the same state for every translation:

$$\text{Coeff}_A(\sigma) = \text{Coeff}_{A^{\tau_{a,b}}}(\sigma).$$

Proof. ✓ Proof Transport gives $\text{Coeff}_{A^{\tau_{a,b}}}(\sigma) = \text{Coeff}_A(\sigma \circ \tau_{a,b})$, and state translation invariance identifies the right-hand side with $\text{Coeff}_A(\sigma)$. □

Definition 24.7.10 (Orientation-uniform bond dimensions on the torus). Lean Lean Lean ✓ Stmt A bond-dimension function on the torus is orientation-uniform with horizontal dimension D_h and vertical dimension D_v if $D_{e_h} = D_h$ for every horizontal edge e_h and $D_{e_v} = D_v$ for every vertical edge e_v . The reference horizontal edge is the right edge at the origin, and the reference vertical edge is the up edge at the origin.

Since Z and G are invertible, so is W . Thus the diagonal entries cannot all vanish; the case $a = b$ gives $\lambda = 1$, and then $W_{a,a} = W_{b,b}$ for all a, b . Hence $W = c\mathbb{1}_4$ and $\lambda = 1$ for some $c \neq 0$, so $Z = cG$.

For a matrix $M \in M_4(\mathbb{C})$, write $M_{rr'} \in M_2(\mathbb{C})$ for the block whose (s, s') entry is $M_{(r,s),(r',s')}$. If Z were a per-edge product, then $Z_{rr'} = (Y_0)_{r,r'} Y_1$ for some $Y_0, Y_1 \in M_2(\mathbb{C})$. In particular, $(Z_{00})_{s,s'}(Z_{01})_{1,1} = (Z_{01})_{s,s'}(Z_{00})_{1,1}$. Since $Z = cG$ and $c \neq 0$, scaling cancels from this block proportionality condition. The corresponding blocks of G are

$$G_{00} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

$$G_{01} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Substitution into $(G_{00})_{00}(G_{01})_{11} = (G_{01})_{00}(G_{00})_{11}$ gives $0 = 1$. Hence no per-edge product conjugator can exist. $\square$

Staircase-window geometry for a reference edge.

Lemma 24.7.15 (Left endpoint of the staircase reference edge). [Lean](#) [✓ Stmt](#) *Let $L, K > 0$, let $s = (a, b)$ be a staircase corner on the torus, and assume $a + 2L \leq n$ and $2K \leq m$. Let e be the horizontal reference edge from $(a + L - 1, b + K - 1)$ to $(a + L, b + K - 1)$, and let W_j be the j th staircase window based at s . Then the left endpoint $u = (a + L - 1, b + K - 1)$ lies in W_j if and only if $1 \leq j$.*

Proof. [✓ Proof](#) The endpoint formula gives the two coordinate distances from the corner: the column distance is $L - 1$ and the row distance is $K - 1$. Substituting these distances into the staircase-window membership criterion gives exactly $1 \leq j$. $\square$

Lemma 24.7.16 (Right endpoint of the staircase reference edge). [Lean](#) [✓ Stmt](#) *In the same notation, the right endpoint $w = (a + L, b + K - 1)$ lies in W_j if and only if $j < L$.*

Proof. [✓ Proof](#) The right endpoint has column distance L and row distance $K - 1$ from s . The row condition is automatic, while the column band contains this distance precisely for $j < L$. $\square$

Lemma 24.7.17 (Boundary windows for the staircase reference edge). [Lean](#) [✓ Stmt](#) *If $j = 0$ or $L \leq j$, then the reference edge e is a boundary edge of W_j : exactly one of its two endpoints lies in the window.*

Proof. [✓ Proof](#) For $j = 0$, Lemmas 24.7.15 and 24.7.16 show that only w lies in W_j . For $L \leq j$, the same two endpoint tests show that only u lies in W_j . $\square$

Lemma 24.7.18 (Interior windows for the staircase reference edge). [Lean](#) [✓ Stmt](#) *If $1 \leq j < L$, then both endpoints of e lie in W_j .*

Proof. [✓ Proof](#) The left-endpoint criterion gives $u \in W_j$ from $1 \leq j$, and the right-endpoint criterion gives $w \in W_j$ from $j < L$. $\square$

Lemma 24.7.19 (Interior windows are not boundary windows). [Lean](#) [✓ Stmt](#) *If $1 \leq j < L$, then e is not a boundary edge of W_j .*

Proof. [✓ Proof](#) By Lemma 24.7.18, both endpoints of e lie in W_j . The definition of a boundary edge requires exactly one endpoint in the region, so e is not a boundary edge of W_j . $\square$

24.7.1 Staircase patch geometry

Definition 24.7.1.1 (Horizontal staircase patch and its corner). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Fix a horizontal reference edge and write the staircase corner as $s = (a, b)$. For coordinate intervals write $I(u, v) = \{t \mid u \leq t < v\}$. The two end windows of the overlapping-window chain are

$$\begin{aligned} W_{\text{left}} &= I(a, a + L) \times I(b, b + K), \\ W_{\text{right}} &= I(a + L, a + 2L) \times I(b + K - 1, b + 2K - 1). \end{aligned}$$

Their end pair is $S = W_{\text{left}} \cup W_{\text{right}}$. The staircase patch is the union

$$P = (I(a, a + 2L) \times I(b + K - 1, b + 2K - 1)) \cup (I(a, a + L) \times I(b, b + 2K - 1)),$$

and the corner cut out by the two end windows is $C = I(a, a + L) \times I(b + K, b + 2K - 1)$. Completing C by one row gives the $L \times K$ rectangle $\widehat{C} = I(a, a + L) \times I(b + K, b + 2K)$. These cyclic rectangles are the regions used in the staircase chaining and corner-stripping step of [186, proof sketch, lines 2320–2445].

Lemma 24.7.1.2 (The staircase end windows are disjoint). [Lean](#) [✓ Stmt](#) If $2L \leq W$, then the two end windows of the horizontal staircase are disjoint: $W_{\text{left}} \cap W_{\text{right}} = \emptyset$.

Proof. [✓ Proof](#) Their horizontal coordinate intervals are $I(a, a + L)$ and $I(a + L, a + 2L)$. The bound $2L \leq W$ prevents wraparound overlap. $\square$

Lemma 24.7.1.3 (Injectivity of the staircase end windows and completed corner). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Under the one-orientation $L \times K$ torus-window injectivity hypotheses, the three regions W_{left} , W_{right} , and $\widehat{C}$ are injective.

Proof. [✓ Proof](#) Each of the three regions is an $L \times K$ cyclic rectangle, so the window-injectivity hypothesis applies directly. $\square$

Lemma 24.7.1.4 (Patch decomposition by the end pair and the corner). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) If $L, K > 0$, $2L \leq W$, and $2K \leq H$, then the corner is disjoint from the end pair and the patch decomposes as $P = S \cup C$, with $P \setminus S = C$.

Proof. [✓ Proof](#) In the no-wraparound rectangle, the rows and columns of the two end windows and the corner are the displayed intervals above. The interval calculation gives the disjointness and identifies every point of P with either an end-window point or a corner point. $\square$

Definition 24.7.1.5 (Torus rectangular injectivity hypotheses). [Lean](#) [✓ Stmt](#)

On the discrete torus, the rectangular injectivity hypotheses assert that every contiguous 2×3 and every contiguous 3×2 coordinate rectangle is injective for the given region-injectivity assignment. This is the torus counterpart of the square-lattice rectangular injectivity hypotheses.

Lemma 24.7.1.6 (Torus reference-edge blocking data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) At a frame origin (x_0, y_0) with at least two columns and one row of margin below and to the left (one column and two rows in the rotated case), and with the removed blocks either touching the seam ($x_0 + 5 = n$, $y_0 + 5 = m$) or keeping a further margin of at least two, the horizontal reference edge e_h is the right edge leaving $(x_0 + 1, y_0 + 2)$, and the vertical reference edge e_v is the up edge leaving $(x_0 + 2, y_0 + 1)$. Around each reference edge the rectangular injectivity hypotheses and union closure supply the one-edge blocking datum: the red block, the blue block,

and the complement are injective, and the reference edge is the only edge crossing between the red and blue blocks,

$$\{g : g \text{ joins the red block to the blue block}\} = \{e_h\},$$

and similarly with e_v at the rotated frame.

Proof. ✓ **Proof** The red and blue blocks are the removed edge blocks of the frame, so rectangular injectivity covers them; union closure then gives the injectivity of the remaining complement. The crossing-edge identity enumerates the edges joining the two blocks. $\square$

Theorem 24.7.1.7 (Bond-dimension equality from normal torus rectangular injectivity). Lean

✓ **Stmnt** Let A and B be translationally invariant PEPS tensors on the $n \times m$ torus with $n, m \geq 7$, positive physical dimension $d > 0$, and positive bond dimensions at every edge, generating the same state. Suppose both tensors satisfy the torus rectangular injectivity hypotheses (every contiguous 2×3 and 3×2 torus rectangle is injective) and the associated region-injectivity predicates are closed under finite unions. Then the bond dimensions of A and B agree on every edge:

$$D_A = D_B.$$

The equality is derived from the one-edge blocking data built from the rectangular injectivity — the source’s Lemma `inj_isomorph` ([186, Section 3, lines 560–583 and 1407–1500]) — together with the translation invariance, same-state, and positive-bond assumptions; the bond dimension is not assumed as a hypothesis.

Proof. ✓ **Proof** Place the frame at $(x_0, y_0) = (n-5, m-5)$; since $n, m \geq 7$ the margins hold with the touching-seam alternative. Let e_h and e_v be the horizontal and vertical reference edges of Lemma 24.7.1.6 at this frame. The blocking data built from A and from B around e_h share the same red and blue blocks, so the one-edge comparison of Theorem 24.8.5 applies at the unique crossing edge and gives

$$D_A(e_h) = D_B(e_h),$$

and the rotated datum at e_v gives

$$D_A(e_v) = D_B(e_v).$$

Every torus edge e is horizontal or vertical. A horizontal e is the right edge leaving some vertex p , and with the translation τ carrying the reference vertex to p , translation invariance (Theorem 24.7.11) gives

$$D_A(e) = D_A(\tau e_h) = D_A(e_h),$$

and the same for B ; hence $D_A(e) = D_B(e)$. A vertical e is treated through e_v , and the equality $D_A = D_B$ follows. $\square$

Theorem 24.7.1.8 (Fundamental Theorem for normal PEPS on the torus). ✓ **Stmnt** Let A and B be translationally invariant normal PEPS on the discrete torus such that every 2×3 and 3×2 region is injective. If they generate the same state on an $n \times m$ region with $n, m \geq 7$, then they are related by horizontal and vertical gauges X, Y , up to a scalar λ satisfying $\lambda^{nm} = 1$, and X and Y are unique up to a multiplicative constant. This is [186, Theorem 3] on the torus geometry, where every edge is interior and translation acts freely. Theorem 24.7.1.9 proves this at the source’s

sizes $n, m \geq 7$ with the gauge family described by its translation covariance, and the uniqueness of X and Y up to a multiplicative constant is proved at the same sizes in Theorem 24.7.1.11 per edge, Theorem 24.7.1.12 on the two orientation classes, and Theorem 24.7.1.13 against the per-vertex relation. In the torus edge convention the two class matrices X and Y are read through this translation-covariant family: seam edges carry the transposed inverse of the corresponding class matrix, so this is the orientation-faithful form of the source's horizontal and vertical gauges.

Theorem 24.7.1.9 (Fundamental Theorem for normal PEPS on the torus, translation-covariant form). Lean ✓ Stmt Let A and B be translationally invariant PEPS tensors on the discrete $n \times m$ torus with $n, m \geq 7$, with positive physical dimension and positive bond dimensions, generating the same state, and such that every contiguous 2×3 and 3×2 coordinate rectangle of either tensor is injective. Then the bond dimensions of A and B agree on every edge, and there are a family of invertible matrices, one on each edge, carried onto each edge by the torus translations from a single matrix on a reference horizontal edge and a single matrix on a reference vertical edge, and a single scalar λ with $\lambda^{nm} = 1$, such that at every site the tensor A equals λ times the gauge action of the family on B . The family also realizes the absorbed inserted-coefficient equality at every edge: inserting any matrix on an edge of A gives the same coefficient as inserting it on the corresponding edge of the gauge-absorbed B . This is [186, Theorem 3] on the torus at the source's sizes $n, m \geq 7$, with the horizontal and vertical gauges X, Y described by translation covariance: the stored orientation of a wraparound edge is reversed, so on the seam the family carries the transposed inverse of the class matrix, and a family that is literally one matrix per orientation class cannot realize the absorbed equality there. The translation covariance of the family, with the scalar λ kept separate rather than absorbed, realizes the remark of [186] following the general normal theorem: for a translationally invariant system the gauges are also translationally invariant, provided the proportionality constants are not absorbed into the gauges. The uniqueness of X and Y up to a multiplicative constant is not part of this statement; the uniqueness clause is Theorem 24.7.1.11 (per edge, against the absorbed equality), Theorem 24.7.1.12 (one constant per orientation class), and Theorem 24.7.1.13 (against the per-vertex relation itself).

Proof. ✓ Proof The union closure of injective regions follows from the positive bond dimensions and Lemma 24.5.10, and the bond-dimension equality $D_A = D_B$ is derived from the rectangular injectivity hypotheses and the same-state hypothesis by Theorem 24.7.1.7; neither is assumed. The reference blockings are anchored against the far seam: the removed blocks keep a margin of at least two rows and columns below and to the left and end exactly at the seam above and to the right, so the complementary region decomposes into wraparound-free rectangular bands on every torus with $n, m \geq 7$. The coherent blocking frames at every edge produce per-edge gauges, whose absorption into B gives the absorbed inserted-coefficient equality at every edge; the absorbing family is built by transporting one reference gauge per orientation class, so it commutes with the torus translations. A gauge preserves blocked-region linear independence, so the region comparison applies to the gauge-absorbed tensor: comparing the corner region with its one-site completion yields proportionality scalars c_R and c_S such that, with $\lambda = c_S/c_R$,

$$A_v(\eta, \sigma) = \lambda(\text{gauge action of } X \text{ on } B)_v(\eta, \sigma).$$

Translation covariance carries the same scalars to every site. Taking the product over all vertices and using the invariance of the closed state coefficient gives $\lambda^{nm} = 1$. □

Corollary 24.7.1.10 (Torus scalar condition from the per-vertex two-block proportionalities).

Lean ✓ Stmt Let A and B be PEPS tensors on the discrete $n \times m$ torus with $n, m > 1$, with equal bond dimensions and positive physical dimension, generating the same state, and let X be a family

of invertible matrices, one on each edge. Suppose that at every site v a region R_v not containing v is supplied together with the two two-block scalar proportionalities of the blocked weights of A and of the gauge-absorbed second tensor over R_v and over $R_v \cup \{v\}$, with ratios $c_R(v)$ and $c_S(v)$ satisfying $c_R(v) \neq 0$ and whose quotient $c_S(v)/c_R(v)$ is a single scalar λ independent of v , and with the gauge-absorbed region block linearly independent. If, in addition, a single comparison region whose block and complement block are both blocked-tensor injective is supplied, then λ raised to the number of torus sites is one: $\lambda^{nm} = 1$.

Proof. ✓ Proof At each of the nm sites v the inserted-site scalar extraction of Corollary 24.11.1.3 turns the two two-block proportionalities into the per-vertex gauge relation

$$A_v(\eta, \sigma) = \lambda \widetilde{B}_v(\eta, \sigma),$$

where $\widetilde{B}_v$ is the gauge action of X on B at v . Contracting both sides over the whole torus pulls one factor of λ out of each of the nm sites, so the closed torus contractions of A and of $\widetilde{B}$ satisfy

$$\langle A \rangle_{\text{torus}} = \lambda^{nm} \langle \widetilde{B} \rangle_{\text{torus}}.$$

The single comparison region and its complement being injective makes this common torus contraction nonzero; by hypothesis the quotient $c_S(v)/c_R(v)$ is the same scalar λ at every site, so $\lambda^{nm} = 1$. □

Theorem 24.7.1.11 (Per-edge uniqueness of the torus gauge up to a constant). Lean ✓ Stmt Let A and B be translationally invariant PEPS tensors on the discrete $n \times m$ torus with $n, m \geq 7$, with positive physical dimension, equal and positive bond dimensions, generating the same state, and such that every contiguous 2×3 and 3×2 coordinate rectangle of either tensor is injective. Let X and X' be two families of invertible matrices, one on each edge of the torus, and let e be an edge at which both families realize the absorbed inserted-coefficient equality: inserting any matrix on the edge e of A gives the same coefficient as inserting it on the edge e of the gauge-absorbed B , for every physical configuration. Then the two families are proportional at e : $X'_e = c \cdot X_e$ for a nonzero scalar c . This is the uniqueness clause of [186, Theorem 3] read at one edge, at the source's sizes $n, m \geq 7$ as in Theorem 24.7.1.9; the source proves it through the determinacy of the conjugator in the isomorphism lemma, whose realizing matrix is unique up to a multiplicative constant.

Proof. ✓ Proof The rectangle hypotheses produce, along the orientation class of e , a comparison region having e as its single boundary edge on which the second tensor and its complement are injective; the union closure of injective regions follows from the positive bond dimensions and Lemma 24.5.10. The bare-edge absorbed equality of each family lifts to the region inserted-coefficient identity over this region, with the inserted matrix conjugated by the orientation-adapted absorption of the family's matrix at e . The region-insertion transfer map is determined by this identity, so both orientation-adapted absorbing matrices induce the same conjugation on every bond matrix. Thus, for every bond matrix N ,

$$X_{\text{abs}} N X_{\text{abs}}^{-1} = X'_{\text{abs}} N (X'_{\text{abs}})^{-1}.$$

A matrix commuting with the full bond matrix algebra is a scalar multiple of the identity, so $X'_{\text{abs}} = c X_{\text{abs}}$ for some $c \neq 0$. Unwinding the orientation adaptation gives $X'_e = c \cdot X_e$. □

Theorem 24.7.1.12 (Class constancy of the gauge comparison scalars). Lean ✓ Stmt Let A, B, X, X' be as in Theorem 24.7.1.11, with both families realizing the absorbed inserted-coefficient equality at every edge, and suppose moreover that both families are translation covariant: the

matrix at any translate of an edge is the matrix at the edge, transposed-inverted exactly when the translation swaps the stored endpoint order. Then there are nonzero scalars c_h and c_v such that on every horizontal edge stored in the natural cyclic order (the stored second endpoint is the cyclic right neighbour of the first) $X'_e = c_h \cdot X_e$, on every horizontal edge wrapping the seam (stored with swapped endpoints) $X'_e = c_h^{-1} \cdot X_e$, and likewise on the vertical class with c_v . The inversion on the seam is forced by the ordered-edge convention: the covariance carries the transposed inverse of the class matrix onto a seam edge, and the transposed inverse of a proportionality inverts its constant. A single constant on the whole class is therefore not the true statement in this convention; this is the class form of the uniqueness clause of [186, Theorem 3].

Proof. ✓ Proof The constants are the per-edge scalars of Theorem 24.7.1.11 at the two reference edges. Every edge of a class is the translate of its reference edge by a unique translation τ . At an interior edge, covariance gives $X_e = \tau \cdot X_{\text{ref}}$, hence

$$X'_e = \tau \cdot X'_{\text{ref}} = \tau \cdot (c_h X_{\text{ref}}) = c_h (\tau \cdot X_{\text{ref}}) = c_h X_e.$$

At a seam edge, covariance gives $X_e = (X_{\text{ref}})^{-\top}$, hence

$$X'_e = (X'_{\text{ref}})^{-\top} = (c_h X_{\text{ref}})^{-\top} = c_h^{-1} (X_{\text{ref}})^{-\top} = c_h^{-1} X_e.$$

The marker $e_{1,x} + 1 = e_{2,x}$ distinguishes the two branches, and the vertical class is identical with c_v in place of c_h . □

Theorem 24.7.1.13 (Uniqueness of the gauges against the per-vertex relation). Lean ✓ Stmt Let A and B be translationally invariant PEPS tensors on the discrete $n \times m$ torus with $n, m \geq 7$, with positive physical dimension, equal and positive bond dimensions, generating the same state, and such that every contiguous 2×3 and 3×2 coordinate rectangle of either tensor is injective. Suppose the families X and X' of invertible matrices, one on each edge, each realize the per-vertex relation of Theorem 24.7.1.9: at every site the tensor A equals λ (respectively λ') times the gauge action of the family on B , with $\lambda^{nm} = 1$ (respectively $\lambda'^{nm} = 1$). Then the two families are proportional at every edge: $X'_e = c \cdot X_e$ for a nonzero scalar c depending on the edge. This is the uniqueness clause of [186, Theorem 3] read against the displayed relation $B = \lambda \cdot (X, Y)A$ itself, at the source's sizes $n, m \geq 7$.

Proof. ✓ Proof The inserted-edge coefficient is multilinear in the site tensors, one factor per site, so the per-vertex relation scales every inserted coefficient by $\lambda^{nm} = 1$: each family realizes the absorbed inserted-coefficient equality at every edge. Theorem 24.7.1.11 then gives the proportionality at each edge. □

24.7.2 Overlapping-window corner cancellation

Definition 24.7.2.1 (Corner extension of a region insert). Lean Lean Lean ✓ Stmt Let $R \subseteq S$ be regions in a finite graph, and let C be a region insert on R . Write $\Phi_{R,S}$ for the coefficient obtained by contracting the added block $S \setminus R$ between the boundary of R and the boundary of S . The bare corner extension is

$$\overline{\text{Ext}}_{R \subseteq S}(C)_{\nu, \sigma} = \sum_{\mu \in \partial R} C_{\mu, \sigma|_R} \Phi_{R,S}(\mu, \sigma|_{S \setminus R}, \nu).$$

The normalized corner extension of C from R to S is the region insert

$$\text{Ext}_{R \subseteq S}(C)_{\nu, \sigma} = P_{V \setminus S}^{-1} \sum_{\mu \in \partial R} C_{\mu, \sigma|_R} \Phi_{R,S}(\mu, \sigma|_{S \setminus R}, \nu),$$

where $P_{V \setminus S}$ is the product of bond dimensions over the edges not crossing the boundary of $V \setminus S$. This is the open-boundary form of the extension used in the overlapping-window proof sketch of [186, Section 4, proof of Theorem 3].

Theorem 24.7.2.2 (State preservation under corner extension). Lean ✓ Stmt *If all bond dimensions are positive, then corner extension from $R \subseteq S$ preserves the deformed region state: $\Psi_{A,R,C} = \Psi_{A,S,\text{Ext}_{R \subseteq S}(C)}$.*

Proof. ✓ Proof Expanding the deformed state on S separates the added block $S \setminus R$ from the complement $V \setminus S$. The corner-extension sum contracts the added block first, giving the pointwise identity

$$\Psi_{A,S,\text{Ext}_{R \subseteq S}(C)}(\tau) = P_{V \setminus S}^{-1} P_{V \setminus S} \Psi_{A,R,C}(\tau) = \Psi_{A,R,C}(\tau).$$

□

Theorem 24.7.2.3 (Consecutive horizontal windows have equal corner extensions). Lean ✓ Stmt *Assume the one-orientation $L \times K$ window-injectivity hypotheses, union closure for injective regions, positive bond dimensions, and the bounds $2 \leq L$, $2 \leq K$, $2L + 1 \leq W$, and $2K + 1 \leq H$. Suppose two horizontally consecutive $L \times K$ windows carry inserts C_1 and C_2 whose deformed states agree. Let U be their $(L + 1) \times K$ union. Then their extensions to U agree:*

$$\text{Ext}_{W_1 \subseteq U}(C_1) = \text{Ext}_{W_2 \subseteq U}(C_2).$$

Proof. ✓ Proof Theorem 24.7.2.2 rewrites each single-window deformed state as the deformed state on U with the corresponding extended insert:

$$\Psi_{A,U,\text{Ext}_{W_1 \subseteq U}(C_1)} = \Psi_{A,W_1,C_1} = \Psi_{A,W_2,C_2} = \Psi_{A,U,\text{Ext}_{W_2 \subseteq U}(C_2)}.$$

Since U is injective, equality of the two U -states implies equality of the two inserts:

$$\Psi_{A,U,\text{Ext}_{W_1 \subseteq U}(C_1)} = \Psi_{A,U,\text{Ext}_{W_2 \subseteq U}(C_2)} \implies \text{Ext}_{W_1 \subseteq U}(C_1) = \text{Ext}_{W_2 \subseteq U}(C_2).$$

□

Uniform interior-bond products for staircase windows. For a PEPS tensor A and a region R , write $P_A(R) = \prod_{e \notin \partial R} D_A(e)$ for the product of bond dimensions over the non-boundary edges of R .

Theorem 24.7.2.4 (Interior-bond product under a graph isomorphism). Lean ✓ Stmt *Let $\phi : G \simeq G'$ be an isomorphism of finite graphs, and let R be a region of G . If A' is a PEPS tensor on G' , then $P_{A'}(\phi(R)) = \prod_{e \notin \partial R} D_{A'}(\phi(e))$.*

Proof. ✓ Proof An edge of $\phi(R)$ is non-boundary exactly when its preimage is non-boundary for R : $e' \notin \partial \phi(R)$ if and only if $\phi^{-1}(e') \notin \partial R$. Reindexing the product over non-boundary edges by the edge bijection $e \mapsto \phi(e)$ gives the stated equality. □

Theorem 24.7.2.5 (Translation invariance of the interior-bond product). Lean ✓ Stmt *Let A be translation invariant on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. For every translation $\tau_{a,b}$ and every region R , one has $P_A(\tau_{a,b}(R)) = P_A(R)$.*

Proof. ✓ Proof Theorem 24.7.2.4 gives $P_A(\tau_{a,b}(R)) = \prod_{e \notin \partial R} D_A(\tau_{a,b}(e))$. Translation invariance gives $D_A(\tau_{a,b}(e)) = D_A(e)$ for every edge e , so the product is $P_A(R)$. □

Theorem 24.7.2.6 (Interior-bond product of a cyclic rectangle is start-independent). [Lean](#) [✓ Stmt](#) Let A be translation invariant on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. If $Q(s; x, y)$ denotes the cyclic $x \times y$ rectangle starting at s , then for any starts s, s' , $P_A(Q(s; x, y)) = P_A(Q(s'; x, y))$.

Proof. [✓ Proof](#) If $s = (s_x, s_y)$ and $s' = (s'_x, s'_y)$, then the translation by $(s_x - s'_x, s_y - s'_y)$ carries one rectangle to the other: $\tau_{s_x - s'_x, s_y - s'_y}(Q(s'; x, y)) = Q(s; x, y)$. Apply Theorem 24.7.2.5. $\square$

Theorem 24.7.2.7 (Uniform interior-bond product of the staircase windows). [Lean](#) [✓ Stmt](#) Let A be translation invariant on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. If W_j and $W_{j'}$ are two windows of the staircase based at the same corner s with window dimensions $L \times K$, then $P_A(W_j) = P_A(W_{j'})$.

Proof. [✓ Proof](#) Each staircase window is an $L \times K$ cyclic rectangle: $W_j = Q(s_j; L, K)$ and $W_{j'} = Q(s_{j'}; L, K)$. Theorem 24.7.2.6 gives $P_A(Q(s_j; L, K)) = P_A(Q(s_{j'}; L, K))$. $\square$

Theorem 24.7.2.8 (Uniform horizontal per-step bond transport). [Lean](#) [✓ Stmt](#) Let B be translation invariant on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. Assume the one-orientation $L \times K$ window-injectivity hypotheses, union closure for injective regions, positive bond dimensions, and the bounds $2 \leq L$, $2 \leq K$, $2L + 1 \leq W$, and $2K + 1 \leq H$. Let $j < L$, let $U_j = W_j \cup W_{j+1}$ be the horizontal consecutive-window union, and let g be a boundary edge of both W_j and W_{j+1} . If the matrices obtained from M_1 and M_2 by the orientation convention at g agree, then, with $P_j = P_B(W_j)$,

$$\text{Ext}_{W_j \subseteq U_j}(P_j I_{B, W_j, g, M_1}) = \text{Ext}_{W_{j+1} \subseteq U_j}(P_j I_{B, W_{j+1}, g, M_2}). \quad (24.11)$$

Proof. [✓ Proof](#) The consecutive-window transport gives the cross-rescaled equality

$$\text{Ext}_{W_j \subseteq U_j}(P_B(W_{j+1}) I_{B, W_j, g, M_1}) = \text{Ext}_{W_{j+1} \subseteq U_j}(P_B(W_j) I_{B, W_{j+1}, g, M_2}). \quad (24.12)$$

By Theorem 24.7.2.7, $P_B(W_{j+1}) = P_B(W_j) = P_j$. Substituting this equality into (24.12) gives (24.11). $\square$

Theorem 24.7.2.9 (Uniform vertical per-step bond transport). [Lean](#) [✓ Stmt](#) Let B be translation invariant on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. Assume the one-orientation $L \times K$ window-injectivity hypotheses, union closure for injective regions, positive bond dimensions, and the bounds $2 \leq L$, $2 \leq K$, $2L + 1 \leq W$, and $2K + 1 \leq H$. Let $L \leq j$ and $j + 1 < L + K$, let $U_j = W_j \cup W_{j+1}$ be the vertical consecutive-window union, and let g be a boundary edge of both W_j and W_{j+1} . If the matrices obtained from M_1 and M_2 by the orientation convention at g agree, then, with $P_j = P_B(W_j)$,

$$\text{Ext}_{W_j \subseteq U_j}(P_j I_{B, W_j, g, M_1}) = \text{Ext}_{W_{j+1} \subseteq U_j}(P_j I_{B, W_{j+1}, g, M_2}). \quad (24.13)$$

Proof. [✓ Proof](#) The descending-arm transport gives the same cross-rescaled equality as in the horizontal case, with U_j now the $L \times (K + 1)$ union:

$$\text{Ext}_{W_j \subseteq U_j}(P_B(W_{j+1}) I_{B, W_j, g, M_1}) = \text{Ext}_{W_{j+1} \subseteq U_j}(P_B(W_j) I_{B, W_{j+1}, g, M_2}). \quad (24.14)$$

The uniformity theorem gives $P_B(W_{j+1}) = P_B(W_j) = P_j$. Substituting this equality into (24.14) gives (24.13). $\square$

Theorem 24.7.2.10 (Patch-extended equality of the staircase end windows). [Lean](#) [✓ Stmt](#) Assume positive bond dimensions and the bounds $0 < L$, $0 < K$, $2L \leq W$, and $2K \leq H$. If the deformed states of the two end-window inserts agree, then their extensions to the staircase patch P have equal deformed states:

$$\Psi_{A,P,\text{Ext}_{W_{\text{right}} \subseteq P}(C_0)} = \Psi_{A,P,\text{Ext}_{W_{\text{left}} \subseteq P}(C_{\text{end}})}.$$

Proof. [✓ Proof](#) Applying Theorem 24.7.2.2 to each end window gives

$$\Psi_{A,P,\text{Ext}_{W_{\text{right}} \subseteq P}(C_0)} = \Psi_{A,W_{\text{right}},C_0} = \Psi_{A,W_{\text{left}},C_{\text{end}}} = \Psi_{A,P,\text{Ext}_{W_{\text{left}} \subseteq P}(C_{\text{end}})},$$

where the middle equality is the hypothesis. $\square$

Lemma 24.7.2.11 (Two-block merge collapse). [Lean](#) [✓ Stmt](#) Let $B_1, K_1 \subseteq V \setminus T$ and $B_2, H_2 \subseteq T$. Fix boundary data c_1 on $V \setminus K_1$ and c_2 on H_2 , and physical labels σ_1 on B_1 and σ_2 on B_2 . Set

$$F_1(\xi) = \prod_{x \in B_1} A_x(\xi|_{\partial x}, \sigma_1(x)),$$

$$F_2(\xi) = \prod_{x \in B_2} A_x(\xi|_{\partial x}, \sigma_2(x)).$$

Then the boundary-compatible double sum collapses to the multiplicity of the interior bonds of T times the single merged sum:

$$\sum_{\substack{\zeta_2|_{\partial(V \setminus K_1)} = c_1, \zeta_1|_{\partial H_2} = c_2 \\ \zeta_1|_{\partial T} = \zeta_2|_{\partial T}}} F_1(\zeta_2) F_2(\zeta_1) = P_T \sum_{\eta|_{\partial(V \setminus K_1)} = c_1, \eta|_{\partial H_2} = c_2} F_1(\eta) F_2(\eta).$$

Proof. [✓ Proof](#) Each merged configuration has exactly P_T preimages, indexed by the virtual bonds strictly inside T . Since B_1 and K_1 lie outside T , while B_2 and H_2 lie inside T , the two products and the two boundary conditions are unchanged under the merge. $\square$

Theorem 24.7.2.12 (Composition of corner-extension coefficient kernels). [Lean](#) [✓ Stmt](#) For nested regions $R \subseteq S \subseteq P$, the coefficient kernels satisfy

$$\sum_{\nu \in \partial S} \Phi_{R,S}(\mu, \sigma|_{S \setminus R}, \nu) \Phi_{S,P}(\nu, \sigma|_{P \setminus S}, \omega) = P_{V \setminus S} \Phi_{R,P}(\mu, \sigma|_{P \setminus R}, \omega).$$

This is the two-block merge collapse in the proof of [186, Section 4, proof of Theorem 3].

Theorem 24.7.2.13 (Linearity and transitivity of corner extension). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The corner extension is linear in the inserted boundary coefficients. For nested regions $R \subseteq S \subseteq P$, the bare extension satisfies

$$\overline{\text{Ext}}_{S \subseteq P}(\overline{\text{Ext}}_{R \subseteq S}(C)) = P_{V \setminus S} \overline{\text{Ext}}_{R \subseteq P}(C)$$

without a positivity assumption. If all bond dimensions are positive, the normalized corner extension is transitive:

$$\text{Ext}_{S \subseteq P}(\text{Ext}_{R \subseteq S}(C)) = \text{Ext}_{R \subseteq P}(C).$$

Proof. [✓ Proof](#) The scalar and additive laws follow from the defining sum. For transitivity, substitute Theorem 24.7.2.12 into the defining sum. This gives the bare formula, and the factors $P_{V \setminus S}$ cancel in the normalized extension. $\square$

Definition 24.7.2.14 (Combination of subregion physical configurations). [Lean](#) [✓ Stmt](#) If $R \subseteq S$, a physical configuration on R and a physical configuration on $S \setminus R$ combine into a physical configuration on S :

$$\text{As}_{R,S}(\sigma_R, \sigma_Q)(v) = \begin{cases} \sigma_R(v), & v \in R, \\ \sigma_Q(v), & v \in S \setminus R. \end{cases}$$

Theorem 24.7.2.15 (Restrictions of combined subregion configurations). [Lean](#) [Lean](#) [✓ Stmt](#) The two restrictions recover the original configurations: $\text{As}_{R,S}(\sigma_R, \sigma_Q)|_R = \sigma_R$ and $\text{As}_{R,S}(\sigma_R, \sigma_Q)|_{S \setminus R} = \sigma_Q$.

Theorem 24.7.2.16 (Kernel triviality of corner extension from the added block). [Lean](#) [✓ Stmt](#) Let $R \subseteq S$ be finite regions. Suppose that all bond dimensions are positive and that the added block $Q = S \setminus R$ is blocked-tensor injective. If a region insert C on R has zero corner extension, $\text{Ext}_{R \subseteq S}(C) = 0$, then $C = 0$.

Proof. [✓ Proof](#) Since all bond dimensions are positive, $P_{V \setminus S} \neq 0$. Thus $\text{Ext}_{R \subseteq S}(C) = 0$ is equivalent to $\overline{\text{Ext}}_{R \subseteq S}(C) = 0$. Fix the physical configuration σ_R on R . Let $\rho : \partial R \simeq \partial(V \setminus R)$ and $\tau : \partial S \simeq \partial(V \setminus S)$ be the complement-boundary bijections, and define $c_\eta = C_{\rho^{-1}(\eta), \sigma_R}$ for $\eta \in \partial(V \setminus R)$. Evaluating the bare zero extension at $\text{As}_{R,S}(\sigma_R, \sigma_Q)$ gives, for every added-block configuration σ_Q and every boundary value $\omega \in \partial(V \setminus S)$,

$$\sum_{\eta \in \partial(V \setminus R)} c_\eta \Phi_{R,S}(\rho^{-1}(\eta), \sigma_Q, \tau^{-1}(\omega)) = 0.$$

For each outer boundary value ω and each boundary value β of Q , define

$$f_\omega(\beta) = \sum_{\eta \in \partial(V \setminus R)} c_\eta \mathbf{1}_{\{\exists q: \partial_{V \setminus R} q = \eta, \partial_{V \setminus S} q = \omega, \partial_Q q = \beta\}}.$$

Let Γ be the product of bond dimensions on the crossing edges separated when the Q -block is read first. Unfolding $\Phi_{R,S}$ as a sum over the blocked weights of Q gives, for every σ_Q ,

$$\begin{aligned} (T_Q f_\omega)(\sigma_Q) &= \sum_{\beta \in \partial Q} f_\omega(\beta) W_Q(\beta, \sigma_Q) \\ &= \sum_{\beta \in \partial Q} \left(\sum_{\eta \in \partial(V \setminus R)} c_\eta \mathbf{1}_{\{\exists q: \partial_{V \setminus R} q = \eta, \partial_{V \setminus S} q = \omega, \partial_Q q = \beta\}} \right) W_Q(\beta, \sigma_Q) \\ &= \Gamma \sum_{\eta \in \partial(V \setminus R)} c_\eta \Phi_{R,S}(\rho^{-1}(\eta), \sigma_Q, \tau^{-1}(\omega)) = 0. \end{aligned}$$

Hence $T_Q(f_\omega) = 0$. Injectivity of the blocked tensor of Q gives $T_Q(f_\omega) = 0 \implies f_\omega = 0$. Now fix $\mu \in \partial R$. Positivity of bond dimensions gives a global virtual configuration q whose residual boundary on $V \setminus R$ is $\rho(\mu)$. Set $\omega = \partial_{V \setminus S} q$ and $\beta = \partial_Q q$. Evaluating $f_\omega = 0$ at β , the residual-boundary uniqueness identity leaves only the term $\eta = \rho(\mu)$:

$$0 = f_\omega(\beta) = c_{\rho(\mu)} \cdot 1 = C_{\mu, \sigma_R}.$$

Since σ_R and μ were arbitrary, $C = 0$. □

Theorem 24.7.2.17 (Cancellation of a shared injective corner). [Lean](#) [✓ Stmt](#) Let $R \subseteq S$, and suppose that all bond dimensions are positive and that $S \setminus R$ is blocked-tensor injective. If two inserts on R have the same corner extension to S , $\text{Ext}_{R \subseteq S}(C_1) = \text{Ext}_{R \subseteq S}(C_2)$, then $C_1 = C_2$.

Proof. [✓ Proof](#) By linearity, $\text{Ext}_{R \subseteq S}(C_1 - C_2) = 0$. Theorem 24.7.2.16 applied to the added block $S \setminus R$ gives $C_1 - C_2 = 0$, hence $C_1 = C_2$. $\square$

Theorem 24.7.2.18 (Staircase-pair cancellation on the torus). [Lean](#) [✓ Stmt](#) *Let the torus have width W and height H , with $1 < W$ and $1 < H$. Let S be the staircase end pair around a horizontal edge, with seam dimensions L and K , and let Q be the completed corner rectangle. Assume the size bounds $2L \leq W$ and $2K \leq H$, that Q is blocked-tensor injective, and that all bond dimensions are positive. If two inserts X and Y on S have equal extensions to $S \cup Q$, $\text{Ext}_{S \subseteq S \cup Q}(X) = \text{Ext}_{S \subseteq S \cup Q}(Y)$, then $X = Y$. Thus the injective completed corner can be cancelled from an open-boundary equality, without assuming injectivity of the torus complement of S .*

Proof. [✓ Proof](#) The size bounds imply that the staircase end pair is disjoint from the completed corner, and therefore $(S \cup Q) \setminus S = Q$. The claimed equality is exactly Theorem 24.7.2.17 applied to the inclusion $S \subseteq S \cup Q$. $\square$

24.7.3 Single-bond peeling and the window witness

Definition 24.7.3.1 (Reference edge of the horizontal staircase). [Lean](#) [✓ Stmt](#) *In the staircase coordinates $s = (a, b)$, the reference edge is the horizontal edge joining $(a + L - 1, b + K - 1)$ to $(a + L, b + K - 1)$. It is the edge between the left and right end windows.*

Theorem 24.7.3.2 (The reference edge is the unique crossing of the two end windows). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Assume $1 < W$, $1 < H$, $0 < L$, $0 < K$, $1 \leq a$, $a + 2L \leq W$, and $b + 2K - 1 \leq H$. Then an edge crosses from the left end window to the right end window if and only if it is the reference edge e :*

$$f \in \partial W_{\text{left}} \cap \partial W_{\text{right}} \iff f = e.$$

Thus e is a boundary edge of each end window, but both endpoints of e lie in the end pair S , so $e \notin \partial S$.

Proof. [✓ Proof](#) The endpoint calculation identifies e with the unique horizontal edge joining the two displayed rectangles. Since one endpoint lies in W_{left} and the other lies in W_{right} , the same edge is internal to their union S . $\square$

Lemma 24.7.3.3 (Boundary edges of the end pair come from the end windows). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Under the bounds $1 < W$, $1 < H$, $0 < L$, $0 < K$, $1 \leq a$, $a + 2L \leq W$, and $b + 2K - 1 \leq H$, a boundary edge of the end pair is a boundary edge of one of the two end windows, and it is not the reference edge. Conversely, any boundary edge of one end window other than e is a boundary edge of S .*

Proof. [✓ Proof](#) A boundary edge of S has one endpoint in S ; that endpoint lies in one of the two end windows. The other endpoint is outside S , hence outside that window. Conversely, a boundary edge of one window which is not e cannot enter the other window, because e is the unique crossing edge. $\square$

Theorem 24.7.3.4 (Boundary of the staircase end pair). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Under the bounds $1 < W$, $1 < H$, $0 < L$, $0 < K$, $1 \leq a$, $a + 2L \leq W$, and $b + 2K - 1 \leq H$, an edge is a boundary edge of the staircase end pair S if and only if it is a boundary edge of one of the two end windows and is not the reference edge:*

$$f \in \partial S \iff (f \in \partial W_{\text{left}} \text{ or } f \in \partial W_{\text{right}}) \text{ and } f \neq e.$$

Moreover each edge in ∂S lies on exactly one of the two window boundaries.

Proof. [✓ Proof](#) Lemma 24.7.3.3 gives both implications. If an edge lay on both end-window boundaries, the unique-crossing theorem would identify it with e , which is not a boundary edge of S . $\square$

Definition 24.7.3.5 (Boundary configurations of the staircase end windows). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the same geometric bounds, write $\partial_{\text{left}} S$ and $\partial_{\text{right}} S$ for the boundary edges of S which lie on the corresponding end-window boundary. Then the boundary-configuration spaces have canonical factorizations

$$\begin{aligned}\text{Conf}(\partial W_{\text{left}}) &\cong \text{Conf}(e) \times \text{Conf}(\partial_{\text{left}} S), \\ \text{Conf}(\partial W_{\text{right}}) &\cong \text{Conf}(e) \times \text{Conf}(\partial_{\text{right}} S), \\ \text{Conf}(\partial S) &\cong \text{Conf}(\partial_{\text{left}} S) \times \text{Conf}(\partial_{\text{right}} S).\end{aligned}$$

These equivalences preserve every underlying lattice-edge label. Thus the reference edge is the only contracted virtual index shared by the two highlighted end windows; every other virtual index remains on the corresponding open boundary of S . The constructions split each end-window configuration at the reference edge and reindex the remaining dependent products along the corresponding boundary-edge identifications.

Lemma 24.7.3.6 (Complement split for an end window). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If $2L \leq W$, the complement of either end window splits as the opposite end window and the complement of the end pair:

$$\begin{aligned}V \setminus W_{\text{left}} &= W_{\text{right}} \cup (V \setminus S), \\ V \setminus W_{\text{right}} &= W_{\text{left}} \cup (V \setminus S).\end{aligned}$$

Proof. [✓ Proof](#) Since the two end windows are disjoint and $S = W_{\text{left}} \cup W_{\text{right}}$, removing one window from S leaves the other. Splitting $V \setminus W$ through the inclusion $W \subseteq S$ gives the displayed decompositions. $\square$

Theorem 24.7.3.7 (The reference edge bounds the two end windows). [Lean](#) [Lean](#) [✓ Stmt](#) Let the torus have width W and height H , with $1 < H$. Let e be the horizontal reference edge of a staircase window family with dimensions L, K and lower-left coordinate (a, b) . Suppose $0 < L$, $0 < K$, $1 \leq a$, $a + 2L \leq W$, and $b + 2K - 1 \leq H$. Then e is a boundary edge of the first staircase window W_0 and of the last staircase window W_{L+K-1} :

$$\begin{aligned}e &\in \partial W_0, \\ e &\in \partial W_{L+K-1}.\end{aligned}$$

Proof. [✓ Proof](#) The first staircase window is the right end window, and the last staircase window is the left end window:

$$\begin{aligned}W_0 &= W_{\text{right}}, \\ W_{L+K-1} &= W_{\text{left}}.\end{aligned}$$

The reference edge is a boundary edge of these two end windows by their defining geometry. $\square$

Theorem 24.7.3.8 (Open-boundary equality on the staircase end pair). [Lean](#) [✓ Stmt](#) Let $S = W_0 \cup W_{L+K-1}$ be the staircase end pair. Suppose that the one-orientation $L \times K$ window injectivity hypotheses hold for a tensor B , unions of injective regions are injective, all bond dimensions of

B are positive, and $2 \leq L$, $2 \leq K$, $2L + 1 \leq W$, and $2K + 1 \leq H$. If C_j are inserts on the staircase windows whose deformed states all equal one common state, then the two end-window inserts have equal corner extensions to S :

$$\text{Ext}_{W_0 \subseteq S}(C_0) = \text{Ext}_{W_{L+K-1} \subseteq S}(C_{L+K-1}).$$

Proof. ✓ Proof The consecutive-window equalities are composed across the staircase patch. Both sides are then extended through the completed corner, and transitivity of corner extension rewrites the two extension chains. The completed corner is an injective $L \times K$ window, so Theorem 24.7.2.18 cancels it and leaves the equality on the end pair S . $\square$

Theorem 24.7.3.9 (Lemma 5 comparison of the two end blocks). Lean Lean Lean ✓ Stmt Let K be the basis of the virtual space on the bond between two injective blocks, and let $\{A_{1,\mu}\}_{\mu \in K}$ and $\{A_{2,\mu}\}_{\mu \in K}$ be linearly independent families. Suppose that the two end physical operations O_1 and O_3 produce families C_1 and C_2 satisfying the open contraction identity

$$\sum_{\mu \in K} C_{1,\mu} \otimes A_{2,\mu} = \sum_{\nu \in K} A_{1,\nu} \otimes C_{2,\nu}. \quad (24.15)$$

Then there is a unique matrix $X \in M_{|K|}(\mathbb{C})$ such that

$$C_{1,\nu} = \sum_{\mu \in K} A_{1,\mu} X_{\mu,\nu}, \quad (24.16)$$

$$C_{2,\mu} = \sum_{\nu \in K} X_{\mu,\nu} A_{2,\nu}. \quad (24.17)$$

Thus the two end operations are obtained by inserting the same virtual operation X on their common bond. No injectivity is required of the modified families C_1 and C_2 . This is the “compare the two ends” step in Lemma 5 of [186].

Proof. ✓ Proof Apply a dual functional to the linearly independent family $\{A_{2,\mu}\}$ in (24.15) to express $C_{1,\nu}$ as a linear combination of the $A_{1,\mu}$ with a coefficient matrix X , yielding (24.16). Substitution into (24.15) and linear independence of $\{A_{1,\mu}\}$ yields (24.17) with the same matrix. Linear independence of $\{A_{1,\mu}\}$ also makes X unique. $\square$

Definition 24.7.3.10 (End operations realizing a virtual matrix). Lean Lean ✓ Stmt Let $\{A_{1,\mu}\}_{\mu \in K}$ and $\{A_{3,\mu}\}_{\mu \in K}$ be two end families. The left physical operation O_1 realizes a virtual matrix X when

$$O_1(A_{1,\nu}) = \sum_{\mu \in K} X_{\mu\nu} A_{1,\mu}.$$

The right physical operation O_3 , read in the transposed orientation $O_3^\top$ of Lemma 5, realizes the same matrix when

$$O_3(A_{3,\mu}) = \sum_{\nu \in K} X_{\mu\nu} A_{3,\nu}.$$

This convention records the orientation of the displayed tensor relation; it does not choose a transpose on the abstract space of physical operations.

Theorem 24.7.3.11 (Multiplication of the end operations in Lemma 5). Lean Lean Lean Lean ✓ Stmt Suppose that the two genuine end families are linearly independent

and that the open contraction identity for O_1 and O_3 holds. Then there is a unique matrix X simultaneously realized by O_1 and by $O_3^\top$.

If O_1 and P_1 realize X and Y , respectively, then $O_1 \circ P_1$ realizes XY . If $O_3^\top$ and $P_3^\top$ realize X and Y , then their product realizes XY ; in terms of the underlying right operations this product is $P_3 \circ O_3$. Uniqueness identifies the virtual matrix of either composite with XY . This is precisely the composition step in the final sentence of Lemma 5 of [186]; it does not by itself identify operations belonging to two different PEPS tensors.

Proof. ✓ Proof The two-end comparison gives existence and uniqueness of X . Applying the first operation to the expansion furnished by the second and interchanging the two finite sums gives matrix multiplication. On the right, transposition reverses the order of the underlying physical operations. □

Theorem 24.7.3.12 (Normalized coupling of the highlighted end windows). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Write the boundary coordinates of the two end windows and their union as

$$\mu_L = (k, \eta_L), \quad \mu_R = (k, \eta_R), \quad \nu = (\eta'_L, \eta'_R),$$

where k is the coordinate on the reference edge. In the extension of a left-window insert to the end pair, the normalized coupling to the right window vanishes unless $\eta_L = \eta'_L$. On that fibre it is

$$\Lambda_L^{-1} A_R(k, \eta'_R),$$

where A_R is the genuine blocked tensor of the right window and Λ_L is the product of the exterior interior-bond factor and the bond dimensions on the external legs of the left window. The analogous statement holds after interchanging left and right. If the tensor is translation invariant and every bond dimension is positive, then $\Lambda_L = \Lambda_R$.

Proof. ✓ Proof The crossing edges are precisely the boundary edges of the host window other than the reference edge. Compatible host and end-pair boundary labels therefore determine one boundary label on the opposite window; the blue coupling reduces to its genuine blocked tensor. Incompatible external labels cannot arise from a global virtual configuration and give zero. Finally, the two end windows are translates. Translation invariance equates their full boundary products, and cancellation of the common positive reference-edge factor equates the two normalizations. □

Theorem 24.7.3.13 (The end-pair extensions contract over one bond). Lean Lean Lean ✓ Stmt Let C_L and C_R be arbitrary inserts on the two end windows. In the boundary coordinates above, their extensions to the end pair are

$$\begin{aligned} \text{Ext}(C_L) &= \Lambda_L^{-1} \sum_k C_L(k, \eta_L) A_R(k, \eta_R), \\ \text{Ext}(C_R) &= \Lambda_R^{-1} \sum_k A_L(k, \eta_L) C_R(k, \eta_R). \end{aligned}$$

Hence, for a translation-invariant tensor with positive bond dimensions, equality of the two extensions implies the single-bond identity

$$\sum_k C_L(k, \eta_L) A_R(k, \eta_R) = \sum_k A_L(k, \eta_L) C_R(k, \eta_R).$$

Proof. [✓ Proof](#) Reindex each extension sum by the reference-edge coordinate and the external boundary data. The normalized-coupling theorem kills every incompatible fibre and identifies the unique surviving fibre with the genuine opposite-window tensor. The common nonzero normalization then cancels from the equality of extensions. $\square$

Theorem 24.7.3.14 (Lemma 5 operation on the staircase reference edge). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the tensor is translation invariant, all bond dimensions are positive, and the two genuine highlighted end-window blocked tensors are injective. If arbitrary inserts C_L and C_R on those windows have equal extensions to the open end pair, then there is a unique matrix X on the reference edge such that

$$\begin{aligned} C_L(k, \eta_L) &= \sum_j A_L(j, \eta_L) X_{jk}, \\ C_R(k, \eta_R) &= \sum_j X_{kj} A_R(j, \eta_R). \end{aligned}$$

No injectivity is assumed for either modified insert, a single vertex, the end pair, or its complement. In particular, when the two inserts are obtained by applying the physical end operations O_1 and O_3 , these relations say that O_1 and $O_3^\top$ realize the same unique matrix X .

Proof. [✓ Proof](#) The preceding theorem turns the open end-pair equality into the exact one-bond contraction identity. Injectivity of each genuine end window implies linear independence after keeping its external boundary and physical coordinates together. The two-end comparison theorem then gives the displayed relations and uniqueness of X . $\square$

Theorem 24.7.3.15 (Single-bond peeling of the staircase end equality). [Lean](#) [✓ Stmt](#) Let the torus have width W and height H . Suppose that the one-orientation $L \times K$ window injectivity hypotheses hold for a tensor B , unions of injective regions are injective, and all bond dimensions of B are positive. Assume

$$\begin{aligned} 2 \leq L, \quad 2 \leq K, \quad 1 \leq a, \quad a + 2L \leq W, \quad b + 2K - 1 \leq H, \\ 2L + 1 \leq W, \quad 2K + 1 \leq H. \end{aligned}$$

Let C_j be inserts on the staircase windows W_j whose deformed states all equal one common state. Suppose that $C_0 = I_{B, W_0, e, M}$ and $C_{L+K-1} = I_{B, W_{L+K-1}, e, M'}$. Then for every global physical configuration σ ,

$$C_B(W_0, e, M; \sigma|_{W_0}, \sigma|_{V \setminus W_0}) = C_B(W_{L+K-1}, e, M'; \sigma|_{W_{L+K-1}}, \sigma|_{V \setminus W_{L+K-1}}).$$

Proof. [✓ Proof](#) Theorem 24.4.10 rewrites the two coefficients as the deformed states on the staircase end pair of the two corner-extended inserts:

$$\begin{aligned} C_B(W_0, e, M; \sigma|_{W_0}, \sigma|_{V \setminus W_0}) &= \Psi_{B, S, \text{Ext}_{W_0 \subseteq S}(C_0)}(\sigma), \\ C_B(W_{L+K-1}, e, M'; \sigma|_{W_{L+K-1}}, \sigma|_{V \setminus W_{L+K-1}}) &= \Psi_{B, S, \text{Ext}_{W_{L+K-1} \subseteq S}(C_{L+K-1})}(\sigma). \end{aligned}$$

The open-boundary equality on the staircase end pair gives $\text{Ext}_{W_0 \subseteq S}(C_0) = \text{Ext}_{W_{L+K-1} \subseteq S}(C_{L+K-1})$, and hence the two coefficients are equal. $\square$

Window-independence of the bond insertion.

Lemma 24.7.3.16 (Restriction and combination of a physical configuration). [Lean](#) [✓ Stmt](#) Let ω be a physical configuration on the full vertex set. If ω_R and $\omega_{V \setminus R}$ are its restrictions to R and to the complement, then $\text{Assemble}_R(\omega_R, \omega_{V \setminus R}) = \omega$.

Proof. [✓ Proof](#) At a vertex in R the combined configuration reads ω_R , and at a vertex outside R it reads $\omega_{V \setminus R}$. These restrictions are both inherited from ω , so the two configurations agree at every vertex. $\square$

Theorem 24.7.3.17 (Bond-inserted state as an edge insertion). [Lean](#) [✓ Stmt](#) Let R be a region, let f be a boundary edge of R , let $M \in M_{D_B(f)}(\mathbb{C})$, and let ω be a physical configuration on the full vertex set. Then the assembled deformed state of the bond-inserted region insert is the non-boundary bond product times the corresponding edge-inserted coefficient:

$$\Psi_{B,R,I_{B,R},f,M}^{\text{as}}(\omega) = p_B(R) E_B(f, \omega, \mathcal{O}_{R,f}(M)).$$

Here $p_B(R) = \prod_{g \notin \partial R} D_B(g)$, and $\mathcal{O}_{R,f}$ is the boundary-edge orientation.

Proof. [✓ Proof](#) Theorem 24.4.8 rewrites the assembled deformed state as $C_B(R, f, M; \omega_R, \omega_{V \setminus R})$. Theorem 24.4.9 gives

$$C_B(R, f, M; \omega_R, \omega_{V \setminus R}) = p_B(R) E_B(f, \text{Assemble}_R(\omega_R, \omega_{V \setminus R}), \mathcal{O}_{R,f}(M)).$$

Lemma 24.7.3.16 identifies the combined configuration with ω . $\square$

Theorem 24.7.3.18 (Window-independence of a bond insertion). [Lean](#) [✓ Stmt](#) Let R_1 and R_2 be two regions for which the same edge e is a boundary edge. Let $M_1, M_2 \in M_{D_B(e)}(\mathbb{C})$, and suppose that the two oriented insertions agree, $\mathcal{O}_{R_1,e}(M_1) = \mathcal{O}_{R_2,e}(M_2)$. Then for every physical configuration ω on the full vertex set,

$$p_B(R_2) \Psi_{B,R_1,I_{B,R_1},e,M_1}^{\text{as}}(\omega) = p_B(R_1) \Psi_{B,R_2,I_{B,R_2},e,M_2}^{\text{as}}(\omega).$$

Proof. [✓ Proof](#) Applying Theorem 24.7.3.17 to each of the two regions gives, for $i = 1, 2$,

$$\Psi_{B,R_i,I_{B,R_i},e,M_i}^{\text{as}}(\omega) = p_B(R_i) E_B(e, \omega, \mathcal{O}_{R_i,e}(M_i)).$$

The oriented insertions are equal by hypothesis, so both sides of the claimed identity are $p_B(R_1) p_B(R_2) E_B(e, \omega, \mathcal{O}_{R_1,e}(M_1))$. $\square$

One virtual operation on every staircase window. The construction below gives the forward virtual/physical-operation correspondence in the proof sketch of the two-dimensional corollary [186, corollary after Lemma 5, lines 2297–2444].

Definition 24.7.3.19 (Partial state across a region cut). [Lean](#) [✓ Stmt](#) For a region R and a physical configuration τ on its complement, define the partial state $\psi_{A,R}^\tau$ on the physical space of R by

$$\psi_{A,R}^\tau(\sigma) = c_A(\omega_{R,\sigma,\tau}).$$

Theorem 24.7.3.20 (Partial states of equal closed states). [Lean](#) [✓ Stmt](#) If A and B generate the same closed state, then, for every region R and every complement configuration τ , $\psi_{A,R}^\tau = \psi_{B,R}^\tau$.

Proof. ✓ **Proof** Evaluate both functions at a physical configuration σ on R and apply the equality of the two closed-state coefficients at $\omega_{R,\sigma,\tau}$. $\square$

Theorem 24.7.3.21 (Blocked-tensor expansion of a partial state). Lean ✓ **Stmt** Let $P_A(R)$ be the product of the dimensions of the bonds internal to R . For every complement configuration τ ,

$$P_A(R)\psi_{A,R}^\tau = \sum_{\mu \in \partial R} T_{A,V \setminus R}(\bar{\mu})(\tau) T_{A,R}(\mu). \quad (24.18)$$

Proof. ✓ **Proof** Split the closed contraction across R . If μ and ν are boundary labels on R , the complement-boundary identification satisfies

$$\bar{\mu} = \bar{\nu} \iff \mu = \nu.$$

Hence the double boundary sum collapses to its diagonal, which gives the sum in (24.18); each bond internal to R contributes its dimension to the factor $P_A(R)$. $\square$

Definition 24.7.3.22 (Physical operator of a region insert). Lean ✓ **Stmt** Let R be an injective region and let $C = \{C(\mu)\}_{\mu \in \partial R}$ be any family in its physical space, indexed by the virtual boundary configurations. If $L_{A,R}$ is a left inverse of the blocked tensor map, define

$$O_C = \left(c \mapsto \sum_{\mu} c_{\mu} C(\mu) \right) L_{A,R}.$$

Theorem 24.7.3.23 (Open-boundary realization of a region insert). Lean ✓ **Stmt** For every virtual boundary configuration μ ,

$$O_C(T_{A,R}(\mu)) = C(\mu).$$

Thus the insert is obtained from the genuine region block by one physical operator before the region is contracted with its complement.

Proof. ✓ **Proof** The left inverse sends $T_{A,R}(\mu)$ to the standard basis vector indexed by μ , and the displayed linear combination sends that basis vector to $C(\mu)$. $\square$

Definition 24.7.3.24 (Region insert of a physical operator). Lean ✓ **Stmt** Let O be a physical operator on the physical space of a region R . Its action on the genuine open-boundary blocks defines the insert

$$C_{A,R}^O(\mu) = O(T_{A,R}(\mu)).$$

Theorem 24.7.3.25 (Closed state of a physical operator). Lean ✓ **Stmt** For a complement physical configuration τ , let $\psi_{A,R}^\tau$ be the partial state across the cut. Then

$$\Psi_{A,R,C_{A,R}^O}(\sigma, \tau) = [O(P_A(R)\psi_{A,R}^\tau)](\sigma). \quad (24.19)$$

Proof. ✓ **Proof** Apply O to (24.18) and evaluate at σ . This proves (24.19). $\square$

Theorem 24.7.3.26 (Physical operator transferred between equal closed states). Lean ✓ **Stmt** If A and B generate the same closed state, then

$$P_B(R)\Psi_{A,R,C_{A,R}^O}(\sigma, \tau) = P_A(R)\Psi_{B,R,C_{B,R}^O}(\sigma, \tau). \quad (24.20)$$

The two virtual boundary spaces need not be identified.

Proof. ✓ Proof Equality of the closed states gives $\psi_{A,R}^\tau = \psi_{B,R}^\tau$. Apply (24.19) to both tensors and cross-multiply the two interior-bond products. $\square$

Definition 24.7.3.27 (Interior-bond region insert). Lean ✓ Stmt Let $e = \{u, w\}$ be an ordered edge whose two endpoints lie in a region R , with u its ordered left endpoint, and let $X \in M_{D_A(e)}(\mathbb{C})$. Write $U = \{u\}$ and let $I_{A,U,e,X}$ be the boundary-bond insert on U . The interior-bond insert on R is

$$I_{A,R,e,X}^{\text{int}} = \frac{P_A(R)}{P_A(U)} \text{Ext}_{U \subseteq R}(I_{A,U,e,X}).$$

Thus the genuine tensors at the vertices of $R \setminus U$ are adjoined to a boundary insertion of X on e , with the normalization changed from the non-boundary bond product of U to that of R .

Theorem 24.7.3.28 (Closed state of an interior-bond insert). Lean ✓ Stmt Suppose all bond dimensions are positive. For every global physical configuration ω ,

$$\Psi_{A,R,I_{A,R,e,X}^{\text{int}}}^{\text{as}}(\omega) = P_A(R)E_A(e, \omega, X). \quad (24.21)$$

Proof. ✓ Proof Corner extension preserves the closed state. On the singleton U , the boundary orientation is the identity because U contains the ordered left endpoint of e . Hence its boundary insert has assembled state $P_A(U)E_A(e, \omega, X)$. The factor $P_A(R)/P_A(U)$ cancels $P_A(U)$ and gives (24.21). $\square$

Definition 24.7.3.29 (Staircase family for one virtual operation). Lean ✓ Stmt Let e be the horizontal reference edge and let $X \in M_{D_B(e)}(\mathbb{C})$. Define an insert $C_j(X)$ on each staircase window W_j , for $0 \leq j < L + K$, by

$$C_j(X) = \begin{cases} I_{B,W_0,e,X^\tau}, & j = 0, \\ I_{B,W_j,e,X}^{\text{int}}, & 1 \leq j < L, \\ I_{B,W_j,e,X}, & L \leq j < L + K. \end{cases}$$

The first window contains only the ordered right endpoint of e , the middle windows contain both endpoints, and the remaining windows contain only the ordered left endpoint. The transpose in the first case makes all three boundary orientations represent the same matrix X .

Theorem 24.7.3.30 (Physical operators on the staircase windows). Lean Lean ✓ Stmt Suppose every $L \times K$ window is injective. For every $0 \leq j < L + K$, there is a physical operator $O_j(X)$ on the physical space of W_j such that

$$O_j(X)(T_{B,W_j}(\mu)) = C_j(X)(\mu) \quad (\mu \in \partial W_j).$$

Hence the virtual operation is realized locally on each window, before any contraction with the complementary tensor network.

Proof. ✓ Proof Each staircase window is an $L \times K$ translate and is therefore injective. Apply the open-boundary realization theorem to $C_j(X)$. $\square$

Lemma 24.7.3.31 (The two end inserts of the staircase family). Lean Lean ✓ Stmt The end members of the family are

$$C_0(X) = I_{B,W_0,e,X^\tau}, \quad (24.22)$$

$$C_{L+K-1}(X) = I_{B,W_{L+K-1},e,X}. \quad (24.23)$$

Proof. [✓ Proof](#) Identity (24.22) is the $j = 0$ case of Definition 24.7.3.29. Since $K > 0$, the last index satisfies $L \leq L + K - 1$, which yields (24.23). $\square$

Theorem 24.7.3.32 (Common state of the staircase virtual-operation family). [Lean](#) [✓ Stmt](#) Let B be translation invariant and suppose all its bond dimensions are positive. Then for $0 \leq j < L + K$ and every global physical configuration ω ,

$$\Psi_{B, W_j, C_j(X)}^{\text{as}}(\omega) = P_B(W_0)E_B(e, \omega, X).$$

In particular, all $L + K$ inserts have one common deformed state.

Proof. [✓ Proof](#) On W_0 , the boundary orientation sends $X^\top$ to X . On $W_1, \dots, W_{L-1}$, apply the interior-bond identity. On the remaining windows the ordered left endpoint lies in the window, so the boundary orientation fixes X . Each case gives $P_B(W_j)E_B(e, \omega, X)$. Translation invariance identifies $P_B(W_j)$ with $P_B(W_0)$. $\square$

Theorem 24.7.3.33 (Common staircase state transferred between tensors). [Lean](#) [✓ Stmt](#) Let $L, K > 0$, and choose staircase coordinates (a, b) with $a + 2L \leq \text{width}$ and $2K \leq \text{height}$. Let A and B be translation invariant tensors with the same physical dimension and the same closed state. Suppose every cyclic $L \times K$ window of A is injective and $D_A(g) > 0$ for every edge g . For $X \in M_{D_A(e)}(\mathbb{C})$, let $O_j^A(X)$ be the physical operators obtained from the staircase insert family of A , and define inserts on the windows of B by

$$C_j^{A \rightarrow B}(X)(\mu) = O_j^A(X)(T_{B, W_j}(\mu)).$$

Then, for all $j, j' \in \mathbb{N}$,

$$\Psi_{B, W_j, C_j^{A \rightarrow B}(X)}^{\text{as}} = \Psi_{B, W_{j'}, C_{j'}^{A \rightarrow B}(X)}^{\text{as}}.$$

The source family has $0 \leq j < L + K$. In the definition used here the truncated offsets stabilize at W_{L+K-1} for $j \geq L + K - 1$; the indices are not read modulo $L + K$. No equality between the edge-dependent bond dimensions of A and B , and no positivity assumption on those of B , is required.

Proof. [✓ Proof](#) For each j , (24.20) and the open-boundary realization of $O_j^A(X)$ give

$$P_B(W_j)\Psi_{A, W_j, C_j^A(X)}^{\text{as}} = P_A(W_j)\Psi_{B, W_j, C_j^{A \rightarrow B}(X)}^{\text{as}}.$$

The staircase family of A has one common deformed state. Translation invariance gives $P_A(W_j) = P_A(W_0)$ and $P_B(W_j) = P_B(W_0)$. Hence the right-hand side is independent of j . Positivity of the bond dimensions makes $P_A(W_0)$ nonzero, so it may be cancelled. $\square$

Definition 24.7.3.34 (The left staircase end operation $O_1^A(X)$). [Lean](#) [✓ Stmt](#) For $X \in M_{D_A(e)}(\mathbb{C})$, define $O_1^A(X) = O_{L+K-1}^A(X)$ after identifying the last staircase window with the left end window.

Definition 24.7.3.35 (The underlying right staircase end operation $O_3^A(X)$). [Lean](#) [✓ Stmt](#) For $X \in M_{D_A(e)}(\mathbb{C})$, define $O_3^A(X) = O_0^A(X)$ after identifying the first staircase window with the right end window. The relation (24.25) has the paper's transposed $O_3^\top$ orientation; no transpose is chosen on the abstract physical endomorphism space.

Theorem 24.7.3.36 (Cross-tensor virtual operation on the staircase edge). [Lean](#) [Stmt](#) Let $L, K \geq 2$, and choose the displayed non-wrapping horizontal staircase coordinates (a, b) with

$$\begin{aligned} 1 &\leq a, & a + 2L &\leq \text{width}, & b + 2K - 1 &\leq \text{height}, \\ 2L + 1 &\leq \text{width}, & 2K + 1 &\leq \text{height}. \end{aligned}$$

Suppose that A and B are translation invariant, generate the same closed state, every cyclic $L \times K$ window of either tensor is injective, and every bond dimension of either tensor is positive. For $X \in M_{D_A(e)}(\mathbb{C})$, there is a unique $Y \in M_{D_B(e)}(\mathbb{C})$ such that, for every external boundary configuration η_L and η_R , and every reference-edge index k ,

$$O_1^A(X)(T_{B, W_L}(k, \eta_L)) = \sum_j Y_{jk} T_{B, W_L}(j, \eta_L), \quad (24.24)$$

$$O_3^A(X)(T_{B, W_R}(k, \eta_R)) = \sum_j Y_{kj} T_{B, W_R}(j, \eta_R). \quad (24.25)$$

Thus the relation (24.25) has the paper's $O_3^\top$ orientation. No equality between $D_A(e)$ and $D_B(e)$ is assumed or concluded, and this statement contains no multiplication or gauge assertion.

Proof. [Proof](#) The transferred physical inserts form one common B -state. The overlapping-union lemma supplies the union closure needed to compare consecutive staircase windows, and corner completion gives equality of the two end inserts with open boundary. At the two ends these inserts are precisely the actions of $O_3^A(X)$ and $O_1^A(X)$ on the genuine B -window tensors. Injectivity of those two windows and the two-end comparison give the displayed relations and the uniqueness of Y . $\square$

Theorem 24.7.3.37 (Composition of the canonical staircase end operations). [Lean](#) [Lean](#) [Stmt](#) Let $L, K > 0$, and choose staircase coordinates (a, b) with $a + 2L \leq \text{width}$ and $2K \leq \text{height}$. Suppose every cyclic $L \times K$ window of A is injective. For matrices $X, Y \in M_{D_A(e)}(\mathbb{C})$, the canonical end operations satisfy

$$\begin{aligned} O_1^A(XY) &= O_1^A(X)O_1^A(Y), \\ O_3^A(XY) &= O_3^A(Y)O_3^A(X). \end{aligned}$$

Thus the right relation preserves multiplication in ordinary order after passing to the orientation $(O_3^A)^\top$ used in Lemma 5.

Proof. [Proof](#) Split the virtual boundary of either end window into the reference-edge coordinate and the remaining external coordinates. Let E be this splitting map, and write C_M for insertion of M on the reference edge. On every fixed external fibre,

$$C_M(E^{-1}(j, \eta)) = \sum_i M_{ij} T_{A, W}(E^{-1}(i, \eta)).$$

The chosen left inverse of the injective blocked tensor is a left inverse on the whole boundary-coordinate space, so this identity gives $O_W(M)O_W(N) = O_W(MN)$. The first window carries $X^\top$; hence $(XY)^\top = Y^\top X^\top$ reverses the order of the underlying right operations. $\square$

Definition 24.7.3.38 (Canonical cross-tensor matrix assignment). [Lean](#) [Lean](#) [Stmt](#) Under the hypotheses of Theorem 24.7.3.36, in the paper's notation the input and uniquely recovered matrices are called X and Y . Writing $Y_{A \rightarrow B}(X)$ only to distinguish the two tensor families, define

$$Y_{A \rightarrow B}(X) \in M_{D_B(e)}(\mathbb{C})$$

to be the unique matrix satisfying (24.24) and (24.25) for every left and right external boundary configuration. This definition makes no choice of an external boundary configuration.

Theorem 24.7.3.39 (Multiplicativity of the cross-tensor assignment). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 24.7.3.36, for all $X_1, X_2 \in M_{D_A(e)}(\mathbb{C})$,*

$$Y_{A \rightarrow B}(X_1 X_2) = Y_{A \rightarrow B}(X_1) Y_{A \rightarrow B}(X_2).$$

Proof. [✓ Proof](#) Set $Y_r = Y_{A \rightarrow B}(X_r)$ for $r = 1, 2$. For every left external boundary configuration, the two O_1 relations compose in ordinary order:

$$O_1^A(X_1) \left(O_1^A(X_2) (T_{B, W_L}(k, \eta_L)) \right) = \sum_i (Y_1 Y_2)_{ik} T_{B, W_L}(i, \eta_L).$$

For every right external boundary configuration, the underlying O_3 operations compose in reverse order:

$$O_3^A(X_2) \left(O_3^A(X_1) (T_{B, W_R}(k, \eta_R)) \right) = \sum_j (Y_1 Y_2)_{kj} T_{B, W_R}(j, \eta_R).$$

This is ordinary multiplication in the $O_3^\top$ orientation. The preceding theorem identifies the left sides with the canonical operations for $X_1 X_2$. Both end relations therefore realize $Y_1 Y_2$, and global uniqueness identifies this product with $Y_{A \rightarrow B}(X_1 X_2)$. $\square$

Theorem 24.7.3.40 (End-window identity for one virtual operation). [Lean](#) [✓ Stmt](#) *Suppose the tensor is translation invariant and the staircase satisfies the stated size bounds. Then for every $X \in M_{D_B(e)}(\mathbb{C})$ and every global physical configuration ω ,*

$$C_B(W_0, e, X^\top; \omega|_{W_0}, \omega|_{V \setminus W_0}) = C_B(W_{L+K-1}, e, X; \omega|_{W_{L+K-1}}, \omega|_{V \setminus W_{L+K-1}}).$$

Proof. [✓ Proof](#) The region-to-edge identity sends both sides to the edge coefficient with X on the same reference edge: the first boundary orientation transposes $X^\top$, while the last fixes X . Translation invariance identifies the two non-boundary bond products. $\square$

Definition 24.7.3.41 (Window-region coefficient-identity witness). [Lean](#) [✓ Stmt](#) Let W be the left end window of a horizontal staircase with parameters (L, K) and lower-left coordinate (a, b) , and let e be its reference edge. Suppose that

$$0 < L, \quad 0 < K, \quad 1 \leq a, \quad a + 2L \leq W_{\text{torus}}, \quad b + 2K - 1 \leq H_{\text{torus}}.$$

Then $e \in \partial W$. Suppose also that the blocked tensors of B on W and $V \setminus W$ are injective, that all bond dimensions of B are positive, and that the bond dimensions of A and B at e are identified by $\iota_e : M_{D_A(e)}(\mathbb{C}) \simeq M_{D_B(e)}(\mathbb{C})$. If $Z \in \text{GL}(D_B(e))$ satisfies, for every matrix M and physical configurations σ, τ ,

$$C_A(W, e, M; \sigma, \tau) = C_B(W, e, Z \iota_e(M) Z^{-1}; \sigma, \tau),$$

then these objects form the coefficient-identity witness at e with witness region W and with both gauge entries equal to Z .

Definition 24.7.3.42 (Window-region witness from window injectivity). [Lean](#) [✓ Stmt](#) Let W be the left end window around the reference edge e , placed with lower-left coordinate (a, b) . Suppose

that every cyclic $L \times K$ window of B is blocked-tensor injective, unions of injective regions are injective, all bond dimensions of B are positive, and

$$\begin{aligned} 2 \leq L, \quad 2 \leq K, \quad 2L + 1 \leq W_{\text{torus}}, \quad 2K + 1 \leq H_{\text{torus}}, \\ 1 \leq a, \quad a + 2L \leq W_{\text{torus}}, \quad b + 2K - 1 \leq H_{\text{torus}}. \end{aligned}$$

Suppose also that the bond dimensions of A and B at e are identified by $\iota_e : M_{D_A(e)}(\mathbb{C}) \simeq M_{D_B(e)}(\mathbb{C})$, and let $Z \in \text{GL}(D_B(e))$. If the conjugation identity

$$C_A(W, e, M; \sigma, \tau) = C_B(W, e, Z \iota_e(M) Z^{-1}; \sigma, \tau)$$

holds for every M, σ, τ , then the coefficient-identity witness of Definition 24.7.3.41 is obtained for the region W . The hypotheses supply the injectivity of both W and its complement at the minimal torus size.

Window realization and witness read-off.

Lemma 24.7.3.43 (Window-level edge-insertion injectivity). Lean ✓ Stmt *Let B be a PEPS tensor on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. Assume the one-orientation $L \times K$ window-injectivity hypotheses for B , union closure for injective regions, positive bond dimensions, and*

$$2 \leq L, \quad 2 \leq K, \quad 1 \leq a, \quad a + 2L \leq W, \quad b + 2K - 1 \leq H. \quad (24.26)$$

Suppose also that

$$2L + 1 \leq W, \quad 2K + 1 \leq H. \quad (24.27)$$

Let W_L be the left end window and let e be the reference edge. If $M, M' \in M_{D_B(e)}(\mathbb{C})$ satisfy

$$C_B(W_L, e, M; \sigma, \tau) = C_B(W_L, e, M'; \sigma, \tau)$$

for every physical configuration σ on W_L and τ on $V \setminus W_L$, then $M = M'$.

Proof. ✓ Proof The window-injectivity hypotheses give injectivity of the blocked tensor map on W_L . The single-window complement is also injective under the size hypotheses (24.26) and (24.27). Since $e \in \partial W_L$, Theorem 24.4.6 applied to the region W_L gives $M = M'$. $\square$

Theorem 24.7.3.44 (Window witness from a region insertion transfer). Lean ✓ Stmt *Let A and B be PEPS tensors on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. Assume the one-orientation $L \times K$ window-injectivity hypotheses for both tensors, union closure for injective regions for both tensors, positive bond dimensions, and $2 \leq L, 2 \leq K, 1 \leq a, a + 2L \leq W$, and $b + 2K - 1 \leq H$, together with $2L + 1 \leq W$ and $2K + 1 \leq H$. Let W_L be the left end window and let e be the reference edge. If $T_e : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$ is a region insertion transfer on W_L at e , then there exist $Z, Z_{\text{ref}} \in \text{GL}(D_B(e))$ and a bond-dimension identification $\iota_e : M_{D_A(e)}(\mathbb{C}) \simeq M_{D_B(e)}(\mathbb{C})$ such that the coefficient-identity witness at e is nonempty. The coefficient identity has the form*

$$C_A(W_L, e, M; \sigma, \tau) = C_B(W_L, e, Z \iota_e(M) Z^{-1}; \sigma, \tau). \quad (24.28)$$

Proof. ✓ Proof The transfer determines a gauge Z by the region insertion algebra: $T_e(M) = Z \iota_e(M) Z^{-1}$. Its coefficient-matching property gives (24.28). The window and complement injectivity hypotheses are those required by Definition 24.7.3.42, which yields the witness with $Z_{\text{ref}} = Z$. $\square$

Theorem 24.7.3.45 (Window witness from single-vertex realization). [Lean](#) [✓ Stmt](#) *Let A and B be PEPS tensors on a torus whose periods W, H satisfy $1 < W$ and $1 < H$. Assume the one-orientation $L \times K$ window-injectivity hypotheses for both tensors, union closure for injective regions for both tensors, matched bond dimensions, equality of the PEPS state, positive bond dimensions, and $2 \leq L$, $2 \leq K$, $1 \leq a$, $a + 2L \leq W$, and $b + 2K - 1 \leq H$, together with $2L + 1 \leq W$ and $2K + 1 \leq H$. Let W_L be the left end window and let e be the reference edge. Suppose the endpoint tensor of each family at the in-region endpoint of e has linearly independent columns, and suppose the boundary matrix transfer is realized on W_L in both directions. Then there exist $Z, Z_{\text{ref}} \in \text{GL}(D_B(e))$ and a bond-dimension identification $\iota_e : M_{D_A(e)}(\mathbb{C}) \simeq M_{D_B(e)}(\mathbb{C})$ such that the coefficient-identity witness at e is nonempty.*

Proof. [✓ Proof](#) The two realized boundary transfers determine a region insertion transfer:

$$\begin{aligned} C_A(W_L, e, M; \sigma, \tau) &= C_B(W_L, e, T_e(M); \sigma, \tau), \\ T_e(MM') &= T_e(M)T_e(M'). \end{aligned}$$

Theorem 24.7.3.44 then gives the two gauges and the nonempty coefficient-identity witness. $\square$

Window witness from bond locality.

Theorem 24.7.3.46 (Forward window coefficient transfer from bond locality). [Lean](#) [✓ Stmt](#) *Let A and B be PEPS tensors on a torus whose periods $W_{\text{torus}}, H_{\text{torus}}$ satisfy $1 < W_{\text{torus}}$ and $1 < H_{\text{torus}}$. Assume the one-orientation $L \times K$ window-injectivity hypotheses and union closure for both tensors, equality of the PEPS state, matched bond dimensions, positive bond dimensions, and $2 \leq L$, $2 \leq K$, $1 \leq a$, $a + 2L \leq W_{\text{torus}}$, and $b + 2K - 1 \leq H_{\text{torus}}$, together with $2L + 1 \leq W_{\text{torus}}$ and $2K + 1 \leq H_{\text{torus}}$. Let W_L be the left end window and let e be the reference edge. Suppose the A -to- B transfer kernel on W_L at e is incident along the boundary leg e . Then for every $M \in M_{D_A(e)}(\mathbb{C})$ there is a matrix $N \in M_{D_B(e)}(\mathbb{C})$ such that $C_A(W_L, e, M; \sigma, \tau) = C_B(W_L, e, N; \sigma, \tau)$ for every physical configuration σ on W_L and τ on $V \setminus W_L$.*

Proof. [✓ Proof](#) The window hypotheses give injectivity of the blocked tensor maps on W_L for A and B , while union closure gives injectivity on the complements. For fixed M , let K_M be the transfer kernel read from the B -blocked left inverses. The double-factorization theorem gives

$$C_A(W_L, e, M; \sigma, \tau) = \sum_{\mu, \nu} K_M(\mu, \nu) T_{B, W_L}^{\mu}(\sigma) T_{B, V \setminus W_L}^{\nu}(\tau).$$

The bond-local incident form supplies a matrix N such that

$$K_M(\mu, \nu) = \mathbf{1}_{\mu|_{\partial W_L \setminus \{e\}} = \nu|_{\partial W_L \setminus \{e\}}} N_{\mu_e, \nu_e}.$$

Substituting this form and summing over the matching residual boundary labels gives $C_A(W_L, e, M; \sigma, \tau) = C_B(W_L, e, N; \sigma, \tau)$. $\square$

Theorem 24.7.3.47 (Backward window coefficient transfer from bond locality). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 24.7.3.46, suppose instead that the B -to- A transfer kernel on W_L at e is incident along the boundary leg e . Then for every $N \in M_{D_B(e)}(\mathbb{C})$ there is a matrix $M \in M_{D_A(e)}(\mathbb{C})$ such that $C_B(W_L, e, N; \sigma, \tau) = C_A(W_L, e, M; \sigma, \tau)$ for every physical configuration σ on W_L and τ on $V \setminus W_L$.*

Proof. ✓ Proof Apply the same coefficient-transfer lemma for bond-local kernels to the ordered pair (B, A) . Equivalently, the double-factorization theorem gives

$$C_B(W_L, e, N; \sigma, \tau) = \sum_{\mu, \nu} K_N(\mu, \nu) T_{A, W_L}^\mu(\sigma) T_{A, V \setminus W_L}^\nu(\tau),$$

while the B -to- A incident form gives a matrix M with

$$K_N(\mu, \nu) = \mathbf{1}_{\mu|_{\partial W_L \setminus \{e\}} = \nu|_{\partial W_L \setminus \{e\}}} M_{\mu_e, \nu_e}.$$

Substitution gives $C_B(W_L, e, N; \sigma, \tau) = C_A(W_L, e, M; \sigma, \tau)$. □

Theorem 24.7.3.48 (Window witness from bond locality and multiplicativity). Lean ✓ Stmt
Assume the hypotheses of Theorem 24.7.3.46 and its backward analog Theorem 24.7.3.47. Let $T_e : M_{D_A(e)}(\mathbb{C}) \rightarrow M_{D_B(e)}(\mathbb{C})$ be the coefficient transfer chosen from the forward theorem. If $T_e(MM') = T_e(M)T_e(M')$ for all $M, M' \in M_{D_A(e)}(\mathbb{C})$, then there exist $Z, Z_{\text{ref}} \in \text{GL}(D_B(e))$ and a bond-dimension identification $\iota_e : M_{D_A(e)}(\mathbb{C}) \simeq M_{D_B(e)}(\mathbb{C})$ such that a coefficient-identity witness at e is nonempty. Thus the witness contains a region R with $e \in \partial R$, and for every $M \in M_{D_A(e)}(\mathbb{C})$ and every physical configuration σ on R and τ on $V \setminus R$, the coefficient identity has the form

$$C_A(R, e, M; \sigma, \tau) = C_B(R, e, Z \iota_e(M) Z^{-1}; \sigma, \tau).$$

Proof. ✓ Proof The two bond-locality hypotheses give the forward and backward coefficient transfers by Theorems 24.7.3.46 and 24.7.3.47. With this multiplicativity, Theorem 24.4.34 gives a conjugating gauge Z for the forward transfer. The window and complement injectivity hypotheses then yield a nonempty coefficient-identity witness with $Z_{\text{ref}} = Z$. □

24.8 Normal PEPS Fundamental Theorem

For a normal PEPS, the tensors become injective after blocking around every edge, and each site separates two injective regions whose complements are also injective. With union closure in hand, these hypotheses yield the Fundamental Theorem for normal PEPS (Theorem 24.8.6) on a connected graph [186, Section 4], with gauges unique modulo balanced edge scalars. Corollaries specialize the result to closed chains, translation-invariant matrix product states, and the square lattice.

Definition 24.8.1 (Normal PEPS blocking hypotheses). Lean Lean Lean Lean Lean ✓ Stmt
The general normal theorem assumes both the edge-centred blockings into three injective regions and the one-site comparison regions with injective complements. These two assumptions are collected as the normal PEPS blocking hypotheses; the same hypotheses give the corresponding edge and one-site consequences without restating the separate edge and one-site assumptions. The following two panels show their square-lattice source realization; they do not restrict the abstract definition to this lattice.

that site. Then their defining tensors are related by a local gauge, and the gauges are unique up to the scalar freedom allowed by the graph. This is the general normal-PEPS theorem in [186, Section 4], following its Theorem 3. A connected-graph form, with the blocking hypothesis stated as the source’s proof uses it (one blocking shared by the two states, each distinguished edge the entire bond of its chain), is Theorem 24.8.6; its per-edge uniqueness clause is Theorem 24.8.7. The source’s remark following the theorem, that for a translationally invariant system the gauges are also translationally invariant when the proportionality constants are not absorbed into them, is realized on the torus by Theorem 24.7.1.9: its gauge family is carried onto every edge by the translations from one reference matrix per orientation class, and its scalar λ is kept separate rather than absorbed.

Definition 24.8.4 (Doubly injective regions). Lean ✓ Stmt For two region-injectivity assignments on the same vertex set, a region is doubly injective when it is injective for each of the two. The general normal theorem blocks two states by one shared geometry, and its final comparison contracts both tensors over the same regions, so its blocking hypotheses are instantiated at the doubly injective assignment of the two tensors.

Theorem 24.8.5 (Equal bond dimensions from the normal blocking hypotheses). Lean ✓ Stmt Lean Let A and B be PEPS tensors on a finite simple graph with the same positive physical dimension and positive bond dimensions, generating the same state and satisfying the normal blocking hypotheses at the doubly injective assignment of the two tensors, with each distinguished edge the only edge of the graph joining its two adjacent blocks. Then the bond dimensions of A and B agree on every edge. In the source the equality is part of the isomorphism lemma of [186, Section 3]: the insertion correspondence $X \mapsto Y$ on the chosen bond is an algebra isomorphism between the two full bond matrix algebras, so the bond dimensions on the two sides agree; the normal case inherits the argument through the blocked three-site chains. The single-crossing condition is the source’s blocking around an edge made explicit: the chosen edge is the entire bond between the first two parties of the blocked chain.

Proof. ✓ Proof Fix an edge. Blocking both networks into the shared three regions around it produces two three-site chains on the three-vertex complete graph whose super-sites carry the blocked-region weight families, injective by hypothesis. Each coarse state coefficient is a positive merge-collapse constant, a product of bond dimensions over the edges not incident to each region, times the corresponding original state coefficient, so the two coarse states are proportional with ratio the two constants. Multiplying one super-site of each chain by the other chain’s constant yields chains with equal states, unchanged bond dimensions, and the super-site injectivities preserved. The matched matrix insertions on the distinguished coarse bond of the two scaled chains form an algebra isomorphism between the two full bond matrix algebras (Theorem 24.3.17), and an isomorphism of full matrix algebras forces equal matrix sizes. Under the single-crossing hypothesis the coarse bond carries the original bond dimension of the distinguished edge for each tensor, so the two original bond dimensions agree. $\square$

Theorem 24.8.6 (Fundamental Theorem for normal PEPS on a connected graph). Lean ✓ Stmt Let A and B be PEPS tensors on a connected finite simple graph with the same positive physical dimension and positive bond dimensions, generating the same state. Suppose the normal blocking hypotheses hold at the doubly injective assignment of the two tensors: around every edge the network blocks into three doubly injective regions with the edge endpoints in the first two, and every site admits two doubly injective regions, with doubly injective complements, differing exactly at that site. Suppose moreover that at every edge the distinguished edge is the only edge of the graph joining its two adjacent blocks. Then the defining tensors are related by a local gauge: there are invertible matrices, one on each edge, whose endpoint actions transform A_v into B_v

at every vertex, with no leftover scalar. The shared blocking gives a common tripartition for the two tensors, with all three regions injective for both. For an edge e between the first two regions, the coefficient transfer T_e supplied by the three-site isomorphism argument satisfies

$$\begin{aligned} C_A(R_1, e, X; \sigma, \tau) &= C_B(R_1, e, T_e(X); \sigma, \tau), \\ T_e(X) &= Z_e \iota_e(X) Z_e^{-1}, \end{aligned}$$

for an invertible Z_e and every virtual insertion X on e . The single-crossing hypothesis identifies e with the entire bond between R_1 and R_2 , so Z_e acts on the original edge. The equality of the bond dimensions is not assumed; it follows from the blocking hypotheses by Theorem 24.8.5. Positive bond dimensions are required in the injectivity argument, while connectivity permits the per-vertex constants to be absorbed into the edge gauges; see Theorem 24.4.1.23. This is the blocking argument of [186, Section 4].

Proof. ✓ Proof The bond dimensions of the two tensors agree on every edge by Theorem 24.8.5. The union closure of injective regions for either tensor follows from the positive bond dimensions and Lemma 24.5.10. At every edge the shared blocking datum restricts to a datum for each tensor over the same three regions, so the per-edge gauge of the region insertion transfer (Theorem 24.4.34) and its orientation-adapted absorption give a family of invertible matrices realizing the absorbed inserted-coefficient equality at every edge. A gauge preserves blocked-region linear independence, so at every site the one-site comparison applies to the gauge-absorbed tensor: comparing the smaller region with its one-site completion yields two proportionality scalars whose ratio is a nonzero scalar λ_v with A_v equal to λ_v times the gauge action on B at v . When the smaller region is empty its block is the count of global virtual configurations, and when the completion is the full vertex set its block is the closed state coefficient; in these degenerate cases the proportionality holds with scalar one, and every other comparison region has a boundary edge by connectivity. The per-vertex relations against the nonzero closed state force $\prod_v \lambda_v = 1$, so on the connected graph a spanning-tree construction realizes the reciprocals of the scalars as oriented incidence products of edge scalars, which fold into the gauges as in the absorption step of the injective Fundamental Theorem (Theorem 24.3.32). □

Theorem 24.8.7 (Uniqueness of the normal-PEPS gauges up to a constant). Lean ✓ Stmt
Let A and B be PEPS tensors on a finite simple graph with equal bond dimensions, the bond dimensions of the first tensor positive, satisfying the normal blocking hypotheses at the doubly injective assignment of the two tensors. Suppose two families of invertible matrices, one on each edge, each realize the scalar-free gauge relation of Theorem 24.8.6: at every vertex the endpoint actions of the family transform A_v into B_v . Then the two families are proportional at every edge: $X'_e = c \cdot X_e$ for a nonzero scalar c depending on the edge. This is the uniqueness clause of the general normal-PEPS theorem in [186, Section 4], following its Theorem 3, read against the scalar-free relation; neither equality of the two states nor connectivity is needed.

Proof. ✓ Proof The inserted-edge coefficient is multilinear in the site tensors, one factor per site, so each scalar-free per-vertex relation makes every inserted coefficient of B equal the corresponding inserted coefficient of the gauge-absorbed A : each family realizes the absorbed inserted-coefficient equality at every edge, with the roles of the two tensors exchanged. On the first block of the edge's frame, whose blocked map and complement blocked map are injective for A by the blocking hypotheses and union closure (Lemma 24.5.10), the region insertion transfer is determined by the absorbed equality (Theorem 24.4.36), so the two orientation-adapted absorbing gauges induce the same conjugation and differ by a nonzero scalar; unwinding the orientation adaptation gives the proportionality of the gauges themselves. □

24.9 Fundamental Theorem for normal matrix product states

The closed chain is the one-dimensional PEPS: the cycle graph on n vertices carrying one tensor of $D \times D$ matrices per site, so that a physical configuration $\sigma = (i_1, \dots, i_n)$ receives the amplitude $\text{tr}(A^{i_1} \dots A^{i_n})$. The Fundamental Theorem for normal PEPS therefore specializes to matrix product states once its blocking hypotheses are read off the chain, yielding the matrix-product-state corollaries of [186, Section 4] and the sharper system-size bound in its closing part *Normal MPS: alternative proof*.

The development proceeds in four steps. The closed chain is first identified with a cycle of tensors, and the normal blocking hypotheses are obtained from the injectivity of every arc of L consecutive sites, giving the Fundamental Theorem for normal matrix product states on a ring of $n \geq 3L$ sites with one invertible gauge per bond. The translation-invariant case then collapses the per-bond gauges to one gauge Z and one scalar λ , with the length- n trace identity forcing $\lambda^n = 1$.

For the sharper bound, the matrix-product-state form of [186, Lemma 5] compares the $L + 1$ overlapping windows around a fixed bond. If C_j is the deformation of the j -th window, then for length- L words a and b the family of consecutive-window trace identities strips to $C_0(a) B^b = B^a C_L(b)$, and hence to $C_0(a) = B^a X$ and $C_L(b) = X B^b$. This is the step that lowers the system size from $n \geq 3L$ to $n \geq 2L + 1$. Finally the same argument is run for site-dependent chains, one tensor per site, which is the setting of [186, Lemma 5].

24.9.1 Matrix product states on the closed chain

Theorem 24.9.1.1 (Cycle blocking hypotheses from arc injectivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) *A closed chain of n sites is the cycle graph on n vertices, and a block of consecutive sites is an arc: the vertices reached clockwise from a starting site. Suppose $0 < L$, $3L \leq n$, every arc of L consecutive sites is injective for a region-injectivity assignment, and injectivity is closed under unions. Then the chain satisfies the normal blocking hypotheses: around every edge it blocks into the L sites on one side, the L sites on the other, and the remaining $n - 2L \geq L$ sites, all injective and covering the chain; and at every site the $L + 1$ sites starting there and the L sites starting at its successor are injective regions with injective complements differing exactly at that site. Moreover the only nearest-neighbour pair joining the two blocks around an edge is the edge itself, since the far ends of the two blocks are separated by the remaining $n - 2L \geq L$ sites.*

Proof. [✓ Proof](#) Membership in an arc is a cyclic-distance inequality, so the disjointness, covering, and unique-crossing claims are modular arithmetic on the distances. Arcs of length between L and n are unions of arcs of length L , hence injective by union closure; this covers the two complements at a site and the third block around an edge. $\square$

Corollary 24.9.1.2 (Fundamental Theorem for normal MPS on a closed chain). [Lean](#) [✓ Stmt](#)

Let A and B be tensors on the closed chain of n sites (the cycle graph on n vertices, one tensor per site with two virtual legs), with the same positive physical dimension and positive bond dimensions, and let $0 < L$ with $n \geq 3L$. Suppose blocking any L consecutive sites gives an injective tensor, for A and for B , and the two MPS generate the same state. Then the defining tensors are related by a local gauge: there are invertible matrices, one on each bond, whose endpoint actions transform A_i into B_i at every site with no leftover scalar. This is the source's family Z_i ($i = 1, \dots, n$, $n + 1 \equiv 1$) with $B_i = Z_i^{-1} A_i Z_{i+1}$, the first corollary following the general normal-PEPS theorem in [186, Section 4], the statement for MPS without translation invariance at a fixed system size. The positive physical and bond dimensions are the counterexample-backed

corrections carried over from the injective Fundamental Theorem (Theorem 24.4.1.23), and $0 < L$ is implicit in the source's blocking of L consecutive sites; the equality of the bond dimensions is not assumed but derived, and the connectivity of the cycle is proved, not assumed.

Proof. ✓ Proof Positive bond dimensions give union closure of injective regions for each tensor (Lemma 24.5.10), hence for the doubly injective assignment. The arc-injectivity hypothesis then yields the normal blocking hypotheses on the cycle with the unique crossing property (Theorem 24.9.1.1), the cycle is connected, and the Fundamental Theorem for normal PEPS on a connected graph (Theorem 24.8.6) supplies the scalar-free local gauge. $\square$

Corollary 24.9.1.3 (Uniqueness of the closed-chain gauges up to a constant). Lean ✓ Stmt *In the setting of Corollary 24.9.1.2, with the bond dimensions of the two tensors equal and positive and the arcs of L consecutive sites injective for each tensor, any two families of invertible bond matrices realizing the scalar-free gauge relation are proportional at every bond: $Z'_i = c \cdot Z_i$ for a nonzero scalar c depending on the bond. This is the uniqueness clause of the first corollary following the general normal-PEPS theorem of [186, Section 4]: the gauges Z_i are unique up to a multiplicative constant. Neither equality of the two states nor connectivity is needed, as in the general uniqueness clause.*

Proof. ✓ Proof The arc-injectivity hypothesis and union closure from the positive bond dimensions (Lemma 24.5.10) establish the normal blocking hypotheses on the cycle (Theorem 24.9.1.1), and the per-edge uniqueness of the general theorem (Theorem 24.8.7) gives the proportionality at every bond. $\square$

Definition 24.9.1.4 (Cycle tensor of a matrix tensor). Lean Lean Lean Lean ✓ Stmt Every edge of the cycle graph on $n \geq 3$ vertices joins a site to its cyclic successor, so sites and bonds are in bijection, and every site has exactly two incident bonds: the bond shared with its cyclic predecessor and the bond shared with its cyclic successor. The *cycle tensor* of a tensor $A = (A^i)_{i=0}^{d-1}$ of $D \times D$ matrices places one copy of A at every site of the closed chain of n sites, every bond of dimension D , with the row index of A^i read from the bond shared with the predecessor and the column index from the bond shared with the successor.

Lemma 24.9.1.5 (State coefficients of the cycle tensor). Lean Lean ✓ Stmt *For $n \geq 3$ and every configuration $\sigma = (i_1, \dots, i_n)$, the state coefficient of the cycle tensor of A at σ is the matrix product vector component $V^{(n)}(A)_\sigma = \text{tr}(A^{i_1} \dots A^{i_n})$.*

Proof. ✓ Proof An entry of a word product is a sum over paths of bond indices pinned at the two ends of products of matrix entries read along the path, by induction on the word; closing the two pinned ends into a loop, the trace of a closed word product is the sum over all cyclic bond configurations of the products of entries around the loop. Contracting the cycle tensor sums the per-site entries over all assignments of bond indices, which is that cyclic sum after matching each bond with the site on its predecessor side. $\square$

Lemma 24.9.1.6 (Arc injectivity of the cycle tensor from block injectivity). Lean Lean Lean ✓ Stmt *Let $0 < L < n$ with $n \geq 3$, let the bond dimension be positive, and suppose A is L -block injective. Then for every arc of L consecutive sites of the closed chain, the blocked tensor of the arc of the cycle tensor of A is injective.*

Proof. ✓ Proof An arc of $L < n$ consecutive sites has exactly two boundary-crossing bonds, the bond entering its first site and the bond leaving its last site, by cyclic-distance arithmetic. The blocked weight of the arc at a boundary assignment (a, b) and a physical word x is $D^{n-(L+1)}(A^{x_1} \dots A^{x_L})_{ab}$: the bonds inside the arc contract to the matrix product, the two

boundary bonds stay pinned, and each bond away from the arc contributes a free factor D . A linear relation among the blocked components over the boundary pairs is a matrix C with $\text{tr}(C^\top A^{x_1} \dots A^{x_L}) = 0$ for every word x ; the word products span the full matrix algebra by L -block injectivity, and the trace pairing is nondegenerate, so $C = 0$. $\square$

24.9.2 Translation-invariant matrix product states

When the same tensor sits at every site, the closed chain ties the per-bond gauges together. If one gauge Z and one scalar λ satisfy $B^i = \lambda Z^{-1} A^i Z$ for every physical index i , then for a configuration $\sigma = (i_1, \dots, i_n)$,

$$\text{tr}(B^{i_1} \dots B^{i_n}) = \lambda^n \text{tr}(Z^{-1} A^{i_1} \dots A^{i_n} Z) = \lambda^n \text{tr}(A^{i_1} \dots A^{i_n}).$$

Equality of the matrix product states therefore forces $\lambda^n = 1$. Thus Z and λ capture precisely the freedom in the translation-invariant corollary of [186, Section 4].

Corollary 24.9.2.1 (Fundamental Theorem for translation-invariant normal MPS on a closed chain, matrix form). [Lean](#) [✓ Stmt](#) *Let n be positive, let $0 < L$ with $2L+1 \leq n$, and let A and B be tensors of $D \times D$ matrices with $D > 0$, over physical dimension d , each L -block injective, whose matrix product vectors agree at system size n : $V^{(n)}(A)_\sigma = V^{(n)}(B)_\sigma$ for every configuration σ . Then there are invertible matrices Z_v , one per bond of the closed chain ($v = 1, \dots, n, n+1 \equiv 1$), with $B^i = Z_v^{-1} A^i Z_{v+1}$ for every site v and every i . This is the first corollary following the general normal-PEPS theorem of [186, Section 4] specialized to one site-independent tensor: the source states the corollary for site-dependent families, and its conclusion is the per-bond gauge family delivered here. The system size is the optimal $n \geq 2L+1$ of the source’s alternative proof (its Section “Normal MPS: alternative proof”), rather than the $n \geq 3L$ of the blocking route. The source’s translation-invariant corollary (a single gauge Z and a constant λ with $\lambda^n = 1$) refines this conclusion and is not part of this statement. A vanishing physical dimension is already excluded by block injectivity.*

Proof. [✓ Proof](#) The per-bond form of the strengthened closed-chain corollary (Corollary 24.9.3.7), proved through the overlapping windows at $n \geq 2L+1$, gives the per-bond gauge family directly. $\square$

Lemma 24.9.2.2 (Two per-bond gauge families are proportional at every bond). [Lean](#) [✓ Stmt](#) *Let n be positive, let A and B be tensors of $D \times D$ matrices over physical dimension d with A being L -block injective, and let Z_v and Z'_v ($v = 1, \dots, n, n+1 \equiv 1$) be two families of invertible matrices with $B^i = Z_v^{-1} A^i Z_{v+1}$ and $B^i = Z'^{-1}_v A^i Z'_{v+1}$ at every site v and every i . Then for every bond v there is a nonzero scalar c with $Z'_v = c Z_v$. No condition on the system size is needed.*

Proof. [✓ Proof](#) Iterating each relation along a word x of length L from the starting site v gives $B^{x_1} \dots B^{x_L} = Z_v^{-1} A^{x_1} \dots A^{x_L} Z_{v+L}$ and the same with Z' . Since the word products $A^{x_1} \dots A^{x_L}$ span the full matrix algebra by L -block injectivity, the comparison extends linearly to every matrix W : $Z_v^{-1} W Z_{v+L} = Z'^{-1}_v W Z'_{v+L}$. Taking W to be the identity in this extended relation pins the two bond transports $Z_v^{-1} Z_{v+L} = Z'^{-1}_v Z'_{v+L}$ to the same invertible matrix; cancelling it leaves $Z_v^{-1} W Z_v = Z'^{-1}_v W Z'_v$ for every W , so $Z'_v Z_v^{-1}$ commutes with the full matrix algebra, hence is a scalar, nonzero by invertibility. $\square$

Corollary 24.9.2.3 (Uniqueness of the per-bond gauges up to one constant, matrix form). [Lean](#) [✓ Stmt](#) *Let n be positive, let $0 < L$ with $2L+1 \leq n$, and let A and B be tensors of $D \times D$ matrices with $D > 0$, each L -block injective. If two families Z and Z' of invertible matrices on the bonds of the closed chain both satisfy $B^i = Z_v^{-1} A^i Z_{v+1}$ at every site, then they are proportional by a*

single constant: there is a nonzero scalar c with $Z'_v = c Z_v$ at every bond. This is the uniqueness clause of the first corollary following the general normal-PEPS theorem of [186, Section 4], for one site-independent tensor. Equality of the two states is not needed.

Proof. ✓ Proof At every bond the two gauges differ by a nonzero scalar μ_v (Lemma 24.9.2.2). The relation $B^i = Z_v^{-1} A^i Z_{v+1}$ for both families forces the scalar $\mu_v^{-1} \mu_{v+1}$ to fix the nonzero tensor B , so consecutive constants agree, and going around the cycle merges them into one. $\square$

Lemma 24.9.2.4 (Consecutive per-bond gauges are proportional). Lean ✓ Stmt *Let n be positive, let A and B be tensors of $D \times D$ matrices over physical dimension d with A being L -block injective, and let Z_v ($v = 1, \dots, n$, $n+1 \equiv 1$) be invertible matrices with $B^i = Z_v^{-1} A^i Z_{v+1}$ at every site v and every i . Then for every site v there is a nonzero scalar c with $Z_{v+1} = c Z_v$.*

Proof. ✓ Proof Iterating the per-bond relation along a word x of length L from the starting site v gives $B^{x_1} \dots B^{x_L} = Z_v^{-1} A^{x_1} \dots A^{x_L} Z_{v+L}$. Comparing the iterations from v and from $v+1$ over the word products $A^{x_1} \dots A^{x_L}$, which span the full matrix algebra by L -block injectivity, the comparison extends linearly to every matrix W : $Z_v^{-1} W Z_{v+L} = Z_{v+1}^{-1} W Z_{v+1+L}$. Taking W to be the identity in this extended relation pins the two bond transports $Z_v^{-1} Z_{v+L} = Z_{v+1}^{-1} Z_{v+1+L}$ to the same invertible matrix; cancelling it leaves $Z_v^{-1} W Z_v = Z_{v+1}^{-1} W Z_{v+1}$ for every W , so $Z_{v+1} Z_v^{-1}$ commutes with the full matrix algebra, hence is a scalar, nonzero by invertibility. $\square$

Corollary 24.9.2.5 (Fundamental Theorem for translation-invariant normal MPS on a closed chain, single-gauge form). Lean ✓ Stmt *Let n be positive, let $0 < L$ with $2L+1 \leq n$, and let A and B be tensors of $D \times D$ matrices with $D > 0$, over physical dimension d , each L -block injective, whose matrix product vectors agree at system size n : $V^{(n)}(A)_\sigma = V^{(n)}(B)_\sigma$ for every configuration σ . Then there are an invertible matrix Z and a constant λ with $\lambda^n = 1$ such that $B^i = \lambda Z^{-1} A^i Z$ for every i . This is the translation-invariant corollary of [186, Section 4], the statement for TI MPS following the first corollary after the general normal-PEPS theorem. The system size is the optimal $n \geq 2L+1$ of the source's alternative proof.*

Proof. ✓ Proof This is the single-gauge form of the strengthened closed-chain corollary (Corollary 24.9.3.6), proved through the overlapping windows at $n \geq 2L+1$. $\square$

Corollary 24.9.2.6 (Uniqueness of the translation-invariant gauge up to a constant). Lean ✓ Stmt *Let $0 < L$ and $0 < D$, and let A and B be tensors of $D \times D$ matrices, each L -block injective. If invertible matrices Z, Z' and constants λ, λ' satisfy $B^i = \lambda Z^{-1} A^i Z$ and $B^i = \lambda' Z'^{-1} A^i Z'$ for every i , then there is a nonzero scalar c with $Z' = c Z$. This is the uniqueness clause of the translation-invariant corollary of [186, Section 4] for TI MPS. No system size enters the statement: equality of the two states is not needed, nor are the root-of-unity conditions on λ and λ' .*

Proof. ✓ Proof The scalar λ is nonzero because B is L -block injective with positive bond dimension, so some $B^{i_0} \neq 0$. Iterating each relation along a word x of length L gives $B^{x_1} \dots B^{x_L} = \lambda^L Z^{-1} A^{x_1} \dots A^{x_L} Z$ and the primed analog; the word products span the full matrix algebra by L -block injectivity, so $\lambda^L Z^{-1} W Z = \lambda'^L Z'^{-1} W Z'$ for every matrix W . The identity pins $\lambda^L = \lambda'^L$; cancelling the nonzero factor leaves $Z^{-1} W Z = Z'^{-1} W Z'$ for every W , so $Z' Z^{-1}$ commutes with the full matrix algebra, hence is a nonzero scalar. $\square$

24.9.3 The optimal system size $n \geq 2L+1$

The bound $n \geq 3L$ is not optimal. A bond of the closed chain is straddled by the $L+1$ overlapping windows of L sites around it. For consecutive windows, the deformed closed-chain coefficients

have the form

$$\mathrm{tr}(B^p C_j(u_1 \cdots u_L) B^{u_{L+1}} B^s) = \mathrm{tr}(B^p B^{u_1} C_{j+1}(u_2 \cdots u_{L+1}) B^s).$$

Injectivity on the complementary arc strips these identities to

$$C_j(u_1 \cdots u_L) B^{u_{L+1}} = B^{u_1} C_{j+1}(u_2 \cdots u_{L+1}),$$

and chaining the relations across the $L+1$ windows gives $C_0(a) B^b = B^a C_L(b)$. This is the overlap mechanism in [186, Lemma 5]. The lemmas below establish that argument: block injectivity extends to every length, word products transport between equal-state chains, and the bond operator is extracted and shown to act by conjugation, lowering the system size to $n \geq 2L + 1$.

Lemma 24.9.3.1 (Block injectivity extends to longer blocks). ✓ Stmt *Let A be a tensor of $D \times D$ matrices over physical dimension d , L -block injective with $0 < L$. Then A is m -block injective for every $m \geq L$. This is the chain form of the statement that any region of at least the blocking size is injective, derived in the closing section of [186] (“Normal MPS: alternative proof”) from its union lemma.*

Proof. ✓ Proof It suffices to pass from length m to $m + 1$. Write the identity as a combination of length- L word products and peel the leading letter of each: any matrix P becomes a combination of terms $A^i(WP)$ with W a word product of length $L - 1$. Each factor WP is a combination of length- m word products by the induction hypothesis, so each term is a combination of length- $(m + 1)$ word products. □

Lemma 24.9.3.2 (Word transport between same-state networks). Lean Lean ✓ Stmt *Let A and B be tensors of $D \times D$ matrices over physical dimension d , let $n = k + q$, and suppose the length- k and length- q word products of A each span the full matrix algebra. If the matrix product vectors of A and B agree at system size n , then there is a linear map Λ of the matrix algebra with $\Lambda(A^{x_1} \cdots A^{x_k}) = B^{x_1} \cdots B^{x_k}$ for every word x of length k . The same holds when, instead of the traces, the word products of the two tensors are themselves equal at length n . This is the operator-transport mechanism underlying Lemma 5 of [186]: an operator expanded over the window products of one network is re-read in the other.*

Proof. ✓ Proof Pair the matrix algebra against the complementary word products: let $\Psi_A(M)$ be the family of traces $\mathrm{tr}(M A^{y_1} \cdots A^{y_q})$ over the length- q words y , and Ψ_B its B -analog. Ψ_A is injective because the length- q products span and the trace pairing is nondegenerate. On the spanning family of length- k products the two pairings match: $\Psi_A(A^x) = \Psi_B(B^x)$, since both compute closed-chain coefficients of total length n , equal by hypothesis. Hence the range of Ψ_A lies in that of Ψ_B , forcing Ψ_B to be injective by dimension count, and a left inverse of Ψ_B composed with Ψ_A is the transport. For the matrix-product version replace the traces by the products themselves. □

Lemma 24.9.3.3 (Bond-operator extraction from overlapping windows). Lean Lean ✓ Stmt *Let $0 < L$ with $2L + 1 \leq n$, let B be a tensor of $D \times D$ matrices over physical dimension d , L -block injective, and let $C_0, \dots, C_L$ be deformed window tensors: each C_j assigns a matrix to every word of length L , read as replacing the blocked tensor of the L -site window starting j sites left of a fixed bond of the closed chain on n sites. Suppose the deformed states agree for consecutive window positions: for every $j < L$, every prefix p of j sites, every assignment u on the $L + 1$ sites covered by the two windows, and every suffix s on the remaining sites,*

$$\mathrm{tr}(B^p C_j(u_1 \cdots u_L) B^{u_{L+1}} B^s) = \mathrm{tr}(B^p B^{u_1} C_{j+1}(u_2 \cdots u_{L+1}) B^s).$$

Then the deformation is a virtual operation on the bond: there is a matrix X such that, for all length- L words a and b ,

$$\begin{aligned} C_0(a) &= B^{a_1} \dots B^{a_L} X, \\ C_L(b) &= X B^{b_1} \dots B^{b_L}, \end{aligned}$$

and X is unique. This is Lemma 5 of [186] (its closing section, “Normal MPS: alternative proof”), specialized to one site-independent tensor; the source states it for site-dependent tensors on five sites with $L = 2$, with the deformed window tensors arising as physical operators contracted with the blocked windows they act on; the site-dependent form, in the setting in which the source states the lemma, is Lemma 24.9.4.4, and this statement is its constant specialization. The algebra-homomorphism clause of the source lemma is established with the insertion correspondence in Lemma 24.9.3.4.

Proof. ✓ **Proof** Place the tensor at every site of the site-dependent form (Lemma 24.9.4.4): the arc products of the constant chain are the word products of the repeated tensor, and L -block injectivity of the tensor is window injectivity of the constant chain. $\square$

Lemma 24.9.3.4 (Conjugate word products at one length from Lemma 5). Lean ✓ **Stmt** Let $0 < L$ with $2L + 1 \leq n$ and $0 < D$, and let A and B be tensors of $D \times D$ matrices over physical dimension d , each L -block injective, whose matrix product vectors agree at system size n . Then there is an invertible matrix Z such that, for every word w of length n ,

$$B^{w_1} \dots B^{w_n} = Z A^{w_1} \dots A^{w_n} Z^{-1}.$$

This captures the insertion correspondence of Lemma 5 of [186] and its algebra-homomorphism clause for one site-independent tensor: the map sending an insertion on one network to the corresponding insertion on the other is an automorphism of the matrix algebra, hence inner.

Proof. ✓ **Proof** Regard A and B as constant site-dependent chains on n sites. Window injectivity of a constant chain is block injectivity of the single tensor, and agreement of the matrix product vectors at size n is the same-state hypothesis, so the site-dependent overlapping-window capstone (Corollary 24.9.4.8) supplies a per-bond gauge family relating the two chains. The arc products of a constant chain are the word products of the tensor, and the gauge conjugation of arc products (Theorem 24.9.4.9), read at the full chain length n where the end bond wraps to the starting bond, is conjugation of the length- n word products by the gauge at the starting bond. $\square$

Lemma 24.9.3.5 (Rigidity of word products at one length). Lean ✓ **Stmt** Let $0 < L$ with $2L + 1 \leq n$ and $0 < D$, and let A and C be tensors of $D \times D$ matrices over physical dimension d , each L -block injective, with equal word products at length n : $C^{w_1} \dots C^{w_n} = A^{w_1} \dots A^{w_n}$ for every word w of length n . Then there is a constant λ with $\lambda^n = 1$ and $C^i = \lambda A^i$ for every i .

Proof. ✓ **Proof** Transport word products of A to those of C at the window length L , at $L + 1$, at the complementary length $n - 1 - L$, and at $n - L$ (Lemma 24.9.3.2, matrix-product version; the lengths lie between L and $n - L$ by $2L + 1 \leq n$). For a transport pair at lengths k and $k + 1$, the transported identity $\Lambda_k(\mathbb{1})$ commutes with every letter of C - its two one-letter extensions agree, $C^i \Lambda_k(\mathbb{1}) = \Lambda_{k+1}(A^i) = \Lambda_k(\mathbb{1}) C^i$ - hence with the full matrix algebra by block injectivity, so it is a scalar $c_k \mathbb{1}$, nonzero because the transport is surjective onto the spanning word products, hence bijective. Peeling one letter off the full-length equality through the spanning windows of lengths L and $n - 1 - L$ then leaves $C^i = (c_L c_{n-1-L})^{-1} A^i =: \lambda A^i$; evaluating any nonzero length- n word product gives $\lambda^n = 1$. $\square$

Corollary 24.9.3.6 (Fundamental Theorem for translation-invariant normal MPS at $n \geq 2L+1$, single-gauge form). [Lean](#) [✓ Stmt](#) Let $0 < L$ with $2L+1 \leq n$ and $0 < D$, and let A and B be tensors of $D \times D$ matrices over physical dimension d , each L -block injective, whose matrix product vectors agree at system size n : $V^{(n)}(A)_\sigma = V^{(n)}(B)_\sigma$ for every configuration σ . Then there are an invertible matrix Z and a constant λ with $\lambda^n = 1$ such that $B^i = \lambda Z^{-1} A^i Z$ for every i . This is the strengthened closed-chain corollary of [186] (the corollary after Lemma 5 in its closing section “Normal MPS: alternative proof”), proved through the overlapping windows at the optimal $n \geq 2L+1$ rather than the $n \geq 3L$ of the blocking route; the single-gauge corollary 24.9.2.5 delegates to it. The uniqueness clause of the source corollary - the gauge Z is unique up to a multiplicative constant - needs no system size and is Corollary 24.9.2.6.

Proof. [✓ Proof](#) By Lemma 24.9.3.4 the word products of B at length n are those of A conjugated by an invertible matrix. Absorbing the conjugation into B - which preserves L -block injectivity - leaves two tensors with equal word products at length n , and Lemma 24.9.3.5 makes them proportional letterwise by an n -th root of unity. Undoing the absorption gives $B^i = \lambda Z^{-1} A^i Z$. $\square$

Corollary 24.9.3.7 (Fundamental Theorem for translation-invariant normal MPS at $n \geq 2L+1$, per-bond form). [Lean](#) [✓ Stmt](#) Let n be positive, let $0 < L$ with $2L+1 \leq n$ and $0 < D$, and let A and B be tensors of $D \times D$ matrices over physical dimension d , each L -block injective, whose matrix product vectors agree at system size n . Then there are invertible matrices Z_v , one per bond of the closed chain ($v = 1, \dots, n, n+1 \equiv 1$), with $B^i = Z_v^{-1} A^i Z_{v+1}$ for every site v and every i . This is the per-bond gauge family of the closed-chain corollary, proved at the source’s strengthened bound $n \geq 2L+1$ ([186], the corollary after Lemma 5 in its closing section “Normal MPS: alternative proof”). The named matrix-level Corollary 24.9.2.1 delegates to it.

Proof. [✓ Proof](#) The single-gauge form (Corollary 24.9.3.6) gives Z and λ with $\lambda^n = 1$ and $B^i = \lambda Z^{-1} A^i Z$. The family $Z_v := \lambda^v Z$ dresses the gauge with powers of the root of unity: away from the seam the consecutive powers contribute the factor λ , and across the seam the wraparound contributes $\lambda^{1-n} = \lambda$ because $\lambda^n = 1$, so $B^i = Z_v^{-1} A^i Z_{v+1}$ at every site. $\square$

24.9.4 Site-dependent chains

The source states its Lemma 5 for site-dependent tensors: five sites carrying tensors $A_1, \dots, A_5$ whose two-site windows are injective. While the previous results focus on a repeated tensor, the site-dependent setting treats chains carrying a distinct tensor per site. Windows are arcs of consecutive sites, and the closed-chain corollary delivers one gauge per bond. Most of the matrix-chain statements below take one physical dimension d and one bond dimension D . For the Applications-section translational-invariance corollary, the cycle-graph formulation instead retains one dimension D_e per bond, derives their equality from the injective Fundamental Theorem, and then identifies the virtual spaces to construct the repeated tensor in the source’s standing regime of at least three sites. The literal two-site reading of the printed corollary is false; this is recorded by Theorem 24.9.4.14 and [149].

Definition 24.9.4.1 (Arc products and window injectivity of a site-dependent chain). [Lean](#) [Lean](#) [✓ Stmt](#) Let a closed chain on $n \geq 1$ sites carry one tensor of $D \times D$ matrices over physical dimension d per site, $A_0, \dots, A_{n-1}$. The *arc product* of a word $x = x_1 \dots x_\ell$ read from site s is

$$A_{(s)}^x = A_s^{x_1} A_{s+1}^{x_2} \dots A_{s+\ell-1}^{x_\ell},$$

with site indices read cyclically. The chain is *window injective at length L* when, for every site s , the arc products of the length- L words read from s span the full matrix algebra. This is

the site-dependent setting of the closing section of [186] (“Normal MPS: alternative proof”): an MPS on five sites in which the blocking of any two consecutive tensors is injective, for a general window length L .

Lemma 24.9.4.2 (Injectivity is window injectivity at length one). Lean ✓ Stmt *If every tensor in a closed chain is injective, then the chain is window injective at length 1. Indeed, the length-one arc products at site s are precisely the matrices of the tensor at site s .*

Proof. ✓ Proof The family of length-one words is identified with the physical alphabet. Under this identification the length-one arc product is exactly the corresponding local matrix, so the two spanning conditions coincide. $\square$

Lemma 24.9.4.3 (Arc spanning from window injectivity). Lean ✓ Stmt *Let a site-dependent chain on $n \geq 1$ sites be window injective at length L with $0 < L$. Then for every $m \geq L$ and every site s , the arc products of the length- m words read from s span the full matrix algebra. This is the site-dependent form of the remark that any region of at least the blocking size is injective ([186], closing section, cf. Lemma 24.9.3.1).*

Proof. ✓ Proof Write the identity as a combination of window products at the starting site and peel the leading letter of each: any matrix becomes a combination of terms $A_s^i(WP)$ with W an arc product of length $L - 1$ read from $s + 1$. Each factor WP is a combination of length- m arc products read from $s + 1$ by induction, so each term is a combination of length- $(m + 1)$ arc products read from s . $\square$

Lemma 24.9.4.4 (Bond-operator extraction from overlapping windows, site-dependent form). Lean Lean ✓ Stmt *Let $0 < L$ with $2L + 1 \leq n$, let the closed chain on n sites carry one tensor of $D \times D$ matrices per site, window injective at length L , and fix a site r . Let $C_0, \dots, C_L$ be deformed window tensors around the bond $(r + L - 1, r + L)$: each C_j assigns a matrix to every word of length L , read as replacing the blocked tensor of the L -site window starting at site $r + j$. Suppose the deformed states agree for consecutive window positions: for every $j < L$, every prefix p of j sites read from r , every assignment u on the $L + 1$ sites covered by the two windows, and every suffix s on the remaining sites,*

$$\begin{aligned} \text{tr}(A_{(r)}^p C_j(u_1 \cdots u_L) A_{r+j+L}^{u_{L+1}} A_{(r+j+L+1)}^s) \\ = \text{tr}(A_{(r)}^p A_{r+j}^{u_1} C_{j+1}(u_2 \cdots u_{L+1}) A_{(r+j+L+1)}^s). \end{aligned}$$

Then the deformation is a virtual operation on the bond: there is a matrix X with $C_0(a) = A_{(r)}^a X$ and $C_L(b) = X A_{(r+L)}^b$ for all length- L words a, b , and X is unique. This is Lemma 5 of [186] in the site-dependent setting in which the source states it, with the deformed window tensors arising as physical operators contracted with the blocked windows they act on; Lemma 24.9.3.3 is its constant specialization.

Proof. ✓ Proof Comparing the windows at j and $j + 1$ through the complementary arc of $n - L - 1 \geq L$ sites, whose arc products span by Lemma 24.9.4.3, strips the state equality to the letter level: $C_j(u_1 \cdots u_L) A_{r+j+L}^{u_{L+1}} = A_{r+j}^{u_1} C_{j+1}(u_2 \cdots u_{L+1})$ on every window of $L + 1$ sites. Chaining the L relations along the $2L$ sites flanking the bond turns the leftmost window into the rightmost one: $C_0(a) A_{(r+L)}^b = A_{(r)}^a C_L(b)$ for all length- L words. Writing the identity as a combination $\sum_b \alpha_b A_{(r+L)}^b = \mathbb{1}$, the matrix $X := \sum_b \alpha_b C_L(b)$ satisfies $A_{(r)}^a X = C_0(a)$; the mirrored combination of the C_0 -family collapses onto the same matrix, giving $X A_{(r+L)}^b = C_L(b)$, and stripping the spanning factors makes X unique. $\square$

Lemma 24.9.4.5 (Arc transport between same-state chains). Lean ✓ Stmt *Let two site-dependent chains A, B on n sites generate the same closed-chain state, let $k + q = n$, and suppose the A -arc products of length k read from a site s and of length q read from $s + k$ each span the full matrix algebra. Then there is a linear map Λ of the matrix algebra with $\Lambda(A_{(s)}^x) = B_{(s)}^x$ for every word x of length k .*

Proof. ✓ Proof As for one repeated tensor (Lemma 24.9.3.2): pair the matrix algebra against the complementary arc products. The pairing is injective on the A -side by spanning and trace nondegeneracy, and the two pairings match on the spanning arc family because both compute closed-chain coefficients of total length n , equal after rotating the trace of the closed chain to the starting site s . □

Lemma 24.9.4.6 (The insertion correspondence is an algebra homomorphism, site-dependent form). Lean Lean ✓ Stmt *Let $0 < L$ with $2L + 1 \leq n$, let A and B be site-dependent chains on n sites, each window injective at length L , generating the same closed-chain state, and fix a site r . Then there is a linear map Φ of the matrix algebra, unital and multiplicative, such that $\text{tr}(XA_{(r+L)}^w) = \text{tr}(\Phi(X)B_{(r+L)}^w)$ for every matrix X and every word w of length n , and Φ is the unique linear map with this pairing property. This is the closing clause of Lemma 5 of [186]—the maps $O_1 \mapsto X$ and $O_3^T \mapsto X$ are uniquely defined and are algebra homomorphisms—in the site-dependent setting of the source, with the algebra structure carried by the insertions: Φ sends an insertion on the bond $(r + L - 1, r + L)$ of the first chain to the bond operator extracted on the second.*

Proof. ✓ Proof A virtual insertion X on the bond is realized on each of the $L + 1$ overlapping windows around it by deformed window tensors placing X after the first $L - j$ letters of the window starting at site $r + j$. Each deformation is transported to the second chain by the arc transport at its own starting site (Lemma 24.9.4.5); the same-state hypothesis makes the transported deformations generate one common state, and the bond-operator extraction (Lemma 24.9.4.4) produces $\Phi(X)$. Uniqueness of the bond operator makes Φ linear, unital and multiplicative; the pairing against all closed-chain words follows by expanding the insertion over the window products, and pins Φ through the spanning arcs of length n (Lemma 24.9.4.3) and nondegeneracy of the trace pairing. □

Lemma 24.9.4.7 (Conjugate arc products at every bond, site-dependent form). Lean ✓ Stmt *Let $0 < L$ with $2L + 1 \leq n$, and let A and B be site-dependent chains on n sites with a common positive bond dimension, each window injective at length L , generating the same closed-chain state. Then for every site p there is an invertible matrix Z with $B_{(p)}^w = ZA_{(p)}^w Z^{-1}$ for every word w of length n . The statement for one repeated tensor is Lemma 24.9.3.4.*

Proof. ✓ Proof The insertion correspondence at the bond (Lemma 24.9.4.6) is unital and nonzero, hence an automorphism of the full matrix algebra, and by Skolem–Noether (Theorem [182, “Skolem–Noether”]) it is conjugation by an invertible Z ; pairing insertions against all closed-chain words read from p and using nondegeneracy of the trace pairing transfers the conjugation to the arc products. □

Corollary 24.9.4.8 (Fundamental Theorem for site-dependent normal closed chains at $n \geq 2L + 1$).

Lean ✓ Stmt *Let $0 < L$ with $2L + 1 \leq n$, and let A and B be site-dependent chains on n sites with a common positive bond dimension, each window injective at length L , generating the same closed-chain state. Then the chains are gauge equivalent: there are invertible matrices Z_v , one per bond of the closed chain, with $B_v^i = Z_v A_v^i Z_{v+1}^{-1}$ for every site v and every i , with indices read cyclically. This is the closed-chain corollary of the closing section of [186] obtained from its*

site-dependent Lemma 5 at every bond; the source displays the corollary for translation-invariant tensors (Corollary 24.9.3.6), with the site-dependent Lemma 5 as its engine.

Proof. [✓ Proof](#) The per-bond conjugations (Lemma 24.9.4.7) give window covariance with nonzero scalars: on every arc of m sites with $L \leq m$ and $m + L \leq n$, the conjugation at the near end intertwines the arc products of the two chains across the far bond, the bond-operator extraction (Lemma 24.9.4.4) produces the connecting matrix, and the conjugation at the far end pins it to a nonzero scalar multiple of the gauge there through the centralizer of the full matrix algebra, so $B_{(p)}^u = c_{p,m} Z_p A_{(p)}^u Z_{p+m}^{-1}$ on all length- m words u . Comparing the covariance on windows of lengths $L + 1$ and L through the spanning window products leaves the letterwise relation $B_v^i = \mu_v Z_v A_v^i Z_{v+1}^{-1}$ with nonzero scalars μ_v ; iterating it once around the closed chain against a nonzero arc product forces $\prod_v \mu_v = 1$, and dressing the gauges with the partial products of the scalars absorbs them, including across the seam. $\square$

Theorem 24.9.4.9 (Cyclic gauges preserve closed-chain states). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{n-1}$ and $B_0, \dots, B_{n-1}$ be two site-dependent closed chains with a cyclic gauge relation $B_v^i = Z_v A_v^i Z_{v+1}^{-1}$ at every site. Then they generate the same closed-chain state. More generally, the product along any arc is conjugated by the two gauges on the boundary bonds of that arc; for a full turn around the chain the two boundary bonds coincide, and cyclicity of trace removes the remaining conjugation.

Proof. [✓ Proof](#) Iterate the sitewise gauge relation along the arc. Consecutive boundary matrices cancel, leaving only the gauges at the two ends. For a word of length n read from the first site, the two ends are the same bond, so the closed trace is invariant under the resulting conjugation. $\square$

Corollary 24.9.4.10 (Fundamental Theorem for injective closed MPS chains). [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{n-1}$ and $B_0, \dots, B_{n-1}$ be two site-dependent closed chains on $n \geq 3$ sites, with common physical dimension and common positive bond dimension. Suppose every local tensor in both chains is injective and the two closed-chain states are equal. Then there are invertible matrices $Z_0, \dots, Z_{n-1}$, one per bond of the closed chain, such that $B_v^i = Z_v A_v^i Z_{v+1}^{-1}$ for every site v and every physical index i , with indices read cyclically. This is the $L = 1$, uniform-dimension instance of [186, Theorem thm:inj_MPS, lines 688–725].

Proof. [✓ Proof](#) Apply Corollary 24.9.4.8 with $L = 1$. The size condition becomes $n \geq 3$, and Lemma 24.9.4.2 converts the sitewise injectivity hypotheses into the required window-injectivity hypotheses. $\square$

Definition 24.9.4.11 (Cyclic shift of a closed MPS chain). [Lean](#) [Lean](#) [✓ Stmt](#) For a closed chain $A_0, \dots, A_{n-1}$, its cyclic shift is the chain whose v -th tensor is A_{v+1} , with indices read modulo n . The generated state is cyclic-shift invariant when the closed-chain state of $A_0, \dots, A_{n-1}$ agrees with that of $A_1, \dots, A_{n-1}, A_0$.

Definition 24.9.4.12 (Cyclic shift of a cycle-graph tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On the cycle graph with $n \geq 3$ sites, let $r(v) = v + 1$ be the one-site rotation. The cyclic shift of a site-dependent tensor family $A_0, \dots, A_{n-1}$ is the transport along r^{-1} , so its tensor at site v is A_{v+1} . Thus the shifted family is $A_1, \dots, A_{n-1}, A_0$, exactly as in [186, lines 1807–1827]. The generated state is cyclic-shift invariant when the original and shifted families generate the same closed-chain state.

Corollary 24.9.4.13 (Bond dimensions of a cyclic-shift-invariant injective MPS). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{n-1}$ be a site-dependent injective closed MPS on $n \geq 3$ sites, with one positive dimension D_e for each bond e . If its generated state is invariant under the cyclic

shift $A_v \mapsto A_{v+1}$, then the injective Fundamental Theorem compares the family with its shift by invertible matrices on the corresponding bonds. Consequently the bond from v to $v+1$ and the bond from $v+1$ to $v+2$ have the same dimension. Iterating around the cycle gives $D_e = D_f$ for every pair of bonds e, f .

This is the positive-bond, at-least-three-site form of the “all bond dimensions are the same” clause of the Applications-section corollary in [186, lines 1803–1842]. Positivity excludes empty virtual spaces, the same counterexample-backed correction as in the injective Fundamental Theorem.

Proof. ✓ Proof Transport along the inverse rotation preserves injectivity and replaces the bond-dimension function D_e by $D_{r(e)}$. Apply the injective Fundamental Theorem to the original and shifted families. Equality of their bond-dimension functions gives $D_e = D_{r(e)}$; rotation sends the successor bond at v to the successor bond at $v+1$. Cyclic induction then proves equality on all bonds. □

Theorem 24.9.4.14 (Two-site obstruction to cyclic MPS bond uniformity). Lean ✓ Stmt Discussion: #3371 The bond-uniformity implication is false for two sites. More precisely, there are injective tensors

$$A_1^i \in \text{Mat}_{1 \times 2}(\mathbb{C}), \quad A_2^i \in \text{Mat}_{2 \times 1}(\mathbb{C}) \quad (i = 0, 1). \quad (24.29)$$

whose closed-chain coefficients are invariant under the cyclic shift, but whose positive virtual dimensions are unequal. Consequently the condition $n \geq 3$ in Corollary 24.9.4.13 cannot be removed. This corrects the unqualified printed corollary in [186, lines 1803–1805]; see [149].

Proof. ✓ Proof Let e_0, e_1 be the standard basis of $\mathbb{C}^2$ and take $A_1^i = e_i^\top$ and $A_2^i = e_i$. Each local tensor map sends a virtual basis to e_0, e_1 , hence is injective. Moreover

$$\text{tr}(A_1^i A_2^j) = \delta_{ij} = \text{tr}(A_2^i A_1^j). \quad (24.30)$$

so the state is cyclically invariant, whereas the adjacent dimensions are 1 and 2. □

Corollary 24.9.4.15 (Gauge comparison with the cyclic shift). Lean ✓ Stmt Let $A_0, \dots, A_{n-1}$ be an injective site-dependent closed chain with positive bond dimension on $n \geq 3$ sites. If its generated state is invariant under the cyclic shift $A_v \mapsto A_{v+1}$, then there are invertible matrices $Z_0, \dots, Z_{n-1}$, one per bond, such that $A_{v+1}^i = Z_v A_v^i Z_{v+1}^{-1}$ for every site v and physical index i . This is the first comparison step in the Applications-section translation-invariant description corollary of [186, lines 1807–1824].

Proof. ✓ Proof Apply Corollary 24.9.4.10 to the chain and its cyclic shift. Injectivity is preserved by the shift, and the state equality is exactly the cyclic-shift invariance hypothesis. □

Corollary 24.9.4.16 (Telescoping against the first tensor). Lean ✓ Stmt Suppose a closed chain $A_0, \dots, A_{n-1}$ is gauge equivalent to its cyclic shift. Then for every site v there are invertible matrices L_v and R_v such that $A_v^i = L_v A_0^i R_v^{-1}$ for every physical index i . This is the telescoping step in [186, lines 1828–1862], where the products denoted L_i and R_i express all tensors through the first one.

Proof. ✓ Proof Write the cyclic-shift gauge relation from one site to the next and iterate it along the chain. At the first site one takes $L_0 = R_0 = \mathbb{1}$; passing from v to $v+1$ multiplies the left and right virtual factors by the adjacent gauges. □

Corollary 24.9.4.17 (Gauge-to-first description from a cyclic-invariant state). [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{n-1}$ be an injective site-dependent closed chain with positive bond dimension on $n \geq 3$ sites. If its generated state is invariant under the cyclic translation of the sites, then for every site v there are invertible matrices L_v and R_v such that $A_v^i = L_v A_0^i R_v^{-1}$ for every physical index i . This is the uniform-bond-dimension form of the L_v, R_v step in the Applications-section proof of [186, lines 1803–1890].

Proof. [✓ Proof](#) Apply the injective chain theorem to the chain and its cyclic shift, then telescope the resulting bond gauges around the chain. $\square$

Corollary 24.9.4.18 (Translation-invariant description of an injective closed MPS chain, uniform bond dimension). [Lean](#) [✓ Stmt](#) Let $A_0, \dots, A_{n-1}$ be an injective site-dependent closed chain on $n \geq 3$ sites, with one physical dimension d and one positive bond dimension D . If the closed-chain state is invariant under the cyclic shift $A_v \mapsto A_{v+1}$, then there is an injective tensor B , with bond dimension D , such that, for every physical configuration σ ,

$$\mathrm{tr}(A_0^{\sigma_0} \dots A_{n-1}^{\sigma_{n-1}}) = \mathrm{tr}(B^{\sigma_0} \dots B^{\sigma_{n-1}}).$$

This is the uniform-bond-dimension target corresponding to the Applications-section corollary of [186, lines 1803–1890]. The source also concludes that all bond dimensions of the original site-dependent representation are equal; in the present uniform-dimension statement that clause has already been built into the setting.

Proof. [✓ Proof](#) Apply Corollary 24.9.4.15 to compare the chain with its cyclic shift, giving one invertible gauge Z_v per bond with $A_{v+1}^i = Z_v A_v^i Z_{v+1}^{-1}$. Telescoping the gauges into the running products $P_m = Z_{m-1} \dots Z_0$ expresses every tensor through the first, $A_m^i = P_m (A_0^i Z_0) P_{m+1}^{-1}$, so the closed trace collapses to $\mathrm{tr}(B^{\sigma_0} \dots B^{\sigma_{n-1}} P_n^{-1})$ with the tentative repeated tensor $B = A_0 Z_0$. Reading the gauge relation across the seam $A_{n+1} \equiv A_0$ shows that the loop product P_n commutes with the full matrix algebra spanned by B , hence is a scalar $\lambda \neq 0$. Since $\mathbb{C}$ is algebraically closed, there is an n -th root $c^n = \lambda^{-1}$; the rescaled tensor cB is still injective and its constant chain generates the same closed state. $\square$

Corollary 24.9.4.19 (Translation-invariant description of an injective closed MPS chain, positive-bond form on at least three sites). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Discussion: #3371](#) Let $A_0, \dots, A_{n-1}$ be a site-dependent injective closed MPS on $n \geq 3$ sites, with positive dimensions D_e on its individual bonds. If the closed-chain state is translationally invariant, in the finite closed-chain sense that it is invariant under the cyclic shift $A_v \mapsto A_{v+1}$ of [186, lines 1807–1827], then there are a positive integer D and an injective tensor B of $D \times D$ matrices such that $D_e = D$ for every bond e and

$$\mathrm{Coeff}_A(\sigma) = \mathrm{tr}(B^{\sigma_0} \dots B^{\sigma_{n-1}}).$$

for every physical configuration σ .

This completes the positive-bond, at-least-three-site form of the translational-invariance corollary and its proof in [186, lines 1803–1890]. The equality $D_e = D$ is Corollary 24.9.4.13. Identifying each virtual space with $\mathbb{C}^D$ gives a uniform closed-chain representation, to which the source’s L_i, R_i telescoping and repeated-tensor construction apply. The source section places its injective Fundamental Theorem in the regime of at least three sites; the omitted two-site case is false by Theorem 24.9.4.14, as recorded in [149].

Proof. ✓ Proof By Corollary 24.9.4.13, every bond has the dimension D of a fixed reference bond. Write e_v for the bond from v to $v + 1$, transport every bond index to $\{0, \dots, D - 1\}$, and let

$$\iota_v(\xi) = (\xi(e_{v-1}), \xi(e_v)), \quad C_v^i(a, b) = A_v'^i(\iota_v^{-1}(a, b)),$$

where A_v' is the transported local tensor. If $g(v) = \eta(e_v)$, the two corresponding changes of finite summation variables give

$$\begin{aligned} \text{Coeff}_A(\sigma) &= \sum_{\eta} \prod_{v=0}^{n-1} A_v'^{\sigma_v}(\eta|_v) \\ &= \sum_{g: \{0, \dots, n-1\} \rightarrow \{0, \dots, D-1\}} \prod_{v=0}^{n-1} C_v^{\sigma_v}(g(v-1), g(v)) = \text{Coeff}_C(\sigma). \end{aligned}$$

Each ι_v is a bijection, so the same coordinate change preserves injectivity of every local tensor. If $r(v) = v + 1$ is the cyclic rotation, then $\text{Coeff}_{\text{shift}(C)}(\sigma) = \text{Coeff}_C(\sigma \circ r^{-1})$; hence cyclic-shift invariance is preserved. Apply Corollary 24.9.4.18 to this uniform chain. Its L_i, R_i telescoping and seam closure produce the required repeated injective tensor B . $\square$

Corollary 24.9.4.20 (Bond uniformity from translation invariance of an injective PEPS state). Lean Lean Lean Lean ✓ Stmt *Let A be a vertex-injective PEPS on a discrete $n \times m$ torus with $n, m > 1$ and positive bond dimensions. If its represented state is translation invariant, then its bond dimensions are orientation-uniform. Explicitly, with*

$$D_h = D_A(e_h), \quad D_v = D_A(e_v),$$

at the reference right and up edges, every horizontal edge has dimension D_h and every vertical edge has dimension D_v .

This is the first conclusion of the two-dimensional Applications-section corollary in [186, lines 1891–1894].

Proof. ✓ Proof State translation invariance identifies A with $A^{\tau_{a,b}}$ at every translation. Transport preserves vertex injectivity and positivity of the virtual spaces, so Theorem 24.4.1.22 gives

$$D_A(\tau_{a,b}(e)) = D_A(e).$$

Translations act transitively on the horizontal edges and separately on the vertical edges. Applying the displayed equality to the reference right and up edges gives the two claimed constants. $\square$

Corollary 24.9.4.21 (Translation-invariant description of an injective 2D PEPS). Not ready *Let A be a vertex-injective PEPS on a discrete $n \times m$ torus with $n, m > 1$ and positive bond dimensions. If the generated state is invariant under all lattice translations $\tau_{a,b}$, that is, $\text{Coeff}_A(\sigma \circ \tau_{a,b}) = \text{Coeff}_A(\sigma)$ for every translation and every physical configuration σ , then there are positive dimensions D_h, D_v such that the bond dimensions of A are orientation-uniform with these values, and there is a vertex-injective translation-invariant torus tensor B with the same bond-dimension function and the same represented state as A .*

This is the full two-dimensional Applications-section corollary of [186, lines 1891–1894].

Proof. The bond-dimension assertion is Corollary 24.9.4.20. The construction of the site-independent injective tensor B is the remaining part of the corollary. $\square$

Theorem 24.9.4.22 (Uniqueness of the injective closed-chain gauge). [Lean](#) [✓ Stmt](#) Under the positive-bond-dimension hypotheses of Corollary 24.9.4.10, suppose Z_v and Z'_v are two families of invertible matrices satisfying $B_v^i = Z_v A_v^i Z_{v+1}^{-1}$ and $B_v^i = Z'_v A_v^i (Z'_{v+1})^{-1}$ for every v, i . Then there is a single nonzero scalar c such that $Z'_v = c Z_v$ for every bond v . This is the uniqueness clause in [186, line 724].

Proof. [✓ Proof](#) For each bond put $C_v = Z_v^{-1} Z'_v$. Comparing the two gauge relations gives $C_v A_v^i = A_v^i C_{v+1}$ for every local matrix. Since the matrices of A_v span the full matrix algebra, evaluating this identity on the identity matrix gives $C_v = C_{v+1}$; hence all C_v are one common matrix. The same relation then says that this common matrix commutes with the full matrix algebra, so it is a scalar matrix. Its invertibility makes the scalar nonzero. $\square$

Two remarks place these statements against [186]. The injective matrix product states of the opening section of [186] are the case $L = 1$, in which a single tensor already spans the matrix algebra; the corollaries above contain them at $n \geq 3$, with no separate argument needed. The site-dependent results are stated for one physical dimension d and one bond dimension D around the chain, while [186] places no restriction on the local dimensions; this uniform-dimension restriction is the only narrowing.

24.10 The vertex-injective case and the residual edge scalars

Theorem 24.10.1 (Fundamental Theorem for normal PEPS, vertex-injective case). [Lean](#) [✓ Stmt](#) Suppose PEPS tensors A and B are vertex injective with positive bond dimensions on a connected graph, generate the same state, and A satisfies the normal blocking hypotheses. Then their defining tensors are related by a local gauge. This is the vertex-injective specialization of the general normal-PEPS theorem: when each single tensor is already injective, the gauge follows from the injective Fundamental Theorem, without requiring the region blocking hypotheses.

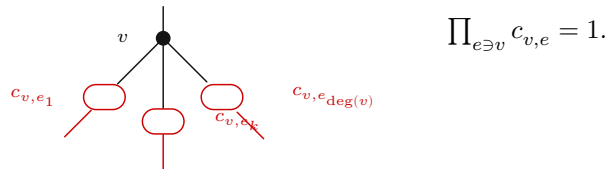
Proof. [✓ Proof](#) Vertex injectivity, positive bond dimensions, same state, and connectivity are exactly the hypotheses of the injective Fundamental Theorem, which supplies the local gauge. $\square$

Definition 24.10.2 (Oriented endpoint scalar). [Lean](#) [✓ Stmt](#) An edge-scalar family $c = (c_e)_{e \in E}$ determines the endpoint scalar $c_{v,e}$ by $c_{v,e} = c_e$ at the lower endpoint of e and $c_{v,e} = c_e^{-1}$ at the upper endpoint.

Definition 24.10.3 (Vertex-balanced edge scalars). [Lean](#) [✓ Stmt](#) An edge-scalar family is vertex-balanced when $\prod_{e \ni v} c_{v,e} = 1$ for every vertex v .

24.11 Balanced Edge Scalars

The edge gauges are determined only up to a scalar on each edge. The residual freedom is the balanced edge scalars: a nonzero scalar c_e on every edge, whose endpoint readings $c_{v,e}$ multiply to one around each vertex, $\prod_{e \ni v} c_{v,e} = 1$.



Two gauge families related by such an assignment give the same gauge equivalence, which is the precise sense in which the gauges of the Fundamental Theorem are unique: modulo balanced edge scalars, not modulo a single global constant.

Definition 24.11.1 (Gauge equivalence modulo balanced edge scalars). Lean ✓ Stmt Two gauge families are equivalent modulo balanced edge scalars if there is a nonzero scalar c_e on each edge such that the corresponding endpoint actions satisfy $X_{v,e} = c_{v,e} Y_{v,e}$, and $\prod_{e \ni v} c_{v,e} = 1$ for every vertex v .

Theorem 24.11.2 (Multiplication of oriented edge scalars). Lean ✓ Stmt *If two edge-scalar families are multiplied, then their endpoint scalars satisfy $(cd)_{v,e} = c_{v,e} d_{v,e}$ on every oriented half-edge.*

Proof. ✓ Proof At the lower endpoint this is ordinary multiplication of the edge scalars. At the upper endpoint, $(c_e d_e)^{-1} = c_e^{-1} d_e^{-1}$ because the edge scalars lie in $\mathbb{C}^\times$. □

Theorem 24.11.3 (Inversion of oriented edge scalars). Lean ✓ Stmt *Inverting an edge-scalar family gives $(c^{-1})_{v,e} = c_{v,e}^{-1}$ on every oriented endpoint.*

Proof. ✓ Proof At the lower endpoint, $(c^{-1})_{v,e} = c_e^{-1} = c_{v,e}^{-1}$. At the upper endpoint, $(c^{-1})_{v,e} = (c_e^{-1})^{-1} = c_{v,e}$. □

Theorem 24.11.4 (Nonzero oriented edge scalars). Lean ✓ Stmt *Every oriented endpoint scalar satisfies $c_{v,e} \neq 0$.*

Proof. ✓ Proof Edge scalars are units, and the inverse of a unit is again a unit. □

Theorem 24.11.5 (Product of balanced edge scalars). Lean ✓ Stmt *The pointwise product of two vertex-balanced edge-scalar families is again vertex-balanced.*

Proof. ✓ Proof For every vertex v , $\prod_{e \ni v} (cd)_{v,e} = (\prod_{e \ni v} c_{v,e})(\prod_{e \ni v} d_{v,e}) = 1 \cdot 1 = 1$. □

Theorem 24.11.6 (Inverse of balanced edge scalars). Lean ✓ Stmt *The pointwise inverse of a vertex-balanced edge-scalar family is again vertex-balanced.*

Proof. ✓ Proof For every vertex v , $\prod_{e \ni v} (c^{-1})_{v,e} = (\prod_{e \ni v} c_{v,e})^{-1} = 1^{-1} = 1$. □

Theorem 24.11.7 (Reflexivity of balanced edge-scalar equivalence). Lean ✓ Stmt *Every edge-gauge family is equivalent to itself modulo balanced edge scalars, witnessed by $c_e = 1$ for every edge e .*

Proof. ✓ Proof If $c_e = 1$, then $c_{v,e} = 1$, $X_{v,e} = c_{v,e} X_{v,e}$, and $\prod_{e \ni v} c_{v,e} = 1$. □

Theorem 24.11.8 (Symmetry of balanced edge-scalar equivalence). Lean ✓ Stmt *If one gauge family differs from another by balanced edge scalars, then the second differs from the first by the inverse balanced edge-scalar family.*

Proof. ✓ Proof If $X_{v,e} = c_{v,e} Y_{v,e}$ with $c_{v,e} \neq 0$, then $Y_{v,e} = c_{v,e}^{-1} X_{v,e}$. The inverse scalar family is balanced by Theorem 24.11.6. □

Theorem 24.11.9 (Transitivity of balanced edge-scalar equivalence). Lean ✓ Stmt *If c and d witness the equivalences of X with Y and of Y with Z modulo balanced edge scalars, then their pointwise product scalar family $(cd)_e = c_e d_e$ witnesses that X is equivalent to Z modulo balanced edge scalars.*

Proof. ✓ Proof From $X_{v,e} = c_{v,e}Y_{v,e}$ and $Y_{v,e} = d_{v,e}Z_{v,e}$ one obtains $X_{v,e} = (c_{v,e}d_{v,e})Z_{v,e}$. The product scalar family is balanced by Theorem 24.11.5. □

Theorem 24.11.10 (The balanced edge-scalar quotient is an equivalence relation). Lean ✓ Stmt *Gauge equivalence modulo balanced edge scalars is reflexive, symmetric, and transitive.*

Proof. ✓ Proof Reflexivity is witnessed by 1, symmetry by c^{-1} , and transitivity by cd . □

Definition 24.11.11 (The balanced edge-scalar quotient relation). Lean ✓ Stmt The preceding equivalence relation is the quotient relation on edge-gauge families modulo vertex-balanced edge scalars.

Theorem 24.11.12 (Balanced edge scalars preserve the local gauge action). Lean ✓ Stmt *If two gauge families X and Y differ by a balanced edge-scalar reweighting, then for every vertex v , virtual configuration η , and physical index σ ,*

$$(A^X)_v(\eta, \sigma) = (A^Y)_v(\eta, \sigma).$$

Proof. ✓ Proof In each summand of the gauged tensor at v , the edge scalars contribute the factor $\prod_{e \ni v} c_{v,e} = 1$. Hence the summand, and therefore the whole local tensor, is unchanged. □

Theorem 24.11.13 (Balanced edge scalars preserve the gauged tensor). Lean ✓ Stmt *If two gauge families X and Y differ by a balanced edge-scalar reweighting, then applying them to the same PEPS tensor A produces the same gauged tensor:*

$$A^X = A^Y.$$

Proof. ✓ Proof For every vertex v , virtual index η , and physical index σ , $(A^X)_v(\eta, \sigma) = (A^Y)_v(\eta, \sigma)$ for the two gauged tensors by Theorem 24.11.12. Hence the two gauged tensors are equal. □

Theorem 24.11.14 (Balanced edge scalars preserve the gauged state). Lean ✓ Stmt *If two gauge families X and Y differ by a balanced edge-scalar reweighting, then the PEPS tensors obtained by applying those gauge families represent the same state: for every physical configuration σ ,*

$$\text{Coeff}_{A^X}(\sigma) = \text{Coeff}_{A^Y}(\sigma).$$

Proof. ✓ Proof The two gauged tensors are equal, hence all of their state coefficients are equal. □

Theorem 24.11.15 (Gauge uniqueness modulo balanced edge scalars). Lean ✓ Stmt *Let every bond space be nonzero. If two gauge families relate the same vertex-injective PEPS tensor to the same target tensor, then their edge gauges differ by nonzero scalars c_e whose oriented product at every vertex is 1. Thus the correct graph quotient is by balanced edge scalars.*

Proof. ✓ Proof Combining the two gauge relations gives, at every vertex, equality of the gauged local tensors. Linear independence of the local tensor family promotes this to equality of the products of incident edge-gauge entries over every pair of virtual configurations. Each oriented endpoint gauge is invertible and, since every bond space is nonzero, indexed by a nonempty type; the many-leg scalar extraction then yields, at each vertex, nonzero proportionality constants between the two families whose product is 1. Reading the lower-endpoint constant of each edge defines a single edge scalar c_e ; the inverse relation transports it to the upper endpoint, and invertibility makes the proportionality constant unique, so the vertex products of the per-vertex constants identify with the oriented products of c_e . These are all 1, so c is vertex-balanced. □

Remark 24.11.16. The local left inverse Ψ_v , the realization map $T \mapsto O_T$, the split at one incident edge, and the partition $V = \{u\} \sqcup (V \setminus \{u, v\}) \sqcup \{v\}$ provide the local structure for the edge-centered reduction of [186, Section 3]. The local gauge then follows from comparison with the corresponding three-site injective chain, with uniqueness determined modulo balanced edge scalars.

24.11.1 Inserted-site scalar extraction

The final comparison of [186, Section 3, proof of Theorem 3] turns two region proportionalities into a single scalar at one site. Fix a finite region R and a site $v \notin R$, and write $S = R \cup \{v\}$ for the region with v adjoined. A local configuration η at v assigns a virtual index to every edge incident to v . The label on its v -incident R -boundary edges reads η on the v -incident edges that bound R , while the base boundary label μ fixes the remaining v -incident edges. Under inserted-site consistency, the R -boundary label and the values fixed by μ on the remaining v -incident edges determine η completely.

Lemma 24.11.1.1 (Inserted-site consistency pins a local configuration to its R -boundary label).

[Lean](#) [✓ Stmt](#) Fix a region R and a site $v \notin R$, and let μ be a boundary label of $S = R \cup \{v\}$. Two local configurations at v that are both consistent with μ and carry the same R -boundary label coincide.

Proof. [✓ Proof](#) Read both configurations on each v -incident edge. On a v -incident edge that bounds R the value is the shared R -boundary label. On a v -incident edge that does not bound R , the edge runs from v to a site outside S , and inserted-site consistency reads the value from μ . Both readings agree edge by edge, so the two configurations are equal. $\square$

Theorem 24.11.1.2 (Inserted-site scalar extraction from two region proportionalities). [Lean](#)

[✓ Stmt](#) Let A and a comparison tensor C share their bond dimensions and have positive bond dimensions, and suppose the R -blocked family of C is linearly independent. If the blocked weights satisfy the two region proportionalities $A_R = c_R C_R$ and $A_S = c_S C_S$ over R and over $S = R \cup \{v\}$, with $c_R \neq 0$, then the inserted-site tensors are proportional with ratio c_S/c_R at every local configuration η at v and every physical index σ :

$$A_v(\eta, \sigma) = \frac{c_S}{c_R} C_v(\eta, \sigma). \quad (24.31)$$

Proof. [✓ Proof](#) Write $b(\mu, \eta)$ for the label that a local configuration η at v , consistent with the base boundary label μ , induces on the v -incident edges that bound R . With m the bond-only inserted-site multiplicity, the inserted-site factorization reads each blocked weight of S as the inserted-site tensor summed against the R -boundary-label blocked weight of R over the local configurations consistent with μ :

$$m A_S(\mu, \sigma) = \sum_{\eta \text{ consistent with } \mu} A_v(\eta, \sigma(v)) A_R(b(\mu, \eta), \sigma|_R), \quad (24.32)$$

and likewise for C . Feeding both proportionalities through (24.32), cancelling the nonzero m , and substituting the R -proportionality leaves, for every base label μ , a vanishing combination of the R -blocked family of C with coefficient $c_R A_v(\eta) - c_S C_v(\eta)$ summed over the consistent local configurations. By Lemma 24.11.1.1 the consistent configurations inject into their R -boundary labels, so linear independence of the R -blocked family separates the combination to one term, forcing $c_R A_v(\eta, \sigma) = c_S C_v(\eta, \sigma)$ and hence (24.31). $\square$

Corollary 24.11.1.3 (Per-vertex gauge relation from two two-block proportionalities). [Lean](#)

[✓ Stmt](#) Suppose A and B have the same bond dimension on every edge, and that these bond dimensions are positive. Let C be the gauge action of X on B , reindexed along this bond-dimension equality, and assume that the R -blocked family of C is linearly independent. If the two two-block scalar proportionalities of the blocked weights of A and C over R and over $S = R \cup \{v\}$ hold, with ratios c_R and c_S and with $c_R \neq 0$, then the per-vertex gauge relation

$$A_v(\eta, \sigma) = \frac{c_S}{c_R} C_v(\eta, \sigma) \quad (24.33)$$

holds at every local configuration η at v and every physical index σ .

Proof. [✓ Proof](#) The two-block proportionalities are exactly the region proportionalities of Theorem 24.11.1.2 for the gauge-absorbed comparison tensor. The reindexed inserted-site tensor of that comparison tensor is the gauge action of X on B at v , so the extraction gives (24.33). $\square$

24.12 Examples of PEPS on the torus

Two-dimensional PEPS on the discrete torus are defined by a four-leg tensor $A_{\alpha\beta\gamma\delta}^i$ placed at every site of a square lattice, with virtual indices ordered top, right, down, left [16, Appendix A, “Two dimensions: PEPS”]. On a torus, the four legs of site $v = (x, y)$ correspond to the edges from v to $(x, y + 1)$ (top), from v to $(x + 1, y)$ (right), from $(x, y - 1)$ to v (down), and from $(x - 1, y)$ to v (left).

Definition 24.12.1 (Torus PEPS of a four-leg tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let $a_{\alpha\beta\gamma\delta}^i$ be a tensor with physical dimension d and bond dimension D . Its torus PEPS on the torus of width w and height h has bond dimension D on every edge and, at every site v , the local tensor a with its four virtual indices read on the top, right, down and left legs of v [16, Appendix A, “Two dimensions: PEPS”]. The right leg of v is the left leg of $v + (1, 0)$, and the top leg of v is the down leg of $v + (0, 1)$.

Lemma 24.12.2 (Shift-invariant functions on the torus are constant). [Lean](#) [✓ Stmt](#) If a function f on the sites of the torus satisfies $f(x + 1, y) = f(x, y)$ and $f(x, y + 1) = f(x, y)$ for all (x, y) , then f is constant.

Proof. [✓ Proof](#) Induction on the representatives $0, \dots, w - 1$ of the first coordinate and $0, \dots, h - 1$ of the second. $\square$

Lemma 24.12.3 (Contraction of a torus PEPS). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let $w, h \geq 3$. The right edges and the up edges of the sites are pairwise distinct and exhaust the edges of the torus, and the coefficient of the torus PEPS of a is

$$\text{Coeff}(\sigma) = \sum_{h_\bullet, v_\bullet} \prod_{(x, y)} a_{v_{(x, y)} h_{(x, y)} v_{(x, y-1)} h_{(x-1, y)}}^{\sigma(x, y)},$$

the sum running over all assignments of bond indices $h_{(x, y)}$ to the horizontal edges from (x, y) to $(x + 1, y)$ and $v_{(x, y)}$ to the vertical edges from (x, y) to $(x, y + 1)$.

Proof. [✓ Proof](#) Every edge is horizontal or vertical, and every horizontal edge is the right edge of one of its endpoints and every vertical edge the up edge of one of its endpoints. For $w \geq 3$ the right edges of distinct sites are distinct: otherwise $p = q + (1, 0)$ and $q = p + (1, 0)$, so $2 = 0$ in the horizontal coordinate. The same holds vertically for $h \geq 3$. A horizontal edge is never vertical. Reindexing the sum over edge labellings along this bijection gives the formula. $\square$

On a torus of width two the right and left neighbours of a site coincide and the two edges joining them are one edge of the lattice graph, so the four legs of a site are not four distinct bonds. The contraction formula, and the cluster, CZX, AKLT, RVB and classical-model statements below, therefore require $w, h \geq 3$; the source constructions are local and apply to every torus [148].

A virtual string on a torus PEPS acts on the bonds it crosses, diagonally in the bond basis: a bond carrying the index α contributes an element X_α of a monoid M , and the contributions are multiplied in the order in which the string crosses the bonds. For a matrix algebra M this is a matrix product operator on the virtual level.

Lemma 24.12.4 (Pulling a virtual string through a rectangle). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let a be a four-leg tensor and $\alpha \mapsto X_\alpha$ a map from the bond indices to a monoid M with $X_t X_r = X_l X_b$ whenever $a_{trbl}^s \neq 0$. Fix a rectangle of m columns and n rows of sites of the torus. Consider the string from its upper left to its lower right corner that crosses the left legs of the leftmost column, from top to bottom, and then the down legs of the bottom row, from left to right, and the string that crosses the top legs of the top row and then the right legs of the rightmost column. On every labelling of the bonds whose term in the contraction of the torus PEPS of a is nonzero the two strings have the same product, and so, for a semiring M with a $\mathbb{C}$ -module structure, the contracted network with the first string inserted equals the contracted network with the second string inserted. This is the pulling-through equation of a matrix product operator symmetry [16, Section “MPO and PEPO”] for strings that are diagonal in the bond basis.

Proof. [✓ Proof](#) Write P_{ij} for the contribution of the left leg of the site in column i and row j of the rectangle, counted from its upper left site, and Q_{ij} for that of its top leg. The right leg of that site is the left leg of the site in column $i + 1$, and its down leg is the top leg of the site in row $j + 1$, so the hypothesis at a site with nonzero entry reads $Q_{ij} P_{i+1,j} = P_{ij} Q_{i,j+1}$. Induction on n gives $P_{i0} \cdots P_{i,n-1} Q_{in} = Q_{i0} P_{i+1,0} \cdots P_{i+1,n-1}$ for one column, and induction on m , moving one column at a time, gives the identity for the rectangle. A nonzero term of the contraction has a nonzero entry at every site, and the terms that vanish contribute zero to both sides. $\square$

24.12.1 The GHZ state

The two-dimensional GHZ state is a superposition of the d constant spin configurations [16, Appendix A, “The GHZ state”]. All bonds are contracted periodically. Separate horizontal and vertical assignments of bond indices retain the self-loops and parallel bonds at small lattice sizes. The simple-graph representation requires $w, h \geq 2$ [148].

Definition 24.12.1.1 (Two-dimensional GHZ tensor). [Lean](#) [Lean](#) [✓ Stmt](#) The GHZ tensor has $D = d$ and $A_{\alpha\beta\gamma\delta}^i = \delta_{i=\alpha=\beta=\gamma=\delta}$ [16, Appendix A, “The GHZ state”].

Theorem 24.12.1.2 (The GHZ PEPS is the GHZ state). [Lean](#) [Lean](#) [✓ Stmt](#) On every torus with $w, h \geq 1$, the GHZ tensor generates the unnormalized state $\sum_{i=0}^{d-1} |i, i, \dots, i\rangle$ [16, Appendix A, “The GHZ state”]. With horizontal and vertical assignments of bond indices $h_{(x,y)}, v_{(x,y)}$ as in Lemma 24.12.3, its coefficient is

$$\sum_{h,v} \prod_{(x,y)} A_{v_{(x,y)}, h_{(x,y)}, v_{(x,y-1)}, h_{(x-1,y)}}^{\sigma(x,y)} = \sum_i \prod_{(x,y)} \delta_{\sigma(x,y), i}. \quad (24.34)$$

For $w, h \geq 2$, the simple-graph GHZ PEPS has the same coefficients.

Proof. ✓ Proof A virtual configuration contributes only if every leg of every site carries the physical index of that site. The right leg of v is the left leg of $v + (1, 0)$ and the top leg of v is the down leg of $v + (0, 1)$, so σ is invariant under both unit shifts and is constant by Lemma 24.12.2. Every edge is then labelled by that constant, so exactly one virtual configuration contributes when σ is constant and none otherwise. $\square$

Theorem 24.12.1.3 (Non-injectivity of the GHZ PEPS). Lean Lean ✓ Stmt For $d \geq 2$, the four-leg GHZ tensor is not injective. For $d \geq 2$ and $w, h \geq 2$, the simple-graph GHZ PEPS is not vertex-injective.

Proof. ✓ Proof The four-leg basis vector with labels $(0, 1, 0, 0)$ has zero image. In the simple graph, give the right bond at one vertex label 1 and every other incident bond label 0. The top and right bonds are distinct, so no physical label can agree with both. This virtual basis vector is mapped to zero. $\square$

Theorem 24.12.1.4 (On-site permutations and their virtual action). Lean Lean Lean Lean ✓ Stmt Every permutation π of the d labels acts on the GHZ site tensor by

$$A_{\pi(\alpha)\pi(\beta)\pi(\gamma)\pi(\delta)}^{\pi(i)} = A_{\alpha\beta\gamma\delta}^i. \quad (24.35)$$

For every positive w, h , the contraction in (24.34) is unchanged when σ is replaced by $\pi \circ \sigma$. For $w, h \geq 2$, the simple-graph GHZ PEPS therefore satisfies $\text{Coeff}(\pi \circ \sigma) = \text{Coeff}(\sigma)$. In particular, for $d \geq 1$ the on-site $\mathbb{Z}_d$ shift is a symmetry, and its virtual action is the same shift on every bond.

Proof. ✓ Proof A permutation preserves equality of the five labels in the site tensor. Reindexing the sum over constant configurations in Theorem 24.12.1.2 proves the state identity. Take $\pi(i) = i + g$ for the cyclic shift. $\square$

24.12.2 The cluster state

Definition 24.12.2.1 (Two-dimensional cluster tensor). Lean Lean ✓ Stmt The cluster tensor is $A = |0\rangle(00++| + |1\rangle(11--|$ with $(\pm) = [(0) \pm (1)]/\sqrt{2}$: the top and right legs copy the physical index s , and the down and left legs carry $(-1)^{sb}/\sqrt{2}$ at bond index b [16, Appendix A, “The cluster state”].

Theorem 24.12.2.2 (The cluster PEPS). Lean ✓ Stmt On a torus with $w, h \geq 3$ and $N = wh$ sites the cluster PEPS has coefficient

$$\text{Coeff}(\sigma) = 2^{-N} \prod_v (-1)^{\sigma_v \sigma_{v+(1,0)} + \sigma_v \sigma_{v+(0,1)}}.$$

Proof. ✓ Proof By Lemma 24.12.3 the top and right legs force every bond index to equal the physical index of the site below or to the left of it. The remaining factor at v is $\frac{1}{2}(-1)^{\sigma_v \sigma_{v-(0,1)} + \sigma_v \sigma_{v-(1,0)}}$; shifting the product over v by one site in each direction gives the formula. $\square$

Definition 24.12.2.3 (Graph-state circuit and stabilizers). Lean Lean Lean Lean Lean Lean ✓ Stmt Let a finite set of bonds have distinct endpoints $a(e), b(e)$ in a finite vertex set V . Parallel bonds are counted with multiplicity. The product of controlled- Z gates has diagonal coefficient $g(\sigma) = \prod_e (-1)^{\sigma_{a(e)} \sigma_{b(e)}}$. The stabilizer K_v flips the qubit at v and applies Z at the opposite endpoint of every incident bond. Thus

$$(K_v \psi)(\sigma) = \prod_{e: a(e)=v} (-1)^{\sigma_{b(e)}} \prod_{e: b(e)=v} (-1)^{\sigma_{a(e)}} \psi(\sigma \oplus \mathbf{1}_v). \quad (24.36)$$

Theorem 24.12.3.2 (The CZX PEPS is a product of plaquette GHZ states). [Lean](#) [Lean](#) [✓ Stmt](#)
On a torus with $w, h \geq 3$ the CZX PEPS has coefficient

$$\text{Coeff}(\sigma) = \prod_v \delta[j_v = i_{v+(1,0)} = k_{v+(0,1)} = l_{v+(1,1)}],$$

where (i_v, j_v, k_v, l_v) are the four qubits of $\sigma(v)$ and the factor at v is the coefficient of $|0000\rangle + |1111\rangle$ on the four qubits around the plaquette at the upper right corner of v [16, Appendix A, “The CZX model”], [10].

Proof. [✓ Proof](#) By Lemma 24.12.3 the top and right legs force the vertical bond above v to carry (i_v, j_v) and the horizontal bond to its right to carry (j_v, k_v) . The down and left legs then require $(i_{v-(0,1)}, j_{v-(0,1)}) = (l_v, k_v)$ and $(j_{v-(1,0)}, k_{v-(1,0)}) = (i_v, l_v)$ at every site, and these conditions, shifted by one site, are equivalent to the plaquette conditions. $\square$

The on-site $\mathbb{Z}_2$ symmetry of the CZX model is generated by $U_{CZX} = U_X U_{CZ}$, where $U_X = X_1 X_2 X_3 X_4$ flips the four qubits of a site and $U_{CZ} = CZ_{12} CZ_{23} CZ_{34} CZ_{41}$ applies controlled-Z gates to its four pairs of neighbouring qubits [10]. On the state $|ijkl\rangle$ it acts by $U_{CZX}|ijkl\rangle = (-1)^{ij+jk+kl+li} |\bar{i}\bar{j}\bar{k}\bar{l}\rangle$.

Definition 24.12.3.3 (CZX on-site symmetry). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The operator $U_{CZX} = U_X U_{CZ}$ on the sixteen states of a site, and its product $U_{CZX}^{\otimes V}$ over the sites V of the lattice, whose entry at (σ', σ) is $\prod_v (U_{CZX})_{\sigma'_v \sigma_v}$ [10].

Theorem 24.12.3.4 (The CZX PEPS is invariant under the on-site symmetry). [Lean](#) [Lean](#)
[✓ Stmt](#) On a torus with $w, h \geq 3$ the CZX PEPS is invariant under $U_{CZX}^{\otimes V}$ [16, Appendix A, “The CZX model”], [10].

Proof. [✓ Proof](#) Flipping every qubit preserves the plaquette constraints of Theorem 24.12.3.2. On a configuration satisfying them, let P_v be the value of the plaquette at the upper right corner of v . The sign of U_{CZX} at v is $(-1)^{e_v}$ with $e_v = P_{v-(1,0)} P_v + P_v P_{v-(0,1)} + P_{v-(0,1)} P_{v-(1,1)} + P_{v-(1,1)} P_{v-(1,0)}$. Shifting the summation variable, the sums over v of the third and fourth terms equal those of the first and second, so $\sum_v e_v$ is even: each controlled-Z gate between two neighbouring plaquettes appears twice. $\square$

Definition 24.12.3.5 (CZX leg operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A virtual leg carries two qubits (a, b) . The operator $V = (X \otimes X) CZ$ acts on a leg by $V|ab\rangle = (-1)^{ab} |\bar{a}\bar{b}\rangle$.

Theorem 24.12.3.6 (Pulling the symmetry through a site). [Lean](#) [✓ Stmt](#) Applying U_{CZX} to the physical index of the CZX tensor equals applying V to each of its four virtual legs [16, Section III.B, “Non-injective PEPS: SPT phases”]: writing A_{trbl}^s for the entry of the CZX tensor with physical index s and top, right, bottom and left legs t, r, b, l ,

$$\sum_s (U_{CZX})_{s's} A_{trbl}^s = \sum_{t', r', b', l'} A_{t'r'b'l'}^{s'} V_{t't} V_{r'r} V_{b'b} V_{l'l}. \quad (24.40)$$

Proof. [✓ Proof](#) Both sides are supported on the flipped legs of the flipped qubits, and the sign $(-1)^{ij+jk+kl+li}$ of U_{CZX} is the product of the signs of V on the four legs (i, j) , (j, k) , (k, l) , (l, i) . $\square$

On the boundary of a region, neighbouring boundary legs carry the same qubit of their common plaquette, one effective spin per boundary plaquette [10].

Definition 24.12.3.7 (Boundary legs of the CZX tensor network). [Lean](#) [Lean](#) [✓ Stmt](#) For $N > 0$ and effective spins $c \in \{0, 1\}^N$, the closed chain of N boundary legs carrying c has the pair (c_m, c_{m+1}) on leg m , indices modulo N . The map ι from N effective spins into the legs is

$$\iota|c\rangle = |(c_0, c_1), (c_1, c_2), \dots, (c_{N-1}, c_0)\rangle. \quad (24.41)$$

Theorem 24.12.3.8 (The boundary action is the CZX matrix product unitary). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $N > 0$, the chain of boundary legs determines c , so ι is an isometry, $\iota^\dagger \iota = \mathbb{1}$. It satisfies $V^{\otimes N} \iota = \iota U_N$ and $\iota^\dagger V^{\otimes N} \iota = U_N$, where $U_N|c\rangle = (-1)^{\sum_m c_m c_{m+1}} |\bar{c}\rangle$ is the CZX matrix product unitary of Theorem 27.3.1.2; in terms of the review’s operator $O(A)$ with the printed matrices, $V^{\otimes N} \iota = (-1)^N \iota O(A)$ [16, Appendix A, “The CZX MPU”, [10]. The identification of the contraction of a region of the torus with a vector in the range of ι is not formalized [43].

Proof. [✓ Proof](#) $V^{\otimes N}$ maps the legs of c to the legs of $\bar{c}$ with the sign $\prod_m (-1)^{c_m c_{m+1}}$, which is the kernel of Theorem 27.3.1.2; the relation to $O(A)$ is Theorem 27.3.2.1. $\square$

The review states that the CZX model lies in the non-trivial symmetry-protected sector of its on-site $\mathbb{Z}_2$ symmetry [16, Appendix A, “The CZX model”]. This classification is not formalized for the two-dimensional state. Its boundary symmetry is a matrix product unitary; the three-cocycle attached to a group of matrix product unitaries is constructed in Theorems 19.5.1.6 and 19.5.1.7, and for the CZX boundary representation its class is computed and shown to be non-trivial in Theorem 27.3.3.8.

24.12.4 The toric code and quantum double models

Let G be a finite group. The quantum double model of G [176] carries a spin with basis $\{|g\rangle\}_{g \in G}$ on every edge of a square lattice; for $G = \mathbb{Z}_2$ it is the toric code. The review gives two PEPS representations [16, Appendix A, “The Toric Code and quantum double models”]. Their symmetries are phrased through the map $\mathcal{P}(A) = \sum A_{\alpha\beta\gamma\delta}^i |i\rangle \langle \alpha\beta\gamma\delta|$ from the virtual to the physical system [203, Section “MPS and PEPS”].

Definition 24.12.4.1 (G -injective and G -isometric tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $g \mapsto U_g$ be a representation of a finite group G on the virtual system of a tensor A , and let Π_U be the projector onto the U_g -invariant subspace. The tensor is G -injective if $\mathcal{P}(A)U_g = \mathcal{P}(A)$ for every g and $\mathcal{P}(A)$ is injective on the invariant subspace [203, Section “Two dimensions: G -injective PEPS”]. It is G -isometric if moreover $\mathcal{P}(A)$ restricted to the invariant subspace is a multiple of an isometry, $\langle \mathcal{P}(A)x, \mathcal{P}(A)y \rangle = c \langle x, y \rangle$ for invariant x, y and one constant $c > 0$ [203, Section “Isometric PEPS”].

Theorem 24.12.4.2 (Left-inverse form of G -injectivity). [Lean](#) [Lean](#) [✓ Stmt](#) A tensor is G -injective if and only if $\mathcal{P}(A)U_g = \mathcal{P}(A)$ for every g and $\mathcal{P}(A)$ has a left inverse L with $L\mathcal{P}(A) = \Pi_U$ [203, Section “Two dimensions: G -injective PEPS”].

Proof. [✓ Proof](#) An invariant map satisfies $\mathcal{P}(A)\Pi_U = \mathcal{P}(A)$, since Π_U is the average of the U_g . If $\mathcal{P}(A)$ is injective on the invariant subspace, choose a linear map L into the invariant subspace with $L\mathcal{P}(A)x = x$ for invariant x ; then $L\mathcal{P}(A) = L\mathcal{P}(A)\Pi_U = \Pi_U$. Conversely, if $L\mathcal{P}(A) = \Pi_U$ and $\mathcal{P}(A)x = 0$ for an invariant x , then $x = \Pi_U x = L\mathcal{P}(A)x = 0$. $\square$

Theorem 24.12.4.3 (G -injectivity for the trivial representation). [Lean](#) [Lean](#) [✓ Stmt](#) For the trivial representation, G -injectivity is injectivity of $\mathcal{P}(A)$, that is, linear independence of the physical vectors $A_{\alpha\beta\gamma\delta}^\bullet$ indexed by the virtual configurations.

Proof. ✓ Proof Every vector is invariant under the trivial representation, and $\mathcal{P}(A)$ is the linear map whose columns are the vectors $A_{\alpha\beta\gamma\delta}^\bullet$. $\square$

The group-theoretic input to G -injectivity is the averaging over the group and an operator Δ that inverts the expansion of an operator in the U_g 's [203, Section “MPS: the G -injective case”]. Throughout, $g \mapsto U_g$ is a representation of a finite group G on a finite-dimensional complex space.

Theorem 24.12.4.4 (Group averaging). Lean Lean Lean Lean Lean ✓ Stmt *The map $\sigma(X) = \frac{1}{|G|} \sum_g U_g X U_g^{-1}$ is a projection onto the operators commuting with every U_g . If the representation is unitary, then $\sigma(X) = \frac{1}{|G|} \sum_g U_g X U_g^\dagger$ and $\text{tr}[Y^\dagger \sigma(X)] = \text{tr}[\sigma(Y)^\dagger X]$, so σ is the orthogonal projector onto these operators for the Hilbert–Schmidt inner product [203, Section “MPS: the G -injective case”].*

Proof. ✓ Proof The map σ is the average of the conjugation action $X \mapsto U_g X U_g^{-1}$, so it is a projection onto the operators fixed by every conjugation, and these are the operators commuting with every U_g . For a unitary representation, $\text{tr}[Y^\dagger U_g X U_g^\dagger] = \text{tr}[(U_g^\dagger Y U_g)^\dagger X]$, and summing over g , with g replaced by g^{-1} , gives the symmetry of σ . $\square$

Definition 24.12.4.5 (Isotypic projectors and the operator Δ). Lean Lean Lean Lean ✓ Stmt Let χ_U be the character of U , and let $\chi_1, \dots, \chi_r$ be the distinct characters of the irreducible subrepresentations of U , with $d_i = \chi_i(1)$ and multiplicities $m_i = \frac{1}{|G|} \sum_g \chi_U(g) \chi_i(g^{-1})$. The isotypic projectors are $P_i = \frac{d_i}{|G|} \sum_g \chi_i(g^{-1}) U_g$, and

$$\Delta = \frac{1}{|G|} \sum_{i=1}^r \frac{d_i}{m_i} P_i$$

[203, Section “MPS: the G -injective case”].

Theorem 24.12.4.6 (Isotypic projectors from characters). Lean Lean Lean Lean Lean ✓ Stmt *For irreducible representations D^i and D^j of G , $\sum_g \chi_i(g^{-1}) D^j(g)$ equals $\frac{|G|}{d_i} \mathbb{1}$ if D^i and D^j are equivalent and 0 otherwise. Consequently P_i acts as the identity on every irreducible subrepresentation of U with character χ_i and as zero on every other irreducible subrepresentation, the number m_i is the dimension of the space of intertwining maps from D^i to U , and*

$$\sum_{i=1}^r P_i = \mathbb{1}, \quad \text{tr}[U_k P_i] = m_i \chi_i(k)$$

[203, Section “MPS: the G -injective case”].

Proof. ✓ Proof The operator $\sum_g \chi_i(g^{-1}) D^j(g)$ commutes with every $D^j(h)$, because characters are constant on conjugacy classes, so by Schur’s lemma it is a multiple of the identity; its trace is $\sum_g \chi_i(g^{-1}) \chi_j(g)$, which orthogonality of characters evaluates. By Maschke’s theorem U is a direct sum of irreducible subrepresentations. On each summand the projectors act as just described, which gives $\sum_i P_i = \mathbb{1}$, and the trace of $U_k P_i$ is the sum of $\chi_i(k)$ over the summands with character χ_i . Orthogonality of characters counts these summands as m_i . $\square$

Theorem 24.12.4.7 (Block form of Δ). Lean ✓ Stmt *On every irreducible subrepresentation of U with character χ_i , Δ acts as the scalar $\frac{d_i}{m_i |G|}$. Thus, in a basis in which $U_g \cong \bigoplus_i D^i(g) \otimes \mathbb{1}_{m_i}$,*

$$\Delta \cong \frac{1}{|G|} \bigoplus_i \frac{d_i}{m_i} \mathbb{1}_{d_i} \otimes \mathbb{1}_{m_i}$$

[203, Section “MPS: the G -injective case”].

Proof. ✓ Proof Only the projector of χ_i acts nontrivially on such a subrepresentation, and it acts as the identity. $\square$

Theorem 24.12.4.8 (Expansion in the representation). Lean Lean Lean ✓ Stmt For every $g \in G$,

$$\sum_h \operatorname{tr}[U_{h^{-1}} \Delta U_g] U_h = \sum_h \operatorname{tr}[U_{gh^{-1}} \Delta] U_h = U_g.$$

For a unitary representation $U_{h^{-1}} = U_h^\dagger$ [203, Section “MPS: the G -injective case”].

Proof. ✓ Proof By the trace formula for the projectors, $\operatorname{tr}[U_k \Delta] = \frac{1}{|G|} \sum_i d_i \chi_i(k)$, so $\sum_k \operatorname{tr}[U_{k^{-1}} \Delta] U_k = \sum_i P_i = \mathbb{1}$. Substituting $h = kg$ gives $\sum_h \operatorname{tr}[U_{gh^{-1}} \Delta] U_h = (\sum_k \operatorname{tr}[U_{k^{-1}} \Delta] U_k) U_g = U_g$, and the first expression equals the second by cyclicity of the trace. $\square$

Definition 24.12.4.9 (Semi-regular representation). Lean Lean ✓ Stmt A representation of G is semi-regular if it contains every irreducible representation of G , that is, if every irreducible representation admits a nonzero intertwining map into it [203, Section “MPS: the G -injective case”]. The left-regular representation is semi-regular.

Theorem 24.12.4.10 (Semi-regularity under representation equivalence). Lean Lean ✓ Stmt Equivalent representations ρ and σ are semi-regular simultaneously. Thus the occurrence condition of [203, Definition 4.5] is independent of coordinates.

Proof. ✓ Proof If e intertwines ρ with σ and is invertible, every nonzero intertwiner f from an irreducible representation into ρ gives a nonzero intertwiner ef into σ . Apply the same argument to e^{-1} for the converse. $\square$

Theorem 24.12.4.11 (Linear independence of a semi-regular representation). Lean Lean Lean ✓ Stmt If U is semi-regular, then $\operatorname{tr}[U_{g^{-1}} U_h \Delta] = \delta_{g,h}$ for all $g, h \in G$, with $U_{g^{-1}} = U_g^\dagger$ for a unitary representation. For the left-regular representation $\Delta = \frac{1}{|G|} \mathbb{1}$ [203, Section “MPS: the G -injective case”].

Proof. ✓ Proof The character of the left-regular representation is $|G| \delta_{k,1}$, so every character χ_i has multiplicity d_i in it, and $\Delta = \frac{1}{|G|} \sum_i P_i = \frac{1}{|G|} \mathbb{1}$. Every irreducible representation occurs in the left-regular representation, and, since U is semi-regular, in U ; hence U and the left-regular representation have the same irreducible characters. Comparing the two expressions for $\operatorname{tr}[U_k \Delta]$ in the left-regular representation gives $\sum_i d_i \chi_i(k) = |G| \delta_{k,1}$, and therefore $\operatorname{tr}[U_{g^{-1}} U_h \Delta] = \frac{1}{|G|} \sum_i d_i \chi_i(g^{-1}h) = \delta_{g,h}$. $\square$

Theorem 24.12.4.12 (Restriction to the physical support). Lean Lean Lean ✓ Stmt Let $T : W \rightarrow H$ be a linear tensor map between Hilbert spaces, with W finite-dimensional. Write $H_T = \operatorname{ran} T$ and let $\tilde{T} : W \rightarrow H_T$ be the same map with its codomain restricted. The inclusion $I : H_T \rightarrow H$ is an isometry and $I\tilde{T} = T$. There is a unique linear map $L : H_T \rightarrow W$ with

$$\tilde{T}L = \operatorname{id}_{H_T}, \quad \operatorname{ran} L = (\ker T)^\perp. \quad (24.42)$$

It satisfies

$$L\tilde{T} = \operatorname{id}_W - P_{\ker T} = P_{(\ker T)^\perp}. \quad (24.43)$$

This is the physical-support reduction of [203, Observation 2.4], stated using the intrinsic initial virtual space. The existence of L requires only a vector space structure on the physical codomain.

Proof. ✓ Proof Choose any right inverse L_0 of $\tilde{T}$ and set $L = (\text{id}_W - P_{\ker T})L_0$. Since T vanishes on its kernel, $\tilde{T}L = \text{id}_{H_T}$. For any $x \in W$, $x - L_0\tilde{T}x \in \ker T$. Orthogonal projection then gives $L\tilde{T}x = x - P_{\ker T}x$, and the range of L is $(\ker T)^\perp$. Any two sections with range in this orthogonal complement differ by a vector in both $\ker T$ and $(\ker T)^\perp$, and hence are equal. The inclusion of a Hilbert subspace preserves its inherited inner product. $\square$

Definition 24.12.4.13 (Finite signed permutation representations). Lean Lean Lean Lean ✓ Stmt For a finite index set I , let G_I be the group of matrices sending each standard basis vector to another standard basis vector with sign $+1$ or -1 . Its defining representation is unitary, and G_I is finite. These groups supply the finite-dimensional matrix representations used in the construction of [203, Theorem 4.1].

Theorem 24.12.4.14 (Matrix units from signed permutations). Lean Lean ✓ Stmt Every matrix unit E_{ij} is of the form $(P - Q)/2$ for two signed permutation matrices $P, Q \in G_I$. Consequently, $\text{span}_{\mathbb{C}} G_I = M_I(\mathbb{C})$.

Proof. ✓ Proof Choose a permutation taking j to i , and let P be its permutation matrix. Negate only its column j to obtain Q . Then $P - Q = 2E_{ij}$. The matrix units span all matrices. The empty index set causes no exception: its matrix space is zero. $\square$

Definition 24.12.4.15 (Independent groups on multiplicity factors). Lean Lean ✓ Stmt For finitely many pairs of dimensions (m_k, d_k) , the finite group $G = \prod_k G_{m_k}$ has the unitary representation $U(g) = \bigoplus_k g_k \otimes I_{d_k}$ on $\bigoplus_k \mathbb{C}^{m_k} \otimes \mathbb{C}^{d_k}$.

Theorem 24.12.4.16 (Commutant of independent multiplicity actions). Lean ✓ Stmt A matrix commutes with every $U(g)$ if and only if it is of the form $\bigoplus_k I_{m_k} \otimes B_k$, with $B_k \in M_{d_k}(\mathbb{C})$.

Proof. ✓ Proof Two group elements which agree on all factors except k produce, by their half-difference, $E_{ij} \otimes I_{d_k}$ supported on block k . Thus the span of $U(G)$ contains every $\bigoplus_k C_k \otimes I_{d_k}$. Commutation passes to this span. Swap the two tensor factors on each block and apply the commutant formula for a direct sum of full matrix algebras. Conversely, $I_{m_k} \otimes B_k$ commutes with $g_k \otimes I_{d_k}$. $\square$

Theorem 24.12.4.17 (Unital matrix algebras as finite-group commutants). Lean ✓ Stmt Let $S \subset M_I(\mathbb{C})$ be a unital C^* -subalgebra, with I finite. There exist a finite group G and a unitary representation U such that $S = \{A \in M_I(\mathbb{C}) : AU(g) = U(g)A \text{ for every } g \in G\}$. This is the unital version of [203, Theorem 4.1]. The source's definition of C^* -algebra does not require the identity; the printed nonunital assertion fails because every commutant contains the identity. See [63].

Proof. ✓ Proof The unitary block-decomposition theorem for matrix C^* -algebras [212, Theorem 6.14] gives a unitary W with $W^*SW = \bigoplus_k I_{m_k} \otimes M_{d_k}(\mathbb{C})$. Apply the independent signed permutation groups to the multiplicity factors. Their commutant is this block algebra. Conjugating their representation by W therefore gives a finite unitary group with commutant S . All decomposition data are obtained from S . $\square$

Theorem 24.12.4.18 (C^* -algebra tensors admit a finite group symmetry). Lean ✓ Stmt Suppose that a family of complex matrices A^i spans a unital C^* -subalgebra $S \subset M_I(\mathbb{C})$, with I finite. There exist a finite group G and a unitary representation U whose commutant is S , such that the tensor A is G -injective. This gives the passage from the unital version of [203, Theorem 4.1] to its Definition 4.2. The unital restriction is recorded in [63].

Proof. ✓ Proof Choose the finite unitary representation with commutant S . Each $A^i \in S$ commutes with this representation, so its site map $X \mapsto (\text{tr}(A^i X))_i$ is invariant under conjugation. If $X \in S$ is in the kernel, linearity gives $\text{tr}(YX) = 0$ for every $Y \in S$. Since S is closed under adjoints, take $Y = X^*$. Then $\text{tr}(X^*X) = 0$, hence $X = 0$. Thus the site map is injective on the invariant subspace S . $\square$

Theorem 24.12.4.19 (Semi-regularity and the group algebra). Lean Lean Lean Lean ✓ Stmt *Let G be a finite group and let ρ be a finite-dimensional complex representation. The following assertions are equivalent: ρ contains every irreducible representation; the operators $(\rho(g))_{g \in G}$ are linearly independent; and the induced algebra homomorphism $\mathbb{C}[G] \rightarrow \text{End}(V)$ is injective. This is an algebraic characterization of the occurrence condition in [203, Definition 4.5 and Lemma 4.6]. More generally, for a representation of a monoid over a commutative semiring, independence of its operators is equivalent to injectivity of its monoid-algebra homomorphism.*

Proof. ✓ Proof The trace-dual identity gives independence for a semi-regular representation. Conversely, if an irreducible character χ is absent, its character projector vanishes:

$$\frac{\chi(1)}{|G|} \sum_{g \in G} \chi(g^{-1}) \rho(g) = 0. \quad (24.44)$$

Indeed, the projector vanishes on every irreducible summand of V . Independence in (24.44) would imply $\chi(1) = 0$, contrary to the positive dimension of the irreducible representation. The standard basis of the group algebra identifies the last assertion with independence of the group operators. $\square$

Theorem 24.12.4.20 (Reversing a semi-regular bond). Lean Lean ✓ Stmt *A finite-dimensional complex representation ρ of a finite group is semi-regular if and only if its contragredient $\rho^*(g) = \rho(g^{-1})^\top$ is semi-regular. This justifies reversing the bond orientation in the contraction convention of [203, Definition 5.1 and Lemma 5.2].*

Proof. ✓ Proof Inversion permutes the group indices, and transposition is injective on endomorphisms of a finite-dimensional vector space. It therefore preserves independence of the group operators. Conversely, independence of their transposes implies independence of the original operators. $\square$

Theorem 24.12.4.21 (Nonzero tensor factors of a semi-regular representation). Lean Lean ✓ Stmt *Let G be a finite group, let ρ be a finite-dimensional semi-regular complex representation, and let σ be any nonzero finite-dimensional complex representation. Then both $\rho \otimes \sigma$ and $\sigma \otimes \rho$ are semi-regular. This strengthens the tensor-product assertion in [203, Lemma 5.2]: the second factor need not contain the trivial representation.*

Proof. ✓ Proof A dependence among the tensor-product operators is a dependence among the tensors $\rho(g) \otimes \sigma(g)$, by the natural identification of their endomorphism spaces. Applying the trace-dual functional at h to the first factor gives $c_h \sigma(h) = 0$. Since $\sigma(h)$ is invertible on a nonzero vector space, $c_h = 0$. The group-algebra criterion proves semi-regularity. Interchanging the tensor factors gives the other order. $\square$

Theorem 24.12.4.22 (Tensor products of semi-regular representations). Lean Lean Lean Lean ✓ Stmt *If U and V are semi-regular representations of G , then so is $U \otimes V$ [203, Section “Two dimensions: G -injective PEPS”].*

Proof. ✓ Proof A one-dimensional representation is irreducible, so the semi-regular representation V contains the trivial representation and has a nonzero invariant vector v . If D is irreducible and f is a nonzero intertwining map from D to U , then $x \mapsto f(x) \otimes v$ is a nonzero intertwining map from D to $U \otimes V$. $\square$

A matrix product tensor is a family of operators A^i on a bond space carrying the representation U . Its virtual system is the space of operators X on the bond space, on which G acts by conjugation $X \mapsto U_g X U_g^{-1}$, and identifying the virtual vector $\sum x_{\alpha\beta} |\alpha, \beta\rangle$ with the operator $X = \sum x_{\alpha\beta} |\beta\rangle\langle\alpha|$ gives $\mathcal{P}(A)X = \sum_i \text{tr}[A^i X] |i\rangle$ [203, Section “MPS and PEPS”].

Definition 24.12.4.23 (G -injective matrix product tensor). Lean Lean ✓ Stmt A matrix product tensor A is G -injective if the map $\mathcal{P}(A)X = \sum_i \text{tr}[A^i X] |i\rangle$ is G -injective for the conjugation action of U on the operators on the bond space.

Theorem 24.12.4.24 (G -injective matrix product tensors). Lean Lean Lean Lean Lean ✓ Stmt A matrix product tensor A is G -injective if and only if

1. $U_g A^i U_g^{-1} = A^i$ for all i and g , and
2. $\mathcal{P}(A)$ has a left inverse on the commutant $\mathcal{S} = \{X \mid [X, U_g] = 0\}$: there is a linear map L with $L \mathcal{P}(A)X = X$ for every $X \in \mathcal{S}$.

For a unitary representation $U_g^{-1} = U_g^\dagger$, and this is the source’s definition [203, Section “MPS: the G -injective case”].

Proof. ✓ Proof Since $\text{tr}[A^i U_g X U_g^{-1}] = \text{tr}[U_g^{-1} A^i U_g X]$, invariance of $\mathcal{P}(A)$ under conjugation by U_g says that $U_g^{-1} A^i U_g$ and A^i have the same trace pairing with every operator X ; pairing with the rank-one operators $|v\rangle\langle f|$ shows that they are equal. The operators fixed by conjugation are the commutant, and a linear map is injective on a subspace if and only if it has a left inverse there. $\square$

Theorem 24.12.4.25 (Stability under concatenation). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let A and B be G -injective matrix product tensors for the same representation, with left inverses L_A and L_B satisfying $L_A \mathcal{P}(A) = L_B \mathcal{P}(B) = \sigma$. Then $C^{ij} = A^i B^j$ is G -injective, and the map

$$L_C |i, j\rangle = |G| (L_B |j\rangle) \Delta (L_A |i\rangle)$$

satisfies $L_C \mathcal{P}(C) = \sigma$ [203, Section “MPS: the G -injective case”].

Proof. ✓ Proof Invariance of C follows from $U_g A^i B^j U_g^{-1} = (U_g A^i U_g^{-1})(U_g B^j U_g^{-1})$. For the left inverse, first $\sum_i \text{tr}[A^i Y] L_A |i\rangle = \sigma(Y)$ for every Y . Second, for every operator M ,

$$\sum_j (L_B |j\rangle) M B^j = \frac{1}{|G|} \sum_h \text{tr}[U_{h^{-1}} M] U_h;$$

writing M in a basis $\{b_k\}$ with coordinate functionals b_k^* , the left-hand side applied to w is $\sum_k \sigma(|w\rangle\langle b_k^*|) M b_k$, which is the right-hand side applied to w . Taking $Y = B^j X$ in the first identity and $M = \Delta U_g$ in the second,

$$L_C \mathcal{P}(C)X = |G| \sum_j (L_B |j\rangle) \Delta \sigma(B^j X) = \sum_g \left(\sum_j (L_B |j\rangle) \Delta U_g B^j \right) X U_g^{-1} = \frac{1}{|G|} \sum_g U_g X U_g^{-1},$$

the last step by the expansion $\sum_h \text{tr}[U_{h^{-1}} \Delta U_g] U_h = U_g$. $\square$

Corollary 24.12.4.26 (Blocking a G -injective tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) If A is G -injective, then the tensor obtained by blocking $L \geq 1$ sites, $(i_1, \dots, i_L) \mapsto A^{i_1} \dots A^{i_L}$, is G -injective.

Proof. [Stmnt](#) Induction on L : the blocked tensor on $L + 1$ sites is the concatenation of A with the blocked tensor on L sites, up to relabelling the physical index, and relabelling preserves G -injectivity. $\square$

For G -injective tensors the parent-Hamiltonian argument runs as in the injective case, with the left inverses replaced by left inverses up to the twirl σ [203, Section “Parent Hamiltonians”]. In the next two theorems A and B are G -injective for the same representation U ; three-leg tensors M , N carry one physical and two virtual legs, and the operators X , M , N , U_g sit on a bond.

Lemma 24.12.4.27 (Twirling the closing operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) If $U_g C^i U_g^{-1} = C^i$ for all i and g , then $\text{tr}[C^i \sigma(X)] = \text{tr}[C^i X]$ for every operator X : an operator closing an invariant tensor may be replaced by its twirl without changing the state. Moreover $\sigma(YZ) = \sigma(Y)Z$ whenever Z commutes with every U_g , and the concatenation of two invariant tensors is invariant [203, Section “Parent Hamiltonians”].

Proof. [Stmnt](#) Invariance of C is invariance of the map $X \mapsto (\text{tr}[C^i X])_i$ under conjugation, and an invariant map does not see the average over the group. For the second claim, $U_g Y Z U_g^{-1} = (U_g Y U_g^{-1}) Z$. $\square$

Lemma 24.12.4.28 (The concatenated left inverse on traces). [Lean](#) [Stmnt](#) Let L_A be any linear map and let $L_B \mathcal{P}(B) = \sigma$. The map $L_C[j, k] = |G| (L_A|k\rangle) \Delta (L_B|j\rangle)$ of the concatenation $C^{jk} = B^j A^k$ sends the vector $(\text{tr}[B^j Y_k])_{j,k}$ to $|G| \sum_k (L_A|k\rangle) \Delta \sigma(Y_k)$, for every family of operators Y_k .

Proof. [Stmnt](#) Summing over j first, $\sum_j \text{tr}[B^j Y_k] L_B|j\rangle = L_B \mathcal{P}(B) Y_k = \sigma(Y_k)$. $\square$

Theorem 24.12.4.29 (Intersection property for G -injective tensors). [Lean](#) [Stmnt](#)

$$\left\{ \sum_{ijk} \text{tr}[A^i B^j M^k] |ijk\rangle \mid M \right\} \cap \left\{ \sum_{ijk} \text{tr}[N^i B^j A^k] |ijk\rangle \mid N \right\} = \left\{ \sum_{ijk} \text{tr}[A^i B^j A^k X] |ijk\rangle \mid X \right\}$$

[203, Section “Parent Hamiltonians”].

Proof. [Stmnt](#) The right-hand side lies in both sets, with $M^k = A^k X$ and $N^i = X A^i$. Conversely, let $\text{tr}[A^i B^j M^k] = \text{tr}[N^i B^j A^k]$ for all i, j, k . Fix left inverses with $L_A \mathcal{P}(A) = L_B \mathcal{P}(B) = \sigma$, and let L_C be the left inverse of $C^{jk} = B^j A^k$, so that $L_C \mathcal{P}(C) = \sigma$. For fixed i the vector $(\text{tr}[B^j A^k N^i])_{j,k}$ equals $(\text{tr}[B^j M^k A^i])_{j,k}$, and applying L_C gives

$$\sigma(N^i) = |G| \sum_k (L_A|k\rangle) \Delta \sigma(M^k A^i) = X A^i, \quad X = |G| \sum_k (L_A|k\rangle) \Delta \sigma(M^k),$$

because A^i commutes with every U_g . Since C is invariant, $\text{tr}[N^i B^j A^k] = \text{tr}[B^j A^k \sigma(N^i)] = \text{tr}[A^i B^j A^k X]$. $\square$

Theorem 24.12.4.30 (Closure property for G -injective tensors). [Lean](#) [Lean](#) [Stmnt](#)

$$\left\{ \sum_{ij} \text{tr}[B^i A^j M] |ij\rangle \mid M \right\} \cap \left\{ \sum_{ij} \text{tr}[B^i N A^j] |ij\rangle \mid N \right\} = \left\{ \sum_g \lambda_g \sum_{ij} \text{tr}[B^i A^j U_g] |ij\rangle \mid \lambda_g \in \mathbb{C} \right\}.$$

A vector in the intersection, given by M and N , has $\lambda_g = \text{tr}[U_{g^{-1}} \sigma(N) \Delta]$ [203, Section “Parent Hamiltonians”].

Proof. ✓ Proof The right-hand side lies in both sets, with $M = N = \sum_g \lambda_g U_g$, since U_g commutes with A^j . Conversely, let $\text{tr}[B^i A^j M] = \text{tr}[B^i N A^j]$ for all i, j , and let L_C be the left inverse of $C^{ij} = B^i A^j$. Applying L_C to the vector $(\text{tr}[B^i N A^j])_{i,j}$, and then the identity $\sum_j (L_A |j\rangle) Y A^j = \frac{1}{|G|} \sum_h \text{tr}[U_{h-1} Y] U_h$ with $Y = \Delta\sigma(N)$,

$$\sigma(M) = |G| \sum_j (L_A |j\rangle) \Delta\sigma(N) A^j = \sum_h \text{tr}[U_{h-1} \Delta\sigma(N)] U_h.$$

Since C is invariant, $\text{tr}[B^i A^j M] = \text{tr}[B^i A^j \sigma(M)]$, and $\text{tr}[U_{h-1} \Delta\sigma(N)] = \text{tr}[U_{h-1} \sigma(N) \Delta]$ because $\sigma(N)$ commutes with U_{h-1} . $\square$

The two properties give the parent Hamiltonian of a G -injective tensor, as the intersection and closure properties of injective tensors give Theorem 3.5 of the source [203, Section “Parent Hamiltonians”]. For a tensor A and a ring of N sites, write $\mathcal{G}_{N,L}(A)$ for the space of Definition 11.9.6, the common kernel of the parent interactions on all windows of L consecutive sites of the ring, and write $|\mathcal{M}(A|K)\rangle = \sum_\sigma \text{tr}[A^{\sigma_1} \dots A^{\sigma_N} K] |\sigma\rangle$ for the matrix product state with closure K . The source’s two-site interaction is $L = 2$. In the rest of this subsection A is G -injective for the representation U .

Lemma 24.12.4.31 (The open chain of a G -injective tensor). Lean Lean ✓ Stmt Let $K \geq 1$. A vector on $K + 2$ sites whose restrictions to the first and to the last $K + 1$ sites lie in $\{\sum_\sigma \text{tr}[A^{\sigma_1} \dots A^{\sigma_{K+1}} X] |\sigma\rangle \mid X\}$ lies in the corresponding space on $K + 2$ sites. Consequently, if $2 \leq L \leq N$, every vector of $\mathcal{G}_{N,L}(A)$ has the form $\sum_\sigma \text{tr}[A^{\sigma_1} \dots A^{\sigma_N} X] |\sigma\rangle$ [203, Section “Parent Hamiltonians”].

Proof. ✓ Proof Fixing the last letter k writes the vector as $\text{tr}[A^i B^w M^k]$ and fixing the first letter i as $\text{tr}[N^i B^w A^k]$, where B is the tensor of K blocked sites, which is G -injective by Corollary 24.12.4.26. The intersection property gives the form $\text{tr}[A^i B^w A^k X]$. Iterating along the non-wrapping windows of the ring gives the second claim, as in the proof of Theorem 3.5 of the source. $\square$

Theorem 24.12.4.32 (Parent Hamiltonians of G -injective tensors). Lean Lean Lean ✓ Stmt If $2 \leq L \leq N$, then

$$\mathcal{G}_{N,L}(A) = \text{span}_{\mathbb{C}} \{ |\mathcal{M}(A|U_g)\rangle \mid g \in G \}.$$

In particular, for the two-site interaction on a ring of $N \geq 2$ sites the frustration-free ground space of the parent Hamiltonian is spanned by the states $|\mathcal{M}(A|U_g)\rangle$ [203, Theorem 4.11].

Proof. ✓ Proof Each U_g commutes with every A^i , so it can be moved to any bond, and every window of $|\mathcal{M}(A|U_g)\rangle$ lies in the local ground space. Conversely, by Lemma 24.12.4.31 a vector of $\mathcal{G}_{N,L}(A)$ is $\text{tr}[A^{\sigma_1} \dots A^{\sigma_N} X]$, and its translate by one site is $\text{tr}[A^{\sigma_1} \dots A^{\sigma_N} Y]$. Hence $\text{tr}[A^a B^w X] = \text{tr}[A^a Y B^w]$, where B is the tensor of the last $N - 1$ blocked sites. The closure property, applied to A and B , writes the vector as a combination of the states $|\mathcal{M}(A|U_g)\rangle$. $\square$

Lemma 24.12.4.33 (Twirled representation operators). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The isotypic projectors P_i commute with every U_g , are linearly independent, and

$$\sigma(U_g) = \frac{1}{|G|} \sum_h U_{hgh^{-1}} = \sum_{i=1}^r \frac{\chi_i(g)}{\chi_i(1)} P_i.$$

If U is semi-regular, then the operators U_g are linearly independent, and so are the twirls $\sigma(U_g)$ of one representative g of each conjugacy class [203, Section “Parent Hamiltonians”].

by contracting the right leg of A with the left leg of B , with the up, down and physical indices blocked:

$$\mathcal{P}(C)(x \otimes y) = \sum_i \mathcal{P}(A)(x \otimes e_i^*) \otimes \mathcal{P}(B)(e_i \otimes y).$$

If A and B are G -injective, then C is G -injective for $\sigma_A \otimes \sigma_B$. Precisely, if $L_A \mathcal{P}(A) = \Pi_{\sigma_A \otimes \tau^*}$ and $L_B \mathcal{P}(B) = \Pi_{\tau \otimes \sigma_B}$, then the map L_C that applies $L_A \otimes L_B$ and contracts the two inner legs through Δ , $(x \otimes f) \otimes (e \otimes y) \mapsto |G| f(\Delta e) x \otimes y$, satisfies $L_C \mathcal{P}(C) = \Pi_{\sigma_A \otimes \sigma_B}$. If moreover σ_A and σ_B are semi-regular, then so is $\sigma_A \otimes \sigma_B$ [203, Lemma 5.2].

Proof. ✓ Proof Invariance of A gives $\mathcal{P}(A)(\sigma_A(g)x \otimes f) = \mathcal{P}(A)(x \otimes \tau^*(g^{-1})f)$ and similarly for B , so acting with $\sigma_A(g) \otimes \sigma_B(g)$ on the virtual system of C acts with $\tau^*(g^{-1}) \otimes \tau(g^{-1})$ on ω , which leaves it invariant by Lemma 24.12.4.35. For the left inverse, $L_A \mathcal{P}(A)(x \otimes f) = \frac{1}{|G|} \sum_g \sigma_A(g)x \otimes \tau^*(g)f$ and $L_B \mathcal{P}(B)(e \otimes y) = \frac{1}{|G|} \sum_h \tau(h)e \otimes \sigma_B(h)y$. Contracting through Δ gives

$$L_C \mathcal{P}(C)(x \otimes y) = \frac{1}{|G|} \sum_{g,h} \text{tr}[U_{g^{-1}} \Delta U_h] \sigma_A(g)x \otimes \sigma_B(h)y = \frac{1}{|G|} \sum_g \sigma_A(g)x \otimes \sigma_B(g)y$$

by Lemma 24.12.4.35. Theorem 24.12.4.2 turns invariance and the left inverse into G -injectivity, and Theorem 24.12.4.22 gives the last claim. $\square$

Theorem 24.12.4.37 (Contracting legs of a connected block). Lean Lean Lean Lean Lean
✓ Stmt Let a tensor have virtual system $W \otimes E^* \otimes E$ with group action $\sigma \otimes \tau^* \otimes \tau$, as for the up leg of A and the down leg of B in the block C above. Contracting the legs E^* and E gives the map $x \mapsto \sum_i \mathcal{P}(x \otimes e_i^* \otimes e_i)$. If $L\mathcal{P} = \Pi_{\sigma \otimes \tau^* \otimes \tau}$ and $\text{tr } M \neq 0$, then contracting the corresponding legs of L through M and dividing by $\text{tr } M$ gives a left inverse of the contracted map up to Π_σ . Consequently, if the tensor is G -injective and τ is semi-regular, the contracted map is G -injective for σ , with $M = \Delta$, whose trace is 1 [203, Observation 5.3].

Proof. ✓ Proof The group elements attached to the two contracted legs cancel, because ω is invariant under $\tau^*(g) \otimes \tau(g)$. Hence $\Pi_{\sigma \otimes \tau^* \otimes \tau}(x \otimes \omega) = \frac{1}{|G|} \sum_g \sigma(g)x \otimes \omega$, and pairing ω through M gives $\text{tr } M$. By Theorem 24.12.4.11 with $g = h = 1$, $\text{tr } \Delta = 1$ for semi-regular τ . $\square$

For a tensor A let $A^\dagger$ be the tensor with complex conjugate entries and the roles of the incoming and outgoing legs exchanged, so that $\mathcal{P}(A^\dagger) = \mathcal{P}(A)^\dagger$. For a matrix product tensor with bond space $\mathbb{C}[G]$, on which the left-regular representation acts by $L_g|h\rangle = |gh\rangle$ and the group elements are orthonormal, $\mathcal{P}(A^\dagger)|i\rangle = (A^i)^\dagger$ under the identification above [203, Section “Isometric PEPS”].

Definition 24.12.4.38 (G -isometric matrix product tensor). Lean Lean Lean Lean Lean Lean
✓ Stmt A matrix product tensor A with bond space $\mathbb{C}[G]$ is G -isometric if $L_g A^i L_g^{-1} = A^i$ for all g and i and $\mathcal{P}(A^\dagger)\mathcal{P}(A) = c\sigma$ for one constant $c > 0$, where σ is the twirl onto the commutant of the left-regular representation. For $c = 1$ this is the condition $\mathcal{P}(A)^{-1} = \mathcal{P}(A^\dagger)$ of the source; the constant absorbs the normalization of the tensor [203, Definition 6.1], [114].

Theorem 24.12.4.39 (G -isometric tensors are G -injective). Lean ✓ Stmt A G -isometric matrix product tensor is G -injective for the left-regular representation, with left inverse $c^{-1}\mathcal{P}(A^\dagger)$ [203, Definition 6.1].

Proof. ✓ Proof Invariance of $\mathcal{P}(A)$ under conjugation is equivalent to $L_g A^i L_g^{-1} = A^i$, and $c^{-1}\mathcal{P}(A^\dagger)\mathcal{P}(A) = \sigma$; the left-inverse form of G -injectivity applies. $\square$

$T|a, b\rangle = |a, ab\rangle$ is unitary and satisfies $T^\dagger(L_g \otimes L_g)T = L_g \otimes \mathbb{1}$, where $L_g|a\rangle = |ga\rangle$. For a finite family of additional labels, the permutation $|a, (b_i)_i\rangle \mapsto |a, (ab_i)_i\rangle$ similarly transforms simultaneous left multiplication into left multiplication of a alone [203, Section 6, “Blocking and renormalization”].

Proof. ✓ Proof The inverse change of coordinates is $(a, (c_i)_i) \mapsto (a, (a^{-1}c_i)_i)$. Thus simultaneous multiplication by g sends $(a, (ab_i)_i)$ to $(ga, (gab_i)_i)$, and in relative coordinates only the first label changes. The permutation matrices are unitary, and the matrix of a product permutation is the Kronecker product of the factor matrices. $\square$

The source shows next that for a G -isometric tensor the virtual level, restricted to the invariant subspace, and the physical level are related by a unitary. Its standing convention is that the physical system of a site is the range of $\mathcal{P}(A)$, which can always be arranged by a local isometry [203, Sections “MPS and PEPS” and “Isometric PEPS”].

Theorem 24.12.4.45 (Virtual unitaries implementable on the physical level). Lean Lean Lean Lean ✓ Stmt Let A be G -isometric with $\mathcal{P}(A)$ onto the physical system. If V is a unitary on the virtual level that commutes with U_g for every g , there is a unitary W on the physical system with $W\mathcal{P}(A) = \mathcal{P}(A)V$. Conversely, for every unitary W on the physical system, $V = \mathcal{P}(A)^{-1}W\mathcal{P}(A)$ satisfies $VU_g = U_gV = V$ for every g , preserves the inner products of invariant vectors, and $\mathcal{P}(A)V = W\mathcal{P}(A)$ [203, Lemma 6.3].

Proof. ✓ Proof Let L be a left inverse with $L\mathcal{P}(A) = \Pi_U$. Since $\mathcal{P}(A)$ is onto, every physical vector is $\mathcal{P}(A)x$ with x invariant, Ly is invariant and $\mathcal{P}(A)Ly = y$. For the first claim let $W = \mathcal{P}(A)VL$. As V commutes with U_g , it preserves the invariant subspace and commutes with Π_U , so $W\mathcal{P}(A) = \mathcal{P}(A)V\Pi_U = \mathcal{P}(A)\Pi_UV = \mathcal{P}(A)V$; and for invariant x, x' , $\langle W\mathcal{P}(A)x, W\mathcal{P}(A)x' \rangle = c\langle Vx, Vx' \rangle = c\langle x, x' \rangle = \langle \mathcal{P}(A)x, \mathcal{P}(A)x' \rangle$, so W is unitary. For the converse, $V = LW\mathcal{P}(A)$ satisfies $VU_g = V$ because $\mathcal{P}(A)U_g = \mathcal{P}(A)$, and $U_gV = V$ because V takes invariant values; $\mathcal{P}(A)V = W\mathcal{P}(A)$ because $\mathcal{P}(A)L$ is the identity, and the same computation as before shows that V preserves inner products of invariant vectors. $\square$

The parent Hamiltonian of a G -isometric matrix product state has commuting local terms [203, Section “Commuting parent Hamiltonians”]. Write $\Gamma_n(X) = \sum_\sigma \text{tr}[A^\sigma X]|\sigma\rangle$ for the parametrization of the local space $\mathcal{S}_n$ on n sites by boundary matrices, where $A^\sigma = A^{\sigma_1} \dots A^{\sigma_n}$.

Definition 24.12.4.46 (The operator $\Gamma_n\Gamma_n^\dagger$). Lean Lean Lean Lean ✓ Stmt For a matrix product tensor A , let $\Gamma_n^\dagger\psi = \sum_\tau \psi(\tau)(A^\tau)^\dagger$, the adjoint of Γ_n for the Hilbert–Schmidt inner product, and let $P_n = \Gamma_n\Gamma_n^\dagger$, the operator on n sites with matrix entries $\text{tr}[A^\sigma(A^\tau)^\dagger]$. For $n = 2$ this is the operator of [203, Lemma 6.11].

Theorem 24.12.4.47 (A multiple of the projector onto the local space). Lean Lean Lean Lean ✓ Stmt The matrix of P_n is Hermitian. If $\Gamma_n\Gamma_n^\dagger\Gamma_n = c\Gamma_n$ for some $c > 0$, then $c^{-1}P_n$ is idempotent with range $\mathcal{S}_n$, so it is the orthogonal projector onto $\mathcal{S}_n$ and the parent interaction is $h = \mathbb{1} - c^{-1}P_n$.

Proof. ✓ Proof The entries satisfy $\overline{\text{tr}[A^\tau(A^\sigma)^\dagger]} = \text{tr}[A^\sigma(A^\tau)^\dagger]$. From $\Gamma_n\Gamma_n^\dagger\Gamma_n = c\Gamma_n$ one gets $P_n^2 = cP_n$, and $P_n\Gamma_nX = c\Gamma_nX$ shows that the range of P_n is that of Γ_n . A Hermitian idempotent is the orthogonal projector onto its range. $\square$

Definition 24.12.4.48 (The matrix tensor of a tensor on $\mathbb{C}[G]$). Lean Lean Lean Lean Lean ✓ Stmt For a matrix product tensor A with bond space $\mathbb{C}[G]$, write each A^i as a matrix in the basis of group elements. In this basis the adjoint $(A^i)^\dagger$ is the conjugate transpose, and the parent Hamiltonian of A is that of these matrices.

legs. For a representation $g \mapsto U_g$ on one bond, U_g acts on the virtual system of a site by U_g on the incoming legs (top and left) and U_g^{-1} on the outgoing legs (right and down), each along the orientation of its edge [203, Section “Two dimensions: G -injective PEPS”].

Lemma 24.12.4.52 (Diagonal operators on the bonds). Lean Lean Lean ✓ Stmt *If every bond carries a diagonal matrix, the network is the sum over one index per bond of the product of the diagonal entries and of the site tensors. In particular, with the identity on every bond it is the contraction of the site tensors, and for $w, h \geq 3$ and $A_v = a^{\sigma(v)}$ it is the coefficient of the torus PEPS of a at σ .*

Proof. ✓ Proof The weight of a labelling vanishes unless the two ends of every bond carry the same index, and the labellings with equal ends correspond to the assignments of one index per bond. □

Lemma 24.12.4.53 (Moving bond operators into the site tensors). Lean ✓ Stmt *Let P_v and Q_v be matrices at every site. Replacing the matrix O of every bond by $P_{\text{head}} O Q_{\text{tail}}$ gives the same network as keeping O and replacing every site tensor A_v by the tensor that has absorbed P_v into its incoming legs and Q_v into its outgoing legs.*

Proof. ✓ Proof Expand each product POQ into its entries. The resulting sum runs over the labellings of the bond ends and over new intermediate indices; exchanging the roles of the two families, the intermediate indices become the labelling of the bond ends and the old labels become the summation indices of the absorbed matrices at the sites. The factors $P_{v+(1,0)}$ and $Q_{v+(0,1)}$ are reindexed by a unit shift of the torus. □

Theorem 24.12.4.54 (Deformation of a string of U_g ’s). Lean Lean Lean Lean Lean ✓ Stmt *Let a be a four-leg tensor with $\mathcal{P}(a)U_g = \mathcal{P}(a)$ for every g , the invariance clause of G -injectivity for the action of Definition 24.12.4.51. Fix a rectangle of $m < w$ columns and $n < h$ rows of sites. A string of U_g ’s carries U_g on the vertical bonds, which it crosses going right, and U_g^{-1} on the horizontal bonds, which it crosses going down. In the torus network of a , the string along the left and bottom sides of the rectangle and the string along its top and right sides give the same network [203, Section “Two dimensions: G -injective PEPS”, eq. (2d:move-strings) and Lemma “Deformations of the string”]. Only the invariance is used: the source assumes that U_g contains every irreducible representation, and its argument, like this one, does not use it.*

Proof. ✓ Proof Let $\Phi_v = g^{-1}$ on the sites of the rectangle and $\Phi_v = 1$ elsewhere, and apply Lemma 24.12.4.53 with $P_v = U_{\Phi_v}$ and $Q_v = U_{\Phi_v}^{-1}$ to the second string. A bond with both ends inside or both outside the rectangle carries the identity in both strings and keeps it. A bond entering the rectangle across its top side is multiplied by U_g^{-1} at its head, which cancels the U_g of the second string; a bond leaving it across its right side is multiplied by U_g at its tail, which cancels its U_g^{-1} . A bond entering across the left side acquires U_g^{-1} and a bond leaving across the bottom side acquires U_g , which is the first string. By the invariance every site tensor absorbs U_{Φ_v} on its incoming and $U_{\Phi_v}^{-1}$ on its outgoing legs unchanged. □

Definition 24.12.4.55 (The genuine open tensor of a finite region). Lean Lean Lean Lean Lean ✓ Stmt *For a region R in a finite PEPS graph, let $T_R(\mu; \sigma)$ be the contraction of its site tensors with physical configuration σ , fixing the crossing-bond labels μ and summing only the internal-bond labels. It defines the linear map from the crossing virtual indices to the physical system of R . No edge wholly outside R enters this contraction. This is the finite-region tensor used when blocking and cutting the PEPS in [203, Section 6, proof of Theorem 6.9].*

Theorem 24.12.4.56 (Exact contraction across a physical cut). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#)

[Learn](#) [Learn](#) [StmT](#) Suppose every bond space has positive dimension. The coefficient of the PEPS at physical configurations σ on R and τ on its complement is exactly

$$\psi(\sigma, \tau) = \sum_{\mu} T_R(\mu; \sigma) T_{R^c}(\mu; \tau).$$

In particular this holds for regular bonds of any finite group. If the region coefficient is instead defined by summing global edge assignments, it equals T_R multiplied by the product of the dimensions of all edges wholly outside R .

This is an exact finite-region contraction identity. Identifying its cut maps with isometries on the regular invariant boundary remains a separate step in [203, Theorem 6.9].

Proof. ✔ **Proof** Split each edge assignment into its incident and exterior parts. Exterior labels do not occur in any tensor of R , so their sum gives the stated multiplicity. Apply the existing coefficient identity for the globally indexed region tensors on the two sides. Their exterior multiplicities multiply to the product of all non-crossing bond dimensions; cancelling this nonzero product gives the formula. Regular bond dimensions are $|G| > 0$. $\square$

Definition 24.12.4.57 (Regular coordinates of the physical cut). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Express every bond in a group basis, and enumerate the $b = n + 1$ crossing bonds. Let A and B be the genuine open tensors of the region and its complement, with their columns indexed by G^b in the same crossing-bond order. Let K be the matrix of physical PEPS coefficients on these two systems. These are the actual cut maps used before the virtual reduction in [203, Proof of Theorem 6.9].

Theorem 24.12.4.58 (Boundary symmetry and projector insertion at the cut). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the coefficient tensor at every site is invariant under simultaneous left translation of all its virtual group labels. Then the genuine open tensor of every region is invariant under simultaneous left translation of its crossing labels. Consequently $AP = A$ and the physical cut satisfies

$$K = AB^\top = APB^\top, \quad ACB^\top = \frac{K}{\sqrt{r}}, \quad C = \frac{P}{\sqrt{r}}, \quad r = |G|^n.$$

In the group bases of regular and dual regular legs, this is the inherited boundary symmetry of the proof of [203, Theorem 6.9]. It identifies the contraction with transport of the virtual boundary vector, up to the displayed scalar. It does not assert isometry of A, B or normalization of the physical vector.

Proof. ✓ Proof Translate every incident-edge assignment by the same group element. This permutes the internal summation and translates the crossing labels; local invariance leaves the product of site coefficients unchanged. Averaging this symmetry gives $AP = A$. The exact cut identity gives $K = AB^T$, and inserting P and then $C = P/\sqrt{r}$ yields the remaining formulas. $\square$

Lemma 24.12.4.59 (Independent finite choices). [Lean] [✓ Stmt] *Let I be a finite set, let J_i be finite sets indexed by $i \in I$, and let P_i be a condition on J_i . In every commutative semiring,*

$$\prod_{i \in I} \sum_{j \in J_i} 1_{P_i(j)} = \sum_{f \in \prod_{i \in I} J_i} 1_{\{P_i(f_i) \text{ for every } i \in I\}}.$$

Thus multiplying the separate numbers of admissible choices counts the admissible simultaneous choices.

Proof. ✓ Proof Distribute the finite product over the finite sums. A product of zero-one indicators is the indicator of the conjunction of their conditions. $\square$

Definition 24.12.4.60 (The canonical regular-projector region). Lean Lean Lean Lean Lean
Lean ✓ Stmt Let R be a finite region of the original graph, E_R its incident edges, and ∂R its crossing edges. Write E_v for the incident edges at v and put $\kappa = |G|$. The canonical physical configuration is $\alpha = (\alpha_v)_{v \in R}$, with $\alpha_v \in G^{E_v}$. The local canonical site map is the regular averaging projector P_v ; it is G -isometric with factor one. Define the canonical open-region coefficient matrix by

$$\begin{aligned} B_R(\alpha, \theta) &= \sum_{\substack{\eta \in G^{E_R} \\ \eta|_{\partial R} = \theta}} \prod_{v \in R} P_v(\alpha_v, \eta|_{E_v}) \\ &= \kappa^{-|R|} \sum_{\eta \in G^{E_R}} \sum_{q \in G^R} \mathbf{1}_{\{\eta|_{\partial R} = \theta\}} \mathbf{1}_{\{\alpha_v = q_v \eta|_{E_v} \ (v \in R)\}}. \end{aligned} \quad (24.46)$$

Every internal bond is summed once, and no exterior bond label occurs. Encoding group labels in finite coordinates preserves the boundary condition. These are the exposed virtual coordinates of [203, Observation after Lemma 6.3].

Theorem 24.12.4.61 (Accessibility of an actual regular region). Lean Lean Lean Lean Lean
Lean Lean ✓ Stmt Suppose the local regular site maps A_v are G -isometric, with positive factors c_v . Let $T_R(\theta)$ be the actual contracted physical column with crossing labels θ , and write $B_R(\theta)$ for the column in Definition 24.12.4.60. With $L_v = c_v^{-1} A_v^\dagger$, one has

$$\begin{aligned} \left(\bigotimes_{v \in R} L_v \right) T_R(\theta) &= B_R(\theta), \\ \left(\bigotimes_{v \in R} A_v \right) B_R(\theta) &= T_R(\theta). \end{aligned} \quad (24.47)$$

The reverse identity needs only the local invariance of the G -injective site maps. More generally, any local matrices F_v satisfying $F_v A_v = P_v$ give the forward identity. Such matrices also exist for arbitrary regular G -injective sites, by their local left-inverse property. These assertions concern the actual contracted columns and do not require the physical maps to be inverse on unused coordinates. No region Gram identity is assumed.

Proof. ✓ Proof The local identities are $c_v^{-1} A_v^\dagger A_v = P_v$ and $A_v P_v = A_v$. The first follows from the local regular Gram kernel; the second follows from virtual invariance by averaging the group action. Apply the physical product maps inside the actual internal-bond contraction. Each local coefficient becomes the corresponding matrix product, giving (24.47). Expanding P_v and distributing the finite products gives the translation formula (24.46). $\square$

Lemma 24.12.4.62 (Physical maps commute with the open-region contraction). Lean Lean
Lean Lean Lean Lean Lean Lean ✓ Stmt For a region R , let F_v be a physical matrix at each vertex, allowing different input and output dimensions at different vertices. Their product matrix is

$$F_R(\tau, \sigma) = \prod_{v \in R} F_v(\tau_v, \sigma_v).$$

Then $(FL)_R = F_R L_R$, $(F^\dagger)_R = F_R^\dagger$, and identity local maps give the identity on the physical space of the region. In particular, local isometries give an isometry F_R .

Applying F_R to the actual open-region tensor is the contraction of the transformed site coefficients:

$$(F_R T_R)(\tau, \mu) = \sum_{\eta: \eta|_{\partial R} = \mu} \prod_{v \in R} \sum_s F_v(\tau_v, s) A_v(s, \eta|_v).$$

This is the physical operation on a block used in [203, §6.1, accessible virtual coordinates].

Proof. ✓ Proof Expand the matrix products and interchange the finite sums over physical and virtual labels. The sum over independent physical labels factors at each vertex. Composition follows from the same finite distributivity; complex conjugation gives the adjoint identity. Applying these identities to $F_v^\dagger F_v = \mathbb{1}$ proves the isometry assertion. $\square$

Definition 24.12.4.63 (Local regular Gram kernel). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let E_v be the set of virtual legs incident to a site v . In the group basis, its coefficient map A_v has entries $A_v(s, \eta) = a_v(\eta; s)$, for $\eta \in G^{E_v}$. Simultaneous left translation acts on functions by $(U_g f)(\eta) = f(g^{-1}\eta)$, and its invariant projector has entries

$$P_v(\eta, \theta) = \frac{1}{|G|} \sum_{g \in G} \mathbf{1}_{\{\eta = g\theta\}}.$$

The local Gram kernel is $\sum_s \overline{a_v(\eta; s)} a_v(\theta; s)$.

Lemma 24.12.4.64 (Gram formula at an isometric regular site). Lean Lean ✓ Stmt If A_v is G -isometric, there is $c_v > 0$ such that

$$\sum_s \overline{a_v(\eta; s)} a_v(\theta; s) = \frac{c_v}{|G|} \sum_{g \in G} \mathbf{1}_{\{\eta = g\theta\}}.$$

Its coefficients are invariant under simultaneous left translation; this invariance already follows from G -injectivity. This is the group-basis form of [203, Definition 6.1], with the positive scalar convention of [114].

Proof. ✓ Proof The defining adjoint identity is $A_v^\dagger A_v = c_v P_v$. Evaluating both sides in the group basis gives the formula. The identity $A_v U_g = A_v$ gives coefficient invariance. $\square$

Lemma 24.12.4.65 (Expansion of the Gram matrix of a region). Lean Lean Lean Lean Lean ✓ Stmt For any finite region R , the entry (μ, ν) of $T_R^\dagger T_R$ is the sum over pairs of incident-edge assignments with crossing labels μ, ν of

$$\prod_{v \in R} \sum_s \overline{a_v(\eta|_{E_v}; s)} a_v(\theta|_{E_v}; s).$$

This identity requires no isometry assumption.

Proof. ✓ Proof Expand both copies of the region tensor. Interchange the finite sums, and distribute the independent physical sums over the vertices. Each factor is the corresponding site Gram kernel. $\square$

Lemma 24.12.4.66 (Synchronization of group labels in a connected region). Lean Lean Lean Lean Lean ✓ Stmt Suppose the induced graph on R is nonempty and connected. The compatibility equations $\eta_e = q_v \theta_e$ at every incident leg hold precisely when $q_v = g$ at every vertex and $\eta = g\theta$, for one $g \in G$. Thus the sum over compatible vertex translations reduces to the sum over simultaneous translations.

Proof. ✓ **Proof** For an internal edge with endpoints u, v , the two equations give $q_u \theta_e = q_v \theta_e$, hence $q_u = q_v$. Equality propagates along paths. Every incident edge has an endpoint in R , so its label is translated by this common element. Nonemptiness makes the assignment $g \mapsto (q_v = g)$ injective. $\square$

Lemma 24.12.4.67 (Counting free internal regular bonds). Lean Lean Lean Lean Lean Lean
✓ **Stmt** Fix the crossing configurations μ, ν and a simultaneous translation g . Pairs of incident configurations satisfying $\eta = g\theta$ and these prescribed boundary values number $|G|^{I_R}$ if $\mu = g\nu$, and zero otherwise, where I_R is the number of edges with both endpoints in R .

Proof. ✓ **Proof** The second configuration has one free label on each internal edge, and fixes the first configuration by translation. An incident edge that is not a crossing edge has both endpoints in R , which identifies these free indices with the internal bonds. $\square$

Theorem 24.12.4.68 (Gram operator of a connected regular region). Lean Lean Lean Lean
✓ **Stmt** Let R induce a nonempty connected graph, with V_R vertices and I_R internal edges. Suppose the local Gram kernels are $A_v^\dagger A_v = c_v P_v$. For unnormalized regular bonds and no inserted bond operators, the genuine open-region tensor satisfies

$$T_R^\dagger T_R = \kappa_R P_{\partial R}, \quad \kappa_R = \left(\prod_{v \in R} c_v \right) \frac{|G|^{I_R+1}}{|G|^{V_R}}.$$

The boundary may be empty. In particular, positive local Gram factors give an isometry on the invariant boundary space after dividing by $\sqrt{\kappa_R}$, also when the region contains cycles. This is the untwisted connected-region contraction underlying [203, Lemma 6.2 and proof of Theorem 6.9].

Proof. ✓ **Proof** Expand the Gram matrix into local pairings. Their averages contribute $|G|^{-V_R}$ and a sum over vertex group labels. Compatibility reduces this to one common translation. For each such translation, the internal labels contribute $|G|^{I_R}$. The remaining boundary translation sum is $|G| P_{\partial R}$. $\square$

Theorem 24.12.4.69 (Physical implementation of a supported canonical unitary). Lean Lean
✓ **Stmt** Let A_v be the physical map of a regular G -isometric site, with arbitrary finite physical dimension, and let P_v be its regular invariant projector. On a finite region R , put $T = \bigotimes_{v \in R} A_v$ and $P = \bigotimes_{v \in R} P_v$. There is $c > 0$ such that $T^\dagger T = cP$, derived from local isometry alone. If a unitary U on the canonical half-edge coordinate space satisfies $UP = PU$, then there is a unitary W on the original physical region satisfying

$$WT = TU. \tag{24.48}$$

No surjectivity of the site maps is required. This auxiliary form of physical accessibility extends the operation to any unused physical directions; compare [203, Lemma 6.3 and §6.2].

Proof. ✓ **Proof** Local isometry gives $A_v^\dagger A_v = c_v P_v$, with $c_v > 0$. Products and adjoints of the product physical maps are computed site by site, so $T^\dagger T = cP$, where $c = \prod_{v \in R} c_v$. Hence

$$(TU)^\dagger (TU) = cU^\dagger PU = cP = T^\dagger T. \tag{24.49}$$

Equal Gram matrices define an isometry between the two ranges. Extend it to a unitary on the finite-dimensional physical space. $\square$

open-region map T_R is G -isometric for simultaneous regular translation on the crossing bonds. Its derived positive factor is

$$c_R = \left(\prod_{v \in R} c_v \right) |G|^{I_R+1-|R|},$$

where I_R is the number of internal edges. This includes blocks with internal cycles and blocks with no crossing bonds. It is the actual finite-region form of the stability and blocking argument in [203, Lemma 6.2 and §6.2], with the positive-factor convention recorded in [114].

Proof. ✓ Proof Translating all virtual labels in the contraction proves boundary invariance. The proved connected-region Gram formula gives $T_R^\dagger T_R = c_R \Pi_{\partial R}$ with $c_R > 0$. Dividing the adjoint by c_R gives a left inverse on the invariant boundary space. The same identity gives the required inner-product identity there. $\square$

Lemma 24.12.4.73 (Removing bond operators on a spanning tree). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let G be finite, choose a spanning tree T of the region R , and fix a root o . Assign a group operator u_e to each oriented edge $e = (s, t)$. There is a unique vertex gauge k with $k_o = 1$ and $k_t k_s^{-1} = u_e$ on T . The internal residual operators $\omega_e = k_t^{-1} u_e k_s$ therefore equal the identity on T ; their values on the remaining internal edges are unrestricted.

Internal references and physical half-edge labels have reversible transports $\eta'_e = k_s^{-1} \eta_e$ and $\alpha'_{v,e} = k_v^{-1} \alpha_{v,e}$. Crossing labels have the reversible transport

$$\theta'_e = \begin{cases} k_s^{-1} \theta_e, & s \in R, t \notin R, \\ k_t^{-1} u_e \theta_e, & s \notin R, t \in R. \end{cases} \quad (24.53)$$

On every internal edge, $\omega_e \eta'_e = k_t^{-1} u_e \eta_e$. More generally, for any ambient vertex labels K , set $\tilde{u}_e = K_t^{-1} u_e K_s$ and $\tilde{\eta}_{v,e} = K_s^{-1} \eta_{v,e}$. If D_u inserts u_e at the head of each edge and leaves its tail label unchanged, then $D_u(\tilde{\eta})_{v,e} = K_v^{-1} D_u(\eta)_{v,e}$. These are group-coordinate identities associated with the reversible regular blocking operations in [203, §6.2–6.3].

Proof. ✓ Proof Invert the tree-gradient bijection at $(u_e)_{e \in T}$ to obtain the normalized k . On a tree edge, $k_t^{-1} u_e k_s = k_t^{-1} (k_t k_s^{-1}) k_s = 1$. The reference identity follows from $(k_t^{-1} u_e k_s)(k_s^{-1} \eta_e) = k_t^{-1} u_e \eta_e$. Each transport is multiplication by a fixed group element at each coordinate, hence is reversible. At a head endpoint the native twisted-label identity is $(K_t^{-1} u_e K_s)(K_s^{-1} \eta_{t,e}) = K_t^{-1} u_e \eta_{t,e}$; at a tail endpoint both expressions are $K_s^{-1} \eta_{s,e}$. $\square$

Theorem 24.12.4.74 (Integer fibers and affine lifts of the additive circle). Lean Lean ✓ Stmt Let $n > 0$ be an integer, let $z \in \mathbb{Z}/n\mathbb{Z}$, and let $\bar{z} \in \{0, \dots, n-1\}$ be its standard representative. If $x \in \mathbb{R}$ has the same image as $\bar{z}$ in $\mathbb{R}/n\mathbb{Z}$, there is an integer j with $j = x$ and $j \equiv z \pmod{n}$.

For any real period p , let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous. If its image in the additive circle satisfies $[f(t)] = [f(0)] + [td]$ for every $t \in [0, 1]$, then $f(t) = f(0) + td$ throughout the interval. These are auxiliary arithmetic and covering-space identities used for the cell and edge paths of the block domains in [203, §6.3].

Proof. ✓ Proof The first equality means that $x - \bar{z}$ is an integer multiple of n , so x is an integer with the stated residue. In the second assertion, f and the affine path have identical projections and agree initially; uniqueness of a covering lift on the connected interval makes them equal. $\square$

Theorem 24.12.4.75 (Finite compact cell images). Lean ✓ Stmt Let K be compact and locally path connected, let Y be Hausdorff, and let $f_i : K \rightarrow Y$ be a finite family of continuous maps. Then $\bigcup_i f_i(K)$ is locally path connected.

Proof. ✓ Proof The finite disjoint union of copies of K is compact and locally path connected. Its natural continuous surjection onto the stated union is closed, hence is a quotient map. Local path connectedness passes through this quotient map. $\square$

Lemma 24.12.4.76 (Closed-cell regions and geometric coordinate lifts). Lean
Lean Lean ✓ Stmt Let $w, H > 0$, let $\pi : \mathbb{R}^2 \rightarrow (\mathbb{R}/w\mathbb{Z}) \times (\mathbb{R}/H\mathbb{Z})$ be the quotient map, and let p_v be the image of the native vertex coordinates. Define the closed cell and the realized region by

$$Q_v = p_v + \pi([-1/2, 1/2]^2), \quad X_R = \bigcup_{v \in R} Q_v. \quad (24.54)$$

The cells and their finite union are compact, and X_R is locally path connected. Every native site in R lies in X_R . The paths $t \mapsto p_v + \pi(t, 0)$ and $t \mapsto p_v + \pi(0, t)$ have the native neighboring sites as their endpoints; whenever both sites belong to R , the whole path lies in X_R , including across a periodic seam.

Any continuous lift $F : X_R \rightarrow \mathbb{R}^2$ of the inclusion yields integer coordinates $L : R \rightarrow \mathbb{Z}^2$ which project to the native vertices and preserve every internal horizontal and vertical unit step. In particular, if X_R is simply connected and a site $o \in R$ is fixed, such a continuous lift and such integer coordinates exist.

This closed-cell convention is an auxiliary geometric realization of the block domains drawn in [203, §6.3]. Simple connectedness is the ordinary topological hypothesis on the actual cell union; no disk predicate, group flatness assumption, or entropy conclusion is introduced. Its relation to arbitrary disk regions in Theorem 6.9 remains recorded in [114].

Proof. ✓ Proof The quotient projection is continuous, so each projected closed square is compact. The finite coproduct of the compact convex squares maps continuously and surjectively onto X_R . This is a closed quotient map into a Hausdorff space, and local path connectedness therefore passes to X_R . Split a native unit path at its midpoint: its first half lies in the initial cell and its second half in the final cell.

Apply covering-space lifting to the two coordinate maps into $\mathbb{R}/w\mathbb{Z}$ and $\mathbb{R}/H\mathbb{Z}$ separately. Every lift of a native site has integer coordinates. Along an internal unit path, uniqueness of the lift identifies the lifted path with the corresponding affine unit segment. Its endpoints thus satisfy the exact integer unit-step equations. $\square$

Theorem 24.12.4.77 (The genuine planar lift of the closed-cell region). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $F : X_R \rightarrow \mathbb{R}^2$ be a continuous lift of the inclusion into the real torus. On every native cell,

$$F(p_v + \pi(d)) = F(p_v) + d, \quad d \in [-1/2, 1/2]^2. \quad (24.55)$$

The lift is a closed embedding. There are integer site coordinates $L(v) = F(p_v)$ which project to the original native vertices and preserve every internal unit step, and

$$F(X_R) = P_L := \bigcup_{v \in R} (L(v) + [-1/2, 1/2]^2). \quad (24.56)$$

Thus F induces a homeomorphism from the actual torus cell union to this finite planar cell union.

This is an auxiliary geometric consequence of the covering lift, supporting the block domains of [203, §6.3]. The homeomorphism transfers simple connectedness to the planar union, which is the domain of the exterior contour argument below.

Proof. ✓ Proof The radial segment from p_v to $p_v + \pi(d)$ stays in the same closed cell. Uniqueness of its covering lift identifies the lifted segment with $t \mapsto F(p_v) + td$. Two neighboring cells have a common side midpoint; evaluating its lift from the two cells shows that their lifted centers differ by the prescribed unit step. The fibers over native sites have integer coordinates. These identities identify the image cell by cell. Since $\pi \circ F$ is the inclusion, F is injective. Compactness of X_R and the Hausdorff property of the plane make this continuous injection a closed embedding, giving the stated homeomorphism. $\square$

Theorem 24.12.4.78 (Separation of distinct period copies of a lifted region). Lean Lean Lean Lean Lean Lean ✓ Stmt Let the torus periods w, H be positive, and let $F : X_R \rightarrow \mathbb{R}^2$ be a continuous lift of the inclusion. For $j = (j_1, j_2) \in \mathbb{Z}^2$, write $\delta_j = (j_1 w, j_2 H)$. Two plane points have the same projection exactly when their difference is some δ_j , and

$$\pi^{-1}(X_R) = \bigcup_{j \in \mathbb{Z}^2} (P_L + \delta_j). \quad (24.57)$$

If $j \neq 0$, then

$$F(X_R) \cap (F(X_R) + \delta_j) = \emptyset, \quad (24.58)$$

$$P_L \cap (P_L + \delta_j) = \emptyset. \quad (24.59)$$

In particular, individual closed cells in distinct period copies cannot meet, even along a side or at a corner. If the actual cell union X_R is simply connected and a site in R is fixed, there exists an integer unit-step lift L satisfying this separation for every nonzero j .

This is an auxiliary geometric statement for the block domains of [203, §6.3]. The exterior path-existence argument remains separate; see [114].

Proof. ✓ Proof The additive-circle fibers are integer multiples of the two periods. Comparing a point with the lift of its projection gives the stated union of all period copies. The projection of δ_j is zero. If $F(q) + \delta_j = F(p)$, projecting to the torus gives $q = p$, because the lift is a section of the projection. Subtracting $F(p)$ then gives $\delta_j = 0$. Positivity of both periods forces $j = 0$, a contradiction. The exact image description in Theorem 24.12.4.77 gives the statement for P_L , and restricting to individual cells gives the last separation assertion. Simple connectedness supplies the continuous lift and hence the required integer coordinates. $\square$

Theorem 24.12.4.79 (No finite holes in simply connected closed-cell regions). Lean Lean Lean Lean Lean Lean ✓ Stmt For a finite set $A \subset \mathbb{Z}^2$, put

$$P_A = \bigcup_{a \in A} (a + [-1/2, 1/2]^2). \quad (24.60)$$

This union is locally path connected. If P_A is simply connected, every component of $\mathbb{Z}^2 \setminus A$ under horizontal and vertical unit steps is infinite. More explicitly, no finite set $K \subset \mathbb{Z}^2$ containing a point outside A can have all four neighbors of each of its points in $K \cup A$.

The cancellation used here holds for arbitrary horizontal and vertical real edge values H_a, V_a : the sum over K of $H_a + V_{a+(1,0)} - H_{a+(0,1)} - V_a$ is the corresponding signed sum over the exposed boundary edges of its cells.

Consequently, if the actual torus realization X_R is simply connected and $o \in R$, there exist integer coordinates L which preserve every native internal unit step and for which all missing cell components of the planar realization P_L are infinite. Here $P_L = P_A$ for $A = L(R)$, so the conclusion concerns the genuine image of the torus covering lift. This is an auxiliary no-hole theorem for [203, Theorem 6.9]. The contour argument below strengthens it to paths in the exterior collar.

Proof. [✓ Proof](#) Local path connectedness follows from the finite compact-cell quotient theorem. Suppose K is such a finite candidate and choose a missing cell center in K . Translate it to the origin. Simple connectedness gives a continuous real lift of the angular map on P_A . Subtract this lift from the principal argument and divide by 2π . At cell corners this gives a real potential whose horizontal increments vanish. Its vertical increment is one precisely when the edge crosses the negative real ray from the missing center, and is zero otherwise.

The potential is defined along every exposed side of K , since its neighboring cell belongs to A . Summing its square curls cancels all interior edges and gives zero. The same boundary increments, evaluated by the branch-cut indicator, give total curl minus one. This contradiction excludes the finite candidate, and hence excludes every finite missing-cell component. Finally, the actual covering lift identifies X_R homeomorphically with its integer cell union, so the torus conclusion follows. $\square$

Definition 24.12.4.80 (Safe graph of an exterior integer-cell collar). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A \subset \mathbb{Z}^2$ be finite, with occupied closed-cell union P_A . Its genuine exterior collar is

$$E_A = (P_A + [-3/4, 3/4]^2) \setminus P_A.$$

The exterior band consists of the unoccupied integer centers q such that $\|q - a\|_\infty \leq 1$ for some $a \in A$. Join two such centers by an edge precisely when they are horizontal or vertical neighbors and one common occupied center $a \in A$ satisfies both $\|q - a\|_\infty \leq 1$ and $\|r - a\|_\infty \leq 1$. The common-center requirement controls the whole segment, rather than its endpoints alone.

Lemma 24.12.4.81 (Integer centers of the actual collar). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The closed cell at a consists exactly of the points whose two coordinate distances from a are at most $1/2$. An integer center belongs to P_A exactly when it belongs to A , and belongs to E_A exactly when it belongs to the exterior band.

Proof. [✓ Proof](#) A distinct integer center cannot be within $1/2$ in both coordinates. In the expanded cells the coordinate distances are at most $5/4$; integer distances are therefore at most one. Conversely, a distance at most one splits into an offset of at most $1/2$ within the cell and an offset of at most $1/2$ within the collar square. $\square$

Theorem 24.12.4.82 (Continuous collar paths from safe exterior walks). [Lean](#) [Lean](#) [✓ Stmt](#) Every edge of the safe exterior graph has its entire straight segment inside E_A . Hence any graph walk between two band centers gives a genuine continuous path between them contained in E_A . This is an auxiliary geometric assertion for [203, proof of Theorem 6.9]. The contour theorem below derives connectivity of the safe graph from simple connectedness of the occupied closed-cell union.

Proof. [✓ Proof](#) Both endpoint centers lie within coordinate distance one of their common occupied center. Convexity preserves these bounds along the segment, which is therefore inside the collar enlargement. If a closed occupied cell met a horizontal or vertical unit segment, integer arithmetic would force its center to be one of the two endpoints. Both endpoints are unoccupied, so the segment is outside P_A . Concatenating the edge segments proves the path assertion. $\square$

Definition 24.12.4.83 (Occupied square contact graph). [Lean](#) [✓ Stmt](#) Let $A \subset \mathbb{Z}^2$ be finite, and let Q_a be the closed unit square centered at a . The contact graph has vertex set A ; distinct a, b are adjacent when $|a_1 - b_1| \leq 1$ and $|a_2 - b_2| \leq 1$. Thus shared corners, as well as shared sides, are retained.

Theorem 24.12.4.84 (Connectivity through occupied square contacts). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Write $P_A = \bigcup_{a \in A} Q_a$. If P_A is preconnected, then every two occupied centers can be joined in the contact graph. In particular, if P_A is simply connected, its contact graph is connected. This auxiliary geometric consequence supports the block decomposition in [203, Theorem 6.9], local source lines 1935–1990. Four-neighbor connectivity of the occupied centers is not assumed.

Proof. [✓ Proof](#) A point in $Q_a \cap Q_b$ places each coordinate of a and b within one half of that point, so their coordinate differences have absolute value at most one. Fix an occupied center u and partition A into the centers reachable from u and the remaining centers. The corresponding square unions are closed, since A is finite. If both are nonempty, preconnectedness of P_A forces an intersection between the unions. The intersecting cells give a contact edge across the partition, a contradiction. Simple connectedness also gives nonemptiness of P_A , hence of A . $\square$

Definition 24.12.4.85 (Horizontal-ray potential of a finite edge flow). [Lean](#) [✓ Stmt](#) Index each oriented lattice edge by its starting corner $a - (1/2, 1/2)$, with horizontal edges pointing east and vertical edges pointing north. For a finitely supported signed vertical flow $V : \mathbb{Z}^2 \rightarrow \mathbb{Z}$, define its ray potential at the cell center q by

$$W(q) = \sum_{\substack{a_2=q_2 \\ q_1 < a_1}} V(a).$$

The sum is finite because V is finitely supported.

Theorem 24.12.4.86 (Discrete gradients of the contour ray potential). [Lean](#) [Lean](#) [✓ Stmt](#) Write $e = (1, 0)$ and $n = (0, 1)$. For any finitely supported signed vertical flow,

$$W(q + e) - W(q) = -V(q + e).$$

If a finitely supported horizontal flow H satisfies conservation

$$H(a) + V(a) - H(a - e) - V(a - n) = 0 \quad (a \in \mathbb{Z}^2),$$

then also $W(q + n) - W(q) = H(q + n)$. These are auxiliary winding-potential identities for the contour argument associated with [203, proof of Theorem 6.9]. Identification of the flow with an actual exposed contour and connectivity of that contour are separate geometric assertions.

Proof. [✓ Proof](#) Moving the ray one cell east removes exactly the vertical edge indexed by $q + e$. Conservation writes the difference between the vertical flow and its north translate as the east translate of the horizontal flow minus the horizontal flow itself. Apply the additive finite ray sum to this equality. Translation of the finite supports and the east-increment identity telescope the horizontal difference to the single term $H(q + n)$. $\square$

Theorem 24.12.4.87 (Connected missing-cell lattice of a simply connected region). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A \subset \mathbb{Z}^2$ be finite, and suppose its closed-cell union $P_A = \bigcup_{a \in A} (a + [-1/2, 1/2]^2)$ is simply connected. Then the four-neighbor lattice on $\mathbb{Z}^2 \setminus A$ is connected. Consequently, a simply connected actual torus region admits an integer unit-step lift whose entire missing-cell lattice is connected.

These are plane-complement walks. They need not stay in the exterior collar, so their torus projections are not yet asserted to avoid all period copies of the region. The following contour theorem restricts the walks to the genuine exterior collar, as required in [203, proof of Theorem 6.9].

Proof. ✓ Proof Enclose A in a finite integer box. All vertices outside that box can be connected to one exterior corner by integer row and column walks avoiding the box. Every missing-cell component is infinite by the preceding no-hole theorem, and hence contains a vertex outside the box. Each component therefore meets the same exterior component, proving connectedness. The genuine torus lift supplied by the no-hole theorem has the same infinitude property, so the argument applies to its integer centers as well. $\square$

Theorem 24.12.4.88 (Connected exterior cell contour). Lean ✓ Stmt Let $A \subset \mathbb{Z}^2$ be finite and suppose that the actual union $P_A = \bigcup_{a \in A} (a + [-1/2, 1/2]^2)$ is simply connected. Let B_A consist of the unoccupied centers whose coordinate distances from some occupied center are at most one. Join four-neighbor centers in B_A whenever one occupied center is within coordinate distance one of both. Then this graph is connected. This is auxiliary exterior-collar geometry for [203, proof of Theorem 6.9], local source lines 1935–2076. Shared corners of occupied cells are allowed; simple connectedness refers to the genuine closed-cell union.

Proof. ✓ Proof Orient each exposed cell edge with its occupied cell on the left. At an edge endpoint, continue with a right turn when the diagonal cell is occupied, straight ahead when only the forward cell is occupied, and a left turn otherwise. The reverse rule makes this successor a permutation of the finite exposed-edge set.

Fix one successor orbit and assign weight one to its edges and zero to the remaining edges. The resulting finite flow is conserved. Its horizontal-ray winding potential is constant on missing cells, since their four-neighbor graph is connected, and vanishes far from the occupied cells. It is also constant on occupied cells: the edge identities give equality across shared sides, and the right-turn rule gives equality across shared corners. The occupied contact graph is connected. Across any exposed edge the winding difference is its weight. Thus every exposed edge has the same weight, and all belong to the chosen successor orbit.

Consecutive contour edges have exterior centers that are equal or joined by one or two four-neighbor steps with a common occupied-cell support. Every missing center in B_A can reach one of these exterior centers using the same local support. These walks prove graph connectivity. Nonemptiness follows by taking an occupied center with maximal first coordinate and the missing center to its east. $\square$

Theorem 24.12.4.89 (Exclusion of foreign period copies from an exterior collar). Lean Lean ✓ Stmt Let $w, H > 0$, and let a genuine continuous lift of the actual closed-cell region X_R have planar image $P = \bigcup_{v \in R} (L(v) + [-1/2, 1/2]^2)$ with integer centers. Put $E = [-3/4, 3/4]^2$, and let $P_j = P + (wj_1, Hj_2)$ be its period copies. Then

$$(P + E) \cap P_j = \emptyset \quad (j \neq 0), \quad (24.61)$$

$$\pi((P + E) \setminus P) \subseteq X_R^c. \quad (24.62)$$

Here addition of sets is Minkowski addition. If X_R is simply connected and contains a native site, there is an integer unit-step lift L with this exterior-collar property; no covering lift is supplied in that conclusion.

This is auxiliary exterior geometry for [203, proof of Theorem 6.9]. Collar paths between exterior integer centers follow from the connected safe exterior graph established above.

Proof. ✓ Proof Disjoint closed unit squares with integer centers differ by at least two in some coordinate; otherwise their midpoint belongs to both. A point in an enlarged cell and in a foreign unit cell would put the two center coordinates within $1/2 + 3/4 + 1/2 = 7/4 < 2$ in both directions, a contradiction. The full preimage of X_R is the union of the period copies P_j . A point

of the exterior collar lies in neither P nor any foreign copy, proving the projection assertion. In the simply connected case, covering-space lifting supplies the genuine lift and its integer centers, so the same argument applies. $\square$

Theorem 24.12.4.90 (Simply connected integer-cell lift with an exterior collar). [Lean](#) [✓ Stmt](#)
Let R be a finite region of a torus with positive integer periods, and assume its actual closed-cell realization X_R is simply connected. There exists an integer lift L of the occupied vertices, preserving each internal unit step. Put $A = L(R)$ and let P_A be the union of the closed unit squares centered at A . Then P_A is simply connected, and every point of its exterior collar projects outside X_R :

$$\pi((P_A + [-\frac{3}{4}, \frac{3}{4}]^2) \setminus P_A) \cap X_R = \emptyset. \quad (24.63)$$

Both conclusions hold for the same integer lift. This is an auxiliary geometric consequence for the block decomposition in [203, Theorem 6.9], local source lines 1935–1990.

Proof. [✓ Proof](#) Simple connectedness gives a continuous lift of the covering projection over X_R . Its image is homeomorphic to X_R and is precisely P_A , so P_A is simply connected. Distinct period copies of the lifted cells are disjoint. The separation of integer centers implies that the enlargement by $[-\frac{3}{4}, \frac{3}{4}]^2$ misses every other period copy. A point in the exterior collar also misses P_A itself; hence it cannot project to X_R , proving (24.63). $\square$

Theorem 24.12.4.91 (Crossing numbers of projected integer steps). [Lean](#) [Lean](#) [✓ Stmt](#) *Let the torus periods w, H be at least three. For $a \in \mathbb{Z}^2$, put*

$$D(a) = (\lfloor a_1/w \rfloor, \lfloor a_2/H \rfloor). \quad (24.64)$$

If a, b differ by one horizontal or vertical unit step, then the oriented crossing pair of their projected native edge is $D(b) - D(a)$. This applies to interior, boundary, and exterior steps alike.

Proof. [✓ Proof](#) In a positive horizontal step, Euclidean division changes precisely when the standard residue crosses the horizontal seam. The vertical case is identical. Reversing a step negates its crossing pair. $\square$

Theorem 24.12.4.92 (Lifted exterior endpoints of a region route). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *A native site center belongs to the closed-cell union X_R exactly when its site belongs to R . For the crossing and walk assertions, take both torus periods at least three. Suppose L is an integer lift preserving the unit steps within R , and put $P = \bigcup_{v \in R} (L(v) + [-1/2, 1/2]^2)$. An edge from $v \in R$ to $t \notin R$ lifts to a unit step from $L(v)$ to an integer point b projecting to t , with*

$$b \in (P + [-3/4, 3/4]^2) \setminus P, \quad (24.65)$$

$$N([v, t]) = D(b) - D(L(v)). \quad (24.66)$$

More generally, let $p : v_0 \rightarrow v$ be any native region walk, and let t_0, t be exterior neighbors of its endpoints. There are lifted exterior integer endpoints b_0, b in the same collar, projecting to t_0, t , such that

$$N([t_0, v_0] * p * [v, t]) = D(b) - D(b_0). \quad (24.67)$$

Concatenation is written in traversal order. This supplies the endpoints and winding pair required in the complement argument of [203, proof of Theorem 6.9]; the connected safe exterior graph gives a collar path between them.

Proof. ✓ Proof A site center in a closed cell differs from its integer center, modulo a period vector, by an integer in each interval $[-1/2, 1/2]$. Both differences vanish, proving the center assertion. Lift a boundary edge by its horizontal or vertical unit direction. Its exterior endpoint is the sum of its midpoint in the interior closed cell and a half-unit displacement, so it belongs to the collar. It cannot belong to P , since projection would put its exterior native center in X_R . The preceding theorem gives the crossing pair of this step. The interior walk has crossing pair $D(L(v)) - D(L(v_0))$; adding the two boundary crossings cancels the interior endpoint coordinates. $\square$

Theorem 24.12.4.93 (Exterior lattice walks from a supplied plane path). Lean ✓ Stmt *Let the torus periods w, H be at least three, and let X_R be the actual closed-cell realization of a finite native region R . Let $a, b \in \mathbb{Z}^2$ project to exterior native vertices v, t . Suppose a continuous plane path γ joins a to b and its torus projection avoids X_R at every time. There is a walk $Q : v \rightarrow t$ in the induced native complement with crossing numbers*

$$N(Q) = \left(\left\lfloor \frac{b_1}{w} \right\rfloor - \left\lfloor \frac{a_1}{w} \right\rfloor, \left\lfloor \frac{b_2}{H} \right\rfloor - \left\lfloor \frac{a_2}{H} \right\rfloor \right). \quad (24.68)$$

Thus the lattice walk has the prescribed lifted endpoint displacement. This is an auxiliary path approximation for the complement argument in [203, proof of Theorem 6.9]. The path γ is supplied in this approximation theorem; the preceding contour construction provides such paths for the required exterior centers of simply connected regions.

Proof. ✓ Proof An exterior point belongs to an open rectangle containing one cell, two cells sharing a side, or four cells sharing a corner, all with exterior lattice centers. At a side or corner every touching closed cell must be exterior, since the point avoids X_R . These rectangles cover the lifted exterior. Subdivide the compact parameter interval so that each subpath lies in one rectangle. Rounded endpoints in that rectangle can be connected through exterior lattice centers by a horizontal and a vertical step. Concatenating gives an integer lattice walk between a and b . Projection gives a native complement walk. The crossing pair of each unit step is the change in its integer deck coordinates; summation telescopes to the stated endpoint difference. $\square$

Lemma 24.12.4.94 (Closure gauges from a supplied integer lift). Lean Lean Lean Lean Lean ✓ Stmt *Let both torus circumferences w, H be at least three, and let R be a set of native torus vertices. Suppose a supplied map $L : R \rightarrow \mathbb{Z}^2$ projects modulo (w, H) to the original vertices and preserves every internal rightward and upward unit step. In particular, L is injective. The arithmetic deck coordinate $\lfloor j/n \rfloor$ increases by one under $j \mapsto j+1$ precisely when $j+1 \equiv 0 \pmod{n}$, and otherwise stays unchanged. For commuting $g, h \in G$, define*

$$k_v = h^{\lfloor L_x(v)/w \rfloor} g^{-\lfloor L_y(v)/H \rfloor}. \quad (24.69)$$

On a native rightward step the gauge changes by left multiplication by h precisely at the horizontal seam; on an upward step it changes by left multiplication by g^{-1} precisely at the vertical seam. For every internal edge, ordered from s to t ,

$$k_t k_s^{-1} = u_e, \quad k_t^{-1} u_e k_s = 1, \quad (24.70)$$

where the native closure assignment is h^{-1} on a horizontal seam edge, g on a vertical seam edge, and the identity elsewhere. Right normalization $k_v \mapsto k_v k_o^{-1}$ preserves the gradient. For finite G , this normalized gauge equals the gauge obtained from any spanning tree rooted at o ; hence all its cycle residuals equal the identity.

This is an auxiliary consequence of the supplied arithmetic lift, supporting the seam and disentangling arguments in [203, §5.3 and Theorem 6.9]. It neither defines a disk nor establishes lifts for arbitrary disk regions. The latter geometric theorem and the removal of sector-dependent crossing transports remain separate, as recorded in [114].

Proof. ✓ Proof Euclidean division shows that the quotient of an integer coordinate increases by one exactly when its residue wraps through zero. The unit-step hypotheses therefore give the two gauge-step identities; commutativity moves the vertical factor past the horizontal factor. A seam reverses the native ordered-edge orientation, yielding h^{-1} and g in the respective gradients. All other internal gradients are the identity. Right normalization cancels in each gradient. Uniqueness in the tree-gradient bijection identifies the normalized gauge with the spanning-tree gauge. $\square$

Theorem 24.12.4.95 (Twisted canonical block in spanning-tree coordinates). Lean ✓ Stmt
Choose a spanning tree T rooted at o in the finite region R , and insert arbitrary operators $u_e \in G$ on its incident bonds. Let $k_o = 1$, $k_t k_s^{-1} = u_e$ on T , and $\omega_e = k_t^{-1} u_e k_s$ on the internal edges. Let θ' be the crossing configuration transported by (24.53). In the original physical coordinates $\alpha(y, a, r, z)$, the actual twisted canonical matrix satisfies

$$M_R^u(\alpha(y, a, r, z), \theta) = |G|^{-|R|} \sum_{x \in G} \mathbf{1}_{\{y=x\theta'\}} \mathbf{1}_{\{z_e=x\omega_e x^{-1} \ (e \in I \setminus T)\}}. \quad (24.71)$$

The reference labels a and rooted vertex labels r do not affect the coefficient. Operators on crossing bonds are allowed, and no condition is imposed on the internal residuals or the size of the boundary. This is an actual finite-contraction form of the regular coordinate changes in [203, §6.2–6.3].

Proof. ✓ Proof Expand the original contraction into its internal bond labels η and local translations q . On the tree put $q'_v = q_v k_v$ and $\eta'_e = k_s^{-1} \eta_e$. The endpoint equations become $r_s a_e = q'_s \eta'_e$ and $r_t a_e = q'_t \eta'_e$. Rooted tree-gradient uniqueness yields $q'_v = r_v x$, with $x = q_o$. Thus $q_v = r_v x k_v^{-1}$, and internal references are uniquely fixed as $\eta_e = k_s x^{-1} a_e$. On each remaining internal edge the head equation is equivalent to $z_e = x \omega_e x^{-1}$.

At a crossing edge, the unique reference equals $k_s x^{-1} y_e$ when $s \in R$, and $u_e^{-1} k_t x^{-1} y_e$ when $t \in R$. Fixing these references to θ is exactly $y = x\theta'$. Conversely, these boundary and cycle equations reconstruct a compatible pair (η, q) . Hence the compatible pairs are in bijection with the surviving $x \in G$, and the original local averages contribute the factor $|G|^{-|R|}$ in (24.71). $\square$

Theorem 24.12.4.96 (Mixed Gram matrices of twisted canonical blocks). Lean Lean ✓ Stmt
Fix a region R , a spanning tree, and a root, and let u, w be arbitrary bond insertions, including crossing operators. Write ω^u, ω^w for their cycle residual tuples and b_u, b_w for their respective boundary transports. Let N be the number of pairs (a, r) of internal references and vertex labels normalized at the root. The actual canonical matrices obey

$$((M_R^u)^\dagger M_R^w)(\theta, \vartheta) = |G|^{-2|R|} N \sum_{x, t \in G} \mathbf{1}_{\{x b_u(\theta) = t b_w(\vartheta)\}} \mathbf{1}_{\{x \omega^u x^{-1} = t \omega^w t^{-1}\}}. \quad (24.72)$$

In particular, if no $s \in G$ satisfies $\omega^u = s \omega^w s^{-1}$ coordinatewise, then $(M_R^u)^\dagger M_R^w = 0$. Thus these two actual blocks have orthogonal physical ranges. The boundary may be empty; neither flatness nor a mixed Gram identity is assumed. This is a finite-region comparison supporting the coherent closure arguments in [203, §6.3–6.4].

Proof. [✓ Proof](#) Reindex the physical row sum by the reversible spanning-tree coordinates and substitute (24.71) for each coefficient. The factors indexed by (a, r) are constant, contributing N . Summing over the shared boundary and cycle coordinates pairs the two indicators, leaving precisely the equalities in (24.72). Any surviving cycle equality implies $\omega^u = (x^{-1}t)\omega^w(x^{-1}t)^{-1}$. If simultaneous conjugacy fails, every summand vanishes. $\square$

Theorem 24.12.4.97 (Vertex gauges in an actual regular-region contraction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose each original site coefficient is invariant under simultaneous left translation of its incident regular labels. Let $k_v \in G$ be arbitrary gauges at the vertices of a finite region R , and let u_e be arbitrary inserted operators, including those on crossing bonds. Put $\tilde{u}_e = k_t^{-1}u_e k_s$ on internal edges $e = (s, t)$, and put $\tilde{u}_e = 1$ on all other edges. With the transported crossing configuration θ' of (24.53), the actual physical columns satisfy

$$T_R^u(\theta) = T_R^{\tilde{u}}(\theta'). \quad (24.73)$$

If every internal residual is the identity, this reduces to $T_R^u(\theta) = T_R(\theta')$ for the original untwisted region. No connectedness or Gram identity is required. These identities implement the regular-coordinate changes in [203, §6.2–6.3] inside the original finite contraction.

Proof. [✓ Proof](#) Reindex the actual sum over incident edge references. On an internal edge multiply the reference by k_s^{-1} ; on a crossing edge use the transport in (24.53). Each change is a bijection. Its boundary restriction is exactly $\theta \mapsto \theta'$. At every region vertex, the resulting twisted half-edge labels are k_v^{-1} times the original twisted labels. Local simultaneous invariance therefore leaves each site coefficient unchanged, and hence leaves their product and the contracted sum unchanged. $\square$

Lemma 24.12.4.98 (The transported boundary projector). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Enumerate the crossing bonds of a region by $\{1, \dots, b\}$, and let F be the resulting permutation of boundary configurations induced by the vertex gauge and the crossing operators. The pulled-back averaging projector is

$$\Pi_{\partial R}^{k,u}(x, y) = \Pi_{\partial R}(F(x), F(y)). \quad (24.74)$$

It is idempotent. If $b \geq 1$, it is self-adjoint, has rank $|G|^{b-1}$, and has trace $|G|^{b-1}$. This coordinate change is part of the boundary disentangling argument of [203, Theorem 6.9]; it imposes no condition on internal twists.

Proof. [✓ Proof](#) Reindex the common summation variable in the square of the matrix by F . Idempotence then follows from that of the original averaging projector. Taking the adjoint commutes with applying the same permutation to both indices. Finally, a basis permutation preserves rank, and reindexing the diagonal sum preserves trace. $\square$

Theorem 24.12.4.99 (The actual Gram matrix with removable internal operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Number the crossing bonds and write F for the boundary permutation induced by an arbitrary vertex gauge k and inserted operators u . If the original sites have simultaneous regular invariance, the actual ordered coefficient matrices satisfy

$$M_R^u(\sigma, x) = M_R^{\tilde{u}}(\sigma, F(x)). \quad (24.75)$$

Suppose further that R is connected, the sites are G -isometric, and every internal residual $k_t^{-1}u_e k_s$ is the identity. Then $M_R^u(\sigma, x) = M_R(\sigma, F(x))$ and there is a positive scalar c_R such that

$$(M_R^u)^\dagger M_R^u = c_R \Pi_{\partial R}^{k,u}. \quad (24.76)$$

If the boundary contains $b > 0$ bonds, the actual matrix has rank $|G|^{b-1}$. The Gram statement allows an empty boundary and derives the block Gram identity from the local sites. It is the removable-operator case of the boundary-isometry argument in [203, Theorem 6.9], not the unrestricted disk theorem [114].

Proof. ✓ **Proof** The ordered coefficient identities follow from the actual contraction identity in Theorem 24.12.4.97. With identity internal residuals, the matrix is the untwisted block with its columns permuted by F . Local regular isometry and connectedness give $M_R^\dagger M_R = c_R \Pi_{\partial R}$ with $c_R > 0$. Pulling both boundary indices back along F therefore gives the claimed Gram matrix. Its rank equals the rank of the actual matrix because $\text{rank}(M^\dagger M) = \text{rank}(M)$; the positive scalar does not change the rank of the boundary projector. $\square$

Theorem 24.12.4.100 (Actual closure-region Grams from an integer lift). Lean Lean Lean
✓ **Stmt** Let the torus circumferences be at least three, and let A be a native regular G -isometric four-leg tensor. Equip a region R with an integer lift preserving all internal unit steps, and let g, h commute. For the actual region matrix $M_R^{g,h}$ of the native closure, the explicit lift gauge k gives

$$M_R^{g,h}(\sigma, x) = M_R(\sigma, F_{g,h}(x)). \quad (24.77)$$

If R is connected, there is a positive scalar c_R such that

$$(M_R^{g,h})^\dagger M_R^{g,h} = c_R \Pi_{\partial R}^{k,u(g,h)}. \quad (24.78)$$

When the boundary has $b > 0$ bonds, $\text{rank } M_R^{g,h} = |G|^{b-1}$. These conclusions follow from the supplied arithmetic lift, without any separate flatness or block Gram hypothesis. They support the seam-deformation argument of [203, Theorem 6.9]; they do not establish lifts for arbitrary disk regions or identify boundary transports for superpositions of distinct closures [114].

Proof. ✓ **Proof** Express the native four-leg tensor on its four incident graph edges. This preserves local regular isometry. The arithmetic lift gauge has the native closure assignment as its internal gradient, so every internal residual equals the identity. Apply Theorem 24.12.4.99 to the resulting actual region matrix, with the boundary transport of this explicit gauge. $\square$

Theorem 24.12.4.101 (Actual open-region rank from geometric simple connectedness). Lean
✓ **Stmt** Let both torus circumferences be at least three and let A be a native regular G -isometric four-leg tensor. Suppose the induced native graph on a finite region R is connected and the actual closed-cell realization of R in the real torus is simply connected. For a commuting closure pair g, h and a numbering of its $b > 0$ crossing bonds, the actual twisted open-region coefficient matrix satisfies

$$\text{rank } M_R^{g,h} = |G|^{b-1}. \quad (24.79)$$

No arithmetic lift, residual-flatness condition, or block Gram identity is supplied. Geometric simple connectedness and native graph connectivity are separate assumptions: closed cells may meet at diagonal points. This is an auxiliary open-region rank consequence of the boundary calculation in [203, Theorem 6.9], not the full disk entropy theorem or a coherent-superposition result [114].

Proof. ✓ **Proof** Graph connectivity supplies a region vertex. Simple connectedness of the actual cell realization gives an integer lift preserving every internal native unit step by Lemma 24.12.4.76. Apply the actual integer-lifted region rank theorem, Theorem 24.12.4.100, to this lift. $\square$

Proof. ✓ Proof Apply the native walk product to the free generators. Evaluation commutes with each product and inverse, giving the residual holonomy. The tree gauge removes each tree operator and retains exactly the prescribed cycle residual on every other edge. Endpoint gauge covariance gives the first two identities. Conjugating every residual operator conjugates the whole ordered product. At the normalized root the endpoint gauges are both the identity. $\square$

Theorem 24.12.4.107 (Cycle-controlled permutation of actual boundary coordinates). Lean
Lean Lean Lean Lean Lean ✓ Stmt Fix spanning-tree physical coordinates (y, a, r, z) of a finite region and any function $W : G^C \rightarrow G^{\partial R}$ of its cycle labels. The permutation $\sigma_W(y, a, r, z) = (W(z)^{-1}y, a, r, z)$ is reversible, with inverse $(y, a, r, z) \mapsto (W(z)y, a, r, z)$, and its matrix U_W on the original half-edge physical basis is unitary. In particular, fixed actual complementary walks define such a matrix by evaluating their cycle words; this construction is independent of all bond assignments.

For the actual canonical block M_R^u , let b_u be its boundary transport and ω its cycle residual tuple. Its transformed coefficients satisfy

$$(U_W M_R^u)(\alpha(y, a, r, z), \theta) = |G|^{-|R|} \sum_{x \in G} \mathbf{1}_{\{y = W(z)^{-1} x b_u(\theta)\}} \mathbf{1}_{\{z = x \omega x^{-1}\}}. \quad (24.89)$$

The cycle constraint is retained without any assumption on the relation between W and the crossing transports. This is an algebraic coordinate operation associated with [203, §6.3].

Proof. ✓ Proof The inverse translates only y , so it leaves its own controlling tuple z fixed. Group cancellation proves the inverse identities. A permutation of the physical group basis is unitary. Its action on coefficient functions is precomposition with σ_W^{-1} ; substituting $W(z)y$ for y in (24.71) gives (24.89). $\square$

Theorem 24.12.4.108 (Combined boundary transport removal from supplied complement walks). Lean ✓ Stmt Let S be the complement of R , and identify their crossing bonds by the same native edges. Choose a region gauge k_R and a rooted spanning tree in S , with normalized gauge k_S and cycle residuals ω_S . Let F_R, F_S be the actual boundary transports and set $M = F_S \circ F_R^{-1}$ under this crossing identification. Write $m_f = M(\mathbf{1})_f$, so that $M(\theta)_f = m_f \theta_f$. Fix a reference crossing bond f_0 and an actual walk p_f entirely in the induced graph on S for each crossing bond f . Assume

$$\text{ev}_{\omega_S} W_T(p_f) = m_f m_{f_0}^{-1}. \quad (24.90)$$

Define $W(z)_f = \text{ev}_z W_T(p_f)$. With the complementary boundary input reread from $F_R^{-1}\theta$, the actual coefficients satisfy

$$(U_W M_S^u)(\alpha_S(y, a, r, z), F_R^{-1}\theta) = |G|^{-|S|} \sum_{x \in G} \mathbf{1}_{\{y_f = x m_{f_0} \theta_f \ (f \in \partial S)\}} \mathbf{1}_{\{z = x \omega_S x^{-1}\}}. \quad (24.91)$$

Here θ_f denotes the label on the corresponding crossing bond of R . This is an auxiliary conditional statement in the complement argument of [203, §6.3]. Constructing the required complement walks with (24.90) remains the geometric step described in [114].

Proof. ✓ Proof At the reindexed input, the actual complementary boundary transport is $M(\theta)_f = m_f \theta_f$. On the cycle support, conjugation of the constructed words gives

$$W(x \omega_S x^{-1})_f = x m_f m_{f_0}^{-1} x^{-1}, \quad (24.92)$$

$$W(z)_f^{-1} x m_f \theta_f = x m_{f_0} \theta_f. \quad (24.93)$$

Substituting (24.93) into (24.89) retains the cycle indicator. $\square$

Theorem 24.12.4.109 (Relative boundary factor of actual controlled complementary blocks).

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Assume the actual complement walks satisfy the relative-word identities of (24.90). Number the $b = n + 1$ crossing bonds and use reference boundary coordinates $y = (t, d)$ and $\theta = (a, r)$, where $d, r \in G^n$. For the actual controlled complementary matrix at the reindexed input $F_R^{-1}\theta$, put $x = ta^{-1}m_{f_0}^{-1}$. Its coefficient is

$$|G|^{-|S|} \mathbf{1}_{\{d=r\}} \mathbf{1}_{\{z=x\omega_S x^{-1}\}}. \quad (24.94)$$

The normalized boundary factor is $|G|^{-n/2} \mathbf{1}_{\{d=r\}}$. Its remaining factor is $|G|^{n/2-|S|} \mathbf{1}_{\{z=x\omega_S x^{-1}\}}$. The trees, numbering, and control walks may be fixed before an arbitrary finite family of bond assignments. The same factor then separates from every coherent linear combination of their actual complementary matrices; the residual coefficients are combined with the same weights.

This statement retains the virtual reference label a in the residual coefficient. It is an auxiliary coefficient statement, rather than a conclusion about the full physical bipartite state; for commuting torus closures, the complementary loops of Theorem 24.12.4.125 realize its geometric walk condition.

Proof. [✓ Proof](#) Equality under a common left translation is equivalent to equality of all relative labels and equality of the reference labels. The latter fixes $x = ta^{-1}m_{f_0}^{-1}$ uniquely. The finite translation sum therefore has one surviving term. Multiplication by $|G|^{n/2}$ isolates the normalized maximally entangled coefficient, and linearity gives the coherent sum identity. $\square$

Theorem 24.12.4.110 (Common boundary factor of actual controlled canonical cuts). [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Retain the same actual complement walk identities, and suppose every actual cycle residual in R is the identity. Contract the two canonical region matrices, applying the fixed controlled physical permutation only in S . Reindexing their common boundary sum by F_R preserves the original contraction.

Let $d_R, d_S \in G^n$ be the relative boundary labels. In the spanning-tree physical coordinates the actual contracted coefficient is

$$|G|^{-|R|-|S|} \mathbf{1}_{\{d_R=d_S\}} \mathbf{1}_{\{z_R=1\}} \sum_{x \in G} \mathbf{1}_{\{z_S=x\omega_S x^{-1}\}}. \quad (24.95)$$

In particular neither reference boundary label nor the common multiplier m_{f_0} occurs in the result.

Let N_R be the number of choices of the reference boundary label, internal reference labels, and normalized vertex labels in R . The fixed region ancillary vector is $\phi_R = N_R^{-1/2} \mathbf{1}_{\{z_R=1\}}$; it has norm one. For any finite sector family obeying the stated actual residual and walk conditions, and any coefficients λ_i , the coherent cut sum has the exact factorization

$$\Omega_{|G|^n}(d_R, d_S) \phi_R \left(|G|^{n/2} \sqrt{N_R} |G|^{-|R|-|S|} \sum_i \lambda_i \sum_{x \in G} \mathbf{1}_{\{z_S=x\omega_{S,i} x^{-1}\}} \right). \quad (24.96)$$

Both normalized factors are fixed before the sector family and its coefficients; all sector dependence is on the complementary cycle coordinates. These are auxiliary canonical-cut statements associated with [203, Theorem 6.9]. The actual geometric conditions and transport to the original physical tensors remain separate steps, as recorded in [114].

Proof. [✓ Proof](#) First change the common boundary summation variable by its reversible region transport. Substitute the two actual spanning-tree coefficient formulas. For each complementary translation, the complementary boundary indicator fixes the shared boundary configuration. Its relative labels equal d_S , and the interior translation sum gives $\mathbf{1}_{\{d_R=d_S\}}$. This proves (24.95).

There are exactly N_R unrestricted region ancillary configurations, while its cycle tuple is fixed, proving normalization. Extract the two square-root normalization factors and use linearity for the finite sector sum. $\square$

Theorem 24.12.4.111 (A common canonical torus cut factor from simple connectedness and winding). [Lean](#) [✓ Stmt](#) *Let both torus dimensions be at least three. Suppose the actual closed-cell realization of R is simply connected. Choose rooted spanning trees in R and S , number their nonempty crossing bonds by $\{0, \dots, n\}$, and choose fixed actual complementary root loops p_f . Assume their ambient torus windings equal those of the native two-tree routes from the reference crossing to f .*

There exists one unitary U on the original complementary half-edge physical basis, chosen before all closure sectors and coefficients, with the following property. For every finite family of commuting pairs (g_i, h_i) and coefficients λ_i , form the actual canonical cut contractions

$$K_i(\alpha, \beta) = \sum_{\theta} M_R^{g_i, h_i}(\alpha, \theta) (U M_S^{g_i, h_i})(\beta, \theta), \quad (24.97)$$

where the same native crossing edges are reread on the two sides. In the fixed spanning-tree and relative boundary coordinates, their coherent sum is

$$\sum_i \lambda_i K_i = \Omega_{|G|^n} \otimes \phi_R \otimes \chi_\lambda, \quad (24.98)$$

with the same normalized maximally entangled factor and the same normalized region ancillary vector as in Theorem 24.12.4.110. The remaining complementary coefficient is

$$\chi_\lambda(z_S) = |G|^{n/2} \sqrt{N_R} |G|^{-|R|-|S|} \sum_i \lambda_i \sum_{x \in G} \mathbf{1}_{\{z_S = x \omega_{S,i} x^{-1}\}}. \quad (24.99)$$

Thus no group-valued flatness, relative-word identity, mixed Gram formula, or tensor factorization is supplied as a hypothesis. This is an auxiliary canonical-cut consequence in the argument of [203, Theorem 6.9]. Existence of the required complementary loops for the paper's disk geometry and transport to the original physical site tensors remain separate steps; see [114].

Proof. [✓ Proof](#) Simple connectedness of the actual realization supplies an integer coordinate lift. For every commuting pair its normalized gauge gives identity region cycle residuals. Equality of the supplied loop windings with the native routes gives the actual complement relative-word identities simultaneously for all commuting pairs. Evaluate the fixed walk words to construct U , then apply the coherent canonical-cut factorization. Reindexing the shared boundary sum gives the original contraction displayed above. $\square$

Definition 24.12.4.112 (Fixed boundary embedding and density). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let D be a nonempty finite set and $\phi \in \mathbb{C}^A$. Define $J_\phi : \mathbb{C}^D \rightarrow \mathbb{C}^{D \times A}$ by $(J_\phi)_{(d,a),s} = \mathbf{1}_{\{d=s\}} \phi(a)$, and set

$$\rho_\phi = |D|^{-1} J_\phi J_\phi^\dagger, \quad K_{\phi,\chi} = |D|^{-1/2} J_\phi J_\chi^\top.$$

Theorem 24.12.4.113 (Reduced matrix of a fixed boundary factor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If $\|\phi\| = 1$, then J_ϕ is an isometry and ρ_ϕ is a positive density of trace one and rank $|D|$, satisfying $\rho_\phi^2 = |D|^{-1} \rho_\phi$. Moreover,*

$$K_{\phi,\chi} K_{\phi,\chi}^\dagger = \|\chi\|^2 \rho_\phi.$$

These identities persist under bijective changes of physical coordinates. A unitary acting on the complementary indices leaves the reduced matrix unchanged. If a nonzero coefficient matrix has reduced matrix $a\rho_\phi$, its trace is positive and trace normalization gives exactly ρ_ϕ .

Proof. [✓ Proof](#) The Gram matrix is $J_\phi^\dagger J_\phi = \|\phi\|^2 I$. Matrix multiplication gives the reduced identity. Cyclicity of trace, the isometry rank identity, and one further multiplication give the density properties. Complementary unitarity cancels between a matrix and its adjoint; positivity and nonvanishing give positive trace. $\square$

Definition 24.12.4.114 (Fixed regular reference-boundary density). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Number the $n + 1$ actual crossing bonds. The spanning-tree physical coordinates identify the region rows with (d, a) , where $d \in G^n$ is the relative boundary label and a consists of its reference label, internal references, normalized vertex labels, and cycle residuals. Pull back ρ_{ϕ_R} to the native half-edge physical rows, with the fixed normalized ancillary vector ϕ_R of Theorem 24.12.4.110. Denote this density by ρ_R^0 .

Theorem 24.12.4.115 (Spectrum of the fixed regular reference density). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) The density ρ_R^0 is positive, has trace one and rank $|G|^n$, and satisfies $(\rho_R^0)^2 = |G|^{-n} \rho_R^0$. Its von Neumann entropy and every nonnegative finite-order Rényi entropy are $n \log |G|$.

Proof. [✓ Proof](#) The fixed ancillary vector has norm one. Apply the fixed-factor density identities and transport them through the actual coordinate bijection. $\square$

Theorem 24.12.4.116 (Common reduced density of actual canonical closure sums). [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) Retain the actual simple connectedness, fixed spanning trees, crossing numbering, and fixed complementary winding loops of Theorem 24.12.4.111. For a finite commuting closure family, let C_λ be its coherent actual canonical cut matrix, before the complementary controlled unitary. Then the fixed density ρ_R^0 , chosen before the sectors and coefficients, satisfies

$$C_\lambda C_\lambda^\dagger = \text{tr}(C_\lambda C_\lambda^\dagger) \rho_R^0.$$

If $C_\lambda \neq 0$, its trace is positive and its normalized reduced density equals ρ_R^0 . Consequently its rank is $|G|^n$, its nonzero spectrum is flat, and its von Neumann and all nonnegative finite-order Rényi entropies are $n \log |G|$.

This is a canonical-cut consequence with supplied fixed complementary winding loops. Their existence for the paper's arbitrary disk geometry and transfer to the original site tensors remain separate steps; see [114]. It is not an unrestricted formalization of [203, Theorem 6.9].

Proof. [✓ Proof](#) The actual controlled cut is K_{ϕ_R, χ_λ} in fixed physical coordinates. Its reduced matrix is $\|\chi_\lambda\|^2 \rho_R^0$. Complementary unitarity gives the same reduced matrix before control. Since $\text{tr} \rho_R^0 = 1$, the scalar is precisely its trace. Nonvanishing then permits normalization; the fixed spectrum gives all stated entropies. $\square$

Theorem 24.12.4.117 (Actual boundary routes through two rooted trees). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Choose rooted spanning trees in a region R and its complement S , with normalized gauges k_R, k_S . Let F_R, F_S be their actual crossing-label transports. After rereading each crossing bond as the same native edge on either side, define $M = F_S \circ F_R^{-1}$. If f joins $v_f \in R$ to $w_f \in S$, write D_f for its actual directed bond transport from v_f to w_f . Then

$$M(\theta)_f = m_f \theta_f, \quad m_f = k_S(w_f)^{-1} D_f k_R(v_f). \quad (24.100)$$

Thus $D_f = u_f$ when the region endpoint is the smaller endpoint, and $D_f = u_f^{-1}$ when it is the larger endpoint.

Fix a reference crossing f_0 . Starting at the complementary root, construct the actual closed walk p_f by following its tree to w_{f_0} , crossing backwards to v_{f_0} , following the region tree to v_f , crossing to w_f , and following the complementary tree back to its root. Its holonomy is

$$H(u, p_f) = m_f m_{f_0}^{-1}. \quad (24.101)$$

The construction and identity permit arbitrary bond operators. Inclusion of an induced-region walk in the ambient graph preserves its holonomy. This supplies actual routes for the cycle argument in [203, Theorem 6.9]; the route passes through R . For commuting torus closures, the complementary loops of Theorem 24.12.4.125 have the same winding, and hence the required transport.

Proof. ✓ Proof The two boundary equivalences multiply at the region and complementary endpoints, giving the displayed crossing multiplier. Along each chosen tree path from v to w , the original bond transport is $k(w)k(v)^{-1}$. Multiplying the five successive portions of p_f in reverse traversal order and cancelling the root gauges gives $m_f m_{f_0}^{-1}$. Induced-walk inclusion retains each ordered native edge and hence each directed factor. □

Theorem 24.12.4.118 (Complement cycle words from actual torus winding data). Lean Lean Lean Lean Lean ✓ Stmt On a native torus of width and height at least three, choose rooted spanning trees in R and $S = R^c$. For each native crossing bond f , membership determines its inside endpoint v_f , its outside endpoint w_f , and its directed step from v_f to w_f . Choose a reference bond f_0 , and form the actual two-tree route q_f of Theorem 24.12.4.117 from these endpoints. For arbitrary native bond operators its holonomy is

$$H(u, q_f) = m_f m_{f_0}^{-1}, \quad m_f = (F_S \circ F_R^{-1})(1)_f. \quad (24.102)$$

Suppose actual loops p_f in the induced complementary graph, based at its normalized root, have been chosen independently of the closure labels. Assume their ambient seam crossing numbers satisfy

$$\text{wind}(p_f) = \text{wind}(q_f). \quad (24.103)$$

For every commuting pair g, h , let $u^{g,h}$ be the actual native closure assignment, $\omega_S^{g,h}$ its complementary tree residuals, and $m_f^{g,h}$ its combined crossing multipliers. Then the genuine complementary walk words satisfy

$$\text{ev}_{\omega_S^{g,h}} W_{T_S}(p_f) = m_f^{g,h} (m_{f_0}^{g,h})^{-1}. \quad (24.104)$$

Thus the same words give the relative identity simultaneously for all commuting closure pairs. This is a geometry-to-group identity for the accessible complement argument in [203, Theorem 6.9], local source lines 1935–1990. It neither assumes a group-valued relative transport equation nor asserts existence of the complementary loops with these winding numbers; the remaining scope is recorded in [114].

Proof. ✓ Proof Region membership selects the two endpoints of every crossing bond. The directed factor is its original operator or its inverse according to that selection, so the actual two-tree route has the stated combined multiplier ratio. Evaluation of a complementary word at its actual tree residual equals the original root-loop holonomy. Inclusion in the ambient graph preserves that holonomy. Equal winding numbers give equal commuting-closure transports, identifying the complementary loop transport with the actual reference-route ratio. □

Theorem 24.12.4.119 (One complementary unitary from fixed winding data). [Lean](#) [✓ Stmt](#)
Choose the two rooted trees, reference crossing bond, and fixed complementary root loops with matching actual winding from Theorem 24.12.4.118. There exists one unitary U on the original complementary half-edge physical space, chosen independently of the commuting closure labels g, h . Write $S = R^c$, let $B_S^{g,h}$ be the actual canonical complementary block, and let $F_R^{g,h}, F_S^{g,h}$ be the actual boundary transports. Reread each crossing label on S under the same native edge, and put $m_f^{g,h} = (F_S^{g,h} \circ (F_R^{g,h})^{-1})(\mathbf{1})_f$. For every commuting pair and every complementary physical coordinate (y, a, r, z) , the actual coefficient formula is

$$(UB_S^{g,h})(\alpha(y, a, r, z), (F_R^{g,h})^{-1}\theta) \quad (24.105)$$

$$= |G|^{-|S|} \sum_{x \in G} \mathbf{1}_{\{y_f = xm_{f_0}^{g,h}\theta_f \text{ for every } f\}} \mathbf{1}_{\{z = x\omega_S^{g,h}x^{-1}\}}. \quad (24.106)$$

Thus the relative crossing multipliers are removed by one common complementary operation, while the reference multiplier and actual cycle constraint remain explicit. This is an auxiliary form of the disentangling calculation in [203, Theorem 6.9], local source lines 1935–1990. Theorem 24.12.4.125 supplies these winding loops from actual simple connectedness.

Proof. [✓ Proof](#) At cycle coordinates z , translate the boundary coordinate y_f by the inverse of the evaluation of the fixed complementary word associated with f . Leave the other coordinates unchanged. This is a permutation of the original half-edge physical basis, hence gives a unitary independent of g, h . Matching winding derives the relative word identity for each commuting closure pair. Applying the exact controlled coefficient formula then cancels the relative multipliers, retaining the displayed reference and cycle factors. $\square$

Theorem 24.12.4.120 (Local invariant support of actual walk-controlled operations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Fix a rooted spanning tree in a finite region. Independent vertex translations λ_v of its actual physical half-edge labels transform the spanning-tree coordinates by

$$(y, a, r, z) \mapsto (\lambda_o y, \lambda_o a, (\lambda_v r_v \lambda_o^{-1})_v, \lambda_o z \lambda_o^{-1}). \quad (24.107)$$

Reconstructing these coordinates gives precisely the original half-edge labels multiplied on the left by λ_v at vertex v . For a fixed family of actual region walks, let U be the native boundary permutation controlled by their constructed cycle words. This matrix commutes with every independent vertex translation. Their finite average is the actual product local invariant projector

$$P = \bigotimes_{v \in R} P_v = |G|^{-|R|} \sum_{\lambda \in G^R} L_\lambda, \quad UP = PU. \quad (24.108)$$

In particular U maps the genuine local invariant support range P into itself.

If the original local physical maps A_v are regular G -isometric, there exists a unitary W on their physical region such that

$$W\left(\bigotimes_{v \in R} A_v\right) = \left(\bigotimes_{v \in R} A_v\right)U. \quad (24.109)$$

The fixed words and their physical implementation are chosen without closure labels. No equivariance premise on an arbitrary control function or surjectivity of the canonical projector is required. These are auxiliary physical support identities for the accessible complement calculation in [203, §6.2–6.3], local source lines 1765–1920 and 1935–1990.

Proof. [✓ Proof](#) Apply the native reconstruction formulas at the two internal endpoints and the region endpoint of each crossing edge. The displayed vertex coordinate cancels its root inverse against the transformed reference or boundary label, giving the independent physical translation. Simultaneous conjugation conjugates evaluation of every actual walk word, so the controlled boundary translation commutes with this action. Taking the product of the local finite averages identifies their independent choices with $\lambda \in G^R$, yielding the displayed product projector and commutation. This also proves range preservation. The supported physical unitary transport theorem then gives W on the original region, including any unused physical directions. $\square$

Theorem 24.12.4.121 (Physical transfer of the actual canonical cut). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For arbitrary bond operators u , contract the two actual canonical projector regions over their common crossing labels:

$$C_u(\alpha, \beta) = \sum_{\theta} B_R^u(\alpha, \theta) B_{R^c}^u(\beta, \theta). \quad (24.110)$$

Here the complement reads the same crossing edge under its native boundary identification. Write $P_R = \bigotimes_{v \in R} P_v$ and $T_R = \bigotimes_{v \in R} A_v$ for the original local projector and physical maps. The actual physical coefficient matrix M_u satisfies

$$P_R C_u = C_u, \quad C_u P_{R^c}^\top = C_u, \quad M_u = T_R C_u T_{R^c}^\top. \quad (24.111)$$

If the local maps are regular G -isometric, choose their derived positive product Gram factors c_R, c_{R^c} . For every finite coherent sum $C = \sum_i \mu_i C_{u_i}$ and $M = \sum_i \mu_i M_{u_i}$, put $\tau_C = C C^\dagger$ and $\tau_M = M M^\dagger$. Then

$$\tau_M = c_{R^c} T_R \tau_C T_R^\dagger, \quad \text{tr } \tau_M = c_R c_{R^c} \text{tr } \tau_C, \quad (24.112)$$

$$(\text{tr } \tau_M)^{-1} \tau_M = c_R^{-1} T_R ((\text{tr } \tau_C)^{-1} \tau_C) T_R^\dagger. \quad (24.113)$$

The inverse is read as zero when its argument is zero. For a nonzero state, the last identity is the transfer of normalized reduced density. This is an auxiliary physical accessibility calculation for [203, Theorem 6.9], local source lines 1765–1820 and 1935–1990; no exterior routing or canonical factorization is assumed.

Proof. [✓ Proof](#) Expand the actual half-edge contraction. Summing the physical row against the product local projector replaces each site factor P_v by $P_v^2 = P_v$. The two reverse accessibility identities give $M_u = T_R C_u T_{R^c}^\top$ directly. These equalities persist under finite coherent sums. Inserting $T_{R^c}^\dagger T_{R^c} = c_{R^c} P_{R^c}$ gives the reduced matrix; cyclicity of trace and $P_R C = C$ give its trace. Cancelling the positive product factors gives the normalized identity. $\square$

Theorem 24.12.4.122 (Spectrum under a supported physical map). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\rho \geq 0$ have trace one, and suppose a rectangular physical map T and a positive scalar c satisfy $T^\dagger T \rho = c \rho$. Define $\hat{\rho} = c^{-1} T \rho T^\dagger$. Then $\hat{\rho} \geq 0$, $\text{tr } \hat{\rho} = 1$, and $\text{rank } \hat{\rho} = \text{rank } \rho$. If $\rho^2 = r^{-1} \rho$, then

$$\hat{\rho}^2 = r^{-1} \hat{\rho}, \quad S(\hat{\rho}) = \log r. \quad (24.114)$$

This auxiliary finite-dimensional calculation expresses physical accessibility on the invariant support in [203, Theorem 6.9], local source lines 1765–1820 and 2043–2072. It makes no assertion that T is surjective.

Proof. [✓ Proof](#) Hermitian conjugation gives $\rho T^\dagger T = c\rho$. Positivity follows by conjugation and positive scaling; cyclicity of trace gives trace one. Moreover

$$T^\dagger \hat{\rho} T = c\rho. \quad (24.115)$$

Applying the rank inequality to each physical image yields equality of ranks. Inserting $T^\dagger T$ between the two densities proves the flat-spectrum identity, which gives the stated entropy. $\square$

Theorem 24.12.4.123 (Common reduced density for original torus closures). [Lean](#) [Lean](#) [✓ Stmt](#)

Work on a native torus with both dimensions at least three, with an original regular G -isometric site map. Assume that the actual closed-cell realization of R is simply connected. Choose rooted spanning trees on R and its complement, a nonempty enumeration of the b crossing bonds, and fixed complement-root loops whose winding equals the corresponding two-tree boundary routes. There exists one density ρ_R on the original physical region, chosen before the closure pairs and their coherent coefficients. For every finite family of commuting pairs (g_i, h_i) and every nonzero coherent closure superposition with actual cut matrix M ,

$$\frac{MM^\dagger}{\text{tr}(MM^\dagger)} = \rho_R, \quad \rho_R \geq 0, \quad \text{tr } \rho_R = 1, \quad (24.116)$$

$$\text{rank } \rho_R = |G|^{b-1}, \quad \rho_R^2 = |G|^{-(b-1)} \rho_R, \quad S(\rho_R) = (b-1) \log |G|. \quad (24.117)$$

For every nonnegative finite Rényi order α , including order zero and the von Neumann value at order one, $S_\alpha(\rho_R) = (b-1) \log |G|$. More generally, the density of every actual coherent twisted cut is the supported physical image of its canonical density, with one positive normalization independent of the closure labels. The first conclusion requires no surjectivity of the original physical map, supplied block Gram, or tensor factorization. This is an auxiliary form of [203, Theorem 6.9], local source lines 1935–1990 and 2027–2072. The fixed exterior winding loops are explicit hypotheses here; Theorem 24.12.4.130 derives them from the contiguous simply connected region.

Proof. [✓ Proof](#) Derive the incident-coordinate local isometries from the original four-leg isometry. Their physical product map has positive scalar Gram $T_R^\dagger T_R = c_R P_R$. The actual cut image shows that a nonzero original coherent cut has a nonzero canonical coherent cut. Its fixed normalized density ρ_R^0 is supported on P_R , so the physical density is the fixed matrix

$$\rho_R = c_R^{-1} T_R \rho_R^0 T_R^\dagger. \quad (24.118)$$

The supported physical density theorem transfers positivity, trace, rank, and the flat spectrum. The spectral entropy calculation gives the stated boundary term at all nonnegative finite Rényi orders. Every physical map and normalization was chosen from the original site tensor before the coherent coefficients. $\square$

Theorem 24.12.4.124 (Exterior replacement with prescribed winding). [Lean](#) [Lean](#) [✓ Stmt](#) *Let both torus periods be at least three. Suppose a region has an integer unit-step lift, its safe exterior graph is connected, and its exterior collar projects outside the actual torus region. Every route which enters through one crossing bond, follows an interior path, and leaves through another crossing bond has a complementary replacement with the same two seam crossing numbers.*

In particular, fix rooted spanning trees in the region and its complement and a reference crossing bond. For each crossing bond there is a loop based at the complementary root, wholly within the induced complement, with the winding of the corresponding two-tree boundary route. The loops are chosen independently of all group-valued closure labels. This is an auxiliary geometric construction for [203, proof of Theorem 6.9].

Proof. [✓ Proof](#) Lift the two exterior crossing endpoints. They belong to the integer band, and the deck-coordinate difference equals the crossing numbers of the interior route. A safe graph walk gives a continuous collar path between them. Projection exclusion and lattice-path approximation give a native complementary walk with that same deck-coordinate difference. Prepend the complementary tree path from the root to the first crossing and append the tree path from the second crossing back to the root. Crossing numbers add under concatenation, proving the loop assertion. $\square$

Theorem 24.12.4.125 (Complementary boundary loops of a simply connected region). [Lean](#)

[✓ Stmt](#) *Let both torus periods be at least three, and suppose the actual closed-cell realization of R is simply connected. Choose rooted spanning trees in R and its induced complement and fix a reference crossing bond. There is a complementary root loop for every crossing bond with the same winding as its two-tree boundary route. These loops are chosen before all group-valued closure labels. No exterior path, collar connectivity, or winding identity is an additional premise. This supplies the exterior geometric construction in [203, proof of Theorem 6.9].*

Proof. [✓ Proof](#) Choose one covering lift whose planar integer-cell union is simply connected and whose exterior collar projects outside the torus region. Its safe exterior graph is connected by the boundary contour theorem. Apply the preceding replacement construction to the chosen two-tree routes. $\square$

Theorem 24.12.4.126 (Connected complement of a contiguous simply connected block). [Lean](#)

[Lean](#) [✓ Stmt](#) *Let both torus circumferences be at least three. Let R be a finite set of sites whose induced nearest-neighbour graph is connected and whose actual closed-cell realization is simply connected. Then the induced graph on the complementary sites is connected, and at least one bond crosses between R and its complement.*

Consequently, spanning trees and roots on both sides of the cut, and a numbering of the crossing bonds by $n + 1$ labels, may be chosen without further hypotheses. Occupied graph connectivity is distinct from simple connectedness of the closed-cell union: cells meeting only at a corner need not be connected by lattice bonds. This is the contiguous-block geometry used in Section 6.3 of [203], rather than a claim that every simply connected closed-cell subset is a contiguous block.

Proof. [✓ Proof](#) Lift the occupied region to a simply connected finite integer-cell union whose exterior collar projects outside the region. The exposed contour joins its nearby missing centres within the safe exterior graph. These joins yield complementary lattice walks between all outside endpoints of crossing bonds. Every complementary site has a walk to an occupied site in the full torus graph; its first crossing joins it, within the complement, to such an endpoint. Hence all complementary sites are connected. The collar also supplies a complementary site, so a walk from an occupied site crosses at least one boundary bond. $\square$

Definition 24.12.4.127 (Normalized pure-state coefficient matrix). [Lean](#) [✓ Stmt](#) For a coefficient matrix M of a pure state on $\mathcal{H}_R \otimes \mathcal{H}_S$, define

$$\widehat{M} = (\text{tr}(MM^\dagger))^{-1/2} M. \quad (24.119)$$

The inverse is taken to be zero when $M = 0$. For nonzero M , $\widehat{M}$ represents a normalized pure state.

Theorem 24.12.4.128 (Complementary unitary from equal reduced densities). [Lean](#) [Lean](#)

[✓ Stmt](#) *Let A and B be coefficient matrices across the same finite bipartition. Their normalized*

matrices satisfy

$$\widehat{MM}^\dagger = \frac{MM^\dagger}{\text{tr}(MM^\dagger)}. \quad (24.120)$$

If the reduced densities on R agree, there exists a unitary U on $\mathcal{H}_S$ such that $\widehat{B} = \widehat{A}U^\top$. Thus the corresponding pure states differ by the unitary U on S . This is the finite-dimensional implication used in [203, Theorem 6.7 and Corollary 6.8], local source lines 1995–2025.

Proof. ✓ Proof The square of the inverse real square root gives (24.120). Equality of the reduced densities is equality of the Gram matrices of $\widehat{A}^\dagger$ and $\widehat{B}^\dagger$. The isometry between their ranges extends to a unitary V on $\mathcal{H}_S$. Taking adjoints gives $\widehat{B} = \widehat{A}V^\dagger$, so $U = (V^\dagger)^\top$ suffices. $\square$

Theorem 24.12.4.129 (Local equivalence of coherent physical torus closures). Lean ✓ Stmt Assume the geometric winding hypotheses of 24.12.4.123. Let A and B be the actual physical coefficient matrices of any two nonzero coherent superpositions of commuting torus closures. Every operator O on the physical region R has the same normalized expectation in the two states:

$$\text{tr}(O\widehat{B}\widehat{B}^\dagger) = \text{tr}(O\widehat{A}\widehat{A}^\dagger). \quad (24.121)$$

There also exists a unitary U on the complementary physical factor such that $\widehat{B} = \widehat{A}U^\top$. This auxiliary statement has the fixed complementary winding loops as explicit hypotheses. It does not assert the width-one stripe construction of [203, Theorem 6.7]. Theorem 24.12.4.125 derives the exterior walks for the actual simply connected regions on tori with periods at least three.

Proof. ✓ Proof The common physical reduced density of 24.12.4.123 gives $\widehat{A}\widehat{A}^\dagger = \widehat{B}\widehat{B}^\dagger$. Multiplying by O and taking traces proves (24.121). Apply 24.12.4.128 for the complementary unitary. $\square$

Theorem 24.12.4.130 (Common physical density with geometrically derived exterior walks). Lean ✓ Stmt Work on a native torus with both dimensions at least three and an original regular G -isometric site map. Assume the actual closed-cell realization X_R is simply connected and the induced occupied graph is connected. Let b be the number of crossing bonds. There exists one physical reduced density ρ_R for every nonzero coherent superposition of commuting torus closures, with coefficient matrix M . It satisfies

$$\frac{MM^\dagger}{\text{tr}(MM^\dagger)} = \rho_R, \quad \rho_R \geq 0, \quad \text{tr } \rho_R = 1, \quad (24.122)$$

$$\text{rank } \rho_R = |G|^{b-1}, \quad \rho_R^2 = |G|^{-(b-1)} \rho_R, \quad (24.123)$$

$$S(\rho_R) = (b-1) \log |G|, \quad S_\alpha(\rho_R) = (b-1) \log |G|. \quad (24.124)$$

The Rényi formula holds for every nonnegative finite order, including zero and the von Neumann value at one. The exterior walks, spanning trees, roots, and boundary enumeration are derived internally. This is a specialization of [203, Theorem 6.9] to regular native tori with periods at least three. Occupied graph connectivity expresses the connected block in the source, lines 1931–1933 and 1349–1354; simple connectedness of the closed-cell union alone does not force four-neighbor connectivity of the occupied sites.

Proof. ✓ Proof The induced complement graph is connected. Choose spanning trees and roots in both regions and enumerate the nonempty boundary. The simply connected planar lift and its exterior collar provide complementary boundary loops with the winding of the two-tree boundary routes. Applying 24.12.4.123 to these derived loops gives (24.124). $\square$

Theorem 24.12.4.131 (Physical local equivalence with geometrically derived exterior walks).

[Lean](#) [✓ Stmt](#) Under the hypotheses of 24.12.4.130, let A and B be the physical coefficient matrices of any two nonzero coherent superpositions of commuting torus closures. Every operator O on the occupied physical region has equal normalized expectation in these states. There exists a unitary U on the complementary physical factor satisfying

$$\widehat{B} = \widehat{A}U^\top, \quad \text{tr}(O\widehat{B}\widehat{B}^\dagger) = \text{tr}(O\widehat{A}\widehat{A}^\dagger). \quad (24.125)$$

This is a specialization of [203, Theorem 6.7 and Corollary 6.8] to connected simply connected regions of regular native tori with periods at least three. The width-one stripe construction is separate.

Proof. [✓ Proof](#) Derive the complementary winding loops from the actual simple connectedness and apply 24.12.4.129. $\square$

Theorem 24.12.4.132 (Canonical coordinates of a supplied lifted torus region). [Lean](#) [Lean](#)

[✓ Stmt](#) Suppose a native torus region has the integer lift described in Lemma 24.12.4.94. Choose a spanning tree and root, and let $k^{g,h}$ be its normalized lift gauge. For commuting closure labels g, h , the actual canonical tensor satisfies

$$B_R^{g,h}(y, a, r, z; \theta) = B_R(y, a, r, z; F_{g,h}(\theta)), \quad (24.126)$$

where $F_{g,h}$ is the native transported crossing configuration. In particular, there are a positive constant γ and one normalized vector $\phi(a, r, z)$, independent of g, h , such that

$$B_R^{g,h}(y, a, r, z; \theta) = \gamma \Pi_{\partial R}(y, F_{g,h}(\theta)) \phi(a, r, z). \quad (24.127)$$

This is an auxiliary lifted-region form of the blocking and boundary-coordinate calculations in [203, §6.2–6.3]. It retains the sector-dependent boundary transport. The covering-lift and complementary-loop theorems supply the geometric choices for actual simply connected regions with torus periods at least three.

Proof. [✓ Proof](#) Every cycle residual is the identity by Lemma 24.12.4.94. The exact twisted coefficient formula therefore reduces to the untwisted coefficient formula at the transported crossing configuration. Apply the normalized factorization of Theorem 24.12.4.71 to this configuration. Its constant and ancillary vector depend only on the region, the group, and the chosen tree and root. $\square$

Theorem 24.12.4.133 (Injectivity of an actual connected regular block). [Lean](#) [✓ Stmt](#) Let R be a nonempty connected finite region whose site tensors are G -injective for the simultaneous regular action on their incident legs. Then the actual open tensor of R is G -injective for the simultaneous regular action on its crossing legs. Internal cycles and an empty boundary are allowed. This is the regular-representation connected-region consequence of [203, Lemma 5.2 and the following observation]; the argument does not assume local isometry.

Proof. [✓ Proof](#) Apply the product of the local inverses to the actual open tensor. The result is its canonical averaging-projector contraction. Choose a spanning tree and a root. In the coordinates of Theorem 24.12.4.71, the row with identity reference, normalized vertex, and residual cycle labels is $|G|^{1-|R|} \Pi_{\partial R}$. This scalar is nonzero, and $\Pi_{\partial R}$ fixes every invariant boundary vector. Thus a vanishing physical output implies a vanishing invariant input. Boundary invariance follows by translating every internal label in the actual contraction. $\square$

has reduced von Neumann entropy $n \log |G|$. For the actual cut maps of Theorem 24.12.4.58, the numerator is the physical PEPS coefficient matrix K .

The Gram identities are explicit assumptions in this statement. Theorem 24.12.4.68 derives them from local G -isometry for connected untwisted regions, including regions containing cycles. The twisted torus contractions of [203, Theorem 6.9] remain separate. The scalar convention is recorded in [114].

Proof. ✓ Proof Since $c_A > 0$, the product of the normalization scalar and its conjugate is c_A^{-1} . Thus $\widehat{M}^\dagger \widehat{M} = P$, and similarly for N . Pulling the scalars out of the coefficient matrix gives the displayed formula. Apply Theorem 24.12.4.142 to the normalized maps. $\square$

Theorem 24.12.4.144 (Entropy of a connected untwisted finite cut). Lean Lean Lean Lean ✓ Stmt Consider a finite graph PEPS with regular bonds and a G -isometric tensor at every site. Assume that both R and its complement induce nonempty connected graphs, and that $b > 0$ bonds cross the cut. For the untwisted contraction there is a positive normalization of the actual physical PEPS vector to a unit vector. Its reduced state on R has rank $r = |G|^{b-1}$ and a flat spectrum: $\rho_R^2 = r^{-1} \rho_R$. Its von Neumann entropy is

$$S(\rho_R) = (b-1) \log |G|.$$

This finite untwisted statement is a restricted consequence of the calculation in [203, Theorem 6.9]; it does not assert the theorem for arbitrary torus closures. The remaining scope is recorded in [114].

Proof. ✓ Proof Each local isometry supplies a positive Gram factor. Apply Theorem 24.12.4.68 on both sides of the cut, and normalize the two maps by their positive factors. The resulting coefficient matrix is a positive scalar multiple of the actual PEPS cut matrix. The regular boundary realization gives its norm and reduced entropy. $\square$

Lemma 24.12.4.145 (A local controlled operation removing a boundary twist). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The permutation

$$F(a, b, c) = (cb^{-1}a, b, c), \quad F^{-1}(a, b, c) = (bc^{-1}a, b, c)$$

acts unitarily in the group basis. If $c = tb$, then $F^{-1}(ta, b, tb) = (a, b, tb)$. In particular this removes the conjugated twist $t = x^{-1}gx$ from the first label using the accessible reference pair b, c .

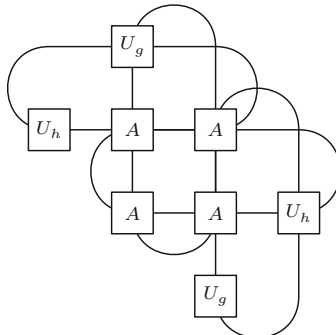
On the bipartite coefficient vector $\mathbf{1}_{\{a=td\}} \mathbf{1}_{\{c=tb\}}$, applying this inverse permutation to the first side gives $\mathbf{1}_{\{a=d\}} \mathbf{1}_{\{c=tb\}}$. All dependence on t is then confined to the reference pair on that side. This proves the local coordinate operation in [203, Equation (iso:tweezer-duplicate-operation)]. Identifying these coordinates with a whole torus contraction is a further step.

Proof. ✓ Proof Group cancellation verifies the two inverse permutations and $cb^{-1} = t$. A permutation of an orthonormal basis is unitary. Substitution in the two indicator functions yields the factorization. $\square$

Definition 24.12.4.146 (Torus PEPS with group closures). Lean Lean Lean ✓ Stmt On a $w \times \ell$ torus let $H_v = U_h$ when $v + (1, 0)$ crosses the horizontal seam and $H_v = \mathbb{1}$ elsewhere; let $V_v = U_g$ when the vertical bond crosses the vertical seam and $V_v = \mathbb{1}$ elsewhere. Define the vector $|\mathcal{M}(A \mid g, h)\rangle$ by contracting the site tensors with these bond operators. Its coefficient at σ is

$$\mathcal{M}(A \mid g, h)_\sigma = \sum_{(t_v, r_v, b_v, l_v)_v} \prod_v A_{t_v, r_v, b_v, l_v}^{\sigma(v)} H_v(l_{v+(1,0)}, r_v) V_v(t_v, b_{v+(0,1)}). \quad (24.128)$$

This is [203, Definition 5.6], which uses a square lattice; the same contraction defines the vector on a rectangle. The four-site diagram suppresses the physical legs at fixed σ . Crossings without a vertex do not denote contractions.



Definition 24.12.4.147 (Pair-conjugacy classes of closures). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Two pairs are equivalent if $(g, h) = (xg'x^{-1}, xh'x^{-1})$ for some $x \in G$. Their equivalence classes are the pair-conjugacy classes of [203, Definition 5.8]. Commutation is constant on each class, so the classes of commuting pairs are well-defined.

Theorem 24.12.4.148 (The torus vector depends only on the pair-conjugacy class). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Suppose A is invariant under the virtual group action. Then $|\mathcal{M}(A \mid xgx^{-1}, xhx^{-1})\rangle = |\mathcal{M}(A \mid g, h)\rangle$ for every $x \in G$. Consequently $|\mathcal{M}(A \mid C)\rangle$ is well-defined for each pair-conjugacy class C [203, Definition 5.8].

Proof. [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) In Lemma 24.12.4.53 take $P_v = U_x$ and $Q_v = U_{x^{-1}}$ at every site. The absorbed site tensor is unchanged by invariance. Each closure matrix U_g or U_h becomes $U_{xgx^{-1}}$ or $U_{xhx^{-1}}$, and each identity bond remains the identity. This proves the assertion for every coefficient in (24.128). $\square$

Theorem 24.12.4.149 (Displacement of commuting closure seams). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let A be invariant under the virtual group action and let $gh = hg$. For any column c and row r , put

$$H_v^c = \begin{cases} U_h, & v_x + 1 = c, \\ \mathbb{1}, & \text{otherwise,} \end{cases} \quad V_v^r = \begin{cases} U_g, & v_y + 1 = r, \\ \mathbb{1}, & \text{otherwise.} \end{cases}$$

Contracting A with these bond operators gives $|\mathcal{M}(A \mid g, h)\rangle$, independently of c and r . This is the displacement of commuting loops described after [203, Equation (2d:move-strings)]; neither isometry nor semi-regularity is needed. The identity holds on every nonempty rectangular torus, including dimensions equal to one.

Proof. [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Choose coordinate representatives between zero and the corresponding circumference minus one. To move the horizontal seam, use the vertex change of basis $U_{h^{-1}}$ on columns preceding c and the identity elsewhere. The adjacent changes cancel on all ordinary bonds, remove the old seam at zero, and insert U_h at the new seam. They commute with every U_g on the transverse seam. Similarly, use U_g on rows preceding r to move the vertical seam, with the inverse convention required by its orientation. Local invariance removes every absorbed change of basis. The network gauge identity therefore preserves each physical coefficient. $\square$

Definition 24.12.4.150 (Regular coordinates of a torus site). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) In the group basis, $L_x|k\rangle = |xk\rangle$. The coordinate map $\sum_k c_k[k] \mapsto (c_k)_k$ intertwines the group-algebra regular representation with these permutation matrices. They are therefore semi-regular by Theorem [24.12.4.10](#). The oriented four-leg action sends (t, r, b, l) to (xt, xr, xb, xl) ; its invariant projector has entries

$$\Pi_{\eta,\theta} = \frac{1}{|G|} \sum_{x \in G} \mathbf{1}_{\eta=x\theta}.$$

Thus the four regular coordinates agree with the oriented virtual action in Definition 6.1 of [203].

Lemma 24.12.4.151 (The local regular Gram identity). [Lean](#) [Lean](#) [✓ Stmt](#) *A regular G -isometric site tensor A is invariant under common translation of its four virtual labels. There is $c > 0$ such that*

$$\sum_s \overline{A_s(\eta)} A_s(\theta) = \frac{c}{|G|} \sum_{x \in G} \mathbf{1}_{\eta=x\theta}.$$

The factor c records the normalization of the site tensor.

Proof. ✓ Proof Express $A^\dagger A = c\Pi$ in the orthonormal group basis and use the matrix entries of the invariant projector. $\square$

Lemma 24.12.4.152 (Access to the virtual invariant space). [Lean] [StmT] If $A^\dagger A = c\Pi$ with $c > 0$, then $J = c^{-1/2}A^\dagger$ satisfies $JA = \sqrt{c}\Pi$ and preserves inner products on $\text{ran } A$. It identifies the physical support with the invariant virtual space, as in Observation 6.4 of [203].

Proof. ✓ Proof For u, v in the virtual space, $JAu = \sqrt{c}\Pi u$ and $\langle Au, Av \rangle = c\langle \Pi u, \Pi v \rangle$.

Lemma 24.12.4.153 (Expansion of a torus overlap). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#)
[Learn](#) [Learn](#) [✔ Stmt](#) Let $W(\beta)$ denote the product of the bond matrix entries and let $\eta_v(\beta)$ be the four labels read by site v . The overlap of two networks with site tensors A_v, B_v is

$$\sum_{\beta, \gamma} \overline{W_A(\beta)} W_B(\gamma) \prod_v \left(\sum_s \overline{A_{v,s}(\eta_v(\beta))} B_{v,s}(\eta_v(\gamma)) \right).$$

For permutation bonds, the head label is the permutation of the tail label. Hence one free label per bond suffices in each network.

Proof. ✓ Proof Expand both contractions, interchange the finite sums, and sum each physical index independently. A permutation matrix has entry one exactly when its row is the image of its column. □

Lemma 24.12.4.154 (Counting compatible regular bonds). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) On a bond e oriented from $s(e)$ to $t(e)$, let u_e, w_e be the group elements in the two networks. Vertex translations q_v are compatible with this bond precisely when

$$u_e q_{s(e)} = q_{t(e)} w_e.$$

If the equations hold, each bond has one independent group label; otherwise there are no compatible labels. For the two bond families on a torus with N sites, the count is $|G|^{2N}$.

Proof. □ Proof The tail labels satisfy $\eta_e = q_{s(e)}\theta_e$. Substituting this into $u_e\eta_e = q_{t(e)}w_e\theta_e$ and cancelling θ_e gives the intertwining equation. Every θ_e is then free. □

Lemma 24.12.4.155 (Translation compatibility across the torus seams). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For closures (g, h) and (g', h') , a compatible vertex translation is constant, $q_v = x$, and its value satisfies*

$$hx = xh', \quad gx = xg'.$$

Summing over compatible vertex translations therefore gives exactly the number of simultaneous intertwiners. This holds also when a torus circumference is one.

Proof. ✓ Proof The nonseam horizontal and vertical bonds identify all translations in a rectangular fundamental domain. The two seams give the displayed equations. For a circumference of one the corresponding seam is already a loop at each site, imposing the same equation. $\square$

Theorem 24.12.4.156 (Overlap of regular isometric torus closures). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

 *Stmnt* Let $A_v^\dagger A_v = c_v \Pi_v$ be the local regular isometry identities on a torus with N sites, with $c_v > 0$. For the actual closure vectors,

$$\langle \psi_{g,h}, \psi_{g',h'} \rangle = \left(\prod_v c_v \right) |G|^N \# \{x \in G : hx = xh', \ gx = xg'\}.$$

For a uniform tensor the prefactor is $c^N |G|^N$. Neither closure pair is required to commute.

Proof.  **Proof** Expand the overlap into local Gram kernels. Their product is the sum over vertex translations with prefactor $(\prod_v c_v)|G|^{-N}$. For a compatible translation, the two bond families contribute $|G|^{2N}$, leaving the stated prefactor. By Lemma 24.12.4.155, the remaining sum counts the simultaneous intertwiners. $\square$

Theorem 24.12.4.157 (Independent regular isometric closure vectors). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a uniform regular G -isometric tensor, the actual vector associated to each pair-conjugacy class is nonzero, and distinct class vectors are orthogonal. Their family is therefore linearly independent. More precisely,

$$\|\psi_{g,h}\|^2 = c^N |G|^N |Z_G(g) \cap Z_G(h)| > 0.$$

Restricting to commuting classes proves the independence assertion underlying [203, Theorem 5.9] in the regular G -isometric case. The all-class assertion is an auxiliary extension: no ground-state claim is made for noncommuting closures. Semi-regular G -injective independence is proved in Theorem 24.12.4.171; ground-space identification remains separate [114].

Proof. ✓ **Proof** A simultaneous intertwiner exists exactly when the two closure pairs are simultaneously conjugate. Thus the overlap between distinct classes vanishes. For a self-overlap the intertwiners form $Z_G(g) \cap Z_G(h)$, which contains the identity. Taking the inner product of a finite linear relation with each nonzero class vector proves independence. Independence is inherited by the commuting subfamily. $\square$

Theorem 24.12.4.158 (Independent simultaneous-conjugacy operator sums). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let U be a semi-regular representation of a finite group G on a finite-dimensional complex vector space E . Define

$$\Omega_{g,h} = \sum_{x \in G} U_{xhx^{-1}} \otimes U_{xgx^{-1}}, \quad (24.129)$$

$$\Phi_{g,h}(A \otimes B) = \text{tr}[U_{h^{-1}} A \Delta] \text{tr}[U_{g^{-1}} B \Delta]. \quad (24.130)$$

The operator $\Omega_{g,h}$ depends only on the simultaneous-conjugacy class C of (g, h) ; denote it by Ω_C . The family $(\Omega_C)_C$ is linearly independent. The detecting functionals satisfy

$$\Phi_{g,h}(\Omega_{g',h'}) = |\{x \in G : g = xg'x^{-1}, h = xh'x^{-1}\}|, \quad (24.131)$$

$$\Phi_{g,h}(\Omega_{g,h}) = |Z_G(g) \cap Z_G(h)| > 0. \quad (24.132)$$

In a coordinate basis, the corresponding matrices $\sum_x U_{xhx^{-1}} \otimes_K U_{xgx^{-1}}$ are likewise well-defined on classes and linearly independent. Here $\otimes_K$ denotes the Kronecker product. Restriction to commuting pair classes gives the operator-independence argument in [203, Theorem 5.9].

Proof. ✓ Proof Replacing (g, h) by (ygy^{-1}, yhy^{-1}) in (24.129) and substituting $x \mapsto xy$ leaves the sum unchanged. By Theorem 24.12.4.11, each factor of (24.130) evaluates to a Kronecker delta on a representation operator, giving (24.131). Its value is zero on distinct classes, while (24.132) is nonzero because the common centralizer contains the identity. Applying $\Phi_{g,h}$ to a finite relation $\sum_C \lambda_C \Omega_C = 0$ therefore gives $\lambda_C |Z_G(g) \cap Z_G(h)| = 0$ for the class of (g, h) , and hence every coefficient vanishes. Finally, the canonical isomorphism $\text{End}(E) \otimes \text{End}(E) \cong \text{End}(E \otimes E)$, followed by passage to the tensor-product coordinate basis, takes simple tensors to Kronecker matrices and preserves linear independence. $\square$

Definition 24.12.4.159 (Products of physical linear maps). Lean Lean Lean Lean Lean Lean
✓ Stmt Let F be a rectangular physical matrix. Its product over a finite site set Λ has coefficients

$$(F^{\otimes \Lambda})_{\tau, \sigma} = \prod_{v \in \Lambda} F_{\tau(v), \sigma(v)}.$$

The transformed site tensor is $(FA)_{t,r,b,l}^q = \sum_s F_{q,s} A_{t,r,b,l}^s$. Input and output physical dimensions may differ.

Lemma 24.12.4.160 (Physical maps commute with torus contraction). Lean Lean Lean ✓ Stmt
Applying F at every physical site composes the local site map with F and preserves every virtual bond operator. In particular,

$$F^{\otimes \Lambda} |\mathcal{M}(A \mid g, h)\rangle = |\mathcal{M}(FA \mid g, h)\rangle.$$

The identity holds for arbitrary torus bond operators, with no invariance or injectivity hypothesis. It is the physical inverse operation used in [203, Theorem 5.9].

Proof. ✓ Proof Expand the physical product map and the actual virtual contraction. Interchange the finite sums and factor the physical sum over sites. Each site coefficient is replaced by $\sum_s F_{q,s} A^s$ while the virtual bond coefficients are unchanged. $\square$

Lemma 24.12.4.161 (Physical product maps and Schmidt rank). Lean Lean Lean Lean Lean
✓ Stmt For a partition $\Lambda = R \sqcup R^c$, let $C_R(\psi)$ be the coefficient matrix of a physical vector in the two sets of configuration coordinates. Then

$$C_R(F^{\otimes \Lambda} \psi) = F^{\otimes R} C_R(\psi) (F^{\otimes R^c})^\top.$$

Consequently its rank cannot increase. If physical product maps send ψ to ϕ and ϕ back to ψ , their Schmidt ranks agree across every partition. Neither product map need be invertible on the full physical space. This is the rank argument used in [203, Corollary 6.10].

Proof. ✓ Proof Split a physical configuration into its restrictions to R and R^c . The product coefficient factors over these restrictions, and the resulting two physical sums give the two matrix products. Matrix multiplication cannot increase rank. Apply this inequality to the forward and reverse actions. $\square$

Proof. ✓ Proof Put $\delta_p(M) = \text{tr}(U_{p^{-1}} M \Delta)$ and $\phi = \sum_p \delta_p$. Trace duality gives $\delta_p(U_z) = \delta_{p,z}$ and $\phi(U_z) = 1$. Retain the horizontal seam bond $(-1, 0)$ and the vertical seam bond $(0, -1)$. Apply δ_1 to every nonseam output bond and ϕ to every other seam output bond. On each retained bond use a matrix entry functional. Explicitly, for fixed rows (i, k) and columns (j, l) , denote these horizontal and vertical functionals by κ_v^h and κ_v^v , respectively. If $E_{a,b}$ is a matrix unit and β assigns four end labels to each pair of bonds, define

$$(Ef)_{(i,k),(j,l)} = \sum_{\beta} f(\mathcal{A}(\beta)) \prod_{v \in \Lambda} \kappa_v^h(E_{\beta_v^h H, \beta_v^h T}) \kappa_v^v(E_{\beta_v^v H, \beta_v^v T}), \quad (24.134)$$

where $\mathcal{A}(\beta)$ is the four-leg configuration read from the bond ends, in top, right, bottom, left order. Finite distributivity evaluates this contraction separately on each bond. For a vertex labelling q , a nonseam horizontal bond gives $\delta_1(U_{q_v + e_1} q_v^{-1}) = \delta_{q_v + e_1, q_v}$; a nonseam vertical bond gives $\delta_1(U_{q_v q_v^{-1} + e_2}^{-1}) = \delta_{q_v, q_v + e_2}$. Lemma 24.12.4.163 forces $q_v = x$ at every vertex. All discarded seam factors then equal one, and the retained factors are $U_{xhx^{-1}}$ and $U_{xgx^{-1}}$. The normalized projector expansion of Lemma 24.12.4.164 supplies the coefficient $|G|^{-|\Lambda|}$, giving (24.133). $\square$

Definition 24.12.4.166 (The averaging-projector site). Lean Lean Lean Lean Lean ✓ Stmt Let U be any representation of a finite group G on a virtual space with finite basis V , and write R_q for its four-leg representation. Take the physical basis to be V^4 and define A_0 by

$$(A_0)_{t,r,b,l}^s = \frac{1}{|G|} \sum_{q \in G} (R_q)_{s,(t,r,b,l)}, \quad (24.135)$$

$$\mathcal{P}(A_0) = \Pi = \frac{1}{|G|} \sum_{q \in G} R_q. \quad (24.136)$$

Then $\Pi R_q = \Pi$ and Π is the identity on invariant vectors, so A_0 is G -injective. Unitarity is not required for these statements [203, Definition 5.1].

Theorem 24.12.4.167 (Isometry of the regular projector site). Lean ✓ Stmt For four regular virtual legs, A_0 is G -isometric with normalization factor one.

Proof. ✓ Proof The unitary group average is self-adjoint and idempotent, so $\Pi^\dagger \Pi = \Pi$. Its invariance was established in Definition 24.12.4.166. $\square$

Theorem 24.12.4.168 (Actual closure vectors and the local inverse). Lean Lean ✓ Stmt Let A be G -injective for any finite virtual representation U and write $T = \mathcal{P}(A)$. There is a local rectangular matrix L with $LT = \Pi$ such that, for every closure pair,

$$L^{\otimes \Lambda} |\mathcal{M}(A \mid g, h)\rangle = |\mathcal{M}(A_0 \mid g, h)\rangle, \quad (24.137)$$

$$T^{\otimes \Lambda} |\mathcal{M}(A_0 \mid g, h)\rangle = |\mathcal{M}(A \mid g, h)\rangle. \quad (24.138)$$

The reverse identity requires only virtual invariance of A . These are actual contraction identities for the inverse argument of [203, Theorem 5.9]; no unitarity or local isometry hypothesis is needed.

Proof. ✓ Proof The local inverse and virtual invariance give $LT = \Pi$ and $T\Pi = T$. By Theorem 24.12.4.162, these are coefficient identities. Apply Lemma 24.12.4.160 at every physical site in both directions. No inverse on the entire physical space is required. $\square$

Theorem 24.12.4.169 (Independent closures of regular G -injective tensors). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) For a regular G -injective tensor on any nonempty rectangular torus, every actual closure vector is nonzero and the vectors indexed by distinct pair-conjugacy classes are linearly independent. In particular the family indexed by commuting classes is independent. The latter is the regular-representation independence component of [203, Theorem 5.9], with no isometry hypothesis. Independence for all pair classes is an auxiliary extension and makes no ground-space assertion. The semi-regular representation result appears in Theorem 24.12.4.171; ground-space identification remains separate [114].

Proof. [✓ Proof](#) The native regular representation is semi-regular by its group-basis equivalence with the group-algebra regular representation. Apply Theorem 24.12.4.171. $\square$

Theorem 24.12.4.170 (Schmidt ranks of regular G -injective closure superpositions). [Lean](#)

[✓ Stmt](#) Let A be regular G -injective. For every finite family of closure pairs and every physical partition R , the two vectors

$$\sum_s \lambda_s |\mathcal{M}(A \mid g_s, h_s)\rangle, \quad \sum_s \lambda_s |\mathcal{M}(A_0 \mid g_s, h_s)\rangle$$

have the same Schmidt rank across R . This is the actual local deformation step of [203, Corollary 6.10]; no particular boundary-rank formula is asserted here.

Proof. [✓ Proof](#) The same forward and reverse physical product maps act on every summand. By linearity they recover the displayed superpositions in both directions. Lemma 24.12.4.161 gives equality of their coefficient-matrix ranks. $\square$

Theorem 24.12.4.171 (Independent closures of semi-regular G -injective tensors). [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) Let U be a semi-regular representation of a finite group G , and let A be G -injective for its four-leg virtual representation. On every nonempty rectangular torus, the actual closure vectors $|\mathcal{M}(A \mid C)\rangle$, indexed by simultaneous-conjugacy classes C of pairs, are linearly independent. Every closure vector is therefore nonzero. In particular, the family indexed by commuting pair classes is independent.

The commuting-family assertion proves the independence component of [203, Theorem 5.9], without a local isometry assumption. Independence for noncommuting classes and size-one tori are auxiliary extensions. No identification with the parent-Hamiltonian ground space is asserted [114].

Proof. [✓ Proof](#) Choose the same local inverse L for every closure vector. With N the number of vertices, the explicit extraction E gives

$$EL^{\otimes \Lambda} |\mathcal{M}(A \mid C)\rangle = |G|^{-N} \Omega_C. \quad (24.139)$$

Thus $|G|^N EL^{\otimes \Lambda}$ sends any proposed linear relation among the physical vectors to a relation among the independent operators Ω_C . Its coefficients vanish by Theorem 24.12.4.158. Each vector is nonzero, and restriction to the commuting classes preserves independence. $\square$

Definition 24.12.4.172 (Regular bonds with inserted group operators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

Order the endpoints of each bond e , writing them as $s(e) < t(e)$. For an assignment $u_e \in G$, the inserted regular matrix makes the label at $t(e)$ equal to u_e times the label at $s(e)$. Equivalently, replace the coefficient at vertex v by

$$A_{v,s}^u(\eta) = A_{v,s}(\eta^u), \quad \eta_e^u = \begin{cases} u_e \eta_e, & v = t(e), \\ \eta_e, & v = s(e). \end{cases}$$

Contracting the internal bonds with a fixed crossing configuration gives the actual open map T_R^u .

Definition 24.12.4.173 (The canonical projector region with bond insertions). [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) For arbitrary inserted group elements u_e , including those on crossing bonds, define

$$\begin{aligned} B_R^u(\alpha, \theta) &= \sum_{\substack{\eta \in G^{E_R} \\ \eta|_{\partial R} = \theta}} \prod_{v \in R} P_v(\alpha_v, \eta^u|_{E_v}) \\ &= \kappa^{-|R|} \sum_{\eta \in G^{E_R}} \sum_{q \in G^R} \mathbf{1}_{\{\eta|_{\partial R} = \theta\}} \mathbf{1}_{\{\alpha_v = q_v \eta^u|_{E_v} \ (v \in R)\}}. \end{aligned}$$

Here the half-edge modification η^u is that of Definition 24.12.4.172. Identity insertions recover B_R . This is the literal finite contraction of the local averaging-projector sites with the inserted regular operators.

Theorem 24.12.4.174 (Accessibility with arbitrary regular bond insertions). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) Let the original regular site maps A_v be G -injective, and choose local matrices F_v satisfying $F_v A_v = P_v$. The same matrices work for every region and every assignment of inserted group elements:

$$\left(\bigotimes_{v \in R} F_v \right) T_R^u(\theta) = B_R^u(\theta), \quad \left(\bigotimes_{v \in R} A_v \right) B_R^u(\theta) = T_R^u(\theta).$$

If the sites are G -isometric with factors $c_v > 0$, one may take $F_v = c_v^{-1} A_v^\dagger$. No connectedness or global Gram formula is required. These are the accessibility identities of [203, Observation after Lemma 6.3], extended algebraically to arbitrary bond insertions.

Proof. [✓ Proof](#) Apply the physical product map inside the actual contraction. The local identity $F_v A_v = P_v$ remains valid at the modified virtual label $\eta^u|_{E_v}$ and proves the forward equality. For the reverse equality, use $A_v P_v = A_v$ and hence $A_v F_v A_v = A_v$, again at the modified label. Expanding the local averages gives the displayed finite-sum formula for B_R^u . $\square$

Lemma 24.12.4.175 (Expansion of a twisted-region cross Gram matrix). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Suppose the original site maps satisfy $A_v^\dagger A_v = c_v \Pi_v$. The entries of $(T_R^u)^\dagger T_R^w$ are obtained by summing over two internal-bond configurations and one translation q_v per vertex. At each incident half-bond the constraint is

$$\eta_{v,e}^u = q_v \theta_{v,e}^w.$$

The prescribed crossing configurations are retained, and the scalar from the local averages is $(\prod_{v \in R} c_v) |G|^{-|R|}$.

Proof. [✓ Proof](#) Expand both genuine open contractions and sum the physical indices independently. Substitute each original local Gram identity at its two transformed virtual configurations. $\square$

Lemma 24.12.4.176 (Synchronization through common identity bonds). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose the region has a connected spanning subgraph on whose bonds both assignments u, w are the identity. Suppose also that both are the identity on crossing bonds. Then compatibility is equivalent to one common translation x , with

$$q_v = x, \quad \eta_e = x \theta_e, \quad u_e x = x w_e \quad \text{for every internal bond } e.$$

Consequently the sum over compatible vertex translations is the corresponding restricted sum over $x \in G$.

Proof. [✓ Proof](#) The tail constraint gives $\eta_e = q_{s(e)}\theta_e$; cancellation in the head constraint gives $u_e q_{s(e)} = q_{t(e)}w_e$. Along each common identity bond this equates the endpoint translations. Connectedness makes them constant. Boundary identity operators then give the common translation of every crossing label. $\square$

Lemma 24.12.4.177 (Counting with a restricted common translation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let I_R count the internal bonds and let $F \subseteq G$ be any allowed set of common translations. With crossing configurations μ, ν , the number of translated pairs of bond assignments is

$$|G|^{I_R} \sum_{x \in F} \mathbf{1}_{\mu=x\nu}.$$

Proof. [✓ Proof](#) For fixed x , the second assignment determines the first. Its crossing labels are fixed; every internal label is free. The boundary conditions agree exactly when $\mu = x\nu$. $\square$

Lemma 24.12.4.178 (Weighted translations on the invariant boundary). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
For weights $a_x \in \mathbb{C}$, let $W_a = \sum_x a_x L_x^{\otimes b}$. Its matrix entries are $(W_a)_{\mu,\nu} = \sum_x a_x \mathbf{1}_{\mu=x\nu}$. If P is the invariant-boundary projector, then

$$W_a P = \left(\sum_x a_x \right) P, \quad P W_a P = \left(\sum_x a_x \right) P.$$

Proof. [✓ Proof](#) Every common translation fixes every vector in the range of P . Sum this identity with the specified weights and use $P^2 = P$. $\square$

Theorem 24.12.4.179 (Actual twisted-region cross Gram matrix). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Under the hypotheses of Lemma 24.12.4.176, put

$$F_{u,w} = \{x \in G : u_e x = x w_e \text{ for every internal bond } e\}, \quad \kappa_R = \left(\prod_{v \in R} c_v \right) \frac{|G|^{I_R+1}}{|G|^{|R|}}.$$

The actual maps satisfy

$$(T_R^u)^\dagger T_R^w = \frac{\kappa_R}{|G|} \sum_{x \in F_{u,w}} L_x^{\otimes b}, \quad P(T_R^u)^\dagger T_R^w P = \kappa_R \frac{|F_{u,w}|}{|G|} P.$$

These are auxiliary finite-region identities with identity operators on crossing bonds. The general torus cut of [203, Theorem 6.9] remains separate [114].

Proof. [✓ Proof](#) Apply the expansion and synchronization lemmas. Counting the free internal labels contributes $|G|^{I_R}$, leaving the prefactor $\kappa_R/|G|$. On the invariant boundary each remaining translation acts identically, giving the projected formula. $\square$

Theorem 24.12.4.180 (Superpositions with a common boundary density). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let $b = n + 1$, let P be the regular boundary projector, and suppose specified cut maps satisfy

$$A^\dagger A = c_A P, \quad c_A > 0, \quad P N_s^\dagger N_t P = k_{s,t} P.$$

For coefficients λ_s , put $B_\lambda = (\sum_s \lambda_s N_s) P$ and $K_\lambda = A B_\lambda^\top$. If $K_\lambda \neq 0$, a positive scalar normalizes its coefficient vector, whose reduced density is

$$\rho_A = \frac{1}{c_A |G|^n} A P A^\dagger.$$

This density is independent of λ . It has rank $|G|^n$, satisfies $\rho_A^2 = |G|^{-n} \rho_A$, and has entropy $n \log |G|$. This is a conditional matrix statement; identification of its Gram hypotheses with a PEPS contraction is separate.

Proof. [✓ Proof](#) Expanding the mixed Gram matrices gives

$$B_\lambda^\dagger B_\lambda = \kappa_\lambda P, \quad \kappa_\lambda = \sum_{s,t} \bar{\lambda}_s \lambda_t k_{s,t}.$$

Nonvanishing of K_λ implies $B_\lambda \neq 0$; positivity of its Gram matrix and its nonzero trace imply $\kappa_\lambda > 0$. Normalize A and B_λ by their positive square-root factors and apply the physical boundary density calculation. $\square$

Theorem 24.12.4.181 (Common density for actual PEPS with complementary twists). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let every site tensor of a finite graph be regular G -isometric. Let R be nonempty and connected, with $b = n + 1$ crossing bonds. Consider a finite family of bond assignments $u^{(s)}$, all equal to the identity on edges incident to R . Suppose their common identity bonds contain a connected spanning subgraph of R^c . Let ψ_s be the actual physical PEPS across the cut with assignment $u^{(s)}$.*

There is one $c_R > 0$ such that the genuine untwisted region map satisfies $T_R^\dagger T_R = c_R P$. Every nonzero superposition $\sum_s \lambda_s \psi_s$ has a positive normalization whose reduced density on R is the same operator

$$\rho_R = \frac{1}{c_R |G|^n} T_R P T_R^\dagger.$$

In particular its rank is $|G|^n$, its nonzero spectrum is flat, and its entropy is $n \log |G|$. This is an auxiliary finite-cut statement with twists confined to the complement, rather than the arbitrary torus cut of [203, Theorem 6.9] [114].

Proof. [✓ Proof](#) Identity operators on edges incident to R leave its actual open map unchanged. The connected-region calculation derives its positive Gram factor from the local isometries. The twisted-region calculation derives each projected complementary mixed Gram matrix as a scalar multiple of P . Exact contraction across the cut identifies the actual superposition with the matrix state of Theorem 24.12.4.180. Apply that theorem; no Gram identity for a whole region is assumed. $\square$

Definition 24.12.4.182 (The rectangular grid obtained by cutting the torus seams). [Lean](#) [Lean](#) [✓ Stmt](#) *On the torus vertices, retain only consecutive coordinate values without wraparound. The result is the ordinary rectangular grid, a subgraph of the periodic square lattice.*

Theorem 24.12.4.183 (Zero-order entropy of regular G -injective torus closures). [Lean](#) [✓ Stmt](#) *Let both torus circumferences be at least three, and let A be a G -injective site tensor with the regular virtual representation. Let R be a contiguous block: its induced nearest-neighbour graph is connected, and its actual closed-cell realization is simply connected. Let b be the number of bonds crossing its boundary. For any nonzero coherent superposition of actual closures with commuting pairs, let M be its physical coefficient matrix across the cut and put*

$$\rho_R = \frac{MM^\dagger}{\text{tr}(MM^\dagger)}. \quad (24.140)$$

The denominator is positive, ρ_R is a density matrix, and

$$\text{rank } \rho_R = |G|^{b-1}, \quad S_0(\rho_R) = \log \text{rank } \rho_R = (b-1) \log |G|. \quad (24.141)$$

Trees, roots, boundary numbering, and complementary paths are not supplied. This is the regular native-torus conclusion of [203, Corollary 6.10], under the stated contiguous closed-cell geometry. It makes no flat-spectrum assertion and does not identify the closure span with a parent-Hamiltonian ground space [114].

Proof. ✓ Proof The products of local inverse maps carry the original state to the canonical closure superposition. The products of the original site maps recover it. Thus the two coefficient matrices have equal rank. Numbering the canonical physical basis preserves its isometry and its cut rank, so the simply connected isometric density theorem gives the canonical rank $|G|^{b-1}$. A matrix and its row Gram matrix have equal rank; division by the positive trace preserves that rank. The definition of zero-order Rényi entropy gives the stated logarithm. $\square$

Lemma 24.12.4.184 (Connectivity of rectangles and their complements). Lean Lean Lean Lean Lean ✓ Stmt A bounded coordinate rectangle with positive side lengths is connected in the ordinary grid, hence also in the periodic lattice. If both side lengths are strictly smaller than the torus dimensions, its complement is connected even without the wrap bonds. This last assertion also holds for coordinate intervals clipped by the bounds of the grid. A nonempty bounded rectangle starting at a positive horizontal coordinate has a crossing bond on its left side.

Proof. ✓ Proof Inside the rectangle, change the two coordinates consecutively. For the complement, choose one row and one column missed by the rectangle. Every outside vertex can reach their intersection along an outside row or column. A vertex at the lower left corner and its immediate left neighbour give the crossing bond. $\square$

Theorem 24.12.4.185 (Simply connected rectangular cell regions). Lean ✓ Stmt Let the torus periods be positive integers w, h . Let R be the native rectangle with coordinate ranges $a \leq x < a + \ell$ and $b \leq y < b + m$, where $\ell, m > 0$, $\ell < w$, $m < h$, $a + \ell \leq w$, and $b + m \leq h$. Then the actual closed-cell realization of R is simply connected. This is auxiliary geometry for the stripe reduction in [203, Theorem 6.7], local source lines 1995–2011.

Proof. ✓ Proof The torus projection maps the planar rectangle $[a - 1/2, a + \ell - 1/2] \times [b - 1/2, b + m - 1/2]$ onto the closed-cell realization of R . To verify surjectivity, choose in each coordinate an integer center in the indicated range at distance at most $1/2$; conversely, every cell centered in R lies in this planar rectangle. Since its side lengths are smaller than the periods, projection is injective there. The continuous bijection from this compact rectangle to its Hausdorff image is a homeomorphism. The planar rectangle is nonempty and convex, hence contractible and simply connected. $\square$

Theorem 24.12.4.186 (Unitary equivalence of closure states on two width-one stripes). Lean Lean ✓ Stmt Consider an original regular G -isometric tensor on a native torus whose two periods are at least three. Let R be the nonwrapping rectangle of sites with both coordinates nonzero. Its complement is the union of two width-one stripes wrapping around the torus:

$$S = \{(x, y) : x = 0 \vee y = 0\}. \quad (24.142)$$

For any two nonzero coherent superpositions of commuting torus closures, let A and B be their physical coefficient matrices across $R \mid S$. There exists a unitary U on the physical factor of S satisfying

$$\widehat{B} = \widehat{A}U^\top. \quad (24.143)$$

Thus U transforms the first normalized physical state into the second while acting only on the stripes (24.142). This is the closure-vector specialization of [203, Theorem 6.7] to regular native

color f_v and the top color u_v of each site. The top left spin gives $u_v = s_1(v)f_v$, the top right spin gives $f_{v+(1,0)} = a_v f_v$, the bottom left spin of the site above gives $f_{v+(0,1)} = b_v f_v$, and the bottom right spin is then equivalent to $s_2 s_1 = s_3 s_4$ at v . So the colorings are counted by the parallel sections of a, b when the local rule holds, and there are none otherwise. Using the local rule, flatness of a, b is the plaquette rule, and Lemma 24.12.4.196 counts the sections. $\square$

Definition 24.12.4.198 (Primal quantum-double tensor). [Lean](#) [Lean](#) [✓ Stmt](#) A tensor carries the four edge spins g_1, g_2, g_3, g_4 of a plaquette (top left, top right, bottom right, bottom left), and is 1 when its virtual labels are $g_1 g_2$ (top), $g_2^{-1} g_3$ (right), $g_4 g_3$ (down) and $g_1 g_4^{-1}$ (left), and 0 otherwise [16, Appendix A, “The Toric Code and quantum double models”].

Theorem 24.12.4.199 (The primal virtual labels fuse to the identity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On the support of the primal tensor the virtual labels satisfy $trb^{-1}l^{-1} = 1$. Hence $\pi(t)\pi(r)\pi(b)^{-1}\pi(l)^{-1} = 1$ for every representation π of G , in particular every irreducible one, so that each entry of the tensor times the operator $\pi(t)\pi(r)\pi(b)^{-1}\pi(l)^{-1}$, acting by π on the top and right legs and by π^{-1} on the down and left legs, equals the same entry times the identity. For a one-dimensional representation χ the tensor is invariant under this virtual action [16, Appendix A, “The Toric Code and quantum double models”]. This is a statement about one tensor; the resulting symmetry of the contracted network is Theorem 24.12.4.200.

Proof. [✓ Proof](#) $(g_1 g_2)(g_2^{-1} g_3)(g_4 g_3)^{-1}(g_1 g_4^{-1})^{-1} = g_1 g_3 g_3^{-1} g_4^{-1} g_4 g_1^{-1} = 1$. Since π is multiplicative,

$$\pi(t)\pi(r)\pi(b)^{-1}\pi(l)^{-1} = \pi(trb^{-1}l^{-1}) = \pi(1) = 1$$

on the support, which gives the operator identity entrywise. A one-dimensional representation is the case of a character χ , where the identity reads $\chi(t)\chi(r)\chi(b)^{-1}\chi(l)^{-1} = 1$. $\square$

Theorem 24.12.4.200 (Virtual symmetry of the primal quantum-double network). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Place the primal quantum-double tensor at every site of the torus, and let π be a representation of G , in particular an irreducible one. The virtual string that multiplies $\pi(g)$ for each bond with label g it crosses can be pulled through every rectangle of sites of the contracted network: inserted along the left and bottom sides of the rectangle it gives the same network as inserted along its top and right sides [16, Appendix A, “The Toric Code and quantum double models”]. Around one site the two paths give $\pi(l)\pi(b)$ and $\pi(t)\pi(r)$, so the closed string around a site is the operator $\pi(t)\pi(r)\pi(b)^{-1}\pi(l)^{-1}$ of Theorem 24.12.4.199. The statement holds on every torus of width and height at least two; the source’s network, with four distinct bonds at each site, is the case of width and height at least three [148].

Proof. [✓ Proof](#) On the support of the primal tensor $trb^{-1}l^{-1} = 1$, so $tr = lb$ and $\pi(t)\pi(r) = \pi(l)\pi(b)$. This is the hypothesis of Lemma 24.12.4 for $X_g = \pi(g)$. $\square$

Theorem 24.12.4.201 (Nonzero regular states with inserted bond operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let G be finite, and let the local tensors on a finite graph be regular G -injective. Insert arbitrary group-valued operators on the oriented bonds. The resulting closed PEPS vector is nonzero. More generally, each open-region vector with identity boundary labels is nonzero, as is the corresponding canonical projector column. In particular, on a torus with both periods at least three, every actual regular G -isometric state with inserted bond operators is nonzero. This is an auxiliary nonvanishing result for the flux measurements of [203, Theorem 6.15], without a parent-Hamiltonian claim.

Proof. [✓ Proof](#) Set all bond labels and vertex translations to the identity, and choose each canonical physical row to be the resulting inserted half-edge labels. This gives a positive term in the

actual canonical coefficient sum. All remaining terms are nonnegative. The derived local physical inverse sends the original contraction to this canonical column, so the original vector cannot vanish. For the full region, the exterior multiplicity is one and the open-region contraction is the closed PEPS vector. The native torus assertion follows from the incident-leg expression of local regular isometry. $\square$

Definition 24.12.4.202 (A cycle conjugacy-class detector). [Lean](#) [✓ Stmt](#) Let G be a finite group and let R be a finite region of an ordered graph. Fix a rooted spanning tree T of the induced graph on R and a non-tree internal edge e . Write the exposed coordinates of a physical half-edge configuration α as (y, a, r, z) . For a conjugacy class C of G , define the diagonal operator

$$D_C|\alpha\rangle = \mathbf{1}_{\{[z_e]=C\}}|\alpha\rangle. \quad (24.144)$$

This is the chosen-cycle form of the conjugacy-class measurement in the proof of [203, Theorem 6.15].

Theorem 24.12.4.203 (Physical projective measurement of a block cycle). [Lean](#) [✓ Stmt](#) In the setting of Definition 24.12.4.202, suppose every original local tensor is regular G -isometric. There is a single projective measurement $(Q_C)_C$, together with an additional outcome $Q_\perp$, on the original physical spins in R . Every operator is Hermitian and positive semidefinite, and the full family satisfies

$$Q_i Q_j = \delta_{ij} Q_i, \quad Q_\perp + \sum_C Q_C = \mathbb{1}. \quad (24.145)$$

For arbitrary inserted regular-bond operators u_f , let $\omega_e(u)$ be the cycle residual obtained from the rooted spanning-tree gauge. Denote the actual twisted open-block column with crossing labels θ by $\psi_{u,\theta}$. Then

$$Q_C \psi_{u,\theta} = \mathbf{1}_{\{[\omega_e(u)]=C\}} \psi_{u,\theta}, \quad Q_\perp \psi_{u,\theta} = 0. \quad (24.146)$$

Thus every nonzero such column yields the conjugacy class of its cycle residual with certainty. This establishes the conjugacy calculation $ab^{-1} = xgx^{-1}$ in [203, proof of Theorem 6.15] for a connected block with a chosen spanning-tree cycle. Its geometric scope and the remaining native plaquette identification are recorded in [114].

Proof. [✓ Proof](#) Independent vertex translations conjugate z_e by their common root translation, so every D_C commutes with the product local averaging projector P . The diagonal operators satisfy $D_C D_D = \delta_{CD} D_C$ and $\sum_C D_C = \mathbb{1}$. Let A be the product of the original local physical maps. Local G -isometry gives $A^\dagger A = cP$ with $c > 0$, while invariance gives $AP = A$. Set

$$Q_C = c^{-1} A D_C A^\dagger, \quad Q_\perp = \mathbb{1} - \sum_C Q_C, \quad Q_C A = A D_C. \quad (24.147)$$

These identities imply (24.145); Hermitian idempotence implies positivity. The extra projection annihilates A . Every nonzero canonical twisted coefficient has $z_e = x\omega_e(u)x^{-1}$ for some $x \in G$. Consequently D_C fixes precisely the columns with $[\omega_e(u)] = C$ and annihilates the others. The exact recovery of the original twisted columns through A , followed by (24.147), gives (24.146). $\square$

Theorem 24.12.4.204 (Physical measurement of a closed-walk holonomy). [Lean](#) [✓ Stmt](#) Let R have connected induced graph and let p be a specified closed walk based at a vertex of R . Suppose the original local tensors are regular G -isometric for a finite group G . There is a complete projective measurement on the original physical spins in R , indexed by the conjugacy

classes of G and one unused outcome, such that every actual twisted open-block column $\psi_{u,\theta}$ has outcome $[\text{Hol}_u(p)]$ with certainty. The operators are Hermitian and positive semidefinite, satisfy $Q_i Q_j = \delta_{ij} Q_i$, and sum to the identity on the whole physical space. This is an auxiliary closed-walk form of the conjugacy calculation in [203, proof of Theorem 6.15].

Proof. ✓ Proof Choose a rooted spanning tree. The actual walk gives a group word w_p in the residual cycles. Evaluation commutes with simultaneous conjugation, and the normalized root gauge gives $w_p(\omega(u)) = \text{Hol}_u(p)$. Replace z_e in (24.144) by $w_p(z)$. This detector still commutes with independent vertex translations. The construction (24.147) therefore gives a complete physical projective measurement with the stated outcome on every actual twisted column. $\square$

Definition 24.12.4.205 (The native plaquette walk and its physical sites). Lean Lean Lean ✓ Stmt On a native torus whose two periods are at least three, let $v = (x, y)$. Define the plaquette walk p_v and its physical region R_v by

$$p_v : (x, y) \longrightarrow (x+1, y) \longrightarrow (x+1, y+1) \longrightarrow (x, y+1) \longrightarrow (x, y), \quad (24.148)$$

$$R_v = \{(x, y), (x+1, y), (x+1, y+1), (x, y+1)\}, \quad |R_v| = 4. \quad (24.149)$$

Coordinates are periodic, so this definition includes plaquettes crossing either seam. It is the four-site region used in [203, proof of Theorem 6.15].

Definition 24.12.4.206 (A one-bond regular flux string). Lean Lean ✓ Stmt For $g \in G$, put directed transport g on the first side of p_v and identity transport on every other bond. In the ordered-edge convention, the distinguished bond therefore carries g when its orientation agrees with the walk and g^{-1} otherwise. Denote this actual assignment by $u^{v,g}$. Its plaquette holonomy is

$$\text{Hol}_{u^{v,g}}(p_v) = g. \quad (24.150)$$

This is the one-bond open regular string of [203, Section 6.5 and proof of Theorem 6.15].

Proof. ✓ Proof The three other sides of the plaquette have at least one endpoint with vertical coordinate $y+1$, whereas both endpoints of the distinguished side have coordinate y . Hence their ordered edges are different. Their transports are the identity, and the orientation convention gives transport g on the first step. The reverse-ordered product along the four steps is therefore g . $\square$

Theorem 24.12.4.207 (Four-site measurement of native plaquette holonomy). Lean Lean ✓ Stmt Let the two torus periods be at least three and let the original local tensor be regular G -isometric for a finite group G . At every vertex v , there is a single complete projective measurement $(Q_C)_C$, with one additional outcome $Q_\perp$, on the four original physical spins in R_v . Every operator is Hermitian and positive semidefinite, the operators satisfy $Q_i Q_j = \delta_{ij} Q_i$, and their sum is the identity. For every actual inserted bond assignment u and every crossing configuration θ , its twisted open-block column satisfies

$$Q_C \psi_{u,\theta} = \mathbf{1}_{\{[\text{Hol}_u(p_v)] = C\}} \psi_{u,\theta}, \quad Q_\perp \psi_{u,\theta} = 0. \quad (24.151)$$

The same operators act on the actual globally contracted coefficient matrix M_u across R_v by

$$Q_C M_u = \mathbf{1}_{\{[\text{Hol}_u(p_v)] = C\}} M_u, \quad Q_\perp M_u = 0. \quad (24.152)$$

This is the native regular torus version of the local conjugacy calculation in [203, proof of Theorem 6.15]; see [114] for its scope.

Proof. [✓ Proof](#) The induced graph on the support of an actual walk is connected. Apply Theorem 24.12.4.204 to p_v in this induced graph. Inclusion of the induced walk in the ambient torus recovers p_v and preserves its holonomy. Incident-edge coordinates preserve the original regular G -isometry, so the resulting operators act on the original four physical spins. All spanning-tree choices are made within this construction. Factoring the global coefficient matrix into its open block and complementary open block transfers the same eigenvalue from every open-block column to the contracted matrix. $\square$

Theorem 24.12.4.208 (Detection of a native one-bond flux string). [Lean](#) [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 24.12.4.207, one fixed complete projective measurement on the four spins of R_v detects the class $[g]$ of every one-bond flux string $u^{v,g}$. Its extra outcome annihilates every such actual open-block column. If $M_{v,g}$ is the physical coefficient matrix of the globally contracted inserted torus PEPS across the cut R_v , then*

$$Q_C M_{v,g} = \mathbf{1}_{\{[g]=C\}} M_{v,g}, \quad Q_\perp M_{v,g} = 0. \quad (24.153)$$

Thus the measurement detects $[g]$ with certainty on every nonzero such contracted state. This is the explicit one-bond string specialization of [203, Theorem 6.15] on native regular tori of periods at least three.

Proof. [✓ Proof](#) Substitute (24.150) into (24.151). Factor the actual global coefficient matrix across the cut into its open block and complementary open block. Each column of the former has the same measured class $[g]$, so left multiplication by Q_C multiplies the contracted matrix by $\mathbf{1}_{\{[g]=C\}}$. $\square$

Definition 24.12.4.209 (Flux strings attached to each plaquette side). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Enumerate the oriented sides of p_v by $i = 0, 1, 2, 3$ in traversal order. For $g \in G$, let $u^{v,i,g}$ put directed transport g on side i and identity transport on all other bonds. The ordered bond receives g or g^{-1} according to its orientation. Then

$$\text{Hol}_{u^{v,i,g}}(p_v) = g. \quad (24.154)$$

This realizes the four attachment directions in the final paragraph of [203, proof of Theorem 6.15].

Proof. [✓ Proof](#) Opposite horizontal sides have distinct vertical coordinates, and opposite vertical sides have distinct horizontal coordinates. Adjacent sides also have distinct ordered endpoint pairs. Hence all four bonds are different. Exactly one factor in the actual four-step holonomy is g , and the other three are the identity. $\square$

Theorem 24.12.4.210 (A single flux measurement for all four attachment directions). [Lean](#) [✓ Stmt](#) *On native regular tori with both periods at least three, one fixed complete positive projective measurement on the four spins in R_v detects the class of every explicitly constructed string $u^{v,i,g}$, for all four attachment sides and all $g \in G$. Writing $M_{v,i,g}$ for its actual globally contracted coefficient matrix, one has*

$$Q_C M_{v,i,g} = \mathbf{1}_{\{[g]=C\}} M_{v,i,g}, \quad Q_\perp M_{v,i,g} = 0. \quad (24.155)$$

The operators are independent of both i and g . This is the explicit four-direction specialization of the final observation in [203, proof of Theorem 6.15]; it does not assert identification with a parent-Hamiltonian excitation space.

Proof. [✓ Proof](#) Substitute (24.154) into (24.152). $\square$

Corollary 24.12.4.211 (Orthogonality of distinct measured plaquette classes). [Lean](#) [✓ Stmt](#) Under the regular native torus hypotheses of Theorem 24.12.4.207, let u, w be two actual bond assignments. If their holonomies around p_v have distinct conjugacy classes, then their globally contracted coefficient matrices satisfy

$$M_u^\dagger M_w = 0. \quad (24.156)$$

Thus every physical coefficient column of one is orthogonal to every coefficient column of the other, without a normalization hypothesis. This is a consequence of the local class measurement in [203, Theorem 6.15].

Proof. [✓ Proof](#) Let Q be the measurement projector for the class of $\text{Hol}_u(p_v)$. Then $QM_u = M_u$ and $QM_w = 0$. Since Q is Hermitian, $M_u^\dagger M_w = M_u^\dagger QM_w = 0$. $\square$

Theorem 24.12.4.212 (Equality on irreducible summands). [Lean](#) [✓ Stmt](#) Let ρ be a finite-dimensional representation of a finite group over a field in which $|G|$ is nonzero. Two endomorphisms are equal if they agree on every irreducible subrepresentation of ρ . This is the decomposition principle behind the block calculations in [203, Lemma 4.4].

Proof. [✓ Proof](#) Maschke's theorem expresses the representation space as an internal direct sum of irreducible subrepresentations. Agreement on these summands implies agreement on their span, which is the whole space. $\square$

Theorem 24.12.4.213 (Scalar calculus of character projectors). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ρ be a finite-dimensional complex representation of a finite group. Given complex weights $f(\chi)$ on the occurring irreducible characters, put $A_f = \sum_\chi f(\chi)P_\chi$. If v lies in an irreducible summand with character ψ , then $A_f v = f(\psi)v$. Moreover, $A_f \rho(g) = \rho(g)A_f$. Each projector P_χ is idempotent. These formulas express the isotypic decomposition used in [203, Lemma 4.4].

Proof. [✓ Proof](#) On the summand with character ψ , the projector P_χ is the identity if $\chi = \psi$ and zero otherwise. Thus only its recorded weight survives. Each summand is preserved by the group action, which commutes with scalar multiplication. The irreducible summands span the whole representation space. $\square$

Definition 24.12.4.214 (Reciprocal weighted trace operator). [Lean](#) [✓ Stmt](#) For the occurring irreducible characters, with dimensions d_χ and multiplicities m_χ , define

$$\Delta^{-1} = \sum_\chi \frac{|G|m_\chi}{d_\chi} P_\chi. \quad (24.157)$$

These are the reciprocal coefficients of the weighted trace operator in [203, Lemma 4.4].

Theorem 24.12.4.215 (Invertibility of the weighted trace operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The operator in (24.157) is the two-sided inverse of Δ . On a summand with character χ , it acts by $|G|m_\chi/d_\chi$. Both Δ and its inverse commute with every group operator. These assertions hold without semi-regularity or unitarity, for every finite-dimensional complex representation of a finite group.

Proof. [✓ Proof](#) The dimensions and multiplicities of the occurring summands are positive. On each irreducible summand the two coefficients multiply to one: $[d_\chi/(|G|m_\chi)][|G|m_\chi/d_\chi] = 1$. Hence $\Delta^{-1}\Delta = I$ on their direct sum. In finite dimension a left inverse is also a right inverse. Commutation follows from the scalar calculus of the character projectors. $\square$

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Proof. ✓ **Proof** The identity $\Theta^4 = |G|\Delta$ makes its fourth power invertible, and therefore makes Θ invertible. For unitary representations, each occurring character projector is an orthogonal projection and each fourth-root coefficient is nonnegative. Their weighted sum is positive. Its invertibility implies positive definiteness. $\square$

Theorem 24.12.4.222 (The square-root weights on bond sectors). Lean Lean ✓ Stmt *On an irreducible sector of dimension d_i and multiplicity m_i , the operator Θ^2 acts by $\sqrt{d_i/m_i}$. In particular, for multiplicity one it acts by $\sqrt{d_i}$. This is the bond weight obtained from two endpoint fourth-root factors in [203, Section 7].*

Proof. ✓ **Proof** The operator Θ acts by $(d_i/m_i)^{1/4}$ on the sector. Applying it twice squares this positive real fourth root and gives $\sqrt{d_i/m_i}$. Set $m_i = 1$ for the second assertion. $\square$

Definition 24.12.4.223 (Averaging sites with fourth-root weights). Lean Lean ✓ Stmt Let U be a representation of a finite group on a finite virtual space, and let Θ be its fourth-root isotypic weight operator. Obtain the weighted site from the canonical averaging site by placing Θ on each of its four virtual legs. For the multiplicity-one representation, this is the tensor displayed in [203, Section 7], divided by $|G|$.

Theorem 24.12.4.224 (Fourth-root site weights combine on the bonds). Lean Lean ✓ Stmt *On a torus with positive circumferences, let N be the number of vertices, and write $s_v = (t_v, r_v, b_v, l_v)$ for the physical labels of the weighted averaging sites. Their actual contraction has coefficient*

$$|G|^{-N} \sum_{q: V_{\text{tor}} \rightarrow G} \prod_v [\Theta^2 U(q_{v+\text{east}} q_v^{-1})]_{l_{v+\text{east}}, r_v} [\Theta^2 U(q_v q_{v+\text{north}}^{-1})]_{t_v, b_{v+\text{north}}}. \quad (24.159)$$

Thus the two fourth-root weights on each bond combine to the square-root weight appearing in [203, Section 7]. The factor $|G|^{-N}$ records the normalized site averages; see [114]. More generally, any matrix W commuting with every $U(g)$ may replace Θ in the dressing; the displayed bond factors then contain W^2 . The coefficient identity itself assumes neither unitarity nor semi-regularity.

Proof. ✓ **Proof** Move the virtual-leg matrices into the incident bonds. Each bond acquires two factors Θ , hence Θ^2 . Expand the average independently at each site to obtain a vertex label q_v . The horizontal bond has matrix $U(q_{v+\text{east}})\Theta^2 U(q_v^{-1})$, and the vertical bond has $U(q_v)\Theta^2 U(q_{v+\text{north}}^{-1})$. Since Θ commutes with U , these become the displayed relative matrices. Reading their actual endpoint labels gives the stated product. $\square$

Definition 24.12.4.225 (Restoring bond multiplicities). Lean Lean Lean Lean Lean ✓ Stmt Let V_i and M_i , indexed by a finite set, be finite-dimensional coordinate spaces, with $m_i = \dim M_i > 0$. On the direct sum of matrix spaces, define

$$J((X_i)_i) = (m_i^{-1/2} X_i \otimes I_{M_i})_i. \quad (24.160)$$

On each matrix unit this reads

$$|k\rangle\langle\ell| \mapsto \frac{1}{\sqrt{m_i}} \sum_{a=1}^{m_i} |k, a\rangle\langle\ell, a|. \quad (24.161)$$

For an irreducible representation of dimension d_i , the regular representation has multiplicity $m_i = d_i$. Thus J is precisely the bond transformation used in the semi-regular bond-dimension reduction of [203, Section 7].

Theorem 24.12.4.226 (The multiplicity-restoring bond isometry). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
The map J preserves the Hilbert–Schmidt inner product:

$$\sum_i \text{tr}[J(X)_i^\dagger J(Y)_i] = \sum_i \text{tr}[X_i^\dagger Y_i]. \quad (24.162)$$

In particular, it is injective.

Proof. [✓ Proof](#) For each block,

$$\begin{aligned} \text{tr}[(m_i^{-1/2} X_i \otimes I)^\dagger (m_i^{-1/2} Y_i \otimes I)] &= m_i^{-1} \text{tr}[X_i^\dagger Y_i] \text{tr}[I_{M_i}] \\ &= \text{tr}[X_i^\dagger Y_i]. \end{aligned}$$

Summing gives the assertion. The Hilbert–Schmidt norm vanishes only on the zero matrix, which proves injectivity. $\square$

Theorem 24.12.4.227 (The weighted semi-regular bond becomes a regular bond). [Lean](#) [Lean](#) [✓ Stmt](#) For any matrices D_i , one has

$$J((\sqrt{m_i} D_i)_i) = (D_i \otimes I_{M_i})_i. \quad (24.163)$$

Taking $m_i = d_i$ and $D_i = D^i(gh^{-1})$ gives the two bond vectors compared in [203, Section 7]. This establishes the isometry on the bond matrix spaces; the equality of complete PEPS contractions after its application on all bonds is a further statement.

Proof. [✓ Proof](#) In each block the factors $m_i^{-1/2}$ and $\sqrt{m_i}$ cancel. $\square$

Definition 24.12.4.228 (The coefficient isometry on a bond). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let I be a finite set of bond labels, and let M_i be a nonempty finite multiplicity set of cardinality m_i . Define

$$J|i\rangle = \frac{1}{\sqrt{m_i}} \sum_{a \in M_i} |i, a, a\rangle. \quad (24.164)$$

Consequently, $J(\sum_i \sqrt{m_i} x_i |i\rangle) = \sum_{i,a} x_i |i, a, a\rangle$. The label i may include an irreducible representation and both matrix indices, as in the bond sums of [203, Section 7].

Theorem 24.12.4.229 (Coherent bond states under restoration of multiplicities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On any finite set E of bonds, $J_E = \bigotimes_{e \in E} J$ is an isometry. For any finite family of coefficients w_q and bond amplitudes $x_e(q, i)$, it carries

$$\psi(\sigma) = \sum_q w_q \prod_{e \in E} \sqrt{m_{\sigma_e}} x_e(q, \sigma_e) \quad (24.165)$$

to the vector with coefficients

$$(J_E \psi)((i_e, a_e, b_e)_e) = \sum_q w_q \prod_{e \in E} \mathbf{1}_{a_e = b_e} x_e(q, i_e). \quad (24.166)$$

Thus it preserves the overlaps and norms of all such coherent states. This proves the transformation of the displayed bond-vector sums in [203, Section 7]; the identification with site tensors carrying the fourth-root weights is a separate assertion.

Proof. ✓ Proof Distinct original labels have disjoint images, and the image of $|i\rangle$ has squared norm $m_i/m_i = 1$. The tensor product is therefore an isometry. Apply its one-bond action to each factor and distribute over the finite coherent sum. Each square-root weight cancels the normalization of its bond map. $\square$

Theorem 24.12.4.230 (Weighted bond coefficients in irreducible-sector coordinates). Lean

✓ Stmt Let a finite-dimensional representation ρ of a finite group have a supplied internal decomposition into irreducible subrepresentations S_i , each with character multiplicity one. Choose coordinates $\psi_i : S_i \cong \mathbb{C}^{d_i}$, and let $D_i(g)$ be the matrices of the restricted representations in these coordinates. There is one coordinate equivalence e for which

$$[\rho(g)]_e = \bigoplus_i D_i(g), \quad [\Theta^2 \rho(g)]_e = \bigoplus_i \sqrt{d_i} D_i(g). \quad (24.167)$$

Apply the normalized multiplicity-restoring bond map to the actual diagonal blocks of the second matrix. Its result is $(D_i(g) \otimes I_{d_i})_i$. This is the coefficient transformation displayed in [203, Section 7]. The assertion concerns the occurring irreducible sectors in the supplied decomposition; it does not require all irreducibles to occur. The coordinates are arbitrary linear coordinates, so the assertion does not claim that e is unitary.

Proof. ✓ Proof Use the coordinates collected from the internal decomposition, simultaneously for the endomorphisms $\rho(g)$ and $\Theta^2 \rho(g)$. Each preserves every irreducible summand. On the summand S_i , the fourth-root square acts by $\sqrt{d_i}$, so the second family has precisely the stated weighted diagonal blocks. Irreducibility gives $d_i > 0$. The normalized bond map sends $\sqrt{d_i} D_i(g)$ to $D_i(g) \otimes I_{d_i}$, cancelling its normalization. $\square$

Theorem 24.12.4.231 (Orthonormal coordinates for weighted bond coefficients). Lean ✓ Stmt

Let a representation ρ of a finite group on a finite-dimensional complex Hilbert space have a supplied orthogonal internal decomposition into irreducible subrepresentations S_i , each with character multiplicity one. Choose orthonormal bases of the S_i , and let $D_i(g)$ be the matrices of the restricted representations in these coordinates. There is one global orthonormal basis e for which

$$[\rho(g)]_e = \bigoplus_i D_i(g), \quad [\Theta^2 \rho(g)]_e = \bigoplus_i \sqrt{d_i} D_i(g). \quad (24.168)$$

Apply the normalized multiplicity-restoring bond map to the actual diagonal blocks of the second matrix. Its result is $(D_i(g) \otimes I_{d_i})_i$. This is the coefficient transformation displayed in [203, Section 7]. The assertion concerns the occurring irreducible sectors in the supplied decomposition; it does not require all irreducibles to occur. Its coordinate identification is unitary. Orthogonality and the decomposition are supplied hypotheses; this auxiliary assertion does not construct them or identify complete PEPS.

Proof. ✓ Proof Collect the sector orthonormal bases into a global orthonormal basis, simultaneously for the endomorphisms $\rho(g)$ and $\Theta^2 \rho(g)$. Each preserves every irreducible summand. On the summand S_i , the fourth-root square acts by $\sqrt{d_i}$, so the second family has precisely the stated weighted diagonal blocks. Irreducibility gives $d_i > 0$. The normalized bond map sends $\sqrt{d_i} D_i(g)$ to $D_i(g) \otimes I_{d_i}$, cancelling its normalization. $\square$

24.12.5 The AKLT state

The two-dimensional AKLT state places singlets on the edges of the lattice and projects the four $\text{spin-}\frac{1}{2}$ at each site onto their symmetric subspace [16, Appendix A, “The AKLT model”]. Label

the basis of four spin- $\frac{1}{2}$ by $s = (s_0, s_1, s_2, s_3) \in \{0, 1\}^4$, write $|s|$ for the number of entries equal to 1, and let P_π be the permutation of the four factors by $\pi \in S_4$.

Definition 24.12.5.1 (Symmetric projector on four spin- $\frac{1}{2}$). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) The projector onto the symmetric subspace of $(\mathbb{C}^2)^{\otimes 4}$ is $\Pi_{\text{sym}} = \frac{1}{24} \sum_{\pi \in S_4} P_\pi$.

Lemma 24.12.5.2 (The symmetric subspace has dimension five). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Π_{sym} is a Hermitian idempotent commuting with $U^{\otimes 4}$ for every 2×2 matrix U . Its entries are $(\Pi_{\text{sym}})_{s,t} = \binom{4}{|s|}^{-1}$ if $|s| = |t|$ and 0 otherwise, and its range, the spin-2 subspace, has dimension five.

Proof. [Stmt](#) $P_\pi P_\rho = P_{\rho\pi}$ and $P_\pi^\dagger = P_{\pi^{-1}}$, so the average over S_4 is a Hermitian idempotent, and $P_\pi U^{\otimes 4} = U^{\otimes 4} P_\pi$ by reordering the tensor factors. The number of π with $s = t \circ \pi$ is $24 / \binom{4}{|t|}$ when $|s| = |t|$ and 0 otherwise, by a finite check. The dimension of the range of an idempotent is its trace, $\sum_s \binom{4}{|s|}^{-1} = \sum_{k=0}^4 \binom{4}{k} \binom{4}{k}^{-1} = 5$. $\square$

Definition 24.12.5.3 (AKLT tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) With the singlet matrix $Y = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the AKLT tensor on the square lattice is $A = \Pi_{\text{sym}}(\mathbb{1} \otimes \mathbb{1} \otimes Y \otimes Y)$, with physical index $s \in \{0, 1\}^4$ (physical dimension 16) and virtual indices ordered top, right, down, left [16, Appendix A, “The AKLT model”].

Lemma 24.12.5.4 (Entries and physical span of the AKLT tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) The entries of the AKLT tensor are

$$A_{\alpha\beta\gamma\delta}^s = \frac{(-1)^{\gamma+\delta}}{\binom{4}{k}} \delta_{|s|,k}, \quad k = \alpha + \beta + (1 - \gamma) + (1 - \delta),$$

and the span of the vectors $s \mapsto A_{\alpha\beta\gamma\delta}^s$ over all virtual indices is the range of Π_{sym} .

Proof. [Stmt](#) $Y|0\rangle = |1\rangle$ and $Y|1\rangle = -|0\rangle$, so the column $(\alpha, \beta, \gamma, \delta)$ of $\mathbb{1} \otimes \mathbb{1} \otimes Y \otimes Y$ is $(-1)^{\gamma+\delta}$ times the basis vector $(\alpha, \beta, 1 - \gamma, 1 - \delta)$. The matrix $L = \mathbb{1} \otimes \mathbb{1} \otimes Y \otimes Y$ satisfies $LL^T = \mathbb{1}$, so the columns of $\Pi_{\text{sym}} L$ span the range of Π_{sym} . $\square$

Theorem 24.12.5.5 ($SU(2)$ covariance of the AKLT tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Regard A as the map from the virtual space $(\mathbb{C}^2)^{\otimes 4}$ to the physical space. For every $U \in SU(2)$,

$$U^{\otimes 4} A = A(U \otimes U \otimes \bar{U} \otimes \bar{U}).$$

For an arbitrary 2×2 matrix U the same identity holds with $\bar{U}$ replaced by $(\text{adj } U)^T$.

Proof. [Stmt](#) For a 2×2 matrix, $UY = Y(\text{adj } U)^T$ by direct computation. Since Π_{sym} commutes with $U^{\otimes 4}$, $U^{\otimes 4} \Pi_{\text{sym}} L = \Pi_{\text{sym}}(U \otimes U \otimes UY \otimes UY) = \Pi_{\text{sym}} L(U \otimes U \otimes V \otimes V)$ with $V = (\text{adj } U)^T$. For $U \in SU(2)$, $U^\dagger U = \mathbb{1}$ and $U \text{adj } U = (\det U) \mathbb{1} = \mathbb{1}$, so $\text{adj } U = U^\dagger U \text{adj } U = U^\dagger$ and $V = \bar{U}$. $\square$

Theorem 24.12.5.6 (The AKLT PEPS as projected singlets). [Lean](#) [Lean](#) [Stmt](#) On a torus with $w, h \geq 3$ the AKLT PEPS has coefficient

$$\text{Coeff}(\sigma) = \sum_{\tau} \prod_v (\Pi_{\text{sym}})_{\sigma_v, \tau_v} \prod_v Y_{\tau_{v+(1,0),3}, \tau_{v,1}} Y_{\tau_{v+(0,1),2}, \tau_{v,0}},$$

the sum running over the spin- $\frac{1}{2}$ labels $\tau_v \in \{0, 1\}^4$ of the four legs of every site. The factor $Y_{b,a}$ is the coefficient of $|a\rangle|b\rangle$ in the singlet $|01\rangle - |10\rangle$, so the state is $\bigotimes_v \Pi_{\text{sym}}$ applied to a singlet on every edge.

Proof. [✓ Proof](#) Expand each $A = \Pi_{\text{sym}} L$ as a sum over the intermediate label τ_v . The identity factors of L force the bond index on the top edge of v to be $\tau_{v,0}$ and on the right edge to be $\tau_{v,1}$, and the Y factors on the down and left legs then give $Y_{\tau_{v,2}, \tau_{v-(0,1),0}} Y_{\tau_{v,3}, \tau_{v-(1,0),1}}$; shifting the product by one site gives the formula. $\square$

The tensor A takes values in $(\mathbb{C}^2)^{\otimes 4}$ with physical support on the five-dimensional symmetric subspace, on which the physical action $U^{\otimes 4}$ acts as the irreducible spin-2 representation [16, Appendix A, “The AKLT model”].

24.12.6 The RVB state

The nearest-neighbour resonating valence bond state is the superposition of all coverings of the lattice by nearest-neighbour singlets. Its PEPS of bond dimension three combines, at each site, the projector

$$P = |0\rangle[(0222| + (2022| + (2202| + (2220|] + |1\rangle[(1222| + (2122| + (2212| + (2221|]$$

with singlet matrices on the edges; on the square lattice with a translation-invariant orientation of the singlets the tensor is $A = P(\mathbb{1} \otimes \mathbb{1} \otimes Y \otimes Y)$ [16, Appendix A, “The RVB state”]. On the bond of dimension three the matrix Y acts on the spin states $0, 1$ and is extended by 1 on the vacuum state 2 , so that the bond state is $|01\rangle - |10\rangle + |22\rangle$. The extension by 0 would make every entry of A vanish, since a leg carrying Y could then never be in the vacuum state [121].

Definition 24.12.6.1 (RVB tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The RVB tensor is $A = P(\mathbb{1} \otimes \mathbb{1} \otimes Y \otimes Y)$ with $Y = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, virtual indices ordered top, right, down, left.

Lemma 24.12.6.2 (Local dimer constraint). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) $A_{urdl}^s \neq 0$ if and only if exactly one of u, r, d, l differs from 2 , that index being s if it is u or r and $1 - s$ if it is d or l . The nonzero entries equal -1 when the down or left index is 1 and $+1$ otherwise.

Proof. [✓ Proof](#) Y maps the bond index d to $1 - d$ on the spin states, with sign -1 at $d = 1$, and fixes the vacuum. The statement is a finite check over the 162 entries. $\square$

Theorem 24.12.6.3 (The RVB PEPS is the superposition of singlet coverings). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $w, h \geq 3$. A dimer covering of the torus is a set c of edges such that every site lies on exactly one edge of c . The RVB PEPS has coefficient

$$\text{Coeff}(\sigma) = \sum_c \prod_{(v, v+e) \in c} Y_{\sigma_{v+e}, \sigma_v},$$

the sum running over all dimer coverings, and each edge of c written as $(v, v + e)$ with $e \in \{(1, 0), (0, 1)\}$. The factor $Y_{\sigma_{v+e}, \sigma_v}$ is the coefficient of the singlet $|01\rangle - |10\rangle$ on the two sites, so the state is the superposition of all nearest-neighbour singlet coverings with the translation-invariant orientation.

Proof. [✓ Proof](#) By Lemma 24.12.3 the coefficient is a sum over bond indices h_v, v_v on the right and up edges. By Lemma 24.12.6.2 a configuration contributes only if the edges carrying a non-vacuum index form a dimer covering and the index on each such edge is the physical index of the site below or to the left of it. Conversely every dimer covering determines one such configuration, and its term is the product over sites of the singlet factors on the down and left legs; shifting the product by one site gives the formula. $\square$

24.12.7 PEPS from classical models

Let $q \geq 1$, let $h: \{0, \dots, q-1\}^2 \rightarrow \mathbb{R}$ be a nearest-neighbour interaction and $\beta \in \mathbb{R}$ an inverse temperature.

Definition 24.12.7.1 (Classical energy). [Lean](#) [✓ Stmt](#) On the torus a configuration σ has energy

$$H(\sigma) = \sum_v (h(\sigma_v, \sigma_{v-(1,0)}) + h(\sigma_v, \sigma_{v-(0,1)})),$$

each nearest-neighbour bond counted once. For symmetric h this is the sum of h over unordered bonds.

Definition 24.12.7.2 (Classical-model tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $M = \sum_{i,j} e^{-\beta h(i,j)/2} |i\rangle\langle j|$. The classical-model tensor is

$$A = \left[\sum_i |i\rangle\langle i i i i| \right] (\mathbb{1} \otimes \mathbb{1} \otimes M \otimes M) :$$

the top and right legs copy the physical index s , and the down and left legs carry M_{sb} and M_{sl} at bond indices b and l [16, Appendix A, “PEPS from classical models”]. The classical-model PEPS places this tensor at every site of the torus.

Theorem 24.12.7.3 (Coefficients of the classical-model PEPS). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On a torus of width and height at least three the classical-model PEPS has coefficient $\text{Coeff}(\sigma) = e^{-\beta H(\sigma)/2}$, so $|\text{Coeff}(\sigma)|^2 = e^{-\beta H(\sigma)}$ is the Gibbs weight of σ [148].

Proof. [✓ Proof](#) By Lemma 24.12.3 the top and right legs force every bond index to equal the physical index of the site below or to the left of it. The remaining factor at v is $M_{\sigma_v \sigma_{v-(0,1)}} M_{\sigma_v \sigma_{v-(1,0)}}$, one factor $e^{-\beta h/2}$ for each bond at v read towards its lower and left neighbours, and the product over v is $e^{-\beta H(\sigma)/2}$. The coefficient is real, so its squared modulus is $e^{-\beta H(\sigma)}$. $\square$

No bond carries a product of two copies of M : the entry $e^{-\beta h/2}$ is not a square root of a transfer matrix, and no symmetry of h is used.

Theorem 24.12.7.4 (Diagonal expectation values are Gibbs averages). [Lean](#) [✓ Stmt](#) Let the torus have width and height at least three and let $O = \sum_{\sigma} O(\sigma) |\sigma\rangle\langle\sigma|$ be a diagonal observable. For the classical-model PEPS state $|\psi\rangle = \sum_{\sigma} \text{Coeff}(\sigma) |\sigma\rangle$,

$$\frac{\sum_{\sigma} \overline{\text{Coeff}(\sigma)} O(\sigma) \text{Coeff}(\sigma)}{\sum_{\sigma} \overline{\text{Coeff}(\sigma)} \text{Coeff}(\sigma)} = \frac{\sum_{\sigma} O(\sigma) e^{-\beta H(\sigma)}}{\sum_{\sigma} e^{-\beta H(\sigma)}},$$

that is, $\langle\psi|O|\psi\rangle/\langle\psi|\psi\rangle$ is the expectation value of O in the Gibbs state $e^{-\beta H}/Z$ [16, Appendix A, “PEPS from classical models”].

Proof. [✓ Proof](#) Replace $\overline{\text{Coeff}(\sigma)} \text{Coeff}(\sigma)$ by $e^{-\beta H(\sigma)}$ in both sums, by Theorem 24.12.7.3. $\square$

In particular, classical correlation functions are expectation values of the PEPS. The review adds that at the critical inverse temperature β_c the PEPS has correlations decaying as a power law, so its parent Hamiltonian must be gapless [16, Appendix A, “PEPS from classical models”], since a gapped local Hamiltonian has exponentially decaying correlations [170, 192].

Theorem 24.12.7.5 (Commuting closure labels and centralizers). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ *Stmt* Let G be a group and choose a representative g_C of each conjugacy class C . Write $\text{Cl}(H)$ for the set of conjugacy classes of a group H . The simultaneous-conjugacy classes of commuting pairs are in bijection with $\coprod_{C \in \text{Cl}(G)} \text{Cl}(Z_G(g_C))$. If G is finite, their number is therefore

$$K = \sum_{C \in \text{Cl}(G)} |\text{Cl}(Z_G(g_C))|. \quad (24.169)$$

This is the group-theoretic classification following [203, Theorem 5.9].

Proof. $\checkmark$ *Proof* Assign to the class of (g, h) the conjugacy class of g . If $ngx^{-1} = g_C$, then $xhx^{-1} \in Z_G(g_C)$ and (g, h) is simultaneously conjugate to (g_C, xhx^{-1}) . For a fixed first element g_C , two such pairs have the same simultaneous-conjugacy class exactly when their second elements are conjugate in $Z_G(g_C)$, by Theorem 24.12.4.191. This proves the stated bijection. Counting its disjoint components gives the formula for K . $\square$

Theorem 24.12.7.6 (Dimension of the commuting torus closure span). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ *Stmt* Let U be a semi-regular representation of a finite group G , and let A be G -injective for its four-leg virtual representation. On a nonempty rectangular torus, let $\mathcal{V}_A$ be the span of the vectors $|\mathcal{M}(A \mid C)\rangle$ indexed by commuting simultaneous-conjugacy classes C . Then

$$\dim \mathcal{V}_A = \sum_{C \in \text{Cl}(G)} |\text{Cl}(Z_G(g_C))|. \quad (24.170)$$

This is the dimension of the commuting closure span appearing in [203, Theorem 5.9].

Proof. $\checkmark$ *Proof* The class-labelled vectors are linearly independent by Theorem 24.12.4.171. They form a basis of their span, whose dimension is therefore the number of commuting pair classes. Apply Theorem 24.12.7.5. $\square$

24.13 Local physical ranges of G -injective tensors

Theorem 24.13.1 (Invariant virtual space and physical range). [Lean](#) [Lean](#) [Lean](#) [Lean](#) $\checkmark$ *Stmt*
Let G be a finite group acting linearly on a complex virtual space $\mathcal{V}$, and let $P_G = |G|^{-1} \sum_{g \in G} U_g$ be the projector onto $\mathcal{V}^G$. If $T : \mathcal{V} \rightarrow \mathcal{P}$ is G -injective, then

$$\ker T = \ker P_G, \quad T|_{\mathcal{V}^G} : \mathcal{V}^G \cong \text{im } T.$$

In finite dimension, $\dim \text{im } T = \dim \mathcal{V}^G$. These are local consequences of [203, Definition 5.1].

Proof. $\checkmark$ *Proof* Invariance gives $TP_G = T$. If $Tx = 0$, then $TP_Gx = 0$; injectivity on $\mathcal{V}^G$ implies $P_Gx = 0$. The converse follows from $TP_G = T$. The restriction is injective by G -injectivity, and every Tx equals $T(P_Gx)$, proving surjectivity onto the physical range. $\square$

Theorem 24.13.2 (Comparison of physical ranges). [Lean](#) [Lean](#) [Lean](#) $\checkmark$ *Stmt* Let $T : \mathcal{V} \rightarrow \mathcal{P}$ and $S : \mathcal{V} \rightarrow \mathcal{Q}$ be G -injective for the same virtual representation. There is a unique linear map $F : \text{im } T \rightarrow \text{im } S$ such that $F(Tx) = Sx$ for every $x \in \mathcal{V}$, and this map is an isomorphism.

Proof. $\checkmark$ *Proof* Compose the inverse of $T|_{\mathcal{V}^G}$ with $S|_{\mathcal{V}^G}$. Since $Tx = T(P_Gx)$ and $Sx = S(P_Gx)$, the resulting map has the required identity. That identity specifies its value at every element of $\text{im } T$, so it also proves uniqueness. $\square$

24.14 Comparison of invariant virtual spaces

Definition 24.14.1 (Virtual comparison). [Lean](#) [✓ Stmt](#) Let G and H be finite groups acting linearly on complex vector spaces $\mathcal{V}$ and $\mathcal{W}$, respectively. Let $T : \mathcal{V} \rightarrow \mathcal{P}$ be G -injective and let $S : \mathcal{W} \rightarrow \mathcal{P}$ be H -injective, with $\text{im } T = \text{im } S$. The virtual comparison is the linear isomorphism

$$\Phi = (S|_{\mathcal{W}^H})^{-1} \circ T|_{\mathcal{V}^G} : \mathcal{V}^G \rightarrow \mathcal{W}^H. \quad (24.171)$$

Both restrictions are isomorphisms onto the common physical range.

Theorem 24.14.2 (Uniqueness of virtual comparison). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Definition 24.14.1, Φ is the unique linear map $F : \mathcal{V}^G \rightarrow \mathcal{W}^H$ satisfying $S(Fx) = Tx$ for every $x \in \mathcal{V}^G$. Reversing T and S gives Φ^{-1} . In finite dimension, $\dim \mathcal{V}^G = \dim \mathcal{W}^H$. This is a local consequence of the invariant-space injectivity in [203, Definition 5.1].

Proof. [✓ Proof](#) The defining composition gives $S(\Phi x) = Tx$. If $S(Fx) = Tx$, then $S(Fx) = S(\Phi x)$, and injectivity of $S|_{\mathcal{W}^H}$ gives $Fx = \Phi x$. Applying the same characterization with the two maps reversed identifies the inverse. The dimension equality follows from the linear isomorphism. $\square$

Lemma 24.14.3 (Physical range recovered from a regular cut). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let G be a finite group and let T and S be G -injective maps from the regular virtual boundary with $b \geq 1$ legs to the two physical spaces of a cut. The coefficient matrix $C = TS^\top$ satisfies

$$\text{im } C = \text{im } T, \quad \text{rank } C = |G|^{b-1}.$$

Consequently, two such factorizations of the same coefficient matrix have the same first physical range, even when their virtual groups or numbers of boundary legs differ. This is a consequence of the regular boundary construction in [203, Definition 5.1 and proof of Theorem 6.9].

Proof. [✓ Proof](#) Write P_G for the regular boundary averaging projector. There is a linear map L with $LS = P_G$. Since $P_G^\top = P_G$ and $TP_G = T$, one has $CL^\top = T$. Thus $\text{im } T \subseteq \text{im } C$, while $C = TS^\top$ gives the reverse inclusion. The rank follows from the dimension of the invariant boundary space. Applying the range identity to each factorization proves the comparison assertion. $\square$

Theorem 24.14.4 (Compatible comparisons across a regular cut). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let T, S and T', S' be two pairs of regular G -injective open maps on the same nonempty virtual boundary. Suppose

$$TS^\top = T'(S')^\top.$$

There are linear boundary maps F, K such that

$$T'F = T, \quad S'K = S, \quad FK^\top = P_G.$$

The maps can be chosen with $P_GF = F$ and $P_GK = K$; with these support conditions, the two physical identities determine each comparison uniquely. The maps act on the whole boundary. This assertion does not give a factorization into maps on individual bonds.

Proof. [✓ Proof](#) The physical ranges agree on the first side by the cut-range identity, and on the second side by applying the same identity to the transposed coefficient matrices. Choose left inverses L and M with $LT' = P_G$ and $MS' = P_G$, and set $F = LT$, $K = MS$. Every vector in the range of T is in the range of T' . Since $T'P_G = T'$, this gives $T'LT = T$; the second-side

identity follows in the same way. Since LT and MS take values in the invariant boundary, the support identities hold. Injectivity of the target open maps on that subspace proves uniqueness. Finally,

$$FK^\top = LTS^\top M^\top = LT'(S')^\top M^\top = P_G P_G^\top = P_G.$$

□

Theorem 24.14.5 (Equality of virtual images under a G -injective map). [Lean](#) [✓ Stmt](#) *Let G be finite, let ρ be a complex representation of G , and let T be G -injective. Write $\Pi = |G|^{-1} \sum_{g \in G} \rho(g)$ for the invariant projector. For virtual vectors x, y , one has $Tx = Ty$ if and only if $\Pi x = \Pi y$. This is a consequence of [203, Definition 5.1].*

Proof. [✓ Proof](#) The equality $Tx = Ty$ is equivalent to $x - y \in \ker T = \ker \Pi$, hence to $\Pi x = \Pi y$. □

Theorem 24.14.6 (A regular G -injective site distinguishes translation orbits). [Lean](#) [✓ Stmt](#) *Let I be a nonempty finite set of incident legs. Let $a(\eta) \in \mathcal{H}$ be the physical vector associated with a configuration $\eta \in G^I$, and suppose that the corresponding site map is G -injective for simultaneous left translation. Then $a(\eta) = a(\theta)$ if and only if $\eta = g\theta$ for some $g \in G$. This is an auxiliary reconstruction consequence of [203, Definition 5.1].*

Proof. [✓ Proof](#) Write δ_η for the coordinate vector at η . Equality of the physical vectors gives $\Pi\delta_\eta = \Pi\delta_\theta$. The matrix entry $\Pi_{\eta,\eta} = |G|^{-1}$ is nonzero, whereas $\Pi_{\eta,\theta} = 0$ if η and θ lie in distinct translation orbits. Thus the configurations lie in the same orbit. The converse is the virtual invariance of the site map. □

Theorem 24.14.7 (Recovering the group of simultaneous virtual relabellings). [Lean](#) [✓ Stmt](#) *In the setting of Theorem 24.14.6, suppose that I has at least two elements. A map $e : G \rightarrow G$ satisfies $a(e \circ \eta) = a(\eta)$ for every configuration η if and only if there exists $g \in G$ with $e(h) = gh$ for every $h \in G$. In particular, the simultaneous permutations of group labels that preserve the site tensor are exactly the left translations.*

Proof. [✓ Proof](#) Choose distinct $i, j \in I$ and fix $h \in G$. Let $\eta_j = h$ and $\eta_k = 1$ for $k \neq j$. By Theorem 24.14.6, there exists $g_h \in G$ with $e \circ \eta = g_h \eta$. Evaluation at i gives $g_h = e(1)$, and evaluation at j gives $e(h) = e(1)h$. Conversely, each left translation preserves the site tensor by its virtual invariance. □

24.15 Ground spaces of finite PEPS regions

Definition 24.15.1 (Ground space of a finite PEPS region). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let R be a region of a finite graph carrying PEPS site tensors. Write $T_R(\mu; \sigma)$ for the contraction over the internal bonds, with crossing-bond configuration μ and physical configuration σ on R . The local ground space is*

$$\mathcal{G}_R = \left\{ \sum_{\mu} x(\mu) T_R(\mu; \cdot) : x \right\} = \text{span}\{T_R(\mu; \cdot) : \mu\}. \quad (24.172)$$

This is the construction of [16, Section IV.C.1], local source lines 2003–2008. It applies to arbitrary finite graphs and site-dependent tensors.

Theorem 24.15.2 (Boundary dimension bound). [Lean](#) [✓ Stmt](#) *If the virtual space on a crossing edge e has dimension D_e , then $\dim \mathcal{G}_R \leq \prod_{e \in \partial R} D_e$.*

Proof. [✓ Proof](#) The contraction is a linear map from the boundary virtual space onto $\mathcal{G}_R$, so its range has dimension at most that of its domain. $\square$

Theorem 24.15.3 (Local slices of a closed PEPS). [Lean](#) [Lean](#) [✓ Stmt](#) *Let ψ be the closed PEPS vector. For each fixed complementary physical configuration τ , the vector $\sigma \mapsto \psi(\sigma, \tau)$ belongs to $\mathcal{G}_R$. Consequently, if a linear operator h_R satisfies $\mathcal{G}_R \subseteq \ker h_R$, then $h_R \psi(\cdot, \tau) = 0$. This is the local contraction assertion underlying frustration freeness in [16, Section IV.C.1], local source lines 2008–2011.*

Proof. [✓ Proof](#) If a global virtual configuration exists, every bond space has positive dimension. The exact contraction across the cut gives $\psi(\sigma, \tau) = \sum_{\mu} T_R(\mu; \sigma) T_{R^c}(\mu; \tau)$, which is a vector of (24.172). If no global virtual configuration exists, the defining sum for ψ is empty and the slice is zero. Applying h_R proves the final assertion. $\square$

24.16 Parent Hamiltonians on finite PEPS graphs

The construction of [16, Section IV.C.1], local source lines 2003–2011, applies to arbitrary finite graphs and site-dependent tensors. Different interaction terms may act on different regions.

Definition 24.16.1 (Regional parent interaction). [Lean](#) [✓ Stmt](#) A parent interaction on a region R is a positive semidefinite operator h_R on its physical space satisfying $\ker h_R = \mathcal{G}_R$. It need not be an orthogonal projector.

Theorem 24.16.2 (Canonical regional parent interaction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The orthogonal projector onto $\mathcal{G}_R^\perp$ is a regional parent interaction.*

Proof. [✓ Proof](#) Orthogonal projections are positive semidefinite, and the kernel of the projector onto $\mathcal{G}_R^\perp$ is $\mathcal{G}_R^{\perp\perp} = \mathcal{G}_R$. $\square$

Definition 24.16.3 (Parent Hamiltonian of a finite family of regions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $(R_i)_{i \in I}$ be a finite family of regions, and let h_i be a parent interaction on R_i . Its extension to the full graph and the parent Hamiltonian are

$$\tilde{h}_i = h_i \otimes \mathbb{1}_{R_i^c}, \quad (24.173)$$

$$H = \sum_{i \in I} \tilde{h}_i. \quad (24.174)$$

The corresponding space of regional boundary conditions is

$$\mathcal{K} = \{\phi : \phi(\cdot, \tau) \in \mathcal{G}_{R_i} \text{ for every } i \in I \text{ and every complementary configuration } \tau\}. \quad (24.175)$$

Theorem 24.16.4 (Positivity and the common kernel). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The parent Hamiltonian is positive semidefinite and satisfies*

$$\ker H = \bigcap_{i \in I} \ker \tilde{h}_i = \mathcal{K}. \quad (24.176)$$

Thus its kernel depends only on the regional PEPS spaces, rather than on the choice of positive interactions with those kernels.

Proof. [✓ Proof](#) Positivity is preserved by tensoring with the identity and by finite sums. For each fixed complementary configuration τ , (24.173) acts on the slice $\phi(\cdot, \tau)$ as h_i . Finally, $\langle \phi, H \phi \rangle = \sum_i \langle \phi, \tilde{h}_i \phi \rangle$ is a sum of nonnegative numbers. If $H \phi = 0$, each summand is zero, and positivity gives $\tilde{h}_i \phi = 0$. The converse follows by summing. $\square$

Theorem 24.16.5 (Frustration freeness of the represented PEPS). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The closed PEPS vector ψ satisfies $\tilde{h}_i \psi = 0$ for every $i \in I$, and hence $H\psi = 0$. Whenever $\psi \neq 0$, it is a ground state of this positive Hamiltonian, and it minimizes every local term simultaneously.*

Proof. [✓ Proof](#) Every complementary slice of ψ belongs to the regional PEPS ground space. Therefore every extended interaction kills ψ , and their sum does as well. $\square$

24.16.1 The one-dimensional specialization

Definition 24.16.1.1 (Physical and boundary coordinates of an arc). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $n \geq 3$ and $0 < L < n$. Enumerate an arc R of L sites in cyclic order. A physical configuration on R is then a word $\sigma = (i_1, \dots, i_L)$. There are exactly two crossing bonds, entering and leaving the arc. If a boundary coefficient is $x(a, b)$ on their ordered bond values, its boundary matrix is $X_{ba} = x(a, b)$. Write E_R for the corresponding ordering of physical coefficients.

Lemma 24.16.1.2 (Arc contraction and the boundary matrix). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be a matrix product tensor of bond dimension D , and let Γ_R be its open contraction on R . Let m be the number of assignments to the cycle bonds wholly outside R . Then*

$$m E_R \Gamma_R(x)(\sigma) = D^{n-L-1} \text{tr}(A^{i_1} \dots A^{i_L} X). \quad (24.177)$$

If $D > 0$, both scalar factors are nonzero, so the regional boundary map differs from the MPS boundary map only by an invertible scalar change of boundary coordinates.

Proof. [✓ Proof](#) Summing a regional contraction over all cycle bonds counts it once for each exterior assignment. At fixed incoming and outgoing bond values (a, b) , the same contraction equals $D^{n-L-1} (A^{i_1} \dots A^{i_L})_{ab}$. Sum against $x(a, b)$ and use $\sum_{a,b} x(a, b) M_{ab} = \text{tr}(MX)$. Positive bond dimension permits cancellation of the exterior assignment count. $\square$

Theorem 24.16.1.3 (The regional ground space on a cycle arc). [Lean](#) [✓ Stmt](#) *For a matrix product tensor A on a cycle of $n \geq 3$ sites, and every arc R of length $0 < L < n$,*

$$E_R \mathcal{G}_R = \{\sigma \mapsto \text{tr}(A^{i_1} \dots A^{i_L} X) : X\} = \mathcal{G}_L(A). \quad (24.178)$$

No injectivity assumption is needed. This identifies the one-dimensional and graph definitions in [16, Section IV.C.1], local source lines 1987–2011.

Proof. [✓ Proof](#) For $D > 0$, the two-bond boundary coefficients range over all matrices, and the nonzero scalar changes of boundary coordinates in (24.177) preserve the range. For $D = 0$, the nonempty arc has an empty boundary-configuration space, while the matrix boundary space has only the zero matrix. Both ground spaces are therefore zero. $\square$

Definition 24.16.1.4 (Physical reduced density of a PEPS region). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ψ be the closed PEPS vector on a finite graph and let R be a region. Its unnormalized reduced density is $\rho_R = \text{tr}_{R^c} |\psi\rangle\langle\psi|$. If $M_R(\sigma, \tau) = \psi(\sigma, \tau)$ is its coefficient matrix across the cut, then $\rho_R = M_R M_R^\dagger$. This is the physical cut associated with the regional construction of [16, Section IV.C.1], local source lines 2003–2011.

Theorem 24.16.1.5 (Contraction across a cut with arbitrary bond dimensions). [Lean](#) [✓ Stmt](#) *For every finite PEPS graph, region R , and pair of physical configurations σ, τ on R, R^c , respectively, $\psi(\sigma, \tau) = \sum_\mu T_R(\mu; \sigma) T_{R^c}(\mu; \tau)$. This identity includes zero-dimensional virtual spaces.*

Proof. [✓ Proof](#) If a global virtual configuration exists, every bond space has positive dimension and the exact contraction across the cut applies. Otherwise, the closed contraction is zero. If incident-edge configurations existed on both R and R^c , choosing the label from either incident region at each edge would produce a global virtual configuration. Thus one regional incident configuration space is empty, and its open contraction vanishes. Both sides of the claimed identity are then zero. $\square$

Theorem 24.16.1.6 (Support of the physical reduced density). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The operator ρ_R is positive semidefinite, and $\text{ran } \rho_R = \text{ran } M_R \subseteq \mathcal{G}_R$. No injectivity of the regional tensor is assumed.*

Proof. [✓ Proof](#) The identity $\rho_R = M_R M_R^\dagger$ gives positivity and $\text{ran } \rho_R \subseteq \text{ran } M_R$. The two operators have the same rank, so these ranges are equal. Each column of M_R is a local slice of ψ and lies in $\mathcal{G}_R$. Their span therefore lies in $\mathcal{G}_R$. $\square$

Theorem 24.16.1.7 (Parent interactions annihilate the local density). [Lean](#) [Lean](#) [✓ Stmt](#) *Every regional parent interaction h_R satisfies $h_R \rho_R = \rho_R h_R = 0$. This is the density-operator form of local frustration freeness in [16, Section IV.C.1], local source lines 2008–2011.*

Proof. [✓ Proof](#) Since $\text{ran } \rho_R \subseteq \mathcal{G}_R = \ker h_R$, the product $h_R \rho_R$ vanishes. Taking adjoints gives $\rho_R h_R = 0$. $\square$

Theorem 24.16.1.8 (Numbering preserves a regular region’s physical range). [Lean](#) [✓ Stmt](#) *For a finite PEPS region with regular group labels, numbering its crossing bonds identifies the columns of its open-region matrix T_R with the open tensors defining $\mathcal{G}_R$. Thus $\text{ran } T_R = \mathcal{G}_R$.*

Proof. [✓ Proof](#) Numbering and converting between group labels and the chosen finite basis are bijections of boundary configurations. They preserve the span of the open-region tensors. $\square$

Theorem 24.16.1.9 (The closed state determines a regular regional ground space). [Lean](#) [Lean](#) [✓ Stmt](#) *Let G be a finite group, and put regular G -injective tensors at the vertices of a finite connected graph. Suppose the subgraphs induced by R and R^c are connected. Then the reduced density of the closed PEPS vector satisfies $\text{ran } \rho_R = \mathcal{G}_R$. The same conclusion holds without connectedness of the ambient graph whenever the cut has a crossing edge. This is an auxiliary consequence of the connected-region construction in [203, Lemma 5.2] and the regular boundary cut in its Theorem 6.9.*

Proof. [✓ Proof](#) Choose one numbering for the crossing bonds on both sides. Let T_R, T_{R^c} denote the resulting open-region matrices. Both maps are injective on the invariant boundary space, and the coefficient matrix is $M_R = T_R T_{R^c}^\top$. A left inverse L of T_{R^c} on that invariant space satisfies $LT_{R^c} = P$, where P is its averaging projector. Since $P^\top = P$ and $T_R P = T_R$, one has $M_R L^\top = T_R$. Hence $\text{ran } M_R = \text{ran } T_R = \mathcal{G}_R$; the reduced density has this same range. In a connected ambient graph, the two nonempty induced regions have a crossing edge, so the required numbering exists. $\square$

Theorem 24.16.1.10 (Equal closed states have equal regular region spaces). [Lean](#) [✓ Stmt](#) *Consider two regular G -injective PEPS tensor families on the same finite connected graph, with the same physical spaces. If their closed coefficient vectors agree, then $\mathcal{G}_R(A) = \mathcal{G}_R(B)$ for every cut whose two induced subgraphs are connected. This is a necessary local relation toward a Fundamental Theorem for regular G -injective PEPS.*

Proof. [✓ Proof](#) Equality of the closed coefficient vectors gives equality of their reduced densities. Each of the two regional ground spaces is the range of that common density. $\square$

Theorem 24.16.1.11 (Dimension and Schmidt rank of a regular connected cut). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Consider regular G -injective tensors on a finite graph and a connected region R with $n + 1$ crossing bonds. Then $\dim \mathcal{G}_R = |G|^n$. If the complementary induced region is also connected, the physical reduced density has rank $|G|^n$, and the closed PEPS vector is nonzero. These are auxiliary untwisted-cut consequences of the invariant boundary calculation in [203, Corollary 6.10].

Proof. [Proof](#) The open-region map is injective on the invariant boundary space and has range $\mathcal{G}_R$. The simultaneous regular action on $n + 1$ boundary labels has invariant dimension $|G|^n$. If the complement is connected, ρ_R has range $\mathcal{G}_R$ and therefore this same rank. The rank is positive; hence the closed vector cannot vanish. $\square$

Theorem 24.16.1.12 (The physical state determines the virtual group order). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let two tensor families be G -injective and H -injective for the regular representations of finite groups G and H , respectively, and represent the same closed PEPS vector on a finite graph. Suppose a cut has connected induced regions on both sides and at least two crossing bonds. Then $|G| = |H|$. Equality of the group orders is asserted; the multiplication laws are not determined by this rank argument.

Proof. [Proof](#) Write the number of crossing bonds as $n + 1$, with $n > 0$. The common physical reduced density has rank both $|G|^n$ and $|H|^n$. The map $k \mapsto k^n$ is injective on the natural numbers, so $|G| = |H|$. $\square$

24.17 Boundary support and Gibbs operators

The bulk–boundary conjecture in [14] asks whether a gap of the parent Hamiltonian is equivalent to a short-range Gibbs representation of the virtual boundary states. The following finite-dimensional results concern the support of such states. They do not assert locality or a uniform gap.

Theorem 24.17.1 (Full-space obstruction for a regular boundary). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let G be a finite group with $|G| > 1$, and let a regular virtual boundary have $b \geq 1$ legs. Its normalized density $\rho = P_G/|G|^{b-1}$ has rank $|G|^{b-1} < |G|^b$. In particular, it is not invertible and cannot equal e^H for any finite matrix H on the full boundary space. More generally, if T is a regular G -injective open map, its virtual Gram operator $T^\dagger T$ has the same rank and cannot be such an exponential.

Proof. [Proof](#) The rank of ρ is the dimension of the invariant boundary space. The rank of $T^\dagger T$ equals the rank of T , which is also that dimension by G -injectivity. Since $|G| > 1$, both ranks are strictly smaller than the full boundary dimension. Every matrix exponential is invertible, its inverse being the exponential of the negative matrix. $\square$

Theorem 24.17.2 (Gibbs representation on the invariant support). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) In relative boundary coordinates the restriction of ρ to its invariant support is

$$\rho_{\text{supp}} = |G|^{-(b-1)} I = e^{-H_{\text{supp}}}, \quad H_{\text{supp}} = \log(|G|^{b-1}) I.$$

This is a statement on the invariant support, whose dimension is $|G|^{b-1}$. It imposes no interaction decomposition on the original b virtual legs.

Proof. [Proof](#) The averaging projector fixes invariant vectors, so the density acts on this space by $|G|^{-(b-1)}$. The exponential of the scalar diagonal matrix $-\log(|G|^{b-1})I$ has precisely that scalar on each diagonal entry. $\square$

Definition 24.17.3 (Normalized invariant boundary coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) For $b = n + 1$ regular boundary legs, let $r \in G^n$ denote relative coordinates. Define the boundary embedding

$$U|r\rangle = |G|^{-1/2} \sum_{g \in G} |g, gr_1, \dots, gr_n\rangle.$$

For an open map T , write $C = TU$ and $Q = C^\dagger C$. Its supported boundary Hamiltonian is $H = -\log Q$, with the logarithm defined by spectral calculus and the value at a zero eigenvalue set to zero. For G -injective T , the operator Q is positive definite and this is the usual spectral logarithm. The logarithm gives the inverse relation below when T is G -injective.

Theorem 24.17.4 (The invariant boundary support). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The embedding U is an isometry onto the invariant boundary:*

$$U^\dagger U = I, \quad UU^\dagger = P_G.$$

Proof. [✓ Proof](#) Distinct relative configurations give disjoint translation orbits, each of cardinality $|G|$. Their normalized orbit sums are therefore orthonormal. The matrix entries of $UU^\dagger$ are $|G|^{-1}$ for configurations in the same orbit and zero otherwise, which are precisely the entries of P_G . $\square$

Theorem 24.17.5 (A finite supported Gibbs representation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) *If T is regular G -injective, then $C = TU$ is injective, $Q = C^\dagger C$ is positive definite, and the supported boundary Hamiltonian is Hermitian with*

$$Q = e^{-H}, \quad T^\dagger T = Ue^{-H}U^\dagger.$$

This establishes a finite Gibbs representation on the invariant support. It does not assert short-range interactions among the original virtual legs.

Proof. [✓ Proof](#) The embedding takes values in the invariant subspace, where T is injective. Hence C is injective, and $\langle x, Qx \rangle = \|Cx\|^2 > 0$ for $x \neq 0$. The spectral logarithm of a positive definite Hermitian operator is Hermitian, and its exponential equals the original operator. $\square$

Definition 24.17.6 (Normalized physical regional density). [Lean](#) [✓ Stmt](#) For a nonzero closed PEPS vector, write $\hat{\rho}_R = \rho_R / \text{tr } \rho_R$ for its normalized physical reduced density. The trace is positive for every region. The normalized density is positive semidefinite, has trace one, and has the same range and rank as ρ_R .

Proof. [✓ Proof](#) [Lean](#) [Lean](#) [Lean](#) The physical coefficient matrix M_R is nonzero because assembly identifies its entries with the coefficients of the closed vector. Thus $\text{tr}(M_R M_R^*) > 0$. Division by this trace preserves positivity, rank, and range and makes the trace equal to one. $\square$

Theorem 24.17.7 (Normalized regular support and local annihilators). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Consider regular G -injective tensors on a finite graph and a cut whose two induced subgraphs are connected and whose boundary is nonempty. Then $\text{ran } \hat{\rho}_R = \mathcal{G}_R$. In particular, for every local linear operator h ,

$$h\hat{\rho}_R = 0 \iff \mathcal{G}_R \subseteq \ker h.$$

Every regional parent interaction annihilates this normalized density. Positivity of h is not needed for the displayed equivalence.

Proof. [✓ Proof](#) The closed vector is nonzero. Normalization preserves the range, which is $\mathcal{G}_R$. The identity $h\hat{\rho}_R = 0$ means precisely that this range is contained in the kernel of h . $\square$

Theorem 24.17.8 (Entropy bound for a regular G-injective cut). [Lean](#) [Lean](#) [✓ Stmt](#) *Let regular G-injective tensors define a PEPS on a finite graph. Suppose that both induced subgraphs of a cut are connected and that the cut has $n + 1$ crossing bonds. Then the closed vector is nonzero and its normalized physical reduced density satisfies*

$$\text{rank } \hat{\rho}_R = |G|^n, \quad 0 \leq S(\hat{\rho}_R) \leq n \log |G|, \quad S_0(\hat{\rho}_R) = n \log |G|.$$

Here S is the von Neumann entropy and S_0 is the zero-order Rényi entropy, with natural logarithms. Neither isometry nor flatness of the physical spectrum is assumed. This is an auxiliary connected regular-cut consequence of the boundary calculation in [203, Corollary 6.10].

Proof. [✓ Proof](#) Normalization preserves the invariant boundary rank $|G|^n$. The von Neumann entropy of a positive trace-one matrix lies between zero and the logarithm of its rank. The zero-order Rényi entropy is that logarithm by definition. $\square$

24.18 Comparison of finite PEPS parent Hamiltonians

The comparison of positive interactions with orthogonal projectors in [16, Section IV.C.3], local source line 2170, applies to the regional PEPS interactions of Section IV.C.1. All collections of regions below are finite.

Theorem 24.18.1 (Comparison with the regional parent projection). [Lean](#) [Lean](#) [✓ Stmt](#) *Let h_R be a positive regional parent interaction, and let P_R be the orthogonal projector onto $\mathcal{G}_R^\perp$. There are strictly positive constants κ_R, C_R such that*

$$\kappa_R P_R \leq h_R \leq C_R P_R. \quad (24.179)$$

Proof. [✓ Proof](#) The kernel is $\mathcal{G}_R$ in the physical Hilbert space. On its orthogonal complement, a positive operator in finite dimension has a positive lower bound. Its operator norm gives an upper bound. On the kernel all three operators vanish. $\square$

Theorem 24.18.2 (Order-preserving regional extension). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Extension by the identity on the complementary physical space is linear and preserves Loewner order. In particular, if two collections $(h_i)_{i \in I}$ and $(k_i)_{i \in I}$ on the same regions satisfy $\kappa k_i \leq h_i \leq C k_i$ for every i , their sums satisfy $\kappa H_k \leq H_h \leq C H_k$.*

Proof. [✓ Proof](#) Tensoring a positive semidefinite difference with the identity preserves positivity. Reordering the physical coordinates also preserves positivity. Sum the resulting inequalities. $\square$

Theorem 24.18.3 (Comparison of finite PEPS parents). [Lean](#) [Lean](#) [✓ Stmt](#) *For a finite family of regional parent interactions h_i , let $H_P = \sum_i \tilde{P}_i$ be the sum of the canonical regional projectors extended by the identity. There are $\kappa, C > 0$ such that $\kappa H_P \leq H_h \leq C H_P$. Consequently, for any two positive parents H_h, H_k on the same family of regions, there are $a, b > 0$ such that $a H_k \leq H_h \leq b H_k$.*

Proof. [✓ Proof](#) For a nonempty family take $\kappa = \min_i \kappa_i$ and $C = \max_i C_i$ in (24.179). The empty family gives zero Hamiltonians, so any positive constants suffice. If the respective comparisons with H_P use constants κ_h, C_h and κ_k, C_k , take $a = \kappa_h / C_k$ and $b = C_h / \kappa_k$. $\square$

Theorem 24.18.4 (Transfer of the finite-volume gap). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let H_h, H_k be positive PEPS parents on the same finite family of regions. They have the same kernel $\mathcal{K}$. Suppose $H_h \geq \kappa H_k$, where $\kappa > 0$, and suppose that for every $\phi \in \mathcal{K}^\perp$, $\gamma \|\phi\| \leq \|H_k \phi\|$, where*

$\gamma \geq 0$. Then every $\phi \in \mathcal{K}^\perp$ satisfies $\kappa\gamma\|\phi\| \leq \|H_h\phi\|$. Existence of a strictly positive bound of this form is therefore independent of the choice of positive regional interactions. The comparison constants may depend on the finite collection.

Proof. ✓ Proof The common regional kernels determine $\mathcal{K}$. Positivity and the spectral theorem convert the norm bound for H_k into $\langle \phi, H_k \phi \rangle \geq \gamma\|\phi\|^2$ on $\mathcal{K}^\perp$. Loewner comparison gives $\langle \phi, H_h \phi \rangle \geq \kappa\gamma\|\phi\|^2$. Cauchy-Schwarz yields the asserted norm bound. Apply the two-sided comparison to obtain the final equivalence. $\square$

24.19 Physical changes of coordinates and regional ground spaces

Definition 24.19.1 (Physical deformation). Lean Lean ✓ Stmt Let A be a tensor family on a finite graph and let $F_v : \mathbb{C}^d \rightarrow \mathbb{C}^e$ be a linear map at every vertex. Its physical deformation A^F keeps the virtual bond dimensions and has components

$$(A^F)_v(\eta, t) = \sum_{s=0}^{d-1} (F_v)_{ts} A_v(\eta, s). \quad (24.180)$$

For a region R , write $F_R = \bigotimes_{v \in R} F_v$. When every F_v is invertible, F_R is invertible with inverse $\bigotimes_{v \in R} F_v^{-1}$.

Theorem 24.19.2 (Covariance of regional ground spaces). Lean Lean Lean Lean Lean Lean ✓ Stmt Let Γ_R^A denote the open-region map and $\mathcal{G}_R(A) = \text{im } \Gamma_R^A$. Then $\Gamma_R^{A^F} = F_R \Gamma_R^A$ and $\mathcal{G}_R(A^F) = F_R \mathcal{G}_R(A)$. If every F_v is invertible, restriction of F_R gives an isomorphism $\mathcal{G}_R(A) \cong \mathcal{G}_R(A^F)$, and $\ker \Gamma_R^{A^F} = \ker \Gamma_R^A$. In particular the regional ground spaces have equal dimensions. No injectivity assumption on A is required. The physical changes of coordinates in [203, §6.1, accessible virtual coordinates] satisfy these identities.

Proof. ✓ Proof Each physical map acts independently of every internal virtual summation. Distributing the products over the finite physical sums gives $\Gamma_R^{A^F} = F_R \Gamma_R^A$ on each boundary basis vector, hence on every boundary vector by linearity. Taking images gives the regional-space identity. The composition rule for product maps gives $F_R F_R^{-1} = F_R^{-1} F_R = \mathbb{1}$. The restriction is therefore a linear isomorphism. Injectivity of F_R gives equality of the virtual kernels. $\square$

Definition 24.19.3 (Transported regional interaction). Lean ✓ Stmt If the physical maps are invertible, the transport of a regional interaction h_R is $h_R^F = (F_R^{-1})^\dagger h_R F_R^{-1}$.

Theorem 24.19.4 (Transport of parent interactions). Lean Lean Lean ✓ Stmt If $h_R \geq 0$ and $\ker h_R = \mathcal{G}_R(A)$, then $h_R^F \geq 0$ and $\ker h_R^F = \mathcal{G}_R(A^F)$. The original interaction may be any positive semidefinite operator with the prescribed kernel.

Proof. ✓ Proof Congruence preserves positive semidefiniteness. Since $(F_R^{-1})^\dagger$ is invertible, $h_R^F \psi = 0$ is equivalent to $h_R F_R^{-1} \psi = 0$. Thus $\ker h_R^F = F_R \ker h_R = F_R \mathcal{G}_R(A) = \mathcal{G}_R(A^F)$. $\square$

24.20 Physical transport of the common parent ground space

Definition 24.20.1 (Product physical map on the full graph). Lean Lean Lean Lean Lean Lean ✓ Stmt For physical maps $F_v : \mathbb{C}^d \rightarrow \mathbb{C}^e$ at the vertices of a finite graph, their product map

Proof. ✓ Proof On a fixed complementary slice, a regional extension acts by its local interaction matrix. The slice of $F_V \psi$ is a linear combination of $F_{R_i} \psi_\beta$. Applying $k_i F_{R_i} = F_{R_i} h_i$ to each summand proves the intertwining identity on every slice, hence on the full vector. Summing the identities gives the full Hamiltonian identity. $\square$

Theorem 24.21.3 (Unitary physical conjugation). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *If every F_v is unitary, then every regional product F_R and the full product F_V are unitary. For any finite family of regional interaction matrices, their inverse-adjoint transports satisfy*

$$h_i^F = F_{R_i} h_i F_{R_i}^\dagger, \quad (24.183)$$

$$H^F = F_V H F_V^\dagger. \quad (24.184)$$

Consequently H^F and H have equal characteristic polynomials and equal spectra, with the same eigenvalue multiplicities.

Proof. ✓ Proof Product composition and adjoint identities give $F_R^\dagger F_R = F_R F_R^\dagger = \mathbb{1}$ from the corresponding vertex identities, and likewise for F_V . Thus $F_R^{-1} = F_R^\dagger$, giving the local conjugation formula. The local identity $h_i^F F_{R_i} = F_{R_i} h_i$ gives $H^F F_V = F_V H$. Multiplication by $F_V^\dagger$ gives the full conjugation formula. Equality of characteristic polynomials follows by exchanging the two factors of the rectangular product $(F_V H) F_V^\dagger$, using equality of the physical dimensions. The spectra are the roots of these characteristic polynomials. $\square$

24.22 Nesting of regional PEPS ground spaces

The regional contraction with arbitrary boundary conditions of [16, Section IV.C.1], local source lines 2003–2011, implies a compatibility between nested regions. No injectivity or nonzero bond dimensions are needed.

Theorem 24.22.1 (Separation of nested incident labels). Lean Lean Lean Lean ✓ Stmt *Let $R \subseteq S$, and write E_R, E_S for their incident edges. Then $E_R \subseteq E_S$, and assignments on E_S split uniquely into assignments on E_R and $E_S \setminus E_R$. For any coefficient f on the crossing-bond labels of R , contraction against its open tensor satisfies*

$$\sum_{\mu_R} f(\mu_R) \psi_{R, \mu_R}(\sigma_R) = \sum_{\eta_R} f(\eta_R|_{\partial R}) \prod_{v \in R} A_v(\eta_R, \sigma_R(v)). \quad (24.185)$$

Proof. ✓ Proof Restriction gives the decomposition of assignments; its inverse joins the two assignments on their disjoint edge sets. Expand the open tensor in (24.185), interchange the finite sums, and retain the unique boundary label $\mu_R = \eta_R|_{\partial R}$. $\square$

Definition 24.22.2 (Physical slice of a nested region). Lean Lean Lean ✓ Stmt For $R \subseteq S$ and fixed physical labels τ on R^c , let $Q_{R, S, \tau}$ send a vector ϕ_S on S to the vector $\sigma_R \mapsto \phi_S(\sigma_R, \tau|_{S \setminus R})$. Fixing all complementary labels of R factors as fixing the labels on S^c and then applying $Q_{R, S, \tau}$.

Theorem 24.22.3 (Nesting of regional ground spaces). Lean Lean ✓ Stmt *If $R \subseteq S$, then every $\phi_S \in \mathcal{G}_S$ and every complementary physical configuration τ satisfy $Q_{R, S, \tau} \phi_S \in \mathcal{G}_R$.*

Proof. ✓ Proof It suffices to consider a boundary-labelled tensor ψ_{S, μ_S} . Separate the incident labels into η_R on E_R and ζ on $E_S \setminus E_R$. Tensor factors at vertices in $S \setminus R$ and the boundary

constraint on S depend on η_R only through its crossing labels μ_R . Equation (24.185) therefore gives

$$Q_{R,S,\tau} \psi_{S,\mu_S} = \sum_{\zeta, \mu_R} c_{\mu_S, \zeta, \mu_R}(\tau) \psi_{R,\mu_R}, \quad (24.186)$$

where each coefficient is the product of the tensor factors on $S \setminus R$ when the assembled boundary labels agree with μ_S , and is zero otherwise. The right side belongs to $\mathcal{G}_R$. Linear combinations give the assertion for every $\phi_S \in \mathcal{G}_S$. $\square$

Theorem 24.22.4 (Monotonicity of common regional conditions). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $(R_i)_{i \in I}$ and $(S_j)_{j \in J}$ be collections of regions such that each R_i is contained in some S_j . Then the common regional ground space for (S_j) is contained in that for (R_i) . In particular, for finite collections, any positive parent Hamiltonians built from these regions satisfy $\ker H_S \subseteq \ker H_R$.*

Proof. [✓ Proof](#) For each R_i , choose a containing S_j . A vector satisfying the S_j condition has every complementary slice in $\mathcal{G}_{S_j}$. Apply the nesting theorem and the factorization of physical slices to obtain the R_i condition. The parent-kernel statement follows from the characterization by common regional conditions. $\square$

Lemma 24.22.5 (Single-site geometry of the native torus). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Suppose both periods of the native square torus are at least three. The graph is connected. For every vertex v , the subgraphs induced by $\{v\}$ and its complement are connected, and the boundary of $\{v\}$ has exactly four bonds. The native top, right, down, and left legs enumerate these incident bonds bijectively.*

Proof. [✓ Proof](#) A singleton is the coordinate rectangle of side lengths one. Its complement is connected by the unused row and column construction. The same rectangular connectivity argument applied to both full coordinate intervals gives connectivity of the torus. The four native legs exhaust the incident bonds and are distinct: horizontal and vertical bonds cannot agree, and opposite bonds have different origins when the corresponding period exceeds two. Every incident bond has just one endpoint at v , so these are precisely the crossing bonds of the singleton. $\square$

Theorem 24.22.6 (Single-site physical support of a regular torus PEPS). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let regular G -injective tensors of physical dimension d be placed at the vertices of the native square torus, with both periods at least three. The tensors may vary with the vertex. Their closed vector is nonzero. For every vertex v ,*

$$\text{ran } \rho_{\{v\}} = \mathcal{G}_{\{v\}}, \quad \text{rank } \rho_{\{v\}} = |G|^3 \leq d.$$

The normalized density satisfies

$$0 \leq S(\hat{\rho}_{\{v\}}) \leq 3 \log |G|, \quad S_0(\hat{\rho}_{\{v\}}) = 3 \log |G|.$$

No isometry or flat spectrum is assumed.

Proof. [✓ Proof](#) Apply the connected regular-cut statements to the singleton and its connected complement. Its four crossing bonds give three relative boundary labels. The density acts on a d -dimensional physical space, so its rank $|G|^3$ is at most d . $\square$

Theorem 24.22.7 (Local rigidity for equal regular torus states). [Lean](#) [Lean](#) [✓ Stmt](#) *On a native square torus with both periods at least three, two regular G -injective descriptions of the same closed physical vector have the same single-site ground spaces at every vertex. If the two descriptions use regular representations of different finite groups G and H , the equality of their closed vectors implies $|G| = |H|$. No assertion about the multiplication laws or individual bond gauges follows from this rank argument.*

Proof. [✓ Proof](#) Equal closed vectors give equal reduced densities. Their ranges recover the local ground spaces. The common single-site density has rank both $|G|^3$ and $|H|^3$, so the group orders agree. $\square$

Theorem 24.22.8 (Physical support of a bounded regular torus rectangle). [Lean](#) [✓ Stmt](#) *Consider regular G -injective tensors on a native square torus with periods at least three. Let R be a coordinate rectangle of positive side lengths, bounded by the coordinate intervals, with each side length strictly smaller than the corresponding torus period. Then $\text{ran } \rho_R = \mathcal{G}_R$.*

Proof. [✓ Proof](#) The rectangle is connected, and its complement is connected by paths along an unused row and an unused column. The torus is connected, so the exact support theorem applies. $\square$

24.23 Entanglement bounds on finite graphs

Theorem 24.23.1 (Crossing-bond rank and entropy bounds). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let a PEPS on a finite graph have virtual dimension D_e on edge e , and let R be any region. Its unnormalized reduced density satisfies*

$$\text{rank } \rho_R \leq \prod_{e \in \partial R} D_e.$$

If the closed vector is nonzero, every virtual bond has positive dimension, and the normalized reduced density $\hat{\rho}_R$ satisfies

$$0 \leq S(\hat{\rho}_R) \leq \sum_{e \in \partial R} \log D_e.$$

For a common crossing-bond dimension D , this gives $S(\hat{\rho}_R) \leq |\partial R| \log D$. This is the finite-graph boundary-dimension argument underlying the PEPS area law in [16, Section II.A.3].

Proof. [✓ Proof](#) The reduced-density range lies in the span of the open-region tensors, indexed by the crossing-bond configurations. That span has dimension at most $\prod_{e \in \partial R} D_e$. Normalization preserves the range when the vector is nonzero. The entropy of a trace-one positive operator is at most the logarithm of its rank. Monotonicity of the logarithm and the logarithm-of-product identity yield the entropy bound. If any bond dimension vanished, every virtual contraction would be an empty sum, contradicting nonvanishing. $\square$

Theorem 24.23.2 (Physical-dimension bounds and trivial cuts). [Lean](#) [Lean](#) [✓ Stmt](#) *If every physical site has dimension d , then*

$$\text{rank } \rho_R \leq \min\{d^{|\mathcal{R}|}, d^{|\mathcal{R}^c|}\}.$$

For a nonzero closed PEPS, a cut whose crossing virtual bonds all have dimension one has zero entanglement entropy. This includes cuts with no crossing edges.

Proof. [✓ Proof](#) The reduced-density rank equals the rank of the physical coefficient matrix across the cut. Its row and column dimensions are $d^{|\mathcal{R}|}$ and $d^{|\mathcal{R}^c|}$. If all crossing dimensions are one, the preceding entropy upper bound is zero; nonnegativity gives equality. $\square$

24.24 Parent spaces determined by the physical ray

Lemma 24.24.1 (Reduced states under rescaling). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose two finite-graph PEPS vectors satisfy $|\Psi_B\rangle = c|\Psi_A\rangle$ with $c \neq 0$. For every region R , their unnormlized reduced densities satisfy $\rho_R(B) = |c|^2 \rho_R(A)$. Their reduced supports and ranks agree, and their normalized reduced densities are equal.

Proof. [✓ Proof](#) The outer product scales by $c\bar{c}$, and partial trace is linear. Multiplication by a nonzero scalar preserves the range and rank. Dividing by the trace cancels the same scalar. The zero-vector convention also respects this identity. $\square$

Lemma 24.24.2 (Parent operators determined by local spaces). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If two tensor families have equal regional spaces on every region in a prescribed collection, they have the same admissible positive parent interactions and the same canonical local projectors on those regions. Their common parent ground spaces agree. For a finite collection, the canonical sums of local projectors are equal as operators.

Proof. [✓ Proof](#) A parent interaction is characterized by positivity and its regional kernel, and the canonical projector is determined by that kernel. The common ground space is the intersection of the corresponding slice conditions. Summing equal lifted projectors gives equal canonical Hamiltonians. $\square$

Theorem 24.24.3 (Regular parent spaces are intrinsic to the physical ray). [Lean](#) [Lean](#) [✓ Stmt](#) Let a finite ambient graph be connected. Suppose two regular group-injective tensor families have proportional closed vectors with nonzero proportionality scalar. Their finite virtual groups may differ. Across any cut with both induced sides connected, their regional PEPS spaces agree. Their common parent ground spaces therefore agree for any family of such cuts.

Proof. [✓ Proof](#) Each regional PEPS space is the support of the corresponding physical reduced density. The rescaling lemma identifies these supports. Equality of the common ground spaces then follows from equality of all regional conditions. $\square$

Theorem 24.24.4 (The physical ray determines the regular group order). [Lean](#) [✓ Stmt](#) Suppose two regular group-injective descriptions have proportional closed vectors with nonzero scalar. If a cut has connected induced sides and $b \geq 2$ crossing edges, the two virtual groups G and H have equal order. This assertion does not identify their multiplication laws.

Proof. [✓ Proof](#) Their reduced ranks agree, while the regular cut-rank formula gives $|G|^{b-1} = |H|^{b-1}$. Since $b-1 > 0$, the power map on natural numbers is injective, and $|G| = |H|$ follows. $\square$

Lemma 24.24.5 (Native regular injectivity in incident-bond coordinates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let G be a finite group and let a four-leg tensor be G -injective for the regular representation. On a native square torus with both periods at least three, identify its four virtual labels with the top, right, down, and left incident bonds. The resulting virtual-to-physical map is G -injective on the actual incident graph coordinates. Its ordinary closed graph contraction equals the native closure with both closure elements equal to the identity.

Proof. [✓ Proof](#) The four actual legs are a bijection. The resulting bijection of virtual configurations intertwines simultaneous regular translation. Changing variables in the finite contraction sum identifies the two site maps. Invariance and injectivity on invariant vectors are therefore preserved. Identity closure elements insert identity operators on every seam bond, so the closed contraction agrees with the ordinary graph contraction. $\square$

Theorem 24.24.6 (Single-site support of a native regular injective tensor). [Lean](#) [Lean](#) [✓ Stmt](#)
Place one regular G -injective four-leg tensor at every site of a native square torus whose periods are at least three. For its untwisted closed vector, the physical density of every site v satisfies

$$\text{ran } \rho_{\{v\}} = \mathcal{G}_{\{v\}}, \quad \text{rank } \rho_{\{v\}} = |G|^3.$$

Proof. [✓ Proof](#) Express native G -injectivity on the actual incident bonds and apply the single-site support and rank statements. $\square$

Theorem 24.24.7 (Local relations forced by equal native regular closures). [Lean](#) [Lean](#) [✓ Stmt](#)
Consider two native four-leg regular G -injective tensors with the same physical space on a square torus whose periods are at least three. If their untwisted closed coefficient vectors agree, their single-site ground spaces agree at every vertex. If the two descriptions instead use regular representations of different finite groups G and H , equality of these untwisted closed vectors forces $|G| = |H|$. These are necessary local relations toward a Fundamental Theorem; neither a group isomorphism nor a product of individual bond gauges is asserted.

Proof. [✓ Proof](#) Identity native closures are the ordinary graph contractions. The incident-bond descriptions remain G -injective, so the local ground-space and group-order rigidity statements apply. $\square$

Lemma 24.24.8 (The crossing bonds of a native torus rectangle). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Suppose the native square torus has periods $W, H \geq 3$. Let R be the coordinate rectangle

$$[x_0, x_0 + w) \times [y_0, y_0 + h), \quad w, h > 0,$$

with $x_0 + w \leq W$, $y_0 + h \leq H$, $w < W$, and $h < H$. Then there are $2h$ horizontal and $2w$ vertical crossing bonds, and hence

$$|\partial R| = 2w + 2h.$$

In particular a bounded square of side length L has $4L$ crossing bonds. The coordinate intervals may start at zero or end at their torus periods; the formula includes the cyclic seam bonds in these cases. These are the actual lattice bonds used in the rectangular boundary construction of [203, Theorem 6.9].

Proof. [✓ Proof](#) For a cyclic coordinate interval $[s, s + \ell)$ shorter than its period N , membership changes between a and $a + 1$ at precisely two origins. Their coordinate values are $s + \ell - 1$ and $s - 1$, where the latter is read as $N - 1$ if $s = 0$. The two origins are distinct because $0 < \ell < N$. A horizontal bond crosses the rectangle exactly when its first coordinate is one of these two origins and its second coordinate lies in the vertical interval. This gives $2h$ bonds. The vertical argument gives $2w$ bonds. On a torus with both periods at least three, right and up bonds enumerate the graph edges bijectively, so these two counts add without multiplicity. $\square$

Lemma 24.24.9 (A perimeter bound including full bands and empty rectangles). [Lean](#) [✓ Stmt](#)
On a native square torus with periods $W, H \geq 3$, every bounded coordinate rectangle of side lengths $w, h \geq 0$ satisfies $|\partial R| \leq 2w + 2h$. No strict upper bound on either side length is required; empty rectangles and full coordinate bands are included.

Proof. [✓ Proof](#) A zero side length gives the empty rectangle. A side length equal to its torus period produces no crossing bond in that coordinate direction. In every other case, the horizontal crossing count is $2h$ and the vertical crossing count is $2w$. The two directional bounds add to the stated perimeter bound. $\square$

Theorem 24.24.10 (The symmetry correction to a rectangular entropy bound). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let regular G -injective tensors be placed at the vertices of the native square torus, with periods $W, H \geq 3$. The tensors may vary with the vertex. Let R be a positive coordinate rectangle of side lengths w, h , bounded by the coordinate intervals, with $w < W$ and $h < H$. For the ordinary untwisted closed vector,

$$\text{rank } \rho_R = |G|^{2w+2h-1}, \quad 0 \leq S(\hat{\rho}_R) \leq (2w+2h-1) \log |G|, \quad S_0(\hat{\rho}_R) = (2w+2h-1) \log |G|.$$

Isometry and flatness of the physical spectrum are not assumed. Coordinate intervals may start at zero or end at their torus periods. This is an auxiliary regular rectangular-cut consequence of the boundary count in [203, Corollary 6.10].

Proof. [Stmt](#) Both the rectangle and its complementary induced graph are connected. The boundary has $2w + 2h$ bonds, hence $2w + 2h - 1$ relative regular labels. The exact regular-cut rank and entropy statements give the asserted formulas and inequality. $\square$

24.25 G-injectivity under physical maps

Theorem 24.25.1 (Physical maps preserving G -injectivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let $T : \mathcal{V} \rightarrow \mathcal{P}$ be G -injective and let $F : \mathcal{P} \rightarrow \mathcal{Q}$ be linear. The composite FT is invariant under the original virtual representation. If $F|_{\text{im } T}$ is injective, then FT is G -injective. For a finite group G , the converse holds: FT is G -injective if and only if $\text{im } T \cap \ker F = \{0\}$. In particular an invertible physical change of coordinates preserves G -injectivity in both directions. These are local consequences of [203, Definition 5.1].

Proof. [Stmt](#) Invariance follows from $FTU_g = FT$. If $x \in \mathcal{V}^G$ and $FTx = 0$, injectivity of F on $\text{im } T$ gives $Tx = 0$, and G -injectivity gives $x = 0$. Conversely, for finite G , every $y \in \text{im } T$ is Tx for an invariant vector x . If $Fy = 0$ and FT is G -injective, then $x = 0$, hence $y = 0$. For an invertible physical map, apply the same implication to its inverse to obtain the reverse direction. $\square$

Theorem 24.25.2 (Physical isometries preserving G -isometry). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let T be G -isometric with normalization factor $c > 0$. A physical linear isometry F makes FT G -isometric with the same factor c . This includes rectangular isometries. An invertible physical isometry preserves G -isometry in both directions.

Proof. [Stmt](#) An isometry is injective, so FT is G -injective. For invariant x, y , $\langle FTx, FTy \rangle = \langle Tx, Ty \rangle = c \langle x, y \rangle$. For a matrix isometry, the first equality follows from $F^\dagger F = \mathbb{1}$. The inverse of an invertible isometry is an isometry, giving the converse. $\square$

Theorem 24.25.3 (Compatibility with virtual comparison). [Lean](#) [Stmt](#) Let $T : \mathcal{V} \rightarrow \mathcal{P}$ and $S : \mathcal{W} \rightarrow \mathcal{P}$ be injective on the invariant spaces of possibly different finite groups, and suppose their physical ranges agree. Applying the same injective physical map $F : \mathcal{P} \rightarrow \mathcal{Q}$ to both tensors leaves their virtual comparison unchanged.

Proof. [Stmt](#) The comparison for FT and FS satisfies $FS\Phi x = FTx$. Injectivity of F gives $S\Phi x = Tx$, which uniquely characterizes the original comparison. $\square$

Definition 24.25.4 (Physical deformation of a regular site). [Lean](#) [Lean](#) [Stmt](#) For a site tensor $a(\eta, s)$ whose virtual labels are group elements on an arbitrary incident-leg set, define $a^F(\eta, t) = \sum_s F_{ts} a(\eta, s)$. Its physical coefficient map is $T_{a^F} = FT_a$.

Theorem 24.25.5 (Physical deformation of G -injective site families). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *An injective physical map preserves G -injectivity of a site tensor, and an isometry preserves its G -isometry. Both statements hold for four virtual legs and for arbitrary incident-leg sets. Consequently invertible physical maps at every vertex preserve regular G -injectivity of a complete finite-graph site family; physical isometries preserve regular G -isometry. In finite bond coordinates the resulting tensor family is exactly the physical deformation of the original group-labelled family.*

Proof. ✔ **Proof** Apply the preceding composition results to each site's coefficient map. The group-labelled coordinates identify the deformed coefficients with the same physical sums in finite bond coordinates. □

24.26 Invertible physical comparison of G-injective tensors

Theorem 24.26.1 (Ambient physical comparison). [Lean] ✓ Stmt *Let a finite group G act on a virtual space $\mathcal{V}$. Suppose $T, S : \mathcal{V} \rightarrow \mathcal{P}$ are G -injective for that same representation and their physical ranges are finite-dimensional. Then an invertible physical map $F : \mathcal{P} \rightarrow \mathcal{P}$ satisfies $FT = S$. This is a local consequence of [203, Definition 5.1].*

Proof. ✔ Proof The physical range isomorphism sends Tx to Sx for every virtual vector x . Extend this isomorphism of finite-dimensional subspaces to an automorphism of $\mathcal{P}$ by choosing complementary subspaces. Its restriction has the required identity. $\square$

Theorem 24.26.2 (Comparison between physical spaces of equal dimension). [Lean] ✓ Stmt *Let $T : \mathcal{V} \rightarrow \mathcal{P}$ and $S : \mathcal{V} \rightarrow \mathcal{Q}$ be G -injective for the same representation of a finite group. If $\mathcal{P}$ and $\mathcal{Q}$ are finite-dimensional and have equal dimensions, there is a linear isomorphism $F : \mathcal{P} \rightarrow \mathcal{Q}$ satisfying $FT = S$.*

Proof. Proof Choose a linear isomorphism $E : \mathcal{P} \rightarrow \mathcal{Q}$. The map $E^{-1}S$ is G -injective. Apply the ambient comparison to T and $E^{-1}S$ to obtain an automorphism L of $\mathcal{P}$ with $LT = E^{-1}S$. Then $F = EL$ satisfies $FT = S$. □

24.27 Normalized local physical form

Theorem 24.27.1 (Normalized invariant-space form). [Lean] [✓ Stmt] *Let a finite group G act linearly on a finite-dimensional complex Hilbert space $\mathcal{V}$, and let $T : \mathcal{V} \rightarrow \mathcal{P}$ be G -injective, where $\mathcal{P}$ is a finite-dimensional complex Hilbert space. There exists an invertible physical map $F : \mathcal{P} \rightarrow \mathcal{P}$ such that FT is invariant and*

$$\langle FTx, FTy \rangle = \langle x, y \rangle \quad (x, y \in \mathcal{V}^G).$$

In particular, for the regular virtual representation this is a normalized G -isometric tensor, with normalization factor one. This local consequence of [203, Definition 5.1 and the definition of G -isometric PEPS] concerns an invertible physical transformation; it asserts no equality of closed states.

Proof. ✓ Proof The invariant virtual space and the physical range have equal dimensions. Choose orthonormal bases to obtain an isometric isomorphism $J : \mathcal{V}^G \rightarrow \text{ran } T$. If $P_G = |G|^{-1} \sum_g \rho(g)$ is group averaging, define $S = JP_G$, with values included in $\mathcal{P}$. Averaging is invariant and restricts to the identity on $\mathcal{V}^G$. Thus S is G -injective and preserves inner products there. The ambient physical comparison gives an invertible F satisfying $FT = S$. $\square$

24.29 Finite-overlap gap criteria

We apply the projector expansion of [16, Section IV.C.3], local source lines 2170–2179, to the canonical regional parent interactions on a finite graph. The anticommutator estimate below is an explicit hypothesis. Deriving it from properties of a boundary state is a separate question.

Theorem 24.29.1 (Disjoint regional interactions commute). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let R, S be disjoint vertex regions and let h_R, h_S be arbitrary linear operators on their respective physical spaces. Their extensions to the full physical space commute.

Proof. ✓ **Proof** The extension of h_R sums only the physical indices in R and retains all complementary labels. Replacements on disjoint regions commute. Interchanging the two finite sums therefore identifies the two compositions. □

Theorem 24.29.2 (Gap from overlapping-pair estimates). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stm](#) Let P_i be the extended canonical parent projector for a region R_i , and let $H = \sum_i P_i$. Suppose that each region overlaps at most m other regions, where $m \geq 1$, and that $\varepsilon \geq 0$ satisfies

$$P_i P_j + P_j P_i \geq -\varepsilon(P_i + P_j)$$

for every distinct overlapping pair. Then

$$H^2 \geq (1 - m\varepsilon)H.$$

If $m\varepsilon < 1$, then every $\xi \perp \ker H$ satisfies

$$\|H\xi\| \geq (1 - m\varepsilon)\|\xi\|.$$

If another positive parent with the same regional kernels satisfies $H' \geq \kappa H$ for some $\kappa > 0$, its norm gap is at least $\kappa(1 - m_\varepsilon)$.

Proof. ✓ Proof Expand H^2 . The diagonal terms sum to H since $P_i^2 = P_i$. A disjoint pair consists of commuting orthogonal projections, so its anticommutator is positive. Sum the assumed lower bounds over the remaining unordered pairs. Each projector occurs at most m times, giving the quadratic-form inequality. The spectral theorem for a positive finite-dimensional operator then gives the asserted norm bound on the complement of its kernel. The comparison assertion follows from the common kernel and the corresponding lower bound between positive operators. $\square$

Corollary 24.29.3 (A gap independent of the volume). [Lean] [✓ Stmt] Consider an arbitrary family of finite graphs, regional canonical PEPS parents, and physical and bond dimensions. If the overlap bound m and anticommutator bound ε hold throughout the family and satisfy $m\varepsilon < 1$, then every parent in the family has norm gap at least the same positive number $1 - m\varepsilon$.

Proof. ✓ Proof Apply the preceding estimate to each member of the family. The resulting constant does not depend on the graph or its volume. □

Theorem 24.29.4 (Unit gap for commuting canonical parents). [Lean] [Lean] [✓ Stmt] *If the extended canonical parent projectors commute pairwise, their sum satisfies $H^2 \geq H$ and has norm gap at least one on $(\ker H)^\perp$. In particular, this holds for pairwise disjoint regions.*

Proof. □ **Proof** Every cross term in the projector expansion is nonnegative. Thus $H^2 \geq H$, and the preceding spectral argument applies. □

Lemma 24.29.5 (Complementary bond insertions preserve a regional ground space). [Lean](#)

[✓ Stmt](#) Consider a finite graph tensor with regular group labels and bond insertions equal to the identity on every edge incident to R . The inserted tensor has the same regional ground space $\mathcal{G}_R$ as the original tensor.

Proof. [✓ Proof](#) All tensor coefficients in the open contraction of R are unchanged. Thus the open-region matrices, and hence their physical ranges, agree. $\square$

Theorem 24.29.6 (Torus closures satisfy interior regional parent constraints). [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let a four-leg tensor with regular group labels be placed on a native square torus with both periods at least three. Let R be a coordinate rectangle whose starting coordinates are positive and whose ending coordinates are strictly below the torus periods. For every pair of closure elements g, h , every physical slice of the native closure vector $\Psi_{g,h}$ across R lies in the original regional ground space $\mathcal{G}_R$. Consequently this holds for every coherent sum $\Psi = \sum_i \mu_i \Psi_{g_i, h_i}$, and every regional parent interaction h_R for the original tensor satisfies

$$h_R^{\text{global}} \Psi = 0.$$

No isometry, G -injectivity, or commutativity of g_i, h_i is assumed. This is an interior local constraint; it does not assert the full torus ground-space classification of [203, Theorem 5.5].

Proof. [✓ Proof](#) The native closure is the graph contraction with its holonomy operators inserted at ordered seam endpoints. No seam bond is incident to the strict interior rectangle. Its open contraction therefore has the original regional ground space. Every slice of the closed inserted tensor belongs to that space. A subspace contains coherent sums of its vectors, so the assertion extends to Ψ . The globally extended interaction vanishes precisely when its regional kernel contains every such slice. $\square$

Theorem 24.29.7 (The rectangular PEPS area law). [Lean](#) [Lean](#) [✓ Stmt](#) Consider a nonzero PEPS on a finite square-lattice torus whose two periods are at least three. Let R be a coordinate rectangle of width w and height h contained in the chosen coordinate domain. Suppose each virtual bond crossing its boundary has dimension D . The normalized reduced density matrix obeys

$$0 \leq S(\rho_R) \leq (2w + 2h) \log D.$$

In particular a square of side length L obeys $S(\rho_R) \leq 4L \log D$, the PEPS area-law bound in [16, Section II.A.3]. The tensors may vary from site to site. Empty rectangles and rectangles spanning a full coordinate period are included. We use the convention $\log 0 = 0$; a nonzero PEPS permits $D = 0$ here only when no bond crosses the cut.

Proof. [✓ Proof](#) A coordinate rectangle has at most $2w + 2h$ crossing bonds. The general entropy bound gives the logarithm of the product of their dimensions. Since $\log D \geq 0$ for natural-number dimensions, the boundary count yields the asserted inequality. When $D = 0$, nonvanishing forces an empty boundary, and the entropy is zero. $\square$

Theorem 24.29.8 (Interior support of coherent regular torus closures). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

Let a native four-leg regular G -injective tensor be placed on a square torus with both periods at least three. Let $\Psi = \sum_i \mu_i \Psi_{g_i, h_i}$ be a nonzero coherent sum of closure vectors, where $g_i h_i = h_i g_i$ for every i . Let R be a positive coordinate rectangle whose starting coordinates are positive and whose ending coordinates are strictly below the torus periods. Then the physical coefficient matrix across R has range exactly the original regional ground space $\mathcal{G}_R$. Equivalently, the actual reduced density of Ψ satisfies

$$\text{ran } \rho_R = \mathcal{G}_R.$$

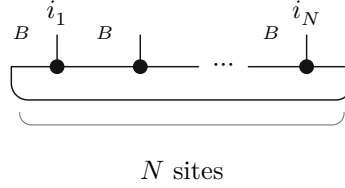
This is an interior support consequence of [203, Theorem 5.5 and Corollary 6.10]; it does not classify the full torus parent-Hamiltonian ground space.

Proof. ✓ Proof Every complementary physical slice of Ψ lies in $\mathcal{G}_R$, so the coefficient range is contained in that space. The simply connected regular Schmidt-rank theorem gives $\text{rank } |G|^{|\partial R|-1}$ for the nonzero coherent sum. The connected open-region contraction has precisely the same dimension. Inclusion and equality of dimensions therefore give equality of the two spaces. The reduced density is the product of the coefficient matrix and its adjoint, which has the same range as the coefficient matrix. $\square$

Chapter 25

The Multi-Block Asymmetric Fundamental Theorem

The asymmetric Fundamental Theorem of matrix product states and operators compares an arbitrary tensor with a weighted direct sum of normal tensors. Equality of all positive-length periodic vectors determines every nonzero simple composition factor of the arbitrary tensor, including its multiplicity; the only undetectable factor is the one-dimensional tensor on which every physical matrix vanishes. The remaining freedom is an upper-triangular, globally nilpotent extension between the factors, explaining the failure of sitewise intertwiners under periodic equality alone (Remark 25.1.2).



25.1 Triangular word products

The distinction between a word-level compression and a sitewise intertwiner is already visible in elementary block matrix multiplication.

Lemma 25.1.1 (Diagonal blocks of block-triangular products). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

Stmt Let o be a finite index set with block sizes n_k , $k \in o$, and let $T^1, \dots, T^d$ be block upper triangular for the resulting decomposition of $M_{\sum_k n_k}(\mathbb{C})$. Write T_{kk}^i for the k -th diagonal block of T^i . Then T^w is block upper triangular for every word w , and its k -th diagonal block is the word evaluation of the diagonal blocks,

$$(T^w)_{kk} = T_{kk}^{i_1} T_{kk}^{i_2} \dots T_{kk}^{i_N} \quad (25.1)$$

for $w = (i_1, \dots, i_N)$. If P_k is the coordinate embedding of the k -th block and Q_k the coordinate projection onto it, then for every matrix M on the same space, $Q_k M P_k = M_{kk}$, and

$$Q_k P_l = \begin{cases} \text{id}, & k = l, \\ 0, & k \neq l. \end{cases} \quad (25.2)$$

Proof. [✓ Proof](#) For two block upper-triangular matrices X, Y , $(XY)_{kk} = \sum_l X_{kl}Y_{lk}$; the terms with $l < k$ vanish because $X_{kl} = 0$, and those with $l > k$ vanish because $Y_{lk} = 0$, leaving $(XY)_{kk} = X_{kk}Y_{kk}$. Induction on the word length gives (25.1) and the persistence of block triangularity. The identity $Q_k M P_k = M_{kk}$ and (25.2) are direct computations from the coordinate description of P_k and $Q_k = P_k^\top$. $\square$

Remark 25.1.2 (Periodic equality does not imply a sitewise intertwiner). Take the one-dimensional normal tensor $A^1 = (1)$, $A^2 = (2)$, and the bond-dimension-three tensor

$$B^1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Both matrices are upper triangular with diagonal blocks $0, A^i, 0$. With $P = e_2$ and $Q = e_2^\top$, Lemma 25.1.1 gives $QB^wP = A^w$ and $QP = 1$ for every nonempty word w , so $\text{tr}(B^w) = A^w = \text{tr}(A^w)$: the periodic vectors of A and B agree at every positive length.

There is nevertheless no nonzero right sitewise intertwiner: a vector $v = (x, y, z)^\top$ with $B^1v = v$ and $B^2v = 2v$ would need $x = y$ and $z = 0$ from $B^1v = v$, and $x = 0$ from the first coordinate of $B^2v = 2v$, forcing $v = 0$. Symmetrically, no nonzero left intertwiner $uB^1 = u$, $uB^2 = 2u$ exists: $uB^1 = u$ gives $u_1 = u_3 = 0$ while the third coordinate of $uB^2 = 2u$ gives $u_2 = 2u_3 = 0$. Periodic equality can therefore force a multiplicative subquotient while admitting neither one-sided sitewise intertwiner, so it cannot imply a biorthogonal zipper pair. This is why the theorem below is stated as a compression $W_s B^w V_s = C_s^w$ rather than as an eigenvector relation.

25.2 The word algebra and normal tensors as simple modules

Definition 25.2.1 (Word algebra and positive ideal). [Lean](#) [Lean](#) [✓ Stmt](#) Fix $d \geq 1$. The *word algebra* is the free unital associative algebra $F = \mathbb{C}\langle x_1, \dots, x_d \rangle$, realized as the algebra of finite $\mathbb{C}$ -linear combinations of words in the d letters under concatenation, so that the words form a basis. The *augmentation* $\text{aug} : F \rightarrow \mathbb{C}$ is the algebra map sending every generator to zero; on a basis word w , $\text{aug}(w) = 1$ if w is empty and 0 otherwise. The *positive ideal* is the two-sided ideal $F_+ = \ker(\text{aug})$ of elements with zero constant term.

Definition 25.2.2 (Word-algebra module of a tensor). [Lean](#) [✓ Stmt](#) A tensor $C = (C^i)_{i=0}^{d-1}$ on $\mathbb{C}^D$ defines an F -module structure on $\mathbb{C}^D$ through the representation $\rho_C(x_i) = C^i$, extended multiplicatively so that a word w acts by $\rho_C(w) = C^w$.

Definition 25.2.3 (Word traces). [Lean](#) [Lean](#) [✓ Stmt](#) For a finite-dimensional F -module M , the *trace character* is the linear functional $p \mapsto \text{tr}(\rho_M(p))$ on F , and the *word trace* of a word w is $\text{tr}(\rho_M(w))$.

Lemma 25.2.4 (Word traces of a tensor). [Lean](#) [✓ Stmt](#) On the word-algebra module of a tensor C , the word trace of w is the periodic coefficient $\text{tr}(C^w)$.

Proof. [✓ Proof](#) The action of w is conjugate to the matrix C^w under the identification of the module with $\mathbb{C}^D$, and the trace of a linear map is the trace of its matrix. $\square$

Lemma 25.2.5 (Normal tensors define simple modules). [Lean](#) [✓ Stmt](#) If C is normal on $\mathbb{C}^D$ with $D > 0$, its word-algebra module is simple.

Proof. ✓ Proof Let $K \subseteq \mathbb{C}^D$ be a nonzero invariant subspace and $u \in K$ nonzero. Normality gives a length L for which the span of the length- L word evaluations is all of $M_D(\mathbb{C})$; since K is invariant under every such word and hence under their span, $Xu \in K$ for every $X \in M_D(\mathbb{C})$. Every vector is Xu for a suitable X because $u \neq 0$, so $K = \mathbb{C}^D$. $\square$

Multiplying every matrix of a tensor by a nonzero scalar does not change its invariant subspaces, so a weighted block $\mu_k A_{(k)}$ built from a normal tensor $A_{(k)}$ and $\mu_k \in \mathbb{C}^\times$ remains a simple F -module.

25.3 Separating simple modules

Lemma 25.3.1 (Positive-word character lemma). Not ready *Let U, V be finite-dimensional F -modules with $\text{tr}(\rho_U(p)) = \text{tr}(\rho_V(p))$ for every $p \in F_+$. Then the semisimplifications of U and V contain every simple F -module with the same multiplicity, except possibly the one-dimensional module $\mathbf{0}$ on which every generator acts as zero: there are integers $a, b \geq 0$ and a semisimple module M without a $\mathbf{0}$ summand such that $U^{\text{ss}} \cong M \oplus \mathbf{0}^{\oplus a}$ and $V^{\text{ss}} \cong M \oplus \mathbf{0}^{\oplus b}$.*

Lemma 25.3.2 (Positive-ideal separation of linear functionals). Lean ✓ Stmt *Two $\mathbb{C}$ -linear functionals on F agreeing on every nonempty word agree on the positive ideal F_+ .*

Proof. ✓ Proof Their difference agrees with $(\chi_1(1) - \chi_2(1)) \cdot \text{aug}$ on every basis word, by induction on the monoid-algebra presentation, hence on all of F by linearity; on F_+ the augmentation vanishes, so the two functionals agree there. $\square$

Lemma 25.3.3 (The zero module is the only simple module with trivial generator action). Lean ✓ Stmt *A finite-dimensional simple F -module on which every generator acts as zero is one-dimensional. It is the module $\mathbf{0}$ of Lemma 25.3.1.*

Proof. ✓ Proof If every generator acts as zero, F acts through the augmentation, so the F -submodules and the $\mathbb{C}$ -subspaces coincide; simplicity as a $\mathbb{C}$ -vector space forces dimension one. $\square$

Theorem 25.3.4 (Burnside's theorem at module level). Lean ✓ Stmt *Every $\mathbb{C}$ -linear endomorphism of a finite-dimensional simple F -module Q is the action of some element of F .*

Proof. ✓ Proof By Schur's lemma over the algebraically closed field $\mathbb{C}$, the commutant $\text{End}_F(Q)$ consists of scalars, so Q is finite-dimensional over its commutant. The Jacobson density theorem then makes the action map $F \rightarrow \text{End}_{\mathbb{C}}(Q)$ surjective. $\square$

Theorem 25.3.5 (Separating a simple module from a finite non-isomorphic family). Lean Lean ✓ Stmt *Let Q be a finite-dimensional simple F -module on which some generator acts nonzero. There is $u \in F_+$ acting as the identity on Q . If T is a finite-dimensional simple module not isomorphic to Q , there is $p \in F_+$ acting as the identity on Q and as zero on T . More generally, if C_s , $s \in S$, is a finite family of word-algebra modules of tensors, each simple and none isomorphic to Q , there is $p \in F_+$ acting as the identity on Q and as zero on every C_s .*

Proof. ✓ Proof For the first claim, Burnside's theorem realizes, for a basis $\{b_j\}$ of Q and a nonzero vector $x_i \cdot q_0$, the maps sending $x_i \cdot q_0 \mapsto b_j$ and $q \mapsto (\text{coordinate } j \text{ of } q) \cdot q_0$; assembling them into $u = \sum_j \alpha_j x_i \beta_j$ gives an element of F_+ acting as the identity on Q . For the pairwise separation, Schur's lemma shows every F -linear endomorphism of $Q \times T$ preserves the two non-isomorphic simple factors, so $Q \times T$ is finite-dimensional over its (scalar) commutant; Jacobson density on $Q \times T$ produces $a \in F$ acting as the identity on the first factor and zero on the second, and $p = au$

acts as the identity on Q and as zero on T . The finite-family statement follows by induction on S , multiplying the separators for the successive members. $\square$

Lemma 25.3.6 (Trace additivity over an invariant submodule). [Lean](#) [✓ Stmt](#) *For a finite-dimensional F -module M and a submodule $K \leq M$, the word trace of M is the sum of the word traces of K and of M/K .*

Proof. [✓ Proof](#) This is the additivity of the ordinary trace of an endomorphism over an invariant subspace, applied to the action of each word on M . $\square$

Theorem 25.3.7 (Nilpotency from vanishing trace powers). [Lean](#) [✓ Stmt](#) *An endomorphism of a finite-dimensional complex vector space, all of whose positive powers have vanishing trace, is nilpotent.*

Proof. [✓ Proof](#) The characteristic polynomial is determined by the power sums of its roots (Newton's identities), so vanishing trace powers force it to equal $X^{\dim}$; the Cayley–Hamilton theorem then gives nilpotency. $\square$

Theorem 25.3.8 (Identification of a simple quotient with a target block). [Lean](#) [✓ Stmt](#) *Let M be a finite-dimensional F -module whose word traces are, on every nonempty word, the sum of the word traces of a finite family of simple tensor modules C_s , $s \in S$. Let $K \leq M$ have simple quotient M/K on which some generator acts nontrivially. Then M/K is isomorphic to C_s for some $s \in S$.*

Proof. [✓ Proof](#) If M/K were isomorphic to no C_s , Theorem 25.3.5 supplies $p \in F_+$ acting as the identity on M/K and as zero on every C_s . Since the word trace of M agrees with the sum of the word traces of the C_s on every nonempty word, and $p \in F_+$, Lemma 25.3.2 shows that every positive power of the action of p on M has vanishing trace, so this action is nilpotent by Theorem 25.3.7. A nilpotent action of p on M descends to a nilpotent action on the quotient M/K , contradicting that p acts there as the identity on a nonzero module. $\square$

25.4 The composition-series flag

Lemma 25.4.1 (Composition-series triangular gauge). [Lean](#) [✓ Stmt](#) *Let $B = (B^i)_i$ act on $\mathbb{C}^{D_B}$, and choose a composition series $0 = H_0 \subset H_1 \subset \dots \subset H_r = \mathbb{C}^{D_B}$ of its word-algebra module. There is $G \in \mathrm{GL}_{D_B}(\mathbb{C})$ such that every $T^i = GB^iG^{-1}$ is block upper triangular for the resulting decomposition into r blocks, and the k -th diagonal block of T^i is the action induced by B^i on the simple subquotient H_k/H_{k-1} .*

Proof. [✓ Proof](#) Choose a basis of H_1 and extend it successively to bases of $H_2, \dots, H_r$. In the resulting basis, every H_k is a coordinate initial segment; since every H_k is invariant under every B^i , all matrices B^i are block upper triangular in this basis, with k -th diagonal block the induced action on H_k/H_{k-1} . $\square$

Definition 25.4.2 (Labelled invariant flag). [Lean](#) [✓ Stmt](#) Let M be a finite-dimensional F -module, S a finite set of target slots, and C_s , $s \in S$, a family of tensors. A *labelled invariant flag* of M over (S, C) is an invariant chain $0 = H_0 \leq H_1 \leq \dots \leq H_r = M$ together with a bijective labelling of the steps $\{1, \dots, r\}$ by $S \sqcup \{1, \dots, z\}$, for some $z \geq 0$, such that a step labelled by $s \in S$ has subquotient H_k/H_{k-1} isomorphic to the word-algebra module of C_s , and a step labelled by an index in $\{1, \dots, z\}$ has one-dimensional subquotient on which every generator acts as zero.

Lemma 25.4.3 (A family of normal blocks with zero total word trace is empty). [Lean](#) [✓ Stmt](#)
If the word traces of a finite family of normal tensors of positive bond dimension sum to zero on every nonempty word, the family is empty.

Proof. [✓ Proof](#) Otherwise Theorem 25.3.5 applied to one member C_{s_0} of the family supplies $u \in F_+$ acting as the identity on C_{s_0} ; vanishing total word trace forces every positive power of the action of u on the product of the modules in the family to have trace zero, so this action is nilpotent by Theorem 25.3.7. A nilpotent action of u contradicts its action as the identity on the nonzero summand C_{s_0} . $\square$

Theorem 25.4.4 (The flag theorem). [Lean](#) [✓ Stmt](#) *Let C_s , $s \in S$, be a finite family of normal tensors of positive bond dimension, and let M be a finite-dimensional F -module whose word traces agree, on every nonempty word, with the sum of the word traces of the C_s . Then M carries a labelled invariant flag over (S, C) .*

Proof. [✓ Proof](#) Induction on $\dim M$. If $M = 0$, Lemma 25.4.3 forces $S = \emptyset$, and the empty chain is the required flag. If $M \neq 0$, choose a maximal proper invariant submodule $K \leq M$ (a coatom of the invariant-submodule lattice, which exists because M is finite-dimensional), so M/K is simple. If some generator acts nontrivially on M/K , Theorem 25.3.8 identifies M/K with some block C_{s_0} ; Lemma 25.3.6 shows the word trace of K agrees with the sum over the remaining blocks $S \setminus \{s_0\}$, and the induction hypothesis (applied to K , of strictly smaller dimension) supplies a flag of K , extended by the matched final step $K \subset M$ labelled s_0 . If instead every generator acts as zero on M/K , Lemma 25.3.3 shows M/K is one-dimensional; Lemma 25.3.6 shows the word trace of K already agrees with the sum over all of S , and the induction hypothesis supplies a flag of K , extended by the unmatched final step $K \subset M$. $\square$

25.5 The multi-block asymmetric compression theorem

Theorem 25.5.1 (Multi-block asymmetric compression theorem). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) *Let $A = \bigoplus_{s \in S} C_s$ be a finite direct sum of nonzero normal tensors $C_s = (C_s^i)_i$ of bond dimension D_s , with repetitions allowed, and let $B = (B^i)_i$ be an arbitrary tensor of bond dimension D_B . Assume that the periodic vectors of A and B agree at every positive length, $|V^{(N)}(B)\rangle = |V^{(N)}(A)\rangle$ for every $N > 0$; equivalently, $\text{tr}(B^w) = \text{tr}(A^w)$ for every nonempty word w . Then there are an integer $z \geq 0$, an integer $r = |S| + z$, an invertible matrix $G \in \text{GL}_{D_B}(\mathbb{C})$, and maps $V_s : \mathbb{C}^{D_s} \rightarrow \mathbb{C}^{D_B}$, $W_s : \mathbb{C}^{D_B} \rightarrow \mathbb{C}^{D_s}$, with the following properties.*

- (i) *Every GB^iG^{-1} is block upper triangular for one common decomposition into r blocks.*
- (ii) *Exactly $|S|$ of the diagonal slots are matched with the target slots: writing $j(s)$ for the position matched with s , there is $X_s \in \text{GL}_{D_s}(\mathbb{C})$ with $(GB^iG^{-1})_{j(s), j(s)} = X_s C_s^i X_s^{-1}$.*
- (iii) *The remaining z diagonal slots are one-dimensional zero tensors.*
- (iv) *The compression pairs are mutually biorthogonal: $W_s V_t = \text{id}_{D_s}$ if $s = t$ and 0 otherwise.*
- (v) *For every nonempty word w , $W_s B^w V_s = C_s^w$.*
- (vi) *Writing $\widehat{C}^i = \sum_{s \in S} V_s C_s^i W_s$ and $R^i = B^i - \widehat{C}^i$, every product $R^{i_1} R^{i_2} \dots R^{i_r} = 0$: the non-unital algebra generated by the R^i is nilpotent.*
- (vii) $D_B = \sum_{s \in S} D_s + z$.

Proposition 25.5.4 (A vanishing remainder gives sitewise intertwiners). [Lean](#) [Lean](#) [✓ Stmt](#) In the setting of Theorem 25.5.1, suppose that the remainder vanishes, $R^i = 0$ for every physical index i . Then for every i and every $s \in S$,

$$B^i V_s = V_s C_s^i, \quad W_s B^i = C_s^i W_s. \quad (25.4)$$

Proof. [✓ Proof](#) Since $R^i = 0$, $B^i = \sum_{t \in S} V_t C_t^i W_t$. Multiplying on the right by V_s , or on the left by W_s , and using the biorthogonality $W_t V_s = \delta_{ts} \text{id}_{D_s}$ of clause (iv) leaves only the term $t = s$. $\square$

Proposition 25.5.5 (Projector-weighted direct sums). [Lean](#) [Lean](#) [✓ Stmt](#) Let the bond space of B be $\mathbb{C}^m \otimes \mathbb{C}^D$ and let $\Pi_1, \dots$ be mutually orthogonal rank-one idempotents of $\mathbb{C}^m$ cut out by an invertible matrix U (the idempotents $U^{-1} E_{jj} U$). If every letter has the form $B^i = \sum_{s \in S} \mu_s \Pi_{j(s)} \otimes C_s^i$ with distinct positions $j(s)$, then B admits the block triangular form of Theorem 25.5.1 onto the weighted blocks $\mu_s C_s$, with $z = (m - |S|)D$ zero factors, the remainder $R^i = 0$, and sitewise intertwiners $B^i V_s = V_s \mu_s C_s^i$ and $W_s B^i = \mu_s C_s^i W_s$ for every slot: the extension splits.

Proof. [✓ Proof](#) Conjugating by $U \otimes \text{id}$ turns each $\Pi_{j(s)}$ into a matrix unit, so every letter becomes block diagonal with the blocks $\mu_s C_s^i$ at the positions $j(s)$ and zero blocks elsewhere; block diagonality gives the triangular form and the vanishing remainder, and Proposition 25.5.4 gives the intertwining relations. $\square$

25.6 Products of matrix product operators

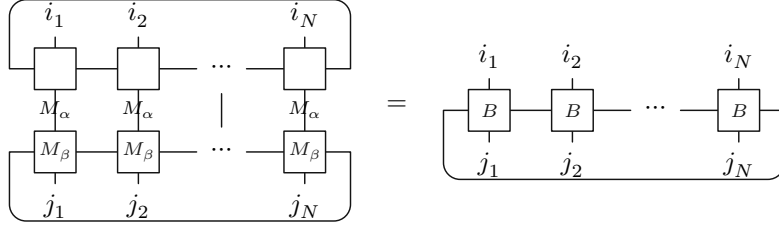
Let M_α and M_β be matrix product operator tensors. Their stacked product is the tensor $B^{ij} = \sum_m M_\alpha^{im} \otimes M_\beta^{mj}$, where the pair (i, j) is treated as one physical index. Write $O_N(M)$ for the periodic operator of length N generated by M .

A word in the paired indices specifies a matrix entry of $O_N(M)$. That entry is the trace of the corresponding product of tensor matrices. Thus identities between word traces at every positive length determine identities between periodic operators.

Lemma 25.6.1 (Word traces determine the periodic operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The trace of a length- N word in the paired indices of M is the matrix entry of $O_N(M)$ at the ket and bra configurations specified by the word. Consequently, if two tensors M and $\tilde{M}$ have equal word traces at every positive length, then $O_N(M) = O_N(\tilde{M})$ for every $N \geq 1$. If instead, for a scalar c , every length- N word trace of M equals c^N times the corresponding trace of $\tilde{M}$, then $O_N(M) = c^N O_N(\tilde{M})$. Rescaling a tensor A by c multiplies each length- N word trace by c^N .

Proof. [✓ Proof](#) A word in paired indices specifies separate ket and bra words, and its trace is the corresponding entry of $O_N(M)$. Equality of these entries for all configurations gives equality of the operators. Rescaling a tensor by c multiplies each letter's matrix by c , hence a length- N word evaluation, and its trace, by c^N . $\square$

Theorem 25.6.2 (Reduction of a product of matrix product operators). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{M_\gamma\}_\gamma$ be a finite family of matrix product operator tensors of positive bond dimension, each normal when its pair of physical indices is treated as one index. Suppose that nonzero scalars λ_γ satisfy $O_N(B) = \sum_\gamma \lambda_\gamma^N O_N(M_\gamma)$ for every $N \geq 1$. Then for every γ there are maps V_γ and W_γ with $W_\gamma V_\gamma = \text{id}$ and $W_\gamma B^{i_1 j_1} \dots B^{i_N j_N} V_\gamma = \lambda_\gamma^N M_\gamma^{i_1 j_1} \dots M_\gamma^{i_N j_N}$ for every $N \geq 1$ and every sequence of index pairs. The pairs may be chosen mutually biorthogonal, and the remainder after all matched sectors are removed is nilpotent.



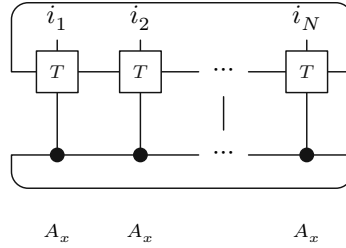
Proof. [✓ Proof](#) Treat each pair (i, j) as one letter. The operator identity is the equality of the periodic vectors of B and of $\bigoplus_{\gamma} \lambda_{\gamma} M_{\gamma}$ on the pair alphabet, so Theorem 25.5.2 applies. $\square$

Corollary 25.6.3 (Fusion tensors of periodic matrix product operator representations). [Lean](#)

[✓ Stmt](#) Let $\{U_g\}_{g \in G}$ be a matrix product operator representation of a group G , not necessarily unitary, by tensors T_g of positive bond dimension whose doubled-index tensors are normal, with $U_g U_h = U_{gh}$ on periodic chains of every positive length. Injective tensors are normal, so this covers the injective representations of [164]; neither unitarity, simplicity, nor $U_e = \text{id}$ is needed. Then for all $g, h \in G$ there are fusion tensors V and W with $WV = \text{id}$ and $W(T_g T_h)^{i_1 j_1} \dots (T_g T_h)^{i_N j_N} V = T_{gh}^{i_1 j_1} \dots T_{gh}^{i_N j_N}$ for every $N \geq 0$, and the remainder is nilpotent of order at most $D_g D_h - D_{gh} + 1$.

Proof. [✓ Proof](#) The stacked tensor of T_g and T_h generates the same periodic operators as T_{gh} , which is normal; apply Corollary 25.9.1. $\square$

Corollary 25.6.4 (Action on matrix product vectors). [Lean](#) [Lean](#) [✓ Stmt](#) Let T be a matrix product operator tensor and A_x a matrix product state tensor, and let $T \cdot A_x$ be the tensor of the operator applied to the state, $(T \cdot A_x)^i = \sum_j T^{ij} \otimes A_x^j$. If the periodic operators of T map the periodic vectors of A_x to those of a normal tensor A_y of positive bond dimension at every positive length, then $T \cdot A_x$ reduces to A_y with a nilpotent remainder. If, at every positive length, they map them to a finite weighted sum $\sum_s \mu_s^N |V^{(N)}(A_s)\rangle$ over normal tensors of positive bond dimension with nonzero weights μ_s , then $T \cdot A_x$ admits a simultaneous reduction to the weighted blocks $\mu_s A_s$.



Proof. [✓ Proof](#) The periodic vectors of $T \cdot A_x$ are the images of those of A_x under the periodic operators of T , so the hypotheses of Corollary 25.9.1 and of Theorem 25.5.2 hold. $\square$

Corollary 25.6.5 (Periodic identities from block reductions). [Lean](#) [Lean](#) [✓ Stmt](#) If the stacked product of M_{α} and M_{β} admits the block triangular form of Theorem 25.5.1 for a family $\{C_s\}_{s \in S}$ of operator tensors, then $O_N(M_{\alpha})O_N(M_{\beta}) = \sum_{s \in S} O_N(C_s)$ for every $N \geq 1$. If the tensor $T \cdot A_x$ admits the same form for a family $\{C_s\}_{s \in S}$ of state tensors, then $O_N(T)|V^{(N)}(A_x)\rangle = \sum_{s \in S} |V^{(N)}(C_s)\rangle$ for every $N \geq 1$.

Proof. [✓ Proof](#) Apply Proposition 25.5.3 on the pair alphabet for the product and on the physical alphabet for $T \cdot A_x$. The definitions of the periodic operator and of $T \cdot A_x$ give the two identities. $\square$

25.6.1 Fusion and action tensors with multiplicity

The fusion rules of a matrix product operator algebra carry nonnegative integer multiplicities: $O_N(M_a)O_N(M_b) = \sum_c N_{ab}^c O_N(M_c)$ for every $N \geq 1$ [164]. Each copy of a channel c is then a separate block of the reduction.

Theorem 25.6.1.1 (Reduction onto blocks with multiplicity). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{C_c\}_c$ be a finite family of normal tensors of positive bond dimension and let $N_c \geq 0$ be integers. If a tensor B of bond dimension D satisfies $|V^{(N)}(B)\rangle = \sum_c N_c |V^{(N)}(C_c)\rangle$ for every $N \geq 1$, then for every pair (c, μ) with $\mu < N_c$ there are maps $V_{c,\mu}$ and $W_{c,\mu}$ such that $V_{c,\mu} W_{d,\nu} = \delta_{cd} \delta_{\mu\nu} \text{id}$, $V_{c,\mu} B^{i_1} \dots B^{i_N} W_{c,\mu} = C_c^{i_1} \dots C_c^{i_N}$ for every $N \geq 0$, and every product of at least D matrices of the remainder $B^i - \sum_{c,\mu} W_{c,\mu} C_c^i V_{c,\mu}$ vanishes.

Proof. [✓ Proof](#) The word traces of B are those of the family in which C_c occurs N_c times, so Theorem 25.5.1 applies to that family. The number of blocks plus the number of zero slots is at most D , which bounds the nilpotency order. $\square$

Corollary 25.6.1.2 (Fusion and action tensors with multiplicity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $\{M_c\}_c$ be matrix product operator tensors of positive bond dimensions χ_c , each normal when its pair of physical indices is treated as one index, with $O_N(M_a)O_N(M_b) = \sum_c N_{ab}^c O_N(M_c)$ for every $N \geq 1$. Then the stacked product of M_a and M_b has fusion tensors $V_{ab}^{c,\mu}$, $W_{ab}^{c,\mu}$, $\mu < N_{ab}^c$, as in Theorem 25.6.1.1, with nilpotency order at most $\chi_a \chi_b$. Likewise, if $O_N(T)|V^{(N)}(A_x)\rangle = \sum_y M_y |V^{(N)}(A_y)\rangle$ for every $N \geq 1$ with normal A_y of positive bond dimension, then $T \cdot A_x$ has action tensors onto M_y copies of each A_y .

Proof. [✓ Proof](#) The periodic vectors of the stacked product on the pair alphabet are the matrix entries of $O_N(M_a)O_N(M_b)$, and those of $T \cdot A_x$ are the entries of $O_N(T)|V^{(N)}(A_x)\rangle$; apply Theorem 25.6.1.1. $\square$

Corollary 25.6.1.3 (Gauge freedom of a multiplicity-free fusion channel). [Lean](#) [Lean](#) [✓ Stmt](#) Let (V, W) and (V', W') be two reductions of the stacked product B of M_a and M_b onto the same normal tensor M_c , each with a remainder all of whose products of at least K factors vanish. Then there is $z \neq 0$ with $VB^w = zV'B^w$ and $B^wW = z^{-1}B^wW'$ for every word w of length greater than $2K$. When $O_N(M_a)O_N(M_b) = O_N(M_c)$ for every $N \geq 1$, Corollary 25.6.1.2 provides such reductions with $K = \chi_a \chi_b$.

Proof. [✓ Proof](#) Apply Theorem 25.11.28 on the pair alphabet. $\square$

25.7 Structure coefficients at every positive length

Structure coefficients in an expansion of an arbitrary tensor over inequivalent normal tensors are power sums of fixed finite multisets of complex weights, at every positive length and without blocking. Normality means injectivity after some blocking (Definition 2.5.5); the tensors need not be normalized. Separation excludes gauge equivalence up to any nonzero complex scalar, not only a unit-modulus phase.

Theorem 25.7.1 (Uniqueness of nonzero weights from tail power sums). [Lean](#) [Lean](#) [✓ Stmt](#) Let Λ and Λ' be finite multisets in $\mathbb{C} \setminus \{0\}$. If their power sums agree for all sufficiently large integers N , then $\Lambda = \Lambda'$, including multiplicities. In particular, if the power sums of Λ vanish for all sufficiently large N , then Λ is empty.

Proof. [✓ Proof](#) List the distinct entries of the two multisets as $z_0, \dots, z_{r-1}$, and let b_j be the difference of their multiplicities. For a threshold N_0 and each $0 \leq i < r$,

$$\sum_{j=0}^{r-1} (b_j z_j^{N_0}) z_j^i = 0. \quad (25.5)$$

The Vandermonde matrix is invertible, so $b_j z_j^{N_0} = 0$. Since $z_j \neq 0$, all multiplicities agree. Comparison with the empty multiset proves the vanishing assertion. $\square$

Theorem 25.7.2 (Power-sum coefficients without blocking). [Lean](#) [Lean](#) [✓ Stmt](#) Let B be an MPS tensor of bond dimension D_B , and let $\{M_\gamma\}_{\gamma \in \Gamma}$ be a finite family of normal MPS tensors on the same physical alphabet, with $D_\gamma > 0$. For distinct γ, δ of equal bond dimension, assume there are no $X \in \text{GL}(D_\gamma, \mathbb{C})$ and $z \in \mathbb{C} \setminus \{0\}$ such that $M_\delta^i = z X M_\gamma^i X^{-1}$ for every physical index i . Suppose that for every $N > 0$ there are coefficients $c_\gamma^{(N)}$ satisfying

$$|V^{(N)}(B)\rangle = \sum_{\gamma \in \Gamma} c_\gamma^{(N)} |V^{(N)}(M_\gamma)\rangle. \quad (25.6)$$

Then there are integers $n_\gamma \geq 0$ and nonzero complex weights $\mu_{\gamma k}$, indexed by $0 \leq k < n_\gamma$, such that for every $N > 0$,

$$|V^{(N)}(B)\rangle = \sum_{\gamma \in \Gamma} \left(\sum_{k=0}^{n_\gamma-1} \mu_{\gamma k}^N \right) |V^{(N)}(M_\gamma)\rangle. \quad (25.7)$$

The multiplicities satisfy $\sum_{\gamma \in \Gamma} n_\gamma D_\gamma \leq D_B$. There is a positive threshold N_0 such that, for every $N \geq N_0$, any coefficients satisfying (25.6) obey $c_\gamma^{(N)} = \sum_{k=0}^{n_\gamma-1} \mu_{\gamma k}^N$. The weight multisets are unique: any finite families of nonzero weights giving an expansion of the form (25.7) at all sufficiently large lengths have, for each γ , exactly the same entries with the same multiplicities. Empty weight multisets are allowed.

Proof. [✓ Proof](#) Apply Theorem 7.1.1.2, normalize each nonzero irreducible factor to a periodic tensor, and group factors whose periodic vectors are proportional by a fixed nonzero scalar raised to the length. The irreducible-form framework is described in Ref. [157]. This gives separated periodic representatives A_j of dimensions d_j , positive multiplicities r_j , and nonzero weights a_{jq} such that, for every $N > 0$,

$$|V^{(N)}(B)\rangle = \sum_j \left(\sum_{q=0}^{r_j-1} a_{jq}^N \right) |V^{(N)}(A_j)\rangle, \quad (25.8)$$

$$\sum_j r_j d_j \leq D_B. \quad (25.9)$$

Grouping preserves the dimension count. Indeed, if periodic tensors A, A' of periods m, m' have such proportional vectors, then $|O_{AA'}(N)|^2 = O_{AA}(N)O_{A'A'}(N)$. Different dimensions would force the left side to tend to zero by Theorem 22.2.3.41, whereas along common multiples of the periods the right side tends to $mm' > 0$.

Normalize the targets to period-one tensors C_γ and combine them with the representatives not related to any target by an invertible gauge and a nonzero scalar. The combined family has no repeated blocks. On sufficiently large multiples pn of a common period, it is linearly independent by Theorem 22.2.3.43. Subtracting (25.6) from (25.8), after absorbing matched terms into the targets, forces every unmatched representative to satisfy $\sum_{q=0}^{r_j-1} (a_{jq}^p)^n = 0$ for all sufficiently large n . Theorem 25.7.1 contradicts $r_j > 0$. Thus every representative matches a target.

Choose its matching label $\gamma(j)$ and a nonzero θ_j so that $|V^{(N)}(M_{\gamma(j)})\rangle = \theta_j^N |V^{(N)}(A_j)\rangle$. For each target, collect the weights a_{jq}/θ_j over its matching representatives. This proves (25.7). Since $d_j = D_{\gamma(j)}$, the dimension bound follows. Normalize the normal targets and use their eventual linear independence to compare arbitrary expansion coefficients with these power sums. The nonzero normalization scalars can be cancelled, giving a positive threshold for coefficient uniqueness. Theorem 25.7.1, applied to each target, then proves uniqueness against any tail expansion. $\square$

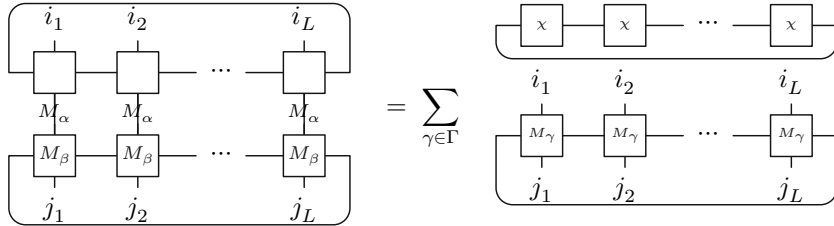
Theorem 25.7.3 (Power-sum structure coefficients for operator closure). Lean Stmt Let $\{M_\gamma\}_{\gamma \in \Gamma}$ be a finite family of MPO tensors of common physical dimension d , with $D_\gamma > 0$. Regard the pair of physical indices (i, j) as one MPS index. Assume these MPS tensors are normal and, for distinct labels of equal bond dimension, are not related by $M_\delta^{ij} = zXM_\gamma^{ij}X^{-1}$ for all i, j , with $z \neq 0$ and X invertible. Fix $\alpha, \beta \in \Gamma$ and write $O_L(M)$ for the periodic operator generated by M . Suppose $O_L(M_\alpha)O_L(M_\beta)$ belongs to $\text{span}_{\mathbb{C}}\{O_L(M_\gamma) : \gamma \in \Gamma\}$ for every $L > 0$.

There are complex diagonal matrices $\chi_{\alpha, \beta, \gamma}$ of sizes $n_\gamma \geq 0$, with nonzero diagonal entries, such that for every $L > 0$,

$$O_L(M_\alpha)O_L(M_\beta) = \sum_{\gamma \in \Gamma} \text{tr}(\chi_{\alpha, \beta, \gamma}^L) O_L(M_\gamma). \quad (25.10)$$

Their sizes satisfy $\sum_{\gamma \in \Gamma} n_\gamma D_\gamma \leq D_\alpha D_\beta$. Beyond a positive threshold, any coefficients of this operator expansion equal $c_{\alpha, \beta, \gamma}^{(L)} = \text{tr}(\chi_{\alpha, \beta, \gamma}^L)$. Any other finite families of nonzero diagonal entries giving such an expansion at all sufficiently large lengths agree target by target, including multiplicities.

This uses the structure-coefficient notation of Ref. [11, Section 4.5].



Proof. Proof Form the product tensor $P^{ij} = \sum_s M_\alpha^{is} \otimes M_\beta^{sj}$, of bond dimension $D_\alpha D_\beta$. Theorem 21.3.9 gives $O_L(P) = O_L(M_\alpha)O_L(M_\beta)$. Equality of operators is equivalent, entry by entry, to equality of MPS coefficients on the combined physical alphabet. Apply Theorem 25.7.2 to P and take $\chi_{\alpha, \beta, \gamma}$ to have diagonal entries $\{\mu_{\gamma k}\}_{k=0}^{n_\gamma-1}$. The same entrywise equivalence transfers the

expansion, eventual coefficient uniqueness, dimension bound, and uniqueness of nonzero entries to operators. $\square$

25.8 The splitting criterion

Periodic vector equality permits upper-triangular nilpotent extensions between composition factors, preventing sitewise intertwining (Remark 25.1.2). Theorem 25.8.1 provides an algebraic condition ensuring that the compressions of Theorem 25.5.1 split into sitewise intertwiners.

Theorem 25.8.1 (Splitting criterion). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *In the setting of Theorem 25.5.1, suppose that the unital algebra generated by the matrices B^i is closed under conjugate transposition, that is, $(B^i)^\dagger$ lies in the unital algebra generated by the B^j for every i . Then the word-algebra module of B is semisimple, and B admits a block triangular form as in Theorem 25.5.1 whose remainder vanishes: the block triangular form is block diagonal.*

Proof. [✓ Proof](#) Since the algebra is closed under conjugate transposition, the orthogonal complement of an invariant subspace is invariant. Thus every invariant subspace has an invariant complement and the module is semisimple. Every step of the flag of Theorem 25.4.4 then has an invariant complement, the module is the direct sum of these complements, and a basis adapted to the direct sum makes every B^i block diagonal with the blocks prescribed by the flag. $\square$

Corollary 25.8.2 (Sitewise fusion tensors). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 25.8.1 there are maps V_s and W_s with $W_s V_t = \delta_{st} \text{id}$, $W_s B^w V_s = C_s^w$ for every word w , and the sitewise relations $B^i V_s = V_s C_s^i$ and $W_s B^i = C_s^i W_s$ for every physical index i and block occurrence s .*

Proof. [✓ Proof](#) With vanishing remainder, $B^i = \sum_s V_s C_s^i W_s$, and the sitewise relations follow from the biorthogonality of the pairs. $\square$

Theorem 25.8.3 (Faithful stationary splitting). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Invariant-projector closure makes the word-algebra module semisimple. In particular, suppose that the source tensor in Theorem 25.5.1 satisfies*

$$\sum_i (B^i)^\dagger B^i = \text{id}, \quad \sum_i B^i \rho (B^i)^\dagger = \rho \quad \text{for some } \rho > 0.$$

Then its compression may be chosen with zero remainder and with two-sided sitewise intertwiners as in Corollary 25.8.2. Normalization and faithful stationarity are sufficient hypotheses here; they are not inferred from periodic equality.

Proof. [✓ Proof](#) An invariant subspace has an orthogonal projection whose lower off-diagonal Kraus blocks vanish. Invariant-projector closure forces the upper blocks to vanish as well, so

its orthogonal complement is invariant. Every invariant subspace therefore has an invariant complement. Semisimplicity splits the composition flag, and the zero-remainder compression gives the asserted intertwiners. $\square$

Corollary 25.8.4 (Obstruction to closure under conjugate transposition). [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ *Stmt* If, in the setting of Theorem 25.5.1, some block occurrence s of positive bond dimension admits no nonzero sitewise right intertwiner X with $B^i X = X C_s^i$ for all i , then every compression of B in that setting has a nonzero remainder, and the unital algebra generated by the B^i is not closed under conjugate transposition. In particular, the stacked product of the anomalous CZX symmetry with itself, for which both sitewise intertwiner spaces vanish, generates a unital algebra that is not closed under conjugate transposition.

Proof. $\checkmark$ *Proof* A vanishing remainder would make the right map V_s a sitewise intertwiner, and $W_s V_s = \text{id}$ forces $V_s \neq 0$. By Theorem 25.8.1, closure under conjugate transposition would give a compression pair whose right map is a nonzero sitewise intertwiner. $\square$

Remark 25.8.5. The physical operator of the CZX symmetry, a product of controlled- Z gates and single-site spin flips, is unitary at every length. The failure of closure under conjugate transposition in Corollary 25.8.4 therefore concerns the virtual algebra of the stacked tensor, rather than the physical operator.

25.8.1 Uniqueness of split compressions

When the remainder vanishes, the compression maps are determined by the blocks up to a change of basis of each multiplicity space. Throughout, the blocks are indexed by labels c with multiplicities N_c , and a split compression of B consists of compressions $V_{c,\mu}$ and embeddings $W_{c,\mu}$, for $0 \leq \mu < N_c$, with

$$V_{c,\mu} W_{c',\nu} = \delta_{cc'} \delta_{\mu\nu} \text{id}, \quad B^i = \sum_c \sum_{\mu=0}^{N_c-1} W_{c,\mu} C_c^i V_{c,\mu} \quad (25.11)$$

for every physical index i . This is the zipper condition of [7]: the block triangular form has no nonzero blocks above the diagonal. The maps $W_{c,\mu}$ need not span the bond space of B , so zero diagonal blocks are allowed. In this subsection the compression is written on the left and denoted V , as for rectangular reductions, so the roles of the letters are exchanged relative to Corollary 25.8.2.

Definition 25.8.1.1 (Gauge-related tensors). [Lean](#) $\checkmark$ *Stmt* Tensors A and A' of bond dimensions D and D' are *gauge related* if there are matrices $X \in \mathbb{C}^{D \times D'}$ and $Y \in \mathbb{C}^{D' \times D}$ with $XY = \text{id}$, $YX = \text{id}$ and $A^i X = X A'^i$ for every physical index i , so that $A'^i = Y A^i X$. The two inverse relations force $D = D'$, and then A' is gauge equivalent to A in the sense of Definition 2.4.1, with gauge matrix $Y = X^{-1}$.

Lemma 25.8.1.2 (Intertwiners of normal tensors). [Lean](#) $\checkmark$ *Stmt* Let A and A' be normal tensors of bond dimensions D and D' , and let $X \in \mathbb{C}^{D \times D'}$ satisfy $A^i X = X A'^i$ for every physical index i . Then $X = 0$, or A and A' are gauge related in the sense of Definition 25.8.1.1.

Proof. $\checkmark$ *Proof* Induction on the length of a word w gives $A^w X = X A'^w$. If $Xv = 0$, then $X(A'^w v) = A^w(Xv) = 0$ for every word w . At a length where the words of A' span the full matrix algebra, this gives $X(Mv) = 0$ for every matrix M , and a nonzero v makes Mv an arbitrary vector, so $X = 0$. Symmetrically, if $X^T u = 0$, then $X^T(A^w)^T u = (X A'^w)^T u = (A'^w)^T X^T u = 0$

for every word w , and the words of A at a spanning length give $X = 0$ unless $u = 0$. If $X \neq 0$, both X and $X^\top$ are injective, so $D' \leq D$ and $D \leq D'$. Then X is invertible, and the pair (X, X^{-1}) witnesses that A and A' are gauge related. $\square$

Theorem 25.8.1.3 (Uniqueness of split compressions up to multiplicity gauge). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let the blocks C_c be normal of positive bond dimension and pairwise not gauge related in the sense of Definition 25.8.1.1, and let (V, W) and $(\widetilde{V}, \widetilde{W})$ be two split compressions of B as in (25.11) with the same blocks and multiplicities. Then $\widetilde{V}_{c,\mu} W_{e,\nu} = 0$ for $e \neq c$, and $\widetilde{V}_{c,\mu} W_{c,\nu} = (Y_c)_{\mu\nu} \text{id}$ for a matrix $Y_c \in \text{GL}_{N_c}(\mathbb{C})$ whose inverse is given by $V_{c,\mu} \widetilde{W}_{c,\nu} = (Y_c^{-1})_{\mu\nu} \text{id}$. The matrices Y_c are the unique ones with

$$\widetilde{V}_{c,\mu} B^i = \sum_\nu (Y_c)_{\mu\nu} V_{c,\nu} B^i, \quad B^i \widetilde{W}_{c,\mu} = \sum_\nu B^i W_{c,\nu} (Y_c^{-1})_{\nu\mu} \quad (25.12)$$

for every physical index i , so the two compressions differ by $\bigoplus_c \text{GL}_{N_c}(\mathbb{C})$ on the support of B , with no dressing by words. This is the uniqueness of fusion tensors under the zipper condition in [7].

Proof. [✓ Proof](#) Multiplying (25.11) on the left by $V_{c,\mu}$, or on the right by $W_{c,\mu}$, gives the sitewise relations $V_{c,\mu} B^i = C_c^i V_{c,\mu}$ and $B^i W_{c,\mu} = W_{c,\mu} C_c^i$, and likewise for the second compression. Hence $C_c^i \widetilde{V}_{c,\mu} W_{e,\nu} = \widetilde{V}_{c,\mu} B^i W_{e,\nu} = \widetilde{V}_{c,\mu} W_{e,\nu} C_e^i$, and Lemma 25.8.1.2 makes this intertwiner zero for $e \neq c$. For $e = c$ it commutes with every C_c^i , and since the words of C_c of a suitable length span the full matrix algebra, it is a scalar $(Y_c)_{\mu\nu}$. Compressing B^i with the second compression on both sides and expanding B^i by the first gives $\delta_{\mu\kappa} C_c^i = \sum_\nu (Y_c)_{\mu\nu} (Y'_c)_{\nu\kappa} C_c^i$, where $V_{c,\nu} \widetilde{W}_{c,\kappa} = (Y'_c)_{\nu\kappa} \text{id}$; a nonzero C_c^i gives $Y_c Y'_c = \text{id}$. Expanding $\widetilde{V}_{c,\mu} B^i$ by the first compression gives (25.12), and multiplying it on the right by $W_{c,\kappa}$ recovers $(Y_c)_{\mu\kappa}$, which gives uniqueness. $\square$

Corollary 25.8.1.4 (Uniqueness of zipper fusion tensors). [Lean](#) [✓ Stmt](#) Let T_a, T_b and T_c be matrix product operator tensors, with the T_c normal as tensors on the pair alphabet, of positive bond dimension, and pairwise not gauge related. Two families of fusion tensors that split the stacked product $T_a T_b$ over the blocks T_c with multiplicities N_{ab}^c , as in (25.11), differ by a unique $Y_{ab}^c \in \text{GL}_{N_{ab}^c}(\mathbb{C})$ for each c in the sense of (25.12).

Proof. [✓ Proof](#) Apply Theorem 25.8.1.3 to the stacked product on the pair alphabet. $\square$

Corollary 25.8.1.5 (Uniqueness of zipper action tensors). [Lean](#) [✓ Stmt](#) Let T_a be a matrix product operator tensor, A_x a tensor, and A_y normal tensors of positive bond dimension that are pairwise not gauge related. Two families of action tensors that split $T_a \cdot A_x$ over the blocks A_y with multiplicities $M_{a,x}^y$, as in (25.11), differ by a unique $X_{ax}^y \in \text{GL}_{M_{a,x}^y}(\mathbb{C})$ for each y in the sense of (25.12). This is the gauge freedom of action tensors in [164, Section 3.1].

Proof. [✓ Proof](#) Apply Theorem 25.8.1.3 to $T_a \cdot A_x$. $\square$

25.9 The single-block reduction of Molnár–Ge–Schuch–Cirac

Corollary 25.9.1 (Single-block case: the normal-target reduction). [Lean](#) [✓ Stmt](#) Let A be a normal tensor (NT) of bond dimension $D_A \geq 1$, let B be an arbitrary tensor of bond dimension D_B , and suppose that $|V^{(N)}(B)\rangle = |V^{(N)}(A)\rangle$ for every $N \geq 1$. Then there are maps $V : \mathbb{C}^{D_A} \rightarrow$

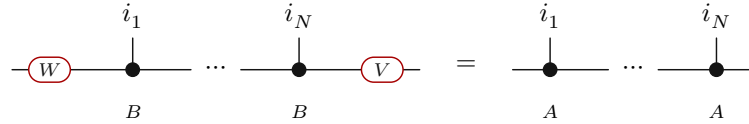
$\mathbb{C}^{D_B}$ and $W : \mathbb{C}^{D_B} \rightarrow \mathbb{C}^{D_A}$ with $WV = \text{id}$ such that, for every $N \geq 0$ and every word of length N ,

$$WB^{i_1} \dots B^{i_N} V = A^{i_1} \dots A^{i_N}, \quad (25.13)$$

and the remainder $R^i = B^i - VA^iW$ generates a nilpotent non-unital algebra. Explicitly, for every word of length $r = D_B - D_A + 1$,

$$R^{i_1} R^{i_2} \dots R^{i_r} = 0. \quad (25.14)$$

In particular $D_B \geq D_A$.



Proof. ✓ Proof Apply Theorem 25.5.1 to the single-block target $C = A$. Clauses (iv) and (v) give the compression pair and (25.13), clause (vii) gives $D_B = D_A + z \geq D_A$, and clause (vi) gives (25.14) with $r = |S| + z = 1 + (D_B - D_A)$. $\square$

The proportional case, in which $|V^{(N)}(B)\rangle = \lambda^N |V^{(N)}(A)\rangle$ for a scalar $\lambda \in \mathbb{C}^\times$, reduces to Corollary 25.9.1 by replacing B with $\lambda^{-1}B$, and then $WB^{i_1} \dots B^{i_N} V = \lambda^N A^{i_1} \dots A^{i_N}$ with remainder $R^i = B^i - \lambda VA^iW$; the compression pair of that case is recorded in Theorem 25.11.7.

Corollary 25.9.1 together with the proportional case corresponds to Propositions 20 and 21 of [187] (where V, W are swapped). The hypothesis $\lambda \neq 0$ is required because the zero tensor $B = 0$ admits no compression onto a nonzero normal target. The non-unital remainder algebra generated by the R^i satisfies the nilpotency length bound $r = D_B - D_A + 1$ from Theorem 25.5.1. The paper-gap note [112] discusses the scalar normalization, and [90] discusses the nilpotency length terminology.

25.10 Normalized split compressions

Definition 25.10.1 (Assembled compression matrices). Lean Lean ✓ Stmt For a multi-block compression with maps V_s and W_s , let $K = \bigoplus_{s \in S} \mathbb{C}^{D_s}$ and define $L : \mathbb{C}^{D_B} \rightarrow K$ and $R : K \rightarrow \mathbb{C}^{D_B}$ by $Lx = (W_s x)_s$ and $R(y_s)_s = \sum_s V_s y_s$.

Theorem 25.10.2 (Similarity from a normalized split compression). Lean Lean Lean Lean ✓ Stmt Suppose that a multi-block compression has zero remainder and that $\sum_i (B^i)^\dagger B^i = \text{id}$. Then $\sum_s V_s W_s = \text{id}$, the assembled matrices satisfy $LR = \text{id}$ and $RL = \text{id}$, and $LB^i R = \bigoplus_s C_s^i$ for every physical index i . In particular, $D_B = \sum_s D_s$ and there are no unmatched zero slots.

Proof. ✓ Proof Put $E = \sum_s V_s W_s$. Zero remainder and biorthogonality give $B^i E = B^i$, hence $B^i(\text{id} - E) = 0$. Normalization yields

$$(\text{id} - E)^\dagger (\text{id} - E) = \sum_i (B^i (\text{id} - E))^\dagger (B^i (\text{id} - E)) = 0. \quad (25.15)$$

Thus $E = \text{id}$. Biorthogonality gives $LR = \text{id}$, and $RL = E = \text{id}$. The sitewise intertwiners give $W_s B^i V_t = \delta_{st} C_s^i$, which is the stated simultaneous similarity. The inverse maps give dimension equality; comparison with $D_B = \sum_s D_s + z$ then gives $z = 0$. $\square$

Theorem 25.10.3 (Faithful stationary asymmetric similarity). [Lean](#) [✓ Stmt](#) Let $A = \bigoplus_{s \in S} C_s$ be a finite direct sum of nonzero normal tensors, and suppose that $V_N(B) = V_N(A)$ for every $N \geq 1$. If $\sum_i (B^i)^\dagger B^i = \text{id}$ and $\sum_i B^i \rho(B^i)^\dagger = \rho$ for some $\rho > 0$, then $D_B = \sum_s D_s$ and there are inverse maps L, R with $LB^i R = \bigoplus_s C_s^i$ for every physical index i .

Proof. [✓ Proof](#) Faithful stationary splitting supplies a compression with zero remainder. The normalized split-compression theorem then supplies the inverse maps, dimension equality, and simultaneous similarity. $\square$

25.11 Rectangular reductions and residual algebras

This section develops the single-block rectangular reductions and residual algebras of [187, Propositions 20 and 21], connecting positive-length periodic vector equality to compression pairs and bounding the nilpotency length of the residual algebra.

Definition 25.11.1 (Coefficient-dual inverse and reduction cross matrix). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\tilde{A} = (\tilde{A}^i)_i$ be injective, with bond dimension D_A , and choose the coefficient dual $C = (C^i)_i$ from a right inverse to the linear-combination map of $\tilde{A}$. Thus $C_{a'a}^i$ is the coefficient of $\tilde{A}^i$ in the chosen expansion of the matrix unit $|a\rangle\langle a'|$. For a tensor $\tilde{B}$ of bond dimension D_B , define the mixed-bond matrix on $\mathbb{C}^{D_B} \otimes \mathbb{C}^{D_A}$ by

$$K(\tilde{B}, C) = \sum_i \tilde{B}^i \otimes (C^i)^T.$$

The superscript T is the ordinary transpose, not the conjugate transpose. This is the matrix drawn in [187, proof of Proposition 20, lines 3817–3838].

$$\begin{array}{c} x \quad \tilde{B} \quad y \\ \bullet \\ \hline \bullet \\ a \quad C \quad a' \end{array}$$

Lemma 25.11.2 (Coefficient dual and reversed-word power formulas). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The coefficient dual obeys the exact matrix-unit identity

$$\sum_i \tilde{A}_{xy}^i C_{a'a}^i = \delta_{x,a} \delta_{y,a'}.$$

Consequently,

$$(K(\tilde{B}, C)^n)_{(b,a),(b',a')} = \sum_{i_1, \dots, i_n} (\tilde{B}^{i_1} \dots \tilde{B}^{i_n})_{bb'} (C^{i_n} \dots C^{i_1})_{a'a}.$$

In particular, pairing $\tilde{A}$ with its own dual gives the rank-one matrix whose entry is $\delta_{x,a} \delta_{y,a'}$. The reversed order $C^{i_n} \dots C^{i_1}$ is forced by the transpose and records the upper-leg orientation in [187, lines 3817–3865].

Proof. ✓ Proof The coefficient identity follows by applying the chosen right inverse to a matrix unit. Expanding the power of the finite sum defining K , multiplicativity of the Kronecker product collects the lower factors in site order. Transposing the upper product reverses its factors, which gives the displayed entry formula. Specializing the lower tensor to $\tilde{A}$ and applying the coefficient identity gives the stated rank-one matrix. $\square$

Theorem 25.11.3 (Cross trace powers and source open-boundary contraction). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Suppose $\tilde{A}$ and $\tilde{B}$ have equal positive-length MPV coefficients and $\tilde{A}$ is injective. This equality hypothesis is the $\lambda = 1$ specialization of the proportional premise printed in Proposition 20; the unrestricted unscaled conclusion is false [112]. Equality transports the trace of every nonempty word. For every $n > 0$,

$$\mathrm{tr} K(\tilde{B}, C)^n = D_A^n.$$

For arbitrary tensors B_0, C_0 , every $n \geq 0$, and every boundary matrix X , define

$$\mathcal{O}_n(B_0, C_0; X)_{a'a} = \sum_{i_1, \dots, i_n} \mathrm{tr}(B_0^{i_1} \dots B_0^{i_n} X) (C_0^{i_n} \dots C_0^{i_1})_{a'a}.$$

If B_0 has bond dimension D_B and C_0 has bond dimension D_A , its exact cross-sum expansion is

$$\mathcal{O}_n(B_0, C_0; X)_{a'a} = \sum_{x,y} K(B_0, C_0)_{(x,a),(y,a')}^n X_{yx}.$$

For $n > 0$, if $B_0 = \tilde{A}$, $C_0 = C$, and $X \in \mathbb{C}^{D_A \times D_A}$, then

$$\mathcal{O}_n(\tilde{A}, C; X)_{a'a} = D_A^{n-1} X_{a'a}.$$

The same-alphabet open-boundary contraction is the specialization $X = B_0^w$. Thus $X = \tilde{A}^w$ gives the target self-evaluation, while equality of the positive-length MPV coefficients gives

$$\mathcal{O}_n(\tilde{B}, C; \tilde{B}^w)_{a'a} = D_A^{n-1} (\tilde{A}^w)_{a'a}.$$

For the padded blocked contraction, suppose additionally that the original tensors A and B have equal positive-length MPV coefficients. If $\tilde{A}$ and $\tilde{B}$ are obtained by blocking $p > 0$ original sites, then for every unblocked lower tail word w ,

$$\mathcal{O}_n(\tilde{B}, C; B^w)_{a'a} = D_A^{n-1} (A^w)_{a'a}.$$

The trace-power identity is the closed contraction at [187, proof of Proposition 20, lines 3839–3865], and the open-boundary identity is the padded contraction at lines 3866–3887. The source next uses a rank-one rewrite and comparison to derive $VB^{i_1} \dots B^{i_m} W = A^{i_1} \dots A^{i_m}$ at lines 3888–3938.

Proof. ✓ Proof Expand K^n by Lemma 25.11.2 and use the trace formula for a Kronecker product. Positive-length MPV equality replaces the trace of each nonempty $\tilde{B}$ word by the corresponding $\tilde{A}$ trace. The resulting scalar sum is therefore the trace of the target cross matrix power. The coefficient-dual identity gives that target cross matrix its rank-one form, whose n th trace is D_A^n .

Expanding the lower trace into matrix entries gives the cross-sum formula for arbitrary X . For the target tensor, the rank-one form of $K(\tilde{A}, C)$ evaluates this sum as $D_A^{n-1} X_{a'a}$. Specializing X to a word evaluation gives the self-evaluation formula.

For blocked tensors, Lemma 2.7.2 identifies each blocked word with the evaluation of its flattened word. Since $n, p > 0$, this flattened word is nonempty. Append the unblocked tail w , apply positive-length MPV equality to the resulting original-site word, and use the target self-evaluation with $X = A^w$. $\square$

Theorem 25.11.4 (Rank-one factorizations). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $f: X \rightarrow Y$ be a linear map over a division ring K whose range has dimension one. Then there are a nonzero linear functional $\varphi: X \rightarrow K$, a nonzero vector $y \in Y$, and a vector $x \in X$ such that

$$\varphi(x) = 1, \quad f(x) = y, \quad f(z) = \varphi(z)y \quad \text{for every } z \in X.$$

Consequently, if I and J are finite index sets, then every rank-one matrix $M \in K^{I \times J}$ over a field is an outer product of two nonzero vectors: there are $w \in K^I$ and $v \in K^J$ such that

$$M_{ij} = w_i v_j.$$

In particular, if both indices of M are the product $D_B \times D_A$, then there are nonzero rectangular matrices $W \in K^{D_B \times D_A}$ and $V \in K^{D_A \times D_B}$ such that

$$M_{(b,a),(b',a')} = W_{ba} V_{a'b'}.$$

This is the rank-one factorization used in [187, proof of Proposition 20, lines 3866–3936].

Proof. [✓ Proof](#) Choose an isomorphism from the scalar division ring to the one-dimensional range of f . Composing its inverse with f gives φ , while the image of 1 gives y . Choose a preimage x of y ; the construction gives $\varphi(x) = 1$, $f(x) = y$, and the stated factorization of f .

In standard bases this factorization is an outer product of two nonzero coordinate vectors. For the product-index specialization, reshape the left vector as a $D_B \times D_A$ matrix and the right vector as a $D_A \times D_B$ matrix, preserving the displayed index order. $\square$

Theorem 25.11.5 (Equality-case existence of rectangular reductions). [Lean](#) [Lean](#) [✓ Stmt](#) Let A and B have the same physical alphabet and positive target bond dimension $D_A > 0$. Suppose that $|V^{(N)}(B)\rangle = |V^{(N)}(A)\rangle$ for every $N > 0$. If A is injective, then there exist $V \in \mathbb{C}^{D_A \times D_B}$ and $W \in \mathbb{C}^{D_B \times D_A}$ forming a reduction from B to A . The same conclusion holds when A is normal.

This is the $\lambda = 1$ equality case of [187, Proposition 20, lines 3815–3938]. The unrestricted proportional premise does not imply the unscaled conclusion [112].

Proof. [✓ Proof](#) First suppose that A is injective; equivalently, use blocking length one. More generally, choose any positive blocking length p for which the blocked target $\tilde{A} = A^{[p]}$ is injective, and put $\tilde{B} = B^{[p]}$. Positive blocking preserves equality of the MPV families. Let C be the coefficient dual of $\tilde{A}$ and set

$$K = \sum_i \tilde{B}^i \otimes (C^i)^\top.$$

Regard K as an endomorphism and take its Jordan–Chevalley decomposition $K = N + S$, where N is nilpotent, S is semisimple, and both belong to the algebra generated by K . Their membership in this commutative generated algebra implies $NS = SN$. The commuting nilpotent part does not change traces of powers. Since $\text{tr } K^n = D_A^n$ for every $n > 0$, the semisimple endomorphism $D_A^{-1}S$ has trace one at every power above one. The semisimple trace-power theorem gives $\text{rank } S = 1$; commutation and nilpotence then imply $NS = SN = 0$ and

$$K^J = S^J$$

for some $J > 0$.

Factor the matrix of S with the mixed-bond orientation

$$S_{(b,a),(b',a')} = W_{ba} V_{a'b'}.$$

Its first trace is D_A , so multiplication of outer products gives

$$S^J = D_A^{J-1} S.$$

Apply the padded open-boundary identity with an arbitrary original-scale word w . Rewriting its cross sum with the preceding factorization yields

$$D_A^{J-1} (VB^w W)_{a'a} = D_A^{J-1} (A^w)_{a'a}.$$

Since $D_A > 0$, cancel the nonzero scalar and conclude $VB^w W = A^w$ for every word w . The empty word gives $VW = 1_{D_A}$. For normal A , choose the positive injective blocking supplied by normality and repeat the same padded argument; the lower tail remains the unblocked word w , so the conclusion is already at the original physical scale. $\square$

Lemma 25.11.6 (Scalar covariance of rectangular reductions). [Lean](#) [Lean](#) [✓ Stmt](#) *If V, W form a reduction from B to A , then they also form a reduction from cB to cA for every $c \in \mathbb{C}$. If instead they form a reduction from B to cA , then*

$$VW = 1_{D_A}, \quad VB^w W = c^{|w|} A^w \quad \text{for every word } w. \quad (25.16)$$

Proof. [✓ Proof](#) Scaling both tensors multiplies each length- $|w|$ word by the same factor $c^{|w|}$, so the reduction identities are preserved. For a reduction to cA , expand $(cA)^w = c^{|w|} A^w$ in the defining word identity. The retraction $VW = 1_{D_A}$ is unchanged. $\square$

Theorem 25.11.7 (Nonzero proportional existence of rectangular reductions). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A and B have the same physical alphabet and positive target bond dimension $D_A > 0$. Let $c \in \mathbb{C}$ be nonzero, and suppose that*

$$V^{(N)}(B)_\sigma = c^N V^{(N)}(A)_\sigma \quad \text{for every } N > 0 \text{ and every configuration } \sigma. \quad (25.17)$$

If A is injective, then there exist $V \in \mathbb{C}^{D_A \times D_B}$ and $W \in \mathbb{C}^{D_B \times D_A}$ such that

$$VW = 1_{D_A}, \quad VB^w W = c^{|w|} A^w \quad \text{for every word } w. \quad (25.18)$$

The same conclusion holds when A is normal.

This is the corrected nonzero-scalar form of [187, Proposition 20]. The unscaled conclusion printed there is false for unrestricted $c \neq 1$, and the case $c = 0$ does not admit this rescaling argument [112].

Proof. [✓ Proof](#) Set $\tilde{B} = c^{-1}B$. Expand its word evaluations with Lemma 2.4.12, take the trace, and use (25.17). The positive-length MPV families of $\tilde{B}$ and A are equal. Apply Theorem 25.11.5 to obtain a reduction from $\tilde{B}$ to A . Scale both tensors by c to obtain a reduction from B to cA , and then apply Lemma 25.11.6 to get (25.18). $\square$

Lemma 25.11.8 (Nonempty products in a generated subsemigroup). [Lean](#) [✓ Stmt](#) *Let M be a monoid and let $S \subseteq M$. Every element $x \in \langle S \rangle_{\text{sg}}$ of the subsemigroup generated by S admits a factorization*

$$x = a_1 \cdots a_n, \quad n \geq 1, \quad a_j \in S \quad (1 \leq j \leq n).$$

Proof. ✓ Proof A generator has the required one-factor expression. The product of two such expressions is represented by their concatenation, which is again nonempty and has every factor in S . □

Definition 25.11.9 (Residual algebra of a selected reduction). Lean Lean Lean ✓ Stmt Let (V, W) be a reduction from B to A . For each physical letter i , define

$$N^i = B^i - WA^iV. \quad (25.19)$$

The *residual algebra* $\mathcal{R}(B, A; V, W)$ is the nonunital subalgebra of $M_{D_B}(\mathbb{C})$ generated by the matrices N^i . This is the residual algebra of Proposition 21 in [187].

Theorem 25.11.10 (Residual letters belong to the residual algebra). Lean ✓ Stmt For every physical letter i , one has $N^i \in \mathcal{R}(B, A; V, W)$.

Proof. ✓ Proof Each residual letter is one of the generators of the residual algebra. □

Definition 25.11.11 (Residual nilpotency length). Lean Lean ✓ Stmt A natural number L is a *residual-word nilpotency bound* if every word $N^{i_1} \dots N^{i_L}$ of exactly L residual letters is zero. The residual nilpotency length $\ell(B, A; V, W)$ is the infimum of the set of exact-length bounds. Definition 8 of [187] instead asks that every residual word of every length $n \geq L$ vanish; the passage from an exact-length bound to this consequence is stated separately in Theorem 25.11.21.

Theorem 25.11.12 (Exterior-word strengthening of the residual sandwich). Lean ✓ Stmt Let (V, W) be a reduction from B to A . For arbitrary exterior words $\mathbf{p}, \mathbf{q}$ and every nonempty word $\mathbf{u}$,

$$VB^{\mathbf{p}}N^{\mathbf{u}}B^{\mathbf{q}}W = 0. \quad (25.20)$$

This strengthens the sandwiched residual identity below by allowing arbitrary exterior B -words.

Proof. ✓ Proof Apply $VB^{\mathbf{w}}W = A^{\mathbf{w}}$ to $\mathbf{p}, \mathbf{q}$, and $\mathbf{p}i\mathbf{q}$. The two expressions for $A^{\mathbf{p}}A^iA^{\mathbf{q}}$ give

$$VB^{\mathbf{p}}N^iB^{\mathbf{q}}W = 0.$$

Induction on the nonempty residual word $\mathbf{u}$ proves (25.20). □

Theorem 25.11.13 (Positive residual words vanish under the reduction). Lean ✓ Stmt Let (V, W) be a reduction from B to A . For every nonempty word $\mathbf{u}$,

$$VN^{\mathbf{u}}W = 0 \quad (|\mathbf{u}| > 0). \quad (25.21)$$

No equality of closed chains is required. This is the sandwiched residual identity of [187].

Proof. ✓ Proof Take both exterior words empty in (25.20). □

Proof. ✓ Proof Induct on the length of the word. The summands with $r = 0$ constitute $\hat{\mathcal{J}}$. By the left all-cut identity,

$$\hat{\mathcal{J}}(i_1, \dots, i_n) = WA^{i_1}VB^{i_2} \dots B^{i_n} = \mathcal{J}(i_1, \dots, i_n), \quad (25.30)$$

where the last equality is the initial-window case of the recursive expansion. Deleting the first letter identifies the summands with $r > 0$ with $N^{i_1}\widehat{\mathcal{W}}(i_2, \dots, i_n)$. $\square$

Theorem 25.11.18 (Corrected strict-window residual expansion). Lean ✓ Stmt Let (V, W) be a reduction from B to A . For every word $\mathbf{i} = (i_1, \dots, i_n)$,

$$B^{i_1} \dots B^{i_n} = N^{i_1} \dots N^{i_n} + \sum_{0 \leq r < s \leq n} N^{i_1} \dots N^{i_r} WA^{i_{r+1}} \dots A^{i_s} VN^{i_{s+1}} \dots N^{i_n}. \quad (25.31)$$

This statement requires only the reduction. The range $0 \leq r < s \leq n$ ensures nonempty central words; see [54].

Proof. ✓ Proof Substitute $\mathcal{W}(\mathbf{i}) = \widehat{\mathcal{W}}(\mathbf{i})$ into (25.26), and then reindex the positive central length by $s = r + m$. $\square$

Theorem 25.11.19 (Residual trace vanishing and elementwise nilpotence). Lean Lean Lean Lean ✓ Stmt Let (V, W) be a reduction from B to A , and assume $\mathcal{V}^+(B) = \mathcal{V}^+(A)$. Then every nonempty residual word has trace zero, every element of $\mathcal{R}(B, A; V, W)$ has trace zero, and every element of this algebra is nilpotent.

This establishes the nil-algebra step of [187, Proposition 21] in the equality case $\lambda = 1$; see [112].

Proof. ✓ Proof Take the trace of the corrected residual expansion. For a mixed term in the explicit strict-window sum, cyclicity gives

$$\mathrm{tr}(N^{\mathbf{p}}WA^{\mathbf{a}}VN^{\mathbf{q}}) = \mathrm{tr}((VN^{\mathbf{q}}N^{\mathbf{p}}W)A^{\mathbf{a}}) = 0,$$

by (25.21). In the recursive sum, the corresponding step is

$$\mathrm{tr}(N^{\mathbf{u}}WA^iVB^{\mathbf{w}}) = \mathrm{tr}((VB^{\mathbf{w}}N^{\mathbf{u}}W)A^i) = 0,$$

by (25.20), for every nonempty residual word $\mathbf{u}$. The unique term with no residual letters is $WA^{\mathbf{i}}V$, whose trace is $\mathrm{tr}(A^{\mathbf{i}})$ because $VW = 1$. Thus

$$\mathrm{tr}(B^{\mathbf{i}}) = \mathrm{tr}(A^{\mathbf{i}}) + \mathrm{tr}(N^{\mathbf{i}}). \quad (25.32)$$

Equality of the two closed chains now gives $\mathrm{tr}(N^{\mathbf{i}}) = 0$. Linearity extends this identity to the generated nonunital algebra. Vanishing traces of positive powers make each algebra element nilpotent. $\square$

Theorem 25.11.20 (Dimension bound for the residual nilpotency length). Lean Lean Lean Lean Lean ✓ Stmt Let (V, W) be a reduction from B to A , and assume $\mathcal{V}^+(B) = \mathcal{V}^+(A)$. Then every mixed product of D_B elements of $\mathcal{R}(B, A; V, W)$ vanishes. Consequently D_B is a residual-word nilpotency bound. The least bound exists, is itself a bound, and satisfies

$$\ell(B, A; V, W) \leq D_B. \quad (25.33)$$

In particular, $N^{\mathbf{i}} = 0$ whenever $|\mathbf{i}| \geq D_B$. The resulting vanishing of all sufficiently long mixed products is the nilpotency conclusion of Proposition 21 in [187]. The specific numerical estimate by D_B follows from the finite-dimensional nil-matrix theorem.

Proof. [✓ Proof](#) Apply the finite-dimensional nil-matrix theorem to the residual algebra. Its dimension-sharp matrix form makes every product of D_B algebra elements zero. Residual letters lie in this algebra, so D_B is a bound. Well-ordering gives the least bound, and prefix factorization gives vanishing for all longer residual words. $\square$

Theorem 25.11.21 (Consequences of an arbitrary residual bound). [Lean](#) [✓ Stmt](#) If L is a residual-word nilpotency bound and $|\mathbf{i}| \geq L$, then

$$N^{\mathbf{i}} = 0. \quad (25.34)$$

This conclusion uses only the assumed bound; it does not require equality of closed chains.

Proof. [✓ Proof](#) Split the residual word after its first L letters. The prefix is zero by the bound, so the full word is zero. $\square$

Theorem 25.11.22 (Bound-truncated contracted residual expansions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let (V, W) be a reduction from B to A , let L be any residual-word nilpotency bound, and write $\mathbf{i} = (i_1, \dots, i_n)$. Then

$$VB^{i_1} \dots B^{i_n} = \sum_{n-L \leq s \leq n} A^{i_1} \dots A^{i_s} V N^{i_{s+1}} \dots N^{i_n}, \quad (25.35)$$

$$B^{i_1} \dots B^{i_n} W = \sum_{0 \leq r \leq \min(L, n)} N^{i_1} \dots N^{i_r} W A^{i_{r+1}} \dots A^{i_n}. \quad (25.36)$$

Here $n - L$ denotes truncated subtraction, equivalently $\max(0, n - L)$ in integer notation. The endpoints with a residual word of length exactly L are retained as zero terms. These are the two contracted formulas of [187].

Proof. [✓ Proof](#) Start from (25.27) and (25.28). In the left sum, every omitted term has residual suffix length at least L . In the right sum, every omitted term has residual prefix length at least L , so (25.34) makes all these terms zero. $\square$

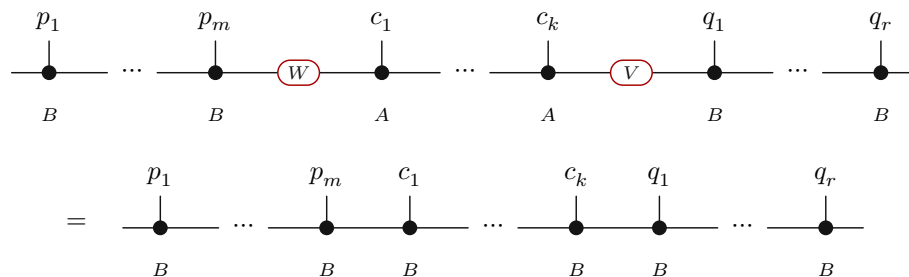
Theorem 25.11.23 (Exterior-buffer identity). [Lean](#) [✓ Stmt](#) Let (V, W) be a reduction from B to A , and let L be any residual-word nilpotency bound. If the central word $\mathbf{c}$ is nonempty and the exterior words $\mathbf{p}, \mathbf{q}$ satisfy

$$L \leq |\mathbf{p}| + 1, \quad L \leq |\mathbf{q}| + 1, \quad (25.37)$$

then

$$B^{\mathbf{p}} W A^{\mathbf{c}} V B^{\mathbf{q}} = B^{\mathbf{p} \mathbf{c} \mathbf{q}}. \quad (25.38)$$

No closed-chain equality is assumed: only the reduction and the supplied residual bound enter this identity. It follows from the corrected residual expansion by eliminating every term with a sufficiently long residual buffer.



Proof. ✓ Proof First use induction along the right exterior word to show that a positive residual block preceded by V vanishes once its length together with the right buffer reaches L . Expand only the left exterior word $B^{\mathbf{p}}$ by Theorem 25.11.15. The left buffer bound removes the pure residual term, while the preceding induction and (25.20) remove the strict-window terms. Hence $B^{\mathbf{p}}N^iB^{\mathbf{q}} = 0$ under the two buffer bounds.

Separately, induction on a word $\mathbf{a}$ gives $VB^{\mathbf{a}\mathbf{q}} = A^{\mathbf{a}}VB^{\mathbf{q}}$ from the right buffer bound. Write the nonempty central word as $\mathbf{c} = i\mathbf{a}$ and decompose the corresponding letter as $B^i = N^i + WA^iV$. The residual term is zero by the first step, and the second induction identifies the remaining term with (25.38). $\square$

Definition 25.11.24 (Exterior buffer length of a reduction). Lean Lean ✓ Stmt Let (V, W) be a reduction from B to A . A natural number m is an *exterior buffer length* for (V, W) if

$$B^{\mathbf{p}}WA^{\mathbf{c}}VB^{\mathbf{q}} = B^{\mathbf{p}\mathbf{c}\mathbf{q}} \quad (25.39)$$

for every nonempty word $\mathbf{c}$ and all words $\mathbf{p}, \mathbf{q}$ with $|\mathbf{p}| \geq m$ and $|\mathbf{q}| \geq m$. Every $m' \geq m$ is then also an exterior buffer length. This is the exterior fusion equation of [163, lines 1409–1498], with exterior length $m \geq \ell$ and central length $|\mathbf{c}| = n + 1$, $n \geq 0$.

Theorem 25.11.25 (Reductions and exterior buffers under blocking). Lean Lean Lean Lean Lean ✓ Stmt Let (V, W) be a reduction from B to A . If N is a residual-word nilpotency bound, then $N - 1$ is an exterior buffer length; in particular, if $\mathcal{V}^+(B) = \mathcal{V}^+(A)$, then $\ell(B, A; V, W) - 1$ is an exterior buffer length.

For every L , the same pair (V, W) is a reduction from $B^{[L]}$ to $A^{[L]}$, and for every map f between physical alphabets it is a reduction from $B \circ f$ to $A \circ f$. If m is an exterior buffer length for (V, W) , $L > 0$, and $m \leq kL$, then k is an exterior buffer length for (V, W) as a reduction from $B^{[L]}$ to $A^{[L]}$; reindexing both tensors by the same map f also preserves exterior buffer lengths.

Proof. ✓ Proof The bounds $N \leq |\mathbf{p}| + 1$ and $N \leq |\mathbf{q}| + 1$ in Theorem 25.11.23 read $N - 1 \leq |\mathbf{p}|$ and $N - 1 \leq |\mathbf{q}|$, and Theorem 25.11.20 makes the least length a bound under equality of the positive-length closed chains. Word evaluation of $B^{[L]}$ on a blocked word is evaluation of B on the flattened word, and evaluation of $B \circ f$ on a word is evaluation of B on the mapped word, so the reduction identities transfer. For the exterior identity, the flattened exterior words have length at least $kL \geq m$ and the flattened central word has positive length $|\mathbf{c}|L$. $\square$

Definition 25.11.26 (Common buffer length of a finite family). Lean Lean Lean Lean ✓ Stmt For a finite family of natural numbers $(N_i)_{i \in I}$, the *common buffer length* is

$$L = \max\left(1, \max_{i \in I}(N_i - 1)\right), \quad (25.40)$$

with $N_i - 1$ read as 0 when $N_i = 0$. It is positive, satisfies $N_i \leq L + 1$ for every $i \in I$, and is the least positive integer with this property.

Theorem 25.11.27 (Common blocking of a finite family of reductions). Lean ✓ Stmt Let $(N_i)_{i \in I}$ be a finite family of natural numbers with common buffer length L , and let (V, W) be a reduction from B to A for which N_i is a residual-word nilpotency bound for some $i \in I$. Then 1 is an exterior buffer length for (V, W) as a reduction from $B^{[L]}$ to $A^{[L]}$: (25.39) holds for the blocked tensors whenever the blocked words $\mathbf{p}, \mathbf{c}, \mathbf{q}$ are nonempty.

This formalizes the reduction of the buffer length to 1 by blocking in [163] and [166, Theorem 1 footnote]; see [90].

Proof. ✓ Proof By Theorem 25.11.25, $N_i - 1$ is an exterior buffer length, and $N_i - 1 \leq L = 1 \cdot L$ with $L > 0$ by (25.40). The blocking statement of the same theorem with $k = 1$ gives the conclusion. $\square$

Theorem 25.11.28 (Boundary-dressed uniqueness of rectangular reductions). Lean Lean
✓ Stmt Discussion: #7395 Let A be normal, let B have the same physical alphabet, and let (V, W) and $(\tilde{V}, \tilde{W})$ be two reductions from B to A . Suppose that $N_0 \in \mathbb{N}$ is a residual-word nilpotency bound for both reductions: every word of length N_0 in either residual tensor vanishes. Then there exists a single scalar $\lambda \in \mathbb{C}$ with $\lambda \neq 0$ such that, for every word $\mathbf{a}$ satisfying

$$|\mathbf{a}| > 2N_0, \quad (25.41)$$

one has

$$VB^{\mathbf{a}} = \lambda \tilde{V}B^{\mathbf{a}}, \quad (25.42)$$

$$B^{\mathbf{a}}W = \lambda^{-1} B^{\mathbf{a}}\tilde{W}. \quad (25.43)$$

If A and B have the same positive-length closed-chain coefficients and the two least residual nilpotency lengths are at most N_0 , then N_0 is such a common bound and the same conclusion follows. The direct-bound statement itself requires no equality of closed-chain coefficients.

Following [187, Theorem 22], these boundary-dressed identities hold for $|\mathbf{a}| > 2N_0$.

The diagram shows two equations. The first equation represents (25.42): a horizontal line with a red oval labeled V on the left, followed by a series of vertical lines labeled $a_1, \dots, a_n$ each connected to a point labeled B . This is equal to λ times a similar diagram where the red oval is labeled $\tilde{V}$. The second equation represents (25.43): a horizontal line with a series of vertical lines labeled $a_1, \dots, a_n$ each connected to a point labeled B , followed by a red oval labeled W . This is equal to λ^{-1} times a similar diagram where the red oval is labeled $\tilde{W}$.

Proof. ✓ Proof If the target bond dimension is zero, both dressed identities hold entrywise for $\lambda = 1$ because the relevant target row or column index is empty. Hence assume that the target bond dimension is positive.

Choose $L > 0$ such that A is L -block injective. For exterior words $\mathbf{p}, \mathbf{q}$ satisfying the N_0 buffer inequalities, compare the two exterior-buffer expansions of the mixed contraction $VB^{\mathbf{p}}\tilde{W}$. Inserting a nonempty central A -word gives

$$(VB^{\mathbf{p}}\tilde{W})A^{\mathbf{c}}A^{\mathbf{q}} = A^{\mathbf{p}}A^{\mathbf{c}}(VB^{\mathbf{q}}\tilde{W}).$$

Since the length- L words span the full target matrix algebra, linearity replaces $A^{\mathbf{c}}$ by an arbitrary matrix. Insert matrix units and choose a nonzero target word at a positive multiple of L . Entrywise cancellation then yields one scalar λ with $VB^{\mathbf{p}}\tilde{W} = \lambda A^{\mathbf{p}}$ for every sufficiently buffered $\mathbf{p}$. Reversing the two reductions gives a scalar μ for the opposite mixed boundary.

Choose a longer nonzero target word, split it into two buffered exterior pieces and a nonempty injective central block, and apply the exterior-buffer identity once more. The two mixed-boundary formulas give $(\lambda\mu)A^{\mathbf{a}} = A^{\mathbf{a}}$; nonvanishing of the chosen word implies $\lambda\mu = 1$, so in particular $\lambda \neq 0$.

Finally, if $|\mathbf{a}| > 2N_0$, split $\mathbf{a}$ into exterior words of length N_0 and a nonempty center. Substitution of the two mixed-boundary formulas into the corresponding exterior-buffer identities gives (25.42) and first gives $B^{\mathbf{a}}\tilde{W} = \lambda B^{\mathbf{a}}W$ on the right. Multiplication by λ^{-1} yields (25.43).

For the least-length corollary, positive-length closed-chain equality makes each least residual nilpotency length an actual bound. Prefix factorization propagates vanishing from that least length to N_0 , so the direct-bound theorem applies. $\square$

Chapter 26

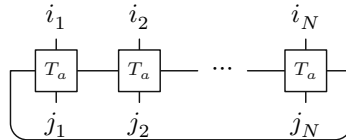
Matrix Product Operator Symmetries and Non-Invertible Symmetries

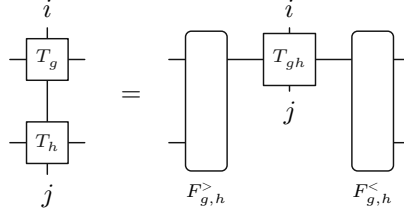
A symmetry of a spin chain is usually a group acting site by site. Two generalizations appear in one-dimensional lattice models. The symmetry operators may be matrix product operators rather than tensor products of one-site unitaries, and they may multiply according to fusion rules $O_a O_b = \sum_c N_{ab}^c O_c$ rather than a group law, so that some of them have no inverse [7, 164, 16]. This chapter introduces both generalizations for a reader who knows matrix product states. Section 26.1 treats invertible symmetries: on-site actions, groups represented by matrix product unitaries, and group operators built from a three-cocycle, whose cohomology class obstructs an on-site form. Section 26.2 defines fusion algebras of matrix product operators and their Perron–Frobenius dimensions, and Section 26.3 presents the non-invertible examples: Fibonacci, Ising, and Kramers–Wannier duality. Section 26.4 derives the integrality consequences for symmetric matrix product states, and Section 26.5 states the local zipper condition. The virtual-level theory, with fusion and action tensors, their gauge freedom, the associator, and the cohomological obstruction, is Chapter 28.

Throughout, T_a denotes a matrix product operator tensor (Definition 16.1.1) of bond dimension χ_a , and $O_a^{(N)}$ its periodic operator on N sites (Definition 16.1.6),

$$O_a^{(N)} = \sum_{i_1, \dots, i_N} \sum_{j_1, \dots, j_N} \text{tr}(T_a^{i_1 j_1} \dots T_a^{i_N j_N}) |i_1, \dots, i_N\rangle \langle j_1, \dots, j_N|, \quad (26.1)$$

where the first physical index is the output. We write $|V^{(N)}(A)\rangle$ for the periodic vector of a matrix product state tensor A . For two operator tensors, $T_a T_b$ denotes the stacked tensor $(T_a T_b)^{ij} = \sum_m T_a^{im} \otimes T_b^{mj}$ of bond dimension $\chi_a \chi_b$, whose periodic operator is $O_a^{(N)} O_b^{(N)}$, and $T_a \cdot A$ denotes the matrix product state tensor $(T_a \cdot A)^i = \sum_j T_a^{ij} \otimes A^j$, whose periodic vector is $O_a^{(N)} |V^{(N)}(A)\rangle$. Chain lengths are positive, $N \geq 1$.





Proof. ✓ Proof The tensor T_e is injective with $\text{tr} T_e^{ij} = \delta_{ij}$ and $\text{tr}(T_e^{ij} T_e^{kl}) = \delta_{ij} \delta_{kl}$, so $\text{tr}(XY) = \text{tr} X \text{tr} Y$ on the whole matrix algebra; testing on two off-diagonal matrix units shows that T_e has bond dimension one and $T_e^{ij} = \delta_{ij}$. Closed fusion then gives $(T_g T_{g^{-1}})^{ij} = \delta_{ij} P$ and $(T_{g^{-1}} T_g)^{ij} = \delta_{ij} Q$ with $\text{tr} P = \text{tr}(F_{g,g^{-1}}^< F_{g,g^{-1}}^>) \neq 0$. Decomposing the triple product $T_g T_{g^{-1}} T_g$ in the two ways gives $P \otimes T_g^{ij} = T_g^{ij} \otimes Q$, hence $P \otimes X = X \otimes Q$ for every matrix X by injectivity. If T_g had bond dimension at least two, testing on $X = E_{cc}$ with $c \neq a$ would kill every diagonal entry of P in the block a , and $\text{tr} P$ would vanish. With every bond dimension one, the periodic operator is the product of the one-site entries, and the fusion tensors are nonzero scalars $v(g, h)$. The two fusion trees are then proportional with ratio $v(gh, k)v(g, h)/(v(g, hk)v(h, k))$, which is $\omega(g, h, k)$ by Theorem 19.5.1.5; this is the coboundary of v^{-1} . A coboundary has trivial class, so the anomaly class vanishes by Corollary 19.5.1.9. $\square$

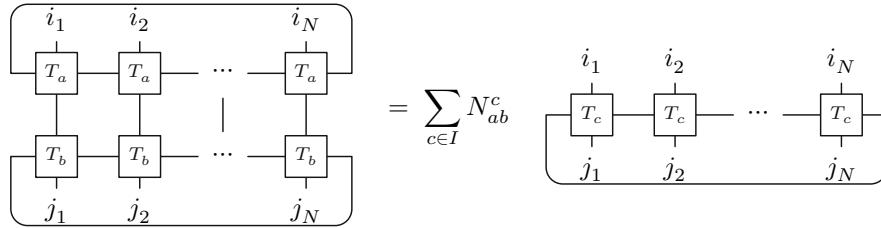
26.2 Fusion algebras of matrix product operators

A matrix product operator symmetry is given by a finite family of operators satisfying the fusion rule (26.2) [164]. The source defines the symmetry by invariance of the states with arbitrary boundary conditions; the statements below use only its consequence for periodic boundary conditions [164, Section 3]; see [86].

Definition 26.2.1 (Matrix product operator fusion algebra). Lean ✓ Stmt A finite family $(T_a)_{a \in I}$ of matrix product operator tensors with structure constants $N_{ab}^c \in \mathbb{N}$ is a *fusion algebra* if, for all $a, b \in I$ and $N \geq 1$,

$$O_a^{(N)} O_b^{(N)} = \sum_{c \in I} N_{ab}^c O_c^{(N)}. \quad (26.2)$$

This is the fusion rule of [164, Section 2] and [7].



Definition 26.2.2 (Unit and invertible labels). Lean Lean ✓ Stmt A label e is a *unit* of the structure constants if $N_{eb}^c = N_{be}^c = \delta_{bc}$ for all b, c . A label a is *invertible* with respect to e if there is a label b with $N_{ab}^c = N_{ba}^c = \delta_{ce}$ for all c . Outside the on-site case where $O_e^{(N)} = \mathbb{1}$, the unit operator is in general a projector $(O_e^{(N)})^2 = O_e^{(N)}$ [164, Section 4].

Definition 26.2.3 (Fusion matrices and Perron–Frobenius dimension). [Lean](#) [Lean](#) [✓ Stmt](#) Let $N_{ab}^c \in \mathbb{N}$ be structure constants on a finite label set I . The *fusion matrix* of a label a is the nonnegative integer matrix N_a with entries $(N_a)_{bc} = N_{ab}^c$ for $b, c \in I$. The *Perron–Frobenius dimension* of a is the spectral radius $d_a = \rho(N_a)$.

Theorem 26.2.4 (A nonnegative eigenvector for the Perron–Frobenius dimension). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every label a , the Perron–Frobenius dimension d_a is an eigenvalue of N_a , and there is a column vector $v \neq 0$ with nonnegative entries such that $N_a v = d_a v$.

Proof. [✓ Proof](#) This is the Perron–Frobenius theorem for nonnegative matrices; we prove it for any square matrix M with nonnegative real entries, and write $\rho = \rho(M)$. Choose a complex eigenvalue μ with $|\mu| = \rho$ and an eigenvector $u \neq 0$, and let $|u|$ be the vector of moduli $|u_i|$. The triangle inequality gives $M|u| \geq \rho|u|$ entrywise. Hence the set K of vectors x with nonnegative entries, $\sum_i x_i = 1$ and $Mx \geq \rho x$ entrywise is nonempty; it is compact and convex. For $x \in K$ the vector $(M+1)x$ is nonnegative with positive entry sum, and the map $x \mapsto (M+1)x / \sum_i ((M+1)x)_i$ sends K into itself, because $M(M+1)x - \rho(M+1)x = (M+1)(Mx - \rho x)$ and $M+1$ has nonnegative entries. The map is continuous on K . By Brouwer’s fixed-point theorem it has a fixed point x , so $Mx = sx$ for some real s , and $x \neq 0$ is nonnegative. From $x \in K$ we get $s \geq \rho$, and s is an eigenvalue of M , so $s \leq \rho$. Thus $Mx = \rho x$ and ρ is an eigenvalue. Applied to $M = N_a$ this gives the statement. $\square$

Lemma 26.2.5 (A positive left eigenvector gives the Perron–Frobenius dimension). [Lean](#) [Lean](#) [✓ Stmt](#) Let a be a label, $\lambda \in \mathbb{R}$, and let $(\delta_b)_{b \in I}$ be real numbers with $\delta_b > 0$ for all b and $\sum_b \delta_b N_{ab}^c = \lambda \delta_c$ for all c . Then λ is the Perron–Frobenius dimension d_a . In particular, if positive numbers δ_b satisfy

$$\sum_b N_{ab}^c \delta_b = \delta_a \delta_c \quad (26.3)$$

for all a and c , then $\delta_a = d_a$ for every label a .

Proof. [✓ Proof](#) Write δ for the row vector with entries δ_b , so that $\delta N_a = \lambda \delta$. Since $\delta \neq 0$, the matrix $\lambda - N_a$ has a nonzero left null vector, so its determinant vanishes and λ is an eigenvalue of N_a . Let μ be any complex eigenvalue of N_a with eigenvector $v \neq 0$, and let $|v|$ be the vector of moduli $|v_b|$. The triangle inequality gives $|\mu| |v| \leq N_a |v|$ entrywise, and pairing with δ gives $|\mu| \delta |v| \leq \delta N_a |v| = \lambda \delta |v|$. Here $\delta |v| > 0$ because every δ_b is positive and $v \neq 0$, so $|\mu| \leq \lambda$. Thus λ is an eigenvalue of largest modulus, and it is nonnegative, so $\lambda = \rho(N_a) = d_a$. The second statement is the first with $\lambda = \delta_a$. $\square$

For matrix product operator representations of a weak Hopf algebra whose unit corresponds to a single irreducible representation, the positive dimensions attached to the labels satisfy (26.3) [188, Section 5], so by Lemma 26.2.5 they are the Perron–Frobenius dimensions. For the Fibonacci and Ising rules below they are the quantum dimensions of the corresponding fusion categories.

26.3 Non-invertible symmetries

A label without an inverse in the sense of Definition 26.2.2 gives an operator that has no inverse within the fusion algebra. In the examples below such operators have a nonzero kernel, so they are not invertible on the chain at all; they are the lattice form of the non-invertible symmetries of Ising and anyon chains [1, 7, 16].

Remark 26.3.1 (The unit is a projector). If e is a unit of a fusion algebra, then $N_{ee}^c = \delta_{ec}$ gives $(O_e^{(N)})^2 = O_e^{(N)}$ and $O_e^{(N)} O_a^{(N)} = O_a^{(N)} O_e^{(N)} = O_a^{(N)}$ for every a . Thus $O_e^{(N)}$ is an idempotent and the algebra acts on its range; it is the identity only in special cases such as on-site symmetries [164, Section 4]. For the Fibonacci operators the unit is the orthogonal projection onto the admissible configurations (Lemma 27.6.1.1), and for the Ising operators it is the orthogonal projection onto the closed fusion paths (Theorem 27.7.1.2).

Example 26.3.2 (Fibonacci fusion rule and the golden ratio). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) The Fibonacci labels are 1 and τ , with unit 1 and $\tau \times \tau = 1 + \tau$. Its fusion matrix N_τ in the order $(1, \tau)$ is (27.134), with characteristic polynomial $\lambda^2 - \lambda - 1$ and eigenvalues $\varphi = (1 + \sqrt{5})/2$ and $-\varphi^{-1}$. Hence $d_\tau = \varphi$, the golden ratio, and $d_1 = 1$; the row vector $(1, \varphi)$ satisfies (26.3). The label τ is not invertible: $\tau \times b$ contains 1 only for $b = \tau$, and $\tau \times \tau$ also contains τ . The operators of Section 27.6 realize this rule at every length (Theorem 27.6.5.3), and the quantum dimensions entering their F-symbols are those of Definition 27.6.6 [7, Appendix D.1].

Example 26.3.3 (Ising fusion rule). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) The Ising labels are 1, ψ and σ , with unit 1 and the fusion rules of Section 27.7, of which the only non-invertible one is $\sigma \times \sigma = 1 + \psi$. In the order $(1, \psi, \sigma)$ the fusion matrices are

$$N_\psi = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad N_\sigma = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}. \quad (26.4)$$

The label ψ is invertible and N_ψ is a permutation matrix, so $d_\psi = 1$. The eigenvalues of N_σ are 0 and $\pm\sqrt{2}$, so $d_\sigma = \sqrt{2}$, and $(1, 1, \sqrt{2})$ satisfies (26.3). The operators of Section 27.7 realize these rules at every length (Theorems 27.7.1.6 and 27.7.1.8). The relation $O_\sigma O_\psi = O_\sigma$ gives $O_\sigma(O_1 - O_\psi) = 0$, so O_σ annihilates the range of $O_1 - O_\psi$.

Remark 26.3.4 (Kramers–Wannier duality). The Kramers–Wannier duality of the transverse-field Ising chain is a matrix product operator of bond dimension two (Definition 27.5.1). With the normalization $\mathcal{D}_N = 2^{-N/2} K_N$ of Section 27.5, its square is $\mathcal{D}_N^2 = (\mathbb{1} + \eta)T$, where η is the global spin flip and T the one-site translation (Corollary 27.5.1.7 and (27.64)), and its kernel is the odd sector of η (Lemma 27.5.1.6). Aasen, Mong and Fendley keep the dual lattice distinct from the original one and obtain the Ising fusion rule $\mathcal{D}_\sigma^2 = 1 + \mathcal{D}_\psi$, with the spin flip as $\mathcal{D}_\psi$ [1]. On one periodic chain, identifying the two lattices produces the translation T , so the duality operator is not a member of a fusion algebra of operators on one chain. It is an intertwiner between the Ising chain and its dual [180], as discussed in Remark 28.5.4.

26.4 Symmetric matrix product states

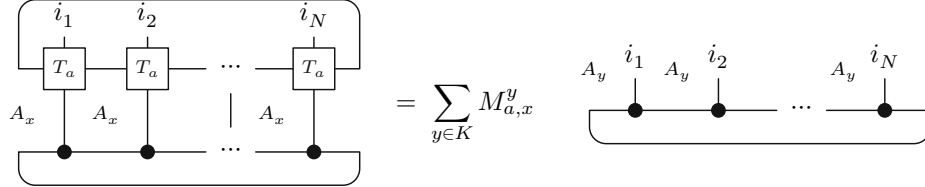
A matrix product state is symmetric when each operator of the algebra maps its periodic vectors into their span with length-independent coefficients. For a single normal tensor the coefficients are eigenvalues; for several normal tensors they form matrices. The results of this section show that these coefficients are nonnegative integers compatible with the fusion rules, which already excludes symmetric normal states for the Fibonacci rule. The integrality rests on a property of normal tensors: a length-independent relation between word traces has nonnegative integer coefficients.

Definition 26.4.1 (Symmetric family of matrix product states). [Lean](#) [Lean](#) [Stmt](#) A finite family $(A_x)_{x \in K}$ of matrix product state tensors is *symmetric* under $(T_a)_{a \in I}$ with coefficients

$M_{a,x}^y \in \mathbb{C}$ if, for all a, x and $N \geq 1$,

$$O_a^{(N)} |V^{(N)}(A_x)\rangle = \sum_{y \in K} M_{a,x}^y |V^{(N)}(A_y)\rangle. \quad (26.5)$$

A single tensor A is symmetric with eigenvalues c_a if $O_a^{(N)} |V^{(N)}(A)\rangle = c_a |V^{(N)}(A)\rangle$ for all a and $N \geq 1$. The coefficients do not depend on N .



Definition 26.4.2 (Fusion character and nonnegative integer representation). [Lean](#) [Lean](#) [✓ Stmt](#) A *fusion character* with unit e is a function m on labels with $m_e = 1$ and $m_a m_b = \sum_c N_{ab}^c m_c$. A *nonnegative integer representation* on a finite set K is a family of matrices $M_{a,x}^y \in \mathbb{N}$ with

$$\sum_c N_{ab}^c M_{c,x}^y = \sum_{z \in K} M_{b,x}^z M_{a,z}^y. \quad (26.6)$$

Theorem 26.4.3 (Word-trace relations against a normal tensor are integral). [Lean](#) [✓ Stmt](#) Let B be a matrix product state tensor and A a word-normal tensor of positive bond dimension, over the same alphabet. If a scalar $c \in \mathbb{C}$ satisfies $\text{tr}(B^w) = c \text{tr}(A^w)$ for every nonempty word w , then c is a nonnegative integer.

Proof. [✓ Proof](#) Normality expresses a diagonal matrix unit as $E_{jj} = \sum_u x_u A^u$, where all words u have the same positive length ℓ . Define $Q = \sum_u x_u B^u$. Expanding powers and applying the trace hypothesis to the concatenated words gives, for every $k \geq 1$,

$$\text{tr}(Q^k) = c \text{tr} \left(\left(\sum_u x_u A^u \right)^k \right) = c \text{tr}(E_{jj}^k) = c. \quad (26.7)$$

A constant trace of all positive powers of a fixed finite-dimensional matrix is a nonnegative integer. Indeed, the power sums of its nonzero eigenvalues, counted with algebraic multiplicity, are constant only when every nonzero eigenvalue is one. The constant is their total multiplicity. $\square$

Theorem 26.4.4 (Rank-one actions on a normal tensor are integral). [Lean](#) [✓ Stmt](#) Let T be a matrix product operator tensor and A a word-normal matrix product state tensor of positive bond dimension, with the same physical alphabet. Suppose there is a single scalar $c \in \mathbb{C}$ such that, for every $N \geq 1$,

$$O_N(T) |V^{(N)}(A)\rangle = c |V^{(N)}(A)\rangle. \quad (26.8)$$

Then c is a nonnegative integer.

Proof. [✓ Proof](#) Corollary 25.6.4 identifies the periodic vector of $T \cdot A$ with $O_N(T) |V^{(N)}(A)\rangle$. Taking components in (26.8) gives, for every nonempty word w ,

$$\text{tr}((T \cdot A)^w) = c \text{tr}(A^w).$$

Apply Theorem 26.4.3. $\square$

Garre-Rubio and Schuch [166, Section II.B] normalize the action of a matrix product unitary on an injective ground state by absorbing a phase into the image tensor. A footnote justifies this: a length-dependent phase of modulus one relating the two states is the N -th power of one phase.

Theorem 26.4.5 (Power sums of modulus one). [Lean](#) [✓ Stmt](#) *Let $\mu_1, \dots, \mu_n$ be nonzero complex numbers with $|\sum_k \mu_k^L| = 1$ for every $L \geq 1$. Then $n = 1$ and $|\mu_1| = 1$.*

Proof. [✓ Proof](#) For $L \geq 1$, $\sum_{k,l} (\mu_k \bar{\mu}_l)^L = |\sum_k \mu_k^L|^2 = 1$, so the n^2 nonzero numbers $\mu_k \bar{\mu}_l$ have the power sums of the single number 1 at every positive exponent. By uniqueness of power sums on a tail they coincide with $\{1\}$ as a multiset, so $n^2 = 1$. The case $L = 1$ gives $|\mu_1| = 1$. $\square$

Theorem 26.4.6 (A unimodular length-dependent phase is a power). [Lean](#) [Lean](#) [✓ Stmt](#) *Let T be a matrix product operator tensor, A a matrix product state tensor, and B an injective matrix product state tensor of positive bond dimension, all with the same physical alphabet. Suppose that for every $N \geq 1$*

$$O_N(T)|V^{(N)}(A)\rangle = c_N|V^{(N)}(B)\rangle, \quad |c_N| = 1.$$

Then there is $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ and $c_N = \lambda^N$ for every $N \geq 1$. In particular this holds when T is a matrix product unitary and A is injective, as in [166, Section II.B].

Proof. [✓ Proof](#) By Corollary 25.6.4 the left-hand side is the periodic vector of the tensor $T \cdot A$. Theorem 25.7.2, applied to $T \cdot A$ and the single normal target B , gives nonzero $\mu_1, \dots, \mu_n$ with $|V^{(N)}(T \cdot A)\rangle = (\sum_k \mu_k^N) |V^{(N)}(B)\rangle$ for every $N \geq 1$. The vector $|V^{(N)}(B)\rangle$ is nonzero: for $N = 1$ because the letters of B span the full matrix algebra and so do not all have trace zero, and for $N \geq 2$ by the nonvanishing of the periodic vectors of an injective tensor. Hence $c_N = \sum_k \mu_k^N$, and Theorem 26.4.5 finishes the proof. $\square$

Theorem 26.4.7 (Integrality of multi-block word-trace relations). [Lean](#) [✓ Stmt](#) *Let $(A_y)_{y \in K}$ be normal tensors of positive bond dimension with pairwise non-isomorphic word modules, let B be a tensor on the same physical alphabet, and let $c_y \in \mathbb{C}$ satisfy $\text{tr}(B^w) = \sum_y c_y \text{tr}(A_y^w)$ for every nonempty word w . Then every c_y is a nonnegative integer.*

Proof. [✓ Proof](#) The word modules of the A_y are simple and pairwise non-isomorphic, so by density the words of one positive length ℓ act on $\bigoplus_y \mathbb{C}^{D_y}$ as the full block-diagonal algebra. Fix y_0 and an index j ; there are coefficients x_u , over words u of length ℓ , with

$$\sum_u x_u A_{y_0}^u = E_{jj}, \quad \sum_u x_u A_y^u = 0 \quad (y \neq y_0). \quad (26.9)$$

Define $Q = \sum_u x_u B^u$. Expanding powers and applying the trace relation to the concatenated nonempty words gives, for every $k \geq 1$,

$$\text{tr}(Q^k) = \sum_y c_y \text{tr} \left(\left(\sum_u x_u A_y^u \right)^k \right) = c_{y_0} \text{tr}(E_{jj}^k) = c_{y_0}. \quad (26.10)$$

Let λ_i be the nonzero eigenvalues of Q , counted with algebraic multiplicity. Since $\text{tr}(Q^k) = \sum_i \lambda_i^k = c_{y_0}$ for all $k \geq 1$, every λ_i equals 1, so c_{y_0} is their total algebraic multiplicity and lies in $\mathbb{N}$. This extends the power-sum argument of Theorem 26.4.3 to multiple blocks. $\square$

Theorem 26.4.8 (The action coefficients form a nonnegative integer representation). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $(A_x)_{x \in K}$ be normal tensors of positive bond dimension whose periodic vectors are linearly independent at some length $N_0 \geq 1$, symmetric under a fusion algebra with coefficients $M_{a,x}^y$. Then every $M_{a,x}^y$ is a nonnegative integer and (26.6) holds. This is the integrality and associativity of the action multiplicities in [164, Section 3].

Proof. [Stmnt](#) Fix a and x . The word traces of $T_a \cdot A_x$ are $\sum_y M_{a,x}^y \text{tr}(A_y^w)$ on nonempty words w . The word modules of the A_y are simple and pairwise non-isomorphic, since isomorphic word modules have equal periodic vectors. An element of the augmentation ideal of the word algebra acts on the block y_0 as a matrix unit and annihilates the other blocks; its powers have constant trace $M_{a,x}^{y_0}$ on $T_a \cdot A_x$, so $M_{a,x}^{y_0}$ is a nonnegative integer (Theorem 26.4.7). Applying (26.2) and (26.5) twice to $O_a^{(N_0)} O_b^{(N_0)} |V^{(N_0)}(A_x)\rangle$ and comparing coefficients of the independent vectors gives (26.6). $\square$

Theorem 26.4.9 (A single symmetric normal tensor gives a fusion character). [Lean](#) [Stmnt](#) If a normal tensor of positive bond dimension is symmetric under a fusion algebra with eigenvalues c_a and $c_e = 1$, then $c_a \in \mathbb{N}$ for every a and $a \mapsto c_a$ is a fusion character.

Proof. [Stmnt](#) Each c_a is a nonnegative integer by Theorem 26.4.4. Applying (26.2) to the nonzero periodic vector at an injective length gives $c_a c_b = \sum_c N_{ab}^c c_c$. $\square$

Corollary 26.4.10 (Rank-one obstruction). [Lean](#) [Stmnt](#) If the structure constants admit no fusion character with values in $\mathbb{N}$, no normal tensor of positive bond dimension is symmetric with $c_e = 1$. This obstruction depends only on the fusion ring; Theorem 28.4.3 gives the cohomological obstruction for group-like algebras with a nontrivial three-cocycle.

Proof. [Stmnt](#) Theorem 26.4.9 produces a fusion character with values in $\mathbb{N}$ from any such tensor. $\square$

Theorem 26.4.11 (Invertible labels). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) A fusion character with values in $\mathbb{N}$ is 1 on every invertible label, so an invertible label fixes the periodic vectors of a single symmetric normal tensor with $c_e = 1$. For a family as in Theorem 26.4.8 on which $O_e^{(N)}$ fixes every periodic vector, each invertible label g permutes the blocks: there is a bijection $x \mapsto g \cdot x$ of the block labels with $O_g^{(N)} |V^{(N)}(A_x)\rangle = |V^{(N)}(A_{g \cdot x})\rangle$ for all $N \geq 1$. This is the permutation action of [164, Section 4].

Proof. [Stmnt](#) From $m_a m_b = m_e = 1$ in $\mathbb{N}$ one gets $m_a = 1$. For the family, the multiplicity matrices satisfy $M_a M_b = M_b M_a = M_e = \mathbb{1}$ with nonnegative integer entries, so each row of M_a is a standard basis vector, and the two relations make the resulting map of blocks a bijection. $\square$

Example 26.4.12 (Fibonacci). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) The Fibonacci operators of Section 27.6 form a fusion algebra with unit 1 and $\tau \times \tau = 1 + \tau$; the two normal states with $\tau \cdot x_1 = x_\tau$ and $\tau \cdot x_\tau = x_1 + x_\tau$ form a symmetric family whose action is the regular representation [164, Section 6]. Since $m^2 = 1 + m$ has no solution in $\mathbb{N}$, no single normal tensor A satisfies $O_1^{(N)} |V^{(N)}(A)\rangle = |V^{(N)}(A)\rangle$ and $O_\tau^{(N)} |V^{(N)}(A)\rangle = c |V^{(N)}(A)\rangle$ for all $N \geq 1$ (Theorem 27.6.5.6).

Theorem 26.4.13 (Fibonacci representations on at most two blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let M be a nonnegative integer representation of the Fibonacci ring on a set K of blocks with $M_{1,x}^y = \delta_{xy}$. If K has one element, no such M exists. If K has two elements, then after labelling the blocks by 1 and τ one has $M_a = N_a$ for both labels, that is $\tau \cdot x_1 = x_\tau$ and $\tau \cdot x_\tau = x_1 + x_\tau$. Consequently a nonempty symmetric family of at most two normal tensors as

in Theorem 26.4.8, on which the unit acts trivially, has exactly two blocks transforming in this way. The source states the uniqueness of this two-block family without a bound on the number of blocks [164, Section 6]; only one and two blocks are treated here [58].

Proof. ✓ Proof Relation (26.6) at $a = b = \tau$ with $\tau \times \tau = 1 + \tau$ gives $M_\tau^2 = 1 + M_\tau$. On one block this is $m^2 = 1 + m$, which has no solution in $\mathbb{N}$. On two blocks write $M_\tau = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The diagonal equations give $a^2 \leq 1 + a$ and $d^2 \leq 1 + d$, so $a, d \in \{0, 1\}$ and $bc = 1$, hence $b = c = 1$. The off-diagonal equation $b = (a + d)b$ gives $a + d = 1$, so M_τ is N_τ or its conjugate by the transposition of the two blocks. For the family, the coefficients form such a representation by Theorem 26.4.8. □

Definition 26.4.14 (The Ising fusion algebra data). Lean Lean Lean ✓ Stmt Index the three simple objects of the Ising category by 0, 1 and 2, corresponding respectively to 1, ψ and σ . The blocks are the matrix product operator tensors A_1 , A_ψ and $\frac{1}{\sqrt{2}}A_\sigma$ of Definition 27.7.4, of bond dimensions 3, 3 and 4; the factor $\frac{1}{\sqrt{2}}$ undoes the scaling of A_σ there. The structure constants N_{ab}^c are those of the Ising fusion ring: $N_{1b}^c = N_{b1}^c = \delta_{bc}$, $\psi \times \psi = 1$, $\psi \times \sigma = \sigma \times \psi = \sigma$ and $\sigma \times \sigma = 1 + \psi$, that is, $N_{ab}^c = 1$ exactly for these products and $N_{ab}^c = 0$ otherwise [7].

Example 26.4.15 (Ising). Lean Lean Lean Lean ✓ Stmt The blocks of Definition 26.4.14 form a fusion algebra with unit 1 and these structure constants [7]. A fusion character in $\mathbb{N}$ would satisfy $m_\psi^2 = 1$, hence $m_\psi = 1$, and then $m_\sigma^2 = 1 + m_\psi = 2$, which has no solution in $\mathbb{N}$. So no normal tensor of positive bond dimension is symmetric under the Ising algebra with the unit acting trivially.

Definition 26.4.16 (The fusion rings of $su(2)_4$ and $\text{Rep}(S_3)$). Lean Lean Lean ✓ Stmt The labels of $su(2)_4$ are $0, \frac{1}{2}, 1, \frac{3}{2}, 2$, with fusion rules $j_1 \times j_2 = \sum j$ over $j = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, \min(j_1 + j_2, 4 - j_1 - j_2)$. The labels of $\text{Rep}(S_3)$ are $1, \pi, \psi$, with $\pi \times \pi = 1 + \pi + \psi$, $\psi \times \psi = 1$ and $\psi \times \pi = \pi \times \psi = \pi$. The labels $1, \pi, \psi$ are assigned to the integer labels $0, 1, 2$ of $su(2)_4$ [164, Section 6].

Theorem 26.4.17 (Units and the embedding of $\text{Rep}(S_3)$ in $su(2)_4$). Lean Lean Lean Lean ✓ Stmt The label 0 is the unit of $su(2)_4$ and 1 is the unit of $\text{Rep}(S_3)$. The assignment $1, \pi, \psi \mapsto 0, 1, 2$ identifies $\text{Rep}(S_3)$ with the fusion subring of integer labels of $su(2)_4$: the fusion rules agree, and products of integer labels contain no half-integer label [164, Section 6].

Proof. ✓ Proof Compare the finitely many structure constants. □

Theorem 26.4.18 (The phases of $su(2)_4$ and $\text{Rep}(S_3)$ represent the fusion rings). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The following actions on blocks, with the unit acting as the identity, satisfy (26.6).

1. For $su(2)_4$: the fusion rules on five blocks, one per label; and the module $\mathcal{M}_{TY}$ on blocks x, y, z, s , where 0 and 2 fix every block, $\frac{1}{2}$ and $\frac{3}{2}$ send x, y, z to s and s to $x + y + z$, and 1 sends x to $y + z$, y to $x + z$, z to $x + y$ and s to $2s$.
2. For $\text{Rep}(S_3)$: on one block s , $\pi \cdot s = 2s$ and $\psi \cdot s = s$ (the subgroup $\mathbb{Z}_1$); on blocks x, y , $\pi \cdot x = \pi \cdot y = x + y$ and ψ exchanging x and y ($\mathbb{Z}_2$); on blocks x, y, z , $\pi \cdot x = y + z$, $\pi \cdot y = x + z$, $\pi \cdot z = x + y$ and ψ fixing every block ($\mathbb{Z}_3$); and the fusion rules on three blocks (S_3).

These are the phases listed in [164, Section 6], except that the source prints the $\mathbb{Z}_3$ action with ψ exchanging y and z ; that action does not satisfy (26.6), since $\pi \cdot (\pi \cdot y) = x + 2y + z$ while $(1 + \pi + \psi) \cdot y = x + y + 2z$ [84]. That the lists are complete is not treated here beyond one block (Theorem 26.4.20; [85]).

Proof. [✓ Proof](#) Each identity (26.6) is a finite check on nonnegative integer matrices. $\square$

Theorem 26.4.19 (The $\text{Rep}(S_3)$ phases inside the $su(2)_4$ phases). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the identification of $\text{Rep}(S_3)$ with the integer labels of $su(2)_4$, the integer labels do not mix the blocks $\{x, y, z\}$ and $\{s\}$ of $\mathcal{M}_{TY}$; they act on $\{x, y, z\}$ as in the $\mathbb{Z}_3$ phase and on $\{s\}$ as in the $\mathbb{Z}_1$ phase. They do not mix the blocks $\frac{1}{2}, \frac{3}{2}$ of the first $su(2)_4$ phase with the other blocks, and act on them as in the $\mathbb{Z}_2$ phase [164, Section 6].

Proof. [✓ Proof](#) Compare the entries of the actions. $\square$

Theorem 26.4.20 (Actions on one block). [Lean](#) [Lean](#) [✓ Stmt](#) The only fusion character of $\text{Rep}(S_3)$ with values in $\mathbb{N}$ is $(m_1, m_\pi, m_\psi) = (1, 2, 1)$, the $\mathbb{Z}_1$ phase; $su(2)_4$ has no fusion character with values in $\mathbb{N}$.

Proof. [✓ Proof](#) For $\text{Rep}(S_3)$, $m_\psi^2 = m_1 = 1$ gives $m_\psi = 1$, and $m_\pi^2 = 1 + m_\pi + m_\psi$ then gives $m_\pi = 2$; conversely $(1, 2, 1)$ satisfies all relations. For $su(2)_4$, $m_2^2 = 1$ gives $m_2 = 1$, $m_1^2 = 2 + m_1$ gives $m_1 = 2$, and then $m_{1/2}^2 = m_0 + m_1 = 3$ has no solution in $\mathbb{N}$. $\square$

Example 26.4.21 (Anomalous cyclic group of order three). [Lean](#) [Lean](#) [✓ Stmt](#) The operators $1, U, U^\dagger$ of Section 27.10 form a group-like fusion algebra in which every label is invertible. The fusion ring does not detect the nontrivial three-cocycle class $[\omega] \neq 1 \in H^3(\mathbb{Z}_3, \mathbb{C}^\times)$ of this representation; the obstruction to invariant normal states comes from Theorem 28.4.3.

The obstruction of Corollary 26.4.10 concerns a single normal tensor. A superposition of the periodic vectors of several normal tensors can still be an eigenvector of every operator. When the algebra is that of a fusion category, such a superposition with positive weights exists for every module [164, Section 5]; it is the boundary state compatible with a two-dimensional bulk carrying the operators as virtual symmetries. The fusion category enters only through its fusion ring.

Definition 26.4.22 (Fusion ring). [Lean](#) [✓ Stmt](#) Structure constants $N_{ab}^c \in \mathbb{N}$ on a finite label set I form a *fusion ring* with unit e and duality $a \mapsto a^*$ if they are associative, $\sum_x N_{ab}^x N_{xc}^f = \sum_x N_{bc}^x N_{ax}^f$ for all a, b, c, f , the label e is a unit in the sense of Definition 26.2.2, the duality is an involution, $N_{ab}^e = \delta_{b, a^*}$, and $N_{b^* a^*}^c = N_{ab}^c$. These are the identity element and dual elements of the fusion category of a physical symmetry [164, Section 5].

Theorem 26.4.23 (Positive regular element of a fusion ring). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a fusion ring there are positive numbers r_b with $\sum_b r_b N_{ab}^c = d_a r_c$ for all labels a and c , where d_a is the Perron–Frobenius dimension of a , and $d_a > 0$ for every a .

Proof. [✓ Proof](#) The coefficient of c in $(b \times b^*) \times c$ is at least $N_{bb}^e N_{ec}^c = 1$, and by associativity it equals the coefficient of c in $b \times (b^* \times c)$; so for all b, c some label a has $N_{ba}^c > 0$. Hence the matrix $P_{bc} = \sum_a N_{ba}^c$ of right multiplication by the sum of all labels has positive entries. By the Perron–Frobenius argument of Theorem 26.2.4, which applies to every nonnegative square matrix, used for the transpose of P , there is a nonzero row vector $r \geq 0$ with $rP = \rho r$, and positivity of P forces every $r_c > 0$. An eigenvector of a positive matrix with positive entries spans its eigenspace: if $wP = \rho w$ and $t = \min_c w_c / r_c$, then $w - tr$ is a nonnegative eigenvector

with a vanishing entry, and positivity of P forces $w = tr$. Left multiplication by a , the matrix N_a , commutes with P by associativity, so rN_a is again an eigenvector of P for ρ and $rN_a = t_a r$. By Lemma 26.2.5, $t_a = d_a$. Finally $d_a r_a = \sum_b r_b N_{ab}^a \geq r_e N_{ae}^a = r_e > 0$. $\square$

Theorem 26.4.24 (A positive common eigenvector of a nonnegative integer representation).

[Lean](#) [Lean](#) [✓ Stmt](#) Let N be a fusion ring with unit e and let $M_{a,x}^y$ be a nonnegative integer representation of it on a finite set K with $M_{e,x}^y = \delta_{xy}$. Then there are positive numbers v_x with

$$\sum_x v_x M_{a,x}^y = d_a v_y \quad (26.11)$$

for all labels a and all $y \in K$. If K is nonempty, d_a is the spectral radius of the matrix M_a . No indecomposability of the representation and no commutativity of the fusion ring is needed.

Proof. [✓ Proof](#) Let r be as in Theorem 26.4.23 and $S = \sum_b r_b M_b$. By (26.6), $M_b M_a = \sum_c N_{ab}^c M_c$, so $S M_a = \sum_c \left(\sum_b r_b N_{ab}^c \right) M_c = d_a S$. The column sums $v_y = \sum_x S_{xy}$ therefore satisfy (26.11), and $v_y \geq r_e M_{e,y}^y = r_e > 0$. A positive left eigenvector of the nonnegative matrix M_a has the spectral radius as its eigenvalue, by the argument of Lemma 26.2.5. $\square$

Theorem 26.4.25 (Positive symmetric boundary state). [Lean](#) [Lean](#) [✓ Stmt](#) Let $(A_x)_{x \in K}$, with K nonempty, be normal tensors of positive bond dimension whose periodic vectors are linearly independent at some length $N_0 \geq 1$, symmetric under a fusion algebra whose structure constants form a fusion ring with unit e , and suppose $M_{e,x}^y = \delta_{xy}$. Then there are positive numbers v_x such that for every label a and every $N \geq 1$,

$$O_a^{(N)} |\psi^{(N)}\rangle = d_a |\psi^{(N)}\rangle, \quad |\psi^{(N)}\rangle = \sum_x v_x |V^{(N)}(A_x)\rangle,$$

with $d_a > 0$, and $|\psi^{(N_0)}\rangle \neq 0$. This is the symmetric boundary state of [164, Section 5], whose eigenvalue r_a is the spectral radius of M_a .

Proof. [✓ Proof](#) By Theorem 26.4.8 the coefficients $M_{a,x}^y$ form a nonnegative integer representation. Take v from Theorem 26.4.24. Then $O_a^{(N)} |\psi^{(N)}\rangle = \sum_x v_x \sum_y M_{a,x}^y |V^{(N)}(A_y)\rangle = \sum_y \left(\sum_x v_x M_{a,x}^y \right) |V^{(N)}(A_y)\rangle = d_a |\psi^{(N)}\rangle$. Since the periodic vectors are linearly independent at length N_0 and $v \neq 0$, $|\psi^{(N_0)}\rangle \neq 0$. $\square$

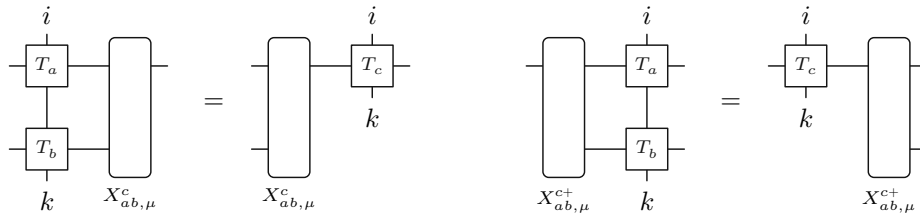
Example 26.4.26 (The Fibonacci boundary state). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Fibonacci structure constants form a fusion ring with unit 1 and both labels self-dual. For the family of Example 26.4.12, $M_\tau = N_\tau$ and $v = (1, \varphi)$, so $|\psi^{(N)}\rangle = |V^{(N)}(A_1)\rangle + \varphi |V^{(N)}(A_\tau)\rangle$ satisfies $O_1^{(N)} |\psi^{(N)}\rangle = |\psi^{(N)}\rangle$ and $O_\tau^{(N)} |\psi^{(N)}\rangle = \varphi |\psi^{(N)}\rangle$ for every $N \geq 1$. The eigenvalue $d_\tau = \varphi$ is not an integer. At $N = 1$ the vector $|\psi^{(1)}\rangle$ vanishes, since the traces of the one-site matrices are 0, 0 on the trivial letter and $1, -\varphi^{-1}$ on τ ; the eigenvector equations carry information only for $N \geq 2$.

26.5 The zipper condition

Remark 26.5.1 (Zipper and pulling-through conditions). The fusion rule (26.2) is an identity of periodic operators. Its local form, the *zipper condition*, asks for maps $X_{ab,\mu}^c : \mathbb{C}^{\chi_c} \rightarrow \mathbb{C}^{\chi_a} \otimes \mathbb{C}^{\chi_b}$ and left inverses $X_{ab,\mu}^{c+}$, for $\mu = 1, \dots, N_{ab}^c$, with $X_{ab,\nu}^{d+} X_{ab,\mu}^c = \delta_{cd} \delta_{\mu\nu} \mathbb{1}_{\chi_c}$ and

$$\begin{aligned} (T_a T_b)^{ik} X_{ab,\mu}^c &= X_{ab,\mu}^c T_c^{ik}, \\ X_{ab,\mu}^{c+} (T_a T_b)^{ik} &= T_c^{ik} X_{ab,\mu}^{c+} \end{aligned} \quad (26.12)$$

for all physical indices i, k [7].



A fusion tensor can then be slid through the product of two operators one site at a time. If in addition the product is reconstructed on the span of the zipper maps, $(T_a T_b)^{ik} = \sum_{c,\mu} X_{ab,\mu}^c T_c^{ik} X_{ab,\mu}^{c+}$, as required in Definition 21.3.85, then closing the chain gives (26.2) at every length. Its pictorial form, in which an operator is pulled through a region of a two-dimensional tensor network, is the *pulling-through* condition of topologically ordered projected entangled pair states [7, 16]. The zipper identities appear as (21.102) and (21.103) in Definition 21.3.85.

The converse fails: equality of periodic operators does not imply a sitewise intertwiner (Remark 25.1.2), and the group operators of the nontrivial three-cocycle of the group of order two satisfy only one of the two identities [164, Section 6]. Chapter 28 therefore obtains fusion and action tensors from equality of periodic traces through reductions (Definitions 28.1.1 and 28.1.3), and shows that fusion tensors with vanishing remainder, the situation of the zipper condition, are unique up to invertible multiplicity matrices (Theorem 28.2.2).

Chapter 27

Physics Examples of the Multi-Block Compression Theorem

Matrix product states and operators describe periodic spin chains by contracting local tensors around a ring [16]. At every positive chain length N , an MPS tensor A and an MPO tensor M give

$$\begin{aligned} |V^{(N)}(A)\rangle &= \sum_{i_1, \dots, i_N} \text{tr}(A^{i_1} \dots A^{i_N}) |i_1, \dots, i_N\rangle, \\ O_N(M) &= \sum_{i_1, \dots, i_N} \sum_{j_1, \dots, j_N} \text{tr}(M^{i_1 j_1} \dots M^{i_N j_N}) |i_1, \dots, i_N\rangle \langle j_1, \dots, j_N|. \end{aligned}$$

The physical indices i, j range over $\{0, \dots, d-1\}$. Each local matrix acts on its tensor's virtual bond space, with rows and columns indexed by $\{0, \dots, D-1\}$ for bond dimension D . The periodic boundary contraction is the ordinary matrix trace, with no boundary insertion. The first MPO physical index is the output and the second is the input; operator products act from right to left. Thus the local tensor $\sum_j M^{ij} \otimes P^{jk}$ represents $O_N(M)O_N(P)$. MPS vectors need not be normalized. A nonzero positive periodic MPDO operator gives a density operator after division by its trace.

Physical examples of matrix product operators, topological symmetries, and fixed points illustrate the multi-block asymmetric compression theorem (Theorem 25.5.1). The cases treated here, in chapter order, are the GHZ and product states [168, 167] in the conventions of Ref. [16, Appendix A]; the CZX symmetry of Chen, Liu, and Wen [10], whose matrix product unitary is displayed in Ref. [16, Appendix A, “The CZX MPU”], together with its decorated form [166], whose anomaly class is shown here to be nontrivial; Kramers–Wannier duality [178, 1, 206, 180]; the Fibonacci [7, Appendix D.1] and Ising [7, Appendix D.2] fusion categories and their anyon chains [161], together with the review’s Fibonacci operator [16, Appendix A, “The MPO for the Fibonacci model”]; the periodic group matrix product operators built from a normalized three-cocycle [164], evaluated for the group of order two, for the cyclic groups with the cocycles of [166], and for $\mathbb{Z}_2 \times \mathbb{Z}_2$; symmetry tensors for $\mathbb{Z}_2 \times \mathbb{Z}_2$ and $\mathbb{Z}_3$ with their condensation defects, constructed here after the anomalous group symmetries discussed in Refs. [164, 166, 197], whose anomaly class this chapter determines for $\mathbb{Z}_3$, where it is nontrivial, and shows to be nontrivial for the full dressed $\mathbb{Z}_2 \times \mathbb{Z}_2$ representation, whose detector values are those of the type-II class; equality with

that class is not claimed [27]; and mixed Bell-bond fixed points addressing the length-dependence question of Ref. [11, Theorem 4.14]. The general notions used by the symmetry examples, namely fusion rules, units, invertible and non-invertible labels, Perron–Frobenius dimensions, symmetric matrix product states with their nonnegative integer representations, and the zipper condition, are defined in Chapter 26; this chapter supplies the tensors, fusion tables, and computations.

27.1 The GHZ state as a compression target

The Greenberger–Horne–Zeilinger (GHZ) state [168, 167] is the equal superposition of the two constant product configurations; for $N > 0$ sites, its unnormalized form on $(\mathbb{C}^2)^{\otimes N}$ is

$$|\text{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

Here $|0\rangle, |1\rangle$ form the orthonormal local basis, following the product-state and GHZ conventions of Ref. [16, Appendix A]. The standard tensor generating this state has bond dimension two and is the direct sum $C_0 \oplus C_1$ of two scalar sectors, as in Ref. [16, Appendix A] and Definition 15.2.7. The upper-triangular source tensor B of bond dimension four (Definition 27.1.1) satisfies $|V^{(N)}(B)\rangle = |\text{GHZ}_N\rangle$ for all $N > 0$, providing an explicit instance of the multi-block compression theorem (Theorem 25.5.1).

Throughout this section and the next, for a tensor $T = \{T^i\}_{i=0}^1$ of bond dimension D , the periodic trace boundary defines

$$\begin{aligned} V^{(N)}(T)_\sigma &= \text{tr}(T^{\sigma(0)} \dots T^{\sigma(N-1)}), \\ |V^{(N)}(T)\rangle &= \sum_{\sigma: \{0, \dots, N-1\} \rightarrow \{0, 1\}} V^{(N)}(T)_\sigma |\sigma\rangle, \end{aligned} \tag{27.1}$$

where $\{0, \dots, N-1\} = \{0, \dots, N-1\}$ for $N > 0$, $\{0, \dots, 0-1\} = \emptyset$, and $|\sigma\rangle = \bigotimes_{n=0}^{N-1} |\sigma(n)\rangle$. At $N = 0$, the matrix product is $\mathbb{1}_D$, so the unique coefficient is D and the physical space is $\mathbb{C}$. For a word $w = (i_1, \dots, i_{|w|})$, write $T^w = T^{i_1} \dots T^{i_{|w|}}$, with $T^\emptyset = \mathbb{1}_D$.

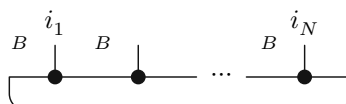
Definition 27.1.1 (The two sectors and the four-dimensional source). [Lean](#) [Lean](#) [✓ Stmt](#) On the alphabet $\{0, 1\}$, define the scalar tensors C_0, C_1 and the standard GHZ tensor G by

$$\begin{aligned} C_0^0 &= (1), & C_0^1 &= (0), \\ C_1^0 &= (0), & C_1^1 &= (1), \\ G^0 &= \text{diag}(1, 0), & G^1 &= \text{diag}(0, 1). \end{aligned}$$

Thus $G^i = C_0^i \oplus C_1^i$, and both scalar tensors are normal. The constructed source is the bond-dimension-four tensor

$$B^0 = \begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad B^1 = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Use the ordinary trace boundary of (27.1). The state and norm agree with the standard GHZ tensor at positive lengths (Theorems 27.1.2 and 27.1.3). The compression is given in Theorem 27.1.4; the intertwiner obstruction is stated in Theorem 27.1.6.



Theorem 27.1.4 (Compression of the two sectors). [Lean](#) [Lean](#) [✓ Stmt](#) *The source of Definition 27.1.1 carries the block triangular form of Theorem 25.5.1 for the slot family (C_0, C_1) with $z = 2$ zero slots, the four diagonal positions carrying, in order, the sector C_1 , a zero slot, the sector C_0 and a second zero slot. The compression pair of each sector satisfies $W_s V_s = \text{id}$ and $W_s B^w V_s = C_s^w$ for every word, and $4 = 1 + 1 + 2$. Here $V_s : \mathbb{C} \rightarrow \mathbb{C}^4$ and $W_s : \mathbb{C}^4 \rightarrow \mathbb{C}$ are the coordinate inclusion and projection at position 2 for $s = 0$ and position 0 for $s = 1$, with coordinates numbered from zero.*

Proof. [✓ Proof](#) The gauge is the relabelling of the four bond coordinates by the four slots, with no change of basis. The three matrix conditions of the compression datum, block triangularity and the identification of the matched and unmatched diagonal blocks, follow by checking the entries of B^0 and B^1 . The compression pairs and the dimension count are clauses (iv), (v) and (vii) of Theorem 25.5.1. $\square$

Theorem 27.1.5 (Periodic traces of the source). [Lean](#) [✓ Stmt](#) *For every nonempty word w , $\text{tr}(B^w) = \text{tr}(C_0^w) + \text{tr}(C_1^w)$.*

Proof. [✓ Proof](#) This is Proposition 25.5.3 applied to the compression datum of Theorem 27.1.4, whose slot set has two members. $\square$

The compression is genuinely a word-level statement. A sitewise intertwiner for a sector would be a vector fixed by one letter of the source and annihilated by the other; the following theorem shows that the sector C_0 admits none in either direction.

Theorem 27.1.6 (The first sector has no sitewise intertwiner). [Lean](#) [Lean](#) [✓ Stmt](#) *If a column vector $v \in \mathbb{C}^4$ satisfies $B^0 v = v$ and $B^1 v = 0$, then $v = 0$; if a row vector $u \in \mathbb{C}^4$ satisfies $u B^0 = u$ and $u B^1 = 0$, then $u = 0$. Both sitewise intertwiner spaces for the sector C_0 therefore vanish, so the word-level compression of Theorem 27.1.4 yields no nonzero sitewise intertwiner in either direction.*

Proof. [✓ Proof](#) The last row of B^0 vanishes, so the last coordinate of $B^0 v = v$ gives $v_3 = 0$; the second coordinate of $B^1 v = 0$ gives $v_2 = 0$; and the second and first coordinates of $B^0 v = v$ then force $v_1 = 0$ and $v_0 = 0$. For the covector, the first and third coordinates of $u B^1 = 0$ give $u_0 = 0$ and $u_1 = 0$, its last coordinate gives $u_2 = 0$, and the last coordinate of $u B^0 = u$ gives $u_3 = 0$. $\square$

27.2 A product state from a repeated block

A product state $|\phi\rangle^{\otimes N}$ is the simplest matrix product state: it is generated by a tensor of bond dimension one whose entries are the local amplitudes of $|\phi\rangle$, as in the product-state construction of Ref. [16, Appendix A]. Here $|\phi\rangle = |0\rangle + 2|1\rangle$, so the bond-one tensor A has local amplitudes $(1, 2)$. The direct sum $A \oplus A$ doubles every amplitude, and the upper-triangular source B of Definition 27.2.1 generates the same vector:

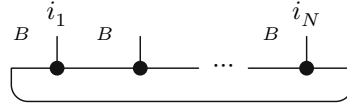
$$|V^{(N)}(B)\rangle = |V^{(N)}(A \oplus A)\rangle = 2|\phi\rangle^{\otimes N} = 2(|0\rangle + 2|1\rangle)^{\otimes N}. \quad (27.3)$$

The tensors A and B in Definition 27.2.1 generate these states via the trace boundary (27.1); at $N = 0$ the tensor power is the scalar 1. The example shows a target with a repeated block, where the compression theorem records the multiplicity of the block, and a source that is not gauge equivalent to the target even though the two generate equal periodic vectors at every length.

Definition 27.2.1 (The scalar block and its Jordan extension). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The target is the scalar normal tensor $A^0 = (1)$, $A^1 = (2)$, taken twice as $C_s = A$ for $s \in \{0, 1\}$, and the source is

$$B^0 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B^1 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Again the boundary is the ordinary trace. The multiplicity factor in the product state and its norm are given in Theorems 27.2.2 and 27.2.3. Theorems 27.2.1.1 and 27.2.1.2 establish the compression and show that it does not arise from a simultaneous similarity to the split tensor.



Theorem 27.2.2 (Product-state amplitudes and multiplicity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Set $a_0 = 1$ and $a_1 = 2$. For every $N \geq 0$ and $\sigma : \{0, \dots, N-1\} \rightarrow \{0, 1\}$,

$$\begin{aligned} V^{(N)}(A)_\sigma &= \prod_{n=0}^{N-1} a_{\sigma(n)}, \\ V^{(N)}(B)_\sigma &= 2V^{(N)}(A)_\sigma = 2 \prod_{n=0}^{N-1} a_{\sigma(n)}. \end{aligned} \quad (27.4)$$

Consequently,

$$|V^{(N)}(A)\rangle = (|0\rangle + 2|1\rangle)^{\otimes N}, \quad (27.5)$$

$$|V^{(N)}(B)\rangle = 2|V^{(N)}(A)\rangle. \quad (27.6)$$

The factor two also holds at $N = 0$, where the coefficients of A and B are 1 and 2, respectively.

Proof. [✓ Proof](#) Induction on N expresses the coefficient of the scalar tensor A as the product of its scalar entries. Substituting $A^0 = (1)$ and $A^1 = (2)$ gives the local amplitudes a_0, a_1 . If $N > 0$, the trace identity in Theorem 27.2.1.1 gives $V^{(N)}(B)_\sigma = 2V^{(N)}(A)_\sigma$. For $N = 0$, this follows from $\text{tr}(\mathbb{1}_2) = 2 \text{tr}(\mathbb{1}_1)$. Equality of the coefficients gives the vector identity. $\square$

Theorem 27.2.3 (Product-state norms). [Lean](#) [Lean](#) [✓ Stmt](#) For every $N \geq 0$,

$$\begin{aligned} \|V^{(N)}(A)\|^2 &= 5^N, \\ \|V^{(N)}(B)\|^2 &= 4 \cdot 5^N. \end{aligned}$$

Both tensors therefore give the same normalized product vector, $((|0\rangle + 2|1\rangle)/\sqrt{5})^{\otimes N}$.

Proof. [✓ Proof](#) Using the amplitudes $a_0 = 1$ and $a_1 = 2$ in (27.4), expand the squared norm in the configuration basis and factor the finite sum:

$$\|V^{(N)}(A)\|^2 = \sum_{\sigma} \prod_{n=0}^{N-1} |a_{\sigma(n)}|^2 = \prod_{n=0}^{N-1} (|a_0|^2 + |a_1|^2) = 5^N.$$

The vector identity in Theorem 27.2.2 multiplies the squared norm by $|2|^2 = 4$. $\square$

27.2.1 Compression and the absence of a gauge

The source B and target $A \oplus A$ satisfy $|V^{(N)}(B)\rangle = |V^{(N)}(A \oplus A)\rangle$ at all lengths $N \geq 0$ by (27.3). However, they are not gauge equivalent: $B^i \neq X(A^i \oplus A^i)X^{-1}$ for any invertible X (Theorem 27.2.1.2), even though $\text{tr}(B^w) = 2 \text{tr}(A^w)$ for all nonempty words w (Theorem 27.2.1.1). The compression theorem acts only on the two diagonal copies of A , yielding the trace identity $\text{tr}(B^w) = 2 \text{tr}(A^w)$ that proves the factor two in (27.4).

Theorem 27.2.1.1 (Compression onto the two copies). [Lean](#) [Lean](#) [✓ Stmt](#) *The source of Definition 27.2.1 carries the block triangular form of Theorem 25.5.1 for the two copies of A with no zero slots. For $s \in \{0, 1\}$, let $V_s : \mathbb{C} \rightarrow \mathbb{C}^2$ and $W_s : \mathbb{C}^2 \rightarrow \mathbb{C}$ be the inclusion and projection of coordinate s . Each coordinate pair compresses its copy: $W_s V_s = \text{id}$ and $W_s B^w V_s = A^w$ for every word. Also, $\text{tr}(B^w) = 2 \text{tr}(A^w)$ for every nonempty word.*

Proof. [✓ Proof](#) Both matrices are upper triangular with both diagonal entries equal to the corresponding scalar of A , which verifies the three matrix conditions. The compression pairs are clauses (iv) and (v) of Theorem 25.5.1, and the trace identity is Proposition 25.5.3 with two equal slots. $\square$

Theorem 27.2.1.2 (No invertible gauge onto the repeated target). [Lean](#) [✓ Stmt](#) *There is no invertible $X \in \text{GL}_2(\mathbb{C})$ with $B^i = X \text{diag}(A^i, A^i)X^{-1}$ for both letters i . Here $\text{diag}(A^i, A^i) = A^i \oplus A^i$ is the repeated target matrix.*

Proof. [✓ Proof](#) For $i = 0$ the matrix $\text{diag}(A^0, A^0)$ is the identity, so any such conjugation would give $B^0 = \mathbb{1}_2$, contradicting the nonzero off-diagonal entry of B^0 . $\square$

27.2.2 A Jordan extension detected by a boundary

A Jordan extension of a repeated block of arbitrary bond dimension shares every periodic trace with the direct sum of two copies of the block, while a single boundary matrix Q detects it:

Definition 27.2.2.1 (Jordan extension and split tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A = \{A^i\}_{i=0}^{d-1}$ be any tensor with $A^i \in M_D(\mathbb{C})$. In the standard two-block coordinates on $\mathbb{C}^{D+D}$, set

$$B^i = \begin{pmatrix} A^i & A^i \\ 0 & A^i \end{pmatrix}, \quad (27.7)$$

$$C^i = \begin{pmatrix} A^i & 0 \\ 0 & A^i \end{pmatrix}, \quad (27.8)$$

$$Q = \begin{pmatrix} 0 & 0 \\ \mathbb{1} & 0 \end{pmatrix}. \quad (27.9)$$

Thus $C = A \oplus A$. For any word w , write A^w for its ordered matrix product, with $A^0 = \mathbb{1}$.

The ordinary trace and the boundary insertion Q are compared in Theorem 27.2.2.2. If some nonempty word of A has nonzero trace, the insertion distinguishes the two tensors despite their identical periodic traces (Theorem 27.2.2.3).

Theorem 27.2.2.2 (Word products and boundary traces). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Every word w , including the empty word, satisfies

$$B^w = \begin{pmatrix} A^w & |w|A^w \\ 0 & A^w \end{pmatrix}, \quad (27.10)$$

$$C^w = \begin{pmatrix} A^w & 0 \\ 0 & A^w \end{pmatrix}, \quad (27.11)$$

$$\mathrm{tr}(B^w) = \mathrm{tr}(C^w) = 2 \mathrm{tr}(A^w), \quad (27.12)$$

$$\mathrm{tr}(QB^w) = |w| \mathrm{tr}(A^w), \quad (27.13)$$

$$\mathrm{tr}(QC^w) = 0. \quad (27.14)$$

Proof. [✓ Proof](#) Induct on w . The empty word gives the identity matrix. Prepending a letter i changes the upper-right block of B^w to $A^i(|w|A^w) + A^iA^w = (|w| + 1)A^iA^w$. The split product retains zero off-diagonal blocks. Taking the sum of the diagonal-block traces proves the periodic formulas. Multiplication by Q puts the upper-right word block in the lower-right block, giving the two boundary formulas. $\square$

Theorem 27.2.2.3 (Equal periodic traces and different boundary families). [Lean](#) [Lean](#) [✓ Stmt](#)

Suppose there is a nonempty word w with $\mathrm{tr}(A^w) \neq 0$, for an arbitrary tensor A . Then B and C have equal periodic traces for every word, but the functions $v \mapsto \mathrm{tr}(QB^v)$ and $v \mapsto \mathrm{tr}(QC^v)$ differ.

Proof. [✓ Proof](#) The periodic traces agree by Theorem 27.2.2.2. At the specified word w , the boundary traces are $|w| \mathrm{tr}(A^w) \neq 0$ and zero, respectively. Equality of the boundary functions would imply equality at w , a contradiction. $\square$

27.3 The CZX symmetry

The CZX model of Chen, Liu and Wen [10] is a two-dimensional symmetry-protected topological phase whose anomalous boundary $\mathbb{Z}_2$ symmetry on a ring of N qubits is the CZX operator: controlled- Z gates on the nearest-neighbor bonds of the ring followed by a Pauli X on every site. The review presents this operator as a bond-two matrix product unitary and its canonical-form reduction [16, Appendix A, “The CZX MPU”]; the symmetry-protected context is [16, Section III.B.1]. The decorated representative with additional Pauli Z factors in [166, Section III.D] is a different tensor.

Definition 27.3.1 (The CZX matrix product unitary of the review). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) The CZX tensor of the review [16, Appendix A, “The CZX MPU”] is the bond-two tensor A , with the output index first:

$$A^{01} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

$$A^{10} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix},$$

and $A^{00} = A^{11} = 0$. Its entries are $A_{\alpha\beta}^{ij} = \delta_{i,1-j} \delta_{\alpha,i} (-1)^{i\beta}$. It is obtained from the bond-two tensor of Definition 27.3.1.1 by transposing every matrix and exchanging the two physical indices: $A^{ij} = (M^{ji})^\top$. The bra-ket form of the same tensor in the review uses normalized virtual vectors and equals $2^{-1/2}A$; its periodic operator is not unitary, and the matrices displayed above are taken as the source tensor [42].

27.3.1 The bond-two tensor and the periodic operator

$$U_N|t\rangle = (-1)^{q(t)}|\bar{t}\rangle. \quad (27.15)$$
$$q(t) = \sum_{j=0}^{N-1} t_j t_{j+1}. \quad (27.16)$$
$$U_N = X^{\otimes N} D_N. \quad (27.17)$$

Definition 27.3.1.1 (The bond-two symmetry tensor and its square). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [StmT](#) The symmetry tensor is $M^{ij} = \delta_{i,1\oplus j} T_j$, where $i, j \in \{0, 1\}$ are respectively the physical output and input, $\delta_{a,b}$ is the Kronecker delta, $\oplus$ denotes addition modulo two, and

$$T_0 = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix},$$

$$T_1 = \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}.$$

$$B^{ij} = \sum_{m=0}^1 M^{im} \otimes M^{mj}. \quad (27.18)$$

The diagram illustrates the contraction of two tensors, M and B . On the left, two rows of M tensors are shown. The top row has inputs $i_1, i_2, \dots, i_N$ and the bottom row has inputs $j_1, j_2, \dots, j_N$. Vertical lines connect corresponding M tensors in the two rows. On the right, a single row of B tensors is shown with inputs $i_1, i_2, \dots, i_N$ and outputs $j_1, j_2, \dots, j_N$. The two diagrams are equated with an equals sign between them.

The periodic boundary is the ordinary, unnormalized trace. Thus the operator generated by M has matrix coefficients

$$U_N(s, t) = \text{tr}(M^{s_0 t_0} \dots M^{s_{N-1} t_{N-1}}). \quad (27.19)$$

There are no open boundary vectors. For bond indices $\alpha, \beta \in \{0, 1\}$, the local entries are

$$M_{\alpha, \beta}^{ij} = \delta_{i, 1-j} \delta_{\beta, j} (-1)^{\alpha j}. \quad (27.20)$$

Theorem 27.3.1.2 (Periodic computational basis kernel). [Lean](#) [Lean](#) [✓ Stmt](#) For every $N > 0$ and every $s, t \in \{0, 1\}^N$, the periodic contraction satisfies

$$U_N(s, t) = \delta_{s, \bar{t}} (-1)^{q(t)}. \quad (27.21)$$

In particular, its basis action is (27.15).

Proof. [✓ Proof](#) Expand the trace over cyclic bond configurations g , using (27.20):

$$U_N(s, t) = \sum_{g \in \{0, 1\}^N} \prod_{j=0}^{N-1} \delta_{s_j, 1-t_j} \delta_{g_{j+1}, t_j} (-1)^{g_j t_j}. \quad (27.22)$$

The bond constraints force $g_j = t_{j-1}$, so exactly one bond configuration can contribute. Its contribution vanishes unless $s = \bar{t}$. Otherwise it equals $(-1)^{\sum_j t_{j-1} t_j} = (-1)^{q(t)}$, by cyclic reindexing. $\square$

Theorem 27.3.1.3 (Positive-length unitarity). [Lean](#) [Lean](#) [✓ Stmt](#) The operator U_N is unitary for every $N > 0$.

Proof. [✓ Proof](#) By Theorem 27.3.1.2, U_N is a monomial matrix. The map $t \mapsto \bar{t}$ is a permutation and every nonzero entry has modulus one, since $\overline{(-1)^{q(t)}} (-1)^{q(t)} = 1$. Its columns and rows are therefore orthonormal. $\square$

Theorem 27.3.1.4 (Square of the periodic operator). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $N > 0$,

$$U_N^2 = (-1)^N \mathbb{1}. \quad (27.23)$$

The periodic operator generated by the stacked tensor B is also $(-1)^N \mathbb{1}$.

Proof. [✓ Proof](#) For bits a, b , direct evaluation gives $(-1)^{(1-a)(1-b)} (-1)^{ab} = -(-1)^a (-1)^b$. Multiplying this identity around the ring, each single-site sign occurs twice, so

$$(-1)^{q(\bar{t})} (-1)^{q(t)} = (-1)^N \prod_{j=0}^{N-1} (-1)^{2t_j} = (-1)^N. \quad (27.24)$$

Apply (27.21) twice and use $\bar{\bar{t}} = t$ to obtain (27.23). The periodic contraction of the stacked product is the product of the two periodic operators, by expansion over the intermediate physical indices in (27.18). $\square$

Unitarity and (27.23) give the inverse and physical adjoint:

$$U_N^{-1} = U_N^\dagger = (-1)^N U_N. \quad (27.25)$$

Consequently $U_N^4 = \mathbb{1}$. On even positive rings, U_N is a self-adjoint involution; on odd rings, $U_N^2 = -\mathbb{1}$. The phase identity (27.24) also gives $D_N X^{\otimes N} = (-1)^N U_N$, explaining the gate-order difference.

At length one, $q(t) = t_0^2 = t_0$, hence $D_1 = Z$ and $U_1 = XZ$, where $Z|a\rangle = (-1)^a|a\rangle$. At length two, $q(t) = 2t_0t_1$, hence $D_2 = \mathbb{1}$ and $U_2 = X \otimes X$. The exponent $q(t)$ includes a self-loop at $N = 1$ and counts the two-site edge twice at $N = 2$.

At length zero the trace convention gives the scalar $\text{tr}(\mathbb{1}_2) = 2$, not a unitary operator. The stacked tensor similarly gives $\text{tr}(\mathbb{1}_4) = 4$, whereas the bond-one target gives $\text{tr}(\mathbb{1}_1) = 1$. All periodic operator and trace comparisons below therefore retain their positive-length or nonempty-word hypotheses.

Remark 27.3.1.5 (The shift matrix product unitary). The shift matrix product unitary of [16, Appendix A, “The shift MPU”] is the right-shift tensor of Definition 18.9.3.9 in Chapter 18. It is not a compression example and is not treated in this chapter.

27.3.2 The tensor printed in the review

The review [16, Appendix A, “The CZX MPU”] prints the CZX matrix product unitary through its two nonzero matrices and reduces its square to canonical form in two steps, following the construction of the canonical form in [16, Section IV]. The presentation below uses the review’s coordinates. Throughout, the first physical index is the output, and for a configuration $s \in \{0, 1\}^N$ the exponent $q(s)$ is (27.16).

Theorem 27.3.2.1 (Periodic kernel of the review’s tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every $N > 0$ and every $s, t \in \{0, 1\}^N$, the periodic operator $O_N(A)$ of the review’s tensor satisfies*

$$O_N(A)(s, t) = \delta_{s, \bar{t}} (-1)^{q(s)}. \quad (27.26)$$

Thus $O_N(A) = D_N X^{\otimes N}$, and

$$O_N(A) = (-1)^N U_N, \quad (27.27)$$

where U_N is the periodic operator of Definition 27.3.1.1.

Proof. [✓ Proof](#) Expand the trace over cyclic bond configurations g . The factor δ_{g_n, s_n} of the entries forces $g = s$, so exactly one bond configuration contributes. Its contribution vanishes unless $s = \bar{t}$ and otherwise equals $\prod_n (-1)^{s_n s_{n+1}} = (-1)^{q(s)}$. Comparing with Theorem 27.3.1.2, the two kernels have the same support and their phases differ by $(-1)^{q(\bar{t})} (-1)^{q(t)} = (-1)^N$, by (27.24). $\square$

The review describes the operator in words as controlled- Z gates followed by a Pauli X on every site, which is $X^{\otimes N} D_N = U_N$; by (27.27) the two readings differ by the global sign $(-1)^N$ [42].

Theorem 27.3.2.2 (Unitarity of the review’s operator). [Lean](#) [Lean](#) [✓ Stmt](#) *The operator $O_N(A)$ is unitary for every $N > 0$.*

Proof. [✓ Proof](#) By (27.26), $O_N(A)$ is the monomial matrix of the permutation $t \mapsto \bar{t}$ with phases of modulus one. $\square$

Theorem 27.3.2.3 (Square law of the review’s operator). [Lean](#) [✓ Stmt](#) *For every $N > 0$,*

$$O_N(A)^2 = (-1)^N \mathbb{1}.$$

$$b^{11} = \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix},$$

and $b^{01} = b^{10} = 0$. For every nonempty word w over the alphabet of index pairs, $\text{tr}(\widetilde{B}^w) = \text{tr}(b^w)$.

Proof. ✓ Proof The identity $P\widetilde{B}^{ij} = \widetilde{B}^{ij}$ says that only the rows 1 and 2 of $\widetilde{B}^{00}$ and $\widetilde{B}^{11}$ are nonzero, which is visible in (27.28) and (27.29); the other two identities follow by multiplying on the right by P and by $\mathbb{1} - P$. Let $W_1 \in M_{2 \times 4}$ select the coordinates 1, 2 and $V_1 = W_1^\top$. Then $V_1 W_1 = P$ and $b^{ij} = W_1 \widetilde{B}^{ij} V_1$, and Lemma 1.1.9 gives the equality of traces. $\square$

Theorem 27.3.2.7 (The second projector and the canonical form). Lean Lean Lean Lean Lean Lean ✓ Stmt The matrix

$$Q = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

is an orthogonal projector with $b^{ij}Q = b^{ij}$ for every index pair. Consequently $Qb^{ij} = Qb^{ij}Q$, so the range of $\mathbb{1} - Q$ is invariant, and

$$Qb^{ij}Q = (-1)\delta_{ij}Q. \quad (27.30)$$

For every nonempty word w over the alphabet of index pairs,

$$\text{tr}(b^w) = (-1)^{|w|} \text{tr}(\delta^w),$$

and hence $\text{tr}(\widetilde{B}^w) = (-1)^{|w|} \text{tr}(\delta^w)$, where δ is the bond-one identity tensor of Definition 27.3.1.1.

Proof. ✓ Proof Both nonzero letters annihilate $(1, 1)^\top$, which spans the range of $\mathbb{1} - Q$; this gives $b^{ij}Q = b^{ij}$, and multiplying on the left by Q gives $Qb^{ij} = Qb^{ij}Q$. Identity (27.30) is a direct computation. Let $W_2 = \frac{1}{2}(1, -1)$ and $V_2 = (1, -1)^\top$. Then $V_2 W_2 = Q$ and $W_2 b^{ij} V_2 = -\delta_{ij}$, and Lemma 1.1.9 gives $\text{tr}(b^w) = \text{tr}((-\delta)^w) = (-1)^{|w|} \text{tr}(\delta^w)$. The last claim combines this with Theorem 27.3.2.6. $\square$

The review says that the projector Q gives an invariant subspace. Read on columns this fails, since $b^{00}(1, -1)^\top = (-2, 0)^\top$, and it is the range of $\mathbb{1} - Q$ that is invariant. The review's discussion of the canonical form notes that a nontrivial invariant subspace exists if and only if one exists for the action on rows, and the identity $Qb^{ij} = Qb^{ij}Q$ says exactly that the range of Q , read on rows, is invariant. In either reading Q cuts out the block $(-1)\delta_{ij}$, which is the canonical form the review states.

Theorem 27.3.2.8 (The global square law from the canonical form). Lean ✓ Stmt For every $N > 0$, the periodic operator generated by the stacked tensor $\widetilde{B}$ is $(-1)^N \mathbb{1}$.

Proof. ✓ Proof By Theorem 27.3.2.7, every word trace of $\widetilde{B}$ of positive length N is $(-1)^N$ times the corresponding word trace of δ . Lemma 25.6.1 converts this into the equality of periodic operators, and δ generates the identity. $\square$

This is the review's conclusion that the canonical form $(-1)\delta_{ij}$ globally means $O(A)^2 = (-1)^N \mathbb{1}$: the stacked product generates the square of the generated operators, so the statement agrees with Theorem 27.3.2.3, reached here through the two reduction steps instead of the explicit kernel. The main text of the review instead describes the one-dimensional block of the square as the identity operator. The block $(-1)\delta_{ij}$ generates $(-1)^N \mathbb{1}$, which is the identity for even N ; rescaling the tensor by a scalar c with $c^2 = -1$ turns the block into δ_{ij} at every length [42].

27.3.3 The decorated representative

Garre-Rubio and Schuch [166, Section III.D] write the CZX symmetry with an additional Pauli Z on every site, $U = \prod_i Z_i \prod_i \text{CZ}_{i,i+1} \prod_i X_i$, as a bond-two tensor built from delta tensors and the unnormalized Hadamard matrix. This representative has order two. Its fusion tensors reduce two stacked layers to one, and the two ways of fusing three layers differ by the sign $\omega = -1$ [166, Section III.A].

Definition 27.3.3.1 (The decorated CZX tensor). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [✔ Stmt](#)
The decorated CZX tensor is the bond-two tensor $\hat{T}$ with

$$\hat{T}^{01} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix},$$

$$\hat{T}^{10} = \begin{pmatrix} 0 & -1 \\ 0 & 1 \end{pmatrix},$$

and $\widehat{T}^{00} = \widehat{T}^{11} = 0$. Its entries are $\widehat{T}_{\ell r}^{ij} = \delta_{i,1-j} \delta_{r,i} (-1)^{\ell+i}$. Its matrices are the transposes of those of the bond-two tensor C of Definition 19.7.2.1, and $C^{ij} = (-1)^i A^{ij}$ with A the tensor of Definition 27.3.1. For a configuration $s \in \{0,1\}^N$ write $|s| = \sum_n s_n$, and let $Z^{\otimes N}$ be the diagonal operator with entries $(-1)^{|s|}$.

Theorem 27.3.3.2 (The decorated operator has order two). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) For every $N > 0$ and every $s, t \in \{0, 1\}^N$, the periodic operator $O_N(\widehat{T})$ satisfies

$$O_N(\widehat{T})(s, t) = \delta_{s, \bar{t}} (-1)^{q(s) + |s|}.$$

Thus $O_N(\widehat{T}) = Z^{\otimes N} O_N(A)$, $O_N(\widehat{T})$ is unitary, and

$$O_N(\hat{T})^2 = \mathbb{1}.$$

Proof. ✓ **Proof** The factor $\delta_{r,i}$ forces the bond configuration $g_n = s_{n-1}$, and the surviving product is $\prod_n (-1)^{s_{n-1}s_n+s_n} = (-1)^{q(s)+|s|}$ when $s = \bar{t}$. Comparison with (27.26) gives $O_N(\widehat{T}) = Z^{\otimes N} O_N(A)$. The operator is a monomial matrix with phases of modulus one, hence unitary. For the square, the phase at t is $(-1)^{q(\bar{t})+q(t)}(-1)^{|\bar{t}|+|t|}$; the first factor is $(-1)^N$ by (27.24), and the second is $(-1)^N$ because $|\bar{t}| + |t| = N$. $\square$

Theorem 27.3.3.3 (The decorated operator is a group-cocycle operator). [Lean] [✓ Stmt] For every $N > 0$, $O_N(\widehat{T})$ is the periodic operator of the group-cocycle construction for the generator s of $\mathbb{Z}_2$ and the nontrivial three-cocycle ω_1 .

Proof. ✓ Proof Both operators have the kernel

$$(s, t) \mapsto \delta_{s, \bar{t}} (-1)^{q(s) + |s|},$$

by Theorems 27.3.3.2 and 27.8.2.1.

Theorem 27.3.3.4 (Normality of the decorated tensor). Lean ✔ Stmt *Read over the alphabet of index pairs, the decorated tensor is normal: its words of length two span $M_5(\mathbb{C})$.*

✓ Proof

$$E_{00} = \frac{1}{2}(\widehat{T}^{01}\widehat{T}^{01} - \widehat{T}^{10}\widehat{T}^{01}).$$

$$E_{01} = -\frac{1}{2}(\widehat{T}^{01}\widehat{T}^{10} + \widehat{T}^{10}\widehat{T}^{01}).$$

$$E_{10} = \frac{1}{2}(\widehat{T}^{01}\widehat{T}^{01} + \widehat{T}^{10}\widehat{T}^{01}).$$

$$E_{11} = \frac{1}{2}(\hat{T}^{10}\hat{T}^{10} - \hat{T}^{01}\hat{T}^{10}).$$

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$$\hat{V}S^wV = \left(\prod_{n=1}^k \delta_{s_n t_n}\right) \mathbb{1}_1, \quad (27.31)$$

$$S^a S^w S^b = \left(\prod_{n=1}^k \delta_{s_n t_n} \right) S^a V \hat{V} S^b. \quad (27.32)$$

In particular $\hat{V}V = 1$ and $S^a S^b = S^a V \hat{V} S^b$.

✓ Proof

The source states (27.32) with at least one intermediate site; the case $k = 0$ is the identity $S^a S^b = S^a V \hat{V} S^b$.

Lean **✓ Stmt**

$$(\hat{V} \otimes \mathbb{1})R^{i_0j_0}R^{i_1j_1}R^{i_2j_2}(\mathbb{1} \otimes V) = -\hat{T}^{i_0j_0}\hat{T}^{i_1j_1}\hat{T}^{i_2j_2}.$$

That is, the two ways of fusing three layers on three sites differ by the sign $\omega = -1$.

✓ Proof

Theorem 27.3.3.10 (Domain walls between the two product states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every choice of action tensors on the product states $|0\rangle^{\otimes N}$ and $|1\rangle^{\otimes N}$, and every pair of domain walls e_{AB} and e_{BA} exchanged by the generator with phases c_{AB} and c_{BA} ,

$$c_{AB} c_{BA} = -1,$$

the anomaly of the decorated CZX symmetry seen on its domain walls [166, Section III]. For the action tensors of Theorem 27.3.3.9, the walls $e_{AB} = |0\rangle$ and $e_{BA} = -|1\rangle$ are exchanged with $c_{AB} = 1$ and $c_{BA} = -1$ (with the corrected left action vectors [80]), and $U|\psi(0-1-0)\rangle = -|\psi(1-0-1)\rangle$ on every chain with long enough regions.

Proof. [✓ Proof](#) By Theorem 19.5.5.15 with the fusion tensors of Theorem 27.3.3.8, $c_{AB}c_{BA} = \omega(s, s, s) \omega(s, e, s) = -1$. For the explicit walls write V_x and W_x for the left and right action vectors of s on $|x\rangle^{\otimes N}$, and $\tilde{A}_x = O_s \cdot A_x$ for the acted letters. For e_{AB} , with $V_0 = \langle 1|$ and $W_1 = -|\hat{+}\rangle$, each letter satisfies

$$\begin{aligned} A_1^j V_0 &= V_0 \tilde{A}_0^j, \\ \tilde{A}_1^j W_1 &= W_1 A_0^j, \\ V_0(O_s \cdot e_{AB})^i W_1 &= e_{BA}^i, \end{aligned}$$

so $V_0 \tilde{A}_0^{\mathbf{u}}(O_s \cdot e_{AB})^i \tilde{A}_1^{\mathbf{v}} W_1 = A_1^{\mathbf{u}} e_{BA}^i A_0^{\mathbf{v}}$ for all words, and $c_{AB} = 1$. For e_{BA} , the right action vector $W_0 = |\hat{+}\rangle$ and $W' = -|0\rangle + |1\rangle$ satisfy $\tilde{A}_0^j W_0 = W' A_1^j$ and $\tilde{A}_0^j W' = W' A_1^j$, so $\tilde{A}_0^j W_0 = W' A_1^j$ for every nonempty word $\mathbf{v}$. Together with $A_0^j V_1 = V_1 \tilde{A}_1^j$ for $V_1 = -\langle 0|$ and $V_1(O_s \cdot e_{BA})^i W' = -e_{AB}^i$, this gives $c_{BA} = -1$ with at least one site on the right of the wall. The last statement is Theorem 19.5.5.3. $\square$

Theorem 27.3.3.11 (Equal domain-wall phases and a string over a pair of walls). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) With the action tensors of Theorem 27.3.3.9 (with the corrected left action vectors [80]), the walls $e_{AB} = |0\rangle$ and $e_{BA} = i|1\rangle$ are exchanged by the generator with $c_{AB} = c_{BA} = i$, and the local action of the generator on an open chain containing the pair $|0\rangle, -|1\rangle$ with long regions is the phase -1 . These are the tensor identities behind the semionic exchange statistics of [166, Section III.F].

Proof. [✓ Proof](#) Rescale the wall $-|1\rangle$ of Theorem 27.3.3.10 by $1/i$, using Theorem 19.5.5.11 with $s = i$; the second statement is Theorem 19.5.5.4 with $c_{AB}c_{BA} = -1$. $\square$

Theorem 27.3.3.12 (Semionic exchange of domain-wall strings). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) With the action tensors and the domain walls $e_{AB} = |0\rangle$, $e_{BA} = -|1\rangle$ of Theorem 27.3.3.10, the vectors $\langle 0|$ and $\langle 1|$ are left inverses of the two product states with separated supports, and the domain-wall string $O^{[i,j]}$ creates the state $|\psi(0-1-0)\rangle$ from $|0\rangle^{\otimes N}$ on the periodic chain of every length $N \geq 2$, for all sites $i < j$. For sites $i_2 < i_1 < j_1 < j_2$ whose three inner regions are longer than a fixed buffer,

$$O^{[i_1, j_1]} O^{[i_2, j_2]} |0\rangle^{\otimes N} = -O^{[i_2, j_2]} O^{[i_1, j_1]} |0\rangle^{\otimes N},$$

the semionic statistics of the domain walls of [166, Section III.F].

Proof. [✓ Proof](#) The left inverses are checked letter by letter. The exchange is Theorem 19.5.5.7 with $c_{AB}c_{BA} = 1 \cdot (-1) = -1$. $\square$

Corollary 27.3.3.13 (No invariant normal state). [Lean](#) [✓ Stmt](#) *There is no normal matrix product state of positive bond dimension whose periodic vector is fixed by the decorated CZX operator on every nonempty chain.*

Proof. [✓ Proof](#) The bond-one identity tensor fixes every periodic vector, so such a state would be invariant under the whole representation of $\mathbb{Z}_2$. By Theorem 19.5.3.4 its anomaly class would then be trivial, contradicting Theorem 27.3.3.8. $\square$

27.3.4 Asymmetric compression of the square

By (27.23), the periodic operator satisfies $O_N(B) = U_N^2 = (-1)^N \mathbb{1}$, and $\text{tr}(B^w) = (-1)^{|w|} \text{tr}(\delta^w)$ for all nonempty words w . This periodic comparison with the bond-one identity tensor with weight -1 lies beyond the unit-weight setting. The sitewise theory of [164, Appendix A] imposes a closedness requirement for arbitrary boundary conditions. The following example distinguishes a word-level compression from a sitewise intertwiner.

For a word $w = ((i_1, j_1), \dots, (i_{|w|}, j_{|w|}))$, write $B^w = B^{i_1 j_1} \dots B^{i_{|w|} j_{|w|}}$, and use the same convention for the other tensors. The empty word evaluates to the identity matrix on the corresponding bond space.

Theorem 27.3.4.1 (Normality of the symmetry tensor and of the target). [Lean](#) [Lean](#) [✓ Stmt](#) *The bond-two symmetry tensor is normal: its length-two words span $M_2(\mathbb{C})$. The target $-\delta$ is normal: its bond dimension is one and one of its matrices is the nonzero scalar -1 .*

Proof. [✓ Proof](#) Let E_{ab} denote the matrix units of $M_2(\mathbb{C})$. Direct multiplication gives

$$E_{00} = \frac{1}{2}(T_0 T_0 + T_1 T_0).$$

$$E_{10} = \frac{1}{2}(T_0 T_0 - T_1 T_0).$$

$$E_{01} = \frac{1}{2}(T_0 T_1 - T_1 T_1).$$

$$E_{11} = \frac{1}{2}(T_0 T_1 + T_1 T_1).$$

Thus every matrix unit lies in the length-two word span. For the target, its matrix at the letter $(0, 0)$ is the nonzero scalar -1 , so Theorem I.2.4 applies with the single position of the one-by-one matrix algebra. $\square$

Theorem 27.3.4.2 (Compression of the squared symmetry). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The stacked product of Definition 27.3.1.1 carries the block triangular form of Theorem 25.5.1 for the single slot $-\delta$ with $z = 3$ zero slots and $4 = 1 + 3$. The compression pair is*

$$W = (0, 0, -1, 0).$$

$$V = (1, 1, -1, -1)^T.$$

It satisfies $WV = 1$ and $WB^wV = (-\delta)^w$ for every word. The remainder $R^{ij} = B^{ij} - V(-\delta)^{ij}W$ satisfies

$$R^{i_1 j_1} R^{i_2 j_2} R^{i_3 j_3} R^{i_4 j_4} = 0$$

for every choice of four index pairs.

Proof. ✓ **Proof** Set $k = (1, 0, 0, -1)^\top$ and $v = (1, 1, -1, -1)^\top$. The only nonzero letters of B are

$$B^{00} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}. \quad (27.33)$$

$$B^{11} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \quad (27.34)$$

Both annihilate k , their images lie in $\text{span}_{\mathbb{C}}\{k, v\}$, and $B^{00}v = -v$, $B^{11}v = 2k - v$. Choose a basis beginning with k, v and complete it with two further vectors. Every letter is upper triangular in this basis, with diagonal $(0, -\delta_{i,j}, 0, 0)$. A completion compatible with the displayed row W gives the compression pair by Theorem 1.1.2 applied to the second coordinate. The vanishing of a product of four remainder matrices is clause (vi) of Theorem 25.5.1 with $|S| + z = 4$. $\square$

Theorem 27.3.4.3 (Periodic data of the squared symmetry). Lean ✓ **Stmt** *For every nonempty word w over the alphabet of index pairs, $\text{tr}(B^w) = (-1)^{|w|} \text{tr}(\delta^w)$. This is $U_N^2 = (-1)^N \mathbb{1}$ at the level of periodic coefficients.*

Proof. ✓ **Proof** Proposition 25.5.3 applied to the compression of Theorem 27.3.4.2 gives $\text{tr}(B^w) = \text{tr}((-\delta)^w)$. Pulling the scalar out of each letter contributes $(-1)^{|w|}$. $\square$

Theorem 27.3.4.4 (Both sitewise intertwiner spaces vanish). Lean Lean ✓ **Stmt** *If $v \in \mathbb{C}^4$ satisfies $B^{ij}v = (-\delta)^{ij}v$ for every index pair, then $v = 0$. If a row vector $u \in \mathbb{C}^4$ satisfies $uB^{ij} = (-\delta)^{ij}u$ for every index pair, then $u = 0$.*

Proof. ✓ **Proof** Use the two nonzero matrices (27.33) and (27.34). The right-intertwiner equations are

$$(v_2, v_2, -v_2, -v_2)^\top = -(v_0, v_1, v_2, v_3)^\top,$$

and

$$(v_1, -v_1, v_1, -v_1)^\top = -(v_0, v_1, v_2, v_3)^\top.$$

They give $v_1 = v_2 = 0$ and then $v_0 = v_3 = 0$. For a left intertwiner, the equations are

$$(0, 0, u_0 + u_1 - u_2 - u_3, 0) = -u,$$

and

$$(0, u_0 - u_1 + u_2 - u_3, 0, 0) = -u.$$

The equation for B^{00} gives $u_0 = u_1 = u_3 = 0$, and the equation for B^{11} gives $u_2 = 0$. $\square$

Remark 27.3.4.5 (Failure of closure under conjugate transposition). Corollary 25.8.4 applies to Theorem 27.3.4.4: the algebra generated by the matrices of B is not closed under conjugate transposition in these coordinates, while the physical operator is unitary (Theorem 27.3.1.3). The absence of a sitewise intertwiner persists under an invertible change of bond coordinates. For the decorated tensor of Section 27.3.3, the anomaly class is computed in Theorem 27.3.3.8: it is nontrivial for every choice of fusion tensors. The same value $\omega(s, s, s) = -1$ is recovered in Theorem 27.3.3.9 as a ratio of the L-symbols of the two product states exchanged by the symmetry.

27.3.5 A non-simple family with an oscillating coefficient

Let M be the bond-two tensor of Definition 27.3.1.1, with the first physical index denoting the output. For a bit configuration $t = (t_0, \dots, t_{N-1})$, write $\bar{t}_n = 1 - t_n$ and $E(t) = \sum_{n=0}^{N-1} t_n t_{n+1}$ (with cyclic subscripts). By Theorem 27.3.1.2, the periodic operator acts as

$$U_N |t\rangle = (-1)^{E(t)} |\bar{t}\rangle. \quad (27.35)$$

In terms of the tensor entries,

$$\text{tr}(M^{s_0, t_0} \dots M^{s_{N-1}, t_{N-1}}) = \left(\prod_{n=0}^{N-1} \delta_{s_n, 1-t_n} \right) (-1)^{\sum_{n=0}^{N-1} t_{n-1} t_n} = \delta_{s, \bar{t}} (-1)^{E(t)}. \quad (27.36)$$

Here $\delta_{s, \bar{t}}$ is one exactly when the two configurations agree, and $I_N = \mathbb{1}^{\otimes N}$ denotes the identity on the N -qubit space. Theorems 27.3.1.3 and 27.3.1.4 give unitarity together with the square and adjoint

$$\begin{aligned} U_N^2 &= (-1)^N I_N, \\ U_N^\dagger &= U_N^{-1} = (-1)^N U_N. \end{aligned} \quad (27.37)$$

Define the operator family $O_N = I_N + U_N$. Its basis action is

$$O_N |t\rangle = |t\rangle + (-1)^{E(t)} |\bar{t}\rangle. \quad (27.38)$$

The identity and the CZX operator are generated by normal, inequivalent blocks of bond dimensions one and two. Their sum therefore has a non-simple matrix product representation of bond dimension three. Before considering its virtual compression, its physical square follows from (27.37):

$$O_N^2 = (1 + (-1)^N) I_N + 2 U_N. \quad (27.39)$$

The identity coefficient is two on even rings and zero on odd rings.

Lemma 27.3.5.1 (Projectors on even rings and inverses on odd rings). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
 $\checkmark$ *Stmnt* For even $N > 0$, $O_N/2$ is an orthogonal projector. For odd $N > 0$, O_N is invertible, with inverse $(I_N - U_N)/2$.

Proof. $\checkmark$ *Proof* For even N , unitarity and (27.37) give $U_N^\dagger = U_N$ and $U_N^2 = I_N$. Consequently,

$$\begin{aligned} \left(\frac{O_N}{2} \right)^\dagger &= \frac{O_N}{2}, \\ \left(\frac{O_N}{2} \right)^2 &= \frac{I_N + 2U_N + U_N^2}{4} = \frac{O_N}{2}. \end{aligned}$$

For odd N , $U_N^2 = -I_N$. Both factors are polynomials in U_N , and

$$(I_N + U_N) \frac{I_N - U_N}{2} = \frac{I_N - U_N^2}{2} = I_N = \frac{I_N - U_N}{2} (I_N + U_N).$$

□

At length zero, the unnormalized trace boundary assigns the scalar value $U_0 = [2]$ to the bond-two tensor and $O_0 = [3]$ to its direct sum with the bond-one identity, so the square law (27.37) holds only for positive lengths.

Definition 27.3.5.2 (The non-simple family and its square). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) The generating tensor is $N^{ij} = \delta^{ij} \oplus M^{ij}$ of bond dimension three, where δ^{ij} is the bond-one identity tensor. Here N^{ij} denotes a local matrix, while the integer N denotes the system size. The ordinary trace boundary gives

$$\mathrm{tr}(N^{s_0, t_0} \dots N^{s_{N-1}, t_{N-1}}) = \prod_{n=0}^{N-1} \delta_{s_n, t_n} + \mathrm{tr}(M^{s_0, t_0} \dots M^{s_{N-1}, t_{N-1}}) = O_N(s, t).$$

The source tensor is the stacked product of bond dimension nine,

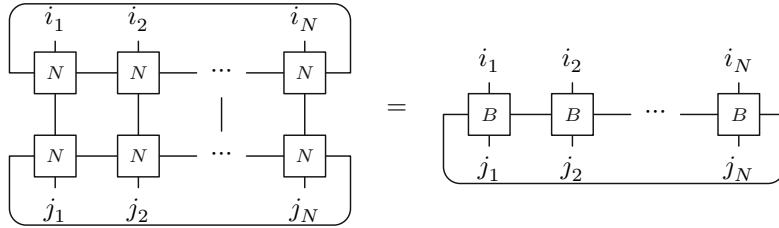
$$B^{ij} = \sum_{m=0}^1 N^{im} \otimes N^{mj}. \quad (27.40)$$

Its four target slots, indexed by $s = 0, 1, 2, 3$, are $C_0 = \delta$, $C_1 = -\delta$, $C_2 = M$, and $C_3 = M$, of bond dimensions 1, 1, 2, and 2, respectively. Each tensor is read over the alphabet of physical index pairs in the order $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$.

Stacking represents physical multiplication: expanding the intermediate physical configuration r and using multiplicativity of the Kronecker product and $\mathrm{tr}(A \otimes D) = \mathrm{tr}(A) \mathrm{tr}(D)$ gives

$$O_N^2(s, t) = \sum_r \mathrm{tr} \left(\prod_{n=0}^{N-1} N^{s_n, r_n} \right) \mathrm{tr} \left(\prod_{n=0}^{N-1} N^{r_n, t_n} \right) = \mathrm{tr} \left(\prod_{n=0}^{N-1} B^{s_n, t_n} \right). \quad (27.41)$$

All matrix products are ordered by increasing site index. Thus the periodic multiplication law is established independently of the asymmetric fundamental theorem.



Theorem 27.3.5.3 (Compression of the squared non-simple family). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
The source of Definition 27.3.5.2 carries the block triangular form of Theorem 25.5.1 for the four slots with $z = 3$ zero slots and $9 = 1 + 1 + 2 + 2 + 3$. Writing d_s for the dimension of C_s , the compression maps $W_s : \mathbb{C}^9 \rightarrow \mathbb{C}^{d_s}$ and $V_s : \mathbb{C}^{d_s} \rightarrow \mathbb{C}^9$ satisfy

$$W_s V_t = \begin{cases} \mathbb{1}_{d_s}, & s = t, \\ 0, & s \neq t. \end{cases}$$

For every word w over the alphabet of physical index pairs, $W_s B^w V_s = C_s^w$. The remainder $R^{ij} = B^{ij} - \sum_{s=0}^3 V_s C_s^{ij} W_s$ satisfies $R^w = 0$ whenever $|w| \geq 7$.

Proof. [✓ Proof](#) The splitting $\mathbb{C}^3 = \mathbb{C} \oplus \mathbb{C}^2$ induces a four-summand decomposition of $\mathbb{C}^3 \otimes \mathbb{C}^3$. After grouping these summands, each stacked letter has the form

$$B^{ij} \cong \delta^{ij} \oplus M^{ij} \oplus M^{ij} \oplus \left(\sum_{m=0}^1 M^{im} \otimes M^{mj} \right).$$

The fourth summand is the squared tensor of Section 27.3. Its flag contributes the target $-\delta$ and three zero slots. The other three summands already carry δ , M , and M .

Assemble the change of bond coordinates from these four summands. Checking block triangularity and the matched and unmatched diagonal blocks is a finite verification over integer matrices. Theorem 1.1.2 then gives the compression pairs. The dimension count and the nilpotency bound are clauses (vii) and (vi) of Theorem 25.5.1, with $|S| + z = 4 + 3 = 7$. $\square$

Theorem 27.3.5.4 (The oscillating structure coefficient). [Lean](#) [✓ Stmt](#) For every nonempty word w over the alphabet of index pairs,

$$\mathrm{tr}(B^w) = (1 + (-1)^{|w|}) \mathrm{tr}(\delta^w) + 2 \mathrm{tr}(M^w). \quad (27.42)$$

Proof. [✓ Proof](#) Proposition 25.5.3 applied to Theorem 27.3.5.3 gives, for nonempty w ,

$$\mathrm{tr}(B^w) = \sum_{s=0}^3 \mathrm{tr}(C_s^w) = \mathrm{tr}(\delta^w) + \mathrm{tr}((-\delta)^w) + 2 \mathrm{tr}(M^w).$$

Each letter of the second slot contributes a factor -1 , so $\mathrm{tr}((-\delta)^w) = (-1)^{|w|} \mathrm{tr}(\delta^w)$. This proves (27.42). $\square$

Taking $|w| = N > 0$, (27.41) identifies (27.42) with the physical square law (27.39). The compression resolves the oscillating integer coefficient into the signed weights $+1$ and -1 : the coefficient $1 + (-1)^N$ is their length- N power sum, as in Theorem 25.7.2. The two copies of M each have unit weight and contribute the constant coefficient two. This is the general power-sum mechanism, not an instance of the positive diagonal matrices in the characterization of matrix product density operator renormalization fixed points of [11, Theorem 4.14(ii)]. In particular, the even-ring projector property does not establish an MPDO renormalization-fixed-point condition.

27.3.6 Scalar three-cocycles on the group of order two

The scalar three-cocycles on the group of order two, whose two gauge classes are distinguished below, are the candidates for the anomaly class of a representation of that group; for the decorated CZX representation the class is the nontrivial one, by Theorem 27.3.3.8. For scalar associators, the three-cocycle and fusion-gauge conventions follow [166].

Definition 27.3.6.1 (Two scalar three-cochains). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Write $\mathbb{Z}_2 = \{e, s\}$ multiplicatively, with $s^2 = e$. Identify e, s with the bits 0, 1 only in exponents. For $p \in \{0, 1\}$, define

$$\omega_p(a, b, c) = (-1)^{pabc}. \quad (27.43)$$

Thus $\omega_p(s, s, s) = (-1)^p$, all other values are one, and ω_0 is the constant cochain one.

Theorem 27.3.6.2 (Normalization, cocycle equation, and unit modulus). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Each ω_p is normalized, has unit modulus, and satisfies, for all $a, b, c, d \in \mathbb{Z}_2$,

$$\omega_p(a, b, c) \omega_p(a, bc, d) \omega_p(b, c, d) = \omega_p(ab, c, d) \omega_p(a, b, cd). \quad (27.44)$$

Normalization means that its value is one whenever any argument is e .

Proof. [✓ Proof](#) An identity argument excludes the triple (s, s, s) , proving normalization. For the cocycle equation, check the two choices for each of a, b, c, d and for p , using $s^2 = e$. Every value is either one or a power of -1 , so its modulus is one. $\square$

Theorem 27.3.6.3 (Distinct scalar gauge classes). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The cochains ω_p and ω_q are cohomologous if and only if $p = q$, even when the scalar two-cochain implementing the gauge change is not required to be normalized. The inversion scalar $\sigma_\omega(g) = \omega(g, g^{-1}, g)$ therefore satisfies $\sigma_{\omega_p}(s) = (-1)^p$. For a normalized scalar two-cochain β , the fusion-gauge transform satisfies

$$(\beta \cdot \omega_p)(s, s, s) = (-1)^p, \quad (27.45)$$

where the convention is

$$(\beta \cdot \omega)(a, b, c) = \frac{\beta(b, c)\beta(a, bc)}{\beta(a, b)\beta(ab, c)}\omega(a, b, c). \quad (27.46)$$

Proof. [✓ Proof](#) Since $s^{-1} = s$, the inversion scalar equals $\omega_p(s, s, s)$. The criterion in Theorem 19.5.10.104 identifies triviality of the scalar gauge class with $\sigma_\omega(s) = 1$. Hence ω_1 is not gauge trivial. If $p \neq q$, one representative is $\omega_0 = 1$ and the other is ω_1 ; cohomology, using symmetry when needed, would contradict this nontriviality. If $p = q$, use reflexivity. Finally, substitute $s^2 = e$ in the fusion-gauge coboundary factor and use $\beta(e, s) = \beta(s, e) = 1$. $\square$

27.4 The anomalous $\mathbb{Z}_2$ symmetry of the Levin–Gu edge

The edge Hamiltonian of the Levin–Gu model, unitarily equivalent to the XX chain, commutes with the $\mathbb{Z}_2$ symmetry $U_N = \prod_n CZ_{n,n+1} \prod_n Z_n X_n$ on a ring of N qubits [179, Section II]. The source writes it as a bond-two matrix product operator, reduces the product of two copies to a bond-three tensor A_0 padded by a zero block, and states that A_0 is not injective, cannot be further decomposed into a direct sum of injective matrix product operators, and generates the identity on periodic chains. This section reads these statements through the multi-block asymmetric compression theorem (Theorem 25.5.1). The associator of the fusion tensors, $\omega(1, 1, 1) = -1$ in [179, Section II], is proved below. The unrestricted scalar relation on the full virtual space is corrected to a relation on the physical images, as explained in [45].

The first physical index is the output and the second the input, as in the source. For a word w in the index pairs, B^w denotes the product of the matrices of B along w , as in Section 27.3.

Definition 27.4.1 (The CZY tensor and its stacked square). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The CZY tensor A_1 has bond dimension two and the only nonzero matrices

$$A_1^{01} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

$$A_1^{10} = \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}.$$

Its stacked square is the bond-four tensor $\tilde{A}_0^{ik} = \sum_j A_1^{ij} \otimes A_1^{jk}$, so that $\tilde{A}_0^{00} = A_1^{01} \otimes A_1^{10}$, $\tilde{A}_0^{11} = A_1^{10} \otimes A_1^{01}$, and the two mixed matrices vanish [179, Section II]. The single target of its compression is the bond-one identity tensor $\delta^{ij} = \delta_{i,j}$.

The periodic operator U_N of A_1 has the matrix coefficients $U_N(s, t) = \text{tr}(A_1^{s_0 t_0} \dots A_1^{s_{N-1} t_{N-1}})$, the ordinary unnormalized trace:

$$\begin{array}{c} i \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} i \\ \text{---} \\ \text{---} \end{array} = \begin{array}{c} i \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{array}{c} i \\ \text{---} \\ \text{---} \end{array}$$

The next theorem states that the two stacked copies of A_1 contract, on every ring of $N \geq 1$ sites, to the identity:

$$\begin{array}{c}
 \begin{array}{c} i_0 \quad i_1 \quad \dots \quad i_{N-1} \\ \hline \begin{array}{c} \boxed{A_1} \quad \boxed{A_1} \quad \dots \quad \boxed{A_1} \\ \hline \boxed{A_1} \quad \boxed{A_1} \quad \dots \quad \boxed{A_1} \\ \hline k_0 \quad k_1 \quad \dots \quad k_{N-1} \end{array} \end{array}
 \end{array}
 =
 \begin{array}{c}
 i_0 \quad i_1 \quad \dots \quad i_{N-1} \\
 \hline
 \begin{array}{c} \vdots \quad \vdots \quad \dots \quad \vdots \\ \hline k_0 \quad k_1 \quad \dots \quad k_{N-1} \end{array}
 \end{array}$$

Theorem 27.4.4 (The stacked operator is the identity at every positive length). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [StmT](#) For every nonempty word w , $\text{tr}(\tilde{A}_0^w) = \text{tr}(\delta^w)$. Consequently $U_N^2 = \mathbb{1}$ for every $N \geq 1$, odd N included. At $N = 0$ the empty periodic contraction of the stacked square is $\text{tr}(\mathbb{1}_4) = 4$, so the restriction $N \geq 1$ is necessary.

Proof.  Proposition 25.5.3 applied to the compression of Theorem 27.4.3 gives the trace identity, and Lemma 25.6.1 turns it into the identity of periodic operators, since δ generates $\mathbb{1}$. The product formula of the operator of a stacked tensor gives $U_N^2 = \mathbb{1}$. $\square$

Theorem 27.4.5 (Nonsplitting of the stacked square). [Lean] [Lean] [Lean] [Lean] [Stmt] If $X \in \mathbb{C}^{4 \times 1}$ satisfies $\tilde{A}_0^{ij} X = X \delta^{ij}$ for every index pair, then $X = 0$; likewise for row vectors Y with $Y \tilde{A}_0^{ij} = \delta^{ij} Y$. Hence every multi-block compression of $\tilde{A}_0$ onto δ has a nonzero remainder, and the algebra generated by the matrices of $\tilde{A}_0$ is not closed under conjugate transposition.

Proof. ✓ Proof From $\widetilde{A}_0^{00}X = X$ the vector X is a multiple of e_1 , and $\widetilde{A}_0^{11}e_1 = -e_2$ forces the multiple to vanish. For rows, $Y\widetilde{A}_0^{00} = Y$ gives $Y = y_1u^\top$, and $Y\widetilde{A}_0^{11} = Y$ then gives $y_1 = 0$. A vanishing remainder would make V a sitewise intertwiner with $WV = 1$; the last assertion is Corollary 25.8.4. $\square$

Definition 27.4.6 (The reduced tensor). [Lean](#) [Lean](#) [✓ Stmt](#) The tensor A_0 of [179, Section II] has bond dimension three, vanishing mixed matrices, and

$$A_0^{00} = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$A_0^{11} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Theorem 27.4.7 (The printed change of basis). Lean Lean Lean Lean ✓ Stmt *The matrix*

$$X_{1,1} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ -1 & 0 & 0 & -1 \end{pmatrix}$$

of [179, Appendix B] is orthogonal, and $X_{1,1}\tilde{A}_0^{ij}X_{1,1}^{-1} = A_0^{ij} \oplus 0$ for every index pair, exactly as printed.

Proof. ✓ Proof Both identities are integer matrix identities after multiplying by $\sqrt{2}$ on each side: $(\sqrt{2}X_{1,1})(\sqrt{2}X_{1,1})^\top = 2I$ and $(\sqrt{2}X_{1,1})\tilde{A}_0^{ij}(\sqrt{2}X_{1,1})^\top = 2(A_0^{ij} \oplus 0)$. □

Theorem 27.4.8 (Compression of the reduced tensor). Lean Lean Lean ✓ Stmt *In its own coordinates, A_0 carries the block triangular form of Theorem 25.5.1 for the single slot δ with $z = 2$ zero slots, ordered as a zero slot, the slot δ , and a zero slot. The remainder R^{ij} of this compression satisfies*

$$R^{i_1j_1}R^{i_2j_2}R^{i_3j_3} = 0$$

for every choice of three index pairs, and the periodic operator of A_0 is the identity at every length $N \geq 1$.

Proof. ✓ Proof The matrices A_0^{00} and A_0^{11} are upper triangular with diagonal $(0, 1, 0)$, and the mixed matrices vanish, so the diagonal is $(0, \delta^{ij}, 0)$. The operator identity follows as in Theorem 27.4.4. □

Theorem 27.4.9 (The reduced tensor does not split). Lean Lean Lean Lean Lean ✓ Stmt *Both sitewise intertwiner spaces between A_0 and δ vanish. Hence every multi-block compression of A_0 onto δ has a nonzero remainder, the algebra generated by the matrices of A_0 is not closed under conjugate transposition, and in particular it is a proper subalgebra of the 3×3 matrices. In particular A_0 is not injective, as stated in [179, Section II].*

Proof. ✓ Proof For $X = (a, b, c)^\top$, $A_0^{00}X = X$ gives $c = 0$ and $a = -b$, and $A_0^{11}X = X$ then gives $b = 0$. For $Y = (p, q, r)$, $YA_0^{00} = Y$ gives $p = 0$ and $r = -q$, and $YA_0^{11} = Y$ gives $q = 0$. The remaining assertions follow as in Theorem 27.4.5; the full matrix algebra is closed under conjugate transposition. □

Theorem 27.4.10 (Indecomposability of the reduced virtual space). Lean Lean Lean ✓ Stmt *Every matrix commuting with A_0^{00} and A_0^{11} has the form $aI + bE_{13}$. The only idempotents in this commutant are 0 and I . Consequently, if $\mathbb{C}^3 = V \oplus W$ and both summands are invariant under the letters of A_0 , then $V = 0$ or $W = 0$. In particular, no simultaneous bond similarity decomposes A_0 into a direct sum of injective matrix product operators, as asserted in [179, Section II].*

Proof. ✓ Proof Comparing the entries of $PA_0^{00} = A_0^{00}P$ and $PA_0^{11} = A_0^{11}P$ gives $P = aI + bE_{13}$. Since $E_{13}^2 = 0$, the equation $P^2 = P$ gives $a^2 = a$ and $2ab = b$. Thus $a \in \{0, 1\}$ and $b = 0$. For an invariant decomposition $V \oplus W$, the projection onto V along W commutes with every letter and is idempotent. Its range is therefore zero or the whole space. Any nontrivial direct sum of injective blocks would give such an invariant decomposition; a single injective block is excluded by Theorem 27.4.9. □

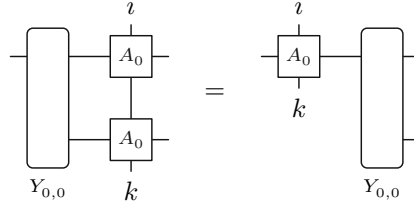
Theorem 27.4.11 (A one-sided fusion identity). Lean ✓ Stmt *Let $Y_{0,0}$ be the first three rows of the matrix $X_{0,0}$ of [179, Appendix B]:*

$$Y_{0,0} = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 2 & 0 \end{pmatrix}.$$

Then $Y_{0,0} \sum_j A_0^{ij} \otimes A_0^{jk} = A_0^{ik} Y_{0,0}$ for all i, k .

Proof. [✓ Proof](#) Direct multiplication of integer matrices. □

The identity of Theorem 27.4.11 moves $Y_{0,0}$ from the west of two stacked copies of A_0 to the east of one copy:



Definition 27.4.12 (Printed fusion maps for the nontrivial element). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) As in [179, Appendix B], $Y_{1,1}$ is the first three rows of $X_{1,1}$, and

$$Y_{0,1} = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 & 0 \end{pmatrix}, \quad Y_{1,0} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

Theorem 27.4.13 (Local fusion and the nontrivial triple). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The printed maps satisfy $Y_{a,b} \sum_j A_a^{ij} \otimes A_b^{jk} = A_{a+b}^{ik} Y_{a,b}$ for $(a,b) = (0,1), (1,0), (1,1)$. At the nontrivial triple, $Y_{0,1}(Y_{1,1} \otimes I_2) = -Y_{1,0}(I_2 \otimes Y_{1,1})$, with the normalization of [179, Section II].

Proof. [✓ Proof](#) Substitute the displayed matrices and the letters of A_0, A_1 . Each equality is a finite matrix identity; the common factor $1/\sqrt{2}$ cancels from the triple comparison. □

Theorem 27.4.14 (Associativity on the physical images). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Put $F_{0,0} = Y_{0,0}$, $F_{0,1} = 2Y_{0,1}$, $F_{1,0} = 2Y_{1,0}$ and $F_{1,1} = 2\sqrt{2}Y_{1,1}$. For $a, b, c \in \mathbb{Z}_2$, define $T^{ik} = \sum_{j,l} A_a^{ij} \otimes A_b^{jl} \otimes A_c^{lk}$, $L = F_{a+b,c}(F_{a,b} \otimes I_c)$ and $R = F_{a,b+c}(I_a \otimes F_{b,c})$, identifying both bond orders. Then $LT^{ik} = (-1)^{abc}RT^{ik}$ for every physical pair (i,k) . More generally, $LT^{p_1} \dots T^{p_n} = (-1)^{abc}RT^{p_1} \dots T^{p_n}$ for every nonempty physical word. This is the corrected relation on the physical images of [45]. On the full virtual space the printed neutral fusions admit no scalar comparison: the entry in row one and column $(3,3,1)$ is two on the left and zero on the right.

Proof. [✓ Proof](#) For the chosen normalization all matrices are integral. Direct multiplication proves the identity for the eight group triples and the four physical pairs. Multiplication on the right by the remaining letters gives the nonempty-word identity. The specified entry proves the full-space obstruction. □

27.5 Kramers–Wannier duality

Kramers–Wannier duality originates in the two-dimensional classical Ising model, where it relates the high- and low-temperature regimes [178]; in the quantum transverse-field Ising chain it exchanges the bond and field terms of the Hamiltonian. On a periodic chain the duality is a noninvertible operator, rather than a unitary change of variables on the full Hilbert space. Aasen, Mong and Fendley realize it as a lattice defect [1]; Seiberg and Shao place it in a noninvertible symmetry algebra together with translation and the spin flip [206]; Lootens, Delcamp, Ortiz and Verstraete write it as a matrix product operator intertwiner between the two Hamiltonians [180]. The review discusses this operator among its matrix product operator symmetries [16,

Section III.A.6]; its relation to fusion algebras is Remark 26.3.4. Here the duality is built from an explicit two-by-two tensor whose periodic operator is unnormalized; it reproduces the primal-to-dual kernel of [1] up to the factor $2^{N/2}$. Its square and its action on ground states are characterized both as operator identities and as asymmetric compressions of stacked tensors.

Definition 27.5.1 (The duality kernel and the two translation tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) With the outgoing label s' written first, the duality tensor is

$$\begin{aligned} W^{00} &= \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, & W^{01} &= \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \\ W^{10} &= \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, & W^{11} &= \begin{pmatrix} 0 & 0 \\ -1 & 1 \end{pmatrix}. \end{aligned}$$

The translation tensor is $T^{s's} = E_{s,s'}$ and the flipped translation tensor is $(\eta T)^{s's} = E_{s,1-s'}$, where E_{pq} is the matrix unit with a single entry 1 in row p and column q . The same symbols T and ηT denote their periodic operators. The stacked product tensor is $B^{s's} = \sum_{m=0}^1 W^{s'm} \otimes W^{ms}$, of bond dimension four. Its two target tensors are $2T$ and $2\eta T$.

Close the virtual bonds by the ordinary trace on a ring of $N > 0$ sites, without dividing by the bond dimension. The resulting operator is $K_N(a, b) = \text{tr}(W^{a_0 b_0} \dots W^{a_{N-1} b_{N-1}})$. Its normalization relative to the source defect follows from Theorem 27.5.1.1 and Corollary 27.5.1.7. Theorem 27.5.3.2 proves that W is normal, and Theorem 27.5.1.12 that T and ηT are; Corollary 27.5.1.4 gives the exchange of Ising couplings, and Lemma 27.5.2.1 describes its action on product and GHZ states. The compression of its square is Theorem 27.5.1.13.

Consider $N \geq 1$ qubits on a periodic chain, with sites $I_N = \{0, \dots, N-1\}$ and all site indices understood modulo N . The Hilbert space $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$ has computational basis $\{|b\rangle : b \in \{0, 1\}^{I_N}\}$. Write $\bar{b}$ for the componentwise complement of b , and $b \oplus e_j$ for the configuration obtained by flipping its bit at site j . The Pauli operators act by

$$\begin{aligned} X_j |b\rangle &= |b \oplus e_j\rangle, \\ Z_j |b\rangle &= (-1)^{b_j} |b\rangle. \end{aligned} \tag{27.48}$$

They are self-adjoint involutions, satisfy $X_j Z_j = -Z_j X_j$, and commute at distinct sites. Define global spin flip η and one-site left translation T by

$$\begin{aligned} \eta |b\rangle &= |\bar{b}\rangle, \\ T |b\rangle &= |\tau b\rangle, \quad (\tau b)_j = b_{j+1}. \end{aligned} \tag{27.49}$$

Thus $\eta = \prod_j X_j$, $\eta^2 = \mathbb{1}$, $T^N = \mathbb{1}$, $T^\dagger = T^{-1}$, and $T\eta = \eta T$. In this convention, $TX_j T^{-1} = X_{j-1}$ and $TZ_j T^{-1} = Z_{j-1}$.

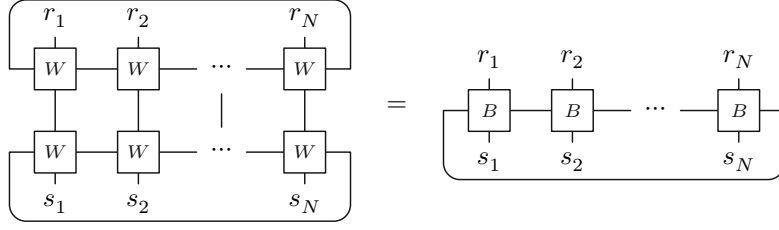
For real couplings J, h , the periodic transverse-field Ising Hamiltonian is

$$H_N(J, h) = -J \sum_{j \in I_N} Z_j Z_{j+1} - h \sum_{j \in I_N} X_j. \tag{27.50}$$

The sum includes one bond per site. In particular, the bond at $N = 1$ is $Z_0^2 = \mathbb{1}$, and at $N = 2$ the two bond terms both equal $Z_0 Z_1$. The duality exchanges the bond terms with the field terms.

27.5.1 The duality kernel and its square

The translation tensor is the shift matrix product unitary of [16, Appendix A, “The shift MPU”], taken with the opposite direction of translation, and the flipped translation tensor composes it with the global spin flip.



The site labels in the diagrams run from 1 to N , a cyclic relabeling of I_N .

Operator identities

Define the raw periodic operator K_N by the ordinary virtual trace, without dividing by the bond dimension:

$$(K_N)_{ab} = \text{tr}(W^{a_0 b_0} \dots W^{a_{N-1} b_{N-1}}). \quad (27.51)$$

We also write $D = K_N$ for this raw operator, retaining D in the square identity below. The normalized operators will be denoted $\mathcal{D}_N$ and U_N , not D .

Theorem 27.5.1.1 (Periodic kernel). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

For all configurations s', s ,

$$\langle s' | D | s \rangle = \prod_{j \in I_N} (-1)^{s'_j(s_j + s_{j+1})}. \quad (27.52)$$

Proof. [✓ Proof](#) The local entries are $(W^{ab})_{uv} = \delta_{u,b}(-1)^{a(b+v)}$. Expanding the trace over closed virtual configurations g gives

$$(K_N)_{ab} = \sum_{g \in \{0,1\}^{I_N}} \prod_{j \in I_N} \delta_{g_j, b_j} (-1)^{a_j(b_j + g_{j+1})} = \prod_{j \in I_N} (-1)^{a_j(b_j + b_{j+1})}.$$

Only the configuration $g = b$ contributes. □

For subsequent calculations, put $q(a, b) = \sum_j a_j(b_j + b_{j+1})$, so $(K_N)_{ab} = (-1)^{q(a,b)}$. Exponents may be calculated modulo two. Unless indicated otherwise, site sums and products run over I_N .

Theorem 27.5.1.2 (Spin-flip absorption and noninvertibility). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) For every $N > 0$, the raw operator satisfies $K_N \eta = \eta K_N = K_N$. If $\eta v = -v$, then $K_N v = 0$. In particular, K_N is not invertible.

Proof. [✓ Proof](#) Complementing either input or output leaves the kernel unchanged: $q(a, \bar{b}) \equiv q(\bar{a}, b) \equiv q(a, b) \pmod{2}$. This proves both absorption identities. For an odd vector, $K_N v = K_N \eta v = -K_N v$, so $K_N v = 0$. The nonzero vector $|0\rangle^{\otimes N} - |1\rangle^{\otimes N}$ has odd spin flip and witnesses noninvertibility. □

Lemma 27.5.1.3 (Local duality relations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every site j ,

$$K_N X_j = Z_{j-1} Z_j K_N, \quad (27.53)$$

$$K_N Z_j Z_{j+1} = X_j K_N. \quad (27.54)$$

Proof. ✓ Proof Flipping one input or output bit gives

$$\begin{aligned} q(a, b \oplus e_j) &\equiv q(a, b) + a_{j-1} + a_j \pmod{2}, \\ q(a \oplus e_j, b) &\equiv q(a, b) + b_j + b_{j+1} \pmod{2}. \end{aligned}$$

These congruences prove (27.53) and (27.54) entrywise. They remain valid when the two indicated sites coincide. $\square$

Under the identification of the dual lattice described after Corollary 27.5.1.7, these are the two local relations of the duality defect in [1].

Corollary 27.5.1.4 (Coupling exchange). Lean Lean Lean ✓ Stmt *The Hamiltonians with exchanged couplings satisfy*

$$K_N H_N(J, h) = H_N(h, J) K_N. \quad (27.55)$$

In particular, K_N commutes with $H_N(J, J)$.

Proof. ✓ Proof Summing the local relations of Lemma 27.5.1.3 and reindexing the bonds yields

$$K_N H_N(J, h) = \left(-J \sum_j X_j - h \sum_j Z_{j-1} Z_j \right) K_N = H_N(h, J) K_N.$$

$\square$

The local identities also exchange strings and endpoint operators. For consecutive sites $j_r = j_0 + r$ and $1 \leq m \leq N$,

$$\begin{aligned} K_N \prod_{r=0}^{m-1} X_{j_r} &= Z_{j_0-1} Z_{j_{m-1}} K_N, \\ K_N Z_{j_0} Z_{j_m} &= \left(\prod_{r=0}^{m-1} X_{j_r} \right) K_N. \end{aligned} \quad (27.56)$$

Indeed, iterating (27.53) and (27.54) cancels every interior factor Z_j^2 . At $m = N$ the endpoints coincide, and these relations give $K_N \eta = \eta K_N = K_N$. They are intertwining identities; no inverse of K_N is involved.

Lemma 27.5.1.5 (Adjoint and Gram matrices). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt *The raw operator satisfies*

$$\begin{aligned} T K_N &= K_N T, \\ K_N^\dagger &= T^{-1} K_N, \end{aligned} \quad (27.57)$$

and

$$K_N^\dagger K_N = K_N K_N^\dagger = 2^N (\mathbb{1} + \eta). \quad (27.58)$$

Proof. ✓ Proof Cyclic reindexing gives $q(\tau a, \tau b) = q(a, b)$ and $q(b, a) = q(\tau a, b)$. Since the kernel is real, these identities imply (27.57).

For a binary configuration c , the character sum factors as

$$\sum_a (-1)^{\sum_j a_j c_j} = \prod_j (1 + (-1)^{c_j}) = \begin{cases} 2^N, & c = \mathbf{0}, \\ 0, & c \neq \mathbf{0}. \end{cases} \quad (27.59)$$

Here $\mathbf{0}$ and $\mathbf{1}$ denote the two constant configurations. For input configurations b, c , put $d = b \oplus c$. Then

$$(K_N^\dagger K_N)_{bc} = \sum_a (-1)^{\sum_j a_j (d_j + d_{j+1})} = 2^N (\delta_{b,c} + \delta_{b,\bar{c}}).$$

Equation (27.59) imposes $d_j = d_{j+1}$ at every site. Such a configuration is constant on the cycle, and the two constant configurations are distinct because $N \geq 1$. For the other Gram matrix, put $d = a \oplus c$ and sum over inputs:

$$(K_N K_N^\dagger)_{ac} = \sum_b (-1)^{\sum_j b_j (d_j + d_{j-1})} = 2^N (\delta_{a,c} + \delta_{a,\bar{c}}).$$

This proves (27.58). $\square$

Define the spin-flip sectors $\mathcal{H}_N^\pm = \{v : \eta v = \pm v\}$ and their orthogonal projections $P_\pm = (\mathbb{1} \pm \eta)/2$. Two useful normalizations of the duality operator are

$$\begin{aligned} \mathcal{D}_N &= 2^{-N/2} K_N, \\ U_N &= 2^{-(N+1)/2} K_N = \mathcal{D}_N / \sqrt{2}. \end{aligned} \quad (27.60)$$

Lemma 27.5.1.6 (Kernel, image, and partial isometry). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) The kernel and image of the raw duality operator are

$$\begin{aligned} \ker K_N &= \mathcal{H}_N^-, \\ \text{ran } K_N &= \mathcal{H}_N^+, \\ \text{rank } K_N &= 2^{N-1}. \end{aligned} \quad (27.61)$$

The normalized operators satisfy

$$\begin{aligned} \mathcal{D}_N^\dagger \mathcal{D}_N &= \mathcal{D}_N \mathcal{D}_N^\dagger = 2P_+, \\ U_N^\dagger U_N &= U_N U_N^\dagger = P_+. \end{aligned} \quad (27.62)$$

Thus U_N is a partial isometry with initial and final projection P_+ . It vanishes on $\mathcal{H}_N^-$ and restricts to a unitary operator on $\mathcal{H}_N^+$.

Proof. [✓ Proof](#) If $K_N v = 0$, then (27.58) gives $0 = K_N^\dagger K_N v = 2^N (v + \eta v)$, so $\eta v = -v$. Conversely, Theorem 27.5.1.2 shows that K_N annihilates $\mathcal{H}_N^-$. The absorption relation $\eta K_N = K_N$ puts the image in $\mathcal{H}_N^+$. Conversely, for $\eta y = y$, $K_N (2^{-(N+1)} K_N^\dagger y) = 2^{-1} (y + \eta y) = y$, so every vector in that sector is in the image. For the rank, $\text{rank } K_N = \text{rank } K_N^\dagger K_N = \text{rank } P_+$, and the rank of the orthogonal projection P_+ is its trace. Since η fixes no basis vector, $\text{tr } \eta = 0$ and $\text{tr } P_+ = 2^N / 2 = 2^{N-1}$. Rescaling (27.58) proves (27.62). $\square$

Each $H_N(J, h)$ preserves both spin-flip sectors: the field terms commute with η , and the two signs acquired by a bond term cancel. Thus U_N implements a unitary equivalence between $H_N(J, h)$ and $H_N(h, J)$ on $\mathcal{H}_N^+$, but not on the full Hilbert space.

and

$$\begin{aligned}
(U_N^\dagger)^2 &= P_+ T^{-1}, \\
U_N^\dagger \eta &= \eta U_N^\dagger = U_N^\dagger, \\
T^{-1} U_N^\dagger &= U_N^\dagger T^{-1}, \\
(U_N^\dagger)^\dagger &= U_N^\dagger T, \\
U_N^\dagger U_N &= U_N U_N^\dagger = P_+, \\
(U_N^\dagger)^{2N} &= P_+.
\end{aligned} \tag{27.67}$$

Proof. [✓ Proof](#) Transposing (27.54) at site j and (27.53) at site $j+1$ gives (27.66), because X_j and $Z_j Z_{j+1}$ are real symmetric. Since the kernel is real, (27.57) gives $K_N^\top = K_N T^{-1} = T^{-1} K_N$, so $(K_N^\top)^2 = K_N^\top K_N T^{-1}$, and (27.58) gives the square. The absorption and commutation relations follow from Theorem 27.5.1.2 and $T K_N = K_N T$, and the adjoint relation from $K_N = K_N^\top T$. The two products with the adjoint are (27.58) rescaled. Finally, P_+ commutes with T , $T^N = \mathbb{1}$ and $P_+^2 = P_+$, so $(U_N^\dagger)^{2N} = P_+^N T^{-N} = P_+$. $\square$

After exchanging the Pauli operators X and Z at every site, the relations (27.66) and (27.67) are those that Ref. [206] prints for its noninvertible translation, with its translation read as T^{-1} and its projection $(1 + \eta)/2$ read as P_+ . The proposition shows that $U_N^\dagger$ satisfies these relations. That reference defines its operator by an explicit product of local factors; Proposition 27.5.1.11 below shows that, after the exchange of X and Z , this product is exactly $U_N^\dagger$, with no scalar factor. The noninvertibility here concerns the periodic bosonic spin chain, not fermionic Majorana translation. The translation in (27.63) must be retained when comparing this lattice relation with a size-independent fusion algebra.

Definition 27.5.1.10 (Circuit of the noninvertible translation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let $H^{\otimes N}$ be the Hadamard gate at every site, $\langle c | H^{\otimes N} | b \rangle = 2^{-N/2} (-1)^{\sum_j c_j b_j}$, and let $\eta_Z = \prod_j Z_j$. Following [206], with its sites $1, \dots, N$ relabeled $0, \dots, N-1$, put $d_j^z = (\mathbb{1} + iZ_j)/\sqrt{2}$, $d_j^x = (\mathbb{1} + iX_j X_{j+1})/\sqrt{2}$, and

$$C_N = e^{-2\pi i N/8} (d_0^z d_0^x) (d_1^z d_1^x) \cdots (d_{N-2}^z d_{N-2}^x) d_{N-1}^z \frac{\mathbb{1} + \eta_Z}{2}. \tag{27.68}$$

The factors d_j^x run over $0 \leq j \leq N-2$; the bond between the sites $N-1$ and 0 does not occur.

Proposition 27.5.1.11 (The circuit is the rotated adjoint duality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $N \geq 1$,

$$H^{\otimes N} C_N H^{\otimes N} = U_N^\dagger. \tag{27.69}$$

Consequently, for every site j , including $j = N-1$, whose bond $X_{N-1} X_0$ does not occur in (27.68),

$$\begin{aligned}
C_N Z_j &= X_j X_{j+1} C_N, \\
C_N X_j X_{j+1} &= Z_{j+1} C_N,
\end{aligned}$$

and

$$\begin{aligned}
C_N \eta_Z &= \eta_Z C_N = C_N, \\
C_N C_N^\dagger &= C_N^\dagger C_N = \frac{\mathbb{1} + \eta_Z}{2}, \\
C_N^2 &= \frac{\mathbb{1} + \eta_Z}{2} T^{-1}.
\end{aligned}$$

Here T^{-1} sends $|b_0, \dots, b_{N-1}\rangle$ to $|b_{N-1}, b_0, \dots, b_{N-2}\rangle$, which is the action of the product of nearest-neighbor swaps by which [206] defines its translation, and T^{-1} commutes with $H^{\otimes N}$. $C_N v = 0$ exactly when $\eta_Z v = -v$, and y lies in the range of C_N exactly when $\eta_Z y = y$: the kernel is the sector $\eta_Z = -1$, and on the sector $\eta_Z = +1$ the circuit acts unitarily, as stated in [206].

Proof. ✓ Proof The Hadamard gate is its own inverse and exchanges X_j and Z_j . Conjugation therefore maps d_j^z to $(\mathbb{1} + iX_j)/\sqrt{2}$, d_j^x to $(\mathbb{1} + iZ_j Z_{j+1})/\sqrt{2}$, and η_Z to η . In this frame the gate $(\mathbb{1} + iX_k)/\sqrt{2}$ changes only the bit at site k , and each site carries exactly one such gate. An entry $\langle c|G|b\rangle$ of the product G of the gates, before the projection, is therefore a single product along one path of configurations; at the diagonal gate of the bond $(k, k+1)$ the two bits are b_k and c_{k+1} . Writing $[x \neq y]$ for the indicator of $x \neq y$, this gives

$$\begin{aligned} \langle c|G|b\rangle &= \frac{i^{[c_{N-1} \neq b_{N-1}]}}{\sqrt{2}} \prod_{k=0}^{N-2} w(c_k, b_k, c_{k+1}), \\ w(x, y, z) &= \frac{i^{[x \neq y]}}{\sqrt{2}} \frac{1 + i(-1)^{y+z}}{\sqrt{2}} = \frac{1+i}{2} i^x (-i)^z (-1)^{y(x+z)}. \end{aligned}$$

The factors $i^{c_k}(-i)^{c_{k+1}}$ telescope to $i^{c_0}(-i)^{c_{N-1}}$. The projection adds the entry at $\bar{b}$, whose open-chain signs differ by $(-1)^{c_0+c_{N-1}}$, and $e^{-2\pi i N/8} = ((1-i)/\sqrt{2})^N$. Checking the eight values of (c_0, b_{N-1}, c_{N-1}) shows that the remaining boundary factor is $(-1)^{b_{N-1}(c_{N-1}+c_0)}$, the sign of the closing bond. By Theorem 27.5.1.1 the entry is $2^{-(N+1)/2} (K_N)_{bc} = \langle c|U_N^\dagger|b\rangle$, which proves (27.69). The remaining relations are those of Proposition 27.5.1.9 conjugated by the Hadamard gate, which commutes with T^{-1} . If $C_N v = 0$, then $C_N^\dagger C_N v = \frac{1}{2}(v + \eta_Z v) = 0$; conversely, if $\eta_Z v = -v$, then $C_N v = C_N \eta_Z v = -C_N v$. The range of C_N lies in the sector $\eta_Z = +1$ because $\eta_Z C_N = C_N$, and for $\eta_Z y = y$ one has $C_N(2C_N^\dagger y) = y + \eta_Z y = 2y$. $\square$

Positive length is essential in these calculations. At $N = 1$, $K_1 = \mathbb{1} + X_0$ and $U_1 = P_+$, consistently with the self-bond convention. The empty virtual word instead has trace $\text{tr } \mathbb{1}_2 = 2$, so extending the raw trace definition to $N = 0$ would give $K_0 = 2$. Its square would be 4, whereas the right-hand side of (27.63) would be 2.

Asymmetric compression

The square identity (27.63) has a virtual decomposition, obtained by applying the multi-block compression theorem to the stacked tensor B of Definition 27.5.1. For the matrix-unit tensors of that definition, expanding the cyclic trace gives

$$\begin{aligned} \text{tr}(T^{a_0 b_0} \dots T^{a_{N-1} b_{N-1}}) &= \prod_j \delta_{a_j, b_{j+1}} = T_{ab}, \\ \text{tr}((\eta T)^{a_0 b_0} \dots (\eta T)^{a_{N-1} b_{N-1}}) &= \prod_j \delta_{a_j, 1-b_{j+1}} = (\eta T)_{ab}. \end{aligned}$$

The stacked tensor B generates K_N^2 : multiplication of periodic operators sums the intermediate physical index at every site.

Normality of a tensor below means that words of some fixed positive length span its virtual matrix algebra. This differs from operator normality in (27.58) and does not impose a normalization on the spectral radius of a transfer map.

Theorem 27.5.1.12 (Normality of the two translation tensors). Lean Lean ✓ Stmt *The translation tensor and the flipped translation tensor are normal: the four matrices of each are the four matrix units of $M_2(\mathbb{C})$, which already span $M_2(\mathbb{C})$ at word length one.*

Proof. [✓ Proof](#) Every matrix unit occurs among the four matrices of each tensor, and the matrix units span $M_2(\mathbb{C})$. Each tensor is therefore injective and hence normal. $\square$

Theorem 27.5.1.13 (Split compression of the squared duality). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The stacked tensor of Definition 27.5.1 has the block triangular form of Theorem 25.5.1 for the two targets $C_1 = 2T$ and $C_2 = 2\eta T$, with no zero slots and $4 = 2 + 2$. There are compression maps $W_s : \mathbb{C}^4 \rightarrow \mathbb{C}^2$ and $V_s : \mathbb{C}^2 \rightarrow \mathbb{C}^4$, indexed here by $s = 1, 2$, such that $W_s V_t = \delta_{st} \mathbb{1}_2$ and $W_s B^w V_s = C_s^w$ for every word w . The remainder vanishes:*

$$R^{ab} = B^{ab} - \sum_{s=1}^2 V_s C_s^{ab} W_s = 0.$$

Thus the extension splits.

Proof. [✓ Proof](#) The change of bond coordinates is

$$G = \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad (27.70)$$

whose inverse has integer entries. For each physical pair (a, b) , direct multiplication gives

$$GB^{ab}G^{-1} = (2T^{ab}) \oplus (2(\eta T)^{ab}).$$

After clearing the denominator, these are equalities of integer matrices. Block diagonality gives the triangularity condition and the vanishing remainder. The coordinate projections and inclusions give the compression pairs by Theorem 1.1.2; the dimension count is clause (vii) of Theorem 25.5.1. $\square$

Theorem 27.5.1.14 (Periodic data of the squared duality). [Lean](#) [✓ Stmt](#) *For every nonempty word w over the alphabet of index pairs,*

$$\mathrm{tr}(B^w) = 2^{|w|} (\mathrm{tr}(T^w) + \mathrm{tr}((\eta T)^w)). \quad (27.71)$$

This is (27.63) at the level of periodic coefficients.

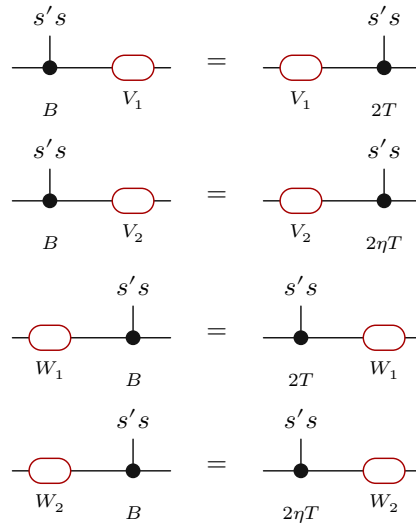
Proof. [✓ Proof](#) Proposition 25.5.3 applied to the compression of Theorem 27.5.1.13 gives $\mathrm{tr}(B^w) = \mathrm{tr}((2T)^w) + \mathrm{tr}((2\eta T)^w)$. Pulling the weight 2 out of each word evaluation contributes the factor $2^{|w|}$. $\square$

Theorem 27.5.1.15 (The sitewise intertwiners of the split case). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Write*

$$\begin{aligned} V_1 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}, & V_2 &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & 0 \\ 0 & 1 \end{pmatrix}, \\ W_1 &= \frac{1}{2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, & W_2 &= \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}. \end{aligned}$$

Then $W_1 V_1 = W_2 V_2 = \mathbb{1}_2$, and for every index pair, $B^{s's} V_1 = V_1 (2T)^{s's}$, $B^{s's} V_2 = V_2 (2\eta T)^{s's}$, $W_1 B^{s's} = (2T)^{s's} W_1$ and $W_2 B^{s's} = (2\eta T)^{s's} W_2$.

Proof. ✓ Proof The matrices W_1, W_2 are the two row blocks of G , and V_1, V_2 are the corresponding column blocks of G^{-1} , as in Theorem I.1.2. Each intertwining identity is verified by multiplying the explicit matrices and clearing the denominator. The two normalizations also follow from clause (iv) of Theorem 25.5.1. $\square$



27.5.2 Action on ground states

The product state and the Greenberger–Horne–Zeiling (GHZ) state [168, 167] give elementary finite-chain examples of the action. Use the unnormalized vectors

$$p_N = (|0\rangle + |1\rangle)^{\otimes N} = \sum_b |b\rangle, \quad (27.72)$$

$$g_N = |\mathbf{0}\rangle + |\mathbf{1}\rangle.$$

For $h > 0$, p_N is proportional to the unique ground state of $H_N(0, h)$. For $J > 0$ and $N \geq 2$, the ground space of $H_N(J, 0)$ is $\text{span}_{\mathbb{C}}\{|\mathbf{0}\rangle, |\mathbf{1}\rangle\}$, and g_N spans its spin-flip-even subspace. These facts follow respectively by minimizing each field term $-hX_j$ and each diagonal bond term $-JZ_jZ_{j+1}$. They do not require a thermodynamic limit.

Lemma 27.5.2.1 (Product-state and GHZ actions). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For every $N \geq 1$,

$$\begin{aligned} K_N p_N &= 2^N g_N, \\ K_N |\mathbf{0}\rangle &= K_N |\mathbf{1}\rangle = p_N, \\ K_N g_N &= 2p_N. \end{aligned} \quad (27.73)$$

Writing $\hat{p}_N = 2^{-N/2} p_N$ and $\hat{g}_N = 2^{-1/2} g_N$, one has $U_N \hat{p}_N = \hat{g}_N$ and $U_N \hat{g}_N = \hat{p}_N$.

Proof. ✓ Proof Reindexing the exponent and using (27.59) gives

$$(K_N p_N)_a = \sum_b (-1)^{\sum_j b_j (a_j + a_{j-1})} = 2^N (\delta_{a, \mathbf{0}} + \delta_{a, \mathbf{1}}).$$

Both constant inputs have phase one against every output configuration, so each maps to p_N . Finally, $\|p_N\|^2 = 2^N$ and $\|g_N\|^2 = 2$ give the normalized statements. $\square$

The finite-chain actions in (27.73) have different virtual decompositions. The product-state action splits into the two GHZ sectors with a local weight 2. The GHZ action has two unweighted copies of the same product-state target, together with a nonzero remainder. The product-state and GHZ tensors are the elementary matrix product state examples of [16, Appendix A]; here their explicit matrices allow the asymmetric compression and its sitewise intertwiners to be compared directly.

Definition 27.5.2.2 (The two action tensors). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) The product-state tensor A_+ has bond dimension one and $A_+^0 = A_+^1 = (1)$. Its periodic vector is p_N . The two ferromagnetic-sector tensors have $C_0^i = (\delta_{i,0})$ and $C_1^i = (\delta_{i,1})$. Their direct sum is the GHZ tensor $A_{\text{GHZ}}^0 = \text{diag}(1, 0)$, $A_{\text{GHZ}}^1 = \text{diag}(0, 1)$, whose periodic vector is g_N .

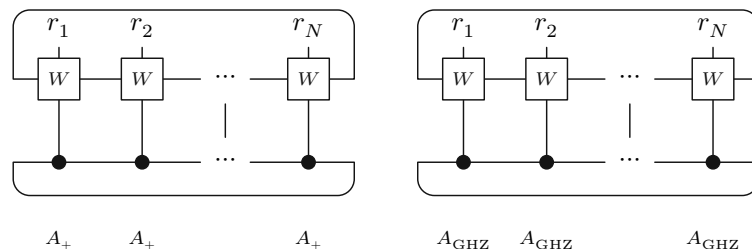
For either input tensor A , the action tensor is $(W \cdot A)^{s'} = \sum_s W^{s's} \otimes A^s$. For $A = A_+$ its matrices are

$$(W \cdot A)^0 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad (W \cdot A)^1 = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

with targets $2C_0$ and $2C_1$. For $A = A_{\text{GHZ}}$ they are

$$(W \cdot A)^0 = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad (W \cdot A)^1 = \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix},$$

with two unweighted copies of A_+ as targets.



For the product-state action, the matrices are $2Q_+$ and $2Q_-$, where

$$Q_{\pm} = \frac{1}{2} \begin{pmatrix} 1 & \pm 1 \\ \pm 1 & 1 \end{pmatrix}.$$

They satisfy $Q_{\pm}^2 = Q_{\pm}$, $Q_+ Q_- = Q_- Q_+ = 0$ and $\text{tr } Q_{\pm} = 1$. Hence a nonempty word of length N has trace 2^N when all letters agree and trace zero otherwise. This recovers $K_N p_N = 2^N g_N$ and identifies the factor 2 as a local tensor weight. By contrast, the factor 2 in $K_N g_N = 2 p_N$ is the multiplicity of two unweighted targets. Replacing p_N or g_N by a normalized vector changes the global scalar as in Lemma 27.5.2.1; it does not change the weights of these fixed tensors.

In the following two results, B denotes the relevant action tensor, rather than the stacked square tensor of Definition 27.5.1.

Theorem 27.5.2.3 (The duality on the product state splits). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The first action tensor of Definition 27.5.2.2 has the block triangular form of Theorem 25.5.1 for the weighted targets $\widehat{C}_s = 2C_s$, $s = 0, 1$, with no zero slots and $2 = 1 + 1$. Its compression pairs satisfy $W_s V_s = \mathbb{1}_1$ and $W_s B^w V_s = \widehat{C}_s^w$ for every word w . The remainder vanishes identically, and for every nonempty word,

$$\mathrm{tr}(B^w) = \mathrm{tr}((2C_0)^w) + \mathrm{tr}((2C_1)^w).$$

Proof. [✓ Proof](#) Let $F = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, so $F^{-1} = F/2$. Direct multiplication gives

$$FB^0 F^{-1} = \mathrm{diag}(2, 0),$$

$$FB^1 F^{-1} = \mathrm{diag}(0, 2).$$

Diagonality gives the triangularity condition, identifies the matched blocks, and makes the remainder zero. The coordinate compression pairs satisfy the stated word identities by Theorem 1.1.2. The trace identity follows from Proposition 25.5.3. $\square$

Theorem 27.5.2.4 (The duality on the ferromagnetic state does not split). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The second action tensor of Definition 27.5.2.2 has the block triangular form of Theorem 25.5.1 for two unweighted copies $C_s = A_+$ of the product-state tensor, with $z = 2$ zero slots and $4 = 1 + 1 + 2$. The compression pair of each copy satisfies $W_s V_s = \mathbb{1}_1$ and $W_s B^w V_s = C_s^w$ for every word. For every nonempty word, $\mathrm{tr}(B^w) = 2$. Every remainder word of length at least four vanishes; in fact,

$$R^0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad R^1 = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},$$

and $R^i R^j = 0$ for all letters i, j .

Proof. [✓ Proof](#) The two copies occupy the first and fourth bond coordinates, and the zero slots occupy the second and third. Ordering the coordinates as $(0, 3, 1, 2)$ makes both letters upper triangular, with diagonal $(1, 1, 0, 0)$. Thus the matched blocks are both A_+ , and the coordinate compression pairs give the asserted reductions. Every word evaluation of A_+ is 1, so Proposition 25.5.3 gives $\mathrm{tr}(B^w) = 1 + 1 = 2$ for nonempty w .

Subtracting the matched diagonal contributions in the original coordinates gives the displayed remainder matrices. Their only nonzero entries lie in positions $(0, 2)$ and $(3, 1)$, so direct multiplication gives $R^i R^j = 0$. The general bound in clause (vi) of Theorem 25.5.1 also gives vanishing at lengths at least $|S| + z = 4$, where S is the set of two matched slots. The obstruction to a sitewise splitting is established in Theorem 27.5.2.5. $\square$

A word reduction need not provide a pair of sitewise intertwiners. Here the periodic coefficients have merged into two copies of the same product state, but the virtual action still has a one-sided obstruction.

Theorem 27.5.2.5 (One-sided sitewise intertwiners for the merged target). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $B = W \cdot A_{\mathrm{GHZ}}$. If a row vector $u \in \mathbb{C}^4$ satisfies $uB^i = u$ for both letters i , then $u = 0$. Thus no nonzero sitewise left intertwiner exists for the product-state target. A nonzero sitewise right intertwiner does exist: the first coordinate vector satisfies $B^i e_0 = e_0$ for both letters.

Proof. ✓ Proof Both target matrices are the scalar 1. In coordinates $u = (u_0, u_1, u_2, u_3)$, the left-intertwiner equations are

$$\begin{aligned} uB^0 &= (u_0, u_3, u_0, u_3) = u, \\ uB^1 &= (u_0, -u_3, -u_0, u_3) = u. \end{aligned}$$

The second coordinates give $u_3 = u_1 = 0$, and the third give $u_0 = u_2 = 0$. For the right intertwiner, the first column of each matrix is e_0 , which is nonzero. $\square$

This obstruction rules out a sitewise split copy of the product-state target: such a splitting would require a nonzero left intertwiner. It does not obstruct the word reductions or the periodic trace formula in Theorem 27.5.2.4. Those statements retain the finite-chain action $K_N g_N = 2p_N$ while distinguishing it from a split virtual direct sum.

27.5.3 The duality as a bimodule over the spin-flip algebra

By (27.63) (Corollary 27.5.1.7), $D^2 = 2^N(\mathbb{1} + \eta)T$, where the non-trivial translation factor T prevents the duality D from closing in a fusion algebra with $\mathbb{1}$ and η . In the framework of [1], it is instead fused with the symmetry operators on either side and with its own return map. These relations are recorded below for the raw operator and derived at the tensor level from the compression theorems, without computing a gauge.

Lemma 27.5.3.1 (Spin-flip algebra and the bimodule relations). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let O_0 and O_1 be the periodic operators of the bond-one tensors $O_0^{s's} = \delta_{s',s}$ and $O_1^{s's} = \delta_{s',1-s}$, and let $W^\top$ be the duality tensor with its physical indices exchanged, $(W^\top)^{s's} = W^{ss'}$. For every $N > 0$, $O_0 = \mathbb{1}$, $O_1 = \eta$, and $O_a O_b = O_{a+b}$ with $a + b$ taken modulo two. The periodic operator of $W^\top$ is $K_N^\top$, and

$$\begin{aligned} K_N O_a &= O_a K_N = K_N \quad (a = 0, 1), \\ K_N K_N^\top &= K_N^\top K_N = 2^N(O_0 + O_1). \end{aligned} \tag{27.74}$$

Proof. ✓ Proof A periodic contraction of bond-one tensors is the product of its one-site entries, which gives $O_0 = \mathbb{1}$ and $O_1 = \eta$; the fusion rule is $\eta^2 = \mathbb{1}$. Exchanging the physical indices of every tensor in the periodic trace transposes the operator. The relations (27.74) are then Theorem 27.5.1.2 and (27.58), using that the kernel is real. $\square$

Theorem 27.5.3.2 (Absorption of the spin flip is a gauge equivalence). Lean Lean Lean ✓ Stmt The duality tensor W is normal. The stacked tensors $\sum_m W^{s'm} \otimes O_1^{ms}$ and $\sum_m O_1^{s'm} \otimes W^{ms}$, of bond dimension two, are each conjugate to W : there are invertible 2×2 matrices X and Y with

$$\sum_m W^{s'm} \otimes O_1^{ms} = X^{-1} W^{s's} X, \quad \sum_m O_1^{s'm} \otimes W^{ms} = Y^{-1} W^{s's} Y$$

for all s', s .

Proof. ✓ Proof The four matrices of W give the four matrix units by

$$\begin{aligned} E_{00} &= \tfrac{1}{2}(W^{00} + W^{10}), & E_{01} &= \tfrac{1}{2}(W^{00} - W^{10}), \\ E_{10} &= \tfrac{1}{2}(W^{01} - W^{11}), & E_{11} &= \tfrac{1}{2}(W^{01} + W^{11}), \end{aligned}$$

so W is injective and hence normal. The periodic operators of both stacked tensors equal K_N for every $N > 0$ by (27.74). Corollary 25.9.1 gives V and W' with $W'V = \mathbb{1}_2$ whose remainder vanishes on every word of length at least $2 - 2 + 1 = 1$. The remainder therefore vanishes letter by letter, and V is square with left inverse W' . $\square$

Explicitly, the stacked tensors have the letters $\sum_m W^{s'm} \otimes O_1^{ms} = W^{s',1-s}$ and $\sum_m O_1^{s'm} \otimes W^{ms} = W^{1-s',s}$, and a direct check on the four matrices of Definition 27.5.1 gives

$$W^{s',1-s} = XW^{s's}X, \quad W^{1-s',s} = YW^{s's}Y, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

with $X^{-1} = X$ and $Y^{-1} = Y$. The proof above does not use these matrices.

Theorem 27.5.3.3 (Fusion of the duality with its adjoint). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The stacked tensor $\sum_m W^{s'm} \otimes (W^\top)^{ms}$, of bond dimension four, has the block triangular form of Theorem 25.5.1 for the two bond-one targets $2O_0$ and $2O_1$, with exactly two zero slots: $4 = 1 + 1 + 2$. The same holds for $\sum_m (W^\top)^{s'm} \otimes W^{ms}$.*

Proof. [✓ Proof](#) Both bond-one tensors are normal, since each has a one-site matrix equal to 1. By (27.74), the periodic operators of the stacked tensor are $\sum_a 2^N O_a$ for every $N > 0$. Theorem 25.6.2 applies with weights 2, and the dimension count of Theorem 25.5.1 gives the number of zero slots. $\square$

The biorthogonal compression pairs of this theorem are fusion tensors for the channels $\mathbb{1}$ and η in the product of the duality with its adjoint. The stacked square of Theorem 27.5.1.13 has no zero slot, whereas the product with the adjoint has two.

27.6 Fibonacci fusion operators

The Fibonacci fusion category has two simple objects, the vacuum 1 and τ , with the fusion rule $\tau \otimes \tau = 1 \oplus \tau$. Its string-net construction produces matrix product operators whose fusion rule, fusion tensors, and F -symbols are given in [7, Appendix D.1]; the review displays the string-net operator built from the same Fibonacci data in [16, Appendix A, “The MPO for the Fibonacci model”], and the general string-net construction of such operator symmetries and their intertwiners is in [181]. The physical space used here is the two-label space of the golden chain [161], one label 1 or τ per site, rather than the full triple-index alphabet of the categorical string-net construction. The entries of the blocks defined below are obtained from the F -symbols of [7, Appendix D.1]. From these data one obtains two operators on the chain whose fusion relation and action on periodic states reduce to finite matrix computations.

Definition 27.6.1 (The string-net tensors of the source and of the review). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $d_1 = 1$, $d_\tau = \varphi = (1 + \sqrt{5})/2$ be the quantum dimensions, and let N_{ab}^c denote the Fibonacci fusion multiplicities. Over $\mathbb{C}$ write $v_a = \sqrt{d_a}$ and $G_{def}^{abc} = v_e^{-1} v_f^{-1} [F_d^{abc}]_e^f$. For a label f , the bond letters of the block f are the pairs (u, l) of labels with $N_{ul}^f > 0$: the letters $(1, 1)$, (τ, τ) for $f = 1$ and $(1, \tau)$, $(\tau, 1)$, (τ, τ) for $f = \tau$. A physical letter of the source’s tensor is a pair (x, e) of a plaquette label and an edge label. The operator tensor of [7, Section 6.2 and Appendix D.1] is drawn as the crossing of the operator line f with an edge line e , with the plaquettes b upper left, q' upper right, c lower left and q'' lower right; its entries are

$$(S_f^{(b,e'),(c,e)})_{p,(q',q'')} = \delta_{e'e} \delta_{p,(b,c)} G_{q''ef}^{q'bc} \sqrt{v_{q'} v_b v_c v_{q''}}, \quad (27.75)$$

and its letters at edge label τ are $S_f^{x'x} = S_f^{(x',\tau),(x,\tau)}$. The periodic operators of S_f act on $(\mathbb{C}^4)^{\otimes N}$, one plaquette label and one edge label per site. The configurations whose edge labels are all τ span a subspace identified with $(\mathbb{C}^2)^{\otimes N}$, and the letters $S_f^{x'x}$ are those of S_f on this subspace. The operator tensor of [16, Appendix A, “The MPO for the Fibonacci model”], at the vertical label τ , has the entries

$$(R_f^{x'x})_{p,(q',q'')} = \delta_{p,(x',x)} \frac{1}{\sqrt{d_{x'}d_{q''}}} [F_{q'}^{fx\tau}]_{x'}^{q''}, \quad (27.76)$$

with the F -symbols of Definition 27.6.6 and their selection rule, in place of the selection rule printed in the review. On configurations $x \in \{1, \tau\}^N$ let $W = \text{diag} \prod_k d_{x_k}$, with square roots $W^{\pm 1/2} = \text{diag} \prod_k d_{x_k}^{\pm 1/2}$, and let $\widetilde{R}_f$ be the tensor with letters $d_x R_f^{x'x}$.

The printed G -symbol tensor, including its quantum-dimension factors, is given in Definition 27.6.1. Its restriction to edge label τ is related to the two-label blocks by the diagonal bond similarity of Theorem 27.6.9. The review uses a different physical normalization; its relation to these blocks is the congruence of Theorem 27.6.11.

For every $N > 0$, close the bond indices by the ordinary, unnormalized trace of the ordered product of local matrices. The first physical index is the output and the second is the input.

The word-span certificates are in Theorem 27.6.12. The admissibility projector and its action as the fusion unit are identified in Lemma 27.6.1.1 and Theorem 27.6.1.2. The fusion relation is proved for the two-label blocks in Theorem 27.6.4.2 and for the source’s blocks on both edge labels in Theorem 27.6.10. Commutation with the golden-chain Hamiltonian is established for $N \geq 2$ in Theorem 27.6.2.4; the regular action on the two explicit state tensors is in Theorem 27.6.3.4.

The physical space at each site is $\mathbb{C}^2$, with computational labels 0 for the vacuum and 1 for τ . At system size $N \geq 1$, write $\mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$ and use the orthonormal basis $|x\rangle = |x_0, \dots, x_{N-1}\rangle$. Site indices are cyclic: $x_N = x_0$. Throughout, the golden-number convention is $\varphi = (1 + \sqrt{5})/2$ and $\sigma = \varphi^{-1/2} = \sqrt{(\sqrt{5} - 1)}/2$. In particular, $\sigma^2 = \varphi^{-1}$ and $\sigma^4 + \sigma^2 = 1$.

For an MPO tensor T^{yx} , the first physical index is the output index. Its periodic operator is defined by

$$\langle y | O_N(T) | x \rangle = \text{tr}(T^{y_0 x_0} \dots T^{y_{N-1} x_{N-1}}). \quad (27.77)$$

For the tensors B_1 and B_τ defined below, write $O_1 = O_N(B_1)$ and $O_\tau = O_N(B_\tau)$ when N is fixed. Their non-invertible fusion relation is

$$O_\tau O_\tau = O_1 + O_\tau, \quad (27.78)$$

as proved below for every positive system size. The label τ is not invertible in the sense of Definition 26.2.2. The vacuum operator O_1 is a projection onto admissible configurations, not the identity on $\mathcal{H}_N$, as for the unit of any fusion algebra (Remark 26.3.1).

Every entry of the tensors below, and of the changes of bond coordinates in the proofs, lies in a quartic ring generated by σ . The identities of this section are first verified in that ring and then embedded into $\mathbb{C}$. The exceptions are the string-net tensors of Definition 27.6.1 and the bond similarity relating them to the blocks, whose entries involve $\varphi^{\pm 1/4}$; those identities are proved over $\mathbb{C}$.

Definition 27.6.2 (The ring of golden integers). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) The ring of *golden integers* is $\mathbb{Z}[\sigma]$, where $\sigma = \varphi^{-1/2} = \sqrt{(\sqrt{5} - 1)}/2$ is the positive real root of $X^4 + X^2 - 1$. It is presented by the four integer coordinates of $c_0 + c_1\sigma + c_2\sigma^2 + c_3\sigma^3$, with

multiplication reduced using $\sigma^4 = 1 - \sigma^2$. It embeds into $\mathbb{C}$ through the quotient $\mathbb{Q}[X]/(X^4 + X^2 - 1)$ evaluated at this real root. For matrices, the embedding is applied entrywise. Word evaluation and the bond-space product of tensors over $\mathbb{Z}[\sigma]$ use the same formulas as over $\mathbb{C}$.

Theorem 27.6.3 (Exact arithmetic in the ring of golden integers). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Stmnt](#) *The generator satisfies $\sigma^4 + \sigma^2 - 1 = 0$ in $\mathbb{Z}[\sigma]$, and so does the real number $\sqrt{(\sqrt{5} - 1)/2}$. In any commutative ring carrying a root s of $X^4 + X^2 - 1$, the product of two integer combinations of $1, s, s^2, s^3$ is the combination prescribed by the multiplication of $\mathbb{Z}[\sigma]$. The entrywise embedding of matrices over $\mathbb{Z}[\sigma]$ into the complex matrices is multiplicative, and it commutes with word evaluation and with the bond-space product of tensors.*

Proof. [Stmnt](#) [Proof](#) Squaring $\sigma^2 = (\sqrt{5} - 1)/2$ gives the quartic for the real root. The multiplication rule is the reduction of a product of two polynomials of degree at most three modulo $X^4 + X^2 - 1$. It is a polynomial identity in their eight integer coefficients. The entrywise embedding is induced by a ring homomorphism, so expansion of a matrix product proves its multiplicativity. Induction on the word proves compatibility with word evaluation. Expansion of the bond-space product proves the remaining compatibility statement. $\square$

Definition 27.6.4 (The two Fibonacci blocks and the stacked square). [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Stmnt](#) Read a physical letter as the pair (x', x) of outgoing and incoming labels, ordered as $(0, 0), (0, 1), (1, 0), (1, 1)$. The reduced vertex weight is

$$F = \begin{pmatrix} 0 & \varphi^{-1} & \varphi^{-1/2} \\ 1 & 0 & 1 \\ 1 & \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}. \quad (27.79)$$

The τ tensor B_τ has bond dimension three and letters

$$B_\tau^{00} = 0. \quad (27.80)$$

$$B_\tau^{01} = \begin{pmatrix} 0 & \sigma^2 & \sigma \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (27.81)$$

$$B_\tau^{10} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}. \quad (27.82)$$

$$B_\tau^{11} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & \sigma & -\sigma^2 \end{pmatrix}. \quad (27.83)$$

Thus each nonzero letter carries one row of (27.79), with the other rows zero. The vacuum tensor B_1 has bond dimension two and letters

$$B_1^{00} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}. \quad (27.84)$$

$$B_1^{11} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}. \quad (27.85)$$

$$B_1^{01} = B_1^{10} = 0. \quad (27.86)$$

The stacked square has bond dimension nine:

$$B^{x'x} = \sum_{m=0}^1 B_\tau^{x'm} \otimes B_\tau^{mx}. \quad (27.87)$$

Viewed as MPS tensors over the four-letter pair alphabet, its two target blocks are B_1 and B_τ , each with multiplicity one and weight one.

The review writes the operator with the prefactor $1/\sqrt{d_A d_D}$ arising from the normalization of the fusion basis, where d_A and d_D are the quantum dimensions of the labels on the adjacent bonds [16, Appendix A, “The MPO for the Fibonacci model”]; that prefactor is omitted in the blocks defined here, and its effect on the periodic operators is Theorem 27.6.11.

Theorem 27.6.5 (The one-site matrices of the τ block). [Lean](#) [Lean](#) [✓ Stmt](#) *The matrix of B_τ at the letter $(0, 1)$ has the first row of (27.79) as its first row and zero elsewhere, and its matrix at the letter $(1, 1)$ has the third row of (27.79) as its third row and zero elsewhere, with $\varphi^{-1/2} = \sqrt{(\sqrt{5} - 1)/2}$.*

Proof. [✓ Proof](#) Both matrices are obtained by embedding their coefficients in $\mathbb{Z}[\sigma]$ into $\mathbb{C}$ entry-wise. Substitution of $\sigma = \sqrt{(\sqrt{5} - 1)/2}$ gives (27.81) and (27.83). $\square$

The blocks above are built from the categorical data of the Fibonacci theory, which [7, Appendix D.1] and the review print in the following form. The former gives the operator tensor only as a diagram and prints no entries of B_1 or B_τ ; the placement of the F -symbols used here, without the closed-loop factors of the source and without the prefactor of the review, is recorded in [59]. Theorem 27.6.9 relates these blocks to the source’s tensor built from the G -symbols, and Theorem 27.6.10 carries their fusion rules to that tensor on both edge labels.

Definition 27.6.6 (Fibonacci fusion data of the review). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Fibonacci theory has the labels 1 and τ , with fusion multiplicities $N_{11}^1 = N_{\tau 1}^\tau = N_{1\tau}^\tau = N_{\tau\tau}^1 = N_{\tau\tau}^\tau = 1$ and all others zero, and quantum dimensions $d_1 = 1$ and $d_\tau = \varphi = 1 + \sigma^2$, the Perron–Frobenius dimensions of Example 26.3.2. Write $v_a = \sqrt{d_a}$, so that $v_1^{-1} = 1$ and $v_\tau^{-1} = \sigma$. For labels a, b, c, d, e, f , with $\delta_{ijk} = 1$ when $N_{ij}^k > 0$ and $\delta_{ijk} = 0$ otherwise, the F -symbols are

$$[F_d^{abc}]_e^f = \delta_{abe} \delta_{cde} \delta_{adf} \delta_{bcf} F_{def}^{abc}, \quad (27.88)$$

where $F_{def}^{abc} = 1$ unless $a = b = c = d = \tau$, and the block at $a = b = c = d = \tau$, indexed by $e, f \in \{1, \tau\}$, is

$$F_\tau^{\tau\tau\tau} = \begin{pmatrix} \sigma^2 & \sigma \\ \sigma & -\sigma^2 \end{pmatrix} = \begin{pmatrix} 1/\varphi & 1/\sqrt{\varphi} \\ 1/\sqrt{\varphi} & -1/\varphi \end{pmatrix}. \quad (27.89)$$

The G -symbols are $G_{def}^{abc} = v_e^{-1} v_f^{-1} [F_d^{abc}]_e^f$. All these quantities are taken in $\mathbb{Z}[\sigma]$. This is the data of [16, Appendix A, “The MPO for the Fibonacci model”] and [7, Appendix D.1].

Proof. ✓ Proof At edge label 1 the selection rule of G_{d1f}^{abc} forces $a = b$ and $c = d$, and $G_{c1f}^{bbc} v_b v_c = 1$ whenever (b, c) is a bond letter of the block f . So the letter of S_f at $((b, 1), (c, 1))$ is the diagonal matrix unit E_{pp} at $p = (b, c)$, which the diagonal matrix h leaves unchanged; together with Theorem 27.6.9 this gives the similarity on all letters, and the periodic operators of S_f and of the extended B_f agree. For each product of two extended blocks, the gauge of the corresponding compression of Theorem 27.6.5.3 makes every one of the sixteen letters of the stacked tensor block upper triangular: at edge label τ the conjugated letters are the block-diagonal ones already recorded, and at edge label 1 they carry the edge-1 letters of the target blocks on the diagonal and further entries only in the columns of the zero slots. These are finite computations in $\mathbb{Z}[\sigma]$. The trace identity of a compression holds for a nonzero nilpotent remainder, which gives the four fusion rules. $\square$

Theorem 27.6.11 (The prefactor of the review). Lean Lean Lean Lean Lean ✓ Stmt For both labels f and every length N , the periodic operator of the tensor (27.76) is the congruence

$$O_N(R_f) = W^{-1/2} O_N(B_f) W^{-1/2}. \quad (27.92)$$

These operators do not satisfy the Fibonacci fusion rules: at $N = 1$, $O_N(R_1) = \text{diag}(0, 1/\varphi)$, whose square has the entry $1/\varphi^2 \neq 1/\varphi$. The operators $O_N(\tilde{R}_f) = O_N(R_f)W = W^{-1/2}O_N(B_f)W^{1/2}$ satisfy them at every positive length.

Proof. ✓ Proof The diagonal bond matrix $g(u, l) = d_l^{-1/2}$ carries $R_f^{x'x}$ to $(d_{x'}d_x)^{-1/2}B_f^{x'x}$, a multiple of each letter by its own scalar. The periodic operator of such a tensor is the operator of B_f with its entry at (x, y) multiplied by $\prod_k (d_{x_k}d_{y_k})^{-1/2}$, which is (27.92). At $N = 1$, $O_N(B_1) = \text{diag}(0, 1)$ gives the stated entry. Finally, $O_N(\tilde{R}_f) = W^{-1/2}O_N(B_f)W^{1/2}$ is similar to $O_N(B_f)$ by one matrix for all labels, so products and linear combinations are carried along and the fusion rules of Theorem 27.6.5.3 transfer. $\square$

While the review states $O_a O_b = \sum_c N_{ab}^c O_c$ for the prefactored review tensor R_f [16, Section 3], that operator already contains the prefactor $1/\sqrt{d_{x'}d_{q''}}$ and fails the fusion rules in the orthonormal plaquette basis. Theorem 27.6.11 shows that the fusion rules hold after composing it with W ; see [59].

Theorem 27.6.12 (Normality of the two Fibonacci blocks). Lean Lean ✓ Stmt Both blocks B_1 and B_τ are normal: their length-three words satisfy $\text{span}_{\mathbb{C}}\{B_1^w : |w| = 3\} = M_2(\mathbb{C})$ and $\text{span}_{\mathbb{C}}\{B_\tau^w : |w| = 3\} = M_3(\mathbb{C})$.

Proof. ✓ Proof For the τ block, let e_i be the i -th coordinate column vector and f_i the i -th row of (27.79). In this proof, superscripts 0, 1, 2, 3 enumerate the ordered pair alphabet. Thus $B_\tau^{i+1} = e_i f_i$ for $i \in \{0, 1, 2\}$, and

$$B_\tau^{i+1} B_\tau^3 B_\tau^{k+1} = F_{i2} F_{2k} e_i f_k. \quad (27.93)$$

All factors $F_{i2} F_{2k}$ are nonzero, and the rows f_k form a basis. Each of the nine matrix units is an explicit combination over $\mathbb{Z}[\sigma]$ of these nine words. For the vacuum block, four length-three words suffice. Both families of coefficients are verified in $\mathbb{Z}[\sigma]$ and transported to complex matrices by Theorem 27.6.3. $\square$

27.6.1 Admissible configurations and the vacuum operator

Define the adjacency matrix $K = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ and, for a cyclic word $x \in \{0, 1\}^N$, set

$$p_N(x) = \prod_{v=0}^{N-1} K_{x_v x_{v+1}}. \quad (27.94)$$

A cyclic word x is admissible if $p_N(x) = 1$, equivalently if it contains no cyclic neighboring pair 00.

Lemma 27.6.1.1 (Vacuum projection and its kernel). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) For $N \geq 1$, the vacuum operator $P_N = O_N(B_1)$ has matrix elements

$$\langle y | P_N | x \rangle = \delta_{yx} p_N(x). \quad (27.95)$$

Thus $P_N | x \rangle = p_N(x) | x \rangle$. It is the orthogonal projection onto the admissible subspace:

$$\text{ran } P_N = \text{span}_{\mathbb{C}} \{ | x \rangle : p_N(x) = 1 \}. \quad (27.96)$$

$$\text{ker } P_N = \text{span}_{\mathbb{C}} \{ | x \rangle : p_N(x) = 0 \}. \quad (27.97)$$

In particular, $P_N^2 = P_N = P_N^\dagger$, but $P_N \neq \mathbb{1}_{\mathcal{H}_N}$.

Proof. [✓ Proof](#) The vacuum letters satisfy $(B_1^{yx})_{ab} = \delta_{yx} \delta_{ax} K_{xb}$. Expanding the periodic trace over bond labels $g = (g_0, \dots, g_{N-1})$, with $g_N = g_0$, gives

$$\langle y | P_N | x \rangle = \sum_g \prod_{v=0}^{N-1} \delta_{y_v x_v} \delta_{g_v x_v} K_{x_v g_{v+1}} = \delta_{yx} \prod_{v=0}^{N-1} K_{x_v x_{v+1}}. \quad (27.98)$$

Only $g = x$ contributes. Since $p_N(x) \in \{0, 1\}$, the operator is diagonal with real idempotent entries. This proves the projection, range, and kernel statements. Since $|0, \dots, 0\rangle \in \text{ker } P_N$, $P_N \neq \mathbb{1}_{\mathcal{H}_N}$, with $P_1 = \text{diag}(0, 1)$ at one site. $\square$

Theorem 27.6.1.2 (Vacuum unit laws). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $N \geq 1$, the periodic operators satisfy

$$O_N(B_1)^2 = O_N(B_1). \quad (27.99)$$

$$O_N(B_1) O_N(B_\tau) = O_N(B_\tau). \quad (27.100)$$

$$O_N(B_\tau) O_N(B_1) = O_N(B_\tau). \quad (27.101)$$

Proof. [✓ Proof](#) For MPO tensors T and U , denote their stacked product by $(TU)^{yx} = \sum_m T^{ym} \otimes U^{mx}$. Write 0_r for the tensor whose letters are zero $r \times r$ matrices, and $\sim$ for simultaneous similarity of all letters. The three stacked products admit the decompositions

$$B_1 B_1 \sim B_1 \oplus 0_2. \quad (27.102)$$

$$B_1 B_\tau \sim B_\tau \oplus 0_3. \quad (27.103)$$

$$B_\tau B_1 \sim B_\tau \oplus 0_3. \quad (27.104)$$

The changes of bond coordinates and their inverses have entries in $\mathbb{Z}[\sigma]$. The decompositions, of dimensions $4 = 2 + 2$ and $6 = 3 + 3$, are verified by matrix multiplication with coefficients reduced using $\sigma^4 = 1 - \sigma^2$. The periodic product formula for a multi-block compression gives the trace of the single nonzero target block. Since $N > 0$, every zero block contributes zero, proving the three identities. $\square$

By (27.100), the range of $O_N(B_\tau)$ lies in the admissible subspace. By (27.101), it vanishes on the orthogonal complement of that subspace. Thus the vacuum acts as the identity on the admissible subspace and is the unit of the algebra generated there by $O_N(B_\tau)$.

This vacuum block is distinct from the weighted string-net projector of [7, Appendix D.1], which combines the vacuum and τ blocks with weights $w_1 = 1/(1 + \varphi^2)$ and $w_\tau = \varphi/(1 + \varphi^2)$.

Definition 27.6.1.3 (The weighted Fibonacci projector). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The weights of the two labels are $w_1 = 1/(1 + \varphi^2)$ and $w_\tau = \varphi/(1 + \varphi^2)$, the quantum dimensions divided by the square of the total quantum dimension. The bond-dimension-five tensor is the direct sum $B_1 \oplus B_\tau$, and its boundary matrix is $\Delta = w_1 \mathbb{1}_2 \oplus w_\tau \mathbb{1}_3$. At every system size N , the weighted projector is

$$\Pi_N = w_1 O_N(B_1) + w_\tau O_N(B_\tau) = \frac{O_N(B_1) + \varphi O_N(B_\tau)}{1 + \varphi^2}, \quad (27.105)$$

following [7, Sections 3.1 and 3.2 and Appendix D.1].

Theorem 27.6.1.4 (The weight relation). [Lean](#) [✓ Stmt](#) For each label $c \in \{1, \tau\}$, $\sum_{a,b \in \{1, \tau\}} N_{ab}^c w_a w_b = w_c$.

Proof. [✓ Proof](#) For $c = 1$ the relation reads $w_1 = w_1^2 + w_\tau^2$, and both sides equal $1/(1 + \varphi^2)$. For $c = \tau$ it reads $w_\tau = 2w_1 w_\tau + w_\tau^2$; after multiplication by $(1 + \varphi^2)^2/\varphi$ it becomes $1 + \varphi^2 = 2 + \varphi$, a consequence of $\varphi^2 = \varphi + 1$. $\square$

Theorem 27.6.1.5 (The projector from the bond-dimension-five tensor). [Lean](#) [Lean](#) [✓ Stmt](#) At every system size N , write $B = B_1 \oplus B_\tau$. Then

$$\langle y | \Pi_N | x \rangle = \text{tr}(\Delta B^{y_0 x_0} \dots B^{y_{N-1} x_{N-1}}), \quad (27.106)$$

and without the boundary matrix the periodic operator is $O_N(B) = O_N(B_1) + O_N(B_\tau)$.

Proof. [✓ Proof](#) Word evaluation of the direct sum is block diagonal by Lemma 16.1.9. Multiplication by the block-diagonal matrix Δ scales the two blocks by w_1 and w_τ , and the trace of a block-diagonal matrix is the sum of the traces of its blocks. The second statement is Theorem 16.1.10. $\square$

Theorem 27.6.1.6 (The weighted Fibonacci projector is idempotent). [Lean](#) [✓ Stmt](#) For every system size $N \geq 1$, $(\Pi_N)^2 = \Pi_N$.

Proof. [✓ Proof](#) Expanding the square and using the unit laws of Theorem 27.6.1.2 and the fusion rule of Theorem 27.6.4.2 gives $(w_1^2 + w_\tau^2)O_N(B_1) + (2w_1 w_\tau + w_\tau^2)O_N(B_\tau)$. Theorem 27.6.1.4 identifies the two coefficients with w_1 and w_τ . $\square$

At $N = 0$ the empty traces give $O_0(B_1) = 2$ and $O_0(B_\tau) = 3$, so $\Pi_0 = (2 + 3\varphi)/(1 + \varphi^2)$, a scalar different from 0 and 1, hence $\Pi_0^2 \neq \Pi_0$; [7, Section 3.2] asks for idempotence at every length, and the empty chain is excluded as a degenerate reading; see [60].

27.6.2 The golden chain

The golden chain of Feiguin et al. [161] acts on the labels $x_0, \dots, x_{N-1} \in \{1, \tau\}$ of a periodic chain. Its three-site term projects the two anyons at links i and $i+1$ onto their trivial fusion channel, with the amplitudes

$$(p_i)_{x_i}^{x'_i} = (F_{x_{i+1}}^{x_{i-1}\tau\tau})_{x'_i}^1 (F_{x_{i+1}}^{x_{i-1}\tau\tau})_{x_i}^1, \quad (27.107)$$

where p_i changes only the label x_i . The Hamiltonian is $H_N = -\sum_i p_i$. The review states that such anyonic chains commute with the whole algebra of periodic operators [16].

Definition 27.6.2.1 (The golden chain). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The golden chain is $H_N = -\sum_i p_i$ with the local terms of (27.107). The topological symmetry has the factors $\left(F_{x_{n+1}}^{\tau x_n \tau}\right)_{x'_n}^{x_{n+1}}$, and the vacuum factor is $\delta_{x_n x_n} \delta_{x'_{n+1} x_{n+1}}$ unless $x_n = x_{n+1} = 1$, where it vanishes.

Theorem 27.6.2.2 (Values of the golden-chain projector). [Lean](#) [Lean](#) [✓ Stmt](#) The local term p_i is the projector onto $|1\tau 1\rangle$ between two trivial neighbours, it vanishes between a trivial and a τ neighbour, and between two τ neighbours it is

$$\begin{pmatrix} \varphi^{-2} & \varphi^{-3/2} \\ \varphi^{-3/2} & \varphi^{-1} \end{pmatrix}$$

in the basis $1, \tau$ of x_i .

Proof. [✓ Proof](#) The entries are products of two F-symbols and are checked entrywise in $\mathbb{Z}[\sigma]$, where $\varphi^{-1} = \sigma^2$. $\square$

Theorem 27.6.2.3 (Site operators and nearest-neighbour kernels). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On a periodic chain of $N \geq 2$ sites, let K have the entries $\langle s|K|t \rangle = \prod_n f(s_n, s_{n+1}, t_n, t_{n+1})$, and let p_i change only the state at site i , with amplitude $h(t_{i-1}, t_{i+1}, s_i, t_i)$. If

$$\sum_y h(a', c', b', y) f(a', y, a, b) f(y, c', b, c) = \sum_y f(a', b', a, y) f(b', c', y, c) h(a, c, y, b)$$

for all labels, then $p_i K = K p_i$.

Proof. [✓ Proof](#) Write $R(s, t) = \prod_{n \neq i-1, i} f(s_n, s_{n+1}, t_n, t_{n+1})$ for the factors of K away from the two bonds at site i ; it does not depend on s_i or t_i . Summing over the label y at site i of the intermediate configuration,

$$\begin{aligned} \langle s|p_i K|t \rangle &= R(s, t) \sum_y h(s_{i-1}, s_{i+1}, s_i, y) f(s_{i-1}, y, t_{i-1}, t_i) f(y, s_{i+1}, t_i, t_{i+1}), \\ \langle s|K p_i|t \rangle &= R(s, t) \sum_y f(s_{i-1}, s_i, t_{i-1}, y) f(s_i, s_{i+1}, y, t_{i+1}) h(t_{i-1}, t_{i+1}, y, t_i). \end{aligned}$$

The hypothesis with $(a', b', c') = (s_{i-1}, s_i, s_{i+1})$ and $(a, b, c) = (t_{i-1}, t_i, t_{i+1})$ equates the two sums. $\square$

Theorem 27.6.2.4 (The golden chain commutes with the Fibonacci operators). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $N \geq 2$, the golden chain H_N commutes with $O_N(B_\tau)$, with $O_N(B_1)$ and with the projector Π_N , and, for each label f , with the periodic operator of the operator tensor S_f restricted to configurations with edge label τ .

Proof. ✓ Proof In the τ block the bond index at site n is determined by the pair (x'_n, x_n) , so the only closed bond configuration that survives gives $\langle x' | O_N(B_\tau) | x \rangle = \prod_n \left(F_{x'_{n+1}}^{\tau x_n} \right)_{x'_n}^{x_{n+1}}$, the topological symmetry of [161]; similarly $O_N(B_1)$ is diagonal with entry 1 exactly on the configurations without two neighbouring trivial labels. In both cases the entry is a product of factors $f(x'_n, x'_{n+1}, x_n, x_{n+1})$ around the ring. The product $p_i O_N(B_\tau)$ and the product $O_N(B_\tau) p_i$ share the factors away from the two bonds at site i , and the remaining sums over the intermediate label at site i agree by a three-site identity, checked on the sixty-four choices of the six outer and inner labels in exact golden arithmetic. The same identity holds for the kernel of $O_N(B_1)$. Summing over i gives the first two statements, and the third follows since $\Pi_N = w_1 O_N(B_1) + w_\tau O_N(B_\tau)$. On edge label τ the periodic operator of S_f equals $O_N(B_f)$ by Theorem 27.6.9, which gives the last statement. $\square$

27.6.3 Two periodic states and the Fibonacci action

For an MPS tensor D , write $|V^{(N)}(D)\rangle = \sum_x \text{tr}(D^{x_0} \dots D^{x_{N-1}}) |x\rangle$. These vectors are not normalized unless this is stated explicitly.

Definition 27.6.3.1 (The all- τ and chain tensors). Lean Lean ✓ Stmt The all- τ tensor A has bond dimension one, with $A^0 = (0)$ and

$$A^1 = (1). \quad (27.108)$$

The chain tensor C has bond dimension two:

$$C^0 = \begin{pmatrix} 0 & \sigma \\ 0 & 0 \end{pmatrix}. \quad (27.109)$$

$$C^1 = \begin{pmatrix} 0 & 0 \\ 1 & -\sigma^2 \end{pmatrix}. \quad (27.110)$$

Lemma 27.6.3.2 (Amplitudes and norms). Lean Lean Lean Lean Lean ✓ Stmt Let $N \geq 1$. The product tensor gives $|V^{(N)}(A)\rangle = |1, \dots, 1\rangle$, of norm one. For a word x , let $n_0(x)$ count its zeros and let $n_{11}(x)$ count its cyclic neighboring pairs 11. Then

$$V^{(N)}(C)_x = p_N(x) \sigma^{n_0(x)} (-\sigma^2)^{n_{11}(x)}. \quad (27.111)$$

Both vectors are fixed by P_N . The chain norm is

$$\|V^{(N)}(C)\|^2 = \sum_{x \in \{0,1\}^N} p_N(x) \sigma^{2n_0(x) + 4n_{11}(x)}. \quad (27.112)$$

This norm is positive.

Proof. ✓ Proof The bond-one trace is $\prod_v \delta_{x_v, 1}$. For the chain, put $H = \begin{pmatrix} 0 & \sigma \\ 1 & -\sigma^2 \end{pmatrix}$. Since $(C^x)_{ab} = \delta_{ax} H_{xb}$, the cyclic contraction gives

$$V^{(N)}(C)_x = \sum_g \prod_v \delta_{g_v x_v} H_{x_v g_{v+1}} = \prod_v H_{x_v x_{v+1}}. \quad (27.113)$$

A pair 00 makes the product zero. In an admissible word, every zero is followed by a one, giving $n_0(x)$ factors of σ . Each pair 10 contributes 1, and each pair 11 contributes $-\sigma^2$. This

proves (27.111). The operator $P_N = O_N(B_1)$ fixes the product vector by Theorem 27.6.3.4. The chain vector is $O_N(B_\tau)|V^{(N)}(A)\rangle$ by the same theorem, and the unit law $O_N(B_1)O_N(B_\tau) = O_N(B_\tau)$ of Theorem 27.6.1.2 shows that P_N fixes it (Theorem 27.6.5.4). Summing squared amplitudes proves (27.112). The all-one amplitude is $(-\sigma^2)^N \neq 0$. $\square$

The normalized chain vector is obtained by dividing by the positive square root of (27.112). For example, $|V^{(1)}(C)\rangle = -\sigma^2|1\rangle$ and its squared norm is σ^4 . In particular, the two periodic vectors are not independent at every length.

Theorem 27.6.3.3 (Normality of the two state tensors). [Lean](#) [Lean](#) [✓ Stmt](#) *The length-one words of A span $M_1(\mathbb{C})$, and the length-three words of C span $M_2(\mathbb{C})$. Thus both tensors are normal.*

Proof. [✓ Proof](#) The letter A^1 spans $M_1(\mathbb{C})$. For C , write E_{ab} for the matrix units of $M_2(\mathbb{C})$. Direct multiplication gives

$$C^0 C^1 C^0 = \sigma^2 E_{01}. \quad (27.114)$$

$$C^0 C^1 C^1 = -\sigma^3 E_{00} + \sigma^5 E_{01}. \quad (27.115)$$

$$C^1 C^0 C^1 = \sigma C^1. \quad (27.116)$$

$$C^1 C^1 C^0 = -\sigma^3 E_{11}. \quad (27.117)$$

These four words span all four matrix units, since $\sigma \neq 0$. The coefficients expressing the units in their span can all be chosen in $\mathbb{Z}[\sigma]$, using $\sigma^{-1} = \sigma + \sigma^3$. Embedding these identities into $\mathbb{C}$ proves that the words span the matrix algebra. $\square$

Theorem 27.6.3.4 (Action on the two periodic states). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every $N \geq 1$,*

$$O_N(B_1)|V^{(N)}(A)\rangle = |V^{(N)}(A)\rangle. \quad (27.118)$$

$$O_N(B_\tau)|V^{(N)}(A)\rangle = |V^{(N)}(C)\rangle. \quad (27.119)$$

$$O_N(B_\tau)|V^{(N)}(C)\rangle = |V^{(N)}(A)\rangle + |V^{(N)}(C)\rangle. \quad (27.120)$$

Proof. [✓ Proof](#) For an MPO tensor T and MPS tensor D , their action tensor has letters $(T \cdot D)^y = \sum_x T^{yx} \otimes D^x$. As in the proof of Theorem 27.6.1.2, let $\sim$ denote simultaneous similarity of letters and let 0_r denote a zero tensor of bond dimension r . The three action tensors have block decompositions

$$B_1 \cdot A \sim A \oplus 0_1. \quad (27.121)$$

$$B_\tau \cdot A \sim C \oplus 0_1. \quad (27.122)$$

$$B_\tau \cdot C \sim A \oplus C \oplus 0_3. \quad (27.123)$$

In lexicographic order on the product bond indices, matrices whose columns give these adapted bases are respectively

$$Q_2 = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}. \quad (27.124)$$

$$Q_3 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & -\sigma \\ 0 & 1 & \sigma^2 \end{pmatrix}. \quad (27.125)$$

$$Q_6 = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & \sigma^2 \\ 1 & 0 & \sigma & 1 & -\sigma & -\sigma^3 \\ 0 & 0 & 0 & 1 + \sigma^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & \sigma^2 & 0 \\ \sigma + \sigma^3 & 0 & -\sigma^2 & 0 & 0 & -\sigma^2 \end{pmatrix}. \quad (27.126)$$

They are invertible, with inverses over $\mathbb{Z}[\sigma]$. For each action tensor, let D_{blk} denote the corresponding block tensor in (27.121), (27.122), or (27.123). Matrix multiplication verifies $(T \cdot D)^y Q = Q D_{\text{blk}}^y$. All off-diagonal blocks vanish, giving the three multi-block compressions. The periodic action formula for each compression sums its target vectors. The zero blocks contribute nothing because $N > 0$. The first two sums have one target each, and the third has the targets A and C . $\square$

The first nontrivial action also has a direct contraction proof. The all-one input selects the letters B_τ^{01} and B_τ^{11} . Both have zero row 1, so a cyclic bond configuration using that coordinate contributes zero. Restricting to the ordered coordinates $(0, 2)$ gives exactly C^0 and C^1 . Hence

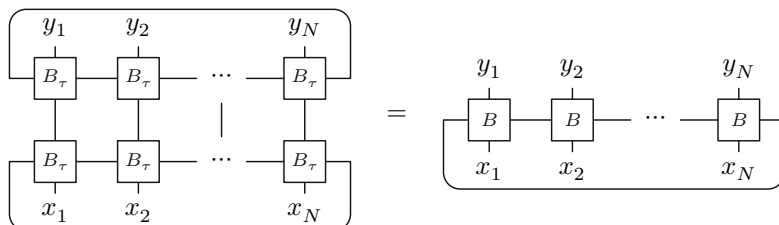
$$\langle x | O_N(B_\tau) | 1, \dots, 1 \rangle = \text{tr}(B_\tau^{x_0 1} \dots B_\tau^{x_{N-1} 1}) = \text{tr}(C^{x_0} \dots C^{x_{N-1}}). \quad (27.127)$$

There is no additional normalization factor.

Relative to the ordered pair of these unnormalized state families, the action coefficients are the Fibonacci matrix $N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, realizing the nonnegative-integer action discussed in [164]. The positive-length hypothesis is essential: at $N = 0$ the periodic trace is the bond dimension, yielding $O_0(B_1) = 2$, $O_0(B_\tau) = 3$, and empty amplitudes 1 and 2 for A and C .

27.6.4 Asymmetric compression of the stacked square

The stacked square in (27.87) contracts the physical index between two copies of B_τ while retaining both virtual indices. Its periodic operator is therefore the square of $O_N(B_\tau)$. The compression maps onto its two target blocks also have entries in $\mathbb{Z}[\sigma]$, so the fusion rule below is again an exact computation in the ring of golden integers.



Theorem 27.6.4.1 (Split compression of the Fibonacci square). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
The source of Definition 27.6.4 carries the block triangular form of Theorem 25.5.1, with target blocks $C_1 = B_1$ and $C_\tau = B_\tau$, and $z = 4$ zero slots. The dimension count is $9 = 2 + 3 + 4$. Here the indexed tensors C_s denote the MPO target blocks, not the chain tensor C . There are compression maps $V_s : \mathbb{C}^{D_s} \rightarrow \mathbb{C}^9$ and $W_s : \mathbb{C}^9 \rightarrow \mathbb{C}^{D_s}$, where $D_1 = 2$ and $D_\tau = 3$, such that $W_s V_s = \mathbb{1}_{D_s}$ and $W_s V_t = 0$ for $s \neq t$. For every word w over the pair alphabet,

$$W_s B^w V_s = C_s^w. \quad (27.128)$$

Every remainder word of length at least six vanishes. In fact, each remainder letter is zero, so the extension splits.

Proof. [✓ Proof](#) The adapted basis has as its first columns the two fusion tensors, followed by a basis of the four-dimensional subspace annihilated by every source letter. Writing Q for this basis matrix, the finite matrix calculation over $\mathbb{Z}[\sigma]$ gives

$$Q^{-1} B^{x'x} Q = B_1^{x'x} \oplus B_\tau^{x'x} \oplus 0_4. \quad (27.129)$$

The corresponding column blocks of Q and row blocks of Q^{-1} give V_s and W_s . Their products give biorthogonality, and multiplication of (27.129) along a word gives (27.128). Block diagonality supplies triangularity, the target diagonal blocks, and the vanishing remainder. The generic nilpotency bound of clause (vi) of Theorem 25.5.1 is $|S| + z = 6$, where S is the two-element set of target blocks; the dimension count is clause (vii). The four zero directions are virtual bond directions, not physical configurations of the chain. $\square$

Theorem 27.6.4.2 (The fusion rule). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every nonempty word w over the alphabet of label pairs,*

$$\mathrm{tr}(B^w) = \mathrm{tr}(B_1^w) + \mathrm{tr}(B_\tau^w). \quad (27.130)$$

Thus at every positive system size N the periodic operators satisfy (27.78), and the periodic vector of the stacked tensor is the sum of the periodic vectors of the two target blocks.

Proof. [✓ Proof](#) Apply Proposition 25.5.3 to Theorem 27.6.4.1. This gives (27.130). For output and input words y, x , the periodic product contracts as

$$\begin{aligned} \langle y | O_N(B_\tau)^2 | x \rangle &= \sum_{m \in \{0,1\}^N} \mathrm{tr}(B_\tau^{y_0 m_0} \dots B_\tau^{y_{N-1} m_{N-1}}) \mathrm{tr}(B_\tau^{m_0 x_0} \dots B_\tau^{m_{N-1} x_{N-1}}) \\ &= \mathrm{tr}(B^{y_0 x_0} \dots B^{y_{N-1} x_{N-1}}). \end{aligned} \quad (27.131)$$

Substitution of (27.130) proves the operator identity entry by entry. The periodic-vector statement is the same trace identity read over the pair alphabet. $\square$

At length zero, the traces of the stacked tensor and the two targets are 9, 2, and 3, respectively. The omitted four-dimensional zero block has empty trace four, although all its positive-length traces vanish. This explains the positive-length restriction in the fusion rule without changing the trace convention. The source asks for the fusion rules at every length. At the empty chain the periodic operators are the scalars $O_0(B_1) = 2$ and $O_0(B_\tau) = 3$, so $O_0(B_1)^2 = 4 \neq 2$ and $O_0(B_1)O_0(B_\tau) = O_0(B_\tau)O_0(B_1) = 6 \neq 3$: the unit laws of Theorem 27.6.1.2 fail there as well, and the empty chain is excluded as a degenerate reading; see [60].

Theorem 27.6.4.3 (The fusion tensors). Lean Lean Lean Lean Lean Lean ✓ Stmt *The compression maps of the two target blocks are the fusion tensors of the Fibonacci algebra. The map V_s is the column block of the adapted basis carrying s , and W_s is the matching row block of its inverse. For every physical letter and $s \in \{1, \tau\}$,*

$$B^{x'x}V_s = V_sC_s^{x'x}. \quad (27.132)$$

$$W_sB^{x'x} = C_s^{x'x}W_s. \quad (27.133)$$

Proof. ✓ Proof Theorem I.1.2 identifies the maps with the row and column blocks of the adapted basis. Each intertwining relation is verified by multiplication over $\mathbb{Z}[\sigma]$ and transported to complex matrices by Theorem 27.6.3. Equivalently, taking the corresponding column or row block of (27.129) gives (27.132) or (27.133). $\square$

$$\begin{array}{ccc}
 \begin{array}{c} x'x \\ | \\ \bullet \\ | \\ B \end{array} \begin{array}{c} \text{---} \end{array} \begin{array}{c} \text{---} \\ | \\ \text{red oval} \\ | \\ V_s \end{array} & = & \begin{array}{c} \text{---} \\ | \\ \text{red oval} \\ | \\ V_s \end{array} \begin{array}{c} \text{---} \\ | \\ \bullet \\ | \\ C_s \end{array} \\
 \begin{array}{c} \text{---} \\ | \\ \text{red oval} \\ | \\ W_s \end{array} \begin{array}{c} \text{---} \\ | \\ \bullet \\ | \\ B \end{array} & = & \begin{array}{c} \text{---} \\ | \\ \bullet \\ | \\ C_s \end{array} \begin{array}{c} \text{---} \\ | \\ \text{red oval} \\ | \\ W_s \end{array}
 \end{array}$$

27.6.5 The categorical anomaly

The Fibonacci fusion ring $\mathbb{Z}[\tau]/(\tau^2 - \tau - 1)$ has no unital ring homomorphism to $\mathbb{Z}$: the two possible images of τ over $\mathbb{C}$, $\varphi = (1 + \sqrt{5})/2$ and $-1/\varphi$, are both irrational. This arithmetic fact is the form in which the anomaly of the Fibonacci fusion rules [16, Section III.A.6] enters the argument below. The general notions used here, fusion algebras with a unit, symmetric families of matrix product states, and their nonnegative integer representations, are those of Chapter 26; the multiplicities with which the normal blocks of an invariant state reappear under the symmetry operators form such a representation (Theorem 26.4.8). In the Fibonacci case the same source exhibits an invariant pair of states carrying the regular action: two blocks x_1, x_τ with $\tau \cdot x_1 = x_\tau$ and $\tau \cdot x_\tau = x_1 + x_\tau$.

Here the two blocks are concrete state tensors on the two-label physical space of this section: the all- τ product tensor and a bond-two tensor generating its image under O_τ . The explicit tensors are defined on the two-label physical alphabet rather than the three-index alphabet of the source. Their raw periodic vectors carry the regular action. The operator O_1 is the admissibility projector introduced earlier in this section, not the identity on the ambient physical space. Its action as the identity on these states is therefore part of the symmetry statement.

The no-go proved below concerns a single word-normal tensor A of positive bond dimension whose periodic vectors satisfy $O_1|V^{(N)}(A)\rangle = |V^{(N)}(A)\rangle$ and $O_\tau|V^{(N)}(A)\rangle = m|V^{(N)}(A)\rangle$ for a length-independent scalar m at every $N > 0$. Then $m \in \mathbb{N}$ must satisfy $m^2 = m + 1$, which has no integer solutions. This is the Fibonacci case of Corollary 26.4.10, a specialization of the statement of [164] to one normal tensor with a length-independent scalar action; the source does not give the proof for this tensor. The specialization does not classify families with more than one block or exclude direct sums of the regular representation. It is a non-invertible analogue of the obstruction to an invariant single-block state from a non-trivial three-cocycle discussed in the same source.

The explicit regular action is recorded in Theorem 27.6.5.4, and the single-block obstruction with the hypotheses just stated is Theorem 27.6.5.6. Neither result classifies all invariant families with several blocks.

Definition 27.6.5.1 (The Fibonacci fusion matrices). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Index the two simple objects of Definition 27.6.4 by 0 and 1, corresponding respectively to the trivial object 1 and the object τ . The fusion matrix

$$N_\tau = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \quad (27.134)$$

represents multiplication by τ in the basis $1, \tau$ of the fusion ring. Together with $N_1 = \mathbb{1}_2$, it gives the regular representation by matrices with nonnegative integer entries. The structure constants are $N_{ab}^c = (N_a)_{bc}$ for $a, b, c \in \{1, \tau\}$, so that N_1 and N_τ are the fusion matrices of Definition 26.2.3. The corresponding matrix product operator tensors are the admissibility block B_1 and the block B_τ of Definition 27.6.4.

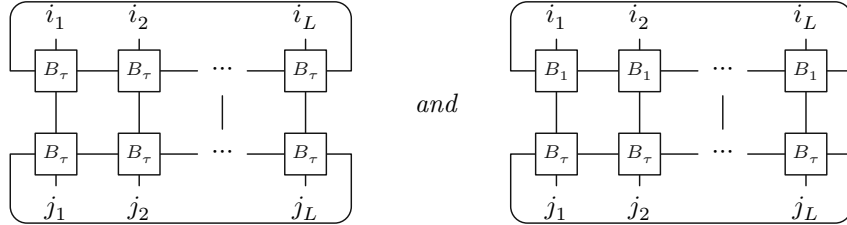
Theorem 27.6.5.2 (The Fibonacci relation for the fusion matrix). [Lean](#) [Stmnt](#) $N_\tau^2 = N_\tau + \mathbb{1}_2$, as matrices with nonnegative integer entries.

Proof. [Proof](#) Direct multiplication gives $N_\tau^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = N_\tau + \mathbb{1}_2$. □

Theorem 27.6.5.3 (The fusion ring on the periodic operators). [Lean](#) [Stmnt](#) For every pair of labels $a, b \in \{1, \tau\}$ and every positive system size N , the periodic operators $O_a = O_a(N) = O_N(B_a)$ satisfy

$$O_a O_b = \sum_{c \in \{1, \tau\}} N_{ab}^c O_c, \quad (27.135)$$

that is, $O_1 O_1 = O_1$, $O_1 O_\tau = O_\tau O_1 = O_\tau$, and $O_\tau O_\tau = O_1 + O_\tau$.



In the diagrams the system size is denoted by L : the first contraction is $O_\tau(L)O_\tau(L)$ and the second is $O_1(L)O_\tau(L)$.

Proof. [Proof](#) The case $a = b = 1$ is the admissibility projector composed with itself. The cases $\{a, b\} = \{1, \tau\}$ are the two unit laws, obtained by block-triangular compression over $\mathbb{Z}[\sigma]$ and transport to complex matrices as in Theorem 27.6.3. The case $a = b = \tau$ is Theorem 27.6.4.2. These four identities give the coefficients of Eq. 27.135. □

The pair of states is the all- τ product tensor and the chain tensor of Definition 27.6.3.1, and the identities $O_1 \Psi_1 = \Psi_1$, $O_\tau \Psi_1 = \Psi_\tau$, and $O_\tau \Psi_\tau = \Psi_1 + \Psi_\tau$ are those of Theorem 27.6.3.4. The content beyond that theorem is the remaining identity $O_1 \Psi_\tau = \Psi_\tau$ and the packaging of all four as the regular representation of the fusion matrices.

Theorem 27.6.5.4 (The regular representation on a pair of normal states). [Lean](#) [Lean](#) [✓ Stmt](#)
Set $s = \varphi^{-1/2}$, the scalar denoted by σ in Definition 27.6.2. On the physical alphabet $\{0, 1\}$, with 0 representing the trivial label and 1 representing τ , define the bond-one tensor A and bond-two tensor C by

$$\begin{aligned} A^0 &= (0), & A^1 &= (1), \\ C^0 &= \begin{pmatrix} 0 & s \\ 0 & 0 \end{pmatrix}, & C^1 &= \begin{pmatrix} 0 & 0 \\ 1 & -s^2 \end{pmatrix}. \end{aligned} \quad (27.136)$$

Both tensors are word-normal: their words of a fixed positive length span their full bond matrix algebra. For A length one suffices, and for C length three suffices.

Use their raw periodic vectors

$$\begin{aligned} \Psi_1(N) &= |V^{(N)}(A)\rangle = |1, \dots, 1\rangle, \\ \Psi_\tau(N) &= |V^{(N)}(C)\rangle = \sum_{x \in \{0,1\}^N} \text{tr}(C^{x_0} \dots C^{x_{N-1}}) |x\rangle. \end{aligned} \quad (27.137)$$

At every positive system size N and for all $a, r \in \{1, \tau\}$, these states satisfy

$$O_a \Psi_r = \sum_{t \in \{1, \tau\}} N_{ar}^t \Psi_t, \quad (27.138)$$

with dependence on N suppressed. Explicitly, $O_\tau \Psi_1 = \Psi_\tau$, $O_\tau \Psi_\tau = \Psi_1 + \Psi_\tau$, and O_1 fixes both states. The multiplicities are the rows of the fusion matrix in Eq. 27.134.

These identities use the tensor normalization in Eq. 27.136; at length $N = 1$, $\Psi_1(1) = |1\rangle$ and $\Psi_\tau(1) = -s^2|1\rangle$.

The following diagrams recall the operator fusion intertwiners of Theorem 27.6.4.3. Here B is the stacked operator square and C_s denotes an operator target block, not the state tensor C of Eq. 27.136. The state action uses the separate action tensors described in the proof.

Proof. [✓ Proof](#) The witnesses are precisely A and C in Eq. 27.136. The nonzero letter of A spans its one-dimensional matrix algebra. For C , the four words 010, 011, 101, and 110 span $M_2(\mathbb{C})$: each matrix unit is a linear combination of these words with coefficients in $\mathbb{Z}[s]$.

For an operator tensor T and a state tensor X , write $(T \cdot X)^i = \sum_j T^{ij} \otimes X^j$ for the action tensor of Corollary 25.6.4. Simultaneous changes of bond coordinates over $\mathbb{Z}[s]$ give

$$\begin{aligned} B_1 \cdot A &\sim A \oplus 0_1, \\ B_\tau \cdot A &\sim C \oplus 0_1, \\ B_\tau \cdot C &\sim A \oplus C \oplus 0_1 \oplus 0_1 \oplus 0_1. \end{aligned} \quad (27.139)$$

Here $\sim$ denotes simultaneous similarity of all letters, and 0_1 is the bond-one tensor whose letters vanish. The source bond dimensions are respectively two, three, and six. All three decompositions are block diagonal, with no remainder.

Taking traces of nonempty words in Eq. 27.139 gives $O_1 \Psi_1 = \Psi_1$, $O_\tau \Psi_1 = \Psi_\tau$, and $O_\tau \Psi_\tau = \Psi_1 + \Psi_\tau$. The remaining case follows from the unit law:

$$O_1 \Psi_\tau = O_1 O_\tau \Psi_1 = O_\tau \Psi_1 = \Psi_\tau. \quad (27.140)$$

This proves Eq. 27.138. The operator fusion intertwiners of Theorem 27.6.4.3 give the compatible operator identity $O_\tau^2 = O_1 + O_\tau$; the state identities above follow from the action decompositions in Eq. 27.139. $\square$

The admissibility projector and the chain amplitudes are derived earlier in this section. The action here uses that same two-label physical space and the same unnormalized contractions. Categorical fusion data alone would not establish these operator and state identities or identify them with a construction on a different physical alphabet. As noted after Theorem 27.6.3.4, the identities do not extend to length zero.

The no-go rests on the integrality of a length-independent scalar action on a normal tensor, Theorem 26.4.4 of Chapter 26.

Theorem 27.6.5.5 (A nonzero length-independent eigenvalue is impossible). Lean ✓ Stmt *Let A be a word-normal tensor of physical dimension two and positive bond dimension, and set $\Psi_A(N) = |V^{(N)}(A)\rangle$. If a single scalar $c \in \mathbb{C}$ satisfies $O_\tau(N)\Psi_A(N) = c\Psi_A(N)$ for every $N \geq 1$, then $c = 0$. No unit-fixing hypothesis is required for this conclusion.*

Proof. ✓ Proof Suppose $c \neq 0$ and suppress $N \geq 1$ in the notation. The unit law $O_1 O_\tau = O_\tau$ of Theorem 27.6.5.3 gives

$$c O_1 \Psi_A = O_1 O_\tau \Psi_A = O_\tau \Psi_A = c \Psi_A. \quad (27.141)$$

Cancelling c yields $O_1 \Psi_A = \Psi_A$ for every $N \geq 1$. Hence A is symmetric under the Fibonacci fusion algebra with eigenvalues $(c_1, c_\tau) = (1, c)$, and Theorem 26.4.9 makes $c \in \mathbb{N}$ with

$$c^2 = c_\tau c_\tau = c_1 + c_\tau = 1 + c. \quad (27.142)$$

No nonnegative integer satisfies this equation (Example 26.4.12), so $c = 0$. $\square$

Theorem 27.6.5.6 (No normal matrix product state is Fibonacci symmetric). Lean ✓ Stmt *There is no positive bond dimension D , word-normal tensor A of physical dimension two and bond dimension D , and single scalar $c \in \mathbb{C}$, such that for every $N \geq 1$,*

$$\begin{aligned} O_1(N)\Psi_A(N) &= \Psi_A(N), \\ O_\tau(N)\Psi_A(N) &= c\Psi_A(N), \end{aligned} \quad (27.143)$$

where $\Psi_A(N) = |V^{(N)}(A)\rangle$. Thus no single normal tensor of positive bond dimension realizes a unital rank-one action of these periodic Fibonacci operators with one length-independent scalar. The obstruction is the absence of a nonnegative integer m satisfying $m^2 = m + 1$.

The explicit pair A, C of Theorem 27.6.5.4 shows that two normal blocks suffice. Among nonempty finite families of word-normal tensors of positive bond dimension, fixed by O_1 and carrying length-independent nonnegative integer action coefficients at every positive length, the minimal number of blocks is therefore two. This counts tensor blocks, not the dimension of their periodic-vector span at each short length.

Proof. [✓ Proof](#) Suppose A and c satisfy Eq. 27.143. Then A is symmetric under the Fibonacci fusion algebra of Theorem 27.6.5.3 with eigenvalues $(c_1, c_\tau) = (1, c)$. A fusion character m with values in $\mathbb{N}$ would satisfy $m_1 = 1$ and

$$m_\tau^2 = m_1 + m_\tau = 1 + m_\tau, \quad (27.144)$$

which no nonnegative integer does (Example 26.4.12). Corollary 26.4.10 therefore excludes A . $\square$

The scalar hypothesis concerns the raw periodic vectors of one fixed tensor at every positive length. The result does not rule out eigenvectors at an individual length, length-dependent eigenvalues, or state tensors that are not word-normal.

27.7 The Ising anyon chain

The Ising fusion category has three simple objects, the vacuum 1 , the fermion ψ , and the non-Abelian anyon σ ; its F -symbols define the string-net operators of Bultinck et al. [7, Appendix D.2]. Chains of interacting anyons were introduced for Fibonacci anyons by Feiguin et al. [161], and the algebra of topological defects of the Ising chain is described by Aasen, Mong, and Fendley [1]. Matrix product operators realize the corresponding topological symmetries of anyon chains, as reviewed in Ref. [16, Section III.A.6]. The matrix product operator tensors for each simple object are taken in the F -symbol convention of Ref. [7, Appendix D.2]; the sigma tensor is scaled by $\sqrt{2}$ so that every entry lies in $\mathbb{Z}[\sqrt{2}]$ and all local computations are exact. Weighted direct sums of these tensors and the weighted Verlinde action on a fusion-tree product state address the question of whether the structure coefficients of a renormalization fixed point can depend on the chain length [11, Theorem 4.14].

Definition 27.7.1 (The Ising string-net tensors of the source). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The quantum dimensions are $d_1 = d_\psi = 1$ and $d_\sigma = \sqrt{2}$; write N_{ab}^c for the Ising fusion multiplicities. Write $v_a = \sqrt{d_a}$ for the Ising labels, so $v_1 = v_\psi = 1$ and $v_\sigma = 2^{1/4}$, and let F_{def}^{abc} be the F -symbols of [7, Appendix D.2], nonzero only when $N_{ab}^e, N_{cd}^e, N_{ad}^f$ and N_{bc}^f are positive, with the values $\pm 1/\sqrt{2}$ on $F_{\sigma\sigma\sigma}^{\sigma\sigma\sigma}$, -1 on $F_{\sigma\sigma\sigma}^{\psi\psi\psi}$ and $F_{\psi\sigma\sigma}^{\sigma\psi\sigma}$, and 1 otherwise. The G -symbols are $G_{def}^{abc} = v_e^{-1} v_f^{-1} F_{def}^{abc}$. For a label f , a basis element (x, y) of P_f in Definition 27.7.4 is read as the pair $(u, l) = (y, x)$ of plaquette labels above and below the operator line; so P_ψ has the bond pairs $[(\psi, 1), (1, \psi), (\sigma, \sigma)]$ and P_σ has $[(\sigma, 1), (\sigma, \psi), (1, \sigma), (\psi, \sigma)]$. The operator tensor of [7, Section 6.2 and Appendix D.1] is the crossing of the operator line f with an edge line e , with the plaquettes b upper left, a upper right, c lower left and d lower right. On the fusion-tree labels $k = (b, e, a)$ and $h = (c, e', d)$ its entries are

$$(S_f^{kh})_{p,q} = \delta_{ee'} \delta_{p,(b,c)} \delta_{q,(a,d)} G_{def}^{abc} \sqrt{v_a v_b v_c v_d}. \quad (27.145)$$

Definition 27.7.1 gives the printed G -symbol tensor with all quantum-dimension factors. The diagonal bond similarity of Theorem 27.7.1.9 relates it to the F -symbol blocks used here. The additional factor $\sqrt{2}$ on the σ block multiplies its periodic operator by $2^{N/2}$.

For every $N > 0$, close the bond indices by the ordinary, unnormalized trace of the ordered product of local matrices. The first physical index is the output and the second is the input.

The vacuum and fermion kernels are identified in Theorem 27.7.1.2; the fermion is a partial isometry whose initial space is the admissible-path subspace (Corollary 27.7.1.3). The fusion identities with the σ block are in Theorems 27.7.1.6 and 27.7.1.8, with the scaling factors retained. The word-span certificates are in Theorem 27.7.2.1. For weighted direct sums, the compression

and periodic trace formula are given in Theorems 27.7.5.1 and 27.7.6.1; the weighted action on the state family is in Theorem 27.7.8.6.

The quantum dimensions of 1 , ψ , and σ are $1, 1, \sqrt{2}$, respectively; they are the Perron–Frobenius dimensions of the fusion rules below (Example 26.3.3). For every simple object a , the vacuum satisfies $1 \otimes a = a \otimes 1 = a$. The remaining fusion rules are

$$\psi \otimes \psi = 1, \quad (27.146)$$

$$\sigma \otimes \psi = \psi \otimes \sigma = \sigma, \quad (27.147)$$

$$\sigma \otimes \sigma = 1 \oplus \psi. \quad (27.148)$$

With rows and columns ordered as $(1, \psi)$, the nontrivial F-symbols are

$$F_{\sigma}^{\sigma\sigma\sigma} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (27.149)$$

$$[F_{\sigma}^{\psi\sigma\psi}]_{\sigma} = -1, \quad (27.150)$$

$$[F_{\psi}^{\sigma\psi\sigma}]_{\sigma} = -1. \quad (27.151)$$

All other allowed components are one, and forbidden components vanish.

The physical site is $\mathbb{C}^{10}$, with orthonormal basis $|h\rangle$, $h = 0, \dots, 9$. A label h is a locally admissible fusion vertex $(\ell(h), \rho(h), r(h))$, meaning that $r(h)$ occurs in $\ell(h) \otimes \rho(h)$. On N sites the ambient Hilbert space is $\mathcal{H}_N = (\mathbb{C}^{10})^{\otimes N}$. For an MPO tensor T , the first physical index is the output index, and its periodic contraction is

$$\langle k | O_N(T) | h \rangle = \text{tr}(T^{k_0 h_0} \dots T^{k_{N-1} h_{N-1}}). \quad (27.152)$$

Here h and k are words of length N , and $|h\rangle$ denotes the corresponding tensor-product basis vector. Unless stated otherwise, $N \geq 1$, and site indices are cyclic.

The entries of the three tensors lie in the ring $\mathbb{Z}[\sqrt{2}]$, so their local products and changes of auxiliary coordinates can be calculated in this ring before embedding into $\mathbb{C}$. We fix the ring and its embedding before writing down the tensors.

Definition 27.7.2 (The ring of $\sqrt{2}$ -integers and its complex embedding). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) The ring $\mathbb{Z}[\sqrt{2}]$ consists of the numbers $a + b\sqrt{2}$ with $a, b \in \mathbb{Z}$, with multiplication determined by $(\sqrt{2})^2 = 2$. It embeds into $\mathbb{C}$ by sending $\sqrt{2}$ to the positive real square root of two. Applying this embedding entrywise gives a complex matrix from a matrix over $\mathbb{Z}[\sqrt{2}]$. For matrix product operator tensors M and N over $\mathbb{Z}[\sqrt{2}]$, their local product is $(M \cdot N)^{ik} = \sum_j M^{ij} \otimes N^{jk}$. For a matrix product operator tensor M and a state tensor A , the local action is $(M \cdot A)^i = \sum_j M^{ij} \otimes A^j$.

Theorem 27.7.3 (Exact arithmetic in $\mathbb{Z}[\sqrt{2}]$). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) The embedding of $\mathbb{Z}[\sqrt{2}]$ into $\mathbb{C}$ is injective and sends every nonzero element to a nonzero complex number. The entrywise embedding of matrices is multiplicative and commutes with the local product and local action of tensors over $\mathbb{Z}[\sqrt{2}]$. If the one-site matrices of a tensor over $\mathbb{Z}[\sqrt{2}]$ include a nonzero $\mathbb{Z}[\sqrt{2}]$ -multiple of every matrix unit, its complex embedding is injective at one site.

Proof. [Stmnt](#) [Proof](#) If $a + b\sqrt{2} = a' + b'\sqrt{2}$ as real numbers, then $(b - b')\sqrt{2} = a' - a$. Irrationality of $\sqrt{2}$ gives $b = b'$, whence $a = a'$. Multiplicativity of the entrywise embedding follows because it is induced by a ring homomorphism. Expanding the finite sums and tensor products proves compatibility with local products and actions. Finally, dividing each designated one-site matrix by its nonzero scalar factor exhibits every matrix unit in the complex span, as in Theorem 1.2.4. $\square$

Definition 27.7.4 (The three tensors A_1, A_ψ, A_σ). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) The ten fusion-tree labels and the functions f and ϵ used in the tensor formulas are

h	$\ell(h)$	$\rho(h)$	$r(h)$	$f(h)$	$\epsilon(h)$
0	1	1	1	3	1
1	1	ψ	ψ	4	1
2	1	σ	σ	5	1
3	ψ	1	ψ	0	1
4	ψ	ψ	1	1	1
5	ψ	σ	σ	2	1
6	σ	1	σ	6	1
7	σ	ψ	σ	7	-1
8	σ	σ	1	9	1
9	σ	σ	ψ	8	1

The auxiliary spaces P_1, P_ψ , and P_σ have ordered bases $[(1, 1), (\psi, \psi), (\sigma, \sigma)], [(1, \psi), (\psi, 1), (\sigma, \sigma)],$ and $[(1, \sigma), (\psi, \sigma), (\sigma, 1), (\sigma, \psi)],$ respectively. Write E_{ab} for the matrix units in these ordered bases, with indices starting at zero. When a matrix-unit subscript is an anyon label, its numerical index is given by the order $(1, \psi, \sigma).$

The complete tensors A_1 and A_ψ are

$$A_1^{kh} = \delta_{kh} E_{\ell(h), r(h)}, \quad (27.153)$$

$$A_\psi^{kh} = \delta_{k, f(h)} \epsilon(h) E_{\ell(h), r(h)}. \quad (27.154)$$

Thus $f = (0\,3)(1\,4)(2\,5)(8\,9)$ fixes 6 and 7, and $\epsilon(h) = -1$ only at $h = 7$. In particular, $A_\psi^{7,7} = -E_{22}$ encodes $[F_\sigma^{\psi\sigma\psi}]_\sigma = -1$.

Put $\widehat{A}_\sigma = \sqrt{2}A_\sigma$. Its complete list of nonzero matrices is

(k, h)	$\widehat{A}_\sigma^{kh}$	(k, h)	$\widehat{A}_\sigma^{kh}$
(0, 6)	$\sqrt{2}E_{22}$	(1, 7)	$\sqrt{2}E_{23}$
(2, 8)	E_{20}	(2, 9)	E_{21}
(3, 6)	$\sqrt{2}E_{33}$	(4, 7)	$\sqrt{2}E_{32}$
(5, 8)	E_{30}	(5, 9)	$-E_{31}$
(6, 0)	$\sqrt{2}E_{00}$	(6, 3)	$\sqrt{2}E_{11}$
(7, 1)	$\sqrt{2}E_{01}$	(7, 4)	$\sqrt{2}E_{10}$
(8, 2)	$\sqrt{2}E_{02}$	(8, 5)	$\sqrt{2}E_{12}$
(9, 2)	$\sqrt{2}E_{03}$	(9, 5)	$-\sqrt{2}E_{13}$

All unlisted matrices vanish. The factors ± 1 in this table are the rescaled $\sigma\sigma\sigma$ F-symbols $\pm 1/\sqrt{2}$. The entries of A_1, A_ψ , and $\widehat{A}_\sigma$ belong to the subring $\mathbb{Z}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Z}\}$ of $\mathbb{C}$.

Theorem 27.7.5 (Sector vanishing of the three tensors). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For each tensor $A \in \{A_1, A_\psi, \widehat{A}_\sigma\}$ and each pair of fusion-tree labels (k, h) , if $\rho(k) \neq \rho(h)$ then $A(k, h) = 0$.

Proof. [✓ Proof](#) Every nonzero matrix in Definition 27.7.4 has $\rho(k) = \rho(h)$, as inspection of the tables shows. $\square$

27.7.1 The periodic operators of the three tensors

The vacuum and fermion tensors are evaluated on closed paths first, where their periodic operators become projections, and the rescaled sigma tensor is evaluated on a product state second.

Local admissibility does not impose compatibility between neighboring vertices. A word h is a closed path when $r(h_v) = \ell(h_{v+1})$ at every site, including the last-to-first bond. Define its closure indicator and sign by

$$\eta_N(h) = \prod_{v=0}^{N-1} \epsilon(h_v) = (-1)^{\#\{v: h_v=7\}}. \quad (27.156)$$

The map f acts on words site by site. The closed-path subspace is $\mathcal{K}_N = \text{span}_{\mathbb{C}}\{|h\rangle : q_N(h) = 1\}$.

Lemma 27.7.1.1 (Cyclic trace of matrix units). Lean ✓ Stmt For $N \geq 1$, scalars c_v , and matrix-unit indices a_v, b_v ,

$$\mathrm{tr} \left(\prod_{v=0}^{N-1} c_v E_{a_v b_v} \right) = \left(\prod_{v=0}^{N-1} c_v \right) \prod_{v=0}^{N-1} \delta_{b_v, a_{v+1}}, \quad (27.157)$$

where $a_N = a_0$.

Proof. ✓ Proof Successive multiplication uses $E_{ab}E_{cd} = \delta_{bc}E_{ad}$. Taking the trace supplies the final factor $\text{tr}(E_{a_0b_{N-1}}) = \delta_{b_{N-1},a_0}$. $\square$

Theorem 27.7.1.2 (Vacuum projection and fermion action on closed paths). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Set $Q_N = O_N(A_1)$ and $S_N = O_N(A_b)$. For $N \geq 1$,

$$\langle k|Q_N|h\rangle = \delta_{kh}q_N(h), \quad (27.158)$$

$$\langle k|S_N|h\rangle = \delta_{k,f(h)}\eta_N(h)q_N(h). \quad (27.159)$$

The operator Q_N is the orthogonal projection onto $\mathcal{K}_N$, and

$$Q_N^2 = Q_N = Q_N^\dagger, \quad (27.160)$$

$$S_N^2 = Q_N, \quad (27.161)$$

$$Q_N S_N = S_N Q_N = S_N, \quad (27.162)$$

$$S_N^\dagger = S_N. \quad (27.163)$$

Proof. ✔ Proof Substituting (27.153) and (27.154) into (27.157) gives the two kernels. Since $q_N(h) \in \{0, 1\}$, the vacuum kernel is the orthogonal projection onto the stated basis subspace.

Let j exchange 1 and ψ and fix σ . The label table gives $\ell(f(h)) = j(\ell(h))$, $r(f(h)) = j(r(h))$, $\rho(f(h)) = \rho(h)$, and $\epsilon(f(h)) = \epsilon(h)$. Since j is bijective, closure is preserved: $q_N(f(h)) = q_N(h)$. Also, $\eta_N(f(h)) = \eta_N(h)$ and $f^2(h) = h$. Therefore

$$S_N^2|h\rangle = q_N(h)q_N(f(h))\eta_N(h)\eta_N(f(h))|f^2(h)\rangle = q_N(h)|h\rangle. \quad (27.164)$$

Closure invariance also gives $Q_N S_N = S_N Q_N = S_N$. The real signs and the involution imply

$$\langle k|S_N^\dagger|h\rangle = \overline{\delta_{h,f(k)}\eta_N(k)q_N(k)} = \delta_{k,f(h)}\eta_N(h)q_N(h). \quad (27.165)$$

Thus $S_N^\dagger = S_N$.

The kernels show $\text{range } S_N = \text{range } Q_N = \mathcal{K}_N$ and $\ker S_N = \ker Q_N = \mathcal{K}_N^\perp$. On $\mathcal{K}_N$, the operator S_N is a self-adjoint signed permutation involution.

Corollary 27.7.1.3 (The fermion operator is a partial isometry). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) For $N \geq 1$, $S_N^\dagger S_N = Q_N$ and $Q_N \neq \mathbb{1}$. Thus S_N is a partial isometry with initial projection Q_N , and it is not unitary on $\mathcal{H}_N$.

Proof. [Proof](#) By Theorem 27.7.1.2, $S_N^\dagger S_N = S_N^2 = Q_N$. The constant word with label $1 = (1, \psi, \psi)$ is not closed, since $r(1) = \psi$ and $\ell(1) = 1$, so the vacuum kernel vanishes on its diagonal entry and $Q_N \neq \mathbb{1}$. A unitary S_N would give $Q_N = S_N^\dagger S_N = \mathbb{1}$. $\square$

At one site, the only closed labels are 0, 3, 6, 7. Using physical matrix units in $M_{10}(\mathbb{C})$,

$$Q_1 = E_{00} + E_{33} + E_{66} + E_{77}, \quad (27.166)$$

$$S_1 = E_{30} + E_{03} + E_{66} - E_{77}. \quad (27.167)$$

Thus the vacuum label acts as the identity on closed paths, not on the entire physical space. These identities follow from the concrete tensor letters rather than from the abstract fusion rules alone.

The sigma normalization and a product-state action

The rescaling $\widehat{A}_\sigma = \sqrt{2}A_\sigma$ occurs at every site. By multilinearity of the periodic contraction,

$$O_N(\widehat{A}_\sigma) = (\sqrt{2})^N O_N(A_\sigma). \quad (27.168)$$

Theorem 27.7.1.4 (Sigma action on the constant closed path). [Lean](#) [Lean](#) [Stmt](#) For $N \geq 1$,

$$O_N(\widehat{A}_\sigma)|6\rangle^{\otimes N} = (\sqrt{2})^N (|0\rangle^{\otimes N} + |3\rangle^{\otimes N}), \quad (27.169)$$

$$O_N(A_\sigma)|6\rangle^{\otimes N} = |0\rangle^{\otimes N} + |3\rangle^{\otimes N}. \quad (27.170)$$

Proof. [Proof](#) At input label $6 = (\sigma, 1, \sigma)$ the only nonzero matrices are $\widehat{A}_\sigma^{0,6} = \sqrt{2}E_{22}$ and $\widehat{A}_\sigma^{3,6} = \sqrt{2}E_{33}$. Products containing both matrix units vanish because $E_{22}E_{33} = E_{33}E_{22} = 0$. Each constant output word has trace $(\sqrt{2})^N$. This proves (27.169); dividing by the factor in (27.168) gives the unscaled formula. $\square$

All three constant paths in this identity lie in $\mathcal{K}_N$. The input is a normalized product state, while the unscaled output has squared norm two. No sigma-square operator identity is needed for this calculation.

Positive length matters here. At length zero, the empty matrix product has trace equal to its auxiliary dimension. Thus $O_0(A_1) = O_0(A_\psi) = 3$ and $O_0(\widehat{A}_\sigma) = 4$, as operators on a one-dimensional physical space. The projection and product-state identities above do not extend to that case.

The fusion algebra of the three periodic operators

The operator blocks of the Fibonacci string-net satisfy the Fibonacci fusion rules [7, Appendix D.1], and the Ising operators are built in the same way from the Ising F-symbols [7, Appendix D.2]. For the Ising category the source prints only the fusion rules themselves. The theorems below prove the corresponding operator identities for the tensors of Definition 27.7.4. These tensors use the F-symbols directly, without the square roots of quantum dimensions that the source divides

Proof. ✓ Proof Apply Lemma 27.7.1.5 with $X = Y = \widehat{A}_\sigma$, $c = 2$, $z = 10$, and $H = G^\top$, where the matrix G of size 16 has fourteen rows of the form $\sqrt{2}e_i$ and the two rows $e_2 \pm e_7$. These two rows carry the F-move of $F_\sigma^{\sigma\sigma\sigma}$, and $GG^\top = G^\top G = 2\mathbb{1}$. The target tensor T is the direct sum of $2A_1$ and $2A_\psi$. As in the proof of Theorem 27.7.1.6, the identity (27.171) is checked in exact arithmetic within each sector of ρ . The periodic operator of T is $2^N(O_N(A_1) + O_N(A_\psi))$ by additivity and the scaling law, which gives (27.176). Dividing by the square of the factor in (27.168) gives (27.177). $\square$

Bond similarity for the source tensors

Theorem 27.7.1.9 (The source's tensors are bond-similar to the formal tensors). Lean Lean Lean Lean Lean Lean ✓ Stmt *The G -symbols satisfy $G_{def}^{abc} = G_{abd}^{fce}$, and the tensors A_1 , A_ψ and A_σ are F -symbol tensors: $(A_f^{kh})_{p,q} = \delta_{ee'}\delta_{p,(b,c)}\delta_{q,(a,d)}F_{abd}^{fce}$. Let g_f be the diagonal matrix on P_f with $g_f(u, l) = (v_u/v_l)^{1/2}$. Then*

$$g_f S_f^{kh} g_f^{-1} = A_f^{kh} \quad (27.178)$$

for $f = 1, \psi, \sigma$ and all labels k, h . Consequently $O_N(S_f) = O_N(A_f)$ at every length, in particular $O_N(S_\sigma) = 2^{-N/2}O_N(\widehat{A}_\sigma)$, and the operators $O_N(S_1)$, $O_N(S_\psi)$, $O_N(S_\sigma)$ satisfy the Ising fusion rules at every positive length.

Proof. ✓ Proof Write $v_a = t^{n_a}$ with $t = 2^{1/4}$, $n_1 = n_\psi = 0$ and $n_\sigma = 1$. Multiplied by $\sqrt{2}t^{n_b+n_d+n_e+n_f}$, the symmetry of the G -symbols becomes an identity between entries of $\mathbb{Z}[\sqrt{2}]$ and powers of $t^2 = \sqrt{2}$, checked in exact arithmetic over all labels; so is the F -symbol form of the three tensors. By the symmetry, $G_{def}^{abc} = v_b^{-1}v_d^{-1}F_{abd}^{fce}$, so the nonzero entries of S_f^{kh} are those of A_f^{kh} multiplied by $\sqrt{v_a v_b v_c v_d}/(v_b v_d) = g_f(a, d)/g_f(b, c)$, which is (27.178). The similarity cancels in the trace defining the periodic operator, so the operators agree, the relation to $\widehat{A}_\sigma$ is the scaling law, and the fusion rules are those of Example 26.4.15. $\square$

27.7.2 One-site injectivity by exact arithmetic

The embedding of $\mathbb{Z}[\sqrt{2}]$ into $\mathbb{C}$ from Theorem 27.7.3 preserves the matrix-unit spanning criterion of Theorem 1.2.4. Here that criterion establishes one-site injectivity, not the spectral-radius-one normalization required of a normal tensor (NT) in the CPSV convention.

Theorem 27.7.2.1 (One-site injectivity of the three tensors). Lean Lean Lean ✓ Stmt *Each of the three tensors is injective at one site: the ten nonzero local matrices of A_1 span $M_3(\mathbb{C})$, those of A_ψ span $M_3(\mathbb{C})$, and the sixteen nonzero local matrices of $\sqrt{2}A_\sigma$ span $M_4(\mathbb{C})$.*

Proof. ✓ Proof Every matrix unit of the relevant auxiliary matrix algebra occurs, up to a nonzero factor in $\mathbb{Z}[\sqrt{2}]$, among the local matrices. This follows directly from the tables. Theorem 27.7.3 gives the spanning statement over $\mathbb{C}$. $\square$

27.7.3 The weighted direct sum and its local product

We now apply asymmetric compression to products of these operators. Consider $\Theta = \lambda_1 A_1 \oplus \lambda_\psi A_\psi \oplus \lambda_\sigma A_\sigma$. First retain the two summands with labels 1 and ψ and rescale $(\lambda_1, \lambda_\psi) = (1/2, 3/10)$ to the integer weights 5, 3. Write $\Theta_2 = 5A_1 \oplus 3A_\psi$.

The local product $\Theta_2 \star (\sqrt{2}A_\sigma)$ represents a product of operators, as in Section 25.6. Here $\star$ denotes the local MPO product of Definition 27.7.2, distinguished from the action of an MPO on

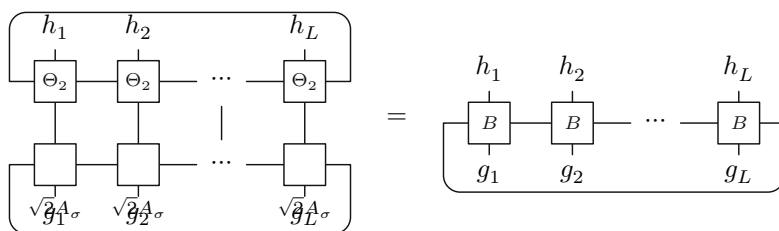
an MPS. Theorem 25.6.2 applies to this product. Section 27.7.8 instead applies the full weighted direct sum to a product state using Corollary 25.6.4. Neither calculation alone establishes a positive vertical decomposition of an MPDO renormalization fixed point.

Definition 27.7.3.1 (The weighted direct sum and the product tensor). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Set $\Theta_2 = 5A_1 \oplus 3A_\psi$, with auxiliary space $\mathbb{C}^6 = P_1 \oplus P_\psi$. The local product tensor $B = \Theta_2 \star (\sqrt{2}A_\sigma)$ has a hundred-letter pair alphabet and auxiliary dimension $24 = 6 \cdot 4$. If $u = 0, \dots, 5$ indexes $P_1 \oplus P_\psi$ and $v = 0, \dots, 3$ indexes P_σ , their tensor-product coordinate is $4u + v$. Explicitly,

$$B^{(h,h')} = \sum_{g=0}^9 \Theta_2^{hg} \otimes (\sqrt{2}A_\sigma)^{gh'}. \quad (27.179)$$

In the periodic diagrams below, L denotes the number of sites.



27.7.4 The gauge and the block decomposition

The change of auxiliary coordinates is a signed permutation of the twenty-four coordinates of B . Its first four rows are the F-move isometry for $1 \otimes \sigma = \sigma$, the next four are the F-move isometry for $\psi \otimes \sigma = \sigma$, and the remaining sixteen form a coordinate complement. The single negative sign is the F-symbol $[F_\sigma^{\psi\sigma\psi}] = -1$.

Definition 27.7.4.1 (Signed permutation matrices and the Ising gauge). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a map π on an index set and a sign vector s , form the matrix with entry $s(x)$ in row x and column $\pi(x)$ and zero elsewhere. When π is a permutation, this is a signed permutation matrix. The Ising gauge is the signed permutation matrix given by the F-move rows described above, in the sense of Definition 1.1.1. There are two nonzero blocks: one from $1 \otimes \sigma = \sigma$, with weight 5, and one from $\psi \otimes \sigma = \sigma$, with weight 3. Both carry $\sqrt{2}A_\sigma$, of auxiliary dimension four. The zero block has dimension $16 = 24 - 2 \cdot 4$.

Theorem 27.7.4.2 (Orthogonality of the Ising gauge). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a permutation π and sign vector s , let $G_{x,\pi(x)} = s(x)$ and let all other entries of G vanish. For every matrix M ,

$$(GMG^\top)_{xy} = s(x) M_{\pi(x)\pi(y)} s(y). \quad (27.180)$$

The Ising gauge G satisfies $GG^\top = G^\top G = \mathbb{1}$, so its transpose is its inverse.

Proof. [✓ Proof](#) Identity (27.180) follows by expanding the matrix product against the two Kronecker deltas of the signed permutation matrices. For the Ising gauge, both products with its transpose are the identity, by finite verification over the twenty-four rows. $\square$

Theorem 27.7.4.3 (The local block identity for the Ising product tensor). [Lean](#) [Lean](#) [✓ Stmt](#) In the block coordinates of Definition 27.7.4.1, conjugation by the Ising gauge gives, for every pair (h, h') ,

$$GB^{(h,h')}G^{-1} = 5(\sqrt{2}A_\sigma)^{(h,h')} \oplus 3(\sqrt{2}A_\sigma)^{(h,h')} \oplus 0_{16}. \quad (27.181)$$

Proof. [✓ Proof](#) If $\rho(h) \neq \rho(h')$, every summand in (27.179) vanishes by Theorem 27.7.5. The right-hand side of (27.181) also vanishes. Otherwise, the sum defining $B^{(h,h')}$ restricts to $\rho(g) = \rho(h)$. Formula (27.180) reduces the claim to the entries in Definitions 27.7.4 and 27.7.4.1, checked separately in the three middle-label sectors. $\square$

27.7.5 Compression onto the σ tensor

Theorem 27.7.5.1 (Compression of the Ising product tensor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The tensor B of Definition 27.7.3.1 has the block triangular form of Theorem 25.5.1 with two weighted copies of $\sqrt{2}A_\sigma$, of weights 5 and 3, and a sixteen-dimensional zero block, $24 = 4 + 4 + 16$. Let V_s include block s in the original auxiliary coordinates, and let W_s project onto it, for $s = 0, 1$. These compression pairs are mutually biorthogonal: $W_s V_t = \delta_{st} \mathbb{1}_4$. For every pair of physical labels,

$$W_s B^{(h,h')} V_s = (\mu_s \sqrt{2} A_\sigma)^{(h,h')}, \quad (27.182)$$

where $\mu_0 = 5$ and $\mu_1 = 3$.

Proof. [✓ Proof](#) Theorem 27.7.4.3 gives the block-diagonal form of every conjugated local matrix over $\mathbb{Z}[\sqrt{2}]$. Theorem 27.7.3 preserves this identity over $\mathbb{C}$. If ι_s and p_s are the coordinate inclusion and projection for block s , set $V_s = G^{-1} \iota_s$ and $W_s = p_s G$. Then $W_s V_t = p_s \iota_t$, and (27.181) gives the compression identity. Block diagonality gives triangularity, as in Theorem I.1.2. The dimension count is clause (vii) of Theorem 25.5.1. $\square$

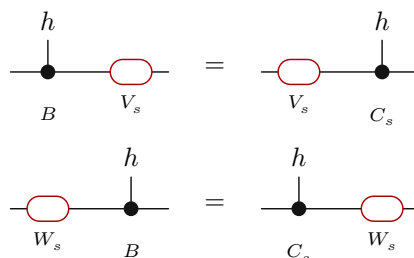
Theorem 27.7.5.2 (The extension splits). [Lean](#) [✓ Stmt](#) The remainder of the compression in Theorem 27.7.5.1 vanishes identically. Thus the extension splits, and each nonzero block has local intertwiners in both directions. For every pair (h, h') ,

$$B^{(h,h')} V_s = V_s (\mu_s \sqrt{2} A_\sigma)^{(h,h')}, \quad (27.183)$$

$$W_s B^{(h,h')} = (\mu_s \sqrt{2} A_\sigma)^{(h,h')} W_s. \quad (27.184)$$

Proof. [✓ Proof](#) Every conjugated local matrix is block diagonal by Theorem 27.7.4.3. Its off-diagonal blocks, which form the conjugated remainder, therefore vanish. Proposition 25.5.4 then gives the two intertwining identities. $\square$

In the following diagram, $C_s = \mu_s \sqrt{2} A_\sigma$, and h denotes a single letter of the pair alphabet.



27.7.6 The length-dependent product coefficient

For a tensor T and a word $w = (i_1, \dots, i_{|w|})$, write $T^w = T^{i_1} \dots T^{i_{|w|}}$.

Theorem 27.7.6.1 (The trace identity for the Ising product tensor). [Lean](#) [Lean](#) [✓ Stmt](#) For every nonempty word w over the alphabet of pair labels,

$$\text{tr}(B^w) = (5^{|w|} + 3^{|w|}) \text{tr}((\sqrt{2}A_\sigma)^w). \quad (27.185)$$

Equivalently, at every positive system size N ,

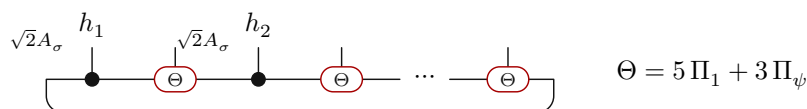
$$O_N(\Theta_2)O_N(\sqrt{2}A_\sigma) = (5^N + 3^N)O_N(\sqrt{2}A_\sigma). \quad (27.186)$$

Proof. [✓ Proof](#) Proposition 25.5.3 applied to the compression in Theorem 27.7.5.1 gives $\text{tr}(B^w)$ as the sum of the two weighted block traces. Pulling each weight out of the matrix product contributes $5^{|w|}$ and $3^{|w|}$. The zero block contributes nothing because w is nonempty. Reading (27.185) at each pair of physical configurations gives the operator identity. $\square$

Remark 27.7.6.2 (Two fusion channels give the same output tensor). The fusions $1 \otimes \sigma = \sigma$ and $\psi \otimes \sigma = \sigma$ give two copies of the same output tensor with different weights. The coefficient in (27.185) is the power sum $5^N + 3^N = \text{tr}(\text{diag}(5, 3)^N)$. This is a length-dependent coefficient for the stated periodic product identity. Its power-sum form alone does not identify these tensors with a basis of normal tensors (BNT) in the positive vertical decomposition of an MPDO renormalization fixed point, nor prove that the length dependence cannot be removed under other choices of tensors.

The two-copy space can be written as $\mathbb{C}^2 \otimes P_\sigma$. Let Π_1 and Π_ψ be the coordinate projections on $\mathbb{C}^2$. In the following diagram, the ring denotes the multiplicity operator $\Theta = 5\Pi_1 + 3\Pi_\psi$, not the direct-sum MPO tensor previously denoted by Θ . Each $\sqrt{2}A_\sigma$ node acts on P_σ . On the combined bond space the ring acts as $\Theta \otimes \mathbb{1}_{P_\sigma}$, and a tensor letter acts as $\mathbb{1}_2 \otimes (\sqrt{2}A_\sigma)^h$. One weight insertion per site therefore gives

$$\text{tr} \left(\prod_{v=0}^{N-1} (\Theta \otimes (\sqrt{2}A_\sigma)^{h_v}) \right) = \text{tr}(\Theta^N) \text{tr} \left(\prod_{v=0}^{N-1} (\sqrt{2}A_\sigma)^{h_v} \right). \quad (27.187)$$



27.7.7 Coefficients from the Ising fusion rules

The power sum $5^N + 3^N$ is one value of a family of structure coefficients built from the Ising fusion multiplicities and one inserted weighted label, which we now define in general. The definition and the theorems of this subsection write L for the chain length that the surrounding prose calls N .

Definition 27.7.7.1 (Structure coefficients from the Ising fusion rules). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let the labels be $1, \psi, \sigma$, and let $N_{ab}^c \in \mathbb{N}$ be the multiplicity of c in the Ising fusion rules. For each label a , these rules are

$$1 \times a = a \times 1 = a, \quad (27.188)$$

$$\psi \times \psi = 1, \quad (27.189)$$

$$\psi \times \sigma = \sigma \times \psi = \sigma, \quad (27.190)$$

$$\sigma \times \sigma = 1 + \psi. \quad (27.191)$$

For arbitrary weights $w_a \in \mathbb{C}$ and $L \in \mathbb{N}$, define

$$c_{a,b,c}^{(L)} = \sum_r \sum_t w_r^L N_{ar}^t N_{tb}^c. \quad (27.192)$$

Both sums run over $\{1, \psi, \sigma\}$. The definition includes $L = 0$ and imposes no positivity or nonvanishing condition on the weights.

Theorem 27.7.7.2 (The three outputs for two sigma labels). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every choice of complex weights and every $L \in \mathbb{N}$,

$$c_{\sigma,\sigma,1}^{(L)} = w_1^L + w_\psi^L, \quad (27.193)$$

$$c_{\sigma,\sigma,\psi}^{(L)} = w_1^L + w_\psi^L, \quad (27.194)$$

$$c_{\sigma,\sigma,\sigma}^{(L)} = 2w_\sigma^L. \quad (27.195)$$

In contrast, the base fusion multiplicity is $N_{\sigma\sigma}^\sigma = 0$.

Proof. [✓ Proof](#) Expand both three-element sums in (27.192) and substitute the fusion table. For output 1 or ψ , the surviving inserted labels are 1 and ψ . For output σ , the inserted label σ contributes through both intermediate labels 1 and ψ . $\square$

Theorem 27.7.7.3 (Associativity of the structure coefficients). [Lean](#) [✓ Stmt](#) For arbitrary complex weights, every $L \in \mathbb{N}$, and all labels a, b, c, d ,

$$\sum_e c_{a,b,e}^{(L)} c_{e,c,d}^{(L)} = \sum_f c_{b,c,f}^{(L)} c_{a,f,d}^{(L)}. \quad (27.196)$$

Proof. [✓ Proof](#) Consider the 3^4 choices of a, b, c, d . In each case, expand the finite sums and substitute the fusion multiplicities. The resulting expressions agree as polynomials in $w_1^L, w_\psi^L, w_\sigma^L$. $\square$

27.7.8 Action on a fusion-tree product state

We next apply the full weighted direct sum $\Theta_3 = 5A_1 \oplus 3A_\psi \oplus 2A_\sigma$ to the constant closed path with label $6 = (\sigma, 1, \sigma)$. This is an operator acting on a state, rather than the operator product of Section 27.7.3. The elementary basis action determines the resulting vector; Corollary 25.6.4 describes its local compression.

Definition 27.7.8.1 (The full weighted direct sum and the fusion-tree product states). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Set $\Theta_3 = 5A_1 \oplus 3A_\psi \oplus 2A_\sigma$, with auxiliary space $\mathbb{C}^{10} = P_1 \oplus P_\psi \oplus P_\sigma$. Its sigma summand is $2A_\sigma = \sqrt{2}\widehat{A}_\sigma$. For a fusion-tree label h_0 , define the bond-dimension-one tensor $\psi_{h_0}^h = \delta_{h,h_0}$. Its periodic vector is $|V^{(N)}(\psi_{h_0})\rangle = |h_0\rangle^{\otimes N}$. In particular, $\psi_{\sigma 1 \sigma} = \psi_6$ generates $|\sigma 1 \sigma\rangle^{\otimes N}$, the fusion tree of the trivial anyon chain. We also write $\psi_{111} = \psi_0$ and $\psi_{\psi 1 \psi} = \psi_3$. The action tensor $\Theta_3 \cdot \psi_{\sigma 1 \sigma}$ is the tensor of Corollary 25.6.4, with one-site matrices $(\Theta_3 \cdot \psi_{\sigma 1 \sigma})^h = \sum_{h'} \Theta_3^{hh'} \otimes \psi_{\sigma 1 \sigma}^{h'}$.

Both Q_N and S_N fix $|6\rangle^{\otimes N}$: label 6 is closed, fixed by f , and has positive sign. The sigma contribution follows from (27.169):

$$O_N(\sqrt{2}\widehat{A}_\sigma)|6\rangle^{\otimes N} = 2^N (|0\rangle^{\otimes N} + |3\rangle^{\otimes N}). \quad (27.197)$$

The factor 2^N comes from two factors of $\sqrt{2}$ at each site, not from a length-independent normalization.

Equivalently, at every positive system size N ,

$$O_N(\Theta_3)|\sigma 1\sigma\rangle^{\otimes N} = (5^N + 3^N)|\sigma 1\sigma\rangle^{\otimes N} + 2^N(|111\rangle^{\otimes N} + |\psi 1\psi\rangle^{\otimes N}). \quad (27.203)$$

The two contributions to $|\sigma 1\sigma\rangle^{\otimes N}$ come from the summands indexed by 1 and ψ , with different weights.

Proof. ✓ Proof Proposition 25.5.3 applied to the compression in Theorem 27.7.8.4 gives the sum of the four weighted block traces. For each target tensor $\mu\psi_{h_0}$, $\text{tr}((\mu\psi_{h_0})^w) = \mu^{|w|} \text{tr}(\psi_{h_0}^w)$. The weights are 5, 3, 2, 2 and the labels are 6, 6, 0, 3, which gives the trace formula. The periodic action identity follows by contracting the action tensor and summing its four target vectors, with the same factors μ^N . This proves (27.203) and agrees with the closed-path basis calculation. $\square$

At length zero the action tensor has trace 10, whereas its four one-dimensional target blocks have total empty trace 4. The difference is the empty trace of the six-dimensional zero block. At positive length that block contributes zero, which explains the nonempty-word and positive-size hypotheses.

27.7.9 The three-object weighted direct sum

Retaining all three hidden bond objects gives four fusion channels, since $\sigma \otimes \sigma = 1 \oplus \psi$. The following result concerns the local product of the weighted direct sum with the σ strand. The literal round-45B boundary tensors require the additional register contraction described in [143].

Definition 27.7.9.1 (The three-object product tensor). Lean Lean Lean Lean ✓ Stmt Put $\widehat{A}_\sigma = \sqrt{2}A_\sigma$, $\Theta_3 = 5A_1 \oplus 3A_\psi \oplus 2A_\sigma$, and $B_3 = \Theta_3 \star \widehat{A}_\sigma$, of auxiliary dimension forty. Its four weighted targets are $5\widehat{A}_\sigma$, $3\widehat{A}_\sigma$, $2\sqrt{2}A_1$, and $2\sqrt{2}A_\psi$, of dimensions 4, 4, 3, 3.

Theorem 27.7.9.2 (Four-channel Ising product). Lean Lean ✓ Stmt At every positive length L ,

$$O_L(\Theta_3)O_L(\widehat{A}_\sigma) = (5^L + 3^L)O_L(\widehat{A}_\sigma) + (2\sqrt{2})^L(O_L(A_1) + O_L(A_\psi)). \quad (27.204)$$

For every nonempty word w , the corresponding trace identity is

$$\text{tr}(B_3^w) = (5^{|w|} + 3^{|w|}) \text{tr}(\widehat{A}_\sigma^w) + (2\sqrt{2})^{|w|}(\text{tr}(A_1^w) + \text{tr}(A_\psi^w)). \quad (27.205)$$

Proof. ✓ Proof The periodic operator of a direct sum is the sum of the periodic operators of its summands. Apply the fusion rules $1 \otimes \sigma = \sigma$, $\psi \otimes \sigma = \sigma$, and $\sigma \otimes \sigma = 1 \oplus \psi$ to obtain (27.204). Every pair-alphabet word is the word of a pair of physical configurations, so taking that matrix element gives (27.205). $\square$

Theorem 27.7.9.3 (Compression of the three-object product). Lean Lean Lean Lean ✓ Stmt The tensor B_3 admits a multi-block asymmetric compression onto the four weighted targets in Definition 27.7.9.1. The auxiliary dimensions satisfy $40 = 4+4+3+3+26$: in an upper-triangular presentation, twenty-six additional one-dimensional diagonal blocks vanish.

Proof. ✓ Proof Each target is a nonzero scalar multiple of a normal tensor of positive auxiliary dimension. Apply Theorem 25.5.1 to (27.205), and subtract the sum of the four target dimensions from forty. $\square$

27.8 Group matrix product operators from a three-cocycle

Garre-Rubio, Lootens and Molnár [164] build, from a finite group G and a three-cocycle ω , a matrix product operator representation of G on a periodic chain whose sites carry the regular representation $\mathbb{C}[G]$. Each periodic operator acts as $U_g = (\otimes_i L_g) \prod_i W_g^{i,i+1}$ (Theorem 27.8.2), combining on-site left shifts L_g with diagonal two-site phase gates W_g determined by ω ; its place among matrix product operator symmetries, and the anomaly carried by the class of ω , are described in Remark 26.1.3. For cyclic groups, Garre-Rubio and Schuch [166] give one explicit cocycle in each cohomology class.

The periodic display of [164] places two left shifts on each pair of neighboring sites, which read literally acts on every site by the shift by g^2 . The local tensor of that construction, and its example for the group of order two, carry one left shift per site; the operator below follows the tensor, as recorded in [96].

Definition 27.8.1 (The tensor of a three-cochain). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) Let G be a finite group and let $\omega: G^3 \rightarrow \mathbb{C}^\times$ be a scalar three-cochain. On $\mathbb{C}[G] \otimes \mathbb{C}[G]$ define the diagonal gate and the left shift by

$$W_a|k, l\rangle = \omega(g, l, l^{-1}k)|k, l\rangle, \quad L_a|k\rangle = |gk\rangle.$$

The tensor $\hat{T}_g$ has physical and bond indices in G , with

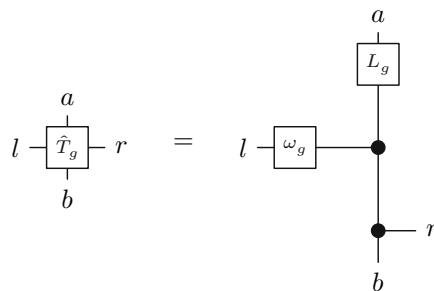
$$(\hat{T}_q^{a,b})_{l,r} = \delta_{a,qb} \delta_{r,b} \omega(g, b, b^{-1}l), \quad (27.206)$$

where a is the output, b the input, and l, r the incoming and outgoing bonds. For a configuration $t = (t_0, \dots, t_{N-1}) \in G^N$ on a periodic chain of $N > 0$ sites, with $t_N = t_0$, write $gt = (gt_0, \dots, gt_{N-1})$ and

$$\phi_g(t) = \prod_{i=0}^{N-1} \omega(g, t_{i+1}, t_{i+1}^{-1} t_i). \quad (27.207)$$

The group elements are identified with the basis labels $\{0, \dots, |G| - 1\}$ by a fixed bijection, and $U_q = O_N(\hat{T}_q)$ denotes the periodic operator.

Following [164], write $\omega_g(l, b) = \omega(g, b, b^{-1}l)$ for the diagonal entries of W_g . By (27.206), the tensor $\hat{T}_g$ copies its input b to the outgoing bond, weights the pair formed by the incoming bond and the input by ω_g , and applies L_g to the input to give the output.



The boundary is the ordinary trace, with no additional boundary matrix. The monomial kernel is given in Theorem 27.8.2. The cocycle equation implies the multiplication law (Theorem 27.8.3), while unit-modulus values imply unitarity (Theorem 27.8.5). For the chosen fusion

tensors, the associator is ω^{-1} (Theorem 27.8.1.3). This does not assert that all the tensors are normal: for a nontrivial group and a normalized cocycle, the identity tensor is not normal (Theorem 27.8.1.4).

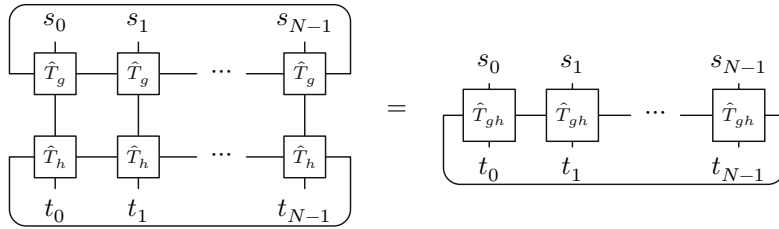
Theorem 27.8.2 (The periodic operator is monomial). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every scalar three-cochain ω , every $g \in G$, every $N > 0$, and all $s, t \in G^N$,*

$$U_g(s, t) = \delta_{s, gt} \phi_g(t).$$

Equivalently, $U_g = (\otimes_i L_g) \prod_i W_g^{i, i+1}$, the gates acting on the neighboring pairs (t_i, t_{i+1}) before the shifts.

Proof. [✓ Proof](#) Expand the trace over cyclic bond configurations b . The factor $\delta_{r, b}$ in (27.206) forces $b_{i+1} = t_i$, so only $b_i = t_{i-1}$ contributes. Its contribution vanishes unless $s_i = gt_i$ at every site, and otherwise equals $\prod_i \omega(g, t_i, t_{i-1}^{-1}t_{i-1})$, which is $\phi_g(t)$ after the cyclic reindexing $i \mapsto i+1$. The gates are diagonal, so the entry of $(\otimes_i L_g) \prod_i W_g^{i, i+1}$ at (s, t) is the same product. $\square$

The product law is the following identity of periodic operators, with U_g applied after U_h .



Theorem 27.8.3 (Product law). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If ω is a three-cocycle, then $U_g U_h = U_{gh}$ for all $g, h \in G$ and every $N > 0$.*

Proof. [✓ Proof](#) Both sides are monomial with the shift by gh , so it suffices that $\phi_g(ht)\phi_h(t) = \phi_{gh}(t)$. The cocycle equation (19.389) at $(g, h, a, a^{-1}x)$ gives, at each site,

$$\omega(g, ha, a^{-1}x) \omega(h, a, a^{-1}x) = \omega(gh, a, a^{-1}x) \frac{\omega(g, h, x)}{\omega(g, h, a)}.$$

With $a = t_{i+1}$ and $x = t_i$ the correction factors telescope around the ring. $\square$

Theorem 27.8.4 (Unit law). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If ω is normalized, then $W_e = 1$ and $U_e = 1$ for every $N > 0$.*

Proof. [✓ Proof](#) Normalization in the first argument makes every factor of W_e and of $\phi_e(t)$ equal to one, and the shift by e is the identity. $\square$

Theorem 27.8.5 (Unitarity and the adjoint law). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If every value of ω has modulus one, then U_g is unitary for every $g \in G$ and every $N > 0$, so $\hat{T}_g$ is a matrix product unitary on nonempty chains. If moreover ω is a normalized three-cocycle, then $U_g^\dagger = U_{g^{-1}}$ for every $N > 0$.*

Proof. [✓ Proof](#) A monomial matrix whose nonzero entries have modulus one is unitary. For the adjoint law, $U_g^\dagger = U_g^\dagger U_g U_{g^{-1}} = U_{g^{-1}}$ by Theorems 27.8.3 and 27.8.4. $\square$

The single-site tensors are not injective: their $|G|^2$ physical matrices span a space of dimension at most $|G|$. The operator laws above are therefore stated directly rather than through an injective representation.

27.8.1 Fusion tensors and the inverse cocycle

The source equips the tensors $\hat{T}_g$ with explicit fusion tensors and states that they realize the inverse cocycle [164]. The bond of $\hat{T}_g$ carries the input label of the preceding site, so along the stacked product $\hat{T}_g \hat{T}_h$ the bond pair over a label l of $\hat{T}_{gh}$ is (hl, l) .

Definition 27.8.1.1 (Fusion tensors of the cocycle construction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $g, h \in G$ let $F_{g,h}^<$ send the bond pair (hl, l) of $\hat{T}_g \hat{T}_h$ to the bond label l of $\hat{T}_{gh}$ with the factor $\omega(g, h, l)^{-1}$, and vanish on every other bond pair, and let $F_{g,h}^>$ send l to (hl, l) with the factor $\omega(g, h, l)$.

Theorem 27.8.1.2 (The fusion tensors reduce the product). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If ω is a three-cocycle, then $F_{g,h}^< F_{g,h}^> = \mathbf{1}$ and $F_{g,h}^< (\hat{T}_g \hat{T}_h)^{ij} = \hat{T}_{gh}^{ij} F_{g,h}^<$ for every pair of physical indices. Hence $F_{g,h}^< (\hat{T}_g \hat{T}_h)^{\mathbf{w}} F_{g,h}^> = \hat{T}_{gh}^{\mathbf{w}}$ for every word $\mathbf{w}$, including the empty word.

Proof. [✓ Proof](#) Both sides of the letter identity vanish unless the output bond pair is (hj, j) and the ket index is ghj . On that support the left side is $\omega(g, h, l)^{-1} \omega(g, hj, j^{-1}l) \omega(h, j, j^{-1}l)$ and the right side is $\omega(gh, j, j^{-1}l) \omega(g, h, j)^{-1}$, which agree by the sitewise cocycle identity in the proof of Theorem 27.8.3. The word identity follows by moving $F_{g,h}^<$ through the word and using $F_{g,h}^< F_{g,h}^> = \mathbf{1}$. $\square$

Theorem 27.8.1.3 (The construction carries the inverse cocycle). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let ω be a three-cocycle. For the fusion tensors of Definition 27.8.1.1, the left boundaries of the two fusion trees of (g, h, k) satisfy $V^L = \omega(g, h, k)^{-1} V^R$, and the anomaly three-cochain of these fusion tensors is ω^{-1} .

Proof. [✓ Proof](#) Both boundaries send the triple bond label $(h k x, k x, x)$ to x and vanish elsewhere. Their factors are $\omega(gh, k, x)^{-1} \omega(g, h, k x)^{-1}$ and $\omega(g, h k, x)^{-1} \omega(h, k, x)^{-1}$, whose ratio is $\omega(g, h, k)^{-1}$ by the cocycle equation at (g, h, k, x) . The scalar is unique because every $\hat{T}_g$ has nonzero words of every length, its periodic operator being a shift with nonzero phases. $\square$

Theorem 27.8.1.4 (The identity member is not normal). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If G is nontrivial and ω is normalized, then $\hat{T}_e$ is not normal. Hence the tensors $\hat{T}_g$ do not form a representation with normal tensors, and the three-cochain of Theorem 27.8.1.3 is that of the fusion tensors of Definition 27.8.1.1.

Proof. [✓ Proof](#) Every letter of $\hat{T}_e$ has all rows equal, so every nonempty product of letters does too, while the matrix unit at a diagonal position has distinct rows. $\square$

27.8.2 The group of order two and the decorated CZX operator

For $G = \mathbb{Z}_2 = \{e, s\}$ with the nontrivial cocycle ω_1 of Definition 27.3.6.1, the source identifies the gate as $CZ(1 \otimes Z)$, the shift as X , and the operator as $\prod_i CZ_{i,i+1} Z_i \prod_i X_i$ [164]. Identify e, s with the qubit labels 0, 1 and write $\bar{t}$ for the bitwise complement of $t \in \{0, 1\}^N$ and $q(s) = \sum_i s_i s_{i+1}$.

Theorem 27.8.2.1 (The decorated CZX operator). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $\omega = \omega_1$ and $g = s$, the gate is $W_s |k, l\rangle = (-1)^{kl+l} |k, l\rangle$, which is $CZ(1 \otimes Z)$. For every $N > 0$,

$$U_s(u, t) = \delta_{u, \bar{t}} (-1)^{q(u) + \sum_i u_i},$$

so that $U_s = \prod_i CZ_{i,i+1} Z_i \prod_i X_i$.

27.8.3 The cyclic groups

For $n \geq 1$ and $j \in \mathbb{N}$, Garre-Rubio and Schuch [166, Section IV.D] write the three-cocycles of $\mathbb{Z}_n$ as ω_j below, with $j = 0, \dots, n-1$ labelling the cohomology classes.

Definition 27.8.3.1 (The cocycles of the cyclic groups). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Represent residues of $\mathbb{Z}_n$ by $\{0, \dots, n-1\}$ and write $[b+c]$ for the representative of $b+c$. Define

$$\omega_j(a, b, c) = \exp\left\{\frac{2\pi i j}{n^2} a(b + c - [b + c])\right\},$$

and let $\zeta = e^{2\pi i/n}$.

Theorem 27.8.3.2 (The cyclic instances). Lean Lean Lean Lean Lean Lean ✓ Stmt For every $n \geq 1$ and $j \in \mathbb{N}$, the value is $\omega_j(a, b, c) = \zeta^{ja \lfloor (b+c)/n \rfloor}$, and ω_j is a normalized three-cocycle of $\mathbb{Z}_n$ with values of modulus one. For every $g \in \mathbb{Z}_n$, every $N > 0$ and all $s, t \in \mathbb{Z}_n^N$,

$$U_g(s, t) = \delta_{s, g+t} \prod_{i=0}^{N-1} \zeta^{jg[(t_{i+1} + [t_i - t_{i+1}])/n]}.$$

Every $\hat{T}_g$ is a matrix product unitary on nonempty chains, and $U_0 = 1$, $U_g U_h = U_{g+h}$ and $U_g^\dagger = U_{-g}$ for every $N > 0$.

Proof. □ **Proof** Since $b + c - [b + c] = n \lfloor (b + c)/n \rfloor$, the exponential is the stated power of ζ . The cocycle equation reduces modulo n to the carry identity

$$\left\lfloor \frac{k+l}{n} \right\rfloor + \left\lfloor \frac{h+[k+l]}{n} \right\rfloor = \left\lfloor \frac{h+k}{n} \right\rfloor + \left\lfloor \frac{[h+k]+l}{n} \right\rfloor,$$

both sides counting the multiples of n in $h + k + l$. An identity argument makes the exponent zero, and every power of ζ has modulus one. The kernel is Theorem 27.8.2 evaluated on these values, and the operator laws are Theorems 27.8.3, 27.8.4 and 27.8.5. $\square$

Garre-Rubio and Schuch [166, Section IV.C] show that interchanging the domain walls created by a symmetry-breaking element g of order $o(g)$ multiplies the state by the phase $\prod_{k=1}^{o(g)} \omega^{-1}(g, g^k, g)$. Since $g^{o(g)} = e$ this phase is the inverse of the cyclic invariant $I_{g, o(g)}(\omega)$ of Definition 19.5.10.42, so it depends only on the class of ω .

Definition 27.8.3.3 (The domain-wall phase). [Lean](#) [Lean](#) [✓ Stmt](#) For a scalar three-cochain ω of a group G and $q \in G$ of order $o(q)$, set

$$P_g(\omega) = \prod_{k=1}^{o(g)} \omega(g, g^k, g)^{-1} = I_{g, o(g)}(\omega)^{-1}.$$

Theorem 27.8.3.4 (Gauge invariance of the domain-wall phase). [\[Lean\]](#) [✓ Stmt](#) *Cochomologous three-cochains have the same phase P_g for every $g \in G$.*

Proof. □ **Proof** Since $P_g(\omega) = I_{g, o(g)}(\omega)^{-1}$, this is the gauge invariance of the cyclic invariant in Theorem 19.5.10.43. □

Theorem 27.8.3.5 (Domain-wall phases of the cyclic cocycles). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [StmT](#) Let $n \geq 1$, $j \in \mathbb{N}$ and $a \in \{0, \dots, n-1\}$, with

Proof. ✓ Proof The exponent $a_1 b_2 c_2$ is trilinear over $\mathbb{Z}_2$, hence an additive three-cocycle, and it vanishes when any argument is zero. The kernel is Theorem 27.8.2 with $t_{i+1}^{-1} t_i = t_i - t_{i+1}$, and the operator laws are Theorems 27.8.3, 27.8.4 and 27.8.5. $\square$

27.9 The anomalous $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry and its condensation defect

The operators $U_{g,L}$ on a periodic chain of L two-qubit sites are defined by (27.208). The construction targets the mixed three-cocycle of type II whose exponent $(-1)^{a_1 b_2 c_2}$ couples the first factor of the group to the second, motivated by anomalous $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry-protected topological phases [9]. The four operators do not form an exact representation. After an on-site dressing they do (Theorem 27.9.1.4), and the anomaly class of the dressed representation has the detector values of the type-II class (Theorem 27.9.3.1); in particular it is nontrivial. Matrix product operator symmetries of this kind are reviewed in [16, Section III.A.6]. The local tensors are assembled from the letters of the CZX matrix product unitary [16, Appendix A, “The CZX MPU”], whose symmetry-protected context is [16, Section III.B.1]. The group matrix product operators of [164] and the anomalous representations of [166] are exact representations, whereas the tensors defined below compose only projectively on odd rings, built from the CZX letters of Section 27.3. The sum $\mathcal{C}_L = \sum_{g \in \mathbb{Z}_2 \times \mathbb{Z}_2} U_{g,L}$ over the group is a condensation defect in the sense of [197]. Throughout this section the chain length is written L rather than N , because N denotes the direct-sum tensor that generates the condensation defect.

On a periodic chain of $L > 0$ two-qubit sites, define

$$U_{g,L}|t\rangle = (-1)^{g_1 \sum_{n=0}^{L-1} t_{n,2} t_{n+1,2}} |g \oplus t\rangle. \quad (27.208)$$

Here $g = (g_1, g_2) \in G = \mathbb{Z}_2 \times \mathbb{Z}_2$, $t = (t_0, \dots, t_{L-1})$ with $t_n = (t_{n,1}, t_{n,2}) \in \{0, 1\}^2$, and $(g \oplus t)_n = g \oplus t_n$ is componentwise addition modulo two. Site indices are cyclic, so $t_L = t_0$. The vectors $|t\rangle$ form the computational basis of $(\mathbb{C}^2 \otimes \mathbb{C}^2)^{\otimes L}$. Write $e = (0, 0)$, $y = (0, 1)$, $x = (1, 0)$, and $xy = (1, 1)$. The product gh denotes $g \oplus h$. The phase exponent in (27.208) is evaluated on the input bits.

Let $X_n^{(r)}$ flip qubit r at site n , and define the diagonal operator $D_{2,L}$ by

$$D_{2,L}|t\rangle = (-1)^{\sum_n t_{n,2} t_{n+1,2}} |t\rangle.$$

Then

$$U_{g,L} = \left(\prod_{n=0}^{L-1} (X_n^{(1)})^{g_1} (X_n^{(2)})^{g_2} \right) D_{2,L}^{g_1}. \quad (27.209)$$

For $L \geq 3$, $D_{2,L}$ is the product of controlled- Z gates between neighboring second qubits. Thus $U_{x,L}$ flips the first bits and applies a controlled phase to the second bits; it does not flip the second bits and is not the CZX operator on those qubits. The operator $U_{xy,L}$ flips both bits, with the diagonal phase acting first. Operator products act from right to left.

The cyclic convention in (27.208) also specifies the short rings. For $L = 1$, the phase exponent is $t_{0,2}^2 = t_{0,2}$, so $D_{2,1} = Z^{(2)}$ and $U_{xy,1} = X^{(1)} X^{(2)} Z^{(2)}$, where $Z|b\rangle = (-1)^b |b\rangle$. For $L = 2$, the exponent is $2t_{0,2} t_{1,2}$, so $D_{2,2} = 1$. There is one self-loop at length one and two cyclic bonds at length two; these are not replaced by the edges of a simple graph.

The local tensors are matrix product operator tensors whose first physical index is the output, or matrix row, and whose second is the input, or matrix column.

Definition 27.9.1 (The four symmetry tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[✓ Stmt](#) Encode a two-qubit label $i = (i_1, i_2)$ by $2i_1 + i_2$. For $g \in G$, let $D_g = 1$ when $g_1 = 0$ and $D_g = 2$ when $g_1 = 1$. Define $M_g^{(o,i)} \in M_{D_g}(\mathbb{C})$ by

$$M_g^{(o,i)}[\alpha, \beta] = \delta_{o, g \oplus i} \delta_{\beta, \pi_g(i)} f_g(\alpha, i), \quad (27.210)$$

where δ is the Kronecker delta and $\alpha, \beta \in \{0, \dots, D_g - 1\}$. For $g \in \{e, y\}$, set $\pi_g(i) = 0$ and $f_g(\alpha, i) = 1$. For $g \in \{x, xy\}$, set $\pi_g(i) = i_2$ and $f_g(\alpha, i) = (-1)^{\alpha i_2}$. Thus the bond-one entries are $\delta_{o,i}$ and $\delta_{o, y \oplus i}$, respectively. Each tensor has four nonzero physical-index pairs. For the bond-two tensors, the nonzero matrix is T_{i_2} , where

$$\begin{aligned} T_0 &= \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \\ T_1 &= \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix}. \end{aligned} \quad (27.211)$$

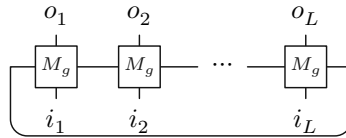
Pairing the physical indices gives an MPS tensor on the sixteen-letter alphabet indexed by $4o + i$, using the integer encodings of o and i .

Every entry lies in $\{0, \pm 1\}$. The matrices T_0, T_1 are those of the bond-two symmetry tensor in Section 27.3, but the physical output constraints are different. The tensor M_{xy} is obtained from M_x by relabeling the outgoing physical index as $o \mapsto o \oplus y$. The boundary convention is the ordinary, unnormalized trace:

$$\rho^{(L)}(M_g)_{s,t} = \text{tr} \left(M_g^{(s_0, t_0)} \dots M_g^{(s_{L-1}, t_{L-1})} \right). \quad (27.212)$$

There are no open boundary vectors.

Theorem 27.9.3 establishes the word spans, with one-site injectivity for the bond-one tensors and two-site injectivity for the bond-two tensors. The multiplication phases are retained in Theorem 27.9.1.2. The dressed exact representation is given in Theorem 27.9.1.4, and its nontrivial anomaly detectors in Theorem 27.9.3.1. These detector values concern the dressed family; they do not identify the full cohomology class of the original family.



The first property is that the periodic contraction of each tensor reproduces the operator of (27.208).

Theorem 27.9.2 (Periodic contraction and unitarity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#)

For every $L > 0$, $\rho^{(L)}(M_g) = U_{g,L}$. Each $U_{g,L}$ is unitary, and

$$U_{g,L}^{-1} = U_{g,L}^\dagger = (-1)^{L g_1 g_2} U_{g,L}. \quad (27.213)$$

Proof. [✓ Proof](#) Expand the trace in (27.212) over bond indices $b_n \in \{0, \dots, D_g - 1\}$, with $b_L = b_0$:

$$\rho^{(L)}(M_g)_{s,t} = \sum_{b_0, \dots, b_{L-1}} \prod_{n=0}^{L-1} M_g^{(s_n, t_n)}[b_n, b_{n+1}]. \quad (27.214)$$

If $s \neq g \oplus t$, the physical constraints make every summand zero. Otherwise the bond constraints force $b_{n+1} = \pi_g(t_n)$, leaving exactly one bond configuration. For $g_1 = 0$ its weight is one. For $g_1 = 1$ its weight is $\prod_n (-1)^{t_{n-1,2}t_{n,2}}$. Reindexing the cyclic sum therefore yields

$$\rho^{(L)}(M_g)_{s,t} = \delta_{s,g \oplus t} (-1)^{g_1 \sum_n t_{n,2} t_{n+1,2}}. \quad (27.215)$$

This is (27.208), including its self-loop convention at $L = 1$.

The map $t \mapsto g \oplus t$ permutes the computational basis, and every coefficient in (27.208) has modulus one. Hence the images form an orthonormal basis, proving unitarity.

Set $E(t_2) = \sum_n t_{n,2} t_{n+1,2}$. Complementing all second bits gives

$$E(1 \oplus t_2) = \sum_n (1 - t_{n,2})(1 - t_{n+1,2}) = L - 2 \sum_n t_{n,2} + E(t_2). \quad (27.216)$$

Thus $E(g_2 \oplus t_2) \equiv E(t_2) + Lg_2 \pmod{2}$. Applying (27.208) twice gives $U_{g,L}^2 = (-1)^{Lg_1g_2} \mathbb{1}$. Together with unitarity, this proves (27.213). $\square$

Equation (27.216) also fixes the gate order: $D_{2,L} \prod_n X_n^{(1)} X_n^{(2)} = (-1)^L U_{xy,L}$. At length zero, the trace of the empty product is D_g . The bond-one tensors therefore give the scalar 1, while the bond-two tensors give 2, not a unitary operator. All unitarity and group-multiplication assertions here concern positive lengths.

The basis action also determines multiplication. For $g, h \in G$,

$$U_{g,L} U_{h,L} |t\rangle = (-1)^{h_1 E(t_2) + g_1 E(h_2 \oplus t_2)} |g \oplus h \oplus t\rangle = (-1)^{Lg_1h_2} U_{gh,L} |t\rangle. \quad (27.217)$$

The last equality uses (27.216) and the fact that the parity of $g_1 + h_1$ is the first bit of gh . Thus $\lambda(g, h) = (-1)^{g_1h_2}$ supplies a length-dependent scalar factor, reflecting the non-trivial projective 2-cocycle class of $\mathbb{Z}_2 \times \mathbb{Z}_2$ [9, 164]. The tensor proof of this law, through word traces, is given in Theorem 27.9.1.2.

The second property concerns the word spans of the four tensors, which the compression results of Section 27.9.2 require.

Theorem 27.9.3 (Injectivity after blocking of the four symmetry tensors). [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The bond-one tensors of U_e and U_y are injective without blocking: a nonzero matrix is the scalar 1 and spans $M_1(\mathbb{C})$. The bond-two tensors of U_x and U_{xy} are not injective without blocking, since their four nonzero matrices span only a two-dimensional space. Their length-two words span $M_2(\mathbb{C})$, so they are injective after blocking two sites. These assertions concern word spans. The tensors are proportional to normal tensors (NT); the spectral-radius normalization required for NT is imposed separately.*

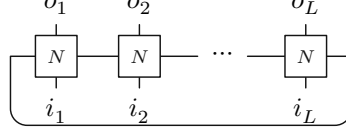
Proof. [✓ Proof](#) For each bond-one tensor, a nonzero scalar matrix spans $M_1(\mathbb{C})$. For each bond-two tensor, all four products $T_a T_b$ occur as length-two words. If E_{ab} denotes a matrix unit, direct multiplication gives

$$\begin{aligned} 2E_{00} &= T_0 T_0 + T_1 T_0, \\ 2E_{10} &= T_0 T_0 - T_1 T_0, \\ 2E_{01} &= T_0 T_1 - T_1 T_1, \\ 2E_{11} &= T_0 T_1 + T_1 T_1. \end{aligned}$$

These integer linear combinations span $M_2(\mathbb{C})$. At length one, T_0, T_1 are linearly independent, but the functional $Q \mapsto Q_{00} - Q_{10}$ vanishes on both and takes the value 1 on E_{00} . Their span is therefore proper. $\square$

27.9.1 The condensation defect and its periodic fusion table

The condensation defect is the operator $A_L = U_{e,L} + U_{x,L} + U_{y,L} + U_{xy,L}$. It is generated by the direct-sum tensor $N = M_e \oplus M_x \oplus M_y \oplus M_{xy}$ of Definition 16.1.8, with bond dimension six.



Definition 27.9.1.1 (The condensation defect tensor). [Lean](#) [Lean](#) [✓ Stmt](#) The condensation defect tensor is the iterated direct sum $N = ((M_e \oplus M_x) \oplus M_y) \oplus M_{xy}$ of bond dimension six. Its periodic operator is the sum of the four periodic operators of Definition 27.9.1:

$$\rho^{(L)}(N) = \rho^{(L)}(M_e) + \rho^{(L)}(M_x) + \rho^{(L)}(M_y) + \rho^{(L)}(M_{xy}). \quad (27.218)$$

This identity holds at every system size, including length zero. For $L > 0$, it reads $\rho^{(L)}(N) = A_L$.

Proof. [✓ Proof](#) Apply the additivity of Theorem 16.1.10 three times. □

The multiplication law (27.217) of the four operators is now derived from the tensors themselves, through word traces of their stacked products.

Theorem 27.9.1.2 (The periodic fusion table). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $L \geq 1$ and every $g, h \in \mathbb{Z}_2 \times \mathbb{Z}_2$, the periodic operators satisfy

$$\rho^{(L)}(M_g) \rho^{(L)}(M_h) = \lambda(g, h)^L \rho^{(L)}(M_{gh}), \quad (27.219)$$

where

$$\lambda(g, h) = \begin{cases} -1 & \text{if } g \in \{x, xy\} \text{ and } h \in \{y, xy\}, \\ +1 & \text{otherwise.} \end{cases}$$

On odd rings, $\lambda(x, y)\lambda(y, x) = -1$, so $\rho^{(L)}(M_x)\rho^{(L)}(M_y) = -\rho^{(L)}(M_y)\rho^{(L)}(M_x)$. The four operators form a projective representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ on odd rings and a linear representation on even rings.

Proof. [✓ Proof](#) The stacked tensor is defined by $(M_g \cdot M_h)^{(o,i)} = \sum_{j=0}^3 M_g^{(o,j)} \otimes M_h^{(j,i)}$. For each of the sixteen pairs, the pair-block compression gives

$$\text{tr}((M_g \cdot M_h)^w) = \lambda(g, h)^{|w|} \text{tr}(M_{gh}^w) \quad (27.220)$$

for every nonempty word w on the paired physical alphabet. The explicit split and non-split compressions are described in Theorems 27.9.2.2 and 27.9.2.3. Lemma 25.6.1 converts (27.220) into the operator identity (27.219). The odd-ring anticommutation follows by substituting $\lambda(x, y) = -1$ and $\lambda(y, x) = +1$ and using $(-1)^L = -1$. □

The four operators are related to the group matrix product operators of Section 27.8 for the mixed cocycle of Definition 27.8.4.1 by an on-site dressing. Identify the two-qubit label $2i_1 + i_2$ with the group element (i_1, i_2) , so that each site carries $\mathbb{C}[\mathbb{Z}_2 \times \mathbb{Z}_2]$, and write $U_{g,L}^\omega$ for the periodic operator of that construction on a ring of L sites.

Theorem 27.9.1.3 (Comparison with the cocycle construction). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Define the diagonal operator $Z_{2,L}$ by $Z_{2,L}|t\rangle = (-1)^{\sum_n t_{n,2}}|t\rangle$, the product of the Pauli Z on the second qubit of every site. For every $g \in \mathbb{Z}_2 \times \mathbb{Z}_2$ and every $L > 0$,

$$U_{g,L} = U_{g,L}^\omega Z_{2,L}^{g_1}. \quad (27.221)$$

The dressing is trivial for $g \in \{e, y\}$ and is a product of single-site unitaries. The cocycle operators form an exact representation, whereas the operators $U_{g,L}$ compose only up to the signs of Theorem 27.9.1.2.

Proof. [✓ Proof](#) Both sides vanish at (s, t) unless $s = g \oplus t$, since the table of the labels $2i_1 + i_2$ is the group law of $\mathbb{Z}_2 \times \mathbb{Z}_2$. On that support, Theorem 27.8.4.2 gives the sign of the right side as $(-1)^e$ with

$$e = g_1 \sum_n (t_{n+1,2}(t_{n,2} - t_{n+1,2}) + t_{n,2}).$$

Reindexing $\sum_n t_{n,2} = \sum_n t_{n+1,2}$ cyclically and using $t_{n+1,2}^2 = t_{n+1,2}$ gives $e \equiv g_1 \sum_n t_{n,2} t_{n+1,2} \pmod{2}$, which is the sign of (27.208). $\square$

Theorem 27.9.1.4 (An exact normal representation by dressed tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For $g \in \mathbb{Z}_2 \times \mathbb{Z}_2$ let M'_g be the tensor M_g of Definition 27.9.1 with the letter of input label i multiplied by $(-1)^{g_1 i_2}$. It has the bond dimension of M_g , its doubled-index tensor is normal, and its periodic operator on every ring of $L > 0$ sites is $U_{g,L}^\omega$. In particular the tensors M'_g form an exact representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ by normal tensors.

Proof. [✓ Proof](#) By Theorem 16.1.7 the periodic operator of M'_g is $U_{g,L} Z_{2,L}^{g_1}$, which is $U_{g,L}^\omega$ by (27.221) and $Z_{2,L}^2 = \mathbb{1}$. The operators $U_{g,L}^\omega$ multiply as the group by Theorem 27.8.4.2. The letters of M'_g are those of the normal tensor M_g of Theorem 27.9.3 multiplied by signs, so Theorem 16.1.7 gives normality. $\square$

Theorem 27.9.1.5 (The condensation defect squares to a length-dependent multiple of itself). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $L \geq 1$,

$$A_L^2 = (3 + (-1)^L)A_L. \quad (27.222)$$

Its all-zero diagonal entry is 1, so $A_L \neq 0$. Consequently $(A_L/4)^2 = A_L/4$ if and only if L is even. Thus the normalization by $\frac{1}{4}$ is not idempotent on odd rings.

Proof. [✓ Proof](#) Expand the square using Theorem 27.9.1.2. For each $k \in G$, the four pairs satisfying $gh = k$ contribute three weights $+1$ and one weight $(-1)^L$. Hence

$$A_L^2 = \sum_{k \in G} \left(\sum_{\substack{g, h \in G \\ gh=k}} \lambda(g, h)^L \right) U_{k,L} = (3 + (-1)^L)A_L.$$

At the all-zero physical-index pair, $M_e^{(0,0)} = 1$ and $M_x^{(0,0)} = M_y^{(0,0)} = M_{xy}^{(0,0)} = 0$. Since $L > 0$, only the identity tensor contributes to the all-zero diagonal entry, giving 1 and proving $A_L \neq 0$. Finally,

$$(A_L/4)^2 - A_L/4 = \frac{(-1)^L - 1}{16} A_L. \quad (27.223)$$

Nonvanishing of A_L reduces idempotence to $(-1)^L = 1$. $\square$

Lemma 27.9.1.6 (Orthogonal projection on even rings and idempotence on odd rings). [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *On every positive even ring, $A_L/4$ is an orthogonal projector. On every odd ring, $A_L/2$ is an idempotent that is not self-adjoint.*

Proof. ✓ Proof Equation (27.213) gives

$$A_L^\dagger = \mathbb{1} + U_{x,L} + U_{y,L} + (-1)^L U_{xy,L}. \quad (27.224)$$

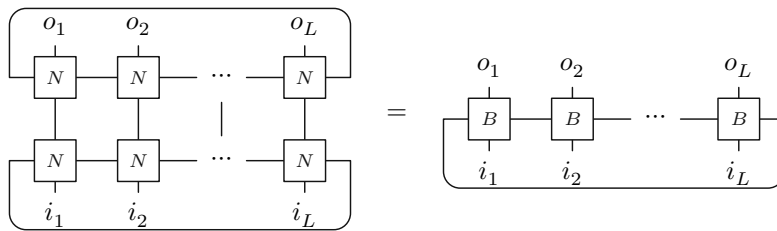
For even L , this equals A_L , and (27.222) gives $(A_L/4)^2 = A_L/4$. A self-adjoint idempotent is an orthogonal projector. For odd L , (27.222) instead gives $(A_L/2)^2 = A_L/2$, while $A_L^\dagger - A_L = -2U_{xy,L} \neq 0$ because $U_{xy,L}$ is unitary. $\square$

27.9.2 The sixteen pair blocks of the stacked defect

The operator identities of the fusion table have word-trace counterparts, to which the multi-block compression theorem applies. The stacked product $B = N \cdot N$ has local matrices

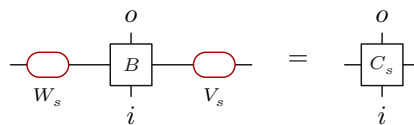
$$B^{(o,i)} = \sum_{j=0}^3 N^{(o,j)} \otimes N^{(j,i)}. \quad (27.225)$$

Its bond dimension is thirty-six, and its periodic operator is A_L^2 for $L > 0$. After grouping bond indices by $(g, h) \in G^2$, it is block diagonal with sixteen blocks $B_{g,h} = M_g \cdot M_h$, each generating $U_{g,L} U_{h,L}$.



Definition 27.9.2.1 (The stacked defect tensor and its sixteen target tensors). [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) Pairing the physical indices of $N \cdot N$ gives an MPS tensor on 16 letters. Its sixteen target tensors are $C_{(g,h)} = \lambda(g,h)M_{gh}$, indexed by the pairs (g,h) . Their bond dimensions are those of M_{gh} .

For twelve of the sixteen pairs, the stacked tensor is conjugate to its target by an invertible integer matrix, with no additional zero diagonal blocks. These are the split pairs. Ten of them, with at most one factor from $\{x, xy\}$ and with $\lambda(g, h) = +1$, are equal to their targets. The two pairs (x, y) and (xy, y) have weight -1 and are conjugate to $-M_{xy}$ and $-M_x$, respectively, by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. In the following diagram, B denotes one such pair block and $s = (g, h)$.



[illegible]

Proof. ✓ Proof Substitution of (27.210) in the stacked product shows that the ten positive-weight blocks equal their targets at every physical-index pair. For the other two, set $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Direct multiplication gives $S^{-1}T_{1-b}S = -T_b$ for $b \in \{0, 1\}$. This proves the conjugation identities for (x, y) and (xy, y) . Theorem I.1.2 gives the compression maps $W = S^{-1}$ and $V = S$ in these two cases, and identity maps in the other ten. In every case $B^{(\circ, i)} = VC_s^{(\circ, i)}W$ on the entire bond space. Thus there are no zero diagonal blocks and the remainder vanishes identically. $\square$

The remaining four pairs are the ones with both factors of bond dimension two. Their stacked blocks have bond dimension four but targets of bond dimension one, so the compression must discard three bond directions.

Theorem 27.9.2.3 (The four non-split pair blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) The four pairs (g, h) with both $g, h \in \{x, xy\}$ do not split. Their targets, in the order $(x, x), (x, xy), (xy, x), (xy, xy)$, are $M_e, -M_y, M_y, -M_e$. Each block has bond dimension four and admits a compression onto its one-dimensional target $\lambda(g, h)M_{gh}$ with $z = 3$ zero diagonal entries, an explicit unimodular integer gauge, and a remainder that vanishes on every word of length at least four. This is the bound $1 + z = 4$ of Theorem 25.5.1. For the block (x, x) , write $\delta^{ij} = \delta_{i,j}$ for the scalar entries of M_e . Its compression pair is

$$\begin{aligned} W &= (0, 0, 0, 1), \\ V &= (1, -1, -1, 1)^T, \end{aligned}$$

with $WV = 1$ and $WB^{ij}V = \delta^{ij}$ for every physical-index pair. The remainder $R^{ij} = B^{ij} - V\delta^{ij}W$ satisfies $R^{i_1j_1}R^{i_2j_2}R^{i_3j_3}R^{i_4j_4} = 0$. The other three blocks admit the same construction, each with its own gauge.

Proof. ✓ Proof For each of the four pairs, a finite calculation verifies an integer gauge and its inverse. This gauge puts every matrix of the stacked tensor in upper triangular form, with the target in the third diagonal position and zero in the other diagonal positions. Theorem [I.1.2](#) gives the compression pair from this gauge. Subtracting $VC^{ij}W$ removes the only potentially nonzero diagonal entry. The remaining matrices are strictly upper triangular in a common four-dimensional basis, so every product of four remainders is zero. Equivalently, this is the nilpotency clause of Theorem [25.5.1](#), with $|S| + z = 4$. The absence of a splitting follows from the intertwiner calculation in Theorem [27.9.2.4](#). □

Theorem 27.9.2.4 (Both sitewise intertwiner spaces vanish for the four non-split blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For each of the four non-split pairs (g, h) , let B denote its stacked block and set $C^{oi} = \lambda(g, h)(M_{gh}^{oi})_{00}$. If a column vector $v \in \mathbb{C}^4$ satisfies $B^{oi}v = C^{oi}v$ for every physical-index pair, then $v = 0$. Likewise, if a row vector u satisfies $uB^{oi} = C^{oi}u$ for every physical-index pair, then $u = 0$.

of the generator tensor. The trees $X = V \otimes \mathbb{1}$ and $X = \mathbb{1} \otimes V$ both intertwine two letters with the generator tensor, $XR^aR^b = M'^aXR^b$, and on single letters $(V \otimes \mathbb{1})R^a = \pm(\mathbb{1} \otimes V)R^a$, with the sign $+$ for x and $-$ for xy . These are integer identities over the few distinct letters, and induction on the word gives $\omega(g, g, g) = +1$ for x and -1 for xy . The value -1 at xy shows that the class is nontrivial, by Theorem 19.5.10.43. Theorem 27.9.4.16 identifies its class from these three products. $\square$

27.9.4 The eight scalar representatives

Definition 27.9.4.1 (Klein-four cocycle representatives). [Lean](#) [✓ Stmt](#) For $p, q, r \in \mathbb{Z}_2$ and $a, b, c \in \mathbb{Z}_2^2$, set

$$\omega_{pqr}(a, b, c) = (-1)^{pa_1b_1c_1 + qa_2b_2c_2 + ra_1b_2c_2}. \quad (27.227)$$

These are the representatives in the $\mathbb{Z}_2 \times \mathbb{Z}_2$ table of [164, Section 6]. The exponent is evaluated in $\mathbb{Z}_2$.

Theorem 27.9.4.2 (Cocycle equation and restriction signs). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Each ω_{pqr} is a normalized three-cocycle. At $a = (1, 0)$, $b = (0, 1)$, and $a + b = (1, 1)$ its diagonal values are

$$(\omega_{pqr}(a, a, a), \omega_{pqr}(b, b, b), \omega_{pqr}(a + b, a + b, a + b)) = ((-1)^p, (-1)^q, (-1)^{p+q+r}). \quad (27.228)$$

Its restriction to the subgroup generated by (x, y) is identically one if and only if $px + qy + rxy = 0$ in $\mathbb{Z}_2$. In particular, for every subgroup $H \leq \mathbb{Z}_2 \times \mathbb{Z}_2$, $\omega_{111}|_H = 1$ if and only if $H = \{0\}$. The representative ω_{001} is the cocycle $(-1)^{a_1b_2c_2}$ of Definition 27.8.4.1.

Proof. [✓ Proof](#) The exponent is additive in each argument, which gives the cocycle equation, and vanishes if any argument is zero. Substitution gives the three signs. On the subgroup generated by $g = (x, y)$, normalization leaves only the value at (g, g, g) to check. Since $x^2 = x$ and $y^2 = y$, this value is $(-1)^{px + qy + rxy}$. For $p = q = r = 1$, the exponent is nonzero at every nonzero g . Hence any subgroup on which the restriction is one contains only zero. $\square$

Theorem 27.9.4.3 (Distinct Klein-four classes and the last-row restriction). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The representatives ω_{pqr} and $\omega_{p'q'r'}$ are cohomologous if and only if $(p, q, r) = (p', q', r')$. For every subgroup $H \leq \mathbb{Z}_2^2$, the class of $\omega_{111}|_H$ is trivial if and only if $H = \{0\}$. These are the distinction of the eight representatives and the last-row subgroup assertion in [164, Section 6].

Proof. [✓ Proof](#) For an element g of order dividing two, the cyclic product $\omega(g, 0, g)\omega(g, g, g)$ is invariant under multiplication by a three-coboundary. Normalization reduces it to $\omega(g, g, g)$. The three signs therefore agree for cohomologous representatives; they recover p, q , and then r . Conversely, equal parameters give the same cocycle. If the restriction of ω_{111} to H has trivial class, Theorem 27.9.4.9 gives diagonal value 1 at every $g \in H$. The last row has diagonal value -1 at every nonzero element, so $H = \{0\}$. On the trivial subgroup the cocycle is identically one. $\square$

Theorem 27.9.4.4 (Four-block orbits for the last Klein-four class). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $G = \mathbb{Z}_2^2$ act on a set X , and let L be compatible with ω_{111} . For each $x \in X$, the stabilizer G_x is trivial and $|Gx| = 4$. In particular, if X is nonempty and the action is transitive, then $|X| = 4$. This gives the block count for the last class in [164, Section 6].

Proof. ✓ Proof Compatibility implies $\omega_{111}|_{G_x} = d(L^x|_{G_x})$. Thus the restricted class is trivial and $G_x = \{0\}$. The orbit-stabilizer identity gives $|Gx||G_x| = |G| = 4$. For a transitive action, $Gx = X$. □

Theorem 27.9.4.5 (L-symbols on a free transitive action). Lean Lean ✓ Stmt *Let G act freely and transitively on a nonempty set X . Any two L -symbols L_1, L_2 compatible with the same three-cochain ω are related by an action-tensor gauge, with the fusion gauge fixed to one. If ω is a three-cocycle, the regular action of G on itself admits the compatible solution $L_{g,h}^x = \omega(g, h, x)^{-1}$. This is the trivial-stabilizer case of [164, Section 4.1], in the compatibility convention of Definition 19.5.10.49.*

Proof. ✓ Proof Fix $x_0 \in X$, and write each $x \in X$ uniquely as $k \cdot x_0$. Set $\gamma_{g,k \cdot x_0} = L_2^{x_0}(g, k)/L_1^{x_0}(g, k)$. Compatibility gives, for $i = 1, 2$,

$$L_i^{k \cdot x_0}(g, h) = \frac{L_i^{x_0}(g, hk)L_i^{x_0}(h, k)}{\omega(g, h, k)L_i^{x_0}(gh, k)}. \quad (27.229)$$

Taking the ratio in (27.229) yields $L_1^x(g, h) = \gamma_{gh,x}L_2^x(g, h)/(\gamma_{g,h,x}\gamma_{h,x})$. For the regular action, the cocycle equation directly verifies compatibility of $\omega(g, h, x)^{-1}$. □

Theorem 27.9.4.6 (Unique four-block L-symbol class). Lean ✓ Stmt *Let $G = \mathbb{Z}_2^2$ act transitively on a nonempty set X , and let L be compatible with ω_{111} . There is an equivariant bijection $e : G \rightarrow X$ under which L is action-gauge equivalent to $L_0^k(g, h) = \omega_{111}(g, h, k)$. Thus there is exactly one compatible L -symbol class, up to equivariant relabeling and action-tensor gauge, and it has four blocks. This is the L -symbol uniqueness assertion of [164, Section 6] for the $(1, 1, 1)$ class.*

Proof. ✓ Proof All stabilizers are trivial, so $e(k) = k \cdot x_0$ is an equivariant bijection. Pull back L along e and apply Theorem 27.9.4.5. Every value of ω_{111} is ± 1 , so its inverse equals itself. The regular solution also proves existence. □

Theorem 27.9.4.7 (Normalization and involution signs). Lean Lean ✓ Stmt *Let L be compatible with a normalized three-cochain ω for an action of G on X . The action-tensor gauge $\gamma_{g,x} = L_{e,e}^x$, with fusion gauge equal to one, makes L normalized. For every three-cochain ω , every normalized L compatible with ω , every $x \in X$, and every $g \in G$ with $g^2 = e$, one has $L_{g,g}^x = \omega(g, g, g)L_{g,g}^{g \cdot x}$. These are the normalization and sign relations used in [164, Section 6].*

Proof. ✓ Proof Compatibility at (g, e, e) and (e, e, g) gives $L_{g,e}^x = L_{e,e}^x/\omega(g, e, e)$ and $L_{e,g}^x = \omega(e, e, g)L_{e,e}^{g \cdot x}$. Since ω is normalized, these are $L_{g,e}^x = L_{e,e}^x$ and $L_{e,g}^x = L_{e,e}^{g \cdot x}$. The gauge formula then makes both identity-axis values equal to one. Compatibility at (g, g, g) reduces to the stated sign relation. □

Theorem 27.9.4.8 (Diagonal L-symbols for the Klein-four representatives). Lean ✓ Stmt *Let L be normalized and compatible with ω_{pqr} , with $p, q, r \in \mathbb{Z}_2$. For $g = (u, v) \in \mathbb{Z}_2^2$ and every block x ,*

$$L_{g,g}^x = (-1)^{pu+qv+ruv}L_{g,g}^{g \cdot x}. \quad (27.230)$$

In particular, for the type-II representative ω_{001} and $c = (1, 1)$, one has $L_{c,c}^x = -L_{c,c}^{c \cdot x}$. If c exchanges two blocks x, y , this is $L_{c,c}^x = -L_{c,c}^y$, the relation for the element ab in [164, Section 6].

Proof. ✓ Proof Every element of $\mathbb{Z}_2^2$ squares to the identity. Substitute $\omega_{pqr}(g, g, g) = (-1)^{pu+qv+ruv}$ into Theorem 27.9.4.7, using $u^2 = u$ and $v^2 = v$ in $\mathbb{Z}_2$. □

Theorem 27.9.4.9 (Trivial restrictions of all Klein-four representatives). [Lean](#) [Lean](#) [✓ Stmt](#)
 Let $H \leq \mathbb{Z}_2^2$, with $a = (1, 0)$ and $b = (0, 1)$. The restriction of ω_{pqr} to H represents the trivial cohomology class if and only if $pu + qv + ruv = 0$ for every $(u, v) \in H$. Equivalently, H must avoid precisely the elements in the right column:

(p, q, r)	forbidden elements of H
$(0, 0, 0)$	none
$(1, 0, 0)$	a, ab
$(0, 1, 0)$	b, ab
$(1, 1, 0)$	a, b
$(0, 0, 1)$	ab
$(0, 1, 1)$	b
$(1, 0, 1)$	a
$(1, 1, 1)$	a, b, ab

This gives the subgroup restrictions in [164, Section 6], including arbitrary changes of fusion gauge.

Proof. [✓ Proof](#) If the restricted class is trivial, its cyclic invariant at every element of H is one. Since each element has square equal to the identity and the representative is normalized, this invariant is $\omega_{pqr}(g, g, g)$; hence $pu + qv + ruv = 0$. Conversely, write $Q(x) = px_1 + qx_2 + rx_1x_2$. A check over the finitely many $p, q, r \in \mathbb{Z}_2$ and $g, h, k \in \mathbb{Z}_2^2$ gives

$$Q(g) = Q(h) = Q(k) = Q(g+h) = Q(g+k) = Q(h+k) = 0 \implies pg_1h_1k_1 + qg_2h_2k_2 + rg_1h_2k_2 = 0.$$

Subgroup closure therefore makes the entire restriction identically one. Testing the identity and the three nonidentity elements gives the table. $\square$

Definition 27.9.4.10 (The one-qubit Klein-four symmetry). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
 On a periodic chain of $N > 0$ qubits, write $\bar{t}_i = 1 - t_i$ and set

$$\begin{aligned} U_a|t\rangle &= (-1)^{\sum_i (t_i t_{i+1} + t_i)} |t\rangle, \\ U_b|t\rangle &= |\bar{t}\rangle. \end{aligned} \tag{27.231}$$

Thus $U_a = \prod_i CZ_{i,i+1}Z_i$ and $U_b = \prod_i X_i$, as in [164, Section 6]. Define $U_{a^p b^q} = U_a^p U_b^q$ for $p, q \in \{0, 1\}$. If $\widehat{T}^{ij}$ is the decorated CZX tensor, the tensors $A^{ij} = \widehat{T}^{i,1-j}$ and $B^{ij} = \delta_{i,1-j}$ have bond dimensions two and one, respectively.

Theorem 27.9.4.11 (The printed Klein-four representation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) [Lean](#)
 At every positive length, $U_a^2 = U_b^2 = I$ and $U_a U_b = U_b U_a$. Every U_g is unitary, $U_e = I$, and $U_g U_h = U_{gh}$, so $g \mapsto U_g$ is a representation of $\mathbb{Z}_2 \times \mathbb{Z}_2$ on the state space. The periodic operators of A and B are U_a and U_b . Their product is the decorated CZX operator.

Proof. [✓ Proof](#) The diagonal entries of U_a are signs, and the spin flip is an involution, so both generators square to the identity and are unitary. By Theorem 27.3.3.2, $O_N(A)(s, t) = O_N(\widehat{T})(s, \bar{t}) = \delta_{s, \bar{t}} (-1)^{\sum_i (s_i s_{i+1} + s_i)} = U_a(s, t)$, and $O_N(B)(s, t) = \mathbb{1}(s, \bar{t}) = \delta_{s, \bar{t}} = U_b(s, t)$. By (27.231), $U_a U_b$ is the decorated CZX operator, whose square is I . Multiplying $(U_a U_b)^2 = I$ on the left by U_a and on the right by U_b proves commutation. The multiplication law follows by adding exponents modulo two. $\square$

Definition 27.9.4.12 (The two invariant state spaces). [Lean](#) [Lean](#) [✓ Stmt](#) Put $z_0 = |0\rangle^{\otimes N}$, $z_1 = |1\rangle^{\otimes N}$, $u = \sum_t |t\rangle$, and $v = U_a u$. The two spaces of [164, Section 6] are $S_z = \text{span}\{z_0, z_1\}$ and $S_c = \text{span}\{u, v\}$. Multiplying u and v by $2^{-N/2}$ gives the normalized uniform and cluster-like vectors used there, without changing S_c .

Theorem 27.9.4.13 (Action on the two state pairs). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Both S_z and S_c are invariant under U_a and U_b . More precisely, for $j \in \{0, 1\}$,

$$\begin{aligned} U_a z_j &= z_j, \\ U_b z_j &= z_{1-j}, \\ U_a u &= u, \\ U_b u &= u, \\ U_b v &= v. \end{aligned} \tag{27.232}$$

Proof. [✓ Proof](#) The exponent in (27.231) is zero on the all-zero configuration and $2N$ on the all-one configuration. Since U_b permutes configurations by the global flip with no sign, it fixes the uniform vector directly: $U_b u = u$. The same flip exchanges the all-zero and all-one configurations, giving $U_b z_j = z_{1-j}$. Since $v = U_a u$, the identities $U_a^2 = I$ and $U_b U_a = U_a U_b$ then give $U_a v = u$ and $U_b v = v$. Linearity proves invariance of both spans. $\square$

Theorem 27.9.4.14 (Normalization of scalar three-cocycles). [Lean](#) [✓ Stmt](#) Let G be a small group, whose underlying set lies in a fixed set-theoretic universe. Every three-cocycle $\omega : G^3 \rightarrow \mathbb{C}^\times$ is cohomologous to a normalized one. One may use the two-cochain $\beta(g, h) = \omega(g, e, e)/\omega(e, e, h)$ and replace ω by $(d\beta)\omega$, where $d\beta(g, h, k) = \beta(g, hk)\beta(h, k)/(\beta(g, h)\beta(gh, k))$.

Proof. [✓ Proof](#) The cocycle equation gives $\omega(e, e, e) = 1$ and

$$\begin{aligned} \omega(g, h, e) &= \frac{\omega(gh, e, e)}{\omega(h, e, e)}, \\ \omega(e, g, h) &= \frac{\omega(e, e, gh)}{\omega(e, e, g)}, \\ \omega(g, e, h) &= \omega(g, e, e)\omega(e, e, h). \end{aligned} \tag{27.233}$$

Substitution of (27.233) into $d\beta$ makes each identity-axis value of $(d\beta)\omega$ equal to one. $\square$

Theorem 27.9.4.15 (Triviality from the Klein diagonal signs). [Lean](#) [✓ Stmt](#) A normalized three-cocycle on $G = \{e, a, b, c = ab\} \cong \mathbb{Z}_2^2$ whose diagonal values $\omega(g, g, g)$ are all one is a coboundary. This is the trivial-class case of the classification in [164, Section 6].

Proof. [✓ Proof](#) Set $(s_1, s_2, s_3, s_4, s_5, s_6) = (\omega(a, a, b), \omega(a, b, a), \omega(a, b, b), \omega(a, b, c), \omega(b, a, a), \omega(b, a, b))$. Choose $t \in \mathbb{C}^\times$ with $t^2 = s_1 s_5 / s_2$ and define a normalized two-cochain by its values on $\{a, b, c\}^2$:

$$(\beta(g, h))_{g, h \in \{a, b, c\}} = \begin{pmatrix} ts_4/(s_3 s_6) & 1/s_3 & ts_1 s_4/s_6 \\ 1/(ts_3) & 1 & s_6/t \\ s_1 s_4/(s_2 s_6) & 1 & 1 \end{pmatrix}. \tag{27.234}$$

The six corresponding values of $d\beta$ are $s_1, \dots, s_6$. Hence $\eta = \omega/(d\beta)$ has those six values equal to one. Its diagonal values remain one because $g^2 = e$ and β is normalized. Applying the cocycle equation successively to the remaining triples gives one for all eighteen other nonidentity entries. Thus $\eta = 1$ and $\omega = d\beta$. $\square$

27.10 The anomalous $\mathbb{Z}_3$ symmetry and its condensation defect

A finite group acting on a spin chain by matrix product operators may be anomalous (Remark 26.1.3): the obstruction to gauging it is a three-cocycle, and for a cyclic group Garre-Rubio and Schuch [166] give the cocycle in closed form together with the interchange statistics of the domain walls it governs. The phase-decorated shift defined below realizes their cocycle formula for $\mathbb{Z}_3$ by an explicit local tensor; its anomaly class is computed in Theorem 27.10.2.4. The general construction of a matrix product operator representation from a finite group is given in Garre-Rubio, Lootens and Molnár [164], and matrix product operator symmetries are reviewed in [16, Section III.A.6]. The sum $\mathcal{C}_N = \mathbb{1} + U_N + U_N^\dagger$ of the three group operators is a condensation defect in the terminology of Roumpedakis, Seifnashri and Shao [197].

Let $\omega = (-1 + i\sqrt{3})/2$ be the primitive cube root of unity. Thus $\omega^2 + \omega + 1 = 0$, $\omega^3 = 1$, and $\bar{\omega} = \omega^{-1} = \omega^2$. On a periodic chain of $N > 0$ qutrits, use basis vectors $|t\rangle$ indexed by $t = (t_0, \dots, t_{N-1}) \in \{0, 1, 2\}^N$. Set $\chi(a) = [a \neq 0]$ and

$$F(t) = \sum_{j=0}^{N-1} \chi(t_j) t_{j+1}, \quad (27.237)$$

where $t_N = t_0$ and the exponent uses the integer representatives 0, 1, 2. Write $X|a\rangle = |a+1\rangle$, with addition modulo three, and $(Xt)_j = t_j + 1$. The phase-decorated shift acts by

$$U_N|t\rangle = \omega^{F(t)}|Xt\rangle. \quad (27.238)$$

Thus $U_N = X^{\otimes N} \text{diag}(\omega^{F(t)})$. Every cyclic bond is included: at $N = 1$ there is a self-loop, and at $N = 2$ both oriented bonds contribute.

The group-cocycle context for anomalous symmetries is discussed in Garre-Rubio and Schuch [166]; see also Sun [208].

Definition 27.10.1 (The tensors of the representation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
The tensor M has bond dimension two, with

$$M^{1,0} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad M^{2,1} = \begin{pmatrix} 0 & 1 \\ 0 & \omega \end{pmatrix}, \quad M^{0,2} = \begin{pmatrix} 0 & 1 \\ 0 & \omega^2 \end{pmatrix},$$

and all other matrices zero. The independently specified tensor denoted by $M^\dagger$ has bond dimension two, with

$$(M^\dagger)^{0,1} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad (M^\dagger)^{1,2} = \begin{pmatrix} 0 & 1 \\ 0 & \omega^2 \end{pmatrix}, \quad (M^\dagger)^{2,0} = \begin{pmatrix} 0 & 1 \\ 0 & \omega \end{pmatrix},$$

and all other matrices zero. The first physical index is the output and the second is the input. The identity tensor δ has bond dimension one and scalar matrices $\delta^{o,i} = \delta_{o,i}$, where $\delta_{o,i}$ is the Kronecker delta. Read over the alphabet of index pairs, M , $M^\dagger$ and δ are matrix product state tensors of physical dimension nine, with the matrix at (o, i) assigned to the letter $3o + i$.

For any such tensor T , its periodic operator uses the ordinary, unnormalized trace:

$$O_N(T)(s, t) = \text{tr}(T^{s_0, t_0} \dots T^{s_{N-1}, t_{N-1}}). \quad (27.239)$$

Set $U_N = O_N(M)$ and $V_N = O_N(M^\dagger)$. Theorem 27.10.2 verifies that this definition of U_N gives (27.238). The contraction below represents U_L on a ring drawn with L sites and physical input labels $s_1, \dots, s_L$.

Proof. [✓ Proof](#) Proposition [25.5.3](#) applied to Theorem [27.10.1.1](#) gives the trace identities, and Theorem [25.6.2](#) gives the stated identities of periodic operators. $\square$

Theorem 27.10.1.3 (Compression of $M \cdot M^\dagger$ and $M^\dagger \cdot M$). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The stacked tensor $B = M \cdot M^\dagger$ has the block triangular form of Theorem [25.5.1](#) for the single target δ with $z = 3$ one-dimensional zero diagonal blocks, $4 = 1 + 3$. The compression pair satisfies $WV = \mathbb{1}$ and $WB^wV = \delta^w$ for every word, and the remainder is nilpotent of order exactly three, one short of the general bound $|S| + z = 4$. Here $|S| = 1$ is the number of target slots. The stacked tensor $M^\dagger \cdot M$ has the analogous block triangular form for the same target δ . Both products use the compression pair

$$V = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \quad W = \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}. \quad (27.250)$$

Proof. [✓ Proof](#) Use the conjugation argument of Theorem [27.10.1.1](#), over $\mathbb{Z}[\omega]$, with two one-dimensional zero diagonal blocks before the target and one after. The target rows and columns give $WV = \mathbb{1}$ and $WB^wV = \delta^w$. Direct multiplication gives zero for every triple product of remainder letters. The products R^0R^4 for $M \cdot M^\dagger$ and R^0R^0 for $M^\dagger \cdot M$ are nonzero, proving that the order is exactly three in both cases. $\square$

Theorem 27.10.1.4 (The exact $\mathbb{Z}_3$ representation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The periodic operator of the identity tensor is the identity map at every length, $O_L(\delta) = \mathbb{1}$. For every nonempty word w , both $\text{tr}((M \cdot M^\dagger)^w)$ and $\text{tr}((M^\dagger \cdot M)^w)$ equal $\text{tr}(\delta^w)$. Thus $O_L(M)O_L(M^\dagger) = O_L(M^\dagger)O_L(M) = \mathbb{1}$ at every positive length. Together with $O_L(M)O_L(M) = O_L(M^\dagger)$, this gives $O_L(M)^3 = \mathbb{1}$: the symmetry gives a representation of $\mathbb{Z}_3$ without any length-dependent phase.

Proof. [✓ Proof](#) The word evaluation of δ is the scalar 1 when the input and output configurations agree, and 0 otherwise. Hence $O_L(\delta) = \mathbb{1}$, including at length zero. Proposition [25.5.3](#) applied to Theorem [27.10.1.3](#) gives the nonempty-word trace identities. By Theorem [25.6.2](#), they imply $O_L(M)O_L(M^\dagger) = \mathbb{1}$ and $O_L(M^\dagger)O_L(M) = \mathbb{1}$. Finally, Theorem [27.10.1.2](#) gives $O_L(M)^3 = O_L(M)^2O_L(M) = O_L(M^\dagger)O_L(M) = \mathbb{1}$. $\square$

Theorem 27.10.1.5 (Both sitewise intertwiner spaces vanish on every nontrivial block). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For each of the four stacked products $M \cdot M$, $M \cdot M^\dagger$, $M^\dagger \cdot M$, $M^\dagger \cdot M^\dagger$, let B denote the stacked tensor and let C denote its respective target $M^\dagger$, δ , δ , M . There is no nonzero matrix X with $B^aX = XC^a$ for every letter a , and no nonzero matrix Y with $YB^a = C^aY$ for every letter a . In particular, the compression identities $WB^wV = C^w$ do not give sitewise intertwiners.

Proof. [✓ Proof](#) Select eight scalar equations for the equal-factor products and four for the inverse-factor products. Write the selected sitewise intertwining equations as $Kx = 0$, where x lists the entries of X or Y . The coefficient matrix K has entries in $\mathbb{Z}[\omega]$ and admits a matrix H with $HK = c\mathbb{1}$ for a nonzero scalar c . These identities are verified over $\mathbb{Z}[\omega]$. Consequently $cx = HKx = 0$, and all entries of the intertwiner vanish. $\square$

Remark 27.10.1.6 (Failure of closure under conjugate transposition). By Corollary [25.8.4](#), Theorem [27.10.1.5](#) shows that the algebra generated by each stacked tensor is not closed under conjugate transposition. This concerns the virtual matrices, rather than the physical operator or a 3-cocycle associator.

27.10.2 The group family

Garre-Rubio and Schuch [166, Section IV.A] define a matrix product unitary representation of a finite group by injective tensors whose periodic operators satisfy $U_g U_h = U_{gh}$ and $U_e = \mathbb{1}$. The three tensors above, indexed by $\mathbb{Z}_3$, satisfy these operator laws; injectivity holds after blocking two sites.

Theorem 27.10.2.1 (The tensors δ , M , $M^\dagger$ as a group family). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) Set $T_0 = \delta$, $T_1 = M$ and $T_2 = M^\dagger$, with indices in $\mathbb{Z}_3$. Every T_g is a matrix product unitary on nonempty chains, and for every $N > 0$ and all $g, h \in \mathbb{Z}_3$,

$$O_N(T_0) = 1, \quad O_N(T_a)O_N(T_b) = O_N(T_{a+b}), \quad O_N(T_a)^\dagger = O_N(T_{-a}).$$

The family is not a raw matrix product unitary representation in the sense of Definition 19.2.4.15, since M is not injective. After common blocking of two sites it is one.

Proof. ✓ Proof The unitarity and the adjoint law are Theorem 27.10.2, and the unit and product laws are Theorems 27.10.1.4 and 27.10.1.2, checked on the nine pairs (g, h) . Theorem 27.10.5 gives the failure of one-site injectivity for M and injectivity after blocking two sites for all three tensors; for δ , injectivity at one site persists at two. Blocking L sites replaces the operator at length N by the operator at length NL , which is positive, so the operator laws persist, and a tensor injective after blocking L sites has an injective blocked tensor. $\square$

The periodic construction of Section 27.8, applied to the cocycle ω_1 of $\mathbb{Z}_3$ from Definition 27.8.3.1, gives a second exact representation of $\mathbb{Z}_3$, of bond dimension three. The two families are conjugate by one diagonal unitary built from on-site and two-site gates (Theorem 27.10.2.3).

Theorem 27.10.2.2 (The cocycle construction for $\mathbb{Z}_3$). [Lean] [Lean] [Stmt] For $n = 3$ and $j = 1$, every tensor $\hat{T}_g$ of the construction is a matrix product unitary on nonempty chains, and $U_0 = \mathbb{1}$, $U_a U_h = U_{a+h}$ and $U_g^\dagger = U_{-g}$ for every $N > 0$.

Proof. **Proof** This is Theorem 27.8.3.2 with $n = 3$ and $j = 1$.

Theorem 27.10.2.3 (The two representations are conjugate). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Stmnt](#) For $N > 0$ define the diagonal unitary V_N by $V_N|t\rangle = \omega^{E(t)}|t\rangle$, where

$$E(t) = \sum_{k=0}^{N-1} ([t_k = 2] + [t_k = 1][t_{k+1} = 2]), \quad (27.251)$$

with $t_N = t_0$. Let $U_g^{\omega_1}$ be the operators of the cocycle construction of Theorem 27.10.2.2. Then for every $g \in \mathbb{Z}_3$ and every $N > 0$,

$$O_N(T_a) V_N = V_N U_g^{\omega_1}, \quad O_N(T_a) = V_N U_g^{\omega_1} V_N^\dagger.$$

Proof. ✓ Proof For $g = 1$ both sides vanish at (s, t) unless $s = Xt$. On that support the left side is $\omega^{F(t)+E(t)}$, by Theorem 27.10.2, and the right side is $\omega^{E(Xt)+c(t)}$, where $c(t)$ counts the bonds with $t_k < t_{k+1}$, by Theorem 27.8.3.2; for $n = 3$ the carry $\lfloor (t_{k+1} + [t_k - t_{k+1}])/3 \rfloor$ is $[t_k < t_{k+1}]$. Both exponents are sums over the bonds (t_k, t_{k+1}) , and the two bond terms agree modulo three for each of the nine pairs. For $g = 0$ both operators are the identity, and for $g = 2$ both operators are the squares of those at $g = 1$, by the product laws of Theorems 27.10.2.1 and 27.10.2.2. The matrix V_N is diagonal with entries of modulus one, hence unitary. $\square$

Reading off the two terms of (27.251), $V_N = D^{\otimes N} \prod_k C_{k,k+1}$, where $D = \text{diag}(1, 1, \omega)$ and the commuting two-site gate C multiplies $|1, 2\rangle$ by ω and fixes the other basis vectors, so V_N is a circuit of depth two. No purely on-site diagonal gauge relates the two families. A diagonal unitary $\prod_k \omega^{w(t_k, t_{k+1})}$ changes the exponent of the shift by $\sum_k (w(t_k + 1, t_{k+1} + 1) - w(t_k, t_{k+1}))$. For an on-site gate $w(a, b) = \alpha(a)$ this is a sum of single-site terms, whereas the difference $[a \neq 0]b - [a < b]$ of the two bond exponents is not, modulo three, the sum of a function of a and a function of b , which it would have to be for the exponents to agree on every chain. What relates the two families is the identity $O_N(T_g) = V_N U_g^{\omega_1} V_N^\dagger$ of Theorem 27.10.2.3, with $V_N = D^{\otimes N} \prod_k C_{k,k+1}$ a product of commuting diagonal gates. The class of Theorem 27.10.2.4 is computed from fusion tensors of the family $\{T_g\}$. The construction for ω_1 has fusion tensors of its own, and they give the same cyclic product (Theorem 27.10.2.6), as invariance of the class under the conjugation by V_N would predict; that invariance is not proved here.

Theorem 27.10.2.4 (The anomaly class of the family). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [✔ Stmt](#) *The tensors $T_0 = \delta$, $T_1 = M$, $T_2 = M^\dagger$ are normal and represent $\mathbb{Z}_3$ exactly on nonempty chains. Choose the fusion tensors to be the identity whenever a factor is T_0 , and the compressions of Theorems 27.10.1.1 and 27.10.1.3 otherwise. Write $\mathbb{Z}_3 = \{e, g, g^2\}$ multiplicatively, with $g^3 = e$, so that M represents g . Then the two fusion trees of (g, g, g) differ by the factor ω^2 and those of (g, g^2, g) and (g, e, g) agree, against every nonempty word, so that*

$$\omega_{\text{an}}(q, e, q) \omega_{\text{an}}(q, g, q) \omega_{\text{an}}(q, g^2, q) = \omega^2,$$

where ω_{an} is the anomaly three-cochain. The same product equals ω^2 for every choice of fusion tensors, and the anomaly three-cocycle is not cohomologous to the constant cochain one: the anomaly class in $H^3(\mathbb{Z}_3, \mathbb{C}_{\text{triv}}^\times)$ is nonzero.

Proof. ✓ **Proof** Each of the two fusion trees X of (g, g, g) and of (g, g^2, g) satisfies $XR^aR^b = T^aXR^b$ for all letters a, b , with R the triple product and T the target tensor, and on single letters the trees differ by ω^2 , respectively agree. These are identities over $\mathbb{Z}[\omega]$ checked letter by letter, and induction on the word gives the dressed identities. For (g, e, g) both trees are the fusion tensor of $M \otimes M$. The product is invariant under the fusion-tensor gauge action because $g^3 = e$, and $\omega^2 \neq 1$. Since the anomaly three-cocycle is not cohomologous to the constant cochain one, its class is nonzero by Corollary 19.5.1.9. $\square$

Theorem 27.10.2.5 (Orientation of the class). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let g be the generator of $\mathbb{Z}_3$. The cocycle ω_1 of [166] has $\omega_1(g, e, g)\omega_1(g, g, g)\omega_1(g, g^2, g) = \omega$, while the anomaly three-cochain of the family has the inverse product ω^2 for every choice of fusion tensors.

Proof. ✓ Proof The cube root $\omega = (-1 + i\sqrt{3})/2$ is $e^{2\pi i/3}$, so $\omega_1(a, b, c) = \omega^{a \lfloor (b+c)/3 \rfloor}$ and only the factor $\omega_1(g, g^2, g) = \omega$ is nontrivial. The product for the family is ω^2 by Theorem 27.10.2.4, and $\omega^2 \cdot \omega = 1$. □

Both [166] and [164] define the anomaly by the same move between the two fusion trees of (g, h, k) as the three-cochain used here. The construction of [164] from a cocycle ω carries ω^{-1} , as that source states for its fusion tensors (Theorem 27.8.1.3). Built from ω_1 it therefore realizes $\omega_2 = \omega_1^{-1}$, the class of the family, and not ω_1 : the orientation is a property of the construction, not a discrepancy between the two definitions of the anomaly.

Theorem 27.10.2.6 (The construction for ω_1 carries ω_2). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
✔ Stmt *With the fusion tensors of Definition 27.8.1.1, the anomaly three-cochain of the construction for ω_1 is ω_1^{-1} , which equals ω_2 at every triple. Its cyclic product at the generator is*

ω^2 , the product of the family for every choice of fusion tensors, and its domain-wall phase at the generator is $e^{2\pi i/3}$, the entry $j = 2$, $a = 1$ of the table of [166]. The construction is not a representation with normal tensors.

Proof. ✓ Proof The first claim is Theorem 27.8.1.3, and $\omega_j^{-1} = \omega_{(n-1)j}$ because ω_j is a power of the primitive root $e^{2\pi i/n}$. The cyclic product of ω_1^{-1} is the inverse of that of ω_1 , which is ω^2 by Theorem 27.10.2.5. The domain-wall phase is the table entry for ω_2 . The last claim is Theorem 27.8.1.4. □

Corollary 27.10.2.7 (The domain-wall phase of the family). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For every choice of fusion tensors, the anomaly three-cochain ω_{an} of the family has, at the generator g , the same cyclic invariant as the cocycle ω_2 of Definition 27.8.3.1 for $n = 3$, and it is not cohomologous to ω_j for $j \not\equiv 2 \pmod{3}$. Its domain-wall phase is $P_g(\omega_{\text{an}}) = e^{2\pi i/3}$, the entry $j = 2$, $a = 1$ of the table of Theorem 27.8.3.5. The three-cochain ω_{an} is cohomologous to ω_2 , so the anomaly class of the family in $H^3(\mathbb{Z}_3, \mathbb{C}_{\text{triv}}^\times)$ is $[\omega_2]$.

Proof. ✓ Proof The primitive cube root ω equals $e^{2\pi i/3} = \zeta$, so the invariant ω^2 of Theorem 27.10.2.4 is ζ^2 , the cyclic invariant of ω_2 at g . The classes ω_j are separated by this invariant, and $P_g = I_{g,3}^{-1} = \zeta^{-2} = \zeta$ because g has order three. The cyclic invariant at the generator determines the class by Theorem 27.8.3.6, so ω_{an} is cohomologous to ω_2 . □

27.10.3 The condensation defect

The condensation defect of the anomalous symmetry is the sum of its three group operators, $\mathbb{1} + U_N + U_N^\dagger$, in the terminology of [197]. It is realized by the direct sum of the three tensors. On the physical side the defect squares to three times itself; on the virtual side the stacked square splits into nine pair blocks, each compressing onto one group tensor.

Definition 27.10.3.1 (The condensation defect). Lean Lean Lean Lean ✓ Stmt The condensation defect is the block-diagonal tensor $A = \delta \oplus M \oplus M^\dagger$ of bond dimension five. Its three diagonal blocks have bond dimensions one, two and two and represent $\mathbb{1}$, U_L and $U_L^\dagger$, respectively, at positive ring length L .

Theorem 27.10.3.2 (The defect is the sum of the three group elements). Lean Lean ✓ Stmt For every word w , $\text{tr}(A^w) = \text{tr}(\delta^w) + \text{tr}(M^w) + \text{tr}((M^\dagger)^w)$; that is, $O_L(A) = O_L(\delta) + O_L(M) + O_L(M^\dagger)$ at every length L .

Proof. ✓ Proof Products of block-diagonal matrices remain block diagonal, so $A^w = \delta^w \oplus M^w \oplus (M^\dagger)^w$. Taking the trace gives the word identity and hence the periodic operator identity. □

Lemma 27.10.3.3 (The normalized defect projects onto invariant states). Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For $L > 0$, the operator $P_L = O_L(A)/3 = (\mathbb{1} + U_L + U_L^\dagger)/3$ is the orthogonal projector onto $\{\psi : U_L \psi = \psi\}$. Its basis action is

$$P_L |t\rangle = \frac{1}{3} \left(|t\rangle + \omega^{F(t)} |Xt\rangle + \omega^{-F(X^{-1}t)} |X^{-1}t\rangle \right). \quad (27.252)$$

In particular, $O_L(A)^2 = 3O_L(A)$ and $\text{rank } P_L = 3^{L-1}$.

Proof. ✓ Proof Theorem 27.10.3.2 and Lemma 27.10.3 give $P_L = (\mathbb{1} + U_L + U_L^\dagger)/3$, and Theorem 27.10.2 gives its basis action. By the operator laws of Theorem 27.10.2.1, the operators $\mathbb{1}, U_L, U_L^\dagger$ form a unitary representation of $\mathbb{Z}_3$ (Definition 19.2.5), and P_L is its normalized group average. By Theorem 19.1.2, P_L is an orthogonal projector. Since $U_L^3 = \mathbb{1}$, $U_L P_L = P_L$, and

The physical identity has a corresponding virtual decomposition. The stacked square has nine pair blocks, each reducing to one of the three group tensors. The asymmetric compression records these nine targets separately.

[illegible]

Proof. ✓ Proof Each letter of A preserves its three summands. The tensor product of two such letters therefore preserves each of the nine summand pairs. In the pair coordinates, its restriction is $M_g^{o,k} \otimes M_h^{k,i}$. Summing over k gives $(M_g \cdot M_h)^{o,i}$, as in (27.248). The specified reindexing and the resulting entry identities are verified over $\mathbb{Z}[\omega]$. $\square$

Proof. ✓ **Proof** By Theorem 27.10.3.4, the trace of $(A \cdot A)^w$ is the sum of the nine pair-block traces. Theorems 27.10.1.2 and 27.10.1.4 give $\text{tr}((M_g \cdot M_h)^w) = \text{tr}(M_{g+h}^w)$ on the four nontrivial pairs. For $g = 0$ or $h = 0$, the pair block equals the target under the natural bond identification. Each group element occurs as $g + h$ exactly three times, so

Theorem 25.6.2 converts this trace identity into the periodic operator identity. $\square$

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the remainder is nilpotent of order exactly three. Explicitly, writing $s = (g, h)$ and $C_s = M_{g+h}$, the pairs satisfy $W_s(A \cdot A)^w V_s = C_s^w$ for every word and $W_s V_t = 0$ for $s \neq t$. The remainder is

$$R^a = (A \cdot A)^a - \sum_s V_s C_s^a W_s, \quad (27.255)$$

and $R^w = 0$ whenever $|w| \geq 3$, while $R^1 R^5 \neq 0$.

Proof. ✓ Proof On the five pairs with $g = 0$ or $h = 0$, the pair block equals its target under the natural bond identification and requires no zero diagonal block. On the other four pairs, use the gauges of Theorem 27.10.1.1 or Theorem 27.10.1.3. Their direct sum, transported along the bond reindexing of Theorem 27.10.3.4, gives the required triangular form (Theorem 1.1.7). Distinct pair blocks have disjoint coordinates, which gives $W_s V_t = 0$ for $s \neq t$.

The original block dimensions sum to $1 + 2 + 2 + 2 + 4 + 4 + 2 + 4 + 4 = 25$. The target dimensions sum to $1 + 2 + 2 + 2 + 2 + 1 + 2 + 1 + 2 = 15$, and the zero diagonal blocks number $0 + 0 + 0 + 0 + 2 + 3 + 0 + 3 + 2 = 10$. Clause (vi) of Theorem 25.5.1 bounds the nilpotency order only by $|S| + z = 9 + 10 = 19$, where S is the set of nine target slots. By Theorem 1.1.7, the remainder is the direct sum of the nine pair remainders. These vanish on the five pairs with $g = 0$ or $h = 0$, and on the four other pairs every word of length three vanishes while $R^1 R^5 \neq 0$ on the pair $(1, 1)$, by Theorems 27.10.1.1 and 27.10.1.3. Hence the order is exactly three. $\square$

27.11 The $\mathbb{Z}_3$ clock symmetry

The anomalous symmetry of Section 27.10 is an exact representation of $\mathbb{Z}_3$, and its condensation defect satisfies $A^2 = 3A$ at every length (Theorem 27.10.3.5). This section treats a second family of $\mathbb{Z}_3$ symmetry operators whose representatives multiply only up to phases depending on the length, and shows that the structure constants of its condensation defect then genuinely depend on the length. The length dependence comes from the per-site phases of the chosen representatives. The expected triviality of the Else–Nayak class remains to be established for these dressed operators; the trivial cocycle in Theorem 27.11.2 alone does not identify that class.

As in Section 27.10, let $\omega = \exp(2\pi i/3)$, let $X|s\rangle = |s+1\rangle$ on a qutrit, and let $CZ|a, b\rangle = \omega^{ab}|a, b\rangle$ and $Z|a\rangle = \omega^a|a\rangle$. On a periodic chain of $N > 0$ qutrits the clock operators are

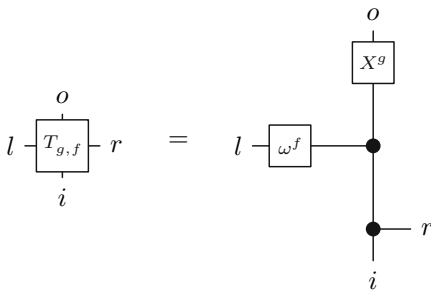
$$U_N = X^{\otimes N} \prod_k CZ_{k, k+1}, \quad V_N = (X^2)^{\otimes N} \prod_k CZ_{k, k+1}^2 Z_k^2. \quad (27.256)$$

Definition 27.11.1 (The clock tensors). Lean Lean Lean Lean ✓ Stmt For $g \in \mathbb{Z}_3$ and a weight $f: \mathbb{Z}_3 \times \mathbb{Z}_3 \rightarrow \mathbb{Z}_3$, the clock tensor of bond dimension three has the entries

$$(T_{g, f}^{o, i})_{l, r} = \begin{cases} \omega^{f(l, i)} & \text{if } o = i + g \text{ and } r = i, \\ 0 & \text{otherwise,} \end{cases} \quad (27.257)$$

with output label o , input label i and bond labels l, r . The tensor U is T_{1, f_U} with $f_U(l, i) = li$, and the tensor V is T_{2, f_V} with $f_V(l, i) = 2li + 2i$.

The clock tensor has the shape of the group tensor of Definition 27.8.1: it copies the input i to the outgoing bond, weights the pair formed by the incoming bond and the input by $\omega^{f(l, i)}$, and applies X^g to the input to give the output.



Both tensors use the ordinary trace on rings of $N > 0$ sites. Theorem 27.11.2 identifies their kernels and proves unitarity. The multiplication table retains its length-dependent phases (Theorem 27.11.3); the resulting square of the condensation defect and the obstruction to length-independent fusion constants are given in Theorems 27.11.6 and 27.11.7. The anomaly-class question remains separate from these operator identities.

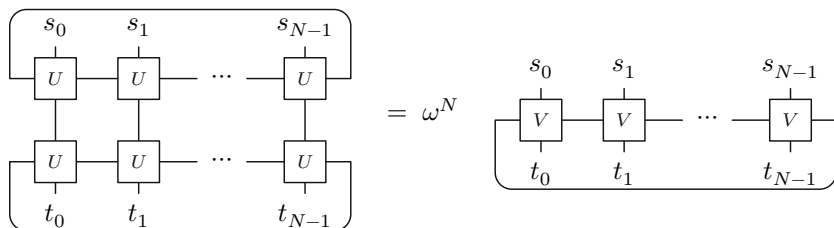
Theorem 27.11.2 (Kernels and the trivial cocycle). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [✓ Stmt](#) On a periodic chain of $N > 0$ qutrits, the periodic operator of $T_{\alpha,f}$ is monomial:

$$\langle s | O_N(T_{g,f}) | t \rangle = \prod_k \delta_{s_k, t_k + g} \omega^{f(t_k, t_{k+1})}. \quad (27.258)$$

In particular the periodic operators of U and V are U_N and V_N . The tensor $T_{g,f}$ is, entrywise, the group tensor of Definition 27.8.1 for the trivial cocycle $\omega_0 = 1$ of $\mathbb{Z}_3$ times the bond weight $\omega^{f(l,i)}$, so that $O_N(T_{g,f})$ is the diagonal circuit with phase $\omega^{\sum_k f(t_k, t_{k+1})}$ on the input configuration t , followed by the plain shift by g . Every clock tensor is a matrix product unitary.

Proof. ✓ Proof The tensor $T_{g,f}$ forces its right bond to carry the input label, so on a closed chain the only surviving bond configuration is $b_k = t_{k-1}$, which gives the kernel. With the trivial cocycle the group tensor has entry one exactly where $T_{g,f}$ is nonzero, and its periodic operator is the shift by g . A permutation matrix times a diagonal matrix of unit-modulus phases is unitary. $\square$

The first relation of the next theorem, $U_N^2 = \omega^N V_N$, reads as follows on a ring of N sites.



Theorem 27.11.3 (Multiplication with length-dependent phases). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [StmT](#) *On every periodic chain of $N > 0$ qutrits,*

$$U_N^2 = \omega^N V_N, \quad U_N V_N = V_N U_N = \omega^N \mathbb{1}, \quad V_N^2 = U_N, \quad U_N^3 = \omega^{2N} \mathbb{1}. \quad (27.259)$$

When $3 \mid N$ these are the relations of $\mathbb{Z}_3$; otherwise the representatives multiply only up to the phases ω^N .

Proof. [✓ Proof](#) The product of $O_N(T_{g,f})$ and $O_N(T_{h,f'})$ is the shift by $g+h$ with phase $\omega^{\sum_k (f(t_k+h, t_{k+1}+h) + f'(t_k, t_{k+1}))}$. If $f(a+h, b+h) + f'(a, b) = f''(a, b) + c + \kappa(a) - \kappa(b)$ for all a, b , the terms $\kappa(t_k) - \kappa(t_{k+1})$ cancel around the chain and the product is $\omega^{cN} O_N(T_{g+h, f''})$. For $U \cdot U$ one has $(a+1)(b+1) + ab = (2ab + 2b) + 1 + (a-b)$; for $U \cdot V$ and $V \cdot U$ the left side is $2a + b + 1 = 0 + 1 + (2a - 2b)$ modulo three; for $V \cdot V$ it is $ab + (a-b)$. These are checked on the nine pairs (a, b) . $\square$

Theorem 27.11.4 (The condensation defect of an exact group representation). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every finite group G and every three-cocycle, the periodic operators U_g of Definition 27.8.1 form a matrix product operator fusion algebra with structure constants $N_{gh}^k = \delta_{k,gh}$, and $A_0 = \sum_g U_g$ satisfies $A_0^2 = |G|A_0$ at every positive length. For $\mathbb{Z}_3$ and the trivial cocycle, $A_0^2 = 3A_0$.

Proof. [✓ Proof](#) The first claim restates $U_g U_h = U_{gh}$. For the second, $A_0^2 = \sum_g \sum_h U_{gh}$, and for each g the map $h \mapsto gh$ is a bijection of G , so every term U_k occurs $|G|$ times. $\square$

Definition 27.11.5 (The clock condensation defect). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The defect $A = \delta \oplus U \oplus V$ is the block-diagonal tensor of bond dimension $1 + 3 + 3$ whose blocks are the identity tensor δ and the clock tensors U and V . Its periodic operator is $O_N(A) = \mathbb{1} + U_N + V_N$.

Theorem 27.11.6 (Length-dependent fusion of the clock defect). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On every periodic chain of length $N > 0$,

$$O_N(A)^2 = (1 + 2\omega^N) \mathbb{1} + 3U_N + (2 + \omega^N) V_N, \quad (27.260)$$

and the operators $\mathbb{1}$, U_N and V_N are linearly independent, so these coefficients are the unique ones. The coefficients are the power sums of the weight multisets $\{1, \omega, \omega\}$, $\{1, 1, 1\}$ and $\{1, 1, \omega\}$. In particular $O_N(A)^2 = 3O_N(A)$ if and only if $3 \mid N$, and there are no coefficients independent of N with $O_N(A)^2 = a\mathbb{1} + bU_N + cV_N$ for every $N > 0$.

Proof. [✓ Proof](#) Expanding $(\mathbb{1} + U_N + V_N)^2$ with Theorem 27.11.3 gives $\mathbb{1} + 3U_N + 2V_N + \omega^N V_N + 2\omega^N \mathbb{1}$. The three operators shift the constant configuration 0 to the constant configurations 0, 1 and 2 respectively, each with phase one, so reading a combination of them in these three entries recovers its coefficients. The coefficient of V_N is $2 + \omega^N$, which equals 3 exactly when $\omega^N = 1$, that is, when $3 \mid N$; comparing the lengths 1 and 3 would force $\omega = 1$. $\square$

Theorem 27.11.7 (The clock family is not a fusion algebra). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For no nonnegative integers m_{ab}^c , independent of the length, do the periodic operators of δ , U and V satisfy $O_N(T_a)O_N(T_b) = \sum_c m_{ab}^c O_N(T_c)$ at every positive length. In contrast, the group operators of the trivial cocycle form such an algebra (Theorem 27.11.4).

Proof. [✓ Proof](#) Such constants would give $\omega^N V_N = U_N^2 = m_{11}^0 \mathbb{1} + m_{11}^1 U_N + m_{11}^2 V_N$, hence $\omega^N = m_{11}^2$ for every $N > 0$ by linear independence, and the lengths 1 and 3 give $\omega = 1$. $\square$

27.12 Mixed Bell bonds and renormalization fixed points

A renormalization fixed point of a matrix product density operator is a tensor whose one-site and two-site closures with an arbitrary boundary matrix are carried into each other by trace-preserving completely positive maps, in both directions [11, Definition 4.1, Example 4.12, Proposition 4.13, Theorem 4.14]. The characterization there expresses the products of the vertical components of such a tensor through structure coefficients, and the question left open after its Theorem 4.14 is whether these coefficients can depend on the length of the chain. The mixed

Bell-bond tensor, the flagged tensor, the one-label tensor, and the dimer twist below address that question. They are not taken from the literature; only the flag factor of the flagged tensor is, namely the normalized tensor of [11, Example 4.12]. The first two are mixed states built from Bell bonds: in the first, incoming bonds carry the mixed state $\beta_R = (25/32)P_+ + (7/32)P_-$; in the second, each site also carries a flag qubit $|f\rangle$ ($f \in \{0, 1\}$), with a controlled flip coupling the Bell sectors to an even-parity flag state. The vertical components of these two tensors yield the one-label tensor and the dimer twist, whose structure coefficients are computed explicitly by the multi-block compression theorem. These results establish tensor decompositions and periodic trace identities for the examples, not a classification of length-dependent fixed points.

The density formulas are (27.268) and Theorem 27.12.2.1. The channel equations, including arbitrary boundary matrices, are proved in Theorem 27.12.3.1; their necessary physical-trace identity is Theorem 27.12.3.2. The one-label and dimer word-span certificates are Theorems 27.12.6.2 and 27.12.7.3, with compression and trace identities in Theorems 27.12.6.3, 27.12.6.4, 27.12.7.5, and 27.12.7.6.

Write $E_{ij} = |i\rangle\langle j|$ and $|\Phi_\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}$, with $P_\pm = |\Phi_\pm\rangle\langle\Phi_\pm|$. For an MPO tensor $M^{ij} \in M_D(\mathbb{C})$, its periodic physical operator is

$$\rho^{(N)}(M) = \sum_{i,j} \text{tr}(M^{i_0 j_0} \dots M^{i_{N-1} j_{N-1}}) |i\rangle\langle j|. \quad (27.261)$$

An MPDO requires positivity for every $N > 0$; trace-one normalization is a separate condition. Throughout the physical state formulas below, $N > 0$ and site indices are cyclic.

27.12.1 Independent mixed Bell bonds

The local physical space is $\mathcal{H}_R = \mathbb{C}_L^2 \otimes \mathbb{C}_R^2$, with basis index $2l + r$. On the incoming bond (R_{n-1}, L_n) , put $\beta_R = (25/32)P_+ + (7/32)P_-$. Let Q_N regroup sites into incoming bonds:

$$Q_N \bigotimes_{n=0}^{N-1} |l_n, r_n\rangle = \bigotimes_{n=0}^{N-1} |r_{n-1}, l_n\rangle. \quad (27.262)$$

Thus $Q_N^\dagger \beta_R^{\otimes N} Q_N$ is a normalized density operator of rank 2^N .

An MPO representation uses four real matrices:

$$\begin{aligned} A^0 &= \frac{4}{5} E_{00}, \\ A^1 &= \frac{4}{5} E_{11}, \\ A^2 &= \frac{3}{5} E_{01}, \\ A^3 &= \frac{3}{5} E_{10}. \end{aligned} \quad (27.263)$$

The auxiliary indices $a, b \in \{0, 1, 2, 3\}$ label these matrices; their ordering is not the physical pair ordering $2l + r$. Define

$$R_{ab}^{(l,r),(l',r')} = \frac{25}{32} A_{ll'}^a A_{rr'}^b. \quad (27.264)$$

To contract this tensor, introduce $w = \begin{pmatrix} 16/25 & 9/25 \\ 9/25 & 16/25 \end{pmatrix}$. The matrix-unit entries give

$$\sum_a A_{uv}^a A_{xy}^a = \delta_{ux} \delta_{vy} w_{uv}. \quad (27.265)$$

For physical strings $p_n = (l_n, r_n)$ and $q_n = (l'_n, r'_n)$, the auxiliary trace pairs the right half of site $n - 1$ with the left half of site n . Consequently,

$$\rho^{(N)}(R)_{p,q} = \left(\frac{25}{32}\right)^N \prod_{n=0}^{N-1} \delta_{r_{n-1}, l'_n} \delta_{r'_{n-1}, l'_n} w_{l_n l'_n}. \quad (27.266)$$

This also gives a support-isometry description. Define $B_N |a_0, \dots, a_{N-1}\rangle = \bigotimes_n |a_n, a_{n+1}\rangle$. Distinct bit strings have distinct images, so $B_N^\dagger B_N = \mathbb{1}$. Set $h = (25/32)w = \begin{pmatrix} 1/2 & 9/32 \\ 9/32 & 1/2 \end{pmatrix}$. Formula (27.266) becomes

$$\rho^{(N)}(R) = B_N h^{\otimes N} B_N^\dagger. \quad (27.267)$$

The eigenvectors of h are $(1, \pm 1)/\sqrt{2}$, with eigenvalues $25/32$ and $7/32$. For $J|a\rangle = |a, a\rangle$, one has $JhJ^\dagger = \beta_R$ and $Q_N B_N = J^{\otimes N}$. Hence

$$Q_N \rho^{(N)}(R) Q_N^\dagger = \beta_R^{\otimes N}. \quad (27.268)$$

This proves positivity, trace one, and rank 2^N directly. At $N = 1$ the bond closes within the site, giving $\rho^{(1)}(R) = \beta_R$. At $N \geq 2$ the two qubits at a site belong to distinct independent bonds. Since both one-qubit marginals of β_R are $\mathbb{1}_2/2$, the one-site marginal is $\mathbb{1}_4/4$.

27.12.2 Mixed Bell bonds with an even-parity flag

The second model has local space $\mathcal{H}_T = \mathbb{C}_L^2 \otimes \mathbb{C}_R^2 \otimes \mathbb{C}_F^2$, with basis index $4l + 2r + f$. Put $x = 7/8$, $y = 1/8$, and define the bond state $\sigma = xP_+ + yP_-$. The flag state is the classical even-parity mixture

$$\eta_N = 2^{-N}(\mathbb{1}_2^{\otimes N} + Z^{\otimes N}) = 2^{1-N} \sum_{\substack{f \in \{0,1\}^N \\ \sum_n f_n \equiv 0 \pmod{2}}} |f\rangle\langle f|, \quad (27.269)$$

where $Z = E_{00} - E_{11}$. There are 2^{N-1} even strings, so η_N is positive and has trace one. For $N \geq 2$ its rank is 2^{N-1} ; it is not a pure GHZ state. At $N = 1$ it is $|0\rangle\langle 0|$.

A cell consists of the incoming bond and its receiving flag. With $X_F = E_{01} + E_{10}$ on the flag, define

$$V = (\mathbb{1}_4 - P_-) \otimes \mathbb{1}_2 + P_- \otimes X_F. \quad (27.270)$$

This gate fixes the flag outside the minus Bell sector and flips it inside that sector. Orthogonality of P_- and $\mathbb{1}_4 - P_-$ gives $V^\dagger = V$ and $V^2 = \mathbb{1}_8$, including on the complement of the two Bell vectors. Let $\tilde{Q}_N$ regroup sites into incoming cells by $\bigotimes_n |l_n, r_n, f_n\rangle \mapsto \bigotimes_n |(r_{n-1}, l_n), f_n\rangle$. The physical state is obtained by applying V on each cell to $\sigma^{\otimes N} \otimes \eta_N$. Here and below, the separated bond and flag factors are identified with cell coordinates by the canonical permutation of tensor factors.

For its MPO representation, set $\tau_k(f) = (-1)^{kf}$ and $C_k(u, v) = \frac{1}{2}(x + (-1)^{k+u+v}y)$. Explicitly,

$$\begin{aligned} C_0 &= \begin{pmatrix} 1/2 & 3/8 \\ 3/8 & 1/2 \end{pmatrix}, \\ C_1 &= \begin{pmatrix} 3/8 & 1/2 \\ 1/2 & 3/8 \end{pmatrix}. \end{aligned} \quad (27.271)$$

An auxiliary index is (p, p', k) , encoded as $4p + 2p' + k$. Define

$$T_{(p,p',k),(q,q',k')}^{(l,r,f),(l',r',g)} = \delta_{kk'} \delta_{fg} \delta_{lp} \delta_{l'p'} \delta_{rq} \delta_{r'q'} \frac{1}{2} C_k(p, p') \tau_k(f). \quad (27.272)$$

Both the physical and auxiliary dimensions are eight.

Theorem 27.12.2.1 (Closed density of the twisted dimer). ✓ Stmt For $N > 0$, write $W_N = V^{\otimes N}$ in incoming-cell coordinates. Then

$$\widetilde{Q}_N \rho^{(N)}(T) \widetilde{Q}_N^\dagger = W_N (\sigma^{\otimes N} \otimes \eta_N) W_N^\dagger. \quad (27.273)$$

In particular, $\rho^{(N)}(T)$ is positive and has trace one.

Proof. ✓ Proof Put $\sigma' = xP_+ - yP_-$. The entries of $xP_+ + (-1)^k yP_-$ on the matched bond subspace are $C_k(u, v)$. In a positive-length auxiliary trace of (27.272), the label k is constant around the ring. The remaining constraints require matching neighboring half-bonds and equal ket and bra flags. Therefore

$$\widetilde{Q}_N \rho^{(N)}(T) \widetilde{Q}_N^\dagger = 2^{-N} [(\sigma \otimes \mathbb{1}_2)^{\otimes N} + (\sigma' \otimes Z)^{\otimes N}]. \quad (27.274)$$

The gate fixes $\sigma \otimes \mathbb{1}_2$ under conjugation and sends $\sigma \otimes Z$ to $\sigma' \otimes Z$, since $X_F Z X_F = -Z$. Applying these identities in each cell proves (27.273). Positivity follows from the positive factor states; their traces are $\text{tr } \sigma = x + y = 1$ and $\text{tr } \eta_N = 1$. Unitary conjugation preserves trace. $\square$

Theorem 27.12.2.2 (The flag factor as a normalized fixed-point state). ✓ Stmt The flag family is represented by the density-normalized tensor $\widehat{M}^{fg} = \frac{1}{2} \delta_{fg} \text{diag}(1, (-1)^f)$ of Example 4.12 of Ref. [11]. This tensor is an MPDO and satisfies the channel fixed-point equations defined below.

Proof. ✓ Proof The two diagonal auxiliary sectors contribute $2^{-N} \mathbb{1}_2^{\otimes N}$ and $2^{-N} Z^{\otimes N}$, respectively. Their sum is (27.269). The channel fixed-point property is that of the normalized example. The normalization includes a factor 1/2 in every local tensor, absent from the unnormalized tensor printed in the example. $\square$

The factorization (27.273) is an identity of closed operators, not an on-site tensor equivalence or an auxiliary gauge equivalence. The positive-length restriction matters for both models: at length zero the auxiliary trace gives $\rho^{(0)}(R) = 4$ and $\rho^{(0)}(T) = 8$, whereas the corresponding empty factor expressions give 1 and 2.

27.12.3 Trace-preserving fixed-point maps

For a tensor M and an auxiliary boundary matrix X , define the one-site and two-site physical closures by

$$\begin{aligned} M_1(X) &= \sum_{i,j} \text{tr}(M^{ij} X) E_{ij}, \\ M_2(X) &= \sum_{i,j,k,l} \text{tr}(M^{ij} M^{kl} X) E_{ij} \otimes E_{kl}. \end{aligned} \quad (27.275)$$

The channel fixed-point condition of [11, Definition 4.1] requires trace-preserving completely positive maps, abbreviated tpCPMs, $\mathcal{S}$ and $\mathcal{T}$ such that, for every X ,

$$\begin{aligned} \mathcal{S}(M_2(X)) &= M_1(X), \\ \mathcal{T}(M_1(X)) &= M_2(X). \end{aligned} \quad (27.276)$$

The symbol T denotes the physical dimer tensor, whereas $\mathcal{T}$ denotes a refinement channel. Closed density factorizations alone do not establish these equations for arbitrary boundary matrices.

Theorem 27.12.3.1 (Channel fixed points of the two physical tensors). ✓ Stmt Both R and T satisfy (27.276) with $tpCPMs$ on the full physical matrix spaces.

Proof. ✓ Proof For R , set $h = (25/32)w$. Its spectral probabilities are $p_0 = 25/32$ and $p_1 = 7/32$. The refinement Kraus operators act by

$$K_s|l, r\rangle = \sqrt{\frac{p_s}{2}} \sum_{b=0}^1 (-1)^{sb} |l, b\rangle \otimes |b, r\rangle. \quad (27.277)$$

Thus $\sum_s K_s^\dagger K_s = \mathbb{1}_4$. Direct contraction of (27.264) gives, for every boundary matrix X ,

$$R_2(X)_{(l,b),(c,r);(l',b'),(c',r')} = \delta_{bc} \delta_{b'c'} h_{bb'} R_1(X)_{(l,r),(l',r')}. \quad (27.278)$$

The spectral identity $\sum_s (p_s/2)(-1)^{s(b+b')} = h_{bb'}$ proves refinement. Coarse-graining contracts the inner bond against the four orthonormal vectors $\Phi_+, \Phi_-, 01, 10$, retaining the outer bits. The four Kraus operators resolve the identity on the full two-site space. The last two annihilate (27.278); summing the first two returns $R_1(X)$ because $\text{tr } h = 1$.

For T , coarse-graining reads the flags f_1, f_2 and measures the inner bond in the basis $\Phi_+, \Phi_-, 01, 10$. It retains the outer qubits and outputs $f_1 + f_2 + s$ modulo two, where $s = 1$ only for outcome Φ_- . Completeness of the bond basis gives the Kraus resolution. On the matched support, the two Bell outcomes have weights x and $(-1)^k y$ in block k . Their flag shifts therefore yield the required factor $\tau_k(f_1 + f_2 + s)$, and $x + y = 1$ gives the coarse-graining equation.

Refinement inserts Φ_+ or Φ_- with probabilities x, y . For outcome s , it splits an input flag f uniformly among the two pairs satisfying $f_1 + f_2 + s = f$. Its boundary equation follows from

$$xP_+ + (-1)^k yP_- = JC_k J^\dagger. \quad (27.279)$$

Each inserted Bell vector is normalized, and the two flag choices each have half the outcome probability, so refinement is trace preserving. These are the coarse-graining and refinement constructions for the respective tensors. $\square$

Theorem 27.12.3.2 (A necessary physical-trace identity). ✓ Stmt If M satisfies (27.276), then $F_M = \sum_i M^{ii}$ satisfies $F_M^2 = F_M$.

Proof. ✓ Proof Trace preservation of $\mathcal{T}$ gives $\text{tr}(F_M^2 X) = \text{tr } M_2(X) = \text{tr } M_1(X) = \text{tr}(F_M X)$ for every X . Nondegeneracy of the matrix trace pairing gives $F_M^2 = F_M$. $\square$

This is a statement about the physical-trace transfer matrix F_M , not the doubled-index completely positive transfer map. For R , the identity also follows directly from $F_R = (25/32)tt^\top$, where $t = (4/5, 4/5, 0, 0)^\top$ and $t^\top t = 32/25$.

27.12.4 Vertical components and their multiplication

Rotation exchanges the physical and auxiliary pairs: $\mathcal{V}(M)_{ij}^{ab} = M_{ab}^{ij}$. For a vertical component K , its periodic operator at length $L > 0$ is

$$O_L(K) = \sum_{a,b} \text{tr}(K^{a_0 b_0} \dots K^{a_{L-1} b_{L-1}}) |a\rangle\langle b|. \quad (27.280)$$

It acts on strings of the original auxiliary indices, rather than on the original physical chain.

For R , the normalized vertical component is $K^{ab} = A^a \otimes A^b$, so $\mathcal{V}(R) = (25/32)K$. This rotation and scalar factor are essential: K is not the physical tensor R . If its indices a, b are instead treated as physical indices, then

$$F_K = \sum_a A^a \otimes A^a = \frac{1}{25} \begin{pmatrix} 16 & 0 & 0 & 9 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 9 & 0 & 0 & 16 \end{pmatrix}. \quad (27.281)$$

Its $(00, 00)$ entry satisfies $(F_K^2 - F_K)_{00,00} = (337 - 400)/625 = -63/625$. Theorem 27.12.3.2 therefore excludes the channel fixed-point property for K in this physical reading.

Writing $|\psi_L\rangle = \sum_a \text{tr}(A^{a_0} \dots A^{a_{L-1}})|a\rangle$ gives $O_L(K) = |\psi_L\rangle\langle\psi_L|$. A nonzero cyclic product of the matrix-unit letters specifies a cyclic bit path, and its squared amplitude is the product of the corresponding entries of w . Thus $\langle\psi_L|\psi_L\rangle = \text{tr}(w^L)$, and

$$O_L(K)^2 = \text{tr}(w^L)O_L(K) = \left(1 + \left(\frac{7}{25}\right)^L\right)O_L(K). \quad (27.282)$$

The eigenvalue $7/25$ controls vertical multiplication; it is not the physical Bell probability $7/32$.

For T , restrict the physical ket and bra flags to the same value f . The unnormalized vertical component D_f is defined by

$$(D_f^{ab})_{(l,r),(l',r')} = T_{ab}^{(l,r,f),(l',r',f)}. \quad (27.283)$$

It satisfies $D_f = \nu \widehat{K}_f$, where $\widehat{K}_f$ is the normalized flag component, $\nu = 5/8$, and $\nu^2 = (x^2 + y^2)/2 = 25/64$. For $a = (p, p', k)$ and $b = (q, q', k')$, its normalized letter is

$$\widehat{K}_f^{ab} = \frac{\delta_{kk'} C_k(p, p') \tau_k(f)}{2\nu} (E_{pp'} \otimes E_{qq'}).$$

The nonzero coefficient magnitudes of D_f are $1/4, 3/16$, whereas those of $\widehat{K}_f$ are $2/5, 3/10$. Neither component is the physical tensor T .

For an MPDO in CPSV canonical form, the fixed-point characterization requires a vertical canonical decomposition with a basis of normal tensors K_α , positive diagonal multiplicity matrices μ_α , and positive diagonal matrices $\chi_{\alpha,\beta,\gamma}$. Set $m_\alpha = \text{tr} \mu_\alpha$ and $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$. The required multiplication identities, for every $L > 0$, are

$$O_L(K_\alpha)O_L(K_\beta) = \sum_\gamma c_{\alpha,\beta,\gamma}^{(L)} O_L(K_\gamma). \quad (27.284)$$

The vector m is an idempotent for the multiplication induced by $c^{(1)}$:

$$m_\gamma = \sum_{\alpha,\beta} c_{\alpha,\beta,\gamma}^{(1)} m_\alpha m_\beta. \quad (27.285)$$

For R , $\mu_0 = (25/32)$ and $\chi_{000} = \text{diag}(1, 7/25)$; indeed $(32/25)(25/32)^2 = 25/32$. For the normalized dimer components, $m_0 = m_1 = 5/8$. The fusion calculation below gives unnormalized weights $7/16, 1/16$, which become $7/10, 1/10$ after division by ν . Each output sector receives two contributions of each kind in (27.285), giving $2(7/10 + 1/10)(5/8)^2 = 5/8$.

Theorem 27.12.4.1 (Fixed points and vertical multiplication). ✓ Stmt Let M generate MPDOs and suppose its doubled-physical-index tensor is in CPSV canonical form (CF), with normalized normal tensor blocks. The channel equations (27.276) are equivalent to the existence of a vertical canonical decomposition of M whose basis of normal tensors (BNT) satisfies (27.284) and (27.285) with positive diagonal χ . They are also equivalent to the existence of active-sector fusion coisometries with exact reconstruction and the same idempotent relation, as specified in Theorem 21.3.30. The first equivalence is Ref. [11, Theorem 4.14(i)–(ii)]; the second is the corrected active-support form of its fusion equivalence. The zero fusion multiplicity of absent product sectors, the empty active support of a fixed pair, and the coisometry orientation of the fusion map are recorded in [20], [101], and [61]. The comparison of one-site and two-site prefixes and the completion of the physical maps on discarded complements are recorded in [89] and [154].

Proof. ✓ Proof The fixed-point channels give an active-sector fusion decomposition, which implies the operator product identities and the idempotent relation. Conversely, these relations for the vertical components of M yield coarse-graining and refinement channels by Theorem 21.3.30. The comparison uses both one-site and two-site prefixes, and the physical maps are completed on discarded complements. These implications also prove the equivalence with the active-sector fusion description. $\square$

27.12.5 Representative rescaling

Scalar rescalings of the normal tensors in a basis change the structure coefficients by geometric factors, so the length-dependence question must be posed modulo this freedom; the entries below record the rescaling rule for the structure coefficients of [11, Theorem 4.14].

Consider the structure coefficients $c_{\alpha,\beta,\gamma}^{(L)} = \text{tr}(\chi_{\alpha,\beta,\gamma}^L)$ and trace scalars m_α of [11, Theorem 4.14(ii)].

Definition 27.12.5.1 (Rescaling the structure coefficients). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let Λ be a label set and let $s_\alpha \in \mathbb{C}$ for every $\alpha \in \Lambda$. In the algebraic formulas below, division is extended by $z/0 = 0$. Their interpretation as changes of tensor representatives requires all scales to be nonzero. Rescale the diagonal entries, coefficients, and trace scalars by

$$\chi'_{\alpha,\beta,\gamma,k} = \frac{s_\alpha s_\beta}{s_\gamma} \chi_{\alpha,\beta,\gamma,k}, \quad (27.286)$$

$$c'^{(L)}_{\alpha,\beta,\gamma} = \left(\frac{s_\alpha s_\beta}{s_\gamma} \right)^L c^{(L)}_{\alpha,\beta,\gamma}, \quad (27.287)$$

$$m'_\alpha = \frac{m_\alpha}{s_\alpha}. \quad (27.288)$$

The multiplicity $r_{\alpha,\beta,\gamma}$ is unchanged, and $0 \leq k < r_{\alpha,\beta,\gamma}$.

Theorem 27.12.5.2 (Trace powers and positivity under rescaling). Lean Lean Lean ✓ Stmt Taking trace-power coefficients commutes with rescaling: for every $L \in \mathbb{N}$, including $L = 0$,

$$\text{tr}((\chi'_{\alpha,\beta,\gamma})^L) = \left(\frac{s_\alpha s_\beta}{s_\gamma} \right)^L \text{tr}(\chi_{\alpha,\beta,\gamma}^L). \quad (27.289)$$

If the original diagonal entries are positive real numbers and every s_α is positive real, the rescaled entries are positive.

Proof. ✓ Proof Expand the trace as the sum of the L th powers of the diagonal entries. Each summand acquires the factor $(s_\alpha s_\beta / s_\gamma)^L$, which factors out of the finite sum. For positive real scales, the ratio $s_\alpha s_\beta / s_\gamma$ is positive, so multiplication by it preserves positivity. $\square$

Theorem 27.12.5.3 (Preservation of the idempotent coefficient equation). Lean Lean ✓ Stmt Suppose Λ is finite and every s_α is nonzero. The vector m' is idempotent for $c'^{(1)}$ if and only if m is idempotent for $c^{(1)}$. Explicitly, the equations

$$m'_\gamma = \sum_{\alpha, \beta} c'^{(1)}_{\alpha, \beta, \gamma} m'_\alpha m'_\beta, \quad (27.290)$$

$$m_\gamma = \sum_{\alpha, \beta} c^{(1)}_{\alpha, \beta, \gamma} m_\alpha m_\beta \quad (27.291)$$

are equivalent when imposed for every $\gamma \in \Lambda$.

Proof. ✓ Proof Substituting (27.287) and (27.288) and cancelling the nonzero scales in each summand gives

$$\sum_{\alpha, \beta} c'^{(1)}_{\alpha, \beta, \gamma} m'_\alpha m'_\beta = \frac{1}{s_\gamma} \sum_{\alpha, \beta} c^{(1)}_{\alpha, \beta, \gamma} m_\alpha m_\beta. \quad (27.292)$$

The left side of the idempotent equation is $m'_\gamma = m_\gamma / s_\gamma$. Cancellation of s_γ proves both implications. $\square$

Scaling a normal-tensor representative by a positive factor $c \neq 1$ alters the transfer map's spectral radius to c^2 , though the rescaled idempotent coefficient equations remain preserved.

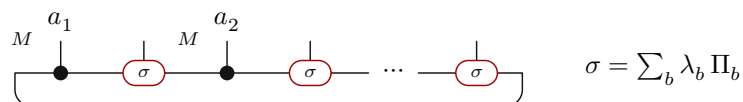
27.12.6 The one-label fixed point

The two remaining subsections apply the multi-block compression theorem to the vertical components of R and T , in the unnormalized representatives fixed above. The adapted basis of the inner bond and the reordering behind the factorization $B^a = \sigma \otimes M^a$ used below are set out in Section I.2, together with the two supporting results the proofs invoke, Theorems I.2.3 and I.2.4.

The vertical component K of R has letters $K^{ij} = A^i \otimes A^j$, with A given in (27.263). Its stacked product has letters $A^i \otimes F_K \otimes A^k$. The inner matrix $F_K = \sum_j A^j \otimes A^j$ has eigenvalues 1, 7/25, 0, 0, with eigenvectors $\Phi_+, \Phi_-, 01, 10$. Thus the stacked tensor contains two weighted copies of the same target, not two inequivalent targets. In this subsection write $M = K$; it remains a vertical component, not the physical tensor R .

Definition 27.12.6.1 (The one-label tensors). Lean Lean Lean Lean Lean ✓ Stmt Set $M^{(i,j)} = A^i \otimes A^j$, of bond dimension four. Its sixteen-letter MPS reading is the target tensor. Let $B^{(i,k)} = \sum_j M^{(i,j)} \otimes M^{(j,k)}$ be the stacked product, of bond dimension sixteen. The two targets are copies of M with scalar weights $\mu_0 = 1$ and $\mu_1 = 7/25$.

The following schematic records the inner spectral factors. Its σ denotes the inner matrix $\Pi_0 + (7/25)\Pi_1$, not a trace-one physical density. After reordering, $B^a = \sigma \otimes M^a$: the spectral factor acts on the inner bond and M^a on the outer bond. In the alternating drawing, each M node means $\mathbb{1}_4 \otimes M^a$ and each spectral ring means $\sigma \otimes \mathbb{1}_4$, both on the full bond space. Their product is B^a . Only the virtual legs are closed; the physical pair legs remain open.



Theorem 27.12.6.2 (Normality of the one-label target). [Lean](#) [✓ Stmt](#) *The vertical tensor is normal at blocking length one in the sense of Definition 2.5.5: its sixteen matrices are scaled matrix units covering every position of $M_4(\mathbb{C})$.*

Proof. [✓ Proof](#) Each position is realized by one of the sixteen letters with a nonzero integer factor. Theorem 1.2.4 gives normality. $\square$

Theorem 27.12.6.3 (Split compression of the one-label fixed point). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The stacked tensor B has the block triangular form of Theorem 25.5.1 for $C_s = \mu_s M$, with an eight-dimensional zero complement, $z = 8$, and $16 = 4 + 4 + 8$. Its compression pairs W_s, V_s are mutually biorthogonal and satisfy $W_s B^w V_s = C_s^w$ for every word. The remainder vanishes identically, and $B^a V_s = V_s(\mu_s M^a)$ and $W_s B^a = (\mu_s M^a) W_s$.*

Proof. [✓ Proof](#) For the integer tensors, twice the stacked letter is the sum of the two adapted-basis blocks with coefficients 25 and 7. This is the integer identity of Theorem 1.2.3, verified over the sixteen physical indices. The scalar identity uses weights 1, 7/25. Proposition 25.5.5 supplies the compression maps, vanishing remainder, and one-site intertwiners, with zero complement of dimension $2 \cdot 4 = 8$. $\square$

Theorem 27.12.6.4 (The one-label structure coefficient). [Lean](#) [✓ Stmt](#) *For every nonempty word w over the sixteen-letter alphabet,*

$$\mathrm{tr}(B^w) = \left(1 + \left(\frac{7}{25}\right)^{|w|}\right) \mathrm{tr}(M^w). \quad (27.293)$$

Proof. [✓ Proof](#) Proposition 25.5.3 applied to Theorem 27.12.6.3 gives the two weighted target traces. Scalar word evaluation contributes 1 and $(7/25)^{|w|}$. $\square$

27.12.7 The graded quantum-dimer twist

For the dimer, write $M_f = D_f$ for the unnormalized vertical components of (27.283). Their matrix space acts on $\mathbb{C}^2 \otimes \mathbb{C}^2$; their physical letter indices are the horizontal auxiliary pairs of T . The matrices C_k are the two-by-two matrices in (27.271), equivalently the restrictions of $xP_+ + (-1)^k yP_-$ to the matched bond subspace.

Stacking sectors f, f' gives targets $M_{f+f'}$ and $M_{f+f'+1}$, with addition modulo two, and weights $x/2 = 7/16$ and $y/2 = 1/16$. Unlike the one-label example, these contributions multiply different targets. For sector zero multiplied by itself, $c_{000}^{(L)} = (7/16)^L$ and $c_{001}^{(L)} = (1/16)^L$ in this unnormalized component convention. These are the trace powers of $\chi_{000} = (7/16)$ and $\chi_{001} = (1/16)$. The normalized BNT coefficients instead use 7/10, 1/10, as explained in Subsection 27.12.4.

Definition 27.12.7.1 (The sector tensors and their stacked product). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [✓ Stmt](#) For $a = (p, p', k)$ and $b = (q, q', k')$, define

$$M_f^{ab} = \delta_{kk'} \frac{1}{2} C_k(p, p') \tau_k(f) (E_{pp'} \otimes E_{qq'}). \quad (27.294)$$

The sixty-four-letter reading is the target tensor, of bond dimension four. The stacked product $B_{f,f'}^{ab} = \sum_c M_f^{ac} \otimes M_{f'}^{cb}$ has bond dimension sixteen. Its target tensors are $M_{f+f'}$ and $M_{f+f'+1}$, with weights $7/16$ and $1/16$.

The inner idempotents select different outer target sectors. The exact reordered letter identity is

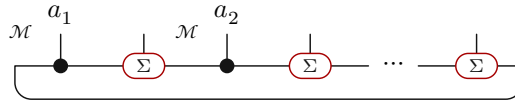
$$B_{f,f'}^a = \frac{7}{16} \Pi_0 \otimes M_{f+f'}^a + \frac{1}{16} \Pi_1 \otimes M_{f+f'+1}^a. \quad (27.295)$$

For fixed f, f' , define matrices on the full inner-outer bond space by

$$\mathcal{M}^a = \sum_{s=0}^1 \Pi_s \otimes M_{f+f'+s}^a.$$

$$\Sigma = \left(\frac{7}{16} \Pi_0 + \frac{1}{16} \Pi_1 \right) \otimes \mathbb{1}_4.$$

Orthogonality of the idempotents gives $B_{f,f'}^a = \mathcal{M}^a \Sigma = \Sigma \mathcal{M}^a$. The alternating ring therefore represents the periodic contraction of these stacked letters. Its spectral factor Σ is not the physical Bell density σ in (27.273).



Lemma 27.12.7.2 (The sectors are rescaled flag sectors). [Lean](#) [✓ Stmt](#) For each $f \in \{0, 1\}$, the sector M_f of (27.294) equals $\frac{5}{8} \widehat{M}_f$, where $\widehat{M}_f$ is the rescaled flag sector of the twisted quantum dimer of Definition 21.2.2.55.

Proof. [✓ Proof](#) Both tensors are supported on the bond pairs with a common label $k = k'$, where both matrices are the unit $E_{pp'} \otimes E_{qq'}$ times a scalar. The scalars agree, $\frac{1}{2} C_k(p, p') \tau_k(f) = \frac{5}{8}$ times the coefficient of $\widehat{M}_f$, as checked over the two sectors and the eight values of (p, p', k) . $\square$

Theorem 27.12.7.3 (Normality of the sectors). [Lean](#) [✓ Stmt](#) Each sector M_f is normal at blocking length one in the sense of Definition 2.5.5.

Proof. [✓ Proof](#) By Lemma 27.12.7.2, $M_f = \frac{5}{8} \widehat{M}_f$. The rescaled flag sector $\widehat{M}_f$ is normal by Theorem 21.2.2.62, and normality is invariant under the nonzero rescaling by $\frac{5}{8}$ (Theorem 12.8.1). $\square$

Lemma 27.12.7.4 (Sign rule of the stacked product). [Lean](#) [Lean](#) [✓ Stmt](#) Write $\widetilde{M}_f = 16M_f$ for the integer sector tensor and

$$\widetilde{B}_{f,f'}^{ab} = \sum_c \widetilde{M}_f^{ac} \otimes \widetilde{M}_{f'}^{cb} = 256 B_{f,f'}^{ab}$$

for the integer stacked product. For all sectors f, f' , $\widetilde{B}_{f,f'} = \widetilde{B}_{0,f+f'}$, and the two integer target tensors $\widetilde{M}_{f+f'}$ and $\widetilde{M}_{f+f'+1}$ of the pair (f, f') are those of the pair $(0, f + f')$.

Proof. [✓ Proof](#) The factor $\widetilde{M}_f^{ac} \otimes \widetilde{M}_{f'}^{cb}$ vanishes unless the three indices a, c, b carry a common label k . On a common label the signs multiply, $\tau_k(f) \tau_k(f') = \tau_k(0) \tau_k(f + f')$, which is checked over the two values of k and the four pairs of sectors. The target tensors depend on (f, f') only through $f + f'$. $\square$

Chapter 28

Matrix Product Operator Algebras

A matrix product operator algebra is a finite family of matrix product operators satisfying the fusion rule (26.2) with structure constants $N_{ab}^c \in \mathbb{N}$ [7, 164, 188]. Chapter 26 introduces these algebras and their non-invertible examples, defines symmetric matrix product states, and proves the integrality of their action coefficients. Transfer of the symmetry to the virtual level rests on the multi-block asymmetric compression theorem (Theorem 25.5.1): the stacked tensor of a product of two matrix product operators, or of an operator acting on a state, need not be in canonical form and in general admits no sitewise intertwiner with the tensors of the fusion products (Remark 25.1.2), so the Fundamental Theorem for tensors in canonical form does not apply to it in general; the asymmetric theorem, which requires equality of periodic traces only, does.

Throughout, T_a denotes a matrix product operator tensor of bond dimension χ_a , $O_a^{(N)}$ its periodic operator on N sites, and $|V^{(N)}(A)\rangle$ the periodic vector of a matrix product state tensor A . For two operator tensors, $T_a T_b$ denotes the stacked tensor $(T_a T_b)^{ij} = \sum_m T_a^{im} \otimes T_b^{mj}$ of bond dimension $\chi_a \chi_b$, and $T_a \cdot A$ denotes the tensor $(T_a \cdot A)^i = \sum_j T_a^{ij} \otimes A^j$. A reduction of a tensor B onto a tensor C is a pair of maps (V, W) with $VW = \mathbb{1}$ and $VB^w W = C^w$ for every word w (Definition 2.4.5), with compression V and embedding W .

28.1 Fusion tensors and action tensors

Equality of periodic traces in the fusion rule (26.2) decomposes the stacked tensor $T_a T_b$ into copies of the blocks T_c .

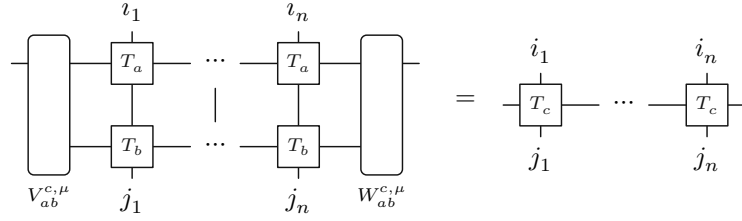
Definition 28.1.1 (Fusion tensors). Not ready Let $(T_a)_{a \in I}$ be a fusion algebra of normal tensors of positive bond dimension. A family of *fusion tensors* for the pair (a, b) consists of maps

$$V_{ab}^{c,\mu} : \mathbb{C}^{\chi_a} \otimes \mathbb{C}^{\chi_b} \rightarrow \mathbb{C}^{\chi_c}, \quad W_{ab}^{c,\mu} : \mathbb{C}^{\chi_c} \rightarrow \mathbb{C}^{\chi_a} \otimes \mathbb{C}^{\chi_b}, \quad (28.1)$$

for $c \in I$ and $1 \leq \mu \leq N_{ab}^c$, such that the pairs are mutually biorthogonal, $V_{ab}^{c,\mu} W_{ab}^{d,\nu} = \delta_{cd} \delta_{\mu\nu} \mathbb{1}_{\chi_c}$, each pair is a reduction of $T_a T_b$ onto T_c ,

$$V_{ab}^{c,\mu} (T_a T_b)^w W_{ab}^{c,\mu} = T_c^w \quad (28.2)$$

for every word $\mathbf{w}$ in the pair alphabet, and the remainder $(T_a T_b)^{ij} - \sum_{c,\mu} W_{ab}^{c,\mu} T_c^{ij} V_{ab}^{c,\mu}$ generates a nilpotent algebra. This is the decomposition of [164, Section 2] and [7], with the nilpotent remainder of [164, Section 4.4] in place of an exact identity.

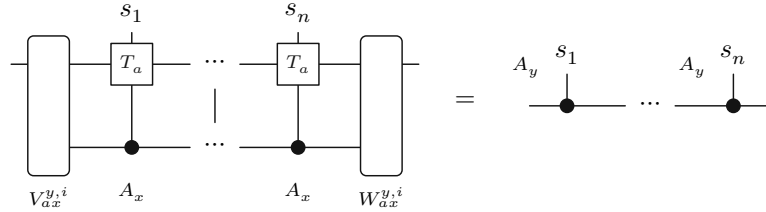


Theorem 28.1.2 (Existence of fusion tensors). *Every pair (a, b) of labels of a fusion algebra of normal tensors of positive bond dimension admits a family of fusion tensors (Corollary 25.6.1.2). This is the existence of fusion tensors in [164, Appendix A], here under the periodic fusion rule (26.2).*

Proof. Apply Theorem 25.6.2 with the target family consisting of N_{ab}^c copies of T_c for each c , all with weight 1. Its compression pairs, indexed by (c, μ) , are mutually biorthogonal reductions with a nilpotent remainder; this is Theorem 25.6.1.1 for the stacked product. $\square$

For a group-like algebra, with labels in a group G and $N_{gh}^k = \delta_{k,gh}$, the fusion tensors reduce to one pair $(V_{g,h}, W_{g,h})$ for each pair of group elements; for representations by normal tensors this is Corollary 25.6.3.

Definition 28.1.3 (Action tensors). Not ready Let $(A_x)_{x \in K}$ be a symmetric family of normal tensors of positive bond dimension whose coefficients $M_{a,x}^y$ are nonnegative integers. A family of *action tensors* for (a, x) consists of maps $V_{ax}^{y,i} : \mathbb{C}^{\chi_a} \otimes \mathbb{C}^{D_x} \rightarrow \mathbb{C}^{D_y}$ and $W_{ax}^{y,i} : \mathbb{C}^{D_y} \rightarrow \mathbb{C}^{\chi_a} \otimes \mathbb{C}^{D_x}$, for $y \in K$ and $1 \leq i \leq M_{a,x}^y$, that are mutually biorthogonal reductions of $T_a \cdot A_x$ onto the blocks A_y with a nilpotent remainder.



Theorem 28.1.4 (Existence of action tensors). *Under the hypotheses of Theorem 26.4.8, every pair (a, x) admits a family of action tensors; for integer coefficients this is the action clause of Corollary 25.6.1.2. This is the existence of action tensors in [164, Section 3].*

Proof. By Theorem 26.4.8 the coefficients are nonnegative integers, so (26.5) writes $O_a^{(N)} |V^{(N)}(A_x)\rangle$ as a sum of periodic vectors of blocks repeated with multiplicities $M_{a,x}^y$. Corollary 25.6.4 applied to this family with unit weights gives the reductions. $\square$

28.2 Gauge freedom of fusion and action tensors

Fusion tensors are unique up to a dressed scalar when the stacked product reduces onto a single normal block, and up to the multiplicity gauge under the zipper condition (Remark 28.2.4).

Corollary 28.2.1 (Uniqueness of a single-target reduction). *Let B be a tensor, C a normal tensor, and $(V, W), (\widetilde{V}, \widetilde{W})$ two reductions of B onto C whose residual tensors $B - WCV$ and $B - \widetilde{W}C\widetilde{V}$ are nilpotent with a common bound N_0 . Then there is a unique $\lambda \in \mathbb{C}^\times$ with*

$$\widetilde{V}B^{\mathbf{w}} = \lambda VB^{\mathbf{w}}, \quad B^{\mathbf{w}}\widetilde{W} = \lambda^{-1}B^{\mathbf{w}}W \quad (28.3)$$

for every word with $|\mathbf{w}| > 2N_0$. This applies to the fusion tensors of a group-like algebra, where $T_g T_h$ reduces onto the single block T_{gh} , to the composite reductions of Theorem 28.3.1, and to the action tensors of an invertible label on a permuted block. It does not apply to a single slot of a reduction onto multiple blocks, whose residual is not nilpotent.

Proof. Existence of λ is Theorem 25.11.28 written for compressions on the left and embeddings on the right, and for fusion tensors Corollary 25.6.1.3. Uniqueness is Lemma 19.5.1.2: $VB^{\mathbf{w}}W = C^{\mathbf{w}}$ is nonzero for arbitrarily long words. $\square$

Theorem 28.2.2 (Uniqueness of zipper fusion tensors up to multiplicity gauge). *Let the blocks T_c of a fusion algebra be normal and pairwise not related by a gauge transformation, and let $(V_{ab}^{c,\mu}, W_{ab}^{c,\mu})$ and $(\widetilde{V}_{ab}^{c,\mu}, \widetilde{W}_{ab}^{c,\mu})$ be two families of fusion tensors for (a, b) with vanishing remainder,*

$$(T_a T_b)^{\mathbf{w}} = \sum_{c,\mu} W_{ab}^{c,\mu} T_c^{\mathbf{w}} V_{ab}^{c,\mu} = \sum_{c,\mu} \widetilde{W}_{ab}^{c,\mu} T_c^{\mathbf{w}} \widetilde{V}_{ab}^{c,\mu} \quad (28.4)$$

for every nonempty word $\mathbf{w}$. Then for each c there is a unique $Y_{ab}^c \in \text{GL}_{N_{ab}^c}(\mathbb{C})$ with $\widetilde{V}_{ab}^{c,\mu} = \sum_{\nu} (Y_{ab}^c)_{\mu\nu} V_{ab}^{c,\nu}$ and $\widetilde{W}_{ab}^{c,\mu} = \sum_{\nu} W_{ab}^{c,\nu} ((Y_{ab}^c)^{-1})_{\nu\mu}$ on the support of the decomposition, so the ambiguity is $\bigoplus_c \text{GL}_{N_{ab}^c}(\mathbb{C})$ with no word dressing. This is the uniqueness of fusion tensors under the zipper condition in [7], and the basis change of [188, Section 4]; it is Corollary 25.8.1.4.

Proof. This is the argument of Theorem 25.8.1.3. Multiply (28.4) by $\widetilde{V}_{ab}^{c,\mu}$ on the left and $W_{ab}^{e,\nu}$ on the right and use biorthogonality of both families: $T_c^{\mathbf{w}} \widetilde{V}_{ab}^{c,\mu} W_{ab}^{e,\nu} = \widetilde{V}_{ab}^{c,\mu} W_{ab}^{e,\nu} T_e^{\mathbf{w}}$ for every nonempty word. For normal blocks, the words of a fixed injective length span the full matrix algebra, so an intertwiner between T_c and T_e is zero unless T_c and T_e are gauge equivalent, and is a scalar when $e = c$. This gives $\widetilde{V}_{ab}^{c,\mu} W_{ab}^{e,\nu} = \delta_{ce} (Y_{ab}^c)_{\mu\nu} \mathbb{1}$, and the same argument with the families exchanged gives the inverse matrix. $\square$

Theorem 28.2.3 (Uniqueness of zipper action tensors up to multiplicity gauge). *Let the blocks A_y be normal and pairwise not related by a gauge transformation. Two families of action tensors for (a, x) with vanishing remainder, $(T_a \cdot A_x)^{\mathbf{w}} = \sum_{y,i} W_{ax}^{y,i} A_y^{\mathbf{w}} V_{ax}^{y,i}$ for every nonempty word, are related by a unique $X_{ax}^y \in \text{GL}_{M_{a,x}^y}(\mathbb{C})$ for each y , in the sense of Theorem 28.2.2. This is the gauge freedom of the action tensors in [164, Section 3.1], and it is Corollary 25.8.1.5.*

Proof. The proof of Theorem 28.2.2 applies to the tensor $T_a \cdot A_x$ and the blocks A_y . $\square$

2. $N^{\mathbf{p}}R^{\mathbf{w}}W_1 = 0$ when $|\mathbf{p}| + |\mathbf{w}| \geq N_1 + N_2$. Prepend a letter x : $N^{\mathbf{p}}R^{x\mathbf{w}}W_1 = N^{\mathbf{p}x}R^{\mathbf{w}}W_1 + N^{\mathbf{p}}W_1M^{x\mathbf{w}}$ by the first step, and the last term vanishes because $|\mathbf{p}| \geq N_1$ or $|x\mathbf{w}| \geq N_2$.
3. $R^{\mathbf{w}}N^{\mathbf{q}} = 0$ when $|\mathbf{w}| + |\mathbf{q}| \geq 2N_1 + N_2$. Append a letter x : $R^{x\mathbf{w}}N^{\mathbf{q}} = R^{\mathbf{w}}N^{x\mathbf{q}} + R^{\mathbf{w}}W_1M^xV_1N^{\mathbf{q}}$, and the last term vanishes because $|\mathbf{q}| \geq N_1$, or else $|\mathbf{w}| \geq N_1 + N_2$ and the second step applies.

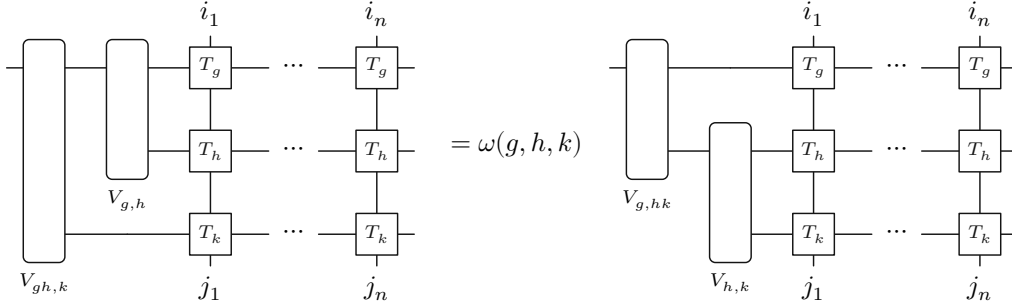
The third step with $\mathbf{q}$ empty is the claim. The residual of $(V \otimes \mathbb{1}, W \otimes \mathbb{1})$ on $T_a T_d$ has words $\sum_{\mathbf{m}} (T_a - WT_c V)^{(\mathbf{i}, \mathbf{m})} \otimes T_d^{(\mathbf{m}, \mathbf{j})}$, which vanish at the same length, and an invertible intertwiner of letters conjugates the residual letters. $\square$

Remark 28.3.2 (Sharpness of the hypotheses and of the bound). In Theorem 28.3.1 the factors must have nilpotent residuals: a single slot of a reduction onto several blocks has a residual that is not nilpotent, and composing it with the identity reduction of $T_c T_d$ for a trivial T_d returns it unchanged. The bound $N_1 + N_2$ fails: for one letter acting as the shift $f_0 \rightarrow f_1 \rightarrow e_1 \rightarrow e_2 \rightarrow f_3 \rightarrow f_4$ on $\mathbb{C}^6$, with (V_1, W_1) the compression onto $\text{span}\{e_1, e_2\}$ and (V_2, W_2) the compression of the shift $e_1 \rightarrow e_2$ onto $\text{span}\{e_2\}$ with $A = 0$, one has $N_1 = 3$, $N_2 = 2$, and $R^5 \neq 0$. Counting the lengths of the nonvanishing terms in the expansion of the proof more carefully gives the bound $\max(N_1, 2N_1 + N_2 - 2)$, which this example attains.

Definition 28.3.3 (Associator). The *associator* of the chosen fusion tensors is the function $\omega: G^3 \rightarrow \mathbb{C}^\times$ defined by

$$V^{(gh)k}(T_g T_h T_k)^{\mathbf{w}} = \omega(g, h, k) V^{g(hk)}(T_g T_h T_k)^{\mathbf{w}} \quad (28.7)$$

for all sufficiently long words $\mathbf{w}$; equivalently, $(T_g T_h T_k)^{\mathbf{w}} W^{g(hk)} = \omega(g, h, k) (T_g T_h T_k)^{\mathbf{w}} W^{(gh)k}$ for the composite embeddings, which is the orientation of the defining display of [163]. It exists and is nonzero by Corollary 28.2.1 applied to the two composite reductions of Theorem 28.3.1, and it is unique because $T_{ghk}^{\mathbf{w}} \neq 0$ for some long word. This is the three-cocycle of [164, Section 4.4], and it is the anomaly three-cochain of Definition 19.5.1.3, characterized by Theorem 19.5.1.5.



Theorem 28.3.4 (The associator is a three-cocycle). *The associator satisfies the three-cocycle relation (19.389),*

$$\omega(gh, k, l) \omega(g, h, kl) = \omega(g, h, k) \omega(g, hk, l) \omega(h, k, l), \quad (28.8)$$

which is the pentagon equation for a group-like algebra [164, Section 4]. This is Theorem 19.5.1.6.

Proof. Write $X = (T_g T_h T_k T_l)^{\mathbf{w}}$ for a long word $\mathbf{w}$ and V^P for the composite compression of the fourfold product onto T_{ghkl} along a bracketing P . By Theorem 28.3.1, applying (28.7) to one

triple inside a bracketing, tensored with identities and followed by the remaining compressions, replaces $(xy)z$ by $x(yz)$ at the cost of $\omega(x, y, z)$. Along the two paths of the pentagon this gives

$$\begin{aligned} V^{((gh)k)l} X &= \omega(gh, k, l) V^{(gh)(kl)} X = \omega(gh, k, l) \omega(g, h, kl) V^{g(h(kl))} X, \\ V^{((gh)k)l} X &= \omega(g, h, k) V^{(g(hk))l} X = \omega(g, h, k) \omega(g, hk, l) V^{g((hk)l)} X \\ &= \omega(g, h, k) \omega(g, hk, l) \omega(h, k, l) V^{g(h(kl))} X. \end{aligned} \quad (28.9)$$

Since $V^{g(h(kl))} X W^{g(h(kl))} = T_{ghkl}^{\mathbf{w}}$ is nonzero for some long word, comparing the two lines of (28.9) gives (28.8). $\square$

Theorem 28.3.5 (Gauge dependence of the associator). *Replacing the fusion tensors by $(\beta_{g,h}^{-1} V_{g,h}, \beta_{g,h} W_{g,h})$, so that $V_{g,h} B^{\mathbf{w}} = \beta_{g,h} V'_{g,h} B^{\mathbf{w}}$ for the new compression $V'_{g,h}$, with $\beta: G^2 \rightarrow \mathbb{C}^\times$ replaces ω by $\beta \triangleright \omega$ of Definition 19.5.10.32. Consequently the class of ω under the relation of Definition 19.5.10.37 does not depend on the choice of fusion tensors, and depends only on the family $(T_g)_{g \in G}$. This is the coboundary freedom of [164, Section 4], and it is Theorem 19.5.1.7.*

Proof. The rescaled pairs are again fusion tensors (Theorem 19.2.4.24). Substituting them into (28.6) multiplies $V^{(gh)k}$ by $(\beta_{g,h} \beta_{gh,k})^{-1}$ and $V^{g(hk)}$ by $(\beta_{g,hk} \beta_{h,k})^{-1}$, so (28.7) holds for the new tensors with ω multiplied by $\beta_{g,hk} \beta_{h,k} / (\beta_{g,h} \beta_{gh,k})$, which is (19.390). By Corollary 28.2.1 every other choice of fusion tensors is boundary-dressed equal to such a rescaling. $\square$

Remark 28.3.6 (A computed anomaly class). For a cyclic group generated by g of order n , the product $\prod_{k < n} \omega(g, g^k, g)$ is unchanged by the gauge action of Theorem 28.3.5, so it depends only on the family $(T_g)_{g \in G}$. For the phase-decorated shift representation of $\mathbb{Z}_3$ it is a primitive cube root of unity (Theorem 27.10.2.4), so its associator does not have trivial gauge class.

Remark 28.3.7 (F-symbols with multiplicity under the zipper condition). For a general fusion algebra the analogue of the associator is a matrix F_d^{abc} relating the two families of composite fusion tensors of $T_a T_b T_c$ onto copies of T_d . Under the zipper condition of [7], which asks that the fusion tensors intertwine the stacked tensor letter by letter, F_d^{abc} is an invertible matrix between multiplicity spaces satisfying the F -move (21.104) of Theorem 21.3.91, and its inverse satisfies the pentagon equation (21.108) of Theorem 21.3.94; both are stated for the complete zipper fusion families of Definition 21.3.85. For a group-like algebra, the scalar playing the role of F_{ghk}^{ghk} in the orientation of (21.104) is $\omega(g, h, k)^{-1}$, and the pentagon equation is (28.8).

Without the zipper condition, the fusion tensors of Theorem 28.1.2 carry a nilpotent remainder and, with several targets, are not determined up to the multiplicity gauge, even after dressing with long words (Remark 28.2.4). No F -move with multiplicity is stated in that setting; for group-like algebras, single-target uniqueness yields (28.7).

28.4 Cohomological obstruction to symmetric states

Definition 28.4.1 (L-symbols of an action). Not ready In the setting of Section 28.3, let $(A_x)_{x \in K}$ be a symmetric family as in Theorem 26.4.11, so that $O_g^{(N)} |V^{(N)}(A_x)\rangle = |V^{(N)}(A_{g \cdot x})\rangle$, with action tensors $(V_{g,x}, W_{g,x})$ reducing $T_g \cdot A_x$ onto $A_{g \cdot x}$. The tensor $T_g T_h \cdot A_x$ has two composite reductions onto $A_{gh \cdot x}$, and the L -symbol $L_{g,h}^x \in \mathbb{C}^\times$ is defined by

$$V_{g,h,x}(1 \otimes V_{h,x})(T_g T_h \cdot A_x)^w = L_{g,h}^x V_{gh,x}(V_{g,h} \otimes 1)(T_g T_h \cdot A_x)^w \quad (28.10)$$

for all sufficiently long words w . This is the L -symbol of [164, Section 4.4].

Then the elements $(B_c^{ik})_c$ span $\bigoplus_c M_{D_c}(\mathbb{C})$, and

$$\Delta((B_c^{ik})_c)_{ab} = \sum_j B_a^{ij} \otimes B_b^{jk}. \quad (28.14)$$

The map Δ is the only linear map with (28.14), so it depends only on the blocks. It is multiplicative, $\Delta(XY) = \Delta(X)\Delta(Y)$, and $\Delta(\mathbb{1})_{ab} = \sum_{c,\mu} X_{ab,\mu}^c X_{ab,\mu}^{c+}$ is the support projector of the fusion tensors. In the notation of [188, Section 4], $X_{ab,\mu}^c$ and $X_{ab,\mu}^{c+}$ are the source's $V_{ab}^{c\mu}$ and $W_{ab}^{c\mu}$, (28.13) is the decomposition of the coproduct of the dual algebra, (28.14) is the coproduct of the matrix coefficients of the physical representation, and multiplicativity is the pre-bialgebra axiom.

Proof. ✓ Proof The simultaneous left inverse of the block letters writes every matrix unit of every block as a combination of the elements $(B_c^{ik})_c$, so they span. Equation (28.14) is the reconstruction of the product letters through the fusion support, and uniqueness follows from spanning. Multiplicativity follows from $X_{ab,\nu}^{d+} X_{ab,\mu}^c = \delta_{cd} \delta_{\mu\nu} \mathbb{1}$, which gives $\Delta(X)\Delta(Y) = \sum_{c,\mu} X_{ab,\mu}^c X_c Y_c X_{ab,\mu}^{c+}$. □

Theorem 28.5.3 (Coassociativity of the fusion coproduct). Lean Lean Lean ✓ Stmt The co-product (28.13) is coassociative, $(\Delta \otimes \mathbb{1}) \circ \Delta = (\mathbb{1} \otimes \Delta) \circ \Delta$, where on the triple bond space

$$\begin{aligned} (\Delta \otimes \mathbb{1})(Y)_{abc} &= \sum_{e,\mu} (X_{ab,\mu}^e \otimes \mathbb{1}) Y_{ec} (X_{ab,\mu}^{e+} \otimes \mathbb{1}), \\ (\mathbb{1} \otimes \Delta)(Y)_{abc} &= \sum_{f,\lambda} (\mathbb{1} \otimes X_{bc,\lambda}^f) Y_{af} (\mathbb{1} \otimes X_{bc,\lambda}^{f+}). \end{aligned}$$

This is the coassociativity axiom of a coalgebra [188, Section 3]. The source assumes it and derives the associativity of fusion trees and the F -symbols; here it is proved from the fusion tensors.

Proof. ✓ Proof Both sides are linear in the argument, and by Theorem 28.5.2 the elements $(B_c^{il})_c$ span the space of tuples $(Z_c)_c$ on which Δ acts. By (28.14) applied twice, both sides send $(B_c^{il})_c$ to $\sum_{j,k} B_a^{ij} \otimes B_b^{jk} \otimes B_c^{kl}$. □

Remark 28.5.4 (Kramers–Wannier duality is an intertwiner). The Kramers–Wannier operator of Section 27.5 does not belong to a fusion algebra on one lattice: by (27.63) its square is $(\mathbb{1} + \eta)T$ up to normalization, which contains the translation T and is not a length-independent combination of the operators of a finite family. It is a matrix product operator intertwiner between two symmetric Hamiltonians, a bimodule between the symmetry algebras of the primal and dual chains, rather than a symmetry of one of them [180, 181]. Fusion and action tensors for such intertwiners come from the same asymmetric compression theorem, applied to products of operators between different chains; the Kramers–Wannier case is worked out in Section 27.5.3.

Remark 28.5.5 (Density operators). Renormalization fixed points of matrix product density operators are described by matrix product operator algebras of C^* -weak Hopf algebras through the pulling-through identity of the canonical regular element [198]; the density-operator setting is that of Chapter 16. The fusion tensors of Section 28.1 apply there without change, since they use only equality of periodic traces.

Chapter 29

Sequential Generation of Matrix Product States

A chain of N sites with local dimension d starts in $|0\rangle^{\otimes N}$, and an ancilla $\mathbb{C}^D$ interacts sequentially with the sites, producing open-boundary matrix product states of bond dimension at most D [193, Section 5.1].

Sites are labelled $k = 0, \dots, N-1$ in ket order, with configuration $\sigma = (\sigma_0, \dots, \sigma_{N-1})$ and A_k the operation acting at site k . For square matrices $Q_0, \dots, Q_{n-1}$ indexed by the physical index, write $Q^\sigma = Q_0^{\sigma_0} \cdots Q_{n-1}^{\sigma_{n-1}}$.

29.1 Sequential schemes

Definition 29.1.1 (Sequentially generated state). [Lean](#) [✓ Stmt](#) Let $|\varphi_I\rangle, |\varphi_F\rangle \in \mathbb{C}^D$ and let $A_k^i \in M_D(\mathbb{C})$ for $k = 0, \dots, N-1$ and $i = 0, \dots, d-1$. Following [193, Section 5.1], the state produced by the operations $\sum_{i,\alpha,\beta} (A_k^i)_{\alpha\beta} |\alpha, i\rangle\langle\beta, 0|$ followed by projecting the ancilla onto $|\varphi_F\rangle$ is

$$\psi(\sigma) = \langle\varphi_F| A_0^{\sigma_0} A_1^{\sigma_1} \cdots A_{N-1}^{\sigma_{N-1}} |\varphi_I\rangle. \quad (29.1)$$

$$\psi(\sigma) = \begin{array}{c} \sigma_0 \quad \sigma_1 \quad \quad \quad \sigma_{N-1} \\ \begin{array}{c} \text{---} \bullet \text{---} \boxed{A_0} \text{---} \boxed{A_1} \text{---} \cdots \text{---} \boxed{A_{N-1}} \text{---} \bullet \text{---} \\ \text{---} \end{array} \\ \langle\varphi_F| \quad \quad \quad A_{N-1} \quad \quad \quad |\varphi_I\rangle \end{array}$$

Definition 29.1.2 (Probabilistic sequential scheme). [Lean](#) [✓ Stmt](#) A vector $\psi \in (\mathbb{C}^d)^{\otimes N}$ is *probabilistically generated with a D -dimensional ancilla* if it satisfies (29.1) for some matrices $A_k^i \in M_D(\mathbb{C})$ and vectors $|\varphi_I\rangle, |\varphi_F\rangle \in \mathbb{C}^D$. The vector is the unnormalized post-measurement vector, as in the source’s “up to normalization”.

Definition 29.1.3 (Operations induced by unitaries). [Lean](#) [Lean](#) [✓ Stmt](#) Let $d \geq 1$. A unitary U on $\mathbb{C}^D \otimes \mathbb{C}^d$ acting on the site state $|0\rangle$ induces the matrices $(A^i)_{\alpha\beta} = \langle\alpha, i|U|\beta, 0\rangle$. For unitaries $U_0, \dots, U_{N-1}$, with U_k applied at step $k+1$ and acting on site $N-1-k$, and an initial ancilla state $|\varphi_I\rangle$, the ancilla component of the joint state at configuration σ is $A_0^{\sigma_0} \cdots A_{N-1}^{\sigma_{N-1}} |\varphi_I\rangle \in \mathbb{C}^D$, where A_k is induced by U_{N-1-k} [193, Section 5.1].

$$\alpha \text{---} \boxed{A} \text{---} \beta \quad = \quad \begin{array}{c} \alpha \quad i \\ \boxed{U} \\ \beta \quad \bullet \end{array} \quad |0\rangle$$

Definition 29.1.4 (Deterministic sequential scheme). [Lean](#) [✓ Stmt](#) Let $d \geq 1$. A vector ψ is *deterministically generated with a D -dimensional ancilla* if there are unitaries $U_0, \dots, U_{N-1}$ on $\mathbb{C}^D \otimes \mathbb{C}^d$ and unit vectors $|\varphi_I\rangle, |\varphi_F\rangle \in \mathbb{C}^D$ such that the ancilla component of the joint state at every configuration σ equals $\psi(\sigma)|\varphi_F\rangle$; that is, the final joint state is $|\varphi_F\rangle \otimes |\psi\rangle$ and the ancilla decouples in the last step without measurement.

$$\psi(\sigma)|\varphi_F\rangle = \begin{array}{c} \sigma_0 \\ \boxed{A_0} \\ \sigma_1 \\ \boxed{A_1} \\ \dots \\ \sigma_{N-1} \\ \boxed{A_{N-1}} \\ \bullet \end{array} \quad |\varphi_I\rangle$$

Definition 29.1.5 (Open-boundary representation with bounded bond dimension). [Lean](#) [✓ Stmt](#) A vector ψ has an *open-boundary representation with bond dimension at most D* if $\psi = c_A$ for some varying-bond open chain A with common bound D , with c_A as in (23.1).

29.2 Isometric decompositions

Definition 29.2.1 (Isometry on a bond block). [Lean](#) [✓ Stmt](#) A family $Q^i \in M_D(\mathbb{C})$, $i = 0, \dots, d-1$, is *isometric on its first b columns* if for all $\beta, \beta' < b$

$$\sum_{i=0}^{d-1} \sum_{\alpha=0}^{D-1} \overline{Q_{\alpha\beta}^i} Q_{\alpha\beta'}^i = \delta_{\beta\beta'}.$$

For $b = D$ this is $\sum_i (Q^i)^\dagger Q^i = \mathbb{1}_D$.

Definition 29.2.2 (Supports below a level). [Lean](#) [Lean](#) [✓ Stmt](#) A vector $v \in \mathbb{C}^D$ is *supported below b* if $v_\beta = 0$ for $\beta \geq b$. A matrix $M \in M_D(\mathbb{C})$ is *row-supported below a* if $M_{\alpha\beta} = 0$ for $\alpha \geq a$.

Theorem 29.2.3 (Propagation of supports). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If M is row-supported below a , then so is MN for every N , and Mv is supported below a for every v . Consequently, if Q_p^i is row-supported below b_p for $p = 0, \dots, n-1$ and v is supported below b_n , then $Q^\sigma v$ is supported below b_0 .

Proof. [✓ Proof](#) The rows $\alpha \geq a$ of MN and of Mv are sums of multiples of vanishing entries. The chain statement follows by induction on n , applying the vector statement to the first factor. $\square$

Definition 29.2.4 (Boundary vector and boundary matrices). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $|0\rangle \in \mathbb{C}^D$ be the first standard basis vector. For $v, r \in \mathbb{C}^D$ let $|0\rangle v^\top$ be the matrix with first row v and all other rows zero, and $r\langle 0|$ the matrix with first column r and all other columns zero.

Theorem 29.2.5 (Coordinates of the boundary vector and matrices). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $D > 0$. The matrix $|0\rangle v^\top$ is row-supported below one; $|0\rangle$ is real and of norm one; $\langle 0|w = w_0$ and $(M|0)_\alpha = M_{\alpha 0}$; the first coordinate of $|0\rangle v^\top Mw$ is $v^\top Mw$; the first column of $Mr\langle 0|$ is Mr and its other columns vanish; a vector supported below one equals $w_0|0\rangle$; and $\|x|0\rangle\|^2 = |x|^2$.

Proof. [Stmnt](#) Direct computation of coordinates. □

Theorem 29.2.6 (Peeling off the last site). [Lean](#) [Stmnt](#) For square matrices $Q_0, \dots, Q_n$ and a configuration σ of $n+1$ sites, $Q^\sigma = (Q_0^{\sigma_0} \dots Q_n^{\sigma_{n-1}}) Q_n^{\sigma_n}$.

Proof. [Stmnt](#) Associativity of the ordered product. □

Theorem 29.2.7 (One decomposition step). [Lean](#) [Stmnt](#) Let $Y^i \in M_D(\mathbb{C})$ be row-supported below a for every i . Then there are $b \leq D$, matrices Q^i and a matrix R with $Y^i = Q^i R$ for every i , such that each Q^i is row-supported below a and vanishes on the columns $\beta \geq b$, the family Q is isometric on its first b columns, and R is row-supported below b . Moreover $b \leq c$ whenever every Y^i vanishes on the columns $\gamma \geq c$.

Proof. [Stmnt](#) Stack the matrices Y^i into one matrix with rows indexed by (α, i) , choose an orthonormal basis $q_0, \dots, q_{b-1}$ of its column space, and let Q^i have columns q_β for $\beta < b$ and zero columns otherwise. The coefficients of the columns of the stacked matrix in this basis form R . The column space is spanned by the at most c nonzero columns, so $b \leq c$. □

Theorem 29.2.8 (Successive decompositions with a matrix remainder). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $a \leq D$, let $R_{-1} \in M_D(\mathbb{C})$ be row-supported below a , and let $P_0, \dots, P_{n-1}$ be families of $D \times D$ matrices. Then there are $b_0 = a$ and $b_1, \dots, b_n \leq D$, families $Q_0, \dots, Q_{n-1}$ such that Q_p^i is row-supported below b_p , vanishes on the columns $\beta \geq b_{p+1}$, and Q_p is isometric on its first b_{p+1} columns, and a matrix R row-supported below b_n , such that for every configuration σ

$$R_{-1} P^\sigma = Q^\sigma R. \quad (29.2)$$

Moreover $b_{p+1} \leq c$ whenever every P_p^i vanishes on the columns $\gamma \geq c$. If Q is isometric on its first b columns and v, w are supported below b , then $\sum_i \langle Q^i v, Q^i w \rangle = \langle v, w \rangle$. Consequently, if Q_p^i is row-supported below b_p , Q_p is isometric on its first b_{p+1} columns, and v, w are supported below b_n , then

$$\sum_\sigma \langle Q^\sigma v, Q^\sigma w \rangle = \langle v, w \rangle. \quad (29.3)$$

Proof. [Stmnt](#) By induction on n . For $n = 0$ take $R = R_{-1}$. Otherwise apply Theorem 29.2.7 to $Y^i = R_{-1} P_0^i$, which gives $Y^i = Q_0^i R_0$ with R_0 row-supported below b_1 , and apply the induction hypothesis to R_0 and $P_1, \dots, P_{n-1}$:

$$R_{-1} P_0^{\sigma_0} P_1^{\sigma_1} \dots P_{n-1}^{\sigma_{n-1}} = Q_0^{\sigma_0} R_0 P_1^{\sigma_1} \dots P_{n-1}^{\sigma_{n-1}} = Q_0^{\sigma_0} Q_1^{\sigma_1} \dots Q_{n-1}^{\sigma_{n-1}} R.$$

For one site, expand

$$\sum_i \langle Q^i v, Q^i w \rangle = \sum_{\beta, \beta' < b} \overline{v_\beta} w_{\beta'} \sum_{i, \alpha} \overline{Q_{\alpha\beta}^i} Q_{\alpha\beta'}^i = \sum_{\beta < b} \overline{v_\beta} w_\beta$$

by the isometry condition. For the chain, write $Q^\sigma = Q_0^{\sigma_0} Q'^{\sigma'}$, where Q' is the chain $Q_1, \dots, Q_{n-1}$ and $\sigma' = (\sigma_1, \dots, \sigma_{n-1})$. The vectors $Q'^{\sigma'} v$ and $Q'^{\sigma'} w$ are supported below b_1 by Theorem 29.2.3, so the one-site identity for Q_0 and the induction hypothesis for Q' give

$$\sum_{\sigma_0, \sigma'} \langle Q_0^{\sigma_0} Q'^{\sigma'} v, Q_0^{\sigma_0} Q'^{\sigma'} w \rangle = \sum_{\sigma'} \langle Q'^{\sigma'} v, Q'^{\sigma'} w \rangle = \langle v, w \rangle.$$

□

Theorem 29.2.9 (Successive decompositions). [Lean](#) [✓ Stmt](#) Let $a \leq D$, let $R_{-1} \in M_D(\mathbb{C})$ be row-supported below a , let $P_0, \dots, P_{n-1}$ be families of $D \times D$ matrices, and let $r \in \mathbb{C}^D$. Then there are $b_0 = a$ and $b_1, \dots, b_n \leq D$, families $Q_0, \dots, Q_{n-1}$ such that Q_p^i is row-supported below b_p , vanishes on the columns $\beta \geq b_{p+1}$, and Q_p is isometric on its first b_{p+1} columns, and a vector r' supported below b_n , such that for every configuration σ

$$R_{-1} P^\sigma r = Q^\sigma r'.$$

Moreover $b_{p+1} \leq c$ whenever every P_p^i vanishes on the columns $\gamma \geq c$.

Proof. [✓ Proof](#) Theorem 29.2.8 gives the bonds b_p , the families Q_p and a matrix R row-supported below b_n with $R_{-1} P^\sigma = Q^\sigma R$ for every σ . Setting $r' = Rr$, which is supported below b_n by Theorem 29.2.3, yields $R_{-1} P^\sigma r = Q^\sigma Rr = Q^\sigma r'$. □

Theorem 29.2.10 (Isometric chains preserve the norm). [Lean](#) [Lean](#) [✓ Stmt](#) If Q is isometric on its first b columns and v is supported below b , then $\sum_i \|Q^i v\|^2 = \|v\|^2$. Consequently, if Q_p^i is row-supported below b_p , Q_p is isometric on its first b_{p+1} columns, and v is supported below b_n , then $\sum_\sigma \|Q^\sigma v\|^2 = \|v\|^2$.

Proof. [✓ Proof](#) Take $w = v$ in the inner-product identities of Theorem 29.2.8: $\sum_i \langle Q^i v, Q^i v \rangle = \langle v, v \rangle$ for one site, and (29.3) with $w = v$ for the chain. □

Theorem 29.2.11 (Changing sites outside the used block). [Lean](#) [Lean](#) [✓ Stmt](#) If $M'_{\alpha\beta} = M_{\alpha\beta}$ for all $\beta < b$ and w is supported below b , then $M'w = Mw$. Consequently, if Q_p^i is row-supported below b_p , the families Q_p^i agree with Q_p on the columns $\beta < b_{p+1}$, and v is supported below b_n , then $Q_p^\sigma v = Q^\sigma v$ for every σ .

Proof. [✓ Proof](#) The first claim holds entrywise. The second follows by induction on the chain from the right, since each partial product applied to v is supported below the next bond level. □

Theorem 29.2.12 (Unitary extension of a block isometry). [Lean](#) [✓ Stmt](#) Let $d \geq 1$ and let Q be isometric on its first b columns. Then there is a unitary U on $\mathbb{C}^D \otimes \mathbb{C}^d$ with $\langle \alpha, i | U | \beta, 0 \rangle = Q_{\alpha\beta}^i$ for all i, α and $\beta < b$.

Proof. [✓ Proof](#) The vectors $\sum_{\alpha, i} Q_{\alpha\beta}^i |\alpha, i\rangle$, $\beta < b$, are orthonormal. Extend them to an orthonormal basis of $\mathbb{C}^D \otimes \mathbb{C}^d$ and let U send the standard basis, with the vectors $|\beta, 0\rangle$, $\beta < b$, first, to this basis. □

Theorem 29.2.13 (Positive bond bound). [Lean](#) [✓ Stmt](#) The common bound D of a varying-bond open chain is positive.

Proof. [✓ Proof](#) $1 = D_0 \leq D$. □

Theorem 29.2.14 (Open-chain coefficient as a corner entry). [Lean](#) [✓ Stmt](#) For a varying-bond open chain with square padding $\tilde{A}$, and every configuration σ of any length $N \geq 0$, $c_A(\sigma) = (\tilde{A}^\sigma)_{00}$.

Proof. [✓ Proof](#) For $N = 0$ both sides are one. Otherwise $c_A(\sigma)$ is the trace of $\tilde{A}^\sigma$, and all diagonal entries except the first vanish because the first padded site matrix is row-supported below $D_0 = 1$. $\square$

Definition 29.2.15 (Cutting a square chain down to its bond blocks). [Lean](#) [✓ Stmt](#) Given $b_0, \dots, b_N \leq D$ with $b_0 = b_N = 1$ and square matrices Q_k^i , the varying-bond open chain with dimensions b_k has as site matrices the upper-left $b_k \times b_{k+1}$ corners of Q_k^i .

Theorem 29.2.16 (Cutting down and padding). [Lean](#) [Lean](#) [✓ Stmt](#) If every Q_k^i is row-supported below b_k and vanishes on the columns $\beta \geq b_{k+1}$, then padding the cut-down chain returns Q . If Q_k^i is row-supported below b_k and Q_k is isometric on its first b_{k+1} columns, then the cut-down site matrices A_k^i satisfy $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}_{b_{k+1}}$.

Proof. [✓ Proof](#) Entrywise comparison; for the second claim the sum over rows α may be restricted to $\alpha < b_k$. $\square$

Theorem 29.2.17 (Assembling an open chain). [Lean](#) [✓ Stmt](#) Let $b_0, \dots, b_{n+1} \leq D$ with $b_0 = 1$, and let $Q_0, \dots, Q_n$ be families with Q_p^i row-supported below b_p , vanishing on the columns $\beta \geq b_{p+1}$, and isometric on the first b_{p+1} columns. For every $r' \in \mathbb{C}^D$ there is a varying-bond open chain A with common bound D such that $c_A(\sigma) = (Q^\sigma r')_0$ for every σ , $D_k = b_k$ for $k \leq n$, and $\sum_i (A_p^i)^\dagger A_p^i = \mathbb{1}$ for $p < n$. If moreover r' is supported below b_{n+1} and $\|r'\| = 1$, then also $\sum_i (A_n^i)^\dagger A_n^i = \mathbb{1}$.

Proof. [✓ Proof](#) Replace the last family by $Q_n^i r' \langle 0|$, set the last bond dimension to one, and cut down. By Theorems 29.2.14 and 29.2.6, the coefficient at σ is the first coordinate of $Q^\sigma r'$. If r' is a unit vector supported below b_{n+1} , the isometry of Q_n on its first b_{n+1} columns gives $\sum_i \|Q_n^i r'\|^2 = \|r'\|^2 = 1$. $\square$

Theorem 29.2.18 (Successive decompositions of an open chain). [Lean](#) [✓ Stmt](#) Let A be a varying-bond open chain of length $n + 1$ with dimensions D_k and common bound D , and let $c \in \mathbb{C}$. There are $b_0 = 1$ and $b_k \leq D_k$, families $Q_0, \dots, Q_n$ as in Theorem 29.2.9, and a vector r' supported below b_{n+1} , such that $Q^\sigma r' = c c_A(\sigma) |0\rangle$ for every σ and $\|r'\|^2 = \sum_\sigma |c c_A(\sigma)|^2$.

Proof. [✓ Proof](#) Apply Theorem 29.2.9 with $a = 1$, $R_{-1} = c|0\rangle\langle 0|$, the square padding of A , and $r = |0\rangle$. The padded site matrix $\tilde{A}_p$ vanishes on the columns $\gamma \geq D_{p+1}$, so $b_{p+1} \leq D_{p+1}$. The vector $Q^\sigma r'$ is supported below $b_0 = 1$, so it is its first coordinate times $|0\rangle$, and that coordinate is $c c_A(\sigma)$ by Theorem 29.2.14. The norm identity is Theorem 29.2.10. $\square$

Theorem 29.2.19 (Left-canonical open-boundary representation). [Lean](#) [✓ Stmt](#) Every varying-bond open chain A of length $n + 1$ has a varying-bond open chain A' with the same common bound, $c_{A'} = c_A$, and $D'_k \leq D_k$ for every k , such that $\sum_i (A_p'^i)^\dagger A_p'^i = \mathbb{1}$ for every site $p < n$.

Proof. [✓ Proof](#) Apply Theorem 29.2.18 with $c = 1$ and assemble the result by Theorem 29.2.17. The last bond has dimension one on both sides, and the others are bounded by Theorem 29.2.18. $\square$

Theorem 29.2.20 (Left-canonical representation of a normalized state). [Lean](#) [✓ Stmt](#) If moreover $\sum_\sigma |c_A(\sigma)|^2 = 1$, the chain A' of Theorem 29.2.19 can be chosen with $\sum_i (A_p'^i)^\dagger A_p'^i = \mathbb{1}$ at every site $p \leq n$.

Proof. [✓ Proof](#) In the construction of Theorem 29.2.19, the final vector r' has norm one by Theorem 29.2.18, so Theorem 29.2.17 makes the last site isometric as well. $\square$

29.3 Resources of the successive decompositions

After the successive decompositions, [201] state that “simple rank considerations” bound the size of the isometries: for qubits, the isometry $V'_{[n-k]}$, which is produced after k earlier decompositions counted from the end at which they start, has dimension $2 \min[D, 2^k] \times \min[D, 2^{k+1}]$. According to [193, Section 5.1], the proof of the characterization by open-boundary representations also provides a recipe for the generation of any given state with minimal resources. The entries below give only the upper bounds on the bond dimensions; the equality of each bond dimension with the Schmidt rank across the corresponding cut is not formalized [32]. In the present labelling the decompositions start at site 0, and the bound concerns the bond k between the sites $k-1$ and k .

Theorem 29.3.1 (Rank bound for one isometric site). Lean Lean ✓ Stmt *If $A^0, \dots, A^{d-1}$ are $a \times b$ matrices with $\sum_i (A^i)^\dagger A^i = \mathbb{1}_b$, then $b \leq ad$. In particular, a site p of a varying-bond open chain with $\sum_i (A_p^i)^\dagger A_p^i = \mathbb{1}$ satisfies $D_{p+1} \leq d D_p$.*

Proof. ✓ Proof The $(ad) \times b$ matrix S obtained by stacking the A^i satisfies $S^\dagger S = \mathbb{1}_b$, so $b = \text{rank}(S^\dagger S) = \text{rank } S \leq ad$. □

Theorem 29.3.2 (Bond growth from the start of the decompositions). Lean ✓ Stmt *Let $d \geq 1$ and let A be a varying-bond open chain of length $n+1$ with $\sum_i (A_p^i)^\dagger A_p^i = \mathbb{1}$ for every site $p < n$. Then $D_k \leq d^k$ for every $k \leq n+1$.*

Proof. ✓ Proof Induction on k : $D_0 = 1$, $D_{p+1} \leq d D_p$ for $p < n$ by Theorem 29.3.1, and $D_{n+1} = 1 \leq d^{n+1}$. □

Theorem 29.3.3 (Resources of the successive decompositions). Lean ✓ Stmt *Let $d \geq 1$. The chain A' of Theorem 29.2.19 can be chosen with $D'_k \leq \min(D_k, d^k)$ for every k ; in particular $D'_k \leq \min(D, d^k)$.*

Proof. ✓ Proof Theorem 29.3.2 applies to the chain of Theorem 29.2.19. □

Definition 29.3.4 (Reversed open chain). Lean ✓ Stmt The reverse of a varying-bond open chain A of length N has dimensions D_{N-k} and site matrices $(A_{N-1-p}^i)^T$.

Theorem 29.3.5 (Coefficients of the reversed chain). Lean Lean ✓ Stmt *The reverse of A has coefficient $\sigma \mapsto c_A(\sigma_{N-1}, \dots, \sigma_0)$.*

Proof. ✓ Proof Relabel the virtual paths $k \mapsto N-k$ and the factors of the product $p \mapsto N-1-p$. □

Theorem 29.3.6 (Successive decompositions from both ends). Lean ✓ Stmt *Let $d \geq 1$ and let A be a varying-bond open chain of length $N = n+1$. There is a varying-bond open chain A' with the same common bound, $c_{A'} = c_A$, $\sum_i (A_p^i)^\dagger A_p^i = \mathbb{1}$ for every site $p < n$, and*

$$D'_k \leq \min(D_k, d^k, d^{N-k})$$

for every $k \leq N$.

Proof. ✓ Proof Apply Theorem 29.3.3 to the reverse of A and reverse the result: by Theorem 29.3.5 this represents c_A with dimensions at most $\min(D_k, d^{N-k})$. Apply Theorem 29.3.3 once more. The bound d^{N-k} is not stated in the source, which bounds only the side at which the decompositions start. □

29.4 Sequential generation with an ancilla

Theorem 29.4.1 (Generated states are open-boundary states). [Lean](#) [✓ Stmt](#) *If a nonzero vector ψ is probabilistically generated with a D -dimensional ancilla, then $c\psi$ has an open-boundary representation with bond dimension at most D for some $c \neq 0$.*

Proof. [✓ Proof](#) Since $\psi \neq 0$, $D > 0$. For $N = 0$ take $c = \psi()^{-1}$ and the empty chain. For $N \geq 1$ apply Theorem 29.2.9 with $R_{-1} = |0\rangle\langle\varphi_F|$, the matrices A_k , and $r = |\varphi_I\rangle$, assemble the result by Theorem 29.2.17, and take $c = 1$. $\square$

Theorem 29.4.2 (Open-boundary states are generated). [Lean](#) [✓ Stmt](#) *If $c \neq 0$ and $c\psi$ has an open-boundary representation with bond dimension at most D , then ψ is probabilistically generated with a D -dimensional ancilla.*

Proof. [✓ Proof](#) Take the square padding $\tilde{A}$ of the representation, $|\varphi_F\rangle = |0\rangle$ and $|\varphi_I\rangle = c^{-1}|0\rangle$; then $\langle 0|\tilde{A}^\sigma c^{-1}|0\rangle = c^{-1}c\psi(\sigma)$ by Theorem 29.2.14. $\square$

Theorem 29.4.3 (Probabilistic sequential generation). [Lean](#) [✓ Stmt](#) *A nonzero vector ψ is probabilistically generated with a D -dimensional ancilla if and only if $c\psi$ has an open-boundary representation with bond dimension at most D for some $c \neq 0$ [193, Section 5, Theorem “Sequential generation with ancilla”].*

Proof. [✓ Proof](#) Theorems 29.4.1 and 29.4.2. $\square$

Theorem 29.4.4 (Unitaries induce isometric operations). [Lean](#) [✓ Stmt](#) *The matrices induced by a unitary on $\mathbb{C}^D \otimes \mathbb{C}^d$ satisfy $\sum_i (A^i)^\dagger A^i = \mathbb{1}_D$.*

Proof. [✓ Proof](#) This is the $(|\beta, 0\rangle, |\beta', 0\rangle)$ entry of $U^\dagger U = \mathbb{1}$. $\square$

Theorem 29.4.5 (Deterministically generated states are normalized). [Lean](#) [✓ Stmt](#) *Every deterministically generated vector ψ satisfies $\sum_\sigma |\psi(\sigma)|^2 = 1$.*

Proof. [✓ Proof](#) By Theorems 29.4.4 and 29.2.10, $\sum_\sigma \|A^\sigma \varphi_I\|^2 = \|\varphi_I\|^2 = 1$, and $\|A^\sigma \varphi_I\|^2 = |\psi(\sigma)|^2 \|\varphi_F\|^2 = |\psi(\sigma)|^2$. $\square$

Theorem 29.4.6 (Deterministic schemes are probabilistic). [Lean](#) [✓ Stmt](#) *Every deterministically generated vector is probabilistically generated with an ancilla of the same dimension.*

Proof. [✓ Proof](#) Use the induced matrices and project onto $|\varphi_F\rangle$: $\langle\varphi_F|A^\sigma|\varphi_I\rangle = \psi(\sigma)\langle\varphi_F|\varphi_F\rangle = \psi(\sigma)$. $\square$

Theorem 29.4.7 (Normalized open-boundary states are deterministically generated). [Lean](#) [✓ Stmt](#) *Let $d \geq 1$. If $\sum_\sigma |\psi(\sigma)|^2 = 1$, $c \neq 0$, and $c\psi$ has an open-boundary representation with bond dimension at most D , then ψ is deterministically generated with a D -dimensional ancilla.*

Proof. ✓ Proof For $N = 0$ take $|\varphi_I\rangle = \psi(|0\rangle)$ and $|\varphi_F\rangle = |0\rangle$. Otherwise apply Theorem 29.2.18 with the scalar c^{-1} : this gives isometric families Q_p and a vector r' with $Q^\sigma r' = \psi(\sigma)|0\rangle$ and $\|r'\|^2 = \sum_\sigma |\psi(\sigma)|^2 = 1$. Extend each Q_p to a unitary U_p by Theorem 29.2.12, and take $|\varphi_I\rangle = r'$ and $|\varphi_F\rangle = |0\rangle$. The induced matrices agree with Q_p on the used columns, so by Theorem 29.2.11 the ancilla component at σ is $Q^\sigma r' = \psi(\sigma)|0\rangle$. The ancilla decouples because the first bond is one-dimensional. $\square$

Theorem 29.4.8 (Deterministic sequential generation). Lean ✓ Stmt Let $d \geq 1$. A vector ψ is deterministically generated with a D -dimensional ancilla if and only if $\sum_\sigma |\psi(\sigma)|^2 = 1$ and $c\psi$ has an open-boundary representation with bond dimension at most D for some $c \neq 0$ [193, Section 5, Theorem “Sequential generation with ancilla”].

Proof. ✓ Proof For the forward direction, ψ is normalized by Theorem 29.4.5, hence nonzero, and probabilistically generated by Theorem 29.4.6; apply Theorem 29.4.1. The converse is Theorem 29.4.7. $\square$

Theorem 29.4.9 (Both schemes agree on normalized states). Lean ✓ Stmt Let $d \geq 1$. A vector ψ with $\sum_\sigma |\psi(\sigma)|^2 = 1$ is deterministically generated with a D -dimensional ancilla if and only if it is probabilistically generated with a D -dimensional ancilla.

Proof. ✓ Proof Theorems 29.4.3 and 29.4.8. $\square$

29.5 The deterministic transition scheme

Definition 29.5.1 (Deterministic transition scheme). Lean Lean Lean ✓ Stmt Let $d = 2$ and let the ancilla be $\mathbb{C}^D \otimes \mathbb{C}^2$, started in a unit vector $|\varphi_I\rangle$. In every step an arbitrary unitary $W^{[k]}$ acts on the ancilla, followed by the fixed interaction T of the ancilla with the current site

$$|\varphi\rangle|1\rangle|0\rangle \mapsto |\varphi\rangle|0\rangle|1\rangle, \quad |\varphi\rangle|0\rangle|0\rangle \mapsto |\varphi\rangle|0\rangle|0\rangle, \quad (29.4)$$

and the ancilla decouples after the last step, that is, the final joint state is $|\varphi_F\rangle \otimes |\psi\rangle$ for a unit vector $|\varphi_F\rangle \in \mathbb{C}^D \otimes \mathbb{C}^2$ [193, Section 5.1]. The step with ancilla unitary W leaving the site in $|i\rangle$ is the matrix $S_i(W)$ on $\mathbb{C}^D \otimes \mathbb{C}^2$ with entries

$$S_i(W)_{x,y} = \delta_{x_2,0} W_{(x_1,i),y}. \quad (29.5)$$

Theorem 29.5.2 (A transition step on a site in $|0\rangle$). Lean ✓ Stmt Let T be an interaction on $\mathbb{C}^D \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$ satisfying (29.4), and let W be a matrix on $\mathbb{C}^D \otimes \mathbb{C}^2$. Then for all x, y and i ,

$$\langle x, i | T(W \otimes \mathbb{1}) | y, 0 \rangle = S_i(W)_{x,y}.$$

Proof. ✓ Proof For $z = (z_1, z_2)$, the relations (29.4) applied to $|\varphi\rangle = |z_1\rangle$ give $T|z_1, z_2, 0\rangle = |z_1, 0, z_2\rangle$, so

$$\langle x, i | T | z, 0 \rangle = \delta_{x_2,0} \delta_{x_1,z_1} \delta_{i,z_2}.$$

Since $(W \otimes \mathbb{1})_{(z,j),(y,0)} = \delta_{j,0} W_{z,y}$,

$$\langle x, i | T(W \otimes \mathbb{1}) | y, 0 \rangle = \sum_z \langle x, i | T | z, 0 \rangle W_{z,y} = \delta_{x_2,0} W_{(x_1,i),y} = S_i(W)_{x,y}.$$

$\square$

Theorem 29.5.3 (Transition schemes generate the open-boundary states). [Lean](#) [Stmnt](#) Let $d = 2$. A vector ψ is generated by a deterministic transition scheme with ancilla $\mathbb{C}^D \otimes \mathbb{C}^2$ if and only if $\sum_{\sigma} |\psi(\sigma)|^2 = 1$ and $c\psi$ has an open-boundary representation with bond dimension at most D for some $c \neq 0$ [193, Section 5, Theorem “Sequential generation with ancilla”].

Proof. [Proof](#) By Theorem 29.4.8 it suffices to show that the transition scheme and the deterministic scheme with ancilla $\mathbb{C}^D$ generate the same vectors. A step of the transition scheme factors as ER_i , where $E|\varphi\rangle = |\varphi\rangle|0\rangle$ and $(R_i)_{\alpha,y} = W_{(\alpha,i),y}$, and R_iE is the operation $\langle\alpha, i|W|\beta, 0\rangle$ of the unitary W on ancilla $\otimes$ site. Given a deterministic scheme with unitaries $U^{[k]}$, take $W^{[k]} = U^{[k]}$ and the initial and final ancilla states $E|\varphi_I\rangle$ and $E|\varphi_F\rangle$; the ancilla components are then E applied to those of the deterministic scheme.

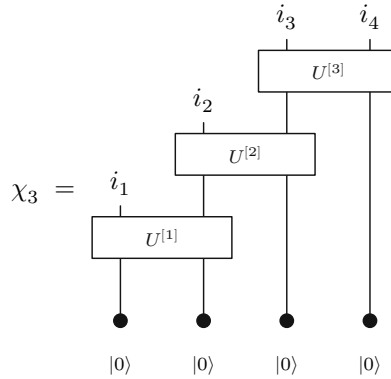
Conversely, after the first step the tag qubit is $|0\rangle$, so the ancilla component at σ is E applied to the product of the matrices R_iE acting on $v_i = R_i|\varphi_I\rangle$, where R_i comes from the first ancilla unitary. Since $\sum_i \|v_i\|^2 = \|\varphi_I\|^2 = 1$, the vectors v_i are the first column of a unitary on $\mathbb{C}^D \otimes \mathbb{C}^2$ by Theorem 29.2.12, and the scheme becomes a deterministic scheme with ancilla $\mathbb{C}^D$ started in $|0\rangle$. The final state is $\psi(\sigma)$ times the tag-zero part of φ_F . The total squared norm is 1 by Theorem 29.2.10, so some $\psi(\sigma)$ is nonzero, the tag-one part of φ_F vanishes, and the tag-zero part is normalized. $\square$

29.6 Sequential generation without an ancilla

Definition 29.6.1 (Sequential scheme without an ancilla). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $d \geq 1$ and $N \geq 2$. Starting from $|0\rangle^{\otimes N}$, an operation $U^{[k]}$ acts on sites k and $k+1$, for $k = 1, \dots, N-1$ in turn. The scheme is deterministic if every $U^{[k]}$ is unitary and probabilistic if the operations are arbitrary [193, Section 5.2]. Here sites carry the source’s labels $1, \dots, N$; source site j is ket position $N-j$ in the labelling of this chapter. After $U^{[k]}$ the scheme is described by the vector χ_k on the first $k+1$ sites, defined by $\chi_1 = U^{[1]}|0, 0\rangle$ and

$$\chi_k(\dots, i, \beta) = \sum_{\alpha} \langle i, \beta | U^{[k]} | \alpha, 0 \rangle \chi_{k-1}(\dots, \alpha),$$

and the generated state is χ_{N-1} .



Remark 29.6.2. Before $U^{[k]}$ acts, sites $k+1, \dots, N$ remain in $|0\rangle$, and site k is unchanged under subsequent operations, so the intermediate chain state after $U^{[k]}$ is $|\chi_k\rangle \otimes |0\rangle^{\otimes(N-k-1)}$.

Theorem 29.6.3 (Sequential generation without an ancilla). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $d \geq 1$ and $N \geq 2$. The states generated by sequential schemes without an ancilla, either deterministically or probabilistically (the latter up to normalization), are exactly the states with an open-boundary representation of bond dimension at most d , normalized in the deterministic case [193, Section 5, Theorem “Sequential generation without ancilla”]. The condition $N \geq 2$ is necessary because the two-site gates require at least two sites [129].

Proof. [✓ Proof](#) With $A_{i,\beta\alpha}^{[k]} = \langle i, \beta | U^{[k]} | \alpha, 0 \rangle$ the recursion for χ_k gives $\psi(\sigma) = (A_{\sigma_1}^{[N-1]} \cdots A_{\sigma_{N-1}}^{[1]} | 0 \rangle)_{\sigma_0}$ in ket positions, an open-boundary representation with bond dimension at most d by Theorems 29.2.9 and 29.2.17. For unitary operations each $A^{[k]}$ satisfies $\sum_i A_i^\dagger A_i = \mathbb{1}$, so the state is normalized by Theorem 29.2.10.

Conversely, write an open-boundary state as $\psi(\sigma) = (R P_1^{\sigma_1} \cdots P_{N-1}^{\sigma_{N-1}} | 0 \rangle)_{\sigma_0}$ with $R_{j\gamma}$ the entries of the first site (Theorem 29.2.14), and run the successive decompositions of Theorem 29.2.9 starting from R : this gives $\psi(\sigma) = (Q_1^{\sigma_1} \cdots Q_{N-1}^{\sigma_{N-1}} r')_{\sigma_0}$ with every Q_p isometric on its bond block, and $\|r'\| = 1$ when ψ is normalized. Take $U^{[1]}$ with $\langle i, \beta | U^{[1]} | 0, 0 \rangle = (Q_{N-1}^i r')_\beta$ and, for $k \geq 2$, $U^{[k]}$ with $\langle i, \beta | U^{[k]} | \alpha, 0 \rangle = (Q_{N-k}^i)_{\beta\alpha}$ on the used columns; in the deterministic case these are unitary extensions by Theorem 29.2.12, and Theorem 29.2.11 identifies the generated state. $\square$

Chapter 30

Preparation of Matrix Product States by Log-Depth Circuits

A translation-invariant matrix product state on N sites generated by a normal tensor is approximated by blocking q sites, replacing the positive part of the polar decomposition of the blocked tensor by its renormalization fixed point, and retaining the isometric part [185]. The fixed point is a product of nearest-neighbor entangled pairs, and the isometric part factorizes into isometries with bounded bond dimension sequentially or along a binary tree. Choosing $q = O(\log(N/\varepsilon))$ yields a quantum circuit of depth $O(\log(N/\varepsilon))$ with error ε , whereas no circuit of depth $o(\log N)$ achieves small error.

Throughout, A is a tensor with physical dimension d and bond dimension D , $\mathcal{E}_A$ is its transfer map (Definition 2.2.1), and

$$|\phi_N(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{Tr}(A^{i_1} \dots A^{i_N}) |i_1 \dots i_N\rangle \quad (30.1)$$

is the translation-invariant vector it generates, with $|\phi_N(A)\rangle = |V^{(N)}(A)\rangle$ in the notation of Definition 2.8.1. For unit vectors ϕ, ψ , the error is $\varepsilon(\phi, \psi) = 1 - |\langle\phi|\psi\rangle|$.

30.1 Polar decomposition of a blocked tensor

The polar decomposition factorizes a rectangular matrix $M \in \mathbb{C}^{m \times k}$ into a positive semidefinite matrix P and a partial isometry V . For a tensor B viewed as a linear map from virtual index pairs to physical configurations, this factorization yields a positive-part tensor P and a partial isometry V on the physical leg, satisfying $\mathcal{E}_B = \mathcal{E}_P$. When $B = A^{[q]}$ is the tensor of q blocked sites of a tensor A , the periodic states satisfy $|V^{(qM)}(A)\rangle = V^{\otimes M} |V^{(M)}(P)\rangle$ for every M [185]. All states are unnormalized.

30.1.1 Polar decomposition of a matrix

Definition 30.1.1.1 (Polar factors of a matrix). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $M \in \mathbb{C}^{m \times k}$. The *positive part* is $P = (M^\dagger M)^{1/2}$, the *support projector* is $\Pi = \chi(M^\dagger M)$ with $\chi(0) = 0$ and $\chi(x) = 1$ for $x \neq 0$, and the *partial isometry* is $V = M g(M^\dagger M)$ with $g(x) = x^{-1/2}$ for $x > 0$ and $g(0) = 0$; the functions χ and g are applied to the positive semidefinite matrix $M^\dagger M$ by

the functional calculus. This is the decomposition $B = VP$ of [185, eq. (8) and Supplemental Material].

Theorem 30.1.1.2 (Positive part). [Lean](#) [Lean](#) [✓ Stmt](#) *The positive part is positive semidefinite and $P^2 = M^\dagger M$.*

Proof. [✓ Proof](#) Properties of the positive square root. □

Theorem 30.1.1.3 (Support projector). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The support projector is Hermitian and idempotent, and $\Pi P = P$ and $M\Pi = M$. The ranges of Π and P coincide, so Π is the orthogonal projector onto the range of P .*

Proof. [✓ Proof](#) The first three identities are the pointwise identities $\chi = \bar{\chi}$, $\chi^2 = \chi$ and $\chi(x)\sqrt{x} = \sqrt{x}$ under the functional calculus. The identity $x\chi(x) = x$ gives $M^\dagger M(\Pi - \mathbb{1}) = 0$, hence $M(\Pi - \mathbb{1}) = 0$. For the ranges, $\Pi = P g(M^\dagger M)$, because $\sqrt{x} g(x) = \chi(x)$ on the nonnegative spectrum, and $P = \Pi P$. □

Theorem 30.1.1.4 (Polar decomposition). [Lean](#) [Lean](#) [✓ Stmt](#) *Every complex matrix satisfies $M = VP$ and $V^\dagger V = \Pi$.*

Proof. [✓ Proof](#) $VP = M g(M^\dagger M)P = M\Pi = M$ by Theorem 30.1.1.3. Since $g(M^\dagger M)$ is Hermitian, $V^\dagger V = g(M^\dagger M) M^\dagger M g(M^\dagger M)$, which is the functional calculus of $x g(x)^2 = \chi(x)$ on the nonnegative spectrum. □

Theorem 30.1.1.5 (Polar decomposition of an injective matrix). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If M is injective, then $\Pi = \mathbb{1}$, $V^\dagger V = \mathbb{1}$, and P is positive definite.*

Proof. [✓ Proof](#) For injective M the matrix $M^\dagger M$ is positive definite, so $\chi = 1$ on its spectrum. Then $V^\dagger V = \Pi = \mathbb{1}$ by Theorem 30.1.1.4, and P is invertible since $P^2 = M^\dagger M$ is. □

Theorem 30.1.1.6 (Existence of the polar decomposition). [Lean](#) [✓ Stmt](#) *Every $M \in \mathbb{C}^{m \times k}$ factors as $M = VP$ with $P \in M_k(\mathbb{C})$ positive semidefinite and $V^\dagger V = \Pi$, where Π is a Hermitian idempotent whose range is the range of P . This is the decomposition in [185, Supplemental Material].*

Proof. [✓ Proof](#) Take the polar factors of Definition 30.1.1.1. □

30.1.2 A tensor as a map on the virtual pair space

Definition 30.1.2.1 (Physical matrix of a tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *The physical matrix of a tensor $B = \{B^i\}_{i=0}^{n-1}$ with $B^i \in M_D(\mathbb{C})$ is the matrix $\widehat{B} \in \mathbb{C}^{n \times D^2}$ with entries $\widehat{B}_{i,(\alpha,\beta)} = B_{\alpha\beta}^i$, the matrix of B as a map from the virtual pair space $\mathbb{C}^{D^2}$ to the physical space $\mathbb{C}^n$. Conversely, every $R \in \mathbb{C}^{n \times D^2}$ is the physical matrix of the tensor $B_{\alpha\beta}^i = R_{i,(\alpha,\beta)}$.*

Lemma 30.1.2.2 (A tensor is its physical matrix). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The two constructions of Definition 30.1.2.1 are mutually inverse; in particular a tensor is determined by its physical matrix.*

Proof. [✓ Proof](#) Both composites are the identity entrywise. □

Theorem 30.1.2.3 (Injective tensors have injective physical matrices). [Lean](#) [✓ Stmt](#) *If B is injective, then $\widehat{B}: \mathbb{C}^{D^2} \rightarrow \mathbb{C}^n$ is injective.*

Proof. ✓ Proof Let $\widehat{B}x = 0$ and consider the linear functional $\varphi(Y) = \sum_{\alpha, \beta} Y_{\alpha\beta} x_{\alpha\beta}$ on $M_D(\mathbb{C})$. Since φ vanishes on every B^i , it vanishes on their span $M_D(\mathbb{C})$; evaluating on matrix units yields $x = 0$. $\square$

Lemma 30.1.2.4 (Physical matrix of a rotated tensor). Lean ✓ Stmt For $W \in \mathbb{C}^{m \times n}$, the physical matrix of B_W is $W\widehat{B}$.

Proof. ✓ Proof Both sides have $(i, (\alpha, \beta))$ entry $\sum_j W_{ij} B_{\alpha\beta}^j$. $\square$

Theorem 30.1.2.5 (Transfer map of a rotated tensor). Lean Lean ✓ Stmt For $W \in \mathbb{C}^{m \times n}$ and every $X \in M_D(\mathbb{C})$,

$$\mathcal{E}_{B_W}(X) = \sum_{l=0}^{n-1} (B_{W^\dagger W})^l X (B^l)^\dagger.$$

Consequently, if $B_{W^\dagger W} = B$, then $\mathcal{E}_{B_W} = \mathcal{E}_B$.

Proof. ✓ Proof Expand $(B_W)^i = \sum_j W_{ij} B^j$ in $\sum_i (B_W)^i X ((B_W)^i)^\dagger$ and collect $\sum_i \overline{W_{il}} W_{ij} = (W^\dagger W)_{lj}$. The case $W^\dagger W = \mathbb{1}$ is the unitary freedom of Kraus representations; the weaker hypothesis $B_{W^\dagger W} = B$ covers partial isometries. $\square$

30.1.3 Polar factors of a tensor

Definition 30.1.3.1 (Polar factors of a tensor). Lean Lean Lean Lean Lean ✓ Stmt Identify $\{0, \dots, D^2 - 1\}$ with the pairs (α, β) of virtual indices, and let V , P and Π be the polar factors of $\widehat{B}$, read as matrices indexed by $\{0, \dots, D^2 - 1\}$ on the virtual side. The *positive-part tensor* of B has physical dimension D^2 and bond dimension D , with $(P^k)_{\alpha\beta} = P_{k, (\alpha, \beta)}$; its matrices are the rows of P . The *isometric factor* is $V \in \mathbb{C}^{n \times D^2}$. This is the tensor P of [185, eqs. (9) and (18)].

Lemma 30.1.3.2 (Physical matrix of the positive-part tensor). Lean ✓ Stmt The physical matrix of the positive-part tensor is P , with its rows indexed by $\{0, \dots, D^2 - 1\}$.

Proof. ✓ Proof By definition. $\square$

Theorem 30.1.3.3 (Polar decomposition of a tensor). Lean Lean Lean Lean Lean Lean ✓ Stmt For every tensor B , the positive-part tensor P and the isometric factor V satisfy $B^i = \sum_k V_{ik} P^k$ for every i , that is, $B = P_V$, with $V^\dagger V = \Pi$, the matrix P positive semidefnite, Π a Hermitian idempotent, and $\sum_l \Pi_{kl} P^l = P^k$ for every k .

$$\alpha \text{---} \boxed{B} \text{---} \beta = \alpha \text{---} \boxed{P} \text{---} \beta$$

$\begin{array}{c} i \\ | \\ \boxed{V} \end{array}$

Proof. ✓ Proof By Lemma 30.1.2.4, the physical matrix of P_V is $VP = \widehat{B}$ by Theorem 30.1.1.4, and a tensor is determined by its physical matrix. The remaining identities are Theorems 30.1.1.2, 30.1.1.3 and 30.1.1.4, reindexed along the identification of $\{0, \dots, D^2 - 1\}$ with the virtual pairs; the last one is $\Pi P = P$ read through Lemma 30.1.2.4. $\square$

Theorem 30.1.3.4 (Transfer map of the positive part). [Lean](#) [✓ Stmt](#) For every tensor B with positive-part tensor P , $\mathcal{E}_P = \mathcal{E}_B$. This is the first equality of [185, eq. (8)], which is stated there for injective B . The Supplemental Material uses the polar decomposition of an arbitrary blocked tensor, with $V^\dagger V = \Pi$; the identities $V^\dagger V = \Pi$ and $\Pi P = P$ give the equality for every tensor.

Proof. [✓ Proof](#) Write $B = P_V$ by Theorem 30.1.3.3. Since $V^\dagger V = \Pi$ and $P_\Pi = P$, Theorem 30.1.2.5 gives $\mathcal{E}_{P_V} = \mathcal{E}_P$. $\square$

Theorem 30.1.3.5 (Polar factors of an injective tensor). [Lean](#) [Lean](#) [✓ Stmt](#) If B is injective, then $V^\dagger V = \mathbb{1}_{D^2}$ and P is positive definite, as in [185, eq. (8)].

Proof. [✓ Proof](#) By Theorem 30.1.2.3, $\widehat{B}$ is injective; apply Theorem 30.1.1.5 and reindex. $\square$

Theorem 30.1.3.6 (Support projector of a matrix supported on a set of columns). [Lean](#) [✓ Stmt](#) Let S be a set of column indices of M , let the columns of M outside S vanish, and let $Mx = 0$ imply $x = 0$ for every vector x supported on S . Then Π is the coordinate projector Π_S onto S .

Proof. [✓ Proof](#) Since $M\Pi_S = M$, the matrix $Q = P(\mathbb{1} - \Pi_S)$ has $Q^\dagger Q = (\mathbb{1} - \Pi_S)M^\dagger M(\mathbb{1} - \Pi_S) = 0$, so $Q = 0$. As $\Pi = PR$ for some R by Theorem 30.1.1.3, and Π and P are Hermitian, $\Pi(\mathbb{1} - \Pi_S) = R^\dagger Q = 0$, that is, $\Pi = \Pi\Pi_S$. Conversely, $M\Pi_S(\mathbb{1} - \Pi) = M(\mathbb{1} - \Pi) = 0$, and every column of $\Pi_S(\mathbb{1} - \Pi)$ is supported on S , so $\Pi_S(\mathbb{1} - \Pi) = 0$. Hence $\Pi = \Pi^\dagger = \Pi_S\Pi = \Pi_S$. $\square$

Definition 30.1.3.7 (Injectivity on a set of bond pairs). [Lean](#) [✓ Stmt](#) Let S be a set of pairs of bond indices. A tensor B of physical dimension n and bond dimension D is *injective on S* if $B_{\alpha\beta}^i = 0$ for all i and $(\alpha, \beta) \notin S$, and the matrices B^i span every matrix unit $|\alpha\rangle\langle\beta|$ with $(\alpha, \beta) \in S$. For the rectangle $S = \{(\alpha, \beta) : \alpha < a, \beta < b\}$ the condition may be read as injectivity of a tensor with bond dimensions a and b whose matrices are padded with zeros; no such rectangular tensor enters the formal statement, and this reading is not proved.

Theorem 30.1.3.8 (Polar factors of a tensor injective on a set of bond pairs). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A tensor is injective on the set of all bond pairs if and only if it is injective. Let B be injective on S . Then $\widehat{B}x = 0$ implies $x = 0$ for every vector x supported on S , the support projector of $B = VP$ is the coordinate projector Π_S , so that $\sum_i \overline{V_{ix}} V_{iy}$ is 1 if $x = y \in S$ and 0 otherwise, and $P^x = 0$ for every pair $x \notin S$. For every tensor B there is a matrix G on the bond pairs with $V_{ix} = \sum_a B_a^i G_{ax}$.

Proof. [✓ Proof](#) If B spans all matrix units, its span is the full matrix algebra, and conversely. For x supported on S with $\widehat{B}x = 0$, the functional $Y \mapsto \sum_{\alpha, \beta} Y_{\alpha\beta} x_{\alpha\beta}$ vanishes on every B^i , hence on every matrix unit in S , so $x = 0$ on S . Then Theorem 30.1.3.6 gives $\Pi = \Pi_S$, the column identity is $V^\dagger V = \Pi_S$, and $P^x = \sum_y (\Pi_S)_{xy} P^y = 0$ for $x \notin S$ because $\Pi P = P$. Finally $V = \widehat{B}R$ for some R by Lemma 30.9.10; reindexing gives G . $\square$

Theorem 30.1.3.9 (Transfer map of the positive part of a blocked tensor). [Lean](#) [✓ Stmt](#) Let A be a tensor and P the positive-part tensor of the blocked tensor $A^{[q]}$. Then $\mathcal{E}_P = \mathcal{E}_A^q$.

Proof. [✓ Proof](#) $\mathcal{E}_P = \mathcal{E}_{A^{[q]}}$ by Theorem 30.1.3.4, and $\mathcal{E}_{A^{[q]}} = \mathcal{E}_A^q$ by Theorem D.4.8 [185, eq. (6)]. $\square$

Theorem 30.1.3.10 (Isometric form of the blocked periodic state). [Lean](#) [✓ Stmt](#) Let A be a tensor, let V and P be the polar factors of the blocked tensor $A^{[q]}$, and let $M \geq 0$. For every

configuration $\sigma = (\sigma_0, \dots, \sigma_{M-1})$ of M blocked sites, with $\tilde{\sigma}$ the corresponding configuration of qM sites,

$$V^{(qM)}(A)_{\tilde{\sigma}} = \sum_{\tau \in \{0, \dots, D^2-1\}^M} \left(\prod_{j=0}^{M-1} V_{\sigma_j \tau_j} \right) V^{(M)}(P)_{\tau},$$

that is, $|V^{(qM)}(A)\rangle = V^{\otimes M}|V^{(M)}(P)\rangle$. Before normalization, this writes the state as $V^{\otimes M}$ applied to the state $|\phi_{\text{pos}}\rangle$ of the positive parts of [185, eq. (18)]. It is the identity $|\phi_N\rangle = \frac{1}{c_N} (\bigotimes_{i=1}^{N/q} V_i) \sum_j \beta_j |v_{\text{pos},j}\rangle$ of [185, Supplemental Material] with the positive part kept whole rather than split into its normal blocks.

Proof. [✓ Proof](#) By Lemma G.1.1.2, the right-hand side is $V^{(M)}(P_V)_{\sigma}$, and $P_V = A^{[q]}$ by Theorem 30.1.3.3. Finally $V^{(M)}(A^{[q]})_{\sigma} = V^{(qM)}(A)_{\tilde{\sigma}}$ by Lemma 2.7.2. $\square$

30.2 The fixed-point pair state of a normal tensor

For a normal tensor A in left-canonical gauge with right fixed point $\sigma > 0$, $\text{tr } \sigma = 1$, the transfer map decomposes as $\mathcal{E}_A = \mathcal{E}_{P_{\infty}} + R$, where $\mathcal{E}_{P_{\infty}}(X) = \text{tr}(X)\sigma$ and R has spectral radius strictly less than one [185]. The fixed-point tensor P_{∞} has transfer map $\mathcal{E}_{P_{\infty}}$, and its periodic state factors into a product of nearest-neighbor entangled pairs.

Definition 30.2.1 (The fixed-point tensor). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\sigma \in M_D(\mathbb{C})$. The *fixed-point tensor* P_{∞} of σ has physical space $\mathbb{C}^D \otimes \mathbb{C}^D$ and bond dimension D ; its matrix at the physical pair (l, r) is

$$P_{\infty}^{(l,r)} = \sqrt{\sigma} |l\rangle\langle r|.$$

This is the tensor P_{∞} of [185].

$$\alpha \begin{array}{c} l \quad r \\ \boxed{P_{\infty}} \\ \beta \end{array} = \alpha \begin{array}{c} l \\ \bullet \\ \sqrt{\sigma} \end{array} \begin{array}{c} r \\ \text{---} \\ \beta \end{array}$$

Theorem 30.2.2 (Positivity of the fixed-point tensor). [Lean](#) [Lean](#) [✓ Stmt](#) Read as the matrix from the virtual pair (α, β) to the physical pair (l, r) , the fixed-point tensor P_{∞} of any $\sigma \in M_D(\mathbb{C})$ is $(\sqrt{\sigma})^T \otimes \mathbb{1}$. It is positive semidefinite.

Proof. [✓ Proof](#) The entry at $((l, r), (\alpha, \beta))$ is $(\sqrt{\sigma} |l\rangle\langle r|)_{\alpha\beta} = (\sqrt{\sigma})_{\alpha l} \delta_{\beta r}$, the entry of $(\sqrt{\sigma})^T \otimes \mathbb{1}$. The transpose of the positive semidefinite matrix $\sqrt{\sigma}$ is positive semidefinite, and so is its Kronecker product with $\mathbb{1}$. $\square$

Definition 30.2.3 (The entangled pair and the pair product state). [Lean](#) [Lean](#) [✓ Stmt](#) Let $\sigma \in M_D(\mathbb{C})$. The *entangled pair* ω on $R \otimes L = \mathbb{C}^D \otimes \mathbb{C}^D$ has coefficient $\omega_{(r,l)} = (\sqrt{\sigma})_{rl}$. For a vector $\omega \in \mathbb{C}^D \otimes \mathbb{C}^D$ and a ring of N sites, where site k carries the physical pair $c_k = (l_k, r_k)$, the *pair product state* is

$$\Omega(c) = \prod_{k=0}^{N-1} \omega_{(r_k, l_{k+1})}, \quad (30.2)$$

with indices of sites taken modulo N . In terms of $\rho = \sigma^T$, ω corresponds to $(\mathbb{1} \otimes \sqrt{\rho}) \sum_i |ii\rangle$, with pair product state $\Omega = \bigotimes_k |\omega\rangle_{R_k L_{k+1}}$ [185].

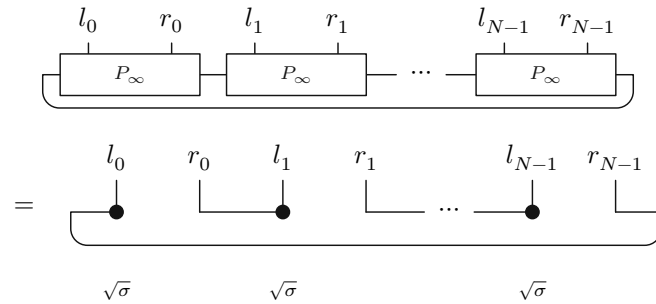
Theorem 30.2.4 (The transfer map of the fixed-point tensor). [Lean](#) [Lean](#) [✓ Stmt](#) Let σ be positive semidefinite. Then $\mathcal{E}_{P_\infty}(X) = \text{tr}(X)\sigma$ for every $X \in M_D(\mathbb{C})$. If moreover $\text{tr}\sigma = 1$, then $\mathcal{E}_{P_\infty}$ is the projection $X \mapsto \text{tr}(X)\sigma$ onto the fixed point σ [185].

Proof. [✓ Proof](#) Write $S = \sqrt{\sigma}$, so $S^\dagger = S$ and $S^2 = \sigma$. Since $|l\rangle\langle r|X|r\rangle\langle l| = X_{rr}|l\rangle\langle l|$,

$$\mathcal{E}_{P_\infty}(X) = \sum_{l,r} X_{rr} S|l\rangle\langle l|S = \text{tr}(X) S \left(\sum_l |l\rangle\langle l| \right) S = \text{tr}(X)\sigma.$$

□

Theorem 30.2.5 (The periodic state of the fixed-point tensor). [Lean](#) [✓ Stmt](#) Let $\sigma \in M_D(\mathbb{C})$ and $N \geq 1$. For every configuration c of the physical pairs $c_k = (l_k, r_k)$ on N sites, $V^{(N)}(P_\infty)_c = \Omega(c)$, where Ω is the pair product state (30.2) of the entangled pair of σ .



Proof. [✓ Proof](#) Expand the trace of the product $\prod_k \sqrt{\sigma} |l_k\rangle\langle r_k|$ as a sum over cyclic index paths $(t_k)_k$, one index per bond. The entry $(\sqrt{\sigma} |l_k\rangle\langle r_k|)_{t_k t_{k+1}}$ vanishes unless $t_{k+1} = r_k$, so only the path $t_{k+1} = r_k$ survives, and its product is $\prod_k (\sqrt{\sigma})_{r_{k-1} l_k}$. Reindexing the product along the cyclic shift of the sites gives (30.2). □

Theorem 30.2.6 (The fixed-point tensor is a renormalization fixed point). [Lean](#) [Lean](#) [✓ Stmt](#) Let σ be positive semidefinite with $\text{tr}\sigma = 1$. Then $\mathcal{E}_{P_\infty}^2 = \mathcal{E}_{P_\infty}$, and P_∞ has a physical blocking isometry.

Proof. [✓ Proof](#) By Theorem 30.2.4, $\mathcal{E}_{P_\infty}$ is the projection $X \mapsto \text{tr}(X)\sigma$, which is idempotent because $\text{tr}\sigma = 1$. The physical blocking isometry follows from Theorem 17.1.6. □

Theorem 30.2.7 (Normalization of the entangled pair). [Lean](#) [✓ Stmt](#) Let σ be positive semidefinite, with entangled pair ω . Then $\langle \omega | \omega \rangle = \text{tr}\sigma$.

Proof. [✓ Proof](#) With $S = \sqrt{\sigma}$, $\langle \omega | \omega \rangle = \sum_{r,l} \overline{S_{rl}} S_{rl} = \text{tr}(S^\dagger S) = \text{tr}\sigma$. □

Definition 30.2.8 (Regrouping of a ring into its bonds). [Lean](#) [✓ Stmt](#) The bond regrouping is the bijection of configurations on a ring of N sites sending $c = (l_k, r_k)_k$ to $(r_k, l_{k+1})_k$, with indices modulo N . Its inverse sends $(p_k)_k = (p_k^1, p_k^2)_k$ to $(p_{k-1}^2, p_k^1)_k$.

Theorem 30.2.9 (Overlap and normalization of pair product states). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For every $\omega, \omega' \in \mathbb{C}^D \otimes \mathbb{C}^D$ and $N \geq 0$, the pair product states (30.2) $|\Omega\rangle$ of ω and $|\Omega'\rangle$ of ω' satisfy $\langle \Omega | \Omega' \rangle = \langle \omega | \omega' \rangle^N$; in particular $\langle \Omega | \Omega \rangle = \langle \omega | \omega \rangle^N$. If ω is the entangled pair of a positive semidefinite σ with $\text{tr}\sigma = 1$, then $\langle \Omega | \Omega \rangle = 1$.

Proof. [✓ Proof](#) Expanding $\langle \omega | \omega' \rangle^N$ as a sum over bond configurations $(p_k)_k$ of $\prod_k \overline{\omega_{p_k}} \omega'_{p_k}$ and reindexing along the bond regrouping of Definition 30.2.8 gives $\sum_c \overline{\Omega(c)} \Omega'(c)$. The last statement follows from Theorem 30.2.7. $\square$

Theorem 30.2.10 (Convergence of the blocked transfer map from a spectral gap). [Lean](#) [✓ Stmt](#) *Let A be left-canonical, let σ be positive semidefinite with $\text{tr } \sigma = 1$ and $\mathcal{E}_A(\sigma) = \sigma$, and suppose that $\mathcal{E}_A - \mathcal{E}_{P_\infty}$ has spectral radius less than one. Then $\mathcal{E}_{A^{[q]}}(X) \rightarrow \mathcal{E}_{P_\infty}(X)$ as $q \rightarrow \infty$, for every $X \in M_D(\mathbb{C})$ [185].*

Proof. [✓ Proof](#) Write $P = \mathcal{E}_{P_\infty}$, the projection $X \mapsto \text{tr}(X)\sigma$ by Theorem 30.2.4. The powers of an operator of spectral radius less than one tend to zero, so $(\mathcal{E}_A - P)^q(X) \rightarrow 0$. Since $\mathcal{E}_A$ is trace preserving and fixes σ , $\mathcal{E}_A^q = P + (\mathcal{E}_A - P)^q$ for $q \geq 1$. Finally $\mathcal{E}_{A^{[q]}} = \mathcal{E}_A^q$ for every tensor by Theorem D.4.8. $\square$

Theorem 30.2.11 (Convergence of the blocked transfer map of a normal tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be irreducible and left-canonical, with 1 the only eigenvalue of $\mathcal{E}_A$ of modulus one. Then A is a normal tensor in the sense of Definition 7.6.1, since $\mathcal{E}_A$ is a channel and so has spectral radius one. Let σ be positive definite with $\text{tr } \sigma = 1$ and $\mathcal{E}_A(\sigma) = \sigma$. Then $\mathcal{E}_{A^{[q]}}(X) \rightarrow \mathcal{E}_{P_\infty}(X)$ as $q \rightarrow \infty$, for every $X \in M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) The transfer map of a left-canonical tensor is a channel, and it is irreducible because A is. By the complementary transfer-map gap for primitive maps [182, “Complementary transfer-map gap for primitive maps”], $\mathcal{E}_A$ minus the projection $X \mapsto \text{tr}(X)\sigma$ has spectral radius less than one. That projection is $\mathcal{E}_{P_\infty}$ by Theorem 30.2.4, and Theorem 30.2.10 applies. $\square$

30.3 The approximating state

Write $\mathcal{B}_q$ for the physical matrix of the q -blocked tensor of A (Definitions 2.7.1 and 30.1.2.1) and $\mathcal{B}_q = VP$ for its polar decomposition (Theorem 30.1.3.3). In this section A is normal (Definition 2.5.5) and in the gauge

$$\sum_{i=0}^{d-1} (A^i)^\dagger A^i = \mathbb{1}, \quad \mathcal{E}_A(\rho) = \rho, \quad \rho > 0, \quad \text{Tr } \rho = 1, \quad (30.3)$$

so that $\mathcal{E}_A^n(X) \rightarrow \text{Tr}(X)\rho$ as $n \rightarrow \infty$ [185, eq. (5)]. Under normality, words of length L_0 span $M_D(\mathbb{C})$, so $\mathcal{E}_A$ is strongly irreducible by Theorem 6.1.1.15: the peripheral spectrum is $\{1\}$, the positive fixed point ρ is unique up to scale, and all remaining eigenvalues have modulus strictly less than 1.

Lemma 30.3.1 (Convergence of the positive part). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $P_q = (\mathcal{B}_q^\dagger \mathcal{B}_q)^{1/2}$. Then $P_q \rightarrow P_\infty$ as $q \rightarrow \infty$, where $P_\infty = (\sqrt{\rho})^\top \otimes \mathbb{1}$ [185, eq. (8)]. More generally, for any tensors B_q whose transfer maps satisfy $\mathcal{E}_{B_q}(X) \rightarrow \text{Tr}(X)\rho$ for every X , where $\rho \geq 0$, the positive parts of the physical matrices of B_q converge to $(\sqrt{\rho})^\top \otimes \mathbb{1}$, and the corresponding tensors converge to P_∞ .*

Proof. [✓ Proof](#) The matrix of $\mathcal{B}_q^\dagger \mathcal{B}_q$ is a rearrangement of $\mathcal{E}_A^q$: $(\mathcal{B}_q^\dagger \mathcal{B}_q)_{(\alpha, \beta), (\alpha', \beta')} = \mathcal{E}_A^q(|\beta'\rangle\langle\beta|)_{\alpha'\alpha}$. By normality and the gauge (30.3), $\mathcal{E}_A^q(X) \rightarrow \text{Tr}(X)\rho$, so the rearranged matrices converge to $\rho^\top \otimes \mathbb{1}$. The square root of $\rho^\top \otimes \mathbb{1}$ is $(\sqrt{\rho})^\top \otimes \mathbb{1}$, since the latter is positive semidefinite and squares to the former, and the square root is continuous on positive semidefinite matrices. $\square$

Theorem 30.3.2 (Scalar lower bound for the square root). [Lean](#) [✓ Stmt](#) Let b be an element of a C^* -algebra and $c \geq 0$ a real number with $b \geq c\mathbb{1}$. Then $\sqrt{b} \geq \sqrt{c}\mathbb{1}$.

Proof. [✓ Proof](#) The element $\sqrt{c}\mathbb{1}$ is positive and its square is $c\mathbb{1} \leq b$, so operator monotonicity of the square root gives $\sqrt{c}\mathbb{1} = \sqrt{c}\mathbb{1} \leq \sqrt{b}$. $\square$

Lemma 30.3.3 (One-sided bounds for the square root above a scalar floor). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let a, b be positive elements of a C^* -algebra, and let $c, s \geq 0$ be real numbers with $b \geq c\mathbb{1}$. Write $\varepsilon = \|a - b\|$.

1. If $\varepsilon \leq 2s\sqrt{c} + s^2$, then $\sqrt{a} - \sqrt{b} \leq s$.
2. If $s \leq \sqrt{c}$ and $\varepsilon + s^2 \leq 2s\sqrt{c}$, then $\sqrt{b} - \sqrt{a} \leq s$.

Proof. [✓ Proof](#) For every real t , $(\sqrt{b} + t)^2 = b + 2t\sqrt{b} + t^2$. By Theorem 30.3.2, $\sqrt{b} \geq \sqrt{c}$, and $-\varepsilon \leq a - b \leq \varepsilon$. For the first bound, $(\sqrt{b} + s)^2 \geq b + 2s\sqrt{c} + s^2 \geq b + \varepsilon \geq a$. For the second, $\sqrt{b} - s \geq \sqrt{c} - s \geq 0$ and $(\sqrt{b} - s)^2 \leq b - 2s\sqrt{c} + s^2 \leq a + \varepsilon + s^2 - 2s\sqrt{c} \leq a$. Operator monotonicity of the square root gives $\sqrt{a} \leq \sqrt{b} + s$ and $\sqrt{b} - s \leq \sqrt{a}$. $\square$

Theorem 30.3.4 (Hölder bound for the square root). [Lean](#) [Lean](#) [✓ Stmt](#) Let a, b be positive elements of a C^* -algebra. Then $\|\sqrt{a} - \sqrt{b}\| \leq \|a - b\|^{1/2}$. This is the operator inequality quoted from Bhatia in [195, Supplemental Material, eq. (26)].

Proof. [✓ Proof](#) Write $\varepsilon = \|a - b\|$ and $s = \sqrt{\varepsilon}$. The first bound of Lemma 30.3.3 with $c = 0$ gives $\sqrt{a} - \sqrt{b} \leq s$, since $\varepsilon = s^2$. Exchanging a and b gives $\sqrt{b} - \sqrt{a} \leq s$. Since $\sqrt{a} - \sqrt{b}$ is self-adjoint and lies between $-s$ and s , its norm is at most s . $\square$

Theorem 30.3.5 (Square root near a positive definite element). [Lean](#) [✓ Stmt](#) Let a, b be positive elements of a C^* -algebra with $b \geq c\mathbb{1}$ for a real number $c > 0$. Then $\|\sqrt{a} - \sqrt{b}\| \leq \|a - b\|/\sqrt{c}$. For matrices and the Frobenius norm, the corresponding bound with the smallest eigenvalue of $\sqrt{b}$ in place of $\sqrt{c}$ is [190, App. B3, eqs. (S27)–(S31)].

Proof. [✓ Proof](#) Write $\varepsilon = \|a - b\|$. If $\varepsilon > c$, the bound $\|\sqrt{a} - \sqrt{b}\| \leq \sqrt{\varepsilon}$ of Theorem 30.3.4 suffices, since $\sqrt{\varepsilon} \leq \varepsilon/\sqrt{c}$. Otherwise let $s = \varepsilon/\sqrt{c} \leq \sqrt{c}$, so that $2s\sqrt{c} = 2\varepsilon$ and $s^2 \leq \varepsilon$. Both hypotheses of Lemma 30.3.3 hold, so $\sqrt{b} - s \leq \sqrt{a} \leq \sqrt{b} + s$, that is, $\|\sqrt{a} - \sqrt{b}\| \leq s$. $\square$

Theorem 30.3.6 (Lipschitz constant of the square root at an invertible positive element). [Lean](#) [✓ Stmt](#) Let b be a positive invertible element of a C^* -algebra. There is a real number $K \geq 0$ such that $\|\sqrt{a} - \sqrt{b}\| \leq K\|a - b\|$ for every positive element a .

Proof. [✓ Proof](#) If the algebra is zero, take $K = 0$. Otherwise the spectrum of b is a compact subset of $(0, \infty)$, so $b \geq c\mathbb{1}$ for some real $c > 0$, and Theorem 30.3.5 gives the bound with $K = 1/\sqrt{c}$. $\square$

Theorem 30.3.7 (Rate of convergence of the positive part). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be normal in the gauge (30.3), let $\lambda_2 \in \mathbb{C}$ satisfy $|\mu| \leq |\lambda_2|$ for every eigenvalue $\mu \neq 1$ of $\mathcal{E}_A$, let ξ be the correlation length at λ_2 of Definition 14.2.4, and let $0 < \gamma < 1$. Write $r = e^{-\gamma/\xi}$, where $-\gamma/\xi$ is read as 0 when $\xi = 0$; thus $r = |\lambda_2|^\gamma$ when $\lambda_2 \neq 0$, and $r = 1$ when $\lambda_2 = 0$. Then $a < r$ for every real number $a < 1$ with $a \leq |\lambda_2|$. There are constants C, K, K' depending only on A, ρ, λ_2 , and γ such that $\|\mathcal{E}_A^n(X) - \text{Tr}(X)\rho\| \leq Cr^n\|X\|$ for all n and X , and

$$\begin{aligned} \|\mathcal{B}_q^\dagger \mathcal{B}_q - \rho^\top \otimes \mathbb{1}\| &\leq Kr^q, \\ \|P_q - P_\infty\| &\leq K'r^q \end{aligned}$$

for all q . In [195, Supplemental Material, eq. (26)] the bound on $P_q - P_\infty$ comes from Theorem 30.3.4, which halves the exponent. This gives the rate of [195, eq. (34)] and the range $0 < \gamma < 1/2$ of [185, Lemma 1 and eq. (S9)]. The Lipschitz bound at the positive definite limit P_∞^2 gives the range $0 < \gamma < 1$. This step is [190, App. B3, eqs. (S27)–(S32)] for diagonalizable transfer matrices, and eqs. (S33)–(S38) there extend it to every normal tensor at each correlation length larger than ξ .

Proof. ✓ Proof Let $a < 1$ with $a \leq |\lambda_2|$. If $0 < |\lambda_2| < 1$, then $r = |\lambda_2|^\gamma > |\lambda_2| \geq a$, because $\gamma < 1$. Otherwise $\lambda_2 = 0$ and $r = 1$, or $|\lambda_2| \geq 1$ and $r = |\lambda_2|^\gamma \geq 1$; in both cases $r \geq 1 > a$.

Write $\mathcal{E}_\infty(X) = \text{Tr}(X)\rho$. The eigenvalues z of $\mathcal{E}_A - \mathcal{E}_\infty$ other than 0 are eigenvalues of $\mathcal{E}_A$ different from 1, so $|z| \leq |\lambda_2|$, and $|z| < 1$ by normality. Taking $a = |z|$ gives $|z| < r$. Hence the spectral radius of $\mathcal{E}_A - \mathcal{E}_\infty$ is less than r , and $\mathcal{E}_A^n - \mathcal{E}_\infty = (\mathcal{E}_A - \mathcal{E}_\infty)^n$ gives the first bound for $n \geq 1$. For $n = 0$ it reads $\|X - \mathcal{E}_\infty(X)\| \leq C\|X\|$, which holds once $C \geq 1 + \|\mathcal{E}_\infty\|$; enlarge C accordingly.

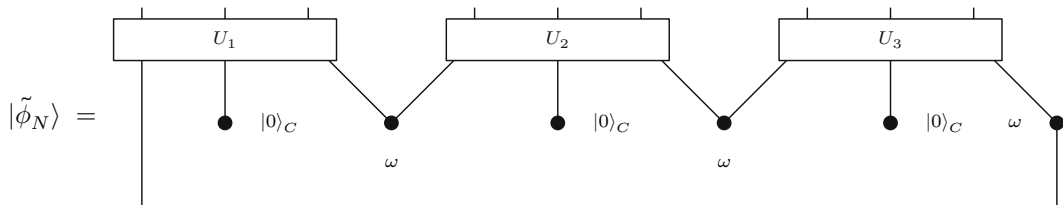
The transfer map of the q -site blocked tensor is $\mathcal{E}_A^q$, so the entries of $\mathcal{B}_q^\dagger \mathcal{B}_q$ are entries of $\mathcal{E}_A^q$ applied to matrix units: $(\mathcal{B}_q^\dagger \mathcal{B}_q)_{(\alpha, \beta), (\alpha', \beta')} = \mathcal{E}_A^q(|\beta'\rangle\langle\beta|)_{\alpha'\alpha}$. In the same way $(\rho^\top \otimes \mathbb{1})_{(\alpha, \beta), (\alpha', \beta')} = \text{Tr}(|\beta'\rangle\langle\beta|) \rho_{\alpha'\alpha}$. Hence every entry of $\mathcal{B}_q^\dagger \mathcal{B}_q - \rho^\top \otimes \mathbb{1}$ is an entry of $\mathcal{E}_A^q(X) - \text{Tr}(X)\rho$ at a matrix unit X , and is at most a fixed multiple of $C r^q$ by the first bound. Bounding the operator norm by the sum of the moduli of the finitely many entries gives $\|P_q^2 - P_\infty^2\| = \|\mathcal{B}_q^\dagger \mathcal{B}_q - \rho^\top \otimes \mathbb{1}\| \leq K r^q$. Since $P_\infty^2 = \rho^\top \otimes \mathbb{1}$ is positive definite, Theorem 30.3.6 gives $L \geq 0$ with $\|P_q - P_\infty\| \leq L K r^q$, so $K' = LK$. $\square$

Definition 30.3.8 (Approximating state). Lean Lean Lean ✓ Stmt With V from Theorem 30.1.3.3 and $N = qM$, the *approximating state* is $|\tilde{\phi}_N\rangle = |\phi_M(VP_\infty)\rangle / \|\phi_M(VP_\infty)\|$. By Lemma G.1.1.2 and Theorem 30.2.5, for $M \geq 1$, $|\phi_M(VP_\infty)\rangle = V^{\otimes M} \otimes_k |\omega\rangle_{R_k L_{k+1}}$. When $\mathcal{B}_q$ is injective, V is an isometry and $\|\phi_M(VP_\infty)\| = 1$, so no normalization is needed. This is [185, eqs. (9) and (10)].

Lemma 30.3.9 (A unitary implementing the isometry). Lean Lean Lean Lean ✓ Stmt Suppose $\mathcal{B}_q$ is injective, and let ι be an injective map from the pairs (γ, δ) with $0 \leq \gamma, \delta < D$ to the computational basis of $(\mathbb{C}^d)^{\otimes q}$. Then there is a unitary U on $(\mathbb{C}^d)^{\otimes q}$ with $U|\iota(\gamma, \delta)\rangle = V|\gamma, \delta\rangle$ for all γ, δ , and for every $M \geq 1$

$$|\tilde{\phi}_N\rangle = \left(\bigotimes_{k=1}^M U_k \right) \sum_c \Omega(c) \bigotimes_{k=1}^M |\iota(c_k)\rangle, \quad (30.4)$$

where c runs over the configurations $c_k = (l_k, r_k)$ and Ω is the pair product state (30.2). The source takes $D = d$, so that L and R below are single sites and the remaining central sites carry $|0\rangle$; the padding of the legs for general D is this chapter's extension. For $d \geq 2$, $r = \lceil \log_d D \rceil$, $q \geq 2r$, and $(\mathbb{C}^d)^{\otimes q} = L \otimes C \otimes R$, where L and R are the first and last r sites of the block and C denotes the remaining $q - 2r$ sites, embed $\mathbb{C}^D$ into L and R as the span of the first D computational basis vectors, and set $\iota(\gamma, \delta) = |\gamma\rangle_L |0\rangle_C |\delta\rangle_R$ with $|0\rangle_C = |0\rangle^{\otimes(q-2r)}$. The right-hand side of (30.4) then evaluates to $(\bigotimes_k U_k) \bigotimes_k (|\omega\rangle_{R_k L_{k+1}} |0\rangle_{C_k})$ [185, eqs. (10) and (11)].



Proof. ✓ Proof Since V is an isometry, its columns $V|\gamma, \delta\rangle$ are orthonormal. Place them at the positions $\iota(\gamma, \delta)$ and extend them to an orthonormal basis of $(\mathbb{C}^d)^{\otimes q}$; the matrix with these columns is the unitary U . Expanding $(\bigotimes_k U_k) \sum_c \Omega(c) \bigotimes_k |\iota(c_k)\rangle$ gives $\sum_c \Omega(c) \bigotimes_k V|c_k\rangle = V^{\otimes M} \bigotimes_k |\omega\rangle_{R_k L_{k+1}}$, which is $|\tilde{\phi}_N\rangle$ by Definition 30.3.8. $\square$

30.4 Exact factorizations of the isometry

Theorem 30.4.1 (Sequential factorization of a matrix product isometry). Lean Lean ✓ Stmt Let $A_1, \dots, A_q$, $q \geq 1$, be tensors of bond dimension D , one on each site, let G be a $D^2 \times D^2$ matrix, and let $V: \mathbb{C}^D \otimes \mathbb{C}^D \rightarrow (\mathbb{C}^d)^{\otimes q}$ be an isometry with matrix elements $\langle i_1 \dots i_q | V | x \rangle = \sum_{\alpha, \beta} (A_1^{i_1} \dots A_q^{i_q})_{\alpha\beta} G_{(\alpha, \beta), x}$. Then there are integers $b_0 = 1$, $b_q = D^2$ and $b_p \leq D^2$, and families of $b_{p-1} \times b_p$ matrices X_p^i , $p = 1, \dots, q$, with $\sum_i (X_p^i)^\dagger X_p^i = \mathbb{1}_{b_p}$ for every p , such that $\langle i_1 \dots i_q | V | x \rangle = (X_1^{i_1} X_2^{i_2} \dots X_q^{i_q})_{1x}$.

Proof. ✓ Proof The matrix elements of V are $(j^\top (\mathbb{1}_D \otimes A_1^{i_1}) \dots (\mathbb{1}_D \otimes A_q^{i_q}) G)_x$ with $j = \sum_\gamma |\gamma\rangle \otimes |\gamma\rangle$. Apply Theorem 29.2.8 to the row matrix of j and the sites $\mathbb{1}_D \otimes A_p^i$; this gives isometric sites $Q_1, \dots, Q_q$ and a remainder $C = RG$ with $V = QC$ in the sense of (29.2). By (29.3), $V^\dagger V = C^\dagger C$, so C is an isometry because V is. Absorb C into the last site, $X_q^i = Q_q^i C$ and $X_p = Q_p$ for $p < q$; the last site is again isometric, $\sum_i C^\dagger (Q_q^i)^\dagger Q_q^i C = C^\dagger C = \mathbb{1}$. $\square$

Theorem 30.4.2 (Sequential factorization). Lean ✓ Stmt Let $A_1, \dots, A_q$, $q \geq 1$, be tensors of bond dimension D , one on each site, suppose their blocked tensor B with $B^{i_1 \dots i_q} = A_1^{i_1} \dots A_q^{i_q}$ is injective, and let $B = VP$ be its polar decomposition as in Theorem 30.1.3.3. For equal tensors $A_p = A$, B is the blocked tensor $\mathcal{B}_q$. There are integers $D'_1 = D^2$, $D'_{q+1} = 1$, and $D'_k \leq D^2$, and matrices $W_k^i \in \mathbb{C}^{D'_{k+1} \times D'_k}$ for $k = 1, \dots, q$ and $i = 1, \dots, d$, with

$$\sum_{i=1}^d (W_k^i)^\dagger W_k^i = \mathbb{1}_{D'_k}, \quad (30.5)$$

$$\langle i_1 \dots i_q | V | x \rangle = (W_q^{i_1} W_{q-1}^{i_2} \dots W_1^{i_q})_{1x}, \quad (30.6)$$

for every k , configuration $i_1 \dots i_q$, and index x , where $\mathbb{C}^{D'_1}$ is identified with $\mathbb{C}^D \otimes \mathbb{C}^D$. This is the matrix product form of [185, eqs. (13)–(15)], where the isometry carrying the input is written $\tilde{C} = V_1$. The block length $q \geq 1$ is implicit in the source, which draws at least one site; the empty block $q = 0$ is excluded, see [128].

Proof. ✓ Proof Since $\mathcal{B}_q$ is injective, P is positive definite and $V = \mathcal{B}_q P^{-1}$, so $\langle i_1 \dots i_q | V | x \rangle = \sum_{\alpha, \beta} (A_1^{i_1} \dots A_q^{i_q})_{\alpha\beta} (P^{-1})_{(\alpha, \beta), x}$. This is an open-boundary matrix product map with bond dimension at most D^2 : the site matrices are $\mathbb{1}_D \otimes A^i$ (the source writes $A^i \otimes \mathbb{1}_D$; the order of the two bond factors only permutes the bond index), the two bond factors are joined at the left end, and P^{-1} is absorbed into the right end [185, eq. (13)]. Theorem 30.4.1 with $G = P^{-1}$ makes

Proof. [✓ Proof](#) Blocking commutes with maps on the physical leg: blocking C_W over m sites gives the tensor C blocked over m sites, rotated by $W^{\otimes m}$ (Theorem [G.1.1.3](#)). Blocking 2^k sites is blocking 2^{k-1} pairs of the two-site blocked tensor, and the outer layer composes with the maps of the inner tree by $(A_{W'})_W = A_{WW'}$ (Lemma [11.7.3](#)), so induction on k gives [\(30.8\)](#). The range of the two-site blocked map of T_j lies in $\text{ran } P_j \otimes \text{ran } P_j$, because Π_j fixes the tensor of P_j ; so $\Pi_j^{\otimes 2^{k-j}}$ fixes the product of the later layers, and the product of all layers is a partial isometry with initial projector the support projector of P_k . Theorem [30.4.4](#) then identifies it with V and P_k with P_q . $\square$

30.5 Approximation error

For a normal tensor A , let λ_2 be an eigenvalue of $\mathcal{E}_A$ of largest modulus other than the Perron eigenvalue 1. By the remark after [\(30.3\)](#), $|\lambda_2| < 1$. The *correlation length* of A is the quantity $\xi = -1/\log|\lambda_2|$ of Definition [14.2.4](#) at this λ_2 . It is finite, it is positive exactly when $\lambda_2 \neq 0$, and $|\lambda_2| = e^{-1/\xi}$ in that case. As in the previous section, A is in the gauge [\(30.3\)](#) throughout this section. The estimates below use λ_2 only through the bound $|\mu| \leq |\lambda_2|$ for the eigenvalues $\mu \neq 1$ of $\mathcal{E}_A$. When $\lambda_2 = 0$, so that $\xi = 0$, they are applied with any λ'_2 satisfying $0 < |\lambda'_2| < 1$ in place of λ_2 , and ξ is read as the correlation length at λ'_2 .

Lemma 30.5.1 (Overlap of the positive part with the fixed point). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let A be normal in the gauge [\(30.3\)](#) with correlation length ξ , and let $0 < \gamma < 1/2$. Then, for $N = qM$,

$$\left| 1 - \langle \phi_M(P_\infty) | \phi_M(P_q) \rangle \right| = O\left(\frac{N}{q} e^{-\gamma q/\xi}\right). \quad (30.9)$$

for $q \geq 1$ [[195](#), Supplemental Material, eq. (S29)], quoted as [[185](#), eq. (S9)]. Explicitly, there is a constant $C > 0$ depending only on A , ρ , and γ such that $|1 - \langle \phi_M(P_\infty) | \phi_M(P_q) \rangle| \leq Cy e^{Cy}$ for all $q \geq 0$ and $M \geq 1$, where $y = M e^{-\gamma q/\xi}$. The same bound holds for the overlap itself: $|\langle \phi_M(P_\infty) | \phi_M(P_q) \rangle - 1| \leq Cy e^{Cy}$.

Proof. [✓ Proof](#) By Theorem [30.3.7](#), $\|P_q - P_\infty\| \leq K_2 e^{-\gamma q/\xi}$.

Let τ_q and τ_∞ be the transfer matrices of the mixed transfer maps of P_q and of P_∞ against P_∞ . Then $\langle \phi_M(P_\infty) | \phi_M(P_q) \rangle = \text{Tr } \tau_q^M$, $1 = \text{Tr } \tau_\infty^M$, $\tau_q^2 = \tau_\infty$, and $\delta = \|\tau_q - \tau_\infty\| \leq K_3 e^{-\gamma q/\xi}$, since $\tau_q - \tau_\infty$ depends linearly on $P_q - P_\infty$. Writing $\tau_q^M - \tau_\infty^M = \sum_{k < M} \tau_q^k (\tau_q - \tau_\infty) \tau_\infty^{M-1-k}$ and bounding $\|\tau_q^k\| \leq c + \|\tau_q^k - \tau_\infty^k\|$ with $c \geq \max(1, \|\tau_\infty\|)$, induction on M gives $\|\tau_q^M - \tau_\infty^M\| \leq c((1 + c\delta)^M - 1) \leq c M c \delta e^{M c \delta}$. Since $\langle \phi_M(P_\infty) | \phi_M(P_q) \rangle - 1 = \text{Tr } \tau_q^M - \text{Tr } \tau_\infty^M$, this bounds the overlap itself, and hence also its modulus, which proves the explicit bounds. When $y \leq 1$ it is at most $C e^{Cy}$. When $y > 1$, the overlap is at most $\|\phi_M(P_q)\| = \|\phi_N(A)\|$, and these norms are bounded because $\|\phi_N(A)\|^2 = \text{Tr } \mathcal{E}_A^N$ converges to 1. $\square$

Theorem 30.5.2 (Block lengths below the injectivity lengths). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $x = e^{-\gamma/\xi} > 0$, so that $e^{-\gamma q/\xi} = x^q$ and $e^{-2\gamma/\xi} = x^2$. Let $L_1, \dots, L_b$ be natural numbers, $C \geq 1 + \sum_j x^{-L_j}$, $M \geq 1$, and $y = Mx^q$. If $Cy < 1$, then $x < 1$ and $L_j \leq q$ for every j . Hence, if $\varepsilon \leq 1$ and $\varepsilon \leq Cy e^{Cy}$ holds whenever $x < 1$, $L_j \leq q$ for every j , and $Cy < 1$, then $\varepsilon \leq Cy e^{Cy}$.

Proof. [Stmnt](#) If $x \geq 1$, then $y \geq 1$ and $Cy \geq 1$. If $x < 1$ and $L_j > q$, then $x^{L_j} \leq x^q \leq y$, so $Cy \geq x^{-L_j} y \geq 1$. When $Cy \geq 1$ the bound follows from $\varepsilon \leq 1 \leq Cy \leq Cy e^{Cy}$. $\square$

Lemma 30.5.3 (From a logarithmic threshold to the error). [Lean](#) [Stmnt](#) Let $K, M, \varepsilon > 0$ and $\log K + \log M - \log \varepsilon \leq x$. Then $KMe^{-x} \leq \varepsilon$.

Proof. [Stmnt](#) $KMe^{-x} = e^{\log K + \log M - x} \leq e^{\log \varepsilon} = \varepsilon$. $\square$

Lemma 30.5.4 (From a block-length threshold to the error). [Lean](#) [Stmnt](#) Let $K, a, \varepsilon > 0$, $M, q \geq 1$, $b \geq a \max(\log K, 0)$, and $a \log(Mq/\varepsilon) + b \leq q$. Then $KMe^{-q/a} \leq \varepsilon$.

Proof. [Stmnt](#) Since $\log q \geq 0$, $q/a \geq \log M + \log q - \log \varepsilon + \log K \geq \log K + \log M - \log \varepsilon$; apply Lemma 30.5.3 with $x = q/a$. $\square$

Theorem 30.5.5 (Powers of a perturbed idempotent without first-order term). [Lean](#) [Lean](#) [Stmnt](#) Let e be an idempotent of a normed ring, $f = 1 - e$, and let Z satisfy $eZe = 0$ and $\|fZe\|, \|eZf\|, \|fZf\| \leq z \leq 1/2$. Let Tr be additive with $\text{Tr}(xy) = \text{Tr}(yx)$ and $|\text{Tr } x| \leq K\|x\|$. Then for $M \geq 2$,

$$|\text{Tr}(e + Z)^M - \text{Tr } e| \leq K(\|e\| + 3\|f\|) 2Mz^2 e^{2Mz^2}.$$

Proof. [Stmnt](#) Put $T = e + Z$, $W = fZf$, $\beta = \|fZe\|$, $\beta' = \|eZf\|$, $\zeta = \|W\|$, and $\rho = 1 + \beta\beta'/(1 - \zeta)$. For $P \in \{e, f\}$ the blocks $x_M = PT^M e$ and $y_M = PT^M f$ satisfy $x_{M+1} = x_M + y_M fZe$ and $y_{M+1} = x_M eZf + y_M W$, because $eTe = e$. Induction on M gives $\|eT^M e - e\| \leq \|e\|(\rho^M - 1)$ and $\|eT^M f\| \leq \|e\|\beta'\rho^M/(1 - \zeta)$, and also $\|fT^M e\| \leq \|f\|\beta(\rho^M - \zeta^M)/(1 - \zeta)$ and $\|fT^M f - fW^M\| \leq \|f\|\beta\beta'(\rho^M - \zeta^M)/(1 - \zeta)^2$. Since $\text{Tr } x = \text{Tr}(exe) + \text{Tr}(fxf)$, it follows that $\text{Tr } T^M - \text{Tr } e$ is the sum of the traces of $eT^M e - e$, of $fT^M f - fW^M$, and of fW^M , the last of norm at most $\|f\|\zeta^M \leq \|f\|z^2$. With $z \leq 1/2$ one has $\rho \leq 1 + 2z^2$ and $\beta\beta'/(1 - \zeta)^2 \leq 4z^2$, and $(1 + 2z^2)^M - 1 \leq 2Mz^2 e^{2Mz^2}$. $\square$

Theorem 30.5.6 (Overlap of the positive part with the fixed point, to second order). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let A be normal in the gauge (30.3) with correlation length ξ , and let $0 < \gamma < 1$. There is $C > 0$ such that for all q and all $M \geq 2$ with $Cy < 1$, where $y = M e^{-2\gamma q/\xi}$,

$$1 - |\langle \phi_M(P_\infty) | \phi_M(P_q) \rangle| \leq Cy e^{Cy}.$$

For $M = 1$ there is K with $\|\phi_1(P_q)\|^2 - |\langle \phi_1(P_\infty) | \phi_1(P_q) \rangle|^2 \leq K e^{-2\gamma q/\xi}$ for all q . These bounds are of second order in $e^{-\gamma q/\xi}$. Lemma 30.5.1, the estimate of [195] quoted as [185, eq. (S9)], is of first order, and so is the bound of [190, App. B4, eq. (S52)], which is linear in the residual. The second-order bounds are new here.

Proof. [Stmnt](#) By Theorem 30.3.7, every linear function of the tensor P_q is within $K' e^{-\gamma q/\xi}$ of its value at P_∞ . Let τ_q and τ_∞ be as in the proof of Lemma 30.5.1, so that the overlap is $\text{Tr } \tau_q^M$. The idempotent τ_∞ is the transfer matrix of the rank-one map $X \mapsto \text{Tr}(X)\rho$; hence $\tau_\infty \tau_q \tau_\infty = \alpha \tau_\infty$ with $\alpha = \text{Tr} \sum_i P_q^i \rho P_\infty^{i\dagger} = \langle v_\infty | v_q \rangle$, where v_q and v_∞ are the vectors $(P^i \sqrt{\rho})_i$. Both are unit vectors, because $\mathcal{E}_{P_q} = \mathcal{E}_A^q$ fixes ρ and $\text{Tr } \rho = 1$. Hence $1 - |\alpha| \leq 1 - \text{Re } \alpha = \frac{1}{2} \|v_q - v_\infty\|^2 = O(e^{-2\gamma q/\xi})$, and $|1 - \alpha| \leq \|v_q - v_\infty\|$. Write $\tau_q = \alpha(\tau_\infty + Z)$. Then $\tau_\infty Z \tau_\infty = 0$

and $\|Z\| = O(e^{-\gamma q/\xi})$, so Theorem 30.5.5 bounds $|\text{Tr}(\tau_\infty + Z)^M - 1|$ by a multiple of $y e^{Cy}$, and $|\alpha|^M \geq 1 - M(1 - |\alpha|)$ is at least 1 minus a multiple of y . For $M = 1$, the vector $\phi_1(P_\infty)$ is a unit vector, and $\|u\|^2 - |\langle w|u\rangle|^2 \leq \|u - w\|^2$ for every unit vector w ; here $\|\phi_1(P_q) - \phi_1(P_\infty)\| = O(e^{-\gamma q/\xi})$, since ϕ_1 is linear in the tensor. $\square$

Theorem 30.5.7 (Approximation error, normal case). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Let A be normal in the gauge (30.3) with correlation length ξ , let $|\phi_N\rangle$ be the normalized vector (30.1), and let $0 < \gamma < 1$. For $N = qM$,

$$\varepsilon(\tilde{\phi}_N, \phi_N) = O\left(\frac{N}{q} e^{-2\gamma q/\xi}\right) \quad (30.10)$$

for $q \geq 1$. Explicitly, there is a constant $C > 0$ depending only on A , ρ , and γ such that $\varepsilon(\tilde{\phi}_N, \phi_N) \leq Cy e^{Cy}$ for all $q \geq 0$ and $M \geq 1$, where $y = M e^{-2\gamma q/\xi}$. The rate $e^{-\gamma q/\xi}$ for $0 < \gamma < 1/2$ is [185, Lemma 1, eq. (17), and Lemma 1'(i)], and [190, App. B4, eqs. (S52) and (S53)] gives the rate $e^{-q/\xi}$. The rate $e^{-2\gamma q/\xi}$ for every $0 < \gamma < 1$ and every normal tensor is new here. Numerically, the error decays as $e^{-\gamma q/\xi}$ with $\gamma \approx 2$ [185, Supplemental Material, discussion of Fig. S1], [211, eq. (4)].

Proof. [✓ Proof](#) Write $c_N = \|\phi_N(A)\|$. For q at least the injectivity length of A , the blocked tensor is injective, V is an isometry, and $|\tilde{\phi}_N\rangle = V^{\otimes M}|\phi_M(P_\infty)\rangle$ is normalized. Since $|\phi_N(A)\rangle = V^{\otimes M}|\phi_M(P_q)\rangle$ by Theorem 30.1.3.10, this gives $c_N \langle \tilde{\phi}_N | \phi_N \rangle = \langle \phi_M(P_\infty) | \phi_M(P_q) \rangle =: a$, and $c_N = \|\phi_M(P_q)\|$. With $t = |a|/c_N \leq 1$,

$$\varepsilon = 1 - t \leq 1 - t^2 = \frac{c_N^2 - |a|^2}{c_N^2},$$

which is at most $2(c_N^2 - |a|^2)$ once $c_N^2 \geq 1/2$. The normalization satisfies $|c_N^2 - 1| = |\text{Tr} \mathcal{E}_A^N - 1| \leq K e^{-\gamma' N/\xi}$ for every $0 < \gamma' < 1$; we use it at $\gamma' = \gamma$. For $M = 1$, Theorem 30.5.6 bounds $c_N^2 - |a|^2$ directly. For $M \geq 2$, $c_N^2 - |a|^2 \leq |c_N^2 - 1| + 2 \max(0, 1 - |a|)$, with $|c_N^2 - 1| \leq K e^{-2\gamma q/\xi}$ because $N \geq 2q$, and $1 - |a|$ is bounded by Theorem 30.5.6. The finitely many q below the injectivity length, and the case $Cy \geq 1$, are covered by $\varepsilon \leq 1$. $\square$

In particular, when $\xi > 0$, the choice $q = \lceil \xi(1 + \eta) \ln N / (2\gamma) \rceil$ gives $e^{-2\gamma q/\xi} \leq N^{-(1+\eta)}$ and hence $\varepsilon = O(N^{-\eta})$ for every $\eta > 0$. The choice $q = \lceil 2\xi(1 + \eta) \ln N \rceil$ of [185, paragraph after Lemma 1] corresponds to $\gamma = 1/4$. Since every $\gamma < 1$ is admissible here, the block length can be taken shorter by any factor less than four.

The proof of the lower bound in the Supplemental Material of [185] cuts the chain into blocks “all of the same size, q_N , except for the last one, which may be larger”. The approximating state and its error extend to blocks of any lengths.

Definition 30.5.8 (Approximating state for blocks of prescribed lengths). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)
Cut the ring of N sites into $M \geq 1$ consecutive blocks of lengths $\ell_1, \dots, \ell_M$ with $\sum_k \ell_k = N$, block k consisting of the sites $o_k + 1, \dots, o_k + \ell_k$ with $o_k = \sum_{i < k} \ell_i$. Let $\mathcal{B}_\ell = V_\ell P_\ell$ be the polar decomposition of the ℓ -site blocked tensor, and let ω be a vector on $\mathbb{C}^D \otimes \mathbb{C}^D$. The state

$$|\psi_{\ell, \omega}\rangle = \left(\bigotimes_{k=1}^M V_{\ell_k} \right) \bigotimes_{k=1}^M |\omega\rangle_{R_k L_{k+1}} \quad (30.11)$$

has the coefficient $\sum_c \prod_k \langle s_k | V_{\ell_k} | c_k \rangle \Omega(c)$ at the configuration $s = (s_1, \dots, s_M)$ read block by block, where $c_k = (l_k, r_k)$ and Ω is the pair product state (30.2) of ω . If the q -site blocked tensor

Lemma 30.6.2 (Light cone of a local circuit). Lean ✓ Stmt *Let U be a local circuit of depth T and let Z act on a set of sites X . Then $U^\dagger Z U$ acts on the sites within ring distance T of X . This is the finite light cone of [185, Theorem 1].*

Proof. □ Proof By induction on the layers it suffices to treat one layer W . Split its gates into those whose pair meets X , with product G , and the others, with product H . The gates commute, so $W = HG$. The operator H is unitary and acts on the complement of X , so it commutes with Z and $W^\dagger ZW = G^\dagger ZG$. The pairs of G lie within distance 1 of X , and products and adjoints of operators acting on a set act on it. □

Lemma 30.6.3 (No connected correlations beyond the light cone). [Lean](#) [Lean](#) [✓ Stmt](#) Let $|\psi\rangle$ be a unit vector prepared in depth T , and let Z, Z' act on sets of sites at distance larger than $2T$. Then $\langle\psi|ZZ'|\psi\rangle = \langle\psi|Z|\psi\rangle\langle\psi|Z'|\psi\rangle$. This is the statement in the proof of [185, Theorem 1] that every connected correlation of such operators vanishes.

Proof. ✓ Proof Write $|\psi\rangle = U|\pi\rangle$ with $|\pi\rangle = \bigotimes_i |v_i\rangle$. Since $UU^\dagger = \mathbb{1}$, $\langle\psi|ZZ'|\psi\rangle = \langle\pi|(U^\dagger ZU)(U^\dagger Z'U)|\pi\rangle$, and by Lemma 30.6.2 the two factors act on disjoint sets of sites. For products $\bigotimes_i m_i$ and $\bigotimes_i m'_i$ with disjoint supports the expectation in $|\pi\rangle$ is $\prod_i \langle v_i|m_i m'_i|v_i\rangle$, and at each site one of m_i, m'_i is the identity; bilinearity then gives $\langle\pi|AB|\pi\rangle\langle\pi|\pi\rangle = \langle\pi|A|\pi\rangle\langle\pi|B|\pi\rangle$ for A, B acting on disjoint sets. $\square$

Theorem 30.6.4 (Unitaries on a bounded number of sites). [Lean] [Lean] [Lean] [Lean] ✓ Stmt For $d \geq 1$ and $n \geq 2$ there is K such that every unitary on $(\mathbb{C}^d)^{\otimes n}$ is a product of at most K unitaries, each acting on two neighbouring sites of the open chain of n sites. This is the step that the unitaries with constant support “can be further expressed with a low-depth circuit of local gates” [185, caption of Fig. 1].

Proof.  **Proof** Dividing by a root of the determinant of order d^n , a global phase and hence a single-site gate, reduces to determinant one. Column-by-column Givens elimination writes such a unitary as a product of two-level operators, each a real rotation or a diagonal phase on the span of two basis vectors. Write $G_{ab}(g)$ for the operator acting as the 2×2 matrix g on the span of the basis vectors $|a\rangle, |b\rangle$ and as the identity on their complement, and $Q = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ for the quarter turn. For pairwise distinct a, b, c ,

$$G_{ab}(g) = G_{cb}(Q)^\dagger G_{ac}(g) G_{cb}(Q), \quad (30.14)$$

and choosing c to differ from a at one site fewer than b does reduces a two-level operator between configurations differing at several sites to one between configurations differing at one site fewer. Between configurations differing at one site t it is a single-site operation at t controlled by the other sites. Every rotation and every phase of determinant one is a group commutator $g_1 g_2 g_1^\dagger g_2^\dagger$. Write $C_S(u)$ for the single-site operation u at t controlled by the sites of S , with $t \notin S$; for unitaries v, w and a site $s \neq t$,

$$C_{S \cup \{s\}}(vwv^\dagger w^\dagger) = C_S(v) C_{\{s\}}(w) C_S(v^\dagger) C_{\{s\}}(w^\dagger), \quad (30.15)$$

so a controlled commutator is a product of two operations with one control fewer and two operations on two sites; induction removes the controls. An operation on two distant sites is conjugated to neighbouring sites by SWAP gates. $\square$

Theorem 30.6.5 (The approximating state in depth $O(q)$). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) There is C , depending only on d and D , such that the following holds. Let A be a tensor, let ω be a unit vector on $\mathbb{C}^D \otimes \mathbb{C}^D$, and cut the ring of $N \geq 1$ sites into $M \geq 1$

blocks of lengths $3D \leq \ell_k \leq L$ for which every blocked tensor is injective. Then the state $|\psi_{\ell,\omega}\rangle$ of Definition 30.5.8 is prepared in depth at most CL . In particular, for $\sigma \geq 0$ with $\text{Tr } \sigma = 1$, a block length $q \geq 3D$ for which the q -site blocked tensor is injective, and $M \geq 1$, the approximating state of Definition 30.3.8 on $N = Mq$ sites is prepared in depth at most Cq . This is the depth count behind [185, eq. (1)]; it does not choose q in terms of N and ε .

Proof. ✓ Proof The case $D = 0$ is trivial. If $d \geq 2$ then $D < d^D$. If $d \leq 1$ and $D \geq 1$, injectivity of the blocked tensor gives $D^2 \leq d^q$, which forces $d = D = 1$. In either case $D \leq d^D$, so each bond index can be encoded in D sites. Encode the two bond indices of a block in its first and last D sites, and write $|l, 0, r\rangle$ for the block with l on the first D sites, r on the last D sites, and $|0\rangle$ elsewhere. A first layer of unitaries, one on the last D sites of each block and the first D sites of the next, sends $|0 \cdots 0\rangle$ to $\bigotimes_k |\omega\rangle_{R_k L_{k+1}} |0\rangle_{C_k}$ [185, eq. (12)]; by Theorem 30.6.4 it has constant depth. On each block k , in parallel, a unitary U_k with $U_k |l, 0, r\rangle = V_{\ell_k} |l, r\rangle$ is applied, as in Lemma 30.3.9. SWAP gates move the first D sites next to the last ones in $O(q)$ gates. By Theorem 30.4.2, V_{ℓ} is the ordered product of ℓ isometries, each from the bond space into the bond space and one site; each is extended to a unitary on the bond register of $2D$ sites and one fresh site, and the register moves one site to the left at each step, so the staircase consists of $q - 2D$ unitaries on $2D + 1$ sites followed by one unitary on the first $2D$ sites [185, paragraph “The sequential-RG circuit”]. Each of them has bounded depth by Theorem 30.6.4, so U_k has depth $O(\ell_k) \leq O(L)$, and circuits on disjoint blocks run in the same layers. A configuration contributes to the pair layer applied to $|0 \cdots 0\rangle$ only if it carries a pair on every window and $|0\rangle$ elsewhere, which identifies the output with (30.11). $\square$

Theorem 30.6.6 (Exact preparation on short chains). Lean Lean Lean Lean ✓ Stmt There is C , depending only on d and D , such that for every tensor A and every $N \geq 3D$ for which the N -site blocked tensor is injective and $\langle \phi_N(A) \rangle \neq 0$, the normalized state $|\phi_N\rangle$ is prepared exactly in depth at most CN . For $d \geq 1$ and every n_0 there is K such that every unit vector on a ring of N sites with $2 \leq N \leq n_0$ is prepared in depth at most K .

Proof. ✓ Proof Let $\mathcal{B}_N = VP$. Then $\phi_N(A)(s) = \text{Tr } \mathcal{B}_N^s = \sum_{l,r} \langle s | V | l, r \rangle \text{Tr } P^{(l,r)}$, so $|\phi_N\rangle$ is the state (30.11) of a single block with the pair $\omega(r, l) = \text{Tr } P^{(l,r)} / \|\phi_N(A)\|$, whose right leg is joined to its own left leg around the ring; ω is a unit vector because $|\phi_N\rangle$ is, by Lemma 30.5.9. Theorem 30.6.5 with $M = 1$ and $L = N$ gives the first statement. For the second, a unit vector $|\psi\rangle$ is the first column of a unitary, which by Theorem 30.6.4 is a product of at most K_N gates on neighbouring sites; take $K = \sum_{N \leq n_0} K_N$. $\square$

Lemmas 30.6.2 and 30.6.3 extend from unitary gates to channels, the operations of a local circuit that may add product ancillas and discard them [195].

Definition 30.6.7 (Local channel circuit of depth T). Lean Lean Lean Lean Lean Lean ✓ Stmt A channel on the pair $\{k, k+1\}$ is a map $\rho \mapsto \sum_j K_j \rho K_j^\dagger$ on density matrices of the ring, where each K_j is an operator on the whole ring acting trivially off $\{k, k+1\}$ and $\sum_j K_j^\dagger K_j = \mathbb{1}$. A channel layer is a composition of channels on disjoint pairs, and a local channel circuit of depth T is a composition $\Phi = \Phi_T \circ \dots \circ \Phi_1$ of T channel layers. Its dual $\Phi^\dagger = \Phi_1^\dagger \circ \dots \circ \Phi_T^\dagger$ replaces each channel on a pair by $A \mapsto \sum_j K_j^\dagger A K_j$. A density matrix is prepared in depth T by local channels if it equals $\Phi(\bigotimes_i \sigma_i)$ for a local channel circuit Φ of depth T and density matrices σ_i on single sites.

Lemma 30.6.17 (Light cone of a local channel protocol). [Lean](#) [✓ Stmt](#) *A local channel protocol Ψ of depth T has a dual $\Psi^\dagger$ with $\text{Tr}(\Psi(\rho)A) = \text{Tr}(\rho \Psi^\dagger(A))$ that maps operators acting on a set of sites X to operators acting on the sites within distance T of X , and that satisfies*

$$\Psi^\dagger(ZZ') = \Psi^\dagger(Z) \Psi^\dagger(Z') \quad (30.17)$$

whenever Z and Z' act on sets whose neighbourhoods of radius T are disjoint. This is the light cone of a depth- T circuit in the proof of [185, Theorem 1], here with channels and onsite operations.

Proof. [✓ Proof](#) By induction on the protocol. For an onsite channel this is Lemma 30.6.13 with radius 0; a channel layer adds one to the radius, by Lemma 30.6.9 for the support and as in the proof of Lemma 30.6.10 for (30.17), and an onsite channel keeps it. $\square$

Lemma 30.6.18 (No connected correlations after a conversion). [Lean](#) [✓ Stmt](#) *Let σ be converted in depth T from a product $\bigotimes_i \sigma_i$ of one-site matrices of trace one, and let Z, Z' act on sets of sites at distance larger than $2T$. Then $\text{Tr}(\sigma ZZ') = \text{Tr}(\sigma Z) \text{Tr}(\sigma Z')$. For pure states prepared from a product state by local unitary circuits this is the necessary condition for preparation by quantum circuits of [195].*

Proof. [✓ Proof](#) Write $\rho = \bigotimes_i \sigma_i$ and $\sigma = \Psi(\rho)$ for a local channel protocol Ψ of depth $T' \leq T$, and let $\Psi^\dagger$ be its dual from Lemma 30.6.17, of radius T' . If Z and Z' act on sets at distance larger than $2T \geq 2T'$, then $\Psi^\dagger(Z)$ and $\Psi^\dagger(Z')$ act on disjoint sets, and (30.17) together with the factorization of the product density ρ gives

$$\begin{aligned} \text{Tr}(\sigma ZZ') &= \text{Tr}(\rho \Psi^\dagger(ZZ')) = \text{Tr}(\rho \Psi^\dagger(Z) \Psi^\dagger(Z')) \\ &= \text{Tr}(\rho \Psi^\dagger(Z)) \text{Tr}(\rho \Psi^\dagger(Z')) = \text{Tr}(\sigma Z) \text{Tr}(\sigma Z'). \end{aligned}$$

$\square$

Theorem 30.6.19 (Preparation in depth $O(\log(N/\varepsilon))$). [Lean](#) [Lean](#) [✓ Stmt](#) *Let A be normal. There is a constant c depending only on A such that, for every $N \geq 2$ and $0 < \varepsilon \leq 1$ for which the periodic state $|\phi_N(A)\rangle$ is nonzero, a state $|\psi_N\rangle$ with $\varepsilon(\psi_N, \phi_N) \leq \varepsilon$ is prepared in depth at most $c \log(N/\varepsilon)$. If $D \geq 1$, there is N_0 such that $|\phi_N(A)\rangle \neq 0$ for every $N \geq N_0$, so for $N \geq N_0$ the condition can be dropped. This is [185, eq. (1)]; the source normalizes $|\phi_N\rangle = |\phi_N(A)\rangle/c_N$ with $c_N > 0$, and for some normal tensors $|\phi_2(A)\rangle = 0$ [109].*

Proof. [✓ Proof](#) Bring A into the gauge (30.3) as in the proof of Theorem 30.6.31, with fixed point σ , a bound $t < 1$ on the moduli of the eigenvalues other than 1, and $r = -\frac{1}{4} \log t$; the error is unchanged by Lemma 30.6.26. Let K be the constant of Theorem 30.5.10 at $\gamma = 1/4$, so the error is at most KMe^{-rq} , and let L_0 be a length at which the blocked tensor is injective. Put $a = 1/r$, $b = \max(\log K, 0)/r + L_0 + 3D + 1$, $Q = a \log(N/\varepsilon) + b$ and $q = \lceil Q \rceil$, so $Q \leq q < 2Q$ and $Q \leq (a + b/\log 2) \log(N/\varepsilon)$.

If $q \leq N$, write $N = (M-1)q + q'$ with $M = \lfloor N/q \rfloor$ and $q' = q + (N \bmod q)$, so $q \leq q' < 2q$, and cut the ring into $M-1$ blocks of length q followed by one of length q' , as in the Supplemental Material of [185], proof of Theorem 1. All blocked tensors are injective since $q \geq L_0$, and Theorem 30.5.10 gives error at most $KMe^{-rq} \leq KN e^{-rQ} = K e^{-rb} \varepsilon \leq \varepsilon$. By Theorem 30.6.5 with $L = 2q$ the depth is at most $2Cq \leq 4CQ$.

If $N < q$, then $N < Q$. If $N \geq L_0 + 3D$, the state $|\phi_N\rangle$ is prepared exactly in depth at most $C'N \leq C'Q$ by Theorem 30.6.6; otherwise N is bounded by a constant depending only on A , and the same theorem prepares $|\phi_N\rangle$ exactly in depth at most $K' \leq (K'/\log 2) \log(N/\varepsilon)$. In every case the depth is at most $c \log(N/\varepsilon)$ with $c = (4C + C')(a + b/\log 2) + K'/\log 2$. The second statement follows from Lemma 30.6.20. $\square$

Lemma 30.6.20 (The periodic state does not vanish on long chains). [Lean](#) [✓ Stmt](#) Let A be normal with $D \geq 1$. There is N_0 such that $|\phi_N(A)\rangle \neq 0$ for every $N \geq N_0$.

Proof. [✓ Proof](#) In the gauge (30.3), which multiplies $|\phi_N(A)\rangle$ by a nonzero scalar, $\|\phi_N(A)\|^2 = \text{Tr } \mathcal{E}_A^N$ differs from 1 by at most $Ke^{-2\gamma N/\varepsilon}$, as in the proof of Theorem 30.5.7, which is less than 1 for large N . $\square$

Lemma 30.6.21 (Inverses and series composition of local circuits). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a layer W with gates g_k , the layer with the gates $g_k^\dagger$ on the same pairs implements $W^\dagger$, and the same holds for the product of the gates of W over any subset of its pairs. If a circuit has the layers $W_1, \dots, W_T$, applied in this order, and implements U , then the circuit with the layers $W_T^\dagger, \dots, W_1^\dagger$ implements $U^\dagger$; the layers of a circuit implementing U followed by those of a circuit implementing U' implement $U'U$. Consequently, a local circuit of depth T is unitary, and its adjoint is a local circuit of depth T . If U and U' are local circuits of depths T and T' , then $U'U$ is a local circuit of depth $T + T'$.

Proof. [✓ Proof](#) If $U = W_T \cdots W_1$ with layers W_k , then $U^\dagger = W_1^\dagger \cdots W_T^\dagger$, and $W_k^\dagger$ is the layer whose gates are the adjoints of the gates of W_k , on the same pairs. The layers of $U'U$ are those of U followed by those of U' . $\square$

Lemma 30.6.22 (Preparation from the all-zero state). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

Let $d \geq 1$. For one-site operators m_i and vectors $|v_i\rangle$ in $\mathbb{C}^d$, $(\bigotimes_i m_i)(\bigotimes_i |v_i\rangle) = \bigotimes_i m_i |v_i\rangle$, and $\bigotimes_i c_i |v_i\rangle = (\prod_i c_i) \bigotimes_i |v_i\rangle$ for scalars c_i . Every nonzero $x \in \mathbb{C}^d$ equals $cu|0\rangle$ for a unitary u and a scalar $c \neq 0$, and every nonzero product vector is a scalar multiple of $(\bigotimes_i u_i)|0 \cdots 0\rangle$ for one-site unitaries u_i . A tensor product $\bigotimes_i u_i$ of one-site unitaries is a local circuit of depth 2. Every nonzero vector prepared in depth T is a scalar multiple of $U|0 \cdots 0\rangle$ for a local circuit U of depth $T + 2$.

Proof. [✓ Proof](#) The first layer applies $u_k \otimes u_{k+1}$ on the pairs $\{k, k+1\}$ with k odd and $k < N$, and the second applies u_N when N is odd. A nonzero product vector $\bigotimes_i |v_i\rangle$ has $v_i \neq 0$ for every i , and $|v_i\rangle = c_i u_i |0\rangle$ for a unitary u_i whose first column is $|v_i\rangle/\|v_i\|$ and $c_i = \|v_i\|$. Hence $\bigotimes_i |v_i\rangle = (\prod_i c_i)(\bigotimes_i u_i)|0 \cdots 0\rangle$, and Lemma 30.6.21 composes the two circuits. $\square$

Lemma 30.6.23 (Triangle step for the error). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For unit vectors a, b in a complex inner product space and every scalar c with $|c| = 1$, $\varepsilon(a, b) \leq \|a - cb\|^2/2$, with equality for some such c . For unit vectors x, y, z , $\varepsilon(x, z) \leq 2(\varepsilon(x, y) + \varepsilon(y, z))$. A unitary matrix U satisfies $\overline{(Ux)} \cdot (Uy) = \bar{x} \cdot y$ for coordinate vectors x, y , and so preserves inner products and norms.

Proof. [✓ Proof](#) For unit vectors a, b and $|c| = 1$, $\|a - cb\|^2 = 2 - 2\text{Re}(c\langle a|b\rangle) \geq 2\varepsilon(a, b)$, with equality for $c = \langle a|b\rangle/|\langle a|b\rangle|$ (any c if $\langle a|b\rangle = 0$). Choose c_1, c_2 attaining equality for (x, y) and (y, z) . Then $\|x - c_1 c_2 z\| \leq \|x - c_1 y\| + \|y - c_2 z\|$, and $(s + t)^2 \leq 2(s^2 + t^2)$ gives $2\varepsilon(x, z) \leq \|x - c_1 c_2 z\|^2 \leq 4(\varepsilon(x, y) + \varepsilon(y, z))$. $\square$

Corollary 30.6.24 (Normal states are transformed into each other in depth $O(\log(N/\varepsilon))$).

[Lean](#) [Lean](#) [✓ Stmt](#) Let A and B be normal tensors with the same physical dimension d . There is a constant c depending only on A and B such that, for every $N \geq 2$ and $0 < \varepsilon \leq 1$ for which $|\phi_N(A)\rangle$ and $|\phi_N(B)\rangle$ are nonzero, some local circuit U of depth at most $c \log(N/\varepsilon)$ satisfies $\varepsilon(U\hat{\phi}_N(A), \hat{\phi}_N(B)) \leq \varepsilon$, where $\hat{\phi}_N(A) = |\phi_N(A)\rangle/\|\phi_N(A)\|$ and $\hat{\phi}_N(B) = |\phi_N(B)\rangle/\|\phi_N(B)\|$ are the normalized states. If both bond dimensions are at least 1, there are c and N_0 such that the same holds for every $N \geq N_0$ and $0 < \varepsilon \leq 1$ without the nonvanishing condition.

The paragraph “Discussion and outlook” of [185] states that its results imply that states in the same phase are transformed into each other by a circuit of logarithmic depth. Normal states lie in the trivial phase of the classification without symmetry; the corollary covers only this case. Non-normal states and phases defined with a symmetry are not treated [35].

Proof. ✓ Proof By Theorem 30.6.19 with error $\varepsilon/4$ and Lemma 30.6.22, there are local circuits U_A , U_B and unit vectors $\psi_A = aU_A|0 \cdots 0\rangle$, $\psi_B = bU_B|0 \cdots 0\rangle$ with $\varepsilon(\psi_A, \hat{\phi}_N(A))$ and $\varepsilon(\psi_B, \hat{\phi}_N(B))$ at most $\varepsilon/4$, of depths $T_A + 2$ and $T_B + 2$ with T_A, T_B at most a constant times $\log(4N/\varepsilon) \leq 3 \log(N/\varepsilon)$. Put $U = U_B U_A^\dagger$, a local circuit of depth $T_A + T_B + 4$ by Lemma 30.6.21; the term 4 is at most $(4/\log 2) \log(N/\varepsilon)$. Then $U\psi_A = (a/b)\psi_B$ is a unit vector, and since U is unitary, $\varepsilon(U\hat{\phi}_N(A), (a/b)\psi_B) = \varepsilon(\hat{\phi}_N(A), \psi_A)$. Lemma 30.6.23 gives $\varepsilon(U\hat{\phi}_N(A), \hat{\phi}_N(B)) \leq 2(\varepsilon/4 + \varepsilon/4) = \varepsilon$. The second statement follows from Lemma 30.6.20. $\square$

The following theorem is the case of equal blocks, for chain lengths that a block length of order $\log(N/\varepsilon)$ divides, as in the N/q equal blocks of [185, eq. (10)]; it was the first formalized case of Theorem 30.6.19 [108].

Lemma 30.6.25 (The approximating state is a unit vector). Lean Lean ✓ Stmt Let $\sigma \geq 0$ with $\text{Tr } \sigma = 1$, let the q -site blocked tensor $\mathcal{B}_q$ be injective, and let $M \geq 1$. Then $\|\phi_M(VP_\infty)\| = 1$, so the approximating state $|\tilde{\phi}_N\rangle$ of Definition 30.3.8 equals $|\phi_M(VP_\infty)\rangle$ and is a unit vector.

Proof. ✓ Proof The squared norm $\|\phi_M(VP_\infty)\|^2$ is the sum of the squared moduli of its coefficients, which equals 1 by the norm identity recorded with Definition 30.3.8. $\square$

Lemma 30.6.26 (Rescaling changes the normalized state by a phase). Lean ✓ Stmt Let A and B be tensors with $|\phi_N(B)\rangle = \zeta^N |\phi_N(A)\rangle$ for some $\zeta \neq 0$, and write $|\hat{\phi}_N(A)\rangle$ for the normalization of $|\phi_N(A)\rangle$ (zero when $|\phi_N(A)\rangle = 0$). Then every vector $|\psi\rangle$ on N sites satisfies $|\langle \psi | \hat{\phi}_N(B) \rangle| = |\langle \psi | \hat{\phi}_N(A) \rangle|$, so the error of $|\psi\rangle$ is the same against both normalized states.

Proof. ✓ Proof $|\hat{\phi}_N(B)\rangle = (\zeta^N / |\zeta^N|) |\hat{\phi}_N(A)\rangle$, and if $|\phi_N(A)\rangle = 0$ both normalized states vanish. $\square$

Lemma 30.6.27 (The error at block length $q \geq a \log(Mq/\varepsilon) + b_0$). Lean ✓ Stmt Let B be a normal tensor in the gauge (30.3), with a positive definite fixed point σ of trace one, and let A be a tensor with $|\phi_N(B)\rangle = \zeta^N |\phi_N(A)\rangle$ for all N and some $\zeta \neq 0$. Let $0 < t < 1$ bound the moduli of the eigenvalues of $\mathcal{E}_B$ other than 1, and let $\xi = -1/\log t$. For every $a > \xi/2$ there is $b_0 \geq 0$ such that for $\varepsilon > 0$, every $q \geq 1$ and every $M \geq 1$ with $q \geq a \log(Mq/\varepsilon) + b_0$, the approximating state of B with blocks of q sites on $N = Mq$ sites has error at most ε against $|\phi_N(A)\rangle$.

Proof. ✓ Proof Theorem 30.5.7 at $\gamma = \xi/(2a) < 1$ gives K with error at most $KMe^{-q/a}$; take $b_0 = a \max(\log K, 0)$ and apply Lemma 30.5.4. The error against $\phi_N(A)$ equals that against $\phi_N(B)$ by Lemma 30.6.26. $\square$

Theorem 30.6.28 (Block length $a \log(N/\varepsilon) + b$ for every slope $a > \xi/2$). Lean ✓ Stmt There is C , depending only on d and D , such that the following holds. Let B be a normal tensor in the gauge (30.3), with a positive definite fixed point σ of trace one, and let A be a tensor with $|\phi_N(B)\rangle = \zeta^N |\phi_N(A)\rangle$ for all N and some $\zeta \neq 0$. Let $0 < t < 1$ bound the moduli of the eigenvalues of $\mathcal{E}_B$ other than 1, and let $\xi = -1/\log t$. For every $a > \xi/2$ there is $b \geq 1$ such that for $0 < \varepsilon \leq 1$, every $N \geq 1$ and every q dividing N with $q \geq a \log(N/\varepsilon) + b$, a unit vector $|\psi\rangle$ with $\varepsilon(\psi, \phi_N(A)) \leq \varepsilon$ is prepared in depth at most Cq . The block length $q = \lceil 2\xi(1 + \eta) \ln N \rceil$ of [185, paragraph after Lemma 1] has slope 2ξ , and [190, App. B4, eq. (S54)] takes $q = O(\xi \log(N/\varepsilon))$. The slopes $a > \xi/2$ are new here.

Proof. [✓ Proof](#) Since $a > \xi/2$, the number $\gamma = \xi/(2a)$ satisfies $0 < \gamma < 1$, and Theorem 30.5.7 at this γ gives K with $\varepsilon(\phi_N, \phi_N(B)) \leq K(N/q)e^{-q/a}$, since $2\gamma/\xi = 1/a$. Take $b = a \max(\log K, 0) + L_0 + 3D + 1$, with L_0 a length at which the blocked tensor is injective. Then $q \geq b$ makes the blocked tensor $\mathcal{B}_q$ injective by Theorem 6.1.1.24 and Lemma 6.5.3, and $q \geq 3D$. Moreover $q/a \geq \log(N/\varepsilon) + \log K \geq \log(N/(q\varepsilon)) + \log K$, so $K(N/q)e^{-q/a} \leq \varepsilon$. The approximating state is a unit vector by Lemma 30.6.25, is prepared in depth at most Cq by Theorem 30.6.5, and has the same error against $\phi_N(A)$ as against $\phi_N(B)$ by Lemma 30.6.26. $\square$

Lemma 30.6.29 (A gap for a normal tensor). [Lean](#) [✓ Stmt](#) For a normal left-canonical tensor A with $D \geq 1$ there is $0 < t < 1$ bounding the moduli of the eigenvalues other than 1 of $\mathcal{E}_A$.

Proof. [✓ Proof](#) The transfer map is a channel whose only eigenvalue of modulus 1 is 1, and it has finitely many eigenvalues, so the others are at most $1 - \delta$ in modulus for some $\delta > 0$; take $t = \max(1 - \delta, 1/2)$. $\square$

Lemma 30.6.30 (Depth from the window of block lengths). [Lean](#) [✓ Stmt](#) Let C, q be natural numbers, $N \geq 2$, $0 < \varepsilon \leq 1$, $b \geq 1$, and $q \leq 2(a \log(N/\varepsilon) + b)$. Then $Cq \leq C(2a + 2b/\log 2) \log(N/\varepsilon)$.

Proof. [✓ Proof](#) Since $N/\varepsilon \geq 2$, $\log(N/\varepsilon) \geq \log 2$, so $b \leq (b/\log 2) \log(N/\varepsilon)$, and $Cq \leq 2C(a \log(N/\varepsilon) + b) \leq C(2a + 2b/\log 2) \log(N/\varepsilon)$. $\square$

Theorem 30.6.31 (Preparation in depth $O(\log(N/\varepsilon))$, block length dividing N). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) There is C , depending only on d and D , such that the following holds. Let A be normal with $D \geq 1$. There are $a > 0$ and $b \geq 1$, depending only on A , such that for $0 < \varepsilon \leq 1$, every $N \geq 1$ and every q dividing N with $q \geq a \log(N/\varepsilon) + b$, a unit vector $|\psi\rangle$ with $\varepsilon(\psi, \phi_N) \leq \varepsilon$ is prepared in depth at most Cq . If moreover $N \geq 2$ and $q \leq 2(a \log(N/\varepsilon) + b)$, for instance $q = \lceil a \log(N/\varepsilon) + b \rceil$ when it divides N , the depth is at most $c \log(N/\varepsilon)$ with c depending only on A .

Proof. [✓ Proof](#) A gauge transformation and a rescaling bring A into the gauge (30.3) by Theorem 12.8.8, and the positive definite fixed point ρ it provides is replaced by $\sigma = \rho/\text{Tr } \rho$, still a fixed point of trace one, by Theorem 6.6.2; they multiply $|\phi_N(A)\rangle$ by a nonzero scalar. The new tensor is normal by Theorems 12.8.1 and 2.5.7. By Theorem 7.6.3 its transfer map is primitive, so 1 is its only eigenvalue of modulus 1. The gauge is left canonical, so the transfer map is a channel by Theorem 2.2.8, and all its eigenvalues have modulus at most 1 by Theorem 5.3.1, and finitely many of them give a bound $0 < t < 1$ on the moduli of the other eigenvalues. With $\xi = -1/\log t > 0$, Theorem 30.6.28 at the slope $a = \xi > \xi/2$ gives $b \geq 1$ and the first statement. For the second statement, Lemma 30.6.30 gives $Cq \leq C(2a + 2b/\log 2) \log(N/\varepsilon)$. $\square$

For an operator X on $(\mathbb{C}^d)^{\otimes L}$, let $\mathcal{E}_X: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be the linear map $Z \mapsto \sum_{w, w'} \langle w' | X | w \rangle A^w Z (A^{w'})^\dagger$, where w, w' run over words of length L and A^w denotes the product of matrices of A along w .

Lemma 30.6.32 (Inserted transfer maps of products and adjoints). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For operators X, X' on $(\mathbb{C}^d)^{\otimes L_1}$ and Y, Y' on $(\mathbb{C}^d)^{\otimes L_2}$, with $X \otimes Y$ acting on $L_1 + L_2$ consecutive sites,

$$\begin{aligned} (X \otimes Y)(X' \otimes Y') &= XX' \otimes YY', & \mathcal{E}_{X \otimes Y} &= \mathcal{E}_X \mathcal{E}_Y, \\ \mathcal{E}_1 &= \mathcal{E}_A^L, & \mathcal{E}_{X^\dagger}(Z) &= \mathcal{E}_X(Z^\dagger)^\dagger. \end{aligned}$$

For an operator O on $(\mathbb{C}^d)^{\otimes N}$ and the unnormalized vector $|\phi_N(A)\rangle$ of (30.1), $\langle \phi_N(A) | O | \phi_N(A) \rangle = \text{tr } \mathcal{E}_O$, the trace of a linear map on $M_D(\mathbb{C})$.

Proof. ✓ Proof The Vandermonde map $v \mapsto (\sum_j v_j \mu_j^u)_{0 \leq u < K}$ is injective, so it is bounded below on $\mathbb{C}^K$. The vector $(a_j \mu_j^t)_j$ has the maximum norm of a . $\square$

Lemma 30.6.37 (Expectations in overlapping states). Lean ✓ Stmt For a Hermitian operator Z and unit vectors ϕ, ψ , with $\sigma_\chi(Z) = \|(Z - \langle Z \rangle_\chi)\chi\|$, the inequality (30.22) holds: $|\langle \phi | \psi \rangle| |\langle Z \rangle_\phi - \langle Z \rangle_\psi| \leq \sigma_\phi(Z) + \sigma_\psi(Z)$.

Proof. ✓ Proof Since $\langle Z \rangle_\phi$ is real, $(\langle Z \rangle_\phi - \langle Z \rangle_\psi) \langle \phi | \psi \rangle = \langle \phi | (Z - \langle Z \rangle_\psi) | \psi \rangle - \langle \phi | (Z - \langle Z \rangle_\phi) | \psi \rangle$; apply the Cauchy–Schwarz inequality to both terms. $\square$

Lemma 30.6.38 (Hermitian observables for the second eigenvalue). Lean Lean ✓ Stmt Suppose the products of L matrices of A span $M_D(\mathbb{C})$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, let ρ be Hermitian with $\text{Tr } \rho = 1$, put $P(Z) = \text{Tr}(Z) \rho$, and let $\lambda \neq 0, 1$ be an eigenvalue of $\mathcal{E}_A$. Then there are Hermitian operators $\mathcal{O}, \mathcal{O}'$ on $(\mathbb{C}^d)^{\otimes L}$ of norm one, an integer $K \in \{1, 2\}$ with $K = 1$ when λ is real, distinct complex numbers $\mu_1, \dots, \mu_K$ of modulus one and a nonzero $a \in \mathbb{C}^K$ such that for every $t \geq 0$

$$\text{Tr} \left(\mathcal{E}_\mathcal{O}((\mathcal{E}_A - P)^t (\mathcal{E}_{\mathcal{O}'}(\rho))) \right) = |\lambda|^t \sum_{j=1}^K a_j \mu_j^t.$$

This replaces the choice of observables in the proof of [185, Lemma 2] by one with Hermitian observables: for non-real λ the eigenvector is not Hermitian, and a Hermitian observable pairs it with its adjoint, an eigenvector for $\bar{\lambda}$.

Proof. ✓ Proof This is the second and third paragraph of the proof of Theorem 30.6.40: a Hermitian vector v and a functional ℓ with $\ell(Z^\dagger) = \overline{\ell(Z)}$ are built from an eigenvector R and its adjoint, the maps $Z \mapsto \text{Tr}(Z) v$ and $Z \mapsto \ell(Z) \rho$ are realized by Hermitian observables through Lemma 30.6.33, and scaling to norm one multiplies a by a positive constant. $\square$

Lemma 30.6.39 (Finite-size corrections). Lean Lean Lean Lean Lean ✓ Stmt Let A be in the gauge (30.3), with the products of $L \geq 1$ matrices spanning $M_D(\mathbb{C})$, let $P(Z) = \text{Tr}(Z) \rho$, and let λ_2 be an eigenvalue of $\mathcal{E}_A$ of largest modulus among those different from 1.

1. For every $r > |\lambda_2|$ there is $C_r > 0$ with $\|(\mathcal{E}_A - P)^n\| \leq C_r r^n$ for all $n \geq 0$.
2. For $t, n \geq 1$ and $N = 2L + t + n$, the connected correlator $G_N(X, Y; L + t + 1)$ of (30.18) equals $(\alpha\beta + G_\infty + \delta_{XY})/(1 + \delta) - (\alpha + \delta_X)(\beta + \delta_Y)/(1 + \delta)^2$, where $\alpha = \text{Tr } \mathcal{E}_X(\rho)$, $\beta = \text{Tr } \mathcal{E}_Y(\rho)$, G_∞ is the limit correlator (30.20) at separation t , and $\delta, \delta_X, \delta_Y, \delta_{XY}$ are traces of linear maps containing $(\mathcal{E}_A - P)^n$, $(\mathcal{E}_A - P)^N$ or a power of $\mathcal{E}_A - P$ of exponent at least n .
3. If $|z - 1|$, $|n_{XY} - (ab + g)|$, $|n_X - a|$ and $|n_Y - b|$ are at most $\varepsilon \leq \frac{1}{2}$ and $|a|, |b|, |g| \leq M$, then $|n_{XY}/z - (n_X/z)(n_Y/z) - g| \leq 4(M + 2)^2 \varepsilon$.
4. The trace of a linear map on $M_D(\mathbb{C})$ is bounded by a constant depending only on D times its operator norm, and $e^{-(s-1)/\xi} \leq |\lambda_2|^t$ whenever $t + 1 \leq s$ and $0 < |\lambda_2| < 1$.

Proof. ✓ Proof A nonzero eigenvalue of $\mathcal{E}_A - P$ is an eigenvalue of $\mathcal{E}_A$ with a traceless eigenvector, hence different from 1 by strong irreducibility (Theorem 6.1.1.15), so the spectral radius of $\mathcal{E}_A - P$ is at most $|\lambda_2|$ and Gelfand’s formula gives (1). Part (2) substitutes $\mathcal{E}_A^k = P + (\mathcal{E}_A - P)^k$ for $k \geq 1$ into the trace formula of Lemma 30.6.34 and into the one-point functions and the normalization. Part (3) is elementary algebra with $|1/z| \leq 2$, and part (4) expands the trace in matrix units and uses $\xi = -1/\log|\lambda_2|$. $\square$

Theorem 30.6.40 (Exponentially decaying correlations). Lean Lean Lean Lean ✔ Stmt Let A be normal and in the gauge (30.3), with correlation length $\xi > 0$, let $L \geq 1$ be a length such that the products of L matrices of A span $M_D(\mathbb{C})$ (one may take $L = 1$ when A is injective), and let $|\phi_N\rangle$ be the normalized vectors (30.1). For operators X, Y on $(\mathbb{C}^d)^{\otimes L}$ let X_1 act on the sites $1, \dots, L$, let Y_s act on the sites $s, \dots, s + L - 1$, and put

$$G_N(X, Y; s) = \langle \phi_N | X_1 Y_s | \phi_N \rangle - \langle \phi_N | X_1 | \phi_N \rangle \langle \phi_N | Y_s | \phi_N \rangle. \quad (30.18)$$

There are Hermitian operators $\mathcal{O}, \mathcal{O}'$ on $(\mathbb{C}^d)^{\otimes L}$ of norm one, a constant $c > 0$, an integer $K \in \{1, 2\}$ and an integer $s_0 \geq L + 2$ such that for every $s \geq s_0$ some $s' \in \{s, s + 1, \dots, s + K - 1\}$ satisfies, for every $N \geq 3s'$,

$$|G_N(\mathcal{O}, \mathcal{O}'; s')| \geq c e^{-(s'-1)/\xi}. \quad (30.19)$$

One may take $K = 1$ when λ_2 is real and $K = 2$ otherwise. This is a Hermitian variant of the correlation estimate of [185, Lemma 2], with the separation s ranging over windows of K consecutive values and the connected correlator. The source's own observables cannot be Hermitian when λ_2 is not real; for them the bound holds in modulus at every separation, while the printed real inequality fails when λ_2 is not real or is negative [46].

Proof. ✔ Proof With the inserted transfer maps $\mathcal{E}_X$ defined before Lemma 30.6.33, and for operators X, Y on consecutive blocks,

$$\mathcal{E}_{\mathbb{1}} = \mathcal{E}_A^L, \quad \mathcal{E}_{X \otimes Y} = \mathcal{E}_X \mathcal{E}_Y.$$

Expanding the traces in (30.1) gives, for $L < s$ and $s + L - 1 \leq N$,

$$\langle \phi_N(A) | X_1 Y_s | \phi_N(A) \rangle = \text{tr}(\mathcal{E}_X \mathcal{E}_A^{s-L-1} \mathcal{E}_Y \mathcal{E}_A^{N-s-L+1}),$$

the trace of a linear map on $M_D(\mathbb{C})$. Let $P(Z) = \text{Tr}(Z) \rho$. In the gauge (30.3), $\mathcal{E}_A^n = P + (\mathcal{E}_A - P)^n$ for $n \geq 1$. An eigenvalue $\nu \neq 0$ of $\mathcal{E}_A - P$ is an eigenvalue of $\mathcal{E}_A$ with a traceless eigenvector, so $\nu \neq 1$ because $\mathcal{E}_A$ is strongly irreducible (Theorem 6.1.1.15), and hence $|\nu| \leq |\lambda_2|$. Fix r with $|\lambda_2| < r < 1$ and $r^2 < |\lambda_2|$; Gelfand's formula gives $\|(\mathcal{E}_A - P)^n\| \leq C_r r^n$. Replacing the last power $\mathcal{E}_A^{N-s-L+1}$ by P , and likewise in the normalization $\text{tr} \mathcal{E}_A^N$ and in the one-point functions, changes $G_N(X, Y; s)$ by at most $C_{X,Y} r^{N-s-L+1}$, which is at most $C_{X,Y} r^{2(s-L-1)}$ when $N \geq 3s$. The remaining term is

$$G_\infty(X, Y; s) = \text{Tr}(\mathcal{E}_X((\mathcal{E}_A - P)^{s-L-1}(\mathcal{E}_Y(\rho))))). \quad (30.20)$$

Let $R \neq 0$ satisfy $\mathcal{E}_A(R) = \lambda_2 R$. Since $\xi > 0$, $\lambda_2 \neq 0$, and trace preservation gives $\text{Tr} R = 0$, so $(\mathcal{E}_A - P)^t R = \lambda_2^t R$; since $\mathcal{E}_A(Z^\dagger) = \mathcal{E}_A(Z)^\dagger$, also $(\mathcal{E}_A - P)^t R^\dagger = \bar{\lambda}_2^t R^\dagger$. If λ_2 is real, one of $R + R^\dagger$ and $i(R - R^\dagger)$ is a nonzero Hermitian eigenvector H ; put $v = H$ and $\ell(Z) = \text{Tr}(HZ)/\text{Tr}(H^2)$, so that $\ell((\mathcal{E}_A - P)^t v) = \lambda_2^t$. If λ_2 is not real, R and $R^\dagger$ are eigenvectors for distinct eigenvalues; take a linear functional f with $f(R) = 1$ and $f(R^\dagger) = 0$, put $v = R + R^\dagger$ and $\ell(Z) = f(Z) + \overline{f(Z^\dagger)}$, so that $\ell((\mathcal{E}_A - P)^t v) = \lambda_2^t + \bar{\lambda}_2^t$. In both cases v is Hermitian, $\ell(Z^\dagger) = \overline{\ell(Z)}$, and $\ell((\mathcal{E}_A - P)^t v) = |\lambda_2|^t \sum_{j=1}^K a_j \mu_j^t$ with $K \leq 2$, distinct μ_j of modulus one and $a \neq 0$.

The maps $Z \mapsto \text{Tr}(Z) v$ and $Z \mapsto \ell(Z) \rho$ preserve Hermitian matrices. By Lemma 30.6.33 they are $\mathcal{E}_Y$ and $\mathcal{E}_X$ for some X, Y , and since $\mathcal{E}_{X^\dagger}(Z) = \mathcal{E}_X(Z^\dagger)^\dagger$ the Hermitian parts of X and Y have the same maps. Then $G_\infty(X, Y; s) = |\lambda_2|^t \sum_j a_j \mu_j^t$ with $t = s - L - 1$, and $X, Y \neq 0$

because this sum does not vanish identically. Take $\mathcal{O}, \mathcal{O}'$ to be X, Y scaled to norm one; this multiplies a by a positive constant.

The Vandermonde matrix $W = (\mu_j^u)_{0 \leq u < K, 1 \leq j \leq K}$ is invertible because the μ_j are distinct. For every t the vector with entries $\sum_j a_j \mu_j^{t+u}$, $0 \leq u < K$, is W applied to $(a_j \mu_j^t)_j$, whose maximum norm equals that of a because $|\mu_j| = 1$. Hence some $u \in \{0, \dots, K-1\}$ satisfies

$$\left| \sum_{j=1}^K a_j \mu_j^{t+u} \right| \geq c_0, \quad c_0 = \frac{\max_j |a_j|}{\|W^{-1}\|_\infty} > 0, \quad (30.21)$$

so $|G_\infty(\mathcal{O}, \mathcal{O}'; s+u)| \geq c_0 |\lambda_2|^{t'}$ with $t' = t+u$. With $s' = s+u$ and $N \geq 3s'$ the finite-size correction is at most $C r^{2t'} = C (r^2/|\lambda_2|)^{t'} |\lambda_2|^{t'}$, which is at most $\frac{c_0}{2} |\lambda_2|^{t'}$ after enlarging s_0 , since $r^2 < |\lambda_2|$. Finally $|\lambda_2|^{t'} = e^{-(s'-L-1)/\xi} \geq e^{-(s'-1)/\xi}$, which gives (30.19) with $c = c_0/2$. $\square$

Theorem 30.6.41 (No preparation in depth $o(\log N)$). Lean Lean Stmt *Let A be normal with correlation length $\xi > 0$. Here $\xi = -1/\log|\lambda_2/\lambda_1|$, where λ_1 is an eigenvalue of $\mathcal{E}_A$ of largest modulus and λ_2 is an eigenvalue of largest modulus among those different from λ_1 ; in the normalization $\lambda_1 = 1$ this is Definition 14.2.4. Let $|\phi_N\rangle$ be the normalized vectors (30.1), and let $|\psi_N\rangle$ be prepared in depth T_N with $T_N = o(\log N)$. Then there is N_0 such that $\varepsilon(\phi_N, \psi_N) > 1/2$ for all $N > N_0$. This is [185, Theorem 1].*

Proof. Proof We may assume that A is in the gauge (30.3). Since A is normal, $\mathcal{E}_A$ is strongly irreducible (Theorem 6.1.1.15), so its spectral radius λ_1 is a positive eigenvalue whose right eigenvector ρ and whose eigenvector σ for the dual map are positive definite. Replacing A^i by $\lambda_1^{-1/2} \sigma^{1/2} A^i \sigma^{-1/2}$ gives (30.3), with fixed point $\sigma^{1/2} \rho \sigma^{1/2}$ scaled to unit trace. The scaling multiplies $|\phi_N\rangle$ by a positive constant and the conjugation leaves every trace in (30.1) unchanged, so $\phi_N/\|\phi_N\|$ is unchanged; the eigenvalues of $\mathcal{E}_A$ are divided by λ_1 , which is the normalization $\lambda_1 = 1$ in which ξ is computed.

Take $L, \mathcal{O}, \mathcal{O}', c, K, s_0$ from Theorem 30.6.40, and suppose $\varepsilon(\phi_N, \psi_N) \leq 1/2$, that is $|\langle \phi_N | \psi_N \rangle| \leq 1/2$. Write $T = T_N$ and $|\psi_N\rangle = U|\pi\rangle$ with U a local circuit of depth T and $|\pi\rangle$ a product state. If Z acts on an interval of sites, then $U^\dagger Z U$ acts on the interval enlarged by T sites on each side, so expectations in $|\psi_N\rangle$ of products of operators whose enlarged intervals are pairwise disjoint factorize.

Let $s_N = \max(s_0, 2T + L + 1)$ and let $s' \in \{s_N, \dots, s_N + K - 1\}$ be given by the lemma; the window changes s' by at most $K - 1$, so still $s' = o(\log N)$ and $N \geq 3s'$ for large N . Put $\Delta = s' + L + 2T \leq 2s'$, $n = \lfloor N/\Delta \rfloor \geq N/(4s')$, and $b_k = 1 + (k-1)\Delta$ for $k = 1, \dots, n$. With $e = \langle \phi_N | \mathcal{O}_1 | \phi_N \rangle$ and $e' = \langle \phi_N | \mathcal{O}'_1 | \phi_N \rangle$ let $\mathcal{O}^{(k)}$ be $\mathcal{O} - e$ on the sites $b_k, \dots, b_k + L - 1$ and $\mathcal{O}'^{(k)}$ be $\mathcal{O}' - e'$ on the sites $b_k + s' - 1, \dots, b_k + s' + L - 2$. These have norm at most 2, zero expectation in $|\phi_N\rangle$ by translation invariance, and $\langle \phi_N | \mathcal{O}^{(k)} \mathcal{O}'^{(k)} | \phi_N \rangle = G_N(\mathcal{O}, \mathcal{O}'; s')$. Since $s' \geq 2T + L + 1$ and $\Delta \geq s' + L + 2T$, the $2n$ enlarged intervals are pairwise disjoint, also around the ring.

Two estimates are used. For Hermitian Z and unit vectors ϕ, ψ , with $\sigma_\chi(Z) = \|(Z - \langle Z \rangle_\chi) \chi\|$,

$$|\langle \phi | \psi \rangle| |\langle Z \rangle_\phi - \langle Z \rangle_\psi| \leq \sigma_\phi(Z) + \sigma_\psi(Z), \quad (30.22)$$

because $(\langle Z \rangle_\phi - \langle Z \rangle_\psi) \langle \phi | \psi \rangle = \langle \phi | (Z - \langle Z \rangle_\psi) | \psi \rangle - \langle \phi | (Z - \langle Z \rangle_\phi) | \psi \rangle$. Secondly, the decomposition $\mathcal{E}_A^n = P + (\mathcal{E}_A - P)^n$ from the proof of Theorem 30.6.40, together with the fact that in the gauge (30.3) the maps $\mathcal{E}_Z$ have norm bounded by a multiple of $\|Z\|$ independent of the size of the support, shows that the connected correlation in $|\phi_N\rangle$ of two operators of norm at most 4 on intervals separated by g and g' sites around the ring is at most $C(r^g + r^{g'})$, with C and $r < 1$ depending only on A . Consecutive intervals $[b_k, b_k + s' + L - 2]$ are separated by at least $2T$

$g \geq l - k$ and $g' \geq n - l + k$, and in $|\psi\rangle$ they vanish for $k \neq l$ by Lemma 30.6.3, since the windows are at least $2T$ sites apart on both sides. $\square$

Theorem 30.6.43 (No preparation in depth $o(\log N)$, gauge form). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let A be normal and in the gauge (30.3), let $L \geq 1$ be a length such that the products of L matrices of A span $M_D(\mathbb{C})$, and let $\lambda_2 \neq 1$ be an eigenvalue of $\mathcal{E}_A$ of largest modulus among those different from 1, with $\xi = -1/\log|\lambda_2| > 0$.

1. There are $B \geq 0$ and $C, b > 0$ such that for every $T \geq 0$ and $N \geq B(T+1)$, if a unit vector $|\psi\rangle$ prepared in depth T satisfies $|\langle\phi_N|\psi\rangle| \geq 1/2$, then $N \leq C(T+1)e^{b(T+1)}$.
2. If $|\psi_N\rangle$ are unit vectors prepared in depth T_N with $T_N = o(\log N)$, then $\varepsilon(\phi_N, \psi_N) > 1/2$ for all sufficiently large N .

Part (2) is Theorem 30.6.41 for a tensor in the gauge (30.3); Lemma 30.6.44 removes the restriction [49].

Proof. [✓ Proof](#) Part (1) is the argument of the proof of Theorem 30.6.41 from its second paragraph on, with $s' \leq s_0 + 2T + L + 2$, $\Delta \leq (s_0 + 2L + 4)(T + 1)$ and $n \geq N/(2\Delta)$, which turn (30.23) into $N \leq C(T+1)e^{b(T+1)}$. For (2), $T_N \leq \log N/(2(b+1))$ for large N gives $C(T_N+1)e^{b(T_N+1)} \leq Ce^{b+1}\sqrt{N} < N$ and $B(T_N+1) \leq N$, so $|\langle\phi_N|\psi_N\rangle| < 1/2$ by (1). $\square$

Lemma 30.6.44 (Reduction to the gauge). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let the products of some fixed number of matrices of A span $M_D(\mathbb{C})$, and let λ_1 be an eigenvalue of $\mathcal{E}_A$ of largest modulus. Then $\lambda_1 \neq 0$, and there are $\zeta \neq 0$, an invertible X and $L \geq 1$ such that $B^i = X\zeta A^i X^{-1}$ satisfies the gauge (30.3) for some ρ , the products of L matrices of B span $M_D(\mathbb{C})$, the eigenvalues of $\mathcal{E}_B$ are those of $\mathcal{E}_A$ divided by λ_1 , and $|\phi_N(B)\rangle = \zeta^N |\phi_N(A)\rangle$ for every N . In particular, whenever $|\phi_N(A)\rangle \neq 0$, $|\langle\phi_N(B)|\psi\rangle|/\|\phi_N(B)\| = |\langle\phi_N(A)|\psi\rangle|/\|\phi_N(A)\|$ for every vector $|\psi\rangle$; when $|\phi_N(A)\rangle = 0$, also $|\phi_N(B)\rangle = 0$. If instead A is irreducible and $\mathcal{E}_A$ has the unique eigenvalue 1 of largest modulus, the same holds with $\zeta = 1$.

Proof. [✓ Proof](#) The eigenvalues of $\mathcal{E}_{\zeta A} = |\zeta|^2 \mathcal{E}_A$ are those of $\mathcal{E}_A$ times $|\zeta|^2$, and a gauge $B^i = X A^i X^{-1}$ maps an eigenvector Y of $\mathcal{E}_A$ to the eigenvector $XYX^\dagger$ of $\mathcal{E}_B$ with the same eigenvalue; it leaves every trace in (30.1) unchanged. Since A is normal, $\mathcal{E}_A$ is irreducible, so it has a positive eigenvector for its spectral radius $r > 0$ and a positive eigenvector σ of its dual; with $\zeta = r^{-1/2}$ and $X = \sigma^{1/2}$, the tensor B satisfies $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ (Theorem 8.2.3). By Theorem 6.1.1.16, $\mathcal{E}_B$ has a fixed point $\rho > 0$, which we scale to unit trace, and 1 is its only eigenvalue of modulus one. All eigenvalues of the trace-preserving map $\mathcal{E}_B$ have modulus at most one, and $|\zeta|^2 \lambda_1$ is one of largest modulus, so $|\zeta|^2 \lambda_1 = 1$. When A is irreducible and 1 is the unique eigenvalue of $\mathcal{E}_A$ of largest modulus, $r = 1$, so $\zeta = 1$; primitivity of $\mathcal{E}_A$ passes to $\mathcal{E}_B$, and some blocking of B is injective by Theorems 8.2.4 and 2.5.7. $\square$

30.7 The tree circuit as a multiscale network

The paragraph “Connection to MERA” of [185] reads the tree circuit of Theorem 30.4.5 as a finite-range multiscale entanglement renormalization ansatz with $O(\log \log N)$ layers: the isometries $V^{(j)}$ are the isometries of the network, and all disentanglers are the identity except those of the first layer, which prepare the fixed-point state. This section states the network, identifies the approximating state with its state, and counts the layers. Up to Theorem 30.7.9 the two-site blocked tensor of A is injective and the chain length is $N = M2^{k+1}$. From Definition 30.7.10 on, the finest isometries write registers of s sites, where A blocked over s sites is injective; this

Lemma 30.7.5 (A unitary preparing the fixed-point pair). Lean ✓ Stmt Let $D \geq 1$ and let $\rho \geq 0$ with $\text{tr } \rho = 1$. Then there is a unitary u on $\mathbb{C}^D \otimes \mathbb{C}^D$ with $u|0\rangle|0\rangle = |\omega\rangle$, the pair of the fixed-point state.

Proof. ✓ Proof The map $V: \mathbb{C} \rightarrow \mathbb{C}^D \otimes \mathbb{C}^D$, $1 \mapsto |\omega\rangle$, satisfies $V^\dagger V = \langle \omega | \omega \rangle = \text{tr } \rho = 1$ (Theorem 30.2.7), so it is an isometry. An isometry defined on the span of some computational basis vectors extends to a unitary, as in the proof of Lemma 30.3.9; applied to the basis vector $|0\rangle|0\rangle$, this gives a unitary u with $u|0\rangle|0\rangle = V1 = |\omega\rangle$. $\square$

Theorem 30.7.6 (The approximating state is a network state). Lean ✓ Stmt Let the two-site blocked tensor of A be injective, let $\rho \geq 0$ with $\text{tr } \rho = 1$, and let u be a unitary with $u|0\rangle|0\rangle = |\omega\rangle$. For $q = 2^{k+1}$ and $M \geq 1$, the approximating state $|\tilde{\phi}_N\rangle = V^{\otimes M}|\Omega\rangle$ on $N = Mq$ sites (Definition 30.3.8) is the state of the tree network of A with $k+1$ isometry layers and unitary u .

Proof. ✓ Proof The blocked tensor over $q = 2^{k+1}$ sites is injective, so the approximating state is the unnormalized one (Lemma 30.6.25). Its isometry V is the product of the layers by Theorem 30.4.5, hence the tree map by Lemma 30.7.3, and the pairs $u|0\rangle|0\rangle$ are the pairs $|\omega\rangle$ of $|\Omega\rangle$. $\square$

Lemma 30.7.7 (From the threshold of the network to the error). Lean ✓ Stmt Let $K, \xi, \varepsilon > 0$, $M, q \geq 1$, and $\xi \log(KMq/\varepsilon) \leq q$. Then $KMe^{-q/\xi} \leq \varepsilon$.

Proof. ✓ Proof Since $M \leq Mq$, $\log K + \log M - \log \varepsilon \leq \log(KMq/\varepsilon) \leq q/\xi$; apply Lemma 30.5.3 with $x = q/\xi$. $\square$

Lemma 30.7.8 (The least number of layers with the threshold). Lean ✓ Stmt Let $\xi, C, M, \varepsilon > 0$ and $s \geq 1$. There is $k \geq 0$ with $x_k \leq s2^{k+1}$ and $k \leq \log_2 \max(1, x_k)$, where $x_k = \xi \log(CMs2^{k+1}/\varepsilon)$.

Proof. ✓ Proof Write $x_k = \xi \log(CMs/\varepsilon) + (k+1)\xi \log 2$. It grows linearly in k while $s2^{k+1} \geq 2^{k+1}$ grows exponentially, so some k satisfies $x_k \leq s2^{k+1}$; let k be the least one. If $k \geq 1$, the inequality fails at $k-1$, so $1 \leq 2^k \leq s2^k < x_{k-1} \leq x_k$ and $k < \log_2 x_k$. If $k = 0$, the bound holds since $\log_2 \max(1, x_0) \geq 0$. $\square$

Theorem 30.7.9 (Networks with $O(\log \log(N/\varepsilon))$ layers, two-site injectivity). Lean Lean ✓ Stmt Let A be normal in the gauge (30.3), let $0 < t < 1$ bound the moduli of the eigenvalues of $\mathcal{E}_A$ other than 1, with $\xi = -1/\log t$, and let the two-site blocked tensor of A be injective. There is $C > 0$ such that for every unitary u with $u|0\rangle|0\rangle = |\omega\rangle$, every $\varepsilon > 0$, and every $k \geq 0$ and $M \geq 1$ with $N = M2^{k+1}$ and

$$x = \xi \log(CN/\varepsilon) \leq 2^{k+1}, \quad (30.27)$$

the state $|\psi\rangle$ of the tree network of A with $k+1$ isometry layers satisfies $\varepsilon(\psi, \phi_N) \leq \varepsilon$. Moreover, for every such u , every $\varepsilon > 0$ and every $M \geq 1$ there is $k \geq 0$ satisfying (30.27) with $k \leq \log_2 \max(1, x)$, so the network has at most $\log_2 \max(1, \xi \log(CN/\varepsilon)) + 1$ isometry layers. Theorem 30.7.13 removes the injectivity hypothesis with registers of s sites. Neither covers every fixed N and ε : for $N = 2p$ with p odd only $k = 0$ is available, and (30.27) fails for small ε [146].

Proof. ✓ Proof By Theorem 30.7.6 the network state is the approximating state, and Theorem 30.5.7 at $\gamma = 1/2$ gives $C > 0$ with $\varepsilon(\tilde{\phi}_N, \phi_N) \leq CMe^{-q/\xi}$. This rate is the strengthening of the first bound of Lemma 1' of [185] proved in Theorem 30.5.7; the bound as printed in the source, for $0 < \gamma < 1/2$ with rate $e^{-\gamma q/\xi}$, gives the threshold $(\xi/\gamma) \log(CN/\varepsilon)$ instead, with the

same count of layers up to a constant. From (30.27), $CMe^{-q/\xi} \leq \varepsilon$ by Lemma 30.7.7. The count is Lemma 30.7.8 with $s = 1$. $\square$

Definition 30.7.10 (The network state on registers). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $s \geq 1$. A binary finite-range network with sites of dimension d^s has its finest isometry $W: \mathbb{C}^{D^2} \rightarrow \mathbb{C}^{d^s} \otimes \mathbb{C}^{d^s}$, which maps one site to two neighbouring *registers* of s sites of dimension d . Its state on $M2^{k+1}$ registers (30.26), read on the $N = Ms2^{k+1}$ sites by splitting every register into its s sites, is its *state on registers*.

Lemma 30.7.11 (The network state on registers is a unit vector). [Lean](#) [✓ Stmt](#) The state on registers of a binary finite-range network with sites of dimension d^s is a unit vector.

Proof. [✓ Proof](#) Splitting the registers into sites permutes the coordinates, so it preserves the norm, and the network state is a unit vector by Lemma 30.7.2. $\square$

Theorem 30.7.12 (The approximating state is a network state on registers). [Lean](#) [✓ Stmt](#) Let A blocked over s sites be injective, let $\rho \geq 0$ with $\text{tr } \rho = 1$, and let u be a unitary with $u|0\rangle|0\rangle = |\omega\rangle$. For $q = s2^{k+1}$ and $M \geq 1$, the approximating state $|\tilde{\phi}_N\rangle = V^{\otimes M}|\Omega\rangle$ on $N = Mq$ sites (Definition 30.3.8) is the state on registers of the tree network, with $k+1$ isometry layers and unitary u , of the tensor B obtained by blocking A over s sites.

Proof. [✓ Proof](#) The two-site blocked tensor of B is A blocked over $2s$ sites, which is injective, so the tree network of B is defined, and A blocked over q sites is injective, so the approximating state is the unnormalized one (Lemma 30.6.25). The tree map of the network of B is the product of the layers by Lemma 30.7.3, hence the isometric factor of B blocked over 2^{k+1} sites by Theorem 30.4.5. Blocking B over 2^{k+1} sites is blocking A over q sites, read through the grouping of the q sites into registers, so the isometric factors agree after the splitting of the registers (Theorem 30.12.6), and the pairs $u|0\rangle|0\rangle$ are the pairs $|\omega\rangle$ of $|\Omega\rangle$. $\square$

Theorem 30.7.13 (Network states on registers within ε). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be normal in the gauge (30.3), and let $0 < t < 1$ bound the moduli of the eigenvalues of $\mathcal{E}_A$ other than 1, with $\xi = -1/\log t$. There is $C > 0$ such that for every $s \geq 1$ with A blocked over s sites injective, every unitary u with $u|0\rangle|0\rangle = |\omega\rangle$, every $\varepsilon > 0$, and every $k \geq 0$ and $M \geq 1$ with $N = Ms2^{k+1}$ and

$$x = \xi \log(CN/\varepsilon) \leq s2^{k+1}, \quad (30.28)$$

the state on registers $|\psi\rangle$ of the tree network, with $k+1$ isometry layers, of A blocked over s sites satisfies $\varepsilon(\psi, \phi_N) \leq \varepsilon$. Moreover, for every such s and u , every $\varepsilon > 0$ and every $M \geq 1$ there is $k \geq 0$ satisfying (30.28) with $k \leq \log_2 \max(1, x)$.

Proof. [✓ Proof](#) By Theorem 30.7.12 the network state is the approximating state with $q = s2^{k+1}$, and the proof of Theorem 30.7.9 applies with 2^{k+1} replaced by q : Theorem 30.5.7 at $\gamma = 1/2$ gives C , which does not depend on s , with $\varepsilon(\tilde{\phi}_N, \phi_N) \leq CMe^{-q/\xi} \leq \varepsilon$ by Lemma 30.7.7. The count is Lemma 30.7.8. $\square$

Theorem 30.7.14 (Normal states as networks, restricted chain lengths). [Lean](#) [✓ Stmt](#) Let A be a normal tensor with $D \geq 1$. There are $s_0 \geq 1$, $\xi > 0$ and $C > 0$ such that for every $s \geq s_0$, every $\varepsilon > 0$ and every $M \geq 1$ there are $k \geq 0$ and a binary finite-range network with sites of dimension d^s and $k+1$ isometry layers with $k \leq \log_2 \max(1, \xi \log(CN/\varepsilon))$, $N = Ms2^{k+1}$, whose state on registers $|\psi\rangle$ satisfies $\varepsilon(\psi, \phi_N) \leq \varepsilon$ against the normalized state $|\phi_N\rangle$ of A . This is a scope restriction, to the chain lengths $N = Ms2^{k+1}$ with k as above, of the containment of normal

translation-invariant matrix product states in finite-range networks with $O(\log \log N)$ layers, within the approximation error, of [185, paragraph “Connection to MERA”]. The containment for every fixed N and ε is not formalized [146].

Proof. ✓ Proof The normal tensor A has a gauge-equivalent rescaling B in the gauge (30.3), with $|\phi_N(B)\rangle = \zeta^N |\phi_N(A)\rangle$, a bound $t < 1$ on the subleading eigenvalues of $\mathcal{E}_B$, and a length L from which on the blocked tensors of B are injective (Theorem 30.6.31). Take $s_0 = L + 1$ and a unitary u preparing the pair of the fixed point of $\mathcal{E}_B$ (Lemma 30.7.5). For $s \geq s_0$, Theorem 30.7.13 applied to B gives k and the tree network of B blocked over s sites, and the error against the normalized state of B equals the error against that of A (Lemma 30.6.26). $\square$

30.8 Inhomogeneous short-range correlated states

The paragraph “Inhomogeneous short-range correlated MPS” of [185] extends the preparation to matrix product states that are not translation invariant. After blocking, the positive parts of the polar decompositions of the blocked tensors generate a state $|\phi_{\text{pos}}\rangle$ [185, eq. (18)]; if it is close to a product of nearest-neighbour pairs $|\Omega\rangle$, then preparing $|\Omega\rangle$ and applying the isometries of the polar decompositions prepares the state with error $\varepsilon(\Omega, \phi_{\text{pos}})$. This section proves the exact identity behind the scheme and the depth and error of the prepared state. The bond dimensions are at most D and may vary along the ring, as in the source; padding the matrices with zeros reduces them to a common bond dimension D . The blocked tensors need not be injective: the isometric factor V_k of a blocked tensor is then a partial isometry with $V_k^\dagger V_k = \Pi_k$, and the block unitary implements an isometry W_k with $W_k \Pi_k = V_k$, an isometric extension of V_k , as the source’s footnote to the paragraph “The sequential-RG circuit” indicates. The preparation theorems assume that every block has at least $3D$ sites, and those with isometric extensions and error exactly $\varepsilon(\Omega, \phi_{\text{pos}})$ also assume $D^2 \leq d^{3D}$, which holds for every D when $d \geq 2$. The source requires $d^q \geq D^2$ for its polar decompositions, which implies $D^2 \leq d^{3D}$ for blocks of $q \geq 3D \geq 1$ sites, and holds the two legs of a block, of dimension D each, on its sites, but states no bound $3D$. The restriction is recorded in [72].

Definition 30.8.1 (Blocked tensors of an inhomogeneous chain). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $A_1, \dots, A_N$ be tensors of physical dimension d and bond dimension D , one on each site of a ring of N sites, and let $|\phi_N\rangle = \sum_s \text{Tr}(A_1^{s_1} \dots A_N^{s_N}) |s\rangle$ be their state. Cut the ring into $M \geq 1$ consecutive blocks of lengths $\ell_1, \dots, \ell_M$ as in Definition 30.5.8. The blocked tensor B_k of block k has physical dimension d^{ℓ_k} and the matrices $B_k^{s_{o_k+1} \dots s_{o_k+\ell_k}} = A_{o_k+1}^{s_{o_k+1}} \dots A_{o_k+\ell_k}^{s_{o_k+\ell_k}}$. When all tensors are equal to A , it is the ℓ_k -site blocked tensor $\mathcal{B}_{\ell_k}$ of A .

Definition 30.8.2 (State of the positive parts and the block isometries). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $B_k = V_k P_k$ be the polar decompositions of the blocked tensors of Definition 30.8.1, with P_k read as a tensor of physical dimension D^2 and bond dimension D as in Theorem 30.1.3.3. The state of the positive parts on the M blocks is

$$|\phi_{\text{pos}}\rangle = \sum_{\tau} \text{Tr}(P_1^{\tau_1} \dots P_M^{\tau_M}) |\tau_1 \dots \tau_M\rangle, \quad (30.29)$$

the state of [185, eq. (18)]. For vectors $\omega^1, \dots, \omega^M$ on $\mathbb{C}^D \otimes \mathbb{C}^D$ let $|\Omega\rangle = \bigotimes_{k=1}^M |\omega^k\rangle_{R_k L_{k+1}}$, the pair ω^k joining the right leg of block k to the left leg of block $k + 1$, cyclically; when all ω^k are equal to ω it is the pair product state of Definition 30.2.3. The state $(\bigotimes_k V_k) |\Omega\rangle$ has the coefficient $\sum_c \prod_k \langle s_k | V_k | c_k \rangle \Omega(c)$ at the configuration s read block by block; for equal tensors and equal pairs it is the state $|\psi_{\ell, \omega}\rangle$ of Definition 30.5.8.

Theorem 30.8.3 (The state is the isometries applied to the positive parts). [Lean](#) [Lean](#) [Stmnt](#)
 For every cutting of the ring into blocks, $|\phi_N\rangle = (\bigotimes_{k=1}^M V_k)|\phi_{\text{pos}}\rangle$.

Proof. [Proof](#) The coefficient of $|\phi_N\rangle$ at s is $\text{Tr}(B_1^{s_1} \dots B_M^{s_M})$ with s_k the configuration of block k . Substituting $B_k^{s_k} = \sum_{\tau_k} \langle s_k | V_k | \tau_k \rangle P_k^{\tau_k}$ and expanding the product gives $\sum_{\tau} \prod_k \langle s_k | V_k | \tau_k \rangle \text{Tr}(P_1^{\tau_1} \dots P_M^{\tau_M})$. $\square$

Theorem 30.8.4 (The isometries preserve overlaps). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) For matrices W_k from $\mathbb{C}^m$ to $\mathbb{C}^{n_k}$, one on each of M sites, and vectors x, y on $(\mathbb{C}^m)^{\otimes M}$,

$$\left\langle \left(\bigotimes_k W_k \right) x, \left(\bigotimes_k W_k \right) y \right\rangle = \sum_{a,b} \overline{x(a)} y(b) \prod_k (W_k^\dagger W_k)_{a_k b_k}. \quad (30.30)$$

For every chain and every cutting into blocks, $\|\phi_N\| = \|\phi_{\text{pos}}\|$; no injectivity is assumed. Suppose every blocked tensor B_k is injective on a set S_k of bond pairs, as in Definition 30.1.3.7. Then $\langle (\bigotimes_k V_k) x, (\bigotimes_k V_k) y \rangle = \langle x, y \rangle$ for all vectors x, y on the M blocks such that $\overline{x(a)} y(a) = 0$ whenever $a_k \notin S_k$ for some k ; if every B_k is injective, this holds for all x, y . More generally, for matrices W_k with $\sum_i (W_k)_{ia} (W_k)_{ib} = 1$ if $a = b \in S_k$ and 0 otherwise, the same identity holds with W_k in place of V_k . The state $|\phi_{\text{pos}}\rangle$ vanishes at every a with some $a_k \notin S_k$. For products $|\Omega\rangle$ and $|\Omega'\rangle$ of pairs ω^k and ω'^k , $\langle \Omega, \Omega' \rangle = \prod_k \langle \omega^k, \omega'^k \rangle$; in particular the norm of $|\Omega\rangle$ is $\prod_k \|\omega^k\|$. For all vectors ω^k ,

$$\left\langle \left(\bigotimes_k V_k \right) \Omega, \frac{\phi_N}{\|\phi_N\|} \right\rangle = \left\langle \Omega, \frac{\phi_{\text{pos}}}{\|\phi_{\text{pos}}\|} \right\rangle. \quad (30.31)$$

Here a vector $x/\|x\|$ with $x = 0$ is read as 0, so the identity holds for every chain, and it is an identity of overlaps with normalized states when $\phi_{\text{pos}} \neq 0$.

Proof. [Proof](#) Expanding both vectors and summing over the configurations $s = (s_k)_k$ factor by factor gives (30.30), since $\sum_{s_k} \overline{(W_k)_{s_k a}} (W_k)_{s_k b} = (W_k^\dagger W_k)_{ab}$. For the norm, $|\phi_N\rangle = (\bigotimes_k V_k)|\phi_{\text{pos}}\rangle$ by Theorem 30.8.3, and $V_k^\dagger V_k = \Pi_k$ by Theorem 30.1.3.3; by (30.30) the squared norm of $|\phi_N\rangle$ is $\sum_{a,b} \overline{\phi_{\text{pos}}(a)} \phi_{\text{pos}}(b) \prod_k (\Pi_k)_{a_k b_k}$, and $\sum_b \prod_k (\Pi_k)_{a_k b_k} \phi_{\text{pos}}(b) = \phi_{\text{pos}}(a)$ because $\sum_l (\Pi_k)_{jl} P_k^l = P_k^j$, expanding the trace as in Lemma 30.5.9. By Theorem 30.1.3.8, $\sum_{s_k} \overline{(V_k)_{s_k a}} (V_k)_{s_k b}$ is 1 if $a = b \in S_k$ and 0 otherwise, for the configurations s_k of block k ; for injective B_k the set S_k contains all pairs. Expanding the inner product over the configurations $s = (s_k)_k$ and applying this identity factor by factor gives

$$\sum_s \overline{\left(\left(\bigotimes_k V_k \right) x \right)(s)} \left(\left(\bigotimes_k V_k \right) y \right)(s) = \sum_{a,b} \overline{x(a)} y(b) \prod_k \sum_{s_k} \overline{(V_k)_{s_k a_k}} (V_k)_{s_k b_k} = \sum_{a \in \prod_k S_k} \overline{x(a)} y(a),$$

which is $\langle x, y \rangle$ under the hypothesis on x and y . The positive part P_k vanishes outside S_k by Theorem 30.1.3.8, so $|\phi_{\text{pos}}\rangle$ vanishes at every a with some $a_k \notin S_k$, and the remaining identities follow from Theorem 30.8.3 with $y = \phi_{\text{pos}}$. For the overlap of the pair products, expanding $\prod_k \langle \omega^k, \omega'^k \rangle$ as a sum over bond configurations $(p_k)_k$ of $\prod_k \overline{\omega_{p_k}^k} \omega_{p_k}'^k$ and reindexing along the bond regrouping of Definition 30.2.8 gives $\sum_c \overline{\Omega(c)} \Omega'(c)$. $\square$

Definition 30.8.5 (Chains with bond dimensions at most D). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let $D_1, \dots, D_N \leq D$ be the dimensions of the bonds

Proof. [✓ Proof](#) By Theorem 30.1.3.8, $V = BG$ for a matrix G on the bond pairs, and the columns $V|\pi(x)\rangle$, $x < |S|$, are orthonormal. Reorder the inputs by π and apply Lemma 30.13.1.6 with $r = |S|$. $\square$

Theorem 30.8.9 (Preparation for site-dependent tensors). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) *There is C , depending only on d and D , such that the following holds. Cut the ring of $N \geq 1$ sites into $M \geq 1$ blocks of lengths $3D \leq \ell_k \leq L$, let $A_1, \dots, A_N$ be tensors of bond dimension D whose blocked tensor B_k is injective on a set S_k of bond pairs for every k , and let ω^k be unit vectors on $\mathbb{C}^D \otimes \mathbb{C}^D$ whose product $|\Omega\rangle$ vanishes at every bond configuration c with some $c_k \notin S_k$. Then $(\bigotimes_k V_k)|\Omega\rangle$ is prepared in depth at most CL . In particular this holds when every B_k is injective, for all unit vectors ω^k . A finite set has an enumeration listing the elements of a given subset first, and for $d \leq 1$ every vector on a ring of $N \geq 1$ sites is prepared in depth 0.*

Proof. [✓ Proof](#) For $d \leq 1$ there is at most one configuration, and a product vector carries any amplitude on it. For $D = 0$ there is no unit pair. Otherwise $d \geq 2$, so $D \leq d^D$, and each bond index is encoded in D sites. The circuit of the proof of Theorem 30.6.5 applies with one change: the first layer prepares the pair ω^k on the window joining block k to block $k+1$, one unitary per window, and on block k the unitary of Theorem 30.13.1.7, built from the sequential factorization of Theorem 30.8.8 with an enumeration listing S_k first, implements V_k on the inputs in S_k , the only ones $|\Omega\rangle$ populates. The depth of that unitary is bounded by a constant times ℓ_k , the constant being the largest over the finitely many enumerations of the bond pairs. $\square$

Lemma 30.8.10 (A basis adapted to a projector). [Lean](#) [Lean](#) [✓ Stmt](#) *For a Hermitian idempotent E on $\mathbb{C}^m$ there are $r \leq m$ and a unitary T with $ET = T\Lambda$, where Λ is the diagonal matrix with $\Lambda_{yy} = 1$ for $y < r$ and $\Lambda_{yy} = 0$ otherwise. For the polar decomposition $B = VP$ of a tensor, with $V^\dagger V = \Pi$ as in Theorem 30.1.3.3, $V\Pi = V$.*

Proof. [✓ Proof](#) By the spectral theorem $EU = U \text{diag}(\mu)$ for a unitary U and real μ_j ; then $U \text{diag}(\mu)^2 = E^2 U = EU$, so $\mu_j^2 = \mu_j$ and $\mu_j \in \{0, 1\}$. Reordering the columns of U so that those with $\mu_j = 1$ come first gives T . For the second claim, $(V(1-\Pi))^\dagger V(1-\Pi) = (1-\Pi)\Pi(1-\Pi) = 0$, so $V(1-\Pi) = 0$. $\square$

Lemma 30.8.11 (A block unitary after a gate on its input). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $D \geq 1$, encode the bond indices injectively in $r_1 \geq 1$ sites, let π enumerate the bond pairs, and let $Q_1, \dots, Q_q$ be an isometric chain with $b_0 = 1$, as in Theorem 30.13.1.7. There is C , independent of $q \geq 3r_1$, of the chain and of G , such that for every unitary G on $\mathbb{C}^{D^2}$ some unitary U on q sites, a product of at most Cq gates on neighbouring sites, and some amplitudes $Z(\sigma, y)$ with $Z(\sigma, y) = (Q_1^{\sigma_1} \dots Q_q^{\sigma_q})_{1y}$ for $y < b_q$ satisfy $\langle \sigma | U | l, 0, r \rangle = \sum_y Z(\sigma, y) G_{yx}$ whenever $\pi(x) = (l, r)$. A unitary on the levels of $\mathbb{C}^m$ encoded injectively in the configurations of a window extends to a unitary on the window.*

Proof. [✓ Proof](#) The encoded levels span an isometric copy of $\mathbb{C}^m$, so G on them is an isometry into the window, which extends to a unitary by Lemma 30.3.9. In the circuit of Theorem 30.13.1.7, after the SWAP gates have brought the two inputs onto the last $2r_1$ sites, apply this unitary for G on those sites, a product of a bounded number of gates by Theorem 30.6.4, and then the staircase; $Z(\sigma, y)$ is the amplitude of the staircase on the encoded input y . $\square$

Theorem 30.8.12 (A block unitary implementing an isometric extension). [Lean](#) [✓ Stmt](#) *Encode the bond indices injectively in $D \geq 1$ sites. There is C such that for every $q \geq 3D$ and all tensors $A_1, \dots, A_q$ of bond dimension D , one on each site, with blocked tensor $B = VP$ and $V^\dagger V = \Pi$, there are a unitary U on the q sites, a product of at most Cq gates on neighbouring sites, and*

that is, $\varepsilon(\psi, \phi_N) = \varepsilon(\Omega, \phi_{\text{pos}})$ for the normalized states when $\phi_{\text{pos}} \neq 0$; with the reading of $x/\|x\|$ for $x = 0$ as in Theorem 30.8.4, the identity of the overlaps holds for every chain. For every d , with C depending only on d and D : if the chain is approximated by pairs with error δ on the given blocks, some unit vector prepared in depth at most CL has error at most δ against the normalized state $|\phi_N\rangle$. No injectivity of the blocked tensors is assumed. This is a scope-restricted variant, for one ring, of the claim of the paragraph “Inhomogeneous short-range correlated MPS” of [185], which states total depth $O(\log(N/\varepsilon))$ with error $\varepsilon(\Omega, \phi_{\text{pos}})$ for matrix product states with bond dimension at most D . The lower bound $3D \leq \ell_k$ is absent from that statement [72], and the unrestricted claim is not formalized. The hypothesis $D^2 \leq d^{3D}$ follows from the source’s requirement $d^q \geq D^2$ for blocks of $q \geq 3D \geq 1$ sites. The bound CL gives the stated depth when $L = O(\log(N/\varepsilon))$; the source’s definition fixes blocks of length $q = O(\log N)$. The choice of L in terms of N and the error is not made here.

Proof. ✓ Proof The padded pairs are unit vectors by Theorem 30.8.6, and Theorem 30.8.14 for the padded chain gives the isometries and the depth. The norm of $|\psi\rangle$ is $\|\Omega\| = 1$, and the error identity is (30.32), since the padded chain has the state $|\phi_N\rangle$. Under the approximation hypothesis with $d \geq 2$, which gives $D^2 \leq d^{3D}$, the error is at most δ . For $d \leq 1$ the ring has at most one configuration; $\phi_N \neq 0$ because $\|\phi_N\| = \|\phi_{\text{pos}}\| \neq 0$ by Theorem 30.8.4, the normalized state $\phi_N/\|\phi_N\|$ is prepared in depth 0 by Theorem 30.8.9, and its error is $0 \leq \delta$ by the Cauchy–Schwarz inequality. $\square$

Corollary 30.8.16 (Preparation with injective blocked tensors). Lean ✓ Stmt There is C , depending only on d and D , such that in the setting of Theorem 30.8.15, for any d , if moreover every blocked tensor is injective in the sense of Definition 30.8.5, the vector $(\bigotimes_k V_k)|\Omega\rangle$ is a unit vector prepared in depth at most CL that satisfies (30.33) in place of $|\psi\rangle$.

Proof. ✓ Proof For $D = 0$ there is no unit pair. Otherwise every block has a site, and by Theorem 30.8.6 the padded blocked tensor B_k is injective on the rectangle R_k , the padded pairs are unit vectors, and $|\Omega\rangle$ vanishes outside $\prod_k R_k$. The depth is Theorem 30.8.9 with $S_k = R_k$. The norm of $|\psi\rangle$ is $\|\Omega\| = \prod_k \|\omega^k\| = 1$ and the error identity is (30.31) by Theorem 30.8.4, since the padded chain has the state $|\phi_N\rangle$ by Theorem 30.8.6. $\square$

30.9 Tensors that are not normal

Definition 30.9.1 (Fixed point of a basis of normal tensors). Let $A^i = \bigoplus_{j=1}^b \text{diag}(\mu_{j,1}, \dots, \mu_{j,m_j}) \otimes A_j^i$ with normal A_j , each in the gauge (30.3), forming a basis of normal tensors, and put $\beta_j = \sum_{k=1}^{m_j} \mu_{j,k}^N$. Consider lengths $N = qM$ with $(\beta_1, \dots, \beta_b) \neq 0$. Write ξ_j for the correlation length of the normal tensor A_j , formed from the subleading eigenvalue of its own transfer map $\mathcal{E}_{A_j}$. With $|\omega_j\rangle$ the pair state of A_j and $\alpha_j^{(N)} = \beta_j / (\sum_l |\beta_l|^2)^{1/2}$, the *fixed-point state* is

$$|\Omega'\rangle = \sum_{j=1}^b \alpha_j^{(N)} \bigotimes_{k=1}^M |\omega_j\rangle_{R_k L_{k+1}}. \quad (30.34)$$

With V from Theorem 30.1.3.3 for the blocked map of A , the *approximating state* is $|\tilde{\phi}_N\rangle = V^{\otimes M} |\Omega'\rangle / \|V^{\otimes M} |\Omega'\rangle\|$, whenever the denominator is nonzero. This is [185, eqs. (19) and (S7)], with the canonical form of [12].

When some block has multiplicity $m_j \geq 2$, the source’s construction fails: no product of pairs carrying the weights β_j gives, after $V^{\otimes M}$, an approximation of $|\phi_N\rangle$. For $A^0 = \text{diag}(1, 1, 0)$

and $A^1 = \text{diag}(0, 0, 1)$ the overlap with $|\phi_N\rangle \propto 2|0 \dots 0\rangle + |1 \dots 1\rangle$ tends to $1/\sqrt{5}$ as $M \rightarrow \infty$, for every q [64]. The failure lies in the coefficients β_j . With the pairs of each block on one of its copies k_j and the coefficients $\beta'_j = \beta_j(c_j/\mu_{j,k_j}^q)^M$ of Definition 30.9.4.3, with the copy norms c_j of Definition 30.9.1.1, in place of β_j , the construction converges for all nonzero complex weights, also when the q -site states of distinct blocks overlap, at a rate that includes the correlation lengths of the mixed transfer maps (Lemma 30.9.4.4 and Theorem 30.9.4.5) [40, 64].

Definition 30.9.2 (Fixed-point state of a sector decomposition). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stm](#)

Let A be given by a sector decomposition with blocks $A_1, \dots, A_b$ of bond dimensions D_j , multiplicities m_j and coefficients $\mu_{j,k}$, whose flattened direct sum $\bigoplus_{(j,k)} \mu_{j,k} A_j$ is the tensor of Definition 30.9.1 up to the order of the bond coordinates, so the bond space of A is $\mathbb{C}^D$ with $D = \sum_j m_j D_j$. The blocks carry no normality, basis-of-normal-tensors or gauge condition. For each j let σ_j be any matrix on $\mathbb{C}^{D_j}$ and choose a copy $k_j \in \{1, \dots, m_j\}$. The pair of block j is $\sum_{r,l} (\sqrt{\sigma_j})_{rl} |r\rangle\langle l|$ placed, on both legs, on the bond coordinates of copy k_j , which gives $|\omega_j\rangle \in \mathbb{C}^D \otimes \mathbb{C}^D$. With β_j and $\alpha_j^{(N)}$ as in Definition 30.9.1, the fixed-point state $|\Omega'\rangle$ is formed from these pairs by (30.34). Here each σ_j is an arbitrary matrix on $\mathbb{C}^{D_j}$, and the copy k_j is specified explicitly.

Theorem 30.9.4 (Local orthogonality of the pairs). [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) *In Definition 30.9.2, let each σ_j be positive semidefinite with $\text{Tr } \sigma_j = 1$. For every choice of the copies k_j the embedded pairs satisfy $\langle \omega_j | \omega_{j'} \rangle = \delta_{jj'}$. The chosen coordinates of block j carry the block $\mu_{j,k_j} A_j$ of A , and embedding the pair embeds its product state. Moreover, for $M \geq 1$, in the bond space $\mathbb{C}^{D_j} \otimes \mathbb{C}^{D_j}$ of block j alone, the tensor $\bigoplus_k \mu_{j,k}^q P_{j,\infty}$ generates on M sites the state $(\sum_k \mu_{j,k}^{qM}) |\Omega_j\rangle = \beta_j |\Omega_j\rangle$ with $N = qM$ and $|\Omega_j\rangle$ formed from the unembedded pair. The orthogonality is asserted after [185, eq. (S7)], where the reason given is that the blocks form a basis of normal tensors; here it follows from the placement of the pairs alone. The identification of these pairs with the fixed points of the positive parts of the blocked tensor is not proved here and can fail when a block has multiplicity greater than one [69].*

$u = (\sqrt{1+s} + \sqrt{1-s})/2$ and $v = (\sqrt{1+s} - \sqrt{1-s})/2$. For every $q \geq 1$ and $M \geq 1$ the approximating state of Definition 30.9.1 satisfies

$$\langle \tilde{\phi}_N | \phi_N \rangle = \frac{u^M + v^M}{(1 + s^M)^{1/2}}, \quad (30.38)$$

and for $M \geq 3$ the error is at least $(9/25)^q/16$, while for every $M \geq 1$ it is at most $M(9/25)^q$, so the construction converges whenever $(N/q)(9/25)^q \rightarrow 0$. The transfer maps of both blocks have no eigenvalue other than one, so every λ_2 bounds their subleading eigenvalues. If $r = e^{-\gamma/\xi_{\text{diag}}} < 9/25$, then for every constant C there is $q \geq 3$ such that, with $M = q$, $N = q^2$ and $y = (N/q)e^{-\gamma q/\xi_{\text{diag}}}$, $\varepsilon(\tilde{\phi}_N, \phi_N) > Cy e^{Cy}$. The sequence $q = M$ satisfies $q = o(N)$, so the bound (30.35) fails even when every multiplicity is one [106]. What fails is the rate: for $M \geq 3$ the error lies between $(9/25)^q/16$ and $M(9/25)^q$, where $9/25 = \tau^2$ is the square of the eigenvalue $\tau = 3/5$ of the mixed transfer map of the two blocks.

Proof. ✓ Proof The hypotheses are checked as in the proof of Theorem 30.9.8: both blocks are one-dimensional with $\sum_i |a_i|^2 = 1$, their states $|0 \cdots 0\rangle$ and $(\frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle)^{\otimes N}$ are linearly independent for $N \geq 1$, and $A_1^1 = 0 \neq A_2^1$.

As in the proof of Theorem 30.9.8, the Gram matrix of the blocked map on the diagonal coordinates is $G = \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix}$, whose positive square root is $Q = \begin{pmatrix} u & v \\ v & u \end{pmatrix}$, since $u^2 + v^2 = 1$ and $2uv = s$. The determinant of Q is $\sqrt{1-s^2} > 0$, so the support projector is $JJ^\dagger$. With $\alpha = (1, 1)/\sqrt{2}$ the overlap with the unnormalized target is $\sqrt{2}(u^M + v^M)$, the approximating vector has norm one, and $\|\phi_N(A)\|^2 = 2 + 2s^M$, which gives (30.38). For $M \geq 3$,

$$u^M + v^M \leq u^3 + v^2 = 1 - u^2(1 - u) \leq 1 - \frac{s^2}{16},$$

because $u^2 = (1 + \sqrt{1-s^2})/2 \geq 1/2$ and $1 - u \geq (1 - u^2)/2 \geq s^2/8$. For the upper bound, the overlap is one when $M = 1$, since $u + v = \sqrt{1+s}$. For $M \geq 2$, $u \geq u^2 = (1 + \sqrt{1-s^2})/2 \geq 1 - s^2/2$, so $u^M \geq 1 - Ms^2/2$, and $(1 + s^M)^{1/2} \leq 1 + s^2/2$; when $Ms^2 < 1$ this gives

$$\frac{u^M + v^M}{(1 + s^M)^{1/2}} \geq \frac{1 - Ms^2/2}{1 + s^2/2} \geq 1 - Ms^2,$$

and when $Ms^2 \geq 1$ the bound $\varepsilon \leq 1$ suffices. Finally $Cy e^{Cy}$ with $y = qr^q$ is eventually below $(9/25)^q/16$, because $q(25r/9)^q \rightarrow 0$. □

Lemma 30.9.10 (Polar decomposition of an orthogonal sum). Lean Lean Lean Lean ✓ Stmt
Let $B = \sum_j c_j B_j K_j^\dagger$ with $c_j > 0$, isometries K_j with orthogonal ranges, so that $K_j^\dagger K_{j'} = \delta_{jj'}$, and $B_j^\dagger B_{j'} = 0$ for $j \neq j'$. With $B_j = V_j P_j$ the polar decompositions of the summands, the polar decomposition of B has

$$V = \sum_j V_j K_j^\dagger, \quad P = \sum_j c_j K_j P_j K_j^\dagger.$$

This is the block form of the positive part asserted in [185, eq. (S5)] (“Since V is an isometry, P inherits the block structure of A ”), for blocks of multiplicity one under the orthogonality of their states, which the source does not state and without which the block form fails [106].

Proof. ✓ Proof Every partial isometry is $V_j = B_j R_j$ for some R_j : the support projector Π_j satisfies $\Pi_j = P_j R_j$ for some R_j , because both have the same range, and $V_j = V_j \Pi_j = B_j R_j$.

30.9.1 Blocks with multiplicities

When a block of the canonical form is repeated, the positive part of the blocked map is not the direct sum printed in the source, and the state built from nearest-neighbour pairs with the source's coefficients does not approximate the target [64]. This subsection gives the correct block form, the corrected approximating state, and the approximation error for it when the states of distinct blocks on q sites are orthogonal [40].

Definition 30.9.1.1 (Direct sum with multiplicities). [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [Learn](#) [✓ Stmt](#) Let $A_1, \dots, A_b$ be tensors, let the isometries $E_{j,k}$, $1 \leq k \leq m_j$, place the copy k of block j in the bond space, with $E_{j,k}^\dagger E_{j',k'} = \delta_{jj'} \delta_{kk'} \mathbf{1}$, and let $\mu_{j,k}$ be nonzero weights, with at least one copy of every block. The direct sum with multiplicities is $A^i = \sum_{j,k} \mu_{j,k} E_{j,k} A_j^i E_{j,k}^\dagger$. For $q \geq 1$ put $K_{i,k} = E_{j,k} \otimes E_{j,k}$,

$$c_j = \left(\sum_k |\mu_{j,k}|^{2q} \right)^{1/2}, \quad L_j = \sum_k \frac{\overline{\mu_{j,k}^q}}{c_j} K_{j,k}.$$

Theorem 30.9.1.2 (Positive part of a blocked tensor with multiplicities). [Lean] [Lean] [Lean] [✓ Stmt] Let A be the direct sum with multiplicities of Definition 30.9.1.1. Its periodic state on $N \geq 1$ sites is $\sum_j \beta_j |\phi_N(A_j)\rangle$ with $\beta_j = \sum_k \mu_{j,k}^N$. For $q \geq 1$ the blocked map of A is $B = \sum_j c_j B_j L_j^\dagger$, with B_j the blocked map of A_j , and $L_j^\dagger L_{j'} = \delta_{jj'} \mathbf{1}$. If $B_j^\dagger B_{j'} = 0$ for $j \neq j'$, then, with $B_j = V_j P_j$,

$$V = \sum_j V_j L_j^\dagger, \quad P = \sum_j c_j L_j P_j L_j^\dagger.$$

On the copies of block j the positive part is therefore of rank one in the copy index, where [185, eq. (S5)] prints $\text{diag}(\mu_{j,1}^q, \dots, \mu_{j,m_j}^q) \otimes P_j$ [40].

Proof. ✓ **Proof** Listing all copies as blocks of multiplicity one, every word of length $q \geq 1$ is a direct sum, so $B = \sum_{j,k} \mu_{j,k}^q B_j K_{j,k}^\dagger$ and $\phi_N(A) = \sum_{j,k} \mu_{j,k}^N \phi_N(A_j)$. Grouping the copies of each block gives $B = \sum_j c_j B_j L_j^\dagger$, since $c_j \mu_{j,k}^q / c_j = \mu_{j,k}^q$. Moreover $L_j^\dagger L_{j'} = \sum_{k,k'} (\mu_{j,k}^q / c_j) (\overline{\mu_{j',k'}^q} / c_{j'}) K_{j,k}^\dagger K_{j',k'}$, which is $\sum_k |\mu_{j,k}|^{2q} / c_j^2 = 1$ times the identity for $j = j'$ and zero otherwise. Lemma 30.9.10 gives the polar decomposition. $\square$

Definition 30.9.1.3 (Corrected approximating state). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) With the notation of Definition [30.9.1.1](#) and Theorem [30.9.1.2](#), fixed points ρ_j of the blocks, and $|\Omega_j\rangle$ the product of the pairs of ρ_j over M sites in the bond space of block j , the corrected approximating state on $N = qM$ sites with coefficients α_j is

$$|\tilde{\phi}_N\rangle \propto V^{\otimes M} \sum_j \alpha_j L_j^{\otimes M} |\Omega_j\rangle.$$

When the weights β_j of Theorem 30.9.1.2 are not all zero, the source’s coefficients are $\alpha_j = \beta_j/(\sum_l |\beta_l|^2)^{1/2}$; if all β_j vanish these coefficients are undefined. It is obtained from the corrected block form of Theorem 30.9.1.2 by replacing each P_j by its fixed point, as [185, eq. (S7)] does with the printed block form. The weights are those of the source. The isometry L_j acts on the two legs of each site, so for a repeated block the state $L_j^{\otimes M}|\Omega_j\rangle$ is not a product of nearest-neighbour pairs of the full bond space; for multiplicity one and $\mu_j > 0$ it is the state of Definition 30.9.1 [40].

Theorem 30.9.1.4 (Approximation error for orthogonal blocks with multiplicities). [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[✓ Stmt\]](#) *With the notation of Theorem 30.9.1.2, let the blocks A_j be normal in the gauge (30.3) with fixed points ρ_j , let λ_2 bound the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$, with correlation length ξ , and let $0 < \gamma < 1/2$. There are constants $C, C' > 0$ such that for every block length $q \geq 1$ at which the q -site states of distinct blocks are orthogonal, $B_j^\dagger B_{j'} = 0$ for $j \neq j'$, and every $M \geq 1$ with $N = qM$ and $(\beta_1, \dots, \beta_b) \neq 0$, the corrected approximating state of Definition 30.9.1.3 satisfies, with $y = (N/q) e^{-\gamma q/\xi}$,*

$$\varepsilon(\tilde{\phi}_N, \phi_N) \leq C'y, \quad \varepsilon(\tilde{\phi}_N, \phi_N) \leq Cy e^{Cy}.$$

This is [185, Lemma 1' (ii)] for the corrected approximating state, with arbitrary multiplicities and complex weights, restricted to blocks whose q -site states are orthogonal and without the condition $q = o(N)$ [40, 106].

Proof. ✓ Proof By Theorem 30.9.1.2, $VL_j = V_j$, so $V^{\otimes M} \sum_j \alpha_j L_j^{\otimes M} |\Omega_j\rangle = \sum_j \alpha_j |s_j\rangle$ with $|s_j\rangle = |\phi_M(V_j P_{j,\infty})\rangle$, while $|\phi_N(A)\rangle = \sum_j \beta_j |t_j\rangle$ with $|t_j\rangle = |\phi_N(A_j)\rangle$. These are the vectors of the proof of Theorem 30.9.11, with the weights β_j in place of μ_j^N , and all cross overlaps vanish. If every blocked tensor B_j is injective, the vectors $|s_j\rangle$ are unit vectors and, with $p_j = |\beta_j|^2$, $z_j = \langle s_j | t_j \rangle$ and $c'_j = \langle t_j | t_j \rangle$,

$$|\langle \tilde{\phi}_N | \phi_N \rangle| = \frac{|\sum_j p_j z_j|}{(\sum_i p_i)^{1/2} (\sum_i p_i c'_i)^{1/2}},$$

because $\overline{\alpha_j} \beta_j = p_j / (\sum_l p_l)^{1/2}$. The remaining estimate is that of Theorem 30.9.11, which uses only $p_j \geq 0$ and $\sum_j p_j > 0$. The finitely many block lengths below the injectivity length of some block are covered by $\varepsilon \leq 1$ after enlarging the constants (Theorem 30.5.2). $\square$

30.9.2 Blocks whose states overlap

When the states of distinct blocks on q sites are not orthogonal, the positive part of the blocked map is not block diagonal, and the rate of [185, Lemma 1'(ii)] fails (Theorem 30.9.9). The construction itself still converges. This subsection proves the approximation error for the source's approximating state, for blocks of multiplicity one with real weights, at the rate set by the larger of the correlation lengths of the blocks and of their mixed transfer maps [106].

Lemma 30.9.2.1 (Decay of mixed transfer maps). [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Stmt\]](#) Let X and Y be tensors with d physical indices and bond dimensions $D_1, D_2 \geq 1$, with mixed transfer map $\mathcal{E}_{XY}(Z) = \sum_i X^i Z (Y^i)^\dagger$ on $D_1 \times D_2$ matrices, and let X_q, Y_q be their blocked tensors on q sites. Then $\mathcal{E}_{X_q Y_q} = \mathcal{E}_{XY}^q$, and for tensors with a common number of physical indices the mixed Gram matrix of their physical matrices, read as maps from bond pairs to physical indices, is a rearrangement of the mixed transfer map, $(\mathcal{B}_X^\dagger \mathcal{B}_Y)_{(a,b),(a',b')} = \mathcal{E}_{YX}(|b'\rangle\langle b|)_{a'a}$. For $\lambda_2 \in \mathbb{C}$ with $|\lambda_2| \leq 1$, correlation length ξ , and $\gamma \geq 0$, $e^{-\gamma/\xi} \leq 1$. Now let $|\lambda_2| < 1$ and $0 < \gamma < 1$, and write $r = e^{-\gamma/\xi}$.

1. If every eigenvalue of $\mathcal{E}_{XY}$ has modulus at most $|\lambda_2|$, there are C, K with $\|\mathcal{E}_{XY}^n(Z)\| \leq Cr^n\|Z\|$ for all n and Z , and $|\sum_s \phi_N(X)_s \overline{\phi_N(Y)_s}| \leq Kr^N$ for all N .
2. If every eigenvalue of $\mathcal{E}_{YX}$ has modulus at most $|\lambda_2|$, there is K' with $\|\mathcal{B}_X^\dagger \mathcal{B}_Y\| \leq K'r^q$ for all q .

For distinct blocks $A_j, A_{j'}$ of a basis of normal tensors the spectral radius $\tau_{jj'}$ of $\mathcal{E}_{jj'} = \mathcal{E}_{A_j A_{j'}}$ is below one, and the source writes $\xi_{jj'} = -1/\ln \tau_{\max}$ with $\tau_{\max} = \tau_{jj'}$ [185, eqs. (S14)–(S15)].

Proof. ✓ Proof The blocked tensors have letters $X^{i_1} \dots X^{i_q}$, so $\mathcal{E}_{X_q Y_q}(Z) = \sum_{i_1, \dots, i_q} X^{i_1} \dots X^{i_q} Z(Y^{i_1} \dots Y^{i_q})^\dagger = \mathcal{E}_{XY}^q(Z)$, and the entry formula is $\sum_i \overline{X_{ab}^i} Y_{a'b'}^i = \mathcal{E}_{YX}(|b'\rangle\langle b|)_{a'a}$. Every point of the spectrum of $\mathcal{E}_{XY}$ is an eigenvalue, of modulus at most $|\lambda_2| < r$ by Theorem 30.3.7, so the spectral radius is below r and Gelfand's formula gives $\|\mathcal{E}_{XY}^n(Z)\| \leq Cr^n \|Z\|$. The overlap is $\text{Tr } \mathcal{E}_{XY}^N = \sum_{p,q} \mathcal{E}_{XY}^N(|p\rangle\langle q|)_{pq}$, and every entry is at most a fixed multiple of the norm. The Gram matrix is bounded by the sum of its entries, each of which is an entry of $\mathcal{E}_{YX}^q$ applied to a matrix unit. Finally $-\gamma/\xi = \gamma \ln |\lambda_2| \leq 0$. □

Lemma 30.9.2.2 (Shared estimates). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let $A_1, \dots, A_b$ be tensors with d physical indices and bond dimensions D_j , normal in the gauge (30.3) with positive definite fixed points ρ_j of trace one, with $\mathcal{B}_j$ the blocked map of A_j on q sites. Let $\lambda_2 \in \mathbb{C}$ with $|\lambda_2| < 1$ bound the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$ and the moduli of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, and let ξ be its correlation length.

1. A matrix L with $L^\dagger L = \mathbb{1}$ has operator norm at most one, and for isometries L_j with $L_j^\dagger L_{j'} = 0$ for $j \neq j'$, $L_j^\dagger \sum_k L_k Y_k = Y_j$ and $(\sum_j L_j Y_j L_j^\dagger)(\sum_j L_j Z_j L_j^\dagger) = \sum_j L_j Y_j Z_j L_j^\dagger$.
2. For matrices W, W' with the same rows and vectors ψ, φ on M sites, $(W^{\otimes M})^\dagger = (W^\dagger)^{\otimes M}$, $\langle W^{\otimes M} \psi | W'^{\otimes M} \varphi \rangle = \langle \psi | (W^\dagger W')^{\otimes M} \varphi \rangle$; if $W^\dagger W$ is idempotent, then $\|W^{\otimes M} \psi\| \leq \|\psi\|$; and on $M \geq 1$ sites $\mathbb{1}^{\otimes M} = \mathbb{1}$ and $0^{\otimes M} = 0$.
3. For a coordinate isometry E , $E^\dagger = E^\top$; for rectangular X and Y , the entry of $X^\top \otimes Y$ at $((l, r), (a, c))$ is that of $X|l\rangle\langle r|Y$ at (a, c) .
4. For $0 < \gamma < 1$ there is K such that for all R , all coefficients $|c_j| \leq R$, all matrices L_j of norm at most one and every q , the matrix $\mathcal{B} = \sum_j c_j \mathcal{B}_j L_j^\dagger$ satisfies $\|\mathcal{B}^\dagger \mathcal{B} - \sum_j |c_j|^2 L_j (\rho_j^\top \otimes \mathbb{1}) L_j^\dagger\| \leq R^2 K e^{-\gamma q/\xi}$.
5. For every matrix $\mathcal{B}$ and positive semidefinite P_∞ , the positive part P of $\mathcal{B}$ satisfies $\|P - P_\infty\| \leq \|\mathcal{B}^\dagger \mathcal{B} - P_\infty^2\|^{1/2}$.
6. For $0 < \gamma < 1/2$ there is K such that for all R , all coefficients $|\beta_j| \leq R$ and every N , $\|\sum_j \beta_j \phi_N(A_j)\|^2 - \sum_j |\beta_j|^2\| \leq R^2 K e^{-\gamma N/\xi}$.
7. For a tensor A with blocked map $\mathcal{B} = VP$ on q sites and any vector F on the bond pairs of M sites, $\langle V^{\otimes M} F | \phi_N(A) \rangle = \langle F | \phi_M(P) \rangle$ with $N = qM$. If $\|v\| \leq 1$ and $t \geq 0$, then $t^{-1} |\langle v | w \rangle| \leq |\langle v | w \rangle| / (\|v\| t)$, the right side read as zero when $v = 0$ or $t = 0$.
8. In a normed ring, if $\|E^k\| \leq c$ for every $k \geq 0$ and $\|X_k - E\| \leq \delta$ for every k , then $\|X_1 \dots X_M - E^M\| \leq c((1 + c\delta)^M - 1)$. Consequently, for a tensor F of bond dimension D_2 and $c \geq 0$ there is $C > 0$ such that for all G, G' from bond pairs of dimension D_1 to bond pairs of dimension D_2 with $\|G - G'\| \leq \delta$, if the powers of the mixed transfer map T' of G' against F satisfy $\|T'^k X\| \leq c\|X\|$, then the overlap of G with F on M sites is $\text{Tr } T'^M$ up to $C(M\delta) e^{CM\delta}$.

Proof. ✓ Proof Part (1) is $\|L\|^2 = \|L^\dagger L\|$. Part (2) is $(W^{\otimes M})^\dagger W'^{\otimes M} = (W^\dagger W')^{\otimes M}$, together with $\mathbb{1} - \Pi \geq 0$ for the Hermitian idempotent $\Pi = (W^\dagger W)^{\otimes M}$. Part (3) is a computation of entries. For part (4), the blocks of $\mathcal{B}^\dagger \mathcal{B}$ are $\overline{c_j} c_{j'} L_j \mathcal{B}_j^\dagger \mathcal{B}_{j'} L_{j'}^\dagger$, with coefficients of modulus at most R^2 and the L_j of norm at most one; the diagonal blocks are estimated by Theorem 30.3.7 and the others by Lemma 30.9.2.1. Part (5) is Theorem 30.3.4 at P_∞^2 , whose square root is P_∞ . For part (6), $\|\sum_j \beta_j \phi_N(A_j)\|^2 = \sum_{j,j'} \beta_j \overline{\beta_{j'}} \langle \phi_N(A_{j'}) | \phi_N(A_j) \rangle$ with coefficients of modulus at most R^2 ; the proof of Theorem 30.5.7 bounds the diagonal terms minus one, and Lemma 30.9.2.1 bounds the others. For part (7), the target is $V^{\otimes M} |\phi_M(P)\rangle$ by $\mathcal{B} = VP$, and $(V^\dagger V)^{\otimes M} |\phi_M(P)\rangle = |\phi_M(P)\rangle$ because $V^\dagger V$ is the projector onto the range of P , so part (2) gives the inner product; the second statement is $\|v\|^{-1} \geq 1$. The first statement of part (8) is the induction in the proof of Lemma 30.5.1, which only uses the bound on the powers of E ; the second is that estimate in the normed algebra of operators on rectangular matrices, with the overlap the trace of the M -th power of the mixed transfer map (Lemma 5.2.2). $\square$

Lemma 30.9.2.3 (Positive part for overlapping blocks). Lean Lean Lean Lean Lean Lean ✓ Stmt
Let $A^i = \bigoplus_{j=1}^b \mu_j A_j^i$ be the direct sum with weights $\mu_j \in \mathbb{C}$ of blocks A_j of bond dimensions D_j , each placed on its own bond coordinates by an isometry E_j , and put $K_j = E_j \otimes E_j$. Let every A_j be normal in the gauge (30.3) with positive definite fixed point ρ_j of trace one. Let $\lambda_2 \in \mathbb{C}$ with $|\lambda_2| < 1$ bound the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$ and the moduli of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, let ξ be its correlation length, and let $0 < \gamma < 1$. For $q \geq 1$ the blocked map of A is $\mathcal{B} = \sum_j \mu_j^q \mathcal{B}_j K_j^\dagger$, with $\mathcal{B}_j$ the blocked map of A_j , so $\mathcal{B}^\dagger \mathcal{B} = \sum_{j,j'} \overline{\mu_j} \mu_{j'}^q K_j \mathcal{B}_j^\dagger \mathcal{B}_{j'} K_{j'}^\dagger$. There is K such that for all weights with $|\mu_j| \leq 1$ and every $q \geq 1$

$$\left\| \mathcal{B}^\dagger \mathcal{B} - \sum_j |\mu_j|^{2q} K_j (\rho_j^\top \otimes \mathbb{1}) K_j^\dagger \right\| \leq K e^{-\gamma q / \xi}.$$

For weights one, $\mathcal{B} = \sum_j \mathcal{B}_j K_j^\dagger$, and with

$$G_\infty = \sum_j K_j (\rho_j^\top \otimes \mathbb{1}) K_j^\dagger, \quad P_\infty = \sum_j K_j (\sqrt{\rho_j}^\top \otimes \mathbb{1}) K_j^\dagger,$$

there are K, K' such that for every $q \geq 1$

$$\|\mathcal{B}^\dagger \mathcal{B} - G_\infty\| \leq K e^{-\gamma q / \xi}, \quad \|P - P_\infty\| \leq K' e^{-\gamma q / \xi},$$

where P is the positive part of $\mathcal{B}$. This replaces the block form asserted in [185, eq. (S5)], which holds only when the states of distinct blocks on q sites are orthogonal [106].

Proof. ✓ Proof Every word of A of length $q \geq 1$ is the direct sum of the words of the blocks, multiplied by μ_j^q , which gives the form of $\mathcal{B}$, as in the proof of Theorem 30.9.11, and of its Gram matrix. The diagonal terms, with coefficients $|\mu_j|^{2q} \leq 1$, satisfy $\|\mathcal{B}_j^\dagger \mathcal{B}_j - \rho_j^\top \otimes \mathbb{1}\| \leq K_j e^{-\gamma q / \xi}$ by Theorem 30.3.7, and the others $\|\mathcal{B}_j^\dagger \mathcal{B}_{j'}\| \leq K_{jj'} e^{-\gamma q / \xi}$ by Lemma 30.9.2.1, with coefficients of modulus at most one; summing gives the first bound. For weights one, let $Q = \mathbb{1} - \sum_j K_j K_j^\dagger$, the projector onto the bond pairs whose two indices lie in different blocks or outside all blocks. Then $\mathcal{B}Q = 0$, so $PQ = V^\dagger \mathcal{B}Q = 0$ and $QP = 0$, whence $(P + Q)^2 = \mathcal{B}^\dagger \mathcal{B} + Q$; likewise $P_\infty Q = QP_\infty = 0$ and $P_\infty^2 = G_\infty$, so $(P_\infty + Q)^2 = G_\infty + Q$. Both $P + Q$ and $P_\infty + Q$ are

positive semidefinite, so they are the square roots. If $\rho_j \geq c_j \mathbb{1}$ and $c = (1 + \sum_j 1/c_j)^{-1}$, then

$$G_\infty + Q - c\mathbb{1} = \sum_j K_j((\rho_j - c\mathbb{1})^\top \otimes \mathbb{1})K_j^\dagger + (1 - c)Q \geq 0,$$

so Theorem 30.3.5 gives $\|P - P_\infty\| = \|\sqrt{\mathcal{B}^\dagger \mathcal{B} + Q} - \sqrt{G_\infty + Q}\| \leq \|\mathcal{B}^\dagger \mathcal{B} - G_\infty\|/\sqrt{c}$. $\square$

Lemma 30.9.2.4 (Positive part for weights of different moduli). [Lean] [✓ Stmt] *In the setting of Lemma 30.9.2.3, let $0 < \gamma < 1/2$. There is K such that for all weights with $|\mu_j| \leq 1$ and every $q \geq 1$ the positive part P of the blocked map of $A = \bigoplus_j \mu_j A_j$ satisfies*

$$\|P - P_\infty^{(q)}\| \leq K e^{-\gamma q/\xi}, \quad P_\infty^{(q)} = \sum_j K_j \left(\sqrt{|\mu_j|^{2q} \rho_j}^\top \otimes \mathbb{1} \right) K_j^\dagger = \sum_j |\mu_j|^q K_j (\sqrt{\rho_j}^\top \otimes \mathbb{1}) K_j^\dagger.$$

Proof. [✓ Proof] The matrix $P_\infty^{(q)}$ is positive semidefinite and its square is $G_\infty^{(q)} = \sum_j |\mu_j|^{2q} K_j (\rho_j^\top \otimes \mathbb{1}) K_j^\dagger$, so it is the square root of $G_\infty^{(q)}$. Lemma 30.9.2.3 with $2\gamma < 1$ in place of γ gives $\|\mathcal{B}^\dagger \mathcal{B} - G_\infty^{(q)}\| \leq K_0 e^{-2\gamma q/\xi}$, and Theorem 30.3.4 gives $\|P - P_\infty^{(q)}\| \leq \|\mathcal{B}^\dagger \mathcal{B} - G_\infty^{(q)}\|^{1/2} \leq K_0^{1/2} e^{-\gamma q/\xi}$. The Lipschitz step of Lemma 30.9.2.3 is not available, since $G_\infty^{(q)}$ has eigenvalues $|\mu_j|^{2q}$ times those of ρ_j , which are not bounded below uniformly in q when $|\mu_j| < 1$. $\square$

Lemma 30.9.2.5 (Overlaps for overlapping blocks). [Lean] [Lean] [Lean] [Lean] [Lean] [Lean] [✓ Stmt]

1. For an isometry E placing the bond coordinates of a block and a positive semidefinite σ , the pair of σ placed along E on both legs is the pair of $\tilde{\sigma} = E\sigma E^\dagger$.
2. For a tensor A , matrices $\sigma'_1, \dots, \sigma'_b$ on the bond space, coefficients α_j , and $M \geq 1$, the state $V^{\otimes M} \sum_j \alpha_j \bigotimes_k |\omega(\sigma'_j)\rangle_{R_k L_{k+1}}$ has inner product $\sum_j \bar{\alpha}_j \langle \phi_M(P_{\infty, \sigma'_j}) | \phi_M(P) \rangle$ with $|\phi_N(A)\rangle$, $N = qM$, where P is the positive part of the blocked map and P_{∞, σ'_j} the fixed-point tensor of σ'_j .
3. For blocks placed as in Lemma 30.9.2.3 and positive semidefinite ρ_j , with $\tilde{\rho}_j = E_j \rho_j E_j^\dagger$ and $\Pi_j = E_j E_j^\dagger$, the mixed transfer map of P_∞ against the fixed-point tensor of $\tilde{\rho}_j$ is $X \mapsto \text{Tr}(\Pi_j X) \tilde{\rho}_j$.
4. In the setting of Lemma 30.9.2.3, for every block j there is $C_j > 0$ such that for every $q \geq 1$ and every $M \geq 1$, with $y = M e^{-\gamma q/\xi}$,

$$|\langle \phi_M(P_{\infty, \tilde{\rho}_j}) | \phi_M(P) \rangle - 1| \leq C_j y e^{C_j y}.$$

5. In the setting of Lemma 30.9.2.3 with $0 < \gamma < 1/2$, there is K with $|\|\phi_N(A)\|^2 - b| \leq K e^{-\gamma N/\xi}$ for every $N \geq 1$.

Proof. [✓ Proof] Part (1) holds because $\sqrt{E\sigma E^\dagger} = E\sqrt{\sigma}E^\dagger$. Part (2) is Lemma 30.9.2.2(7) for the product of pairs. For part (3), the letter of P_∞ at the physical pair (l, r) is $\sum_k \sqrt{\tilde{\rho}_k} |l\rangle \langle r| \Pi_k$, that of the fixed-point tensor is $\sqrt{\tilde{\rho}_j} |l\rangle \langle r|$, and $\sum_{l,r} |l\rangle \langle r| Y |r\rangle \langle l| = \text{Tr}(Y) \mathbb{1}$, while $\sqrt{\tilde{\rho}_k} \sqrt{\tilde{\rho}_j} = \delta_{kj} \tilde{\rho}_j$. Since $\text{Tr}(\Pi_j \tilde{\rho}_j) = \text{Tr} \rho_j = 1$, this map is idempotent with trace one, and the telescoping estimate in the proof of Lemma 30.5.1, applied with $\|P - P_\infty\| \leq K' e^{-\gamma q/\xi}$ from Lemma 30.9.2.3, gives part (4). Part (5) is Lemma 30.9.2.2(6) with $\beta_j = 1$ and $R = 1$, since $\phi_N(A) = \sum_j \phi_N(A_j)$ for $N \geq 1$. $\square$

Lemma 30.9.2.6 (Overlaps for weights of different moduli). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

1. For blocks placed as in Lemma 30.9.2.3, positive semidefinite ρ_k and reals $t_k \geq 0$, the mixed transfer map of $P_\infty^t = \sum_k K_k(\sqrt{t_k^2 \rho_k}^\top \otimes \mathbb{1})K_k^\dagger$ against the fixed-point tensor of $\tilde{\rho}_j$ is $X \mapsto t_j \text{Tr}(\Pi_j X) \tilde{\rho}_j$.
2. For blocks placed as there, positive semidefinite ρ_k of trace one, and a block j , there is $C > 0$ such that for all reals $t_k \geq 0$ with $t_j \leq 1$, all δ , all $M \geq 1$, and every matrix P on the bond pairs with $\|P - P_\infty^t\| \leq \delta$,

$$|\langle \phi_M(P_{\infty, \tilde{\rho}_j}) | \phi_M(P) \rangle - t_j^M| \leq C(M\delta) e^{CM\delta}.$$

3. In the setting of Lemma 30.9.2.3 with $0 < \gamma < 1/2$, there is K such that for all weights with $|\mu_j| \leq 1$ and every $N \geq 1$, $\|\phi_N(A)\|^2 - \sum_j |\mu_j^N|^2 \leq K e^{-\gamma N/\xi}$ for $A = \bigoplus_j \mu_j A_j$.

Proof. [✓ Proof](#) For part (1), $\sqrt{t_k^2 \rho_k} = t_k \sqrt{\rho_k}$, so the letter of P_∞^t at (l, r) is $\sum_k t_k \sqrt{\rho_k} |l\rangle \langle r| \Pi_k$, and the computation of Lemma 30.9.2.5(3) carries the factor t_j . For part (2), the mixed transfer map of P_∞^t is $t_j R_j$ with R_j the idempotent of trace one of Lemma 30.9.2.5(3); its powers $t_j^k R_j$, $k \geq 1$, have norm at most $\|R_j\|$, and its M -th power has trace t_j^M . Lemma 30.9.2.2(8) with $D_1 = D_2 = D$, the fixed-point tensor of $\tilde{\rho}_j$, and $c = 1 + \|R_j\|$ gives the estimate. Part (3) is Lemma 30.9.2.2(6) with $R = 1$, with $\|\phi_N(A)\|^2 = \sum_{j,j'} \mu_j^N \overline{\mu_{j'}^N} \langle \phi_N(A_{j'}) | \phi_N(A_j) \rangle$ and coefficients of modulus at most one. $\square$

Theorem 30.9.2.7 (Approximation error for overlapping blocks). [Lean](#) [Lean](#) [✓ Stmt](#) Let $A^i = \bigoplus_{j=1}^b A_j^i$ with $b \geq 1$ be the direct sum with weights one of normal blocks A_j in the gauge (30.3), with positive definite fixed points ρ_j of trace one, each placed on its own bond coordinates. Let $\lambda_2 \in \mathbb{C}$ bound the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$ and the moduli of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, so that, when $|\lambda_2| < 1$, its correlation length ξ bounds every ξ_j and every $\xi_{jj'}$, and let $0 < \gamma < 1/2$. Form the approximating state $\tilde{\phi}_N$ of Definition 30.9.1 with the pairs of the ρ_j on the bond coordinates of their blocks and $\beta_j = 1$. There are constants $C, C' > 0$ such that for every block length q and every $M \geq 1$, with $N = qM$ and $y = M e^{-\gamma q/\xi}$, which is $(N/q) e^{-\gamma q/\xi}$ for $q \geq 1$,

$$\varepsilon(\tilde{\phi}_N, \phi_N) \leq C' y \quad \text{and} \quad \varepsilon(\tilde{\phi}_N, \phi_N) \leq C y e^{C y}. \quad (30.40)$$

This is [185, Lemma 1' (ii)] for blocks of multiplicity one and weight one, with ξ_{diag} replaced by $\max(\xi_{\text{diag}}, \xi_{\text{off-diag}})$. The states of distinct blocks need not be orthogonal, and no condition $q = o(N)$ is needed. With ξ_{diag} alone the bound is false (Theorem 30.9.9) [106].

Proof. [✓ Proof](#) This is the case $w_j = 1$ of Theorem 30.9.2.8, where $\beta_j = 1$ and $\alpha_j = b^{-1/2}$. $\square$

Theorem 30.9.2.8 (Approximation error for overlapping blocks with weights). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A^i = \bigoplus_{j=1}^b w_j A_j^i$ be the direct sum with real weights $0 \leq w_j \leq 1$, at least one equal to one, of normal blocks A_j in the gauge (30.3), with positive definite fixed points ρ_j of trace one, each placed on its own bond coordinates. Let $\lambda_2 \in \mathbb{C}$ bound the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$ and the moduli of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, let ξ be its correlation length, and let $0 < \gamma < 1/2$. Form the approximating state $\tilde{\phi}_N$ of Definition 30.9.1 with the pairs of the ρ_j on the bond coordinates of

their blocks and $\beta_j = w_j^N$. There are constants $C, C' > 0$ such that for every block length q and every $M \geq 1$, with $N = qM$ and $y = Me^{-\gamma q/\xi}$,

$$\varepsilon(\tilde{\phi}_N, \phi_N) \leq C'y \quad \text{and} \quad \varepsilon(\tilde{\phi}_N, \phi_N) \leq Cy e^{C'y}.$$

This is [185, Lemma 1' (ii)] for blocks of multiplicity one with weights of possibly different moduli, normalized as in the source so that the largest is one, and with phases absorbed into the blocks, with ξ_{diag} replaced by $\max(\xi_{\text{diag}}, \xi_{\text{off-diag}})$ [106].

Proof. ✓ Proof If $|\lambda_2| \geq 1$, or $q = 0$, then $y \geq 1$ and the bound holds with $C = 1$, since $\varepsilon \leq 1$. Otherwise let $q \geq 1$. By Lemma 30.9.2.5(1) and (2), with $\alpha_j = \beta_j/\sqrt{b}$ and $b = \sum_j \beta_j^2 \geq 1$, the unnormalized approximating vector has inner product $S/\sqrt{b}$ with $|\phi_N(A)\rangle$, where $S = \sum_j \beta_j z_j$ and $z_j = \langle \phi_M(P_{\infty, \tilde{\rho}_j}) | \phi_M(P) \rangle$; its norm is at most one, so $|\langle \tilde{\phi}_N | \phi_N \rangle| \geq |S|/(\sqrt{b}t)$ with $t = \|\phi_N(A)\|$. Lemma 30.9.2.4 gives $\|P - P_\infty^t\| \leq Ke^{-\gamma q/\xi}$ with $t_k = w_k^q$, so Lemma 30.9.2.6(2) gives $|z_j - w_j^N| \leq C_j y e^{C_j y}$ and $|S - b| \leq \sum_j \beta_j |z_j - \beta_j| \leq \sum_j C_j y e^{C_j y}$. Lemma 30.9.2.6(3) gives $|t^2 - b| \leq K'e^{-\gamma N/\xi} \leq K'y$. For $b \geq 1$, $|S - b| \leq \delta$ and $|t^2 - b| \leq \eta$ give $1 - |S|/(\sqrt{b}t) \leq \delta + \eta$; this is trivial when $\delta + \eta \geq 1$, and otherwise $t > 0$, $|S| \geq b - \delta$ and $\sqrt{b}t \leq (b + t^2)/2 \leq b + \eta/2$, so $|S|/(\sqrt{b}t) \geq 1 - \delta - \eta$. Two elementary inequalities finish the proof, here with $s = 1$. Let $0 < s \leq 1$. If $r \geq 0$, $C \geq 1$ and $u \geq 1$, then $1 - r \leq Cu e^{Cu}/s$. If $0 \leq x \leq 1$, $M \geq 1$, $u = Mx^q$, $K' \geq 0$, $C_j \geq 0$, $e_j \leq C_j u e^{C_j u}$ and $1 - r \leq s^{-1} \sum_j e_j + K'x^{Mq}$, then $1 - r \leq Cu e^{Cu}/s$ with $C = \sum_j C_j + K' + 1$. With $x = e^{-\gamma/\xi}$, $e_j = |z_j - \beta_j|$ and $r = |\langle \tilde{\phi}_N | \phi_N \rangle|$ this gives the second bound; the first follows because $\varepsilon \leq 1$. $\square$

30.9.3 Repeated blocks whose states overlap

The two corrections of the preceding subsections combine: for blocks with multiplicities whose states on q sites overlap, the corrected approximating state of Definition 30.9.1.3 converges at the rate set by the correlation lengths of the blocks and of their mixed transfer maps [40, 106]. The estimates of Lemma 30.9.2.2 use only that the embeddings are isometries with orthogonal ranges, and apply with the isometries L_j of Theorem 30.9.1.2.

Lemma 30.9.3.1 (Positive part and overlaps for repeated overlapping blocks). Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt With the notation of Theorem 30.9.1.2, let the blocks be as in Lemma 30.9.2.3, with c_j and L_j the norms and isometries of the copies on q sites, and put

$$P_\infty = \sum_j c_j L_j (\sqrt{\rho_j}^\top \otimes \mathbb{1}) L_j^\dagger, \quad L_j^\dagger P_\infty = \sum_k \mu_{j,k}^q (\sqrt{\rho_j}^\top \otimes \mathbb{1}) K_{j,k}^\dagger.$$

1. P_∞ is positive semidefinite, its square is $\sum_j c_j^2 L_j (\rho_j^\top \otimes \mathbb{1}) L_j^\dagger$, and for $0 < \gamma < 1/2$ there is K such that for all weights with $|\mu_{j,k}| \leq 1$ and every $q \geq 1$ the positive part P of the blocked map satisfies $\|P - P_\infty\| \leq Ke^{-\gamma q/\xi}$.
2. The mixed transfer map of $L_j^\dagger P_\infty$, read as a tensor whose physical leg is a bond pair of block j , against the fixed-point tensor of ρ_j is $X \mapsto \sum_k \mu_{j,k}^q \text{Tr}(E_{j,k}^\dagger X) E_{j,k} \rho_j$.
3. For a block j with at least one copy there is $C_j > 0$ such that for all weights with $|\mu_{j,k}| \leq 1$, all q , δ , $M \geq 1$ and every G with $\|G - L_j^\dagger P_\infty\| \leq \delta$, the overlap of G with the fixed-point tensor of ρ_j on M sites is $\beta_j = \sum_k \mu_{j,k}^{qM}$ up to $C_j(M\delta) e^{C_j M\delta}$.

4. The unnormalized corrected state has inner product $\sum_j \overline{\alpha_j} \langle \phi_M(P_{\infty, \rho_j}) | \phi_M(L_j^\dagger P) \rangle$ with $|\phi_N(A)\rangle$, and for isometries L_j with orthogonal ranges, $\beta \neq 0$ and $\alpha_j = \beta_j / (\sum_l |\beta_l|^2)^{1/2}$ it has norm at most one.

Proof. ✓ Proof Part (1): $c_j L_j^\dagger = \sum_k \mu_{j,k}^q K_{j,k}^\dagger$ and Lemma 30.9.2.2(1) give the form of $L_j^\dagger P_\infty$ and of P_∞^2 ; Lemma 30.9.2.2(4) with $c_j \leq (\sum_l m_l)^{1/2}$ and 2γ in place of γ , and Lemma 30.9.2.2(5), give the rate. Part (2): the letter of $(\sqrt{\rho}^\top \otimes \mathbb{1})K^\dagger$ at (l, r) is $E\sqrt{\rho}|l\rangle\langle r|E^\dagger$, and $\mathcal{E}_{P_\infty, \rho}(Y) = \text{Tr}(Y)\rho$. Part (3): the maps $R_k(X) = \text{Tr}(E_{j,k}^\dagger X)E_{j,k}\rho_j$ satisfy $R_k R_{k'} = \delta_{kk'} R_k$ and $\text{Tr} R_k = 1$, so $(\sum_k \mu_{j,k}^q R_k)^n = \sum_k \mu_{j,k}^{qn} R_k$ for $n \geq 1$ has norm at most $\sum_k \|R_k\|$ and its M -th power has trace β_j ; Lemma 30.9.2.2(8) applies. Part (4) follows from Lemma 30.9.2.2(7), $\langle L_j^{\otimes M} \Omega_j | \psi \rangle = \langle \Omega_j | (L_j^\dagger)^{\otimes M} \psi \rangle$, and the orthonormality of the states $L_j^{\otimes M} |\Omega_j\rangle$. $\square$

Lemma 30.9.3.2 (A Sylvester equation with positive coefficients). Lean ✓ Stmt Let P and S be positive semidefinite $n \times n$ matrices, let $\lambda > 0$ and R satisfy $\lambda R R^\dagger \leq P^2$, and let $X = P W_1 + W_2 S$ satisfy $PX + XS = RZ$. Then $\|X\| \leq n^2 \lambda^{-1/2} \|Z\|$.

Proof. ✓ Proof In orthonormal eigenbases $P = \sum_a p_a f_a f_a^\dagger$ and $S = \sum_i s_i e_i e_i^\dagger$ the equation reads $(p_a + s_i) \langle f_a | X e_i \rangle = \langle R^\dagger f_a | Z e_i \rangle$. From $\lambda R R^\dagger \leq P^2$, $\|R^\dagger f_a\| \leq \lambda^{-1/2} p_a$, so $|\langle f_a | X e_i \rangle| \leq \lambda^{-1/2} \|Z\|$ whenever $p_a + s_i > 0$, since $p_a / (p_a + s_i) \leq 1$. When $p_a = s_i = 0$ the entry vanishes, because $f_a^\dagger P = 0$ and $S e_i = 0$. The operator norm is at most the sum of the moduli of the n^2 entries. $\square$

Lemma 30.9.3.3 (Positive part relative to the block weights). Lean Lean Lean Lean Lean Lean ✓ Stmt

1. Let L_j be isometries with orthogonal ranges in dimension n , $c_j > 0$, $C = \sum_j c_j L_j L_j^\dagger$ and $C^{-1} = \sum_j c_j^{-1} L_j L_j^\dagger$, let $h_j \geq 0$ with $h_j^2 = g_j \geq \lambda \mathbb{1}$ for some $\lambda > 0$, and let $\mathcal{B} = \widetilde{\mathcal{B}} C$. If $\Delta = \widetilde{\mathcal{B}}^\dagger \widetilde{\mathcal{B}} - \sum_j L_j g_j L_j^\dagger$ satisfies $\|\Delta\| \leq \lambda/2$, the positive part P of $\mathcal{B}$ satisfies $\|(P - S)C^{-1}\| \leq n^2 (\lambda/2)^{-1/2} \|\Delta\|$ with $S = \sum_j c_j L_j h_j L_j^\dagger$.
2. With the notation of Theorem 30.9.1.2 and Lemma 30.9.3.1, $C^{-1} L_j = c_j^{-1} L_j$, the block $L_j^\dagger P C^{-1} L_{j'}$ equals $c_{j'}^{-1} L_j^\dagger P L_{j'}$, and the block $L_j^\dagger P_\infty C^{-1} L_{j'}$ is $\sqrt{\rho_j}^\top \otimes \mathbb{1}$ for $j = j'$ and zero otherwise.
3. Let the blocks be normal in the gauge (30.3) with positive definite fixed points ρ_j of trace one, let λ_2 with $|\lambda_2| < 1$ bound the moduli of the eigenvalues other than one of every $\mathcal{E}_{A_j}$ and of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, and let $0 < \gamma < 1$ with $e^{-\gamma/\xi} < 1$. There are K and q_0 such that for all nonzero weights and all $q \geq \max(q_0, 1)$, $\|(P - P_\infty)C^{-1}\| \leq K e^{-\gamma q/\xi}$, and every block $L_j^\dagger P C^{-1} L_{j'}$ is $L_j^\dagger P_\infty C^{-1} L_{j'}$ up to $K e^{-\gamma q/\xi}$.

Proof. ✓ Proof (1) The positive part satisfies $P^2 = \mathcal{B}^\dagger \mathcal{B} = C \widetilde{\mathcal{B}}^\dagger \widetilde{\mathcal{B}} C \geq (\lambda/2) C^2$, because $\widetilde{\mathcal{B}}^\dagger \widetilde{\mathcal{B}} \geq \sum_j L_j g_j L_j^\dagger - (\lambda/2) \mathbb{1}$ and $g_j \geq \lambda \mathbb{1}$. The matrix $X = (P - S)C^{-1}$ solves $PX + XS = C \Delta \Pi$ with $\Pi = C C^{-1}$, since $S^2 = C(\sum_j L_j g_j L_j^\dagger)C$ and S commutes with C^{-1} . It has the form $P W_1 + W_2 S$ with $W_1 = C^{-1}$ and $W_2 = -C^{-1}$, and $\|\Pi\| \leq 1$; Lemma 30.9.3.2 applies with $R = C$ and $Z = \Delta \Pi$. (2) follows from the orthogonality of the ranges of the L_j and the form of P_∞ in Lemma 30.9.3.1. (3) By Theorem 30.9.1.2 the blocked map is $\mathcal{B} = \widetilde{\mathcal{B}} C$ with $\widetilde{\mathcal{B}} = \sum_j \mathcal{B}_j L_j^\dagger$, and Lemma 30.9.2.2(4) with unit coefficients gives $\|\Delta\| \leq K e^{-\gamma q/\xi}$ for $g_j = \rho_j^\top \otimes \mathbb{1}$ and $h_j = \sqrt{\rho_j}^\top \otimes \mathbb{1}$.

which is at most $\lambda/2$ for $q \geq q_0$, where $\lambda > 0$ bounds every $\rho_j^\top \otimes \mathbb{1}$ from below. Part (1) gives the first bound, and the blocks follow from part (2) since the L_j have norm one. $\square$

Lemma 30.9.3.4 (Overlaps relative to the block weights). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Stmt](#) With the notation of Lemma 30.9.3.3, let $q \geq 1$ and $M \geq 1$, and put $N = qM$.

1. The letters of the tensor read from $YK^\dagger$, for a coordinate isometry K placing a bond pair, are those of the tensor read from Y , placed on the bond coordinates of K .
2. $L_j^\dagger P = \sum_{j'} \sum_k \mu_{j',k}^q (L_j^\dagger P C^{-1} L_{j'}) K_{j',k}^\dagger$, so the tensor read from $L_j^\dagger P$ is the direct sum over the copies (j', k) , with the weights $\mu_{j',k}^q$, of the tensors read from the blocks $L_j^\dagger P C^{-1} L_{j'}$.
3. For every tensor F on the bond pairs of block j , $\langle \phi_M(F) | \phi_M(L_j^\dagger P) \rangle = \sum_{j'} \beta_{j'} \langle \phi_M(F) | \phi_M(L_j^\dagger P C^{-1} L_{j'}) \rangle$ with $\beta_{j'} = \sum_k \mu_{j',k}^N$.
4. For positive semidefinite ρ_j of trace one and blocks j, j' there is $C > 0$ such that for all nonzero weights, all $q, \delta, M \geq 1$, and every G with $\|G - L_j^\dagger P_\infty C^{-1} L_{j'}\| \leq \delta$, the overlap of the tensor read from G with the fixed-point tensor of ρ_j on M sites is $\delta_{jj'}$ up to $C(M\delta) e^{CM\delta}$.

Proof. [Stmt](#) (1) is a computation of entries. (2) P vanishes off the ranges of the L_j , so $L_j^\dagger P = \sum_{j'} L_j^\dagger P L_{j'} L_{j'}^\dagger$, and $c_{j'} L_{j'}^\dagger = \sum_k \mu_{j',k}^q K_{j',k}^\dagger$; part (1) gives the direct sum. (3) The state of a direct sum of copies with weights $\mu_{j',k}^q$ on M sites is $\sum_{j'} (\sum_k \mu_{j',k}^{qM})$ times the state of block j' . (4) By Lemma 30.9.3.3(2) the limit is $\sqrt{\rho_j}^\top \otimes \mathbb{1}$ for $j = j'$, read as the fixed-point tensor of ρ_j , whose mixed transfer map against itself is the idempotent $X \mapsto \text{Tr}(X) \rho_j$ of trace one, and zero otherwise, whose mixed transfer map vanishes. Lemma 30.9.2.2(8) applies. $\square$

Theorem 30.9.3.5 (Approximation error for repeated overlapping blocks). [Lean](#) [Lean](#) [Stmt](#)
 With the notation of Theorem 30.9.1.2, let the blocks A_j be normal in the gauge (30.3) with positive definite fixed points ρ_j of trace one, let $\lambda_2 \in \mathbb{C}$ bound the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$ and the moduli of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, let ξ be its correlation length, and let $0 < \gamma < 1/2$. There are $C, C' > 0$ such that for all nonzero complex weights, every block length q and every $M \geq 1$ with $N = qM$ and $(\beta_1, \dots, \beta_b) \neq 0$, the corrected approximating state of Definition 30.9.1.3 satisfies, with $y = M e^{-\gamma q / \xi}$,

$$\varepsilon(\tilde{\phi}_N, \phi_N) \leq C y e^{C y}, \quad \varepsilon(\tilde{\phi}_N, \phi_N) \leq C' y.$$

This is [185, Lemma 1' (ii)] for the corrected state, with ξ_{diag} replaced by $\max(\xi_{\text{diag}}, \xi_{\text{off-diag}})$. The constants do not depend on the weights: the sums β_j may cancel and $\sum_j |\beta_j|^2$ may be arbitrarily small, and the normalization $|\mu_{j,k}| \leq 1$ of the source is not used [118]. The states of distinct blocks need not be orthogonal, and no condition $q = o(N)$ is needed.

Proof. [Stmt](#) Let q_0 be the threshold of Lemma 30.9.3.3(3). If $e^{-\gamma/\xi} \geq 1$, or $q = 0$, or $q < q_0$, then C can be chosen with $C y \geq 1$, and $\varepsilon \leq 1$ gives the bound. Otherwise $|\lambda_2| < 1$. By Lemma 30.9.3.1(4) the unnormalized corrected state has inner product $S/\sqrt{b}$ with $|\phi_N(A)\rangle = \sum_j \beta_j |\phi_N(A_j)\rangle$, where $b = \sum_j |\beta_j|^2$, $S = \sum_j \bar{\beta}_j z_j$ and $z_j = \langle \phi_M(P_{\infty, \rho_j}) | \phi_M(L_j^\dagger P) \rangle$, and it has norm at most one, so $|\langle \tilde{\phi}_N | \phi_N \rangle| \geq |S|/(\sqrt{b} t)$ with $t = \|\phi_N(A)\|$. By Lemma 30.9.3.4(3), $z_j = \sum_{j'} \beta_{j'} w_{jj'}$, and by Lemma 30.9.3.3(3) and Lemma 30.9.3.4(4), $|w_{jj'} - \delta_{jj'}| \leq C_{jj'} y e^{C_{jj'} y}$. Hence

$$\frac{S}{b} - 1 = \frac{1}{b} \sum_{j,j'} \bar{\beta}_j \beta_{j'} (w_{jj'} - \delta_{jj'}), \quad \left| \frac{S}{b} - 1 \right| \leq \sum_{j,j'} C_{jj'} y e^{C_{jj'} y},$$

since $|\beta_j \beta_{j'}| \leq b$. Lemma 30.9.2.2(6) with $R = \sqrt{b}$ gives $|t^2/b - 1| \leq Ke^{-\gamma N/\xi} \leq Ky$. The inequalities in the proof of Theorem 30.9.2.8, applied to S/b and $t/\sqrt{b}$, give $\varepsilon \leq Cy e^{Cy}$, and the second form follows because $\varepsilon \leq 1$. $\square$

30.9.4 Weights with a phase

For blocks of multiplicity one write $\mu_j = |\mu_j|u_j$ with $|u_j| = 1$. The blocked map is $B = B_0U$, where B_0 is the blocked map for the weights $|\mu_j|$ and the diagonal unitary U multiplies the bond coordinates of block j by u_j^q . Hence $B^\dagger B = U^\dagger B_0^\dagger B_0 U$, and the polar decompositions satisfy

$$P = U^\dagger P_0 U, \quad V = V_0 U.$$

The diagonal blocks of P are those of P_0 and do not depend on the phases, contrary to the block form [185, eq. (S5)]; only the mixed blocks of P , which vanish for orthogonal blocks, carry the phases $\bar{u}_j^q u_j^q$. The product $|\Omega_j\rangle = \bigotimes_{k=1}^M |\omega_j\rangle_{R_k L_{k+1}}$ of the pairs of block j lies on the bond coordinates of block j , so $V^{\otimes M} |\Omega_j\rangle = u_j^N V_0^{\otimes M} |\Omega_j\rangle$ on $N = qM$ sites. The coefficients $\beta_j = \mu_j^N$ of [185, eq. (S4)] are correct in the expansion [185, eq. (S3)] of $|\phi_N\rangle$, but under $V^{\otimes M}$ in [185, eqs. (S6) and (S7)] they count the phase u_j^N twice, and already for orthogonal blocks the approximating state does not converge (Theorem 30.9.4.1). With $\beta_j = |\mu_j|^N$ it converges at the rate of Theorem 30.9.2.8 (Theorem 30.9.4.2) [106].

[illegible]

$$\langle \tilde{\phi}_N | \phi_N \rangle = \frac{1 + \bar{\mu}^N}{2} \quad \text{for } \beta = (1, \mu^N), \quad \langle \tilde{\phi}_N | \phi_N \rangle = 1 \quad \text{for } \beta = (1, |\mu|^N) = (1, 1). \quad (30.41)$$

More generally, for coefficients α with $|\alpha_1|^2 + |\alpha_2|^2 = 1$ the overlap is $(\bar{\alpha}_1 + \bar{\alpha}_2)/\sqrt{2}$. For $\mu = -1$ and odd q and M the approximating state with $\beta = (1, \mu^N)$ is orthogonal to $|\phi_N\rangle$, so $\varepsilon(\tilde{\phi}_N, \phi_N) = 1$. The transfer maps of both blocks have no eigenvalue other than one, so every λ_2 bounds their subleading eigenvalues. For $\mu = -1$, every λ_2 with $0 < |\lambda_2| < 1$, every $\gamma > 0$ and every constant C there is an odd M such that, with $q = M$, $N = M^2$ and $y = (N/q)e^{-\gamma q/\xi_{\text{diag}}}$, $\varepsilon(\tilde{\phi}_N, \phi_N) > Cy e^{Cy}$. The sequence $q = M$ satisfies $q = o(N)$, so the bound (30.35) fails for orthogonal blocks of multiplicity one when a weight is not a nonnegative real number [106].

Proof. ✓ Proof The hypotheses are checked as in the proof of Theorem 30.9.9: both blocks are one-dimensional with $\sum_i |a_i|^2 = 1$, their states $|0 \cdots 0\rangle$ and $|1 \cdots 1\rangle$ are linearly independent for $N \geq 1$, and $A_1^1 = 0 \neq A_2^1$. These checks hold for the blocks A_1 and A_2 with any multiplicities and any nonzero weights $|\mu_{i,k}| \leq 1$, one of modulus one, and are shared with Theorem 30.9.8.

For $q \geq 1$ the Gram matrix of the blocked map on the diagonal coordinates is the identity, since $|\mu|^{2q} = 1$ and the off-diagonal entries are $0^q = 0$. So P and the support projector both equal $JJ^\dagger$, where J places a coordinate e on the pair (e, e) . As in the proof of Theorem 30.9.9, the

overlap of the unnormalized approximating vector with the unnormalized target is $\overline{\alpha_1} + \overline{\alpha_2}$, the vector has squared norm $|\alpha_1|^2 + |\alpha_2|^2 = 1$, and $\|\phi_N(A)\|^2 = 2$. For $\beta = (1, \mu^N)$, $\alpha = (1, \mu^N)/\sqrt{2}$, and for $\beta = (1, 1)$, $\alpha = (1, 1)/\sqrt{2}$, which gives (30.41). For $\mu = -1$ and N odd, $1 + \bar{\mu}^N = 0$. Finally Cye^{Cy} with $y = Mr^M$ and $r = e^{-\gamma/\xi_{\text{diag}}} < 1$ is eventually below one. $\square$

Theorem 30.9.4.2 (Approximation error for complex weights). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) Let $A^i = \bigoplus_{j=1}^b \mu_j A_j^i$ be the direct sum with complex weights $|\mu_j| \leq 1$, at least one of modulus one, of normal blocks A_j in the gauge (30.3), with positive definite fixed points ρ_j of trace one, each placed on its own bond coordinates. Let $\lambda_2 \in \mathbb{C}$ bound the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$ and the moduli of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, let ξ be its correlation length, and let $0 < \gamma < 1/2$. Form the approximating state $\tilde{\phi}_N$ of Definition 30.9.1 with the pairs of the ρ_j on the bond coordinates of their blocks and $\beta_j = |\mu_j|^N$. There are constants $C, C' > 0$ such that for every block length q and every $M \geq 1$, with $N = qM$ and $y = Me^{-\gamma q/\xi}$,

$$\varepsilon(\tilde{\phi}_N, \phi_N) \leq C'y \quad \text{and} \quad \varepsilon(\tilde{\phi}_N, \phi_N) \leq Cye^{Cy}.$$

This is [185, Lemma 1'(ii)] for blocks of multiplicity one, with $\beta_j = |\mu_j|^N$ in place of the coefficients $\beta_j = \mu_j^N$ that [185, eqs. (S6) and (S7)] take from [185, eq. (S4)], and with ξ_{diag} replaced by $\max(\xi_{\text{diag}}, \xi_{\text{off-diag}})$. With $\beta_j = \mu_j^N$ the bound is false (Theorem 30.9.4.1) [106].

Proof. [Proof](#) Let $u_j = e^{i \arg \mu_j}$, so that $\mu_j = |\mu_j|u_j$ and $|u_j| = 1$. Then $\bigoplus_j \mu_j A_j = \bigoplus_j |\mu_j|(u_j A_j)$. The blocks $u_j A_j$ are normal, satisfy the gauge (30.3), and have the transfer maps $\mathcal{E}_{u_j A_j} = |u_j|^2 \mathcal{E}_{A_j} = \mathcal{E}_{A_j}$, hence the same fixed points and eigenvalues. Their mixed transfer maps are $u_j \overline{u_{j'}} \mathcal{E}_{jj'}$, whose eigenvalues are those of $\mathcal{E}_{jj'}$ multiplied by a number of modulus one. Theorem 30.9.2.8, applied to the blocks $u_j A_j$ with the real weights $w_j = |\mu_j|$, gives the bounds with $\beta_j = w_j^N = |\mu_j|^N$. $\square$

Copies with weights of different phases

When a block has several copies whose weights have different phases, the phases cannot be moved into the block. For the corrected state of Definition 30.9.1.3 the coefficients are those of [185, eq. (S4)], $\beta_j = \sum_k \mu_{j,k}^N$, because the isometry L_j of Definition 30.9.1.1 carries the phases $\overline{\mu_{j,k}^q}/c_j$ of the copies: Theorems 30.9.1.4 and 30.9.3.5 hold for complex weights. Neither $\sum_k |\mu_{j,k}|^N$, the coefficient of Theorem 30.9.4.2 for one copy, nor c_j^M works in their place [40]. The source's construction itself, with the pairs of each block on one of its copies, converges once its coefficients are changed: the partial isometry V sends the pairs placed on one copy to a multiple of the pairs placed by L_j .

Definition 30.9.4.3 (Coefficients for the pairs on one copy). [Lean](#) [Lean](#) [Lean](#) [Stmt](#) With the notation of Definition 30.9.1.1, choose a copy k_j of every block j , and let $N = qM$. The coefficient of block j is

$$\beta'_j = \beta_j \left(\frac{c_j}{\mu_{j,k_j}^q} \right)^M, \quad \beta_j = \sum_k \mu_{j,k}^N.$$

If block j has the single copy k_j , then $\beta'_j = |\mu_{j,k_j}|^N$, the coefficient of Theorem 30.9.4.2. Since the weights are nonzero, $\beta' \neq 0$ whenever $\beta \neq 0$.

This is [185, Lemma 1' (ii)] for the source's construction with the coefficients β'_j in place of the coefficients β_j that [185, eq. (S7)] takes from [185, eq. (S4)], for weights of any phase, in the scopes of Theorems 30.9.1.4 and 30.9.3.5. With the coefficients β_j the bound is false (Theorems 30.9.8 and 30.9.4.1) [40].

Proof. ✓ Proof By Lemma 30.9.4.4(6) the overlap is that of the corrected state with the coefficients β_j , which Theorems 30.9.1.4 and 30.9.3.5 bound. □

30.9.5 Overlapping blocks at the second-order rate

For blocks of multiplicity one and weight one whose states overlap, Theorem 30.9.2.7 bounds the error at the rate $e^{-\gamma q/\xi}$ with $\gamma < 1/2$. The positive part P converges to P_∞ only to first order in the overlaps of distinct blocks, but the overlaps $z_j = \langle \phi_M(P_{\infty, \tilde{\rho}_j}) | \phi_M(P) \rangle$ depend on P only to second order around P_∞ , as in the normal case (Theorem 30.5.6). This subsection proves the rate $e^{-2\gamma q/\xi}$ for every $0 < \gamma < 1$, the rate of the normal case (Theorem 30.5.7). For the blocks of Theorem 30.9.9 the error is at least $(9/25)^q/16$ for $M \geq 3$, with $3/5$ the eigenvalue of their mixed transfer map; since $\lambda_2 = 3/5$ is admissible there, the exponent $2\gamma q/\xi$ cannot be replaced by cq/ξ with $c > 2$ [106].

Theorem 30.9.5.1 (Powers of an element compressed by an idempotent). Lean ✓ Stmt Let e be an idempotent of a normed algebra over $\mathbb{C}$, and let Tr be linear with $\text{Tr}(xy) = \text{Tr}(yx)$, $|\text{Tr } x| \leq K\|x\|$ and $\text{Tr } e = 1$. There is $C > 0$ such that every T with $eTe = \alpha e$ for a real $\alpha \leq 1$, $\|T - e\| \leq \delta$ and $1 - \alpha \leq \delta^2$ satisfies, for $M \geq 2$ and $CM\delta^2 < 1$,

$$|\text{Tr } T^M - 1| \leq C(M\delta^2) e^{CM\delta^2}.$$

Proof. ✓ Proof From $CM\delta^2 < 1$ with $C \geq 4$ one gets $\delta \leq 1$ and $\alpha \geq 3/4$. Put $Z = \alpha^{-1}T - e$. Then $eZe = 0$, $T = \alpha(e + Z)$, and $\|Z\| \leq \alpha^{-1}(\|T - e\| + (1 - \alpha)\|e\|) \leq 2(1 + \|e\|)\delta$, so the blocks of Z are bounded by a multiple z of δ with $z \leq 1/2$. Theorem 30.5.5 bounds $|\text{Tr}(e + Z)^M - 1|$ by a multiple of $M\delta^2 e^{cM\delta^2}$. Finally $\text{Tr } T^M - 1 = \alpha^M(\text{Tr}(e + Z)^M - 1) + (\alpha^M - 1)$, with $0 \leq \alpha^M \leq 1$ and $1 - \alpha^M \leq M(1 - \alpha) \leq M\delta^2$. □

Lemma 30.9.5.2 (Overlaps of overlapping blocks, to second order). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt

1. For a $D^2 \times D^2$ matrix H , read as a tensor X with physical index the output pair, and a $D \times D$ matrix σ , the mixed transfer map of X against the fixed-point tensor $P_{\infty, \sigma}$ satisfies $\text{Tr} \sum_i X^i \sigma (P_{\infty, \sigma}^i)^\dagger = \text{Tr}(H(\sqrt{\sigma}^\top \otimes \sigma))$, which is real when H and σ are Hermitian.
2. For an isometry E placing a block, $\sqrt{E\sigma E^\dagger} = E\sqrt{\sigma}E^\dagger$ for $\sigma \geq 0$, $\text{Tr}(E\sigma E^\dagger) = \text{Tr}(\Pi E\sigma E^\dagger) = \text{Tr } \sigma$, and $EXE^\dagger \Pi = EXE^\dagger$ for $\Pi = EE^\dagger$.
3. For blocks placed as in Lemma 30.9.2.3 with weights one, $A^i E_j = E_j A_j^i$, and the transfer map of A satisfies $\mathcal{E}_A(E_j X E_j^\dagger) = E_j \mathcal{E}_{A_j}(X) E_j^\dagger$; in particular $\tilde{\rho}_j = E_j \rho_j E_j^\dagger$ is a fixed point of $\mathcal{E}_A$.
4. A linear map $R : X \mapsto \text{Tr}(QX)\sigma$ has trace $\text{Tr}(Q\sigma)$ and is idempotent when $\text{Tr}(Q\sigma) = 1$, and $RTR = \text{Tr}(QT(\sigma))R$ for every linear map T . If $\sqrt{\sigma}Q = \sqrt{\sigma}$, then $\text{Tr}(QY) = \text{Tr } Y$ for every value Y of a mixed transfer map against $P_{\infty, \sigma}$.

5. The mixed transfer matrix R_j of P_∞ against $P_{\infty, \tilde{\rho}_j}$ is idempotent with trace one, and for every $D^2 \times D^2$ matrix H , read as a tensor X , the mixed transfer matrix τ of X against $P_{\infty, \tilde{\rho}_j}$ satisfies $R_j \tau R_j = \alpha R_j$ with $\alpha = \text{Tr} \sum_i X^i \tilde{\rho}_j (P_{\infty, \tilde{\rho}_j}^i)^\dagger$.
6. For a state σ , the vector $(P_{\infty, \sigma}^i \sqrt{\sigma})_i$ has norm one, and so has $(X^i \sqrt{\sigma})_i$ whenever $\text{Tr} \mathcal{E}_X(\sigma) = 1$. If $\|u_q\| - c \leq K' x^q$ and $\|u_q - v\| \leq K x^q$ for every $q \geq 1$, with $0 \leq x < 1$, then $\|v\| = c$.
7. In the setting of Lemma 30.9.2.3 with weights one, if $\lambda_2 \neq 0$, then for every block j there is $C_j > 0$ such that for every $q \geq 1$ and every $M \geq 2$ with $C_j y < 1$, where $y = M e^{-2\gamma q/\xi}$,

$$|\langle \phi_M(P_{\infty, \tilde{\rho}_j}) | \phi_M(P) \rangle - 1| \leq C_j y e^{C_j y}.$$

Proof. ✓ Proof Part (1) is the entrywise computation: the letter of $P_{\infty, \sigma}$ at the pair (l, r) is $\sqrt{\sigma} |l\rangle \langle r|$, and the conjugate of $\text{Tr}(HW)$ is $\text{Tr}((HW)^\dagger) = \text{Tr}(WH)$ for Hermitian H and W . Part (2) follows from $E^\dagger E = \mathbb{1}$ and the uniqueness of the positive square root, and part (3) from $E_k^\dagger E_j = \delta_{kj} \mathbb{1}$. In part (4), $R(T(R(X))) = \text{Tr}(QX) \text{Tr}(QT(\sigma)) \sigma$, and the values of a mixed transfer map against $P_{\infty, \sigma}$ are sums of terms $Y \sqrt{\sigma}$, on which $\text{Tr}(QY \sqrt{\sigma}) = \text{Tr}(Y \sqrt{\sigma} Q)$. Part (5) is part (4) with the map $X \mapsto \text{Tr}(\Pi_j X) \tilde{\rho}_j$ of Lemma 30.9.2.5(3) and $\text{Tr}(\Pi_j \tilde{\rho}_j) = 1$, since $\sqrt{\tilde{\rho}_j} \Pi_j = \sqrt{\tilde{\rho}_j}$ by part (2). In part (6), the squared norm of $(X^i \sqrt{\sigma})_i$ is $\text{Tr} \mathcal{E}_X(\sigma)$, and $\|v\| - c \leq (K + K') x^q$ for every q .

For part (7), $\lambda_2 \neq 0$ gives $0 < e^{-\gamma/\xi} < 1$. Let $\iota(X) = (X^i \sqrt{\tilde{\rho}_j})_i$. By part (3) and $\mathcal{E}_P = \mathcal{E}_A^q$, $\text{Tr} \mathcal{E}_P(\tilde{\rho}_j) = 1$, so $\iota(P)$ is a unit vector, as is $\iota(P_{\infty, \tilde{\rho}_j})$. The inner product $\langle \iota(P_{\infty, \tilde{\rho}_j}) | \iota(P_\infty) \rangle$ is the trace of $\text{Tr}(\Pi_j \tilde{\rho}_j) \tilde{\rho}_j$, which is one, and $\iota(P_\infty)$ is a unit vector by part (6) with $u_q = \iota(P)$; hence $\iota(P_\infty) = \iota(P_{\infty, \tilde{\rho}_j})$ and $\|\iota(P) - \iota(P_{\infty, \tilde{\rho}_j})\| \leq K'' e^{-\gamma q/\xi}$ by Lemma 30.9.2.3. The coefficient α of part (5) is $\langle \iota(P_{\infty, \tilde{\rho}_j}) | \iota(P) \rangle$, real by part (1) since P and $\tilde{\rho}_j$ are Hermitian, at most one, and $1 - \alpha = \frac{1}{2} \|\iota(P) - \iota(P_{\infty, \tilde{\rho}_j})\|^2$. With $\|\tau - R_j\| \leq K_3 e^{-\gamma q/\xi}$ and $z_j = \text{Tr} \tau^M$, Theorem 30.9.5.1 with δ a multiple of $e^{-\gamma q/\xi}$ gives the bound. □

Lemma 30.9.5.3 (The norm of the target and one block of sites). Lean Lean Lean Lean ✓ Stmt
In the setting of Lemma 30.9.2.3 with weights one, let ϕ_N be the periodic state of A .

1. There is K with $\|\phi_N\|^2 - b \leq K e^{-\gamma N/\xi}$ for every $N \geq 1$.
2. The states $\phi_N(P_{\infty, \tilde{\rho}_j})$ are orthonormal for every $N \geq 1$.
3. For a vector W with $\|W\|^2 = c > 0$ and every u , $\|u\|^2 - |\langle W | u \rangle|^2 / c \leq \|u - W\|^2$.
4. If $\lambda_2 \neq 0$, there is K' such that for every $q \geq 1$

$$\|\phi_q\|^2 - \frac{1}{b} \left| \sum_j \langle \phi_1(P_{\infty, \tilde{\rho}_j}) | \phi_1(P) \rangle \right|^2 \leq K' e^{-2\gamma q/\xi}.$$

Proof. ✓ Proof For part (1), $\|\phi_N\|^2 = \sum_{j, j'} \langle \phi_N(A_j) | \phi_N(A_{j'}) \rangle$; the diagonal terms minus one are bounded by $K_j e^{-\gamma N/\xi}$ by the normalization in the proof of Theorem 30.5.7 at $\gamma/2$, and the others by $K_{jj'} e^{-\gamma N/\xi}$ by Lemma 30.9.2.1. Part (2) holds because the pairs of the $\tilde{\rho}_j$ are orthonormal and the overlap of products of pairs is the N -th power of the overlap of the pairs. Part (3) is $c - 2 \text{Re } z + |z|^2 / c = |c - z|^2 / c \geq 0$ with $z = \langle W | u \rangle$. For part (4) put $u = \phi_1(P)$ and $W = \sum_j \phi_1(P_{\infty, \tilde{\rho}_j})$. Then $\|u\| = \|\phi_q\|$, the sum is $\langle W | u \rangle$, and $\|W\|^2 = b$ by part (2), so

part (3) bounds the left side by $\|u - W\|^2$. The overlaps of $\phi_1(P_\infty)$ with the $\phi_1(P_{\infty, \tilde{\rho}_j})$ are the traces of the idempotents R_j of Lemma 30.9.5.2(5), which are one, and $\|\phi_1(P_\infty)\|^2 = b$ by Lemma 30.9.5.2(6) and part (1). Hence $\phi_1(P_\infty) = W$, and $\|u - W\|$ is at most a multiple of $\|P - P_\infty\| \leq K'' e^{-\gamma q/\xi}$. $\square$

Theorem 30.9.5.4 (Approximation error for overlapping blocks at the second-order rate). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) In the setting of Theorem 30.9.2.7, let $0 < \gamma < 1$ in place of $0 < \gamma < 1/2$. There are constants $C, C' > 0$ such that for every block length q and every $M \geq 1$, with $N = qM$ and $y = M e^{-2\gamma q/\xi}$, which is $(N/q) e^{-2\gamma q/\xi}$ for $q \geq 1$,

$$\varepsilon(\tilde{\phi}_N, \phi_N) \leq C' y \quad \text{and} \quad \varepsilon(\tilde{\phi}_N, \phi_N) \leq C y e^{C y}.$$

This is [185, Lemma 1' (ii)] for blocks of multiplicity one and weight one, with ξ_{diag} replaced by $\max(\xi_{\text{diag}}, \xi_{\text{off-diag}})$ and the exponent $\gamma q/\xi$ by $2\gamma q/\xi$ [106].

Proof. [Proof](#) If $e^{-\gamma/\xi} \geq 1$, if $q = 0$, or if $Cy \geq 1$, the bound holds because $\varepsilon \leq 1$; otherwise $0 < |\lambda_2| < 1$ and $q \geq 1$. With $z_j = \langle \phi_M(P_{\infty, \tilde{\rho}_j}) | \phi_M(P) \rangle$, $s = |\sum_j z_j|/\sqrt{b}$ and $t = \|\phi_N(A)\|$, the proof of Theorem 30.9.2.8 gives $\varepsilon \leq 1 - s/t$. If $t^2 \geq 1/2$ and $t^2 - s^2 \leq X$ with $X \geq 0$, then $1 - s/t \leq 2X$: for $s < t$, $1 - s/t \leq 1 - s^2/t^2 = (t^2 - s^2)/t^2$. For $M = 1$, Lemma 30.9.5.3(4) gives $X = K'y$, and $t^2 \geq b - K e^{-\gamma q/\xi} \geq 1/2$ once $4K^2 y < 1$. For $M \geq 2$, $N \geq 2q$, so $|t^2 - b| \leq Ky$ by Lemma 30.9.5.3(1), and $|\sum_j z_j - b| \leq \delta = \sum_j C_j y e^{C y}$ by Lemma 30.9.5.2(7), so $s^2 \geq b - 2\delta$ and $X = Ky + 2\delta$. The first bound follows from the second because $\varepsilon \leq 1$. $\square$

30.10 Preparation with measurements

Measurements followed by corrections chosen from their outcomes shorten state preparation [185], which states the idea without fixing a model. The model used here is that of [195]: a local circuit, measurements, and local unitaries depending on the outcomes of all the measurements, which are communicated to every site at no cost. Only the layers of the circuit are counted. In this model the states $|\chi_M\rangle = \sum_{i \in \mathbb{Z}_b} \alpha_i |i\rangle^{\otimes M}$ of Definition 30.9.3 are prepared in a depth independent of M , although their connected correlations do not decay with distance, while by Lemma 30.6.3 those of a state prepared by a local circuit of depth T vanish beyond distance $2T$.

Definition 30.10.1 (Preparation with measurements). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) The local labels are $\{0, \dots, d-1\}$. For a set S of sites of the ring of N sites and a string $m: S \rightarrow \{0, \dots, d-1\}$, let P_m be the projection onto the span of the basis vectors $|x\rangle$ with $x_i = m_i$ for every $i \in S$; these are the outcome projections of a measurement of the sites of S in the computational basis, and P_m is the projection onto the configurations agreeing on S with any configuration extending m . A *protocol* consists of a product vector $|\pi\rangle$, a local circuit U of depth T_1 , a set S of measured sites, and for every m unitaries $v_{m,i}$ on the sites i , with product $V_m = \bigotimes_i v_{m,i}$. It *prepares* a vector $|\psi\rangle$ if $|\pi\rangle \neq 0$ and for every m with $P_m U |\pi\rangle \neq 0$ there is $c \in \mathbb{C}$ with $V_m P_m U |\pi\rangle = c |\psi\rangle$. A vector is *prepared with measurements in depth T* if a protocol with $T_1 \leq T$ prepares it.

The model is that of [195], where the class of such preparations is called QCcc. The correction V_m may depend on the whole string m : classical communication is global and free, and the single-site corrections are not counted. Ancillas are sites of the ring, and the local unitaries between the layers of the circuit, acting on a site and its ancillas and free in [195], are counted here as gates. The correction V_m is a local unitary in the sense of [195]. So every preparation in this sense is one in the sense of [195], of no larger depth.

4. gates on pairwise disjoint neighbouring pairs, given by a list, form a layer whose operator is the product of the gates along the list;
5. a product vector whose site vectors at the sites of Z are multiples of $|0\rangle$ has $|0\rangle$ at the sites of Z .

Proof. ✓ Proof These are entrywise computations. For the third item, $|v\rangle$ has $|0\rangle$ on Z exactly when the diagonal projection onto the configurations vanishing on Z fixes $|v\rangle$, and that projection commutes with an operator acting outside Z . The product vector vanishes at x as soon as one factor $\pi_i(x_i)$ with $i \in Z$ and $x_i \neq 0$ vanishes. $\square$

Lemma 30.11.8 (One teleportation hop). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean
Lean Lean Lean Lean ✓ Stmt A hop consists of three distinct sites c, e, f such that $\{e, f\}$ and $\{c, e\}$ are neighbouring pairs. Its first gate, on $\{e, f\}$, applies F at e and then $|a\rangle_e |b\rangle_f \mapsto |a\rangle_e |b + a\rangle_f$; its second gate, on $\{c, e\}$, applies $|a\rangle_c |b\rangle_e \mapsto |a\rangle_c |b - a\rangle_e$ and then F at c . Both are unitary and act on their pairs. Let $|v\rangle$ have $|0\rangle$ at e and f , and let $z_c, z_e \in \mathbb{Z}_d$. Applying the two gates and then the projection onto $|z_c\rangle$ at c and $|z_e\rangle$ at e gives

$$\frac{1}{d} (s_{z_c})_c (s_{z_e})_e (X^{z_e} Z^{z_c})_f P_{(cf)} |v\rangle,$$

where $s_z |q\rangle = |q + z\rangle$ and (cf) is the transposition of the sites c and f . So the content of c is found at f , up to the unitary $X^{z_e} Z^{z_c}$, and c and e carry the outcomes. The operator acts on the sites c, e, f .

Proof. ✓ Proof On a vector with $|0\rangle$ at e and f , the Fourier matrix at e gives the amplitude $d^{-1/2}$ to every value x_e , and the shift copies it to f : this prepares $d^{-1/2} \sum_j |j\rangle_e |j\rangle_f$. The second gate and the projections compute, configuration by configuration, the amplitude $d^{-1/2} F_{z_c, x_f - z_e}$ for the content $x_f - z_e$ moved from c to f , which is $d^{-1} \zeta^{z_c(x_f - z_e)}$, the amplitude of the right-hand side. Configurations with $x_c \neq z_c$ or $x_e \neq z_e$ vanish on both sides. $\square$

Theorem 30.11.9 (Teleportation along chains of hops in one round). Lean
Lean ✓ Stmt A list $h_1, \dots, h_H$ of hops is valid if every hop h_j meets the hops $h_{j+1}, \dots, h_H$ after it in the list only through its site c : its sites e and f are not sites of $h_{j+1}, \dots, h_H$, and its site c is not a site c or e of one of them. Its site c may be the site f of one of them, so the hops form chains $c_0 \rightarrow f_0 = c_1 \rightarrow f_1 = \dots$, the hops after h_j in the list acting before it. For a valid list and a local circuit U , the teleportation round applies U , the first gates of all hops in one layer, the second gates of all hops in one layer, measures the sites c and e of all hops, and corrects by the inverse of the product of the single-site unitaries that the hops leave. Its depth is the depth of U plus 2. It implements $P_{h_1} \dots P_{h_H} U$, with P_h the transposition of the sites c and f of h , on the vectors $|v\rangle$ such that $U|v\rangle$ has $|0\rangle$ at the sites e and f of every hop, with the scalar d^{-H} for every outcome.

Proof. ✓ Proof The first gates act on pairwise disjoint pairs $\{e, f\}$, and the second gates on pairwise disjoint pairs $\{c, e\}$, so they form two layers. By induction over the list, the projections and the two layers factor into the hops one after the other: the first gate of the head h of the list commutes with the second gates of the other hops because its sites e, f are not theirs, and the projections at the sites of the other hops commute with the operator of h . Lemma 30.11.8 for h applies because the other hops leave $|0\rangle$ at e and f . A single-site unitary at the site c of h , left by another hop whose site f is c , reappears at the site f of h after the transposition; so the hops leave a product of single-site unitaries, depending on the outcomes, after $d^{-H} P_{h_1} \dots P_{h_H}$. The correction is its inverse. $\square$

block of n sites carry a tree layout. Let the node p of depth j cover $\ell_{j,p}$ sites. The unitary of a node of depth $j \leq h$ extends the isometry W of Lemma 30.13.1 joining its two halves, from its input to its two registers; the input of the root is placed on its two registers by a given injective map, and below the root the input is a register on the first register of an even node and on the last register of an odd one, the other in $|0\rangle$. The unitary of a leaf of ℓ sites extends V_ℓ from its input to all its configurations. The tree T of these unitaries, the depths $0, \dots, h$ applied first and the leaves last, satisfies

$$\langle \tau | T | \iota_0(x) \rangle = \langle \tau | V_n | x \rangle$$

for every configuration τ of the block, with $\iota_0(x)$ the input x on the two registers of the root and $|0\rangle$ elsewhere.

Proof. ✓ Proof By induction from the leaves: after the leaves and the depths $h, \dots, j$, the amplitude of τ on the inputs x_p of the nodes of depth j is $\prod_p \langle \tau_{j,p} | V_{\ell_{j,p}} | x_p \rangle$, with $\tau_{j,p}$ the part of τ on the node. At the leaves this is the definition of their unitaries, since the leaves cover the block. A unitary of depth j maps its input to its two registers with the amplitudes of W , and the registers it writes are the inputs of the two nodes below it, by the second item of Lemma 30.12.5; summing over these inputs and applying (30.42) at every node gives the claim for depth j . The root covers the block. □

Lemma 30.13.6 (Leaves of a block of any length). Lean Lean Lean Lean Lean Lean ✓ Stmt Let $L \geq 2^{h+2}s$. The widths $w_i = 2\lfloor L/2 \rfloor / 2^{h+1}$, increased by 2 for the first $\lfloor L/2 \rfloor \bmod 2^{h+1}$ leaves, are a tree layout of a block of L sites with $h+1$ coarse depths, whose leaves cover all the sites but at most one. If $L \leq 2^{h+1}c$, every leaf has at most $c+3$ sites.

Proof. ✓ Proof The widths sum to $2\lfloor L/2 \rfloor$, and $\lfloor \lfloor L/2 \rfloor / 2^{h+1} \rfloor \geq s$. If $L \leq 2^{h+1}c$, then $2\lfloor \lfloor L/2 \rfloor / 2^{h+1} \rfloor \leq c$; the last leaf has one more site when L is odd. □

Theorem 30.13.7 (The state of blocks of unequal lengths with measurements). Lean Lean Lean Lean Lean Lean ✓ Stmt For every $s \geq 2$ and c there is C , depending only on d , s and c , with the following property. Let A be a tensor whose blocked tensors over $m \geq s$ sites are injective, and ω a unit vector on the pair space. For every $h \geq 0$ and every cutting of a ring into $M \geq 1$ blocks of lengths $2^{h+2}s \leq \ell_k \leq 2^{h+1}c$, the state (30.11) of these blocks with the pair ω is prepared with measurement rounds in depth at most $C(h+1)$. No outcome is post-selected.

Proof. ✓ Proof Injectivity over s sites gives $D^2 \leq d^s$, hence $D \leq d^{s-1}$ since $s \geq 2$, so a leg of a pair is written on the first $s-1$ sites of a register, the last site in $|0\rangle$, and a register carries $\mathbb{C}^{D^2}$. Cut every block into leaves as in Lemma 30.13.6. The input $|l, 0, \dots, 0, r\rangle$ of a block, with l on its first and r on its last s sites, lies on the two registers of its root: when the length is odd, the last site of the block lies after the leaves, and it carries the last site of the leg r , which is $|0\rangle$. A local circuit of bounded depth, applied in one round, prepares the pairs from the all- $|0\rangle$ product state; Theorem 30.13.4 applies the depths $0, \dots, h$ of the trees with the unitaries of Theorem 30.13.5; and a local circuit of bounded depth, by Theorem 30.6.4 for the boundedly many lengths of the leaves, applies the leaves in one round. The output is the state of the blocks as in Theorem 30.6.5. □

Theorem 30.13.8 (Preparation with measurements in depth $O(\log \log(N/\varepsilon))$, every chain length). Lean Lean ✓ Stmt Let A be normal. There is a constant c depending only on A such that, for every $N \geq 2$ and $0 < \varepsilon \leq 1$ for which the periodic state $|\phi_N(A)\rangle$ is nonzero, a unit vector $|\psi\rangle$ with $\varepsilon(\psi, \phi_N) \leq \varepsilon$ is prepared with measurement rounds in depth at most $c \log(\log(N/\varepsilon) + 1)$,

which is $O(\log \log(N/\varepsilon))$. No outcome is post-selected. If $D \geq 1$, there is N_0 such that the condition $|\phi_N(A)\rangle \neq 0$ holds for every $N \geq N_0$. This is the depth of the paragraph “Tree-RG circuit with measurements” of [185] for normal tensors; the source normalizes $|\phi_N\rangle$, which presupposes $|\phi_N(A)\rangle \neq 0$ [109].

Proof. ✓ Proof Bring A into the gauge (30.3), which multiplies $|\phi_N\rangle$ by a nonzero scalar, with a bound $t < 1$ on the moduli of the subleading eigenvalues, $r = -\frac{1}{4} \log t$, and an injectivity length L ; put $s = L + 2$, $a = 1/r$, $b = \max(\log K, 0)/r + 4s + 1$ with K the constant of Theorem 30.5.10 at $\gamma = 1/4$, and $q = \lceil a \log(N/\varepsilon) + b \rceil \geq 4s$. If $q \leq N$, write $N = (M-1)q + q'$ with $q \leq q' < 2q$; the state of these blocks with the pair of the fixed point has error at most $KMe^{-rq} \leq \varepsilon$ by Theorem 30.5.10, and with $2^{h+2}s \leq q < 2^{h+3}s$ it is prepared in depth $C(h+1)$ by Theorem 30.13.7 with $c = 8s$. If $4s \leq N < q$, the chain is one block, and by Theorem 30.6.6 $|\phi_N\rangle$ is the state of this block with the pair $\omega \propto P|\mathbb{1}\rangle$, prepared in depth $C(h+1)$ with $2^{h+2}s \leq N < 2^{h+3}s$. If $N < 4s$, $|\phi_N\rangle$ is prepared by a local circuit of bounded depth. In each case $2^h \leq a \log(N/\varepsilon) + b + 1 \leq (a+b+1)(\log(N/\varepsilon) + 1)$, and $\log(\log(N/\varepsilon) + 1) \geq \log(1 + \log 2) > 0$, so the depth is at most a constant times $\log(\log(N/\varepsilon) + 1)$. $\square$

30.13.1 Tensors that are not normal, with measurements

For a tensor that is not normal, [185] prepares the fixed point $|\Omega'\rangle = W^{\otimes N/q} |\chi_{N/q}\rangle$ of Theorem 30.9.5 by first preparing $|\chi_{N/q}\rangle$ with measurements and then applying the isometries $W: |j\rangle \mapsto |\omega_j\rangle$ and the isometries of the blocked tensor as circuits. With the block length $q \propto \log(N/\varepsilon)$, “If instead measurements are only used for the preparation of $|\chi_{N/q}\rangle$, the depth is $O(\log(N/\varepsilon))$ ”. This subsection proves this for direct sums of normal blocks whose q -site states are orthogonal and for block lengths dividing N [81]; Section 30.13.2 removes both restrictions. On the chain of N sites the label of each block is stored in the last sites of the block, and the GHZ-type state is prepared in depth $O(q)$; the isometries of the blocked tensor take depth $O(q)$ as well.

Definition 30.13.1.1 (Preparation with measurements and a circuit). Lean ✓ Stmt A vector $|\psi\rangle$ is prepared with measurements and a circuit in depth T if $|\psi\rangle = U|\varphi\rangle$ for a vector $|\varphi\rangle$ prepared with measurements in depth T_1 and a local circuit U of depth T_2 , with $T_1 + T_2 \leq T$. The circuit U does not depend on the outcomes of the measurements.

Lemma 30.13.1.2 (Layers acting on configurations). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Let X_k be operators on pairwise disjoint sets S_k of sites, each acting by a map f_k of the configurations of S_k , that is $X_k v = v \circ f_k$ for every vector v ; then X_k has the entry 1 at $(u, f_k(u))$ and 0 elsewhere. The product of the X_k acts by the map ρ with $\rho(x)|_{S_k} = f_k(x|_{S_k})$ for every k and $\rho(x)_i = x_i$ at every site i outside all the S_k . A permutation τ of the sites acts as $v \mapsto v \circ (x \mapsto x \circ \tau)$. If X acts on the sites of S and the product vector $|\pi\rangle$ is $|z\rangle$ on every site of S , then $(X|\pi\rangle)(y) = X_{y|S, (z, \dots, z)} \pi'(y)$, where $|\pi'\rangle$ is the product vector with the all-ones vector at the sites of S and the vectors of $|\pi\rangle$ elsewhere.

Proof. ✓ Proof Applying X_k to the basis vector $|u\rangle$ gives the entries. The entry of the product at (x, y) is $\prod_k [y|_{S_k} = f_k(x|_{S_k})]$ when x and y agree outside the S_k , and 0 otherwise; this is 1 exactly when $y = \rho(x)$. For the product vector, only the configuration y' agreeing with y outside S and equal to $(z, \dots, z)$ on S contributes. $\square$

Definition 30.13.1.3 (Registers and the GHZ-type state on them). Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean Lean ✓ Stmt Cut the ring of N sites into $M \geq 1$ blocks of lengths $\ell_k \geq 2r_1$. The register R_k consists of the last

r_1 sites of block k , and its *ancilla* of the first r_1 sites of block $k + 1$, cyclically; together they form the pair window k . For a vector α on $(\mathbb{C}^d)^{\otimes r_1}$, the *GHZ-type state on the registers* has the amplitude $\alpha(u)$ at a configuration carrying the same configuration u on every register and 0 at every other site, and the amplitude 0 at every other configuration. If the labels j of b blocks are encoded injectively as configurations $\delta(j)$ of r_1 sites and $\alpha(\delta(j)) = \alpha_j$, $\alpha(u) = 0$ for every other u , it is $\sum_j \alpha_j |x_j\rangle$, where x_j carries $\delta(j)$ on every register and 0 elsewhere: the state $|\chi_M\rangle$ of Definition 30.9.3, with the label of each block stored in its register. The configuration x_j carries $(\delta(j), 0, \dots, 0)$ on every pair window and 0 outside the windows.

Lemma 30.13.1.4 (A controlled shift between the ends of a block). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[✓ Stmt](#) Let $r_1 \geq 1$ and values be taken in $\mathbb{Z}_d$. There is C such that for every $q \geq 3r_1$ some unitary on q sites, a product of at most Cq gates on neighbouring sites, acts by the map of configurations that adds the last r_1 sites to the first r_1 , site by site, and leaves the other sites unchanged.

Proof. [✓ Proof](#) SWAP gates move the first r_1 sites next to the last r_1 in at most $2qr_1$ gates; one unitary on the last $2r_1$ sites, of bounded depth by Theorem 30.6.4, adds the second half to the first; and the same SWAP gates move the sites back. By Lemma 30.13.1.2 the composite acts by the stated map. $\square$

Theorem 30.13.1.5 (The GHZ-type state on the registers with measurements). [Lean](#) [✓ Stmt](#)

Let $r_1 \geq 2$. There is C such that for every ring of N sites cut into $M \geq 1$ blocks of lengths $3r_1 \leq \ell_k \leq L$ and every unit vector α on $(\mathbb{C}^d)^{\otimes r_1}$, the GHZ-type state on the registers of Definition 30.13.1.3 is prepared with measurements in depth at most CL . This is the preparation of $|\chi_{N/q}\rangle$ in the paragraph “Long-range MPS using measurements” of [185], by the construction of [195, Example 1] for registers of r_1 sites; the depth is $O(L)$ rather than constant, because the register of a block and the ancilla at its start are $\ell_k - r_1$ sites apart [81].

Proof. [✓ Proof](#) Start from the product vector with the all-ones vector on the sites of the registers R_k , $k \neq 0$, and $|0\rangle$ elsewhere. One unitary on the sites of R_0 , of bounded depth, sends $|0 \dots 0\rangle$ to $\sum_u \alpha(u) |u\rangle$. One layer of gates on the pair windows adds each register to its ancilla, and on every block the unitary of Lemma 30.13.1.4 subtracts the register of the block from the ancilla at its start, which belongs to the register of the previous block. The circuit has depth $O(L)$. Measure every ancilla, with outcomes $m_k \in \mathbb{Z}_d^{r_1}$, and let $p_k = m_0 + \dots + m_{k-1}$. Correct by adding m_k on the ancilla of R_k and $-p_k$ on R_k , site by site. By Lemma 30.13.1.2, the corrected vector has the amplitude $\alpha(x|_{R_0})$ at a configuration x that is 0 outside the registers and satisfies $m_k + (x|_{R_{k+1}} - p_{k+1}) - (x|_{R_k} - p_k) = 0$ for every k , cyclically, and the amplitude 0 elsewhere. These conditions hold exactly when $\sum_k m_k = 0$ and all the registers of x agree. So every outcome with $\sum_k m_k = 0$ gives the GHZ-type state, and every other outcome gives the zero vector, which has probability zero. $\square$

Lemma 30.13.1.6 (Sequential factorization on the first inputs). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#)

In Theorem 30.4.1, let the matrix elements of V have the stated form only for the first r inputs x , and let these $r \leq D^2$ columns of V be orthonormal. Then the same conclusion holds with $b_q = r$, for these r inputs. For an injective blocked tensor $B = VP$ of tensors $A_1, \dots, A_q$, one on each site, $V = BP^{-1}$ has the matrix elements $\langle i_1 \dots i_q | V | x \rangle = \sum_{\alpha, \beta} (A_1^{i_1} \dots A_q^{i_q})_{\alpha\beta} (P^{-1})_{(\alpha, \beta), x}$; for equal tensors $A_p = A$ this reads $\sum_{\alpha, \beta} (A^{i_1} \dots A^{i_q})_{\alpha\beta} (P^{-1})_{(\alpha, \beta), x}$ with $B = \mathcal{B}_q$.

Proof. [✓ Proof](#) The proof of Theorem 30.4.1 applies to the first r columns of G , the others being replaced by zero: the remainder of the sweep has orthonormal first r columns because those of V are, and absorbing it into the last site keeps that site isometric on its first r levels. $\square$

makes the physical matrices of the approximating tensors of A_j and $A_{j'}$ orthogonal, so that the overlap of the two states on M blocks vanishes. Hence $\langle \psi_j | \psi_{j'} \rangle = \delta_{jj'}$ and

$$\left\| \sum_j \alpha_j |\psi_j\rangle \right\|^2 = \sum_{j,j'} \overline{\alpha_j} \alpha_{j'} \langle \psi_j | \psi_{j'} \rangle = \sum_j |\alpha_j|^2 = 1.$$

Theorem 30.13.1.8 with $\ell_k = q$ prepares the sum. $\square$

Lemma 30.13.1.10 (A common gap for normal blocks). Lean ✓ Stmt *For finitely many normal left-canonical tensors $A_1, \dots, A_b$, $b \geq 1$, of positive bond dimensions, there is $0 < t < 1$ bounding the moduli of the eigenvalues other than 1 of every transfer map $\mathcal{E}_{A_j}$.*

Proof. ✓ Proof Lemma 30.6.29 gives such a bound t_j for every block; take $t = \max_j t_j$. $\square$

Theorem 30.13.1.11 (Tensors that are not normal, with measurements, in depth $O(\log(N/\varepsilon))$).

Lean ✓ Stmt *Let A be the direct sum with multiplicities of Definition 30.9.1.1, with $b \geq 1$ blocks, every block A_j normal in the gauge (30.3) with a positive definite fixed point σ_j of trace one. There are $a > 0$, $b' \geq 1$ and c , depending only on A , with the following property. Let $N \geq 2$, $0 < \varepsilon \leq 1$, and let q be a block length dividing N with $a \log(N/\varepsilon) + b' \leq q \leq 2(a \log(N/\varepsilon) + b')$, at which $B_j^\dagger B_{j'} = 0$ for $j \neq j'$, such that the weights $\beta_j = \sum_k \mu_{j,k}^N$ are not all zero. Then a unit vector $|\psi\rangle$ with $\varepsilon(\psi, \phi_N) \leq \varepsilon$ is prepared with measurements and a circuit in depth at most $c \log(N/\varepsilon)$. This is the depth $O(\log(N/\varepsilon))$ of the paragraph “Long-range MPS using measurements” of [185], with measurements used only for the GHZ-type state, for the corrected approximating state, restricted to blocks whose q -site states are orthogonal and to block lengths dividing N [81]. Because q must divide N and lie in the window above, the statement applies only to chain lengths with a divisor in the window and is vacuous for the others, for instance for every large prime N .*

Proof. ✓ Proof Let t be the common gap of Lemma 30.13.1.10, $r = -\frac{1}{4} \log t$, and K the constant of Theorem 30.9.1.4 at $\gamma = 1/4$, so the error of the corrected approximating state with the amplitudes $\alpha_j = \beta_j / (\sum_l |\beta_l|^2)^{1/2}$ is at most KMe^{-rq} . Put $a = 1/r$ and $b' = a \max(\log K, 0) + L_0 + L_1 + 1$, with L_0 the length of Theorem 30.13.1.9 and L_1 the sum of the injectivity lengths of the blocks. Then $q \geq L_0$, every blocked tensor is injective, and the state is prepared in depth at most Cq . Since $N = Mq$, $rq = q/a$ and $a \log(N/\varepsilon) + b' \leq q$, the error is at most ε by Lemma 30.5.4, and $Cq \leq c \log(N/\varepsilon)$ with $c = C(2a + 2b'/\log 2)$ by Lemma 30.6.30. $\square$

Theorem 30.13.1.12 (All translation-invariant MPS with measurements). *Every translation-invariant MPS of constant bond dimension, short- or long-range correlated, is prepared with error ε in depth $O(\log(N/\varepsilon))$ by local circuits assisted by measurements, the measurements being used only for the preparation of the GHZ-type state [185, paragraph “Long-range MPS using measurements”]. Theorem 30.13.1.11 proves this for direct sums of normal blocks whose q -site states are orthogonal and for block lengths dividing N . Theorem 30.13.2.8 proves it for direct sums of normal blocks in the canonical form [185, eq. (S2)] whose q -site states may overlap, with every multiplicity and every nonzero complex weight, and for every chain length $N \geq 2$ at which $|\phi_N(A)\rangle \neq 0$, with the mixed transfer maps of distinct blocks of spectral radius below one, the isometries of the direct sum with weights one, and the GHZ-type state prepared in depth $O(q)$ rather than in constant depth [81]. The same paragraph claims the depth $O(\log \log(N/\varepsilon))$ when the tree-RG circuit is also assisted by measurements; for normal tensors and the chain lengths divisible by $s2^{k+1}$ for some k with $a \log(N/\varepsilon) + b \leq s2^{k+1} \leq 2(a \log(N/\varepsilon) + b)$ this is Theorem 30.12.8, and for tensors that are not normal it is not formalized.*

30.13.2 Overlapping blocks with measurements, for every chain length

The preparation of Theorem 30.13.1.11 applies the isometry of the blocked tensor of every block separately, which needs the q -site states of distinct blocks to be orthogonal, and cuts the chain into equal blocks. This subsection removes both restrictions. It applies, on every block, the isometric factor V of the polar decomposition of the blocked map of the direct sum $\bigoplus_j A_j$ with weights one and one copy of each block. For large block lengths this map is injective on the bond pairs of the blocks even when their states overlap, so V is an isometry there, and the weights of the canonical form enter only through the amplitudes of the GHZ-type state. The chain is cut into blocks “all of the same size, q_N , except for the last one, which may be larger”, as in the proof of Theorem 1 of [185]. The bound on the error is relative to the norm of the weights, so no factor $\min(1, \sum_j |\beta_j|^2)^{-1/2}$ enters [81].

Throughout, $A_1, \dots, A_b$ are normal tensors in the gauge (30.3) with positive definite fixed points ρ_j of trace one, placed on disjoint bond coordinates by isometries E_j , $K_j = E_j \otimes E_j$, and $\lambda_2 \in \mathbb{C}$ bounds the moduli of the eigenvalues other than one of every transfer map $\mathcal{E}_{A_j}$ and the moduli of all eigenvalues of the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, with correlation length ξ . For a block length ℓ , $\mathcal{B}^{(\ell)} = V^{(\ell)} P^{(\ell)}$ is the polar decomposition of the ℓ -site blocked map of $\bigoplus_j A_j$.

Lemma 30.13.2.1 (Injectivity on the bond pairs of the blocks). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#)
[Lean](#) [✓ Stmt](#) *Let S be the set of bond pairs $(E_j a, E_j c)$ whose two indices lie in the same block.*

1. If the matrices of a tensor B vanish outside a set S of bond pairs and $\mathcal{B}x = 0$ has no solution $x \neq 0$ supported on S , then B is injective on S .
2. With $|\lambda_2| < 1$ there is L_0 such that for every $\ell \geq L_0$ the ℓ -site blocked tensor of $\bigoplus_j A_j$ is injective on S .

Proof. ✓ Proof For part (1), if a matrix unit $|\alpha\rangle\langle\beta|$ with $(\alpha, \beta) \in S$ were not in the span of the matrices of B , a linear functional f vanishing on that span but not on $|\alpha\rangle\langle\beta|$ would give the vector $x_p = f(|p_1\rangle\langle p_2|)$ for $p \in S$, and $x_p = 0$ otherwise, with $\mathcal{B}x = 0$ and $x \neq 0$. For part (2), let $Q = \mathbb{1} - \sum_j K_j K_j^\dagger$ and G_∞ be as in Lemma 30.9.2.3; that lemma gives $c > 0$ with $G_\infty + Q \geq c\mathbb{1}$ and $\|\mathcal{B}^\dagger B - G_\infty\| \leq Ke^{-\ell/(2\xi_t)}$, where ξ_t is the correlation length at $t = \max(|\lambda_2|, 1/2)$, which satisfies $0 < t < 1$ and bounds the same eigenvalues as $|\lambda_2|$. The matrices of the blocked tensor vanish outside S , and $Qx = 0$ for x supported on S . For ℓ with $Ke^{-\ell/(2\xi_t)} < c$ and such x with $\mathcal{B}x = 0$,

$$0 = x^\dagger (\mathcal{B}^\dagger \mathcal{B} + Q)x \geq (c - \|\mathcal{B}^\dagger \mathcal{B} - G_\infty\|)x^\dagger x,$$

so $x = 0$, and part (1) applies.

Lemma 30.13.2.2 (The states of the blocks through the isometries of the direct sum). [\[Lean\]](#)
[\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[Lean\]](#) [\[✓ Stmt\]](#) Cut a ring of $N \geq 1$ sites into $M \geq 1$ blocks of lengths $\ell_k \geq 1$, and let ω_i be the pair of ρ_i placed along E_i .

1. The ℓ -site blocked map of A_j , placed along E_j , is $V^{(\ell)} P^{(\ell)} K_j K_j^\dagger$, so $|\phi_N(A_j)\rangle = (\bigotimes_k V^{(\ell_k)}) |\phi_M(P^{(\ell_k)} K_j K_j^\dagger)\rangle$, where the second factor is the periodic state of the family of tensors $P^{(\ell_k)} K_j K_j^\dagger$ on the blocks.
2. If every blocked map $\mathcal{B}^{(\ell_k)}$ is injective on the bond pairs of the blocks, then the states $|\psi_j\rangle = (\bigotimes_k V^{(\ell_k)}) \bigotimes_k |\omega_j\rangle$ are orthonormal, so $\sum_j \alpha_j |\psi_j\rangle$ is a unit vector for every unit vector α , and $z_{jj'} = \langle \psi_j | \phi_N(A_{j'}) \rangle$ is the trace of the ordered product over the blocks of the mixed transfer matrices of $P^{(\ell_k)} K_{j'} K_{j'}^\dagger$, against the fixed-point tensor of $E_j \rho_j E_j^\dagger$.

Proof. ✓ Proof For part (1), the blocked map of A_j placed along E_j is $\mathcal{B}_j K_j^\dagger$, and that of the direct sum is $\sum_{j'} \mathcal{B}_{j'} K_{j'}^\dagger$; since $K_{j'}^\dagger K_j = \delta_{jj'} \mathbb{1}$, the first is the second followed by $K_j K_j^\dagger$. Every word of length N is a product of the blocked words. For part (2), the product of the pairs ω_j vanishes on every configuration of bond pairs leaving the pairs of the blocks, and on these $V^{(\ell_k)}$ is an isometry, so the inner products are those of the products of pairs, which are $\delta_{jj'}$ because the ω_j are unit vectors on disjoint coordinates. The same isometry property and part (1) give $z_{jj'}$ as the overlap of $\bigotimes_k |\omega_j\rangle$ with the periodic state of the family $P^{(\ell_k)} K_{j'} K_j^\dagger$, which is the stated trace by Lemma 30.9.2.5(1). $\square$

Lemma 30.13.2.3 (The overlaps of the blocks near the limit). Lean Lean Lean ✓ Stmt

1. For a finite index set there is $K \geq 0$ such that for every matrix E whose powers have norm at most c and all matrices $X_0, \dots, X_{M-1}$ within δ of E , $|\text{Tr}(X_0 \cdots X_{M-1}) - \text{Tr } E^M| \leq K c^2 (M\delta) e^{cM\delta}$.
2. The mixed transfer map of $P_\infty K_{j'} K_j^\dagger$ against the fixed-point tensor of $E_j \rho_j E_j^\dagger$ is $X \mapsto \text{Tr}(E_j E_j^\dagger X) E_j \rho_j E_j^\dagger$ for $j = j'$ and zero for $j \neq j'$.
3. With $|\lambda_2| < 1$ and $0 < \gamma < 1$, for all j, j' there is $C > 0$ such that for every cut into $M \geq 1$ blocks of lengths $\ell_k \geq q \geq 1$, with $y = M e^{-\gamma q/\xi}$, the trace $z_{jj'}$ of Lemma 30.13.2.2(2) satisfies $|z_{jj'} - \delta_{jj'}| \leq C y e^{C y}$.

Proof. ✓ Proof Part (1) is the telescoping estimate of Lemma 30.9.2.2(8) for a product of different matrices, followed by the bound on the trace; it follows the argument of [195, Supplemental Material]. Part (2) is Lemma 30.9.2.6(1) with $t = e_{j'}$, since $P_\infty K_{j'} K_j^\dagger$ is the limit for these weights. For part (3), the limit E of the mixed transfer matrices is, by part (2), an idempotent of trace one for $j = j'$ and zero otherwise, so its powers are bounded and $\text{Tr } E^M = \delta_{jj'}$. By Lemma 30.9.2.3, $\|P^{(\ell_k)} - P_\infty\| \leq K' e^{-\gamma \ell_k/\xi} \leq K' e^{-\gamma q/\xi}$, and part (1) applies. $\square$

Theorem 30.13.2.4 (Error for overlapping blocks of unequal lengths and all weights). Lean ✓ Stmt Let $0 < \gamma < 1/2$. There is $C > 0$ such that the following holds for all weights $\beta \in \mathbb{C}^b \setminus \{0\}$ and every cut of a ring of N sites into $M \geq 1$ blocks of lengths $\ell_k \geq q$ at which every blocked map $\mathcal{B}^{(\ell_k)}$ is injective on the bond pairs of the blocks. With $\alpha_j = \beta_j / (\sum_l |\beta_l|^2)^{1/2}$ and $|\phi\rangle = \sum_j \beta_j |\phi_N(A_j)\rangle$,

$$1 - \left| \left\langle \sum_j \alpha_j (\bigotimes_k V^{(\ell_k)}) \bigotimes_k \omega_j \mid \frac{\phi}{\|\phi\|} \right\rangle \right| \leq C M e^{-\gamma q/\xi}.$$

Here $-\gamma q/\xi$ is read as 0 when $\xi = 0$, that is when $\lambda_2 = 0$; the bound then gives no decay in q . The constant does not depend on the weights. For the canonical form $A^i = \bigoplus_j \text{diag}(\mu_{j,1}, \dots, \mu_{j,m_j}) \otimes A_j^i$ of [185, eq. (S2)], $|\phi_N(A)\rangle$ is $|\phi\rangle$ with $\beta_j = \sum_k \mu_{j,k}^N$, so this bounds the error for every multiplicity and every complex weight. This is [185, Lemma 1' (ii)] for the state of the paragraph “Long-range MPS using measurements”, with blocks of unequal lengths, with ξ_{diag} replaced by $\max(\xi_{\text{diag}}, \xi_{\text{off-diag}})$ [106], and with the isometries of the direct sum with weights one in place of that of the blocked map of A [81]. Unlike Theorem 30.9.3.5, it has no factor $\min(1, \sum_j |\beta_j|^2)^{-1/2}$.

Proof. ✓ Proof If $\lambda_2 = 0$, $|\lambda_2| \geq 1$ or $q = 0$, then $M e^{-\gamma q/\xi} \geq 1$ and the bound holds with $C = 1$. Otherwise write $w = \sum_l |\beta_l|^2$ and $|\psi\rangle = \sum_j \alpha_j |\psi_j\rangle$. By Lemma 30.13.2.2(2), $\langle \psi | \phi \rangle = w^{-1/2} S$ with $S = \sum_{j,j'} \overline{\beta_j} \beta_{j'} z_{jj'}$. Since $|\beta_j \beta_{j'}| \leq w$, Lemma 30.13.2.3(3) gives $|S/w - 1| \leq b^2 C_0 y e^{C_0 y}$, a

bound relative to w , with $y = Me^{-\gamma/\xi}$. Lemma 30.9.2.2(6) with $R = \sqrt{w}$ gives $|\|\phi\|^2/w - 1| \leq Ke^{-\gamma N/\xi} \leq Ky$. The inequality in the proof of Theorem 30.9.2.8, applied to S/w and $\|\phi\|/\sqrt{w}$, and the two elementary inequalities there with $s = 1$ give the bound. $\square$

Theorem 30.13.2.5 (The state of a tensor injective on a set of bond pairs, with measurements).

[Lean](#) [Lean](#) [✓ Stmt](#) For $d \geq 2$, every bond dimension D and every number b of labels, there are C and L_0 such that the following holds. Let A be a tensor of bond dimension D , S a set of bond pairs, $\omega_1, \dots, \omega_b$ orthonormal pairs whose products vanish on every configuration of bond pairs leaving S , and α a unit vector. Cut the ring of N sites into $M \geq 1$ blocks of lengths $L_0 \leq \ell_k \leq L$ at which every blocked tensor of A is injective on S . Then $\sum_j \alpha_j (\bigotimes_k V^{(\ell_k)} \bigotimes_k |\omega_j\rangle)$ is prepared with measurements and a circuit in depth at most CL . A nonzero vector prepared by a local circuit of depth T is prepared with measurements and a circuit in depth T .

Proof. [✓ Proof](#) As in the proof of Theorem 30.13.1.8, encode the labels and the bond indices in $r_1 = b + D + 2$ sites, prepare the GHZ-type state with amplitudes α on the registers by Theorem 30.13.1.5, and apply on every pair window one unitary sending the encoded label j to $|\omega_j\rangle$. On every block apply the unitary of Theorem 30.13.1.7 for an ordering of the bond pairs in which those of S come first; by Theorem 30.8.8 it implements the isometric factor $V^{(\ell_k)}$ on the pairs of S , which suffices because the products of the pairs are supported there. For the second statement, the measurement stage measures no site and the circuit after it is empty, by Lemma 30.10.3. $\square$

Lemma 30.13.2.6 (A chain of one block). [Lean](#) [✓ Stmt](#) Let the N -site blocked map of $\bigoplus_j A_j$ be injective on the bond pairs of the blocks, and let $|\phi\rangle = \sum_j \beta_j |\phi_N(A_j)\rangle \neq 0$. Then $|\phi\rangle/\|\phi\| = V^{(N)}|\omega\rangle$ for a unit pair ω supported on the bond pairs of the blocks.

Proof. [✓ Proof](#) By Lemma 30.13.2.2(1) with one block, $|\phi\rangle = V^{(N)}|Y\rangle$ with $Y = \sum_j \beta_j \text{Tr}(P^{(N)} K_j K_j^\dagger)$, and the rows of $P^{(N)}$ at bond pairs outside the blocks vanish by Theorem 30.1.3.8, so Y is supported on the bond pairs of the blocks. There $V^{(N)}$ is an isometry, so $\|\phi\| = \|Y\|$ and $\omega = Y/\|Y\|$. $\square$

Lemma 30.13.2.7 (A common rate for the blocks and their mixed transfer maps). [Lean](#) [✓ Stmt](#) For finitely many normal left-canonical tensors $A_1, \dots, A_b$, $b \geq 1$, of positive bond dimensions, whose mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, have all eigenvalues of modulus below one, there is $0 < t < 1$ bounding the moduli of the eigenvalues other than one of every $\mathcal{E}_{A_j}$ and of all eigenvalues of every $\mathcal{E}_{jj'}$.

Proof. [✓ Proof](#) Lemma 30.13.1.10 bounds the transfer maps. Each $\mathcal{E}_{jj'}$ has finitely many eigenvalues, all of modulus below one, so their largest modulus $t_{jj'}$ is below one; take the maximum of these bounds. $\square$

Theorem 30.13.2.8 (Every translation-invariant MPS with measurements, in depth $O(\log(N/\varepsilon))$).

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be the direct sum with multiplicities of Definition 30.9.1.1, with nonzero complex weights, every block A_j normal in the gauge (30.3) with a positive definite fixed point ρ_j of trace one, and let the mixed transfer maps $\mathcal{E}_{jj'}$, $j \neq j'$, have all eigenvalues of modulus below one. There is c , depending only on A , such that for every $N \geq 2$ and $0 < \varepsilon \leq 1$ at which $|\phi_N(A)\rangle \neq 0$, a unit vector $|\psi\rangle$ with $\varepsilon(\psi, \phi_N) \leq \varepsilon$ is prepared with measurements and a circuit in depth at most $c \log(N/\varepsilon)$. There is N_0 such that $|\phi_N(A)\rangle \neq 0$ for every $N \geq N_0$ at which the weights $\beta_j = \sum_k \mu_{j,k}^N$ are not all zero, so the same holds for these N . This is the depth $O(\log(N/\varepsilon))$ of the paragraph “Long-range MPS using measurements” of [185], with

measurements used only for the GHZ-type state, for blocks whose q -site states may overlap, every multiplicity, every nonzero complex weight and every chain length, under the readings recorded in [81, 106, 109].

Proof. ✓ Proof Let t be the rate of Lemma 30.13.2.7, $r = -\frac{1}{4} \log t$, K the constant of Theorem 30.13.2.4 at $\gamma = 1/4$, and $L_{\min}$ larger than the lengths of Lemma 30.13.2.1(2) and Theorem 30.13.2.5. Put $Q = a \log(N/\varepsilon) + b'$ with $a = 1/r$ and $b' = \max(\log K, 0)/r + L_{\min}$, and $q = \lceil Q \rceil$. If $q \leq N$, cut the chain into $\lfloor N/q \rfloor - 1$ blocks of length q and one of length $q + (N \bmod q) < 2q$. Every blocked map is injective on the bond pairs of the blocks, the state of Theorem 30.13.2.4 has error at most $KN e^{-rq} \leq \varepsilon$ by Lemma 30.5.3, and it is prepared in depth $O(q) = O(\log(N/\varepsilon))$ by Theorem 30.13.2.5. If $L_{\min} \leq N < q$, the normalized target itself is $V^{(N)}|\omega\rangle$ by Lemma 30.13.2.6 and is prepared by a circuit of depth $O(N) = O(\log(N/\varepsilon))$ by Theorem 30.8.9. If $N < L_{\min}$, it is prepared in bounded depth by Theorem 30.6.6. For the second statement, $|\|\phi_N(A)\|^2/w - 1| \leq K e^{-N/(4\xi_t)}$, with ξ_t the correlation length at t and $w = \sum_j |\beta_j|^2$ by Lemma 30.9.2.2(6), which is below one for large N . $\square$

Part I

Appendices

Appendix A

Perron–Frobenius Theory for Channels and Transfer Maps: Supporting Results

Canonical-gauge calculations supporting Chapter 4 verify the normalization of the transformed Kraus operators directly.

A.1 Canonical-gauge algebra

For the right-canonical gauge in Theorem [182, “Right-canonical (unital) gauge”], each summand becomes

$$\begin{aligned}(S^{-1}A^iS)(S^{-1}A^iS)^\dagger &= S^{-1}A^iSS^\dagger(A^i)^\dagger(S^\dagger)^{-1} \\ &= S^{-1}A^i\rho(A^i)^\dagger(S^\dagger)^{-1}.\end{aligned}$$

Hence

$$\sum_i A'^i(A'^i)^\dagger = S^{-1}\mathcal{E}_A(\rho)(S^\dagger)^{-1} = S^{-1}\rho(S^\dagger)^{-1} = \mathbb{1}.$$

For the left-canonical gauge in Theorem 4.4.2,

$$\begin{aligned}(SA^iS^{-1})^\dagger(SA^iS^{-1}) &= (S^\dagger)^{-1}(A^i)^\dagger S^\dagger SA^iS^{-1} \\ &= (S^\dagger)^{-1}(A^i)^\dagger \sigma A^iS^{-1},\end{aligned}$$

and therefore

$$\sum_i (A'^i)^\dagger A'^i = (S^\dagger)^{-1} \left(\sum_i (A^i)^\dagger \sigma A^i \right) S^{-1} = (S^\dagger)^{-1} \sigma S^{-1} = \mathbb{1}.$$

Finally, for the square-root gauge (4.1), self-adjointness of $\sigma^{1/2}$ gives

$$\begin{aligned}(B^i)^\dagger B^i &= \sigma^{-1/2}(A^i)^\dagger \sigma^{1/2} \sigma^{1/2} A^i \sigma^{-1/2} \\ &= \sigma^{-1/2}(A^i)^\dagger \sigma A^i \sigma^{-1/2}.\end{aligned}$$

Summing, substituting the adjoint fixed-point equation, and cancelling the outer square-root factors yields

$$\sum_i (B^i)^\dagger B^i = \sigma^{-1/2} \left(\sum_i (A^i)^\dagger \sigma A^i \right) \sigma^{-1/2} = \sigma^{-1/2} \sigma \sigma^{-1/2} = \mathbb{1}.$$

Lemma A.1.1 (Square-root trace-preserving gauge is a gauge equivalence). [Lean](#) [✓ Stmt](#) *If σ is positive definite and B is defined by (4.1), then A and B are gauge equivalent.*

Proof. [✓ Proof](#) The gauge matrix is $\sigma^{1/2}$, which is invertible because σ is positive definite. Thus (4.1) has the form of the gauge relation (2.8). $\square$

Appendix B

Peripheral Channel Structure and Transfer-Operator Gaps: Supporting Results

The spectral theory of MPS transfer maps in Chapter 5 relies on several structural steps: algebraic mixed-transfer identities, peripheral intertwiners for gauge rigidity, Banach-algebra decay bounds for overlaps, the rank-one Perron projection for primitive channels, and cyclic-sector decompositions.

B.1 Mixed-transfer powers and overlap traces

The mixed transfer operator F_{AB} of Definition 5.1.1 and its rectangular form F_{AB}^{rect} of Definition 5.1.2 satisfy exact word expansions and trace identities.

Theorem B.1.1 (Self-transfer). Lean ✓ Stmt *Setting $B = A$ recovers the ordinary transfer map: $F_{AA} = \mathcal{E}_A$.*

Proof. ✓ Proof Substituting $B = A$ into (5.2) gives $F_{AA}(X) = \sum_i A^i X (A^i)^\dagger = \mathcal{E}_A(X)$ by (2.4). □

Lemma B.1.2 (Iterated mixed transfer). Lean ✓ Stmt *For every $X \in M_D(\mathbb{C})$ and $N \geq 0$,*

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X (B^\sigma)^\dagger, \quad (\text{B.1})$$

where $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$ denotes the word evaluation of Definition 2.1.2.

Proof. ✓ Proof Induct on N . The base case is $F_{AB}^0(X) = X$. For the inductive step, apply F_{AB} to each summand in (B.1) and reindex using $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. □

Theorem 5.2.2 combines Lemma B.1.2 with Lemma [182, “Trace expansion over matrix units”] over the rectangular matrix-unit basis: expanding $\text{Tr}(T)$ for $T = (F_{AB}^{\text{rect}})^N$ by the trace-expansion formula of [182, “Trace expansion over matrix units”], applying (B.1) to each $T(E_{pq})$, and re-assembling the sums over matrix-unit indices gives

$$\text{Tr}(T) = \sum_{\sigma} \text{tr}(A^\sigma) \overline{\text{tr}(B^\sigma)} = \sum_{\sigma} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N).$$

Theorem 5.2.1 is the square case $D_1 = D_2 = D$ of this statement, so it needs no separate expansion.

B.2 Auxiliary trace positivity

B.3 Rectangular spectral-radius formulation

Lemma B.3.1 (Rectangular spectral radius bound). Lean ✓ Stmt For normalized tensors of bond dimensions D_1 and D_2 , $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) \leq 1$.

Proof. ✓ Proof This is the MPS formulation of the corresponding theorem in the companion quantum-channel volume [182, “Spectral-radius bound for a trace-preserving mixed Kraus map”]: the rectangular mixed transfer is the mixed Kraus map, and the two tensor normalization identities are its trace-preserving hypotheses. $\square$

B.4 Peripheral intertwiners and gauge rigidity

Transporting a modulus-one mixed-transfer eigenvector through separate unital gauges produces Kraus-level intertwiners that force equal bond dimension once invertible, establishing the gauge rigidity of Theorem 5.6.1 and Theorem 5.6.4.

Lemma B.4.1 (Gauged peripheral eigenvector yields Kraus intertwining). Lean ✓ Stmt Let A be a normalized tensor of bond dimension D_1 and B a normalized tensor of bond dimension D_2 . Suppose ρ_A, ρ_B are fixed points of the respective transfer maps and S_A, S_B are invertible matrices with $S_A S_A^\dagger = \rho_A$ and $S_B S_B^\dagger = \rho_B$ (for instance the positive definite Perron–Frobenius fixed points of Theorem 4.3.2 and their square roots, as in the right-canonical gauge of Theorem [182, “Right-canonical (unital) gauge”]). Gauge to unital families $A^i = S_A^{-1} A^i S_A$ and $B^i = S_B^{-1} B^i S_B$. If $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ satisfies $F_{AB}^{\text{rect}}(X) = \mu X$ with $|\mu| = 1$, then $X' = S_A^{-1} X (S_B^\dagger)^{-1}$ is nonzero, the families A' and B' are unital, and, for every i ,

$$X'(B'^i)^\dagger = \mu(A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i. \quad (\text{B.2})$$

Proof. ✓ Proof Embed the gauged families and transported eigenvector into the block matrices

$$C^i := \begin{pmatrix} A'^i & 0 \\ 0 & B'^i \end{pmatrix}, \quad Y := \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}, \quad (\text{B.3})$$

on $M_{D_1+D_2}(\mathbb{C})$; the matrices $\{C^i\}$ form a unital Kraus family, and $E_C(Y) = \mu Y$. The positive definite fixed points of the adjoint maps give a faithful weighted trace for the block map. Applying the Kadison–Schwarz inequality to Y and pairing its positive semidefinite gap with this fixed point shows that the gap vanishes. The Kraus relation of the corresponding theorem in the companion quantum-channel volume [182, “KS equality implies Kraus commutation”] then gives $Y(C^i)^\dagger = \mu(C^i)^\dagger Y$; applying the same argument to the complementary product gives $C^i Y = \mu Y C^i$. Expanding these block relations gives (B.2). $\square$

Lemma B.4.2 (Intertwining yields gauge-phase equivalence). Lean ✓ Stmt Let A and B be tensors of the same bond dimension. Suppose invertible gauges take them to A' and B' , and suppose there are an invertible matrix X' and a scalar μ with $|\mu| = 1$ such that $A'^i X' = \mu X' B'^i$ for every physical index i . Then A and B are gauge-phase equivalent.

Proof. ✓ Proof If the bond dimension is zero, the bond spaces are empty, and the unit scalar together with the unique identity gauge satisfies the relation. Now suppose the bond dimension is positive, and write S_A, S_B for the two gauges. The corresponding gauged-intertwining theorem in the companion quantum-channel volume [182, the square gauge construction] forms the inverse pair

$$X := S_A X' S_B^{-1}, \quad X^{-1} = S_B (X')^{-1} S_A^{-1}. \quad (\text{B.4})$$

Transporting the assumed relation through S_A and S_B gives $A^i X = \mu X B^i$. Multiplication by X^{-1} therefore yields

$$B^i = \mu^{-1} X^{-1} A^i X. \quad (\text{B.5})$$

Since $|\mu| = 1$, the scalar μ^{-1} is nonzero, so this is the required gauge-phase equivalence. $\square$

Combining these two lemmas proves Theorem 5.6.1: Lemma B.4.1 supplies the intertwining relations (B.2) for the separately gauged, unital families. The Kadison–Schwarz equality used inside its own proof makes $X' X'^\dagger$ and $X'^\dagger X'$ nonzero positive semidefinite fixed points of the gauged transfer maps; under irreducibility, Theorem 4.1.1 upgrades these to positive definite matrices, so the transported intertwiner X' is invertible. Lemma B.4.2 then converts the invertible intertwiner, together with the two gauge matrices, into a single invertible matrix realizing the gauge-phase relation $B^i = \bar{\mu} X A^i X^{-1}$ via the Skolem–Noether theorem (Theorem [182, “Skolem–Noether”]).

Lemma 5.4.1 gives the companion kernel-invariance argument, independent of irreducibility: if a modulus-one rectangular intertwiner satisfying (5.6) and (5.7) exists between injective tensors of bond dimensions D_1 and D_2 , then (5.6) shows that $\ker(X)$ is invariant under every $(B^i)^\dagger$; since the $\{B^i\}$ span the full matrix algebra, so do the $\{(B^i)^\dagger\}$, so $\ker(X)$ is trivial and $D_2 \leq D_1$. Taking adjoints of (5.7) and applying the same argument to $X^\dagger$ shows that $\ker(X^\dagger)$ is invariant under every $(A^i)^\dagger$ and hence trivial, giving $D_1 \leq D_2$. Combined with Lemma B.4.1, this proves $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ whenever A and B are injective tensors of unequal bond dimension, which is exactly Theorem 5.4.2.

B.5 Spectral-radius decay and overlap limits

The spectral-radius gap $\rho_{\text{spec}}(F_{AB}) < 1$ translates into operator-norm decay and asymptotic orthogonality of matrix product vectors through standard Banach-algebra convergence.

Theorem 5.4.3 applies Lemma [182, “Powers vanish below unit spectral radius”] to the rectangular mixed transfer operator and then evaluates its powers at a fixed matrix. Theorem 5.5.4 combines the gap $\rho_{\text{spec}}(F_{AB}) < 1$ of Theorem 5.3.6 with the equal-bond-dimension specialization of this rectangular decay theorem. Since Tr is continuous on the finite-dimensional matrix-endomorphism algebra, this gives $\text{Tr}(F_{AB}^N) \rightarrow 0$ term by term over the finitely many matrix-unit entries in the trace-expansion formula of [182, “Trace expansion over matrix units”]; combined with the overlap–trace identity (5.4), this proves Theorem 5.5.5.

Theorem B.5.1 (Cross-correlation decay). Lean ✓ Stmt *If A and B are injective normalized tensors that are not gauge-phase equivalent, then $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ as $N \rightarrow \infty$ for every $X \in M_D(\mathbb{C})$.*

Proof. ✓ Proof Theorem 5.5.4 gives $F_{AB}^N(X) \rightarrow 0$ in operator norm, and continuity of the trace gives the conclusion. $\square$

Theorem B.5.2 (Self-correlation persists). [Lean](#) [✓ Stmt](#) *If ρ is a fixed point of $\mathcal{E}_A$, then $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ for every $N \geq 0$. In particular, this applies to the distinguished Perron–Frobenius fixed point from Theorem 4.3.2.*

Proof. [✓ Proof](#) Since $\mathcal{E}_A(\rho) = \rho$, one has $\mathcal{E}_A^N(\rho) = \rho$ for every N . Taking the trace proves the claim. $\square$

B.6 Rank-one Perron projection

Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a trace-preserving linear map with a fixed point $\rho \neq 0$, $\text{tr}(\rho) \neq 0$, underlying the primitive overlap convergence in Theorem 5.10.2. Write $P := P_\rho$ for the fixed-point projection of the corresponding definition in the companion quantum-channel volume [182, “Fixed-point projection”],

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho,$$

and $N := E - P$ for the complementary part. The power decomposition (the corresponding theorem in the companion quantum-channel volume [182, “Power decomposition”]) gives $E^n = P + N^n$ for $n \geq 1$.

Under the hypothesis $\rho_{\text{spec}}(N) < 1$ of Theorem 5.10.1, Lemma [182, “Powers vanish below unit spectral radius”] gives $N^n \rightarrow 0$, so $\text{Tr}(E^n) = \text{Tr}(P) + \text{Tr}(N^n) \rightarrow \text{Tr}(P) = 1$ by Lemma [182, “Trace of the fixed-point projection”]. This is exactly the mechanism recorded in Theorem 5.10.1.

When $E = \mathcal{E}_A$ is the transfer map of a normalized MPS tensor and ρ is its Perron–Frobenius fixed point (Theorem 4.3.2, Chapter 4), the overlap–trace identity (5.4) with $A = B$ gives $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$, so Theorem 5.10.1 yields the primitive self-overlap limit of Theorem 5.10.2. The hypothesis $\rho_{\text{spec}}(N) < 1$ is exactly the complementary transfer-map gap that the complementary spectral-radius theorem for primitive maps (the corresponding theorem in the companion quantum-channel volume [182, “Complementary transfer-map gap for primitive maps”]) supplies: for a primitive E all eigenvalues bounded by 1 in modulus and no trace-zero fixed point, every eigenvalue of $N = E - P$ has modulus strictly less than 1, and since $M_D(\mathbb{C})$ is finite-dimensional this bounds the spectral radius $\rho_{\text{spec}}(N)$ itself.

Theorem B.6.1 (Complementary spectral-radius gap for irreducible primitive tensors). [Lean](#) [✓ Stmt](#) *Let A be an irreducible normalized MPS tensor whose transfer map $\mathcal{E}_A$ is primitive. Then there is a nonzero positive semidefinite fixed point ρ of $\mathcal{E}_A$ with $\text{tr}(\rho) \neq 0$ such that, for $P := P_\rho$ and $N := \mathcal{E}_A - P$, $\rho_{\text{spec}}(N) < 1$.*

Proof. [✓ Proof](#) The channel fixed-point existence theorem (Theorem 4.3.1) supplies a nonzero positive semidefinite fixed point ρ . Apply Theorem [182, “Complementary gap at a prescribed fixed point”] to $\mathcal{E}_A$ and ρ . $\square$

This concretely instantiates the hypothesis of Theorem 5.10.1 and, via Theorem 5.10.2, gives the primitive self-overlap limit for every irreducible peripherally primitive normalized MPS tensor.

B.7 Periodicity removal

The cyclic decomposition for irreducible finite Kraus maps proceeds by normalizing a peripheral eigenvector to a unitary, forming its discrete Fourier spectral projections, and tracking the cyclic shift of the resulting corners under the channel.

Combining these lemmas proves Theorem [182, “Cyclic decomposition for irreducible finite Kraus maps”]: Lemma [182, “Normalized peripheral unitary”] supplies a unitary generator of exact order m for a primitive m -th peripheral root, and Lemma [182, “Cyclic projections from a peripheral unitary”] builds the m Fourier projections out of its powers, using Lemma [182, “Powers on a peripheral-unitary orbit”] to identify how the channel shifts them.

B.7.1 Cyclic corners and restricted primitivity

Corner preservation, the associated corner subspace, irreducibility of the restricted map, and corner rank are defined in the companion quantum-channel volume [182, “Corner preservation” and the following corner definitions]. We use those definitions here in the MPS periodicity-removal argument.

B.7.2 Group structure and divisibility

Theorem [182, “Peripheral eigenspaces are one-dimensional”] follows the same pattern: given a peripheral unitary eigenvector U for γ (Lemma [182, “Peripheral unitary eigenvector”]), the multiplicative-domain identity gives $E(XU^\dagger) = XU^\dagger$ for any γ -eigenvector X , so the scalar fixed-point lemma for irreducible unital maps (Lemma [182, “Scalar fixed points for irreducible unital maps”]) gives $XU^\dagger = c\mathbb{1}$, hence $X = cU$ and the eigenspace is one-dimensional.

The divisibility statements of Theorem [182, “Transfer-map peripheral eigenvalues form a cyclic group”] and Theorem [182, “Transfer-map period divides bond dimension”] follow from Lemma [182, “Cyclic projections from a peripheral unitary”] and Theorem [182, “Rank equals trace for an orthogonal projection”]: the m cyclic Fourier projections $P_0, \dots, P_{m-1}$ sum to $\mathbb{1}$ and are cyclically permuted by E , so trace preservation gives $\text{tr}(P_{k+1}) = \text{tr}(P_k)$ for every k , hence $D = \text{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \text{tr}(P_k) = m \text{tr}(P_0)$. Since the trace of an orthogonal projection is a natural number (Theorem [182, “Rank equals trace for an orthogonal projection”]), $m \mid D$.

This periodicity-removal chain — unitary normalization, Fourier projections, cyclic Kraus shift (Lemma 5.8.1.1), the blocked cyclic corners of this section, and restricted primitivity — is exactly the machinery behind the period-removing blocking theorems of Chapter 7: [212, Theorem 6.6] supplies the cyclic peripheral spectrum of the adjoint transfer map, and Theorem 7.3.1.1 (Section 7.3.1) and Theorem 7.11.1 apply it to realize the cyclic-sector decomposition of a blocked irreducible tensor with rectangular support isometries.

Appendix C

Wielandt Bound: Supporting Results

The quantitative Wielandt argument of Chapter 6 rests on structural properties of exact-word spans, cumulative spans, one-step augmentations, blocking, and complementary transfer-map gaps.

C.1 Exact-word and block-injectivity support

Exact-word products satisfy direct concatenation factorizations and dimension bounds in low dimensions.

Lemma C.1.1 (Word span products). [Lean](#) [✓ Stmt](#) *One has $S_m(A)S_n(A) \subseteq S_{m+n}(A)$: the product of word spans at lengths m and n is contained in the word span at length $m+n$.*

Proof. [✓ Proof](#) Every product $A^{w_1}A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1w_2}$, where $|w_1w_2| = m+n$. $\square$

Theorem C.1.2 (Exact factorization of word spans). [Lean](#) [✓ Stmt](#) *For all $m, n \geq 0$, one has $S_{m+n}(A) = S_m(A)S_n(A)$.*

Proof. [✓ Proof](#) For words w_1, w_2 of lengths m, n , concatenation satisfies

$$A^{w_1w_2} = A^{w_1}A^{w_2}.$$

Split each word of length $m+n$ after its first m letters for one inclusion, and concatenate words of lengths m and n for the other. $\square$

Lemma C.1.3 (Zero-length word span). [Lean](#) [✓ Stmt](#) *If $D \geq 2$, then $S_0(A) \neq M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) The zero-length products consist only of the identity, so $S_0(A) = \text{span}_{\mathbb{C}}\{1\}$ has dimension 1. Since $\dim M_D(\mathbb{C}) = D^2 \geq 4$, this span cannot equal $M_D(\mathbb{C})$. $\square$

Corollary C.1.4 (0-block injectivity). [Lean](#) [✓ Stmt](#) *If $D \geq 1$ and A is 0-block injective, then $D = 1$.*

Proof. [✓ Proof](#) By Lemma 6.1.1.23, 0-block injectivity gives $S_0(A) = M_D(\mathbb{C})$. Lemma C.1.3 excludes $D \geq 2$, so the nonzero bond dimension must be 1. $\square$

Lemma C.1.5 (Exact words around a nonzero middle matrix). [Lean](#) [✓ Stmt](#) *If A is L -block injective and $X \neq 0$, then*

$$\text{span}_{\mathbb{C}}\{A^u X A^v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

Proof. [✓ Proof](#) By L -block injectivity, the length- L word products span $M_D(\mathbb{C})$. Bilinearity of $(R, S) \mapsto RXS$ reduces the claim to Lemma [182, “Two-sided span from one nonzero matrix”], with R and S chosen among those word products. $\square$

C.2 Cumulative-span and spectral linear algebra

Cumulative word spans form an ascending chain of subspaces whose stabilization is governed by finite dimensionality and Burnside’s theorem.

The next two lemmas connect irreducible matrix actions with the cumulative word span used throughout Chapter 6.

Remark C.2.1 (Burnside input for cumulative spanning). The passage from irreducible tensors to full algebra spanning first passes through the irreducible action on $\mathbb{C}^D$. Burnside’s theorem then says that the resulting irreducible subalgebra of $M_D(\mathbb{C})$ is all of $M_D(\mathbb{C})$. Once the algebra span is full, finite-dimensionality gives a single cumulative span that already equals $M_D(\mathbb{C})$.

Lemma C.2.2 (Irreducible action gives an irreducible tensor). [Lean](#) [✓ Stmt](#) *If the letters of A act irreducibly on $\mathbb{C}^D$, then A has no nontrivial invariant orthogonal projection.*

Proof. [✓ Proof](#) If P were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of A . Irreducibility of the action forces this range to be either 0 or all of $\mathbb{C}^D$, so P must be 0 or $\mathbb{1}$, a contradiction. $\square$

Lemma C.2.3 (The algebra span reaches a finite cumulative span). [Lean](#) [✓ Stmt](#) *If the generated unital algebra satisfies $\text{alg}(A) = M_D(\mathbb{C})$, then there exists N such that $T_N(A) = M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) Every element of $\text{alg}(A)$ lies in some cumulative span $T_n(A)$: the generators lie in $T_1(A)$, scalar multiples of the identity lie in $T_0(A)$, and the family $\{T_n(A)\}_{n \geq 0}$ is closed under addition and under multiplication with lengths adding. Since $M_D(\mathbb{C})$ is finite-dimensional, the ascending chain $T_0(A) \subseteq T_1(A) \subseteq \dots$ stabilizes, so one can choose a single N with $T_N(A) = M_D(\mathbb{C})$. $\square$

Theorem C.2.4 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). [Lean](#) [✓ Stmt](#) *Let $D > 0$. If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. [✓ Proof](#) By Theorem 6.1.1.22, if $T_n \neq T_{n+1}$, then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ by Lemma 6.1.1.20, after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality, which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0 and hence $T_{L_0} = M_D(\mathbb{C})$. $\square$

Theorem C.2.5 (Nonzero trace product). [Lean](#) [✓ Stmt](#) *Let $D > 0$. If A is normal, then there exists a word w of length at most D^2 such that $\text{tr}(A^w) \neq 0$. This is the weaker D^2 corollary of [200, Lemma 1]. The sharp source statement is Theorem 6.2.4.*

Proof. [✓ Proof](#) By Theorem 6.2.1, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length at most D^2 . At least one has nonzero trace because $\text{tr}(\mathbb{1}) = D \neq 0$. $\square$

Theorem C.2.6 (Eigenvalue extraction). [Lean](#) [✓ Stmt](#) Let $D > 0$. If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.

Proof. [✓ Proof](#) By Theorem C.2.5, there is a word w_0 with $\text{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem [182, “Nonzero trace implies eigenvector”], A^{w_0} has a nonzero eigenvector. $\square$

Theorem C.2.7 (Vector span stabilisation). [Lean](#) [✓ Stmt](#) If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for every $m \geq n$.

Proof. [✓ Proof](#) The argument is the same as for Theorem 6.1.1.21: left multiplication by A^i cannot escape the stabilised span. $\square$

Lemma C.2.8 (Vector span from matrix span). [Lean](#) [✓ Stmt](#) If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.

Proof. [✓ Proof](#) Every matrix $M \in M_D(\mathbb{C})$ lies in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v lies in $T_N(A)$. Applying such matrices to φ recovers all of $\mathbb{C}^D$. $\square$

C.3 One-step augmentation

Augmenting a Kraus family by a matrix already in its one-step span preserves all exact word spans while allowing the adjunction of an invertible generator or an eigenvector.

Definition C.3.1 (One-step augmentation). [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with Kraus family $\{A^i\}_{i=0}^{d-1}$, and let $X \in M_D(\mathbb{C})$. The *one-step augmentation* of A by X is the tensor $\tilde{A}$ with physical dimension $d + 1$ whose first Kraus operator is X and whose remaining Kraus operators are those of A :

$$\begin{aligned}\tilde{A}^0 &= X, \\ \tilde{A}^{i+1} &= A^i.\end{aligned}$$

Lemma C.3.2 (Kraus operators lie in the one-step span). [Lean](#) [✓ Stmt](#) Every Kraus operator A^i belongs to the one-step span $S_1(A)$.

Proof. [✓ Proof](#) This is the length-one word with letter i . $\square$

Lemma C.3.3 (Redundant one-step augmentation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . Then $S_n(\tilde{A}) = S_n(A)$ for every $n \geq 0$.

Proof. [✓ Proof](#) The new generator X is already in $S_1(A)$, and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length. $\square$

Lemma C.3.4 (Normality and Kraus rank under one-step augmentation). [Lean](#) [Lean](#) [✓ Stmt](#) Let $X \in S_1(A)$, and let $\tilde{A}$ be the one-step augmentation of A by X . If A is normal, then $\tilde{A}$ is normal, and $\text{kr}(\tilde{A}) = \text{kr}(A)$.

Proof. [✓ Proof](#) Since all exact word spans are unchanged by Lemma C.3.3, eventual full word span for A gives eventual full word span for $\tilde{A}$. The equality of Kraus ranks is the equality of the dimensions of the common one-step span. $\square$

Theorem C.3.5 (Wielandt case (2), invertible generator). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be a normal MPS tensor with bond dimension $D > 0$, and suppose that some Kraus operator A^{i_0} is invertible. Then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}),$$

$$\iota(A) \leq D^2 - \text{kr}(A) + 1.$$

Proof. [✓ Proof](#) The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument at every later length would contradict normality. Since $\dim S_1(A) = \text{kr}(A)$ and $\dim M_D(\mathbb{C}) = D^2$, the full algebra is reached by length $D^2 - \text{kr}(A) + 1$. The index bound follows from the definition of $\iota(A)$. $\square$

Corollary C.3.6 (Wielandt case (2), invertible Kraus operator). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 6.1.1.3. If some Kraus operator A^{i_0} is invertible, then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}),$$

$$\iota(A) \leq D^2 - \text{kr}(A) + 1.$$

Proof. [✓ Proof](#) Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma C.3.2. $\square$

Corollary C.3.7 (Wielandt case (3), non-invertible Kraus operator). [Lean](#) [Lean](#) [✓ Stmt](#) Let A be an MPS tensor with bond dimension $D \geq 1$, satisfying $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, and suppose that A is primitive in the sense of Definition 6.1.1.3. If some Kraus operator A^{i_0} is non-invertible and has an eigenvector $\varphi \neq 0$ with nonzero eigenvalue μ , then

$$S_{D^2}(A) = M_D(\mathbb{C}),$$

$$\iota(A) \leq D^2.$$

Proof. [✓ Proof](#) Apply the one-step case to $X = A^{i_0}$, which belongs to $S_1(A)$ by Lemma C.3.2. $\square$

C.4 Blocking and fixed-length spanning

Blocking preserves tensor normality and embeds exact word spans, allowing word eigenvectors to be transferred to single-index eigenvectors of blocked tensors.

Theorem C.4.1 (Blocking preserves normality). [Lean](#) [✓ Stmt](#) If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.

Proof. [✓ Proof](#) Choose N with $S_N(A) = M_D(\mathbb{C})$. Then $\mathbb{1} \in S_N(A)$. For any $M \in S_N(A)$, write $\mathbb{1} = \sum_r c_r A^{w_r}$ with each $|w_r| = N$. Then

$$M = M\mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) S_N(A) \subseteq S_{2N}(A). \quad (\text{C.1})$$

By Lemma C.1.1, iterating (C.1) gives $S_N(A) \subseteq S_{kN}(A)$ for every $k \geq 1$. Taking $k = L$ yields $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$. By Lemma C.4.2, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

Lemma C.4.2 (Chunking: word span embeds into blocked word span). [Lean](#) [✓ Stmt](#) One has $S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L .

Proof. ✓ Proof Every length- nL word in the original alphabet can be split into n consecutive chunks of length L , each of which is a single blocked Kraus operator by (2.18). The claim follows by induction on n , using Lemma C.1.1. $\square$

Theorem C.4.3 (Word eigenvector gives blocked eigenvector). Lean ✓ Stmt If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L , then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0 .

Proof. ✓ Proof By (2.18), $(A^{[L]})^{j_0} = A^{w_0}$. $\square$

Theorem C.4.4 (Blocked fixed-length matrix spanning). Lean ✓ Stmt Suppose A has bond dimension $D > 0$, is normal, and $L > 0$, and we have:

1. a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$),
2. a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$),
3. a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m .

Then there exists N such that $S_N(A) = M_D(\mathbb{C})$.

Remark. The proof gives the explicit bound $N = ((D-1) + m + (D-1))L$; the theorem states only the existential conclusion.

Proof. ✓ Proof Set $B = A^{[L]}$, which is normal by Theorem C.4.1. The word eigenvectors transfer to single-index eigenvectors of B and B^T by Theorem C.4.3. Eigenvector spreading and padding then give $H_{D-1}(B, \varphi) = H_{D-1}(B^T, \psi) = \mathbb{C}^D$.

For each standard basis vector e_j , choose $R_j \in S_{D-1}(B^T)$ with $R_j\psi = e_j$. Reversing each word after transposition shows that $R_j^T \in S_{D-1}(B)$, and therefore $(\varphi\psi^T)R_j^T = \varphi e_j^T$ belongs to $S_{m+D-1}(B)$. Theorem 6.5.2, applied with $n = D-1$ and $m' = m + D-1$, now gives $S_{(D-1)+m+(D-1)}(B) = M_D(\mathbb{C})$. Finally, Lemma 6.5.4 transfers this identity to $S_{((D-1)+m+(D-1))L}(A) = M_D(\mathbb{C})$. $\square$

Theorem C.4.5 (Rank-one extraction for blocked tensor). Lean ✓ Stmt Suppose A has bond dimension $D > 0$, is normal, and $N_0 \geq 1$ with $S_{N_0}(A) = M_D(\mathbb{C})$. Then there exist words σ_0, τ_0 of length N_0 , nonzero vectors φ, ψ , nonzero scalars μ, ν , and $m \in \mathbb{N}$ such that:

1. $A^{\sigma_0}\varphi = \mu\varphi$,
2. $(A^{\tau_0})^T\psi = \nu\psi$,
3. $\varphi\psi^T \in S_m(A^{[N_0]})$.

Proof. ✓ Proof Set $B = A^{[N_0]}$. By (2.18) and $S_{N_0}(A) = M_D(\mathbb{C})$, one has $S_1(B) = M_D(\mathbb{C})$. Hence $\mathbb{1} \in S_1(B)$, so not all blocked generators can have trace zero. Choose j_0 with $\text{tr}(B^{j_0}) \neq 0$. By Theorem [182, “Nonzero trace implies eigenvector”], B^{j_0} has a nonzero eigenvector φ with eigenvalue $\mu \neq 0$. Writing j_0 as the blocked index corresponding to a word σ_0 of length N_0 gives $A^{\sigma_0}\varphi = \mu\varphi$. Applying the same argument to the transpose blocked tensor gives a word τ_0 of length N_0 , a nonzero vector ψ , and $\nu \neq 0$ with $(A^{\tau_0})^T\psi = \nu\psi$. Since A is normal and $N_0 \geq 1$, the blocked tensor B is normal by Theorem C.4.1. Therefore there exists M with $S_M(B) = M_D(\mathbb{C})$. In particular, $\varphi\psi^T \in S_M(B)$. Taking $m = M$ proves (3). $\square$

Theorem C.4.6 (Fixed-length word-span saturation via blocking). Lean ✓ Stmt If A is a normal MPS tensor with bond dimension $D > 0$, then there exists N such that $S_N(A) = M_D(\mathbb{C})$. This is a blocked fixed-length construction under the stated normality hypothesis. The quantitative result of [200, Lemma 2(b)] is Theorem 6.4.5.

Proof. ✓ Proof By normality, some $N_0 > 0$ has $S_{N_0}(A) = M_D(\mathbb{C})$. Theorem C.4.5 produces eigenvectors and a rank-one element $\varphi\psi^T \in S_m(A^{[N_0]})$. Theorem C.4.4 then gives $S_{((D-1)+m+(D-1))N_0}(A) = M_D(\mathbb{C})$. $\square$

C.5 Complementary-gap consequences

A strict spectral-radius bound on the complementary transfer map implies overlap convergence, non-vanishing of iterated transfer maps, and fixed-point uniqueness.

Theorem C.5.1 (Primitive overlap convergence). Lean ✓ Stmt *If A is a primitive MPS tensor, then $O_{AA}(N) \rightarrow 1$.*

Proof. ✓ Proof The claim follows from the self-overlap convergence under a complementary transfer-map gap (Theorem 5.10.2). $\square$

Theorem C.5.2 (Peripheral primitivity and irreducibility give normalized overlap). Lean ✓ Stmt *Let A have positive bond dimension and satisfy $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. If A is tensor-irreducible and the transfer map is primitive, then $O_{AA}(N) \rightarrow 1$.*

Proof. ✓ Proof Irreducible Perron–Frobenius theory supplies a nonzero positive fixed point. Peripheral primitivity gives a strict spectral-radius bound on the complementary transfer map. Thus A is a primitive MPS tensor, and Theorem C.5.1 applies. $\square$

Lemma C.5.3 (Primitive MPS tensors have nonzero word products). Lean ✓ Stmt *If A is a primitive MPS tensor, then for every $n \geq 0$ there exists a word w of length exactly n with $A^w \neq 0$.*

Proof. ✓ Proof The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. $\square$

Lemma C.5.4 (Iterated transfer does not vanish). Lean ✓ Stmt *If A is a primitive MPS tensor, then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. ✓ Proof By Lemma C.5.3, there exists a word w of length n with $A^w \neq 0$. The expansion $\mathcal{E}_A^n(\mathbb{1}) = \sum_{|u|=n} A^u (A^u)^\dagger$ contains the nonzero positive semidefinite summand $A^w (A^w)^\dagger$, so $\mathcal{E}_A^n(\mathbb{1}) \neq 0$. $\square$

Lemma C.5.5 (Fixed-point uniqueness from a complementary transfer-map gap). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ , then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ : $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$.*

Proof. ✓ Proof Set $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$. Then $\text{tr}(\sigma') = 0$ and $(\mathcal{E}_A - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $\mathcal{E}_A - P_\rho$, contradicting the complementary transfer-map gap $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. $\square$

Lemma C.5.6 (Complement powers tend to zero). Lean ✓ Stmt *If A is a primitive MPS tensor with PSD fixed point ρ , then $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$ in operator norm.*

Proof. ✓ Proof The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the complementary transfer-map gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. $\square$

Lemma C.5.7 (Primitive tensors with positive-definite fixed point have irreducible transfer map). [Lean](#) [✓ Stmt](#) *If A is a primitive MPS tensor with positive-definite fixed point ρ , then the transfer map $\mathcal{E}_A$ is irreducible.*

Proof. [✓ Proof](#) By Lemma C.5.5, if $\sigma \geq 0$ and $\mathcal{E}_A(\sigma) = \sigma$, then $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$. Since $\rho > 0$, Wolf's uniqueness criterion for irreducibility applies. $\square$

C.6 Identity in the one-step span

When the identity matrix lies in the one-step span $S_1(A)$, exact word spans grow monotonically and coincide with cumulative spans.

Remark C.6.1 (Identity-in-the-one-step-span condition). [✓ Stmt](#) The *one-step padding condition* is $\mathbb{1} \in S_1(A)$: the identity matrix lies in the span of the Kraus operators $\{A^i\}_{i=0}^{d-1}$. While tensor normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ is quadratic, the one-step condition $\mathbb{1} \in S_1(A)$ is linear in the Kraus operators. Theorem 6.7.4.4 obtains normality from a positive-definite fixed point and a complementary gap without this padding condition.

Remark C.6.2 (Counterexample: cumulative spanning does not imply normality). [✓ Stmt](#) In $M_2(\mathbb{C})$, let $A^0 = E_{12}$ and $A^1 = E_{21}$ be the off-diagonal matrix units. Then $\text{alg}(A) = M_2(\mathbb{C})$ and $T_2(A) = M_2(\mathbb{C})$, but the word spans alternate: $S_n(A)$ contains only off-diagonal matrices for odd n and only diagonal matrices for even n . Hence $S_n(A) \neq M_2(\mathbb{C})$ for every n , and A is not normal. The one-step padding condition fails because $\mathbb{1} \notin S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$.

Theorem C.6.3 (Word-span monotonicity from one-step padding). [Lean](#) [✓ Stmt](#) *If $\mathbb{1} \in S_1(A)$, then $S_n(A) \leq S_{n+1}(A)$ for every n .*

Proof. [✓ Proof](#) For any $M \in S_n(A)$, one has $M = M\mathbb{1} \in S_n(A)S_1(A) \subseteq S_{n+1}(A)$ by Lemma C.1.1. $\square$

Theorem C.6.4 (Cumulative span equals word span under one-step padding). [Lean](#) [✓ Stmt](#) *If $\mathbb{1} \in S_1(A)$, then $T_n(A) = S_n(A)$ for every n .*

Proof. [✓ Proof](#) By Theorem C.6.3, the word spans form the chain $S_0(A) \leq S_1(A) \leq \dots \leq S_n(A)$. Therefore their sum $T_n(A) = S_0(A) + \dots + S_n(A)$ equals the largest term $S_n(A)$. $\square$

Theorem C.6.5 (Cumulative spanning plus one-step padding implies normality). [Lean](#) [✓ Stmt](#) *If $T_N(A) = M_D(\mathbb{C})$ and $\mathbb{1} \in S_1(A)$, then A is normal.*

Proof. [✓ Proof](#) Theorem C.6.4 gives $S_N(A) = T_N(A) = M_D(\mathbb{C})$. $\square$

Appendix D

Canonical Form Reduction: Supporting Results

The canonical form reductions of Chapter 7 rely on the algebraic and spectral behavior of projection, gauge transformation, blocking, and cyclic-sector reindexing developed here.

Throughout, $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$, $A^{[p]}$ denotes blocking by p , and $V^{(N)}(A)_\sigma = \text{tr}(A^{\sigma_1} \dots A^{\sigma_N})$. Weighted direct sums are written $\bigoplus_k \mu_k A_k$.

D.1 Similarities, strict splitting, and zero blocks

Similarity invariance and strict dimension decrease justify the induction used in Theorem 7.1.1.2; the final result removes blocks that vanish at every positive length.

Theorem D.1.1 (Unitary conjugation preserves the MPV family). [Lean](#) [✓ Stmt](#) *If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family.*

Proof. [✓ Proof](#) By cyclicity of the trace, $\text{tr}((U^\dagger A^{i_1} U) \dots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \dots A^{i_N})$. □

Theorem D.1.2 (Nontrivial invariant projection gives strict two-block reduction). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 7.1.2, assume in addition that $P \neq 0$ and $P \neq \mathbb{1}$. Then there exist n, m with $n + m = D$, $n < D$, and $m < D$, together with block tensors A_1, A_2 , such that A and $A_1 \oplus A_2$ generate the same MPV family in the sense of Definition 2.4.3.*

Proof. [✓ Proof](#) Apply Theorem 7.1.2. In the diagonal form of P , the 1-eigenspace is nonempty because $P \neq 0$, and the 0-eigenspace is nonempty because $P \neq \mathbb{1}$. Hence both block sizes are positive. Since $n + m = D$, this implies $n < D$ and $m < D$. □

Theorem D.1.3 (Nonzero irreducible block decomposition). [Lean](#) [✓ Stmt](#) *Every MPS tensor A admits a finite family of nonzero irreducible blocks $(B_k)_{k=1}^r$, each with at least one nonzero Kraus operator and positive bond dimension. For every $N \geq 1$ and every σ ,*

$$V^{(N)}(A)_\sigma = V^{(N)}\left(\bigoplus_{k=1}^r B_k\right)_\sigma. \quad (\text{D.1})$$

Moreover, $\sum_{k=1}^r D_k \leq D$.

Proof. ✓ Proof Apply the irreducible block decomposition (Theorem 7.1.1.2) and retain the blocks having a nonzero Kraus operator. For every all-zero irreducible block C_j and every $N \geq 1$, $V^{(N)}(C_j)_\sigma = 0$, so (D.1) holds. The dimension inequality follows by restricting the structural identity $\sum_j D_j = D$ to the retained subset. $\square$

D.2 Projector closure and isometric corner decompositions

Reducing orthogonal projections yield support isometries that compress tensors to corner blocks, preserving matrix product vectors across direct sums (supporting Theorems 7.7.3 and 7.7.5).

Theorem D.2.1 (Invariant projections are reducing under projector closure). Lean ✓ Stmt Suppose that A has invariant-projector closure and that P is an orthogonal projection satisfying $(\mathbb{1} - P)A^iP = 0$ for all i . Then P reduces every tensor matrix: $PA^i = A^iP$ for all i .

Proof. ✓ Proof The lower-corner relation gives $(\mathbb{1} - P)A^i = (\mathbb{1} - P)A^i(\mathbb{1} - P)$. Applying projector closure to $\mathbb{1} - P$ gives $A^i(\mathbb{1} - P) = (\mathbb{1} - P)A^i(\mathbb{1} - P)$, hence the upper corner $PA^i(\mathbb{1} - P)$ also vanishes. Therefore $PA^i = PA^iP = A^iP$. $\square$

Theorem D.2.2 (Invariant-projector closure gives a reducing projection). Lean ✓ Stmt If A has invariant-projector closure and admits a nontrivial invariant projection, then it admits a nontrivial orthogonal projection commuting with every A^i .

Proof. ✓ Proof Apply Theorem D.2.1 to the projection supplied by Definition 7.1.3. $\square$

Lemma D.2.3 (Support isometry of an orthogonal projection). Lean ✓ Stmt

Every orthogonal projection P on $\mathbb{C}^D$ factors through an isometry: there are $n \in \mathbb{N}$ and $V \in \mathbb{C}^{D \times n}$ such that

$$\begin{aligned} V^\dagger V &= \mathbb{1}, \\ VV^\dagger &= P, \\ n &= \text{tr}(P). \end{aligned} \tag{D.2}$$

Proof. ✓ Proof Since P is Hermitian and idempotent, its eigenvalues are 0 and 1. Writing $n = \text{tr}(P)$, unitary diagonalization gives

$$P = U \text{diag}(\underbrace{1, \dots, 1}_n, 0, \dots, 0)U^\dagger.$$

The matrix $V \in \mathbb{C}^{D \times n}$ formed by the first n columns of U satisfies (D.2). $\square$

The rectangular Gram-equality criterion that turns $B^\dagger B = A^\dagger A$ into a unitary U with $B = UA$ is a generic algebra fact, proved in the companion quantum-channel volume [182, “Rectangular Gram equality gives a unitary”].

Lemma D.2.4 (Corner tensors inherit projector closure). Lean ✓ Stmt Let A have invariant-projector closure, and let V be an isometry, $V^\dagger V = \mathbb{1}$, whose range projection $VV^\dagger$ commutes with every A^i . Then the corner tensor $B^i = V^\dagger A^i V$ again has invariant-projector closure.

Proof. ✓ Proof Let Q be an orthogonal projection with $QB^i = QB^iQ$ for all i . Then $VQV^\dagger$ is an orthogonal projection. Commutation of $VV^\dagger$ with each A^i gives $V^\dagger A^i = B^i V^\dagger$, so left-invariance of Q yields

$$(VQV^\dagger)A^i = VQB^iV^\dagger = VQB^iQV^\dagger = (VQV^\dagger)A^i(VQV^\dagger).$$

Projector closure for A yields $A^i(VQV^\dagger) = (VQV^\dagger)A^i(VQV^\dagger)$, and compressing both sides between $V^\dagger$ and V returns $B^iQ = QB^iQ$. $\square$

Lemma D.2.5 (Transfer map of an intertwined corner). [Lean](#) [Lean](#) [✓ Stmt](#) *Let V satisfy $A^iV = VB^i$ for all i . Then, for every X ,*

$$\mathcal{E}_A(VXV^\dagger) = V\mathcal{E}_B(X)V^\dagger. \quad (\text{D.3})$$

If moreover $V^\dagger V = \mathbb{1}$, then every eigenvalue of $\mathcal{E}_B$ is an eigenvalue of $\mathcal{E}_A$.

Proof. [✓ Proof](#) Expanding the left-hand side of (D.3) termwise, the intertwining $A^iV = VB^i$ and its adjoint $V^\dagger(A^i)^\dagger = (B^i)^\dagger V^\dagger$ move both factors past V and $V^\dagger$. For the eigenvalue statement, if $\mathcal{E}_B(X) = \mu X$ with $X \neq 0$, then $\mathcal{E}_A(VXV^\dagger) = \mu VXV^\dagger$, and $VXV^\dagger \neq 0$ because compression with the isometry gives $V^\dagger(VXV^\dagger)V = X$. $\square$

Lemma D.2.6 (Matrix product vectors split along an isometric partition). [Lean](#) [✓ Stmt](#) *Let $V_1, \dots, V_r$ be isometries, $V_k^\dagger V_k = \mathbb{1}$, with $\sum_k V_k V_k^\dagger = \mathbb{1}$, and suppose $A^i V_k = V_k B_k^i$ for some tensors $B_1, \dots, B_r$. Then $\mathcal{V}(A) = \mathcal{V}(\bigoplus_{k=1}^r B_k)$.*

Proof. [✓ Proof](#) The intertwining relation propagates from letters to words, so $A^w V_k = V_k B_k^w$ for every word w . Inserting the partition of unity $\sum_k V_k V_k^\dagger = \mathbb{1}$ into the trace and cycling gives

$$\text{tr}(A^w) = \sum_{k=1}^r \text{tr}(V_k^\dagger A^w V_k) = \sum_{k=1}^r \text{tr}(B_k^w),$$

which is the matrix product vector of the direct sum. $\square$

Lemma D.2.7 (Canonical inclusion of a dependent direct summand). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let $E_k : H_k \rightarrow \bigoplus_l H_l$ be the canonical inclusion and let $B = \bigoplus_l B_l$. For distinct k and ℓ , the corresponding inclusions have orthogonal ranges. The identities are*

$$\begin{aligned} E_k^\dagger E_k &= \mathbb{1}, \\ E_k^\dagger E_\ell &= 0, \\ B E_k &= E_k B_k, \\ E_k^\dagger B &= B_k E_k^\dagger, \\ E_k^\dagger B E_k &= B_k, \\ B &= \sum_k E_k B_k E_k^\dagger. \end{aligned} \quad (\text{D.4})$$

In particular, B commutes with the range projection $E_k E_k^\dagger$.

Proof. [✓ Proof](#) The identities in (D.4) follow directly from the defining block-diagonal entries. The relations $B E_k = E_k B_k$ and $E_k^\dagger B = B_k E_k^\dagger$ imply that B commutes with $E_k E_k^\dagger$. $\square$

Lemma D.2.8 (Canonical hypotheses descend to dependent direct summands). [Lean](#) [✓ Stmt](#) *Suppose that, after a change of coordinates, A is the dependent direct sum of tensors B_k . If A has invariant-projector closure and no nontrivial periodic vectors, then every B_k has both properties. The assertion also applies to summands of dimension zero.*

Proof. ✓ Proof Conjugate the canonical inclusion of the k -th summand back to the original bond coordinates of A . This is an isometry which intertwines B_k with A on both sides, and its range projection commutes with every letter of A . Projector closure therefore descends by exact compression. A periodic vector of B_k would embed through the same isometry as a periodic vector of A . $\square$

Lemma D.2.9 (Closed-chain equality from an exact isometric decomposition). Lean ✓ Stmt Suppose that isometries V_k satisfy

$$\begin{aligned} V_k^\dagger A^i &= (\mu_k B_k^i) V_k^\dagger, \\ A^i &= \sum_k V_k (\mu_k B_k^i) V_k^\dagger. \end{aligned} \tag{D.5}$$

Then A and $\bigoplus_k \mu_k B_k$ have the same matrix product vectors at every positive length.

Proof. ✓ Proof Expand the first letter by the reconstruction formula in (D.5). In each summand, move the remaining word through $V_k^\dagger$ using the intertwining relation. Cyclicity of the trace places $V_k^\dagger V_k = \mathbb{1}$ next to the weighted block word, leaving precisely its closed-chain trace. $\square$

D.3 Perron gauges and preservation under conjugation

Positive Perron eigenpairs determine the spectral normalization and unital gauge transformations of irreducible blocks.

Lemma D.3.1 (Perron eigenpair of a nonzero irreducible tensor). Lean ✓ Stmt Let B be an irreducible MPS tensor of positive bond dimension, not all of whose matrices vanish. Then there are $\rho > 0$ and $r > 0$ with $\mathcal{E}_B(\rho) = r\rho$.

Proof. ✓ Proof

Irreducibility of the tensor makes the transfer map an irreducible completely positive map, and a nonzero matrix makes it nonzero, so it does not annihilate any nonzero PSD matrix. The Perron–Frobenius theorem for positive maps (Theorem [182, “Positive-eigenvalue specialization”]) gives a nonzero PSD eigenvector with eigenvalue $r > 0$, and irreducibility upgrades it to a positive-definite one ([212, Theorem 6.3(2)–(3)]). $\square$

Lemma D.3.2 (Spectral normalization of an irreducible block). Lean ✓ Stmt Let B be an irreducible MPS tensor of positive bond dimension with $\mathcal{E}_B(\rho) = r\rho$ for some $\rho > 0$ and $r > 0$, and suppose every eigenvalue of $\mathcal{E}_B$ of modulus r equals r . Then $r^{-1/2}B$ is a normal tensor.

Proof. ✓ Proof The rescaling identity $\mathcal{E}_{r^{-1/2}B}(X) = r^{-1}\mathcal{E}_B(X)$ gives $\mathcal{E}_{r^{-1/2}B}(\rho) = r^{-1}r\rho = \rho$, so ρ is a positive-definite fixed point of $\mathcal{E}_{r^{-1/2}B}$. The rescaled map remains irreducible and completely positive, so Theorem [182, “Spectral radius is the Perron eigenvalue”] identifies its spectral radius with the eigenvalue one. The eigenvalues of modulus one of $\mathcal{E}_{r^{-1/2}B}$ are the eigenvalues of $\mathcal{E}_B$ of modulus r , divided by r ; by hypothesis the only such eigenvalue is one, so the peripheral eigenvalue set is $\{1\}$. Irreducibility is unchanged by a nonzero rescaling. $\square$

Lemma D.3.3 (Range factorization of the support projection). Lean Lean Lean ✓ Stmt

The support projection π of a positive semidefinite matrix ρ factors as $\pi = \rho W$ for a spectral pseudo-inverse W , witnessing that the range of π is the range of ρ . For a rectangular matrix Y , the support projection π of $YY^\dagger$ fixes Y , $\pi Y = Y$; and whenever $AY = YG$ for some matrix G —that is, A maps the range of Y into itself—the one-sided invariance $(\mathbb{1} - \pi)A\pi = 0$ holds.

Proof. ✓ Proof

In the spectral decomposition $\rho = U \text{diag}(\lambda) U^\dagger$, take $W = U \text{diag}(\lambda^+) U^\dagger$, where λ^+ inverts the positive eigenvalues and vanishes on the kernel. Then $\rho W = U \text{diag}(\mathbf{1}_{\lambda>0}) U^\dagger = \pi$. From $\pi(YY^\dagger) = YY^\dagger$, the matrix $((\mathbf{1} - \pi)Y)((\mathbf{1} - \pi)Y)^\dagger$ vanishes, hence $\pi Y = Y$. Finally,

$$(\mathbf{1} - \pi)A\pi = (\mathbf{1} - \pi)AY(Y^\dagger W) = (\mathbf{1} - \pi)Y G(Y^\dagger W) = 0.$$

□

Theorem D.3.4 (Irreducibility under rescaled conjugation). Lean ✓ Stmt *Let B be an irreducible tensor, let X be invertible, and let $c \neq 0$. Then the rescaled conjugated tensor with letters $cX^{-1}B^vX$ is irreducible. In particular, the rescaled unital gauge of an irreducible tensor is irreducible; this is the normalization of [11, Section 2.3] applied to an irreducible corner.*

Proof. ✓ Proof An invariant orthogonal projection $P \notin \{0, \mathbf{1}\}$ of the conjugated tensor makes the columns of $Y = XP$ span a subspace invariant under every letter of B : $B^v Y = Y G_v$, where $G_v = X^{-1} B^v X P$. The support projection of $Y Y^\dagger$ is then a nontrivial invariant orthogonal projection of B by Lemma D.3.3: it is nonzero because $Y \neq 0$, and it is not the identity because any nonzero vector in the kernel of P transports through $(X^\dagger)^{-1}$ to the kernel of $Y Y^\dagger$. This contradicts irreducibility of B . □

Theorem D.3.5 (Unital Schwarz data for an irreducible corner). Lean ✓ Stmt *Let B be an irreducible tensor with a positive definite transfer eigenvector, $\mathcal{E}_B(\rho) = r\rho$, where $\rho > 0$ and $r > 0$. Then the rescaled unital gauge $K^v = r^{-1/2} \rho^{-1/2} B^v \rho^{1/2}$ is a unital Kraus family, is an irreducible tensor with an irreducible transfer map, and its adjoint map has a positive definite fixed point. These are the hypotheses of the cyclic decomposition of the peripheral spectrum, [212, Theorem 6.6].*

Proof. ✓ Proof Unitality is Theorem 7.8.2; irreducibility is Theorem D.3.4. The adjoint of a unital family is trace-preserving, so its transfer map is a channel and has a positive semidefinite fixed point by the Cesàro argument; irreducibility of the adjoint map upgrades the fixed point to a positive definite one. □

Lemma D.3.6 (Eigenvalue transport to the unital gauge). Lean ✓ Stmt *In the setting of Theorem D.3.5, every transfer eigenvalue μ of B becomes the transfer eigenvalue μ/r of the rescaled unital gauge: the gauge conjugates the transfer map by the congruence with $\rho^{1/2}$ and divides it by r . In particular, the peripheral eigenvalues $e^{2\pi i q/p}$ described in [11, Section 2.3] arise from the transfer eigenvalues of modulus r .*

Proof. ✓ Proof If $\mathcal{E}_B(Z) = \mu Z$, then $Z' = \rho^{-1/2} Z \rho^{-1/2}$ is nonzero and satisfies $\mathcal{E}_K(Z') = (\mu/r) Z'$ by direct computation. □

D.4 Blocking identities and primitive stability

Transfer-map powers, weighted direct sums, and matrix product vectors commute with blocking, enabling the common-length reduction of Theorem 7.3.1.4.

Blocking by a length p commutes with weighted direct sums up to the p th power of each weight: for nonzero weights (μ_k) and blocks (B_k) ,

$$\left(\bigoplus_k \mu_k B_k \right)^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]}, \quad (\text{D.6})$$

where $\sim$ is equality of MPV families. When two block families are reblocked to a common length $p = m_k e_k$, the p -blocked alphabet is identified with the e_k -fold tensor power of the m_k -blocked alphabet, so each sector reblocked through its own period agrees with the directly p -blocked block.

Theorem D.4.1 (Common blocking period for a block family). [Lean] [✓ Stmt] *Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension MPS tensors, each admitting a blocking period p_k making its transfer map primitive. Then $p = \text{lcm}(p_1, \dots, p_r)$ is a common period: $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k .*

Proof. [✓ Proof] Each p_k divides p , so primitivity of $\mathcal{E}_{B_k^{[p_k]}}$ carries over to $\mathcal{E}_{B_k^{[p]}}$ via the divisibility identity for transfer-map powers. □

Theorem D.4.2 (Common primitive blocking for irreducible left-canonical blocks). [Lean] [✓ Stmt] *Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension, left-canonical, tensor-irreducible MPS tensors. Then there is a single blocking length $p > 0$ such that $\mathcal{E}_{B_k^{[p]}}$ is primitive for every k .*

Proof. [✓ Proof] Apply Theorem 7.9.2.1 to each block, then apply Theorem D.4.1 to the resulting finite family of positive periods. Their least common multiple is positive and makes every blocked transfer map primitive. □

Theorem D.4.3 (Quasi-idempotence under physical blocking). [Lean] [✓ Stmt] *Let A be an MPS tensor and let $c \in \mathbb{C}$. If*

$$\mathcal{E}_A^2 = c\mathcal{E}_A,$$

then, for every integer $L \geq 0$,

$$\mathcal{E}_{A^{[L]}}^2 = c^L \mathcal{E}_{A^{[L]}}.$$

Proof. [✓ Proof] The transfer map of $A^{[L]}$ is $\mathcal{E}_A^L$ (Theorem D.4.8). Therefore

$$(\mathcal{E}_A^L)^2 = (\mathcal{E}_A^2)^L = (c\mathcal{E}_A)^L = c^L \mathcal{E}_A^L,$$

which is the required identity. □

Lemma D.4.4 (Adjoint primitivity equivalence). [Lean] [Lean] [✓ Stmt] *Let E be a linear endomorphism of a finite-dimensional complex inner product space. Then E is primitive if and only if its adjoint $E^\dagger$ is primitive. Equivalently, the eigenvalues λ of E with $|\lambda| = 1$ form the singleton $\{1\}$ if and only if the same holds for $E^\dagger$, because adjunction sends every eigenvalue to its complex conjugate.*

Lemma D.4.5 (Primitivity under blocking by a multiple). [Lean] [✓ Stmt] *Let A have positive bond dimension and let $p_0, p \geq 1$ with $p_0 \mid p$. If $\mathcal{E}_{A^{[p_0]}}$ is primitive, then $\mathcal{E}_{A^{[p]}}$ is primitive.*

Theorem D.4.6 (Left-canonical normalization under blocking). [Lean] [✓ Stmt] *If a finite matrix family satisfies*

$$\sum_i (A^i)^\dagger A^i = \mathbb{1},$$

then every blocked family satisfies the same identity.

Proof. [✓ Proof](#) Expand the blocked matrices as products indexed by words and sum one letter at a time. $\square$

Theorem D.4.7 (Irreducibility from a faithful fixed point). [Lean](#) [✓ Stmt](#)

Let E be a channel with a positive-definite fixed point. If all nonzero positive semidefinite fixed points of E are proportional, then E is irreducible.

Proof. [✓ Proof](#) A nontrivial invariant projection would produce a positive semidefinite fixed point not proportional to the faithful one. $\square$

Theorem D.4.8 (Transfer map of a blocked tensor). [Lean](#) [Lean](#) [✓ Stmt](#) For every tensor A and every L , $\mathcal{E}_{A^{[L]}} = \mathcal{E}_A^L$.

Proof. [✓ Proof](#) A letter of $A^{[L]}$ is a word w of length L with matrix A^w , so $\mathcal{E}_{A^{[L]}}(X) = \sum_{|w|=L} A^w X (A^w)^\dagger$, which is the L -fold iterate of $\mathcal{E}_A$ by induction on L . $\square$

Lemma D.4.9 (Fixed-point criteria for irreducibility after primitive blocking). [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [✓ Stmt](#) Let A be left-canonical, tensor-irreducible, and have primitive transfer map. Then A has primitive fixed-point data (ρ, P_ρ, N) with $\rho > 0$ and $N^n \rightarrow 0$. For every blocking length P , the tensor $A^{[P]}$ is left-canonical, its transfer map is $\mathcal{E}_A^P$, and ρ is fixed by this blocked map. For every $n \geq 1$,

$$\mathcal{E}_A^n = P_\rho + N^n. \quad (\text{D.7})$$

Finally, a channel with a positive-definite fixed point whose positive semidefinite fixed points are all proportional is irreducible as a map; irreducibility of a transfer map implies tensor irreducibility.

Proof. [✓ Proof](#) Peripheral primitivity, tensor irreducibility, and left-canonical normalization first give primitive fixed-point data, and tensor irreducibility makes its positive semidefinite fixed point positive definite. Blocking preserves the left-canonical identity, replaces the transfer map by its positive power (Theorem D.4.8), and therefore preserves the fixed point. The defining complementary-power identity gives (D.7). The channel fixed-point criterion then gives map irreducibility from uniqueness of positive semidefinite fixed points, and the final implication converts map irreducibility to tensor irreducibility. $\square$

Theorem D.4.10 (Blocking preserves irreducibility for primitive irreducible blocks). [Lean](#)

[✓ Stmt](#) Let A be left-canonical, irreducible, primitive, of positive bond dimension. For every $P > 0$, the blocked tensor $A^{[P]}$ is irreducible: there is no orthogonal projection Q with $Q \neq 0$, $Q \neq \mathbb{1}$ such that $(\mathbb{1} - Q)(A^{[P]})^\alpha Q = 0$ for every P -blocked index α .

Proof. [✓ Proof](#) Lemma D.4.9 gives primitive fixed-point data with $\rho > 0$. In particular, if $P_\rho(X) = \text{tr}(X)\rho / \text{tr}(\rho)$ and $N := \mathcal{E}_A - P_\rho$, then

$$\begin{aligned} \mathcal{E}_A &= P_\rho + N, \\ N^n &\rightarrow 0. \end{aligned}$$

The primitive fixed-point theorem supplies this complementary-power convergence.

The blocked transfer map is the corresponding positive power:

$$\mathcal{E}_{A^{[P]}}(X) = \mathcal{E}_A^P(X). \quad (\text{D.8})$$

Hence ρ remains a positive-definite fixed point after blocking. Let $\sigma \geq 0$ be any fixed point of $\mathcal{E}_A^P$, set

$$c = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)},$$

$$\sigma' = \sigma - c\rho.$$

Then $\text{tr}(\sigma') = 0$, so $P_\rho(\sigma') = 0$, while $\mathcal{E}_A^P(\sigma') = \sigma'$. Since the fixed-point projection and the complement annihilate one another, iteration gives, for every $k \geq 1$,

$$\sigma' = \mathcal{E}_A^{Pk}(\sigma') = N^{Pk}(\sigma').$$

But $N^{Pk}(\sigma') \rightarrow 0$ as $k \rightarrow \infty$, hence $\sigma' = 0$. Thus every positive semidefinite fixed point of the blocked map is proportional to ρ .

Blocking preserves the left-canonical identity, so Theorem 2.2.8 makes the blocked transfer map a channel. Its positive-definite fixed point and the uniqueness just proved imply map irreducibility by Lemma D.4.9. The final implication in that lemma then shows that $A^{[P]}$ is tensor-irreducible. $\square$

Theorem D.4.11 (Extra blocking of primitive irreducible sector blocks). [Lean](#) [✓ Stmt](#) *Let A have positive bond dimension and be trace-preserving, primitive, and tensor-irreducible. For every $k > 0$,*

$$\sum_{\alpha} ((A^{[k]})^{\alpha})^{\dagger} (A^{[k]})^{\alpha} = \mathbb{1},$$

$$\sigma_{\partial}(\mathcal{E}_{A^{[k]}}) = \{1\}, \tag{D.9}$$

and $A^{[k]}$ is tensor-irreducible. This k is a later common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.

Proof. [✓ Proof](#) Trace preservation follows from blocking a trace-preserving tensor, primitivity from the transfer-power identity (D.8), and irreducibility from Theorem D.4.10. $\square$

Lemma D.4.12 (Blocking the weighted block sum). [Lean](#) [✓ Stmt](#) *For nonzero weights $(\mu_k)_{k=1}^r$, blocks $(B_k)_{k=1}^r$, and any $p \geq 1$, the blocking identity (D.6) holds, where $\sim$ denotes generation of the same MPV family at every length.*

Theorem D.4.13 (Blocking a weighted block-sum presentation). [Lean](#) [✓ Stmt](#) *If A generates the same MPV family as $\bigoplus_k \mu_k B_k$ at every length, then for every $p \geq 0$ the blocked tensor $A^{[p]}$ generates the same MPV family as $\bigoplus_k \mu_k^p B_k^{[p]}$ at every length.*

Proof. [✓ Proof](#) For a blocked configuration σ of length N with unblocked configuration σ' of length pN ,

$$V^{(N)}(A^{[p]})_{\sigma} = V^{(pN)}(A)_{\sigma'} = \sum_k \mu_k^{pN} V^{(pN)}(B_k)_{\sigma'} = \sum_k (\mu_k^p)^N V^{(N)}(B_k^{[p]})_{\sigma}.$$

$\square$

Lemma D.4.14 (Positive-length equality survives blocking). [Lean](#) [✓ Stmt](#) *If two tensors have the same MPV coefficients at every positive length, then so do their blocked tensors after any positive blocking length.*

Theorem D.4.15 (Eventual nonzero proportionality survives blocking). [Lean](#) [✓ Stmt](#) *If two tensors have eventually nonzero proportional MPV coefficients, then so do their blocked tensors after any positive blocking length.*

Proof. [✓ Proof](#) If proportionality holds from length N_0 onward, a blocked chain of length $N \geq N_0$ expands to an original chain of length $Np \geq N_0$. Use the original proportionality scalar at length Np . $\square$

Lemma D.4.16 (Positive-length blocking of a weighted block sum). [Lean](#) [✓ Stmt](#) *If A and $\bigoplus_k \mu_k B_k$ have the same MPV coefficients at every positive length, then after any positive blocking length p , $A^{[p]}$ and $\bigoplus_k \mu_k^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma D.4.17 (Positive-length equality of powered block sums). [Lean](#) [✓ Stmt](#) *If $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ have the same MPV coefficients at every positive length, then after any positive common blocking length p , $\bigoplus_j (\mu_j^A)^p A_j^{[p]}$ and $\bigoplus_k (\mu_k^B)^p B_k^{[p]}$ have the same MPV coefficients at every positive length.*

Lemma D.4.18 (Nonzero weights survive blocking). [Lean](#) [✓ Stmt](#) *If $\mu \in \mathbb{C}$ is nonzero and $p \geq 1$, then $\mu^p \neq 0$; blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.*

Lemma D.4.19 (Iterated blocking at the transfer-map level). [Lean](#) [✓ Stmt](#) *For any MPS tensor A and $m, n \geq 0$,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}. \quad (\text{D.10})$$

D.5 Support compression and cyclic-sector families

Support isometries project a blocked tensor onto its cyclic corners, embedding multiple cyclic-sector families into a single common blocked alphabet.

Theorem D.5.1 (Supported corner compression with a support isometry). [Lean](#) [✓ Stmt](#)

Let A be a tensor normalized on the support of an orthogonal projection P , so that $PA^iP = A^i$ and $\sum_i (A^i)^\dagger A^i = P$. Then there are a corner tensor C , a linear isomorphism $\varphi : M_{\dim C}(\mathbb{C}) \xrightarrow{\sim} PM_D(\mathbb{C})P$, and a rectangular isometry $V : \mathbb{C}^{\dim C} \rightarrow \mathbb{C}^D$ such that

$$\begin{aligned} V^\dagger V &= \mathbb{1}, \\ VV^\dagger &= P, \\ \varphi(X) &= VXV^\dagger. \end{aligned} \quad (\text{D.11})$$

Moreover, $\varphi(C^i) = A^i$ for every physical letter, and for every word $\sigma = (\sigma_1, \dots, \sigma_N)$,

$$V^{(N)}(C)_\sigma = \text{tr}(PA^{\sigma_1} \dots A^{\sigma_N}). \quad (\text{D.12})$$

Proof. [✓ Proof](#)

Apply Theorem [182, “Compression of a supported finite Kraus family”] to the family $(A^i)_i$. It gives C , φ , and V satisfying (D.11) and $\varphi(C^i) = A^i$. It remains to identify the coefficients of the associated words. For each word $\sigma = (\sigma_1, \dots, \sigma_N)$, cyclicity of the trace and $VV^\dagger = P$ give

$$V^{(N)}(C)_\sigma = \text{tr}(C^{\sigma_1} \dots C^{\sigma_N}) = \text{tr}(V^\dagger A^{\sigma_1} \dots A^{\sigma_N} V) = \text{tr}(PA^{\sigma_1} \dots A^{\sigma_N}),$$

which proves (D.12). $\square$

Theorem D.5.2 (Commuting projections give sector isometries). [Lean](#) [✓ Stmt](#) Let a left-canonical tensor A commute with a family of orthogonal projections (P_k) summing to $\mathbb{1}$. Then A decomposes into compressed sector tensors (C_k) , with a unit-weight direct-sum MPV decomposition, corner isomorphisms $\varphi_k : M_{\dim C_k}(\mathbb{C}) \rightarrow P_k M_D(\mathbb{C}) P_k$, and rectangular isometries V_k satisfying

$$\begin{aligned} V_k^\dagger V_k &= \mathbb{1}, \\ V_k V_k^\dagger &= P_k, \\ \varphi_k(X) &= V_k X V_k^\dagger. \end{aligned} \tag{D.13}$$

The sector letters obey $\varphi_k(C_k^i) = P_k A^i P_k$.

Proof. [✓ Proof](#) Apply Theorem D.5.1 to the compression of A on each sector P_k . Orthogonality and $\sum_k P_k = \mathbb{1}$ give the direct-sum decomposition. For every word $\sigma = (\sigma_1, \dots, \sigma_N)$, the partition of unity and (D.12) give

$$V^{(N)}(A)_\sigma = \text{tr} \left(\left(\sum_k P_k \right) A^{\sigma_1} \dots A^{\sigma_N} \right) = \sum_k \text{tr}(P_k A^{\sigma_1} \dots A^{\sigma_N}) = \sum_k V^{(N)}(C_k)_\sigma.$$

This is the unit-weight matrix-product-vector identity. $\square$

Theorem D.5.3 (Adjoint-fixed projections give sector isometries). [Lean](#) [✓ Stmt](#) Let A be left-canonical and let (P_k) be orthogonal projections summing to $\mathbb{1}$ and fixed by the adjoint transfer map. Then the same sector decomposition and support isometries of Theorem D.5.2 exist. In particular, the adjoint fixed-point condition $\mathcal{E}_A^\dagger(P_k) = P_k$ gives $P_k A^i = A^i P_k$, which gives the corner-letter identity $\varphi_k(C_k^i) = P_k A^i P_k$.

Proof. [✓ Proof](#) The fixed-point identity $\mathcal{E}_A^\dagger(P_k) = P_k$, together with left-canonicity $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, places P_k in the multiplicative domain of $\mathcal{E}_A^\dagger$, forcing $P_k A^i = A^i P_k$ for every letter A^i . The result then follows from Theorem D.5.2. $\square$

Theorem D.5.4 (Cyclic sectors after removing zero blocks). [Lean](#) [✓ Stmt](#) If A and B generate the same MPV family at every positive length, then after removing the all-zero blocks and applying the trace-preserving gauge there are weighted nonzero-block tensors $\bigoplus_j \mu_j^A A_j$ and $\bigoplus_k \mu_k^B B_k$ such that, for every $N \geq 1$ and σ ,

$$V^{(N)}(A)_\sigma = V^{(N)} \left(\bigoplus_j \mu_j^A A_j \right)_\sigma, \tag{D.14}$$

$$V^{(N)}(B)_\sigma = V^{(N)} \left(\bigoplus_k \mu_k^B B_k \right)_\sigma. \tag{D.15}$$

The two nonzero parts agree:

$$V^{(N)} \left(\bigoplus_j \mu_j^A A_j \right)_\sigma = V^{(N)} \left(\bigoplus_k \mu_k^B B_k \right)_\sigma. \tag{D.16}$$

Each nonzero block has primitive irreducible cyclic sectors: for each j , there is $m_j \geq 1$ with a unit-weight sector decomposition $A_j^{[m_j]} \sim \bigoplus_{u=1}^{m_j} C_{j,u}^A$, and similarly for B_k .

Proof. ✓ Proof Apply Theorem 7.2.1.1 separately to A and B . Its positive-length conclusion gives (D.14) and (D.15). Chaining these identities through $\mathcal{V}^+(A) = \mathcal{V}^+(B)$ gives (D.16). Finally apply Theorem 7.3.1.3 to every trace-preserving irreducible nonzero-weight block. $\square$

Lemma D.5.5 (Common reblocking of cyclic-sector families). Lean ✓ Stmt *Given a finite family $(B_k)_{k=1}^r$, each with primitive irreducible cyclic sectors, there is a common blocking length p such that each $B_k^{[p]}$ admits a unit-weight direct-sum decomposition over a single common blocked alphabet $d^{[p]}$ into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.*

Lemma D.5.6 (Common reblocking at a prescribed multiple). Lean ✓ Stmt *With hypotheses as in Lemma D.5.5, and given any common multiple p of the per-block period-removal lengths (m_k) , the common reblocking can be performed at the prescribed length p .*

Lemma D.5.7 (Structural properties of the common reblocked sectors). Lean ✓ Stmt *Each sector tensor in the common reblocked cyclic-sector family of Lemma D.5.5 is trace-preserving, $\sum_i (B^i)^\dagger B^i = \mathbb{1}$, has primitive transfer map, $\sigma_\partial(\mathcal{E}_B) = \{1\}$, is tensor-irreducible, and has positive bond dimension.*

Lemma D.5.8 (Common-alphabet flattening). Lean ✓ Stmt *With $p = m_k e_k$ for each k , the iterated blocked tensor $(B_k^{[m_k]})^{[e_k]}$ agrees with $B_k^{[p]}$ in MPV family, after the canonical identification of the p -blocked alphabet $d^{[p]}$ with the e_k -fold tensor power of the m_k -blocked alphabet $d^{[m_k]}$.*

Lemma D.5.9 (Direct blocking in the common alphabet). Lean ✓ Stmt *For every original block B_k , direct blocking at the common length $p = m_k e_k$ agrees, as an MPV family, with the same block transported to the common blocked alphabet. Write $\widetilde{B}_k$ for this common-alphabet image. Then*

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}(\widetilde{B}_k)_\sigma.$$

Lemma D.5.10 (Reindexed nonzero-part identity). Lean ✓ Stmt *The powered nonzero-part decomposition $\bigoplus_k \mu_k^p B_k^{[p]}$ carries over under the common-alphabet identification of Lemma D.5.8: the common-alphabet sectors have the same nonzero-part decomposition as the direct p -blocked tensors.*

D.6 Prepared normal-form consequences

For families of irreducible, trace-preserving, primitive blocks with non-zero weights and positive bond dimensions, the asymptotic overlap condition $O_{A_k A_k}(N) \rightarrow 1$ is a derived consequence, and ordering the blocks by non-increasing weight modulus produces a normal canonical form.

Theorem D.6.1 (Self-overlap is derived in normal canonical form). Lean ✓ Stmt *If (μ_k, A_k) satisfies the normal canonical form conditions, then $O_{A_k A_k}(N) \rightarrow 1$ for every block k .*

Proof. ✓ Proof Fix a block k . Irreducibility and the TP normalization give a nonzero positive semidefinite fixed point ρ of $\mathcal{E}_{A_k}$ with fixed-point projector P ; primitivity gives the complementary gap $\rho_{\text{spec}}(\mathcal{E}_{A_k} - P) < 1$ (the corresponding theorem in the companion quantum-channel volume [182, “Complementary transfer-map gap for primitive maps”]). Theorem 5.10.2 then yields $O_{A_k A_k}(N) \rightarrow 1$. $\square$

Theorem D.6.2 (Reordering prepared primitive blocks into normal canonical form). [Lean](#)
 $\checkmark$ *Stmt* Suppose A generates the same MPV family as $\bigoplus_k \mu_k A_k$, and that each A_k is irreducible, satisfies $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$, and has primitive transfer map, all $\mu_k \neq 0$, and all bond dimensions are positive. Then there exist a blocking length $p > 0$, weights (ν_k) , and blocked tensors (B_k) such that $A^{[p]}$ generates the same MPV family as $\bigoplus_k \nu_k B_k$, and (ν_k, B_k) satisfies the normal canonical form conditions. The weight moduli need not be distinct; equal-modulus blocks are allowed.

Proof. $\checkmark$ *Proof* The given blocks are already primitive, so take $p = 1$. Reorder the blocks by non-increasing weight modulus, with ties allowed, and set $\nu_k := \mu_k$ and $B_k := A_k$ in that order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are preserved, so the reordered family satisfies Definition 7.9.2 and generates the same MPV family as A . $\square$

Remark D.6.3 (Scope of Theorem D.6.2). Theorem D.6.2 reorganizes an already reduced family of primitive, irreducible blocks into normal canonical form by sorting their weight moduli. The reduction from an arbitrary MPS tensor, including the structural bond-dimension bound, is established by Theorem 7.11.1.2 in Section 7.11.1 following [11, Section 2.3].

Appendix E

Basis of Normal Tensors: Supporting Results

The classification of matrix product states in a basis of normal tensors (Chapter 8) relies on algebraic properties of normal tensors, Gram-matrix criteria, change-of-basis relations, sector-blocking identities, and direct-sum block separation.

E.1 Elementary normal-tensor and Gram-matrix support

Throughout this section, $V^{(N)}(A)$ denotes the length- N matrix product vector of a tensor A , and $O_{AB}(N)$ denotes the corresponding mixed overlap. Asymptotic Gram-matrix criteria govern the linear independence of matrix product vector families in the BNT characterization and matching arguments.

Lemma E.1.1 (Positive bond dimensions in a coefficient-form BNT). [Lean](#) [✓ Stmt](#) *Every block in a coefficient-form basis of normal tensors has positive bond dimension.*

Proof. [✓ Proof](#) Eventual linear independence makes the matrix product vector of each block nonzero at some positive length. A tensor with zero bond dimension has zero matrix product vector at every length, which is a contradiction. $\square$

Lemma E.1.2 (Bond dimension one implies irreducibility). [Lean](#) [✓ Stmt](#) *Every tensor with a one-dimensional bond space is irreducible.*

Proof. [✓ Proof](#) The only orthogonal projections on a one-dimensional complex vector space are zero and the identity. $\square$

Lemma E.1.3 (Normality in bond dimension one). [Lean](#) [✓ Stmt](#) *A tensor with one-dimensional bond space and identity transfer map is a normal tensor.*

Proof. [✓ Proof](#) Irreducibility follows from Lemma E.1.2. The identity transfer map has spectral radius one, and its only eigenvalue is one. $\square$

Lemma E.1.4 (Perron gauges preserve primitivity). [Lean](#) [✓ Stmt](#) *Let σ be positive definite. The transfer map of the Perron gauge determined by σ is primitive if and only if the original transfer map is primitive.*

Proof. ✓ Proof The two transfer maps are similar through the invertible congruence $X \mapsto \sigma^{-1/2} X \sigma^{-1/2}$. Similar linear maps have the same spectrum, so the condition that 1 is the only eigenvalue on the unit circle is unchanged. $\square$

Lemma E.1.5 (Preservation of an algebraic BNT under positive-length MPV equality). Lean ✓ Stmt *Let A and A' have the same matrix product vectors at every positive length. If $(A_j)_{j=1}^g$ is a basis of normal tensors for A , then the same family is a basis of normal tensors for A' .*

Proof. ✓ Proof Substitute the positive-length MPV equality into the spanning identity for A . Normality and eventual linear independence depend only on the family $(A_j)_{j=1}^g$. $\square$

Corollary E.1.6 (Positive-length geometric proportionality phase). Lean ✓ Stmt *Let A and B be normal tensors of positive, possibly different, bond dimensions. Suppose that, for every positive N , their matrix product vectors are proportional by a nonzero scalar. Then there is $a \in \mathbb{C}$, with $|a| = 1$, such that $|V^{(N)}(A)\rangle = a^N |V^{(N)}(B)\rangle$ for every positive N . This is the positive-length one-normal-block consequence of [11, Corollary A.3 and Theorem 2.10].*

Proof. ✓ Proof Positive-length nonzero proportionality implies eventual nonzero proportionality. Apply Theorem 8.2.12. $\square$

Lemma E.1.7 (Eventual linear independence from finite-index overlap orthonormality). Lean ✓ Stmt *Let I be a finite index set and let $(A_i)_{i \in I}$ be a finite family of tensors such that $O_{A_i A_i}(N) \rightarrow 1$ for each i , and $O_{A_i A_j}(N) \rightarrow 0$ whenever $i \neq j$. Then the MPV states $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$ are linearly independent for all sufficiently large N .*

Proof. ✓ Proof The Gram matrix of the family $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$ converges entrywise to the identity matrix. Hence for all sufficiently large N it is invertible, so the MPV states are linearly independent. $\square$

Stating the threshold explicitly yields an existential formulation.

Theorem E.1.8 (Existential BNT independence from overlap limits). Lean ✓ Stmt *If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.*

Proof. ✓ Proof This is Lemma E.1.7, restated with an explicit threshold N_0 . $\square$

Lemma E.1.9 (Eventual linear independence for two families). Lean ✓ Stmt *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap $O_{A_j B_k}(N)$ tends to 0. Then the combined MPV family*

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large N .

Proof. ✓ Proof Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks. $\square$

Eventual linear independence converts equality of linear combinations of MPV states into equality of their scalar coefficients, supplying the identification used in Chapter 9.

Lemma E.1.10 (Eventual coefficient extraction from linear independence). [Lean](#) [✓ Stmt](#) *If a finite family of vectors is linearly independent for all sufficiently large N , and two finite linear combinations of that family are equal for all sufficiently large N , then the corresponding coefficients are equal for all sufficiently large N .*

Proof. [✓ Proof](#) Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length. $\square$

When two BNT families must be compared against each other, the same Gram-matrix criterion applies to their disjoint union, provided the mixed overlaps also decay. This is the form used in the exact sector matching of Section 8.8.

E.2 Change of basis, power sums, and sector bookkeeping

Linear-algebraic change of basis, geometric power-sum extrapolation, and sector bookkeeping provide the tools for comparing alternative BNT decompositions. A sector has copy weights $\mu_{j,q} \neq 0$, coefficient $c_{N,j} = \sum_q \mu_{j,q}^N$, representative bond dimension D_j , and total bond dimension $D_{\text{tot}} = \sum_j r_j D_j$.

Lemma E.2.1 (Invertible change of basis). [Lean](#) [✓ Stmt](#) *Let $v_1, \dots, v_g$ and $w_1, \dots, w_g$ be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix $U \in \text{GL}_g(\mathbb{C})$ such that $w_j = \sum_{i=1}^g U_{ij} v_i$ for every j .*

Proof. [✓ Proof](#) Because the two families span the same subspace, each w_j has a unique expansion in the basis $\{v_i\}_{i=1}^g$; these coefficients form the columns of U . If $Ua = 0$, then the corresponding linear relation among the w_j is trivial because the w_j are linearly independent. Thus U is injective, hence invertible. $\square$

Lemma E.2.2 (Eventual change of basis from equal spans). [Lean](#) [✓ Stmt](#) *If two finite families of blocks have asymptotically orthonormal overlaps within each family, and if their MPV spans agree for all sufficiently large system sizes N , then for all sufficiently large N there exists an invertible matrix U_N expressing the MPV states of one family as linear combinations of the MPV states of the other.*

Proof. [✓ Proof](#) By Lemma E.1.7, both families are linearly independent for all sufficiently large N . Hence, for all sufficiently large N , both families are linearly independent and the assumed span equality holds. Lemma E.2.1 applies at each such N . $\square$

Lemma E.2.3 (Eventual vanishing of a finite geometric sum). [Lean](#) [✓ Stmt](#) *Let $w_1, \dots, w_n$ be nonzero complex numbers. If $\sum_i c_i w_i^N = 0$ for every sufficiently large N , then the same equality holds for every non-negative exponent.*

Proof. [✓ Proof](#) Induct on the number of terms. Subtracting w_1 times the equality at exponent N from the equality at exponent $N + 1$ gives $\sum_{i=2}^n c_i (w_i - w_1) w_i^N = 0$. The induction hypothesis yields this equality at every exponent, so the original sum $S_N = \sum_i c_i w_i^N$ satisfies $S_{N+1} = w_1 S_N$ for every N . Since S_N eventually vanishes and $w_1 \neq 0$, it follows that $S_0 = 0$ and hence $S_N = 0$ for every N . $\square$

Lemma E.2.4 (Eventual power-sum equality holds at every exponent). [Lean](#) [✓ Stmt](#) *Let $a_1, \dots, a_m$ and $b_1, \dots, b_n$ be nonzero complex numbers. If their power sums agree for every sufficiently large exponent, then $\sum_i a_i^k = \sum_j b_j^k$ for every non-negative exponent k .*

Proof. ✓ Proof Set $w_l = a_l$ for $l \leq m$ and $w_{m+l} = b_l$ for $l \leq n$, with coefficients $c_l = 1$ for $l \leq m$ and $c_l = -1$ for $l > m$. Then $\sum_{l=1}^{m+n} c_l w_l^k = \sum_i a_i^k - \sum_j b_j^k$. The hypothesis makes this expression zero for every sufficiently large k . Lemma E.2.3 makes it zero for every k , which is the required power-sum equality. $\square$

Theorem E.2.5 (Equal power sums imply equal multisets). Lean ✓ Stmt Let $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.

Proof. ✓ Proof Construct diagonal matrices $\text{diag}(\alpha)$ and $\text{diag}(\beta)$. The power-sum hypothesis gives $\text{tr}(\text{diag}(\alpha)^k) = \text{tr}(\text{diag}(\beta)^k)$. By the corresponding theorem in the companion quantum-channel volume [182, “Newton–Girard trace recursion”], the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. $\square$

Theorem E.2.6 (Representative dimensions are bounded by the total bond dimension). Lean Lean ✓ Stmt Let the representative A_j have bond dimension D_j , and let D be the bond dimension of the flattened sector tensor. Then $D_j \leq D$. If every D_j is positive and there are g representatives, then also $g \leq D$.

Proof. ✓ Proof Since $r_j > 0$, choose one copy of A_j . Its contribution D_j is one non-negative summand in the total dimension of the flattened copy family, so $D_j \leq D$.

For the second assertion, every representative has multiplicity $r_j \geq 1$, and each copy has positive dimension $D_j > 0$. Therefore

$$g = \sum_j 1 \leq \sum_j r_j \leq \sum_j r_j D_j = D.$$

$\square$

Theorem E.2.7 (Blocked sector coefficients and assembled tensor). Lean Lean ✓ Stmt The coefficient of the blocked decomposition satisfies

$$c_{P^{[p]},j}^{(N)} = \sum_q (\mu_{j,q}^p)^N = c_{P,j}^{(Np)}. \quad (\text{E.1})$$

Moreover, blocking the tensor assembled from P gives the same MPV family as the tensor assembled from the blocked sector decomposition.

Proof. ✓ Proof The coefficient identity in (E.1) is $(\mu^p)^N = \mu^{pN}$. The tensor identity is Theorem D.4.12 applied to the flattened copy family. $\square$

Lemma E.2.8 (Existence of an MPV phase class). Lean ✓ Stmt A nonempty finite family of blocks has at least one MPV phase class.

Proof. ✓ Proof Choose one original block index. Its occurrence in the phase-class enumeration supplies a phase class containing it. $\square$

Lemma E.2.9 (Regrouping a matrix sum by MPV phase classes). Lean ✓ Stmt Let F_k be a finite family of square matrices indexed by the original blocks. If $k = (j, q)$ enumerates the copies in the MPV phase classes, then $\sum_j \sum_q F_{j,q} = \sum_k F_k$.

Proof. ✓ Proof Evaluate both sides at a pair of matrix indices and apply the bijective enumeration of the finite phase-class quotient. $\square$

Definition E.2.10 (Single-copy sector decomposition). [Lean](#) [✓ Stmt](#) Given nonzero weights $(\mu_k)_{k=1}^r$ and blocks $(B_k)_{k=1}^r$, the *single-copy sector decomposition* has r sectors, one basis block B_k per sector, multiplicity 1, and sector weight μ_k . As a finite sum,

$$\sum_{k=1}^r \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)}\left(\bigoplus_k \mu_k B_k\right)_\sigma. \quad (\text{E.2})$$

Lemma E.2.11 (Single-copy sector decomposition preserves the represented family). [Lean](#) [✓ Stmt](#) The single-copy sector decomposition of $(\mu_k, B_k)_k$ represents the same MPS family as $\bigoplus_k \mu_k B_k$.

Proof. [✓ Proof](#) Expanding the sector decomposition gives the left-hand side of (E.2), which is the finite block-sum expression for $\bigoplus_k \mu_k B_k$. $\square$

Lemma E.2.12 (Sector-decomposition coefficient is not eventually zero). [Lean](#) [Lean](#) [✓ Stmt](#) For any sector decomposition P and any sector j , the coefficient $c_{P,j}^{(N)} = \sum_q \mu_{j,q}^N$ is not eventually zero in N . More generally, for every positive integer p , the subsequence $c_{P,j}^{(pN)}$ is not eventually zero.

Proof. [✓ Proof](#) If the coefficient vanished for every $N \geq M$, the finite geometric-sum extrapolation would make it vanish at every non-negative exponent. However, at exponent zero, $c_{P,j}^{(0)} = \sum_q \mu_{j,q}^0 = r_j > 0$, which is impossible. For the subsequence, apply the same argument to the nonzero bases $\mu_{j,q}^p$. $\square$

E.2.1 Proportional matching and finite bijection extraction

The equal-MPV matching theorems in Chapter 8 specialize an eventual proportionality argument; matching in both directions produces a bijection between the sector sets.

Theorem E.2.1.1 (Exact sector matching under eventual proportionality). [Lean](#) [✓ Stmt](#) Let P and Q be BNT canonical forms whose total MPV families are eventually nonzero and proportional. For every sector j of P , there is a sector k of Q with equal bond dimension, gauge-phase-equivalent basis tensor after the dimension identification, and non-decaying cross-overlap.

Proof. [✓ Proof](#) Suppose the chosen sector decays against every sector of Q , and separate all such P -sectors from the remaining ones. For each remaining sector, choose a Q -sector with non-decaying overlap. Both blocks are normal by the canonical-form hypotheses. Theorem 8.2.9 rules out its decay alternative and supplies equal bond dimensions and a gauge-phase relation with a unit phase. Gauge-phase covariance therefore expresses that P -sector MPV as a geometric scalar multiple of the chosen Q -sector MPV.

Substitute these relations into the eventual proportionality identity. The combined-family Gram criterion makes the decaying residue together with the Q family eventually linearly independent. Exact coefficient extraction then makes the chosen sector coefficient eventually zero, contradicting its power-sum nonvanishing. Thus a non-decaying partner exists. A second application of Theorem 8.2.9 rules out decay and gives the required dimension equality and gauge-phase equivalence; no modulus-one overlap limit is assumed. $\square$

Theorem E.2.1.2 (Full-basis matching under eventual proportionality). [Lean](#) [✓ Stmt](#) Under the same hypotheses, every sector of Q has an equal-dimensional, gauge-phase-equivalent partner in P with non-decaying cross-overlap.

Proof. ✓ **Proof** Reverse the eventual proportionality and apply Theorem E.2.1.1 to each sector of Q ; symmetry of the overlap converts the resulting witness to the stated orientation. $\square$

Lemma E.2.1.3 (Bijection from matches in both directions). Lean ✓ **Stmt** Suppose every sector of Q has an equal-dimensional, gauge-phase-equivalent partner in P with non-decaying overlap, and the converse holds from P to Q . Then the forward partners may be chosen as a bijection between the two finite sector sets, preserving all three properties.

Proof. ✓ **Proof** Basis separation makes each partner map injective: two sectors with the same partner would be gauge-phase equivalent to one another. The injections in both directions force equal finite cardinalities, so the forward map is bijective. $\square$

E.3 Direct-sum separation and prepared-family bookkeeping

Following Pérez-García et al. [193], disjoint local MPS spaces yield pair-trace selectors whose products isolate individual BNT blocks. Grouping phase-equivalent blocks into sectors preserves total bond dimension and allows projective weight normalization.

Lemma E.3.1 (Three-block separation of distinct BNT representatives). Lean Lean Lean ✓ **Stmt** Let $A^1, \dots, A^r$ be irreducible left-canonical representatives with normalized self-overlaps, and suppose that distinct same-dimensional representatives are not gauge-phase equivalent. Fix $L_0 > 0$ such that every block is injective at lengths L_0 , $L_0 + 1$, and $3(L_0 + 1)$. Then each ordered pair $k \neq j$ is trace-separated at the common length $S = 3(L_0 + 1)$: the map

$$(c_\omega)_{|\omega|=S} \mapsto \left(\sum_{|\omega|=S} c_\omega A_\omega^k, \sum_{|\omega|=S} c_\omega A_\omega^j \right)$$

is onto $M_{D_k}(\mathbb{C}) \oplus M_{D_j}(\mathbb{C})$. In particular, one may prescribe an arbitrary matrix on either block while making the same word combination vanish on the other block.

Proof. ✓ **Proof** Put $L = L_0 + 1$ and $S = 3L$. For a block A , write

$$\Gamma_n^A(X) = \sum_{|w|=n} \text{tr}(XA_w)|w\rangle, \quad \mathcal{G}_n(A) = \text{im } \Gamma_n^A.$$

Block injectivity at L and S makes both boundary maps injective; in particular,

$$\dim \mathcal{G}_L(A^q) = D_q^2 \quad (1 \leq q \leq r). \quad (\text{E.3})$$

Fix $k \neq j$ and first order the pair so that $D_j \leq D_k$. Suppose that a nonzero vector belongs to $\mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j)$. There are boundary matrices X and Y such that

$$0 \neq \psi = \Gamma_S^{A^k}(X) = \Gamma_S^{A^j}(Y). \quad (\text{E.4})$$

Injectivity at S gives $X \neq 0$ and $Y \neq 0$. Split every length- S word as uvw with $|u| = |v| = |w| = L$. Relation (E.4) gives

$$\text{tr}(XA_u^k A_v^k A_w^k) = \text{tr}(YA_u^j A_v^j A_w^j). \quad (\text{E.5})$$

Because the length- L words of A^k span $M_{D_k}(\mathbb{C})$ and $X \neq 0$, the two-sided ideal generated by X is all of $M_{D_k}(\mathbb{C})$:

$$\text{span}\{A_u^k X A_v^k : |u| = |v| = L\} = M_{D_k}(\mathbb{C}). \quad (\text{E.6})$$

Indeed, expand arbitrary left and right matrix units in length- L words; a nonzero entry of X then produces every matrix unit. Thus, for each $Z \in M_{D_k}(\mathbb{C})$, choose coefficients a_{uv} with $Z = \sum_{u,v} a_{uv} A_u^k X A_v^k$ and set $R(Z) = \sum_{u,v} a_{uv} A_u^j Y A_v^j$. Cyclicity of the trace and (E.5), applied to the word vwu , yield

$$\begin{aligned} \text{tr}(Z A_w^k) &= \sum_{u,v} a_{uv} \text{tr}(X A_v^k A_w^k A_u^k) \\ &= \sum_{u,v} a_{uv} \text{tr}(Y A_v^j A_w^j A_u^j) = \text{tr}(R(Z) A_w^j). \end{aligned}$$

Hence $\Gamma_L^{A^k}(Z) = \Gamma_L^{A^j}(R(Z))$ for every Z , and therefore

$$\mathcal{G}_L(A^k) \subseteq \mathcal{G}_L(A^j). \quad (\text{E.7})$$

Equations (E.3) and (E.7) give

$$D_k^2 = \dim \mathcal{G}_L(A^k) \leq \dim \mathcal{G}_L(A^j) = D_j^2.$$

Together with $D_j \leq D_k$, this forces $D_j = D_k$ and equality in (E.7). This proves the unequal-dimension branch immediately: if $D_j < D_k$, a nonzero intersection is impossible. Reversing the initial ordering treats $D_k < D_j$ in the same way.

It remains to exclude the intersection when $D_j = D_k$. Gauge-phase separation and Theorem 8.4.3 give, beyond any prescribed threshold, a chain length $N > L$ for which the two periodic MPVs are not proportional. On the other hand, equality $\mathcal{G}_L(A^k) = \mathcal{G}_L(A^j)$ gives identical translated length- L constraints on the N -site ring. The parent-Hamiltonian uniqueness mechanism identifies the corresponding chain ground spaces with the one-dimensional MPV spans:

$$\mathcal{G}_{N,L}(A^k) = \mathbb{C} V^{(N)}(A^k), \quad \mathcal{G}_{N,L}(A^j) = \mathbb{C} V^{(N)}(A^j).$$

Equality of the local spaces therefore forces these two MPV lines to be equal, contradicting the choice of N . Consequently, for every distinct pair, including the unequal-dimensional pairs,

$$\mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j) = \{0\}. \quad (\text{E.8})$$

Finally, let Φ_{kj} be the displayed length- S word-evaluation map from coefficient families to $M_{D_k}(\mathbb{C}) \oplus M_{D_j}(\mathbb{C})$. Pair the target by

$$\langle (X, Y), (M, N) \rangle = \text{tr}(XM) + \text{tr}(YN).$$

A pair (X, Y) annihilates $\text{im}(\Phi_{kj})$ precisely when

$$\text{tr}(X A_w^k) + \text{tr}(Y A_w^j) = 0 \quad (|w| = S),$$

or equivalently $\Gamma_S^{A^k}(X) = \Gamma_S^{A^j}(-Y)$. There is therefore a linear map

$$\Theta : \text{im}(\Phi_{kj})^\perp \longrightarrow \mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j), \quad \Theta(X, Y) = \Gamma_S^{A^k}(X) = \Gamma_S^{A^j}(-Y).$$

It is injective because both boundary maps are injective. Conversely, if ψ lies in the intersection, there are unique X and Z with $\Gamma_S^{A^k}(X) = \psi = \Gamma_S^{A^j}(Z)$; the pair $(X, -Z)$ annihilates $\text{im}(\Phi_{kj})$ and maps to ψ . This construction is the inverse of Θ , so the two spaces are isomorphic:

$$\text{im}(\Phi_{kj})^\perp \cong \mathcal{G}_S(A^k) \cap \mathcal{G}_S(A^j). \quad (\text{E.9})$$

The right-hand side vanishes by (E.8), and hence so does the annihilator. The trace pairing is nondegenerate and all spaces are finite-dimensional, so $\text{im}(\Phi_{kj})^\perp = \{0\}$ implies that Φ_{kj} is surjective. Surjectivity first gives an arbitrary value on one block and zero on an anchor block; the selector construction then chooses $(1, 0)$ for the remaining pairs, and products of these homogeneous selectors give the simultaneous word-tuple span in Lemma 8.3.3. $\square$

Theorem E.3.2 (Common injective blocking for primitive irreducible families). [Lean](#) [✓ Stmt](#) *Let $(B_k)_{k=1}^r$ be a finite family of positive-bond-dimension MPS tensors, each trace-preserving, primitive, and tensor-irreducible. Then there is a single $L \geq 1$ such that each $B_k^{[L]}$ is one-site injective, and this blocking preserves trace preservation, primitivity, and irreducibility for every block.*

Proof. [✓ Proof](#) For each block, Theorem 6.9.1 gives a positive length at which it becomes one-site injective; the product of these lengths is a common length. Injectivity persists under positive further blocking, and Theorem D.4.11 preserves the other hypotheses. $\square$

Lemma E.3.3 (Total dimension after grouping by BNT representatives). [Lean](#) [✓ Stmt](#) *Suppose CF summands are grouped by BNT representative, every original copy weight is nonzero, and every summand assigned to a representative has that representative's bond dimension. Then the total bond dimension of the resulting BNT sector decomposition is the sum of the original block dimensions.*

Proof. [✓ Proof](#) The collapsed decomposition counts one representative dimension for each assigned copy. Let $e(j, q)$ denote the original CF summand assigned to the q th copy of representative P_j . Since $(j, q) \mapsto e(j, q)$ reindexes the original summands and each assigned summand has the representative's bond dimension,

$$\sum_{j=1}^g \sum_{q=1}^{r_j} \dim(P_j) = \sum_{j=1}^g \sum_{q=1}^{r_j} D_{e(j,q)} = \sum_{k=1}^r D_k.$$

This is the equality between the collapsed total dimension and the original direct-sum dimension. $\square$

Lemma E.3.4 (Prepared phase-equivalent NTs have the same bond dimension). [Lean](#) [✓ Stmt](#) *Let X and Y be trace-preserving, primitive, irreducible NTs of positive bond dimension. If their MPV families differ by a nonzero scalar power, then X and Y have the same bond dimension.*

Proof. [✓ Proof](#) Choose $\zeta \neq 0$ with $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$ for every positive length N . The normalized self-overlaps give $|O_{XX}(N)| \rightarrow 1$, and the phase relation gives

$$\begin{aligned} O_{YX}(N) &= \zeta^N O_{XX}(N), \\ |O_{YX}(N)| &= |\zeta|^N |O_{XX}(N)|. \end{aligned}$$

Since the same normalization forces $|\zeta| = 1$, this gives $|O_{YX}(N)| \rightarrow 1$. If the two bond dimensions differed, the rectangular overlap-decay theorem would force $O_{YX}(N) \rightarrow 0$, contradicting the preceding limit. $\square$

Lemma E.3.5 (Prepared BNT representative classes preserve bond dimension). [Lean](#) [✓ Stmt](#) *In a prepared family of positive-bond-dimension trace-preserving primitive irreducible NTs, every CF summand assigned to a BNT representative has that representative's bond dimension.*

Proof. [✓ Proof](#) Apply Lemma E.3.4 to each representative and each assigned summand. $\square$

Lemma E.3.6 (Prepared collapsed BNT total dimension). [Lean](#) [✓ Stmt](#) *For prepared trace-preserving primitive irreducible NTs of positive bond dimension with all copy weights nonzero, grouping CF summands by BNT representative preserves the total bond dimension: the collapsed sector decomposition has total bond dimension equal to the sum of the original block dimensions.*

Proof. [✓ Proof](#) Every assigned summand shares the bond dimension of its representative, by Lemma E.3.5. The displayed regrouping identity in Lemma E.3.3 then identifies the collapsed sum over pairs (j, q) with $\sum_{k=1}^r D_k$, the original direct-sum dimension. $\square$

Theorem E.3.7 (Projective weight normalization). [Lean](#) [✓ Stmt](#) *For nonzero weights $\mu_1, \dots, \mu_r$ with $r \geq 1$ there is a positive real m such that every normalized weight satisfies*

$$\left| \frac{\mu_k}{m} \right| \leq 1, \\ \frac{\mu_k}{m} \neq 0,$$

and at least one normalized weight has unit modulus, $|\mu_{k_0}/m| = 1$. This is the normalization that [11, line 246] chooses for the canonical-form weights.

Proof. [✓ Proof](#) Take $m = \max_k |\mu_k|$; the maximum over a nonempty finite family is attained, positive because every weight is nonzero, an upper bound for every modulus, and equal to the modulus of the attaining weight. $\square$

Lemma E.3.8 (Common scalar rescaling of block weights). [Lean](#) [✓ Stmt](#) *Multiplying every block weight in a block-diagonal MPS tensor by a scalar c multiplies every length- N MPV coefficient by c^N .*

Proof. [✓ Proof](#) Expand the block-diagonal MPV coefficient as a finite sum over blocks and factor c^N out of the sum. $\square$

Appendix F

Proof of the Fundamental Theorem: Supporting Results

The global proof of the Fundamental Theorem in Chapter 9 combines several algebraic components: phase normalization, coefficient extraction from matched gauges, direct-sum conjugation, coordinate reindexing, and unitary refinements.

F.1 Matched phases and coefficient extraction

Sector pairings in both directions determine a finite bijection between BNT canonical forms, and normalization forces every intertwining gauge scalar to be a unit phase.

Theorem F.1.1 (Bijective matching under eventual proportionality). [Lean] [✓ Stmt] *Let P and Q be BNT canonical forms whose total MPV families are eventually nonzero and proportional. There is a bijection $\beta : J(Q) \simeq J(P)$ such that every Q_k has an equal-dimensional, gauge-phase-equivalent partner $P_{\beta(k)}$ satisfying*

$$O_{P_{\beta(k)} Q_k}(N) \not\rightarrow 0.$$

Proof. [✓ Proof] Apply the full-basis matching theorem first from Q to P and then to the reverse proportionality from P to Q . If P_j and $P_{j'}$ had the same partner Q_k , then symmetry and transitivity would give

$$P_j \sim_{\text{gp}} Q_k \sim_{\text{gp}} P_{j'} \implies P_j \sim_{\text{gp}} P_{j'},$$

contradicting BNT basis separation. Both partner maps are therefore injective. The two finite sector sets have equal cardinality, so the forward map is bijective and retains the displayed overlap limit together with the dimension and gauge data. $\square$

Lemma F.1.2 (Unit phase from matched normalized BNT blocks). [Lean] [✓ Stmt] *Let P and Q be BNT canonical forms. If, for every positive length N and word σ , a sector P_j and a sector Q_k have MPV families related by*

$$V^{(N)}(Q_k)_\sigma = \zeta^N V^{(N)}(P_j)_\sigma, \tag{F.1}$$

then $|\zeta| = 1$.

Proof. ✓ Proof The BNT normalization gives self-overlap limits of modulus 1 for both sectors. The MPV identity (F.1) gives, for every positive N ,

$$O_{Q_k Q_k}(N) = |\zeta|^{2N} O_{P_j P_j}(N).$$

Since $O_{Q_k Q_k}(N) \rightarrow 1$ and $O_{P_j P_j}(N) \rightarrow 1$, taking limits forces $|\zeta|^{2N} \rightarrow 1$ and hence $|\zeta| = 1$. $\square$

F.2 Coefficient identities from gauge-phase equalities

Gauge-phase equivalences select unit phases for each sector, allowing eventual coefficient identification from the linear independence of the BNT basis.

Lemma F.2.1 (Full-basis coefficient identity from matched sectors). Lean ✓ Stmt *Let P and Q be BNT canonical forms with the same MPV family at every positive length, matched by a bijection $\beta : J(Q) \simeq J(P)$ with bond-dimension equalities and a gauge-phase equivalence between Q_k and $P_{\beta(k)}$ for every k . Then each sector carries a unit-modulus phase ζ_k , and eventually*

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q). \quad (\text{F.2})$$

This is the equal-MPV coefficient comparison of [11, Appendix MPV proof, lines 1187–1188].

Proof. ✓ Proof For each matched sector, the gauge-phase equivalence gives $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$. Lemma F.1.2 gives $|\zeta_k| = 1$. Applying Lemma 9.2.2 to the same phases gives $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N . $\square$

F.3 The scalar-threaded proportional identity

For proportional total MPVs, coefficient extraction retains the single scalar that relates the two total vectors at each length. This identity is exact, but it does not imply the scalar-free identity needed to recover copy weights; the counterexample in Remark 9.3.6 shows that no such implication is possible.

Lemma F.3.1 (Proportional matched-phase coefficient identity, scalar-threaded). Lean ✓ Stmt *Let P be a BNT canonical form and Q a sector decomposition whose total tensors generate eventually nonzero proportional MPV families. Suppose a bijection $\beta : J(Q) \simeq J(P)$ and scalars ζ_k satisfy the matched MPV phase identities (9.2) for every sector k . Then there is a single eventually-nonzero scalar sequence c_N and a threshold N_0 such that, for all $N > N_0$ and every $k \in J(Q)$,*

$$c_N^{(\beta(k))}(P) = c_N \zeta_k^N c_N^{(k)}(Q). \quad (\text{F.3})$$

The scalar c_N is the same for every sector k : it is the single global proportionality scalar of [11, Theorem thm1, lines 1167–1170] at length N .

Proof. ✓ Proof For each N on the proportionality tail, expand both total MPVs in their sector bases and use the global proportionality scalar c_N :

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = c_N \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

The matched phase identities (9.2) give $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$; because every sector is matched, reindexing the Q -sum by β rewrites the right-hand side entirely in the P -basis. The BNT eventual linear independence of the P -basis and Lemma E.1.10 then extract the coefficient identities (F.3) for all sufficiently large N . $\square$

F.4 Direct-sum conjugation

Block-diagonal gauge conjugations assemble into a global similarity transformation on the direct-sum tensor (Section 9.3).

Definition F.4.1 (Invertible matrix associated to a unitary). [Lean](#) [✓ Stmt](#) A unitary matrix U determines an invertible matrix whose inverse is $U^\dagger$.

Lemma F.4.2 (Unitary block-diagonal gauges). [Lean](#) [✓ Stmt](#) If every block gauge U_k is unitary, then the flattened block-diagonal gauge $U = \bigoplus_k U_k$ is unitary.

Proof. [✓ Proof](#) The products are computed blockwise:

$$UU^\dagger = \bigoplus_k U_k U_k^\dagger = \bigoplus_k \mathbb{1} = \mathbb{1}.$$

□

Lemma F.4.3 (Direct-sum conjugation by the flattened gauge). [Lean](#) [Lean](#) [✓ Stmt](#) If, for every block k and physical index i , $B_k^i = X_k A_k^i X_k^{-1}$, then the weighted direct sums satisfy

$$\bigoplus_k \mu_k B_k^i = X \left(\bigoplus_k \mu_k A_k^i \right) X^{-1},$$

where X is the flattened block-diagonal gauge from Definition 9.3.1. Hence the two assembled tensors are gauge equivalent.

Proof. [✓ Proof](#) Multiplication of block-diagonal matrices is computed blockwise. The scalar weight μ_k commutes with the conjugation on the k th block, and the reindexing from dependent block coordinates to the flattened bond index preserves products and inverses. □

Theorem F.4.4 (Global gauge from a copy-weight matching). [Lean](#) [✓ Stmt](#) Suppose that the sectors of P and Q are matched, with bond-dimension equalities, nonzero phases, and block gauges satisfying $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. If the same phases also give a copy-weight matching, then the flattened Q -tensor is obtained from the matched-coordinate P -tensor by the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$. This reformulates Theorem 9.3.2 in terms of copy-weight matchings.

Proof. [✓ Proof](#) The copy-weight matching supplies $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k),\tau_k(q)}$. Together with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$, these are exactly the two hypotheses (9.4) of Theorem 9.3.2. □

F.5 Additional proportional consequences

The following implications are useful when copy weights satisfy extra relations. They are not conclusions of CPSV16 Theorem 2.10. In particular, their additional hypotheses fail for the proportional BNT forms in Remark 9.3.6.

Theorem F.5.1 (Additional proportional global gauge under copy-weight matching). [Lean](#) [✓ Stmt](#) Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then the proportional sector-matching theorem gives β , the bond-dimension equalities, the unit phases ζ_k , and the block gauges X_k . If one additionally assumes a copy-weight matching for these same phases, then the flattened Q -tensor is obtained from the matched-coordinate P -tensor by the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$.

Proof. [✓ Proof](#) The proportional sector-matching theorem supplies β , X_k , and unit phases ζ_k satisfying $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $|\zeta_k| = 1$, each ζ_k is nonzero. A copy-weight matching gives $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$, so Theorem F.4.4 gives the direct-sum gauge. $\square$

Theorem F.5.2 (Additional proportional global gauge under coefficient identities). [Lean](#) [✓ Stmt](#) *Let P and Q be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection $\beta : J(Q) \simeq J(P)$, bond-dimension equalities, unit-modulus phases ζ_k , and block gauges X_k such that $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ for every sector k and physical index i . If one additionally assumes that these same phases satisfy the matched-sector coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ for all sufficiently large N , then they determine a copy-weight matching and hence the flattened Q -tensor is obtained from the matched-coordinate P -tensor by the direct-sum gauge $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$.*

Proof. [✓ Proof](#) The sector-matching theorem gives β , X_k , and $|\zeta_k| = 1$ with $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$. Since $\zeta_k \neq 0$, the coefficient identities $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ produce $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ by Definition 9.2.4. Theorem F.5.1 then applies to this copy-weight matching. $\square$

F.6 Matched flattened coordinates

Let $\beta : J(Q) \simeq J(P)$ match the basis sectors and let τ_k match the copies in sector k . Flatten the pair (k, q) to the copy index of Q . The matched P -data on that index are the weight and tensor

$$\tilde{\mu}_{k,q} = \mu_{\beta(k), \tau_k(q)}, \quad \tilde{P}_{k,q} = P_{\beta(k)}, \quad (\text{F.4})$$

where the second expression is read after the matched bond-dimension identification. The block gauge repeats X_k on every copy q .

Definition F.6.1 (Matched weights, tensors, and repeated gauges). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The matched flattened data are the functions $\tilde{\mu}$, $\tilde{P}$, and $(k, q) \mapsto X_k$ defined in (F.4) on the flattened copy coordinates of Q .

Lemma F.6.2 (Equality of the matched total bond dimensions). [Lean](#) [✓ Stmt](#) *If matched sectors have equal bond dimensions and matched copy sets are bijective, then P and Q have the same total bond dimension.*

Proof. [✓ Proof](#) Reindex first by the sector bijection and then by each copy bijection. Every summand $\dim P_{\beta(k)}$ becomes $\dim Q_k$, so

$$\sum_{j \in J(P)} r_j(P) \dim P_j = \sum_{k \in J(Q)} r_k(Q) \dim Q_k.$$

$\square$

Definition F.6.3 (Sector and bond-coordinate equivalences). [Lean](#) [Lean](#) [✓ Stmt](#) The sector bijection, copy bijections, and bond-dimension equalities induce an equivalence of the dependent triples (j, q, a) and (k, q', a') . After flattening the dependent bond coordinate, this gives an equivalence of the standard total bond-index sets.

Proof. [✓ Proof](#) Send (k, q, a) to $(\beta(k), \tau_k(q), a)$, using the bond-dimension equality to identify the last coordinate. The inverse uses β^{-1} and τ_k^{-1} . Both composites are the identity componentwise. Conjugating by the standard equivalence between dependent triples and a finite interval gives the flattened bond-coordinate equivalence. $\square$

Theorem F.6.4 (Equal-MPV gauge with all matched witnesses). [Lean](#) [✓ Stmt](#) *Under the hypotheses of the equal-MPV theorem, one may choose simultaneously the sector bijection, bond-dimension equalities, copy bijections, phases, block gauges, total-dimension equality, and the flattened global gauge. They satisfy the matched block and weight identities of Theorem 9.3.4, together with the conjugation of Q by the matched-coordinate P -tensor.*

Proof. [✓ Proof](#) Theorem 9.3.4 supplies all witnesses except the displayed equality of total dimensions. Apply Lemma F.6.2 to its sector and copy bijections. The resulting equality identifies the total bond spaces, while the global conjugation equation is unchanged. $\square$

F.7 Permutation gauges and literal coordinates

A permutation matrix relates the matched-coordinate ordering to the literal block ordering of P in Theorem 9.3.5.

Definition F.7.1 (Invertible permutation gauge). [Lean](#) [✓ Stmt](#) For a permutation ρ of a finite bond-index set, let Π_ρ be its permutation matrix. The corresponding element of the general linear group is denoted G_ρ .

Lemma F.7.2 (Matrix and inverse identities for a permutation gauge). [Lean](#) [Lean](#) [✓ Stmt](#) *The matrix of G_ρ and the matrix of its inverse satisfy*

$$G_\rho = \Pi_\rho, \tag{F.5}$$

$$G_\rho^{-1} = \Pi_{\rho^{-1}}. \tag{F.6}$$

Proof. [✓ Proof](#) Identity (F.5) is the definition of the permutation gauge. Since the inverse permutation reverses the coordinate relabelling, $G_\rho^{-1} = G_{\rho^{-1}}$, which yields (F.6). $\square$

Every row and every column of Π_ρ contains exactly one nonzero entry, equal to 1. Hence

$$\Pi_\rho \Pi_{\rho^{-1}} = \mathbb{1}, \quad \Pi_\rho^\dagger = \Pi_{\rho^{-1}}.$$

Together with Lemma F.7.2, these identities show that $G_\rho^\dagger = G_\rho^{-1}$, so the permutation gauge is unitary.

Lemma F.7.3 (Permutation conjugation reindexes both coordinates). [Lean](#) [✓ Stmt](#) *For a permutation ρ and a square matrix M , conjugation by the permutation matrix reindexes both coordinates by ρ :*

$$\Pi_\rho M \Pi_{\rho^{-1}} = (M_{\rho(i), \rho(j)})_{i,j}. \tag{F.7}$$

Equivalently, the right-hand side is the submatrix of M along $\rho \times \rho$.

Proof. [✓ Proof](#) Left multiplication by Π_ρ selects row $\rho(i)$ in row i , and right multiplication by $\Pi_{\rho^{-1}}$ selects column $\rho(j)$ in column j . The two selections give the displayed submatrix. $\square$

The coordinate equivalence of Definition F.6.3 induces the permutation ρ on the total bond space. Applying (F.7) to P_{tot}^i produces the matched-coordinate tensor, and composing G_ρ with the block-diagonal gauge yields the literal gauge transformation of Theorem 9.3.5.

F.8 Unitary witness refinements

Between left-canonical irreducible sectors, any gauge-phase equivalence can be refined to a unitary gauge with the same phase, extending across all copies in a sector.

Lemma F.8.1 (Unitary gauge for left-canonical irreducible tensors). Lean ✓ Stmt *Let A and B be left-canonical irreducible tensors of common positive bond dimension that are gauge-phase equivalent. Then the gauge can be taken unitary: there are a unitary matrix U and a scalar ζ with $|\zeta| = 1$ such that, for every physical index i ,*

$$B^i = \zeta U A^i U^\dagger.$$

Proof. ✓ Proof Unpack the gauge-phase equivalence as $B^i = \zeta X A^i X^{-1}$ with X invertible and $\zeta \neq 0$. Lemma 9.4.1.2 replaces X by a unitary U while retaining this same phase and proves $|\zeta| = 1$. □

Lemma F.8.2 (Unitary gauge in matched sector coordinates). Lean ✓ Stmt *Let P and Q be BNT canonical forms whose total tensors generate the same MPV family at every positive length. There are a sector bijection β , copy bijections τ_k , phases ζ_k , sector unitaries U_k , and a unitary block-diagonal matrix X such that*

$$\begin{aligned} Q_k^i &= \zeta_k U_k P_{\beta(k)}^i U_k^\dagger, \\ \nu_{k,q} &= \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}, \\ |\zeta_k| &= 1. \end{aligned}$$

The matched sectors and copies have equal bond dimensions and multiplicities. If P_{match} denotes the corresponding reordered block tensor, then $Q_{\text{flat}}^i = X P_{\text{match}}^i X^{-1}$.

Proof. ✓ Proof Apply Lemma 9.4.1.2 to each matched sector, retaining the phase in the copy-weight identity. Repeating U_k on every copy gives $X = \bigoplus_{k,q} U_k$, which is unitary by Lemma F.4.2. For every matched copy (k, q) ,

$$\nu_{k,q} Q_k^i = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)} \zeta_k U_k P_{\beta(k)}^i U_k^\dagger = \mu_{\beta(k), \tau_k(q)} U_k P_{\beta(k)}^i U_k^\dagger.$$

These are precisely the blocks of $Q_{\text{flat}}^i = X P_{\text{match}}^i X^{-1}$. □

Appendix G

Symmetries and Virtual-Boundary Nondecay: Supporting Results

The classification of matrix product state symmetries and boundary nondecay in Chapter 10 rests on permutation gauges, factor-system cocycle identities, stationary-boundary twists, and asymptotic transfer-operator limits.

Let $I = \{0, \dots, d-1\}$ be the physical index set, $M_D(\mathbb{C})$ the bond algebra, and G a group. All sums below are finite. We treat permutation symmetries on I , gauge scalar uniqueness for injective tensors, and factor-system cocycle relations.

G.1 Permutation reindexing and identity bookkeeping

The linear-combination twist $\tilde{A}_g^i = \sum_j U(g)_{ij} A^j$ is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is useful for symmetries that act by reordering basis states rather than by a general linear map.

Definition G.1.1 (Action on the physical index). ✓ ML stmt An *action* of a monoid G on the physical index set I is a map $\sigma : G \rightarrow \text{End}(I)$ with $\sigma(1) = \text{id}$ and $\sigma(gh) = \sigma(g) \circ \sigma(h)$ for all $g, h \in G$. When G is a group, each $\sigma(g)$ is a permutation of I , with inverse $\sigma(g^{-1})$.

Definition G.1.2 (Permutation-twisted tensor). Lean ✓ Stmt Given an action σ of G on I , the *permutation-twisted tensor* at the element g is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}. \quad (\text{G.1})$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:

$$\begin{array}{ccc} \begin{array}{c} i \\ | \\ \bullet \\ | \\ A \end{array} & \xrightarrow{\sigma_g} & \begin{array}{c} \sigma(g)(i) \\ | \\ \bullet \\ | \\ A \end{array} . \end{array}$$

Lemma G.1.3 (Identity twist). [Lean](#) [✓ Stmt](#) For an action σ , $(A^{\sigma_1})^i = A^i$ for all i .

Proof. [✓ Proof](#) The claim follows from $\sigma(1)(i) = i$ for all i . $\square$

Composition order. The linear-combination twist (Lemma 10.1.4) composes as $\widetilde{(\widetilde{A}_h)}_g = \widetilde{A}_{gh}$, applying h first then g . The action twist below uses the same left-action convention $\sigma(gh) = \sigma(g) \circ \sigma(h)$, so $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$; the inner map $\sigma(h)$ acts first on the index.

Lemma G.1.4 (Composition of action twists). [Lean](#) [✓ Stmt](#) For an action σ and all $g, h \in G$,

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}. \quad (\text{G.2})$$

Proof. [✓ Proof](#) Expanding both sides gives $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$. Diagrammatically this is just two consecutive relabellings of the same local tensor:

$$\begin{array}{ccc} \begin{array}{c} i \\ | \\ \bullet \\ \hline A \end{array} & \xrightarrow{\sigma_h} & \begin{array}{c} \sigma(h)(i) \\ | \\ \bullet \\ \hline A \end{array} & \xrightarrow{\sigma_g} & \begin{array}{c} \sigma(g)(\sigma(h)(i)) \\ | \\ \bullet \\ \hline A \end{array} . \end{array}$$

$\square$

G.1.1 Linear physical-index rotations

The same bookkeeping applies to a general matrix acting on the physical index.

Definition G.1.1.1 (Physical-index rotation). [Lean](#) [Lean](#) [✓ Stmt](#) Given a matrix $M \in \mathbb{C}^{m \times d}$ and an MPS tensor A with physical dimension d , the *physical-index rotation* is the tensor with physical dimension m given by $(A_M)^i := \sum_j M_{ij} A^j$ for $i = 0, \dots, m-1$. When the matrix is a unitary u , the rotation is written A_u .

Lemma G.1.1.2 (Matrix product vector under physical rotation). [Lean](#) [✓ Stmt](#) For $M \in \mathbb{C}^{m \times d}$ and a length- N physical configuration $s = (s_0, \dots, s_{N-1}) \in \{0, \dots, m-1\}^N$, physical rotation satisfies

$$\text{tr}((A_M)^{s_0} \dots (A_M)^{s_{N-1}}) = \sum_{t \in \{0, \dots, d-1\}^N} \left(\prod_{n=0}^{N-1} M_{s_n, t_n} \right) \text{tr}(A^{t_0} \dots A^{t_{N-1}}). \quad (\text{G.3})$$

Proof. [✓ Proof](#) Substitute $(A_M)^{s_n} = \sum_{t_n} M_{s_n, t_n} A^{t_n}$ at each site, distribute the finite sums through the ordered matrix product, and use linearity of the trace to obtain (G.3). $\square$

Theorem G.1.1.3 (Blocking commutes with physical maps). [Lean](#) [✓ Stmt](#) For $W \in \mathbb{C}^{m \times n}$, a tensor C with physical dimension n , and $L \geq 0$, blocking the rotated tensor over L sites is rotating the blocked tensor by the L -fold Kronecker power of W :

$$(C_W)^{[L]} = (C^{[L]})_{W^{\otimes L}}. \quad (\text{G.4})$$

Proof. [✓ Proof](#) Decode a blocked physical index as a word b of length L . Substituting $(C_W)^{b_k} = \sum_{v_k} W_{b_k v_k} C^{v_k}$ into the ordered product and distributing gives, for each word v , the summand $(\prod_k W_{b_k v_k}) C^{v_0} \dots C^{v_{L-1}}$. Reindexing the sum over words by the blocked index gives the right-hand side of (G.4), whose coefficients are the entries of $W^{\otimes L}$. $\square$

G.2 Gauge ratios and scalar uniqueness

For injective tensors, any two similarity transformations yielding the same target tensor differ by a scalar multiple.

Lemma G.2.1 (Gauge ratio commutes with tensor matrices). [Lean](#) [Stmnt](#) Diagrammatically, the hypothesis is that the two local gauges

$$\begin{array}{c}
 \begin{array}{c} i \\ | \\ \bullet \\ B \end{array} = \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \begin{array}{c} i \\ | \\ \bullet \\ A \end{array} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \\
 \text{and} \\
 \begin{array}{c} i \\ | \\ \bullet \\ B \end{array} = \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \begin{array}{c} i \\ | \\ \bullet \\ A \end{array} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array}
 \end{array}$$

produce the same target tensor B^i from the same source tensor A^i . If $X, Y \in \text{GL}_D(\mathbb{C})$ both conjugate an MPS tensor A to the same tensor B (i.e. $B^i = X A^i X^{-1} = Y A^i Y^{-1}$ for all i), then $Y^{-1}X$ commutes with every A^i .

Proof. [Proof](#) Direct matrix calculation: conjugating both gauge relations and composing gives $(Y^{-1}X)A^i = A^i(Y^{-1}X)$. $\square$

Lemma G.2.2 (Gauge ratio is scalar for injective tensors). [Lean](#) [Stmnt](#) Under the same hypotheses, if A is additionally injective, then $Y^{-1}X = \lambda \mathbb{1}_D$ for some scalar $\lambda \in \mathbb{C}$.

Proof. [Proof](#) By Lemma G.2.1, $Y^{-1}X$ commutes with all A^i . Since $\{A^i\}$ spans $M_D(\mathbb{C})$, the matrix $Y^{-1}X$ commutes with all of $M_D(\mathbb{C})$. By Lemma 2.1.5, it is a scalar matrix. $\square$

Theorem G.2.3 (Projective multiplication law for gauge matrices). [Lean](#) [Stmnt](#) Let A be an injective MPS tensor with on-site symmetry under a group representation $U : G \rightarrow \text{GL}_d(\mathbb{C})$. For each $g \in G$, let $X(g) \in \text{GL}_D(\mathbb{C})$ satisfy $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$ for all i . Then for all $g, h \in G$, there exists $c(g, h) \in \mathbb{C}$ such that $X(h) X(g) = c(g, h) X(g \cdot h)$. Invertibility of both sides forces $c(g, h) \in \mathbb{C}^\times$. Diagrammatically, both candidate gauges realize the twisted tensor $\tilde{A}_{gh}^i$:

$$\begin{array}{c}
 \begin{array}{c} i \\ | \\ \bullet \\ A \end{array} = \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \begin{array}{c} i \\ | \\ \bullet \\ A \end{array} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \\
 \text{and} \\
 \begin{array}{c} i \\ | \\ \bullet \\ A \end{array} = \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \begin{array}{c} i \\ | \\ \bullet \\ A \end{array} \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array}
 \end{array}$$

and both sides conjugate A^i to the same twisted tensor $\tilde{A}_{gh}^i$.

Proof. [Proof](#) By the composition law $\tilde{A}_{gh} = \widetilde{(\tilde{A}_h)_g}$ (Lemma 10.1.4), both $X(h) X(g)$ and $X(gh)$ conjugate A to $\tilde{A}_{gh}$. Lemma G.2.2 then gives the scalar factor. $\square$

G.3 Cocycle equivalence and coboundaries

Let $\omega_1, \omega_2 : G \times G \rightarrow \mathbb{C}^\times$ be arbitrary scalar 2-cochains. The definitions of coboundary and cohomology are Definitions 10.2.1 and 10.2.2.

Theorem G.3.1 (Cohomology relation is an equivalence relation). [Lean](#) [✓ Stmt](#) *The relation “cohomologous to” is reflexive, symmetric, and transitive on the set of $\mathbb{C}^\times$ -valued scalar 2-cochains, and therefore also on the subset of cocycles.*

Proof. [✓ Proof](#) Reflexivity: take $\varphi = 1$. Symmetry: replace φ by φ^{-1} . Transitivity: compose the witnesses φ and ψ pointwise in (10.11). $\square$

Lemma G.3.2 (Coboundary iff cohomologous to the trivial cocycle). [Lean](#) [✓ Stmt](#) *A scalar 2-cochain ω is a coboundary if and only if $\omega \sim 1$, where $1(g, h) = 1$ for all g, h .*

Proof. [✓ Proof](#) Both directions follow by absorbing or introducing the factor $1(g, h) = 1$ via $x \cdot 1 = x$ in (10.11). $\square$

G.4 Stationary-boundary twist support

Fix an MPS tensor $A = (A^i)_{i=0}^{d-1}$, a physical matrix u , and a stationary boundary matrix $\Lambda \in M_D(\mathbb{C})$. The twisted transfer map $\mathcal{E}_u$ (Definition 10.3.1) acts on virtual-boundary functionals, complementing the physical endpoint correlators of Definition 10.3.4.

Definition G.4.1 (Twisted companion tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor A and a physical-index matrix u , define the companion tensor B_u by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}} A^{n'}. \quad (\text{G.5})$$

Lemma G.4.2 (Twisted transfer as a mixed transfer operator). [Lean](#) [✓ Stmt](#) *The twisted transfer map of A is the mixed transfer operator of the pair (A, B_u) :*

$$\mathcal{E}_u = F_{A, B_u}. \quad (\text{G.6})$$

Proof. [✓ Proof](#) Expand both definitions and rearrange the double sum. $\square$

Lemma G.4.3 (The companion tensor has the same transfer map). [Lean](#) [✓ Stmt](#) *If $uu^\dagger = 1_d$, then for every $X \in M_D(\mathbb{C})$ the companion family B_u satisfies*

$$\mathcal{E}_{B_u}(X) = \mathcal{E}_A(X).$$

Proof. [✓ Proof](#) The corresponding theorem in the companion quantum-channel volume [182, “Unitary freedom in Kraus operators”] applies to the unitary combination defining B_u . Equivalently, expand $B_u^n = \sum_{n'} \overline{u_{n'n}} A^{n'}$. The coefficient of $A^j X (A^k)^\dagger$ in $\mathcal{E}_{B_u}(X)$ is

$$\sum_n \overline{u_{jn}} u_{kn} = (uu^\dagger)_{kj} = \delta_{kj}.$$

All off-diagonal terms vanish and the diagonal terms sum to $\mathcal{E}_A(X)$. $\square$

Definition G.4.4 (Virtual-boundary twist functional). [Lean](#) [✓ Stmt](#) For a boundary state Λ and boundary matrices $X, Y \in M_D(\mathbb{C})$, the *virtual-boundary twist functional* is

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)). \quad (\text{G.7})$$

The special case $X = Y = 1$ is the stationary-boundary block-twist functional of Definition 10.3.2. The matrices X, Y act on the virtual level. In the physical string correlator of [194]

the string of twists is instead flanked by local operators x, y acting on physical sites; expanding such operators in the transfer picture produces particular virtual matrices, and [194] argues that for injective tensors, physical operators on sufficiently many sites produce every virtual matrix. The results below are stated directly in this virtual-boundary form.

Lemma G.4.5 (Virtual-boundary nondecay prevents convergence to zero). Lean ✓ Stmt *If virtual-boundary nondecay holds for A, u, Λ , then for some boundary matrices X, Y the sequence $R_L(u; X, Y)$ does not tend to 0 as $L \rightarrow \infty$.*

Proof. ✓ Proof A uniform lower bound $c \leq \|R_L(u; X, Y)\|$ is incompatible with convergence to 0. □

G.5 Kraus mixing, transfer expansions, and a common TP gauge

Because A and its companion B_u share the same transfer map, they admit a common trace-preserving gauge.

Lemma G.5.1 (Unitary Kraus mixing). Lean ✓ Stmt *If u is unitary, then replacing Kraus operators $\{A_i\}$ by their unitary mixtures $\{\sum_j u_{ij} A_j\}$ preserves the channel:*

$$\sum_i \left(\sum_j u_{ij} A_j \right) Y \left(\sum_j u_{ij} A_j \right)^\dagger = \sum_i A_i Y A_i^\dagger. \quad (\text{G.8})$$

Proof. ✓ Proof This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see the corresponding theorem in the companion quantum-channel volume [182, “Unitary freedom in Kraus operators”]. □

Definition G.5.2 (Common trace-preserving gauge for a twisted pair). Lean ✓ Stmt A *common trace-preserving gauge setup* for (A, u) consists of a tensor B equal to the companion B_u and satisfying $\mathcal{E}_B(X) = \mathcal{E}_A(X)$ for every $X \in M_D(\mathbb{C})$. It also includes a positive definite matrix σ fixed by the adjoint transfer map of B , its positive square root S , and gauged tensors A', B' satisfying

$$\begin{aligned} B &= B_u, & \mathcal{E}_B(X) &= \mathcal{E}_A(X), & \mathcal{E}_{B^\dagger}(\sigma) &= \sigma, & \sigma &> 0, \\ S &= \sigma^{1/2}, & S^\dagger &= S, & A'^i &= S A^i S^{-1}, & B'^i &= S B^i S^{-1}. \end{aligned} \quad (\text{G.9})$$

In addition,

$$\begin{aligned} S S^{-1} &= S^{-1} S = \mathbb{1}, & S^\dagger (S^\dagger)^{-1} &= \mathbb{1}, & (S^{-1})^\dagger &= S^{-1}, \\ \sum_i (A'^i)^\dagger A'^i &= \mathbb{1}, & \sum_i (B'^i)^\dagger B'^i &= \mathbb{1}, \end{aligned} \quad (\text{G.10})$$

together with irreducibility of both A' and B' . These identities complete the common trace-preserving gauge data.

Definition G.5.3 (Construction of the common trace-preserving gauge). Lean ✓ Stmt Let $D > 0$. If A is injective, $u u^\dagger = \mathbb{1}_d$, and $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, define the common trace-preserving gauge setup for (A, u) by choosing the unique positive definite stationary state of the adjoint channel and using the data in Definition G.5.2.

Proof. ✓ Proof Injectivity makes $\mathcal{E}_A$ irreducible. The adjoint channel therefore has a positive definite stationary state σ . By Lemma G.4.3, A and B_u have the same transfer map, hence the same adjoint fixed point. Put $S = \sigma^{1/2}$. Positivity makes S invertible, and direct substitution gives

$$\begin{aligned} \sum_i (SA^i S^{-1})^\dagger (SA^i S^{-1}) &= \mathbb{1}, \\ \sum_i (SB_u^i S^{-1})^\dagger (SB_u^i S^{-1}) &= \mathbb{1}. \end{aligned} \tag{G.11}$$

Similarity preserves irreducibility, so the two gauged families have all the asserted properties. $\square$

Theorem G.5.4 (Twisted eigenvalues survive the common gauge). Lean ✓ Stmt *If $V \neq 0$ and $\mathcal{E}_u(V) = \lambda V$, then λ is an eigenvalue of the mixed transfer map $F_{A',B'}$ in any common trace-preserving gauge setup.*

Proof. ✓ Proof By Lemma G.4.2, the original equation is $F_{A,B_u}(V) = \lambda V$. Multiplying on the left and right by S and using $S^\dagger = S$ gives

$$F_{A',B'}(SVS) = \lambda SVS.$$

Since S is invertible, $SVS \neq 0$. $\square$

G.6 Boundary and limit support

Lemma G.6.1 (Spectral radius < 1 forces virtual-boundary decay). Lean ✓ Stmt *If $\rho(\mathcal{E}_u) < 1$, then, as $L \rightarrow \infty$, for all boundary matrices $X, Y, \Lambda \in M_D(\mathbb{C})$ one has $R_L(u; X, Y) \rightarrow 0$.*

Proof. ✓ Proof In finite dimension, $\rho(\mathcal{E}_u) < 1$ implies $\mathcal{E}_u^L \rightarrow 0$ in operator norm. Composing with the fixed trace pairing $Z \mapsto \text{tr}(\Lambda X Z)$ gives the stated scalar convergence. $\square$

G.7 Universality of virtual-boundary nondecay

For injective on-site symmetric tensors with canonical normalizations, virtual-boundary nondecay is invariant under symmetry-preserving equivalences.

Theorem G.7.1 (Virtual-boundary nondecay agrees for two injective MPV-symmetric tensors).

Lean ✓ Stmt *If A and B are injective, both on-site symmetric under a unitary representation U , and if $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$, $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$, Λ_A and Λ_B are positive definite, $\text{tr}(\Lambda_A) = 1$, $\text{tr}(\Lambda_B) = 1$, $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$, and $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$, then for every $g \in G$, virtual-boundary nondecay holds for A with twist $U(g)$ if and only if it holds for B . In particular, since Definition 10.4.1 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase; it applies equally to symmetric tensors in different SPT phases.*

Proof. ✓ Proof Both directions follow from Theorem 10.4.4, which shows that virtual-boundary nondecay holds universally for injective symmetric tensors with canonical normalisations. $\square$

Appendix H

Relative Commutants of Full Matrix Algebras: Supporting Results

The index argument for matrix product unitaries and cellular automata in Chapter 18 rests on a finite-dimensional relative-commutant property.

Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a unital homomorphism of complex algebras. Its *relative commutant* is the subalgebra

$$\Phi(M_p(\mathbb{C}))' \cap M_q(\mathbb{C}) = \{T \in M_q(\mathbb{C}) : T\Phi(A) = \Phi(A)T \ \forall A \in M_p(\mathbb{C})\}, \quad (\text{H.1})$$

which contains the scalar matrices $\mathbb{C}\mathbf{1}_q$. When this relative commutant reduces to the scalars, Φ is an isomorphism; simplicity of $M_p(\mathbb{C})$ ensures injectivity, while representation theory forces $p = q$ and surjectivity.

These results isolate the finite-dimensional matrix-algebra steps in [169] for Chapter 18: the inclusion $\alpha(\mathcal{A}_{2x} \otimes \mathcal{A}_{2x+1}) \subseteq \mathcal{R}_{2x} \otimes \mathcal{R}_{2x+1}$ [169, lines 1276–1287] is not strict because a strict inclusion would produce a non-scalar element of the relative commutant, and the inclusion $\mathcal{R}_{2x+1} \otimes \mathcal{R}_{2x+2} \subseteq \mathcal{A}_{2x+1} \otimes \mathcal{A}_{2x+2}$ [169, lines 1306–1315] is not strict because α is surjective.

Lemma H.1 (Simplicity of the defining module). [Lean] [✓ Stmt] *Let K be a field and let n be a nonempty finite index set. The defining module K^n of $M_n(K)$, with A acting by $v \mapsto Av$, is a simple $M_n(K)$ -module.*

Proof. [✓ Proof] Sending a matrix to the endomorphism $v \mapsto Av$ is an isomorphism of K -algebras $M_n(K) \rightarrow \text{End}_K(K^n)$, and the identity map of K^n is semilinear along it, since it carries the matrix action to the endomorphism action. A nonzero vector space over a division ring is a simple module over its own endomorphism ring, and simplicity transports along that semilinear bijection. Schumacher and Werner call K^n the defining representation of $M_n(K)$ at [205, lines 2112–2114]. □

Lemma H.2 (Uniqueness of the simple module over a simple Artinian ring). [Lean] [✓ Stmt] *Let R be a simple Artinian ring and let M and N be simple R -modules. Then $M \cong N$ as R -modules.*

Proof. [✓ Proof] A simple Artinian ring is semisimple, so a simple module over it is isomorphic to a simple left ideal: $M \cong I$ and $N \cong J$ with $I, J \subseteq R$ simple left ideals. The regular module of

a simple Artinian ring is isotypic, so $I \cong J$, and composing gives $M \cong N$. This is the general form of the property of $M_d(\mathbb{C})$ that all its irreducible representations are unitarily equivalent to the defining one, quoted at [205, lines 2112–2114]. $\square$

Theorem H.3 (Equal matrix sizes from a scalar relative commutant). [Lean] ✓ Stmt *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a homomorphism of complex algebras with $\Phi(1_p) = 1_q$. If $\Phi(M_p(\mathbb{C}))' \cap M_q(\mathbb{C}) = \mathbb{C}1_q$, then $p = q$.*

Proof. ✓ Proof Give $\mathbb{C}^q$ the $M_p(\mathbb{C})$ -action $A \cdot v = \Phi(A)v$; it is compatible with the complex scalars because Φ is complex linear. Let $W \subseteq \mathbb{C}^q$ be an $M_p(\mathbb{C})$ -submodule, choose an $M_p(\mathbb{C})$ -submodule complement W' with $\mathbb{C}^q = W \oplus W'$, and let P be the projection onto W along W' , so that $W = \text{ran } P$. Since P is $M_p(\mathbb{C})$ -linear, $P\Phi(A) = \Phi(A)P$ for every $A \in M_p(\mathbb{C})$, so P lies in (H.1) and hence $P = c1_q$ for some $c \in \mathbb{C}$. If $c = 0$ then $P = 0$ and $W = 0$; if $c \neq 0$ then $v = P(c^{-1}v)$ for every $v \in \mathbb{C}^q$ and $W = \mathbb{C}^q$. So $\mathbb{C}^q$ is a simple $M_p(\mathbb{C})$ -module. The ring $M_p(\mathbb{C})$ is simple Artinian and $\mathbb{C}^p$ is a simple $M_p(\mathbb{C})$ -module by Lemma H.1, so Lemma H.2 gives an $M_p(\mathbb{C})$ -linear isomorphism $\mathbb{C}^q \cong \mathbb{C}^p$. Reading it as an isomorphism of complex vector spaces and comparing dimensions gives $q = p$. This is the single-summand, multiplicity-one case of the representation argument at [205, lines 2101–2116]. $\square$

Theorem H.4 (Scalar relative commutant forces surjectivity). [Lean] ✓ Stmt *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a homomorphism of complex algebras with $\Phi(1_p) = 1_q$. If $\Phi(M_p(\mathbb{C}))' \cap M_q(\mathbb{C}) = \mathbb{C}1_q$, then Φ is onto.*

Proof. ✓ Proof The kernel of Φ is a two-sided ideal of the simple algebra $M_p(\mathbb{C})$, and it is not all of $M_p(\mathbb{C})$ because $\Phi(1_p) = 1_q \neq 0$; hence Φ is injective. By Theorem H.3, $p = q$, so $M_p(\mathbb{C})$ and $M_q(\mathbb{C})$ have the same finite complex dimension, and an injective complex-linear map between complex vector spaces of equal finite dimension is onto. $\square$

Theorem H.5 (Scalar relative commutant forces surjectivity of a star homomorphism). [Lean] ✓ Stmt *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a homomorphism of complex matrix algebras with $\Phi(A^\dagger) = \Phi(A)^\dagger$ and $\Phi(1_p) = 1_q$. If $\Phi(M_p(\mathbb{C}))' \cap M_q(\mathbb{C}) = \mathbb{C}1_q$, then Φ is onto. This is the form matching the unital star homomorphisms of [205, lines 2101–2105].*

Proof. ✓ Proof The involution plays no part in the argument, so Theorem H.4 applies to Φ read as a homomorphism of complex algebras. $\square$

Theorem H.6 (Non-scalar relative commutant of a proper image). [Lean] ✓ Stmt *Let $p, q \geq 1$ and let $\Phi : M_p(\mathbb{C}) \rightarrow M_q(\mathbb{C})$ be a homomorphism of complex algebras with $\Phi(1_p) = 1_q$ that is not onto. Then there is $T \in M_q(\mathbb{C})$ with $T\Phi(A) = \Phi(A)T$ for every $A \in M_p(\mathbb{C})$ and $T \notin \mathbb{C}1_q$.*

Proof. ✓ Proof If every T commuting with the whole image of Φ were scalar, then (H.1) would equal $\mathbb{C}1_q$ and Theorem H.4 would make Φ onto. $\square$

Appendix I

Physics Examples of the Multi-Block Compression Theorem: Supporting Results

The compression theorems of Chapter 27 rely on several explicit coordinate constructions: explicit gauge transformations and integer arithmetic certificates, adapted bases and scaled matrix units for normality certificates, and a sign-weighted pair testing the sharpness of the remainder bound.

I.1 Explicit gauges and exact arithmetic

The compression datum of Theorem 25.5.1 carries its change of bond coordinates as an invertible map onto a graded coordinate space, one summand per slot. For explicit bond coordinates, this map is given by an invertible matrix and a labelling of coordinates by slots.

Definition I.1.1 (Explicit gauge and entrywise integer coercion). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let τ be a bijection between the graded coordinate space of a slot family and the bond index set $\{0, \dots, D_B - 1\}$, and let G and G^{-1} be mutually inverse matrices in $M_{D_B}(\mathbb{C})$. The *explicit gauge* attached to (τ, G) is the invertible map sending a bond vector v to the graded vector $b \mapsto (Gv)(\tau b)$. For an integer matrix X , its *entrywise coercion* is the complex matrix with the same entries; for integer matrix product operator tensors M and N , their integer bond-space product is $(M \cdot N)^{ik} = \sum_j M^{ij} \otimes N^{jk}$.

Theorem I.1.2 (Conjugation, coercion and the compression pairs of a slot). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Conjugation by the explicit gauge attached to (τ, G) sends a bond matrix A to the matrix GAG^{-1} relabelled along τ . Entrywise coercion is multiplicative, and for integer tensors M, N and $c \in \mathbb{C}$,

$$(cM) \cdot (cN) = c^2(M \cdot N).$$

If a compression datum uses the explicit gauge (τ, G) , then the compression maps for slot s are given in coordinates by $W_s = \pi_s G$ and $V_s = G^{-1} \iota_s$, where π_s selects the rows and ι_s embeds the columns belonging to slot s .

Proof. ✓ Proof The matrix of the explicit gauge is G relabelled along τ in its rows, and the matrix of its inverse is G^{-1} relabelled along τ in its columns, so conjugation is the stated matrix conjugation followed by the relabelling. Multiplicativity of the coercion is the fact that a sum of products of integers is the same computed in $\mathbb{Z}$ or in $\mathbb{C}$, and the statement about the bond-space product follows by expanding both sides entrywise. The two compression maps are the composites of the coordinate embedding and projection of the slot with the gauge and its inverse; evaluating the matrix products at a single graded coordinate leaves one row of G and one column of G^{-1} . $\square$

Definition I.1.3 (One-slot compression data). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt For a stacked product of two integer matrix product operator tensors, read over the pair alphabet, the letter at $a = do + i$ is $(M \cdot N)^{oi}$. A *one-slot compression datum* for a tensor B onto a single target C is the data of Theorem 25.5.1 for the one-element slot set: an explicit gauge (τ, G) conjugating B into a matrix that is block upper triangular for a flag with C at one diagonal position. Two shapes recur in the examples below: a conjugation $GB^iG^{-1} = C^i$ for an invertible G gives such a datum with no zero slots ($z = 0$); and, for a four-dimensional B and a one-dimensional C , an integer gauge sending every letter to a matrix with C at the third diagonal position of a four-step flag and zero at the other three gives such a datum with three zero slots ($z = 3$).

Theorem I.1.4 (The remainder of a conjugate datum vanishes; word traces of a one-slot datum). Lean Lean Lean Lean Lean ✓ Stmt *The remainder of a conjugate one-slot datum vanishes identically: the compression is an exact change of coordinates. For every one-slot datum with target tensor C and nonempty word $w \neq \emptyset$,*

$$\text{tr}(B^w) = \text{tr}(C^w),$$

or $\text{tr}(B^w) = c^{|w|} \text{tr}(C^w)$ when the target is carried with scalar weight c .

Proof. ✓ Proof Conjugation preserves the trace, which decomposes as the sum of traces of the diagonal blocks. For the upper block-triangular gauge, the diagonal blocks of a word product are the word products of the individual diagonal blocks; the zero-slot diagonal block is strictly upper triangular (or zero) and therefore has vanishing trace on any nonempty word $w \neq \emptyset$, leaving only the contribution of the target block C . The scalar factor carries across words by the length $|w|$. $\square$

Theorem I.1.5 (Stacked products of block-diagonal tensors). Lean Lean Lean ✓ Stmt *Let M and N be matrix product operator tensors whose letters are, after relabellings of their bond coordinates, block diagonal with blocks M_k and N_l . Then every letter of the stacked product $M \cdot N$ is, after a relabelling of its bond coordinates, block diagonal over the pairs (k, l) , with the block $(M_k \cdot N_l)^{ij}$ at the pair (k, l) . The bond coordinates of the block (k, l) are the pairs of a bond coordinate of M_k and a bond coordinate of N_l .*

Proof. ✓ Proof The entry of $M^{ij} \otimes N^{jk}$ at a pair of pairs of coordinates is the product of an entry of M^{ij} and an entry of N^{jk} . It vanishes unless both row coordinates lie in the same block as the corresponding column coordinates, and it then equals the corresponding entry of $M_k^{ij} \otimes N_l^{jk}$. Summing over j gives the claim. $\square$

Lemma I.1.6 (Direct sums over a subring of the complex numbers). Lean ✓ Stmt *Let $f: R \rightarrow \mathbb{C}$ be a ring homomorphism and let M and N be matrix product operator tensors with entries in R . The direct sum of the entrywise images of M and N under f is the entrywise image of the direct sum of M and N formed over R .*

Proof. [✓ Proof](#) Both sides are block diagonal with the images of M^{ij} and N^{ij} on the diagonal, and $f(0) = 0$ off the diagonal blocks. $\square$

Theorem I.1.7 (Direct sums of one-slot compression data). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let the letters of a tensor B be, after a relabelling of the bond coordinates, block diagonal with blocks B_k , k in a finite set K , and let every B_k carry a one-slot compression datum onto a target C_k with z_k zero slots and remainder R_k . Then the direct sum of the gauges of the blocks is a compression datum of B onto the family $(C_k)_{k \in K}$ in the sense of Theorem 25.5.1, with $z = \sum_k z_k$ zero slots, and its remainder is

$$R^a = \bigoplus_{k \in K} R_k^a, \quad (\text{I.1})$$

relabelled in the same way. Consequently a word w satisfies $R^w = 0$ if and only if $R_k^w = 0$ for every k : the nilpotency order of the remainder of B is the largest nilpotency order of the remainders of the blocks.

Proof. [✓ Proof](#) In the direct sum of the gauges, the letters of B are block diagonal with the letters of the blocks in their own block coordinates. Ordering the blocks of B block by block, each in its own order, makes every letter block upper triangular, with the targets and the zero diagonal blocks of the summands on the diagonal. Removing the diagonal blocks of a block-diagonal matrix removes the diagonal blocks of each summand, so the remainder is the direct sum of the remainders, and so are its words. $\square$

Theorem I.1.8 (Integer certificates for normality and for vanishing sitewise intertwiners). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $A \in (M_D(\mathbb{C}))^d$ have integer entries $A_i \in M_D(\mathbb{Z})$. If there exist index pairs (a_k, b_k) , integer coefficients $T_{xyk} \in \mathbb{Z}$, and a nonzero integer $m \neq 0$ such that for all matrix units E_{xy} ,

$$\sum_k T_{xyk} A_{a_k} A_{b_k} = m E_{xy},$$

then $\text{span}_{\mathbb{C}}\{A_i A_j : i, j \in \{0, \dots, d-1\}\} = M_D(\mathbb{C})$. Similarly, let $B \in (M_D(\mathbb{Z}))^d$ and $c \in (M_1(\mathbb{Z}))^d$. If a selection of rows from the matrices $B_i - c_i \mathbb{1}_D$ forms an integer matrix $K \in M_D(\mathbb{Z})$ admitting an integer left inverse $K'K = m \mathbb{1}_D$ with $m \neq 0$, then every $v \in \mathbb{C}^D$ satisfying $B_i v = c_i v$ for all i satisfies $v = 0$. Likewise, if a selection of columns from $B_i - c_i \mathbb{1}_D$ admits an integer left inverse up to $m \neq 0$, then every $u \in \mathbb{C}^D$ with $u B_i = c_i u$ satisfies $u = 0$.

Proof. [✓ Proof](#) Dividing the linear combination by $m \neq 0$ exhibits each matrix unit E_{xy} in the span of length-two words of A , showing they span $M_D(\mathbb{C})$. For the sitewise intertwiner, $B_i v = c_i v$ implies $Kv = 0$, whence $mv = K'Kv = 0$, so $v = 0$ since $m \neq 0$ in $\mathbb{C}$. The column argument for u is identical. $\square$

The reduction of the square of the CZX tensor printed in the review (Section 27.3.2) replaces the letters by a diagonal block of an invariant subspace, following the construction of the canonical form in [16, Section IV]. When the complementary diagonal block vanishes, the replacement is a compression by rectangular matrices, and the following lemma shows that it preserves every nonempty word.

Lemma I.1.9 (Compression onto an absorbing corner). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let $B^0, \dots, B^{d-1} \in M_D(\mathbb{C})$, $W \in M_{k \times D}(\mathbb{C})$ and $V \in M_{D \times k}(\mathbb{C})$, and write $\widetilde{B}^i = WB^iV$. If $VWB^i = B^i$ for every i , then every nonempty word w satisfies $VWB^w = B^w$,

$$\widetilde{B}^w = WB^wV,$$

and $\text{tr}(\widetilde{B}^w) = \text{tr}(B^w)$. The same conclusions hold, with $B^w V W = B^w$ in place of $V W B^w = B^w$, if $B^i V W = B^i$ for every i .

Proof. [✓ Proof](#) For the left condition, $V W B^w = B^w$ follows from the first letter of w . By induction on the length, for $w = a w'$ with w' nonempty,

$$\widetilde{B}^a \widetilde{B}^{w'} = W B^a (V W B^{w'}) V = W B^a B^{w'} V.$$

Cyclicity of the trace then gives $\text{tr}(W B^w V) = \text{tr}(V W B^w) = \text{tr}(B^w)$. The right condition is symmetric: $B^w V W = B^w$ follows from the last letter, the induction step uses $B^a V W = B^a$, and $\text{tr}(W B^w V) = \text{tr}(B^w V W) = \text{tr}(B^w)$. $\square$

I.2 Adapted bases and scaled matrix units

The stacked bond spaces of vertical components are tensor squares. Reordering the four bond bits separates the two middle bits, which form the inner bond, from the two outer bits; the intermediate physical-index sum acts on the inner bond. The Bell vectors together with the two remaining coordinate vectors form the adapted basis used throughout. Each weighted copy of a target contributes its weight raised to the number of sites N , and weights are summed when they multiply the same target as in Theorem 25.7.2. The results below establish tensor decompositions and periodic trace identities, not a classification of length-dependent renormalization fixed points.

Definition I.2.1 (The idempotents of an adapted basis). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For mutually inverse matrices U and U^{-1} , define $\Pi_j = U^{-1} E_{jj} U$, the product of the j -th column of U^{-1} with the j -th row of U . Reorder a stacked bond quadruple into its two middle bits and its two outer bits. On the inner bond choose the adapted basis with first two vectors $(1, 0, 0, \pm 1)$ and last two vectors the remaining coordinate vectors. The Bell idempotents carry the weighted targets. Their tensor products with target matrices are expressed in the original stacked coordinates by undoing the reordering.

Theorem I.2.2 (The idempotents of an adapted basis are mutually orthogonal). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The idempotents satisfy $\Pi_j^2 = \Pi_j$, $\Pi_j \Pi_k = 0$ for $j \neq k$, and $U \Pi_j U^{-1} = E_{jj}$.

Proof. [✓ Proof](#) Conjugation by U returns E_{jj} , and products reduce to $E_{jj} E_{kk} = \delta_{jk} E_{jj}$. $\square$

Theorem I.2.3 (From an integer matrix identity to a weighted direct sum). [Lean](#) [✓ Stmt](#) Suppose $B = c_B \widehat{B}$ and $C_s = c_C \widehat{C}_s$, where the tensors with hats have integer entries and $s \in \{0, 1\}$. Suppose a nonzero integer n and integers k_s satisfy, for each s and each physical index a ,

$$n \mu_s \frac{c_C}{2} = k_s c_B, \tag{I.2}$$

$$n \widehat{B}^a = \sum_s k_s (\widehat{\Pi}_s \otimes \widehat{C}_s^a), \tag{I.3}$$

where $\widehat{\Pi}_s = 2 \Pi_s$ and the right-hand side of (I.3) is in stacked bond coordinates. Then $B^a = \sum_s \mu_s (\Pi_s \otimes C_s^a)$ in those coordinates, the hypothesis of Proposition 25.5.5. Here μ_s denotes a scalar fusion weight, not a vertical multiplicity matrix.

Proof. [✓ Proof](#) Using the idempotents of Theorem I.2.2, the s -th summand is $\mu_s c_C (\widehat{\Pi}_s \otimes \widehat{C}_s^a)/2$. Multiplication by n and substitution of (I.2) and (I.3) give $n B^a$. Cancel the nonzero scalar n . $\square$

Theorem I.2.4 (Normality from scaled matrix units). [Lean](#) [✓ Stmt](#) *Let a tensor be one nonzero scalar times an integer tensor. Suppose that for every bond position (x, y) a letter of the integer tensor is a nonzero integer multiple of E_{xy} . Then its one-site matrices span the full bond matrix algebra, so it is normal at blocking length one in the sense of Definition 2.5.5. This establishes injectivity, not transfer spectral-radius normalization; a normal tensor (NT) in the RFP canonical form additionally requires that normalization.*

Proof. [✓ Proof](#) Dividing each designated letter by its nonzero factor exhibits every matrix unit in the one-site span. $\square$

I.3 A sign-weighted pair with a non-split extension

The sign-weighted pair $(A, -A)$ of bond dimension two and a bond-dimension-five source B carrying both targets together with a one-dimensional zero slot illustrate the sharpness of Theorem 25.5.1: the remainder bound is attained (Theorem I.3.2.1), the periodic trace identity $\text{tr}(B^w) = (1 + (-1)^{|w|}) \text{tr}(A^w)$ (I.22) acts as an even-length filter (Theorem I.3.2.2), and a word-level compression need not arise from sitewise intertwiners $(B^i V = V(-A^i) \implies V = 0$, Theorem I.3.2.3).

Consider a two-state physical site with basis $\{|0\rangle, |1\rangle\}$. Define the bond-two MPS tensor by

$$A^0 = P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A^1 = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (\text{I.4})$$

For $N \geq 1$, write its periodic amplitudes as

$$a_N(x) = \text{tr}(A^{x_0} \cdots A^{x_{N-1}}) \quad (x \in \{0, 1\}^N). \quad (\text{I.5})$$

The associated unnormalized state is

$$|V^{(N)}(A)\rangle = \sum_{x \in \{0, 1\}^N} a_N(x) |x\rangle, \quad (\text{I.6})$$

where $|x\rangle = |x_0\rangle \otimes \cdots \otimes |x_{N-1}\rangle$. As in the elementary MPS examples of Refs. [194, Section II] and [16, Appendix A], the state can be understood directly from its tensor letters, with an on-site $\mathbb{Z}_2$ symmetry reflected on the virtual bond space: $P = |0\rangle\langle 0|$ projects onto virtual coordinate 0, while $X = |0\rangle\langle 1| + |1\rangle\langle 0|$ exchanges the two virtual coordinates.

Multiplying both letters of A by -1 gives one minus sign per site, whatever the physical labels. For a word $w = (i_1, \dots, i_{|w|})$, write $A^w = A^{i_1} \cdots A^{i_{|w|}}$, with the identity matrix as the empty product. Then

$$(-A)^w = (-1)^{|w|} A^w. \quad (\text{I.7})$$

The sum of periodic amplitudes $\text{tr}(A^w) + \text{tr}((-A)^w) = (1 + (-1)^{|w|}) \text{tr}(A^w)$ therefore vanishes on all odd lengths $|w|$. The following five-dimensional tensor realizes that sum without being a direct sum in its displayed coordinates.

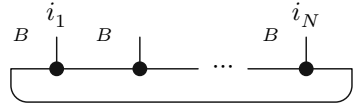
Definition I.3.1 (The graded pair and the five-dimensional source). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let A be the tensor in (I.4). The two target tensors are $C_+ = A$ and $C_- = -A$. The source is the bond-dimension-five tensor with letters

$$B^0 = \begin{pmatrix} -1 & 1 & -1 & 1 & 3 \\ 2 & 2 & -2 & -1 & 0 \\ 1 & 2 & -2 & 0 & 2 \\ 1 & 3 & -3 & -1 & 2 \\ -1 & 0 & 0 & 1 & 2 \end{pmatrix} \quad (\text{I.8})$$

and

$$B^1 = \begin{pmatrix} 3 & 1 & 0 & -2 & -3 \\ -3 & 1 & -1 & 2 & 5 \\ -1 & 2 & -1 & 1 & 3 \\ 0 & 1 & -1 & -1 & 1 \\ 2 & 1 & 0 & -1 & -2 \end{pmatrix}. \quad (\text{I.9})$$

Neither source letter has a zero row or a zero column. Nevertheless, a simultaneous change of bond coordinates reveals diagonal blocks A , $-A$, and a one-dimensional zero block. The latter contributes nothing to positive-length traces.



$$\text{tr}(B^w), \text{ computed in Theorem I.3.2.2}$$

Theorem I.3.2 (Normality of the graded block). [Lean](#) [✓ Stmt](#) *The block A is normal: its length-two words span $M_2(\mathbb{C})$.*

Proof. [✓ Proof](#) Let E_{ab} denote the matrix unit with its sole nonzero entry equal to one in row a , column b . Direct multiplication gives $E_{00} = A^0 A^0$, $E_{01} = A^0 A^1$, and $E_{10} = A^1 A^0$. The remaining unit is $E_{11} = A^1 A^1 - A^0 A^0$, so all four matrix units lie in the length-two word span. $\square$

I.3.1 Occupation parity and periodic amplitudes

Call $n_1(x) = \sum_{v=0}^{N-1} x_v$ the occupation of the configuration x . Let $z = \text{diag}(1, -1)$ act on the physical site, and let $Z = \text{diag}(1, -1)$ act on the bond space. Although their matrices are equal, these operators act on different spaces. The physical parity operator has basis action

$$z^{\otimes N} |x\rangle = (-1)^{n_1(x)} |x\rangle. \quad (\text{I.10})$$

Theorem I.3.1.1 (Occupation parity). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For each physical label $i \in \{0, 1\}$, the tensor satisfies*

$$Z A^i Z = (-1)^i A^i. \quad (\text{I.11})$$

Consequently, $a_N(x) = 0$ whenever $n_1(x)$ is odd, and

$$z^{\otimes N} |V^{(N)}(A)\rangle = |V^{(N)}(A)\rangle. \quad (\text{I.12})$$

Proof. [✓ Proof](#) Direct multiplication gives $Z P Z = P$ and $Z X Z = -X$. Since $Z^2 = \mathbb{1}_2$, multiplying (I.11) along the word yields

$$Z(A^{x_0} \dots A^{x_{N-1}})Z = (-1)^{n_1(x)} A^{x_0} \dots A^{x_{N-1}}.$$

Taking the trace and using cyclicity gives $a_N(x) = (-1)^{n_1(x)} a_N(x)$. Thus odd occupation forces $a_N(x) = 0$. Applying (I.10) to the expansion (I.6) proves (I.12). $\square$

The support of the state is further constrained by the relative positions of the zeros.

Theorem I.3.1.2 (Amplitudes from cyclic runs). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Suppose x contains $m \geq 1$ zeros. Read the sites cyclically, and let $r_1, \dots, r_m$ be the numbers of ones between successive zeros. Define $b(r) = 1$ for even r and $b(r) = 0$ for odd r . Then

$$a_N(x) = \prod_{j=1}^m b(r_j). \quad (\text{I.13})$$

For the word with no zeros,

$$a_N(1, \dots, 1) = 1 + (-1)^N. \quad (\text{I.14})$$

Proof. [✓ Proof](#) The identities $X^2 = \mathbb{1}_2$ and $PXP = 0$ imply $PX^rP = b(r)P$ and $\text{tr}(PX^r) = b(r)$. Rotate the word to start at a zero, using cyclicity of the trace. Successive contractions give

$$\begin{aligned} a_N(x) &= \text{tr}(PX^{r_1}PX^{r_2} \dots PX^{r_m}) \\ &= \left(\prod_{j=1}^{m-1} b(r_j) \right) \text{tr}(PX^{r_m}) = \prod_{j=1}^m b(r_j). \end{aligned}$$

For the all-one word, $X^N = \mathbb{1}_2$ when N is even and $X^N = X$ when N is odd. Their traces are 2 and 0, respectively, which proves (I.14). $\square$

Thus a word containing a zero has amplitude one exactly when every cyclic run of ones has even length. For example, 0101 has even occupation but zero amplitude, since both runs have length one. The all-zero word has amplitude one at every positive length, so $|V^{(N)}(A)\rangle$ is always nonzero in the specified computational basis.

I.3.2 Asymmetric compression and length cancellation

Theorem I.3.2.1 (Compression of the graded pair). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The source of Definition I.3.1 carries the block triangular form of Theorem 25.5.1 for $S = \{+, -\}$, with targets $C_+ = A$, $C_- = -A$, one zero slot $z = 1$, and dimension count $5 = 2 + 2 + 1$. There are compression maps $V_s \in \mathbb{C}^{5 \times 2}$ and $W_s \in \mathbb{C}^{2 \times 5}$ such that, for every $s, t \in S$,

$$W_s V_t = \delta_{st} \mathbb{1}_2. \quad (\text{I.15})$$

For every word w and each $s \in S$,

$$W_s B^w V_s = C_s^w. \quad (\text{I.16})$$

Define the remainder letters by

$$R^i = B^i - V_+ A^i W_+ - V_- (-A^i) W_-. \quad (\text{I.17})$$

Every product R^w with $|w| \geq 3$ vanishes, whereas $R^0 R^0 \neq 0$. Thus the bound $|S| + z = 3$ in clause (vi) of Theorem 25.5.1 is attained.

Proof. [✓ Proof](#) Use the integer change of coordinates

$$G = \begin{pmatrix} 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & -1 & -1 \\ -1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 & 1 \end{pmatrix}. \quad (\text{I.18})$$

Its inverse is

$$G^{-1} = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{I.19})$$

Direct multiplication verifies both inverse identities and gives

$$GB^0G^{-1} = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{I.20})$$

For the second letter,

$$GB^1G^{-1} = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{I.21})$$

With block sizes $(2, 2, 1)$, the matrices in (I.20) and (I.21) are block upper triangular, with diagonal blocks A^i , $-A^i$, and 0. As in Theorem I.1.2, take W_+ and W_- to be rows $(0, 1)$ and $(2, 3)$ of G , respectively, and V_+ and V_- to be the corresponding columns of G^{-1} . The inverse identities give (I.15). Diagonal blocks multiply under word evaluation, giving (I.16).

Subtracting the two diagonal target blocks leaves a strictly upper triangular three-block remainder in these coordinates. The compression remainder bound therefore gives $R^w = 0$ for $|w| \geq 3$. To check sharpness in the original coordinates, substitution into (I.17) gives

$$R^0 = \begin{pmatrix} -1 & 2 & -2 & 1 & 3 \\ 1 & 2 & -2 & 0 & 1 \\ 0 & 3 & -3 & 1 & 3 \\ 0 & 3 & -3 & 0 & 3 \\ -1 & 1 & -1 & 1 & 2 \end{pmatrix}.$$

Its square has $(0, 1)$ entry 2, so $R^0R^0 \neq 0$. □

Theorem I.3.2.2 (Periodic trace of the sign-weighted pair). Lean ✓ Stmt For every nonempty word w ,

$$\text{tr}(B^w) = (1 + (-1)^{|w|}) \text{tr}(A^w). \quad (\text{I.22})$$

Proof. ✓ Proof Proposition 25.5.3 applied to the datum of Theorem I.3.2.1 gives $\text{tr}(B^w) = \text{tr}(A^w) + \text{tr}((-A)^w)$. By (I.7), the second summand is $(-1)^{|w|} \text{tr}(A^w)$, proving (I.22). □

Define $|V^{(N)}(B)\rangle$ by the same periodic contraction as in (I.6), with A replaced by B . Applying (I.22) to each basis coefficient gives, for $N \geq 1$,

$$|V^{(N)}(B)\rangle = (1 + (-1)^N) |V^{(N)}(A)\rangle. \quad (\text{I.23})$$

The source acts as a length filter: at odd length the periodic vector vanishes, while at even length every amplitude of A is doubled. For instance, at odd length the all-zero amplitude of A is one but its amplitude in B vanishes; at even length the run restrictions of Theorem I.3.1.2 apply with doubled amplitudes.

The positive-length hypothesis is necessary. The empty-word traces are $\text{tr}(\mathbb{1}_5) = 5$ and $\text{tr}(\mathbb{1}_2) = 2$, whereas substituting $|w| = 0$ into (I.22) would give $5 = 4$. The missing contribution is the empty trace of the one-dimensional zero block, which is one; its traces vanish only at positive length.

Theorem I.3.2.3 (The weight -1 block has no sitewise intertwiner). [Lean](#) [Lean](#) [✓ Stmt](#) *If $V \in \mathbb{C}^{5 \times 2}$ satisfies $B^i V = V(-A^i)$ for both letters, then $V = 0$; if $W \in \mathbb{C}^{2 \times 5}$ satisfies $W B^i = (-A^i) W$ for both letters, then $W = 0$.*

Proof. [✓ Proof](#) Each condition is a homogeneous linear system in ten unknown entries. For the right intertwiner, select entries $(3, 0)$, $(4, 0)$, $(2, 1)$, $(3, 1)$, and $(4, 1)$ of $B^0 V = V(-A^0)$, and all five entries of column 0 of $B^1 V = V(-A^1)$. Rational linear combinations of these ten equations force all entries of column 0 of V to vanish. The selected equations for letter 1 then force column 1 to vanish.

For the left intertwiner, select entries $(0, 4)$, $(1, 0)$, $(1, 2)$, $(1, 3)$, and $(1, 4)$ of $W B^0 = (-A^0) W$, and all five entries of row 0 of $W B^1 = (-A^1) W$. Rational linear combinations force row 0 of W to vanish, after which the letter-1 equations force row 1 to vanish. $\square$

The nonzero maps W_- and V_- in (I.16) therefore give a word-level compression onto $-A$ without either map being a sitewise intertwiner. The trace identity and the sharp remainder bound hold despite this obstruction.

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